REVERSE ENGINEERING REPORT

Reverse Engineering Report: RNC.dll

Portable Executable (PE) | 1,327,104 bytes (1.2 MiB) | SHA-256
23e91fa2482cfa4cac9b4c8ec599c0080a828e12c9d0b21df94a03e775b22260

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12.1328 sub_100f1c20 at 0x100f1c20 (67 instructions) 1752
12.1329 sub_100f1d50 at 0x100f1d50 (31 instructions) 1753
12.1330 sub_100f3040 at 0x100f3040 (68 instructions) 1753
12.1331 sub_100f3240 at 0x100f3240 (11 instructions) 1754
12.1332 sub_100f6390 at 0x100f6390 (31 instructions) 1755
12.1333 j_strlen at 0x100fc10e (1 instructions) 1755

12.1334 j_strcpy at 0x100fc114 (1 instructions) 1755
12.1335 j_pow at 0x100fc11a (1 instructions) 1755
12.1336 sub_100fc230 at 0x100fc230 (15 instructions) 1755
12.1337 sub_100fc385 at 0x100fc385 (8 instructions) 1756
12.1338 sub_100fc3a8 at 0x100fc3a8 (8 instructions) 1756
12.1339 sub_100fc3f1 at 0x100fc3f1 (12 instructions) 1756
12.1340 sub_100fc4db at 0x100fc4db (12 instructions) 1756
12.1341 sub_100fc510 at 0x100fc510 (8 instructions) 1757
12.1342 sub_100fc530 at 0x100fc530 (8 instructions) 1757
12.1343 sub_100fc550 at 0x100fc550 (8 instructions) 1757
12.1344 sub_100fc570 at 0x100fc570 (11 instructions) 1757
12.1345 sub_100fc5a0 at 0x100fc5a0 (8 instructions) 1758
12.1346 sub_100fc5c0 at 0x100fc5c0 (11 instructions) 1758
12.1347 sub_100fc5f0 at 0x100fc5f0 (8 instructions) 1758
12.1348 sub_100fc610 at 0x100fc610 (8 instructions) 1758
12.1349 sub_100fc630 at 0x100fc630 (8 instructions) 1759
12.1350 sub_100fc658 at 0x100fc658 (8 instructions) 1759
12.1351 sub_100fc6d1 at 0x100fc6d1 (12 instructions) 1759
12.1352 sub_100fc70b at 0x100fc70b (8 instructions) 1759
12.1353 sub_100fc768 at 0x100fc768 (11 instructions) 1760
12.1354 sub_100fc7a6 at 0x100fc7a6 (12 instructions) 1760
12.1355 sub_100fc7e0 at 0x100fc7e0 (8 instructions) 1760
12.1356 sub_100fc839 at 0x100fc839 (8 instructions) 1760
12.1357 sub_100fc868 at 0x100fc868 (8 instructions) 1761
12.1358 sub_100fc899 at 0x100fc899 (11 instructions) 1761
12.1359 sub_100fc94b at 0x100fc94b (12 instructions) 1761
12.1360 sub_100fc998 at 0x100fc998 (8 instructions) 1761
12.1361 sub_100fc9c5 at 0x100fc9c5 (8 instructions) 1762
12.1362 sub_100fc9e9 at 0x100fc9e9 (8 instructions) 1762
12.1363 sub_100fca18 at 0x100fca18 (8 instructions) 1762
12.1364 sub_100fca58 at 0x100fca58 (8 instructions) 1762
12.1365 sub_100fca89 at 0x100fca89 (8 instructions) 1763
12.1366 sub_100fcb18 at 0x100fcb18 (8 instructions) 1763
12.1367 sub_100fcb4b at 0x100fcb4b (8 instructions) 1763
12.1368 sub_100fcb75 at 0x100fcb75 (8 instructions) 1763
12.1369 sub_100fcb98 at 0x100fcb98 (8 instructions) 1763
12.1370 sub_100fcbd9 at 0x100fcbd9 (8 instructions) 1764
12.1371 sub_100fcc17 at 0x100fcc17 (12 instructions) 1764
12.1372 sub_100fcca8 at 0x100fcca8 (8 instructions) 1764
12.1373 sub_100fccd8 at 0x100fccd8 (8 instructions) 1764
12.1374 sub_100fcd60 at 0x100fcd60 (8 instructions) 1765
12.1375 sub_100fcd88 at 0x100fcd88 (8 instructions) 1765
12.1376 sub_100fcdb9 at 0x100fcdb9 (8 instructions) 1765
12.1377 sub_100fcdf1 at 0x100fcdf1 (8 instructions) 1765
12.1378 sub_100fce18 at 0x100fce18 (8 instructions) 1766
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12.1381 sub_100fcf69 at 0x100fcf69 (8 instructions) 1766
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12.1389 sub_100fd198 at 0x100fd198 (8 instructions) 1768
12.1390 sub_100fd205 at 0x100fd205 (12 instructions) 1768
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12.1392 sub_100fd38b at 0x100fd38b (8 instructions) 1769
12.1393 sub_100fd3c0 at 0x100fd3c0 (8 instructions) 1769
12.1394 sub_100fd418 at 0x100fd418 (8 instructions) 1769
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1 Executive Summary

Observed: RNC.dll is a 32-bit Windows dynamic-link library for X86, a Portable Executable (PE) file of 1,327,104 bytes (1.2 MiB).

- Derived: imports support File system, Anti-analysis, Dynamic API resolution, Input capture, System reconnaissance.
- Inferred: the file is not packed (packing score 0).
- Observed: missing mitigations: ASLR, DEP / NX, Control Flow Guard, Forced integrity checks, Authenticode signature.
- Observed: 7 indicators extracted, including 1 URL.
- Derived: recovered 1,396 functions (105,458 instructions).
- Inferred: start with sub_100eb620 at 0x100eb620: References text "CASIO".

Evidence labels: Observed facts are read directly from the file; Derived facts are computed deterministically from observed data; Inferred statements are interpretations that an analyst should confirm.

2 What This Program Does

RNC.dll is a 32-bit X86 dynamic-link library.

2.1 Execution Lifecycle & Boot Sequence

Under Windows X86, the operating system loader maps executable image sections at 0x10000000 and transfers control to the process entry point at 0x10022894.

The C runtime startup routines initialize the process environment, register Structured Exception Handling (SEH) or TLS callbacks, initialize standard I/O, and transfer execution to the application entry.

2.2 Variable Storage & Compiler Optimization

Variable Storage Analysis: Read-only constants and string literals reside in the `.rdata` section, initialized global variables are stored in the `.data` section, and uninitialized globals reside in `.bss` allocated in virtual memory.

Local function variables and call arguments are allocated dynamically within processor registers and thread stack frames.

2.3 Binary Patching & Reverse Engineering Vectors

Binary Patching: Analysis revealed no direct immediate constants in the main execution loop.

3 Identity and Fingerprints

Property	Value
File name	RNC.dll
Size	1,327,104 bytes (1.2 MiB)
Format	Portable Executable (PE)
Entropy (bits/byte)	6.54
MD5	0aa6e39d7868cc74618acdc28b38d4bb
SHA-1	55f991d8ad9eb53e601e3db0e2e4d3673c851c8e
SHA-256	23e91fa2482cfa4cac9b4c8ec599c0080a828e12c9d0b21df94a03e775b22260
Import hash (imphash)	03d88063ce9dbfb22859a424925dad38
Image kind	Dynamic-link library
Architecture	X86
Address width	32 bits
Entry point	0x10022894
Image base	0x10000000

3.1 Build and Metadata Facts

Fact	Value
Link timestamp	2023-01-19 08:52:02 UTC (raw 0x63c904b2)
Subsystem	Windows GUI (2)
Linker version	8.0
Debug database (PDB)	d:\RLD\pro\RNF_X\release\RNC.pdb (GUID c391d7b56202554ba6f3797fa10b9244, age 2)
Rich header	Present (Microsoft toolchain fingerprint, key 0xb10f064f)
Internal module name	RNC.dll

4 Sections and Memory Layout

Observed: 5 sections (1 executable, 1 writable, 0 both).

Name	Address	Virtual size	Raw size	Entropy	Access
.text	0x10001000	0xfc541	0xfd000	6.66	R-X
.rdata	0x100fe000	0x37601	0x38000	4.85	R--
.data	0x10136000	0xf47c	0x3000	6.06	RW-
.rsrc	0x10146000	0x25e8	0x3000	5.66	R--
.reloc	0x10149000	0x7dee	0x8000	5.61	R--

5 Security Mitigations

Observed: each verdict cites the header field or import that decides it.

Mitigation	Status	Evidence
ASLR	Missing	OptionalHeader.DllCharacteristics DYNAMIC_BASE
DEP / NX	Missing	OptionalHeader.DllCharacteristics NX_COMPAT
Control Flow Guard	Missing	OptionalHeader.DllCharacteristics GUARD_CF
Forced integrity checks	Missing	OptionalHeader.DllCharacteristics FORCE_INTEGRITY
Authenticode signature	Missing	Certificate table

6 Capabilities

Derived: capabilities come from the APIs the program imports; an import shows ability, not proof of use.

Capability	What it enables	Imported APIs
File system	Creates, modifies, deletes, or enumerates files	_unlink, fopen
Anti-analysis	Detects debuggers or hides from analysis	IsDebuggerPresent
Dynamic API resolution	Loads libraries and resolves functions at run time	GetProcAddress, LoadLibraryW
Input capture	Installs input hooks or polls keyboard state	SetWindowsHookExW
System reconnaissance	Collects host, user, or operating system details	GetVersionExA

7 Imports

Observed: 428 imported functions from 7 libraries.

Library	Imported functions
GDI32.dll	39
KERNEL32.dll	32
MFC80U.DLL	157
MSVCP80.dll	3
MSVCR80.dll	122

Library	Imported functions
USER32.dll	71
WS2_32.dll	4

7.1 Import Table

Library	Function	Slot
MFC80U.DLL	ORDINAL 760	0x100fe124
MFC80U.DLL	ORDINAL 2384	0x100fe128
MFC80U.DLL	ORDINAL 931	0x100fe12c
MFC80U.DLL	ORDINAL 927	0x100fe130
MFC80U.DLL	ORDINAL 3678	0x100fe134
MFC80U.DLL	ORDINAL 929	0x100fe138
MFC80U.DLL	ORDINAL 925	0x100fe13c
MFC80U.DLL	ORDINAL 920	0x100fe140
MFC80U.DLL	ORDINAL 5229	0x100fe144
MFC80U.DLL	ORDINAL 5231	0x100fe148
MFC80U.DLL	ORDINAL 5956	0x100fe14c
MFC80U.DLL	ORDINAL 1591	0x100fe150
MFC80U.DLL	ORDINAL 4716	0x100fe154
MFC80U.DLL	ORDINAL 3397	0x100fe158
MFC80U.DLL	ORDINAL 1155	0x100fe15c
MFC80U.DLL	ORDINAL 1183	0x100fe160
MFC80U.DLL	ORDINAL 3331	0x100fe164
MFC80U.DLL	ORDINAL 283	0x100fe168
MFC80U.DLL	ORDINAL 3752	0x100fe16c
MFC80U.DLL	ORDINAL 3198	0x100fe170
MFC80U.DLL	ORDINAL 567	0x100fe174
MFC80U.DLL	ORDINAL 758	0x100fe178
MFC80U.DLL	ORDINAL 3395	0x100fe17c
MFC80U.DLL	ORDINAL 5609	0x100fe180
MFC80U.DLL	ORDINAL 2361	0x100fe184
MFC80U.DLL	ORDINAL 6140	0x100fe188
MFC80U.DLL	ORDINAL 4109	0x100fe18c
MFC80U.DLL	ORDINAL 6059	0x100fe190
MFC80U.DLL	ORDINAL 2366	0x100fe194
MFC80U.DLL	ORDINAL 774	0x100fe198
MFC80U.DLL	ORDINAL 563	0x100fe19c
MFC80U.DLL	ORDINAL 5630	0x100fe1a0
MFC80U.DLL	ORDINAL 6061	0x100fe1a4
MFC80U.DLL	ORDINAL 2362	0x100fe1a8
MFC80U.DLL	ORDINAL 1472	0x100fe1ac
MFC80U.DLL	ORDINAL 2066	0x100fe1b0
MFC80U.DLL	ORDINAL 2404	0x100fe1b4
MFC80U.DLL	ORDINAL 5398	0x100fe1b8
MFC80U.DLL	ORDINAL 2365	0x100fe1bc
MFC80U.DLL	ORDINAL 6058	0x100fe1c0
MFC80U.DLL	ORDINAL 776	0x100fe1c4
MFC80U.DLL	ORDINAL 5638	0x100fe1c8
MFC80U.DLL	ORDINAL 6033	0x100fe1cc
MFC80U.DLL	ORDINAL 5727	0x100fe1d0
MFC80U.DLL	ORDINAL 3756	0x100fe1d4
MFC80U.DLL	ORDINAL 416	0x100fe1d8

Library	Function	Slot
MFC80U.DLL	ORDINAL 651	0x100fe1dc
MFC80U.DLL	ORDINAL 4094	0x100fe1e0
MFC80U.DLL	ORDINAL 2085	0x100fe1e4
MFC80U.DLL	ORDINAL 2444	0x100fe1e8
MFC80U.DLL	ORDINAL 3238	0x100fe1ec
MFC80U.DLL	ORDINAL 1946	0x100fe1f0
MFC80U.DLL	ORDINAL 6086	0x100fe1f4
MFC80U.DLL	ORDINAL 5829	0x100fe1f8
MFC80U.DLL	ORDINAL 6749	0x100fe1fc
MFC80U.DLL	ORDINAL 4119	0x100fe200
MFC80U.DLL	ORDINAL 1921	0x100fe204
MFC80U.DLL	ORDINAL 1555	0x100fe208
MFC80U.DLL	ORDINAL 1176	0x100fe20c
MFC80U.DLL	ORDINAL 1274	0x100fe210
MFC80U.DLL	ORDINAL 2121	0x100fe214
MFC80U.DLL	ORDINAL 2932	0x100fe218
MFC80U.DLL	ORDINAL 6751	0x100fe21c
MFC80U.DLL	ORDINAL 280	0x100fe220
MFC80U.DLL	ORDINAL 502	0x100fe224
MFC80U.DLL	ORDINAL 3281	0x100fe228
MFC80U.DLL	ORDINAL 5637	0x100fe22c
MFC80U.DLL	ORDINAL 5636	0x100fe230
MFC80U.DLL	ORDINAL 6167	0x100fe234
MFC80U.DLL	ORDINAL 6173	0x100fe238
MFC80U.DLL	ORDINAL 4117	0x100fe23c
MFC80U.DLL	ORDINAL 3995	0x100fe240
MFC80U.DLL	ORDINAL 572	0x100fe244
MFC80U.DLL	ORDINAL 2388	0x100fe248
MFC80U.DLL	ORDINAL 2394	0x100fe24c
MFC80U.DLL	ORDINAL 2392	0x100fe250
MFC80U.DLL	ORDINAL 2390	0x100fe254
MFC80U.DLL	ORDINAL 2407	0x100fe258
MFC80U.DLL	ORDINAL 2402	0x100fe25c
MFC80U.DLL	ORDINAL 2386	0x100fe260
MFC80U.DLL	ORDINAL 2409	0x100fe264
MFC80U.DLL	ORDINAL 2397	0x100fe268
MFC80U.DLL	ORDINAL 2379	0x100fe26c
MFC80U.DLL	ORDINAL 2381	0x100fe270
MFC80U.DLL	ORDINAL 2399	0x100fe274
MFC80U.DLL	ORDINAL 2169	0x100fe278
MFC80U.DLL	ORDINAL 2163	0x100fe27c
MFC80U.DLL	ORDINAL 1513	0x100fe280
MFC80U.DLL	ORDINAL 6273	0x100fe284
MFC80U.DLL	ORDINAL 3796	0x100fe288
MFC80U.DLL	ORDINAL 6275	0x100fe28c
MFC80U.DLL	ORDINAL 3339	0x100fe290
MFC80U.DLL	ORDINAL 1894	0x100fe294
MFC80U.DLL	ORDINAL 4961	0x100fe298
MFC80U.DLL	ORDINAL 4276	0x100fe29c
MFC80U.DLL	ORDINAL 1353	0x100fe2a0
MFC80U.DLL	ORDINAL 5171	0x100fe2a4

Library	Function	Slot
MFC80U.DLL	ORDINAL 1955	0x100fe2a8
MFC80U.DLL	ORDINAL 1647	0x100fe2ac
MFC80U.DLL	ORDINAL 1590	0x100fe2b0
MFC80U.DLL	ORDINAL 5196	0x100fe2b4
MFC80U.DLL	ORDINAL 2531	0x100fe2b8
MFC80U.DLL	ORDINAL 764	0x100fe2bc
MFC80U.DLL	ORDINAL 2725	0x100fe2c0
MFC80U.DLL	ORDINAL 2829	0x100fe2c4
MFC80U.DLL	ORDINAL 4301	0x100fe2c8
MFC80U.DLL	ORDINAL 6721	0x100fe2cc
MFC80U.DLL	ORDINAL 2708	0x100fe2d0
MFC80U.DLL	ORDINAL 5911	0x100fe2d4
MFC80U.DLL	ORDINAL 2856	0x100fe2d8
MFC80U.DLL	ORDINAL 1611	0x100fe2dc
MFC80U.DLL	ORDINAL 2534	0x100fe2e0
MFC80U.DLL	ORDINAL 1608	0x100fe2e4
MFC80U.DLL	ORDINAL 2640	0x100fe2e8
MFC80U.DLL	ORDINAL 3940	0x100fe2ec
MFC80U.DLL	ORDINAL 2527	0x100fe2f0
MFC80U.DLL	ORDINAL 1393	0x100fe2f4
MFC80U.DLL	ORDINAL 3712	0x100fe2f8
MFC80U.DLL	ORDINAL 4238	0x100fe2fc
MFC80U.DLL	ORDINAL 3713	0x100fe300
MFC80U.DLL	ORDINAL 5148	0x100fe304
MFC80U.DLL	ORDINAL 3703	0x100fe308
MFC80U.DLL	ORDINAL 1899	0x100fe30c
MFC80U.DLL	ORDINAL 2638	0x100fe310
MFC80U.DLL	ORDINAL 5067	0x100fe314
MFC80U.DLL	ORDINAL 3943	0x100fe318
MFC80U.DLL	ORDINAL 6271	0x100fe31c
MFC80U.DLL	ORDINAL 4480	0x100fe320
MFC80U.DLL	ORDINAL 4179	0x100fe324
MFC80U.DLL	ORDINAL 4255	0x100fe328
MFC80U.DLL	ORDINAL 5210	0x100fe32c
MFC80U.DLL	ORDINAL 1646	0x100fe330
MFC80U.DLL	ORDINAL 3157	0x100fe334
MFC80U.DLL	ORDINAL 6013	0x100fe338
MFC80U.DLL	ORDINAL 5633	0x100fe33c
MFC80U.DLL	ORDINAL 1270	0x100fe340
MFC80U.DLL	ORDINAL 602	0x100fe344
MFC80U.DLL	ORDINAL 347	0x100fe348
MFC80U.DLL	ORDINAL 327	0x100fe34c
MFC80U.DLL	ORDINAL 5723	0x100fe350
MFC80U.DLL	ORDINAL 3155	0x100fe354
MFC80U.DLL	ORDINAL 266	0x100fe358
MFC80U.DLL	ORDINAL 3204	0x100fe35c
MFC80U.DLL	ORDINAL 2255	0x100fe360
MFC80U.DLL	ORDINAL 1959	0x100fe364
MFC80U.DLL	ORDINAL 293	0x100fe368
MFC80U.DLL	ORDINAL 1271	0x100fe36c
MFC80U.DLL	ORDINAL 1925	0x100fe370

Library	Function	Slot
MFC80U.DLL	ORDINAL 577	0x100fe374
MFC80U.DLL	ORDINAL 2311	0x100fe378
MFC80U.DLL	ORDINAL 1118	0x100fe37c
MFC80U.DLL	ORDINAL 1168	0x100fe380
MFC80U.DLL	ORDINAL 371	0x100fe384
MFC80U.DLL	ORDINAL 762	0x100fe388
MFC80U.DLL	ORDINAL 1093	0x100fe38c
MFC80U.DLL	ORDINAL 1079	0x100fe390
MFC80U.DLL	ORDINAL 2460	0x100fe394
MSVCR80.dll	_adjust_fdiv	0x100fe3ac
MSVCR80.dll	_amsg_exit	0x100fe3b0
MSVCR80.dll	_initterm_e	0x100fe3b4
MSVCR80.dll	_initterm	0x100fe3b8
MSVCR80.dll	_encoded_null	0x100fe3bc
MSVCR80.dll	_malloc_crt	0x100fe3c0
MSVCR80.dll	_decode_pointer	0x100fe3c4
MSVCR80.dll	_onexit	0x100fe3c8
MSVCR80.dll	_lock	0x100fe3cc
MSVCR80.dll	_encode_pointer	0x100fe3d0
MSVCR80.dll	__dllonexit	0x100fe3d4
MSVCR80.dll	_unlock	0x100fe3d8
MSVCR80.dll	?terminate@@YAXXZ	0x100fe3dc
MSVCR80.dll	printf	0x100fe3e0
MSVCR80.dll	isprint	0x100fe3e4
MSVCR80.dll	_CIlog	0x100fe3e8
MSVCR80.dll	_open	0x100fe3ec
MSVCR80.dll	_close	0x100fe3f0
MSVCR80.dll	_unlink	0x100fe3f4
MSVCR80.dll	_write	0x100fe3f8
MSVCR80.dll	_read	0x100fe3fc
MSVCR80.dll	ceil	0x100fe400
MSVCR80.dll	rand	0x100fe404
MSVCR80.dll	qsort	0x100fe408
MSVCR80.dll	_CIexp	0x100fe40c
MSVCR80.dll	realloc	0x100fe410
MSVCR80.dll	strncmp	0x100fe414
MSVCR80.dll	_strnicmp	0x100fe418
MSVCR80.dll	isupper	0x100fe41c
MSVCR80.dll	tmpfile	0x100fe420
MSVCR80.dll	strstr	0x100fe424
MSVCR80.dll	perror	0x100fe428
MSVCR80.dll	fgetc	0x100fe42c
MSVCR80.dll	fopen	0x100fe430
MSVCR80.dll	__CppXcptFilter	0x100fe434
MSVCR80.dll	_getcwd	0x100fe438
MSVCR80.dll	putc	0x100fe43c
MSVCR80.dll	_stricmp	0x100fe440
MSVCR80.dll	_mktime64	0x100fe444
MSVCR80.dll	strtok	0x100fe448
MSVCR80.dll	vfprintf	0x100fe44c
MSVCR80.dll	isalpha	0x100fe450

Library	Function	Slot
MSVCR80.dll	strchr	0x100fe454
MSVCR80.dll	atoi	0x100fe458
MSVCR80.dll	isdigit	0x100fe45c
MSVCR80.dll	isspace	0x100fe460
MSVCR80.dll	strrchr	0x100fe464
MSVCR80.dll	vsprintf	0x100fe468
MSVCR80.dll	tmpnam	0x100fe46c
MSVCR80.dll	_lseek	0x100fe470
MSVCR80.dll	abort	0x100fe474
MSVCR80.dll	memcmp	0x100fe478
MSVCR80.dll	atan2	0x100fe47c
MSVCR80.dll	fabs	0x100fe480
MSVCR80.dll	sqrt	0x100fe484
MSVCR80.dll	log	0x100fe488
MSVCR80.dll	exp	0x100fe48c
MSVCR80.dll	bsearch	0x100fe490
MSVCR80.dll	pow	0x100fe494
MSVCR80.dll	memmove	0x100fe498
MSVCR80.dll	strcpy	0x100fe49c
MSVCR80.dll	strlen	0x100fe4a0
MSVCR80.dll	getenv	0x100fe4a4
MSVCR80.dll	sscanf	0x100fe4a8
MSVCR80.dll	exit	0x100fe4ac
MSVCR80.dll	_gmtime64	0x100fe4b0
MSVCR80.dll	atof	0x100fe4b4
MSVCR80.dll	frexp	0x100fe4b8
MSVCR80.dll	modf	0x100fe4bc
MSVCR80.dll	_snprintf	0x100fe4c0
MSVCR80.dll	__iob_func	0x100fe4c4
MSVCR80.dll	fprintf	0x100fe4c8
MSVCR80.dll	_crt_debugger_hook	0x100fe4cc
MSVCR80.dll	?_type_info_dtor_internal_method@type_info@@QAEXXZ	0x100fe4d0
MSVCR80.dll	_except_handler4_common	0x100fe4d4
MSVCR80.dll	_errno	0x100fe4d8
MSVCR80.dll	__clean_type_info_names_internal	0x100fe4dc
MSVCR80.dll	longjmp	0x100fe4e0
MSVCR80.dll	__CxxLongjmpUnwind	0x100fe4e4
MSVCR80.dll	_setjmp3	0x100fe4e8
MSVCR80.dll	_swab	0x100fe4ec
MSVCR80.dll	_CxxThrowException	0x100fe4f0
MSVCR80.dll	_CIpow	0x100fe4f4
MSVCR80.dll	sprintf	0x100fe4f8
MSVCR80.dll	calloc	0x100fe4fc
MSVCR80.dll	ldiv	0x100fe500
MSVCR80.dll	_CIsqrt	0x100fe504
MSVCR80.dll	_CIsin	0x100fe508
MSVCR80.dll	_CIfmod	0x100fe50c
MSVCR80.dll	_CIcos	0x100fe510
MSVCR80.dll	_wfopen	0x100fe514
MSVCR80.dll	fgets	0x100fe518
MSVCR80.dll	getc	0x100fe51c

Library	Function	Slot
MSVCR80.dll	fputc	0x100fe520
MSVCR80.dll	ferror	0x100fe524
MSVCR80.dll	feof	0x100fe528
MSVCR80.dll	fflush	0x100fe52c
MSVCR80.dll	ftell	0x100fe530
MSVCR80.dll	fseek	0x100fe534
MSVCR80.dll	fwrite	0x100fe538
MSVCR80.dll	fread	0x100fe53c
MSVCR80.dll	fclose	0x100fe540
MSVCR80.dll	__CxxFrameHandler3	0x100fe544
MSVCR80.dll	floor	0x100fe548
MSVCR80.dll	memcpy	0x100fe54c
MSVCR80.dll	malloc	0x100fe550
MSVCR80.dll	free	0x100fe554
MSVCR80.dll	memset	0x100fe558
MSVCR80.dll	wcstok	0x100fe55c
MSVCR80.dll	toupper	0x100fe560
MSVCR80.dll	wcsncpy	0x100fe564
MSVCR80.dll	strncpy	0x100fe568
MSVCR80.dll	?what@exception@std@@UBEPBDXZ	0x100fe56c
MSVCR80.dll	??0exception@std@@QAE@XZ	0x100fe570
MSVCR80.dll	??1exception@std@@UAE@XZ	0x100fe574
MSVCR80.dll	??0exception@std@@QAE@ABQBD@Z	0x100fe578
MSVCR80.dll	memmove_s	0x100fe57c
MSVCR80.dll	??0exception@std@@QAE@ABV01@@Z	0x100fe580
MSVCR80.dll	_invalid_parameter_noinfo	0x100fe584
MSVCR80.dll	_purecall	0x100fe588
MSVCR80.dll	fscanf	0x100fe58c
MSVCR80.dll	_setmode	0x100fe590
KERNEL32.dll	GetLocaleInfoA	0x100fe0a0
KERNEL32.dll	GetThreadLocale	0x100fe0a4
KERNEL32.dll	GetVersionExA	0x100fe0a8
KERNEL32.dll	ExitProcess	0x100fe0ac
KERNEL32.dll	lstrlenA	0x100fe0b0
KERNEL32.dll	GetSystemTimeAsFileTime	0x100fe0b4
KERNEL32.dll	GetCurrentProcessId	0x100fe0b8
KERNEL32.dll	QueryPerformanceCounter	0x100fe0bc
KERNEL32.dll	IsDebuggerPresent	0x100fe0c0
KERNEL32.dll	SetUnhandledExceptionFilter	0x100fe0c4
KERNEL32.dll	UnhandledExceptionFilter	0x100fe0c8
KERNEL32.dll	GetCurrentProcess	0x100fe0cc
KERNEL32.dll	TerminateProcess	0x100fe0d0
KERNEL32.dll	InterlockedCompareExchange	0x100fe0d4
KERNEL32.dll	InterlockedExchange	0x100fe0d8
KERNEL32.dll	ResumeThread	0x100fe0dc
KERNEL32.dll	WaitForSingleObject	0x100fe0e0
KERNEL32.dll	CloseHandle	0x100fe0e4
KERNEL32.dll	GlobalFree	0x100fe0e8
KERNEL32.dll	GlobalAlloc	0x100fe0ec
KERNEL32.dll	GetLastError	0x100fe0f0
KERNEL32.dll	SetLastError	0x100fe0f4

Library	Function	Slot
KERNEL32.dll	lstrlenW	0x100fe0f8
KERNEL32.dll	Sleep	0x100fe0fc
KERNEL32.dll	WideCharToMultiByte	0x100fe100
KERNEL32.dll	lstrcpyW	0x100fe104
KERNEL32.dll	GetCurrentThreadId	0x100fe108
KERNEL32.dll	GetTickCount	0x100fe10c
KERNEL32.dll	LoadLibraryW	0x100fe110
KERNEL32.dll	GetModuleHandleW	0x100fe114
KERNEL32.dll	GetACP	0x100fe118
KERNEL32.dll	GetProcAddress	0x100fe11c
USER32.dll	GetDC	0x100fe598
USER32.dll	GetDesktopWindow	0x100fe59c
USER32.dll	DrawIconEx	0x100fe5a0
USER32.dll	LoadImageW	0x100fe5a4
USER32.dll	ReleaseDC	0x100fe5a8
USER32.dll	DestroyIcon	0x100fe5ac
USER32.dll	AppendMenuW	0x100fe5b0
USER32.dll	CreatePopupMenu	0x100fe5b4
USER32.dll	UnionRect	0x100fe5b8
USER32.dll	DestroyMenu	0x100fe5bc
USER32.dll	GetSubMenu	0x100fe5c0
USER32.dll	GetSysColor	0x100fe5c4
USER32.dll	SetRectEmpty	0x100fe5c8
USER32.dll	DispatchMessageW	0x100fe5cc
USER32.dll	GetMessageW	0x100fe5d0
USER32.dll	RemoveMenu	0x100fe5d4
USER32.dll	GetMenuItemID	0x100fe5d8
USER32.dll	GetMenuItemInfoW	0x100fe5dc
USER32.dll	LoadBitmapW	0x100fe5e0
USER32.dll	KillTimer	0x100fe5e4
USER32.dll	GetFocus	0x100fe5e8
USER32.dll	CallNextHookEx	0x100fe5ec
USER32.dll	GetSystemMenu	0x100fe5f0
USER32.dll	GetWindow	0x100fe5f4
USER32.dll	GetParent	0x100fe5f8
USER32.dll	UnhookWindowsHookEx	0x100fe5fc
USER32.dll	MessageBeep	0x100fe600
USER32.dll	SetWindowsHookExW	0x100fe604
USER32.dll	PostMessageW	0x100fe608
USER32.dll	SetTimer	0x100fe60c
USER32.dll	PtInRect	0x100fe610
USER32.dll	SetWindowPos	0x100fe614
USER32.dll	SetWindowLongW	0x100fe618
USER32.dll	SetPropW	0x100fe61c
USER32.dll	SendMessageW	0x100fe620
USER32.dll	GetClientRect	0x100fe624
USER32.dll	GetKeyState	0x100fe628
USER32.dll	IsWindow	0x100fe62c
USER32.dll	InvalidateRect	0x100fe630
USER32.dll	SetMenu	0x100fe634
USER32.dll	GetClassNameW	0x100fe638

Library	Function	Slot
USER32.dll	IsMenu	0x100fe63c
USER32.dll	ShowOwnedPopups	0x100fe640
USER32.dll	GetMenu	0x100fe644
USER32.dll	TrackMouseEvent	0x100fe648
USER32.dll	SetCapture	0x100fe64c
USER32.dll	InflateRect	0x100fe650
USER32.dll	ReleaseCapture	0x100fe654
USER32.dll	EnableMenuItem	0x100fe658
USER32.dll	IsIconic	0x100fe65c
USER32.dll	SetMenuDefaultItem	0x100fe660
USER32.dll	CopyRect	0x100fe664
USER32.dll	GetPropW	0x100fe668
USER32.dll	LockWindowUpdate	0x100fe66c
USER32.dll	GetWindowInfo	0x100fe670
USER32.dll	GetWindowRect	0x100fe674
USER32.dll	RemovePropW	0x100fe678
USER32.dll	CallWindowProcW	0x100fe67c
USER32.dll	GetMenuState	0x100fe680
USER32.dll	GetMenuItemCount	0x100fe684
USER32.dll	EnableWindow	0x100fe688
USER32.dll	LoadCursorW	0x100fe68c
USER32.dll	DefWindowProcW	0x100fe690
USER32.dll	GetClassInfoW	0x100fe694
USER32.dll	ClientToScreen	0x100fe698
USER32.dll	SetRect	0x100fe69c
USER32.dll	SystemParametersInfoW	0x100fe6a0
USER32.dll	FillRect	0x100fe6a4
USER32.dll	RegisterWindowMessageW	0x100fe6a8
USER32.dll	IsRectEmpty	0x100fe6ac
USER32.dll	OffsetRect	0x100fe6b0
GDI32.dll	GetDeviceCaps	0x100fe000
GDI32.dll	SetBkColor	0x100fe004
GDI32.dll	SaveDC	0x100fe008
GDI32.dll	GetClipBox	0x100fe00c
GDI32.dll	SetStretchBltMode	0x100fe010
GDI32.dll	StretchDIBits	0x100fe014
GDI32.dll	SetDIBitsToDevice	0x100fe018
GDI32.dll	RestoreDC	0x100fe01c
GDI32.dll	RealizePalette	0x100fe020
GDI32.dll	GetDIBits	0x100fe024
GDI32.dll	CreateDIBSection	0x100fe028
GDI32.dll	DeleteDC	0x100fe02c
GDI32.dll	SelectObject	0x100fe030
GDI32.dll	GetTextColor	0x100fe034
GDI32.dll	GetStockObject	0x100fe038
GDI32.dll	GetTextExtentPoint32W	0x100fe03c
GDI32.dll	CreateCompatibleBitmap	0x100fe040
GDI32.dll	OffsetRgn	0x100fe044
GDI32.dll	DeleteObject	0x100fe048
GDI32.dll	CombineRgn	0x100fe04c
GDI32.dll	CreateRectRgnIndirect	0x100fe050

Library	Function	Slot
GDI32.dll	ExtSelectClipRgn	0x100fe054
GDI32.dll	CreateBitmap	0x100fe058
GDI32.dll	StretchBlt	0x100fe05c
GDI32.dll	CreateCompatibleDC	0x100fe060
GDI32.dll	CreateFontIndirectW	0x100fe064
GDI32.dll	SetBrushOrgEx	0x100fe068
GDI32.dll	BitBlt	0x100fe06c
GDI32.dll	GetPixel	0x100fe070
GDI32.dll	GetObjectW	0x100fe074
GDI32.dll	DeleteEnhMetaFile	0x100fe078
GDI32.dll	GetEnhMetaFileHeader	0x100fe07c
GDI32.dll	ExtTextOutW	0x100fe080
GDI32.dll	GetEnhMetaFilePaletteEntries	0x100fe084
GDI32.dll	CreatePalette	0x100fe088
GDI32.dll	SelectPalette	0x100fe08c
GDI32.dll	PlayEnhMetaFile	0x100fe090
GDI32.dll	SetEnhMetaFileBits	0x100fe094
GDI32.dll	SetWinMetaFileBits	0x100fe098
MSVCP80.dll	??1?$basic_string@DU?$char_traits@D@std@@V?$allocator@D@2@@std@@QAE@XZ	0x100fe39c
MSVCP80.dll	??0?$basic_string@DU?$char_traits@D@std@@V?$allocator@D@2@@std@@QAE@ABV01@@Z	0x100fe3a0
MSVCP80.dll	??0?$basic_string@DU?$char_traits@D@std@@V?$allocator@D@2@@std@@QAE@PBD@Z	0x100fe3a4
WS2_32.dll	ORDINAL 15	0x100fe6b8
WS2_32.dll	ORDINAL 9	0x100fe6bc
WS2_32.dll	ORDINAL 8	0x100fe6c0
WS2_32.dll	ORDINAL 14	0x100fe6c4

8 Exports

Export	Address
??0CCustomTitle@@QAE@ABV0@@Z	0x100010c0
??0CCustomTitle@@QAE@W4WTitleTheme@@@Z	0x10001530
??0CCustomTitle@@QAE@XZ	0x100014b0
??1CCustomTitle@@UAE@XZ	0x100015b0
??4CCustomTitle@@QAEAAV0@ABV0@@Z	0x100010e0
??_7CCustomTitle@@6B@	0x1012f3d4
?AddQuickButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0KKKK@Z	0x10001710
?AddQuickButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0PB_W@Z	0x10001690
?AddToolButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0KKKK@Z	0x10001b20
?AddToolButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0PB_W@Z	0x10001aa0
?AddToolSeparator@CCustomTitle@@QAEHK@Z	0x10001b90
?CheckToolButton@CCustomTitle@@QAEXI@Z	0x10001c60
?DeleteQuickButton@CCustomTitle@@QAEHH@Z	0x100017b0
?GetMenu@CCustomTitle@@QAEPAVCMenu@@XZ	0x10001660
?GetQuickButtonCount@CCustomTitle@@QAEHXZ	0x10001670
?GetToolButtonCount@CCustomTitle@@QAEHXZ	0x10001a70
?IsSubClassed@CCustomTitle@@QBEHXZ	0x10001600
?IsToolButtonChecked@CCustomTitle@@QAEHI@Z	0x10001c40

Export	Address
?RefreshFrame@CCustomTitle@@QAEHXZ	0x10001620
?RemoveAllQuickButtons@CCustomTitle@@QAEXXZ	0x10001800
?RemoveAllToolButtons@CCustomTitle@@QAEXXZ	0x10001c30
?SetCaptionBkImage@CCustomTitle@@QAEHPAVCBitmap@@H@Z	0x10001a30
?SetCaptionText@CCustomTitle@@QAEXABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@@Z	0x100019b0
?SetCaptionTextColor@CCustomTitle@@QAEXK@Z	0x10001980
?SetCaptionTextMargin@CCustomTitle@@QAEXHH@Z	0x10001a10
?SetControlBtnsBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z	0x10001840
?SetDefaultTheme@CCustomTitle@@SAXW4WTitleTheme@@@Z	0x100014a0
?SetMX_NCTITLE@CCustomTitle@@QAEXH@Z	0x10001610
?SetMenu@CCustomTitle@@QAEHPAVCMenu@@@Z	0x10001640
?SetMenuBarBreakImage@CCustomTitle@@QAEHPAVCBitmap@@K@Z	0x10001900
?SetMenuBarDownBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z	0x10001920
?SetMenuLeftMargin@CCustomTitle@@QAEXH@Z	0x10001950
?SetMenuTextColor@CCustomTitle@@QAEXKKK@Z	0x10001960
?SetPageInfoText@CCustomTitle@@QAEXABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@@Z	0x100019f0
?SetQuickBarBreakImage@CCustomTitle@@QAEHPAVCBitmap@@K@Z	0x100018e0
?SetQuickButtonTextColor@CCustomTitle@@QAEXIKKKK@Z	0x10001780
?SetSysmenuBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z	0x100018c0
?SetSysmenuIcon@CCustomTitle@@QAEHPAUHICON__@@@Z	0x10001860
?SetToolBarBkImage@CCustomTitle@@QAEHPAVCBitmap@@H@Z	0x10001a50
?SetToolBarLeftMargin@CCustomTitle@@QAEXH@Z	0x10001a90
?SetToolButtonTextColor@CCustomTitle@@QAEXIKKKK@Z	0x10001be0
?SetToolSeparatorColor@CCustomTitle@@QAEXK@Z	0x10001c10
?ShowTitle@CCustomTitle@@QAEXH@Z	0x10001ce0
?SubClassWnd@CCustomTitle@@QAEHPAVCWnd@@HH@Z	0x100015d0
?UncheckAllToolButton@CCustomTitle@@QAEXXZ	0x10001ca0
?UnsubClassWnd@CCustomTitle@@QAEXXZ	0x100015f0
?m_wThemeDefault@CCustomTitle@@1W4WTitleTheme@@A	0x1013f370
png_access_version_number	0x10023c20
png_benign_error	0x10023210
png_build_grayscale_palette	0x10033760
png_calloc	0x1002cdd0
png_chunk_benign_error	0x10023270
png_chunk_error	0x10023240
png_chunk_warning	0x10023150
png_convert_from_struct_tm	0x1002a310
png_convert_from_time_t	0x1002a350
png_convert_to_rfc1123	0x10023a40
png_create_info_struct	0x10025000
png_create_read_struct	0x10028570
png_create_read_struct_2	0x10025260
png_create_write_struct	0x1002b420
png_create_write_struct_2	0x1002af50
png_data_freer	0x10023590
png_destroy_info_struct	0x10025040
png_destroy_read_struct	0x100285a0
png_destroy_write_struct	0x1002b450
png_error	0x100231e0

Export	Address
png_free	0x1002cd70
png_free_data	0x100235e0
png_free_default	0x1002cc80
png_get_IHDR	0x100295f0
png_get_PLTE	0x10029940
png_get_bKGD	0x100292e0
png_get_bit_depth	0x10028da0
png_get_cHRM	0x10029310
png_get_cHRM_fixed	0x100293e0
png_get_channels	0x100292a0
png_get_chunk_cache_max	0x10029b80
png_get_chunk_malloc_max	0x10029ba0
png_get_color_type	0x10028dc0
png_get_compression_buffer_size	0x10029b20
png_get_compression_type	0x10028e20
png_get_copyright	0x10023ae0
png_get_error_ptr	0x100230e0
png_get_filter_type	0x10028de0
png_get_gAMA	0x100294c0
png_get_gAMA_fixed	0x10029490
png_get_hIST	0x100295c0
png_get_header_ver	0x10023af0
png_get_header_version	0x10023b00
png_get_iCCP	0x10029520
png_get_image_height	0x10028d80
png_get_image_width	0x10028d60
png_get_int_32	0x100a3680
png_get_interlace_type	0x10028e00
png_get_io_chunk_name	0x10029c00
png_get_io_chunk_type	0x10029bd0
png_get_io_ptr	0x10023a20
png_get_io_state	0x10029bc0
png_get_libpng_ver	0x10025090
png_get_mem_ptr	0x1002ccd0
png_get_oFFs	0x100296b0
png_get_pCAL	0x10029700
png_get_pHYs	0x100298f0
png_get_pHYs_dpi	0x100291a0
png_get_pixel_aspect_ratio	0x10028ed0
png_get_pixel_aspect_ratio_fixed	0x10028f30
png_get_pixels_per_inch	0x10029080
png_get_pixels_per_meter	0x10028ea0
png_get_rgb_to_gray_status	0x10029ae0
png_get_rowbytes	0x10028d20
png_get_rows	0x10028d40
png_get_sBIT	0x10029980
png_get_sCAL	0x10029840
png_get_sCAL_fixed	0x100297c0
png_get_sCAL_s	0x100298a0
png_get_sPLT	0x10029590
png_get_sRGB	0x100294f0

Export	Address
png_get_signature	0x100292c0
png_get_tIME	0x10029a00
png_get_tRNS	0x10029a30
png_get_text	0x100299b0
png_get_uint_16	0x1002ce80
png_get_uint_31	0x1002ce00
png_get_uint_32	0x100a3680
png_get_unknown_chunks	0x10029ab0
png_get_user_chunk_ptr	0x10029b00
png_get_user_height_max	0x10029b60
png_get_user_transform_ptr	0x10028cf0
png_get_user_width_max	0x10029b40
png_get_valid	0x10028d00
png_get_x_offset_inches	0x10029140
png_get_x_offset_inches_fixed	0x10029100
png_get_x_offset_microns	0x10028f90
png_get_x_offset_pixels	0x10028ff0
png_get_x_pixels_per_inch	0x100290a0
png_get_x_pixels_per_meter	0x10028e40
png_get_y_offset_inches	0x10029170
png_get_y_offset_inches_fixed	0x10029120
png_get_y_offset_microns	0x10028fc0
png_get_y_offset_pixels	0x10029020
png_get_y_pixels_per_inch	0x100290c0
png_get_y_pixels_per_meter	0x10028e70
png_handle_as_unknown	0x10023b10
png_info_init_3	0x10023550
png_init_io	0x10023a30
png_longjmp	0x10023050
png_malloc	0x1002cd20
png_malloc_default	0x1002cc60
png_malloc_warn	0x1002cda0
png_permit_mng_features	0x1002c760
png_read_end	0x10026f60
png_read_image	0x10026eb0
png_read_info	0x10025580
png_read_png	0x100283a0
png_read_row	0x100268d0
png_read_rows	0x10026e10
png_read_update_info	0x10026850
png_reset_zstream	0x10023c00
png_save_int_32	0x10035980
png_save_uint_16	0x100359b0
png_save_uint_32	0x10035950
png_set_IHDR	0x1002b7e0
png_set_PLTE	0x1002be10
png_set_add_alpha	0x100287a0
png_set_bKGD	0x1002b4f0
png_set_background	0x10030e80
png_set_background_fixed	0x10030e00
png_set_benign_errors	0x1002c9e0

Export	Address
png_set_bgr	0x10028670
png_set_cHRM	0x1002b5c0
png_set_cHRM_fixed	0x1002b520
png_set_chunk_cache_max	0x1002c9a0
png_set_chunk_malloc_max	0x1002c9c0
png_set_compression_buffer_size	0x1002c8f0
png_set_compression_level	0x1002ae20
png_set_compression_mem_level	0x1002ae40
png_set_compression_method	0x1002aee0
png_set_compression_strategy	0x1002ae60
png_set_compression_window_bits	0x1002ae80
png_set_crc_action	0x10030d30
png_set_error_fn	0x100230c0
png_set_expand	0x10031860
png_set_expand_gray_1_2_4_to_8	0x10031880
png_set_filler	0x10028740
png_set_filter	0x1002a6c0
png_set_filter_heuristics	0x1002aa70
png_set_filter_heuristics_fixed	0x1002ac00
png_set_flush	0x1002a5c0
png_set_gAMA	0x1002b6f0
png_set_gAMA_fixed	0x1002b6a0
png_set_gamma	0x10031810
png_set_gamma_fixed	0x100317c0
png_set_gray_to_rgb	0x100318b0
png_set_hIST	0x1002b720
png_set_iCCP	0x1002bf60
png_set_interlace_handling	0x10028710
png_set_invalid	0x1002c960
png_set_invert_alpha	0x100287e0
png_set_invert_mono	0x100287f0
png_set_keep_unknown_chunks	0x1002c780
png_set_longjmp_fn	0x10023030
png_set_mem_fn	0x1002cca0
png_set_oFFs	0x1002b8c0
png_set_pCAL	0x1002b8f0
png_set_pHYs	0x1002bde0
png_set_packing	0x100286a0
png_set_packswap	0x100286c0
png_set_palette_to_rgb	0x10031860
png_set_quantize	0x10030ee0
png_set_read_fn	0x10029c70
png_set_read_status_fn	0x10028380
png_set_read_user_chunk_fn	0x1002c880
png_set_read_user_transform_fn	0x10031a00
png_set_rgb_to_gray	0x100319b0
png_set_rgb_to_gray_fixed	0x100318d0
png_set_rows	0x1002c8a0
png_set_sBIT	0x1002beb0
png_set_sCAL	0x1002bc50
png_set_sCAL_fixed	0x1002bd30

Export	Address
png_set_sCAL_s	0x1002bae0
png_set_sPLT	0x1002c430
png_set_sRGB	0x1002bee0
png_set_sRGB_gAMA_and_cHRM	0x1002bf00
png_set_shift	0x100286e0
png_set_sig_bytes	0x10023320
png_set_strip_16	0x10030ec0
png_set_strip_alpha	0x10030ed0
png_set_swap	0x10028680
png_set_swap_alpha	0x100287d0
png_set_tIME	0x1002c310
png_set_tRNS	0x1002c340
png_set_tRNS_to_alpha	0x100318a0
png_set_text	0x1002ca00
png_set_unknown_chunk_location	0x1002c720
png_set_unknown_chunks	0x1002c5c0
png_set_user_limits	0x1002c980
png_set_user_transform_info	0x10028cd0
png_set_write_fn	0x1002cad0
png_set_write_status_fn	0x1002af10
png_set_write_user_transform_fn	0x1002af30
png_sig_cmp	0x10023350
png_start_read_image	0x100268a0
png_warning	0x100230f0
png_write_chunk	0x10036bf0
png_write_chunk_data	0x10035ae0
png_write_chunk_end	0x10035b20
png_write_chunk_start	0x10035a60
png_write_end	0x1002a190
png_write_flush	0x1002a5e0
png_write_image	0x1002b1c0
png_write_info	0x10029e90
png_write_info_before_PLTE	0x10029cc0
png_write_png	0x1002b320
png_write_row	0x1002a370
png_write_rows	0x1002b190
png_write_sig	0x100359d0

9 Strings and Indicators

Observed: 6,252 printable strings (6213 ASCII, 39 UTF-16LE).

9.1 Indicators

Type	Value	File offset
Windows path	C:\Program Files (x86)\Microsoft Visual Studio 8\VC\atlmfc\include\afxwin1.inl	0x12f3d8
Windows path	d:\RLD\pro\RNF_X\release\RNC.pdb	0x12f870
URL	http://www.iec.ch	0x1373a8
IPv4 address	1.0.0.1	0x13b218
Windows path	:,:0:4:8:<:P:T:X:\:`:t:x:|:	0x13ebd3
Windows path	9R:\:	0x13f687
Windows path	: :$:,:0:4:8:<:D:L:P:\:d:h:t:x:|:	0x141b81

9.2 String Sample

```
000fe728  CxImage::Create : width and height must be greater than zero
000fe790  CXIMAGE_MAX_MEMORY exceeded
000fe8b8  null file handler
000fe8cc  Decode: Unknown or wrong format
000fea18  Unable to allocate TIFF palette
000fea38  Too big palette to handle
000fea6c  No space for raster buffer
000feaa8  Cancelled
000feadc  Error: page not present in TIFF file
000feb04  Error encountered while opening TIFF file
000feb30  error: too many components
000feb5c  error: cannot allocate memory
000febc4  error: cannot load image data
000febe4  error: unknowm format
000febfc  error: cannot open standard input
000fec20  cannot initialize jasper
000fec6c  CxImagePSD: psd_status_malloc_failed
000fec94  CxImageRAW: missing raw decoder
000fecb4  CxImageRAW: not a raw image
000fecd0  CxImageRAW: unknown option
000fecec  CxImageRAW
000fed04  corrupted WMF
000fed60  PCX_PlanesToPixels: more than 4 planes or more than 1 bit per pixel
000fedec  bad color map signature
000fee04  corrupted PCX
000fee14  repeat count spans end of image
000fee60  PCX file has unknown encoding scheme
000fee88  Error: Not a PCX file
000feeb8  Unknown TGA image type
000feed0  corrupted TGA
000feee0  TGA Create failed
000feef4  bad TGA header
000fef04  Not a TGA
000fef18  CxImageMNG::Decode cannot create image
000fef40  Error: frame not present in MNG file
000fef68  Error in MNG file
000fefb0  unknown PNG color type
000fefcf  AFailed to initialize PNG info structure
000feff8  Failed to create PNG structure
000ff0c0  unknown compression
000ff0d4  Error reading BMP info
000ff0ec  Not a BMP
000ff3b4  TIFF IMAGE
000ff3f4  Illegal subdirectory link
000ff410  Illegal pointer offset value in EXIF.
000ff438  Illegal format code in EXIF dir
000ff488  Illegally sized directory
000ff4e0  Invalid Exif alignment marker.
000ff518  Incorrect Exif header
000ff54c  could not read the rest of the image
000ff574  could not allocate data for entire image
000ff5a0  Premature end of file?
000ff5b8  Could not allocate memory
000ff5d4  invalid marker
000ff5e4  too many padding bytes
000ff5fc  Too many sections in jpg file
000ff64c  NoRemove
000ff660  ForceRemove
000ff6c4  0123456789ABCDEF
000ff6d8  libpng warning: %s
000ff6ec  libpng error: %s
000ff700  fixed point overflow in
000ff74c  Too many bytes for PNG signature
000ff794  Unknown freer parameter in png_data_freer
000ff8d4  Invalid IHDR data
000ff8e8  Invalid filter method in IHDR
```

```
000ff908  Unknown filter method in IHDR
000ff928  MNG features are not allowed in a PNG datastream
000ff95c  Unknown compression method in IHDR
000ff980  Unknown interlace method in IHDR
000ff9a4  Invalid color type/bit depth combination in IHDR
000ff9d8  Invalid color type in IHDR
000ff9f4  Invalid bit depth in IHDR
000ffa40  Invalid image height in IHDR
000ffa60  Invalid image width in IHDR
000ffa7c  Image height exceeds user limit in IHDR
000ffaa4  Image width exceeds user limit in IHDR
000ffacc  Image height is zero in IHDR
000ffaec  Image width is zero in IHDR
000ffc4c  Ignoring attempt to set chromaticity value exceeding 21474.83
000ffc8c  Ignoring attempt to set negative chromaticity value
000ffcc0  Unknown zlib error
000ffcd4  zlib version error
000ffce8  zlib stream error
000ffcfc  zlib memory error
000ffd18  Application built with libpng-%.20s but running with %.20s
000ffd54  Missing PLTE before IDAT
000ffd70  Missing IHDR before IDAT
000ffe20  Decompression error
000ffe34  Extra compressed data
000ffe4c  Not enough image data
000ffe64  Invalid attempt to read row data
000ffe88  Interlace handling should be turned on when using png_read_image
000ffecc  Too many IDATs found
00100227  ?sCAL height
00100240  Call to NULL read function
001002c8  Valid palette required for paletted images
001002f4  No IDATs written into file
00100310  png_write_info was never called before png_write_row
00100348  zlib error
00100354  Unknown custom filter method
001003e8  Unknown row filter for method 0
00100408  Unknown filter heuristic method
00100428  Compression window is being reset to 512
001004bc  Only compression method 8 is supported by PNG
0010055c  Negative or zero gamma ignored
0010057c  Gamma too large, ignored
00100598  png_set_gAMA
001005a8  Invalid palette size, hIST allocation skipped
001005d8  Insufficient memory for hIST chunk data
00100600  Invalid pCAL equation type
0010061c  Insufficient memory for pCAL parameter
00100644  Insufficient memory for pCAL params
00100668  Insufficient memory for pCAL units
0010068c  Invalid format for pCAL parameter
001006b0  Insufficient memory for pCAL purpose
001006d8  Invalid sCAL width
001006ec  Invalid sCAL height
00100700  Memory allocation failed while processing sCAL
00100730  Invalid sCAL unit
00100744  Invalid sCAL height ignored
00100760  Invalid sCAL width ignored
00100794  Insufficient memory to process iCCP profile
001007c0  Insufficient memory to process iCCP chunk
001007ec  text compression mode is out of range
00100890  Out of memory while processing unknown chunk
001008c0  Attempt to set buffer size beyond max ignored
001008f0  Insufficient memory to store text
00100914  Call to NULL write function
00100930  Out of Memory
00100940  PNG unsigned integer out of range
00100964  PNG fixed point integer out of range
0010098c  PNG file corrupted by ASCII conversion
```

```
001009b4  Not a PNG file
001009c4  Buffer error in compressed datastream in %s chunk
001009f8  Data error in compressed datastream in %s chunk
00100a28  Incomplete compressed datastream in %s chunk
00100a58  Unknown zTXt compression type %d
00100a7c  Not enough memory to decompress chunk
00100aa4  png_inflate logic error
00100abc  Exceeded size limit while expanding chunk
00100afc  invalid chunk type
00100b10  Ignoring bad adaptive filter type
00100b34  Row has too many bytes to allocate in memory
00100b70  Invalid IHDR chunk
00100b84  Out of place IHDR
00100b98  Invalid palette chunk
00100c04  Ignoring PLTE chunk in grayscale PNG
00100c2c  Duplicate PLTE chunk
00100c44  Invalid PLTE after IDAT
```

10 Code Analysis

Derived: 32-bit recursive descent recovered 1,396 functions and 105,458 instructions.

10.1 Largest Functions

Function	Address	Instructions	Callers	Callees	Imports used
sub_100eb620	0x100eb620	4010	1	30	__iob_func, _stricmp, _strnicmp, atoi (+6 more)
png_read_info	0x10025580	1677	2	26	
png_read_end	0x10026f60	1541	1	26	
sub_100143f0	0x100143f0	1383	1	34	_CIpow, _CxxThrowException, calloc, free (+2 more)
sub_100e2b50	0x100e2b50	1345	2	11	_CIpow, isdigit, strncmp, strstr
sub_100e8a00	0x100e8a00	1116	3	13	free, malloc, sprintf, sscanf (+3 more)
sub_100e0950	0x100e0950	1062	1	6	_CIpow, __iob_func, fprintf, free (+2 more)
sub_10011fb0	0x10011fb0	1058	1	9	BitBlt, CreateCompatibleDC, CreateDIBSection, CreateRectRgnIndirect (+11 more)
sub_1007171b	0x1007171b	1041	2	32	free
sub_100380e0	0x100380e0	899	1	1	
sub_100dea30	0x100dea30	876	1	4	__iob_func, fprintf, fputc, free (+2 more)
sub_1000f760	0x1000f760	835	2	34	ORDINAL 762
sub_1001fd10	0x1001fd10	750	1	12	
sub_100e2010	0x100e2010	733	1	4	_CIpow, __iob_func, calloc, fprintf (+2 more)
sub_1001a5b0	0x1001a5b0	701	1	38	ORDINAL 762, ORDINAL 764, memcpy, memset (+1 more)
sub_10062a80	0x10062a80	690	4	9	memset
sub_1001b970	0x1001b970	658	1	25	free, malloc, memset
sub_10031a20	0x10031a20	640	2	8	
png_set_quantize	0x10030ee0	631	0	5	memset
sub_100de200	0x100de200	604	1	3	_CIsqrt, __iob_func, fprintf, free (+1 more)
sub_10016040	0x10016040	550	1	34	ORDINAL 9, _CxxThrowException, _setjmp3, _swab (+4 more)
sub_100e4a30	0x100e4a30	539	3	6	_CIpow, strstr
sub_1008b280	0x1008b280	532	2	3	
sub_100156b0	0x100156b0	529	1	20	_CxxThrowException, calloc, free, strncpy
sub_100e67b0	0x100e67b0	528	1	5	ORDINAL 8, _CIpow, __iob_func, calloc (+2 more)

11 Key Functions

Inferred: functions are ranked by the capability-relevant APIs they call and the indicators they reference. Listings use Intel syntax with imports, functions, and strings resolved in comments.

11.1 sub_100eb620 at 0x100eb620

- References text "CASIO"
- References text "Minolta"
- References text "MINOLTA"
- References text "KODAK"
- References text "NIKON"
- References text "Canon"
- References text "Sinar"
- References text "Valeo 6"

```
100eb620  sub esp, 0x7c
100eb623  mov eax, [0x10136220]
100eb628  xor eax, esp
100eb62a  mov [esp+0x78], eax
100eb62e  fldz
100eb630  push ebx
100eb631  xor ebx, ebx
100eb633  push ebp
100eb634  mov ebp, [esp+0x88]
100eb63b  or eax, 0xffffffff
100eb63e  fst dword ptr [ebp+0x659c], st
100eb644  push 0x80
100eb649  fst dword ptr [ebp+0x6598], st
100eb64f  mov [ebp+0x65b0], eax
100eb655  fst dword ptr [ebp+0x6594], st
100eb65b  mov [ebp+0x66c8], eax
100eb661  fstp dword ptr [ebp+0x6590], st
100eb667  mov [ebp+0x66cc], eax
100eb66d  lea eax, [ebp+0x662c]
100eb673  push ebx
100eb674  push eax
100eb675  mov [ebp+0xe766], bx
100eb67c  mov [ebp+0x65e4], ebx
100eb682  mov [ebp+0x66c2], bx
100eb689  mov [ebp+0x66b2], bx
100eb690  mov [ebp+0x66b0], bx
100eb697  mov [ebp+0x66ba], bx
100eb69e  mov [ebp+0x66b8], bx
100eb6a5  mov [ebp+0x66b6], bx
100eb6ac  mov [ebp+0x66b4], bx
100eb6b3  mov [ebp+0x65fc], ebx
100eb6b9  mov [ebp+0x6505], bl
100eb6bf  mov [ebp+0x64c5], bl
100eb6c5  mov [ebp+0x6485], bl
100eb6cb  mov [ebp+0x6545], bl
100eb6d1  mov [ebp+0x6285], bl
100eb6d7  mov [ebp+0x6280], bl
100eb6dd  mov [ebp+0x65b8], ebx
100eb6e3  call 0x100228c0  ; j_memset
100eb6e8  push 0x80
100eb6ed  lea ecx, [ebp+0x66e4]
100eb6f3  push ebx
100eb6f4  push ecx
100eb6f5  call 0x100228c0  ; j_memset
100eb6fa  fld1
100eb6fc  mov eax, 1
100eb701  fstp qword ptr [ebp+0x66d8], st
100eb707  fldz
100eb709  add esp, 0x18
100eb70c  mov [ebp+0x6608], eax
100eb712  mov [ebp+0x6604], eax
100eb718  mov [ebp+0x6614], eax
100eb71e  mov eax, 0x7fffffff
100eb723  push esi
100eb724  mov [ebp+0x6628], eax
```

```
100eb72a  mov [ebp+0x6624], eax
100eb730  mov [ebp+0x66c6], bx
100eb737  mov [ebp+0x66c4], bx
100eb73e  mov [ebp+0x65d0], ebx
100eb744  mov [ebp+0x65c4], ebx
```

11.2 sub_1006daaf at 0x1006daaf

- References text "PhotometricInterpretation"
- References text "Sorry, can not handle images with %d-bit samples"
- References text "Sorry, LogL data must have %s=%d"
- References text "Sorry, can not handle LogLuv images with %s=%d"
- References text "Planarconfiguration"
- References text "Sorry, LogLuv data must have %s=%d or %d"
- References text "Compression"
- References text "Sorry, can not handle image with %s=%d"

```
1006daaf  push ebp
1006dab0  mov ebp, esp
1006dab2  mov eax, [ebp+0xc]
1006dab5  mov ecx, [ebp+0x10]
1006dab8  sub esp, 0x18
1006dabb  push ebx
1006dabc  push esi
1006dabd  mov esi, [ebp+8]
1006dac0  push edi
1006dac1  lea edi, [esi+0x18]
1006dac4  xor ebx, ebx
1006dac6  push edi
1006dac7  push 0x102
1006dacc  push eax
1006dacd  mov [esi+0x4c], ebx
1006dad0  mov [esi+0x50], ebx
1006dad3  mov [esi+0x24], ebx
1006dad6  mov [esi+0x28], ebx
1006dad9  mov [esi+0x2c], ebx
1006dadc  mov word ptr [esi+0x1e], 4
1006dae2  mov [esi], eax
1006dae4  mov [esi+4], ecx
1006dae7  call 0x1006e579
1006daec  movzx eax, word ptr [edi]
1006daef  add esp, 0xc
1006daf2  cmp eax, ebx
1006daf4  jle 0x1006df13
1006dafa  cmp eax, 2
1006dafd  jle short 0x1006db12
1006daff  cmp eax, 4
1006db02  je short 0x1006db12
1006db04  cmp eax, 8
1006db07  je short 0x1006db12
1006db09  cmp eax, 0x10
1006db0c  jne 0x1006df13
1006db12  lea edi, [esi+0x1a]
1006db15  push edi
1006db16  push 0x115
1006db1b  push dword ptr [ebp+0xc]
1006db1e  mov [esi+0xc], ebx
1006db21  call 0x1006e579
1006db26  lea eax, [ebp-0xc]
1006db29  push eax
1006db2a  lea eax, [ebp-4]
1006db2d  push eax
1006db2e  push 0x152
1006db33  push dword ptr [ebp+0xc]
```

```
1006db36  call 0x1006e579
1006db3b  xor ecx, ecx
1006db3d  inc ecx
1006db3e  add esp, 0x1c
1006db41  cmp [ebp-4], cx
1006db45  jb short 0x1006db66
1006db47  mov eax, [ebp-0xc]
1006db4a  movzx eax, word ptr [eax]
1006db4d  cmp eax, ebx
1006db4f  je short 0x1006db5d
1006db51  lea edx, [eax-1]
1006db54  cmp edx, ecx
1006db56  ja short 0x1006db66
```

11.3 sub_10023c30 at 0x10023c30

- References text "Invalid IHDR data"
- References text "Invalid filter method in IHDR"
- References text "Unknown filter method in IHDR"
- References text "MNG features are not allowed in a PNG datastream"
- References text "Unknown compression method in IHDR"
- References text "Unknown interlace method in IHDR"
- References text "Invalid color type/bit depth combination in IHDR"
- References text "Invalid color type in IHDR"

```
10023c30  push ebx
10023c31  mov ebx, [esp+0xc]
10023c35  push ebp
10023c36  push esi
10023c37  mov esi, [esp+0x10]
10023c3b  push edi
10023c3c  xor edi, edi
10023c3e  test ebx, ebx
10023c40  jne short 0x10023c55
10023c42  push 0x100ffaec  ; "Image width is zero in IHDR"
10023c47  push esi
10023c48  call 0x100230f0  ; png_warning
10023c4d  add esp, 8
10023c50  mov edi, 1
10023c55  mov ebp, [esp+0x1c]
10023c59  test ebp, ebp
10023c5b  jne short 0x10023c6e
10023c5d  push 0x100ffacc  ; "Image height is zero in IHDR"
10023c62  push esi
10023c63  call 0x100230f0  ; png_warning
10023c68  add esp, 8
10023c6b  lea edi, [ebp+1]
10023c6e  cmp ebx, [esi+0x258]
10023c74  ja short 0x10023c7e
10023c76  cmp ebx, 0xf4240
10023c7c  jbe short 0x10023c91
10023c7e  push 0x100ffaa4  ; "Image width exceeds user limit in IHDR"
10023c83  push esi
10023c84  call 0x100230f0  ; png_warning
10023c89  add esp, 8
10023c8c  mov edi, 1
10023c91  cmp ebp, [esi+0x25c]
10023c97  ja short 0x10023ca1
10023c99  cmp ebp, 0xf4240
10023c9f  jbe short 0x10023cb4
10023ca1  push 0x100ffa7c  ; "Image height exceeds user limit in IHDR"
10023ca6  push esi
10023ca7  call 0x100230f0  ; png_warning
10023cac  add esp, 8
```

```
10023caf  mov edi, 1
10023cb4  cmp ebx, 0x7fffffff
10023cba  jbe short 0x10023ccf
10023cbc  push 0x100ffa60  ; "Invalid image width in IHDR"
10023cc1  push esi
10023cc2  call 0x100230f0  ; png_warning
10023cc7  add esp, 8
10023cca  mov edi, 1
10023ccf  cmp ebp, 0x7fffffff
10023cd5  jbe short 0x10023cea
10023cd7  push 0x100ffa40  ; "Invalid image height in IHDR"
10023cdc  push esi
10023cdd  call 0x100230f0  ; png_warning
10023ce2  add esp, 8
10023ce5  mov edi, 1
10023cea  cmp ebx, 0x1fffff8e
10023cf0  jbe short 0x10023d00
10023cf2  push 0x100ffa10  ; "Width is too large for libpng to process pixels"
10023cf7  push esi
10023cf8  call 0x100230f0  ; png_warning
10023cfd  add esp, 8
```

11.4 sub_10068d67 at 0x10068d67

- References text "PhotometricInterpretation"
- References text "Sorry, can not handle images with %d-bit samples"
- References text "Sorry, LogL data must have %s=%d"
- References text "Sorry, can not handle LogLuv images with %s=%d"
- References text "Planarconfiguration"
- References text "Sorry, LogLuv data must have %s=%d or %d"
- References text "Compression"
- References text "Sorry, can not handle image with %s=%d"

```
10068d67  push ebp
10068d68  mov ebp, esp
10068d6a  push ecx
10068d6b  push esi
10068d6c  mov esi, [ebp+8]
10068d6f  cmp dword ptr [esi+0x200], 0
10068d76  push edi
10068d77  jne short 0x10068d90
10068d79  push 0x10108cb0  ; "Sorry, requested compression method is not configured"
10068d7e  push dword ptr [ebp+0xc]
10068d81  call dword ptr [0x100fe4f8]  ; MSVCR80.dll!sprintf
10068d87  pop ecx
10068d88  pop ecx
10068d89  xor eax, eax
10068d8b  jmp 0x10068f96
10068d90  movzx eax, word ptr [esi+0x54]
10068d94  test eax, eax
10068d96  push ebx
10068d97  jle 0x10068f81
10068d9d  push 2
10068d9f  pop ebx
10068da0  cmp eax, ebx
10068da2  jle short 0x10068db7
10068da4  cmp eax, 4
10068da7  je short 0x10068db7
10068da9  cmp eax, 8
10068dac  je short 0x10068db7
10068dae  cmp eax, 0x10
10068db1  jne 0x10068f81
10068db7  movzx eax, word ptr [esi+0xa8]
10068dbe  movzx edi, word ptr [esi+0x62]
```

```
10068dc2  sub edi, eax
10068dc4  lea eax, [ebp-4]
10068dc7  push eax
10068dc8  push 0x106
10068dcd  push esi
10068dce  call 0x10067985
10068dd3  add esp, 0xc
10068dd6  test eax, eax
10068dd8  jne short 0x10068dfe
10068dda  mov eax, edi
10068ddc  dec eax
10068ddd  je short 0x10068df7
10068ddf  sub eax, ebx
10068de1  je short 0x10068df2
10068de3  push 0x10108aa8  ; "PhotometricInterpretation"
10068de8  push 0x10108c98  ; "Missing needed %s tag"
10068ded  jmp 0x10068f87
10068df2  mov [ebp-4], ebx
10068df5  jmp short 0x10068dfe
10068df7  mov dword ptr [ebp-4], 1
10068dfe  movzx eax, word ptr [ebp-4]
10068e02  cmp eax, 6
10068e05  jg 0x10068eeb
10068e0b  je 0x10068f7c
10068e11  test eax, eax
10068e13  jl 0x10068f01
10068e19  cmp eax, 1
10068e1c  jle short 0x10068e98
10068e1e  cmp eax, ebx
```

11.5 sub_1007171b at 0x1007171b

- References text "TileOffsets"
- References text "StripOffsets"
- References text "TIFFReadDirectory"
- References text "%s: cannot handle zero strip size"
- References text "%s: cannot handle zero tile size"
- References text "%s: cannot handle zero scanline size"
- References text "SamplesPerPixel tag is missing, assuming correct SamplesPerP..."
- References text "Colormap"

```
1007171b  push ebp
1007171c  mov ebp, esp
1007171e  sub esp, 0x28
10071721  push ebx
10071722  mov ebx, [ebp+8]
10071725  push esi
10071726  xor eax, eax
10071728  push edi
10071729  mov edi, [ebx+0x14]
1007172c  mov esi, ebx
1007172e  mov [ebp-8], eax
10071731  mov [ebp-0x1c], eax
10071734  mov [ebp-0x24], eax
10071737  mov [ebx+0x10], edi
1007173a  call 0x100704ba
1007173f  test eax, eax
10071741  jne short 0x1007174a
10071743  xor eax, eax
10071745  jmp 0x100724ea
1007174a  push ebx
1007174b  call dword ptr [ebx+0x23c]
10071751  inc word ptr [ebx+0x1dc]
10071758  lea edi, [ebx+0x14]
```

```
1007175b  lea eax, [ebp-8]
1007175e  push edi
1007175f  push eax
10071760  mov eax, [edi]
10071762  call 0x10070584
10071767  movzx eax, ax
1007176a  add esp, 0xc
1007176d  test ax, ax
10071770  mov [ebp-0x14], eax
10071773  jne short 0x10071793
10071775  push dword ptr [edi]
10071777  push dword ptr [ebx]
10071779  push 0x1010bb44  ; "%s: Failed to read directory at offset %u"
1007177e  push 0x1010b308  ; "TIFFReadDirectory"
10071783  push dword ptr [ebx+0x274]
10071789  call 0x1006faf1
1007178e  add esp, 0x14
10071791  jmp short 0x10071743
10071793  and dword ptr [ebx+0xc], 0xffffffbf
10071797  push ebx
10071798  call 0x10067617
1007179d  push ebx
1007179e  call 0x1006799b
100717a3  push 1
100717a5  push 0x11c
100717aa  push ebx
100717ab  call 0x1006796f
100717b0  movzx eax, word ptr [ebp-0x14]
100717b4  mov edi, [ebp-8]
100717b7  add esp, 0x14
100717ba  mov esi, eax
100717bc  jmp short 0x100717fa
100717be  test byte ptr [ebx+0xc], 0x80
100717c2  je short 0x100717da
100717c4  push 2
100717c6  push edi
100717c7  call 0x1006fb53
```

11.6 sub_100e2b50 at 0x100e2b50

- References text "PENTAX"
- References text "EPSON"
- References text "NIKON"
- References text "Canon"
- References text "Ricoh"
- References text "LEICA"
- References text "Panasonic"
- References text "FUJIFILM"

```
100e2b50  sub esp, 0x198
100e2b56  mov eax, [0x10136220]
100e2b5b  xor eax, esp
100e2b5d  mov [esp+0x194], eax
100e2b64  push ebx
100e2b65  push ebp
100e2b66  mov ebp, [esp+0x1a4]
100e2b6d  movzx eax, word ptr [ebp+0x6274]
100e2b74  mov ecx, [ebp]
100e2b77  mov ecx, [ecx]
100e2b79  push esi
100e2b7a  push edi
100e2b7b  push 0xa
100e2b7d  push 1
100e2b7f  lea edx, [esp+0x5c]
```

```
100e2b83  mov [esp+0x44], eax
100e2b87  mov eax, [ebp+4]
100e2b8a  xor esi, esi
100e2b8c  push edx
100e2b8d  push eax
100e2b8e  mov [esp+0x38], esi
100e2b92  mov [esp+0x40], esi
100e2b96  mov [esp+0x3c], esi
100e2b9a  xor edi, edi
100e2b9c  xor ebx, ebx
100e2b9e  call ecx
100e2ba0  push 3
100e2ba2  lea edx, [esp+0x68]
100e2ba6  push 0x1012e5ac
100e2bab  push edx
100e2bac  call dword ptr [0x100fe414]  ; MSVCR80.dll!strncmp
100e2bb2  add esp, 0x1c
100e2bb5  test eax, eax
100e2bb7  je 0x100e3d46
100e2bbd  push 3
100e2bbf  lea eax, [esp+0x58]
100e2bc3  push 0x1012e5a8
100e2bc8  push eax
100e2bc9  call dword ptr [0x100fe414]  ; MSVCR80.dll!strncmp
100e2bcf  add esp, 0xc
100e2bd2  test eax, eax
100e2bd4  je 0x100e3d46
100e2bda  push 4
100e2bdc  lea ecx, [esp+0x58]
100e2be0  push 0x1012e5a0
100e2be5  push ecx
100e2be6  call dword ptr [0x100fe414]  ; MSVCR80.dll!strncmp
100e2bec  add esp, 0xc
100e2bef  test eax, eax
100e2bf1  je 0x100e3d46
100e2bf7  push 4
100e2bf9  lea edx, [esp+0x58]
100e2bfd  push 0x1012e598
100e2c02  push edx
100e2c03  call dword ptr [0x100fe414]  ; MSVCR80.dll!strncmp
100e2c09  add esp, 0xc
100e2c0c  test eax, eax
100e2c0e  je 0x100e3d46
100e2c14  push 2
100e2c16  lea eax, [esp+0x58]
```

11.7 sub_100e8a00 at 0x100e8a00

- References text "Hasselblad"
- References text "%f %f %f"
- References text "Neutral "
- References text "Ixpress %d-Mp"
- References text "Imacon"
- References text "Digital Photo Professional"
- References text "Nikon Scan"
- References text "UFRaw"

```
100e8a00  push ebp
100e8a01  mov ebp, esp
100e8a03  and esp, 0xfffffff8
100e8a06  sub esp, 0x34c
100e8a0c  mov eax, [0x10136220]
100e8a11  xor eax, esp
100e8a13  mov [esp+0x348], eax
```

```
100e8a1a  fld1
100e8a1c  xor eax, eax
100e8a1e  fst qword ptr [esp+0x94], st
100e8a25  push ebx
100e8a26  mov ebx, [ebp+8]
100e8a29  fst qword ptr [esp+0xa0], st
100e8a30  fst qword ptr [esp+0xa8], st
100e8a37  push esi
100e8a38  fst qword ptr [esp+0xb4], st
100e8a3f  push edi
100e8a40  fldz
100e8a42  mov edi, [ebx+0x65e8]
100e8a48  cmp edi, 0xa
100e8a4b  fst qword ptr [esp+0x80], st
100e8a52  fst qword ptr [esp+0x88], st
100e8a59  mov dword ptr [esp+0x3c], 0x10
100e8a61  fst qword ptr [esp+0x90], st
100e8a68  mov [esp+0x44], eax
100e8a6c  fstp qword ptr [esp+0x98], st
100e8a73  mov [esp+0x2c], eax
100e8a77  mov [esp+0x24], al
100e8a7b  mov byte ptr [esp+0x25], 1
100e8a80  mov byte ptr [esp+0x26], 2
100e8a85  fst qword ptr [esp+0x68], st
100e8a89  mov byte ptr [esp+0x27], 3
100e8a8e  fst qword ptr [esp+0x70], st
100e8a92  mov [esp+0x50], eax
100e8a96  fstp qword ptr [esp+0x78], st
100e8a9a  mov [esp+0x54], eax
100e8a9e  mov [esp+0x58], eax
100e8aa2  mov [esp+0x5c], eax
100e8aa6  mov [esp+0x60], eax
100e8aaa  mov dword ptr [esp+0x64], 0xfff
100e8ab2  mov [esp+0x30], eax
100e8ab6  mov [esp+0x34], eax
100e8aba  mov [esp+0x38], eax
100e8abe  jb short 0x100e8ada
100e8ac0  mov eax, 1
100e8ac5  pop edi
100e8ac6  pop esi
100e8ac7  pop ebx
100e8ac8  mov ecx, [esp+0x348]
100e8acf  xor ecx, esp
100e8ad1  call 0x10022326
100e8ad6  mov esp, ebp
100e8ad8  pop ebp
100e8ad9  ret
100e8ada  lea ecx, [edi+1]
100e8add  mov [esp+0x40], edi
100e8ae1  mov [ebx+0x65e8], ecx
100e8ae7  mov [esp+0x10], eax
100e8aeb  lea esi, [esp+0x120]
100e8af2  jmp short 0x100e8af6
```

11.8 sub_1002ec10 at 0x1002ec10

- References text "Profile size field missing from iCCP chunk"
- References text "Ignoring iCCP chunk with declared size = %u and actual lengt..."
- References text "Ignoring nonzero compression type in iCCP chunk"
- References text "Malformed iCCP chunk"
- References text "Duplicate iCCP chunk"
- References text "Out of place iCCP chunk"
- References text "Invalid iCCP after IDAT"
- References text "Missing IHDR before iCCP"

```
1002ec10  sub esp, 0x5c
1002ec13  mov eax, [0x10136220]
1002ec18  xor eax, esp
1002ec1a  mov [esp+0x58], eax
1002ec1e  push esi
1002ec1f  mov esi, [esp+0x64]
1002ec23  mov eax, [esi+0x6c]
1002ec26  test al, 1
1002ec28  push edi
1002ec29  mov edi, [esp+0x6c]
1002ec2d  mov [esp+8], edi
1002ec31  jne short 0x1002ec3e
1002ec33  push 0x101010fc  ; "Missing IHDR before iCCP"
1002ec38  push esi
1002ec39  call 0x100231e0  ; png_error
1002ec3e  test al, 4
1002ec40  je short 0x1002ec6c
1002ec42  push 0x101010e4  ; "Invalid iCCP after IDAT"
1002ec47  push esi
1002ec48  call 0x100230f0  ; png_warning
1002ec4d  mov eax, [esp+0x78]
1002ec51  push eax
1002ec52  push esi
1002ec53  call 0x1002e050
1002ec58  add esp, 0x10
1002ec5b  pop edi
1002ec5c  pop esi
1002ec5d  mov ecx, [esp+0x58]
1002ec61  xor ecx, esp
1002ec63  call 0x10022326
1002ec68  add esp, 0x5c
1002ec6b  ret
1002ec6c  test al, 2
1002ec6e  je short 0x1002ec7e
1002ec70  push 0x101010cc  ; "Out of place iCCP chunk"
1002ec75  push esi
1002ec76  call 0x100230f0  ; png_warning
1002ec7b  add esp, 8
1002ec7e  push ebx
1002ec7f  xor ebx, ebx
1002ec81  cmp edi, ebx
1002ec83  je short 0x1002ecb9
1002ec85  test dword ptr [edi+8], 0x1000
1002ec8c  je short 0x1002ecb9
1002ec8e  push 0x101010b4  ; "Duplicate iCCP chunk"
1002ec93  push esi
1002ec94  call 0x100230f0  ; png_warning
1002ec99  mov ecx, [esp+0x7c]
1002ec9d  push ecx
1002ec9e  push esi
1002ec9f  call 0x1002e050
1002eca4  add esp, 0x10
1002eca7  pop ebx
1002eca8  pop edi
1002eca9  pop esi
1002ecaa  mov ecx, [esp+0x58]
1002ecae  xor ecx, esp
1002ecb0  call 0x10022326
1002ecb5  add esp, 0x5c
1002ecb8  ret
```

12 Assembly Walkthrough & Instruction Analysis

Observed: complete disassembly of every recovered function. Each function shows the full raw assembler listing first, followed by a plain-language summary of what the function actually does, and dual code/data analysis of any embedded literal pools.

12.1 sub_10001030 at 0x10001030 (28 instructions)

```
0x10001030: mov eax,[esp+8]
0x10001034: cmp eax,1
0x10001037: jne short 10001073h
0x10001039: mov eax,[esp+4]
0x1000103d: push eax
0x1000103e: push 1013F35Ch
0x10001043: call 10021FD8h
0x10001048: test eax,eax
0x1000104a: jne short 1000104Fh
0x1000104c: ret 0Ch
0x1000104f: push 40h
0x10001051: call 10021FDEh
0x10001056: add esp,4
0x10001059: test eax,eax
0x1000105b: je short 10001082h
0x1000105d: push 0
0x1000105f: push 1013F35Ch
0x10001064: mov ecx,eax
0x10001066: call 10021FE4h
0x1000106b: mov eax,1
0x10001070: ret 0Ch
0x10001073: test eax,eax
0x10001075: jne short 10001082h
0x10001077: push eax
0x10001078: push 1013F35Ch
0x1000107d: call 10021FEAh
0x10001082: mov eax,1
0x10001087: ret 0Ch
```

`sub_10001030` is called by sub_1002277e. It delegates to j_ORDINAL 1093, j_ORDINAL 762, j_ORDINAL 371 (4 callees).

12.2 ??0CCustomTitle@@QAE@ABV0@@Z at 0x100010c0 (6 instructions)

```
0x100010c0: mov eax,ecx
0x100010c2: mov ecx,[esp+4]
0x100010c6: mov dword ptr [eax],1012F3D4h
0x100010cc: mov edx,[ecx+4]
0x100010cf: mov [eax+4],edx
0x100010d2: ret 4
```

`??0CCustomTitle@@QAE@ABV0@@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.3 ??4CCustomTitle@@QAEAAV0@ABV0@@Z at 0x100010e0 (5 instructions)

```
0x100010e0: mov eax,ecx
0x100010e2: mov ecx,[esp+4]
0x100010e6: mov edx,[ecx+4]
0x100010e9: mov [eax+4],edx
0x100010ec: ret 4
```

`??4CCustomTitle@@QAEAAV0@ABV0@@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.4 sub_10001370 at 0x10001370 (34 instructions)

```
0x10001370: push ebp
0x10001371: lea ebp,[esp-404h]
0x10001378: sub esp,404h
0x1000137e: push 0FFFFFFFFh
0x10001380: push 100FC899h
0x10001385: mov eax,fs:[0]
0x1000138b: push eax
0x1000138c: sub esp,0Ch
0x1000138f: mov eax,ds:[10136220h]
0x10001394: xor eax,ebp
0x10001396: mov [ebp+400h],eax
0x1000139c: push ebx
0x1000139d: push esi
0x1000139e: push edi
0x1000139f: push eax
0x100013a0: lea eax,[ebp-0Ch]
0x100013a3: mov fs:[0],eax
0x100013a9: mov [ebp-10h],esp
0x100013ac: mov dword ptr [ecx],1012F480h
0x100013b2: mov dword ptr [ebp-4],0
0x100013b9: call 10021FF6h
0x100013be: mov ecx,[ebp-0Ch]
0x100013c1: mov fs:[0],ecx
0x100013c8: pop ecx
0x100013c9: pop edi
0x100013ca: pop esi
0x100013cb: pop ebx
0x100013cc: mov ecx,[ebp+400h]
0x100013d2: xor ecx,ebp
0x100013d4: call 10022326h
0x100013d9: add ebp,404h
0x100013df: mov esp,ebp
0x100013e1: pop ebp
0x100013e2: ret
```

`sub_10001370` is called by sub_10001d00, sub_10005fd0, sub_10008290, sub_1000a3c0, sub_1000a510, sub_1000a680, sub_1000aa40, sub_1000cd90. It delegates to j_ORDINAL 1925, sub_10022326 (2 callees).

12.5 ?SetDefaultTheme@CCustomTitle@@SAXW4WTitleTheme@@@Z at 0x100014a0 (3 instructions)

```
0x100014a0: mov eax,[esp+4]
0x100014a4: mov ds:[1013F370h],eax
0x100014a9: ret
```

`?SetDefaultTheme@CCustomTitle@@SAXW4WTitleTheme@@@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.6 ??0CCustomTitle@@QAE@XZ at 0x100014b0 (36 instructions)

```
0x100014b0: push 0FFFFFFFFh
0x100014b2: push 100FD38Bh
0x100014b7: mov eax,fs:[0]
0x100014bd: push eax
0x100014be: push ecx
0x100014bf: push esi
0x100014c0: push edi
0x100014c1: mov eax,ds:[10136220h]
0x100014c6: xor eax,esp
0x100014c8: push eax
0x100014c9: lea eax,[esp+10h]
0x100014cd: mov fs:[0],eax
```

```
0x100014d3: mov esi,ecx
0x100014d5: push 2A8h
0x100014da: mov dword ptr [esi],1012F3D4h
0x100014e0: call 10021FDEh
0x100014e5: add esp,4
0x100014e8: mov [esp+0Ch],eax
0x100014ec: test eax,eax
0x100014ee: mov dword ptr [esp+18h],0
0x100014f6: je short 10001508h
0x100014f8: mov ecx,ds:[1013F370h]
0x100014fe: push ecx
0x100014ff: mov edi,eax
0x10001501: call 100032E0h
0x10001506: jmp short 1000150Ah
0x10001508: xor eax,eax
0x1000150a: mov [esi+4],eax
0x1000150d: mov eax,esi
0x1000150f: mov ecx,[esp+10h]
0x10001513: mov fs:[0],ecx
0x1000151a: pop ecx
0x1000151b: pop edi
0x1000151c: pop esi
0x1000151d: add esp,10h
0x10001520: ret
```

`??0CCustomTitle@@QAE@XZ` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100032e0, j_ORDINAL 762 (2 callees).

12.7 ??0CCustomTitle@@QAE@W4WTitleTheme@@@Z at 0x10001530 (36 instructions)

```
0x10001530: push 0FFFFFFFFh
0x10001532: push 100FD38Bh
0x10001537: mov eax,fs:[0]
0x1000153d: push eax
0x1000153e: push ecx
0x1000153f: push esi
0x10001540: push edi
0x10001541: mov eax,ds:[10136220h]
0x10001546: xor eax,esp
0x10001548: push eax
0x10001549: lea eax,[esp+10h]
0x1000154d: mov fs:[0],eax
0x10001553: mov esi,ecx
0x10001555: push 2A8h
0x1000155a: mov dword ptr [esi],1012F3D4h
0x10001560: call 10021FDEh
0x10001565: add esp,4
0x10001568: mov [esp+0Ch],eax
0x1000156c: test eax,eax
0x1000156e: mov dword ptr [esp+18h],0
0x10001576: je short 10001586h
0x10001578: mov ecx,[esp+20h]
0x1000157c: push ecx
0x1000157d: mov edi,eax
0x1000157f: call 100032E0h
0x10001584: jmp short 10001588h
0x10001586: xor eax,eax
0x10001588: mov [esi+4],eax
0x1000158b: mov eax,esi
0x1000158d: mov ecx,[esp+10h]
0x10001591: mov fs:[0],ecx
0x10001598: pop ecx
0x10001599: pop edi
0x1000159a: pop esi
```

```
0x1000159b: add esp,10h
0x1000159e: ret 4
```

`??0CCustomTitle@@QAE@W4WTitleTheme@@@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100032e0, j_ORDINAL 762 (2 callees).

12.8 ??1CCustomTitle@@UAE@XZ at 0x100015b0 (9 instructions)

```
0x100015b0: mov dword ptr [ecx],1012F3D4h
0x100015b6: mov ecx,[ecx+4]
0x100015b9: test ecx,ecx
0x100015bb: je short 100015C5h
0x100015bd: mov eax,[ecx]
0x100015bf: mov edx,[eax]
0x100015c1: push 1
0x100015c3: call edx
0x100015c5: ret
```

`??1CCustomTitle@@UAE@XZ` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.9 ?SubClassWnd@CCustomTitle@@QAEHPAVCWnd@@HH@Z at 0x100015d0 (8 instructions)

```
0x100015d0: mov eax,[esp+0Ch]
0x100015d4: mov edx,[esp+8]
0x100015d8: mov ecx,[ecx+4]
0x100015db: push eax
0x100015dc: mov eax,[esp+8]
0x100015e0: push edx
0x100015e1: call 10003670h
0x100015e6: ret 0Ch
```

`?SubClassWnd@CCustomTitle@@QAEHPAVCWnd@@HH@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10003670 (1 callees).

12.10 ?UnsubClassWnd@CCustomTitle@@QAEXXZ at 0x100015f0 (5 instructions)

```
0x100015f0: push esi
0x100015f1: mov esi,[ecx+4]
0x100015f4: call 10003A40h
0x100015f9: pop esi
0x100015fa: ret
```

`?UnsubClassWnd@CCustomTitle@@QAEXXZ` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10003a40 (1 callees).

12.11 ?IsSubClassed@CCustomTitle@@QBEHXZ at 0x10001600 (6 instructions)

```
0x10001600: mov eax,[ecx+4]
0x10001603: xor ecx,ecx
0x10001605: cmp [eax+0Ch],ecx
0x10001608: setne cl
0x1000160b: mov eax,ecx
0x1000160d: ret
```

`?IsSubClassed@CCustomTitle@@QBEHXZ` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.12 ?SetMX_NCTITLE@CCustomTitle@@QAEXH@Z at 0x10001610 (4 instructions)

```
0x10001610: mov eax,[ecx+4]
0x10001613: mov ecx,[esp+4]
0x10001617: mov [eax+4],ecx
0x1000161a: ret 4
```

`?SetMX_NCTITLE@CCustomTitle@@QAEXH@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.13 ?RefreshFrame@CCustomTitle@@QAEHXZ at 0x10001620 (10 instructions)

```
0x10001620: mov ecx,[ecx+4]
0x10001623: mov ecx,[ecx+8]
0x10001626: push 37h
0x10001628: push 0
0x1000162a: push 0
0x1000162c: push 0
0x1000162e: push 0
0x10001630: push 0
0x10001632: call 1002226Ch
0x10001637: ret
```

`?RefreshFrame@CCustomTitle@@QAEHXZ` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to j_ORDINAL 6061 (1 callees).

12.14 ?SetMenu@CCustomTitle@@QAEHPAVCMenu@@@Z at 0x10001640 (7 instructions)

```
0x10001640: mov eax,[esp+4]
0x10001644: push esi
0x10001645: mov esi,[ecx+4]
0x10001648: push eax
0x10001649: call 10003B30h
0x1000164e: pop esi
0x1000164f: ret 4
```

`?SetMenu@CCustomTitle@@QAEHPAVCMenu@@@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10003b30 (1 callees).

12.15 ?GetMenu@CCustomTitle@@QAEPAVCMenu@@XZ at 0x10001660 (5 instructions)

```
0x10001660: mov ecx,[ecx+4]
0x10001663: mov eax,[ecx+48h]
0x10001666: push eax
0x10001667: call 1002227Eh
0x1000166c: ret
```

`?GetMenu@CCustomTitle@@QAEPAVCMenu@@XZ` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to j_ORDINAL 2365 (1 callees).

12.16 ?GetQuickButtonCount@CCustomTitle@@QAEHXZ at 0x10001670 (11 instructions)

```
0x10001670: mov eax,[ecx+4]
0x10001673: mov ecx,[eax+0E0h]
0x10001679: add eax,0DCh
0x1000167e: test ecx,ecx
```

```
0x10001680: jne short 10001685h
0x10001682: xor eax,eax
0x10001684: ret
0x10001685: mov eax,[eax+8]
0x10001688: sub eax,ecx
0x1000168a: sar eax,2
0x1000168d: ret
```

`?GetQuickButtonCount@CCustomTitle@@QAEHXZ` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.17 ?AddQuickButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0PB_W@Z at 0x10001690 (48 instructions)

```
0x10001690: push ebp
0x10001691: mov ebp,esp
0x10001693: and esp,0FFFFFFF8h
0x10001696: mov eax,[ebp+10h]
0x10001699: sub esp,10h
0x1000169c: push esi
0x1000169d: mov esi,[ecx+4]
0x100016a0: mov ecx,[ebp+14h]
0x100016a3: test ecx,ecx
0x100016a5: push edi
0x100016a6: je short 100016BCh
0x100016a8: mov edx,[ebp+0Ch]
0x100016ab: push 1
0x100016ad: push eax
0x100016ae: mov eax,[ebp+8]
0x100016b1: push edx
0x100016b2: push eax
0x100016b3: lea edx,[esi+0D8h]
0x100016b9: push edx
0x100016ba: jmp short 100016D0h
0x100016bc: mov ecx,[ebp+0Ch]
0x100016bf: mov edx,[ebp+8]
0x100016c2: push 0
0x100016c4: push eax
0x100016c5: push ecx
0x100016c6: push edx
0x100016c7: lea eax,[esi+0D8h]
0x100016cd: push eax
0x100016ce: xor ecx,ecx
0x100016d0: call 1000ACA0h
0x100016d5: mov edi,eax
0x100016d7: test edi,edi
0x100016d9: jl short 100016F7h
0x100016db: mov edx,[esi+8]
0x100016de: mov eax,[edx+20h]
0x100016e1: lea ecx,[esp+8]
0x100016e5: push ecx
0x100016e6: push eax
0x100016e7: call dword ptr ds:[100FE674h]
0x100016ed: push esi
0x100016ee: lea eax,[esp+0Ch]
0x100016f2: call 100056C0h
0x100016f7: mov eax,edi
0x100016f9: pop edi
0x100016fa: pop esi
0x100016fb: mov esp,ebp
0x100016fd: pop ebp
0x100016fe: ret 10h
```

`?AddQuickButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0PB_W@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100056c0, sub_1000aca0 (2 callees).

12.18 ?AddQuickButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0KKKK@Z at 0x10001710 (41 instructions)

```
0x10001710: push ebp
0x10001711: mov ebp,esp
0x10001713: and esp,0FFFFFFF8h
0x10001716: mov eax,[ebp+20h]
0x10001719: mov edx,[ebp+18h]
0x1000171c: sub esp,10h
0x1000171f: push esi
0x10001720: mov esi,[ecx+4]
0x10001723: mov ecx,[ebp+1Ch]
0x10001726: push edi
0x10001727: push eax
0x10001728: mov eax,[ebp+14h]
0x1000172b: push ecx
0x1000172c: mov ecx,[ebp+10h]
0x1000172f: push edx
0x10001730: mov edx,[ebp+0Ch]
0x10001733: push eax
0x10001734: mov eax,[ebp+8]
0x10001737: push ecx
0x10001738: push edx
0x10001739: push eax
0x1000173a: lea ecx,[esi+0D8h]
0x10001740: call 1000AD90h
0x10001745: mov edi,eax
0x10001747: test edi,edi
0x10001749: jl short 10001769h
0x1000174b: mov edx,[esi+8]
0x1000174e: mov eax,[edx+20h]
0x10001751: lea ecx,[esp+8]
0x10001755: push ecx
0x10001756: push eax
0x10001757: call dword ptr ds:[100FE674h]
0x1000175d: push esi
0x1000175e: lea eax,[esp+0Ch]
0x10001762: call 100056C0h
0x10001767: mov eax,edi
0x10001769: pop edi
0x1000176a: pop esi
0x1000176b: mov esp,ebp
0x1000176d: pop ebp
0x1000176e: ret 1Ch
```

`?AddQuickButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0KKKK@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100056c0, sub_1000ad90 (2 callees).

12.19 ?SetQuickButtonTextColor@CCustomTitle@@QAEXIKKKK@Z at 0x10001780 (17 instructions)

```
0x10001780: mov eax,[esp+14h]
0x10001784: mov edx,[esp+10h]
0x10001788: push ebx
0x10001789: mov ebx,[esp+8]
0x1000178d: push esi
```

```
0x1000178e: mov esi,[ecx+4]
0x10001791: push eax
0x10001792: mov eax,[esp+18h]
0x10001796: push edx
0x10001797: mov edx,[esp+18h]
0x1000179b: push eax
0x1000179c: push edx
0x1000179d: add esi,0D8h
0x100017a3: call 1000AE90h
0x100017a8: pop esi
0x100017a9: pop ebx
0x100017aa: ret 14h
```

`?SetQuickButtonTextColor@CCustomTitle@@QAEXIKKKK@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1000ae90 (1 callees).

12.20 ?DeleteQuickButton@CCustomTitle@@QAEHH@Z at 0x100017b0 (30 instructions)

```
0x100017b0: push ebp
0x100017b1: mov ebp,esp
0x100017b3: and esp,0FFFFFFF8h
0x100017b6: sub esp,14h
0x100017b9: push ebx
0x100017ba: mov ebx,[ebp+8]
0x100017bd: push esi
0x100017be: push edi
0x100017bf: mov edi,[ecx+4]
0x100017c2: lea esi,[edi+0D8h]
0x100017c8: call 1000AF00h
0x100017cd: mov esi,eax
0x100017cf: test esi,esi
0x100017d1: je short 100017F1h
0x100017d3: mov ecx,[edi+8]
0x100017d6: mov edx,[ecx+20h]
0x100017d9: lea eax,[esp+10h]
0x100017dd: push eax
0x100017de: push edx
0x100017df: call dword ptr ds:[100FE674h]
0x100017e5: push edi
0x100017e6: lea eax,[esp+14h]
0x100017ea: call 100056C0h
0x100017ef: mov eax,esi
0x100017f1: pop edi
0x100017f2: pop esi
0x100017f3: pop ebx
0x100017f4: mov esp,ebp
0x100017f6: pop ebp
0x100017f7: ret 4
```

`?DeleteQuickButton@CCustomTitle@@QAEHH@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100056c0, sub_1000af00 (2 callees).

12.21 ?RemoveAllQuickButtons@CCustomTitle@@QAEXXZ at 0x10001800 (21 instructions)

```
0x10001800: push ebp
0x10001801: mov ebp,esp
0x10001803: and esp,0FFFFFFF8h
0x10001806: sub esp,14h
0x10001809: push esi
0x1000180a: mov esi,[ecx+4]
0x1000180d: lea ecx,[esi+0D8h]
0x10001813: call 1000AFB0h
```

```
0x10001818: mov ecx,[esi+8]
0x1000181b: mov edx,[ecx+20h]
0x1000181e: lea eax,[esp+8]
0x10001822: push eax
0x10001823: push edx
0x10001824: call dword ptr ds:[100FE674h]
0x1000182a: push esi
0x1000182b: lea eax,[esp+0Ch]
0x1000182f: call 100056C0h
0x10001834: pop esi
0x10001835: mov esp,ebp
0x10001837: pop ebp
0x10001838: ret
```

`?RemoveAllQuickButtons@CCustomTitle@@QAEXXZ` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100056c0, sub_1000afb0 (2 callees).

12.22 ?SetControlBtnsBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z at 0x10001840 (9 instructions)

```
0x10001840: mov ecx,[ecx+4]
0x10001843: push ebx
0x10001844: mov ebx,[esp+8]
0x10001848: push esi
0x10001849: lea esi,[ecx+124h]
0x1000184f: call 10007670h
0x10001854: pop esi
0x10001855: pop ebx
0x10001856: ret 4
```

`?SetControlBtnsBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10007670 (1 callees).

12.23 ?SetSysmenuIcon@CCustomTitle@@QAEHPAUHICON__@@@Z at 0x10001860 (31 instructions)

```
0x10001860: push esi
0x10001861: mov esi,[ecx+4]
0x10001864: push edi
0x10001865: mov edi,[esp+0Ch]
0x10001869: add esi,18h
0x1000186c: test edi,edi
0x1000186e: jne short 10001877h
0x10001870: pop edi
0x10001871: xor eax,eax
0x10001873: pop esi
0x10001874: ret 4
0x10001877: mov eax,[esi+8]
0x1000187a: test eax,eax
0x1000187c: je short 10001885h
0x1000187e: push eax
0x1000187f: call dword ptr ds:[100FE048h]
0x10001885: mov eax,[esi+4]
0x10001888: test eax,eax
0x1000188a: mov dword ptr [esi+8],0
0x10001891: je short 1000189Ah
0x10001893: push eax
0x10001894: call dword ptr ds:[100FE5ACh]
0x1000189a: mov eax,10h
0x1000189f: mov [esi+4],edi
0x100018a2: mov ecx,eax
0x100018a4: mov [esi+10h],eax
0x100018a7: pop edi
0x100018a8: mov [esi+14h],ecx
```

```
0x100018ab: lea eax,[ecx-0Fh]
0x100018ae: pop esi
0x100018af: ret 4
```

`?SetSysmenuIcon@CCustomTitle@@QAEHPAUHICON__@@@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.24 ?SetSysmenuBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z at 0x100018c0 (9 instructions)

```
0x100018c0: mov ecx,[ecx+4]
0x100018c3: push esi
0x100018c4: push edi
0x100018c5: mov edi,[esp+0Ch]
0x100018c9: lea esi,[ecx+18h]
0x100018cc: call 10009FC0h
0x100018d1: pop edi
0x100018d2: pop esi
0x100018d3: ret 4
```

`?SetSysmenuBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10009fc0 (1 callees).

12.25 ?SetQuickBarBreakImage@CCustomTitle@@QAEHPAVCBitmap@@K@Z at 0x100018e0 (11 instructions)

```
0x100018e0: mov ecx,[ecx+4]
0x100018e3: mov eax,[esp+8]
0x100018e7: push esi
0x100018e8: push edi
0x100018e9: mov edi,[esp+0Ch]
0x100018ed: push eax
0x100018ee: lea esi,[ecx+0D8h]
0x100018f4: call 1000AB10h
0x100018f9: pop edi
0x100018fa: pop esi
0x100018fb: ret 8
```

`?SetQuickBarBreakImage@CCustomTitle@@QAEHPAVCBitmap@@K@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1000ab10 (1 callees).

12.26 ?SetMenuBarBreakImage@CCustomTitle@@QAEHPAVCBitmap@@K@Z at 0x10001900 (11 instructions)

```
0x10001900: mov ecx,[ecx+4]
0x10001903: mov eax,[esp+8]
0x10001907: push esi
0x10001908: push edi
0x10001909: mov edi,[esp+0Ch]
0x1000190d: push eax
0x1000190e: lea esi,[ecx+40h]
0x10001911: call 100083E0h
0x10001916: pop edi
0x10001917: pop esi
0x10001918: ret 8
```

`?SetMenuBarBreakImage@CCustomTitle@@QAEHPAVCBitmap@@K@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100083e0 (1 callees).

12.27 ?SetMenuBarDownBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z at 0x10001920 (14 instructions)

```
0x10001920: mov eax,[esp+4]
0x10001924: test eax,eax
0x10001926: mov ecx,[ecx+4]
0x10001929: je short 10001932h
0x1000192b: mov eax,[eax+4]
0x1000192e: test eax,eax
0x10001930: jne short 10001937h
0x10001932: xor eax,eax
0x10001934: ret 4
0x10001937: push eax
0x10001938: add ecx,9Ch
0x1000193e: call 10021FFCh
0x10001943: mov eax,1
0x10001948: ret 4
```

`?SetMenuBarDownBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to j_ORDINAL 1271 (1 callees).

12.28 ?SetMenuLeftMargin@CCustomTitle@@QAEXH@Z at 0x10001950 (4 instructions)

```
0x10001950: mov eax,[ecx+4]
0x10001953: mov ecx,[esp+4]
0x10001957: mov [eax+60h],ecx
0x1000195a: ret 4
```

`?SetMenuLeftMargin@CCustomTitle@@QAEXH@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.29 ?SetMenuTextColor@CCustomTitle@@QAEXKKK@Z at 0x10001960 (7 instructions)

```
0x10001960: mov eax,[esp+4]
0x10001964: mov ecx,[esp+8]
0x10001968: mov edx,[esp+0Ch]
0x1000196c: mov ds:[1013F380h],eax
0x10001971: mov ds:[1013F384h],ecx
0x10001977: mov ds:[10138C10h],edx
0x1000197d: ret 0Ch
```

`?SetMenuTextColor@CCustomTitle@@QAEXKKK@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.30 ?SetCaptionTextColor@CCustomTitle@@QAEXK@Z at 0x10001980 (8 instructions)

```
0x10001980: mov eax,[ecx+4]
0x10001983: mov ecx,[esp+4]
0x10001987: mov [eax+17Ch],ecx
0x1000198d: mov [eax+180h],ecx
0x10001993: mov [eax+184h],ecx
0x10001999: mov [eax+188h],ecx
0x1000199f: mov [eax+21Ch],ecx
0x100019a5: ret 4
```

`?SetCaptionTextColor@CCustomTitle@@QAEXK@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.31 ?SetCaptionText@CCustomTitle@@QAEXABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@@Z at 0x100019b0 (16 instructions)

```
0x100019b0: push esi
0x100019b1: mov esi,[ecx+4]
0x100019b4: push edi
0x100019b5: mov edi,[esp+0Ch]
0x100019b9: push edi
0x100019ba: lea ecx,[esi+1F4h]
0x100019c0: call dword ptr ds:[100FE198h]
0x100019c6: add esi,178h
0x100019cc: push edi
0x100019cd: lea ecx,[esi+24h]
0x100019d0: call dword ptr ds:[100FE198h]
0x100019d6: mov ecx,esi
0x100019d8: call 1000C0F0h
0x100019dd: pop edi
0x100019de: pop esi
0x100019df: ret 4
```

`?SetCaptionText@CCustomTitle@@QAEXABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1000c0f0 (1 callees).

12.32 ?SetPageInfoText@CCustomTitle@@QAEXABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@@Z at 0x100019f0 (6 instructions)

```
0x100019f0: mov eax,[esp+4]
0x100019f4: mov edx,[eax]
0x100019f6: mov ecx,[ecx+4]
0x100019f9: mov [esp+4],edx
0x100019fd: add ecx,200h
0x10001a03: jmp dword ptr ds:[100FE1C4h]
```

`?SetPageInfoText@CCustomTitle@@QAEXABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.33 ?SetCaptionTextMargin@CCustomTitle@@QAEXHH@Z at 0x10001a10 (6 instructions)

```
0x10001a10: mov eax,[ecx+4]
0x10001a13: mov ecx,[esp+4]
0x10001a17: mov edx,[esp+8]
0x10001a1b: mov [eax+1F8h],ecx
0x10001a21: mov [eax+1FCh],edx
0x10001a27: ret 8
```

`?SetCaptionTextMargin@CCustomTitle@@QAEXHH@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.34 ?SetCaptionBkImage@CCustomTitle@@QAEHPAVCBitmap@@H@Z at 0x10001a30 (8 instructions)

```
0x10001a30: mov eax,[esp+4]
0x10001a34: mov edx,[esp+8]
0x10001a38: push esi
0x10001a39: mov esi,[ecx+4]
0x10001a3c: push eax
```

```
0x10001a3d: call 10003C30h
0x10001a42: pop esi
0x10001a43: ret 8
```

`?SetCaptionBkImage@CCustomTitle@@QAEHPAVCBitmap@@H@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10003c30 (1 callees).

12.35 ?SetToolBarBkImage@CCustomTitle@@QAEHPAVCBitmap@@H@Z at 0x10001a50 (9 instructions)

```
0x10001a50: mov eax,[esp+8]
0x10001a54: push ebx
0x10001a55: mov ebx,[ecx+4]
0x10001a58: push edi
0x10001a59: mov edi,[esp+0Ch]
0x10001a5d: call 10003CA0h
0x10001a62: pop edi
0x10001a63: pop ebx
0x10001a64: ret 8
```

`?SetToolBarBkImage@CCustomTitle@@QAEHPAVCBitmap@@H@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10003ca0 (1 callees).

12.36 ?GetToolButtonCount@CCustomTitle@@QAEHXZ at 0x10001a70 (11 instructions)

```
0x10001a70: mov eax,[ecx+4]
0x10001a73: mov ecx,[eax+1C8h]
0x10001a79: add eax,1C4h
0x10001a7e: test ecx,ecx
0x10001a80: jne short 10001A85h
0x10001a82: xor eax,eax
0x10001a84: ret
0x10001a85: mov eax,[eax+8]
0x10001a88: sub eax,ecx
0x10001a8a: sar eax,2
0x10001a8d: ret
```

`?GetToolButtonCount@CCustomTitle@@QAEHXZ` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.37 ?SetToolBarLeftMargin@CCustomTitle@@QAEXH@Z at 0x10001a90 (4 instructions)

```
0x10001a90: mov eax,[ecx+4]
0x10001a93: mov ecx,[esp+4]
0x10001a97: mov [eax+1DCh],ecx
0x10001a9d: ret 4
```

`?SetToolBarLeftMargin@CCustomTitle@@QAEXH@Z` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.38 ?AddToolButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0PB_W@Z at 0x10001aa0 (48 instructions)

```
0x10001aa0: push ebp
0x10001aa1: mov ebp,esp
```

```
0x10001aa3: and esp,0FFFFFFF8h
0x10001aa6: mov eax,[ebp+10h]
0x10001aa9: sub esp,10h
0x10001aac: push esi
0x10001aad: mov esi,[ecx+4]
0x10001ab0: mov ecx,[ebp+14h]
0x10001ab3: test ecx,ecx
0x10001ab5: push edi
0x10001ab6: je short 10001ACCh
0x10001ab8: mov edx,[ebp+0Ch]
0x10001abb: push 1
0x10001abd: push eax
0x10001abe: mov eax,[ebp+8]
0x10001ac1: push edx
0x10001ac2: push eax
0x10001ac3: lea edx,[esi+1C0h]
0x10001ac9: push edx
0x10001aca: jmp short 10001AE0h
0x10001acc: mov ecx,[ebp+0Ch]
0x10001acf: mov edx,[ebp+8]
0x10001ad2: push 0
0x10001ad4: push eax
0x10001ad5: push ecx
0x10001ad6: push edx
0x10001ad7: lea eax,[esi+1C0h]
0x10001add: push eax
0x10001ade: xor ecx,ecx
0x10001ae0: call 1000D330h
0x10001ae5: mov edi,eax
0x10001ae7: test edi,edi
0x10001ae9: jl short 10001B07h
0x10001aeb: mov edx,[esi+8]
0x10001aee: mov eax,[edx+20h]
0x10001af1: lea ecx,[esp+8]
0x10001af5: push ecx
0x10001af6: push eax
0x10001af7: call dword ptr ds:[100FE674h]
0x10001afd: push esi
0x10001afe: lea eax,[esp+0Ch]
0x10001b02: call 100056C0h
0x10001b07: mov eax,edi
0x10001b09: pop edi
0x10001b0a: pop esi
0x10001b0b: mov esp,ebp
0x10001b0d: pop ebp
0x10001b0e: ret 10h
```

`?AddToolButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0PB_W@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100056c0, sub_1000d330 (2 callees).

12.39 ?AddToolButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0KKKK@Z at 0x10001b20 (41 instructions)

```
0x10001b20: push ebp
0x10001b21: mov ebp,esp
0x10001b23: and esp,0FFFFFFF8h
0x10001b26: mov eax,[ebp+20h]
0x10001b29: mov edx,[ebp+18h]
0x10001b2c: sub esp,10h
0x10001b2f: push esi
0x10001b30: mov esi,[ecx+4]
```

```
0x10001b33: mov ecx,[ebp+1Ch]
0x10001b36: push edi
0x10001b37: push eax
0x10001b38: mov eax,[ebp+14h]
0x10001b3b: push ecx
0x10001b3c: mov ecx,[ebp+10h]
0x10001b3f: push edx
0x10001b40: mov edx,[ebp+0Ch]
0x10001b43: push eax
0x10001b44: mov eax,[ebp+8]
0x10001b47: push ecx
0x10001b48: push edx
0x10001b49: push eax
0x10001b4a: lea ecx,[esi+1C0h]
0x10001b50: call 1000D430h
0x10001b55: mov edi,eax
0x10001b57: test edi,edi
0x10001b59: jl short 10001B79h
0x10001b5b: mov edx,[esi+8]
0x10001b5e: mov eax,[edx+20h]
0x10001b61: lea ecx,[esp+8]
0x10001b65: push ecx
0x10001b66: push eax
0x10001b67: call dword ptr ds:[100FE674h]
0x10001b6d: push esi
0x10001b6e: lea eax,[esp+0Ch]
0x10001b72: call 100056C0h
0x10001b77: mov eax,edi
0x10001b79: pop edi
0x10001b7a: pop esi
0x10001b7b: mov esp,ebp
0x10001b7d: pop ebp
0x10001b7e: ret 1Ch
```

`?AddToolButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@ATL@@0KKKK@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100056c0, sub_1000d430 (2 callees).

12.40 ?AddToolSeparator@CCustomTitle@@QAEHK@Z at 0x10001b90 (30 instructions)

```
0x10001b90: push ebp
0x10001b91: mov ebp,esp
0x10001b93: and esp,0FFFFFFF8h
0x10001b96: mov eax,[ebp+8]
0x10001b99: sub esp,10h
0x10001b9c: push esi
0x10001b9d: mov esi,[ecx+4]
0x10001ba0: push edi
0x10001ba1: push eax
0x10001ba2: lea ecx,[esi+1C0h]
0x10001ba8: push ecx
0x10001ba9: call 1000D550h
0x10001bae: mov edi,eax
0x10001bb0: test edi,edi
0x10001bb2: jl short 10001BD2h
0x10001bb4: mov eax,[esi+8]
0x10001bb7: mov ecx,[eax+20h]
0x10001bba: lea edx,[esp+8]
0x10001bbe: push edx
0x10001bbf: push ecx
0x10001bc0: call dword ptr ds:[100FE674h]
0x10001bc6: push esi
0x10001bc7: lea eax,[esp+0Ch]
0x10001bcb: call 100056C0h
```

```
0x10001bd0: mov eax,edi
0x10001bd2: pop edi
0x10001bd3: pop esi
0x10001bd4: mov esp,ebp
0x10001bd6: pop ebp
0x10001bd7: ret 4
```

`?AddToolSeparator@CCustomTitle@@QAEHK@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100056c0, sub_1000d550 (2 callees).

12.41 ?SetToolButtonTextColor@CCustomTitle@@QAEXIKKKK@Z at 0x10001be0 (17 instructions)

```
0x10001be0: mov eax,[esp+14h]
0x10001be4: mov edx,[esp+10h]
0x10001be8: push ebx
0x10001be9: mov ebx,[esp+8]
0x10001bed: push esi
0x10001bee: mov esi,[ecx+4]
0x10001bf1: push eax
0x10001bf2: mov eax,[esp+18h]
0x10001bf6: push edx
0x10001bf7: mov edx,[esp+18h]
0x10001bfb: push eax
0x10001bfc: push edx
0x10001bfd: add esi,1C0h
0x10001c03: call 1000D650h
0x10001c08: pop esi
0x10001c09: pop ebx
0x10001c0a: ret 14h
```

`?SetToolButtonTextColor@CCustomTitle@@QAEXIKKKK@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1000d650 (1 callees).

12.42 ?SetToolSeparatorColor@CCustomTitle@@QAEXK@Z at 0x10001c10 (9 instructions)

```
0x10001c10: push ebx
0x10001c11: mov ebx,[esp+8]
0x10001c15: push esi
0x10001c16: mov esi,[ecx+4]
0x10001c19: add esi,1C0h
0x10001c1f: call 1000D6D0h
0x10001c24: pop esi
0x10001c25: pop ebx
0x10001c26: ret 4
```

`?SetToolSeparatorColor@CCustomTitle@@QAEXK@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1000d6d0 (1 callees).

12.43 ?RemoveAllToolButtons@CCustomTitle@@QAEXXZ at 0x10001c30 (69 instructions)

```
0x10001c30: mov ecx,[ecx+4]
0x10001c33: add ecx,1C0h
0x10001c39: jmp near ptr 1000D730h
0x1000d730: push 0FFFFFFFFh
0x1000d732: push 100FCCA8h
0x1000d737: mov eax,fs:[0]
0x1000d73d: push eax
0x1000d73e: push ecx
0x1000d73f: push ebx
```

```
0x1000d740: push esi
0x1000d741: push edi
0x1000d742: mov eax,ds:[10136220h]
0x1000d747: xor eax,esp
0x1000d749: push eax
0x1000d74a: lea eax,[esp+14h]
0x1000d74e: mov fs:[0],eax
0x1000d754: mov ebx,ecx
0x1000d756: xor edi,edi
0x1000d758: add ebx,4
0x1000d75b: jmp short 1000D760h
0x1000d760: mov ecx,[ebx+4]
0x1000d763: test ecx,ecx
0x1000d765: jne short 1000D76Bh
0x1000d767: xor eax,eax
0x1000d769: jmp short 1000D773h
0x1000d76b: mov eax,[ebx+8]
0x1000d76e: sub eax,ecx
0x1000d770: sar eax,2
0x1000d773: cmp edi,eax
0x1000d775: jge short 1000D7E0h
0x1000d777: test ecx,ecx
0x1000d779: je short 1000D787h
0x1000d77b: mov eax,[ebx+8]
0x1000d77e: sub eax,ecx
0x1000d780: sar eax,2
0x1000d783: cmp edi,eax
0x1000d785: jb short 1000D78Dh
0x1000d787: call dword ptr ds:[100FE584h]
0x1000d78d: mov eax,[ebx+4]
0x1000d790: mov esi,[eax+edi*4]
0x1000d793: test esi,esi
0x1000d795: mov [esp+10h],esi
0x1000d799: je short 1000D7DBh
0x1000d79b: push 1000C970h
0x1000d7a0: push 4
0x1000d7a2: push 3D0h
0x1000d7a7: lea ecx,[esi+0Ch]
0x1000d7aa: push ecx
0x1000d7ab: mov dword ptr [esp+2Ch],1
0x1000d7b3: call 100223DCh
0x1000d7b8: lea ecx,[esi+8]
0x1000d7bb: call dword ptr ds:[100FE374h]
0x1000d7c1: lea ecx,[esi+4]
0x1000d7c4: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000d7cc: call dword ptr ds:[100FE374h]
0x1000d7d2: push esi
0x1000d7d3: call 10021FCCh
0x1000d7d8: add esp,4
0x1000d7db: add edi,1
0x1000d7de: jmp short 1000D760h
0x1000d7e0: call 1000BA40h
0x1000d7e5: mov ecx,[esp+14h]
0x1000d7e9: mov fs:[0],ecx
0x1000d7f0: pop ecx
0x1000d7f1: pop edi
0x1000d7f2: pop esi
0x1000d7f3: pop ebx
0x1000d7f4: add esp,10h
0x1000d7f7: ret
```

`?RemoveAllToolButtons@CCustomTitle@@QAEXXZ` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1000ba40, j_ORDINAL 764, sub_100223dc (3 callees).

12.44 ?IsToolButtonChecked@CCustomTitle@@QAEHI@Z at 0x10001c40 (9 instructions)

```
0x10001c40: mov ecx,[ecx+4]
0x10001c43: push ebx
0x10001c44: mov ebx,[esp+8]
0x10001c48: push esi
0x10001c49: lea esi,[ecx+1C0h]
0x10001c4f: call 1000DF60h
0x10001c54: pop esi
0x10001c55: pop ebx
0x10001c56: ret 4
```

`?IsToolButtonChecked@CCustomTitle@@QAEHI@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1000df60 (1 callees).

12.45 ?CheckToolButton@CCustomTitle@@QAEXI@Z at 0x10001c60 (24 instructions)

```
0x10001c60: push ebp
0x10001c61: mov ebp,esp
0x10001c63: and esp,0FFFFFFF8h
0x10001c66: mov eax,[ebp+8]
0x10001c69: sub esp,14h
0x10001c6c: push ebx
0x10001c6d: push esi
0x10001c6e: push edi
0x10001c6f: mov edi,[ecx+4]
0x10001c72: lea esi,[edi+1C0h]
0x10001c78: push eax
0x10001c79: call 1000DFE0h
0x10001c7e: test eax,eax
0x10001c80: je short 10001C92h
0x10001c82: lea ebx,[esp+10h]
0x10001c86: call 1000D8E0h
0x10001c8b: mov ecx,eax
0x10001c8d: call 10006310h
0x10001c92: pop edi
0x10001c93: pop esi
0x10001c94: pop ebx
0x10001c95: mov esp,ebp
0x10001c97: pop ebp
0x10001c98: ret 4
```

`?CheckToolButton@CCustomTitle@@QAEXI@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10006310, sub_1000d8e0, sub_1000dfe0 (3 callees).

12.46 ?UncheckAllToolButton@CCustomTitle@@QAEXXZ at 0x10001ca0 (20 instructions)

```
0x10001ca0: push ebp
0x10001ca1: mov ebp,esp
0x10001ca3: and esp,0FFFFFFF8h
0x10001ca6: sub esp,14h
0x10001ca9: push ebx
0x10001caa: push esi
0x10001cab: push edi
0x10001cac: mov edi,[ecx+4]
0x10001caf: lea esi,[edi+1C0h]
0x10001cb5: call 1000D2D0h
0x10001cba: lea ebx,[esp+10h]
0x10001cbe: call 1000D8E0h
0x10001cc3: mov ecx,eax
```

```
0x10001cc5: call 10006310h
0x10001cca: pop edi
0x10001ccb: pop esi
0x10001ccc: pop ebx
0x10001ccd: mov esp,ebp
0x10001ccf: pop ebp
0x10001cd0: ret
```

`?UncheckAllToolButton@CCustomTitle@@QAEXXZ` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10006310, sub_1000d2d0, sub_1000d8e0 (3 callees).

12.47 ?ShowTitle@CCustomTitle@@QAEXH@Z at 0x10001ce0 (6 instructions)

```
0x10001ce0: mov eax,[esp+4]
0x10001ce4: push esi
0x10001ce5: mov esi,[ecx+4]
0x10001ce8: call 10003AC0h
0x10001ced: pop esi
0x10001cee: ret 4
```

`?ShowTitle@CCustomTitle@@QAEXH@Z` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10003ac0 (1 callees).

12.48 sub_10001d00 at 0x10001d00 (268 instructions)

```
0x10001d00: push 0FFFFFFFFh
0x10001d02: push 100FCFD8h
0x10001d07: mov eax,fs:[0]
0x10001d0d: push eax
0x10001d0e: sub esp,40h
0x10001d11: push ebx
0x10001d12: push ebp
0x10001d13: push esi
0x10001d14: mov eax,ds:[10136220h]
0x10001d19: xor eax,esp
0x10001d1b: push eax
0x10001d1c: lea eax,[esp+50h]
0x10001d20: mov fs:[0],eax
0x10001d26: mov eax,[esp+64h]
0x10001d2a: mov esi,ecx
0x10001d2c: mov ecx,[eax]
0x10001d2e: mov ebx,[eax+4]
0x10001d31: mov edx,[eax+8]
0x10001d34: mov eax,[eax+0Ch]
0x10001d37: xor ebp,ebp
0x10001d39: cmp ecx,ebp
0x10001d3b: mov [esp+20h],ecx
0x10001d3f: mov [esp+24h],ebx
0x10001d43: mov [esp+28h],edx
0x10001d47: mov [esp+2Ch],eax
0x10001d4b: jne short 10001D73h
0x10001d4d: cmp edx,ebp
0x10001d4f: jne short 10001D73h
0x10001d51: cmp ebx,ebp
0x10001d53: jne short 10001D73h
0x10001d55: cmp eax,ebp
0x10001d57: jne short 10001D73h
0x10001d59: mov eax,[esi+0Ch]
0x10001d5c: sub eax,[esi+4]
0x10001d5f: mov ecx,[esi+8]
0x10001d62: sub ecx,[esi]
0x10001d64: push eax
0x10001d65: push ecx
0x10001d66: push ebp
0x10001d67: push ebp
```

```
0x10001d68: lea eax,[esp+30h]
0x10001d6c: push eax
0x10001d6d: call dword ptr ds:[100FE69Ch]
0x10001d73: lea ecx,[esp+30h]
0x10001d77: call 1002202Ch
0x10001d7c: cmp edi,ebp
0x10001d7e: mov [esp+58h],ebp
0x10001d82: jne short 10001D88h
0x10001d84: xor eax,eax
0x10001d86: jmp short 10001D8Bh
0x10001d88: mov eax,[edi+4]
0x10001d8b: mov ebx,ds:[100FE060h]
0x10001d91: push eax
0x10001d92: call ebx
0x10001d94: push eax
0x10001d95: lea ecx,[esp+34h]
0x10001d99: call 10022038h
0x10001d9e: mov eax,[esp+60h]
0x10001da2: cmp eax,ebp
0x10001da4: jne short 10001DAAh
0x10001da6: xor eax,eax
0x10001da8: jmp short 10001DADh
0x10001daa: mov eax,[eax+4]
0x10001dad: mov ecx,[esp+34h]
0x10001db1: push eax
0x10001db2: push ecx
0x10001db3: call 1002203Eh
0x10001db8: lea ecx,[esp+40h]
0x10001dbc: mov [esp+14h],eax
0x10001dc0: call 1002202Ch
0x10001dc5: cmp edi,ebp
0x10001dc7: mov byte ptr [esp+58h],1
0x10001dcc: jne short 10001DD2h
0x10001dce: xor eax,eax
0x10001dd0: jmp short 10001DD5h
0x10001dd2: mov eax,[edi+4]
0x10001dd5: push eax
0x10001dd6: call ebx
0x10001dd8: push eax
0x10001dd9: lea ecx,[esp+44h]
0x10001ddd: call 10022038h
0x10001de2: mov [esp+1Ch],ebp
0x10001de6: mov dword ptr [esp+18h],1012F498h
0x10001dee: mov byte ptr [esp+58h],2
0x10001df3: mov eax,[esi+0Ch]
0x10001df6: sub eax,[esi+4]
0x10001df9: mov ecx,[esi+8]
0x10001dfc: sub ecx,[esi]
0x10001dfe: push ebp
0x10001dff: push 1
0x10001e01: push 1
0x10001e03: push eax
0x10001e04: push ecx
0x10001e05: call dword ptr ds:[100FE058h]
0x10001e0b: push eax
0x10001e0c: lea ecx,[esp+1Ch]
0x10001e10: call 10021FFCh
0x10001e15: mov edx,[esp+1Ch]
0x10001e19: mov eax,[esp+44h]
0x10001e1d: push edx
0x10001e1e: push eax
0x10001e1f: call 1002203Eh
0x10001e24: mov ecx,[esp+34h]
0x10001e28: push ebp
0x10001e29: push ebp
0x10001e2a: push ecx
0x10001e2b: mov [esp+1Ch],eax
```

```
0x10001e2f: call dword ptr ds:[100FE070h]
0x10001e35: push 3
0x10001e37: mov ecx,edi
0x10001e39: mov ebp,eax
0x10001e3b: call 10022044h
0x10001e40: push ebp
0x10001e41: lea ecx,[esp+34h]
0x10001e45: call 10022020h
0x10001e4a: mov ebx,[esp+24h]
0x10001e4e: mov eax,[esp+2Ch]
0x10001e52: mov ecx,[esp+28h]
0x10001e56: sub ecx,[esp+20h]
0x10001e5a: mov edx,[esi+0Ch]
0x10001e5d: sub edx,[esi+4]
0x10001e60: mov ebp,[esi+8]
0x10001e63: sub ebp,[esi]
0x10001e65: push 0CC0020h
0x10001e6a: sub eax,ebx
0x10001e6c: push eax
0x10001e6d: mov eax,[esp+28h]
0x10001e71: push ecx
0x10001e72: mov ecx,[esp+40h]
0x10001e76: push ebx
0x10001e77: push eax
0x10001e78: push ecx
0x10001e79: push edx
0x10001e7a: mov edx,[esp+60h]
0x10001e7e: push ebp
0x10001e7f: push 0
0x10001e81: push 0
0x10001e83: push edx
0x10001e84: call dword ptr ds:[100FE05Ch]
0x10001e8a: mov eax,[edi]
0x10001e8c: mov edx,[eax+34h]
0x10001e8f: push 0FFFFFFh
0x10001e94: mov ecx,edi
0x10001e96: call edx
0x10001e98: mov [esp+64h],eax
0x10001e9c: mov eax,[edi]
0x10001e9e: mov edx,[eax+38h]
0x10001ea1: push 0
0x10001ea3: mov ecx,edi
0x10001ea5: call edx
0x10001ea7: mov ecx,[esp+28h]
0x10001eab: mov ebx,[esi+4]
0x10001eae: mov [esp+60h],eax
0x10001eb2: mov eax,[esp+2Ch]
0x10001eb6: sub eax,[esp+24h]
0x10001eba: sub ecx,[esp+20h]
0x10001ebe: mov edx,[esi+0Ch]
0x10001ec1: mov ebp,[esi]
0x10001ec3: push 660046h
0x10001ec8: push eax
0x10001ec9: mov eax,[esp+2Ch]
0x10001ecd: push ecx
0x10001ece: mov ecx,[esp+2Ch]
0x10001ed2: push eax
0x10001ed3: mov eax,[esp+44h]
0x10001ed7: push ecx
0x10001ed8: mov ecx,[edi+4]
0x10001edb: push eax
0x10001edc: mov eax,[esi+4]
0x10001edf: sub edx,ebx
0x10001ee1: mov ebx,[esi+8]
0x10001ee4: push edx
0x10001ee5: sub ebx,ebp
0x10001ee7: push ebx
```

```
0x10001ee8: push eax
0x10001ee9: push ebp
0x10001eea: push ecx
0x10001eeb: call dword ptr ds:[100FE05Ch]
0x10001ef1: mov edx,[esi+4]
0x10001ef4: mov eax,[esi+0Ch]
0x10001ef7: mov ebp,[esp+44h]
0x10001efb: mov ebx,[esi]
0x10001efd: mov ecx,[esi+8]
0x10001f00: push 8800C6h
0x10001f05: push 0
0x10001f07: push 0
0x10001f09: push ebp
0x10001f0a: sub eax,edx
0x10001f0c: push eax
0x10001f0d: sub ecx,ebx
0x10001f0f: push ecx
0x10001f10: push edx
0x10001f11: mov edx,[edi+4]
0x10001f14: push ebx
0x10001f15: push edx
0x10001f16: call dword ptr ds:[100FE06Ch]
0x10001f1c: mov eax,[esp+2Ch]
0x10001f20: sub eax,[esp+24h]
0x10001f24: mov ecx,[esp+28h]
0x10001f28: sub ecx,[esp+20h]
0x10001f2c: mov ebx,[esi+4]
0x10001f2f: mov edx,[esi+0Ch]
0x10001f32: mov ebp,[esi]
0x10001f34: mov esi,[esi+8]
0x10001f37: push 660046h
0x10001f3c: push eax
0x10001f3d: mov eax,[esp+2Ch]
0x10001f41: push ecx
0x10001f42: mov ecx,[esp+2Ch]
0x10001f46: push eax
0x10001f47: mov eax,[esp+44h]
0x10001f4b: push ecx
0x10001f4c: mov ecx,[edi+4]
0x10001f4f: push eax
0x10001f50: sub edx,ebx
0x10001f52: push edx
0x10001f53: sub esi,ebp
0x10001f55: push esi
0x10001f56: push ebx
0x10001f57: push ebp
0x10001f58: push ecx
0x10001f59: call dword ptr ds:[100FE05Ch]
0x10001f5f: mov edx,[edi]
0x10001f61: mov eax,[esp+64h]
0x10001f65: mov edx,[edx+34h]
0x10001f68: push eax
0x10001f69: mov ecx,edi
0x10001f6b: call edx
0x10001f6d: mov ecx,[esp+60h]
0x10001f71: mov eax,[edi]
0x10001f73: mov edx,[eax+38h]
0x10001f76: push ecx
0x10001f77: mov ecx,edi
0x10001f79: call edx
0x10001f7b: mov eax,[esp+10h]
0x10001f7f: test eax,eax
0x10001f81: je short 10001F86h
0x10001f83: mov eax,[eax+4]
0x10001f86: push eax
0x10001f87: mov eax,[esp+48h]
0x10001f8b: push eax
```

```
0x10001f8c: call 1002203Eh
0x10001f91: mov eax,[esp+14h]
0x10001f95: test eax,eax
0x10001f97: je short 10001F9Ch
0x10001f99: mov eax,[eax+4]
0x10001f9c: mov ecx,[esp+34h]
0x10001fa0: push eax
0x10001fa1: push ecx
0x10001fa2: call 1002203Eh
0x10001fa7: mov byte ptr [esp+58h],1
0x10001fac: lea ecx,[esp+18h]
0x10001fb0: mov dword ptr [esp+18h],1012F498h
0x10001fb8: call 10001370h
0x10001fbd: lea ecx,[esp+40h]
0x10001fc1: mov byte ptr [esp+58h],0
0x10001fc6: call 10022032h
0x10001fcb: lea ecx,[esp+30h]
0x10001fcf: mov dword ptr [esp+58h],0FFFFFFFFh
0x10001fd7: call 10022032h
0x10001fdc: mov ecx,[esp+50h]
0x10001fe0: mov fs:[0],ecx
0x10001fe7: pop ecx
0x10001fe8: pop esi
0x10001fe9: pop ebp
0x10001fea: pop ebx
0x10001feb: add esp,4Ch
0x10001fee: ret
```

`sub_10001d00` is called by sub_10001ff0, sub_1000a0c0. It delegates to sub_10001370, j_ORDINAL 1271, j_ORDINAL 5723 (8 callees).

12.49 sub_10001ff0 at 0x10001ff0 (44 instructions)

```
0x10001ff0: sub esp,18h
0x10001ff3: push edi
0x10001ff4: push esi
0x10001ff5: call dword ptr ds:[100FE6ACh]
0x10001ffb: test eax,eax
0x10001ffd: jne short 10002068h
0x10001fff: mov ecx,[esi+4]
0x10002002: mov eax,[esi]
0x10002004: mov edx,[esi+8]
0x10002007: mov edi,ds:[100FE6B0h]
0x1000200d: mov [esp+0Ch],ecx
0x10002011: mov [esp+8],eax
0x10002015: neg ecx
0x10002017: push ecx
0x10002018: neg eax
0x1000201a: push eax
0x1000201b: mov [esp+18h],edx
0x1000201f: mov edx,[esi+0Ch]
0x10002022: lea eax,[esp+10h]
0x10002026: push eax
0x10002027: mov [esp+20h],edx
0x1000202b: call edi
0x1000202d: mov ecx,[esp+14h]
0x10002031: sub ecx,[esp+0Ch]
0x10002035: mov edx,[esp+10h]
0x10002039: sub edx,[esp+8]
0x1000203d: imul ecx,[esp+2Ch]
0x10002042: imul edx,[esp+28h]
0x10002047: push ecx
0x10002048: push edx
0x10002049: lea eax,[esp+10h]
0x1000204d: push eax
0x1000204e: call edi
```

```
0x10002050: mov edx,[esp+24h]
0x10002054: mov edi,[esp+20h]
0x10002058: lea ecx,[esp+8]
0x1000205c: push ecx
0x1000205d: push edx
0x1000205e: mov ecx,esi
0x10002060: call 10001D00h
0x10002065: add esp,8
0x10002068: pop edi
0x10002069: add esp,18h
0x1000206c: ret
```

`sub_10001ff0` is called by sub_100078e0, sub_100079a0, sub_10007b10. It delegates to sub_10001d00 (1 callees).

12.50 sub_10002070 at 0x10002070 (32 instructions)

```
0x10002070: sub esp,1F8h
0x10002076: mov eax,ds:[10136220h]
0x1000207b: xor eax,esp
0x1000207d: mov [esp+1F4h],eax
0x10002084: push edi
0x10002085: push 1F4h
0x1000208a: lea eax,[esp+8]
0x1000208e: push 0
0x10002090: push eax
0x10002091: mov edi,ecx
0x10002093: call 100228C0h
0x10002098: add esp,0Ch
0x1000209b: push 0
0x1000209d: lea ecx,[esp+8]
0x100020a1: push ecx
0x100020a2: push 1F4h
0x100020a7: push 29h
0x100020a9: mov dword ptr [esp+14h],1F4h
0x100020b1: call dword ptr ds:[100FE6A0h]
0x100020b7: test eax,eax
0x100020b9: je short 100020CBh
0x100020bb: push esi
0x100020bc: mov ecx,17h
0x100020c1: lea esi,[esp+0E8h]
0x100020c8: rep movsd
0x100020ca: pop esi
0x100020cb: mov ecx,[esp+1F8h]
0x100020d2: pop edi
0x100020d3: xor ecx,esp
0x100020d5: call 10022326h
0x100020da: add esp,1F8h
0x100020e0: ret
```

`sub_10002070` is called by sub_10008050, sub_1000a890. It delegates to sub_10022326, j_memset (2 callees).

12.51 sub_100020f0 at 0x100020f0 (83 instructions)

```
0x100020f0: push 0FFFFFFFFh
0x100020f2: push 100FCB18h
0x100020f7: mov eax,fs:[0]
0x100020fd: push eax
0x100020fe: sub esp,30h
0x10002101: push ebx
0x10002102: push ebp
0x10002103: push esi
0x10002104: push edi
0x10002105: mov eax,ds:[10136220h]
0x1000210a: xor eax,esp
0x1000210c: push eax
```

```
0x1000210d: lea eax,[esp+44h]
0x10002111: mov fs:[0],eax
0x10002117: mov ebp,[esp+58h]
0x1000211b: mov ebx,ecx
0x1000211d: mov esi,edx
0x1000211f: lea ecx,[esp+30h]
0x10002123: call 1002202Ch
0x10002128: mov edi,[esp+54h]
0x1000212c: xor eax,eax
0x1000212e: cmp edi,eax
0x10002130: mov [esp+4Ch],eax
0x10002134: je short 10002139h
0x10002136: mov eax,[edi+4]
0x10002139: push eax
0x1000213a: call dword ptr ds:[100FE060h]
0x10002140: push eax
0x10002141: lea ecx,[esp+34h]
0x10002145: call 10022038h
0x1000214a: test ebx,ebx
0x1000214c: jne short 10002152h
0x1000214e: xor eax,eax
0x10002150: jmp short 10002155h
0x10002152: mov eax,[ebx+4]
0x10002155: push eax
0x10002156: mov eax,[esp+38h]
0x1000215a: push eax
0x1000215b: call 1002203Eh
0x10002160: mov ecx,[esp+5Ch]
0x10002164: mov [esp+1Ch],eax
0x10002168: lea eax,[ecx-1]
0x1000216b: cmp eax,4
0x1000216e: ja near ptr 1000232Eh
0x10002174: jmp dword ptr [eax*4+10002394h]
0x1000232e: mov ecx,[esi+4]
0x10002331: mov eax,[esi+0Ch]
0x10002334: mov edx,[esi]
0x10002336: mov ebx,[esp+34h]
0x1000233a: mov esi,[esi+8]
0x1000233d: push 0CC0020h
0x10002342: push 0
0x10002344: push 0
0x10002346: push ebx
0x10002347: sub eax,ecx
0x10002349: push eax
0x1000234a: sub esi,edx
0x1000234c: push esi
0x1000234d: push ecx
0x1000234e: mov ecx,[edi+4]
0x10002351: push edx
0x10002352: push ecx
0x10002353: call dword ptr ds:[100FE06Ch]
0x10002359: mov eax,[esp+1Ch]
0x1000235d: test eax,eax
0x1000235f: je short 10002364h
0x10002361: mov eax,[eax+4]
0x10002364: mov edx,[esp+34h]
0x10002368: push eax
0x10002369: push edx
0x1000236a: call 1002203Eh
0x1000236f: lea ecx,[esp+30h]
0x10002373: mov dword ptr [esp+4Ch],0FFFFFFFFh
0x1000237b: call 10022032h
0x10002380: mov ecx,[esp+44h]
0x10002384: mov fs:[0],ecx
0x1000238b: pop ecx
0x1000238c: pop edi
0x1000238d: pop esi
```

```
0x1000238e: pop ebp
0x1000238f: pop ebx
0x10002390: add esp,3Ch
0x10002393: ret
```

`sub_100020f0` is called by sub_10005fd0. It delegates to j_ORDINAL 347, j_ORDINAL 602, j_ORDINAL 1270 (4 callees).

12.52 sub_100023f0 at 0x100023f0 (21 instructions)

```
0x100023f0: test eax,eax
0x100023f2: je short 100023F7h
0x100023f4: mov eax,[eax+4]
0x100023f7: push ecx
0x100023f8: mov ecx,[esp+1Ch]
0x100023fc: push edx
0x100023fd: mov edx,[esp+1Ch]
0x10002401: push ecx
0x10002402: mov ecx,[esp+18h]
0x10002406: push eax
0x10002407: mov eax,[esp+20h]
0x1000240b: push edx
0x1000240c: mov edx,[esp+1Ch]
0x10002410: push eax
0x10002411: mov eax,[esp+1Ch]
0x10002415: push ecx
0x10002416: mov ecx,[eax+4]
0x10002419: push edx
0x1000241a: push ecx
0x1000241b: call dword ptr ds:[100FE06Ch]
0x10002421: ret 18h
```

`sub_100023f0` is called by sub_10007dd0. It calls nothing else.

12.53 sub_10002480 at 0x10002480 (15 instructions)

```
0x10002480: test eax,eax
0x10002482: jne short 10002495h
0x10002484: push eax
0x10002485: mov eax,[esp+8]
0x10002489: mov ecx,[eax+4]
0x1000248c: push ecx
0x1000248d: call 1002203Eh
0x10002492: ret 4
0x10002495: mov eax,[eax+4]
0x10002498: push eax
0x10002499: mov eax,[esp+8]
0x1000249d: mov ecx,[eax+4]
0x100024a0: push ecx
0x100024a1: call 1002203Eh
0x100024a6: ret 4
```

`sub_10002480` is called by sub_10007dd0. It delegates to j_ORDINAL 5633 (1 callees).

12.54 sub_100024b0 at 0x100024b0 (15 instructions)

```
0x100024b0: test eax,eax
0x100024b2: jne short 100024C8h
0x100024b4: push eax
0x100024b5: call dword ptr ds:[100FE060h]
0x100024bb: mov ecx,[esp+4]
0x100024bf: push eax
0x100024c0: call 10022038h
0x100024c5: ret 4
```

```
0x100024c8: mov eax,[eax+4]
0x100024cb: push eax
0x100024cc: call dword ptr ds:[100FE060h]
0x100024d2: mov ecx,[esp+4]
0x100024d6: push eax
0x100024d7: call 10022038h
0x100024dc: ret 4
```

`sub_100024b0` is called by sub_10007dd0. It delegates to j_ORDINAL 1270 (1 callees).

12.55 sub_100024e0 at 0x100024e0 (25 instructions)

```
0x100024e0: push 0FFFFFFFFh
0x100024e2: push 100FC868h
0x100024e7: mov eax,fs:[0]
0x100024ed: push eax
0x100024ee: push esi
0x100024ef: mov eax,ds:[10136220h]
0x100024f4: xor eax,esp
0x100024f6: push eax
0x100024f7: lea eax,[esp+8]
0x100024fb: mov fs:[0],eax
0x10002501: mov esi,[esp+18h]
0x10002505: mov ecx,esi
0x10002507: call 100221ACh
0x1000250c: mov dword ptr [esp+10h],0
0x10002514: mov dword ptr [esi],1012F5BCh
0x1000251a: call 10002580h
0x1000251f: mov eax,[esp+1Ch]
0x10002523: mov [esi+54h],eax
0x10002526: mov eax,esi
0x10002528: mov ecx,[esp+8]
0x1000252c: mov fs:[0],ecx
0x10002533: pop ecx
0x10002534: pop esi
0x10002535: add esp,0Ch
0x10002538: ret 8
```

`sub_100024e0` is called by sub_10009450. It delegates to sub_10002580, j_ORDINAL 572 (2 callees).

12.56 sub_10002580 at 0x10002580 (43 instructions)

```
0x10002580: sub esp,2Ch
0x10002583: push esi
0x10002584: push edi
0x10002585: call 10021FD2h
0x1000258a: mov esi,[eax+8]
0x1000258d: lea eax,[esp+8]
0x10002591: push eax
0x10002592: push 1012F38Ch
0x10002597: push esi
0x10002598: call dword ptr ds:[100FE694h]
0x1000259e: test eax,eax
0x100025a0: jne short 10002605h
0x100025a2: mov ecx,ds:[100FE690h]
0x100025a8: xor edi,edi
0x100025aa: push 7F00h
0x100025af: push edi
0x100025b0: mov dword ptr [esp+10h],800h
0x100025b8: mov [esp+14h],ecx
0x100025bc: mov [esp+1Ch],edi
0x100025c0: mov [esp+18h],edi
0x100025c4: mov [esp+20h],esi
0x100025c8: mov [esp+24h],edi
0x100025cc: call dword ptr ds:[100FE68Ch]
```

```
0x100025d2: lea edx,[esp+8]
0x100025d6: push edx
0x100025d7: mov [esp+24h],eax
0x100025db: mov dword ptr [esp+28h],10h
0x100025e3: mov [esp+2Ch],edi
0x100025e7: mov dword ptr [esp+30h],1012F38Ch
0x100025ef: call 1002220Ch
0x100025f4: test eax,eax
0x100025f6: jne short 10002605h
0x100025f8: call 10022212h
0x100025fd: xor eax,eax
0x100025ff: pop edi
0x10002600: pop esi
0x10002601: add esp,2Ch
0x10002604: ret
0x10002605: pop edi
0x10002606: mov eax,1
0x1000260b: pop esi
0x1000260c: add esp,2Ch
0x1000260f: ret
```

`sub_10002580` is called by sub_100024e0. It works with the text "_MenuHost_A$". It delegates to j_ORDINAL 1079, j_ORDINAL 1155, j_ORDINAL 1183 (3 callees).

12.57 sub_10002910 at 0x10002910 (27 instructions)

```
0x10002910: push 0FFFFFFFFh
0x10002912: push 100FC839h
0x10002917: mov eax,fs:[0]
0x1000291d: push eax
0x1000291e: push ecx
0x1000291f: push esi
0x10002920: mov eax,ds:[10136220h]
0x10002925: xor eax,esp
0x10002927: push eax
0x10002928: lea eax,[esp+0Ch]
0x1000292c: mov fs:[0],eax
0x10002932: mov esi,ecx
0x10002934: mov [esp+8],esi
0x10002938: call dword ptr ds:[100FE570h]
0x1000293e: mov eax,[esp+1Ch]
0x10002942: mov dword ptr [esp+14h],0
0x1000294a: push eax
0x1000294b: lea ecx,[esi+0Ch]
0x1000294e: mov dword ptr [esi],100FF3A0h
0x10002954: call dword ptr ds:[100FE3A0h]
0x1000295a: mov eax,esi
0x1000295c: mov ecx,[esp+0Ch]
0x10002960: mov fs:[0],ecx
0x10002967: pop ecx
0x10002968: pop esi
0x10002969: add esp,10h
0x1000296c: ret 4
```

`sub_10002910` is called by sub_10009b40. It calls nothing else.

12.58 sub_100029d0 at 0x100029d0 (28 instructions)

```
0x100029d0: push 0FFFFFFFFh
0x100029d2: push 100FCF9Bh
0x100029d7: mov eax,fs:[0]
0x100029dd: push eax
0x100029de: push esi
0x100029df: push edi
0x100029e0: mov eax,ds:[10136220h]
```

```
0x100029e5: xor eax,esp
0x100029e7: push eax
0x100029e8: lea eax,[esp+0Ch]
0x100029ec: mov fs:[0],eax
0x100029f2: mov edi,[esp+1Ch]
0x100029f6: lea esi,[edi+4]
0x100029f9: xor eax,eax
0x100029fb: mov dword ptr [edi],1012F710h
0x10002a01: mov [esi+4],eax
0x10002a04: mov [esi+8],eax
0x10002a07: mov [esi+0Ch],eax
0x10002a0a: mov [esp+14h],eax
0x10002a0e: call 10002C40h
0x10002a13: mov eax,edi
0x10002a15: mov ecx,[esp+0Ch]
0x10002a19: mov fs:[0],ecx
0x10002a20: pop ecx
0x10002a21: pop edi
0x10002a22: pop esi
0x10002a23: add esp,0Ch
0x10002a26: ret 4
```

`sub_100029d0` is called by sub_10008050. It delegates to sub_10002c40 (1 callees).

12.59 sub_10002b80 at 0x10002b80 (74 instructions)

```
0x10002b80: sub esp,8
0x10002b83: mov ecx,[eax+8]
0x10002b86: push ebx
0x10002b87: push ebp
0x10002b88: add eax,4
0x10002b8b: push esi
0x10002b8c: xor esi,esi
0x10002b8e: cmp ecx,esi
0x10002b90: push edi
0x10002b91: mov [esp+10h],esi
0x10002b95: je near ptr 10002C2Eh
0x10002b9b: mov edx,[eax+8]
0x10002b9e: sub edx,ecx
0x10002ba0: sar edx,2
0x10002ba3: je near ptr 10002C2Eh
0x10002ba9: cmp [esp+1Ch],esi
0x10002bad: je short 10002BD6h
0x10002baf: cmp ecx,esi
0x10002bb1: je short 10002C2Eh
0x10002bb3: mov eax,[eax+8]
0x10002bb6: sub eax,ecx
0x10002bb8: sar eax,2
0x10002bbb: cmp eax,1
0x10002bbe: jle short 10002C2Eh
0x10002bc0: mov dword ptr [esp+10h],1
0x10002bc8: mov eax,[esp+10h]
0x10002bcc: pop edi
0x10002bcd: pop esi
0x10002bce: pop ebp
0x10002bcf: pop ebx
0x10002bd0: add esp,8
0x10002bd3: ret 4
0x10002bd6: call 10002CF0h
0x10002bdb: mov edi,[eax]
0x10002bdd: mov ebp,ds:[100FE684h]
0x10002be3: push edi
0x10002be4: call ebp
0x10002be6: test eax,eax
0x10002be8: jle short 10002C2Eh
0x10002bea: mov ebx,ds:[100FE680h]
```

```
0x10002bf0: push 400h
0x10002bf5: push esi
0x10002bf6: push edi
0x10002bf7: call ebx
0x10002bf9: test al,al
0x10002bfb: js short 10002C15h
0x10002bfd: push edi
0x10002bfe: add esi,1
0x10002c01: call ebp
0x10002c03: cmp esi,eax
0x10002c05: jl short 10002BF0h
0x10002c07: mov eax,[esp+10h]
0x10002c0b: pop edi
0x10002c0c: pop esi
0x10002c0d: pop ebp
0x10002c0e: pop ebx
0x10002c0f: add esp,8
0x10002c12: ret 4
0x10002c15: test esi,esi
0x10002c17: jl short 10002C2Eh
0x10002c19: push 400h
0x10002c1e: push esi
0x10002c1f: push edi
0x10002c20: call ebx
0x10002c22: test al,10h
0x10002c24: je short 10002C2Eh
0x10002c26: mov dword ptr [esp+10h],1
0x10002c2e: mov eax,[esp+10h]
0x10002c32: pop edi
0x10002c33: pop esi
0x10002c34: pop ebp
0x10002c35: pop ebx
0x10002c36: add esp,8
0x10002c39: ret 4
```

`sub_10002b80` is called by sub_10007040. It delegates to sub_10002cf0 (1 callees).

12.60 sub_10002c40 at 0x10002c40 (61 instructions)

```
0x10002c40: push ebp
0x10002c41: mov ebp,esp
0x10002c43: and esp,0FFFFFFF8h
0x10002c46: mov ecx,[esi+4]
0x10002c49: sub esp,8
0x10002c4c: test ecx,ecx
0x10002c4e: push ebx
0x10002c4f: push edi
0x10002c50: je short 10002C5Fh
0x10002c52: mov eax,[esi+0Ch]
0x10002c55: sub eax,ecx
0x10002c57: sar eax,2
0x10002c5a: cmp eax,0Ah
0x10002c5d: jae short 10002CDDh
0x10002c5f: mov ecx,0Ah
0x10002c64: call 10009BB0h
0x10002c69: mov edi,[esi+8]
0x10002c6c: cmp [esi+4],edi
0x10002c6f: mov [esp+0Ch],eax
0x10002c73: jbe short 10002C7Bh
0x10002c75: call dword ptr ds:[100FE584h]
0x10002c7b: mov ebx,[esi+4]
0x10002c7e: cmp ebx,[esi+8]
0x10002c81: jbe short 10002C89h
0x10002c83: call dword ptr ds:[100FE584h]
0x10002c89: sub edi,ebx
0x10002c8b: sar edi,2
```

```
0x10002c8e: je short 10002CAAh
0x10002c90: lea eax,[edi*4]
0x10002c97: push eax
0x10002c98: push ebx
0x10002c99: mov ebx,[esp+14h]
0x10002c9d: push eax
0x10002c9e: push ebx
0x10002c9f: call dword ptr ds:[100FE57Ch]
0x10002ca5: add esp,10h
0x10002ca8: jmp short 10002CAEh
0x10002caa: mov ebx,[esp+0Ch]
0x10002cae: mov eax,[esi+4]
0x10002cb1: test eax,eax
0x10002cb3: jne short 10002CB9h
0x10002cb5: xor edi,edi
0x10002cb7: jmp short 10002CC1h
0x10002cb9: mov edi,[esi+8]
0x10002cbc: sub edi,eax
0x10002cbe: sar edi,2
0x10002cc1: test eax,eax
0x10002cc3: je short 10002CCEh
0x10002cc5: push eax
0x10002cc6: call 10021FCCh
0x10002ccb: add esp,4
0x10002cce: lea eax,[ebx+28h]
0x10002cd1: lea ecx,[ebx+edi*4]
0x10002cd4: mov [esi+0Ch],eax
0x10002cd7: mov [esi+8],ecx
0x10002cda: mov [esi+4],ebx
0x10002cdd: pop edi
0x10002cde: pop ebx
0x10002cdf: mov esp,ebp
0x10002ce1: pop ebp
0x10002ce2: ret
```

`sub_10002c40` is called by sub_100029d0. It delegates to sub_10009bb0, j_ORDINAL 764 (2 callees).

12.61 sub_10002cf0 at 0x10002cf0 (30 instructions)

```
0x10002cf0: sub esp,0Ch
0x10002cf3: push ebx
0x10002cf4: push ebp
0x10002cf5: mov ebp,ds:[100FE584h]
0x10002cfb: push esi
0x10002cfc: mov esi,eax
0x10002cfe: push edi
0x10002cff: mov edi,[esi+8]
0x10002d02: cmp [esi+4],edi
0x10002d05: jbe short 10002D09h
0x10002d07: call ebp
0x10002d09: lea eax,[edi-4]
0x10002d0c: cmp eax,[esi+8]
0x10002d0f: mov ebx,esi
0x10002d11: mov [esp+18h],edi
0x10002d15: ja short 10002D1Ch
0x10002d17: cmp eax,[esi+4]
0x10002d1a: jae short 10002D1Eh
0x10002d1c: call ebp
0x10002d1e: lea esi,[edi-4]
0x10002d21: cmp esi,[ebx+8]
0x10002d24: jb short 10002D28h
0x10002d26: call ebp
0x10002d28: pop edi
0x10002d29: mov eax,esi
0x10002d2b: pop esi
0x10002d2c: pop ebp
```

```
0x10002d2d: pop ebx
0x10002d2e: add esp,0Ch
0x10002d31: ret
```

`sub_10002cf0` is called by sub_10002b80. It calls nothing else.

12.62 sub_10002eb0 at 0x10002eb0 (15 instructions)

```
0x10002eb0: test esi,esi
0x10002eb2: mov eax,esi
0x10002eb4: mov ecx,edi
0x10002eb6: jbe short 10002ECFh
0x10002eb8: push ebx
0x10002eb9: lea esp,[esp]
0x10002ec0: mov ebx,[edx]
0x10002ec2: mov [ecx],ebx
0x10002ec4: sub eax,1
0x10002ec7: add ecx,4
0x10002eca: test eax,eax
0x10002ecc: ja short 10002EC0h
0x10002ece: pop ebx
0x10002ecf: lea eax,[edi+esi*4]
0x10002ed2: ret
```

`sub_10002eb0` is called by sub_100072e0, sub_10009950. It calls nothing else.

12.63 sub_100031f0 at 0x100031f0 (15 instructions)

```
0x100031f0: sub eax,edx
0x100031f2: sar eax,2
0x100031f5: lea eax,[eax*4]
0x100031fc: push esi
0x100031fd: lea esi,[eax+ecx]
0x10003200: je short 1000320Fh
0x10003202: push eax
0x10003203: push edx
0x10003204: push eax
0x10003205: push ecx
0x10003206: call dword ptr ds:[100FE57Ch]
0x1000320c: add esp,10h
0x1000320f: mov eax,esi
0x10003211: pop esi
0x10003212: ret
```

`sub_100031f0` is called by sub_100072e0, sub_10009950. It calls nothing else.

12.64 sub_10003240 at 0x10003240 (18 instructions)

```
0x10003240: mov ecx,[edi+8]
0x10003243: mov edx,[ecx+20h]
0x10003246: sub esp,14h
0x10003249: lea eax,[esp]
0x1000324c: push eax
0x1000324d: push edx
0x1000324e: call dword ptr ds:[100FE674h]
0x10003254: mov eax,[edi+8]
0x10003257: mov ecx,[eax+20h]
0x1000325a: push esi
0x1000325b: push ecx
0x1000325c: call dword ptr ds:[100FE698h]
0x10003262: mov edx,[esp]
0x10003265: mov eax,[esp+4]
0x10003269: sub [esi],edx
0x1000326b: sub [esi+4],eax
```

```
0x1000326e: add esp,14h
0x10003271: ret
```

`sub_10003240` is called by sub_100067c0. It calls nothing else.

12.65 sub_10003280 at 0x10003280 (16 instructions)

```
0x10003280: mov edx,[ecx+8]
0x10003283: sub esp,10h
0x10003286: lea eax,[esp]
0x10003289: push eax
0x1000328a: mov eax,[edx+20h]
0x1000328d: push eax
0x1000328e: call dword ptr ds:[100FE674h]
0x10003294: mov ecx,[esp+4]
0x10003298: mov edx,[esp]
0x1000329b: mov eax,[esp+14h]
0x1000329f: push ecx
0x100032a0: push edx
0x100032a1: push eax
0x100032a2: call dword ptr ds:[100FE6B0h]
0x100032a8: add esp,10h
0x100032ab: ret 4
```

`sub_10003280` is called by sub_100051b0. It calls nothing else.

12.66 sub_100032b0 at 0x100032b0 (13 instructions)

```
0x100032b0: mov edx,[ecx+8]
0x100032b3: sub esp,14h
0x100032b6: lea eax,[esp]
0x100032b9: push eax
0x100032ba: mov eax,[edx+20h]
0x100032bd: push eax
0x100032be: call dword ptr ds:[100FE674h]
0x100032c4: mov ecx,[esp]
0x100032c7: mov edx,[esp+4]
0x100032cb: sub [esi],ecx
0x100032cd: sub [esi+4],edx
0x100032d0: add esp,14h
0x100032d3: ret
```

`sub_100032b0` is called by sub_100067c0. It calls nothing else.

12.67 sub_100032e0 at 0x100032e0 (141 instructions)

```
0x100032e0: push 0FFFFFFFFh
0x100032e2: push 100FD34Ah
0x100032e7: mov eax,fs:[0]
0x100032ed: push eax
0x100032ee: sub esp,64h
0x100032f1: mov eax,ds:[10136220h]
0x100032f6: xor eax,esp
0x100032f8: mov [esp+60h],eax
0x100032fc: push ebx
0x100032fd: push ebp
0x100032fe: push esi
0x100032ff: mov eax,ds:[10136220h]
0x10003304: xor eax,esp
0x10003306: push eax
0x10003307: lea eax,[esp+74h]
0x1000330b: mov fs:[0],eax
0x10003311: mov eax,[esp+84h]
0x10003318: mov ebp,ds:[100FE5C8h]
```

```
0x1000331e: mov [edi+10h],eax
0x10003321: mov dword ptr [edi],1012F738h
0x10003327: xor ebx,ebx
0x10003329: lea ecx,[edi+30h]
0x1000332c: xor eax,eax
0x1000332e: push ecx
0x1000332f: mov [esp+14h],edi
0x10003333: mov dword ptr [edi+18h],1012F7B0h
0x1000333a: mov dword ptr [edi+24h],1
0x10003341: mov [edi+1Ch],ebx
0x10003344: mov [edi+20h],ebx
0x10003347: mov [edi+28h],eax
0x1000334a: mov [edi+2Ch],ebx
0x1000334d: call ebp
0x1000334f: lea esi,[edi+40h]
0x10003352: mov [esp+7Ch],ebx
0x10003356: call 10008050h
0x1000335b: lea esi,[edi+0D8h]
0x10003361: mov byte ptr [esp+7Ch],1
0x10003366: call 1000A890h
0x1000336b: mov dword ptr [edi+124h],1012F75Ch
0x10003375: mov [edi+12Ch],ebx
0x1000337b: mov dword ptr [edi+128h],1012F498h
0x10003385: lea edx,[edi+140h]
0x1000338b: xor eax,eax
0x1000338d: push edx
0x1000338e: mov dword ptr [edi+130h],1
0x10003398: mov [edi+134h],eax
0x1000339e: mov [edi+138h],ebx
0x100033a4: mov [edi+13Ch],ebx
0x100033aa: call ebp
0x100033ac: lea eax,[edi+154h]
0x100033b2: push eax
0x100033b3: mov [edi+150h],ebx
0x100033b9: call ebp
0x100033bb: lea ecx,[edi+168h]
0x100033c1: push ecx
0x100033c2: mov [edi+164h],ebx
0x100033c8: call ebp
0x100033ca: lea eax,[edi+178h]
0x100033d0: push eax
0x100033d1: mov byte ptr [esp+80h],3
0x100033d9: call 1000BB00h
0x100033de: lea esi,[edi+1C0h]
0x100033e4: mov byte ptr [esp+7Ch],4
0x100033e9: call 1000D0A0h
0x100033ee: lea ecx,[edi+1F4h]
0x100033f4: call dword ptr ds:[100FE368h]
0x100033fa: lea ecx,[edi+200h]
0x10003400: call dword ptr ds:[100FE368h]
0x10003406: mov [edi+218h],ebx
0x1000340c: mov dword ptr [edi+214h],1012F744h
0x10003416: mov byte ptr [esp+7Ch],8
0x1000341b: lea edx,[edi+258h]
0x10003421: push edx
0x10003422: mov [edi+8],ebx
0x10003425: mov [edi+0Ch],ebx
0x10003428: mov [edi+14h],ebx
0x1000342b: mov dword ptr [edi+4],1Fh
0x10003432: call ebp
0x10003434: lea eax,[edi+268h]
0x1000343a: push eax
0x1000343b: call ebp
0x1000343d: mov eax,3
0x10003442: mov [edi+27Ch],eax
0x10003448: mov [edi+278h],eax
0x1000344e: mov [edi+280h],ebx
```

```
0x10003454: mov [edi+284h],ebx
0x1000345a: mov [edi+288h],ebx
0x10003460: mov [edi+2A4h],ebx
0x10003466: push 30h
0x10003468: or eax,0FFFFFFFFh
0x1000346b: lea ecx,[edi+228h]
0x10003471: push ebx
0x10003472: push ecx
0x10003473: mov [edi+1F8h],eax
0x10003479: mov [edi+1FCh],eax
0x1000347f: mov [edi+290h],ebx
0x10003485: mov [edi+28Ch],ebx
0x1000348b: mov [edi+2A0h],ebx
0x10003491: mov [edi+220h],ebx
0x10003497: mov [edi+224h],eax
0x1000349d: call 100228C0h
0x100034a2: add esp,0Ch
0x100034a5: lea esi,[edi+178h]
0x100034ab: call 1000BC60h
0x100034b0: push 1012F714h
0x100034b5: lea edx,[esp+34h]
0x100034b9: push edx
0x100034ba: mov [esp+20h],ebx
0x100034be: mov [esp+24h],ebx
0x100034c2: mov [esp+28h],ebx
0x100034c6: mov [esp+2Ch],ebx
0x100034ca: mov [esp+30h],bl
0x100034ce: mov [esp+31h],bl
0x100034d2: mov [esp+32h],bl
0x100034d6: mov byte ptr [esp+33h],1
0x100034db: call dword ptr ds:[100FE104h]
0x100034e1: lea eax,[esp+14h]
0x100034e5: push eax
0x100034e6: mov [esp+30h],bl
0x100034ea: mov [esp+31h],bl
0x100034ee: mov [esp+32h],bl
0x100034f2: mov [esp+33h],bl
0x100034f6: mov dword ptr [esp+18h],13h
0x100034fe: mov [esp+1Ch],ebx
0x10003502: mov dword ptr [esp+28h],190h
0x1000350a: call dword ptr ds:[100FE064h]
0x10003510: push eax
0x10003511: lea ecx,[edi+214h]
0x10003517: call 10021FFCh
0x1000351c: mov eax,edi
0x1000351e: mov ecx,[esp+74h]
0x10003522: mov fs:[0],ecx
0x10003529: pop ecx
0x1000352a: pop esi
0x1000352b: pop ebp
0x1000352c: pop ebx
0x1000352d: mov ecx,[esp+60h]
0x10003531: xor ecx,esp
0x10003533: call 10022326h
0x10003538: add esp,70h
0x1000353b: ret 4
```

`sub_100032e0` is called by ??0CCustomTitle@@QAE@XZ, ??0CCustomTitle@@QAE@W4WTitleTheme@@@Z. It works with the text "Microsoft Yahei". It delegates to sub_10008050, sub_1000a890, sub_1000bb00 (8 callees).

12.68 sub_10003670 at 0x10003670 (141 instructions)

```
0x10003670: push ebp
0x10003671: mov ebp,esp
0x10003673: and esp,0FFFFFFF8h
0x10003676: sub esp,24h
```

```
0x10003679: push ebx
0x1000367a: push esi
0x1000367b: xor esi,esi
0x1000367d: cmp eax,esi
0x1000367f: push edi
0x10003680: mov edi,ecx
0x10003682: mov [edi+280h],esi
0x10003688: mov [edi+8],eax
0x1000368b: mov [edi+0Ch],esi
0x1000368e: jne short 10003694h
0x10003690: xor eax,eax
0x10003692: jmp short 10003697h
0x10003694: mov eax,[eax+20h]
0x10003697: push eax
0x10003698: call dword ptr ds:[100FE62Ch]
0x1000369e: test eax,eax
0x100036a0: jne short 100036B0h
0x100036a2: mov [edi+8],esi
0x100036a5: xor eax,eax
0x100036a7: pop edi
0x100036a8: pop esi
0x100036a9: pop ebx
0x100036aa: mov esp,ebp
0x100036ac: pop ebp
0x100036ad: ret 8
0x100036b0: mov ecx,[edi+8]
0x100036b3: mov edx,[ecx+20h]
0x100036b6: lea eax,[esp+20h]
0x100036ba: push eax
0x100036bb: push edx
0x100036bc: call dword ptr ds:[100FE624h]
0x100036c2: push edi
0x100036c3: call 10003940h
0x100036c8: test eax,eax
0x100036ca: je short 100036A5h
0x100036cc: mov ecx,edi
0x100036ce: call 100039B0h
0x100036d3: test eax,eax
0x100036d5: je short 100036A5h
0x100036d7: cmp [edi+12Ch],esi
0x100036dd: jne short 100036F3h
0x100036df: push 138Fh
0x100036e4: lea esi,[edi+124h]
0x100036ea: call 100075D0h
0x100036ef: test eax,eax
0x100036f1: je short 100036A5h
0x100036f3: mov ecx,[ebp+8]
0x100036f6: call 10003820h
0x100036fb: test eax,eax
0x100036fd: je short 100036A5h
0x100036ff: mov esi,edi
0x10003701: call 10003880h
0x10003706: test eax,eax
0x10003708: je short 100036A5h
0x1000370a: mov eax,[edi+8]
0x1000370d: mov ecx,[eax+20h]
0x10003710: push edi
0x10003711: push 10138C18h
0x10003716: push ecx
0x10003717: call dword ptr ds:[100FE61Ch]
0x1000371d: mov edx,[edi+8]
0x10003720: mov eax,[edx+20h]
0x10003723: push 10003DD0h
0x10003728: push 0FFFFFFFCh
0x1000372a: push eax
0x1000372b: call dword ptr ds:[100FE618h]
0x10003731: mov ecx,[edi+8]
```

```
0x10003734: mov [edi+0Ch],eax
0x10003737: call 10022236h
0x1000373c: mov ebx,eax
0x1000373e: test ebx,0C00000h
0x10003744: jne short 10003750h
0x10003746: mov dword ptr [edi+2A4h],1
0x10003750: call 10003D20h
0x10003755: mov eax,edi
0x10003757: call 10003D00h
0x1000375c: test ebx,40000h
0x10003762: jne short 100037D2h
0x10003764: mov ecx,[esp+20h]
0x10003768: mov edx,[esp+24h]
0x1000376c: mov eax,[esp+28h]
0x10003770: mov [esp+10h],ecx
0x10003774: mov ecx,[esp+2Ch]
0x10003778: mov [esp+1Ch],ecx
0x1000377c: mov ecx,[edi+8]
0x1000377f: mov [esp+14h],edx
0x10003783: mov [esp+18h],eax
0x10003787: mov edx,[ecx]
0x10003789: mov edx,[edx+70h]
0x1000378c: push 1
0x1000378e: lea eax,[esp+14h]
0x10003792: push eax
0x10003793: call edx
0x10003795: cmp dword ptr [edi+2A4h],0
0x1000379c: jne short 100037B0h
0x1000379e: lea ecx,[edi+1C0h]
0x100037a4: call 1000E360h
0x100037a9: add eax,[edi+4]
0x100037ac: sub [esp+14h],eax
0x100037b0: mov eax,[esp+1Ch]
0x100037b4: sub eax,[esp+14h]
0x100037b8: mov ecx,[esp+18h]
0x100037bc: sub ecx,[esp+10h]
0x100037c0: push 6
0x100037c2: push eax
0x100037c3: push ecx
0x100037c4: mov ecx,[edi+8]
0x100037c7: push 0
0x100037c9: push 0
0x100037cb: push 0
0x100037cd: call 1002226Ch
0x100037d2: cmp dword ptr [ebp+0Ch],0
0x100037d6: je short 1000380Ch
0x100037d8: mov eax,[edi+8]
0x100037db: mov edx,[eax+20h]
0x100037de: push 0
0x100037e0: push 0C8h
0x100037e5: push 1FCCh
0x100037ea: push edx
0x100037eb: call dword ptr ds:[100FE60Ch]
0x100037f1: push 0
0x100037f3: push eax
0x100037f4: mov [edi+288h],eax
0x100037fa: mov eax,[edi+8]
0x100037fd: mov ecx,[eax+20h]
0x10003800: push 113h
0x10003805: push ecx
0x10003806: call dword ptr ds:[100FE608h]
0x1000380c: push edi
0x1000380d: call 10006C70h
0x10003812: pop edi
0x10003813: pop esi
0x10003814: mov eax,1
0x10003819: pop ebx
```

```
0x1000381a: mov esp,ebp
0x1000381c: pop ebp
0x1000381d: ret 8
```

`sub_10003670` is called by ?SubClassWnd@CCustomTitle@@QAEHPAVCWnd@@HH@Z. It delegates to sub_10003820, sub_10003880, sub_10003940 (11 callees).

12.69 sub_10003820 at 0x10003820 (33 instructions)

```
0x10003820: mov eax,[edi+8]
0x10003823: push esi
0x10003824: lea esi,[edi+40h]
0x10003827: mov [esi+4],eax
0x1000382a: mov [esi+0Ch],ecx
0x1000382d: cmp dword ptr [edi+68h],0
0x10003831: jne short 10003840h
0x10003833: call 10008290h
0x10003838: test eax,eax
0x1000383a: jne short 10003840h
0x1000383c: xor eax,eax
0x1000383e: pop esi
0x1000383f: ret
0x10003840: cmp dword ptr [edi+0A0h],0
0x10003847: jne short 10003879h
0x10003849: add esi,5Ch
0x1000384c: je short 1000385Bh
0x1000384e: cmp dword ptr [esi+4],0
0x10003852: je short 1000385Bh
0x10003854: mov ecx,esi
0x10003856: call 10021FF6h
0x1000385b: mov edx,ds:[1013F360h]
0x10003861: push 138Dh
0x10003866: push edx
0x10003867: call dword ptr ds:[100FE5E0h]
0x1000386d: push eax
0x1000386e: mov ecx,esi
0x10003870: call 10021FFCh
0x10003875: test eax,eax
0x10003877: je short 1000383Ch
0x10003879: mov eax,1
0x1000387e: pop esi
0x1000387f: ret
```

`sub_10003820` is called by sub_10003670. It delegates to sub_10008290, j_ORDINAL 1925, j_ORDINAL 1271 (3 callees).

12.70 sub_10003880 at 0x10003880 (38 instructions)

```
0x10003880: push ecx
0x10003881: cmp dword ptr [esi+0F8h],0
0x10003888: jne short 1000389Ch
0x1000388a: lea eax,[esi+0D8h]
0x10003890: push eax
0x10003891: call 1000AA40h
0x10003896: test eax,eax
0x10003898: jne short 1000389Ch
0x1000389a: pop ecx
0x1000389b: ret
0x1000389c: cmp dword ptr [esi+10h],0
0x100038a0: je short 100038F4h
0x100038a2: mov eax,808080h
0x100038a7: mov ds:[10138C08h],eax
0x100038ac: mov ds:[10138C04h],eax
0x100038b1: mov dword ptr ds:[10138BECh],484848h
0x100038bb: mov dword ptr ds:[10138BF0h],601010h
```

```
0x100038c5: mov dword ptr ds:[10138BF4h],4040C0h
0x100038cf: mov dword ptr ds:[10138BF8h],0C0h
0x100038d9: mov dword ptr ds:[10138BFCh],404080h
0x100038e3: mov dword ptr ds:[10138C00h],80h
0x100038ed: mov eax,1
0x100038f2: pop ecx
0x100038f3: ret
0x100038f4: mov eax,0E8FFFFh
0x100038f9: mov ds:[10138BF8h],eax
0x100038fe: mov ds:[10138C00h],eax
0x10003903: mov eax,0C0C0C0h
0x10003908: mov ecx,0E0E0E0h
0x1000390d: mov ds:[10138C08h],eax
0x10003912: mov ds:[10138C04h],eax
0x10003917: mov dword ptr ds:[10138BECh],0D0D0D0h
0x10003921: mov dword ptr ds:[10138BF0h],0FFE8E8h
0x1000392b: mov ds:[10138BF4h],ecx
0x10003931: mov ds:[10138BFCh],ecx
0x10003937: mov eax,1
0x1000393c: pop ecx
0x1000393d: ret
```

`sub_10003880` is called by sub_10003670. It delegates to sub_1000aa40 (1 callees).

12.71 sub_10003940 at 0x10003940 (34 instructions)

```
0x10003940: push esi
0x10003941: push 30h
0x10003943: call 10021FDEh
0x10003948: mov esi,eax
0x1000394a: add esp,4
0x1000394d: test esi,esi
0x1000394f: je short 1000399Ch
0x10003951: lea eax,[esi+4]
0x10003954: push eax
0x10003955: mov dword ptr [esi],1012F3B0h
0x1000395b: call dword ptr ds:[100FE5C8h]
0x10003961: mov ecx,[esp+8]
0x10003965: mov dword ptr [esi],1012F3C0h
0x1000396b: mov dword ptr [esi+18h],0
0x10003972: mov eax,1012F498h
0x10003977: mov [esi+14h],eax
0x1000397a: mov [esi+1Ch],eax
0x1000397d: mov dword ptr [esi+20h],0
0x10003984: mov [esi+24h],eax
0x10003987: xor eax,eax
0x10003989: mov dword ptr [esi+28h],0
0x10003990: test esi,esi
0x10003992: mov [ecx+14h],esi
0x10003995: setne al
0x10003998: pop esi
0x10003999: ret 4
0x1000399c: mov ecx,[esp+8]
0x100039a0: xor esi,esi
0x100039a2: xor eax,eax
0x100039a4: test esi,esi
0x100039a6: mov [ecx+14h],esi
0x100039a9: setne al
0x100039ac: pop esi
0x100039ad: ret 4
```

`sub_10003940` is called by sub_10003670. It delegates to j_ORDINAL 762 (1 callees).

12.72 sub_100039b0 at 0x100039b0 (39 instructions)

```
0x100039b0: push 0FFFFFFFFh
0x100039b2: push 100FCF69h
0x100039b7: mov eax,fs:[0]
0x100039bd: push eax
0x100039be: sub esp,8
0x100039c1: push esi
0x100039c2: mov eax,ds:[10136220h]
0x100039c7: xor eax,esp
0x100039c9: push eax
0x100039ca: lea eax,[esp+10h]
0x100039ce: mov fs:[0],eax
0x100039d4: mov esi,ecx
0x100039d6: mov eax,[esi+1F4h]
0x100039dc: mov eax,[eax-0Ch]
0x100039df: test eax,eax
0x100039e1: jne short 10003A27h
0x100039e3: lea ecx,[esp+0Ch]
0x100039e7: call dword ptr ds:[100FE368h]
0x100039ed: lea ecx,[esp+0Ch]
0x100039f1: mov dword ptr [esp+18h],0
0x100039f9: push ecx
0x100039fa: mov ecx,[esi+8]
0x100039fd: call 1002229Ch
0x10003a02: lea edx,[esp+0Ch]
0x10003a06: add esi,178h
0x10003a0c: push edx
0x10003a0d: lea ecx,[esi+24h]
0x10003a10: call dword ptr ds:[100FE198h]
0x10003a16: mov ecx,esi
0x10003a18: call 1000C0F0h
0x10003a1d: lea ecx,[esp+0Ch]
0x10003a21: call dword ptr ds:[100FE374h]
0x10003a27: mov eax,1
0x10003a2c: mov ecx,[esp+10h]
0x10003a30: mov fs:[0],ecx
0x10003a37: pop ecx
0x10003a38: pop esi
0x10003a39: add esp,14h
0x10003a3c: ret
```

`sub_100039b0` is called by sub_10003670. It delegates to sub_1000c0f0, j_ORDINAL 3756 (2 callees).

12.73 sub_10003a40 at 0x10003a40 (44 instructions)

```
0x10003a40: push ebx
0x10003a41: push edi
0x10003a42: push esi
0x10003a43: call 10006D30h
0x10003a48: mov ecx,[esi+8]
0x10003a4b: mov ebx,[esi+0Ch]
0x10003a4e: xor edi,edi
0x10003a50: cmp ecx,edi
0x10003a52: je short 10003A9Eh
0x10003a54: cmp ebx,edi
0x10003a56: je short 10003A9Eh
0x10003a58: mov eax,[esi+288h]
0x10003a5e: cmp eax,edi
0x10003a60: je short 10003A73h
0x10003a62: push eax
0x10003a63: mov eax,[ecx+20h]
0x10003a66: push eax
0x10003a67: call dword ptr ds:[100FE5E4h]
0x10003a6d: mov [esi+288h],edi
0x10003a73: mov ecx,[esi+0Ch]
0x10003a76: mov edx,[esi+8]
```

```
0x10003a79: mov eax,[edx+20h]
0x10003a7c: push ecx
0x10003a7d: push 0FFFFFFFCh
0x10003a7f: push eax
0x10003a80: call dword ptr ds:[100FE618h]
0x10003a86: mov ecx,[esi+8]
0x10003a89: mov edx,[ecx+20h]
0x10003a8c: push 10138C18h
0x10003a91: push edx
0x10003a92: call dword ptr ds:[100FE678h]
0x10003a98: mov [esi+8],edi
0x10003a9b: mov [esi+0Ch],edi
0x10003a9e: mov eax,[esi+14h]
0x10003aa1: cmp eax,edi
0x10003aa3: je short 10003AB1h
0x10003aa5: push eax
0x10003aa6: call 10021FCCh
0x10003aab: add esp,4
0x10003aae: mov [esi+14h],edi
0x10003ab1: pop edi
0x10003ab2: mov eax,ebx
0x10003ab4: pop ebx
0x10003ab5: ret
```

`sub_10003a40` is called by ?UnsubClassWnd@CCustomTitle@@QAEXXZ. It delegates to sub_10006d30, j_ORDINAL 764 (2 callees).

12.74 sub_10003ac0 at 0x10003ac0 (37 instructions)

```
0x10003ac0: push ebp
0x10003ac1: mov ebp,esp
0x10003ac3: and esp,0FFFFFFF8h
0x10003ac6: sub esp,10h
0x10003ac9: test eax,eax
0x10003acb: jne short 10003AD9h
0x10003acd: cmp [esi+2A4h],eax
0x10003ad3: je short 10003AE2h
0x10003ad5: mov esp,ebp
0x10003ad7: pop ebp
0x10003ad8: ret
0x10003ad9: cmp dword ptr [esi+2A4h],0
0x10003ae0: je short 10003B1Eh
0x10003ae2: xor ecx,ecx
0x10003ae4: test eax,eax
0x10003ae6: mov eax,[esi+8]
0x10003ae9: sete cl
0x10003aec: lea edx,[esp]
0x10003aef: push edx
0x10003af0: mov [esi+2A4h],ecx
0x10003af6: mov ecx,[eax+20h]
0x10003af9: push ecx
0x10003afa: call dword ptr ds:[100FE674h]
0x10003b00: push esi
0x10003b01: lea eax,[esp+4]
0x10003b05: call 100056C0h
0x10003b0a: mov ecx,[esi+8]
0x10003b0d: push 37h
0x10003b0f: push 0
0x10003b11: push 0
0x10003b13: push 0
0x10003b15: push 0
0x10003b17: push 0
0x10003b19: call 1002226Ch
0x10003b1e: mov esp,ebp
0x10003b20: pop ebp
0x10003b21: ret
```

`sub_10003ac0` is called by ?ShowTitle@CCustomTitle@@QAEXH@Z. It delegates to sub_100056c0, j_ORDINAL 6061 (2 callees).

12.75 sub_10003b30 at 0x10003b30 (51 instructions)

```
0x10003b30: push ebp
0x10003b31: mov ebp,esp
0x10003b33: and esp,0FFFFFFF8h
0x10003b36: sub esp,10h
0x10003b39: cmp dword ptr [esi+288h],0
0x10003b40: push ebx
0x10003b41: push edi
0x10003b42: je short 10003B4Eh
0x10003b44: xor eax,eax
0x10003b46: pop edi
0x10003b47: pop ebx
0x10003b48: mov esp,ebp
0x10003b4a: pop ebp
0x10003b4b: ret 4
0x10003b4e: mov eax,[esi+8]
0x10003b51: mov ecx,[eax+20h]
0x10003b54: push 0
0x10003b56: push 0
0x10003b58: push 1Fh
0x10003b5a: push ecx
0x10003b5b: call dword ptr ds:[100FE620h]
0x10003b61: mov edx,[ebp+8]
0x10003b64: mov ecx,[edx+4]
0x10003b67: lea ebx,[esi+40h]
0x10003b6a: call 10008DF0h
0x10003b6f: mov edi,eax
0x10003b71: test edi,edi
0x10003b73: je short 10003BA5h
0x10003b75: mov ecx,[esi+8]
0x10003b78: mov edx,[ecx+20h]
0x10003b7b: lea eax,[esp+8]
0x10003b7f: push eax
0x10003b80: push edx
0x10003b81: call dword ptr ds:[100FE674h]
0x10003b87: push esi
0x10003b88: lea eax,[esp+0Ch]
0x10003b8c: call 100056C0h
0x10003b91: mov ecx,[esi+8]
0x10003b94: push 37h
0x10003b96: push 0
0x10003b98: push 0
0x10003b9a: push 0
0x10003b9c: push 0
0x10003b9e: push 0
0x10003ba0: call 1002226Ch
0x10003ba5: mov eax,edi
0x10003ba7: pop edi
0x10003ba8: pop ebx
0x10003ba9: mov esp,ebp
0x10003bab: pop ebp
0x10003bac: ret 4
```

`sub_10003b30` is called by ?SetMenu@CCustomTitle@@QAEHPAVCMenu@@@Z. It delegates to sub_100056c0, sub_10008df0, j_ORDINAL 6061 (3 callees).

12.76 sub_10003c30 at 0x10003c30 (48 instructions)

```
0x10003c30: push ebp
0x10003c31: mov ebp,esp
0x10003c33: and esp,0FFFFFFF8h
0x10003c36: sub esp,14h
```

```
0x10003c39: cmp dword ptr [esi+14h],0
0x10003c3d: push edi
0x10003c3e: jne short 10003C49h
0x10003c40: xor eax,eax
0x10003c42: pop edi
0x10003c43: mov esp,ebp
0x10003c45: pop ebp
0x10003c46: ret 4
0x10003c49: mov ecx,[esi+14h]
0x10003c4c: mov eax,[ecx]
0x10003c4e: mov eax,[eax]
0x10003c50: push edx
0x10003c51: mov edx,[ebp+8]
0x10003c54: push edx
0x10003c55: call eax
0x10003c57: mov edi,eax
0x10003c59: test edi,edi
0x10003c5b: je short 10003C97h
0x10003c5d: mov eax,[esi+8]
0x10003c60: test eax,eax
0x10003c62: je short 10003C97h
0x10003c64: cmp dword ptr [eax+20h],0
0x10003c68: je short 10003C97h
0x10003c6a: mov edx,[eax+20h]
0x10003c6d: lea ecx,[esp+8]
0x10003c71: push ecx
0x10003c72: push edx
0x10003c73: call dword ptr ds:[100FE674h]
0x10003c79: push esi
0x10003c7a: lea eax,[esp+0Ch]
0x10003c7e: call 100056C0h
0x10003c83: mov ecx,[esi+8]
0x10003c86: push 37h
0x10003c88: push 0
0x10003c8a: push 0
0x10003c8c: push 0
0x10003c8e: push 0
0x10003c90: push 0
0x10003c92: call 1002226Ch
0x10003c97: mov eax,edi
0x10003c99: pop edi
0x10003c9a: mov esp,ebp
0x10003c9c: pop ebp
0x10003c9d: ret 4
```

`sub_10003c30` is called by ?SetCaptionBkImage@CCustomTitle@@QAEHPAVCBitmap@@H@Z. It delegates to sub_100056c0, j_ORDINAL 6061 (2 callees).

12.77 sub_10003ca0 at 0x10003ca0 (37 instructions)

```
0x10003ca0: push ebp
0x10003ca1: mov ebp,esp
0x10003ca3: and esp,0FFFFFFF8h
0x10003ca6: sub esp,14h
0x10003ca9: push esi
0x10003caa: push eax
0x10003cab: lea esi,[ebx+1C0h]
0x10003cb1: call 1000E2F0h
0x10003cb6: mov esi,eax
0x10003cb8: test esi,esi
0x10003cba: je short 10003CF8h
0x10003cbc: mov eax,[ebx+8]
0x10003cbf: test eax,eax
0x10003cc1: je short 10003CF6h
0x10003cc3: cmp dword ptr [eax+20h],0
0x10003cc7: je short 10003CF6h
```

```
0x10003cc9: mov edx,[eax+20h]
0x10003ccc: lea ecx,[esp+8]
0x10003cd0: push ecx
0x10003cd1: push edx
0x10003cd2: call dword ptr ds:[100FE674h]
0x10003cd8: push ebx
0x10003cd9: lea eax,[esp+0Ch]
0x10003cdd: call 100056C0h
0x10003ce2: mov ecx,[ebx+8]
0x10003ce5: push 37h
0x10003ce7: push 0
0x10003ce9: push 0
0x10003ceb: push 0
0x10003ced: push 0
0x10003cef: push 0
0x10003cf1: call 1002226Ch
0x10003cf6: mov eax,esi
0x10003cf8: pop esi
0x10003cf9: mov esp,ebp
0x10003cfb: pop ebp
0x10003cfc: ret
```

`sub_10003ca0` is called by ?SetToolBarBkImage@CCustomTitle@@QAEHPAVCBitmap@@H@Z. It delegates to sub_100056c0, sub_1000e2f0, j_ORDINAL 6061 (3 callees).

12.78 sub_10003d00 at 0x10003d00 (9 instructions)

```
0x10003d00: mov ecx,[eax+8]
0x10003d03: push 37h
0x10003d05: push 0
0x10003d07: push 0
0x10003d09: push 0
0x10003d0b: push 0
0x10003d0d: push 0
0x10003d0f: call 1002226Ch
0x10003d14: ret
```

`sub_10003d00` is called by sub_10003670. It delegates to j_ORDINAL 6061 (1 callees).

12.79 sub_10003d20 at 0x10003d20 (46 instructions)

```
0x10003d20: sub esp,68h
0x10003d23: mov ecx,[esi+8]
0x10003d26: xor eax,eax
0x10003d28: mov [esp+4],eax
0x10003d2c: mov [esp+8],eax
0x10003d30: mov [esp+0Ch],eax
0x10003d34: mov [esp+10h],eax
0x10003d38: mov [esp+14h],eax
0x10003d3c: mov [esp+18h],eax
0x10003d40: mov [esp+1Ch],eax
0x10003d44: mov [esp+20h],eax
0x10003d48: mov [esp+24h],eax
0x10003d4c: mov [esp+28h],eax
0x10003d50: lea eax,[esp]
0x10003d53: push eax
0x10003d54: mov dword ptr [esp+4],2Ch
0x10003d5c: call 1002221Eh
0x10003d61: mov eax,3
0x10003d66: cmp [esp+8],eax
0x10003d6a: jne short 10003DB3h
0x10003d6c: push 3Ch
0x10003d6e: lea ecx,[esp+30h]
0x10003d72: push 0
0x10003d74: push ecx
```

```
0x10003d75: call 100228C0h
0x10003d7a: mov eax,[esi+8]
0x10003d7d: add esp,0Ch
0x10003d80: lea edx,[esp+2Ch]
0x10003d84: mov dword ptr [esp+2Ch],3Ch
0x10003d8c: mov ecx,[eax+20h]
0x10003d8f: push edx
0x10003d90: push ecx
0x10003d91: call dword ptr ds:[100FE670h]
0x10003d97: test eax,eax
0x10003d99: je short 10003DC4h
0x10003d9b: mov edx,[esp+5Ch]
0x10003d9f: mov ecx,[esp+60h]
0x10003da3: mov [esi+278h],edx
0x10003da9: mov [esi+27Ch],ecx
0x10003daf: add esp,68h
0x10003db2: ret
0x10003db3: mov [esi+278h],eax
0x10003db9: mov [esi+27Ch],eax
0x10003dbf: mov eax,1
0x10003dc4: add esp,68h
0x10003dc7: ret
```

`sub_10003d20` is called by sub_10003670. It delegates to j_ORDINAL 3752, j_memset (2 callees).

12.80 sub_10003dd0 at 0x10003dd0 (26 instructions)

```
0x10003dd0: push esi
0x10003dd1: mov esi,[esp+8]
0x10003dd5: push 10138C18h
0x10003dda: push esi
0x10003ddb: call dword ptr ds:[100FE668h]
0x10003de1: mov edx,[esp+0Ch]
0x10003de5: mov ecx,eax
0x10003de7: test ecx,ecx
0x10003de9: je short 10003DFFh
0x10003deb: mov eax,[esp+10h]
0x10003def: push eax
0x10003df0: mov eax,[esp+18h]
0x10003df4: push edx
0x10003df5: push esi
0x10003df6: call 10003E20h
0x10003dfb: pop esi
0x10003dfc: ret 10h
0x10003dff: mov eax,[esp+14h]
0x10003e03: mov ecx,[esp+10h]
0x10003e07: push eax
0x10003e08: push ecx
0x10003e09: push edx
0x10003e0a: push esi
0x10003e0b: call dword ptr ds:[100FE690h]
0x10003e11: pop esi
0x10003e12: ret 10h
```

`sub_10003dd0` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10003e20 (1 callees).

12.81 sub_10003e20 at 0x10003e20 (114 instructions)

```
0x10003e20: push ebp
0x10003e21: mov ebp,esp
0x10003e23: and esp,0FFFFFFF8h
0x10003e26: push ecx
0x10003e27: push ebx
0x10003e28: mov ebx,[ebp+10h]
```

```
0x10003e2b: push esi
0x10003e2c: push edi
0x10003e2d: mov edi,eax
0x10003e2f: mov eax,[ebp+0Ch]
0x10003e32: push edi
0x10003e33: push ebx
0x10003e34: mov esi,ecx
0x10003e36: mov ecx,[ebp+8]
0x10003e39: push eax
0x10003e3a: push ecx
0x10003e3b: mov ecx,esi
0x10003e3d: call 100067C0h
0x10003e42: mov eax,[ebp+0Ch]
0x10003e45: cmp eax,112h
0x10003e4a: ja near ptr 10003F55h
0x10003e50: je near ptr 10003F41h
0x10003e56: lea ecx,[eax-6]
0x10003e59: cmp ecx,9Eh
0x10003e5f: ja near ptr 10003FEBh
0x10003e65: movzx edx,byte ptr [ecx+10004080h]
0x10003e6c: jmp dword ptr [edx*4+10004050h]
0x10003f41: mov eax,edi
0x10003f43: mov ecx,ebx
0x10003f45: mov edx,esi
0x10003f47: call 100051B0h
0x10003f4c: pop edi
0x10003f4d: pop esi
0x10003f4e: pop ebx
0x10003f4f: mov esp,ebp
0x10003f51: pop ebp
0x10003f52: ret 0Ch
0x10003f55: cmp eax,202h
0x10003f5a: ja short 10003FD4h
0x10003f5c: je short 10003FC1h
0x10003f5e: lea ecx,[eax-113h]
0x10003f64: cmp ecx,0EDh
0x10003f6a: ja short 10003FEBh
0x10003f6c: movzx ecx,byte ptr [ecx+10004134h]
0x10003f73: jmp dword ptr [ecx*4+10004120h]
0x10003fc1: push esi
0x10003fc2: mov eax,edi
0x10003fc4: mov edx,ebx
0x10003fc6: call 10004910h
0x10003fcb: pop edi
0x10003fcc: pop esi
0x10003fcd: pop ebx
0x10003fce: mov esp,ebp
0x10003fd0: pop ebp
0x10003fd1: ret 0Ch
0x10003fd4: mov ecx,eax
0x10003fd6: sub ecx,205h
0x10003fdc: je short 10004040h
0x10003fde: sub ecx,9Dh
0x10003fe4: je short 1000402Fh
0x10003fe6: sub ecx,75h
0x10003fe9: je short 1000401Ch
0x10003feb: mov esi,[esi+0Ch]
0x10003fee: test esi,esi
0x10003ff0: push edi
0x10003ff1: push ebx
0x10003ff2: push eax
0x10003ff3: je short 10004009h
0x10003ff5: mov edx,[ebp+8]
0x10003ff8: push edx
0x10003ff9: push esi
0x10003ffa: call dword ptr ds:[100FE67Ch]
0x10004000: pop edi
```

```
0x10004001: pop esi
0x10004002: pop ebx
0x10004003: mov esp,ebp
0x10004005: pop ebp
0x10004006: ret 0Ch
0x10004009: mov eax,[ebp+8]
0x1000400c: push eax
0x1000400d: call dword ptr ds:[100FE690h]
0x10004013: pop edi
0x10004014: pop esi
0x10004015: pop ebx
0x10004016: mov esp,ebp
0x10004018: pop ebp
0x10004019: ret 0Ch
0x1000401c: mov ebx,edi
0x1000401e: mov edi,[ebp+10h]
0x10004021: call 10005680h
0x10004026: pop edi
0x10004027: pop esi
0x10004028: pop ebx
0x10004029: mov esp,ebp
0x1000402b: pop ebp
0x1000402c: ret 0Ch
0x1000402f: push edi
0x10004030: push ebx
0x10004031: push esi
0x10004032: call 10004D40h
0x10004037: pop edi
0x10004038: pop esi
0x10004039: pop ebx
0x1000403a: mov esp,ebp
0x1000403c: pop ebp
0x1000403d: ret 0Ch
0x10004040: mov eax,edi
0x10004042: call 100050F0h
0x10004047: pop edi
0x10004048: pop esi
0x10004049: pop ebx
0x1000404a: mov esp,ebp
0x1000404c: pop ebp
0x1000404d: ret 0Ch
```

`sub_10003e20` is called by sub_10003dd0. It delegates to sub_10004910, sub_10004d40, sub_100050f0 (6 callees).

12.82 sub_10004910 at 0x10004910 (182 instructions)

```
0x10004910: sub esp,18h
0x10004913: movsx ecx,ax
0x10004916: mov [esp],ecx
0x10004919: push ebx
0x1000491a: mov ecx,eax
0x1000491c: shr ecx,10h
0x1000491f: push ebp
0x10004920: mov ebp,[esp+24h]
0x10004924: cmp dword ptr [ebp+28Ch],0
0x1000492b: movsx ecx,cx
0x1000492e: push esi
0x1000492f: push edi
0x10004930: mov [esp+14h],ecx
0x10004934: je near ptr 10004B09h
0x1000493a: mov dword ptr [ebp+28Ch],0
0x10004944: call dword ptr ds:[100FE654h]
0x1000494a: mov eax,[ebp+8]
0x1000494d: mov ecx,[eax+20h]
0x10004950: lea edx,[esp+18h]
0x10004954: push edx
```

```
0x10004955: push ecx
0x10004956: call dword ptr ds:[100FE674h]
0x1000495c: mov eax,[ebp+8]
0x1000495f: mov ecx,[eax+20h]
0x10004962: lea edx,[esp+10h]
0x10004966: push edx
0x10004967: push ecx
0x10004968: call dword ptr ds:[100FE698h]
0x1000496e: mov eax,[esp+1Ch]
0x10004972: sub [esp+14h],eax
0x10004976: mov edx,[esp+18h]
0x1000497a: mov eax,[ebp+29Ch]
0x10004980: sub [esp+10h],edx
0x10004984: cmp eax,8
0x10004987: je near ptr 10004A5Ch
0x1000498d: cmp eax,9
0x10004990: je near ptr 10004A5Ch
0x10004996: cmp eax,14h
0x10004999: je near ptr 10004A5Ch
0x1000499f: cmp eax,12h
0x100049a2: jne near ptr 10004AC9h
0x100049a8: lea esi,[ebp+0D8h]
0x100049ae: call 1000B7C0h
0x100049b3: mov ebx,eax
0x100049b5: test ebx,ebx
0x100049b7: jl short 100049EFh
0x100049b9: push 0
0x100049bb: push 0FFFFFFFFh
0x100049bd: mov edi,esi
0x100049bf: call 1000B660h
0x100049c4: lea edi,[esp+18h]
0x100049c8: mov eax,esi
0x100049ca: call 1000B140h
0x100049cf: mov ecx,eax
0x100049d1: mov edi,ebp
0x100049d3: call 10006310h
0x100049d8: mov ecx,[ebp+8]
0x100049db: push ecx
0x100049dc: mov eax,ebx
0x100049de: call 1000B820h
0x100049e3: xor eax,eax
0x100049e5: pop edi
0x100049e6: pop esi
0x100049e7: pop ebp
0x100049e8: pop ebx
0x100049e9: add esp,18h
0x100049ec: ret 4
0x100049ef: lea edx,[esp+2Ch]
0x100049f3: push edx
0x100049f4: lea eax,[ebp+294h]
0x100049fa: lea esi,[ebp+1C0h]
0x10004a00: push eax
0x10004a01: call 1000D800h
0x10004a06: test eax,eax
0x10004a08: jl near ptr 10004AC9h
0x10004a0e: lea ecx,[esp+2Ch]
0x10004a12: push ecx
0x10004a13: lea edx,[esp+14h]
0x10004a17: push edx
0x10004a18: call 1000D800h
0x10004a1d: mov [esp+2Ch],eax
0x10004a21: push eax
0x10004a22: mov eax,esi
0x10004a24: call 1000DB80h
0x10004a29: test eax,eax
0x10004a2b: je near ptr 10004AC9h
0x10004a31: lea ebx,[esp+18h]
```

```
0x10004a35: call 1000D8E0h
0x10004a3a: mov ecx,eax
0x10004a3c: mov edi,ebp
0x10004a3e: call 10006310h
0x10004a43: mov eax,[ebp+8]
0x10004a46: mov edi,[esp+2Ch]
0x10004a4a: push eax
0x10004a4b: call 1000DED0h
0x10004a50: xor eax,eax
0x10004a52: pop edi
0x10004a53: pop esi
0x10004a54: pop ebp
0x10004a55: pop ebx
0x10004a56: add esp,18h
0x10004a59: ret 4
0x10004a5c: mov ecx,[ebp+8]
0x10004a5f: push ecx
0x10004a60: push 14h
0x10004a62: lea edx,[ebp+124h]
0x10004a68: xor ecx,ecx
0x10004a6a: call 100079A0h
0x10004a6f: lea ebx,[esp+2Ch]
0x10004a73: lea eax,[esp+10h]
0x10004a77: mov edi,ebp
0x10004a79: call 10005B50h
0x10004a7e: test eax,eax
0x10004a80: je short 10004AC9h
0x10004a82: mov esi,[esp+2Ch]
0x10004a86: cmp esi,[ebp+29Ch]
0x10004a8c: jne short 10004AC9h
0x10004a8e: mov ecx,[ebp+8]
0x10004a91: call 10022236h
0x10004a96: cmp esi,8
0x10004a99: jne short 10004AD5h
0x10004a9b: test eax,20000h
0x10004aa0: je short 10004AC9h
0x10004aa2: shr eax,15h
0x10004aa5: and eax,100h
0x10004aaa: or eax,0F020h
0x10004aaf: cmp eax,0FFFFFFFFh
0x10004ab2: je short 10004AC9h
0x10004ab4: mov edx,[ebp+8]
0x10004ab7: push 0
0x10004ab9: push eax
0x10004aba: mov eax,[edx+20h]
0x10004abd: push 112h
0x10004ac2: push eax
0x10004ac3: call dword ptr ds:[100FE620h]
0x10004ac9: xor eax,eax
0x10004acb: pop edi
0x10004acc: pop esi
0x10004acd: pop ebp
0x10004ace: pop ebx
0x10004acf: add esp,18h
0x10004ad2: ret 4
0x10004ad5: cmp esi,9
0x10004ad8: jne short 10004AF6h
0x10004ada: test eax,10000h
0x10004adf: je short 10004AC9h
0x10004ae1: and eax,1000000h
0x10004ae6: neg eax
0x10004ae8: sbb eax,eax
0x10004aea: and eax,0F0h
0x10004aef: add eax,0F030h
0x10004af4: jmp short 10004AAFh
0x10004af6: cmp esi,14h
0x10004af9: jne short 10004AC9h
```

```
0x10004afb: test eax,80000h
0x10004b00: je short 10004AC9h
0x10004b02: mov eax,0F060h
0x10004b07: jmp short 10004AB4h
0x10004b09: mov ecx,[ebp+8]
0x10004b0c: mov ebp,[ebp+0Ch]
0x10004b0f: test ebp,ebp
0x10004b11: mov ecx,[ecx+20h]
0x10004b14: push eax
0x10004b15: push edx
0x10004b16: push 202h
0x10004b1b: push ecx
0x10004b1c: je short 10004B2Fh
0x10004b1e: push ebp
0x10004b1f: call dword ptr ds:[100FE67Ch]
0x10004b25: pop edi
0x10004b26: pop esi
0x10004b27: pop ebp
0x10004b28: pop ebx
0x10004b29: add esp,18h
0x10004b2c: ret 4
0x10004b2f: call dword ptr ds:[100FE690h]
0x10004b35: pop edi
0x10004b36: pop esi
0x10004b37: pop ebp
0x10004b38: pop ebx
0x10004b39: add esp,18h
0x10004b3c: ret 4
```

`sub_10004910` is called by sub_10003e20. It delegates to sub_10005b50, sub_10006310, sub_100079a0 (12 callees).

12.83 sub_10004d40 at 0x10004d40 (89 instructions)

```
0x10004d40: sub esp,10h
0x10004d43: push ebx
0x10004d44: push ebp
0x10004d45: mov ebp,[esp+1Ch]
0x10004d49: cmp dword ptr [ebp+280h],0
0x10004d50: push esi
0x10004d51: push edi
0x10004d52: je near ptr 10004E11h
0x10004d58: cmp dword ptr [ebp+28Ch],0
0x10004d5f: jne near ptr 10004E11h
0x10004d65: cmp dword ptr [ebp+0A8h],0
0x10004d6c: jne short 10004D99h
0x10004d6e: lea esi,[ebp+40h]
0x10004d71: call 10008C00h
0x10004d76: test eax,eax
0x10004d78: jl short 10004D99h
0x10004d7a: push 0
0x10004d7c: push 0FFFFFFFFh
0x10004d7e: xor eax,eax
0x10004d80: call 10008B20h
0x10004d85: lea edi,[esp+10h]
0x10004d89: mov eax,esi
0x10004d8b: call 100086B0h
0x10004d90: mov ecx,eax
0x10004d92: mov edi,ebp
0x10004d94: call 10006310h
0x10004d99: lea ecx,[ebp+124h]
0x10004d9f: mov eax,1
0x10004da4: cmp [ecx+18h],eax
0x10004da7: je short 10004DB3h
0x10004da9: cmp [ecx+2Ch],eax
0x10004dac: je short 10004DB3h
0x10004dae: cmp [ecx+40h],eax
```

```
0x10004db1: jne short 10004DC0h
0x10004db3: mov eax,[ebp+8]
0x10004db6: push eax
0x10004db7: push 0
0x10004db9: push 14h
0x10004dbb: call 10007B10h
0x10004dc0: lea esi,[ebp+0D8h]
0x10004dc6: call 1000B760h
0x10004dcb: test eax,eax
0x10004dcd: jl short 10004DEEh
0x10004dcf: push 0FFFFFFFFh
0x10004dd1: xor eax,eax
0x10004dd3: mov edi,esi
0x10004dd5: call 1000B6E0h
0x10004dda: lea edi,[esp+10h]
0x10004dde: mov eax,esi
0x10004de0: call 1000B140h
0x10004de5: mov ecx,eax
0x10004de7: mov edi,ebp
0x10004de9: call 10006310h
0x10004dee: lea esi,[ebp+1C0h]
0x10004df4: push 0FFFFFFFFh
0x10004df6: call 1000DC90h
0x10004dfb: test eax,eax
0x10004dfd: je short 10004E11h
0x10004dff: lea ebx,[esp+10h]
0x10004e03: call 1000D8E0h
0x10004e08: mov ecx,eax
0x10004e0a: mov edi,ebp
0x10004e0c: call 10006310h
0x10004e11: mov ecx,[ebp+8]
0x10004e14: mov eax,[ecx+20h]
0x10004e17: mov edx,[esp+2Ch]
0x10004e1b: mov ecx,[esp+28h]
0x10004e1f: mov ebp,[ebp+0Ch]
0x10004e22: test ebp,ebp
0x10004e24: push edx
0x10004e25: push ecx
0x10004e26: push 2A2h
0x10004e2b: push eax
0x10004e2c: je short 10004E3Fh
0x10004e2e: push ebp
0x10004e2f: call dword ptr ds:[100FE67Ch]
0x10004e35: pop edi
0x10004e36: pop esi
0x10004e37: pop ebp
0x10004e38: pop ebx
0x10004e39: add esp,10h
0x10004e3c: ret 0Ch
0x10004e3f: call dword ptr ds:[100FE690h]
0x10004e45: pop edi
0x10004e46: pop esi
0x10004e47: pop ebp
0x10004e48: pop ebx
0x10004e49: add esp,10h
0x10004e4c: ret 0Ch
```

`sub_10004d40` is called by sub_10003e20. It delegates to sub_10006310, sub_10007b10, sub_100086b0 (10 callees).

12.84 sub_100050f0 at 0x100050f0 (56 instructions)

```
0x100050f0: sub esp,28h
0x100050f3: movsx ecx,ax
0x100050f6: shr eax,10h
0x100050f9: cmp dword ptr [esi+290h],0
0x10005100: movsx edx,ax
```

```
0x10005103: push edi
0x10005104: mov [esp+10h],ecx
0x10005108: mov [esp+14h],edx
0x1000510c: je near ptr 1000519Eh
0x10005112: mov dword ptr [esi+290h],0
0x1000511c: call dword ptr ds:[100FE654h]
0x10005122: mov eax,[esp+10h]
0x10005126: mov ecx,[esp+14h]
0x1000512a: mov [esp+8],eax
0x1000512e: mov eax,[esi+8]
0x10005131: lea edx,[esp+18h]
0x10005135: mov [esp+0Ch],ecx
0x10005139: mov ecx,[eax+20h]
0x1000513c: push edx
0x1000513d: push ecx
0x1000513e: call dword ptr ds:[100FE674h]
0x10005144: mov eax,[esi+8]
0x10005147: mov ecx,[eax+20h]
0x1000514a: mov edi,ds:[100FE698h]
0x10005150: lea edx,[esp+8]
0x10005154: push edx
0x10005155: push ecx
0x10005156: call edi
0x10005158: mov ecx,[esp+0Ch]
0x1000515c: mov eax,[esp+8]
0x10005160: sub ecx,[esp+1Ch]
0x10005164: sub eax,[esp+18h]
0x10005168: push ecx
0x10005169: push eax
0x1000516a: lea edx,[esi+268h]
0x10005170: push edx
0x10005171: mov [esp+14h],eax
0x10005175: mov [esp+18h],ecx
0x10005179: call dword ptr ds:[100FE610h]
0x1000517f: test eax,eax
0x10005181: je short 1000519Eh
0x10005183: mov ecx,[esi+8]
0x10005186: mov edx,[ecx+20h]
0x10005189: lea eax,[esp+10h]
0x1000518d: push eax
0x1000518e: push edx
0x1000518f: call edi
0x10005191: push 0
0x10005193: lea eax,[esp+14h]
0x10005197: push eax
0x10005198: push esi
0x10005199: call 100063A0h
0x1000519e: xor eax,eax
0x100051a0: pop edi
0x100051a1: add esp,28h
0x100051a4: ret
```

`sub_100050f0` is called by sub_10003e20. It delegates to sub_100063a0 (1 callees).

12.85 sub_100051b0 at 0x100051b0 (141 instructions)

```
0x100051b0: sub esp,1Ch
0x100051b3: push esi
0x100051b4: mov esi,edx
0x100051b6: mov edx,[esi+8]
0x100051b9: test edx,edx
0x100051bb: push edi
0x100051bc: jne short 100051ECh
0x100051be: mov esi,[esi+0Ch]
0x100051c1: test esi,esi
0x100051c3: mov edx,ds:[20h]
```

```
0x100051c9: push eax
0x100051ca: push ecx
0x100051cb: push 112h
0x100051d0: push edx
0x100051d1: je short 100051E0h
0x100051d3: push esi
0x100051d4: call dword ptr ds:[100FE67Ch]
0x100051da: pop edi
0x100051db: pop esi
0x100051dc: add esp,1Ch
0x100051df: ret
0x100051e0: call dword ptr ds:[100FE690h]
0x100051e6: pop edi
0x100051e7: pop esi
0x100051e8: add esp,1Ch
0x100051eb: ret
0x100051ec: mov edi,ecx
0x100051ee: and edi,0FFF0h
0x100051f4: cmp edi,0F100h
0x100051fa: je short 1000521Bh
0x100051fc: mov esi,[esi+0Ch]
0x100051ff: test esi,esi
0x10005201: mov edx,[edx+20h]
0x10005204: push eax
0x10005205: push ecx
0x10005206: push 112h
0x1000520b: push edx
0x1000520c: je short 100051E0h
0x1000520e: push esi
0x1000520f: call dword ptr ds:[100FE67Ch]
0x10005215: pop edi
0x10005216: pop esi
0x10005217: add esp,1Ch
0x1000521a: ret
0x1000521b: cmp ax,20h
0x1000521f: jne near ptr 100052DAh
0x10005225: cmp dword ptr [esi+1Ch],0
0x10005229: mov edi,ds:[100FE65Ch]
0x1000522f: lea ecx,[esi+18h]
0x10005232: jne short 1000523Ah
0x10005234: cmp dword ptr [ecx+8],0
0x10005238: je short 1000524Fh
0x1000523a: lea edx,[esp+10h]
0x1000523e: call 1000A150h
0x10005243: mov eax,edx
0x10005245: push eax
0x10005246: mov ecx,esi
0x10005248: call 10003280h
0x1000524d: jmp short 10005293h
0x1000524f: mov edx,[edx+20h]
0x10005252: lea ecx,[esp+10h]
0x10005256: push ecx
0x10005257: push edx
0x10005258: call dword ptr ds:[100FE674h]
0x1000525e: mov eax,[esi+27Ch]
0x10005264: mov ecx,[esi+278h]
0x1000526a: neg eax
0x1000526c: push eax
0x1000526d: neg ecx
0x1000526f: push ecx
0x10005270: lea edx,[esp+18h]
0x10005274: push edx
0x10005275: call dword ptr ds:[100FE650h]
0x1000527b: mov eax,[esi+8]
0x1000527e: mov eax,[eax+20h]
0x10005281: push eax
0x10005282: call edi
```

```
0x10005284: test eax,eax
0x10005286: jne short 10005293h
0x10005288: mov ecx,[esi+4]
0x1000528b: add ecx,[esp+14h]
0x1000528f: mov [esp+1Ch],ecx
0x10005293: mov eax,[esi+8]
0x10005296: mov edx,[eax+20h]
0x10005299: push edx
0x1000529a: call edi
0x1000529c: push 1
0x1000529e: lea edx,[esp+0Ch]
0x100052a2: test eax,eax
0x100052a4: mov eax,[esp+14h]
0x100052a8: push edx
0x100052a9: push esi
0x100052aa: mov [esp+14h],eax
0x100052ae: je short 100052C5h
0x100052b0: mov ecx,[esp+20h]
0x100052b4: mov [esp+18h],ecx
0x100052b8: call 100063A0h
0x100052bd: xor eax,eax
0x100052bf: pop edi
0x100052c0: pop esi
0x100052c1: add esp,1Ch
0x100052c4: ret
0x100052c5: mov ecx,[esp+28h]
0x100052c9: mov [esp+18h],ecx
0x100052cd: call 100063A0h
0x100052d2: xor eax,eax
0x100052d4: pop edi
0x100052d5: pop esi
0x100052d6: add esp,1Ch
0x100052d9: ret
0x100052da: jbe short 1000530Ch
0x100052dc: add esi,40h
0x100052df: push eax
0x100052e0: call 10008CC0h
0x100052e5: test eax,eax
0x100052e7: jl short 100052FCh
0x100052e9: mov ecx,eax
0x100052eb: mov edx,esi
0x100052ed: push 1
0x100052ef: call 10009450h
0x100052f4: xor eax,eax
0x100052f6: pop edi
0x100052f7: pop esi
0x100052f8: add esp,1Ch
0x100052fb: ret
0x100052fc: push 40h
0x100052fe: call dword ptr ds:[100FE600h]
0x10005304: xor eax,eax
0x10005306: pop edi
0x10005307: pop esi
0x10005308: add esp,1Ch
0x1000530b: ret
0x1000530c: lea edx,[esi+40h]
0x1000530f: or ecx,0FFFFFFFFh
0x10005312: push 1
0x10005314: call 10009450h
0x10005319: pop edi
0x1000531a: xor eax,eax
0x1000531c: pop esi
0x1000531d: add esp,1Ch
0x10005320: ret
```

`sub_100051b0` is called by sub_10003e20. It delegates to sub_10003280, sub_100063a0, sub_10008cc0 (5 callees).

12.86 sub_10005680 at 0x10005680 (26 instructions)

```
0x10005680: push ecx
0x10005681: test bl,2
0x10005684: je short 10005697h
0x10005686: push 1
0x10005688: push 0
0x1000568a: push edi
0x1000568b: call 10022242h
0x10005690: push eax
0x10005691: push esi
0x10005692: call 10005FD0h
0x10005697: mov eax,[esi+8]
0x1000569a: mov ecx,[esi+0Ch]
0x1000569d: test ecx,ecx
0x1000569f: mov eax,[eax+20h]
0x100056a2: push ebx
0x100056a3: push edi
0x100056a4: push 317h
0x100056a9: push eax
0x100056aa: je short 100056B5h
0x100056ac: push ecx
0x100056ad: call dword ptr ds:[100FE67Ch]
0x100056b3: pop ecx
0x100056b4: ret
0x100056b5: call dword ptr ds:[100FE690h]
0x100056bb: pop ecx
0x100056bc: ret
```

`sub_10005680` is called by sub_10003e20. It delegates to sub_10005fd0, j_ORDINAL 2361 (2 callees).

12.87 sub_100056c0 at 0x100056c0 (332 instructions)

```
0x100056c0: sub esp,48h
0x100056c3: mov edx,[eax+4]
0x100056c6: mov ecx,[eax]
0x100056c8: push ebx
0x100056c9: push ebp
0x100056ca: mov ebp,[esp+54h]
0x100056ce: push esi
0x100056cf: mov esi,[eax+8]
0x100056d2: mov eax,[eax+0Ch]
0x100056d5: push edi
0x100056d6: mov [esp+1Ch],edx
0x100056da: mov [esp+18h],ecx
0x100056de: neg edx
0x100056e0: push edx
0x100056e1: neg ecx
0x100056e3: push ecx
0x100056e4: lea ecx,[esp+20h]
0x100056e8: push ecx
0x100056e9: mov [esp+2Ch],esi
0x100056ed: mov [esp+30h],eax
0x100056f1: call dword ptr ds:[100FE6B0h]
0x100056f7: cmp dword ptr [ebp+2A4h],0
0x100056fe: mov edx,[esp+18h]
0x10005702: mov eax,[esp+1Ch]
0x10005706: mov ecx,[esp+20h]
0x1000570a: mov [ebp+258h],edx
0x10005710: mov edx,[esp+24h]
0x10005714: mov [ebp+25Ch],eax
0x1000571a: mov [ebp+260h],ecx
0x10005720: mov [ebp+264h],edx
0x10005726: je short 1000573Fh
0x10005728: add ebp,268h
0x1000572e: push ebp
0x1000572f: call dword ptr ds:[100FE5C8h]
```

```
0x10005735: pop edi
0x10005736: pop esi
0x10005737: pop ebp
0x10005738: pop ebx
0x10005739: add esp,48h
0x1000573c: ret 4
0x1000573f: mov eax,[ebp+8]
0x10005742: mov eax,[eax+20h]
0x10005745: push eax
0x10005746: call dword ptr ds:[100FE65Ch]
0x1000574c: mov ecx,[ebp+27Ch]
0x10005752: mov edx,[ebp+278h]
0x10005758: neg ecx
0x1000575a: push ecx
0x1000575b: mov esi,eax
0x1000575d: neg edx
0x1000575f: push edx
0x10005760: lea eax,[esp+20h]
0x10005764: push eax
0x10005765: mov [esp+68h],esi
0x10005769: call dword ptr ds:[100FE650h]
0x1000576f: test esi,esi
0x10005771: mov edx,[esp+18h]
0x10005775: mov [ebp+268h],edx
0x1000577b: jne short 100057E4h
0x1000577d: mov ecx,[esp+1Ch]
0x10005781: mov eax,[ebp+4]
0x10005784: add eax,ecx
0x10005786: mov [ebp+26Ch],ecx
0x1000578c: mov ecx,[esp+20h]
0x10005790: mov [ebp+270h],ecx
0x10005796: mov [ebp+274h],eax
0x1000579c: cmp [ebp+1E4h],esi
0x100057a2: mov [esp+24h],eax
0x100057a6: je short 100057B6h
0x100057a8: mov eax,[ebp+1ECh]
0x100057ae: add [ebp+274h],eax
0x100057b4: jmp short 10005802h
0x100057b6: mov eax,[ebp+1C8h]
0x100057bc: test eax,eax
0x100057be: je short 100057DAh
0x100057c0: mov ecx,[ebp+1CCh]
0x100057c6: sub ecx,eax
0x100057c8: sar ecx,2
0x100057cb: je short 100057DAh
0x100057cd: mov eax,19h
0x100057d2: add [ebp+274h],eax
0x100057d8: jmp short 10005802h
0x100057da: xor eax,eax
0x100057dc: add [ebp+274h],eax
0x100057e2: jmp short 10005802h
0x100057e4: mov eax,[esp+1Ch]
0x100057e8: mov ecx,[esp+20h]
0x100057ec: mov [ebp+26Ch],eax
0x100057f2: mov eax,[esp+24h]
0x100057f6: mov [ebp+270h],ecx
0x100057fc: mov [ebp+274h],eax
0x10005802: mov eax,[ebp+14h]
0x10005805: mov [eax+4],edx
0x10005808: mov ecx,[esp+1Ch]
0x1000580c: add eax,4
0x1000580f: mov [eax+4],ecx
0x10005812: mov edx,[esp+20h]
0x10005816: mov [eax+8],edx
0x10005819: mov ecx,[esp+24h]
0x1000581d: mov [eax+0Ch],ecx
0x10005820: mov ecx,[ebp+8]
```

```
0x10005823: call 10022236h
0x10005828: lea edx,[esp+18h]
0x1000582c: mov edi,eax
0x1000582e: push edx
0x1000582f: push edi
0x10005830: lea ebx,[ebp+18h]
0x10005833: lea esi,[esp+30h]
0x10005837: call 1000A030h
0x1000583c: mov ebx,ds:[100FE6ACh]
0x10005842: mov eax,esi
0x10005844: push eax
0x10005845: call ebx
0x10005847: test eax,eax
0x10005849: jne short 10005853h
0x1000584b: mov ecx,[esp+30h]
0x1000584f: mov [esp+18h],ecx
0x10005853: lea edx,[esp+18h]
0x10005857: push edx
0x10005858: push edi
0x10005859: lea edi,[ebp+124h]
0x1000585f: lea esi,[esp+30h]
0x10005863: call 100077A0h
0x10005868: mov eax,esi
0x1000586a: push eax
0x1000586b: call ebx
0x1000586d: test eax,eax
0x1000586f: jne short 10005879h
0x10005871: mov ecx,[esp+28h]
0x10005875: mov [esp+20h],ecx
0x10005879: cmp dword ptr [esp+5Ch],0
0x1000587e: jne near ptr 10005949h
0x10005884: mov eax,[esp+18h]
0x10005888: mov edx,[esp+1Ch]
0x1000588c: mov ecx,[esp+20h]
0x10005890: mov [esp+38h],eax
0x10005894: add eax,2
0x10005897: lea edi,[ebp+0D8h]
0x1000589d: mov [esp+38h],eax
0x100058a1: mov eax,[edi+8]
0x100058a4: test eax,eax
0x100058a6: mov [esp+3Ch],edx
0x100058aa: mov edx,[esp+24h]
0x100058ae: mov [esp+44h],edx
0x100058b2: je short 100058BEh
0x100058b4: mov edx,[edi+0Ch]
0x100058b7: sub edx,eax
0x100058b9: sar edx,2
0x100058bc: jne short 100058C2h
0x100058be: xor eax,eax
0x100058c0: jmp short 100058C5h
0x100058c2: mov eax,[edi+24h]
0x100058c5: mov edx,0FFFFFFF6h
0x100058ca: sub edx,eax
0x100058cc: add ecx,edx
0x100058ce: mov [esp+40h],ecx
0x100058d2: lea ecx,[esp+38h]
0x100058d6: lea eax,[ebp+40h]
0x100058d9: push ecx
0x100058da: lea ebx,[esp+2Ch]
0x100058de: call 100087C0h
0x100058e3: mov esi,ds:[100FE6ACh]
0x100058e9: mov edx,ebx
0x100058eb: push edx
0x100058ec: call esi
0x100058ee: test eax,eax
0x100058f0: jne short 100058FCh
0x100058f2: mov eax,[esp+30h]
```

```
0x100058f6: mov [esp+18h],eax
0x100058fa: jmp short 10005900h
0x100058fc: mov eax,[esp+18h]
0x10005900: mov ecx,[esp+1Ch]
0x10005904: mov edx,[esp+24h]
0x10005908: mov [esp+38h],eax
0x1000590c: add eax,50h
0x1000590f: mov [esp+3Ch],ecx
0x10005913: mov ecx,[esp+20h]
0x10005917: mov [esp+38h],eax
0x1000591b: lea eax,[esp+38h]
0x1000591f: sub ecx,0Ah
0x10005922: push eax
0x10005923: lea eax,[esp+2Ch]
0x10005927: mov [esp+48h],edx
0x1000592b: mov [esp+44h],ecx
0x1000592f: call 1000B250h
0x10005934: lea ecx,[esp+28h]
0x10005938: push ecx
0x10005939: call esi
0x1000593b: test eax,eax
0x1000593d: jne short 1000599Eh
0x1000593f: mov eax,[esp+28h]
0x10005943: mov [esp+20h],eax
0x10005947: jmp short 100059A2h
0x10005949: lea esi,[ebp+40h]
0x1000594c: mov eax,esi
0x1000594e: call 100084A0h
0x10005953: call 10008510h
0x10005958: or ebx,0FFFFFFFFh
0x1000595b: cmp dword ptr [esi+34h],0
0x1000595f: jl short 1000596Bh
0x10005961: mov edi,esi
0x10005963: mov [esi+34h],ebx
0x10005966: call 100091B0h
0x1000596b: lea esi,[ebp+0D8h]
0x10005971: mov eax,esi
0x10005973: call 1000ABD0h
0x10005978: call 1000AC40h
0x1000597d: cmp dword ptr [esi+2Ch],0
0x10005981: jl short 1000599Eh
0x10005983: lea ecx,[ebp+108h]
0x10005989: test ecx,ecx
0x1000598b: mov [ebp+104h],ebx
0x10005991: je short 1000599Eh
0x10005993: cmp dword ptr [ecx+4],0
0x10005997: je short 1000599Eh
0x10005999: call 100222C0h
0x1000599e: mov eax,[esp+20h]
0x100059a2: mov ecx,[ebp+1F8h]
0x100059a8: test ecx,ecx
0x100059aa: jge short 100059ECh
0x100059ac: mov ecx,[esp+18h]
0x100059b0: sub eax,0Ah
0x100059b3: add ecx,50h
0x100059b6: mov edx,eax
0x100059b8: sub edx,ecx
0x100059ba: mov [esp+18h],ecx
0x100059be: mov [esp+20h],eax
0x100059c2: jns short 100059CAh
0x100059c4: mov eax,ecx
0x100059c6: mov [esp+20h],eax
0x100059ca: mov edx,[esp+24h]
0x100059ce: mov [ebp+18Ch],ecx
0x100059d4: mov ecx,[esp+1Ch]
0x100059d8: mov [ebp+190h],ecx
0x100059de: mov [ebp+194h],eax
```

```
0x100059e4: mov [ebp+198h],edx
0x100059ea: jmp short 10005A56h
0x100059ec: mov edx,[esp+18h]
0x100059f0: mov esi,[esp+24h]
0x100059f4: mov [esp+38h],edx
0x100059f8: mov edx,[esp+1Ch]
0x100059fc: mov ebx,eax
0x100059fe: add ecx,8
0x10005a01: lea edi,[esp+48h]
0x10005a05: lea eax,[ebp+40h]
0x10005a08: mov [esp+3Ch],edx
0x10005a0c: mov [esp+38h],ecx
0x10005a10: call 100086B0h
0x10005a15: mov eax,edi
0x10005a17: push eax
0x10005a18: call dword ptr ds:[100FE6ACh]
0x10005a1e: test eax,eax
0x10005a20: jne short 10005A29h
0x10005a22: mov ebx,[esp+48h]
0x10005a26: add ebx,0FFFFFFF8h
0x10005a29: mov ecx,[esp+38h]
0x10005a2d: mov edx,[esp+3Ch]
0x10005a31: sub esi,[ebp+1FCh]
0x10005a37: mov [ebp+18Ch],ecx
0x10005a3d: mov [ebp+190h],edx
0x10005a43: mov [ebp+194h],ebx
0x10005a49: mov [ebp+198h],esi
0x10005a4f: mov byte ptr [ebp+1B4h],1
0x10005a56: cmp dword ptr [esp+5Ch],0
0x10005a5b: jne short 10005AC0h
0x10005a5d: cmp dword ptr [ebp+1E4h],0
0x10005a64: lea eax,[ebp+1C0h]
0x10005a6a: je short 10005AAFh
0x10005a6c: mov ecx,[eax+2Ch]
0x10005a6f: test ecx,ecx
0x10005a71: jle short 10005AC0h
0x10005a73: mov ecx,[ebp+268h]
0x10005a79: mov edx,[ebp+270h]
0x10005a7f: mov [esp+38h],ecx
0x10005a83: mov ecx,[ebp+26Ch]
0x10005a89: add ecx,[ebp+4]
0x10005a8c: mov [esp+40h],edx
0x10005a90: mov edx,[ebp+274h]
0x10005a96: mov [esp+3Ch],ecx
0x10005a9a: lea ecx,[esp+38h]
0x10005a9e: push ecx
0x10005a9f: push eax
0x10005aa0: lea esi,[esp+30h]
0x10005aa4: mov [esp+4Ch],edx
0x10005aa8: call 1000D9B0h
0x10005aad: jmp short 10005ACCh
0x10005aaf: mov ecx,[eax+8]
0x10005ab2: test ecx,ecx
0x10005ab4: je short 10005AC0h
0x10005ab6: mov edx,[eax+0Ch]
0x10005ab9: sub edx,ecx
0x10005abb: sar edx,2
0x10005abe: jne short 10005A73h
0x10005ac0: lea edx,[ebp+1C0h]
0x10005ac6: push edx
0x10005ac7: call 1000D260h
0x10005acc: mov eax,[esp+18h]
0x10005ad0: mov ecx,[esp+1Ch]
0x10005ad4: mov edx,[esp+20h]
0x10005ad8: mov [ebp+204h],eax
0x10005ade: mov eax,[esp+24h]
0x10005ae2: mov [ebp+208h],ecx
```

```
0x10005ae8: mov [ebp+20Ch],edx
0x10005aee: mov [ebp+210h],eax
0x10005af4: add dword ptr [ebp+208h],8
0x10005afb: lea edi,[esp+48h]
0x10005aff: lea eax,[ebp+40h]
0x10005b02: call 100086B0h
0x10005b07: mov esi,ds:[100FE6ACh]
0x10005b0d: mov ecx,edi
0x10005b0f: push ecx
0x10005b10: call esi
0x10005b12: test eax,eax
0x10005b14: jne short 10005B20h
0x10005b16: mov edx,[esp+50h]
0x10005b1a: mov [ebp+204h],edx
0x10005b20: lea edi,[esp+38h]
0x10005b24: lea eax,[ebp+0D8h]
0x10005b2a: call 1000B140h
0x10005b2f: mov eax,edi
0x10005b31: push eax
0x10005b32: call esi
0x10005b34: test eax,eax
0x10005b36: jne short 10005B42h
0x10005b38: mov ecx,[esp+38h]
0x10005b3c: mov [ebp+20Ch],ecx
0x10005b42: pop edi
0x10005b43: pop esi
0x10005b44: pop ebp
0x10005b45: pop ebx
0x10005b46: add esp,48h
0x10005b49: ret 4
```

`sub_100056c0`
is called by ?AddQuickButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0PB_W@Z, ?AddQuickButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0KKKK@Z, ?DeleteQuickButton@CCustomTitle@@QAEHH@Z, ?RemoveAllQuickButtons@CCustomTitle@@QAEXXZ, ?AddToolButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0PB_W@Z, ?AddToolButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0KKKK@Z, ?AddToolSeparator@CCustomTitle@@QAEHK@Z, sub_10003ac0, sub_10003b30, sub_10003c30, sub_10003ca0. It delegates to sub_100077a0, sub_100084a0, sub_10008510 (15 callees).

12.88 sub_10005b50 at 0x10005b50 (61 instructions)

```
0x10005b50: push ecx
0x10005b51: push ebp
0x10005b52: mov ebp,ds:[100FE610h]
0x10005b58: push esi
0x10005b59: mov esi,eax
0x10005b5b: mov eax,[esi+4]
0x10005b5e: mov ecx,[esi]
0x10005b60: push eax
0x10005b61: push ecx
0x10005b62: lea edx,[edi+268h]
0x10005b68: push edx
0x10005b69: call ebp
0x10005b6b: test eax,eax
0x10005b6d: je short 10005BDAh
0x10005b6f: mov eax,[esi+4]
0x10005b72: mov ecx,[esi]
0x10005b74: push eax
0x10005b75: push ecx
0x10005b76: lea edx,[edi+30h]
0x10005b79: push edx
```

```
0x10005b7a: mov dword ptr [ebx],2
0x10005b80: call ebp
0x10005b82: test eax,eax
0x10005b84: je short 10005B95h
0x10005b86: mov dword ptr [ebx],3
0x10005b8c: mov eax,1
0x10005b91: pop esi
0x10005b92: pop ebp
0x10005b93: pop ecx
0x10005b94: ret
0x10005b95: push ebx
0x10005b96: push esi
0x10005b97: lea eax,[edi+40h]
0x10005b9a: call 100085E0h
0x10005b9f: test eax,eax
0x10005ba1: jge short 10005B8Ch
0x10005ba3: push ebx
0x10005ba4: push esi
0x10005ba5: lea eax,[edi+0D8h]
0x10005bab: call 1000B070h
0x10005bb0: test eax,eax
0x10005bb2: jge short 10005B8Ch
0x10005bb4: push ebx
0x10005bb5: lea eax,[edi+124h]
0x10005bbb: call 10007710h
0x10005bc0: test eax,eax
0x10005bc2: jne short 10005B8Ch
0x10005bc4: push ebx
0x10005bc5: push esi
0x10005bc6: lea esi,[edi+1C0h]
0x10005bcc: call 1000D800h
0x10005bd1: mov eax,1
0x10005bd6: pop esi
0x10005bd7: pop ebp
0x10005bd8: pop ecx
0x10005bd9: ret
0x10005bda: pop esi
0x10005bdb: xor eax,eax
0x10005bdd: pop ebp
0x10005bde: pop ecx
0x10005bdf: ret
```

`sub_10005b50` is called by sub_10004910, sub_10006f30. It delegates to sub_10007710, sub_100085e0, sub_1000b070 (4 callees).

12.89 sub_10005fd0 at 0x10005fd0 (248 instructions)

```
0x10005fd0: push 0FFFFFFFFh
0x10005fd2: push 100FD3C0h
0x10005fd7: mov eax,fs:[0]
0x10005fdd: push eax
0x10005fde: sub esp,2Ch
0x10005fe1: push ebx
0x10005fe2: push ebp
0x10005fe3: push esi
0x10005fe4: push edi
0x10005fe5: mov eax,ds:[10136220h]
0x10005fea: xor eax,esp
0x10005fec: push eax
0x10005fed: lea eax,[esp+40h]
0x10005ff1: mov fs:[0],eax
0x10005ff7: mov ebp,[esp+50h]
0x10005ffb: mov eax,[esp+58h]
0x10005fff: push eax
0x10006000: call 10022272h
0x10006005: mov esi,[esp+54h]
```

```
0x10006009: push eax
0x1000600a: mov ecx,esi
0x1000600c: call 10022266h
0x10006011: lea ecx,[esp+30h]
0x10006015: call 1002202Ch
0x1000601a: xor edi,edi
0x1000601c: cmp esi,edi
0x1000601e: mov [esp+48h],edi
0x10006022: jne short 10006028h
0x10006024: xor eax,eax
0x10006026: jmp short 1000602Bh
0x10006028: mov eax,[esi+4]
0x1000602b: push eax
0x1000602c: call dword ptr ds:[100FE060h]
0x10006032: push eax
0x10006033: lea ecx,[esp+34h]
0x10006037: call 10022038h
0x1000603c: mov [esp+1Ch],edi
0x10006040: mov dword ptr [esp+18h],1012F498h
0x10006048: mov byte ptr [esp+48h],1
0x1000604d: mov eax,[ebp+274h]
0x10006053: sub eax,[ebp+26Ch]
0x10006059: mov ecx,[ebp+270h]
0x1000605f: sub ecx,[ebp+268h]
0x10006065: push eax
0x10006066: push ecx
0x10006067: mov ecx,[esi+4]
0x1000606a: push ecx
0x1000606b: call dword ptr ds:[100FE040h]
0x10006071: push eax
0x10006072: lea ecx,[esp+1Ch]
0x10006076: call 10021FFCh
0x1000607b: mov edx,[esp+1Ch]
0x1000607f: mov eax,[esp+34h]
0x10006083: push edx
0x10006084: push eax
0x10006085: call 1002203Eh
0x1000608a: mov ecx,[ebp+26Ch]
0x10006090: mov edx,[ebp+268h]
0x10006096: push ecx
0x10006097: mov [esp+54h],eax
0x1000609b: push edx
0x1000609c: lea eax,[esp+28h]
0x100060a0: push eax
0x100060a1: lea ecx,[esp+3Ch]
0x100060a5: call 10022284h
0x100060aa: mov ecx,[ebp+14h]
0x100060ad: mov edi,[esp+5Ch]
0x100060b1: mov edx,[ecx]
0x100060b3: mov edx,[edx+4]
0x100060b6: push edi
0x100060b7: lea eax,[esp+34h]
0x100060bb: push eax
0x100060bc: call edx
0x100060be: lea ebx,[ebp+18h]
0x100060c1: lea eax,[esp+30h]
0x100060c5: call 1000A0C0h
0x100060ca: push edi
0x100060cb: lea eax,[esp+34h]
0x100060cf: push eax
0x100060d0: lea eax,[ebp+40h]
0x100060d3: call 100089A0h
0x100060d8: push edi
0x100060d9: lea ecx,[esp+34h]
0x100060dd: push ecx
0x100060de: lea eax,[ebp+0D8h]
0x100060e4: call 1000B5A0h
```

```
0x100060e9: mov ecx,[ebp+8]
0x100060ec: call 10022236h
0x100060f1: push edi
0x100060f2: lea edx,[esp+34h]
0x100060f6: push edx
0x100060f7: lea ecx,[ebp+124h]
0x100060fd: call 100078E0h
0x10006102: lea esi,[ebp+178h]
0x10006108: push edi
0x10006109: lea eax,[esi+14h]
0x1000610c: push eax
0x1000610d: lea edi,[esp+38h]
0x10006111: call 1000BDE0h
0x10006116: lea ecx,[ebp+214h]
0x1000611c: push ecx
0x1000611d: mov ecx,edi
0x1000611f: call 1002228Ah
0x10006124: mov edx,[ebp+21Ch]
0x1000612a: push edx
0x1000612b: mov ecx,edi
0x1000612d: mov esi,eax
0x1000612f: call 10022290h
0x10006134: mov edi,eax
0x10006136: push 1
0x10006138: lea ecx,[esp+34h]
0x1000613c: call 10022296h
0x10006141: mov ebx,eax
0x10006143: mov eax,[ebp+200h]
0x10006149: mov ecx,[eax-0Ch]
0x1000614c: lea edx,[esp+20h]
0x10006150: push edx
0x10006151: push ecx
0x10006152: push eax
0x10006153: mov eax,[esp+44h]
0x10006157: push eax
0x10006158: call dword ptr ds:[100FE03Ch]
0x1000615e: mov ecx,[ebp+20Ch]
0x10006164: sub ecx,[ebp+204h]
0x1000616a: lea eax,[ebp+204h]
0x10006170: cmp [esp+20h],ecx
0x10006174: jge short 10006191h
0x10006176: mov ecx,[ebp+200h]
0x1000617c: mov edx,[ecx-0Ch]
0x1000617f: push 21h
0x10006181: push eax
0x10006182: push edx
0x10006183: mov edx,[esp+3Ch]
0x10006187: mov eax,[edx+70h]
0x1000618a: push ecx
0x1000618b: lea ecx,[esp+40h]
0x1000618f: call eax
0x10006191: push esi
0x10006192: lea ecx,[esp+34h]
0x10006196: call 1002228Ah
0x1000619b: push ebx
0x1000619c: lea ecx,[esp+34h]
0x100061a0: call 10022296h
0x100061a5: push edi
0x100061a6: lea ecx,[esp+34h]
0x100061aa: call 10022290h
0x100061af: push esi
0x100061b0: lea ecx,[esp+34h]
0x100061b4: call 1002228Ah
0x100061b9: mov eax,[ebp+8]
0x100061bc: mov ecx,[eax+20h]
0x100061bf: push ecx
0x100061c0: call dword ptr ds:[100FE65Ch]
```

```
0x100061c6: test eax,eax
0x100061c8: jne near ptr 1000626Eh
0x100061ce: cmp [ebp+1E4h],eax
0x100061d4: lea edi,[ebp+1C0h]
0x100061da: je short 10006236h
0x100061dc: mov eax,[edi+2Ch]
0x100061df: test eax,eax
0x100061e1: je near ptr 1000626Eh
0x100061e7: mov eax,[ebp+268h]
0x100061ed: mov ecx,[ebp+270h]
0x100061f3: mov edx,[ebp+274h]
0x100061f9: mov [esp+20h],eax
0x100061fd: mov eax,[ebp+26Ch]
0x10006203: add eax,[ebp+4]
0x10006206: cmp dword ptr [edi+24h],0
0x1000620a: mov [esp+28h],ecx
0x1000620e: mov [esp+2Ch],edx
0x10006212: mov [esp+24h],eax
0x10006216: je short 10006249h
0x10006218: mov eax,[edi+30h]
0x1000621b: push eax
0x1000621c: lea ecx,[edi+28h]
0x1000621f: push ecx
0x10006220: lea edx,[esp+38h]
0x10006224: push edx
0x10006225: lea ecx,[edi+20h]
0x10006228: lea edx,[esp+2Ch]
0x1000622c: call 100020F0h
0x10006231: add esp,0Ch
0x10006234: jmp short 10006260h
0x10006236: mov eax,[edi+8]
0x10006239: test eax,eax
0x1000623b: je short 1000626Eh
0x1000623d: mov edx,[edi+0Ch]
0x10006240: sub edx,eax
0x10006242: sar edx,2
0x10006245: jne short 100061E7h
0x10006247: jmp short 1000626Eh
0x10006249: push 0Fh
0x1000624b: call dword ptr ds:[100FE5C4h]
0x10006251: push eax
0x10006252: lea eax,[esp+24h]
0x10006256: push eax
0x10006257: lea ecx,[esp+38h]
0x1000625b: call 10022008h
0x10006260: mov ecx,[esp+5Ch]
0x10006264: push ecx
0x10006265: lea eax,[esp+34h]
0x10006269: call 1000DAC0h
0x1000626e: mov edx,[ebp+26Ch]
0x10006274: mov esi,[ebp+268h]
0x1000627a: mov eax,[ebp+274h]
0x10006280: mov edi,[esp+34h]
0x10006284: mov ecx,[ebp+270h]
0x1000628a: push 0CC0020h
0x1000628f: push edx
0x10006290: push esi
0x10006291: push edi
0x10006292: sub eax,edx
0x10006294: push eax
0x10006295: sub ecx,esi
0x10006297: push ecx
0x10006298: push edx
0x10006299: push esi
0x1000629a: mov esi,[esp+74h]
0x1000629e: mov edx,[esi+4]
0x100062a1: push edx
```

```
0x100062a2: call dword ptr ds:[100FE06Ch]
0x100062a8: mov eax,[esp+50h]
0x100062ac: test eax,eax
0x100062ae: je short 100062B3h
0x100062b0: mov eax,[eax+4]
0x100062b3: push eax
0x100062b4: mov eax,[esp+38h]
0x100062b8: push eax
0x100062b9: call 1002203Eh
0x100062be: push 0
0x100062c0: mov ecx,esi
0x100062c2: call 10022266h
0x100062c7: mov byte ptr [esp+48h],0
0x100062cc: lea ecx,[esp+18h]
0x100062d0: mov dword ptr [esp+18h],1012F498h
0x100062d8: call 10001370h
0x100062dd: lea ecx,[esp+30h]
0x100062e1: mov dword ptr [esp+48h],0FFFFFFFFh
0x100062e9: call 10022032h
0x100062ee: mov ecx,[esp+40h]
0x100062f2: mov fs:[0],ecx
0x100062f9: pop ecx
0x100062fa: pop edi
0x100062fb: pop esi
0x100062fc: pop ebp
0x100062fd: pop ebx
0x100062fe: add esp,38h
0x10006301: ret 10h
```

`sub_10005fd0` is called by sub_10005680, sub_10006310. It delegates to sub_10001370, sub_100020f0, sub_100078e0 (21 callees).

12.90 sub_10006310 at 0x10006310 (41 instructions)

```
0x10006310: push 0FFFFFFFFh
0x10006312: push 100FD478h
0x10006317: mov eax,fs:[0]
0x1000631d: push eax
0x1000631e: sub esp,18h
0x10006321: push esi
0x10006322: mov eax,ds:[10136220h]
0x10006327: xor eax,esp
0x10006329: push eax
0x1000632a: lea eax,[esp+20h]
0x1000632e: mov fs:[0],eax
0x10006334: mov esi,ecx
0x10006336: push esi
0x10006337: call dword ptr ds:[100FE6ACh]
0x1000633d: test eax,eax
0x1000633f: jne short 10006385h
0x10006341: push esi
0x10006342: call dword ptr ds:[100FE050h]
0x10006348: mov esi,eax
0x1000634a: mov eax,[edi+8]
0x1000634d: push eax
0x1000634e: lea ecx,[esp+10h]
0x10006352: call 1002222Ah
0x10006357: push 1
0x10006359: push esi
0x1000635a: lea ecx,[esp+14h]
0x1000635e: push ecx
0x1000635f: push edi
0x10006360: mov dword ptr [esp+38h],0
0x10006368: call 10005FD0h
0x1000636d: push esi
0x1000636e: call dword ptr ds:[100FE048h]
```

```
0x10006374: lea ecx,[esp+0Ch]
0x10006378: mov dword ptr [esp+28h],0FFFFFFFFh
0x10006380: call 10022230h
0x10006385: mov ecx,[esp+20h]
0x10006389: mov fs:[0],ecx
0x10006390: pop ecx
0x10006391: pop esi
0x10006392: add esp,24h
0x10006395: ret
```

`sub_10006310` is called by ?CheckToolButton@CCustomTitle@@QAEXI@Z, ?UncheckAllToolButton@CCustomTitle@@QAEXXZ, sub_10004910, sub_10004d40. It delegates to sub_10005fd0, j_ORDINAL 567, j_ORDINAL 758 (3 callees).

12.91 sub_100063a0 at 0x100063a0 (135 instructions)

```
0x100063a0: push ecx
0x100063a1: push ebx
0x100063a2: push esi
0x100063a3: mov esi,[esp+10h]
0x100063a7: mov eax,[esi+8]
0x100063aa: mov eax,[eax+20h]
0x100063ad: push 0
0x100063af: push eax
0x100063b0: call dword ptr ds:[100FE5F0h]
0x100063b6: push eax
0x100063b7: call 1002227Eh
0x100063bc: mov ebx,eax
0x100063be: test ebx,ebx
0x100063c0: mov [esp+8],ebx
0x100063c4: je near ptr 10006524h
0x100063ca: mov ecx,[esi+8]
0x100063cd: push ebp
0x100063ce: push edi
0x100063cf: call 10022236h
0x100063d4: mov edx,[ebx+4]
0x100063d7: mov esi,eax
0x100063d9: mov edi,esi
0x100063db: and edi,20000000h
0x100063e1: mov ecx,edi
0x100063e3: neg ecx
0x100063e5: sbb ecx,ecx
0x100063e7: and ecx,0C0h
0x100063ed: push 0
0x100063ef: add ecx,0F060h
0x100063f5: push ecx
0x100063f6: push edx
0x100063f7: call dword ptr ds:[100FE660h]
0x100063fd: mov eax,esi
0x100063ff: and eax,21000000h
0x10006404: mov ebx,0
0x10006409: sete bl
0x1000640c: test eax,eax
0x1000640e: jne short 1000641Dh
0x10006410: test esi,40000h
0x10006416: je short 1000641Dh
0x10006418: lea ebp,[eax+1]
0x1000641b: jmp short 1000641Fh
0x1000641d: xor ebp,ebp
0x1000641f: test edi,edi
0x10006421: jne short 10006432h
0x10006423: test esi,20000h
0x10006429: je short 10006432h
0x1000642b: mov edi,1
0x10006430: jmp short 10006434h
0x10006432: xor edi,edi
```

```
0x10006434: test esi,1000000h
0x1000643a: jne short 1000644Bh
0x1000643c: test esi,10000h
0x10006442: je short 1000644Bh
0x10006444: mov esi,1
0x10006449: jmp short 1000644Dh
0x1000644b: xor esi,esi
0x1000644d: neg eax
0x1000644f: sbb eax,eax
0x10006451: and eax,0FFFFFFFDh
0x10006454: add eax,3
0x10006457: push eax
0x10006458: mov eax,[esp+14h]
0x1000645c: mov ecx,[eax+4]
0x1000645f: push 0F120h
0x10006464: push ecx
0x10006465: call dword ptr ds:[100FE658h]
0x1000646b: neg ebx
0x1000646d: sbb ebx,ebx
0x1000646f: and ebx,0FFFFFFFDh
0x10006472: add ebx,3
0x10006475: push ebx
0x10006476: mov ebx,[esp+14h]
0x1000647a: mov edx,[ebx+4]
0x1000647d: push 0F010h
0x10006482: push edx
0x10006483: call dword ptr ds:[100FE658h]
0x10006489: mov eax,[ebx+4]
0x1000648c: neg ebp
0x1000648e: sbb ebp,ebp
0x10006490: and ebp,0FFFFFFFDh
0x10006493: add ebp,3
0x10006496: push ebp
0x10006497: mov ebp,ds:[100FE658h]
0x1000649d: push 0F000h
0x100064a2: push eax
0x100064a3: call ebp
0x100064a5: mov ecx,[ebx+4]
0x100064a8: neg edi
0x100064aa: sbb edi,edi
0x100064ac: and edi,0FFFFFFFDh
0x100064af: add edi,3
0x100064b2: push edi
0x100064b3: push 0F020h
0x100064b8: push ecx
0x100064b9: call ebp
0x100064bb: mov edx,[ebx+4]
0x100064be: neg esi
0x100064c0: sbb esi,esi
0x100064c2: and esi,0FFFFFFFDh
0x100064c5: add esi,3
0x100064c8: push esi
0x100064c9: push 0F030h
0x100064ce: push edx
0x100064cf: call ebp
0x100064d1: mov eax,[esp+20h]
0x100064d5: mov esi,[esp+18h]
0x100064d9: mov ecx,[esi+8]
0x100064dc: mov [esi+2A0h],eax
0x100064e2: mov eax,[esp+1Ch]
0x100064e6: mov edx,[eax+4]
0x100064e9: mov eax,[eax]
0x100064eb: push 0
0x100064ed: push ecx
0x100064ee: push edx
0x100064ef: push eax
0x100064f0: push 100h
```

```
0x100064f5: mov ecx,ebx
0x100064f7: call 10022248h
0x100064fc: cmp eax,0F000h
0x10006501: pop edi
0x10006502: mov dword ptr [esi+2A0h],0
0x1000650c: pop ebp
0x1000650d: jb short 10006524h
0x1000650f: mov ecx,[esi+8]
0x10006512: mov edx,[ecx+20h]
0x10006515: push 0
0x10006517: push eax
0x10006518: push 112h
0x1000651d: push edx
0x1000651e: call dword ptr ds:[100FE620h]
0x10006524: pop esi
0x10006525: pop ebx
0x10006526: pop ecx
0x10006527: ret 0Ch
```

`sub_100063a0` is called by sub_100050f0, sub_100051b0. It delegates to j_ORDINAL 3395, j_ORDINAL 6140, j_ORDINAL 2365 (3 callees).

12.92 sub_100067c0 at 0x100067c0 (322 instructions)

```
0x100067c0: push 0FFFFFFFFh
0x100067c2: push 100FCEDBh
0x100067c7: mov eax,fs:[0]
0x100067cd: push eax
0x100067ce: sub esp,0ACh
0x100067d4: push ebx
0x100067d5: push ebp
0x100067d6: push esi
0x100067d7: push edi
0x100067d8: mov eax,ds:[10136220h]
0x100067dd: xor eax,esp
0x100067df: push eax
0x100067e0: lea eax,[esp+0C0h]
0x100067e7: mov fs:[0],eax
0x100067ed: mov ebx,ecx
0x100067ef: mov edi,[esp+0D4h]
0x100067f6: cmp edi,200h
0x100067fc: je short 10006822h
0x100067fe: cmp edi,0A0h
0x10006804: je short 10006822h
0x10006806: cmp edi,202h
0x1000680c: je short 10006822h
0x1000680e: cmp edi,205h
0x10006814: je short 10006822h
0x10006816: cmp edi,208h
0x1000681c: jne near ptr 10006BA9h
0x10006822: mov esi,ds:[100FE628h]
0x10006828: push 1
0x1000682a: call esi
0x1000682c: test ax,ax
0x1000682f: jl near ptr 10006BA9h
0x10006835: push 2
0x10006837: call esi
0x10006839: test ax,ax
0x1000683c: jl near ptr 10006BA9h
0x10006842: push 4
0x10006844: call esi
0x10006846: test ax,ax
0x10006849: jl near ptr 10006BA9h
0x1000684f: mov eax,[esp+0D0h]
0x10006856: push eax
0x10006857: call 1002225Ah
```

```
0x1000685c: cmp eax,[ebx+8]
0x1000685f: jne near ptr 10006C4Dh
0x10006865: mov ebp,ds:[100FE620h]
0x1000686b: xor esi,esi
0x1000686d: cmp [ebx+220h],esi
0x10006873: jne short 100068F0h
0x10006875: push 70h
0x10006877: call 10021FDEh
0x1000687c: add esp,4
0x1000687f: mov [esp+18h],eax
0x10006883: cmp eax,esi
0x10006885: mov [esp+0C8h],esi
0x1000688c: je short 10006897h
0x1000688e: mov ecx,eax
0x10006890: call 10022260h
0x10006895: jmp short 10006899h
0x10006897: xor eax,eax
0x10006899: mov dword ptr [esp+0C8h],0FFFFFFFFh
0x100068a4: mov ecx,[ebx+8]
0x100068a7: mov [ebx+220h],eax
0x100068ad: mov edx,[eax]
0x100068af: mov edx,[edx+13Ch]
0x100068b5: push esi
0x100068b6: push ecx
0x100068b7: mov ecx,eax
0x100068b9: call edx
0x100068bb: test eax,eax
0x100068bd: jne short 100068DDh
0x100068bf: mov ecx,[ebx+220h]
0x100068c5: cmp ecx,esi
0x100068c7: je short 100068D2h
0x100068c9: mov eax,[ecx]
0x100068cb: mov edx,[eax+4]
0x100068ce: push 1
0x100068d0: call edx
0x100068d2: mov [ebx+220h],esi
0x100068d8: jmp near ptr 10006C4Dh
0x100068dd: mov eax,[ebx+220h]
0x100068e3: mov ecx,[eax+20h]
0x100068e6: push esi
0x100068e7: push esi
0x100068e8: push 401h
0x100068ed: push ecx
0x100068ee: call ebp
0x100068f0: mov eax,[esp+0DCh]
0x100068f7: movsx edx,ax
0x100068fa: shr eax,10h
0x100068fd: cmp edi,0A0h
0x10006903: movsx eax,ax
0x10006906: mov [esp+1Ch],edx
0x1000690a: mov [esp+20h],eax
0x1000690e: lea esi,[esp+1Ch]
0x10006912: jne short 1000691Dh
0x10006914: mov ecx,ebx
0x10006916: call 100032B0h
0x1000691b: jmp short 1000692Bh
0x1000691d: mov edi,ebx
0x1000691f: call 10003240h
0x10006924: mov edi,[esp+0D4h]
0x1000692b: push 30h
0x1000692d: lea ecx,[esp+54h]
0x10006931: push 0
0x10006933: push ecx
0x10006934: call 100228C0h
0x10006939: mov edx,[ebx+8]
0x1000693c: mov eax,[edx+20h]
0x1000693f: mov ecx,[esp+2Ch]
```

```
0x10006943: mov edx,[esp+28h]
0x10006947: add esp,0Ch
0x1000694a: push ecx
0x1000694b: mov [esp+5Ch],eax
0x1000694f: push edx
0x10006950: lea eax,[esp+58h]
0x10006954: push eax
0x10006955: lea eax,[ebx+0D8h]
0x1000695b: mov dword ptr [esp+5Ch],30h
0x10006963: call 1000B8D0h
0x10006968: mov esi,eax
0x1000696a: test esi,esi
0x1000696c: mov [esp+18h],esi
0x10006970: jge short 10006998h
0x10006972: mov ecx,[esp+20h]
0x10006976: mov edx,[esp+1Ch]
0x1000697a: push ecx
0x1000697b: push edx
0x1000697c: lea eax,[ebx+1C0h]
0x10006982: lea edi,[esp+58h]
0x10006986: call 1000E130h
0x1000698b: mov edi,[esp+0D4h]
0x10006992: mov [esp+18h],eax
0x10006996: mov esi,eax
0x10006998: cmp [ebx+224h],esi
0x1000699e: je near ptr 10006B40h
0x100069a4: cmp esi,0FFFFFFFFh
0x100069a7: je near ptr 10006A97h
0x100069ad: mov ecx,0Ch
0x100069b2: lea esi,[esp+50h]
0x100069b6: lea edi,[esp+90h]
0x100069bd: lea eax,[esp+0A0h]
0x100069c4: rep movsd
0x100069c6: push eax
0x100069c7: lea ecx,[esp+28h]
0x100069cb: push ecx
0x100069cc: call dword ptr ds:[100FE664h]
0x100069d2: mov eax,[ebx+8]
0x100069d5: mov ecx,[eax+20h]
0x100069d8: lea edx,[esp+80h]
0x100069df: push edx
0x100069e0: push ecx
0x100069e1: call dword ptr ds:[100FE674h]
0x100069e7: mov edx,[esp+84h]
0x100069ee: mov eax,[esp+80h]
0x100069f5: push edx
0x100069f6: push eax
0x100069f7: lea ecx,[esp+2Ch]
0x100069fb: push ecx
0x100069fc: call dword ptr ds:[100FE6B0h]
0x10006a02: mov ecx,[ebx+8]
0x10006a05: lea edx,[esp+24h]
0x10006a09: push edx
0x10006a0a: call 1002223Ch
0x10006a0f: mov ecx,[esp+28h]
0x10006a13: mov eax,[esp+24h]
0x10006a17: mov edx,[esp+2Ch]
0x10006a1b: mov [esp+0A4h],ecx
0x10006a22: lea ecx,[esp+90h]
0x10006a29: push ecx
0x10006a2a: mov [esp+0A4h],eax
0x10006a31: mov eax,[esp+34h]
0x10006a35: mov [esp+0ACh],edx
0x10006a3c: mov edx,[ebx+220h]
0x10006a42: push 0
0x10006a44: mov [esp+0B4h],eax
0x10006a4b: mov eax,[edx+20h]
```

```
0x10006a4e: push 432h
0x10006a53: push eax
0x10006a54: call ebp
0x10006a56: mov ecx,[ebx+220h]
0x10006a5c: mov edx,[ecx+20h]
0x10006a5f: push 0
0x10006a61: push 1
0x10006a63: push 401h
0x10006a68: push edx
0x10006a69: call ebp
0x10006a6b: mov eax,[ebx+220h]
0x10006a71: mov ecx,[eax+20h]
0x10006a74: push 213h
0x10006a79: push 0
0x10006a7b: push 0
0x10006a7d: push 0
0x10006a7f: push 0
0x10006a81: push 0
0x10006a83: push ecx
0x10006a84: call dword ptr ds:[100FE614h]
0x10006a8a: mov esi,[esp+18h]
0x10006a8e: mov edi,[esp+0D4h]
0x10006a95: jmp short 10006AACh
0x10006a97: mov edx,[ebx+220h]
0x10006a9d: mov eax,[edx+20h]
0x10006aa0: push 0
0x10006aa2: push 0
0x10006aa4: push 401h
0x10006aa9: push eax
0x10006aaa: call ebp
0x10006aac: mov ecx,[esp+0D0h]
0x10006ab3: mov edx,[esp+0D8h]
0x10006aba: xor eax,eax
0x10006abc: mov [esp+34h],eax
0x10006ac0: mov [esp+34h],ecx
0x10006ac4: mov [esp+38h],eax
0x10006ac8: mov [esp+3Ch],eax
0x10006acc: mov [esp+40h],eax
0x10006ad0: lea ecx,[esp+34h]
0x10006ad4: push ecx
0x10006ad5: mov [esp+48h],eax
0x10006ad9: mov [esp+4Ch],eax
0x10006add: mov [esp+50h],eax
0x10006ae1: mov eax,[esp+0E0h]
0x10006ae8: mov [esp+40h],edx
0x10006aec: mov edx,[ebx+220h]
0x10006af2: push 0
0x10006af4: mov [esp+40h],edi
0x10006af8: mov [esp+48h],eax
0x10006afc: mov eax,[edx+20h]
0x10006aff: push 407h
0x10006b04: push eax
0x10006b05: call ebp
0x10006b07: cmp dword ptr [ebx+228h],30h
0x10006b0e: lea edi,[ebx+228h]
0x10006b14: jb short 10006B2Ah
0x10006b16: mov ecx,[ebx+220h]
0x10006b1c: mov edx,[ecx+20h]
0x10006b1f: push edi
0x10006b20: push 0
0x10006b22: push 433h
0x10006b27: push edx
0x10006b28: call ebp
0x10006b2a: mov [ebx+224h],esi
0x10006b30: mov ecx,0Ch
0x10006b35: lea esi,[esp+50h]
0x10006b39: rep movsd
```

```
0x10006b3b: jmp near ptr 10006C4Dh
0x10006b40: cmp esi,0FFFFFFFFh
0x10006b43: je near ptr 10006C4Dh
0x10006b49: mov ecx,[esp+0D8h]
0x10006b50: mov edx,[esp+0DCh]
0x10006b57: xor eax,eax
0x10006b59: mov [esp+50h],eax
0x10006b5d: mov [esp+54h],eax
0x10006b61: mov [esp+58h],eax
0x10006b65: mov [esp+5Ch],eax
0x10006b69: mov [esp+60h],eax
0x10006b6d: mov [esp+64h],eax
0x10006b71: mov [esp+68h],eax
0x10006b75: mov eax,[esp+0D0h]
0x10006b7c: mov [esp+50h],eax
0x10006b80: lea eax,[esp+50h]
0x10006b84: push eax
0x10006b85: mov [esp+5Ch],ecx
0x10006b89: mov ecx,[ebx+220h]
0x10006b8f: push 0
0x10006b91: mov [esp+5Ch],edi
0x10006b95: mov [esp+64h],edx
0x10006b99: mov edx,[ecx+20h]
0x10006b9c: push 407h
0x10006ba1: push edx
0x10006ba2: call ebp
0x10006ba4: jmp near ptr 10006C4Dh
0x10006ba9: mov eax,[esp+0D0h]
0x10006bb0: push eax
0x10006bb1: call 1002225Ah
0x10006bb6: cmp eax,[ebx+8]
0x10006bb9: jne near ptr 10006C4Dh
0x10006bbf: lea ecx,[edi-100h]
0x10006bc5: cmp ecx,9
0x10006bc8: jbe short 10006C2Ah
0x10006bca: cmp edi,201h
0x10006bd0: je short 10006C2Ah
0x10006bd2: cmp edi,203h
0x10006bd8: je short 10006C2Ah
0x10006bda: cmp edi,204h
0x10006be0: je short 10006C2Ah
0x10006be2: cmp edi,206h
0x10006be8: je short 10006C2Ah
0x10006bea: cmp edi,207h
0x10006bf0: je short 10006C2Ah
0x10006bf2: cmp edi,209h
0x10006bf8: je short 10006C2Ah
0x10006bfa: cmp edi,0A1h
0x10006c00: je short 10006C2Ah
0x10006c02: cmp edi,0A3h
0x10006c08: je short 10006C2Ah
0x10006c0a: cmp edi,0A4h
0x10006c10: je short 10006C2Ah
0x10006c12: cmp edi,0A6h
0x10006c18: je short 10006C2Ah
0x10006c1a: cmp edi,0A7h
0x10006c20: je short 10006C2Ah
0x10006c22: cmp edi,0A9h
0x10006c28: jne short 10006C4Dh
0x10006c2a: mov ebx,[ebx+220h]
0x10006c30: test ebx,ebx
0x10006c32: je short 10006C4Dh
0x10006c34: cmp dword ptr [ebx+20h],0
0x10006c38: je short 10006C4Dh
0x10006c3a: mov edx,[ebx+20h]
0x10006c3d: push 0
0x10006c3f: push 0
```

```
0x10006c41: push 401h
0x10006c46: push edx
0x10006c47: call dword ptr ds:[100FE620h]
0x10006c4d: mov ecx,[esp+0C0h]
0x10006c54: mov fs:[0],ecx
0x10006c5b: pop ecx
0x10006c5c: pop edi
0x10006c5d: pop esi
0x10006c5e: pop ebp
0x10006c5f: pop ebx
0x10006c60: add esp,0B8h
0x10006c66: ret 10h
```

`sub_100067c0` is called by sub_10003e20. It delegates to sub_10003240, sub_100032b0, sub_1000b8d0 (9 callees).

12.93 sub_10006c70 at 0x10006c70 (70 instructions)

```
0x10006c70: cmp dword ptr ds:[1013F374h],0
0x10006c77: push ebx
0x10006c78: mov ebx,ds:[100FE108h]
0x10006c7e: push ebp
0x10006c7f: mov ebp,[esp+0Ch]
0x10006c83: push esi
0x10006c84: push edi
0x10006c85: mov edi,ds:[100FE604h]
0x10006c8b: mov esi,1
0x10006c90: jne short 10006CB2h
0x10006c92: call ebx
0x10006c94: push eax
0x10006c95: push 0
0x10006c97: push 10006EA0h
0x10006c9c: push 7
0x10006c9e: call edi
0x10006ca0: xor ecx,ecx
0x10006ca2: test eax,eax
0x10006ca4: setne cl
0x10006ca7: mov ds:[1013F374h],eax
0x10006cac: mov esi,ecx
0x10006cae: test esi,esi
0x10006cb0: je short 10006D1Ah
0x10006cb2: cmp dword ptr ds:[1013F378h],0
0x10006cb9: jne short 10006CD7h
0x10006cbb: call ebx
0x10006cbd: push eax
0x10006cbe: push 0
0x10006cc0: push 10006EE0h
0x10006cc5: push 2
0x10006cc7: call edi
0x10006cc9: xor edx,edx
0x10006ccb: test eax,eax
0x10006ccd: setne dl
0x10006cd0: mov ds:[1013F378h],eax
0x10006cd5: mov esi,edx
0x10006cd7: test esi,esi
0x10006cd9: je short 10006D1Ah
0x10006cdb: cmp dword ptr ds:[1013F37Ch],0
0x10006ce2: jne short 10006D00h
0x10006ce4: call ebx
0x10006ce6: push eax
0x10006ce7: push 0
0x10006ce9: push 100070A0h
0x10006cee: push 5
0x10006cf0: call edi
0x10006cf2: xor ecx,ecx
0x10006cf4: test eax,eax
0x10006cf6: setne cl
```

```
0x10006cf9: mov ds:[1013F37Ch],eax
0x10006cfe: mov esi,ecx
0x10006d00: test esi,esi
0x10006d02: je short 10006D1Ah
0x10006d04: lea ebx,[esp+14h]
0x10006d08: mov [esp+14h],ebp
0x10006d0c: call 100071C0h
0x10006d11: mov eax,esi
0x10006d13: pop edi
0x10006d14: pop esi
0x10006d15: pop ebp
0x10006d16: pop ebx
0x10006d17: ret 4
0x10006d1a: push ebp
0x10006d1b: call 10006D30h
0x10006d20: pop edi
0x10006d21: mov eax,esi
0x10006d23: pop esi
0x10006d24: pop ebp
0x10006d25: pop ebx
0x10006d26: ret 4
```

`sub_10006c70` is called by sub_10003670. It delegates to sub_10006d30, sub_100071c0 (2 callees).

12.94 sub_10006d30 at 0x10006d30 (91 instructions)

```
0x10006d30: push ebp
0x10006d31: mov ebp,esp
0x10006d33: and esp,0FFFFFFF8h
0x10006d36: mov ecx,ds:[1013F398h]
0x10006d3c: sub esp,0Ch
0x10006d3f: cmp ds:[1013F394h],ecx
0x10006d45: push ebx
0x10006d46: push esi
0x10006d47: push edi
0x10006d48: mov edi,ecx
0x10006d4a: jbe short 10006D58h
0x10006d4c: call dword ptr ds:[100FE584h]
0x10006d52: mov ecx,ds:[1013F398h]
0x10006d58: mov esi,ds:[1013F394h]
0x10006d5e: cmp esi,ecx
0x10006d60: jbe short 10006D6Eh
0x10006d62: call dword ptr ds:[100FE584h]
0x10006d68: mov ecx,ds:[1013F398h]
0x10006d6e: cmp esi,edi
0x10006d70: mov eax,1013F390h
0x10006d75: mov ebx,eax
0x10006d77: mov [esp+14h],esi
0x10006d7b: je short 10006D8Eh
0x10006d7d: lea ecx,[ecx]
0x10006d80: mov eax,[ebp+8]
0x10006d83: cmp [esi],eax
0x10006d85: je short 10006D8Eh
0x10006d87: add esi,4
0x10006d8a: cmp esi,edi
0x10006d8c: jne short 10006D80h
0x10006d8e: cmp ds:[1013F394h],ecx
0x10006d94: mov edi,ecx
0x10006d96: jbe short 10006DA4h
0x10006d98: call dword ptr ds:[100FE584h]
0x10006d9e: mov ecx,ds:[1013F398h]
0x10006da4: test ebx,ebx
0x10006da6: je short 10006DB0h
0x10006da8: cmp ebx,1013F390h
0x10006dae: je short 10006DBCh
0x10006db0: call dword ptr ds:[100FE584h]
```

```
0x10006db6: mov ecx,ds:[1013F398h]
0x10006dbc: cmp esi,edi
0x10006dbe: je short 10006DEEh
0x10006dc0: lea edx,[esi+4]
0x10006dc3: mov eax,ecx
0x10006dc5: sub eax,edx
0x10006dc7: sar eax,2
0x10006dca: test eax,eax
0x10006dcc: jle short 10006DE5h
0x10006dce: add eax,eax
0x10006dd0: add eax,eax
0x10006dd2: push eax
0x10006dd3: push edx
0x10006dd4: push eax
0x10006dd5: push esi
0x10006dd6: call dword ptr ds:[100FE57Ch]
0x10006ddc: mov ecx,ds:[1013F398h]
0x10006de2: add esp,10h
0x10006de5: sub ecx,4
0x10006de8: mov ds:[1013F398h],ecx
0x10006dee: mov eax,ds:[1013F394h]
0x10006df3: test eax,eax
0x10006df5: je short 10006DFEh
0x10006df7: sub ecx,eax
0x10006df9: sar ecx,2
0x10006dfc: jne short 10006E46h
0x10006dfe: mov eax,ds:[1013F374h]
0x10006e03: test eax,eax
0x10006e05: mov esi,ds:[100FE5FCh]
0x10006e0b: je short 10006E1Ah
0x10006e0d: push eax
0x10006e0e: call esi
0x10006e10: mov dword ptr ds:[1013F374h],0
0x10006e1a: mov eax,ds:[1013F378h]
0x10006e1f: test eax,eax
0x10006e21: je short 10006E30h
0x10006e23: push eax
0x10006e24: call esi
0x10006e26: mov dword ptr ds:[1013F378h],0
0x10006e30: mov eax,ds:[1013F37Ch]
0x10006e35: test eax,eax
0x10006e37: je short 10006E46h
0x10006e39: push eax
0x10006e3a: call esi
0x10006e3c: mov dword ptr ds:[1013F37Ch],0
0x10006e46: pop edi
0x10006e47: pop esi
0x10006e48: pop ebx
0x10006e49: mov esp,ebp
0x10006e4b: pop ebp
0x10006e4c: ret 4
```

`sub_10006d30` is called by sub_10003a40, sub_10006c70. It calls nothing else.

12.95 sub_10006e50 at 0x10006e50 (33 instructions)

```
0x10006e50: push esi
0x10006e51: mov esi,eax
0x10006e53: xor eax,eax
0x10006e55: test esi,esi
0x10006e57: je short 10006E93h
0x10006e59: push ebx
0x10006e5a: mov ebx,ds:[100FE668h]
0x10006e60: push ebp
0x10006e61: mov ebp,ds:[100FE5F4h]
0x10006e67: push edi
```

```
0x10006e68: push 10138C18h
0x10006e6d: push esi
0x10006e6e: call ebx
0x10006e70: mov edi,eax
0x10006e72: test edi,edi
0x10006e74: jne short 10006E8Eh
0x10006e76: push esi
0x10006e77: call dword ptr ds:[100FE5F8h]
0x10006e7d: mov esi,eax
0x10006e7f: test esi,esi
0x10006e81: jne short 10006E68h
0x10006e83: push 4
0x10006e85: push eax
0x10006e86: call ebp
0x10006e88: mov esi,eax
0x10006e8a: test esi,esi
0x10006e8c: jne short 10006E68h
0x10006e8e: mov eax,edi
0x10006e90: pop edi
0x10006e91: pop ebp
0x10006e92: pop ebx
0x10006e93: pop esi
0x10006e94: ret
```

`sub_10006e50` is called by sub_10006ea0, sub_10006ee0. It calls nothing else.

12.96 sub_10006ea0 at 0x10006ea0 (26 instructions)

```
0x10006ea0: push ebx
0x10006ea1: mov ebx,[esp+8]
0x10006ea5: test ebx,ebx
0x10006ea7: push esi
0x10006ea8: mov esi,[esp+14h]
0x10006eac: push edi
0x10006ead: mov edi,[esp+14h]
0x10006eb1: jne short 10006ECAh
0x10006eb3: mov eax,[esi+8]
0x10006eb6: call 10006E50h
0x10006ebb: mov edx,eax
0x10006ebd: test edx,edx
0x10006ebf: je short 10006ECAh
0x10006ec1: mov eax,esi
0x10006ec3: mov ecx,edi
0x10006ec5: call 10006F30h
0x10006eca: mov eax,ds:[1013F374h]
0x10006ecf: push esi
0x10006ed0: push edi
0x10006ed1: push ebx
0x10006ed2: push eax
0x10006ed3: call dword ptr ds:[100FE5ECh]
0x10006ed9: pop edi
0x10006eda: pop esi
0x10006edb: pop ebx
0x10006edc: ret 0Ch
```

`sub_10006ea0` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10006e50, sub_10006f30 (2 callees).

12.97 sub_10006ee0 at 0x10006ee0 (30 instructions)

```
0x10006ee0: push ecx
0x10006ee1: push ebx
0x10006ee2: mov ebx,[esp+0Ch]
0x10006ee6: test ebx,ebx
0x10006ee8: push esi
```

```
0x10006ee9: mov esi,[esp+18h]
0x10006eed: push edi
0x10006eee: mov edi,[esp+18h]
0x10006ef2: jne short 10006F12h
0x10006ef4: call dword ptr ds:[100FE5E8h]
0x10006efa: call 10006E50h
0x10006eff: mov ecx,eax
0x10006f01: test ecx,ecx
0x10006f03: je short 10006F12h
0x10006f05: mov eax,esi
0x10006f07: shr eax,10h
0x10006f0a: push eax
0x10006f0b: mov eax,edi
0x10006f0d: call 10007040h
0x10006f12: mov ecx,ds:[1013F378h]
0x10006f18: push esi
0x10006f19: push edi
0x10006f1a: push ebx
0x10006f1b: push ecx
0x10006f1c: call dword ptr ds:[100FE5ECh]
0x10006f22: pop edi
0x10006f23: pop esi
0x10006f24: pop ebx
0x10006f25: pop ecx
0x10006f26: ret 0Ch
```

`sub_10006ee0` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10006e50, sub_10007040 (2 callees).

12.98 sub_10006f30 at 0x10006f30 (97 instructions)

```
0x10006f30: push ebp
0x10006f31: mov ebp,esp
0x10006f33: and esp,0FFFFFFF8h
0x10006f36: sub esp,1Ch
0x10006f39: cmp ecx,203h
0x10006f3f: push ebx
0x10006f40: push esi
0x10006f41: push edi
0x10006f42: mov edi,edx
0x10006f44: jne near ptr 10006FFBh
0x10006f4a: mov ecx,[edi+8]
0x10006f4d: test ecx,ecx
0x10006f4f: jne short 10006F55h
0x10006f51: xor edx,edx
0x10006f53: jmp short 10006F58h
0x10006f55: mov edx,[ecx+20h]
0x10006f58: cmp [eax+8],edx
0x10006f5b: jne near ptr 10007037h
0x10006f61: mov esi,[eax]
0x10006f63: mov ebx,[eax+4]
0x10006f66: mov ecx,[ecx+20h]
0x10006f69: lea eax,[esp+18h]
0x10006f6d: push eax
0x10006f6e: push ecx
0x10006f6f: call dword ptr ds:[100FE674h]
0x10006f75: sub ebx,[esp+1Ch]
0x10006f79: sub esi,[esp+18h]
0x10006f7d: mov [esp+14h],ebx
0x10006f81: lea ebx,[esp+0Ch]
0x10006f85: lea eax,[esp+10h]
0x10006f89: mov [esp+10h],esi
0x10006f8d: call 10005B50h
0x10006f92: test eax,eax
0x10006f94: je short 10006FC7h
0x10006f96: mov eax,[edi+8]
```

```
0x10006f99: mov edx,[eax+20h]
0x10006f9c: push edx
0x10006f9d: call dword ptr ds:[100FE65Ch]
0x10006fa3: test eax,eax
0x10006fa5: je short 10006FC7h
0x10006fa7: mov eax,[edi+8]
0x10006faa: mov ecx,[eax+20h]
0x10006fad: push 0
0x10006faf: push 0F120h
0x10006fb4: push 112h
0x10006fb9: push ecx
0x10006fba: call dword ptr ds:[100FE608h]
0x10006fc0: pop edi
0x10006fc1: pop esi
0x10006fc2: pop ebx
0x10006fc3: mov esp,ebp
0x10006fc5: pop ebp
0x10006fc6: ret
0x10006fc7: mov edx,[esp+14h]
0x10006fcb: push edx
0x10006fcc: push esi
0x10006fcd: lea eax,[edi+30h]
0x10006fd0: push eax
0x10006fd1: call dword ptr ds:[100FE610h]
0x10006fd7: test eax,eax
0x10006fd9: je short 10007037h
0x10006fdb: mov ecx,[edi+8]
0x10006fde: mov edx,[ecx+20h]
0x10006fe1: push 0
0x10006fe3: push 0F060h
0x10006fe8: push 112h
0x10006fed: push edx
0x10006fee: call dword ptr ds:[100FE608h]
0x10006ff4: pop edi
0x10006ff5: pop esi
0x10006ff6: pop ebx
0x10006ff7: mov esp,ebp
0x10006ff9: pop ebp
0x10006ffa: ret
0x10006ffb: cmp ecx,200h
0x10007001: jne short 10007037h
0x10007003: cmp dword ptr [edi+0A8h],0
0x1000700a: lea ecx,[edi+40h]
0x1000700d: je short 10007037h
0x1000700f: cmp dword ptr [ecx+70h],0
0x10007013: je short 10007037h
0x10007015: cmp dword ptr [ecx+68h],0
0x10007019: mov edx,[eax]
0x1000701b: mov eax,[eax+4]
0x1000701e: mov [esp+10h],edx
0x10007022: mov [esp+14h],eax
0x10007026: je short 10007037h
0x10007028: cmp dword ptr [ecx+70h],0
0x1000702c: je short 10007037h
0x1000702e: lea eax,[esp+10h]
0x10007032: call 10009740h
0x10007037: pop edi
0x10007038: pop esi
0x10007039: pop ebx
0x1000703a: mov esp,ebp
0x1000703c: pop ebp
0x1000703d: ret
```

`sub_10006f30` is called by sub_10006ea0. It delegates to sub_10005b50, sub_10009740 (2 callees).

12.99 sub_10007040 at 0x10007040 (33 instructions)

```
0x10007040: push esi
0x10007041: mov esi,ecx
0x10007043: add esi,40h
0x10007046: cmp dword ptr [esi+68h],0
0x1000704a: push edi
0x1000704b: je short 10007095h
0x1000704d: cmp dword ptr [esi+70h],0
0x10007051: je short 10007095h
0x10007053: cmp eax,25h
0x10007056: je short 1000705Dh
0x10007058: cmp eax,27h
0x1000705b: jne short 10007095h
0x1000705d: test dword ptr [esp+0Ch],8000h
0x10007065: jne short 10007095h
0x10007067: xor ecx,ecx
0x10007069: cmp eax,25h
0x1000706c: sete cl
0x1000706f: cmp dword ptr [esi+68h],0
0x10007073: mov edi,ecx
0x10007075: je short 10007095h
0x10007077: cmp dword ptr [esi+70h],0
0x1000707b: je short 10007095h
0x1000707d: push edi
0x1000707e: lea eax,[esi+84h]
0x10007084: call 10002B80h
0x10007089: test eax,eax
0x1000708b: jne short 10007095h
0x1000708d: push edi
0x1000708e: mov eax,esi
0x10007090: call 10009630h
0x10007095: pop edi
0x10007096: pop esi
0x10007097: ret 4
```

`sub_10007040` is called by sub_10006ee0. It delegates to sub_10002b80, sub_10009630 (2 callees).

12.100 sub_100070a0 at 0x100070a0 (38 instructions)

```
0x100070a0: push ebx
0x100070a1: mov ebx,[esp+8]
0x100070a5: cmp ebx,1
0x100070a8: push esi
0x100070a9: mov esi,[esp+14h]
0x100070ad: push edi
0x100070ae: mov edi,[esp+14h]
0x100070b2: jne short 100070EBh
0x100070b4: push 10138C18h
0x100070b9: push edi
0x100070ba: call dword ptr ds:[100FE668h]
0x100070c0: test eax,eax
0x100070c2: je short 100070EBh
0x100070c4: movzx ecx,si
0x100070c7: call 10007110h
0x100070cc: cmp eax,1
0x100070cf: jne short 100070EBh
0x100070d1: push esi
0x100070d2: push edi
0x100070d3: push eax
0x100070d4: mov eax,ds:[1013F37Ch]
0x100070d9: push eax
0x100070da: call dword ptr ds:[100FE5ECh]
0x100070e0: pop edi
0x100070e1: pop esi
0x100070e2: mov eax,1
0x100070e7: pop ebx
```

```
0x100070e8: ret 0Ch
0x100070eb: mov ecx,ds:[1013F37Ch]
0x100070f1: push esi
0x100070f2: push edi
0x100070f3: push ebx
0x100070f4: push ecx
0x100070f5: call dword ptr ds:[100FE5ECh]
0x100070fb: pop edi
0x100070fc: pop esi
0x100070fd: pop ebx
0x100070fe: ret 0Ch
```

`sub_100070a0` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10007110 (1 callees).

12.101 sub_10007110 at 0x10007110 (14 instructions)

```
0x10007110: push edi
0x10007111: mov edi,eax
0x10007113: xor eax,eax
0x10007115: cmp ds:[1013F3A4h],eax
0x1000711b: jne near ptr 100071A1h
0x10007121: lea edx,[ecx-1]
0x10007124: cmp edx,8
0x10007127: mov dword ptr ds:[1013F3A4h],1
0x10007131: ja short 10007197h
0x10007133: movzx edx,byte ptr [edx+100071B4h]
0x1000713a: jmp dword ptr [edx*4+100071A4h]
0x10007197: mov dword ptr ds:[1013F3A4h],0
0x100071a1: pop edi
0x100071a2: ret
```

`sub_10007110` is called by sub_100070a0. It calls nothing else.

12.102 sub_100071c0 at 0x100071c0 (39 instructions)

```
0x100071c0: mov edx,ds:[1013F398h]
0x100071c6: sub esp,0Ch
0x100071c9: push esi
0x100071ca: mov esi,ds:[1013F394h]
0x100071d0: test esi,esi
0x100071d2: push edi
0x100071d3: je short 100071FFh
0x100071d5: mov ecx,ds:[1013F39Ch]
0x100071db: mov eax,edx
0x100071dd: sub eax,esi
0x100071df: sub ecx,esi
0x100071e1: sar eax,2
0x100071e4: sar ecx,2
0x100071e7: cmp eax,ecx
0x100071e9: jae short 100071FFh
0x100071eb: mov ecx,[ebx]
0x100071ed: mov eax,edx
0x100071ef: add eax,4
0x100071f2: mov [edx],ecx
0x100071f4: mov ds:[1013F398h],eax
0x100071f9: pop edi
0x100071fa: pop esi
0x100071fb: add esp,0Ch
0x100071fe: ret
0x100071ff: cmp esi,edx
0x10007201: mov edi,edx
0x10007203: jbe short 1000720Bh
0x10007205: call dword ptr ds:[100FE584h]
0x1000720b: push edi
```

```
0x1000720c: mov eax,1013F390h
0x10007211: push eax
0x10007212: push ebx
0x10007213: lea edx,[esp+14h]
0x10007217: push edx
0x10007218: call 10007230h
0x1000721d: pop edi
0x1000721e: pop esi
0x1000721f: add esp,0Ch
0x10007222: ret
```

`sub_100071c0` is called by sub_10006c70. It delegates to sub_10007230 (1 callees).

12.103 sub_10007230 at 0x10007230 (57 instructions)

```
0x10007230: push ecx
0x10007231: mov eax,ds:[1013F394h]
0x10007236: test eax,eax
0x10007238: push ebx
0x10007239: push ebp
0x1000723a: mov ebp,[esp+1Ch]
0x1000723e: push esi
0x1000723f: mov esi,[esp+1Ch]
0x10007243: push edi
0x10007244: je short 10007255h
0x10007246: mov ecx,ds:[1013F398h]
0x1000724c: mov edx,ecx
0x1000724e: sub edx,eax
0x10007250: sar edx,2
0x10007253: jne short 10007259h
0x10007255: xor edi,edi
0x10007257: jmp short 1000727Eh
0x10007259: cmp eax,ecx
0x1000725b: mov ebx,eax
0x1000725d: jbe short 10007265h
0x1000725f: call dword ptr ds:[100FE584h]
0x10007265: test esi,esi
0x10007267: je short 10007271h
0x10007269: cmp esi,1013F390h
0x1000726f: je short 10007277h
0x10007271: call dword ptr ds:[100FE584h]
0x10007277: mov edi,ebp
0x10007279: sub edi,ebx
0x1000727b: sar edi,2
0x1000727e: mov eax,[esp+1Ch]
0x10007282: push ebp
0x10007283: push esi
0x10007284: call 100072E0h
0x10007289: mov ecx,ds:[1013F394h]
0x1000728f: cmp ecx,ds:[1013F398h]
0x10007295: mov esi,ecx
0x10007297: jbe short 100072A5h
0x10007299: call dword ptr ds:[100FE584h]
0x1000729f: mov ecx,ds:[1013F394h]
0x100072a5: mov [esp+24h],esi
0x100072a9: lea esi,[esi+edi*4]
0x100072ac: cmp esi,ds:[1013F398h]
0x100072b2: mov eax,1013F390h
0x100072b7: mov ebx,eax
0x100072b9: ja short 100072BFh
0x100072bb: cmp esi,ecx
0x100072bd: jae short 100072C5h
0x100072bf: call dword ptr ds:[100FE584h]
0x100072c5: mov eax,[esp+18h]
0x100072c9: pop edi
0x100072ca: mov [eax+4],esi
```

```
0x100072cd: pop esi
0x100072ce: pop ebp
0x100072cf: mov [eax],ebx
0x100072d1: pop ebx
0x100072d2: pop ecx
0x100072d3: ret 10h
```

`sub_10007230` is called by sub_100071c0. It delegates to sub_100072e0 (1 callees).

12.104 sub_100072e0 at 0x100072e0 (186 instructions)

```
0x100072e0: push ecx
0x100072e1: mov ecx,[eax]
0x100072e3: mov edx,ds:[1013F394h]
0x100072e9: test edx,edx
0x100072eb: mov eax,ds:[1013F398h]
0x100072f0: push ebx
0x100072f1: push ebp
0x100072f2: push esi
0x100072f3: push edi
0x100072f4: mov [esp+10h],ecx
0x100072f8: jne short 10007300h
0x100072fa: xor ebp,ebp
0x100072fc: xor ecx,ecx
0x100072fe: jmp short 10007312h
0x10007300: mov ebp,ds:[1013F39Ch]
0x10007306: sub ebp,edx
0x10007308: mov ecx,eax
0x1000730a: sub ecx,edx
0x1000730c: sar ebp,2
0x1000730f: sar ecx,2
0x10007312: mov esi,3FFFFFFFh
0x10007317: sub esi,ecx
0x10007319: cmp esi,1
0x1000731c: jae short 10007323h
0x1000731e: call 10009B40h
0x10007323: test edx,edx
0x10007325: jne short 1000732Bh
0x10007327: xor ecx,ecx
0x10007329: jmp short 10007332h
0x1000732b: mov ecx,eax
0x1000732d: sub ecx,edx
0x1000732f: sar ecx,2
0x10007332: add ecx,1
0x10007335: cmp ebp,ecx
0x10007337: jae near ptr 1000741Ah
0x1000733d: mov ecx,ebp
0x1000733f: shr ecx,1
0x10007341: mov esi,3FFFFFFFh
0x10007346: sub esi,ecx
0x10007348: cmp esi,ebp
0x1000734a: jae short 10007350h
0x1000734c: xor ebp,ebp
0x1000734e: jmp short 10007352h
0x10007350: add ebp,ecx
0x10007352: test edx,edx
0x10007354: jne short 1000735Ah
0x10007356: xor ecx,ecx
0x10007358: jmp short 10007361h
0x1000735a: mov ecx,eax
0x1000735c: sub ecx,edx
0x1000735e: sar ecx,2
0x10007361: add ecx,1
0x10007364: cmp ebp,ecx
0x10007366: jae short 10007378h
0x10007368: test edx,edx
```

```
0x1000736a: jne short 10007370h
0x1000736c: xor eax,eax
0x1000736e: jmp short 10007375h
0x10007370: sub eax,edx
0x10007372: sar eax,2
0x10007375: lea ebp,[eax+1]
0x10007378: mov ecx,ebp
0x1000737a: call 10009BB0h
0x1000737f: mov ebx,[esp+1Ch]
0x10007383: mov ecx,ds:[1013F394h]
0x10007389: mov edi,eax
0x1000738b: mov eax,ebx
0x1000738d: sub eax,ecx
0x1000738f: sar eax,2
0x10007392: lea esi,[eax*4]
0x10007399: je short 100073A8h
0x1000739b: push esi
0x1000739c: push ecx
0x1000739d: push esi
0x1000739e: push edi
0x1000739f: call dword ptr ds:[100FE57Ch]
0x100073a5: add esp,10h
0x100073a8: mov edx,[esp+10h]
0x100073ac: mov [esi+edi],edx
0x100073af: mov ecx,ds:[1013F398h]
0x100073b5: mov eax,ecx
0x100073b7: sub eax,ebx
0x100073b9: sar eax,2
0x100073bc: lea edx,[esi+edi+4]
0x100073c0: je short 100073D9h
0x100073c2: add eax,eax
0x100073c4: add eax,eax
0x100073c6: push eax
0x100073c7: push ebx
0x100073c8: push eax
0x100073c9: push edx
0x100073ca: call dword ptr ds:[100FE57Ch]
0x100073d0: mov ecx,ds:[1013F398h]
0x100073d6: add esp,10h
0x100073d9: mov eax,ds:[1013F394h]
0x100073de: test eax,eax
0x100073e0: jne short 100073E6h
0x100073e2: xor esi,esi
0x100073e4: jmp short 100073EDh
0x100073e6: sub ecx,eax
0x100073e8: sar ecx,2
0x100073eb: mov esi,ecx
0x100073ed: test eax,eax
0x100073ef: je short 100073FAh
0x100073f1: push eax
0x100073f2: call 10021FCCh
0x100073f7: add esp,4
0x100073fa: lea eax,[edi+ebp*4]
0x100073fd: lea ecx,[edi+esi*4+4]
0x10007401: mov ds:[1013F39Ch],eax
0x10007406: mov ds:[1013F398h],ecx
0x1000740c: mov ds:[1013F394h],edi
0x10007412: pop edi
0x10007413: pop esi
0x10007414: pop ebp
0x10007415: pop ebx
0x10007416: pop ecx
0x10007417: ret 8
0x1000741a: mov ebx,[esp+1Ch]
0x1000741e: mov edx,eax
0x10007420: sub edx,ebx
0x10007422: sar edx,2
```

```
0x10007425: cmp edx,1
0x10007428: jae short 1000747Ch
0x1000742a: lea ecx,[ebx+4]
0x1000742d: mov edx,ebx
0x1000742f: call 100031F0h
0x10007434: mov edi,ds:[1013F398h]
0x1000743a: mov eax,edi
0x1000743c: sub eax,ebx
0x1000743e: sar eax,2
0x10007441: mov esi,1
0x10007446: sub esi,eax
0x10007448: lea edx,[esp+10h]
0x1000744c: call 10002EB0h
0x10007451: mov eax,ds:[1013F398h]
0x10007456: add eax,4
0x10007459: lea ecx,[eax-4]
0x1000745c: cmp ebx,ecx
0x1000745e: mov ds:[1013F398h],eax
0x10007463: mov eax,ebx
0x10007465: je short 100074C4h
0x10007467: mov edx,[esp+10h]
0x1000746b: mov [eax],edx
0x1000746d: add eax,4
0x10007470: cmp eax,ecx
0x10007472: jne short 10007467h
0x10007474: pop edi
0x10007475: pop esi
0x10007476: pop ebp
0x10007477: pop ebx
0x10007478: pop ecx
0x10007479: ret 8
0x1000747c: mov edi,eax
0x1000747e: lea esi,[edi-4]
0x10007481: mov ecx,eax
0x10007483: mov edx,esi
0x10007485: call 100031F0h
0x1000748a: sub esi,ebx
0x1000748c: sar esi,2
0x1000748f: test esi,esi
0x10007491: mov ds:[1013F398h],eax
0x10007496: jle short 100074AEh
0x10007498: lea eax,[esi*4]
0x1000749f: push eax
0x100074a0: push ebx
0x100074a1: push eax
0x100074a2: sub edi,eax
0x100074a4: push edi
0x100074a5: call dword ptr ds:[100FE57Ch]
0x100074ab: add esp,10h
0x100074ae: lea ecx,[ebx+4]
0x100074b1: cmp ebx,ecx
0x100074b3: mov eax,ebx
0x100074b5: je short 100074C4h
0x100074b7: mov edx,[esp+10h]
0x100074bb: mov [eax],edx
0x100074bd: add eax,4
0x100074c0: cmp eax,ecx
0x100074c2: jne short 100074B7h
0x100074c4: pop edi
0x100074c5: pop esi
0x100074c6: pop ebp
0x100074c7: pop ebx
0x100074c8: pop ecx
0x100074c9: ret 8
```

`sub_100072e0` is called by sub_10007230. It delegates to sub_10002eb0, sub_100031f0, sub_10009b40 (5 callees).

12.105 sub_100074d0 at 0x100074d0 (16 instructions)

```
0x100074d0: mov eax,[eax]
0x100074d2: mov edx,[eax-0Ch]
0x100074d5: push esi
0x100074d6: mov esi,[ecx]
0x100074d8: push edi
0x100074d9: mov edi,[esp+10h]
0x100074dd: push edi
0x100074de: mov edi,[esp+10h]
0x100074e2: push edi
0x100074e3: push edx
0x100074e4: mov edx,[esi+70h]
0x100074e7: push eax
0x100074e8: call edx
0x100074ea: pop edi
0x100074eb: pop esi
0x100074ec: ret 8
```

`sub_100074d0` is called by sub_1000cd90. It calls nothing else.

12.106 sub_100075d0 at 0x100075d0 (55 instructions)

```
0x100075d0: sub esp,18h
0x100075d3: push ebx
0x100075d4: push edi
0x100075d5: lea edi,[esi+4]
0x100075d8: test edi,edi
0x100075da: je short 100075E9h
0x100075dc: cmp dword ptr [edi+4],0
0x100075e0: je short 100075E9h
0x100075e2: mov ecx,edi
0x100075e4: call 10021FF6h
0x100075e9: xor eax,eax
0x100075eb: xor ecx,ecx
0x100075ed: mov [esi+10h],eax
0x100075f0: mov eax,[esp+24h]
0x100075f4: mov dword ptr [esi+0Ch],1
0x100075fb: mov [esi+14h],ecx
0x100075fe: mov ecx,ds:[1013F360h]
0x10007604: push eax
0x10007605: push ecx
0x10007606: xor ebx,ebx
0x10007608: call dword ptr ds:[100FE5E0h]
0x1000760e: push eax
0x1000760f: mov ecx,edi
0x10007611: call 10021FFCh
0x10007616: test eax,eax
0x10007618: je short 10007658h
0x1000761a: mov eax,[esi+8]
0x1000761d: lea edx,[esp+8]
0x10007621: push edx
0x10007622: push 18h
0x10007624: push eax
0x10007625: call dword ptr ds:[100FE074h]
0x1000762b: mov eax,[esp+0Ch]
0x1000762f: cdq
0x10007630: and edx,3
0x10007633: add eax,edx
0x10007635: mov ecx,eax
0x10007637: mov eax,[esp+10h]
0x1000763b: cdq
0x1000763c: and edx,3
0x1000763f: add eax,edx
0x10007641: sar ecx,2
0x10007644: sar eax,2
0x10007647: mov [esi+14h],eax
```

```
0x1000764a: mov [esi+10h],ecx
0x1000764d: lea eax,[ebx+1]
0x10007650: pop edi
0x10007651: pop ebx
0x10007652: add esp,18h
0x10007655: ret 4
0x10007658: pop edi
0x10007659: mov eax,ebx
0x1000765b: pop ebx
0x1000765c: add esp,18h
0x1000765f: ret 4
```

`sub_100075d0` is called by sub_10003670. It delegates to j_ORDINAL 1925, j_ORDINAL 1271 (2 callees).

12.107 sub_10007670 at 0x10007670 (56 instructions)

```
0x10007670: sub esp,18h
0x10007673: test ebx,ebx
0x10007675: je short 1000767Dh
0x10007677: cmp dword ptr [ebx+4],0
0x1000767b: jne short 10007683h
0x1000767d: xor eax,eax
0x1000767f: add esp,18h
0x10007682: ret
0x10007683: push edi
0x10007684: lea edi,[esi+4]
0x10007687: test edi,edi
0x10007689: je short 10007698h
0x1000768b: cmp dword ptr [edi+4],0
0x1000768f: je short 10007698h
0x10007691: mov ecx,edi
0x10007693: call 10021FF6h
0x10007698: xor eax,eax
0x1000769a: xor ecx,ecx
0x1000769c: mov [esi+14h],ecx
0x1000769f: mov dword ptr [esi+0Ch],0
0x100076a6: mov [esi+10h],eax
0x100076a9: mov eax,[ebx+4]
0x100076ac: push eax
0x100076ad: mov ecx,edi
0x100076af: call 10021FFCh
0x100076b4: mov edx,[esi+8]
0x100076b7: lea ecx,[esp+4]
0x100076bb: push ecx
0x100076bc: push 18h
0x100076be: push edx
0x100076bf: call dword ptr ds:[100FE074h]
0x100076c5: test eax,eax
0x100076c7: jne short 100076D7h
0x100076c9: mov ecx,edi
0x100076cb: call 10022002h
0x100076d0: xor eax,eax
0x100076d2: pop edi
0x100076d3: add esp,18h
0x100076d6: ret
0x100076d7: mov eax,[esp+8]
0x100076db: cdq
0x100076dc: and edx,3
0x100076df: add eax,edx
0x100076e1: mov ecx,eax
0x100076e3: mov eax,[esp+0Ch]
0x100076e7: cdq
0x100076e8: and edx,3
0x100076eb: add eax,edx
0x100076ed: sar ecx,2
0x100076f0: sar eax,2
```

```
0x100076f3: mov [esi+14h],eax
0x100076f6: mov [esi+10h],ecx
0x100076f9: mov eax,1
0x100076fe: pop edi
0x100076ff: add esp,18h
0x10007702: ret
```

`sub_10007670` is called by ?SetControlBtnsBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z. It delegates to j_ORDINAL 1925, j_ORDINAL 1271, j_ORDINAL 1959 (3 callees).

12.108 sub_10007710 at 0x10007710 (64 instructions)

```
0x10007710: push ecx
0x10007711: mov ecx,[esi]
0x10007713: push ebx
0x10007714: mov ebx,ds:[100FE610h]
0x1000771a: push ebp
0x1000771b: push edi
0x1000771c: mov edi,eax
0x1000771e: mov eax,[esi+4]
0x10007721: push eax
0x10007722: push ecx
0x10007723: lea edx,[edi+1Ch]
0x10007726: push edx
0x10007727: mov ebp,1
0x1000772c: call ebx
0x1000772e: test eax,eax
0x10007730: je short 10007745h
0x10007732: mov eax,[esp+14h]
0x10007736: mov dword ptr [eax],14h
0x1000773c: mov eax,ebp
0x1000773e: pop edi
0x1000773f: pop ebp
0x10007740: pop ebx
0x10007741: pop ecx
0x10007742: ret 4
0x10007745: mov ecx,[esi+4]
0x10007748: mov edx,[esi]
0x1000774a: push ecx
0x1000774b: push edx
0x1000774c: lea eax,[edi+30h]
0x1000774f: push eax
0x10007750: call ebx
0x10007752: test eax,eax
0x10007754: je short 10007769h
0x10007756: mov ecx,[esp+14h]
0x1000775a: mov dword ptr [ecx],9
0x10007760: mov eax,ebp
0x10007762: pop edi
0x10007763: pop ebp
0x10007764: pop ebx
0x10007765: pop ecx
0x10007766: ret 4
0x10007769: mov edx,[esi+4]
0x1000776c: mov eax,[esi]
0x1000776e: push edx
0x1000776f: push eax
0x10007770: add edi,44h
0x10007773: push edi
0x10007774: call ebx
0x10007776: test eax,eax
0x10007778: je short 1000778Dh
0x1000777a: mov ecx,[esp+14h]
0x1000777e: mov dword ptr [ecx],8
0x10007784: mov eax,ebp
0x10007786: pop edi
```

```
0x10007787: pop ebp
0x10007788: pop ebx
0x10007789: pop ecx
0x1000778a: ret 4
0x1000778d: pop edi
0x1000778e: pop ebp
0x1000778f: xor eax,eax
0x10007791: pop ebx
0x10007792: pop ecx
0x10007793: ret 4
```

`sub_10007710` is called by sub_10005b50. It calls nothing else.

12.109 sub_100077a0 at 0x100077a0 (107 instructions)

```
0x100077a0: push ebx
0x100077a1: push ebp
0x100077a2: lea ebx,[edi+1Ch]
0x100077a5: xor ebp,ebp
0x100077a7: push ebx
0x100077a8: mov [edi+18h],ebp
0x100077ab: call dword ptr ds:[100FE5C8h]
0x100077b1: mov [edi+2Ch],ebp
0x100077b4: lea ebp,[edi+30h]
0x100077b7: push ebp
0x100077b8: call dword ptr ds:[100FE5C8h]
0x100077be: lea eax,[edi+44h]
0x100077c1: push eax
0x100077c2: mov dword ptr [edi+40h],0
0x100077c9: call dword ptr ds:[100FE5C8h]
0x100077cf: mov eax,[esp+10h]
0x100077d3: mov ecx,[eax]
0x100077d5: mov [esi],ecx
0x100077d7: mov edx,[eax+4]
0x100077da: mov [esi+4],edx
0x100077dd: mov ecx,[eax+8]
0x100077e0: mov [esi+8],ecx
0x100077e3: mov edx,[eax+0Ch]
0x100077e6: mov [esi+0Ch],edx
0x100077e9: mov eax,[esi+8]
0x100077ec: mov [esi],eax
0x100077ee: cmp dword ptr [edi+10h],0
0x100077f2: jle near ptr 100078D8h
0x100077f8: mov edx,[edi+14h]
0x100077fb: test edx,edx
0x100077fd: jle near ptr 100078D8h
0x10007803: mov eax,3
0x10007808: mov [edi+40h],eax
0x1000780b: mov [edi+2Ch],eax
0x1000780e: mov [edi+18h],eax
0x10007811: mov ecx,[esi+4]
0x10007814: mov eax,[esi+0Ch]
0x10007817: sub eax,ecx
0x10007819: sub eax,edx
0x1000781b: add eax,1
0x1000781e: cdq
0x1000781f: sub eax,edx
0x10007821: mov edx,[esp+0Ch]
0x10007825: sar eax,1
0x10007827: add eax,ecx
0x10007829: mov [esi+4],eax
0x1000782c: mov ecx,[edi+14h]
0x1000782f: add ecx,eax
0x10007831: test edx,80000h
0x10007837: mov [esi+0Ch],ecx
0x1000783a: je near ptr 100078D8h
```

```
0x10007840: xor ecx,ecx
0x10007842: mov [edi+18h],ecx
0x10007845: mov eax,[esi]
0x10007847: mov [ebx],eax
0x10007849: mov eax,[esi+4]
0x1000784c: mov [ebx+4],eax
0x1000784f: mov eax,[esi+8]
0x10007852: mov [ebx+8],eax
0x10007855: mov eax,[esi+0Ch]
0x10007858: mov [ebx+0Ch],eax
0x1000785b: mov eax,[esi]
0x1000785d: add eax,0FFFFFFF4h
0x10007860: mov [edi+24h],eax
0x10007863: sub eax,[edi+10h]
0x10007866: test edx,30000h
0x1000786c: mov [ebx],eax
0x1000786e: mov [esi],eax
0x10007870: je short 100078D8h
0x10007872: test edx,10000h
0x10007878: je short 1000787Dh
0x1000787a: mov [edi+2Ch],ecx
0x1000787d: mov eax,[esi]
0x1000787f: mov [ebp],eax
0x10007882: mov eax,[esi+4]
0x10007885: mov [ebp+4],eax
0x10007888: mov eax,[esi+8]
0x1000788b: mov [ebp+8],eax
0x1000788e: mov eax,[esi+0Ch]
0x10007891: mov [ebp+0Ch],eax
0x10007894: mov eax,[esi]
0x10007896: add eax,0FFFFFFF4h
0x10007899: mov [edi+38h],eax
0x1000789c: sub eax,[edi+10h]
0x1000789f: test edx,20000h
0x100078a5: mov [ebp],eax
0x100078a8: mov [esi],eax
0x100078aa: je short 100078AFh
0x100078ac: mov [edi+40h],ecx
0x100078af: mov ecx,[esi]
0x100078b1: mov [edi+44h],ecx
0x100078b4: mov edx,[esi+4]
0x100078b7: lea eax,[edi+44h]
0x100078ba: mov [eax+4],edx
0x100078bd: mov ecx,[esi+8]
0x100078c0: mov [eax+8],ecx
0x100078c3: mov edx,[esi+0Ch]
0x100078c6: mov [eax+0Ch],edx
0x100078c9: mov ecx,[esi]
0x100078cb: add ecx,0FFFFFFF4h
0x100078ce: mov [edi+4Ch],ecx
0x100078d1: sub ecx,[edi+10h]
0x100078d4: mov [eax],ecx
0x100078d6: mov [esi],ecx
0x100078d8: pop ebp
0x100078d9: pop ebx
0x100078da: ret 8
```

`sub_100077a0` is called by sub_100056c0. It calls nothing else.

12.110 sub_100078e0 at 0x100078e0 (73 instructions)

```
0x100078e0: push ecx
0x100078e1: push ebx
0x100078e2: push ebp
0x100078e3: mov ebp,[esp+14h]
0x100078e7: push esi
```

```
0x100078e8: push edi
0x100078e9: mov edi,ecx
0x100078eb: lea esi,[edi+1Ch]
0x100078ee: push esi
0x100078ef: mov ebx,eax
0x100078f1: call dword ptr ds:[100FE6ACh]
0x100078f7: test eax,eax
0x100078f9: jne short 1000791Bh
0x100078fb: test ebp,ebp
0x100078fd: mov eax,3
0x10007902: je short 10007907h
0x10007904: mov eax,[edi+18h]
0x10007907: mov ecx,[esp+18h]
0x1000790b: push eax
0x1000790c: push 3
0x1000790e: lea eax,[edi+4]
0x10007911: push eax
0x10007912: push ecx
0x10007913: call 10001FF0h
0x10007918: add esp,10h
0x1000791b: lea esi,[edi+30h]
0x1000791e: push esi
0x1000791f: call dword ptr ds:[100FE6ACh]
0x10007925: test eax,eax
0x10007927: jne short 10007958h
0x10007929: mov eax,ebx
0x1000792b: and eax,1000000h
0x10007930: neg eax
0x10007932: sbb eax,eax
0x10007934: neg eax
0x10007936: add eax,1
0x10007939: test ebp,ebp
0x1000793b: mov ecx,3
0x10007940: je short 10007945h
0x10007942: mov ecx,[edi+2Ch]
0x10007945: push ecx
0x10007946: push eax
0x10007947: mov eax,[esp+20h]
0x1000794b: lea edx,[edi+4]
0x1000794e: push edx
0x1000794f: push eax
0x10007950: call 10001FF0h
0x10007955: add esp,10h
0x10007958: lea esi,[edi+44h]
0x1000795b: push esi
0x1000795c: call dword ptr ds:[100FE6ACh]
0x10007962: test eax,eax
0x10007964: jne short 1000798Bh
0x10007966: shr ebx,1Ch
0x10007969: and ebx,2
0x1000796c: test ebp,ebp
0x1000796e: mov eax,3
0x10007973: je short 10007978h
0x10007975: mov eax,[edi+40h]
0x10007978: mov ecx,[esp+18h]
0x1000797c: push eax
0x1000797d: push ebx
0x1000797e: add edi,4
0x10007981: push edi
0x10007982: push ecx
0x10007983: call 10001FF0h
0x10007988: add esp,10h
0x1000798b: pop edi
0x1000798c: pop esi
0x1000798d: pop ebp
0x1000798e: pop ebx
```

```
0x1000798f: pop ecx
0x10007990: ret 8
```

`sub_100078e0` is called by sub_10005fd0. It delegates to sub_10001ff0 (1 callees).

12.111 sub_100079a0 at 0x100079a0 (129 instructions)

```
0x100079a0: push 0FFFFFFFFh
0x100079a2: push 100FD448h
0x100079a7: mov eax,fs:[0]
0x100079ad: push eax
0x100079ae: sub esp,14h
0x100079b1: push ebx
0x100079b2: push ebp
0x100079b3: push esi
0x100079b4: push edi
0x100079b5: mov eax,ds:[10136220h]
0x100079ba: xor eax,esp
0x100079bc: push eax
0x100079bd: lea eax,[esp+28h]
0x100079c1: mov fs:[0],eax
0x100079c7: mov ebx,ecx
0x100079c9: mov edi,edx
0x100079cb: mov eax,[esp+38h]
0x100079cf: sub eax,8
0x100079d2: je short 100079FBh
0x100079d4: sub eax,1
0x100079d7: je short 100079ECh
0x100079d9: sub eax,0Bh
0x100079dc: jne near ptr 10007AF5h
0x100079e2: lea esi,[edi+18h]
0x100079e5: mov eax,esi
0x100079e7: lea edx,[edi+40h]
0x100079ea: jmp short 10007A03h
0x100079ec: lea esi,[edi+18h]
0x100079ef: lea ebp,[edi+2Ch]
0x100079f2: mov ecx,esi
0x100079f4: mov eax,ebp
0x100079f6: lea edx,[edi+40h]
0x100079f9: jmp short 10007A08h
0x100079fb: lea esi,[edi+18h]
0x100079fe: mov edx,esi
0x10007a00: lea eax,[edi+40h]
0x10007a03: lea ebp,[edi+2Ch]
0x10007a06: mov ecx,ebp
0x10007a08: cmp dword ptr [eax],3
0x10007a0b: je short 10007A16h
0x10007a0d: neg ebx
0x10007a0f: sbb ebx,ebx
0x10007a11: and ebx,2
0x10007a14: mov [eax],ebx
0x10007a16: mov eax,[ecx]
0x10007a18: cmp eax,3
0x10007a1b: je short 10007A27h
0x10007a1d: test eax,eax
0x10007a1f: je short 10007A27h
0x10007a21: mov dword ptr [ecx],0
0x10007a27: mov eax,[edx]
0x10007a29: cmp eax,3
0x10007a2c: je short 10007A38h
0x10007a2e: test eax,eax
0x10007a30: je short 10007A38h
0x10007a32: mov dword ptr [edx],0
0x10007a38: mov ebx,[esp+3Ch]
0x10007a3c: test ebx,ebx
0x10007a3e: je near ptr 10007AF5h
```

```
0x10007a44: push ebx
0x10007a45: lea ecx,[esp+18h]
0x10007a49: call 1002222Ah
0x10007a4e: mov ecx,ebx
0x10007a50: mov dword ptr [esp+30h],0
0x10007a58: call 10022236h
0x10007a5d: mov ebx,eax
0x10007a5f: lea eax,[edi+1Ch]
0x10007a62: push eax
0x10007a63: call dword ptr ds:[100FE6ACh]
0x10007a69: test eax,eax
0x10007a6b: jne short 10007A86h
0x10007a6d: mov esi,[esi]
0x10007a6f: push esi
0x10007a70: push 3
0x10007a72: lea eax,[edi+4]
0x10007a75: push eax
0x10007a76: lea ecx,[esp+20h]
0x10007a7a: push ecx
0x10007a7b: lea esi,[edi+1Ch]
0x10007a7e: call 10001FF0h
0x10007a83: add esp,10h
0x10007a86: lea esi,[edi+30h]
0x10007a89: push esi
0x10007a8a: call dword ptr ds:[100FE6ACh]
0x10007a90: test eax,eax
0x10007a92: jne short 10007ABAh
0x10007a94: mov ebp,[ebp]
0x10007a97: mov eax,ebx
0x10007a99: and eax,1000000h
0x10007a9e: neg eax
0x10007aa0: sbb eax,eax
0x10007aa2: neg eax
0x10007aa4: add eax,1
0x10007aa7: push ebp
0x10007aa8: push eax
0x10007aa9: lea edx,[edi+4]
0x10007aac: push edx
0x10007aad: lea eax,[esp+20h]
0x10007ab1: push eax
0x10007ab2: call 10001FF0h
0x10007ab7: add esp,10h
0x10007aba: lea esi,[edi+44h]
0x10007abd: push esi
0x10007abe: call dword ptr ds:[100FE6ACh]
0x10007ac4: test eax,eax
0x10007ac6: jne short 10007AE4h
0x10007ac8: mov eax,[edi+40h]
0x10007acb: push eax
0x10007acc: shr ebx,1Ch
0x10007acf: and ebx,2
0x10007ad2: push ebx
0x10007ad3: add edi,4
0x10007ad6: lea ecx,[esp+1Ch]
0x10007ada: push edi
0x10007adb: push ecx
0x10007adc: call 10001FF0h
0x10007ae1: add esp,10h
0x10007ae4: lea ecx,[esp+14h]
0x10007ae8: mov dword ptr [esp+30h],0FFFFFFFFh
0x10007af0: call 10022230h
0x10007af5: mov ecx,[esp+28h]
0x10007af9: mov fs:[0],ecx
0x10007b00: pop ecx
0x10007b01: pop edi
0x10007b02: pop esi
0x10007b03: pop ebp
```

```
0x10007b04: pop ebx
0x10007b05: add esp,20h
0x10007b08: ret 8
```

`sub_100079a0` is called by sub_10004910. It delegates to sub_10001ff0, j_ORDINAL 567, j_ORDINAL 758 (4 callees).

12.112 sub_10007b10 at 0x10007b10 (125 instructions)

```
0x10007b10: push 0FFFFFFFFh
0x10007b12: push 100FD418h
0x10007b17: mov eax,fs:[0]
0x10007b1d: push eax
0x10007b1e: sub esp,18h
0x10007b21: push ebx
0x10007b22: push ebp
0x10007b23: push esi
0x10007b24: push edi
0x10007b25: mov eax,ds:[10136220h]
0x10007b2a: xor eax,esp
0x10007b2c: push eax
0x10007b2d: lea eax,[esp+2Ch]
0x10007b31: mov fs:[0],eax
0x10007b37: mov edi,ecx
0x10007b39: mov eax,[esp+3Ch]
0x10007b3d: sub eax,8
0x10007b40: je short 10007B67h
0x10007b42: sub eax,1
0x10007b45: je short 10007B5Ah
0x10007b47: sub eax,0Bh
0x10007b4a: jne near ptr 10007C60h
0x10007b50: lea ebx,[edi+18h]
0x10007b53: mov eax,ebx
0x10007b55: lea ebp,[edi+40h]
0x10007b58: jmp short 10007B6Fh
0x10007b5a: lea ebx,[edi+18h]
0x10007b5d: lea eax,[edi+2Ch]
0x10007b60: mov esi,ebx
0x10007b62: lea ebp,[edi+40h]
0x10007b65: jmp short 10007B72h
0x10007b67: lea ebx,[edi+18h]
0x10007b6a: lea eax,[edi+40h]
0x10007b6d: mov ebp,ebx
0x10007b6f: lea esi,[edi+2Ch]
0x10007b72: cmp dword ptr [eax],3
0x10007b75: je short 10007B82h
0x10007b77: xor ecx,ecx
0x10007b79: cmp [esp+40h],ecx
0x10007b7d: setne cl
0x10007b80: mov [eax],ecx
0x10007b82: mov eax,[esi]
0x10007b84: cmp eax,3
0x10007b87: je short 10007B93h
0x10007b89: test eax,eax
0x10007b8b: je short 10007B93h
0x10007b8d: mov dword ptr [esi],0
0x10007b93: mov eax,[ebp]
0x10007b96: cmp eax,3
0x10007b99: je short 10007BA6h
0x10007b9b: test eax,eax
0x10007b9d: je short 10007BA6h
0x10007b9f: mov dword ptr [ebp],0
0x10007ba6: mov esi,[esp+44h]
0x10007baa: test esi,esi
0x10007bac: je near ptr 10007C60h
0x10007bb2: push esi
0x10007bb3: lea ecx,[esp+1Ch]
```

```
0x10007bb7: call 1002222Ah
0x10007bbc: mov ecx,esi
0x10007bbe: mov dword ptr [esp+34h],0
0x10007bc6: call 10022236h
0x10007bcb: lea esi,[edi+1Ch]
0x10007bce: push esi
0x10007bcf: mov ebp,eax
0x10007bd1: call dword ptr ds:[100FE6ACh]
0x10007bd7: test eax,eax
0x10007bd9: jne short 10007BF1h
0x10007bdb: mov ebx,[ebx]
0x10007bdd: push ebx
0x10007bde: push 3
0x10007be0: lea edx,[edi+4]
0x10007be3: push edx
0x10007be4: lea eax,[esp+24h]
0x10007be8: push eax
0x10007be9: call 10001FF0h
0x10007bee: add esp,10h
0x10007bf1: lea esi,[edi+30h]
0x10007bf4: push esi
0x10007bf5: call dword ptr ds:[100FE6ACh]
0x10007bfb: test eax,eax
0x10007bfd: jne short 10007C25h
0x10007bff: mov ecx,[edi+2Ch]
0x10007c02: mov eax,ebp
0x10007c04: and eax,1000000h
0x10007c09: neg eax
0x10007c0b: sbb eax,eax
0x10007c0d: push ecx
0x10007c0e: neg eax
0x10007c10: add eax,1
0x10007c13: push eax
0x10007c14: lea ecx,[edi+4]
0x10007c17: push ecx
0x10007c18: lea edx,[esp+24h]
0x10007c1c: push edx
0x10007c1d: call 10001FF0h
0x10007c22: add esp,10h
0x10007c25: lea esi,[edi+44h]
0x10007c28: push esi
0x10007c29: call dword ptr ds:[100FE6ACh]
0x10007c2f: test eax,eax
0x10007c31: jne short 10007C4Fh
0x10007c33: mov eax,[edi+40h]
0x10007c36: push eax
0x10007c37: shr ebp,1Ch
0x10007c3a: and ebp,2
0x10007c3d: push ebp
0x10007c3e: add edi,4
0x10007c41: lea eax,[esp+20h]
0x10007c45: push edi
0x10007c46: push eax
0x10007c47: call 10001FF0h
0x10007c4c: add esp,10h
0x10007c4f: lea ecx,[esp+18h]
0x10007c53: mov dword ptr [esp+34h],0FFFFFFFFh
0x10007c5b: call 10022230h
0x10007c60: mov ecx,[esp+2Ch]
0x10007c64: mov fs:[0],ecx
0x10007c6b: pop ecx
0x10007c6c: pop edi
0x10007c6d: pop esi
0x10007c6e: pop ebp
0x10007c6f: pop ebx
0x10007c70: add esp,24h
0x10007c73: ret 0Ch
```

`sub_10007b10` is called by sub_10004d40. It delegates to sub_10001ff0, j_ORDINAL 567, j_ORDINAL 758 (4 callees).

12.113 sub_10007c80 at 0x10007c80 (41 instructions)

```
0x10007c80: sub esp,18h
0x10007c83: mov ecx,esi
0x10007c85: call dword ptr ds:[100FE368h]
0x10007c8b: xor eax,eax
0x10007c8d: mov [esi+8],eax
0x10007c90: lea eax,[esi+10h]
0x10007c93: xor ecx,ecx
0x10007c95: push eax
0x10007c96: mov word ptr [esi+4],0
0x10007c9c: mov [esi+0Ch],ecx
0x10007c9f: call dword ptr ds:[100FE5C8h]
0x10007ca5: test edi,edi
0x10007ca7: mov ecx,[esp+1Ch]
0x10007cab: mov dword ptr [esi+20h],0
0x10007cb2: mov [esi+24h],ecx
0x10007cb5: je short 10007CC2h
0x10007cb7: cmp dword ptr [edi+4],0
0x10007cbb: je short 10007CC2h
0x10007cbd: mov [esi+28h],edi
0x10007cc0: jmp short 10007CC9h
0x10007cc2: mov dword ptr [esi+28h],0
0x10007cc9: xor eax,eax
0x10007ccb: mov [esi+2Ch],eax
0x10007cce: mov eax,[esi+28h]
0x10007cd1: xor ecx,ecx
0x10007cd3: test eax,eax
0x10007cd5: mov [esi+30h],ecx
0x10007cd8: je short 10007CF8h
0x10007cda: mov eax,[eax+4]
0x10007cdd: lea edx,[esp]
0x10007ce0: push edx
0x10007ce1: push 18h
0x10007ce3: push eax
0x10007ce4: call dword ptr ds:[100FE074h]
0x10007cea: mov eax,[esp+4]
0x10007cee: mov ecx,[esp+8]
0x10007cf2: mov [esi+2Ch],eax
0x10007cf5: mov [esi+30h],ecx
0x10007cf8: mov eax,esi
0x10007cfa: add esp,18h
0x10007cfd: ret 4
```

`sub_10007c80` is called by sub_10008df0. It calls nothing else.

12.114 sub_10007d00 at 0x10007d00 (80 instructions)

```
0x10007d00: sub esp,18h
0x10007d03: push ebx
0x10007d04: push ebp
0x10007d05: push edi
0x10007d06: mov edi,ecx
0x10007d08: push eax
0x10007d09: mov ecx,esi
0x10007d0b: call dword ptr ds:[100FE198h]
0x10007d11: xor eax,eax
0x10007d13: xor ecx,ecx
0x10007d15: mov [esi+8],eax
0x10007d18: mov eax,[esi+24h]
0x10007d1b: mov [esi+0Ch],ecx
0x10007d1e: mov edx,[edi]
0x10007d20: mov edx,[edx+30h]
0x10007d23: push eax
```

```
0x10007d24: mov ecx,edi
0x10007d26: xor ebx,ebx
0x10007d28: call edx
0x10007d2a: mov edx,[edi]
0x10007d2c: mov edx,[edx+70h]
0x10007d2f: push 420h
0x10007d34: mov ebp,eax
0x10007d36: mov eax,[esi]
0x10007d38: lea ecx,[esp+14h]
0x10007d3c: push ecx
0x10007d3d: mov [esp+18h],ebx
0x10007d41: mov [esp+1Ch],ebx
0x10007d45: mov [esp+20h],ebx
0x10007d49: mov [esp+24h],ebx
0x10007d4d: mov ecx,[eax-0Ch]
0x10007d50: push ecx
0x10007d51: push eax
0x10007d52: mov ecx,edi
0x10007d54: call edx
0x10007d56: mov eax,[edi]
0x10007d58: mov edx,[eax+30h]
0x10007d5b: push ebp
0x10007d5c: mov ecx,edi
0x10007d5e: call edx
0x10007d60: mov eax,[esp+18h]
0x10007d64: sub eax,[esp+10h]
0x10007d68: mov ecx,[esi]
0x10007d6a: add eax,0Ch
0x10007d6d: mov [esi+8],eax
0x10007d70: mov dword ptr [esi+0Ch],14h
0x10007d77: mov [esi+4],bx
0x10007d7b: mov ebp,[ecx-0Ch]
0x10007d7e: xor edi,edi
0x10007d80: cmp ebp,ebx
0x10007d82: jle short 10007DC1h
0x10007d84: push edi
0x10007d85: mov ecx,esi
0x10007d87: call dword ptr ds:[100FE1E8h]
0x10007d8d: cmp ax,26h
0x10007d91: jne short 10007DAFh
0x10007d93: lea edx,[ebp-1]
0x10007d96: cmp edi,edx
0x10007d98: jge short 10007DAFh
0x10007d9a: add edi,1
0x10007d9d: push edi
0x10007d9e: mov ecx,esi
0x10007da0: call dword ptr ds:[100FE1E8h]
0x10007da6: movzx eax,ax
0x10007da9: cmp ax,26h
0x10007dad: jne short 10007DBDh
0x10007daf: add edi,1
0x10007db2: cmp edi,ebp
0x10007db4: jl short 10007D84h
0x10007db6: pop edi
0x10007db7: pop ebp
0x10007db8: pop ebx
0x10007db9: add esp,18h
0x10007dbc: ret
0x10007dbd: mov [esi+4],ax
0x10007dc1: pop edi
0x10007dc2: pop ebp
0x10007dc3: pop ebx
0x10007dc4: add esp,18h
0x10007dc7: ret
```

`sub_10007d00` is called by sub_10008df0. It calls nothing else.

12.115 sub_10007dd0 at 0x10007dd0 (214 instructions)

```
0x10007dd0: push 0FFFFFFFFh
0x10007dd2: push 100FCEA8h
0x10007dd7: mov eax,fs:[0]
0x10007ddd: push eax
0x10007dde: sub esp,34h
0x10007de1: push ebp
0x10007de2: push esi
0x10007de3: push edi
0x10007de4: mov eax,ds:[10136220h]
0x10007de9: xor eax,esp
0x10007deb: push eax
0x10007dec: lea eax,[esp+44h]
0x10007df0: mov fs:[0],eax
0x10007df6: mov edi,ecx
0x10007df8: lea esi,[edi+10h]
0x10007dfb: push esi
0x10007dfc: call dword ptr ds:[100FE6ACh]
0x10007e02: test eax,eax
0x10007e04: jne near ptr 1000802Fh
0x10007e0a: cmp [edi+8],eax
0x10007e0d: jle near ptr 1000802Fh
0x10007e13: cmp [edi+0Ch],eax
0x10007e16: jle near ptr 1000802Fh
0x10007e1c: mov eax,[edi+20h]
0x10007e1f: cmp eax,1
0x10007e22: mov ebp,ds:[1013F380h]
0x10007e28: jne short 10007E30h
0x10007e2a: mov ebp,ds:[1013F384h]
0x10007e30: cmp dword ptr [esp+54h],0
0x10007e35: je short 10007E3Ch
0x10007e37: cmp eax,3
0x10007e3a: jne short 10007E42h
0x10007e3c: mov ebp,ds:[10138C10h]
0x10007e42: mov eax,[esi]
0x10007e44: mov edx,[esi+8]
0x10007e47: mov ecx,[esi+4]
0x10007e4a: mov [esp+30h],eax
0x10007e4e: mov eax,[esi+0Ch]
0x10007e51: mov [esp+38h],edx
0x10007e55: mov edx,[ebx]
0x10007e57: mov edx,[edx+30h]
0x10007e5a: mov [esp+3Ch],eax
0x10007e5e: mov eax,[edi+24h]
0x10007e61: mov [esp+34h],ecx
0x10007e65: push eax
0x10007e66: mov ecx,ebx
0x10007e68: call edx
0x10007e6a: mov [esp+1Ch],eax
0x10007e6e: mov eax,[ebx]
0x10007e70: mov edx,[eax+38h]
0x10007e73: push ebp
0x10007e74: mov ecx,ebx
0x10007e76: call edx
0x10007e78: push 1
0x10007e7a: mov ecx,ebx
0x10007e7c: mov [esp+1Ch],eax
0x10007e80: call 10022296h
0x10007e85: mov [esp+14h],eax
0x10007e89: mov eax,[edi+20h]
0x10007e8c: cmp eax,2
0x10007e8f: jne short 10007EACh
0x10007e91: push 808080h
0x10007e96: push 0FFFFFFh
0x10007e9b: lea eax,[esp+38h]
0x10007e9f: push eax
0x10007ea0: mov ecx,ebx
```

```
0x10007ea2: call 10022278h
0x10007ea7: jmp near ptr 10007FF0h
0x10007eac: cmp eax,1
0x10007eaf: jne near ptr 10007FF0h
0x10007eb5: cmp dword ptr [edi+28h],0
0x10007eb9: jne short 10007ED6h
0x10007ebb: push 0FFFFFFh
0x10007ec0: push 808080h
0x10007ec5: lea ecx,[esp+38h]
0x10007ec9: push ecx
0x10007eca: mov ecx,ebx
0x10007ecc: call 10022278h
0x10007ed1: jmp near ptr 10007FE1h
0x10007ed6: lea ecx,[esp+20h]
0x10007eda: call 1002202Ch
0x10007edf: lea edx,[esp+20h]
0x10007ee3: push edx
0x10007ee4: mov eax,ebx
0x10007ee6: mov dword ptr [esp+50h],0
0x10007eee: call 100024B0h
0x10007ef3: lea eax,[esp+20h]
0x10007ef7: push eax
0x10007ef8: mov eax,[edi+28h]
0x10007efb: call 10002480h
0x10007f00: mov ecx,[esi+8]
0x10007f03: sub ecx,[esi]
0x10007f05: mov [esp+10h],eax
0x10007f09: mov eax,[edi+2Ch]
0x10007f0c: cmp ecx,eax
0x10007f0e: jg short 10007F15h
0x10007f10: mov eax,[esi+8]
0x10007f13: sub eax,[esi]
0x10007f15: mov ebp,[esi+8]
0x10007f18: sub ebp,[esi]
0x10007f1a: mov ecx,[esi]
0x10007f1c: cdq
0x10007f1d: sub eax,edx
0x10007f1f: mov edx,[esi+0Ch]
0x10007f22: sub edx,[esi+4]
0x10007f25: push 0
0x10007f27: sar eax,1
0x10007f29: push edx
0x10007f2a: push eax
0x10007f2b: mov [esp+60h],eax
0x10007f2f: sub ebp,eax
0x10007f31: mov eax,[edi+14h]
0x10007f34: push eax
0x10007f35: push ecx
0x10007f36: push ebx
0x10007f37: mov ecx,0CC0020h
0x10007f3c: xor edx,edx
0x10007f3e: lea eax,[esp+38h]
0x10007f42: call 100023F0h
0x10007f47: mov eax,[esp+54h]
0x10007f4b: mov edx,[edi+2Ch]
0x10007f4e: mov ecx,[esi+0Ch]
0x10007f51: sub ecx,[esi+4]
0x10007f54: sub edx,eax
0x10007f56: push edx
0x10007f57: mov edx,[edi+14h]
0x10007f5a: push ecx
0x10007f5b: mov ecx,[edi+18h]
0x10007f5e: push eax
0x10007f5f: push edx
0x10007f60: sub ecx,eax
0x10007f62: push ecx
0x10007f63: push ebx
```

```
0x10007f64: mov ecx,0CC0020h
0x10007f69: xor edx,edx
0x10007f6b: lea eax,[esp+38h]
0x10007f6f: call 100023F0h
0x10007f74: mov eax,[esp+54h]
0x10007f78: cmp eax,ebp
0x10007f7a: mov [esp+54h],eax
0x10007f7e: jge short 10007FC2h
0x10007f80: mov eax,[edi+2Ch]
0x10007f83: mov ecx,[esi+0Ch]
0x10007f86: sub ecx,[esi+4]
0x10007f89: cdq
0x10007f8a: push 0CC0020h
0x10007f8f: sub eax,edx
0x10007f91: mov edx,[esi]
0x10007f93: add edx,[esp+58h]
0x10007f97: push 0
0x10007f99: sar eax,1
0x10007f9b: push eax
0x10007f9c: mov eax,[esp+30h]
0x10007fa0: push eax
0x10007fa1: mov eax,[edi+14h]
0x10007fa4: push ecx
0x10007fa5: mov ecx,[ebx+4]
0x10007fa8: push 1
0x10007faa: push eax
0x10007fab: push edx
0x10007fac: push ecx
0x10007fad: call dword ptr ds:[100FE06Ch]
0x10007fb3: mov eax,[esp+54h]
0x10007fb7: add eax,1
0x10007fba: cmp eax,ebp
0x10007fbc: mov [esp+54h],eax
0x10007fc0: jl short 10007F80h
0x10007fc2: mov eax,[esp+10h]
0x10007fc6: lea edx,[esp+20h]
0x10007fca: push edx
0x10007fcb: call 10002480h
0x10007fd0: lea ecx,[esp+20h]
0x10007fd4: mov dword ptr [esp+4Ch],0FFFFFFFFh
0x10007fdc: call 10022032h
0x10007fe1: push 1
0x10007fe3: push 1
0x10007fe5: lea eax,[esp+38h]
0x10007fe9: push eax
0x10007fea: call dword ptr ds:[100FE6B0h]
0x10007ff0: mov edi,[edi]
0x10007ff2: mov ecx,[edi-0Ch]
0x10007ff5: mov edx,[ebx]
0x10007ff7: mov edx,[edx+70h]
0x10007ffa: push 25h
0x10007ffc: lea eax,[esp+34h]
0x10008000: push eax
0x10008001: push ecx
0x10008002: push edi
0x10008003: mov ecx,ebx
0x10008005: call edx
0x10008007: mov eax,[esp+14h]
0x1000800b: push eax
0x1000800c: mov ecx,ebx
0x1000800e: call 10022296h
0x10008013: mov edx,[ebx]
0x10008015: mov eax,[esp+18h]
0x10008019: mov edx,[edx+38h]
0x1000801c: push eax
0x1000801d: mov ecx,ebx
0x1000801f: call edx
```

```
0x10008021: mov ecx,[esp+1Ch]
0x10008025: mov eax,[ebx]
0x10008027: mov edx,[eax+30h]
0x1000802a: push ecx
0x1000802b: mov ecx,ebx
0x1000802d: call edx
0x1000802f: mov ecx,[esp+44h]
0x10008033: mov fs:[0],ecx
0x1000803a: pop ecx
0x1000803b: pop edi
0x1000803c: pop esi
0x1000803d: pop ebp
0x1000803e: add esp,40h
0x10008041: ret 4
```

`sub_10007dd0` is called by sub_100089a0. It delegates to sub_100023f0, sub_10002480, sub_100024b0 (7 callees).

12.116 sub_10008050 at 0x10008050 (83 instructions)

```
0x10008050: push 0FFFFFFFFh
0x10008052: push 100FD205h
0x10008057: mov eax,fs:[0]
0x1000805d: push eax
0x1000805e: sub esp,64h
0x10008061: mov eax,ds:[10136220h]
0x10008066: xor eax,esp
0x10008068: mov [esp+60h],eax
0x1000806c: push ebx
0x1000806d: push edi
0x1000806e: mov eax,ds:[10136220h]
0x10008073: xor eax,esp
0x10008075: push eax
0x10008076: lea eax,[esp+70h]
0x1000807a: mov fs:[0],eax
0x10008080: xor edi,edi
0x10008082: mov dword ptr [esi],1012F764h
0x10008088: mov [esp+0Ch],esi
0x1000808c: mov [esi+14h],edi
0x1000808f: mov [esi+18h],edi
0x10008092: mov [esi+1Ch],edi
0x10008095: lea ecx,[esi+24h]
0x10008098: mov [esp+78h],edi
0x1000809c: call 100222A2h
0x100080a1: mov dword ptr [esi+38h],1012F790h
0x100080a8: mov [esi+3Ch],edi
0x100080ab: lea ebx,[esi+54h]
0x100080ae: mov [ebx+4],edi
0x100080b1: mov dword ptr [ebx],1012F744h
0x100080b7: mov [esi+60h],edi
0x100080ba: mov dword ptr [esi+5Ch],1012F498h
0x100080c1: lea eax,[esi+84h]
0x100080c7: push eax
0x100080c8: mov byte ptr [esp+7Ch],4
0x100080cd: call 100029D0h
0x100080d2: mov byte ptr [esp+78h],5
0x100080d7: lea ecx,[esp+10h]
0x100080db: mov [esi+4],edi
0x100080de: mov [esi+8],edi
0x100080e1: mov [esi+0Ch],edi
0x100080e4: call 10002070h
0x100080e9: test eax,eax
0x100080eb: je short 10008110h
0x100080ed: cmp ebx,edi
0x100080ef: je short 100080FDh
0x100080f1: cmp [ebx+4],edi
0x100080f4: je short 100080FDh
```

```
0x100080f6: mov ecx,ebx
0x100080f8: call 10021FF6h
0x100080fd: lea ecx,[esp+10h]
0x10008101: push ecx
0x10008102: call dword ptr ds:[100FE064h]
0x10008108: push eax
0x10008109: mov ecx,ebx
0x1000810b: call 10021FFCh
0x10008110: lea edx,[esi+40h]
0x10008113: or ebx,0FFFFFFFFh
0x10008116: xor eax,eax
0x10008118: xor ecx,ecx
0x1000811a: push edx
0x1000811b: mov [esi+34h],ebx
0x1000811e: mov [esi+2Ch],eax
0x10008121: mov [esi+30h],ecx
0x10008124: call dword ptr ds:[100FE5C8h]
0x1000812a: mov [esi+50h],edi
0x1000812d: mov [esi+68h],edi
0x10008130: mov [esi+70h],edi
0x10008133: mov [esi+74h],ebx
0x10008136: mov [esi+7Ch],edi
0x10008139: mov [esi+80h],edi
0x1000813f: mov dword ptr [esi+64h],1
0x10008146: mov dword ptr [esi+20h],8
0x1000814d: mov eax,esi
0x1000814f: mov ecx,[esp+70h]
0x10008153: mov fs:[0],ecx
0x1000815a: pop ecx
0x1000815b: pop edi
0x1000815c: pop ebx
0x1000815d: mov ecx,[esp+60h]
0x10008161: xor ecx,esp
0x10008163: call 10022326h
0x10008168: add esp,70h
0x1000816b: ret
```

`sub_10008050` is called by sub_100032e0. It delegates to sub_10002070, sub_100029d0, j_ORDINAL 1925 (6 callees).

12.117 sub_10008290 at 0x10008290 (98 instructions)

```
0x10008290: push 0FFFFFFFFh
0x10008292: push 100FCE18h
0x10008297: mov eax,fs:[0]
0x1000829d: push eax
0x1000829e: sub esp,24h
0x100082a1: push ebx
0x100082a2: push edi
0x100082a3: mov eax,ds:[10136220h]
0x100082a8: xor eax,esp
0x100082aa: push eax
0x100082ab: lea eax,[esp+30h]
0x100082af: mov fs:[0],eax
0x100082b5: cmp dword ptr [esi+28h],0
0x100082b9: je short 100082C3h
0x100082bb: lea ecx,[esi+24h]
0x100082be: call 100222DEh
0x100082c3: mov ebx,1012F498h
0x100082c8: mov dword ptr [esp+14h],0
0x100082d0: mov [esp+10h],ebx
0x100082d4: mov dword ptr [esp+38h],0
0x100082dc: mov eax,ds:[1013F360h]
0x100082e1: push 138Ch
0x100082e6: push eax
0x100082e7: call dword ptr ds:[100FE5E0h]
0x100082ed: push eax
```

```
0x100082ee: lea ecx,[esp+14h]
0x100082f2: call 10021FFCh
0x100082f7: test eax,eax
0x100082f9: je short 10008366h
0x100082fb: mov edx,[esp+14h]
0x100082ff: lea ecx,[esp+18h]
0x10008303: push ecx
0x10008304: push 18h
0x10008306: push edx
0x10008307: call dword ptr ds:[100FE074h]
0x1000830d: movzx eax,word ptr [esp+2Ah]
0x10008312: mov ecx,[esp+20h]
0x10008316: push 0
0x10008318: or eax,1
0x1000831b: push 4
0x1000831d: push eax
0x1000831e: mov eax,[esp+28h]
0x10008322: cdq
0x10008323: and edx,3
0x10008326: add eax,edx
0x10008328: push ecx
0x10008329: sar eax,2
0x1000832c: lea edi,[esi+24h]
0x1000832f: push eax
0x10008330: mov ecx,edi
0x10008332: call 100222E4h
0x10008337: test eax,eax
0x10008339: je short 10008366h
0x1000833b: mov edx,[esp+14h]
0x1000833f: mov eax,[esi+28h]
0x10008342: push 0FF000000h
0x10008347: push edx
0x10008348: push eax
0x10008349: call 10021FD2h
0x1000834e: mov eax,[eax+94h]
0x10008354: mov ecx,[eax]
0x10008356: call 10009DB0h
0x1000835b: test eax,eax
0x1000835d: jge short 1000838Fh
0x1000835f: mov ecx,edi
0x10008361: call 100222DEh
0x10008366: mov dword ptr [esp+38h],0FFFFFFFFh
0x1000836e: lea ecx,[esp+10h]
0x10008372: mov [esp+10h],ebx
0x10008376: call 10001370h
0x1000837b: xor eax,eax
0x1000837d: mov ecx,[esp+30h]
0x10008381: mov fs:[0],ecx
0x10008388: pop ecx
0x10008389: pop edi
0x1000838a: pop ebx
0x1000838b: add esp,30h
0x1000838e: ret
0x1000838f: mov eax,[esp+1Ch]
0x10008393: mov ecx,[esp+20h]
0x10008397: cdq
0x10008398: and edx,3
0x1000839b: add eax,edx
0x1000839d: sar eax,2
0x100083a0: mov [esi+30h],ecx
0x100083a3: mov dword ptr [esp+38h],0FFFFFFFFh
0x100083ab: lea ecx,[esp+10h]
0x100083af: mov [esi+2Ch],eax
0x100083b2: mov [esp+10h],ebx
0x100083b6: call 10001370h
0x100083bb: mov eax,1
0x100083c0: mov ecx,[esp+30h]
```

```
0x100083c4: mov fs:[0],ecx
0x100083cb: pop ecx
0x100083cc: pop edi
0x100083cd: pop ebx
0x100083ce: add esp,30h
0x100083d1: ret
```

`sub_10008290` is called by sub_10003820. It delegates to sub_10001370, sub_10009db0, j_ORDINAL 1079 (6 callees).

12.118 sub_100083e0 at 0x100083e0 (69 instructions)

```
0x100083e0: sub esp,18h
0x100083e3: test edi,edi
0x100083e5: je short 100083EDh
0x100083e7: cmp dword ptr [edi+4],0
0x100083eb: jne short 100083F5h
0x100083ed: xor eax,eax
0x100083ef: add esp,18h
0x100083f2: ret 4
0x100083f5: cmp dword ptr [esi+28h],0
0x100083f9: je short 10008403h
0x100083fb: lea ecx,[esi+24h]
0x100083fe: call 100222DEh
0x10008403: mov ecx,[edi+4]
0x10008406: lea eax,[esp]
0x10008409: push eax
0x1000840a: push 18h
0x1000840c: push ecx
0x1000840d: call dword ptr ds:[100FE074h]
0x10008413: test eax,eax
0x10008415: je short 100083EDh
0x10008417: movzx edx,word ptr [esp+12h]
0x1000841c: mov eax,[esp+8]
0x10008420: push ebx
0x10008421: push 0
0x10008423: push 4
0x10008425: or edx,1
0x10008428: push edx
0x10008429: push eax
0x1000842a: mov eax,[esp+18h]
0x1000842e: cdq
0x1000842f: and edx,3
0x10008432: add eax,edx
0x10008434: sar eax,2
0x10008437: lea ebx,[esi+24h]
0x1000843a: push eax
0x1000843b: mov ecx,ebx
0x1000843d: call 100222E4h
0x10008442: test eax,eax
0x10008444: je short 10008470h
0x10008446: mov edx,[esp+20h]
0x1000844a: mov eax,[edi+4]
0x1000844d: mov ecx,[esi+28h]
0x10008450: push edx
0x10008451: push eax
0x10008452: push ecx
0x10008453: call 10021FD2h
0x10008458: mov eax,[eax+94h]
0x1000845e: mov ecx,[eax]
0x10008460: call 10009DB0h
0x10008465: test eax,eax
0x10008467: jge short 10008479h
0x10008469: mov ecx,ebx
0x1000846b: call 100222DEh
0x10008470: xor eax,eax
0x10008472: pop ebx
```

```
0x10008473: add esp,18h
0x10008476: ret 4
0x10008479: mov eax,[esp+8]
0x1000847d: mov ecx,[esp+0Ch]
0x10008481: cdq
0x10008482: and edx,3
0x10008485: add eax,edx
0x10008487: sar eax,2
0x1000848a: mov [esi+2Ch],eax
0x1000848d: mov [esi+30h],ecx
0x10008490: mov eax,1
0x10008495: pop ebx
0x10008496: add esp,18h
0x10008499: ret 4
```

`sub_100083e0` is called by ?SetMenuBarBreakImage@CCustomTitle@@QAEHPAVCBitmap@@K@Z. It delegates to sub_10009db0, j_ORDINAL 1079, j_ORDINAL 1921 (4 callees).

12.119 sub_100084a0 at 0x100084a0 (45 instructions)

```
0x100084a0: push ebx
0x100084a1: push ebp
0x100084a2: push esi
0x100084a3: push edi
0x100084a4: mov edi,eax
0x100084a6: xor esi,esi
0x100084a8: mov ecx,[edi+14h]
0x100084ab: test ecx,ecx
0x100084ad: jne short 100084B3h
0x100084af: xor eax,eax
0x100084b1: jmp short 100084BBh
0x100084b3: mov eax,[edi+18h]
0x100084b6: sub eax,ecx
0x100084b8: sar eax,2
0x100084bb: cmp esi,eax
0x100084bd: jge short 100084F6h
0x100084bf: test ecx,ecx
0x100084c1: je short 100084CFh
0x100084c3: mov eax,[edi+18h]
0x100084c6: sub eax,ecx
0x100084c8: sar eax,2
0x100084cb: cmp esi,eax
0x100084cd: jb short 100084D5h
0x100084cf: call dword ptr ds:[100FE584h]
0x100084d5: mov eax,[edi+14h]
0x100084d8: mov eax,[eax+esi*4]
0x100084db: add eax,10h
0x100084de: xor ecx,ecx
0x100084e0: xor edx,edx
0x100084e2: mov [eax],ecx
0x100084e4: xor ebx,ebx
0x100084e6: xor ebp,ebp
0x100084e8: mov [eax+4],edx
0x100084eb: mov [eax+8],ebx
0x100084ee: mov [eax+0Ch],ebp
0x100084f1: add esi,1
0x100084f4: jmp short 100084A8h
0x100084f6: add edi,40h
0x100084f9: push edi
0x100084fa: call dword ptr ds:[100FE5C8h]
0x10008500: pop edi
0x10008501: pop esi
0x10008502: pop ebp
0x10008503: pop ebx
0x10008504: ret
```

`sub_100084a0` is called by sub_100056c0, sub_100087c0. It calls nothing else.

12.120 sub_10008510 at 0x10008510 (32 instructions)

```
0x10008510: push ebp
0x10008511: mov ebp,ds:[100FE584h]
0x10008517: push edi
0x10008518: xor edi,edi
0x1000851a: lea ebx,[ebx]
0x10008520: mov ecx,[esi+14h]
0x10008523: test ecx,ecx
0x10008525: jne short 1000852Bh
0x10008527: xor eax,eax
0x10008529: jmp short 10008533h
0x1000852b: mov eax,[esi+18h]
0x1000852e: sub eax,ecx
0x10008530: sar eax,2
0x10008533: cmp edi,eax
0x10008535: jge short 1000855Bh
0x10008537: test ecx,ecx
0x10008539: je short 10008547h
0x1000853b: mov eax,[esi+18h]
0x1000853e: sub eax,ecx
0x10008540: sar eax,2
0x10008543: cmp edi,eax
0x10008545: jb short 10008549h
0x10008547: call ebp
0x10008549: mov eax,[esi+14h]
0x1000854c: mov ecx,[eax+edi*4]
0x1000854f: mov dword ptr [ecx+20h],0
0x10008556: add edi,1
0x10008559: jmp short 10008520h
0x1000855b: pop edi
0x1000855c: mov dword ptr [esi+50h],0
0x10008563: pop ebp
0x10008564: ret
```

`sub_10008510` is called by sub_100056c0, sub_100087c0. It calls nothing else.

12.121 sub_10008570 at 0x10008570 (43 instructions)

```
0x10008570: push ebx
0x10008571: push ebp
0x10008572: mov ebp,ds:[100FE584h]
0x10008578: push esi
0x10008579: push edi
0x1000857a: mov ebx,eax
0x1000857c: xor edi,edi
0x1000857e: add ebx,10h
0x10008581: mov ecx,[ebx+4]
0x10008584: test ecx,ecx
0x10008586: jne short 1000858Ch
0x10008588: xor eax,eax
0x1000858a: jmp short 10008594h
0x1000858c: mov eax,[ebx+8]
0x1000858f: sub eax,ecx
0x10008591: sar eax,2
0x10008594: cmp edi,eax
0x10008596: jge short 100085CAh
0x10008598: test ecx,ecx
0x1000859a: je short 100085A8h
0x1000859c: mov eax,[ebx+8]
0x1000859f: sub eax,ecx
0x100085a1: sar eax,2
0x100085a4: cmp edi,eax
0x100085a6: jb short 100085AAh
```

```
0x100085a8: call ebp
0x100085aa: mov eax,[ebx+4]
0x100085ad: mov esi,[eax+edi*4]
0x100085b0: test esi,esi
0x100085b2: je short 100085C5h
0x100085b4: mov ecx,esi
0x100085b6: call dword ptr ds:[100FE374h]
0x100085bc: push esi
0x100085bd: call 10021FCCh
0x100085c2: add esp,4
0x100085c5: add edi,1
0x100085c8: jmp short 10008581h
0x100085ca: call 1000BA40h
0x100085cf: pop edi
0x100085d0: pop esi
0x100085d1: pop ebp
0x100085d2: pop ebx
0x100085d3: ret
```

`sub_10008570` is called by sub_10008df0. It delegates to sub_1000ba40, j_ORDINAL 764 (2 callees).

12.122 sub_100085e0 at 0x100085e0 (81 instructions)

```
0x100085e0: sub esp,14h
0x100085e3: push ebx
0x100085e4: push ebp
0x100085e5: push esi
0x100085e6: push edi
0x100085e7: mov esi,eax
0x100085e9: xor edi,edi
0x100085eb: jmp short 100085F0h
0x100085f0: mov ecx,[esi+14h]
0x100085f3: test ecx,ecx
0x100085f5: jne short 100085FBh
0x100085f7: xor eax,eax
0x100085f9: jmp short 10008603h
0x100085fb: mov eax,[esi+18h]
0x100085fe: sub eax,ecx
0x10008600: sar eax,2
0x10008603: cmp edi,eax
0x10008605: jge near ptr 100086A0h
0x1000860b: mov eax,[esi+34h]
0x1000860e: test eax,eax
0x10008610: jl short 10008616h
0x10008612: cmp edi,eax
0x10008614: je short 1000866Dh
0x10008616: test ecx,ecx
0x10008618: mov eax,[esp+28h]
0x1000861c: mov ebx,[eax]
0x1000861e: mov ebp,[eax+4]
0x10008621: je short 1000862Fh
0x10008623: mov eax,[esi+18h]
0x10008626: sub eax,ecx
0x10008628: sar eax,2
0x1000862b: cmp edi,eax
0x1000862d: jb short 10008635h
0x1000862f: call dword ptr ds:[100FE584h]
0x10008635: mov eax,[esi+14h]
0x10008638: mov eax,[eax+edi*4]
0x1000863b: mov ecx,[eax+10h]
0x1000863e: mov [esp+10h],ecx
0x10008642: mov edx,[eax+14h]
0x10008645: mov [esp+14h],edx
0x10008649: mov ecx,[eax+18h]
0x1000864c: push ebp
0x1000864d: mov [esp+1Ch],ecx
```

```
0x10008651: mov edx,[eax+1Ch]
0x10008654: push ebx
0x10008655: lea eax,[esp+18h]
0x10008659: push eax
0x1000865a: mov [esp+28h],edx
0x1000865e: call dword ptr ds:[100FE610h]
0x10008664: test eax,eax
0x10008666: jne short 10008686h
0x10008668: add edi,1
0x1000866b: jmp short 100085F0h
0x1000866d: mov eax,[esp+28h]
0x10008671: mov ecx,[eax+4]
0x10008674: mov edx,[eax]
0x10008676: push ecx
0x10008677: push edx
0x10008678: add esi,40h
0x1000867b: push esi
0x1000867c: call dword ptr ds:[100FE610h]
0x10008682: test eax,eax
0x10008684: je short 100086A0h
0x10008686: test edi,edi
0x10008688: mov eax,edi
0x1000868a: jl short 100086A3h
0x1000868c: mov ecx,[esp+2Ch]
0x10008690: mov dword ptr [ecx],5
0x10008696: pop edi
0x10008697: pop esi
0x10008698: pop ebp
0x10008699: pop ebx
0x1000869a: add esp,14h
0x1000869d: ret 8
0x100086a0: or eax,0FFFFFFFFh
0x100086a3: pop edi
0x100086a4: pop esi
0x100086a5: pop ebp
0x100086a6: pop ebx
0x100086a7: add esp,14h
0x100086aa: ret 8
```

`sub_100085e0` is called by sub_10005b50, sub_10009740. It calls nothing else.

12.123 sub_100086b0 at 0x100086b0 (101 instructions)

```
0x100086b0: sub esp,18h
0x100086b3: push ebx
0x100086b4: push ebp
0x100086b5: push esi
0x100086b6: push edi
0x100086b7: mov esi,eax
0x100086b9: call dword ptr ds:[100FE5C8h]
0x100086bf: mov ebp,ds:[100FE6ACh]
0x100086c5: xor ebx,ebx
0x100086c7: mov ecx,[esi+14h]
0x100086ca: test ecx,ecx
0x100086cc: jne short 100086D2h
0x100086ce: xor eax,eax
0x100086d0: jmp short 100086DAh
0x100086d2: mov eax,[esi+18h]
0x100086d5: sub eax,ecx
0x100086d7: sar eax,2
0x100086da: cmp ebx,eax
0x100086dc: jge near ptr 100087ACh
0x100086e2: mov eax,[esi+34h]
0x100086e5: test eax,eax
0x100086e7: jl short 100086F1h
0x100086e9: cmp ebx,eax
```

```
0x100086eb: je near ptr 10008773h
0x100086f1: test ecx,ecx
0x100086f3: je short 10008701h
0x100086f5: mov eax,[esi+18h]
0x100086f8: sub eax,ecx
0x100086fa: sar eax,2
0x100086fd: cmp ebx,eax
0x100086ff: jb short 10008707h
0x10008701: call dword ptr ds:[100FE584h]
0x10008707: mov eax,[esi+14h]
0x1000870a: mov eax,[eax+ebx*4]
0x1000870d: mov ecx,[eax+10h]
0x10008710: mov [esp+10h],ecx
0x10008714: mov edx,[eax+14h]
0x10008717: mov [esp+14h],edx
0x1000871b: mov ecx,[eax+18h]
0x1000871e: mov [esp+18h],ecx
0x10008722: mov edx,[eax+1Ch]
0x10008725: lea eax,[esp+10h]
0x10008729: push eax
0x1000872a: mov [esp+20h],edx
0x1000872e: call ebp
0x10008730: test eax,eax
0x10008732: jne short 1000876Bh
0x10008734: push edi
0x10008735: call ebp
0x10008737: test eax,eax
0x10008739: je short 1000875Eh
0x1000873b: mov ecx,[esp+10h]
0x1000873f: mov edx,[esp+14h]
0x10008743: mov eax,[esp+18h]
0x10008747: mov [edi],ecx
0x10008749: mov ecx,[esp+1Ch]
0x1000874d: mov [edi+4],edx
0x10008750: mov [edi+8],eax
0x10008753: mov [edi+0Ch],ecx
0x10008756: add ebx,1
0x10008759: jmp near ptr 100086C7h
0x1000875e: lea edx,[esp+10h]
0x10008762: push edx
0x10008763: push edi
0x10008764: push edi
0x10008765: call dword ptr ds:[100FE5B8h]
0x1000876b: add ebx,1
0x1000876e: jmp near ptr 100086C7h
0x10008773: add esi,40h
0x10008776: push esi
0x10008777: call ebp
0x10008779: test eax,eax
0x1000877b: jne short 100087ACh
0x1000877d: push edi
0x1000877e: call ebp
0x10008780: test eax,eax
0x10008782: je short 100087A3h
0x10008784: mov eax,[esi]
0x10008786: mov [edi],eax
0x10008788: mov ecx,[esi+4]
0x1000878b: mov [edi+4],ecx
0x1000878e: mov edx,[esi+8]
0x10008791: mov [edi+8],edx
0x10008794: mov eax,[esi+0Ch]
0x10008797: mov [edi+0Ch],eax
0x1000879a: mov eax,edi
0x1000879c: pop esi
0x1000879d: pop ebp
0x1000879e: pop ebx
0x1000879f: add esp,18h
```

```
0x100087a2: ret
0x100087a3: push esi
0x100087a4: push edi
0x100087a5: push edi
0x100087a6: call dword ptr ds:[100FE5B8h]
0x100087ac: pop esi
0x100087ad: pop ebp
0x100087ae: mov eax,edi
0x100087b0: pop ebx
0x100087b1: add esp,18h
0x100087b4: ret
```

`sub_100086b0` is called by sub_10004d40, sub_100056c0, sub_10008b20. It calls nothing else.

12.124 sub_100087c0 at 0x100087c0 (164 instructions)

```
0x100087c0: sub esp,1Ch
0x100087c3: push ebp
0x100087c4: push esi
0x100087c5: push edi
0x100087c6: mov esi,eax
0x100087c8: call 100084A0h
0x100087cd: call 10008510h
0x100087d2: cmp dword ptr [esi+34h],0
0x100087d6: jl short 100087E6h
0x100087d8: mov edi,esi
0x100087da: mov dword ptr [esi+34h],0FFFFFFFFh
0x100087e1: call 100091B0h
0x100087e6: mov edi,[esp+2Ch]
0x100087ea: mov eax,[edi]
0x100087ec: mov ecx,[edi+4]
0x100087ef: mov edx,[edi+8]
0x100087f2: mov [ebx],eax
0x100087f4: mov eax,[edi+0Ch]
0x100087f7: mov [ebx+4],ecx
0x100087fa: mov [ebx+8],edx
0x100087fd: mov [ebx+0Ch],eax
0x10008800: mov ecx,[esi+20h]
0x10008803: add ecx,[ebx]
0x10008805: xor ebp,ebp
0x10008807: mov [ebx+8],ecx
0x1000880a: lea ebx,[ebx]
0x10008810: mov ecx,[esi+14h]
0x10008813: test ecx,ecx
0x10008815: jne short 1000881Bh
0x10008817: xor eax,eax
0x10008819: jmp short 10008823h
0x1000881b: mov eax,[esi+18h]
0x1000881e: sub eax,ecx
0x10008820: sar eax,2
0x10008823: cmp ebp,eax
0x10008825: jge near ptr 1000896Ah
0x1000882b: test ecx,ecx
0x1000882d: je short 1000883Bh
0x1000882f: mov eax,[esi+18h]
0x10008832: sub eax,ecx
0x10008834: sar eax,2
0x10008837: cmp ebp,eax
0x10008839: jb short 10008841h
0x1000883b: call dword ptr ds:[100FE584h]
0x10008841: mov edx,[esi+14h]
0x10008844: mov ecx,[edx+ebp*4]
0x10008847: mov eax,[ecx+0Ch]
0x1000884a: mov edx,[ecx+8]
0x1000884d: mov [esp+14h],eax
0x10008851: mov eax,[ebx+8]
```

```
0x10008854: add edx,eax
0x10008856: cmp edx,[edi+8]
0x10008859: jg short 100088CEh
0x1000885b: mov edx,[ebx]
0x1000885d: mov [esp+18h],edx
0x10008861: mov edx,eax
0x10008863: mov edi,[ebx+4]
0x10008866: mov [esp+20h],edx
0x1000886a: mov edx,[ebx+0Ch]
0x1000886d: mov [esp+24h],edx
0x10008871: mov edx,[ecx+0Ch]
0x10008874: mov [esp+14h],edx
0x10008878: mov edx,[ecx+8]
0x1000887b: add edx,eax
0x1000887d: mov eax,[ecx+8]
0x10008880: mov [esp+10h],eax
0x10008884: mov [esp+20h],edx
0x10008888: mov edx,[esp+2Ch]
0x1000888c: mov eax,[edx+0Ch]
0x1000888f: sub eax,[edx+4]
0x10008892: sub eax,[ecx+0Ch]
0x10008895: add eax,1
0x10008898: cdq
0x10008899: sub eax,edx
0x1000889b: mov edx,[ecx+8]
0x1000889e: sar eax,1
0x100088a0: add edi,eax
0x100088a2: mov eax,[ecx+0Ch]
0x100088a5: mov [esp+10h],edx
0x100088a9: lea edx,[eax+edi]
0x100088ac: mov eax,[ebx+8]
0x100088af: mov [ecx+10h],eax
0x100088b2: mov eax,[esp+20h]
0x100088b6: mov [ecx+14h],edi
0x100088b9: mov edi,[esp+2Ch]
0x100088bd: mov [ecx+18h],eax
0x100088c0: mov [ecx+1Ch],edx
0x100088c3: mov [ebx+8],eax
0x100088c6: add ebp,1
0x100088c9: jmp near ptr 10008810h
0x100088ce: test ebp,ebp
0x100088d0: jle short 1000891Eh
0x100088d2: mov eax,[esi+2Ch]
0x100088d5: add eax,[ebx+8]
0x100088d8: cmp eax,[edi+8]
0x100088db: jle short 1000891Eh
0x100088dd: mov ecx,[esi+14h]
0x100088e0: sub ebp,1
0x100088e3: test ecx,ecx
0x100088e5: je short 100088F3h
0x100088e7: mov eax,[esi+18h]
0x100088ea: sub eax,ecx
0x100088ec: sar eax,2
0x100088ef: cmp ebp,eax
0x100088f1: jb short 100088F9h
0x100088f3: call dword ptr ds:[100FE584h]
0x100088f9: test ebp,ebp
0x100088fb: mov ecx,[esi+14h]
0x100088fe: mov eax,[ecx+ebp*4]
0x10008901: mov edx,[eax+14h]
0x10008904: mov ecx,[eax+10h]
0x10008907: mov [esp+1Ch],edx
0x1000890b: mov edx,[eax+18h]
0x1000890e: mov eax,[eax+1Ch]
0x10008911: mov [esp+20h],edx
0x10008915: mov [esp+24h],eax
0x10008919: mov [ebx+8],ecx
```

```
0x1000891c: jg short 100088D2h
0x1000891e: mov [esi+34h],ebp
0x10008921: mov edx,[ebx+8]
0x10008924: mov eax,[ebx+0Ch]
0x10008927: mov ecx,[ebx]
0x10008929: mov ebp,[ebx+4]
0x1000892c: mov [esp+24h],eax
0x10008930: mov edi,edx
0x10008932: mov [esp+20h],edx
0x10008936: mov edx,[esp+2Ch]
0x1000893a: mov eax,[edx+0Ch]
0x1000893d: sub eax,[edx+4]
0x10008940: mov [esp+18h],ecx
0x10008944: sub eax,[esi+30h]
0x10008947: mov ecx,[esi+2Ch]
0x1000894a: add eax,1
0x1000894d: cdq
0x1000894e: sub eax,edx
0x10008950: sar eax,1
0x10008952: add ebp,eax
0x10008954: mov eax,[esi+30h]
0x10008957: mov [esi+40h],edi
0x1000895a: add ecx,edi
0x1000895c: mov [esi+44h],ebp
0x1000895f: add eax,ebp
0x10008961: mov [esi+48h],ecx
0x10008964: mov [esi+4Ch],eax
0x10008967: mov [ebx+8],ecx
0x1000896a: cmp dword ptr [esi+34h],0
0x1000896e: jl short 10008995h
0x10008970: lea edi,[esi+38h]
0x10008973: call dword ptr ds:[100FE5B4h]
0x10008979: push eax
0x1000897a: mov ecx,edi
0x1000897c: call 100222F0h
0x10008981: mov ecx,[esi+8]
0x10008984: push edi
0x10008985: push ecx
0x10008986: call 1002227Eh
0x1000898b: mov ecx,[esi+34h]
0x1000898e: mov edi,eax
0x10008990: call 10008FF0h
0x10008995: pop edi
0x10008996: pop esi
0x10008997: pop ebp
0x10008998: add esp,1Ch
0x1000899b: ret 4
```

`sub_100087c0` is called by sub_100056c0. It delegates to sub_100084a0, sub_10008510, sub_10008ff0 (6 callees).

12.125 sub_100089a0 at 0x100089a0 (77 instructions)

```
0x100089a0: push ecx
0x100089a1: push ebx
0x100089a2: push ebp
0x100089a3: mov ebp,[esp+10h]
0x100089a7: push esi
0x100089a8: push edi
0x100089a9: mov esi,eax
0x100089ab: xor edi,edi
0x100089ad: lea ecx,[ecx]
0x100089b0: mov ecx,[esi+14h]
0x100089b3: test ecx,ecx
0x100089b5: jne short 100089BBh
0x100089b7: xor eax,eax
0x100089b9: jmp short 100089C3h
```

```
0x100089bb: mov eax,[esi+18h]
0x100089be: sub eax,ecx
0x100089c0: sar eax,2
0x100089c3: cmp edi,eax
0x100089c5: jge near ptr 10008A57h
0x100089cb: mov eax,[esi+34h]
0x100089ce: test eax,eax
0x100089d0: jl short 100089D6h
0x100089d2: cmp edi,eax
0x100089d4: je short 10008A03h
0x100089d6: test ecx,ecx
0x100089d8: je short 100089E6h
0x100089da: mov eax,[esi+18h]
0x100089dd: sub eax,ecx
0x100089df: sar eax,2
0x100089e2: cmp edi,eax
0x100089e4: jb short 100089ECh
0x100089e6: call dword ptr ds:[100FE584h]
0x100089ec: mov eax,[esp+1Ch]
0x100089f0: mov ecx,[esi+14h]
0x100089f3: mov ecx,[ecx+edi*4]
0x100089f6: push eax
0x100089f7: mov ebx,ebp
0x100089f9: call 10007DD0h
0x100089fe: add edi,1
0x10008a01: jmp short 100089B0h
0x10008a03: lea edi,[esi+40h]
0x10008a06: push edi
0x10008a07: call dword ptr ds:[100FE6ACh]
0x10008a0d: test eax,eax
0x10008a0f: jne short 10008A57h
0x10008a11: lea eax,[esi+24h]
0x10008a14: test eax,eax
0x10008a16: je short 10008A57h
0x10008a18: cmp dword ptr [eax+4],0
0x10008a1c: je short 10008A57h
0x10008a1e: cmp dword ptr [esp+1Ch],0
0x10008a23: mov ebx,[esi+50h]
0x10008a26: jne short 10008A2Dh
0x10008a28: mov ebx,3
0x10008a2d: test ebp,ebp
0x10008a2f: mov ecx,[edi]
0x10008a31: mov edx,[edi+4]
0x10008a34: jne short 10008A3Ah
0x10008a36: xor eax,eax
0x10008a38: jmp short 10008A3Dh
0x10008a3a: mov eax,[ebp+4]
0x10008a3d: mov esi,[esi+28h]
0x10008a40: push edx
0x10008a41: push ecx
0x10008a42: push eax
0x10008a43: push ebx
0x10008a44: push esi
0x10008a45: call 10021FD2h
0x10008a4a: mov edx,[eax+94h]
0x10008a50: mov ecx,[edx]
0x10008a52: call 10009C50h
0x10008a57: pop edi
0x10008a58: pop esi
0x10008a59: pop ebp
0x10008a5a: pop ebx
0x10008a5b: pop ecx
0x10008a5c: ret 8
```

`sub_100089a0` is called by sub_10005fd0. It delegates to sub_10007dd0, sub_10009c50, j_ORDINAL 1079 (3 callees).

12.126 sub_10008b20 at 0x10008b20 (86 instructions)

```
0x10008b20: push ebp
0x10008b21: mov ebp,esp
0x10008b23: and esp,0FFFFFFF8h
0x10008b26: sub esp,20h
0x10008b29: neg eax
0x10008b2b: sbb eax,eax
0x10008b2d: push ebx
0x10008b2e: and eax,2
0x10008b31: push edi
0x10008b32: mov ebx,eax
0x10008b34: xor edi,edi
0x10008b36: mov ecx,[esi+14h]
0x10008b39: test ecx,ecx
0x10008b3b: jne short 10008B41h
0x10008b3d: xor eax,eax
0x10008b3f: jmp short 10008B49h
0x10008b41: mov eax,[esi+18h]
0x10008b44: sub eax,ecx
0x10008b46: sar eax,2
0x10008b49: cmp edi,eax
0x10008b4b: jge short 10008B99h
0x10008b4d: mov eax,[esi+34h]
0x10008b50: test eax,eax
0x10008b52: jl short 10008B58h
0x10008b54: cmp edi,eax
0x10008b56: je short 10008B89h
0x10008b58: test ecx,ecx
0x10008b5a: je short 10008B68h
0x10008b5c: mov eax,[esi+18h]
0x10008b5f: sub eax,ecx
0x10008b61: sar eax,2
0x10008b64: cmp edi,eax
0x10008b66: jb short 10008B6Eh
0x10008b68: call dword ptr ds:[100FE584h]
0x10008b6e: mov ecx,[esi+14h]
0x10008b71: mov edx,[ecx+edi*4]
0x10008b74: xor eax,eax
0x10008b76: cmp edi,[ebp+8]
0x10008b79: setne al
0x10008b7c: sub eax,1
0x10008b7f: and eax,ebx
0x10008b81: mov [edx+20h],eax
0x10008b84: add edi,1
0x10008b87: jmp short 10008B36h
0x10008b89: xor eax,eax
0x10008b8b: cmp edi,[ebp+8]
0x10008b8e: setne al
0x10008b91: sub eax,1
0x10008b94: and eax,ebx
0x10008b96: mov [esi+50h],eax
0x10008b99: cmp dword ptr [ebp+0Ch],0
0x10008b9d: je short 10008BF6h
0x10008b9f: lea edi,[esp+18h]
0x10008ba3: mov eax,esi
0x10008ba5: call 100086B0h
0x10008baa: mov edx,[esi+4]
0x10008bad: mov eax,[edx+20h]
0x10008bb0: lea ecx,[esp+8]
0x10008bb4: push ecx
0x10008bb5: push eax
0x10008bb6: call dword ptr ds:[100FE674h]
0x10008bbc: mov ecx,[esp+0Ch]
0x10008bc0: mov edx,[esp+8]
0x10008bc4: push ecx
0x10008bc5: push edx
0x10008bc6: mov eax,edi
```

```
0x10008bc8: push eax
0x10008bc9: call dword ptr ds:[100FE6B0h]
0x10008bcf: mov ecx,edi
0x10008bd1: push ecx
0x10008bd2: call dword ptr ds:[100FE050h]
0x10008bd8: mov edx,[esi+4]
0x10008bdb: push 0
0x10008bdd: mov edi,eax
0x10008bdf: mov eax,[edx+20h]
0x10008be2: push edi
0x10008be3: push 85h
0x10008be8: push eax
0x10008be9: call dword ptr ds:[100FE620h]
0x10008bef: push edi
0x10008bf0: call dword ptr ds:[100FE048h]
0x10008bf6: pop edi
0x10008bf7: pop ebx
0x10008bf8: mov esp,ebp
0x10008bfa: pop ebp
0x10008bfb: ret 8
```

`sub_10008b20` is called by sub_10004d40, sub_10009450, sub_10009630, sub_10009740. It delegates to sub_100086b0 (1 callees).

12.127 sub_10008c00 at 0x10008c00 (43 instructions)

```
0x10008c00: push ebp
0x10008c01: push edi
0x10008c02: or ebp,0FFFFFFFFh
0x10008c05: xor edi,edi
0x10008c07: mov ecx,[esi+14h]
0x10008c0a: test ecx,ecx
0x10008c0c: jne short 10008C12h
0x10008c0e: xor eax,eax
0x10008c10: jmp short 10008C1Ah
0x10008c12: mov eax,[esi+18h]
0x10008c15: sub eax,ecx
0x10008c17: sar eax,2
0x10008c1a: cmp edi,eax
0x10008c1c: jge short 10008C5Bh
0x10008c1e: mov eax,[esi+34h]
0x10008c21: test eax,eax
0x10008c23: jl short 10008C29h
0x10008c25: cmp edi,eax
0x10008c27: je short 10008C50h
0x10008c29: test ecx,ecx
0x10008c2b: je short 10008C39h
0x10008c2d: mov eax,[esi+18h]
0x10008c30: sub eax,ecx
0x10008c32: sar eax,2
0x10008c35: cmp edi,eax
0x10008c37: jb short 10008C3Fh
0x10008c39: call dword ptr ds:[100FE584h]
0x10008c3f: mov eax,[esi+14h]
0x10008c42: mov ecx,[eax+edi*4]
0x10008c45: cmp dword ptr [ecx+20h],2
0x10008c49: je short 10008C56h
0x10008c4b: add edi,1
0x10008c4e: jmp short 10008C07h
0x10008c50: cmp dword ptr [esi+50h],2
0x10008c54: jne short 10008C5Bh
0x10008c56: mov eax,edi
0x10008c58: pop edi
0x10008c59: pop ebp
0x10008c5a: ret
0x10008c5b: pop edi
```

```
0x10008c5c: mov eax,ebp
0x10008c5e: pop ebp
0x10008c5f: ret
```

`sub_10008c00` is called by sub_10004d40, sub_10009630, sub_10009740. It calls nothing else.

12.128 sub_10008c60 at 0x10008c60 (43 instructions)

```
0x10008c60: push ebp
0x10008c61: push edi
0x10008c62: or ebp,0FFFFFFFFh
0x10008c65: xor edi,edi
0x10008c67: mov ecx,[esi+14h]
0x10008c6a: test ecx,ecx
0x10008c6c: jne short 10008C72h
0x10008c6e: xor eax,eax
0x10008c70: jmp short 10008C7Ah
0x10008c72: mov eax,[esi+18h]
0x10008c75: sub eax,ecx
0x10008c77: sar eax,2
0x10008c7a: cmp edi,eax
0x10008c7c: jge short 10008CBBh
0x10008c7e: mov eax,[esi+34h]
0x10008c81: test eax,eax
0x10008c83: jl short 10008C89h
0x10008c85: cmp edi,eax
0x10008c87: je short 10008CB0h
0x10008c89: test ecx,ecx
0x10008c8b: je short 10008C99h
0x10008c8d: mov eax,[esi+18h]
0x10008c90: sub eax,ecx
0x10008c92: sar eax,2
0x10008c95: cmp edi,eax
0x10008c97: jb short 10008C9Fh
0x10008c99: call dword ptr ds:[100FE584h]
0x10008c9f: mov eax,[esi+14h]
0x10008ca2: mov ecx,[eax+edi*4]
0x10008ca5: cmp dword ptr [ecx+20h],1
0x10008ca9: je short 10008CB6h
0x10008cab: add edi,1
0x10008cae: jmp short 10008C67h
0x10008cb0: cmp dword ptr [esi+50h],1
0x10008cb4: jne short 10008CBBh
0x10008cb6: mov eax,edi
0x10008cb8: pop edi
0x10008cb9: pop ebp
0x10008cba: ret
0x10008cbb: pop edi
0x10008cbc: mov eax,ebp
0x10008cbe: pop ebp
0x10008cbf: ret
```

`sub_10008c60` is called by sub_10009630, sub_10009740. It calls nothing else.

12.129 sub_10008cc0 at 0x10008cc0 (56 instructions)

```
0x10008cc0: push ebx
0x10008cc1: mov ebx,ds:[100FE560h]
0x10008cc7: push ebp
0x10008cc8: push edi
0x10008cc9: xor edi,edi
0x10008ccb: jmp short 10008CD0h
0x10008cd0: mov ecx,[esi+14h]
0x10008cd3: test ecx,ecx
0x10008cd5: jne short 10008CDBh
```

```
0x10008cd7: xor eax,eax
0x10008cd9: jmp short 10008CE3h
0x10008cdb: mov eax,[esi+18h]
0x10008cde: sub eax,ecx
0x10008ce0: sar eax,2
0x10008ce3: cmp edi,eax
0x10008ce5: jge short 10008D3Ah
0x10008ce7: test ecx,ecx
0x10008ce9: je short 10008CF7h
0x10008ceb: mov eax,[esi+18h]
0x10008cee: sub eax,ecx
0x10008cf0: sar eax,2
0x10008cf3: cmp edi,eax
0x10008cf5: jb short 10008CFDh
0x10008cf7: call dword ptr ds:[100FE584h]
0x10008cfd: mov eax,[esi+14h]
0x10008d00: mov ecx,[eax+edi*4]
0x10008d03: movzx eax,word ptr [ecx+4]
0x10008d07: movzx edx,ax
0x10008d0a: push edx
0x10008d0b: call ebx
0x10008d0d: mov ebp,eax
0x10008d0f: movzx eax,word ptr [esp+14h]
0x10008d14: push eax
0x10008d15: call ebx
0x10008d17: add esp,8
0x10008d1a: cmp ebp,eax
0x10008d1c: je short 10008D23h
0x10008d1e: add edi,1
0x10008d21: jmp short 10008CD0h
0x10008d23: test edi,edi
0x10008d25: jl short 10008D32h
0x10008d27: mov eax,[esi+34h]
0x10008d2a: test eax,eax
0x10008d2c: jl short 10008D32h
0x10008d2e: cmp edi,eax
0x10008d30: jge short 10008D3Dh
0x10008d32: mov eax,edi
0x10008d34: pop edi
0x10008d35: pop ebp
0x10008d36: pop ebx
0x10008d37: ret 4
0x10008d3a: or eax,0FFFFFFFFh
0x10008d3d: pop edi
0x10008d3e: pop ebp
0x10008d3f: pop ebx
0x10008d40: ret 4
```

`sub_10008cc0` is called by sub_100051b0. It calls nothing else.

12.130 sub_10008df0 at 0x10008df0 (162 instructions)

```
0x10008df0: push 0FFFFFFFFh
0x10008df2: push 100FCDF1h
0x10008df7: mov eax,fs:[0]
0x10008dfd: push eax
0x10008dfe: sub esp,28h
0x10008e01: push ebp
0x10008e02: push esi
0x10008e03: push edi
0x10008e04: mov eax,ds:[10136220h]
0x10008e09: xor eax,esp
0x10008e0b: push eax
0x10008e0c: lea eax,[esp+38h]
0x10008e10: mov fs:[0],eax
0x10008e16: mov esi,ecx
```

```
0x10008e18: xor ebp,ebp
0x10008e1a: cmp esi,ebp
0x10008e1c: je short 10008E3Ch
0x10008e1e: push esi
0x10008e1f: call dword ptr ds:[100FE63Ch]
0x10008e25: test eax,eax
0x10008e27: jne short 10008E3Ch
0x10008e29: mov ecx,[esp+38h]
0x10008e2d: mov fs:[0],ecx
0x10008e34: pop ecx
0x10008e35: pop edi
0x10008e36: pop esi
0x10008e37: pop ebp
0x10008e38: add esp,34h
0x10008e3b: ret
0x10008e3c: mov eax,[ebx+8]
0x10008e3f: cmp eax,esi
0x10008e41: je short 10008E59h
0x10008e43: cmp eax,ebp
0x10008e45: je short 10008E56h
0x10008e47: cmp [ebx+0Ch],ebp
0x10008e4a: jne short 10008E53h
0x10008e4c: push eax
0x10008e4d: call dword ptr ds:[100FE5BCh]
0x10008e53: mov [ebx+8],ebp
0x10008e56: mov [ebx+8],esi
0x10008e59: mov eax,[ebx+8]
0x10008e5c: cmp eax,ebp
0x10008e5e: jne short 10008E68h
0x10008e60: xor esi,esi
0x10008e62: mov [esp+14h],esi
0x10008e66: jmp short 10008E74h
0x10008e68: push eax
0x10008e69: call 1002227Eh
0x10008e6e: mov [esp+14h],eax
0x10008e72: mov esi,eax
0x10008e74: mov eax,ebx
0x10008e76: call 10008570h
0x10008e7b: lea eax,[ebx+40h]
0x10008e7e: push eax
0x10008e7f: mov dword ptr [ebx+34h],0FFFFFFFFh
0x10008e86: call dword ptr ds:[100FE5C8h]
0x10008e8c: mov edi,ebx
0x10008e8e: mov [ebx+50h],ebp
0x10008e91: call 100091B0h
0x10008e96: cmp esi,ebp
0x10008e98: je near ptr 10008FCCh
0x10008e9e: lea ecx,[esp+18h]
0x10008ea2: call dword ptr ds:[100FE368h]
0x10008ea8: lea ecx,[esp+28h]
0x10008eac: mov [esp+40h],ebp
0x10008eb0: call 1002202Ch
0x10008eb5: push ebp
0x10008eb6: mov byte ptr [esp+44h],1
0x10008ebb: call dword ptr ds:[100FE060h]
0x10008ec1: push eax
0x10008ec2: lea ecx,[esp+2Ch]
0x10008ec6: call 10022038h
0x10008ecb: mov ecx,[esi+4]
0x10008ece: xor edi,edi
0x10008ed0: push ecx
0x10008ed1: mov [esp+20h],edi
0x10008ed5: call dword ptr ds:[100FE684h]
0x10008edb: test eax,eax
0x10008edd: jle near ptr 10008FB4h
0x10008ee3: lea ebp,[ebx+10h]
0x10008ee6: jmp short 10008EF4h
```

```
0x10008ef0: mov esi,[esp+14h]
0x10008ef4: lea ecx,[esp+18h]
0x10008ef8: call dword ptr ds:[100FE214h]
0x10008efe: push 400h
0x10008f03: lea edx,[esp+1Ch]
0x10008f07: push edx
0x10008f08: push edi
0x10008f09: mov ecx,esi
0x10008f0b: call 100222F6h
0x10008f10: push 34h
0x10008f12: call 10021FDEh
0x10008f17: add esp,4
0x10008f1a: test eax,eax
0x10008f1c: je short 10008F34h
0x10008f1e: lea ecx,[ebx+54h]
0x10008f21: lea edi,[ebx+5Ch]
0x10008f24: push ecx
0x10008f25: mov esi,eax
0x10008f27: call 10007C80h
0x10008f2c: mov edi,[esp+1Ch]
0x10008f30: mov esi,eax
0x10008f32: jmp short 10008F36h
0x10008f34: xor esi,esi
0x10008f36: lea eax,[esp+18h]
0x10008f3a: lea ecx,[esp+28h]
0x10008f3e: mov [esp+1Ch],esi
0x10008f42: call 10007D00h
0x10008f47: mov ecx,[ebp+4]
0x10008f4a: test ecx,ecx
0x10008f4c: jne short 10008F52h
0x10008f4e: xor edx,edx
0x10008f50: jmp short 10008F5Ah
0x10008f52: mov edx,[ebp+8]
0x10008f55: sub edx,ecx
0x10008f57: sar edx,2
0x10008f5a: test ecx,ecx
0x10008f5c: je short 10008F77h
0x10008f5e: mov eax,[ebp+0Ch]
0x10008f61: sub eax,ecx
0x10008f63: sar eax,2
0x10008f66: cmp edx,eax
0x10008f68: jae short 10008F77h
0x10008f6a: mov eax,[ebp+8]
0x10008f6d: mov [eax],esi
0x10008f6f: add eax,4
0x10008f72: mov [ebp+8],eax
0x10008f75: jmp short 10008F97h
0x10008f77: mov esi,[ebp+8]
0x10008f7a: cmp ecx,esi
0x10008f7c: jbe short 10008F84h
0x10008f7e: call dword ptr ds:[100FE584h]
0x10008f84: push esi
0x10008f85: push ebp
0x10008f86: lea edx,[esp+24h]
0x10008f8a: push edx
0x10008f8b: lea eax,[esp+2Ch]
0x10008f8f: push eax
0x10008f90: mov eax,ebp
0x10008f92: call 100098C0h
0x10008f97: mov ecx,[esp+14h]
0x10008f9b: mov edx,[ecx+4]
0x10008f9e: add edi,1
0x10008fa1: push edx
0x10008fa2: mov [esp+20h],edi
0x10008fa6: call dword ptr ds:[100FE684h]
0x10008fac: cmp edi,eax
0x10008fae: jl near ptr 10008EF0h
```

```
0x10008fb4: lea ecx,[esp+28h]
0x10008fb8: mov byte ptr [esp+40h],0
0x10008fbd: call 10022032h
0x10008fc2: lea ecx,[esp+18h]
0x10008fc6: call dword ptr ds:[100FE374h]
0x10008fcc: mov eax,1
0x10008fd1: mov ecx,[esp+38h]
0x10008fd5: mov fs:[0],ecx
0x10008fdc: pop ecx
0x10008fdd: pop edi
0x10008fde: pop esi
0x10008fdf: pop ebp
0x10008fe0: add esp,34h
0x10008fe3: ret
```

`sub_10008df0` is called by sub_10003b30. It delegates to sub_10007c80, sub_10007d00, sub_10008570 (11 callees).

12.131 sub_10008ff0 at 0x10008ff0 (144 instructions)

```
0x10008ff0: push 0FFFFFFFFh
0x10008ff2: push 100FCA89h
0x10008ff7: mov eax,fs:[0]
0x10008ffd: push eax
0x10008ffe: sub esp,3Ch
0x10009001: push ebx
0x10009002: push ebp
0x10009003: push esi
0x10009004: mov eax,ds:[10136220h]
0x10009009: xor eax,esp
0x1000900b: push eax
0x1000900c: lea eax,[esp+4Ch]
0x10009010: mov fs:[0],eax
0x10009016: mov esi,ecx
0x10009018: lea ecx,[esp+14h]
0x1000901c: call dword ptr ds:[100FE368h]
0x10009022: mov dword ptr [esp+54h],0
0x1000902a: mov eax,[edi+4]
0x1000902d: push eax
0x1000902e: call dword ptr ds:[100FE684h]
0x10009034: cmp esi,eax
0x10009036: jae near ptr 1000915Ch
0x1000903c: mov ecx,[edi+4]
0x1000903f: push 400h
0x10009044: push esi
0x10009045: push ecx
0x10009046: call dword ptr ds:[100FE680h]
0x1000904c: mov ebx,eax
0x1000904e: test bl,10h
0x10009051: je short 100090A7h
0x10009053: push 400h
0x10009058: lea edx,[esp+18h]
0x1000905c: push edx
0x1000905d: push esi
0x1000905e: mov ecx,edi
0x10009060: and ebx,0FFh
0x10009066: call 100222F6h
0x1000906b: mov eax,[edi+4]
0x1000906e: mov ebp,[esp+14h]
0x10009072: push esi
0x10009073: push eax
0x10009074: call dword ptr ds:[100FE5C0h]
0x1000907a: push eax
0x1000907b: call 1002227Eh
0x10009080: test eax,eax
0x10009082: jne short 10009094h
0x10009084: mov ecx,[esp+5Ch]
```

```
0x10009088: mov edx,[ecx+4]
0x1000908b: push ebp
0x1000908c: push eax
0x1000908d: push ebx
0x1000908e: push edx
0x1000908f: jmp near ptr 1000913Dh
0x10009094: mov eax,[eax+4]
0x10009097: mov ecx,[esp+5Ch]
0x1000909b: mov edx,[ecx+4]
0x1000909e: push ebp
0x1000909f: push eax
0x100090a0: push ebx
0x100090a1: push edx
0x100090a2: jmp near ptr 1000913Dh
0x100090a7: test ebx,800h
0x100090ad: je short 100090BAh
0x100090af: push 0
0x100090b1: push 0
0x100090b3: push 800h
0x100090b8: jmp short 10009135h
0x100090ba: test ebx,100h
0x100090c0: je short 10009111h
0x100090c2: push 30h
0x100090c4: lea edx,[esp+1Ch]
0x100090c8: push 0
0x100090ca: push edx
0x100090cb: call 100228C0h
0x100090d0: mov ecx,[edi+4]
0x100090d3: add esp,0Ch
0x100090d6: lea eax,[esp+18h]
0x100090da: push eax
0x100090db: push 1
0x100090dd: push esi
0x100090de: push ecx
0x100090df: mov dword ptr [esp+28h],30h
0x100090e7: mov dword ptr [esp+2Ch],20h
0x100090ef: call dword ptr ds:[100FE5DCh]
0x100090f5: mov edx,[edi+4]
0x100090f8: push esi
0x100090f9: push edx
0x100090fa: call dword ptr ds:[100FE5D8h]
0x10009100: mov ecx,[esp+38h]
0x10009104: mov edx,[esp+5Ch]
0x10009108: push ecx
0x10009109: push eax
0x1000910a: mov eax,[edx+4]
0x1000910d: push ebx
0x1000910e: push eax
0x1000910f: jmp short 1000913Dh
0x10009111: push 400h
0x10009116: lea ecx,[esp+18h]
0x1000911a: push ecx
0x1000911b: push esi
0x1000911c: mov ecx,edi
0x1000911e: call 100222F6h
0x10009123: mov edx,[edi+4]
0x10009126: mov ebp,[esp+14h]
0x1000912a: push esi
0x1000912b: push edx
0x1000912c: call dword ptr ds:[100FE5D8h]
0x10009132: push ebp
0x10009133: push eax
0x10009134: push ebx
0x10009135: mov eax,[esp+68h]
0x10009139: mov ecx,[eax+4]
0x1000913c: push ecx
0x1000913d: call dword ptr ds:[100FE5B0h]
```

```
0x10009143: test eax,eax
0x10009145: je short 10009180h
0x10009147: mov edx,[edi+4]
0x1000914a: push edx
0x1000914b: add esi,1
0x1000914e: call dword ptr ds:[100FE684h]
0x10009154: cmp esi,eax
0x10009156: jb near ptr 1000903Ch
0x1000915c: lea ecx,[esp+14h]
0x10009160: call dword ptr ds:[100FE374h]
0x10009166: mov eax,1
0x1000916b: mov ecx,[esp+4Ch]
0x1000916f: mov fs:[0],ecx
0x10009176: pop ecx
0x10009177: pop esi
0x10009178: pop ebp
0x10009179: pop ebx
0x1000917a: add esp,48h
0x1000917d: ret 4
0x10009180: lea ecx,[esp+14h]
0x10009184: call dword ptr ds:[100FE374h]
0x1000918a: xor eax,eax
0x1000918c: mov ecx,[esp+4Ch]
0x10009190: mov fs:[0],ecx
0x10009197: pop ecx
0x10009198: pop esi
0x10009199: pop ebp
0x1000919a: pop ebx
0x1000919b: add esp,48h
0x1000919e: ret 4
```

`sub_10008ff0` is called by sub_100087c0. It delegates to j_ORDINAL 2365, j_ORDINAL 2932, j_memset (3 callees).

12.132 sub_100091b0 at 0x100091b0 (30 instructions)

```
0x100091b0: push ebp
0x100091b1: lea ebp,[edi+38h]
0x100091b4: test ebp,ebp
0x100091b6: je short 100091F2h
0x100091b8: cmp dword ptr [ebp+4],0
0x100091bc: je short 100091F2h
0x100091be: mov eax,[edi+3Ch]
0x100091c1: push esi
0x100091c2: push eax
0x100091c3: call dword ptr ds:[100FE684h]
0x100091c9: mov esi,eax
0x100091cb: sub esi,1
0x100091ce: js short 100091E9h
0x100091d0: push ebx
0x100091d1: mov ebx,ds:[100FE5D4h]
0x100091d7: mov ecx,[edi+3Ch]
0x100091da: push 400h
0x100091df: push esi
0x100091e0: push ecx
0x100091e1: call ebx
0x100091e3: sub esi,1
0x100091e6: jns short 100091D7h
0x100091e8: pop ebx
0x100091e9: pop esi
0x100091ea: mov ecx,ebp
0x100091ec: pop ebp
0x100091ed: jmp near ptr 100222C0h
0x100091f2: pop ebp
0x100091f3: ret
0x100222c0: jmp dword ptr ds:[100FE1F0h]
```

`sub_100091b0` is called by sub_100056c0, sub_100087c0, sub_10008df0. It calls nothing else.

12.133 sub_10009450 at 0x10009450 (154 instructions)

```
0x10009450: push 0FFFFFFFFh
0x10009452: push 100FD198h
0x10009457: mov eax,fs:[0]
0x1000945d: push eax
0x1000945e: sub esp,38h
0x10009461: push ebx
0x10009462: push ebp
0x10009463: push esi
0x10009464: push edi
0x10009465: mov eax,ds:[10136220h]
0x1000946a: xor eax,esp
0x1000946c: push eax
0x1000946d: lea eax,[esp+4Ch]
0x10009471: mov fs:[0],eax
0x10009477: mov ebx,ecx
0x10009479: mov esi,edx
0x1000947b: xor edi,edi
0x1000947d: cmp [esi+68h],edi
0x10009480: je short 10009489h
0x10009482: xor eax,eax
0x10009484: jmp near ptr 100095EAh
0x10009489: or ebp,0FFFFFFFFh
0x1000948c: push 58h
0x1000948e: mov dword ptr [esi+68h],1
0x10009495: mov [esi+70h],edi
0x10009498: mov [esi+74h],ebp
0x1000949b: call 10021FDEh
0x100094a0: add esp,4
0x100094a3: mov [esp+18h],eax
0x100094a7: cmp eax,edi
0x100094a9: mov [esp+54h],edi
0x100094ad: je short 100094B8h
0x100094af: push esi
0x100094b0: push eax
0x100094b1: call 100024E0h
0x100094b6: jmp short 100094BAh
0x100094b8: xor eax,eax
0x100094ba: mov [esp+54h],ebp
0x100094be: mov edx,[esi+4]
0x100094c1: cmp edx,edi
0x100094c3: mov [esi+7Ch],eax
0x100094c6: je short 100094FCh
0x100094c8: cmp [edx+20h],edi
0x100094cb: je short 100094FCh
0x100094cd: push edi
0x100094ce: push edi
0x100094cf: push edx
0x100094d0: lea ecx,[esp+28h]
0x100094d4: push ecx
0x100094d5: push 80000000h
0x100094da: push edi
0x100094db: push 1012F38Ch
0x100094e0: push edi
0x100094e1: mov ecx,eax
0x100094e3: mov [esp+3Ch],edi
0x100094e7: mov [esp+40h],edi
0x100094eb: mov [esp+44h],edi
0x100094ef: mov [esp+48h],edi
0x100094f3: call 10022050h
0x100094f8: cmp eax,edi
0x100094fa: jne short 10009513h
0x100094fc: mov ecx,[esi+7Ch]
0x100094ff: cmp ecx,edi
```

```
0x10009501: je short 1000950Ch
0x10009503: mov edx,[ecx]
0x10009505: mov eax,[edx+4]
0x10009508: push 1
0x1000950a: call eax
0x1000950c: xor eax,eax
0x1000950e: jmp near ptr 100095E7h
0x10009513: mov eax,[esi+7Ch]
0x10009516: mov ecx,[eax+20h]
0x10009519: push ecx
0x1000951a: call dword ptr ds:[100FE64Ch]
0x10009520: push eax
0x10009521: call 1002225Ah
0x10009526: mov ecx,[esi+7Ch]
0x10009529: push 1
0x1000952b: push edi
0x1000952c: push edi
0x1000952d: push edi
0x1000952e: push edi
0x1000952f: call 100222D8h
0x10009534: mov ecx,[esi+7Ch]
0x10009537: push 8
0x10009539: call 100222C6h
0x1000953e: mov ecx,[esi+7Ch]
0x10009541: call 100222CCh
0x10009546: xor edx,edx
0x10009548: cmp ebx,edi
0x1000954a: setl dl
0x1000954d: push 1
0x1000954f: mov eax,1
0x10009554: sub edx,1
0x10009557: and edx,ebx
0x10009559: push edx
0x1000955a: call 10008B20h
0x1000955f: cmp ebx,edi
0x10009561: jl short 10009588h
0x10009563: movzx eax,word ptr [esp+5Ch]
0x10009568: mov edx,[esi+7Ch]
0x1000956b: movzx ecx,bx
0x1000956e: shl eax,10h
0x10009571: or eax,ecx
0x10009573: push eax
0x10009574: mov eax,[edx+20h]
0x10009577: push 903Ch
0x1000957c: push 111h
0x10009581: push eax
0x10009582: call dword ptr ds:[100FE608h]
0x10009588: mov ebp,ds:[100FE5D0h]
0x1000958e: mov ebx,ds:[100FE5CCh]
0x10009594: mov dword ptr [esi+6Ch],1
0x1000959b: jmp short 100095A0h
0x100095a0: push edi
0x100095a1: push edi
0x100095a2: push edi
0x100095a3: lea ecx,[esp+38h]
0x100095a7: push ecx
0x100095a8: call ebp
0x100095aa: test eax,eax
0x100095ac: je short 100095BAh
0x100095ae: lea edx,[esp+2Ch]
0x100095b2: push edx
0x100095b3: call ebx
0x100095b5: cmp [esi+6Ch],edi
0x100095b8: jne short 100095A0h
0x100095ba: mov ecx,[esi+7Ch]
0x100095bd: mov [esi+6Ch],edi
0x100095c0: mov eax,[ecx]
```

```
0x100095c2: mov edx,[eax+68h]
0x100095c5: call edx
0x100095c7: mov ecx,[esi+7Ch]
0x100095ca: cmp ecx,edi
0x100095cc: je short 100095D7h
0x100095ce: mov eax,[ecx]
0x100095d0: mov edx,[eax+4]
0x100095d3: push 1
0x100095d5: call edx
0x100095d7: push 1
0x100095d9: push 0FFFFFFFFh
0x100095db: xor eax,eax
0x100095dd: call 10008B20h
0x100095e2: mov eax,1
0x100095e7: mov [esi+68h],edi
0x100095ea: mov ecx,[esp+4Ch]
0x100095ee: mov fs:[0],ecx
0x100095f5: pop ecx
0x100095f6: pop edi
0x100095f7: pop esi
0x100095f8: pop ebp
0x100095f9: pop ebx
0x100095fa: add esp,44h
0x100095fd: ret 4
```

`sub_10009450` is called by sub_100051b0. It works with the text "_MenuHost_A$". It accesses memory-mapped peripheral registers at 0x80000000. It delegates to sub_100024e0, sub_10008b20, j_ORDINAL 762 (8 callees).

12.134 sub_10009630 at 0x10009630 (52 instructions)

```
0x10009630: push esi
0x10009631: mov esi,eax
0x10009633: cmp dword ptr [esi+68h],0
0x10009637: je near ptr 100096BFh
0x1000963d: cmp dword ptr [esi+70h],0
0x10009641: je short 1000964Ah
0x10009643: call 10008C60h
0x10009648: jmp short 1000964Fh
0x1000964a: call 10008C00h
0x1000964f: mov ecx,[esi+34h]
0x10009652: test ecx,ecx
0x10009654: jge short 1000966Ch
0x10009656: mov edx,[esi+14h]
0x10009659: test edx,edx
0x1000965b: jne short 10009661h
0x1000965d: xor ecx,ecx
0x1000965f: jmp short 10009669h
0x10009661: mov ecx,[esi+18h]
0x10009664: sub ecx,edx
0x10009666: sar ecx,2
0x10009669: add ecx,0FFFFFFFFh
0x1000966c: cmp dword ptr [esp+8],0
0x10009671: je short 1000967Ch
0x10009673: sub eax,1
0x10009676: jns short 10009685h
0x10009678: mov eax,ecx
0x1000967a: jmp short 10009685h
0x1000967c: add eax,1
0x1000967f: cmp eax,ecx
0x10009681: jle short 10009685h
0x10009683: xor eax,eax
0x10009685: cmp dword ptr [esi+70h],0
0x10009689: je short 100096B2h
0x1000968b: cmp dword ptr [esi+68h],0
0x1000968f: je short 1000969Bh
0x10009691: mov [esi+74h],eax
```

```
0x10009694: mov dword ptr [esi+78h],1
0x1000969b: mov eax,[esi+4]
0x1000969e: mov ecx,[eax+20h]
0x100096a1: push 0
0x100096a3: push 0
0x100096a5: push 1Fh
0x100096a7: push ecx
0x100096a8: call dword ptr ds:[100FE620h]
0x100096ae: pop esi
0x100096af: ret 4
0x100096b2: push 1
0x100096b4: push eax
0x100096b5: mov eax,1
0x100096ba: call 10008B20h
0x100096bf: pop esi
0x100096c0: ret 4
```

`sub_10009630` is called by sub_10007040. It delegates to sub_10008b20, sub_10008c00, sub_10008c60 (3 callees).

12.135 sub_10009740 at 0x10009740 (68 instructions)

```
0x10009740: sub esp,24h
0x10009743: push esi
0x10009744: mov esi,ecx
0x10009746: cmp dword ptr [esi+80h],0
0x1000974d: push edi
0x1000974e: mov edi,eax
0x10009750: je short 10009762h
0x10009752: mov dword ptr [esi+80h],0
0x1000975c: pop edi
0x1000975d: pop esi
0x1000975e: add esp,24h
0x10009761: ret
0x10009762: cmp dword ptr [esi+68h],0
0x10009766: je near ptr 100097FFh
0x1000976c: mov ecx,[esi+4]
0x1000976f: mov edx,[ecx+20h]
0x10009772: lea eax,[esp+18h]
0x10009776: push eax
0x10009777: push edx
0x10009778: call dword ptr ds:[100FE674h]
0x1000977e: mov eax,[edi]
0x10009780: sub eax,[esp+18h]
0x10009784: mov ecx,[edi+4]
0x10009787: sub ecx,[esp+1Ch]
0x1000978b: mov [esp+10h],eax
0x1000978f: lea eax,[esp+0Ch]
0x10009793: mov [esp+14h],ecx
0x10009797: push eax
0x10009798: lea ecx,[esp+14h]
0x1000979c: push ecx
0x1000979d: mov eax,esi
0x1000979f: call 100085E0h
0x100097a4: cmp dword ptr [esp+0Ch],5
0x100097a9: mov edi,eax
0x100097ab: jne short 100097FFh
0x100097ad: test edi,edi
0x100097af: jl short 100097FFh
0x100097b1: call 10008C00h
0x100097b6: test eax,eax
0x100097b8: jge short 100097BFh
0x100097ba: call 10008C60h
0x100097bf: cmp edi,eax
0x100097c1: je short 100097FFh
0x100097c3: cmp dword ptr [esi+70h],0
0x100097c7: je short 100097F2h
```

```
0x100097c9: cmp dword ptr [esi+68h],0
0x100097cd: je short 100097D9h
0x100097cf: mov [esi+74h],edi
0x100097d2: mov dword ptr [esi+78h],0
0x100097d9: mov edx,[esi+4]
0x100097dc: mov eax,[edx+20h]
0x100097df: push 0
0x100097e1: push 0
0x100097e3: push 1Fh
0x100097e5: push eax
0x100097e6: call dword ptr ds:[100FE620h]
0x100097ec: pop edi
0x100097ed: pop esi
0x100097ee: add esp,24h
0x100097f1: ret
0x100097f2: push 1
0x100097f4: push edi
0x100097f5: mov eax,1
0x100097fa: call 10008B20h
0x100097ff: pop edi
0x10009800: pop esi
0x10009801: add esp,24h
0x10009804: ret
```

`sub_10009740` is called by sub_10006f30. It delegates to sub_100085e0, sub_10008b20, sub_10008c00 (4 callees).

12.136 sub_100098c0 at 0x100098c0 (54 instructions)

```
0x100098c0: push ecx
0x100098c1: push ebx
0x100098c2: push ebp
0x100098c3: mov ebp,[esp+18h]
0x100098c7: push esi
0x100098c8: mov esi,eax
0x100098ca: push edi
0x100098cb: mov edi,[esi+4]
0x100098ce: test edi,edi
0x100098d0: je short 100098DEh
0x100098d2: mov eax,[esi+8]
0x100098d5: mov ecx,eax
0x100098d7: sub ecx,edi
0x100098d9: sar ecx,2
0x100098dc: jne short 100098E2h
0x100098de: xor ebx,ebx
0x100098e0: jmp short 10009903h
0x100098e2: cmp edi,eax
0x100098e4: jbe short 100098ECh
0x100098e6: call dword ptr ds:[100FE584h]
0x100098ec: test ebp,ebp
0x100098ee: je short 100098F4h
0x100098f0: cmp ebp,esi
0x100098f2: je short 100098FAh
0x100098f4: call dword ptr ds:[100FE584h]
0x100098fa: mov ebx,[esp+24h]
0x100098fe: sub ebx,edi
0x10009900: sar ebx,2
0x10009903: mov edx,[esp+24h]
0x10009907: mov eax,[esp+1Ch]
0x1000990b: push edx
0x1000990c: push ebp
0x1000990d: mov ecx,esi
0x1000990f: call 10009950h
0x10009914: mov edi,[esi+4]
0x10009917: cmp edi,[esi+8]
0x1000991a: jbe short 10009922h
0x1000991c: call dword ptr ds:[100FE584h]
```

```
0x10009922: mov [esp+24h],edi
0x10009926: lea edi,[edi+ebx*4]
0x10009929: cmp edi,[esi+8]
0x1000992c: ja short 10009933h
0x1000992e: cmp edi,[esi+4]
0x10009931: jae short 10009939h
0x10009933: call dword ptr ds:[100FE584h]
0x10009939: mov eax,[esp+18h]
0x1000993d: mov [eax+4],edi
0x10009940: pop edi
0x10009941: mov [eax],esi
0x10009943: pop esi
0x10009944: pop ebp
0x10009945: pop ebx
0x10009946: pop ecx
0x10009947: ret 10h
```

`sub_100098c0` is called by sub_10008df0, sub_1000b9e0, sub_1000d550. It delegates to sub_10009950 (1 callees).

12.137 sub_10009950 at 0x10009950 (190 instructions)

```
0x10009950: sub esp,8
0x10009953: push ebx
0x10009954: mov ebx,ecx
0x10009956: mov ecx,[eax]
0x10009958: mov eax,[ebx+4]
0x1000995b: test eax,eax
0x1000995d: push ebp
0x1000995e: push esi
0x1000995f: push edi
0x10009960: mov [esp+14h],ecx
0x10009964: jne short 1000996Ah
0x10009966: xor ecx,ecx
0x10009968: jmp short 10009972h
0x1000996a: mov ecx,[ebx+0Ch]
0x1000996d: sub ecx,eax
0x1000996f: sar ecx,2
0x10009972: test eax,eax
0x10009974: jne short 1000997Ah
0x10009976: xor edx,edx
0x10009978: jmp short 10009982h
0x1000997a: mov edx,[ebx+8]
0x1000997d: sub edx,eax
0x1000997f: sar edx,2
0x10009982: mov esi,3FFFFFFFh
0x10009987: sub esi,edx
0x10009989: cmp esi,1
0x1000998c: jae short 10009993h
0x1000998e: call 10009B40h
0x10009993: test eax,eax
0x10009995: jne short 1000999Bh
0x10009997: xor edx,edx
0x10009999: jmp short 100099A3h
0x1000999b: mov edx,[ebx+8]
0x1000999e: sub edx,eax
0x100099a0: sar edx,2
0x100099a3: add edx,1
0x100099a6: cmp ecx,edx
0x100099a8: jae near ptr 10009A8Eh
0x100099ae: mov edx,ecx
0x100099b0: shr edx,1
0x100099b2: mov esi,3FFFFFFFh
0x100099b7: sub esi,edx
0x100099b9: cmp esi,ecx
0x100099bb: jae short 100099CBh
0x100099bd: mov dword ptr [esp+10h],0
```

```
0x100099c5: mov ecx,[esp+10h]
0x100099c9: jmp short 100099D1h
0x100099cb: add ecx,edx
0x100099cd: mov [esp+10h],ecx
0x100099d1: test eax,eax
0x100099d3: jne short 100099D9h
0x100099d5: xor edx,edx
0x100099d7: jmp short 100099E1h
0x100099d9: mov edx,[ebx+8]
0x100099dc: sub edx,eax
0x100099de: sar edx,2
0x100099e1: add edx,1
0x100099e4: cmp ecx,edx
0x100099e6: jae short 100099FFh
0x100099e8: test eax,eax
0x100099ea: jne short 100099F0h
0x100099ec: xor ecx,ecx
0x100099ee: jmp short 100099F8h
0x100099f0: mov ecx,[ebx+8]
0x100099f3: sub ecx,eax
0x100099f5: sar ecx,2
0x100099f8: add ecx,1
0x100099fb: mov [esp+10h],ecx
0x100099ff: call 10009BB0h
0x10009a04: mov ebp,[esp+20h]
0x10009a08: mov ecx,[ebx+4]
0x10009a0b: mov esi,eax
0x10009a0d: mov eax,ebp
0x10009a0f: sub eax,ecx
0x10009a11: sar eax,2
0x10009a14: lea edi,[eax*4]
0x10009a1b: je short 10009A2Ah
0x10009a1d: push edi
0x10009a1e: push ecx
0x10009a1f: push edi
0x10009a20: push esi
0x10009a21: call dword ptr ds:[100FE57Ch]
0x10009a27: add esp,10h
0x10009a2a: mov edx,[esp+14h]
0x10009a2e: mov [edi+esi],edx
0x10009a31: mov eax,[ebx+8]
0x10009a34: sub eax,ebp
0x10009a36: sar eax,2
0x10009a39: lea ecx,[edi+esi+4]
0x10009a3d: je short 10009A50h
0x10009a3f: add eax,eax
0x10009a41: add eax,eax
0x10009a43: push eax
0x10009a44: push ebp
0x10009a45: push eax
0x10009a46: push ecx
0x10009a47: call dword ptr ds:[100FE57Ch]
0x10009a4d: add esp,10h
0x10009a50: mov eax,[ebx+4]
0x10009a53: test eax,eax
0x10009a55: jne short 10009A5Bh
0x10009a57: xor edi,edi
0x10009a59: jmp short 10009A63h
0x10009a5b: mov edi,[ebx+8]
0x10009a5e: sub edi,eax
0x10009a60: sar edi,2
0x10009a63: test eax,eax
0x10009a65: je short 10009A70h
0x10009a67: push eax
0x10009a68: call 10021FCCh
0x10009a6d: add esp,4
0x10009a70: mov eax,[esp+10h]
```

```
0x10009a74: lea ecx,[esi+eax*4]
0x10009a77: lea edx,[esi+edi*4+4]
0x10009a7b: mov [ebx+0Ch],ecx
0x10009a7e: mov [ebx+8],edx
0x10009a81: mov [ebx+4],esi
0x10009a84: pop edi
0x10009a85: pop esi
0x10009a86: pop ebp
0x10009a87: pop ebx
0x10009a88: add esp,8
0x10009a8b: ret 8
0x10009a8e: mov esi,[ebx+8]
0x10009a91: mov ebp,[esp+20h]
0x10009a95: mov eax,esi
0x10009a97: sub eax,ebp
0x10009a99: sar eax,2
0x10009a9c: cmp eax,1
0x10009a9f: mov eax,esi
0x10009aa1: jae short 10009AEEh
0x10009aa3: lea ecx,[ebp+4]
0x10009aa6: mov edx,ebp
0x10009aa8: call 100031F0h
0x10009aad: mov edi,[ebx+8]
0x10009ab0: mov ecx,edi
0x10009ab2: sub ecx,ebp
0x10009ab4: sar ecx,2
0x10009ab7: mov esi,1
0x10009abc: sub esi,ecx
0x10009abe: lea edx,[esp+14h]
0x10009ac2: call 10002EB0h
0x10009ac7: add dword ptr [ebx+8],4
0x10009acb: mov ebx,[ebx+8]
0x10009ace: lea ecx,[ebx-4]
0x10009ad1: cmp ebp,ecx
0x10009ad3: mov eax,ebp
0x10009ad5: je short 10009B31h
0x10009ad7: mov edx,[esp+14h]
0x10009adb: mov [eax],edx
0x10009add: add eax,4
0x10009ae0: cmp eax,ecx
0x10009ae2: jne short 10009AD7h
0x10009ae4: pop edi
0x10009ae5: pop esi
0x10009ae6: pop ebp
0x10009ae7: pop ebx
0x10009ae8: add esp,8
0x10009aeb: ret 8
0x10009aee: lea edi,[esi-4]
0x10009af1: mov ecx,esi
0x10009af3: mov edx,edi
0x10009af5: call 100031F0h
0x10009afa: sub edi,ebp
0x10009afc: sar edi,2
0x10009aff: test edi,edi
0x10009b01: mov [ebx+8],eax
0x10009b04: jle short 10009B1Ch
0x10009b06: lea eax,[edi*4]
0x10009b0d: push eax
0x10009b0e: push ebp
0x10009b0f: push eax
0x10009b10: sub esi,eax
0x10009b12: push esi
0x10009b13: call dword ptr ds:[100FE57Ch]
0x10009b19: add esp,10h
0x10009b1c: lea eax,[ebp+4]
0x10009b1f: cmp ebp,eax
0x10009b21: je short 10009B31h
```

```
0x10009b23: mov ecx,[esp+14h]
0x10009b27: mov [ebp],ecx
0x10009b2a: add ebp,4
0x10009b2d: cmp ebp,eax
0x10009b2f: jne short 10009B23h
0x10009b31: pop edi
0x10009b32: pop esi
0x10009b33: pop ebp
0x10009b34: pop ebx
0x10009b35: add esp,8
0x10009b38: ret 8
```

`sub_10009950` is called by sub_100098c0. It delegates to sub_10002eb0, sub_100031f0, sub_10009b40 (5 callees).

12.138 sub_10009b40 at 0x10009b40 (27 instructions)

```
0x10009b40: push ebp
0x10009b41: mov ebp,esp
0x10009b43: and esp,0FFFFFFF8h
0x10009b46: push 0FFFFFFFFh
0x10009b48: push 100FCDB9h
0x10009b4d: mov eax,fs:[0]
0x10009b53: push eax
0x10009b54: sub esp,4Ch
0x10009b57: mov eax,ds:[10136220h]
0x10009b5c: xor eax,esp
0x10009b5e: push eax
0x10009b5f: lea eax,[esp+50h]
0x10009b63: mov fs:[0],eax
0x10009b69: push 1012F6F8h
0x10009b6e: lea ecx,[esp+0Ch]
0x10009b72: call dword ptr ds:[100FE3A4h]
0x10009b78: lea eax,[esp+8]
0x10009b7c: push eax
0x10009b7d: lea ecx,[esp+28h]
0x10009b81: mov dword ptr [esp+5Ch],0
0x10009b89: call 10002910h
0x10009b8e: push 10130E80h
0x10009b93: lea ecx,[esp+28h]
0x10009b97: push ecx
0x10009b98: mov dword ptr [esp+2Ch],100FF3ACh
0x10009ba0: call 100229A6h
0x10009ba5: int 3
```

`sub_10009b40` is called by sub_100072e0, sub_10009950, sub_1000c480. It delegates to sub_10002910, j__CxxThrowException (2 callees).

12.139 sub_10009bb0 at 0x10009bb0 (26 instructions)

```
0x10009bb0: sub esp,10h
0x10009bb3: test ecx,ecx
0x10009bb5: ja short 10009BCDh
0x10009bb7: xor ecx,ecx
0x10009bb9: lea edx,[ecx*4]
0x10009bc0: push edx
0x10009bc1: call 10021FDEh
0x10009bc6: add esp,4
0x10009bc9: add esp,10h
0x10009bcc: ret
0x10009bcd: or eax,0FFFFFFFFh
0x10009bd0: xor edx,edx
0x10009bd2: div ecx
0x10009bd4: cmp eax,4
0x10009bd7: jae short 10009BB9h
0x10009bd9: lea eax,[esp]
```

```
0x10009bdc: push eax
0x10009bdd: lea ecx,[esp+8]
0x10009be1: mov dword ptr [esp+4],0
0x10009be9: call dword ptr ds:[100FE578h]
0x10009bef: push 10130EB8h
0x10009bf4: lea ecx,[esp+8]
0x10009bf8: push ecx
0x10009bf9: mov dword ptr [esp+0Ch],100FF394h
0x10009c01: call 100229A6h
0x10009c06: int 3
```

`sub_10009bb0` is called by sub_10002c40, sub_100072e0, sub_10009950. It delegates to j_ORDINAL 762, j__CxxThrowException (2 callees).

12.140 sub_10009c10 at 0x10009c10 (20 instructions)

```
0x10009c10: test eax,eax
0x10009c12: jne short 10009C18h
0x10009c14: xor ecx,ecx
0x10009c16: jmp short 10009C1Bh
0x10009c18: mov ecx,[eax+4]
0x10009c1b: mov edx,[esp+10h]
0x10009c1f: mov eax,[esp+4]
0x10009c23: mov eax,[eax+4]
0x10009c26: push edx
0x10009c27: mov edx,[esp+10h]
0x10009c2b: push edx
0x10009c2c: push ecx
0x10009c2d: mov ecx,[esp+14h]
0x10009c31: push ecx
0x10009c32: push eax
0x10009c33: call 10021FD2h
0x10009c38: mov edx,[eax+94h]
0x10009c3e: mov ecx,[edx]
0x10009c40: call 10009C50h
0x10009c45: ret 10h
```

`sub_10009c10` is called by sub_1000a680. It delegates to sub_10009c50, j_ORDINAL 1079 (2 callees).

12.141 sub_10009c50 at 0x10009c50 (77 instructions)

```
0x10009c50: push ebp
0x10009c51: mov ebp,esp
0x10009c53: push 0FFFFFFFEh
0x10009c55: push 10130EC8h
0x10009c5a: push 10022B59h
0x10009c5f: mov eax,fs:[0]
0x10009c65: push eax
0x10009c66: sub esp,18h
0x10009c69: push ebx
0x10009c6a: push esi
0x10009c6b: push edi
0x10009c6c: mov eax,ds:[10136220h]
0x10009c71: xor [ebp-8],eax
0x10009c74: xor eax,ebp
0x10009c76: push eax
0x10009c77: lea eax,[ebp-10h]
0x10009c7a: mov fs:[0],eax
0x10009c80: mov esi,ecx
0x10009c82: xor edi,edi
0x10009c84: mov [ebp-20h],edi
0x10009c87: lea eax,[ebp-20h]
0x10009c8a: push eax
0x10009c8b: call 10021FD2h
0x10009c90: mov ecx,[eax+9Ch]
```

```
0x10009c96: push ecx
0x10009c97: call 100222D2h
0x10009c9c: mov ebx,eax
0x10009c9e: mov [ebp-24h],ebx
0x10009ca1: mov [ebp-1Ch],edi
0x10009ca4: cmp ebx,edi
0x10009ca6: jne short 10009CBEh
0x10009ca8: xor eax,eax
0x10009caa: mov ecx,[ebp-10h]
0x10009cad: mov fs:[0],ecx
0x10009cb4: pop ecx
0x10009cb5: pop edi
0x10009cb6: pop esi
0x10009cb7: pop ebx
0x10009cb8: mov esp,ebp
0x10009cba: pop ebp
0x10009cbb: ret 14h
0x10009cbe: mov [ebp-4],edi
0x10009cc1: lea edi,[ebp-28h]
0x10009cc4: call 10009D60h
0x10009cc9: mov eax,[esi+3Ch]
0x10009ccc: xor ecx,ecx
0x10009cce: test eax,eax
0x10009cd0: setne cl
0x10009cd3: test ecx,ecx
0x10009cd5: jne short 10009CDCh
0x10009cd7: call 100222EAh
0x10009cdc: push 1
0x10009cde: mov edx,[ebp+18h]
0x10009ce1: push edx
0x10009ce2: mov ecx,[ebp+14h]
0x10009ce5: push ecx
0x10009ce6: mov edx,[ebp+10h]
0x10009ce9: push edx
0x10009cea: mov ecx,[ebp+0Ch]
0x10009ced: push ecx
0x10009cee: mov edx,[ebp+8]
0x10009cf1: push edx
0x10009cf2: call eax
0x10009cf4: mov esi,eax
0x10009cf6: mov [ebp-1Ch],esi
0x10009cf9: mov dword ptr [ebp-4],0FFFFFFFEh
0x10009d00: call 10009D21h
0x10009d05: mov eax,esi
0x10009d07: mov ecx,[ebp-10h]
0x10009d0a: mov fs:[0],ecx
0x10009d11: pop ecx
0x10009d12: pop edi
0x10009d13: pop esi
0x10009d14: pop ebx
0x10009d15: mov esp,ebp
0x10009d17: pop ebp
0x10009d18: ret 14h
```

`sub_10009c50` is called by sub_100089a0, sub_10009c10, sub_1000b5a0. It delegates to sub_10009d21, sub_10009d60, j_ORDINAL 1079 (5 callees).

12.142 sub_10009d21 at 0x10009d21 (21 instructions)

```
0x10009d21: cmp ebx,2
0x10009d24: je short 10009D55h
0x10009d26: xor eax,eax
0x10009d28: test esi,esi
0x10009d2a: sete al
0x10009d2d: mov edi,eax
0x10009d2f: test edi,edi
```

```
0x10009d31: je short 10009D3Dh
0x10009d33: call dword ptr ds:[100FE0F0h]
0x10009d39: mov ebx,eax
0x10009d3b: jmp short 10009D3Fh
0x10009d3d: xor ebx,ebx
0x10009d3f: mov ecx,[ebp-20h]
0x10009d42: push ecx
0x10009d43: push 0
0x10009d45: call 100222FCh
0x10009d4a: test edi,edi
0x10009d4c: je short 10009D55h
0x10009d4e: push ebx
0x10009d4f: call dword ptr ds:[100FE0F4h]
0x10009d55: ret
```

`sub_10009d21` is called by sub_10009c50. It delegates to j_ORDINAL 6751 (1 callees).

12.143 sub_10009d60 at 0x10009d60 (26 instructions)

```
0x10009d60: cmp dword ptr [esi+3Ch],0
0x10009d64: jne short 10009DA4h
0x10009d66: cmp dword ptr [esi+4],0
0x10009d6a: jne short 10009D92h
0x10009d6c: mov eax,[esi+0Ch]
0x10009d6f: push eax
0x10009d70: call dword ptr ds:[100FE114h]
0x10009d76: test eax,eax
0x10009d78: mov [esi+4],eax
0x10009d7b: jne short 10009D92h
0x10009d7d: mov eax,[esi+0Ch]
0x10009d80: push eax
0x10009d81: call dword ptr ds:[100FE110h]
0x10009d87: test eax,eax
0x10009d89: mov [esi+4],eax
0x10009d8c: setne al
0x10009d8f: mov [esi+8],al
0x10009d92: mov eax,[esi+4]
0x10009d95: push 1012F768h
0x10009d9a: push eax
0x10009d9b: call dword ptr ds:[100FE11Ch]
0x10009da1: mov [esi+3Ch],eax
0x10009da4: mov ecx,[esi+3Ch]
0x10009da7: mov [edi],ecx
0x10009da9: mov eax,edi
0x10009dab: ret
```

`sub_10009d60` is called by sub_10009c50. It works with the text "ImageList_Draw". It calls nothing else.

12.144 sub_10009db0 at 0x10009db0 (63 instructions)

```
0x10009db0: push ebp
0x10009db1: mov ebp,esp
0x10009db3: push 0FFFFFFFEh
0x10009db5: push 10131180h
0x10009dba: push 10022B59h
0x10009dbf: mov eax,fs:[0]
0x10009dc5: push eax
0x10009dc6: sub esp,18h
0x10009dc9: push ebx
0x10009dca: push esi
0x10009dcb: push edi
0x10009dcc: mov eax,ds:[10136220h]
0x10009dd1: xor [ebp-8],eax
0x10009dd4: xor eax,ebp
0x10009dd6: push eax
```

```
0x10009dd7: lea eax,[ebp-10h]
0x10009dda: mov fs:[0],eax
0x10009de0: mov esi,ecx
0x10009de2: xor edi,edi
0x10009de4: mov [ebp-20h],edi
0x10009de7: lea eax,[ebp-20h]
0x10009dea: push eax
0x10009deb: call 10021FD2h
0x10009df0: mov ecx,[eax+9Ch]
0x10009df6: push ecx
0x10009df7: call 100222D2h
0x10009dfc: mov ebx,eax
0x10009dfe: mov [ebp-24h],ebx
0x10009e01: or eax,0FFFFFFFFh
0x10009e04: mov [ebp-1Ch],eax
0x10009e07: cmp ebx,edi
0x10009e09: je short 10009E4Ah
0x10009e0b: mov [ebp-4],edi
0x10009e0e: lea edi,[ebp-28h]
0x10009e11: call 10009EA0h
0x10009e16: mov eax,[esi+44h]
0x10009e19: xor ecx,ecx
0x10009e1b: test eax,eax
0x10009e1d: setne cl
0x10009e20: test ecx,ecx
0x10009e22: jne short 10009E29h
0x10009e24: call 100222EAh
0x10009e29: mov edx,[ebp+10h]
0x10009e2c: push edx
0x10009e2d: mov ecx,[ebp+0Ch]
0x10009e30: push ecx
0x10009e31: mov edx,[ebp+8]
0x10009e34: push edx
0x10009e35: call eax
0x10009e37: mov edi,eax
0x10009e39: mov [ebp-1Ch],edi
0x10009e3c: mov dword ptr [ebp-4],0FFFFFFFEh
0x10009e43: call 10009E64h
0x10009e48: mov eax,edi
0x10009e4a: mov ecx,[ebp-10h]
0x10009e4d: mov fs:[0],ecx
0x10009e54: pop ecx
0x10009e55: pop edi
0x10009e56: pop esi
0x10009e57: pop ebx
0x10009e58: mov esp,ebp
0x10009e5a: pop ebp
0x10009e5b: ret 0Ch
```

`sub_10009db0` is called by sub_10008290, sub_100083e0, sub_1000a3c0, sub_1000a510, sub_1000ab10. It delegates to sub_10009e64, sub_10009ea0, j_ORDINAL 1079 (5 callees).

12.145 sub_10009e64 at 0x10009e64 (21 instructions)

```
0x10009e64: cmp ebx,2
0x10009e67: je short 10009E99h
0x10009e69: xor eax,eax
0x10009e6b: cmp edi,0FFFFFFFFh
0x10009e6e: sete al
0x10009e71: mov esi,eax
0x10009e73: test esi,esi
0x10009e75: je short 10009E81h
0x10009e77: call dword ptr ds:[100FE0F0h]
0x10009e7d: mov ebx,eax
0x10009e7f: jmp short 10009E83h
0x10009e81: xor ebx,ebx
```

```
0x10009e83: mov ecx,[ebp-20h]
0x10009e86: push ecx
0x10009e87: push 0
0x10009e89: call 100222FCh
0x10009e8e: test esi,esi
0x10009e90: je short 10009E99h
0x10009e92: push ebx
0x10009e93: call dword ptr ds:[100FE0F4h]
0x10009e99: ret
```

`sub_10009e64` is called by sub_10009db0. It delegates to j_ORDINAL 6751 (1 callees).

12.146 sub_10009ea0 at 0x10009ea0 (26 instructions)

```
0x10009ea0: cmp dword ptr [esi+44h],0
0x10009ea4: jne short 10009EE4h
0x10009ea6: cmp dword ptr [esi+4],0
0x10009eaa: jne short 10009ED2h
0x10009eac: mov eax,[esi+0Ch]
0x10009eaf: push eax
0x10009eb0: call dword ptr ds:[100FE114h]
0x10009eb6: test eax,eax
0x10009eb8: mov [esi+4],eax
0x10009ebb: jne short 10009ED2h
0x10009ebd: mov eax,[esi+0Ch]
0x10009ec0: push eax
0x10009ec1: call dword ptr ds:[100FE110h]
0x10009ec7: test eax,eax
0x10009ec9: mov [esi+4],eax
0x10009ecc: setne al
0x10009ecf: mov [esi+8],al
0x10009ed2: mov eax,[esi+4]
0x10009ed5: push 1012F778h
0x10009eda: push eax
0x10009edb: call dword ptr ds:[100FE11Ch]
0x10009ee1: mov [esi+44h],eax
0x10009ee4: mov ecx,[esi+44h]
0x10009ee7: mov [edi],ecx
0x10009ee9: mov eax,edi
0x10009eeb: ret
```

`sub_10009ea0` is called by sub_10009db0. It works with the text "ImageList_AddMasked". It calls nothing else.

12.147 sub_10009fc0 at 0x10009fc0 (36 instructions)

```
0x10009fc0: sub esp,18h
0x10009fc3: test edi,edi
0x10009fc5: je short 1000A00Ah
0x10009fc7: cmp dword ptr [edi+4],0
0x10009fcb: je short 1000A00Ah
0x10009fcd: mov eax,[esi+4]
0x10009fd0: test eax,eax
0x10009fd2: je short 10009FDBh
0x10009fd4: push eax
0x10009fd5: call dword ptr ds:[100FE5ACh]
0x10009fdb: mov eax,[esi+8]
0x10009fde: test eax,eax
0x10009fe0: mov dword ptr [esi+4],0
0x10009fe7: je short 10009FF0h
0x10009fe9: push eax
0x10009fea: call dword ptr ds:[100FE048h]
0x10009ff0: mov eax,[edi+4]
0x10009ff3: lea ecx,[esp]
0x10009ff6: push ecx
0x10009ff7: push 18h
```

```
0x10009ff9: push eax
0x10009ffa: mov [esi+8],eax
0x10009ffd: call dword ptr ds:[100FE074h]
0x1000a003: test eax,eax
0x1000a005: jne short 1000A010h
0x1000a007: mov [esi+8],eax
0x1000a00a: xor eax,eax
0x1000a00c: add esp,18h
0x1000a00f: ret
0x1000a010: mov eax,[esp+4]
0x1000a014: mov ecx,[esp+8]
0x1000a018: mov [esi+10h],eax
0x1000a01b: mov [esi+14h],ecx
0x1000a01e: mov eax,1
0x1000a023: add esp,18h
0x1000a026: ret
```

`sub_10009fc0` is called by ?SetSysmenuBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z. It calls nothing else.

12.148 sub_1000a030 at 0x1000a030 (54 instructions)

```
0x1000a030: push ebp
0x1000a031: mov ebp,[esp+0Ch]
0x1000a035: push edi
0x1000a036: lea edi,[ebx+18h]
0x1000a039: push edi
0x1000a03a: call dword ptr ds:[100FE5C8h]
0x1000a040: mov eax,[ebp]
0x1000a043: mov [esi],eax
0x1000a045: mov ecx,[ebp+4]
0x1000a048: mov [esi+4],ecx
0x1000a04b: mov edx,[ebp+8]
0x1000a04e: mov [esi+8],edx
0x1000a051: mov eax,[ebp+0Ch]
0x1000a054: mov [esi+0Ch],eax
0x1000a057: mov ecx,[esi]
0x1000a059: mov [esi+8],ecx
0x1000a05c: cmp dword ptr [ebx+10h],0
0x1000a060: jle short 1000A0BBh
0x1000a062: mov edx,[ebx+14h]
0x1000a065: test edx,edx
0x1000a067: jle short 1000A0BBh
0x1000a069: mov ecx,[esi+4]
0x1000a06c: mov eax,[esi+0Ch]
0x1000a06f: sub eax,ecx
0x1000a071: sub eax,edx
0x1000a073: add eax,1
0x1000a076: cdq
0x1000a077: sub eax,edx
0x1000a079: sar eax,1
0x1000a07b: add eax,ecx
0x1000a07d: mov [esi+4],eax
0x1000a080: mov edx,[ebx+14h]
0x1000a083: add edx,eax
0x1000a085: test dword ptr [esp+0Ch],80000h
0x1000a08d: mov [esi+0Ch],edx
0x1000a090: je short 1000A0BBh
0x1000a092: mov eax,[esi]
0x1000a094: mov [edi],eax
0x1000a096: mov ecx,[esi+4]
0x1000a099: mov [edi+4],ecx
0x1000a09c: mov edx,[esi+8]
0x1000a09f: mov ecx,[ebx+10h]
0x1000a0a2: mov [edi+8],edx
0x1000a0a5: mov eax,[esi+0Ch]
```

```
0x1000a0a8: mov [edi+0Ch],eax
0x1000a0ab: mov eax,[esi+8]
0x1000a0ae: add eax,2
0x1000a0b1: mov [edi],eax
0x1000a0b3: add eax,ecx
0x1000a0b5: mov [ebx+20h],eax
0x1000a0b8: mov [esi+8],eax
0x1000a0bb: pop edi
0x1000a0bc: pop ebp
0x1000a0bd: ret 8
```

`sub_1000a030` is called by sub_100056c0. It calls nothing else.

12.149 sub_1000a0c0 at 0x1000a0c0 (57 instructions)

```
0x1000a0c0: sub esp,14h
0x1000a0c3: push esi
0x1000a0c4: push edi
0x1000a0c5: lea esi,[ebx+18h]
0x1000a0c8: push esi
0x1000a0c9: mov edi,eax
0x1000a0cb: call dword ptr ds:[100FE6ACh]
0x1000a0d1: test eax,eax
0x1000a0d3: jne short 1000A13Eh
0x1000a0d5: mov ecx,[ebx+4]
0x1000a0d8: xor edx,edx
0x1000a0da: cmp ecx,edx
0x1000a0dc: je short 1000A10Bh
0x1000a0de: cmp edi,edx
0x1000a0e0: je short 1000A0E5h
0x1000a0e2: mov eax,[edi+4]
0x1000a0e5: push 3
0x1000a0e7: push edx
0x1000a0e8: push edx
0x1000a0e9: mov edx,[esi+0Ch]
0x1000a0ec: sub edx,[esi+4]
0x1000a0ef: push edx
0x1000a0f0: mov edx,[esi+8]
0x1000a0f3: sub edx,[esi]
0x1000a0f5: push edx
0x1000a0f6: mov edx,[esi]
0x1000a0f8: push ecx
0x1000a0f9: mov ecx,[ebx+1Ch]
0x1000a0fc: push ecx
0x1000a0fd: push edx
0x1000a0fe: push eax
0x1000a0ff: call dword ptr ds:[100FE5A0h]
0x1000a105: pop edi
0x1000a106: pop esi
0x1000a107: add esp,14h
0x1000a10a: ret
0x1000a10b: mov eax,[ebx+8]
0x1000a10e: cmp eax,edx
0x1000a110: je short 1000A13Eh
0x1000a112: mov ecx,[ebx+10h]
0x1000a115: mov [esp+10h],ecx
0x1000a119: lea ecx,[esp+8]
0x1000a11d: mov [esp+8],edx
0x1000a121: mov [esp+0Ch],edx
0x1000a125: mov edx,[ebx+14h]
0x1000a128: push ecx
0x1000a129: push eax
0x1000a12a: mov [esp+1Ch],edx
0x1000a12e: call 10022272h
0x1000a133: push eax
0x1000a134: mov ecx,esi
```

```
0x1000a136: call 10001D00h
0x1000a13b: add esp,8
0x1000a13e: pop edi
0x1000a13f: pop esi
0x1000a140: add esp,14h
0x1000a143: ret
```

`sub_1000a0c0` is called by sub_10005fd0. It delegates to sub_10001d00, j_ORDINAL 2362 (2 callees).

12.150 sub_1000a150 at 0x1000a150 (18 instructions)

```
0x1000a150: cmp dword ptr [ecx+4],0
0x1000a154: jne short 1000A15Fh
0x1000a156: cmp dword ptr [ecx+8],0
0x1000a15a: jne short 1000A15Fh
0x1000a15c: xor eax,eax
0x1000a15e: ret
0x1000a15f: push esi
0x1000a160: mov esi,[ecx+18h]
0x1000a163: mov [edx],esi
0x1000a165: mov esi,[ecx+1Ch]
0x1000a168: mov [edx+4],esi
0x1000a16b: mov esi,[ecx+20h]
0x1000a16e: mov [edx+8],esi
0x1000a171: mov ecx,[ecx+24h]
0x1000a174: mov eax,1
0x1000a179: mov [edx+0Ch],ecx
0x1000a17c: pop esi
0x1000a17d: ret
```

`sub_1000a150` is called by sub_100051b0. It calls nothing else.

12.151 sub_1000a180 at 0x1000a180 (9 instructions)

```
0x1000a180: push ecx
0x1000a181: add eax,4
0x1000a184: push eax
0x1000a185: mov ecx,esi
0x1000a187: mov dword ptr [esp+4],0
0x1000a18f: call dword ptr ds:[100FE220h]
0x1000a195: mov eax,esi
0x1000a197: pop ecx
0x1000a198: ret
```

`sub_1000a180` is called by sub_1000b250. It calls nothing else.

12.152 sub_1000a1c0 at 0x1000a1c0 (44 instructions)

```
0x1000a1c0: push 0FFFFFFFFh
0x1000a1c2: push 100FCA58h
0x1000a1c7: mov eax,fs:[0]
0x1000a1cd: push eax
0x1000a1ce: push esi
0x1000a1cf: mov eax,ds:[10136220h]
0x1000a1d4: xor eax,esp
0x1000a1d6: push eax
0x1000a1d7: lea eax,[esp+8]
0x1000a1db: mov fs:[0],eax
0x1000a1e1: mov esi,[esp+18h]
0x1000a1e5: lea ecx,[esi+4]
0x1000a1e8: call dword ptr ds:[100FE368h]
0x1000a1ee: lea ecx,[esi+8]
0x1000a1f1: mov dword ptr [esp+10h],0
0x1000a1f9: call dword ptr ds:[100FE368h]
```

```
0x1000a1ff: lea ecx,[esi+0Ch]
0x1000a202: mov byte ptr [esp+10h],1
0x1000a207: call 100222A2h
0x1000a20c: xor eax,eax
0x1000a20e: mov [esi+14h],eax
0x1000a211: lea eax,[esi+1Ch]
0x1000a214: xor ecx,ecx
0x1000a216: push eax
0x1000a217: mov dword ptr [esi],0
0x1000a21d: mov [esi+18h],ecx
0x1000a220: call dword ptr ds:[100FE5C8h]
0x1000a226: mov ecx,ds:[10138BF0h]
0x1000a22c: mov eax,ds:[10138BF8h]
0x1000a231: mov edx,ds:[10138C08h]
0x1000a237: mov [esi+34h],ecx
0x1000a23a: mov ecx,ds:[10138C00h]
0x1000a240: mov [esi+3Ch],eax
0x1000a243: mov dword ptr [esi+2Ch],0
0x1000a24a: mov dword ptr [esi+30h],0
0x1000a251: mov [esi+38h],edx
0x1000a254: mov [esi+40h],ecx
0x1000a257: mov eax,esi
0x1000a259: mov ecx,[esp+8]
0x1000a25d: mov fs:[0],ecx
0x1000a264: pop ecx
0x1000a265: pop esi
0x1000a266: add esp,0Ch
0x1000a269: ret 4
```

`sub_1000a1c0` is called by sub_1000aca0, sub_1000ad90. It delegates to j_ORDINAL 416 (1 callees).

12.153 sub_1000a270 at 0x1000a270 (24 instructions)

```
0x1000a270: push 0FFFFFFFFh
0x1000a272: push 100FCA58h
0x1000a277: mov eax,fs:[0]
0x1000a27d: push eax
0x1000a27e: push esi
0x1000a27f: mov eax,ds:[10136220h]
0x1000a284: xor eax,esp
0x1000a286: push eax
0x1000a287: lea eax,[esp+8]
0x1000a28b: mov fs:[0],eax
0x1000a291: mov esi,[esp+18h]
0x1000a295: lea ecx,[esi+0Ch]
0x1000a298: mov dword ptr [esp+10h],1
0x1000a2a0: call 100222A8h
0x1000a2a5: lea ecx,[esi+8]
0x1000a2a8: call dword ptr ds:[100FE374h]
0x1000a2ae: lea ecx,[esi+4]
0x1000a2b1: call dword ptr ds:[100FE374h]
0x1000a2b7: mov ecx,[esp+8]
0x1000a2bb: mov fs:[0],ecx
0x1000a2c2: pop ecx
0x1000a2c3: pop esi
0x1000a2c4: add esp,0Ch
0x1000a2c7: ret 4
```

`sub_1000a270` is called by sub_1000aca0, sub_1000ad90, sub_1000af00. It delegates to j_ORDINAL 651 (1 callees).

12.154 sub_1000a2d0 at 0x1000a2d0 (71 instructions)

```
0x1000a2d0: push 0FFFFFFFFh
0x1000a2d2: push 100FCA18h
0x1000a2d7: mov eax,fs:[0]
```

```
0x1000a2dd: push eax
0x1000a2de: sub esp,24h
0x1000a2e1: push edi
0x1000a2e2: mov eax,ds:[10136220h]
0x1000a2e7: xor eax,esp
0x1000a2e9: push eax
0x1000a2ea: lea eax,[esp+2Ch]
0x1000a2ee: mov fs:[0],eax
0x1000a2f4: lea ecx,[esi+0Ch]
0x1000a2f7: test ecx,ecx
0x1000a2f9: je short 1000A306h
0x1000a2fb: cmp dword ptr [ecx+4],0
0x1000a2ff: je short 1000A306h
0x1000a301: call 100222DEh
0x1000a306: xor eax,eax
0x1000a308: xor ecx,ecx
0x1000a30a: mov [esi+14h],eax
0x1000a30d: mov [esi+18h],ecx
0x1000a310: call dword ptr ds:[100FE59Ch]
0x1000a316: push eax
0x1000a317: call 1002225Ah
0x1000a31c: test eax,eax
0x1000a31e: jne short 1000A333h
0x1000a320: mov ecx,[esp+2Ch]
0x1000a324: mov fs:[0],ecx
0x1000a32b: pop ecx
0x1000a32c: pop edi
0x1000a32d: add esp,30h
0x1000a330: ret 4
0x1000a333: mov ecx,[esp+3Ch]
0x1000a337: mov [esi+30h],ecx
0x1000a33a: push eax
0x1000a33b: lea ecx,[esp+18h]
0x1000a33f: call 1002222Ah
0x1000a344: mov dword ptr [esp+34h],0
0x1000a34c: mov edx,[esi+30h]
0x1000a34f: push edx
0x1000a350: lea ecx,[esp+18h]
0x1000a354: call 1002228Ah
0x1000a359: mov edi,eax
0x1000a35b: mov eax,[esi+4]
0x1000a35e: mov ecx,[eax-0Ch]
0x1000a361: lea edx,[esp+0Ch]
0x1000a365: push edx
0x1000a366: push ecx
0x1000a367: push eax
0x1000a368: mov eax,[esp+28h]
0x1000a36c: push eax
0x1000a36d: call dword ptr ds:[100FE03Ch]
0x1000a373: mov ecx,[esp+0Ch]
0x1000a377: mov edx,[esp+10h]
0x1000a37b: mov [esi+14h],ecx
0x1000a37e: push edi
0x1000a37f: lea ecx,[esp+18h]
0x1000a383: mov [esi+18h],edx
0x1000a386: call 1002228Ah
0x1000a38b: add dword ptr [esi+14h],8
0x1000a38f: add dword ptr [esi+18h],4
0x1000a393: lea ecx,[esp+14h]
0x1000a397: mov dword ptr [esp+34h],0FFFFFFFFh
0x1000a39f: call 10022230h
0x1000a3a4: mov eax,1
0x1000a3a9: mov ecx,[esp+2Ch]
0x1000a3ad: mov fs:[0],ecx
0x1000a3b4: pop ecx
0x1000a3b5: pop edi
```

```
0x1000a3b6: add esp,30h
0x1000a3b9: ret 4
```

`sub_1000a2d0` is called by sub_1000aca0, sub_1000ad90. It delegates to j_ORDINAL 567, j_ORDINAL 758, j_ORDINAL 2366 (5 callees).

12.155 sub_1000a3c0 at 0x1000a3c0 (104 instructions)

```
0x1000a3c0: push 0FFFFFFFFh
0x1000a3c2: push 100FCD88h
0x1000a3c7: mov eax,fs:[0]
0x1000a3cd: push eax
0x1000a3ce: sub esp,28h
0x1000a3d1: push ebx
0x1000a3d2: push edi
0x1000a3d3: mov eax,ds:[10136220h]
0x1000a3d8: xor eax,esp
0x1000a3da: push eax
0x1000a3db: lea eax,[esp+34h]
0x1000a3df: mov fs:[0],eax
0x1000a3e5: lea edi,[esi+0Ch]
0x1000a3e8: test edi,edi
0x1000a3ea: je short 1000A3F9h
0x1000a3ec: cmp dword ptr [edi+4],0
0x1000a3f0: je short 1000A3F9h
0x1000a3f2: mov ecx,edi
0x1000a3f4: call 100222DEh
0x1000a3f9: xor eax,eax
0x1000a3fb: xor ecx,ecx
0x1000a3fd: mov ebx,1012F498h
0x1000a402: mov [esi+14h],eax
0x1000a405: mov [esi+18h],ecx
0x1000a408: mov [esp+14h],eax
0x1000a40c: mov [esp+10h],ebx
0x1000a410: mov eax,[esp+44h]
0x1000a414: mov [esp+3Ch],ecx
0x1000a418: mov ecx,ds:[1013F360h]
0x1000a41e: push eax
0x1000a41f: push ecx
0x1000a420: call dword ptr ds:[100FE5E0h]
0x1000a426: push eax
0x1000a427: lea ecx,[esp+14h]
0x1000a42b: call 10021FFCh
0x1000a430: test eax,eax
0x1000a432: je short 1000A496h
0x1000a434: mov eax,[esp+14h]
0x1000a438: lea edx,[esp+18h]
0x1000a43c: push edx
0x1000a43d: push 18h
0x1000a43f: push eax
0x1000a440: call dword ptr ds:[100FE074h]
0x1000a446: movzx ecx,word ptr [esp+2Ah]
0x1000a44b: mov edx,[esp+20h]
0x1000a44f: mov eax,[esp+1Ch]
0x1000a453: push 0
0x1000a455: push 4
0x1000a457: push ecx
0x1000a458: push edx
0x1000a459: cdq
0x1000a45a: and edx,3
0x1000a45d: add eax,edx
0x1000a45f: sar eax,2
0x1000a462: push eax
0x1000a463: mov ecx,edi
0x1000a465: call 100222E4h
0x1000a46a: test eax,eax
```

```
0x1000a46c: je short 1000A496h
0x1000a46e: mov ecx,[esp+14h]
0x1000a472: mov eax,[esi+10h]
0x1000a475: push 0FFFFFFFFh
0x1000a477: push ecx
0x1000a478: push eax
0x1000a479: call 10021FD2h
0x1000a47e: mov edx,[eax+94h]
0x1000a484: mov ecx,[edx]
0x1000a486: call 10009DB0h
0x1000a48b: test eax,eax
0x1000a48d: jge short 1000A4C1h
0x1000a48f: mov ecx,edi
0x1000a491: call 100222DEh
0x1000a496: mov dword ptr [esp+3Ch],0FFFFFFFFh
0x1000a49e: lea ecx,[esp+10h]
0x1000a4a2: mov [esp+10h],ebx
0x1000a4a6: call 10001370h
0x1000a4ab: xor eax,eax
0x1000a4ad: mov ecx,[esp+34h]
0x1000a4b1: mov fs:[0],ecx
0x1000a4b8: pop ecx
0x1000a4b9: pop edi
0x1000a4ba: pop ebx
0x1000a4bb: add esp,34h
0x1000a4be: ret 4
0x1000a4c1: mov eax,[esp+1Ch]
0x1000a4c5: mov ecx,[esp+20h]
0x1000a4c9: cdq
0x1000a4ca: and edx,3
0x1000a4cd: add eax,edx
0x1000a4cf: sar eax,2
0x1000a4d2: mov [esi+18h],ecx
0x1000a4d5: mov dword ptr [esp+3Ch],0FFFFFFFFh
0x1000a4dd: lea ecx,[esp+10h]
0x1000a4e1: mov [esi+14h],eax
0x1000a4e4: mov [esp+10h],ebx
0x1000a4e8: call 10001370h
0x1000a4ed: mov eax,1
0x1000a4f2: mov ecx,[esp+34h]
0x1000a4f6: mov fs:[0],ecx
0x1000a4fd: pop ecx
0x1000a4fe: pop edi
0x1000a4ff: pop ebx
0x1000a500: add esp,34h
0x1000a503: ret 4
```

`sub_1000a3c0` is called by sub_1000aca0. It delegates to sub_10001370, sub_10009db0, j_ORDINAL 1079 (6 callees).

12.156 sub_1000a510 at 0x1000a510 (118 instructions)

```
0x1000a510: push 0FFFFFFFFh
0x1000a512: push 100FCD88h
0x1000a517: mov eax,fs:[0]
0x1000a51d: push eax
0x1000a51e: sub esp,28h
0x1000a521: push ebx
0x1000a522: push edi
0x1000a523: mov eax,ds:[10136220h]
0x1000a528: xor eax,esp
0x1000a52a: push eax
0x1000a52b: lea eax,[esp+34h]
0x1000a52f: mov fs:[0],eax
0x1000a535: lea edi,[esi+0Ch]
0x1000a538: test edi,edi
0x1000a53a: je short 1000A549h
```

```
0x1000a53c: cmp dword ptr [edi+4],0
0x1000a540: je short 1000A549h
0x1000a542: mov ecx,edi
0x1000a544: call 100222DEh
0x1000a549: xor ecx,ecx
0x1000a54b: push 2050h
0x1000a550: push ecx
0x1000a551: xor eax,eax
0x1000a553: push ecx
0x1000a554: mov [esi+14h],eax
0x1000a557: mov eax,[esp+50h]
0x1000a55b: push ecx
0x1000a55c: push eax
0x1000a55d: push ecx
0x1000a55e: mov [esi+18h],ecx
0x1000a561: call dword ptr ds:[100FE5A4h]
0x1000a567: test eax,eax
0x1000a569: je short 1000A5DDh
0x1000a56b: mov ebx,1012F498h
0x1000a570: mov dword ptr [esp+14h],0
0x1000a578: mov [esp+10h],ebx
0x1000a57c: push eax
0x1000a57d: lea ecx,[esp+14h]
0x1000a581: mov dword ptr [esp+40h],0
0x1000a589: call 10021FFCh
0x1000a58e: mov edx,[esp+14h]
0x1000a592: lea ecx,[esp+18h]
0x1000a596: push ecx
0x1000a597: push 18h
0x1000a599: push edx
0x1000a59a: call dword ptr ds:[100FE074h]
0x1000a5a0: movzx eax,word ptr [esp+2Ah]
0x1000a5a5: mov ecx,[esp+20h]
0x1000a5a9: push 0
0x1000a5ab: push 4
0x1000a5ad: push eax
0x1000a5ae: mov eax,[esp+28h]
0x1000a5b2: cdq
0x1000a5b3: and edx,3
0x1000a5b6: add eax,edx
0x1000a5b8: push ecx
0x1000a5b9: sar eax,2
0x1000a5bc: push eax
0x1000a5bd: mov ecx,edi
0x1000a5bf: call 100222E4h
0x1000a5c4: test eax,eax
0x1000a5c6: jne short 1000A5F3h
0x1000a5c8: mov dword ptr [esp+3Ch],0FFFFFFFFh
0x1000a5d0: lea ecx,[esp+10h]
0x1000a5d4: mov [esp+10h],ebx
0x1000a5d8: call 10001370h
0x1000a5dd: xor eax,eax
0x1000a5df: mov ecx,[esp+34h]
0x1000a5e3: mov fs:[0],ecx
0x1000a5ea: pop ecx
0x1000a5eb: pop edi
0x1000a5ec: pop ebx
0x1000a5ed: add esp,34h
0x1000a5f0: ret 4
0x1000a5f3: mov edx,[esp+14h]
0x1000a5f7: mov eax,[esi+10h]
0x1000a5fa: or edi,0FFFFFFFFh
0x1000a5fd: push edi
0x1000a5fe: push edx
0x1000a5ff: push eax
0x1000a600: call 10021FD2h
0x1000a605: mov eax,[eax+94h]
```

```
0x1000a60b: mov ecx,[eax]
0x1000a60d: call 10009DB0h
0x1000a612: test eax,eax
0x1000a614: jge short 1000A63Dh
0x1000a616: mov [esp+3Ch],edi
0x1000a61a: lea ecx,[esp+10h]
0x1000a61e: mov [esp+10h],ebx
0x1000a622: call 10001370h
0x1000a627: xor eax,eax
0x1000a629: mov ecx,[esp+34h]
0x1000a62d: mov fs:[0],ecx
0x1000a634: pop ecx
0x1000a635: pop edi
0x1000a636: pop ebx
0x1000a637: add esp,34h
0x1000a63a: ret 4
0x1000a63d: mov eax,[esp+1Ch]
0x1000a641: mov ecx,[esp+20h]
0x1000a645: cdq
0x1000a646: and edx,3
0x1000a649: add eax,edx
0x1000a64b: sar eax,2
0x1000a64e: mov [esi+18h],ecx
0x1000a651: mov [esp+3Ch],edi
0x1000a655: lea ecx,[esp+10h]
0x1000a659: mov [esi+14h],eax
0x1000a65c: mov [esp+10h],ebx
0x1000a660: call 10001370h
0x1000a665: mov eax,1
0x1000a66a: mov ecx,[esp+34h]
0x1000a66e: mov fs:[0],ecx
0x1000a675: pop ecx
0x1000a676: pop edi
0x1000a677: pop ebx
0x1000a678: add esp,34h
0x1000a67b: ret 4
```

`sub_1000a510` is called by sub_1000aca0. It delegates to sub_10001370, sub_10009db0, j_ORDINAL 1079 (6 callees).

12.157 sub_1000a680 at 0x1000a680 (169 instructions)

```
0x1000a680: push 0FFFFFFFFh
0x1000a682: push 100FCD60h
0x1000a687: mov eax,fs:[0]
0x1000a68d: push eax
0x1000a68e: sub esp,24h
0x1000a691: push ebx
0x1000a692: push ebp
0x1000a693: push edi
0x1000a694: mov eax,ds:[10136220h]
0x1000a699: xor eax,esp
0x1000a69b: push eax
0x1000a69c: lea eax,[esp+34h]
0x1000a6a0: mov fs:[0],eax
0x1000a6a6: mov edi,ecx
0x1000a6a8: lea ebp,[edi+1Ch]
0x1000a6ab: push ebp
0x1000a6ac: call dword ptr ds:[100FE6ACh]
0x1000a6b2: test eax,eax
0x1000a6b4: jne near ptr 1000A870h
0x1000a6ba: cmp [edi+14h],eax
0x1000a6bd: jle near ptr 1000A870h
0x1000a6c3: cmp [edi+18h],eax
0x1000a6c6: jle near ptr 1000A870h
0x1000a6cc: lea eax,[edi+0Ch]
0x1000a6cf: test eax,eax
```

```
0x1000a6d1: je short 1000A6DDh
0x1000a6d3: cmp dword ptr [eax+4],0
0x1000a6d7: jne near ptr 1000A850h
0x1000a6dd: cmp dword ptr [esp+44h],0
0x1000a6e2: mov eax,ds:[10138BECh]
0x1000a6e7: mov ebx,[edi+34h]
0x1000a6ea: je short 1000A712h
0x1000a6ec: mov ecx,[edi+2Ch]
0x1000a6ef: cmp ecx,3
0x1000a6f2: je short 1000A712h
0x1000a6f4: cmp ecx,2
0x1000a6f7: jne short 1000A703h
0x1000a6f9: mov eax,ds:[10138BFCh]
0x1000a6fe: mov ebx,[edi+40h]
0x1000a701: jmp short 1000A71Ah
0x1000a703: cmp ecx,1
0x1000a706: jne short 1000A71Ah
0x1000a708: mov eax,ds:[10138BF4h]
0x1000a70d: mov ebx,[edi+3Ch]
0x1000a710: jmp short 1000A71Ah
0x1000a712: mov eax,ds:[10138C04h]
0x1000a717: mov ebx,[edi+38h]
0x1000a71a: push eax
0x1000a71b: push 1
0x1000a71d: push 0
0x1000a71f: lea ecx,[esp+24h]
0x1000a723: call 10022302h
0x1000a728: xor eax,eax
0x1000a72a: mov [esp+3Ch],eax
0x1000a72e: mov [esp+14h],eax
0x1000a732: mov dword ptr [esp+10h],1012F4F0h
0x1000a73a: push 5
0x1000a73c: mov byte ptr [esp+40h],1
0x1000a741: call dword ptr ds:[100FE038h]
0x1000a747: mov [esp+14h],eax
0x1000a74b: lea eax,[esp+18h]
0x1000a74f: push eax
0x1000a750: mov ecx,esi
0x1000a752: call 1002230Eh
0x1000a757: lea ecx,[esp+10h]
0x1000a75b: push ecx
0x1000a75c: mov ecx,esi
0x1000a75e: mov [esp+48h],eax
0x1000a762: call 10022314h
0x1000a767: push eax
0x1000a768: mov ecx,esi
0x1000a76a: call 10022314h
0x1000a76f: mov edx,[esp+44h]
0x1000a773: push edx
0x1000a774: mov ecx,esi
0x1000a776: call 1002230Eh
0x1000a77b: cmp dword ptr [edi+2Ch],1
0x1000a77f: mov eax,[ebp]
0x1000a782: mov ecx,[ebp+4]
0x1000a785: mov edx,[ebp+8]
0x1000a788: mov [esp+20h],eax
0x1000a78c: mov eax,[ebp+0Ch]
0x1000a78f: mov [esp+24h],ecx
0x1000a793: mov [esp+28h],edx
0x1000a797: mov [esp+2Ch],eax
0x1000a79b: jne short 1000A7ACh
0x1000a79d: push 1
0x1000a79f: push 1
0x1000a7a1: lea ecx,[esp+28h]
0x1000a7a5: push ecx
0x1000a7a6: call dword ptr ds:[100FE6B0h]
0x1000a7ac: mov edx,[esi]
```

```
0x1000a7ae: mov eax,[edi+30h]
0x1000a7b1: mov edx,[edx+30h]
0x1000a7b4: push eax
0x1000a7b5: mov ecx,esi
0x1000a7b7: call edx
0x1000a7b9: mov ebp,eax
0x1000a7bb: mov eax,[esi]
0x1000a7bd: mov edx,[eax+38h]
0x1000a7c0: push ebx
0x1000a7c1: mov ecx,esi
0x1000a7c3: call edx
0x1000a7c5: push 1
0x1000a7c7: mov ecx,esi
0x1000a7c9: mov ebx,eax
0x1000a7cb: call 10022296h
0x1000a7d0: mov edi,[edi+4]
0x1000a7d3: mov edx,[esi]
0x1000a7d5: mov edx,[edx+70h]
0x1000a7d8: push 25h
0x1000a7da: lea ecx,[esp+24h]
0x1000a7de: push ecx
0x1000a7df: mov [esp+4Ch],eax
0x1000a7e3: mov eax,[edi-0Ch]
0x1000a7e6: push eax
0x1000a7e7: push edi
0x1000a7e8: mov ecx,esi
0x1000a7ea: call edx
0x1000a7ec: mov eax,[esp+44h]
0x1000a7f0: push eax
0x1000a7f1: mov ecx,esi
0x1000a7f3: call 10022296h
0x1000a7f8: mov edx,[esi]
0x1000a7fa: mov eax,[edx+38h]
0x1000a7fd: push ebx
0x1000a7fe: mov ecx,esi
0x1000a800: call eax
0x1000a802: mov edx,[esi]
0x1000a804: mov eax,[edx+30h]
0x1000a807: push ebp
0x1000a808: mov ecx,esi
0x1000a80a: call eax
0x1000a80c: mov byte ptr [esp+3Ch],0
0x1000a811: lea ecx,[esp+10h]
0x1000a815: mov dword ptr [esp+10h],1012F4F0h
0x1000a81d: call 10001370h
0x1000a822: mov dword ptr [esp+3Ch],0FFFFFFFFh
0x1000a82a: lea ecx,[esp+18h]
0x1000a82e: mov dword ptr [esp+18h],1012F7C0h
0x1000a836: call 10001370h
0x1000a83b: mov ecx,[esp+34h]
0x1000a83f: mov fs:[0],ecx
0x1000a846: pop ecx
0x1000a847: pop edi
0x1000a848: pop ebp
0x1000a849: pop ebx
0x1000a84a: add esp,30h
0x1000a84d: ret 4
0x1000a850: cmp dword ptr [esp+44h],0
0x1000a855: mov edi,[edi+2Ch]
0x1000a858: jne short 1000A85Fh
0x1000a85a: mov edi,3
0x1000a85f: mov ecx,[ebp+4]
0x1000a862: mov edx,[ebp]
0x1000a865: push ecx
0x1000a866: push edx
0x1000a867: push edi
0x1000a868: push eax
```

```
0x1000a869: mov eax,esi
0x1000a86b: call 10009C10h
0x1000a870: mov ecx,[esp+34h]
0x1000a874: mov fs:[0],ecx
0x1000a87b: pop ecx
0x1000a87c: pop edi
0x1000a87d: pop ebp
0x1000a87e: pop ebx
0x1000a87f: add esp,30h
0x1000a882: ret 4
```

`sub_1000a680` is called by sub_1000b5a0. It delegates to sub_10001370, sub_10009c10, j_ORDINAL 5727 (6 callees).

12.158 sub_1000a890 at 0x1000a890 (67 instructions)

```
0x1000a890: push 0FFFFFFFFh
0x1000a892: push 100FD15Ch
0x1000a897: mov eax,fs:[0]
0x1000a89d: push eax
0x1000a89e: sub esp,64h
0x1000a8a1: mov eax,ds:[10136220h]
0x1000a8a6: xor eax,esp
0x1000a8a8: mov [esp+60h],eax
0x1000a8ac: push ebx
0x1000a8ad: push edi
0x1000a8ae: mov eax,ds:[10136220h]
0x1000a8b3: xor eax,esp
0x1000a8b5: push eax
0x1000a8b6: lea eax,[esp+70h]
0x1000a8ba: mov fs:[0],eax
0x1000a8c0: xor ebx,ebx
0x1000a8c2: mov dword ptr [esi],1012F7B8h
0x1000a8c8: mov [esp+0Ch],esi
0x1000a8cc: mov [esi+8],ebx
0x1000a8cf: mov [esi+0Ch],ebx
0x1000a8d2: mov [esi+10h],ebx
0x1000a8d5: lea edi,[esi+14h]
0x1000a8d8: mov [esp+78h],ebx
0x1000a8dc: mov [edi+4],ebx
0x1000a8df: mov dword ptr [edi],1012F744h
0x1000a8e5: lea ecx,[esi+1Ch]
0x1000a8e8: mov byte ptr [esp+78h],1
0x1000a8ed: call 100222A2h
0x1000a8f2: mov dword ptr [esi+30h],1012F790h
0x1000a8f9: mov [esi+34h],ebx
0x1000a8fc: lea ecx,[esp+10h]
0x1000a900: mov byte ptr [esp+78h],3
0x1000a905: call 10002070h
0x1000a90a: test eax,eax
0x1000a90c: je short 1000A931h
0x1000a90e: cmp edi,ebx
0x1000a910: je short 1000A91Eh
0x1000a912: cmp [edi+4],ebx
0x1000a915: je short 1000A91Eh
0x1000a917: mov ecx,edi
0x1000a919: call 10021FF6h
0x1000a91e: lea eax,[esp+10h]
0x1000a922: push eax
0x1000a923: call dword ptr ds:[100FE064h]
0x1000a929: push eax
0x1000a92a: mov ecx,edi
0x1000a92c: call 10021FFCh
0x1000a931: xor ecx,ecx
0x1000a933: mov [esi+28h],ecx
0x1000a936: lea ecx,[esi+38h]
0x1000a939: xor eax,eax
```

```
0x1000a93b: push ecx
0x1000a93c: mov dword ptr [esi+2Ch],0FFFFFFFFh
0x1000a943: mov [esi+24h],eax
0x1000a946: call dword ptr ds:[100FE5C8h]
0x1000a94c: mov [esi+48h],ebx
0x1000a94f: mov eax,esi
0x1000a951: mov ecx,[esp+70h]
0x1000a955: mov fs:[0],ecx
0x1000a95c: pop ecx
0x1000a95d: pop edi
0x1000a95e: pop ebx
0x1000a95f: mov ecx,[esp+60h]
0x1000a963: xor ecx,esp
0x1000a965: call 10022326h
0x1000a96a: add esp,70h
0x1000a96d: ret
```

`sub_1000a890` is called by sub_100032e0. It delegates to sub_10002070, j_ORDINAL 1925, j_ORDINAL 1271 (5 callees).

12.159 sub_1000aa40 at 0x1000aa40 (54 instructions)

```
0x1000aa40: push 0FFFFFFFFh
0x1000aa42: push 100FCCD8h
0x1000aa47: mov eax,fs:[0]
0x1000aa4d: push eax
0x1000aa4e: sub esp,10h
0x1000aa51: push esi
0x1000aa52: push edi
0x1000aa53: mov eax,ds:[10136220h]
0x1000aa58: xor eax,esp
0x1000aa5a: push eax
0x1000aa5b: lea eax,[esp+1Ch]
0x1000aa5f: mov fs:[0],eax
0x1000aa65: xor eax,eax
0x1000aa67: mov [esp+14h],eax
0x1000aa6b: mov dword ptr [esp+10h],1012F498h
0x1000aa73: mov [esp+24h],eax
0x1000aa77: mov eax,ds:[1013F360h]
0x1000aa7c: push 138Ch
0x1000aa81: push eax
0x1000aa82: call dword ptr ds:[100FE5E0h]
0x1000aa88: push eax
0x1000aa89: lea ecx,[esp+14h]
0x1000aa8d: call 10021FFCh
0x1000aa92: test eax,eax
0x1000aa94: jne short 1000AAC5h
0x1000aa96: mov dword ptr [esp+24h],0FFFFFFFFh
0x1000aa9e: lea ecx,[esp+10h]
0x1000aaa2: mov dword ptr [esp+10h],1012F498h
0x1000aaaa: call 10001370h
0x1000aaaf: xor eax,eax
0x1000aab1: mov ecx,[esp+1Ch]
0x1000aab5: mov fs:[0],ecx
0x1000aabc: pop ecx
0x1000aabd: pop edi
0x1000aabe: pop esi
0x1000aabf: add esp,1Ch
0x1000aac2: ret 4
0x1000aac5: mov esi,[esp+2Ch]
0x1000aac9: push 0FF000000h
0x1000aace: lea edi,[esp+14h]
0x1000aad2: call 1000AB10h
0x1000aad7: mov dword ptr [esp+24h],0FFFFFFFFh
0x1000aadf: mov ecx,edi
0x1000aae1: mov esi,eax
```

```
0x1000aae3: mov dword ptr [esp+10h],1012F498h
0x1000aaeb: call 10001370h
0x1000aaf0: mov eax,esi
0x1000aaf2: mov ecx,[esp+1Ch]
0x1000aaf6: mov fs:[0],ecx
0x1000aafd: pop ecx
0x1000aafe: pop edi
0x1000aaff: pop esi
0x1000ab00: add esp,1Ch
0x1000ab03: ret 4
```

`sub_1000aa40` is called by sub_10003880. It delegates to sub_10001370, sub_1000ab10, j_ORDINAL 1271 (3 callees).

12.160 sub_1000ab10 at 0x1000ab10 (66 instructions)

```
0x1000ab10: sub esp,1Ch
0x1000ab13: test edi,edi
0x1000ab15: push ebx
0x1000ab16: je near ptr 1000AB9Dh
0x1000ab1c: cmp dword ptr [edi+4],0
0x1000ab20: je short 1000AB9Dh
0x1000ab22: cmp dword ptr [esi+20h],0
0x1000ab26: je short 1000AB30h
0x1000ab28: lea ecx,[esi+1Ch]
0x1000ab2b: call 100222DEh
0x1000ab30: mov ecx,[edi+4]
0x1000ab33: lea eax,[esp+8]
0x1000ab37: push eax
0x1000ab38: push 18h
0x1000ab3a: push ecx
0x1000ab3b: call dword ptr ds:[100FE074h]
0x1000ab41: test eax,eax
0x1000ab43: je short 1000AB9Dh
0x1000ab45: movzx edx,word ptr [esp+1Ah]
0x1000ab4a: mov eax,[esp+10h]
0x1000ab4e: push 0
0x1000ab50: push 4
0x1000ab52: or edx,1
0x1000ab55: push edx
0x1000ab56: push eax
0x1000ab57: mov eax,[esp+1Ch]
0x1000ab5b: cdq
0x1000ab5c: and edx,3
0x1000ab5f: add eax,edx
0x1000ab61: sar eax,2
0x1000ab64: lea ebx,[esi+1Ch]
0x1000ab67: push eax
0x1000ab68: mov ecx,ebx
0x1000ab6a: call 100222E4h
0x1000ab6f: test eax,eax
0x1000ab71: je short 1000AB9Dh
0x1000ab73: mov edx,[esp+24h]
0x1000ab77: mov eax,[edi+4]
0x1000ab7a: mov ecx,[esi+20h]
0x1000ab7d: push edx
0x1000ab7e: push eax
0x1000ab7f: push ecx
0x1000ab80: call 10021FD2h
0x1000ab85: mov eax,[eax+94h]
0x1000ab8b: mov ecx,[eax]
0x1000ab8d: call 10009DB0h
0x1000ab92: test eax,eax
0x1000ab94: jge short 1000ABA6h
0x1000ab96: mov ecx,ebx
0x1000ab98: call 100222DEh
0x1000ab9d: xor eax,eax
```

```
0x1000ab9f: pop ebx
0x1000aba0: add esp,1Ch
0x1000aba3: ret 4
0x1000aba6: mov eax,[esp+0Ch]
0x1000abaa: mov ecx,[esp+10h]
0x1000abae: cdq
0x1000abaf: and edx,3
0x1000abb2: add eax,edx
0x1000abb4: sar eax,2
0x1000abb7: mov [esi+24h],eax
0x1000abba: mov [esi+28h],ecx
0x1000abbd: mov eax,1
0x1000abc2: pop ebx
0x1000abc3: add esp,1Ch
0x1000abc6: ret 4
```

`sub_1000ab10` is called by ?SetQuickBarBreakImage@CCustomTitle@@QAEHPAVCBitmap@@K@Z, sub_1000aa40. It delegates to sub_10009db0, j_ORDINAL 1079, j_ORDINAL 1921 (4 callees).

12.161 sub_1000abd0 at 0x1000abd0 (45 instructions)

```
0x1000abd0: push ebx
0x1000abd1: push ebp
0x1000abd2: push esi
0x1000abd3: push edi
0x1000abd4: mov edi,eax
0x1000abd6: xor esi,esi
0x1000abd8: mov ecx,[edi+8]
0x1000abdb: test ecx,ecx
0x1000abdd: jne short 1000ABE3h
0x1000abdf: xor eax,eax
0x1000abe1: jmp short 1000ABEBh
0x1000abe3: mov eax,[edi+0Ch]
0x1000abe6: sub eax,ecx
0x1000abe8: sar eax,2
0x1000abeb: cmp esi,eax
0x1000abed: jge short 1000AC26h
0x1000abef: test ecx,ecx
0x1000abf1: je short 1000ABFFh
0x1000abf3: mov eax,[edi+0Ch]
0x1000abf6: sub eax,ecx
0x1000abf8: sar eax,2
0x1000abfb: cmp esi,eax
0x1000abfd: jb short 1000AC05h
0x1000abff: call dword ptr ds:[100FE584h]
0x1000ac05: mov eax,[edi+8]
0x1000ac08: mov eax,[eax+esi*4]
0x1000ac0b: add eax,1Ch
0x1000ac0e: xor ecx,ecx
0x1000ac10: xor edx,edx
0x1000ac12: mov [eax],ecx
0x1000ac14: xor ebx,ebx
0x1000ac16: xor ebp,ebp
0x1000ac18: mov [eax+4],edx
0x1000ac1b: mov [eax+8],ebx
0x1000ac1e: mov [eax+0Ch],ebp
0x1000ac21: add esi,1
0x1000ac24: jmp short 1000ABD8h
0x1000ac26: add edi,38h
0x1000ac29: push edi
0x1000ac2a: call dword ptr ds:[100FE5C8h]
0x1000ac30: pop edi
0x1000ac31: pop esi
0x1000ac32: pop ebp
0x1000ac33: pop ebx
0x1000ac34: ret
```

`sub_1000abd0` is called by sub_100056c0, sub_1000b250. It calls nothing else.

12.162 sub_1000ac40 at 0x1000ac40 (32 instructions)

```
0x1000ac40: push ebp
0x1000ac41: mov ebp,ds:[100FE584h]
0x1000ac47: push edi
0x1000ac48: xor edi,edi
0x1000ac4a: lea ebx,[ebx]
0x1000ac50: mov ecx,[esi+8]
0x1000ac53: test ecx,ecx
0x1000ac55: jne short 1000AC5Bh
0x1000ac57: xor eax,eax
0x1000ac59: jmp short 1000AC63h
0x1000ac5b: mov eax,[esi+0Ch]
0x1000ac5e: sub eax,ecx
0x1000ac60: sar eax,2
0x1000ac63: cmp edi,eax
0x1000ac65: jge short 1000AC8Bh
0x1000ac67: test ecx,ecx
0x1000ac69: je short 1000AC77h
0x1000ac6b: mov eax,[esi+0Ch]
0x1000ac6e: sub eax,ecx
0x1000ac70: sar eax,2
0x1000ac73: cmp edi,eax
0x1000ac75: jb short 1000AC79h
0x1000ac77: call ebp
0x1000ac79: mov eax,[esi+8]
0x1000ac7c: mov ecx,[eax+edi*4]
0x1000ac7f: mov dword ptr [ecx+2Ch],0
0x1000ac86: add edi,1
0x1000ac89: jmp short 1000AC50h
0x1000ac8b: pop edi
0x1000ac8c: mov dword ptr [esi+48h],0
0x1000ac93: pop ebp
0x1000ac94: ret
```

`sub_1000ac40` is called by sub_100056c0, sub_1000b250. It calls nothing else.

12.163 sub_1000aca0 at 0x1000aca0 (82 instructions)

```
0x1000aca0: push 0FFFFFFFFh
0x1000aca2: push 100FD10Bh
0x1000aca7: mov eax,fs:[0]
0x1000acad: push eax
0x1000acae: push ebx
0x1000acaf: push ebp
0x1000acb0: push esi
0x1000acb1: push edi
0x1000acb2: mov eax,ds:[10136220h]
0x1000acb7: xor eax,esp
0x1000acb9: push eax
0x1000acba: lea eax,[esp+14h]
0x1000acbe: mov fs:[0],eax
0x1000acc4: mov ebx,ecx
0x1000acc6: mov ebp,[esp+24h]
0x1000acca: push 44h
0x1000accc: or edi,0FFFFFFFFh
0x1000accf: call 10021FDEh
0x1000acd4: add esp,4
0x1000acd7: mov [esp+24h],eax
0x1000acdb: xor esi,esi
0x1000acdd: cmp eax,esi
0x1000acdf: mov [esp+1Ch],esi
0x1000ace3: je short 1000ACEDh
0x1000ace5: push eax
```

```
0x1000ace6: call 1000A1C0h
0x1000aceb: mov esi,eax
0x1000aced: mov ecx,[esp+2Ch]
0x1000acf1: mov eax,[esp+28h]
0x1000acf5: mov [esp+1Ch],edi
0x1000acf9: push ecx
0x1000acfa: lea ecx,[esi+4]
0x1000acfd: mov [esp+28h],esi
0x1000ad01: mov [esi],eax
0x1000ad03: call dword ptr ds:[100FE198h]
0x1000ad09: mov edx,[esp+30h]
0x1000ad0d: push edx
0x1000ad0e: lea ecx,[esi+8]
0x1000ad11: call dword ptr ds:[100FE198h]
0x1000ad17: cmp dword ptr [esp+34h],0
0x1000ad1c: je short 1000AD36h
0x1000ad1e: push ebx
0x1000ad1f: call 1000A510h
0x1000ad24: test eax,eax
0x1000ad26: je short 1000AD4Fh
0x1000ad28: mov eax,[ebp+8]
0x1000ad2b: test eax,eax
0x1000ad2d: lea esi,[ebp+4]
0x1000ad30: jne short 1000AD60h
0x1000ad32: xor edi,edi
0x1000ad34: jmp short 1000AD68h
0x1000ad36: test ebx,ebx
0x1000ad38: jne short 1000AD45h
0x1000ad3a: lea eax,[ebp+14h]
0x1000ad3d: push eax
0x1000ad3e: call 1000A2D0h
0x1000ad43: jmp short 1000AD24h
0x1000ad45: push ebx
0x1000ad46: call 1000A3C0h
0x1000ad4b: test eax,eax
0x1000ad4d: jne short 1000AD28h
0x1000ad4f: push esi
0x1000ad50: call 1000A270h
0x1000ad55: push esi
0x1000ad56: call 10021FCCh
0x1000ad5b: add esp,4
0x1000ad5e: jmp short 1000AD71h
0x1000ad60: mov edi,[esi+8]
0x1000ad63: sub edi,eax
0x1000ad65: sar edi,2
0x1000ad68: lea ebx,[esp+24h]
0x1000ad6c: call 1000B9E0h
0x1000ad71: mov eax,edi
0x1000ad73: mov ecx,[esp+14h]
0x1000ad77: mov fs:[0],ecx
0x1000ad7e: pop ecx
0x1000ad7f: pop edi
0x1000ad80: pop esi
0x1000ad81: pop ebp
0x1000ad82: pop ebx
0x1000ad83: add esp,0Ch
0x1000ad86: ret 14h
```

`sub_1000aca0` is called by ?AddQuickButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0PB_W@Z. It delegates to sub_1000a1c0, sub_1000a270, sub_1000a2d0 (8 callees).

12.164 sub_1000ad90 at 0x1000ad90 (86 instructions)

```
0x1000ad90: push 0FFFFFFFFh
0x1000ad92: push 100FC70Bh
```

```
0x1000ad97: mov eax,fs:[0]
0x1000ad9d: push eax
0x1000ad9e: push ecx
0x1000ad9f: push ebx
0x1000ada0: push esi
0x1000ada1: push edi
0x1000ada2: mov eax,ds:[10136220h]
0x1000ada7: xor eax,esp
0x1000ada9: push eax
0x1000adaa: lea eax,[esp+14h]
0x1000adae: mov fs:[0],eax
0x1000adb4: mov edi,ecx
0x1000adb6: push 44h
0x1000adb8: or ebx,0FFFFFFFFh
0x1000adbb: call 10021FDEh
0x1000adc0: add esp,4
0x1000adc3: mov [esp+10h],eax
0x1000adc7: test eax,eax
0x1000adc9: mov dword ptr [esp+1Ch],0
0x1000add1: je short 1000ADDDh
0x1000add3: push eax
0x1000add4: call 1000A1C0h
0x1000add9: mov esi,eax
0x1000addb: jmp short 1000ADDFh
0x1000addd: xor esi,esi
0x1000addf: mov ecx,[esp+28h]
0x1000ade3: mov eax,[esp+24h]
0x1000ade7: mov [esp+1Ch],ebx
0x1000adeb: push ecx
0x1000adec: lea ecx,[esi+4]
0x1000adef: mov [esp+14h],esi
0x1000adf3: mov [esi],eax
0x1000adf5: call dword ptr ds:[100FE198h]
0x1000adfb: mov edx,[esp+2Ch]
0x1000adff: push edx
0x1000ae00: lea ecx,[esi+8]
0x1000ae03: call dword ptr ds:[100FE198h]
0x1000ae09: mov eax,[esp+30h]
0x1000ae0d: mov ecx,[esp+34h]
0x1000ae11: mov edx,[esp+38h]
0x1000ae15: mov [esi+34h],eax
0x1000ae18: mov eax,[esp+3Ch]
0x1000ae1c: mov [esi+38h],ecx
0x1000ae1f: lea ecx,[edi+14h]
0x1000ae22: mov [esi+3Ch],edx
0x1000ae25: push ecx
0x1000ae26: mov [esi+40h],eax
0x1000ae29: call 1000A2D0h
0x1000ae2e: test eax,eax
0x1000ae30: jne short 1000AE58h
0x1000ae32: push esi
0x1000ae33: call 1000A270h
0x1000ae38: push esi
0x1000ae39: call 10021FCCh
0x1000ae3e: add esp,4
0x1000ae41: mov eax,ebx
0x1000ae43: mov ecx,[esp+14h]
0x1000ae47: mov fs:[0],ecx
0x1000ae4e: pop ecx
0x1000ae4f: pop edi
0x1000ae50: pop esi
0x1000ae51: pop ebx
0x1000ae52: add esp,10h
0x1000ae55: ret 1Ch
0x1000ae58: mov eax,[edi+8]
0x1000ae5b: test eax,eax
0x1000ae5d: lea esi,[edi+4]
```

```
0x1000ae60: jne short 1000AE66h
0x1000ae62: xor edi,edi
0x1000ae64: jmp short 1000AE6Eh
0x1000ae66: mov edi,[esi+8]
0x1000ae69: sub edi,eax
0x1000ae6b: sar edi,2
0x1000ae6e: lea ebx,[esp+10h]
0x1000ae72: call 1000B9E0h
0x1000ae77: mov eax,edi
0x1000ae79: mov ecx,[esp+14h]
0x1000ae7d: mov fs:[0],ecx
0x1000ae84: pop ecx
0x1000ae85: pop edi
0x1000ae86: pop esi
0x1000ae87: pop ebx
0x1000ae88: add esp,10h
0x1000ae8b: ret 1Ch
```

`sub_1000ad90` is called by ?AddQuickButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0KKKK@Z. It delegates to sub_1000a1c0, sub_1000a270, sub_1000a2d0 (6 callees).

12.165 sub_1000ae90 at 0x1000ae90 (38 instructions)

```
0x1000ae90: push ebp
0x1000ae91: mov ebp,ds:[100FE584h]
0x1000ae97: push edi
0x1000ae98: xor edi,edi
0x1000ae9a: lea ebx,[ebx]
0x1000aea0: mov ecx,[esi+8]
0x1000aea3: test ecx,ecx
0x1000aea5: je short 1000AEF0h
0x1000aea7: mov eax,[esi+0Ch]
0x1000aeaa: sub eax,ecx
0x1000aeac: sar eax,2
0x1000aeaf: cmp edi,eax
0x1000aeb1: jae short 1000AEF0h
0x1000aeb3: test ecx,ecx
0x1000aeb5: je short 1000AEC3h
0x1000aeb7: mov eax,[esi+0Ch]
0x1000aeba: sub eax,ecx
0x1000aebc: sar eax,2
0x1000aebf: cmp edi,eax
0x1000aec1: jb short 1000AEC5h
0x1000aec3: call ebp
0x1000aec5: mov eax,[esi+8]
0x1000aec8: mov eax,[eax+edi*4]
0x1000aecb: cmp [eax],ebx
0x1000aecd: je short 1000AED4h
0x1000aecf: add edi,1
0x1000aed2: jmp short 1000AEA0h
0x1000aed4: mov ecx,[esp+0Ch]
0x1000aed8: mov edx,[esp+10h]
0x1000aedc: mov [eax+34h],ecx
0x1000aedf: mov ecx,[esp+14h]
0x1000aee3: mov [eax+38h],edx
0x1000aee6: mov edx,[esp+18h]
0x1000aeea: mov [eax+3Ch],ecx
0x1000aeed: mov [eax+40h],edx
0x1000aef0: pop edi
0x1000aef1: pop ebp
0x1000aef2: ret 10h
```

`sub_1000ae90` is called by ?SetQuickButtonTextColor@CCustomTitle@@QAEXIKKKK@Z. It calls nothing else.

12.166 sub_1000af00 at 0x1000af00 (68 instructions)

```
0x1000af00: push ecx
0x1000af01: test ebx,ebx
0x1000af03: push ebp
0x1000af04: push edi
0x1000af05: jge short 1000AF24h
0x1000af07: mov ecx,[esi+8]
0x1000af0a: test ecx,ecx
0x1000af0c: jne short 1000AF12h
0x1000af0e: xor eax,eax
0x1000af10: jmp short 1000AF1Ah
0x1000af12: mov eax,[esi+0Ch]
0x1000af15: sub eax,ecx
0x1000af17: sar eax,2
0x1000af1a: cmp ebx,eax
0x1000af1c: jl short 1000AF24h
0x1000af1e: xor eax,eax
0x1000af20: pop edi
0x1000af21: pop ebp
0x1000af22: pop ecx
0x1000af23: ret
0x1000af24: mov ecx,[esi+8]
0x1000af27: test ecx,ecx
0x1000af29: je short 1000AF37h
0x1000af2b: mov eax,[esi+0Ch]
0x1000af2e: sub eax,ecx
0x1000af30: sar eax,2
0x1000af33: cmp ebx,eax
0x1000af35: jb short 1000AF3Dh
0x1000af37: call dword ptr ds:[100FE584h]
0x1000af3d: mov eax,[esi+8]
0x1000af40: mov ebp,[eax+ebx*4]
0x1000af43: mov edi,eax
0x1000af45: cmp edi,[esi+0Ch]
0x1000af48: jbe short 1000AF50h
0x1000af4a: call dword ptr ds:[100FE584h]
0x1000af50: lea edi,[edi+ebx*4]
0x1000af53: cmp edi,[esi+0Ch]
0x1000af56: ja short 1000AF5Dh
0x1000af58: cmp edi,[esi+8]
0x1000af5b: jae short 1000AF63h
0x1000af5d: call dword ptr ds:[100FE584h]
0x1000af63: mov eax,[esi+0Ch]
0x1000af66: lea ecx,[edi+4]
0x1000af69: sub eax,ecx
0x1000af6b: sar eax,2
0x1000af6e: test eax,eax
0x1000af70: jle short 1000AF83h
0x1000af72: add eax,eax
0x1000af74: add eax,eax
0x1000af76: push eax
0x1000af77: push ecx
0x1000af78: push eax
0x1000af79: push edi
0x1000af7a: call dword ptr ds:[100FE57Ch]
0x1000af80: add esp,10h
0x1000af83: add dword ptr [esi+0Ch],0FFFFFFFCh
0x1000af87: test ebp,ebp
0x1000af89: je short 1000AF9Ah
0x1000af8b: push ebp
0x1000af8c: call 1000A270h
0x1000af91: push ebp
0x1000af92: call 10021FCCh
0x1000af97: add esp,4
0x1000af9a: pop edi
0x1000af9b: mov eax,1
0x1000afa0: pop ebp
```

```
0x1000afa1: pop ecx
0x1000afa2: ret
```

`sub_1000af00` is called by ?DeleteQuickButton@CCustomTitle@@QAEHH@Z. It delegates to sub_1000a270, j_ORDINAL 764 (2 callees).

12.167 sub_1000afb0 at 0x1000afb0 (62 instructions)

```
0x1000afb0: push 0FFFFFFFFh
0x1000afb2: push 100FCCA8h
0x1000afb7: mov eax,fs:[0]
0x1000afbd: push eax
0x1000afbe: push ecx
0x1000afbf: push ebx
0x1000afc0: push esi
0x1000afc1: push edi
0x1000afc2: mov eax,ds:[10136220h]
0x1000afc7: xor eax,esp
0x1000afc9: push eax
0x1000afca: lea eax,[esp+14h]
0x1000afce: mov fs:[0],eax
0x1000afd4: mov ebx,ecx
0x1000afd6: xor edi,edi
0x1000afd8: add ebx,4
0x1000afdb: jmp short 1000AFE0h
0x1000afe0: mov ecx,[ebx+4]
0x1000afe3: test ecx,ecx
0x1000afe5: jne short 1000AFEBh
0x1000afe7: xor eax,eax
0x1000afe9: jmp short 1000AFF3h
0x1000afeb: mov eax,[ebx+8]
0x1000afee: sub eax,ecx
0x1000aff0: sar eax,2
0x1000aff3: cmp edi,eax
0x1000aff5: jge short 1000B053h
0x1000aff7: test ecx,ecx
0x1000aff9: je short 1000B007h
0x1000affb: mov eax,[ebx+8]
0x1000affe: sub eax,ecx
0x1000b000: sar eax,2
0x1000b003: cmp edi,eax
0x1000b005: jb short 1000B00Dh
0x1000b007: call dword ptr ds:[100FE584h]
0x1000b00d: mov eax,[ebx+4]
0x1000b010: mov esi,[eax+edi*4]
0x1000b013: test esi,esi
0x1000b015: mov [esp+10h],esi
0x1000b019: je short 1000B04Eh
0x1000b01b: lea ecx,[esi+0Ch]
0x1000b01e: mov dword ptr [esp+1Ch],1
0x1000b026: call 100222A8h
0x1000b02b: lea ecx,[esi+8]
0x1000b02e: call dword ptr ds:[100FE374h]
0x1000b034: lea ecx,[esi+4]
0x1000b037: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000b03f: call dword ptr ds:[100FE374h]
0x1000b045: push esi
0x1000b046: call 10021FCCh
0x1000b04b: add esp,4
0x1000b04e: add edi,1
0x1000b051: jmp short 1000AFE0h
0x1000b053: call 1000BA40h
0x1000b058: mov ecx,[esp+14h]
0x1000b05c: mov fs:[0],ecx
0x1000b063: pop ecx
0x1000b064: pop edi
```

```
0x1000b065: pop esi
0x1000b066: pop ebx
0x1000b067: add esp,10h
0x1000b06a: ret
```

`sub_1000afb0` is called by ?RemoveAllQuickButtons@CCustomTitle@@QAEXXZ. It delegates to sub_1000ba40, j_ORDINAL 764, j_ORDINAL 651 (3 callees).

12.168 sub_1000b070 at 0x1000b070 (81 instructions)

```
0x1000b070: sub esp,14h
0x1000b073: push ebx
0x1000b074: push ebp
0x1000b075: push esi
0x1000b076: push edi
0x1000b077: mov esi,eax
0x1000b079: xor edi,edi
0x1000b07b: jmp short 1000B080h
0x1000b080: mov ecx,[esi+8]
0x1000b083: test ecx,ecx
0x1000b085: jne short 1000B08Bh
0x1000b087: xor eax,eax
0x1000b089: jmp short 1000B093h
0x1000b08b: mov eax,[esi+0Ch]
0x1000b08e: sub eax,ecx
0x1000b090: sar eax,2
0x1000b093: cmp edi,eax
0x1000b095: jge near ptr 1000B130h
0x1000b09b: mov eax,[esi+2Ch]
0x1000b09e: test eax,eax
0x1000b0a0: jl short 1000B0A6h
0x1000b0a2: cmp edi,eax
0x1000b0a4: je short 1000B0FDh
0x1000b0a6: test ecx,ecx
0x1000b0a8: mov eax,[esp+28h]
0x1000b0ac: mov ebx,[eax]
0x1000b0ae: mov ebp,[eax+4]
0x1000b0b1: je short 1000B0BFh
0x1000b0b3: mov eax,[esi+0Ch]
0x1000b0b6: sub eax,ecx
0x1000b0b8: sar eax,2
0x1000b0bb: cmp edi,eax
0x1000b0bd: jb short 1000B0C5h
0x1000b0bf: call dword ptr ds:[100FE584h]
0x1000b0c5: mov eax,[esi+8]
0x1000b0c8: mov eax,[eax+edi*4]
0x1000b0cb: mov ecx,[eax+1Ch]
0x1000b0ce: mov [esp+10h],ecx
0x1000b0d2: mov edx,[eax+20h]
0x1000b0d5: mov [esp+14h],edx
0x1000b0d9: mov ecx,[eax+24h]
0x1000b0dc: push ebp
0x1000b0dd: mov [esp+1Ch],ecx
0x1000b0e1: mov edx,[eax+28h]
0x1000b0e4: push ebx
0x1000b0e5: lea eax,[esp+18h]
0x1000b0e9: push eax
0x1000b0ea: mov [esp+28h],edx
0x1000b0ee: call dword ptr ds:[100FE610h]
0x1000b0f4: test eax,eax
0x1000b0f6: jne short 1000B116h
0x1000b0f8: add edi,1
0x1000b0fb: jmp short 1000B080h
0x1000b0fd: mov eax,[esp+28h]
0x1000b101: mov ecx,[eax+4]
0x1000b104: mov edx,[eax]
```

```
0x1000b106: push ecx
0x1000b107: push edx
0x1000b108: add esi,38h
0x1000b10b: push esi
0x1000b10c: call dword ptr ds:[100FE610h]
0x1000b112: test eax,eax
0x1000b114: je short 1000B130h
0x1000b116: test edi,edi
0x1000b118: mov eax,edi
0x1000b11a: jl short 1000B133h
0x1000b11c: mov ecx,[esp+2Ch]
0x1000b120: mov dword ptr [ecx],12h
0x1000b126: pop edi
0x1000b127: pop esi
0x1000b128: pop ebp
0x1000b129: pop ebx
0x1000b12a: add esp,14h
0x1000b12d: ret 8
0x1000b130: or eax,0FFFFFFFFh
0x1000b133: pop edi
0x1000b134: pop esi
0x1000b135: pop ebp
0x1000b136: pop ebx
0x1000b137: add esp,14h
0x1000b13a: ret 8
```

`sub_1000b070` is called by sub_10005b50. It calls nothing else.

12.169 sub_1000b140 at 0x1000b140 (101 instructions)

```
0x1000b140: sub esp,18h
0x1000b143: push ebx
0x1000b144: push ebp
0x1000b145: push esi
0x1000b146: push edi
0x1000b147: mov esi,eax
0x1000b149: call dword ptr ds:[100FE5C8h]
0x1000b14f: mov ebp,ds:[100FE6ACh]
0x1000b155: xor ebx,ebx
0x1000b157: mov ecx,[esi+8]
0x1000b15a: test ecx,ecx
0x1000b15c: jne short 1000B162h
0x1000b15e: xor eax,eax
0x1000b160: jmp short 1000B16Ah
0x1000b162: mov eax,[esi+0Ch]
0x1000b165: sub eax,ecx
0x1000b167: sar eax,2
0x1000b16a: cmp ebx,eax
0x1000b16c: jge near ptr 1000B23Ch
0x1000b172: mov eax,[esi+2Ch]
0x1000b175: test eax,eax
0x1000b177: jl short 1000B181h
0x1000b179: cmp ebx,eax
0x1000b17b: je near ptr 1000B203h
0x1000b181: test ecx,ecx
0x1000b183: je short 1000B191h
0x1000b185: mov eax,[esi+0Ch]
0x1000b188: sub eax,ecx
0x1000b18a: sar eax,2
0x1000b18d: cmp ebx,eax
0x1000b18f: jb short 1000B197h
0x1000b191: call dword ptr ds:[100FE584h]
0x1000b197: mov eax,[esi+8]
0x1000b19a: mov eax,[eax+ebx*4]
0x1000b19d: mov ecx,[eax+1Ch]
0x1000b1a0: mov [esp+10h],ecx
```

```
0x1000b1a4: mov edx,[eax+20h]
0x1000b1a7: mov [esp+14h],edx
0x1000b1ab: mov ecx,[eax+24h]
0x1000b1ae: mov [esp+18h],ecx
0x1000b1b2: mov edx,[eax+28h]
0x1000b1b5: lea eax,[esp+10h]
0x1000b1b9: push eax
0x1000b1ba: mov [esp+20h],edx
0x1000b1be: call ebp
0x1000b1c0: test eax,eax
0x1000b1c2: jne short 1000B1FBh
0x1000b1c4: push edi
0x1000b1c5: call ebp
0x1000b1c7: test eax,eax
0x1000b1c9: je short 1000B1EEh
0x1000b1cb: mov ecx,[esp+10h]
0x1000b1cf: mov edx,[esp+14h]
0x1000b1d3: mov eax,[esp+18h]
0x1000b1d7: mov [edi],ecx
0x1000b1d9: mov ecx,[esp+1Ch]
0x1000b1dd: mov [edi+4],edx
0x1000b1e0: mov [edi+8],eax
0x1000b1e3: mov [edi+0Ch],ecx
0x1000b1e6: add ebx,1
0x1000b1e9: jmp near ptr 1000B157h
0x1000b1ee: lea edx,[esp+10h]
0x1000b1f2: push edx
0x1000b1f3: push edi
0x1000b1f4: push edi
0x1000b1f5: call dword ptr ds:[100FE5B8h]
0x1000b1fb: add ebx,1
0x1000b1fe: jmp near ptr 1000B157h
0x1000b203: add esi,38h
0x1000b206: push esi
0x1000b207: call ebp
0x1000b209: test eax,eax
0x1000b20b: jne short 1000B23Ch
0x1000b20d: push edi
0x1000b20e: call ebp
0x1000b210: test eax,eax
0x1000b212: je short 1000B233h
0x1000b214: mov eax,[esi]
0x1000b216: mov [edi],eax
0x1000b218: mov ecx,[esi+4]
0x1000b21b: mov [edi+4],ecx
0x1000b21e: mov edx,[esi+8]
0x1000b221: mov [edi+8],edx
0x1000b224: mov eax,[esi+0Ch]
0x1000b227: mov [edi+0Ch],eax
0x1000b22a: mov eax,edi
0x1000b22c: pop esi
0x1000b22d: pop ebp
0x1000b22e: pop ebx
0x1000b22f: add esp,18h
0x1000b232: ret
0x1000b233: push esi
0x1000b234: push edi
0x1000b235: push edi
0x1000b236: call dword ptr ds:[100FE5B8h]
0x1000b23c: pop esi
0x1000b23d: pop ebp
0x1000b23e: mov eax,edi
0x1000b240: pop ebx
0x1000b241: add esp,18h
0x1000b244: ret
```

`sub_1000b140` is called by sub_10004910, sub_10004d40, sub_100056c0. It calls nothing else.

12.170 sub_1000b250 at 0x1000b250 (288 instructions)

```
0x1000b250: sub esp,24h
0x1000b253: push ebx
0x1000b254: push ebp
0x1000b255: mov ebx,eax
0x1000b257: push esi
0x1000b258: mov eax,edi
0x1000b25a: call 1000ABD0h
0x1000b25f: mov esi,edi
0x1000b261: call 1000AC40h
0x1000b266: cmp dword ptr [edi+2Ch],0
0x1000b26a: jl short 1000B285h
0x1000b26c: lea ecx,[edi+30h]
0x1000b26f: test ecx,ecx
0x1000b271: mov dword ptr [edi+2Ch],0FFFFFFFFh
0x1000b278: je short 1000B285h
0x1000b27a: cmp dword ptr [ecx+4],0
0x1000b27e: je short 1000B285h
0x1000b280: call 100222C0h
0x1000b285: mov ebp,[esp+34h]
0x1000b289: mov eax,[ebp]
0x1000b28c: mov ecx,[ebp+4]
0x1000b28f: mov edx,[ebp+8]
0x1000b292: mov [ebx],eax
0x1000b294: mov eax,[ebp+0Ch]
0x1000b297: mov [ebx+4],ecx
0x1000b29a: mov [ebx+8],edx
0x1000b29d: mov [ebx+0Ch],eax
0x1000b2a0: mov ecx,[ebx]
0x1000b2a2: mov [ebx+8],ecx
0x1000b2a5: xor esi,esi
0x1000b2a7: mov ecx,[edi+8]
0x1000b2aa: test ecx,ecx
0x1000b2ac: jne short 1000B2B2h
0x1000b2ae: xor eax,eax
0x1000b2b0: jmp short 1000B2BAh
0x1000b2b2: mov eax,[edi+0Ch]
0x1000b2b5: sub eax,ecx
0x1000b2b7: sar eax,2
0x1000b2ba: cmp esi,eax
0x1000b2bc: jge near ptr 1000B43Bh
0x1000b2c2: test ecx,ecx
0x1000b2c4: je short 1000B2D2h
0x1000b2c6: mov eax,[edi+0Ch]
0x1000b2c9: sub eax,ecx
0x1000b2cb: sar eax,2
0x1000b2ce: cmp esi,eax
0x1000b2d0: jb short 1000B2D8h
0x1000b2d2: call dword ptr ds:[100FE584h]
0x1000b2d8: mov edx,[edi+8]
0x1000b2db: mov ecx,[edx+esi*4]
0x1000b2de: mov eax,[ecx+18h]
0x1000b2e1: mov edx,[ecx+14h]
0x1000b2e4: mov [esp+14h],eax
0x1000b2e8: mov eax,[ebx+8]
0x1000b2eb: add edx,eax
0x1000b2ed: cmp edx,[ebp+8]
0x1000b2f0: jg near ptr 1000B388h
0x1000b2f6: mov edx,[ebx]
0x1000b2f8: mov ebp,[ebx+4]
0x1000b2fb: mov [esp+20h],edx
0x1000b2ff: mov edx,eax
0x1000b301: mov [esp+28h],edx
0x1000b305: mov edx,[ebx+0Ch]
0x1000b308: add eax,1
0x1000b30b: mov [esp+20h],eax
0x1000b30f: mov [esp+2Ch],edx
```

```
0x1000b313: mov [esp+24h],ebp
0x1000b317: mov edx,[ecx+18h]
0x1000b31a: mov [esp+14h],edx
0x1000b31e: mov edx,[ecx+14h]
0x1000b321: add edx,eax
0x1000b323: mov [esp+28h],edx
0x1000b327: mov eax,[ecx+14h]
0x1000b32a: mov edx,[esp+34h]
0x1000b32e: mov [esp+10h],eax
0x1000b332: mov eax,[edx+0Ch]
0x1000b335: sub eax,[edx+4]
0x1000b338: sub eax,[ecx+18h]
0x1000b33b: add eax,1
0x1000b33e: cdq
0x1000b33f: sub eax,edx
0x1000b341: sar eax,1
0x1000b343: add ebp,eax
0x1000b345: mov [esp+24h],ebp
0x1000b349: mov eax,[ecx+18h]
0x1000b34c: mov edx,[ecx+14h]
0x1000b34f: add eax,ebp
0x1000b351: mov ebp,[esp+34h]
0x1000b355: mov [esp+2Ch],eax
0x1000b359: mov eax,[esp+20h]
0x1000b35d: mov [ecx+1Ch],eax
0x1000b360: mov [esp+10h],edx
0x1000b364: mov edx,[esp+24h]
0x1000b368: mov [ecx+20h],edx
0x1000b36b: mov eax,[esp+28h]
0x1000b36f: mov [ecx+24h],eax
0x1000b372: mov edx,[esp+2Ch]
0x1000b376: mov [ecx+28h],edx
0x1000b379: mov eax,[esp+28h]
0x1000b37d: mov [ebx+8],eax
0x1000b380: add esi,1
0x1000b383: jmp near ptr 1000B2A7h
0x1000b388: test esi,esi
0x1000b38a: jle short 1000B3DBh
0x1000b38c: mov ecx,[edi+24h]
0x1000b38f: add ecx,[ebx+8]
0x1000b392: cmp ecx,[ebp+8]
0x1000b395: jle short 1000B3DBh
0x1000b397: mov ecx,[edi+8]
0x1000b39a: sub esi,1
0x1000b39d: test ecx,ecx
0x1000b39f: je short 1000B3ADh
0x1000b3a1: mov eax,[edi+0Ch]
0x1000b3a4: sub eax,ecx
0x1000b3a6: sar eax,2
0x1000b3a9: cmp esi,eax
0x1000b3ab: jb short 1000B3B3h
0x1000b3ad: call dword ptr ds:[100FE584h]
0x1000b3b3: mov edx,[edi+8]
0x1000b3b6: mov eax,[edx+esi*4]
0x1000b3b9: mov edx,[eax+20h]
0x1000b3bc: mov ecx,[eax+1Ch]
0x1000b3bf: mov [esp+14h],edx
0x1000b3c3: mov edx,[eax+24h]
0x1000b3c6: mov eax,[eax+28h]
0x1000b3c9: add ecx,0FFFFFFFFh
0x1000b3cc: test esi,esi
0x1000b3ce: mov [esp+18h],edx
0x1000b3d2: mov [esp+1Ch],eax
0x1000b3d6: mov [ebx+8],ecx
0x1000b3d9: jg short 1000B38Ch
0x1000b3db: mov [edi+2Ch],esi
0x1000b3de: mov edx,[ebx+8]
```

```
0x1000b3e1: mov eax,[ebx+0Ch]
0x1000b3e4: mov ecx,[ebx]
0x1000b3e6: mov esi,[ebx+4]
0x1000b3e9: mov [esp+2Ch],eax
0x1000b3ed: mov ebp,edx
0x1000b3ef: mov [esp+28h],edx
0x1000b3f3: mov edx,[esp+34h]
0x1000b3f7: mov eax,[edx+0Ch]
0x1000b3fa: sub eax,[edx+4]
0x1000b3fd: mov [esp+20h],ecx
0x1000b401: sub eax,[edi+28h]
0x1000b404: mov ecx,[edi+24h]
0x1000b407: add eax,1
0x1000b40a: cdq
0x1000b40b: sub eax,edx
0x1000b40d: sar eax,1
0x1000b40f: add esi,eax
0x1000b411: mov eax,[edi+28h]
0x1000b414: add ecx,ebp
0x1000b416: mov [edi+38h],ebp
0x1000b419: mov [edi+3Ch],esi
0x1000b41c: add eax,esi
0x1000b41e: mov [edi+40h],ecx
0x1000b421: mov [esp+20h],ebp
0x1000b425: mov ebp,[esp+34h]
0x1000b429: mov [edi+44h],eax
0x1000b42c: mov [esp+28h],ecx
0x1000b430: mov [esp+24h],esi
0x1000b434: mov [esp+2Ch],eax
0x1000b438: mov [ebx+8],ecx
0x1000b43b: mov ecx,[ebx+8]
0x1000b43e: mov eax,[ebp+8]
0x1000b441: cmp ecx,eax
0x1000b443: je near ptr 1000B51Ah
0x1000b449: sub eax,ecx
0x1000b44b: push 0
0x1000b44d: mov ebp,eax
0x1000b44f: push ebp
0x1000b450: push ebx
0x1000b451: mov ebx,ds:[100FE6B0h]
0x1000b457: call ebx
0x1000b459: xor esi,esi
0x1000b45b: jmp short 1000B460h
0x1000b460: mov ecx,[edi+8]
0x1000b463: test ecx,ecx
0x1000b465: jne short 1000B46Bh
0x1000b467: xor eax,eax
0x1000b469: jmp short 1000B473h
0x1000b46b: mov eax,[edi+0Ch]
0x1000b46e: sub eax,ecx
0x1000b470: sar eax,2
0x1000b473: cmp esi,eax
0x1000b475: jge near ptr 1000B51Ah
0x1000b47b: mov eax,[edi+2Ch]
0x1000b47e: test eax,eax
0x1000b480: jl short 1000B48Ah
0x1000b482: cmp esi,eax
0x1000b484: je near ptr 1000B511h
0x1000b48a: test ecx,ecx
0x1000b48c: je short 1000B49Ah
0x1000b48e: mov eax,[edi+0Ch]
0x1000b491: sub eax,ecx
0x1000b493: sar eax,2
0x1000b496: cmp esi,eax
0x1000b498: jb short 1000B4A0h
0x1000b49a: call dword ptr ds:[100FE584h]
0x1000b4a0: mov ecx,[edi+8]
```

```
0x1000b4a3: mov eax,[ecx+esi*4]
0x1000b4a6: mov edx,[eax+1Ch]
0x1000b4a9: mov [esp+20h],edx
0x1000b4ad: mov ecx,[eax+20h]
0x1000b4b0: mov [esp+24h],ecx
0x1000b4b4: mov edx,[eax+24h]
0x1000b4b7: push 0
0x1000b4b9: mov [esp+2Ch],edx
0x1000b4bd: mov eax,[eax+28h]
0x1000b4c0: push ebp
0x1000b4c1: lea ecx,[esp+28h]
0x1000b4c5: push ecx
0x1000b4c6: mov [esp+38h],eax
0x1000b4ca: call ebx
0x1000b4cc: mov ecx,[edi+8]
0x1000b4cf: test ecx,ecx
0x1000b4d1: je short 1000B4DFh
0x1000b4d3: mov eax,[edi+0Ch]
0x1000b4d6: sub eax,ecx
0x1000b4d8: sar eax,2
0x1000b4db: cmp esi,eax
0x1000b4dd: jb short 1000B4E5h
0x1000b4df: call dword ptr ds:[100FE584h]
0x1000b4e5: mov edx,[edi+8]
0x1000b4e8: mov eax,[edx+esi*4]
0x1000b4eb: mov ecx,[esp+20h]
0x1000b4ef: add eax,1Ch
0x1000b4f2: mov [eax],ecx
0x1000b4f4: mov edx,[esp+24h]
0x1000b4f8: mov [eax+4],edx
0x1000b4fb: mov ecx,[esp+28h]
0x1000b4ff: mov [eax+8],ecx
0x1000b502: mov edx,[esp+2Ch]
0x1000b506: mov [eax+0Ch],edx
0x1000b509: add esi,1
0x1000b50c: jmp near ptr 1000B460h
0x1000b511: push 0
0x1000b513: push ebp
0x1000b514: lea eax,[edi+38h]
0x1000b517: push eax
0x1000b518: call ebx
0x1000b51a: cmp dword ptr [edi+2Ch],0
0x1000b51e: jl short 1000B58Fh
0x1000b520: call dword ptr ds:[100FE5B4h]
0x1000b526: push eax
0x1000b527: lea ecx,[edi+30h]
0x1000b52a: call 100222F0h
0x1000b52f: mov ebp,[edi+2Ch]
0x1000b532: mov ecx,[edi+8]
0x1000b535: test ecx,ecx
0x1000b537: jne short 1000B53Dh
0x1000b539: xor eax,eax
0x1000b53b: jmp short 1000B545h
0x1000b53d: mov eax,[edi+0Ch]
0x1000b540: sub eax,ecx
0x1000b542: sar eax,2
0x1000b545: cmp ebp,eax
0x1000b547: jge short 1000B58Fh
0x1000b549: test ecx,ecx
0x1000b54b: je short 1000B559h
0x1000b54d: mov eax,[edi+0Ch]
0x1000b550: sub eax,ecx
0x1000b552: sar eax,2
0x1000b555: cmp ebp,eax
0x1000b557: jb short 1000B55Fh
0x1000b559: call dword ptr ds:[100FE584h]
0x1000b55f: mov ecx,[edi+8]
```

```
0x1000b562: mov ebx,[ecx+ebp*4]
0x1000b565: lea esi,[esp+34h]
0x1000b569: mov eax,ebx
0x1000b56b: call 1000A180h
0x1000b570: mov edx,[eax]
0x1000b572: mov eax,[ebx]
0x1000b574: mov ecx,[edi+34h]
0x1000b577: push edx
0x1000b578: push eax
0x1000b579: push 0
0x1000b57b: push ecx
0x1000b57c: call dword ptr ds:[100FE5B0h]
0x1000b582: mov ecx,esi
0x1000b584: call dword ptr ds:[100FE374h]
0x1000b58a: add ebp,1
0x1000b58d: jmp short 1000B532h
0x1000b58f: pop esi
0x1000b590: pop ebp
0x1000b591: pop ebx
0x1000b592: add esp,24h
0x1000b595: ret 4
```

`sub_1000b250` is called by sub_100056c0. It delegates to sub_1000a180, sub_1000abd0, sub_1000ac40 (5 callees).

12.171 sub_1000b5a0 at 0x1000b5a0 (77 instructions)

```
0x1000b5a0: push ecx
0x1000b5a1: push ebx
0x1000b5a2: push ebp
0x1000b5a3: mov ebp,[esp+10h]
0x1000b5a7: push esi
0x1000b5a8: push edi
0x1000b5a9: mov edi,eax
0x1000b5ab: xor ebx,ebx
0x1000b5ad: lea ecx,[ecx]
0x1000b5b0: mov ecx,[edi+8]
0x1000b5b3: test ecx,ecx
0x1000b5b5: jne short 1000B5BBh
0x1000b5b7: xor eax,eax
0x1000b5b9: jmp short 1000B5C3h
0x1000b5bb: mov eax,[edi+0Ch]
0x1000b5be: sub eax,ecx
0x1000b5c0: sar eax,2
0x1000b5c3: cmp ebx,eax
0x1000b5c5: jge near ptr 1000B657h
0x1000b5cb: mov eax,[edi+2Ch]
0x1000b5ce: test eax,eax
0x1000b5d0: jl short 1000B5D6h
0x1000b5d2: cmp ebx,eax
0x1000b5d4: je short 1000B603h
0x1000b5d6: test ecx,ecx
0x1000b5d8: je short 1000B5E6h
0x1000b5da: mov eax,[edi+0Ch]
0x1000b5dd: sub eax,ecx
0x1000b5df: sar eax,2
0x1000b5e2: cmp ebx,eax
0x1000b5e4: jb short 1000B5ECh
0x1000b5e6: call dword ptr ds:[100FE584h]
0x1000b5ec: mov eax,[esp+1Ch]
0x1000b5f0: mov ecx,[edi+8]
0x1000b5f3: mov ecx,[ecx+ebx*4]
0x1000b5f6: push eax
0x1000b5f7: mov esi,ebp
0x1000b5f9: call 1000A680h
0x1000b5fe: add ebx,1
0x1000b601: jmp short 1000B5B0h
```

```
0x1000b603: lea ebx,[edi+38h]
0x1000b606: push ebx
0x1000b607: call dword ptr ds:[100FE6ACh]
0x1000b60d: test eax,eax
0x1000b60f: jne short 1000B657h
0x1000b611: lea eax,[edi+1Ch]
0x1000b614: test eax,eax
0x1000b616: je short 1000B657h
0x1000b618: cmp dword ptr [eax+4],0
0x1000b61c: je short 1000B657h
0x1000b61e: cmp dword ptr [esp+1Ch],0
0x1000b623: mov esi,[edi+48h]
0x1000b626: jne short 1000B62Dh
0x1000b628: mov esi,3
0x1000b62d: test ebp,ebp
0x1000b62f: mov ecx,[ebx]
0x1000b631: mov edx,[ebx+4]
0x1000b634: jne short 1000B63Ah
0x1000b636: xor eax,eax
0x1000b638: jmp short 1000B63Dh
0x1000b63a: mov eax,[ebp+4]
0x1000b63d: mov edi,[edi+20h]
0x1000b640: push edx
0x1000b641: push ecx
0x1000b642: push eax
0x1000b643: push esi
0x1000b644: push edi
0x1000b645: call 10021FD2h
0x1000b64a: mov edx,[eax+94h]
0x1000b650: mov ecx,[edx]
0x1000b652: call 10009C50h
0x1000b657: pop edi
0x1000b658: pop esi
0x1000b659: pop ebp
0x1000b65a: pop ebx
0x1000b65b: pop ecx
0x1000b65c: ret 8
```

`sub_1000b5a0` is called by sub_10005fd0. It delegates to sub_10009c50, sub_1000a680, j_ORDINAL 1079 (3 callees).

12.172 sub_1000b660 at 0x1000b660 (51 instructions)

```
0x1000b660: push ebx
0x1000b661: xor ebx,ebx
0x1000b663: cmp [esp+0Ch],ebx
0x1000b667: push ebp
0x1000b668: mov ebp,[esp+0Ch]
0x1000b66c: push esi
0x1000b66d: setne bl
0x1000b670: xor esi,esi
0x1000b672: mov ecx,[edi+8]
0x1000b675: test ecx,ecx
0x1000b677: jne short 1000B67Dh
0x1000b679: xor eax,eax
0x1000b67b: jmp short 1000B685h
0x1000b67d: mov eax,[edi+0Ch]
0x1000b680: sub eax,ecx
0x1000b682: sar eax,2
0x1000b685: cmp esi,eax
0x1000b687: jge short 1000B6D3h
0x1000b689: mov eax,[edi+2Ch]
0x1000b68c: test eax,eax
0x1000b68e: jl short 1000B694h
0x1000b690: cmp esi,eax
0x1000b692: je short 1000B6C4h
0x1000b694: test ecx,ecx
```

```
0x1000b696: je short 1000B6A4h
0x1000b698: mov eax,[edi+0Ch]
0x1000b69b: sub eax,ecx
0x1000b69d: sar eax,2
0x1000b6a0: cmp esi,eax
0x1000b6a2: jb short 1000B6AAh
0x1000b6a4: call dword ptr ds:[100FE584h]
0x1000b6aa: mov ecx,[edi+8]
0x1000b6ad: mov edx,[ecx+esi*4]
0x1000b6b0: xor eax,eax
0x1000b6b2: cmp esi,ebp
0x1000b6b4: setne al
0x1000b6b7: sub eax,1
0x1000b6ba: and eax,ebx
0x1000b6bc: mov [edx+2Ch],eax
0x1000b6bf: add esi,1
0x1000b6c2: jmp short 1000B672h
0x1000b6c4: xor eax,eax
0x1000b6c6: cmp esi,ebp
0x1000b6c8: setne al
0x1000b6cb: sub eax,1
0x1000b6ce: and eax,ebx
0x1000b6d0: mov [edi+48h],eax
0x1000b6d3: pop esi
0x1000b6d4: pop ebp
0x1000b6d5: pop ebx
0x1000b6d6: ret 8
```

`sub_1000b660` is called by sub_10004910. It calls nothing else.

12.173 sub_1000b6e0 at 0x1000b6e0 (52 instructions)

```
0x1000b6e0: neg eax
0x1000b6e2: push ebx
0x1000b6e3: sbb eax,eax
0x1000b6e5: push ebp
0x1000b6e6: mov ebp,[esp+0Ch]
0x1000b6ea: and eax,2
0x1000b6ed: push esi
0x1000b6ee: mov ebx,eax
0x1000b6f0: xor esi,esi
0x1000b6f2: mov ecx,[edi+8]
0x1000b6f5: test ecx,ecx
0x1000b6f7: jne short 1000B6FDh
0x1000b6f9: xor eax,eax
0x1000b6fb: jmp short 1000B705h
0x1000b6fd: mov eax,[edi+0Ch]
0x1000b700: sub eax,ecx
0x1000b702: sar eax,2
0x1000b705: cmp esi,eax
0x1000b707: jge short 1000B753h
0x1000b709: mov eax,[edi+2Ch]
0x1000b70c: test eax,eax
0x1000b70e: jl short 1000B714h
0x1000b710: cmp esi,eax
0x1000b712: je short 1000B744h
0x1000b714: test ecx,ecx
0x1000b716: je short 1000B724h
0x1000b718: mov eax,[edi+0Ch]
0x1000b71b: sub eax,ecx
0x1000b71d: sar eax,2
0x1000b720: cmp esi,eax
0x1000b722: jb short 1000B72Ah
0x1000b724: call dword ptr ds:[100FE584h]
0x1000b72a: mov ecx,[edi+8]
0x1000b72d: mov edx,[ecx+esi*4]
```

```
0x1000b730: xor eax,eax
0x1000b732: cmp esi,ebp
0x1000b734: setne al
0x1000b737: sub eax,1
0x1000b73a: and eax,ebx
0x1000b73c: mov [edx+2Ch],eax
0x1000b73f: add esi,1
0x1000b742: jmp short 1000B6F2h
0x1000b744: xor eax,eax
0x1000b746: cmp esi,ebp
0x1000b748: setne al
0x1000b74b: sub eax,1
0x1000b74e: and eax,ebx
0x1000b750: mov [edi+48h],eax
0x1000b753: pop esi
0x1000b754: pop ebp
0x1000b755: pop ebx
0x1000b756: ret 4
```

`sub_1000b6e0` is called by sub_10004d40. It calls nothing else.

12.174 sub_1000b760 at 0x1000b760 (43 instructions)

```
0x1000b760: push ebp
0x1000b761: push edi
0x1000b762: or ebp,0FFFFFFFFh
0x1000b765: xor edi,edi
0x1000b767: mov ecx,[esi+8]
0x1000b76a: test ecx,ecx
0x1000b76c: jne short 1000B772h
0x1000b76e: xor eax,eax
0x1000b770: jmp short 1000B77Ah
0x1000b772: mov eax,[esi+0Ch]
0x1000b775: sub eax,ecx
0x1000b777: sar eax,2
0x1000b77a: cmp edi,eax
0x1000b77c: jge short 1000B7BBh
0x1000b77e: mov eax,[esi+2Ch]
0x1000b781: test eax,eax
0x1000b783: jl short 1000B789h
0x1000b785: cmp edi,eax
0x1000b787: je short 1000B7B0h
0x1000b789: test ecx,ecx
0x1000b78b: je short 1000B799h
0x1000b78d: mov eax,[esi+0Ch]
0x1000b790: sub eax,ecx
0x1000b792: sar eax,2
0x1000b795: cmp edi,eax
0x1000b797: jb short 1000B79Fh
0x1000b799: call dword ptr ds:[100FE584h]
0x1000b79f: mov eax,[esi+8]
0x1000b7a2: mov ecx,[eax+edi*4]
0x1000b7a5: cmp dword ptr [ecx+2Ch],2
0x1000b7a9: je short 1000B7B6h
0x1000b7ab: add edi,1
0x1000b7ae: jmp short 1000B767h
0x1000b7b0: cmp dword ptr [esi+48h],2
0x1000b7b4: jne short 1000B7BBh
0x1000b7b6: mov eax,edi
0x1000b7b8: pop edi
0x1000b7b9: pop ebp
0x1000b7ba: ret
0x1000b7bb: pop edi
0x1000b7bc: mov eax,ebp
0x1000b7be: pop ebp
0x1000b7bf: ret
```

`sub_1000b760` is called by sub_10004d40. It calls nothing else.

12.175 sub_1000b7c0 at 0x1000b7c0 (43 instructions)

```
0x1000b7c0: push ebp
0x1000b7c1: push edi
0x1000b7c2: or ebp,0FFFFFFFFh
0x1000b7c5: xor edi,edi
0x1000b7c7: mov ecx,[esi+8]
0x1000b7ca: test ecx,ecx
0x1000b7cc: jne short 1000B7D2h
0x1000b7ce: xor eax,eax
0x1000b7d0: jmp short 1000B7DAh
0x1000b7d2: mov eax,[esi+0Ch]
0x1000b7d5: sub eax,ecx
0x1000b7d7: sar eax,2
0x1000b7da: cmp edi,eax
0x1000b7dc: jge short 1000B81Bh
0x1000b7de: mov eax,[esi+2Ch]
0x1000b7e1: test eax,eax
0x1000b7e3: jl short 1000B7E9h
0x1000b7e5: cmp edi,eax
0x1000b7e7: je short 1000B810h
0x1000b7e9: test ecx,ecx
0x1000b7eb: je short 1000B7F9h
0x1000b7ed: mov eax,[esi+0Ch]
0x1000b7f0: sub eax,ecx
0x1000b7f2: sar eax,2
0x1000b7f5: cmp edi,eax
0x1000b7f7: jb short 1000B7FFh
0x1000b7f9: call dword ptr ds:[100FE584h]
0x1000b7ff: mov eax,[esi+8]
0x1000b802: mov ecx,[eax+edi*4]
0x1000b805: cmp dword ptr [ecx+2Ch],1
0x1000b809: je short 1000B816h
0x1000b80b: add edi,1
0x1000b80e: jmp short 1000B7C7h
0x1000b810: cmp dword ptr [esi+48h],1
0x1000b814: jne short 1000B81Bh
0x1000b816: mov eax,edi
0x1000b818: pop edi
0x1000b819: pop ebp
0x1000b81a: ret
0x1000b81b: pop edi
0x1000b81c: mov eax,ebp
0x1000b81e: pop ebp
0x1000b81f: ret
```

`sub_1000b7c0` is called by sub_10004910. It calls nothing else.

12.176 sub_1000b820 at 0x1000b820 (66 instructions)

```
0x1000b820: sub esp,14h
0x1000b823: push ebx
0x1000b824: mov ebx,ds:[1013F38Ch]
0x1000b82a: push ebp
0x1000b82b: mov ebp,[esp+20h]
0x1000b82f: push edi
0x1000b830: mov edi,eax
0x1000b832: mov eax,[esi+2Ch]
0x1000b835: test eax,eax
0x1000b837: mov [esp+24h],ebx
0x1000b83b: jl short 1000B88Fh
0x1000b83d: cmp edi,eax
0x1000b83f: jne short 1000B88Fh
0x1000b841: mov ecx,[ebp+20h]
```

```
0x1000b844: mov edi,[esi+40h]
0x1000b847: mov ebx,[esi+44h]
0x1000b84a: lea eax,[esp+10h]
0x1000b84e: push eax
0x1000b84f: push ecx
0x1000b850: call dword ptr ds:[100FE674h]
0x1000b856: add ebx,[esp+14h]
0x1000b85a: add edi,[esp+10h]
0x1000b85e: push 0
0x1000b860: push ebp
0x1000b861: push ebx
0x1000b862: push edi
0x1000b863: push 180h
0x1000b868: lea ecx,[esi+30h]
0x1000b86b: call 10022248h
0x1000b870: test eax,eax
0x1000b872: je short 1000B8BEh
0x1000b874: mov edx,[esp+24h]
0x1000b878: push 0
0x1000b87a: push eax
0x1000b87b: mov eax,[ebp+20h]
0x1000b87e: push edx
0x1000b87f: push eax
0x1000b880: call dword ptr ds:[100FE620h]
0x1000b886: pop edi
0x1000b887: pop ebp
0x1000b888: pop ebx
0x1000b889: add esp,14h
0x1000b88c: ret 4
0x1000b88f: mov ecx,[esi+8]
0x1000b892: test ecx,ecx
0x1000b894: je short 1000B8A2h
0x1000b896: mov eax,[esi+0Ch]
0x1000b899: sub eax,ecx
0x1000b89b: sar eax,2
0x1000b89e: cmp edi,eax
0x1000b8a0: jb short 1000B8A8h
0x1000b8a2: call dword ptr ds:[100FE584h]
0x1000b8a8: mov ecx,[esi+8]
0x1000b8ab: mov edx,[ecx+edi*4]
0x1000b8ae: mov eax,[edx]
0x1000b8b0: mov ecx,[ebp+20h]
0x1000b8b3: push 0
0x1000b8b5: push eax
0x1000b8b6: push ebx
0x1000b8b7: push ecx
0x1000b8b8: call dword ptr ds:[100FE620h]
0x1000b8be: pop edi
0x1000b8bf: pop ebp
0x1000b8c0: pop ebx
0x1000b8c1: add esp,14h
0x1000b8c4: ret 4
```

`sub_1000b820` is called by sub_10004910. It delegates to j_ORDINAL 6140 (1 callees).

12.177 sub_1000b8d0 at 0x1000b8d0 (99 instructions)

```
0x1000b8d0: sub esp,10h
0x1000b8d3: push ebx
0x1000b8d4: push ebp
0x1000b8d5: mov ebp,ds:[100FE584h]
0x1000b8db: push esi
0x1000b8dc: push edi
0x1000b8dd: mov esi,eax
0x1000b8df: xor ebx,ebx
0x1000b8e1: mov ecx,[esi+8]
```

```
0x1000b8e4: test ecx,ecx
0x1000b8e6: jne short 1000B8ECh
0x1000b8e8: xor eax,eax
0x1000b8ea: jmp short 1000B8F4h
0x1000b8ec: mov eax,[esi+0Ch]
0x1000b8ef: sub eax,ecx
0x1000b8f1: sar eax,2
0x1000b8f4: cmp ebx,eax
0x1000b8f6: jge near ptr 1000B9C8h
0x1000b8fc: mov eax,[esi+2Ch]
0x1000b8ff: test eax,eax
0x1000b901: jl short 1000B907h
0x1000b903: cmp ebx,eax
0x1000b905: je short 1000B95Bh
0x1000b907: test ecx,ecx
0x1000b909: je short 1000B917h
0x1000b90b: mov eax,[esi+0Ch]
0x1000b90e: sub eax,ecx
0x1000b910: sar eax,2
0x1000b913: cmp ebx,eax
0x1000b915: jb short 1000B919h
0x1000b917: call ebp
0x1000b919: mov eax,[esi+8]
0x1000b91c: lea edi,[eax+ebx*4]
0x1000b91f: mov eax,[edi]
0x1000b921: mov ecx,[eax+1Ch]
0x1000b924: mov [esp+10h],ecx
0x1000b928: mov edx,[eax+20h]
0x1000b92b: mov [esp+14h],edx
0x1000b92f: mov ecx,[eax+24h]
0x1000b932: mov [esp+18h],ecx
0x1000b936: mov edx,[eax+28h]
0x1000b939: mov eax,[esp+2Ch]
0x1000b93d: mov ecx,[esp+28h]
0x1000b941: push eax
0x1000b942: mov [esp+20h],edx
0x1000b946: push ecx
0x1000b947: lea edx,[esp+18h]
0x1000b94b: push edx
0x1000b94c: call dword ptr ds:[100FE610h]
0x1000b952: test eax,eax
0x1000b954: jne short 1000B985h
0x1000b956: add ebx,1
0x1000b959: jmp short 1000B8E1h
0x1000b95b: mov ecx,[esp+2Ch]
0x1000b95f: mov edx,[esp+28h]
0x1000b963: mov edi,[esi+3Ch]
0x1000b966: mov ebx,[esi+40h]
0x1000b969: mov ebp,[esi+44h]
0x1000b96c: lea eax,[esi+38h]
0x1000b96f: mov esi,[eax]
0x1000b971: push ecx
0x1000b972: push edx
0x1000b973: push eax
0x1000b974: call dword ptr ds:[100FE610h]
0x1000b97a: test eax,eax
0x1000b97c: je short 1000B9C8h
0x1000b97e: mov eax,1000h
0x1000b983: jmp short 1000B99Ch
0x1000b985: mov eax,[edi]
0x1000b987: mov esi,[eax+1Ch]
0x1000b98a: mov edi,[eax+20h]
0x1000b98d: mov ebx,[eax+24h]
0x1000b990: mov ebp,[eax+28h]
0x1000b993: mov eax,[eax]
0x1000b995: add eax,1000h
0x1000b99a: js short 1000B9CBh
```

```
0x1000b99c: mov ecx,[esp+24h]
0x1000b9a0: mov [ecx+10h],esi
0x1000b9a3: xor edx,edx
0x1000b9a5: mov [ecx+14h],edi
0x1000b9a8: mov [ecx+18h],ebx
0x1000b9ab: mov [ecx+0Ch],eax
0x1000b9ae: mov [ecx+4],edx
0x1000b9b1: mov [ecx+20h],edx
0x1000b9b4: mov dword ptr [ecx+24h],0FFFFFFFFh
0x1000b9bb: mov [ecx+1Ch],ebp
0x1000b9be: pop edi
0x1000b9bf: pop esi
0x1000b9c0: pop ebp
0x1000b9c1: pop ebx
0x1000b9c2: add esp,10h
0x1000b9c5: ret 0Ch
0x1000b9c8: or eax,0FFFFFFFFh
0x1000b9cb: pop edi
0x1000b9cc: pop esi
0x1000b9cd: pop ebp
0x1000b9ce: pop ebx
0x1000b9cf: add esp,10h
0x1000b9d2: ret 0Ch
```

`sub_1000b8d0` is called by sub_100067c0. It calls nothing else.

12.178 sub_1000b9e0 at 0x1000b9e0 (39 instructions)

```
0x1000b9e0: mov ecx,[esi+4]
0x1000b9e3: sub esp,10h
0x1000b9e6: test ecx,ecx
0x1000b9e8: push edi
0x1000b9e9: jne short 1000B9EFh
0x1000b9eb: xor edx,edx
0x1000b9ed: jmp short 1000B9F7h
0x1000b9ef: mov edx,[esi+8]
0x1000b9f2: sub edx,ecx
0x1000b9f4: sar edx,2
0x1000b9f7: test ecx,ecx
0x1000b9f9: je short 1000BA19h
0x1000b9fb: mov eax,[esi+0Ch]
0x1000b9fe: sub eax,ecx
0x1000ba00: sar eax,2
0x1000ba03: cmp edx,eax
0x1000ba05: jae short 1000BA19h
0x1000ba07: mov eax,[esi+8]
0x1000ba0a: mov ecx,[ebx]
0x1000ba0c: mov [eax],ecx
0x1000ba0e: add eax,4
0x1000ba11: mov [esi+8],eax
0x1000ba14: pop edi
0x1000ba15: add esp,10h
0x1000ba18: ret
0x1000ba19: mov edi,[esi+8]
0x1000ba1c: cmp ecx,edi
0x1000ba1e: jbe short 1000BA26h
0x1000ba20: call dword ptr ds:[100FE584h]
0x1000ba26: push edi
0x1000ba27: push esi
0x1000ba28: push ebx
0x1000ba29: lea edx,[esp+14h]
0x1000ba2d: push edx
0x1000ba2e: mov eax,esi
0x1000ba30: call 100098C0h
0x1000ba35: pop edi
```

```
0x1000ba36: add esp,10h
0x1000ba39: ret
```

`sub_1000b9e0` is called by sub_1000aca0, sub_1000ad90, sub_1000d330, sub_1000d430. It delegates to sub_100098c0 (1 callees).

12.179 sub_1000ba40 at 0x1000ba40 (37 instructions)

```
0x1000ba40: push ebp
0x1000ba41: mov ebp,esp
0x1000ba43: and esp,0FFFFFFF8h
0x1000ba46: sub esp,8
0x1000ba49: push esi
0x1000ba4a: mov esi,[ebx+8]
0x1000ba4d: cmp [ebx+4],esi
0x1000ba50: push edi
0x1000ba51: jbe short 1000BA59h
0x1000ba53: call dword ptr ds:[100FE584h]
0x1000ba59: mov edi,[ebx+4]
0x1000ba5c: cmp edi,[ebx+8]
0x1000ba5f: jbe short 1000BA67h
0x1000ba61: call dword ptr ds:[100FE584h]
0x1000ba67: cmp edi,esi
0x1000ba69: je short 1000BA99h
0x1000ba6b: mov eax,[ebx+8]
0x1000ba6e: sub eax,esi
0x1000ba70: sar eax,2
0x1000ba73: test eax,eax
0x1000ba75: lea ecx,[eax*4]
0x1000ba7c: lea edx,[ecx+edi]
0x1000ba7f: mov [esp+0Ch],edx
0x1000ba83: jle short 1000BA92h
0x1000ba85: push ecx
0x1000ba86: push esi
0x1000ba87: push ecx
0x1000ba88: push edi
0x1000ba89: call dword ptr ds:[100FE57Ch]
0x1000ba8f: add esp,10h
0x1000ba92: mov eax,[esp+0Ch]
0x1000ba96: mov [ebx+8],eax
0x1000ba99: pop edi
0x1000ba9a: pop esi
0x1000ba9b: mov esp,ebp
0x1000ba9d: pop ebp
0x1000ba9e: ret
```

`sub_1000ba40` is called by ?RemoveAllToolButtons@CCustomTitle@@QAEXXZ, sub_10008570, sub_1000afb0. It calls nothing else.

12.180 sub_1000baa0 at 0x1000baa0 (9 instructions)

```
0x1000baa0: push 5
0x1000baa2: call dword ptr ds:[100FE038h]
0x1000baa8: mov ecx,[esp+4]
0x1000baac: xor edx,edx
0x1000baae: test eax,eax
0x1000bab0: setne dl
0x1000bab3: mov [ecx+4],eax
0x1000bab6: mov eax,edx
0x1000bab8: ret 4
```

`sub_1000baa0` is called by sub_1000cd90. It calls nothing else.

12.181 sub_1000bb00 at 0x1000bb00 (43 instructions)

```
0x1000bb00: push 0FFFFFFFFh
0x1000bb02: push 100FD0DCh
0x1000bb07: mov eax,fs:[0]
0x1000bb0d: push eax
0x1000bb0e: push ebx
0x1000bb0f: push esi
0x1000bb10: push edi
0x1000bb11: mov eax,ds:[10136220h]
0x1000bb16: xor eax,esp
0x1000bb18: push eax
0x1000bb19: lea eax,[esp+10h]
0x1000bb1d: mov fs:[0],eax
0x1000bb23: mov esi,[esp+20h]
0x1000bb27: lea edi,[esi+24h]
0x1000bb2a: mov ecx,edi
0x1000bb2c: mov dword ptr [esi],1012F804h
0x1000bb32: call dword ptr ds:[100FE368h]
0x1000bb38: xor ebx,ebx
0x1000bb3a: mov [esi+2Ch],ebx
0x1000bb3d: mov [esi+30h],ebx
0x1000bb40: mov [esi+34h],ebx
0x1000bb43: mov eax,1C2DFFh
0x1000bb48: mov [esi+44h],ebx
0x1000bb4b: mov dword ptr [esi+40h],1012F744h
0x1000bb52: mov [esi+4],eax
0x1000bb55: mov [esi+8],eax
0x1000bb58: mov [esi+0Ch],eax
0x1000bb5b: mov [esi+10h],eax
0x1000bb5e: lea eax,[esi+14h]
0x1000bb61: push eax
0x1000bb62: call dword ptr ds:[100FE5C8h]
0x1000bb68: mov ecx,edi
0x1000bb6a: call dword ptr ds:[100FE214h]
0x1000bb70: mov [esi+3Ch],bl
0x1000bb73: mov eax,esi
0x1000bb75: mov ecx,[esp+10h]
0x1000bb79: mov fs:[0],ecx
0x1000bb80: pop ecx
0x1000bb81: pop edi
0x1000bb82: pop esi
0x1000bb83: pop ebx
0x1000bb84: add esp,0Ch
0x1000bb87: ret 4
```

`sub_1000bb00` is called by sub_100032e0. It calls nothing else.

12.182 sub_1000bc60 at 0x1000bc60 (101 instructions)

```
0x1000bc60: push ebp
0x1000bc61: mov ebp,esp
0x1000bc63: and esp,0FFFFFFF8h
0x1000bc66: push 0FFFFFFFFh
0x1000bc68: push 100FCC17h
0x1000bc6d: mov eax,fs:[0]
0x1000bc73: push eax
0x1000bc74: sub esp,94h
0x1000bc7a: mov eax,ds:[10136220h]
0x1000bc7f: xor eax,esp
0x1000bc81: mov [esp+8Ch],eax
0x1000bc88: push ebx
0x1000bc89: push edi
0x1000bc8a: mov eax,ds:[10136220h]
0x1000bc8f: xor eax,esp
0x1000bc91: push eax
0x1000bc92: lea eax,[esp+0A0h]
```

```
0x1000bc99: mov fs:[0],eax
0x1000bc9f: xor ebx,ebx
0x1000bca1: push 1012F7D4h
0x1000bca6: lea eax,[esp+54h]
0x1000bcaa: push eax
0x1000bcab: mov [esp+40h],ebx
0x1000bcaf: mov [esp+44h],ebx
0x1000bcb3: mov [esp+48h],ebx
0x1000bcb7: mov [esp+4Ch],ebx
0x1000bcbb: mov [esp+50h],bl
0x1000bcbf: mov [esp+51h],bl
0x1000bcc3: mov [esp+52h],bl
0x1000bcc7: mov byte ptr [esp+53h],1
0x1000bccc: call dword ptr ds:[100FE104h]
0x1000bcd2: lea ecx,[esp+34h]
0x1000bcd6: push ecx
0x1000bcd7: mov [esp+50h],bl
0x1000bcdb: mov [esp+51h],bl
0x1000bcdf: mov [esp+52h],bl
0x1000bce3: mov [esp+53h],bl
0x1000bce7: mov dword ptr [esp+38h],0Dh
0x1000bcef: mov [esp+3Ch],ebx
0x1000bcf3: mov dword ptr [esp+48h],190h
0x1000bcfb: lea edi,[esi+40h]
0x1000bcfe: call dword ptr ds:[100FE064h]
0x1000bd04: push eax
0x1000bd05: mov ecx,edi
0x1000bd07: call 10021FFCh
0x1000bd0c: lea ecx,[esp+14h]
0x1000bd10: call 1002202Ch
0x1000bd15: push ebx
0x1000bd16: mov [esp+0ACh],ebx
0x1000bd1d: call dword ptr ds:[100FE060h]
0x1000bd23: push eax
0x1000bd24: lea ecx,[esp+18h]
0x1000bd28: call 10022038h
0x1000bd2d: push edi
0x1000bd2e: lea ecx,[esp+18h]
0x1000bd32: call 1002228Ah
0x1000bd37: push 1012F7F4h
0x1000bd3c: lea ecx,[esp+10h]
0x1000bd40: mov [esp+14h],eax
0x1000bd44: mov [esi+38h],ebx
0x1000bd47: call dword ptr ds:[100FE168h]
0x1000bd4d: mov byte ptr [esp+0A8h],1
0x1000bd55: mov eax,[esp+0Ch]
0x1000bd59: mov ecx,[eax-0Ch]
0x1000bd5c: push 420h
0x1000bd61: lea edx,[esp+28h]
0x1000bd65: push edx
0x1000bd66: push ecx
0x1000bd67: push eax
0x1000bd68: mov eax,[esp+24h]
0x1000bd6c: mov edx,[eax+70h]
0x1000bd6f: lea ecx,[esp+24h]
0x1000bd73: call edx
0x1000bd75: lea ecx,[esp+0Ch]
0x1000bd79: mov edi,eax
0x1000bd7b: mov [esp+0A8h],bl
0x1000bd82: call dword ptr ds:[100FE374h]
0x1000bd88: cmp edi,ebx
0x1000bd8a: je short 1000BD97h
0x1000bd8c: mov eax,[esp+2Ch]
0x1000bd90: sub eax,[esp+24h]
0x1000bd94: mov [esi+38h],eax
0x1000bd97: mov ecx,[esp+10h]
0x1000bd9b: push ecx
```

```
0x1000bd9c: lea ecx,[esp+18h]
0x1000bda0: call 1002228Ah
0x1000bda5: lea ecx,[esp+14h]
0x1000bda9: mov dword ptr [esp+0A8h],0FFFFFFFFh
0x1000bdb4: call 10022032h
0x1000bdb9: mov al,1
0x1000bdbb: mov ecx,[esp+0A0h]
0x1000bdc2: mov fs:[0],ecx
0x1000bdc9: pop ecx
0x1000bdca: pop edi
0x1000bdcb: pop ebx
0x1000bdcc: mov ecx,[esp+8Ch]
0x1000bdd3: xor ecx,esp
0x1000bdd5: call 10022326h
0x1000bdda: mov esp,ebp
0x1000bddc: pop ebp
0x1000bddd: ret
```

`sub_1000bc60` is called by sub_100032e0. It works with the text "Roboto-Regular". It delegates to j_ORDINAL 1271, j_ORDINAL 347, j_ORDINAL 602 (6 callees).

12.183 sub_1000bde0 at 0x1000bde0 (263 instructions)

```
0x1000bde0: push 0FFFFFFFFh
0x1000bde2: push 100FC9E9h
0x1000bde7: mov eax,fs:[0]
0x1000bded: push eax
0x1000bdee: sub esp,2Ch
0x1000bdf1: push ebx
0x1000bdf2: push ebp
0x1000bdf3: mov eax,ds:[10136220h]
0x1000bdf8: xor eax,esp
0x1000bdfa: push eax
0x1000bdfb: lea eax,[esp+38h]
0x1000bdff: mov fs:[0],eax
0x1000be05: mov ebx,[esp+48h]
0x1000be09: mov eax,[ebx+8]
0x1000be0c: sub eax,[ebx]
0x1000be0e: test eax,eax
0x1000be10: jle near ptr 1000C0D3h
0x1000be16: mov eax,[esi+2Ch]
0x1000be19: xor ebp,ebp
0x1000be1b: cmp eax,ebp
0x1000be1d: je near ptr 1000C0D3h
0x1000be23: mov ecx,[esi+30h]
0x1000be26: sub ecx,eax
0x1000be28: sar ecx,3
0x1000be2b: je near ptr 1000C0D3h
0x1000be31: mov [esp+10h],ebp
0x1000be35: mov dword ptr [esp+14h],0FFFFFFFFh
0x1000be3d: lea ecx,[ecx]
0x1000be40: mov ecx,[esi+2Ch]
0x1000be43: test ecx,ecx
0x1000be45: jne short 1000BE4Bh
0x1000be47: xor eax,eax
0x1000be49: jmp short 1000BE53h
0x1000be4b: mov eax,[esi+30h]
0x1000be4e: sub eax,ecx
0x1000be50: sar eax,3
0x1000be53: cmp ebp,eax
0x1000be55: jge short 1000BED6h
0x1000be57: test ebp,ebp
0x1000be59: jne short 1000BE61h
0x1000be5b: mov [esp+18h],ebp
0x1000be5f: jmp short 1000BE68h
0x1000be61: mov edx,[esi+38h]
```

```
0x1000be64: mov [esp+18h],edx
0x1000be68: test ecx,ecx
0x1000be6a: je short 1000BE78h
0x1000be6c: mov eax,[esi+30h]
0x1000be6f: sub eax,ecx
0x1000be71: sar eax,3
0x1000be74: cmp ebp,eax
0x1000be76: jb short 1000BE7Eh
0x1000be78: call dword ptr ds:[100FE584h]
0x1000be7e: mov eax,[esi+2Ch]
0x1000be81: mov ecx,[eax+ebp*8+4]
0x1000be85: add ecx,[esp+18h]
0x1000be89: mov edx,[ebx+8]
0x1000be8c: add ecx,[esp+10h]
0x1000be90: sub edx,[ebx]
0x1000be92: cmp ecx,edx
0x1000be94: jg short 1000BED6h
0x1000be96: test ebp,ebp
0x1000be98: jne short 1000BE9Eh
0x1000be9a: xor ebx,ebx
0x1000be9c: jmp short 1000BEA1h
0x1000be9e: mov ebx,[esi+38h]
0x1000bea1: mov ecx,eax
0x1000bea3: test ecx,ecx
0x1000bea5: je short 1000BEB3h
0x1000bea7: mov eax,[esi+30h]
0x1000beaa: sub eax,ecx
0x1000beac: sar eax,3
0x1000beaf: cmp ebp,eax
0x1000beb1: jb short 1000BEB9h
0x1000beb3: call dword ptr ds:[100FE584h]
0x1000beb9: mov eax,[esi+2Ch]
0x1000bebc: mov ecx,[eax+ebp*8+4]
0x1000bec0: add ecx,ebx
0x1000bec2: add [esp+10h],ecx
0x1000bec6: mov ebx,[esp+48h]
0x1000beca: mov [esp+14h],ebp
0x1000bece: add ebp,1
0x1000bed1: jmp near ptr 1000BE40h
0x1000bed6: cmp dword ptr [esp+14h],0
0x1000bedb: jl near ptr 1000C0D3h
0x1000bee1: mov edx,[edi]
0x1000bee3: mov edx,[edx+30h]
0x1000bee6: lea eax,[esi+40h]
0x1000bee9: push eax
0x1000beea: mov ecx,edi
0x1000beec: call edx
0x1000beee: push 1
0x1000bef0: mov ecx,edi
0x1000bef2: mov [esp+28h],eax
0x1000bef6: call 10022296h
0x1000befb: mov [esp+20h],eax
0x1000beff: mov eax,[edi+8]
0x1000bf02: push eax
0x1000bf03: call dword ptr ds:[100FE034h]
0x1000bf09: cmp dword ptr [esp+4Ch],0
0x1000bf0e: mov [esp+1Ch],eax
0x1000bf12: jne short 1000BF21h
0x1000bf14: mov edx,[edi]
0x1000bf16: mov eax,[esi+10h]
0x1000bf19: mov edx,[edx+38h]
0x1000bf1c: push eax
0x1000bf1d: mov ecx,edi
0x1000bf1f: call edx
0x1000bf21: mov eax,[ebx+4]
0x1000bf24: mov edx,[ebx+8]
0x1000bf27: mov ecx,[ebx]
```

```
0x1000bf29: mov [esp+2Ch],eax
0x1000bf2d: mov eax,[ebx+0Ch]
0x1000bf30: mov [esp+34h],eax
0x1000bf34: mov eax,edx
0x1000bf36: sub eax,ecx
0x1000bf38: sub eax,[esp+10h]
0x1000bf3c: mov [esp+30h],edx
0x1000bf40: cdq
0x1000bf41: sub eax,edx
0x1000bf43: sar eax,1
0x1000bf45: add ecx,eax
0x1000bf47: xor ebx,ebx
0x1000bf49: cmp [esp+14h],ebx
0x1000bf4d: mov [esp+28h],ecx
0x1000bf51: jl near ptr 1000C0ABh
0x1000bf57: test ebx,ebx
0x1000bf59: je short 1000BFC4h
0x1000bf5b: cmp dword ptr [esp+4Ch],0
0x1000bf60: je short 1000BF6Fh
0x1000bf62: mov edx,[edi]
0x1000bf64: mov eax,[esi+0Ch]
0x1000bf67: mov edx,[edx+38h]
0x1000bf6a: push eax
0x1000bf6b: mov ecx,edi
0x1000bf6d: call edx
0x1000bf6f: mov eax,[esi+38h]
0x1000bf72: add eax,[esp+28h]
0x1000bf76: push 1012F7F4h
0x1000bf7b: lea ecx,[esp+1Ch]
0x1000bf7f: mov [esp+34h],eax
0x1000bf83: call dword ptr ds:[100FE168h]
0x1000bf89: mov dword ptr [esp+40h],0
0x1000bf91: mov eax,[esp+18h]
0x1000bf95: mov ecx,[eax-0Ch]
0x1000bf98: mov edx,[edi]
0x1000bf9a: push 24h
0x1000bf9c: lea ebp,[esp+2Ch]
0x1000bfa0: push ebp
0x1000bfa1: push ecx
0x1000bfa2: push eax
0x1000bfa3: mov eax,[edx+70h]
0x1000bfa6: mov ecx,edi
0x1000bfa8: call eax
0x1000bfaa: lea ecx,[esp+18h]
0x1000bfae: mov dword ptr [esp+40h],0FFFFFFFFh
0x1000bfb6: call dword ptr ds:[100FE374h]
0x1000bfbc: mov ecx,[esp+30h]
0x1000bfc0: mov [esp+28h],ecx
0x1000bfc4: cmp dword ptr [esp+4Ch],0
0x1000bfc9: je short 1000BFF1h
0x1000bfcb: test ebx,ebx
0x1000bfcd: jne short 1000BFDAh
0x1000bfcf: mov edx,[edi]
0x1000bfd1: mov eax,[esi+4]
0x1000bfd4: mov edx,[edx+38h]
0x1000bfd7: push eax
0x1000bfd8: jmp short 1000BFEDh
0x1000bfda: cmp ebx,1
0x1000bfdd: mov eax,[edi]
0x1000bfdf: mov edx,[eax+38h]
0x1000bfe2: jne short 1000BFE9h
0x1000bfe4: mov ecx,[esi+8]
0x1000bfe7: jmp short 1000BFECh
0x1000bfe9: mov ecx,[esi+0Ch]
0x1000bfec: push ecx
0x1000bfed: mov ecx,edi
0x1000bfef: call edx
```

```
0x1000bff1: mov ecx,[esi+2Ch]
0x1000bff4: test ecx,ecx
0x1000bff6: je short 1000C004h
0x1000bff8: mov eax,[esi+30h]
0x1000bffb: sub eax,ecx
0x1000bffd: sar eax,3
0x1000c000: cmp ebx,eax
0x1000c002: jb short 1000C00Ah
0x1000c004: call dword ptr ds:[100FE584h]
0x1000c00a: mov eax,[esi+2Ch]
0x1000c00d: mov ecx,[eax+ebx*8+4]
0x1000c011: add ecx,[esp+28h]
0x1000c015: cmp byte ptr [esi+3Ch],0
0x1000c019: lea ebp,[ebx*8]
0x1000c020: mov [esp+30h],ecx
0x1000c024: je short 1000C062h
0x1000c026: mov eax,[esp+48h]
0x1000c02a: mov edx,[eax]
0x1000c02c: mov ecx,[eax+4]
0x1000c02f: mov [esp+28h],edx
0x1000c033: mov edx,[eax+8]
0x1000c036: mov eax,[eax+0Ch]
0x1000c039: mov [esp+2Ch],ecx
0x1000c03d: mov ecx,[esi+2Ch]
0x1000c040: test ecx,ecx
0x1000c042: mov [esp+30h],edx
0x1000c046: mov [esp+34h],eax
0x1000c04a: je short 1000C058h
0x1000c04c: mov eax,[esi+30h]
0x1000c04f: sub eax,ecx
0x1000c051: sar eax,3
0x1000c054: cmp ebx,eax
0x1000c056: jb short 1000C05Eh
0x1000c058: call dword ptr ds:[100FE584h]
0x1000c05e: push 28h
0x1000c060: jmp short 1000C07Ch
0x1000c062: mov ecx,eax
0x1000c064: test ecx,ecx
0x1000c066: je short 1000C074h
0x1000c068: mov eax,[esi+30h]
0x1000c06b: sub eax,ecx
0x1000c06d: sar eax,3
0x1000c070: cmp ebx,eax
0x1000c072: jb short 1000C07Ah
0x1000c074: call dword ptr ds:[100FE584h]
0x1000c07a: push 24h
0x1000c07c: mov eax,[esi+2Ch]
0x1000c07f: mov edx,[edi]
0x1000c081: mov edx,[edx+70h]
0x1000c084: add eax,ebp
0x1000c086: mov eax,[eax]
0x1000c088: lea ecx,[esp+2Ch]
0x1000c08c: push ecx
0x1000c08d: mov ecx,[eax-0Ch]
0x1000c090: push ecx
0x1000c091: push eax
0x1000c092: mov ecx,edi
0x1000c094: call edx
0x1000c096: mov eax,[esp+30h]
0x1000c09a: add ebx,1
0x1000c09d: cmp ebx,[esp+14h]
0x1000c0a1: mov [esp+28h],eax
0x1000c0a5: jle near ptr 1000BF57h
0x1000c0ab: mov edx,[edi]
0x1000c0ad: mov eax,[esp+1Ch]
0x1000c0b1: mov edx,[edx+38h]
0x1000c0b4: push eax
```

```
0x1000c0b5: mov ecx,edi
0x1000c0b7: call edx
0x1000c0b9: mov eax,[esp+20h]
0x1000c0bd: push eax
0x1000c0be: mov ecx,edi
0x1000c0c0: call 10022296h
0x1000c0c5: mov edx,[edi]
0x1000c0c7: mov eax,[esp+24h]
0x1000c0cb: mov edx,[edx+30h]
0x1000c0ce: push eax
0x1000c0cf: mov ecx,edi
0x1000c0d1: call edx
0x1000c0d3: mov ecx,[esp+38h]
0x1000c0d7: mov fs:[0],ecx
0x1000c0de: pop ecx
0x1000c0df: pop ebp
0x1000c0e0: pop ebx
0x1000c0e1: add esp,38h
0x1000c0e4: ret 8
```

`sub_1000bde0` is called by sub_10005fd0. It delegates to j_ORDINAL 5727 (1 callees).

12.184 sub_1000c0f0 at 0x1000c0f0 (163 instructions)

```
0x1000c0f0: push ebp
0x1000c0f1: mov ebp,esp
0x1000c0f3: and esp,0FFFFFFF8h
0x1000c0f6: push 0FFFFFFFFh
0x1000c0f8: push 100FCBD9h
0x1000c0fd: mov eax,fs:[0]
0x1000c103: push eax
0x1000c104: sub esp,48h
0x1000c107: push ebx
0x1000c108: push esi
0x1000c109: push edi
0x1000c10a: mov eax,ds:[10136220h]
0x1000c10f: xor eax,esp
0x1000c111: push eax
0x1000c112: lea eax,[esp+58h]
0x1000c116: mov fs:[0],eax
0x1000c11c: mov edi,ecx
0x1000c11e: mov ebx,[edi+30h]
0x1000c121: cmp [edi+2Ch],ebx
0x1000c124: lea esi,[edi+28h]
0x1000c127: jbe short 1000C12Fh
0x1000c129: call dword ptr ds:[100FE584h]
0x1000c12f: mov eax,[esi+4]
0x1000c132: cmp eax,[esi+8]
0x1000c135: mov [esp+18h],eax
0x1000c139: jbe short 1000C145h
0x1000c13b: call dword ptr ds:[100FE584h]
0x1000c141: mov eax,[esp+18h]
0x1000c145: push ebx
0x1000c146: push esi
0x1000c147: push eax
0x1000c148: push esi
0x1000c149: lea eax,[esp+2Ch]
0x1000c14d: push eax
0x1000c14e: push esi
0x1000c14f: call 1000C3F0h
0x1000c154: lea ecx,[esp+24h]
0x1000c158: call dword ptr ds:[100FE368h]
0x1000c15e: lea ecx,[edi+24h]
0x1000c161: push ecx
0x1000c162: lea ecx,[esp+18h]
0x1000c166: mov dword ptr [esp+64h],0
```

```
0x1000c16e: call dword ptr ds:[100FE220h]
0x1000c174: push 0
0x1000c176: lea ecx,[esp+18h]
0x1000c17a: mov byte ptr [esp+64h],1
0x1000c17f: call dword ptr ds:[100FE394h]
0x1000c185: lea ecx,[esp+34h]
0x1000c189: mov ebx,eax
0x1000c18b: call 1002202Ch
0x1000c190: push 0
0x1000c192: mov byte ptr [esp+64h],2
0x1000c197: call dword ptr ds:[100FE060h]
0x1000c19d: push eax
0x1000c19e: lea ecx,[esp+38h]
0x1000c1a2: call 10022038h
0x1000c1a7: add edi,40h
0x1000c1aa: push edi
0x1000c1ab: lea ecx,[esp+38h]
0x1000c1af: call 1002228Ah
0x1000c1b4: push 1012F7FCh
0x1000c1b9: push ebx
0x1000c1ba: mov ebx,ds:[100FE55Ch]
0x1000c1c0: mov [esp+24h],eax
0x1000c1c4: call ebx
0x1000c1c6: add esp,8
0x1000c1c9: test eax,eax
0x1000c1cb: je near ptr 1000C2ACh
0x1000c1d1: push eax
0x1000c1d2: lea ecx,[esp+28h]
0x1000c1d6: call dword ptr ds:[100FE1C4h]
0x1000c1dc: lea ecx,[esp+24h]
0x1000c1e0: call dword ptr ds:[100FE234h]
0x1000c1e6: lea ecx,[esp+24h]
0x1000c1ea: call dword ptr ds:[100FE238h]
0x1000c1f0: mov eax,[esp+24h]
0x1000c1f4: push 420h
0x1000c1f9: lea edx,[esp+48h]
0x1000c1fd: push edx
0x1000c1fe: mov dword ptr [esp+30h],0
0x1000c206: mov ecx,[eax-0Ch]
0x1000c209: push ecx
0x1000c20a: push eax
0x1000c20b: mov eax,[esp+44h]
0x1000c20f: mov edx,[eax+70h]
0x1000c212: lea ecx,[esp+44h]
0x1000c216: call edx
0x1000c218: test eax,eax
0x1000c21a: je short 1000C228h
0x1000c21c: mov eax,[esp+4Ch]
0x1000c220: sub eax,[esp+44h]
0x1000c224: mov [esp+28h],eax
0x1000c228: mov edx,[esi+4]
0x1000c22b: test edx,edx
0x1000c22d: jne short 1000C233h
0x1000c22f: xor ecx,ecx
0x1000c231: jmp short 1000C23Bh
0x1000c233: mov ecx,[esi+8]
0x1000c236: sub ecx,edx
0x1000c238: sar ecx,3
0x1000c23b: test edx,edx
0x1000c23d: je short 1000C278h
0x1000c23f: mov eax,[esi+0Ch]
0x1000c242: sub eax,edx
0x1000c244: sar eax,3
0x1000c247: cmp ecx,eax
0x1000c249: jae short 1000C278h
0x1000c24b: mov edx,[esp+1Ch]
0x1000c24f: mov edi,[esi+8]
```

```
0x1000c252: mov byte ptr [esp+18h],0
0x1000c257: mov ecx,[esp+18h]
0x1000c25b: push ecx
0x1000c25c: push edx
0x1000c25d: lea eax,[esp+2Ch]
0x1000c261: push eax
0x1000c262: push edi
0x1000c263: mov ecx,1
0x1000c268: call 1000C830h
0x1000c26d: add esp,10h
0x1000c270: add edi,8
0x1000c273: mov [esi+8],edi
0x1000c276: jmp short 1000C298h
0x1000c278: mov edi,[esi+8]
0x1000c27b: cmp edx,edi
0x1000c27d: jbe short 1000C285h
0x1000c27f: call dword ptr ds:[100FE584h]
0x1000c285: push edi
0x1000c286: push esi
0x1000c287: lea ecx,[esp+2Ch]
0x1000c28b: push ecx
0x1000c28c: lea edx,[esp+38h]
0x1000c290: push edx
0x1000c291: mov eax,esi
0x1000c293: call 1000C360h
0x1000c298: push 1012F7FCh
0x1000c29d: push 0
0x1000c29f: call ebx
0x1000c2a1: add esp,8
0x1000c2a4: test eax,eax
0x1000c2a6: jne near ptr 1000C1D1h
0x1000c2ac: mov eax,[esp+1Ch]
0x1000c2b0: push eax
0x1000c2b1: lea ecx,[esp+38h]
0x1000c2b5: call 1002228Ah
0x1000c2ba: push 0FFFFFFFFh
0x1000c2bc: lea ecx,[esp+18h]
0x1000c2c0: call dword ptr ds:[100FE1B8h]
0x1000c2c6: lea ecx,[esp+34h]
0x1000c2ca: mov byte ptr [esp+60h],1
0x1000c2cf: call 10022032h
0x1000c2d4: lea ecx,[esp+14h]
0x1000c2d8: call dword ptr ds:[100FE374h]
0x1000c2de: lea ecx,[esp+24h]
0x1000c2e2: call dword ptr ds:[100FE374h]
0x1000c2e8: mov ecx,[esp+58h]
0x1000c2ec: mov fs:[0],ecx
0x1000c2f3: pop ecx
0x1000c2f4: pop edi
0x1000c2f5: pop esi
0x1000c2f6: pop ebx
0x1000c2f7: mov esp,ebp
0x1000c2f9: pop ebp
0x1000c2fa: ret
```

`sub_1000c0f0` is called by ?SetCaptionText@CCustomTitle@@QAEXABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@@Z, sub_100039b0. It delegates to sub_1000c360, sub_1000c3f0, sub_1000c830 (7 callees).

12.185 sub_1000c360 at 0x1000c360 (54 instructions)

```
0x1000c360: push ecx
0x1000c361: push ebx
0x1000c362: push ebp
0x1000c363: mov ebp,[esp+18h]
0x1000c367: push esi
```

```
0x1000c368: mov esi,eax
0x1000c36a: push edi
0x1000c36b: mov edi,[esi+4]
0x1000c36e: test edi,edi
0x1000c370: je short 1000C37Eh
0x1000c372: mov eax,[esi+8]
0x1000c375: mov ecx,eax
0x1000c377: sub ecx,edi
0x1000c379: sar ecx,3
0x1000c37c: jne short 1000C382h
0x1000c37e: xor ebx,ebx
0x1000c380: jmp short 1000C3A3h
0x1000c382: cmp edi,eax
0x1000c384: jbe short 1000C38Ch
0x1000c386: call dword ptr ds:[100FE584h]
0x1000c38c: test ebp,ebp
0x1000c38e: je short 1000C394h
0x1000c390: cmp ebp,esi
0x1000c392: je short 1000C39Ah
0x1000c394: call dword ptr ds:[100FE584h]
0x1000c39a: mov ebx,[esp+24h]
0x1000c39e: sub ebx,edi
0x1000c3a0: sar ebx,3
0x1000c3a3: mov edx,[esp+24h]
0x1000c3a7: mov ecx,[esp+1Ch]
0x1000c3ab: push edx
0x1000c3ac: push ebp
0x1000c3ad: push esi
0x1000c3ae: call 1000C480h
0x1000c3b3: mov edi,[esi+4]
0x1000c3b6: cmp edi,[esi+8]
0x1000c3b9: jbe short 1000C3C1h
0x1000c3bb: call dword ptr ds:[100FE584h]
0x1000c3c1: mov [esp+24h],edi
0x1000c3c5: lea edi,[edi+ebx*8]
0x1000c3c8: cmp edi,[esi+8]
0x1000c3cb: ja short 1000C3D2h
0x1000c3cd: cmp edi,[esi+4]
0x1000c3d0: jae short 1000C3D8h
0x1000c3d2: call dword ptr ds:[100FE584h]
0x1000c3d8: mov eax,[esp+18h]
0x1000c3dc: mov [eax+4],edi
0x1000c3df: pop edi
0x1000c3e0: mov [eax],esi
0x1000c3e2: pop esi
0x1000c3e3: pop ebp
0x1000c3e4: pop ebx
0x1000c3e5: pop ecx
0x1000c3e6: ret 10h
```

`sub_1000c360` is called by sub_1000c0f0. It delegates to sub_1000c480 (1 callees).

12.186 sub_1000c3f0 at 0x1000c3f0 (40 instructions)

```
0x1000c3f0: push ebp
0x1000c3f1: mov ebp,[esp+8]
0x1000c3f5: push esi
0x1000c3f6: mov esi,[esp+14h]
0x1000c3fa: test esi,esi
0x1000c3fc: je short 1000C404h
0x1000c3fe: cmp esi,[esp+1Ch]
0x1000c402: je short 1000C40Ah
0x1000c404: call dword ptr ds:[100FE584h]
0x1000c40a: mov ecx,[esp+20h]
0x1000c40e: mov eax,[esp+18h]
0x1000c412: cmp eax,ecx
```

```
0x1000c414: je short 1000C44Ch
0x1000c416: push ebx
0x1000c417: mov ebx,[ebp+8]
0x1000c41a: push edi
0x1000c41b: call 1000C760h
0x1000c420: mov edi,[ebp+8]
0x1000c423: mov ebx,eax
0x1000c425: cmp ebx,edi
0x1000c427: mov esi,ebx
0x1000c429: je short 1000C43Fh
0x1000c42b: jmp short 1000C430h
0x1000c430: mov ecx,esi
0x1000c432: call dword ptr ds:[100FE374h]
0x1000c438: add esi,8
0x1000c43b: cmp esi,edi
0x1000c43d: jne short 1000C430h
0x1000c43f: mov eax,[esp+20h]
0x1000c443: mov esi,[esp+1Ch]
0x1000c447: pop edi
0x1000c448: mov [ebp+8],ebx
0x1000c44b: pop ebx
0x1000c44c: mov ecx,[esp+10h]
0x1000c450: mov [ecx],esi
0x1000c452: pop esi
0x1000c453: mov [ecx+4],eax
0x1000c456: mov eax,ecx
0x1000c458: pop ebp
0x1000c459: ret 18h
```

`sub_1000c3f0` is called by sub_1000c0f0. It delegates to sub_1000c760 (1 callees).

12.187 sub_1000c460 at 0x1000c460 (11 instructions)

```
0x1000c460: push esi
0x1000c461: mov esi,eax
0x1000c463: cmp esi,edi
0x1000c465: je short 1000C476h
0x1000c467: mov ecx,esi
0x1000c469: call dword ptr ds:[100FE374h]
0x1000c46f: add esi,8
0x1000c472: cmp esi,edi
0x1000c474: jne short 1000C467h
0x1000c476: pop esi
0x1000c477: ret
```

`sub_1000c460` is called by sub_1000c480. It calls nothing else.

12.188 sub_1000c480 at 0x1000c480 (212 instructions)

```
0x1000c480: push ebp
0x1000c481: mov ebp,esp
0x1000c483: push 0FFFFFFFFh
0x1000c485: push 100FCB98h
0x1000c48a: mov eax,fs:[0]
0x1000c490: push eax
0x1000c491: sub esp,20h
0x1000c494: push ebx
0x1000c495: push esi
0x1000c496: push edi
0x1000c497: mov eax,ds:[10136220h]
0x1000c49c: xor eax,ebp
0x1000c49e: push eax
0x1000c49f: lea eax,[ebp-0Ch]
0x1000c4a2: mov fs:[0],eax
0x1000c4a8: mov [ebp-10h],esp
```

```
0x1000c4ab: mov esi,ecx
0x1000c4ad: mov ebx,[ebp+8]
0x1000c4b0: push esi
0x1000c4b1: lea ecx,[ebp-28h]
0x1000c4b4: call dword ptr ds:[100FE220h]
0x1000c4ba: mov eax,[esi+4]
0x1000c4bd: mov [ebp-24h],eax
0x1000c4c0: xor esi,esi
0x1000c4c2: mov [ebp-4],esi
0x1000c4c5: mov eax,[ebx+4]
0x1000c4c8: cmp eax,esi
0x1000c4ca: jne short 1000C4D0h
0x1000c4cc: xor edx,edx
0x1000c4ce: jmp short 1000C4DAh
0x1000c4d0: mov ecx,[ebx+0Ch]
0x1000c4d3: sub ecx,eax
0x1000c4d5: sar ecx,3
0x1000c4d8: mov edx,ecx
0x1000c4da: cmp eax,esi
0x1000c4dc: jne short 1000C4E2h
0x1000c4de: xor ecx,ecx
0x1000c4e0: jmp short 1000C4EAh
0x1000c4e2: mov ecx,[ebx+8]
0x1000c4e5: sub ecx,eax
0x1000c4e7: sar ecx,3
0x1000c4ea: mov edi,1FFFFFFFh
0x1000c4ef: sub edi,ecx
0x1000c4f1: cmp edi,1
0x1000c4f4: jae short 1000C4FBh
0x1000c4f6: call 10009B40h
0x1000c4fb: cmp eax,esi
0x1000c4fd: jne short 1000C503h
0x1000c4ff: xor ecx,ecx
0x1000c501: jmp short 1000C50Bh
0x1000c503: mov ecx,[ebx+8]
0x1000c506: sub ecx,eax
0x1000c508: sar ecx,3
0x1000c50b: add ecx,1
0x1000c50e: cmp edx,ecx
0x1000c510: jae near ptr 1000C63Dh
0x1000c516: mov ecx,edx
0x1000c518: shr ecx,1
0x1000c51a: mov edi,1FFFFFFFh
0x1000c51f: sub edi,ecx
0x1000c521: cmp edi,edx
0x1000c523: jae short 1000C52Ch
0x1000c525: mov [ebp-18h],esi
0x1000c528: mov edx,esi
0x1000c52a: jmp short 1000C531h
0x1000c52c: add edx,ecx
0x1000c52e: mov [ebp-18h],edx
0x1000c531: cmp eax,esi
0x1000c533: jne short 1000C539h
0x1000c535: xor ecx,ecx
0x1000c537: jmp short 1000C541h
0x1000c539: mov ecx,[ebx+8]
0x1000c53c: sub ecx,eax
0x1000c53e: sar ecx,3
0x1000c541: add ecx,1
0x1000c544: cmp edx,ecx
0x1000c546: jae short 1000C560h
0x1000c548: cmp eax,esi
0x1000c54a: jne short 1000C550h
0x1000c54c: xor ecx,ecx
0x1000c54e: jmp short 1000C558h
0x1000c550: mov ecx,[ebx+8]
0x1000c553: sub ecx,eax
```

```
0x1000c555: sar ecx,3
0x1000c558: add ecx,1
0x1000c55b: mov [ebp-18h],ecx
0x1000c55e: mov edx,ecx
0x1000c560: mov ecx,edx
0x1000c562: call 1000C700h
0x1000c567: mov byte ptr [ebp-4],1
0x1000c56b: mov esi,[ebp+10h]
0x1000c56e: mov ecx,[ebx+4]
0x1000c571: mov byte ptr [ebp-20h],0
0x1000c575: mov edx,[ebp-20h]
0x1000c578: push edx
0x1000c579: mov [ebp+8],eax
0x1000c57c: mov edx,[ebp+8]
0x1000c57f: push edx
0x1000c580: push eax
0x1000c581: push esi
0x1000c582: mov [ebp-1Ch],eax
0x1000c585: call 1000C8D0h
0x1000c58a: mov edi,eax
0x1000c58c: add esp,10h
0x1000c58f: mov byte ptr [ebp+10h],0
0x1000c593: mov eax,[ebp+10h]
0x1000c596: push eax
0x1000c597: mov [ebp+8],edi
0x1000c59a: mov ecx,[ebp+8]
0x1000c59d: push ecx
0x1000c59e: lea edx,[ebp-28h]
0x1000c5a1: push edx
0x1000c5a2: push edi
0x1000c5a3: mov ecx,1
0x1000c5a8: call 1000C830h
0x1000c5ad: mov eax,[ebx+8]
0x1000c5b0: add esp,10h
0x1000c5b3: add edi,8
0x1000c5b6: mov byte ptr [ebp+10h],0
0x1000c5ba: mov ecx,[ebp+10h]
0x1000c5bd: push ecx
0x1000c5be: mov [ebp+8],edi
0x1000c5c1: mov edx,[ebp+8]
0x1000c5c4: push edx
0x1000c5c5: push edi
0x1000c5c6: push eax
0x1000c5c7: mov ecx,esi
0x1000c5c9: call 1000C8D0h
0x1000c5ce: mov dword ptr [ebp-4],0
0x1000c5d5: mov eax,[ebx+4]
0x1000c5d8: add esp,10h
0x1000c5db: test eax,eax
0x1000c5dd: jne short 1000C5E3h
0x1000c5df: xor esi,esi
0x1000c5e1: jmp short 1000C5EBh
0x1000c5e3: mov esi,[ebx+8]
0x1000c5e6: sub esi,eax
0x1000c5e8: sar esi,3
0x1000c5eb: test eax,eax
0x1000c5ed: je short 1000C603h
0x1000c5ef: mov edi,[ebx+8]
0x1000c5f2: call 1000C460h
0x1000c5f7: mov eax,[ebx+4]
0x1000c5fa: push eax
0x1000c5fb: call 10021FCCh
0x1000c600: add esp,4
0x1000c603: mov eax,[ebp-1Ch]
0x1000c606: mov ecx,[ebp-18h]
0x1000c609: lea edx,[eax+ecx*8]
0x1000c60c: lea ecx,[eax+esi*8+8]
```

```
0x1000c610: mov [ebx+0Ch],edx
0x1000c613: mov [ebx+8],ecx
0x1000c616: mov [ebx+4],eax
0x1000c619: jmp near ptr 1000C6E1h
0x1000c63d: mov edi,[ebx+8]
0x1000c640: mov esi,[ebp+10h]
0x1000c643: mov edx,edi
0x1000c645: sub edx,esi
0x1000c647: sar edx,3
0x1000c64a: cmp edx,1
0x1000c64d: push edi
0x1000c64e: jae short 1000C6B7h
0x1000c650: lea edx,[esi+8]
0x1000c653: push esi
0x1000c654: call 1000C7A0h
0x1000c659: mov byte ptr [ebp-4],3
0x1000c65d: mov eax,[ebx+8]
0x1000c660: mov edx,eax
0x1000c662: sub edx,esi
0x1000c664: sar edx,3
0x1000c667: mov ecx,1
0x1000c66c: sub ecx,edx
0x1000c66e: mov byte ptr [ebp-20h],0
0x1000c672: mov edx,[ebp-20h]
0x1000c675: push edx
0x1000c676: mov edx,[ebp+8]
0x1000c679: push edx
0x1000c67a: lea edx,[ebp-28h]
0x1000c67d: push edx
0x1000c67e: push eax
0x1000c67f: call 1000C830h
0x1000c684: mov dword ptr [ebp-4],0
0x1000c68b: add dword ptr [ebx+8],8
0x1000c68f: mov ebx,[ebx+8]
0x1000c692: add esp,10h
0x1000c695: add ebx,0FFFFFFF8h
0x1000c698: jmp short 1000C6D7h
0x1000c6b7: lea eax,[edi-8]
0x1000c6ba: push eax
0x1000c6bb: mov edx,edi
0x1000c6bd: mov [ebp+8],eax
0x1000c6c0: call 1000C7A0h
0x1000c6c5: mov ecx,[ebp+8]
0x1000c6c8: mov [ebx+8],eax
0x1000c6cb: mov eax,edi
0x1000c6cd: mov ebx,esi
0x1000c6cf: call 1000C7F0h
0x1000c6d4: lea ebx,[esi+8]
0x1000c6d7: lea edi,[ebp-28h]
0x1000c6da: mov eax,esi
0x1000c6dc: call 1000C7D0h
0x1000c6e1: lea ecx,[ebp-28h]
0x1000c6e4: call dword ptr ds:[100FE374h]
0x1000c6ea: mov ecx,[ebp-0Ch]
0x1000c6ed: mov fs:[0],ecx
0x1000c6f4: pop ecx
0x1000c6f5: pop edi
0x1000c6f6: pop esi
0x1000c6f7: pop ebx
0x1000c6f8: mov esp,ebp
0x1000c6fa: pop ebp
0x1000c6fb: ret 0Ch
```

`sub_1000c480` is called by sub_1000c360. It delegates to sub_10009b40, sub_1000c460, sub_1000c700 (9 callees).

12.189 sub_1000c700 at 0x1000c700 (26 instructions)

```
0x1000c700: sub esp,10h
0x1000c703: test ecx,ecx
0x1000c705: ja short 1000C71Dh
0x1000c707: xor ecx,ecx
0x1000c709: lea edx,[ecx*8]
0x1000c710: push edx
0x1000c711: call 10021FDEh
0x1000c716: add esp,4
0x1000c719: add esp,10h
0x1000c71c: ret
0x1000c71d: or eax,0FFFFFFFFh
0x1000c720: xor edx,edx
0x1000c722: div ecx
0x1000c724: cmp eax,8
0x1000c727: jae short 1000C709h
0x1000c729: lea eax,[esp]
0x1000c72c: push eax
0x1000c72d: lea ecx,[esp+8]
0x1000c731: mov dword ptr [esp+4],0
0x1000c739: call dword ptr ds:[100FE578h]
0x1000c73f: push 10130EB8h
0x1000c744: lea ecx,[esp+8]
0x1000c748: push ecx
0x1000c749: mov dword ptr [esp+0Ch],100FF394h
0x1000c751: call 100229A6h
0x1000c756: int 3
```

`sub_1000c700` is called by sub_1000c480. It delegates to j_ORDINAL 762, j__CxxThrowException (2 callees).

12.190 sub_1000c760 at 0x1000c760 (26 instructions)

```
0x1000c760: push ebp
0x1000c761: push esi
0x1000c762: push edi
0x1000c763: mov edi,eax
0x1000c765: mov esi,ecx
0x1000c767: mov eax,ebx
0x1000c769: sub eax,esi
0x1000c76b: sar eax,3
0x1000c76e: cmp esi,ebx
0x1000c770: lea ebp,[edi+eax*8]
0x1000c773: je short 1000C798h
0x1000c775: sub edi,esi
0x1000c777: jmp short 1000C780h
0x1000c780: push esi
0x1000c781: lea ecx,[edi+esi]
0x1000c784: call dword ptr ds:[100FE198h]
0x1000c78a: mov ecx,[esi+4]
0x1000c78d: mov [edi+esi+4],ecx
0x1000c791: add esi,8
0x1000c794: cmp esi,ebx
0x1000c796: jne short 1000C780h
0x1000c798: pop edi
0x1000c799: pop esi
0x1000c79a: mov eax,ebp
0x1000c79c: pop ebp
0x1000c79d: ret
```

`sub_1000c760` is called by sub_1000c3f0. It calls nothing else.

12.191 sub_1000c7a0 at 0x1000c7a0 (13 instructions)

```
0x1000c7a0: push ecx
0x1000c7a1: mov ecx,[esp+0Ch]
```

```
0x1000c7a5: mov byte ptr [esp],0
0x1000c7a9: mov eax,[esp]
0x1000c7ac: push eax
0x1000c7ad: mov eax,[esp+10h]
0x1000c7b1: push ecx
0x1000c7b2: mov ecx,[esp+10h]
0x1000c7b6: push edx
0x1000c7b7: push eax
0x1000c7b8: call 1000C8D0h
0x1000c7bd: add esp,14h
0x1000c7c0: ret 8
```

`sub_1000c7a0` is called by sub_1000c480. It delegates to sub_1000c8d0 (1 callees).

12.192 sub_1000c7d0 at 0x1000c7d0 (14 instructions)

```
0x1000c7d0: push esi
0x1000c7d1: mov esi,eax
0x1000c7d3: cmp esi,ebx
0x1000c7d5: je short 1000C7EDh
0x1000c7d7: push edi
0x1000c7d8: mov ecx,esi
0x1000c7da: call dword ptr ds:[100FE198h]
0x1000c7e0: mov eax,[edi+4]
0x1000c7e3: mov [esi+4],eax
0x1000c7e6: add esi,8
0x1000c7e9: cmp esi,ebx
0x1000c7eb: jne short 1000C7D7h
0x1000c7ed: pop esi
0x1000c7ee: ret
```

`sub_1000c7d0` is called by sub_1000c480. It calls nothing else.

12.193 sub_1000c7f0 at 0x1000c7f0 (30 instructions)

```
0x1000c7f0: push ebp
0x1000c7f1: push esi
0x1000c7f2: mov esi,ecx
0x1000c7f4: push edi
0x1000c7f5: mov edi,eax
0x1000c7f7: mov eax,esi
0x1000c7f9: sub eax,ebx
0x1000c7fb: sar eax,3
0x1000c7fe: add eax,eax
0x1000c800: add eax,eax
0x1000c802: add eax,eax
0x1000c804: mov ebp,edi
0x1000c806: sub ebp,eax
0x1000c808: cmp ebx,esi
0x1000c80a: je short 1000C828h
0x1000c80c: sub edi,esi
0x1000c80e: mov edi,edi
0x1000c810: sub esi,8
0x1000c813: push esi
0x1000c814: lea ecx,[edi+esi]
0x1000c817: call dword ptr ds:[100FE198h]
0x1000c81d: cmp esi,ebx
0x1000c81f: mov ecx,[esi+4]
0x1000c822: mov [edi+esi+4],ecx
0x1000c826: jne short 1000C810h
0x1000c828: pop edi
0x1000c829: pop esi
0x1000c82a: mov eax,ebp
0x1000c82c: pop ebp
0x1000c82d: ret
```

`sub_1000c7f0` is called by sub_1000c480. It calls nothing else.

12.194 sub_1000c830 at 0x1000c830 (47 instructions)

```
0x1000c830: push ebp
0x1000c831: mov ebp,esp
0x1000c833: push 0FFFFFFFFh
0x1000c835: push 100FC9C5h
0x1000c83a: mov eax,fs:[0]
0x1000c840: push eax
0x1000c841: sub esp,8
0x1000c844: push ebx
0x1000c845: push esi
0x1000c846: push edi
0x1000c847: mov eax,ds:[10136220h]
0x1000c84c: xor eax,ebp
0x1000c84e: push eax
0x1000c84f: lea eax,[ebp-0Ch]
0x1000c852: mov fs:[0],eax
0x1000c858: mov [ebp-10h],esp
0x1000c85b: mov edi,ecx
0x1000c85d: mov esi,[ebp+8]
0x1000c860: xor ebx,ebx
0x1000c862: mov [ebp-14h],esi
0x1000c865: mov [ebp-4],ebx
0x1000c868: cmp edi,ebx
0x1000c86a: jbe short 1000C8B9h
0x1000c86c: cmp esi,ebx
0x1000c86e: mov byte ptr [ebp-4],1
0x1000c872: je short 1000C889h
0x1000c874: mov eax,[ebp+0Ch]
0x1000c877: push eax
0x1000c878: mov ecx,esi
0x1000c87a: call dword ptr ds:[100FE220h]
0x1000c880: mov ecx,[ebp+0Ch]
0x1000c883: mov edx,[ecx+4]
0x1000c886: mov [esi+4],edx
0x1000c889: sub edi,1
0x1000c88c: add esi,8
0x1000c88f: mov [ebp-4],bl
0x1000c892: mov [ebp+8],esi
0x1000c895: jmp short 1000C868h
0x1000c8b9: mov ecx,[ebp-0Ch]
0x1000c8bc: mov fs:[0],ecx
0x1000c8c3: pop ecx
0x1000c8c4: pop edi
0x1000c8c5: pop esi
0x1000c8c6: pop ebx
0x1000c8c7: mov esp,ebp
0x1000c8c9: pop ebp
0x1000c8ca: ret
```

`sub_1000c830` is called by sub_1000c0f0, sub_1000c480. It calls nothing else.

12.195 sub_1000c8d0 at 0x1000c8d0 (46 instructions)

```
0x1000c8d0: push ebp
0x1000c8d1: mov ebp,esp
0x1000c8d3: push 0FFFFFFFFh
0x1000c8d5: push 100FCB75h
0x1000c8da: mov eax,fs:[0]
0x1000c8e0: push eax
0x1000c8e1: sub esp,8
0x1000c8e4: push ebx
0x1000c8e5: push esi
0x1000c8e6: push edi
```

```
0x1000c8e7: mov eax,ds:[10136220h]
0x1000c8ec: xor eax,ebp
0x1000c8ee: push eax
0x1000c8ef: lea eax,[ebp-0Ch]
0x1000c8f2: mov fs:[0],eax
0x1000c8f8: mov [ebp-10h],esp
0x1000c8fb: mov esi,[ebp+0Ch]
0x1000c8fe: mov edi,ecx
0x1000c900: xor ebx,ebx
0x1000c902: mov [ebp-14h],esi
0x1000c905: mov [ebp-4],ebx
0x1000c908: cmp edi,[ebp+8]
0x1000c90b: je short 1000C958h
0x1000c90d: cmp esi,ebx
0x1000c90f: mov byte ptr [ebp-4],1
0x1000c913: je short 1000C924h
0x1000c915: push edi
0x1000c916: mov ecx,esi
0x1000c918: call dword ptr ds:[100FE220h]
0x1000c91e: mov eax,[edi+4]
0x1000c921: mov [esi+4],eax
0x1000c924: add esi,8
0x1000c927: mov [ebp-4],bl
0x1000c92a: mov [ebp+0Ch],esi
0x1000c92d: add edi,8
0x1000c930: jmp short 1000C908h
0x1000c958: mov eax,esi
0x1000c95a: mov ecx,[ebp-0Ch]
0x1000c95d: mov fs:[0],ecx
0x1000c964: pop ecx
0x1000c965: pop edi
0x1000c966: pop esi
0x1000c967: pop ebx
0x1000c968: mov esp,ebp
0x1000c96a: pop ebp
0x1000c96b: ret
```

`sub_1000c8d0` is called by sub_1000c480, sub_1000c7a0. It calls nothing else.

12.196 sub_1000c970 at 0x1000c970 (67 instructions)

```
0x1000c970: push esi
0x1000c971: mov esi,ecx
0x1000c973: mov dword ptr [esi],100FE724h
0x1000c979: call 1000E550h
0x1000c97e: mov ecx,esi
0x1000c980: pop esi
0x1000c981: jmp near ptr 1000E4A0h
0x1000e4a0: push ebx
0x1000e4a1: push esi
0x1000e4a2: mov esi,ecx
0x1000e4a4: xor ebx,ebx
0x1000e4a6: cmp [esi+38h],ebx
0x1000e4a9: jne near ptr 1000E545h
0x1000e4af: cmp [esi+3C8h],ebx
0x1000e4b5: push edi
0x1000e4b6: je short 1000E4FDh
0x1000e4b8: xor edi,edi
0x1000e4ba: cmp [esi+1E4h],ebx
0x1000e4c0: jle short 1000E4E2h
0x1000e4c2: mov eax,[esi+3C8h]
0x1000e4c8: mov ecx,[eax+edi*4]
0x1000e4cb: cmp ecx,ebx
0x1000e4cd: je short 1000E4D7h
0x1000e4cf: mov edx,[ecx]
0x1000e4d1: mov eax,[edx]
```

```
0x1000e4d3: push 1
0x1000e4d5: call eax
0x1000e4d7: add edi,1
0x1000e4da: cmp edi,[esi+1E4h]
0x1000e4e0: jl short 1000E4C2h
0x1000e4e2: mov ecx,[esi+3C8h]
0x1000e4e8: push ecx
0x1000e4e9: call 10021FCCh
0x1000e4ee: add esp,4
0x1000e4f1: mov [esi+3C8h],ebx
0x1000e4f7: mov [esi+1E4h],ebx
0x1000e4fd: mov eax,[esi+3C0h]
0x1000e503: cmp eax,ebx
0x1000e505: mov edi,ds:[100FE554h]
0x1000e50b: je short 1000E519h
0x1000e50d: push eax
0x1000e50e: call edi
0x1000e510: add esp,4
0x1000e513: mov [esi+3C0h],ebx
0x1000e519: mov eax,[esi+3C4h]
0x1000e51f: cmp eax,ebx
0x1000e521: je short 1000E52Fh
0x1000e523: push eax
0x1000e524: call edi
0x1000e526: add esp,4
0x1000e529: mov [esi+3C4h],ebx
0x1000e52f: mov eax,[esi+4]
0x1000e532: cmp eax,ebx
0x1000e534: je short 1000E53Fh
0x1000e536: push eax
0x1000e537: call edi
0x1000e539: add esp,4
0x1000e53c: mov [esi+4],ebx
0x1000e53f: pop edi
0x1000e540: pop esi
0x1000e541: mov al,1
0x1000e543: pop ebx
0x1000e544: ret
0x1000e545: pop esi
0x1000e546: xor al,al
0x1000e548: pop ebx
0x1000e549: ret
```

`sub_1000c970` is called by sub_100103d0, sub_1001b970. It delegates to sub_1000e550, j_ORDINAL 764 (2 callees).

12.197 sub_1000c9c0 at 0x1000c9c0 (46 instructions)

```
0x1000c9c0: push 0FFFFFFFFh
0x1000c9c2: push 100FC998h
0x1000c9c7: mov eax,fs:[0]
0x1000c9cd: push eax
0x1000c9ce: push ebx
0x1000c9cf: push esi
0x1000c9d0: mov eax,ds:[10136220h]
0x1000c9d5: xor eax,esp
0x1000c9d7: push eax
0x1000c9d8: lea eax,[esp+0Ch]
0x1000c9dc: mov fs:[0],eax
0x1000c9e2: mov esi,[esp+1Ch]
0x1000c9e6: lea ecx,[esi+4]
0x1000c9e9: call dword ptr ds:[100FE368h]
0x1000c9ef: xor ebx,ebx
0x1000c9f1: lea ecx,[esi+8]
0x1000c9f4: mov [esp+14h],ebx
0x1000c9f8: call dword ptr ds:[100FE368h]
0x1000c9fe: push 1000C970h
```

```
0x1000ca03: push 1000CA70h
0x1000ca08: push 4
0x1000ca0a: push 3D0h
0x1000ca0f: lea eax,[esi+0Ch]
0x1000ca12: push eax
0x1000ca13: mov byte ptr [esp+28h],1
0x1000ca18: call 10022509h
0x1000ca1d: xor ecx,ecx
0x1000ca1f: mov [esi+0F50h],ecx
0x1000ca25: lea ecx,[esi+0F54h]
0x1000ca2b: xor eax,eax
0x1000ca2d: push ecx
0x1000ca2e: mov [esi],ebx
0x1000ca30: mov [esi+0F4Ch],eax
0x1000ca36: call dword ptr ds:[100FE5C8h]
0x1000ca3c: mov [esi+0F68h],ebx
0x1000ca42: mov [esi+0F6Ch],ebx
0x1000ca48: mov [esi+0F70h],ebx
0x1000ca4e: mov [esi+0F64h],bl
0x1000ca54: mov eax,esi
0x1000ca56: mov ecx,[esp+0Ch]
0x1000ca5a: mov fs:[0],ecx
0x1000ca61: pop ecx
0x1000ca62: pop esi
0x1000ca63: pop ebx
0x1000ca64: add esp,0Ch
0x1000ca67: ret 4
```

`sub_1000c9c0` is called by sub_1000d330, sub_1000d430, sub_1000d550. It delegates to sub_10022509 (1 callees).

12.198 sub_1000ca70 at 0x1000ca70 (3 instructions)

```
0x1000ca70: push 0
0x1000ca72: call 1000E480h
0x1000ca77: ret
```

`sub_1000ca70` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1000e480 (1 callees).

12.199 sub_1000ca80 at 0x1000ca80 (28 instructions)

```
0x1000ca80: push 0FFFFFFFFh
0x1000ca82: push 100FCA58h
0x1000ca87: mov eax,fs:[0]
0x1000ca8d: push eax
0x1000ca8e: push esi
0x1000ca8f: mov eax,ds:[10136220h]
0x1000ca94: xor eax,esp
0x1000ca96: push eax
0x1000ca97: lea eax,[esp+8]
0x1000ca9b: mov fs:[0],eax
0x1000caa1: mov esi,[esp+18h]
0x1000caa5: push 1000C970h
0x1000caaa: push 4
0x1000caac: push 3D0h
0x1000cab1: lea eax,[esi+0Ch]
0x1000cab4: push eax
0x1000cab5: mov dword ptr [esp+20h],1
0x1000cabd: call 100223DCh
0x1000cac2: lea ecx,[esi+8]
0x1000cac5: call dword ptr ds:[100FE374h]
0x1000cacb: lea ecx,[esi+4]
0x1000cace: call dword ptr ds:[100FE374h]
0x1000cad4: mov ecx,[esp+8]
0x1000cad8: mov fs:[0],ecx
```

```
0x1000cadf: pop ecx
0x1000cae0: pop esi
0x1000cae1: add esp,0Ch
0x1000cae4: ret 4
```

`sub_1000ca80` is called by sub_1000d330, sub_1000d430. It delegates to sub_100223dc (1 callees).

12.200 sub_1000caf0 at 0x1000caf0 (81 instructions)

```
0x1000caf0: push 0FFFFFFFFh
0x1000caf2: push 100FCA18h
0x1000caf7: mov eax,fs:[0]
0x1000cafd: push eax
0x1000cafe: sub esp,20h
0x1000cb01: push ebx
0x1000cb02: push edi
0x1000cb03: mov eax,ds:[10136220h]
0x1000cb08: xor eax,esp
0x1000cb0a: push eax
0x1000cb0b: lea eax,[esp+2Ch]
0x1000cb0f: mov fs:[0],eax
0x1000cb15: lea edi,[esi+0Ch]
0x1000cb18: mov ebx,4
0x1000cb1d: lea ecx,[ecx]
0x1000cb20: mov ecx,edi
0x1000cb22: call 1000F190h
0x1000cb27: test al,al
0x1000cb29: je short 1000CB32h
0x1000cb2b: mov ecx,edi
0x1000cb2d: call 1000E4A0h
0x1000cb32: add edi,3D0h
0x1000cb38: sub ebx,1
0x1000cb3b: jne short 1000CB20h
0x1000cb3d: xor eax,eax
0x1000cb3f: xor ecx,ecx
0x1000cb41: mov [esi+0F4Ch],eax
0x1000cb47: mov [esi+0F50h],ecx
0x1000cb4d: call dword ptr ds:[100FE59Ch]
0x1000cb53: push eax
0x1000cb54: call 1002225Ah
0x1000cb59: test eax,eax
0x1000cb5b: jne short 1000CB71h
0x1000cb5d: mov ecx,[esp+2Ch]
0x1000cb61: mov fs:[0],ecx
0x1000cb68: pop ecx
0x1000cb69: pop edi
0x1000cb6a: pop ebx
0x1000cb6b: add esp,2Ch
0x1000cb6e: ret 4
0x1000cb71: mov ecx,[esp+3Ch]
0x1000cb75: mov [esi+0F70h],ecx
0x1000cb7b: push eax
0x1000cb7c: lea ecx,[esp+18h]
0x1000cb80: call 1002222Ah
0x1000cb85: mov dword ptr [esp+34h],0
0x1000cb8d: mov edx,[esi+0F70h]
0x1000cb93: push edx
0x1000cb94: lea ecx,[esp+18h]
0x1000cb98: call 1002228Ah
0x1000cb9d: mov edi,eax
0x1000cb9f: mov eax,[esi+4]
0x1000cba2: mov ecx,[eax-0Ch]
0x1000cba5: lea edx,[esp+0Ch]
0x1000cba9: push edx
0x1000cbaa: push ecx
0x1000cbab: push eax
```

```
0x1000cbac: mov eax,[esp+28h]
0x1000cbb0: push eax
0x1000cbb1: call dword ptr ds:[100FE03Ch]
0x1000cbb7: mov ecx,[esp+0Ch]
0x1000cbbb: mov edx,[esp+10h]
0x1000cbbf: mov [esi+0F4Ch],ecx
0x1000cbc5: push edi
0x1000cbc6: lea ecx,[esp+18h]
0x1000cbca: mov [esi+0F50h],edx
0x1000cbd0: call 1002228Ah
0x1000cbd5: mov eax,8
0x1000cbda: add [esi+0F4Ch],eax
0x1000cbe0: add [esi+0F50h],eax
0x1000cbe6: lea ecx,[esp+14h]
0x1000cbea: mov dword ptr [esp+34h],0FFFFFFFFh
0x1000cbf2: call 10022230h
0x1000cbf7: mov eax,1
0x1000cbfc: mov ecx,[esp+2Ch]
0x1000cc00: mov fs:[0],ecx
0x1000cc07: pop ecx
0x1000cc08: pop edi
0x1000cc09: pop ebx
0x1000cc0a: add esp,2Ch
0x1000cc0d: ret 4
```

`sub_1000caf0` is called by sub_1000d330, sub_1000d430. It delegates to sub_1000e4a0, sub_1000f190, j_ORDINAL 567 (6 callees).

12.201 sub_1000cc10 at 0x1000cc10 (102 instructions)

```
0x1000cc10: push 0FFFFFFFFh
0x1000cc12: push 100FC94Bh
0x1000cc17: mov eax,fs:[0]
0x1000cc1d: push eax
0x1000cc1e: sub esp,3ECh
0x1000cc24: mov eax,ds:[10136220h]
0x1000cc29: xor eax,esp
0x1000cc2b: mov [esp+3E4h],eax
0x1000cc32: push ebp
0x1000cc33: push esi
0x1000cc34: push edi
0x1000cc35: mov eax,ds:[10136220h]
0x1000cc3a: xor eax,esp
0x1000cc3c: push eax
0x1000cc3d: lea eax,[esp+3FCh]
0x1000cc44: mov fs:[0],eax
0x1000cc4a: mov eax,[esp+40Ch]
0x1000cc51: lea edi,[ebx+0Ch]
0x1000cc54: mov [esp+10h],eax
0x1000cc58: mov esi,edi
0x1000cc5a: mov ebp,4
0x1000cc5f: nop
0x1000cc60: mov ecx,esi
0x1000cc62: call 1000F190h
0x1000cc67: test al,al
0x1000cc69: je short 1000CC72h
0x1000cc6b: mov ecx,esi
0x1000cc6d: call 1000E4A0h
0x1000cc72: add esi,3D0h
0x1000cc78: sub ebp,1
0x1000cc7b: jne short 1000CC60h
0x1000cc7d: xor ecx,ecx
0x1000cc7f: xor eax,eax
0x1000cc81: mov [ebx+0F50h],ecx
0x1000cc87: push ebp
0x1000cc88: lea ecx,[esp+28h]
```

```
0x1000cc8c: mov [ebx+0F4Ch],eax
0x1000cc92: call 1000E480h
0x1000cc97: mov ecx,[esp+10h]
0x1000cc9b: push ebp
0x1000cc9c: push ecx
0x1000cc9d: lea ecx,[esp+2Ch]
0x1000cca1: mov [esp+40Ch],ebp
0x1000cca8: call 100102B0h
0x1000ccad: xor esi,esi
0x1000ccaf: nop
0x1000ccb0: lea ecx,[esp+24h]
0x1000ccb4: call 1000EFF0h
0x1000ccb9: shr eax,2
0x1000ccbc: imul eax,esi
0x1000ccbf: lea ecx,[esp+24h]
0x1000ccc3: mov [esp+14h],eax
0x1000ccc7: call 1000EFF0h
0x1000cccc: shr eax,2
0x1000cccf: add eax,[esp+14h]
0x1000ccd3: lea ecx,[esp+24h]
0x1000ccd7: mov [esp+1Ch],eax
0x1000ccdb: mov [esp+18h],ebp
0x1000ccdf: call 1000EFE0h
0x1000cce4: push 1
0x1000cce6: push 1
0x1000cce8: push 1
0x1000ccea: lea edx,[esp+30h]
0x1000ccee: push edx
0x1000ccef: mov ecx,edi
0x1000ccf1: mov [esp+30h],eax
0x1000ccf5: call 1000ECF0h
0x1000ccfa: push ebp
0x1000ccfb: lea eax,[esp+18h]
0x1000ccff: push eax
0x1000cd00: mov ecx,edi
0x1000cd02: call 10011F00h
0x1000cd07: add esi,1
0x1000cd0a: add edi,3D0h
0x1000cd10: cmp esi,4
0x1000cd13: jl short 1000CCB0h
0x1000cd15: lea ecx,[esp+24h]
0x1000cd19: call 1000EFE0h
0x1000cd1e: lea ecx,[esp+24h]
0x1000cd22: mov esi,eax
0x1000cd24: call 1000EFF0h
0x1000cd29: shr eax,2
0x1000cd2c: mov dword ptr [esp+404h],0FFFFFFFFh
0x1000cd37: lea ecx,[esp+24h]
0x1000cd3b: mov [ebx+0F4Ch],eax
0x1000cd41: mov [ebx+0F50h],esi
0x1000cd47: mov dword ptr [esp+24h],100FE724h
0x1000cd4f: call 1000E550h
0x1000cd54: lea ecx,[esp+24h]
0x1000cd58: call 1000E4A0h
0x1000cd5d: mov eax,1
0x1000cd62: mov ecx,[esp+3FCh]
0x1000cd69: mov fs:[0],ecx
0x1000cd70: pop ecx
0x1000cd71: pop edi
0x1000cd72: pop esi
0x1000cd73: pop ebp
0x1000cd74: mov ecx,[esp+3E4h]
0x1000cd7b: xor ecx,esp
0x1000cd7d: call 10022326h
0x1000cd82: add esp,3F8h
0x1000cd88: ret 4
```

`sub_1000cc10` is called by sub_1000d330. It delegates to sub_1000e480, sub_1000e4a0, sub_1000e550 (10 callees).

12.202 sub_1000cd90 at 0x1000cd90 (244 instructions)

```
0x1000cd90: push 0FFFFFFFFh
0x1000cd92: push 100FD0A8h
0x1000cd97: mov eax,fs:[0]
0x1000cd9d: push eax
0x1000cd9e: sub esp,34h
0x1000cda1: push ebx
0x1000cda2: push ebp
0x1000cda3: push edi
0x1000cda4: mov eax,ds:[10136220h]
0x1000cda9: xor eax,esp
0x1000cdab: push eax
0x1000cdac: lea eax,[esp+44h]
0x1000cdb0: mov fs:[0],eax
0x1000cdb6: mov ebx,ecx
0x1000cdb8: mov ebp,[esp+54h]
0x1000cdbc: lea edi,[ebp+0F54h]
0x1000cdc2: push edi
0x1000cdc3: call dword ptr ds:[100FE6ACh]
0x1000cdc9: test eax,eax
0x1000cdcb: jne near ptr 1000D088h
0x1000cdd1: cmp [ebp+0F4Ch],eax
0x1000cdd7: jle near ptr 1000D088h
0x1000cddd: cmp [ebp+0F50h],eax
0x1000cde3: jle near ptr 1000D088h
0x1000cde9: cmp [ebp+0F64h],al
0x1000cdef: je near ptr 1000CEA0h
0x1000cdf5: mov eax,[ebp+0F74h]
0x1000cdfb: push eax
0x1000cdfc: push 1
0x1000cdfe: push 0
0x1000ce00: lea ecx,[esp+24h]
0x1000ce04: call 10022302h
0x1000ce09: lea ecx,[esp+18h]
0x1000ce0d: push ecx
0x1000ce0e: mov ecx,esi
0x1000ce10: mov dword ptr [esp+50h],0
0x1000ce18: call 1002230Eh
0x1000ce1d: mov ebx,eax
0x1000ce1f: mov eax,[edi+0Ch]
0x1000ce22: add eax,[edi+4]
0x1000ce25: cdq
0x1000ce26: sub eax,edx
0x1000ce28: mov ecx,eax
0x1000ce2a: mov eax,[edi+8]
0x1000ce2d: add eax,[edi]
0x1000ce2f: sar ecx,1
0x1000ce31: cdq
0x1000ce32: sub eax,edx
0x1000ce34: add ecx,0FFFFFFF9h
0x1000ce37: push ecx
0x1000ce38: sar eax,1
0x1000ce3a: push eax
0x1000ce3b: lea edx,[esp+30h]
0x1000ce3f: push edx
0x1000ce40: mov ecx,esi
0x1000ce42: call 1002231Ah
0x1000ce47: mov eax,[edi+0Ch]
0x1000ce4a: add eax,[edi+4]
0x1000ce4d: cdq
0x1000ce4e: sub eax,edx
0x1000ce50: mov ecx,eax
0x1000ce52: mov eax,[edi+8]
0x1000ce55: add eax,[edi]
```

```
0x1000ce57: sar ecx,1
0x1000ce59: cdq
0x1000ce5a: add ecx,7
0x1000ce5d: sub eax,edx
0x1000ce5f: push ecx
0x1000ce60: sar eax,1
0x1000ce62: push eax
0x1000ce63: mov ecx,esi
0x1000ce65: call 10022320h
0x1000ce6a: push ebx
0x1000ce6b: mov ecx,esi
0x1000ce6d: call 1002230Eh
0x1000ce72: mov dword ptr [esp+4Ch],0FFFFFFFFh
0x1000ce7a: lea ecx,[esp+18h]
0x1000ce7e: mov dword ptr [esp+18h],1012F7C0h
0x1000ce86: call 10001370h
0x1000ce8b: mov ecx,[esp+44h]
0x1000ce8f: mov fs:[0],ecx
0x1000ce96: pop ecx
0x1000ce97: pop edi
0x1000ce98: pop ebp
0x1000ce99: pop ebx
0x1000ce9a: add esp,40h
0x1000ce9d: ret 4
0x1000cea0: lea ecx,[ebp+0Ch]
0x1000cea3: call 1000F190h
0x1000cea8: test al,al
0x1000ceaa: jne near ptr 1000D040h
0x1000ceb0: test ebx,ebx
0x1000ceb2: mov ecx,[ebp+0F74h]
0x1000ceb8: mov eax,484848h
0x1000cebd: mov [esp+54h],ecx
0x1000cec1: je short 1000CEF6h
0x1000cec3: mov ecx,[ebp+0F68h]
0x1000cec9: test cl,4
0x1000cecc: jne short 1000CEF6h
0x1000cece: test cl,3
0x1000ced1: je short 1000CEE0h
0x1000ced3: mov edx,[ebp+0F7Ch]
0x1000ced9: mov eax,4040C0h
0x1000cede: jmp short 1000CF01h
0x1000cee0: test cl,8
0x1000cee3: je short 1000CF05h
0x1000cee5: mov ecx,[ebp+0F80h]
0x1000ceeb: mov eax,404080h
0x1000cef0: mov [esp+54h],ecx
0x1000cef4: jmp short 1000CF05h
0x1000cef6: mov edx,[ebp+0F78h]
0x1000cefc: mov eax,808080h
0x1000cf01: mov [esp+54h],edx
0x1000cf05: push eax
0x1000cf06: push 1
0x1000cf08: push 0
0x1000cf0a: lea ecx,[esp+2Ch]
0x1000cf0e: call 10022302h
0x1000cf13: mov dword ptr [esp+4Ch],1
0x1000cf1b: mov dword ptr [esp+14h],0
0x1000cf23: mov dword ptr [esp+10h],1012F4F0h
0x1000cf2b: lea eax,[esp+10h]
0x1000cf2f: push eax
0x1000cf30: mov byte ptr [esp+50h],2
0x1000cf35: call 1000BAA0h
0x1000cf3a: lea ecx,[esp+20h]
0x1000cf3e: push ecx
0x1000cf3f: mov ecx,esi
0x1000cf41: call 1002230Eh
0x1000cf46: lea edx,[esp+10h]
```

```
0x1000cf4a: push edx
0x1000cf4b: mov ecx,esi
0x1000cf4d: mov ebx,eax
0x1000cf4f: call 10022314h
0x1000cf54: push eax
0x1000cf55: mov ecx,esi
0x1000cf57: call 10022314h
0x1000cf5c: push ebx
0x1000cf5d: mov ecx,esi
0x1000cf5f: call 1002230Eh
0x1000cf64: mov eax,[edi]
0x1000cf66: mov ecx,[edi+4]
0x1000cf69: mov edx,[edi+8]
0x1000cf6c: mov [esp+30h],eax
0x1000cf70: mov eax,[edi+0Ch]
0x1000cf73: mov [esp+3Ch],eax
0x1000cf77: mov eax,[ebp+0F68h]
0x1000cf7d: test al,4
0x1000cf7f: mov [esp+34h],ecx
0x1000cf83: mov [esp+38h],edx
0x1000cf87: jne short 1000CF9Ch
0x1000cf89: test al,3
0x1000cf8b: je short 1000CF9Ch
0x1000cf8d: push 1
0x1000cf8f: push 1
0x1000cf91: lea ecx,[esp+38h]
0x1000cf95: push ecx
0x1000cf96: call dword ptr ds:[100FE6B0h]
0x1000cf9c: mov edx,[esi]
0x1000cf9e: mov eax,[ebp+0F70h]
0x1000cfa4: mov edx,[edx+30h]
0x1000cfa7: push eax
0x1000cfa8: mov ecx,esi
0x1000cfaa: call edx
0x1000cfac: mov ecx,[esp+54h]
0x1000cfb0: mov edi,eax
0x1000cfb2: mov eax,[esi]
0x1000cfb4: mov edx,[eax+38h]
0x1000cfb7: push ecx
0x1000cfb8: mov ecx,esi
0x1000cfba: call edx
0x1000cfbc: push 1
0x1000cfbe: mov ecx,esi
0x1000cfc0: mov ebx,eax
0x1000cfc2: call 10022296h
0x1000cfc7: mov [esp+54h],eax
0x1000cfcb: push 25h
0x1000cfcd: lea eax,[esp+34h]
0x1000cfd1: push eax
0x1000cfd2: lea eax,[ebp+4]
0x1000cfd5: mov ecx,esi
0x1000cfd7: call 100074D0h
0x1000cfdc: mov ecx,[esp+54h]
0x1000cfe0: push ecx
0x1000cfe1: mov ecx,esi
0x1000cfe3: call 10022296h
0x1000cfe8: mov edx,[esi]
0x1000cfea: mov eax,[edx+38h]
0x1000cfed: push ebx
0x1000cfee: mov ecx,esi
0x1000cff0: call eax
0x1000cff2: mov edx,[esi]
0x1000cff4: mov eax,[edx+30h]
0x1000cff7: push edi
0x1000cff8: mov ecx,esi
0x1000cffa: call eax
0x1000cffc: mov byte ptr [esp+4Ch],1
```

```
0x1000d001: lea ecx,[esp+10h]
0x1000d005: mov dword ptr [esp+10h],1012F4F0h
0x1000d00d: call 10001370h
0x1000d012: mov dword ptr [esp+4Ch],0FFFFFFFFh
0x1000d01a: lea ecx,[esp+20h]
0x1000d01e: mov dword ptr [esp+20h],1012F7C0h
0x1000d026: call 10001370h
0x1000d02b: mov ecx,[esp+44h]
0x1000d02f: mov fs:[0],ecx
0x1000d036: pop ecx
0x1000d037: pop edi
0x1000d038: pop ebp
0x1000d039: pop ebx
0x1000d03a: add esp,40h
0x1000d03d: ret 4
0x1000d040: xor eax,eax
0x1000d042: test ebx,ebx
0x1000d044: je short 1000D069h
0x1000d046: mov ecx,[ebp+0F68h]
0x1000d04c: test cl,4
0x1000d04f: jne short 1000D069h
0x1000d051: test cl,3
0x1000d054: je short 1000D05Dh
0x1000d056: mov eax,2
0x1000d05b: jmp short 1000D06Eh
0x1000d05d: test cl,8
0x1000d060: je short 1000D06Eh
0x1000d062: mov eax,1
0x1000d067: jmp short 1000D06Eh
0x1000d069: mov eax,3
0x1000d06e: mov ecx,[esi+4]
0x1000d071: imul eax,3D0h
0x1000d077: push 0
0x1000d079: push 0
0x1000d07b: push 0
0x1000d07d: push edi
0x1000d07e: push ecx
0x1000d07f: lea ecx,[eax+ebp+0Ch]
0x1000d083: call 10012E00h
0x1000d088: mov ecx,[esp+44h]
0x1000d08c: mov fs:[0],ecx
0x1000d093: pop ecx
0x1000d094: pop edi
0x1000d095: pop ebp
0x1000d096: pop ebx
0x1000d097: add esp,40h
0x1000d09a: ret 4
```

`sub_1000cd90` is called by sub_1000dac0. It delegates to sub_10001370, sub_100074d0, sub_1000baa0 (11 callees).

12.203 sub_1000d0a0 at 0x1000d0a0 (70 instructions)

```
0x1000d0a0: push 0FFFFFFFFh
0x1000d0a2: push 100FD061h
0x1000d0a7: mov eax,fs:[0]
0x1000d0ad: push eax
0x1000d0ae: sub esp,64h
0x1000d0b1: mov eax,ds:[10136220h]
0x1000d0b6: xor eax,esp
0x1000d0b8: mov [esp+60h],eax
0x1000d0bc: push ebx
0x1000d0bd: push edi
0x1000d0be: mov eax,ds:[10136220h]
0x1000d0c3: xor eax,esp
0x1000d0c5: push eax
0x1000d0c6: lea eax,[esp+70h]
```

```
0x1000d0ca: mov fs:[0],eax
0x1000d0d0: xor ebx,ebx
0x1000d0d2: mov dword ptr [esi],1012F80Ch
0x1000d0d8: mov [esp+0Ch],esi
0x1000d0dc: mov [esi+8],ebx
0x1000d0df: mov [esi+0Ch],ebx
0x1000d0e2: mov [esi+10h],ebx
0x1000d0e5: lea edi,[esi+14h]
0x1000d0e8: mov [esp+78h],ebx
0x1000d0ec: mov [edi+4],ebx
0x1000d0ef: mov dword ptr [edi],1012F744h
0x1000d0f5: mov [esi+24h],ebx
0x1000d0f8: mov dword ptr [esi+20h],1012F498h
0x1000d0ff: mov byte ptr [esp+78h],2
0x1000d104: push 1012F714h
0x1000d109: lea eax,[esp+30h]
0x1000d10d: push eax
0x1000d10e: mov [esp+1Ch],ebx
0x1000d112: mov [esp+20h],ebx
0x1000d116: mov [esp+24h],ebx
0x1000d11a: mov [esp+28h],ebx
0x1000d11e: mov [esp+2Ch],bl
0x1000d122: mov [esp+2Dh],bl
0x1000d126: mov [esp+2Eh],bl
0x1000d12a: mov byte ptr [esp+2Fh],1
0x1000d12f: call dword ptr ds:[100FE104h]
0x1000d135: lea ecx,[esp+10h]
0x1000d139: push ecx
0x1000d13a: mov [esp+2Ch],bl
0x1000d13e: mov [esp+2Dh],bl
0x1000d142: mov [esp+2Eh],bl
0x1000d146: mov [esp+2Fh],bl
0x1000d14a: mov dword ptr [esp+14h],13h
0x1000d152: mov [esp+18h],ebx
0x1000d156: mov dword ptr [esp+24h],190h
0x1000d15e: call dword ptr ds:[100FE064h]
0x1000d164: push eax
0x1000d165: mov ecx,edi
0x1000d167: call 10021FFCh
0x1000d16c: xor eax,eax
0x1000d16e: xor ecx,ecx
0x1000d170: mov [esi+28h],eax
0x1000d173: mov dword ptr [esi+1Ch],8
0x1000d17a: mov [esi+2Ch],ecx
0x1000d17d: mov [esi+30h],ebx
0x1000d180: mov eax,esi
0x1000d182: mov ecx,[esp+70h]
0x1000d186: mov fs:[0],ecx
0x1000d18d: pop ecx
0x1000d18e: pop edi
0x1000d18f: pop ebx
0x1000d190: mov ecx,[esp+60h]
0x1000d194: xor ecx,esp
0x1000d196: call 10022326h
0x1000d19b: add esp,70h
0x1000d19e: ret
```

`sub_1000d0a0` is called by sub_100032e0. It works with the text "Microsoft Yahei". It delegates to j_ORDINAL 1271, sub_10022326 (2 callees).

12.204 sub_1000d260 at 0x1000d260 (43 instructions)

```
0x1000d260: push ebx
0x1000d261: push ebp
0x1000d262: mov ebp,[esp+0Ch]
0x1000d266: push esi
```

```
0x1000d267: push edi
0x1000d268: xor esi,esi
0x1000d26a: lea ebx,[ebx]
0x1000d270: mov ecx,[ebp+8]
0x1000d273: test ecx,ecx
0x1000d275: jne short 1000D27Bh
0x1000d277: xor eax,eax
0x1000d279: jmp short 1000D283h
0x1000d27b: mov eax,[ebp+0Ch]
0x1000d27e: sub eax,ecx
0x1000d280: sar eax,2
0x1000d283: cmp esi,eax
0x1000d285: jge short 1000D2C0h
0x1000d287: test ecx,ecx
0x1000d289: je short 1000D297h
0x1000d28b: mov eax,[ebp+0Ch]
0x1000d28e: sub eax,ecx
0x1000d290: sar eax,2
0x1000d293: cmp esi,eax
0x1000d295: jb short 1000D29Dh
0x1000d297: call dword ptr ds:[100FE584h]
0x1000d29d: mov eax,[ebp+8]
0x1000d2a0: mov eax,[eax+esi*4]
0x1000d2a3: add eax,0F54h
0x1000d2a8: xor ecx,ecx
0x1000d2aa: xor edx,edx
0x1000d2ac: mov [eax],ecx
0x1000d2ae: xor edi,edi
0x1000d2b0: xor ebx,ebx
0x1000d2b2: mov [eax+4],edx
0x1000d2b5: mov [eax+8],edi
0x1000d2b8: mov [eax+0Ch],ebx
0x1000d2bb: add esi,1
0x1000d2be: jmp short 1000D270h
0x1000d2c0: pop edi
0x1000d2c1: pop esi
0x1000d2c2: pop ebp
0x1000d2c3: pop ebx
0x1000d2c4: ret 4
```

`sub_1000d260` is called by sub_100056c0, sub_1000d9b0. It calls nothing else.

12.205 sub_1000d2d0 at 0x1000d2d0 (31 instructions)

```
0x1000d2d0: push ebp
0x1000d2d1: mov ebp,ds:[100FE584h]
0x1000d2d7: push edi
0x1000d2d8: xor edi,edi
0x1000d2da: lea ebx,[ebx]
0x1000d2e0: mov ecx,[esi+8]
0x1000d2e3: test ecx,ecx
0x1000d2e5: jne short 1000D2EBh
0x1000d2e7: xor eax,eax
0x1000d2e9: jmp short 1000D2F3h
0x1000d2eb: mov eax,[esi+0Ch]
0x1000d2ee: sub eax,ecx
0x1000d2f0: sar eax,2
0x1000d2f3: cmp edi,eax
0x1000d2f5: jge short 1000D31Eh
0x1000d2f7: test ecx,ecx
0x1000d2f9: je short 1000D307h
0x1000d2fb: mov eax,[esi+0Ch]
0x1000d2fe: sub eax,ecx
0x1000d300: sar eax,2
0x1000d303: cmp edi,eax
0x1000d305: jb short 1000D309h
```

```
0x1000d307: call ebp
0x1000d309: mov eax,[esi+8]
0x1000d30c: mov ecx,[eax+edi*4]
0x1000d30f: mov dword ptr [ecx+0F68h],0
0x1000d319: add edi,1
0x1000d31c: jmp short 1000D2E0h
0x1000d31e: pop edi
0x1000d31f: pop ebp
0x1000d320: ret
```

`sub_1000d2d0` is called by ?UncheckAllToolButton@CCustomTitle@@QAEXXZ. It calls nothing else.

12.206 sub_1000d330 at 0x1000d330 (81 instructions)

```
0x1000d330: push 0FFFFFFFFh
0x1000d332: push 100FC70Bh
0x1000d337: mov eax,fs:[0]
0x1000d33d: push eax
0x1000d33e: push ecx
0x1000d33f: push ebx
0x1000d340: push esi
0x1000d341: push edi
0x1000d342: mov eax,ds:[10136220h]
0x1000d347: xor eax,esp
0x1000d349: push eax
0x1000d34a: lea eax,[esp+14h]
0x1000d34e: mov fs:[0],eax
0x1000d354: mov esi,ecx
0x1000d356: push 0F84h
0x1000d35b: or edi,0FFFFFFFFh
0x1000d35e: call 10021FDEh
0x1000d363: add esp,4
0x1000d366: mov [esp+10h],eax
0x1000d36a: xor ebx,ebx
0x1000d36c: cmp eax,ebx
0x1000d36e: mov [esp+1Ch],ebx
0x1000d372: je short 1000D37Ch
0x1000d374: push eax
0x1000d375: call 1000C9C0h
0x1000d37a: mov ebx,eax
0x1000d37c: mov ecx,[esp+2Ch]
0x1000d380: mov eax,[esp+28h]
0x1000d384: mov [esp+1Ch],edi
0x1000d388: push ecx
0x1000d389: lea ecx,[ebx+4]
0x1000d38c: mov [esp+14h],ebx
0x1000d390: mov [ebx],eax
0x1000d392: call dword ptr ds:[100FE198h]
0x1000d398: mov edx,[esp+30h]
0x1000d39c: push edx
0x1000d39d: lea ecx,[ebx+8]
0x1000d3a0: call dword ptr ds:[100FE198h]
0x1000d3a6: cmp dword ptr [esp+34h],0
0x1000d3ab: mov dword ptr [ebx+0F6Ch],7
0x1000d3b5: jne short 1000D3E0h
0x1000d3b7: test esi,esi
0x1000d3b9: jne short 1000D3E0h
0x1000d3bb: mov eax,[esp+24h]
0x1000d3bf: add eax,14h
0x1000d3c2: push eax
0x1000d3c3: mov esi,ebx
0x1000d3c5: call 1000CAF0h
0x1000d3ca: test eax,eax
0x1000d3cc: je short 1000D3EAh
0x1000d3ce: mov esi,[esp+24h]
0x1000d3d2: mov eax,[esi+8]
```

```
0x1000d3d5: add esi,4
0x1000d3d8: test eax,eax
0x1000d3da: jne short 1000D3FBh
0x1000d3dc: xor edi,edi
0x1000d3de: jmp short 1000D403h
0x1000d3e0: push esi
0x1000d3e1: call 1000CC10h
0x1000d3e6: test eax,eax
0x1000d3e8: jne short 1000D3CEh
0x1000d3ea: push ebx
0x1000d3eb: call 1000CA80h
0x1000d3f0: push ebx
0x1000d3f1: call 10021FCCh
0x1000d3f6: add esp,4
0x1000d3f9: jmp short 1000D40Ch
0x1000d3fb: mov edi,[esi+8]
0x1000d3fe: sub edi,eax
0x1000d400: sar edi,2
0x1000d403: lea ebx,[esp+10h]
0x1000d407: call 1000B9E0h
0x1000d40c: mov eax,edi
0x1000d40e: mov ecx,[esp+14h]
0x1000d412: mov fs:[0],ecx
0x1000d419: pop ecx
0x1000d41a: pop edi
0x1000d41b: pop esi
0x1000d41c: pop ebx
0x1000d41d: add esp,10h
0x1000d420: ret 14h
```

`sub_1000d330` is called by ?AddToolButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0PB_W@Z. It delegates to sub_1000b9e0, sub_1000c9c0, sub_1000ca80 (7 callees).

12.207 sub_1000d430 at 0x1000d430 (87 instructions)

```
0x1000d430: push 0FFFFFFFFh
0x1000d432: push 100FC70Bh
0x1000d437: mov eax,fs:[0]
0x1000d43d: push eax
0x1000d43e: push ecx
0x1000d43f: push ebx
0x1000d440: push esi
0x1000d441: push edi
0x1000d442: mov eax,ds:[10136220h]
0x1000d447: xor eax,esp
0x1000d449: push eax
0x1000d44a: lea eax,[esp+14h]
0x1000d44e: mov fs:[0],eax
0x1000d454: mov edi,ecx
0x1000d456: push 0F84h
0x1000d45b: or ebx,0FFFFFFFFh
0x1000d45e: call 10021FDEh
0x1000d463: add esp,4
0x1000d466: mov [esp+10h],eax
0x1000d46a: test eax,eax
0x1000d46c: mov dword ptr [esp+1Ch],0
0x1000d474: je short 1000D480h
0x1000d476: push eax
0x1000d477: call 1000C9C0h
0x1000d47c: mov esi,eax
0x1000d47e: jmp short 1000D482h
0x1000d480: xor esi,esi
0x1000d482: mov ecx,[esp+28h]
0x1000d486: mov eax,[esp+24h]
0x1000d48a: mov [esp+1Ch],ebx
```

```
0x1000d48e: push ecx
0x1000d48f: lea ecx,[esi+4]
0x1000d492: mov [esp+14h],esi
0x1000d496: mov [esi],eax
0x1000d498: call dword ptr ds:[100FE198h]
0x1000d49e: mov edx,[esp+2Ch]
0x1000d4a2: push edx
0x1000d4a3: lea ecx,[esi+8]
0x1000d4a6: call dword ptr ds:[100FE198h]
0x1000d4ac: mov eax,[esp+30h]
0x1000d4b0: mov ecx,[esp+34h]
0x1000d4b4: mov edx,[esp+38h]
0x1000d4b8: mov [esi+0F74h],eax
0x1000d4be: mov eax,[esp+3Ch]
0x1000d4c2: mov [esi+0F78h],ecx
0x1000d4c8: mov [esi+0F7Ch],edx
0x1000d4ce: lea ecx,[edi+14h]
0x1000d4d1: mov [esi+0F80h],eax
0x1000d4d7: push ecx
0x1000d4d8: mov dword ptr [esi+0F6Ch],7
0x1000d4e2: call 1000CAF0h
0x1000d4e7: test eax,eax
0x1000d4e9: jne short 1000D511h
0x1000d4eb: push esi
0x1000d4ec: call 1000CA80h
0x1000d4f1: push esi
0x1000d4f2: call 10021FCCh
0x1000d4f7: add esp,4
0x1000d4fa: mov eax,ebx
0x1000d4fc: mov ecx,[esp+14h]
0x1000d500: mov fs:[0],ecx
0x1000d507: pop ecx
0x1000d508: pop edi
0x1000d509: pop esi
0x1000d50a: pop ebx
0x1000d50b: add esp,10h
0x1000d50e: ret 1Ch
0x1000d511: mov eax,[edi+8]
0x1000d514: test eax,eax
0x1000d516: lea esi,[edi+4]
0x1000d519: jne short 1000D51Fh
0x1000d51b: xor edi,edi
0x1000d51d: jmp short 1000D527h
0x1000d51f: mov edi,[esi+8]
0x1000d522: sub edi,eax
0x1000d524: sar edi,2
0x1000d527: lea ebx,[esp+10h]
0x1000d52b: call 1000B9E0h
0x1000d530: mov eax,edi
0x1000d532: mov ecx,[esp+14h]
0x1000d536: mov fs:[0],ecx
0x1000d53d: pop ecx
0x1000d53e: pop edi
0x1000d53f: pop esi
0x1000d540: pop ebx
0x1000d541: add esp,10h
0x1000d544: ret 1Ch
```

`sub_1000d430` is called by ?AddToolButton@CCustomTitle@@QAEHIABV?$CStringT@_WV?$StrTraitMFC_DLL@_WV?$ChTraitsCRT@_W@ATL@@@@@ATL@@0KKKK@Z. It delegates to sub_1000b9e0, sub_1000c9c0, sub_1000ca80 (6 callees).

12.208 sub_1000d550 at 0x1000d550 (85 instructions)

```
0x1000d550: push 0FFFFFFFFh
0x1000d552: push 100FCB4Bh
```

```
0x1000d557: mov eax,fs:[0]
0x1000d55d: push eax
0x1000d55e: sub esp,0Ch
0x1000d561: push ebx
0x1000d562: push ebp
0x1000d563: push esi
0x1000d564: push edi
0x1000d565: mov eax,ds:[10136220h]
0x1000d56a: xor eax,esp
0x1000d56c: push eax
0x1000d56d: lea eax,[esp+20h]
0x1000d571: mov fs:[0],eax
0x1000d577: push 0F84h
0x1000d57c: call 10021FDEh
0x1000d581: add esp,4
0x1000d584: mov [esp+14h],eax
0x1000d588: test eax,eax
0x1000d58a: mov dword ptr [esp+28h],0
0x1000d592: je short 1000D59Ch
0x1000d594: push eax
0x1000d595: call 1000C9C0h
0x1000d59a: jmp short 1000D59Eh
0x1000d59c: xor eax,eax
0x1000d59e: mov ecx,[esp+34h]
0x1000d5a2: mov esi,[esp+30h]
0x1000d5a6: mov dword ptr [esp+28h],0FFFFFFFFh
0x1000d5ae: mov byte ptr [eax+0F64h],1
0x1000d5b5: mov [eax+0F74h],ecx
0x1000d5bb: mov ecx,8
0x1000d5c0: mov [eax+0F4Ch],ecx
0x1000d5c6: add esi,4
0x1000d5c9: mov [eax+0F50h],ecx
0x1000d5cf: mov ecx,[esi+4]
0x1000d5d2: test ecx,ecx
0x1000d5d4: mov [esp+14h],eax
0x1000d5d8: jne short 1000D5DEh
0x1000d5da: xor ebx,ebx
0x1000d5dc: jmp short 1000D5E6h
0x1000d5de: mov ebx,[esi+8]
0x1000d5e1: sub ebx,ecx
0x1000d5e3: sar ebx,2
0x1000d5e6: mov edi,ecx
0x1000d5e8: test edi,edi
0x1000d5ea: jne short 1000D5F0h
0x1000d5ec: xor edx,edx
0x1000d5ee: jmp short 1000D5F8h
0x1000d5f0: mov edx,[esi+8]
0x1000d5f3: sub edx,edi
0x1000d5f5: sar edx,2
0x1000d5f8: test edi,edi
0x1000d5fa: je short 1000D615h
0x1000d5fc: mov ecx,[esi+0Ch]
0x1000d5ff: sub ecx,edi
0x1000d601: sar ecx,2
0x1000d604: cmp edx,ecx
0x1000d606: jae short 1000D615h
0x1000d608: mov ecx,[esi+8]
0x1000d60b: mov [ecx],eax
0x1000d60d: add ecx,4
0x1000d610: mov [esi+8],ecx
0x1000d613: jmp short 1000D635h
0x1000d615: mov ebp,[esi+8]
0x1000d618: cmp edi,ebp
0x1000d61a: jbe short 1000D622h
0x1000d61c: call dword ptr ds:[100FE584h]
0x1000d622: push ebp
0x1000d623: push esi
```

```
0x1000d624: lea edx,[esp+1Ch]
0x1000d628: push edx
0x1000d629: lea eax,[esp+24h]
0x1000d62d: push eax
0x1000d62e: mov eax,esi
0x1000d630: call 100098C0h
0x1000d635: mov eax,ebx
0x1000d637: mov ecx,[esp+20h]
0x1000d63b: mov fs:[0],ecx
0x1000d642: pop ecx
0x1000d643: pop edi
0x1000d644: pop esi
0x1000d645: pop ebp
0x1000d646: pop ebx
0x1000d647: add esp,18h
0x1000d64a: ret 8
```

`sub_1000d550` is called by ?AddToolSeparator@CCustomTitle@@QAEHK@Z. It delegates to sub_100098c0, sub_1000c9c0, j_ORDINAL 762 (3 callees).

12.209 sub_1000d650 at 0x1000d650 (38 instructions)

```
0x1000d650: push ebp
0x1000d651: mov ebp,ds:[100FE584h]
0x1000d657: push edi
0x1000d658: xor edi,edi
0x1000d65a: lea ebx,[ebx]
0x1000d660: mov ecx,[esi+8]
0x1000d663: test ecx,ecx
0x1000d665: je short 1000D6BCh
0x1000d667: mov eax,[esi+0Ch]
0x1000d66a: sub eax,ecx
0x1000d66c: sar eax,2
0x1000d66f: cmp edi,eax
0x1000d671: jae short 1000D6BCh
0x1000d673: test ecx,ecx
0x1000d675: je short 1000D683h
0x1000d677: mov eax,[esi+0Ch]
0x1000d67a: sub eax,ecx
0x1000d67c: sar eax,2
0x1000d67f: cmp edi,eax
0x1000d681: jb short 1000D685h
0x1000d683: call ebp
0x1000d685: mov eax,[esi+8]
0x1000d688: mov eax,[eax+edi*4]
0x1000d68b: cmp [eax],ebx
0x1000d68d: je short 1000D694h
0x1000d68f: add edi,1
0x1000d692: jmp short 1000D660h
0x1000d694: mov ecx,[esp+0Ch]
0x1000d698: mov edx,[esp+10h]
0x1000d69c: mov [eax+0F74h],ecx
0x1000d6a2: mov ecx,[esp+14h]
0x1000d6a6: mov [eax+0F78h],edx
0x1000d6ac: mov edx,[esp+18h]
0x1000d6b0: mov [eax+0F7Ch],ecx
0x1000d6b6: mov [eax+0F80h],edx
0x1000d6bc: pop edi
0x1000d6bd: pop ebp
0x1000d6be: ret 10h
```

`sub_1000d650` is called by ?SetToolButtonTextColor@CCustomTitle@@QAEXIKKKK@Z. It calls nothing else.

12.210 sub_1000d6d0 at 0x1000d6d0 (31 instructions)

```
0x1000d6d0: push ebp
0x1000d6d1: mov ebp,ds:[100FE584h]
0x1000d6d7: push edi
0x1000d6d8: xor edi,edi
0x1000d6da: lea ebx,[ebx]
0x1000d6e0: mov ecx,[esi+8]
0x1000d6e3: test ecx,ecx
0x1000d6e5: je short 1000D71Fh
0x1000d6e7: mov eax,[esi+0Ch]
0x1000d6ea: sub eax,ecx
0x1000d6ec: sar eax,2
0x1000d6ef: cmp edi,eax
0x1000d6f1: jae short 1000D71Fh
0x1000d6f3: test ecx,ecx
0x1000d6f5: je short 1000D703h
0x1000d6f7: mov eax,[esi+0Ch]
0x1000d6fa: sub eax,ecx
0x1000d6fc: sar eax,2
0x1000d6ff: cmp edi,eax
0x1000d701: jb short 1000D705h
0x1000d703: call ebp
0x1000d705: mov eax,[esi+8]
0x1000d708: mov eax,[eax+edi*4]
0x1000d70b: cmp byte ptr [eax+0F64h],0
0x1000d712: je short 1000D71Ah
0x1000d714: mov [eax+0F74h],ebx
0x1000d71a: add edi,1
0x1000d71d: jmp short 1000D6E0h
0x1000d71f: pop edi
0x1000d720: pop ebp
0x1000d721: ret
```

`sub_1000d6d0` is called by ?SetToolSeparatorColor@CCustomTitle@@QAEXK@Z. It calls nothing else.

12.211 sub_1000d800 at 0x1000d800 (74 instructions)

```
0x1000d800: sub esp,18h
0x1000d803: push ebx
0x1000d804: push ebp
0x1000d805: push edi
0x1000d806: xor edi,edi
0x1000d808: mov ecx,[esi+8]
0x1000d80b: test ecx,ecx
0x1000d80d: jne short 1000D813h
0x1000d80f: xor eax,eax
0x1000d811: jmp short 1000D81Bh
0x1000d813: mov eax,[esi+0Ch]
0x1000d816: sub eax,ecx
0x1000d818: sar eax,2
0x1000d81b: cmp edi,eax
0x1000d81d: jge near ptr 1000D8C7h
0x1000d823: test ecx,ecx
0x1000d825: mov eax,[esp+28h]
0x1000d829: mov ebx,[eax]
0x1000d82b: mov ebp,[eax+4]
0x1000d82e: je short 1000D83Ch
0x1000d830: mov eax,[esi+0Ch]
0x1000d833: sub eax,ecx
0x1000d835: sar eax,2
0x1000d838: cmp edi,eax
0x1000d83a: jb short 1000D842h
0x1000d83c: call dword ptr ds:[100FE584h]
0x1000d842: mov eax,[esi+8]
0x1000d845: mov eax,[eax+edi*4]
0x1000d848: mov ecx,[eax+0F54h]
```

```
0x1000d84e: mov [esp+10h],ecx
0x1000d852: mov edx,[eax+0F58h]
0x1000d858: mov [esp+14h],edx
0x1000d85c: mov ecx,[eax+0F5Ch]
0x1000d862: push ebp
0x1000d863: mov [esp+1Ch],ecx
0x1000d867: mov edx,[eax+0F60h]
0x1000d86d: push ebx
0x1000d86e: lea eax,[esp+18h]
0x1000d872: push eax
0x1000d873: mov [esp+28h],edx
0x1000d877: call dword ptr ds:[100FE610h]
0x1000d87d: test eax,eax
0x1000d87f: jne short 1000D886h
0x1000d881: add edi,1
0x1000d884: jmp short 1000D808h
0x1000d886: mov ecx,[esi+8]
0x1000d889: test ecx,ecx
0x1000d88b: je short 1000D899h
0x1000d88d: mov eax,[esi+0Ch]
0x1000d890: sub eax,ecx
0x1000d892: sar eax,2
0x1000d895: cmp edi,eax
0x1000d897: jb short 1000D89Fh
0x1000d899: call dword ptr ds:[100FE584h]
0x1000d89f: mov ecx,[esi+8]
0x1000d8a2: mov edx,[ecx+edi*4]
0x1000d8a5: cmp byte ptr [edx+0F64h],0
0x1000d8ac: jne short 1000D8C7h
0x1000d8ae: test edi,edi
0x1000d8b0: jl short 1000D8BCh
0x1000d8b2: mov eax,[esp+2Ch]
0x1000d8b6: mov dword ptr [eax],12h
0x1000d8bc: mov eax,edi
0x1000d8be: pop edi
0x1000d8bf: pop ebp
0x1000d8c0: pop ebx
0x1000d8c1: add esp,18h
0x1000d8c4: ret 8
0x1000d8c7: pop edi
0x1000d8c8: pop ebp
0x1000d8c9: or eax,0FFFFFFFFh
0x1000d8cc: pop ebx
0x1000d8cd: add esp,18h
0x1000d8d0: ret 8
```

`sub_1000d800` is called by sub_10004910, sub_10005b50. It calls nothing else.

12.212 sub_1000d8e0 at 0x1000d8e0 (66 instructions)

```
0x1000d8e0: sub esp,14h
0x1000d8e3: push ebp
0x1000d8e4: push edi
0x1000d8e5: push ebx
0x1000d8e6: call dword ptr ds:[100FE5C8h]
0x1000d8ec: mov ebp,ds:[100FE6ACh]
0x1000d8f2: xor edi,edi
0x1000d8f4: mov ecx,[esi+8]
0x1000d8f7: test ecx,ecx
0x1000d8f9: jne short 1000D8FFh
0x1000d8fb: xor eax,eax
0x1000d8fd: jmp short 1000D907h
0x1000d8ff: mov eax,[esi+0Ch]
0x1000d902: sub eax,ecx
0x1000d904: sar eax,2
0x1000d907: cmp edi,eax
```

```
0x1000d909: jge near ptr 1000D99Dh
0x1000d90f: test ecx,ecx
0x1000d911: je short 1000D91Fh
0x1000d913: mov eax,[esi+0Ch]
0x1000d916: sub eax,ecx
0x1000d918: sar eax,2
0x1000d91b: cmp edi,eax
0x1000d91d: jb short 1000D925h
0x1000d91f: call dword ptr ds:[100FE584h]
0x1000d925: mov eax,[esi+8]
0x1000d928: mov eax,[eax+edi*4]
0x1000d92b: mov ecx,[eax+0F54h]
0x1000d931: mov [esp+8],ecx
0x1000d935: mov edx,[eax+0F58h]
0x1000d93b: mov [esp+0Ch],edx
0x1000d93f: mov ecx,[eax+0F5Ch]
0x1000d945: mov [esp+10h],ecx
0x1000d949: mov edx,[eax+0F60h]
0x1000d94f: lea eax,[esp+8]
0x1000d953: push eax
0x1000d954: mov [esp+18h],edx
0x1000d958: call ebp
0x1000d95a: test eax,eax
0x1000d95c: jne short 1000D995h
0x1000d95e: push ebx
0x1000d95f: call ebp
0x1000d961: test eax,eax
0x1000d963: je short 1000D988h
0x1000d965: mov ecx,[esp+8]
0x1000d969: mov edx,[esp+0Ch]
0x1000d96d: mov eax,[esp+10h]
0x1000d971: mov [ebx],ecx
0x1000d973: mov ecx,[esp+14h]
0x1000d977: mov [ebx+4],edx
0x1000d97a: mov [ebx+8],eax
0x1000d97d: mov [ebx+0Ch],ecx
0x1000d980: add edi,1
0x1000d983: jmp near ptr 1000D8F4h
0x1000d988: lea edx,[esp+8]
0x1000d98c: push edx
0x1000d98d: push ebx
0x1000d98e: push ebx
0x1000d98f: call dword ptr ds:[100FE5B8h]
0x1000d995: add edi,1
0x1000d998: jmp near ptr 1000D8F4h
0x1000d99d: pop edi
0x1000d99e: mov eax,ebx
0x1000d9a0: pop ebp
0x1000d9a1: add esp,14h
0x1000d9a4: ret
```

`sub_1000d8e0` is called by ?CheckToolButton@CCustomTitle@@QAEXI@Z, ?UncheckAllToolButton@CCustomTitle@@QAEXXZ, sub_10004910, sub_10004d40. It calls nothing else.

12.213 sub_1000d9b0 at 0x1000d9b0 (83 instructions)

```
0x1000d9b0: sub esp,18h
0x1000d9b3: push ebx
0x1000d9b4: push ebp
0x1000d9b5: push edi
0x1000d9b6: mov edi,[esp+28h]
0x1000d9ba: push edi
0x1000d9bb: call 1000D260h
0x1000d9c0: mov eax,[esp+2Ch]
0x1000d9c4: mov ecx,[eax]
0x1000d9c6: mov edx,[eax+4]
```

```
0x1000d9c9: mov [esi],ecx
0x1000d9cb: mov ecx,[eax+8]
0x1000d9ce: mov [esi+4],edx
0x1000d9d1: mov edx,[eax+0Ch]
0x1000d9d4: mov [esi+8],ecx
0x1000d9d7: mov [esi+0Ch],edx
0x1000d9da: mov eax,[edi+1Ch]
0x1000d9dd: add eax,[esi]
0x1000d9df: xor ebx,ebx
0x1000d9e1: mov [esi+8],eax
0x1000d9e4: mov [esp+10h],ebx
0x1000d9e8: mov ecx,[edi+8]
0x1000d9eb: test ecx,ecx
0x1000d9ed: jne short 1000D9F3h
0x1000d9ef: xor eax,eax
0x1000d9f1: jmp short 1000D9FBh
0x1000d9f3: mov eax,[edi+0Ch]
0x1000d9f6: sub eax,ecx
0x1000d9f8: sar eax,2
0x1000d9fb: cmp ebx,eax
0x1000d9fd: jge near ptr 1000DAB0h
0x1000da03: test ecx,ecx
0x1000da05: je short 1000DA13h
0x1000da07: mov eax,[edi+0Ch]
0x1000da0a: sub eax,ecx
0x1000da0c: sar eax,2
0x1000da0f: cmp ebx,eax
0x1000da11: jb short 1000DA19h
0x1000da13: call dword ptr ds:[100FE584h]
0x1000da19: mov ecx,[edi+8]
0x1000da1c: mov ecx,[ecx+ebx*4]
0x1000da1f: mov edx,[esi]
0x1000da21: mov eax,[esi+8]
0x1000da24: mov edi,[esi+4]
0x1000da27: mov ebx,eax
0x1000da29: mov ebp,[ecx+0F4Ch]
0x1000da2f: mov [esp+1Ch],eax
0x1000da33: mov eax,[ecx+0F50h]
0x1000da39: mov [esp+14h],edx
0x1000da3d: mov edx,[esi+0Ch]
0x1000da40: mov [esp+18h],eax
0x1000da44: mov [esp+20h],edx
0x1000da48: mov edx,[ecx+0F4Ch]
0x1000da4e: mov [esp+14h],edx
0x1000da52: mov edx,[esp+2Ch]
0x1000da56: mov eax,[edx+0Ch]
0x1000da59: sub eax,[edx+4]
0x1000da5c: mov [ecx+0F54h],ebx
0x1000da62: sub eax,[ecx+0F50h]
0x1000da68: add ebp,ebx
0x1000da6a: add eax,1
0x1000da6d: cdq
0x1000da6e: sub eax,edx
0x1000da70: sar eax,1
0x1000da72: add edi,eax
0x1000da74: mov eax,[ecx+0F4Ch]
0x1000da7a: mov [ecx+0F58h],edi
0x1000da80: mov [esp+14h],eax
0x1000da84: mov eax,[ecx+0F50h]
0x1000da8a: mov [ecx+0F5Ch],ebp
0x1000da90: add eax,edi
0x1000da92: mov edi,[esp+28h]
0x1000da96: add ebp,1
0x1000da99: add dword ptr [esp+10h],1
0x1000da9e: mov ebx,[esp+10h]
0x1000daa2: mov [ecx+0F60h],eax
0x1000daa8: mov [esi+8],ebp
```

```
0x1000daab: jmp near ptr 1000D9E8h
0x1000dab0: pop edi
0x1000dab1: pop ebp
0x1000dab2: pop ebx
0x1000dab3: add esp,18h
0x1000dab6: ret 8
```

`sub_1000d9b0` is called by sub_100056c0. It delegates to sub_1000d260 (1 callees).

12.214 sub_1000dac0 at 0x1000dac0 (38 instructions)

```
0x1000dac0: push ecx
0x1000dac1: push ebx
0x1000dac2: push ebp
0x1000dac3: mov ebp,[esp+10h]
0x1000dac7: push esi
0x1000dac8: mov esi,eax
0x1000daca: xor ebx,ebx
0x1000dacc: lea esp,[esp]
0x1000dad0: mov ecx,[edi+8]
0x1000dad3: test ecx,ecx
0x1000dad5: jne short 1000DADBh
0x1000dad7: xor eax,eax
0x1000dad9: jmp short 1000DAE3h
0x1000dadb: mov eax,[edi+0Ch]
0x1000dade: sub eax,ecx
0x1000dae0: sar eax,2
0x1000dae3: cmp ebx,eax
0x1000dae5: jge short 1000DB10h
0x1000dae7: test ecx,ecx
0x1000dae9: je short 1000DAF7h
0x1000daeb: mov eax,[edi+0Ch]
0x1000daee: sub eax,ecx
0x1000daf0: sar eax,2
0x1000daf3: cmp ebx,eax
0x1000daf5: jb short 1000DAFDh
0x1000daf7: call dword ptr ds:[100FE584h]
0x1000dafd: mov eax,[edi+8]
0x1000db00: mov ecx,[eax+ebx*4]
0x1000db03: push ecx
0x1000db04: mov ecx,ebp
0x1000db06: call 1000CD90h
0x1000db0b: add ebx,1
0x1000db0e: jmp short 1000DAD0h
0x1000db10: pop esi
0x1000db11: pop ebp
0x1000db12: pop ebx
0x1000db13: pop ecx
0x1000db14: ret 4
```

`sub_1000dac0` is called by sub_10005fd0. It delegates to sub_1000cd90 (1 callees).

12.215 sub_1000db80 at 0x1000db80 (96 instructions)

```
0x1000db80: push esi
0x1000db81: push edi
0x1000db82: mov edi,[esp+0Ch]
0x1000db86: test edi,edi
0x1000db88: jl short 1000DBCFh
0x1000db8a: mov esi,eax
0x1000db8c: mov ecx,[esi+8]
0x1000db8f: add esi,4
0x1000db92: test ecx,ecx
0x1000db94: jne short 1000DB9Ah
0x1000db96: xor eax,eax
```

```
0x1000db98: jmp short 1000DBA2h
0x1000db9a: mov eax,[esi+8]
0x1000db9d: sub eax,ecx
0x1000db9f: sar eax,2
0x1000dba2: cmp edi,eax
0x1000dba4: jge short 1000DBCFh
0x1000dba6: test ecx,ecx
0x1000dba8: je short 1000DBB6h
0x1000dbaa: mov eax,[esi+8]
0x1000dbad: sub eax,ecx
0x1000dbaf: sar eax,2
0x1000dbb2: cmp edi,eax
0x1000dbb4: jb short 1000DBBCh
0x1000dbb6: call dword ptr ds:[100FE584h]
0x1000dbbc: mov eax,[esi+4]
0x1000dbbf: lea edx,[eax+edi*4]
0x1000dbc2: mov eax,[edx]
0x1000dbc4: mov ecx,[eax+0F68h]
0x1000dbca: test cl,4
0x1000dbcd: je short 1000DBD6h
0x1000dbcf: pop edi
0x1000dbd0: xor eax,eax
0x1000dbd2: pop esi
0x1000dbd3: ret 4
0x1000dbd6: test cl,2
0x1000dbd9: je short 1000DBCFh
0x1000dbdb: push ebx
0x1000dbdc: mov ebx,1
0x1000dbe1: test bl,cl
0x1000dbe3: je short 1000DBEDh
0x1000dbe5: pop ebx
0x1000dbe6: pop edi
0x1000dbe7: xor eax,eax
0x1000dbe9: pop esi
0x1000dbea: ret 4
0x1000dbed: and dword ptr [eax+0F68h],0FFFFFFFDh
0x1000dbf4: mov eax,[edx]
0x1000dbf6: mov ecx,[eax+0F6Ch]
0x1000dbfc: test bl,cl
0x1000dbfe: je short 1000DC14h
0x1000dc00: test cl,2
0x1000dc03: jne short 1000DC2Ah
0x1000dc05: test [eax+0F68h],bl
0x1000dc0b: je short 1000DC1Ch
0x1000dc0d: and dword ptr [eax+0F68h],0FFFFFFFEh
0x1000dc14: mov eax,ebx
0x1000dc16: pop ebx
0x1000dc17: pop edi
0x1000dc18: pop esi
0x1000dc19: ret 4
0x1000dc1c: or [eax+0F68h],ebx
0x1000dc22: mov eax,ebx
0x1000dc24: pop ebx
0x1000dc25: pop edi
0x1000dc26: pop esi
0x1000dc27: ret 4
0x1000dc2a: test [eax+0F68h],bl
0x1000dc30: jne short 1000DC3Ah
0x1000dc32: or [eax+0F68h],ebx
0x1000dc38: jmp short 1000DC46h
0x1000dc3a: test cl,4
0x1000dc3d: je short 1000DC46h
0x1000dc3f: and dword ptr [eax+0F68h],0FFFFFFFEh
0x1000dc46: xor edi,edi
0x1000dc48: mov ecx,[esi+4]
0x1000dc4b: test ecx,ecx
0x1000dc4d: jne short 1000DC53h
```

```
0x1000dc4f: xor eax,eax
0x1000dc51: jmp short 1000DC5Bh
0x1000dc53: mov eax,[esi+8]
0x1000dc56: sub eax,ecx
0x1000dc58: sar eax,2
0x1000dc5b: cmp edi,eax
0x1000dc5d: jge short 1000DC14h
0x1000dc5f: cmp edi,[esp+10h]
0x1000dc63: je short 1000DC82h
0x1000dc65: call 1000E3C0h
0x1000dc6a: mov eax,[eax]
0x1000dc6c: test [eax+0F68h],bl
0x1000dc72: je short 1000DC82h
0x1000dc74: call 1000E3C0h
0x1000dc79: mov eax,[eax]
0x1000dc7b: and dword ptr [eax+0F68h],0FFFFFFFEh
0x1000dc82: add edi,ebx
0x1000dc84: jmp short 1000DC48h
```

`sub_1000db80` is called by sub_10004910. It delegates to sub_1000e3c0 (1 callees).

12.216 sub_1000dc90 at 0x1000dc90 (118 instructions)

```
0x1000dc90: push ebx
0x1000dc91: push ebp
0x1000dc92: xor ebp,ebp
0x1000dc94: push edi
0x1000dc95: xor edi,edi
0x1000dc97: lea ebx,[ebp+8]
0x1000dc9a: lea ebx,[ebx]
0x1000dca0: mov ecx,[esi+8]
0x1000dca3: test ecx,ecx
0x1000dca5: jne short 1000DCABh
0x1000dca7: xor eax,eax
0x1000dca9: jmp short 1000DCB3h
0x1000dcab: mov eax,[esi+0Ch]
0x1000dcae: sub eax,ecx
0x1000dcb0: sar eax,2
0x1000dcb3: cmp edi,eax
0x1000dcb5: jge near ptr 1000DDEFh
0x1000dcbb: test ecx,ecx
0x1000dcbd: je short 1000DCCBh
0x1000dcbf: mov eax,[esi+0Ch]
0x1000dcc2: sub eax,ecx
0x1000dcc4: sar eax,2
0x1000dcc7: cmp edi,eax
0x1000dcc9: jb short 1000DCD1h
0x1000dccb: call dword ptr ds:[100FE584h]
0x1000dcd1: mov eax,[esi+8]
0x1000dcd4: mov ecx,[eax+edi*4]
0x1000dcd7: test byte ptr [ecx+0F68h],2
0x1000dcde: je short 1000DD0Ah
0x1000dce0: mov ecx,eax
0x1000dce2: test ecx,ecx
0x1000dce4: je short 1000DCF2h
0x1000dce6: mov eax,[esi+0Ch]
0x1000dce9: sub eax,ecx
0x1000dceb: sar eax,2
0x1000dcee: cmp edi,eax
0x1000dcf0: jb short 1000DCF8h
0x1000dcf2: call dword ptr ds:[100FE584h]
0x1000dcf8: mov edx,[esi+8]
0x1000dcfb: mov eax,[edx+edi*4]
0x1000dcfe: and dword ptr [eax+0F68h],0FFFFFFFDh
0x1000dd05: mov ebp,1
0x1000dd0a: cmp edi,[esp+10h]
```

```
0x1000dd0e: jne near ptr 1000DD96h
0x1000dd14: mov ecx,[esi+8]
0x1000dd17: test ecx,ecx
0x1000dd19: je short 1000DD27h
0x1000dd1b: mov eax,[esi+0Ch]
0x1000dd1e: sub eax,ecx
0x1000dd20: sar eax,2
0x1000dd23: cmp edi,eax
0x1000dd25: jb short 1000DD2Dh
0x1000dd27: call dword ptr ds:[100FE584h]
0x1000dd2d: mov eax,[esi+8]
0x1000dd30: mov ecx,[eax+edi*4]
0x1000dd33: test byte ptr [ecx+0F68h],4
0x1000dd3a: jne short 1000DD96h
0x1000dd3c: mov ecx,eax
0x1000dd3e: test ecx,ecx
0x1000dd40: je short 1000DD4Eh
0x1000dd42: mov eax,[esi+0Ch]
0x1000dd45: sub eax,ecx
0x1000dd47: sar eax,2
0x1000dd4a: cmp edi,eax
0x1000dd4c: jb short 1000DD54h
0x1000dd4e: call dword ptr ds:[100FE584h]
0x1000dd54: mov edx,[esi+8]
0x1000dd57: mov eax,[edx+edi*4]
0x1000dd5a: test [eax+0F68h],bl
0x1000dd60: jne near ptr 1000DDE7h
0x1000dd66: mov ecx,edx
0x1000dd68: test ecx,ecx
0x1000dd6a: je short 1000DD78h
0x1000dd6c: mov eax,[esi+0Ch]
0x1000dd6f: sub eax,ecx
0x1000dd71: sar eax,2
0x1000dd74: cmp edi,eax
0x1000dd76: jb short 1000DD7Eh
0x1000dd78: call dword ptr ds:[100FE584h]
0x1000dd7e: mov ecx,[esi+8]
0x1000dd81: mov eax,[ecx+edi*4]
0x1000dd84: or [eax+0F68h],ebx
0x1000dd8a: mov ebp,1
0x1000dd8f: add edi,ebp
0x1000dd91: jmp near ptr 1000DCA0h
0x1000dd96: mov ecx,[esi+8]
0x1000dd99: test ecx,ecx
0x1000dd9b: je short 1000DDA9h
0x1000dd9d: mov eax,[esi+0Ch]
0x1000dda0: sub eax,ecx
0x1000dda2: sar eax,2
0x1000dda5: cmp edi,eax
0x1000dda7: jb short 1000DDAFh
0x1000dda9: call dword ptr ds:[100FE584h]
0x1000ddaf: mov edx,[esi+8]
0x1000ddb2: mov eax,[edx+edi*4]
0x1000ddb5: test [eax+0F68h],bl
0x1000ddbb: je short 1000DDE7h
0x1000ddbd: mov ecx,edx
0x1000ddbf: test ecx,ecx
0x1000ddc1: je short 1000DDCFh
0x1000ddc3: mov eax,[esi+0Ch]
0x1000ddc6: sub eax,ecx
0x1000ddc8: sar eax,2
0x1000ddcb: cmp edi,eax
0x1000ddcd: jb short 1000DDD5h
0x1000ddcf: call dword ptr ds:[100FE584h]
0x1000ddd5: mov ecx,[esi+8]
0x1000ddd8: mov eax,[ecx+edi*4]
0x1000dddb: and dword ptr [eax+0F68h],0FFFFFFF7h
```

```
0x1000dde2: mov ebp,1
0x1000dde7: add edi,1
0x1000ddea: jmp near ptr 1000DCA0h
0x1000ddef: pop edi
0x1000ddf0: mov eax,ebp
0x1000ddf2: pop ebp
0x1000ddf3: pop ebx
0x1000ddf4: ret 4
```

`sub_1000dc90` is called by sub_10004d40. It calls nothing else.

12.217 sub_1000ded0 at 0x1000ded0 (51 instructions)

```
0x1000ded0: mov ecx,[esi+8]
0x1000ded3: test ecx,ecx
0x1000ded5: push ebp
0x1000ded6: mov ebp,ds:[1013F38Ch]
0x1000dedc: je short 1000DEEAh
0x1000dede: mov eax,[esi+0Ch]
0x1000dee1: sub eax,ecx
0x1000dee3: sar eax,2
0x1000dee6: cmp edi,eax
0x1000dee8: jb short 1000DEF0h
0x1000deea: call dword ptr ds:[100FE584h]
0x1000def0: mov eax,[esi+8]
0x1000def3: mov ecx,[eax+edi*4]
0x1000def6: cmp byte ptr [ecx+0F64h],0
0x1000defd: jne short 1000DF59h
0x1000deff: mov ecx,eax
0x1000df01: test ecx,ecx
0x1000df03: je short 1000DF11h
0x1000df05: mov eax,[esi+0Ch]
0x1000df08: sub eax,ecx
0x1000df0a: sar eax,2
0x1000df0d: cmp edi,eax
0x1000df0f: jb short 1000DF17h
0x1000df11: call dword ptr ds:[100FE584h]
0x1000df17: mov edx,[esi+8]
0x1000df1a: mov eax,[edx+edi*4]
0x1000df1d: push ebx
0x1000df1e: mov ebx,[eax+0F68h]
0x1000df24: mov ecx,edx
0x1000df26: and ebx,1
0x1000df29: test ecx,ecx
0x1000df2b: je short 1000DF39h
0x1000df2d: mov eax,[esi+0Ch]
0x1000df30: sub eax,ecx
0x1000df32: sar eax,2
0x1000df35: cmp edi,eax
0x1000df37: jb short 1000DF3Fh
0x1000df39: call dword ptr ds:[100FE584h]
0x1000df3f: mov ecx,[esi+8]
0x1000df42: mov edx,[ecx+edi*4]
0x1000df45: mov eax,[edx]
0x1000df47: mov ecx,[esp+0Ch]
0x1000df4b: mov edx,[ecx+20h]
0x1000df4e: push ebx
0x1000df4f: push eax
0x1000df50: push ebp
0x1000df51: push edx
0x1000df52: call dword ptr ds:[100FE620h]
0x1000df58: pop ebx
0x1000df59: pop ebp
0x1000df5a: ret 4
```

`sub_1000ded0` is called by sub_10004910. It calls nothing else.

12.218 sub_1000df60 at 0x1000df60 (48 instructions)

```
0x1000df60: push ebp
0x1000df61: push edi
0x1000df62: xor ebp,ebp
0x1000df64: xor edi,edi
0x1000df66: mov ecx,[esi+8]
0x1000df69: test ecx,ecx
0x1000df6b: jne short 1000DF71h
0x1000df6d: xor eax,eax
0x1000df6f: jmp short 1000DF79h
0x1000df71: mov eax,[esi+0Ch]
0x1000df74: sub eax,ecx
0x1000df76: sar eax,2
0x1000df79: cmp edi,eax
0x1000df7b: jge short 1000DFCCh
0x1000df7d: test ecx,ecx
0x1000df7f: je short 1000DF8Dh
0x1000df81: mov eax,[esi+0Ch]
0x1000df84: sub eax,ecx
0x1000df86: sar eax,2
0x1000df89: cmp edi,eax
0x1000df8b: jb short 1000DF93h
0x1000df8d: call dword ptr ds:[100FE584h]
0x1000df93: mov eax,[esi+8]
0x1000df96: mov ecx,[eax+edi*4]
0x1000df99: cmp [ecx],ebx
0x1000df9b: je short 1000DFA2h
0x1000df9d: add edi,1
0x1000dfa0: jmp short 1000DF66h
0x1000dfa2: mov ecx,eax
0x1000dfa4: test ecx,ecx
0x1000dfa6: je short 1000DFB4h
0x1000dfa8: mov eax,[esi+0Ch]
0x1000dfab: sub eax,ecx
0x1000dfad: sar eax,2
0x1000dfb0: cmp edi,eax
0x1000dfb2: jb short 1000DFBAh
0x1000dfb4: call dword ptr ds:[100FE584h]
0x1000dfba: mov edx,[esi+8]
0x1000dfbd: mov eax,[edx+edi*4]
0x1000dfc0: mov eax,[eax+0F68h]
0x1000dfc6: pop edi
0x1000dfc7: and eax,1
0x1000dfca: pop ebp
0x1000dfcb: ret
0x1000dfcc: pop edi
0x1000dfcd: mov eax,ebp
0x1000dfcf: pop ebp
0x1000dfd0: ret
```

`sub_1000df60` is called by ?IsToolButtonChecked@CCustomTitle@@QAEHI@Z. It calls nothing else.

12.219 sub_1000dfe0 at 0x1000dfe0 (117 instructions)

```
0x1000dfe0: push ebp
0x1000dfe1: mov ebp,[esp+8]
0x1000dfe5: push edi
0x1000dfe6: mov dword ptr [esp+0Ch],0FFFFFFFFh
0x1000dfee: xor edi,edi
0x1000dff0: mov ecx,[esi+8]
0x1000dff3: test ecx,ecx
0x1000dff5: jne short 1000DFFBh
0x1000dff7: xor eax,eax
0x1000dff9: jmp short 1000E003h
0x1000dffb: mov eax,[esi+0Ch]
0x1000dffe: sub eax,ecx
```

```
0x1000e000: sar eax,2
0x1000e003: cmp edi,eax
0x1000e005: jge short 1000E05Eh
0x1000e007: test ecx,ecx
0x1000e009: je short 1000E017h
0x1000e00b: mov eax,[esi+0Ch]
0x1000e00e: sub eax,ecx
0x1000e010: sar eax,2
0x1000e013: cmp edi,eax
0x1000e015: jb short 1000E01Dh
0x1000e017: call dword ptr ds:[100FE584h]
0x1000e01d: mov eax,[esi+8]
0x1000e020: mov ecx,[eax+edi*4]
0x1000e023: cmp [ecx],ebp
0x1000e025: je short 1000E02Ch
0x1000e027: add edi,1
0x1000e02a: jmp short 1000DFF0h
0x1000e02c: mov ecx,eax
0x1000e02e: test ecx,ecx
0x1000e030: je short 1000E03Eh
0x1000e032: mov eax,[esi+0Ch]
0x1000e035: sub eax,ecx
0x1000e037: sar eax,2
0x1000e03a: cmp edi,eax
0x1000e03c: jb short 1000E044h
0x1000e03e: call dword ptr ds:[100FE584h]
0x1000e044: mov edx,[esi+8]
0x1000e047: mov eax,[edx+edi*4]
0x1000e04a: test byte ptr [eax+0F68h],4
0x1000e051: je short 1000E05Ah
0x1000e053: pop edi
0x1000e054: xor eax,eax
0x1000e056: pop ebp
0x1000e057: ret 4
0x1000e05a: mov [esp+0Ch],edi
0x1000e05e: xor ebp,ebp
0x1000e060: xor edi,edi
0x1000e062: mov ecx,[esi+8]
0x1000e065: test ecx,ecx
0x1000e067: jne short 1000E06Dh
0x1000e069: xor eax,eax
0x1000e06b: jmp short 1000E075h
0x1000e06d: mov eax,[esi+0Ch]
0x1000e070: sub eax,ecx
0x1000e072: sar eax,2
0x1000e075: cmp edi,eax
0x1000e077: jge near ptr 1000E128h
0x1000e07d: cmp edi,[esp+0Ch]
0x1000e081: jne short 1000E0D3h
0x1000e083: test ecx,ecx
0x1000e085: je short 1000E093h
0x1000e087: mov eax,[esi+0Ch]
0x1000e08a: sub eax,ecx
0x1000e08c: sar eax,2
0x1000e08f: cmp edi,eax
0x1000e091: jb short 1000E099h
0x1000e093: call dword ptr ds:[100FE584h]
0x1000e099: mov ecx,[esi+8]
0x1000e09c: mov edx,[ecx+edi*4]
0x1000e09f: test byte ptr [edx+0F68h],1
0x1000e0a6: jne short 1000E120h
0x1000e0a8: test ecx,ecx
0x1000e0aa: je short 1000E0B8h
0x1000e0ac: mov eax,[esi+0Ch]
0x1000e0af: sub eax,ecx
0x1000e0b1: sar eax,2
0x1000e0b4: cmp edi,eax
```

```
0x1000e0b6: jb short 1000E0BEh
0x1000e0b8: call dword ptr ds:[100FE584h]
0x1000e0be: mov eax,[esi+8]
0x1000e0c1: mov eax,[eax+edi*4]
0x1000e0c4: mov ebp,1
0x1000e0c9: or [eax+0F68h],ebp
0x1000e0cf: add edi,ebp
0x1000e0d1: jmp short 1000E062h
0x1000e0d3: test ecx,ecx
0x1000e0d5: je short 1000E0E3h
0x1000e0d7: mov eax,[esi+0Ch]
0x1000e0da: sub eax,ecx
0x1000e0dc: sar eax,2
0x1000e0df: cmp edi,eax
0x1000e0e1: jb short 1000E0E9h
0x1000e0e3: call dword ptr ds:[100FE584h]
0x1000e0e9: mov ecx,[esi+8]
0x1000e0ec: mov edx,[ecx+edi*4]
0x1000e0ef: test byte ptr [edx+0F68h],1
0x1000e0f6: je short 1000E120h
0x1000e0f8: test ecx,ecx
0x1000e0fa: je short 1000E108h
0x1000e0fc: mov eax,[esi+0Ch]
0x1000e0ff: sub eax,ecx
0x1000e101: sar eax,2
0x1000e104: cmp edi,eax
0x1000e106: jb short 1000E10Eh
0x1000e108: call dword ptr ds:[100FE584h]
0x1000e10e: mov eax,[esi+8]
0x1000e111: mov eax,[eax+edi*4]
0x1000e114: and dword ptr [eax+0F68h],0FFFFFFFEh
0x1000e11b: mov ebp,1
0x1000e120: add edi,1
0x1000e123: jmp near ptr 1000E062h
0x1000e128: pop edi
0x1000e129: mov eax,ebp
0x1000e12b: pop ebp
0x1000e12c: ret 4
```

`sub_1000dfe0` is called by ?CheckToolButton@CCustomTitle@@QAEXI@Z. It calls nothing else.

12.220 sub_1000e130 at 0x1000e130 (75 instructions)

```
0x1000e130: sub esp,18h
0x1000e133: push ebx
0x1000e134: push ebp
0x1000e135: push esi
0x1000e136: mov ebx,eax
0x1000e138: xor ebp,ebp
0x1000e13a: lea ebx,[ebx]
0x1000e140: mov ecx,[ebx+8]
0x1000e143: test ecx,ecx
0x1000e145: jne short 1000E14Bh
0x1000e147: xor eax,eax
0x1000e149: jmp short 1000E153h
0x1000e14b: mov eax,[ebx+0Ch]
0x1000e14e: sub eax,ecx
0x1000e150: sar eax,2
0x1000e153: cmp ebp,eax
0x1000e155: jge near ptr 1000E209h
0x1000e15b: test ecx,ecx
0x1000e15d: je short 1000E16Bh
0x1000e15f: mov eax,[ebx+0Ch]
0x1000e162: sub eax,ecx
0x1000e164: sar eax,2
0x1000e167: cmp ebp,eax
```

```
0x1000e169: jb short 1000E171h
0x1000e16b: call dword ptr ds:[100FE584h]
0x1000e171: mov eax,[ebx+8]
0x1000e174: lea esi,[eax+ebp*4]
0x1000e177: mov eax,[esi]
0x1000e179: mov ecx,[eax+0F54h]
0x1000e17f: mov [esp+10h],ecx
0x1000e183: mov edx,[eax+0F58h]
0x1000e189: mov [esp+14h],edx
0x1000e18d: mov ecx,[eax+0F5Ch]
0x1000e193: mov [esp+18h],ecx
0x1000e197: mov edx,[eax+0F60h]
0x1000e19d: mov eax,[esp+2Ch]
0x1000e1a1: mov ecx,[esp+28h]
0x1000e1a5: push eax
0x1000e1a6: mov [esp+20h],edx
0x1000e1aa: push ecx
0x1000e1ab: lea edx,[esp+18h]
0x1000e1af: push edx
0x1000e1b0: call dword ptr ds:[100FE610h]
0x1000e1b6: test eax,eax
0x1000e1b8: jne short 1000E1BFh
0x1000e1ba: add ebp,1
0x1000e1bd: jmp short 1000E140h
0x1000e1bf: mov eax,[esi]
0x1000e1c1: mov ecx,[eax+0F54h]
0x1000e1c7: mov edx,[eax+0F58h]
0x1000e1cd: mov esi,[eax+0F5Ch]
0x1000e1d3: mov ebx,[eax+0F60h]
0x1000e1d9: mov eax,[eax]
0x1000e1db: add eax,1000h
0x1000e1e0: js short 1000E20Ch
0x1000e1e2: mov [edi+10h],ecx
0x1000e1e5: xor ebp,ebp
0x1000e1e7: mov [edi+14h],edx
0x1000e1ea: mov [edi+18h],esi
0x1000e1ed: mov [edi+0Ch],eax
0x1000e1f0: mov [edi+4],ebp
0x1000e1f3: mov [edi+20h],ebp
0x1000e1f6: mov dword ptr [edi+24h],0FFFFFFFFh
0x1000e1fd: mov [edi+1Ch],ebx
0x1000e200: pop esi
0x1000e201: pop ebp
0x1000e202: pop ebx
0x1000e203: add esp,18h
0x1000e206: ret 8
0x1000e209: or eax,0FFFFFFFFh
0x1000e20c: pop esi
0x1000e20d: pop ebp
0x1000e20e: pop ebx
0x1000e20f: add esp,18h
0x1000e212: ret 8
```

`sub_1000e130` is called by sub_100067c0. It calls nothing else.

12.221 sub_1000e2f0 at 0x1000e2f0 (33 instructions)

```
0x1000e2f0: sub esp,18h
0x1000e2f3: cmp dword ptr [esi+24h],0
0x1000e2f7: je short 1000E301h
0x1000e2f9: lea ecx,[esi+20h]
0x1000e2fc: call 10021FF6h
0x1000e301: xor eax,eax
0x1000e303: mov [esi+28h],eax
0x1000e306: xor ecx,ecx
0x1000e308: lea eax,[esp]
```

```
0x1000e30b: push eax
0x1000e30c: mov [esi+2Ch],ecx
0x1000e30f: mov ecx,[edi+4]
0x1000e312: push 18h
0x1000e314: push ecx
0x1000e315: call dword ptr ds:[100FE074h]
0x1000e31b: test eax,eax
0x1000e31d: jne short 1000E325h
0x1000e31f: add esp,18h
0x1000e322: ret 4
0x1000e325: mov ecx,edi
0x1000e327: call 10022002h
0x1000e32c: push eax
0x1000e32d: lea ecx,[esi+20h]
0x1000e330: call 10021FFCh
0x1000e335: mov eax,[esp+4]
0x1000e339: mov ecx,[esp+8]
0x1000e33d: mov edx,[esp+1Ch]
0x1000e341: mov [esi+28h],eax
0x1000e344: mov [esi+2Ch],ecx
0x1000e347: mov [esi+30h],edx
0x1000e34a: mov eax,1
0x1000e34f: add esp,18h
0x1000e352: ret 4
```

`sub_1000e2f0` is called by sub_10003ca0. It delegates to j_ORDINAL 1925, j_ORDINAL 1271, j_ORDINAL 1959 (3 callees).

12.222 sub_1000e360 at 0x1000e360 (19 instructions)

```
0x1000e360: push ecx
0x1000e361: cmp dword ptr [ecx+24h],0
0x1000e365: je short 1000E36Ch
0x1000e367: mov eax,[ecx+2Ch]
0x1000e36a: pop ecx
0x1000e36b: ret
0x1000e36c: mov eax,[ecx+8]
0x1000e36f: test eax,eax
0x1000e371: je short 1000E384h
0x1000e373: mov ecx,[ecx+0Ch]
0x1000e376: sub ecx,eax
0x1000e378: sar ecx,2
0x1000e37b: je short 1000E384h
0x1000e37d: mov eax,19h
0x1000e382: pop ecx
0x1000e383: ret
0x1000e384: xor eax,eax
0x1000e386: pop ecx
0x1000e387: ret
```

`sub_1000e360` is called by sub_10003670. It calls nothing else.

12.223 sub_1000e3c0 at 0x1000e3c0 (12 instructions)

```
0x1000e3c0: mov ecx,[esi+4]
0x1000e3c3: test ecx,ecx
0x1000e3c5: je short 1000E3D3h
0x1000e3c7: mov eax,[esi+8]
0x1000e3ca: sub eax,ecx
0x1000e3cc: sar eax,2
0x1000e3cf: cmp edi,eax
0x1000e3d1: jb short 1000E3D9h
0x1000e3d3: call dword ptr ds:[100FE584h]
0x1000e3d9: mov eax,[esi+4]
```

```
0x1000e3dc: lea eax,[eax+edi*4]
0x1000e3df: ret
```

`sub_1000e3c0` is called by sub_1000db80. It calls nothing else.

12.224 sub_1000e3e0 at 0x1000e3e0 (41 instructions)

```
0x1000e3e0: xor eax,eax
0x1000e3e2: push esi
0x1000e3e3: mov esi,ecx
0x1000e3e5: mov [esi+3C4h],eax
0x1000e3eb: mov [esi+3C0h],eax
0x1000e3f1: mov [esi+4],eax
0x1000e3f4: mov [esi+3CCh],eax
0x1000e3fa: mov [esi+3C8h],eax
0x1000e400: mov [esi+8],eax
0x1000e403: mov [esi+0Ch],eax
0x1000e406: mov [esi+10h],eax
0x1000e409: mov [esi+14h],eax
0x1000e40c: mov [esi+18h],eax
0x1000e40f: mov [esi+1Ch],eax
0x1000e412: mov [esi+20h],eax
0x1000e415: mov [esi+24h],eax
0x1000e418: mov [esi+28h],eax
0x1000e41b: push 390h
0x1000e420: push eax
0x1000e421: mov [esi+2Ch],eax
0x1000e424: lea eax,[esi+30h]
0x1000e427: push eax
0x1000e428: call 100228C0h
0x1000e42d: fld dword ptr ds:[100FE71Ch]
0x1000e433: mov ecx,[esp+14h]
0x1000e437: fstp dword ptr [esi+154h]
0x1000e43d: add esp,0Ch
0x1000e440: mov [esi+40h],ecx
0x1000e443: push 60h
0x1000e445: mov ecx,esi
0x1000e447: mov byte ptr [esi+180h],0FFh
0x1000e44e: mov dword ptr [esi+14Ch],0FFFFFFFFh
0x1000e458: mov byte ptr [esi+182h],1
0x1000e45f: mov byte ptr [esi+158h],1
0x1000e466: call 1000EF10h
0x1000e46b: push 60h
0x1000e46d: mov ecx,esi
0x1000e46f: call 1000EF70h
0x1000e474: mov byte ptr [esi+1EEh],1
0x1000e47b: pop esi
0x1000e47c: ret 4
```

`sub_1000e3e0` is called by sub_1000e480, sub_1000ecb0, sub_1000ee20. It delegates to sub_1000ef10, sub_1000ef70, j_memset (3 callees).

12.225 sub_1000e480 at 0x1000e480 (9 instructions)

```
0x1000e480: mov eax,[esp+4]
0x1000e484: push esi
0x1000e485: mov esi,ecx
0x1000e487: push eax
0x1000e488: mov dword ptr [esi],100FE724h
0x1000e48e: call 1000E3E0h
0x1000e493: mov eax,esi
0x1000e495: pop esi
0x1000e496: ret 4
```

`sub_1000e480` is called by sub_1000ca70, sub_1000cc10, sub_1000f5d0, sub_1000f5f0, sub_1000f610, sub_1000f660, sub_1000f680, sub_1000f6a0, sub_1000f6c0, sub_1000f6e0, sub_1000f700, sub_1000f720, sub_1000f740, sub_1000f760, sub_100103d0, sub_10011b00, sub_10018670, sub_10019590, sub_1001b160, sub_1001b970. It delegates to sub_1000e3e0 (1 callees).

12.226 sub_1000e4a0 at 0x1000e4a0 (60 instructions)

```
0x1000e4a0: push ebx
0x1000e4a1: push esi
0x1000e4a2: mov esi,ecx
0x1000e4a4: xor ebx,ebx
0x1000e4a6: cmp [esi+38h],ebx
0x1000e4a9: jne near ptr 1000E545h
0x1000e4af: cmp [esi+3C8h],ebx
0x1000e4b5: push edi
0x1000e4b6: je short 1000E4FDh
0x1000e4b8: xor edi,edi
0x1000e4ba: cmp [esi+1E4h],ebx
0x1000e4c0: jle short 1000E4E2h
0x1000e4c2: mov eax,[esi+3C8h]
0x1000e4c8: mov ecx,[eax+edi*4]
0x1000e4cb: cmp ecx,ebx
0x1000e4cd: je short 1000E4D7h
0x1000e4cf: mov edx,[ecx]
0x1000e4d1: mov eax,[edx]
0x1000e4d3: push 1
0x1000e4d5: call eax
0x1000e4d7: add edi,1
0x1000e4da: cmp edi,[esi+1E4h]
0x1000e4e0: jl short 1000E4C2h
0x1000e4e2: mov ecx,[esi+3C8h]
0x1000e4e8: push ecx
0x1000e4e9: call 10021FCCh
0x1000e4ee: add esp,4
0x1000e4f1: mov [esi+3C8h],ebx
0x1000e4f7: mov [esi+1E4h],ebx
0x1000e4fd: mov eax,[esi+3C0h]
0x1000e503: cmp eax,ebx
0x1000e505: mov edi,ds:[100FE554h]
0x1000e50b: je short 1000E519h
0x1000e50d: push eax
0x1000e50e: call edi
0x1000e510: add esp,4
0x1000e513: mov [esi+3C0h],ebx
0x1000e519: mov eax,[esi+3C4h]
0x1000e51f: cmp eax,ebx
0x1000e521: je short 1000E52Fh
0x1000e523: push eax
0x1000e524: call edi
0x1000e526: add esp,4
0x1000e529: mov [esi+3C4h],ebx
0x1000e52f: mov eax,[esi+4]
0x1000e532: cmp eax,ebx
0x1000e534: je short 1000E53Fh
0x1000e536: push eax
0x1000e537: call edi
0x1000e539: add esp,4
0x1000e53c: mov [esi+4],ebx
0x1000e53f: pop edi
0x1000e540: pop esi
0x1000e541: mov al,1
0x1000e543: pop ebx
0x1000e544: ret
0x1000e545: pop esi
0x1000e546: xor al,al
0x1000e548: pop ebx
0x1000e549: ret
```

`sub_1000e4a0` is called by sub_1000caf0, sub_1000cc10, sub_1000e6d0, sub_1000ead0, sub_100103d0, sub_10011b00, sub_10019690. It delegates to j_ORDINAL 764 (1 callees).

12.227 sub_1000e550 at 0x1000e550 (35 instructions)

```
0x1000e550: push esi
0x1000e551: mov esi,ecx
0x1000e553: cmp dword ptr [esi+38h],0
0x1000e557: jne short 1000E5B8h
0x1000e559: cmp dword ptr [esi+3CCh],0
0x1000e560: je short 1000E5B4h
0x1000e562: push edi
0x1000e563: xor edi,edi
0x1000e565: cmp [esi+160h],edi
0x1000e56b: jle short 1000E590h
0x1000e56d: lea ecx,[ecx]
0x1000e570: mov eax,[esi+3CCh]
0x1000e576: mov ecx,[eax+edi*4]
0x1000e579: test ecx,ecx
0x1000e57b: je short 1000E585h
0x1000e57d: mov edx,[ecx]
0x1000e57f: mov eax,[edx]
0x1000e581: push 1
0x1000e583: call eax
0x1000e585: add edi,1
0x1000e588: cmp edi,[esi+160h]
0x1000e58e: jl short 1000E570h
0x1000e590: mov ecx,[esi+3CCh]
0x1000e596: push ecx
0x1000e597: call 10021FCCh
0x1000e59c: add esp,4
0x1000e59f: mov dword ptr [esi+3CCh],0
0x1000e5a9: mov dword ptr [esi+160h],0
0x1000e5b3: pop edi
0x1000e5b4: mov al,1
0x1000e5b6: pop esi
0x1000e5b7: ret
0x1000e5b8: xor al,al
0x1000e5ba: pop esi
0x1000e5bb: ret
```

`sub_1000e550` is called by sub_1000c970, sub_1000cc10, sub_1000e6d0, sub_100103d0, sub_10011b00, sub_10019690, sub_1001a5b0. It delegates to j_ORDINAL 764 (1 callees).

12.228 sub_1000e5c0 at 0x1000e5c0 (12 instructions)

```
0x1000e5c0: cmp dword ptr [ecx+4],0
0x1000e5c4: jne short 1000E5DBh
0x1000e5c6: push esi
0x1000e5c7: mov esi,[esp+8]
0x1000e5cb: push edi
0x1000e5cc: lea edi,[ecx+30h]
0x1000e5cf: add esi,30h
0x1000e5d2: mov ecx,0E4h
0x1000e5d7: rep movsd
0x1000e5d9: pop edi
0x1000e5da: pop esi
0x1000e5db: ret 4
```

`sub_1000e5c0` is called by sub_1000f760, sub_100103d0, sub_10011b00, sub_1001a5b0. It calls nothing else.

12.229 sub_1000e5e0 at 0x1000e5e0 (38 instructions)

```
0x1000e5e0: push esi
0x1000e5e1: mov esi,ecx
```

```
0x1000e5e3: cmp dword ptr [esi+4],0
0x1000e5e7: je short 1000E62Ch
0x1000e5e9: push ebx
0x1000e5ea: mov ebx,[esp+0Ch]
0x1000e5ee: test ebx,ebx
0x1000e5f0: push edi
0x1000e5f1: je short 1000E61Ah
0x1000e5f3: cmp ebx,[esi+10h]
0x1000e5f6: jae short 1000E612h
0x1000e5f8: mov edi,[esi+4]
0x1000e5fb: call 10012E40h
0x1000e600: mov ecx,[esi+30h]
0x1000e603: imul ecx,ebx
0x1000e606: add eax,ecx
0x1000e608: add eax,[edi]
0x1000e60a: add eax,edi
0x1000e60c: pop edi
0x1000e60d: pop ebx
0x1000e60e: pop esi
0x1000e60f: ret 4
0x1000e612: pop edi
0x1000e613: pop ebx
0x1000e614: xor eax,eax
0x1000e616: pop esi
0x1000e617: ret 4
0x1000e61a: mov edi,[esi+4]
0x1000e61d: call 10012E40h
0x1000e622: add eax,[edi]
0x1000e624: add eax,edi
0x1000e626: pop edi
0x1000e627: pop ebx
0x1000e628: pop esi
0x1000e629: ret 4
0x1000e62c: xor eax,eax
0x1000e62e: pop esi
0x1000e62f: ret 4
```

`sub_1000e5e0` is called by sub_1000ead0, sub_10010810, sub_10011f20, sub_10013780, sub_10016040, sub_10016840, sub_10016cd0, sub_10017120, sub_100187f0, sub_10018d20, sub_10019a40, sub_1001b350, sub_1001b970. It delegates to sub_10012e40 (1 callees).

12.230 sub_1000e640 at 0x1000e640 (7 instructions)

```
0x1000e640: push esi
0x1000e641: mov esi,ecx
0x1000e643: call 10012E40h
0x1000e648: add eax,[esi+1Ch]
0x1000e64b: add eax,[esi+8]
0x1000e64e: pop esi
0x1000e64f: ret
```

`sub_1000e640` is called by sub_1000ead0, sub_1000ecf0, sub_1001a5b0. It delegates to sub_10012e40 (1 callees).

12.231 sub_1000e650 at 0x1000e650 (14 instructions)

```
0x1000e650: mov eax,[esp+8]
0x1000e654: test eax,eax
0x1000e656: jl short 1000E672h
0x1000e658: cmp eax,[ecx+10h]
0x1000e65b: jge short 1000E672h
0x1000e65d: mov eax,[esp+4]
0x1000e661: test eax,eax
0x1000e663: jl short 1000E672h
0x1000e665: cmp eax,[ecx+0Ch]
0x1000e668: jge short 1000E672h
```

```
0x1000e66a: mov eax,1
0x1000e66f: ret 8
0x1000e672: xor eax,eax
0x1000e674: ret 8
```

`sub_1000e650` is called by sub_10013af0, sub_10013b70, sub_10019a40, sub_1001d050. It calls nothing else.

12.232 sub_1000e680 at 0x1000e680 (32 instructions)

```
0x1000e680: push esi
0x1000e681: mov esi,ecx
0x1000e683: cmp dword ptr [esi+4],0
0x1000e687: je short 1000E6CCh
0x1000e689: push ebx
0x1000e68a: call 1000F020h
0x1000e68f: cmp ax,1
0x1000e693: mov bl,[esp+0Ch]
0x1000e697: jne short 1000E6A0h
0x1000e699: test bl,bl
0x1000e69b: jbe short 1000E6A0h
0x1000e69d: or bl,0FFh
0x1000e6a0: mov ecx,esi
0x1000e6a2: call 1000F020h
0x1000e6a7: cmp ax,4
0x1000e6ab: jne short 1000E6B7h
0x1000e6ad: mov al,bl
0x1000e6af: and al,0Fh
0x1000e6b1: mov cl,11h
0x1000e6b3: imul cl
0x1000e6b5: mov bl,al
0x1000e6b7: mov edx,[esi+1Ch]
0x1000e6ba: mov ecx,[esi+34h]
0x1000e6bd: movzx eax,bl
0x1000e6c0: push edx
0x1000e6c1: push eax
0x1000e6c2: push ecx
0x1000e6c3: call 100228C0h
0x1000e6c8: add esp,0Ch
0x1000e6cb: pop ebx
0x1000e6cc: pop esi
0x1000e6cd: ret 4
```

`sub_1000e680` is called by sub_1001a5b0. It delegates to sub_1000f020, j_memset (2 callees).

12.233 sub_1000e6d0 at 0x1000e6d0 (63 instructions)

```
0x1000e6d0: push ebx
0x1000e6d1: mov ebx,ecx
0x1000e6d3: call 1000E4A0h
0x1000e6d8: test al,al
0x1000e6da: jne short 1000E6E0h
0x1000e6dc: pop ebx
0x1000e6dd: ret 8
0x1000e6e0: push ebp
0x1000e6e1: mov ebp,[esp+0Ch]
0x1000e6e5: lea eax,[ebp+8]
0x1000e6e8: push esi
0x1000e6e9: push edi
0x1000e6ea: mov esi,eax
0x1000e6ec: lea edx,[ebp+30h]
0x1000e6ef: lea edi,[ebx+8]
0x1000e6f2: mov ecx,0Ah
0x1000e6f7: rep movsd
0x1000e6f9: mov esi,edx
0x1000e6fb: lea edi,[ebx+30h]
```

```
0x1000e6fe: mov ecx,0E4h
0x1000e703: rep movsd
0x1000e705: mov ecx,[ebp+4]
0x1000e708: mov [ebx+4],ecx
0x1000e70b: mov ecx,[ebp+3C0h]
0x1000e711: mov [ebx+3C0h],ecx
0x1000e717: mov ecx,[ebp+3C4h]
0x1000e71d: mov [ebx+3C4h],ecx
0x1000e723: mov ecx,[ebp+3C8h]
0x1000e729: mov [ebx+3C8h],ecx
0x1000e72f: xor ecx,ecx
0x1000e731: mov [eax],ecx
0x1000e733: mov [eax+4],ecx
0x1000e736: mov [eax+8],ecx
0x1000e739: mov [eax+0Ch],ecx
0x1000e73c: mov [eax+10h],ecx
0x1000e73f: mov [eax+14h],ecx
0x1000e742: mov [eax+18h],ecx
0x1000e745: push 390h
0x1000e74a: xor esi,esi
0x1000e74c: mov [eax+1Ch],ecx
0x1000e74f: push esi
0x1000e750: mov [eax+20h],ecx
0x1000e753: push edx
0x1000e754: mov [eax+24h],ecx
0x1000e757: call 100228C0h
0x1000e75c: add esp,0Ch
0x1000e75f: cmp byte ptr [esp+18h],0
0x1000e764: mov [ebp+3C4h],esi
0x1000e76a: mov [ebp+3C0h],esi
0x1000e770: mov [ebp+4],esi
0x1000e773: mov [ebp+3C8h],esi
0x1000e779: je short 1000E794h
0x1000e77b: mov ecx,ebx
0x1000e77d: call 1000E550h
0x1000e782: mov edx,[ebp+3CCh]
0x1000e788: mov [ebx+3CCh],edx
0x1000e78e: mov [ebp+3CCh],esi
0x1000e794: pop edi
0x1000e795: pop esi
0x1000e796: pop ebp
0x1000e797: mov al,1
0x1000e799: pop ebx
0x1000e79a: ret 8
```

`sub_1000e6d0` is called by sub_1000f760, sub_100103d0, sub_10011b00, sub_1001a5b0, sub_1001b970. It delegates to sub_1000e4a0, sub_1000e550, j_memset (3 callees).

12.234 sub_1000e7a0 at 0x1000e7a0 (28 instructions)

```
0x1000e7a0: mov edx,[esp+4]
0x1000e7a4: test edx,edx
0x1000e7a6: mov eax,ecx
0x1000e7a8: je short 1000E801h
0x1000e7aa: push esi
0x1000e7ab: push edi
0x1000e7ac: lea esi,[edx+8]
0x1000e7af: lea edi,[eax+8]
0x1000e7b2: mov ecx,0Ah
0x1000e7b7: rep movsd
0x1000e7b9: lea esi,[edx+30h]
0x1000e7bc: lea edi,[eax+30h]
0x1000e7bf: mov ecx,0E4h
0x1000e7c4: rep movsd
0x1000e7c6: mov ecx,[edx+4]
0x1000e7c9: mov [eax+4],ecx
```

```
0x1000e7cc: mov ecx,[edx+3C0h]
0x1000e7d2: mov [eax+3C0h],ecx
0x1000e7d8: mov ecx,[edx+3C4h]
0x1000e7de: mov [eax+3C4h],ecx
0x1000e7e4: mov ecx,[edx+3C8h]
0x1000e7ea: mov [eax+3C8h],ecx
0x1000e7f0: mov ecx,[edx+3CCh]
0x1000e7f6: pop edi
0x1000e7f7: mov [eax+3CCh],ecx
0x1000e7fd: mov [eax+38h],edx
0x1000e800: pop esi
0x1000e801: ret 4
```

`sub_1000e7a0` is called by sub_1000ecf0. It calls nothing else.

12.235 sub_1000ead0 at 0x1000ead0 (155 instructions)

```
0x1000ead0: push ebx
0x1000ead1: mov ebx,ecx
0x1000ead3: call 1000E4A0h
0x1000ead8: test al,al
0x1000eada: jne short 1000EAE2h
0x1000eadc: xor eax,eax
0x1000eade: pop ebx
0x1000eadf: ret 10h
0x1000eae2: push esi
0x1000eae3: push edi
0x1000eae4: mov edi,[esp+10h]
0x1000eae8: test edi,edi
0x1000eaea: je near ptr 1000EC91h
0x1000eaf0: mov edx,[esp+14h]
0x1000eaf4: test edx,edx
0x1000eaf6: je near ptr 1000EC91h
0x1000eafc: mov eax,[esp+18h]
0x1000eb00: cmp eax,1
0x1000eb03: ja short 1000EB0Ch
0x1000eb05: mov ecx,1
0x1000eb0a: jmp short 1000EB27h
0x1000eb0c: cmp eax,4
0x1000eb0f: ja short 1000EB18h
0x1000eb11: mov ecx,4
0x1000eb16: jmp short 1000EB27h
0x1000eb18: mov ecx,8
0x1000eb1d: cmp ecx,eax
0x1000eb1f: sbb ecx,ecx
0x1000eb21: and ecx,10h
0x1000eb24: add ecx,8
0x1000eb27: mov eax,edi
0x1000eb29: test eax,eax
0x1000eb2b: mov [esp+10h],eax
0x1000eb2f: fild dword ptr [esp+10h]
0x1000eb33: jge short 1000EB3Bh
0x1000eb35: fadd dword ptr ds:[100FE7C0h]
0x1000eb3b: mov eax,edx
0x1000eb3d: test eax,eax
0x1000eb3f: mov [esp+10h],eax
0x1000eb43: fild dword ptr [esp+10h]
0x1000eb47: jge short 1000EB4Fh
0x1000eb49: fadd dword ptr ds:[100FE7C0h]
0x1000eb4f: mov eax,ecx
0x1000eb51: fmulp
0x1000eb53: test eax,eax
0x1000eb55: mov [esp+10h],eax
0x1000eb59: fild dword ptr [esp+10h]
0x1000eb5d: jge short 1000EB65h
0x1000eb5f: fadd dword ptr ds:[100FE7C0h]
```

```
0x1000eb65: fmulp
0x1000eb67: fmul qword ptr ds:[100FE7B8h]
0x1000eb6d: fcomp qword ptr ds:[100FE7B0h]
0x1000eb73: fnstsw ax
0x1000eb75: test ah,41h
0x1000eb78: jne short 1000EB91h
0x1000eb7a: lea edi,[ebx+44h]
0x1000eb7d: mov ecx,7
0x1000eb82: mov esi,100FE790h
0x1000eb87: rep movsd
0x1000eb89: pop edi
0x1000eb8a: pop esi
0x1000eb8b: xor eax,eax
0x1000eb8d: pop ebx
0x1000eb8e: ret 10h
0x1000eb91: mov eax,ecx
0x1000eb93: sub eax,1
0x1000eb96: je short 1000EBBDh
0x1000eb98: sub eax,3
0x1000eb9b: je short 1000EBB4h
0x1000eb9d: sub eax,4
0x1000eba0: je short 1000EBABh
0x1000eba2: mov dword ptr [ebx+28h],0
0x1000eba9: jmp short 1000EBC4h
0x1000ebab: mov dword ptr [ebx+28h],100h
0x1000ebb2: jmp short 1000EBC4h
0x1000ebb4: mov dword ptr [ebx+28h],10h
0x1000ebbb: jmp short 1000EBC4h
0x1000ebbd: mov dword ptr [ebx+28h],2
0x1000ebc4: mov esi,[esp+1Ch]
0x1000ebc8: mov eax,edi
0x1000ebca: imul eax,ecx
0x1000ebcd: add eax,1Fh
0x1000ebd0: shr eax,5
0x1000ebd3: add eax,eax
0x1000ebd5: add eax,eax
0x1000ebd7: mov [ebx+30h],eax
0x1000ebda: imul eax,edx
0x1000ebdd: mov [ebx+40h],esi
0x1000ebe0: mov [ebx+16h],cx
0x1000ebe4: lea esi,[ebx+8]
0x1000ebe7: mov ecx,ebx
0x1000ebe9: mov dword ptr [esi],28h
0x1000ebef: mov [ebx+0Ch],edi
0x1000ebf2: mov [ebx+10h],edx
0x1000ebf5: mov word ptr [ebx+14h],1
0x1000ebfb: mov dword ptr [ebx+18h],0
0x1000ec02: mov [ebx+1Ch],eax
0x1000ec05: call 1000E640h
0x1000ec0a: push eax
0x1000ec0b: call dword ptr ds:[100FE550h]
0x1000ec11: add esp,4
0x1000ec14: test eax,eax
0x1000ec16: mov [ebx+4],eax
0x1000ec19: jne short 1000EC32h
0x1000ec1b: lea edi,[ebx+44h]
0x1000ec1e: mov ecx,9
0x1000ec23: mov esi,100FE768h
0x1000ec28: rep movsd
0x1000ec2a: movsw
0x1000ec2c: pop edi
0x1000ec2d: pop esi
0x1000ec2e: pop ebx
0x1000ec2f: ret 10h
0x1000ec32: mov ecx,ebx
0x1000ec34: call 10012EA0h
0x1000ec39: mov edi,eax
```

```
0x1000ec3b: test edi,edi
0x1000ec3d: je short 1000EC52h
0x1000ec3f: mov ecx,ebx
0x1000ec41: call 10012E40h
0x1000ec46: push eax
0x1000ec47: push 0
0x1000ec49: push edi
0x1000ec4a: call 100228C0h
0x1000ec4f: add esp,0Ch
0x1000ec52: cmp dword ptr [ebx+3C0h],0
0x1000ec59: je short 1000EC62h
0x1000ec5b: mov ecx,ebx
0x1000ec5d: call 10013DA0h
0x1000ec62: cmp dword ptr [ebx+3C4h],0
0x1000ec69: je short 1000EC72h
0x1000ec6b: mov ecx,ebx
0x1000ec6d: call 10013A10h
0x1000ec72: mov edi,[ebx+4]
0x1000ec75: mov ecx,0Ah
0x1000ec7a: rep movsd
0x1000ec7c: push 0
0x1000ec7e: mov ecx,ebx
0x1000ec80: call 1000E5E0h
0x1000ec85: pop edi
0x1000ec86: mov [ebx+34h],eax
0x1000ec89: mov eax,[ebx+4]
0x1000ec8c: pop esi
0x1000ec8d: pop ebx
0x1000ec8e: ret 10h
0x1000ec91: lea edi,[ebx+44h]
0x1000ec94: mov ecx,0Fh
0x1000ec99: mov esi,100FE728h
0x1000ec9e: rep movsd
0x1000eca0: movsb
0x1000eca1: pop edi
0x1000eca2: pop esi
0x1000eca3: xor eax,eax
0x1000eca5: pop ebx
0x1000eca6: ret 10h
```

`sub_1000ead0` is called by sub_1000ecb0, sub_1000ecf0, sub_100103d0, sub_10011b00, sub_100143f0, sub_100156b0, sub_10015d20, sub_10016040, sub_10016840, sub_10016cd0, sub_100171e0, sub_10017680, sub_100187f0, sub_1001a5b0, sub_1001b350, sub_1001b970, sub_1001c370. It works with the text "CxImage::Create : width and height must be greater than zero", "CXIMAGE_MAX_MEMORY exceeded". It delegates to sub_1000e4a0, sub_1000e5e0, sub_1000e640 (8 callees).

12.236 sub_1000ecb0 at 0x1000ecb0 (20 instructions)

```
0x1000ecb0: push esi
0x1000ecb1: push edi
0x1000ecb2: mov edi,[esp+18h]
0x1000ecb6: mov esi,ecx
0x1000ecb8: push edi
0x1000ecb9: mov dword ptr [esi],100FE724h
0x1000ecbf: call 1000E3E0h
0x1000ecc4: mov eax,[esp+14h]
0x1000ecc8: mov ecx,[esp+10h]
0x1000eccc: mov edx,[esp+0Ch]
0x1000ecd0: push edi
0x1000ecd1: push eax
0x1000ecd2: push ecx
0x1000ecd3: push edx
0x1000ecd4: mov ecx,esi
0x1000ecd6: call 1000EAD0h
0x1000ecdb: pop edi
```

```
0x1000ecdc: mov eax,esi
0x1000ecde: pop esi
0x1000ecdf: ret 10h
```

`sub_1000ecb0` is called by sub_1001a5b0. It delegates to sub_1000e3e0, sub_1000ead0 (2 callees).

12.237 sub_1000ecf0 at 0x1000ecf0 (104 instructions)

```
0x1000ecf0: push ebx
0x1000ecf1: push ebp
0x1000ecf2: mov ebp,[esp+0Ch]
0x1000ecf6: cmp dword ptr [ebp+38h],0
0x1000ecfa: mov ebx,ecx
0x1000ecfc: je short 1000ED09h
0x1000ecfe: push ebp
0x1000ecff: call 1000E7A0h
0x1000ed04: pop ebp
0x1000ed05: pop ebx
0x1000ed06: ret 10h
0x1000ed09: push esi
0x1000ed0a: push edi
0x1000ed0b: lea esi,[ebp+30h]
0x1000ed0e: lea edi,[ebx+30h]
0x1000ed11: mov ecx,0E4h
0x1000ed16: rep movsd
0x1000ed18: lea esi,[ebp+8]
0x1000ed1b: lea edi,[ebx+8]
0x1000ed1e: mov ecx,0Ah
0x1000ed23: rep movsd
0x1000ed25: mov ecx,ebp
0x1000ed27: call 1000F030h
0x1000ed2c: push eax
0x1000ed2d: mov ecx,ebp
0x1000ed2f: call 1000F020h
0x1000ed34: movzx eax,ax
0x1000ed37: push eax
0x1000ed38: mov ecx,ebp
0x1000ed3a: call 1000EFE0h
0x1000ed3f: push eax
0x1000ed40: mov ecx,ebp
0x1000ed42: call 1000EFF0h
0x1000ed47: push eax
0x1000ed48: mov ecx,ebx
0x1000ed4a: call 1000EAD0h
0x1000ed4f: cmp byte ptr [esp+18h],0
0x1000ed54: je short 1000ED7Ch
0x1000ed56: cmp dword ptr [ebx+4],0
0x1000ed5a: je short 1000ED7Ch
0x1000ed5c: cmp dword ptr [ebp+4],0
0x1000ed60: je short 1000ED7Ch
0x1000ed62: mov ecx,ebx
0x1000ed64: call 1000E640h
0x1000ed69: mov ecx,[ebp+4]
0x1000ed6c: mov edx,[ebx+4]
0x1000ed6f: push eax
0x1000ed70: push ecx
0x1000ed71: push edx
0x1000ed72: call 100228C6h
0x1000ed77: add esp,0Ch
0x1000ed7a: jmp short 1000ED90h
0x1000ed7c: push 100h
0x1000ed81: mov ecx,ebp
0x1000ed83: call 10012EA0h
0x1000ed88: push eax
0x1000ed89: mov ecx,ebx
0x1000ed8b: call 10013690h
```

```
0x1000ed90: mov esi,[ebx+10h]
0x1000ed93: imul esi,[ebx+0Ch]
0x1000ed97: cmp byte ptr [esp+1Ch],0
0x1000ed9c: mov edi,ds:[100FE550h]
0x1000eda2: je short 1000EDDBh
0x1000eda4: cmp dword ptr [ebp+3C0h],0
0x1000edab: je short 1000EDDBh
0x1000edad: mov eax,[ebx+3C0h]
0x1000edb3: test eax,eax
0x1000edb5: je short 1000EDC1h
0x1000edb7: push eax
0x1000edb8: call dword ptr ds:[100FE554h]
0x1000edbe: add esp,4
0x1000edc1: push esi
0x1000edc2: call edi
0x1000edc4: mov [ebx+3C0h],eax
0x1000edca: mov ecx,[ebp+3C0h]
0x1000edd0: push esi
0x1000edd1: push ecx
0x1000edd2: push eax
0x1000edd3: call 100228C6h
0x1000edd8: add esp,10h
0x1000eddb: cmp byte ptr [esp+20h],0
0x1000ede0: je short 1000EE19h
0x1000ede2: cmp dword ptr [ebp+3C4h],0
0x1000ede9: je short 1000EE19h
0x1000edeb: mov eax,[ebx+3C4h]
0x1000edf1: test eax,eax
0x1000edf3: je short 1000EDFFh
0x1000edf5: push eax
0x1000edf6: call dword ptr ds:[100FE554h]
0x1000edfc: add esp,4
0x1000edff: push esi
0x1000ee00: call edi
0x1000ee02: mov [ebx+3C4h],eax
0x1000ee08: mov edx,[ebp+3C4h]
0x1000ee0e: push esi
0x1000ee0f: push edx
0x1000ee10: push eax
0x1000ee11: call 100228C6h
0x1000ee16: add esp,10h
0x1000ee19: pop edi
0x1000ee1a: pop esi
0x1000ee1b: pop ebp
0x1000ee1c: pop ebx
0x1000ee1d: ret 10h
```

`sub_1000ecf0` is called by sub_1000cc10, sub_1000ee20, sub_1001a5b0. It delegates to sub_1000e640, sub_1000e7a0, sub_1000ead0 (10 callees).

12.238 sub_1000ee20 at 0x1000ee20 (23 instructions)

```
0x1000ee20: push esi
0x1000ee21: push edi
0x1000ee22: mov edi,[esp+0Ch]
0x1000ee26: mov esi,ecx
0x1000ee28: mov ecx,edi
0x1000ee2a: mov dword ptr [esi],100FE724h
0x1000ee30: call 1000F030h
0x1000ee35: push eax
0x1000ee36: mov ecx,esi
0x1000ee38: call 1000E3E0h
0x1000ee3d: mov eax,[esp+18h]
0x1000ee41: mov ecx,[esp+14h]
0x1000ee45: mov edx,[esp+10h]
0x1000ee49: push eax
```

```
0x1000ee4a: push ecx
0x1000ee4b: push edx
0x1000ee4c: push edi
0x1000ee4d: mov ecx,esi
0x1000ee4f: call 1000ECF0h
0x1000ee54: pop edi
0x1000ee55: mov eax,esi
0x1000ee57: pop esi
0x1000ee58: ret 10h
```

`sub_1000ee20` is called by sub_1001a5b0. It delegates to sub_1000e3e0, sub_1000ecf0, sub_1000f030 (3 callees).

12.239 sub_1000ee60 at 0x1000ee60 (17 instructions)

```
0x1000ee60: cmp word ptr [ecx+16h],18h
0x1000ee65: jae short 1000EE89h
0x1000ee67: cmp dword ptr [ecx+14Ch],0
0x1000ee6e: jl short 1000EE89h
0x1000ee70: movzx eax,byte ptr [ecx+14Ch]
0x1000ee77: push esi
0x1000ee78: mov esi,[esp+8]
0x1000ee7c: push eax
0x1000ee7d: push esi
0x1000ee7e: call 10012EC0h
0x1000ee83: mov eax,esi
0x1000ee85: pop esi
0x1000ee86: ret 4
0x1000ee89: mov ecx,[ecx+150h]
0x1000ee8f: mov eax,[esp+4]
0x1000ee93: mov [eax],ecx
0x1000ee95: ret 4
```

`sub_1000ee60` is called by sub_10011fb0. It delegates to sub_10012ec0 (1 callees).

12.240 sub_1000eea0 at 0x1000eea0 (2 instructions)

```
0x1000eea0: mov eax,[ecx+14Ch]
0x1000eea6: ret
```

`sub_1000eea0` is called by sub_10011fb0, sub_1001a270. It calls nothing else.

12.241 sub_1000eeb0 at 0x1000eeb0 (8 instructions)

```
0x1000eeb0: mov eax,[esp+4]
0x1000eeb4: xor edx,edx
0x1000eeb6: cmp eax,[ecx+28h]
0x1000eeb9: setge dl
0x1000eebc: sub edx,1
0x1000eebf: and edx,eax
0x1000eec1: mov [ecx+14Ch],edx
0x1000eec7: ret 4
```

`sub_1000eeb0` is called by sub_1001a5b0, sub_1001b970. It calls nothing else.

12.242 sub_1000eed0 at 0x1000eed0 (4 instructions)

```
0x1000eed0: mov byte ptr [esp+7],0
0x1000eed5: mov eax,[esp+4]
0x1000eed9: mov [ecx+150h],eax
0x1000eedf: ret 4
```

`sub_1000eed0` is called by sub_1001a5b0, sub_1001b970. It calls nothing else.

12.243 sub_1000ef00 at 0x1000ef00 (2 instructions)

```
0x1000ef00: mov eax,[ecx+168h]
0x1000ef06: ret
```

`sub_1000ef00` is called by sub_1001b350. It calls nothing else.

12.244 sub_1000ef10 at 0x1000ef10 (24 instructions)

```
0x1000ef10: mov eax,[esp+4]
0x1000ef14: test eax,eax
0x1000ef16: push esi
0x1000ef17: mov esi,ecx
0x1000ef19: jg short 1000EF24h
0x1000ef1b: mov eax,60h
0x1000ef20: mov [esp+8],eax
0x1000ef24: fild dword ptr [esp+8]
0x1000ef28: sub esp,8
0x1000ef2b: mov [esi+168h],eax
0x1000ef31: fmul qword ptr ds:[100FE7D8h]
0x1000ef37: fdiv qword ptr ds:[100FE7D0h]
0x1000ef3d: fadd qword ptr ds:[100FE7C8h]
0x1000ef43: fstp qword ptr [esp]
0x1000ef46: call 1002297Ch
0x1000ef4b: add esp,8
0x1000ef4e: call 100228D0h
0x1000ef53: mov [esi+20h],eax
0x1000ef56: mov esi,[esi+4]
0x1000ef59: test esi,esi
0x1000ef5b: je short 1000EF60h
0x1000ef5d: mov [esi+18h],eax
0x1000ef60: pop esi
0x1000ef61: ret 4
```

`sub_1000ef10` is called by sub_1000e3e0, sub_100143f0, sub_1001b350, sub_1001c370. It delegates to sub_100228d0, j_floor (2 callees).

12.245 sub_1000ef70 at 0x1000ef70 (24 instructions)

```
0x1000ef70: mov eax,[esp+4]
0x1000ef74: test eax,eax
0x1000ef76: push esi
0x1000ef77: mov esi,ecx
0x1000ef79: jg short 1000EF84h
0x1000ef7b: mov eax,60h
0x1000ef80: mov [esp+8],eax
0x1000ef84: fild dword ptr [esp+8]
0x1000ef88: sub esp,8
0x1000ef8b: mov [esi+16Ch],eax
0x1000ef91: fmul qword ptr ds:[100FE7D8h]
0x1000ef97: fdiv qword ptr ds:[100FE7D0h]
0x1000ef9d: fadd qword ptr ds:[100FE7C8h]
0x1000efa3: fstp qword ptr [esp]
0x1000efa6: call 1002297Ch
0x1000efab: add esp,8
0x1000efae: call 100228D0h
0x1000efb3: mov [esi+24h],eax
0x1000efb6: mov esi,[esi+4]
0x1000efb9: test esi,esi
0x1000efbb: je short 1000EFC0h
0x1000efbd: mov [esi+1Ch],eax
0x1000efc0: pop esi
0x1000efc1: ret 4
```

`sub_1000ef70` is called by sub_1000e3e0, sub_100143f0, sub_1001b350, sub_1001c370. It delegates to sub_100228d0, j_floor (2 callees).

12.246 sub_1000efd0 at 0x1000efd0 (2 instructions)

```
0x1000efd0: mov eax,[ecx+4]
0x1000efd3: ret
```

`sub_1000efd0` is called by sub_1001a5b0. It calls nothing else.

12.247 sub_1000efe0 at 0x1000efe0 (2 instructions)

```
0x1000efe0: mov eax,[ecx+10h]
0x1000efe3: ret
```

`sub_1000efe0` is called by sub_1000cc10, sub_1000ecf0, sub_10017120, sub_10019a40, sub_1001a270, sub_1001a5b0. It calls nothing else.

12.248 sub_1000eff0 at 0x1000eff0 (2 instructions)

```
0x1000eff0: mov eax,[ecx+0Ch]
0x1000eff3: ret
```

`sub_1000eff0` is called by sub_1000cc10, sub_1000ecf0, sub_1001a270, sub_1001a5b0. It calls nothing else.

12.249 sub_1000f000 at 0x1000f000 (2 instructions)

```
0x1000f000: mov eax,[ecx+30h]
0x1000f003: ret
```

`sub_1000f000` is called by sub_10016040, sub_10016840, sub_10017120, sub_10017170, sub_10018be0, sub_10019a40, sub_1001b350. It calls nothing else.

12.250 sub_1000f010 at 0x1000f010 (2 instructions)

```
0x1000f010: mov eax,[ecx+28h]
0x1000f013: ret
```

`sub_1000f010` is called by sub_100103d0. It calls nothing else.

12.251 sub_1000f020 at 0x1000f020 (2 instructions)

```
0x1000f020: mov ax,[ecx+16h]
0x1000f024: ret
```

`sub_1000f020` is called by sub_1000e680, sub_1000ecf0, sub_1001a5b0. It calls nothing else.

12.252 sub_1000f030 at 0x1000f030 (2 instructions)

```
0x1000f030: mov eax,[ecx+40h]
0x1000f033: ret
```

`sub_1000f030` is called by sub_1000ecf0, sub_1000ee20, sub_1000f2e0, sub_1000f310. It calls nothing else.

12.253 sub_1000f040 at 0x1000f040 (9 instructions)

```
0x1000f040: mov edx,[esp+4]
0x1000f044: lea eax,[edx-1]
0x1000f047: cmp eax,13h
0x1000f04a: ja short 1000F062h
```

```
0x1000f04c: movzx eax,byte ptr [eax+1000F078h]
0x1000f053: jmp dword ptr [eax*4+1000F070h]
0x1000f062: mov dword ptr [ecx+40h],0
0x1000f069: xor al,al
0x1000f06b: ret 4
```

`sub_1000f040` is called by sub_1001b970. It calls nothing else.

12.254 sub_1000f090 at 0x1000f090 (68 instructions)

```
0x1000f090: mov eax,[esp+4]
0x1000f094: test eax,eax
0x1000f096: jne short 1000F099h
0x1000f098: ret
0x1000f099: cmp eax,1
0x1000f09c: jne short 1000F09Fh
0x1000f09e: ret
0x1000f09f: cmp eax,2
0x1000f0a2: jne short 1000F0A5h
0x1000f0a4: ret
0x1000f0a5: cmp eax,3
0x1000f0a8: jne short 1000F0ABh
0x1000f0aa: ret
0x1000f0ab: cmp eax,4
0x1000f0ae: jne short 1000F0B1h
0x1000f0b0: ret
0x1000f0b1: cmp eax,5
0x1000f0b4: jne short 1000F0B7h
0x1000f0b6: ret
0x1000f0b7: cmp eax,6
0x1000f0ba: jne short 1000F0BDh
0x1000f0bc: ret
0x1000f0bd: cmp eax,7
0x1000f0c0: jne short 1000F0C3h
0x1000f0c2: ret
0x1000f0c3: cmp eax,8
0x1000f0c6: jne short 1000F0C9h
0x1000f0c8: ret
0x1000f0c9: cmp eax,9
0x1000f0cc: jne short 1000F0CFh
0x1000f0ce: ret
0x1000f0cf: cmp eax,0Ah
0x1000f0d2: jne short 1000F0D5h
0x1000f0d4: ret
0x1000f0d5: cmp eax,0Bh
0x1000f0d8: jne short 1000F0DBh
0x1000f0da: ret
0x1000f0db: cmp eax,0Ch
0x1000f0de: jne short 1000F0E1h
0x1000f0e0: ret
0x1000f0e1: cmp eax,0Dh
0x1000f0e4: jne short 1000F0E7h
0x1000f0e6: ret
0x1000f0e7: cmp eax,0Eh
0x1000f0ea: jne short 1000F0EDh
0x1000f0ec: ret
0x1000f0ed: cmp eax,0Fh
0x1000f0f0: jne short 1000F0F3h
0x1000f0f2: ret
0x1000f0f3: cmp eax,11h
0x1000f0f6: jne short 1000F0FEh
0x1000f0f8: mov eax,10h
0x1000f0fd: ret
0x1000f0fe: cmp eax,12h
0x1000f101: jne short 1000F109h
0x1000f103: mov eax,11h
```

```
0x1000f108: ret
0x1000f109: cmp eax,13h
0x1000f10c: jne short 1000F114h
0x1000f10e: mov eax,12h
0x1000f113: ret
0x1000f114: xor ecx,ecx
0x1000f116: cmp eax,14h
0x1000f119: setne cl
0x1000f11c: sub ecx,1
0x1000f11f: and ecx,13h
0x1000f122: mov eax,ecx
0x1000f124: ret
```

`sub_1000f090` is called by sub_1000f2e0, sub_1000f310, sub_100102b0. It calls nothing else.

12.255 sub_1000f130 at 0x1000f130 (13 instructions)

```
0x1000f130: sub esp,8
0x1000f133: fld dword ptr [ecx+154h]
0x1000f139: fadd qword ptr ds:[100FE7C8h]
0x1000f13f: fnstcw [esp+2]
0x1000f143: movzx eax,word ptr [esp+2]
0x1000f148: or eax,0C00h
0x1000f14d: mov [esp+4],eax
0x1000f151: fldcw [esp+4]
0x1000f155: fistp dword ptr [esp+4]
0x1000f159: mov al,[esp+4]
0x1000f15d: fldcw [esp+2]
0x1000f161: add esp,8
0x1000f164: ret
```

`sub_1000f130` is called by sub_1001b350. It calls nothing else.

12.256 sub_1000f170 at 0x1000f170 (2 instructions)

```
0x1000f170: mov al,[ecx+158h]
0x1000f176: ret
```

`sub_1000f170` is called by sub_1001b350. It calls nothing else.

12.257 sub_1000f180 at 0x1000f180 (3 instructions)

```
0x1000f180: mov eax,[esp+4]
0x1000f184: mov [ecx+148h],eax
0x1000f18a: ret 4
```

`sub_1000f180` is called by sub_1001b970. It calls nothing else.

12.258 sub_1000f190 at 0x1000f190 (4 instructions)

```
0x1000f190: xor eax,eax
0x1000f192: cmp [ecx+4],eax
0x1000f195: setne al
0x1000f198: ret
```

`sub_1000f190` is called by sub_1000caf0, sub_1000cc10, sub_1000cd90, sub_100103d0, sub_10011b00, sub_10016840, sub_100171e0. It calls nothing else.

12.259 sub_1000f1a0 at 0x1000f1a0 (3 instructions)

```
0x1000f1a0: mov al,[esp+4]
0x1000f1a4: mov [ecx+1ECh],al
0x1000f1aa: ret 4
```

`sub_1000f1a0` is called by sub_10019770. It calls nothing else.

12.260 sub_1000f1b0 at 0x1000f1b0 (3 instructions)

```
0x1000f1b0: mov al,[esp+4]
0x1000f1b4: mov [ecx+1EDh],al
0x1000f1ba: ret 4
```

`sub_1000f1b0` is called by sub_1001a5b0. It calls nothing else.

12.261 sub_1000f1c0 at 0x1000f1c0 (10 instructions)

```
0x1000f1c0: cmp byte ptr [ecx+1EEh],0
0x1000f1c7: mov ax,[esp+4]
0x1000f1cc: jne short 1000F1DDh
0x1000f1ce: xor ecx,ecx
0x1000f1d0: mov ch,al
0x1000f1d2: movzx edx,ax
0x1000f1d5: shr edx,8
0x1000f1d8: or ecx,edx
0x1000f1da: mov ax,cx
0x1000f1dd: ret 4
```

`sub_1000f1c0` is called by sub_1000f220, sub_10016840, sub_100175f0, sub_10017dd0, sub_10019770, sub_1001a380, sub_1001a5b0, sub_1001b970, sub_1001c370. It calls nothing else.

12.262 sub_1000f1e0 at 0x1000f1e0 (18 instructions)

```
0x1000f1e0: cmp byte ptr [ecx+1EEh],0
0x1000f1e7: je short 1000F1F0h
0x1000f1e9: mov eax,[esp+4]
0x1000f1ed: ret 4
0x1000f1f0: mov ecx,[esp+4]
0x1000f1f4: mov eax,ecx
0x1000f1f6: mov edx,ecx
0x1000f1f8: and eax,0FF00h
0x1000f1fd: shl edx,10h
0x1000f200: or eax,edx
0x1000f202: xor edx,edx
0x1000f204: mov dh,[esp+6]
0x1000f208: shl eax,8
0x1000f20b: sar ecx,18h
0x1000f20e: and ecx,0FFh
0x1000f214: or eax,edx
0x1000f216: or eax,ecx
0x1000f218: ret 4
```

`sub_1000f1e0` is called by sub_1000f220, sub_10016840, sub_1001b970, sub_1001c370. It calls nothing else.

12.263 sub_1000f220 at 0x1000f220 (61 instructions)

```
0x1000f220: push esi
0x1000f221: mov esi,[esp+8]
0x1000f225: mov eax,[esi]
0x1000f227: push edi
0x1000f228: push eax
0x1000f229: mov edi,ecx
```

```
0x1000f22b: call 1000F1E0h
0x1000f230: mov ecx,[esi+4]
0x1000f233: push ecx
0x1000f234: mov ecx,edi
0x1000f236: mov [esi],eax
0x1000f238: call 1000F1E0h
0x1000f23d: mov edx,[esi+8]
0x1000f240: push edx
0x1000f241: mov ecx,edi
0x1000f243: mov [esi+4],eax
0x1000f246: call 1000F1E0h
0x1000f24b: mov [esi+8],eax
0x1000f24e: movzx eax,word ptr [esi+0Ch]
0x1000f252: push eax
0x1000f253: mov ecx,edi
0x1000f255: call 1000F1C0h
0x1000f25a: movzx ecx,word ptr [esi+0Eh]
0x1000f25e: push ecx
0x1000f25f: mov ecx,edi
0x1000f261: mov [esi+0Ch],ax
0x1000f265: call 1000F1C0h
0x1000f26a: mov edx,[esi+10h]
0x1000f26d: push edx
0x1000f26e: mov ecx,edi
0x1000f270: mov [esi+0Eh],ax
0x1000f274: call 1000F1E0h
0x1000f279: mov [esi+10h],eax
0x1000f27c: mov eax,[esi+14h]
0x1000f27f: push eax
0x1000f280: mov ecx,edi
0x1000f282: call 1000F1E0h
0x1000f287: mov ecx,[esi+18h]
0x1000f28a: push ecx
0x1000f28b: mov ecx,edi
0x1000f28d: mov [esi+14h],eax
0x1000f290: call 1000F1E0h
0x1000f295: mov edx,[esi+1Ch]
0x1000f298: push edx
0x1000f299: mov ecx,edi
0x1000f29b: mov [esi+18h],eax
0x1000f29e: call 1000F1E0h
0x1000f2a3: mov [esi+1Ch],eax
0x1000f2a6: mov eax,[esi+20h]
0x1000f2a9: push eax
0x1000f2aa: mov ecx,edi
0x1000f2ac: call 1000F1E0h
0x1000f2b1: mov ecx,[esi+24h]
0x1000f2b4: push ecx
0x1000f2b5: mov ecx,edi
0x1000f2b7: mov [esi+20h],eax
0x1000f2ba: call 1000F1E0h
0x1000f2bf: pop edi
0x1000f2c0: mov [esi+24h],eax
0x1000f2c3: pop esi
0x1000f2c4: ret 4
```

`sub_1000f220` is called by sub_1001b970, sub_1001c190. It delegates to sub_1000f1c0, sub_1000f1e0 (2 callees).

12.264 sub_1000f2d0 at 0x1000f2d0 (2 instructions)

```
0x1000f2d0: lea eax,[ecx+44h]
0x1000f2d3: ret
```

`sub_1000f2d0` is called by sub_1000f760, sub_100103d0, sub_10011b00. It calls nothing else.

12.265 sub_1000f2e0 at 0x1000f2e0 (14 instructions)

```
0x1000f2e0: mov eax,[esp+4]
0x1000f2e4: push eax
0x1000f2e5: mov edx,ecx
0x1000f2e7: call 1000F090h
0x1000f2ec: add esp,4
0x1000f2ef: test eax,eax
0x1000f2f1: jne short 1000F303h
0x1000f2f3: mov ecx,edx
0x1000f2f5: call 1000F030h
0x1000f2fa: push eax
0x1000f2fb: call 1000F090h
0x1000f300: add esp,4
0x1000f303: mov eax,[edx+eax*4+18Ch]
0x1000f30a: ret 4
```

`sub_1000f2e0` is called by sub_10016040, sub_1001b350. It delegates to sub_1000f030, sub_1000f090 (2 callees).

12.266 sub_1000f310 at 0x1000f310 (16 instructions)

```
0x1000f310: mov eax,[esp+8]
0x1000f314: push eax
0x1000f315: mov edx,ecx
0x1000f317: call 1000F090h
0x1000f31c: add esp,4
0x1000f31f: test eax,eax
0x1000f321: jne short 1000F333h
0x1000f323: mov ecx,edx
0x1000f325: call 1000F030h
0x1000f32a: push eax
0x1000f32b: call 1000F090h
0x1000f330: add esp,4
0x1000f333: mov ecx,[esp+4]
0x1000f337: mov [edx+eax*4+18Ch],ecx
0x1000f33e: mov al,1
0x1000f340: ret 8
```

`sub_1000f310` is called by sub_100143f0. It delegates to sub_1000f030, sub_1000f090 (2 callees).

12.267 sub_1000f5d0 at 0x1000f5d0 (8 instructions)

```
0x1000f5d0: push esi
0x1000f5d1: push 4
0x1000f5d3: mov esi,ecx
0x1000f5d5: call 1000E480h
0x1000f5da: mov dword ptr [esi],100FE85Ch
0x1000f5e0: mov eax,esi
0x1000f5e2: pop esi
0x1000f5e3: ret
```

`sub_1000f5d0` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.268 sub_1000f5f0 at 0x1000f5f0 (9 instructions)

```
0x1000f5f0: push esi
0x1000f5f1: push 5
0x1000f5f3: mov esi,ecx
0x1000f5f5: call 1000E480h
0x1000f5fa: mov dword ptr [esi],100FE86Ch
0x1000f600: mov dword ptr [esi+3D0h],0
0x1000f60a: mov eax,esi
0x1000f60c: pop esi
0x1000f60d: ret
```

`sub_1000f5f0` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.269 sub_1000f610 at 0x1000f610 (12 instructions)

```
0x1000f610: push esi
0x1000f611: push 6
0x1000f613: mov esi,ecx
0x1000f615: call 1000E480h
0x1000f61a: xor eax,eax
0x1000f61c: mov [esi+3D0h],eax
0x1000f622: mov [esi+3D4h],al
0x1000f628: mov [esi+3D8h],eax
0x1000f62e: mov dword ptr [esi],100FE874h
0x1000f634: mov eax,esi
0x1000f636: pop esi
0x1000f637: ret
```

`sub_1000f610` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.270 sub_1000f660 at 0x1000f660 (8 instructions)

```
0x1000f660: push esi
0x1000f661: push 7
0x1000f663: mov esi,ecx
0x1000f665: call 1000E480h
0x1000f66a: mov dword ptr [esi],100FE87Ch
0x1000f670: mov eax,esi
0x1000f672: pop esi
0x1000f673: ret
```

`sub_1000f660` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.271 sub_1000f680 at 0x1000f680 (8 instructions)

```
0x1000f680: push esi
0x1000f681: push 8
0x1000f683: mov esi,ecx
0x1000f685: call 1000E480h
0x1000f68a: mov dword ptr [esi],100FE884h
0x1000f690: mov eax,esi
0x1000f692: pop esi
0x1000f693: ret
```

`sub_1000f680` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.272 sub_1000f6a0 at 0x1000f6a0 (8 instructions)

```
0x1000f6a0: push esi
0x1000f6a1: push 9
0x1000f6a3: mov esi,ecx
0x1000f6a5: call 1000E480h
0x1000f6aa: mov dword ptr [esi],100FE88Ch
0x1000f6b0: mov eax,esi
0x1000f6b2: pop esi
0x1000f6b3: ret
```

`sub_1000f6a0` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.273 sub_1000f6c0 at 0x1000f6c0 (8 instructions)

```
0x1000f6c0: push esi
0x1000f6c1: push 0Ah
0x1000f6c3: mov esi,ecx
0x1000f6c5: call 1000E480h
```

```
0x1000f6ca: mov dword ptr [esi],100FE894h
0x1000f6d0: mov eax,esi
0x1000f6d2: pop esi
0x1000f6d3: ret
```

`sub_1000f6c0` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.274 sub_1000f6e0 at 0x1000f6e0 (8 instructions)

```
0x1000f6e0: push esi
0x1000f6e1: push 0
0x1000f6e3: mov esi,ecx
0x1000f6e5: call 1000E480h
0x1000f6ea: mov dword ptr [esi],100FE89Ch
0x1000f6f0: mov eax,esi
0x1000f6f2: pop esi
0x1000f6f3: ret
```

`sub_1000f6e0` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.275 sub_1000f700 at 0x1000f700 (8 instructions)

```
0x1000f700: push esi
0x1000f701: push 12h
0x1000f703: mov esi,ecx
0x1000f705: call 1000E480h
0x1000f70a: mov dword ptr [esi],100FE8A4h
0x1000f710: mov eax,esi
0x1000f712: pop esi
0x1000f713: ret
```

`sub_1000f700` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.276 sub_1000f720 at 0x1000f720 (8 instructions)

```
0x1000f720: push esi
0x1000f721: push 13h
0x1000f723: mov esi,ecx
0x1000f725: call 1000E480h
0x1000f72a: mov dword ptr [esi],100FE8ACh
0x1000f730: mov eax,esi
0x1000f732: pop esi
0x1000f733: ret
```

`sub_1000f720` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.277 sub_1000f740 at 0x1000f740 (8 instructions)

```
0x1000f740: push esi
0x1000f741: push 14h
0x1000f743: mov esi,ecx
0x1000f745: call 1000E480h
0x1000f74a: mov dword ptr [esi],100FE8B4h
0x1000f750: mov eax,esi
0x1000f752: pop esi
0x1000f753: ret
```

`sub_1000f740` is called by sub_1000f760. It delegates to sub_1000e480 (1 callees).

12.278 sub_1000f760 at 0x1000f760 (835 instructions)

```
0x1000f760: push 0FFFFFFFFh
0x1000f762: push 100FC385h
0x1000f767: mov eax,fs:[0]
0x1000f76d: push eax
0x1000f76e: push ebx
0x1000f76f: push ebp
0x1000f770: push esi
0x1000f771: push edi
0x1000f772: mov eax,ds:[10136220h]
0x1000f777: xor eax,esp
0x1000f779: push eax
0x1000f77a: lea eax,[esp+14h]
0x1000f77e: mov fs:[0],eax
0x1000f784: mov ebx,ecx
0x1000f786: mov esi,[esp+24h]
0x1000f78a: test esi,esi
0x1000f78c: jne short 1000F7C0h
0x1000f78e: mov eax,ds:[100FE8B8h]
0x1000f793: mov [ebx+44h],eax
0x1000f796: mov ecx,ds:[100FE8BCh]
0x1000f79c: mov [ebx+48h],ecx
0x1000f79f: mov edx,ds:[100FE8C0h]
0x1000f7a5: mov [ebx+4Ch],edx
0x1000f7a8: mov eax,ds:[100FE8C4h]
0x1000f7ad: mov [ebx+50h],eax
0x1000f7b0: mov cx,ds:[100FE8C8h]
0x1000f7b7: mov [ebx+54h],cx
0x1000f7bb: jmp near ptr 1001014Dh
0x1000f7c0: mov edx,[esi]
0x1000f7c2: mov eax,[edx+14h]
0x1000f7c5: mov ecx,esi
0x1000f7c7: call eax
0x1000f7c9: mov ebp,[esp+28h]
0x1000f7cd: test ebp,ebp
0x1000f7cf: mov [esp+24h],eax
0x1000f7d3: je short 1000F7DEh
0x1000f7d5: cmp ebp,1
0x1000f7d8: jne near ptr 1000F891h
0x1000f7de: push 3D0h
0x1000f7e3: call 10021FDEh
0x1000f7e8: mov edi,eax
0x1000f7ea: add esp,4
0x1000f7ed: mov [esp+28h],edi
0x1000f7f1: test edi,edi
0x1000f7f3: mov dword ptr [esp+1Ch],0
0x1000f7fb: je short 1000F80Eh
0x1000f7fd: push 1
0x1000f7ff: mov ecx,edi
0x1000f801: call 1000E480h
0x1000f806: mov dword ptr [edi],100FE864h
0x1000f80c: jmp short 1000F810h
0x1000f80e: xor edi,edi
0x1000f810: test edi,edi
0x1000f812: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000f81a: je near ptr 1001014Dh
0x1000f820: push ebx
0x1000f821: mov ecx,edi
0x1000f823: call 1000E5C0h
0x1000f828: push esi
0x1000f829: mov ecx,edi
0x1000f82b: call 1001C370h
0x1000f830: test al,al
0x1000f832: je short 1000F84Fh
0x1000f834: push 1
0x1000f836: push edi
0x1000f837: mov ecx,ebx
```

```
0x1000f839: call 1000E6D0h
0x1000f83e: mov edx,[edi]
0x1000f840: mov eax,[edx]
0x1000f842: push 1
0x1000f844: mov ecx,edi
0x1000f846: call eax
0x1000f848: mov al,1
0x1000f84a: jmp near ptr 1001014Fh
0x1000f84f: mov ecx,edi
0x1000f851: call 1000F2D0h
0x1000f856: lea edx,[ebx+44h]
0x1000f859: lea esp,[esp]
0x1000f860: mov cl,[eax]
0x1000f862: mov [edx],cl
0x1000f864: add eax,1
0x1000f867: add edx,1
0x1000f86a: test cl,cl
0x1000f86c: jne short 1000F860h
0x1000f86e: mov edx,[esi]
0x1000f870: mov eax,[esp+24h]
0x1000f874: mov edx,[edx+10h]
0x1000f877: push 0
0x1000f879: push eax
0x1000f87a: mov ecx,esi
0x1000f87c: call edx
0x1000f87e: mov eax,[edi]
0x1000f880: mov edx,[eax]
0x1000f882: push 1
0x1000f884: mov ecx,edi
0x1000f886: call edx
0x1000f888: test ebp,ebp
0x1000f88a: je short 1000F89Ah
0x1000f88c: jmp near ptr 1001014Dh
0x1000f891: cmp ebp,3
0x1000f894: jne near ptr 1000F941h
0x1000f89a: push 3E8h
0x1000f89f: call 10021FDEh
0x1000f8a4: add esp,4
0x1000f8a7: mov [esp+28h],eax
0x1000f8ab: test eax,eax
0x1000f8ad: mov dword ptr [esp+1Ch],1
0x1000f8b5: je short 1000F8C2h
0x1000f8b7: mov ecx,eax
0x1000f8b9: call 1001B160h
0x1000f8be: mov edi,eax
0x1000f8c0: jmp short 1000F8C4h
0x1000f8c2: xor edi,edi
0x1000f8c4: test edi,edi
0x1000f8c6: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000f8ce: je near ptr 1001014Dh
0x1000f8d4: push ebx
0x1000f8d5: mov ecx,edi
0x1000f8d7: call 1000E5C0h
0x1000f8dc: push esi
0x1000f8dd: mov ecx,edi
0x1000f8df: call 1001B350h
0x1000f8e4: test al,al
0x1000f8e6: je short 1000F903h
0x1000f8e8: push 1
0x1000f8ea: push edi
0x1000f8eb: mov ecx,ebx
0x1000f8ed: call 1000E6D0h
0x1000f8f2: mov eax,[edi]
0x1000f8f4: mov edx,[eax]
0x1000f8f6: push 1
0x1000f8f8: mov ecx,edi
0x1000f8fa: call edx
```

```
0x1000f8fc: mov al,1
0x1000f8fe: jmp near ptr 1001014Fh
0x1000f903: mov ecx,edi
0x1000f905: call 1000F2D0h
0x1000f90a: lea edx,[ebx+44h]
0x1000f90d: lea ecx,[ecx]
0x1000f910: mov cl,[eax]
0x1000f912: mov [edx],cl
0x1000f914: add eax,1
0x1000f917: add edx,1
0x1000f91a: test cl,cl
0x1000f91c: jne short 1000F910h
0x1000f91e: mov ecx,[esp+24h]
0x1000f922: mov eax,[esi]
0x1000f924: mov edx,[eax+10h]
0x1000f927: push 0
0x1000f929: push ecx
0x1000f92a: mov ecx,esi
0x1000f92c: call edx
0x1000f92e: mov eax,[edi]
0x1000f930: mov edx,[eax]
0x1000f932: push 1
0x1000f934: mov ecx,edi
0x1000f936: call edx
0x1000f938: test ebp,ebp
0x1000f93a: je short 1000F94Ah
0x1000f93c: jmp near ptr 1001014Dh
0x1000f941: cmp ebp,5
0x1000f944: jne near ptr 1000F9E3h
0x1000f94a: push 3D4h
0x1000f94f: call 10021FDEh
0x1000f954: add esp,4
0x1000f957: mov [esp+28h],eax
0x1000f95b: test eax,eax
0x1000f95d: mov dword ptr [esp+1Ch],2
0x1000f965: je short 1000F972h
0x1000f967: mov ecx,eax
0x1000f969: call 1000F5F0h
0x1000f96e: mov edi,eax
0x1000f970: jmp short 1000F974h
0x1000f972: xor edi,edi
0x1000f974: test edi,edi
0x1000f976: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000f97e: je near ptr 1001014Dh
0x1000f984: push ebx
0x1000f985: mov ecx,edi
0x1000f987: call 1000E5C0h
0x1000f98c: push esi
0x1000f98d: mov ecx,edi
0x1000f98f: call 1001B970h
0x1000f994: test al,al
0x1000f996: jne near ptr 1000F8E8h
0x1000f99c: mov eax,[edi+160h]
0x1000f9a2: mov ecx,edi
0x1000f9a4: mov [ebx+160h],eax
0x1000f9aa: call 1000F2D0h
0x1000f9af: lea edx,[ebx+44h]
0x1000f9b2: mov cl,[eax]
0x1000f9b4: mov [edx],cl
0x1000f9b6: add eax,1
0x1000f9b9: add edx,1
0x1000f9bc: test cl,cl
0x1000f9be: jne short 1000F9B2h
0x1000f9c0: mov edx,[esi]
0x1000f9c2: mov eax,[esp+24h]
0x1000f9c6: mov edx,[edx+10h]
0x1000f9c9: push 0
```

```
0x1000f9cb: push eax
0x1000f9cc: mov ecx,esi
0x1000f9ce: call edx
0x1000f9d0: mov eax,[edi]
0x1000f9d2: mov edx,[eax]
0x1000f9d4: push 1
0x1000f9d6: mov ecx,edi
0x1000f9d8: call edx
0x1000f9da: test ebp,ebp
0x1000f9dc: je short 1000F9ECh
0x1000f9de: jmp near ptr 1001014Dh
0x1000f9e3: cmp ebp,2
0x1000f9e6: jne near ptr 1000FA85h
0x1000f9ec: push 464h
0x1000f9f1: call 10021FDEh
0x1000f9f6: add esp,4
0x1000f9f9: mov [esp+28h],eax
0x1000f9fd: test eax,eax
0x1000f9ff: mov dword ptr [esp+1Ch],3
0x1000fa07: je short 1000FA14h
0x1000fa09: mov ecx,eax
0x1000fa0b: call 10019590h
0x1000fa10: mov edi,eax
0x1000fa12: jmp short 1000FA16h
0x1000fa14: xor edi,edi
0x1000fa16: test edi,edi
0x1000fa18: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000fa20: je near ptr 1001014Dh
0x1000fa26: push ebx
0x1000fa27: mov ecx,edi
0x1000fa29: call 1000E5C0h
0x1000fa2e: push esi
0x1000fa2f: mov ecx,edi
0x1000fa31: call 1001A5B0h
0x1000fa36: test al,al
0x1000fa38: jne near ptr 1000F8E8h
0x1000fa3e: mov eax,[edi+160h]
0x1000fa44: mov ecx,edi
0x1000fa46: mov [ebx+160h],eax
0x1000fa4c: call 1000F2D0h
0x1000fa51: lea edx,[ebx+44h]
0x1000fa54: mov cl,[eax]
0x1000fa56: mov [edx],cl
0x1000fa58: add eax,1
0x1000fa5b: add edx,1
0x1000fa5e: test cl,cl
0x1000fa60: jne short 1000FA54h
0x1000fa62: mov edx,[esi]
0x1000fa64: mov eax,[esp+24h]
0x1000fa68: mov edx,[edx+10h]
0x1000fa6b: push 0
0x1000fa6d: push eax
0x1000fa6e: mov ecx,esi
0x1000fa70: call edx
0x1000fa72: mov eax,[edi]
0x1000fa74: mov edx,[eax]
0x1000fa76: push 1
0x1000fa78: mov ecx,edi
0x1000fa7a: call edx
0x1000fa7c: test ebp,ebp
0x1000fa7e: je short 1000FA8Eh
0x1000fa80: jmp near ptr 1001014Dh
0x1000fa85: cmp ebp,4
0x1000fa88: jne near ptr 1000FB21h
0x1000fa8e: push 3D0h
0x1000fa93: call 10021FDEh
0x1000fa98: add esp,4
```

```
0x1000fa9b: mov [esp+28h],eax
0x1000fa9f: test eax,eax
0x1000faa1: mov dword ptr [esp+1Ch],4
0x1000faa9: je short 1000FAB6h
0x1000faab: mov ecx,eax
0x1000faad: call 1000F5D0h
0x1000fab2: mov edi,eax
0x1000fab4: jmp short 1000FAB8h
0x1000fab6: xor edi,edi
0x1000fab8: test edi,edi
0x1000faba: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000fac2: je near ptr 1001014Dh
0x1000fac8: push ebx
0x1000fac9: mov ecx,edi
0x1000facb: call 1000E5C0h
0x1000fad0: push esi
0x1000fad1: mov ecx,edi
0x1000fad3: call 10018D20h
0x1000fad8: test al,al
0x1000fada: jne near ptr 1000F8E8h
0x1000fae0: mov ecx,edi
0x1000fae2: call 1000F2D0h
0x1000fae7: lea edx,[ebx+44h]
0x1000faea: lea ebx,[ebx]
0x1000faf0: mov cl,[eax]
0x1000faf2: mov [edx],cl
0x1000faf4: add eax,1
0x1000faf7: add edx,1
0x1000fafa: test cl,cl
0x1000fafc: jne short 1000FAF0h
0x1000fafe: mov ecx,[esp+24h]
0x1000fb02: mov eax,[esi]
0x1000fb04: mov edx,[eax+10h]
0x1000fb07: push 0
0x1000fb09: push ecx
0x1000fb0a: mov ecx,esi
0x1000fb0c: call edx
0x1000fb0e: mov eax,[edi]
0x1000fb10: mov edx,[eax]
0x1000fb12: push 1
0x1000fb14: mov ecx,edi
0x1000fb16: call edx
0x1000fb18: test ebp,ebp
0x1000fb1a: je short 1000FB2Ah
0x1000fb1c: jmp near ptr 1001014Dh
0x1000fb21: cmp ebp,6
0x1000fb24: jne near ptr 1000FBC3h
0x1000fb2a: push 3DCh
0x1000fb2f: call 10021FDEh
0x1000fb34: add esp,4
0x1000fb37: mov [esp+28h],eax
0x1000fb3b: test eax,eax
0x1000fb3d: mov dword ptr [esp+1Ch],5
0x1000fb45: je short 1000FB52h
0x1000fb47: mov ecx,eax
0x1000fb49: call 1000F610h
0x1000fb4e: mov edi,eax
0x1000fb50: jmp short 1000FB54h
0x1000fb52: xor edi,edi
0x1000fb54: test edi,edi
0x1000fb56: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000fb5e: je near ptr 1001014Dh
0x1000fb64: push ebx
0x1000fb65: mov ecx,edi
0x1000fb67: call 1000E5C0h
0x1000fb6c: push esi
0x1000fb6d: mov ecx,edi
```

```
0x1000fb6f: call 100143F0h
0x1000fb74: test al,al
0x1000fb76: jne near ptr 1000F8E8h
0x1000fb7c: mov eax,[edi+160h]
0x1000fb82: mov ecx,edi
0x1000fb84: mov [ebx+160h],eax
0x1000fb8a: call 1000F2D0h
0x1000fb8f: lea edx,[ebx+44h]
0x1000fb92: mov cl,[eax]
0x1000fb94: mov [edx],cl
0x1000fb96: add eax,1
0x1000fb99: add edx,1
0x1000fb9c: test cl,cl
0x1000fb9e: jne short 1000FB92h
0x1000fba0: mov edx,[esi]
0x1000fba2: mov eax,[esp+24h]
0x1000fba6: mov edx,[edx+10h]
0x1000fba9: push 0
0x1000fbab: push eax
0x1000fbac: mov ecx,esi
0x1000fbae: call edx
0x1000fbb0: mov eax,[edi]
0x1000fbb2: mov edx,[eax]
0x1000fbb4: push 1
0x1000fbb6: mov ecx,edi
0x1000fbb8: call edx
0x1000fbba: test ebp,ebp
0x1000fbbc: je short 1000FBCCh
0x1000fbbe: jmp near ptr 1001014Dh
0x1000fbc3: cmp ebp,11h
0x1000fbc6: jne near ptr 1000FC65h
0x1000fbcc: push 404h
0x1000fbd1: call 10021FDEh
0x1000fbd6: add esp,4
0x1000fbd9: mov [esp+28h],eax
0x1000fbdd: test eax,eax
0x1000fbdf: mov dword ptr [esp+1Ch],6
0x1000fbe7: je short 1000FBF4h
0x1000fbe9: mov ecx,eax
0x1000fbeb: call 10018670h
0x1000fbf0: mov edi,eax
0x1000fbf2: jmp short 1000FBF6h
0x1000fbf4: xor edi,edi
0x1000fbf6: test edi,edi
0x1000fbf8: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000fc00: je near ptr 1001014Dh
0x1000fc06: push ebx
0x1000fc07: mov ecx,edi
0x1000fc09: call 1000E5C0h
0x1000fc0e: push esi
0x1000fc0f: mov ecx,edi
0x1000fc11: call 100187F0h
0x1000fc16: test al,al
0x1000fc18: jne near ptr 1000F8E8h
0x1000fc1e: mov eax,[edi+160h]
0x1000fc24: mov ecx,edi
0x1000fc26: mov [ebx+160h],eax
0x1000fc2c: call 1000F2D0h
0x1000fc31: lea edx,[ebx+44h]
0x1000fc34: mov cl,[eax]
0x1000fc36: mov [edx],cl
0x1000fc38: add eax,1
0x1000fc3b: add edx,1
0x1000fc3e: test cl,cl
0x1000fc40: jne short 1000FC34h
0x1000fc42: mov edx,[esi]
0x1000fc44: mov eax,[esp+24h]
```

```
0x1000fc48: mov edx,[edx+10h]
0x1000fc4b: push 0
0x1000fc4d: push eax
0x1000fc4e: mov ecx,esi
0x1000fc50: call edx
0x1000fc52: mov eax,[edi]
0x1000fc54: mov edx,[eax]
0x1000fc56: push 1
0x1000fc58: mov ecx,edi
0x1000fc5a: call edx
0x1000fc5c: test ebp,ebp
0x1000fc5e: je short 1000FC6Eh
0x1000fc60: jmp near ptr 1001014Dh
0x1000fc65: cmp ebp,7
0x1000fc68: jne near ptr 1000FD01h
0x1000fc6e: push 3D0h
0x1000fc73: call 10021FDEh
0x1000fc78: add esp,4
0x1000fc7b: mov [esp+28h],eax
0x1000fc7f: test eax,eax
0x1000fc81: mov dword ptr [esp+1Ch],7
0x1000fc89: je short 1000FC96h
0x1000fc8b: mov ecx,eax
0x1000fc8d: call 1000F660h
0x1000fc92: mov edi,eax
0x1000fc94: jmp short 1000FC98h
0x1000fc96: xor edi,edi
0x1000fc98: test edi,edi
0x1000fc9a: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000fca2: je near ptr 1001014Dh
0x1000fca8: push ebx
0x1000fca9: mov ecx,edi
0x1000fcab: call 1000E5C0h
0x1000fcb0: push esi
0x1000fcb1: mov ecx,edi
0x1000fcb3: call 10018120h
0x1000fcb8: test al,al
0x1000fcba: jne near ptr 1000F8E8h
0x1000fcc0: mov ecx,edi
0x1000fcc2: call 1000F2D0h
0x1000fcc7: lea edx,[ebx+44h]
0x1000fcca: lea ebx,[ebx]
0x1000fcd0: mov cl,[eax]
0x1000fcd2: mov [edx],cl
0x1000fcd4: add eax,1
0x1000fcd7: add edx,1
0x1000fcda: test cl,cl
0x1000fcdc: jne short 1000FCD0h
0x1000fcde: mov ecx,[esp+24h]
0x1000fce2: mov eax,[esi]
0x1000fce4: mov edx,[eax+10h]
0x1000fce7: push 0
0x1000fce9: push ecx
0x1000fcea: mov ecx,esi
0x1000fcec: call edx
0x1000fcee: mov eax,[edi]
0x1000fcf0: mov edx,[eax]
0x1000fcf2: push 1
0x1000fcf4: mov ecx,edi
0x1000fcf6: call edx
0x1000fcf8: test ebp,ebp
0x1000fcfa: je short 1000FD0Ah
0x1000fcfc: jmp near ptr 1001014Dh
0x1000fd01: cmp ebp,8
0x1000fd04: jne near ptr 1000FD97h
0x1000fd0a: push 3D0h
0x1000fd0f: call 10021FDEh
```

```
0x1000fd14: add esp,4
0x1000fd17: mov [esp+28h],eax
0x1000fd1b: test eax,eax
0x1000fd1d: mov dword ptr [esp+1Ch],8
0x1000fd25: je short 1000FD32h
0x1000fd27: mov ecx,eax
0x1000fd29: call 1000F680h
0x1000fd2e: mov edi,eax
0x1000fd30: jmp short 1000FD34h
0x1000fd32: xor edi,edi
0x1000fd34: test edi,edi
0x1000fd36: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000fd3e: je near ptr 1001014Dh
0x1000fd44: push ebx
0x1000fd45: mov ecx,edi
0x1000fd47: call 1000E5C0h
0x1000fd4c: push esi
0x1000fd4d: mov ecx,edi
0x1000fd4f: call 10017680h
0x1000fd54: test al,al
0x1000fd56: jne near ptr 1000F8E8h
0x1000fd5c: mov ecx,edi
0x1000fd5e: call 1000F2D0h
0x1000fd63: lea edx,[ebx+44h]
0x1000fd66: mov cl,[eax]
0x1000fd68: mov [edx],cl
0x1000fd6a: add eax,1
0x1000fd6d: add edx,1
0x1000fd70: test cl,cl
0x1000fd72: jne short 1000FD66h
0x1000fd74: mov ecx,[esp+24h]
0x1000fd78: mov eax,[esi]
0x1000fd7a: mov edx,[eax+10h]
0x1000fd7d: push 0
0x1000fd7f: push ecx
0x1000fd80: mov ecx,esi
0x1000fd82: call edx
0x1000fd84: mov eax,[edi]
0x1000fd86: mov edx,[eax]
0x1000fd88: push 1
0x1000fd8a: mov ecx,edi
0x1000fd8c: call edx
0x1000fd8e: test ebp,ebp
0x1000fd90: je short 1000FDA0h
0x1000fd92: jmp near ptr 1001014Dh
0x1000fd97: cmp ebp,9
0x1000fd9a: jne near ptr 1000FE31h
0x1000fda0: push 3D0h
0x1000fda5: call 10021FDEh
0x1000fdaa: add esp,4
0x1000fdad: mov [esp+28h],eax
0x1000fdb1: test eax,eax
0x1000fdb3: mov dword ptr [esp+1Ch],9
0x1000fdbb: je short 1000FDC8h
0x1000fdbd: mov ecx,eax
0x1000fdbf: call 1000F6A0h
0x1000fdc4: mov edi,eax
0x1000fdc6: jmp short 1000FDCAh
0x1000fdc8: xor edi,edi
0x1000fdca: test edi,edi
0x1000fdcc: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000fdd4: je near ptr 1001014Dh
0x1000fdda: push ebx
0x1000fddb: mov ecx,edi
0x1000fddd: call 1000E5C0h
0x1000fde2: push esi
0x1000fde3: mov ecx,edi
```

```
0x1000fde5: call 100171E0h
0x1000fdea: test al,al
0x1000fdec: jne near ptr 1000F8E8h
0x1000fdf2: mov ecx,edi
0x1000fdf4: call 1000F2D0h
0x1000fdf9: lea edx,[ebx+44h]
0x1000fdfc: lea esp,[esp]
0x1000fe00: mov cl,[eax]
0x1000fe02: mov [edx],cl
0x1000fe04: add eax,1
0x1000fe07: add edx,1
0x1000fe0a: test cl,cl
0x1000fe0c: jne short 1000FE00h
0x1000fe0e: mov ecx,[esp+24h]
0x1000fe12: mov eax,[esi]
0x1000fe14: mov edx,[eax+10h]
0x1000fe17: push 0
0x1000fe19: push ecx
0x1000fe1a: mov ecx,esi
0x1000fe1c: call edx
0x1000fe1e: mov eax,[edi]
0x1000fe20: mov edx,[eax]
0x1000fe22: push 1
0x1000fe24: mov ecx,edi
0x1000fe26: call edx
0x1000fe28: test ebp,ebp
0x1000fe2a: je short 1000FE3Ah
0x1000fe2c: jmp near ptr 1001014Dh
0x1000fe31: cmp ebp,0Ah
0x1000fe34: jne near ptr 1000FED1h
0x1000fe3a: push 3D0h
0x1000fe3f: call 10021FDEh
0x1000fe44: add esp,4
0x1000fe47: mov [esp+28h],eax
0x1000fe4b: test eax,eax
0x1000fe4d: mov dword ptr [esp+1Ch],0Ah
0x1000fe55: je short 1000FE62h
0x1000fe57: mov ecx,eax
0x1000fe59: call 1000F6C0h
0x1000fe5e: mov edi,eax
0x1000fe60: jmp short 1000FE64h
0x1000fe62: xor edi,edi
0x1000fe64: test edi,edi
0x1000fe66: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000fe6e: je near ptr 1001014Dh
0x1000fe74: push ebx
0x1000fe75: mov ecx,edi
0x1000fe77: call 1000E5C0h
0x1000fe7c: push 0
0x1000fe7e: push 0
0x1000fe80: push esi
0x1000fe81: mov ecx,edi
0x1000fe83: call 10016CD0h
0x1000fe88: test al,al
0x1000fe8a: jne near ptr 1000F8E8h
0x1000fe90: mov ecx,edi
0x1000fe92: call 1000F2D0h
0x1000fe97: lea edx,[ebx+44h]
0x1000fe9a: lea ebx,[ebx]
0x1000fea0: mov cl,[eax]
0x1000fea2: mov [edx],cl
0x1000fea4: add eax,1
0x1000fea7: add edx,1
0x1000feaa: test cl,cl
0x1000feac: jne short 1000FEA0h
0x1000feae: mov ecx,[esp+24h]
0x1000feb2: mov eax,[esi]
```

```
0x1000feb4: mov edx,[eax+10h]
0x1000feb7: push 0
0x1000feb9: push ecx
0x1000feba: mov ecx,esi
0x1000febc: call edx
0x1000febe: mov eax,[edi]
0x1000fec0: mov edx,[eax]
0x1000fec2: push 1
0x1000fec4: mov ecx,edi
0x1000fec6: call edx
0x1000fec8: test ebp,ebp
0x1000feca: je short 1000FEEEh
0x1000fecc: jmp near ptr 1001014Dh
0x1000fed1: cmp ebp,0Bh
0x1000fed4: je short 1000FEEEh
0x1000fed6: cmp ebp,0Ch
0x1000fed9: je short 1000FEEEh
0x1000fedb: cmp ebp,0Dh
0x1000fede: je short 1000FEEEh
0x1000fee0: cmp ebp,0Eh
0x1000fee3: je short 1000FEEEh
0x1000fee5: cmp ebp,0Fh
0x1000fee8: jne near ptr 1000FF81h
0x1000feee: push 3D0h
0x1000fef3: call 10021FDEh
0x1000fef8: add esp,4
0x1000fefb: mov [esp+28h],eax
0x1000feff: test eax,eax
0x1000ff01: mov dword ptr [esp+1Ch],0Bh
0x1000ff09: je short 1000FF16h
0x1000ff0b: mov ecx,eax
0x1000ff0d: call 1000F6E0h
0x1000ff12: mov edi,eax
0x1000ff14: jmp short 1000FF18h
0x1000ff16: xor edi,edi
0x1000ff18: test edi,edi
0x1000ff1a: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000ff22: je near ptr 1001014Dh
0x1000ff28: push ebx
0x1000ff29: mov ecx,edi
0x1000ff2b: call 1000E5C0h
0x1000ff30: push 0
0x1000ff32: push esi
0x1000ff33: mov ecx,edi
0x1000ff35: call 100156B0h
0x1000ff3a: test al,al
0x1000ff3c: jne near ptr 1000F8E8h
0x1000ff42: mov ecx,edi
0x1000ff44: call 1000F2D0h
0x1000ff49: lea edx,[ebx+44h]
0x1000ff4c: lea esp,[esp]
0x1000ff50: mov cl,[eax]
0x1000ff52: mov [edx],cl
0x1000ff54: add eax,1
0x1000ff57: add edx,1
0x1000ff5a: test cl,cl
0x1000ff5c: jne short 1000FF50h
0x1000ff5e: mov ecx,[esp+24h]
0x1000ff62: mov eax,[esi]
0x1000ff64: mov edx,[eax+10h]
0x1000ff67: push 0
0x1000ff69: push ecx
0x1000ff6a: mov ecx,esi
0x1000ff6c: call edx
0x1000ff6e: mov eax,[edi]
0x1000ff70: mov edx,[eax]
0x1000ff72: push 1
```

```
0x1000ff74: mov ecx,edi
0x1000ff76: call edx
0x1000ff78: test ebp,ebp
0x1000ff7a: je short 1000FF8Ah
0x1000ff7c: jmp near ptr 1001014Dh
0x1000ff81: cmp ebp,12h
0x1000ff84: jne near ptr 10010017h
0x1000ff8a: push 3D0h
0x1000ff8f: call 10021FDEh
0x1000ff94: add esp,4
0x1000ff97: mov [esp+28h],eax
0x1000ff9b: test eax,eax
0x1000ff9d: mov dword ptr [esp+1Ch],0Ch
0x1000ffa5: je short 1000FFB2h
0x1000ffa7: mov ecx,eax
0x1000ffa9: call 1000F700h
0x1000ffae: mov edi,eax
0x1000ffb0: jmp short 1000FFB4h
0x1000ffb2: xor edi,edi
0x1000ffb4: test edi,edi
0x1000ffb6: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1000ffbe: je near ptr 1001014Dh
0x1000ffc4: push ebx
0x1000ffc5: mov ecx,edi
0x1000ffc7: call 1000E5C0h
0x1000ffcc: push esi
0x1000ffcd: mov ecx,edi
0x1000ffcf: call 10016840h
0x1000ffd4: test al,al
0x1000ffd6: jne near ptr 1000F8E8h
0x1000ffdc: mov ecx,edi
0x1000ffde: call 1000F2D0h
0x1000ffe3: lea edx,[ebx+44h]
0x1000ffe6: mov cl,[eax]
0x1000ffe8: mov [edx],cl
0x1000ffea: add eax,1
0x1000ffed: add edx,1
0x1000fff0: test cl,cl
0x1000fff2: jne short 1000FFE6h
0x1000fff4: mov ecx,[esp+24h]
0x1000fff8: mov eax,[esi]
0x1000fffa: mov edx,[eax+10h]
0x1000fffd: push 0
0x1000ffff: push ecx
0x10010000: mov ecx,esi
0x10010002: call edx
0x10010004: mov eax,[edi]
0x10010006: mov edx,[eax]
0x10010008: push 1
0x1001000a: mov ecx,edi
0x1001000c: call edx
0x1001000e: test ebp,ebp
0x10010010: je short 10010020h
0x10010012: jmp near ptr 1001014Dh
0x10010017: cmp ebp,13h
0x1001001a: jne near ptr 100100B1h
0x10010020: push 3D0h
0x10010025: call 10021FDEh
0x1001002a: add esp,4
0x1001002d: mov [esp+28h],eax
0x10010031: test eax,eax
0x10010033: mov dword ptr [esp+1Ch],0Dh
0x1001003b: je short 10010048h
0x1001003d: mov ecx,eax
0x1001003f: call 1000F720h
0x10010044: mov edi,eax
0x10010046: jmp short 1001004Ah
```

```
0x10010048: xor edi,edi
0x1001004a: test edi,edi
0x1001004c: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x10010054: je near ptr 1001014Dh
0x1001005a: push ebx
0x1001005b: mov ecx,edi
0x1001005d: call 1000E5C0h
0x10010062: push esi
0x10010063: mov ecx,edi
0x10010065: call 10016040h
0x1001006a: test al,al
0x1001006c: jne near ptr 1000F8E8h
0x10010072: mov ecx,edi
0x10010074: call 1000F2D0h
0x10010079: lea edx,[ebx+44h]
0x1001007c: lea esp,[esp]
0x10010080: mov cl,[eax]
0x10010082: mov [edx],cl
0x10010084: add eax,1
0x10010087: add edx,1
0x1001008a: test cl,cl
0x1001008c: jne short 10010080h
0x1001008e: mov ecx,[esp+24h]
0x10010092: mov eax,[esi]
0x10010094: mov edx,[eax+10h]
0x10010097: push 0
0x10010099: push ecx
0x1001009a: mov ecx,esi
0x1001009c: call edx
0x1001009e: mov eax,[edi]
0x100100a0: mov edx,[eax]
0x100100a2: push 1
0x100100a4: mov ecx,edi
0x100100a6: call edx
0x100100a8: test ebp,ebp
0x100100aa: je short 100100BAh
0x100100ac: jmp near ptr 1001014Dh
0x100100b1: cmp ebp,14h
0x100100b4: jne near ptr 1001013Eh
0x100100ba: push 3D0h
0x100100bf: call 10021FDEh
0x100100c4: add esp,4
0x100100c7: mov [esp+28h],eax
0x100100cb: test eax,eax
0x100100cd: mov dword ptr [esp+1Ch],0Eh
0x100100d5: je short 100100E2h
0x100100d7: mov ecx,eax
0x100100d9: call 1000F740h
0x100100de: mov edi,eax
0x100100e0: jmp short 100100E4h
0x100100e2: xor edi,edi
0x100100e4: test edi,edi
0x100100e6: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x100100ee: je short 1001014Dh
0x100100f0: push ebx
0x100100f1: mov ecx,edi
0x100100f3: call 1000E5C0h
0x100100f8: push esi
0x100100f9: mov ecx,edi
0x100100fb: call 10015D20h
0x10010100: test al,al
0x10010102: jne near ptr 1000F8E8h
0x10010108: mov ecx,edi
0x1001010a: call 1000F2D0h
0x1001010f: lea edx,[ebx+44h]
0x10010112: mov cl,[eax]
0x10010114: mov [edx],cl
```

```
0x10010116: add eax,1
0x10010119: add edx,1
0x1001011c: test cl,cl
0x1001011e: jne short 10010112h
0x10010120: mov ecx,[esp+24h]
0x10010124: mov eax,[esi]
0x10010126: mov edx,[eax+10h]
0x10010129: push 0
0x1001012b: push ecx
0x1001012c: mov ecx,esi
0x1001012e: call edx
0x10010130: mov eax,[edi]
0x10010132: mov edx,[eax]
0x10010134: push 1
0x10010136: mov ecx,edi
0x10010138: call edx
0x1001013a: test ebp,ebp
0x1001013c: jne short 1001014Dh
0x1001013e: lea edi,[ebx+44h]
0x10010141: mov ecx,8
0x10010146: mov esi,100FE8CCh
0x1001014b: rep movsd
0x1001014d: xor al,al
0x1001014f: mov ecx,[esp+14h]
0x10010153: mov fs:[0],ecx
0x1001015a: pop ecx
0x1001015b: pop edi
0x1001015c: pop esi
0x1001015d: pop ebp
0x1001015e: pop ebx
0x1001015f: add esp,0Ch
0x10010162: ret 8
```

`sub_1000f760` is called by sub_10010220, sub_1001b970. It works with the text "null file handler", "Decode: Unknown or wrong format". It delegates to sub_1000e480, sub_1000e5c0, sub_1000e6d0 (34 callees).

12.279 sub_10010220 at 0x10010220 (40 instructions)

```
0x10010220: push 0FFFFFFFFh
0x10010222: push 100FC3A8h
0x10010227: mov eax,fs:[0]
0x1001022d: push eax
0x1001022e: sub esp,0Ch
0x10010231: push ebx
0x10010232: mov eax,ds:[10136220h]
0x10010237: xor eax,esp
0x10010239: push eax
0x1001023a: lea eax,[esp+14h]
0x1001023e: mov fs:[0],eax
0x10010244: mov eax,[esp+24h]
0x10010248: test eax,eax
0x1001024a: mov [esp+0Ch],eax
0x1001024e: sete al
0x10010251: mov dword ptr [esp+8],100FE820h
0x10010259: mov [esp+10h],al
0x1001025d: mov edx,[esp+28h]
0x10010261: push edx
0x10010262: lea eax,[esp+0Ch]
0x10010266: push eax
0x10010267: mov dword ptr [esp+24h],0
0x1001026f: call 1000F760h
0x10010274: mov bl,al
0x10010276: mov eax,[esp+0Ch]
0x1001027a: test eax,eax
0x1001027c: mov dword ptr [esp+8],100FE820h
0x10010284: je short 10010297h
```

```
0x10010286: cmp byte ptr [esp+10h],0
0x1001028b: je short 10010297h
0x1001028d: push eax
0x1001028e: call dword ptr ds:[100FE540h]
0x10010294: add esp,4
0x10010297: mov al,bl
0x10010299: mov ecx,[esp+14h]
0x1001029d: mov fs:[0],ecx
0x100102a4: pop ecx
0x100102a5: pop ebx
0x100102a6: add esp,18h
0x100102a9: ret 8
```

`sub_10010220` is called by sub_100102b0. It delegates to sub_1000f760 (1 callees).

12.280 sub_100102b0 at 0x100102b0 (82 instructions)

```
0x100102b0: sub esp,108h
0x100102b6: mov eax,ds:[10136220h]
0x100102bb: xor eax,esp
0x100102bd: mov [esp+104h],eax
0x100102c4: push ebx
0x100102c5: mov ebx,[esp+114h]
0x100102cc: push ebp
0x100102cd: mov ebp,[esp+114h]
0x100102d4: push esi
0x100102d5: push edi
0x100102d6: push ebx
0x100102d7: mov edi,ecx
0x100102d9: call 1000F090h
0x100102de: add esp,4
0x100102e1: test eax,eax
0x100102e3: je short 1001031Ah
0x100102e5: push 100FE8ECh
0x100102ea: push ebp
0x100102eb: call dword ptr ds:[100FE514h]
0x100102f1: mov esi,eax
0x100102f3: add esp,8
0x100102f6: test esi,esi
0x100102f8: jne short 10010301h
0x100102fa: xor al,al
0x100102fc: jmp near ptr 10010386h
0x10010301: push ebx
0x10010302: push esi
0x10010303: mov ecx,edi
0x10010305: call 10010220h
0x1001030a: push esi
0x1001030b: mov bl,al
0x1001030d: call dword ptr ds:[100FE540h]
0x10010313: add esp,4
0x10010316: test bl,bl
0x10010318: jne short 10010384h
0x1001031a: lea eax,[edi+44h]
0x1001031d: lea edx,[esp+14h]
0x10010321: mov [esp+10h],eax
0x10010325: sub edx,eax
0x10010327: mov cl,[eax]
0x10010329: mov [edx+eax],cl
0x1001032c: add eax,1
0x1001032f: test cl,cl
0x10010331: jne short 10010327h
0x10010333: push 100FE8ECh
0x10010338: push ebp
0x10010339: call dword ptr ds:[100FE514h]
0x1001033f: mov esi,eax
0x10010341: add esp,8
```

```
0x10010344: test esi,esi
0x10010346: je short 100102FAh
0x10010348: push 0
0x1001034a: push esi
0x1001034b: mov ecx,edi
0x1001034d: call 10010220h
0x10010352: push esi
0x10010353: mov bl,al
0x10010355: call dword ptr ds:[100FE540h]
0x1001035b: add esp,4
0x1001035e: test bl,bl
0x10010360: jne short 10010384h
0x10010362: cmp dword ptr [esp+120h],0
0x1001036a: jbe short 10010384h
0x1001036c: mov edx,[esp+10h]
0x10010370: lea eax,[esp+14h]
0x10010374: mov ecx,eax
0x10010376: sub edx,ecx
0x10010378: mov cl,[eax]
0x1001037a: mov [edx+eax],cl
0x1001037d: add eax,1
0x10010380: test cl,cl
0x10010382: jne short 10010378h
0x10010384: mov al,bl
0x10010386: mov ecx,[esp+114h]
0x1001038d: pop edi
0x1001038e: pop esi
0x1001038f: pop ebp
0x10010390: pop ebx
0x10010391: xor ecx,esp
0x10010393: call 10022326h
0x10010398: add esp,108h
0x1001039e: ret 8
```

`sub_100102b0` is called by sub_1000cc10. It delegates to sub_1000f090, sub_10010220, sub_10022326 (3 callees).

12.281 sub_100103b0 at 0x100103b0 (8 instructions)

```
0x100103b0: fld dword ptr [esp+4]
0x100103b4: sub esp,8
0x100103b7: fstp qword ptr [esp]
0x100103ba: call 1002297Ch
0x100103bf: fstp dword ptr [esp+0Ch]
0x100103c3: add esp,8
0x100103c6: fld dword ptr [esp+4]
0x100103ca: ret
```

`sub_100103b0` is called by sub_10011fb0. It delegates to j_floor (1 callees).

12.282 sub_100103d0 at 0x100103d0 (303 instructions)

```
0x100103d0: push 0FFFFFFFFh
0x100103d2: push 100FC3F1h
0x100103d7: mov eax,fs:[0]
0x100103dd: push eax
0x100103de: sub esp,7ACh
0x100103e4: mov eax,ds:[10136220h]
0x100103e9: xor eax,esp
0x100103eb: mov [esp+7A8h],eax
0x100103f2: push ebx
0x100103f3: push esi
0x100103f4: push edi
0x100103f5: mov eax,ds:[10136220h]
0x100103fa: xor eax,esp
0x100103fc: push eax
```

```
0x100103fd: lea eax,[esp+7BCh]
0x10010404: mov fs:[0],eax
0x1001040a: mov esi,ecx
0x1001040c: cmp dword ptr [esi+4],0
0x10010410: je near ptr 10010752h
0x10010416: mov eax,[esp+7CCh]
0x1001041d: cmp eax,4
0x10010420: je near ptr 100106B1h
0x10010426: cmp eax,8
0x10010429: je near ptr 100105C7h
0x1001042f: cmp eax,18h
0x10010432: jne near ptr 10010752h
0x10010438: movzx eax,word ptr [esi+16h]
0x1001043c: cmp ax,18h
0x10010440: je near ptr 100107D8h
0x10010446: ja near ptr 10010752h
0x1001044c: push 0
0x1001044e: lea ecx,[esp+3ECh]
0x10010455: call 1000E480h
0x1001045a: push esi
0x1001045b: lea ecx,[esp+3ECh]
0x10010462: mov dword ptr [esp+7C8h],2
0x1001046d: call 1000E5C0h
0x10010472: mov eax,[esi+40h]
0x10010475: mov ecx,[esi+10h]
0x10010478: mov edx,[esi+0Ch]
0x1001047b: push eax
0x1001047c: push 18h
0x1001047e: push ecx
0x1001047f: push edx
0x10010480: lea ecx,[esp+3F8h]
0x10010487: call 1000EAD0h
0x1001048c: lea ecx,[esp+3E8h]
0x10010493: call 1000F190h
0x10010498: test al,al
0x1001049a: jne short 100104DCh
0x1001049c: lea ecx,[esp+3E8h]
0x100104a3: call 1000F2D0h
0x100104a8: lea edx,[esi+44h]
0x100104ab: jmp short 100104B0h
0x100104b0: mov cl,[eax]
0x100104b2: mov [edx],cl
0x100104b4: add eax,1
0x100104b7: add edx,1
0x100104ba: test cl,cl
0x100104bc: jne short 100104B0h
0x100104be: lea ecx,[esp+3E8h]
0x100104c5: mov dword ptr [esp+7C4h],0FFFFFFFFh
0x100104d0: call 1000C970h
0x100104d5: xor al,al
0x100104d7: jmp near ptr 100107DAh
0x100104dc: cmp dword ptr [esi+14Ch],0
0x100104e3: jl short 10010502h
0x100104e5: movzx eax,byte ptr [esi+14Ch]
0x100104ec: push eax
0x100104ed: lea ecx,[esp+18h]
0x100104f1: push ecx
0x100104f2: mov ecx,esi
0x100104f4: call 10012EC0h
0x100104f9: mov edx,[eax]
0x100104fb: mov [esp+538h],edx
0x10010502: push esi
0x10010503: lea ecx,[esp+3ECh]
0x1001050a: call 10013E20h
0x1001050f: push esi
0x10010510: lea ecx,[esp+3ECh]
0x10010517: call 10013A70h
```

```
0x1001051c: mov ecx,esi
0x1001051e: call 10013BD0h
0x10010523: test al,al
0x10010525: je short 1001053Eh
0x10010527: mov ecx,esi
0x10010529: call 100139B0h
0x1001052e: test al,al
0x10010530: jne short 1001053Eh
0x10010532: lea ecx,[esp+3E8h]
0x10010539: call 100139C0h
0x1001053e: xor ebx,ebx
0x10010540: cmp [esi+10h],ebx
0x10010543: jle short 10010588h
0x10010545: cmp dword ptr [esi+148h],0
0x1001054c: jne short 10010588h
0x1001054e: xor edi,edi
0x10010550: cmp [esi+0Ch],edi
0x10010553: jle short 10010580h
0x10010555: push 1
0x10010557: push ebx
0x10010558: push edi
0x10010559: lea eax,[esp+1Ch]
0x1001055d: push eax
0x1001055e: mov ecx,esi
0x10010560: call 100131F0h
0x10010565: mov ecx,[eax]
0x10010567: push 1
0x10010569: push ecx
0x1001056a: push ebx
0x1001056b: push edi
0x1001056c: lea ecx,[esp+3F8h]
0x10010573: call 10013890h
0x10010578: add edi,1
0x1001057b: cmp edi,[esi+0Ch]
0x1001057e: jl short 10010555h
0x10010580: add ebx,1
0x10010583: cmp ebx,[esi+10h]
0x10010586: jl short 10010545h
0x10010588: push 1
0x1001058a: lea edx,[esp+3ECh]
0x10010591: push edx
0x10010592: mov ecx,esi
0x10010594: call 1000E6D0h
0x10010599: lea ecx,[esp+3E8h]
0x100105a0: mov dword ptr [esp+7C4h],0FFFFFFFFh
0x100105ab: mov dword ptr [esp+3E8h],100FE724h
0x100105b6: call 1000E550h
0x100105bb: lea ecx,[esp+3E8h]
0x100105c2: jmp near ptr 100107D3h
0x100105c7: movzx eax,word ptr [esi+16h]
0x100105cb: cmp ax,8
0x100105cf: je near ptr 100107D8h
0x100105d5: ja near ptr 10010752h
0x100105db: push 0
0x100105dd: lea ecx,[esp+1Ch]
0x100105e1: call 1000E480h
0x100105e6: push esi
0x100105e7: lea ecx,[esp+1Ch]
0x100105eb: mov dword ptr [esp+7C8h],1
0x100105f6: call 1000E5C0h
0x100105fb: mov eax,[esi+40h]
0x100105fe: mov ecx,[esi+10h]
0x10010601: mov edx,[esi+0Ch]
0x10010604: push eax
0x10010605: push 8
0x10010607: push ecx
0x10010608: push edx
```

```
0x10010609: lea ecx,[esp+28h]
0x1001060d: call 1000EAD0h
0x10010612: mov ecx,esi
0x10010614: call 1000F010h
0x10010619: push eax
0x1001061a: mov ecx,esi
0x1001061c: call 10012EA0h
0x10010621: push eax
0x10010622: lea ecx,[esp+20h]
0x10010626: call 10013690h
0x1001062b: lea ecx,[esp+18h]
0x1001062f: call 1000F190h
0x10010634: test al,al
0x10010636: lea ecx,[esp+18h]
0x1001063a: jne short 10010657h
0x1001063c: call 1000F2D0h
0x10010641: add esi,44h
0x10010644: mov cl,[eax]
0x10010646: mov [esi],cl
0x10010648: add eax,1
0x1001064b: add esi,1
0x1001064e: test cl,cl
0x10010650: jne short 10010644h
0x10010652: jmp near ptr 1001073Eh
0x10010657: push esi
0x10010658: call 10013E20h
0x1001065d: push esi
0x1001065e: lea ecx,[esp+1Ch]
0x10010662: call 10013A70h
0x10010667: xor ebx,ebx
0x10010669: cmp [esi+10h],ebx
0x1001066c: jle short 100106A5h
0x1001066e: mov edi,edi
0x10010670: cmp dword ptr [esi+148h],0
0x10010677: jne short 100106A5h
0x10010679: xor edi,edi
0x1001067b: cmp [esi+0Ch],edi
0x1001067e: jle short 1001069Dh
0x10010680: push ebx
0x10010681: push edi
0x10010682: mov ecx,esi
0x10010684: call 10013030h
0x10010689: lea ecx,[esp+18h]
0x1001068d: push eax
0x1001068e: push ebx
0x1001068f: push edi
0x10010690: call 10013280h
0x10010695: add edi,1
0x10010698: cmp edi,[esi+0Ch]
0x1001069b: jl short 10010680h
0x1001069d: add ebx,1
0x100106a0: cmp ebx,[esi+10h]
0x100106a3: jl short 10010670h
0x100106a5: push 1
0x100106a7: lea eax,[esp+1Ch]
0x100106ab: push eax
0x100106ac: jmp near ptr 100107ACh
0x100106b1: movzx eax,word ptr [esi+16h]
0x100106b5: cmp ax,4
0x100106b9: je near ptr 100107D8h
0x100106bf: ja near ptr 10010752h
0x100106c5: push 0
0x100106c7: lea ecx,[esp+1Ch]
0x100106cb: call 1000E480h
0x100106d0: push esi
0x100106d1: lea ecx,[esp+1Ch]
0x100106d5: mov dword ptr [esp+7C8h],0
```

```
0x100106e0: call 1000E5C0h
0x100106e5: mov ecx,[esi+40h]
0x100106e8: mov edx,[esi+10h]
0x100106eb: mov eax,[esi+0Ch]
0x100106ee: push ecx
0x100106ef: push 4
0x100106f1: push edx
0x100106f2: push eax
0x100106f3: lea ecx,[esp+28h]
0x100106f7: call 1000EAD0h
0x100106fc: mov ecx,esi
0x100106fe: call 1000F010h
0x10010703: push eax
0x10010704: mov ecx,esi
0x10010706: call 10012EA0h
0x1001070b: push eax
0x1001070c: lea ecx,[esp+20h]
0x10010710: call 10013690h
0x10010715: lea ecx,[esp+18h]
0x10010719: call 1000F190h
0x1001071e: test al,al
0x10010720: lea ecx,[esp+18h]
0x10010724: jne short 10010759h
0x10010726: call 1000F2D0h
0x1001072b: lea edx,[esi+44h]
0x1001072e: mov edi,edi
0x10010730: mov cl,[eax]
0x10010732: mov [edx],cl
0x10010734: add eax,1
0x10010737: add edx,1
0x1001073a: test cl,cl
0x1001073c: jne short 10010730h
0x1001073e: lea ecx,[esp+18h]
0x10010742: mov dword ptr [esp+7C4h],0FFFFFFFFh
0x1001074d: call 1000C970h
0x10010752: xor al,al
0x10010754: jmp near ptr 100107DAh
0x10010759: push esi
0x1001075a: call 10013E20h
0x1001075f: push esi
0x10010760: lea ecx,[esp+1Ch]
0x10010764: call 10013A70h
0x10010769: xor ebx,ebx
0x1001076b: cmp [esi+10h],ebx
0x1001076e: jle short 100107A5h
0x10010770: cmp dword ptr [esi+148h],0
0x10010777: jne short 100107A5h
0x10010779: xor edi,edi
0x1001077b: cmp [esi+0Ch],edi
0x1001077e: jle short 1001079Dh
0x10010780: push ebx
0x10010781: push edi
0x10010782: mov ecx,esi
0x10010784: call 10013030h
0x10010789: lea ecx,[esp+18h]
0x1001078d: push eax
0x1001078e: push ebx
0x1001078f: push edi
0x10010790: call 10013280h
0x10010795: add edi,1
0x10010798: cmp edi,[esi+0Ch]
0x1001079b: jl short 10010780h
0x1001079d: add ebx,1
0x100107a0: cmp ebx,[esi+10h]
0x100107a3: jl short 10010770h
0x100107a5: push 1
0x100107a7: lea ecx,[esp+1Ch]
```

```
0x100107ab: push ecx
0x100107ac: mov ecx,esi
0x100107ae: call 1000E6D0h
0x100107b3: lea ecx,[esp+18h]
0x100107b7: mov dword ptr [esp+7C4h],0FFFFFFFFh
0x100107c2: mov dword ptr [esp+18h],100FE724h
0x100107ca: call 1000E550h
0x100107cf: lea ecx,[esp+18h]
0x100107d3: call 1000E4A0h
0x100107d8: mov al,1
0x100107da: mov ecx,[esp+7BCh]
0x100107e1: mov fs:[0],ecx
0x100107e8: pop ecx
0x100107e9: pop edi
0x100107ea: pop esi
0x100107eb: pop ebx
0x100107ec: mov ecx,[esp+7A8h]
0x100107f3: xor ecx,esp
0x100107f5: call 10022326h
0x100107fa: add esp,7B8h
0x10010800: ret 4
```

`sub_100103d0` is called by sub_1001a5b0. It delegates to sub_1000c970, sub_1000e480, sub_1000e4a0 (23 callees).

12.283 sub_10010810 at 0x10010810 (90 instructions)

```
0x10010810: sub esp,8
0x10010813: push esi
0x10010814: mov esi,ecx
0x10010816: cmp dword ptr [esi+4],0
0x1001081a: jne short 10010825h
0x1001081c: xor al,al
0x1001081e: pop esi
0x1001081f: add esp,8
0x10010822: ret 8
0x10010825: mov eax,[esi+30h]
0x10010828: push ebp
0x10010829: push eax
0x1001082a: call dword ptr ds:[100FE550h]
0x10010830: mov ebp,eax
0x10010832: add esp,4
0x10010835: test ebp,ebp
0x10010837: mov [esp+8],ebp
0x1001083b: jne short 10010847h
0x1001083d: pop ebp
0x1001083e: xor al,al
0x10010840: pop esi
0x10010841: add esp,8
0x10010844: ret 8
0x10010847: mov ecx,[esi+10h]
0x1001084a: push ebx
0x1001084b: sub ecx,1
0x1001084e: push edi
0x1001084f: push ecx
0x10010850: mov ecx,esi
0x10010852: call 1000E5E0h
0x10010857: push 0
0x10010859: mov ecx,esi
0x1001085b: mov edi,eax
0x1001085d: call 1000E5E0h
0x10010862: mov ebx,eax
0x10010864: mov eax,[esi+10h]
0x10010867: cdq
0x10010868: sub eax,edx
0x1001086a: sar eax,1
0x1001086c: test eax,eax
```

```
0x1001086e: mov dword ptr [esp+14h],0
0x10010876: jle short 100108C3h
0x10010878: mov ebp,[esi+30h]
0x1001087b: mov edx,[esp+10h]
0x1001087f: push ebp
0x10010880: push edi
0x10010881: push edx
0x10010882: call 100228C6h
0x10010887: push ebp
0x10010888: push ebx
0x10010889: push edi
0x1001088a: call 100228C6h
0x1001088f: mov eax,[esi+30h]
0x10010892: mov ecx,[esp+28h]
0x10010896: push eax
0x10010897: push ecx
0x10010898: push ebx
0x10010899: call 100228C6h
0x1001089e: mov eax,[esi+30h]
0x100108a1: mov ecx,[esp+38h]
0x100108a5: sub edi,eax
0x100108a7: add ebx,eax
0x100108a9: mov eax,[esi+10h]
0x100108ac: cdq
0x100108ad: sub eax,edx
0x100108af: add ecx,1
0x100108b2: sar eax,1
0x100108b4: add esp,24h
0x100108b7: cmp ecx,eax
0x100108b9: mov [esp+14h],ecx
0x100108bd: jl short 10010878h
0x100108bf: mov ebp,[esp+10h]
0x100108c3: push ebp
0x100108c4: call dword ptr ds:[100FE554h]
0x100108ca: add esp,4
0x100108cd: cmp byte ptr [esp+1Ch],0
0x100108d2: pop edi
0x100108d3: pop ebx
0x100108d4: je short 100108DDh
0x100108d6: mov ecx,esi
0x100108d8: call 10013F20h
0x100108dd: cmp byte ptr [esp+18h],0
0x100108e2: je short 100108EBh
0x100108e4: mov ecx,esi
0x100108e6: call 10013C10h
0x100108eb: pop ebp
0x100108ec: mov al,1
0x100108ee: pop esi
0x100108ef: add esp,8
0x100108f2: ret 8
```

`sub_10010810` is called by sub_10016840, sub_1001c370. It delegates to sub_1000e5e0, sub_10013c10, sub_10013f20 (4 callees).

12.284 sub_100118f0 at 0x100118f0 (173 instructions)

```
0x100118f0: sub esp,8
0x100118f3: push esi
0x100118f4: mov esi,ecx
0x100118f6: cmp dword ptr [esi+4],0
0x100118fa: jne short 10011903h
0x100118fc: xor al,al
0x100118fe: pop esi
0x100118ff: add esp,8
0x10011902: ret
0x10011903: cmp word ptr [esi+16h],8
```

```
0x10011908: push ebx
0x10011909: push edi
0x1001190a: ja near ptr 10011A03h
0x10011910: call 10013730h
0x10011915: test al,al
0x10011917: je near ptr 100119BFh
0x1001191d: cmp dword ptr [esi+3C0h],0
0x10011924: je short 10011997h
0x10011926: mov ebx,[esi+17Ch]
0x1001192c: cmp ebx,[esi+174h]
0x10011932: jge near ptr 10011AF6h
0x10011938: jmp short 10011940h
0x10011940: mov edi,[esi+170h]
0x10011946: cmp edi,[esi+178h]
0x1001194c: jge short 10011983h
0x1001194e: mov edi,edi
0x10011950: push ebx
0x10011951: push edi
0x10011952: mov ecx,esi
0x10011954: call 10013DF0h
0x10011959: test al,al
0x1001195b: je short 10011978h
0x1001195d: push ebx
0x1001195e: push edi
0x1001195f: mov ecx,esi
0x10011961: call 10013030h
0x10011966: or ecx,0FFh
0x1001196c: sub cl,al
0x1001196e: push ecx
0x1001196f: push ebx
0x10011970: push edi
0x10011971: mov ecx,esi
0x10011973: call 10013280h
0x10011978: add edi,1
0x1001197b: cmp edi,[esi+178h]
0x10011981: jl short 10011950h
0x10011983: add ebx,1
0x10011986: cmp ebx,[esi+174h]
0x1001198c: jl short 10011940h
0x1001198e: pop edi
0x1001198f: pop ebx
0x10011990: mov al,1
0x10011992: pop esi
0x10011993: add esp,8
0x10011996: ret
0x10011997: mov eax,[esi+34h]
0x1001199a: xor ecx,ecx
0x1001199c: cmp [esi+1Ch],ecx
0x1001199f: jbe near ptr 10011AF6h
0x100119a5: mov dl,[eax]
0x100119a7: not dl
0x100119a9: mov [eax],dl
0x100119ab: add ecx,1
0x100119ae: add eax,1
0x100119b1: cmp ecx,[esi+1Ch]
0x100119b4: jb short 100119A5h
0x100119b6: pop edi
0x100119b7: pop ebx
0x100119b8: mov al,1
0x100119ba: pop esi
0x100119bb: add esp,8
0x100119be: ret
0x100119bf: mov ecx,esi
0x100119c1: call 10012EA0h
0x100119c6: xor ecx,ecx
0x100119c8: cmp [esi+28h],ecx
0x100119cb: jbe near ptr 10011AF6h
```

```
0x100119d1: add eax,2
0x100119d4: or dl,0FFh
0x100119d7: sub dl,[eax-2]
0x100119da: add ecx,1
0x100119dd: mov [eax-2],dl
0x100119e0: or dl,0FFh
0x100119e3: sub dl,[eax-1]
0x100119e6: add eax,4
0x100119e9: mov [eax-5],dl
0x100119ec: or dl,0FFh
0x100119ef: sub dl,[eax-4]
0x100119f2: mov [eax-4],dl
0x100119f5: cmp ecx,[esi+28h]
0x100119f8: jb short 100119D4h
0x100119fa: pop edi
0x100119fb: pop ebx
0x100119fc: mov al,1
0x100119fe: pop esi
0x100119ff: add esp,8
0x10011a02: ret
0x10011a03: cmp dword ptr [esi+3C0h],0
0x10011a0a: jne short 10011A36h
0x10011a0c: mov eax,[esi+34h]
0x10011a0f: xor ecx,ecx
0x10011a11: cmp [esi+1Ch],ecx
0x10011a14: jbe near ptr 10011ACAh
0x10011a1a: lea ebx,[ebx]
0x10011a20: mov dl,[eax]
0x10011a22: not dl
0x10011a24: mov [eax],dl
0x10011a26: add ecx,1
0x10011a29: add eax,1
0x10011a2c: cmp ecx,[esi+1Ch]
0x10011a2f: jb short 10011A20h
0x10011a31: jmp near ptr 10011ACAh
0x10011a36: mov ebx,[esi+17Ch]
0x10011a3c: cmp ebx,[esi+174h]
0x10011a42: jge near ptr 10011ACAh
0x10011a48: jmp short 10011A50h
0x10011a50: mov edi,[esi+170h]
0x10011a56: cmp edi,[esi+178h]
0x10011a5c: jge short 10011ABFh
0x10011a5e: mov edi,edi
0x10011a60: push ebx
0x10011a61: push edi
0x10011a62: mov ecx,esi
0x10011a64: call 10013DF0h
0x10011a69: test al,al
0x10011a6b: je short 10011AB4h
0x10011a6d: push 1
0x10011a6f: push ebx
0x10011a70: push edi
0x10011a71: lea eax,[esp+1Ch]
0x10011a75: push eax
0x10011a76: mov ecx,esi
0x10011a78: call 100131F0h
0x10011a7d: mov eax,[eax]
0x10011a7f: mov [esp+0Ch],eax
0x10011a83: shr eax,10h
0x10011a86: or cl,0FFh
0x10011a89: sub cl,al
0x10011a8b: or dl,0FFh
0x10011a8e: sub dl,[esp+0Dh]
0x10011a92: or al,0FFh
0x10011a94: sub al,[esp+0Ch]
0x10011a98: mov [esp+0Eh],cl
0x10011a9c: push 0
```

```
0x10011a9e: mov [esp+11h],dl
0x10011aa2: mov [esp+10h],al
0x10011aa6: mov ecx,[esp+10h]
0x10011aaa: push ecx
0x10011aab: push ebx
0x10011aac: push edi
0x10011aad: mov ecx,esi
0x10011aaf: call 10013890h
0x10011ab4: add edi,1
0x10011ab7: cmp edi,[esi+178h]
0x10011abd: jl short 10011A60h
0x10011abf: add ebx,1
0x10011ac2: cmp ebx,[esi+174h]
0x10011ac8: jl short 10011A50h
0x10011aca: or dl,0FFh
0x10011acd: sub dl,[esi+150h]
0x10011ad3: or al,0FFh
0x10011ad5: sub al,[esi+151h]
0x10011adb: or cl,0FFh
0x10011ade: sub cl,[esi+152h]
0x10011ae4: mov [esi+150h],dl
0x10011aea: mov [esi+151h],al
0x10011af0: mov [esi+152h],cl
0x10011af6: pop edi
0x10011af7: pop ebx
0x10011af8: mov al,1
0x10011afa: pop esi
0x10011afb: add esp,8
0x10011afe: ret
```

`sub_100118f0` is called by sub_1001b970. It delegates to sub_10012ea0, sub_10013030, sub_100131f0 (7 callees).

12.285 sub_10011b00 at 0x10011b00 (218 instructions)

```
0x10011b00: push 0FFFFFFFFh
0x10011b02: push 100FC4DBh
0x10011b07: mov eax,fs:[0]
0x10011b0d: push eax
0x10011b0e: sub esp,3F8h
0x10011b14: mov eax,ds:[10136220h]
0x10011b19: xor eax,esp
0x10011b1b: mov [esp+3F4h],eax
0x10011b22: push ebx
0x10011b23: push ebp
0x10011b24: push esi
0x10011b25: push edi
0x10011b26: mov eax,ds:[10136220h]
0x10011b2b: xor eax,esp
0x10011b2d: push eax
0x10011b2e: lea eax,[esp+40Ch]
0x10011b35: mov fs:[0],eax
0x10011b3b: mov eax,[esp+42Ch]
0x10011b42: mov esi,ecx
0x10011b44: cmp dword ptr [esi+4],0
0x10011b48: mov [esp+30h],eax
0x10011b4c: je near ptr 10011CC3h
0x10011b52: mov ecx,[esp+41Ch]
0x10011b59: mov eax,[esi+0Ch]
0x10011b5c: cmp ecx,eax
0x10011b5e: mov edx,ecx
0x10011b60: jl short 10011B64h
0x10011b62: mov edx,eax
0x10011b64: test edx,edx
0x10011b66: jge short 10011B72h
0x10011b68: mov dword ptr [esp+14h],0
0x10011b70: jmp short 10011B7Eh
```

```
0x10011b72: cmp ecx,eax
0x10011b74: mov [esp+14h],ecx
0x10011b78: jl short 10011B7Eh
0x10011b7a: mov [esp+14h],eax
0x10011b7e: mov edx,[esp+424h]
0x10011b85: cmp edx,eax
0x10011b87: mov ecx,edx
0x10011b89: jl short 10011B8Dh
0x10011b8b: mov ecx,eax
0x10011b8d: test ecx,ecx
0x10011b8f: jge short 10011B99h
0x10011b91: xor ebp,ebp
0x10011b93: mov [esp+18h],ebp
0x10011b97: jmp short 10011BABh
0x10011b99: cmp edx,eax
0x10011b9b: jge short 10011BA5h
0x10011b9d: mov [esp+18h],edx
0x10011ba1: mov ebp,edx
0x10011ba3: jmp short 10011BABh
0x10011ba5: mov [esp+18h],eax
0x10011ba9: mov ebp,eax
0x10011bab: mov ecx,[esp+420h]
0x10011bb2: mov eax,[esi+10h]
0x10011bb5: cmp ecx,eax
0x10011bb7: mov edx,ecx
0x10011bb9: jl short 10011BBDh
0x10011bbb: mov edx,eax
0x10011bbd: test edx,edx
0x10011bbf: jge short 10011BC5h
0x10011bc1: xor ecx,ecx
0x10011bc3: jmp short 10011BCBh
0x10011bc5: cmp ecx,eax
0x10011bc7: jl short 10011BCBh
0x10011bc9: mov ecx,eax
0x10011bcb: mov ebx,eax
0x10011bcd: sub ebx,ecx
0x10011bcf: mov ecx,[esp+428h]
0x10011bd6: cmp ecx,eax
0x10011bd8: mov [esp+20h],ebx
0x10011bdc: mov edx,ecx
0x10011bde: jl short 10011BE2h
0x10011be0: mov edx,eax
0x10011be2: test edx,edx
0x10011be4: jge short 10011BEAh
0x10011be6: xor ecx,ecx
0x10011be8: jmp short 10011BF0h
0x10011bea: cmp ecx,eax
0x10011bec: jl short 10011BF0h
0x10011bee: mov ecx,eax
0x10011bf0: sub eax,ecx
0x10011bf2: mov edi,eax
0x10011bf4: mov eax,[esp+14h]
0x10011bf8: cmp eax,ebp
0x10011bfa: mov [esp+24h],edi
0x10011bfe: je near ptr 10011CC3h
0x10011c04: cmp ebx,edi
0x10011c06: je near ptr 10011CC3h
0x10011c0c: cmp eax,ebp
0x10011c0e: jle short 10011C1Ah
0x10011c10: mov [esp+14h],ebp
0x10011c14: mov [esp+18h],eax
0x10011c18: mov ebp,eax
0x10011c1a: cmp ebx,edi
0x10011c1c: jle short 10011C2Ch
0x10011c1e: mov [esp+24h],ebx
0x10011c22: mov [esp+20h],edi
0x10011c26: mov ebx,edi
```

```
0x10011c28: mov edi,[esp+24h]
0x10011c2c: push 0
0x10011c2e: lea ecx,[esp+3Ch]
0x10011c32: call 1000E480h
0x10011c37: push esi
0x10011c38: lea ecx,[esp+3Ch]
0x10011c3c: mov dword ptr [esp+418h],0
0x10011c47: call 1000E5C0h
0x10011c4c: mov ecx,[esi+40h]
0x10011c4f: movzx edx,word ptr [esi+16h]
0x10011c53: push ecx
0x10011c54: push edx
0x10011c55: sub edi,ebx
0x10011c57: mov eax,ebp
0x10011c59: sub eax,[esp+1Ch]
0x10011c5d: push edi
0x10011c5e: push eax
0x10011c5f: lea ecx,[esp+48h]
0x10011c63: mov [esp+3Ch],edi
0x10011c67: mov [esp+44h],eax
0x10011c6b: call 1000EAD0h
0x10011c70: lea ecx,[esp+38h]
0x10011c74: call 1000F190h
0x10011c79: test al,al
0x10011c7b: jne short 10011CEFh
0x10011c7d: lea ecx,[esp+38h]
0x10011c81: call 1000F2D0h
0x10011c86: lea edx,[esi+44h]
0x10011c89: lea esp,[esp]
0x10011c90: mov cl,[eax]
0x10011c92: mov [edx],cl
0x10011c94: add eax,1
0x10011c97: add edx,1
0x10011c9a: test cl,cl
0x10011c9c: jne short 10011C90h
0x10011c9e: lea ecx,[esp+38h]
0x10011ca2: mov dword ptr [esp+414h],0FFFFFFFFh
0x10011cad: mov dword ptr [esp+38h],100FE724h
0x10011cb5: call 1000E550h
0x10011cba: lea ecx,[esp+38h]
0x10011cbe: call 1000E4A0h
0x10011cc3: xor al,al
0x10011cc5: mov ecx,[esp+40Ch]
0x10011ccc: mov fs:[0],ecx
0x10011cd3: pop ecx
0x10011cd4: pop edi
0x10011cd5: pop esi
0x10011cd6: pop ebp
0x10011cd7: pop ebx
0x10011cd8: mov ecx,[esp+3F4h]
0x10011cdf: xor ecx,esp
0x10011ce1: call 10022326h
0x10011ce6: add esp,404h
0x10011cec: ret 14h
0x10011cef: mov eax,[esi+28h]
0x10011cf2: push eax
0x10011cf3: mov ecx,esi
0x10011cf5: call 10012EA0h
0x10011cfa: push eax
0x10011cfb: lea ecx,[esp+40h]
0x10011cff: call 10013690h
0x10011d04: mov ecx,[esi+14Ch]
0x10011d0a: movzx eax,word ptr [esi+16h]
0x10011d0e: mov edx,[esi+150h]
0x10011d14: mov [esp+184h],ecx
0x10011d1b: lea ecx,[eax-1]
0x10011d1e: cmp ecx,17h
```

```
0x10011d21: mov [esp+188h],edx
0x10011d28: ja near ptr 10011E24h
0x10011d2e: movzx ecx,byte ptr [ecx+10011EE0h]
0x10011d35: jmp dword ptr [ecx*4+10011ED4h]
0x10011e24: mov ecx,esi
0x10011e26: call 100139B0h
0x10011e2b: test al,al
0x10011e2d: je short 10011E8Bh
0x10011e2f: lea ecx,[esp+38h]
0x10011e33: call 100139C0h
0x10011e38: lea ecx,[esp+38h]
0x10011e3c: call 100139B0h
0x10011e41: test al,al
0x10011e43: je near ptr 10011C9Eh
0x10011e49: mov edi,[esi+0Ch]
0x10011e4c: mov ebp,[esp+3FCh]
0x10011e53: imul edi,ebx
0x10011e56: add edi,[esi+3C4h]
0x10011e5c: add edi,[esp+14h]
0x10011e60: cmp ebx,[esp+24h]
0x10011e64: jge short 10011E8Bh
0x10011e66: mov ebx,[esp+2Ch]
0x10011e6a: lea ebx,[ebx]
0x10011e70: mov ecx,[esp+34h]
0x10011e74: push ecx
0x10011e75: push edi
0x10011e76: push ebp
0x10011e77: call 100228C6h
0x10011e7c: add ebp,[esp+50h]
0x10011e80: add edi,[esi+0Ch]
0x10011e83: add esp,0Ch
0x10011e86: sub ebx,1
0x10011e89: jne short 10011E70h
0x10011e8b: mov ecx,[esp+30h]
0x10011e8f: test ecx,ecx
0x10011e91: push 1
0x10011e93: je short 10011E9Ch
0x10011e95: lea edx,[esp+3Ch]
0x10011e99: push edx
0x10011e9a: jmp short 10011EA3h
0x10011e9c: lea eax,[esp+3Ch]
0x10011ea0: push eax
0x10011ea1: mov ecx,esi
0x10011ea3: call 1000E6D0h
0x10011ea8: lea ecx,[esp+38h]
0x10011eac: mov dword ptr [esp+414h],0FFFFFFFFh
0x10011eb7: mov dword ptr [esp+38h],100FE724h
0x10011ebf: call 1000E550h
0x10011ec4: lea ecx,[esp+38h]
0x10011ec8: call 1000E4A0h
0x10011ecd: mov al,1
0x10011ecf: jmp near ptr 10011CC5h
```

`sub_10011b00` is called by sub_10011f00. It delegates to sub_1000e480, sub_1000e4a0, sub_1000e550 (14 callees).

12.286 sub_10011f00 at 0x10011f00 (13 instructions)

```
0x10011f00: mov eax,[esp+8]
0x10011f04: push eax
0x10011f05: mov eax,[esp+8]
0x10011f09: mov edx,[eax+0Ch]
0x10011f0c: push edx
0x10011f0d: mov edx,[eax+8]
0x10011f10: push edx
0x10011f11: mov edx,[eax+4]
0x10011f14: mov eax,[eax]
```

```
0x10011f16: push edx
0x10011f17: push eax
0x10011f18: call 10011B00h
0x10011f1d: ret 8
```

`sub_10011f00` is called by sub_1000cc10. It delegates to sub_10011b00 (1 callees).

12.287 sub_10011f20 at 0x10011f20 (16 instructions)

```
0x10011f20: push esi
0x10011f21: mov esi,ecx
0x10011f23: mov ecx,[esp+8]
0x10011f27: test ecx,ecx
0x10011f29: mov [esi+14h],ecx
0x10011f2c: je short 10011F38h
0x10011f2e: push 0
0x10011f30: call 1000E5E0h
0x10011f35: mov [esi+10h],eax
0x10011f38: mov dword ptr [esi+4],0
0x10011f3f: mov dword ptr [esi],0
0x10011f45: mov dword ptr [esi+0Ch],0
0x10011f4c: mov dword ptr [esi+8],0
0x10011f53: mov eax,esi
0x10011f55: pop esi
0x10011f56: ret 4
```

`sub_10011f20` is called by sub_100171e0, sub_1001a5b0. It delegates to sub_1000e5e0 (1 callees).

12.288 sub_10011fb0 at 0x10011fb0 (1058 instructions)

```
0x10011fb0: push ebp
0x10011fb1: mov ebp,esp
0x10011fb3: and esp,0FFFFFFC0h
0x10011fb6: sub esp,0F4h
0x10011fbc: push ebx
0x10011fbd: push esi
0x10011fbe: push edi
0x10011fbf: mov edi,ecx
0x10011fc1: cmp dword ptr [edi+4],0
0x10011fc5: je near ptr 10012DEFh
0x10011fcb: mov esi,[ebp+8]
0x10011fce: test esi,esi
0x10011fd0: je near ptr 10012DEFh
0x10011fd6: mov eax,[ebp+14h]
0x10011fd9: test eax,eax
0x10011fdb: je near ptr 10012DEFh
0x10011fe1: mov ecx,[ebp+18h]
0x10011fe4: test ecx,ecx
0x10011fe6: je near ptr 10012DEFh
0x10011fec: cmp byte ptr [edi+182h],0
0x10011ff3: je near ptr 10012DEFh
0x10011ff9: test eax,eax
0x10011ffb: jge short 10012003h
0x10011ffd: mov eax,[edi+0Ch]
0x10012000: mov [ebp+14h],eax
0x10012003: test ecx,ecx
0x10012005: jge short 1001200Dh
0x10012007: mov ecx,[edi+10h]
0x1001200a: mov [ebp+18h],ecx
0x1001200d: cmp dword ptr [edi+14Ch],0
0x10012014: push esi
0x10012015: setge byte ptr [esp+33h]
0x1001201a: cmp dword ptr [edi+3C4h],0
0x10012021: setne byte ptr [esp+32h]
0x10012026: call dword ptr ds:[100FE008h]
```

```
0x1001202c: test eax,eax
0x1001202e: mov [esp+0D8h],eax
0x10012035: je near ptr 10012DEFh
0x1001203b: mov ebx,[ebp+1Ch]
0x1001203e: test ebx,ebx
0x10012040: je short 1001206Bh
0x10012042: lea edx,[esp+0F0h]
0x10012049: push edx
0x1001204a: push esi
0x1001204b: call dword ptr ds:[100FE00Ch]
0x10012051: push ebx
0x10012052: call dword ptr ds:[100FE050h]
0x10012058: push 1
0x1001205a: mov ebx,eax
0x1001205c: push ebx
0x1001205d: push esi
0x1001205e: call dword ptr ds:[100FE054h]
0x10012064: push ebx
0x10012065: call dword ptr ds:[100FE048h]
0x1001206b: lea eax,[esp+0C8h]
0x10012072: push eax
0x10012073: push esi
0x10012074: call dword ptr ds:[100FE00Ch]
0x1001207a: mov ecx,[esp+0CCh]
0x10012081: mov edx,[ebp+10h]
0x10012084: cmp ecx,edx
0x10012086: mov eax,ecx
0x10012088: jg short 1001208Ch
0x1001208a: mov eax,edx
0x1001208c: mov ebx,[esp+0D4h]
0x10012093: cmp ebx,eax
0x10012095: jge short 100120A0h
0x10012097: mov [esp+84h],ebx
0x1001209e: jmp short 100120B2h
0x100120a0: cmp ecx,edx
0x100120a2: mov [esp+84h],ecx
0x100120a9: jg short 100120B2h
0x100120ab: mov [esp+84h],edx
0x100120b2: mov ebx,[esp+0C8h]
0x100120b9: mov ecx,[ebp+0Ch]
0x100120bc: cmp ebx,ecx
0x100120be: mov edx,ebx
0x100120c0: jg short 100120C4h
0x100120c2: mov edx,ecx
0x100120c4: mov eax,[esp+0D0h]
0x100120cb: cmp eax,edx
0x100120cd: jge short 100120D8h
0x100120cf: mov [esp+80h],eax
0x100120d6: jmp short 100120EAh
0x100120d8: cmp ebx,ecx
0x100120da: mov [esp+80h],ebx
0x100120e1: jg short 100120EAh
0x100120e3: mov [esp+80h],ecx
0x100120ea: mov edx,[ebp+14h]
0x100120ed: add ecx,edx
0x100120ef: cmp eax,ecx
0x100120f1: mov edx,eax
0x100120f3: jl short 100120F7h
0x100120f5: mov edx,ecx
0x100120f7: cmp [esp+0C8h],edx
0x100120fe: jle short 10012109h
0x10012100: mov eax,[esp+0C8h]
0x10012107: jmp short 1001210Fh
0x10012109: cmp eax,ecx
0x1001210b: jl short 1001210Fh
0x1001210d: mov eax,ecx
0x1001210f: mov edx,[ebp+10h]
```

```
0x10012112: mov ecx,[ebp+18h]
0x10012115: mov ebx,[esp+0D4h]
0x1001211c: add ecx,edx
0x1001211e: cmp ebx,ecx
0x10012120: mov [esp+58h],ecx
0x10012124: mov edx,ebx
0x10012126: jl short 1001212Ah
0x10012128: mov edx,ecx
0x1001212a: cmp [esp+0CCh],edx
0x10012131: jle short 1001213Ch
0x10012133: mov ebx,[esp+0CCh]
0x1001213a: jmp short 10012144h
0x1001213c: cmp ebx,ecx
0x1001213e: jl short 10012144h
0x10012140: mov ebx,[esp+58h]
0x10012144: sub eax,[esp+80h]
0x1001214b: mov [esp+3Ch],eax
0x1001214f: mov eax,ebx
0x10012151: sub eax,[esp+84h]
0x10012158: cmp byte ptr [esp+2Fh],0
0x1001215d: mov [esp+40h],eax
0x10012161: jne near ptr 1001258Dh
0x10012167: cmp byte ptr [esp+2Eh],0
0x1001216c: jne near ptr 1001258Dh
0x10012172: cmp byte ptr [edi+181h],0
0x10012179: jne near ptr 1001258Dh
0x1001217f: mov eax,[ebp+14h]
0x10012182: cmp eax,[edi+0Ch]
0x10012185: jne short 10012205h
0x10012187: mov ecx,[ebp+18h]
0x1001218a: cmp ecx,[edi+10h]
0x1001218d: jne short 10012205h
0x1001218f: push 3
0x10012191: push esi
0x10012192: call dword ptr ds:[100FE010h]
0x10012198: cmp byte ptr [ebp+24h],0
0x1001219c: je short 100121D8h
0x1001219e: mov edx,[edi+4]
0x100121a1: mov eax,[edi+34h]
0x100121a4: mov ecx,[ebp+14h]
0x100121a7: push 0CC0020h
0x100121ac: push 0
0x100121ae: push edx
0x100121af: mov edx,[ebp+0Ch]
0x100121b2: push eax
0x100121b3: mov eax,[ebp+18h]
0x100121b6: push eax
0x100121b7: push ecx
0x100121b8: push 0
0x100121ba: push 0
0x100121bc: neg eax
0x100121be: push eax
0x100121bf: push ecx
0x100121c0: mov ecx,[esp+80h]
0x100121c7: add ecx,0FFFFFFFFh
0x100121ca: push ecx
0x100121cb: push edx
0x100121cc: push esi
0x100121cd: call dword ptr ds:[100FE014h]
0x100121d3: jmp near ptr 10012DA5h
0x100121d8: mov eax,[edi+4]
0x100121db: mov ecx,[edi+34h]
0x100121de: mov edx,[ebp+14h]
0x100121e1: push 0
0x100121e3: push eax
0x100121e4: mov eax,[ebp+18h]
0x100121e7: push ecx
```

```
0x100121e8: mov ecx,[ebp+0Ch]
0x100121eb: push eax
0x100121ec: push 0
0x100121ee: push 0
0x100121f0: push 0
0x100121f2: push eax
0x100121f3: mov eax,[ebp+10h]
0x100121f6: push edx
0x100121f7: push eax
0x100121f8: push ecx
0x100121f9: push esi
0x100121fa: call dword ptr ds:[100FE018h]
0x10012200: jmp near ptr 10012DA5h
0x10012205: mov edx,[esp+3Ch]
0x10012209: xor eax,eax
0x1001220b: mov [esp+0A4h],eax
0x10012212: mov [esp+98h],eax
0x10012219: mov [esp+9Ch],eax
0x10012220: mov [esp+0A0h],eax
0x10012227: mov [esp+0A8h],eax
0x1001222e: mov [esp+0ACh],eax
0x10012235: mov [esp+0B0h],eax
0x1001223c: mov [esp+0B4h],eax
0x10012243: mov [esp+0B8h],eax
0x1001224a: mov [esp+0BCh],eax
0x10012251: mov eax,[esp+40h]
0x10012255: push esi
0x10012256: mov dword ptr [esp+9Ch],28h
0x10012261: mov [esp+0A0h],edx
0x10012268: mov [esp+0A4h],eax
0x1001226f: mov word ptr [esp+0A8h],1
0x10012279: mov word ptr [esp+0AAh],18h
0x10012283: call dword ptr ds:[100FE060h]
0x10012289: push 0
0x1001228b: push 0
0x1001228d: lea ecx,[esp+38h]
0x10012291: push ecx
0x10012292: push 0
0x10012294: lea edx,[esp+0A8h]
0x1001229b: push edx
0x1001229c: push esi
0x1001229d: mov [esp+80h],eax
0x100122a4: call dword ptr ds:[100FE028h]
0x100122aa: push eax
0x100122ab: mov eax,[esp+6Ch]
0x100122af: push eax
0x100122b0: call dword ptr ds:[100FE030h]
0x100122b6: cmp dword ptr [esp+30h],0
0x100122bb: mov [esp+70h],eax
0x100122bf: je near ptr 10012545h
0x100122c5: fild dword ptr [edi+0Ch]
0x100122c8: mov eax,[esp+3Ch]
0x100122cc: lea eax,[eax+eax*2]
0x100122cf: lea eax,[eax*8+1Fh]
0x100122d6: fidiv dword ptr [ebp+14h]
0x100122d9: cdq
0x100122da: and edx,1Fh
0x100122dd: add eax,edx
0x100122df: sar eax,5
0x100122e2: add eax,eax
0x100122e4: add eax,eax
0x100122e6: cmp dword ptr [esp+40h],0
0x100122eb: mov [esp+4Ch],eax
0x100122ef: fstp dword ptr [esp+58h]
0x100122f3: fild dword ptr [edi+10h]
0x100122f6: fidiv dword ptr [ebp+18h]
0x100122f9: fstp dword ptr [esp+54h]
```

```
0x100122fd: jle near ptr 10012545h
0x10012303: mov ecx,[esp+40h]
0x10012307: sub ebx,[ebp+10h]
0x1001230a: mov edx,eax
0x1001230c: neg eax
0x1001230e: mov [esp+7Ch],eax
0x10012312: lea eax,[ecx-1]
0x10012315: imul eax,edx
0x10012318: mov dword ptr [esp+44h],0
0x10012320: mov [esp+48h],ebx
0x10012324: mov [esp+38h],eax
0x10012328: mov [esp+50h],ecx
0x1001232c: fild dword ptr [edi+10h]
0x1001232f: push ecx
0x10012330: fild dword ptr [esp+4Ch]
0x10012334: fmul dword ptr [esp+58h]
0x10012338: fsubp
0x1001233a: fstp dword ptr [esp+64h]
0x1001233e: fld dword ptr [esp+64h]
0x10012342: fstp dword ptr [esp]
0x10012345: call 100103B0h
0x1001234a: add esp,4
0x1001234d: call 100228D0h
0x10012352: xor ecx,ecx
0x10012354: test eax,eax
0x10012356: setl cl
0x10012359: sub ecx,1
0x1001235c: and ecx,eax
0x1001235e: mov eax,[edi+30h]
0x10012361: imul eax,ecx
0x10012364: add eax,[edi+34h]
0x10012367: cmp byte ptr [ebp+24h],0
0x1001236b: mov [esp+78h],ecx
0x1001236f: mov [esp+74h],eax
0x10012373: je short 10012382h
0x10012375: mov ecx,[esp+38h]
0x10012379: mov edx,[esp+30h]
0x1001237d: lea esi,[ecx+edx]
0x10012380: jmp short 1001238Dh
0x10012382: mov eax,[esp+44h]
0x10012386: mov ecx,[esp+30h]
0x1001238a: lea esi,[eax+ecx]
0x1001238d: xor ebx,ebx
0x1001238f: cmp [esp+3Ch],ebx
0x10012393: jle near ptr 10012522h
0x10012399: mov eax,[esp+80h]
0x100123a0: sub eax,[ebp+0Ch]
0x100123a3: mov [esp+6Ch],eax
0x100123a7: jmp short 100123B0h
0x100123b0: mov edx,[esp+6Ch]
0x100123b4: add edx,ebx
0x100123b6: mov [esp+34h],edx
0x100123ba: fild dword ptr [esp+34h]
0x100123be: push ecx
0x100123bf: fmul dword ptr [esp+5Ch]
0x100123c3: fstp dword ptr [esp+60h]
0x100123c7: fld dword ptr [esp+60h]
0x100123cb: fstp dword ptr [esp]
0x100123ce: call 100103B0h
0x100123d3: add esp,4
0x100123d6: call 100228D0h
0x100123db: xor ecx,ecx
0x100123dd: test eax,eax
0x100123df: setl cl
0x100123e2: sub ecx,1
0x100123e5: and ecx,eax
0x100123e7: cmp byte ptr [ebp+20h],0
```

```
0x100123eb: je near ptr 100124B2h
0x100123f1: fld1
0x100123f3: fld dword ptr [esp+58h]
0x100123f7: fcom
0x100123f9: fnstsw ax
0x100123fb: test ah,41h
0x100123fe: jne short 10012465h
0x10012400: fld dword ptr [esp+54h]
0x10012404: fcom st(2)
0x10012406: fnstsw ax
0x10012408: fstp st(2)
0x1001240a: test ah,41h
0x1001240d: jne short 10012465h
0x1001240f: push 0
0x10012411: fxch
0x10012413: push 5
0x10012415: push 2
0x10012417: sub esp,10h
0x1001241a: fstp dword ptr [esp+0Ch]
0x1001241e: lea eax,[esp+0E0h]
0x10012425: mov ecx,edi
0x10012427: fstp dword ptr [esp+8]
0x1001242b: fld dword ptr [esp+7Ch]
0x1001242f: fld qword ptr ds:[100FE7C8h]
0x10012435: fsub st(1),st
0x10012437: fxch
0x10012439: fstp dword ptr [esp+50h]
0x1001243d: fld dword ptr [esp+50h]
0x10012441: fstp dword ptr [esp+4]
0x10012445: fsubr dword ptr [esp+78h]
0x10012449: fstp dword ptr [esp+50h]
0x1001244d: fld dword ptr [esp+50h]
0x10012451: fstp dword ptr [esp]
0x10012454: push eax
0x10012455: call 1001FD10h
0x1001245a: mov ecx,[eax]
0x1001245c: mov [esp+24h],ecx
0x10012460: jmp near ptr 100124FAh
0x10012465: fstp st(1)
0x10012467: push 0
0x10012469: fstp st(0)
0x1001246b: push 5
0x1001246d: fld dword ptr [esp+68h]
0x10012471: push 2
0x10012473: fld qword ptr ds:[100FE7C8h]
0x10012479: sub esp,8
0x1001247c: fsub st(1),st
0x1001247e: lea edx,[esp+0A4h]
0x10012485: fxch
0x10012487: mov ecx,edi
0x10012489: fstp dword ptr [esp+48h]
0x1001248d: fld dword ptr [esp+48h]
0x10012491: fstp dword ptr [esp+4]
0x10012495: fsubr dword ptr [esp+70h]
0x10012499: fstp dword ptr [esp+48h]
0x1001249d: fld dword ptr [esp+48h]
0x100124a1: fstp dword ptr [esp]
0x100124a4: push edx
0x100124a5: call 1001DDC0h
0x100124aa: mov eax,[eax]
0x100124ac: mov [esp+24h],eax
0x100124b0: jmp short 100124FAh
0x100124b2: cmp dword ptr [edi+28h],0
0x100124b6: je short 100124DAh
0x100124b8: mov edx,[esp+78h]
0x100124bc: push edx
0x100124bd: push ecx
```

```
0x100124be: mov ecx,edi
0x100124c0: call 10012F20h
0x100124c5: mov ecx,edi
0x100124c7: push eax
0x100124c8: lea eax,[esp+2Ch]
0x100124cc: push eax
0x100124cd: call 10012EC0h
0x100124d2: mov ecx,[eax]
0x100124d4: mov [esp+24h],ecx
0x100124d8: jmp short 100124FAh
0x100124da: mov edx,[esp+74h]
0x100124de: lea eax,[edx+ecx*2]
0x100124e1: add eax,ecx
0x100124e3: mov cl,[eax]
0x100124e5: mov dl,[eax+1]
0x100124e8: add eax,1
0x100124eb: mov al,[eax+1]
0x100124ee: mov [esp+24h],cl
0x100124f2: mov [esp+25h],dl
0x100124f6: mov [esp+26h],al
0x100124fa: mov cl,[esp+24h]
0x100124fe: mov dl,[esp+25h]
0x10012502: mov al,[esp+26h]
0x10012506: mov [esi],cl
0x10012508: add esi,1
0x1001250b: mov [esi],dl
0x1001250d: add esi,1
0x10012510: mov [esi],al
0x10012512: add ebx,1
0x10012515: add esi,1
0x10012518: cmp ebx,[esp+3Ch]
0x1001251c: jl near ptr 100123B0h
0x10012522: mov ecx,[esp+7Ch]
0x10012526: mov edx,[esp+4Ch]
0x1001252a: add [esp+38h],ecx
0x1001252e: add [esp+44h],edx
0x10012532: mov eax,1
0x10012537: sub [esp+48h],eax
0x1001253b: sub [esp+50h],eax
0x1001253f: jne near ptr 1001232Ch
0x10012545: mov ecx,[esp+30h]
0x10012549: mov edx,[esp+3Ch]
0x1001254d: push 0
0x1001254f: lea eax,[esp+9Ch]
0x10012556: push eax
0x10012557: mov eax,[esp+48h]
0x1001255b: push ecx
0x1001255c: mov ecx,[esp+8Ch]
0x10012563: push eax
0x10012564: push 0
0x10012566: push 0
0x10012568: push 0
0x1001256a: push eax
0x1001256b: mov eax,[esp+0A4h]
0x10012572: push edx
0x10012573: mov edx,[ebp+8]
0x10012576: push eax
0x10012577: push ecx
0x10012578: push edx
0x10012579: call dword ptr ds:[100FE018h]
0x1001257f: mov eax,[esp+70h]
0x10012583: mov esi,[esp+68h]
0x10012587: push eax
0x10012588: jmp near ptr 10012D90h
0x1001258d: lea ecx,[esp+78h]
0x10012591: push ecx
0x10012592: mov ecx,edi
```

```
0x10012594: mov byte ptr [esp+28h],0
0x10012599: mov byte ptr [esp+29h],0
0x1001259e: mov byte ptr [esp+2Ah],0
0x100125a3: mov byte ptr [esp+2Bh],0
0x100125a8: call 1000EE60h
0x100125ad: mov ecx,edi
0x100125af: call 1000EEA0h
0x100125b4: mov edx,[esp+3Ch]
0x100125b8: mov [esp+74h],eax
0x100125bc: xor eax,eax
0x100125be: mov [esp+0A4h],eax
0x100125c5: mov [esp+98h],eax
0x100125cc: mov [esp+9Ch],eax
0x100125d3: mov [esp+0A0h],eax
0x100125da: mov [esp+5Ch],eax
0x100125de: mov [esp+0A8h],eax
0x100125e5: mov [esp+0ACh],eax
0x100125ec: mov [esp+0B0h],eax
0x100125f3: mov [esp+0B4h],eax
0x100125fa: mov [esp+0B8h],eax
0x10012601: mov [esp+0BCh],eax
0x10012608: mov eax,[esp+40h]
0x1001260c: push esi
0x1001260d: mov dword ptr [esp+9Ch],28h
0x10012618: mov [esp+0A0h],edx
0x1001261f: mov [esp+0A4h],eax
0x10012626: mov word ptr [esp+0A8h],1
0x10012630: mov word ptr [esp+0AAh],18h
0x1001263a: call dword ptr ds:[100FE060h]
0x10012640: push 0
0x10012642: push 0
0x10012644: lea ecx,[esp+50h]
0x10012648: push ecx
0x10012649: push 0
0x1001264b: lea edx,[esp+0A8h]
0x10012652: push edx
0x10012653: push esi
0x10012654: mov [esp+70h],eax
0x10012658: call dword ptr ds:[100FE028h]
0x1001265e: push eax
0x1001265f: mov eax,[esp+5Ch]
0x10012663: push eax
0x10012664: call dword ptr ds:[100FE030h]
0x1001266a: mov ecx,[esp+84h]
0x10012671: mov edx,[esp+80h]
0x10012678: push 0CC0020h
0x1001267d: push ecx
0x1001267e: mov ecx,[esp+44h]
0x10012682: push edx
0x10012683: mov edx,[esp+64h]
0x10012687: push esi
0x10012688: mov [esp+0D4h],eax
0x1001268f: mov eax,[esp+50h]
0x10012693: push eax
0x10012694: push ecx
0x10012695: push 0
0x10012697: push 0
0x10012699: push edx
0x1001269a: call dword ptr ds:[100FE06Ch]
0x100126a0: cmp dword ptr [esp+48h],0
0x100126a5: je near ptr 10012D4Ah
0x100126ab: mov eax,[esp+3Ch]
0x100126af: lea eax,[eax+eax*2]
0x100126b2: lea eax,[eax*8+1Fh]
0x100126b9: cdq
0x100126ba: and edx,1Fh
0x100126bd: add eax,edx
```

```
0x100126bf: sar eax,5
0x100126c2: add eax,eax
0x100126c4: add eax,eax
0x100126c6: mov [esp+60h],eax
0x100126ca: mov eax,[edi+0Ch]
0x100126cd: cmp [ebp+14h],eax
0x100126d0: mov [esp+34h],eax
0x100126d4: jne near ptr 10012905h
0x100126da: mov eax,[edi+10h]
0x100126dd: cmp [ebp+18h],eax
0x100126e0: jne near ptr 10012905h
0x100126e6: sub eax,ebx
0x100126e8: add eax,[ebp+10h]
0x100126eb: cmp dword ptr [esp+40h],0
0x100126f0: mov [esp+38h],eax
0x100126f4: jle near ptr 10012D4Ah
0x100126fa: mov eax,[esp+80h]
0x10012701: sub eax,[ebp+0Ch]
0x10012704: mov ecx,[esp+60h]
0x10012708: mov edx,[esp+40h]
0x1001270c: mov [esp+6Ch],eax
0x10012710: mov eax,ecx
0x10012712: neg eax
0x10012714: mov [esp+68h],eax
0x10012718: lea eax,[edx-1]
0x1001271b: imul eax,ecx
0x1001271e: mov dword ptr [esp+64h],0
0x10012726: mov [esp+44h],eax
0x1001272a: mov [esp+50h],edx
0x1001272e: mov eax,[edi+0Ch]
0x10012731: mov edx,[esp+38h]
0x10012735: mov ecx,[esp+6Ch]
0x10012739: imul eax,edx
0x1001273c: cmp byte ptr [ebp+24h],0
0x10012740: mov esi,[esp+48h]
0x10012744: mov [esp+94h],eax
0x1001274b: mov eax,[esp+44h]
0x1001274f: mov [esp+30h],ecx
0x10012753: jne short 10012759h
0x10012755: mov eax,[esp+64h]
0x10012759: add esi,eax
0x1001275b: mov eax,[edi+30h]
0x1001275e: imul eax,edx
0x10012761: lea ecx,[ecx+ecx*2]
0x10012764: add eax,ecx
0x10012766: add eax,[edi+34h]
0x10012769: cmp dword ptr [esp+3Ch],0
0x1001276e: mov [esp+54h],eax
0x10012772: jle near ptr 100128DDh
0x10012778: mov edx,[esp+3Ch]
0x1001277c: mov [esp+4Ch],edx
0x10012780: cmp byte ptr [esp+2Eh],0
0x10012785: je short 1001279Dh
0x10012787: mov eax,[edi+3C4h]
0x1001278d: add eax,[esp+30h]
0x10012791: mov ecx,[esp+94h]
0x10012798: mov al,[eax+ecx]
0x1001279b: jmp short 1001279Fh
0x1001279d: or al,0FFh
0x1001279f: movzx ebx,byte ptr [edi+180h]
0x100127a6: add ebx,1
0x100127a9: movzx edx,al
0x100127ac: imul ebx,edx
0x100127af: sar ebx,8
0x100127b2: cmp dword ptr [edi+28h],0
0x100127b6: je short 10012804h
0x100127b8: mov eax,[esp+38h]
```

```
0x100127bc: mov ecx,[esp+30h]
0x100127c0: push eax
0x100127c1: push ecx
0x100127c2: mov ecx,edi
0x100127c4: call 10012F20h
0x100127c9: movzx eax,al
0x100127cc: push eax
0x100127cd: lea edx,[esp+94h]
0x100127d4: push edx
0x100127d5: mov ecx,edi
0x100127d7: mov [esp+64h],eax
0x100127db: call 10012EC0h
0x100127e0: cmp byte ptr [edi+181h],0
0x100127e7: mov eax,[eax]
0x100127e9: mov [esp+24h],eax
0x100127ed: je short 1001282Eh
0x100127ef: mov ecx,eax
0x100127f1: shr ecx,18h
0x100127f4: movzx edx,bl
0x100127f7: add ecx,1
0x100127fa: imul ecx,edx
0x100127fd: sar ecx,8
0x10012800: mov bl,cl
0x10012802: jmp short 1001282Eh
0x10012804: mov eax,[esp+54h]
0x10012808: movzx ecx,byte ptr [eax]
0x1001280b: mov dl,[eax+1]
0x1001280e: add eax,1
0x10012811: add eax,1
0x10012814: mov [esp+24h],cl
0x10012818: movzx ecx,byte ptr [eax]
0x1001281b: add eax,1
0x1001281e: mov [esp+25h],dl
0x10012822: mov [esp+26h],cl
0x10012826: mov [esp+54h],eax
0x1001282a: mov eax,[esp+24h]
0x1001282e: cmp dword ptr [edi+28h],0
0x10012832: je short 1001283Eh
0x10012834: mov edx,[esp+74h]
0x10012838: cmp [esp+5Ch],edx
0x1001283c: jmp short 10012842h
0x1001283e: cmp eax,[esp+78h]
0x10012842: jne short 1001284Bh
0x10012844: cmp byte ptr [esp+2Fh],0
0x10012849: jne short 100128C7h
0x1001284b: test bl,bl
0x1001284d: je short 100128C7h
0x1001284f: cmp bl,0FFh
0x10012852: jne short 10012871h
0x10012854: mov al,[esp+24h]
0x10012858: mov cl,[esp+25h]
0x1001285c: mov dl,[esp+26h]
0x10012860: mov [esi],al
0x10012862: add esi,1
0x10012865: mov [esi],cl
0x10012867: add esi,1
0x1001286a: mov [esi],dl
0x1001286c: add esi,1
0x1001286f: jmp short 100128CAh
0x10012871: movzx edx,byte ptr [esp+24h]
0x10012876: mov al,bl
0x10012878: movzx ecx,bl
0x1001287b: movzx ebx,byte ptr [esi]
0x1001287e: imul edx,ecx
0x10012881: not al
0x10012883: movzx eax,al
0x10012886: imul ebx,eax
```

```
0x10012889: add edx,ebx
0x1001288b: sar edx,8
0x1001288e: mov [esi],dl
0x10012890: movzx ebx,byte ptr [esi+1]
0x10012894: movzx edx,byte ptr [esp+25h]
0x10012899: imul ebx,eax
0x1001289c: add esi,1
0x1001289f: imul edx,ecx
0x100128a2: add edx,ebx
0x100128a4: sar edx,8
0x100128a7: mov [esi],dl
0x100128a9: movzx edx,byte ptr [esp+26h]
0x100128ae: imul edx,ecx
0x100128b1: movzx ecx,byte ptr [esi+1]
0x100128b5: add esi,1
0x100128b8: imul ecx,eax
0x100128bb: add edx,ecx
0x100128bd: sar edx,8
0x100128c0: mov [esi],dl
0x100128c2: add esi,1
0x100128c5: jmp short 100128CAh
0x100128c7: add esi,3
0x100128ca: mov eax,1
0x100128cf: add [esp+30h],eax
0x100128d3: sub [esp+4Ch],eax
0x100128d7: jne near ptr 10012780h
0x100128dd: mov eax,[esp+60h]
0x100128e1: add [esp+64h],eax
0x100128e5: mov edx,[esp+68h]
0x100128e9: add [esp+44h],edx
0x100128ed: mov eax,1
0x100128f2: add [esp+38h],eax
0x100128f6: sub [esp+50h],eax
0x100128fa: jne near ptr 1001272Eh
0x10012900: jmp near ptr 10012D4Ah
0x10012905: cmp dword ptr [esp+40h],0
0x1001290a: fild dword ptr [esp+34h]
0x1001290e: fidiv dword ptr [ebp+14h]
0x10012911: fstp dword ptr [esp+64h]
0x10012915: fild dword ptr [edi+10h]
0x10012918: fidiv dword ptr [ebp+18h]
0x1001291b: fstp dword ptr [esp+44h]
0x1001291f: jle near ptr 10012D4Ah
0x10012925: mov ecx,[esp+60h]
0x10012929: mov edx,[esp+40h]
0x1001292d: sub ebx,[ebp+10h]
0x10012930: mov eax,ecx
0x10012932: neg eax
0x10012934: mov [esp+68h],eax
0x10012938: lea eax,[edx-1]
0x1001293b: imul eax,ecx
0x1001293e: mov dword ptr [esp+54h],0
0x10012946: mov [esp+4Ch],ebx
0x1001294a: mov [esp+50h],eax
0x1001294e: mov [esp+34h],edx
0x10012952: fild dword ptr [edi+10h]
0x10012955: push ecx
0x10012956: fild dword ptr [esp+50h]
0x1001295a: fmul dword ptr [esp+48h]
0x1001295e: fsubp
0x10012960: fstp dword ptr [esp+3Ch]
0x10012964: fld dword ptr [esp+3Ch]
0x10012968: fstp dword ptr [esp]
0x1001296b: call 100103B0h
0x10012970: add esp,4
0x10012973: call 100228D0h
0x10012978: xor ecx,ecx
```

```
0x1001297a: test eax,eax
0x1001297c: mov edx,[esp+50h]
0x10012980: setl cl
0x10012983: sub ecx,1
0x10012986: and ecx,eax
0x10012988: mov eax,[edi+0Ch]
0x1001298b: imul eax,ecx
0x1001298e: cmp byte ptr [ebp+24h],0
0x10012992: mov [esp+94h],eax
0x10012999: mov eax,[esp+48h]
0x1001299d: mov [esp+7Ch],ecx
0x100129a1: jne short 100129A7h
0x100129a3: mov edx,[esp+54h]
0x100129a7: lea esi,[edx+eax]
0x100129aa: mov eax,[edi+30h]
0x100129ad: imul eax,ecx
0x100129b0: add eax,[edi+34h]
0x100129b3: cmp dword ptr [esp+3Ch],0
0x100129b8: mov [esp+90h],eax
0x100129bf: mov dword ptr [esp+70h],0
0x100129c7: jle near ptr 10012D27h
0x100129cd: mov eax,[esp+80h]
0x100129d4: sub eax,[ebp+0Ch]
0x100129d7: mov [esp+6Ch],eax
0x100129db: jmp short 100129E0h
0x100129e0: mov edx,[esp+70h]
0x100129e4: mov ecx,[esp+6Ch]
0x100129e8: add ecx,edx
0x100129ea: mov [esp+28h],ecx
0x100129ee: fild dword ptr [esp+28h]
0x100129f2: sub esp,8
0x100129f5: fmul dword ptr [esp+6Ch]
0x100129f9: fstp dword ptr [esp+38h]
0x100129fd: fld dword ptr [esp+38h]
0x10012a01: fstp qword ptr [esp]
0x10012a04: call 1002297Ch
0x10012a09: fstp dword ptr [esp+30h]
0x10012a0d: add esp,8
0x10012a10: fld dword ptr [esp+28h]
0x10012a14: call 100228D0h
0x10012a19: xor ecx,ecx
0x10012a1b: test eax,eax
0x10012a1d: setl cl
0x10012a20: sub ecx,1
0x10012a23: and eax,ecx
0x10012a25: cmp byte ptr [esp+2Eh],0
0x10012a2a: mov [esp+28h],eax
0x10012a2e: je short 10012A44h
0x10012a30: mov edx,[edi+3C4h]
0x10012a36: add edx,eax
0x10012a38: mov eax,[esp+94h]
0x10012a3f: mov al,[edx+eax]
0x10012a42: jmp short 10012A46h
0x10012a44: or al,0FFh
0x10012a46: movzx ebx,byte ptr [edi+180h]
0x10012a4d: add ebx,1
0x10012a50: movzx ecx,al
0x10012a53: imul ebx,ecx
0x10012a56: sar ebx,8
0x10012a59: cmp dword ptr [edi+28h],0
0x10012a5d: je near ptr 10012B83h
0x10012a63: mov edx,[esp+7Ch]
0x10012a67: mov eax,[esp+28h]
0x10012a6b: push edx
0x10012a6c: push eax
0x10012a6d: mov ecx,edi
0x10012a6f: call 10012F20h
```

```
0x10012a74: cmp byte ptr [ebp+20h],0
0x10012a78: movzx ecx,al
0x10012a7b: mov [esp+5Ch],ecx
0x10012a7f: je near ptr 10012B37h
0x10012a85: fld1
0x10012a87: fld dword ptr [esp+64h]
0x10012a8b: fcom
0x10012a8d: fnstsw ax
0x10012a8f: test ah,41h
0x10012a92: jne short 10012AF0h
0x10012a94: fld dword ptr [esp+44h]
0x10012a98: fcom st(2)
0x10012a9a: fnstsw ax
0x10012a9c: fstp st(2)
0x10012a9e: test ah,41h
0x10012aa1: jne short 10012AF0h
0x10012aa3: push 0
0x10012aa5: fxch
0x10012aa7: push 5
0x10012aa9: push 2
0x10012aab: sub esp,10h
0x10012aae: fstp dword ptr [esp+0Ch]
0x10012ab2: lea edx,[esp+0F8h]
0x10012ab9: mov ecx,edi
0x10012abb: fstp dword ptr [esp+8]
0x10012abf: fld dword ptr [esp+54h]
0x10012ac3: fld qword ptr ds:[100FE7C8h]
0x10012ac9: fsub st(1),st
0x10012acb: fxch
0x10012acd: fstp dword ptr [esp+44h]
0x10012ad1: fld dword ptr [esp+44h]
0x10012ad5: fstp dword ptr [esp+4]
0x10012ad9: fsubr dword ptr [esp+4Ch]
0x10012add: fstp dword ptr [esp+44h]
0x10012ae1: fld dword ptr [esp+44h]
0x10012ae5: fstp dword ptr [esp]
0x10012ae8: push edx
0x10012ae9: call 1001FD10h
0x10012aee: jmp short 10012B58h
0x10012af0: fstp st(1)
0x10012af2: push 0
0x10012af4: fstp st(0)
0x10012af6: push 5
0x10012af8: fld dword ptr [esp+40h]
0x10012afc: push 2
0x10012afe: fld qword ptr ds:[100FE7C8h]
0x10012b04: sub esp,8
0x10012b07: fsub st(1),st
0x10012b09: lea eax,[esp+100h]
0x10012b10: fxch
0x10012b12: mov ecx,edi
0x10012b14: fstp dword ptr [esp+3Ch]
0x10012b18: fld dword ptr [esp+3Ch]
0x10012b1c: fstp dword ptr [esp+4]
0x10012b20: fsubr dword ptr [esp+44h]
0x10012b24: fstp dword ptr [esp+3Ch]
0x10012b28: fld dword ptr [esp+3Ch]
0x10012b2c: fstp dword ptr [esp]
0x10012b2f: push eax
0x10012b30: call 1001DDC0h
0x10012b35: jmp short 10012B58h
0x10012b37: mov ecx,[esp+7Ch]
0x10012b3b: mov edx,[esp+28h]
0x10012b3f: push ecx
0x10012b40: push edx
0x10012b41: mov ecx,edi
0x10012b43: call 10012F20h
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0x10012b48: mov ecx,edi
0x10012b4a: push eax
0x10012b4b: lea eax,[esp+0E4h]
0x10012b52: push eax
0x10012b53: call 10012EC0h
0x10012b58: cmp byte ptr [edi+181h],0
0x10012b5f: mov eax,[eax]
0x10012b61: mov [esp+24h],eax
0x10012b65: je near ptr 10012C76h
0x10012b6b: mov ecx,eax
0x10012b6d: shr ecx,18h
0x10012b70: movzx edx,bl
0x10012b73: add ecx,1
0x10012b76: imul ecx,edx
0x10012b79: sar ecx,8
0x10012b7c: mov bl,cl
0x10012b7e: jmp near ptr 10012C76h
0x10012b83: cmp byte ptr [ebp+20h],0
0x10012b87: je near ptr 10012C4Bh
0x10012b8d: fld1
0x10012b8f: fld dword ptr [esp+64h]
0x10012b93: fcom
0x10012b95: fnstsw ax
0x10012b97: test ah,41h
0x10012b9a: jne short 10012BFEh
0x10012b9c: fld dword ptr [esp+44h]
0x10012ba0: fcom st(2)
0x10012ba2: fnstsw ax
0x10012ba4: fstp st(2)
0x10012ba6: test ah,41h
0x10012ba9: jne short 10012BFEh
0x10012bab: push 0
0x10012bad: fxch
0x10012baf: push 5
0x10012bb1: push 2
0x10012bb3: sub esp,10h
0x10012bb6: fstp dword ptr [esp+0Ch]
0x10012bba: lea eax,[esp+104h]
0x10012bc1: mov ecx,edi
0x10012bc3: fstp dword ptr [esp+8]
0x10012bc7: fld dword ptr [esp+54h]
0x10012bcb: fld qword ptr ds:[100FE7C8h]
0x10012bd1: fsub st(1),st
0x10012bd3: fxch
0x10012bd5: fstp dword ptr [esp+44h]
0x10012bd9: fld dword ptr [esp+44h]
0x10012bdd: fstp dword ptr [esp+4]
0x10012be1: fsubr dword ptr [esp+4Ch]
0x10012be5: fstp dword ptr [esp+44h]
0x10012be9: fld dword ptr [esp+44h]
0x10012bed: fstp dword ptr [esp]
0x10012bf0: push eax
0x10012bf1: call 1001FD10h
0x10012bf6: mov eax,[eax]
0x10012bf8: mov [esp+24h],eax
0x10012bfc: jmp short 10012C76h
0x10012bfe: fstp st(1)
0x10012c00: push 0
0x10012c02: fstp st(0)
0x10012c04: push 5
0x10012c06: fld dword ptr [esp+40h]
0x10012c0a: push 2
0x10012c0c: fld qword ptr ds:[100FE7C8h]
0x10012c12: sub esp,8
0x10012c15: fsub st(1),st
0x10012c17: lea ecx,[esp+0F8h]
0x10012c1e: fxch
```

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0x10012c20: fstp dword ptr [esp+3Ch]
0x10012c24: fld dword ptr [esp+3Ch]
0x10012c28: fstp dword ptr [esp+4]
0x10012c2c: fsubr dword ptr [esp+44h]
0x10012c30: fstp dword ptr [esp+3Ch]
0x10012c34: fld dword ptr [esp+3Ch]
0x10012c38: fstp dword ptr [esp]
0x10012c3b: push ecx
0x10012c3c: mov ecx,edi
0x10012c3e: call 1001DDC0h
0x10012c43: mov eax,[eax]
0x10012c45: mov [esp+24h],eax
0x10012c49: jmp short 10012C76h
0x10012c4b: mov ecx,[esp+28h]
0x10012c4f: mov edx,[esp+90h]
0x10012c56: lea eax,[edx+ecx*2]
0x10012c59: add eax,ecx
0x10012c5b: mov cl,[eax]
0x10012c5d: mov dl,[eax+1]
0x10012c60: add eax,1
0x10012c63: mov al,[eax+1]
0x10012c66: mov [esp+24h],cl
0x10012c6a: mov [esp+25h],dl
0x10012c6e: mov [esp+26h],al
0x10012c72: mov eax,[esp+24h]
0x10012c76: cmp dword ptr [edi+28h],0
0x10012c7a: je short 10012C86h
0x10012c7c: mov ecx,[esp+5Ch]
0x10012c80: cmp ecx,[esp+74h]
0x10012c84: jmp short 10012C8Ah
0x10012c86: cmp eax,[esp+78h]
0x10012c8a: jne short 10012C93h
0x10012c8c: cmp byte ptr [esp+2Fh],0
0x10012c91: jne short 10012D0Fh
0x10012c93: test bl,bl
0x10012c95: je short 10012D0Fh
0x10012c97: cmp bl,0FFh
0x10012c9a: jne short 10012CB9h
0x10012c9c: mov dl,[esp+24h]
0x10012ca0: mov al,[esp+25h]
0x10012ca4: mov cl,[esp+26h]
0x10012ca8: mov [esi],dl
0x10012caa: add esi,1
0x10012cad: mov [esi],al
0x10012caf: add esi,1
0x10012cb2: mov [esi],cl
0x10012cb4: add esi,1
0x10012cb7: jmp short 10012D12h
0x10012cb9: movzx edx,byte ptr [esp+24h]
0x10012cbe: mov al,bl
0x10012cc0: movzx ecx,bl
0x10012cc3: movzx ebx,byte ptr [esi]
0x10012cc6: imul edx,ecx
0x10012cc9: not al
0x10012ccb: movzx eax,al
0x10012cce: imul ebx,eax
0x10012cd1: add edx,ebx
0x10012cd3: sar edx,8
0x10012cd6: mov [esi],dl
0x10012cd8: movzx ebx,byte ptr [esi+1]
0x10012cdc: movzx edx,byte ptr [esp+25h]
0x10012ce1: imul ebx,eax
0x10012ce4: add esi,1
0x10012ce7: imul edx,ecx
0x10012cea: add edx,ebx
0x10012cec: sar edx,8
0x10012cef: mov [esi],dl
```

```
0x10012cf1: movzx edx,byte ptr [esp+26h]
0x10012cf6: imul edx,ecx
0x10012cf9: movzx ecx,byte ptr [esi+1]
0x10012cfd: add esi,1
0x10012d00: imul ecx,eax
0x10012d03: add edx,ecx
0x10012d05: sar edx,8
0x10012d08: mov [esi],dl
0x10012d0a: add esi,1
0x10012d0d: jmp short 10012D12h
0x10012d0f: add esi,3
0x10012d12: mov eax,[esp+70h]
0x10012d16: add eax,1
0x10012d19: cmp eax,[esp+3Ch]
0x10012d1d: mov [esp+70h],eax
0x10012d21: jl near ptr 100129E0h
0x10012d27: mov edx,[esp+68h]
0x10012d2b: mov ecx,[esp+60h]
0x10012d2f: add [esp+50h],edx
0x10012d33: add [esp+54h],ecx
0x10012d37: mov eax,1
0x10012d3c: sub [esp+4Ch],eax
0x10012d40: sub [esp+34h],eax
0x10012d44: jne near ptr 10012952h
0x10012d4a: mov eax,[esp+48h]
0x10012d4e: mov ecx,[esp+3Ch]
0x10012d52: push 0
0x10012d54: lea edx,[esp+9Ch]
0x10012d5b: push edx
0x10012d5c: mov edx,[esp+8Ch]
0x10012d63: push eax
0x10012d64: mov eax,[esp+4Ch]
0x10012d68: push eax
0x10012d69: push 0
0x10012d6b: push 0
0x10012d6d: push 0
0x10012d6f: push eax
0x10012d70: mov eax,[esp+0A0h]
0x10012d77: push ecx
0x10012d78: mov ecx,[ebp+8]
0x10012d7b: push edx
0x10012d7c: push eax
0x10012d7d: push ecx
0x10012d7e: call dword ptr ds:[100FE018h]
0x10012d84: mov edx,[esp+0C4h]
0x10012d8b: mov esi,[esp+58h]
0x10012d8f: push edx
0x10012d90: push esi
0x10012d91: call dword ptr ds:[100FE030h]
0x10012d97: push eax
0x10012d98: call dword ptr ds:[100FE048h]
0x10012d9e: push esi
0x10012d9f: call dword ptr ds:[100FE02Ch]
0x10012da5: cmp dword ptr [ebp+1Ch],0
0x10012da9: je short 10012DCFh
0x10012dab: lea eax,[esp+0F0h]
0x10012db2: push eax
0x10012db3: call dword ptr ds:[100FE050h]
0x10012db9: mov ecx,[ebp+8]
0x10012dbc: push 2
0x10012dbe: mov esi,eax
0x10012dc0: push esi
0x10012dc1: push ecx
0x10012dc2: call dword ptr ds:[100FE054h]
0x10012dc8: push esi
0x10012dc9: call dword ptr ds:[100FE048h]
0x10012dcf: mov edx,[esp+0D8h]
```

```
0x10012dd6: mov eax,[ebp+8]
0x10012dd9: push edx
0x10012dda: push eax
0x10012ddb: call dword ptr ds:[100FE01Ch]
0x10012de1: mov eax,1
0x10012de6: pop edi
0x10012de7: pop esi
0x10012de8: pop ebx
0x10012de9: mov esp,ebp
0x10012deb: pop ebp
0x10012dec: ret 20h
0x10012def: pop edi
0x10012df0: pop esi
0x10012df1: xor eax,eax
0x10012df3: pop ebx
0x10012df4: mov esp,ebp
0x10012df6: pop ebp
0x10012df7: ret 20h
```

`sub_10011fb0` is called by sub_10012e00. It delegates to sub_1000ee60, sub_1000eea0, sub_100103b0 (9 callees).

12.289 sub_10012e00 at 0x10012e00 (25 instructions)

```
0x10012e00: mov eax,[esp+8]
0x10012e04: mov edx,[eax+4]
0x10012e07: push esi
0x10012e08: mov esi,[eax]
0x10012e0a: push edi
0x10012e0b: mov edi,[esp+1Ch]
0x10012e0f: push edi
0x10012e10: mov edi,[esp+1Ch]
0x10012e14: push edi
0x10012e15: mov edi,[esp+1Ch]
0x10012e19: push edi
0x10012e1a: mov edi,[eax+0Ch]
0x10012e1d: mov eax,[eax+8]
0x10012e20: sub edi,edx
0x10012e22: push edi
0x10012e23: sub eax,esi
0x10012e25: push eax
0x10012e26: push edx
0x10012e27: mov edx,[esp+24h]
0x10012e2b: push esi
0x10012e2c: push edx
0x10012e2d: call 10011FB0h
0x10012e32: pop edi
0x10012e33: pop esi
0x10012e34: ret 14h
```

`sub_10012e00` is called by sub_1000cd90. It delegates to sub_10011fb0 (1 callees).

12.290 sub_10012e40 at 0x10012e40 (4 instructions)

```
0x10012e40: mov eax,[ecx+28h]
0x10012e43: add eax,eax
0x10012e45: add eax,eax
0x10012e47: ret
```

`sub_10012e40` is called by sub_1000e5e0, sub_1000e640, sub_1000ead0, sub_10013690. It calls nothing else.

12.291 sub_10012e50 at 0x10012e50 (26 instructions)

```
0x10012e50: push esi
0x10012e51: mov esi,ecx
```

```
0x10012e53: mov edx,[esi+4]
0x10012e56: test edx,edx
0x10012e58: je short 10012E9Bh
0x10012e5a: mov ecx,[esi+28h]
0x10012e5d: test ecx,ecx
0x10012e5f: je short 10012E9Bh
0x10012e61: movzx eax,byte ptr [esp+8]
0x10012e66: cmp eax,ecx
0x10012e68: jae short 10012E9Bh
0x10012e6a: mov cl,[esp+14h]
0x10012e6e: add eax,eax
0x10012e70: add eax,eax
0x10012e72: mov [eax+edx+28h],cl
0x10012e76: mov cl,[esp+10h]
0x10012e7a: mov [eax+edx+29h],cl
0x10012e7e: mov cl,[esp+0Ch]
0x10012e82: add eax,1
0x10012e85: add eax,1
0x10012e88: mov [eax+edx+28h],cl
0x10012e8c: mov cl,[esp+18h]
0x10012e90: mov [eax+edx+29h],cl
0x10012e94: mov byte ptr [esi+1E1h],0
0x10012e9b: pop esi
0x10012e9c: ret 14h
```

`sub_10012e50` is called by sub_10017680. It calls nothing else.

12.292 sub_10012ea0 at 0x10012ea0 (9 instructions)

```
0x10012ea0: mov eax,[ecx+4]
0x10012ea3: test eax,eax
0x10012ea5: je short 10012EB1h
0x10012ea7: cmp dword ptr [ecx+28h],0
0x10012eab: je short 10012EB1h
0x10012ead: add eax,28h
0x10012eb0: ret
0x10012eb1: xor eax,eax
0x10012eb3: ret
```

`sub_10012ea0` is called by sub_1000ead0, sub_1000ecf0, sub_100103d0, sub_100118f0, sub_10011b00, sub_100135b0, sub_10013630, sub_10013690, sub_100136e0, sub_10013730, sub_10013780, sub_1001a5b0, sub_1001c370. It calls nothing else.

12.293 sub_10012ec0 at 0x10012ec0 (29 instructions)

```
0x10012ec0: mov eax,[esp+4]
0x10012ec4: xor edx,edx
0x10012ec6: push esi
0x10012ec7: mov esi,[ecx+4]
0x10012eca: cmp esi,edx
0x10012ecc: mov [eax],dl
0x10012ece: mov [eax+1],dl
0x10012ed1: mov [eax+2],dl
0x10012ed4: mov [eax+3],dl
0x10012ed7: je short 10012F14h
0x10012ed9: mov ecx,[ecx+28h]
0x10012edc: cmp ecx,edx
0x10012ede: je short 10012F14h
0x10012ee0: movzx edx,byte ptr [esp+0Ch]
0x10012ee5: cmp edx,ecx
0x10012ee7: jae short 10012F14h
0x10012ee9: lea ecx,[edx*4]
0x10012ef0: movzx edx,byte ptr [ecx+esi+28h]
0x10012ef5: add ecx,1
0x10012ef8: mov [eax],dl
```

```
0x10012efa: movzx edx,byte ptr [ecx+esi+28h]
0x10012eff: add ecx,1
0x10012f02: mov [eax+1],dl
0x10012f05: movzx edx,byte ptr [ecx+esi+28h]
0x10012f0a: mov cl,[ecx+esi+29h]
0x10012f0e: mov [eax+2],dl
0x10012f11: mov [eax+3],cl
0x10012f14: pop esi
0x10012f15: ret 8
```

`sub_10012ec0` is called by sub_1000ee60, sub_100103d0, sub_10011fb0, sub_100130e0, sub_100131f0, sub_10013bd0, sub_1001a5b0. It calls nothing else.

12.294 sub_10012f20 at 0x10012f20 (100 instructions)

```
0x10012f20: cmp dword ptr [ecx+4],0
0x10012f24: je near ptr 10013023h
0x10012f2a: cmp dword ptr [ecx+28h],0
0x10012f2e: je near ptr 10013023h
0x10012f34: mov eax,[esp+4]
0x10012f38: test eax,eax
0x10012f3a: push esi
0x10012f3b: jl near ptr 10013007h
0x10012f41: mov esi,[esp+0Ch]
0x10012f45: test esi,esi
0x10012f47: jl near ptr 10013007h
0x10012f4d: cmp eax,[ecx+0Ch]
0x10012f50: jge near ptr 10013007h
0x10012f56: cmp esi,[ecx+10h]
0x10012f59: jge near ptr 10013007h
0x10012f5f: push edi
0x10012f60: movzx edi,word ptr [ecx+16h]
0x10012f64: cmp di,8
0x10012f68: jne short 10012F7Bh
0x10012f6a: mov edx,[ecx+30h]
0x10012f6d: imul edx,esi
0x10012f70: add edx,[ecx+34h]
0x10012f73: pop edi
0x10012f74: mov al,[edx+eax]
0x10012f77: pop esi
0x10012f78: ret 8
0x10012f7b: movzx edx,word ptr [ecx+16h]
0x10012f7f: push ebx
0x10012f80: movzx ebx,dx
0x10012f83: imul ebx,eax
0x10012f86: push ebp
0x10012f87: mov ebp,[ecx+30h]
0x10012f8a: mov ecx,[ecx+34h]
0x10012f8d: imul ebp,esi
0x10012f90: sar ebx,3
0x10012f93: add ebx,ebp
0x10012f95: cmp dx,4
0x10012f99: mov bl,[ebx+ecx]
0x10012f9c: jne short 10012FCDh
0x10012f9e: mov edx,eax
0x10012fa0: and edx,80000001h
0x10012fa6: jns short 10012FADh
0x10012fa8: dec edx
0x10012fa9: or edx,0FFFFFFFEh
0x10012fac: inc edx
0x10012fad: mov al,1
0x10012faf: sub al,dl
0x10012fb1: add al,al
0x10012fb3: add al,al
0x10012fb5: movzx ecx,al
0x10012fb8: mov al,0Fh
```

```
0x10012fba: shl al,cl
0x10012fbc: pop ebp
0x10012fbd: movzx edx,bl
0x10012fc0: movzx eax,al
0x10012fc3: pop ebx
0x10012fc4: and eax,edx
0x10012fc6: pop edi
0x10012fc7: shr eax,cl
0x10012fc9: pop esi
0x10012fca: ret 8
0x10012fcd: cmp di,1
0x10012fd1: jne short 10012FFEh
0x10012fd3: mov ecx,eax
0x10012fd5: and ecx,80000007h
0x10012fdb: jns short 10012FE2h
0x10012fdd: dec ecx
0x10012fde: or ecx,0FFFFFFF8h
0x10012fe1: inc ecx
0x10012fe2: mov al,7
0x10012fe4: sub al,cl
0x10012fe6: movzx ecx,al
0x10012fe9: mov dl,1
0x10012feb: shl dl,cl
0x10012fed: pop ebp
0x10012fee: movzx eax,dl
0x10012ff1: movzx edx,bl
0x10012ff4: pop ebx
0x10012ff5: and eax,edx
0x10012ff7: pop edi
0x10012ff8: shr eax,cl
0x10012ffa: pop esi
0x10012ffb: ret 8
0x10012ffe: pop ebp
0x10012fff: pop ebx
0x10013000: pop edi
0x10013001: xor al,al
0x10013003: pop esi
0x10013004: ret 8
0x10013007: cmp dword ptr [ecx+14Ch],0
0x1001300e: jl short 1001301Ah
0x10013010: mov al,[ecx+14Ch]
0x10013016: pop esi
0x10013017: ret 8
0x1001301a: mov eax,[ecx+34h]
0x1001301d: mov al,[eax]
0x1001301f: pop esi
0x10013020: ret 8
0x10013023: xor al,al
0x10013025: ret 8
```

`sub_10012f20` is called by sub_10011fb0, sub_1001a270, sub_1001b970. It accesses memory-mapped peripheral registers at 0x80000001. It calls nothing else.

12.295 sub_10013030 at 0x10013030 (70 instructions)

```
0x10013030: push esi
0x10013031: movzx esi,word ptr [ecx+16h]
0x10013035: cmp si,8
0x10013039: jne short 10013051h
0x1001303b: mov eax,[ecx+30h]
0x1001303e: imul eax,[esp+0Ch]
0x10013043: add eax,[ecx+34h]
0x10013046: mov ecx,[esp+8]
0x1001304a: mov al,[eax+ecx]
0x1001304d: pop esi
0x1001304e: ret 8
```

```
0x10013051: movzx eax,word ptr [ecx+16h]
0x10013055: mov edx,[esp+8]
0x10013059: push ebx
0x1001305a: mov ebx,[ecx+30h]
0x1001305d: imul ebx,[esp+10h]
0x10013062: mov ecx,[ecx+34h]
0x10013065: push edi
0x10013066: movzx edi,ax
0x10013069: imul edi,edx
0x1001306c: sar edi,3
0x1001306f: add edi,ebx
0x10013071: cmp ax,4
0x10013075: mov bl,[edi+ecx]
0x10013078: jne short 100130A6h
0x1001307a: and edx,80000001h
0x10013080: jns short 10013087h
0x10013082: dec edx
0x10013083: or edx,0FFFFFFEh
0x10013086: inc edx
0x10013087: mov al,1
0x10013089: sub al,dl
0x1001308b: add al,al
0x1001308d: add al,al
0x1001308f: movzx ecx,al
0x10013092: mov dl,0Fh
0x10013094: shl dl,cl
0x10013096: pop edi
0x10013097: movzx eax,dl
0x1001309a: movzx edx,bl
0x1001309d: and eax,edx
0x1001309f: pop ebx
0x100130a0: shr eax,cl
0x100130a2: pop esi
0x100130a3: ret 8
0x100130a6: cmp si,1
0x100130aa: jne short 100130D4h
0x100130ac: and edx,80000007h
0x100130b2: jns short 100130B9h
0x100130b4: dec edx
0x100130b5: or edx,0FFFFFFF8h
0x100130b8: inc edx
0x100130b9: mov al,7
0x100130bb: sub al,dl
0x100130bd: movzx ecx,al
0x100130c0: mov al,1
0x100130c2: shl al,cl
0x100130c4: movzx edx,bl
0x100130c7: movzx eax,al
0x100130ca: pop edi
0x100130cb: and eax,edx
0x100130cd: pop ebx
0x100130ce: shr eax,cl
0x100130d0: pop esi
0x100130d1: ret 8
0x100130d4: pop edi
0x100130d5: pop ebx
0x100130d6: xor al,al
0x100130d8: pop esi
0x100130d9: ret 8
```

`sub_10013030` is called by sub_100103d0, sub_100118f0, sub_100130e0, sub_100131f0, sub_1001b970. It accesses memory-mapped peripheral registers at 0x80000001. It calls nothing else.

12.296 sub_100130e0 at 0x100130e0 (94 instructions)

```
0x100130e0: push ecx
0x100130e1: push ebx
0x100130e2: push esi
0x100130e3: mov esi,ecx
0x100130e5: mov eax,[esi+4]
0x100130e8: test eax,eax
0x100130ea: mov ecx,[esi+150h]
0x100130f0: push edi
0x100130f1: mov [esp+0Ch],ecx
0x100130f5: je near ptr 10013195h
0x100130fb: mov edi,[esp+18h]
0x100130ff: test edi,edi
0x10013101: jl near ptr 10013195h
0x10013107: mov ebx,[esp+1Ch]
0x1001310b: test ebx,ebx
0x1001310d: jl near ptr 10013195h
0x10013113: cmp edi,[esi+0Ch]
0x10013116: jge short 10013195h
0x10013118: cmp ebx,[esi+10h]
0x1001311b: jge short 10013195h
0x1001311d: cmp dword ptr [esi+28h],0
0x10013121: je short 10013141h
0x10013123: push ebx
0x10013124: push edi
0x10013125: mov ecx,esi
0x10013127: call 10013030h
0x1001312c: mov ecx,esi
0x1001312e: push eax
0x1001312f: lea eax,[esp+1Ch]
0x10013133: push eax
0x10013134: call 10012EC0h
0x10013139: mov ecx,[eax]
0x1001313b: mov [esp+0Ch],ecx
0x1001313f: jmp short 10013167h
0x10013141: mov eax,[esi+30h]
0x10013144: imul eax,ebx
0x10013147: add eax,[esi+34h]
0x1001314a: lea edx,[edi+edi*2]
0x1001314d: mov cl,[eax+edx]
0x10013150: add eax,edx
0x10013152: mov dl,[eax+1]
0x10013155: add eax,1
0x10013158: mov al,[eax+1]
0x1001315b: mov [esp+0Ch],cl
0x1001315f: mov [esp+0Dh],dl
0x10013163: mov [esp+0Eh],al
0x10013167: cmp dword ptr [esi+3C4h],0
0x1001316e: je short 10013184h
0x10013170: cmp byte ptr [esp+20h],0
0x10013175: je short 10013184h
0x10013177: push ebx
0x10013178: push edi
0x10013179: mov ecx,esi
0x1001317b: call 10013BB0h
0x10013180: mov [esp+0Fh],al
0x10013184: mov ecx,[esp+0Ch]
0x10013188: mov eax,[esp+14h]
0x1001318c: pop edi
0x1001318d: pop esi
0x1001318e: mov [eax],ecx
0x10013190: pop ebx
0x10013191: pop ecx
0x10013192: ret 10h
0x10013195: cmp dword ptr [esi+14Ch],0
0x1001319c: jl short 100131C2h
0x1001319e: cmp word ptr [esi+16h],18h
```

```
0x100131a3: jae short 10013188h
0x100131a5: movzx edx,byte ptr [esi+14Ch]
0x100131ac: mov edi,[esp+14h]
0x100131b0: push edx
0x100131b1: push edi
0x100131b2: mov ecx,esi
0x100131b4: call 10012EC0h
0x100131b9: mov eax,edi
0x100131bb: pop edi
0x100131bc: pop esi
0x100131bd: pop ebx
0x100131be: pop ecx
0x100131bf: ret 10h
0x100131c2: test eax,eax
0x100131c4: je short 10013188h
0x100131c6: mov edi,[esp+14h]
0x100131ca: push 1
0x100131cc: push 0
0x100131ce: push 0
0x100131d0: push edi
0x100131d1: mov ecx,esi
0x100131d3: call 100130E0h
0x100131d8: mov eax,edi
0x100131da: pop edi
0x100131db: pop esi
0x100131dc: pop ebx
0x100131dd: pop ecx
0x100131de: ret 10h
```

`sub_100130e0` is called by sub_100130e0, sub_1001a5b0, sub_1001b970. It delegates to sub_10012ec0, sub_10013030, sub_100130e0 (4 callees).

12.297 sub_100131f0 at 0x100131f0 (50 instructions)

```
0x100131f0: push ebx
0x100131f1: mov ebx,[esp+0Ch]
0x100131f5: push ebp
0x100131f6: mov ebp,[esp+14h]
0x100131fa: push esi
0x100131fb: push edi
0x100131fc: mov edi,ecx
0x100131fe: cmp dword ptr [edi+28h],0
0x10013202: je short 10013222h
0x10013204: push ebp
0x10013205: push ebx
0x10013206: call 10013030h
0x1001320b: mov ecx,edi
0x1001320d: push eax
0x1001320e: lea eax,[esp+20h]
0x10013212: push eax
0x10013213: call 10012EC0h
0x10013218: mov ecx,[eax]
0x1001321a: mov esi,[esp+14h]
0x1001321e: mov [esi],ecx
0x10013220: jmp short 1001324Ch
0x10013222: mov eax,[edi+30h]
0x10013225: mov esi,[esp+14h]
0x10013229: imul eax,ebp
0x1001322c: add eax,[edi+34h]
0x1001322f: lea edx,[ebx+ebx*2]
0x10013232: mov cl,[eax+edx]
0x10013235: add eax,edx
0x10013237: mov dl,[eax+1]
0x1001323a: add eax,1
0x1001323d: mov al,[eax+1]
0x10013240: mov [esi],cl
```

```
0x10013242: mov [esi+1],dl
0x10013245: mov [esi+2],al
0x10013248: mov byte ptr [esi+3],0
0x1001324c: cmp dword ptr [edi+3C4h],0
0x10013253: je short 10013268h
0x10013255: cmp byte ptr [esp+20h],0
0x1001325a: je short 10013268h
0x1001325c: push ebp
0x1001325d: push ebx
0x1001325e: mov ecx,edi
0x10013260: call 10013BB0h
0x10013265: mov [esi+3],al
0x10013268: pop edi
0x10013269: mov eax,esi
0x1001326b: pop esi
0x1001326c: pop ebp
0x1001326d: pop ebx
0x1001326e: ret 10h
```

`sub_100131f0` is called by sub_100103d0, sub_100118f0, sub_1001d050. It delegates to sub_10012ec0, sub_10013030, sub_10013bb0 (3 callees).

12.298 sub_10013280 at 0x10013280 (65 instructions)

```
0x10013280: push edi
0x10013281: movzx edi,word ptr [ecx+16h]
0x10013285: cmp di,8
0x10013289: jne short 100132A5h
0x1001328b: mov eax,[ecx+30h]
0x1001328e: imul eax,[esp+0Ch]
0x10013293: add eax,[ecx+34h]
0x10013296: mov cl,[esp+10h]
0x1001329a: mov edx,[esp+8]
0x1001329e: mov [eax+edx],cl
0x100132a1: pop edi
0x100132a2: ret 0Ch
0x100132a5: movzx edx,word ptr [ecx+16h]
0x100132a9: mov eax,[esp+8]
0x100132ad: push ebx
0x100132ae: mov ebx,[ecx+30h]
0x100132b1: imul ebx,[esp+10h]
0x100132b6: push esi
0x100132b7: movzx esi,dx
0x100132ba: imul esi,eax
0x100132bd: sar esi,3
0x100132c0: add esi,ebx
0x100132c2: add esi,[ecx+34h]
0x100132c5: cmp dx,4
0x100132c9: jne short 100132EFh
0x100132cb: mov ecx,eax
0x100132cd: and ecx,80000001h
0x100132d3: jns short 100132DAh
0x100132d5: dec ecx
0x100132d6: or ecx,0FFFFFFFEh
0x100132d9: inc ecx
0x100132da: mov al,1
0x100132dc: sub al,cl
0x100132de: add al,al
0x100132e0: add al,al
0x100132e2: movzx edi,al
0x100132e5: mov al,[esp+18h]
0x100132e9: mov dl,0Fh
0x100132eb: and al,dl
0x100132ed: jmp short 10013313h
0x100132ef: cmp di,1
0x100132f3: jne short 10013321h
```

```
0x100132f5: mov ecx,eax
0x100132f7: and ecx,80000007h
0x100132fd: jns short 10013304h
0x100132ff: dec ecx
0x10013300: or ecx,0FFFFFFF8h
0x10013303: inc ecx
0x10013304: mov al,7
0x10013306: sub al,cl
0x10013308: movzx edi,al
0x1001330b: mov al,[esp+18h]
0x1001330f: mov dl,1
0x10013311: and al,dl
0x10013313: mov ecx,edi
0x10013315: shl dl,cl
0x10013317: shl al,cl
0x10013319: not dl
0x1001331b: and dl,[esi]
0x1001331d: or dl,al
0x1001331f: mov [esi],dl
0x10013321: pop esi
0x10013322: pop ebx
0x10013323: pop edi
0x10013324: ret 0Ch
```

`sub_10013280` is called by sub_100103d0, sub_100118f0, sub_10013890, sub_10013900. It accesses memory-mapped peripheral registers at 0x80000001. It calls nothing else.

12.299 sub_10013330 at 0x10013330 (91 instructions)

```
0x10013330: cmp dword ptr [ecx+4],0
0x10013334: je near ptr 1001341Dh
0x1001333a: cmp dword ptr [ecx+28h],0
0x1001333e: je near ptr 1001341Dh
0x10013344: mov eax,[esp+4]
0x10013348: test eax,eax
0x1001334a: jl near ptr 1001341Dh
0x10013350: push edi
0x10013351: mov edi,[esp+0Ch]
0x10013355: test edi,edi
0x10013357: jl near ptr 1001341Ch
0x1001335d: cmp eax,[ecx+0Ch]
0x10013360: jge near ptr 1001341Ch
0x10013366: cmp edi,[ecx+10h]
0x10013369: jge near ptr 1001341Ch
0x1001336f: push ebx
0x10013370: movzx ebx,word ptr [ecx+16h]
0x10013374: cmp bx,8
0x10013378: jne short 1001338Fh
0x1001337a: mov edx,[ecx+30h]
0x1001337d: imul edx,edi
0x10013380: add edx,[ecx+34h]
0x10013383: mov cl,[esp+14h]
0x10013387: pop ebx
0x10013388: mov [edx+eax],cl
0x1001338b: pop edi
0x1001338c: ret 0Ch
0x1001338f: movzx edx,word ptr [ecx+16h]
0x10013393: push ebp
0x10013394: mov ebp,[ecx+30h]
0x10013397: push esi
0x10013398: imul ebp,edi
0x1001339b: movzx esi,dx
0x1001339e: imul esi,eax
0x100133a1: sar esi,3
0x100133a4: add esi,ebp
0x100133a6: add esi,[ecx+34h]
```

```
0x100133a9: cmp dx,4
0x100133ad: jne short 100133E7h
0x100133af: mov edx,eax
0x100133b1: and edx,80000001h
0x100133b7: jns short 100133BEh
0x100133b9: dec edx
0x100133ba: or edx,0FFFFFFFEh
0x100133bd: inc edx
0x100133be: mov al,1
0x100133c0: sub al,dl
0x100133c2: mov dl,[esp+1Ch]
0x100133c6: add al,al
0x100133c8: add al,al
0x100133ca: movzx edi,al
0x100133cd: mov ecx,edi
0x100133cf: mov al,0Fh
0x100133d1: shl al,cl
0x100133d3: and dl,0Fh
0x100133d6: shl dl,cl
0x100133d8: not al
0x100133da: and al,[esi]
0x100133dc: or al,dl
0x100133de: mov [esi],al
0x100133e0: pop esi
0x100133e1: pop ebp
0x100133e2: pop ebx
0x100133e3: pop edi
0x100133e4: ret 0Ch
0x100133e7: cmp bx,1
0x100133eb: jne short 10013419h
0x100133ed: mov ecx,eax
0x100133ef: and ecx,80000007h
0x100133f5: jns short 100133FCh
0x100133f7: dec ecx
0x100133f8: or ecx,0FFFFFFF8h
0x100133fb: inc ecx
0x100133fc: mov al,7
0x100133fe: sub al,cl
0x10013400: movzx edi,al
0x10013403: mov al,[esp+1Ch]
0x10013407: mov ecx,edi
0x10013409: mov dl,1
0x1001340b: shl dl,cl
0x1001340d: and al,1
0x1001340f: shl al,cl
0x10013411: not dl
0x10013413: and dl,[esi]
0x10013415: or dl,al
0x10013417: mov [esi],dl
0x10013419: pop esi
0x1001341a: pop ebp
0x1001341b: pop ebx
0x1001341c: pop edi
0x1001341d: ret 0Ch
```

`sub_10013330` is called by sub_100143f0, sub_100156b0, sub_10017680, sub_1001a270, sub_1001b970. It accesses memory-mapped peripheral registers at 0x80000001. It calls nothing else.

12.300 sub_10013420 at 0x10013420 (88 instructions)

```
0x10013420: sub esp,14h
0x10013423: push ebx
0x10013424: mov ebx,[ecx+4]
0x10013427: test ebx,ebx
0x10013429: push edi
0x1001342a: je near ptr 10013521h
```

```
0x10013430: mov edi,[ecx+28h]
0x10013433: test edi,edi
0x10013435: je near ptr 10013521h
0x1001343b: cmp byte ptr [ecx+1E1h],0
0x10013442: mov edx,[esp+20h]
0x10013446: je short 1001345Eh
0x10013448: cmp [ecx+1DCh],edx
0x1001344e: jne short 1001345Eh
0x10013450: mov al,[ecx+1E0h]
0x10013456: pop edi
0x10013457: pop ebx
0x10013458: add esp,14h
0x1001345b: ret 4
0x1001345e: push ebp
0x1001345f: push esi
0x10013460: mov esi,[ecx+2Ch]
0x10013463: xor eax,eax
0x10013465: test esi,esi
0x10013467: mov [ecx+1DCh],edx
0x1001346d: mov byte ptr [ecx+1E1h],1
0x10013474: mov dword ptr [esp+10h],30D40h
0x1001347c: mov [esp+14h],eax
0x10013480: jne short 1001348Ah
0x10013482: mov esi,edi
0x10013484: mov [esp+18h],edi
0x10013488: jmp short 1001348Eh
0x1001348a: mov [esp+18h],esi
0x1001348e: xor ebp,ebp
0x10013490: test esi,esi
0x10013492: jle short 10013511h
0x10013494: movzx eax,dh
0x10013497: movzx edx,dl
0x1001349a: mov [esp+1Ch],eax
0x1001349e: mov [esp+20h],edx
0x100134a2: add ebx,29h
0x100134a5: jmp short 100134B0h
0x100134b0: movzx edi,byte ptr [ebx-1]
0x100134b4: movzx eax,byte ptr [esp+2Ah]
0x100134b9: movzx esi,byte ptr [ebx+1]
0x100134bd: movzx edx,byte ptr [ebx]
0x100134c0: sub edi,[esp+20h]
0x100134c4: sub edx,[esp+1Ch]
0x100134c8: sub esi,eax
0x100134ca: mov eax,edi
0x100134cc: imul edx,edx
0x100134cf: imul eax,edi
0x100134d2: add edx,eax
0x100134d4: mov eax,esi
0x100134d6: imul eax,esi
0x100134d9: add edx,eax
0x100134db: je short 1001350Fh
0x100134dd: cmp edx,[esp+10h]
0x100134e1: jge short 100134EFh
0x100134e3: mov eax,ebp
0x100134e5: mov [esp+10h],edx
0x100134e9: mov [esp+14h],eax
0x100134ed: jmp short 100134F3h
0x100134ef: mov eax,[esp+14h]
0x100134f3: add ebp,1
0x100134f6: add ebx,4
0x100134f9: cmp ebp,[esp+18h]
0x100134fd: jl short 100134B0h
0x100134ff: pop esi
0x10013500: pop ebp
0x10013501: pop edi
0x10013502: mov [ecx+1E0h],al
0x10013508: pop ebx
```

```
0x10013509: add esp,14h
0x1001350c: ret 4
0x1001350f: mov eax,ebp
0x10013511: pop esi
0x10013512: pop ebp
0x10013513: pop edi
0x10013514: mov [ecx+1E0h],al
0x1001351a: pop ebx
0x1001351b: add esp,14h
0x1001351e: ret 4
0x10013521: pop edi
0x10013522: xor al,al
0x10013524: pop ebx
0x10013525: add esp,14h
0x10013528: ret 4
```

`sub_10013420` is called by sub_10013890, sub_10013900. It calls nothing else.

12.301 sub_10013530 at 0x10013530 (36 instructions)

```
0x10013530: push esi
0x10013531: mov esi,[esp+8]
0x10013535: test esi,esi
0x10013537: je short 10013583h
0x10013539: cmp dword ptr [ecx+28h],0
0x1001353d: jne short 10013583h
0x1001353f: mov edx,[ecx+30h]
0x10013542: mov eax,[esp+0Ch]
0x10013546: cmp eax,edx
0x10013548: jl short 1001354Ch
0x1001354a: mov eax,edx
0x1001354c: mov ecx,[ecx+0Ch]
0x1001354f: lea ecx,[ecx+ecx*2]
0x10013552: cmp eax,ecx
0x10013554: jl short 10013558h
0x10013556: mov eax,ecx
0x10013558: test eax,eax
0x1001355a: jle short 10013583h
0x1001355c: lea edx,[eax-1]
0x1001355f: mov eax,0AAAAAAABh
0x10013564: mul edx
0x10013566: shr edx,1
0x10013568: lea ecx,[esi+2]
0x1001356b: add edx,1
0x1001356e: push ebx
0x1001356f: nop
0x10013570: mov al,[ecx-2]
0x10013573: mov bl,[ecx]
0x10013575: mov [ecx-2],bl
0x10013578: mov [ecx],al
0x1001357a: add ecx,3
0x1001357d: sub edx,1
0x10013580: jne short 10013570h
0x10013582: pop ebx
0x10013583: pop esi
0x10013584: ret 8
```

`sub_10013530` is called by sub_10013780, sub_10016040, sub_1001b350. It accesses memory-mapped peripheral registers at 0xaaaaaaab. It calls nothing else.

12.302 sub_10013590 at 0x10013590 (8 instructions)

```
0x10013590: mov eax,[esp+4]
0x10013594: mov [esp+6],al
0x10013598: mov [esp+5],ah
```

```
0x1001359c: shr eax,10h
0x1001359f: mov [esp+4],al
0x100135a3: mov byte ptr [esp+7],0
0x100135a8: mov eax,[esp+4]
0x100135ac: ret
```

`sub_10013590` is called by sub_10013980. It calls nothing else.

12.303 sub_100135b0 at 0x100135b0 (47 instructions)

```
0x100135b0: push esi
0x100135b1: mov esi,[esp+0Ch]
0x100135b5: test esi,esi
0x100135b7: je short 10013628h
0x100135b9: cmp dword ptr [ecx+4],0
0x100135bd: je short 10013628h
0x100135bf: push ebx
0x100135c0: mov ebx,[ecx+28h]
0x100135c3: test ebx,ebx
0x100135c5: je short 10013627h
0x100135c7: mov edx,[esp+14h]
0x100135cb: test edx,edx
0x100135cd: jne short 100135D1h
0x100135cf: mov edx,esi
0x100135d1: push ebp
0x100135d2: push edi
0x100135d3: mov edi,[esp+20h]
0x100135d7: test edi,edi
0x100135d9: jne short 100135DDh
0x100135db: mov edi,edx
0x100135dd: call 10012EA0h
0x100135e2: mov ebp,[esp+14h]
0x100135e6: cmp ebp,ebx
0x100135e8: jb short 100135ECh
0x100135ea: mov ebp,ebx
0x100135ec: test ebp,ebp
0x100135ee: jbe short 1001361Eh
0x100135f0: add eax,1
0x100135f3: sub esi,edx
0x100135f5: sub edi,edx
0x100135f7: jmp short 10013600h
0x10013600: movzx ebx,byte ptr [esi+edx]
0x10013604: mov [eax+1],bl
0x10013607: movzx ebx,byte ptr [edx]
0x1001360a: mov [eax],bl
0x1001360c: movzx ebx,byte ptr [edi+edx]
0x10013610: mov [eax-1],bl
0x10013613: add edx,1
0x10013616: add eax,4
0x10013619: sub ebp,1
0x1001361c: jne short 10013600h
0x1001361e: pop edi
0x1001361f: mov byte ptr [ecx+1E1h],0
0x10013626: pop ebp
0x10013627: pop ebx
0x10013628: pop esi
0x10013629: ret 10h
```

`sub_100135b0` is called by sub_1001a5b0, sub_1001b350. It delegates to sub_10012ea0 (1 callees).

12.304 sub_10013630 at 0x10013630 (36 instructions)

```
0x10013630: push ebp
0x10013631: push edi
0x10013632: mov edi,[esp+0Ch]
```

```
0x10013636: test edi,edi
0x10013638: mov ebp,ecx
0x1001363a: je short 10013689h
0x1001363c: cmp dword ptr [ebp+4],0
0x10013640: je short 10013689h
0x10013642: mov edx,[ebp+28h]
0x10013645: test edx,edx
0x10013647: je short 10013689h
0x10013649: push esi
0x1001364a: call 10012EA0h
0x1001364f: mov esi,[esp+14h]
0x10013653: cmp esi,edx
0x10013655: jb short 10013659h
0x10013657: mov esi,edx
0x10013659: test esi,esi
0x1001365b: jbe short 10013681h
0x1001365d: lea edx,[edi+2]
0x10013660: add eax,1
0x10013663: movzx ecx,byte ptr [edx-2]
0x10013667: mov [eax+1],cl
0x1001366a: movzx ecx,byte ptr [edx-1]
0x1001366e: mov [eax],cl
0x10013670: movzx ecx,byte ptr [edx]
0x10013673: mov [eax-1],cl
0x10013676: add edx,3
0x10013679: add eax,4
0x1001367c: sub esi,1
0x1001367f: jne short 10013663h
0x10013681: mov byte ptr [ebp+1E1h],0
0x10013688: pop esi
0x10013689: pop edi
0x1001368a: pop ebp
0x1001368b: ret 8
```

`sub_10013630` is called by sub_10016840. It delegates to sub_10012ea0 (1 callees).

12.305 sub_10013690 at 0x10013690 (30 instructions)

```
0x10013690: push esi
0x10013691: push edi
0x10013692: mov edi,[esp+0Ch]
0x10013696: test edi,edi
0x10013698: mov esi,ecx
0x1001369a: je short 100136DBh
0x1001369c: cmp dword ptr [esi+4],0
0x100136a0: je short 100136DBh
0x100136a2: cmp dword ptr [esi+28h],0
0x100136a6: je short 100136DBh
0x100136a8: mov edx,[esp+10h]
0x100136ac: add edx,edx
0x100136ae: add edx,edx
0x100136b0: call 10012E40h
0x100136b5: cmp eax,edx
0x100136b7: jae short 100136C0h
0x100136b9: call 10012E40h
0x100136be: jmp short 100136C2h
0x100136c0: mov eax,edx
0x100136c2: push eax
0x100136c3: push edi
0x100136c4: mov ecx,esi
0x100136c6: call 10012EA0h
0x100136cb: push eax
0x100136cc: call 100228C6h
0x100136d1: add esp,0Ch
0x100136d4: mov byte ptr [esi+1E1h],0
0x100136db: pop edi
```

```
0x100136dc: pop esi
0x100136dd: ret 8
```

`sub_10013690` is called by sub_1000ecf0, sub_100103d0, sub_10011b00, sub_100143f0, sub_1001a5b0, sub_1001b970. It delegates to sub_10012e40, sub_10012ea0, j_memcpy (3 callees).

12.306 sub_100136e0 at 0x100136e0 (31 instructions)

```
0x100136e0: cmp dword ptr [ecx+4],0
0x100136e4: je short 10013727h
0x100136e6: mov edx,[ecx+28h]
0x100136e9: test edx,edx
0x100136eb: je short 10013727h
0x100136ed: push ebx
0x100136ee: call 10012EA0h
0x100136f3: xor ebx,ebx
0x100136f5: test edx,edx
0x100136f7: jbe short 10013726h
0x100136f9: push esi
0x100136fa: push edi
0x100136fb: lea esi,[eax+1]
0x100136fe: mov edi,edi
0x10013700: lea edi,[edx-1]
0x10013703: xor edx,edx
0x10013705: mov eax,0FFh
0x1001370a: div edi
0x1001370c: add esi,4
0x1001370f: imul bl
0x10013711: mov [esi-3],al
0x10013714: mov [esi-4],al
0x10013717: mov [esi-5],al
0x1001371a: mov edx,[ecx+28h]
0x1001371d: add ebx,1
0x10013720: cmp ebx,edx
0x10013722: jb short 10013700h
0x10013724: pop edi
0x10013725: pop esi
0x10013726: pop ebx
0x10013727: ret
```

`sub_100136e0` is called by sub_100156b0, sub_100171e0, sub_1001b350. It delegates to sub_10012ea0 (1 callees).

12.307 sub_10013730 at 0x10013730 (34 instructions)

```
0x10013730: call 10012EA0h
0x10013735: cmp dword ptr [ecx+4],0
0x10013739: je short 10013779h
0x1001373b: test eax,eax
0x1001373d: je short 10013779h
0x1001373f: mov edx,[ecx+28h]
0x10013742: test edx,edx
0x10013744: je short 10013779h
0x10013746: xor ecx,ecx
0x10013748: test edx,edx
0x1001374a: push esi
0x1001374b: jbe short 10013771h
0x1001374d: add eax,2
0x10013750: movzx esi,byte ptr [eax-2]
0x10013754: cmp esi,ecx
0x10013756: jne short 10013775h
0x10013758: movzx esi,byte ptr [eax-1]
0x1001375c: cmp esi,ecx
0x1001375e: jne short 10013775h
0x10013760: movzx esi,byte ptr [eax]
0x10013763: cmp esi,ecx
```

```
0x10013765: jne short 10013775h
0x10013767: add ecx,1
0x1001376a: add eax,4
0x1001376d: cmp ecx,edx
0x1001376f: jb short 10013750h
0x10013771: mov al,1
0x10013773: pop esi
0x10013774: ret
0x10013775: xor al,al
0x10013777: pop esi
0x10013778: ret
0x10013779: xor al,al
0x1001377b: ret
```

`sub_10013730` is called by sub_100118f0. It delegates to sub_10012ea0 (1 callees).

12.308 sub_10013780 at 0x10013780 (51 instructions)

```
0x10013780: push ebp
0x10013781: push esi
0x10013782: mov esi,ecx
0x10013784: cmp dword ptr [esi+4],0
0x10013788: je short 100137F0h
0x1001378a: mov edx,[esi+28h]
0x1001378d: test edx,edx
0x1001378f: push edi
0x10013790: je short 100137C4h
0x10013792: call 10012EA0h
0x10013797: test eax,eax
0x10013799: je short 100137EFh
0x1001379b: xor ebp,ebp
0x1001379d: test edx,edx
0x1001379f: jbe short 100137EFh
0x100137a1: xor ecx,ecx
0x100137a3: push ebx
0x100137a4: mov dl,[eax+ecx*4]
0x100137a7: mov bl,[eax+ecx*4+2]
0x100137ab: lea edi,[eax+ecx*4+2]
0x100137af: add ebp,1
0x100137b2: mov [eax+ecx*4],bl
0x100137b5: movzx ecx,bp
0x100137b8: mov [edi],dl
0x100137ba: cmp ecx,[esi+28h]
0x100137bd: jb short 100137A4h
0x100137bf: pop ebx
0x100137c0: pop edi
0x100137c1: pop esi
0x100137c2: pop ebp
0x100137c3: ret
0x100137c4: xor edi,edi
0x100137c6: cmp [esi+10h],edi
0x100137c9: jle short 100137EFh
0x100137cb: jmp short 100137D0h
0x100137d0: mov eax,[esi+0Ch]
0x100137d3: lea eax,[eax+eax*2]
0x100137d6: push eax
0x100137d7: push edi
0x100137d8: mov ecx,esi
0x100137da: call 1000E5E0h
0x100137df: push eax
0x100137e0: mov ecx,esi
0x100137e2: call 10013530h
0x100137e7: add edi,1
0x100137ea: cmp edi,[esi+10h]
0x100137ed: jl short 100137D0h
0x100137ef: pop edi
```

```
0x100137f0: pop esi
0x100137f1: pop ebp
0x100137f2: ret
```

`sub_10013780` is called by sub_100187f0. It delegates to sub_1000e5e0, sub_10012ea0, sub_10013530 (3 callees).

12.309 sub_10013890 at 0x10013890 (41 instructions)

```
0x10013890: push ebx
0x10013891: push esi
0x10013892: mov esi,ecx
0x10013894: cmp dword ptr [esi+28h],0
0x10013898: push edi
0x10013899: je short 100138B7h
0x1001389b: mov eax,[esp+18h]
0x1001389f: push eax
0x100138a0: call 10013420h
0x100138a5: mov ebx,[esp+14h]
0x100138a9: mov edi,[esp+10h]
0x100138ad: push eax
0x100138ae: push ebx
0x100138af: push edi
0x100138b0: call 10013280h
0x100138b5: jmp short 100138DFh
0x100138b7: mov eax,[esi+30h]
0x100138ba: mov ebx,[esp+14h]
0x100138be: mov edi,[esp+10h]
0x100138c2: imul eax,ebx
0x100138c5: mov dl,[esp+1Ah]
0x100138c9: lea ecx,[edi+edi*2]
0x100138cc: add eax,ecx
0x100138ce: add eax,[esi+34h]
0x100138d1: mov ecx,[esp+18h]
0x100138d5: mov [eax],cl
0x100138d7: add eax,1
0x100138da: mov [eax],ch
0x100138dc: mov [eax+1],dl
0x100138df: cmp byte ptr [esp+1Ch],0
0x100138e4: je short 100138F4h
0x100138e6: mov eax,[esp+1Bh]
0x100138ea: push eax
0x100138eb: push ebx
0x100138ec: push edi
0x100138ed: mov ecx,esi
0x100138ef: call 10013AF0h
0x100138f4: pop edi
0x100138f5: pop esi
0x100138f6: pop ebx
0x100138f7: ret 10h
```

`sub_10013890` is called by sub_100103d0, sub_100118f0. It delegates to sub_10013280, sub_10013420, sub_10013af0 (3 callees).

12.310 sub_10013900 at 0x10013900 (49 instructions)

```
0x10013900: push esi
0x10013901: mov esi,ecx
0x10013903: cmp dword ptr [esi+4],0
0x10013907: je short 10013976h
0x10013909: push edi
0x1001390a: mov edi,[esp+0Ch]
0x1001390e: test edi,edi
0x10013910: jl short 10013975h
0x10013912: push ebx
0x10013913: mov ebx,[esp+14h]
```

```
0x10013917: test ebx,ebx
0x10013919: jl short 10013974h
0x1001391b: cmp edi,[esi+0Ch]
0x1001391e: jge short 10013974h
0x10013920: cmp ebx,[esi+10h]
0x10013923: jge short 10013974h
0x10013925: cmp dword ptr [esi+28h],0
0x10013929: je short 1001393Fh
0x1001392b: mov eax,[esp+18h]
0x1001392f: push eax
0x10013930: call 10013420h
0x10013935: push eax
0x10013936: push ebx
0x10013937: push edi
0x10013938: call 10013280h
0x1001393d: jmp short 1001395Fh
0x1001393f: mov eax,[esi+30h]
0x10013942: mov dl,[esp+1Ah]
0x10013946: imul eax,ebx
0x10013949: lea ecx,[edi+edi*2]
0x1001394c: add eax,ecx
0x1001394e: add eax,[esi+34h]
0x10013951: mov ecx,[esp+18h]
0x10013955: mov [eax],cl
0x10013957: add eax,1
0x1001395a: mov [eax],ch
0x1001395c: mov [eax+1],dl
0x1001395f: cmp byte ptr [esp+1Ch],0
0x10013964: je short 10013974h
0x10013966: mov eax,[esp+1Bh]
0x1001396a: push eax
0x1001396b: push ebx
0x1001396c: push edi
0x1001396d: mov ecx,esi
0x1001396f: call 10013AF0h
0x10013974: pop ebx
0x10013975: pop edi
0x10013976: pop esi
0x10013977: ret 10h
```

`sub_10013900` is called by sub_10013980, sub_100143f0, sub_100156b0, sub_10015d20, sub_1001a5b0. It delegates to sub_10013280, sub_10013420, sub_10013af0 (3 callees).

12.311 sub_10013980 at 0x10013980 (12 instructions)

```
0x10013980: mov eax,[esp+0Ch]
0x10013984: push eax
0x10013985: call 10013590h
0x1001398a: mov edx,[esp+0Ch]
0x1001398e: add esp,4
0x10013991: push 0
0x10013993: push eax
0x10013994: mov eax,[esp+0Ch]
0x10013998: push edx
0x10013999: push eax
0x1001399a: call 10013900h
0x1001399f: ret 0Ch
```

`sub_10013980` is called by sub_10017680. It delegates to sub_10013590, sub_10013900 (2 callees).

12.312 sub_100139b0 at 0x100139b0 (4 instructions)

```
0x100139b0: xor eax,eax
0x100139b2: cmp [ecx+3C4h],eax
```

```
0x100139b8: setne al
0x100139bb: ret
```

`sub_100139b0` is called by sub_100103d0, sub_10011b00, sub_100187f0, sub_1001b970. It calls nothing else.

12.313 sub_100139c0 at 0x100139c0 (24 instructions)

```
0x100139c0: push esi
0x100139c1: mov esi,ecx
0x100139c3: cmp dword ptr [esi+3C4h],0
0x100139ca: jne short 100139FDh
0x100139cc: mov eax,[esi+0Ch]
0x100139cf: imul eax,[esi+10h]
0x100139d3: push eax
0x100139d4: call dword ptr ds:[100FE550h]
0x100139da: add esp,4
0x100139dd: test eax,eax
0x100139df: mov [esi+3C4h],eax
0x100139e5: je short 100139FDh
0x100139e7: mov ecx,[esi+0Ch]
0x100139ea: imul ecx,[esi+10h]
0x100139ee: push ecx
0x100139ef: push 0FFh
0x100139f4: push eax
0x100139f5: call 100228C0h
0x100139fa: add esp,0Ch
0x100139fd: xor eax,eax
0x100139ff: cmp [esi+3C4h],eax
0x10013a05: pop esi
0x10013a06: setne al
0x10013a09: ret
```

`sub_100139c0` is called by sub_100103d0, sub_10011b00, sub_100143f0, sub_10015d20, sub_10017680, sub_100187f0, sub_1001b970. It delegates to j_memset (1 callees).

12.314 sub_10013a10 at 0x10013a10 (11 instructions)

```
0x10013a10: push esi
0x10013a11: mov esi,ecx
0x10013a13: mov eax,[esi+3C4h]
0x10013a19: test eax,eax
0x10013a1b: je short 10013A31h
0x10013a1d: push eax
0x10013a1e: call dword ptr ds:[100FE554h]
0x10013a24: add esp,4
0x10013a27: mov dword ptr [esi+3C4h],0
0x10013a31: pop esi
0x10013a32: ret
```

`sub_10013a10` is called by sub_1000ead0, sub_1001b970. It calls nothing else.

12.315 sub_10013a70 at 0x10013a70 (40 instructions)

```
0x10013a70: push esi
0x10013a71: push edi
0x10013a72: mov edi,[esp+0Ch]
0x10013a76: cmp dword ptr [edi+3C4h],0
0x10013a7d: mov esi,ecx
0x10013a7f: je short 10013AE2h
0x10013a81: mov ecx,[esi+0Ch]
0x10013a84: cmp ecx,[edi+0Ch]
0x10013a87: jne short 10013AE2h
0x10013a89: mov eax,[esi+10h]
0x10013a8c: cmp eax,[edi+10h]
```

```
0x10013a8f: jne short 10013AE2h
0x10013a91: cmp dword ptr [esi+3C4h],0
0x10013a98: jne short 10013AADh
0x10013a9a: imul eax,ecx
0x10013a9d: push eax
0x10013a9e: call dword ptr ds:[100FE550h]
0x10013aa4: add esp,4
0x10013aa7: mov [esi+3C4h],eax
0x10013aad: mov eax,[esi+3C4h]
0x10013ab3: test eax,eax
0x10013ab5: je short 10013AE2h
0x10013ab7: mov ecx,[esi+0Ch]
0x10013aba: imul ecx,[esi+10h]
0x10013abe: mov edx,[edi+3C4h]
0x10013ac4: push ecx
0x10013ac5: push edx
0x10013ac6: push eax
0x10013ac7: call 100228C6h
0x10013acc: mov al,[edi+180h]
0x10013ad2: add esp,0Ch
0x10013ad5: mov [esi+180h],al
0x10013adb: pop edi
0x10013adc: mov al,1
0x10013ade: pop esi
0x10013adf: ret 4
0x10013ae2: pop edi
0x10013ae3: xor al,al
0x10013ae5: pop esi
0x10013ae6: ret 4
```

`sub_10013a70` is called by sub_100103d0. It delegates to j_memcpy (1 callees).

12.316 sub_10013af0 at 0x10013af0 (23 instructions)

```
0x10013af0: push esi
0x10013af1: mov esi,ecx
0x10013af3: cmp dword ptr [esi+3C4h],0
0x10013afa: je short 10013B28h
0x10013afc: push ebx
0x10013afd: mov ebx,[esp+0Ch]
0x10013b01: push edi
0x10013b02: mov edi,[esp+14h]
0x10013b06: push edi
0x10013b07: push ebx
0x10013b08: call 1000E650h
0x10013b0d: test al,al
0x10013b0f: je short 10013B26h
0x10013b11: mov eax,[esi+0Ch]
0x10013b14: mov ecx,[esi+3C4h]
0x10013b1a: imul eax,edi
0x10013b1d: mov dl,[esp+18h]
0x10013b21: add eax,ebx
0x10013b23: mov [eax+ecx],dl
0x10013b26: pop edi
0x10013b27: pop ebx
0x10013b28: pop esi
0x10013b29: ret 0Ch
```

`sub_10013af0` is called by sub_10013890, sub_10013900, sub_100143f0, sub_10015d20, sub_10017680, sub_1001b970. It delegates to sub_1000e650 (1 callees).

12.317 sub_10013b70 at 0x10013b70 (26 instructions)

```
0x10013b70: push ebx
0x10013b71: push esi
```

```
0x10013b72: mov esi,ecx
0x10013b74: cmp dword ptr [esi+3C4h],0
0x10013b7b: push edi
0x10013b7c: je short 10013BA5h
0x10013b7e: mov edi,[esp+14h]
0x10013b82: mov ebx,[esp+10h]
0x10013b86: push edi
0x10013b87: push ebx
0x10013b88: call 1000E650h
0x10013b8d: test al,al
0x10013b8f: je short 10013BA5h
0x10013b91: mov eax,[esi+0Ch]
0x10013b94: imul eax,edi
0x10013b97: add eax,ebx
0x10013b99: add eax,[esi+3C4h]
0x10013b9f: pop edi
0x10013ba0: pop esi
0x10013ba1: pop ebx
0x10013ba2: ret 8
0x10013ba5: pop edi
0x10013ba6: pop esi
0x10013ba7: xor eax,eax
0x10013ba9: pop ebx
0x10013baa: ret 8
```

`sub_10013b70` is called by sub_100187f0, sub_1001ddc0. It delegates to sub_1000e650 (1 callees).

12.318 sub_10013bb0 at 0x10013bb0 (6 instructions)

```
0x10013bb0: mov eax,[ecx+0Ch]
0x10013bb3: imul eax,[esp+8]
0x10013bb8: add eax,[ecx+3C4h]
0x10013bbe: mov ecx,[esp+4]
0x10013bc2: mov al,[eax+ecx]
0x10013bc5: ret 8
```

`sub_10013bb0` is called by sub_100130e0, sub_100131f0. It calls nothing else.

12.319 sub_10013bd0 at 0x10013bd0 (31 instructions)

```
0x10013bd0: push ecx
0x10013bd1: push esi
0x10013bd2: push edi
0x10013bd3: mov edi,ecx
0x10013bd5: xor esi,esi
0x10013bd7: cmp [edi+28h],esi
0x10013bda: jbe short 10013C01h
0x10013bdc: lea esp,[esp]
0x10013be0: push esi
0x10013be1: lea eax,[esp+0Ch]
0x10013be5: push eax
0x10013be6: mov ecx,edi
0x10013be8: call 10012EC0h
0x10013bed: mov ecx,[eax]
0x10013bef: shr ecx,18h
0x10013bf2: test cl,cl
0x10013bf4: jne short 10013C07h
0x10013bf6: add esi,1
0x10013bf9: movzx edx,si
0x10013bfc: cmp edx,[edi+28h]
0x10013bff: jb short 10013BE0h
0x10013c01: pop edi
0x10013c02: xor al,al
0x10013c04: pop esi
0x10013c05: pop ecx
```

```
0x10013c06: ret
0x10013c07: pop edi
0x10013c08: mov al,1
0x10013c0a: pop esi
0x10013c0b: pop ecx
0x10013c0c: ret
```

`sub_10013bd0` is called by sub_100103d0. It delegates to sub_10012ec0 (1 callees).

12.320 sub_10013c10 at 0x10013c10 (71 instructions)

```
0x10013c10: sub esp,8
0x10013c13: push esi
0x10013c14: mov esi,ecx
0x10013c16: cmp dword ptr [esi+3C4h],0
0x10013c1d: jne short 10013C26h
0x10013c1f: xor al,al
0x10013c21: pop esi
0x10013c22: add esp,8
0x10013c25: ret
0x10013c26: mov eax,[esi+0Ch]
0x10013c29: push eax
0x10013c2a: call dword ptr ds:[100FE550h]
0x10013c30: mov ecx,eax
0x10013c32: add esp,4
0x10013c35: test ecx,ecx
0x10013c37: mov [esp+4],ecx
0x10013c3b: je short 10013C1Fh
0x10013c3d: mov eax,[esi+10h]
0x10013c40: push ebx
0x10013c41: mov ebx,[esi+3C4h]
0x10013c47: push edi
0x10013c48: lea edi,[eax-1]
0x10013c4b: imul edi,[esi+0Ch]
0x10013c4f: cdq
0x10013c50: sub eax,edx
0x10013c52: sar eax,1
0x10013c54: add edi,ebx
0x10013c56: test eax,eax
0x10013c58: mov dword ptr [esp+10h],0
0x10013c60: jle short 10013CAFh
0x10013c62: push ebp
0x10013c63: mov ebp,[esi+0Ch]
0x10013c66: mov ecx,[esp+10h]
0x10013c6a: push ebp
0x10013c6b: push edi
0x10013c6c: push ecx
0x10013c6d: call 100228C6h
0x10013c72: push ebp
0x10013c73: push ebx
0x10013c74: push edi
0x10013c75: call 100228C6h
0x10013c7a: mov edx,[esi+0Ch]
0x10013c7d: mov eax,[esp+28h]
0x10013c81: push edx
0x10013c82: push eax
0x10013c83: push ebx
0x10013c84: call 100228C6h
0x10013c89: mov eax,[esi+0Ch]
0x10013c8c: mov ecx,[esp+38h]
0x10013c90: sub edi,eax
0x10013c92: add ebx,eax
0x10013c94: mov eax,[esi+10h]
0x10013c97: cdq
0x10013c98: sub eax,edx
0x10013c9a: add ecx,1
```

```
0x10013c9d: sar eax,1
0x10013c9f: add esp,24h
0x10013ca2: cmp ecx,eax
0x10013ca4: mov [esp+14h],ecx
0x10013ca8: jl short 10013C63h
0x10013caa: mov ecx,[esp+10h]
0x10013cae: pop ebp
0x10013caf: push ecx
0x10013cb0: call dword ptr ds:[100FE554h]
0x10013cb6: add esp,4
0x10013cb9: pop edi
0x10013cba: pop ebx
0x10013cbb: mov al,1
0x10013cbd: pop esi
0x10013cbe: add esp,8
0x10013cc1: ret
```

`sub_10013c10` is called by sub_10010810. It delegates to j_memcpy (1 callees).

12.321 sub_10013da0 at 0x10013da0 (18 instructions)

```
0x10013da0: push esi
0x10013da1: mov esi,ecx
0x10013da3: mov eax,[esi+3C0h]
0x10013da9: test eax,eax
0x10013dab: je short 10013DC1h
0x10013dad: push eax
0x10013dae: call dword ptr ds:[100FE554h]
0x10013db4: add esp,4
0x10013db7: mov dword ptr [esi+3C0h],0
0x10013dc1: mov eax,[esi+0Ch]
0x10013dc4: mov ecx,[esi+10h]
0x10013dc7: mov [esi+170h],eax
0x10013dcd: mov [esi+17Ch],ecx
0x10013dd3: mov dword ptr [esi+174h],0
0x10013ddd: mov dword ptr [esi+178h],0
0x10013de7: mov al,1
0x10013de9: pop esi
0x10013dea: ret
```

`sub_10013da0` is called by sub_1000ead0. It calls nothing else.

12.322 sub_10013df0 at 0x10013df0 (13 instructions)

```
0x10013df0: mov eax,[ecx+3C0h]
0x10013df6: test eax,eax
0x10013df8: jne short 10013DFFh
0x10013dfa: mov al,1
0x10013dfc: ret 8
0x10013dff: mov ecx,[ecx+0Ch]
0x10013e02: imul ecx,[esp+8]
0x10013e07: mov edx,[esp+4]
0x10013e0b: add ecx,eax
0x10013e0d: xor eax,eax
0x10013e0f: cmp [ecx+edx],al
0x10013e12: setne al
0x10013e15: ret 8
```

`sub_10013df0` is called by sub_100118f0. It calls nothing else.

12.323 sub_10013e20 at 0x10013e20 (48 instructions)

```
0x10013e20: push esi
0x10013e21: push edi
```

```
0x10013e22: mov edi,[esp+0Ch]
0x10013e26: cmp dword ptr [edi+3C0h],0
0x10013e2d: mov esi,ecx
0x10013e2f: je short 10013EACh
0x10013e31: mov ecx,[esi+0Ch]
0x10013e34: cmp ecx,[edi+0Ch]
0x10013e37: jne short 10013EACh
0x10013e39: mov eax,[esi+10h]
0x10013e3c: cmp eax,[edi+10h]
0x10013e3f: jne short 10013EACh
0x10013e41: cmp dword ptr [esi+3C0h],0
0x10013e48: jne short 10013E5Dh
0x10013e4a: imul eax,ecx
0x10013e4d: push eax
0x10013e4e: call dword ptr ds:[100FE550h]
0x10013e54: add esp,4
0x10013e57: mov [esi+3C0h],eax
0x10013e5d: mov eax,[esi+3C0h]
0x10013e63: test eax,eax
0x10013e65: je short 10013EACh
0x10013e67: mov ecx,[esi+0Ch]
0x10013e6a: imul ecx,[esi+10h]
0x10013e6e: mov edx,[edi+3C0h]
0x10013e74: push ecx
0x10013e75: push edx
0x10013e76: push eax
0x10013e77: call 100228C6h
0x10013e7c: mov edx,[edi+170h]
0x10013e82: lea eax,[edi+170h]
0x10013e88: lea ecx,[esi+170h]
0x10013e8e: mov [ecx],edx
0x10013e90: mov edx,[eax+4]
0x10013e93: mov [ecx+4],edx
0x10013e96: mov edx,[eax+8]
0x10013e99: add esp,0Ch
0x10013e9c: mov [ecx+8],edx
0x10013e9f: mov eax,[eax+0Ch]
0x10013ea2: pop edi
0x10013ea3: mov [ecx+0Ch],eax
0x10013ea6: mov al,1
0x10013ea8: pop esi
0x10013ea9: ret 4
0x10013eac: pop edi
0x10013ead: xor al,al
0x10013eaf: pop esi
0x10013eb0: ret 4
```

`sub_10013e20` is called by sub_100103d0. It delegates to j_memcpy (1 callees).

12.324 sub_10013f20 at 0x10013f20 (77 instructions)

```
0x10013f20: sub esp,8
0x10013f23: push esi
0x10013f24: mov esi,ecx
0x10013f26: cmp dword ptr [esi+3C0h],0
0x10013f2d: jne short 10013F36h
0x10013f2f: xor al,al
0x10013f31: pop esi
0x10013f32: add esp,8
0x10013f35: ret
0x10013f36: mov eax,[esi+0Ch]
0x10013f39: push eax
0x10013f3a: call dword ptr ds:[100FE550h]
0x10013f40: mov ecx,eax
0x10013f42: add esp,4
0x10013f45: test ecx,ecx
```

```
0x10013f47: mov [esp+4],ecx
0x10013f4b: je short 10013F2Fh
0x10013f4d: mov eax,[esi+10h]
0x10013f50: push ebx
0x10013f51: push ebp
0x10013f52: mov ebp,[esi+3C0h]
0x10013f58: lea ebx,[eax-1]
0x10013f5b: push edi
0x10013f5c: mov edi,[esi+0Ch]
0x10013f5f: imul ebx,edi
0x10013f62: cdq
0x10013f63: sub eax,edx
0x10013f65: sar eax,1
0x10013f67: add ebx,ebp
0x10013f69: test eax,eax
0x10013f6b: mov dword ptr [esp+14h],0
0x10013f73: jle short 10013FB9h
0x10013f75: push edi
0x10013f76: push ebx
0x10013f77: push ecx
0x10013f78: call 100228C6h
0x10013f7d: push edi
0x10013f7e: push ebp
0x10013f7f: push ebx
0x10013f80: call 100228C6h
0x10013f85: mov ecx,[esi+0Ch]
0x10013f88: mov edx,[esp+28h]
0x10013f8c: push ecx
0x10013f8d: push edx
0x10013f8e: push ebp
0x10013f8f: call 100228C6h
0x10013f94: mov eax,[esi+10h]
0x10013f97: mov ecx,[esp+38h]
0x10013f9b: mov edi,[esi+0Ch]
0x10013f9e: cdq
0x10013f9f: sub eax,edx
0x10013fa1: add ecx,1
0x10013fa4: sar eax,1
0x10013fa6: add esp,24h
0x10013fa9: sub ebx,edi
0x10013fab: add ebp,edi
0x10013fad: mov [esp+14h],ecx
0x10013fb1: cmp ecx,eax
0x10013fb3: mov ecx,[esp+10h]
0x10013fb7: jl short 10013F75h
0x10013fb9: push ecx
0x10013fba: call dword ptr ds:[100FE554h]
0x10013fc0: mov eax,[esi+10h]
0x10013fc3: mov ecx,[esi+174h]
0x10013fc9: add esp,4
0x10013fcc: mov edx,eax
0x10013fce: sub edx,[esi+17Ch]
0x10013fd4: pop edi
0x10013fd5: pop ebp
0x10013fd6: sub eax,ecx
0x10013fd8: mov [esi+17Ch],eax
0x10013fde: pop ebx
0x10013fdf: mov [esi+174h],edx
0x10013fe5: mov al,1
0x10013fe7: pop esi
0x10013fe8: add esp,8
0x10013feb: ret
```

`sub_10013f20` is called by sub_10010810. It delegates to j_memcpy (1 callees).

12.325 sub_10014140 at 0x10014140 (33 instructions)

```
0x10014140: push edi
0x10014141: mov edi,[esp+10h]
0x10014145: test edi,edi
0x10014147: jbe short 1001418Eh
0x10014149: mov edx,[esp+1Ch]
0x1001414d: mov ecx,[esp+8]
0x10014151: push ebx
0x10014152: push ebp
0x10014153: mov ebp,[esp+1Ch]
0x10014157: push esi
0x10014158: mov esi,[esp+18h]
0x1001415c: lea esp,[esp]
0x10014160: sub edi,1
0x10014163: test ebp,ebp
0x10014165: mov eax,ebp
0x10014167: jbe short 10014181h
0x10014169: lea esp,[esp]
0x10014170: mov bl,[esi]
0x10014172: sub eax,1
0x10014175: mov [ecx],bl
0x10014177: add ecx,1
0x1001417a: add esi,1
0x1001417d: test eax,eax
0x1001417f: ja short 10014170h
0x10014181: add ecx,[esp+24h]
0x10014185: add esi,edx
0x10014187: test edi,edi
0x10014189: ja short 10014160h
0x1001418b: pop esi
0x1001418c: pop ebp
0x1001418d: pop ebx
0x1001418e: pop edi
0x1001418f: ret 18h
```

`sub_10014140` is called by sub_100143f0. It calls nothing else.

12.326 sub_100141a0 at 0x100141a0 (152 instructions)

```
0x100141a0: mov edx,[esp+10h]
0x100141a4: push ebx
0x100141a5: push ebp
0x100141a6: push esi
0x100141a7: xor eax,eax
0x100141a9: cmp edx,8
0x100141ac: push edi
0x100141ad: jg near ptr 10014249h
0x100141b3: mov ebp,[esp+1Ch]
0x100141b7: test ebp,ebp
0x100141b9: jle near ptr 10014328h
0x100141bf: mov esi,[esp+18h]
0x100141c3: mov bl,1
0x100141c5: mov ecx,edx
0x100141c7: shl bl,cl
0x100141c9: sub bl,1
0x100141cc: mov [esp+1Ch],bl
0x100141d0: lea edi,[eax+edx]
0x100141d3: sub ebp,1
0x100141d6: cmp edi,8
0x100141d9: jg short 100141F2h
0x100141db: mov cl,8
0x100141dd: sub cl,dl
0x100141df: sub cl,al
0x100141e1: mov al,[esi]
0x100141e3: shr al,cl
0x100141e5: movzx cx,al
```

```
0x100141e9: movzx eax,cx
0x100141ec: mov [esp+18h],eax
0x100141f0: jmp short 1001421Eh
0x100141f2: movzx bx,byte ptr [esi]
0x100141f6: lea ecx,[eax+edx-8]
0x100141fa: shl bx,cl
0x100141fd: add esi,1
0x10014200: movzx ecx,bx
0x10014203: mov bl,[esp+1Ch]
0x10014207: mov [esp+18h],ecx
0x1001420b: mov cl,10h
0x1001420d: sub cl,dl
0x1001420f: sub cl,al
0x10014211: mov al,[esi]
0x10014213: shr al,cl
0x10014215: movzx cx,al
0x10014219: or [esp+18h],cx
0x1001421e: mov eax,edi
0x10014220: cmp eax,8
0x10014223: jl short 1001422Dh
0x10014225: sub eax,8
0x10014228: jne short 1001422Dh
0x1001422a: add esi,1
0x1001422d: mov edi,[esp+14h]
0x10014231: add dword ptr [esp+14h],1
0x10014236: mov cl,bl
0x10014238: and cl,[esp+18h]
0x1001423c: test ebp,ebp
0x1001423e: mov [edi],cl
0x10014240: jg short 100141D0h
0x10014242: pop edi
0x10014243: pop esi
0x10014244: pop ebp
0x10014245: pop ebx
0x10014246: ret 10h
0x10014249: cmp edx,10h
0x1001424c: jge short 100142ABh
0x1001424e: mov ebp,[esp+1Ch]
0x10014252: test ebp,ebp
0x10014254: jle near ptr 10014328h
0x1001425a: mov esi,[esp+18h]
0x1001425e: mov edi,[esp+14h]
0x10014262: movzx ecx,byte ptr [esi+2]
0x10014266: xor ebx,ebx
0x10014268: mov bh,[esi]
0x1001426a: sub ebp,1
0x1001426d: add edi,1
0x10014270: mov bl,[esi+1]
0x10014273: shl ebx,8
0x10014276: or ebx,ecx
0x10014278: movzx ecx,byte ptr [esi+3]
0x1001427c: shl ebx,8
0x1001427f: or ebx,ecx
0x10014281: mov ecx,18h
0x10014286: sub ecx,eax
0x10014288: shr ebx,cl
0x1001428a: add eax,edx
0x1001428c: cmp eax,8
0x1001428f: mov [edi-1],bl
0x10014292: jl short 100142A0h
0x10014294: mov ecx,eax
0x10014296: shr ecx,3
0x10014299: add esi,ecx
0x1001429b: neg ecx
0x1001429d: lea eax,[eax+ecx*8]
0x100142a0: test ebp,ebp
0x100142a2: jg short 10014262h
```

```
0x100142a4: pop edi
0x100142a5: pop esi
0x100142a6: pop ebp
0x100142a7: pop ebx
0x100142a8: ret 10h
0x100142ab: cmp edx,20h
0x100142ae: jge short 10014306h
0x100142b0: mov ebp,[esp+1Ch]
0x100142b4: test ebp,ebp
0x100142b6: jle short 10014328h
0x100142b8: mov esi,[esp+18h]
0x100142bc: mov edi,[esp+14h]
0x100142c0: movzx ecx,byte ptr [esi+2]
0x100142c4: xor ebx,ebx
0x100142c6: mov bh,[esi]
0x100142c8: sub ebp,1
0x100142cb: add edi,1
0x100142ce: mov bl,[esi+1]
0x100142d1: shl ebx,8
0x100142d4: or ebx,ecx
0x100142d6: movzx ecx,byte ptr [esi+3]
0x100142da: shl ebx,8
0x100142dd: or ebx,ecx
0x100142df: lea ecx,[eax+edx-8]
0x100142e3: shr ebx,cl
0x100142e5: add eax,edx
0x100142e7: cmp eax,8
0x100142ea: mov [edi-1],bl
0x100142ed: jl short 100142FBh
0x100142ef: mov ecx,eax
0x100142f1: shr ecx,3
0x100142f4: add esi,ecx
0x100142f6: neg ecx
0x100142f8: lea eax,[eax+ecx*8]
0x100142fb: test ebp,ebp
0x100142fd: jg short 100142C0h
0x100142ff: pop edi
0x10014300: pop esi
0x10014301: pop ebp
0x10014302: pop ebx
0x10014303: ret 10h
0x10014306: mov eax,[esp+1Ch]
0x1001430a: test eax,eax
0x1001430c: jle short 10014328h
0x1001430e: mov esi,[esp+18h]
0x10014312: mov ecx,[esp+14h]
0x10014316: mov dl,[esi+3]
0x10014319: sub eax,1
0x1001431c: mov [ecx],dl
0x1001431e: add ecx,1
0x10014321: add esi,4
0x10014324: test eax,eax
0x10014326: jg short 10014316h
0x10014328: pop edi
0x10014329: pop esi
0x1001432a: pop ebp
0x1001432b: pop ebx
0x1001432c: ret 10h
```

`sub_100141a0` is called by sub_100143f0. It calls nothing else.

12.327 sub_10014330 at 0x10014330 (64 instructions)

```
0x10014330: push esi
0x10014331: xor esi,esi
0x10014333: cmp [esp+10h],esi
```

```
0x10014337: jle near ptr 100143E1h
0x1001433d: mov edx,[esp+8]
0x10014341: push ebx
0x10014342: push ebp
0x10014343: mov ebp,[esp+1Ch]
0x10014347: push edi
0x10014348: mov edi,[esp+18h]
0x1001434c: lea esp,[esp]
0x10014350: movzx ecx,byte ptr [edi+2]
0x10014354: xor eax,eax
0x10014356: mov ah,[edi]
0x10014358: sub dword ptr [esp+1Ch],1
0x1001435d: mov al,[edi+1]
0x10014360: shl eax,8
0x10014363: or eax,ecx
0x10014365: movzx ecx,byte ptr [edi+3]
0x10014369: shl eax,8
0x1001436c: or eax,ecx
0x1001436e: mov ecx,20h
0x10014373: sub ecx,esi
0x10014375: sub ecx,ebp
0x10014377: shr eax,cl
0x10014379: cmp ebp,10h
0x1001437c: movzx eax,ax
0x1001437f: jge short 10014396h
0x10014381: mov ebx,10h
0x10014386: sub ebx,ebp
0x10014388: mov cx,bx
0x1001438b: shl eax,cl
0x1001438d: rol ax,8
0x10014391: shr ax,cl
0x10014394: jmp short 1001439Ah
0x10014396: rol ax,8
0x1001439a: mov ecx,[esp+24h]
0x1001439e: movzx eax,ax
0x100143a1: movzx eax,ax
0x100143a4: lea eax,[ecx+eax*4]
0x100143a7: movzx ecx,byte ptr [eax]
0x100143aa: mov [edx],cl
0x100143ac: movzx ecx,byte ptr [eax+1]
0x100143b0: add edx,1
0x100143b3: mov [edx],cl
0x100143b5: mov al,[eax+2]
0x100143b8: add edx,1
0x100143bb: mov [edx],al
0x100143bd: add esi,ebp
0x100143bf: add edx,1
0x100143c2: cmp esi,8
0x100143c5: jl short 100143D3h
0x100143c7: mov eax,esi
0x100143c9: shr eax,3
0x100143cc: add edi,eax
0x100143ce: neg eax
0x100143d0: lea esi,[esi+eax*8]
0x100143d3: cmp dword ptr [esp+1Ch],0
0x100143d8: jg near ptr 10014350h
0x100143de: pop edi
0x100143df: pop ebp
0x100143e0: pop ebx
0x100143e1: pop esi
0x100143e2: ret 14h
```

`sub_10014330` is called by sub_100143f0. It calls nothing else.

12.328 sub_100143f0 at 0x100143f0 (1383 instructions)

```
0x100143f0: push ebp
0x100143f1: lea ebp,[esp-74h]
0x100143f5: sub esp,74h
0x100143f8: push 0FFFFFFFFh
0x100143fa: push 100FC510h
0x100143ff: mov eax,fs:[0]
0x10014405: push eax
0x10014406: sub esp,58h
0x10014409: push ebx
0x1001440a: push esi
0x1001440b: push edi
0x1001440c: mov eax,ds:[10136220h]
0x10014411: xor eax,ebp
0x10014413: push eax
0x10014414: lea eax,[ebp-0Ch]
0x10014417: mov fs:[0],eax
0x1001441d: mov [ebp-10h],esp
0x10014420: mov ebx,ecx
0x10014422: mov [ebp+20h],ebx
0x10014425: mov eax,[ebp+7Ch]
0x10014428: push 100FE928h
0x1001442d: push eax
0x1001442e: call 100209C0h
0x10014433: mov edi,eax
0x10014435: xor eax,eax
0x10014437: mov [ebp+60h],eax
0x1001443a: mov [ebp+6Ch],eax
0x1001443d: mov eax,1
0x10014442: or esi,0FFFFFFFFh
0x10014445: add esp,8
0x10014448: test edi,edi
0x1001444a: mov [ebp+40h],edi
0x1001444d: mov [ebp+5Ch],eax
0x10014450: mov [ebp+48h],eax
0x10014453: mov [ebp+28h],esi
0x10014456: mov dword ptr [ebp+10h],0
0x1001445d: mov [ebp-24h],eax
0x10014460: mov [ebp-28h],eax
0x10014463: mov dword ptr [ebp-4],0
0x1001446a: jne short 10014481h
0x1001446c: push 1013083Ch
0x10014471: lea ecx,[ebp-60h]
0x10014474: push ecx
0x10014475: mov dword ptr [ebp-60h],100FEB04h
0x1001447c: call 100229A6h
0x10014481: push edi
0x10014482: call 100678CFh
0x10014487: movzx edx,ax
0x1001448a: movzx eax,word ptr [ebx+15Ch]
0x10014491: push eax
0x10014492: push edi
0x10014493: mov [ebx+160h],edx
0x10014499: call 10067905h
0x1001449e: add esp,0Ch
0x100144a1: test eax,eax
0x100144a3: jne short 100144BAh
0x100144a5: push 1013083Ch
0x100144aa: lea ecx,[ebp-5Ch]
0x100144ad: push ecx
0x100144ae: mov dword ptr [ebp-5Ch],100FEADCh
0x100144b5: call 100229A6h
0x100144ba: lea edx,[ebp+6Ch]
0x100144bd: push edx
0x100144be: push 100h
0x100144c3: push edi
0x100144c4: call 10067985h
```

```
0x100144c9: lea eax,[ebp+60h]
0x100144cc: push eax
0x100144cd: push 101h
0x100144d2: push edi
0x100144d3: call 10067985h
0x100144d8: lea ecx,[ebp+48h]
0x100144db: push ecx
0x100144dc: push 115h
0x100144e1: push edi
0x100144e2: call 10067985h
0x100144e7: lea edx,[ebp+5Ch]
0x100144ea: push edx
0x100144eb: push 102h
0x100144f0: push edi
0x100144f1: call 10067985h
0x100144f6: lea eax,[ebp+28h]
0x100144f9: push eax
0x100144fa: push 116h
0x100144ff: push edi
0x10014500: call 10067985h
0x10014505: lea ecx,[ebp+10h]
0x10014508: push ecx
0x10014509: push 106h
0x1001450e: push edi
0x1001450f: call 10067985h
0x10014514: add esp,48h
0x10014517: lea edx,[ebp-28h]
0x1001451a: push edx
0x1001451b: push 112h
0x10014520: push edi
0x10014521: call 10067985h
0x10014526: add esp,0Ch
0x10014529: cmp [ebx+148h],esi
0x1001452f: jne short 10014559h
0x10014531: mov eax,[ebp+6Ch]
0x10014534: mov ecx,[ebp+60h]
0x10014537: push 1013083Ch
0x1001453c: lea edx,[ebp-38h]
0x1001453f: push edx
0x10014540: mov [ebx+0Ch],eax
0x10014543: mov [ebx+10h],ecx
0x10014546: mov dword ptr [ebx+40h],6
0x1001454d: mov dword ptr [ebp-38h],100FEAC0h
0x10014554: call 100229A6h
0x10014559: lea eax,[ebp+7Ch]
0x1001455c: push eax
0x1001455d: push 128h
0x10014562: push edi
0x10014563: call 1006E579h
0x10014568: lea ecx,[ebp+1Ch]
0x1001456b: push ecx
0x1001456c: push 11Ah
0x10014571: push edi
0x10014572: call 10067985h
0x10014577: add esp,18h
0x1001457a: test eax,eax
0x1001457c: je short 100145A7h
0x1001457e: cmp word ptr [ebp+7Ch],3
0x10014583: jne short 10014597h
0x10014585: fld dword ptr [ebp+1Ch]
0x10014588: fmul qword ptr ds:[100FEAB8h]
0x1001458e: fadd qword ptr ds:[100FE7C8h]
0x10014594: fstp dword ptr [ebp+1Ch]
0x10014597: fld dword ptr [ebp+1Ch]
0x1001459a: call 100228D0h
0x1001459f: push eax
0x100145a0: mov ecx,ebx
```

```
0x100145a2: call 1000EF10h
0x100145a7: lea edx,[ebp+1Ch]
0x100145aa: push edx
0x100145ab: push 11Bh
0x100145b0: push edi
0x100145b1: call 10067985h
0x100145b6: add esp,0Ch
0x100145b9: test eax,eax
0x100145bb: je short 100145E6h
0x100145bd: cmp word ptr [ebp+7Ch],3
0x100145c2: jne short 100145D6h
0x100145c4: fld dword ptr [ebp+1Ch]
0x100145c7: fmul qword ptr ds:[100FEAB8h]
0x100145cd: fadd qword ptr ds:[100FE7C8h]
0x100145d3: fstp dword ptr [ebp+1Ch]
0x100145d6: fld dword ptr [ebp+1Ch]
0x100145d9: call 100228D0h
0x100145de: push eax
0x100145df: mov ecx,ebx
0x100145e1: call 1000EF70h
0x100145e6: lea eax,[ebp]
0x100145e9: push eax
0x100145ea: push 11Eh
0x100145ef: push edi
0x100145f0: call 10067985h
0x100145f5: add esp,0Ch
0x100145f8: test eax,eax
0x100145fa: je short 1001460Ah
0x100145fc: fld dword ptr [ebp]
0x100145ff: call 100228D0h
0x10014604: mov [ebx+184h],eax
0x1001460a: lea ecx,[ebp]
0x1001460d: push ecx
0x1001460e: push 11Fh
0x10014613: push edi
0x10014614: call 10067985h
0x10014619: add esp,0Ch
0x1001461c: test eax,eax
0x1001461e: je short 1001462Eh
0x10014620: fld dword ptr [ebp]
0x10014623: call 100228D0h
0x10014628: mov [ebx+188h],eax
0x1001462e: mov edx,[ebp+60h]
0x10014631: cmp [ebp+28h],edx
0x10014634: mov dword ptr [ebx+28h],0
0x1001463b: mov [ebx+14Ch],esi
0x10014641: jbe short 10014658h
0x10014643: push edx
0x10014644: push 116h
0x10014649: push edi
0x1001464a: mov [ebp+28h],edx
0x1001464d: call 1006796Fh
0x10014652: mov edx,[ebp+60h]
0x10014655: add esp,0Ch
0x10014658: mov eax,[ebp+10h]
0x1001465b: cmp ax,2
0x1001465f: je short 100146A2h
0x10014661: cmp ax,6
0x10014665: je short 100146A2h
0x10014667: cmp ax,5
0x1001466b: je short 100146A2h
0x1001466d: cmp ax,804Ch
0x10014671: je short 100146A2h
0x10014673: cmp ax,804Dh
0x10014677: je short 100146A2h
0x10014679: xor esi,esi
0x1001467b: cmp ax,1
```

```
0x1001467f: je short 100146A9h
0x10014681: test ax,ax
0x10014684: je short 100146A9h
0x10014686: cmp ax,3
0x1001468a: je short 100146A9h
0x1001468c: mov ecx,[ebp+5Ch]
0x1001468f: cmp cx,4
0x10014693: jne short 100146BFh
0x10014695: mov dword ptr [ebx+28h],10h
0x1001469c: mov [ebx+16h],cx
0x100146a0: jmp short 100146E1h
0x100146a2: mov esi,1
0x100146a7: jmp short 100146DBh
0x100146a9: mov ecx,[ebp+5Ch]
0x100146ac: cmp cx,1
0x100146b0: jne short 1001468Fh
0x100146b2: mov [ebx+16h],cx
0x100146b6: mov dword ptr [ebx+28h],2
0x100146bd: jmp short 100146E1h
0x100146bf: cmp cx,8
0x100146c3: mov dword ptr [ebx+28h],100h
0x100146ca: mov word ptr [ebx+16h],8
0x100146d0: jbe short 100146E1h
0x100146d2: cmp ax,3
0x100146d6: jne short 100146E1h
0x100146d8: mov [ebx+28h],esi
0x100146db: mov word ptr [ebx+16h],18h
0x100146e1: cmp dword ptr [ebx+148h],0
0x100146e8: je short 100146FFh
0x100146ea: push 1013083Ch
0x100146ef: lea edx,[ebp-50h]
0x100146f2: push edx
0x100146f3: mov dword ptr [ebp-50h],100FEAA8h
0x100146fa: call 100229A6h
0x100146ff: movzx eax,word ptr [ebx+16h]
0x10014703: mov ecx,[ebp+6Ch]
0x10014706: push 6
0x10014708: push eax
0x10014709: push edx
0x1001470a: push ecx
0x1001470b: mov ecx,ebx
0x1001470d: call 1000EAD0h
0x10014712: cmp dword ptr [ebx+4],0
0x10014716: jne short 1001472Dh
0x10014718: push 1013083Ch
0x1001471d: lea edx,[ebp-40h]
0x10014720: push edx
0x10014721: mov dword ptr [ebp-40h],100FEA88h
0x10014728: call 100229A6h
0x1001472d: cmp word ptr [ebp+48h],4
0x10014732: jne short 1001473Bh
0x10014734: mov ecx,ebx
0x10014736: call 100139C0h
0x1001473b: cmp word ptr [ebp+48h],2
0x10014740: jne short 10014750h
0x10014742: cmp word ptr [ebp+5Ch],8
0x10014747: jne short 10014750h
0x10014749: mov ecx,ebx
0x1001474b: call 100139C0h
0x10014750: lea eax,[ebp-24h]
0x10014753: push eax
0x10014754: push 103h
0x10014759: push edi
0x1001475a: call 10067985h
0x1001475f: movzx ecx,word ptr [ebp-24h]
0x10014763: add esp,0Ch
0x10014766: push 0
```

```
0x10014768: push ecx
0x10014769: mov ecx,ebx
0x1001476b: call 1000F310h
0x10014770: test esi,esi
0x10014772: je near ptr 100148ABh
0x10014778: mov edx,[ebp+6Ch]
0x1001477b: imul edx,[ebp+60h]
0x1001477f: add edx,edx
0x10014781: add edx,edx
0x10014783: push edx
0x10014784: call 100EFDD0h
0x10014789: mov esi,eax
0x1001478b: add esp,4
0x1001478e: test esi,esi
0x10014790: mov [ebp+8],esi
0x10014793: jne short 100147AAh
0x10014795: push 1013083Ch
0x1001479a: lea eax,[ebp-30h]
0x1001479d: push eax
0x1001479e: mov dword ptr [ebp-30h],100FEA6Ch
0x100147a5: call 100229A6h
0x100147aa: mov ecx,[ebp+60h]
0x100147ad: mov edx,[ebp+6Ch]
0x100147b0: push 1
0x100147b2: push esi
0x100147b3: push ecx
0x100147b4: push edx
0x100147b5: push edi
0x100147b6: call 1006E00Bh
0x100147bb: add esp,14h
0x100147be: test eax,eax
0x100147c0: jne short 100147E0h
0x100147c2: push esi
0x100147c3: call 100209E0h
0x100147c8: add esp,4
0x100147cb: push 1013083Ch
0x100147d0: lea eax,[ebp-48h]
0x100147d3: push eax
0x100147d4: mov dword ptr [ebp-48h],100FEA54h
0x100147db: call 100229A6h
0x100147e0: mov ecx,[ebx+34h]
0x100147e3: mov [ebp+70h],ecx
0x100147e6: mov ecx,[ebp+6Ch]
0x100147e9: mov [ebp+3Ch],esi
0x100147ec: mov dword ptr [ebp+64h],0
0x100147f3: mov edx,[ebp+64h]
0x100147f6: cmp edx,[ebp+60h]
0x100147f9: jae near ptr 1001488Fh
0x100147ff: cmp dword ptr [ebx+148h],0
0x10014806: je short 10014826h
0x10014808: push esi
0x10014809: call 100209E0h
0x1001480e: add esp,4
0x10014811: push 1013083Ch
0x10014816: lea eax,[ebp-58h]
0x10014819: push eax
0x1001481a: mov dword ptr [ebp-58h],100FEAA8h
0x10014821: call 100229A6h
0x10014826: mov edi,[ebp+70h]
0x10014829: mov eax,[ebp+3Ch]
0x1001482c: xor esi,esi
0x1001482e: mov edi,edi
0x10014830: cmp esi,ecx
0x10014832: jae short 10014877h
0x10014834: movzx ecx,byte ptr [eax+esi*4+2]
0x10014839: mov [edi],cl
0x1001483b: mov dl,[eax+esi*4+1]
```

```
0x1001483f: add edi,1
0x10014842: mov [edi],dl
0x10014844: movzx ecx,byte ptr [eax+esi*4]
0x10014848: add edi,1
0x1001484b: mov [edi],cl
0x1001484d: add edi,1
0x10014850: cmp word ptr [ebp+48h],4
0x10014855: jne short 1001486Fh
0x10014857: mov edx,[ebp+3Ch]
0x1001485a: movzx eax,byte ptr [edx+esi*4+3]
0x1001485f: mov ecx,[ebp+64h]
0x10014862: push eax
0x10014863: push ecx
0x10014864: push esi
0x10014865: mov ecx,ebx
0x10014867: call 10013AF0h
0x1001486c: mov eax,[ebp+3Ch]
0x1001486f: mov ecx,[ebp+6Ch]
0x10014872: add esi,1
0x10014875: jmp short 10014830h
0x10014877: mov esi,[ebp+8]
0x1001487a: lea edx,[eax+ecx*4]
0x1001487d: mov eax,[ebx+30h]
0x10014880: add [ebp+70h],eax
0x10014883: add dword ptr [ebp+64h],1
0x10014887: mov [ebp+3Ch],edx
0x1001488a: jmp near ptr 100147F3h
0x1001488f: push esi
0x10014890: call 100209E0h
0x10014895: mov eax,[ebp+40h]
0x10014898: add esp,4
0x1001489b: push eax
0x1001489c: call 10065D71h
0x100148a1: add esp,4
0x100148a4: mov al,1
0x100148a6: jmp near ptr 10015604h
0x100148ab: mov ecx,[ebp+5Ch]
0x100148ae: cmp cx,8
0x100148b2: jbe short 100148DEh
0x100148b4: cmp word ptr [ebp+10h],3
0x100148b9: jne short 100148DEh
0x100148bb: cmp cx,18h
0x100148bf: mov eax,1
0x100148c4: mov [ebp+24h],eax
0x100148c7: jbe short 10014914h
0x100148c9: push 1013083Ch
0x100148ce: lea ecx,[ebp-34h]
0x100148d1: push ecx
0x100148d2: mov dword ptr [ebp-34h],100FEA38h
0x100148d9: call 100229A6h
0x100148de: mov dword ptr [ebp+24h],0
0x100148e5: mov eax,100h
0x100148ea: push 4
0x100148ec: push eax
0x100148ed: call dword ptr ds:[100FE4FCh]
0x100148f3: mov edi,eax
0x100148f5: add esp,8
0x100148f8: test edi,edi
0x100148fa: mov [ebp+3Ch],eax
0x100148fd: jne short 10014918h
0x100148ff: push 1013083Ch
0x10014904: lea edx,[ebp-3Ch]
0x10014907: push edx
0x10014908: mov dword ptr [ebp-3Ch],100FEA18h
0x1001490f: call 100229A6h
0x10014914: shl eax,cl
0x10014916: jmp short 100148EAh
```

```
0x10014918: mov edx,[ebp+5Ch]
0x1001491b: cmp dx,8
0x1001491f: ja short 10014929h
0x10014921: movzx eax,dx
0x10014924: mov [ebp+54h],eax
0x10014927: jmp short 10014930h
0x10014929: mov dword ptr [ebp+54h],8
0x10014930: mov ecx,[ebp+10h]
0x10014933: movzx eax,cx
0x10014936: test eax,eax
0x10014938: jl near ptr 10014A87h
0x1001493e: cmp eax,1
0x10014941: jle near ptr 10014A68h
0x10014947: cmp eax,3
0x1001494a: jne near ptr 10014A87h
0x10014950: lea ecx,[ebp+68h]
0x10014953: push ecx
0x10014954: mov ecx,[ebp+40h]
0x10014957: lea edx,[ebp+30h]
0x1001495a: push edx
0x1001495b: lea eax,[ebp+70h]
0x1001495e: push eax
0x1001495f: push 140h
0x10014964: push ecx
0x10014965: call 10067985h
0x1001496a: mov eax,[ebp+24h]
0x1001496d: add esp,14h
0x10014970: test eax,eax
0x10014972: mov [ebp+14h],eax
0x10014975: jne short 100149B8h
0x10014977: mov ecx,[ebp+54h]
0x1001497a: mov edx,[ebp+68h]
0x1001497d: mov eax,1
0x10014982: shl eax,cl
0x10014984: mov ecx,[ebp+30h]
0x10014987: mov esi,eax
0x10014989: sub eax,1
0x1001498c: test esi,esi
0x1001498e: jle short 100149B2h
0x10014990: mov esi,[ebp+70h]
0x10014993: cmp word ptr [esi+eax*2],100h
0x10014999: jae short 100149ABh
0x1001499b: cmp word ptr [ecx+eax*2],100h
0x100149a1: jae short 100149ABh
0x100149a3: cmp word ptr [edx+eax*2],100h
0x100149a9: jb short 10014987h
0x100149ab: mov dword ptr [ebp+14h],1
0x100149b2: cmp dword ptr [ebp+24h],0
0x100149b6: je short 100149BEh
0x100149b8: movzx ecx,word ptr [ebp+5Ch]
0x100149bc: jmp short 100149C1h
0x100149be: mov ecx,[ebp+54h]
0x100149c1: mov esi,1
0x100149c6: shl esi,cl
0x100149c8: sub esi,1
0x100149cb: test esi,esi
0x100149cd: jl near ptr 10014A87h
0x100149d3: cmp dword ptr [ebp+14h],0
0x100149d7: je short 10014A46h
0x100149d9: mov edx,[ebp+70h]
0x100149dc: movzx ecx,word ptr [edx+esi*2]
0x100149e0: imul ecx,0FFh
0x100149e6: mov eax,80008001h
0x100149eb: imul ecx
0x100149ed: add edx,ecx
0x100149ef: mov ecx,[ebp+30h]
0x100149f2: sar edx,0Fh
```

```
0x100149f5: mov eax,edx
0x100149f7: shr eax,1Fh
0x100149fa: add eax,edx
0x100149fc: mov [edi+esi*4+2],al
0x10014a00: movzx ecx,word ptr [ecx+esi*2]
0x10014a04: imul ecx,0FFh
0x10014a0a: mov eax,80008001h
0x10014a0f: imul ecx
0x10014a11: add edx,ecx
0x10014a13: mov ecx,[ebp+68h]
0x10014a16: sar edx,0Fh
0x10014a19: mov eax,edx
0x10014a1b: shr eax,1Fh
0x10014a1e: add eax,edx
0x10014a20: mov [edi+esi*4+1],al
0x10014a24: movzx ecx,word ptr [ecx+esi*2]
0x10014a28: imul ecx,0FFh
0x10014a2e: mov eax,80008001h
0x10014a33: imul ecx
0x10014a35: add edx,ecx
0x10014a37: sar edx,0Fh
0x10014a3a: mov eax,edx
0x10014a3c: shr eax,1Fh
0x10014a3f: add eax,edx
0x10014a41: mov [edi+esi*4],al
0x10014a44: jmp short 100149C8h
0x10014a46: mov ecx,[ebp+70h]
0x10014a49: mov dl,[ecx+esi*2]
0x10014a4c: mov eax,[ebp+30h]
0x10014a4f: mov [edi+esi*4+2],dl
0x10014a53: mov cl,[eax+esi*2]
0x10014a56: mov edx,[ebp+68h]
0x10014a59: mov [edi+esi*4+1],cl
0x10014a5d: mov al,[edx+esi*2]
0x10014a60: mov [edi+esi*4],al
0x10014a63: jmp near ptr 100149C8h
0x10014a68: cmp dx,1
0x10014a6c: jne near ptr 10014C42h
0x10014a72: cmp cx,dx
0x10014a75: jne near ptr 10014C32h
0x10014a7b: mov byte ptr [edi+4],0FFh
0x10014a7f: mov byte ptr [edi+5],0FFh
0x10014a83: mov byte ptr [edi+6],0FFh
0x10014a87: cmp dword ptr [ebp+24h],0
0x10014a8b: jne short 10014AAFh
0x10014a8d: mov ecx,[ebp+54h]
0x10014a90: mov edx,1
0x10014a95: shl edx,cl
0x10014a97: mov ecx,ebx
0x10014a99: push edx
0x10014a9a: push edi
0x10014a9b: call 10013690h
0x10014aa0: push edi
0x10014aa1: call dword ptr ds:[100FE554h]
0x10014aa7: add esp,4
0x10014aaa: xor edi,edi
0x10014aac: mov [ebp+3Ch],edi
0x10014aaf: movzx eax,word ptr [ebp+48h]
0x10014ab3: movzx ecx,word ptr [ebp+5Ch]
0x10014ab7: mov edx,[ebp+40h]
0x10014aba: imul eax,ecx
0x10014abd: imul eax,[ebp+6Ch]
0x10014ac1: add eax,7
0x10014ac4: shr eax,3
0x10014ac7: push edx
0x10014ac8: mov [ebp+2Ch],eax
0x10014acb: call 1006634Ch
```

```
0x10014ad0: movzx ecx,word ptr [ebp+48h]
0x10014ad4: mov esi,eax
0x10014ad6: mov eax,[ebx+1Ch]
0x10014ad9: imul eax,ecx
0x10014adc: add esp,4
0x10014adf: cmp esi,eax
0x10014ae1: jle short 10014AE5h
0x10014ae3: mov esi,eax
0x10014ae5: mov eax,[ebx+30h]
0x10014ae8: imul eax,[ebp+28h]
0x10014aec: cmp esi,eax
0x10014aee: jge short 10014AF2h
0x10014af0: mov esi,eax
0x10014af2: mov eax,[ebp+5Ch]
0x10014af5: cmp ax,8
0x10014af9: jbe short 10014B13h
0x10014afb: cmp ax,10h
0x10014aff: je short 10014B13h
0x10014b01: movzx eax,ax
0x10014b04: add eax,7
0x10014b07: cdq
0x10014b08: and edx,7
0x10014b0b: add eax,edx
0x10014b0d: sar eax,3
0x10014b10: imul esi,eax
0x10014b13: mov edx,[ebp+40h]
0x10014b16: push edx
0x10014b17: call 10068D59h
0x10014b1c: xor ecx,ecx
0x10014b1e: add esp,4
0x10014b21: cmp eax,ecx
0x10014b23: mov [ebp+14h],eax
0x10014b26: mov [ebp+0Ch],ecx
0x10014b29: mov [ebp-2Ch],ecx
0x10014b2c: mov [ebp+54h],ecx
0x10014b2f: je short 10014B84h
0x10014b31: mov esi,[ebp+40h]
0x10014b34: lea eax,[ebp+0Ch]
0x10014b37: push eax
0x10014b38: push 142h
0x10014b3d: push esi
0x10014b3e: call 10067985h
0x10014b43: lea ecx,[ebp-2Ch]
0x10014b46: push ecx
0x10014b47: push 143h
0x10014b4c: push esi
0x10014b4d: call 10067985h
0x10014b52: mov edx,[ebp-2Ch]
0x10014b55: push esi
0x10014b56: mov [ebp+28h],edx
0x10014b59: call 1006883Eh
0x10014b5e: mov esi,eax
0x10014b60: mov eax,[ebp+6Ch]
0x10014b63: xor edx,edx
0x10014b65: div dword ptr [ebp+0Ch]
0x10014b68: add eax,1
0x10014b6b: imul esi,eax
0x10014b6e: mov eax,[ebp+40h]
0x10014b71: push eax
0x10014b72: call 1006883Eh
0x10014b77: push eax
0x10014b78: call dword ptr ds:[100FE550h]
0x10014b7e: add esp,24h
0x10014b81: mov [ebp+54h],eax
0x10014b84: cmp word ptr [ebp+5Ch],10h
0x10014b89: lea eax,[esi+esi]
0x10014b8c: je short 10014B90h
```

```
0x10014b8e: mov eax,esi
0x10014b90: push eax
0x10014b91: call dword ptr ds:[100FE550h]
0x10014b97: mov [ebp+58h],eax
0x10014b9a: xor eax,eax
0x10014b9c: add esp,4
0x10014b9f: cmp [ebp+14h],eax
0x10014ba2: mov [ebp+44h],eax
0x10014ba5: mov [ebp+30h],eax
0x10014ba8: jne short 10014BDFh
0x10014baa: cmp word ptr [ebp+5Ch],10h
0x10014baf: jne short 10014BDFh
0x10014bb1: movzx edx,word ptr [ebp+48h]
0x10014bb5: imul edx,[ebp+6Ch]
0x10014bb9: mov ecx,[ebp+2Ch]
0x10014bbc: lea eax,[edx*8+7]
0x10014bc3: mov [ebp+30h],ecx
0x10014bc6: mov ecx,[ebp+58h]
0x10014bc9: shr eax,3
0x10014bcc: push esi
0x10014bcd: mov [ebp+2Ch],eax
0x10014bd0: mov [ebp+44h],ecx
0x10014bd3: call dword ptr ds:[100FE550h]
0x10014bd9: add esp,4
0x10014bdc: mov [ebp+58h],eax
0x10014bdf: cmp dword ptr [ebp+58h],0
0x10014be3: jne near ptr 10014CC1h
0x10014be9: cmp dword ptr [ebp+44h],0
0x10014bed: je short 10014BFCh
0x10014bef: mov edx,[ebp+44h]
0x10014bf2: push edx
0x10014bf3: call dword ptr ds:[100FE554h]
0x10014bf9: add esp,4
0x10014bfc: test edi,edi
0x10014bfe: je short 10014C0Ah
0x10014c00: push edi
0x10014c01: call dword ptr ds:[100FE554h]
0x10014c07: add esp,4
0x10014c0a: cmp dword ptr [ebp+54h],0
0x10014c0e: je short 10014C1Dh
0x10014c10: mov eax,[ebp+54h]
0x10014c13: push eax
0x10014c14: call dword ptr ds:[100FE554h]
0x10014c1a: add esp,4
0x10014c1d: push 1013083Ch
0x10014c22: lea ecx,[ebp-44h]
0x10014c25: push ecx
0x10014c26: mov dword ptr [ebp-44h],100FE9F4h
0x10014c2d: call 100229A6h
0x10014c32: mov byte ptr [edi],0FFh
0x10014c35: mov byte ptr [edi+1],0FFh
0x10014c39: mov byte ptr [edi+2],0FFh
0x10014c3d: jmp near ptr 10014A87h
0x10014c42: cmp cx,1
0x10014c46: mov dword ptr [ebp+70h],0
0x10014c4d: jne short 10014C86h
0x10014c4f: nop
0x10014c50: mov ecx,[ebp+54h]
0x10014c53: mov eax,1
0x10014c58: shl eax,cl
0x10014c5a: cmp [ebp+70h],eax
0x10014c5d: jge near ptr 10014A87h
0x10014c63: lea ecx,[eax-1]
0x10014c66: mov eax,0FFh
0x10014c6b: cdq
0x10014c6c: idiv ecx
0x10014c6e: mov ecx,[ebp+70h]
```

```
0x10014c71: imul cl
0x10014c73: mov [edi+ecx*4],al
0x10014c76: mov [edi+ecx*4+1],al
0x10014c7a: mov [edi+ecx*4+2],al
0x10014c7e: add ecx,1
0x10014c81: mov [ebp+70h],ecx
0x10014c84: jmp short 10014C50h
0x10014c86: mov ecx,[ebp+54h]
0x10014c89: mov eax,1
0x10014c8e: shl eax,cl
0x10014c90: cmp [ebp+70h],eax
0x10014c93: jge near ptr 10014A87h
0x10014c99: lea ecx,[eax-1]
0x10014c9c: mov eax,0FFh
0x10014ca1: cdq
0x10014ca2: idiv ecx
0x10014ca4: mov edx,[ebp+70h]
0x10014ca7: or cl,0FFh
0x10014caa: imul dl
0x10014cac: sub cl,al
0x10014cae: mov [edi+edx*4],cl
0x10014cb1: mov [edi+edx*4+1],cl
0x10014cb5: mov [edi+edx*4+2],cl
0x10014cb9: add edx,1
0x10014cbc: mov [ebp+70h],edx
0x10014cbf: jmp short 10014C86h
0x10014cc1: xor esi,esi
0x10014cc3: cmp [ebp+44h],esi
0x10014cc6: mov [ebp+18h],esi
0x10014cc9: je short 10014CDDh
0x10014ccb: mov edx,[ebp+60h]
0x10014cce: push edx
0x10014ccf: call dword ptr ds:[100FE550h]
0x10014cd5: add esp,4
0x10014cd8: mov [ebp+18h],eax
0x10014cdb: mov esi,eax
0x10014cdd: mov ecx,[ebp+28h]
0x10014ce0: xor eax,eax
0x10014ce2: mov [ebp+4Ch],eax
0x10014ce5: mov edx,[ebp+60h]
0x10014ce8: cmp eax,edx
0x10014cea: jae near ptr 10014E49h
0x10014cf0: cmp dword ptr [ebx+148h],0
0x10014cf7: je short 10014D1Bh
0x10014cf9: mov eax,[ebp+58h]
0x10014cfc: push eax
0x10014cfd: call dword ptr ds:[100FE554h]
0x10014d03: add esp,4
0x10014d06: push 1013083Ch
0x10014d0b: lea ecx,[ebp-4Ch]
0x10014d0e: push ecx
0x10014d0f: mov dword ptr [ebp-4Ch],100FEAA8h
0x10014d16: call 100229A6h
0x10014d1b: lea esi,[eax+ecx]
0x10014d1e: cmp esi,edx
0x10014d20: jbe short 10014D29h
0x10014d22: sub edx,eax
0x10014d24: mov [ebp+50h],edx
0x10014d27: jmp short 10014D2Ch
0x10014d29: mov [ebp+50h],ecx
0x10014d2c: cmp dword ptr [ebp+14h],0
0x10014d30: je near ptr 10014E0Dh
0x10014d36: mov edi,[ebp+40h]
0x10014d39: push edi
0x10014d3a: call 10065FAFh
0x10014d3f: mov esi,eax
0x10014d41: push edi
```

```
0x10014d42: mov [ebp+38h],esi
0x10014d45: call 10068686h
0x10014d4a: mov edi,eax
0x10014d4c: mov eax,esi
0x10014d4e: sub eax,edi
0x10014d50: add esp,8
0x10014d53: xor esi,esi
0x10014d55: mov [ebp+70h],eax
0x10014d58: mov [ebp+68h],esi
0x10014d5b: mov eax,esi
0x10014d5d: lea ecx,[ecx]
0x10014d60: cmp eax,[ebp+6Ch]
0x10014d63: jae near ptr 10014EBBh
0x10014d69: mov edx,[ebp+4Ch]
0x10014d6c: mov eax,[ebp+68h]
0x10014d6f: mov ecx,[ebp+54h]
0x10014d72: push 0
0x10014d74: push 0
0x10014d76: push edx
0x10014d77: mov edx,[ebp+40h]
0x10014d7a: push eax
0x10014d7b: push ecx
0x10014d7c: push edx
0x10014d7d: call 100683BAh
0x10014d82: add esp,18h
0x10014d85: test eax,eax
0x10014d87: jge short 10014DB3h
0x10014d89: mov eax,[ebp+54h]
0x10014d8c: mov esi,ds:[100FE554h]
0x10014d92: push eax
0x10014d93: call esi
0x10014d95: mov ecx,[ebp+58h]
0x10014d98: push ecx
0x10014d99: call esi
0x10014d9b: add esp,8
0x10014d9e: push 1013083Ch
0x10014da3: lea edx,[ebp-54h]
0x10014da6: push edx
0x10014da7: mov dword ptr [ebp-54h],100FE9D8h
0x10014dae: call 100229A6h
0x10014db3: mov eax,[ebp+0Ch]
0x10014db6: lea ecx,[esi+eax]
0x10014db9: cmp ecx,[ebp+38h]
0x10014dbc: jbe short 10014DE0h
0x10014dbe: mov eax,[ebp+38h]
0x10014dc1: mov edx,[ebp+70h]
0x10014dc4: sub eax,esi
0x10014dc6: mov ecx,edi
0x10014dc8: sub ecx,eax
0x10014dca: push ecx
0x10014dcb: add ecx,edx
0x10014dcd: mov edx,[ebp+58h]
0x10014dd0: push ecx
0x10014dd1: mov ecx,[ebp+54h]
0x10014dd4: push eax
0x10014dd5: mov eax,[ebp+50h]
0x10014dd8: push eax
0x10014dd9: push ecx
0x10014dda: lea eax,[esi+edx]
0x10014ddd: push eax
0x10014dde: jmp short 10014DF6h
0x10014de0: mov ecx,[ebp+70h]
0x10014de3: mov edx,[ebp+50h]
0x10014de6: mov eax,[ebp+54h]
0x10014de9: push 0
0x10014deb: push ecx
0x10014dec: mov ecx,[ebp+58h]
```

```
0x10014def: push edi
0x10014df0: push edx
0x10014df1: push eax
0x10014df2: lea edx,[esi+ecx]
0x10014df5: push edx
0x10014df6: mov ecx,ebx
0x10014df8: call 10014140h
0x10014dfd: mov eax,[ebp+68h]
0x10014e00: add esi,edi
0x10014e02: add eax,[ebp+0Ch]
0x10014e05: mov [ebp+68h],eax
0x10014e08: jmp near ptr 10014D60h
0x10014e0d: mov ecx,[ebp+44h]
0x10014e10: test ecx,ecx
0x10014e12: je short 10014E19h
0x10014e14: mov eax,[ebp+30h]
0x10014e17: jmp short 10014E1Fh
0x10014e19: mov eax,[ebp+2Ch]
0x10014e1c: mov ecx,[ebp+58h]
0x10014e1f: imul eax,[ebp+50h]
0x10014e23: mov esi,[ebp+40h]
0x10014e26: push eax
0x10014e27: mov eax,[ebp+4Ch]
0x10014e2a: push ecx
0x10014e2b: push 0
0x10014e2d: push eax
0x10014e2e: push esi
0x10014e2f: call 10065F13h
0x10014e34: add esp,0Ch
0x10014e37: push eax
0x10014e38: push esi
0x10014e39: call 10068247h
0x10014e3e: add esp,10h
0x10014e41: cmp eax,0FFFFFFFFh
0x10014e44: jne short 10014EBEh
0x10014e46: mov esi,[ebp+18h]
0x10014e49: mov ecx,[ebp+58h]
0x10014e4c: push ecx
0x10014e4d: call dword ptr ds:[100FE554h]
0x10014e53: mov eax,[ebp+44h]
0x10014e56: add esp,4
0x10014e59: test eax,eax
0x10014e5b: je short 10014E67h
0x10014e5d: push eax
0x10014e5e: call dword ptr ds:[100FE554h]
0x10014e64: add esp,4
0x10014e67: test esi,esi
0x10014e69: je near ptr 1001548Fh
0x10014e6f: cmp word ptr [ebp+48h],1
0x10014e74: jne near ptr 10015481h
0x10014e7a: cmp word ptr [ebp+5Ch],10h
0x10014e7f: jne near ptr 10015481h
0x10014e85: cmp dword ptr [ebp+24h],0
0x10014e89: jne near ptr 10015481h
0x10014e8f: mov edx,8
0x10014e94: mov [ebp+38h],edx
0x10014e97: xor ecx,ecx
0x10014e99: lea esp,[esp]
0x10014ea0: cmp ecx,[ebp+60h]
0x10014ea3: jae near ptr 10015441h
0x10014ea9: movzx eax,byte ptr [esi+ecx]
0x10014ead: cmp edx,eax
0x10014eaf: jle short 10014EB6h
0x10014eb1: mov [ebp+38h],eax
0x10014eb4: mov edx,eax
0x10014eb6: add ecx,1
0x10014eb9: jmp short 10014EA0h
```

```
0x10014ebb: mov edi,[ebp+3Ch]
0x10014ebe: mov dword ptr [ebp+64h],0
0x10014ec5: jmp short 10014ED0h
0x10014ed0: mov ecx,[ebp+64h]
0x10014ed3: cmp ecx,[ebp+50h]
0x10014ed6: jae near ptr 10015430h
0x10014edc: mov ecx,[ebp+50h]
0x10014edf: sub ecx,[ebp+64h]
0x10014ee2: add ecx,0FFFFFFFFh
0x10014ee5: mov edi,ecx
0x10014ee7: imul edi,[ebp+2Ch]
0x10014eeb: cmp word ptr [ebp+5Ch],10h
0x10014ef0: mov [ebp+38h],edi
0x10014ef3: jne near ptr 10014FBAh
0x10014ef9: cmp dword ptr [ebp+24h],0
0x10014efd: jne near ptr 10014FBAh
0x10014f03: imul ecx,[ebp+30h]
0x10014f07: cmp dword ptr [ebp+44h],0
0x10014f0b: mov [ebp-14h],ecx
0x10014f0e: je near ptr 10014F9Eh
0x10014f14: xor dl,dl
0x10014f16: xor eax,eax
0x10014f18: cmp eax,[ebp+2Ch]
0x10014f1b: mov [ebp+70h],eax
0x10014f1e: jae short 10014F38h
0x10014f20: mov esi,[ebp+44h]
0x10014f23: lea eax,[ecx+eax*2]
0x10014f26: mov al,[eax+esi+1]
0x10014f2a: cmp al,dl
0x10014f2c: jbe short 10014F30h
0x10014f2e: mov dl,al
0x10014f30: mov eax,[ebp+70h]
0x10014f33: add eax,1
0x10014f36: jmp short 10014F18h
0x10014f38: mov al,dl
0x10014f3a: neg al
0x10014f3c: sbb eax,eax
0x10014f3e: and eax,0FFFFFFF8h
0x10014f41: add eax,8
0x10014f44: jne short 10014F51h
0x10014f46: test dl,dl
0x10014f48: js short 10014F51h
0x10014f4a: add eax,1
0x10014f4d: add dl,dl
0x10014f4f: jmp short 10014F46h
0x10014f51: mov edx,[ebp+18h]
0x10014f54: sub edx,[ebp+50h]
0x10014f57: mov esi,[ebp+60h]
0x10014f5a: sub edx,[ebp+4Ch]
0x10014f5d: add edx,[ebp+64h]
0x10014f60: mov [edx+esi],al
0x10014f63: mov esi,8
0x10014f68: sub esi,eax
0x10014f6a: xor eax,eax
0x10014f6c: lea esp,[esp]
0x10014f70: cmp eax,[ebp+2Ch]
0x10014f73: jae short 10014FBAh
0x10014f75: mov edx,[ebp+44h]
0x10014f78: lea ecx,[ecx+eax*2]
0x10014f7b: add edx,ecx
0x10014f7d: xor ecx,ecx
0x10014f7f: mov ch,[edx+1]
0x10014f82: mov cl,[edx]
0x10014f84: mov edx,ecx
0x10014f86: mov ecx,esi
0x10014f88: sar edx,cl
0x10014f8a: lea ecx,[eax+edi]
```

```
0x10014f8d: mov edi,[ebp+58h]
0x10014f90: add eax,1
0x10014f93: mov [ecx+edi],dl
0x10014f96: mov edi,[ebp+38h]
0x10014f99: mov ecx,[ebp-14h]
0x10014f9c: jmp short 10014F70h
0x10014f9e: mov esi,[ebp+58h]
0x10014fa1: xor eax,eax
0x10014fa3: cmp eax,[ebp+6Ch]
0x10014fa6: jae short 10014FBAh
0x10014fa8: lea edx,[edi+eax*2]
0x10014fab: mov dl,[edx+esi+1]
0x10014faf: lea ecx,[eax+edi]
0x10014fb2: mov [ecx+esi],dl
0x10014fb5: add eax,1
0x10014fb8: jmp short 10014FA3h
0x10014fba: mov eax,[ebp+48h]
0x10014fbd: cmp ax,1
0x10014fc1: jne near ptr 100150BBh
0x10014fc7: cmp dword ptr [ebp+24h],0
0x10014fcb: je short 10015037h
0x10014fcd: cmp dword ptr [ebp+44h],0
0x10014fd1: mov eax,[ebp+3Ch]
0x10014fd4: movzx ecx,word ptr [ebp+5Ch]
0x10014fd8: mov edx,[ebp+6Ch]
0x10014fdb: push eax
0x10014fdc: push ecx
0x10014fdd: push edx
0x10014fde: je short 1001501Bh
0x10014fe0: mov ecx,[ebp+50h]
0x10014fe3: mov eax,[ebp+64h]
0x10014fe6: mov edx,ecx
0x10014fe8: sub edx,eax
0x10014fea: sub eax,ecx
0x10014fec: sub eax,[ebp+4Ch]
0x10014fef: sub edx,1
0x10014ff2: add eax,[ebp+60h]
0x10014ff5: imul edx,[ebp+30h]
0x10014ff9: imul eax,[ebx+30h]
0x10014ffd: add edx,[ebp+44h]
0x10015000: add eax,[ebx+34h]
0x10015003: push edx
0x10015004: push eax
0x10015005: mov ecx,ebx
0x10015007: call 10014330h
0x1001500c: add dword ptr [ebp+64h],1
0x10015010: mov ebx,[ebp+20h]
0x10015013: mov edi,[ebp+3Ch]
0x10015016: jmp near ptr 10014ED0h
0x1001501b: mov ecx,[ebp+64h]
0x1001501e: sub ecx,[ebp+50h]
0x10015021: mov eax,[ebp+58h]
0x10015024: sub ecx,[ebp+4Ch]
0x10015027: add edi,eax
0x10015029: add ecx,[ebp+60h]
0x1001502c: push edi
0x1001502d: imul ecx,[ebx+30h]
0x10015031: add ecx,[ebx+34h]
0x10015034: push ecx
0x10015035: jmp short 10015005h
0x10015037: mov eax,[ebp+5Ch]
0x1001503a: cmp ax,[ebx+16h]
0x1001503e: je short 1001507Eh
0x10015040: cmp ax,10h
0x10015044: je short 1001507Eh
0x10015046: mov ecx,[ebp+58h]
0x10015049: movzx edx,ax
```

```
0x1001504c: mov eax,[ebp+6Ch]
0x1001504f: push edx
0x10015050: mov edx,[ebp+64h]
0x10015053: sub edx,[ebp+50h]
0x10015056: push eax
0x10015057: sub edx,[ebp+4Ch]
0x1001505a: add edi,ecx
0x1001505c: add edx,[ebp+60h]
0x1001505f: push edi
0x10015060: imul edx,[ebx+30h]
0x10015064: add edx,[ebx+34h]
0x10015067: mov ecx,ebx
0x10015069: push edx
0x1001506a: call 100141A0h
0x1001506f: add dword ptr [ebp+64h],1
0x10015073: mov ebx,[ebp+20h]
0x10015076: mov edi,[ebp+3Ch]
0x10015079: jmp near ptr 10014ED0h
0x1001507e: mov eax,[ebx+30h]
0x10015081: mov ecx,[ebp+2Ch]
0x10015084: cmp ecx,eax
0x10015086: jb short 1001508Ah
0x10015088: mov ecx,eax
0x1001508a: mov edx,[ebp+64h]
0x1001508d: sub edx,[ebp+50h]
0x10015090: push ecx
0x10015091: sub edx,[ebp+4Ch]
0x10015094: mov ecx,[ebp+58h]
0x10015097: add edx,[ebp+60h]
0x1001509a: add edi,ecx
0x1001509c: imul edx,eax
0x1001509f: add edx,[ebx+34h]
0x100150a2: push edi
0x100150a3: push edx
0x100150a4: call 100228C6h
0x100150a9: mov ebx,[ebp+20h]
0x100150ac: mov edi,[ebp+3Ch]
0x100150af: add esp,0Ch
0x100150b2: add dword ptr [ebp+64h],1
0x100150b6: jmp near ptr 10014ED0h
0x100150bb: cmp ax,2
0x100150bf: jne short 1001512Ah
0x100150c1: mov esi,[ebp+64h]
0x100150c4: sub esi,[ebp+50h]
0x100150c7: xor edi,edi
0x100150c9: sub esi,[ebp+4Ch]
0x100150cc: mov [ebp+68h],edi
0x100150cf: add esi,[ebp+60h]
0x100150d2: cmp [ebx+3C4h],edi
0x100150d8: jne short 100150E1h
0x100150da: mov ecx,ebx
0x100150dc: call 100139C0h
0x100150e1: mov eax,[ebp+68h]
0x100150e4: cmp eax,[ebp+2Ch]
0x100150e7: jge near ptr 1001500Ch
0x100150ed: mov edx,[ebp+38h]
0x100150f0: mov ecx,eax
0x100150f2: mov eax,[ebp+58h]
0x100150f5: add ecx,edx
0x100150f7: lea ebx,[ecx+eax]
0x100150fa: movzx ecx,byte ptr [ebx]
0x100150fd: push ecx
0x100150fe: mov ecx,[ebp+20h]
0x10015101: push esi
0x10015102: push edi
0x10015103: call 10013330h
0x10015108: movzx edx,byte ptr [ebx+1]
```

```
0x1001510c: mov ecx,[ebp+20h]
0x1001510f: push edx
0x10015110: push esi
0x10015111: push edi
0x10015112: call 10013AF0h
0x10015117: add dword ptr [ebp+68h],2
0x1001511b: add edi,1
0x1001511e: cmp edi,[ebp+6Ch]
0x10015121: jl short 100150E1h
0x10015123: sub esi,1
0x10015126: xor edi,edi
0x10015128: jmp short 100150E1h
0x1001512a: cmp word ptr [ebx+16h],18h
0x1001512f: je short 10015152h
0x10015131: mov eax,[ebp+60h]
0x10015134: mov ecx,[ebp+6Ch]
0x10015137: push 6
0x10015139: push 18h
0x1001513b: push eax
0x1001513c: push ecx
0x1001513d: mov ecx,ebx
0x1001513f: call 1000EAD0h
0x10015144: cmp word ptr [ebp+48h],4
0x10015149: jne short 10015152h
0x1001514b: mov ecx,ebx
0x1001514d: call 100139C0h
0x10015152: mov ebx,[ebp+64h]
0x10015155: sub ebx,[ebp+50h]
0x10015158: mov edi,[ebp+58h]
0x1001515b: sub ebx,[ebp+4Ch]
0x1001515e: mov ecx,[ebp+6Ch]
0x10015161: xor eax,eax
0x10015163: add ebx,[ebp+60h]
0x10015166: mov [ebp+68h],eax
0x10015169: mov [ebp+70h],eax
0x1001516c: cmp [ebp+70h],ecx
0x1001516f: jae near ptr 1001500Ch
0x10015175: movzx esi,word ptr [ebp+48h]
0x10015179: imul esi,[ebp+70h]
0x1001517d: add esi,[ebp+38h]
0x10015180: movzx eax,byte ptr [esi+edi+1]
0x10015185: cmp eax,7Fh
0x10015188: movzx edx,byte ptr [esi+edi]
0x1001518c: movzx ecx,byte ptr [esi+edi+2]
0x10015191: mov [ebp-1Ch],edx
0x10015194: mov [ebp-14h],eax
0x10015197: mov [ebp+8],ecx
0x1001519a: jle short 100151A4h
0x1001519c: sub eax,100h
0x100151a1: mov [ebp-14h],eax
0x100151a4: cmp ecx,7Fh
0x100151a7: jle short 100151B2h
0x100151a9: sub ecx,100h
0x100151af: mov [ebp+8],ecx
0x100151b2: fild dword ptr [ebp-1Ch]
0x100151b5: mov eax,3
0x100151ba: fdiv qword ptr ds:[100FE9D0h]
0x100151c0: fadd qword ptr ds:[100FE9C8h]
0x100151c6: fdiv qword ptr ds:[100FE9C0h]
0x100151cc: fild dword ptr [ebp-14h]
0x100151cf: fmul qword ptr ds:[100FE9B8h]
0x100151d5: fadd st,st(1)
0x100151d7: fld1
0x100151d9: fld st(0)
0x100151db: test al,1
0x100151dd: je short 100151E1h
0x100151df: fmul st,st(2)
```

```
0x100151e1: shr eax,1
0x100151e3: jne near ptr 10015427h
0x100151e9: fstp st(2)
0x100151eb: mov eax,3
0x100151f0: fld st(2)
0x100151f2: fld st(1)
0x100151f4: test al,1
0x100151f6: je short 100151FAh
0x100151f8: fmul st,st(1)
0x100151fa: shr eax,1
0x100151fc: jne near ptr 1001541Eh
0x10015202: fstp st(1)
0x10015204: mov eax,3
0x10015209: fild dword ptr [ebp+8]
0x1001520c: fmul qword ptr ds:[100FE9B0h]
0x10015212: fsubp st(4),st
0x10015214: test al,1
0x10015216: je short 1001521Eh
0x10015218: fxch
0x1001521a: fmul st,st(3)
0x1001521c: fxch
0x1001521e: shr eax,1
0x10015220: jne near ptr 10015415h
0x10015226: fstp st(3)
0x10015228: fxch
0x1001522a: fmul qword ptr ds:[100FE9A8h]
0x10015230: fxch
0x10015232: fmul qword ptr ds:[100FE9A0h]
0x10015238: fld st(1)
0x1001523a: fmul qword ptr ds:[100FE998h]
0x10015240: fld st(3)
0x10015242: fmul qword ptr ds:[100FE990h]
0x10015248: fsubp
0x1001524a: fld st(1)
0x1001524c: fmul qword ptr ds:[100FE988h]
0x10015252: fsubp
0x10015254: fld st(3)
0x10015256: fmul qword ptr ds:[100FE980h]
0x1001525c: fld st(3)
0x1001525e: fmul qword ptr ds:[100FE978h]
0x10015264: fsubp
0x10015266: fld st(2)
0x10015268: fmul qword ptr ds:[100FE970h]
0x1001526e: faddp
0x10015270: fst qword ptr [ebp-20h]
0x10015273: fxch st(3)
0x10015275: fmul qword ptr ds:[100FE968h]
0x1001527b: fxch st(4)
0x1001527d: fmul qword ptr ds:[100FE960h]
0x10015283: fsubp st(4),st
0x10015285: fxch
0x10015287: fmul qword ptr ds:[100FE958h]
0x1001528d: faddp st(3),st
0x1001528f: fxch st(2)
0x10015291: fst qword ptr [ebp+4]
0x10015294: fld qword ptr ds:[100FE950h]
0x1001529a: fcomp st(3)
0x1001529c: fnstsw ax
0x1001529e: test ah,5
0x100152a1: jp short 100152C9h
0x100152a3: fstp st(0)
0x100152a5: fstp st(0)
0x100152a7: fld qword ptr ds:[100FE948h]
0x100152ad: call 100229A0h
0x100152b2: fmul qword ptr ds:[100FE940h]
0x100152b8: fsub qword ptr ds:[100FE938h]
0x100152be: fstp qword ptr [ebp-18h]
```

```
0x100152c1: fld qword ptr [ebp+4]
0x100152c4: fld qword ptr [ebp-20h]
0x100152c7: jmp short 100152D4h
0x100152c9: fxch st(2)
0x100152cb: fmul qword ptr ds:[100FE930h]
0x100152d1: fstp qword ptr [ebp-18h]
0x100152d4: fld qword ptr ds:[100FE950h]
0x100152da: fcomp
0x100152dc: fnstsw ax
0x100152de: test ah,5
0x100152e1: jp short 10015304h
0x100152e3: fstp st(1)
0x100152e5: fld qword ptr ds:[100FE948h]
0x100152eb: call 100229A0h
0x100152f0: fmul qword ptr ds:[100FE940h]
0x100152f6: fsub qword ptr ds:[100FE938h]
0x100152fc: fstp qword ptr [ebp-20h]
0x100152ff: fld qword ptr [ebp+4]
0x10015302: jmp short 1001530Dh
0x10015304: fmul qword ptr ds:[100FE930h]
0x1001530a: fstp qword ptr [ebp-20h]
0x1001530d: fld qword ptr ds:[100FE950h]
0x10015313: fcomp
0x10015315: fnstsw ax
0x10015317: test ah,5
0x1001531a: jp short 10015335h
0x1001531c: fld qword ptr ds:[100FE948h]
0x10015322: call 100229A0h
0x10015327: fmul qword ptr ds:[100FE940h]
0x1001532d: fsub qword ptr ds:[100FE938h]
0x10015333: jmp short 1001533Bh
0x10015335: fmul qword ptr ds:[100FE930h]
0x1001533b: fld qword ptr [ebp-18h]
0x1001533e: fld qword ptr ds:[100FE918h]
0x10015344: fmul st(1),st
0x10015346: fxch
0x10015348: call 100228D0h
0x1001534d: cmp eax,0FFh
0x10015352: jg short 10015365h
0x10015354: test eax,eax
0x10015356: jge short 1001535Eh
0x10015358: mov byte ptr [ebp+36h],0
0x1001535c: jmp short 1001536Eh
0x1001535e: cmp eax,0FFh
0x10015363: jle short 1001536Bh
0x10015365: mov byte ptr [ebp+36h],0FFh
0x10015369: jmp short 1001536Eh
0x1001536b: mov [ebp+36h],al
0x1001536e: fld qword ptr [ebp-20h]
0x10015371: fmul st,st(1)
0x10015373: call 100228D0h
0x10015378: cmp eax,0FFh
0x1001537d: jg short 10015390h
0x1001537f: test eax,eax
0x10015381: jge short 10015389h
0x10015383: mov byte ptr [ebp+35h],0
0x10015387: jmp short 10015399h
0x10015389: cmp eax,0FFh
0x1001538e: jle short 10015396h
0x10015390: mov byte ptr [ebp+35h],0FFh
0x10015394: jmp short 10015399h
0x10015396: mov [ebp+35h],al
0x10015399: fmulp
0x1001539b: call 100228D0h
0x100153a0: cmp eax,0FFh
0x100153a5: jg short 100153B8h
0x100153a7: test eax,eax
```

```
0x100153a9: jge short 100153B1h
0x100153ab: mov byte ptr [ebp+34h],0
0x100153af: jmp short 100153C1h
0x100153b1: cmp eax,0FFh
0x100153b6: jle short 100153BEh
0x100153b8: mov byte ptr [ebp+34h],0FFh
0x100153bc: jmp short 100153C1h
0x100153be: mov [ebp+34h],al
0x100153c1: mov eax,[ebp+34h]
0x100153c4: mov ecx,[ebp+68h]
0x100153c7: push 0
0x100153c9: push eax
0x100153ca: push ebx
0x100153cb: push ecx
0x100153cc: mov ecx,[ebp+20h]
0x100153cf: call 10013900h
0x100153d4: cmp word ptr [ebp+48h],4
0x100153d9: jne short 100153EEh
0x100153db: movzx edx,byte ptr [esi+edi+3]
0x100153e0: mov eax,[ebp+68h]
0x100153e3: mov ecx,[ebp+20h]
0x100153e6: push edx
0x100153e7: push ebx
0x100153e8: push eax
0x100153e9: call 10013AF0h
0x100153ee: mov eax,[ebp+68h]
0x100153f1: mov ecx,[ebp+6Ch]
0x100153f4: add dword ptr [ebp+70h],1
0x100153f8: add eax,1
0x100153fb: cmp eax,ecx
0x100153fd: mov [ebp+68h],eax
0x10015400: jl near ptr 1001516Ch
0x10015406: sub ebx,1
0x10015409: mov dword ptr [ebp+68h],0
0x10015410: jmp near ptr 1001516Ch
0x10015415: fld st(3)
0x10015417: fmulp st(4),st
0x10015419: jmp near ptr 10015214h
0x1001541e: fld st(1)
0x10015420: fmulp st(2),st
0x10015422: jmp near ptr 100151F4h
0x10015427: fld st(2)
0x10015429: fmulp st(3),st
0x1001542b: jmp near ptr 100151DBh
0x10015430: mov ecx,[ebp+28h]
0x10015433: add [ebp+4Ch],ecx
0x10015436: mov esi,[ebp+18h]
0x10015439: mov eax,[ebp+4Ch]
0x1001543c: jmp near ptr 10014CE5h
0x10015441: xor ecx,ecx
0x10015443: cmp ecx,[ebp+60h]
0x10015446: mov [ebp+64h],ecx
0x10015449: jae short 10015481h
0x1001544b: movzx eax,byte ptr [esi+ecx]
0x1001544f: cmp edx,eax
0x10015451: jge short 10015479h
0x10015453: sub eax,edx
0x10015455: mov edx,eax
0x10015457: mov eax,[ebx+30h]
0x1001545a: imul eax,ecx
0x1001545d: add eax,[ebx+34h]
0x10015460: xor esi,esi
0x10015462: cmp esi,[ebp+6Ch]
0x10015465: jae short 10015476h
0x10015467: mov cl,dl
0x10015469: shr byte ptr [eax],cl
0x1001546b: mov ecx,[ebp+64h]
```

```
0x1001546e: add esi,1
0x10015471: add eax,1
0x10015474: jmp short 10015462h
0x10015476: mov edx,[ebp+38h]
0x10015479: mov esi,[ebp+18h]
0x1001547c: add ecx,1
0x1001547f: jmp short 10015443h
0x10015481: test esi,esi
0x10015483: je short 1001548Fh
0x10015485: push esi
0x10015486: call dword ptr ds:[100FE554h]
0x1001548c: add esp,4
0x1001548f: cmp dword ptr [ebp+14h],0
0x10015493: je short 100154A2h
0x10015495: mov edx,[ebp+54h]
0x10015498: push edx
0x10015499: call dword ptr ds:[100FE554h]
0x1001549f: add esp,4
0x100154a2: test edi,edi
0x100154a4: je short 100154B0h
0x100154a6: push edi
0x100154a7: call dword ptr ds:[100FE554h]
0x100154ad: add esp,4
0x100154b0: movzx eax,word ptr [ebp-28h]
0x100154b4: add eax,0FFFFFFFEh
0x100154b7: cmp eax,6
0x100154ba: ja near ptr 1001559Ch
0x100154c0: jmp dword ptr [eax*4+1001561Ch]
0x1001559c: mov eax,[ebp+40h]
0x1001559f: push eax
0x100155a0: call 10065D71h
0x100155a5: add esp,4
0x100155a8: mov al,1
0x100155aa: jmp short 10015604h
0x10015604: mov ecx,[ebp-0Ch]
0x10015607: mov fs:[0],ecx
0x1001560e: pop ecx
0x1001560f: pop edi
0x10015610: pop esi
0x10015611: pop ebx
0x10015612: add ebp,74h
0x10015615: mov esp,ebp
0x10015617: pop ebp
0x10015618: ret 4
```

`sub_100143f0` is called by sub_1000f760. It works with the text "Unable to allocate TIFF palette", "Too big palette to handle", "No space for raster buffer". It accesses memory-mapped peripheral registers at 0x80008001. It delegates to sub_1000ead0, sub_1000ef10, sub_1000ef70 (34 callees).

12.329 sub_10015660 at 0x10015660 (21 instructions)

```
0x10015660: push esi
0x10015661: push edi
0x10015662: mov edi,[esp+10h]
0x10015666: mov eax,[edi+38h]
0x10015669: test eax,eax
0x1001566b: mov esi,ecx
0x1001566d: je short 10015678h
0x1001566f: push eax
0x10015670: call 100209E0h
0x10015675: add esp,4
0x10015678: mov eax,[esp+0Ch]
0x1001567c: mov [edi+38h],eax
0x1001567f: mov dword ptr [esi+0Ch],10015F70h
0x10015686: mov dword ptr [esi],100208E0h
0x1001568c: mov dword ptr [esi+8],10015640h
```

```
0x10015693: mov dword ptr [esi+4],10020900h
0x1001569a: mov [edi+34h],esi
0x1001569d: pop edi
0x1001569e: mov eax,esi
0x100156a0: pop esi
0x100156a1: ret 8
```

`sub_10015660` is called by sub_100156b0. It delegates to j_free (1 callees).

12.330 sub_100156b0 at 0x100156b0 (529 instructions)

```
0x100156b0: push ebp
0x100156b1: mov ebp,esp
0x100156b3: push 0FFFFFFFFh
0x100156b5: push 100FC530h
0x100156ba: mov eax,fs:[0]
0x100156c0: push eax
0x100156c1: sub esp,70h
0x100156c4: push ebx
0x100156c5: push esi
0x100156c6: push edi
0x100156c7: mov eax,ds:[10136220h]
0x100156cc: xor eax,ebp
0x100156ce: push eax
0x100156cf: lea eax,[ebp-0Ch]
0x100156d2: mov fs:[0],eax
0x100156d8: mov [ebp-10h],esp
0x100156db: mov [ebp-18h],ecx
0x100156de: mov esi,[ebp+8]
0x100156e1: xor ebx,ebx
0x100156e3: cmp esi,ebx
0x100156e5: jne short 100156FDh
0x100156e7: xor al,al
0x100156e9: mov ecx,[ebp-0Ch]
0x100156ec: mov fs:[0],ecx
0x100156f3: pop ecx
0x100156f4: pop edi
0x100156f5: pop esi
0x100156f6: pop ebx
0x100156f7: mov esp,ebp
0x100156f9: pop ebp
0x100156fa: ret 8
0x100156fd: mov [ebp+8],ebx
0x10015700: mov [ebp-38h],ebx
0x10015703: mov [ebp-30h],ebx
0x10015706: mov [ebp-34h],ebx
0x10015709: mov [ebp-4],ebx
0x1001570c: call 100AE1A0h
0x10015711: test eax,eax
0x10015713: je short 1001572Ah
0x10015715: push 1013083Ch
0x1001571a: lea eax,[ebp-3Ch]
0x1001571d: push eax
0x1001571e: mov dword ptr [ebp-3Ch],100FEC20h
0x10015725: call 100229A6h
0x1001572a: push 100FE928h
0x1001572f: push ebx
0x10015730: call 100ADCA0h
0x10015735: mov edi,eax
0x10015737: add esp,8
0x1001573a: cmp edi,ebx
0x1001573c: mov [ebp-38h],edi
0x1001573f: jne short 10015756h
0x10015741: push 1013083Ch
0x10015746: lea ecx,[ebp-40h]
0x10015749: push ecx
```

```
0x1001574a: mov dword ptr [ebp-40h],100FEBFCh
0x10015751: call 100229A6h
0x10015756: push edi
0x10015757: push esi
0x10015758: lea ecx,[ebp-7Ch]
0x1001575b: call 10015660h
0x10015760: push edi
0x10015761: call 100AE9C0h
0x10015766: mov esi,eax
0x10015768: add esp,4
0x1001576b: cmp esi,ebx
0x1001576d: mov [ebp-14h],esi
0x10015770: jge short 10015787h
0x10015772: push 1013083Ch
0x10015777: lea edx,[ebp-44h]
0x1001577a: push edx
0x1001577b: mov dword ptr [ebp-44h],100FEBE4h
0x10015782: call 100229A6h
0x10015787: push ebx
0x10015788: push esi
0x10015789: push edi
0x1001578a: call 100AF270h
0x1001578f: add esp,0Ch
0x10015792: cmp eax,ebx
0x10015794: mov [ebp+8],eax
0x10015797: jne short 100157B5h
0x10015799: push 1013083Ch
0x1001579e: lea eax,[ebp-48h]
0x100157a1: push eax
0x100157a2: mov dword ptr [ebp-14h],0FFFFFFFFh
0x100157a9: mov dword ptr [ebp-48h],100FEBC4h
0x100157b0: call 100229A6h
0x100157b5: push 3
0x100157b7: push esi
0x100157b8: mov byte ptr [ebp-1Ch],0
0x100157bc: call 100AF750h
0x100157c1: add esp,4
0x100157c4: push eax
0x100157c5: lea ecx,[ebp-1Ch]
0x100157c8: push ecx
0x100157c9: call dword ptr ds:[100FE568h]
0x100157cf: or edx,0FFFFFFFFh
0x100157d2: add esp,0Ch
0x100157d5: mov edi,100FEBC0h
0x100157da: lea esi,[ebp-1Ch]
0x100157dd: mov ecx,4
0x100157e2: xor eax,eax
0x100157e4: mov byte ptr [ebp-19h],0
0x100157e8: repe cmpsb
0x100157ea: mov [ebp-14h],edx
0x100157ed: jne short 100157F6h
0x100157ef: mov dword ptr [ebp-14h],0Bh
0x100157f6: mov edi,100FEBBCh
0x100157fb: lea esi,[ebp-1Ch]
0x100157fe: mov ecx,4
0x10015803: xor eax,eax
0x10015805: repe cmpsb
0x10015807: jne short 10015810h
0x10015809: mov dword ptr [ebp-14h],0Ch
0x10015810: mov edi,100FEBB8h
0x10015815: lea esi,[ebp-1Ch]
0x10015818: mov ecx,4
0x1001581d: xor eax,eax
0x1001581f: repe cmpsb
0x10015821: jne short 1001582Ah
0x10015823: mov dword ptr [ebp-14h],0Fh
0x1001582a: mov edi,100FEBB4h
```

```
0x1001582f: lea esi,[ebp-1Ch]
0x10015832: mov ecx,4
0x10015837: xor eax,eax
0x10015839: repe cmpsb
0x1001583b: jne short 10015844h
0x1001583d: mov dword ptr [ebp-14h],0Eh
0x10015844: mov edi,100FEBB0h
0x10015849: lea esi,[ebp-1Ch]
0x1001584c: mov ecx,4
0x10015851: xor eax,eax
0x10015853: repe cmpsb
0x10015855: jne short 1001585Eh
0x10015857: mov dword ptr [ebp-14h],0Dh
0x1001585e: mov edi,[ebp+8]
0x10015861: mov ecx,[edi+18h]
0x10015864: mov eax,[ecx]
0x10015866: mov ecx,[eax+10h]
0x10015869: mov esi,[eax+14h]
0x1001586c: mov ebx,[eax+18h]
0x1001586f: mov eax,[ebp-18h]
0x10015872: cmp [eax+148h],edx
0x10015878: mov [ebp-20h],ecx
0x1001587b: mov [ebp-24h],esi
0x1001587e: jne short 100158ADh
0x10015880: mov [eax+0Ch],ecx
0x10015883: mov ecx,[ebp-14h]
0x10015886: xor edx,edx
0x10015888: test ecx,ecx
0x1001588a: setl dl
0x1001588d: mov [eax+10h],esi
0x10015890: push 1013083Ch
0x10015895: mov dword ptr [ebp-4Ch],100FEAC0h
0x1001589c: sub edx,1
0x1001589f: and edx,ecx
0x100158a1: mov [eax+40h],edx
0x100158a4: lea eax,[ebp-4Ch]
0x100158a7: push eax
0x100158a8: call 100229A6h
0x100158ad: mov eax,[edi+10h]
0x100158b0: cmp eax,40h
0x100158b3: jg near ptr 10015C2Ah
0x100158b9: test eax,eax
0x100158bb: jl near ptr 10015C2Ah
0x100158c1: cmp eax,3
0x100158c4: jl short 1001593Ah
0x100158c6: cmp ebx,8
0x100158c9: jg short 1001593Ah
0x100158cb: push 401h
0x100158d0: call 100B1DF0h
0x100158d5: mov esi,eax
0x100158d7: add esp,4
0x100158da: test esi,esi
0x100158dc: jne short 100158F3h
0x100158de: push 1013083Ch
0x100158e3: lea ecx,[ebp-50h]
0x100158e6: push ecx
0x100158e7: mov dword ptr [ebp-50h],100FEB94h
0x100158ee: call 100229A6h
0x100158f3: push 0
0x100158f5: mov edx,edi
0x100158f7: push esi
0x100158f8: push edx
0x100158f9: call 100B0040h
0x100158fe: mov edi,eax
0x10015900: add esp,0Ch
0x10015903: test edi,edi
0x10015905: jne short 10015925h
```

```
0x10015907: push esi
0x10015908: call 100B1420h
0x1001590d: add esp,4
0x10015910: push 1013083Ch
0x10015915: lea eax,[ebp-54h]
0x10015918: push eax
0x10015919: mov dword ptr [ebp-54h],100FEB7Ch
0x10015920: call 100229A6h
0x10015925: mov ecx,[ebp+8]
0x10015928: push ecx
0x10015929: call 100AF0F0h
0x1001592e: push esi
0x1001592f: call 100B1420h
0x10015934: add esp,8
0x10015937: mov [ebp+8],edi
0x1001593a: mov edx,[ebp+8]
0x1001593d: mov eax,[edx+10h]
0x10015940: push 4
0x10015942: push eax
0x10015943: call dword ptr ds:[100FE4FCh]
0x10015949: mov edi,eax
0x1001594b: add esp,8
0x1001594e: mov [ebp-30h],edi
0x10015951: xor esi,esi
0x10015953: mov edx,[ebp+8]
0x10015956: cmp esi,[edx+10h]
0x10015959: jge short 1001598Ah
0x1001595b: mov ecx,[ebp-20h]
0x1001595e: push ecx
0x1001595f: push 1
0x10015961: call 100AE620h
0x10015966: add esp,8
0x10015969: test eax,eax
0x1001596b: mov [edi+esi*4],eax
0x1001596e: jne short 10015985h
0x10015970: push 1013083Ch
0x10015975: lea edx,[ebp-58h]
0x10015978: push edx
0x10015979: mov dword ptr [ebp-58h],100FEB5Ch
0x10015980: call 100229A6h
0x10015985: add esi,1
0x10015988: jmp short 10015953h
0x1001598a: cmp ebx,8
0x1001598d: jle short 10015997h
0x1001598f: add ebx,0FFFFFFF8h
0x10015992: mov [ebp-28h],ebx
0x10015995: jmp short 1001599Eh
0x10015997: mov dword ptr [ebp-28h],0
0x1001599e: mov ecx,[edx+10h]
0x100159a1: cmp ecx,3
0x100159a4: jne near ptr 10015B1Ah
0x100159aa: mov eax,[edx+18h]
0x100159ad: mov esi,[eax+4]
0x100159b0: mov eax,[eax]
0x100159b2: mov eax,[eax+10h]
0x100159b5: cmp eax,[esi+10h]
0x100159b8: jne near ptr 10015B1Ah
0x100159be: mov eax,[edx+18h]
0x100159c1: mov esi,[eax+8]
0x100159c4: mov eax,[eax+4]
0x100159c7: mov eax,[eax+10h]
0x100159ca: cmp eax,[esi+10h]
0x100159cd: jne near ptr 10015B1Ah
0x100159d3: mov eax,[edx+18h]
0x100159d6: mov esi,[eax+4]
0x100159d9: mov eax,[eax]
0x100159db: mov eax,[eax+14h]
```

```
0x100159de: cmp eax,[esi+14h]
0x100159e1: jne near ptr 10015B1Ah
0x100159e7: mov eax,[edx+18h]
0x100159ea: mov esi,[eax+8]
0x100159ed: mov eax,[eax+4]
0x100159f0: mov eax,[eax+14h]
0x100159f3: cmp eax,[esi+14h]
0x100159f6: jne near ptr 10015B1Ah
0x100159fc: mov eax,[edx+18h]
0x100159ff: mov esi,[eax+4]
0x10015a02: mov eax,[eax]
0x10015a04: mov eax,[eax+18h]
0x10015a07: cmp eax,[esi+18h]
0x10015a0a: jne near ptr 10015B1Ah
0x10015a10: mov eax,[edx+18h]
0x10015a13: mov edx,[eax+8]
0x10015a16: mov eax,[eax+4]
0x10015a19: mov eax,[eax+18h]
0x10015a1c: cmp eax,[edx+18h]
0x10015a1f: jne near ptr 10015B1Ah
0x10015a25: mov ecx,[ebp-14h]
0x10015a28: mov edx,[ebp-24h]
0x10015a2b: mov eax,[ebp-20h]
0x10015a2e: push ecx
0x10015a2f: mov ecx,[ebp-18h]
0x10015a32: push 18h
0x10015a34: push edx
0x10015a35: push eax
0x10015a36: call 1000EAD0h
0x10015a3b: test eax,eax
0x10015a3d: jne short 10015A54h
0x10015a3f: push 1013083Ch
0x10015a44: lea ecx,[ebp-5Ch]
0x10015a47: push ecx
0x10015a48: mov dword ptr [ebp-5Ch],100FE92Bh
0x10015a4f: call 100229A6h
0x10015a54: xor ebx,ebx
0x10015a56: cmp ebx,[ebp-24h]
0x10015a59: jge near ptr 10015C90h
0x10015a5f: xor esi,esi
0x10015a61: mov edx,[ebp+8]
0x10015a64: cmp esi,[edx+10h]
0x10015a67: jge short 10015A85h
0x10015a69: mov eax,[edi+esi*4]
0x10015a6c: mov ecx,[ebp-20h]
0x10015a6f: push eax
0x10015a70: push 1
0x10015a72: push ecx
0x10015a73: push ebx
0x10015a74: push 0
0x10015a76: push esi
0x10015a77: push edx
0x10015a78: call 100AF330h
0x10015a7d: add esp,1Ch
0x10015a80: add esi,1
0x10015a83: jmp short 10015A61h
0x10015a85: xor esi,esi
0x10015a87: cmp esi,[ebp-20h]
0x10015a8a: jge near ptr 10015B12h
0x10015a90: mov eax,[edi]
0x10015a92: cmp dword ptr [eax+14h],1
0x10015a96: jne short 10015AA2h
0x10015a98: mov eax,[eax+1Ch]
0x10015a9b: mov ecx,[eax]
0x10015a9d: mov eax,[ecx+esi*4]
0x10015aa0: jmp short 10015AAAh
0x10015aa2: mov edx,[eax+1Ch]
```

```
0x10015aa5: mov eax,[edx+esi*4]
0x10015aa8: mov eax,[eax]
0x10015aaa: mov ecx,[ebp-28h]
0x10015aad: sar eax,cl
0x10015aaf: mov [ebp+0Eh],al
0x10015ab2: mov eax,[edi+4]
0x10015ab5: cmp dword ptr [eax+14h],1
0x10015ab9: mov edx,[eax+1Ch]
0x10015abc: jne short 10015AC5h
0x10015abe: mov eax,[edx]
0x10015ac0: mov eax,[eax+esi*4]
0x10015ac3: jmp short 10015ACAh
0x10015ac5: mov eax,[edx+esi*4]
0x10015ac8: mov eax,[eax]
0x10015aca: sar eax,cl
0x10015acc: mov [ebp+0Dh],al
0x10015acf: mov eax,[edi+8]
0x10015ad2: cmp dword ptr [eax+14h],1
0x10015ad6: jne short 10015AE2h
0x10015ad8: mov ecx,[eax+1Ch]
0x10015adb: mov edx,[ecx]
0x10015add: mov eax,[edx+esi*4]
0x10015ae0: jmp short 10015AEAh
0x10015ae2: mov eax,[eax+1Ch]
0x10015ae5: mov ecx,[eax+esi*4]
0x10015ae8: mov eax,[ecx]
0x10015aea: mov ecx,[ebp-28h]
0x10015aed: sar eax,cl
0x10015aef: mov ecx,[ebp-18h]
0x10015af2: push 0
0x10015af4: mov [ebp+0Ch],al
0x10015af7: mov eax,[ebp-24h]
0x10015afa: mov edx,[ebp+0Ch]
0x10015afd: sub eax,ebx
0x10015aff: push edx
0x10015b00: sub eax,1
0x10015b03: push eax
0x10015b04: push esi
0x10015b05: call 10013900h
0x10015b0a: add esi,1
0x10015b0d: jmp near ptr 10015A87h
0x10015b12: add ebx,1
0x10015b15: jmp near ptr 10015A56h
0x10015b1a: mov edx,[ebp-18h]
0x10015b1d: mov eax,[edx+15Ch]
0x10015b23: test eax,eax
0x10015b25: mov [edx+160h],ecx
0x10015b2b: jl near ptr 10015C15h
0x10015b31: cmp eax,ecx
0x10015b33: jge near ptr 10015C15h
0x10015b39: xor eax,eax
0x10015b3b: mov [ebp-2Ch],eax
0x10015b3e: mov edi,edi
0x10015b40: cmp eax,[edx+15Ch]
0x10015b46: jg near ptr 10015C90h
0x10015b4c: mov ecx,[ebp+8]
0x10015b4f: mov edx,[ecx+18h]
0x10015b52: mov eax,[edx+eax*4]
0x10015b55: mov ecx,[eax+10h]
0x10015b58: mov edx,[eax+14h]
0x10015b5b: mov eax,[eax+18h]
0x10015b5e: cmp eax,8
0x10015b61: mov [ebp-20h],ecx
0x10015b64: mov [ebp-24h],edx
0x10015b67: jle short 10015B6Eh
0x10015b69: mov eax,8
0x10015b6e: mov esi,[ebp+0Ch]
```

```
0x10015b71: push esi
0x10015b72: mov esi,[ebp-18h]
0x10015b75: push eax
0x10015b76: push edx
0x10015b77: push ecx
0x10015b78: mov ecx,esi
0x10015b7a: call 1000EAD0h
0x10015b7f: test eax,eax
0x10015b81: jne short 10015B98h
0x10015b83: push 1013083Ch
0x10015b88: lea eax,[ebp-60h]
0x10015b8b: push eax
0x10015b8c: mov dword ptr [ebp-60h],100FE92Bh
0x10015b93: call 100229A6h
0x10015b98: mov ecx,esi
0x10015b9a: call 100136E0h
0x10015b9f: xor ebx,ebx
0x10015ba1: cmp ebx,[ebp-24h]
0x10015ba4: jge short 10015C06h
0x10015ba6: mov ecx,[ebp-30h]
0x10015ba9: mov edi,[ecx]
0x10015bab: mov edx,[ebp-20h]
0x10015bae: mov eax,[ebp-2Ch]
0x10015bb1: mov ecx,[ebp+8]
0x10015bb4: push edi
0x10015bb5: push 1
0x10015bb7: push edx
0x10015bb8: push ebx
0x10015bb9: push 0
0x10015bbb: push eax
0x10015bbc: push ecx
0x10015bbd: call 100AF330h
0x10015bc2: add esp,1Ch
0x10015bc5: xor esi,esi
0x10015bc7: cmp esi,[ebp-20h]
0x10015bca: jge short 10015C01h
0x10015bcc: cmp dword ptr [edi+14h],1
0x10015bd0: jne short 10015BDCh
0x10015bd2: mov edx,[edi+1Ch]
0x10015bd5: mov eax,[edx]
0x10015bd7: mov eax,[eax+esi*4]
0x10015bda: jmp short 10015BE4h
0x10015bdc: mov ecx,[edi+1Ch]
0x10015bdf: mov edx,[ecx+esi*4]
0x10015be2: mov eax,[edx]
0x10015be4: mov ecx,[ebp-28h]
0x10015be7: sar eax,cl
0x10015be9: mov ecx,[ebp-18h]
0x10015bec: push eax
0x10015bed: mov eax,[ebp-24h]
0x10015bf0: sub eax,ebx
0x10015bf2: sub eax,1
0x10015bf5: push eax
0x10015bf6: push esi
0x10015bf7: call 10013330h
0x10015bfc: add esi,1
0x10015bff: jmp short 10015BC7h
0x10015c01: add ebx,1
0x10015c04: jmp short 10015BA1h
0x10015c06: add dword ptr [ebp-2Ch],1
0x10015c0a: mov eax,[ebp-2Ch]
0x10015c0d: mov edx,[ebp-18h]
0x10015c10: jmp near ptr 10015B40h
0x10015c15: push 1013083Ch
0x10015c1a: lea ecx,[ebp-64h]
0x10015c1d: push ecx
0x10015c1e: mov dword ptr [ebp-64h],100FEB4Ch
```

```
0x10015c25: call 100229A6h
0x10015c2a: push 1013083Ch
0x10015c2f: lea edx,[ebp-68h]
0x10015c32: push edx
0x10015c33: mov dword ptr [ebp-68h],100FEB30h
0x10015c3a: call 100229A6h
0x10015c3f: mov eax,[ebp-6Ch]
0x10015c42: mov edi,100FE92Bh
0x10015c47: mov esi,eax
0x10015c49: mov ecx,1
0x10015c4e: xor edx,edx
0x10015c50: repe cmpsb
0x10015c52: je short 10015C6Ah
0x10015c54: push 0FFh
0x10015c59: push eax
0x10015c5a: mov eax,[ebp-18h]
0x10015c5d: add eax,44h
0x10015c60: push eax
0x10015c61: call dword ptr ds:[100FE568h]
0x10015c67: add esp,0Ch
0x10015c6a: mov ecx,[ebp-18h]
0x10015c6d: cmp dword ptr [ecx+148h],0FFFFFFFFh
0x10015c74: jne short 10015C83h
0x10015c76: cmp dword ptr [ebp-14h],0
0x10015c7a: mov dword ptr [ebp-34h],0
0x10015c81: jg short 10015C8Ah
0x10015c83: mov dword ptr [ebp-34h],1
0x10015c8a: mov eax,10015C90h
0x10015c8f: ret
0x10015c90: mov ebx,[ebp-30h]
0x10015c93: test ebx,ebx
0x10015c95: mov edi,[ebp+8]
0x10015c98: je short 10015CC3h
0x10015c9a: xor esi,esi
0x10015c9c: cmp [edi+10h],esi
0x10015c9f: jle short 10015CB9h
0x10015ca1: mov eax,[ebx+esi*4]
0x10015ca4: test eax,eax
0x10015ca6: je short 10015CB1h
0x10015ca8: push eax
0x10015ca9: call 100AE350h
0x10015cae: add esp,4
0x10015cb1: add esi,1
0x10015cb4: cmp esi,[edi+10h]
0x10015cb7: jl short 10015CA1h
0x10015cb9: push ebx
0x10015cba: call dword ptr ds:[100FE554h]
0x10015cc0: add esp,4
0x10015cc3: call 100AE190h
0x10015cc8: test edi,edi
0x10015cca: je short 10015CD5h
0x10015ccc: push edi
0x10015ccd: call 100AF0F0h
0x10015cd2: add esp,4
0x10015cd5: mov eax,[ebp-38h]
0x10015cd8: test eax,eax
0x10015cda: je short 10015CE5h
0x10015cdc: push eax
0x10015cdd: call 100ADF90h
0x10015ce2: add esp,4
0x10015ce5: xor eax,eax
0x10015ce7: cmp [ebp-34h],eax
0x10015cea: sete al
0x10015ced: mov ecx,[ebp-0Ch]
0x10015cf0: mov fs:[0],ecx
0x10015cf7: pop ecx
0x10015cf8: pop edi
```

```
0x10015cf9: pop esi
0x10015cfa: pop ebx
0x10015cfb: mov esp,ebp
0x10015cfd: pop ebp
0x10015cfe: ret 8
```

`sub_100156b0` is called by sub_1000f760. It works with the text "error: too many components", "error: cannot allocate memory", "error: cannot load image data". It delegates to sub_1000ead0, sub_10013330, sub_100136e0 (20 callees).

12.331 sub_10015d10 at 0x10015d10 (4 instructions)

```
0x10015d10: mov ecx,[esp+4]
0x10015d14: mov eax,[ecx]
0x10015d16: mov edx,[eax+18h]
0x10015d19: jmp edx
```

`sub_10015d10` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.332 sub_10015d20 at 0x10015d20 (149 instructions)

```
0x10015d20: push ebp
0x10015d21: mov ebp,esp
0x10015d23: push 0FFFFFFFFh
0x10015d25: push 100FC550h
0x10015d2a: mov eax,fs:[0]
0x10015d30: push eax
0x10015d31: sub esp,28h
0x10015d34: push ebx
0x10015d35: push esi
0x10015d36: push edi
0x10015d37: mov eax,ds:[10136220h]
0x10015d3c: xor eax,ebp
0x10015d3e: push eax
0x10015d3f: lea eax,[ebp-0Ch]
0x10015d42: mov fs:[0],eax
0x10015d48: mov [ebp-10h],esp
0x10015d4b: mov [ebp-14h],ecx
0x10015d4e: mov esi,[ebp+8]
0x10015d51: xor ebx,ebx
0x10015d53: cmp esi,ebx
0x10015d55: je near ptr 10015F1Ah
0x10015d5b: push 2C8h
0x10015d60: mov [ebp-4],ebx
0x10015d63: call dword ptr ds:[100FE550h]
0x10015d69: mov edi,eax
0x10015d6b: add esp,4
0x10015d6e: cmp edi,ebx
0x10015d70: mov [ebp-1Ch],edi
0x10015d73: jne short 10015D8Ah
0x10015d75: push 1013083Ch
0x10015d7a: lea eax,[ebp-20h]
0x10015d7d: push eax
0x10015d7e: mov dword ptr [ebp-20h],100FEC6Ch
0x10015d85: call 100229A6h
0x10015d8a: push 2C8h
0x10015d8f: push ebx
0x10015d90: push edi
0x10015d91: call 100228C0h
0x10015d96: mov [edi+4],esi
0x10015d99: mov dword ptr [ebp-34h],10015D10h
0x10015da0: mov dword ptr [ebp-30h],10015640h
0x10015da7: mov dword ptr [ebp-2Ch],10015F30h
0x10015dae: lea ecx,[ebp-34h]
0x10015db1: mov [edi+8],ecx
0x10015db4: mov [edi+20h],ebx
```

```
0x10015db7: mov edx,[esi]
0x10015db9: mov eax,[edx+18h]
0x10015dbc: add esp,0Ch
0x10015dbf: mov ecx,esi
0x10015dc1: call eax
0x10015dc3: push edi
0x10015dc4: mov [edi+18h],eax
0x10015dc7: mov [edi+24h],ebx
0x10015dca: call 100EFCB0h
0x10015dcf: add esp,4
0x10015dd2: cmp eax,ebx
0x10015dd4: je short 10015DEBh
0x10015dd6: push 1013083Ch
0x10015ddb: lea ecx,[ebp-24h]
0x10015dde: push ecx
0x10015ddf: mov dword ptr [ebp-24h],100FEC48h
0x10015de6: call 100229A6h
0x10015deb: mov edx,[edi+2Ch]
0x10015dee: mov eax,[edi+28h]
0x10015df1: mov ecx,[ebp-14h]
0x10015df4: push 14h
0x10015df6: push 18h
0x10015df8: push edx
0x10015df9: push eax
0x10015dfa: call 1000EAD0h
0x10015dff: mov eax,[edi+168h]
0x10015e05: cmp eax,ebx
0x10015e07: mov esi,[edi+2A0h]
0x10015e0d: mov [ebp-18h],ebx
0x10015e10: je short 10015E27h
0x10015e12: mov eax,[eax+10Ch]
0x10015e18: cmp eax,ebx
0x10015e1a: mov [ebp-18h],eax
0x10015e1d: je short 10015E27h
0x10015e1f: mov ecx,[ebp-14h]
0x10015e22: call 100139C0h
0x10015e27: cmp esi,ebx
0x10015e29: mov byte ptr [ebp+0Bh],0
0x10015e2d: je short 10015E94h
0x10015e2f: mov ebx,[edi+2Ch]
0x10015e32: sub ebx,1
0x10015e35: mov ecx,ebx
0x10015e37: sub ebx,1
0x10015e3a: test ecx,ecx
0x10015e3c: je short 10015E94h
0x10015e3e: xor edi,edi
0x10015e40: mov edx,[ebp-1Ch]
0x10015e43: cmp edi,[edx+28h]
0x10015e46: jge short 10015E90h
0x10015e48: mov al,[esi]
0x10015e4a: mov cl,[esi+1]
0x10015e4d: add esi,1
0x10015e50: mov dl,[esi+1]
0x10015e53: add esi,1
0x10015e56: mov [ebp+9],cl
0x10015e59: mov ecx,[ebp-14h]
0x10015e5c: push 0
0x10015e5e: mov [ebp+8],al
0x10015e61: mov [ebp+0Ah],dl
0x10015e64: mov eax,[ebp+8]
0x10015e67: push eax
0x10015e68: push ebx
0x10015e69: push edi
0x10015e6a: add esi,2
0x10015e6d: call 10013900h
0x10015e72: mov eax,[ebp-18h]
0x10015e75: test eax,eax
```

```
0x10015e77: je short 10015E8Bh
0x10015e79: movzx eax,byte ptr [eax]
0x10015e7c: mov ecx,[ebp-14h]
0x10015e7f: push eax
0x10015e80: push ebx
0x10015e81: push edi
0x10015e82: call 10013AF0h
0x10015e87: add dword ptr [ebp-18h],1
0x10015e8b: add edi,1
0x10015e8e: jmp short 10015E40h
0x10015e90: mov edi,edx
0x10015e92: jmp short 10015E35h
0x10015e94: push edi
0x10015e95: call 100EFC70h
0x10015e9a: push edi
0x10015e9b: call dword ptr ds:[100FE554h]
0x10015ea1: add esp,8
0x10015ea4: mov al,1
0x10015ea6: mov ecx,[ebp-0Ch]
0x10015ea9: mov fs:[0],ecx
0x10015eb0: pop ecx
0x10015eb1: pop edi
0x10015eb2: pop esi
0x10015eb3: pop ebx
0x10015eb4: mov esp,ebp
0x10015eb6: pop ebp
0x10015eb7: ret 4
0x10015f1a: xor al,al
0x10015f1c: mov ecx,[ebp-0Ch]
0x10015f1f: mov fs:[0],ecx
0x10015f26: pop ecx
0x10015f27: pop edi
0x10015f28: pop esi
0x10015f29: pop ebx
0x10015f2a: mov esp,ebp
0x10015f2c: pop ebp
0x10015f2d: ret 4
```

`sub_10015d20` is called by sub_1000f760. It works with the text "CxImagePSD: psd_status_malloc_failed". It delegates to sub_1000ead0, sub_10013900, sub_100139c0 (8 callees).

12.333 sub_10015ff0 at 0x10015ff0 (15 instructions)

```
0x10015ff0: mov edx,[esp+4]
0x10015ff4: mov eax,ecx
0x10015ff6: mov ecx,[esp+8]
0x10015ffa: mov [ecx+4],edx
0x10015ffd: mov dword ptr [eax],10015F30h
0x10016003: mov dword ptr [eax+4],10015F50h
0x1001600a: mov dword ptr [eax+8],10015640h
0x10016011: mov dword ptr [eax+0Ch],10015F70h
0x10016018: mov dword ptr [eax+10h],10015F80h
0x1001601f: mov dword ptr [eax+14h],10015FA0h
0x10016026: mov dword ptr [eax+18h],10015FB0h
0x1001602d: mov dword ptr [eax+1Ch],10015FC0h
0x10016034: mov dword ptr [eax+20h],10015FD0h
0x1001603b: mov [ecx],eax
0x1001603d: ret 8
```

`sub_10015ff0` is called by sub_10016040. It calls nothing else.

12.334 sub_10016040 at 0x10016040 (550 instructions)

```
0x10016040: push ebp
0x10016041: lea ebp,[esp-3E858h]
```

```
0x10016048: mov eax,3E858h
0x1001604d: call 100229B0h
0x10016052: push 0FFFFFFFFh
0x10016054: push 100FC570h
0x10016059: mov eax,fs:[0]
0x1001605f: push eax
0x10016060: sub esp,5Ch
0x10016063: mov eax,ds:[10136220h]
0x10016068: xor eax,ebp
0x1001606a: mov [ebp+3E854h],eax
0x10016070: push ebx
0x10016071: push esi
0x10016072: push edi
0x10016073: push eax
0x10016074: lea eax,[ebp-0Ch]
0x10016077: mov fs:[0],eax
0x1001607d: mov [ebp-10h],esp
0x10016080: mov eax,[ebp+3E860h]
0x10016086: test eax,eax
0x10016088: mov esi,ecx
0x1001608a: mov [ebp-14h],esi
0x1001608d: mov [ebp-40h],eax
0x10016090: je near ptr 10016816h
0x10016096: lea eax,[ebp]
0x10016099: push eax
0x1001609a: mov dword ptr [ebp-4],0
0x100160a1: call 100E80F0h
0x100160a6: add esp,4
0x100160a9: push 13h
0x100160ab: mov ecx,esi
0x100160ad: call 1000F2E0h
0x100160b2: mov ecx,ds:[100FECECh]
0x100160b8: mov edx,ds:[100FECF0h]
0x100160be: and eax,3
0x100160c1: mov [ebp+18h],eax
0x100160c4: mov ax,ds:[100FECF4h]
0x100160ca: push 0
0x100160cc: push 0
0x100160ce: mov [ebp+3E850h],ax
0x100160d5: mov [ebp+3E848h],ecx
0x100160db: mov cl,ds:[100FECF6h]
0x100160e1: mov [ebp+3E84Ch],edx
0x100160e7: lea eax,[ebp]
0x100160ea: lea edx,[ebp+3E848h]
0x100160f0: push 0
0x100160f2: add esi,44h
0x100160f5: push eax
0x100160f6: mov [ebp+3E852h],cl
0x100160fc: mov [ebp+6278h],edx
0x10016102: mov [ebp+2E844h],esi
0x10016108: call 100E81E0h
0x1001610d: add esp,10h
0x10016110: test eax,eax
0x10016112: je short 10016129h
0x10016114: push 1013083Ch
0x10016119: lea ecx,[ebp-30h]
0x1001611c: push ecx
0x1001611d: mov dword ptr [ebp-30h],100FECD0h
0x10016124: call 100229A6h
0x10016129: mov eax,[ebp-4]
0x1001612c: push 101309CCh
0x10016131: push eax
0x10016132: push 100229E2h
0x10016137: lea edx,[ebp+2E804h]
0x1001613d: push 3
0x1001613f: push edx
0x10016140: call 100229DCh
```

```
0x10016145: add esp,14h
0x10016148: test eax,eax
0x1001614a: je short 10016161h
0x1001614c: push 1013083Ch
0x10016151: lea ecx,[ebp-38h]
0x10016154: push ecx
0x10016155: mov dword ptr [ebp-38h],100FE92Bh
0x1001615c: call 100229A6h
0x10016161: mov ebx,[ebp-40h]
0x10016164: lea edx,[ebp]
0x10016167: push edx
0x10016168: push ebx
0x10016169: lea ecx,[ebp-68h]
0x1001616c: call 10015FF0h
0x10016171: lea eax,[ebp]
0x10016174: push eax
0x10016175: call 100EB620h
0x1001617a: add esp,4
0x1001617d: cmp dword ptr [ebp+6614h],0
0x10016184: jne short 1001619Bh
0x10016186: push 1013083Ch
0x1001618b: lea ecx,[ebp-2Ch]
0x1001618e: push ecx
0x1001618f: mov dword ptr [ebp-2Ch],100FECB4h
0x10016196: call 100229A6h
0x1001619b: mov eax,[ebp+2E7FCh]
0x100161a1: test eax,eax
0x100161a3: jne short 100161BAh
0x100161a5: push 1013083Ch
0x100161aa: lea edx,[ebp-28h]
0x100161ad: push edx
0x100161ae: mov dword ptr [ebp-28h],100FEC94h
0x100161b5: call 100229A6h
0x100161ba: cmp eax,100DB9C0h
0x100161bf: jne short 100161E3h
0x100161c1: movzx eax,byte ptr [ebp+66B4h]
0x100161c8: movzx ecx,byte ptr [ebp+66B6h]
0x100161cf: and eax,1
0x100161d2: add [ebp+66B4h],ax
0x100161d9: and ecx,1
0x100161dc: add [ebp+66B6h],cx
0x100161e3: mov edi,[ebp-14h]
0x100161e6: cmp dword ptr [edi+148h],0FFFFFFFFh
0x100161ed: jne short 1001621Fh
0x100161ef: movzx edx,word ptr [ebp+66B6h]
0x100161f6: mov [edi+0Ch],edx
0x100161f9: movzx eax,word ptr [ebp+66B4h]
0x10016200: push 1013083Ch
0x10016205: lea ecx,[ebp-20h]
0x10016208: mov [edi+10h],eax
0x1001620b: push ecx
0x1001620c: mov dword ptr [edi+40h],13h
0x10016213: mov dword ptr [ebp-20h],100FEAC0h
0x1001621a: call 100229A6h
0x1001621f: cmp dword ptr [ebp+65B0h],0
0x10016226: je short 1001625Fh
0x10016228: cmp dword ptr [ebp+0B4h],0
0x1001622f: jne near ptr 100162DEh
0x10016235: fldz
0x10016237: fcomp dword ptr [ebp+34h]
0x1001623a: fnstsw ax
0x1001623c: test ah,44h
0x1001623f: jp near ptr 100162DEh
0x10016245: fld1
0x10016247: fcom qword ptr [ebp+50h]
0x1001624a: fnstsw ax
0x1001624c: test ah,44h
```

```
0x1001624f: jp near ptr 100162DCh
0x10016255: fcomp qword ptr [ebp+60h]
0x10016258: fnstsw ax
0x1001625a: test ah,44h
0x1001625d: jp short 100162DEh
0x1001625f: xor ax,ax
0x10016262: movzx esi,word ptr [ebp+66B4h]
0x10016269: movzx edx,word ptr [ebp+66B6h]
0x10016270: movzx ecx,ax
0x10016273: add esi,ecx
0x10016275: add edx,ecx
0x10016277: sar esi,cl
0x10016279: sar edx,cl
0x1001627b: cmp dword ptr [ebp+0E0h],0
0x10016282: mov [ebp+66BCh],ax
0x10016289: mov [ebp+66BEh],si
0x10016290: mov [ebp+66C0h],dx
0x10016297: je short 100162E7h
0x10016299: fld dword ptr ds:[100FE910h]
0x1001629f: fcomp dword ptr [ebp+0E794h]
0x100162a5: fnstsw ax
0x100162a7: test ah,5
0x100162aa: jp short 100162E7h
0x100162ac: mov ecx,0Ch
0x100162b1: lea esi,[ebp+0E794h]
0x100162b7: lea edi,[ebp+0E7C4h]
0x100162bd: rep movsd
0x100162bf: mov dx,[ebp+66C0h]
0x100162c6: mov si,[ebp+66BEh]
0x100162cd: mov edi,[ebp-14h]
0x100162d0: mov dword ptr [ebp+6604h],0
0x100162da: jmp short 100162F1h
0x100162dc: fstp st(0)
0x100162de: mov ax,1
0x100162e2: jmp near ptr 10016262h
0x100162e7: mov dword ptr [ebp+0DCh],1
0x100162f1: movzx eax,si
0x100162f4: mov esi,ds:[100FE4FCh]
0x100162fa: movzx edx,dx
0x100162fd: imul edx,eax
0x10016300: push 8
0x10016302: push edx
0x10016303: call esi
0x10016305: lea ecx,[ebp+3E848h]
0x1001630b: push ecx
0x1001630c: push eax
0x1001630d: lea edx,[ebp]
0x10016310: push edx
0x10016311: mov [ebp+66E0h],eax
0x10016317: call 100D2E20h
0x1001631c: mov eax,[ebp+65D4h]
0x10016322: add esp,14h
0x10016325: test eax,eax
0x10016327: je short 1001634Ah
0x10016329: push eax
0x1001632a: call dword ptr ds:[100FE550h]
0x10016330: lea ecx,[ebp+3E848h]
0x10016336: push ecx
0x10016337: push eax
0x10016338: lea edx,[ebp]
0x1001633b: push edx
0x1001633c: mov [ebp+627Ch],eax
0x10016342: call 100D2E20h
0x10016347: add esp,10h
0x1001634a: mov ecx,[ebp+65C0h]
0x10016350: mov eax,[ebx]
0x10016352: mov edx,[eax+10h]
```

```
0x10016355: push 0
0x10016357: push ecx
0x10016358: mov ecx,ebx
0x1001635a: call edx
0x1001635c: lea eax,[ebp]
0x1001635f: push eax
0x10016360: call dword ptr [ebp+2E7FCh]
0x10016366: add esp,4
0x10016369: cmp dword ptr [ebp+660Ch],0
0x10016370: je short 1001637Eh
0x10016372: lea ecx,[ebp]
0x10016375: push ecx
0x10016376: call 100D40C0h
0x1001637b: add esp,4
0x1001637e: mov edx,[ebp+0Ch]
0x10016381: push edx
0x10016382: lea eax,[ebp]
0x10016385: push eax
0x10016386: call 100DD260h
0x1001638b: mov eax,[ebp+8]
0x1001638e: add esp,8
0x10016391: test eax,eax
0x10016393: je short 100163A2h
0x10016395: push eax
0x10016396: lea ecx,[ebp]
0x10016399: push ecx
0x1001639a: call 100DD690h
0x1001639f: add esp,8
0x100163a2: mov eax,[ebp+18h]
0x100163a5: xor edx,edx
0x100163a7: cmp [ebp+66C2h],dx
0x100163ae: mov ebx,2
0x100163b3: sete dl
0x100163b6: add edx,ebx
0x100163b8: test eax,eax
0x100163ba: mov [ebp+66D4h],edx
0x100163c0: jl short 100163C8h
0x100163c2: mov [ebp+66D4h],eax
0x100163c8: mov eax,[ebp+14h]
0x100163cb: test eax,eax
0x100163cd: jl short 100163D5h
0x100163cf: mov [ebp+65F8h],eax
0x100163d5: mov eax,[ebp+1Ch]
0x100163d8: test eax,eax
0x100163da: jl short 100163E2h
0x100163dc: mov [ebp+65FCh],eax
0x100163e2: cmp dword ptr [ebp+661Ch],0
0x100163e9: jne short 100163FFh
0x100163eb: cmp [ebp+0C0h],ebx
0x100163f1: jge short 100163FFh
0x100163f3: lea eax,[ebp]
0x100163f6: push eax
0x100163f7: call 100DEA30h
0x100163fc: add esp,4
0x100163ff: lea ecx,[ebp]
0x10016402: push ecx
0x10016403: call 100DF640h
0x10016408: add esp,4
0x1001640b: cmp dword ptr [ebp+65B0h],0
0x10016412: je short 10016466h
0x10016414: cmp dword ptr [ebp+0C0h],0
0x1001641b: jne short 10016466h
0x1001641d: mov eax,[ebp+66D4h]
0x10016423: test eax,eax
0x10016425: jne short 10016432h
0x10016427: lea edx,[ebp]
0x1001642a: push edx
```

```
0x1001642b: call 100DF9E0h
0x10016430: jmp short 10016463h
0x10016432: cmp eax,1
0x10016435: je short 1001645Ah
0x10016437: cmp dword ptr [ebp+66D0h],3
0x1001643e: jg short 1001645Ah
0x10016440: cmp eax,ebx
0x10016442: jne short 1001644Fh
0x10016444: lea eax,[ebp]
0x10016447: push eax
0x10016448: call 100E03B0h
0x1001644d: jmp short 10016463h
0x1001644f: lea ecx,[ebp]
0x10016452: push ecx
0x10016453: call 100E0950h
0x10016458: jmp short 10016463h
0x1001645a: lea edx,[ebp]
0x1001645d: push edx
0x1001645e: call 100DFCA0h
0x10016463: add esp,4
0x10016466: cmp dword ptr [ebp+6600h],0
0x1001646d: je short 100164BAh
0x1001646f: mov dword ptr [ebp+66D0h],3
0x10016479: xor eax,eax
0x1001647b: jmp short 10016480h
0x10016480: movzx ecx,word ptr [ebp+66B6h]
0x10016487: movzx edx,word ptr [ebp+66B4h]
0x1001648e: imul ecx,edx
0x10016491: cmp eax,ecx
0x10016493: jge short 100164B5h
0x10016495: mov edx,[ebp+66E0h]
0x1001649b: movzx ebx,word ptr [edx+eax*8+2]
0x100164a0: lea ecx,[edx+eax*8+2]
0x100164a4: movzx edx,word ptr [edx+eax*8+6]
0x100164a9: add edx,ebx
0x100164ab: sar edx,1
0x100164ad: mov [ecx],dx
0x100164b0: add eax,1
0x100164b3: jmp short 10016480h
0x100164b5: mov ebx,2
0x100164ba: cmp dword ptr [ebp+661Ch],0
0x100164c1: jne short 10016512h
0x100164c3: cmp dword ptr [ebp+66D0h],3
0x100164ca: jne short 100164D8h
0x100164cc: lea eax,[ebp]
0x100164cf: push eax
0x100164d0: call 100E1850h
0x100164d5: add esp,4
0x100164d8: cmp dword ptr [ebp+661Ch],0
0x100164df: jne short 10016512h
0x100164e1: cmp [ebp+9Ch],ebx
0x100164e7: jne short 100164F5h
0x100164e9: lea ecx,[ebp]
0x100164ec: push ecx
0x100164ed: call 100E1A50h
0x100164f2: add esp,4
0x100164f5: cmp dword ptr [ebp+661Ch],0
0x100164fc: jne short 10016512h
0x100164fe: cmp [ebp+9Ch],ebx
0x10016504: jle short 10016512h
0x10016506: lea edx,[ebp]
0x10016509: push edx
0x1001650a: call 100E2010h
0x1001650f: add esp,4
0x10016512: cmp dword ptr [ebp+2Ch],0
0x10016516: je short 10016524h
0x10016518: lea eax,[ebp]
```

```
0x1001651b: push eax
0x1001651c: call 100E6FF0h
0x10016521: add esp,4
0x10016524: lea ecx,[ebp]
0x10016527: push ecx
0x10016528: call 100E67B0h
0x1001652d: add esp,4
0x10016530: cmp dword ptr [ebp+2Ch],0
0x10016534: je short 10016542h
0x10016536: lea edx,[ebp]
0x10016539: push edx
0x1001653a: call 100E7350h
0x1001653f: add esp,4
0x10016542: test byte ptr [ebp+66C8h],4
0x10016549: mov ax,[ebp+66B4h]
0x10016550: mov cx,[ebp+66B6h]
0x10016557: mov [ebp+66BEh],ax
0x1001655e: mov [ebp+66C0h],cx
0x10016565: je short 1001657Eh
0x10016567: xor ax,cx
0x1001656a: xor cx,ax
0x1001656d: xor ax,cx
0x10016570: mov [ebp+66B6h],cx
0x10016577: mov [ebp+66B4h],ax
0x1001657e: push 13h
0x10016580: movzx eax,ax
0x10016583: push 18h
0x10016585: movzx ecx,cx
0x10016588: push eax
0x10016589: push ecx
0x1001658a: mov ecx,edi
0x1001658c: call 1000EAD0h
0x10016591: test eax,eax
0x10016593: jne short 100165AAh
0x10016595: push 1013083Ch
0x1001659a: lea edx,[ebp-3Ch]
0x1001659d: push edx
0x1001659e: mov dword ptr [ebp-3Ch],100FE92Bh
0x100165a5: call 100229A6h
0x100165aa: mov eax,[ebp+0ACh]
0x100165b0: imul eax,[ebp+66D0h]
0x100165b7: cdq
0x100165b8: and edx,7
0x100165bb: add eax,edx
0x100165bd: sar eax,3
0x100165c0: push eax
0x100165c1: movzx eax,word ptr [ebp+66B6h]
0x100165c8: push eax
0x100165c9: call esi
0x100165cb: lea ecx,[ebp+3E848h]
0x100165d1: push ecx
0x100165d2: mov esi,eax
0x100165d4: lea edx,[ebp]
0x100165d7: push esi
0x100165d8: push edx
0x100165d9: call 100D2E20h
0x100165de: add esp,14h
0x100165e1: cmp dword ptr [ebp+0ACh],8
0x100165e8: jne short 100165FDh
0x100165ea: lea eax,[ebp+2E848h]
0x100165f0: push eax
0x100165f1: lea ecx,[ebp]
0x100165f4: push ecx
0x100165f5: call 100E77A0h
0x100165fa: add esp,8
0x100165fd: push 0
0x100165ff: lea edx,[ebp]
```

```
0x10016602: push 0
0x10016604: push edx
0x10016605: call 100E7740h
0x1001660a: mov edi,eax
0x1001660c: push 1
0x1001660e: lea eax,[ebp]
0x10016611: push 0
0x10016613: push eax
0x10016614: mov [ebp-18h],edi
0x10016617: call 100E7740h
0x1001661c: movzx ecx,word ptr [ebp+66B6h]
0x10016623: push ecx
0x10016624: lea edx,[ebp]
0x10016627: sub eax,edi
0x10016629: push 0
0x1001662b: push edx
0x1001662c: mov [ebp-34h],eax
0x1001662f: call 100E7740h
0x10016634: push 0
0x10016636: mov ebx,eax
0x10016638: lea eax,[ebp]
0x1001663b: push 1
0x1001663d: push eax
0x1001663e: call 100E7740h
0x10016643: add esp,30h
0x10016646: sub eax,ebx
0x10016648: mov [ebp-24h],eax
0x1001664b: mov dword ptr [ebp-1Ch],0
0x10016652: jmp short 10016660h
0x10016660: movzx ecx,word ptr [ebp+66B4h]
0x10016667: cmp [ebp-1Ch],ecx
0x1001666a: jge near ptr 100167ADh
0x10016670: mov ecx,[ebp+66D0h]
0x10016676: xor edx,edx
0x10016678: jmp short 10016680h
0x10016680: movzx eax,word ptr [ebp+66B6h]
0x10016687: cmp edx,eax
0x10016689: jge short 100166FBh
0x1001668b: xor eax,eax
0x1001668d: cmp dword ptr [ebp+0ACh],8
0x10016694: jne short 100166C6h
0x10016696: cmp eax,ecx
0x10016698: jge short 100166F0h
0x1001669a: mov ebx,[ebp+66E0h]
0x100166a0: lea edi,[eax+edi*4]
0x100166a3: movzx edi,word ptr [ebx+edi*2]
0x100166a7: mov ebx,edx
0x100166a9: imul ebx,ecx
0x100166ac: mov cl,[ebp+edi+2E848h]
0x100166b3: mov edi,[ebp-18h]
0x100166b6: add ebx,eax
0x100166b8: mov [ebx+esi],cl
0x100166bb: mov ecx,[ebp+66D0h]
0x100166c1: add eax,1
0x100166c4: jmp short 10016696h
0x100166c6: cmp eax,ecx
0x100166c8: jge short 100166F0h
0x100166ca: mov ebx,edx
0x100166cc: imul ebx,ecx
0x100166cf: mov ecx,[ebp+66E0h]
0x100166d5: lea edi,[eax+edi*4]
0x100166d8: mov cx,[ecx+edi*2]
0x100166dc: mov edi,[ebp-18h]
0x100166df: add ebx,eax
0x100166e1: mov [esi+ebx*2],cx
0x100166e5: mov ecx,[ebp+66D0h]
0x100166eb: add eax,1
```

```
0x100166ee: jmp short 100166C6h
0x100166f0: add edx,1
0x100166f3: add edi,[ebp-34h]
0x100166f6: mov [ebp-18h],edi
0x100166f9: jmp short 10016680h
0x100166fb: cmp dword ptr [ebp+0ACh],10h
0x10016702: jne short 1001673Fh
0x10016704: cmp dword ptr [ebp+0B0h],0
0x1001670b: jne short 1001673Fh
0x1001670d: push 55AAh
0x10016712: call 10022F7Ch
0x10016717: cmp ax,55AAh
0x1001671b: je short 10016739h
0x1001671d: movzx edx,word ptr [ebp+66B6h]
0x10016724: imul edx,[ebp+66D0h]
0x1001672b: add edx,edx
0x1001672d: push edx
0x1001672e: push esi
0x1001672f: push esi
0x10016730: call dword ptr ds:[100FE4ECh]
0x10016736: add esp,0Ch
0x10016739: mov ecx,[ebp+66D0h]
0x1001673f: mov eax,[ebp+0ACh]
0x10016745: imul eax,ecx
0x10016748: mov ecx,[ebp-14h]
0x1001674b: cdq
0x1001674c: and edx,7
0x1001674f: add eax,edx
0x10016751: mov ebx,eax
0x10016753: movzx eax,word ptr [ebp+66B6h]
0x1001675a: sar ebx,3
0x1001675d: imul ebx,eax
0x10016760: push ebx
0x10016761: push esi
0x10016762: call 10013530h
0x10016767: mov ecx,[ebp-14h]
0x1001676a: call 1000F000h
0x1001676f: cmp ebx,eax
0x10016771: jb short 1001677Dh
0x10016773: mov ecx,[ebp-14h]
0x10016776: call 1000F000h
0x1001677b: mov ebx,eax
0x1001677d: movzx ecx,word ptr [ebp+66B4h]
0x10016784: sub ecx,[ebp-1Ch]
0x10016787: sub ecx,1
0x1001678a: push ecx
0x1001678b: mov ecx,[ebp-14h]
0x1001678e: call 1000E5E0h
0x10016793: push ebx
0x10016794: push esi
0x10016795: push eax
0x10016796: call 100228C6h
0x1001679b: add dword ptr [ebp-1Ch],1
0x1001679f: add esp,0Ch
0x100167a2: add edi,[ebp-24h]
0x100167a5: mov [ebp-18h],edi
0x100167a8: jmp near ptr 10016660h
0x100167ad: push esi
0x100167ae: call dword ptr ds:[100FE554h]
0x100167b4: lea edx,[ebp]
0x100167b7: push edx
0x100167b8: call 100E81A0h
0x100167bd: add esp,8
0x100167c0: mov al,1
0x100167c2: jmp short 10016818h
0x10016816: xor al,al
0x10016818: mov ecx,[ebp-0Ch]
```

```
0x1001681b: mov fs:[0],ecx
0x10016822: pop ecx
0x10016823: pop edi
0x10016824: pop esi
0x10016825: pop ebx
0x10016826: mov ecx,[ebp+3E854h]
0x1001682c: xor ecx,ebp
0x1001682e: call 10022326h
0x10016833: add ebp,3E858h
0x10016839: mov esp,ebp
0x1001683b: pop ebp
0x1001683c: ret 4
```

`sub_10016040` is called by sub_1000f760. It works with the text "CxImageRAW: missing raw decoder", "CxImageRAW: not a raw image", "CxImageRAW: unknown option". It delegates to sub_1000e5e0, sub_1000ead0, sub_1000f000 (34 callees).

12.335 sub_10016840 at 0x10016840 (158 instructions)

```
0x10016840: sub esp,0Ch
0x10016843: push esi
0x10016844: push edi
0x10016845: mov edi,[esp+18h]
0x10016849: test edi,edi
0x1001684b: mov esi,ecx
0x1001684d: jne short 10016859h
0x1001684f: pop edi
0x10016850: xor al,al
0x10016852: pop esi
0x10016853: add esp,0Ch
0x10016856: ret 4
0x10016859: mov eax,[edi]
0x1001685b: mov edx,[eax+8]
0x1001685e: push 1
0x10016860: push 9
0x10016862: lea ecx,[esp+10h]
0x10016866: push ecx
0x10016867: mov ecx,edi
0x10016869: call edx
0x1001686b: mov eax,[esp+8]
0x1001686f: push eax
0x10016870: mov ecx,esi
0x10016872: call 1000F1C0h
0x10016877: mov ecx,[esp+0Ah]
0x1001687b: push ecx
0x1001687c: mov ecx,esi
0x1001687e: mov [esp+0Ch],ax
0x10016883: call 1000F1C0h
0x10016888: mov edx,[esp+0Dh]
0x1001688c: push edx
0x1001688d: mov ecx,esi
0x1001688f: mov [esp+0Eh],ax
0x10016894: call 1000F1E0h
0x10016899: cmp eax,1000000h
0x1001689e: mov [esp+0Dh],eax
0x100168a2: jne short 1001684Fh
0x100168a4: mov cx,[esp+8]
0x100168a9: cmp cx,7FFFh
0x100168ae: ja short 1001684Fh
0x100168b0: mov ax,[esp+0Ah]
0x100168b5: cmp ax,7FFFh
0x100168b9: ja short 1001684Fh
0x100168bb: cmp byte ptr [esp+0Ch],3
0x100168c0: jne short 1001684Fh
0x100168c2: cmp dword ptr [esi+148h],0FFFFFFFFh
0x100168c9: movzx ecx,cx
```

```
0x100168cc: jne short 100168E8h
0x100168ce: movzx edx,ax
0x100168d1: pop edi
0x100168d2: mov [esi+0Ch],ecx
0x100168d5: mov [esi+10h],edx
0x100168d8: mov dword ptr [esi+40h],12h
0x100168df: mov al,1
0x100168e1: pop esi
0x100168e2: add esp,0Ch
0x100168e5: ret 4
0x100168e8: push 12h
0x100168ea: movzx eax,ax
0x100168ed: push 8
0x100168ef: push eax
0x100168f0: push ecx
0x100168f1: mov ecx,esi
0x100168f3: call 1000EAD0h
0x100168f8: mov ecx,esi
0x100168fa: call 1000F190h
0x100168ff: test al,al
0x10016901: je near ptr 1001684Fh
0x10016907: push ebx
0x10016908: push 300h
0x1001690d: call dword ptr ds:[100FE550h]
0x10016913: mov ebx,eax
0x10016915: add esp,4
0x10016918: test ebx,ebx
0x1001691a: jne short 10016927h
0x1001691c: pop ebx
0x1001691d: pop edi
0x1001691e: xor al,al
0x10016920: pop esi
0x10016921: add esp,0Ch
0x10016924: ret 4
0x10016927: mov edx,[edi]
0x10016929: mov eax,[edx+8]
0x1001692c: push 1
0x1001692e: push 300h
0x10016933: push ebx
0x10016934: mov ecx,edi
0x10016936: call eax
0x10016938: push 100h
0x1001693d: push ebx
0x1001693e: mov ecx,esi
0x10016940: call 10013630h
0x10016945: push ebx
0x10016946: call dword ptr ds:[100FE554h]
0x1001694c: movzx ecx,word ptr [esp+12h]
0x10016951: movzx edx,word ptr [esp+10h]
0x10016956: mov ebx,[edi]
0x10016958: imul ecx,edx
0x1001695b: add esp,4
0x1001695e: push 1
0x10016960: push ecx
0x10016961: push 0
0x10016963: mov ecx,esi
0x10016965: add ebx,8
0x10016968: call 1000E5E0h
0x1001696d: push eax
0x1001696e: mov eax,[ebx]
0x10016970: mov ecx,edi
0x10016972: call eax
0x10016974: mov ecx,esi
0x10016976: call 1000F000h
0x1001697b: movzx ecx,word ptr [esp+0Ch]
0x10016980: cmp eax,ecx
0x10016982: je short 100169E5h
```

```
0x10016984: movzx ebx,word ptr [esp+0Eh]
0x10016989: push ebp
0x1001698a: push 0
0x1001698c: mov ecx,esi
0x1001698e: call 1000E5E0h
0x10016993: mov edi,eax
0x10016995: movzx eax,word ptr [esp+10h]
0x1001699a: lea edx,[ebx-1]
0x1001699d: imul edx,eax
0x100169a0: add ebx,0FFFFFFFFh
0x100169a3: push ebx
0x100169a4: mov ecx,esi
0x100169a6: add edi,edx
0x100169a8: call 1000E5E0h
0x100169ad: xor ebp,ebp
0x100169af: cmp [esp+12h],bp
0x100169b4: mov ebx,eax
0x100169b6: jbe short 100169E4h
0x100169b8: movzx ecx,word ptr [esp+10h]
0x100169bd: push ecx
0x100169be: push edi
0x100169bf: push ebx
0x100169c0: call 100228C6h
0x100169c5: movzx edx,word ptr [esp+1Ch]
0x100169ca: add esp,0Ch
0x100169cd: mov ecx,esi
0x100169cf: sub edi,edx
0x100169d1: call 1000F000h
0x100169d6: sub ebx,eax
0x100169d8: movzx eax,word ptr [esp+12h]
0x100169dd: add ebp,1
0x100169e0: cmp ebp,eax
0x100169e2: jl short 100169B8h
0x100169e4: pop ebp
0x100169e5: push 1
0x100169e7: push 0
0x100169e9: mov ecx,esi
0x100169eb: call 10010810h
0x100169f0: pop ebx
0x100169f1: pop edi
0x100169f2: mov al,1
0x100169f4: pop esi
0x100169f5: add esp,0Ch
0x100169f8: ret 4
```

`sub_10016840` is called by sub_1000f760. It delegates to sub_1000e5e0, sub_1000ead0, sub_1000f000 (9 callees).

12.336 sub_10016a00 at 0x10016a00 (26 instructions)

```
0x10016a00: mov eax,[esp+4]
0x10016a04: cmp dword ptr [eax],9AC6CDD7h
0x10016a0a: jne short 10016A57h
0x10016a0c: mov dx,[eax+12h]
0x10016a10: xor dx,[eax+10h]
0x10016a14: movzx ecx,word ptr [eax]
0x10016a17: xor dx,[eax+0Eh]
0x10016a1b: mov [esp+4],ecx
0x10016a1f: xor dx,[eax+0Ch]
0x10016a23: xor dx,[eax+0Ah]
0x10016a27: xor dx,[eax+8]
0x10016a2b: xor dx,[eax+6]
0x10016a2f: xor dx,[eax+4]
0x10016a33: xor dx,[eax+2]
0x10016a37: xor cx,dx
0x10016a3a: cmp cx,[eax+14h]
0x10016a3e: jne short 10016A57h
```

```
0x10016a40: movzx eax,word ptr [eax+0Eh]
0x10016a44: test ax,ax
0x10016a47: jbe short 10016A57h
0x10016a49: cmp ax,9ECh
0x10016a4d: ja short 10016A57h
0x10016a4f: mov eax,1
0x10016a54: ret 4
0x10016a57: xor eax,eax
0x10016a59: ret 4
```

`sub_10016a00` is called by sub_10016a60. It accesses memory-mapped peripheral registers at 0x9ac6cdd7. It calls nothing else.

12.337 sub_10016a60 at 0x10016a60 (127 instructions)

```
0x10016a60: sub esp,24h
0x10016a63: push ebx
0x10016a64: push ebp
0x10016a65: push esi
0x10016a66: mov esi,[esp+34h]
0x10016a6a: mov eax,[esi]
0x10016a6c: mov edx,[eax+18h]
0x10016a6f: push edi
0x10016a70: mov edi,ecx
0x10016a72: mov ecx,esi
0x10016a74: call edx
0x10016a76: mov ebx,[esp+3Ch]
0x10016a7a: push 16h
0x10016a7c: mov ebp,eax
0x10016a7e: mov eax,[esi]
0x10016a80: mov edx,[eax+8]
0x10016a83: push 1
0x10016a85: push ebx
0x10016a86: mov ecx,esi
0x10016a88: call edx
0x10016a8a: cmp eax,16h
0x10016a8d: jb short 10016B05h
0x10016a8f: push ebx
0x10016a90: mov ecx,edi
0x10016a92: call 10016A00h
0x10016a97: test eax,eax
0x10016a99: je short 10016B05h
0x10016a9b: mov eax,[esi]
0x10016a9d: mov edx,[eax+8]
0x10016aa0: push 12h
0x10016aa2: push 1
0x10016aa4: lea ecx,[esp+28h]
0x10016aa8: push ecx
0x10016aa9: mov ecx,esi
0x10016aab: call edx
0x10016aad: cmp eax,12h
0x10016ab0: jb short 10016B05h
0x10016ab2: mov ax,[esp+20h]
0x10016ab7: cmp ax,1
0x10016abb: je short 10016AC3h
0x10016abd: cmp ax,2
0x10016ac1: jne short 10016B05h
0x10016ac3: mov edi,[esp+26h]
0x10016ac7: add edi,edi
0x10016ac9: cmp edi,ebp
0x10016acb: ja short 10016B05h
0x10016acd: push edi
0x10016ace: call dword ptr ds:[100FE550h]
0x10016ad4: mov ebx,eax
0x10016ad6: add esp,4
0x10016ad9: test ebx,ebx
```

```
0x10016adb: je short 10016B05h
0x10016add: mov eax,[esi]
0x10016adf: mov edx,[eax+10h]
0x10016ae2: push 0
0x10016ae4: push 16h
0x10016ae6: mov ecx,esi
0x10016ae8: call edx
0x10016aea: mov eax,[esi]
0x10016aec: mov edx,[eax+8]
0x10016aef: push edi
0x10016af0: push 1
0x10016af2: push ebx
0x10016af3: mov ecx,esi
0x10016af5: call edx
0x10016af7: cmp eax,edi
0x10016af9: je short 10016B11h
0x10016afb: push ebx
0x10016afc: call dword ptr ds:[100FE554h]
0x10016b02: add esp,4
0x10016b05: pop edi
0x10016b06: pop esi
0x10016b07: pop ebp
0x10016b08: xor eax,eax
0x10016b0a: pop ebx
0x10016b0b: add esp,24h
0x10016b0e: ret 8
0x10016b11: push 0
0x10016b13: call dword ptr ds:[100FE598h]
0x10016b19: mov ebp,ds:[100FE000h]
0x10016b1f: mov esi,eax
0x10016b21: push 58h
0x10016b23: push esi
0x10016b24: call ebp
0x10016b26: push 5Ah
0x10016b28: push esi
0x10016b29: call ebp
0x10016b2b: xor eax,eax
0x10016b2d: mov [esp+10h],eax
0x10016b31: mov [esp+14h],eax
0x10016b35: mov [esp+18h],eax
0x10016b39: mov [esp+1Ch],eax
0x10016b3d: mov eax,2710h
0x10016b42: mov [esp+14h],eax
0x10016b46: mov [esp+18h],eax
0x10016b4a: lea eax,[esp+10h]
0x10016b4e: push eax
0x10016b4f: push esi
0x10016b50: push ebx
0x10016b51: push edi
0x10016b52: mov dword ptr [esp+20h],8
0x10016b5a: mov dword ptr [esp+2Ch],0
0x10016b62: call dword ptr ds:[100FE098h]
0x10016b68: mov ebp,eax
0x10016b6a: test ebp,ebp
0x10016b6c: jne short 10016B86h
0x10016b6e: lea ecx,[esp+10h]
0x10016b72: push ecx
0x10016b73: push esi
0x10016b74: push ebx
0x10016b75: push edi
0x10016b76: mov dword ptr [esp+20h],1
0x10016b7e: call dword ptr ds:[100FE098h]
0x10016b84: mov ebp,eax
0x10016b86: push esi
0x10016b87: push 0
0x10016b89: call dword ptr ds:[100FE5A8h]
0x10016b8f: push ebx
```

```
0x10016b90: call dword ptr ds:[100FE554h]
0x10016b96: add esp,4
0x10016b99: pop edi
0x10016b9a: pop esi
0x10016b9b: mov eax,ebp
0x10016b9d: pop ebp
0x10016b9e: pop ebx
0x10016b9f: add esp,24h
0x10016ba2: ret 8
```

`sub_10016a60` is called by sub_10016cd0. It delegates to sub_10016a00 (1 callees).

12.338 sub_10016bb0 at 0x10016bb0 (87 instructions)

```
0x10016bb0: push ebx
0x10016bb1: push ebp
0x10016bb2: push esi
0x10016bb3: mov esi,[esp+10h]
0x10016bb7: mov eax,[esi]
0x10016bb9: mov edx,[eax+18h]
0x10016bbc: push edi
0x10016bbd: mov ecx,esi
0x10016bbf: call edx
0x10016bc1: mov ebx,eax
0x10016bc3: mov eax,[esi]
0x10016bc5: mov edx,[eax+14h]
0x10016bc8: mov ecx,esi
0x10016bca: call edx
0x10016bcc: mov ebp,[esp+18h]
0x10016bd0: push 6Ch
0x10016bd2: mov edi,eax
0x10016bd4: mov eax,[esi]
0x10016bd6: mov edx,[eax+8]
0x10016bd9: push 1
0x10016bdb: push ebp
0x10016bdc: mov ecx,esi
0x10016bde: call edx
0x10016be0: cmp eax,6Ch
0x10016be3: jb short 10016C2Bh
0x10016be5: cmp dword ptr [ebp],1
0x10016be9: jne short 10016C2Bh
0x10016beb: cmp dword ptr [ebp+28h],464D4520h
0x10016bf2: jne short 10016C2Bh
0x10016bf4: mov eax,[esi]
0x10016bf6: mov edx,[eax+10h]
0x10016bf9: push 0
0x10016bfb: push edi
0x10016bfc: mov ecx,esi
0x10016bfe: call edx
0x10016c00: push ebx
0x10016c01: call dword ptr ds:[100FE550h]
0x10016c07: mov edi,eax
0x10016c09: add esp,4
0x10016c0c: test edi,edi
0x10016c0e: je short 10016C2Bh
0x10016c10: mov eax,[esi]
0x10016c12: mov edx,[eax+8]
0x10016c15: push ebx
0x10016c16: push 1
0x10016c18: push edi
0x10016c19: mov ecx,esi
0x10016c1b: call edx
0x10016c1d: cmp eax,ebx
0x10016c1f: push edi
0x10016c20: je short 10016C34h
0x10016c22: call dword ptr ds:[100FE554h]
```

```
0x10016c28: add esp,4
0x10016c2b: pop edi
0x10016c2c: pop esi
0x10016c2d: pop ebp
0x10016c2e: xor eax,eax
0x10016c30: pop ebx
0x10016c31: ret 8
0x10016c34: push ebx
0x10016c35: call dword ptr ds:[100FE094h]
0x10016c3b: push edi
0x10016c3c: mov esi,eax
0x10016c3e: call dword ptr ds:[100FE554h]
0x10016c44: add esp,4
0x10016c47: test esi,esi
0x10016c49: je short 10016C2Bh
0x10016c4b: push ebp
0x10016c4c: push 6Ch
0x10016c4e: push esi
0x10016c4f: call dword ptr ds:[100FE07Ch]
0x10016c55: test eax,eax
0x10016c57: jne short 10016C69h
0x10016c59: push esi
0x10016c5a: call dword ptr ds:[100FE078h]
0x10016c60: pop edi
0x10016c61: pop esi
0x10016c62: pop ebp
0x10016c63: xor eax,eax
0x10016c65: pop ebx
0x10016c66: ret 8
0x10016c69: pop edi
0x10016c6a: mov eax,esi
0x10016c6c: pop esi
0x10016c6d: pop ebp
0x10016c6e: pop ebx
0x10016c6f: ret 8
```

`sub_10016bb0` is called by sub_10016cd0. It accesses memory-mapped peripheral registers at 0x464d4520. It calls nothing else.

12.339 sub_10016c80 at 0x10016c80 (25 instructions)

```
0x10016c80: mov ecx,[esp+8]
0x10016c84: push esi
0x10016c85: mov esi,[esp+8]
0x10016c89: push edi
0x10016c8a: mov edi,[esi]
0x10016c8c: cmp edi,0FA0h
0x10016c92: jle short 10016CA7h
0x10016c94: mov eax,[ecx]
0x10016c96: imul eax,0FA0h
0x10016c9c: cdq
0x10016c9d: idiv edi
0x10016c9f: mov [ecx],eax
0x10016ca1: mov dword ptr [esi],0FA0h
0x10016ca7: mov edi,[ecx]
0x10016ca9: cmp edi,0FA0h
0x10016caf: jle short 10016CC4h
0x10016cb1: mov eax,[esi]
0x10016cb3: imul eax,0FA0h
0x10016cb9: cdq
0x10016cba: idiv edi
0x10016cbc: mov [esi],eax
0x10016cbe: mov dword ptr [ecx],0FA0h
0x10016cc4: pop edi
0x10016cc5: pop esi
0x10016cc6: ret 8
```

`sub_10016c80` is called by sub_10016cd0. It calls nothing else.

12.340 sub_10016cd0 at 0x10016cd0 (364 instructions)

```
0x10016cd0: sub esp,0ACh
0x10016cd6: push esi
0x10016cd7: mov esi,[esp+0B4h]
0x10016cde: test esi,esi
0x10016ce0: push edi
0x10016ce1: mov edi,ecx
0x10016ce3: jne short 10016CF2h
0x10016ce5: pop edi
0x10016ce6: xor al,al
0x10016ce8: pop esi
0x10016ce9: add esp,0ACh
0x10016cef: ret 0Ch
0x10016cf2: mov eax,[esi]
0x10016cf4: mov edx,[eax+14h]
0x10016cf7: push ebx
0x10016cf8: push ebp
0x10016cf9: mov ecx,esi
0x10016cfb: call edx
0x10016cfd: mov ebp,eax
0x10016cff: lea eax,[esp+38h]
0x10016d03: push eax
0x10016d04: push esi
0x10016d05: mov ecx,edi
0x10016d07: call 10016A60h
0x10016d0c: mov ebx,eax
0x10016d0e: test ebx,ebx
0x10016d10: push 0
0x10016d12: je short 10016D74h
0x10016d14: call dword ptr ds:[100FE598h]
0x10016d1a: mov esi,eax
0x10016d1c: push 58h
0x10016d1e: push esi
0x10016d1f: call dword ptr ds:[100FE000h]
0x10016d25: push 5Ah
0x10016d27: push esi
0x10016d28: mov ebp,eax
0x10016d2a: call dword ptr ds:[100FE000h]
0x10016d30: push esi
0x10016d31: push 0
0x10016d33: mov [esp+24h],eax
0x10016d37: call dword ptr ds:[100FE5A8h]
0x10016d3d: movsx edx,word ptr [esp+3Eh]
0x10016d42: movsx eax,word ptr [esp+42h]
0x10016d47: movzx ecx,word ptr [esp+46h]
0x10016d4c: sub eax,edx
0x10016d4e: imul eax,ebp
0x10016d51: mov esi,ecx
0x10016d53: shr esi,1
0x10016d55: add eax,esi
0x10016d57: cdq
0x10016d58: idiv ecx
0x10016d5a: movsx edx,word ptr [esp+40h]
0x10016d5f: mov ebp,eax
0x10016d61: movsx eax,word ptr [esp+44h]
0x10016d66: sub eax,edx
0x10016d68: imul eax,[esp+1Ch]
0x10016d6d: add eax,esi
0x10016d6f: cdq
0x10016d70: idiv ecx
0x10016d72: jmp short 10016DD5h
0x10016d74: mov eax,[esi]
0x10016d76: mov edx,[eax+10h]
0x10016d79: push ebp
```

```
0x10016d7a: mov ecx,esi
0x10016d7c: call edx
0x10016d7e: lea eax,[esp+50h]
0x10016d82: push eax
0x10016d83: push esi
0x10016d84: mov ecx,edi
0x10016d86: call 10016BB0h
0x10016d8b: mov ebx,eax
0x10016d8d: test ebx,ebx
0x10016d8f: jne short 10016DC5h
0x10016d91: mov ecx,ds:[100FED04h]
0x10016d97: mov [edi+44h],ecx
0x10016d9a: mov edx,ds:[100FED08h]
0x10016da0: mov [edi+48h],edx
0x10016da3: mov eax,ds:[100FED0Ch]
0x10016da8: mov [edi+4Ch],eax
0x10016dab: mov cx,ds:[100FED10h]
0x10016db2: pop ebp
0x10016db3: pop ebx
0x10016db4: mov [edi+50h],cx
0x10016db8: pop edi
0x10016db9: xor al,al
0x10016dbb: pop esi
0x10016dbc: add esp,0ACh
0x10016dc2: ret 0Ch
0x10016dc5: mov ebp,[esp+60h]
0x10016dc9: mov eax,[esp+64h]
0x10016dcd: sub ebp,[esp+58h]
0x10016dd1: sub eax,[esp+5Ch]
0x10016dd5: cmp dword ptr [edi+148h],0FFFFFFFFh
0x10016ddc: mov [esp+18h],eax
0x10016de0: mov [esp+10h],ebp
0x10016de4: jne short 10016E1Bh
0x10016de6: push ebx
0x10016de7: mov [edi+0Ch],ebp
0x10016dea: mov [edi+10h],eax
0x10016ded: mov dword ptr [edi+40h],0Ah
0x10016df4: call dword ptr ds:[100FE078h]
0x10016dfa: add edi,44h
0x10016dfd: mov ecx,6
0x10016e02: mov esi,100FEAC0h
0x10016e07: rep movsd
0x10016e09: movsw
0x10016e0b: pop ebp
0x10016e0c: pop ebx
0x10016e0d: movsb
0x10016e0e: pop edi
0x10016e0f: mov al,1
0x10016e11: pop esi
0x10016e12: add esp,0ACh
0x10016e18: ret 0Ch
0x10016e1b: test ebp,ebp
0x10016e1d: je near ptr 100170DFh
0x10016e23: test eax,eax
0x10016e25: je near ptr 100170DFh
0x10016e2b: mov eax,[esp+0C4h]
0x10016e32: test eax,eax
0x10016e34: je short 10016E3Ah
0x10016e36: mov [esp+10h],eax
0x10016e3a: mov eax,[esp+0C8h]
0x10016e41: test eax,eax
0x10016e43: je short 10016E49h
0x10016e45: mov [esp+18h],eax
0x10016e49: lea edx,[esp+18h]
0x10016e4d: push edx
0x10016e4e: lea eax,[esp+14h]
0x10016e52: push eax
```

```
0x10016e53: mov ecx,edi
0x10016e55: call 10016C80h
0x10016e5a: push 0
0x10016e5c: call dword ptr ds:[100FE598h]
0x10016e62: mov ecx,[esp+18h]
0x10016e66: mov edx,[esp+10h]
0x10016e6a: push ecx
0x10016e6b: mov ebp,eax
0x10016e6d: push edx
0x10016e6e: push ebp
0x10016e6f: call dword ptr ds:[100FE040h]
0x10016e75: push ebp
0x10016e76: mov [esp+18h],eax
0x10016e7a: call dword ptr ds:[100FE060h]
0x10016e80: push ebp
0x10016e81: push 0
0x10016e83: mov esi,eax
0x10016e85: call dword ptr ds:[100FE5A8h]
0x10016e8b: test esi,esi
0x10016e8d: je near ptr 100170BAh
0x10016e93: mov ebp,[esp+14h]
0x10016e97: xor eax,eax
0x10016e99: cmp ebp,eax
0x10016e9b: je near ptr 1001709Dh
0x10016ea1: mov ecx,[esp+18h]
0x10016ea5: mov [esp+28h],eax
0x10016ea9: mov [esp+2Ch],eax
0x10016ead: mov eax,[esp+10h]
0x10016eb1: push 0Ch
0x10016eb3: push esi
0x10016eb4: mov [esp+38h],eax
0x10016eb8: mov [esp+3Ch],ecx
0x10016ebc: call dword ptr ds:[100FE000h]
0x10016ec2: push ebp
0x10016ec3: push esi
0x10016ec4: mov [esp+2Ch],eax
0x10016ec8: call dword ptr ds:[100FE030h]
0x10016ece: push 5
0x10016ed0: mov [esp+24h],eax
0x10016ed4: call dword ptr ds:[100FE5C4h]
0x10016eda: mov ebp,ds:[100FE004h]
0x10016ee0: push eax
0x10016ee1: push esi
0x10016ee2: call ebp
0x10016ee4: push 0
0x10016ee6: push 0
0x10016ee8: push 0
0x10016eea: lea edx,[esp+34h]
0x10016eee: push edx
0x10016eef: push 2
0x10016ef1: push 0
0x10016ef3: push 0
0x10016ef5: push esi
0x10016ef6: mov [esp+3Ch],eax
0x10016efa: call dword ptr ds:[100FE080h]
0x10016f00: mov eax,[esp+1Ch]
0x10016f04: push eax
0x10016f05: push esi
0x10016f06: call ebp
0x10016f08: push 0
0x10016f0a: push 0
0x10016f0c: push ebx
0x10016f0d: call dword ptr ds:[100FE084h]
0x10016f13: mov ebp,eax
0x10016f15: test ebp,ebp
0x10016f17: je near ptr 10016FB7h
0x10016f1d: lea ecx,[ebp*4+4]
```

```
0x10016f24: push ecx
0x10016f25: push 40h
0x10016f27: call dword ptr ds:[100FE0ECh]
0x10016f2d: test eax,eax
0x10016f2f: mov [esp+1Ch],eax
0x10016f33: jne short 10016F79h
0x10016f35: mov edx,[esp+14h]
0x10016f39: push edx
0x10016f3a: call dword ptr ds:[100FE048h]
0x10016f40: push esi
0x10016f41: call dword ptr ds:[100FE02Ch]
0x10016f47: push ebx
0x10016f48: call dword ptr ds:[100FE078h]
0x10016f4e: mov eax,ds:[100FEAA8h]
0x10016f53: mov [edi+44h],eax
0x10016f56: mov ecx,ds:[100FEAACh]
0x10016f5c: mov [edi+48h],ecx
0x10016f5f: mov dx,ds:[100FEAB0h]
0x10016f66: pop ebp
0x10016f67: pop ebx
0x10016f68: mov [edi+4Ch],dx
0x10016f6c: pop edi
0x10016f6d: xor al,al
0x10016f6f: pop esi
0x10016f70: add esp,0ACh
0x10016f76: ret 0Ch
0x10016f79: mov word ptr [eax],300h
0x10016f7e: mov [eax+2],bp
0x10016f82: add eax,4
0x10016f85: push eax
0x10016f86: push ebp
0x10016f87: push ebx
0x10016f88: call dword ptr ds:[100FE084h]
0x10016f8e: mov eax,[esp+1Ch]
0x10016f92: push eax
0x10016f93: call dword ptr ds:[100FE088h]
0x10016f99: mov ecx,[esp+1Ch]
0x10016f9d: push ecx
0x10016f9e: mov ebp,eax
0x10016fa0: call dword ptr ds:[100FE0E8h]
0x10016fa6: push 0
0x10016fa8: push ebp
0x10016fa9: push esi
0x10016faa: call dword ptr ds:[100FE08Ch]
0x10016fb0: push esi
0x10016fb1: call dword ptr ds:[100FE020h]
0x10016fb7: lea edx,[esp+28h]
0x10016fbb: push edx
0x10016fbc: push ebx
0x10016fbd: push esi
0x10016fbe: call dword ptr ds:[100FE090h]
0x10016fc4: mov eax,[esp+20h]
0x10016fc8: push eax
0x10016fc9: push esi
0x10016fca: call dword ptr ds:[100FE030h]
0x10016fd0: push ebx
0x10016fd1: call dword ptr ds:[100FE078h]
0x10016fd7: cmp dword ptr [edi+148h],0
0x10016fde: je short 1001701Dh
0x10016fe0: mov ecx,[esp+14h]
0x10016fe4: push ecx
0x10016fe5: call dword ptr ds:[100FE048h]
0x10016feb: push esi
0x10016fec: call dword ptr ds:[100FE02Ch]
0x10016ff2: mov edx,ds:[100FEAA8h]
0x10016ff8: mov [edi+44h],edx
0x10016ffb: mov eax,ds:[100FEAACh]
```

```
0x10017000: mov [edi+48h],eax
0x10017003: mov cx,ds:[100FEAB0h]
0x1001700a: pop ebp
0x1001700b: pop ebx
0x1001700c: mov [edi+4Ch],cx
0x10017010: pop edi
0x10017011: xor al,al
0x10017013: pop esi
0x10017014: add esp,0ACh
0x1001701a: ret 0Ch
0x1001701d: mov edx,[esp+24h]
0x10017021: mov ebx,[esp+18h]
0x10017025: mov eax,[esp+10h]
0x10017029: push 0Ah
0x1001702b: push edx
0x1001702c: push ebx
0x1001702d: push eax
0x1001702e: mov ecx,edi
0x10017030: call 1000EAD0h
0x10017035: test eax,eax
0x10017037: jne short 1001705Ah
0x10017039: mov ecx,[esp+14h]
0x1001703d: push ecx
0x1001703e: call dword ptr ds:[100FE048h]
0x10017044: push esi
0x10017045: call dword ptr ds:[100FE02Ch]
0x1001704b: pop ebp
0x1001704c: pop ebx
0x1001704d: pop edi
0x1001704e: xor al,al
0x10017050: pop esi
0x10017051: add esp,0ACh
0x10017057: ret 0Ch
0x1001705a: mov edx,[edi+4]
0x1001705d: push 0
0x1001705f: push edx
0x10017060: push 0
0x10017062: mov ecx,edi
0x10017064: call 1000E5E0h
0x10017069: push eax
0x1001706a: push ebx
0x1001706b: mov ebx,[esp+24h]
0x1001706f: push 0
0x10017071: push ebx
0x10017072: push esi
0x10017073: call dword ptr ds:[100FE024h]
0x10017079: push ebx
0x1001707a: mov edi,eax
0x1001707c: call dword ptr ds:[100FE048h]
0x10017082: push esi
0x10017083: call dword ptr ds:[100FE02Ch]
0x10017089: pop ebp
0x1001708a: xor eax,eax
0x1001708c: pop ebx
0x1001708d: test edi,edi
0x1001708f: pop edi
0x10017090: setne al
0x10017093: pop esi
0x10017094: add esp,0ACh
0x1001709a: ret 0Ch
0x1001709d: push esi
0x1001709e: call dword ptr ds:[100FE02Ch]
0x100170a4: push ebx
0x100170a5: call dword ptr ds:[100FE078h]
0x100170ab: pop ebp
0x100170ac: pop ebx
0x100170ad: pop edi
```

```
0x100170ae: xor al,al
0x100170b0: pop esi
0x100170b1: add esp,0ACh
0x100170b7: ret 0Ch
0x100170ba: mov eax,[esp+14h]
0x100170be: test eax,eax
0x100170c0: je short 100170C9h
0x100170c2: push eax
0x100170c3: call dword ptr ds:[100FE048h]
0x100170c9: push ebx
0x100170ca: call dword ptr ds:[100FE078h]
0x100170d0: pop ebp
0x100170d1: pop ebx
0x100170d2: pop edi
0x100170d3: xor al,al
0x100170d5: pop esi
0x100170d6: add esp,0ACh
0x100170dc: ret 0Ch
0x100170df: push ebx
0x100170e0: call dword ptr ds:[100FE078h]
0x100170e6: mov eax,ds:[100FECF8h]
0x100170eb: mov [edi+44h],eax
0x100170ee: mov ecx,ds:[100FECFCh]
0x100170f4: mov [edi+48h],ecx
0x100170f7: mov dx,ds:[100FED00h]
0x100170fe: pop ebp
0x100170ff: pop ebx
0x10017100: mov [edi+4Ch],dx
0x10017104: pop edi
0x10017105: xor al,al
0x10017107: pop esi
0x10017108: add esp,0ACh
0x1001710e: ret 0Ch
```

`sub_10016cd0` is called by sub_1000f760. It works with the text "Cancelled", "corrupted WMF". It delegates to sub_1000e5e0, sub_1000ead0, sub_10016a60 (5 callees).

12.341 sub_10017120 at 0x10017120 (25 instructions)

```
0x10017120: push ebx
0x10017121: push esi
0x10017122: mov esi,ecx
0x10017124: mov ecx,[esi+14h]
0x10017127: push edi
0x10017128: mov dword ptr [esi],0
0x1001712e: call 1000EFE0h
0x10017133: mov ecx,[esi+14h]
0x10017136: sub eax,1
0x10017139: mov [esi+4],eax
0x1001713c: call 1000EFE0h
0x10017141: mov ecx,[esi+14h]
0x10017144: lea ebx,[eax-1]
0x10017147: call 1000F000h
0x1001714c: mov ecx,[esi+14h]
0x1001714f: mov edi,eax
0x10017151: push 0
0x10017153: imul edi,ebx
0x10017156: call 1000E5E0h
0x1001715b: add eax,edi
0x1001715d: pop edi
0x1001715e: mov [esi+10h],eax
0x10017161: pop esi
0x10017162: pop ebx
0x10017163: ret
```

`sub_10017120` is called by sub_100171e0, sub_1001a5b0, sub_1001b350. It delegates to sub_1000e5e0, sub_1000efe0, sub_1000f000 (3 callees).

12.342 sub_10017170 at 0x10017170 (13 instructions)

```
0x10017170: push esi
0x10017171: mov esi,ecx
0x10017173: add dword ptr [esi+4],0FFFFFFFFh
0x10017177: jns short 1001717Dh
0x10017179: xor eax,eax
0x1001717b: pop esi
0x1001717c: ret
0x1001717d: mov ecx,[esi+14h]
0x10017180: call 1000F000h
0x10017185: sub [esi+10h],eax
0x10017188: mov eax,1
0x1001718d: pop esi
0x1001718e: ret
```

`sub_10017170` is called by sub_100171e0. It delegates to sub_1000f000 (1 callees).

12.343 sub_10017190 at 0x10017190 (31 instructions)

```
0x10017190: push esi
0x10017191: mov esi,[esp+8]
0x10017195: push edi
0x10017196: mov edi,[esp+10h]
0x1001719a: mov dword ptr [edi],0
0x100171a0: mov eax,[esi]
0x100171a2: mov edx,[eax+20h]
0x100171a5: mov ecx,esi
0x100171a7: call edx
0x100171a9: test al,al
0x100171ab: jne short 100171D0h
0x100171ad: mov eax,[esi]
0x100171af: mov edx,[eax+2Ch]
0x100171b2: mov ecx,esi
0x100171b4: call edx
0x100171b6: mov edx,[edi]
0x100171b8: movzx ecx,al
0x100171bb: and ecx,7Fh
0x100171be: shl edx,7
0x100171c1: or ecx,edx
0x100171c3: test al,al
0x100171c5: mov [edi],ecx
0x100171c7: js short 100171A0h
0x100171c9: pop edi
0x100171ca: mov al,1
0x100171cc: pop esi
0x100171cd: ret 8
0x100171d0: pop edi
0x100171d1: xor al,al
0x100171d3: pop esi
0x100171d4: ret 8
```

`sub_10017190` is called by sub_100171e0. It calls nothing else.

12.344 sub_100171e0 at 0x100171e0 (158 instructions)

```
0x100171e0: push ebp
0x100171e1: mov ebp,esp
0x100171e3: push 0FFFFFFFFh
0x100171e5: push 100FC5A0h
0x100171ea: mov eax,fs:[0]
0x100171f0: push eax
```

```
0x100171f1: sub esp,34h
0x100171f4: push ebx
0x100171f5: push esi
0x100171f6: push edi
0x100171f7: mov eax,ds:[10136220h]
0x100171fc: xor eax,ebp
0x100171fe: push eax
0x100171ff: lea eax,[ebp-0Ch]
0x10017202: mov fs:[0],eax
0x10017208: mov [ebp-10h],esp
0x1001720b: mov esi,ecx
0x1001720d: mov [ebp-24h],esi
0x10017210: mov edi,[ebp+8]
0x10017213: test edi,edi
0x10017215: je near ptr 1001739Eh
0x1001721b: lea eax,[ebp-38h]
0x1001721e: push eax
0x1001721f: push edi
0x10017220: mov dword ptr [ebp-4],0
0x10017227: call 10017190h
0x1001722c: lea ecx,[ebp+8]
0x1001722f: push ecx
0x10017230: push edi
0x10017231: mov ecx,esi
0x10017233: call 10017190h
0x10017238: mov dl,[ebp+8]
0x1001723b: lea eax,[ebp-33h]
0x1001723e: push eax
0x1001723f: push edi
0x10017240: mov ecx,esi
0x10017242: mov [ebp-34h],dl
0x10017245: call 10017190h
0x1001724a: lea ecx,[ebp-2Fh]
0x1001724d: push ecx
0x1001724e: push edi
0x1001724f: mov ecx,esi
0x10017251: call 10017190h
0x10017256: mov edx,[edi]
0x10017258: mov eax,[edx+20h]
0x1001725b: mov ecx,edi
0x1001725d: call eax
0x1001725f: test al,al
0x10017261: je short 10017278h
0x10017263: push 1013083Ch
0x10017268: lea ecx,[ebp-14h]
0x1001726b: push ecx
0x1001726c: mov dword ptr [ebp-14h],100FED50h
0x10017273: call 100229A6h
0x10017278: cmp dword ptr [ebp-38h],0
0x1001727c: je short 10017293h
0x1001727e: push 1013083Ch
0x10017283: lea edx,[ebp-18h]
0x10017286: push edx
0x10017287: mov dword ptr [ebp-18h],100FED38h
0x1001728e: call 100229A6h
0x10017293: mov eax,[ebp-33h]
0x10017296: test eax,eax
0x10017298: mov ecx,[ebp-2Fh]
0x1001729b: mov [esi+0Ch],eax
0x1001729e: mov [esi+10h],ecx
0x100172a1: jle near ptr 10017358h
0x100172a7: test ecx,ecx
0x100172a9: jle near ptr 10017358h
0x100172af: cmp dword ptr [esi+148h],0FFFFFFFFh
0x100172b6: jne short 100172D5h
0x100172b8: mov dword ptr [esi+40h],9
0x100172bf: mov al,1
```

```
0x100172c1: mov ecx,[ebp-0Ch]
0x100172c4: mov fs:[0],ecx
0x100172cb: pop ecx
0x100172cc: pop edi
0x100172cd: pop esi
0x100172ce: pop ebx
0x100172cf: mov esp,ebp
0x100172d1: pop ebp
0x100172d2: ret 4
0x100172d5: push 9
0x100172d7: push 1
0x100172d9: push ecx
0x100172da: push eax
0x100172db: mov ecx,esi
0x100172dd: call 1000EAD0h
0x100172e2: mov ecx,esi
0x100172e4: call 1000F190h
0x100172e9: test al,al
0x100172eb: jne short 10017302h
0x100172ed: push 1013083Ch
0x100172f2: lea eax,[ebp-1Ch]
0x100172f5: push eax
0x100172f6: mov dword ptr [ebp-1Ch],100FED24h
0x100172fd: call 100229A6h
0x10017302: mov ecx,esi
0x10017304: call 100136E0h
0x10017309: mov eax,[esi+0Ch]
0x1001730c: add eax,7
0x1001730f: cdq
0x10017310: and edx,7
0x10017313: add eax,edx
0x10017315: mov ebx,eax
0x10017317: push esi
0x10017318: lea ecx,[ebp-40h]
0x1001731b: sar ebx,3
0x1001731e: call 10011F20h
0x10017323: lea ecx,[ebp-40h]
0x10017326: call 10017120h
0x1001732b: mov dword ptr [ebp+8],0
0x10017332: mov ecx,[ebp+8]
0x10017335: cmp ecx,[esi+10h]
0x10017338: jge short 100172BFh
0x1001733a: mov edx,[edi]
0x1001733c: mov eax,[ebp-30h]
0x1001733f: mov edx,[edx+8]
0x10017342: push 1
0x10017344: push ebx
0x10017345: push eax
0x10017346: mov ecx,edi
0x10017348: call edx
0x1001734a: lea ecx,[ebp-40h]
0x1001734d: call 10017170h
0x10017352: add dword ptr [ebp+8],1
0x10017356: jmp short 10017332h
0x10017358: push 1013083Ch
0x1001735d: lea eax,[ebp-20h]
0x10017360: push eax
0x10017361: mov dword ptr [ebp-20h],100FED14h
0x10017368: call 100229A6h
0x1001736d: mov eax,[ebp-28h]
0x10017370: mov edi,100FE92Bh
0x10017375: mov esi,eax
0x10017377: mov ecx,1
0x1001737c: xor edx,edx
0x1001737e: repe cmpsb
0x10017380: je short 10017398h
0x10017382: push 0FFh
```

```
0x10017387: push eax
0x10017388: mov eax,[ebp-24h]
0x1001738b: add eax,44h
0x1001738e: push eax
0x1001738f: call dword ptr ds:[100FE568h]
0x10017395: add esp,0Ch
0x10017398: mov eax,1001739Eh
0x1001739d: ret
0x1001739e: xor al,al
0x100173a0: mov ecx,[ebp-0Ch]
0x100173a3: mov fs:[0],ecx
0x100173aa: pop ecx
0x100173ab: pop edi
0x100173ac: pop esi
0x100173ad: pop ebx
0x100173ae: mov esp,ebp
0x100173b0: pop ebp
0x100173b1: ret 4
```

`sub_100171e0` is called by sub_1000f760. It delegates to sub_1000ead0, sub_1000f190, sub_10011f20 (8 callees).

12.345 sub_100173c0 at 0x100173c0 (58 instructions)

```
0x100173c0: push esi
0x100173c1: mov si,[esp+14h]
0x100173c6: cmp si,4
0x100173ca: jle short 100173D2h
0x100173cc: xor al,al
0x100173ce: pop esi
0x100173cf: ret 14h
0x100173d2: cmp word ptr [esp+18h],1
0x100173d8: jne short 100173CCh
0x100173da: push ebp
0x100173db: movsx ebp,word ptr [esp+14h]
0x100173e0: lea eax,[ebp*8-1]
0x100173e7: test eax,eax
0x100173e9: jl short 100173FEh
0x100173eb: add eax,1
0x100173ee: push eax
0x100173ef: mov eax,[esp+10h]
0x100173f3: push 0
0x100173f5: push eax
0x100173f6: call 100228C0h
0x100173fb: add esp,0Ch
0x100173fe: movsx eax,si
0x10017401: xor ecx,ecx
0x10017403: test eax,eax
0x10017405: mov [esp+1Ch],eax
0x10017409: jle short 10017453h
0x1001740b: push ebx
0x1001740c: push edi
0x1001740d: mov edi,[esp+18h]
0x10017411: mov edx,[esp+14h]
0x10017415: mov ebx,1
0x1001741a: shl ebx,cl
0x1001741c: test ebp,ebp
0x1001741e: jle short 1001744Ah
0x10017420: mov [esp+20h],ebp
0x10017424: movzx esi,byte ptr [edi]
0x10017427: add edi,1
0x1001742a: mov eax,80h
0x1001742f: nop
0x10017430: test esi,eax
0x10017432: je short 10017436h
0x10017434: or [edx],bl
0x10017436: sar eax,1
```

```
0x10017438: add edx,1
0x1001743b: test eax,eax
0x1001743d: jne short 10017430h
0x1001743f: sub dword ptr [esp+20h],1
0x10017444: jne short 10017424h
0x10017446: mov eax,[esp+24h]
0x1001744a: add ecx,1
0x1001744d: cmp ecx,eax
0x1001744f: jl short 10017411h
0x10017451: pop edi
0x10017452: pop ebx
0x10017453: pop ebp
0x10017454: mov al,1
0x10017456: pop esi
0x10017457: ret 14h
```

`sub_100173c0` is called by sub_10017680. It delegates to j_memset (1 callees).

12.346 sub_10017460 at 0x10017460 (138 instructions)

```
0x10017460: cmp word ptr [esp+10h],1
0x10017466: je short 1001746Dh
0x10017468: xor al,al
0x1001746a: ret 14h
0x1001746d: mov ax,[esp+14h]
0x10017472: cmp ax,8
0x10017476: push esi
0x10017477: jne short 100174A8h
0x10017479: mov eax,[esp+10h]
0x1001747d: test ax,ax
0x10017480: jle near ptr 100175E1h
0x10017486: mov esi,[esp+0Ch]
0x1001748a: mov ecx,[esp+8]
0x1001748e: mov edi,edi
0x10017490: mov dl,[esi]
0x10017492: sub eax,1
0x10017495: mov [ecx],dl
0x10017497: add ecx,1
0x1001749a: add esi,1
0x1001749d: test ax,ax
0x100174a0: jg short 10017490h
0x100174a2: mov al,1
0x100174a4: pop esi
0x100174a5: ret 14h
0x100174a8: cmp ax,4
0x100174ac: push edi
0x100174ad: jne short 100174EDh
0x100174af: mov esi,[esp+14h]
0x100174b3: test si,si
0x100174b6: jle near ptr 100175E0h
0x100174bc: mov edi,[esp+10h]
0x100174c0: mov ecx,[esp+0Ch]
0x100174c4: movzx eax,byte ptr [edi]
0x100174c7: mov edx,eax
0x100174c9: sar edx,4
0x100174cc: and dl,0Fh
0x100174cf: mov [ecx],dl
0x100174d1: add ecx,1
0x100174d4: and al,0Fh
0x100174d6: sub esi,1
0x100174d9: mov [ecx],al
0x100174db: add edi,1
0x100174de: add ecx,1
0x100174e1: test si,si
0x100174e4: jg short 100174C4h
0x100174e6: pop edi
```

```
0x100174e7: mov al,1
0x100174e9: pop esi
0x100174ea: ret 14h
0x100174ed: cmp ax,2
0x100174f1: jne short 1001754Ch
0x100174f3: mov esi,[esp+14h]
0x100174f7: test si,si
0x100174fa: jle near ptr 100175E0h
0x10017500: mov edi,[esp+10h]
0x10017504: mov eax,[esp+0Ch]
0x10017508: movzx ecx,byte ptr [edi]
0x1001750b: mov edx,ecx
0x1001750d: sar edx,6
0x10017510: and dl,3
0x10017513: mov [eax],dl
0x10017515: mov edx,ecx
0x10017517: sar edx,4
0x1001751a: add eax,1
0x1001751d: and dl,3
0x10017520: mov [eax],dl
0x10017522: mov edx,ecx
0x10017524: add eax,1
0x10017527: sar edx,2
0x1001752a: and dl,3
0x1001752d: mov [eax],dl
0x1001752f: add eax,1
0x10017532: and cl,3
0x10017535: sub esi,1
0x10017538: mov [eax],cl
0x1001753a: add edi,1
0x1001753d: add eax,1
0x10017540: test si,si
0x10017543: jg short 10017508h
0x10017545: pop edi
0x10017546: mov al,1
0x10017548: pop esi
0x10017549: ret 14h
0x1001754c: cmp ax,1
0x10017550: jne near ptr 100175E0h
0x10017556: mov esi,[esp+14h]
0x1001755a: test si,si
0x1001755d: jle near ptr 100175E0h
0x10017563: mov edi,[esp+10h]
0x10017567: mov eax,[esp+0Ch]
0x1001756b: jmp short 10017570h
0x10017570: movzx ecx,byte ptr [edi]
0x10017573: mov edx,ecx
0x10017575: shr edx,7
0x10017578: and dl,1
0x1001757b: mov [eax],dl
0x1001757d: add eax,1
0x10017580: mov edx,ecx
0x10017582: shr edx,6
0x10017585: and dl,1
0x10017588: mov [eax],dl
0x1001758a: add eax,1
0x1001758d: mov edx,ecx
0x1001758f: shr edx,5
0x10017592: and dl,1
0x10017595: mov [eax],dl
0x10017597: add eax,1
0x1001759a: mov edx,ecx
0x1001759c: shr edx,4
0x1001759f: and dl,1
0x100175a2: mov [eax],dl
0x100175a4: add eax,1
0x100175a7: mov edx,ecx
```

```
0x100175a9: shr edx,3
0x100175ac: and dl,1
0x100175af: mov [eax],dl
0x100175b1: mov edx,ecx
0x100175b3: shr edx,2
0x100175b6: add eax,1
0x100175b9: and dl,1
0x100175bc: mov [eax],dl
0x100175be: mov edx,ecx
0x100175c0: add eax,1
0x100175c3: shr edx,1
0x100175c5: and dl,1
0x100175c8: mov [eax],dl
0x100175ca: add eax,1
0x100175cd: and cl,1
0x100175d0: sub esi,1
0x100175d3: mov [eax],cl
0x100175d5: add edi,1
0x100175d8: add eax,1
0x100175db: test si,si
0x100175de: jg short 10017570h
0x100175e0: pop edi
0x100175e1: mov al,1
0x100175e3: pop esi
0x100175e4: ret 14h
```

`sub_10017460` is called by sub_10017680. It calls nothing else.

12.347 sub_100175f0 at 0x100175f0 (46 instructions)

```
0x100175f0: push esi
0x100175f1: mov esi,[esp+8]
0x100175f5: movzx eax,word ptr [esi+4]
0x100175f9: push edi
0x100175fa: push eax
0x100175fb: mov edi,ecx
0x100175fd: call 1000F1C0h
0x10017602: movzx ecx,word ptr [esi+6]
0x10017606: push ecx
0x10017607: mov ecx,edi
0x10017609: mov [esi+4],ax
0x1001760d: call 1000F1C0h
0x10017612: movzx edx,word ptr [esi+8]
0x10017616: push edx
0x10017617: mov ecx,edi
0x10017619: mov [esi+6],ax
0x1001761d: call 1000F1C0h
0x10017622: mov [esi+8],ax
0x10017626: movzx eax,word ptr [esi+0Ah]
0x1001762a: push eax
0x1001762b: mov ecx,edi
0x1001762d: call 1000F1C0h
0x10017632: movzx ecx,word ptr [esi+0Ch]
0x10017636: push ecx
0x10017637: mov ecx,edi
0x10017639: mov [esi+0Ah],ax
0x1001763d: call 1000F1C0h
0x10017642: movzx edx,word ptr [esi+0Eh]
0x10017646: push edx
0x10017647: mov ecx,edi
0x10017649: mov [esi+0Ch],ax
0x1001764d: call 1000F1C0h
0x10017652: mov [esi+0Eh],ax
0x10017656: movzx eax,word ptr [esi+42h]
0x1001765a: push eax
0x1001765b: mov ecx,edi
```

```
0x1001765d: call 1000F1C0h
0x10017662: movzx ecx,word ptr [esi+44h]
0x10017666: push ecx
0x10017667: mov ecx,edi
0x10017669: mov [esi+42h],ax
0x1001766d: call 1000F1C0h
0x10017672: pop edi
0x10017673: mov [esi+44h],ax
0x10017677: pop esi
0x10017678: ret 4
```

`sub_100175f0` is called by sub_10017680. It delegates to sub_1000f1c0 (1 callees).

12.348 sub_10017680 at 0x10017680 (450 instructions)

```
0x10017680: push ebp
0x10017681: lea ebp,[esp-384h]
0x10017688: sub esp,384h
0x1001768e: push 0FFFFFFFFh
0x10017690: push 100FC5C0h
0x10017695: mov eax,fs:[0]
0x1001769b: push eax
0x1001769c: sub esp,50h
0x1001769f: mov eax,ds:[10136220h]
0x100176a4: xor eax,ebp
0x100176a6: mov [ebp+380h],eax
0x100176ac: push ebx
0x100176ad: push esi
0x100176ae: push edi
0x100176af: push eax
0x100176b0: lea eax,[ebp-0Ch]
0x100176b3: mov fs:[0],eax
0x100176b9: mov [ebp-10h],esp
0x100176bc: mov ebx,[ebp+38Ch]
0x100176c2: xor edi,edi
0x100176c4: cmp ebx,edi
0x100176c6: mov esi,ecx
0x100176c8: mov [ebp-20h],esi
0x100176cb: je near ptr 10017C33h
0x100176d1: mov eax,[ebx]
0x100176d3: mov edx,[eax+8]
0x100176d6: push 1
0x100176d8: push 80h
0x100176dd: lea ecx,[ebp]
0x100176e0: push ecx
0x100176e1: mov ecx,ebx
0x100176e3: mov [ebp-24h],edi
0x100176e6: mov [ebp-1Ch],edi
0x100176e9: mov [ebp-4],edi
0x100176ec: call edx
0x100176ee: cmp eax,edi
0x100176f0: jne short 10017707h
0x100176f2: push 1013083Ch
0x100176f7: lea eax,[ebp-38h]
0x100176fa: push eax
0x100176fb: mov dword ptr [ebp-38h],100FEEA0h
0x10017702: call 100229A6h
0x10017707: lea ecx,[ebp]
0x1001770a: push ecx
0x1001770b: mov ecx,esi
0x1001770d: call 100175F0h
0x10017712: cmp byte ptr [ebp],0Ah
0x10017716: je short 1001772Dh
0x10017718: push 1013083Ch
0x1001771d: lea edx,[ebp-58h]
0x10017720: push edx
```

```
0x10017721: mov dword ptr [ebp-58h],100FEE88h
0x10017728: call 100229A6h
0x1001772d: cmp byte ptr [ebp+2],1
0x10017731: je short 10017748h
0x10017733: push 1013083Ch
0x10017738: lea eax,[ebp-54h]
0x1001773b: push eax
0x1001773c: mov dword ptr [ebp-54h],100FEE60h
0x10017743: call 100229A6h
0x10017748: movzx ecx,word ptr [ebp+4]
0x1001774c: movzx edx,word ptr [ebp+6]
0x10017750: movzx eax,word ptr [ebp+8]
0x10017754: movzx edi,word ptr [ebp+0Ah]
0x10017758: sub eax,ecx
0x1001775a: movzx ecx,word ptr [ebp+0Ch]
0x1001775e: sub edi,edx
0x10017760: movzx edx,word ptr [ebp+0Eh]
0x10017764: add eax,1
0x10017767: add edi,1
0x1001776a: cmp dword ptr [esi+148h],0FFFFFFFFh
0x10017771: mov [ebp-28h],eax
0x10017774: mov [ebp-2Ch],edi
0x10017777: mov [esi+168h],ecx
0x1001777d: mov [esi+16Ch],edx
0x10017783: jne short 10017799h
0x10017785: mov [esi+0Ch],eax
0x10017788: mov [esi+10h],edi
0x1001778b: mov dword ptr [esi+40h],8
0x10017792: mov al,1
0x10017794: jmp near ptr 10017C35h
0x10017799: mov dl,[ebp+41h]
0x1001779c: cmp dl,4
0x1001779f: jle short 100177B6h
0x100177a1: push 1013083Ch
0x100177a6: lea eax,[ebp-30h]
0x100177a9: push eax
0x100177aa: mov dword ptr [ebp-30h],100FEE34h
0x100177b1: call 100229A6h
0x100177b6: cmp dl,3
0x100177b9: mov cl,[ebp+3]
0x100177bc: jl short 100177DFh
0x100177be: cmp cl,8
0x100177c1: jne short 100177DFh
0x100177c3: push 8
0x100177c5: push 18h
0x100177c7: push edi
0x100177c8: push eax
0x100177c9: mov ecx,esi
0x100177cb: call 1000EAD0h
0x100177d0: cmp byte ptr [ebp+41h],4
0x100177d4: jne short 100177FEh
0x100177d6: mov ecx,esi
0x100177d8: call 100139C0h
0x100177dd: jmp short 100177FEh
0x100177df: cmp dl,4
0x100177e2: jne short 100177EFh
0x100177e4: cmp cl,1
0x100177e7: jne short 100177EFh
0x100177e9: push 8
0x100177eb: push 4
0x100177ed: jmp short 100177F5h
0x100177ef: movsx ecx,cl
0x100177f2: push 8
0x100177f4: push ecx
0x100177f5: push edi
0x100177f6: push eax
0x100177f7: mov ecx,esi
```

```
0x100177f9: call 1000EAD0h
0x100177fe: cmp dword ptr [esi+148h],0
0x10017805: je short 1001781Ch
0x10017807: push 1013083Ch
0x1001780c: lea edx,[ebp-4Ch]
0x1001780f: push edx
0x10017810: mov dword ptr [ebp-4Ch],100FEAA8h
0x10017817: call 100229A6h
0x1001781c: movzx esi,word ptr [ebp+42h]
0x10017820: movsx eax,byte ptr [ebp+41h]
0x10017824: imul esi,eax
0x10017827: imul esi,edi
0x1001782a: push esi
0x1001782b: call dword ptr ds:[100FE550h]
0x10017831: mov edi,eax
0x10017833: add esp,4
0x10017836: mov [ebp-24h],edi
0x10017839: lea esp,[esp]
0x10017840: test esi,esi
0x10017842: jle near ptr 100178DAh
0x10017848: test ebx,ebx
0x1001784a: je short 100178C5h
0x1001784c: mov edx,[ebx]
0x1001784e: mov eax,[edx+20h]
0x10017851: mov ecx,ebx
0x10017853: call eax
0x10017855: test al,al
0x10017857: jne short 100178C5h
0x10017859: mov edx,[ebx]
0x1001785b: mov edx,[edx+8]
0x1001785e: push 1
0x10017860: push 1
0x10017862: lea eax,[ebp-11h]
0x10017865: push eax
0x10017866: mov ecx,ebx
0x10017868: call edx
0x1001786a: mov al,[ebp-11h]
0x1001786d: mov cl,al
0x1001786f: and cl,0C0h
0x10017872: cmp cl,0C0h
0x10017875: je short 10017881h
0x10017877: mov [edi],al
0x10017879: add edi,1
0x1001787c: sub esi,1
0x1001787f: jmp short 10017840h
0x10017881: mov edx,[ebx]
0x10017883: mov edx,[edx+8]
0x10017886: and eax,3Fh
0x10017889: push 1
0x1001788b: mov [ebp-18h],eax
0x1001788e: push 1
0x10017890: lea eax,[ebp-11h]
0x10017893: push eax
0x10017894: mov ecx,ebx
0x10017896: call edx
0x10017898: mov eax,[ebp-18h]
0x1001789b: cmp eax,esi
0x1001789d: jle short 100178B4h
0x1001789f: push 1013083Ch
0x100178a4: lea eax,[ebp-40h]
0x100178a7: push eax
0x100178a8: mov dword ptr [ebp-40h],100FEE14h
0x100178af: call 100229A6h
0x100178b4: sub esi,eax
0x100178b6: sub eax,1
0x100178b9: js short 10017840h
0x100178bb: mov cl,[ebp-11h]
```

```
0x100178be: mov [edi],cl
0x100178c0: add edi,1
0x100178c3: jmp short 100178B6h
0x100178c5: push 1013083Ch
0x100178ca: lea edx,[ebp-50h]
0x100178cd: push edx
0x100178ce: mov dword ptr [ebp-50h],100FEE04h
0x100178d5: call 100229A6h
0x100178da: xor esi,esi
0x100178dc: lea esp,[esp]
0x100178e0: cmp esi,10h
0x100178e3: jge short 10017910h
0x100178e5: lea eax,[esi+esi*2]
0x100178e8: movzx ecx,byte ptr [ebp+eax+10h]
0x100178ed: mov dl,[ebp+eax+11h]
0x100178f1: mov [ebp+eax+80h],cl
0x100178f8: movzx ecx,byte ptr [ebp+eax+12h]
0x100178fd: mov [ebp+eax+81h],dl
0x10017904: mov [ebp+eax+82h],cl
0x1001790b: add esi,1
0x1001790e: jmp short 100178E0h
0x10017910: cmp byte ptr [ebp+3],8
0x10017914: jne near ptr 100179A1h
0x1001791a: cmp byte ptr [ebp+41h],1
0x1001791e: jne near ptr 100179D6h
0x10017924: mov edx,[ebx]
0x10017926: mov edx,[edx+8]
0x10017929: push 1
0x1001792b: push 1
0x1001792d: lea eax,[ebp-11h]
0x10017930: push eax
0x10017931: mov ecx,ebx
0x10017933: call edx
0x10017935: cmp byte ptr [ebp-11h],0Ch
0x10017939: je short 10017950h
0x1001793b: push 1013083Ch
0x10017940: lea eax,[ebp-48h]
0x10017943: push eax
0x10017944: mov dword ptr [ebp-48h],100FEDECh
0x1001794b: call 100229A6h
0x10017950: xor edi,edi
0x10017952: cmp edi,100h
0x10017958: jge short 100179A1h
0x1001795a: mov edx,[ebx]
0x1001795c: mov edx,[edx+8]
0x1001795f: push 1
0x10017961: lea esi,[edi+edi*2]
0x10017964: push 1
0x10017966: lea eax,[ebp+esi+80h]
0x1001796d: push eax
0x1001796e: mov ecx,ebx
0x10017970: call edx
0x10017972: mov eax,[ebx]
0x10017974: mov edx,[eax+8]
0x10017977: push 1
0x10017979: push 1
0x1001797b: lea ecx,[ebp+esi+81h]
0x10017982: push ecx
0x10017983: mov ecx,ebx
0x10017985: call edx
0x10017987: mov eax,[ebx]
0x10017989: mov edx,[eax+8]
0x1001798c: push 1
0x1001798e: push 1
0x10017990: lea ecx,[ebp+esi+82h]
0x10017997: push ecx
0x10017998: mov ecx,ebx
```

```
0x1001799a: call edx
0x1001799c: add edi,1
0x1001799f: jmp short 10017952h
0x100179a1: cmp byte ptr [ebp+3],1
0x100179a5: jne short 100179D6h
0x100179a7: cmp byte ptr [ebp+41h],1
0x100179ab: jne short 100179D6h
0x100179ad: mov al,0FFh
0x100179af: mov byte ptr [ebp+82h],0
0x100179b6: mov byte ptr [ebp+81h],0
0x100179bd: mov byte ptr [ebp+80h],0
0x100179c4: mov [ebp+85h],al
0x100179ca: mov [ebp+84h],al
0x100179d0: mov [ebp+83h],al
0x100179d6: mov edi,[ebp-20h]
0x100179d9: xor esi,esi
0x100179db: jmp short 100179E0h
0x100179e0: cmp esi,[edi+28h]
0x100179e3: jae short 10017A12h
0x100179e5: lea eax,[esi+esi*2]
0x100179e8: movzx ecx,byte ptr [ebp+eax+82h]
0x100179f0: movzx edx,byte ptr [ebp+eax+81h]
0x100179f8: movzx eax,byte ptr [ebp+eax+80h]
0x10017a00: push 0
0x10017a02: push ecx
0x10017a03: push edx
0x10017a04: push eax
0x10017a05: push esi
0x10017a06: mov ecx,edi
0x10017a08: call 10012E50h
0x10017a0d: add esi,1
0x10017a10: jmp short 100179E0h
0x10017a12: movzx ecx,word ptr [ebp+42h]
0x10017a16: mov edx,[ebp-28h]
0x10017a19: lea eax,[edx+ecx*8]
0x10017a1c: push eax
0x10017a1d: call dword ptr ds:[100FE550h]
0x10017a23: mov [ebp-1Ch],eax
0x10017a26: mov eax,[ebp+42h]
0x10017a29: add esp,4
0x10017a2c: mov dword ptr [ebp-18h],0
0x10017a33: mov ebx,[ebp-2Ch]
0x10017a36: cmp [ebp-18h],ebx
0x10017a39: jge near ptr 10017BBAh
0x10017a3f: cmp dword ptr [edi+148h],0
0x10017a46: je short 10017A5Dh
0x10017a48: push 1013083Ch
0x10017a4d: lea ecx,[ebp-34h]
0x10017a50: push ecx
0x10017a51: mov dword ptr [ebp-34h],100FEAA8h
0x10017a58: call 100229A6h
0x10017a5d: movsx edx,byte ptr [ebp+41h]
0x10017a61: mov ecx,[ebp-18h]
0x10017a64: movzx esi,ax
0x10017a67: imul esi,edx
0x10017a6a: imul esi,ecx
0x10017a6d: add esi,[ebp-24h]
0x10017a70: sub ebx,ecx
0x10017a72: mov cl,[ebp+41h]
0x10017a75: sub ebx,1
0x10017a78: cmp cl,3
0x10017a7b: jne short 10017AC5h
0x10017a7d: cmp byte ptr [ebp+3],8
0x10017a81: jne near ptr 10017B56h
0x10017a87: xor edi,edi
0x10017a89: lea esp,[esp]
0x10017a90: cmp edi,[ebp-28h]
```

```
0x10017a93: jge near ptr 10017BAEh
0x10017a99: movzx eax,ax
0x10017a9c: lea ecx,[edi+eax*2]
0x10017a9f: xor edx,edx
0x10017aa1: mov dh,[ecx+esi]
0x10017aa4: mov ecx,[ebp-20h]
0x10017aa7: add eax,edi
0x10017aa9: mov dl,[eax+esi]
0x10017aac: movzx eax,byte ptr [edi+esi]
0x10017ab0: shl edx,8
0x10017ab3: or edx,eax
0x10017ab5: push edx
0x10017ab6: push ebx
0x10017ab7: push edi
0x10017ab8: call 10013980h
0x10017abd: mov eax,[ebp+42h]
0x10017ac0: add edi,1
0x10017ac3: jmp short 10017A90h
0x10017ac5: cmp cl,4
0x10017ac8: jne short 10017B23h
0x10017aca: cmp byte ptr [ebp+3],8
0x10017ace: jne near ptr 10017B56h
0x10017ad4: xor edi,edi
0x10017ad6: cmp edi,[ebp-28h]
0x10017ad9: jge near ptr 10017BAEh
0x10017adf: movzx eax,ax
0x10017ae2: lea ecx,[edi+eax*2]
0x10017ae5: xor edx,edx
0x10017ae7: mov dh,[ecx+esi]
0x10017aea: mov ecx,[ebp-20h]
0x10017aed: add eax,edi
0x10017aef: mov dl,[eax+esi]
0x10017af2: movzx eax,byte ptr [edi+esi]
0x10017af6: shl edx,8
0x10017af9: or edx,eax
0x10017afb: push edx
0x10017afc: push ebx
0x10017afd: push edi
0x10017afe: call 10013980h
0x10017b03: movzx eax,word ptr [ebp+42h]
0x10017b07: lea ecx,[edi+eax*2]
0x10017b0a: add ecx,eax
0x10017b0c: movzx edx,byte ptr [ecx+esi]
0x10017b10: mov ecx,[ebp-20h]
0x10017b13: push edx
0x10017b14: push ebx
0x10017b15: push edi
0x10017b16: call 10013AF0h
0x10017b1b: mov eax,[ebp+42h]
0x10017b1e: add edi,1
0x10017b21: jmp short 10017AD6h
0x10017b23: cmp cl,1
0x10017b26: jne short 10017B56h
0x10017b28: movsx cx,byte ptr [ebp+3]
0x10017b2d: mov edx,[ebp-1Ch]
0x10017b30: push ecx
0x10017b31: push 1
0x10017b33: push eax
0x10017b34: push esi
0x10017b35: push edx
0x10017b36: mov ecx,edi
0x10017b38: call 10017460h
0x10017b3d: test al,al
0x10017b3f: jne short 10017B87h
0x10017b41: push 1013083Ch
0x10017b46: lea eax,[ebp-3Ch]
0x10017b49: push eax
```

```
0x10017b4a: mov dword ptr [ebp-3Ch],100FEDA8h
0x10017b51: call 100229A6h
0x10017b56: movsx dx,byte ptr [ebp+3]
0x10017b5b: movsx cx,cl
0x10017b5f: push edx
0x10017b60: mov edx,[ebp-1Ch]
0x10017b63: push ecx
0x10017b64: push eax
0x10017b65: push esi
0x10017b66: push edx
0x10017b67: mov ecx,edi
0x10017b69: call 100173C0h
0x10017b6e: test al,al
0x10017b70: jne short 10017B87h
0x10017b72: push 1013083Ch
0x10017b77: lea eax,[ebp-44h]
0x10017b7a: push eax
0x10017b7b: mov dword ptr [ebp-44h],100FED60h
0x10017b82: call 100229A6h
0x10017b87: xor esi,esi
0x10017b89: lea esp,[esp]
0x10017b90: cmp esi,[ebp-28h]
0x10017b93: jge short 10017BABh
0x10017b95: mov ecx,[ebp-1Ch]
0x10017b98: movzx edx,byte ptr [esi+ecx]
0x10017b9c: push edx
0x10017b9d: push ebx
0x10017b9e: push esi
0x10017b9f: mov ecx,edi
0x10017ba1: call 10013330h
0x10017ba6: add esi,1
0x10017ba9: jmp short 10017B90h
0x10017bab: mov eax,[ebp+42h]
0x10017bae: add dword ptr [ebp-18h],1
0x10017bb2: mov edi,[ebp-20h]
0x10017bb5: jmp near ptr 10017A33h
0x10017bba: mov eax,[ebp-24h]
0x10017bbd: test eax,eax
0x10017bbf: je short 10017BCBh
0x10017bc1: push eax
0x10017bc2: call dword ptr ds:[100FE554h]
0x10017bc8: add esp,4
0x10017bcb: mov eax,[ebp-1Ch]
0x10017bce: test eax,eax
0x10017bd0: je short 10017BDCh
0x10017bd2: push eax
0x10017bd3: call dword ptr ds:[100FE554h]
0x10017bd9: add esp,4
0x10017bdc: mov al,1
0x10017bde: jmp short 10017C35h
0x10017c33: xor al,al
0x10017c35: mov ecx,[ebp-0Ch]
0x10017c38: mov fs:[0],ecx
0x10017c3f: pop ecx
0x10017c40: pop edi
0x10017c41: pop esi
0x10017c42: pop ebx
0x10017c43: mov ecx,[ebp+380h]
0x10017c49: xor ecx,ebp
0x10017c4b: call 10022326h
0x10017c50: add ebp,384h
0x10017c56: mov esp,ebp
0x10017c58: pop ebp
0x10017c59: ret 4
```

`sub_10017680` is called by sub_1000f760. It works with the text "Cancelled", "PCX_PlanesToPixels: more than 4 planes or more than 1 bit per pixel", "bad color map signature". It delegates to sub_1000ead0, sub_10012e50, sub_10013330 (11 callees).

12.349 sub_10017dd0 at 0x10017dd0 (36 instructions)

```
0x10017dd0: push esi
0x10017dd1: mov esi,[esp+8]
0x10017dd5: movzx eax,word ptr [esi+3]
0x10017dd9: push edi
0x10017dda: push eax
0x10017ddb: mov edi,ecx
0x10017ddd: call 1000F1C0h
0x10017de2: movzx ecx,word ptr [esi+5]
0x10017de6: push ecx
0x10017de7: mov ecx,edi
0x10017de9: mov [esi+3],ax
0x10017ded: call 1000F1C0h
0x10017df2: movzx edx,word ptr [esi+8]
0x10017df6: push edx
0x10017df7: mov ecx,edi
0x10017df9: mov [esi+5],ax
0x10017dfd: call 1000F1C0h
0x10017e02: mov [esi+8],ax
0x10017e06: movzx eax,word ptr [esi+0Ah]
0x10017e0a: push eax
0x10017e0b: mov ecx,edi
0x10017e0d: call 1000F1C0h
0x10017e12: movzx ecx,word ptr [esi+0Ch]
0x10017e16: push ecx
0x10017e17: mov ecx,edi
0x10017e19: mov [esi+0Ah],ax
0x10017e1d: call 1000F1C0h
0x10017e22: movzx edx,word ptr [esi+0Eh]
0x10017e26: push edx
0x10017e27: mov ecx,edi
0x10017e29: mov [esi+0Ch],ax
0x10017e2d: call 1000F1C0h
0x10017e32: pop edi
0x10017e33: mov [esi+0Eh],ax
0x10017e37: pop esi
0x10017e38: ret 4
```

`sub_10017dd0` is called by sub_10018120. It delegates to sub_1000f1c0 (1 callees).

12.350 sub_10018120 at 0x10018120 (78 instructions)

```
0x10018120: push ebp
0x10018121: mov ebp,esp
0x10018123: push 0FFFFFFFFh
0x10018125: push 100FC5F0h
0x1001812a: mov eax,fs:[0]
0x10018130: push eax
0x10018131: sub esp,360h
0x10018137: push ebx
0x10018138: push esi
0x10018139: push edi
0x1001813a: mov eax,ds:[10136220h]
0x1001813f: xor eax,ebp
0x10018141: push eax
0x10018142: lea eax,[ebp-0Ch]
0x10018145: mov fs:[0],eax
0x1001814b: mov [ebp-10h],esp
0x1001814e: mov edi,ecx
0x10018150: mov [ebp-3Ch],edi
0x10018153: mov ebx,[ebp+8]
```

```
0x10018156: test ebx,ebx
0x10018158: je near ptr 1001849Dh
0x1001815e: mov eax,[ebx]
0x10018160: mov edx,[eax+8]
0x10018163: push 1
0x10018165: push 12h
0x10018167: lea ecx,[ebp-28h]
0x1001816a: push ecx
0x1001816b: mov ecx,ebx
0x1001816d: mov dword ptr [ebp-4],0
0x10018174: call edx
0x10018176: test eax,eax
0x10018178: jne short 1001818Fh
0x1001817a: push 1013083Ch
0x1001817f: lea eax,[ebp-40h]
0x10018182: push eax
0x10018183: mov dword ptr [ebp-40h],100FEF04h
0x1001818a: call 100229A6h
0x1001818f: lea ecx,[ebp-28h]
0x10018192: push ecx
0x10018193: mov ecx,edi
0x10018195: call 10017DD0h
0x1001819a: movzx eax,byte ptr [ebp-26h]
0x1001819e: add eax,0FFFFFFFFh
0x100181a1: cmp eax,0Ah
0x100181a4: ja near ptr 10018457h
0x100181aa: movzx edx,byte ptr [eax+100184C0h]
0x100181b1: jmp dword ptr [edx*4+100184B4h]
0x10018457: push 1013083Ch
0x1001845c: lea edx,[ebp-34h]
0x1001845f: push edx
0x10018460: mov dword ptr [ebp-34h],100FEEB8h
0x10018467: call 100229A6h
0x1001846c: mov eax,[ebp-54h]
0x1001846f: mov edi,100FE92Bh
0x10018474: mov esi,eax
0x10018476: mov ecx,1
0x1001847b: xor edx,edx
0x1001847d: repe cmpsb
0x1001847f: je short 10018497h
0x10018481: push 0FFh
0x10018486: push eax
0x10018487: mov eax,[ebp-3Ch]
0x1001848a: add eax,44h
0x1001848d: push eax
0x1001848e: call dword ptr ds:[100FE568h]
0x10018494: add esp,0Ch
0x10018497: mov eax,1001849Dh
0x1001849c: ret
0x1001849d: xor al,al
0x1001849f: mov ecx,[ebp-0Ch]
0x100184a2: mov fs:[0],ecx
0x100184a9: pop ecx
0x100184aa: pop edi
0x100184ab: pop esi
0x100184ac: pop ebx
0x100184ad: mov esp,ebp
0x100184af: pop ebp
0x100184b0: ret 4
```

`sub_10018120` is called by sub_1000f760. It works with the text "Unknown TGA image type", "Not a TGA". It delegates to sub_10017dd0, j__CxxThrowException (2 callees).

12.351 sub_100184d0 at 0x100184d0 (6 instructions)

```
0x100184d0: mov eax,[esp+4]
0x100184d4: push eax
0x100184d5: push 1
0x100184d7: call dword ptr ds:[100FE4FCh]
0x100184dd: add esp,8
0x100184e0: ret
```

`sub_100184d0` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.352 sub_100184f0 at 0x100184f0 (5 instructions)

```
0x100184f0: mov eax,[esp+4]
0x100184f4: push eax
0x100184f5: call dword ptr ds:[100FE554h]
0x100184fb: pop ecx
0x100184fc: ret
```

`sub_100184f0` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.353 sub_10018500 at 0x10018500 (2 instructions)

```
0x10018500: mov al,1
0x10018502: ret
```

`sub_10018500` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.354 sub_10018510 at 0x10018510 (17 instructions)

```
0x10018510: mov eax,[esp+4]
0x10018514: push eax
0x10018515: call 1007F360h
0x1001851a: mov ecx,[eax]
0x1001851c: mov eax,[esp+10h]
0x10018520: mov edx,[ecx]
0x10018522: mov edx,[edx+8]
0x10018525: add esp,4
0x10018528: push eax
0x10018529: mov eax,[esp+0Ch]
0x1001852d: push 1
0x1001852f: push eax
0x10018530: call edx
0x10018532: mov ecx,[esp+10h]
0x10018536: mov [ecx],eax
0x10018538: mov al,1
0x1001853a: ret
```

`sub_10018510` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1007f360 (1 callees).

12.355 sub_10018540 at 0x10018540 (46 instructions)

```
0x10018540: push ebx
0x10018541: mov ebx,[esp+8]
0x10018545: push ebp
0x10018546: push esi
0x10018547: push edi
0x10018548: push ebx
0x10018549: call 1007F360h
0x1001854e: mov edi,[esp+1Ch]
0x10018552: mov ebp,[esp+20h]
0x10018556: mov esi,eax
```

```
0x10018558: lea eax,[edi+edi*2]
0x1001855b: lea eax,[eax*8+1Fh]
0x10018562: shr eax,3
0x10018565: and eax,1FFFFFFCh
0x1001856a: add esp,4
0x1001856d: mov [esi+14h],edi
0x10018570: mov [esi+18h],ebp
0x10018573: mov word ptr [esi+20h],18h
0x10018579: mov [esi+1Ch],eax
0x1001857c: cmp byte ptr [ebx+56h],0
0x10018580: je short 1001859Dh
0x10018582: mov dword ptr [esi+28h],0
0x10018589: movzx ecx,byte ptr [ebx+51h]
0x1001858d: mov [esi+2Eh],cl
0x10018590: mov dl,[ebx+53h]
0x10018593: mov [esi+2Dh],dl
0x10018596: movzx ecx,byte ptr [ebx+55h]
0x1001859a: mov [esi+2Ch],cl
0x1001859d: imul eax,ebp
0x100185a0: push eax
0x100185a1: call dword ptr ds:[100FE550h]
0x100185a7: push 5000h
0x100185ac: push ebx
0x100185ad: mov [esi+4],eax
0x100185b0: call 1007F1A0h
0x100185b5: imul edi,ebp
0x100185b8: push edi
0x100185b9: call dword ptr ds:[100FE550h]
0x100185bf: add esp,10h
0x100185c2: pop edi
0x100185c3: mov [esi+8],eax
0x100185c6: pop esi
0x100185c7: pop ebp
0x100185c8: mov al,1
0x100185ca: pop ebx
0x100185cb: ret
```

`sub_10018540` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1007f1a0, sub_1007f360 (2 callees).

12.356 sub_100185d0 at 0x100185d0 (11 instructions)

```
0x100185d0: mov eax,[esp+4]
0x100185d4: push eax
0x100185d5: call 1007F360h
0x100185da: mov ecx,eax
0x100185dc: mov eax,[ecx+18h]
0x100185df: sub eax,[esp+0Ch]
0x100185e3: add esp,4
0x100185e6: sub eax,1
0x100185e9: imul eax,[ecx+1Ch]
0x100185ed: add eax,[ecx+4]
0x100185f0: ret
```

`sub_100185d0` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1007f360 (1 callees).

12.357 sub_10018600 at 0x10018600 (11 instructions)

```
0x10018600: mov eax,[esp+4]
0x10018604: push eax
0x10018605: call 1007F360h
0x1001860a: mov ecx,eax
0x1001860c: mov eax,[ecx+18h]
0x1001860f: sub eax,[esp+0Ch]
```

```
0x10018613: add esp,4
0x10018616: sub eax,1
0x10018619: imul eax,[ecx+14h]
0x1001861d: add eax,[ecx+8]
0x10018620: ret
```

`sub_10018600` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1007f360 (1 callees).

12.358 j_GetTickCount at 0x10018630 (1 instructions)

```
0x10018630: jmp dword ptr ds:[100FE10Ch]
```

`j_GetTickCount` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.359 sub_10018640 at 0x10018640 (8 instructions)

```
0x10018640: mov eax,[esp+4]
0x10018644: push eax
0x10018645: call 1007F360h
0x1001864a: mov ecx,[esp+0Ch]
0x1001864e: mov [eax+10h],ecx
0x10018651: add esp,4
0x10018654: mov al,1
0x10018656: ret
```

`sub_10018640` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1007f360 (1 callees).

12.360 sub_10018660 at 0x10018660 (5 instructions)

```
0x10018660: lea eax,[esp+4]
0x10018664: push eax
0x10018665: call 1007FF10h
0x1001866a: add esp,4
0x1001866d: ret
```

`sub_10018660` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1007ff10 (1 callees).

12.361 sub_10018670 at 0x10018670 (18 instructions)

```
0x10018670: push esi
0x10018671: push 11h
0x10018673: mov esi,ecx
0x10018675: call 1000E480h
0x1001867a: push 30h
0x1001867c: lea eax,[esi+3D4h]
0x10018682: push 0
0x10018684: push eax
0x10018685: mov dword ptr [esi],100FEF14h
0x1001868b: mov dword ptr [esi+3D0h],0
0x10018695: call 100228C0h
0x1001869a: fld1
0x1001869c: add esp,0Ch
0x1001869f: fstp dword ptr [esi+3F8h]
0x100186a5: mov dword ptr [esi+3FCh],0FFFFFFFFh
0x100186af: mov eax,esi
0x100186b1: pop esi
0x100186b2: ret
```

`sub_10018670` is called by sub_1000f760. It delegates to sub_1000e480, j_memset (2 callees).

12.362 sub_10018760 at 0x10018760 (39 instructions)

```
0x10018760: push esi
0x10018761: mov esi,[esp+8]
0x10018765: push 10018660h
0x1001876a: push esi
0x1001876b: call 100805D0h
0x10018770: push 10018500h
0x10018775: push esi
0x10018776: call 10080540h
0x1001877b: push 10018500h
0x10018780: push esi
0x10018781: call 10080570h
0x10018786: push 10018510h
0x1001878b: push esi
0x1001878c: call 100805A0h
0x10018791: push 10018540h
0x10018796: push esi
0x10018797: call 10080600h
0x1001879c: push 100185D0h
0x100187a1: push esi
0x100187a2: call 10080630h
0x100187a7: push 10018500h
0x100187ac: push esi
0x100187ad: call 10080690h
0x100187b2: push 10018630h
0x100187b7: push esi
0x100187b8: call 100806C0h
0x100187bd: add esp,40h
0x100187c0: push 10018640h
0x100187c5: push esi
0x100187c6: call 100806F0h
0x100187cb: push 10018500h
0x100187d0: push esi
0x100187d1: call 10080690h
0x100187d6: push 10018600h
0x100187db: push esi
0x100187dc: call 10080660h
0x100187e1: add esp,18h
0x100187e4: pop esi
0x100187e5: ret 4
```

`sub_10018760` is called by sub_100187f0. It delegates to sub_10080540, sub_10080570, sub_100805a0 (10 callees).

12.363 sub_100187f0 at 0x100187f0 (244 instructions)

```
0x100187f0: push ebp
0x100187f1: mov ebp,esp
0x100187f3: push 0FFFFFFFFh
0x100187f5: push 100FC610h
0x100187fa: mov eax,fs:[0]
0x10018800: push eax
0x10018801: sub esp,20h
0x10018804: push ebx
0x10018805: push esi
0x10018806: push edi
0x10018807: mov eax,ds:[10136220h]
0x1001880c: xor eax,ebp
0x1001880e: push eax
0x1001880f: lea eax,[ebp-0Ch]
0x10018812: mov fs:[0],eax
0x10018818: mov [ebp-10h],esp
0x1001881b: mov esi,ecx
0x1001881d: mov [ebp-28h],esi
0x10018820: mov ebx,[ebp+8]
0x10018823: test ebx,ebx
0x10018825: je near ptr 10018B2Bh
```

```
0x1001882b: push 0
0x1001882d: push 100184F0h
0x10018832: lea edi,[esi+3D4h]
0x10018838: push 100184D0h
0x1001883d: push edi
0x1001883e: mov dword ptr [ebp-4],0
0x10018845: call 10080300h
0x1001884a: add esp,10h
0x1001884d: test eax,eax
0x1001884f: mov [esi+3D0h],eax
0x10018855: jne short 1001886Ch
0x10018857: push 1013083Ch
0x1001885c: lea eax,[ebp-14h]
0x1001885f: push eax
0x10018860: mov dword ptr [ebp-14h],100FEF7Ch
0x10018867: call 100229A6h
0x1001886c: push 1
0x1001886e: push eax
0x1001886f: mov [edi],ebx
0x10018871: call 1007F2F0h
0x10018876: mov ecx,[esi+3D0h]
0x1001887c: push 0FFFFh
0x10018881: push 0FFFFh
0x10018886: push 0FFFFh
0x1001888b: push ecx
0x1001888c: call 1007F270h
0x10018891: mov edx,[esi+3D0h]
0x10018897: push 1
0x10018899: push edx
0x1001889a: call 1007F2B0h
0x1001889f: mov eax,[esi+3D0h]
0x100188a5: push 0
0x100188a7: push eax
0x100188a8: call 1007F2D0h
0x100188ad: mov ecx,[esi+3D0h]
0x100188b3: push 0
0x100188b5: push ecx
0x100188b6: call 1007F310h
0x100188bb: mov edx,[esi+3D0h]
0x100188c1: add esp,30h
0x100188c4: push edx
0x100188c5: mov ecx,esi
0x100188c7: call 10018760h
0x100188cc: mov edi,[esi+3D0h]
0x100188d2: push edi
0x100188d3: mov dword ptr [esi+160h],0
0x100188dd: call 1007FF70h
0x100188e2: add esp,4
0x100188e5: test eax,eax
0x100188e7: je short 10018938h
0x100188e9: cmp eax,0Fh
0x100188ec: je short 10018938h
0x100188ee: push 0
0x100188f0: push 0
0x100188f2: push eax
0x100188f3: mov eax,[esi+3D0h]
0x100188f9: push eax
0x100188fa: call 10080720h
0x100188ff: mov esi,[esi+3D0h]
0x10018905: add esp,10h
0x10018908: cmp dword ptr [esi+94h],0
0x1001890f: push 1013083Ch
0x10018914: je short 10018928h
0x10018916: mov ecx,[esi+94h]
0x1001891c: lea edx,[ebp-18h]
0x1001891f: push edx
0x10018920: mov [ebp-18h],ecx
```

```
0x10018923: call 100229A6h
0x10018928: lea eax,[ebp-1Ch]
0x1001892b: push eax
0x1001892c: mov dword ptr [ebp-1Ch],100FEF68h
0x10018933: call 100229A6h
0x10018938: or ebx,0FFFFFFFFh
0x1001893b: cmp [esi+148h],ebx
0x10018941: jne short 10018972h
0x10018943: mov eax,[esi+3D0h]
0x10018949: mov ecx,[eax+10h]
0x1001894c: mov [esi+0Ch],ecx
0x1001894f: mov edx,[eax+14h]
0x10018952: mov [esi+10h],edx
0x10018955: mov dword ptr [esi+40h],11h
0x1001895c: mov al,1
0x1001895e: mov ecx,[ebp-0Ch]
0x10018961: mov fs:[0],ecx
0x10018968: pop ecx
0x10018969: pop edi
0x1001896a: pop esi
0x1001896b: pop ebx
0x1001896c: mov esp,ebp
0x1001896e: pop ebp
0x1001896f: ret 4
0x10018972: cmp byte ptr [edi+158h],0
0x10018979: je short 10018993h
0x1001897b: mov eax,[esi+3D0h]
0x10018981: push eax
0x10018982: call 10080150h
0x10018987: add esp,4
0x1001898a: add dword ptr [esi+160h],1
0x10018991: jmp short 10018972h
0x10018993: cmp eax,0Fh
0x10018996: je short 100189A0h
0x10018998: add [esi+160h],ebx
0x1001899e: jmp short 100189A7h
0x100189a0: mov byte ptr [esi+3F6h],1
0x100189a7: cmp dword ptr [esi+160h],0
0x100189ae: jg short 100189BAh
0x100189b0: mov dword ptr [esi+160h],1
0x100189ba: cmp byte ptr [esi+3F7h],0
0x100189c1: jne short 10018A08h
0x100189c3: mov eax,[esi+15Ch]
0x100189c9: test eax,eax
0x100189cb: jl short 100189F3h
0x100189cd: cmp eax,[esi+160h]
0x100189d3: jge short 100189F3h
0x100189d5: xor edi,edi
0x100189d7: cmp edi,[esi+15Ch]
0x100189dd: jge short 10018A08h
0x100189df: mov ecx,[esi+3D0h]
0x100189e5: push ecx
0x100189e6: call 10080150h
0x100189eb: add esp,4
0x100189ee: add edi,1
0x100189f1: jmp short 100189D7h
0x100189f3: push 1013083Ch
0x100189f8: lea edx,[ebp-20h]
0x100189fb: push edx
0x100189fc: mov dword ptr [ebp-20h],100FEF40h
0x10018a03: call 100229A6h
0x10018a08: mov eax,[esi+3FCh]
0x10018a0e: test eax,eax
0x10018a10: jl short 10018A3Ch
0x10018a12: mov cl,[esi+401h]
0x10018a18: mov dl,[esi+400h]
0x10018a1e: mov [esi+14Ch],eax
```

```
0x10018a24: mov al,[esi+402h]
0x10018a2a: mov [esi+152h],al
0x10018a30: mov [esi+151h],cl
0x10018a36: mov [esi+150h],dl
0x10018a3c: movsx eax,word ptr [esi+3F4h]
0x10018a43: mov ecx,[esi+3ECh]
0x10018a49: mov edx,[esi+3E8h]
0x10018a4f: push 11h
0x10018a51: push eax
0x10018a52: push ecx
0x10018a53: push edx
0x10018a54: mov ecx,esi
0x10018a56: call 1000EAD0h
0x10018a5b: test eax,eax
0x10018a5d: je near ptr 10018AE5h
0x10018a63: push 0
0x10018a65: mov ecx,esi
0x10018a67: call 1000E5E0h
0x10018a6c: mov ecx,[esi+30h]
0x10018a6f: imul ecx,[esi+10h]
0x10018a73: mov edx,[esi+3D8h]
0x10018a79: push ecx
0x10018a7a: push edx
0x10018a7b: push eax
0x10018a7c: call 100228C6h
0x10018a81: add esp,0Ch
0x10018a84: mov ecx,esi
0x10018a86: call 10013780h
0x10018a8b: mov ecx,esi
0x10018a8d: call 100139C0h
0x10018a92: mov ecx,esi
0x10018a94: call 100139B0h
0x10018a99: test al,al
0x10018a9b: je short 10018ACFh
0x10018a9d: cmp dword ptr [esi+3DCh],0
0x10018aa4: je short 10018ACFh
0x10018aa6: push 0
0x10018aa8: push 0
0x10018aaa: mov ecx,esi
0x10018aac: call 10013B70h
0x10018ab1: mov ecx,[esi+3ECh]
0x10018ab7: imul ecx,[esi+3E8h]
0x10018abe: mov edx,[esi+3DCh]
0x10018ac4: push ecx
0x10018ac5: push edx
0x10018ac6: push eax
0x10018ac7: call 100228C6h
0x10018acc: add esp,0Ch
0x10018acf: mov al,1
0x10018ad1: mov ecx,[ebp-0Ch]
0x10018ad4: mov fs:[0],ecx
0x10018adb: pop ecx
0x10018adc: pop edi
0x10018add: pop esi
0x10018ade: pop ebx
0x10018adf: mov esp,ebp
0x10018ae1: pop ebp
0x10018ae2: ret 4
0x10018ae5: push 1013083Ch
0x10018aea: lea eax,[ebp-24h]
0x10018aed: push eax
0x10018aee: mov dword ptr [ebp-24h],100FEF18h
0x10018af5: call 100229A6h
0x10018afa: mov eax,[ebp-2Ch]
0x10018afd: mov edi,100FE92Bh
0x10018b02: mov esi,eax
0x10018b04: mov ecx,1
```

```
0x10018b09: xor edx,edx
0x10018b0b: repe cmpsb
0x10018b0d: je short 10018B25h
0x10018b0f: push 0FFh
0x10018b14: push eax
0x10018b15: mov eax,[ebp-28h]
0x10018b18: add eax,44h
0x10018b1b: push eax
0x10018b1c: call dword ptr ds:[100FE568h]
0x10018b22: add esp,0Ch
0x10018b25: mov eax,10018B2Bh
0x10018b2a: ret
0x10018b2b: xor al,al
0x10018b2d: mov ecx,[ebp-0Ch]
0x10018b30: mov fs:[0],ecx
0x10018b37: pop ecx
0x10018b38: pop edi
0x10018b39: pop esi
0x10018b3a: pop ebx
0x10018b3b: mov esp,ebp
0x10018b3d: pop ebp
0x10018b3e: ret 4
```

`sub_100187f0` is called by sub_1000f760. It works with the text "CxImageMNG::Decode cannot create image", "Error: frame not present in MNG file", "Error in MNG file". It delegates to sub_1000e5e0, sub_1000ead0, sub_10013780 (18 callees).

12.364 sub_10018b70 at 0x10018b70 (25 instructions)

```
0x10018b70: push edi
0x10018b71: mov edi,[esp+8]
0x10018b75: push edi
0x10018b76: call 10023A20h
0x10018b7b: add esp,4
0x10018b7e: test eax,eax
0x10018b80: je short 10018B9Dh
0x10018b82: mov ecx,[esp+0Ch]
0x10018b86: mov edx,[eax]
0x10018b88: mov edx,[edx+8]
0x10018b8b: push esi
0x10018b8c: mov esi,[esp+14h]
0x10018b90: push esi
0x10018b91: push 1
0x10018b93: push ecx
0x10018b94: mov ecx,eax
0x10018b96: call edx
0x10018b98: cmp eax,esi
0x10018b9a: pop esi
0x10018b9b: je short 10018BA8h
0x10018b9d: push 100FEF98h
0x10018ba2: push edi
0x10018ba3: call 100231E0h
0x10018ba8: pop edi
0x10018ba9: ret
```

`sub_10018b70` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_error, png_get_io_ptr (2 callees).

12.365 sub_10018bb0 at 0x10018bb0 (14 instructions)

```
0x10018bb0: mov eax,[esp+8]
0x10018bb4: push esi
0x10018bb5: mov esi,[esp+8]
0x10018bb9: mov ecx,[esi+4Ch]
```

```
0x10018bbc: push 0FFh
0x10018bc1: push eax
0x10018bc2: push ecx
0x10018bc3: call dword ptr ds:[100FE568h]
0x10018bc9: add esp,0Ch
0x10018bcc: push 1
0x10018bce: push esi
0x10018bcf: call dword ptr ds:[100FE4E0h]
0x10018bd5: pop esi
0x10018bd6: int 3
```

`sub_10018bb0` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.366 sub_10018be0 at 0x10018be0 (29 instructions)

```
0x10018be0: push esi
0x10018be1: mov esi,ecx
0x10018be3: mov ecx,[esi+14h]
0x10018be6: push edi
0x10018be7: mov edi,[esp+10h]
0x10018beb: call 1000F000h
0x10018bf0: test edi,edi
0x10018bf2: jl short 10018C00h
0x10018bf4: cmp edi,eax
0x10018bf6: jl short 10018C02h
0x10018bf8: mov ecx,[esi+14h]
0x10018bfb: call 1000F000h
0x10018c00: mov edi,eax
0x10018c02: mov esi,[esi+10h]
0x10018c05: test esi,esi
0x10018c07: je short 10018C20h
0x10018c09: mov eax,[esp+0Ch]
0x10018c0d: test eax,eax
0x10018c0f: je short 10018C20h
0x10018c11: test edi,edi
0x10018c13: jle short 10018C20h
0x10018c15: push edi
0x10018c16: push eax
0x10018c17: push esi
0x10018c18: call 100228C6h
0x10018c1d: add esp,0Ch
0x10018c20: pop edi
0x10018c21: pop esi
0x10018c22: ret 8
```

`sub_10018be0` is called by sub_10019a40, sub_1001b350. It delegates to sub_1000f000, j_memcpy (2 callees).

12.367 sub_10018d20 at 0x10018d20 (157 instructions)

```
0x10018d20: push ebp
0x10018d21: mov ebp,esp
0x10018d23: push 0FFFFFFFFh
0x10018d25: push 100FC630h
0x10018d2a: mov eax,fs:[0]
0x10018d30: push eax
0x10018d31: sub esp,64h
0x10018d34: push ebx
0x10018d35: push esi
0x10018d36: push edi
0x10018d37: mov eax,ds:[10136220h]
0x10018d3c: xor eax,ebp
0x10018d3e: push eax
0x10018d3f: lea eax,[ebp-0Ch]
0x10018d42: mov fs:[0],eax
0x10018d48: mov [ebp-10h],esp
```

```
0x10018d4b: mov [ebp-1Ch],ecx
0x10018d4e: xor esi,esi
0x10018d50: cmp ecx,esi
0x10018d52: mov [ebp-30h],esi
0x10018d55: mov [ebp-5Ch],ecx
0x10018d58: je short 10018D63h
0x10018d5a: push esi
0x10018d5b: call 1000E5E0h
0x10018d60: mov [ebp-60h],eax
0x10018d63: push esi
0x10018d64: push esi
0x10018d65: push esi
0x10018d66: push 100FF018h
0x10018d6b: mov [ebp-6Ch],esi
0x10018d6e: mov [ebp-70h],esi
0x10018d71: mov [ebp-64h],esi
0x10018d74: mov [ebp-68h],esi
0x10018d77: mov [ebp-4],esi
0x10018d7a: call 10028570h
0x10018d7f: add esp,10h
0x10018d82: cmp eax,esi
0x10018d84: mov [ebp-18h],eax
0x10018d87: jne short 10018D9Eh
0x10018d89: push 1013083Ch
0x10018d8e: lea eax,[ebp-40h]
0x10018d91: push eax
0x10018d92: mov dword ptr [ebp-40h],100FEFF8h
0x10018d99: call 100229A6h
0x10018d9e: push eax
0x10018d9f: call 10025000h
0x10018da4: add esp,4
0x10018da7: cmp eax,esi
0x10018da9: mov [ebp-14h],eax
0x10018dac: jne short 10018DD1h
0x10018dae: push esi
0x10018daf: lea ecx,[ebp-18h]
0x10018db2: push esi
0x10018db3: push ecx
0x10018db4: call 100285A0h
0x10018db9: add esp,0Ch
0x10018dbc: push 1013083Ch
0x10018dc1: lea edx,[ebp-44h]
0x10018dc4: push edx
0x10018dc5: mov dword ptr [ebp-44h],100FEFD0h
0x10018dcc: call 100229A6h
0x10018dd1: mov eax,[ebp-4]
0x10018dd4: mov ecx,[ebp-18h]
0x10018dd7: push 10130B84h
0x10018ddc: push eax
0x10018ddd: push 100229E2h
0x10018de2: push 3
0x10018de4: push ecx
0x10018de5: call 100229DCh
0x10018dea: add esp,14h
0x10018ded: test eax,eax
0x10018def: je short 10018E23h
0x10018df1: mov edx,[ebp-30h]
0x10018df4: push edx
0x10018df5: call 10021FCCh
0x10018dfa: add esp,4
0x10018dfd: push esi
0x10018dfe: lea eax,[ebp-14h]
0x10018e01: push eax
0x10018e02: lea ecx,[ebp-18h]
0x10018e05: push ecx
0x10018e06: call 100285A0h
0x10018e0b: add esp,0Ch
```

```
0x10018e0e: push 1013083Ch
0x10018e13: lea edx,[ebp-48h]
0x10018e16: push edx
0x10018e17: mov dword ptr [ebp-48h],100FE92Bh
0x10018e1e: call 100229A6h
0x10018e23: mov eax,[ebp+8]
0x10018e26: mov ecx,[ebp-18h]
0x10018e29: push 10018B70h
0x10018e2e: push eax
0x10018e2f: push ecx
0x10018e30: call 10029C70h
0x10018e35: mov edi,[ebp-1Ch]
0x10018e38: mov edx,[ebp-18h]
0x10018e3b: push esi
0x10018e3c: push 10018BB0h
0x10018e41: lea ebx,[edi+44h]
0x10018e44: push ebx
0x10018e45: push edx
0x10018e46: call 100230C0h
0x10018e4b: mov eax,[ebp-14h]
0x10018e4e: mov ecx,[ebp-18h]
0x10018e51: push eax
0x10018e52: push ecx
0x10018e53: call 10025580h
0x10018e58: add esp,24h
0x10018e5b: cmp dword ptr [edi+148h],0FFFFFFFFh
0x10018e62: jne short 10018E85h
0x10018e64: mov eax,[ebp-14h]
0x10018e67: mov edx,[eax]
0x10018e69: mov ecx,[ebp-18h]
0x10018e6c: mov [edi+0Ch],edx
0x10018e6f: mov eax,[eax+4]
0x10018e72: push 1
0x10018e74: push ecx
0x10018e75: mov [edi+10h],eax
0x10018e78: mov dword ptr [edi+40h],4
0x10018e7f: call dword ptr ds:[100FE4E0h]
0x10018e85: mov ecx,[ebp-14h]
0x10018e88: movzx eax,byte ptr [ecx+19h]
0x10018e8c: cmp eax,6
0x10018e8f: ja near ptr 100194F2h
0x10018e95: jmp dword ptr [eax*4+1001956Ch]
0x100194f2: mov ecx,5
0x100194f7: mov esi,100FEFB0h
0x100194fc: mov edi,ebx
0x100194fe: rep movsd
0x10019500: mov ecx,[ebp-18h]
0x10019503: movsw
0x10019505: push 1
0x10019507: push ecx
0x10019508: movsb
0x10019509: call dword ptr ds:[100FE4E0h]
0x1001950f: mov eax,[ebp-58h]
0x10019512: mov edi,100FE92Bh
0x10019517: mov esi,eax
0x10019519: mov ecx,1
0x1001951e: xor edx,edx
0x10019520: repe cmpsb
0x10019522: mov esi,[ebp-1Ch]
0x10019525: je short 1001953Ah
0x10019527: push 0FFh
0x1001952c: push eax
0x1001952d: lea eax,[esi+44h]
0x10019530: push eax
0x10019531: call dword ptr ds:[100FE568h]
0x10019537: add esp,0Ch
0x1001953a: cmp dword ptr [esi+148h],0FFFFFFFFh
```

```
0x10019541: jne short 1001954Fh
0x10019543: cmp dword ptr [esi+40h],4
0x10019547: jne short 1001954Fh
0x10019549: mov eax,100194DCh
0x1001954e: ret
0x1001954f: mov eax,10019555h
0x10019554: ret
```

`sub_10018d20` is called by sub_1000f760. It works with the text "unknown PNG color type", "Failed to create PNG structure", "1.5.1". It delegates to sub_1000e5e0, j_ORDINAL 764, j__CxxThrowException (10 callees).

12.368 sub_10019590 at 0x10019590 (55 instructions)

```
0x10019590: push 0FFFFFFFFh
0x10019592: push 100FC658h
0x10019597: mov eax,fs:[0]
0x1001959d: push eax
0x1001959e: push ecx
0x1001959f: push esi
0x100195a0: mov eax,ds:[10136220h]
0x100195a5: xor eax,esp
0x100195a7: push eax
0x100195a8: lea eax,[esp+0Ch]
0x100195ac: mov fs:[0],eax
0x100195b2: mov esi,ecx
0x100195b4: mov [esp+8],esi
0x100195b8: push 2
0x100195ba: call 1000E480h
0x100195bf: push 4000h
0x100195c4: mov dword ptr [esp+18h],0
0x100195cc: mov dword ptr [esi],100FF068h
0x100195d2: call 10021FDEh
0x100195d7: push 1000h
0x100195dc: mov [esi+404h],eax
0x100195e2: call 10021FDEh
0x100195e7: push 1000h
0x100195ec: mov [esi+424h],eax
0x100195f2: call 10021FDEh
0x100195f7: push 2000h
0x100195fc: mov [esi+428h],eax
0x10019602: call 10021FDEh
0x10019607: push 4E2Ch
0x1001960c: mov [esi+42Ch],eax
0x10019612: call 10021FDEh
0x10019617: push 2716h
0x1001961c: mov [esi+430h],eax
0x10019622: call 10021FDEh
0x10019627: push 101h
0x1001962c: mov [esi+434h],eax
0x10019632: call 10021FDEh
0x10019637: push 100h
0x1001963c: mov [esi+41Ch],eax
0x10019642: call 10021FDEh
0x10019647: push 100h
0x1001964c: mov [esi+458h],eax
0x10019652: call 10021FDEh
0x10019657: mov [esi+45Ch],eax
0x1001965d: mov dword ptr [esi+460h],0
0x10019667: mov byte ptr [esi+1ECh],0
0x1001966e: mov byte ptr [eax],0
0x10019671: add esp,24h
0x10019674: mov eax,esi
0x10019676: mov ecx,[esp+0Ch]
0x1001967a: mov fs:[0],ecx
0x10019681: pop ecx
0x10019682: pop esi
```

```
0x10019683: add esp,10h
0x10019686: ret
```

`sub_10019590` is called by sub_1000f760, sub_1001a5b0. It delegates to sub_1000e480, j_ORDINAL 762 (2 callees).

12.369 sub_10019690 at 0x10019690 (55 instructions)

```
0x10019690: push 0FFFFFFFFh
0x10019692: push 100FC658h
0x10019697: mov eax,fs:[0]
0x1001969d: push eax
0x1001969e: push ecx
0x1001969f: push esi
0x100196a0: mov eax,ds:[10136220h]
0x100196a5: xor eax,esp
0x100196a7: push eax
0x100196a8: lea eax,[esp+0Ch]
0x100196ac: mov fs:[0],eax
0x100196b2: mov esi,ecx
0x100196b4: mov [esp+8],esi
0x100196b8: mov dword ptr [esi],100FF068h
0x100196be: mov eax,[esi+404h]
0x100196c4: push eax
0x100196c5: mov dword ptr [esp+18h],0
0x100196cd: call 10021FCCh
0x100196d2: mov eax,[esi+424h]
0x100196d8: push eax
0x100196d9: call 10021FCCh
0x100196de: mov eax,[esi+428h]
0x100196e4: push eax
0x100196e5: call 10021FCCh
0x100196ea: mov eax,[esi+42Ch]
0x100196f0: push eax
0x100196f1: call 10021FCCh
0x100196f6: mov eax,[esi+430h]
0x100196fc: push eax
0x100196fd: call 10021FCCh
0x10019702: mov eax,[esi+434h]
0x10019708: push eax
0x10019709: call 10021FCCh
0x1001970e: mov eax,[esi+41Ch]
0x10019714: push eax
0x10019715: call 10021FCCh
0x1001971a: mov eax,[esi+458h]
0x10019720: push eax
0x10019721: call 10021FCCh
0x10019726: mov eax,[esi+45Ch]
0x1001972c: push eax
0x1001972d: call 10021FCCh
0x10019732: add esp,24h
0x10019735: mov ecx,esi
0x10019737: mov dword ptr [esp+14h],0FFFFFFFFh
0x1001973f: mov dword ptr [esi],100FE724h
0x10019745: call 1000E550h
0x1001974a: mov ecx,esi
0x1001974c: call 1000E4A0h
0x10019751: mov ecx,[esp+0Ch]
0x10019755: mov fs:[0],ecx
0x1001975c: pop ecx
0x1001975d: pop esi
0x1001975e: add esp,10h
0x10019761: ret
```

`sub_10019690` is called by sub_1001a5b0. It delegates to sub_1000e4a0, sub_1000e550, j_ORDINAL 764 (3 callees).

12.370 sub_10019770 at 0x10019770 (211 instructions)

```
0x10019770: sub esp,14h
0x10019773: mov eax,ds:[10136220h]
0x10019778: xor eax,esp
0x1001977a: mov [esp+10h],eax
0x1001977e: push ebx
0x1001977f: push esi
0x10019780: mov esi,[esp+20h]
0x10019784: mov eax,[esi]
0x10019786: mov edx,[eax+8]
0x10019789: push edi
0x1001978a: push 1
0x1001978c: mov edi,ecx
0x1001978e: push 1
0x10019790: lea ecx,[esp+17h]
0x10019794: push ecx
0x10019795: mov ecx,esi
0x10019797: call edx
0x10019799: cmp eax,1
0x1001979c: sete bl
0x1001979f: test bl,bl
0x100197a1: je near ptr 100199AAh
0x100197a7: cmp byte ptr [esp+0Fh],0F9h
0x100197ac: push ebp
0x100197ad: jne near ptr 1001983Bh
0x100197b3: mov eax,[esi]
0x100197b5: mov edx,[eax+8]
0x100197b8: push 1
0x100197ba: push 1
0x100197bc: lea ecx,[esp+1Ah]
0x100197c0: push ecx
0x100197c1: mov ecx,esi
0x100197c3: call edx
0x100197c5: cmp eax,1
0x100197c8: sete bl
0x100197cb: test bl,bl
0x100197cd: je short 1001983Bh
0x100197cf: mov eax,[esi]
0x100197d1: mov edx,[eax+8]
0x100197d4: push 4
0x100197d6: push 1
0x100197d8: lea ebp,[edi+3D0h]
0x100197de: push ebp
0x100197df: mov ecx,esi
0x100197e1: call edx
0x100197e3: movzx ecx,byte ptr [esp+12h]
0x100197e8: movzx edx,word ptr [edi+3D1h]
0x100197ef: cmp ecx,eax
0x100197f1: push edx
0x100197f2: mov ecx,edi
0x100197f4: sete bl
0x100197f7: call 1000F1C0h
0x100197fc: test bl,bl
0x100197fe: mov [edi+3D1h],ax
0x10019805: je short 1001983Bh
0x10019807: mov cl,[ebp]
0x1001980a: test cl,1
0x1001980d: je short 10019818h
0x1001980f: movzx edx,byte ptr [edi+3D3h]
0x10019816: jmp short 1001981Bh
0x10019818: or edx,0FFFFFFFFh
0x1001981b: shr cl,2
0x1001981e: movzx eax,ax
0x10019821: mov [edi+14Ch],edx
0x10019827: mov [edi+164h],eax
0x1001982d: and ecx,0FFFFFF07h
0x10019833: push ecx
```

```
0x10019834: mov ecx,edi
0x10019836: call 1000F1A0h
0x1001983b: cmp byte ptr [esp+13h],0FEh
0x10019840: jne short 1001988Bh
0x10019842: mov edx,[esi]
0x10019844: mov edx,[edx+8]
0x10019847: push 1
0x10019849: push 1
0x1001984b: lea eax,[esp+1Ah]
0x1001984f: push eax
0x10019850: mov ecx,esi
0x10019852: call edx
0x10019854: cmp eax,1
0x10019857: sete bl
0x1001985a: test bl,bl
0x1001985c: je short 1001988Bh
0x1001985e: movzx ecx,byte ptr [esp+12h]
0x10019863: mov eax,[esi]
0x10019865: mov edx,[edi+45Ch]
0x1001986b: mov eax,[eax+8]
0x1001986e: push 1
0x10019870: push ecx
0x10019871: push edx
0x10019872: mov ecx,esi
0x10019874: call eax
0x10019876: movzx ecx,byte ptr [esp+12h]
0x1001987b: mov edx,[edi+45Ch]
0x10019881: cmp eax,1
0x10019884: sete bl
0x10019887: mov byte ptr [ecx+edx],0
0x1001988b: cmp byte ptr [esp+13h],0FFh
0x10019890: jne near ptr 10019960h
0x10019896: mov eax,[esi]
0x10019898: mov edx,[eax+8]
0x1001989b: push 1
0x1001989d: push 1
0x1001989f: lea ecx,[esp+1Ah]
0x100198a3: push ecx
0x100198a4: mov ecx,esi
0x100198a6: call edx
0x100198a8: cmp eax,1
0x100198ab: sete al
0x100198ae: test al,al
0x100198b0: je near ptr 10019995h
0x100198b6: mov cl,[esp+12h]
0x100198ba: cmp cl,0Bh
0x100198bd: sete al
0x100198c0: test al,al
0x100198c2: je near ptr 10019995h
0x100198c8: mov eax,[esi]
0x100198ca: mov eax,[eax+8]
0x100198cd: movzx ecx,cl
0x100198d0: push 1
0x100198d2: push ecx
0x100198d3: lea edx,[esp+1Ch]
0x100198d7: push edx
0x100198d8: mov ecx,esi
0x100198da: call eax
0x100198dc: cmp eax,1
0x100198df: sete al
0x100198e2: test al,al
0x100198e4: je near ptr 10019995h
0x100198ea: mov edx,[esi]
0x100198ec: mov edx,[edx+8]
0x100198ef: push 1
0x100198f1: push 1
0x100198f3: lea eax,[esp+1Ah]
```

```
0x100198f7: push eax
0x100198f8: mov ecx,esi
0x100198fa: call edx
0x100198fc: cmp eax,1
0x100198ff: sete al
0x10019902: test al,al
0x10019904: je near ptr 10019995h
0x1001990a: movzx eax,byte ptr [esp+12h]
0x1001990f: push eax
0x10019910: call dword ptr ds:[100FE550h]
0x10019916: add esp,4
0x10019919: mov ebp,eax
0x1001991b: test ebp,ebp
0x1001991d: setne bl
0x10019920: test bl,bl
0x10019922: je short 10019956h
0x10019924: movzx eax,byte ptr [esp+12h]
0x10019929: mov edx,[esi]
0x1001992b: mov edx,[edx+8]
0x1001992e: push 1
0x10019930: push eax
0x10019931: push ebp
0x10019932: mov ecx,esi
0x10019934: call edx
0x10019936: cmp eax,1
0x10019939: sete bl
0x1001993c: cmp byte ptr [esp+12h],2
0x10019941: jbe short 10019956h
0x10019943: movzx eax,byte ptr [ebp+2]
0x10019947: movzx ecx,byte ptr [ebp+1]
0x1001994b: shl eax,8
0x1001994e: add eax,ecx
0x10019950: mov [edi+460h],eax
0x10019956: push ebp
0x10019957: call dword ptr ds:[100FE554h]
0x1001995d: add esp,4
0x10019960: test bl,bl
0x10019962: je short 10019993h
0x10019964: mov edx,[esi]
0x10019966: mov edx,[edx+8]
0x10019969: push 1
0x1001996b: push 1
0x1001996d: lea eax,[esp+1Ah]
0x10019971: push eax
0x10019972: mov ecx,esi
0x10019974: call edx
0x10019976: test eax,eax
0x10019978: je short 10019993h
0x1001997a: mov al,[esp+12h]
0x1001997e: test al,al
0x10019980: je short 10019993h
0x10019982: mov edx,[esi]
0x10019984: mov edx,[edx+10h]
0x10019987: movzx eax,al
0x1001998a: push 1
0x1001998c: push eax
0x1001998d: mov ecx,esi
0x1001998f: call edx
0x10019991: jmp short 10019964h
0x10019993: mov al,bl
0x10019995: pop ebp
0x10019996: pop edi
0x10019997: pop esi
0x10019998: pop ebx
0x10019999: mov ecx,[esp+10h]
0x1001999d: xor ecx,esp
0x1001999f: call 10022326h
```

```
0x100199a4: add esp,14h
0x100199a7: ret 4
0x100199aa: mov ecx,[esp+1Ch]
0x100199ae: pop edi
0x100199af: pop esi
0x100199b0: mov al,bl
0x100199b2: pop ebx
0x100199b3: xor ecx,esp
0x100199b5: call 10022326h
0x100199ba: add esp,14h
0x100199bd: ret 4
```

`sub_10019770` is called by sub_1001a380, sub_1001a5b0. It delegates to sub_1000f1a0, sub_1000f1c0, sub_10022326 (3 callees).

12.371 sub_100199c0 at 0x100199c0 (30 instructions)

```
0x100199c0: push esi
0x100199c1: mov esi,ecx
0x100199c3: cmp dword ptr [esi+3FCh],3FFFh
0x100199cd: jl short 10019A09h
0x100199cf: mov ecx,[esp+8]
0x100199d3: mov eax,[ecx]
0x100199d5: mov edx,[esi+404h]
0x100199db: mov eax,[eax+8]
0x100199de: push 3FFFh
0x100199e3: push 1
0x100199e5: push edx
0x100199e6: call eax
0x100199e8: cmp eax,3FFFh
0x100199ed: mov [esi+400h],eax
0x100199f3: jge short 100199FFh
0x100199f5: mov ecx,[esi+404h]
0x100199fb: mov byte ptr [eax+ecx],0FFh
0x100199ff: mov dword ptr [esi+3FCh],0
0x10019a09: mov ecx,[esi+3FCh]
0x10019a0f: cmp ecx,[esi+400h]
0x10019a15: jl short 10019A1Eh
0x10019a17: or eax,0FFFFFFFFh
0x10019a1a: pop esi
0x10019a1b: ret 4
0x10019a1e: mov edx,[esi+404h]
0x10019a24: movzx eax,byte ptr [edx+ecx]
0x10019a28: add ecx,1
0x10019a2b: mov [esi+3FCh],ecx
0x10019a31: pop esi
0x10019a32: ret 4
```

`sub_100199c0` is called by sub_10019cb0, sub_10019e70. It calls nothing else.

12.372 sub_10019a40 at 0x10019a40 (162 instructions)

```
0x10019a40: push ebx
0x10019a41: push ebp
0x10019a42: mov ebp,[esp+0Ch]
0x10019a46: test ebp,ebp
0x10019a48: push edi
0x10019a49: mov edi,ecx
0x10019a4b: je near ptr 10019BF9h
0x10019a51: mov ebx,[esp+14h]
0x10019a55: test ebx,ebx
0x10019a57: je near ptr 10019BF9h
0x10019a5d: cmp word ptr [edi+16h],8
0x10019a62: push esi
0x10019a63: jae near ptr 10019AF9h
```

```
0x10019a69: xor esi,esi
0x10019a6b: cmp [edi+0Ch],esi
0x10019a6e: jle near ptr 10019AF9h
0x10019a74: movzx ecx,word ptr [edi+16h]
0x10019a78: movzx eax,cx
0x10019a7b: imul eax,esi
0x10019a7e: sar eax,3
0x10019a81: add eax,ebx
0x10019a83: cmp cx,4
0x10019a87: jne short 10019AB5h
0x10019a89: mov edx,esi
0x10019a8b: and edx,80000001h
0x10019a91: jns short 10019A98h
0x10019a93: dec edx
0x10019a94: or edx,0FFFFFFFEh
0x10019a97: inc edx
0x10019a98: mov cl,1
0x10019a9a: sub cl,dl
0x10019a9c: add cl,cl
0x10019a9e: add cl,cl
0x10019aa0: movzx ebp,cl
0x10019aa3: mov dl,0Fh
0x10019aa5: mov ecx,ebp
0x10019aa7: shl dl,cl
0x10019aa9: not dl
0x10019aab: and [eax],dl
0x10019aad: mov dl,[esi+ebx]
0x10019ab0: and dl,0Fh
0x10019ab3: jmp short 10019AE1h
0x10019ab5: cmp cx,1
0x10019ab9: jne short 10019AEDh
0x10019abb: mov edx,esi
0x10019abd: and edx,80000007h
0x10019ac3: jns short 10019ACAh
0x10019ac5: dec edx
0x10019ac6: or edx,0FFFFFFF8h
0x10019ac9: inc edx
0x10019aca: mov cl,7
0x10019acc: sub cl,dl
0x10019ace: movzx ebp,cl
0x10019ad1: mov dl,1
0x10019ad3: mov ecx,ebp
0x10019ad5: shl dl,cl
0x10019ad7: not dl
0x10019ad9: and [eax],dl
0x10019adb: mov dl,[esi+ebx]
0x10019ade: and dl,1
0x10019ae1: mov ecx,ebp
0x10019ae3: mov ebp,[esp+14h]
0x10019ae7: shl dl,cl
0x10019ae9: or dl,[eax]
0x10019aeb: mov [eax],dl
0x10019aed: add esi,1
0x10019af0: cmp esi,[edi+0Ch]
0x10019af3: jl near ptr 10019A74h
0x10019af9: cmp dword ptr [edi+3E8h],0
0x10019b00: je near ptr 10019BADh
0x10019b06: mov esi,[edi+3F4h]
0x10019b0c: sub esi,[edi+3ECh]
0x10019b12: sub esi,1
0x10019b15: js short 10019B46h
0x10019b17: mov ecx,[ebp+14h]
0x10019b1a: call 1000EFE0h
0x10019b1f: cmp esi,eax
0x10019b21: jg short 10019B46h
0x10019b23: mov ecx,[ebp+14h]
0x10019b26: mov [ebp+4],esi
```

```
0x10019b29: call 1000F000h
0x10019b2e: imul eax,esi
0x10019b31: mov ecx,[ebp+14h]
0x10019b34: push 0
0x10019b36: mov [esp+18h],eax
0x10019b3a: call 1000E5E0h
0x10019b3f: add eax,[esp+14h]
0x10019b43: mov [ebp+10h],eax
0x10019b46: mov eax,[esp+1Ch]
0x10019b4a: push eax
0x10019b4b: push ebx
0x10019b4c: mov ecx,ebp
0x10019b4e: call 10018BE0h
0x10019b53: mov ecx,[edi+3F0h]
0x10019b59: add [edi+3ECh],ecx
0x10019b5f: mov eax,[edi+3ECh]
0x10019b65: mov esi,[edi+3F4h]
0x10019b6b: cmp eax,esi
0x10019b6d: jl short 10019BA4h
0x10019b6f: mov ebx,[edi+3F8h]
0x10019b75: mov edx,ebx
0x10019b77: add ebx,1
0x10019b7a: test edx,edx
0x10019b7c: jle short 10019B87h
0x10019b7e: mov eax,ecx
0x10019b80: cdq
0x10019b81: sub eax,edx
0x10019b83: sar eax,1
0x10019b85: mov ecx,eax
0x10019b87: mov eax,ecx
0x10019b89: cdq
0x10019b8a: sub eax,edx
0x10019b8c: sar eax,1
0x10019b8e: cmp eax,esi
0x10019b90: jg short 10019B75h
0x10019b92: mov [edi+3ECh],eax
0x10019b98: mov [edi+3F0h],ecx
0x10019b9e: mov [edi+3F8h],ebx
0x10019ba4: pop esi
0x10019ba5: pop edi
0x10019ba6: pop ebp
0x10019ba7: xor eax,eax
0x10019ba9: pop ebx
0x10019baa: ret 0Ch
0x10019bad: mov ecx,[ebp+14h]
0x10019bb0: test ecx,ecx
0x10019bb2: je short 10019BEFh
0x10019bb4: mov eax,[ebp+4]
0x10019bb7: mov edx,[ebp]
0x10019bba: push eax
0x10019bbb: push edx
0x10019bbc: call 1000E650h
0x10019bc1: movzx eax,al
0x10019bc4: test eax,eax
0x10019bc6: je short 10019BEFh
0x10019bc8: mov eax,[esp+1Ch]
0x10019bcc: push eax
0x10019bcd: push ebx
0x10019bce: mov ecx,ebp
0x10019bd0: call 10018BE0h
0x10019bd5: add dword ptr [ebp+4],0FFFFFFFFh
0x10019bd9: js short 10019BA4h
0x10019bdb: mov ecx,[ebp+14h]
0x10019bde: call 1000F000h
0x10019be3: sub [ebp+10h],eax
0x10019be6: pop esi
0x10019be7: pop edi
```

```
0x10019be8: pop ebp
0x10019be9: xor eax,eax
0x10019beb: pop ebx
0x10019bec: ret 0Ch
0x10019bef: pop esi
0x10019bf0: pop edi
0x10019bf1: pop ebp
0x10019bf2: or eax,0FFFFFFFFh
0x10019bf5: pop ebx
0x10019bf6: ret 0Ch
0x10019bf9: pop edi
0x10019bfa: pop ebp
0x10019bfb: or eax,0FFFFFFFFh
0x10019bfe: pop ebx
0x10019bff: ret 0Ch
```

`sub_10019a40` is called by sub_10019e70. It accesses memory-mapped peripheral registers at 0x80000001. It delegates to sub_1000e5e0, sub_1000e650, sub_1000efe0 (5 callees).

12.373 sub_10019c10 at 0x10019c10 (38 instructions)

```
0x10019c10: mov edx,[esp+4]
0x10019c14: push esi
0x10019c15: mov esi,ecx
0x10019c17: lea ecx,[edx+1]
0x10019c1a: mov eax,1
0x10019c1f: shl eax,cl
0x10019c21: mov [esi+408h],cx
0x10019c28: mov cl,dl
0x10019c2a: push 1000h
0x10019c2f: mov [esi+410h],ax
0x10019c36: mov eax,1
0x10019c3b: shl eax,cl
0x10019c3d: mov ecx,[esi+424h]
0x10019c43: push 0
0x10019c45: push ecx
0x10019c46: mov [esi+40Ah],ax
0x10019c4d: add eax,1
0x10019c50: mov [esi+40Ch],ax
0x10019c57: add eax,1
0x10019c5a: mov [esi+40Eh],ax
0x10019c61: mov [esi+412h],ax
0x10019c68: mov word ptr [esi+416h],0
0x10019c71: mov word ptr [esi+414h],0
0x10019c7a: call 100228C0h
0x10019c7f: mov edx,[esi+42Ch]
0x10019c85: push 1000h
0x10019c8a: push 0
0x10019c8c: push edx
0x10019c8d: call 100228C0h
0x10019c92: mov eax,[esi+428h]
0x10019c98: push 1000h
0x10019c9d: push 0
0x10019c9f: push eax
0x10019ca0: call 100228C0h
0x10019ca5: add esp,24h
0x10019ca8: xor ax,ax
0x10019cab: pop esi
0x10019cac: ret 4
```

`sub_10019c10` is called by sub_10019e70. It delegates to j_memset (1 callees).

12.374 sub_10019cb0 at 0x10019cb0 (107 instructions)

```
0x10019cb0: push ebx
0x10019cb1: mov ebx,[esp+8]
0x10019cb5: push esi
0x10019cb6: mov esi,ecx
0x10019cb8: cmp word ptr [esi+416h],0
0x10019cc0: push edi
0x10019cc1: jne near ptr 10019D54h
0x10019cc7: cmp word ptr [esi+414h],0
0x10019ccf: jg short 10019D2Ch
0x10019cd1: mov eax,[esi+41Ch]
0x10019cd7: push ebx
0x10019cd8: mov [esi+420h],eax
0x10019cde: call 100199C0h
0x10019ce3: test ax,ax
0x10019ce6: mov [esi+414h],ax
0x10019ced: jl near ptr 10019E5Dh
0x10019cf3: je short 10019D2Ch
0x10019cf5: xor edi,edi
0x10019cf7: test ax,ax
0x10019cfa: jle short 10019D2Ch
0x10019cfc: lea esp,[esp]
0x10019d00: push ebx
0x10019d01: mov ecx,esi
0x10019d03: call 100199C0h
0x10019d08: movzx eax,ax
0x10019d0b: test ax,ax
0x10019d0e: jl near ptr 10019E5Dh
0x10019d14: mov edx,[esi+41Ch]
0x10019d1a: movsx ecx,di
0x10019d1d: add edi,1
0x10019d20: mov [ecx+edx],al
0x10019d23: cmp di,[esi+414h]
0x10019d2a: jl short 10019D00h
0x10019d2c: mov eax,[esi+420h]
0x10019d32: mov cl,[eax]
0x10019d34: add eax,1
0x10019d37: add word ptr [esi+414h],0FFFFh
0x10019d3f: mov [esi+418h],cl
0x10019d45: mov [esi+420h],eax
0x10019d4b: mov word ptr [esi+416h],8
0x10019d54: cmp word ptr [esi+414h],0
0x10019d5c: jge short 10019D6Bh
0x10019d5e: mov ax,[esi+40Ch]
0x10019d65: pop edi
0x10019d66: pop esi
0x10019d67: pop ebx
0x10019d68: ret 4
0x10019d6b: movzx eax,word ptr [esi+416h]
0x10019d72: push ebp
0x10019d73: movzx ebp,byte ptr [esi+418h]
0x10019d7a: mov ecx,8
0x10019d7f: sub cl,al
0x10019d81: shr ebp,cl
0x10019d83: cmp [esi+408h],ax
0x10019d8a: jle near ptr 10019E40h
0x10019d90: cmp word ptr [esi+414h],0
0x10019d98: jg short 10019DF3h
0x10019d9a: mov edx,[esi+41Ch]
0x10019da0: push ebx
0x10019da1: mov ecx,esi
0x10019da3: mov [esi+420h],edx
0x10019da9: call 100199C0h
0x10019dae: test ax,ax
0x10019db1: mov [esi+414h],ax
0x10019db8: jl near ptr 10019E5Ch
0x10019dbe: je short 10019DF3h
```

```
0x10019dc0: xor edi,edi
0x10019dc2: test ax,ax
0x10019dc5: jle short 10019DF3h
0x10019dc7: push ebx
0x10019dc8: mov ecx,esi
0x10019dca: call 100199C0h
0x10019dcf: movzx eax,ax
0x10019dd2: test ax,ax
0x10019dd5: jl near ptr 10019E5Ch
0x10019ddb: mov edx,[esi+41Ch]
0x10019de1: movsx ecx,di
0x10019de4: add edi,1
0x10019de7: mov [ecx+edx],al
0x10019dea: cmp di,[esi+414h]
0x10019df1: jl short 10019DC7h
0x10019df3: mov eax,[esi+420h]
0x10019df9: mov cl,[eax]
0x10019dfb: add word ptr [esi+414h],0FFFFh
0x10019e03: add eax,1
0x10019e06: mov [esi+420h],eax
0x10019e0c: movzx eax,word ptr [esi+416h]
0x10019e13: mov [esi+418h],cl
0x10019e19: movzx edx,cl
0x10019e1c: mov cl,al
0x10019e1e: shl edx,cl
0x10019e20: add eax,8
0x10019e23: mov [esi+416h],ax
0x10019e2a: mov ax,[esi+408h]
0x10019e31: or ebp,edx
0x10019e33: cmp ax,[esi+416h]
0x10019e3a: jg near ptr 10019D90h
0x10019e40: movzx eax,word ptr [esi+408h]
0x10019e47: sub [esi+416h],ax
0x10019e4e: movsx ecx,ax
0x10019e51: mov ax,[ecx*4+100FF020h]
0x10019e59: and ax,bp
0x10019e5c: pop ebp
0x10019e5d: pop edi
0x10019e5e: pop esi
0x10019e5f: pop ebx
0x10019e60: ret 4
```

`sub_10019cb0` is called by sub_10019e70. It delegates to sub_100199c0 (1 callees).

12.375 sub_10019e70 at 0x10019e70 (260 instructions)

```
0x10019e70: sub esp,18h
0x10019e73: push ebp
0x10019e74: mov bp,[esp+28h]
0x10019e79: push esi
0x10019e7a: push edi
0x10019e7b: xor edi,edi
0x10019e7d: cmp bp,di
0x10019e80: mov esi,ecx
0x10019e82: jg short 10019E91h
0x10019e84: pop edi
0x10019e85: pop esi
0x10019e86: mov ax,0FFFBh
0x10019e8a: pop ebp
0x10019e8b: add esp,18h
0x10019e8e: ret 10h
0x10019e91: mov ecx,[esp+28h]
0x10019e95: mov eax,[esp+34h]
0x10019e99: push ecx
0x10019e9a: mov ecx,esi
0x10019e9c: mov [eax],edi
```

```
0x10019e9e: call 100199C0h
0x10019ea3: movzx eax,ax
0x10019ea6: cmp ax,di
0x10019ea9: mov [esp+20h],eax
0x10019ead: jl near ptr 1001A1A7h
0x10019eb3: lea edx,[eax-2]
0x10019eb6: cmp dx,7
0x10019eba: ja near ptr 1001A1E0h
0x10019ec0: push eax
0x10019ec1: mov ecx,esi
0x10019ec3: call 10019C10h
0x10019ec8: movsx eax,bp
0x10019ecb: mov [esp+1Ch],eax
0x10019ecf: add eax,1
0x10019ed2: push eax
0x10019ed3: mov [esp+1Ch],edi
0x10019ed7: mov [esp+18h],edi
0x10019edb: call 10021FDEh
0x10019ee0: add esp,4
0x10019ee3: cmp eax,edi
0x10019ee5: mov [esp+0Ch],eax
0x10019ee9: jne short 10019EF8h
0x10019eeb: pop edi
0x10019eec: pop esi
0x10019eed: mov ax,0FFF6h
0x10019ef1: pop ebp
0x10019ef2: add esp,18h
0x10019ef5: ret 10h
0x10019ef8: mov edi,[esi+424h]
0x10019efe: push ebx
0x10019eff: mov ebx,eax
0x10019f01: mov eax,[esp+2Ch]
0x10019f05: push eax
0x10019f06: mov ecx,esi
0x10019f08: mov [esp+18h],ebx
0x10019f0c: movzx ebp,bp
0x10019f0f: call 10019CB0h
0x10019f14: movzx ecx,ax
0x10019f17: cmp cx,[esi+40Ch]
0x10019f1e: je near ptr 1001A16Bh
0x10019f24: test cx,cx
0x10019f27: jl near ptr 1001A1B0h
0x10019f2d: cmp cx,[esi+40Ah]
0x10019f34: jne near ptr 10019FEBh
0x10019f3a: mov ecx,[esp+24h]
0x10019f3e: mov dx,[esi+40Eh]
0x10019f45: add ecx,1
0x10019f48: mov eax,1
0x10019f4d: shl eax,cl
0x10019f4f: mov [esi+408h],cx
0x10019f56: mov [esi+412h],dx
0x10019f5d: mov [esi+410h],ax
0x10019f64: mov ecx,[esp+2Ch]
0x10019f68: push ecx
0x10019f69: mov ecx,esi
0x10019f6b: call 10019CB0h
0x10019f70: movzx eax,ax
0x10019f73: cmp ax,[esi+40Ah]
0x10019f7a: je short 10019F64h
0x10019f7c: cmp ax,[esi+40Ch]
0x10019f83: je near ptr 1001A16Bh
0x10019f89: cmp ax,[esi+412h]
0x10019f90: jl short 10019F94h
0x10019f92: xor eax,eax
0x10019f94: movzx edx,ax
0x10019f97: movzx ecx,ax
0x10019f9a: mov [ebx],al
```

```
0x10019f9c: add ebx,1
0x10019f9f: sub ebp,1
0x10019fa2: test bp,bp
0x10019fa5: mov [esp+1Ch],edx
0x10019fa9: mov [esp+18h],ecx
0x10019fad: mov [esp+14h],ebx
0x10019fb1: jne near ptr 1001A14Fh
0x10019fb7: mov eax,[esp+30h]
0x10019fbb: test eax,eax
0x10019fbd: mov ebx,[esp+10h]
0x10019fc1: je short 10019FDDh
0x10019fc3: mov edx,[esp+20h]
0x10019fc7: push edx
0x10019fc8: push ebx
0x10019fc9: push eax
0x10019fca: mov ecx,esi
0x10019fcc: call 10019A40h
0x10019fd1: movzx ebp,ax
0x10019fd4: test bp,bp
0x10019fd7: jl near ptr 1001A1CAh
0x10019fdd: movzx ebp,word ptr [esp+34h]
0x10019fe2: mov [esp+14h],ebx
0x10019fe6: jmp near ptr 1001A14Fh
0x10019feb: movzx edx,word ptr [esi+412h]
0x10019ff2: cmp cx,dx
0x10019ff5: movzx eax,cx
0x10019ff8: jl short 1001A02Bh
0x10019ffa: mov edx,[esi+424h]
0x1001a000: add edx,0FFEh
0x1001a006: cmp edi,edx
0x1001a008: jae short 1001A02Bh
0x1001a00a: movzx eax,word ptr [esi+412h]
0x1001a011: cmp cx,ax
0x1001a014: jle short 1001A01Dh
0x1001a016: mov eax,[esp+38h]
0x1001a01a: add dword ptr [eax],1
0x1001a01d: mov dl,[esp+1Ch]
0x1001a021: movzx eax,word ptr [esp+18h]
0x1001a026: mov [edi],dl
0x1001a028: add edi,1
0x1001a02b: cmp ax,[esi+40Eh]
0x1001a032: jl short 1001A074h
0x1001a034: jmp short 1001A040h
0x1001a040: mov edx,[esi+424h]
0x1001a046: add edx,0FFEh
0x1001a04c: cmp edi,edx
0x1001a04e: jae short 1001A074h
0x1001a050: mov edx,[esi+428h]
0x1001a056: movsx eax,ax
0x1001a059: mov dl,[eax+edx]
0x1001a05c: mov [edi],dl
0x1001a05e: mov edx,[esi+42Ch]
0x1001a064: movzx eax,word ptr [edx+eax*2]
0x1001a068: add edi,1
0x1001a06b: cmp ax,[esi+40Eh]
0x1001a072: jge short 1001A040h
0x1001a074: mov [edi],al
0x1001a076: movzx edx,word ptr [esi+412h]
0x1001a07d: add edi,1
0x1001a080: cmp dx,[esi+410h]
0x1001a087: jge short 1001A0C9h
0x1001a089: mov ebx,[esi+428h]
0x1001a08f: movzx ax,al
0x1001a093: movzx eax,ax
0x1001a096: movsx edx,dx
0x1001a099: mov [edx+ebx],al
0x1001a09c: mov bx,[esp+18h]
```

```
0x1001a0a1: mov edx,[esi+42Ch]
0x1001a0a7: mov [esp+1Ch],eax
0x1001a0ab: movsx eax,word ptr [esi+412h]
0x1001a0b2: mov [edx+eax*2],bx
0x1001a0b6: add word ptr [esi+412h],1
0x1001a0be: mov ebx,[esp+14h]
0x1001a0c2: movzx eax,cx
0x1001a0c5: mov [esp+18h],eax
0x1001a0c9: movzx ecx,word ptr [esi+410h]
0x1001a0d0: cmp [esi+412h],cx
0x1001a0d7: jl short 1001A0F9h
0x1001a0d9: movzx eax,word ptr [esi+408h]
0x1001a0e0: cmp ax,0Ch
0x1001a0e4: jge short 1001A0F9h
0x1001a0e6: add ecx,ecx
0x1001a0e8: add eax,1
0x1001a0eb: mov [esi+410h],cx
0x1001a0f2: mov [esi+408h],ax
0x1001a0f9: cmp edi,[esi+424h]
0x1001a0ff: jbe short 1001A14Fh
0x1001a101: mov dl,[edi-1]
0x1001a104: sub edi,1
0x1001a107: mov [ebx],dl
0x1001a109: add ebx,1
0x1001a10c: sub ebp,1
0x1001a10f: test bp,bp
0x1001a112: mov [esp+14h],ebx
0x1001a116: jne short 1001A147h
0x1001a118: mov eax,[esp+30h]
0x1001a11c: test eax,eax
0x1001a11e: mov ebx,[esp+10h]
0x1001a122: je short 1001A13Eh
0x1001a124: mov ecx,[esp+20h]
0x1001a128: push ecx
0x1001a129: push ebx
0x1001a12a: push eax
0x1001a12b: mov ecx,esi
0x1001a12d: call 10019A40h
0x1001a132: movzx ebp,ax
0x1001a135: test bp,bp
0x1001a138: jl near ptr 1001A1CAh
0x1001a13e: movzx ebp,word ptr [esp+34h]
0x1001a143: mov [esp+14h],ebx
0x1001a147: cmp edi,[esi+424h]
0x1001a14d: ja short 1001A101h
0x1001a14f: mov edx,[esp+2Ch]
0x1001a153: push edx
0x1001a154: mov ecx,esi
0x1001a156: call 10019CB0h
0x1001a15b: movzx ecx,ax
0x1001a15e: cmp cx,[esi+40Ch]
0x1001a165: jne near ptr 10019F24h
0x1001a16b: xor edi,edi
0x1001a16d: cmp bp,[esp+34h]
0x1001a172: je short 1001A196h
0x1001a174: mov ecx,[esp+30h]
0x1001a178: test ecx,ecx
0x1001a17a: je short 1001A196h
0x1001a17c: mov eax,[esp+20h]
0x1001a180: movsx edx,bp
0x1001a183: sub eax,edx
0x1001a185: push eax
0x1001a186: mov eax,[esp+14h]
0x1001a18a: push eax
0x1001a18b: push ecx
0x1001a18c: mov ecx,esi
0x1001a18e: call 10019A40h
```

```
0x1001a193: movzx edi,ax
0x1001a196: mov ecx,[esp+10h]
0x1001a19a: push ecx
0x1001a19b: call 10021FCCh
0x1001a1a0: add esp,4
0x1001a1a3: mov ax,di
0x1001a1a6: pop ebx
0x1001a1a7: pop edi
0x1001a1a8: pop esi
0x1001a1a9: pop ebp
0x1001a1aa: add esp,18h
0x1001a1ad: ret 10h
0x1001a1b0: mov eax,[esp+10h]
0x1001a1b4: push eax
0x1001a1b5: call 10021FCCh
0x1001a1ba: add esp,4
0x1001a1bd: pop ebx
0x1001a1be: pop edi
0x1001a1bf: pop esi
0x1001a1c0: xor ax,ax
0x1001a1c3: pop ebp
0x1001a1c4: add esp,18h
0x1001a1c7: ret 10h
0x1001a1ca: push ebx
0x1001a1cb: call 10021FCCh
0x1001a1d0: add esp,4
0x1001a1d3: pop ebx
0x1001a1d4: pop edi
0x1001a1d5: pop esi
0x1001a1d6: mov ax,bp
0x1001a1d9: pop ebp
0x1001a1da: add esp,18h
0x1001a1dd: ret 10h
0x1001a1e0: pop edi
0x1001a1e1: pop esi
0x1001a1e2: mov ax,0FFECh
0x1001a1e6: pop ebp
0x1001a1e7: add esp,18h
0x1001a1ea: ret 10h
```

`sub_10019e70` is called by sub_1001a380, sub_1001a5b0. It delegates to sub_100199c0, sub_10019a40, sub_10019c10 (6 callees).

12.376 sub_1001a1f0 at 0x1001a1f0 (54 instructions)

```
0x1001a1f0: mov ecx,[esp+8]
0x1001a1f4: push ebx
0x1001a1f5: push esi
0x1001a1f6: mov esi,[esp+0Ch]
0x1001a1fa: mov eax,[esi]
0x1001a1fc: mov edx,[eax+10h]
0x1001a1ff: push 0
0x1001a201: push ecx
0x1001a202: mov ecx,esi
0x1001a204: call edx
0x1001a206: mov eax,[esi]
0x1001a208: mov edx,[eax+8]
0x1001a20b: push 1
0x1001a20d: push 1
0x1001a20f: lea ecx,[esp+14h]
0x1001a213: push ecx
0x1001a214: mov ecx,esi
0x1001a216: mov byte ptr [esp+18h],0
0x1001a21b: xor bl,bl
0x1001a21d: call edx
0x1001a21f: test eax,eax
```

```
0x1001a221: jbe short 1001A246h
0x1001a223: test bl,bl
0x1001a225: mov bl,[esp+0Ch]
0x1001a229: jne short 1001A230h
0x1001a22b: cmp bl,2Ch
0x1001a22e: je short 1001A24Eh
0x1001a230: mov eax,[esi]
0x1001a232: mov edx,[eax+8]
0x1001a235: push 1
0x1001a237: push 1
0x1001a239: lea ecx,[esp+14h]
0x1001a23d: push ecx
0x1001a23e: mov ecx,esi
0x1001a240: call edx
0x1001a242: test eax,eax
0x1001a244: ja short 1001A223h
0x1001a246: pop esi
0x1001a247: or eax,0FFFFFFFFh
0x1001a24a: pop ebx
0x1001a24b: ret 8
0x1001a24e: mov eax,[esi]
0x1001a250: mov edx,[eax+10h]
0x1001a253: push 1
0x1001a255: push 0FFFFFFFFh
0x1001a257: mov ecx,esi
0x1001a259: call edx
0x1001a25b: mov eax,[esi]
0x1001a25d: mov edx,[eax+14h]
0x1001a260: mov ecx,esi
0x1001a262: call edx
0x1001a264: pop esi
0x1001a265: pop ebx
0x1001a266: ret 8
```

`sub_1001a1f0` is called by sub_1001a380, sub_1001a5b0. It calls nothing else.

12.377 sub_1001a270 at 0x1001a270 (83 instructions)

```
0x1001a270: sub esp,14h
0x1001a273: push ebx
0x1001a274: push ebp
0x1001a275: push esi
0x1001a276: mov esi,[esp+28h]
0x1001a27a: mov ebp,ecx
0x1001a27c: push edi
0x1001a27d: mov [esp+1Ch],ebp
0x1001a281: call 1000EFE0h
0x1001a286: movzx ecx,word ptr [esi+2]
0x1001a28a: movzx edx,word ptr [esi+6]
0x1001a28e: sub eax,ecx
0x1001a290: sub eax,edx
0x1001a292: jns short 1001A298h
0x1001a294: xor edi,edi
0x1001a296: jmp short 1001A2ADh
0x1001a298: mov ecx,ebp
0x1001a29a: call 1000EFE0h
0x1001a29f: movzx ecx,word ptr [esi+6]
0x1001a2a3: mov edi,eax
0x1001a2a5: movzx eax,word ptr [esi+2]
0x1001a2a9: sub edi,eax
0x1001a2ab: sub edi,ecx
0x1001a2ad: mov ecx,ebp
0x1001a2af: call 1000EFE0h
0x1001a2b4: movzx edx,word ptr [esi+2]
0x1001a2b8: movzx ecx,word ptr [esi+4]
0x1001a2bc: mov ebx,eax
```

```
0x1001a2be: movzx eax,word ptr [esi]
0x1001a2c1: sub ebx,edx
0x1001a2c3: movzx edx,word ptr [esi]
0x1001a2c6: lea esi,[ecx+edx]
0x1001a2c9: mov ecx,ebp
0x1001a2cb: mov [esp+20h],ebx
0x1001a2cf: mov [esp+10h],eax
0x1001a2d3: call 1000EFF0h
0x1001a2d8: cmp eax,esi
0x1001a2da: jae short 1001A2E9h
0x1001a2dc: mov ecx,ebp
0x1001a2de: call 1000EFF0h
0x1001a2e3: mov [esp+2Ch],eax
0x1001a2e7: jmp short 1001A2EDh
0x1001a2e9: mov [esp+2Ch],esi
0x1001a2ed: mov ecx,[esp+28h]
0x1001a2f1: call 1000EEA0h
0x1001a2f6: cmp edi,ebx
0x1001a2f8: mov [esp+14h],eax
0x1001a2fc: mov ebp,edi
0x1001a2fe: jge short 1001A350h
0x1001a300: xor ebx,ebx
0x1001a302: mov esi,[esp+10h]
0x1001a306: cmp esi,[esp+2Ch]
0x1001a30a: jge short 1001A344h
0x1001a30c: xor edi,edi
0x1001a30e: mov edi,edi
0x1001a310: mov ecx,[esp+28h]
0x1001a314: push ebx
0x1001a315: push edi
0x1001a316: call 10012F20h
0x1001a31b: mov [esp+18h],al
0x1001a31f: movzx eax,al
0x1001a322: cmp eax,[esp+14h]
0x1001a326: je short 1001A338h
0x1001a328: mov ecx,[esp+18h]
0x1001a32c: push ecx
0x1001a32d: mov ecx,[esp+20h]
0x1001a331: push ebp
0x1001a332: push esi
0x1001a333: call 10013330h
0x1001a338: add esi,1
0x1001a33b: add edi,1
0x1001a33e: cmp esi,[esp+2Ch]
0x1001a342: jl short 1001A310h
0x1001a344: add ebp,1
0x1001a347: add ebx,1
0x1001a34a: cmp ebp,[esp+20h]
0x1001a34e: jl short 1001A302h
0x1001a350: pop edi
0x1001a351: pop esi
0x1001a352: pop ebp
0x1001a353: pop ebx
0x1001a354: add esp,14h
0x1001a357: ret 8
```

`sub_1001a270` is called by sub_1001a5b0. It delegates to sub_1000eea0, sub_1000efe0, sub_1000eff0 (5 callees).

12.378 sub_1001a380 at 0x1001a380 (173 instructions)

```
0x1001a380: push ebp
0x1001a381: mov ebp,esp
0x1001a383: and esp,0FFFFFFF8h
0x1001a386: sub esp,324h
0x1001a38c: push ebx
0x1001a38d: mov ebx,ecx
```

```
0x1001a38f: mov ecx,[ebp+8]
0x1001a392: mov eax,[ecx]
0x1001a394: mov edx,[eax+14h]
0x1001a397: push esi
0x1001a398: push edi
0x1001a399: call edx
0x1001a39b: mov esi,[ebp+0Ch]
0x1001a39e: mov ecx,0C1h
0x1001a3a3: lea edi,[esp+28h]
0x1001a3a7: mov [esp+24h],eax
0x1001a3ab: mov dword ptr [esp+1Ch],0
0x1001a3b3: rep movsd
0x1001a3b5: mov byte ptr [esp+0Eh],1
0x1001a3ba: lea ebx,[ebx]
0x1001a3c0: mov esi,[ebp+8]
0x1001a3c3: mov eax,[esi]
0x1001a3c5: mov edx,[eax+8]
0x1001a3c8: push 1
0x1001a3ca: push 1
0x1001a3cc: lea ecx,[esp+17h]
0x1001a3d0: push ecx
0x1001a3d1: mov ecx,esi
0x1001a3d3: call edx
0x1001a3d5: cmp eax,1
0x1001a3d8: jne near ptr 1001A587h
0x1001a3de: cmp byte ptr [esp+0Eh],0
0x1001a3e3: mov al,[esp+0Fh]
0x1001a3e7: jne short 1001A3EDh
0x1001a3e9: test al,al
0x1001a3eb: jne short 1001A3C0h
0x1001a3ed: cmp al,21h
0x1001a3ef: je near ptr 1001A57Ah
0x1001a3f5: cmp al,2Ch
0x1001a3f7: je short 1001A40Ah
0x1001a3f9: cmp al,3Bh
0x1001a3fb: je near ptr 1001A587h
0x1001a401: test al,al
0x1001a403: sete byte ptr [esp+0Eh]
0x1001a408: jmp short 1001A3C0h
0x1001a40a: mov eax,[esi]
0x1001a40c: mov edx,[eax+8]
0x1001a40f: push 1
0x1001a411: push 9
0x1001a413: lea ecx,[esp+18h]
0x1001a417: push ecx
0x1001a418: mov ecx,esi
0x1001a41a: call edx
0x1001a41c: mov eax,[esp+10h]
0x1001a420: push eax
0x1001a421: mov ecx,ebx
0x1001a423: call 1000F1C0h
0x1001a428: mov ecx,[esp+12h]
0x1001a42c: push ecx
0x1001a42d: mov ecx,ebx
0x1001a42f: mov [esp+14h],ax
0x1001a434: call 1000F1C0h
0x1001a439: mov edx,[esp+14h]
0x1001a43d: push edx
0x1001a43e: mov ecx,ebx
0x1001a440: mov [esp+16h],ax
0x1001a445: call 1000F1C0h
0x1001a44a: mov [esp+14h],ax
0x1001a44f: mov eax,[esp+16h]
0x1001a453: push eax
0x1001a454: mov ecx,ebx
0x1001a456: call 1000F1C0h
0x1001a45b: mov ecx,[ebp+10h]
```

```
0x1001a45e: cmp word ptr [ecx+6],0
0x1001a463: mov dx,[esp+14h]
0x1001a468: mov [esp+16h],ax
0x1001a46d: jne short 1001A47Eh
0x1001a46f: cmp word ptr [ecx+8],0
0x1001a474: jne short 1001A47Eh
0x1001a476: mov [ecx+6],dx
0x1001a47a: mov [ecx+8],ax
0x1001a47e: movzx edi,word ptr [esp+10h]
0x1001a483: movzx ecx,word ptr [ecx+6]
0x1001a487: movzx edx,dx
0x1001a48a: add edx,edi
0x1001a48c: cmp edx,ecx
0x1001a48e: jg near ptr 1001A3C0h
0x1001a494: movzx ecx,word ptr [esp+12h]
0x1001a499: movzx edx,ax
0x1001a49c: add edx,ecx
0x1001a49e: mov ecx,[ebp+10h]
0x1001a4a1: movzx ecx,word ptr [ecx+8]
0x1001a4a5: cmp edx,ecx
0x1001a4a7: jg near ptr 1001A3C0h
0x1001a4ad: movzx ecx,byte ptr [esp+18h]
0x1001a4b2: add dword ptr [esp+1Ch],1
0x1001a4b7: test cl,cl
0x1001a4b9: jns short 1001A4EEh
0x1001a4bb: mov edx,[esi]
0x1001a4bd: mov edx,[edx+8]
0x1001a4c0: and ecx,7
0x1001a4c3: add ecx,1
0x1001a4c6: mov eax,1
0x1001a4cb: shl eax,cl
0x1001a4cd: push 1
0x1001a4cf: lea ecx,[esp+30h]
0x1001a4d3: mov [esp+2Eh],ax
0x1001a4d8: movsx eax,ax
0x1001a4db: lea eax,[eax+eax*2]
0x1001a4de: push eax
0x1001a4df: push ecx
0x1001a4e0: mov ecx,esi
0x1001a4e2: call edx
0x1001a4e4: movzx ecx,byte ptr [esp+18h]
0x1001a4e9: mov ax,[esp+16h]
0x1001a4ee: xor edx,edx
0x1001a4f0: movzx eax,ax
0x1001a4f3: and ecx,40h
0x1001a4f6: mov [ebx+3E8h],ecx
0x1001a4fc: mov [ebx+3F4h],eax
0x1001a502: mov [esp+20h],edx
0x1001a506: mov dword ptr [ebx+3FCh],4000h
0x1001a510: mov dword ptr [ebx+3F0h],8
0x1001a51a: mov [ebx+3ECh],edx
0x1001a520: mov [ebx+3F8h],edx
0x1001a526: mov edx,[esi]
0x1001a528: mov eax,[edx+14h]
0x1001a52b: mov ecx,esi
0x1001a52d: call eax
0x1001a52f: mov edx,[esp+14h]
0x1001a533: lea ecx,[esp+20h]
0x1001a537: push ecx
0x1001a538: push edx
0x1001a539: push 0
0x1001a53b: push esi
0x1001a53c: mov ecx,ebx
0x1001a53e: mov edi,eax
0x1001a540: call 10019E70h
0x1001a545: cmp dword ptr [esp+20h],0
0x1001a54a: je short 1001A55Ah
```

```
0x1001a54c: push edi
0x1001a54d: push esi
0x1001a54e: mov ecx,ebx
0x1001a550: call 1001A1F0h
0x1001a555: jmp near ptr 1001A3C0h
0x1001a55a: mov ecx,[ebx+3FCh]
0x1001a560: sub ecx,[ebx+400h]
0x1001a566: mov eax,[esi]
0x1001a568: mov edx,[eax+10h]
0x1001a56b: add ecx,1
0x1001a56e: push 1
0x1001a570: push ecx
0x1001a571: mov ecx,esi
0x1001a573: call edx
0x1001a575: jmp near ptr 1001A3C0h
0x1001a57a: push esi
0x1001a57b: mov ecx,ebx
0x1001a57d: call 10019770h
0x1001a582: jmp near ptr 1001A3C0h
0x1001a587: mov ecx,[esp+24h]
0x1001a58b: mov eax,[esi]
0x1001a58d: mov edx,[eax+10h]
0x1001a590: push 0
0x1001a592: push ecx
0x1001a593: mov ecx,esi
0x1001a595: call edx
0x1001a597: mov eax,[esp+1Ch]
0x1001a59b: pop edi
0x1001a59c: pop esi
0x1001a59d: pop ebx
0x1001a59e: mov esp,ebp
0x1001a5a0: pop ebp
0x1001a5a1: ret 0Ch
```

`sub_1001a380` is called by sub_1001a5b0. It delegates to sub_1000f1c0, sub_10019770, sub_10019e70 (4 callees).

12.379 sub_1001a5b0 at 0x1001a5b0 (701 instructions)

```
0x1001a5b0: push 0FFFFFFFFh
0x1001a5b2: push 100FC6D1h
0x1001a5b7: mov eax,fs:[0]
0x1001a5bd: push eax
0x1001a5be: sub esp,0DC0h
0x1001a5c4: mov eax,ds:[10136220h]
0x1001a5c9: xor eax,esp
0x1001a5cb: mov [esp+0DBCh],eax
0x1001a5d2: push ebx
0x1001a5d3: push ebp
0x1001a5d4: push esi
0x1001a5d5: push edi
0x1001a5d6: mov eax,ds:[10136220h]
0x1001a5db: xor eax,esp
0x1001a5dd: push eax
0x1001a5de: lea eax,[esp+0DD4h]
0x1001a5e5: mov fs:[0],eax
0x1001a5eb: mov edi,[esp+0DE4h]
0x1001a5f2: xor ebp,ebp
0x1001a5f4: cmp edi,ebp
0x1001a5f6: mov esi,ecx
0x1001a5f8: mov [esp+28h],edi
0x1001a5fc: je near ptr 1001AF32h
0x1001a602: mov eax,[edi]
0x1001a604: mov edx,[eax+8]
0x1001a607: mov ebx,1
0x1001a60c: push ebx
0x1001a60d: push 0Dh
```

```
0x1001a60f: lea ecx,[esp+664h]
0x1001a616: push ecx
0x1001a617: mov ecx,edi
0x1001a619: call edx
0x1001a61b: push 4
0x1001a61d: lea eax,[esp+660h]
0x1001a624: push 100FF06Ch
0x1001a629: push eax
0x1001a62a: call dword ptr ds:[100FE414h]
0x1001a630: add esp,0Ch
0x1001a633: test eax,eax
0x1001a635: jne near ptr 1001AF32h
0x1001a63b: mov ecx,[esp+664h]
0x1001a642: push ecx
0x1001a643: mov ecx,esi
0x1001a645: call 1000F1C0h
0x1001a64a: mov [esp+664h],ax
0x1001a652: mov edx,[esp+662h]
0x1001a659: push edx
0x1001a65a: mov ecx,esi
0x1001a65c: call 1000F1C0h
0x1001a661: cmp dword ptr [esi+148h],0FFFFFFFFh
0x1001a668: mov [esp+662h],ax
0x1001a670: jne short 1001A691h
0x1001a672: movzx ecx,word ptr [esp+664h]
0x1001a67a: movzx eax,ax
0x1001a67d: mov [esi+0Ch],eax
0x1001a680: mov [esi+10h],ecx
0x1001a683: mov dword ptr [esi+40h],2
0x1001a68a: mov al,bl
0x1001a68c: jmp near ptr 1001AF34h
0x1001a691: mov dl,[esp+666h]
0x1001a698: movzx ecx,dl
0x1001a69b: and ecx,7
0x1001a69e: add ecx,ebx
0x1001a6a0: mov eax,ebx
0x1001a6a2: shl eax,cl
0x1001a6a4: mov cl,dl
0x1001a6a6: sar cl,4
0x1001a6a9: and cl,7
0x1001a6ac: movsx cx,cl
0x1001a6b0: add cx,bx
0x1001a6b3: test dl,dl
0x1001a6b5: mov [esp+5Ah],ax
0x1001a6ba: mov [esp+58h],cx
0x1001a6bf: mov [esp+2Ch],ebp
0x1001a6c3: mov [esp+30h],ebp
0x1001a6c7: jns short 1001A6E1h
0x1001a6c9: mov edx,[edi]
0x1001a6cb: mov edx,[edx+8]
0x1001a6ce: movsx eax,ax
0x1001a6d1: push ebx
0x1001a6d2: lea eax,[eax+eax*2]
0x1001a6d5: push eax
0x1001a6d6: lea ecx,[esp+64h]
0x1001a6da: push ecx
0x1001a6db: mov ecx,edi
0x1001a6dd: call edx
0x1001a6df: jmp short 1001A6E5h
0x1001a6e1: mov [esp+2Ch],ebx
0x1001a6e5: lea eax,[esp+65Ch]
0x1001a6ec: push eax
0x1001a6ed: lea ecx,[esp+5Ch]
0x1001a6f1: push ecx
0x1001a6f2: push edi
0x1001a6f3: mov ecx,esi
0x1001a6f5: mov [esp+58h],ebp
```

```
0x1001a6f9: mov [esp+50h],ebp
0x1001a6fd: call 1001A380h
0x1001a702: mov ecx,[esi+15Ch]
0x1001a708: cmp ecx,ebp
0x1001a70a: mov [esi+160h],eax
0x1001a710: jl near ptr 1001AF32h
0x1001a716: cmp ecx,eax
0x1001a718: jge near ptr 1001AF32h
0x1001a71e: cmp eax,ebx
0x1001a720: jne short 1001A726h
0x1001a722: mov [esp+2Ch],ebp
0x1001a726: mov [esp+17h],bl
0x1001a72a: mov [esp+40h],ebp
0x1001a72e: mov [esp+3Ch],ebp
0x1001a732: mov [esp+50h],ebx
0x1001a736: mov ecx,[esp+28h]
0x1001a73a: mov edx,[ecx]
0x1001a73c: mov edx,[edx+8]
0x1001a73f: push 1
0x1001a741: push 1
0x1001a743: lea eax,[esp+2Fh]
0x1001a747: push eax
0x1001a748: call edx
0x1001a74a: cmp eax,1
0x1001a74d: jne near ptr 1001AEB0h
0x1001a753: cmp dword ptr [esi+148h],0
0x1001a75a: jg near ptr 1001AF32h
0x1001a760: cmp byte ptr [esp+17h],0
0x1001a765: mov al,[esp+27h]
0x1001a769: jne short 1001A773h
0x1001a76b: test al,al
0x1001a76d: jne near ptr 1001AEA5h
0x1001a773: cmp al,21h
0x1001a775: je near ptr 1001AE92h
0x1001a77b: cmp al,2Ch
0x1001a77d: je short 1001A793h
0x1001a77f: cmp al,3Bh
0x1001a781: je near ptr 1001AEB0h
0x1001a787: test al,al
0x1001a789: sete byte ptr [esp+17h]
0x1001a78e: jmp near ptr 1001AEA5h
0x1001a793: mov ecx,[esp+28h]
0x1001a797: mov eax,[ecx]
0x1001a799: mov eax,[eax+8]
0x1001a79c: push 1
0x1001a79e: push 9
0x1001a7a0: lea edx,[esp+20h]
0x1001a7a4: push edx
0x1001a7a5: call eax
0x1001a7a7: mov ecx,[esp+18h]
0x1001a7ab: push ecx
0x1001a7ac: mov ecx,esi
0x1001a7ae: call 1000F1C0h
0x1001a7b3: mov edx,[esp+1Ah]
0x1001a7b7: push edx
0x1001a7b8: mov ecx,esi
0x1001a7ba: mov [esp+1Ch],ax
0x1001a7bf: call 1000F1C0h
0x1001a7c4: mov [esp+1Ah],ax
0x1001a7c9: mov eax,[esp+1Ch]
0x1001a7cd: push eax
0x1001a7ce: mov ecx,esi
0x1001a7d0: call 1000F1C0h
0x1001a7d5: mov ecx,[esp+1Eh]
0x1001a7d9: push ecx
0x1001a7da: mov ecx,esi
0x1001a7dc: mov [esp+20h],ax
```

```
0x1001a7e1: call 1000F1C0h
0x1001a7e6: mov cx,[esp+1Ch]
0x1001a7eb: movzx edx,word ptr [esp+18h]
0x1001a7f0: movzx edi,cx
0x1001a7f3: add edi,edx
0x1001a7f5: mov dx,[esp+662h]
0x1001a7fd: movzx ebx,dx
0x1001a800: cmp edi,ebx
0x1001a802: mov [esp+1Eh],ax
0x1001a807: jg near ptr 1001AEA5h
0x1001a80d: movzx edi,word ptr [esp+1Ah]
0x1001a812: movzx ebx,ax
0x1001a815: add ebx,edi
0x1001a817: mov di,[esp+664h]
0x1001a81f: movzx ebp,di
0x1001a822: cmp ebx,ebp
0x1001a824: jg near ptr 1001AEA5h
0x1001a82a: mov ebp,[esp+44h]
0x1001a82e: test ebp,ebp
0x1001a830: jne short 1001A841h
0x1001a832: cmp cx,dx
0x1001a835: je short 1001A841h
0x1001a837: cmp ax,di
0x1001a83a: je short 1001A841h
0x1001a83c: add dword ptr [esp+2Ch],1
0x1001a841: movzx ecx,byte ptr [esp+20h]
0x1001a846: test cl,cl
0x1001a848: movzx ebx,word ptr [esp+5Ah]
0x1001a84d: lea eax,[esp+5Ch]
0x1001a851: mov [esp+38h],eax
0x1001a855: jns short 1001A88Eh
0x1001a857: and ecx,7
0x1001a85a: add ecx,1
0x1001a85d: mov edx,1
0x1001a862: shl edx,cl
0x1001a864: mov ecx,[esp+28h]
0x1001a868: push 1
0x1001a86a: movzx ebx,dx
0x1001a86d: mov edx,[ecx]
0x1001a86f: mov edx,[edx+8]
0x1001a872: movsx eax,bx
0x1001a875: lea eax,[eax+eax*2]
0x1001a878: push eax
0x1001a879: lea eax,[esp+364h]
0x1001a880: push eax
0x1001a881: call edx
0x1001a883: lea eax,[esp+35Ch]
0x1001a88a: mov [esp+38h],eax
0x1001a88e: cmp bx,2
0x1001a892: jg short 1001A89Fh
0x1001a894: mov edi,1
0x1001a899: mov [esp+48h],edi
0x1001a89d: jmp short 1001A8B5h
0x1001a89f: xor ecx,ecx
0x1001a8a1: cmp bx,10h
0x1001a8a5: setg cl
0x1001a8a8: lea ecx,[ecx*4+4]
0x1001a8af: mov edi,ecx
0x1001a8b1: mov [esp+48h],ecx
0x1001a8b5: lea ecx,[esp+66Ch]
0x1001a8bc: call 10019590h
0x1001a8c1: push esi
0x1001a8c2: lea ecx,[esp+670h]
0x1001a8c9: mov dword ptr [esp+0DE0h],0
0x1001a8d4: call 1000E5C0h
0x1001a8d9: test ebp,ebp
0x1001a8db: lea ecx,[esp+66Ch]
```

```
0x1001a8e2: jne near ptr 1001A970h
0x1001a8e8: movzx edx,word ptr [esp+664h]
0x1001a8f0: movzx eax,word ptr [esp+662h]
0x1001a8f8: push 2
0x1001a8fa: push edi
0x1001a8fb: push edx
0x1001a8fc: push eax
0x1001a8fd: call 1000EAD0h
0x1001a902: mov ecx,[esi+14Ch]
0x1001a908: movzx edx,byte ptr [esi+3D3h]
0x1001a90f: mov [esp+4Ch],ecx
0x1001a913: push edx
0x1001a914: lea ecx,[esp+670h]
0x1001a91b: call 1000E680h
0x1001a920: push 3D0h
0x1001a925: call 10021FDEh
0x1001a92a: add esp,4
0x1001a92d: mov [esp+34h],eax
0x1001a931: test eax,eax
0x1001a933: mov byte ptr [esp+0DDCh],1
0x1001a93b: je short 1001A954h
0x1001a93d: push 1
0x1001a93f: push 1
0x1001a941: push 1
0x1001a943: lea ecx,[esp+678h]
0x1001a94a: push ecx
0x1001a94b: mov ecx,eax
0x1001a94d: call 1000EE20h
0x1001a952: jmp short 1001A956h
0x1001a954: xor eax,eax
0x1001a956: push 0
0x1001a958: mov ecx,eax
0x1001a95a: mov byte ptr [esp+0DE0h],0
0x1001a962: mov [esp+40h],eax
0x1001a966: call 1000F1B0h
0x1001a96b: jmp near ptr 1001AA0Eh
0x1001a970: mov eax,[esp+40h]
0x1001a974: cmp eax,2
0x1001a977: jne short 1001A998h
0x1001a979: push 0
0x1001a97b: push 0
0x1001a97d: push 0
0x1001a97f: push esi
0x1001a980: call 1000ECF0h
0x1001a985: mov edx,[esp+4Ch]
0x1001a989: push edx
0x1001a98a: lea ecx,[esp+670h]
0x1001a991: call 1000E680h
0x1001a996: jmp short 1001AA0Eh
0x1001a998: cmp eax,3
0x1001a99b: jne short 1001AA02h
0x1001a99d: push 0
0x1001a99f: push 0
0x1001a9a1: push 0
0x1001a9a3: push esi
0x1001a9a4: call 1000ECF0h
0x1001a9a9: mov edi,[esp+3Ch]
0x1001a9ad: push 2
0x1001a9af: mov ecx,edi
0x1001a9b1: call 1000F020h
0x1001a9b6: movzx eax,ax
0x1001a9b9: push eax
0x1001a9ba: mov ecx,edi
0x1001a9bc: call 1000EFE0h
0x1001a9c1: push eax
0x1001a9c2: mov ecx,edi
0x1001a9c4: call 1000EFF0h
```

```
0x1001a9c9: push eax
0x1001a9ca: lea ecx,[esp+67Ch]
0x1001a9d1: call 1000EAD0h
0x1001a9d6: lea ecx,[esp+66Ch]
0x1001a9dd: call 1000E640h
0x1001a9e2: push eax
0x1001a9e3: mov ecx,edi
0x1001a9e5: call 1000EFD0h
0x1001a9ea: push eax
0x1001a9eb: lea ecx,[esp+674h]
0x1001a9f2: call 1000EFD0h
0x1001a9f7: push eax
0x1001a9f8: call 100228C6h
0x1001a9fd: add esp,0Ch
0x1001aa00: jmp short 1001AA0Eh
0x1001aa02: push 1
0x1001aa04: push 1
0x1001aa06: push 1
0x1001aa08: push esi
0x1001aa09: call 1000ECF0h
0x1001aa0e: mov ecx,[esp+48h]
0x1001aa12: movzx edx,word ptr [esp+1Eh]
0x1001aa17: movzx eax,word ptr [esp+1Ch]
0x1001aa1c: push 2
0x1001aa1e: push ecx
0x1001aa1f: push edx
0x1001aa20: push eax
0x1001aa21: mov ecx,esi
0x1001aa23: call 1000EAD0h
0x1001aa28: test byte ptr [esp+20h],80h
0x1001aa2d: jne short 1001AA3Dh
0x1001aa2f: test byte ptr [esp+666h],80h
0x1001aa37: je near ptr 1001AB03h
0x1001aa3d: movsx edi,bx
0x1001aa40: xor eax,eax
0x1001aa42: test edi,edi
0x1001aa44: mov [esp+40h],edi
0x1001aa48: jle short 1001AA7Bh
0x1001aa4a: mov ecx,[esp+38h]
0x1001aa4e: add ecx,2
0x1001aa51: movzx edx,byte ptr [ecx-2]
0x1001aa55: mov [esp+eax+0BD0h],dl
0x1001aa5c: movzx edx,byte ptr [ecx-1]
0x1001aa60: mov [esp+eax+0CD0h],dl
0x1001aa67: movzx edx,byte ptr [ecx]
0x1001aa6a: mov [esp+eax+0AD0h],dl
0x1001aa71: add eax,1
0x1001aa74: add ecx,3
0x1001aa77: cmp eax,edi
0x1001aa79: jl short 1001AA51h
0x1001aa7b: cmp dword ptr [esi+14Ch],0
0x1001aa82: jl short 1001AAD2h
0x1001aa84: cmp eax,100h
0x1001aa89: jge short 1001AAD2h
0x1001aa8b: mov edi,100h
0x1001aa90: sub edi,eax
0x1001aa92: push edi
0x1001aa93: lea ecx,[esp+eax+0BD4h]
0x1001aa9a: push 0FFh
0x1001aa9f: push ecx
0x1001aaa0: lea ebp,[esp+eax+0CDCh]
0x1001aaa7: lea ebx,[esp+eax+0ADCh]
0x1001aaae: call 100228C0h
0x1001aab3: push edi
0x1001aab4: push 0FFh
0x1001aab9: push ebp
0x1001aaba: call 100228C0h
```

```
0x1001aabf: push edi
0x1001aac0: push 0FFh
0x1001aac5: push ebx
0x1001aac6: call 100228C0h
0x1001aacb: mov edi,[esp+64h]
0x1001aacf: add esp,24h
0x1001aad2: xor ebp,ebp
0x1001aad4: cmp [esi+14Ch],ebp
0x1001aada: jl short 1001AAE1h
0x1001aadc: mov edi,100h
0x1001aae1: lea eax,[esp+0AD0h]
0x1001aae8: push eax
0x1001aae9: lea ecx,[esp+0CD4h]
0x1001aaf0: push ecx
0x1001aaf1: lea edx,[esp+0BD8h]
0x1001aaf8: push edx
0x1001aaf9: push edi
0x1001aafa: mov ecx,esi
0x1001aafc: call 100135B0h
0x1001ab01: jmp short 1001AB05h
0x1001ab03: xor ebp,ebp
0x1001ab05: push 18h
0x1001ab07: call 10021FDEh
0x1001ab0c: add esp,4
0x1001ab0f: mov [esp+34h],eax
0x1001ab13: cmp eax,ebp
0x1001ab15: mov byte ptr [esp+0DDCh],2
0x1001ab1d: je short 1001AB2Bh
0x1001ab1f: push esi
0x1001ab20: mov ecx,eax
0x1001ab22: call 10011F20h
0x1001ab27: mov edi,eax
0x1001ab29: jmp short 1001AB2Dh
0x1001ab2b: xor edi,edi
0x1001ab2d: mov ecx,edi
0x1001ab2f: mov byte ptr [esp+0DDCh],0
0x1001ab37: call 10017120h
0x1001ab3c: movzx eax,byte ptr [esp+20h]
0x1001ab41: movzx ecx,word ptr [esp+1Eh]
0x1001ab46: mov ebx,[esp+28h]
0x1001ab4a: and eax,40h
0x1001ab4d: mov [esi+3E8h],eax
0x1001ab53: mov [esi+3F4h],ecx
0x1001ab59: mov dword ptr [esi+3FCh],4000h
0x1001ab63: mov dword ptr [esi+3F0h],8
0x1001ab6d: mov [esi+3ECh],ebp
0x1001ab73: mov [esi+3F8h],ebp
0x1001ab79: mov edx,[ebx]
0x1001ab7b: mov eax,[edx+14h]
0x1001ab7e: mov ecx,ebx
0x1001ab80: mov [esp+38h],ebp
0x1001ab84: call eax
0x1001ab86: mov edx,[esp+1Ch]
0x1001ab8a: lea ecx,[esp+38h]
0x1001ab8e: push ecx
0x1001ab8f: push edx
0x1001ab90: push edi
0x1001ab91: push ebx
0x1001ab92: mov ecx,esi
0x1001ab94: mov [esp+44h],eax
0x1001ab98: call 10019E70h
0x1001ab9d: push edi
0x1001ab9e: call 10021FCCh
0x1001aba3: add esp,4
0x1001aba6: cmp [esi+148h],ebp
0x1001abac: jne near ptr 1001AF1Bh
0x1001abb2: cmp dword ptr [esp+2Ch],2
```

```
0x1001abb7: jge short 1001AC1Bh
0x1001abb9: mov eax,[esp+48h]
0x1001abbd: push eax
0x1001abbe: lea ecx,[esp+670h]
0x1001abc5: call 100103D0h
0x1001abca: lea ecx,[esp+18h]
0x1001abce: push ecx
0x1001abcf: push esi
0x1001abd0: lea ecx,[esp+674h]
0x1001abd7: call 1001A270h
0x1001abdc: mov edx,[esp+4Ch]
0x1001abe0: push edx
0x1001abe1: lea ecx,[esp+670h]
0x1001abe8: call 1000EEB0h
0x1001abed: push 100h
0x1001abf2: mov ecx,esi
0x1001abf4: call 10012EA0h
0x1001abf9: push eax
0x1001abfa: lea ecx,[esp+674h]
0x1001ac01: call 10013690h
0x1001ac06: push ebp
0x1001ac07: lea eax,[esp+670h]
0x1001ac0e: push eax
0x1001ac0f: mov ecx,esi
0x1001ac11: call 1000E6D0h
0x1001ac16: jmp near ptr 1001ACEDh
0x1001ac1b: mov ecx,[esi+160h]
0x1001ac21: sub ecx,1
0x1001ac24: cmp [esp+30h],ebp
0x1001ac28: mov [esi+15Ch],ecx
0x1001ac2e: jne short 1001AC7Ah
0x1001ac30: push 3D0h
0x1001ac35: call 10021FDEh
0x1001ac3a: add esp,4
0x1001ac3d: mov [esp+40h],eax
0x1001ac41: cmp eax,ebp
0x1001ac43: mov byte ptr [esp+0DDCh],3
0x1001ac4b: je short 1001AC6Ch
0x1001ac4d: movzx edx,word ptr [esp+664h]
0x1001ac55: movzx ecx,word ptr [esp+662h]
0x1001ac5d: push 2
0x1001ac5f: push 18h
0x1001ac61: push edx
0x1001ac62: push ecx
0x1001ac63: mov ecx,eax
0x1001ac65: call 1000ECB0h
0x1001ac6a: jmp short 1001AC6Eh
0x1001ac6c: xor eax,eax
0x1001ac6e: mov byte ptr [esp+0DDCh],0
0x1001ac76: mov [esp+30h],eax
0x1001ac7a: xor ebp,ebp
0x1001ac7c: cmp [esp+1Eh],bp
0x1001ac81: jbe short 1001ACEDh
0x1001ac83: movzx eax,word ptr [esp+1Ch]
0x1001ac88: xor edi,edi
0x1001ac8a: test eax,eax
0x1001ac8c: jle short 1001ACE1h
0x1001ac8e: mov ebx,[esp+30h]
0x1001ac92: movzx edx,word ptr [esp+1Eh]
0x1001ac97: push 1
0x1001ac99: sub edx,ebp
0x1001ac9b: sub edx,1
0x1001ac9e: push edx
0x1001ac9f: push edi
0x1001aca0: lea eax,[esp+60h]
0x1001aca4: push eax
0x1001aca5: mov ecx,esi
```

```
0x1001aca7: call 100130E0h
0x1001acac: mov ecx,[eax]
0x1001acae: movzx edx,word ptr [esp+1Ah]
0x1001acb3: movzx eax,word ptr [esp+664h]
0x1001acbb: push 0
0x1001acbd: push ecx
0x1001acbe: movzx ecx,word ptr [esp+20h]
0x1001acc3: sub eax,edx
0x1001acc5: sub eax,ebp
0x1001acc7: sub eax,1
0x1001acca: add ecx,edi
0x1001accc: push eax
0x1001accd: push ecx
0x1001acce: mov ecx,ebx
0x1001acd0: call 10013900h
0x1001acd5: movzx eax,word ptr [esp+1Ch]
0x1001acda: add edi,1
0x1001acdd: cmp edi,eax
0x1001acdf: jl short 1001AC92h
0x1001ace1: movzx edx,word ptr [esp+1Eh]
0x1001ace6: add ebp,1
0x1001ace9: cmp ebp,edx
0x1001aceb: jl short 1001AC88h
0x1001aced: movzx eax,byte ptr [esi+3D0h]
0x1001acf4: shr eax,2
0x1001acf7: and eax,7
0x1001acfa: cmp dword ptr [esp+38h],0
0x1001acff: mov ebp,eax
0x1001ad01: mov [esp+40h],ebp
0x1001ad05: je short 1001AD1Ah
0x1001ad07: mov ecx,[esp+34h]
0x1001ad0b: mov edx,[esp+28h]
0x1001ad0f: push ecx
0x1001ad10: push edx
0x1001ad11: mov ecx,esi
0x1001ad13: call 1001A1F0h
0x1001ad18: jmp short 1001AD37h
0x1001ad1a: mov edx,[esi+3FCh]
0x1001ad20: sub edx,[esi+400h]
0x1001ad26: mov ecx,[esp+28h]
0x1001ad2a: mov eax,[ecx]
0x1001ad2c: mov eax,[eax+10h]
0x1001ad2f: push 1
0x1001ad31: add edx,1
0x1001ad34: push edx
0x1001ad35: call eax
0x1001ad37: cmp byte ptr [esi+1EDh],0
0x1001ad3e: je near ptr 1001ADFAh
0x1001ad44: cmp dword ptr [esp+30h],0
0x1001ad49: jne near ptr 1001ADFAh
0x1001ad4f: mov edi,[esp+44h]
0x1001ad53: test edi,edi
0x1001ad55: jne short 1001ADA8h
0x1001ad57: mov ecx,esi
0x1001ad59: call 1000E550h
0x1001ad5e: mov eax,[esi+160h]
0x1001ad64: xor ecx,ecx
0x1001ad66: mov edx,4
0x1001ad6b: mul edx
0x1001ad6d: seto cl
0x1001ad70: neg ecx
0x1001ad72: or ecx,eax
0x1001ad74: push ecx
0x1001ad75: call 10021FDEh
0x1001ad7a: mov [esi+3CCh],eax
0x1001ad80: xor eax,eax
0x1001ad82: add esp,4
```

```
0x1001ad85: cmp [esi+160h],eax
0x1001ad8b: jle short 1001ADA8h
0x1001ad8d: lea ecx,[ecx]
0x1001ad90: mov ecx,[esi+3CCh]
0x1001ad96: mov dword ptr [ecx+eax*4],0
0x1001ad9d: add eax,1
0x1001ada0: cmp eax,[esi+160h]
0x1001ada6: jl short 1001AD90h
0x1001ada8: push 3D0h
0x1001adad: call 10021FDEh
0x1001adb2: add esp,4
0x1001adb5: mov [esp+34h],eax
0x1001adb9: test eax,eax
0x1001adbb: mov byte ptr [esp+0DDCh],4
0x1001adc3: je short 1001ADD5h
0x1001adc5: push 1
0x1001adc7: push 1
0x1001adc9: push 1
0x1001adcb: push esi
0x1001adcc: mov ecx,eax
0x1001adce: call 1000EE20h
0x1001add3: jmp short 1001ADD7h
0x1001add5: xor eax,eax
0x1001add7: mov edx,[esi+3CCh]
0x1001addd: mov [edx+edi*4],eax
0x1001ade0: mov eax,[esi+3CCh]
0x1001ade6: mov ecx,[eax+edi*4]
0x1001ade9: push 0
0x1001adeb: mov byte ptr [esp+0DE0h],0
0x1001adf3: call 1000F1B0h
0x1001adf8: jmp short 1001ADFEh
0x1001adfa: mov edi,[esp+44h]
0x1001adfe: cmp ebp,1
0x1001ae01: jg short 1001AE57h
0x1001ae03: mov ecx,[esp+3Ch]
0x1001ae07: test ecx,ecx
0x1001ae09: je short 1001AE13h
0x1001ae0b: mov edx,[ecx]
0x1001ae0d: mov eax,[edx]
0x1001ae0f: push 1
0x1001ae11: call eax
0x1001ae13: push 3D0h
0x1001ae18: call 10021FDEh
0x1001ae1d: add esp,4
0x1001ae20: mov [esp+34h],eax
0x1001ae24: test eax,eax
0x1001ae26: mov byte ptr [esp+0DDCh],5
0x1001ae2e: je short 1001AE40h
0x1001ae30: push 1
0x1001ae32: push 1
0x1001ae34: push 1
0x1001ae36: push esi
0x1001ae37: mov ecx,eax
0x1001ae39: call 1000EE20h
0x1001ae3e: jmp short 1001AE42h
0x1001ae40: xor eax,eax
0x1001ae42: push 0
0x1001ae44: mov ecx,eax
0x1001ae46: mov byte ptr [esp+0DE0h],0
0x1001ae4e: mov [esp+40h],eax
0x1001ae52: call 1000F1B0h
0x1001ae57: cmp [esi+15Ch],edi
0x1001ae5d: jne short 1001AE72h
0x1001ae5f: cmp byte ptr [esi+1EDh],0
0x1001ae66: jne short 1001AE72h
0x1001ae68: mov dword ptr [esp+50h],0
0x1001ae70: jmp short 1001AE79h
```

```
0x1001ae72: add edi,1
0x1001ae75: mov [esp+44h],edi
0x1001ae79: lea ecx,[esp+66Ch]
0x1001ae80: mov dword ptr [esp+0DDCh],0FFFFFFFFh
0x1001ae8b: call 10019690h
0x1001ae90: jmp short 1001AEA5h
0x1001ae92: mov ecx,[esp+28h]
0x1001ae96: push ecx
0x1001ae97: mov ecx,esi
0x1001ae99: call 10019770h
0x1001ae9e: movzx edx,al
0x1001aea1: mov [esp+50h],edx
0x1001aea5: cmp dword ptr [esp+50h],0
0x1001aeaa: jne near ptr 1001A736h
0x1001aeb0: cmp dword ptr [esp+2Ch],2
0x1001aeb5: mov edi,[esp+30h]
0x1001aeb9: jl short 1001AEF9h
0x1001aebb: test edi,edi
0x1001aebd: je short 1001AF07h
0x1001aebf: test byte ptr [esi+3D0h],1
0x1001aec6: je short 1001AEEFh
0x1001aec8: movzx eax,byte ptr [esi+14Ch]
0x1001aecf: push eax
0x1001aed0: lea ecx,[esp+38h]
0x1001aed4: push ecx
0x1001aed5: mov ecx,esi
0x1001aed7: call 10012EC0h
0x1001aedc: mov edx,[eax]
0x1001aede: push edx
0x1001aedf: mov ecx,edi
0x1001aee1: call 1000EED0h
0x1001aee6: push 0
0x1001aee8: mov ecx,edi
0x1001aeea: call 1000EEB0h
0x1001aeef: push 1
0x1001aef1: push edi
0x1001aef2: mov ecx,esi
0x1001aef4: call 1000E6D0h
0x1001aef9: test edi,edi
0x1001aefb: je short 1001AF07h
0x1001aefd: mov eax,[edi]
0x1001aeff: mov edx,[eax]
0x1001af01: push 1
0x1001af03: mov ecx,edi
0x1001af05: call edx
0x1001af07: mov ecx,[esp+3Ch]
0x1001af0b: test ecx,ecx
0x1001af0d: je short 1001AF17h
0x1001af0f: mov eax,[ecx]
0x1001af11: mov edx,[eax]
0x1001af13: push 1
0x1001af15: call edx
0x1001af17: mov al,1
0x1001af19: jmp short 1001AF34h
0x1001af1b: lea ecx,[esp+66Ch]
0x1001af22: mov dword ptr [esp+0DDCh],0FFFFFFFFh
0x1001af2d: call 10019690h
0x1001af32: xor al,al
0x1001af34: mov ecx,[esp+0DD4h]
0x1001af3b: mov fs:[0],ecx
0x1001af42: pop ecx
0x1001af43: pop edi
0x1001af44: pop esi
0x1001af45: pop ebp
0x1001af46: pop ebx
0x1001af47: mov ecx,[esp+0DBCh]
0x1001af4e: xor ecx,esp
```

```
0x1001af50: call 10022326h
0x1001af55: add esp,0DCCh
0x1001af5b: ret 4
```

`sub_1001a5b0` is called by sub_1000f760. It delegates to sub_1000e550, sub_1000e5c0, sub_1000e640 (38 callees).

12.380 sub_1001b160 at 0x1001b160 (15 instructions)

```
0x1001b160: push esi
0x1001b161: push 3
0x1001b163: mov esi,ecx
0x1001b165: call 1000E480h
0x1001b16a: push 1D0h
0x1001b16f: lea eax,[esi+1F0h]
0x1001b175: push 0
0x1001b177: push eax
0x1001b178: mov dword ptr [esi],100FF078h
0x1001b17e: mov dword ptr [esi+3D0h],0
0x1001b188: call 100228C0h
0x1001b18d: add esp,0Ch
0x1001b190: mov eax,esi
0x1001b192: pop esi
0x1001b193: ret
```

`sub_1001b160` is called by sub_1000f760. It delegates to sub_1000e480, j_memset (2 callees).

12.381 sub_1001b1a0 at 0x1001b1a0 (67 instructions)

```
0x1001b1a0: push 0FFFFFFFFh
0x1001b1a2: push 100FC70Bh
0x1001b1a7: mov eax,fs:[0]
0x1001b1ad: push eax
0x1001b1ae: push ecx
0x1001b1af: push ebx
0x1001b1b0: push esi
0x1001b1b1: push edi
0x1001b1b2: mov eax,ds:[10136220h]
0x1001b1b7: xor eax,esp
0x1001b1b9: push eax
0x1001b1ba: lea eax,[esp+14h]
0x1001b1be: mov fs:[0],eax
0x1001b1c4: mov edi,ecx
0x1001b1c6: push 204h
0x1001b1cb: call 10021FDEh
0x1001b1d0: add esp,4
0x1001b1d3: mov [esp+10h],eax
0x1001b1d7: test eax,eax
0x1001b1d9: mov dword ptr [esp+1Ch],0
0x1001b1e1: je short 1001B1F3h
0x1001b1e3: lea ecx,[edi+1F0h]
0x1001b1e9: push ecx
0x1001b1ea: mov ecx,eax
0x1001b1ec: call 10020AA0h
0x1001b1f1: jmp short 1001B1F5h
0x1001b1f3: xor eax,eax
0x1001b1f5: test eax,eax
0x1001b1f7: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x1001b1ff: mov [edi+3D0h],eax
0x1001b205: je short 1001B253h
0x1001b207: mov esi,[esp+24h]
0x1001b20b: mov edx,[esi]
0x1001b20d: mov eax,[edx+14h]
0x1001b210: mov ecx,esi
0x1001b212: call eax
0x1001b214: mov ecx,[edi+3D0h]
```

```
0x1001b21a: push 1
0x1001b21c: push esi
0x1001b21d: mov ebx,eax
0x1001b21f: call 10021AF0h
0x1001b224: mov edx,[esi]
0x1001b226: mov eax,[edx+10h]
0x1001b229: push 0
0x1001b22b: push ebx
0x1001b22c: mov ecx,esi
0x1001b22e: call eax
0x1001b230: mov ecx,[edi+3D0h]
0x1001b236: mov edx,[ecx]
0x1001b238: mov al,[edx+1CCh]
0x1001b23e: mov ecx,[esp+14h]
0x1001b242: mov fs:[0],ecx
0x1001b249: pop ecx
0x1001b24a: pop edi
0x1001b24b: pop esi
0x1001b24c: pop ebx
0x1001b24d: add esp,10h
0x1001b250: ret 4
0x1001b253: xor al,al
0x1001b255: mov ecx,[esp+14h]
0x1001b259: mov fs:[0],ecx
0x1001b260: pop ecx
0x1001b261: pop edi
0x1001b262: pop esi
0x1001b263: pop ebx
0x1001b264: add esp,10h
0x1001b267: ret 4
```

`sub_1001b1a0` is called by sub_1001b350. It delegates to sub_10020aa0, sub_10021af0, j_ORDINAL 762 (3 callees).

12.382 sub_1001b270 at 0x1001b270 (22 instructions)

```
0x1001b270: mov eax,[esp+4]
0x1001b274: push esi
0x1001b275: mov esi,ecx
0x1001b277: mov [esi+30h],eax
0x1001b27a: xor eax,eax
0x1001b27c: push 1000h
0x1001b281: mov dword ptr [esi+8],1001AF70h
0x1001b288: mov dword ptr [esi+0Ch],1001AF90h
0x1001b28f: mov dword ptr [esi+10h],1001AFE0h
0x1001b296: mov dword ptr [esi+1Ch],1001B050h
0x1001b29d: mov dword ptr [esi+20h],1001B070h
0x1001b2a4: mov dword ptr [esi+24h],1001B0F0h
0x1001b2ab: mov dword ptr [esi+28h],1003B070h
0x1001b2b2: mov dword ptr [esi+2Ch],100B3980h
0x1001b2b9: mov [esi+14h],eax
0x1001b2bc: mov [esi+18h],eax
0x1001b2bf: call 10021FDEh
0x1001b2c4: mov [esi+34h],eax
0x1001b2c7: add esp,4
0x1001b2ca: mov eax,esi
0x1001b2cc: pop esi
0x1001b2cd: ret 4
```

`sub_1001b270` is called by sub_1001b350. It delegates to j_ORDINAL 762 (1 callees).

12.383 sub_1001b350 at 0x1001b350 (416 instructions)

```
0x1001b350: push ebp
0x1001b351: lea ebp,[esp-28Ch]
0x1001b358: sub esp,28Ch
```

```
0x1001b35e: push 0FFFFFFFFh
0x1001b360: push 100FC768h
0x1001b365: mov eax,fs:[0]
0x1001b36b: push eax
0x1001b36c: sub esp,68h
0x1001b36f: mov eax,ds:[10136220h]
0x1001b374: xor eax,ebp
0x1001b376: mov [ebp+288h],eax
0x1001b37c: push ebx
0x1001b37d: push esi
0x1001b37e: push edi
0x1001b37f: push eax
0x1001b380: lea eax,[ebp-0Ch]
0x1001b383: mov fs:[0],eax
0x1001b389: mov ebx,[ebp+294h]
0x1001b38f: mov edi,ecx
0x1001b391: push ebx
0x1001b392: mov [ebp-18h],edi
0x1001b395: call 1001B1A0h
0x1001b39a: xor esi,esi
0x1001b39c: cmp edi,esi
0x1001b39e: mov [ebp-11h],al
0x1001b3a1: mov [ebp-24h],edi
0x1001b3a4: je short 1001B3B1h
0x1001b3a6: push esi
0x1001b3a7: mov ecx,edi
0x1001b3a9: call 1000E5E0h
0x1001b3ae: mov [ebp-28h],eax
0x1001b3b1: lea eax,[ebp]
0x1001b3b4: add edi,44h
0x1001b3b7: push eax
0x1001b3b8: mov [ebp-34h],esi
0x1001b3bb: mov [ebp-38h],esi
0x1001b3be: mov [ebp-2Ch],esi
0x1001b3c1: mov [ebp-30h],esi
0x1001b3c4: mov [ebp+0C4h],edi
0x1001b3ca: call 1003C3D0h
0x1001b3cf: add esp,4
0x1001b3d2: push ebx
0x1001b3d3: lea ecx,[ebp-74h]
0x1001b3d6: mov [ebp+0C8h],eax
0x1001b3dc: mov dword ptr [ebp],1001B130h
0x1001b3e3: call 1001B270h
0x1001b3e8: push 10130C88h
0x1001b3ed: mov [ebp-4],esi
0x1001b3f0: mov edx,[ebp-4]
0x1001b3f3: push edx
0x1001b3f4: push 100229E2h
0x1001b3f9: lea ecx,[ebp+84h]
0x1001b3ff: push 3
0x1001b401: push ecx
0x1001b402: call 100229DCh
0x1001b407: add esp,14h
0x1001b40a: test eax,eax
0x1001b40c: je short 1001B434h
0x1001b40e: lea eax,[ebp+0C8h]
0x1001b414: push eax
0x1001b415: call 1003C530h
0x1001b41a: mov ecx,[ebp-40h]
0x1001b41d: push ecx
0x1001b41e: mov dword ptr [ebp-4],0FFFFFFFFh
0x1001b425: call 10021FCCh
0x1001b42a: add esp,8
0x1001b42d: xor al,al
0x1001b42f: jmp near ptr 1001B90Fh
0x1001b434: push 1C0h
0x1001b439: lea edx,[ebp+0C8h]
```

```
0x1001b43f: push 50h
0x1001b441: push edx
0x1001b442: call 1003B220h
0x1001b447: lea ecx,[ebp+0C8h]
0x1001b44d: lea eax,[ebp-60h]
0x1001b450: push 1
0x1001b452: push ecx
0x1001b453: mov [ebp+0E0h],eax
0x1001b459: call 1003B6E0h
0x1001b45e: mov edi,[ebp-18h]
0x1001b461: add esp,14h
0x1001b464: push 3
0x1001b466: mov ecx,edi
0x1001b468: call 1000F2E0h
0x1001b46d: mov ebx,eax
0x1001b46f: test bl,bl
0x1001b471: jns short 1001B47Dh
0x1001b473: mov dword ptr [ebp+0F4h],1
0x1001b47d: test ebx,100h
0x1001b483: je short 1001B49Ch
0x1001b485: mov ecx,edi
0x1001b487: mov byte ptr [ebp+112h],1
0x1001b48e: call 1000F130h
0x1001b493: movzx edx,al
0x1001b496: mov [ebp+11Ch],edx
0x1001b49c: test ebx,200h
0x1001b4a2: je short 1001B4B0h
0x1001b4a4: mov eax,[edi+3E4h]
0x1001b4aa: mov [ebp+114h],eax
0x1001b4b0: test ebx,400h
0x1001b4b6: je short 1001B4BFh
0x1001b4b8: mov byte ptr [ebp+118h],0
0x1001b4bf: test ebx,800h
0x1001b4c5: je short 1001B4CEh
0x1001b4c7: mov byte ptr [ebp+110h],0
0x1001b4ce: mov ecx,edi
0x1001b4d0: mov dword ptr [ebp+0F8h],1
0x1001b4da: call 1000F170h
0x1001b4df: movzx ecx,al
0x1001b4e2: or ebx,0FFFFFFFFh
0x1001b4e5: mov [ebp+0FCh],ecx
0x1001b4eb: cmp [edi+148h],ebx
0x1001b4f1: jne short 1001B538h
0x1001b4f3: lea edx,[ebp+0C8h]
0x1001b4f9: push edx
0x1001b4fa: call 1003BC10h
0x1001b4ff: mov eax,[ebp+124h]
0x1001b505: mov [edi+0Ch],eax
0x1001b508: mov ecx,[ebp+128h]
0x1001b50e: lea edx,[ebp+0C8h]
0x1001b514: mov [edi+10h],ecx
0x1001b517: push edx
0x1001b518: mov dword ptr [edi+40h],3
0x1001b51f: call 1003C530h
0x1001b524: mov eax,[ebp-40h]
0x1001b527: push eax
0x1001b528: mov [ebp-4],ebx
0x1001b52b: call 10021FCCh
0x1001b530: add esp,0Ch
0x1001b533: jmp near ptr 1001B90Dh
0x1001b538: lea ecx,[ebp+0C8h]
0x1001b53e: push ecx
0x1001b53f: call 1003BA90h
0x1001b544: mov edx,[ebp+130h]
0x1001b54a: mov ecx,[ebp+128h]
0x1001b550: add esp,4
0x1001b553: push 3
```

```
0x1001b555: lea eax,[edx*8]
0x1001b55c: mov edx,[ebp+124h]
0x1001b562: push eax
0x1001b563: push ecx
0x1001b564: push edx
0x1001b565: mov ecx,edi
0x1001b567: call 1000EAD0h
0x1001b56c: cmp [edi+4],esi
0x1001b56f: jne short 1001B580h
0x1001b571: push 1
0x1001b573: lea eax,[ebp+84h]
0x1001b579: push eax
0x1001b57a: call dword ptr ds:[100FE4E0h]
0x1001b580: cmp byte ptr [ebp-11h],0
0x1001b584: je short 1001B5FFh
0x1001b586: fldz
0x1001b588: fcom dword ptr [edi+2A4h]
0x1001b58e: fnstsw ax
0x1001b590: test ah,44h
0x1001b593: jnp short 1001B5BFh
0x1001b595: fcom dword ptr [edi+2ACh]
0x1001b59b: fnstsw ax
0x1001b59d: test ah,44h
0x1001b5a0: jnp short 1001B5BFh
0x1001b5a2: fstp st(0)
0x1001b5a4: fld dword ptr [edi+2A4h]
0x1001b5aa: fdiv dword ptr [edi+2ACh]
0x1001b5b0: call 100228D0h
0x1001b5b5: push eax
0x1001b5b6: mov ecx,edi
0x1001b5b8: call 1000EF10h
0x1001b5bd: fldz
0x1001b5bf: fcom dword ptr [edi+2A8h]
0x1001b5c5: fnstsw ax
0x1001b5c7: test ah,44h
0x1001b5ca: jnp near ptr 1001B6ECh
0x1001b5d0: fcomp dword ptr [edi+2ACh]
0x1001b5d6: fnstsw ax
0x1001b5d8: test ah,44h
0x1001b5db: jnp near ptr 1001B6EEh
0x1001b5e1: fld dword ptr [edi+2A8h]
0x1001b5e7: fdiv dword ptr [edi+2ACh]
0x1001b5ed: call 100228D0h
0x1001b5f2: push eax
0x1001b5f3: mov ecx,edi
0x1001b5f5: call 1000EF70h
0x1001b5fa: jmp near ptr 1001B6EEh
0x1001b5ff: movzx eax,byte ptr [ebp+1CBh]
0x1001b606: sub eax,esi
0x1001b608: je near ptr 1001B6A0h
0x1001b60e: sub eax,2
0x1001b611: je short 1001B636h
0x1001b613: movzx ecx,word ptr [ebp+1CCh]
0x1001b61a: push ecx
0x1001b61b: mov ecx,edi
0x1001b61d: call 1000EF10h
0x1001b622: movzx edx,word ptr [ebp+1CEh]
0x1001b629: push edx
0x1001b62a: mov ecx,edi
0x1001b62c: call 1000EF70h
0x1001b631: jmp near ptr 1001B6EEh
0x1001b636: movzx eax,word ptr [ebp+1CCh]
0x1001b63d: mov [ebp-10h],eax
0x1001b640: sub esp,8
0x1001b643: fild dword ptr [ebp-10h]
0x1001b646: fmul qword ptr ds:[100FF080h]
0x1001b64c: fadd qword ptr ds:[100FE7C8h]
```

```
0x1001b652: fstp qword ptr [esp]
0x1001b655: call 1002297Ch
0x1001b65a: add esp,8
0x1001b65d: call 100228D0h
0x1001b662: push eax
0x1001b663: mov ecx,edi
0x1001b665: call 1000EF10h
0x1001b66a: movzx ecx,word ptr [ebp+1CEh]
0x1001b671: mov [ebp-10h],ecx
0x1001b674: sub esp,8
0x1001b677: fild dword ptr [ebp-10h]
0x1001b67a: fmul qword ptr ds:[100FF080h]
0x1001b680: fadd qword ptr ds:[100FE7C8h]
0x1001b686: fstp qword ptr [esp]
0x1001b689: call 1002297Ch
0x1001b68e: add esp,8
0x1001b691: call 100228D0h
0x1001b696: push eax
0x1001b697: mov ecx,edi
0x1001b699: call 1000EF70h
0x1001b69e: jmp short 1001B6EEh
0x1001b6a0: cmp [ebp+1CEh],si
0x1001b6a7: jbe short 1001B6EEh
0x1001b6a9: cmp [ebp+1CCh],si
0x1001b6b0: jbe short 1001B6EEh
0x1001b6b2: mov ecx,edi
0x1001b6b4: call 1000EF00h
0x1001b6b9: movzx edx,word ptr [ebp+1CEh]
0x1001b6c0: mov [ebp-10h],eax
0x1001b6c3: movzx eax,word ptr [ebp+1CCh]
0x1001b6ca: fild dword ptr [ebp-10h]
0x1001b6cd: mov [ebp-10h],edx
0x1001b6d0: fild dword ptr [ebp-10h]
0x1001b6d3: mov [ebp-10h],eax
0x1001b6d6: fild dword ptr [ebp-10h]
0x1001b6d9: fdivp
0x1001b6db: fmulp
0x1001b6dd: call 100228D0h
0x1001b6e2: push eax
0x1001b6e3: mov ecx,edi
0x1001b6e5: call 1000EF70h
0x1001b6ea: jmp short 1001B6EEh
0x1001b6ec: fstp st(0)
0x1001b6ee: cmp dword ptr [ebp+0F4h],1
0x1001b6f5: jne short 1001B707h
0x1001b6f7: mov ecx,edi
0x1001b6f9: call 100136E0h
0x1001b6fe: mov dword ptr [edi+28h],100h
0x1001b705: jmp short 1001B73Dh
0x1001b707: cmp byte ptr [ebp+112h],0
0x1001b70e: je short 1001B73Ah
0x1001b710: mov eax,[ebp+13Ch]
0x1001b716: mov ecx,[eax+8]
0x1001b719: mov edx,[eax+4]
0x1001b71c: mov eax,[eax]
0x1001b71e: push ecx
0x1001b71f: mov ecx,[ebp+138h]
0x1001b725: push edx
0x1001b726: push eax
0x1001b727: push ecx
0x1001b728: mov ecx,edi
0x1001b72a: call 100135B0h
0x1001b72f: mov edx,[ebp+138h]
0x1001b735: mov [edi+28h],edx
0x1001b738: jmp short 1001B73Dh
0x1001b73a: mov [edi+28h],esi
0x1001b73d: mov esi,[ebp+130h]
```

```
0x1001b743: imul esi,[ebp+124h]
0x1001b74a: mov ecx,[ebp+0CCh]
0x1001b750: mov edx,[ecx+8]
0x1001b753: push 1
0x1001b755: push esi
0x1001b756: lea eax,[ebp+0C8h]
0x1001b75c: push 1
0x1001b75e: push eax
0x1001b75f: mov [ebp-1Ch],esi
0x1001b762: call edx
0x1001b764: mov ebx,eax
0x1001b766: add esp,10h
0x1001b769: lea ecx,[ebp-38h]
0x1001b76c: mov [ebp-10h],ebx
0x1001b76f: call 10017120h
0x1001b774: mov eax,[ebp+140h]
0x1001b77a: cmp eax,[ebp+128h]
0x1001b780: jae near ptr 1001B894h
0x1001b786: jmp short 1001B790h
0x1001b790: cmp dword ptr [edi+148h],0
0x1001b797: push 1
0x1001b799: jne near ptr 1001B573h
0x1001b79f: lea ecx,[ebp+0C8h]
0x1001b7a5: push ebx
0x1001b7a6: push ecx
0x1001b7a7: call 1003B830h
0x1001b7ac: add esp,0Ch
0x1001b7af: cmp dword ptr [ebp+0ECh],4
0x1001b7b6: jne near ptr 1001B936h
0x1001b7bc: cmp byte ptr [ebp+112h],0
0x1001b7c3: jne near ptr 1001B936h
0x1001b7c9: mov ebx,[ebx]
0x1001b7cb: xor ecx,ecx
0x1001b7cd: xor esi,esi
0x1001b7cf: cmp [edi+30h],ecx
0x1001b7d2: jle near ptr 1001B86Bh
0x1001b7d8: jmp short 1001B7E0h
0x1001b7e0: cmp ecx,[ebp-1Ch]
0x1001b7e3: jge near ptr 1001B86Bh
0x1001b7e9: movzx eax,byte ptr [ecx+ebx+3]
0x1001b7ee: movzx edi,byte ptr [ecx+ebx+2]
0x1001b7f3: imul edi,eax
0x1001b7f6: mov [ebp-20h],eax
0x1001b7f9: mov eax,80808081h
0x1001b7fe: imul edi
0x1001b800: add edx,edi
0x1001b802: sar edx,7
0x1001b805: mov eax,edx
0x1001b807: shr eax,1Fh
0x1001b80a: add eax,edx
0x1001b80c: mov edx,[ebp-28h]
0x1001b80f: mov [edx+esi],al
0x1001b812: movzx edi,byte ptr [ecx+ebx+1]
0x1001b817: imul edi,[ebp-20h]
0x1001b81b: mov eax,80808081h
0x1001b820: imul edi
0x1001b822: add edx,edi
0x1001b824: sar edx,7
0x1001b827: mov eax,edx
0x1001b829: shr eax,1Fh
0x1001b82c: add eax,edx
0x1001b82e: mov edx,[ebp-28h]
0x1001b831: mov [edx+esi+1],al
0x1001b835: movzx edi,byte ptr [ecx+ebx]
0x1001b839: imul edi,[ebp-20h]
0x1001b83d: mov eax,80808081h
0x1001b842: imul edi
```

```
0x1001b844: add edx,edi
0x1001b846: sar edx,7
0x1001b849: mov eax,edx
0x1001b84b: shr eax,1Fh
0x1001b84e: add eax,edx
0x1001b850: mov edx,[ebp-28h]
0x1001b853: mov [edx+esi+2],al
0x1001b857: mov eax,[ebp-18h]
0x1001b85a: add esi,3
0x1001b85d: add ecx,4
0x1001b860: cmp esi,[eax+30h]
0x1001b863: mov edi,eax
0x1001b865: jl near ptr 1001B7E0h
0x1001b86b: mov ebx,[ebp-10h]
0x1001b86e: mov esi,[ebp-1Ch]
0x1001b871: sub dword ptr [ebp-34h],1
0x1001b875: js short 1001B882h
0x1001b877: mov ecx,[ebp-24h]
0x1001b87a: call 1000F000h
0x1001b87f: sub [ebp-28h],eax
0x1001b882: mov edx,[ebp+140h]
0x1001b888: cmp edx,[ebp+128h]
0x1001b88e: jb near ptr 1001B790h
0x1001b894: lea ecx,[ebp+0C8h]
0x1001b89a: push ecx
0x1001b89b: call 1003B610h
0x1001b8a0: add esp,4
0x1001b8a3: cmp dword ptr [ebp+0ECh],3
0x1001b8aa: jne short 1001B8EEh
0x1001b8ac: cmp byte ptr [ebp+112h],0
0x1001b8b3: jne short 1001B8EEh
0x1001b8b5: push 0
0x1001b8b7: mov ecx,edi
0x1001b8b9: call 1000E5E0h
0x1001b8be: xor esi,esi
0x1001b8c0: cmp [edi+10h],esi
0x1001b8c3: mov ebx,eax
0x1001b8c5: jle short 1001B8EEh
0x1001b8c7: cmp dword ptr [edi+148h],0
0x1001b8ce: jne near ptr 1001B571h
0x1001b8d4: mov eax,[edi+0Ch]
0x1001b8d7: lea edx,[eax+eax*2]
0x1001b8da: push edx
0x1001b8db: push ebx
0x1001b8dc: mov ecx,edi
0x1001b8de: call 10013530h
0x1001b8e3: add ebx,[edi+30h]
0x1001b8e6: add esi,1
0x1001b8e9: cmp esi,[edi+10h]
0x1001b8ec: jl short 1001B8C7h
0x1001b8ee: lea ecx,[ebp+0C8h]
0x1001b8f4: push ecx
0x1001b8f5: call 1003C530h
0x1001b8fa: mov edx,[ebp-40h]
0x1001b8fd: push edx
0x1001b8fe: mov dword ptr [ebp-4],0FFFFFFFFh
0x1001b905: call 10021FCCh
0x1001b90a: add esp,8
0x1001b90d: mov al,1
0x1001b90f: mov ecx,[ebp-0Ch]
0x1001b912: mov fs:[0],ecx
0x1001b919: pop ecx
0x1001b91a: pop edi
0x1001b91b: pop esi
0x1001b91c: pop ebx
0x1001b91d: mov ecx,[ebp+288h]
0x1001b923: xor ecx,ebp
```

```
0x1001b925: call 10022326h
0x1001b92a: add ebp,28Ch
0x1001b930: mov esp,ebp
0x1001b932: pop ebp
0x1001b933: ret 4
0x1001b936: mov ecx,[ebx]
0x1001b938: push esi
0x1001b939: push ecx
0x1001b93a: lea ecx,[ebp-38h]
0x1001b93d: call 10018BE0h
0x1001b942: jmp near ptr 1001B871h
```

`sub_1001b350` is called by sub_1000f760. It accesses memory-mapped peripheral registers at 0x80808081. It delegates to sub_1000e5e0, sub_1000ead0, sub_1000ef00 (29 callees).

12.384 sub_1001b970 at 0x1001b970 (658 instructions)

```
0x1001b970: push 0FFFFFFFFh
0x1001b972: push 100FC7A6h
0x1001b977: mov eax,fs:[0]
0x1001b97d: push eax
0x1001b97e: sub esp,934h
0x1001b984: mov eax,ds:[10136220h]
0x1001b989: xor eax,esp
0x1001b98b: mov [esp+930h],eax
0x1001b992: push ebx
0x1001b993: push ebp
0x1001b994: push esi
0x1001b995: push edi
0x1001b996: mov eax,ds:[10136220h]
0x1001b99b: xor eax,esp
0x1001b99d: push eax
0x1001b99e: lea eax,[esp+948h]
0x1001b9a5: mov fs:[0],eax
0x1001b9ab: mov ebx,[esp+958h]
0x1001b9b2: test ebx,ebx
0x1001b9b4: mov esi,ecx
0x1001b9b6: je near ptr 1001C160h
0x1001b9bc: mov eax,[ebx]
0x1001b9be: mov edx,[eax+14h]
0x1001b9c1: mov ecx,ebx
0x1001b9c3: call edx
0x1001b9c5: mov ebp,[esi+15Ch]
0x1001b9cb: push 1
0x1001b9cd: mov [esp+28h],eax
0x1001b9d1: mov eax,[ebx]
0x1001b9d3: mov edx,[eax+8]
0x1001b9d6: push 6
0x1001b9d8: lea ecx,[esp+38h]
0x1001b9dc: push ecx
0x1001b9dd: mov ecx,ebx
0x1001b9df: call edx
0x1001b9e1: mov eax,[esp+32h]
0x1001b9e5: push eax
0x1001b9e6: mov ecx,esi
0x1001b9e8: call 1000F1C0h
0x1001b9ed: mov ecx,[esp+34h]
0x1001b9f1: push ecx
0x1001b9f2: mov ecx,esi
0x1001b9f4: mov [esp+36h],ax
0x1001b9f9: call 1000F1C0h
0x1001b9fe: cmp word ptr [esp+30h],0
0x1001ba04: mov [esp+34h],ax
0x1001ba09: jne near ptr 1001C160h
0x1001ba0f: mov cx,[esp+32h]
0x1001ba14: cmp cx,1
```

```
0x1001ba18: je short 1001BA24h
0x1001ba1a: cmp cx,2
0x1001ba1e: jne near ptr 1001C160h
0x1001ba24: movzx eax,ax
0x1001ba27: mov [esi+160h],eax
0x1001ba2d: shl eax,4
0x1001ba30: push eax
0x1001ba31: call dword ptr ds:[100FE550h]
0x1001ba37: mov [esp+2Ch],eax
0x1001ba3b: mov ax,[esp+38h]
0x1001ba40: add esp,4
0x1001ba43: test ax,ax
0x1001ba46: mov dword ptr [esp+2Ch],0
0x1001ba4e: jbe short 1001BABCh
0x1001ba50: mov edi,[esp+28h]
0x1001ba54: add edi,6
0x1001ba57: mov edx,[ebx]
0x1001ba59: mov edx,[edx+8]
0x1001ba5c: push 1
0x1001ba5e: push 10h
0x1001ba60: lea eax,[edi-6]
0x1001ba63: push eax
0x1001ba64: mov ecx,ebx
0x1001ba66: call edx
0x1001ba68: movzx eax,word ptr [edi-2]
0x1001ba6c: push eax
0x1001ba6d: mov ecx,esi
0x1001ba6f: call 1000F1C0h
0x1001ba74: movzx ecx,word ptr [edi]
0x1001ba77: push ecx
0x1001ba78: mov ecx,esi
0x1001ba7a: mov [edi-2],ax
0x1001ba7e: call 1000F1C0h
0x1001ba83: mov edx,[edi+2]
0x1001ba86: push edx
0x1001ba87: mov ecx,esi
0x1001ba89: mov [edi],ax
0x1001ba8c: call 1000F1E0h
0x1001ba91: mov [edi+2],eax
0x1001ba94: mov eax,[edi+6]
0x1001ba97: push eax
0x1001ba98: mov ecx,esi
0x1001ba9a: call 1000F1E0h
0x1001ba9f: mov ecx,[esp+2Ch]
0x1001baa3: mov [edi+6],eax
0x1001baa6: mov ax,[esp+34h]
0x1001baab: movzx edx,ax
0x1001baae: add ecx,1
0x1001bab1: add edi,10h
0x1001bab4: cmp ecx,edx
0x1001bab6: mov [esp+2Ch],ecx
0x1001baba: jl short 1001BA57h
0x1001babc: test ebp,ebp
0x1001babe: jl near ptr 1001C152h
0x1001bac4: movzx eax,ax
0x1001bac7: cmp ebp,eax
0x1001bac9: jge near ptr 1001C152h
0x1001bacf: shl ebp,4
0x1001bad2: cmp dword ptr [esi+148h],0FFFFFFFFh
0x1001bad9: jne near ptr 1001BB9Bh
0x1001badf: mov edi,[esp+28h]
0x1001bae3: movzx ecx,byte ptr [ebp+edi]
0x1001bae8: add ebp,edi
0x1001baea: test ecx,ecx
0x1001baec: mov [esi+0Ch],ecx
0x1001baef: movzx eax,byte ptr [ebp+1]
0x1001baf3: mov [esi+10h],eax
```

```
0x1001baf6: jne near ptr 1001BB83h
0x1001bafc: test eax,eax
0x1001bafe: jne near ptr 1001BB83h
0x1001bb04: mov edx,[ebx]
0x1001bb06: mov edx,[edx+10h]
0x1001bb09: push eax
0x1001bb0a: mov eax,[ebp+0Ch]
0x1001bb0d: add eax,[esp+28h]
0x1001bb11: mov ecx,ebx
0x1001bb13: push eax
0x1001bb14: call edx
0x1001bb16: push 0
0x1001bb18: lea ecx,[esp+478h]
0x1001bb1f: call 1000E480h
0x1001bb24: push 0FFFFFFFFh
0x1001bb26: lea ecx,[esp+478h]
0x1001bb2d: mov dword ptr [esp+954h],0
0x1001bb38: call 1000F180h
0x1001bb3d: push 4
0x1001bb3f: push ebx
0x1001bb40: lea ecx,[esp+47Ch]
0x1001bb47: call 1000F760h
0x1001bb4c: test al,al
0x1001bb4e: je short 1001BB6Ch
0x1001bb50: push 1
0x1001bb52: lea eax,[esp+478h]
0x1001bb59: push eax
0x1001bb5a: mov ecx,esi
0x1001bb5c: call 1000E6D0h
0x1001bb61: movzx ecx,word ptr [esp+34h]
0x1001bb66: mov [esi+160h],ecx
0x1001bb6c: lea ecx,[esp+474h]
0x1001bb73: mov dword ptr [esp+950h],0FFFFFFFFh
0x1001bb7e: call 1000C970h
0x1001bb83: push edi
0x1001bb84: call dword ptr ds:[100FE554h]
0x1001bb8a: add esp,4
0x1001bb8d: mov dword ptr [esi+40h],5
0x1001bb94: mov al,1
0x1001bb96: jmp near ptr 1001C162h
0x1001bb9b: mov edx,[esp+28h]
0x1001bb9f: mov ecx,[edx+ebp+0Ch]
0x1001bba3: add ecx,[esp+24h]
0x1001bba7: mov eax,[ebx]
0x1001bba9: lea edi,[edx+ebp]
0x1001bbac: mov edx,[eax+10h]
0x1001bbaf: push 0
0x1001bbb1: push ecx
0x1001bbb2: mov ecx,ebx
0x1001bbb4: call edx
0x1001bbb6: cmp byte ptr [edi],0
0x1001bbb9: jne near ptr 1001BC42h
0x1001bbbf: cmp byte ptr [edi+1],0
0x1001bbc3: jne short 1001BC42h
0x1001bbc5: push 0
0x1001bbc7: lea ecx,[esp+478h]
0x1001bbce: call 1000E480h
0x1001bbd3: push 4
0x1001bbd5: push ebx
0x1001bbd6: lea ecx,[esp+47Ch]
0x1001bbdd: mov dword ptr [esp+958h],1
0x1001bbe8: call 1000F760h
0x1001bbed: test al,al
0x1001bbef: je short 1001BC0Dh
0x1001bbf1: push 1
0x1001bbf3: lea eax,[esp+478h]
0x1001bbfa: push eax
```

```
0x1001bbfb: mov ecx,esi
0x1001bbfd: call 1000E6D0h
0x1001bc02: movzx ecx,word ptr [esp+34h]
0x1001bc07: mov [esi+160h],ecx
0x1001bc0d: push 5
0x1001bc0f: mov ecx,esi
0x1001bc11: call 1000F040h
0x1001bc16: lea ecx,[esp+474h]
0x1001bc1d: mov dword ptr [esp+950h],0FFFFFFFFh
0x1001bc28: call 1000C970h
0x1001bc2d: mov eax,[esp+28h]
0x1001bc31: push eax
0x1001bc32: call dword ptr ds:[100FE554h]
0x1001bc38: add esp,4
0x1001bc3b: mov al,1
0x1001bc3d: jmp near ptr 1001C162h
0x1001bc42: mov edx,[ebx]
0x1001bc44: mov edx,[edx+8]
0x1001bc47: push 1
0x1001bc49: push 28h
0x1001bc4b: lea eax,[esp+54h]
0x1001bc4f: push eax
0x1001bc50: mov ecx,ebx
0x1001bc52: call edx
0x1001bc54: lea eax,[esp+4Ch]
0x1001bc58: push eax
0x1001bc59: mov ecx,esi
0x1001bc5b: call 1000F220h
0x1001bc60: movzx ebp,word ptr [esp+5Ah]
0x1001bc65: movzx ecx,byte ptr [edi+1]
0x1001bc69: movzx edx,byte ptr [edi]
0x1001bc6c: push 5
0x1001bc6e: push ebp
0x1001bc6f: push ecx
0x1001bc70: push edx
0x1001bc71: mov ecx,esi
0x1001bc73: mov [esp+3Ch],ebp
0x1001bc77: call 1000EAD0h
0x1001bc7c: mov eax,[esp+6Ch]
0x1001bc80: test eax,eax
0x1001bc82: push 1
0x1001bc84: je short 1001BC9Bh
0x1001bc86: mov edx,[ebx]
0x1001bc88: mov edx,[edx+8]
0x1001bc8b: add eax,eax
0x1001bc8d: add eax,eax
0x1001bc8f: push eax
0x1001bc90: lea ecx,[esp+7Ch]
0x1001bc94: push ecx
0x1001bc95: mov ecx,ebx
0x1001bc97: call edx
0x1001bc99: jmp short 1001BCB1h
0x1001bc9b: mov ecx,[esi+28h]
0x1001bc9e: mov eax,[ebx]
0x1001bca0: mov eax,[eax+8]
0x1001bca3: add ecx,ecx
0x1001bca5: add ecx,ecx
0x1001bca7: push ecx
0x1001bca8: lea edx,[esp+7Ch]
0x1001bcac: push edx
0x1001bcad: mov ecx,ebx
0x1001bcaf: call eax
0x1001bcb1: mov ecx,[esi+28h]
0x1001bcb4: push ecx
0x1001bcb5: lea edx,[esp+78h]
0x1001bcb9: push edx
0x1001bcba: mov ecx,esi
```

```
0x1001bcbc: call 10013690h
0x1001bcc1: cmp ebp,18h
0x1001bcc4: jg short 1001BCDEh
0x1001bcc6: mov ecx,[esi+1Ch]
0x1001bcc9: mov eax,[ebx]
0x1001bccb: mov edx,[esi+34h]
0x1001bcce: mov eax,[eax+8]
0x1001bcd1: push 1
0x1001bcd3: push ecx
0x1001bcd4: push edx
0x1001bcd5: mov ecx,ebx
0x1001bcd7: call eax
0x1001bcd9: jmp near ptr 1001BDB1h
0x1001bcde: mov ecx,[esi+0Ch]
0x1001bce1: imul ecx,[esi+10h]
0x1001bce5: add ecx,ecx
0x1001bce7: add ecx,ecx
0x1001bce9: push ecx
0x1001bcea: call dword ptr ds:[100FE550h]
0x1001bcf0: mov ecx,[esi+0Ch]
0x1001bcf3: imul ecx,[esi+10h]
0x1001bcf7: mov edx,[ebx]
0x1001bcf9: mov edx,[edx+8]
0x1001bcfc: add esp,4
0x1001bcff: add ecx,ecx
0x1001bd01: push 1
0x1001bd03: add ecx,ecx
0x1001bd05: push ecx
0x1001bd06: push eax
0x1001bd07: mov ecx,ebx
0x1001bd09: mov [esp+2Ch],eax
0x1001bd0d: mov ebp,eax
0x1001bd0f: call edx
0x1001bd11: mov ecx,esi
0x1001bd13: call 100139B0h
0x1001bd18: test al,al
0x1001bd1a: jne short 1001BD23h
0x1001bd1c: mov ecx,esi
0x1001bd1e: call 100139C0h
0x1001bd23: cmp dword ptr [esi+10h],0
0x1001bd27: mov dword ptr [esp+18h],0
0x1001bd2f: jle short 1001BDA3h
0x1001bd31: mov eax,[esp+18h]
0x1001bd35: push eax
0x1001bd36: mov ecx,esi
0x1001bd38: call 1000E5E0h
0x1001bd3d: cmp dword ptr [esi+0Ch],0
0x1001bd41: mov edi,eax
0x1001bd43: mov dword ptr [esp+1Ch],0
0x1001bd4b: jle short 1001BD93h
0x1001bd4d: lea ecx,[ecx]
0x1001bd50: movzx ecx,byte ptr [ebp]
0x1001bd54: mov [edi],cl
0x1001bd56: mov dl,[ebp+1]
0x1001bd59: mov [edi+1],dl
0x1001bd5c: mov al,[ebp+2]
0x1001bd5f: mov edx,[esp+18h]
0x1001bd63: add edi,1
0x1001bd66: add edi,1
0x1001bd69: mov [edi],al
0x1001bd6b: movzx ecx,byte ptr [ebp+3]
0x1001bd6f: mov eax,[esp+1Ch]
0x1001bd73: push ecx
0x1001bd74: push edx
0x1001bd75: push eax
0x1001bd76: mov ecx,esi
0x1001bd78: add edi,1
```

```
0x1001bd7b: call 10013AF0h
0x1001bd80: mov eax,[esp+1Ch]
0x1001bd84: add eax,1
0x1001bd87: add ebp,4
0x1001bd8a: cmp eax,[esi+0Ch]
0x1001bd8d: mov [esp+1Ch],eax
0x1001bd91: jl short 1001BD50h
0x1001bd93: mov eax,[esp+18h]
0x1001bd97: add eax,1
0x1001bd9a: cmp eax,[esi+10h]
0x1001bd9d: mov [esp+18h],eax
0x1001bda1: jl short 1001BD31h
0x1001bda3: mov ecx,[esp+20h]
0x1001bda7: push ecx
0x1001bda8: call dword ptr ds:[100FE554h]
0x1001bdae: add esp,4
0x1001bdb1: mov eax,[esi+0Ch]
0x1001bdb4: mov edi,[esi+10h]
0x1001bdb7: add eax,1Fh
0x1001bdba: cdq
0x1001bdbb: and edx,1Fh
0x1001bdbe: add eax,edx
0x1001bdc0: sar eax,5
0x1001bdc3: add eax,eax
0x1001bdc5: add eax,eax
0x1001bdc7: imul edi,eax
0x1001bdca: push edi
0x1001bdcb: mov [esp+28h],eax
0x1001bdcf: call dword ptr ds:[100FE550h]
0x1001bdd5: mov edx,[ebx]
0x1001bdd7: add esp,4
0x1001bdda: push 1
0x1001bddc: push edi
0x1001bddd: mov [esp+20h],eax
0x1001bde1: push eax
0x1001bde2: mov eax,[edx+8]
0x1001bde5: mov ecx,ebx
0x1001bde7: call eax
0x1001bde9: test eax,eax
0x1001bdeb: je short 1001BE14h
0x1001bded: xor eax,eax
0x1001bdef: test edi,edi
0x1001bdf1: jle short 1001BE04h
0x1001bdf3: mov ecx,[esp+18h]
0x1001bdf7: cmp byte ptr [eax+ecx],0FFh
0x1001bdfb: jne short 1001BE37h
0x1001bdfd: add eax,1
0x1001be00: cmp eax,edi
0x1001be02: jl short 1001BDF7h
0x1001be04: push 0
0x1001be06: mov ecx,esi
0x1001be08: call 1000EEB0h
0x1001be0d: mov ecx,esi
0x1001be0f: call 100118F0h
0x1001be14: mov edx,[esp+18h]
0x1001be18: push edx
0x1001be19: call dword ptr ds:[100FE554h]
0x1001be1f: mov eax,[esp+2Ch]
0x1001be23: add esp,4
0x1001be26: push eax
0x1001be27: call dword ptr ds:[100FE554h]
0x1001be2d: add esp,4
0x1001be30: mov al,1
0x1001be32: jmp near ptr 1001C162h
0x1001be37: mov ecx,esi
0x1001be39: mov byte ptr [esp+17h],0
0x1001be3e: call 100139B0h
```

```
0x1001be43: test al,al
0x1001be45: jne short 1001BE50h
0x1001be47: mov ecx,esi
0x1001be49: call 100139C0h
0x1001be4e: jmp short 1001BE55h
0x1001be50: mov byte ptr [esp+17h],1
0x1001be55: xor ebx,ebx
0x1001be57: cmp [esi+10h],ebx
0x1001be5a: jle short 1001BEB0h
0x1001be5c: xor ebp,ebp
0x1001be5e: xor edi,edi
0x1001be60: cmp [esi+0Ch],edi
0x1001be63: jle short 1001BEA4h
0x1001be65: mov edx,edi
0x1001be67: and edx,80000007h
0x1001be6d: jns short 1001BE74h
0x1001be6f: dec edx
0x1001be70: or edx,0FFFFFFF8h
0x1001be73: inc edx
0x1001be74: mov eax,edi
0x1001be76: sar eax,3
0x1001be79: mov cl,7
0x1001be7b: sub cl,dl
0x1001be7d: mov edx,[esp+18h]
0x1001be81: add eax,ebp
0x1001be83: mov al,[eax+edx]
0x1001be86: shr al,cl
0x1001be88: test al,1
0x1001be8a: je short 1001BE9Ch
0x1001be8c: push 0
0x1001be8e: push ebx
0x1001be8f: push edi
0x1001be90: mov ecx,esi
0x1001be92: call 10013AF0h
0x1001be97: mov byte ptr [esp+17h],1
0x1001be9c: add edi,1
0x1001be9f: cmp edi,[esi+0Ch]
0x1001bea2: jl short 1001BE65h
0x1001bea4: add ebp,[esp+24h]
0x1001bea8: add ebx,1
0x1001beab: cmp ebx,[esi+10h]
0x1001beae: jl short 1001BE5Eh
0x1001beb0: cmp byte ptr [esp+17h],0
0x1001beb5: jne short 1001BEBEh
0x1001beb7: mov ecx,esi
0x1001beb9: call 10013A10h
0x1001bebe: xor ebp,ebp
0x1001bec0: cmp [esi+10h],ebp
0x1001bec3: mov [esp+1Ch],ebp
0x1001bec7: mov [esp+20h],ebp
0x1001becb: mov [esp+3Ch],ebp
0x1001becf: jle near ptr 1001C041h
0x1001bed5: mov [esp+38h],ebp
0x1001bed9: lea esp,[esp]
0x1001bee0: xor ebx,ebx
0x1001bee2: cmp [esi+0Ch],ebx
0x1001bee5: jle near ptr 1001BFF8h
0x1001beeb: jmp short 1001BEF0h
0x1001bef0: mov edx,ebx
0x1001bef2: and edx,80000007h
0x1001bef8: jns short 1001BEFFh
0x1001befa: dec edx
0x1001befb: or edx,0FFFFFFF8h
0x1001befe: inc edx
0x1001beff: mov eax,ebx
0x1001bf01: sar eax,3
0x1001bf04: add eax,[esp+38h]
```

```
0x1001bf08: mov cl,7
0x1001bf0a: sub cl,dl
0x1001bf0c: mov edx,[esp+18h]
0x1001bf10: mov al,[eax+edx]
0x1001bf13: shr al,cl
0x1001bf15: test al,1
0x1001bf17: je near ptr 1001BFECh
0x1001bf1d: push 0
0x1001bf1f: push ebp
0x1001bf20: push ebx
0x1001bf21: lea ecx,[esp+54h]
0x1001bf25: push ecx
0x1001bf26: mov ecx,esi
0x1001bf28: call 100130E0h
0x1001bf2d: cmp dword ptr [esp+1Ch],0
0x1001bf32: mov edi,[eax]
0x1001bf34: mov [esp+44h],edi
0x1001bf38: jne short 1001BF5Bh
0x1001bf3a: push ebp
0x1001bf3b: push ebx
0x1001bf3c: mov ecx,esi
0x1001bf3e: call 10012F20h
0x1001bf43: movzx edx,al
0x1001bf46: mov [esp+20h],edx
0x1001bf4a: mov dword ptr [esp+1Ch],1
0x1001bf52: mov [esp+40h],edi
0x1001bf56: jmp near ptr 1001BFECh
0x1001bf5b: mov eax,4
0x1001bf60: lea ecx,[esp+40h]
0x1001bf64: lea edx,[esp+44h]
0x1001bf68: mov edi,[edx]
0x1001bf6a: cmp edi,[ecx]
0x1001bf6c: jne short 1001BF80h
0x1001bf6e: sub eax,4
0x1001bf71: add ecx,4
0x1001bf74: add edx,4
0x1001bf77: cmp eax,4
0x1001bf7a: jae short 1001BF68h
0x1001bf7c: test eax,eax
0x1001bf7e: je short 1001BFDDh
0x1001bf80: movzx edi,byte ptr [edx]
0x1001bf83: movzx ebp,byte ptr [ecx]
0x1001bf86: sub edi,ebp
0x1001bf88: jne short 1001BFCFh
0x1001bf8a: sub eax,1
0x1001bf8d: add ecx,1
0x1001bf90: add edx,1
0x1001bf93: test eax,eax
0x1001bf95: je short 1001BFDDh
0x1001bf97: movzx edi,byte ptr [edx]
0x1001bf9a: movzx ebp,byte ptr [ecx]
0x1001bf9d: sub edi,ebp
0x1001bf9f: jne short 1001BFCFh
0x1001bfa1: sub eax,1
0x1001bfa4: add ecx,1
0x1001bfa7: add edx,1
0x1001bfaa: test eax,eax
0x1001bfac: je short 1001BFDDh
0x1001bfae: movzx edi,byte ptr [edx]
0x1001bfb1: movzx ebp,byte ptr [ecx]
0x1001bfb4: sub edi,ebp
0x1001bfb6: jne short 1001BFCFh
0x1001bfb8: sub eax,1
0x1001bfbb: add ecx,1
0x1001bfbe: add edx,1
0x1001bfc1: test eax,eax
0x1001bfc3: je short 1001BFDDh
```

```
0x1001bfc5: movzx edi,byte ptr [edx]
0x1001bfc8: movzx eax,byte ptr [ecx]
0x1001bfcb: sub edi,eax
0x1001bfcd: je short 1001BFDDh
0x1001bfcf: test edi,edi
0x1001bfd1: mov eax,1
0x1001bfd6: jg short 1001BFDFh
0x1001bfd8: or eax,0FFFFFFFFh
0x1001bfdb: jmp short 1001BFDFh
0x1001bfdd: xor eax,eax
0x1001bfdf: test eax,eax
0x1001bfe1: mov ebp,[esp+3Ch]
0x1001bfe5: je short 1001BFECh
0x1001bfe7: add dword ptr [esp+1Ch],1
0x1001bfec: add ebx,1
0x1001bfef: cmp ebx,[esi+0Ch]
0x1001bff2: jl near ptr 1001BEF0h
0x1001bff8: mov ecx,[esp+24h]
0x1001bffc: add [esp+38h],ecx
0x1001c000: add ebp,1
0x1001c003: cmp ebp,[esi+10h]
0x1001c006: mov [esp+3Ch],ebp
0x1001c00a: jl near ptr 1001BEE0h
0x1001c010: cmp dword ptr [esp+1Ch],1
0x1001c015: jne short 1001C041h
0x1001c017: cmp dword ptr [esp+2Ch],8
0x1001c01c: jg near ptr 1001BE14h
0x1001c022: mov edx,[esp+40h]
0x1001c026: push edx
0x1001c027: mov ecx,esi
0x1001c029: call 1000EED0h
0x1001c02e: mov eax,[esp+20h]
0x1001c032: push eax
0x1001c033: mov ecx,esi
0x1001c035: call 1000EEB0h
0x1001c03a: mov ecx,esi
0x1001c03c: call 10013A10h
0x1001c041: cmp dword ptr [esp+2Ch],8
0x1001c046: jg near ptr 1001BE14h
0x1001c04c: push 100h
0x1001c051: lea ecx,[esp+848h]
0x1001c058: push 0
0x1001c05a: push ecx
0x1001c05b: call 100228C0h
0x1001c060: xor ebx,ebx
0x1001c062: add esp,0Ch
0x1001c065: cmp [esi+10h],ebx
0x1001c068: jle short 1001C09Bh
0x1001c06a: lea ebx,[ebx]
0x1001c070: xor edi,edi
0x1001c072: cmp [esi+0Ch],edi
0x1001c075: jle short 1001C093h
0x1001c077: push ebx
0x1001c078: push edi
0x1001c079: mov ecx,esi
0x1001c07b: call 10013030h
0x1001c080: movzx edx,al
0x1001c083: add edi,1
0x1001c086: cmp edi,[esi+0Ch]
0x1001c089: mov byte ptr [esp+edx+844h],1
0x1001c091: jl short 1001C077h
0x1001c093: add ebx,1
0x1001c096: cmp ebx,[esi+10h]
0x1001c099: jl short 1001C070h
0x1001c09b: mov eax,[esi+28h]
0x1001c09e: sub eax,1
0x1001c0a1: js near ptr 1001BE14h
```

```
0x1001c0a7: jmp short 1001C0B0h
0x1001c0b0: cmp byte ptr [esp+eax+844h],0
0x1001c0b8: je short 1001C0C4h
0x1001c0ba: sub eax,1
0x1001c0bd: jns short 1001C0B0h
0x1001c0bf: jmp near ptr 1001BE14h
0x1001c0c4: test eax,eax
0x1001c0c6: mov [esp+20h],eax
0x1001c0ca: jl near ptr 1001BE14h
0x1001c0d0: xor ebx,ebx
0x1001c0d2: cmp [esi+10h],ebx
0x1001c0d5: mov byte ptr [esp+17h],0
0x1001c0da: jle short 1001C146h
0x1001c0dc: xor ebp,ebp
0x1001c0de: xor edi,edi
0x1001c0e0: cmp [esi+0Ch],edi
0x1001c0e3: jle short 1001C127h
0x1001c0e5: mov eax,edi
0x1001c0e7: and eax,80000007h
0x1001c0ec: jns short 1001C0F3h
0x1001c0ee: dec eax
0x1001c0ef: or eax,0FFFFFFF8h
0x1001c0f2: inc eax
0x1001c0f3: mov edx,edi
0x1001c0f5: sar edx,3
0x1001c0f8: mov cl,7
0x1001c0fa: sub cl,al
0x1001c0fc: mov eax,[esp+18h]
0x1001c100: add edx,ebp
0x1001c102: mov dl,[edx+eax]
0x1001c105: shr dl,cl
0x1001c107: test dl,1
0x1001c10a: je short 1001C11Fh
0x1001c10c: mov eax,[esp+20h]
0x1001c110: push eax
0x1001c111: push ebx
0x1001c112: push edi
0x1001c113: mov ecx,esi
0x1001c115: call 10013330h
0x1001c11a: mov byte ptr [esp+17h],1
0x1001c11f: add edi,1
0x1001c122: cmp edi,[esi+0Ch]
0x1001c125: jl short 1001C0E5h
0x1001c127: add ebp,[esp+24h]
0x1001c12b: add ebx,1
0x1001c12e: cmp ebx,[esi+10h]
0x1001c131: jl short 1001C0DEh
0x1001c133: cmp byte ptr [esp+17h],0
0x1001c138: je short 1001C146h
0x1001c13a: mov ecx,[esp+20h]
0x1001c13e: push ecx
0x1001c13f: mov ecx,esi
0x1001c141: call 1000EEB0h
0x1001c146: mov ecx,esi
0x1001c148: call 10013A10h
0x1001c14d: jmp near ptr 1001BE14h
0x1001c152: mov ecx,[esp+28h]
0x1001c156: push ecx
0x1001c157: call dword ptr ds:[100FE554h]
0x1001c15d: add esp,4
0x1001c160: xor al,al
0x1001c162: mov ecx,[esp+948h]
0x1001c169: mov fs:[0],ecx
0x1001c170: pop ecx
0x1001c171: pop edi
0x1001c172: pop esi
0x1001c173: pop ebp
```

```
0x1001c174: pop ebx
0x1001c175: mov ecx,[esp+930h]
0x1001c17c: xor ecx,esp
0x1001c17e: call 10022326h
0x1001c183: add esp,940h
0x1001c189: ret 4
```

`sub_1001b970` is called by sub_1000f760. It accesses memory-mapped peripheral registers at 0x80000007. It delegates to sub_1000c970, sub_1000e480, sub_1000e5e0 (25 callees).

12.385 sub_1001c190 at 0x1001c190 (94 instructions)

```
0x1001c190: push ebx
0x1001c191: push ebp
0x1001c192: push esi
0x1001c193: push edi
0x1001c194: mov edi,[esp+14h]
0x1001c198: xor ebx,ebx
0x1001c19a: cmp edi,ebx
0x1001c19c: mov ebp,ecx
0x1001c19e: je near ptr 1001C2DCh
0x1001c1a4: mov esi,[esp+18h]
0x1001c1a8: cmp esi,ebx
0x1001c1aa: je near ptr 1001C2DCh
0x1001c1b0: mov eax,[edi]
0x1001c1b2: mov edx,[eax+8]
0x1001c1b5: push 1
0x1001c1b7: push 28h
0x1001c1b9: push esi
0x1001c1ba: mov ecx,edi
0x1001c1bc: call edx
0x1001c1be: test eax,eax
0x1001c1c0: je near ptr 1001C2DCh
0x1001c1c6: push esi
0x1001c1c7: mov ecx,ebp
0x1001c1c9: call 1000F220h
0x1001c1ce: mov ecx,[esi]
0x1001c1d0: lea eax,[ecx-0Ch]
0x1001c1d3: cmp eax,70h
0x1001c1d6: ja short 1001C243h
0x1001c1d8: movzx eax,byte ptr [eax+1001C2FCh]
0x1001c1df: jmp dword ptr [eax*4+1001C2E8h]
0x1001c243: cmp ecx,28h
0x1001c246: jbe near ptr 1001C2DCh
0x1001c24c: movzx eax,word ptr [esi+0Eh]
0x1001c250: imul eax,[esi+4]
0x1001c254: add eax,1Fh
0x1001c257: cdq
0x1001c258: and edx,1Fh
0x1001c25b: add eax,edx
0x1001c25d: sar eax,5
0x1001c260: imul eax,[esi+8]
0x1001c264: add eax,eax
0x1001c266: add eax,eax
0x1001c268: cmp [esi+14h],eax
0x1001c26b: jb short 1001C2DCh
0x1001c26d: cmp word ptr [esi+0Ch],1
0x1001c272: jne short 1001C2DCh
0x1001c274: cmp [esi+20h],ebx
0x1001c277: jne short 1001C2DCh
0x1001c279: cmp [esi+10h],ebx
0x1001c27c: jne short 1001C28Dh
0x1001c27e: push 1
0x1001c280: add ecx,0FFFFFFD8h
0x1001c283: push ecx
0x1001c284: mov eax,[edi]
```

```
0x1001c286: mov edx,[eax+10h]
0x1001c289: mov ecx,edi
0x1001c28b: call edx
0x1001c28d: cmp [esi+14h],ebx
0x1001c290: jne short 1001C2ACh
0x1001c292: movzx eax,word ptr [esi+0Eh]
0x1001c296: imul eax,[esi+4]
0x1001c29a: add eax,1Fh
0x1001c29d: shr eax,3
0x1001c2a0: and eax,1FFFFFFCh
0x1001c2a5: imul eax,[esi+8]
0x1001c2a9: mov [esi+14h],eax
0x1001c2ac: cmp [esi+20h],ebx
0x1001c2af: jne short 1001C2D3h
0x1001c2b1: movzx ecx,word ptr [esi+0Eh]
0x1001c2b5: cmp cx,8
0x1001c2b9: ja short 1001C2CEh
0x1001c2bb: mov eax,1
0x1001c2c0: shl eax,cl
0x1001c2c2: pop edi
0x1001c2c3: mov [esi+20h],eax
0x1001c2c6: pop esi
0x1001c2c7: pop ebp
0x1001c2c8: mov al,1
0x1001c2ca: pop ebx
0x1001c2cb: ret 8
0x1001c2ce: xor eax,eax
0x1001c2d0: mov [esi+20h],eax
0x1001c2d3: pop edi
0x1001c2d4: pop esi
0x1001c2d5: pop ebp
0x1001c2d6: mov al,1
0x1001c2d8: pop ebx
0x1001c2d9: ret 8
0x1001c2dc: pop edi
0x1001c2dd: pop esi
0x1001c2de: pop ebp
0x1001c2df: xor al,al
0x1001c2e1: pop ebx
0x1001c2e2: ret 8
```

`sub_1001c190` is called by sub_1001c370. It delegates to sub_1000f220 (1 callees).

12.386 sub_1001c370 at 0x1001c370 (250 instructions)

```
0x1001c370: push ebp
0x1001c371: lea ebp,[esp-74h]
0x1001c375: sub esp,74h
0x1001c378: push 0FFFFFFFFh
0x1001c37a: push 100FC7E0h
0x1001c37f: mov eax,fs:[0]
0x1001c385: push eax
0x1001c386: sub esp,3Ch
0x1001c389: push ebx
0x1001c38a: push esi
0x1001c38b: push edi
0x1001c38c: mov eax,ds:[10136220h]
0x1001c391: xor eax,ebp
0x1001c393: push eax
0x1001c394: lea eax,[ebp-0Ch]
0x1001c397: mov fs:[0],eax
0x1001c39d: mov [ebp-10h],esp
0x1001c3a0: mov ebx,ecx
0x1001c3a2: mov [ebp+60h],ebx
0x1001c3a5: mov edi,[ebp+7Ch]
0x1001c3a8: test edi,edi
```

```
0x1001c3aa: je near ptr 1001CCA8h
0x1001c3b0: mov eax,[edi]
0x1001c3b2: mov edx,[eax+14h]
0x1001c3b5: mov ecx,edi
0x1001c3b7: call edx
0x1001c3b9: push 1
0x1001c3bb: mov esi,eax
0x1001c3bd: mov eax,[edi]
0x1001c3bf: mov edx,[eax+8]
0x1001c3c2: push 0Eh
0x1001c3c4: lea ecx,[ebp+44h]
0x1001c3c7: push ecx
0x1001c3c8: mov ecx,edi
0x1001c3ca: mov [ebp+6Ch],esi
0x1001c3cd: mov dword ptr [ebp-4],0
0x1001c3d4: call edx
0x1001c3d6: test eax,eax
0x1001c3d8: jne short 1001C3EFh
0x1001c3da: push 1013083Ch
0x1001c3df: lea eax,[ebp-1Ch]
0x1001c3e2: push eax
0x1001c3e3: mov dword ptr [ebp-1Ch],100FF0ECh
0x1001c3ea: call 100229A6h
0x1001c3ef: mov ecx,[ebp+46h]
0x1001c3f2: push ecx
0x1001c3f3: mov ecx,ebx
0x1001c3f5: call 1000F1E0h
0x1001c3fa: mov edx,[ebp+4Eh]
0x1001c3fd: push edx
0x1001c3fe: mov ecx,ebx
0x1001c400: mov [ebp+46h],eax
0x1001c403: call 1000F1E0h
0x1001c408: mov [ebp+4Eh],eax
0x1001c40b: mov eax,[ebp+44h]
0x1001c40e: push eax
0x1001c40f: mov ecx,ebx
0x1001c411: call 1000F1C0h
0x1001c416: movzx eax,ax
0x1001c419: cmp ax,4D42h
0x1001c41d: je short 1001C432h
0x1001c41f: mov edx,[edi]
0x1001c421: mov eax,[edx+10h]
0x1001c424: push 0
0x1001c426: push esi
0x1001c427: mov ecx,edi
0x1001c429: mov dword ptr [ebp+4Eh],0
0x1001c430: call eax
0x1001c432: lea ecx,[ebp-44h]
0x1001c435: push ecx
0x1001c436: push edi
0x1001c437: mov ecx,ebx
0x1001c439: call 1001C190h
0x1001c43e: test al,al
0x1001c440: jne short 1001C457h
0x1001c442: push 1013083Ch
0x1001c447: lea edx,[ebp]
0x1001c44a: push edx
0x1001c44b: mov dword ptr [ebp],100FF0D4h
0x1001c452: call 100229A6h
0x1001c457: mov eax,[ebp-34h]
0x1001c45a: cmp dword ptr [ebp-44h],0Ch
0x1001c45e: movzx edx,word ptr [ebp-36h]
0x1001c462: mov [ebp+64h],eax
0x1001c465: mov eax,[ebp-3Ch]
0x1001c468: sete byte ptr [ebp+72h]
0x1001c46c: test eax,eax
0x1001c46e: setl cl
```

```
0x1001c471: test cl,cl
0x1001c473: mov [ebp+54h],edx
0x1001c476: mov [ebp+5Bh],cl
0x1001c479: je short 1001C480h
0x1001c47b: neg eax
0x1001c47d: mov [ebp-3Ch],eax
0x1001c480: cmp dword ptr [ebx+148h],0FFFFFFFFh
0x1001c487: jne short 1001C4AEh
0x1001c489: mov ecx,[ebp-40h]
0x1001c48c: push 1013083Ch
0x1001c491: lea edx,[ebp+1Ch]
0x1001c494: push edx
0x1001c495: mov [ebx+0Ch],ecx
0x1001c498: mov [ebx+10h],eax
0x1001c49b: mov dword ptr [ebx+40h],1
0x1001c4a2: mov dword ptr [ebp+1Ch],100FEAC0h
0x1001c4a9: call 100229A6h
0x1001c4ae: push 1
0x1001c4b0: push edx
0x1001c4b1: push eax
0x1001c4b2: mov eax,[ebp-40h]
0x1001c4b5: push eax
0x1001c4b6: mov ecx,ebx
0x1001c4b8: call 1000EAD0h
0x1001c4bd: test eax,eax
0x1001c4bf: jne short 1001C4D6h
0x1001c4c1: push 1013083Ch
0x1001c4c6: lea ecx,[ebp-18h]
0x1001c4c9: push ecx
0x1001c4ca: mov dword ptr [ebp-18h],100FE92Bh
0x1001c4d1: call 100229A6h
0x1001c4d6: fild dword ptr [ebp-2Ch]
0x1001c4d9: sub esp,8
0x1001c4dc: fmul qword ptr ds:[100FE7D0h]
0x1001c4e2: fdiv qword ptr ds:[100FE7D8h]
0x1001c4e8: fadd qword ptr ds:[100FE7C8h]
0x1001c4ee: fstp qword ptr [esp]
0x1001c4f1: call 1002297Ch
0x1001c4f6: add esp,8
0x1001c4f9: call 100228D0h
0x1001c4fe: push eax
0x1001c4ff: mov ecx,ebx
0x1001c501: call 1000EF10h
0x1001c506: fild dword ptr [ebp-28h]
0x1001c509: sub esp,8
0x1001c50c: fmul qword ptr ds:[100FE7D0h]
0x1001c512: fdiv qword ptr ds:[100FE7D8h]
0x1001c518: fadd qword ptr ds:[100FE7C8h]
0x1001c51e: fstp qword ptr [esp]
0x1001c521: call 1002297Ch
0x1001c526: add esp,8
0x1001c529: call 100228D0h
0x1001c52e: push eax
0x1001c52f: mov ecx,ebx
0x1001c531: call 1000EF70h
0x1001c536: cmp dword ptr [ebx+148h],0
0x1001c53d: je short 1001C554h
0x1001c53f: push 1013083Ch
0x1001c544: lea edx,[ebp-14h]
0x1001c547: push edx
0x1001c548: mov dword ptr [ebp-14h],100FEAA8h
0x1001c54f: call 100229A6h
0x1001c554: mov ecx,ebx
0x1001c556: call 10012EA0h
0x1001c55b: mov esi,eax
0x1001c55d: test esi,esi
0x1001c55f: je near ptr 1001C60Fh
```

```
0x1001c565: cmp byte ptr [ebp+72h],0
0x1001c569: mov eax,[ebp-24h]
0x1001c56c: je short 1001C5D9h
0x1001c56e: test eax,eax
0x1001c570: jne short 1001C582h
0x1001c572: mov ecx,[ebp-36h]
0x1001c575: cmp cx,8
0x1001c579: ja short 1001C582h
0x1001c57b: mov eax,1
0x1001c580: shl eax,cl
0x1001c582: mov edx,[edi]
0x1001c584: mov edx,[edx+8]
0x1001c587: push 1
0x1001c589: lea eax,[eax+eax*2]
0x1001c58c: push eax
0x1001c58d: push esi
0x1001c58e: mov ecx,edi
0x1001c590: call edx
0x1001c592: mov eax,[ebx+28h]
0x1001c595: test eax,eax
0x1001c597: jne short 1001C5AAh
0x1001c599: movzx ecx,word ptr [ebx+16h]
0x1001c59d: cmp cx,8
0x1001c5a1: ja short 1001C5AAh
0x1001c5a3: mov eax,1
0x1001c5a8: shl eax,cl
0x1001c5aa: add eax,0FFFFFFFFh
0x1001c5ad: lea ecx,[ecx]
0x1001c5b0: test eax,eax
0x1001c5b2: jl short 1001C60Fh
0x1001c5b4: lea ecx,[esi+eax*2]
0x1001c5b7: movzx edx,byte ptr [ecx+eax+2]
0x1001c5bc: add ecx,eax
0x1001c5be: mov [esi+eax*4+2],dl
0x1001c5c2: movzx edx,byte ptr [ecx]
0x1001c5c5: mov [esi+eax*4],dl
0x1001c5c8: mov cl,[ecx+1]
0x1001c5cb: mov [esi+eax*4+1],cl
0x1001c5cf: mov byte ptr [esi+eax*4+3],0
0x1001c5d4: sub eax,1
0x1001c5d7: jmp short 1001C5B0h
0x1001c5d9: test eax,eax
0x1001c5db: jne short 1001C5EDh
0x1001c5dd: mov ecx,[ebp-36h]
0x1001c5e0: cmp cx,8
0x1001c5e4: ja short 1001C5EDh
0x1001c5e6: mov eax,1
0x1001c5eb: shl eax,cl
0x1001c5ed: mov edx,[edi]
0x1001c5ef: mov edx,[edx+8]
0x1001c5f2: add eax,eax
0x1001c5f4: push 1
0x1001c5f6: add eax,eax
0x1001c5f8: push eax
0x1001c5f9: push esi
0x1001c5fa: mov ecx,edi
0x1001c5fc: call edx
0x1001c5fe: xor eax,eax
0x1001c600: cmp eax,[ebx+28h]
0x1001c603: jae short 1001C60Fh
0x1001c605: mov byte ptr [esi+eax*4+3],0
0x1001c60a: add eax,1
0x1001c60d: jmp short 1001C600h
0x1001c60f: cmp dword ptr [ebx+148h],0
0x1001c616: je short 1001C62Dh
0x1001c618: push 1013083Ch
0x1001c61d: lea eax,[ebp+10h]
```

```
0x1001c620: push eax
0x1001c621: mov dword ptr [ebp+10h],100FEAA8h
0x1001c628: call 100229A6h
0x1001c62d: mov eax,[ebp+54h]
0x1001c630: add eax,0FFFFFFFFh
0x1001c633: cmp eax,1Fh
0x1001c636: ja near ptr 1001CC4Ch
0x1001c63c: movzx ecx,byte ptr [eax+1001CCD8h]
0x1001c643: jmp dword ptr [ecx*4+1001CCC4h]
0x1001cc4c: cmp byte ptr [ebp+5Bh],0
0x1001cc50: je short 1001CC5Eh
0x1001cc52: mov ecx,[ebp+60h]
0x1001cc55: push 1
0x1001cc57: push 0
0x1001cc59: call 10010810h
0x1001cc5e: mov al,1
0x1001cc60: jmp short 1001CCAAh
0x1001cca8: xor al,al
0x1001ccaa: mov ecx,[ebp-0Ch]
0x1001ccad: mov fs:[0],ecx
0x1001ccb4: pop ecx
0x1001ccb5: pop edi
0x1001ccb6: pop esi
0x1001ccb7: pop ebx
0x1001ccb8: add ebp,74h
0x1001ccbb: mov esp,ebp
0x1001ccbd: pop ebp
0x1001ccbe: ret 4
```

`sub_1001c370` is called by sub_1000f760. It works with the text "Cancelled", "Error reading BMP info", "Not a BMP". It delegates to sub_1000ead0, sub_1000ef10, sub_1000ef70 (11 callees).

12.387 sub_1001ce00 at 0x1001ce00 (212 instructions)

```
0x1001ce00: fldz
0x1001ce02: sub esp,8
0x1001ce05: push ebx
0x1001ce06: mov ebx,[esp+14h]
0x1001ce0a: push esi
0x1001ce0b: mov esi,[esp+14h]
0x1001ce0f: fcom dword ptr [esi]
0x1001ce11: push edi
0x1001ce12: mov edi,ecx
0x1001ce14: fnstsw ax
0x1001ce16: test ah,41h
0x1001ce19: jp short 1001CE44h
0x1001ce1b: fld dword ptr [esi]
0x1001ce1d: fild dword ptr [edi+0Ch]
0x1001ce20: fcompp
0x1001ce22: fnstsw ax
0x1001ce24: test ah,41h
0x1001ce27: jne short 1001CE44h
0x1001ce29: fcom dword ptr [ebx]
0x1001ce2b: fnstsw ax
0x1001ce2d: test ah,41h
0x1001ce30: jp short 1001CE44h
0x1001ce32: fld dword ptr [ebx]
0x1001ce34: fild dword ptr [edi+10h]
0x1001ce37: fcompp
0x1001ce39: fnstsw ax
0x1001ce3b: test ah,41h
0x1001ce3e: je near ptr 1001D036h
0x1001ce44: mov eax,[esp+20h]
0x1001ce48: sub eax,4
0x1001ce4b: je near ptr 1001CFE1h
0x1001ce51: sub eax,1
```

```
0x1001ce54: je near ptr 1001CF37h
0x1001ce5a: sub eax,1
0x1001ce5d: jne near ptr 1001D036h
0x1001ce63: fcom dword ptr [esi]
0x1001ce65: fnstsw ax
0x1001ce67: test ah,41h
0x1001ce6a: jne short 1001CE88h
0x1001ce6c: fstp st(0)
0x1001ce6e: fld dword ptr [esi]
0x1001ce70: fchs
0x1001ce72: fild dword ptr [edi+0Ch]
0x1001ce75: call 1002298Eh
0x1001ce7a: fstp dword ptr [esp+18h]
0x1001ce7e: fld dword ptr [esp+18h]
0x1001ce82: fstp dword ptr [esi]
0x1001ce84: fldz
0x1001ce86: jmp short 1001CEC9h
0x1001ce88: fild dword ptr [edi+0Ch]
0x1001ce8b: fstp dword ptr [esp+18h]
0x1001ce8f: fld dword ptr [esi]
0x1001ce91: fld dword ptr [esp+18h]
0x1001ce95: fcom
0x1001ce97: fnstsw ax
0x1001ce99: fstp st(1)
0x1001ce9b: test ah,41h
0x1001ce9e: jp short 1001CEC7h
0x1001cea0: fstp st(1)
0x1001cea2: fst qword ptr [esp+0Ch]
0x1001cea6: fld dword ptr [esi]
0x1001cea8: fxch
0x1001ceaa: call 1002298Eh
0x1001ceaf: fstp dword ptr [esp+18h]
0x1001ceb3: fld dword ptr [esp+18h]
0x1001ceb7: fadd qword ptr ds:[100FE8F8h]
0x1001cebd: fsubr qword ptr [esp+0Ch]
0x1001cec1: fstp dword ptr [esi]
0x1001cec3: fldz
0x1001cec5: jmp short 1001CEC9h
0x1001cec7: fstp st(0)
0x1001cec9: fcomp dword ptr [ebx]
0x1001cecb: fnstsw ax
0x1001cecd: test ah,41h
0x1001ced0: jne short 1001CEF1h
0x1001ced2: fld dword ptr [ebx]
0x1001ced4: fchs
0x1001ced6: fild dword ptr [edi+10h]
0x1001ced9: call 1002298Eh
0x1001cede: fstp dword ptr [esp+18h]
0x1001cee2: fld dword ptr [esp+18h]
0x1001cee6: pop edi
0x1001cee7: pop esi
0x1001cee8: fstp dword ptr [ebx]
0x1001ceea: pop ebx
0x1001ceeb: add esp,8
0x1001ceee: ret 0Ch
0x1001cef1: fild dword ptr [edi+10h]
0x1001cef4: fstp dword ptr [esp+18h]
0x1001cef8: fld dword ptr [ebx]
0x1001cefa: fld dword ptr [esp+18h]
0x1001cefe: fcom
0x1001cf00: fnstsw ax
0x1001cf02: fstp st(1)
0x1001cf04: test ah,41h
0x1001cf07: jp near ptr 1001D036h
0x1001cf0d: fst qword ptr [esp+0Ch]
0x1001cf11: fld dword ptr [ebx]
0x1001cf13: fxch
```

```
0x1001cf15: call 1002298Eh
0x1001cf1a: fstp dword ptr [esp+18h]
0x1001cf1e: fld dword ptr [esp+18h]
0x1001cf22: pop edi
0x1001cf23: fadd qword ptr ds:[100FE8F8h]
0x1001cf29: pop esi
0x1001cf2a: fsubr qword ptr [esp+4]
0x1001cf2e: fstp dword ptr [ebx]
0x1001cf30: pop ebx
0x1001cf31: add esp,8
0x1001cf34: ret 0Ch
0x1001cf37: fcom dword ptr [esi]
0x1001cf39: fnstsw ax
0x1001cf3b: test ah,5
0x1001cf3e: jp short 1001CF48h
0x1001cf40: fld dword ptr [esi]
0x1001cf42: fstp dword ptr [esp+18h]
0x1001cf46: jmp short 1001CF4Ch
0x1001cf48: fst dword ptr [esp+18h]
0x1001cf4c: fld dword ptr [esp+18h]
0x1001cf50: fst dword ptr [esi]
0x1001cf52: mov eax,[edi+0Ch]
0x1001cf55: sub eax,1
0x1001cf58: mov [esp+18h],eax
0x1001cf5c: fild dword ptr [esp+18h]
0x1001cf60: fstp dword ptr [esp+18h]
0x1001cf64: fld dword ptr [esp+18h]
0x1001cf68: fcom
0x1001cf6a: fnstsw ax
0x1001cf6c: test ah,41h
0x1001cf6f: jne short 1001CF75h
0x1001cf71: fstp st(0)
0x1001cf73: jmp short 1001CF77h
0x1001cf75: fstp st(1)
0x1001cf77: fstp dword ptr [esp+18h]
0x1001cf7b: fld dword ptr [esp+18h]
0x1001cf7f: fstp dword ptr [esi]
0x1001cf81: fcom dword ptr [ebx]
0x1001cf83: fnstsw ax
0x1001cf85: test ah,5
0x1001cf88: jp short 1001CF8Eh
0x1001cf8a: fstp st(0)
0x1001cf8c: fld dword ptr [ebx]
0x1001cf8e: fstp dword ptr [esp+18h]
0x1001cf92: fld dword ptr [esp+18h]
0x1001cf96: fst dword ptr [ebx]
0x1001cf98: mov ecx,[edi+10h]
0x1001cf9b: sub ecx,1
0x1001cf9e: mov [esp+18h],ecx
0x1001cfa2: fild dword ptr [esp+18h]
0x1001cfa6: fstp dword ptr [esp+18h]
0x1001cfaa: fld dword ptr [esp+18h]
0x1001cfae: fcom
0x1001cfb0: fnstsw ax
0x1001cfb2: test ah,41h
0x1001cfb5: jne short 1001CFCCh
0x1001cfb7: fstp st(0)
0x1001cfb9: pop edi
0x1001cfba: fstp dword ptr [esp+14h]
0x1001cfbe: pop esi
0x1001cfbf: fld dword ptr [esp+10h]
0x1001cfc3: fstp dword ptr [ebx]
0x1001cfc5: pop ebx
0x1001cfc6: add esp,8
0x1001cfc9: ret 0Ch
0x1001cfcc: fstp st(1)
0x1001cfce: pop edi
```

```
0x1001cfcf: fstp dword ptr [esp+14h]
0x1001cfd3: pop esi
0x1001cfd4: fld dword ptr [esp+10h]
0x1001cfd8: fstp dword ptr [ebx]
0x1001cfda: pop ebx
0x1001cfdb: add esp,8
0x1001cfde: ret 0Ch
0x1001cfe1: fstp st(0)
0x1001cfe3: fld dword ptr [esi]
0x1001cfe5: fild dword ptr [edi+0Ch]
0x1001cfe8: call 1002298Eh
0x1001cfed: fstp dword ptr [esp+18h]
0x1001cff1: fld dword ptr [esp+18h]
0x1001cff5: fstp dword ptr [esi]
0x1001cff7: fld dword ptr [ebx]
0x1001cff9: fild dword ptr [edi+10h]
0x1001cffc: call 1002298Eh
0x1001d001: fstp dword ptr [esp+18h]
0x1001d005: fld dword ptr [esp+18h]
0x1001d009: fstp dword ptr [ebx]
0x1001d00b: fldz
0x1001d00d: fcom dword ptr [esi]
0x1001d00f: fnstsw ax
0x1001d011: test ah,41h
0x1001d014: jne short 1001D01Dh
0x1001d016: fild dword ptr [edi+0Ch]
0x1001d019: fadd dword ptr [esi]
0x1001d01b: fstp dword ptr [esi]
0x1001d01d: fcomp dword ptr [ebx]
0x1001d01f: fnstsw ax
0x1001d021: test ah,41h
0x1001d024: jne short 1001D038h
0x1001d026: fild dword ptr [edi+10h]
0x1001d029: pop edi
0x1001d02a: pop esi
0x1001d02b: fadd dword ptr [ebx]
0x1001d02d: fstp dword ptr [ebx]
0x1001d02f: pop ebx
0x1001d030: add esp,8
0x1001d033: ret 0Ch
0x1001d036: fstp st(0)
0x1001d038: pop edi
0x1001d039: pop esi
0x1001d03a: pop ebx
0x1001d03b: add esp,8
0x1001d03e: ret 0Ch
```

`sub_1001ce00` is called by sub_1001ddc0. It delegates to j__CIfmod (1 callees).

12.388 sub_1001d050 at 0x1001d050 (53 instructions)

```
0x1001d050: push ecx
0x1001d051: mov eax,[esp+10h]
0x1001d055: push esi
0x1001d056: push edi
0x1001d057: mov esi,ecx
0x1001d059: mov ecx,[esp+14h]
0x1001d05d: push eax
0x1001d05e: push ecx
0x1001d05f: mov ecx,esi
0x1001d061: call 1000E650h
0x1001d066: test al,al
0x1001d068: je short 1001D074h
0x1001d06a: cmp dword ptr [esi+4],0
0x1001d06e: jne near ptr 1001D14Fh
0x1001d074: mov eax,[esp+20h]
```

```
0x1001d078: test eax,eax
0x1001d07a: je short 1001D084h
0x1001d07c: mov edi,[eax]
0x1001d07e: mov [esp+20h],edi
0x1001d082: jmp short 1001D09Bh
0x1001d084: mov al,0FFh
0x1001d086: mov [esp+20h],al
0x1001d08a: mov [esp+21h],al
0x1001d08e: mov [esp+22h],al
0x1001d092: mov byte ptr [esp+23h],0
0x1001d097: mov edi,[esp+20h]
0x1001d09b: cmp dword ptr [esi+4],0
0x1001d09f: je short 1001D0EDh
0x1001d0a1: mov eax,[esp+1Ch]
0x1001d0a5: lea ecx,[eax-2]
0x1001d0a8: cmp ecx,4
0x1001d0ab: ja short 1001D0EDh
0x1001d0ad: jmp dword ptr [ecx*4+1001D170h]
0x1001d0ed: mov eax,[esp+10h]
0x1001d0f1: mov [eax],edi
0x1001d0f3: pop edi
0x1001d0f4: pop esi
0x1001d0f5: pop ecx
0x1001d0f6: ret 14h
0x1001d14f: mov ecx,[esp+18h]
0x1001d153: mov edx,[esp+14h]
0x1001d157: mov edi,[esp+10h]
0x1001d15b: push 1
0x1001d15d: push ecx
0x1001d15e: push edx
0x1001d15f: push edi
0x1001d160: mov ecx,esi
0x1001d162: call 100131F0h
0x1001d167: mov eax,edi
0x1001d169: pop edi
0x1001d16a: pop esi
0x1001d16b: pop ecx
0x1001d16c: ret 14h
```

`sub_1001d050` is called by sub_1001ddc0, sub_1001fd10. It delegates to sub_1000e650, sub_100131f0 (2 callees).

12.389 sub_1001d190 at 0x1001d190 (34 instructions)

```
0x1001d190: mov eax,[esp+4]
0x1001d194: movzx ecx,byte ptr [eax+2]
0x1001d198: mov [esp+4],ecx
0x1001d19c: mov ecx,[esp+0Ch]
0x1001d1a0: fild dword ptr [esp+4]
0x1001d1a4: fld dword ptr [esp+8]
0x1001d1a8: fld st(0)
0x1001d1aa: fmulp st(2),st
0x1001d1ac: fld dword ptr [ecx]
0x1001d1ae: faddp st(2),st
0x1001d1b0: fxch
0x1001d1b2: fstp dword ptr [ecx]
0x1001d1b4: movzx edx,byte ptr [eax+1]
0x1001d1b8: mov ecx,[esp+10h]
0x1001d1bc: mov [esp+4],edx
0x1001d1c0: fild dword ptr [esp+4]
0x1001d1c4: fmul st,st(1)
0x1001d1c6: fadd dword ptr [ecx]
0x1001d1c8: fstp dword ptr [ecx]
0x1001d1ca: movzx ecx,byte ptr [eax]
0x1001d1cd: mov [esp+4],ecx
0x1001d1d1: mov ecx,[esp+14h]
0x1001d1d5: fild dword ptr [esp+4]
```

```
0x1001d1d9: fmul st,st(1)
0x1001d1db: fadd dword ptr [ecx]
0x1001d1dd: fstp dword ptr [ecx]
0x1001d1df: movzx edx,byte ptr [eax+3]
0x1001d1e3: mov eax,[esp+18h]
0x1001d1e7: mov [esp+4],edx
0x1001d1eb: fild dword ptr [esp+4]
0x1001d1ef: fmulp
0x1001d1f1: fadd dword ptr [eax]
0x1001d1f3: fstp dword ptr [eax]
0x1001d1f5: ret 18h
```

`sub_1001d190` is called by sub_1001fd10. It calls nothing else.

12.390 sub_1001ddc0 at 0x1001ddc0 (465 instructions)

```
0x1001ddc0: sub esp,88h
0x1001ddc6: fld dword ptr [esp+90h]
0x1001ddcd: push ebx
0x1001ddce: push ebp
0x1001ddcf: fld st(0)
0x1001ddd1: push esi
0x1001ddd2: push edi
0x1001ddd3: mov ebp,ecx
0x1001ddd5: call 100228D0h
0x1001ddda: fldz
0x1001dddc: fcom
0x1001ddde: mov edi,eax
0x1001dde0: mov [esp+34h],eax
0x1001dde4: fnstsw ax
0x1001dde6: test ah,41h
0x1001dde9: jne short 1001DDF2h
0x1001ddeb: sub edi,1
0x1001ddee: mov [esp+34h],edi
0x1001ddf2: fld dword ptr [esp+0A4h]
0x1001ddf9: fld st(0)
0x1001ddfb: call 100228D0h
0x1001de00: fcom
0x1001de02: mov esi,eax
0x1001de04: mov [esp+28h],eax
0x1001de08: fnstsw ax
0x1001de0a: test ah,5
0x1001de0d: jp short 1001DE16h
0x1001de0f: sub esi,1
0x1001de12: mov [esp+28h],esi
0x1001de16: mov ebx,[esp+0A8h]
0x1001de1d: lea eax,[ebx-1]
0x1001de20: cmp eax,11h
0x1001de23: ja near ptr 1001F706h
0x1001de29: movzx eax,byte ptr [eax+1001FCB4h]
0x1001de30: jmp dword ptr [eax*4+1001FCA4h]
0x1001f706: cmp edi,0FFFFFFFFh
0x1001f709: fstp st(1)
0x1001f70b: mov ebx,[esp+0ACh]
0x1001f712: jl short 1001F727h
0x1001f714: cmp edi,[ebp+0Ch]
0x1001f717: jge short 1001F727h
0x1001f719: cmp esi,0FFFFFFFFh
0x1001f71c: jl short 1001F727h
0x1001f71e: cmp esi,[ebp+10h]
0x1001f721: jl near ptr 1001F7C9h
0x1001f727: test ebx,ebx
0x1001f729: fstp st(1)
0x1001f72b: fstp st(0)
0x1001f72d: jle short 1001F765h
0x1001f72f: cmp ebx,3
```

```
0x1001f732: jg short 1001F765h
0x1001f734: mov edx,[esp+0B0h]
0x1001f73b: push edx
0x1001f73c: push ebx
0x1001f73d: mov esi,[esp+0A4h]
0x1001f744: push 0FFFFFC19h
0x1001f749: push 0FFFFFC19h
0x1001f74e: push esi
0x1001f74f: mov ecx,ebp
0x1001f751: call 1001D050h
0x1001f756: pop edi
0x1001f757: mov eax,esi
0x1001f759: pop esi
0x1001f75a: pop ebp
0x1001f75b: pop ebx
0x1001f75c: add esp,88h
0x1001f762: ret 18h
0x1001f765: push ebx
0x1001f766: lea eax,[esp+0A8h]
0x1001f76d: push eax
0x1001f76e: lea ecx,[esp+0A8h]
0x1001f775: push ecx
0x1001f776: mov ecx,ebp
0x1001f778: call 1001CE00h
0x1001f77d: fld dword ptr [esp+0A0h]
0x1001f784: fld st(0)
0x1001f786: call 100228D0h
0x1001f78b: fldz
0x1001f78d: fcom
0x1001f78f: mov [esp+34h],eax
0x1001f793: fnstsw ax
0x1001f795: test ah,41h
0x1001f798: jne short 1001F79Fh
0x1001f79a: sub dword ptr [esp+34h],1
0x1001f79f: fld dword ptr [esp+0A4h]
0x1001f7a6: fld st(0)
0x1001f7a8: call 100228D0h
0x1001f7ad: fcom
0x1001f7af: mov edi,[esp+34h]
0x1001f7b3: mov esi,eax
0x1001f7b5: mov [esp+28h],esi
0x1001f7b9: fnstsw ax
0x1001f7bb: fstp st(1)
0x1001f7bd: test ah,5
0x1001f7c0: jp short 1001F7C9h
0x1001f7c2: sub esi,1
0x1001f7c5: mov [esp+28h],esi
0x1001f7c9: lea eax,[edi+1]
0x1001f7cc: cmp eax,[ebp+0Ch]
0x1001f7cf: mov [esp+30h],eax
0x1001f7d3: jge near ptr 1001FA0Dh
0x1001f7d9: test edi,edi
0x1001f7db: jl near ptr 1001FA0Dh
0x1001f7e1: lea edx,[esi+1]
0x1001f7e4: cmp edx,[ebp+10h]
0x1001f7e7: jge near ptr 1001FA0Dh
0x1001f7ed: test esi,esi
0x1001f7ef: jl near ptr 1001FA0Dh
0x1001f7f5: cmp dword ptr [ebp+28h],0
0x1001f7f9: jne near ptr 1001FA0Dh
0x1001f7ff: fild dword ptr [esp+34h]
0x1001f803: fnstcw [esp+18h]
0x1001f807: movzx eax,word ptr [esp+18h]
0x1001f80c: fsubp st(2),st
0x1001f80e: fld qword ptr ds:[100FE920h]
0x1001f814: or eax,0C00h
0x1001f819: mov [esp+10h],eax
```

```
0x1001f81d: fmul st(2),st
0x1001f81f: fldcw [esp+10h]
0x1001f823: fxch st(2)
0x1001f825: fistp dword ptr [esp+10h]
0x1001f829: mov ax,[esp+10h]
0x1001f82e: movzx ecx,ax
0x1001f831: movzx edi,cx
0x1001f834: fldcw [esp+18h]
0x1001f838: fisub dword ptr [esp+28h]
0x1001f83c: fnstcw [esp+18h]
0x1001f840: movzx eax,word ptr [esp+18h]
0x1001f845: or eax,0C00h
0x1001f84a: mov [esp+10h],eax
0x1001f84e: fmulp
0x1001f850: fldcw [esp+10h]
0x1001f854: fistp dword ptr [esp+10h]
0x1001f858: movzx edx,word ptr [esp+10h]
0x1001f85d: movzx eax,dx
0x1001f860: fldcw [esp+18h]
0x1001f864: movzx edx,ax
0x1001f867: imul edx,edi
0x1001f86a: sar edx,8
0x1001f86d: movzx edi,dx
0x1001f870: sub eax,edi
0x1001f872: movzx ebx,ax
0x1001f875: mov edx,ecx
0x1001f877: sub edx,edi
0x1001f879: movzx edx,dx
0x1001f87c: mov [esp+2Ch],edx
0x1001f880: mov edx,[esp+34h]
0x1001f884: mov eax,100h
0x1001f889: sub eax,ebx
0x1001f88b: sub eax,ecx
0x1001f88d: movzx ecx,ax
0x1001f890: mov eax,[ebp+30h]
0x1001f893: imul eax,esi
0x1001f896: lea edx,[edx+edx*2]
0x1001f899: add eax,edx
0x1001f89b: add eax,[ebp+34h]
0x1001f89e: mov [esp+10h],ecx
0x1001f8a2: movzx edx,byte ptr [eax]
0x1001f8a5: imul dx,cx
0x1001f8a9: movzx edx,dx
0x1001f8ac: mov [esp+28h],edx
0x1001f8b0: movzx edx,byte ptr [eax+1]
0x1001f8b4: imul dx,cx
0x1001f8b8: add eax,1
0x1001f8bb: movzx edx,dx
0x1001f8be: mov [esp+24h],edx
0x1001f8c2: movzx edx,byte ptr [eax+1]
0x1001f8c6: add eax,1
0x1001f8c9: imul dx,cx
0x1001f8cd: movzx ecx,dx
0x1001f8d0: movzx edx,byte ptr [eax+1]
0x1001f8d4: imul dx,[esp+2Ch]
0x1001f8da: add [esp+28h],dx
0x1001f8df: add eax,1
0x1001f8e2: mov [esp+30h],ecx
0x1001f8e6: movzx ecx,byte ptr [eax+1]
0x1001f8ea: imul cx,[esp+2Ch]
0x1001f8f0: add [esp+24h],cx
0x1001f8f5: mov ecx,[ebp+30h]
0x1001f8f8: add eax,1
0x1001f8fb: movzx edx,byte ptr [eax+1]
0x1001f8ff: imul dx,[esp+2Ch]
0x1001f905: add [esp+30h],dx
0x1001f90a: add eax,1
```

```
0x1001f90d: lea eax,[eax+ecx-5]
0x1001f911: movzx ecx,byte ptr [eax]
0x1001f914: movzx edx,byte ptr [eax+1]
0x1001f918: imul cx,bx
0x1001f91c: add [esp+28h],cx
0x1001f921: add eax,1
0x1001f924: imul dx,bx
0x1001f928: movzx ecx,byte ptr [eax+1]
0x1001f92c: add [esp+24h],dx
0x1001f931: imul cx,bx
0x1001f935: add [esp+30h],cx
0x1001f93a: add eax,1
0x1001f93d: movzx edx,byte ptr [eax+1]
0x1001f941: add eax,1
0x1001f944: imul dx,di
0x1001f948: add [esp+28h],dx
0x1001f94d: movzx ecx,byte ptr [eax+2]
0x1001f951: movzx edx,byte ptr [eax+1]
0x1001f955: imul cx,di
0x1001f959: add cx,[esp+30h]
0x1001f95e: imul dx,di
0x1001f962: add dx,[esp+24h]
0x1001f967: add eax,1
0x1001f96a: cmp dword ptr [ebp+3C4h],0
0x1001f971: mov al,[esp+29h]
0x1001f975: mov [esp+1Eh],ch
0x1001f979: mov [esp+1Dh],dh
0x1001f97d: mov [esp+1Ch],al
0x1001f981: je short 1001F9EEh
0x1001f983: mov ecx,[esp+34h]
0x1001f987: push esi
0x1001f988: push ecx
0x1001f989: mov ecx,ebp
0x1001f98b: call 10013B70h
0x1001f990: movzx edx,byte ptr [eax]
0x1001f993: imul dx,[esp+10h]
0x1001f999: movzx ecx,dx
0x1001f99c: movzx edx,byte ptr [eax+1]
0x1001f9a0: imul dx,[esp+2Ch]
0x1001f9a6: add eax,1
0x1001f9a9: mov si,cx
0x1001f9ac: mov [esp+10h],ecx
0x1001f9b0: mov ecx,[ebp+0Ch]
0x1001f9b3: lea eax,[eax+ecx-1]
0x1001f9b7: add si,dx
0x1001f9ba: movzx edx,byte ptr [eax+1]
0x1001f9be: movzx ax,byte ptr [eax]
0x1001f9c2: imul dx,di
0x1001f9c6: imul ax,bx
0x1001f9ca: add dx,ax
0x1001f9cd: mov eax,[esp+9Ch]
0x1001f9d4: pop edi
0x1001f9d5: add dx,si
0x1001f9d8: pop esi
0x1001f9d9: mov [esp+17h],dh
0x1001f9dd: mov ecx,[esp+14h]
0x1001f9e1: pop ebp
0x1001f9e2: mov [eax],ecx
0x1001f9e4: pop ebx
0x1001f9e5: add esp,88h
0x1001f9eb: ret 18h
0x1001f9ee: mov eax,[esp+9Ch]
0x1001f9f5: pop edi
0x1001f9f6: pop esi
0x1001f9f7: mov byte ptr [esp+17h],0
0x1001f9fc: mov ecx,[esp+14h]
0x1001fa00: pop ebp
```

```
0x1001fa01: mov [eax],ecx
0x1001fa03: pop ebx
0x1001fa04: add esp,88h
0x1001fa0a: ret 18h
0x1001fa0d: fild dword ptr [esp+34h]
0x1001fa11: mov edi,[esp+0B0h]
0x1001fa18: mov edx,[esp+34h]
0x1001fa1c: push edi
0x1001fa1d: fsubp st(2),st
0x1001fa1f: push ebx
0x1001fa20: fxch
0x1001fa22: push esi
0x1001fa23: push edx
0x1001fa24: fstp dword ptr [esp+48h]
0x1001fa28: lea eax,[esp+0A4h]
0x1001fa2f: push eax
0x1001fa30: fisub dword ptr [esp+3Ch]
0x1001fa34: mov ecx,ebp
0x1001fa36: fstp dword ptr [esp+24h]
0x1001fa3a: fld dword ptr [esp+24h]
0x1001fa3e: fld st(0)
0x1001fa40: fld dword ptr [esp+4Ch]
0x1001fa44: fld st(0)
0x1001fa46: fmulp st(2),st
0x1001fa48: fxch
0x1001fa4a: fstp dword ptr [esp+38h]
0x1001fa4e: fld st(0)
0x1001fa50: fld dword ptr [esp+38h]
0x1001fa54: fld st(0)
0x1001fa56: fsubp st(2),st
0x1001fa58: fxch
0x1001fa5a: fstp dword ptr [esp+60h]
0x1001fa5e: fsubp st(2),st
0x1001fa60: fxch
0x1001fa62: fstp dword ptr [esp+30h]
0x1001fa66: fld1
0x1001fa68: fsubrp
0x1001fa6a: fsub dword ptr [esp+30h]
0x1001fa6e: fstp dword ptr [esp+4Ch]
0x1001fa72: call 1001D050h
0x1001fa77: mov ecx,[eax]
0x1001fa79: mov edx,[esp+30h]
0x1001fa7d: push edi
0x1001fa7e: push ebx
0x1001fa7f: push esi
0x1001fa80: push edx
0x1001fa81: lea eax,[esp+98h]
0x1001fa88: mov [esp+24h],ecx
0x1001fa8c: push eax
0x1001fa8d: mov ecx,ebp
0x1001fa8f: call 1001D050h
0x1001fa94: mov ecx,[eax]
0x1001fa96: mov edx,[esp+34h]
0x1001fa9a: push edi
0x1001fa9b: push ebx
0x1001fa9c: add esi,1
0x1001fa9f: push esi
0x1001faa0: push edx
0x1001faa1: lea eax,[esp+90h]
0x1001faa8: mov [esp+30h],ecx
0x1001faac: push eax
0x1001faad: mov ecx,ebp
0x1001faaf: call 1001D050h
0x1001fab4: mov ecx,[eax]
0x1001fab6: mov edx,[esp+30h]
0x1001faba: push edi
0x1001fabb: push ebx
```

```
0x1001fabc: push esi
0x1001fabd: push edx
0x1001fabe: lea eax,[esp+94h]
0x1001fac5: mov [esp+38h],ecx
0x1001fac9: push eax
0x1001faca: mov ecx,ebp
0x1001facc: call 1001D050h
0x1001fad1: movzx edx,byte ptr [esp+16h]
0x1001fad6: mov ecx,[eax]
0x1001fad8: movzx eax,byte ptr [esp+22h]
0x1001fadd: mov [esp+10h],edx
0x1001fae1: fild dword ptr [esp+10h]
0x1001fae5: mov [esp+10h],eax
0x1001fae9: fld dword ptr [esp+38h]
0x1001faed: fld st(0)
0x1001faef: fmulp st(2),st
0x1001faf1: fild dword ptr [esp+10h]
0x1001faf5: fld dword ptr [esp+4Ch]
0x1001faf9: fld st(0)
0x1001fafb: movzx edx,byte ptr [esp+2Ah]
0x1001fb00: fmulp st(2),st
0x1001fb02: mov [esp+10h],edx
0x1001fb06: fxch st(3)
0x1001fb08: mov eax,ecx
0x1001fb0a: shr eax,10h
0x1001fb0d: fnstcw [esp+18h]
0x1001fb11: faddp
0x1001fb13: movzx edx,al
0x1001fb16: fild dword ptr [esp+10h]
0x1001fb1a: fld dword ptr [esp+1Ch]
0x1001fb1e: fld st(0)
0x1001fb20: movzx eax,word ptr [esp+18h]
0x1001fb25: fmulp st(2),st
0x1001fb27: fxch st(2)
0x1001fb29: mov [esp+10h],edx
0x1001fb2d: mov edx,[esp+14h]
0x1001fb31: faddp
0x1001fb33: or eax,0C00h
0x1001fb38: fild dword ptr [esp+10h]
0x1001fb3c: mov [esp+10h],eax
0x1001fb40: fld dword ptr [esp+24h]
0x1001fb44: mov ebx,[esp+20h]
0x1001fb48: fld st(0)
0x1001fb4a: fmulp st(2),st
0x1001fb4c: fxch st(2)
0x1001fb4e: faddp
0x1001fb50: fldcw [esp+10h]
0x1001fb54: fistp dword ptr [esp+10h]
0x1001fb58: movzx eax,byte ptr [esp+10h]
0x1001fb5d: mov [esp+1Eh],al
0x1001fb61: movzx eax,dh
0x1001fb64: fldcw [esp+18h]
0x1001fb68: mov [esp+10h],eax
0x1001fb6c: movzx eax,bh
0x1001fb6f: movzx edx,dl
0x1001fb72: fild dword ptr [esp+10h]
0x1001fb76: fnstcw [esp+18h]
0x1001fb7a: mov [esp+10h],eax
0x1001fb7e: movzx eax,byte ptr [esp+29h]
0x1001fb83: fmul st,st(3)
0x1001fb85: fild dword ptr [esp+10h]
0x1001fb89: mov [esp+10h],eax
0x1001fb8d: mov eax,ecx
0x1001fb8f: shr eax,8
0x1001fb92: fmul st,st(5)
0x1001fb94: movzx eax,al
0x1001fb97: faddp
```

```
0x1001fb99: fild dword ptr [esp+10h]
0x1001fb9d: mov [esp+10h],eax
0x1001fba1: movzx eax,word ptr [esp+18h]
0x1001fba6: fmul st,st(3)
0x1001fba8: or eax,0C00h
0x1001fbad: faddp
0x1001fbaf: fild dword ptr [esp+10h]
0x1001fbb3: mov [esp+10h],eax
0x1001fbb7: fmul st,st(2)
0x1001fbb9: faddp
0x1001fbbb: fldcw [esp+10h]
0x1001fbbf: fistp dword ptr [esp+10h]
0x1001fbc3: movzx eax,byte ptr [esp+10h]
0x1001fbc8: mov [esp+10h],edx
0x1001fbcc: movzx edx,byte ptr [esp+28h]
0x1001fbd1: fldcw [esp+18h]
0x1001fbd5: mov [esp+1Dh],al
0x1001fbd9: movzx eax,bl
0x1001fbdc: fild dword ptr [esp+10h]
0x1001fbe0: mov [esp+10h],eax
0x1001fbe4: movzx eax,cl
0x1001fbe7: fmul st,st(3)
0x1001fbe9: fild dword ptr [esp+10h]
0x1001fbed: mov [esp+10h],edx
0x1001fbf1: fmul st,st(5)
0x1001fbf3: faddp
0x1001fbf5: fild dword ptr [esp+10h]
0x1001fbf9: mov [esp+10h],eax
0x1001fbfd: fmul st,st(3)
0x1001fbff: faddp
0x1001fc01: fild dword ptr [esp+10h]
0x1001fc05: fmul st,st(2)
0x1001fc07: faddp
0x1001fc09: shr ecx,18h
0x1001fc0c: fnstcw [esp+18h]
0x1001fc10: pop edi
0x1001fc11: movzx eax,word ptr [esp+14h]
0x1001fc16: or eax,0C00h
0x1001fc1b: mov [esp+0Ch],eax
0x1001fc1f: movzx eax,byte ptr [esp+13h]
0x1001fc24: fldcw [esp+0Ch]
0x1001fc28: pop esi
0x1001fc29: pop ebp
0x1001fc2a: pop ebx
0x1001fc2b: fistp dword ptr [esp]
0x1001fc2e: mov dl,[esp]
0x1001fc31: mov [esp],eax
0x1001fc34: movzx eax,byte ptr [esp+1Bh]
0x1001fc39: fldcw [esp+8]
0x1001fc3d: mov [esp+0Ch],dl
0x1001fc41: movzx edx,byte ptr [esp+13h]
0x1001fc46: fild dword ptr [esp]
0x1001fc49: fnstcw [esp+8]
0x1001fc4d: mov [esp],edx
0x1001fc50: fmulp st(3),st
0x1001fc52: fild dword ptr [esp]
0x1001fc55: mov [esp],eax
0x1001fc58: movzx eax,word ptr [esp+8]
0x1001fc5d: or eax,0C00h
0x1001fc62: fmulp st(4),st
0x1001fc64: fxch st(2)
0x1001fc66: faddp st(3),st
0x1001fc68: fild dword ptr [esp]
0x1001fc6b: mov [esp],ecx
0x1001fc6e: fmulp
0x1001fc70: faddp st(2),st
0x1001fc72: fild dword ptr [esp]
```

```
0x1001fc75: mov [esp],eax
0x1001fc78: mov eax,[esp+8Ch]
0x1001fc7f: fmulp
0x1001fc81: faddp
0x1001fc83: fldcw [esp]
0x1001fc86: fistp dword ptr [esp]
0x1001fc89: mov cl,[esp]
0x1001fc8c: mov [esp+0Fh],cl
0x1001fc90: mov edx,[esp+0Ch]
0x1001fc94: fldcw [esp+8]
0x1001fc98: mov [eax],edx
0x1001fc9a: add esp,88h
0x1001fca0: ret 18h
```

`sub_1001ddc0` is called by sub_10011fb0, sub_1001fd10. It delegates to sub_10013b70, sub_1001ce00, sub_1001d050 (4 callees).

12.391 sub_1001fd10 at 0x1001fd10 (750 instructions)

```
0x1001fd10: fld1
0x1001fd12: sub esp,8Ch
0x1001fd18: fld dword ptr [esp+0A0h]
0x1001fd1f: push esi
0x1001fd20: fcom
0x1001fd22: push edi
0x1001fd23: mov esi,ecx
0x1001fd25: fnstsw ax
0x1001fd27: fld dword ptr [esp+0A4h]
0x1001fd2e: test ah,41h
0x1001fd31: jp short 1001FD8Eh
0x1001fd33: fcom st(2)
0x1001fd35: fnstsw ax
0x1001fd37: fstp st(2)
0x1001fd39: test ah,41h
0x1001fd3c: jp short 1001FD90h
0x1001fd3e: mov eax,[esp+0B4h]
0x1001fd45: fstp st(0)
0x1001fd47: mov ecx,[esp+0B0h]
0x1001fd4e: fstp st(0)
0x1001fd50: mov edx,[esp+0ACh]
0x1001fd57: fld dword ptr [esp+0A0h]
0x1001fd5e: mov edi,[esp+98h]
0x1001fd65: push eax
0x1001fd66: push ecx
0x1001fd67: push edx
0x1001fd68: sub esp,8
0x1001fd6b: fstp dword ptr [esp+4]
0x1001fd6f: mov ecx,esi
0x1001fd71: fld dword ptr [esp+0B0h]
0x1001fd78: fstp dword ptr [esp]
0x1001fd7b: push edi
0x1001fd7c: call 1001DDC0h
0x1001fd81: mov eax,edi
0x1001fd83: pop edi
0x1001fd84: pop esi
0x1001fd85: add esp,8Ch
0x1001fd8b: ret 20h
0x1001fd8e: fstp st(2)
0x1001fd90: fld qword ptr ds:[100FE7C8h]
0x1001fd96: push ebx
0x1001fd97: fmul st(1),st
0x1001fd99: push ebp
0x1001fd9a: fmulp st(2),st
0x1001fd9c: sub esp,10h
0x1001fd9f: fld dword ptr [esp+0B8h]
0x1001fda6: lea ecx,[esp+6Ch]
```

```
0x1001fdaa: fld st(0)
0x1001fdac: fadd st,st(2)
0x1001fdae: fstp dword ptr [esp+20h]
0x1001fdb2: fld dword ptr [esp+20h]
0x1001fdb6: fstp dword ptr [esp+0Ch]
0x1001fdba: fld dword ptr [esp+0B4h]
0x1001fdc1: fld st(0)
0x1001fdc3: fadd st,st(4)
0x1001fdc5: fstp dword ptr [esp+20h]
0x1001fdc9: fld dword ptr [esp+20h]
0x1001fdcd: fstp dword ptr [esp+8]
0x1001fdd1: fxch
0x1001fdd3: fsubrp st(2),st
0x1001fdd5: fxch
0x1001fdd7: fstp dword ptr [esp+20h]
0x1001fddb: fld dword ptr [esp+20h]
0x1001fddf: fstp dword ptr [esp+4]
0x1001fde3: fsubrp
0x1001fde5: fstp dword ptr [esp+20h]
0x1001fde9: fld dword ptr [esp+20h]
0x1001fded: fstp dword ptr [esp]
0x1001fdf0: call 10020710h
0x1001fdf5: fld dword ptr [esp+5Ch]
0x1001fdf9: fld qword ptr ds:[100FE7C8h]
0x1001fdff: fadd st(1),st
0x1001fe01: fxch
0x1001fe03: call 100228D0h
0x1001fe08: fld dword ptr [esp+60h]
0x1001fe0c: fadd st,st(1)
0x1001fe0e: mov [esp+3Ch],eax
0x1001fe12: call 100228D0h
0x1001fe17: fld dword ptr [esp+64h]
0x1001fe1b: mov ebx,eax
0x1001fe1d: fadd st,st(1)
0x1001fe1f: mov [esp+2Ch],ebx
0x1001fe23: call 100228D0h
0x1001fe28: fadd dword ptr [esp+68h]
0x1001fe2c: mov edi,eax
0x1001fe2e: mov [esp+40h],edi
0x1001fe32: call 100228D0h
0x1001fe37: fldz
0x1001fe39: fst dword ptr [esp+28h]
0x1001fe3d: mov ebp,eax
0x1001fe3f: fst dword ptr [esp+24h]
0x1001fe43: mov [esp+34h],ebp
0x1001fe47: fst dword ptr [esp+20h]
0x1001fe4b: fstp dword ptr [esp+14h]
0x1001fe4f: fld1
0x1001fe51: fcom dword ptr [esp+0B0h]
0x1001fe58: fnstsw ax
0x1001fe5a: test ah,5
0x1001fe5d: jp near ptr 10020331h
0x1001fe63: fcomp dword ptr [esp+0ACh]
0x1001fe6a: fnstsw ax
0x1001fe6c: test ah,5
0x1001fe6f: jp near ptr 10020333h
0x1001fe75: lea ecx,[esp+7Ch]
0x1001fe79: call 100206F0h
0x1001fe7e: lea ecx,[esp+4Ch]
0x1001fe82: call 100206F0h
0x1001fe87: fild dword ptr [esp+2Ch]
0x1001fe8b: sub esp,10h
0x1001fe8e: lea ecx,[esp+7Ch]
0x1001fe92: fstp dword ptr [esp+20h]
0x1001fe96: fild dword ptr [esp+4Ch]
0x1001fe9a: fstp dword ptr [esp+48h]
0x1001fe9e: fld dword ptr [esp+20h]
```

```
0x1001fea2: fld st(0)
0x1001fea4: fld qword ptr ds:[100FE7C8h]
0x1001feaa: fadd st(1),st
0x1001feac: fxch
0x1001feae: fstp dword ptr [esp+20h]
0x1001feb2: fld dword ptr [esp+20h]
0x1001feb6: fstp dword ptr [esp+0Ch]
0x1001feba: fld dword ptr [esp+48h]
0x1001febe: fld st(0)
0x1001fec0: fadd st,st(2)
0x1001fec2: fstp dword ptr [esp+20h]
0x1001fec6: fld dword ptr [esp+20h]
0x1001feca: fstp dword ptr [esp+8]
0x1001fece: fxch st(2)
0x1001fed0: fsub st,st(1)
0x1001fed2: fstp dword ptr [esp+20h]
0x1001fed6: fld dword ptr [esp+20h]
0x1001feda: fstp dword ptr [esp+4]
0x1001fede: fsubp
0x1001fee0: fstp dword ptr [esp+20h]
0x1001fee4: fld dword ptr [esp+20h]
0x1001fee8: fstp dword ptr [esp]
0x1001feeb: call 10020710h
0x1001fef0: push eax
0x1001fef1: lea eax,[esp+90h]
0x1001fef8: push eax
0x1001fef9: lea ecx,[esp+64h]
0x1001fefd: call 10020790h
0x1001ff02: fild dword ptr [esp+34h]
0x1001ff06: mov ecx,[eax]
0x1001ff08: sub esp,10h
0x1001ff0b: fstp dword ptr [esp+20h]
0x1001ff0f: mov [esp+8Ch],ecx
0x1001ff16: fild dword ptr [esp+50h]
0x1001ff1a: mov edx,[eax+4]
0x1001ff1d: mov [esp+90h],edx
0x1001ff24: mov ecx,[eax+8]
0x1001ff27: fstp dword ptr [esp+48h]
0x1001ff2b: mov [esp+94h],ecx
0x1001ff32: fld dword ptr [esp+20h]
0x1001ff36: mov edx,[eax+0Ch]
0x1001ff39: fld st(0)
0x1001ff3b: lea ecx,[esp+9Ch]
0x1001ff42: fld qword ptr ds:[100FE7C8h]
0x1001ff48: mov [esp+98h],edx
0x1001ff4f: fadd st(1),st
0x1001ff51: fxch
0x1001ff53: fstp dword ptr [esp+20h]
0x1001ff57: fld dword ptr [esp+20h]
0x1001ff5b: fstp dword ptr [esp+0Ch]
0x1001ff5f: fld dword ptr [esp+48h]
0x1001ff63: fld st(0)
0x1001ff65: fadd st,st(2)
0x1001ff67: fstp dword ptr [esp+20h]
0x1001ff6b: fld dword ptr [esp+20h]
0x1001ff6f: fstp dword ptr [esp+8]
0x1001ff73: fxch st(2)
0x1001ff75: fsub st,st(1)
0x1001ff77: fstp dword ptr [esp+20h]
0x1001ff7b: fld dword ptr [esp+20h]
0x1001ff7f: fstp dword ptr [esp+4]
0x1001ff83: fsubp
0x1001ff85: fstp dword ptr [esp+20h]
0x1001ff89: fld dword ptr [esp+20h]
0x1001ff8d: fstp dword ptr [esp]
0x1001ff90: call 10020710h
0x1001ff95: push eax
```

```
0x1001ff96: lea eax,[esp+70h]
0x1001ff9a: push eax
0x1001ff9b: lea ecx,[esp+64h]
0x1001ff9f: call 10020790h
0x1001ffa4: mov ecx,[eax]
0x1001ffa6: mov [esp+4Ch],ecx
0x1001ffaa: mov edx,[eax+4]
0x1001ffad: mov [esp+50h],edx
0x1001ffb1: mov ecx,[eax+8]
0x1001ffb4: mov [esp+54h],ecx
0x1001ffb8: mov edx,[eax+0Ch]
0x1001ffbb: lea ecx,[esp+7Ch]
0x1001ffbf: mov [esp+58h],edx
0x1001ffc3: call 100208C0h
0x1001ffc8: fstp dword ptr [esp+38h]
0x1001ffcc: lea ecx,[esp+7Ch]
0x1001ffd0: call 100208D0h
0x1001ffd5: fstp dword ptr [esp+30h]
0x1001ffd9: lea ecx,[esp+4Ch]
0x1001ffdd: call 100208C0h
0x1001ffe2: fstp dword ptr [esp+10h]
0x1001ffe6: lea ecx,[esp+4Ch]
0x1001ffea: call 100208D0h
0x1001ffef: fstp dword ptr [esp+40h]
0x1001fff3: fld dword ptr [esp+30h]
0x1001fff7: mov ebx,[esp+0BCh]
0x1001fffe: fmul dword ptr [esp+38h]
0x10020002: mov ebp,[esp+0B8h]
0x10020009: lea eax,[esp+28h]
0x1002000d: push eax
0x1002000e: lea ecx,[esp+28h]
0x10020012: fstp dword ptr [esp+20h]
0x10020016: fld dword ptr [esp+20h]
0x1002001a: push ecx
0x1002001b: lea edx,[esp+28h]
0x1002001f: push edx
0x10020020: mov edx,[esp+48h]
0x10020024: lea eax,[esp+20h]
0x10020028: push eax
0x10020029: push ecx
0x1002002a: mov ecx,[esp+40h]
0x1002002e: fstp dword ptr [esp]
0x10020031: push ebx
0x10020032: push ebp
0x10020033: push ecx
0x10020034: push edx
0x10020035: lea eax,[esp+3Ch]
0x10020039: push eax
0x1002003a: mov ecx,esi
0x1002003c: call 1001D050h
0x10020041: push eax
0x10020042: mov ecx,esi
0x10020044: call 1001D190h
0x10020049: fld dword ptr [esp+10h]
0x1002004d: fmul dword ptr [esp+30h]
0x10020051: lea ecx,[esp+28h]
0x10020055: push ecx
0x10020056: lea edx,[esp+28h]
0x1002005a: push edx
0x1002005b: fstp dword ptr [esp+24h]
0x1002005f: fld dword ptr [esp+24h]
0x10020063: mov edx,[esp+34h]
0x10020067: lea eax,[esp+28h]
0x1002006b: push eax
0x1002006c: lea ecx,[esp+20h]
0x10020070: push ecx
0x10020071: push ecx
```

```
0x10020072: fstp dword ptr [esp]
0x10020075: push ebx
0x10020076: push ebp
0x10020077: push edx
0x10020078: push edi
0x10020079: lea eax,[esp+3Ch]
0x1002007d: push eax
0x1002007e: mov ecx,esi
0x10020080: call 1001D050h
0x10020085: push eax
0x10020086: mov ecx,esi
0x10020088: call 1001D190h
0x1002008d: lea ecx,[esp+28h]
0x10020091: push ecx
0x10020092: lea edx,[esp+28h]
0x10020096: push edx
0x10020097: lea eax,[esp+28h]
0x1002009b: fld dword ptr [esp+48h]
0x1002009f: mov edx,[esp+3Ch]
0x100200a3: fmul dword ptr [esp+40h]
0x100200a7: push eax
0x100200a8: mov eax,[esp+48h]
0x100200ac: lea ecx,[esp+20h]
0x100200b0: push ecx
0x100200b1: fstp dword ptr [esp+2Ch]
0x100200b5: fld dword ptr [esp+2Ch]
0x100200b9: push ecx
0x100200ba: fstp dword ptr [esp]
0x100200bd: push ebx
0x100200be: push ebp
0x100200bf: push edx
0x100200c0: push eax
0x100200c1: lea ecx,[esp+3Ch]
0x100200c5: push ecx
0x100200c6: mov ecx,esi
0x100200c8: call 1001D050h
0x100200cd: push eax
0x100200ce: mov ecx,esi
0x100200d0: call 1001D190h
0x100200d5: fld dword ptr [esp+40h]
0x100200d9: fmul dword ptr [esp+10h]
0x100200dd: lea edx,[esp+28h]
0x100200e1: push edx
0x100200e2: lea eax,[esp+28h]
0x100200e6: push eax
0x100200e7: fstp dword ptr [esp+24h]
0x100200eb: fld dword ptr [esp+24h]
0x100200ef: mov eax,[esp+3Ch]
0x100200f3: lea ecx,[esp+28h]
0x100200f7: push ecx
0x100200f8: lea edx,[esp+20h]
0x100200fc: push edx
0x100200fd: push ecx
0x100200fe: fstp dword ptr [esp]
0x10020101: push ebx
0x10020102: push ebp
0x10020103: push eax
0x10020104: push edi
0x10020105: lea ecx,[esp+3Ch]
0x10020109: push ecx
0x1002010a: mov ecx,esi
0x1002010c: call 1001D050h
0x10020111: push eax
0x10020112: mov ecx,esi
0x10020114: call 1001D190h
0x10020119: mov eax,[esp+3Ch]
0x1002011d: add eax,1
```

```
0x10020120: cmp eax,edi
0x10020122: mov [esp+18h],eax
0x10020126: mov [esp+1Ch],eax
0x1002012a: jge near ptr 100201C0h
0x10020130: jmp short 10020136h
0x10020132: mov eax,[esp+1Ch]
0x10020136: fld dword ptr [esp+30h]
0x1002013a: lea edx,[esp+28h]
0x1002013e: push edx
0x1002013f: lea ecx,[esp+28h]
0x10020143: push ecx
0x10020144: lea edx,[esp+28h]
0x10020148: push edx
0x10020149: mov edx,[esp+38h]
0x1002014d: lea ecx,[esp+20h]
0x10020151: push ecx
0x10020152: push ecx
0x10020153: fstp dword ptr [esp]
0x10020156: push ebx
0x10020157: push ebp
0x10020158: push edx
0x10020159: push eax
0x1002015a: lea eax,[esp+90h]
0x10020161: push eax
0x10020162: mov ecx,esi
0x10020164: call 1001D050h
0x10020169: push eax
0x1002016a: mov ecx,esi
0x1002016c: call 1001D190h
0x10020171: fld dword ptr [esp+40h]
0x10020175: lea ecx,[esp+28h]
0x10020179: push ecx
0x1002017a: lea edx,[esp+28h]
0x1002017e: push edx
0x1002017f: mov edx,[esp+3Ch]
0x10020183: lea eax,[esp+28h]
0x10020187: push eax
0x10020188: mov eax,[esp+28h]
0x1002018c: lea ecx,[esp+20h]
0x10020190: push ecx
0x10020191: push ecx
0x10020192: fstp dword ptr [esp]
0x10020195: push ebx
0x10020196: push ebp
0x10020197: push edx
0x10020198: push eax
0x10020199: lea ecx,[esp+6Ch]
0x1002019d: push ecx
0x1002019e: mov ecx,esi
0x100201a0: call 1001D050h
0x100201a5: push eax
0x100201a6: mov ecx,esi
0x100201a8: call 1001D190h
0x100201ad: mov eax,[esp+1Ch]
0x100201b1: add eax,1
0x100201b4: cmp eax,edi
0x100201b6: mov [esp+1Ch],eax
0x100201ba: jl near ptr 10020132h
0x100201c0: mov eax,[esp+2Ch]
0x100201c4: add eax,1
0x100201c7: cmp eax,[esp+34h]
0x100201cb: mov [esp+40h],eax
0x100201cf: mov [esp+30h],eax
0x100201d3: jge near ptr 10020270h
0x100201d9: jmp short 100201E4h
0x100201e0: mov eax,[esp+30h]
0x100201e4: fld dword ptr [esp+38h]
```

```
0x100201e8: lea edx,[esp+28h]
0x100201ec: push edx
0x100201ed: lea ecx,[esp+28h]
0x100201f1: push ecx
0x100201f2: lea edx,[esp+28h]
0x100201f6: push edx
0x100201f7: mov edx,[esp+48h]
0x100201fb: lea ecx,[esp+20h]
0x100201ff: push ecx
0x10020200: push ecx
0x10020201: fstp dword ptr [esp]
0x10020204: push ebx
0x10020205: push ebp
0x10020206: push eax
0x10020207: push edx
0x10020208: lea eax,[esp+6Ch]
0x1002020c: push eax
0x1002020d: mov ecx,esi
0x1002020f: call 1001D050h
0x10020214: push eax
0x10020215: mov ecx,esi
0x10020217: call 1001D190h
0x1002021c: fld dword ptr [esp+10h]
0x10020220: lea ecx,[esp+28h]
0x10020224: push ecx
0x10020225: lea edx,[esp+28h]
0x10020229: push edx
0x1002022a: mov edx,[esp+38h]
0x1002022e: lea eax,[esp+28h]
0x10020232: push eax
0x10020233: lea ecx,[esp+20h]
0x10020237: push ecx
0x10020238: push ecx
0x10020239: fstp dword ptr [esp]
0x1002023c: push ebx
0x1002023d: push ebp
0x1002023e: push edx
0x1002023f: push edi
0x10020240: lea eax,[esp+90h]
0x10020247: push eax
0x10020248: mov ecx,esi
0x1002024a: call 1001D050h
0x1002024f: push eax
0x10020250: mov ecx,esi
0x10020252: call 1001D190h
0x10020257: mov eax,[esp+30h]
0x1002025b: add eax,1
0x1002025e: cmp eax,[esp+34h]
0x10020262: mov [esp+30h],eax
0x10020266: jl near ptr 100201E0h
0x1002026c: mov eax,[esp+40h]
0x10020270: mov ecx,eax
0x10020272: cmp ecx,[esp+34h]
0x10020276: mov [esp+30h],eax
0x1002027a: jge near ptr 10020505h
0x10020280: mov eax,[esp+18h]
0x10020284: cmp eax,edi
0x10020286: mov [esp+1Ch],eax
0x1002028a: jge near ptr 1002031Bh
0x10020290: jmp short 10020296h
0x10020292: mov eax,[esp+1Ch]
0x10020296: push ebx
0x10020297: push ebp
0x10020298: push ecx
0x10020299: push eax
0x1002029a: lea ecx,[esp+58h]
0x1002029e: push ecx
```

```
0x1002029f: mov ecx,esi
0x100202a1: call 1001D050h
0x100202a6: mov eax,[eax]
0x100202a8: mov edx,eax
0x100202aa: shr edx,10h
0x100202ad: movzx ecx,dl
0x100202b0: mov [esp+10h],ecx
0x100202b4: mov edx,eax
0x100202b6: shr edx,8
0x100202b9: fild dword ptr [esp+10h]
0x100202bd: movzx ecx,dl
0x100202c0: fadd dword ptr [esp+14h]
0x100202c4: mov [esp+10h],ecx
0x100202c8: mov ecx,[esp+30h]
0x100202cc: mov [esp+2Ch],eax
0x100202d0: movzx edx,byte ptr [esp+2Ch]
0x100202d5: fstp dword ptr [esp+14h]
0x100202d9: fild dword ptr [esp+10h]
0x100202dd: mov [esp+10h],edx
0x100202e1: fadd dword ptr [esp+20h]
0x100202e5: shr eax,18h
0x100202e8: fstp dword ptr [esp+20h]
0x100202ec: fild dword ptr [esp+10h]
0x100202f0: mov [esp+10h],eax
0x100202f4: mov eax,[esp+1Ch]
0x100202f8: add eax,1
0x100202fb: cmp eax,edi
0x100202fd: fadd dword ptr [esp+24h]
0x10020301: mov [esp+1Ch],eax
0x10020305: fstp dword ptr [esp+24h]
0x10020309: fild dword ptr [esp+10h]
0x1002030d: fadd dword ptr [esp+28h]
0x10020311: fstp dword ptr [esp+28h]
0x10020315: jl near ptr 10020292h
0x1002031b: add ecx,1
0x1002031e: cmp ecx,[esp+34h]
0x10020322: mov [esp+30h],ecx
0x10020326: jl near ptr 10020280h
0x1002032c: jmp near ptr 10020505h
0x10020331: fstp st(0)
0x10020333: lea ecx,[esp+4Ch]
0x10020337: call 100206F0h
0x1002033c: lea ecx,[esp+6Ch]
0x10020340: call 100206C0h
0x10020345: cmp ebx,ebp
0x10020347: mov [esp+30h],ebx
0x1002034b: jg near ptr 10020505h
0x10020351: mov ebp,[esp+0BCh]
0x10020358: jmp short 10020360h
0x10020360: mov ebx,[esp+3Ch]
0x10020364: cmp ebx,edi
0x10020366: mov [esp+1Ch],ebx
0x1002036a: jg near ptr 100204F0h
0x10020370: fild dword ptr [esp+30h]
0x10020374: fstp dword ptr [esp+18h]
0x10020378: fld dword ptr [esp+18h]
0x1002037c: fld st(0)
0x1002037e: fld qword ptr ds:[100FE7C8h]
0x10020384: fadd st(1),st
0x10020386: fxch
0x10020388: fstp dword ptr [esp+18h]
0x1002038c: fsub st(1),st
0x1002038e: fxch
0x10020390: fstp dword ptr [esp+38h]
0x10020394: jmp short 100203A6h
0x100203a0: fld qword ptr ds:[100FE7C8h]
0x100203a6: fild dword ptr [esp+1Ch]
```

```
0x100203aa: sub esp,10h
0x100203ad: lea ecx,[esp+9Ch]
0x100203b4: fstp dword ptr [esp+20h]
0x100203b8: fld dword ptr [esp+28h]
0x100203bc: fstp dword ptr [esp+0Ch]
0x100203c0: fld dword ptr [esp+20h]
0x100203c4: fld st(0)
0x100203c6: fadd st,st(2)
0x100203c8: fstp dword ptr [esp+20h]
0x100203cc: fld dword ptr [esp+20h]
0x100203d0: fstp dword ptr [esp+8]
0x100203d4: fld dword ptr [esp+48h]
0x100203d8: fstp dword ptr [esp+4]
0x100203dc: fsubrp
0x100203de: fstp dword ptr [esp+20h]
0x100203e2: fld dword ptr [esp+20h]
0x100203e6: fstp dword ptr [esp]
0x100203e9: call 10020710h
0x100203ee: push eax
0x100203ef: lea eax,[esp+80h]
0x100203f6: push eax
0x100203f7: lea ecx,[esp+64h]
0x100203fb: call 10020790h
0x10020400: mov ecx,[eax]
0x10020402: mov [esp+4Ch],ecx
0x10020406: mov edx,[eax+4]
0x10020409: mov [esp+50h],edx
0x1002040d: mov ecx,[eax+8]
0x10020410: mov [esp+54h],ecx
0x10020414: mov edx,[eax+0Ch]
0x10020417: lea eax,[esp+40h]
0x1002041b: push eax
0x1002041c: lea ecx,[esp+50h]
0x10020420: mov [esp+5Ch],edx
0x10020424: call 10020870h
0x10020429: mov ecx,[eax]
0x1002042b: mov [esp+6Ch],ecx
0x1002042f: mov edx,[eax+4]
0x10020432: mov eax,[esp+0B8h]
0x10020439: mov ecx,[esp+0B4h]
0x10020440: push ebp
0x10020441: push eax
0x10020442: push ecx
0x10020443: mov [esp+7Ch],edx
0x10020447: fld dword ptr [esp+7Ch]
0x1002044b: sub esp,8
0x1002044e: fstp dword ptr [esp+4]
0x10020452: lea edx,[esp+5Ch]
0x10020456: fld dword ptr [esp+80h]
0x1002045d: mov ecx,esi
0x1002045f: fstp dword ptr [esp]
0x10020462: push edx
0x10020463: call 1001DDC0h
0x10020468: mov eax,[eax]
0x1002046a: lea ecx,[esp+4Ch]
0x1002046e: mov [esp+2Ch],eax
0x10020472: call 10020770h
0x10020477: fstp dword ptr [esp+10h]
0x1002047b: fld dword ptr [esp+10h]
0x1002047f: movzx ecx,byte ptr [esp+2Eh]
0x10020484: movzx edx,byte ptr [esp+2Dh]
0x10020489: fld st(0)
0x1002048b: movzx eax,byte ptr [esp+2Ch]
0x10020490: mov [esp+10h],ecx
0x10020494: fild dword ptr [esp+10h]
0x10020498: mov [esp+10h],edx
0x1002049c: fmulp
```

```
0x1002049e: fadd dword ptr [esp+14h]
0x100204a2: fstp dword ptr [esp+14h]
0x100204a6: fld st(0)
0x100204a8: fild dword ptr [esp+10h]
0x100204ac: mov [esp+10h],eax
0x100204b0: fmulp
0x100204b2: fadd dword ptr [esp+20h]
0x100204b6: fstp dword ptr [esp+20h]
0x100204ba: fld st(0)
0x100204bc: fild dword ptr [esp+10h]
0x100204c0: fmulp
0x100204c2: fadd dword ptr [esp+24h]
0x100204c6: fstp dword ptr [esp+24h]
0x100204ca: movzx ecx,byte ptr [esp+2Fh]
0x100204cf: mov [esp+10h],ecx
0x100204d3: add ebx,1
0x100204d6: cmp ebx,edi
0x100204d8: fild dword ptr [esp+10h]
0x100204dc: mov [esp+1Ch],ebx
0x100204e0: fmulp
0x100204e2: fadd dword ptr [esp+28h]
0x100204e6: fstp dword ptr [esp+28h]
0x100204ea: jle near ptr 100203A0h
0x100204f0: mov eax,[esp+30h]
0x100204f4: add eax,1
0x100204f7: cmp eax,[esp+34h]
0x100204fb: mov [esp+30h],eax
0x100204ff: jle near ptr 10020360h
0x10020505: lea ecx,[esp+5Ch]
0x10020509: call 10020770h
0x1002050e: fstp dword ptr [esp+18h]
0x10020512: fld dword ptr [esp+14h]
0x10020516: pop ebp
0x10020517: fld dword ptr [esp+14h]
0x1002051b: pop ebx
0x1002051c: fld st(0)
0x1002051e: fdivp st(2),st
0x10020520: fxch
0x10020522: fstp dword ptr [esp+0Ch]
0x10020526: fld dword ptr [esp+18h]
0x1002052a: fdiv st,st(1)
0x1002052c: fstp dword ptr [esp+18h]
0x10020530: fld dword ptr [esp+1Ch]
0x10020534: fdiv st,st(1)
0x10020536: fstp dword ptr [esp+1Ch]
0x1002053a: fdivr dword ptr [esp+20h]
0x1002053e: fstp dword ptr [esp+20h]
0x10020542: fld dword ptr [esp+0Ch]
0x10020546: fld qword ptr ds:[100FE918h]
0x1002054c: fcom
0x1002054e: fnstsw ax
0x10020550: fld dword ptr ds:[100FF364h]
0x10020556: test ah,5
0x10020559: jp short 10020567h
0x1002055b: fstp st(2)
0x1002055d: fxch
0x1002055f: fst dword ptr [esp+0Ch]
0x10020563: fldz
0x10020565: jmp short 10020580h
0x10020567: fldz
0x10020569: fcom st(3)
0x1002056b: fnstsw ax
0x1002056d: fstp st(3)
0x1002056f: test ah,41h
0x10020572: jne short 100205C1h
0x10020574: fxch st(2)
0x10020576: fst dword ptr [esp+0Ch]
```

```
0x1002057a: fxch
0x1002057c: fxch st(2)
0x1002057e: fxch
0x10020580: fld dword ptr [esp+0Ch]
0x10020584: fnstcw [esp+0Ch]
0x10020588: movzx eax,word ptr [esp+0Ch]
0x1002058d: or eax,0C00h
0x10020592: mov [esp+10h],eax
0x10020596: fldcw [esp+10h]
0x1002059a: fistp dword ptr [esp+10h]
0x1002059e: mov dl,[esp+10h]
0x100205a2: mov [esp+26h],dl
0x100205a6: fldcw [esp+0Ch]
0x100205aa: fld dword ptr [esp+18h]
0x100205ae: fcom st(3)
0x100205b0: fnstsw ax
0x100205b2: test ah,41h
0x100205b5: jne short 100205C7h
0x100205b7: fstp st(0)
0x100205b9: fxch
0x100205bb: fst dword ptr [esp+18h]
0x100205bf: jmp short 100205D6h
0x100205c1: fxch
0x100205c3: fxch st(2)
0x100205c5: jmp short 10020580h
0x100205c7: fcomp
0x100205c9: fnstsw ax
0x100205cb: test ah,5
0x100205ce: jp short 100205D4h
0x100205d0: fst dword ptr [esp+18h]
0x100205d4: fxch
0x100205d6: fld dword ptr [esp+18h]
0x100205da: fnstcw [esp+0Ch]
0x100205de: movzx eax,word ptr [esp+0Ch]
0x100205e3: or eax,0C00h
0x100205e8: mov [esp+10h],eax
0x100205ec: fldcw [esp+10h]
0x100205f0: fistp dword ptr [esp+10h]
0x100205f4: mov al,[esp+10h]
0x100205f8: mov [esp+25h],al
0x100205fc: fldcw [esp+0Ch]
0x10020600: fld dword ptr [esp+1Ch]
0x10020604: fcom st(3)
0x10020606: fnstsw ax
0x10020608: test ah,41h
0x1002060b: jne short 10020615h
0x1002060d: fstp st(0)
0x1002060f: fst dword ptr [esp+1Ch]
0x10020613: jmp short 10020626h
0x10020615: fcomp st(2)
0x10020617: fnstsw ax
0x10020619: test ah,5
0x1002061c: jp short 10020626h
0x1002061e: fxch
0x10020620: fst dword ptr [esp+1Ch]
0x10020624: fxch
0x10020626: fld dword ptr [esp+1Ch]
0x1002062a: fnstcw [esp+0Ch]
0x1002062e: movzx eax,word ptr [esp+0Ch]
0x10020633: or eax,0C00h
0x10020638: mov [esp+10h],eax
0x1002063c: fldcw [esp+10h]
0x10020640: fistp dword ptr [esp+10h]
0x10020644: mov cl,[esp+10h]
0x10020648: mov [esp+24h],cl
0x1002064c: fldcw [esp+0Ch]
0x10020650: fld dword ptr [esp+20h]
```

```
0x10020654: fcom st(3)
0x10020656: fnstsw ax
0x10020658: fstp st(3)
0x1002065a: test ah,41h
0x1002065d: jne short 10020669h
0x1002065f: fstp st(2)
0x10020661: fstp st(0)
0x10020663: fstp dword ptr [esp+20h]
0x10020667: jmp short 10020678h
0x10020669: fstp st(0)
0x1002066b: fcom
0x1002066d: fnstsw ax
0x1002066f: fstp st(1)
0x10020671: test ah,41h
0x10020674: je short 10020663h
0x10020676: fstp st(0)
0x10020678: fld dword ptr [esp+20h]
0x1002067c: pop edi
0x1002067d: fnstcw [esp+8]
0x10020681: movzx eax,word ptr [esp+8]
0x10020686: or eax,0C00h
0x1002068b: mov [esp+0Ch],eax
0x1002068f: mov eax,[esp+94h]
0x10020696: fldcw [esp+0Ch]
0x1002069a: pop esi
0x1002069b: fistp dword ptr [esp+8]
0x1002069f: mov dl,[esp+8]
0x100206a3: mov [esp+1Fh],dl
0x100206a7: mov ecx,[esp+1Ch]
0x100206ab: mov [eax],ecx
0x100206ad: fldcw [esp+4]
0x100206b1: add esp,8Ch
0x100206b7: ret 20h
```

`sub_1001fd10` is called by sub_10011fb0. It delegates to sub_1001d050, sub_1001d190, sub_1001ddc0 (12 callees).

12.392 sub_100206c0 at 0x100206c0 (5 instructions)

```
0x100206c0: fldz
0x100206c2: mov eax,ecx
0x100206c4: fst dword ptr [eax+4]
0x100206c7: fstp dword ptr [eax]
0x100206c9: ret
```

`sub_100206c0` is called by sub_1001fd10, sub_100206f0, sub_10020710, sub_10020740. It calls nothing else.

12.393 sub_100206d0 at 0x100206d0 (6 instructions)

```
0x100206d0: fld dword ptr [esp+4]
0x100206d4: mov eax,ecx
0x100206d6: fstp dword ptr [eax]
0x100206d8: fld dword ptr [esp+8]
0x100206dc: fstp dword ptr [eax+4]
0x100206df: ret 8
```

`sub_100206d0` is called by sub_10020870. It calls nothing else.

12.394 sub_100206f0 at 0x100206f0 (6 instructions)

```
0x100206f0: mov edx,ecx
0x100206f2: call 100206C0h
0x100206f7: lea ecx,[edx+8]
0x100206fa: call 100206C0h
```

```
0x100206ff: mov eax,edx
0x10020701: ret
```

`sub_100206f0` is called by sub_1001fd10, sub_10020790. It delegates to sub_100206c0 (1 callees).

12.395 sub_10020710 at 0x10020710 (14 instructions)

```
0x10020710: mov edx,ecx
0x10020712: call 100206C0h
0x10020717: lea ecx,[edx+8]
0x1002071a: call 100206C0h
0x1002071f: fld dword ptr [esp+4]
0x10020723: fstp dword ptr [edx]
0x10020725: mov eax,edx
0x10020727: fld dword ptr [esp+8]
0x1002072b: fstp dword ptr [edx+4]
0x1002072e: fld dword ptr [esp+0Ch]
0x10020732: fstp dword ptr [ecx]
0x10020734: fld dword ptr [esp+10h]
0x10020738: fstp dword ptr [edx+0Ch]
0x1002073b: ret 10h
```

`sub_10020710` is called by sub_1001fd10, sub_10020790. It delegates to sub_100206c0 (1 callees).

12.396 sub_10020740 at 0x10020740 (17 instructions)

```
0x10020740: push esi
0x10020741: mov edx,ecx
0x10020743: call 100206C0h
0x10020748: lea ecx,[edx+8]
0x1002074b: call 100206C0h
0x10020750: mov eax,[esp+8]
0x10020754: mov esi,[eax]
0x10020756: mov [edx],esi
0x10020758: mov esi,[eax+4]
0x1002075b: mov [edx+4],esi
0x1002075e: mov esi,[eax+8]
0x10020761: mov [ecx],esi
0x10020763: mov eax,[eax+0Ch]
0x10020766: mov [ecx+4],eax
0x10020769: mov eax,edx
0x1002076b: pop esi
0x1002076c: ret 4
```

`sub_10020740` is called by sub_10020790. It delegates to sub_100206c0 (1 callees).

12.397 sub_10020770 at 0x10020770 (10 instructions)

```
0x10020770: push ecx
0x10020771: fld dword ptr [ecx+0Ch]
0x10020774: fsub dword ptr [ecx+4]
0x10020777: fld dword ptr [ecx+8]
0x1002077a: fsub dword ptr [ecx]
0x1002077c: fmulp
0x1002077e: fstp dword ptr [esp]
0x10020781: fld dword ptr [esp]
0x10020784: pop ecx
0x10020785: ret
```

`sub_10020770` is called by sub_1001fd10. It calls nothing else.

12.398 sub_10020790 at 0x10020790 (80 instructions)

```
0x10020790: sub esp,10h
0x10020793: push esi
0x10020794: mov esi,ecx
0x10020796: lea ecx,[esp+4]
0x1002079a: call 100206F0h
0x1002079f: fld dword ptr [esi]
0x100207a1: mov ecx,[esp+1Ch]
0x100207a5: fld dword ptr [ecx]
0x100207a7: fcompp
0x100207a9: fnstsw ax
0x100207ab: test ah,5
0x100207ae: jp short 100207B4h
0x100207b0: fld dword ptr [esi]
0x100207b2: jmp short 100207B6h
0x100207b4: fld dword ptr [ecx]
0x100207b6: fstp dword ptr [esp+4]
0x100207ba: fld dword ptr [esi+4]
0x100207bd: fld dword ptr [ecx+4]
0x100207c0: fcompp
0x100207c2: fnstsw ax
0x100207c4: test ah,5
0x100207c7: jp short 100207CEh
0x100207c9: fld dword ptr [esi+4]
0x100207cc: jmp short 100207D1h
0x100207ce: fld dword ptr [ecx+4]
0x100207d1: fstp dword ptr [esp+8]
0x100207d5: fld dword ptr [esi+8]
0x100207d8: fld dword ptr [ecx+8]
0x100207db: fcompp
0x100207dd: fnstsw ax
0x100207df: test ah,41h
0x100207e2: jne short 100207E9h
0x100207e4: fld dword ptr [esi+8]
0x100207e7: jmp short 100207ECh
0x100207e9: fld dword ptr [ecx+8]
0x100207ec: fstp dword ptr [esp+0Ch]
0x100207f0: fld dword ptr [esi+0Ch]
0x100207f3: fld dword ptr [ecx+0Ch]
0x100207f6: fcompp
0x100207f8: fnstsw ax
0x100207fa: test ah,41h
0x100207fd: jne short 10020804h
0x100207ff: fld dword ptr [esi+0Ch]
0x10020802: jmp short 10020807h
0x10020804: fld dword ptr [ecx+0Ch]
0x10020807: fstp dword ptr [esp+10h]
0x1002080b: fld dword ptr [esp+4]
0x1002080f: fld dword ptr [esp+0Ch]
0x10020813: fcompp
0x10020815: fnstsw ax
0x10020817: test ah,1
0x1002081a: jne short 10020846h
0x1002081c: fld dword ptr [esp+8]
0x10020820: fld dword ptr [esp+10h]
0x10020824: fcompp
0x10020826: fnstsw ax
0x10020828: test ah,1
0x1002082b: jne short 10020846h
0x1002082d: mov esi,[esp+18h]
0x10020831: lea eax,[esp+4]
0x10020835: push eax
0x10020836: mov ecx,esi
0x10020838: call 10020740h
0x1002083d: mov eax,esi
0x1002083f: pop esi
0x10020840: add esp,10h
```

```
0x10020843: ret 8
0x10020846: fldz
0x10020848: mov esi,[esp+18h]
0x1002084c: sub esp,10h
0x1002084f: fst dword ptr [esp+0Ch]
0x10020853: fst dword ptr [esp+8]
0x10020857: mov ecx,esi
0x10020859: fst dword ptr [esp+4]
0x1002085d: fstp dword ptr [esp]
0x10020860: call 10020710h
0x10020865: mov eax,esi
0x10020867: pop esi
0x10020868: add esp,10h
0x1002086b: ret 8
```

`sub_10020790` is called by sub_1001fd10. It delegates to sub_100206f0, sub_10020710, sub_10020740 (3 callees).

12.399 sub_10020870 at 0x10020870 (21 instructions)

```
0x10020870: push ecx
0x10020871: fld dword ptr [ecx+0Ch]
0x10020874: sub esp,8
0x10020877: fadd dword ptr [ecx+4]
0x1002087a: fld qword ptr ds:[100FE7C8h]
0x10020880: fmul st(1),st
0x10020882: fxch
0x10020884: fstp dword ptr [esp+8]
0x10020888: fld dword ptr [esp+8]
0x1002088c: fstp dword ptr [esp+4]
0x10020890: fld dword ptr [ecx+8]
0x10020893: fadd dword ptr [ecx]
0x10020895: mov ecx,[esp+10h]
0x10020899: fmulp
0x1002089b: fstp dword ptr [esp+8]
0x1002089f: fld dword ptr [esp+8]
0x100208a3: fstp dword ptr [esp]
0x100208a6: call 100206D0h
0x100208ab: mov eax,ecx
0x100208ad: pop ecx
0x100208ae: ret 4
```

`sub_10020870` is called by sub_1001fd10. It delegates to sub_100206d0 (1 callees).

12.400 sub_100208c0 at 0x100208c0 (7 instructions)

```
0x100208c0: push ecx
0x100208c1: fld dword ptr [ecx+8]
0x100208c4: fsub dword ptr [ecx]
0x100208c6: fstp dword ptr [esp]
0x100208c9: fld dword ptr [esp]
0x100208cc: pop ecx
0x100208cd: ret
```

`sub_100208c0` is called by sub_1001fd10. It calls nothing else.

12.401 sub_100208d0 at 0x100208d0 (7 instructions)

```
0x100208d0: push ecx
0x100208d1: fld dword ptr [ecx+0Ch]
0x100208d4: fsub dword ptr [ecx+4]
0x100208d7: fstp dword ptr [esp]
0x100208da: fld dword ptr [esp]
0x100208dd: pop ecx
0x100208de: ret
```

`sub_100208d0` is called by sub_1001fd10. It calls nothing else.

12.402 sub_100208e0 at 0x100208e0 (10 instructions)

```
0x100208e0: mov edx,[esp+0Ch]
0x100208e4: mov ecx,[esp+4]
0x100208e8: mov eax,[ecx]
0x100208ea: mov eax,[eax+8]
0x100208ed: push edx
0x100208ee: mov edx,[esp+0Ch]
0x100208f2: push 1
0x100208f4: push edx
0x100208f5: call eax
0x100208f7: ret
```

`sub_100208e0` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.403 sub_10020900 at 0x10020900 (10 instructions)

```
0x10020900: mov edx,[esp+0Ch]
0x10020904: mov ecx,[esp+4]
0x10020908: mov eax,[ecx]
0x1002090a: mov eax,[eax+0Ch]
0x1002090d: push edx
0x1002090e: mov edx,[esp+0Ch]
0x10020912: push 1
0x10020914: push edx
0x10020915: call eax
0x10020917: ret
```

`sub_10020900` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.404 sub_10020920 at 0x10020920 (38 instructions)

```
0x10020920: push edi
0x10020921: mov edi,[esp+0Ch]
0x10020925: cmp edi,0FFFFFFFFh
0x10020928: jne short 1002092Eh
0x1002092a: or eax,edi
0x1002092c: pop edi
0x1002092d: ret
0x1002092e: push ebx
0x1002092f: mov ebx,[esp+14h]
0x10020933: push esi
0x10020934: mov esi,[esp+10h]
0x10020938: mov eax,[esi]
0x1002093a: mov edx,[eax+10h]
0x1002093d: push ebx
0x1002093e: push edi
0x1002093f: mov ecx,esi
0x10020941: call edx
0x10020943: test al,al
0x10020945: jne short 1002094Eh
0x10020947: pop esi
0x10020948: pop ebx
0x10020949: or eax,0FFFFFFFFh
0x1002094c: pop edi
0x1002094d: ret
0x1002094e: test ebx,ebx
0x10020950: jne short 10020958h
0x10020952: pop esi
0x10020953: pop ebx
0x10020954: mov eax,edi
0x10020956: pop edi
0x10020957: ret
0x10020958: mov eax,[esi]
```

```
0x1002095a: mov edx,[eax+14h]
0x1002095d: mov ecx,esi
0x1002095f: pop esi
0x10020960: pop ebx
0x10020961: pop edi
0x10020962: jmp edx
```

`sub_10020920` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.405 sub_10020970 at 0x10020970 (21 instructions)

```
0x10020970: mov eax,[esp+0Ch]
0x10020974: mov ecx,[esp+8]
0x10020978: push esi
0x10020979: mov esi,[esp+8]
0x1002097d: push 100B3980h
0x10020982: push 100B8550h
0x10020987: push 10015D10h
0x1002098c: push 100B8550h
0x10020991: push 10020920h
0x10020996: push 10020900h
0x1002099b: push 100208E0h
0x100209a0: push esi
0x100209a1: push eax
0x100209a2: push ecx
0x100209a3: call 1006891Dh
0x100209a8: add esp,28h
0x100209ab: test eax,eax
0x100209ad: je short 100209B2h
0x100209af: mov [eax+4],esi
0x100209b2: pop esi
0x100209b3: ret
```

`sub_10020970` is called by sub_100209c0. It delegates to sub_1006891d (1 callees).

12.406 sub_100209c0 at 0x100209c0 (8 instructions)

```
0x100209c0: mov eax,[esp+8]
0x100209c4: mov ecx,[esp+4]
0x100209c8: push eax
0x100209c9: push 100FF3B4h
0x100209ce: push ecx
0x100209cf: call 10020970h
0x100209d4: add esp,0Ch
0x100209d7: ret
```

`sub_100209c0` is called by sub_100143f0. It works with the text "TIFF IMAGE". It delegates to sub_10020970 (1 callees).

12.407 j_free at 0x100209e0 (1 instructions)

```
0x100209e0: jmp dword ptr ds:[100FE554h]
```

`j_free` is called by sub_100143f0, sub_10015660, sub_10065c7b, sub_10067617, sub_10067caa, sub_10067f4a, sub_100680ec, sub_10068f9a, sub_1006cf39, sub_1006d21a, sub_1006e09e, sub_1006ea55, sub_1006eba4, sub_1006f20b, sub_1006f2c7, sub_1006f391, sub_1006fea1, sub_1006ffe0, sub_100702f4, sub_10070584, sub_10071192, sub_1007127c, sub_10071346, sub_10071411, sub_10071523, sub_100715da, sub_1007171b, sub_100ad500, sub_100ae190, sub_100ae350, sub_100ae880, sub_100ae910, sub_100af0f0, sub_100b0040, sub_100b05b0, sub_100b05f0, sub_100b0810, sub_100b13e0, sub_100b1420, sub_100b1470, sub_100bff80, sub_100c07b0, sub_100c1f70, sub_100c1fb0, sub_100f0400, sub_100f16f0. It calls nothing else.

12.408 j_memset at 0x100209f0 (2 instructions)

```
0x100209f0: jmp near ptr 100228C0h
0x100228c0: jmp dword ptr ds:[100FE558h]
```

`j_memset` is called by sub_10067617, sub_1006799b, sub_1006891d, sub_1006fea1, sub_10071411, sub_100729b6. It calls nothing else.

12.409 j_memcpy at 0x10020a00 (2 instructions)

```
0x10020a00: jmp near ptr 100228C6h
0x100228c6: jmp dword ptr ds:[100FE54Ch]
```

`j_memcpy` is called by sub_10067774, sub_10067aa8, sub_10067b9e, sub_1006daaf, sub_1006e09e, sub_1006f391, sub_10070584, sub_10070799, sub_100727aa. It calls nothing else.

12.410 sub_10020a10 at 0x10020a10 (59 instructions)

```
0x10020a10: mov eax,[esp+0Ch]
0x10020a14: cmp eax,4
0x10020a17: mov ecx,[esp+8]
0x10020a1b: mov edx,[esp+4]
0x10020a1f: push esi
0x10020a20: push edi
0x10020a21: jb short 10020A37h
0x10020a23: mov esi,[edx]
0x10020a25: cmp esi,[ecx]
0x10020a27: jne short 10020A3Bh
0x10020a29: sub eax,4
0x10020a2c: add ecx,4
0x10020a2f: add edx,4
0x10020a32: cmp eax,4
0x10020a35: jae short 10020A23h
0x10020a37: test eax,eax
0x10020a39: je short 10020A99h
0x10020a3b: movzx esi,byte ptr [edx]
0x10020a3e: movzx edi,byte ptr [ecx]
0x10020a41: sub esi,edi
0x10020a43: jne short 10020A8Ah
0x10020a45: sub eax,1
0x10020a48: add ecx,1
0x10020a4b: add edx,1
0x10020a4e: test eax,eax
0x10020a50: je short 10020A99h
0x10020a52: movzx esi,byte ptr [edx]
0x10020a55: movzx edi,byte ptr [ecx]
0x10020a58: sub esi,edi
0x10020a5a: jne short 10020A8Ah
0x10020a5c: sub eax,1
0x10020a5f: add ecx,1
0x10020a62: add edx,1
0x10020a65: test eax,eax
0x10020a67: je short 10020A99h
0x10020a69: movzx esi,byte ptr [edx]
0x10020a6c: movzx edi,byte ptr [ecx]
0x10020a6f: sub esi,edi
0x10020a71: jne short 10020A8Ah
0x10020a73: sub eax,1
0x10020a76: add ecx,1
0x10020a79: add edx,1
0x10020a7c: test eax,eax
0x10020a7e: je short 10020A99h
0x10020a80: movzx esi,byte ptr [edx]
0x10020a83: movzx eax,byte ptr [ecx]
0x10020a86: sub esi,eax
```

```
0x10020a88: je short 10020A99h
0x10020a8a: test esi,esi
0x10020a8c: mov eax,1
0x10020a91: jg short 10020A9Bh
0x10020a93: pop edi
0x10020a94: or eax,0FFFFFFFFh
0x10020a97: pop esi
0x10020a98: ret
0x10020a99: xor eax,eax
0x10020a9b: pop edi
0x10020a9c: pop esi
0x10020a9d: ret
```

`sub_10020a10` is called by sub_1006edb9. It calls nothing else.

12.411 sub_10020aa0 at 0x10020aa0 (33 instructions)

```
0x10020aa0: mov eax,[esp+4]
0x10020aa4: push ebx
0x10020aa5: xor ebx,ebx
0x10020aa7: cmp eax,ebx
0x10020aa9: push esi
0x10020aaa: mov esi,ecx
0x10020aac: je short 10020AB8h
0x10020aae: mov [esi],eax
0x10020ab0: mov [esi+200h],bl
0x10020ab6: jmp short 10020ADAh
0x10020ab8: push 1D0h
0x10020abd: call 10021FDEh
0x10020ac2: push 1D0h
0x10020ac7: push ebx
0x10020ac8: push eax
0x10020ac9: mov [esi],eax
0x10020acb: call 100228C0h
0x10020ad0: add esp,10h
0x10020ad3: mov byte ptr [esi+200h],1
0x10020ada: push 0F0h
0x10020adf: lea eax,[esi+10Ch]
0x10020ae5: push ebx
0x10020ae6: push eax
0x10020ae7: mov [esi+4],bl
0x10020aea: mov [esi+108h],ebx
0x10020af0: mov [esi+104h],ebx
0x10020af6: mov [esi+1FCh],ebx
0x10020afc: call 100228C0h
0x10020b01: add esp,0Ch
0x10020b04: mov eax,esi
0x10020b06: pop esi
0x10020b07: pop ebx
0x10020b08: ret 4
```

`sub_10020aa0` is called by sub_1001b1a0. It delegates to j_ORDINAL 762, j_memset (2 callees).

12.412 sub_10021af0 at 0x10021af0 (183 instructions)

```
0x10021af0: sub esp,0Ch
0x10021af3: push ebx
0x10021af4: mov ebx,[esp+14h]
0x10021af8: mov eax,[ebx]
0x10021afa: mov edx,[eax+2Ch]
0x10021afd: push esi
0x10021afe: mov esi,ecx
0x10021b00: mov ecx,ebx
0x10021b02: mov dword ptr [esp+8],0
0x10021b0a: call edx
```

```
0x10021b0c: cmp eax,0FFh
0x10021b11: jne near ptr 10021F65h
0x10021b17: mov eax,[ebx]
0x10021b19: mov edx,[eax+2Ch]
0x10021b1c: mov ecx,ebx
0x10021b1e: call edx
0x10021b20: cmp eax,0D8h
0x10021b25: jne near ptr 10021F65h
0x10021b2b: cmp dword ptr [esi+1FCh],14h
0x10021b32: push ebp
0x10021b33: push edi
0x10021b34: jge near ptr 10021DD5h
0x10021b3a: lea ebx,[ebx]
0x10021b40: xor edi,edi
0x10021b42: mov eax,[ebx]
0x10021b44: mov edx,[eax+2Ch]
0x10021b47: mov ecx,ebx
0x10021b49: call edx
0x10021b4b: cmp eax,0FFh
0x10021b50: mov [esp+18h],eax
0x10021b54: jne short 10021BA7h
0x10021b56: cmp edi,6
0x10021b59: jge near ptr 10021DF2h
0x10021b5f: add edi,1
0x10021b62: cmp edi,7
0x10021b65: jl short 10021B42h
0x10021b67: mov edx,ds:[100FF624h]
0x10021b6d: mov [esi+4],edx
0x10021b70: mov eax,ds:[100FF628h]
0x10021b75: mov [esi+8],eax
0x10021b78: mov ecx,ds:[100FF62Ch]
0x10021b7e: mov [esi+0Ch],ecx
0x10021b81: mov edx,ds:[100FF630h]
0x10021b87: mov [esi+10h],edx
0x10021b8a: mov eax,ds:[100FF634h]
0x10021b8f: mov [esi+14h],eax
0x10021b92: mov ecx,ds:[100FF638h]
0x10021b98: pop edi
0x10021b99: pop ebp
0x10021b9a: mov [esi+18h],ecx
0x10021b9d: pop esi
0x10021b9e: xor al,al
0x10021ba0: pop ebx
0x10021ba1: add esp,0Ch
0x10021ba4: ret 8
0x10021ba7: mov ecx,[esi+1FCh]
0x10021bad: lea ecx,[ecx+ecx*2]
0x10021bb0: mov [esi+ecx*4+110h],eax
0x10021bb7: mov edx,[ebx]
0x10021bb9: mov eax,[edx+2Ch]
0x10021bbc: mov ecx,ebx
0x10021bbe: call eax
0x10021bc0: mov edx,[ebx]
0x10021bc2: mov [esp+20h],eax
0x10021bc6: mov eax,[edx+2Ch]
0x10021bc9: mov ecx,ebx
0x10021bcb: call eax
0x10021bcd: mov edi,[esp+20h]
0x10021bd1: shl edi,8
0x10021bd4: or edi,eax
0x10021bd6: cmp edi,2
0x10021bd9: mov [esp+14h],eax
0x10021bdd: jl near ptr 10021E0Ch
0x10021be3: mov eax,[esi+1FCh]
0x10021be9: add eax,17h
0x10021bec: lea ecx,[eax+eax*2]
0x10021bef: push edi
```

```
0x10021bf0: mov [esi+ecx*4],edi
0x10021bf3: call dword ptr ds:[100FE550h]
0x10021bf9: mov ebp,eax
0x10021bfb: add esp,4
0x10021bfe: test ebp,ebp
0x10021c00: je near ptr 10021E45h
0x10021c06: mov eax,[esi+1FCh]
0x10021c0c: mov cl,[esp+14h]
0x10021c10: lea edx,[eax+eax*2]
0x10021c13: mov al,[esp+20h]
0x10021c17: mov [esi+edx*4+10Ch],ebp
0x10021c1e: mov [ebp],al
0x10021c21: lea eax,[edi-2]
0x10021c24: push eax
0x10021c25: mov [ebp+1],cl
0x10021c28: mov edx,[ebx]
0x10021c2a: lea ecx,[ebp+2]
0x10021c2d: push 1
0x10021c2f: mov [esp+1Ch],eax
0x10021c33: mov eax,[edx+8]
0x10021c36: mov [esp+28h],ecx
0x10021c3a: push ecx
0x10021c3b: mov ecx,ebx
0x10021c3d: call eax
0x10021c3f: lea ecx,[edi-2]
0x10021c42: cmp eax,ecx
0x10021c44: jne near ptr 10021E62h
0x10021c4a: mov edx,[esp+18h]
0x10021c4e: add dword ptr [esi+1FCh],1
0x10021c55: mov ecx,[esi+1FCh]
0x10021c5b: lea eax,[edx-0C0h]
0x10021c61: cmp eax,3Eh
0x10021c64: ja near ptr 10021DC8h
0x10021c6a: movzx eax,byte ptr [eax+10021F8Ch]
0x10021c71: jmp dword ptr [eax*4+10021F70h]
0x10021dc8: cmp dword ptr [esi+1FCh],14h
0x10021dcf: jl near ptr 10021B40h
0x10021dd5: lea edi,[esi+4]
0x10021dd8: mov ecx,7
0x10021ddd: mov esi,100FF5FCh
0x10021de2: rep movsd
0x10021de4: movsw
0x10021de6: pop edi
0x10021de7: pop ebp
0x10021de8: pop esi
0x10021de9: xor al,al
0x10021deb: pop ebx
0x10021dec: add esp,0Ch
0x10021def: ret 8
0x10021df2: push 100FF5E4h
0x10021df7: call dword ptr ds:[100FE3E0h]
0x10021dfd: add esp,4
0x10021e00: pop edi
0x10021e01: pop ebp
0x10021e02: pop esi
0x10021e03: xor al,al
0x10021e05: pop ebx
0x10021e06: add esp,0Ch
0x10021e09: ret 8
0x10021e0c: mov edx,ds:[100FF5D4h]
0x10021e12: mov [esi+4],edx
0x10021e15: mov eax,ds:[100FF5D8h]
0x10021e1a: mov [esi+8],eax
0x10021e1d: mov ecx,ds:[100FF5DCh]
0x10021e23: mov [esi+0Ch],ecx
0x10021e26: mov dx,ds:[100FF5E0h]
0x10021e2d: mov [esi+10h],dx
```

```
0x10021e31: mov al,ds:[100FF5E2h]
0x10021e36: pop edi
0x10021e37: pop ebp
0x10021e38: mov [esi+12h],al
0x10021e3b: pop esi
0x10021e3c: xor al,al
0x10021e3e: pop ebx
0x10021e3f: add esp,0Ch
0x10021e42: ret 8
0x10021e45: lea edi,[esi+4]
0x10021e48: mov ecx,6
0x10021e4d: mov esi,100FF5B8h
0x10021e52: rep movsd
0x10021e54: movsw
0x10021e56: pop edi
0x10021e57: pop ebp
0x10021e58: pop esi
0x10021e59: xor al,al
0x10021e5b: pop ebx
0x10021e5c: add esp,0Ch
0x10021e5f: ret 8
0x10021e62: lea edi,[esi+4]
0x10021e65: mov ecx,5
0x10021e6a: mov esi,100FF5A0h
0x10021e6f: rep movsd
0x10021e71: movsw
0x10021e73: movsb
0x10021e74: pop edi
0x10021e75: pop ebp
0x10021e76: pop esi
0x10021e77: xor al,al
0x10021e79: pop ebx
0x10021e7a: add esp,0Ch
0x10021e7d: ret 8
0x10021f65: pop esi
0x10021f66: xor al,al
0x10021f68: pop ebx
0x10021f69: add esp,0Ch
0x10021f6c: ret 8
```

`sub_10021af0` is called by sub_1001b1a0. It works with the text "Premature end of file?", "Could not allocate memory", "invalid marker". It calls nothing else.

12.413 j_ORDINAL 764 at 0x10021fcc (1 instructions)

```
0x10021fcc: jmp dword ptr ds:[100FE2BCh]
```

`j_ORDINAL 764` is called by ?RemoveAllToolButtons@CCustomTitle@@QAEXXZ, sub_10002c40, sub_10003a40, sub_100072e0, sub_10008570, sub_10009950, sub_1000aca0, sub_1000ad90, sub_1000af00, sub_1000afb0, sub_1000c480, sub_1000c970, sub_1000d330, sub_1000d430, sub_1000e4a0, sub_1000e550, sub_10018d20, sub_10019690, sub_10019e70, sub_1001a5b0, sub_1001b350. It calls nothing else.

12.414 j_ORDINAL 1079 at 0x10021fd2 (1 instructions)

```
0x10021fd2: jmp dword ptr ds:[100FE390h]
```

`j_ORDINAL 1079` is called by sub_10002580, sub_10008290, sub_100083e0, sub_100089a0, sub_10009c10, sub_10009c50, sub_10009db0, sub_1000a3c0, sub_1000a510, sub_1000ab10, sub_1000b5a0. It calls nothing else.

12.415 j_ORDINAL 1093 at 0x10021fd8 (1 instructions)

```
0x10021fd8: jmp dword ptr ds:[100FE38Ch]
```

`j_ORDINAL 1093` is called by sub_10001030. It calls nothing else.

12.416 j_ORDINAL 762 at 0x10021fde (1 instructions)

```
0x10021fde: jmp dword ptr ds:[100FE388h]
```

`j_ORDINAL 762` is called by sub_10001030, ??0CCustomTitle@@QAE@XZ, ??0CCustomTitle@@QAE@W4WTitleTheme@@@Z, sub_10003940, sub_100067c0, sub_10008df0, sub_10009450, sub_10009bb0, sub_1000aca0, sub_1000ad90, sub_1000c700, sub_1000d330, sub_1000d430, sub_1000d550, sub_1000f760, sub_10019590, sub_10019e70, sub_1001a5b0, sub_1001b1a0, sub_1001b270, sub_10020aa0. It calls nothing else.

12.417 j_ORDINAL 371 at 0x10021fe4 (1 instructions)

```
0x10021fe4: jmp dword ptr ds:[100FE384h]
```

`j_ORDINAL 371` is called by sub_10001030. It calls nothing else.

12.418 j_ORDINAL 1168 at 0x10021fea (1 instructions)

```
0x10021fea: jmp dword ptr ds:[100FE380h]
```

`j_ORDINAL 1168` is called by sub_10001030. It calls nothing else.

12.419 j_ORDINAL 1925 at 0x10021ff6 (1 instructions)

```
0x10021ff6: jmp dword ptr ds:[100FE370h]
```

`j_ORDINAL 1925` is called by sub_10001370, sub_10003820, sub_100075d0, sub_10007670, sub_10008050, sub_1000a890, sub_1000e2f0. It calls nothing else.

12.420 j_ORDINAL 1271 at 0x10021ffc (1 instructions)

```
0x10021ffc: jmp dword ptr ds:[100FE36Ch]
```

`j_ORDINAL 1271` is called by ?SetMenuBarDownBitmap@CCustomTitle@@QAEHPAVCBitmap@@@Z, sub_10001d00, sub_100032e0, sub_10003820, sub_10005fd0, sub_100075d0, sub_10007670, sub_10008050, sub_10008290, sub_1000a3c0, sub_1000a510, sub_1000a890, sub_1000aa40, sub_1000bc60, sub_1000d0a0, sub_1000e2f0. It calls nothing else.

12.421 j_ORDINAL 1959 at 0x10022002 (1 instructions)

```
0x10022002: jmp dword ptr ds:[100FE364h]
```

`j_ORDINAL 1959` is called by sub_10007670, sub_1000e2f0. It calls nothing else.

12.422 j_ORDINAL 2255 at 0x10022008 (1 instructions)

```
0x10022008: jmp dword ptr ds:[100FE360h]
```

`j_ORDINAL 2255` is called by sub_10005fd0. It calls nothing else.

12.423 j_ORDINAL 5723 at 0x10022020 (1 instructions)

```
0x10022020: jmp dword ptr ds:[100FE350h]
```

`j_ORDINAL 5723` is called by sub_10001d00. It calls nothing else.

12.424 j_ORDINAL 347 at 0x1002202c (1 instructions)

```
0x1002202c: jmp dword ptr ds:[100FE348h]
```

`j_ORDINAL 347` is called by sub_10001d00, sub_100020f0, sub_10005fd0, sub_10007dd0, sub_10008df0, sub_1000bc60, sub_1000c0f0. It calls nothing else.

12.425 j_ORDINAL 602 at 0x10022032 (1 instructions)

```
0x10022032: jmp dword ptr ds:[100FE344h]
```

`j_ORDINAL 602` is called by sub_10001d00, sub_100020f0, sub_10005fd0, sub_10007dd0, sub_10008df0, sub_1000bc60, sub_1000c0f0. It calls nothing else.

12.426 j_ORDINAL 1270 at 0x10022038 (1 instructions)

```
0x10022038: jmp dword ptr ds:[100FE340h]
```

`j_ORDINAL 1270` is called by sub_10001d00, sub_100020f0, sub_100024b0, sub_10005fd0, sub_10008df0, sub_1000bc60, sub_1000c0f0. It calls nothing else.

12.427 j_ORDINAL 5633 at 0x1002203e (1 instructions)

```
0x1002203e: jmp dword ptr ds:[100FE33Ch]
```

`j_ORDINAL 5633` is called by sub_10001d00, sub_100020f0, sub_10002480, sub_10005fd0. It calls nothing else.

12.428 j_ORDINAL 6013 at 0x10022044 (1 instructions)

```
0x10022044: jmp dword ptr ds:[100FE338h]
```

`j_ORDINAL 6013` is called by sub_10001d00. It calls nothing else.

12.429 j_ORDINAL 1646 at 0x10022050 (1 instructions)

```
0x10022050: jmp dword ptr ds:[100FE330h]
```

`j_ORDINAL 1646` is called by sub_10009450. It calls nothing else.

12.430 j_ORDINAL 572 at 0x100221ac (1 instructions)

```
0x100221ac: jmp dword ptr ds:[100FE244h]
```

`j_ORDINAL 572` is called by sub_100024e0. It calls nothing else.

12.431 j_ORDINAL 1155 at 0x1002220c (1 instructions)

```
0x1002220c: jmp dword ptr ds:[100FE15Ch]
```

`j_ORDINAL 1155` is called by sub_10002580. It calls nothing else.

12.432 j_ORDINAL 1183 at 0x10022212 (1 instructions)

```
0x10022212: jmp dword ptr ds:[100FE160h]
```

`j_ORDINAL 1183` is called by sub_10002580. It calls nothing else.

12.433 j_ORDINAL 3752 at 0x1002221e (1 instructions)

```
0x1002221e: jmp dword ptr ds:[100FE16Ch]
```

`j_ORDINAL 3752` is called by sub_10003d20. It calls nothing else.

12.434 j_ORDINAL 567 at 0x1002222a (1 instructions)

```
0x1002222a: jmp dword ptr ds:[100FE174h]
```

`j_ORDINAL 567` is called by sub_10006310, sub_100079a0, sub_10007b10, sub_1000a2d0, sub_1000caf0. It calls nothing else.

12.435 j_ORDINAL 758 at 0x10022230 (1 instructions)

```
0x10022230: jmp dword ptr ds:[100FE178h]
```

`j_ORDINAL 758` is called by sub_10006310, sub_100079a0, sub_10007b10, sub_1000a2d0, sub_1000caf0. It calls nothing else.

12.436 j_ORDINAL 3395 at 0x10022236 (1 instructions)

```
0x10022236: jmp dword ptr ds:[100FE17Ch]
```

`j_ORDINAL 3395` is called by sub_10003670, sub_10004910, sub_100056c0, sub_10005fd0, sub_100063a0, sub_100079a0, sub_10007b10. It calls nothing else.

12.437 j_ORDINAL 5609 at 0x1002223c (1 instructions)

```
0x1002223c: jmp dword ptr ds:[100FE180h]
```

`j_ORDINAL 5609` is called by sub_100067c0. It calls nothing else.

12.438 j_ORDINAL 2361 at 0x10022242 (1 instructions)

```
0x10022242: jmp dword ptr ds:[100FE184h]
```

`j_ORDINAL 2361` is called by sub_10005680. It calls nothing else.

12.439 j_ORDINAL 6140 at 0x10022248 (1 instructions)

```
0x10022248: jmp dword ptr ds:[100FE188h]
```

`j_ORDINAL 6140` is called by sub_100063a0, sub_1000b820. It calls nothing else.

12.440 j_ORDINAL 2366 at 0x1002225a (1 instructions)

```
0x1002225a: jmp dword ptr ds:[100FE194h]
```

`j_ORDINAL 2366` is called by sub_100067c0, sub_10009450, sub_1000a2d0, sub_1000caf0. It calls nothing else.

12.441 j_ORDINAL 563 at 0x10022260 (1 instructions)

```
0x10022260: jmp dword ptr ds:[100FE19Ch]
```

`j_ORDINAL 563` is called by sub_100067c0. It calls nothing else.

12.442 j_ORDINAL 5630 at 0x10022266 (1 instructions)

```
0x10022266: jmp dword ptr ds:[100FE1A0h]
```

`j_ORDINAL 5630` is called by sub_10005fd0. It calls nothing else.

12.443 j_ORDINAL 6061 at 0x1002226c (1 instructions)

```
0x1002226c: jmp dword ptr ds:[100FE1A4h]
```

`j_ORDINAL 6061` is called by ?RefreshFrame@CCustomTitle@@QAEHXZ, sub_10003670, sub_10003ac0, sub_10003b30, sub_10003c30, sub_10003ca0, sub_10003d00. It calls nothing else.

12.444 j_ORDINAL 2362 at 0x10022272 (1 instructions)

```
0x10022272: jmp dword ptr ds:[100FE1A8h]
```

`j_ORDINAL 2362` is called by sub_10005fd0, sub_1000a0c0. It calls nothing else.

12.445 j_ORDINAL 2066 at 0x10022278 (1 instructions)

```
0x10022278: jmp dword ptr ds:[100FE1B0h]
```

`j_ORDINAL 2066` is called by sub_10007dd0. It calls nothing else.

12.446 j_ORDINAL 2365 at 0x1002227e (1 instructions)

```
0x1002227e: jmp dword ptr ds:[100FE1BCh]
```

`j_ORDINAL 2365` is called by ?GetMenu@CCustomTitle@@QAEPAVCMenu@@XZ, sub_100063a0, sub_100087c0, sub_10008df0, sub_10008ff0. It calls nothing else.

12.447 j_ORDINAL 6058 at 0x10022284 (1 instructions)

```
0x10022284: jmp dword ptr ds:[100FE1C0h]
```

`j_ORDINAL 6058` is called by sub_10005fd0. It calls nothing else.

12.448 j_ORDINAL 5638 at 0x1002228a (1 instructions)

```
0x1002228a: jmp dword ptr ds:[100FE1C8h]
```

`j_ORDINAL 5638` is called by sub_10005fd0, sub_1000a2d0, sub_1000bc60, sub_1000c0f0, sub_1000caf0. It calls nothing else.

12.449 j_ORDINAL 6033 at 0x10022290 (1 instructions)

```
0x10022290: jmp dword ptr ds:[100FE1CCh]
```

`j_ORDINAL 6033` is called by sub_10005fd0. It calls nothing else.

12.450 j_ORDINAL 5727 at 0x10022296 (1 instructions)

```
0x10022296: jmp dword ptr ds:[100FE1D0h]
```

`j_ORDINAL 5727` is called by sub_10005fd0, sub_10007dd0, sub_1000a680, sub_1000bde0, sub_1000cd90. It calls nothing else.

12.451 j_ORDINAL 3756 at 0x1002229c (1 instructions)

```
0x1002229c: jmp dword ptr ds:[100FE1D4h]
```

`j_ORDINAL 3756` is called by sub_100039b0. It calls nothing else.

12.452 j_ORDINAL 416 at 0x100222a2 (1 instructions)

```
0x100222a2: jmp dword ptr ds:[100FE1D8h]
```

`j_ORDINAL 416` is called by sub_10008050, sub_1000a1c0, sub_1000a890. It calls nothing else.

12.453 j_ORDINAL 651 at 0x100222a8 (1 instructions)

```
0x100222a8: jmp dword ptr ds:[100FE1DCh]
```

`j_ORDINAL 651` is called by sub_1000a270, sub_1000afb0. It calls nothing else.

12.454 j_ORDINAL 1946 at 0x100222c0 (1 instructions)

```
0x100222c0: jmp dword ptr ds:[100FE1F0h]
```

`j_ORDINAL 1946` is called by sub_100056c0, sub_1000b250. It calls nothing else.

12.455 j_ORDINAL 6086 at 0x100222c6 (1 instructions)

```
0x100222c6: jmp dword ptr ds:[100FE1F4h]
```

`j_ORDINAL 6086` is called by sub_10009450. It calls nothing else.

12.456 j_ORDINAL 5829 at 0x100222cc (1 instructions)

```
0x100222cc: jmp dword ptr ds:[100FE1F8h]
```

`j_ORDINAL 5829` is called by sub_10009450. It calls nothing else.

12.457 j_ORDINAL 6749 at 0x100222d2 (1 instructions)

```
0x100222d2: jmp dword ptr ds:[100FE1FCh]
```

`j_ORDINAL 6749` is called by sub_10009c50, sub_10009db0. It calls nothing else.

12.458 j_ORDINAL 4119 at 0x100222d8 (1 instructions)

```
0x100222d8: jmp dword ptr ds:[100FE200h]
```

`j_ORDINAL 4119` is called by sub_10009450. It calls nothing else.

12.459 j_ORDINAL 1921 at 0x100222de (1 instructions)

```
0x100222de: jmp dword ptr ds:[100FE204h]
```

`j_ORDINAL 1921` is called by sub_10008290, sub_100083e0, sub_1000a2d0, sub_1000a3c0, sub_1000a510, sub_1000ab10. It calls nothing else.

12.460 j_ORDINAL 1555 at 0x100222e4 (1 instructions)

```
0x100222e4: jmp dword ptr ds:[100FE208h]
```

`j_ORDINAL_1555` is called by sub_10008290, sub_100083e0, sub_1000a3c0, sub_1000a510, sub_1000ab10. It calls nothing else.

12.461 j_ORDINAL 1176 at 0x100222ea (1 instructions)

```
0x100222ea: jmp dword ptr ds:[100FE20Ch]
```

`j_ORDINAL_1176` is called by sub_10009c50, sub_10009db0. It calls nothing else.

12.462 j_ORDINAL 1274 at 0x100222f0 (1 instructions)

```
0x100222f0: jmp dword ptr ds:[100FE210h]
```

`j_ORDINAL_1274` is called by sub_100087c0, sub_1000b250. It calls nothing else.

12.463 j_ORDINAL 2932 at 0x100222f6 (1 instructions)

```
0x100222f6: jmp dword ptr ds:[100FE218h]
```

`j_ORDINAL_2932` is called by sub_10008df0, sub_10008ff0. It calls nothing else.

12.464 j_ORDINAL 6751 at 0x100222fc (1 instructions)

```
0x100222fc: jmp dword ptr ds:[100FE21Ch]
```

`j_ORDINAL_6751` is called by sub_10009d21, sub_10009e64. It calls nothing else.

12.465 j_ORDINAL 502 at 0x10022302 (1 instructions)

```
0x10022302: jmp dword ptr ds:[100FE224h]
```

`j_ORDINAL_502` is called by sub_1000a680, sub_1000cd90. It calls nothing else.

12.466 j_ORDINAL 5637 at 0x1002230e (1 instructions)

```
0x1002230e: jmp dword ptr ds:[100FE22Ch]
```

`j_ORDINAL_5637` is called by sub_1000a680, sub_1000cd90. It calls nothing else.

12.467 j_ORDINAL 5636 at 0x10022314 (1 instructions)

```
0x10022314: jmp dword ptr ds:[100FE230h]
```

`j_ORDINAL_5636` is called by sub_1000a680, sub_1000cd90. It calls nothing else.

12.468 j_ORDINAL 4117 at 0x1002231a (1 instructions)

```
0x1002231a: jmp dword ptr ds:[100FE23Ch]
```

`j_ORDINAL_4117` is called by sub_1000cd90. It calls nothing else.

12.469 j_ORDINAL 3995 at 0x10022320 (1 instructions)

```
0x10022320: jmp dword ptr ds:[100FE240h]
```

`j_ORDINAL_3995` is called by sub_1000cd90. It calls nothing else.

12.470 sub_10022326 at 0x10022326 (57 instructions)

```
0x10022326: cmp ecx,ds:[10136220h]
0x1002232c: jne short 10022330h
0x1002232e: rep ret
0x10022330: jmp near ptr 100229F4h
0x100229f4: push ebp
0x100229f5: mov ebp,esp
0x100229f7: sub esp,328h
0x100229fd: mov ds:[10138F78h],eax
0x10022a02: mov ds:[10138F74h],ecx
0x10022a08: mov ds:[10138F70h],edx
0x10022a0e: mov ds:[10138F6Ch],ebx
0x10022a14: mov ds:[10138F68h],esi
0x10022a1a: mov ds:[10138F64h],edi
0x10022a20: mov ds:[10138F90h],ss
0x10022a27: mov ds:[10138F84h],cs
0x10022a2e: mov ds:[10138F60h],ds
0x10022a35: mov ds:[10138F5Ch],es
0x10022a3c: mov ds:[10138F58h],fs
0x10022a43: mov ds:[10138F54h],gs
0x10022a4a: pushfd
0x10022a4b: pop dword ptr ds:[10138F88h]
0x10022a51: mov eax,[ebp]
0x10022a54: mov ds:[10138F7Ch],eax
0x10022a59: mov eax,[ebp+4]
0x10022a5c: mov ds:[10138F80h],eax
0x10022a61: lea eax,[ebp+8]
0x10022a64: mov ds:[10138F8Ch],eax
0x10022a69: mov eax,[ebp-320h]
0x10022a6f: mov dword ptr ds:[10138EC8h],10001h
0x10022a79: mov eax,ds:[10138F80h]
0x10022a7e: mov ds:[10138E7Ch],eax
0x10022a83: mov dword ptr ds:[10138E70h],0C0000409h
0x10022a8d: mov dword ptr ds:[10138E74h],1
0x10022a97: mov eax,ds:[10136220h]
0x10022a9c: mov [ebp-328h],eax
0x10022aa2: mov eax,ds:[10136224h]
0x10022aa7: mov [ebp-324h],eax
0x10022aad: call dword ptr ds:[100FE0C0h]
0x10022ab3: mov ds:[10138EC0h],eax
0x10022ab8: push 1
0x10022aba: call 10022E96h
0x10022abf: pop ecx
0x10022ac0: push 0
0x10022ac2: call dword ptr ds:[100FE0C4h]
0x10022ac8: push 100FF6BCh
0x10022acd: call dword ptr ds:[100FE0C8h]
0x10022ad3: cmp dword ptr ds:[10138EC0h],0
0x10022ada: jne short 10022AE4h
0x10022adc: push 1
0x10022ade: call 10022E96h
0x10022ae3: pop ecx
0x10022ae4: push 0C0000409h
0x10022ae9: call dword ptr ds:[100FE0CCh]
0x10022aef: push eax
0x10022af0: call dword ptr ds:[100FE0D0h]
0x10022af6: leave
0x10022af7: ret
```

`sub_10022326` is called by sub_10001370, sub_10002070, sub_100032e0, sub_10008050, sub_1000a890, sub_1000bc60, sub_1000cc10, sub_1000d0a0, sub_100102b0, sub_100103d0, sub_10011b00, sub_10016040, sub_10017680, sub_10019770, sub_1001a5b0, sub_1001b350, sub_1001b970, png_chunk_warning, png_chunk_benign_error, png_sig_cmp, sub_10024050, sub_100245f0, png_create_read_struct_2, png_read_info, png_read_row, png_read_end, png_create_write_struct_2, png_set_sCAL, png_set_sCAL_fixed, sub_1002cb30, sub_1002cbe0, sub_1002cff0, sub_1002d150, sub_1002d670, sub_1002dfd0, sub_1002e100, sub_1002e700, sub_1002ec10,

sub_1002f170, sub_1002f400, sub_1002f650, sub_1002f7e0, sub_1002f930, sub_1002ffb0, sub_10030750, sub_100309c0, sub_100354b0, png_write_sig, png_write_chunk_start, sub_10035b60, sub_10035e80, sub_10035f80, sub_10036050, sub_10036250, sub_10036310, sub_10036450, sub_10036610, sub_10036c30, sub_10037020, sub_10037160, sub_100371c0, sub_10037220, sub_10037290, sub_10037430, sub_100375c0, sub_100376c0, sub_100377c0, sub_10037930, sub_10037ab0, sub_10037b40, sub_10037c20, sub_10037cb0, sub_10065490, sub_1006df2c, sub_1006eba4, sub_1006f20b, sub_10071192, sub_100d3060, sub_100d4ce0, sub_100d5e70, sub_100d64f0, sub_100d6630, sub_100d66e0, sub_100d6900, sub_100dd260, sub_100e2b50, sub_100e3d60, sub_100e42b0, sub_100e5130, sub_100e53f0, sub_100e5c30, sub_100e64c0, sub_100e66a0, sub_100e8a00, sub_100eb170, sub_100eb620, sub_100fc385, sub_100fc3a8, sub_100fc3f1, sub_100fc4db, sub_100fc510, sub_100fc530, sub_100fc550, sub_100fc570, sub_100fc5a0, sub_100fc5c0, sub_100fc5f0, sub_100fc610, sub_100fc630, sub_100fc658, sub_100fc6d1, sub_100fc70b, sub_100fc768, sub_100fc7a6, sub_100fc7e0, sub_100fc839, sub_100fc868, sub_100fc899, sub_100fc94b, sub_100fc998, sub_100fc9c5, sub_100fc9e9, sub_100fca18, sub_100fca58, sub_100fca89, sub_100fcb18, sub_100fcb4b, sub_100fcb75, sub_100fcb98, sub_100fcbd9, sub_100fcc17, sub_100fcca8, sub_100fccd8, sub_100fcd60, sub_100fcd88, sub_100fcdb9, sub_100fcdf1, sub_100fce18, sub_100fcea8, sub_100fcedb, sub_100fcf69, sub_100fcf9b, sub_100fcfd8, sub_100fd061, sub_100fd0a8, sub_100fd0dc, sub_100fd10b, sub_100fd15c, sub_100fd198, sub_100fd205, sub_100fd34a, sub_100fd38b, sub_100fd3c0, sub_100fd418, sub_100fd448, sub_100fd478. It accesses memory-mapped peripheral registers at 0xc0000409. It delegates to j__crt_debugger_hook (1 callees).

12.471 sub_1002237e at 0x1002237e (14 instructions)

```
0x1002237e: push 14h
0x10022380: push 10130D38h
0x10022385: call 10022B00h
0x1002238a: and dword ptr [ebp-4],0
0x1002238e: dec dword ptr [ebp+10h]
0x10022391: js short 100223CDh
0x10022393: mov ecx,[ebp+8]
0x10022396: sub ecx,[ebp+0Ch]
0x10022399: mov [ebp+8],ecx
0x1002239c: call dword ptr [ebp+14h]
0x1002239f: jmp short 1002238Eh
0x100223cd: mov dword ptr [ebp-4],0FFFFFFFEh
0x100223d4: call 10022B45h
0x100223d9: ret 10h
```

`sub_1002237e` is called by sub_10022427, sub_10022556. It delegates to sub_10022b00, sub_10022b45 (2 callees).

12.472 sub_100223dc at 0x100223dc (20 instructions)

```
0x100223dc: push 0Ch
0x100223de: push 10130D58h
0x100223e3: call 10022B00h
0x100223e8: and dword ptr [ebp-1Ch],0
0x100223ec: mov esi,[ebp+0Ch]
0x100223ef: mov eax,esi
0x100223f1: imul eax,[ebp+10h]
0x100223f5: add [ebp+8],eax
0x100223f8: and dword ptr [ebp-4],0
0x100223fc: dec dword ptr [ebp+10h]
0x100223ff: js short 1002240Ch
0x10022401: sub [ebp+8],esi
0x10022404: mov ecx,[ebp+8]
0x10022407: call dword ptr [ebp+14h]
0x1002240a: jmp short 100223FCh
0x1002240c: mov dword ptr [ebp-1Ch],1
0x10022413: mov dword ptr [ebp-4],0FFFFFFFEh
0x1002241a: call 10022427h
0x1002241f: call 10022B45h
0x10022424: ret 10h
```

`sub_100223dc` is called by ?RemoveAllToolButtons@CCustomTitle@@QAEXXZ, sub_1000ca80. It delegates to sub_10022427, sub_10022b00, sub_10022b45 (3 callees).

12.473 sub_10022427 at 0x10022427 (8 instructions)

```
0x10022427: cmp dword ptr [ebp-1Ch],0
0x1002242b: jne short 1002243Eh
0x1002242d: push dword ptr [ebp+14h]
0x10022430: push dword ptr [ebp+10h]
0x10022433: push dword ptr [ebp+0Ch]
0x10022436: push dword ptr [ebp+8]
0x10022439: call 1002237Eh
0x1002243e: ret
```

`sub_10022427` is called by sub_100223dc. It delegates to sub_1002237e (1 callees).

12.474 sub_10022452 at 0x10022452 (49 instructions)

```
0x10022452: push 14h
0x10022454: push 10130D78h
0x10022459: call 10022B00h
0x1002245e: push dword ptr ds:[10145478h]
0x10022464: mov esi,ds:[100FE3C4h]
0x1002246a: call esi
0x1002246c: pop ecx
0x1002246d: mov [ebp-1Ch],eax
0x10022470: cmp eax,0FFFFFFFFh
0x10022473: jne short 10022481h
0x10022475: push dword ptr [ebp+8]
0x10022478: call dword ptr ds:[100FE3C8h]
0x1002247e: pop ecx
0x1002247f: jmp short 100224E8h
0x10022481: push 8
0x10022483: call 10022B88h
0x10022488: pop ecx
0x10022489: and dword ptr [ebp-4],0
0x1002248d: push dword ptr ds:[10145478h]
0x10022493: call esi
0x10022495: mov [ebp-1Ch],eax
0x10022498: push dword ptr ds:[10145474h]
0x1002249e: call esi
0x100224a0: pop ecx
0x100224a1: pop ecx
0x100224a2: mov [ebp-20h],eax
0x100224a5: lea eax,[ebp-20h]
0x100224a8: push eax
0x100224a9: lea eax,[ebp-1Ch]
0x100224ac: push eax
0x100224ad: push dword ptr [ebp+8]
0x100224b0: mov esi,ds:[100FE3D0h]
0x100224b6: call esi
0x100224b8: pop ecx
0x100224b9: push eax
0x100224ba: call 10022B82h
0x100224bf: mov [ebp-24h],eax
0x100224c2: push dword ptr [ebp-1Ch]
0x100224c5: call esi
0x100224c7: mov ds:[10145478h],eax
0x100224cc: push dword ptr [ebp-20h]
0x100224cf: call esi
0x100224d1: add esp,14h
0x100224d4: mov ds:[10145474h],eax
0x100224d9: mov dword ptr [ebp-4],0FFFFFFFEh
0x100224e0: call 100224EEh
0x100224e5: mov eax,[ebp-24h]
```

```
0x100224e8: call 10022B45h
0x100224ed: ret
```

`sub_10022452` is called by sub_100224f7. It delegates to sub_100224ee, sub_10022b00, sub_10022b45 (5 callees).

12.475 sub_100224ee at 0x100224ee (4 instructions)

```
0x100224ee: push 8
0x100224f0: call 10022B7Ch
0x100224f5: pop ecx
0x100224f6: ret
```

`sub_100224ee` is called by sub_10022452. It delegates to j__unlock (1 callees).

12.476 sub_100224f7 at 0x100224f7 (8 instructions)

```
0x100224f7: push dword ptr [esp+4]
0x100224fb: call 10022452h
0x10022500: neg eax
0x10022502: sbb eax,eax
0x10022504: neg eax
0x10022506: pop ecx
0x10022507: dec eax
0x10022508: ret
```

`sub_100224f7` is called by sub_100ae1a0. It delegates to sub_10022452 (1 callees).

12.477 sub_10022509 at 0x10022509 (22 instructions)

```
0x10022509: push 10h
0x1002250b: push 10130D98h
0x10022510: call 10022B00h
0x10022515: xor eax,eax
0x10022517: mov [ebp-20h],eax
0x1002251a: mov [ebp-4],eax
0x1002251d: mov [ebp-1Ch],eax
0x10022520: mov eax,[ebp-1Ch]
0x10022523: cmp eax,[ebp+10h]
0x10022526: jge short 1002253Bh
0x10022528: mov esi,[ebp+8]
0x1002252b: mov ecx,esi
0x1002252d: call dword ptr [ebp+14h]
0x10022530: add esi,[ebp+0Ch]
0x10022533: mov [ebp+8],esi
0x10022536: inc dword ptr [ebp-1Ch]
0x10022539: jmp short 10022520h
0x1002253b: mov dword ptr [ebp-20h],1
0x10022542: mov dword ptr [ebp-4],0FFFFFFFEh
0x10022549: call 10022556h
0x1002254e: call 10022B45h
0x10022553: ret 14h
```

`sub_10022509` is called by sub_1000c9c0. It delegates to sub_10022556, sub_10022b00, sub_10022b45 (3 callees).

12.478 sub_10022556 at 0x10022556 (8 instructions)

```
0x10022556: cmp dword ptr [ebp-20h],0
0x1002255a: jne short 1002256Dh
0x1002255c: push dword ptr [ebp+18h]
0x1002255f: push dword ptr [ebp-1Ch]
0x10022562: push dword ptr [ebp+0Ch]
0x10022565: push dword ptr [ebp+8]
```

```
0x10022568: call 1002237Eh
0x1002256d: ret
```

`sub_10022556` is called by sub_10022509. It delegates to sub_1002237e (1 callees).

12.479 sub_100225bb at 0x100225bb (141 instructions)

```
0x100225bb: mov eax,[esp+8]
0x100225bf: push ebp
0x100225c0: xor ebp,ebp
0x100225c2: cmp eax,ebp
0x100225c4: jne short 100225D4h
0x100225c6: cmp ds:[10138E68h],ebp
0x100225cc: jle short 10022608h
0x100225ce: dec dword ptr ds:[10138E68h]
0x100225d4: cmp eax,1
0x100225d7: mov ecx,ds:[100FE3ACh]
0x100225dd: mov ecx,[ecx]
0x100225df: push ebx
0x100225e0: push esi
0x100225e1: push edi
0x100225e2: mov ds:[10145464h],ecx
0x100225e8: jne near ptr 100226C2h
0x100225ee: mov eax,fs:[18h]
0x100225f4: mov esi,[eax+4]
0x100225f7: mov ebx,ds:[100FE0D4h]
0x100225fd: mov [esp+18h],ebp
0x10022601: mov edi,1014546Ch
0x10022606: jmp short 1002261Eh
0x10022608: xor eax,eax
0x1002260a: jmp near ptr 1002277Ah
0x1002260f: cmp eax,esi
0x10022611: je short 10022629h
0x10022613: push 3E8h
0x10022618: call dword ptr ds:[100FE0FCh]
0x1002261e: push ebp
0x1002261f: push esi
0x10022620: push edi
0x10022621: call ebx
0x10022623: cmp eax,ebp
0x10022625: jne short 1002260Fh
0x10022627: jmp short 10022631h
0x10022629: mov dword ptr [esp+18h],1
0x10022631: mov eax,ds:[10145468h]
0x10022636: test eax,eax
0x10022638: push 2
0x1002263a: pop esi
0x1002263b: je short 10022646h
0x1002263d: push 1Fh
0x1002263f: call 10022D38h
0x10022644: jmp short 10022682h
0x10022646: push 100FE6F0h
0x1002264b: push 100FE6E4h
0x10022650: mov dword ptr ds:[10145468h],1
0x1002265a: call 10022D32h
0x1002265f: test eax,eax
0x10022661: pop ecx
0x10022662: pop ecx
0x10022663: je short 1002266Ch
0x10022665: xor eax,eax
0x10022667: jmp near ptr 10022777h
0x1002266c: push 100FE6E0h
0x10022671: push 100FE6CCh
0x10022676: call 10022D2Ch
0x1002267b: pop ecx
0x1002267c: mov ds:[10145468h],esi
```

```
0x10022682: cmp [esp+1Ch],ebp
0x10022686: pop ecx
0x10022687: jne short 10022691h
0x10022689: push ebp
0x1002268a: push edi
0x1002268b: call dword ptr ds:[100FE0D8h]
0x10022691: cmp ds:[10145470h],ebp
0x10022697: je short 100226B7h
0x10022699: push 10145470h
0x1002269e: call 10022C70h
0x100226a3: test eax,eax
0x100226a5: pop ecx
0x100226a6: je short 100226B7h
0x100226a8: push dword ptr [esp+1Ch]
0x100226ac: push esi
0x100226ad: push dword ptr [esp+1Ch]
0x100226b1: call dword ptr ds:[10145470h]
0x100226b7: inc dword ptr ds:[10138E68h]
0x100226bd: jmp near ptr 10022774h
0x100226c2: cmp eax,ebp
0x100226c4: jne near ptr 10022774h
0x100226ca: mov esi,ds:[100FE0D4h]
0x100226d0: mov edi,1014546Ch
0x100226d5: jmp short 100226E2h
0x100226d7: push 3E8h
0x100226dc: call dword ptr ds:[100FE0FCh]
0x100226e2: push ebp
0x100226e3: push 1
0x100226e5: push edi
0x100226e6: call esi
0x100226e8: test eax,eax
0x100226ea: jne short 100226D7h
0x100226ec: mov eax,ds:[10145468h]
0x100226f1: cmp eax,2
0x100226f4: je short 10022700h
0x100226f6: push 1Fh
0x100226f8: call 10022D38h
0x100226fd: pop ecx
0x100226fe: jmp short 10022774h
0x10022700: push dword ptr ds:[10145478h]
0x10022706: mov ebx,ds:[100FE3C4h]
0x1002270c: call ebx
0x1002270e: mov ebp,eax
0x10022710: test ebp,ebp
0x10022712: pop ecx
0x10022713: je short 10022761h
0x10022715: push dword ptr ds:[10145474h]
0x1002271b: call ebx
0x1002271d: pop ecx
0x1002271e: mov esi,eax
0x10022720: jmp short 10022742h
0x10022722: cmp dword ptr [esi],0
0x10022725: je short 10022742h
0x10022727: mov eax,[esi]
0x10022729: mov [esp+18h],eax
0x1002272d: call dword ptr ds:[100FE3BCh]
0x10022733: cmp [esp+18h],eax
0x10022737: je short 10022742h
0x10022739: push dword ptr [esp+18h]
0x1002273d: call ebx
0x1002273f: pop ecx
0x10022740: call eax
0x10022742: sub esi,4
0x10022745: cmp esi,ebp
0x10022747: jae short 10022722h
0x10022749: push ebp
0x1002274a: call dword ptr ds:[100FE554h]
```

```
0x10022750: pop ecx
0x10022751: call dword ptr ds:[100FE3BCh]
0x10022757: mov ds:[10145474h],eax
0x1002275c: mov ds:[10145478h],eax
0x10022761: push 0
0x10022763: push edi
0x10022764: mov dword ptr ds:[10145468h],0
0x1002276e: call dword ptr ds:[100FE0D8h]
0x10022774: xor eax,eax
0x10022776: inc eax
0x10022777: pop edi
0x10022778: pop esi
0x10022779: pop ebx
0x1002277a: pop ebp
0x1002277b: ret 0Ch
```

`sub_100225bb` is called by sub_1002277e. It delegates to sub_10022c70, j__initterm, j__initterm_e (4 callees).

12.480 sub_1002277e at 0x1002277e (92 instructions)

```
0x1002277e: push 10h
0x10022780: push 10130DB8h
0x10022785: call 10022B00h
0x1002278a: mov edi,ecx
0x1002278c: mov esi,edx
0x1002278e: mov ebx,[ebp+8]
0x10022791: xor eax,eax
0x10022793: inc eax
0x10022794: mov [ebp-1Ch],eax
0x10022797: xor ecx,ecx
0x10022799: mov [ebp-4],ecx
0x1002279c: mov ds:[10136230h],esi
0x100227a2: mov [ebp-4],eax
0x100227a5: cmp esi,ecx
0x100227a7: jne short 100227B9h
0x100227a9: cmp ds:[10138E68h],ecx
0x100227af: jne short 100227B9h
0x100227b1: mov [ebp-1Ch],ecx
0x100227b4: jmp near ptr 10022870h
0x100227b9: cmp esi,eax
0x100227bb: je short 100227C2h
0x100227bd: cmp esi,2
0x100227c0: jne short 100227F0h
0x100227c2: mov eax,ds:[1012F3A8h]
0x100227c7: cmp eax,ecx
0x100227c9: je short 100227D3h
0x100227cb: push edi
0x100227cc: push esi
0x100227cd: push ebx
0x100227ce: call eax
0x100227d0: mov [ebp-1Ch],eax
0x100227d3: cmp dword ptr [ebp-1Ch],0
0x100227d7: je near ptr 10022870h
0x100227dd: push edi
0x100227de: push esi
0x100227df: push ebx
0x100227e0: call 100225BBh
0x100227e5: mov [ebp-1Ch],eax
0x100227e8: test eax,eax
0x100227ea: je near ptr 10022870h
0x100227f0: push edi
0x100227f1: push esi
0x100227f2: push ebx
0x100227f3: call 10001030h
0x100227f8: mov [ebp-1Ch],eax
0x100227fb: cmp esi,1
```

```
0x100227fe: jne short 10022824h
0x10022800: test eax,eax
0x10022802: jne short 10022824h
0x10022804: push edi
0x10022805: push eax
0x10022806: push ebx
0x10022807: call 10001030h
0x1002280c: push edi
0x1002280d: push 0
0x1002280f: push ebx
0x10022810: call 100225BBh
0x10022815: mov eax,ds:[1012F3A8h]
0x1002281a: test eax,eax
0x1002281c: je short 10022824h
0x1002281e: push edi
0x1002281f: push 0
0x10022821: push ebx
0x10022822: call eax
0x10022824: test esi,esi
0x10022826: je short 1002282Dh
0x10022828: cmp esi,3
0x1002282b: jne short 10022870h
0x1002282d: push edi
0x1002282e: push esi
0x1002282f: push ebx
0x10022830: call 100225BBh
0x10022835: test eax,eax
0x10022837: jne short 1002283Ch
0x10022839: and [ebp-1Ch],eax
0x1002283c: cmp dword ptr [ebp-1Ch],0
0x10022840: je short 10022870h
0x10022842: mov eax,ds:[1012F3A8h]
0x10022847: test eax,eax
0x10022849: je short 10022870h
0x1002284b: push edi
0x1002284c: push esi
0x1002284d: push ebx
0x1002284e: call eax
0x10022850: mov [ebp-1Ch],eax
0x10022853: jmp short 10022870h
0x10022870: and dword ptr [ebp-4],0
0x10022874: mov dword ptr [ebp-4],0FFFFFFFEh
0x1002287b: call 10022889h
0x10022880: mov eax,[ebp-1Ch]
0x10022883: call 10022B45h
0x10022888: ret
```

`sub_1002277e` is called by entry. It delegates to sub_10001030, sub_100225bb, sub_10022889 (5 callees).

12.481 sub_10022889 at 0x10022889 (2 instructions)

```
0x10022889: mov dword ptr ds:[10136230h],0FFFFFFFFh
0x10022893: ret
```

`sub_10022889` is called by sub_1002277e. It calls nothing else.

12.482 entry at 0x10022894 (9 instructions)

```
0x10022894: cmp dword ptr [esp+8],1
0x10022899: jne short 100228A0h
0x1002289b: call 10022D44h
0x100228a0: push dword ptr [esp+4]
0x100228a4: mov ecx,[esp+10h]
0x100228a8: mov edx,[esp+0Ch]
0x100228ac: call 1002277Eh
```

```
0x100228b1: pop ecx
0x100228b2: ret 0Ch
```

`entry` is the reset handler: the first code that runs after boot. It delegates to sub_1002277e, sub_10022d44 (2 callees).

12.483 j_memset at 0x100228c0 (1 instructions)

```
0x100228c0: jmp dword ptr ds:[100FE558h]
```

`j_memset` is called by sub_10002070, sub_100032e0, sub_10003d20, sub_100067c0, sub_10008ff0, sub_1000e3e0, sub_1000e680, sub_1000e6d0, sub_1000ead0, sub_100139c0, sub_10015d20, sub_100173c0, sub_10018670, sub_10019c10, sub_1001a5b0, sub_1001b160, sub_1001b970, sub_10020aa0, png_info_init_3, sub_10024050, sub_100245f0, png_start_read_image, sub_100280e0, sub_1002b220, png_set_pCAL, sub_1002cb30, png_calloc, sub_1002dcc0, sub_100309c0, png_set_quantize, sub_10037d80, sub_1003b220, sub_1003be10, sub_100436f0, sub_10061af0, sub_10061d00, sub_10062a80, sub_1006df2c, sub_100ae7b0, sub_100b04b0, sub_100b0570, sub_100bffb0, sub_100c0b60, sub_100c22e0, sub_100d4660, sub_100d4ce0, sub_100d6590, sub_100dea30, sub_100e0950, sub_100e2010, sub_100e67b0, sub_100e80f0, sub_100ea5a0, sub_100eb620. It calls nothing else.

12.484 j_memcpy at 0x100228c6 (1 instructions)

```
0x100228c6: jmp dword ptr ds:[100FE54Ch]
```

`j_memcpy` is called by sub_1000ecf0, sub_10010810, sub_10011b00, sub_10013690, sub_10013a70, sub_10013c10, sub_10013e20, sub_10013f20, sub_100143f0, sub_10016040, sub_10016840, sub_100187f0, sub_10018be0, sub_1001a5b0, png_read_row, png_write_row, png_set_pCAL, png_set_sCAL_s, png_set_PLTE, png_set_iCCP, sub_1002c050, png_set_tRNS, png_set_sPLT, png_set_unknown_chunks, png_set_keep_unknown_chunks, sub_1002cff0, sub_1002d150, sub_1002d370, sub_1002d670, sub_10035b60, sub_10036940, sub_10037b40, sub_10061950, sub_10061a80, sub_10061d00, sub_10082e60, sub_10087190, sub_10089620, sub_1008ae80, sub_1008af60, sub_100dfca0. It calls nothing else.

12.485 sub_100228d0 at 0x100228d0 (47 instructions)

```
0x100228d0: cmp dword ptr ds:[10145460h],0
0x100228d7: je short 10022906h
0x100228d9: push ebp
0x100228da: mov ebp,esp
0x100228dc: sub esp,8
0x100228df: and esp,0FFFFFFF8h
0x100228e2: fstp qword ptr [esp]
0x100228e5: cvttsd2si eax,qword ptr [esp]
0x100228ea: leave
0x100228eb: ret
0x10022906: push ebp
0x10022907: mov ebp,esp
0x10022909: sub esp,20h
0x1002290c: and esp,0FFFFFFF0h
0x1002290f: fld st(0)
0x10022911: fst dword ptr [esp+18h]
0x10022915: fistp qword ptr [esp+10h]
0x10022919: fild qword ptr [esp+10h]
0x1002291d: mov edx,[esp+18h]
0x10022921: mov eax,[esp+10h]
0x10022925: test eax,eax
0x10022927: je short 10022965h
0x10022929: fsubp
0x1002292b: test edx,edx
0x1002292d: jns short 1002294Dh
0x1002292f: fstp dword ptr [esp]
0x10022932: mov ecx,[esp]
0x10022935: xor ecx,80000000h
```

```
0x1002293b: add ecx,7FFFFFFFh
0x10022941: adc eax,0
0x10022944: mov edx,[esp+14h]
0x10022948: adc edx,0
0x1002294b: jmp short 10022979h
0x1002294d: fstp dword ptr [esp]
0x10022950: mov ecx,[esp]
0x10022953: add ecx,7FFFFFFFh
0x10022959: sbb eax,0
0x1002295c: mov edx,[esp+14h]
0x10022960: sbb edx,0
0x10022963: jmp short 10022979h
0x10022965: mov edx,[esp+14h]
0x10022969: test edx,7FFFFFFFh
0x1002296f: jne short 10022929h
0x10022971: fstp dword ptr [esp+18h]
0x10022975: fstp dword ptr [esp+18h]
0x10022979: leave
0x1002297a: ret
```

`sub_100228d0` is called by sub_1000ef10, sub_1000ef70, sub_10011fb0, sub_100143f0, sub_1001b350, sub_1001c370, sub_1001ddc0, sub_1001fd10, sub_10024050, sub_10024770, sub_1006eba4, sub_100729b6, sub_100b0c80, sub_100dea30, sub_100e0950, sub_100e1a50, sub_100e2010, sub_100e67b0, sub_100e7350, sub_100e77a0. It accesses memory-mapped peripheral registers at 0x7fffffff. It calls nothing else.

12.486 sub_10022906 at 0x10022906 (37 instructions)

```
0x10022906: push ebp
0x10022907: mov ebp,esp
0x10022909: sub esp,20h
0x1002290c: and esp,0FFFFFFF0h
0x1002290f: fld st(0)
0x10022911: fst dword ptr [esp+18h]
0x10022915: fistp qword ptr [esp+10h]
0x10022919: fild qword ptr [esp+10h]
0x1002291d: mov edx,[esp+18h]
0x10022921: mov eax,[esp+10h]
0x10022925: test eax,eax
0x10022927: je short 10022965h
0x10022929: fsubp
0x1002292b: test edx,edx
0x1002292d: jns short 1002294Dh
0x1002292f: fstp dword ptr [esp]
0x10022932: mov ecx,[esp]
0x10022935: xor ecx,80000000h
0x1002293b: add ecx,7FFFFFFFh
0x10022941: adc eax,0
0x10022944: mov edx,[esp+14h]
0x10022948: adc edx,0
0x1002294b: jmp short 10022979h
0x1002294d: fstp dword ptr [esp]
0x10022950: mov ecx,[esp]
0x10022953: add ecx,7FFFFFFFh
0x10022959: sbb eax,0
0x1002295c: mov edx,[esp+14h]
0x10022960: sbb edx,0
0x10022963: jmp short 10022979h
0x10022965: mov edx,[esp+14h]
0x10022969: test edx,7FFFFFFFh
0x1002296f: jne short 10022929h
0x10022971: fstp dword ptr [esp+18h]
0x10022975: fstp dword ptr [esp+18h]
0x10022979: leave
0x1002297a: ret
```

`sub_10022906` is called by sub_1006e09e, sub_1006ea55, sub_1006eba4. It accesses memory-mapped peripheral registers at 0x7fffffff. It calls nothing else.

12.487 j_floor at 0x1002297c (1 instructions)

```
0x1002297c: jmp dword ptr ds:[100FE548h]
```

`j_floor` is called by sub_1000ef10, sub_1000ef70, sub_100103b0, sub_10011fb0, sub_1001b350, sub_1001c370, sub_10024050, sub_10024710, sub_10024770, sub_10024830, sub_10024880, sub_100248d0, sub_100249a0, sub_10024a20, sub_10024b00. It calls nothing else.

12.488 j__CIfmod at 0x1002298e (1 instructions)

```
0x1002298e: jmp dword ptr ds:[100FE50Ch]
```

`j__CIfmod` is called by sub_1001ce00. It calls nothing else.

12.489 j__CIsqrt at 0x1002299a (1 instructions)

```
0x1002299a: jmp dword ptr ds:[100FE504h]
```

`j__CIsqrt` is called by sub_100de200, sub_100e1a50, sub_100e6ff0. It calls nothing else.

12.490 j__CIpow at 0x100229a0 (1 instructions)

```
0x100229a0: jmp dword ptr ds:[100FE4F4h]
```

`j__CIpow` is called by sub_100143f0, sub_100249a0, sub_10024a20, sub_10024b00, sub_100e0950, sub_100e2010, sub_100e2b50, sub_100e3e60, sub_100e4a30, sub_100e67b0, sub_100e77a0. It calls nothing else.

12.491 j__CxxThrowException at 0x100229a6 (1 instructions)

```
0x100229a6: jmp dword ptr ds:[100FE4F0h]
```

`j__CxxThrowException` is called by sub_10009b40, sub_10009bb0, sub_1000c700, sub_100143f0, sub_100156b0, sub_10015d20, sub_10016040, sub_100171e0, sub_10017680, sub_10018120, sub_100187f0, sub_10018d20, sub_1001c370. It calls nothing else.

12.492 sub_100229b0 at 0x100229b0 (19 instructions)

```
0x100229b0: push ecx
0x100229b1: lea ecx,[esp+4]
0x100229b5: sub ecx,eax
0x100229b7: sbb eax,eax
0x100229b9: not eax
0x100229bb: and ecx,eax
0x100229bd: mov eax,esp
0x100229bf: and eax,0FFFFF000h
0x100229c4: cmp ecx,eax
0x100229c6: jb short 100229D2h
0x100229c8: mov eax,ecx
0x100229ca: pop ecx
0x100229cb: xchg esp,eax
0x100229cc: mov eax,[eax]
0x100229ce: mov [esp],eax
0x100229d1: ret
0x100229d2: sub eax,1000h
0x100229d7: test [eax],eax
0x100229d9: jmp short 100229C4h
```

`sub_100229b0` is called by sub_10016040, sub_100b0c80, sub_100d4ce0, sub_100df9e0, sub_100e0950. It calls nothing else.

12.493 j__setjmp3 at 0x100229dc (1 instructions)

```
0x100229dc: jmp dword ptr ds:[100FE4E8h]
```

`j__setjmp3` is called by sub_10016040, sub_10018d20, sub_1001b350, png_create_read_struct_2, png_create_write_struct_2, sub_100a0020. It calls nothing else.

12.494 j___CxxLongjmpUnwind at 0x100229e2 (1 instructions)

```
0x100229e2: jmp dword ptr ds:[100FE4E4h]
```

`j___CxxLongjmpUnwind` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.495 sub_10022b00 at 0x10022b00 (21 instructions)

```
0x10022b00: push 10022B59h
0x10022b05: push dword ptr fs:[0]
0x10022b0c: mov eax,[esp+10h]
0x10022b10: mov [esp+10h],ebp
0x10022b14: lea ebp,[esp+10h]
0x10022b18: sub esp,eax
0x10022b1a: push ebx
0x10022b1b: push esi
0x10022b1c: push edi
0x10022b1d: mov eax,ds:[10136220h]
0x10022b22: xor [ebp-4],eax
0x10022b25: xor eax,ebp
0x10022b27: push eax
0x10022b28: mov [ebp-18h],esp
0x10022b2b: push dword ptr [ebp-8]
0x10022b2e: mov eax,[ebp-4]
0x10022b31: mov dword ptr [ebp-4],0FFFFFFFEh
0x10022b38: mov [ebp-8],eax
0x10022b3b: lea eax,[ebp-10h]
0x10022b3e: mov fs:[0],eax
0x10022b44: ret
```

`sub_10022b00` is called by sub_1002237e, sub_100223dc, sub_10022452, sub_10022509, sub_1002277e. It calls nothing else.

12.496 sub_10022b45 at 0x10022b45 (11 instructions)

```
0x10022b45: mov ecx,[ebp-10h]
0x10022b48: mov fs:[0],ecx
0x10022b4f: pop ecx
0x10022b50: pop edi
0x10022b51: pop edi
0x10022b52: pop esi
0x10022b53: pop ebx
0x10022b54: mov esp,ebp
0x10022b56: pop ebp
0x10022b57: push ecx
0x10022b58: ret
```

`sub_10022b45` is called by sub_1002237e, sub_100223dc, sub_10022452, sub_10022509, sub_1002277e. It calls nothing else.

12.497 sub_10022b59 at 0x10022b59 (9 instructions)

```
0x10022b59: push dword ptr [esp+10h]
0x10022b5d: push dword ptr [esp+10h]
0x10022b61: push dword ptr [esp+10h]
0x10022b65: push dword ptr [esp+10h]
0x10022b69: push 10022326h
0x10022b6e: push 10136220h
0x10022b73: call 10022EA2h
0x10022b78: add esp,18h
0x10022b7b: ret
```

`sub_10022b59` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to j__except_handler4_common (1 callees).

12.498 j__unlock at 0x10022b7c (1 instructions)

```
0x10022b7c: jmp dword ptr ds:[100FE3D8h]
```

`j__unlock` is called by sub_100224ee. It calls nothing else.

12.499 j___dllonexit at 0x10022b82 (1 instructions)

```
0x10022b82: jmp dword ptr ds:[100FE3D4h]
```

`j___dllonexit` is called by sub_10022452. It calls nothing else.

12.500 j__lock at 0x10022b88 (1 instructions)

```
0x10022b88: jmp dword ptr ds:[100FE3CCh]
```

`j__lock` is called by sub_10022452. It calls nothing else.

12.501 sub_10022bf0 at 0x10022bf0 (14 instructions)

```
0x10022bf0: mov ecx,[esp+4]
0x10022bf4: cmp word ptr [ecx],5A4Dh
0x10022bf9: je short 10022BFEh
0x10022bfb: xor eax,eax
0x10022bfd: ret
0x10022bfe: mov eax,[ecx+3Ch]
0x10022c01: add eax,ecx
0x10022c03: cmp dword ptr [eax],4550h
0x10022c09: jne short 10022BFBh
0x10022c0b: xor ecx,ecx
0x10022c0d: cmp word ptr [eax+18h],10Bh
0x10022c13: sete cl
0x10022c16: mov eax,ecx
0x10022c18: ret
```

`sub_10022bf0` is called by sub_10022c70. It calls nothing else.

12.502 sub_10022c20 at 0x10022c20 (29 instructions)

```
0x10022c20: mov eax,[esp+4]
0x10022c24: mov ecx,[eax+3Ch]
0x10022c27: add ecx,eax
0x10022c29: movzx eax,word ptr [ecx+14h]
0x10022c2d: push ebx
0x10022c2e: push esi
0x10022c2f: movzx esi,word ptr [ecx+6]
0x10022c33: xor edx,edx
```

```
0x10022c35: test esi,esi
0x10022c37: push edi
0x10022c38: lea eax,[eax+ecx+18h]
0x10022c3c: jbe short 10022C5Ch
0x10022c3e: mov edi,[esp+14h]
0x10022c42: mov ecx,[eax+0Ch]
0x10022c45: cmp edi,ecx
0x10022c47: jb short 10022C52h
0x10022c49: mov ebx,[eax+8]
0x10022c4c: add ebx,ecx
0x10022c4e: cmp edi,ebx
0x10022c50: jb short 10022C5Eh
0x10022c52: add edx,1
0x10022c55: add eax,28h
0x10022c58: cmp edx,esi
0x10022c5a: jb short 10022C42h
0x10022c5c: xor eax,eax
0x10022c5e: pop edi
0x10022c5f: pop esi
0x10022c60: pop ebx
0x10022c61: ret
```

`sub_10022c20` is called by sub_10022c70. It calls nothing else.

12.503 sub_10022c70 at 0x10022c70 (57 instructions)

```
0x10022c70: push ebp
0x10022c71: mov ebp,esp
0x10022c73: push 0FFFFFFFEh
0x10022c75: push 10130DE0h
0x10022c7a: push 10022B59h
0x10022c7f: mov eax,fs:[0]
0x10022c85: push eax
0x10022c86: sub esp,8
0x10022c89: push ebx
0x10022c8a: push esi
0x10022c8b: push edi
0x10022c8c: mov eax,ds:[10136220h]
0x10022c91: xor [ebp-8],eax
0x10022c94: xor eax,ebp
0x10022c96: push eax
0x10022c97: lea eax,[ebp-10h]
0x10022c9a: mov fs:[0],eax
0x10022ca0: mov [ebp-18h],esp
0x10022ca3: mov dword ptr [ebp-4],0
0x10022caa: push 10000000h
0x10022caf: call 10022BF0h
0x10022cb4: add esp,4
0x10022cb7: test eax,eax
0x10022cb9: je short 10022D10h
0x10022cbb: mov eax,[ebp+8]
0x10022cbe: sub eax,10000000h
0x10022cc3: push eax
0x10022cc4: push 10000000h
0x10022cc9: call 10022C20h
0x10022cce: add esp,8
0x10022cd1: test eax,eax
0x10022cd3: je short 10022D10h
0x10022cd5: mov eax,[eax+24h]
0x10022cd8: shr eax,1Fh
0x10022cdb: not eax
0x10022cdd: and eax,1
0x10022ce0: mov dword ptr [ebp-4],0FFFFFFFEh
0x10022ce7: mov ecx,[ebp-10h]
0x10022cea: mov fs:[0],ecx
0x10022cf1: pop ecx
```

```
0x10022cf2: pop edi
0x10022cf3: pop esi
0x10022cf4: pop ebx
0x10022cf5: mov esp,ebp
0x10022cf7: pop ebp
0x10022cf8: ret
0x10022d10: mov dword ptr [ebp-4],0FFFFFFFEh
0x10022d17: xor eax,eax
0x10022d19: mov ecx,[ebp-10h]
0x10022d1c: mov fs:[0],ecx
0x10022d23: pop ecx
0x10022d24: pop edi
0x10022d25: pop esi
0x10022d26: pop ebx
0x10022d27: mov esp,ebp
0x10022d29: pop ebp
0x10022d2a: ret
```

`sub_10022c70` is called by sub_100225bb. It delegates to sub_10022bf0, sub_10022c20 (2 callees).

12.504 j__initterm at 0x10022d2c (1 instructions)

```
0x10022d2c: jmp dword ptr ds:[100FE3B8h]
```

`j__initterm` is called by sub_100225bb. It calls nothing else.

12.505 j__initterm_e at 0x10022d32 (1 instructions)

```
0x10022d32: jmp dword ptr ds:[100FE3B4h]
```

`j__initterm_e` is called by sub_100225bb. It calls nothing else.

12.506 j__amsg_exit at 0x10022d38 (1 instructions)

```
0x10022d38: jmp dword ptr ds:[100FE3B0h]
```

`j__amsg_exit` is called by sub_100225bb. It calls nothing else.

12.507 sub_10022d44 at 0x10022d44 (52 instructions)

```
0x10022d44: push ebp
0x10022d45: mov ebp,esp
0x10022d47: sub esp,10h
0x10022d4a: mov eax,ds:[10136220h]
0x10022d4f: and dword ptr [ebp-8],0
0x10022d53: and dword ptr [ebp-4],0
0x10022d57: push ebx
0x10022d58: push edi
0x10022d59: mov edi,0BB40E64Eh
0x10022d5e: cmp eax,edi
0x10022d60: mov ebx,0FFFF0000h
0x10022d65: je short 10022D74h
0x10022d67: test ebx,eax
0x10022d69: je short 10022D74h
0x10022d6b: not eax
0x10022d6d: mov ds:[10136224h],eax
0x10022d72: jmp short 10022DD4h
0x10022d74: push esi
0x10022d75: lea eax,[ebp-8]
0x10022d78: push eax
0x10022d79: call dword ptr ds:[100FE0B4h]
0x10022d7f: mov esi,[ebp-4]
0x10022d82: xor esi,[ebp-8]
```

```
0x10022d85: call dword ptr ds:[100FE0B8h]
0x10022d8b: xor esi,eax
0x10022d8d: call dword ptr ds:[100FE108h]
0x10022d93: xor esi,eax
0x10022d95: call dword ptr ds:[100FE10Ch]
0x10022d9b: xor esi,eax
0x10022d9d: lea eax,[ebp-10h]
0x10022da0: push eax
0x10022da1: call dword ptr ds:[100FE0BCh]
0x10022da7: mov eax,[ebp-0Ch]
0x10022daa: xor eax,[ebp-10h]
0x10022dad: xor esi,eax
0x10022daf: cmp esi,edi
0x10022db1: jne short 10022DBAh
0x10022db3: mov esi,0BB40E64Fh
0x10022db8: jmp short 10022DC5h
0x10022dba: test ebx,esi
0x10022dbc: jne short 10022DC5h
0x10022dbe: mov eax,esi
0x10022dc0: shl eax,10h
0x10022dc3: or esi,eax
0x10022dc5: mov ds:[10136220h],esi
0x10022dcb: not esi
0x10022dcd: mov ds:[10136224h],esi
0x10022dd3: pop esi
0x10022dd4: pop edi
0x10022dd5: pop ebx
0x10022dd6: leave
0x10022dd7: ret
```

`sub_10022d44` is called by entry. It accesses memory-mapped peripheral registers at 0xbb40e64e. It calls nothing else.

12.508 j__crt_debugger_hook at 0x10022e96 (1 instructions)

```
0x10022e96: jmp dword ptr ds:[100FE4CCh]
```

`j__crt_debugger_hook` is called by sub_10022326. It calls nothing else.

12.509 j__except_handler4_common at 0x10022ea2 (1 instructions)

```
0x10022ea2: jmp dword ptr ds:[100FE4D4h]
```

`j__except_handler4_common` is called by sub_10022b59. It calls nothing else.

12.510 j_ORDINAL 9 at 0x10022f7c (1 instructions)

```
0x10022f7c: jmp dword ptr ds:[100FE6BCh]
```

`j_ORDINAL 9` is called by sub_10016040. It calls nothing else.

12.511 sub_10022f90 at 0x10022f90 (59 instructions)

```
0x10022f90: push ebx
0x10022f91: push ebp
0x10022f92: push esi
0x10022f93: push edi
0x10022f94: mov edi,eax
0x10022f96: xor eax,eax
0x10022f98: mov esi,ecx
0x10022f9a: xor ebp,ebp
0x10022f9c: lea esp,[esp]
0x10022fa0: movzx ecx,byte ptr [esi+ebp+11Ch]
0x10022fa8: lea ebx,[ecx-41h]
```

```
0x10022fab: add ebp,1
0x10022fae: cmp ebx,39h
0x10022fb1: ja short 10022FC2h
0x10022fb3: cmp ecx,5Ah
0x10022fb6: jle short 10022FBDh
0x10022fb8: cmp ecx,61h
0x10022fbb: jl short 10022FC2h
0x10022fbd: mov [eax+edi],cl
0x10022fc0: jmp short 10022FF0h
0x10022fc2: mov byte ptr [eax+edi],5Bh
0x10022fc6: mov ebx,ecx
0x10022fc8: sar ebx,4
0x10022fcb: and ebx,0Fh
0x10022fce: mov bl,[ebx+100FF6C4h]
0x10022fd4: add eax,1
0x10022fd7: mov [eax+edi],bl
0x10022fda: and ecx,0Fh
0x10022fdd: mov cl,[ecx+100FF6C4h]
0x10022fe3: add eax,1
0x10022fe6: mov [eax+edi],cl
0x10022fe9: add eax,1
0x10022fec: mov byte ptr [eax+edi],5Dh
0x10022ff0: add eax,1
0x10022ff3: cmp ebp,4
0x10022ff6: jl short 10022FA0h
0x10022ff8: test edx,edx
0x10022ffa: jne short 10023004h
0x10022ffc: mov [eax+edi],dl
0x10022fff: pop edi
0x10023000: pop esi
0x10023001: pop ebp
0x10023002: pop ebx
0x10023003: ret
0x10023004: mov byte ptr [eax+edi],3Ah
0x10023008: mov byte ptr [eax+edi+1],20h
0x1002300d: add eax,1
0x10023010: add eax,1
0x10023013: add eax,edi
0x10023015: mov ecx,10h
0x1002301a: mov esi,edx
0x1002301c: mov edi,eax
0x1002301e: rep movsd
0x10023020: pop edi
0x10023021: pop esi
0x10023022: pop ebp
0x10023023: mov byte ptr [eax+3Fh],0
0x10023027: pop ebx
0x10023028: ret
```

`sub_10022f90` is called by png_chunk_warning, png_chunk_error, png_chunk_benign_error. It calls nothing else.

12.512 png_set_longjmp_fn at 0x10023030 (10 instructions)

```
0x10023030: mov eax,[esp+4]
0x10023034: test eax,eax
0x10023036: je short 10023047h
0x10023038: cmp dword ptr [esp+0Ch],40h
0x1002303d: jne short 10023047h
0x1002303f: mov ecx,[esp+8]
0x10023043: mov [eax+40h],ecx
0x10023046: ret
0x10023047: xor eax,eax
0x10023049: ret
```

`png_set_longjmp_fn` is called by png_create_read_struct_2, png_create_write_struct_2. It calls nothing else.

12.513 png_longjmp at 0x10023050 (14 instructions)

```
0x10023050: mov ecx,[esp+4]
0x10023054: test ecx,ecx
0x10023056: je short 1002306Ah
0x10023058: mov eax,[ecx+40h]
0x1002305b: test eax,eax
0x1002305d: je short 1002306Ah
0x1002305f: mov edx,[esp+8]
0x10023063: push edx
0x10023064: push ecx
0x10023065: call eax
0x10023067: add esp,8
0x1002306a: push 0
0x1002306c: call dword ptr ds:[100FE0ACh]
0x10023072: int 3
```

`png_longjmp` is called by sub_100231a0. It calls nothing else.

12.514 png_set_error_fn at 0x100230c0 (10 instructions)

```
0x100230c0: mov eax,[esp+4]
0x100230c4: test eax,eax
0x100230c6: je short 100230DDh
0x100230c8: mov ecx,[esp+8]
0x100230cc: mov edx,[esp+0Ch]
0x100230d0: mov [eax+4Ch],ecx
0x100230d3: mov ecx,[esp+10h]
0x100230d7: mov [eax+44h],edx
0x100230da: mov [eax+48h],ecx
0x100230dd: ret
```

`png_set_error_fn` is called by sub_10018d20, png_create_read_struct_2, png_create_write_struct_2. It calls nothing else.

12.515 png_get_error_ptr at 0x100230e0 (6 instructions)

```
0x100230e0: mov eax,[esp+4]
0x100230e4: test eax,eax
0x100230e6: jne short 100230E9h
0x100230e8: ret
0x100230e9: mov eax,[eax+4Ch]
0x100230ec: ret
```

`png_get_error_ptr` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.516 png_warning at 0x100230f0 (53 instructions)

```
0x10023080: push esi
0x10023081: mov esi,ds:[100FE4C4h]
0x10023087: push edi
0x10023088: push eax
0x10023089: push 100FF6D8h
0x1002308e: call esi
0x10023090: mov edi,ds:[100FE4C8h]
0x10023096: add eax,40h
0x10023099: push eax
0x1002309a: call edi
0x1002309c: add esp,0Ch
0x1002309f: push 100FF6D4h
0x100230a4: call esi
0x100230a6: add eax,40h
0x100230a9: push eax
```

```
0x100230aa: call edi
0x100230ac: add esp,8
0x100230af: pop edi
0x100230b0: pop esi
0x100230b1: ret
0x100230f0: mov ecx,[esp+8]
0x100230f4: push esi
0x100230f5: mov esi,[esp+8]
0x100230f9: xor eax,eax
0x100230fb: test esi,esi
0x100230fd: je short 1002313Ah
0x100230ff: cmp byte ptr [ecx],23h
0x10023102: jne short 10023128h
0x10023104: mov eax,1
0x10023109: mov dl,20h
0x1002310b: jmp short 10023110h
0x10023110: cmp [eax+ecx],dl
0x10023113: je short 10023128h
0x10023115: cmp [eax+ecx+1],dl
0x10023119: je short 10023125h
0x1002311b: add eax,2
0x1002311e: cmp eax,0Fh
0x10023121: jl short 10023110h
0x10023123: jmp short 10023128h
0x10023125: add eax,1
0x10023128: mov edx,[esi+48h]
0x1002312b: test edx,edx
0x1002312d: je short 1002313Ah
0x1002312f: add eax,ecx
0x10023131: push eax
0x10023132: push esi
0x10023133: call edx
0x10023135: add esp,8
0x10023138: pop esi
0x10023139: ret
0x1002313a: add eax,ecx
0x1002313c: pop esi
0x1002313d: jmp near ptr 10023080h
```

`png_warning` is called by png_chunk_warning, png_chunk_benign_error, png_data_freer, sub_10023c30, sub_100247f0, sub_100250a0, png_create_read_struct_2, png_read_update_info, png_start_read_image, png_read_image, png_set_read_fn, png_write_info_before_PLTE, png_set_filter, sub_1002a910, png_set_compression_window_bits, png_set_compression_method, png_create_write_struct_2, png_set_gAMA_fixed, png_set_hIST, png_set_pCAL, png_set_sCAL_s, png_set_sCAL, png_set_sCAL_fixed, png_set_PLTE, png_set_iCCP, sub_1002c050, png_set_tRNS, png_set_sPLT, png_set_unknown_chunks, png_set_compression_buffer_size, png_set_write_fn, sub_1002ce40, sub_1002cff0, sub_1002d150, sub_1002e260, sub_1002e400, sub_1002e450, sub_1002e590, sub_1002e700, sub_1002ea20, sub_1002ec10, sub_1002ee90, sub_1002f170, sub_1002f400, sub_1002f650, sub_1002f7e0, sub_1002f930, sub_1002faa0, sub_1002fdd0, sub_1002ffb0, sub_100300d0, sub_10030280, sub_10030490, sub_10030750, sub_100309c0, png_set_background_fixed, png_set_rgb_to_gray_fixed, sub_100354b0, sub_10035b60, sub_10035e80, sub_10035f80, sub_10036050, sub_10036450, sub_10036610, sub_10037220, sub_10037290, sub_100375c0, sub_100377c0, sub_10037930, sub_10037ab0, sub_10037b40, sub_10037c20, sub_10037cb0. It works with the text "libpng warning: %s". It calls nothing else.

12.517 png_chunk_warning at 0x10023150 (26 instructions)

```
0x10023150: sub esp,58h
0x10023153: mov eax,ds:[10136220h]
0x10023158: xor eax,esp
0x1002315a: mov [esp+54h],eax
0x1002315e: mov edx,[esp+60h]
0x10023162: push esi
```

```
0x10023163: mov esi,[esp+60h]
0x10023167: test esi,esi
0x10023169: jne short 1002316Fh
0x1002316b: push edx
0x1002316c: push esi
0x1002316d: jmp short 10023180h
0x1002316f: lea eax,[esp+4]
0x10023173: mov ecx,esi
0x10023175: call 10022F90h
0x1002317a: lea eax,[esp+4]
0x1002317e: push eax
0x1002317f: push esi
0x10023180: call 100230F0h
0x10023185: mov ecx,[esp+60h]
0x10023189: add esp,8
0x1002318c: pop esi
0x1002318d: xor ecx,esp
0x1002318f: call 10022326h
0x10023194: add esp,58h
0x10023197: ret
```

`png_chunk_warning` is called by sub_1002e050. It delegates to sub_10022326, sub_10022f90, png_warning (3 callees).

12.518 sub_100231a0 at 0x100231a0 (24 instructions)

```
0x100231a0: push esi
0x100231a1: mov esi,ds:[100FE4C4h]
0x100231a7: push edi
0x100231a8: push eax
0x100231a9: push 100FF6ECh
0x100231ae: call esi
0x100231b0: mov edi,ds:[100FE4C8h]
0x100231b6: add eax,40h
0x100231b9: push eax
0x100231ba: call edi
0x100231bc: add esp,0Ch
0x100231bf: push 100FF6D4h
0x100231c4: call esi
0x100231c6: add eax,40h
0x100231c9: push eax
0x100231ca: call edi
0x100231cc: mov ecx,[esp+14h]
0x100231d0: add esp,8
0x100231d3: push 1
0x100231d5: push ecx
0x100231d6: call 10023050h
0x100231db: pop edi
0x100231dc: pop esi
0x100231dd: int 3
```

`sub_100231a0` is called by png_error. It works with the text "libpng error: %s". It delegates to png_longjmp (1 callees).

12.519 png_error at 0x100231e0 (19 instructions)

```
0x100231e0: push esi
0x100231e1: mov esi,[esp+8]
0x100231e5: test esi,esi
0x100231e7: push edi
0x100231e8: mov edi,[esp+10h]
0x100231ec: je short 100231FCh
0x100231ee: mov eax,[esi+44h]
0x100231f1: test eax,eax
0x100231f3: je short 100231FCh
0x100231f5: push edi
0x100231f6: push esi
```

```
0x100231f7: call eax
0x100231f9: add esp,8
0x100231fc: push esi
0x100231fd: mov eax,edi
0x100231ff: call 100231A0h
0x10023204: pop edi
0x10023205: pop esi
0x10023206: int 3
```

`png_error` is called by sub_10018b70, png_benign_error, png_chunk_error, sub_100232a0, png_set_sig_bytes, sub_10023c30, sub_10024050, sub_100245f0, png_read_info, png_start_read_image, png_read_row, png_read_png, sub_10029c10, png_write_info, png_write_end, png_write_row, png_write_flush, png_set_filter, png_set_pCAL, png_set_sCAL_s, png_set_PLTE, png_set_text, sub_1002ca30, png_malloc, png_get_uint_31, sub_1002cea0, sub_1002dae0, sub_1002dcc0, sub_1002e100, sub_1002e260, sub_1002e400, sub_1002e450, sub_1002e590, sub_1002e700, sub_1002ea20, sub_1002ec10, sub_1002ee90, sub_1002f170, sub_1002f400, sub_1002f650, sub_1002f7e0, sub_1002f930, sub_1002faa0, sub_1002fdd0, sub_1002ffb0, sub_100300d0, sub_10030280, sub_10030490, sub_100309c0, png_set_rgb_to_gray_fixed, sub_10031a20, sub_100354b0, sub_10035b60, sub_10035e80, sub_10036c30, sub_10037020, sub_10037d80, sub_10037fe0. It delegates to sub_100231a0 (1 callees).

12.520 png_benign_error at 0x10023210 (63 instructions)

```
0x10023080: push esi
0x10023081: mov esi,ds:[100FE4C4h]
0x10023087: push edi
0x10023088: push eax
0x10023089: push 100FF6D8h
0x1002308e: call esi
0x10023090: mov edi,ds:[100FE4C8h]
0x10023096: add eax,40h
0x10023099: push eax
0x1002309a: call edi
0x1002309c: add esp,0Ch
0x1002309f: push 100FF6D4h
0x100230a4: call esi
0x100230a6: add eax,40h
0x100230a9: push eax
0x100230aa: call edi
0x100230ac: add esp,8
0x100230af: pop edi
0x100230b0: pop esi
0x100230b1: ret
0x100230f0: mov ecx,[esp+8]
0x100230f4: push esi
0x100230f5: mov esi,[esp+8]
0x100230f9: xor eax,eax
0x100230fb: test esi,esi
0x100230fd: je short 1002313Ah
0x100230ff: cmp byte ptr [ecx],23h
0x10023102: jne short 10023128h
0x10023104: mov eax,1
0x10023109: mov dl,20h
0x1002310b: jmp short 10023110h
0x10023110: cmp [eax+ecx],dl
0x10023113: je short 10023128h
0x10023115: cmp [eax+ecx+1],dl
0x10023119: je short 10023125h
0x1002311b: add eax,2
0x1002311e: cmp eax,0Fh
0x10023121: jl short 10023110h
0x10023123: jmp short 10023128h
0x10023125: add eax,1
0x10023128: mov edx,[esi+48h]
```

```
0x1002312b: test edx,edx
0x1002312d: je short 1002313Ah
0x1002312f: add eax,ecx
0x10023131: push eax
0x10023132: push esi
0x10023133: call edx
0x10023135: add esp,8
0x10023138: pop esi
0x10023139: ret
0x1002313a: add eax,ecx
0x1002313c: pop esi
0x1002313d: jmp near ptr 10023080h
0x10023210: mov eax,[esp+4]
0x10023214: test dword ptr [eax+70h],800000h
0x1002321b: je short 10023226h
0x1002321d: mov [esp+4],eax
0x10023221: jmp near ptr 100230F0h
0x10023226: mov edx,[esp+8]
0x1002322a: push edx
0x1002322b: push eax
0x1002322c: call 100231E0h
0x10023231: int 3
```

`png_benign_error` is called by png_read_row, png_read_end. It works with the text "libpng warning: %s". It delegates to png_error (1 callees).

12.521 png_chunk_error at 0x10023240 (17 instructions)

```
0x10023240: mov edx,[esp+8]
0x10023244: sub esp,54h
0x10023247: push esi
0x10023248: mov esi,[esp+5Ch]
0x1002324c: test esi,esi
0x1002324e: jne short 10023257h
0x10023250: push edx
0x10023251: push esi
0x10023252: call 100231E0h
0x10023257: lea eax,[esp+4]
0x1002325b: mov ecx,esi
0x1002325d: call 10022F90h
0x10023262: lea eax,[esp+4]
0x10023266: push eax
0x10023267: push esi
0x10023268: call 100231E0h
0x1002326d: int 3
```

`png_chunk_error` is called by png_chunk_benign_error, sub_1002d310, sub_10030750. It delegates to sub_10022f90, png_error (2 callees).

12.522 png_chunk_benign_error at 0x10023270 (36 instructions)

```
0x10023150: sub esp,58h
0x10023153: mov eax,ds:[10136220h]
0x10023158: xor eax,esp
0x1002315a: mov [esp+54h],eax
0x1002315e: mov edx,[esp+60h]
0x10023162: push esi
0x10023163: mov esi,[esp+60h]
0x10023167: test esi,esi
0x10023169: jne short 1002316Fh
0x1002316b: push edx
0x1002316c: push esi
0x1002316d: jmp short 10023180h
0x1002316f: lea eax,[esp+4]
0x10023173: mov ecx,esi
```

```
0x10023175: call 10022F90h
0x1002317a: lea eax,[esp+4]
0x1002317e: push eax
0x1002317f: push esi
0x10023180: call 100230F0h
0x10023185: mov ecx,[esp+60h]
0x10023189: add esp,8
0x1002318c: pop esi
0x1002318d: xor ecx,esp
0x1002318f: call 10022326h
0x10023194: add esp,58h
0x10023197: ret
0x10023270: mov eax,[esp+4]
0x10023274: test dword ptr [eax+70h],800000h
0x1002327b: je short 10023286h
0x1002327d: mov [esp+4],eax
0x10023281: jmp near ptr 10023150h
0x10023286: mov edx,[esp+8]
0x1002328a: push edx
0x1002328b: push eax
0x1002328c: call 10023240h
0x10023291: int 3
```

`png_chunk_benign_error` is called by sub_1002e050. It delegates to sub_10022326, sub_10022f90, png_warning (4 callees).

12.523 sub_100232a0 at 0x100232a0 (38 instructions)

```
0x100232a0: sub esp,58h
0x100232a3: mov eax,ds:[100FF700h]
0x100232a8: mov ecx,ds:[100FF704h]
0x100232ae: mov edx,ds:[100FF708h]
0x100232b4: mov [esp],eax
0x100232b7: mov eax,ds:[100FF70Ch]
0x100232bc: mov [esp+4],ecx
0x100232c0: mov ecx,ds:[100FF710h]
0x100232c6: mov [esp+0Ch],eax
0x100232ca: mov eax,[esp+60h]
0x100232ce: mov [esp+8],edx
0x100232d2: mov edx,ds:[100FF714h]
0x100232d8: mov [esp+10h],ecx
0x100232dc: xor ecx,ecx
0x100232de: test eax,eax
0x100232e0: push edi
0x100232e1: mov edi,[esp+60h]
0x100232e5: mov [esp+18h],edx
0x100232e9: je short 10023307h
0x100232eb: push esi
0x100232ec: lea esi,[esp+20h]
0x100232f0: sub esi,eax
0x100232f2: mov dl,[eax]
0x100232f4: test dl,dl
0x100232f6: je short 10023306h
0x100232f8: mov [esi+eax],dl
0x100232fb: add ecx,1
0x100232fe: add eax,1
0x10023301: cmp ecx,3Fh
0x10023304: jl short 100232F2h
0x10023306: pop esi
0x10023307: lea eax,[esp+4]
0x1002330b: push eax
0x1002330c: push edi
0x1002330d: mov byte ptr [esp+ecx+24h],0
0x10023312: call 100231E0h
0x10023317: pop edi
0x10023318: int 3
```

`sub_100232a0` is called by sub_10024710. It works with the text "fixed point overflow in ". It delegates to png_error (1 callees).

12.524 png_set_sig_bytes at 0x10023320 (15 instructions)

```
0x10023320: mov ecx,[esp+4]
0x10023324: test ecx,ecx
0x10023326: je short 1002334Ch
0x10023328: mov eax,[esp+8]
0x1002332c: cmp eax,8
0x1002332f: jle short 1002333Ch
0x10023331: push 100FF74Ch
0x10023336: push ecx
0x10023337: call 100231E0h
0x1002333c: test eax,eax
0x1002333e: setl dl
0x10023341: sub dl,1
0x10023344: and edx,eax
0x10023346: mov [ecx+12Ch],dl
0x1002334c: ret
```

`png_set_sig_bytes` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "Too many bytes for PNG signature". It delegates to png_error (1 callees).

12.525 png_sig_cmp at 0x10023350 (100 instructions)

```
0x10023350: sub esp,0Ch
0x10023353: mov eax,ds:[10136220h]
0x10023358: xor eax,esp
0x1002335a: mov [esp+8],eax
0x1002335e: mov al,0Ah
0x10023360: mov [esp+5],al
0x10023364: mov [esp+7],al
0x10023368: mov eax,[esp+18h]
0x1002336c: cmp eax,8
0x1002336f: mov byte ptr [esp],89h
0x10023373: mov byte ptr [esp+1],50h
0x10023378: mov byte ptr [esp+2],4Eh
0x1002337d: mov byte ptr [esp+3],47h
0x10023382: mov byte ptr [esp+4],0Dh
0x10023387: mov byte ptr [esp+6],1Ah
0x1002338c: jbe near ptr 1002344Ch
0x10023392: mov eax,8
0x10023397: mov edx,[esp+14h]
0x1002339b: cmp edx,7
0x1002339e: ja near ptr 10023455h
0x100233a4: lea ecx,[edx+eax]
0x100233a7: cmp ecx,8
0x100233aa: jbe short 100233B3h
0x100233ac: mov eax,8
0x100233b1: sub eax,edx
0x100233b3: push esi
0x100233b4: mov esi,[esp+14h]
0x100233b8: lea ecx,[esp+edx+4]
0x100233bc: add edx,esi
0x100233be: cmp eax,4
0x100233c1: push edi
0x100233c2: jb short 100233D8h
0x100233c4: mov esi,[edx]
0x100233c6: cmp esi,[ecx]
0x100233c8: jne short 100233E0h
0x100233ca: sub eax,4
0x100233cd: add ecx,4
0x100233d0: add edx,4
0x100233d3: cmp eax,4
0x100233d6: jae short 100233C4h
```

```
0x100233d8: test eax,eax
0x100233da: je near ptr 10023467h
0x100233e0: movzx esi,byte ptr [edx]
0x100233e3: movzx edi,byte ptr [ecx]
0x100233e6: sub esi,edi
0x100233e8: jne short 1002342Fh
0x100233ea: sub eax,1
0x100233ed: add ecx,1
0x100233f0: add edx,1
0x100233f3: test eax,eax
0x100233f5: je short 10023467h
0x100233f7: movzx esi,byte ptr [edx]
0x100233fa: movzx edi,byte ptr [ecx]
0x100233fd: sub esi,edi
0x100233ff: jne short 1002342Fh
0x10023401: sub eax,1
0x10023404: add ecx,1
0x10023407: add edx,1
0x1002340a: test eax,eax
0x1002340c: je short 10023467h
0x1002340e: movzx esi,byte ptr [edx]
0x10023411: movzx edi,byte ptr [ecx]
0x10023414: sub esi,edi
0x10023416: jne short 1002342Fh
0x10023418: sub eax,1
0x1002341b: add ecx,1
0x1002341e: add edx,1
0x10023421: test eax,eax
0x10023423: je short 10023467h
0x10023425: movzx esi,byte ptr [edx]
0x10023428: movzx edx,byte ptr [ecx]
0x1002342b: sub esi,edx
0x1002342d: je short 10023467h
0x1002342f: test esi,esi
0x10023431: mov eax,1
0x10023436: jg short 10023469h
0x10023438: pop edi
0x10023439: or eax,0FFFFFFFFh
0x1002343c: pop esi
0x1002343d: mov ecx,[esp+8]
0x10023441: xor ecx,esp
0x10023443: call 10022326h
0x10023448: add esp,0Ch
0x1002344b: ret
0x1002344c: cmp eax,1
0x1002344f: jae near ptr 10023397h
0x10023455: or eax,0FFFFFFFFh
0x10023458: mov ecx,[esp+8]
0x1002345c: xor ecx,esp
0x1002345e: call 10022326h
0x10023463: add esp,0Ch
0x10023466: ret
0x10023467: xor eax,eax
0x10023469: mov ecx,[esp+10h]
0x1002346d: pop edi
0x1002346e: pop esi
0x1002346f: xor ecx,esp
0x10023471: call 10022326h
0x10023476: add esp,0Ch
0x10023479: ret
```

`png_sig_cmp` is called by sub_1002cea0. It delegates to sub_10022326 (1 callees).

12.526 sub_100234d0 at 0x100234d0 (24 instructions)

```
0x100234d0: jmp near ptr 1002CD70h
0x1002cc80: cmp dword ptr [esp+4],0
0x1002cc85: je short 1002CC97h
0x1002cc87: mov eax,[esp+8]
0x1002cc8b: test eax,eax
0x1002cc8d: je short 1002CC97h
0x1002cc8f: push eax
0x1002cc90: call dword ptr ds:[100FE554h]
0x1002cc96: pop ecx
0x1002cc97: ret
0x1002cd70: mov ecx,[esp+4]
0x1002cd74: test ecx,ecx
0x1002cd76: je short 1002CD99h
0x1002cd78: mov edx,[esp+8]
0x1002cd7c: test edx,edx
0x1002cd7e: je short 1002CD99h
0x1002cd80: mov eax,[ecx+240h]
0x1002cd86: test eax,eax
0x1002cd88: mov [esp+8],edx
0x1002cd8c: mov [esp+4],ecx
0x1002cd90: je short 1002CD94h
0x1002cd92: jmp eax
0x1002cd94: jmp near ptr 1002CC80h
0x1002cd99: ret
```

`sub_100234d0` is called by png_free_data, sub_100280e0. It calls nothing else.

12.527 sub_100234e0 at 0x100234e0 (8 instructions)

```
0x100234e0: push 0
0x100234e2: push 0
0x100234e4: push 0
0x100234e6: call 1005FDF0h
0x100234eb: mov ecx,[esp+10h]
0x100234ef: add esp,0Ch
0x100234f2: mov [ecx+110h],eax
0x100234f8: ret
```

`sub_100234e0` is called by sub_1002dfd0, png_write_chunk_start. It delegates to sub_1005fdf0 (1 callees).

12.528 sub_10023500 at 0x10023500 (23 instructions)

```
0x10023500: push esi
0x10023501: mov esi,[esp+8]
0x10023505: test byte ptr [esi+11Ch],20h
0x1002350c: je short 1002351Fh
0x1002350e: mov eax,[esi+70h]
0x10023511: and eax,300h
0x10023516: cmp eax,300h
0x1002351b: jne short 10023528h
0x1002351d: pop esi
0x1002351e: ret
0x1002351f: test dword ptr [esi+70h],800h
0x10023526: jne short 10023547h
0x10023528: mov ecx,[esp+10h]
0x1002352c: mov edx,[esp+0Ch]
0x10023530: mov eax,[esi+110h]
0x10023536: push ecx
0x10023537: push edx
0x10023538: push eax
0x10023539: call 1005FDF0h
0x1002353e: add esp,0Ch
0x10023541: mov [esi+110h],eax
```

```
0x10023547: pop esi
0x10023548: ret
```

`sub_10023500` is called by sub_1002cf30, sub_1002dfd0, png_write_chunk_start, png_write_chunk_data. It delegates to sub_1005fdf0 (1 callees).

12.529 png_info_init_3 at 0x10023550 (20 instructions)

```
0x10023550: push esi
0x10023551: mov esi,[esp+8]
0x10023555: mov eax,[esi]
0x10023557: test eax,eax
0x10023559: je short 10023587h
0x1002355b: cmp dword ptr [esp+0Ch],0F0h
0x10023563: jae short 10023577h
0x10023565: push eax
0x10023566: call 1002CD00h
0x1002356b: push 2
0x1002356d: call 1002CCE0h
0x10023572: add esp,8
0x10023575: mov [esi],eax
0x10023577: push 0F0h
0x1002357c: push 0
0x1002357e: push eax
0x1002357f: call 100228C0h
0x10023584: add esp,0Ch
0x10023587: pop esi
0x10023588: ret
```

`png_info_init_3` is called by sub_100239c0, png_create_info_struct. It delegates to j_memset, sub_1002cce0, sub_1002cd00 (3 callees).

12.530 png_data_freer at 0x10023590 (23 instructions)

```
0x10023590: mov edx,[esp+4]
0x10023594: test edx,edx
0x10023596: je short 100235D4h
0x10023598: mov eax,[esp+8]
0x1002359c: test eax,eax
0x1002359e: je short 100235D4h
0x100235a0: mov ecx,[esp+0Ch]
0x100235a4: cmp ecx,1
0x100235a7: jne short 100235B4h
0x100235a9: mov ecx,[esp+10h]
0x100235ad: or [eax+0BCh],ecx
0x100235b3: ret
0x100235b4: cmp ecx,2
0x100235b7: jne short 100235C6h
0x100235b9: mov edx,[esp+10h]
0x100235bd: not edx
0x100235bf: and [eax+0BCh],edx
0x100235c5: ret
0x100235c6: push 100FF794h
0x100235cb: push edx
0x100235cc: call 100230F0h
0x100235d1: add esp,8
0x100235d4: ret
```

`png_data_freer` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "Unknown freer parameter in png_data_freer". It delegates to png_warning (1 callees).

12.531 png_free_data at 0x100235e0 (293 instructions)

```
0x100235e0: push ebx
0x100235e1: push edi
0x100235e2: mov edi,[esp+0Ch]
0x100235e6: xor ebx,ebx
0x100235e8: cmp edi,ebx
0x100235ea: je near ptr 100239B1h
0x100235f0: push esi
0x100235f1: mov esi,[esp+14h]
0x100235f5: cmp esi,ebx
0x100235f7: je near ptr 100239B0h
0x100235fd: mov eax,[esi+0BCh]
0x10023603: push ebp
0x10023604: mov ebp,[esp+1Ch]
0x10023608: and eax,ebp
0x1002360a: test eax,4000h
0x1002360f: je short 1002367Fh
0x10023611: mov ecx,[esp+20h]
0x10023615: cmp ecx,0FFFFFFFFh
0x10023618: je short 10023646h
0x1002361a: mov eax,[esi+38h]
0x1002361d: cmp eax,ebx
0x1002361f: je short 1002367Fh
0x10023621: lea ebp,[ecx*8]
0x10023628: sub ebp,ecx
0x1002362a: add ebp,ebp
0x1002362c: add ebp,ebp
0x1002362e: mov eax,[eax+ebp+4]
0x10023632: cmp eax,ebx
0x10023634: je short 1002367Bh
0x10023636: push eax
0x10023637: push edi
0x10023638: call 1002CD70h
0x1002363d: mov ecx,[esi+38h]
0x10023640: mov [ecx+ebp+4],ebx
0x10023644: jmp short 10023678h
0x10023646: xor ebp,ebp
0x10023648: cmp [esi+30h],ebx
0x1002364b: jle short 10023668h
0x1002364d: lea ecx,[ecx]
0x10023650: push ebp
0x10023651: push 4000h
0x10023656: push esi
0x10023657: push edi
0x10023658: call 100235E0h
0x1002365d: add ebp,1
0x10023660: add esp,10h
0x10023663: cmp ebp,[esi+30h]
0x10023666: jl short 10023650h
0x10023668: mov edx,[esi+38h]
0x1002366b: push edx
0x1002366c: push edi
0x1002366d: call 1002CD70h
0x10023672: mov [esi+38h],ebx
0x10023675: mov [esi+30h],ebx
0x10023678: add esp,8
0x1002367b: mov ebp,[esp+1Ch]
0x1002367f: mov eax,[esi+0BCh]
0x10023685: and eax,ebp
0x10023687: test eax,2000h
0x1002368c: je short 100236A2h
0x1002368e: mov ecx,[esi+50h]
0x10023691: push ecx
0x10023692: push edi
0x10023693: call 1002CD70h
0x10023698: add esp,8
0x1002369b: and dword ptr [esi+8],0FFFFFFEFh
```

```
0x1002369f: mov [esi+50h],ebx
0x100236a2: mov eax,[esi+0BCh]
0x100236a8: and eax,ebp
0x100236aa: test eax,100h
0x100236af: je short 100236B8h
0x100236b1: and dword ptr [esi+8],0FFFFBFFFh
0x100236b8: test al,al
0x100236ba: jns near ptr 1002374Bh
0x100236c0: mov edx,[esi+0A4h]
0x100236c6: push edx
0x100236c7: push edi
0x100236c8: call 1002CD70h
0x100236cd: mov eax,[esi+0B0h]
0x100236d3: push eax
0x100236d4: push edi
0x100236d5: call 1002CD70h
0x100236da: add esp,10h
0x100236dd: cmp [esi+0B4h],ebx
0x100236e3: mov [esi+0A4h],ebx
0x100236e9: mov [esi+0B0h],ebx
0x100236ef: je short 10023744h
0x100236f1: xor ebp,ebp
0x100236f3: cmp [esi+0B9h],bl
0x100236f9: jbe short 1002372Ah
0x100236fb: jmp short 10023700h
0x10023700: mov ecx,[esi+0B4h]
0x10023706: mov edx,[ecx+ebp*4]
0x10023709: push edx
0x1002370a: push edi
0x1002370b: call 1002CD70h
0x10023710: mov eax,[esi+0B4h]
0x10023716: mov [eax+ebp*4],ebx
0x10023719: movzx ecx,byte ptr [esi+0B9h]
0x10023720: add ebp,1
0x10023723: add esp,8
0x10023726: cmp ebp,ecx
0x10023728: jl short 10023700h
0x1002372a: mov edx,[esi+0B4h]
0x10023730: push edx
0x10023731: push edi
0x10023732: call 1002CD70h
0x10023737: mov ebp,[esp+24h]
0x1002373b: add esp,8
0x1002373e: mov [esi+0B4h],ebx
0x10023744: and dword ptr [esi+8],0FFFFFBFFh
0x1002374b: mov eax,[esi+0BCh]
0x10023751: and eax,ebp
0x10023753: test al,10h
0x10023755: je short 10023787h
0x10023757: mov ecx,[esi+0C8h]
0x1002375d: push ecx
0x1002375e: push edi
0x1002375f: call 1002CD70h
0x10023764: mov edx,[esi+0CCh]
0x1002376a: push edx
0x1002376b: push edi
0x1002376c: call 1002CD70h
0x10023771: add esp,10h
0x10023774: and dword ptr [esi+8],0FFFFEFFFh
0x1002377b: mov [esi+0C8h],ebx
0x10023781: mov [esi+0CCh],ebx
0x10023787: mov eax,[esi+0BCh]
0x1002378d: and eax,ebp
0x1002378f: test al,20h
0x10023791: je near ptr 1002383Fh
0x10023797: mov ecx,[esp+20h]
0x1002379b: cmp ecx,0FFFFFFFFh
```

```
0x1002379e: je short 100237EAh
0x100237a0: mov eax,[esi+0D8h]
0x100237a6: cmp eax,ebx
0x100237a8: je near ptr 1002383Fh
0x100237ae: mov ebp,ecx
0x100237b0: shl ebp,4
0x100237b3: mov ecx,[eax+ebp]
0x100237b6: push ecx
0x100237b7: push edi
0x100237b8: call 1002CD70h
0x100237bd: mov edx,[esi+0D8h]
0x100237c3: mov eax,[edx+ebp+8]
0x100237c7: push eax
0x100237c8: push edi
0x100237c9: call 1002CD70h
0x100237ce: mov ecx,[esi+0D8h]
0x100237d4: mov [ecx+ebp],ebx
0x100237d7: mov edx,[esi+0D8h]
0x100237dd: mov [edx+ebp+8],ebx
0x100237e1: mov ebp,[esp+2Ch]
0x100237e5: add esp,10h
0x100237e8: jmp short 1002383Fh
0x100237ea: mov eax,[esi+0DCh]
0x100237f0: cmp eax,ebx
0x100237f2: je short 10023838h
0x100237f4: xor ebp,ebp
0x100237f6: cmp eax,ebx
0x100237f8: jle short 10023818h
0x100237fa: lea ebx,[ebx]
0x10023800: push ebp
0x10023801: push 20h
0x10023803: push esi
0x10023804: push edi
0x10023805: call 100235E0h
0x1002380a: add ebp,1
0x1002380d: add esp,10h
0x10023810: cmp ebp,[esi+0DCh]
0x10023816: jl short 10023800h
0x10023818: mov eax,[esi+0D8h]
0x1002381e: push eax
0x1002381f: push edi
0x10023820: call 1002CD70h
0x10023825: mov ebp,[esp+24h]
0x10023829: add esp,8
0x1002382c: mov [esi+0D8h],ebx
0x10023832: mov [esi+0DCh],ebx
0x10023838: and dword ptr [esi+8],0FFFFDFFFh
0x1002383f: mov eax,[edi+270h]
0x10023845: cmp eax,ebx
0x10023847: je short 10023859h
0x10023849: push eax
0x1002384a: push edi
0x1002384b: call 1002CD70h
0x10023850: add esp,8
0x10023853: mov [edi+270h],ebx
0x10023859: mov ecx,[esi+0BCh]
0x1002385f: and ecx,ebp
0x10023861: test ecx,200h
0x10023867: je near ptr 100238EBh
0x1002386d: mov ecx,[esp+20h]
0x10023871: cmp ecx,0FFFFFFFFh
0x10023874: je short 1002389Eh
0x10023876: mov eax,[esi+0C0h]
0x1002387c: cmp eax,ebx
0x1002387e: je short 100238EBh
0x10023880: lea ebp,[ecx+ecx*4]
0x10023883: add ebp,ebp
```

```
0x10023885: add ebp,ebp
0x10023887: mov edx,[eax+ebp+8]
0x1002388b: push edx
0x1002388c: push edi
0x1002388d: call 1002CD70h
0x10023892: mov eax,[esi+0C0h]
0x10023898: mov [eax+ebp+8],ebx
0x1002389c: jmp short 100238E4h
0x1002389e: mov eax,[esi+0C4h]
0x100238a4: cmp eax,ebx
0x100238a6: je short 100238EBh
0x100238a8: xor ebp,ebp
0x100238aa: cmp eax,ebx
0x100238ac: jle short 100238CBh
0x100238ae: mov edi,edi
0x100238b0: push ebp
0x100238b1: push 200h
0x100238b6: push esi
0x100238b7: push edi
0x100238b8: call 100235E0h
0x100238bd: add ebp,1
0x100238c0: add esp,10h
0x100238c3: cmp ebp,[esi+0C4h]
0x100238c9: jl short 100238B0h
0x100238cb: mov ecx,[esi+0C0h]
0x100238d1: push ecx
0x100238d2: push edi
0x100238d3: call 1002CD70h
0x100238d8: mov [esi+0C0h],ebx
0x100238de: mov [esi+0C4h],ebx
0x100238e4: mov ebp,[esp+24h]
0x100238e8: add esp,8
0x100238eb: mov edx,[esi+0BCh]
0x100238f1: and edx,ebp
0x100238f3: test dl,8
0x100238f6: je short 10023912h
0x100238f8: mov eax,[esi+80h]
0x100238fe: push eax
0x100238ff: push edi
0x10023900: call 1002CD70h
0x10023905: add esp,8
0x10023908: and dword ptr [esi+8],0FFFFFFBFh
0x1002390c: mov [esi+80h],ebx
0x10023912: mov ecx,[esi+0BCh]
0x10023918: and ecx,ebp
0x1002391a: test ecx,1000h
0x10023920: je short 1002393Ah
0x10023922: mov edx,[esi+10h]
0x10023925: push edx
0x10023926: push edi
0x10023927: call 100234D0h
0x1002392c: add esp,8
0x1002392f: and dword ptr [esi+8],0FFFFFFF7h
0x10023933: mov [esi+10h],ebx
0x10023936: mov [esi+14h],bx
0x1002393a: mov eax,[esi+0BCh]
0x10023940: and eax,ebp
0x10023942: test al,40h
0x10023944: je short 1002399Ah
0x10023946: cmp [esi+0ECh],ebx
0x1002394c: je short 10023993h
0x1002394e: xor ebp,ebp
0x10023950: cmp [esi+4],ebx
0x10023953: jle short 10023979h
0x10023955: mov ecx,[esi+0ECh]
0x1002395b: mov edx,[ecx+ebp*4]
0x1002395e: push edx
```

```
0x1002395f: push edi
0x10023960: call 1002CD70h
0x10023965: mov eax,[esi+0ECh]
0x1002396b: mov [eax+ebp*4],ebx
0x1002396e: add ebp,1
0x10023971: add esp,8
0x10023974: cmp ebp,[esi+4]
0x10023977: jl short 10023955h
0x10023979: mov ecx,[esi+0ECh]
0x1002397f: push ecx
0x10023980: push edi
0x10023981: call 1002CD70h
0x10023986: mov ebp,[esp+24h]
0x1002398a: add esp,8
0x1002398d: mov [esi+0ECh],ebx
0x10023993: and dword ptr [esi+8],0FFFF7FFFh
0x1002399a: cmp dword ptr [esp+20h],0FFFFFFFFh
0x1002399f: je short 100239A7h
0x100239a1: and ebp,0FFFFBDDFh
0x100239a7: not ebp
0x100239a9: and [esi+0BCh],ebp
0x100239af: pop ebp
0x100239b0: pop esi
0x100239b1: pop edi
0x100239b2: pop ebx
0x100239b3: ret
```

`png_free_data` is called by png_free_data, sub_100239c0, png_read_png, png_destroy_read_struct, png_destroy_write_struct, png_set_hIST, png_set_PLTE, png_set_iCCP, png_set_tRNS, png_set_rows. It delegates to sub_100234d0, png_free_data, png_free (3 callees).

12.532 sub_100239c0 at 0x100239c0 (25 instructions)

```
0x100239c0: mov eax,[esp+8]
0x100239c4: push esi
0x100239c5: mov esi,[esp+8]
0x100239c9: push 0FFFFFFFFh
0x100239cb: push 7FFFh
0x100239d0: push eax
0x100239d1: push esi
0x100239d2: call 100235E0h
0x100239d7: add esp,10h
0x100239da: cmp dword ptr [esi+220h],0
0x100239e1: je short 10023A07h
0x100239e3: mov ecx,[esi+224h]
0x100239e9: push ecx
0x100239ea: push esi
0x100239eb: call 1002CD70h
0x100239f0: add esp,8
0x100239f3: mov dword ptr [esi+224h],0
0x100239fd: mov dword ptr [esi+220h],0
0x10023a07: lea edx,[esp+0Ch]
0x10023a0b: push 0F0h
0x10023a10: push edx
0x10023a11: call 10023550h
0x10023a16: add esp,8
0x10023a19: pop esi
0x10023a1a: ret
```

`sub_100239c0` is called by png_destroy_info_struct, sub_100280e0. It delegates to png_info_init_3, png_free_data, png_free (3 callees).

12.533 png_get_io_ptr at 0x10023a20 (6 instructions)

```
0x10023a20: mov eax,[esp+4]
0x10023a24: test eax,eax
0x10023a26: jne short 10023A29h
0x10023a28: ret
0x10023a29: mov eax,[eax+58h]
0x10023a2c: ret
```

`png_get_io_ptr` is called by sub_10018b70. It calls nothing else.

12.534 png_init_io at 0x10023a30 (6 instructions)

```
0x10023a30: mov eax,[esp+4]
0x10023a34: test eax,eax
0x10023a36: je short 10023A3Fh
0x10023a38: mov ecx,[esp+8]
0x10023a3c: mov [eax+58h],ecx
0x10023a3f: ret
```

`png_init_io` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.535 png_convert_to_rfc1123 at 0x10023a40 (57 instructions)

```
0x10023a40: push esi
0x10023a41: mov esi,[esp+8]
0x10023a45: test esi,esi
0x10023a47: jne short 10023A4Dh
0x10023a49: xor eax,eax
0x10023a4b: pop esi
0x10023a4c: ret
0x10023a4d: cmp dword ptr [esi+210h],0
0x10023a54: jne short 10023A67h
0x10023a56: push 1Dh
0x10023a58: push esi
0x10023a59: call 1002CD20h
0x10023a5e: add esp,8
0x10023a61: mov [esi+210h],eax
0x10023a67: mov ecx,[esp+0Ch]
0x10023a6b: movzx eax,byte ptr [ecx+6]
0x10023a6f: cdq
0x10023a70: push edi
0x10023a71: mov edi,3Dh
0x10023a76: idiv edi
0x10023a78: movzx eax,byte ptr [ecx+5]
0x10023a7c: mov edi,3Ch
0x10023a81: push edx
0x10023a82: cdq
0x10023a83: idiv edi
0x10023a85: movzx eax,byte ptr [ecx+4]
0x10023a89: mov edi,18h
0x10023a8e: push edx
0x10023a8f: cdq
0x10023a90: idiv edi
0x10023a92: movzx eax,word ptr [ecx]
0x10023a95: mov edi,0Ch
0x10023a9a: push edx
0x10023a9b: push eax
0x10023a9c: movzx eax,byte ptr [ecx+2]
0x10023aa0: sub eax,1
0x10023aa3: cdq
0x10023aa4: idiv edi
0x10023aa6: movzx eax,byte ptr [ecx+3]
0x10023aaa: and eax,8000001Fh
0x10023aaf: lea edx,[edx*4+100FF71Ch]
0x10023ab6: push edx
```

```
0x10023ab7: jns short 10023ABEh
0x10023ab9: dec eax
0x10023aba: or eax,0FFFFFFE0h
0x10023abd: inc eax
0x10023abe: mov ecx,[esi+210h]
0x10023ac4: push eax
0x10023ac5: push 100FF7C0h
0x10023aca: push 1Dh
0x10023acc: push ecx
0x10023acd: call dword ptr ds:[100FE4C0h]
0x10023ad3: mov eax,[esi+210h]
0x10023ad9: add esp,24h
0x10023adc: pop edi
0x10023add: pop esi
0x10023ade: ret
```

`png_convert_to_rfc1123` has no recorded callers; it is likely reached through the vector table or an indirect jump. It accesses memory-mapped peripheral registers at 0x8000001f. It delegates to png_malloc (1 callees).

12.536 png_get_copyright at 0x10023ae0 (2 instructions)

```
0x10023ae0: mov eax,100FF7E0h
0x10023ae5: ret
```

`png_get_copyright` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.537 png_get_header_ver at 0x10023af0 (2 instructions)

```
0x10023af0: mov eax,100FF018h
0x10023af5: ret
```

`png_get_header_ver` is called by png_create_read_struct_2, png_create_write_struct_2. It works with the text "1.5.1". It calls nothing else.

12.538 png_get_header_version at 0x10023b00 (2 instructions)

```
0x10023b00: mov eax,100FF8A8h
0x10023b05: ret
```

`png_get_header_version` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.539 png_handle_as_unknown at 0x10023b10 (88 instructions)

```
0x10023b10: mov ecx,[esp+4]
0x10023b14: test ecx,ecx
0x10023b16: je near ptr 10023BF9h
0x10023b1c: cmp dword ptr [esp+8],0
0x10023b21: je near ptr 10023BF9h
0x10023b27: cmp dword ptr [ecx+220h],0
0x10023b2e: jle near ptr 10023BF9h
0x10023b34: mov eax,[ecx+220h]
0x10023b3a: mov ecx,[ecx+224h]
0x10023b40: push ebx
0x10023b41: push ebp
0x10023b42: push esi
0x10023b43: mov ebx,eax
0x10023b45: test ebx,ebx
0x10023b47: lea edx,[eax+eax*4]
0x10023b4a: push edi
0x10023b4b: lea edi,[edx+ecx-5]
```

```
0x10023b4f: je near ptr 10023BE9h
0x10023b55: mov edx,[esp+18h]
0x10023b59: mov eax,4
0x10023b5e: mov ecx,edi
0x10023b60: mov esi,[edx]
0x10023b62: cmp esi,[ecx]
0x10023b64: jne short 10023B78h
0x10023b66: sub eax,4
0x10023b69: add ecx,4
0x10023b6c: add edx,4
0x10023b6f: cmp eax,4
0x10023b72: jae short 10023B60h
0x10023b74: test eax,eax
0x10023b76: je short 10023BD5h
0x10023b78: movzx esi,byte ptr [edx]
0x10023b7b: movzx ebp,byte ptr [ecx]
0x10023b7e: sub esi,ebp
0x10023b80: jne short 10023BC7h
0x10023b82: sub eax,1
0x10023b85: add ecx,1
0x10023b88: add edx,1
0x10023b8b: test eax,eax
0x10023b8d: je short 10023BD5h
0x10023b8f: movzx esi,byte ptr [edx]
0x10023b92: movzx ebp,byte ptr [ecx]
0x10023b95: sub esi,ebp
0x10023b97: jne short 10023BC7h
0x10023b99: sub eax,1
0x10023b9c: add ecx,1
0x10023b9f: add edx,1
0x10023ba2: test eax,eax
0x10023ba4: je short 10023BD5h
0x10023ba6: movzx esi,byte ptr [edx]
0x10023ba9: movzx ebp,byte ptr [ecx]
0x10023bac: sub esi,ebp
0x10023bae: jne short 10023BC7h
0x10023bb0: sub eax,1
0x10023bb3: add ecx,1
0x10023bb6: add edx,1
0x10023bb9: test eax,eax
0x10023bbb: je short 10023BD5h
0x10023bbd: movzx esi,byte ptr [edx]
0x10023bc0: movzx edx,byte ptr [ecx]
0x10023bc3: sub esi,edx
0x10023bc5: je short 10023BD5h
0x10023bc7: test esi,esi
0x10023bc9: mov eax,1
0x10023bce: jg short 10023BD7h
0x10023bd0: or eax,0FFFFFFFFh
0x10023bd3: jmp short 10023BD7h
0x10023bd5: xor eax,eax
0x10023bd7: test eax,eax
0x10023bd9: je short 10023BF0h
0x10023bdb: sub ebx,1
0x10023bde: sub edi,5
0x10023be1: test ebx,ebx
0x10023be3: jne near ptr 10023B55h
0x10023be9: pop edi
0x10023bea: pop esi
0x10023beb: pop ebp
0x10023bec: xor eax,eax
0x10023bee: pop ebx
0x10023bef: ret
0x10023bf0: movzx eax,byte ptr [edi+4]
0x10023bf4: pop edi
0x10023bf5: pop esi
0x10023bf6: pop ebp
```

```
0x10023bf7: pop ebx
0x10023bf8: ret
0x10023bf9: xor eax,eax
0x10023bfb: ret
```

`png_handle_as_unknown` is called by png_read_info, png_read_end, png_write_info_before_PLTE, png_write_info, png_write_end, sub_10030750. It calls nothing else.

12.540 png_reset_zstream at 0x10023c00 (40 instructions)

```
0x10023c00: mov eax,[esp+4]
0x10023c04: test eax,eax
0x10023c06: jne short 10023C0Eh
0x10023c08: mov eax,0FFFFFFFEh
0x10023c0d: ret
0x10023c0e: add eax,78h
0x10023c11: mov [esp+4],eax
0x10023c15: jmp near ptr 1005FE10h
0x1005fe10: mov edx,[esp+4]
0x1005fe14: xor ecx,ecx
0x1005fe16: cmp edx,ecx
0x1005fe18: je short 1005FE7Bh
0x1005fe1a: mov eax,[edx+1Ch]
0x1005fe1d: cmp eax,ecx
0x1005fe1f: je short 1005FE7Bh
0x1005fe21: mov [eax+1Ch],ecx
0x1005fe24: mov [edx+14h],ecx
0x1005fe27: mov [edx+8],ecx
0x1005fe2a: mov [edx+18h],ecx
0x1005fe2d: mov dword ptr [edx+30h],1
0x1005fe34: mov [eax],ecx
0x1005fe36: mov [eax+4],ecx
0x1005fe39: mov [eax+0Ch],ecx
0x1005fe3c: mov [eax+20h],ecx
0x1005fe3f: mov [eax+28h],ecx
0x1005fe42: mov [eax+2Ch],ecx
0x1005fe45: mov [eax+30h],ecx
0x1005fe48: mov [eax+38h],ecx
0x1005fe4b: mov [eax+3Ch],ecx
0x1005fe4e: lea ecx,[eax+530h]
0x1005fe54: mov dword ptr [eax+14h],8000h
0x1005fe5b: mov [eax+6Ch],ecx
0x1005fe5e: mov [eax+50h],ecx
0x1005fe61: mov [eax+4Ch],ecx
0x1005fe64: mov dword ptr [eax+1BC0h],1
0x1005fe6e: mov dword ptr [eax+1BC4h],0FFFFFFFFh
0x1005fe78: xor eax,eax
0x1005fe7a: ret
0x1005fe7b: mov eax,0FFFFFFFEh
0x1005fe80: ret
```

`png_reset_zstream` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.541 png_access_version_number at 0x10023c20 (2 instructions)

```
0x10023c20: mov eax,2905h
0x10023c25: ret
```

`png_access_version_number` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.542 sub_10023c30 at 0x10023c30 (166 instructions)

```
0x10023c30: push ebx
0x10023c31: mov ebx,[esp+0Ch]
0x10023c35: push ebp
0x10023c36: push esi
0x10023c37: mov esi,[esp+10h]
0x10023c3b: push edi
0x10023c3c: xor edi,edi
0x10023c3e: test ebx,ebx
0x10023c40: jne short 10023C55h
0x10023c42: push 100FFAECh
0x10023c47: push esi
0x10023c48: call 100230F0h
0x10023c4d: add esp,8
0x10023c50: mov edi,1
0x10023c55: mov ebp,[esp+1Ch]
0x10023c59: test ebp,ebp
0x10023c5b: jne short 10023C6Eh
0x10023c5d: push 100FFACCh
0x10023c62: push esi
0x10023c63: call 100230F0h
0x10023c68: add esp,8
0x10023c6b: lea edi,[ebp+1]
0x10023c6e: cmp ebx,[esi+258h]
0x10023c74: ja short 10023C7Eh
0x10023c76: cmp ebx,0F4240h
0x10023c7c: jbe short 10023C91h
0x10023c7e: push 100FFAA4h
0x10023c83: push esi
0x10023c84: call 100230F0h
0x10023c89: add esp,8
0x10023c8c: mov edi,1
0x10023c91: cmp ebp,[esi+25Ch]
0x10023c97: ja short 10023CA1h
0x10023c99: cmp ebp,0F4240h
0x10023c9f: jbe short 10023CB4h
0x10023ca1: push 100FFA7Ch
0x10023ca6: push esi
0x10023ca7: call 100230F0h
0x10023cac: add esp,8
0x10023caf: mov edi,1
0x10023cb4: cmp ebx,7FFFFFFFh
0x10023cba: jbe short 10023CCFh
0x10023cbc: push 100FFA60h
0x10023cc1: push esi
0x10023cc2: call 100230F0h
0x10023cc7: add esp,8
0x10023cca: mov edi,1
0x10023ccf: cmp ebp,7FFFFFFFh
0x10023cd5: jbe short 10023CEAh
0x10023cd7: push 100FFA40h
0x10023cdc: push esi
0x10023cdd: call 100230F0h
0x10023ce2: add esp,8
0x10023ce5: mov edi,1
0x10023cea: cmp ebx,1FFFFF8Eh
0x10023cf0: jbe short 10023D00h
0x10023cf2: push 100FFA10h
0x10023cf7: push esi
0x10023cf8: call 100230F0h
0x10023cfd: add esp,8
0x10023d00: mov ebx,[esp+20h]
0x10023d04: cmp ebx,1
0x10023d07: je short 10023D30h
0x10023d09: cmp ebx,2
0x10023d0c: je short 10023D30h
0x10023d0e: cmp ebx,4
```

```
0x10023d11: je short 10023D30h
0x10023d13: cmp ebx,8
0x10023d16: je short 10023D30h
0x10023d18: cmp ebx,10h
0x10023d1b: je short 10023D30h
0x10023d1d: push 100FF9F4h
0x10023d22: push esi
0x10023d23: call 100230F0h
0x10023d28: add esp,8
0x10023d2b: mov edi,1
0x10023d30: mov ebp,[esp+24h]
0x10023d34: test ebp,ebp
0x10023d36: jl short 10023D47h
0x10023d38: cmp ebp,1
0x10023d3b: je short 10023D47h
0x10023d3d: cmp ebp,5
0x10023d40: je short 10023D47h
0x10023d42: cmp ebp,6
0x10023d45: jle short 10023D5Ah
0x10023d47: push 100FF9D8h
0x10023d4c: push esi
0x10023d4d: call 100230F0h
0x10023d52: add esp,8
0x10023d55: mov edi,1
0x10023d5a: cmp ebp,3
0x10023d5d: jne short 10023D66h
0x10023d5f: cmp ebx,8
0x10023d62: jg short 10023D7Ah
0x10023d64: jmp short 10023D8Dh
0x10023d66: cmp ebp,2
0x10023d69: je short 10023D75h
0x10023d6b: cmp ebp,4
0x10023d6e: je short 10023D75h
0x10023d70: cmp ebp,6
0x10023d73: jne short 10023D8Dh
0x10023d75: cmp ebx,8
0x10023d78: jge short 10023D8Dh
0x10023d7a: push 100FF9A4h
0x10023d7f: push esi
0x10023d80: call 100230F0h
0x10023d85: add esp,8
0x10023d88: mov edi,1
0x10023d8d: cmp dword ptr [esp+28h],2
0x10023d92: jl short 10023DA7h
0x10023d94: push 100FF980h
0x10023d99: push esi
0x10023d9a: call 100230F0h
0x10023d9f: add esp,8
0x10023da2: mov edi,1
0x10023da7: cmp dword ptr [esp+2Ch],0
0x10023dac: je short 10023DC1h
0x10023dae: push 100FF95Ch
0x10023db3: push esi
0x10023db4: call 100230F0h
0x10023db9: add esp,8
0x10023dbc: mov edi,1
0x10023dc1: mov ebx,1000h
0x10023dc6: test [esi+6Ch],ebx
0x10023dc9: je short 10023DE2h
0x10023dcb: cmp dword ptr [esi+230h],0
0x10023dd2: je short 10023DE2h
0x10023dd4: push 100FF928h
0x10023dd9: push esi
0x10023dda: call 100230F0h
0x10023ddf: add esp,8
0x10023de2: mov eax,[esp+30h]
0x10023de6: test eax,eax
```

```
0x10023de8: je short 10023E2Fh
0x10023dea: test byte ptr [esi+230h],4
0x10023df1: je short 10023E07h
0x10023df3: cmp eax,40h
0x10023df6: jne short 10023E07h
0x10023df8: test [esi+6Ch],ebx
0x10023dfb: jne short 10023E07h
0x10023dfd: cmp ebp,2
0x10023e00: je short 10023E1Ah
0x10023e02: cmp ebp,6
0x10023e05: je short 10023E1Ah
0x10023e07: push 100FF908h
0x10023e0c: push esi
0x10023e0d: call 100230F0h
0x10023e12: add esp,8
0x10023e15: mov edi,1
0x10023e1a: test [esi+6Ch],ebx
0x10023e1d: je short 10023E2Fh
0x10023e1f: push 100FF8E8h
0x10023e24: push esi
0x10023e25: call 100230F0h
0x10023e2a: add esp,8
0x10023e2d: jmp short 10023E34h
0x10023e2f: cmp edi,1
0x10023e32: jne short 10023E3Fh
0x10023e34: push 100FF8D4h
0x10023e39: push esi
0x10023e3a: call 100231E0h
0x10023e3f: pop edi
0x10023e40: pop esi
0x10023e41: pop ebp
0x10023e42: pop ebx
0x10023e43: ret
```

`sub_10023c30` is called by png_get_IHDR, png_set_IHDR. It works with the text "Invalid IHDR data", "Invalid filter method in IHDR", "Unknown filter method in IHDR". It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to png_warning, png_error (2 callees).

12.543 sub_10023e50 at 0x10023e50 (41 instructions)

```
0x10023e50: mov ecx,[esp+10h]
0x10023e54: mov edx,[esp+0Ch]
0x10023e58: mov eax,[edx]
0x10023e5a: push ebp
0x10023e5b: mov ebp,[esp+0Ch]
0x10023e5f: push esi
0x10023e60: mov esi,[ecx]
0x10023e62: cmp esi,ebp
0x10023e64: jae near ptr 10023EF8h
0x10023e6a: push edi
0x10023e6b: mov edi,[esp+10h]
0x10023e6f: push ebx
0x10023e70: mov cl,[esi+edi]
0x10023e73: mov bl,cl
0x10023e75: sub bl,30h
0x10023e78: cmp bl,9
0x10023e7b: ja short 10023E84h
0x10023e7d: mov ecx,8
0x10023e82: jmp short 10023EADh
0x10023e84: movsx ecx,cl
0x10023e87: add ecx,0FFFFFFD5h
0x10023e8a: cmp ecx,3Ah
0x10023e8d: ja short 10023EF6h
0x10023e8f: movzx ecx,byte ptr [ecx+10023F28h]
0x10023e96: jmp dword ptr [ecx*4+10023F18h]
0x10023ead: mov ebx,eax
```

```
0x10023eaf: and ebx,3
0x10023eb2: add ecx,ebx
0x10023eb4: cmp ecx,0Dh
0x10023eb7: ja short 10023EF6h
0x10023eb9: jmp dword ptr [ecx*4+10023F64h]
0x10023ef6: pop ebx
0x10023ef7: pop edi
0x10023ef8: mov [edx],eax
0x10023efa: mov edx,[esp+18h]
0x10023efe: mov [edx],esi
0x10023f00: shr eax,3
0x10023f03: pop esi
0x10023f04: and eax,1
0x10023f07: pop ebp
0x10023f08: ret
```

`sub_10023e50` is called by sub_10023fa0, sub_1002fdd0. It calls nothing else.

12.544 sub_10023fa0 at 0x10023fa0 (32 instructions)

```
0x10023fa0: sub esp,8
0x10023fa3: push esi
0x10023fa4: mov esi,[esp+14h]
0x10023fa8: push edi
0x10023fa9: mov edi,[esp+14h]
0x10023fad: lea eax,[esp+8]
0x10023fb1: push eax
0x10023fb2: lea ecx,[esp+10h]
0x10023fb6: push ecx
0x10023fb7: push esi
0x10023fb8: push edi
0x10023fb9: mov dword ptr [esp+1Ch],0
0x10023fc1: mov dword ptr [esp+18h],0
0x10023fc9: call 10023E50h
0x10023fce: add esp,10h
0x10023fd1: test eax,eax
0x10023fd3: je short 10023FEEh
0x10023fd5: mov eax,[esp+8]
0x10023fd9: cmp eax,esi
0x10023fdb: je short 10023FE3h
0x10023fdd: cmp byte ptr [eax+edi],0
0x10023fe1: jne short 10023FEEh
0x10023fe3: pop edi
0x10023fe4: mov eax,1
0x10023fe9: pop esi
0x10023fea: add esp,8
0x10023fed: ret
0x10023fee: pop edi
0x10023fef: xor eax,eax
0x10023ff1: pop esi
0x10023ff2: add esp,8
0x10023ff5: ret
```

`sub_10023fa0` is called by png_set_pCAL, png_set_sCAL_s. It delegates to sub_10023e50 (1 callees).

12.545 sub_10024000 at 0x10024000 (31 instructions)

```
0x10024000: fld1
0x10024002: xor ecx,ecx
0x10024004: test eax,eax
0x10024006: fld st(0)
0x10024008: jge short 10024021h
0x1002400a: cmp eax,0FFFFFECDh
0x1002400f: jge short 10024018h
0x10024011: fstp st(0)
```

```
0x10024013: fstp st(0)
0x10024015: fldz
0x10024017: ret
0x10024018: neg eax
0x1002401a: mov ecx,1
0x1002401f: test eax,eax
0x10024021: jle short 10024042h
0x10024023: fld qword ptr ds:[100FFB08h]
0x10024029: test al,1
0x1002402b: je short 10024031h
0x1002402d: fld st(0)
0x1002402f: fmulp st(2),st
0x10024031: sar eax,1
0x10024033: fmul st(0),st
0x10024035: test eax,eax
0x10024037: jg short 10024029h
0x10024039: test ecx,ecx
0x1002403b: fstp st(0)
0x1002403d: je short 10024042h
0x1002403f: fdivp
0x10024041: ret
0x10024042: fstp st(1)
0x10024044: ret
```

`sub_10024000` is called by sub_10024050. It calls nothing else.

12.546 sub_10024050 at 0x10024050 (486 instructions)

```
0x10024050: push ebp
0x10024051: mov ebp,esp
0x10024053: and esp,0FFFFFFC0h
0x10024056: sub esp,34h
0x10024059: mov eax,ds:[10136220h]
0x1002405e: xor eax,esp
0x10024060: mov [esp+30h],eax
0x10024064: mov eax,[ebp+8]
0x10024067: push ebx
0x10024068: mov [esp+1Ch],eax
0x1002406c: mov eax,[ebp+1Ch]
0x1002406f: cmp eax,1
0x10024072: push esi
0x10024073: mov esi,[ebp+0Ch]
0x10024076: push edi
0x10024077: jae short 10024080h
0x10024079: mov eax,0Fh
0x1002407e: jmp short 1002408Ah
0x10024080: cmp eax,10h
0x10024083: jbe short 1002408Dh
0x10024085: mov eax,10h
0x1002408a: mov [ebp+1Ch],eax
0x1002408d: add eax,5
0x10024090: cmp [ebp+10h],eax
0x10024093: jb near ptr 100245DCh
0x10024099: fldz
0x1002409b: fld qword ptr [ebp+14h]
0x1002409e: fcom
0x100240a0: fnstsw ax
0x100240a2: fstp st(1)
0x100240a4: test ah,5
0x100240a7: jp short 100240B8h
0x100240a9: fchs
0x100240ab: mov byte ptr [esi],2Dh
0x100240ae: add esi,1
0x100240b1: fst qword ptr [ebp+14h]
0x100240b4: sub dword ptr [ebp+10h],1
0x100240b8: fld qword ptr ds:[100FFB48h]
```

```
0x100240be: fcomp
0x100240c0: fnstsw ax
0x100240c2: test ah,41h
0x100240c5: jp near ptr 1002459Ah
0x100240cb: fld qword ptr ds:[100FFB40h]
0x100240d1: fcomp
0x100240d3: fnstsw ax
0x100240d5: test ah,1
0x100240d8: jne near ptr 100245B5h
0x100240de: lea ecx,[esp+18h]
0x100240e2: push ecx
0x100240e3: sub esp,8
0x100240e6: fstp qword ptr [esp]
0x100240e9: call dword ptr ds:[100FE4B8h]
0x100240ef: mov ebx,[esp+24h]
0x100240f3: fstp st(0)
0x100240f5: imul ebx,4Dh
0x100240f8: sar ebx,8
0x100240fb: add esp,0Ch
0x100240fe: mov eax,ebx
0x10024100: mov [esp+18h],ebx
0x10024104: call 10024000h
0x10024109: fst qword ptr [esp+28h]
0x1002410d: fld qword ptr ds:[100FFB48h]
0x10024113: fcomp
0x10024115: fnstsw ax
0x10024117: test ah,41h
0x1002411a: je short 10024128h
0x1002411c: fld qword ptr [ebp+14h]
0x1002411f: fcomp
0x10024121: fnstsw ax
0x10024123: test ah,41h
0x10024126: jne short 1002414Dh
0x10024128: lea edx,[ebx+1]
0x1002412b: fstp st(0)
0x1002412d: mov eax,edx
0x1002412f: call 10024000h
0x10024134: fld qword ptr ds:[100FFB40h]
0x1002413a: fcomp
0x1002413c: fnstsw ax
0x1002413e: test ah,1
0x10024141: jne short 10024147h
0x10024143: mov ebx,edx
0x10024145: jmp short 10024109h
0x10024147: fstp st(0)
0x10024149: fld qword ptr [esp+28h]
0x1002414d: fdivr qword ptr [ebp+14h]
0x10024150: mov [esp+18h],ebx
0x10024154: fld1
0x10024156: fcom
0x10024158: fnstsw ax
0x1002415a: fld qword ptr ds:[100FFB08h]
0x10024160: test ah,41h
0x10024163: jnp short 1002416Bh
0x10024165: fstp st(1)
0x10024167: jmp short 10024183h
0x10024169: fxch st(2)
0x1002416b: fdiv st(2),st
0x1002416d: add ebx,1
0x10024170: fxch st(2)
0x10024172: fcom
0x10024174: fnstsw ax
0x10024176: test ah,1
0x10024179: je short 10024169h
0x1002417b: fstp st(1)
0x1002417d: mov [esp+18h],ebx
0x10024181: fxch
```

```
0x10024183: lea edx,[ebx+2]
0x10024186: cmp edx,1
0x10024189: ja short 10024197h
0x1002418b: mov eax,ebx
0x1002418d: neg eax
0x1002418f: xor ebx,ebx
0x10024191: mov [esp+18h],ebx
0x10024195: jmp short 10024199h
0x10024197: xor eax,eax
0x10024199: mov edi,[ebp+10h]
0x1002419c: mov [esp+1Ch],eax
0x100241a0: mov [esp+20h],eax
0x100241a4: mov dword ptr [esp+14h],0
0x100241ac: jmp short 100241B4h
0x100241ae: fld qword ptr ds:[100FFB08h]
0x100241b4: mov eax,[esp+14h]
0x100241b8: fmulp
0x100241ba: sub eax,[esp+20h]
0x100241be: mov ecx,[esp+1Ch]
0x100241c2: lea edx,[eax+ecx+1]
0x100241c6: cmp edx,[ebp+1Ch]
0x100241c9: jge short 100241F4h
0x100241cb: lea eax,[esp+28h]
0x100241cf: push eax
0x100241d0: sub esp,8
0x100241d3: fstp qword ptr [esp]
0x100241d6: call dword ptr ds:[100FE4BCh]
0x100241dc: mov ebx,[esp+24h]
0x100241e0: fld qword ptr [esp+34h]
0x100241e4: mov ecx,[esp+20h]
0x100241e8: mov edx,[esp+28h]
0x100241ec: add esp,0Ch
0x100241ef: jmp near ptr 10024433h
0x100241f4: fadd qword ptr ds:[100FE7C8h]
0x100241fa: sub esp,8
0x100241fd: fstp qword ptr [esp]
0x10024200: call 1002297Ch
0x10024205: mov edx,[esp+24h]
0x10024209: fst qword ptr [esp+30h]
0x1002420d: fld qword ptr ds:[100FFB38h]
0x10024213: mov ecx,[esp+1Ch]
0x10024217: fcom
0x10024219: add esp,8
0x1002421c: fnstsw ax
0x1002421e: test ah,5
0x10024221: jp near ptr 10024419h
0x10024227: test edx,edx
0x10024229: jle short 1002424Eh
0x1002422b: fstp st(1)
0x1002422d: sub edx,1
0x10024230: test ecx,ecx
0x10024232: fstp st(0)
0x10024234: fld1
0x10024236: mov [esp+1Ch],edx
0x1002423a: fst qword ptr [esp+28h]
0x1002423e: jne near ptr 1002442Fh
0x10024244: sub dword ptr [esp+20h],1
0x10024249: jmp near ptr 1002442Fh
0x1002424e: cmp ecx,4
0x10024251: jge short 10024286h
0x10024253: fxch
0x10024255: test ecx,ecx
0x10024257: jle near ptr 100243E4h
0x1002425d: fcom
0x1002425f: fnstsw ax
0x10024261: test ah,41h
0x10024264: jne near ptr 10024421h
```

```
0x1002426a: movsx eax,byte ptr [esi-1]
0x1002426e: fstp st(0)
0x10024270: sub esi,1
0x10024273: cmp ebx,0FFFFFFFFh
0x10024276: je near ptr 1002439Fh
0x1002427c: add ebx,1
0x1002427f: jmp near ptr 100243B3h
0x10024284: fxch
0x10024286: fcom
0x10024288: fnstsw ax
0x1002428a: test ah,5
0x1002428d: jp near ptr 1002441Dh
0x10024293: movsx eax,byte ptr [esi-1]
0x10024297: fstp st(1)
0x10024299: sub esi,1
0x1002429c: cmp ebx,0FFFFFFFFh
0x1002429f: je short 100242A6h
0x100242a1: add ebx,1
0x100242a4: jmp short 100242BAh
0x100242a6: cmp eax,2Eh
0x100242a9: jne short 100242BAh
0x100242ab: movsx eax,byte ptr [esi-1]
0x100242af: sub esi,1
0x100242b2: add edi,1
0x100242b5: mov ebx,1
0x100242ba: add eax,0FFFFFFD1h
0x100242bd: mov [esp+14h],eax
0x100242c1: fild dword ptr [esp+14h]
0x100242c5: fcom
0x100242c7: fnstsw ax
0x100242c9: test ah,41h
0x100242cc: jne near ptr 100243CBh
0x100242d2: movsx eax,byte ptr [esi-1]
0x100242d6: fstp st(0)
0x100242d8: sub esi,1
0x100242db: cmp ebx,0FFFFFFFFh
0x100242de: je short 100242E5h
0x100242e0: add ebx,1
0x100242e3: jmp short 100242F9h
0x100242e5: cmp eax,2Eh
0x100242e8: jne short 100242F9h
0x100242ea: movsx eax,byte ptr [esi-1]
0x100242ee: sub esi,1
0x100242f1: add edi,1
0x100242f4: mov ebx,1
0x100242f9: add eax,0FFFFFFD1h
0x100242fc: mov [esp+14h],eax
0x10024300: fild dword ptr [esp+14h]
0x10024304: fcom
0x10024306: fnstsw ax
0x10024308: test ah,41h
0x1002430b: jne near ptr 100243D0h
0x10024311: movsx eax,byte ptr [esi-1]
0x10024315: fstp st(0)
0x10024317: sub esi,1
0x1002431a: cmp ebx,0FFFFFFFFh
0x1002431d: je short 10024324h
0x1002431f: add ebx,1
0x10024322: jmp short 10024338h
0x10024324: cmp eax,2Eh
0x10024327: jne short 10024338h
0x10024329: movsx eax,byte ptr [esi-1]
0x1002432d: sub esi,1
0x10024330: add edi,1
0x10024333: mov ebx,1
0x10024338: add eax,0FFFFFFD1h
0x1002433b: mov [esp+14h],eax
```

```
0x1002433f: fild dword ptr [esp+14h]
0x10024343: fcom
0x10024345: fnstsw ax
0x10024347: test ah,41h
0x1002434a: jne near ptr 100243D5h
0x10024350: movsx eax,byte ptr [esi-1]
0x10024354: fstp st(0)
0x10024356: sub esi,1
0x10024359: cmp ebx,0FFFFFFFFh
0x1002435c: je short 10024363h
0x1002435e: add ebx,1
0x10024361: jmp short 10024377h
0x10024363: cmp eax,2Eh
0x10024366: jne short 10024377h
0x10024368: movsx eax,byte ptr [esi-1]
0x1002436c: sub esi,1
0x1002436f: add edi,1
0x10024372: mov ebx,1
0x10024377: add eax,0FFFFFFD1h
0x1002437a: sub ecx,4
0x1002437d: cmp ecx,3
0x10024380: mov [esp+14h],eax
0x10024384: fild dword ptr [esp+14h]
0x10024388: jg near ptr 10024284h
0x1002438e: fst qword ptr [esp+28h]
0x10024392: mov [esp+14h],ecx
0x10024396: mov [esp+18h],ebx
0x1002439a: jmp near ptr 10024255h
0x1002439f: cmp eax,2Eh
0x100243a2: jne short 100243B3h
0x100243a4: movsx eax,byte ptr [esi-1]
0x100243a8: sub esi,1
0x100243ab: add edi,1
0x100243ae: mov ebx,1
0x100243b3: add eax,0FFFFFFD1h
0x100243b6: sub ecx,1
0x100243b9: test ecx,ecx
0x100243bb: mov [esp+14h],eax
0x100243bf: fild dword ptr [esp+14h]
0x100243c3: jg near ptr 1002425Dh
0x100243c9: jmp short 100243D8h
0x100243cb: sub ecx,1
0x100243ce: jmp short 100243D8h
0x100243d0: sub ecx,2
0x100243d3: jmp short 100243D8h
0x100243d5: sub ecx,3
0x100243d8: fst qword ptr [esp+28h]
0x100243dc: mov [esp+14h],ecx
0x100243e0: mov [esp+18h],ebx
0x100243e4: fcom
0x100243e6: fnstsw ax
0x100243e8: fstp st(1)
0x100243ea: test ah,41h
0x100243ed: jne short 1002442Fh
0x100243ef: cmp ebx,0FFFFFFFFh
0x100243f2: fstp st(0)
0x100243f4: jne short 1002440Eh
0x100243f6: sub esi,1
0x100243f9: cmp byte ptr [esi],2Eh
0x100243fc: jne short 10024415h
0x100243fe: fld1
0x10024400: mov ebx,1
0x10024405: add edi,1
0x10024408: mov [esp+18h],ebx
0x1002440c: jmp short 1002442Bh
0x1002440e: add ebx,1
0x10024411: mov [esp+18h],ebx
```

```
0x10024415: fld1
0x10024417: jmp short 1002442Bh
0x10024419: fstp st(0)
0x1002441b: jmp short 1002442Fh
0x1002441d: fstp st(0)
0x1002441f: jmp short 10024423h
0x10024421: fstp st(1)
0x10024423: mov [esp+18h],ebx
0x10024427: mov [esp+14h],ecx
0x1002442b: fst qword ptr [esp+28h]
0x1002442f: fldz
0x10024431: fxch
0x10024433: fldz
0x10024435: fucomp
0x10024437: fnstsw ax
0x10024439: test ah,44h
0x1002443c: jp short 10024452h
0x1002443e: add edx,1
0x10024441: fstp st(0)
0x10024443: test ecx,ecx
0x10024445: mov [esp+1Ch],edx
0x10024449: jne short 100244C6h
0x1002444b: add dword ptr [esp+20h],1
0x10024450: jmp short 100244C6h
0x10024452: mov eax,edx
0x10024454: sub eax,[esp+20h]
0x10024458: mov dword ptr [esp+20h],0
0x10024460: add ecx,eax
0x10024462: test edx,edx
0x10024464: mov [esp+14h],ecx
0x10024468: jle short 10024494h
0x1002446a: cmp ebx,0FFFFFFFFh
0x1002446d: je short 1002447Fh
0x1002446f: test ebx,ebx
0x10024471: jne short 1002447Ch
0x10024473: mov byte ptr [esi],2Eh
0x10024476: add esi,1
0x10024479: sub edi,1
0x1002447c: sub ebx,1
0x1002447f: mov byte ptr [esi],30h
0x10024482: sub edx,1
0x10024485: add esi,1
0x10024488: test edx,edx
0x1002448a: jg short 1002446Ah
0x1002448c: mov [esp+1Ch],edx
0x10024490: mov [esp+18h],ebx
0x10024494: cmp ebx,0FFFFFFFFh
0x10024497: je short 100244ADh
0x10024499: test ebx,ebx
0x1002449b: jne short 100244A6h
0x1002449d: mov byte ptr [esi],2Eh
0x100244a0: add esi,1
0x100244a3: sub edi,1
0x100244a6: sub ebx,1
0x100244a9: mov [esp+18h],ebx
0x100244ad: call 100228D0h
0x100244b2: mov edx,[esp+1Ch]
0x100244b6: add al,30h
0x100244b8: mov [esi],al
0x100244ba: add esi,1
0x100244bd: add dword ptr [esp+14h],1
0x100244c2: mov ecx,[esp+14h]
0x100244c6: mov eax,ecx
0x100244c8: sub eax,[esp+20h]
0x100244cc: add eax,edx
0x100244ce: cmp eax,[ebp+1Ch]
0x100244d1: jge short 100244E6h
```

```
0x100244d3: fld qword ptr ds:[100FFB48h]
0x100244d9: fcomp
0x100244db: fnstsw ax
0x100244dd: test ah,5
0x100244e0: jnp near ptr 100241AEh
0x100244e6: lea edx,[ebx+1]
0x100244e9: fstp st(0)
0x100244eb: cmp edx,3
0x100244ee: ja short 1002451Bh
0x100244f0: sub ebx,1
0x100244f3: js short 10024506h
0x100244f5: lea edi,[ebx+1]
0x100244f8: push edi
0x100244f9: push 30h
0x100244fb: push esi
0x100244fc: call 100228C0h
0x10024501: add esp,0Ch
0x10024504: add esi,edi
0x10024506: mov byte ptr [esi],0
0x10024509: pop edi
0x1002450a: pop esi
0x1002450b: pop ebx
0x1002450c: mov ecx,[esp+30h]
0x10024510: xor ecx,esp
0x10024512: call 10022326h
0x10024517: mov esp,ebp
0x10024519: pop ebp
0x1002451a: ret
0x1002451b: or eax,0FFFFFFFFh
0x1002451e: sub eax,ecx
0x10024520: mov byte ptr [esi],45h
0x10024523: add edi,eax
0x10024525: add esi,1
0x10024528: test ebx,ebx
0x1002452a: mov [ebp+10h],edi
0x1002452d: jge short 1002453Dh
0x1002452f: sub edi,1
0x10024532: mov byte ptr [esi],2Dh
0x10024535: add esi,1
0x10024538: mov [ebp+10h],edi
0x1002453b: neg ebx
0x1002453d: xor edi,edi
0x1002453f: test ebx,ebx
0x10024541: jle short 1002456Ch
0x10024543: mov eax,66666667h
0x10024548: imul ebx
0x1002454a: sar edx,2
0x1002454d: mov ecx,edx
0x1002454f: shr ecx,1Fh
0x10024552: add ecx,edx
0x10024554: mov al,cl
0x10024556: mov dl,0Ah
0x10024558: imul dl
0x1002455a: sub bl,al
0x1002455c: add bl,30h
0x1002455f: mov [esp+edi+28h],bl
0x10024563: mov ebx,ecx
0x10024565: add edi,1
0x10024568: test ebx,ebx
0x1002456a: jg short 10024543h
0x1002456c: cmp [ebp+10h],edi
0x1002456f: jle short 100245DCh
0x10024571: test edi,edi
0x10024573: jle short 10024506h
0x10024575: mov al,[esp+edi+27h]
0x10024579: sub edi,1
0x1002457c: mov [esi],al
```

```
0x1002457e: add esi,1
0x10024581: test edi,edi
0x10024583: jg short 10024575h
0x10024585: mov byte ptr [esi],0
0x10024588: pop edi
0x10024589: pop esi
0x1002458a: pop ebx
0x1002458b: mov ecx,[esp+30h]
0x1002458f: xor ecx,esp
0x10024591: call 10022326h
0x10024596: mov esp,ebp
0x10024598: pop ebp
0x10024599: ret
0x1002459a: mov byte ptr [esi],30h
0x1002459d: fstp st(0)
0x1002459f: mov byte ptr [esi+1],0
0x100245a3: pop edi
0x100245a4: pop esi
0x100245a5: pop ebx
0x100245a6: mov ecx,[esp+30h]
0x100245aa: xor ecx,esp
0x100245ac: call 10022326h
0x100245b1: mov esp,ebp
0x100245b3: pop ebp
0x100245b4: ret
0x100245b5: mov ecx,[esp+3Ch]
0x100245b9: fstp st(0)
0x100245bb: mov byte ptr [esi],69h
0x100245be: add esi,1
0x100245c1: mov byte ptr [esi],6Eh
0x100245c4: add esi,1
0x100245c7: mov byte ptr [esi],66h
0x100245ca: pop edi
0x100245cb: mov byte ptr [esi+1],0
0x100245cf: pop esi
0x100245d0: pop ebx
0x100245d1: xor ecx,esp
0x100245d3: call 10022326h
0x100245d8: mov esp,ebp
0x100245da: pop ebp
0x100245db: ret
0x100245dc: mov ecx,[esp+24h]
0x100245e0: push 100FFB10h
0x100245e5: push ecx
0x100245e6: call 100231E0h
0x100245eb: int 3
```

`sub_10024050` is called by png_set_sCAL. It accesses memory-mapped peripheral registers at 0x66666667. It delegates to sub_10022326, j_memset, sub_100228d0 (6 callees).

12.547 sub_100245f0 at 0x100245f0 (105 instructions)

```
0x100245f0: sub esp,10h
0x100245f3: mov eax,ds:[10136220h]
0x100245f8: xor eax,esp
0x100245fa: mov [esp+0Ch],eax
0x100245fe: cmp dword ptr [esp+1Ch],0Ch
0x10024603: mov eax,[esp+14h]
0x10024607: push edi
0x10024608: mov edi,[esp+1Ch]
0x1002460c: jbe near ptr 10024700h
0x10024612: mov ecx,[esp+24h]
0x10024616: test ecx,ecx
0x10024618: jge short 10024622h
0x1002461a: mov byte ptr [edi],2Dh
0x1002461d: add edi,1
```

```
0x10024620: neg ecx
0x10024622: cmp ecx,80000000h
0x10024628: ja near ptr 10024700h
0x1002462e: push ebp
0x1002462f: push esi
0x10024630: xor esi,esi
0x10024632: test ecx,ecx
0x10024634: lea ebp,[esi+10h]
0x10024637: je near ptr 100246E5h
0x1002463d: lea ecx,[ecx]
0x10024640: mov eax,0CCCCCCCDh
0x10024645: mul ecx
0x10024647: shr edx,3
0x1002464a: mov eax,edx
0x1002464c: neg eax
0x1002464e: add eax,eax
0x10024650: add eax,eax
0x10024652: sub eax,edx
0x10024654: lea eax,[ecx+eax*2]
0x10024657: mov cl,al
0x10024659: add cl,30h
0x1002465c: mov [esp+esi+0Ch],cl
0x10024660: add esi,1
0x10024663: cmp ebp,10h
0x10024666: jne short 1002466Eh
0x10024668: test eax,eax
0x1002466a: jbe short 1002466Eh
0x1002466c: mov ebp,esi
0x1002466e: test edx,edx
0x10024670: mov ecx,edx
0x10024672: jne short 10024640h
0x10024674: test esi,esi
0x10024676: jbe short 100246E5h
0x10024678: cmp esi,5
0x1002467b: jbe short 10024691h
0x1002467d: lea ecx,[ecx]
0x10024680: mov dl,[esp+esi+0Bh]
0x10024684: sub esi,1
0x10024687: mov [edi],dl
0x10024689: add edi,1
0x1002468c: cmp esi,5
0x1002468f: ja short 10024680h
0x10024691: cmp ebp,5
0x10024694: ja short 100246EBh
0x10024696: mov byte ptr [edi],2Eh
0x10024699: add edi,1
0x1002469c: cmp esi,5
0x1002469f: jae short 100246B8h
0x100246a1: push ebx
0x100246a2: mov ebx,5
0x100246a7: sub ebx,esi
0x100246a9: push ebx
0x100246aa: push 30h
0x100246ac: push edi
0x100246ad: call 100228C0h
0x100246b2: add esp,0Ch
0x100246b5: add edi,ebx
0x100246b7: pop ebx
0x100246b8: cmp esi,ebp
0x100246ba: jb short 100246EBh
0x100246bc: lea esp,[esp]
0x100246c0: mov al,[esp+esi+0Bh]
0x100246c4: sub esi,1
0x100246c7: mov [edi],al
0x100246c9: add edi,1
0x100246cc: cmp esi,ebp
0x100246ce: jae short 100246C0h
```

```
0x100246d0: pop esi
0x100246d1: pop ebp
0x100246d2: mov byte ptr [edi],0
0x100246d5: pop edi
0x100246d6: mov ecx,[esp+0Ch]
0x100246da: xor ecx,esp
0x100246dc: call 10022326h
0x100246e1: add esp,10h
0x100246e4: ret
0x100246e5: mov byte ptr [edi],30h
0x100246e8: add edi,1
0x100246eb: mov ecx,[esp+18h]
0x100246ef: pop esi
0x100246f0: pop ebp
0x100246f1: mov byte ptr [edi],0
0x100246f4: pop edi
0x100246f5: xor ecx,esp
0x100246f7: call 10022326h
0x100246fc: add esp,10h
0x100246ff: ret
0x10024700: push 100FFB10h
0x10024705: push eax
0x10024706: call 100231E0h
0x1002470b: int 3
```

`sub_100245f0` is called by png_set_sCAL_fixed. It accesses memory-mapped peripheral registers at 0x80000000. It delegates to sub_10022326, j_memset, png_error (3 callees).

12.548 sub_10024710 at 0x10024710 (72 instructions)

```
0x100228d0: cmp dword ptr ds:[10145460h],0
0x100228d7: je short 10022906h
0x100228d9: push ebp
0x100228da: mov ebp,esp
0x100228dc: sub esp,8
0x100228df: and esp,0FFFFFFF8h
0x100228e2: fstp qword ptr [esp]
0x100228e5: cvttsd2si eax,qword ptr [esp]
0x100228ea: leave
0x100228eb: ret
0x10022906: push ebp
0x10022907: mov ebp,esp
0x10022909: sub esp,20h
0x1002290c: and esp,0FFFFFFF0h
0x1002290f: fld st(0)
0x10022911: fst dword ptr [esp+18h]
0x10022915: fistp qword ptr [esp+10h]
0x10022919: fild qword ptr [esp+10h]
0x1002291d: mov edx,[esp+18h]
0x10022921: mov eax,[esp+10h]
0x10022925: test eax,eax
0x10022927: je short 10022965h
0x10022929: fsubp
0x1002292b: test edx,edx
0x1002292d: jns short 1002294Dh
0x1002292f: fstp dword ptr [esp]
0x10022932: mov ecx,[esp]
0x10022935: xor ecx,80000000h
0x1002293b: add ecx,7FFFFFFFh
0x10022941: adc eax,0
0x10022944: mov edx,[esp+14h]
0x10022948: adc edx,0
0x1002294b: jmp short 10022979h
0x1002294d: fstp dword ptr [esp]
0x10022950: mov ecx,[esp]
0x10022953: add ecx,7FFFFFFFh
```

```
0x10022959: sbb eax,0
0x1002295c: mov edx,[esp+14h]
0x10022960: sbb edx,0
0x10022963: jmp short 10022979h
0x10022965: mov edx,[esp+14h]
0x10022969: test edx,7FFFFFFFh
0x1002296f: jne short 10022929h
0x10022971: fstp dword ptr [esp+18h]
0x10022975: fstp dword ptr [esp+18h]
0x10022979: leave
0x1002297a: ret
0x10024710: fld qword ptr [esp+8]
0x10024714: sub esp,8
0x10024717: fmul qword ptr ds:[100FFB60h]
0x1002471d: fadd qword ptr ds:[100FE7C8h]
0x10024723: fstp qword ptr [esp]
0x10024726: call 1002297Ch
0x1002472b: fld qword ptr ds:[100FFB58h]
0x10024731: add esp,8
0x10024734: fcomp
0x10024736: fnstsw ax
0x10024738: test ah,5
0x1002473b: jnp short 10024751h
0x1002473d: fld qword ptr ds:[100FFB50h]
0x10024743: fcomp
0x10024745: fnstsw ax
0x10024747: test ah,41h
0x1002474a: je short 10024751h
0x1002474c: jmp near ptr 100228D0h
0x10024751: mov eax,[esp+10h]
0x10024755: fstp st(0)
0x10024757: mov ecx,[esp+4]
0x1002475b: push eax
0x1002475c: push ecx
0x1002475d: call 100232A0h
0x10024762: int 3
```

`sub_10024710` is called by png_get_sCAL_fixed, png_set_cHRM, png_set_gAMA, png_set_background, png_set_gamma, png_set_rgb_to_gray. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to j_floor, sub_100232a0 (2 callees).

12.549 sub_10024770 at 0x10024770 (36 instructions)

```
0x10024770: cmp dword ptr [esp+10h],0
0x10024775: je short 100247E6h
0x10024777: cmp dword ptr [esp+8],0
0x1002477c: je short 100247D4h
0x1002477e: cmp dword ptr [esp+0Ch],0
0x10024783: je short 100247D4h
0x10024785: fild dword ptr [esp+8]
0x10024789: sub esp,8
0x1002478c: fimul dword ptr [esp+14h]
0x10024790: fidiv dword ptr [esp+18h]
0x10024794: fadd qword ptr ds:[100FE7C8h]
0x1002479a: fstp qword ptr [esp]
0x1002479d: call 1002297Ch
0x100247a2: fld qword ptr ds:[100FFB58h]
0x100247a8: add esp,8
0x100247ab: fcomp
0x100247ad: fnstsw ax
0x100247af: test ah,1
0x100247b2: jne short 100247E4h
0x100247b4: fld qword ptr ds:[100FFB50h]
0x100247ba: fcomp
0x100247bc: fnstsw ax
0x100247be: test ah,41h
```

```
0x100247c1: jp short 100247E4h
0x100247c3: call 100228D0h
0x100247c8: mov ecx,[esp+4]
0x100247cc: mov [ecx],eax
0x100247ce: mov eax,1
0x100247d3: ret
0x100247d4: mov edx,[esp+4]
0x100247d8: mov dword ptr [edx],0
0x100247de: mov eax,1
0x100247e3: ret
0x100247e4: fstp st(0)
0x100247e6: xor eax,eax
0x100247e8: ret
```

`sub_10024770` is called by sub_100247f0, png_get_pixel_aspect_ratio_fixed, png_get_pixels_per_inch, png_get_x_pixels_per_inch, png_get_y_pixels_per_inch, sub_10031780. It delegates to sub_100228d0, j_floor (2 callees).

12.550 sub_100247f0 at 0x100247f0 (24 instructions)

```
0x100247f0: push ecx
0x100247f1: mov eax,[esp+14h]
0x100247f5: mov ecx,[esp+10h]
0x100247f9: mov edx,[esp+0Ch]
0x100247fd: push eax
0x100247fe: push ecx
0x100247ff: push edx
0x10024800: lea eax,[esp+0Ch]
0x10024804: push eax
0x10024805: call 10024770h
0x1002480a: add esp,10h
0x1002480d: test eax,eax
0x1002480f: je short 10024816h
0x10024811: mov eax,[esp]
0x10024814: pop ecx
0x10024815: ret
0x10024816: mov ecx,[esp+8]
0x1002481a: push 100FFB68h
0x1002481f: push ecx
0x10024820: call 100230F0h
0x10024825: add esp,8
0x10024828: xor eax,eax
0x1002482a: pop ecx
0x1002482b: ret
```

`sub_100247f0` is called by png_get_x_offset_inches_fixed, png_get_y_offset_inches_fixed. It delegates to png_warning, sub_10024770 (2 callees).

12.551 sub_10024830 at 0x10024830 (68 instructions)

```
0x100228d0: cmp dword ptr ds:[10145460h],0
0x100228d7: je short 10022906h
0x100228d9: push ebp
0x100228da: mov ebp,esp
0x100228dc: sub esp,8
0x100228df: and esp,0FFFFFFF8h
0x100228e2: fstp qword ptr [esp]
0x100228e5: cvttsd2si eax,qword ptr [esp]
0x100228ea: leave
0x100228eb: ret
0x10022906: push ebp
0x10022907: mov ebp,esp
0x10022909: sub esp,20h
0x1002290c: and esp,0FFFFFFF0h
```

```
0x1002290f: fld st(0)
0x10022911: fst dword ptr [esp+18h]
0x10022915: fistp qword ptr [esp+10h]
0x10022919: fild qword ptr [esp+10h]
0x1002291d: mov edx,[esp+18h]
0x10022921: mov eax,[esp+10h]
0x10022925: test eax,eax
0x10022927: je short 10022965h
0x10022929: fsubp
0x1002292b: test edx,edx
0x1002292d: jns short 1002294Dh
0x1002292f: fstp dword ptr [esp]
0x10022932: mov ecx,[esp]
0x10022935: xor ecx,80000000h
0x1002293b: add ecx,7FFFFFFFh
0x10022941: adc eax,0
0x10022944: mov edx,[esp+14h]
0x10022948: adc edx,0
0x1002294b: jmp short 10022979h
0x1002294d: fstp dword ptr [esp]
0x10022950: mov ecx,[esp]
0x10022953: add ecx,7FFFFFFFh
0x10022959: sbb eax,0
0x1002295c: mov edx,[esp+14h]
0x10022960: sbb edx,0
0x10022963: jmp short 10022979h
0x10022965: mov edx,[esp+14h]
0x10022969: test edx,7FFFFFFFh
0x1002296f: jne short 10022929h
0x10022971: fstp dword ptr [esp+18h]
0x10022975: fstp dword ptr [esp+18h]
0x10022979: leave
0x1002297a: ret
0x10024830: fild dword ptr [esp+4]
0x10024834: sub esp,8
0x10024837: fdivr qword ptr ds:[100FFB88h]
0x1002483d: fadd qword ptr ds:[100FE7C8h]
0x10024843: fstp qword ptr [esp]
0x10024846: call 1002297Ch
0x1002484b: fld qword ptr ds:[100FFB58h]
0x10024851: add esp,8
0x10024854: fcomp
0x10024856: fnstsw ax
0x10024858: test ah,1
0x1002485b: jne short 10024871h
0x1002485d: fld qword ptr ds:[100FFB50h]
0x10024863: fcomp
0x10024865: fnstsw ax
0x10024867: test ah,41h
0x1002486a: jp short 10024871h
0x1002486c: jmp near ptr 100228D0h
0x10024871: fstp st(0)
0x10024873: xor eax,eax
0x10024875: ret
```

`sub_10024830` is called by sub_10024e30, sub_10031a20. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to j_floor (1 callees).

12.552 sub_10024880 at 0x10024880 (69 instructions)

```
0x100228d0: cmp dword ptr ds:[10145460h],0
0x100228d7: je short 10022906h
0x100228d9: push ebp
0x100228da: mov ebp,esp
0x100228dc: sub esp,8
0x100228df: and esp,0FFFFFFF8h
```

```
0x100228e2: fstp qword ptr [esp]
0x100228e5: cvttsd2si eax,qword ptr [esp]
0x100228ea: leave
0x100228eb: ret
0x10022906: push ebp
0x10022907: mov ebp,esp
0x10022909: sub esp,20h
0x1002290c: and esp,0FFFFFFF0h
0x1002290f: fld st(0)
0x10022911: fst dword ptr [esp+18h]
0x10022915: fistp qword ptr [esp+10h]
0x10022919: fild qword ptr [esp+10h]
0x1002291d: mov edx,[esp+18h]
0x10022921: mov eax,[esp+10h]
0x10022925: test eax,eax
0x10022927: je short 10022965h
0x10022929: fsubp
0x1002292b: test edx,edx
0x1002292d: jns short 1002294Dh
0x1002292f: fstp dword ptr [esp]
0x10022932: mov ecx,[esp]
0x10022935: xor ecx,80000000h
0x1002293b: add ecx,7FFFFFFFh
0x10022941: adc eax,0
0x10022944: mov edx,[esp+14h]
0x10022948: adc edx,0
0x1002294b: jmp short 10022979h
0x1002294d: fstp dword ptr [esp]
0x10022950: mov ecx,[esp]
0x10022953: add ecx,7FFFFFFFh
0x10022959: sbb eax,0
0x1002295c: mov edx,[esp+14h]
0x10022960: sbb edx,0
0x10022963: jmp short 10022979h
0x10022965: mov edx,[esp+14h]
0x10022969: test edx,7FFFFFFFh
0x1002296f: jne short 10022929h
0x10022971: fstp dword ptr [esp+18h]
0x10022975: fstp dword ptr [esp+18h]
0x10022979: leave
0x1002297a: ret
0x10024880: fild dword ptr [esp+4]
0x10024884: sub esp,8
0x10024887: fmul qword ptr ds:[100FFB90h]
0x1002488d: fimul dword ptr [esp+10h]
0x10024891: fadd qword ptr ds:[100FE7C8h]
0x10024897: fstp qword ptr [esp]
0x1002489a: call 1002297Ch
0x1002489f: fld qword ptr ds:[100FFB58h]
0x100248a5: add esp,8
0x100248a8: fcomp
0x100248aa: fnstsw ax
0x100248ac: test ah,1
0x100248af: jne short 100248C5h
0x100248b1: fld qword ptr ds:[100FFB50h]
0x100248b7: fcomp
0x100248b9: fnstsw ax
0x100248bb: test ah,41h
0x100248be: jp short 100248C5h
0x100248c0: jmp near ptr 100228D0h
0x100248c5: fstp st(0)
0x100248c7: xor eax,eax
0x100248c9: ret
```

`sub_10024880` is called by sub_10024e30. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to j_floor (1 callees).

12.553 sub_100248d0 at 0x100248d0 (69 instructions)

```
0x100228d0: cmp dword ptr ds:[10145460h],0
0x100228d7: je short 10022906h
0x100228d9: push ebp
0x100228da: mov ebp,esp
0x100228dc: sub esp,8
0x100228df: and esp,0FFFFFFF8h
0x100228e2: fstp qword ptr [esp]
0x100228e5: cvttsd2si eax,qword ptr [esp]
0x100228ea: leave
0x100228eb: ret
0x10022906: push ebp
0x10022907: mov ebp,esp
0x10022909: sub esp,20h
0x1002290c: and esp,0FFFFFFF0h
0x1002290f: fld st(0)
0x10022911: fst dword ptr [esp+18h]
0x10022915: fistp qword ptr [esp+10h]
0x10022919: fild qword ptr [esp+10h]
0x1002291d: mov edx,[esp+18h]
0x10022921: mov eax,[esp+10h]
0x10022925: test eax,eax
0x10022927: je short 10022965h
0x10022929: fsubp
0x1002292b: test edx,edx
0x1002292d: jns short 1002294Dh
0x1002292f: fstp dword ptr [esp]
0x10022932: mov ecx,[esp]
0x10022935: xor ecx,80000000h
0x1002293b: add ecx,7FFFFFFFh
0x10022941: adc eax,0
0x10022944: mov edx,[esp+14h]
0x10022948: adc edx,0
0x1002294b: jmp short 10022979h
0x1002294d: fstp dword ptr [esp]
0x10022950: mov ecx,[esp]
0x10022953: add ecx,7FFFFFFFh
0x10022959: sbb eax,0
0x1002295c: mov edx,[esp+14h]
0x10022960: sbb edx,0
0x10022963: jmp short 10022979h
0x10022965: mov edx,[esp+14h]
0x10022969: test edx,7FFFFFFFh
0x1002296f: jne short 10022929h
0x10022971: fstp dword ptr [esp+18h]
0x10022975: fstp dword ptr [esp+18h]
0x10022979: leave
0x1002297a: ret
0x100248d0: fild dword ptr [esp+4]
0x100248d4: sub esp,8
0x100248d7: fdivr qword ptr ds:[100FFB98h]
0x100248dd: fidiv dword ptr [esp+10h]
0x100248e1: fadd qword ptr ds:[100FE7C8h]
0x100248e7: fstp qword ptr [esp]
0x100248ea: call 1002297Ch
0x100248ef: fld qword ptr ds:[100FFB58h]
0x100248f5: add esp,8
0x100248f8: fcomp
0x100248fa: fnstsw ax
0x100248fc: test ah,1
0x100248ff: jne short 10024915h
0x10024901: fld qword ptr ds:[100FFB50h]
0x10024907: fcomp
0x10024909: fnstsw ax
0x1002490b: test ah,41h
0x1002490e: jp short 10024915h
0x10024910: jmp near ptr 100228D0h
```

```
0x10024915: fstp st(0)
0x10024917: xor eax,eax
0x10024919: ret
```

`sub_100248d0` is called by sub_10024e30, sub_10031a20. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to j_floor (1 callees).

12.554 sub_10024920 at 0x10024920 (40 instructions)

```
0x10024920: mov eax,[esp+4]
0x10024924: push ebx
0x10024925: push esi
0x10024926: mov esi,[esp+10h]
0x1002492a: mov ecx,esi
0x1002492c: mov edx,eax
0x1002492e: and eax,0FFFFh
0x10024933: mov ebx,eax
0x10024935: sar edx,10h
0x10024938: sar ecx,10h
0x1002493b: and edx,0FFFFh
0x10024941: and esi,0FFFFh
0x10024947: and ecx,0FFFFh
0x1002494d: push edi
0x1002494e: mov edi,esi
0x10024950: imul esi,edx
0x10024953: mov eax,ecx
0x10024955: imul edi,ebx
0x10024958: imul ecx,edx
0x1002495b: imul eax,ebx
0x1002495e: add eax,esi
0x10024960: mov esi,edi
0x10024962: sar esi,10h
0x10024965: and esi,0FFFFh
0x1002496b: add eax,esi
0x1002496d: mov edx,eax
0x1002496f: sar edx,10h
0x10024972: and edx,0FFFFh
0x10024978: add ecx,edx
0x1002497a: mov edx,[esp+18h]
0x1002497e: mov [edx],ecx
0x10024980: mov edx,[esp+1Ch]
0x10024984: movzx ecx,di
0x10024987: pop edi
0x10024988: shl eax,10h
0x1002498b: or eax,ecx
0x1002498d: pop esi
0x1002498e: mov [edx],eax
0x10024990: pop ebx
0x10024991: ret
```

`sub_10024920` is called by sub_100250a0. It calls nothing else.

12.555 sub_100249a0 at 0x100249a0 (30 instructions)

```
0x100249a0: push ecx
0x100249a1: mov eax,[esp+8]
0x100249a5: lea ecx,[eax-1]
0x100249a8: cmp ecx,0FDh
0x100249ae: ja short 10024A0Fh
0x100249b0: test eax,eax
0x100249b2: mov [esp+8],eax
0x100249b6: fild dword ptr [esp+8]
0x100249ba: jge short 100249C2h
0x100249bc: fadd qword ptr ds:[100FEFC8h]
0x100249c2: fdiv qword ptr ds:[100FE918h]
```

```
0x100249c8: fild dword ptr [esp+0Ch]
0x100249cc: fmul qword ptr ds:[100FFB90h]
0x100249d2: call 100229A0h
0x100249d7: fmul qword ptr ds:[100FE918h]
0x100249dd: sub esp,8
0x100249e0: fadd qword ptr ds:[100FE7C8h]
0x100249e6: fstp qword ptr [esp]
0x100249e9: call 1002297Ch
0x100249ee: fnstcw [esp+10h]
0x100249f2: add esp,8
0x100249f5: movzx eax,word ptr [esp+8]
0x100249fa: or eax,0C00h
0x100249ff: mov [esp],eax
0x10024a02: fldcw [esp]
0x10024a05: fistp dword ptr [esp]
0x10024a08: mov al,[esp]
0x10024a0b: fldcw [esp+8]
0x10024a0f: pop ecx
0x10024a10: ret
```

`sub_100249a0` is called by sub_10024aa0, sub_10024de0, sub_10031a20. It delegates to j_floor, j__CIpow (2 callees).

12.556 sub_10024a20 at 0x10024a20 (30 instructions)

```
0x10024a20: push ecx
0x10024a21: mov eax,[esp+8]
0x10024a25: lea ecx,[eax-1]
0x10024a28: cmp ecx,0FFFDh
0x10024a2e: ja short 10024A90h
0x10024a30: test eax,eax
0x10024a32: mov [esp+8],eax
0x10024a36: fild dword ptr [esp+8]
0x10024a3a: jge short 10024A42h
0x10024a3c: fadd qword ptr ds:[100FEFC8h]
0x10024a42: fdiv qword ptr ds:[100FFBA0h]
0x10024a48: fild dword ptr [esp+0Ch]
0x10024a4c: fmul qword ptr ds:[100FFB90h]
0x10024a52: call 100229A0h
0x10024a57: fmul qword ptr ds:[100FFBA0h]
0x10024a5d: sub esp,8
0x10024a60: fadd qword ptr ds:[100FE7C8h]
0x10024a66: fstp qword ptr [esp]
0x10024a69: call 1002297Ch
0x10024a6e: fnstcw [esp+10h]
0x10024a72: add esp,8
0x10024a75: movzx eax,word ptr [esp+8]
0x10024a7a: or eax,0C00h
0x10024a7f: mov [esp],eax
0x10024a82: fldcw [esp]
0x10024a85: fistp dword ptr [esp]
0x10024a88: mov ax,[esp]
0x10024a8c: fldcw [esp+8]
0x10024a90: pop ecx
0x10024a91: ret
```

`sub_10024a20` is called by sub_10024aa0, sub_10024c90. It delegates to j_floor, j__CIpow (2 callees).

12.557 sub_10024aa0 at 0x10024aa0 (18 instructions)

```
0x10024aa0: mov eax,[esp+4]
0x10024aa4: cmp byte ptr [eax+127h],8
0x10024aab: jne short 10024AC4h
0x10024aad: mov ecx,[esp+0Ch]
0x10024ab1: mov edx,[esp+8]
0x10024ab5: push ecx
```

```
0x10024ab6: push edx
0x10024ab7: call 100249A0h
0x10024abc: movzx ax,al
0x10024ac0: add esp,8
0x10024ac3: ret
0x10024ac4: mov eax,[esp+0Ch]
0x10024ac8: mov ecx,[esp+8]
0x10024acc: push eax
0x10024acd: push ecx
0x10024ace: call 10024A20h
0x10024ad3: add esp,8
0x10024ad6: ret
```

`sub_10024aa0` is called by sub_10031a20. It delegates to sub_100249a0, sub_10024a20 (2 callees).

12.558 sub_10024ae0 at 0x10024ae0 (7 instructions)

```
0x10024ae0: mov eax,[esp+4]
0x10024ae4: add eax,0FFFE8CE8h
0x10024ae9: mov ecx,2710h
0x10024aee: cmp ecx,eax
0x10024af0: sbb eax,eax
0x10024af2: neg eax
0x10024af4: ret
```

`sub_10024ae0` is called by sub_10024b00, sub_10024de0, sub_10031780, sub_10031a20. It calls nothing else.

12.559 sub_10024b00 at 0x10024b00 (119 instructions)

```
0x10024b00: push ebp
0x10024b01: mov ebp,esp
0x10024b03: and esp,0FFFFFFC0h
0x10024b06: sub esp,34h
0x10024b09: mov edx,[ebp+10h]
0x10024b0c: push ebx
0x10024b0d: push esi
0x10024b0e: mov ebx,8
0x10024b13: sub ebx,edx
0x10024b15: mov ecx,ebx
0x10024b17: mov esi,1
0x10024b1c: shl esi,cl
0x10024b1e: mov ecx,10h
0x10024b23: sub ecx,edx
0x10024b25: mov eax,1
0x10024b2a: shl eax,cl
0x10024b2c: mov ecx,0Fh
0x10024b31: sub ecx,edx
0x10024b33: push edi
0x10024b34: sub eax,1
0x10024b37: mov [esp+14h],eax
0x10024b3b: mov eax,1
0x10024b40: shl eax,cl
0x10024b42: mov ecx,[ebp+8]
0x10024b45: mov [esp+2Ch],esi
0x10024b49: mov [esp+28h],eax
0x10024b4d: lea eax,[esi*4]
0x10024b54: push eax
0x10024b55: push ecx
0x10024b56: call 1002CDD0h
0x10024b5b: mov edx,[ebp+0Ch]
0x10024b5e: add esp,8
0x10024b61: xor edi,edi
0x10024b63: test esi,esi
0x10024b65: mov [esp+1Ch],eax
0x10024b69: mov [edx],eax
```

```
0x10024b6b: jbe near ptr 10024C86h
0x10024b71: mov eax,[ebp+14h]
0x10024b74: push eax
0x10024b75: call 10024AE0h
0x10024b7a: add esp,4
0x10024b7d: mov [esp+20h],eax
0x10024b81: mov ecx,[ebp+8]
0x10024b84: push 200h
0x10024b89: push ecx
0x10024b8a: call 1002CD20h
0x10024b8f: mov edx,eax
0x10024b91: mov eax,[esp+24h]
0x10024b95: xor esi,esi
0x10024b97: add esp,8
0x10024b9a: cmp [esp+20h],esi
0x10024b9e: mov [esp+18h],edx
0x10024ba2: mov [eax+edi*4],edx
0x10024ba5: je near ptr 10024C48h
0x10024bab: mov ecx,[esp+14h]
0x10024baf: fild dword ptr [esp+14h]
0x10024bb3: test ecx,ecx
0x10024bb5: jge short 10024BBDh
0x10024bb7: fadd qword ptr ds:[100FEFC8h]
0x10024bbd: fstp qword ptr [esp+30h]
0x10024bc1: fild dword ptr [ebp+14h]
0x10024bc4: fmul qword ptr ds:[100FFB90h]
0x10024bca: fstp qword ptr [esp+38h]
0x10024bce: mov edx,esi
0x10024bd0: mov ecx,ebx
0x10024bd2: shl edx,cl
0x10024bd4: add edx,edi
0x10024bd6: test edx,edx
0x10024bd8: mov [esp+24h],edx
0x10024bdc: fild dword ptr [esp+24h]
0x10024be0: jge short 10024BE8h
0x10024be2: fadd qword ptr ds:[100FEFC8h]
0x10024be8: fdiv qword ptr [esp+30h]
0x10024bec: fld qword ptr [esp+38h]
0x10024bf0: call 100229A0h
0x10024bf5: fmul qword ptr ds:[100FFBA0h]
0x10024bfb: sub esp,8
0x10024bfe: fadd qword ptr ds:[100FE7C8h]
0x10024c04: fstp qword ptr [esp]
0x10024c07: call 1002297Ch
0x10024c0c: mov ecx,[esp+20h]
0x10024c10: fnstcw [esp+1Ah]
0x10024c14: movzx eax,word ptr [esp+1Ah]
0x10024c19: or eax,0C00h
0x10024c1e: mov [esp+2Ch],eax
0x10024c22: add esi,1
0x10024c25: add esp,8
0x10024c28: cmp esi,100h
0x10024c2e: fldcw [esp+24h]
0x10024c32: fistp dword ptr [esp+24h]
0x10024c36: mov ax,[esp+24h]
0x10024c3b: mov [ecx+esi*2-2],ax
0x10024c40: fldcw [esp+12h]
0x10024c44: jb short 10024BCEh
0x10024c46: jmp short 10024C79h
0x10024c48: mov eax,esi
0x10024c4a: mov ecx,ebx
0x10024c4c: shl eax,cl
0x10024c4e: add eax,edi
0x10024c50: cmp dword ptr [ebp+10h],0
0x10024c54: je short 10024C6Ah
0x10024c56: imul eax,0FFFFh
0x10024c5c: add eax,[esp+28h]
```

```
0x10024c60: xor edx,edx
0x10024c62: div dword ptr [esp+14h]
0x10024c66: mov edx,[esp+18h]
0x10024c6a: mov [edx+esi*2],ax
0x10024c6e: add esi,1
0x10024c71: cmp esi,100h
0x10024c77: jb short 10024C48h
0x10024c79: add edi,1
0x10024c7c: cmp edi,[esp+2Ch]
0x10024c80: jb near ptr 10024B81h
0x10024c86: pop edi
0x10024c87: pop esi
0x10024c88: pop ebx
0x10024c89: mov esp,ebp
0x10024c8b: pop ebp
0x10024c8c: ret
```

`sub_10024b00` is called by sub_10024e30. It delegates to j_floor, j__CIpow, sub_10024ae0 (5 callees).

12.560 sub_10024c90 at 0x10024c90 (105 instructions)

```
0x10024c90: sub esp,10h
0x10024c93: push ebx
0x10024c94: push ebp
0x10024c95: push esi
0x10024c96: push edi
0x10024c97: mov edi,eax
0x10024c99: mov eax,[esp+28h]
0x10024c9d: mov ecx,8
0x10024ca2: sub ecx,eax
0x10024ca4: mov [esp+10h],ecx
0x10024ca8: mov ebp,1
0x10024cad: shl ebp,cl
0x10024caf: mov ecx,10h
0x10024cb4: sub ecx,eax
0x10024cb6: mov eax,1
0x10024cbb: shl eax,cl
0x10024cbd: mov [esp+1Ch],ebp
0x10024cc1: sub eax,1
0x10024cc4: mov [esp+14h],eax
0x10024cc8: lea eax,[ebp*4]
0x10024ccf: push eax
0x10024cd0: push edi
0x10024cd1: call 1002CDD0h
0x10024cd6: mov ecx,[esp+2Ch]
0x10024cda: add esp,8
0x10024cdd: mov ebx,eax
0x10024cdf: xor esi,esi
0x10024ce1: test ebp,ebp
0x10024ce3: mov [ecx],ebx
0x10024ce5: jbe short 10024D08h
0x10024ce7: jmp short 10024CF0h
0x10024cf0: push 200h
0x10024cf5: push edi
0x10024cf6: call 1002CD20h
0x10024cfb: mov [ebx+esi*4],eax
0x10024cfe: add esi,1
0x10024d01: add esp,8
0x10024d04: cmp esi,ebp
0x10024d06: jb short 10024CF0h
0x10024d08: xor esi,esi
0x10024d0a: mov [esp+24h],esi
0x10024d0e: mov edi,edi
0x10024d10: mov edx,[esp+24h]
0x10024d14: mov eax,[esp+2Ch]
0x10024d18: imul edx,101h
```

```
0x10024d1e: movzx edi,dx
0x10024d21: movzx ecx,di
0x10024d24: push eax
0x10024d25: add ecx,80h
0x10024d2b: push ecx
0x10024d2c: call 10024A20h
0x10024d31: movzx edx,ax
0x10024d34: imul edx,[esp+1Ch]
0x10024d39: add edx,8000h
0x10024d3f: mov eax,80008001h
0x10024d44: mul edx
0x10024d46: shr edx,0Fh
0x10024d49: add edx,1
0x10024d4c: add esp,8
0x10024d4f: cmp esi,edx
0x10024d51: jae short 10024D84h
0x10024d53: mov ecx,[esp+28h]
0x10024d57: mov eax,0FFh
0x10024d5c: shr eax,cl
0x10024d5e: mov [esp+18h],eax
0x10024d62: jmp short 10024D68h
0x10024d64: mov eax,[esp+18h]
0x10024d68: mov ecx,[esp+10h]
0x10024d6c: mov ebp,esi
0x10024d6e: and eax,esi
0x10024d70: mov eax,[ebx+eax*4]
0x10024d73: shr ebp,cl
0x10024d75: add esi,1
0x10024d78: cmp esi,edx
0x10024d7a: mov [eax+ebp*2],di
0x10024d7e: jb short 10024D64h
0x10024d80: mov ebp,[esp+1Ch]
0x10024d84: mov eax,[esp+24h]
0x10024d88: add eax,1
0x10024d8b: cmp eax,0FFh
0x10024d90: mov [esp+24h],eax
0x10024d94: jb near ptr 10024D10h
0x10024d9a: shl ebp,8
0x10024d9d: cmp esi,ebp
0x10024d9f: jae short 10024DCCh
0x10024da1: mov ecx,[esp+28h]
0x10024da5: mov eax,0FFh
0x10024daa: sar eax,cl
0x10024dac: lea esp,[esp]
0x10024db0: mov ecx,[esp+10h]
0x10024db4: mov edi,esi
0x10024db6: shr edi,cl
0x10024db8: mov edx,eax
0x10024dba: and edx,esi
0x10024dbc: mov ecx,[ebx+edx*4]
0x10024dbf: add esi,1
0x10024dc2: cmp esi,ebp
0x10024dc4: mov word ptr [ecx+edi*2],0FFFFh
0x10024dca: jb short 10024DB0h
0x10024dcc: pop edi
0x10024dcd: pop esi
0x10024dce: pop ebp
0x10024dcf: pop ebx
0x10024dd0: add esp,10h
0x10024dd3: ret
```

`sub_10024c90` is called by sub_10024e30. It accesses memory-mapped peripheral registers at 0x80008001. It delegates to sub_10024a20, png_malloc, png_calloc (3 callees).

12.561 sub_10024de0 at 0x10024de0 (32 instructions)

```
0x10024de0: push edi
0x10024de1: push 100h
0x10024de6: push eax
0x10024de7: call 1002CD20h
0x10024dec: mov ecx,[esp+10h]
0x10024df0: mov edi,eax
0x10024df2: push ebx
0x10024df3: mov [ecx],edi
0x10024df5: call 10024AE0h
0x10024dfa: add esp,0Ch
0x10024dfd: test eax,eax
0x10024dff: je short 10024E1Fh
0x10024e01: push esi
0x10024e02: xor esi,esi
0x10024e04: push ebx
0x10024e05: push esi
0x10024e06: call 100249A0h
0x10024e0b: mov [esi+edi],al
0x10024e0e: add esi,1
0x10024e11: add esp,8
0x10024e14: cmp esi,100h
0x10024e1a: jb short 10024E04h
0x10024e1c: pop esi
0x10024e1d: pop edi
0x10024e1e: ret
0x10024e1f: xor eax,eax
0x10024e21: mov [eax+edi],al
0x10024e24: add eax,1
0x10024e27: cmp eax,100h
0x10024e2c: jb short 10024E21h
0x10024e2e: pop edi
0x10024e2f: ret
```

`sub_10024de0` is called by sub_10024e30. It delegates to sub_100249a0, sub_10024ae0, png_malloc (3 callees).

12.562 sub_10024e30 at 0x10024e30 (149 instructions)

```
0x10024e30: cmp dword ptr [esp+8],8
0x10024e35: push esi
0x10024e36: mov esi,[esp+8]
0x10024e3a: jg near ptr 10024ECEh
0x10024e40: mov eax,[esi+160h]
0x10024e46: test eax,eax
0x10024e48: jle short 10024E5Ch
0x10024e4a: push eax
0x10024e4b: mov eax,[esi+15Ch]
0x10024e51: push eax
0x10024e52: call 100248D0h
0x10024e57: add esp,8
0x10024e5a: jmp short 10024E61h
0x10024e5c: mov eax,186A0h
0x10024e61: push ebx
0x10024e62: lea ecx,[esi+164h]
0x10024e68: mov ebx,eax
0x10024e6a: push ecx
0x10024e6b: mov eax,esi
0x10024e6d: call 10024DE0h
0x10024e72: add esp,4
0x10024e75: test dword ptr [esi+74h],600080h
0x10024e7c: je short 10024ECBh
0x10024e7e: mov edx,[esi+15Ch]
0x10024e84: push edx
0x10024e85: call 10024830h
0x10024e8a: lea ecx,[esi+16Ch]
0x10024e90: mov ebx,eax
```

```
0x10024e92: push ecx
0x10024e93: mov eax,esi
0x10024e95: call 10024DE0h
0x10024e9a: mov eax,[esi+160h]
0x10024ea0: add esp,8
0x10024ea3: test eax,eax
0x10024ea5: jle short 10024EB2h
0x10024ea7: push eax
0x10024ea8: call 10024830h
0x10024ead: add esp,4
0x10024eb0: jmp short 10024EB8h
0x10024eb2: mov eax,[esi+15Ch]
0x10024eb8: lea edx,[esi+168h]
0x10024ebe: mov ebx,eax
0x10024ec0: push edx
0x10024ec1: mov eax,esi
0x10024ec3: call 10024DE0h
0x10024ec8: add esp,4
0x10024ecb: pop ebx
0x10024ecc: pop esi
0x10024ecd: ret
0x10024ece: test byte ptr [esi+126h],2
0x10024ed5: je short 10024EF7h
0x10024ed7: mov cl,[esi+17Ch]
0x10024edd: mov al,[esi+17Dh]
0x10024ee3: cmp al,cl
0x10024ee5: jbe short 10024EE9h
0x10024ee7: mov cl,al
0x10024ee9: mov al,[esi+17Eh]
0x10024eef: cmp al,cl
0x10024ef1: jbe short 10024EFDh
0x10024ef3: mov cl,al
0x10024ef5: jmp short 10024EFDh
0x10024ef7: mov cl,[esi+17Fh]
0x10024efd: mov al,cl
0x10024eff: sub al,1
0x10024f01: cmp al,0Eh
0x10024f03: ja short 10024F0Bh
0x10024f05: mov al,10h
0x10024f07: sub al,cl
0x10024f09: jmp short 10024F0Dh
0x10024f0b: xor al,al
0x10024f0d: mov ecx,[esi+74h]
0x10024f10: test ecx,400h
0x10024f16: je short 10024F20h
0x10024f18: cmp al,5
0x10024f1a: jae short 10024F20h
0x10024f1c: mov al,5
0x10024f1e: jmp short 10024F26h
0x10024f20: cmp al,8
0x10024f22: jbe short 10024F26h
0x10024f24: mov al,8
0x10024f26: test ecx,480h
0x10024f2c: push edi
0x10024f2d: movzx edi,al
0x10024f30: mov eax,[esi+160h]
0x10024f36: mov [esi+158h],edi
0x10024f3c: je short 10024F6Eh
0x10024f3e: test eax,eax
0x10024f40: jle short 10024F54h
0x10024f42: mov ecx,[esi+15Ch]
0x10024f48: push eax
0x10024f49: push ecx
0x10024f4a: call 10024880h
0x10024f4f: add esp,8
0x10024f52: jmp short 10024F59h
0x10024f54: mov eax,186A0h
```

```
0x10024f59: push eax
0x10024f5a: lea edx,[esi+170h]
0x10024f60: push edi
0x10024f61: push edx
0x10024f62: mov eax,esi
0x10024f64: call 10024C90h
0x10024f69: add esp,0Ch
0x10024f6c: jmp short 10024F9Bh
0x10024f6e: test eax,eax
0x10024f70: jle short 10024F84h
0x10024f72: push eax
0x10024f73: mov eax,[esi+15Ch]
0x10024f79: push eax
0x10024f7a: call 100248D0h
0x10024f7f: add esp,8
0x10024f82: jmp short 10024F89h
0x10024f84: mov eax,186A0h
0x10024f89: push eax
0x10024f8a: push edi
0x10024f8b: lea ecx,[esi+170h]
0x10024f91: push ecx
0x10024f92: push esi
0x10024f93: call 10024B00h
0x10024f98: add esp,10h
0x10024f9b: test dword ptr [esi+74h],600080h
0x10024fa2: je short 10024FEFh
0x10024fa4: mov edx,[esi+15Ch]
0x10024faa: push edx
0x10024fab: call 10024830h
0x10024fb0: push eax
0x10024fb1: push edi
0x10024fb2: lea eax,[esi+178h]
0x10024fb8: push eax
0x10024fb9: push esi
0x10024fba: call 10024B00h
0x10024fbf: mov eax,[esi+160h]
0x10024fc5: add esp,14h
0x10024fc8: test eax,eax
0x10024fca: jle short 10024FD7h
0x10024fcc: push eax
0x10024fcd: call 10024830h
0x10024fd2: add esp,4
0x10024fd5: jmp short 10024FDDh
0x10024fd7: mov eax,[esi+15Ch]
0x10024fdd: push eax
0x10024fde: push edi
0x10024fdf: lea ecx,[esi+174h]
0x10024fe5: push ecx
0x10024fe6: push esi
0x10024fe7: call 10024B00h
0x10024fec: add esp,10h
0x10024fef: pop edi
0x10024ff0: pop esi
0x10024ff1: ret
```

`sub_10024e30` is called by sub_10031a20. It delegates to sub_10024830, sub_10024880, sub_100248d0 (6 callees).

12.563 png_create_info_struct at 0x10025000 (21 instructions)

```
0x10025000: mov eax,[esp+4]
0x10025004: test eax,eax
0x10025006: jne short 10025009h
0x10025008: ret
0x10025009: mov ecx,[eax+238h]
0x1002500f: mov edx,[eax+23Ch]
0x10025015: push ecx
```

```
0x10025016: push edx
0x10025017: push 2
0x10025019: call 1002CB30h
0x1002501e: add esp,0Ch
0x10025021: test eax,eax
0x10025023: mov [esp+4],eax
0x10025027: je short 1002503Fh
0x10025029: lea eax,[esp+4]
0x1002502d: push 0F0h
0x10025032: push eax
0x10025033: call 10023550h
0x10025038: mov eax,[esp+0Ch]
0x1002503c: add esp,8
0x1002503f: ret
```

`png_create_info_struct` is called by sub_10018d20. It delegates to png_info_init_3, sub_1002cb30 (2 callees).

12.564 png_destroy_info_struct at 0x10025040 (27 instructions)

```
0x10025040: push edi
0x10025041: mov edi,[esp+8]
0x10025045: test edi,edi
0x10025047: je short 1002507Fh
0x10025049: push ebx
0x1002504a: mov ebx,[esp+10h]
0x1002504e: test ebx,ebx
0x10025050: je short 1002507Eh
0x10025052: push esi
0x10025053: mov esi,[ebx]
0x10025055: test esi,esi
0x10025057: je short 1002507Dh
0x10025059: push esi
0x1002505a: push edi
0x1002505b: call 100239C0h
0x10025060: mov eax,[edi+238h]
0x10025066: mov ecx,[edi+240h]
0x1002506c: push eax
0x1002506d: push ecx
0x1002506e: push esi
0x1002506f: call 1002CBE0h
0x10025074: add esp,14h
0x10025077: mov dword ptr [ebx],0
0x1002507d: pop esi
0x1002507e: pop ebx
0x1002507f: pop edi
0x10025080: ret
```

`png_destroy_info_struct` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100239c0, sub_1002cbe0 (2 callees).

12.565 png_get_libpng_ver at 0x10025090 (3 instructions)

```
0x10023af0: mov eax,100FF018h
0x10023af5: ret
0x10025090: jmp near ptr 10023AF0h
```

`png_get_libpng_ver` has no recorded callers; it is a leaf stub, likely reached through the vector table. It works with the text "1.5.1". It calls nothing else.

12.566 sub_100250a0 at 0x100250a0 (137 instructions)

```
0x100250a0: sub esp,14h
0x100250a3: mov eax,[esp+18h]
0x100250a7: test eax,eax
```

```
0x100250a9: mov dword ptr [esp],1
0x100250b0: je near ptr 1002524Eh
0x100250b6: cmp dword ptr [esp+1Ch],0
0x100250bb: push ebx
0x100250bc: mov ebx,[esp+34h]
0x100250c0: push ebp
0x100250c1: mov ebp,[esp+34h]
0x100250c5: push esi
0x100250c6: mov esi,[esp+44h]
0x100250ca: push edi
0x100250cb: mov edi,[esp+44h]
0x100250cf: jl short 100250F6h
0x100250d1: cmp dword ptr [esp+30h],0
0x100250d6: jle short 100250F6h
0x100250d8: cmp dword ptr [esp+34h],0
0x100250dd: jl short 100250F6h
0x100250df: cmp dword ptr [esp+38h],0
0x100250e4: jl short 100250F6h
0x100250e6: test ebp,ebp
0x100250e8: jl short 100250F6h
0x100250ea: test ebx,ebx
0x100250ec: jl short 100250F6h
0x100250ee: test edi,edi
0x100250f0: jl short 100250F6h
0x100250f2: test esi,esi
0x100250f4: jge short 10025110h
0x100250f6: push 100FFC8Ch
0x100250fb: push eax
0x100250fc: call 100230F0h
0x10025101: mov eax,[esp+30h]
0x10025105: add esp,8
0x10025108: mov dword ptr [esp+10h],0
0x10025110: mov ecx,7FFFFFFFh
0x10025115: cmp [esp+2Ch],ecx
0x10025119: jg short 1002513Dh
0x1002511b: cmp [esp+30h],ecx
0x1002511f: jg short 1002513Dh
0x10025121: cmp [esp+34h],ecx
0x10025125: jg short 1002513Dh
0x10025127: cmp [esp+38h],ecx
0x1002512b: jg short 1002513Dh
0x1002512d: cmp ebp,ecx
0x1002512f: jg short 1002513Dh
0x10025131: cmp ebx,ecx
0x10025133: jg short 1002513Dh
0x10025135: cmp edi,ecx
0x10025137: jg short 1002513Dh
0x10025139: cmp esi,ecx
0x1002513b: jle short 10025157h
0x1002513d: push 100FFC4Ch
0x10025142: push eax
0x10025143: call 100230F0h
0x10025148: mov eax,[esp+30h]
0x1002514c: add esp,8
0x1002514f: mov dword ptr [esp+10h],0
0x10025157: mov ecx,186A0h
0x1002515c: sub ecx,[esp+30h]
0x10025160: cmp [esp+2Ch],ecx
0x10025164: jle short 10025180h
0x10025166: push 100FFC30h
0x1002516b: push eax
0x1002516c: call 100230F0h
0x10025171: mov eax,[esp+30h]
0x10025175: add esp,8
0x10025178: mov dword ptr [esp+10h],0
0x10025180: mov edx,186A0h
0x10025185: sub edx,[esp+38h]
```

```
0x10025189: cmp [esp+34h],edx
0x1002518d: jle short 100251A9h
0x1002518f: push 100FFC18h
0x10025194: push eax
0x10025195: call 100230F0h
0x1002519a: mov eax,[esp+30h]
0x1002519e: add esp,8
0x100251a1: mov dword ptr [esp+10h],0
0x100251a9: mov ecx,186A0h
0x100251ae: sub ecx,ebx
0x100251b0: cmp ebp,ecx
0x100251b2: jle short 100251CEh
0x100251b4: push 100FFBFCh
0x100251b9: push eax
0x100251ba: call 100230F0h
0x100251bf: mov eax,[esp+30h]
0x100251c3: add esp,8
0x100251c6: mov dword ptr [esp+10h],0
0x100251ce: mov edx,186A0h
0x100251d3: sub edx,esi
0x100251d5: cmp edi,edx
0x100251d7: jle short 100251EFh
0x100251d9: push 100FFBE4h
0x100251de: push eax
0x100251df: call 100230F0h
0x100251e4: add esp,8
0x100251e7: mov dword ptr [esp+10h],0
0x100251ef: sub esi,[esp+38h]
0x100251f3: sub ebp,[esp+34h]
0x100251f7: lea eax,[esp+1Ch]
0x100251fb: push eax
0x100251fc: lea ecx,[esp+18h]
0x10025200: push ecx
0x10025201: push esi
0x10025202: push ebp
0x10025203: call 10024920h
0x10025208: sub edi,[esp+44h]
0x1002520c: sub ebx,[esp+48h]
0x10025210: lea edx,[esp+30h]
0x10025214: push edx
0x10025215: lea eax,[esp+2Ch]
0x10025219: push eax
0x1002521a: push edi
0x1002521b: push ebx
0x1002521c: call 10024920h
0x10025221: mov ecx,[esp+34h]
0x10025225: add esp,20h
0x10025228: cmp ecx,[esp+18h]
0x1002522c: pop edi
0x1002522d: pop esi
0x1002522e: pop ebp
0x1002522f: pop ebx
0x10025230: jne short 10025254h
0x10025232: mov edx,[esp+0Ch]
0x10025236: cmp edx,[esp+10h]
0x1002523a: jne short 10025254h
0x1002523c: mov eax,[esp+18h]
0x10025240: push 100FFBA8h
0x10025245: push eax
0x10025246: call 100230F0h
0x1002524b: add esp,8
0x1002524e: xor eax,eax
0x10025250: add esp,14h
0x10025253: ret
0x10025254: mov eax,[esp]
0x10025257: add esp,14h
0x1002525a: ret
```

`sub_100250a0` is called by png_set_cHRM_fixed, sub_100376c0. It works with the text "Ignoring attempt to set chromaticity value exceeding 21474.83", "Ignoring attempt to set negative chromaticity value". It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to png_warning, sub_10024920 (2 callees).

12.567 png_create_read_struct_2 at 0x10025260 (233 instructions)

```
0x10025260: push ebp
0x10025261: mov ebp,esp
0x10025263: sub esp,78h
0x10025266: mov eax,ds:[10136220h]
0x1002526b: xor eax,ebp
0x1002526d: mov [ebp-4],eax
0x10025270: mov eax,[ebp+8]
0x10025273: mov ecx,[ebp+0Ch]
0x10025276: mov edx,[ebp+10h]
0x10025279: mov [ebp-60h],eax
0x1002527c: mov eax,[ebp+14h]
0x1002527f: mov [ebp-78h],eax
0x10025282: mov eax,[ebp+18h]
0x10025285: push esi
0x10025286: mov [ebp-70h],ecx
0x10025289: mov ecx,[ebp+1Ch]
0x1002528c: push eax
0x1002528d: mov [ebp-6Ch],edx
0x10025290: mov edx,[ebp+20h]
0x10025293: push ecx
0x10025294: mov [ebp-68h],eax
0x10025297: mov [ebp-74h],ecx
0x1002529a: mov [ebp-64h],edx
0x1002529d: mov dword ptr [ebp-5Ch],0
0x100252a4: push 1
0x100252a6: call 1002CB30h
0x100252ab: mov [ebp-58h],eax
0x100252ae: mov eax,[ebp-58h]
0x100252b1: add esp,0Ch
0x100252b4: test eax,eax
0x100252b6: je near ptr 10025501h
0x100252bc: mov ecx,[ebp-58h]
0x100252bf: mov edx,[ebp-58h]
0x100252c2: mov eax,0F4240h
0x100252c7: mov [ecx+258h],eax
0x100252cd: mov ecx,ds:[100FE4E0h]
0x100252d3: mov [edx+25Ch],eax
0x100252d9: mov eax,[ebp-58h]
0x100252dc: mov edx,[ebp-58h]
0x100252df: push 40h
0x100252e1: push ecx
0x100252e2: push edx
0x100252e3: mov dword ptr [eax+260h],0
0x100252ed: call 10023030h
0x100252f2: push 0
0x100252f4: push eax
0x100252f5: call 100229DCh
0x100252fa: add esp,14h
0x100252fd: test eax,eax
0x100252ff: je short 10025309h
0x10025301: push 0
0x10025303: call dword ptr ds:[100FE0ACh]
0x10025309: mov eax,[ebp-64h]
0x1002530c: mov ecx,[ebp-74h]
0x1002530f: mov edx,[ebp-68h]
0x10025312: push eax
0x10025313: mov eax,[ebp-58h]
0x10025316: push ecx
0x10025317: push edx
0x10025318: push eax
0x10025319: call 1002CCA0h
```

```
0x1002531e: mov ecx,[ebp-78h]
0x10025321: mov edx,[ebp-6Ch]
0x10025324: mov eax,[ebp-70h]
0x10025327: push ecx
0x10025328: mov ecx,[ebp-58h]
0x1002532b: push edx
0x1002532c: push eax
0x1002532d: push ecx
0x1002532e: call 100230C0h
0x10025333: add esp,20h
0x10025336: cmp dword ptr [ebp-60h],0
0x1002533a: je short 10025375h
0x1002533c: xor esi,esi
0x1002533e: mov edi,edi
0x10025340: push 0
0x10025342: call 10023AF0h
0x10025347: mov edx,[ebp-60h]
0x1002534a: mov cl,[edx+esi]
0x1002534d: add esp,4
0x10025350: cmp cl,[eax+esi]
0x10025353: je short 1002535Fh
0x10025355: mov eax,[ebp-58h]
0x10025358: or dword ptr [eax+70h],20000h
0x1002535f: push 0
0x10025361: call 10023AF0h
0x10025366: mov dl,[eax+esi]
0x10025369: add esp,4
0x1002536c: add esi,1
0x1002536f: test dl,dl
0x10025371: jne short 10025340h
0x10025373: jmp short 1002537Fh
0x10025375: mov eax,[ebp-58h]
0x10025378: or dword ptr [eax+70h],20000h
0x1002537f: mov eax,[ebp-58h]
0x10025382: test dword ptr [eax+70h],20000h
0x10025389: je short 100253FCh
0x1002538b: mov esi,[ebp-60h]
0x1002538e: test esi,esi
0x10025390: je short 100253F2h
0x10025392: push 0
0x10025394: call 10023AF0h
0x10025399: mov cl,[esi]
0x1002539b: add esp,4
0x1002539e: cmp cl,[eax]
0x100253a0: jne short 100253C4h
0x100253a2: cmp cl,31h
0x100253a5: jne short 100253B9h
0x100253a7: push 0
0x100253a9: call 10023AF0h
0x100253ae: mov cl,[esi+2]
0x100253b1: add esp,4
0x100253b4: cmp cl,[eax+2]
0x100253b7: jne short 100253C4h
0x100253b9: cmp byte ptr [esi],30h
0x100253bc: jne short 100253FCh
0x100253be: cmp byte ptr [esi+2],39h
0x100253c2: jge short 100253FCh
0x100253c4: test esi,esi
0x100253c6: je short 100253F2h
0x100253c8: push 0
0x100253ca: call 10023AF0h
0x100253cf: push eax
0x100253d0: push esi
0x100253d1: push 100FFD18h
0x100253d6: lea edx,[ebp-54h]
0x100253d9: push 50h
0x100253db: push edx
```

```
0x100253dc: call dword ptr ds:[100FE4C0h]
0x100253e2: mov ecx,[ebp-58h]
0x100253e5: lea eax,[ebp-54h]
0x100253e8: push eax
0x100253e9: push ecx
0x100253ea: call 100230F0h
0x100253ef: add esp,20h
0x100253f2: mov esi,1
0x100253f7: mov [ebp-5Ch],esi
0x100253fa: jmp short 10025401h
0x100253fc: mov esi,1
0x10025401: mov edx,[ebp-5Ch]
0x10025404: test edx,edx
0x10025406: jne short 10025443h
0x10025408: mov eax,[ebp-58h]
0x1002540b: mov ecx,[ebp-58h]
0x1002540e: mov dword ptr [eax+0B4h],2000h
0x10025418: mov edx,[ecx+0B4h]
0x1002541e: mov eax,[ebp-58h]
0x10025421: push edx
0x10025422: push eax
0x10025423: call 1002CDA0h
0x10025428: mov ecx,[ebp-58h]
0x1002542b: mov edx,[ebp-58h]
0x1002542e: add esp,8
0x10025431: mov [ecx+0B0h],eax
0x10025437: cmp dword ptr [edx+0B0h],0
0x1002543e: jne short 10025443h
0x10025440: mov [ebp-5Ch],esi
0x10025443: mov eax,[ebp-58h]
0x10025446: mov ecx,[ebp-58h]
0x10025449: mov edx,[ebp-58h]
0x1002544c: mov dword ptr [eax+98h],10023480h
0x10025456: mov eax,[ebp-58h]
0x10025459: mov dword ptr [ecx+9Ch],100234D0h
0x10025463: mov ecx,[ebp-5Ch]
0x10025466: test ecx,ecx
0x10025468: mov [eax+0A0h],edx
0x1002546e: jne short 100254C6h
0x10025470: mov edx,[ebp-58h]
0x10025473: push 38h
0x10025475: add edx,78h
0x10025478: push 100FFD10h
0x1002547d: push edx
0x1002547e: call 1005FFD0h
0x10025483: add eax,6
0x10025486: add esp,0Ch
0x10025489: cmp eax,6
0x1002548c: ja short 100254B2h
0x1002548e: jmp dword ptr [eax*4+10025558h]
0x100254b2: push 100FFCC0h
0x100254b7: mov eax,[ebp-58h]
0x100254ba: push eax
0x100254bb: call 100230F0h
0x100254c0: mov [ebp-5Ch],esi
0x100254c3: add esp,8
0x100254c6: mov ecx,[ebp-5Ch]
0x100254c9: test ecx,ecx
0x100254cb: je short 10025512h
0x100254cd: mov edx,[ebp-58h]
0x100254d0: mov eax,[edx+0B0h]
0x100254d6: mov ecx,[ebp-58h]
0x100254d9: push eax
0x100254da: push ecx
0x100254db: call 1002CD70h
0x100254e0: mov edx,[ebp-58h]
0x100254e3: mov eax,[ebp-68h]
```

```
0x100254e6: mov ecx,[ebp-64h]
0x100254e9: push eax
0x100254ea: mov dword ptr [edx+0B0h],0
0x100254f4: mov edx,[ebp-58h]
0x100254f7: push ecx
0x100254f8: push edx
0x100254f9: call 1002CBE0h
0x100254fe: add esp,14h
0x10025501: xor eax,eax
0x10025503: pop esi
0x10025504: mov ecx,[ebp-4]
0x10025507: xor ecx,ebp
0x10025509: call 10022326h
0x1002550e: mov esp,ebp
0x10025510: pop ebp
0x10025511: ret
0x10025512: mov eax,[ebp-58h]
0x10025515: mov ecx,[ebp-58h]
0x10025518: mov edx,[eax+0B0h]
0x1002551e: mov eax,[ebp-58h]
0x10025521: mov [ecx+84h],edx
0x10025527: mov ecx,[ebp-58h]
0x1002552a: mov edx,[eax+0B4h]
0x10025530: mov eax,[ebp-58h]
0x10025533: push 0
0x10025535: push 0
0x10025537: push eax
0x10025538: mov [ecx+88h],edx
0x1002553e: call 10029C70h
0x10025543: mov eax,[ebp-58h]
0x10025546: mov ecx,[ebp-4]
0x10025549: add esp,0Ch
0x1002554c: xor ecx,ebp
0x1002554e: pop esi
0x1002554f: call 10022326h
0x10025554: mov esp,ebp
0x10025556: pop ebp
0x10025557: ret
```

`png_create_read_struct_2` is called by png_create_read_struct. It works with the text "Unknown zlib error", "1.2.5", "Application built with libpng-%.20s but running with %.20s". It delegates to sub_10022326, j__setjmp3, png_set_longjmp_fn (13 callees).

12.568 png_read_info at 0x10025580 (1677 instructions)

```
0x10025580: sub esp,0B4h
0x10025586: mov eax,ds:[10136220h]
0x1002558b: xor eax,esp
0x1002558d: mov [esp+0B0h],eax
0x10025594: mov eax,[esp+0BCh]
0x1002559b: push ebp
0x1002559c: mov ebp,[esp+0BCh]
0x100255a3: test ebp,ebp
0x100255a5: mov [esp+8],eax
0x100255a9: je near ptr 10026834h
0x100255af: test eax,eax
0x100255b1: je near ptr 10026834h
0x100255b7: push ebx
0x100255b8: push edi
0x100255b9: push eax
0x100255ba: push ebp
0x100255bb: call 1002CEA0h
0x100255c0: mov cl,44h
0x100255c2: mov [esp+96h],cl
0x100255c9: mov [esp+1Dh],cl
0x100255cd: mov [esp+0B7h],cl
```

```
0x100255d4: mov [esp+47h],cl
0x100255d8: mov cl,69h
0x100255da: mov [esp+5Ch],cl
0x100255de: mov [esp+0BCh],cl
0x100255e5: mov cl,46h
0x100255e7: mov bl,49h
0x100255e9: mov al,54h
0x100255eb: mov [esp+65h],cl
0x100255ef: mov [esp+66h],cl
0x100255f3: mov cl,73h
0x100255f5: mov dl,74h
0x100255f7: add esp,8
0x100255fa: mov [esp+8Ch],bl
0x10025601: mov byte ptr [esp+8Dh],48h
0x10025609: mov byte ptr [esp+8Fh],52h
0x10025611: mov byte ptr [esp+90h],0
0x10025619: mov [esp+14h],bl
0x1002561d: mov byte ptr [esp+16h],41h
0x10025622: mov [esp+17h],al
0x10025626: mov byte ptr [esp+18h],0
0x1002562b: mov [esp+0ACh],bl
0x10025632: mov byte ptr [esp+0ADh],45h
0x1002563a: mov byte ptr [esp+0AEh],4Eh
0x10025642: mov byte ptr [esp+0B0h],0
0x1002564a: mov byte ptr [esp+1Ch],50h
0x1002564f: mov byte ptr [esp+1Dh],4Ch
0x10025654: mov [esp+1Eh],al
0x10025658: mov byte ptr [esp+1Fh],45h
0x1002565d: mov byte ptr [esp+20h],0
0x10025662: mov byte ptr [esp+3Ch],62h
0x10025667: mov byte ptr [esp+3Dh],4Bh
0x1002566c: mov byte ptr [esp+3Eh],47h
0x10025671: mov byte ptr [esp+40h],0
0x10025676: mov byte ptr [esp+7Ch],63h
0x1002567b: mov byte ptr [esp+7Dh],48h
0x10025680: mov byte ptr [esp+7Eh],52h
0x10025685: mov byte ptr [esp+7Fh],4Dh
0x1002568a: mov byte ptr [esp+80h],0
0x10025692: mov byte ptr [esp+4Ch],67h
0x10025697: mov byte ptr [esp+4Dh],41h
0x1002569c: mov byte ptr [esp+4Eh],4Dh
0x100256a1: mov byte ptr [esp+4Fh],41h
0x100256a6: mov byte ptr [esp+50h],0
0x100256ab: mov byte ptr [esp+9Ch],68h
0x100256b3: mov [esp+9Dh],bl
0x100256ba: mov byte ptr [esp+9Eh],53h
0x100256c2: mov [esp+9Fh],al
0x100256c9: mov byte ptr [esp+0A0h],0
0x100256d1: mov byte ptr [esp+55h],43h
0x100256d6: mov byte ptr [esp+56h],43h
0x100256db: mov byte ptr [esp+57h],50h
0x100256e0: mov byte ptr [esp+58h],0
0x100256e5: mov [esp+0B5h],al
0x100256ec: mov byte ptr [esp+0B6h],58h
0x100256f4: mov [esp+0B7h],dl
0x100256fb: mov byte ptr [esp+0B8h],0
0x10025703: mov byte ptr [esp+5Ch],6Fh
0x10025708: mov [esp+5Fh],cl
0x1002570c: mov byte ptr [esp+60h],0
0x10025711: mov byte ptr [esp+24h],70h
0x10025716: mov byte ptr [esp+25h],43h
0x1002571b: mov byte ptr [esp+26h],41h
0x10025720: mov byte ptr [esp+27h],4Ch
0x10025725: mov byte ptr [esp+28h],0
0x1002572a: mov byte ptr [esp+2Ch],70h
0x1002572f: mov byte ptr [esp+2Dh],48h
0x10025734: mov byte ptr [esp+2Eh],59h
```

```
0x10025739: mov [esp+2Fh],cl
0x1002573d: mov byte ptr [esp+30h],0
0x10025742: mov [esp+34h],cl
0x10025746: mov byte ptr [esp+35h],42h
0x1002574b: mov [esp+36h],bl
0x1002574f: mov [esp+37h],al
0x10025753: mov byte ptr [esp+38h],0
0x10025758: mov [esp+6Ch],cl
0x1002575c: mov byte ptr [esp+6Dh],43h
0x10025761: mov byte ptr [esp+6Eh],41h
0x10025766: mov byte ptr [esp+6Fh],4Ch
0x1002576b: mov byte ptr [esp+70h],0
0x10025770: mov [esp+64h],cl
0x10025774: mov byte ptr [esp+65h],50h
0x10025779: mov byte ptr [esp+66h],4Ch
0x1002577e: mov [esp+67h],al
0x10025782: mov byte ptr [esp+68h],0
0x10025787: mov [esp+44h],cl
0x1002578b: mov byte ptr [esp+45h],52h
0x10025790: mov byte ptr [esp+46h],47h
0x10025795: mov byte ptr [esp+47h],42h
0x1002579a: mov byte ptr [esp+48h],0
0x1002579f: mov [esp+74h],dl
0x100257a3: mov byte ptr [esp+75h],45h
0x100257a8: mov byte ptr [esp+76h],58h
0x100257ad: mov [esp+77h],dl
0x100257b1: mov byte ptr [esp+78h],0
0x100257b6: mov [esp+84h],dl
0x100257bd: mov [esp+85h],bl
0x100257c4: mov byte ptr [esp+86h],4Dh
0x100257cc: mov byte ptr [esp+87h],45h
0x100257d4: mov byte ptr [esp+88h],0
0x100257dc: mov [esp+94h],dl
0x100257e3: mov byte ptr [esp+95h],52h
0x100257eb: mov byte ptr [esp+96h],4Eh
0x100257f3: mov byte ptr [esp+97h],53h
0x100257fb: mov byte ptr [esp+98h],0
0x10025803: mov byte ptr [esp+0A4h],7Ah
0x1002580b: mov [esp+0A5h],al
0x10025812: mov byte ptr [esp+0A6h],58h
0x1002581a: mov [esp+0A7h],dl
0x10025821: mov byte ptr [esp+0A8h],0
0x10025829: lea edi,[ebp+11Ch]
0x1002582f: push esi
0x10025830: push ebp
0x10025831: call 1002DFD0h
0x10025836: mov [esp+14h],eax
0x1002583a: add esp,4
0x1002583d: mov eax,4
0x10025842: lea ecx,[esp+18h]
0x10025846: mov edx,edi
0x10025848: mov esi,[edx]
0x1002584a: cmp esi,[ecx]
0x1002584c: jne short 10025860h
0x1002584e: sub eax,4
0x10025851: add ecx,4
0x10025854: add edx,4
0x10025857: cmp eax,4
0x1002585a: jae short 10025848h
0x1002585c: test eax,eax
0x1002585e: je short 100258BDh
0x10025860: movzx esi,byte ptr [edx]
0x10025863: movzx ebx,byte ptr [ecx]
0x10025866: sub esi,ebx
0x10025868: jne short 100258AFh
0x1002586a: sub eax,1
0x1002586d: add ecx,1
```

```
0x10025870: add edx,1
0x10025873: test eax,eax
0x10025875: je short 100258BDh
0x10025877: movzx esi,byte ptr [edx]
0x1002587a: movzx ebx,byte ptr [ecx]
0x1002587d: sub esi,ebx
0x1002587f: jne short 100258AFh
0x10025881: sub eax,1
0x10025884: add ecx,1
0x10025887: add edx,1
0x1002588a: test eax,eax
0x1002588c: je short 100258BDh
0x1002588e: movzx esi,byte ptr [edx]
0x10025891: movzx ebx,byte ptr [ecx]
0x10025894: sub esi,ebx
0x10025896: jne short 100258AFh
0x10025898: sub eax,1
0x1002589b: add ecx,1
0x1002589e: add edx,1
0x100258a1: test eax,eax
0x100258a3: je short 100258BDh
0x100258a5: movzx esi,byte ptr [edx]
0x100258a8: movzx eax,byte ptr [ecx]
0x100258ab: sub esi,eax
0x100258ad: je short 100258BDh
0x100258af: test esi,esi
0x100258b1: mov eax,1
0x100258b6: jg short 100258BFh
0x100258b8: or eax,0FFFFFFFFh
0x100258bb: jmp short 100258BFh
0x100258bd: xor eax,eax
0x100258bf: test eax,eax
0x100258c1: jne short 100258D2h
0x100258c3: mov eax,[ebp+6Ch]
0x100258c6: test al,8
0x100258c8: je short 100258D2h
0x100258ca: or eax,2000h
0x100258cf: mov [ebp+6Ch],eax
0x100258d2: mov eax,4
0x100258d7: lea ecx,[esp+90h]
0x100258de: mov edx,edi
0x100258e0: mov esi,[edx]
0x100258e2: cmp esi,[ecx]
0x100258e4: jne short 100258F8h
0x100258e6: sub eax,4
0x100258e9: add ecx,4
0x100258ec: add edx,4
0x100258ef: cmp eax,4
0x100258f2: jae short 100258E0h
0x100258f4: test eax,eax
0x100258f6: je short 10025955h
0x100258f8: movzx esi,byte ptr [edx]
0x100258fb: movzx ebx,byte ptr [ecx]
0x100258fe: sub esi,ebx
0x10025900: jne short 10025947h
0x10025902: sub eax,1
0x10025905: add ecx,1
0x10025908: add edx,1
0x1002590b: test eax,eax
0x1002590d: je short 10025955h
0x1002590f: movzx esi,byte ptr [edx]
0x10025912: movzx ebx,byte ptr [ecx]
0x10025915: sub esi,ebx
0x10025917: jne short 10025947h
0x10025919: sub eax,1
0x1002591c: add ecx,1
0x1002591f: add edx,1
```

```
0x10025922: test eax,eax
0x10025924: je short 10025955h
0x10025926: movzx esi,byte ptr [edx]
0x10025929: movzx ebx,byte ptr [ecx]
0x1002592c: sub esi,ebx
0x1002592e: jne short 10025947h
0x10025930: sub eax,1
0x10025933: add ecx,1
0x10025936: add edx,1
0x10025939: test eax,eax
0x1002593b: je short 10025955h
0x1002593d: movzx esi,byte ptr [edx]
0x10025940: movzx ecx,byte ptr [ecx]
0x10025943: sub esi,ecx
0x10025945: je short 10025955h
0x10025947: test esi,esi
0x10025949: mov eax,1
0x1002594e: jg short 10025957h
0x10025950: or eax,0FFFFFFFFh
0x10025953: jmp short 10025957h
0x10025955: xor eax,eax
0x10025957: test eax,eax
0x10025959: jne short 10025973h
0x1002595b: mov edx,[esp+10h]
0x1002595f: mov eax,[esp+14h]
0x10025963: push edx
0x10025964: push eax
0x10025965: push ebp
0x10025966: call 1002E100h
0x1002596b: add esp,0Ch
0x1002596e: jmp near ptr 10025830h
0x10025973: mov eax,4
0x10025978: lea ecx,[esp+0B0h]
0x1002597f: mov edx,edi
0x10025981: mov esi,[edx]
0x10025983: cmp esi,[ecx]
0x10025985: jne short 10025999h
0x10025987: sub eax,4
0x1002598a: add ecx,4
0x1002598d: add edx,4
0x10025990: cmp eax,4
0x10025993: jae short 10025981h
0x10025995: test eax,eax
0x10025997: je short 100259F6h
0x10025999: movzx esi,byte ptr [edx]
0x1002599c: movzx ebx,byte ptr [ecx]
0x1002599f: sub esi,ebx
0x100259a1: jne short 100259E8h
0x100259a3: sub eax,1
0x100259a6: add ecx,1
0x100259a9: add edx,1
0x100259ac: test eax,eax
0x100259ae: je short 100259F6h
0x100259b0: movzx esi,byte ptr [edx]
0x100259b3: movzx ebx,byte ptr [ecx]
0x100259b6: sub esi,ebx
0x100259b8: jne short 100259E8h
0x100259ba: sub eax,1
0x100259bd: add ecx,1
0x100259c0: add edx,1
0x100259c3: test eax,eax
0x100259c5: je short 100259F6h
0x100259c7: movzx esi,byte ptr [edx]
0x100259ca: movzx ebx,byte ptr [ecx]
0x100259cd: sub esi,ebx
0x100259cf: jne short 100259E8h
0x100259d1: sub eax,1
```

```
0x100259d4: add ecx,1
0x100259d7: add edx,1
0x100259da: test eax,eax
0x100259dc: je short 100259F6h
0x100259de: movzx esi,byte ptr [edx]
0x100259e1: movzx ecx,byte ptr [ecx]
0x100259e4: sub esi,ecx
0x100259e6: je short 100259F6h
0x100259e8: test esi,esi
0x100259ea: mov eax,1
0x100259ef: jg short 100259F8h
0x100259f1: or eax,0FFFFFFFFh
0x100259f4: jmp short 100259F8h
0x100259f6: xor eax,eax
0x100259f8: test eax,eax
0x100259fa: jne short 10025A14h
0x100259fc: mov edx,[esp+10h]
0x10025a00: mov eax,[esp+14h]
0x10025a04: push edx
0x10025a05: push eax
0x10025a06: push ebp
0x10025a07: call 1002E400h
0x10025a0c: add esp,0Ch
0x10025a0f: jmp near ptr 10025830h
0x10025a14: push edi
0x10025a15: push ebp
0x10025a16: call 10023B10h
0x10025a1b: add esp,8
0x10025a1e: test eax,eax
0x10025a20: mov edx,edi
0x10025a22: mov eax,4
0x10025a27: je near ptr 10025BF5h
0x10025a2d: lea ecx,[esp+18h]
0x10025a31: mov esi,[edx]
0x10025a33: cmp esi,[ecx]
0x10025a35: jne short 10025A49h
0x10025a37: sub eax,4
0x10025a3a: add ecx,4
0x10025a3d: add edx,4
0x10025a40: cmp eax,4
0x10025a43: jae short 10025A31h
0x10025a45: test eax,eax
0x10025a47: je short 10025AA6h
0x10025a49: movzx esi,byte ptr [edx]
0x10025a4c: movzx ebx,byte ptr [ecx]
0x10025a4f: sub esi,ebx
0x10025a51: jne short 10025A98h
0x10025a53: sub eax,1
0x10025a56: add ecx,1
0x10025a59: add edx,1
0x10025a5c: test eax,eax
0x10025a5e: je short 10025AA6h
0x10025a60: movzx esi,byte ptr [edx]
0x10025a63: movzx ebx,byte ptr [ecx]
0x10025a66: sub esi,ebx
0x10025a68: jne short 10025A98h
0x10025a6a: sub eax,1
0x10025a6d: add ecx,1
0x10025a70: add edx,1
0x10025a73: test eax,eax
0x10025a75: je short 10025AA6h
0x10025a77: movzx esi,byte ptr [edx]
0x10025a7a: movzx ebx,byte ptr [ecx]
0x10025a7d: sub esi,ebx
0x10025a7f: jne short 10025A98h
0x10025a81: sub eax,1
0x10025a84: add ecx,1
```

```
0x10025a87: add edx,1
0x10025a8a: test eax,eax
0x10025a8c: je short 10025AA6h
0x10025a8e: movzx esi,byte ptr [edx]
0x10025a91: movzx ecx,byte ptr [ecx]
0x10025a94: sub esi,ecx
0x10025a96: je short 10025AA6h
0x10025a98: test esi,esi
0x10025a9a: mov eax,1
0x10025a9f: jg short 10025AA8h
0x10025aa1: or eax,0FFFFFFFFh
0x10025aa4: jmp short 10025AA8h
0x10025aa6: xor eax,eax
0x10025aa8: test eax,eax
0x10025aaa: jne short 10025AB0h
0x10025aac: or dword ptr [ebp+6Ch],4
0x10025ab0: mov edx,[esp+10h]
0x10025ab4: mov eax,[esp+14h]
0x10025ab8: push edx
0x10025ab9: push eax
0x10025aba: push ebp
0x10025abb: call 10030750h
0x10025ac0: add esp,0Ch
0x10025ac3: mov eax,4
0x10025ac8: lea ecx,[esp+20h]
0x10025acc: mov edx,edi
0x10025ace: mov edi,edi
0x10025ad0: mov esi,[edx]
0x10025ad2: cmp esi,[ecx]
0x10025ad4: jne short 10025AE8h
0x10025ad6: sub eax,4
0x10025ad9: add ecx,4
0x10025adc: add edx,4
0x10025adf: cmp eax,4
0x10025ae2: jae short 10025AD0h
0x10025ae4: test eax,eax
0x10025ae6: je short 10025B45h
0x10025ae8: movzx esi,byte ptr [edx]
0x10025aeb: movzx ebx,byte ptr [ecx]
0x10025aee: sub esi,ebx
0x10025af0: jne short 10025B37h
0x10025af2: sub eax,1
0x10025af5: add ecx,1
0x10025af8: add edx,1
0x10025afb: test eax,eax
0x10025afd: je short 10025B45h
0x10025aff: movzx esi,byte ptr [edx]
0x10025b02: movzx ebx,byte ptr [ecx]
0x10025b05: sub esi,ebx
0x10025b07: jne short 10025B37h
0x10025b09: sub eax,1
0x10025b0c: add ecx,1
0x10025b0f: add edx,1
0x10025b12: test eax,eax
0x10025b14: je short 10025B45h
0x10025b16: movzx esi,byte ptr [edx]
0x10025b19: movzx ebx,byte ptr [ecx]
0x10025b1c: sub esi,ebx
0x10025b1e: jne short 10025B37h
0x10025b20: sub eax,1
0x10025b23: add ecx,1
0x10025b26: add edx,1
0x10025b29: test eax,eax
0x10025b2b: je short 10025B45h
0x10025b2d: movzx esi,byte ptr [edx]
0x10025b30: movzx ecx,byte ptr [ecx]
0x10025b33: sub esi,ecx
```

```
0x10025b35: je short 10025B45h
0x10025b37: test esi,esi
0x10025b39: mov eax,1
0x10025b3e: jg short 10025B47h
0x10025b40: or eax,0FFFFFFFFh
0x10025b43: jmp short 10025B47h
0x10025b45: xor eax,eax
0x10025b47: test eax,eax
0x10025b49: jne short 10025B54h
0x10025b4b: or dword ptr [ebp+6Ch],2
0x10025b4f: jmp near ptr 10025830h
0x10025b54: mov eax,4
0x10025b59: lea ecx,[esp+18h]
0x10025b5d: mov edx,edi
0x10025b5f: nop
0x10025b60: mov esi,[edx]
0x10025b62: cmp esi,[ecx]
0x10025b64: jne short 10025B78h
0x10025b66: sub eax,4
0x10025b69: add ecx,4
0x10025b6c: add edx,4
0x10025b6f: cmp eax,4
0x10025b72: jae short 10025B60h
0x10025b74: test eax,eax
0x10025b76: je short 10025BD5h
0x10025b78: movzx esi,byte ptr [edx]
0x10025b7b: movzx ebx,byte ptr [ecx]
0x10025b7e: sub esi,ebx
0x10025b80: jne short 10025BC7h
0x10025b82: sub eax,1
0x10025b85: add ecx,1
0x10025b88: add edx,1
0x10025b8b: test eax,eax
0x10025b8d: je short 10025BD5h
0x10025b8f: movzx esi,byte ptr [edx]
0x10025b92: movzx ebx,byte ptr [ecx]
0x10025b95: sub esi,ebx
0x10025b97: jne short 10025BC7h
0x10025b99: sub eax,1
0x10025b9c: add ecx,1
0x10025b9f: add edx,1
0x10025ba2: test eax,eax
0x10025ba4: je short 10025BD5h
0x10025ba6: movzx esi,byte ptr [edx]
0x10025ba9: movzx ebx,byte ptr [ecx]
0x10025bac: sub esi,ebx
0x10025bae: jne short 10025BC7h
0x10025bb0: sub eax,1
0x10025bb3: add ecx,1
0x10025bb6: add edx,1
0x10025bb9: test eax,eax
0x10025bbb: je short 10025BD5h
0x10025bbd: movzx esi,byte ptr [edx]
0x10025bc0: movzx edx,byte ptr [ecx]
0x10025bc3: sub esi,edx
0x10025bc5: je short 10025BD5h
0x10025bc7: test esi,esi
0x10025bc9: mov eax,1
0x10025bce: jg short 10025BD7h
0x10025bd0: or eax,0FFFFFFFFh
0x10025bd3: jmp short 10025BD7h
0x10025bd5: xor eax,eax
0x10025bd7: test eax,eax
0x10025bd9: jne near ptr 10025830h
0x10025bdf: mov eax,[ebp+6Ch]
0x10025be2: test al,1
0x10025be4: jne near ptr 100267DFh
```

```
0x10025bea: push 100FFD70h
0x10025bef: push ebp
0x10025bf0: call 100231E0h
0x10025bf5: lea ecx,[esp+20h]
0x10025bf9: lea esp,[esp]
0x10025c00: mov esi,[edx]
0x10025c02: cmp esi,[ecx]
0x10025c04: jne short 10025C18h
0x10025c06: sub eax,4
0x10025c09: add ecx,4
0x10025c0c: add edx,4
0x10025c0f: cmp eax,4
0x10025c12: jae short 10025C00h
0x10025c14: test eax,eax
0x10025c16: je short 10025C75h
0x10025c18: movzx esi,byte ptr [edx]
0x10025c1b: movzx ebx,byte ptr [ecx]
0x10025c1e: sub esi,ebx
0x10025c20: jne short 10025C67h
0x10025c22: sub eax,1
0x10025c25: add ecx,1
0x10025c28: add edx,1
0x10025c2b: test eax,eax
0x10025c2d: je short 10025C75h
0x10025c2f: movzx esi,byte ptr [edx]
0x10025c32: movzx ebx,byte ptr [ecx]
0x10025c35: sub esi,ebx
0x10025c37: jne short 10025C67h
0x10025c39: sub eax,1
0x10025c3c: add ecx,1
0x10025c3f: add edx,1
0x10025c42: test eax,eax
0x10025c44: je short 10025C75h
0x10025c46: movzx esi,byte ptr [edx]
0x10025c49: movzx ebx,byte ptr [ecx]
0x10025c4c: sub esi,ebx
0x10025c4e: jne short 10025C67h
0x10025c50: sub eax,1
0x10025c53: add ecx,1
0x10025c56: add edx,1
0x10025c59: test eax,eax
0x10025c5b: je short 10025C75h
0x10025c5d: movzx esi,byte ptr [edx]
0x10025c60: movzx eax,byte ptr [ecx]
0x10025c63: sub esi,eax
0x10025c65: je short 10025C75h
0x10025c67: test esi,esi
0x10025c69: mov eax,1
0x10025c6e: jg short 10025C77h
0x10025c70: or eax,0FFFFFFFFh
0x10025c73: jmp short 10025C77h
0x10025c75: xor eax,eax
0x10025c77: test eax,eax
0x10025c79: jne short 10025C93h
0x10025c7b: mov ecx,[esp+10h]
0x10025c7f: mov edx,[esp+14h]
0x10025c83: push ecx
0x10025c84: push edx
0x10025c85: push ebp
0x10025c86: call 1002E260h
0x10025c8b: add esp,0Ch
0x10025c8e: jmp near ptr 10025830h
0x10025c93: mov eax,4
0x10025c98: lea ecx,[esp+18h]
0x10025c9c: mov edx,edi
0x10025c9e: mov edi,edi
0x10025ca0: mov esi,[edx]
```

```
0x10025ca2: cmp esi,[ecx]
0x10025ca4: jne short 10025CB8h
0x10025ca6: sub eax,4
0x10025ca9: add ecx,4
0x10025cac: add edx,4
0x10025caf: cmp eax,4
0x10025cb2: jae short 10025CA0h
0x10025cb4: test eax,eax
0x10025cb6: je short 10025D15h
0x10025cb8: movzx esi,byte ptr [edx]
0x10025cbb: movzx ebx,byte ptr [ecx]
0x10025cbe: sub esi,ebx
0x10025cc0: jne short 10025D07h
0x10025cc2: sub eax,1
0x10025cc5: add ecx,1
0x10025cc8: add edx,1
0x10025ccb: test eax,eax
0x10025ccd: je short 10025D15h
0x10025ccf: movzx esi,byte ptr [edx]
0x10025cd2: movzx ebx,byte ptr [ecx]
0x10025cd5: sub esi,ebx
0x10025cd7: jne short 10025D07h
0x10025cd9: sub eax,1
0x10025cdc: add ecx,1
0x10025cdf: add edx,1
0x10025ce2: test eax,eax
0x10025ce4: je short 10025D15h
0x10025ce6: movzx esi,byte ptr [edx]
0x10025ce9: movzx ebx,byte ptr [ecx]
0x10025cec: sub esi,ebx
0x10025cee: jne short 10025D07h
0x10025cf0: sub eax,1
0x10025cf3: add ecx,1
0x10025cf6: add edx,1
0x10025cf9: test eax,eax
0x10025cfb: je short 10025D15h
0x10025cfd: movzx esi,byte ptr [edx]
0x10025d00: movzx eax,byte ptr [ecx]
0x10025d03: sub esi,eax
0x10025d05: je short 10025D15h
0x10025d07: test esi,esi
0x10025d09: mov eax,1
0x10025d0e: jg short 10025D17h
0x10025d10: or eax,0FFFFFFFFh
0x10025d13: jmp short 10025D17h
0x10025d15: xor eax,eax
0x10025d17: test eax,eax
0x10025d19: je near ptr 100267F7h
0x10025d1f: mov eax,4
0x10025d24: lea ecx,[esp+40h]
0x10025d28: mov edx,edi
0x10025d2a: lea ebx,[ebx]
0x10025d30: mov esi,[edx]
0x10025d32: cmp esi,[ecx]
0x10025d34: jne short 10025D48h
0x10025d36: sub eax,4
0x10025d39: add ecx,4
0x10025d3c: add edx,4
0x10025d3f: cmp eax,4
0x10025d42: jae short 10025D30h
0x10025d44: test eax,eax
0x10025d46: je short 10025DA5h
0x10025d48: movzx esi,byte ptr [edx]
0x10025d4b: movzx ebx,byte ptr [ecx]
0x10025d4e: sub esi,ebx
0x10025d50: jne short 10025D97h
0x10025d52: sub eax,1
```

```
0x10025d55: add ecx,1
0x10025d58: add edx,1
0x10025d5b: test eax,eax
0x10025d5d: je short 10025DA5h
0x10025d5f: movzx esi,byte ptr [edx]
0x10025d62: movzx ebx,byte ptr [ecx]
0x10025d65: sub esi,ebx
0x10025d67: jne short 10025D97h
0x10025d69: sub eax,1
0x10025d6c: add ecx,1
0x10025d6f: add edx,1
0x10025d72: test eax,eax
0x10025d74: je short 10025DA5h
0x10025d76: movzx esi,byte ptr [edx]
0x10025d79: movzx ebx,byte ptr [ecx]
0x10025d7c: sub esi,ebx
0x10025d7e: jne short 10025D97h
0x10025d80: sub eax,1
0x10025d83: add ecx,1
0x10025d86: add edx,1
0x10025d89: test eax,eax
0x10025d8b: je short 10025DA5h
0x10025d8d: movzx esi,byte ptr [edx]
0x10025d90: movzx ecx,byte ptr [ecx]
0x10025d93: sub esi,ecx
0x10025d95: je short 10025DA5h
0x10025d97: test esi,esi
0x10025d99: mov eax,1
0x10025d9e: jg short 10025DA7h
0x10025da0: or eax,0FFFFFFFFh
0x10025da3: jmp short 10025DA7h
0x10025da5: xor eax,eax
0x10025da7: test eax,eax
0x10025da9: jne short 10025DC3h
0x10025dab: mov edx,[esp+10h]
0x10025daf: mov eax,[esp+14h]
0x10025db3: push edx
0x10025db4: push eax
0x10025db5: push ebp
0x10025db6: call 1002F400h
0x10025dbb: add esp,0Ch
0x10025dbe: jmp near ptr 10025830h
0x10025dc3: mov eax,4
0x10025dc8: lea ecx,[esp+80h]
0x10025dcf: mov edx,edi
0x10025dd1: mov esi,[edx]
0x10025dd3: cmp esi,[ecx]
0x10025dd5: jne short 10025DE9h
0x10025dd7: sub eax,4
0x10025dda: add ecx,4
0x10025ddd: add edx,4
0x10025de0: cmp eax,4
0x10025de3: jae short 10025DD1h
0x10025de5: test eax,eax
0x10025de7: je short 10025E46h
0x10025de9: movzx esi,byte ptr [edx]
0x10025dec: movzx ebx,byte ptr [ecx]
0x10025def: sub esi,ebx
0x10025df1: jne short 10025E38h
0x10025df3: sub eax,1
0x10025df6: add ecx,1
0x10025df9: add edx,1
0x10025dfc: test eax,eax
0x10025dfe: je short 10025E46h
0x10025e00: movzx esi,byte ptr [edx]
0x10025e03: movzx ebx,byte ptr [ecx]
0x10025e06: sub esi,ebx
```

```
0x10025e08: jne short 10025E38h
0x10025e0a: sub eax,1
0x10025e0d: add ecx,1
0x10025e10: add edx,1
0x10025e13: test eax,eax
0x10025e15: je short 10025E46h
0x10025e17: movzx esi,byte ptr [edx]
0x10025e1a: movzx ebx,byte ptr [ecx]
0x10025e1d: sub esi,ebx
0x10025e1f: jne short 10025E38h
0x10025e21: sub eax,1
0x10025e24: add ecx,1
0x10025e27: add edx,1
0x10025e2a: test eax,eax
0x10025e2c: je short 10025E46h
0x10025e2e: movzx esi,byte ptr [edx]
0x10025e31: movzx ecx,byte ptr [ecx]
0x10025e34: sub esi,ecx
0x10025e36: je short 10025E46h
0x10025e38: test esi,esi
0x10025e3a: mov eax,1
0x10025e3f: jg short 10025E48h
0x10025e41: or eax,0FFFFFFFFh
0x10025e44: jmp short 10025E48h
0x10025e46: xor eax,eax
0x10025e48: test eax,eax
0x10025e4a: jne short 10025E64h
0x10025e4c: mov edx,[esp+10h]
0x10025e50: mov eax,[esp+14h]
0x10025e54: push edx
0x10025e55: push eax
0x10025e56: push ebp
0x10025e57: call 1002E700h
0x10025e5c: add esp,0Ch
0x10025e5f: jmp near ptr 10025830h
0x10025e64: mov eax,4
0x10025e69: lea ecx,[esp+50h]
0x10025e6d: mov edx,edi
0x10025e6f: nop
0x10025e70: mov esi,[edx]
0x10025e72: cmp esi,[ecx]
0x10025e74: jne short 10025E88h
0x10025e76: sub eax,4
0x10025e79: add ecx,4
0x10025e7c: add edx,4
0x10025e7f: cmp eax,4
0x10025e82: jae short 10025E70h
0x10025e84: test eax,eax
0x10025e86: je short 10025EE5h
0x10025e88: movzx esi,byte ptr [edx]
0x10025e8b: movzx ebx,byte ptr [ecx]
0x10025e8e: sub esi,ebx
0x10025e90: jne short 10025ED7h
0x10025e92: sub eax,1
0x10025e95: add ecx,1
0x10025e98: add edx,1
0x10025e9b: test eax,eax
0x10025e9d: je short 10025EE5h
0x10025e9f: movzx esi,byte ptr [edx]
0x10025ea2: movzx ebx,byte ptr [ecx]
0x10025ea5: sub esi,ebx
0x10025ea7: jne short 10025ED7h
0x10025ea9: sub eax,1
0x10025eac: add ecx,1
0x10025eaf: add edx,1
0x10025eb2: test eax,eax
0x10025eb4: je short 10025EE5h
```

```
0x10025eb6: movzx esi,byte ptr [edx]
0x10025eb9: movzx ebx,byte ptr [ecx]
0x10025ebc: sub esi,ebx
0x10025ebe: jne short 10025ED7h
0x10025ec0: sub eax,1
0x10025ec3: add ecx,1
0x10025ec6: add edx,1
0x10025ec9: test eax,eax
0x10025ecb: je short 10025EE5h
0x10025ecd: movzx esi,byte ptr [edx]
0x10025ed0: movzx ecx,byte ptr [ecx]
0x10025ed3: sub esi,ecx
0x10025ed5: je short 10025EE5h
0x10025ed7: test esi,esi
0x10025ed9: mov eax,1
0x10025ede: jg short 10025EE7h
0x10025ee0: or eax,0FFFFFFFFh
0x10025ee3: jmp short 10025EE7h
0x10025ee5: xor eax,eax
0x10025ee7: test eax,eax
0x10025ee9: jne short 10025F03h
0x10025eeb: mov edx,[esp+10h]
0x10025eef: mov eax,[esp+14h]
0x10025ef3: push edx
0x10025ef4: push eax
0x10025ef5: push ebp
0x10025ef6: call 1002E450h
0x10025efb: add esp,0Ch
0x10025efe: jmp near ptr 10025830h
0x10025f03: mov eax,4
0x10025f08: lea ecx,[esp+0A0h]
0x10025f0f: mov edx,edi
0x10025f11: mov esi,[edx]
0x10025f13: cmp esi,[ecx]
0x10025f15: jne short 10025F29h
0x10025f17: sub eax,4
0x10025f1a: add ecx,4
0x10025f1d: add edx,4
0x10025f20: cmp eax,4
0x10025f23: jae short 10025F11h
0x10025f25: test eax,eax
0x10025f27: je short 10025F86h
0x10025f29: movzx esi,byte ptr [edx]
0x10025f2c: movzx ebx,byte ptr [ecx]
0x10025f2f: sub esi,ebx
0x10025f31: jne short 10025F78h
0x10025f33: sub eax,1
0x10025f36: add ecx,1
0x10025f39: add edx,1
0x10025f3c: test eax,eax
0x10025f3e: je short 10025F86h
0x10025f40: movzx esi,byte ptr [edx]
0x10025f43: movzx ebx,byte ptr [ecx]
0x10025f46: sub esi,ebx
0x10025f48: jne short 10025F78h
0x10025f4a: sub eax,1
0x10025f4d: add ecx,1
0x10025f50: add edx,1
0x10025f53: test eax,eax
0x10025f55: je short 10025F86h
0x10025f57: movzx esi,byte ptr [edx]
0x10025f5a: movzx ebx,byte ptr [ecx]
0x10025f5d: sub esi,ebx
0x10025f5f: jne short 10025F78h
0x10025f61: sub eax,1
0x10025f64: add ecx,1
0x10025f67: add edx,1
```

```
0x10025f6a: test eax,eax
0x10025f6c: je short 10025F86h
0x10025f6e: movzx esi,byte ptr [edx]
0x10025f71: movzx ecx,byte ptr [ecx]
0x10025f74: sub esi,ecx
0x10025f76: je short 10025F86h
0x10025f78: test esi,esi
0x10025f7a: mov eax,1
0x10025f7f: jg short 10025F88h
0x10025f81: or eax,0FFFFFFFFh
0x10025f84: jmp short 10025F88h
0x10025f86: xor eax,eax
0x10025f88: test eax,eax
0x10025f8a: jne short 10025FA4h
0x10025f8c: mov edx,[esp+10h]
0x10025f90: mov eax,[esp+14h]
0x10025f94: push edx
0x10025f95: push eax
0x10025f96: push ebp
0x10025f97: call 1002F650h
0x10025f9c: add esp,0Ch
0x10025f9f: jmp near ptr 10025830h
0x10025fa4: mov eax,4
0x10025fa9: lea ecx,[esp+60h]
0x10025fad: mov edx,edi
0x10025faf: nop
0x10025fb0: mov esi,[edx]
0x10025fb2: cmp esi,[ecx]
0x10025fb4: jne short 10025FC8h
0x10025fb6: sub eax,4
0x10025fb9: add ecx,4
0x10025fbc: add edx,4
0x10025fbf: cmp eax,4
0x10025fc2: jae short 10025FB0h
0x10025fc4: test eax,eax
0x10025fc6: je short 10026025h
0x10025fc8: movzx esi,byte ptr [edx]
0x10025fcb: movzx ebx,byte ptr [ecx]
0x10025fce: sub esi,ebx
0x10025fd0: jne short 10026017h
0x10025fd2: sub eax,1
0x10025fd5: add ecx,1
0x10025fd8: add edx,1
0x10025fdb: test eax,eax
0x10025fdd: je short 10026025h
0x10025fdf: movzx esi,byte ptr [edx]
0x10025fe2: movzx ebx,byte ptr [ecx]
0x10025fe5: sub esi,ebx
0x10025fe7: jne short 10026017h
0x10025fe9: sub eax,1
0x10025fec: add ecx,1
0x10025fef: add edx,1
0x10025ff2: test eax,eax
0x10025ff4: je short 10026025h
0x10025ff6: movzx esi,byte ptr [edx]
0x10025ff9: movzx ebx,byte ptr [ecx]
0x10025ffc: sub esi,ebx
0x10025ffe: jne short 10026017h
0x10026000: sub eax,1
0x10026003: add ecx,1
0x10026006: add edx,1
0x10026009: test eax,eax
0x1002600b: je short 10026025h
0x1002600d: movzx esi,byte ptr [edx]
0x10026010: movzx ecx,byte ptr [ecx]
0x10026013: sub esi,ecx
0x10026015: je short 10026025h
```

```
0x10026017: test esi,esi
0x10026019: mov eax,1
0x1002601e: jg short 10026027h
0x10026020: or eax,0FFFFFFFFh
0x10026023: jmp short 10026027h
0x10026025: xor eax,eax
0x10026027: test eax,eax
0x10026029: jne short 10026043h
0x1002602b: mov edx,[esp+10h]
0x1002602f: mov eax,[esp+14h]
0x10026033: push edx
0x10026034: push eax
0x10026035: push ebp
0x10026036: call 1002F930h
0x1002603b: add esp,0Ch
0x1002603e: jmp near ptr 10025830h
0x10026043: mov eax,4
0x10026048: lea ecx,[esp+28h]
0x1002604c: mov edx,edi
0x1002604e: mov edi,edi
0x10026050: mov esi,[edx]
0x10026052: cmp esi,[ecx]
0x10026054: jne short 10026068h
0x10026056: sub eax,4
0x10026059: add ecx,4
0x1002605c: add edx,4
0x1002605f: cmp eax,4
0x10026062: jae short 10026050h
0x10026064: test eax,eax
0x10026066: je short 100260C5h
0x10026068: movzx esi,byte ptr [edx]
0x1002606b: movzx ebx,byte ptr [ecx]
0x1002606e: sub esi,ebx
0x10026070: jne short 100260B7h
0x10026072: sub eax,1
0x10026075: add ecx,1
0x10026078: add edx,1
0x1002607b: test eax,eax
0x1002607d: je short 100260C5h
0x1002607f: movzx esi,byte ptr [edx]
0x10026082: movzx ebx,byte ptr [ecx]
0x10026085: sub esi,ebx
0x10026087: jne short 100260B7h
0x10026089: sub eax,1
0x1002608c: add ecx,1
0x1002608f: add edx,1
0x10026092: test eax,eax
0x10026094: je short 100260C5h
0x10026096: movzx esi,byte ptr [edx]
0x10026099: movzx ebx,byte ptr [ecx]
0x1002609c: sub esi,ebx
0x1002609e: jne short 100260B7h
0x100260a0: sub eax,1
0x100260a3: add ecx,1
0x100260a6: add edx,1
0x100260a9: test eax,eax
0x100260ab: je short 100260C5h
0x100260ad: movzx esi,byte ptr [edx]
0x100260b0: movzx ecx,byte ptr [ecx]
0x100260b3: sub esi,ecx
0x100260b5: je short 100260C5h
0x100260b7: test esi,esi
0x100260b9: mov eax,1
0x100260be: jg short 100260C7h
0x100260c0: or eax,0FFFFFFFFh
0x100260c3: jmp short 100260C7h
0x100260c5: xor eax,eax
```

```
0x100260c7: test eax,eax
0x100260c9: jne short 100260E3h
0x100260cb: mov edx,[esp+10h]
0x100260cf: mov eax,[esp+14h]
0x100260d3: push edx
0x100260d4: push eax
0x100260d5: push ebp
0x100260d6: call 1002FAA0h
0x100260db: add esp,0Ch
0x100260de: jmp near ptr 10025830h
0x100260e3: mov eax,4
0x100260e8: lea ecx,[esp+70h]
0x100260ec: mov edx,edi
0x100260ee: mov edi,edi
0x100260f0: mov esi,[edx]
0x100260f2: cmp esi,[ecx]
0x100260f4: jne short 10026108h
0x100260f6: sub eax,4
0x100260f9: add ecx,4
0x100260fc: add edx,4
0x100260ff: cmp eax,4
0x10026102: jae short 100260F0h
0x10026104: test eax,eax
0x10026106: je short 10026165h
0x10026108: movzx esi,byte ptr [edx]
0x1002610b: movzx ebx,byte ptr [ecx]
0x1002610e: sub esi,ebx
0x10026110: jne short 10026157h
0x10026112: sub eax,1
0x10026115: add ecx,1
0x10026118: add edx,1
0x1002611b: test eax,eax
0x1002611d: je short 10026165h
0x1002611f: movzx esi,byte ptr [edx]
0x10026122: movzx ebx,byte ptr [ecx]
0x10026125: sub esi,ebx
0x10026127: jne short 10026157h
0x10026129: sub eax,1
0x1002612c: add ecx,1
0x1002612f: add edx,1
0x10026132: test eax,eax
0x10026134: je short 10026165h
0x10026136: movzx esi,byte ptr [edx]
0x10026139: movzx ebx,byte ptr [ecx]
0x1002613c: sub esi,ebx
0x1002613e: jne short 10026157h
0x10026140: sub eax,1
0x10026143: add ecx,1
0x10026146: add edx,1
0x10026149: test eax,eax
0x1002614b: je short 10026165h
0x1002614d: movzx esi,byte ptr [edx]
0x10026150: movzx ecx,byte ptr [ecx]
0x10026153: sub esi,ecx
0x10026155: je short 10026165h
0x10026157: test esi,esi
0x10026159: mov eax,1
0x1002615e: jg short 10026167h
0x10026160: or eax,0FFFFFFFFh
0x10026163: jmp short 10026167h
0x10026165: xor eax,eax
0x10026167: test eax,eax
0x10026169: jne short 10026183h
0x1002616b: mov edx,[esp+10h]
0x1002616f: mov eax,[esp+14h]
0x10026173: push edx
0x10026174: push eax
```

```
0x10026175: push ebp
0x10026176: call 1002FDD0h
0x1002617b: add esp,0Ch
0x1002617e: jmp near ptr 10025830h
0x10026183: mov eax,4
0x10026188: lea ecx,[esp+30h]
0x1002618c: mov edx,edi
0x1002618e: mov edi,edi
0x10026190: mov esi,[edx]
0x10026192: cmp esi,[ecx]
0x10026194: jne short 100261A8h
0x10026196: sub eax,4
0x10026199: add ecx,4
0x1002619c: add edx,4
0x1002619f: cmp eax,4
0x100261a2: jae short 10026190h
0x100261a4: test eax,eax
0x100261a6: je short 10026205h
0x100261a8: movzx esi,byte ptr [edx]
0x100261ab: movzx ebx,byte ptr [ecx]
0x100261ae: sub esi,ebx
0x100261b0: jne short 100261F7h
0x100261b2: sub eax,1
0x100261b5: add ecx,1
0x100261b8: add edx,1
0x100261bb: test eax,eax
0x100261bd: je short 10026205h
0x100261bf: movzx esi,byte ptr [edx]
0x100261c2: movzx ebx,byte ptr [ecx]
0x100261c5: sub esi,ebx
0x100261c7: jne short 100261F7h
0x100261c9: sub eax,1
0x100261cc: add ecx,1
0x100261cf: add edx,1
0x100261d2: test eax,eax
0x100261d4: je short 10026205h
0x100261d6: movzx esi,byte ptr [edx]
0x100261d9: movzx ebx,byte ptr [ecx]
0x100261dc: sub esi,ebx
0x100261de: jne short 100261F7h
0x100261e0: sub eax,1
0x100261e3: add ecx,1
0x100261e6: add edx,1
0x100261e9: test eax,eax
0x100261eb: je short 10026205h
0x100261ed: movzx esi,byte ptr [edx]
0x100261f0: movzx ecx,byte ptr [ecx]
0x100261f3: sub esi,ecx
0x100261f5: je short 10026205h
0x100261f7: test esi,esi
0x100261f9: mov eax,1
0x100261fe: jg short 10026207h
0x10026200: or eax,0FFFFFFFFh
0x10026203: jmp short 10026207h
0x10026205: xor eax,eax
0x10026207: test eax,eax
0x10026209: jne short 10026223h
0x1002620b: mov edx,[esp+10h]
0x1002620f: mov eax,[esp+14h]
0x10026213: push edx
0x10026214: push eax
0x10026215: push ebp
0x10026216: call 1002F7E0h
0x1002621b: add esp,0Ch
0x1002621e: jmp near ptr 10025830h
0x10026223: mov eax,4
0x10026228: lea ecx,[esp+38h]
```

```
0x1002622c: mov edx,edi
0x1002622e: mov edi,edi
0x10026230: mov esi,[edx]
0x10026232: cmp esi,[ecx]
0x10026234: jne short 10026248h
0x10026236: sub eax,4
0x10026239: add ecx,4
0x1002623c: add edx,4
0x1002623f: cmp eax,4
0x10026242: jae short 10026230h
0x10026244: test eax,eax
0x10026246: je short 100262A5h
0x10026248: movzx esi,byte ptr [edx]
0x1002624b: movzx ebx,byte ptr [ecx]
0x1002624e: sub esi,ebx
0x10026250: jne short 10026297h
0x10026252: sub eax,1
0x10026255: add ecx,1
0x10026258: add edx,1
0x1002625b: test eax,eax
0x1002625d: je short 100262A5h
0x1002625f: movzx esi,byte ptr [edx]
0x10026262: movzx ebx,byte ptr [ecx]
0x10026265: sub esi,ebx
0x10026267: jne short 10026297h
0x10026269: sub eax,1
0x1002626c: add ecx,1
0x1002626f: add edx,1
0x10026272: test eax,eax
0x10026274: je short 100262A5h
0x10026276: movzx esi,byte ptr [edx]
0x10026279: movzx ebx,byte ptr [ecx]
0x1002627c: sub esi,ebx
0x1002627e: jne short 10026297h
0x10026280: sub eax,1
0x10026283: add ecx,1
0x10026286: add edx,1
0x10026289: test eax,eax
0x1002628b: je short 100262A5h
0x1002628d: movzx esi,byte ptr [edx]
0x10026290: movzx ecx,byte ptr [ecx]
0x10026293: sub esi,ecx
0x10026295: je short 100262A5h
0x10026297: test esi,esi
0x10026299: mov eax,1
0x1002629e: jg short 100262A7h
0x100262a0: or eax,0FFFFFFFFh
0x100262a3: jmp short 100262A7h
0x100262a5: xor eax,eax
0x100262a7: test eax,eax
0x100262a9: jne short 100262C3h
0x100262ab: mov edx,[esp+10h]
0x100262af: mov eax,[esp+14h]
0x100262b3: push edx
0x100262b4: push eax
0x100262b5: push ebp
0x100262b6: call 1002E590h
0x100262bb: add esp,0Ch
0x100262be: jmp near ptr 10025830h
0x100262c3: mov eax,4
0x100262c8: lea ecx,[esp+48h]
0x100262cc: mov edx,edi
0x100262ce: mov edi,edi
0x100262d0: mov esi,[edx]
0x100262d2: cmp esi,[ecx]
0x100262d4: jne short 100262E8h
0x100262d6: sub eax,4
```

```
0x100262d9: add ecx,4
0x100262dc: add edx,4
0x100262df: cmp eax,4
0x100262e2: jae short 100262D0h
0x100262e4: test eax,eax
0x100262e6: je short 10026345h
0x100262e8: movzx esi,byte ptr [edx]
0x100262eb: movzx ebx,byte ptr [ecx]
0x100262ee: sub esi,ebx
0x100262f0: jne short 10026337h
0x100262f2: sub eax,1
0x100262f5: add ecx,1
0x100262f8: add edx,1
0x100262fb: test eax,eax
0x100262fd: je short 10026345h
0x100262ff: movzx esi,byte ptr [edx]
0x10026302: movzx ebx,byte ptr [ecx]
0x10026305: sub esi,ebx
0x10026307: jne short 10026337h
0x10026309: sub eax,1
0x1002630c: add ecx,1
0x1002630f: add edx,1
0x10026312: test eax,eax
0x10026314: je short 10026345h
0x10026316: movzx esi,byte ptr [edx]
0x10026319: movzx ebx,byte ptr [ecx]
0x1002631c: sub esi,ebx
0x1002631e: jne short 10026337h
0x10026320: sub eax,1
0x10026323: add ecx,1
0x10026326: add edx,1
0x10026329: test eax,eax
0x1002632b: je short 10026345h
0x1002632d: movzx esi,byte ptr [edx]
0x10026330: movzx ecx,byte ptr [ecx]
0x10026333: sub esi,ecx
0x10026335: je short 10026345h
0x10026337: test esi,esi
0x10026339: mov eax,1
0x1002633e: jg short 10026347h
0x10026340: or eax,0FFFFFFFFh
0x10026343: jmp short 10026347h
0x10026345: xor eax,eax
0x10026347: test eax,eax
0x10026349: jne short 10026363h
0x1002634b: mov edx,[esp+10h]
0x1002634f: mov eax,[esp+14h]
0x10026353: push edx
0x10026354: push eax
0x10026355: push ebp
0x10026356: call 1002EA20h
0x1002635b: add esp,0Ch
0x1002635e: jmp near ptr 10025830h
0x10026363: mov eax,4
0x10026368: lea ecx,[esp+58h]
0x1002636c: mov edx,edi
0x1002636e: mov edi,edi
0x10026370: mov esi,[edx]
0x10026372: cmp esi,[ecx]
0x10026374: jne short 10026388h
0x10026376: sub eax,4
0x10026379: add ecx,4
0x1002637c: add edx,4
0x1002637f: cmp eax,4
0x10026382: jae short 10026370h
0x10026384: test eax,eax
0x10026386: je short 100263E5h
```

```
0x10026388: movzx esi,byte ptr [edx]
0x1002638b: movzx ebx,byte ptr [ecx]
0x1002638e: sub esi,ebx
0x10026390: jne short 100263D7h
0x10026392: sub eax,1
0x10026395: add ecx,1
0x10026398: add edx,1
0x1002639b: test eax,eax
0x1002639d: je short 100263E5h
0x1002639f: movzx esi,byte ptr [edx]
0x100263a2: movzx ebx,byte ptr [ecx]
0x100263a5: sub esi,ebx
0x100263a7: jne short 100263D7h
0x100263a9: sub eax,1
0x100263ac: add ecx,1
0x100263af: add edx,1
0x100263b2: test eax,eax
0x100263b4: je short 100263E5h
0x100263b6: movzx esi,byte ptr [edx]
0x100263b9: movzx ebx,byte ptr [ecx]
0x100263bc: sub esi,ebx
0x100263be: jne short 100263D7h
0x100263c0: sub eax,1
0x100263c3: add ecx,1
0x100263c6: add edx,1
0x100263c9: test eax,eax
0x100263cb: je short 100263E5h
0x100263cd: movzx esi,byte ptr [edx]
0x100263d0: movzx ecx,byte ptr [ecx]
0x100263d3: sub esi,ecx
0x100263d5: je short 100263E5h
0x100263d7: test esi,esi
0x100263d9: mov eax,1
0x100263de: jg short 100263E7h
0x100263e0: or eax,0FFFFFFFFh
0x100263e3: jmp short 100263E7h
0x100263e5: xor eax,eax
0x100263e7: test eax,eax
0x100263e9: jne short 10026403h
0x100263eb: mov edx,[esp+10h]
0x100263ef: mov eax,[esp+14h]
0x100263f3: push edx
0x100263f4: push eax
0x100263f5: push ebp
0x100263f6: call 1002EC10h
0x100263fb: add esp,0Ch
0x100263fe: jmp near ptr 10025830h
0x10026403: mov eax,4
0x10026408: lea ecx,[esp+68h]
0x1002640c: mov edx,edi
0x1002640e: mov edi,edi
0x10026410: mov esi,[edx]
0x10026412: cmp esi,[ecx]
0x10026414: jne short 10026428h
0x10026416: sub eax,4
0x10026419: add ecx,4
0x1002641c: add edx,4
0x1002641f: cmp eax,4
0x10026422: jae short 10026410h
0x10026424: test eax,eax
0x10026426: je short 10026485h
0x10026428: movzx esi,byte ptr [edx]
0x1002642b: movzx ebx,byte ptr [ecx]
0x1002642e: sub esi,ebx
0x10026430: jne short 10026477h
0x10026432: sub eax,1
0x10026435: add ecx,1
```

```
0x10026438: add edx,1
0x1002643b: test eax,eax
0x1002643d: je short 10026485h
0x1002643f: movzx esi,byte ptr [edx]
0x10026442: movzx ebx,byte ptr [ecx]
0x10026445: sub esi,ebx
0x10026447: jne short 10026477h
0x10026449: sub eax,1
0x1002644c: add ecx,1
0x1002644f: add edx,1
0x10026452: test eax,eax
0x10026454: je short 10026485h
0x10026456: movzx esi,byte ptr [edx]
0x10026459: movzx ebx,byte ptr [ecx]
0x1002645c: sub esi,ebx
0x1002645e: jne short 10026477h
0x10026460: sub eax,1
0x10026463: add ecx,1
0x10026466: add edx,1
0x10026469: test eax,eax
0x1002646b: je short 10026485h
0x1002646d: movzx esi,byte ptr [edx]
0x10026470: movzx ecx,byte ptr [ecx]
0x10026473: sub esi,ecx
0x10026475: je short 10026485h
0x10026477: test esi,esi
0x10026479: mov eax,1
0x1002647e: jg short 10026487h
0x10026480: or eax,0FFFFFFFFh
0x10026483: jmp short 10026487h
0x10026485: xor eax,eax
0x10026487: test eax,eax
0x10026489: jne short 100264A3h
0x1002648b: mov edx,[esp+10h]
0x1002648f: mov eax,[esp+14h]
0x10026493: push edx
0x10026494: push eax
0x10026495: push ebp
0x10026496: call 1002EE90h
0x1002649b: add esp,0Ch
0x1002649e: jmp near ptr 10025830h
0x100264a3: mov eax,4
0x100264a8: lea ecx,[esp+78h]
0x100264ac: mov edx,edi
0x100264ae: mov edi,edi
0x100264b0: mov esi,[edx]
0x100264b2: cmp esi,[ecx]
0x100264b4: jne short 100264C8h
0x100264b6: sub eax,4
0x100264b9: add ecx,4
0x100264bc: add edx,4
0x100264bf: cmp eax,4
0x100264c2: jae short 100264B0h
0x100264c4: test eax,eax
0x100264c6: je short 10026525h
0x100264c8: movzx esi,byte ptr [edx]
0x100264cb: movzx ebx,byte ptr [ecx]
0x100264ce: sub esi,ebx
0x100264d0: jne short 10026517h
0x100264d2: sub eax,1
0x100264d5: add ecx,1
0x100264d8: add edx,1
0x100264db: test eax,eax
0x100264dd: je short 10026525h
0x100264df: movzx esi,byte ptr [edx]
0x100264e2: movzx ebx,byte ptr [ecx]
0x100264e5: sub esi,ebx
```

```
0x100264e7: jne short 10026517h
0x100264e9: sub eax,1
0x100264ec: add ecx,1
0x100264ef: add edx,1
0x100264f2: test eax,eax
0x100264f4: je short 10026525h
0x100264f6: movzx esi,byte ptr [edx]
0x100264f9: movzx ebx,byte ptr [ecx]
0x100264fc: sub esi,ebx
0x100264fe: jne short 10026517h
0x10026500: sub eax,1
0x10026503: add ecx,1
0x10026506: add edx,1
0x10026509: test eax,eax
0x1002650b: je short 10026525h
0x1002650d: movzx esi,byte ptr [edx]
0x10026510: movzx ecx,byte ptr [ecx]
0x10026513: sub esi,ecx
0x10026515: je short 10026525h
0x10026517: test esi,esi
0x10026519: mov eax,1
0x1002651e: jg short 10026527h
0x10026520: or eax,0FFFFFFFFh
0x10026523: jmp short 10026527h
0x10026525: xor eax,eax
0x10026527: test eax,eax
0x10026529: jne short 10026543h
0x1002652b: mov edx,[esp+10h]
0x1002652f: mov eax,[esp+14h]
0x10026533: push edx
0x10026534: push eax
0x10026535: push ebp
0x10026536: call 100300D0h
0x1002653b: add esp,0Ch
0x1002653e: jmp near ptr 10025830h
0x10026543: mov eax,4
0x10026548: lea ecx,[esp+88h]
0x1002654f: mov edx,edi
0x10026551: mov esi,[edx]
0x10026553: cmp esi,[ecx]
0x10026555: jne short 10026569h
0x10026557: sub eax,4
0x1002655a: add ecx,4
0x1002655d: add edx,4
0x10026560: cmp eax,4
0x10026563: jae short 10026551h
0x10026565: test eax,eax
0x10026567: je short 100265C6h
0x10026569: movzx esi,byte ptr [edx]
0x1002656c: movzx ebx,byte ptr [ecx]
0x1002656f: sub esi,ebx
0x10026571: jne short 100265B8h
0x10026573: sub eax,1
0x10026576: add ecx,1
0x10026579: add edx,1
0x1002657c: test eax,eax
0x1002657e: je short 100265C6h
0x10026580: movzx esi,byte ptr [edx]
0x10026583: movzx ebx,byte ptr [ecx]
0x10026586: sub esi,ebx
0x10026588: jne short 100265B8h
0x1002658a: sub eax,1
0x1002658d: add ecx,1
0x10026590: add edx,1
0x10026593: test eax,eax
0x10026595: je short 100265C6h
0x10026597: movzx esi,byte ptr [edx]
```

```
0x1002659a: movzx ebx,byte ptr [ecx]
0x1002659d: sub esi,ebx
0x1002659f: jne short 100265B8h
0x100265a1: sub eax,1
0x100265a4: add ecx,1
0x100265a7: add edx,1
0x100265aa: test eax,eax
0x100265ac: je short 100265C6h
0x100265ae: movzx esi,byte ptr [edx]
0x100265b1: movzx ecx,byte ptr [ecx]
0x100265b4: sub esi,ecx
0x100265b6: je short 100265C6h
0x100265b8: test esi,esi
0x100265ba: mov eax,1
0x100265bf: jg short 100265C8h
0x100265c1: or eax,0FFFFFFFFh
0x100265c4: jmp short 100265C8h
0x100265c6: xor eax,eax
0x100265c8: test eax,eax
0x100265ca: jne short 100265E4h
0x100265cc: mov edx,[esp+10h]
0x100265d0: mov eax,[esp+14h]
0x100265d4: push edx
0x100265d5: push eax
0x100265d6: push ebp
0x100265d7: call 1002FFB0h
0x100265dc: add esp,0Ch
0x100265df: jmp near ptr 10025830h
0x100265e4: mov eax,4
0x100265e9: lea ecx,[esp+98h]
0x100265f0: mov edx,edi
0x100265f2: mov esi,[edx]
0x100265f4: cmp esi,[ecx]
0x100265f6: jne short 1002660Ah
0x100265f8: sub eax,4
0x100265fb: add ecx,4
0x100265fe: add edx,4
0x10026601: cmp eax,4
0x10026604: jae short 100265F2h
0x10026606: test eax,eax
0x10026608: je short 10026667h
0x1002660a: movzx esi,byte ptr [edx]
0x1002660d: movzx ebx,byte ptr [ecx]
0x10026610: sub esi,ebx
0x10026612: jne short 10026659h
0x10026614: sub eax,1
0x10026617: add ecx,1
0x1002661a: add edx,1
0x1002661d: test eax,eax
0x1002661f: je short 10026667h
0x10026621: movzx esi,byte ptr [edx]
0x10026624: movzx ebx,byte ptr [ecx]
0x10026627: sub esi,ebx
0x10026629: jne short 10026659h
0x1002662b: sub eax,1
0x1002662e: add ecx,1
0x10026631: add edx,1
0x10026634: test eax,eax
0x10026636: je short 10026667h
0x10026638: movzx esi,byte ptr [edx]
0x1002663b: movzx ebx,byte ptr [ecx]
0x1002663e: sub esi,ebx
0x10026640: jne short 10026659h
0x10026642: sub eax,1
0x10026645: add ecx,1
0x10026648: add edx,1
0x1002664b: test eax,eax
```

```
0x1002664d: je short 10026667h
0x1002664f: movzx esi,byte ptr [edx]
0x10026652: movzx ecx,byte ptr [ecx]
0x10026655: sub esi,ecx
0x10026657: je short 10026667h
0x10026659: test esi,esi
0x1002665b: mov eax,1
0x10026660: jg short 10026669h
0x10026662: or eax,0FFFFFFFFh
0x10026665: jmp short 10026669h
0x10026667: xor eax,eax
0x10026669: test eax,eax
0x1002666b: jne short 10026685h
0x1002666d: mov edx,[esp+10h]
0x10026671: mov eax,[esp+14h]
0x10026675: push edx
0x10026676: push eax
0x10026677: push ebp
0x10026678: call 1002F170h
0x1002667d: add esp,0Ch
0x10026680: jmp near ptr 10025830h
0x10026685: mov eax,4
0x1002668a: lea ecx,[esp+0A8h]
0x10026691: mov edx,edi
0x10026693: mov esi,[edx]
0x10026695: cmp esi,[ecx]
0x10026697: jne short 100266ABh
0x10026699: sub eax,4
0x1002669c: add ecx,4
0x1002669f: add edx,4
0x100266a2: cmp eax,4
0x100266a5: jae short 10026693h
0x100266a7: test eax,eax
0x100266a9: je short 10026708h
0x100266ab: movzx esi,byte ptr [edx]
0x100266ae: movzx ebx,byte ptr [ecx]
0x100266b1: sub esi,ebx
0x100266b3: jne short 100266FAh
0x100266b5: sub eax,1
0x100266b8: add ecx,1
0x100266bb: add edx,1
0x100266be: test eax,eax
0x100266c0: je short 10026708h
0x100266c2: movzx esi,byte ptr [edx]
0x100266c5: movzx ebx,byte ptr [ecx]
0x100266c8: sub esi,ebx
0x100266ca: jne short 100266FAh
0x100266cc: sub eax,1
0x100266cf: add ecx,1
0x100266d2: add edx,1
0x100266d5: test eax,eax
0x100266d7: je short 10026708h
0x100266d9: movzx esi,byte ptr [edx]
0x100266dc: movzx ebx,byte ptr [ecx]
0x100266df: sub esi,ebx
0x100266e1: jne short 100266FAh
0x100266e3: sub eax,1
0x100266e6: add ecx,1
0x100266e9: add edx,1
0x100266ec: test eax,eax
0x100266ee: je short 10026708h
0x100266f0: movzx esi,byte ptr [edx]
0x100266f3: movzx ecx,byte ptr [ecx]
0x100266f6: sub esi,ecx
0x100266f8: je short 10026708h
0x100266fa: test esi,esi
0x100266fc: mov eax,1
```

```
0x10026701: jg short 1002670Ah
0x10026703: or eax,0FFFFFFFFh
0x10026706: jmp short 1002670Ah
0x10026708: xor eax,eax
0x1002670a: test eax,eax
0x1002670c: jne short 10026726h
0x1002670e: mov edx,[esp+10h]
0x10026712: mov eax,[esp+14h]
0x10026716: push edx
0x10026717: push eax
0x10026718: push ebp
0x10026719: call 10030280h
0x1002671e: add esp,0Ch
0x10026721: jmp near ptr 10025830h
0x10026726: mov eax,4
0x1002672b: lea ecx,[esp+0B8h]
0x10026732: mov edx,edi
0x10026734: mov esi,[edx]
0x10026736: cmp esi,[ecx]
0x10026738: jne short 1002674Ch
0x1002673a: sub eax,4
0x1002673d: add ecx,4
0x10026740: add edx,4
0x10026743: cmp eax,4
0x10026746: jae short 10026734h
0x10026748: test eax,eax
0x1002674a: je short 100267A9h
0x1002674c: movzx esi,byte ptr [edx]
0x1002674f: movzx ebx,byte ptr [ecx]
0x10026752: sub esi,ebx
0x10026754: jne short 1002679Bh
0x10026756: sub eax,1
0x10026759: add ecx,1
0x1002675c: add edx,1
0x1002675f: test eax,eax
0x10026761: je short 100267A9h
0x10026763: movzx esi,byte ptr [edx]
0x10026766: movzx ebx,byte ptr [ecx]
0x10026769: sub esi,ebx
0x1002676b: jne short 1002679Bh
0x1002676d: sub eax,1
0x10026770: add ecx,1
0x10026773: add edx,1
0x10026776: test eax,eax
0x10026778: je short 100267A9h
0x1002677a: movzx esi,byte ptr [edx]
0x1002677d: movzx ebx,byte ptr [ecx]
0x10026780: sub esi,ebx
0x10026782: jne short 1002679Bh
0x10026784: sub eax,1
0x10026787: add ecx,1
0x1002678a: add edx,1
0x1002678d: test eax,eax
0x1002678f: je short 100267A9h
0x10026791: movzx esi,byte ptr [edx]
0x10026794: movzx ecx,byte ptr [ecx]
0x10026797: sub esi,ecx
0x10026799: je short 100267A9h
0x1002679b: test esi,esi
0x1002679d: mov eax,1
0x100267a2: jg short 100267ABh
0x100267a4: or eax,0FFFFFFFFh
0x100267a7: jmp short 100267ABh
0x100267a9: xor eax,eax
0x100267ab: test eax,eax
0x100267ad: jne short 100267C7h
0x100267af: mov edx,[esp+10h]
```

```
0x100267b3: mov eax,[esp+14h]
0x100267b7: push edx
0x100267b8: push eax
0x100267b9: push ebp
0x100267ba: call 10030490h
0x100267bf: add esp,0Ch
0x100267c2: jmp near ptr 10025830h
0x100267c7: mov ecx,[esp+10h]
0x100267cb: mov edx,[esp+14h]
0x100267cf: push ecx
0x100267d0: push edx
0x100267d1: push ebp
0x100267d2: call 10030750h
0x100267d7: add esp,0Ch
0x100267da: jmp near ptr 10025830h
0x100267df: cmp byte ptr [ebp+126h],3
0x100267e6: jne short 10026831h
0x100267e8: test al,2
0x100267ea: jne short 10026831h
0x100267ec: push 100FFD54h
0x100267f1: push ebp
0x100267f2: call 100231E0h
0x100267f7: mov eax,[ebp+6Ch]
0x100267fa: test al,1
0x100267fc: jne short 10026809h
0x100267fe: push 100FFD70h
0x10026803: push ebp
0x10026804: call 100231E0h
0x10026809: cmp byte ptr [ebp+126h],3
0x10026810: jne short 10026821h
0x10026812: test al,2
0x10026814: jne short 10026821h
0x10026816: push 100FFD54h
0x1002681b: push ebp
0x1002681c: call 100231E0h
0x10026821: mov ecx,[esp+10h]
0x10026825: or eax,4
0x10026828: mov [ebp+10Ch],ecx
0x1002682e: mov [ebp+6Ch],eax
0x10026831: pop esi
0x10026832: pop edi
0x10026833: pop ebx
0x10026834: mov ecx,[esp+0B4h]
0x1002683b: pop ebp
0x1002683c: xor ecx,esp
0x1002683e: call 10022326h
0x10026843: add esp,0B4h
0x10026849: ret
```

`png_read_info` is called by sub_10018d20, png_read_png. It works with the text "Missing PLTE before IDAT", "Missing IHDR before IDAT". It delegates to sub_10022326, png_error, png_handle_as_unknown (26 callees).

12.569 png_read_update_info at 0x10026850 (27 instructions)

```
0x10026850: push esi
0x10026851: mov esi,[esp+8]
0x10026855: test esi,esi
0x10026857: je short 10026894h
0x10026859: test byte ptr [esi+70h],40h
0x1002685d: jne short 10026878h
0x1002685f: push esi
0x10026860: call 1002DCC0h
0x10026865: mov eax,[esp+10h]
0x10026869: add esp,4
0x1002686c: push eax
0x1002686d: push esi
```

```
0x1002686e: call 10032400h
0x10026873: add esp,8
0x10026876: pop esi
0x10026877: ret
0x10026878: push 100FFD90h
0x1002687d: push esi
0x1002687e: call 100230F0h
0x10026883: mov eax,[esp+14h]
0x10026887: add esp,8
0x1002688a: push eax
0x1002688b: push esi
0x1002688c: call 10032400h
0x10026891: add esp,8
0x10026894: pop esi
0x10026895: ret
```

`png_read_update_info` is called by png_read_png. It delegates to png_warning, sub_1002dcc0, sub_10032400 (3 callees).

12.570 png_start_read_image at 0x100268a0 (244 instructions)

```
0x100268a0: mov eax,[esp+4]
0x100268a4: test eax,eax
0x100268a6: je short 100268C5h
0x100268a8: test byte ptr [eax+70h],40h
0x100268ac: jne short 100268B7h
0x100268ae: mov [esp+4],eax
0x100268b2: jmp near ptr 1002DCC0h
0x100268b7: push 100FFDD8h
0x100268bc: push eax
0x100268bd: call 100230F0h
0x100268c2: add esp,8
0x100268c5: ret
0x1002dcc0: sub esp,38h
0x1002dcc3: push ebp
0x1002dcc4: push esi
0x1002dcc5: mov esi,1
0x1002dcca: xor edx,edx
0x1002dccc: mov eax,4
0x1002dcd1: mov ecx,2
0x1002dcd6: push edi
0x1002dcd7: mov [esp+3Ch],esi
0x1002dcdb: mov [esp+24h],esi
0x1002dcdf: mov esi,[esp+48h]
0x1002dce3: mov edi,8
0x1002dce8: push esi
0x1002dce9: mov [esp+2Ch],edx
0x1002dced: mov [esp+30h],eax
0x1002dcf1: mov [esp+34h],edx
0x1002dcf5: mov [esp+38h],ecx
0x1002dcf9: mov [esp+3Ch],edx
0x1002dcfd: mov [esp+44h],edx
0x1002dd01: mov [esp+10h],edi
0x1002dd05: mov [esp+14h],edi
0x1002dd09: mov [esp+18h],eax
0x1002dd0d: mov [esp+1Ch],eax
0x1002dd11: mov [esp+20h],ecx
0x1002dd15: mov [esp+24h],ecx
0x1002dd19: mov [esi+7Ch],edx
0x1002dd1c: call 10031A20h
0x1002dd21: add esp,4
0x1002dd24: cmp byte ptr [esi+123h],0
0x1002dd2b: je short 1002DD7Eh
0x1002dd2d: test byte ptr [esi+74h],2
0x1002dd31: jne short 1002DD47h
0x1002dd33: mov eax,[esi+0D0h]
```

```
0x1002dd39: add eax,7
0x1002dd3c: shr eax,3
0x1002dd3f: mov [esi+0D4h],eax
0x1002dd45: jmp short 1002DD53h
0x1002dd47: mov ecx,[esi+0D0h]
0x1002dd4d: mov [esi+0D4h],ecx
0x1002dd53: movzx eax,byte ptr [esi+124h]
0x1002dd5a: mov ebp,[esi+0CCh]
0x1002dd60: add eax,eax
0x1002dd62: add eax,eax
0x1002dd64: mov ecx,[esp+eax+0Ch]
0x1002dd68: mov edx,ebp
0x1002dd6a: sub edx,[esp+eax+28h]
0x1002dd6e: lea eax,[edx+ecx-1]
0x1002dd72: xor edx,edx
0x1002dd74: div ecx
0x1002dd76: mov [esi+0E0h],eax
0x1002dd7c: jmp short 1002DD96h
0x1002dd7e: mov eax,[esi+0D0h]
0x1002dd84: mov ebp,[esi+0CCh]
0x1002dd8a: mov [esi+0D4h],eax
0x1002dd90: mov [esi+0E0h],ebp
0x1002dd96: movzx eax,byte ptr [esi+129h]
0x1002dd9d: push ebx
0x1002dd9e: mov ebx,[esi+74h]
0x1002dda1: test bl,4
0x1002dda4: je short 1002DDB1h
0x1002dda6: cmp byte ptr [esi+127h],8
0x1002ddad: jae short 1002DDB1h
0x1002ddaf: mov eax,edi
0x1002ddb1: mov edi,ebx
0x1002ddb3: and edi,1000h
0x1002ddb9: je short 1002DE1Dh
0x1002ddbb: mov cl,[esi+126h]
0x1002ddc1: cmp cl,3
0x1002ddc4: jne short 1002DDDBh
0x1002ddc6: xor eax,eax
0x1002ddc8: cmp [esi+11Ah],ax
0x1002ddcf: setne al
0x1002ddd2: lea eax,[eax*8+18h]
0x1002ddd9: jmp short 1002DE1Dh
0x1002dddb: test cl,cl
0x1002dddd: jne short 1002DDF7h
0x1002dddf: cmp eax,8
0x1002dde2: jge short 1002DDE9h
0x1002dde4: mov eax,8
0x1002dde9: cmp word ptr [esi+11Ah],0
0x1002ddf1: je short 1002DE1Dh
0x1002ddf3: add eax,eax
0x1002ddf5: jmp short 1002DE1Dh
0x1002ddf7: cmp cl,2
0x1002ddfa: jne short 1002DE1Dh
0x1002ddfc: cmp word ptr [esi+11Ah],0
0x1002de04: je short 1002DE1Dh
0x1002de06: lea ecx,[eax*4]
0x1002de0d: mov eax,55555556h
0x1002de12: imul ecx
0x1002de14: mov ecx,edx
0x1002de16: shr ecx,1Fh
0x1002de19: add ecx,edx
0x1002de1b: mov eax,ecx
0x1002de1d: mov edx,ebx
0x1002de1f: and edx,8000h
0x1002de25: je short 1002DE68h
0x1002de27: mov cl,[esi+126h]
0x1002de2d: cmp cl,3
0x1002de30: jne short 1002DE39h
```

```
0x1002de32: mov eax,20h
0x1002de37: jmp short 1002DE68h
0x1002de39: test cl,cl
0x1002de3b: jne short 1002DE50h
0x1002de3d: xor ecx,ecx
0x1002de3f: cmp eax,8
0x1002de42: setg cl
0x1002de45: sub ecx,1
0x1002de48: and ecx,0FFFFFFF0h
0x1002de4b: add ecx,20h
0x1002de4e: jmp short 1002DE66h
0x1002de50: cmp cl,2
0x1002de53: jne short 1002DE68h
0x1002de55: xor ecx,ecx
0x1002de57: cmp eax,20h
0x1002de5a: setg cl
0x1002de5d: sub ecx,1
0x1002de60: and ecx,0FFFFFFE0h
0x1002de63: add ecx,40h
0x1002de66: mov eax,ecx
0x1002de68: test ebx,4000h
0x1002de6e: je short 1002DEC7h
0x1002de70: cmp word ptr [esi+11Ah],0
0x1002de78: je short 1002DE7Eh
0x1002de7a: test edi,edi
0x1002de7c: jne short 1002DEB4h
0x1002de7e: test edx,edx
0x1002de80: jne short 1002DEB4h
0x1002de82: mov cl,[esi+126h]
0x1002de88: cmp cl,4
0x1002de8b: je short 1002DEB4h
0x1002de8d: cmp eax,8
0x1002de90: jg short 1002DEA3h
0x1002de92: xor eax,eax
0x1002de94: cmp cl,6
0x1002de97: sete al
0x1002de9a: lea eax,[eax*8+18h]
0x1002dea1: jmp short 1002DEC7h
0x1002dea3: sub cl,6
0x1002dea6: neg cl
0x1002dea8: sbb ecx,ecx
0x1002deaa: and ecx,0FFFFFFF0h
0x1002dead: add ecx,40h
0x1002deb0: mov eax,ecx
0x1002deb2: jmp short 1002DEC7h
0x1002deb4: xor edx,edx
0x1002deb6: cmp eax,10h
0x1002deb9: setg dl
0x1002debc: sub edx,1
0x1002debf: and edx,0FFFFFFE0h
0x1002dec2: add edx,40h
0x1002dec5: mov eax,edx
0x1002dec7: test ebx,100000h
0x1002decd: pop ebx
0x1002dece: je short 1002DEE1h
0x1002ded0: movzx ecx,byte ptr [esi+69h]
0x1002ded4: movzx edx,byte ptr [esi+68h]
0x1002ded8: imul ecx,edx
0x1002dedb: cmp ecx,eax
0x1002dedd: jle short 1002DEE1h
0x1002dedf: mov eax,ecx
0x1002dee1: lea ecx,[ebp+7]
0x1002dee4: and ecx,0FFFFFFF8h
0x1002dee7: cmp eax,8
0x1002deea: jl short 1002DEF6h
0x1002deec: mov edx,eax
0x1002deee: shr edx,3
```

```
0x1002def1: imul edx,ecx
0x1002def4: jmp short 1002DF01h
0x1002def6: imul ecx,eax
0x1002def9: add ecx,7
0x1002defc: shr ecx,3
0x1002deff: mov edx,ecx
0x1002df01: add eax,7
0x1002df04: sar eax,3
0x1002df07: lea eax,[eax+edx+1]
0x1002df0b: lea edi,[eax+30h]
0x1002df0e: cmp edi,[esi+27Ch]
0x1002df14: jbe short 1002DF57h
0x1002df16: mov eax,[esi+244h]
0x1002df1c: push eax
0x1002df1d: push esi
0x1002df1e: call 1002CD70h
0x1002df23: add esp,8
0x1002df26: cmp byte ptr [esi+123h],0
0x1002df2d: push edi
0x1002df2e: push esi
0x1002df2f: je short 1002DF38h
0x1002df31: call 1002CDD0h
0x1002df36: jmp short 1002DF3Dh
0x1002df38: call 1002CD20h
0x1002df3d: mov ecx,eax
0x1002df3f: add esp,8
0x1002df42: add ecx,20h
0x1002df45: mov [esi+244h],eax
0x1002df4b: mov [esi+0ECh],ecx
0x1002df51: mov [esi+27Ch],edi
0x1002df57: mov eax,[esi+0DCh]
0x1002df5d: cmp eax,0FFFFFFFEh
0x1002df60: jbe short 1002DF6Dh
0x1002df62: push 10100B34h
0x1002df67: push esi
0x1002df68: call 100231E0h
0x1002df6d: add eax,1
0x1002df70: cmp eax,[esi+280h]
0x1002df76: jbe short 1002DFADh
0x1002df78: mov edx,[esi+0E8h]
0x1002df7e: push edx
0x1002df7f: push esi
0x1002df80: call 1002CD70h
0x1002df85: mov eax,[esi+0DCh]
0x1002df8b: add eax,1
0x1002df8e: push eax
0x1002df8f: push esi
0x1002df90: call 1002CD20h
0x1002df95: mov [esi+0E8h],eax
0x1002df9b: mov eax,[esi+0DCh]
0x1002dfa1: add esp,10h
0x1002dfa4: add eax,1
0x1002dfa7: mov [esi+280h],eax
0x1002dfad: mov ecx,[esi+0E8h]
0x1002dfb3: push eax
0x1002dfb4: push 0
0x1002dfb6: push ecx
0x1002dfb7: call 100228C0h
0x1002dfbc: add esp,0Ch
0x1002dfbf: or dword ptr [esi+70h],40h
0x1002dfc3: pop edi
0x1002dfc4: pop esi
0x1002dfc5: pop ebp
0x1002dfc6: add esp,38h
0x1002dfc9: ret
```

`png_start_read_image` is called by png_read_image. It works with the text "Row has too many bytes to allocate in memory". It accesses memory-mapped peripheral registers at 0x55555556. It delegates to j_memset, png_warning, png_error (7 callees).

12.571 png_read_row at 0x100268d0 (326 instructions)

```
0x100268d0: sub esp,4Ch
0x100268d3: mov eax,ds:[10136220h]
0x100268d8: xor eax,esp
0x100268da: mov [esp+48h],eax
0x100268de: mov eax,[esp+54h]
0x100268e2: push ebx
0x100268e3: mov ebx,[esp+5Ch]
0x100268e7: push ebp
0x100268e8: push esi
0x100268e9: mov esi,[esp+5Ch]
0x100268ed: test esi,esi
0x100268ef: mov ebp,0FFh
0x100268f4: push edi
0x100268f5: mov edi,55h
0x100268fa: mov [esp+10h],eax
0x100268fe: mov [esp+14h],ebx
0x10026902: mov byte ptr [esp+50h],49h
0x10026907: mov byte ptr [esp+51h],44h
0x1002690c: mov byte ptr [esp+52h],41h
0x10026911: mov byte ptr [esp+53h],54h
0x10026916: mov byte ptr [esp+54h],0
0x1002691b: mov [esp+34h],ebp
0x1002691f: mov dword ptr [esp+38h],0Fh
0x10026927: mov [esp+3Ch],ebp
0x1002692b: mov dword ptr [esp+40h],33h
0x10026933: mov [esp+44h],ebp
0x10026937: mov [esp+48h],edi
0x1002693b: mov [esp+4Ch],ebp
0x1002693f: mov dword ptr [esp+18h],80h
0x10026947: mov dword ptr [esp+1Ch],8
0x1002694f: mov dword ptr [esp+20h],88h
0x10026957: mov dword ptr [esp+24h],22h
0x1002695f: mov dword ptr [esp+28h],0AAh
0x10026967: mov [esp+2Ch],edi
0x1002696b: mov [esp+30h],ebp
0x1002696f: je near ptr 10026DE2h
0x10026975: test byte ptr [esi+70h],40h
0x10026979: jne short 10026984h
0x1002697b: push esi
0x1002697c: call 1002DCC0h
0x10026981: add esp,4
0x10026984: cmp byte ptr [esi+123h],0
0x1002698b: je near ptr 10026A89h
0x10026991: test byte ptr [esi+74h],2
0x10026995: je near ptr 10026A89h
0x1002699b: movzx eax,byte ptr [esi+124h]
0x100269a2: cmp eax,5
0x100269a5: ja near ptr 10026A7Ch
0x100269ab: jmp dword ptr [eax*4+10026DF8h]
0x100269ce: push esi
0x100269cf: call 100309C0h
0x100269d4: add esp,4
0x100269d7: pop edi
0x100269d8: pop esi
0x100269d9: pop ebp
0x100269da: pop ebx
0x100269db: mov ecx,[esp+48h]
0x100269df: xor ecx,esp
0x100269e1: call 10022326h
0x100269e6: add esp,4Ch
0x100269e9: ret
```

```
0x10026a7c: test byte ptr [esi+0E4h],1
0x10026a83: je near ptr 100269CEh
0x10026a89: test byte ptr [esi+6Ch],4
0x10026a8d: jne short 10026A9Ah
0x10026a8f: push 100FFE64h
0x10026a94: push esi
0x10026a95: call 100231E0h
0x10026a9a: mov eax,[esi+0ECh]
0x10026aa0: mov [esi+84h],eax
0x10026aa6: mov al,[esi+129h]
0x10026aac: cmp al,8
0x10026aae: movzx eax,al
0x10026ab1: jb short 10026ABFh
0x10026ab3: shr eax,3
0x10026ab6: imul eax,[esi+0E0h]
0x10026abd: jmp short 10026ACCh
0x10026abf: imul eax,[esi+0E0h]
0x10026ac6: add eax,7
0x10026ac9: shr eax,3
0x10026acc: add eax,1
0x10026acf: mov [esi+88h],eax
0x10026ad5: cmp dword ptr [esi+7Ch],0
0x10026ad9: jne near ptr 10026BE2h
0x10026adf: cmp dword ptr [esi+10Ch],0
0x10026ae6: jne near ptr 10026BACh
0x10026aec: lea ebp,[esi+11Ch]
0x10026af2: push 0
0x10026af4: push esi
0x10026af5: call 1002E050h
0x10026afa: push esi
0x10026afb: call 1002DFD0h
0x10026b00: mov [esi+10Ch],eax
0x10026b06: add esp,0Ch
0x10026b09: mov eax,4
0x10026b0e: lea ecx,[esp+50h]
0x10026b12: mov edx,ebp
0x10026b14: mov edi,[edx]
0x10026b16: cmp edi,[ecx]
0x10026b18: jne short 10026B2Ch
0x10026b1a: sub eax,4
0x10026b1d: add ecx,4
0x10026b20: add edx,4
0x10026b23: cmp eax,4
0x10026b26: jae short 10026B14h
0x10026b28: test eax,eax
0x10026b2a: je short 10026B91h
0x10026b2c: movzx edi,byte ptr [edx]
0x10026b2f: movzx ebx,byte ptr [ecx]
0x10026b32: sub edi,ebx
0x10026b34: jne short 10026B7Bh
0x10026b36: sub eax,1
0x10026b39: add ecx,1
0x10026b3c: add edx,1
0x10026b3f: test eax,eax
0x10026b41: je short 10026B8Dh
0x10026b43: movzx edi,byte ptr [edx]
0x10026b46: movzx ebx,byte ptr [ecx]
0x10026b49: sub edi,ebx
0x10026b4b: jne short 10026B7Bh
0x10026b4d: sub eax,1
0x10026b50: add ecx,1
0x10026b53: add edx,1
0x10026b56: test eax,eax
0x10026b58: je short 10026B8Dh
0x10026b5a: movzx edi,byte ptr [edx]
0x10026b5d: movzx ebx,byte ptr [ecx]
0x10026b60: sub edi,ebx
```

```
0x10026b62: jne short 10026B7Bh
0x10026b64: sub eax,1
0x10026b67: add ecx,1
0x10026b6a: add edx,1
0x10026b6d: test eax,eax
0x10026b6f: je short 10026B8Dh
0x10026b71: movzx edi,byte ptr [edx]
0x10026b74: movzx ecx,byte ptr [ecx]
0x10026b77: sub edi,ecx
0x10026b79: je short 10026B8Dh
0x10026b7b: test edi,edi
0x10026b7d: mov ebx,[esp+14h]
0x10026b81: mov eax,1
0x10026b86: jg short 10026B93h
0x10026b88: or eax,0FFFFFFFFh
0x10026b8b: jmp short 10026B93h
0x10026b8d: mov ebx,[esp+14h]
0x10026b91: xor eax,eax
0x10026b93: test eax,eax
0x10026b95: jne near ptr 10026C58h
0x10026b9b: cmp [esi+10Ch],eax
0x10026ba1: je near ptr 10026AF2h
0x10026ba7: mov ebp,0FFh
0x10026bac: mov eax,[esi+0B4h]
0x10026bb2: mov ecx,[esi+10Ch]
0x10026bb8: cmp eax,ecx
0x10026bba: mov edx,[esi+0B0h]
0x10026bc0: mov [esi+7Ch],eax
0x10026bc3: mov [esi+78h],edx
0x10026bc6: jbe short 10026BCBh
0x10026bc8: mov [esi+7Ch],ecx
0x10026bcb: mov eax,[esi+7Ch]
0x10026bce: push eax
0x10026bcf: push edx
0x10026bd0: push esi
0x10026bd1: call 1002CF30h
0x10026bd6: mov ecx,[esi+7Ch]
0x10026bd9: add esp,0Ch
0x10026bdc: sub [esi+10Ch],ecx
0x10026be2: lea edx,[esi+78h]
0x10026be5: push 1
0x10026be7: push edx
0x10026be8: call 10060100h
0x10026bed: add esp,8
0x10026bf0: cmp eax,1
0x10026bf3: je short 10026C63h
0x10026bf5: test eax,eax
0x10026bf7: jne near ptr 10026C96h
0x10026bfd: cmp [esi+88h],eax
0x10026c03: jne near ptr 10026AD5h
0x10026c09: movzx eax,byte ptr [esi+126h]
0x10026c10: mov ecx,[esi+0E0h]
0x10026c16: mov dl,[esi+12Ah]
0x10026c1c: mov [esi+108h],al
0x10026c22: movzx eax,byte ptr [esi+127h]
0x10026c29: mov [esi+109h],al
0x10026c2f: mov al,[esi+129h]
0x10026c35: lea edi,[esi+100h]
0x10026c3b: mov [esi+10Bh],al
0x10026c41: cmp al,8
0x10026c43: mov [edi],ecx
0x10026c45: mov [esi+10Ah],dl
0x10026c4b: movzx eax,al
0x10026c4e: jb short 10026CACh
0x10026c50: shr eax,3
0x10026c53: imul eax,ecx
0x10026c56: jmp short 10026CB5h
```

```
0x10026c58: push 100FFE4Ch
0x10026c5d: push esi
0x10026c5e: call 100231E0h
0x10026c63: cmp dword ptr [esi+88h],0
0x10026c6a: jne short 10026C7Bh
0x10026c6c: cmp dword ptr [esi+7Ch],0
0x10026c70: jne short 10026C7Bh
0x10026c72: cmp dword ptr [esi+10Ch],0
0x10026c79: je short 10026C89h
0x10026c7b: push 100FFE34h
0x10026c80: push esi
0x10026c81: call 10023210h
0x10026c86: add esp,8
0x10026c89: or dword ptr [esi+6Ch],8
0x10026c8d: or dword ptr [esi+70h],20h
0x10026c91: jmp near ptr 10026C09h
0x10026c96: mov eax,[esi+90h]
0x10026c9c: test eax,eax
0x10026c9e: jne short 10026CA5h
0x10026ca0: mov eax,100FFE20h
0x10026ca5: push eax
0x10026ca6: push esi
0x10026ca7: call 100231E0h
0x10026cac: imul eax,ecx
0x10026caf: add eax,7
0x10026cb2: shr eax,3
0x10026cb5: mov [esi+104h],eax
0x10026cbb: mov eax,[esi+0ECh]
0x10026cc1: mov cl,[eax]
0x10026cc3: test cl,cl
0x10026cc5: je short 10026CE3h
0x10026cc7: mov edx,[esi+0E8h]
0x10026ccd: movzx ecx,cl
0x10026cd0: push ecx
0x10026cd1: add edx,1
0x10026cd4: push edx
0x10026cd5: add eax,1
0x10026cd8: push eax
0x10026cd9: push edi
0x10026cda: push esi
0x10026cdb: call 1002DAE0h
0x10026ce0: add esp,14h
0x10026ce3: mov eax,[esi+0DCh]
0x10026ce9: mov ecx,[esi+0ECh]
0x10026cef: mov edx,[esi+0E8h]
0x10026cf5: add eax,1
0x10026cf8: push eax
0x10026cf9: push ecx
0x10026cfa: push edx
0x10026cfb: call 100228C6h
0x10026d00: add esp,0Ch
0x10026d03: test byte ptr [esi+230h],4
0x10026d0a: je short 10026D28h
0x10026d0c: cmp byte ptr [esi+234h],40h
0x10026d13: jne short 10026D28h
0x10026d15: mov eax,[esi+0ECh]
0x10026d1b: add eax,1
0x10026d1e: push eax
0x10026d1f: push edi
0x10026d20: call 100353E0h
0x10026d25: add esp,8
0x10026d28: cmp dword ptr [esi+74h],0
0x10026d2c: jne short 10026D37h
0x10026d2e: test dword ptr [esi+70h],400000h
0x10026d35: je short 10026D40h
0x10026d37: push esi
0x10026d38: call 100354B0h
```

```
0x10026d3d: add esp,4
0x10026d40: cmp byte ptr [esi+123h],0
0x10026d47: je short 10026D95h
0x10026d49: test byte ptr [esi+74h],2
0x10026d4d: je short 10026D95h
0x10026d4f: cmp byte ptr [esi+124h],6
0x10026d56: jae short 10026D61h
0x10026d58: push esi
0x10026d59: call 1002D670h
0x10026d5e: add esp,4
0x10026d61: test ebx,ebx
0x10026d63: je short 10026D7Bh
0x10026d65: movzx ecx,byte ptr [esi+124h]
0x10026d6c: mov edx,[esp+ecx*4+34h]
0x10026d70: push edx
0x10026d71: push ebx
0x10026d72: push esi
0x10026d73: call 1002D370h
0x10026d78: add esp,0Ch
0x10026d7b: cmp dword ptr [esp+10h],0
0x10026d80: je short 10026DBAh
0x10026d82: movzx eax,byte ptr [esi+124h]
0x10026d89: mov ecx,[esp+eax*4+18h]
0x10026d8d: mov edx,[esp+10h]
0x10026d91: push ecx
0x10026d92: push edx
0x10026d93: jmp short 10026DB1h
0x10026d95: cmp dword ptr [esp+10h],0
0x10026d9a: je short 10026DABh
0x10026d9c: mov eax,[esp+10h]
0x10026da0: push ebp
0x10026da1: push eax
0x10026da2: push esi
0x10026da3: call 1002D370h
0x10026da8: add esp,0Ch
0x10026dab: test ebx,ebx
0x10026dad: je short 10026DBAh
0x10026daf: push ebp
0x10026db0: push ebx
0x10026db1: push esi
0x10026db2: call 1002D370h
0x10026db7: add esp,0Ch
0x10026dba: push esi
0x10026dbb: call 100309C0h
0x10026dc0: mov eax,[esi+198h]
0x10026dc6: add esp,4
0x10026dc9: test eax,eax
0x10026dcb: je short 10026DE2h
0x10026dcd: movzx ecx,byte ptr [esi+124h]
0x10026dd4: mov edx,[esi+0E4h]
0x10026dda: push ecx
0x10026ddb: push edx
0x10026ddc: push esi
0x10026ddd: call eax
0x10026ddf: add esp,0Ch
0x10026de2: mov ecx,[esp+58h]
0x10026de6: pop edi
0x10026de7: pop esi
0x10026de8: pop ebp
0x10026de9: pop ebx
0x10026dea: xor ecx,esp
0x10026dec: call 10022326h
0x10026df1: add esp,4Ch
0x10026df4: ret
```

`png_read_row` is called by png_read_rows, png_read_image. It works with the text "Decompression error", "Extra compressed data", "Not enough image data". It delegates to sub_10022326, j_memcpy, png_error (15 callees).

12.572 png_read_rows at 0x10026e10 (69 instructions)

```
0x10026e10: push ebx
0x10026e11: mov ebx,[esp+8]
0x10026e15: test ebx,ebx
0x10026e17: push ebp
0x10026e18: je near ptr 10026E9Fh
0x10026e1e: push esi
0x10026e1f: mov esi,[esp+14h]
0x10026e23: test esi,esi
0x10026e25: push edi
0x10026e26: mov edi,[esp+1Ch]
0x10026e2a: je short 10026E7Bh
0x10026e2c: test edi,edi
0x10026e2e: je short 10026E57h
0x10026e30: mov ebp,[esp+20h]
0x10026e34: test ebp,ebp
0x10026e36: jbe short 10026E9Dh
0x10026e38: mov ecx,[edi]
0x10026e3a: mov eax,[esi]
0x10026e3c: push ecx
0x10026e3d: push eax
0x10026e3e: push ebx
0x10026e3f: add esi,4
0x10026e42: add edi,4
0x10026e45: call 100268D0h
0x10026e4a: add esp,0Ch
0x10026e4d: sub ebp,1
0x10026e50: jne short 10026E38h
0x10026e52: pop edi
0x10026e53: pop esi
0x10026e54: pop ebp
0x10026e55: pop ebx
0x10026e56: ret
0x10026e57: mov edi,[esp+20h]
0x10026e5b: test edi,edi
0x10026e5d: jbe short 10026E9Dh
0x10026e5f: nop
0x10026e60: mov eax,[esi]
0x10026e62: push 0
0x10026e64: push eax
0x10026e65: push ebx
0x10026e66: call 100268D0h
0x10026e6b: add esp,0Ch
0x10026e6e: add esi,4
0x10026e71: sub edi,1
0x10026e74: jne short 10026E60h
0x10026e76: pop edi
0x10026e77: pop esi
0x10026e78: pop ebp
0x10026e79: pop ebx
0x10026e7a: ret
0x10026e7b: test edi,edi
0x10026e7d: je short 10026E9Dh
0x10026e7f: mov esi,[esp+20h]
0x10026e83: test esi,esi
0x10026e85: jbe short 10026E9Dh
0x10026e87: mov ecx,[edi]
0x10026e89: push ecx
0x10026e8a: push 0
0x10026e8c: push ebx
0x10026e8d: call 100268D0h
0x10026e92: add esp,0Ch
0x10026e95: add edi,4
0x10026e98: sub esi,1
0x10026e9b: jne short 10026E87h
0x10026e9d: pop edi
0x10026e9e: pop esi
```

```
0x10026e9f: pop ebp
0x10026ea0: pop ebx
0x10026ea1: ret
```

`png_read_rows` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_read_row (1 callees).

12.573 png_read_image at 0x10026eb0 (56 instructions)

```
0x10026eb0: push edi
0x10026eb1: mov edi,[esp+8]
0x10026eb5: test edi,edi
0x10026eb7: je near ptr 10026F50h
0x10026ebd: test byte ptr [edi+70h],40h
0x10026ec1: push esi
0x10026ec2: jne short 10026ED7h
0x10026ec4: push edi
0x10026ec5: call 10028710h
0x10026eca: push edi
0x10026ecb: mov esi,eax
0x10026ecd: call 100268A0h
0x10026ed2: add esp,8
0x10026ed5: jmp short 10026F0Bh
0x10026ed7: cmp byte ptr [edi+123h],0
0x10026ede: je short 10026F00h
0x10026ee0: test byte ptr [edi+74h],2
0x10026ee4: jne short 10026F00h
0x10026ee6: push 100FFE88h
0x10026eeb: push edi
0x10026eec: call 100230F0h
0x10026ef1: mov eax,[edi+0D0h]
0x10026ef7: add esp,8
0x10026efa: mov [edi+0D4h],eax
0x10026f00: push edi
0x10026f01: call 10028710h
0x10026f06: add esp,4
0x10026f09: mov esi,eax
0x10026f0b: test esi,esi
0x10026f0d: push ebp
0x10026f0e: mov ebp,[edi+0D0h]
0x10026f14: jle short 10026F4Eh
0x10026f16: mov [esp+10h],esi
0x10026f1a: push ebx
0x10026f1b: jmp short 10026F20h
0x10026f20: test ebp,ebp
0x10026f22: mov esi,[esp+18h]
0x10026f26: jbe short 10026F46h
0x10026f28: mov ebx,ebp
0x10026f2a: lea ebx,[ebx]
0x10026f30: mov ecx,[esi]
0x10026f32: push 0
0x10026f34: push ecx
0x10026f35: push edi
0x10026f36: call 100268D0h
0x10026f3b: add esp,0Ch
0x10026f3e: add esi,4
0x10026f41: sub ebx,1
0x10026f44: jne short 10026F30h
0x10026f46: sub dword ptr [esp+14h],1
0x10026f4b: jne short 10026F20h
0x10026f4d: pop ebx
0x10026f4e: pop ebp
0x10026f4f: pop esi
0x10026f50: pop edi
0x10026f51: ret
```

`png_read_image` is called by png_read_png. It works with the text "Interlace handling should be turned on when using png_read_image". It delegates to png_warning, png_start_read_image, png_read_row (4 callees).

12.574 png_read_end at 0x10026f60 (1541 instructions)

```
0x10026f60: sub esp,0B4h
0x10026f66: mov eax,ds:[10136220h]
0x10026f6b: xor eax,esp
0x10026f6d: mov [esp+0B0h],eax
0x10026f74: mov eax,[esp+0BCh]
0x10026f7b: push ebp
0x10026f7c: mov ebp,[esp+0BCh]
0x10026f83: test ebp,ebp
0x10026f85: mov [esp+8],eax
0x10026f89: je near ptr 100280BDh
0x10026f8f: push ebx
0x10026f90: push esi
0x10026f91: push edi
0x10026f92: push 0
0x10026f94: push ebp
0x10026f95: call 1002E050h
0x10026f9a: mov cl,44h
0x10026f9c: mov [esp+9Ah],cl
0x10026fa3: mov [esp+21h],cl
0x10026fa7: mov [esp+0BBh],cl
0x10026fae: mov [esp+4Bh],cl
0x10026fb2: mov cl,69h
0x10026fb4: mov [esp+60h],cl
0x10026fb8: mov [esp+0C0h],cl
0x10026fbf: mov cl,46h
0x10026fc1: mov bl,49h
0x10026fc3: mov al,54h
0x10026fc5: mov [esp+69h],cl
0x10026fc9: mov [esp+6Ah],cl
0x10026fcd: mov cl,73h
0x10026fcf: mov dl,74h
0x10026fd1: add esp,8
0x10026fd4: mov [esp+90h],bl
0x10026fdb: mov byte ptr [esp+91h],48h
0x10026fe3: mov byte ptr [esp+93h],52h
0x10026feb: mov byte ptr [esp+94h],0
0x10026ff3: mov [esp+18h],bl
0x10026ff7: mov byte ptr [esp+1Ah],41h
0x10026ffc: mov [esp+1Bh],al
0x10027000: mov byte ptr [esp+1Ch],0
0x10027005: mov [esp+0B0h],bl
0x1002700c: mov byte ptr [esp+0B1h],45h
0x10027014: mov byte ptr [esp+0B2h],4Eh
0x1002701c: mov byte ptr [esp+0B4h],0
0x10027024: mov byte ptr [esp+20h],50h
0x10027029: mov byte ptr [esp+21h],4Ch
0x1002702e: mov [esp+22h],al
0x10027032: mov byte ptr [esp+23h],45h
0x10027037: mov byte ptr [esp+24h],0
0x1002703c: mov byte ptr [esp+40h],62h
0x10027041: mov byte ptr [esp+41h],4Bh
0x10027046: mov byte ptr [esp+42h],47h
0x1002704b: mov byte ptr [esp+44h],0
0x10027050: mov byte ptr [esp+80h],63h
0x10027058: mov byte ptr [esp+81h],48h
0x10027060: mov byte ptr [esp+82h],52h
0x10027068: mov byte ptr [esp+83h],4Dh
0x10027070: mov byte ptr [esp+84h],0
0x10027078: mov byte ptr [esp+50h],67h
0x1002707d: mov byte ptr [esp+51h],41h
0x10027082: mov byte ptr [esp+52h],4Dh
0x10027087: mov byte ptr [esp+53h],41h
```

```
0x1002708c: mov byte ptr [esp+54h],0
0x10027091: mov byte ptr [esp+0A0h],68h
0x10027099: mov [esp+0A1h],bl
0x100270a0: mov byte ptr [esp+0A2h],53h
0x100270a8: mov [esp+0A3h],al
0x100270af: mov byte ptr [esp+0A4h],0
0x100270b7: mov byte ptr [esp+59h],43h
0x100270bc: mov byte ptr [esp+5Ah],43h
0x100270c1: mov byte ptr [esp+5Bh],50h
0x100270c6: mov byte ptr [esp+5Ch],0
0x100270cb: mov [esp+0B9h],al
0x100270d2: mov byte ptr [esp+0BAh],58h
0x100270da: mov [esp+0BBh],dl
0x100270e1: mov byte ptr [esp+0BCh],0
0x100270e9: mov byte ptr [esp+60h],6Fh
0x100270ee: mov [esp+63h],cl
0x100270f2: mov byte ptr [esp+64h],0
0x100270f7: mov byte ptr [esp+28h],70h
0x100270fc: mov byte ptr [esp+29h],43h
0x10027101: mov byte ptr [esp+2Ah],41h
0x10027106: mov byte ptr [esp+2Bh],4Ch
0x1002710b: mov byte ptr [esp+2Ch],0
0x10027110: mov byte ptr [esp+30h],70h
0x10027115: mov byte ptr [esp+31h],48h
0x1002711a: mov byte ptr [esp+32h],59h
0x1002711f: mov [esp+33h],cl
0x10027123: mov byte ptr [esp+34h],0
0x10027128: mov [esp+38h],cl
0x1002712c: mov byte ptr [esp+39h],42h
0x10027131: mov [esp+3Ah],bl
0x10027135: mov [esp+3Bh],al
0x10027139: mov byte ptr [esp+3Ch],0
0x1002713e: mov [esp+70h],cl
0x10027142: mov byte ptr [esp+71h],43h
0x10027147: mov byte ptr [esp+72h],41h
0x1002714c: mov byte ptr [esp+73h],4Ch
0x10027151: mov byte ptr [esp+74h],0
0x10027156: mov [esp+68h],cl
0x1002715a: mov byte ptr [esp+69h],50h
0x1002715f: mov byte ptr [esp+6Ah],4Ch
0x10027164: mov [esp+6Bh],al
0x10027168: mov byte ptr [esp+6Ch],0
0x1002716d: mov [esp+48h],cl
0x10027171: mov byte ptr [esp+49h],52h
0x10027176: mov byte ptr [esp+4Ah],47h
0x1002717b: mov byte ptr [esp+4Bh],42h
0x10027180: mov byte ptr [esp+4Ch],0
0x10027185: mov [esp+78h],dl
0x10027189: mov byte ptr [esp+79h],45h
0x1002718e: mov byte ptr [esp+7Ah],58h
0x10027193: mov [esp+7Bh],dl
0x10027197: mov byte ptr [esp+7Ch],0
0x1002719c: mov [esp+88h],dl
0x100271a3: mov [esp+89h],bl
0x100271aa: mov byte ptr [esp+8Ah],4Dh
0x100271b2: mov byte ptr [esp+8Bh],45h
0x100271ba: mov byte ptr [esp+8Ch],0
0x100271c2: mov [esp+98h],dl
0x100271c9: mov byte ptr [esp+99h],52h
0x100271d1: mov byte ptr [esp+9Ah],4Eh
0x100271d9: mov byte ptr [esp+9Bh],53h
0x100271e1: mov byte ptr [esp+9Ch],0
0x100271e9: mov byte ptr [esp+0A8h],7Ah
0x100271f1: mov [esp+0A9h],al
0x100271f8: mov byte ptr [esp+0AAh],58h
0x10027200: mov [esp+0ABh],dl
0x10027207: mov byte ptr [esp+0ACh],0
```

```
0x1002720f: lea edi,[ebp+11Ch]
0x10027215: push ebp
0x10027216: call 1002DFD0h
0x1002721b: mov [esp+14h],eax
0x1002721f: add esp,4
0x10027222: mov eax,4
0x10027227: lea ecx,[esp+90h]
0x1002722e: mov edx,edi
0x10027230: mov esi,[edx]
0x10027232: cmp esi,[ecx]
0x10027234: jne short 10027248h
0x10027236: sub eax,4
0x10027239: add ecx,4
0x1002723c: add edx,4
0x1002723f: cmp eax,4
0x10027242: jae short 10027230h
0x10027244: test eax,eax
0x10027246: je short 100272A5h
0x10027248: movzx ebx,byte ptr [ecx]
0x1002724b: movzx esi,byte ptr [edx]
0x1002724e: sub esi,ebx
0x10027250: jne short 10027297h
0x10027252: sub eax,1
0x10027255: add ecx,1
0x10027258: add edx,1
0x1002725b: test eax,eax
0x1002725d: je short 100272A5h
0x1002725f: movzx ebx,byte ptr [ecx]
0x10027262: movzx esi,byte ptr [edx]
0x10027265: sub esi,ebx
0x10027267: jne short 10027297h
0x10027269: sub eax,1
0x1002726c: add ecx,1
0x1002726f: add edx,1
0x10027272: test eax,eax
0x10027274: je short 100272A5h
0x10027276: movzx ebx,byte ptr [ecx]
0x10027279: movzx esi,byte ptr [edx]
0x1002727c: sub esi,ebx
0x1002727e: jne short 10027297h
0x10027280: sub eax,1
0x10027283: add ecx,1
0x10027286: add edx,1
0x10027289: test eax,eax
0x1002728b: je short 100272A5h
0x1002728d: movzx ecx,byte ptr [ecx]
0x10027290: movzx esi,byte ptr [edx]
0x10027293: sub esi,ecx
0x10027295: je short 100272A5h
0x10027297: test esi,esi
0x10027299: mov eax,1
0x1002729e: jg short 100272A7h
0x100272a0: or eax,0FFFFFFFFh
0x100272a3: jmp short 100272A7h
0x100272a5: xor eax,eax
0x100272a7: test eax,eax
0x100272a9: jne short 100272C0h
0x100272ab: mov edx,[esp+10h]
0x100272af: mov eax,[esp+14h]
0x100272b3: push edx
0x100272b4: push eax
0x100272b5: push ebp
0x100272b6: call 1002E100h
0x100272bb: jmp near ptr 100280ADh
0x100272c0: mov eax,4
0x100272c5: lea ecx,[esp+0B0h]
0x100272cc: mov edx,edi
```

```
0x100272ce: mov edi,edi
0x100272d0: mov esi,[edx]
0x100272d2: cmp esi,[ecx]
0x100272d4: jne short 100272E8h
0x100272d6: sub eax,4
0x100272d9: add ecx,4
0x100272dc: add edx,4
0x100272df: cmp eax,4
0x100272e2: jae short 100272D0h
0x100272e4: test eax,eax
0x100272e6: je short 10027345h
0x100272e8: movzx esi,byte ptr [edx]
0x100272eb: movzx ebx,byte ptr [ecx]
0x100272ee: sub esi,ebx
0x100272f0: jne short 10027337h
0x100272f2: sub eax,1
0x100272f5: add ecx,1
0x100272f8: add edx,1
0x100272fb: test eax,eax
0x100272fd: je short 10027345h
0x100272ff: movzx esi,byte ptr [edx]
0x10027302: movzx ebx,byte ptr [ecx]
0x10027305: sub esi,ebx
0x10027307: jne short 10027337h
0x10027309: sub eax,1
0x1002730c: add ecx,1
0x1002730f: add edx,1
0x10027312: test eax,eax
0x10027314: je short 10027345h
0x10027316: movzx esi,byte ptr [edx]
0x10027319: movzx ebx,byte ptr [ecx]
0x1002731c: sub esi,ebx
0x1002731e: jne short 10027337h
0x10027320: sub eax,1
0x10027323: add ecx,1
0x10027326: add edx,1
0x10027329: test eax,eax
0x1002732b: je short 10027345h
0x1002732d: movzx esi,byte ptr [edx]
0x10027330: movzx ecx,byte ptr [ecx]
0x10027333: sub esi,ecx
0x10027335: je short 10027345h
0x10027337: test esi,esi
0x10027339: mov eax,1
0x1002733e: jg short 10027347h
0x10027340: or eax,0FFFFFFFFh
0x10027343: jmp short 10027347h
0x10027345: xor eax,eax
0x10027347: test eax,eax
0x10027349: jne short 10027360h
0x1002734b: mov edx,[esp+10h]
0x1002734f: mov eax,[esp+14h]
0x10027353: push edx
0x10027354: push eax
0x10027355: push ebp
0x10027356: call 1002E400h
0x1002735b: jmp near ptr 100280ADh
0x10027360: push edi
0x10027361: push ebp
0x10027362: call 10023B10h
0x10027367: add esp,8
0x1002736a: test eax,eax
0x1002736c: lea ecx,[esp+18h]
0x10027370: mov edx,edi
0x10027372: mov eax,4
0x10027377: je near ptr 100274C0h
0x1002737d: lea ecx,[ecx]
```

```
0x10027380: mov esi,[edx]
0x10027382: cmp esi,[ecx]
0x10027384: jne short 10027398h
0x10027386: sub eax,4
0x10027389: add ecx,4
0x1002738c: add edx,4
0x1002738f: cmp eax,4
0x10027392: jae short 10027380h
0x10027394: test eax,eax
0x10027396: je short 100273F5h
0x10027398: movzx ebx,byte ptr [ecx]
0x1002739b: movzx esi,byte ptr [edx]
0x1002739e: sub esi,ebx
0x100273a0: jne short 100273E7h
0x100273a2: sub eax,1
0x100273a5: add ecx,1
0x100273a8: add edx,1
0x100273ab: test eax,eax
0x100273ad: je short 100273F5h
0x100273af: movzx ebx,byte ptr [ecx]
0x100273b2: movzx esi,byte ptr [edx]
0x100273b5: sub esi,ebx
0x100273b7: jne short 100273E7h
0x100273b9: sub eax,1
0x100273bc: add ecx,1
0x100273bf: add edx,1
0x100273c2: test eax,eax
0x100273c4: je short 100273F5h
0x100273c6: movzx ebx,byte ptr [ecx]
0x100273c9: movzx esi,byte ptr [edx]
0x100273cc: sub esi,ebx
0x100273ce: jne short 100273E7h
0x100273d0: sub eax,1
0x100273d3: add ecx,1
0x100273d6: add edx,1
0x100273d9: test eax,eax
0x100273db: je short 100273F5h
0x100273dd: movzx ecx,byte ptr [ecx]
0x100273e0: movzx esi,byte ptr [edx]
0x100273e3: sub esi,ecx
0x100273e5: je short 100273F5h
0x100273e7: test esi,esi
0x100273e9: mov eax,1
0x100273ee: jg short 100273F7h
0x100273f0: or eax,0FFFFFFFFh
0x100273f3: jmp short 100273F7h
0x100273f5: xor eax,eax
0x100273f7: test eax,eax
0x100273f9: jne short 10027418h
0x100273fb: cmp [esp+10h],eax
0x100273ff: ja short 1002740Ah
0x10027401: test dword ptr [ebp+6Ch],2000h
0x10027408: je short 10027418h
0x1002740a: push 100FFECCh
0x1002740f: push ebp
0x10027410: call 10023210h
0x10027415: add esp,8
0x10027418: mov edx,[esp+10h]
0x1002741c: mov eax,[esp+14h]
0x10027420: push edx
0x10027421: push eax
0x10027422: push ebp
0x10027423: call 10030750h
0x10027428: add esp,0Ch
0x1002742b: mov eax,4
0x10027430: lea ecx,[esp+20h]
0x10027434: mov edx,edi
```

```
0x10027436: mov esi,[edx]
0x10027438: cmp esi,[ecx]
0x1002743a: jne short 1002744Eh
0x1002743c: sub eax,4
0x1002743f: add ecx,4
0x10027442: add edx,4
0x10027445: cmp eax,4
0x10027448: jae short 10027436h
0x1002744a: test eax,eax
0x1002744c: je short 100274ABh
0x1002744e: movzx esi,byte ptr [edx]
0x10027451: movzx ebx,byte ptr [ecx]
0x10027454: sub esi,ebx
0x10027456: jne short 1002749Dh
0x10027458: sub eax,1
0x1002745b: add ecx,1
0x1002745e: add edx,1
0x10027461: test eax,eax
0x10027463: je short 100274ABh
0x10027465: movzx esi,byte ptr [edx]
0x10027468: movzx ebx,byte ptr [ecx]
0x1002746b: sub esi,ebx
0x1002746d: jne short 1002749Dh
0x1002746f: sub eax,1
0x10027472: add ecx,1
0x10027475: add edx,1
0x10027478: test eax,eax
0x1002747a: je short 100274ABh
0x1002747c: movzx esi,byte ptr [edx]
0x1002747f: movzx ebx,byte ptr [ecx]
0x10027482: sub esi,ebx
0x10027484: jne short 1002749Dh
0x10027486: sub eax,1
0x10027489: add ecx,1
0x1002748c: add edx,1
0x1002748f: test eax,eax
0x10027491: je short 100274ABh
0x10027493: movzx esi,byte ptr [edx]
0x10027496: movzx ecx,byte ptr [ecx]
0x10027499: sub esi,ecx
0x1002749b: je short 100274ABh
0x1002749d: test esi,esi
0x1002749f: mov eax,1
0x100274a4: jg short 100274ADh
0x100274a6: or eax,0FFFFFFFFh
0x100274a9: jmp short 100274ADh
0x100274ab: xor eax,eax
0x100274ad: test eax,eax
0x100274af: jne near ptr 100280B0h
0x100274b5: or dword ptr [ebp+6Ch],2
0x100274b9: jmp near ptr 100280B0h
0x100274c0: mov esi,[edx]
0x100274c2: cmp esi,[ecx]
0x100274c4: jne short 100274D8h
0x100274c6: sub eax,4
0x100274c9: add ecx,4
0x100274cc: add edx,4
0x100274cf: cmp eax,4
0x100274d2: jae short 100274C0h
0x100274d4: test eax,eax
0x100274d6: je short 10027535h
0x100274d8: movzx ebx,byte ptr [ecx]
0x100274db: movzx esi,byte ptr [edx]
0x100274de: sub esi,ebx
0x100274e0: jne short 10027527h
0x100274e2: sub eax,1
0x100274e5: add ecx,1
```

```
0x100274e8: add edx,1
0x100274eb: test eax,eax
0x100274ed: je short 10027535h
0x100274ef: movzx ebx,byte ptr [ecx]
0x100274f2: movzx esi,byte ptr [edx]
0x100274f5: sub esi,ebx
0x100274f7: jne short 10027527h
0x100274f9: sub eax,1
0x100274fc: add ecx,1
0x100274ff: add edx,1
0x10027502: test eax,eax
0x10027504: je short 10027535h
0x10027506: movzx ebx,byte ptr [ecx]
0x10027509: movzx esi,byte ptr [edx]
0x1002750c: sub esi,ebx
0x1002750e: jne short 10027527h
0x10027510: sub eax,1
0x10027513: add ecx,1
0x10027516: add edx,1
0x10027519: test eax,eax
0x1002751b: je short 10027535h
0x1002751d: movzx eax,byte ptr [ecx]
0x10027520: movzx esi,byte ptr [edx]
0x10027523: sub esi,eax
0x10027525: je short 10027535h
0x10027527: test esi,esi
0x10027529: mov eax,1
0x1002752e: jg short 10027537h
0x10027530: or eax,0FFFFFFFFh
0x10027533: jmp short 10027537h
0x10027535: xor eax,eax
0x10027537: test eax,eax
0x10027539: jne short 10027569h
0x1002753b: mov esi,[esp+10h]
0x1002753f: test esi,esi
0x10027541: ja short 1002754Ch
0x10027543: test dword ptr [ebp+6Ch],2000h
0x1002754a: je short 1002755Ah
0x1002754c: push 100FFECCh
0x10027551: push ebp
0x10027552: call 10023210h
0x10027557: add esp,8
0x1002755a: push esi
0x1002755b: push ebp
0x1002755c: call 1002E050h
0x10027561: add esp,8
0x10027564: jmp near ptr 100280B0h
0x10027569: mov eax,4
0x1002756e: lea ecx,[esp+20h]
0x10027572: mov edx,edi
0x10027574: mov esi,[edx]
0x10027576: cmp esi,[ecx]
0x10027578: jne short 1002758Ch
0x1002757a: sub eax,4
0x1002757d: add ecx,4
0x10027580: add edx,4
0x10027583: cmp eax,4
0x10027586: jae short 10027574h
0x10027588: test eax,eax
0x1002758a: je short 100275E9h
0x1002758c: movzx esi,byte ptr [edx]
0x1002758f: movzx ebx,byte ptr [ecx]
0x10027592: sub esi,ebx
0x10027594: jne short 100275DBh
0x10027596: sub eax,1
0x10027599: add ecx,1
0x1002759c: add edx,1
```

```
0x1002759f: test eax,eax
0x100275a1: je short 100275E9h
0x100275a3: movzx esi,byte ptr [edx]
0x100275a6: movzx ebx,byte ptr [ecx]
0x100275a9: sub esi,ebx
0x100275ab: jne short 100275DBh
0x100275ad: sub eax,1
0x100275b0: add ecx,1
0x100275b3: add edx,1
0x100275b6: test eax,eax
0x100275b8: je short 100275E9h
0x100275ba: movzx esi,byte ptr [edx]
0x100275bd: movzx ebx,byte ptr [ecx]
0x100275c0: sub esi,ebx
0x100275c2: jne short 100275DBh
0x100275c4: sub eax,1
0x100275c7: add ecx,1
0x100275ca: add edx,1
0x100275cd: test eax,eax
0x100275cf: je short 100275E9h
0x100275d1: movzx esi,byte ptr [edx]
0x100275d4: movzx ecx,byte ptr [ecx]
0x100275d7: sub esi,ecx
0x100275d9: je short 100275E9h
0x100275db: test esi,esi
0x100275dd: mov eax,1
0x100275e2: jg short 100275EBh
0x100275e4: or eax,0FFFFFFFFh
0x100275e7: jmp short 100275EBh
0x100275e9: xor eax,eax
0x100275eb: test eax,eax
0x100275ed: jne short 10027604h
0x100275ef: mov edx,[esp+10h]
0x100275f3: mov eax,[esp+14h]
0x100275f7: push edx
0x100275f8: push eax
0x100275f9: push ebp
0x100275fa: call 1002E260h
0x100275ff: jmp near ptr 100280ADh
0x10027604: mov eax,4
0x10027609: lea ecx,[esp+40h]
0x1002760d: mov edx,edi
0x1002760f: nop
0x10027610: mov esi,[edx]
0x10027612: cmp esi,[ecx]
0x10027614: jne short 10027628h
0x10027616: sub eax,4
0x10027619: add ecx,4
0x1002761c: add edx,4
0x1002761f: cmp eax,4
0x10027622: jae short 10027610h
0x10027624: test eax,eax
0x10027626: je short 10027685h
0x10027628: movzx ebx,byte ptr [ecx]
0x1002762b: movzx esi,byte ptr [edx]
0x1002762e: sub esi,ebx
0x10027630: jne short 10027677h
0x10027632: sub eax,1
0x10027635: add ecx,1
0x10027638: add edx,1
0x1002763b: test eax,eax
0x1002763d: je short 10027685h
0x1002763f: movzx ebx,byte ptr [ecx]
0x10027642: movzx esi,byte ptr [edx]
0x10027645: sub esi,ebx
0x10027647: jne short 10027677h
0x10027649: sub eax,1
```

```
0x1002764c: add ecx,1
0x1002764f: add edx,1
0x10027652: test eax,eax
0x10027654: je short 10027685h
0x10027656: movzx ebx,byte ptr [ecx]
0x10027659: movzx esi,byte ptr [edx]
0x1002765c: sub esi,ebx
0x1002765e: jne short 10027677h
0x10027660: sub eax,1
0x10027663: add ecx,1
0x10027666: add edx,1
0x10027669: test eax,eax
0x1002766b: je short 10027685h
0x1002766d: movzx ecx,byte ptr [ecx]
0x10027670: movzx esi,byte ptr [edx]
0x10027673: sub esi,ecx
0x10027675: je short 10027685h
0x10027677: test esi,esi
0x10027679: mov eax,1
0x1002767e: jg short 10027687h
0x10027680: or eax,0FFFFFFFFh
0x10027683: jmp short 10027687h
0x10027685: xor eax,eax
0x10027687: test eax,eax
0x10027689: jne short 100276A0h
0x1002768b: mov edx,[esp+10h]
0x1002768f: mov eax,[esp+14h]
0x10027693: push edx
0x10027694: push eax
0x10027695: push ebp
0x10027696: call 1002F400h
0x1002769b: jmp near ptr 100280ADh
0x100276a0: mov eax,4
0x100276a5: lea ecx,[esp+80h]
0x100276ac: mov edx,edi
0x100276ae: mov edi,edi
0x100276b0: mov esi,[edx]
0x100276b2: cmp esi,[ecx]
0x100276b4: jne short 100276C8h
0x100276b6: sub eax,4
0x100276b9: add ecx,4
0x100276bc: add edx,4
0x100276bf: cmp eax,4
0x100276c2: jae short 100276B0h
0x100276c4: test eax,eax
0x100276c6: je short 10027725h
0x100276c8: movzx esi,byte ptr [edx]
0x100276cb: movzx ebx,byte ptr [ecx]
0x100276ce: sub esi,ebx
0x100276d0: jne short 10027717h
0x100276d2: sub eax,1
0x100276d5: add ecx,1
0x100276d8: add edx,1
0x100276db: test eax,eax
0x100276dd: je short 10027725h
0x100276df: movzx esi,byte ptr [edx]
0x100276e2: movzx ebx,byte ptr [ecx]
0x100276e5: sub esi,ebx
0x100276e7: jne short 10027717h
0x100276e9: sub eax,1
0x100276ec: add ecx,1
0x100276ef: add edx,1
0x100276f2: test eax,eax
0x100276f4: je short 10027725h
0x100276f6: movzx esi,byte ptr [edx]
0x100276f9: movzx ebx,byte ptr [ecx]
0x100276fc: sub esi,ebx
```

```
0x100276fe: jne short 10027717h
0x10027700: sub eax,1
0x10027703: add ecx,1
0x10027706: add edx,1
0x10027709: test eax,eax
0x1002770b: je short 10027725h
0x1002770d: movzx esi,byte ptr [edx]
0x10027710: movzx ecx,byte ptr [ecx]
0x10027713: sub esi,ecx
0x10027715: je short 10027725h
0x10027717: test esi,esi
0x10027719: mov eax,1
0x1002771e: jg short 10027727h
0x10027720: or eax,0FFFFFFFFh
0x10027723: jmp short 10027727h
0x10027725: xor eax,eax
0x10027727: test eax,eax
0x10027729: jne short 10027740h
0x1002772b: mov edx,[esp+10h]
0x1002772f: mov eax,[esp+14h]
0x10027733: push edx
0x10027734: push eax
0x10027735: push ebp
0x10027736: call 1002E700h
0x1002773b: jmp near ptr 100280ADh
0x10027740: mov eax,4
0x10027745: lea ecx,[esp+50h]
0x10027749: mov edx,edi
0x1002774b: jmp short 10027750h
0x10027750: mov esi,[edx]
0x10027752: cmp esi,[ecx]
0x10027754: jne short 10027768h
0x10027756: sub eax,4
0x10027759: add ecx,4
0x1002775c: add edx,4
0x1002775f: cmp eax,4
0x10027762: jae short 10027750h
0x10027764: test eax,eax
0x10027766: je short 100277C5h
0x10027768: movzx ebx,byte ptr [ecx]
0x1002776b: movzx esi,byte ptr [edx]
0x1002776e: sub esi,ebx
0x10027770: jne short 100277B7h
0x10027772: sub eax,1
0x10027775: add ecx,1
0x10027778: add edx,1
0x1002777b: test eax,eax
0x1002777d: je short 100277C5h
0x1002777f: movzx ebx,byte ptr [ecx]
0x10027782: movzx esi,byte ptr [edx]
0x10027785: sub esi,ebx
0x10027787: jne short 100277B7h
0x10027789: sub eax,1
0x1002778c: add ecx,1
0x1002778f: add edx,1
0x10027792: test eax,eax
0x10027794: je short 100277C5h
0x10027796: movzx ebx,byte ptr [ecx]
0x10027799: movzx esi,byte ptr [edx]
0x1002779c: sub esi,ebx
0x1002779e: jne short 100277B7h
0x100277a0: sub eax,1
0x100277a3: add ecx,1
0x100277a6: add edx,1
0x100277a9: test eax,eax
0x100277ab: je short 100277C5h
0x100277ad: movzx ecx,byte ptr [ecx]
```

```
0x100277b0: movzx esi,byte ptr [edx]
0x100277b3: sub esi,ecx
0x100277b5: je short 100277C5h
0x100277b7: test esi,esi
0x100277b9: mov eax,1
0x100277be: jg short 100277C7h
0x100277c0: or eax,0FFFFFFFFh
0x100277c3: jmp short 100277C7h
0x100277c5: xor eax,eax
0x100277c7: test eax,eax
0x100277c9: jne short 100277E0h
0x100277cb: mov edx,[esp+10h]
0x100277cf: mov eax,[esp+14h]
0x100277d3: push edx
0x100277d4: push eax
0x100277d5: push ebp
0x100277d6: call 1002E450h
0x100277db: jmp near ptr 100280ADh
0x100277e0: mov eax,4
0x100277e5: lea ecx,[esp+0A0h]
0x100277ec: mov edx,edi
0x100277ee: mov edi,edi
0x100277f0: mov esi,[edx]
0x100277f2: cmp esi,[ecx]
0x100277f4: jne short 10027808h
0x100277f6: sub eax,4
0x100277f9: add ecx,4
0x100277fc: add edx,4
0x100277ff: cmp eax,4
0x10027802: jae short 100277F0h
0x10027804: test eax,eax
0x10027806: je short 10027865h
0x10027808: movzx esi,byte ptr [edx]
0x1002780b: movzx ebx,byte ptr [ecx]
0x1002780e: sub esi,ebx
0x10027810: jne short 10027857h
0x10027812: sub eax,1
0x10027815: add ecx,1
0x10027818: add edx,1
0x1002781b: test eax,eax
0x1002781d: je short 10027865h
0x1002781f: movzx esi,byte ptr [edx]
0x10027822: movzx ebx,byte ptr [ecx]
0x10027825: sub esi,ebx
0x10027827: jne short 10027857h
0x10027829: sub eax,1
0x1002782c: add ecx,1
0x1002782f: add edx,1
0x10027832: test eax,eax
0x10027834: je short 10027865h
0x10027836: movzx esi,byte ptr [edx]
0x10027839: movzx ebx,byte ptr [ecx]
0x1002783c: sub esi,ebx
0x1002783e: jne short 10027857h
0x10027840: sub eax,1
0x10027843: add ecx,1
0x10027846: add edx,1
0x10027849: test eax,eax
0x1002784b: je short 10027865h
0x1002784d: movzx esi,byte ptr [edx]
0x10027850: movzx ecx,byte ptr [ecx]
0x10027853: sub esi,ecx
0x10027855: je short 10027865h
0x10027857: test esi,esi
0x10027859: mov eax,1
0x1002785e: jg short 10027867h
0x10027860: or eax,0FFFFFFFFh
```

```
0x10027863: jmp short 10027867h
0x10027865: xor eax,eax
0x10027867: test eax,eax
0x10027869: jne short 10027880h
0x1002786b: mov edx,[esp+10h]
0x1002786f: mov eax,[esp+14h]
0x10027873: push edx
0x10027874: push eax
0x10027875: push ebp
0x10027876: call 1002F650h
0x1002787b: jmp near ptr 100280ADh
0x10027880: mov eax,4
0x10027885: lea ecx,[esp+60h]
0x10027889: mov edx,edi
0x1002788b: jmp short 10027890h
0x10027890: mov esi,[edx]
0x10027892: cmp esi,[ecx]
0x10027894: jne short 100278A8h
0x10027896: sub eax,4
0x10027899: add ecx,4
0x1002789c: add edx,4
0x1002789f: cmp eax,4
0x100278a2: jae short 10027890h
0x100278a4: test eax,eax
0x100278a6: je short 10027905h
0x100278a8: movzx ebx,byte ptr [ecx]
0x100278ab: movzx esi,byte ptr [edx]
0x100278ae: sub esi,ebx
0x100278b0: jne short 100278F7h
0x100278b2: sub eax,1
0x100278b5: add ecx,1
0x100278b8: add edx,1
0x100278bb: test eax,eax
0x100278bd: je short 10027905h
0x100278bf: movzx ebx,byte ptr [ecx]
0x100278c2: movzx esi,byte ptr [edx]
0x100278c5: sub esi,ebx
0x100278c7: jne short 100278F7h
0x100278c9: sub eax,1
0x100278cc: add ecx,1
0x100278cf: add edx,1
0x100278d2: test eax,eax
0x100278d4: je short 10027905h
0x100278d6: movzx ebx,byte ptr [ecx]
0x100278d9: movzx esi,byte ptr [edx]
0x100278dc: sub esi,ebx
0x100278de: jne short 100278F7h
0x100278e0: sub eax,1
0x100278e3: add ecx,1
0x100278e6: add edx,1
0x100278e9: test eax,eax
0x100278eb: je short 10027905h
0x100278ed: movzx ecx,byte ptr [ecx]
0x100278f0: movzx esi,byte ptr [edx]
0x100278f3: sub esi,ecx
0x100278f5: je short 10027905h
0x100278f7: test esi,esi
0x100278f9: mov eax,1
0x100278fe: jg short 10027907h
0x10027900: or eax,0FFFFFFFFh
0x10027903: jmp short 10027907h
0x10027905: xor eax,eax
0x10027907: test eax,eax
0x10027909: jne short 10027920h
0x1002790b: mov edx,[esp+10h]
0x1002790f: mov eax,[esp+14h]
0x10027913: push edx
```

```
0x10027914: push eax
0x10027915: push ebp
0x10027916: call 1002F930h
0x1002791b: jmp near ptr 100280ADh
0x10027920: mov eax,4
0x10027925: lea ecx,[esp+28h]
0x10027929: mov edx,edi
0x1002792b: jmp short 10027930h
0x10027930: mov esi,[edx]
0x10027932: cmp esi,[ecx]
0x10027934: jne short 10027948h
0x10027936: sub eax,4
0x10027939: add ecx,4
0x1002793c: add edx,4
0x1002793f: cmp eax,4
0x10027942: jae short 10027930h
0x10027944: test eax,eax
0x10027946: je short 100279A5h
0x10027948: movzx esi,byte ptr [edx]
0x1002794b: movzx ebx,byte ptr [ecx]
0x1002794e: sub esi,ebx
0x10027950: jne short 10027997h
0x10027952: sub eax,1
0x10027955: add ecx,1
0x10027958: add edx,1
0x1002795b: test eax,eax
0x1002795d: je short 100279A5h
0x1002795f: movzx esi,byte ptr [edx]
0x10027962: movzx ebx,byte ptr [ecx]
0x10027965: sub esi,ebx
0x10027967: jne short 10027997h
0x10027969: sub eax,1
0x1002796c: add ecx,1
0x1002796f: add edx,1
0x10027972: test eax,eax
0x10027974: je short 100279A5h
0x10027976: movzx esi,byte ptr [edx]
0x10027979: movzx ebx,byte ptr [ecx]
0x1002797c: sub esi,ebx
0x1002797e: jne short 10027997h
0x10027980: sub eax,1
0x10027983: add ecx,1
0x10027986: add edx,1
0x10027989: test eax,eax
0x1002798b: je short 100279A5h
0x1002798d: movzx esi,byte ptr [edx]
0x10027990: movzx ecx,byte ptr [ecx]
0x10027993: sub esi,ecx
0x10027995: je short 100279A5h
0x10027997: test esi,esi
0x10027999: mov eax,1
0x1002799e: jg short 100279A7h
0x100279a0: or eax,0FFFFFFFFh
0x100279a3: jmp short 100279A7h
0x100279a5: xor eax,eax
0x100279a7: test eax,eax
0x100279a9: jne short 100279C0h
0x100279ab: mov edx,[esp+10h]
0x100279af: mov eax,[esp+14h]
0x100279b3: push edx
0x100279b4: push eax
0x100279b5: push ebp
0x100279b6: call 1002FAA0h
0x100279bb: jmp near ptr 100280ADh
0x100279c0: mov eax,4
0x100279c5: lea ecx,[esp+70h]
0x100279c9: mov edx,edi
```

```
0x100279cb: jmp short 100279D0h
0x100279d0: mov esi,[edx]
0x100279d2: cmp esi,[ecx]
0x100279d4: jne short 100279E8h
0x100279d6: sub eax,4
0x100279d9: add ecx,4
0x100279dc: add edx,4
0x100279df: cmp eax,4
0x100279e2: jae short 100279D0h
0x100279e4: test eax,eax
0x100279e6: je short 10027A45h
0x100279e8: movzx ebx,byte ptr [ecx]
0x100279eb: movzx esi,byte ptr [edx]
0x100279ee: sub esi,ebx
0x100279f0: jne short 10027A37h
0x100279f2: sub eax,1
0x100279f5: add ecx,1
0x100279f8: add edx,1
0x100279fb: test eax,eax
0x100279fd: je short 10027A45h
0x100279ff: movzx ebx,byte ptr [ecx]
0x10027a02: movzx esi,byte ptr [edx]
0x10027a05: sub esi,ebx
0x10027a07: jne short 10027A37h
0x10027a09: sub eax,1
0x10027a0c: add ecx,1
0x10027a0f: add edx,1
0x10027a12: test eax,eax
0x10027a14: je short 10027A45h
0x10027a16: movzx ebx,byte ptr [ecx]
0x10027a19: movzx esi,byte ptr [edx]
0x10027a1c: sub esi,ebx
0x10027a1e: jne short 10027A37h
0x10027a20: sub eax,1
0x10027a23: add ecx,1
0x10027a26: add edx,1
0x10027a29: test eax,eax
0x10027a2b: je short 10027A45h
0x10027a2d: movzx ecx,byte ptr [ecx]
0x10027a30: movzx esi,byte ptr [edx]
0x10027a33: sub esi,ecx
0x10027a35: je short 10027A45h
0x10027a37: test esi,esi
0x10027a39: mov eax,1
0x10027a3e: jg short 10027A47h
0x10027a40: or eax,0FFFFFFFFh
0x10027a43: jmp short 10027A47h
0x10027a45: xor eax,eax
0x10027a47: test eax,eax
0x10027a49: jne short 10027A60h
0x10027a4b: mov edx,[esp+10h]
0x10027a4f: mov eax,[esp+14h]
0x10027a53: push edx
0x10027a54: push eax
0x10027a55: push ebp
0x10027a56: call 1002FDD0h
0x10027a5b: jmp near ptr 100280ADh
0x10027a60: mov eax,4
0x10027a65: lea ecx,[esp+30h]
0x10027a69: mov edx,edi
0x10027a6b: jmp short 10027A70h
0x10027a70: mov esi,[edx]
0x10027a72: cmp esi,[ecx]
0x10027a74: jne short 10027A88h
0x10027a76: sub eax,4
0x10027a79: add ecx,4
0x10027a7c: add edx,4
```

```
0x10027a7f: cmp eax,4
0x10027a82: jae short 10027A70h
0x10027a84: test eax,eax
0x10027a86: je short 10027AE5h
0x10027a88: movzx esi,byte ptr [edx]
0x10027a8b: movzx ebx,byte ptr [ecx]
0x10027a8e: sub esi,ebx
0x10027a90: jne short 10027AD7h
0x10027a92: sub eax,1
0x10027a95: add ecx,1
0x10027a98: add edx,1
0x10027a9b: test eax,eax
0x10027a9d: je short 10027AE5h
0x10027a9f: movzx esi,byte ptr [edx]
0x10027aa2: movzx ebx,byte ptr [ecx]
0x10027aa5: sub esi,ebx
0x10027aa7: jne short 10027AD7h
0x10027aa9: sub eax,1
0x10027aac: add ecx,1
0x10027aaf: add edx,1
0x10027ab2: test eax,eax
0x10027ab4: je short 10027AE5h
0x10027ab6: movzx esi,byte ptr [edx]
0x10027ab9: movzx ebx,byte ptr [ecx]
0x10027abc: sub esi,ebx
0x10027abe: jne short 10027AD7h
0x10027ac0: sub eax,1
0x10027ac3: add ecx,1
0x10027ac6: add edx,1
0x10027ac9: test eax,eax
0x10027acb: je short 10027AE5h
0x10027acd: movzx esi,byte ptr [edx]
0x10027ad0: movzx ecx,byte ptr [ecx]
0x10027ad3: sub esi,ecx
0x10027ad5: je short 10027AE5h
0x10027ad7: test esi,esi
0x10027ad9: mov eax,1
0x10027ade: jg short 10027AE7h
0x10027ae0: or eax,0FFFFFFFFh
0x10027ae3: jmp short 10027AE7h
0x10027ae5: xor eax,eax
0x10027ae7: test eax,eax
0x10027ae9: jne short 10027B00h
0x10027aeb: mov edx,[esp+10h]
0x10027aef: mov eax,[esp+14h]
0x10027af3: push edx
0x10027af4: push eax
0x10027af5: push ebp
0x10027af6: call 1002F7E0h
0x10027afb: jmp near ptr 100280ADh
0x10027b00: mov eax,4
0x10027b05: lea ecx,[esp+38h]
0x10027b09: mov edx,edi
0x10027b0b: jmp short 10027B10h
0x10027b10: mov esi,[edx]
0x10027b12: cmp esi,[ecx]
0x10027b14: jne short 10027B28h
0x10027b16: sub eax,4
0x10027b19: add ecx,4
0x10027b1c: add edx,4
0x10027b1f: cmp eax,4
0x10027b22: jae short 10027B10h
0x10027b24: test eax,eax
0x10027b26: je short 10027B85h
0x10027b28: movzx ebx,byte ptr [ecx]
0x10027b2b: movzx esi,byte ptr [edx]
0x10027b2e: sub esi,ebx
```

```
0x10027b30: jne short 10027B77h
0x10027b32: sub eax,1
0x10027b35: add ecx,1
0x10027b38: add edx,1
0x10027b3b: test eax,eax
0x10027b3d: je short 10027B85h
0x10027b3f: movzx ebx,byte ptr [ecx]
0x10027b42: movzx esi,byte ptr [edx]
0x10027b45: sub esi,ebx
0x10027b47: jne short 10027B77h
0x10027b49: sub eax,1
0x10027b4c: add ecx,1
0x10027b4f: add edx,1
0x10027b52: test eax,eax
0x10027b54: je short 10027B85h
0x10027b56: movzx ebx,byte ptr [ecx]
0x10027b59: movzx esi,byte ptr [edx]
0x10027b5c: sub esi,ebx
0x10027b5e: jne short 10027B77h
0x10027b60: sub eax,1
0x10027b63: add ecx,1
0x10027b66: add edx,1
0x10027b69: test eax,eax
0x10027b6b: je short 10027B85h
0x10027b6d: movzx ecx,byte ptr [ecx]
0x10027b70: movzx esi,byte ptr [edx]
0x10027b73: sub esi,ecx
0x10027b75: je short 10027B85h
0x10027b77: test esi,esi
0x10027b79: mov eax,1
0x10027b7e: jg short 10027B87h
0x10027b80: or eax,0FFFFFFFFh
0x10027b83: jmp short 10027B87h
0x10027b85: xor eax,eax
0x10027b87: test eax,eax
0x10027b89: jne short 10027BA0h
0x10027b8b: mov edx,[esp+10h]
0x10027b8f: mov eax,[esp+14h]
0x10027b93: push edx
0x10027b94: push eax
0x10027b95: push ebp
0x10027b96: call 1002E590h
0x10027b9b: jmp near ptr 100280ADh
0x10027ba0: mov eax,4
0x10027ba5: lea ecx,[esp+48h]
0x10027ba9: mov edx,edi
0x10027bab: jmp short 10027BB0h
0x10027bb0: mov esi,[edx]
0x10027bb2: cmp esi,[ecx]
0x10027bb4: jne short 10027BC8h
0x10027bb6: sub eax,4
0x10027bb9: add ecx,4
0x10027bbc: add edx,4
0x10027bbf: cmp eax,4
0x10027bc2: jae short 10027BB0h
0x10027bc4: test eax,eax
0x10027bc6: je short 10027C25h
0x10027bc8: movzx esi,byte ptr [edx]
0x10027bcb: movzx ebx,byte ptr [ecx]
0x10027bce: sub esi,ebx
0x10027bd0: jne short 10027C17h
0x10027bd2: sub eax,1
0x10027bd5: add ecx,1
0x10027bd8: add edx,1
0x10027bdb: test eax,eax
0x10027bdd: je short 10027C25h
0x10027bdf: movzx esi,byte ptr [edx]
```

```
0x10027be2: movzx ebx,byte ptr [ecx]
0x10027be5: sub esi,ebx
0x10027be7: jne short 10027C17h
0x10027be9: sub eax,1
0x10027bec: add ecx,1
0x10027bef: add edx,1
0x10027bf2: test eax,eax
0x10027bf4: je short 10027C25h
0x10027bf6: movzx esi,byte ptr [edx]
0x10027bf9: movzx ebx,byte ptr [ecx]
0x10027bfc: sub esi,ebx
0x10027bfe: jne short 10027C17h
0x10027c00: sub eax,1
0x10027c03: add ecx,1
0x10027c06: add edx,1
0x10027c09: test eax,eax
0x10027c0b: je short 10027C25h
0x10027c0d: movzx esi,byte ptr [edx]
0x10027c10: movzx ecx,byte ptr [ecx]
0x10027c13: sub esi,ecx
0x10027c15: je short 10027C25h
0x10027c17: test esi,esi
0x10027c19: mov eax,1
0x10027c1e: jg short 10027C27h
0x10027c20: or eax,0FFFFFFFFh
0x10027c23: jmp short 10027C27h
0x10027c25: xor eax,eax
0x10027c27: test eax,eax
0x10027c29: jne short 10027C40h
0x10027c2b: mov edx,[esp+10h]
0x10027c2f: mov eax,[esp+14h]
0x10027c33: push edx
0x10027c34: push eax
0x10027c35: push ebp
0x10027c36: call 1002EA20h
0x10027c3b: jmp near ptr 100280ADh
0x10027c40: mov eax,4
0x10027c45: lea ecx,[esp+58h]
0x10027c49: mov edx,edi
0x10027c4b: jmp short 10027C50h
0x10027c50: mov esi,[edx]
0x10027c52: cmp esi,[ecx]
0x10027c54: jne short 10027C68h
0x10027c56: sub eax,4
0x10027c59: add ecx,4
0x10027c5c: add edx,4
0x10027c5f: cmp eax,4
0x10027c62: jae short 10027C50h
0x10027c64: test eax,eax
0x10027c66: je short 10027CC5h
0x10027c68: movzx ebx,byte ptr [ecx]
0x10027c6b: movzx esi,byte ptr [edx]
0x10027c6e: sub esi,ebx
0x10027c70: jne short 10027CB7h
0x10027c72: sub eax,1
0x10027c75: add ecx,1
0x10027c78: add edx,1
0x10027c7b: test eax,eax
0x10027c7d: je short 10027CC5h
0x10027c7f: movzx ebx,byte ptr [ecx]
0x10027c82: movzx esi,byte ptr [edx]
0x10027c85: sub esi,ebx
0x10027c87: jne short 10027CB7h
0x10027c89: sub eax,1
0x10027c8c: add ecx,1
0x10027c8f: add edx,1
0x10027c92: test eax,eax
```

```
0x10027c94: je short 10027CC5h
0x10027c96: movzx ebx,byte ptr [ecx]
0x10027c99: movzx esi,byte ptr [edx]
0x10027c9c: sub esi,ebx
0x10027c9e: jne short 10027CB7h
0x10027ca0: sub eax,1
0x10027ca3: add ecx,1
0x10027ca6: add edx,1
0x10027ca9: test eax,eax
0x10027cab: je short 10027CC5h
0x10027cad: movzx ecx,byte ptr [ecx]
0x10027cb0: movzx esi,byte ptr [edx]
0x10027cb3: sub esi,ecx
0x10027cb5: je short 10027CC5h
0x10027cb7: test esi,esi
0x10027cb9: mov eax,1
0x10027cbe: jg short 10027CC7h
0x10027cc0: or eax,0FFFFFFFFh
0x10027cc3: jmp short 10027CC7h
0x10027cc5: xor eax,eax
0x10027cc7: test eax,eax
0x10027cc9: jne short 10027CE0h
0x10027ccb: mov edx,[esp+10h]
0x10027ccf: mov eax,[esp+14h]
0x10027cd3: push edx
0x10027cd4: push eax
0x10027cd5: push ebp
0x10027cd6: call 1002EC10h
0x10027cdb: jmp near ptr 100280ADh
0x10027ce0: mov eax,4
0x10027ce5: lea ecx,[esp+68h]
0x10027ce9: mov edx,edi
0x10027ceb: jmp short 10027CF0h
0x10027cf0: mov esi,[edx]
0x10027cf2: cmp esi,[ecx]
0x10027cf4: jne short 10027D08h
0x10027cf6: sub eax,4
0x10027cf9: add ecx,4
0x10027cfc: add edx,4
0x10027cff: cmp eax,4
0x10027d02: jae short 10027CF0h
0x10027d04: test eax,eax
0x10027d06: je short 10027D65h
0x10027d08: movzx esi,byte ptr [edx]
0x10027d0b: movzx ebx,byte ptr [ecx]
0x10027d0e: sub esi,ebx
0x10027d10: jne short 10027D57h
0x10027d12: sub eax,1
0x10027d15: add ecx,1
0x10027d18: add edx,1
0x10027d1b: test eax,eax
0x10027d1d: je short 10027D65h
0x10027d1f: movzx esi,byte ptr [edx]
0x10027d22: movzx ebx,byte ptr [ecx]
0x10027d25: sub esi,ebx
0x10027d27: jne short 10027D57h
0x10027d29: sub eax,1
0x10027d2c: add ecx,1
0x10027d2f: add edx,1
0x10027d32: test eax,eax
0x10027d34: je short 10027D65h
0x10027d36: movzx esi,byte ptr [edx]
0x10027d39: movzx ebx,byte ptr [ecx]
0x10027d3c: sub esi,ebx
0x10027d3e: jne short 10027D57h
0x10027d40: sub eax,1
0x10027d43: add ecx,1
```

```
0x10027d46: add edx,1
0x10027d49: test eax,eax
0x10027d4b: je short 10027D65h
0x10027d4d: movzx esi,byte ptr [edx]
0x10027d50: movzx ecx,byte ptr [ecx]
0x10027d53: sub esi,ecx
0x10027d55: je short 10027D65h
0x10027d57: test esi,esi
0x10027d59: mov eax,1
0x10027d5e: jg short 10027D67h
0x10027d60: or eax,0FFFFFFFFh
0x10027d63: jmp short 10027D67h
0x10027d65: xor eax,eax
0x10027d67: test eax,eax
0x10027d69: jne short 10027D80h
0x10027d6b: mov edx,[esp+10h]
0x10027d6f: mov eax,[esp+14h]
0x10027d73: push edx
0x10027d74: push eax
0x10027d75: push ebp
0x10027d76: call 1002EE90h
0x10027d7b: jmp near ptr 100280ADh
0x10027d80: mov eax,4
0x10027d85: lea ecx,[esp+78h]
0x10027d89: mov edx,edi
0x10027d8b: jmp short 10027D90h
0x10027d90: mov esi,[edx]
0x10027d92: cmp esi,[ecx]
0x10027d94: jne short 10027DA8h
0x10027d96: sub eax,4
0x10027d99: add ecx,4
0x10027d9c: add edx,4
0x10027d9f: cmp eax,4
0x10027da2: jae short 10027D90h
0x10027da4: test eax,eax
0x10027da6: je short 10027E05h
0x10027da8: movzx ebx,byte ptr [ecx]
0x10027dab: movzx esi,byte ptr [edx]
0x10027dae: sub esi,ebx
0x10027db0: jne short 10027DF7h
0x10027db2: sub eax,1
0x10027db5: add ecx,1
0x10027db8: add edx,1
0x10027dbb: test eax,eax
0x10027dbd: je short 10027E05h
0x10027dbf: movzx ebx,byte ptr [ecx]
0x10027dc2: movzx esi,byte ptr [edx]
0x10027dc5: sub esi,ebx
0x10027dc7: jne short 10027DF7h
0x10027dc9: sub eax,1
0x10027dcc: add ecx,1
0x10027dcf: add edx,1
0x10027dd2: test eax,eax
0x10027dd4: je short 10027E05h
0x10027dd6: movzx ebx,byte ptr [ecx]
0x10027dd9: movzx esi,byte ptr [edx]
0x10027ddc: sub esi,ebx
0x10027dde: jne short 10027DF7h
0x10027de0: sub eax,1
0x10027de3: add ecx,1
0x10027de6: add edx,1
0x10027de9: test eax,eax
0x10027deb: je short 10027E05h
0x10027ded: movzx ecx,byte ptr [ecx]
0x10027df0: movzx esi,byte ptr [edx]
0x10027df3: sub esi,ecx
0x10027df5: je short 10027E05h
```

```
0x10027df7: test esi,esi
0x10027df9: mov eax,1
0x10027dfe: jg short 10027E07h
0x10027e00: or eax,0FFFFFFFFh
0x10027e03: jmp short 10027E07h
0x10027e05: xor eax,eax
0x10027e07: test eax,eax
0x10027e09: jne short 10027E20h
0x10027e0b: mov edx,[esp+10h]
0x10027e0f: mov eax,[esp+14h]
0x10027e13: push edx
0x10027e14: push eax
0x10027e15: push ebp
0x10027e16: call 100300D0h
0x10027e1b: jmp near ptr 100280ADh
0x10027e20: mov eax,4
0x10027e25: lea ecx,[esp+88h]
0x10027e2c: mov edx,edi
0x10027e2e: mov edi,edi
0x10027e30: mov esi,[edx]
0x10027e32: cmp esi,[ecx]
0x10027e34: jne short 10027E48h
0x10027e36: sub eax,4
0x10027e39: add ecx,4
0x10027e3c: add edx,4
0x10027e3f: cmp eax,4
0x10027e42: jae short 10027E30h
0x10027e44: test eax,eax
0x10027e46: je short 10027EA5h
0x10027e48: movzx esi,byte ptr [edx]
0x10027e4b: movzx ebx,byte ptr [ecx]
0x10027e4e: sub esi,ebx
0x10027e50: jne short 10027E97h
0x10027e52: sub eax,1
0x10027e55: add ecx,1
0x10027e58: add edx,1
0x10027e5b: test eax,eax
0x10027e5d: je short 10027EA5h
0x10027e5f: movzx esi,byte ptr [edx]
0x10027e62: movzx ebx,byte ptr [ecx]
0x10027e65: sub esi,ebx
0x10027e67: jne short 10027E97h
0x10027e69: sub eax,1
0x10027e6c: add ecx,1
0x10027e6f: add edx,1
0x10027e72: test eax,eax
0x10027e74: je short 10027EA5h
0x10027e76: movzx esi,byte ptr [edx]
0x10027e79: movzx ebx,byte ptr [ecx]
0x10027e7c: sub esi,ebx
0x10027e7e: jne short 10027E97h
0x10027e80: sub eax,1
0x10027e83: add ecx,1
0x10027e86: add edx,1
0x10027e89: test eax,eax
0x10027e8b: je short 10027EA5h
0x10027e8d: movzx esi,byte ptr [edx]
0x10027e90: movzx ecx,byte ptr [ecx]
0x10027e93: sub esi,ecx
0x10027e95: je short 10027EA5h
0x10027e97: test esi,esi
0x10027e99: mov eax,1
0x10027e9e: jg short 10027EA7h
0x10027ea0: or eax,0FFFFFFFFh
0x10027ea3: jmp short 10027EA7h
0x10027ea5: xor eax,eax
0x10027ea7: test eax,eax
```

```
0x10027ea9: jne short 10027EC0h
0x10027eab: mov edx,[esp+10h]
0x10027eaf: mov eax,[esp+14h]
0x10027eb3: push edx
0x10027eb4: push eax
0x10027eb5: push ebp
0x10027eb6: call 1002FFB0h
0x10027ebb: jmp near ptr 100280ADh
0x10027ec0: mov eax,4
0x10027ec5: lea ecx,[esp+98h]
0x10027ecc: mov edx,edi
0x10027ece: mov edi,edi
0x10027ed0: mov esi,[edx]
0x10027ed2: cmp esi,[ecx]
0x10027ed4: jne short 10027EE8h
0x10027ed6: sub eax,4
0x10027ed9: add ecx,4
0x10027edc: add edx,4
0x10027edf: cmp eax,4
0x10027ee2: jae short 10027ED0h
0x10027ee4: test eax,eax
0x10027ee6: je short 10027F45h
0x10027ee8: movzx ebx,byte ptr [ecx]
0x10027eeb: movzx esi,byte ptr [edx]
0x10027eee: sub esi,ebx
0x10027ef0: jne short 10027F37h
0x10027ef2: sub eax,1
0x10027ef5: add ecx,1
0x10027ef8: add edx,1
0x10027efb: test eax,eax
0x10027efd: je short 10027F45h
0x10027eff: movzx ebx,byte ptr [ecx]
0x10027f02: movzx esi,byte ptr [edx]
0x10027f05: sub esi,ebx
0x10027f07: jne short 10027F37h
0x10027f09: sub eax,1
0x10027f0c: add ecx,1
0x10027f0f: add edx,1
0x10027f12: test eax,eax
0x10027f14: je short 10027F45h
0x10027f16: movzx ebx,byte ptr [ecx]
0x10027f19: movzx esi,byte ptr [edx]
0x10027f1c: sub esi,ebx
0x10027f1e: jne short 10027F37h
0x10027f20: sub eax,1
0x10027f23: add ecx,1
0x10027f26: add edx,1
0x10027f29: test eax,eax
0x10027f2b: je short 10027F45h
0x10027f2d: movzx ecx,byte ptr [ecx]
0x10027f30: movzx esi,byte ptr [edx]
0x10027f33: sub esi,ecx
0x10027f35: je short 10027F45h
0x10027f37: test esi,esi
0x10027f39: mov eax,1
0x10027f3e: jg short 10027F47h
0x10027f40: or eax,0FFFFFFFFh
0x10027f43: jmp short 10027F47h
0x10027f45: xor eax,eax
0x10027f47: test eax,eax
0x10027f49: jne short 10027F60h
0x10027f4b: mov edx,[esp+10h]
0x10027f4f: mov eax,[esp+14h]
0x10027f53: push edx
0x10027f54: push eax
0x10027f55: push ebp
0x10027f56: call 1002F170h
```

```
0x10027f5b: jmp near ptr 100280ADh
0x10027f60: mov eax,4
0x10027f65: lea ecx,[esp+0A8h]
0x10027f6c: mov edx,edi
0x10027f6e: mov edi,edi
0x10027f70: mov esi,[edx]
0x10027f72: cmp esi,[ecx]
0x10027f74: jne short 10027F88h
0x10027f76: sub eax,4
0x10027f79: add ecx,4
0x10027f7c: add edx,4
0x10027f7f: cmp eax,4
0x10027f82: jae short 10027F70h
0x10027f84: test eax,eax
0x10027f86: je short 10027FE5h
0x10027f88: movzx esi,byte ptr [edx]
0x10027f8b: movzx ebx,byte ptr [ecx]
0x10027f8e: sub esi,ebx
0x10027f90: jne short 10027FD7h
0x10027f92: sub eax,1
0x10027f95: add ecx,1
0x10027f98: add edx,1
0x10027f9b: test eax,eax
0x10027f9d: je short 10027FE5h
0x10027f9f: movzx esi,byte ptr [edx]
0x10027fa2: movzx ebx,byte ptr [ecx]
0x10027fa5: sub esi,ebx
0x10027fa7: jne short 10027FD7h
0x10027fa9: sub eax,1
0x10027fac: add ecx,1
0x10027faf: add edx,1
0x10027fb2: test eax,eax
0x10027fb4: je short 10027FE5h
0x10027fb6: movzx esi,byte ptr [edx]
0x10027fb9: movzx ebx,byte ptr [ecx]
0x10027fbc: sub esi,ebx
0x10027fbe: jne short 10027FD7h
0x10027fc0: sub eax,1
0x10027fc3: add ecx,1
0x10027fc6: add edx,1
0x10027fc9: test eax,eax
0x10027fcb: je short 10027FE5h
0x10027fcd: movzx esi,byte ptr [edx]
0x10027fd0: movzx ecx,byte ptr [ecx]
0x10027fd3: sub esi,ecx
0x10027fd5: je short 10027FE5h
0x10027fd7: test esi,esi
0x10027fd9: mov eax,1
0x10027fde: jg short 10027FE7h
0x10027fe0: or eax,0FFFFFFFFh
0x10027fe3: jmp short 10027FE7h
0x10027fe5: xor eax,eax
0x10027fe7: test eax,eax
0x10027fe9: jne short 10028000h
0x10027feb: mov edx,[esp+10h]
0x10027fef: mov eax,[esp+14h]
0x10027ff3: push edx
0x10027ff4: push eax
0x10027ff5: push ebp
0x10027ff6: call 10030280h
0x10027ffb: jmp near ptr 100280ADh
0x10028000: mov eax,4
0x10028005: lea ecx,[esp+0B8h]
0x1002800c: mov edx,edi
0x1002800e: mov edi,edi
0x10028010: mov esi,[edx]
0x10028012: cmp esi,[ecx]
```

```
0x10028014: jne short 10028028h
0x10028016: sub eax,4
0x10028019: add ecx,4
0x1002801c: add edx,4
0x1002801f: cmp eax,4
0x10028022: jae short 10028010h
0x10028024: test eax,eax
0x10028026: je short 10028085h
0x10028028: movzx ebx,byte ptr [ecx]
0x1002802b: movzx esi,byte ptr [edx]
0x1002802e: sub esi,ebx
0x10028030: jne short 10028077h
0x10028032: sub eax,1
0x10028035: add ecx,1
0x10028038: add edx,1
0x1002803b: test eax,eax
0x1002803d: je short 10028085h
0x1002803f: movzx ebx,byte ptr [ecx]
0x10028042: movzx esi,byte ptr [edx]
0x10028045: sub esi,ebx
0x10028047: jne short 10028077h
0x10028049: sub eax,1
0x1002804c: add ecx,1
0x1002804f: add edx,1
0x10028052: test eax,eax
0x10028054: je short 10028085h
0x10028056: movzx ebx,byte ptr [ecx]
0x10028059: movzx esi,byte ptr [edx]
0x1002805c: sub esi,ebx
0x1002805e: jne short 10028077h
0x10028060: sub eax,1
0x10028063: add ecx,1
0x10028066: add edx,1
0x10028069: test eax,eax
0x1002806b: je short 10028085h
0x1002806d: movzx ecx,byte ptr [ecx]
0x10028070: movzx esi,byte ptr [edx]
0x10028073: sub esi,ecx
0x10028075: je short 10028085h
0x10028077: test esi,esi
0x10028079: mov eax,1
0x1002807e: jg short 10028087h
0x10028080: or eax,0FFFFFFFFh
0x10028083: jmp short 10028087h
0x10028085: xor eax,eax
0x10028087: test eax,eax
0x10028089: jne short 1002809Dh
0x1002808b: mov edx,[esp+10h]
0x1002808f: mov eax,[esp+14h]
0x10028093: push edx
0x10028094: push eax
0x10028095: push ebp
0x10028096: call 10030490h
0x1002809b: jmp short 100280ADh
0x1002809d: mov ecx,[esp+10h]
0x100280a1: mov edx,[esp+14h]
0x100280a5: push ecx
0x100280a6: push edx
0x100280a7: push ebp
0x100280a8: call 10030750h
0x100280ad: add esp,0Ch
0x100280b0: test byte ptr [ebp+6Ch],10h
0x100280b4: je near ptr 10027215h
0x100280ba: pop edi
0x100280bb: pop esi
0x100280bc: pop ebx
0x100280bd: mov ecx,[esp+0B4h]
```

```
0x100280c4: pop ebp
0x100280c5: xor ecx,esp
0x100280c7: call 10022326h
0x100280cc: add esp,0B4h
0x100280d2: ret
```

`png_read_end` is called by png_read_png. It works with the text "Too many IDATs found". It delegates to sub_10022326, png_benign_error, png_handle_as_unknown (26 callees).

12.575 sub_100280e0 at 0x100280e0 (199 instructions)

```
0x100280e0: mov eax,[esp+8]
0x100280e4: sub esp,40h
0x100280e7: test eax,eax
0x100280e9: push ebx
0x100280ea: mov ebx,[esp+48h]
0x100280ee: push ebp
0x100280ef: push esi
0x100280f0: push edi
0x100280f1: je short 100280FDh
0x100280f3: push eax
0x100280f4: push ebx
0x100280f5: call 100239C0h
0x100280fa: add esp,8
0x100280fd: mov eax,[esp+5Ch]
0x10028101: test eax,eax
0x10028103: je short 1002810Fh
0x10028105: push eax
0x10028106: push ebx
0x10028107: call 100239C0h
0x1002810c: add esp,8
0x1002810f: mov eax,[ebx+0B0h]
0x10028115: push eax
0x10028116: push ebx
0x10028117: call 1002CD70h
0x1002811c: mov ecx,[ebx+244h]
0x10028122: push ecx
0x10028123: push ebx
0x10028124: call 1002CD70h
0x10028129: mov edx,[ebx+0E8h]
0x1002812f: push edx
0x10028130: push ebx
0x10028131: call 1002CD70h
0x10028136: mov eax,[ebx+284h]
0x1002813c: push eax
0x1002813d: push ebx
0x1002813e: call 1002CD70h
0x10028143: mov ecx,[ebx+1ECh]
0x10028149: push ecx
0x1002814a: push ebx
0x1002814b: call 1002CD70h
0x10028150: mov edx,[ebx+1F0h]
0x10028156: push edx
0x10028157: push ebx
0x10028158: call 1002CD70h
0x1002815d: mov eax,[ebx+164h]
0x10028163: push eax
0x10028164: push ebx
0x10028165: call 1002CD70h
0x1002816a: mov ecx,[ebx+168h]
0x10028170: push ecx
0x10028171: push ebx
0x10028172: call 1002CD70h
0x10028177: mov edx,[ebx+16Ch]
0x1002817d: add esp,40h
0x10028180: push edx
```

```
0x10028181: push ebx
0x10028182: call 1002CD70h
0x10028187: add esp,8
0x1002818a: test dword ptr [ebx+214h],1000h
0x10028194: je short 100281A6h
0x10028196: mov eax,[ebx+114h]
0x1002819c: push eax
0x1002819d: push ebx
0x1002819e: call 100234D0h
0x100281a3: add esp,8
0x100281a6: and dword ptr [ebx+214h],0FFFFEFFFh
0x100281b0: test dword ptr [ebx+214h],2000h
0x100281ba: je short 100281CCh
0x100281bc: mov ecx,[ebx+188h]
0x100281c2: push ecx
0x100281c3: push ebx
0x100281c4: call 1002CD70h
0x100281c9: add esp,8
0x100281cc: and dword ptr [ebx+214h],0FFFFDFFFh
0x100281d6: mov eax,[ebx+214h]
0x100281dc: test al,8
0x100281de: je short 100281F0h
0x100281e0: mov edx,[ebx+1F4h]
0x100281e6: push edx
0x100281e7: push ebx
0x100281e8: call 1002CD70h
0x100281ed: add esp,8
0x100281f0: and dword ptr [ebx+214h],0FFFFFFF7h
0x100281f7: cmp dword ptr [ebx+170h],0
0x100281fe: je short 1002824Ah
0x10028200: mov ecx,8
0x10028205: sub ecx,[ebx+158h]
0x1002820b: mov esi,1
0x10028210: shl esi,cl
0x10028212: xor edi,edi
0x10028214: test esi,esi
0x10028216: jle short 1002823Ah
0x10028218: jmp short 10028220h
0x10028220: mov eax,[ebx+170h]
0x10028226: mov ecx,[eax+edi*4]
0x10028229: push ecx
0x1002822a: push ebx
0x1002822b: call 1002CD70h
0x10028230: add edi,1
0x10028233: add esp,8
0x10028236: cmp edi,esi
0x10028238: jl short 10028220h
0x1002823a: mov edx,[ebx+170h]
0x10028240: push edx
0x10028241: push ebx
0x10028242: call 1002CD70h
0x10028247: add esp,8
0x1002824a: cmp dword ptr [ebx+174h],0
0x10028251: je short 1002829Ah
0x10028253: mov ecx,8
0x10028258: sub ecx,[ebx+158h]
0x1002825e: mov esi,1
0x10028263: shl esi,cl
0x10028265: xor edi,edi
0x10028267: test esi,esi
0x10028269: jle short 1002828Ah
0x1002826b: jmp short 10028270h
0x10028270: mov eax,[ebx+174h]
0x10028276: mov ecx,[eax+edi*4]
0x10028279: push ecx
0x1002827a: push ebx
0x1002827b: call 1002CD70h
```

```
0x10028280: add edi,1
0x10028283: add esp,8
0x10028286: cmp edi,esi
0x10028288: jl short 10028270h
0x1002828a: mov edx,[ebx+174h]
0x10028290: push edx
0x10028291: push ebx
0x10028292: call 1002CD70h
0x10028297: add esp,8
0x1002829a: cmp dword ptr [ebx+178h],0
0x100282a1: je short 100282EAh
0x100282a3: mov ecx,8
0x100282a8: sub ecx,[ebx+158h]
0x100282ae: mov esi,1
0x100282b3: shl esi,cl
0x100282b5: xor edi,edi
0x100282b7: test esi,esi
0x100282b9: jle short 100282DAh
0x100282bb: jmp short 100282C0h
0x100282c0: mov eax,[ebx+178h]
0x100282c6: mov ecx,[eax+edi*4]
0x100282c9: push ecx
0x100282ca: push ebx
0x100282cb: call 1002CD70h
0x100282d0: add edi,1
0x100282d3: add esp,8
0x100282d6: cmp edi,esi
0x100282d8: jl short 100282C0h
0x100282da: mov edx,[ebx+178h]
0x100282e0: push edx
0x100282e1: push ebx
0x100282e2: call 1002CD70h
0x100282e7: add esp,8
0x100282ea: mov eax,[ebx+210h]
0x100282f0: push eax
0x100282f1: push ebx
0x100282f2: call 1002CD70h
0x100282f7: lea ecx,[ebx+78h]
0x100282fa: push ecx
0x100282fb: call 10061790h
0x10028300: mov edx,[ebx+1B0h]
0x10028306: push edx
0x10028307: push ebx
0x10028308: call 1002CD70h
0x1002830d: mov eax,[ebx+1E4h]
0x10028313: push eax
0x10028314: push ebx
0x10028315: call 1002CD70h
0x1002831a: mov ebp,[ebx+4Ch]
0x1002831d: push 28Ch
0x10028322: mov ecx,10h
0x10028327: mov esi,ebx
0x10028329: lea edi,[esp+30h]
0x1002832d: rep movsd
0x1002832f: mov ecx,[ebx+240h]
0x10028335: mov esi,[ebx+44h]
0x10028338: mov edi,[ebx+48h]
0x1002833b: push 0
0x1002833d: push ebx
0x1002833e: mov [esp+80h],ecx
0x10028345: call 100228C0h
0x1002834a: mov edx,[esp+80h]
0x10028351: mov [ebx+44h],esi
0x10028354: mov [ebx+48h],edi
0x10028357: add esp,28h
0x1002835a: mov edi,ebx
0x1002835c: mov ecx,10h
```

```
0x10028361: lea esi,[esp+10h]
0x10028365: rep movsd
0x10028367: pop edi
0x10028368: pop esi
0x10028369: mov [ebx+4Ch],ebp
0x1002836c: pop ebp
0x1002836d: mov [ebx+240h],edx
0x10028373: pop ebx
0x10028374: add esp,40h
0x10028377: ret
```

`sub_100280e0` is called by png_destroy_read_struct. It delegates to j_memset, sub_100234d0, sub_100239c0 (5 callees).

12.576 png_set_read_status_fn at 0x10028380 (6 instructions)

```
0x10028380: mov eax,[esp+4]
0x10028384: test eax,eax
0x10028386: je short 10028392h
0x10028388: mov ecx,[esp+8]
0x1002838c: mov [eax+198h],ecx
0x10028392: ret
```

`png_set_read_status_fn` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.577 png_read_png at 0x100283a0 (154 instructions)

```
0x100283a0: push edi
0x100283a1: mov edi,[esp+8]
0x100283a5: test edi,edi
0x100283a7: je near ptr 10028565h
0x100283ad: push esi
0x100283ae: mov esi,[esp+10h]
0x100283b2: push esi
0x100283b3: push edi
0x100283b4: call 10025580h
0x100283b9: add esp,8
0x100283bc: cmp dword ptr [esi+4],3FFFFFFFh
0x100283c3: jbe short 100283D0h
0x100283c5: push 100FFEE4h
0x100283ca: push edi
0x100283cb: call 100231E0h
0x100283d0: push ebx
0x100283d1: mov ebx,[esp+18h]
0x100283d5: test bl,1
0x100283d8: je short 100283E3h
0x100283da: push edi
0x100283db: call 10030EC0h
0x100283e0: add esp,4
0x100283e3: test bl,2
0x100283e6: je short 100283F1h
0x100283e8: push edi
0x100283e9: call 10030ED0h
0x100283ee: add esp,4
0x100283f1: test bl,8
0x100283f4: je short 100283FFh
0x100283f6: push edi
0x100283f7: call 100286C0h
0x100283fc: add esp,4
0x100283ff: test bl,10h
0x10028402: je short 1002842Fh
0x10028404: cmp byte ptr [edi+127h],8
0x1002840b: jb short 10028426h
0x1002840d: cmp byte ptr [edi+126h],3
```

```
0x10028414: je short 10028426h
0x10028416: push 10h
0x10028418: push esi
0x10028419: push edi
0x1002841a: call 10028D00h
0x1002841f: add esp,0Ch
0x10028422: test eax,eax
0x10028424: je short 1002842Fh
0x10028426: push edi
0x10028427: call 10031860h
0x1002842c: add esp,4
0x1002842f: test bl,20h
0x10028432: je short 1002843Dh
0x10028434: push edi
0x10028435: call 100287F0h
0x1002843a: add esp,4
0x1002843d: test bl,40h
0x10028440: je short 1002846Ch
0x10028442: push 2
0x10028444: push esi
0x10028445: push edi
0x10028446: call 10028D00h
0x1002844b: add esp,0Ch
0x1002844e: test eax,eax
0x10028450: je short 1002846Ch
0x10028452: lea eax,[esp+10h]
0x10028456: push eax
0x10028457: push esi
0x10028458: push edi
0x10028459: call 10029980h
0x1002845e: mov ecx,[esp+1Ch]
0x10028462: push ecx
0x10028463: push edi
0x10028464: call 100286E0h
0x10028469: add esp,14h
0x1002846c: test bl,bl
0x1002846e: jns short 10028479h
0x10028470: push edi
0x10028471: call 10028670h
0x10028476: add esp,4
0x10028479: test ebx,100h
0x1002847f: je short 1002848Ah
0x10028481: push edi
0x10028482: call 100287D0h
0x10028487: add esp,4
0x1002848a: test ebx,200h
0x10028490: je short 1002849Bh
0x10028492: push edi
0x10028493: call 10028680h
0x10028498: add esp,4
0x1002849b: test ebx,400h
0x100284a1: je short 100284ACh
0x100284a3: push edi
0x100284a4: call 100287E0h
0x100284a9: add esp,4
0x100284ac: test ebx,2000h
0x100284b2: je short 100284BDh
0x100284b4: push edi
0x100284b5: call 100318B0h
0x100284ba: add esp,4
0x100284bd: push esi
0x100284be: push edi
0x100284bf: call 10026850h
0x100284c4: push 0
0x100284c6: push 40h
0x100284c8: push esi
0x100284c9: push edi
```

```
0x100284ca: call 100235E0h
0x100284cf: add esp,18h
0x100284d2: cmp dword ptr [esi+0ECh],0
0x100284d9: jne short 10028545h
0x100284db: mov edx,[esi+4]
0x100284de: add edx,edx
0x100284e0: add edx,edx
0x100284e2: push edx
0x100284e3: push edi
0x100284e4: call 1002CD20h
0x100284e9: mov [esi+0ECh],eax
0x100284ef: xor eax,eax
0x100284f1: add esp,8
0x100284f4: cmp [esi+4],eax
0x100284f7: jbe short 10028515h
0x100284f9: lea esp,[esp]
0x10028500: mov ecx,[esi+0ECh]
0x10028506: mov dword ptr [ecx+eax*4],0
0x1002850d: add eax,1
0x10028510: cmp eax,[esi+4]
0x10028513: jb short 10028500h
0x10028515: or dword ptr [esi+0BCh],40h
0x1002851c: xor ebx,ebx
0x1002851e: cmp [esi+4],ebx
0x10028521: jle short 10028545h
0x10028523: push esi
0x10028524: push edi
0x10028525: call 10028D20h
0x1002852a: push eax
0x1002852b: push edi
0x1002852c: call 1002CD20h
0x10028531: mov edx,[esi+0ECh]
0x10028537: mov [edx+ebx*4],eax
0x1002853a: add ebx,1
0x1002853d: add esp,10h
0x10028540: cmp ebx,[esi+4]
0x10028543: jl short 10028523h
0x10028545: mov eax,[esi+0ECh]
0x1002854b: push eax
0x1002854c: push edi
0x1002854d: call 10026EB0h
0x10028552: or dword ptr [esi+8],8000h
0x10028559: push esi
0x1002855a: push edi
0x1002855b: call 10026F60h
0x10028560: add esp,10h
0x10028563: pop ebx
0x10028564: pop esi
0x10028565: pop edi
0x10028566: ret
```

`png_read_png` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_error, png_free_data, png_read_info (21 callees).

12.578 png_create_read_struct at 0x10028570 (14 instructions)

```
0x10028570: mov eax,[esp+10h]
0x10028574: mov ecx,[esp+0Ch]
0x10028578: mov edx,[esp+8]
0x1002857c: push 0
0x1002857e: push 0
0x10028580: push 0
0x10028582: push eax
0x10028583: mov eax,[esp+14h]
0x10028587: push ecx
0x10028588: push edx
```

```
0x10028589: push eax
0x1002858a: call 10025260h
0x1002858f: add esp,1Ch
0x10028592: ret
```

`png_create_read_struct` is called by sub_10018d20. It delegates to png_create_read_struct_2 (1 callees).

12.579 png_destroy_read_struct at 0x100285a0 (73 instructions)

```
0x100285a0: push ecx
0x100285a1: mov eax,[esp+8]
0x100285a5: push ebx
0x100285a6: push edi
0x100285a7: xor edi,edi
0x100285a9: xor ebx,ebx
0x100285ab: test eax,eax
0x100285ad: je near ptr 1002865Fh
0x100285b3: push esi
0x100285b4: mov esi,[eax]
0x100285b6: test esi,esi
0x100285b8: je near ptr 1002865Eh
0x100285be: mov eax,[esi+238h]
0x100285c4: mov [esp+0Ch],eax
0x100285c8: mov eax,[esp+18h]
0x100285cc: test eax,eax
0x100285ce: push ebp
0x100285cf: mov ebp,[esi+240h]
0x100285d5: je short 100285D9h
0x100285d7: mov edi,[eax]
0x100285d9: mov eax,[esp+20h]
0x100285dd: test eax,eax
0x100285df: je short 100285E3h
0x100285e1: mov ebx,[eax]
0x100285e3: push ebx
0x100285e4: push edi
0x100285e5: push esi
0x100285e6: call 100280E0h
0x100285eb: add esp,0Ch
0x100285ee: test edi,edi
0x100285f0: je short 10028619h
0x100285f2: push 0FFFFFFFFh
0x100285f4: push 4000h
0x100285f9: push edi
0x100285fa: push esi
0x100285fb: call 100235E0h
0x10028600: mov ecx,[esp+20h]
0x10028604: push ecx
0x10028605: push ebp
0x10028606: push edi
0x10028607: call 1002CBE0h
0x1002860c: mov edx,[esp+38h]
0x10028610: add esp,1Ch
0x10028613: mov dword ptr [edx],0
0x10028619: test ebx,ebx
0x1002861b: je short 10028644h
0x1002861d: push 0FFFFFFFFh
0x1002861f: push 4000h
0x10028624: push ebx
0x10028625: push esi
0x10028626: call 100235E0h
0x1002862b: mov eax,[esp+20h]
0x1002862f: push eax
0x10028630: push ebp
0x10028631: push ebx
0x10028632: call 1002CBE0h
0x10028637: mov ecx,[esp+3Ch]
```

```
0x1002863b: add esp,1Ch
0x1002863e: mov dword ptr [ecx],0
0x10028644: mov edx,[esp+10h]
0x10028648: push edx
0x10028649: push ebp
0x1002864a: push esi
0x1002864b: call 1002CBE0h
0x10028650: mov eax,[esp+24h]
0x10028654: add esp,0Ch
0x10028657: mov dword ptr [eax],0
0x1002865d: pop ebp
0x1002865e: pop esi
0x1002865f: pop edi
0x10028660: pop ebx
0x10028661: pop ecx
0x10028662: ret
```

`png_destroy_read_struct` is called by sub_10018d20. It delegates to png_free_data, sub_100280e0, sub_1002cbe0 (3 callees).

12.580 png_set_bgr at 0x10028670 (5 instructions)

```
0x10028670: mov eax,[esp+4]
0x10028674: test eax,eax
0x10028676: je short 1002867Ch
0x10028678: or dword ptr [eax+74h],1
0x1002867c: ret
```

`png_set_bgr` is called by png_read_png, png_write_png. It calls nothing else.

12.581 png_set_swap at 0x10028680 (8 instructions)

```
0x10028680: mov eax,[esp+4]
0x10028684: test eax,eax
0x10028686: je short 10028698h
0x10028688: mov ecx,10h
0x1002868d: cmp [eax+127h],cl
0x10028693: jne short 10028698h
0x10028695: or [eax+74h],ecx
0x10028698: ret
```

`png_set_swap` is called by png_read_png, png_write_png. It calls nothing else.

12.582 png_set_packing at 0x100286a0 (9 instructions)

```
0x100286a0: mov eax,[esp+4]
0x100286a4: test eax,eax
0x100286a6: je short 100286BCh
0x100286a8: mov cl,8
0x100286aa: cmp [eax+127h],cl
0x100286b0: jae short 100286BCh
0x100286b2: or dword ptr [eax+74h],4
0x100286b6: mov [eax+128h],cl
0x100286bc: ret
```

`png_set_packing` is called by png_write_png. It calls nothing else.

12.583 png_set_packswap at 0x100286c0 (7 instructions)

```
0x100286c0: mov eax,[esp+4]
0x100286c4: test eax,eax
0x100286c6: je short 100286D8h
0x100286c8: cmp byte ptr [eax+127h],8
```

```
0x100286cf: jae short 100286D8h
0x100286d1: or dword ptr [eax+74h],10000h
0x100286d8: ret
```

`png_set_packswap` is called by png_read_png, png_write_png. It calls nothing else.

12.584 png_set_shift at 0x100286e0 (10 instructions)

```
0x100286e0: mov eax,[esp+4]
0x100286e4: test eax,eax
0x100286e6: je short 10028701h
0x100286e8: mov ecx,[esp+8]
0x100286ec: or dword ptr [eax+74h],8
0x100286f0: mov edx,[ecx]
0x100286f2: mov [eax+181h],edx
0x100286f8: mov cl,[ecx+4]
0x100286fb: mov [eax+185h],cl
0x10028701: ret
```

`png_set_shift` is called by png_read_png, png_write_png. It calls nothing else.

12.585 png_set_interlace_handling at 0x10028710 (10 instructions)

```
0x10028710: mov eax,[esp+4]
0x10028714: test eax,eax
0x10028716: je short 1002872Bh
0x10028718: cmp byte ptr [eax+123h],0
0x1002871f: je short 1002872Bh
0x10028721: or dword ptr [eax+74h],2
0x10028725: mov eax,7
0x1002872a: ret
0x1002872b: mov eax,1
0x10028730: ret
```

`png_set_interlace_handling` is called by png_read_image, png_write_image. It calls nothing else.

12.586 png_set_filler at 0x10028740 (21 instructions)

```
0x10028740: mov eax,[esp+4]
0x10028744: test eax,eax
0x10028746: je short 10028798h
0x10028748: or dword ptr [eax+74h],8000h
0x1002874f: cmp dword ptr [esp+0Ch],1
0x10028754: mov cx,[esp+8]
0x10028759: mov [eax+12Eh],cx
0x10028760: jne short 1002876Bh
0x10028762: or dword ptr [eax+70h],80h
0x10028769: jmp short 10028772h
0x1002876b: and dword ptr [eax+70h],0FFFFFF7Fh
0x10028772: mov cl,[eax+126h]
0x10028778: cmp cl,2
0x1002877b: jne short 10028784h
0x1002877d: mov byte ptr [eax+12Bh],4
0x10028784: test cl,cl
0x10028786: jne short 10028798h
0x10028788: cmp byte ptr [eax+127h],8
0x1002878f: jb short 10028798h
0x10028791: mov byte ptr [eax+12Bh],2
0x10028798: ret
```

`png_set_filler` is called by png_set_add_alpha, png_write_png. It calls nothing else.

12.587 png_set_add_alpha at 0x100287a0 (12 instructions)

```
0x100287a0: mov edx,[esp+4]
0x100287a4: test edx,edx
0x100287a6: je short 100287C2h
0x100287a8: mov eax,[esp+0Ch]
0x100287ac: mov ecx,[esp+8]
0x100287b0: push eax
0x100287b1: push ecx
0x100287b2: push edx
0x100287b3: call 10028740h
0x100287b8: add esp,0Ch
0x100287bb: or dword ptr [edx+74h],1000000h
0x100287c2: ret
```

`png_set_add_alpha` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_set_filler (1 callees).

12.588 png_set_swap_alpha at 0x100287d0 (5 instructions)

```
0x100287d0: mov eax,[esp+4]
0x100287d4: test eax,eax
0x100287d6: je short 100287DFh
0x100287d8: or dword ptr [eax+74h],20000h
0x100287df: ret
```

`png_set_swap_alpha` is called by png_read_png, png_write_png. It calls nothing else.

12.589 png_set_invert_alpha at 0x100287e0 (5 instructions)

```
0x100287e0: mov eax,[esp+4]
0x100287e4: test eax,eax
0x100287e6: je short 100287EFh
0x100287e8: or dword ptr [eax+74h],80000h
0x100287ef: ret
```

`png_set_invert_alpha` is called by png_read_png, png_write_png. It calls nothing else.

12.590 png_set_invert_mono at 0x100287f0 (5 instructions)

```
0x100287f0: mov eax,[esp+4]
0x100287f4: test eax,eax
0x100287f6: je short 100287FCh
0x100287f8: or dword ptr [eax+74h],20h
0x100287fc: ret
```

`png_set_invert_mono` is called by png_read_png, png_write_png. It calls nothing else.

12.591 sub_10028800 at 0x10028800 (55 instructions)

```
0x10028800: mov ecx,[esp+4]
0x10028804: mov al,[ecx+8]
0x10028807: test al,al
0x10028809: jne short 10028825h
0x1002880b: mov ecx,[ecx+4]
0x1002880e: test ecx,ecx
0x10028810: mov eax,[esp+8]
0x10028814: jbe short 1002888Ah
0x10028816: mov dl,[eax]
0x10028818: not dl
0x1002881a: mov [eax],dl
0x1002881c: add eax,1
0x1002881f: sub ecx,1
```

```
0x10028822: jne short 10028816h
0x10028824: ret
0x10028825: cmp al,4
0x10028827: jne short 1002888Ah
0x10028829: cmp byte ptr [ecx+9],8
0x1002882d: jne short 10028851h
0x1002882f: mov ecx,[ecx+4]
0x10028832: test ecx,ecx
0x10028834: mov eax,[esp+8]
0x10028838: jbe short 1002888Ah
0x1002883a: add ecx,0FFFFFFFFh
0x1002883d: shr ecx,1
0x1002883f: add ecx,1
0x10028842: mov dl,[eax]
0x10028844: not dl
0x10028846: mov [eax],dl
0x10028848: add eax,2
0x1002884b: sub ecx,1
0x1002884e: jne short 10028842h
0x10028850: ret
0x10028851: cmp al,4
0x10028853: jne short 1002888Ah
0x10028855: cmp byte ptr [ecx+9],10h
0x10028859: jne short 1002888Ah
0x1002885b: mov ecx,[ecx+4]
0x1002885e: test ecx,ecx
0x10028860: jbe short 1002888Ah
0x10028862: mov eax,[esp+8]
0x10028866: add ecx,0FFFFFFFFh
0x10028869: shr ecx,2
0x1002886c: add eax,1
0x1002886f: add ecx,1
0x10028872: movzx edx,byte ptr [eax-1]
0x10028876: not dl
0x10028878: mov [eax-1],dl
0x1002887b: movzx edx,byte ptr [eax]
0x1002887e: not dl
0x10028880: mov [eax],dl
0x10028882: add eax,4
0x10028885: sub ecx,1
0x10028888: jne short 10028872h
0x1002888a: ret
```

`sub_10028800` is called by sub_100354b0, sub_10039380. It calls nothing else.

12.592 sub_10028890 at 0x10028890 (20 instructions)

```
0x10028890: mov edx,[esp+4]
0x10028894: cmp byte ptr [edx+9],10h
0x10028898: jne short 100288C3h
0x1002889a: movzx ecx,byte ptr [edx+0Ah]
0x1002889e: imul ecx,[edx]
0x100288a1: test ecx,ecx
0x100288a3: mov eax,[esp+8]
0x100288a7: jbe short 100288C3h
0x100288a9: push esi
0x100288aa: mov esi,ecx
0x100288ac: lea esp,[esp]
0x100288b0: mov cl,[eax]
0x100288b2: mov dl,[eax+1]
0x100288b5: mov [eax],dl
0x100288b7: mov [eax+1],cl
0x100288ba: add eax,2
0x100288bd: sub esi,1
0x100288c0: jne short 100288B0h
```

```
0x100288c2: pop esi
0x100288c3: ret
```

`sub_10028890` is called by sub_100354b0, sub_10039380. It calls nothing else.

12.593 sub_100288d0 at 0x100288d0 (30 instructions)

```
0x100288d0: mov eax,[esp+4]
0x100288d4: mov dl,[eax+9]
0x100288d7: cmp dl,8
0x100288da: jae short 10028920h
0x100288dc: mov ecx,[eax+4]
0x100288df: mov eax,[esp+8]
0x100288e3: add ecx,eax
0x100288e5: cmp dl,1
0x100288e8: push esi
0x100288e9: jne short 100288F2h
0x100288eb: mov esi,100FFF18h
0x100288f0: jmp short 10028908h
0x100288f2: cmp dl,2
0x100288f5: jne short 100288FEh
0x100288f7: mov esi,10100018h
0x100288fc: jmp short 10028908h
0x100288fe: cmp dl,4
0x10028901: jne short 1002891Fh
0x10028903: mov esi,10100118h
0x10028908: cmp eax,ecx
0x1002890a: jae short 1002891Fh
0x1002890c: lea esp,[esp]
0x10028910: movzx edx,byte ptr [eax]
0x10028913: mov dl,[edx+esi]
0x10028916: mov [eax],dl
0x10028918: add eax,1
0x1002891b: cmp eax,ecx
0x1002891d: jb short 10028910h
0x1002891f: pop esi
0x10028920: ret
```

`sub_100288d0` is called by sub_100354b0, sub_10039380. It calls nothing else.

12.594 sub_10028930 at 0x10028930 (228 instructions)

```
0x10028930: push ebx
0x10028931: mov ebx,[esp+10h]
0x10028935: push ebp
0x10028936: mov ebp,[esp+0Ch]
0x1002893a: mov dl,[ebp+8]
0x1002893d: cmp dl,2
0x10028940: push esi
0x10028941: mov esi,[esp+14h]
0x10028945: push edi
0x10028946: mov edi,[ebp]
0x10028949: mov eax,esi
0x1002894b: mov ecx,esi
0x1002894d: je short 10028964h
0x1002894f: cmp dl,6
0x10028952: jne near ptr 10028AE7h
0x10028958: test ebx,400000h
0x1002895e: je near ptr 10028AE7h
0x10028964: cmp byte ptr [ebp+0Ah],4
0x10028968: jne near ptr 10028AE7h
0x1002896e: cmp byte ptr [ebp+9],8
0x10028972: jne near ptr 10028A10h
0x10028978: test bl,bl
0x1002897a: jns short 100289CBh
```

```
0x1002897c: cmp edi,1
0x1002897f: lea edx,[esi+3]
0x10028982: lea eax,[esi+4]
0x10028985: jbe short 100289FDh
0x10028987: lea esi,[edi-1]
0x1002898a: lea ebx,[ebx]
0x10028990: movzx ecx,byte ptr [eax]
0x10028993: mov [edx],cl
0x10028995: movzx ecx,byte ptr [eax+1]
0x10028999: add eax,1
0x1002899c: add edx,1
0x1002899f: mov [edx],cl
0x100289a1: movzx ecx,byte ptr [eax+1]
0x100289a5: add eax,1
0x100289a8: add edx,1
0x100289ab: mov [edx],cl
0x100289ad: add edx,1
0x100289b0: add eax,2
0x100289b3: sub esi,1
0x100289b6: jne short 10028990h
0x100289b8: lea eax,[edi+edi*2]
0x100289bb: mov byte ptr [ebp+0Bh],18h
0x100289bf: mov [ebp+4],eax
0x100289c2: mov byte ptr [ebp+0Ah],3
0x100289c6: jmp near ptr 10028BCEh
0x100289cb: test edi,edi
0x100289cd: jbe short 100289FDh
0x100289cf: mov esi,edi
0x100289d1: movzx edx,byte ptr [eax+1]
0x100289d5: add eax,1
0x100289d8: mov [ecx],dl
0x100289da: movzx edx,byte ptr [eax+1]
0x100289de: add eax,1
0x100289e1: add ecx,1
0x100289e4: mov [ecx],dl
0x100289e6: movzx edx,byte ptr [eax+1]
0x100289ea: add eax,1
0x100289ed: add ecx,1
0x100289f0: mov [ecx],dl
0x100289f2: add ecx,1
0x100289f5: add eax,1
0x100289f8: sub esi,1
0x100289fb: jne short 100289D1h
0x100289fd: lea eax,[edi+edi*2]
0x10028a00: mov byte ptr [ebp+0Bh],18h
0x10028a04: mov [ebp+4],eax
0x10028a07: mov byte ptr [ebp+0Ah],3
0x10028a0b: jmp near ptr 10028BCEh
0x10028a10: test bl,bl
0x10028a12: jns short 10028A76h
0x10028a14: cmp edi,1
0x10028a17: lea edx,[esi+8]
0x10028a1a: lea eax,[esi+6]
0x10028a1d: jbe near ptr 10028AD2h
0x10028a23: lea esi,[edi-1]
0x10028a26: movzx ecx,byte ptr [edx]
0x10028a29: mov [eax],cl
0x10028a2b: movzx ecx,byte ptr [edx+1]
0x10028a2f: add edx,1
0x10028a32: mov [eax+1],cl
0x10028a35: movzx ecx,byte ptr [edx+1]
0x10028a39: add eax,1
0x10028a3c: add edx,1
0x10028a3f: mov [eax+1],cl
0x10028a42: movzx ecx,byte ptr [edx+1]
0x10028a46: add eax,1
0x10028a49: add edx,1
```

```
0x10028a4c: add eax,1
0x10028a4f: mov [eax],cl
0x10028a51: movzx ecx,byte ptr [edx+1]
0x10028a55: add edx,1
0x10028a58: add eax,1
0x10028a5b: mov [eax],cl
0x10028a5d: movzx ecx,byte ptr [edx+1]
0x10028a61: add edx,1
0x10028a64: add eax,1
0x10028a67: mov [eax],cl
0x10028a69: add eax,1
0x10028a6c: add edx,3
0x10028a6f: sub esi,1
0x10028a72: jne short 10028A26h
0x10028a74: jmp short 10028AD2h
0x10028a76: test edi,edi
0x10028a78: jbe short 10028AD2h
0x10028a7a: mov esi,edi
0x10028a7c: lea esp,[esp]
0x10028a80: movzx edx,byte ptr [eax+2]
0x10028a84: mov [ecx],dl
0x10028a86: add eax,2
0x10028a89: movzx edx,byte ptr [eax+1]
0x10028a8d: add eax,1
0x10028a90: mov [ecx+1],dl
0x10028a93: movzx edx,byte ptr [eax+1]
0x10028a97: add ecx,1
0x10028a9a: add eax,1
0x10028a9d: mov [ecx+1],dl
0x10028aa0: movzx edx,byte ptr [eax+1]
0x10028aa4: add ecx,1
0x10028aa7: add eax,1
0x10028aaa: add ecx,1
0x10028aad: mov [ecx],dl
0x10028aaf: movzx edx,byte ptr [eax+1]
0x10028ab3: add eax,1
0x10028ab6: add ecx,1
0x10028ab9: mov [ecx],dl
0x10028abb: movzx edx,byte ptr [eax+1]
0x10028abf: add eax,1
0x10028ac2: add ecx,1
0x10028ac5: mov [ecx],dl
0x10028ac7: add ecx,1
0x10028aca: add eax,1
0x10028acd: sub esi,1
0x10028ad0: jne short 10028A80h
0x10028ad2: lea eax,[edi+edi*2]
0x10028ad5: add eax,eax
0x10028ad7: mov byte ptr [ebp+0Bh],30h
0x10028adb: mov [ebp+4],eax
0x10028ade: mov byte ptr [ebp+0Ah],3
0x10028ae2: jmp near ptr 10028BCEh
0x10028ae7: test dl,dl
0x10028ae9: je short 10028B00h
0x10028aeb: cmp dl,4
0x10028aee: jne near ptr 10028BCEh
0x10028af4: test ebx,400000h
0x10028afa: je near ptr 10028BDAh
0x10028b00: cmp byte ptr [ebp+0Ah],2
0x10028b04: jne near ptr 10028BCEh
0x10028b0a: mov dl,8
0x10028b0c: cmp [ebp+9],dl
0x10028b0f: jne short 10028B63h
0x10028b11: test bl,bl
0x10028b13: jns short 10028B3Eh
0x10028b15: test edi,edi
0x10028b17: jbe short 10028B5Bh
```

```
0x10028b19: mov esi,edi
0x10028b1b: jmp short 10028B20h
0x10028b20: mov bl,[eax]
0x10028b22: mov [ecx],bl
0x10028b24: add ecx,1
0x10028b27: add eax,2
0x10028b2a: sub esi,1
0x10028b2d: jne short 10028B20h
0x10028b2f: mov ebx,[esp+1Ch]
0x10028b33: mov [ebp+0Bh],dl
0x10028b36: mov [ebp+4],edi
0x10028b39: jmp near ptr 10028BCAh
0x10028b3e: test edi,edi
0x10028b40: jbe short 10028B5Bh
0x10028b42: mov esi,edi
0x10028b44: mov bl,[eax+1]
0x10028b47: add eax,1
0x10028b4a: mov [ecx],bl
0x10028b4c: add ecx,1
0x10028b4f: add eax,1
0x10028b52: sub esi,1
0x10028b55: jne short 10028B44h
0x10028b57: mov ebx,[esp+1Ch]
0x10028b5b: mov [ebp+0Bh],dl
0x10028b5e: mov [ebp+4],edi
0x10028b61: jmp short 10028BCAh
0x10028b63: test bl,bl
0x10028b65: jns short 10028B93h
0x10028b67: cmp edi,1
0x10028b6a: lea edx,[esi+4]
0x10028b6d: lea eax,[esi+2]
0x10028b70: jbe short 10028BC0h
0x10028b72: lea esi,[edi-1]
0x10028b75: movzx ecx,byte ptr [edx]
0x10028b78: mov [eax],cl
0x10028b7a: movzx ecx,byte ptr [edx+1]
0x10028b7e: add edx,1
0x10028b81: add eax,1
0x10028b84: mov [eax],cl
0x10028b86: add eax,1
0x10028b89: add edx,3
0x10028b8c: sub esi,1
0x10028b8f: jne short 10028B75h
0x10028b91: jmp short 10028BC0h
0x10028b93: test edi,edi
0x10028b95: jbe short 10028BC0h
0x10028b97: mov esi,edi
0x10028b99: lea esp,[esp]
0x10028ba0: movzx edx,byte ptr [eax+2]
0x10028ba4: add eax,2
0x10028ba7: mov [ecx],dl
0x10028ba9: movzx edx,byte ptr [eax+1]
0x10028bad: add eax,1
0x10028bb0: add ecx,1
0x10028bb3: mov [ecx],dl
0x10028bb5: add ecx,1
0x10028bb8: add eax,1
0x10028bbb: sub esi,1
0x10028bbe: jne short 10028BA0h
0x10028bc0: lea eax,[edi+edi]
0x10028bc3: mov byte ptr [ebp+0Bh],10h
0x10028bc7: mov [ebp+4],eax
0x10028bca: mov byte ptr [ebp+0Ah],1
0x10028bce: test ebx,400000h
0x10028bd4: je short 10028BDAh
0x10028bd6: and byte ptr [ebp+8],0FBh
0x10028bda: pop edi
```

```
0x10028bdb: pop esi
0x10028bdc: pop ebp
0x10028bdd: pop ebx
0x10028bde: ret
```

`sub_10028930` is called by sub_100354b0, sub_10039380. It calls nothing else.

12.595 sub_10028be0 at 0x10028be0 (84 instructions)

```
0x10028be0: mov eax,[esp+4]
0x10028be4: mov dl,[eax+8]
0x10028be7: test dl,2
0x10028bea: je near ptr 10028CC5h
0x10028bf0: mov ecx,[eax]
0x10028bf2: mov al,[eax+9]
0x10028bf5: cmp al,8
0x10028bf7: push esi
0x10028bf8: jne short 10028C54h
0x10028bfa: cmp dl,2
0x10028bfd: jne short 10028C24h
0x10028bff: test ecx,ecx
0x10028c01: mov eax,[esp+0Ch]
0x10028c05: jbe near ptr 10028CC4h
0x10028c0b: mov esi,ecx
0x10028c0d: lea ecx,[ecx]
0x10028c10: mov cl,[eax]
0x10028c12: mov dl,[eax+2]
0x10028c15: mov [eax],dl
0x10028c17: mov [eax+2],cl
0x10028c1a: add eax,3
0x10028c1d: sub esi,1
0x10028c20: jne short 10028C10h
0x10028c22: pop esi
0x10028c23: ret
0x10028c24: cmp dl,6
0x10028c27: jne near ptr 10028CC4h
0x10028c2d: test ecx,ecx
0x10028c2f: mov eax,[esp+0Ch]
0x10028c33: jbe near ptr 10028CC4h
0x10028c39: mov esi,ecx
0x10028c3b: jmp short 10028C40h
0x10028c40: mov cl,[eax]
0x10028c42: mov dl,[eax+2]
0x10028c45: mov [eax],dl
0x10028c47: mov [eax+2],cl
0x10028c4a: add eax,4
0x10028c4d: sub esi,1
0x10028c50: jne short 10028C40h
0x10028c52: pop esi
0x10028c53: ret
0x10028c54: cmp al,10h
0x10028c56: jne short 10028CC4h
0x10028c58: cmp dl,2
0x10028c5b: jne short 10028C92h
0x10028c5d: test ecx,ecx
0x10028c5f: jbe short 10028CC4h
0x10028c61: mov eax,[esp+0Ch]
0x10028c65: add eax,1
0x10028c68: mov esi,ecx
0x10028c6a: lea ebx,[ebx]
0x10028c70: movzx edx,byte ptr [eax+3]
0x10028c74: mov cl,[eax-1]
0x10028c77: mov [eax-1],dl
0x10028c7a: movzx edx,byte ptr [eax+4]
0x10028c7e: mov [eax+3],cl
0x10028c81: mov cl,[eax]
```

```
0x10028c83: mov [eax],dl
0x10028c85: mov [eax+4],cl
0x10028c88: add eax,6
0x10028c8b: sub esi,1
0x10028c8e: jne short 10028C70h
0x10028c90: pop esi
0x10028c91: ret
0x10028c92: cmp dl,6
0x10028c95: jne short 10028CC4h
0x10028c97: test ecx,ecx
0x10028c99: jbe short 10028CC4h
0x10028c9b: mov eax,[esp+0Ch]
0x10028c9f: add eax,1
0x10028ca2: mov esi,ecx
0x10028ca4: movzx edx,byte ptr [eax+3]
0x10028ca8: mov cl,[eax-1]
0x10028cab: mov [eax-1],dl
0x10028cae: movzx edx,byte ptr [eax+4]
0x10028cb2: mov [eax+3],cl
0x10028cb5: mov cl,[eax]
0x10028cb7: mov [eax],dl
0x10028cb9: mov [eax+4],cl
0x10028cbc: add eax,8
0x10028cbf: sub esi,1
0x10028cc2: jne short 10028CA4h
0x10028cc4: pop esi
0x10028cc5: ret
```

`sub_10028be0` is called by sub_100354b0, sub_10039380. It calls nothing else.

12.596 png_set_user_transform_info at 0x10028cd0 (10 instructions)

```
0x10028cd0: mov eax,[esp+4]
0x10028cd4: test eax,eax
0x10028cd6: je short 10028CEDh
0x10028cd8: mov ecx,[esp+8]
0x10028cdc: mov dl,[esp+0Ch]
0x10028ce0: mov [eax+64h],ecx
0x10028ce3: mov cl,[esp+10h]
0x10028ce7: mov [eax+68h],dl
0x10028cea: mov [eax+69h],cl
0x10028ced: ret
```

`png_set_user_transform_info` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.597 png_get_user_transform_ptr at 0x10028cf0 (6 instructions)

```
0x10028cf0: mov eax,[esp+4]
0x10028cf4: test eax,eax
0x10028cf6: jne short 10028CF9h
0x10028cf8: ret
0x10028cf9: mov eax,[eax+64h]
0x10028cfc: ret
```

`png_get_user_transform_ptr` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.598 png_get_valid at 0x10028d00 (10 instructions)

```
0x10028d00: cmp dword ptr [esp+4],0
0x10028d05: je short 10028D17h
0x10028d07: mov eax,[esp+8]
0x10028d0b: test eax,eax
```

```
0x10028d0d: je short 10028D17h
0x10028d0f: mov eax,[eax+8]
0x10028d12: and eax,[esp+0Ch]
0x10028d16: ret
0x10028d17: xor eax,eax
0x10028d19: ret
```

`png_get_valid` is called by png_read_png. It calls nothing else.

12.599 png_get_rowbytes at 0x10028d20 (9 instructions)

```
0x10028d20: cmp dword ptr [esp+4],0
0x10028d25: je short 10028D33h
0x10028d27: mov eax,[esp+8]
0x10028d2b: test eax,eax
0x10028d2d: je short 10028D33h
0x10028d2f: mov eax,[eax+0Ch]
0x10028d32: ret
0x10028d33: xor eax,eax
0x10028d35: ret
```

`png_get_rowbytes` is called by png_read_png. It calls nothing else.

12.600 png_get_rows at 0x10028d40 (9 instructions)

```
0x10028d40: cmp dword ptr [esp+4],0
0x10028d45: je short 10028D56h
0x10028d47: mov eax,[esp+8]
0x10028d4b: test eax,eax
0x10028d4d: je short 10028D56h
0x10028d4f: mov eax,[eax+0ECh]
0x10028d55: ret
0x10028d56: xor eax,eax
0x10028d58: ret
```

`png_get_rows` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.601 png_get_image_width at 0x10028d60 (9 instructions)

```
0x10028d60: cmp dword ptr [esp+4],0
0x10028d65: je short 10028D72h
0x10028d67: mov eax,[esp+8]
0x10028d6b: test eax,eax
0x10028d6d: je short 10028D72h
0x10028d6f: mov eax,[eax]
0x10028d71: ret
0x10028d72: xor eax,eax
0x10028d74: ret
```

`png_get_image_width` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.602 png_get_image_height at 0x10028d80 (9 instructions)

```
0x10028d80: cmp dword ptr [esp+4],0
0x10028d85: je short 10028D93h
0x10028d87: mov eax,[esp+8]
0x10028d8b: test eax,eax
0x10028d8d: je short 10028D93h
0x10028d8f: mov eax,[eax+4]
0x10028d92: ret
0x10028d93: xor eax,eax
0x10028d95: ret
```

`png_get_image_height` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.603 png_get_bit_depth at 0x10028da0 (9 instructions)

```
0x10028da0: cmp dword ptr [esp+4],0
0x10028da5: je short 10028DB3h
0x10028da7: mov eax,[esp+8]
0x10028dab: test eax,eax
0x10028dad: je short 10028DB3h
0x10028daf: mov al,[eax+18h]
0x10028db2: ret
0x10028db3: xor al,al
0x10028db5: ret
```

`png_get_bit_depth` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.604 png_get_color_type at 0x10028dc0 (9 instructions)

```
0x10028dc0: cmp dword ptr [esp+4],0
0x10028dc5: je short 10028DD3h
0x10028dc7: mov eax,[esp+8]
0x10028dcb: test eax,eax
0x10028dcd: je short 10028DD3h
0x10028dcf: mov al,[eax+19h]
0x10028dd2: ret
0x10028dd3: xor al,al
0x10028dd5: ret
```

`png_get_color_type` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.605 png_get_filter_type at 0x10028de0 (9 instructions)

```
0x10028de0: cmp dword ptr [esp+4],0
0x10028de5: je short 10028DF3h
0x10028de7: mov eax,[esp+8]
0x10028deb: test eax,eax
0x10028ded: je short 10028DF3h
0x10028def: mov al,[eax+1Bh]
0x10028df2: ret
0x10028df3: xor al,al
0x10028df5: ret
```

`png_get_filter_type` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.606 png_get_interlace_type at 0x10028e00 (9 instructions)

```
0x10028e00: cmp dword ptr [esp+4],0
0x10028e05: je short 10028E13h
0x10028e07: mov eax,[esp+8]
0x10028e0b: test eax,eax
0x10028e0d: je short 10028E13h
0x10028e0f: mov al,[eax+1Ch]
0x10028e12: ret
0x10028e13: xor al,al
0x10028e15: ret
```

`png_get_interlace_type` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.607 png_get_compression_type at 0x10028e20 (9 instructions)

```
0x10028e20: cmp dword ptr [esp+4],0
0x10028e25: je short 10028E33h
0x10028e27: mov eax,[esp+8]
0x10028e2b: test eax,eax
0x10028e2d: je short 10028E33h
0x10028e2f: mov al,[eax+1Ah]
0x10028e32: ret
0x10028e33: xor al,al
0x10028e35: ret
```

`png_get_compression_type` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.608 png_get_x_pixels_per_meter at 0x10028e40 (13 instructions)

```
0x10028e40: cmp dword ptr [esp+4],0
0x10028e45: je short 10028E5Fh
0x10028e47: mov eax,[esp+8]
0x10028e4b: test eax,eax
0x10028e4d: je short 10028E5Fh
0x10028e4f: test byte ptr [eax+8],80h
0x10028e53: je short 10028E5Fh
0x10028e55: cmp byte ptr [eax+7Ch],1
0x10028e59: jne short 10028E5Fh
0x10028e5b: mov eax,[eax+74h]
0x10028e5e: ret
0x10028e5f: xor eax,eax
0x10028e61: ret
```

`png_get_x_pixels_per_meter` is called by png_get_x_pixels_per_inch. It calls nothing else.

12.609 png_get_y_pixels_per_meter at 0x10028e70 (13 instructions)

```
0x10028e70: cmp dword ptr [esp+4],0
0x10028e75: je short 10028E8Fh
0x10028e77: mov eax,[esp+8]
0x10028e7b: test eax,eax
0x10028e7d: je short 10028E8Fh
0x10028e7f: test byte ptr [eax+8],80h
0x10028e83: je short 10028E8Fh
0x10028e85: cmp byte ptr [eax+7Ch],1
0x10028e89: jne short 10028E8Fh
0x10028e8b: mov eax,[eax+78h]
0x10028e8e: ret
0x10028e8f: xor eax,eax
0x10028e91: ret
```

`png_get_y_pixels_per_meter` is called by png_get_y_pixels_per_inch. It calls nothing else.

12.610 png_get_pixels_per_meter at 0x10028ea0 (14 instructions)

```
0x10028ea0: cmp dword ptr [esp+4],0
0x10028ea5: je short 10028EC3h
0x10028ea7: mov ecx,[esp+8]
0x10028eab: test ecx,ecx
0x10028ead: je short 10028EC3h
0x10028eaf: test byte ptr [ecx+8],80h
0x10028eb3: je short 10028EC3h
0x10028eb5: cmp byte ptr [ecx+7Ch],1
0x10028eb9: jne short 10028EC3h
0x10028ebb: mov eax,[ecx+74h]
0x10028ebe: cmp eax,[ecx+78h]
```

```
0x10028ec1: je short 10028EC5h
0x10028ec3: xor eax,eax
0x10028ec5: ret
```

`png_get_pixels_per_meter` is called by png_get_pixels_per_inch. It calls nothing else.

12.611 png_get_pixel_aspect_ratio at 0x10028ed0 (27 instructions)

```
0x10028ed0: cmp dword ptr [esp+4],0
0x10028ed5: je short 10028F1Eh
0x10028ed7: mov ecx,[esp+8]
0x10028edb: test ecx,ecx
0x10028edd: je short 10028F1Eh
0x10028edf: test byte ptr [ecx+8],80h
0x10028ee3: je short 10028F1Eh
0x10028ee5: mov eax,[ecx+74h]
0x10028ee8: test eax,eax
0x10028eea: je short 10028F1Eh
0x10028eec: mov ecx,[ecx+78h]
0x10028eef: test ecx,ecx
0x10028ef1: mov [esp+4],ecx
0x10028ef5: fild dword ptr [esp+4]
0x10028ef9: jge short 10028F01h
0x10028efb: fadd dword ptr ds:[100FE7C0h]
0x10028f01: test eax,eax
0x10028f03: mov [esp+4],eax
0x10028f07: fild dword ptr [esp+4]
0x10028f0b: jge short 10028F13h
0x10028f0d: fadd dword ptr ds:[100FE7C0h]
0x10028f13: fdivp
0x10028f15: fstp dword ptr [esp+4]
0x10028f19: fld dword ptr [esp+4]
0x10028f1d: ret
0x10028f1e: fldz
0x10028f20: ret
```

`png_get_pixel_aspect_ratio` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.612 png_get_pixel_aspect_ratio_fixed at 0x10028f30 (30 instructions)

```
0x10028f30: cmp dword ptr [esp+4],0
0x10028f35: je short 10028F7Fh
0x10028f37: mov eax,[esp+8]
0x10028f3b: test eax,eax
0x10028f3d: je short 10028F7Fh
0x10028f3f: test byte ptr [eax+8],80h
0x10028f43: je short 10028F7Fh
0x10028f45: mov ecx,[eax+74h]
0x10028f48: test ecx,ecx
0x10028f4a: jbe short 10028F7Fh
0x10028f4c: mov eax,[eax+78h]
0x10028f4f: test eax,eax
0x10028f51: jbe short 10028F7Fh
0x10028f53: cmp ecx,7FFFFFFFh
0x10028f59: ja short 10028F7Fh
0x10028f5b: cmp eax,7FFFFFFFh
0x10028f60: ja short 10028F7Fh
0x10028f62: push ecx
0x10028f63: push 186A0h
0x10028f68: push eax
0x10028f69: lea eax,[esp+10h]
0x10028f6d: push eax
0x10028f6e: call 10024770h
0x10028f73: add esp,10h
```

```
0x10028f76: test eax,eax
0x10028f78: je short 10028F7Fh
0x10028f7a: mov eax,[esp+4]
0x10028f7e: ret
0x10028f7f: xor eax,eax
0x10028f81: ret
```

`png_get_pixel_aspect_ratio_fixed` has no recorded callers; it is likely reached through the vector table or an indirect jump. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to sub_10024770 (1 callees).

12.613 png_get_x_offset_microns at 0x10028f90 (13 instructions)

```
0x10028f90: cmp dword ptr [esp+4],0
0x10028f95: je short 10028FB2h
0x10028f97: mov eax,[esp+8]
0x10028f9b: test eax,eax
0x10028f9d: je short 10028FB2h
0x10028f9f: test dword ptr [eax+8],100h
0x10028fa6: je short 10028FB2h
0x10028fa8: cmp byte ptr [eax+70h],1
0x10028fac: jne short 10028FB2h
0x10028fae: mov eax,[eax+68h]
0x10028fb1: ret
0x10028fb2: xor eax,eax
0x10028fb4: ret
```

`png_get_x_offset_microns` is called by png_get_x_offset_inches_fixed, png_get_x_offset_inches. It calls nothing else.

12.614 png_get_y_offset_microns at 0x10028fc0 (13 instructions)

```
0x10028fc0: cmp dword ptr [esp+4],0
0x10028fc5: je short 10028FE2h
0x10028fc7: mov eax,[esp+8]
0x10028fcb: test eax,eax
0x10028fcd: je short 10028FE2h
0x10028fcf: test dword ptr [eax+8],100h
0x10028fd6: je short 10028FE2h
0x10028fd8: cmp byte ptr [eax+70h],1
0x10028fdc: jne short 10028FE2h
0x10028fde: mov eax,[eax+6Ch]
0x10028fe1: ret
0x10028fe2: xor eax,eax
0x10028fe4: ret
```

`png_get_y_offset_microns` is called by png_get_y_offset_inches_fixed, png_get_y_offset_inches. It calls nothing else.

12.615 png_get_x_offset_pixels at 0x10028ff0 (13 instructions)

```
0x10028ff0: cmp dword ptr [esp+4],0
0x10028ff5: je short 10029012h
0x10028ff7: mov eax,[esp+8]
0x10028ffb: test eax,eax
0x10028ffd: je short 10029012h
0x10028fff: test dword ptr [eax+8],100h
0x10029006: je short 10029012h
0x10029008: cmp byte ptr [eax+70h],0
0x1002900c: jne short 10029012h
0x1002900e: mov eax,[eax+68h]
0x10029011: ret
0x10029012: xor eax,eax
0x10029014: ret
```

`png_get_x_offset_pixels` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.616 png_get_y_offset_pixels at 0x10029020 (13 instructions)

```
0x10029020: cmp dword ptr [esp+4],0
0x10029025: je short 10029042h
0x10029027: mov eax,[esp+8]
0x1002902b: test eax,eax
0x1002902d: je short 10029042h
0x1002902f: test dword ptr [eax+8],100h
0x10029036: je short 10029042h
0x10029038: cmp byte ptr [eax+70h],0
0x1002903c: jne short 10029042h
0x1002903e: mov eax,[eax+6Ch]
0x10029041: ret
0x10029042: xor eax,eax
0x10029044: ret
```

`png_get_y_offset_pixels` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.617 png_get_pixels_per_inch at 0x10029080 (25 instructions)

```
0x10029050: push ecx
0x10029051: cmp eax,7FFFFFFFh
0x10029056: ja short 10029076h
0x10029058: push 1388h
0x1002905d: push 7Fh
0x1002905f: push eax
0x10029060: lea eax,[esp+0Ch]
0x10029064: push eax
0x10029065: call 10024770h
0x1002906a: add esp,10h
0x1002906d: test eax,eax
0x1002906f: je short 10029076h
0x10029071: mov eax,[esp]
0x10029074: pop ecx
0x10029075: ret
0x10029076: xor eax,eax
0x10029078: pop ecx
0x10029079: ret
0x10029080: mov eax,[esp+8]
0x10029084: mov ecx,[esp+4]
0x10029088: push eax
0x10029089: push ecx
0x1002908a: call 10028EA0h
0x1002908f: add esp,8
0x10029092: jmp near ptr 10029050h
```

`png_get_pixels_per_inch` has no recorded callers; it is likely reached through the vector table or an indirect jump. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to sub_10024770, png_get_pixels_per_meter (2 callees).

12.618 png_get_x_pixels_per_inch at 0x100290a0 (25 instructions)

```
0x10029050: push ecx
0x10029051: cmp eax,7FFFFFFFh
0x10029056: ja short 10029076h
0x10029058: push 1388h
0x1002905d: push 7Fh
0x1002905f: push eax
0x10029060: lea eax,[esp+0Ch]
0x10029064: push eax
```

```
0x10029065: call 10024770h
0x1002906a: add esp,10h
0x1002906d: test eax,eax
0x1002906f: je short 10029076h
0x10029071: mov eax,[esp]
0x10029074: pop ecx
0x10029075: ret
0x10029076: xor eax,eax
0x10029078: pop ecx
0x10029079: ret
0x100290a0: mov eax,[esp+8]
0x100290a4: mov ecx,[esp+4]
0x100290a8: push eax
0x100290a9: push ecx
0x100290aa: call 10028E40h
0x100290af: add esp,8
0x100290b2: jmp near ptr 10029050h
```

`png_get_x_pixels_per_inch` has no recorded callers; it is likely reached through the vector table or an indirect jump. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to sub_10024770, png_get_x_pixels_per_meter (2 callees).

12.619 png_get_y_pixels_per_inch at 0x100290c0 (25 instructions)

```
0x10029050: push ecx
0x10029051: cmp eax,7FFFFFFFh
0x10029056: ja short 10029076h
0x10029058: push 1388h
0x1002905d: push 7Fh
0x1002905f: push eax
0x10029060: lea eax,[esp+0Ch]
0x10029064: push eax
0x10029065: call 10024770h
0x1002906a: add esp,10h
0x1002906d: test eax,eax
0x1002906f: je short 10029076h
0x10029071: mov eax,[esp]
0x10029074: pop ecx
0x10029075: ret
0x10029076: xor eax,eax
0x10029078: pop ecx
0x10029079: ret
0x100290c0: mov eax,[esp+8]
0x100290c4: mov ecx,[esp+4]
0x100290c8: push eax
0x100290c9: push ecx
0x100290ca: call 10028E70h
0x100290cf: add esp,8
0x100290d2: jmp near ptr 10029050h
```

`png_get_y_pixels_per_inch` has no recorded callers; it is likely reached through the vector table or an indirect jump. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to sub_10024770, png_get_y_pixels_per_meter (2 callees).

12.620 png_get_x_offset_inches_fixed at 0x10029100 (14 instructions)

```
0x100290e0: push 7Fh
0x100290e2: push 1F4h
0x100290e7: push eax
0x100290e8: push ecx
0x100290e9: call 100247F0h
0x100290ee: add esp,10h
0x100290f1: ret
0x10029100: mov eax,[esp+8]
```

```
0x10029104: mov ecx,[esp+4]
0x10029108: push eax
0x10029109: push ecx
0x1002910a: call 10028F90h
0x1002910f: add esp,8
0x10029112: jmp near ptr 100290E0h
```

`png_get_x_offset_inches_fixed` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100247f0, png_get_x_offset_microns (2 callees).

12.621 png_get_y_offset_inches_fixed at 0x10029120 (14 instructions)

```
0x100290e0: push 7Fh
0x100290e2: push 1F4h
0x100290e7: push eax
0x100290e8: push ecx
0x100290e9: call 100247F0h
0x100290ee: add esp,10h
0x100290f1: ret
0x10029120: mov eax,[esp+8]
0x10029124: mov ecx,[esp+4]
0x10029128: push eax
0x10029129: push ecx
0x1002912a: call 10028FC0h
0x1002912f: add esp,8
0x10029132: jmp near ptr 100290E0h
```

`png_get_y_offset_inches_fixed` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_100247f0, png_get_y_offset_microns (2 callees).

12.622 png_get_x_offset_inches at 0x10029140 (13 instructions)

```
0x10029140: push ecx
0x10029141: mov eax,[esp+0Ch]
0x10029145: mov ecx,[esp+8]
0x10029149: push eax
0x1002914a: push ecx
0x1002914b: call 10028F90h
0x10029150: mov [esp+8],eax
0x10029154: fild dword ptr [esp+8]
0x10029158: fmul qword ptr ds:[10100218h]
0x1002915e: fstp dword ptr [esp+8]
0x10029162: fld dword ptr [esp+8]
0x10029166: add esp,0Ch
0x10029169: ret
```

`png_get_x_offset_inches` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_get_x_offset_microns (1 callees).

12.623 png_get_y_offset_inches at 0x10029170 (13 instructions)

```
0x10029170: push ecx
0x10029171: mov eax,[esp+0Ch]
0x10029175: mov ecx,[esp+8]
0x10029179: push eax
0x1002917a: push ecx
0x1002917b: call 10028FC0h
0x10029180: mov [esp+8],eax
0x10029184: fild dword ptr [esp+8]
0x10029188: fmul qword ptr ds:[10100218h]
0x1002918e: fstp dword ptr [esp+8]
0x10029192: fld dword ptr [esp+8]
```

```
0x10029196: add esp,0Ch
0x10029199: ret
```

`png_get_y_offset_inches` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_get_y_offset_microns (1 callees).

12.624 png_get_pHYs_dpi at 0x100291a0 (81 instructions)

```
0x100291a0: xor ecx,ecx
0x100291a2: sub esp,8
0x100291a5: cmp [esp+0Ch],ecx
0x100291a9: je near ptr 10029298h
0x100291af: mov eax,[esp+10h]
0x100291b3: test eax,eax
0x100291b5: je near ptr 10029298h
0x100291bb: test byte ptr [eax+8],80h
0x100291bf: je near ptr 10029298h
0x100291c5: push edi
0x100291c6: mov edi,[esp+18h]
0x100291ca: test edi,edi
0x100291cc: je short 100291D8h
0x100291ce: mov ecx,[eax+74h]
0x100291d1: mov [edi],ecx
0x100291d3: mov ecx,80h
0x100291d8: mov edx,[esp+1Ch]
0x100291dc: test edx,edx
0x100291de: push esi
0x100291df: je short 100291ECh
0x100291e1: mov esi,[eax+78h]
0x100291e4: mov [edx],esi
0x100291e6: or ecx,80h
0x100291ec: mov esi,[esp+24h]
0x100291f0: test esi,esi
0x100291f2: je near ptr 10029296h
0x100291f8: movzx eax,byte ptr [eax+7Ch]
0x100291fc: or ecx,80h
0x10029202: cmp eax,1
0x10029205: mov [esi],eax
0x10029207: jne near ptr 10029296h
0x1002920d: test edi,edi
0x1002920f: fld qword ptr ds:[10100220h]
0x10029215: fld qword ptr ds:[100FE7C8h]
0x1002921b: je short 10029253h
0x1002921d: mov eax,[edi]
0x1002921f: fild dword ptr [edi]
0x10029221: test eax,eax
0x10029223: jge short 1002922Bh
0x10029225: fadd qword ptr ds:[100FEFC8h]
0x1002922b: fnstcw [esp+14h]
0x1002922f: movzx eax,word ptr [esp+14h]
0x10029234: fmul st,st(2)
0x10029236: or eax,0C00h
0x1002923b: mov [esp+8],eax
0x1002923f: fadd st,st(1)
0x10029241: fldcw [esp+8]
0x10029245: fistp qword ptr [esp+8]
0x10029249: mov eax,[esp+8]
0x1002924d: mov [edi],eax
0x1002924f: fldcw [esp+14h]
0x10029253: test edx,edx
0x10029255: je short 10029292h
0x10029257: mov eax,[edx]
0x10029259: fild dword ptr [edx]
0x1002925b: test eax,eax
0x1002925d: jge short 10029265h
0x1002925f: fadd qword ptr ds:[100FEFC8h]
```

```
0x10029265: fnstcw [esp+14h]
0x10029269: pop esi
0x1002926a: movzx eax,word ptr [esp+10h]
0x1002926f: fmulp st(2),st
0x10029271: or eax,0C00h
0x10029276: mov [esp+4],eax
0x1002927a: pop edi
0x1002927b: faddp
0x1002927d: fldcw [esp]
0x10029280: fistp qword ptr [esp]
0x10029283: mov eax,[esp]
0x10029286: mov [edx],eax
0x10029288: mov eax,ecx
0x1002928a: fldcw [esp+0Ch]
0x1002928e: add esp,8
0x10029291: ret
0x10029292: fstp st(1)
0x10029294: fstp st(0)
0x10029296: pop esi
0x10029297: pop edi
0x10029298: mov eax,ecx
0x1002929a: add esp,8
0x1002929d: ret
```

`png_get_pHYs_dpi` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.625 png_get_channels at 0x100292a0 (9 instructions)

```
0x100292a0: cmp dword ptr [esp+4],0
0x100292a5: je short 100292B3h
0x100292a7: mov eax,[esp+8]
0x100292ab: test eax,eax
0x100292ad: je short 100292B3h
0x100292af: mov al,[eax+1Dh]
0x100292b2: ret
0x100292b3: xor al,al
0x100292b5: ret
```

`png_get_channels` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.626 png_get_signature at 0x100292c0 (9 instructions)

```
0x100292c0: cmp dword ptr [esp+4],0
0x100292c5: je short 100292D3h
0x100292c7: mov eax,[esp+8]
0x100292cb: test eax,eax
0x100292cd: je short 100292D3h
0x100292cf: add eax,20h
0x100292d2: ret
0x100292d3: xor eax,eax
0x100292d5: ret
```

`png_get_signature` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.627 png_get_bKGD at 0x100292e0 (16 instructions)

```
0x100292e0: cmp dword ptr [esp+4],0
0x100292e5: je short 10029308h
0x100292e7: mov eax,[esp+8]
0x100292eb: test eax,eax
```

```
0x100292ed: je short 10029308h
0x100292ef: test byte ptr [eax+8],20h
0x100292f3: je short 10029308h
0x100292f5: mov ecx,[esp+0Ch]
0x100292f9: test ecx,ecx
0x100292fb: je short 10029308h
0x100292fd: add eax,5Eh
0x10029300: mov [ecx],eax
0x10029302: mov eax,20h
0x10029307: ret
0x10029308: xor eax,eax
0x1002930a: ret
```

`png_get_bKGD` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.628 png_get_cHRM at 0x10029310 (62 instructions)

```
0x10029310: cmp dword ptr [esp+4],0
0x10029315: je near ptr 100293D3h
0x1002931b: mov eax,[esp+8]
0x1002931f: test eax,eax
0x10029321: je near ptr 100293D3h
0x10029327: test byte ptr [eax+8],4
0x1002932b: je near ptr 100293D3h
0x10029331: mov ecx,[esp+0Ch]
0x10029335: fld qword ptr ds:[100FFB90h]
0x1002933b: test ecx,ecx
0x1002933d: je short 10029349h
0x1002933f: fild dword ptr [eax+84h]
0x10029345: fmul st,st(1)
0x10029347: fstp qword ptr [ecx]
0x10029349: mov ecx,[esp+10h]
0x1002934d: test ecx,ecx
0x1002934f: je short 1002935Bh
0x10029351: fild dword ptr [eax+88h]
0x10029357: fmul st,st(1)
0x10029359: fstp qword ptr [ecx]
0x1002935b: mov ecx,[esp+14h]
0x1002935f: test ecx,ecx
0x10029361: je short 1002936Dh
0x10029363: fild dword ptr [eax+8Ch]
0x10029369: fmul st,st(1)
0x1002936b: fstp qword ptr [ecx]
0x1002936d: mov ecx,[esp+18h]
0x10029371: test ecx,ecx
0x10029373: je short 1002937Fh
0x10029375: fild dword ptr [eax+90h]
0x1002937b: fmul st,st(1)
0x1002937d: fstp qword ptr [ecx]
0x1002937f: mov ecx,[esp+1Ch]
0x10029383: test ecx,ecx
0x10029385: je short 10029391h
0x10029387: fild dword ptr [eax+94h]
0x1002938d: fmul st,st(1)
0x1002938f: fstp qword ptr [ecx]
0x10029391: mov ecx,[esp+20h]
0x10029395: test ecx,ecx
0x10029397: je short 100293A3h
0x10029399: fild dword ptr [eax+98h]
0x1002939f: fmul st,st(1)
0x100293a1: fstp qword ptr [ecx]
0x100293a3: mov ecx,[esp+24h]
0x100293a7: test ecx,ecx
0x100293a9: je short 100293B5h
0x100293ab: fild dword ptr [eax+9Ch]
0x100293b1: fmul st,st(1)
```

```
0x100293b3: fstp qword ptr [ecx]
0x100293b5: mov ecx,[esp+28h]
0x100293b9: test ecx,ecx
0x100293bb: je short 100293CBh
0x100293bd: fimul dword ptr [eax+0A0h]
0x100293c3: mov eax,4
0x100293c8: fstp qword ptr [ecx]
0x100293ca: ret
0x100293cb: fstp st(0)
0x100293cd: mov eax,4
0x100293d2: ret
0x100293d3: xor eax,eax
0x100293d5: ret
```

`png_get_cHRM` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.629 png_get_cHRM_fixed at 0x100293e0 (51 instructions)

```
0x100293e0: cmp dword ptr [esp+4],0
0x100293e5: je near ptr 10029487h
0x100293eb: mov eax,[esp+8]
0x100293ef: test eax,eax
0x100293f1: je near ptr 10029487h
0x100293f7: test byte ptr [eax+8],4
0x100293fb: je near ptr 10029487h
0x10029401: mov ecx,[esp+0Ch]
0x10029405: test ecx,ecx
0x10029407: je short 10029411h
0x10029409: mov edx,[eax+84h]
0x1002940f: mov [ecx],edx
0x10029411: mov ecx,[esp+10h]
0x10029415: test ecx,ecx
0x10029417: je short 10029421h
0x10029419: mov edx,[eax+88h]
0x1002941f: mov [ecx],edx
0x10029421: mov ecx,[esp+14h]
0x10029425: test ecx,ecx
0x10029427: je short 10029431h
0x10029429: mov edx,[eax+8Ch]
0x1002942f: mov [ecx],edx
0x10029431: mov ecx,[esp+18h]
0x10029435: test ecx,ecx
0x10029437: je short 10029441h
0x10029439: mov edx,[eax+90h]
0x1002943f: mov [ecx],edx
0x10029441: mov ecx,[esp+1Ch]
0x10029445: test ecx,ecx
0x10029447: je short 10029451h
0x10029449: mov edx,[eax+94h]
0x1002944f: mov [ecx],edx
0x10029451: mov ecx,[esp+20h]
0x10029455: test ecx,ecx
0x10029457: je short 10029461h
0x10029459: mov edx,[eax+98h]
0x1002945f: mov [ecx],edx
0x10029461: mov ecx,[esp+24h]
0x10029465: test ecx,ecx
0x10029467: je short 10029471h
0x10029469: mov edx,[eax+9Ch]
0x1002946f: mov [ecx],edx
0x10029471: mov ecx,[esp+28h]
0x10029475: test ecx,ecx
0x10029477: je short 10029481h
0x10029479: mov eax,[eax+0A0h]
0x1002947f: mov [ecx],eax
0x10029481: mov eax,4
```

```
0x10029486: ret
0x10029487: xor eax,eax
0x10029489: ret
```

`png_get_cHRM_fixed` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.630 png_get_gAMA_fixed at 0x10029490 (16 instructions)

```
0x10029490: cmp dword ptr [esp+4],0
0x10029495: je short 100294B8h
0x10029497: mov eax,[esp+8]
0x1002949b: test eax,eax
0x1002949d: je short 100294B8h
0x1002949f: test byte ptr [eax+8],1
0x100294a3: je short 100294B8h
0x100294a5: mov ecx,[esp+0Ch]
0x100294a9: test ecx,ecx
0x100294ab: je short 100294B8h
0x100294ad: mov eax,[eax+28h]
0x100294b0: mov [ecx],eax
0x100294b2: mov eax,1
0x100294b7: ret
0x100294b8: xor eax,eax
0x100294ba: ret
```

`png_get_gAMA_fixed` is called by png_get_gAMA. It calls nothing else.

12.631 png_get_gAMA at 0x100294c0 (17 instructions)

```
0x100294c0: push ecx
0x100294c1: mov ecx,[esp+0Ch]
0x100294c5: mov edx,[esp+8]
0x100294c9: lea eax,[esp]
0x100294cc: push eax
0x100294cd: push ecx
0x100294ce: push edx
0x100294cf: call 10029490h
0x100294d4: add esp,0Ch
0x100294d7: test eax,eax
0x100294d9: je short 100294EAh
0x100294db: fild dword ptr [esp]
0x100294de: mov ecx,[esp+10h]
0x100294e2: fmul qword ptr ds:[100FFB90h]
0x100294e8: fstp qword ptr [ecx]
0x100294ea: pop ecx
0x100294eb: ret
```

`png_get_gAMA` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_get_gAMA_fixed (1 callees).

12.632 png_get_sRGB at 0x100294f0 (16 instructions)

```
0x100294f0: cmp dword ptr [esp+4],0
0x100294f5: je short 1002951Ch
0x100294f7: mov eax,[esp+8]
0x100294fb: test eax,eax
0x100294fd: je short 1002951Ch
0x100294ff: test dword ptr [eax+8],800h
0x10029506: je short 1002951Ch
0x10029508: mov ecx,[esp+0Ch]
0x1002950c: test ecx,ecx
0x1002950e: je short 1002951Ch
0x10029510: movzx eax,byte ptr [eax+2Ch]
```

```
0x10029514: mov [ecx],eax
0x10029516: mov eax,800h
0x1002951b: ret
0x1002951c: xor eax,eax
0x1002951e: ret
```

`png_get_sRGB` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.633 png_get_iCCP at 0x10029520 (34 instructions)

```
0x10029520: cmp dword ptr [esp+4],0
0x10029525: push esi
0x10029526: je short 1002957Fh
0x10029528: mov eax,[esp+0Ch]
0x1002952c: test eax,eax
0x1002952e: je short 1002957Fh
0x10029530: test dword ptr [eax+8],1000h
0x10029537: je short 1002957Fh
0x10029539: mov esi,[esp+10h]
0x1002953d: test esi,esi
0x1002953f: je short 1002957Fh
0x10029541: mov edx,[esp+18h]
0x10029545: test edx,edx
0x10029547: je short 1002957Fh
0x10029549: mov ecx,[esp+1Ch]
0x1002954d: test ecx,ecx
0x1002954f: je short 1002957Fh
0x10029551: push edi
0x10029552: mov edi,[eax+0C8h]
0x10029558: mov [esi],edi
0x1002955a: mov esi,[eax+0CCh]
0x10029560: mov [edx],esi
0x10029562: mov edx,[eax+0D0h]
0x10029568: mov [ecx],edx
0x1002956a: movzx eax,byte ptr [eax+0D4h]
0x10029571: mov ecx,[esp+18h]
0x10029575: pop edi
0x10029576: mov [ecx],eax
0x10029578: mov eax,1000h
0x1002957d: pop esi
0x1002957e: ret
0x1002957f: xor eax,eax
0x10029581: pop esi
0x10029582: ret
```

`png_get_iCCP` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.634 png_get_sPLT at 0x10029590 (14 instructions)

```
0x10029590: cmp dword ptr [esp+4],0
0x10029595: je short 100295B6h
0x10029597: mov eax,[esp+8]
0x1002959b: test eax,eax
0x1002959d: je short 100295B6h
0x1002959f: mov ecx,[esp+0Ch]
0x100295a3: test ecx,ecx
0x100295a5: je short 100295B6h
0x100295a7: mov edx,[eax+0D8h]
0x100295ad: mov [ecx],edx
0x100295af: mov eax,[eax+0DCh]
0x100295b5: ret
0x100295b6: xor eax,eax
0x100295b8: ret
```

`png_get_sPLT` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.635 png_get_hIST at 0x100295c0 (16 instructions)

```
0x100295c0: cmp dword ptr [esp+4],0
0x100295c5: je short 100295EBh
0x100295c7: mov eax,[esp+8]
0x100295cb: test eax,eax
0x100295cd: je short 100295EBh
0x100295cf: test byte ptr [eax+8],40h
0x100295d3: je short 100295EBh
0x100295d5: mov ecx,[esp+0Ch]
0x100295d9: test ecx,ecx
0x100295db: je short 100295EBh
0x100295dd: mov eax,[eax+80h]
0x100295e3: mov [ecx],eax
0x100295e5: mov eax,40h
0x100295ea: ret
0x100295eb: xor eax,eax
0x100295ed: ret
```

`png_get_hIST` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.636 png_get_IHDR at 0x100295f0 (73 instructions)

```
0x100295f0: push ebx
0x100295f1: mov ebx,[esp+8]
0x100295f5: test ebx,ebx
0x100295f7: push esi
0x100295f8: push edi
0x100295f9: je near ptr 100296A6h
0x100295ff: mov eax,[esp+14h]
0x10029603: test eax,eax
0x10029605: je near ptr 100296A6h
0x1002960b: mov edi,[esp+18h]
0x1002960f: test edi,edi
0x10029611: je near ptr 100296A6h
0x10029617: mov esi,[esp+1Ch]
0x1002961b: test esi,esi
0x1002961d: je near ptr 100296A6h
0x10029623: mov edx,[esp+20h]
0x10029627: test edx,edx
0x10029629: je short 100296A6h
0x1002962b: mov ecx,[esp+24h]
0x1002962f: test ecx,ecx
0x10029631: je short 100296A6h
0x10029633: push ebp
0x10029634: mov ebp,[eax]
0x10029636: mov [edi],ebp
0x10029638: mov edi,[eax+4]
0x1002963b: mov [esi],edi
0x1002963d: movzx esi,byte ptr [eax+18h]
0x10029641: mov [edx],esi
0x10029643: movzx edx,byte ptr [eax+19h]
0x10029647: mov [ecx],edx
0x10029649: mov ecx,[esp+30h]
0x1002964d: test ecx,ecx
0x1002964f: pop ebp
0x10029650: je short 10029658h
0x10029652: movzx edx,byte ptr [eax+1Ah]
0x10029656: mov [ecx],edx
0x10029658: mov ecx,[esp+30h]
0x1002965c: test ecx,ecx
0x1002965e: je short 10029666h
0x10029660: movzx edx,byte ptr [eax+1Bh]
0x10029664: mov [ecx],edx
0x10029666: mov ecx,[esp+28h]
0x1002966a: test ecx,ecx
0x1002966c: je short 10029674h
```

```
0x1002966e: movzx edx,byte ptr [eax+1Ch]
0x10029672: mov [ecx],edx
0x10029674: movzx ecx,byte ptr [eax+1Bh]
0x10029678: movzx edx,byte ptr [eax+1Ah]
0x1002967c: push ecx
0x1002967d: movzx ecx,byte ptr [eax+1Ch]
0x10029681: push edx
0x10029682: movzx edx,byte ptr [eax+19h]
0x10029686: push ecx
0x10029687: movzx ecx,byte ptr [eax+18h]
0x1002968b: push edx
0x1002968c: mov edx,[eax+4]
0x1002968f: mov eax,[eax]
0x10029691: push ecx
0x10029692: push edx
0x10029693: push eax
0x10029694: push ebx
0x10029695: call 10023C30h
0x1002969a: add esp,20h
0x1002969d: pop edi
0x1002969e: pop esi
0x1002969f: mov eax,1
0x100296a4: pop ebx
0x100296a5: ret
0x100296a6: pop edi
0x100296a7: pop esi
0x100296a8: xor eax,eax
0x100296aa: pop ebx
0x100296ab: ret
```

`png_get_IHDR` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10023c30 (1 callees).

12.637 png_get_oFFs at 0x100296b0 (31 instructions)

```
0x100296b0: cmp dword ptr [esp+4],0
0x100296b5: push esi
0x100296b6: je short 100296FAh
0x100296b8: mov eax,[esp+0Ch]
0x100296bc: test eax,eax
0x100296be: je short 100296FAh
0x100296c0: test dword ptr [eax+8],100h
0x100296c7: je short 100296FAh
0x100296c9: mov esi,[esp+10h]
0x100296cd: test esi,esi
0x100296cf: je short 100296FAh
0x100296d1: mov edx,[esp+14h]
0x100296d5: test edx,edx
0x100296d7: je short 100296FAh
0x100296d9: mov ecx,[esp+18h]
0x100296dd: test ecx,ecx
0x100296df: je short 100296FAh
0x100296e1: push edi
0x100296e2: mov edi,[eax+68h]
0x100296e5: mov [esi],edi
0x100296e7: mov esi,[eax+6Ch]
0x100296ea: mov [edx],esi
0x100296ec: movzx eax,byte ptr [eax+70h]
0x100296f0: pop edi
0x100296f1: mov [ecx],eax
0x100296f3: mov eax,100h
0x100296f8: pop esi
0x100296f9: ret
0x100296fa: xor eax,eax
0x100296fc: pop esi
0x100296fd: ret
```

`png_get_oFFs` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.638 png_get_pCAL at 0x10029700 (57 instructions)

```
0x10029700: cmp dword ptr [esp+4],0
0x10029705: push ebx
0x10029706: push esi
0x10029707: push edi
0x10029708: je near ptr 100297ACh
0x1002970e: mov eax,[esp+14h]
0x10029712: test eax,eax
0x10029714: je near ptr 100297ACh
0x1002971a: test dword ptr [eax+8],400h
0x10029721: je near ptr 100297ACh
0x10029727: mov ebx,[esp+18h]
0x1002972b: test ebx,ebx
0x1002972d: je short 100297ACh
0x1002972f: cmp dword ptr [esp+1Ch],0
0x10029734: je short 100297ACh
0x10029736: cmp dword ptr [esp+20h],0
0x1002973b: je short 100297ACh
0x1002973d: mov edi,[esp+24h]
0x10029741: test edi,edi
0x10029743: je short 100297ACh
0x10029745: mov esi,[esp+28h]
0x10029749: test esi,esi
0x1002974b: je short 100297ACh
0x1002974d: mov edx,[esp+2Ch]
0x10029751: test edx,edx
0x10029753: je short 100297ACh
0x10029755: mov ecx,[esp+30h]
0x10029759: test ecx,ecx
0x1002975b: je short 100297ACh
0x1002975d: push ebp
0x1002975e: mov ebp,[eax+0A4h]
0x10029764: mov [ebx],ebp
0x10029766: mov ebx,[eax+0A8h]
0x1002976c: mov ebp,[esp+20h]
0x10029770: mov [ebp],ebx
0x10029773: mov ebx,[eax+0ACh]
0x10029779: mov ebp,[esp+24h]
0x1002977d: mov [ebp],ebx
0x10029780: movzx ebx,byte ptr [eax+0B8h]
0x10029787: mov [edi],ebx
0x10029789: movzx edi,byte ptr [eax+0B9h]
0x10029790: mov [esi],edi
0x10029792: mov esi,[eax+0B0h]
0x10029798: pop ebp
0x10029799: mov [edx],esi
0x1002979b: mov eax,[eax+0B4h]
0x100297a1: pop edi
0x100297a2: pop esi
0x100297a3: mov [ecx],eax
0x100297a5: mov eax,400h
0x100297aa: pop ebx
0x100297ab: ret
0x100297ac: pop edi
0x100297ad: pop esi
0x100297ae: xor eax,eax
0x100297b0: pop ebx
0x100297b1: ret
```

`png_get_pCAL` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.639 png_get_sCAL_fixed at 0x100297c0 (43 instructions)

```
0x100297c0: push esi
0x100297c1: push edi
0x100297c2: mov edi,[esp+0Ch]
0x100297c6: test edi,edi
0x100297c8: je short 1002983Ah
0x100297ca: mov esi,[esp+10h]
0x100297ce: test esi,esi
0x100297d0: je short 1002983Ah
0x100297d2: test dword ptr [esi+8],4000h
0x100297d9: je short 1002983Ah
0x100297db: movzx eax,byte ptr [esi+0E0h]
0x100297e2: mov ecx,[esp+14h]
0x100297e6: mov [ecx],eax
0x100297e8: mov edx,[esi+0E4h]
0x100297ee: push 10100234h
0x100297f3: push edx
0x100297f4: call dword ptr ds:[100FE4B4h]
0x100297fa: push ecx
0x100297fb: fstp qword ptr [esp]
0x100297fe: push edi
0x100297ff: call 10024710h
0x10029804: mov ecx,[esp+28h]
0x10029808: add esp,10h
0x1002980b: mov [ecx],eax
0x1002980d: mov edx,[esi+0E8h]
0x10029813: push 10100228h
0x10029818: push edx
0x10029819: call dword ptr ds:[100FE4B4h]
0x1002981f: push ecx
0x10029820: fstp qword ptr [esp]
0x10029823: push edi
0x10029824: call 10024710h
0x10029829: mov ecx,[esp+2Ch]
0x1002982d: add esp,10h
0x10029830: pop edi
0x10029831: mov [ecx],eax
0x10029833: mov eax,4000h
0x10029838: pop esi
0x10029839: ret
0x1002983a: pop edi
0x1002983b: xor eax,eax
0x1002983d: pop esi
0x1002983e: ret
```

`png_get_sCAL_fixed` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10024710 (1 callees).

12.640 png_get_sCAL at 0x10029840 (28 instructions)

```
0x10029840: cmp dword ptr [esp+4],0
0x10029845: push esi
0x10029846: je short 10029896h
0x10029848: mov esi,[esp+0Ch]
0x1002984c: test esi,esi
0x1002984e: je short 10029896h
0x10029850: test dword ptr [esi+8],4000h
0x10029857: je short 10029896h
0x10029859: movzx eax,byte ptr [esi+0E0h]
0x10029860: mov ecx,[esp+10h]
0x10029864: mov [ecx],eax
0x10029866: mov edx,[esi+0E4h]
0x1002986c: push edx
0x1002986d: call dword ptr ds:[100FE4B4h]
0x10029873: mov eax,[esp+18h]
```

```
0x10029877: fstp qword ptr [eax]
0x10029879: mov ecx,[esi+0E8h]
0x1002987f: push ecx
0x10029880: call dword ptr ds:[100FE4B4h]
0x10029886: mov edx,[esp+20h]
0x1002988a: add esp,8
0x1002988d: fstp qword ptr [edx]
0x1002988f: mov eax,4000h
0x10029894: pop esi
0x10029895: ret
0x10029896: xor eax,eax
0x10029898: pop esi
0x10029899: ret
```

`png_get_sCAL` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.641 png_get_sCAL_s at 0x100298a0 (20 instructions)

```
0x100298a0: cmp dword ptr [esp+4],0
0x100298a5: je short 100298E3h
0x100298a7: mov eax,[esp+8]
0x100298ab: test eax,eax
0x100298ad: je short 100298E3h
0x100298af: test dword ptr [eax+8],4000h
0x100298b6: je short 100298E3h
0x100298b8: movzx ecx,byte ptr [eax+0E0h]
0x100298bf: mov edx,[esp+0Ch]
0x100298c3: mov [edx],ecx
0x100298c5: mov ecx,[eax+0E4h]
0x100298cb: mov edx,[esp+10h]
0x100298cf: mov [edx],ecx
0x100298d1: mov eax,[eax+0E8h]
0x100298d7: mov ecx,[esp+14h]
0x100298db: mov [ecx],eax
0x100298dd: mov eax,4000h
0x100298e2: ret
0x100298e3: xor eax,eax
0x100298e5: ret
```

`png_get_sCAL_s` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.642 png_get_pHYs at 0x100298f0 (29 instructions)

```
0x100298f0: xor eax,eax
0x100298f2: cmp [esp+4],eax
0x100298f6: je short 1002993Fh
0x100298f8: mov ecx,[esp+8]
0x100298fc: test ecx,ecx
0x100298fe: je short 1002993Fh
0x10029900: test byte ptr [ecx+8],80h
0x10029904: je short 1002993Fh
0x10029906: mov edx,[esp+0Ch]
0x1002990a: test edx,edx
0x1002990c: je short 10029918h
0x1002990e: mov eax,[ecx+74h]
0x10029911: mov [edx],eax
0x10029913: mov eax,80h
0x10029918: mov edx,[esp+10h]
0x1002991c: test edx,edx
0x1002991e: je short 1002992Ch
0x10029920: push esi
0x10029921: mov esi,[ecx+78h]
0x10029924: mov [edx],esi
0x10029926: or eax,80h
0x1002992b: pop esi
```

```
0x1002992c: mov edx,[esp+14h]
0x10029930: test edx,edx
0x10029932: je short 1002993Fh
0x10029934: movzx ecx,byte ptr [ecx+7Ch]
0x10029938: mov [edx],ecx
0x1002993a: or eax,80h
0x1002993f: ret
```

`png_get_pHYs` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.643 png_get_PLTE at 0x10029940 (19 instructions)

```
0x10029940: cmp dword ptr [esp+4],0
0x10029945: je short 10029972h
0x10029947: mov eax,[esp+8]
0x1002994b: test eax,eax
0x1002994d: je short 10029972h
0x1002994f: test byte ptr [eax+8],8
0x10029953: je short 10029972h
0x10029955: mov ecx,[esp+0Ch]
0x10029959: test ecx,ecx
0x1002995b: je short 10029972h
0x1002995d: mov edx,[eax+10h]
0x10029960: mov [ecx],edx
0x10029962: movzx eax,word ptr [eax+14h]
0x10029966: mov ecx,[esp+10h]
0x1002996a: mov [ecx],eax
0x1002996c: mov eax,8
0x10029971: ret
0x10029972: xor eax,eax
0x10029974: ret
```

`png_get_PLTE` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.644 png_get_sBIT at 0x10029980 (16 instructions)

```
0x10029980: cmp dword ptr [esp+4],0
0x10029985: je short 100299A8h
0x10029987: mov eax,[esp+8]
0x1002998b: test eax,eax
0x1002998d: je short 100299A8h
0x1002998f: test byte ptr [eax+8],2
0x10029993: je short 100299A8h
0x10029995: mov ecx,[esp+0Ch]
0x10029999: test ecx,ecx
0x1002999b: je short 100299A8h
0x1002999d: add eax,44h
0x100299a0: mov [ecx],eax
0x100299a2: mov eax,2
0x100299a7: ret
0x100299a8: xor eax,eax
0x100299aa: ret
```

`png_get_sBIT` is called by png_read_png. It calls nothing else.

12.645 png_get_text at 0x100299b0 (25 instructions)

```
0x100299b0: cmp dword ptr [esp+4],0
0x100299b5: je short 100299E3h
0x100299b7: mov eax,[esp+8]
0x100299bb: test eax,eax
0x100299bd: je short 100299E3h
0x100299bf: cmp dword ptr [eax+30h],0
0x100299c3: jle short 100299E3h
```

```
0x100299c5: mov ecx,[esp+0Ch]
0x100299c9: test ecx,ecx
0x100299cb: je short 100299D2h
0x100299cd: mov edx,[eax+38h]
0x100299d0: mov [ecx],edx
0x100299d2: mov ecx,[esp+10h]
0x100299d6: test ecx,ecx
0x100299d8: je short 100299DFh
0x100299da: mov edx,[eax+30h]
0x100299dd: mov [ecx],edx
0x100299df: mov eax,[eax+30h]
0x100299e2: ret
0x100299e3: mov eax,[esp+10h]
0x100299e7: test eax,eax
0x100299e9: je short 100299F1h
0x100299eb: mov dword ptr [eax],0
0x100299f1: xor eax,eax
0x100299f3: ret
```

`png_get_text` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.646 png_get_tIME at 0x10029a00 (16 instructions)

```
0x10029a00: cmp dword ptr [esp+4],0
0x10029a05: je short 10029A2Bh
0x10029a07: mov eax,[esp+8]
0x10029a0b: test eax,eax
0x10029a0d: je short 10029A2Bh
0x10029a0f: test dword ptr [eax+8],200h
0x10029a16: je short 10029A2Bh
0x10029a18: mov ecx,[esp+0Ch]
0x10029a1c: test ecx,ecx
0x10029a1e: je short 10029A2Bh
0x10029a20: add eax,3Ch
0x10029a23: mov [ecx],eax
0x10029a25: mov eax,200h
0x10029a2a: ret
0x10029a2b: xor eax,eax
0x10029a2d: ret
```

`png_get_tIME` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.647 png_get_tRNS at 0x10029a30 (41 instructions)

```
0x10029a30: xor eax,eax
0x10029a32: cmp [esp+4],eax
0x10029a36: je short 10029AA0h
0x10029a38: mov ecx,[esp+8]
0x10029a3c: test ecx,ecx
0x10029a3e: je short 10029AA0h
0x10029a40: test byte ptr [ecx+8],10h
0x10029a44: je short 10029AA0h
0x10029a46: cmp byte ptr [ecx+19h],3
0x10029a4a: jne short 10029A6Fh
0x10029a4c: mov edx,[esp+0Ch]
0x10029a50: test edx,edx
0x10029a52: je short 10029A5Eh
0x10029a54: mov eax,[ecx+50h]
0x10029a57: mov [edx],eax
0x10029a59: mov eax,10h
0x10029a5e: mov edx,[esp+14h]
0x10029a62: test edx,edx
0x10029a64: je short 10029A8Fh
0x10029a66: push esi
0x10029a67: lea esi,[ecx+54h]
```

```
0x10029a6a: mov [edx],esi
0x10029a6c: pop esi
0x10029a6d: jmp short 10029A8Fh
0x10029a6f: mov edx,[esp+14h]
0x10029a73: test edx,edx
0x10029a75: je short 10029A81h
0x10029a77: lea eax,[ecx+54h]
0x10029a7a: mov [edx],eax
0x10029a7c: mov eax,10h
0x10029a81: mov edx,[esp+0Ch]
0x10029a85: test edx,edx
0x10029a87: je short 10029A8Fh
0x10029a89: mov dword ptr [edx],0
0x10029a8f: mov edx,[esp+10h]
0x10029a93: test edx,edx
0x10029a95: je short 10029AA0h
0x10029a97: movzx ecx,word ptr [ecx+16h]
0x10029a9b: mov [edx],ecx
0x10029a9d: or eax,10h
0x10029aa0: ret
```

`png_get_tRNS` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.648 png_get_unknown_chunks at 0x10029ab0 (14 instructions)

```
0x10029ab0: cmp dword ptr [esp+4],0
0x10029ab5: je short 10029AD6h
0x10029ab7: mov eax,[esp+8]
0x10029abb: test eax,eax
0x10029abd: je short 10029AD6h
0x10029abf: mov ecx,[esp+0Ch]
0x10029ac3: test ecx,ecx
0x10029ac5: je short 10029AD6h
0x10029ac7: mov edx,[eax+0C0h]
0x10029acd: mov [ecx],edx
0x10029acf: mov eax,[eax+0C4h]
0x10029ad5: ret
0x10029ad6: xor eax,eax
0x10029ad8: ret
```

`png_get_unknown_chunks` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.649 png_get_rgb_to_gray_status at 0x10029ae0 (7 instructions)

```
0x10029ae0: mov eax,[esp+4]
0x10029ae4: test eax,eax
0x10029ae6: je short 10029AF0h
0x10029ae8: movzx eax,byte ptr [eax+228h]
0x10029aef: ret
0x10029af0: xor eax,eax
0x10029af2: ret
```

`png_get_rgb_to_gray_status` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.650 png_get_user_chunk_ptr at 0x10029b00 (7 instructions)

```
0x10029b00: mov eax,[esp+4]
0x10029b04: test eax,eax
0x10029b06: je short 10029B0Fh
0x10029b08: mov eax,[eax+218h]
0x10029b0e: ret
```

```
0x10029b0f: xor eax,eax
0x10029b11: ret
```

`png_get_user_chunk_ptr` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.651 png_get_compression_buffer_size at 0x10029b20 (7 instructions)

```
0x10029b20: mov eax,[esp+4]
0x10029b24: test eax,eax
0x10029b26: je short 10029B2Fh
0x10029b28: mov eax,[eax+0B4h]
0x10029b2e: ret
0x10029b2f: xor eax,eax
0x10029b31: ret
```

`png_get_compression_buffer_size` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.652 png_get_user_width_max at 0x10029b40 (7 instructions)

```
0x10029b40: mov eax,[esp+4]
0x10029b44: test eax,eax
0x10029b46: je short 10029B4Fh
0x10029b48: mov eax,[eax+258h]
0x10029b4e: ret
0x10029b4f: xor eax,eax
0x10029b51: ret
```

`png_get_user_width_max` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.653 png_get_user_height_max at 0x10029b60 (7 instructions)

```
0x10029b60: mov eax,[esp+4]
0x10029b64: test eax,eax
0x10029b66: je short 10029B6Fh
0x10029b68: mov eax,[eax+25Ch]
0x10029b6e: ret
0x10029b6f: xor eax,eax
0x10029b71: ret
```

`png_get_user_height_max` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.654 png_get_chunk_cache_max at 0x10029b80 (7 instructions)

```
0x10029b80: mov eax,[esp+4]
0x10029b84: test eax,eax
0x10029b86: je short 10029B8Fh
0x10029b88: mov eax,[eax+260h]
0x10029b8e: ret
0x10029b8f: xor eax,eax
0x10029b91: ret
```

`png_get_chunk_cache_max` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.655 png_get_chunk_malloc_max at 0x10029ba0 (7 instructions)

```
0x10029ba0: mov eax,[esp+4]
0x10029ba4: test eax,eax
0x10029ba6: je short 10029BAFh
0x10029ba8: mov eax,[eax+264h]
0x10029bae: ret
0x10029baf: xor eax,eax
0x10029bb1: ret
```

`png_get_chunk_malloc_max` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.656 png_get_io_state at 0x10029bc0 (3 instructions)

```
0x10029bc0: mov eax,[esp+4]
0x10029bc4: mov eax,[eax+288h]
0x10029bca: ret
```

`png_get_io_state` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.657 png_get_io_chunk_type at 0x10029bd0 (12 instructions)

```
0x10029bd0: mov ecx,[esp+4]
0x10029bd4: movzx eax,byte ptr [ecx+11Ch]
0x10029bdb: movzx edx,byte ptr [ecx+11Dh]
0x10029be2: shl eax,8
0x10029be5: add eax,edx
0x10029be7: movzx edx,byte ptr [ecx+11Eh]
0x10029bee: movzx ecx,byte ptr [ecx+11Fh]
0x10029bf5: shl eax,8
0x10029bf8: add eax,edx
0x10029bfa: shl eax,8
0x10029bfd: add eax,ecx
0x10029bff: ret
```

`png_get_io_chunk_type` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.658 png_get_io_chunk_name at 0x10029c00 (3 instructions)

```
0x10029c00: mov eax,[esp+4]
0x10029c04: add eax,11Ch
0x10029c09: ret
```

`png_get_io_chunk_name` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.659 sub_10029c10 at 0x10029c10 (10 instructions)

```
0x10029c10: mov ecx,[esp+4]
0x10029c14: mov eax,[ecx+54h]
0x10029c17: test eax,eax
0x10029c19: je short 10029C21h
0x10029c1b: mov [esp+4],ecx
0x10029c1f: jmp eax
0x10029c21: push 10100240h
0x10029c26: push ecx
0x10029c27: call 100231E0h
0x10029c2c: int 3
```

`sub_10029c10` is called by sub_1002cea0, sub_1002cf30, sub_1002cf60, sub_1002dfd0. It works with the text "Call to NULL read function". It delegates to png_error (1 callees).

12.660 png_set_read_fn at 0x10029c70 (22 instructions)

```
0x10029c70: push esi
0x10029c71: mov esi,[esp+8]
0x10029c75: test esi,esi
0x10029c77: je short 10029CB9h
0x10029c79: mov eax,[esp+0Ch]
0x10029c7d: mov [esi+58h],eax
0x10029c80: mov eax,[esp+10h]
0x10029c84: test eax,eax
0x10029c86: je short 10029C8Dh
0x10029c88: mov [esi+54h],eax
0x10029c8b: jmp short 10029C94h
0x10029c8d: mov dword ptr [esi+54h],10029C30h
0x10029c94: cmp dword ptr [esi+50h],0
0x10029c98: je short 10029CAFh
0x10029c9a: push 10100260h
0x10029c9f: push esi
0x10029ca0: mov dword ptr [esi+50h],0
0x10029ca7: call 100230F0h
0x10029cac: add esp,8
0x10029caf: mov dword ptr [esi+14Ch],0
0x10029cb9: pop esi
0x10029cba: ret
```

`png_set_read_fn` is called by sub_10018d20, png_create_read_struct_2. It delegates to png_warning (1 callees).

12.661 png_write_info_before_PLTE at 0x10029cc0 (148 instructions)

```
0x10029cc0: push ebp
0x10029cc1: mov ebp,[esp+8]
0x10029cc5: test ebp,ebp
0x10029cc7: je near ptr 10029E88h
0x10029ccd: push esi
0x10029cce: mov esi,[esp+10h]
0x10029cd2: test esi,esi
0x10029cd4: je near ptr 10029E87h
0x10029cda: test dword ptr [ebp+6Ch],400h
0x10029ce1: jne near ptr 10029E87h
0x10029ce7: push edi
0x10029ce8: push ebp
0x10029ce9: call 100359D0h
0x10029cee: mov edi,1000h
0x10029cf3: add esp,4
0x10029cf6: test [ebp+6Ch],edi
0x10029cf9: je short 10029D1Ch
0x10029cfb: cmp dword ptr [ebp+230h],0
0x10029d02: je short 10029D1Ch
0x10029d04: push 100FF928h
0x10029d09: push ebp
0x10029d0a: call 100230F0h
0x10029d0f: add esp,8
0x10029d12: mov dword ptr [ebp+230h],0
0x10029d1c: movzx eax,byte ptr [esi+1Ch]
0x10029d20: movzx ecx,byte ptr [esi+1Bh]
0x10029d24: movzx edx,byte ptr [esi+1Ah]
0x10029d28: push eax
0x10029d29: movzx eax,byte ptr [esi+19h]
0x10029d2d: push ecx
0x10029d2e: movzx ecx,byte ptr [esi+18h]
0x10029d32: push edx
0x10029d33: mov edx,[esi+4]
0x10029d36: push eax
```

```
0x10029d37: mov eax,[esi]
0x10029d39: push ecx
0x10029d3a: push edx
0x10029d3b: push eax
0x10029d3c: push ebp
0x10029d3d: call 10036C30h
0x10029d42: add esp,20h
0x10029d45: test byte ptr [esi+8],1
0x10029d49: je short 10029D58h
0x10029d4b: mov ecx,[esi+28h]
0x10029d4e: push ecx
0x10029d4f: push ebp
0x10029d50: call 100371C0h
0x10029d55: add esp,8
0x10029d58: test dword ptr [esi+8],800h
0x10029d5f: je short 10029D6Fh
0x10029d61: movzx edx,byte ptr [esi+2Ch]
0x10029d65: push edx
0x10029d66: push ebp
0x10029d67: call 10037220h
0x10029d6c: add esp,8
0x10029d6f: test [esi+8],edi
0x10029d72: je short 10029D94h
0x10029d74: mov eax,[esi+0D0h]
0x10029d7a: mov ecx,[esi+0CCh]
0x10029d80: mov edx,[esi+0C8h]
0x10029d86: push eax
0x10029d87: push ecx
0x10029d88: push 0
0x10029d8a: push edx
0x10029d8b: push ebp
0x10029d8c: call 10037290h
0x10029d91: add esp,14h
0x10029d94: test byte ptr [esi+8],2
0x10029d98: je short 10029DACh
0x10029d9a: movzx eax,byte ptr [esi+19h]
0x10029d9e: push eax
0x10029d9f: lea ecx,[esi+44h]
0x10029da2: push ecx
0x10029da3: push ebp
0x10029da4: call 100375C0h
0x10029da9: add esp,0Ch
0x10029dac: test byte ptr [esi+8],4
0x10029db0: je short 10029DF3h
0x10029db2: mov edx,[esi+0A0h]
0x10029db8: mov eax,[esi+9Ch]
0x10029dbe: mov ecx,[esi+98h]
0x10029dc4: push edx
0x10029dc5: mov edx,[esi+94h]
0x10029dcb: push eax
0x10029dcc: mov eax,[esi+90h]
0x10029dd2: push ecx
0x10029dd3: mov ecx,[esi+8Ch]
0x10029dd9: push edx
0x10029dda: mov edx,[esi+88h]
0x10029de0: push eax
0x10029de1: mov eax,[esi+84h]
0x10029de7: push ecx
0x10029de8: push edx
0x10029de9: push eax
0x10029dea: push ebp
0x10029deb: call 100376C0h
0x10029df0: add esp,24h
0x10029df3: mov eax,[esi+0C4h]
0x10029df9: test eax,eax
0x10029dfb: je near ptr 10029E7Fh
0x10029e01: mov edi,[esi+0C0h]
```

```
0x10029e07: lea ecx,[eax+eax*4]
0x10029e0a: lea edx,[edi+ecx*4]
0x10029e0d: cmp edi,edx
0x10029e0f: jae short 10029E7Fh
0x10029e11: push edi
0x10029e12: push ebp
0x10029e13: call 10023B10h
0x10029e18: add esp,8
0x10029e1b: cmp eax,1
0x10029e1e: je short 10029E66h
0x10029e20: mov cl,[edi+10h]
0x10029e23: test cl,cl
0x10029e25: je short 10029E66h
0x10029e27: test cl,6
0x10029e2a: jne short 10029E66h
0x10029e2c: test byte ptr [edi+3],20h
0x10029e30: jne short 10029E40h
0x10029e32: cmp eax,3
0x10029e35: je short 10029E40h
0x10029e37: test dword ptr [ebp+70h],10000h
0x10029e3e: je short 10029E66h
0x10029e40: cmp dword ptr [edi+0Ch],0
0x10029e44: jne short 10029E54h
0x10029e46: push 101002A4h
0x10029e4b: push ebp
0x10029e4c: call 100230F0h
0x10029e51: add esp,8
0x10029e54: mov eax,[edi+0Ch]
0x10029e57: mov ecx,[edi+8]
0x10029e5a: push eax
0x10029e5b: push ecx
0x10029e5c: push edi
0x10029e5d: push ebp
0x10029e5e: call 10036BF0h
0x10029e63: add esp,10h
0x10029e66: mov eax,[esi+0C4h]
0x10029e6c: lea edx,[eax+eax*4]
0x10029e6f: mov eax,[esi+0C0h]
0x10029e75: add edi,14h
0x10029e78: lea ecx,[eax+edx*4]
0x10029e7b: cmp edi,ecx
0x10029e7d: jb short 10029E11h
0x10029e7f: or dword ptr [ebp+6Ch],400h
0x10029e86: pop edi
0x10029e87: pop esi
0x10029e88: pop ebp
0x10029e89: ret
```

`png_write_info_before_PLTE` is called by png_write_info. It works with the text "MNG features are not allowed in a PNG datastream". It delegates to png_warning, png_handle_as_unknown, png_write_sig (10 callees).

12.662 png_write_info at 0x10029e90 (264 instructions)

```
0x10029e90: push ebp
0x10029e91: mov ebp,[esp+8]
0x10029e95: test ebp,ebp
0x10029e97: je near ptr 1002A18Eh
0x10029e9d: push esi
0x10029e9e: mov esi,[esp+10h]
0x10029ea2: test esi,esi
0x10029ea4: je near ptr 1002A18Dh
0x10029eaa: push esi
0x10029eab: push ebp
0x10029eac: call 10029CC0h
0x10029eb1: add esp,8
0x10029eb4: test byte ptr [esi+8],8
```

```
0x10029eb8: je near ptr 1002A098h
0x10029ebe: movzx eax,word ptr [esi+14h]
0x10029ec2: mov ecx,[esi+10h]
0x10029ec5: push eax
0x10029ec6: push ecx
0x10029ec7: push ebp
0x10029ec8: call 10035E80h
0x10029ecd: add esp,0Ch
0x10029ed0: test byte ptr [esi+8],10h
0x10029ed4: je short 10029F23h
0x10029ed6: test dword ptr [ebp+74h],80000h
0x10029edd: je short 10029F08h
0x10029edf: cmp byte ptr [esi+19h],3
0x10029ee3: jne short 10029F08h
0x10029ee5: xor eax,eax
0x10029ee7: cmp [esi+16h],ax
0x10029eeb: jbe short 10029F08h
0x10029eed: lea ecx,[ecx]
0x10029ef0: mov edx,[esi+50h]
0x10029ef3: lea ecx,[edx+eax]
0x10029ef6: or dl,0FFh
0x10029ef9: sub dl,[ecx]
0x10029efb: add eax,1
0x10029efe: mov [ecx],dl
0x10029f00: movzx ecx,word ptr [esi+16h]
0x10029f04: cmp eax,ecx
0x10029f06: jl short 10029EF0h
0x10029f08: movzx edx,byte ptr [esi+19h]
0x10029f0c: movzx eax,word ptr [esi+16h]
0x10029f10: push edx
0x10029f11: mov edx,[esi+50h]
0x10029f14: push eax
0x10029f15: lea ecx,[esi+54h]
0x10029f18: push ecx
0x10029f19: push edx
0x10029f1a: push ebp
0x10029f1b: call 100377C0h
0x10029f20: add esp,14h
0x10029f23: test byte ptr [esi+8],20h
0x10029f27: je short 10029F3Bh
0x10029f29: movzx eax,byte ptr [esi+19h]
0x10029f2d: push eax
0x10029f2e: lea ecx,[esi+5Eh]
0x10029f31: push ecx
0x10029f32: push ebp
0x10029f33: call 10037930h
0x10029f38: add esp,0Ch
0x10029f3b: test byte ptr [esi+8],40h
0x10029f3f: je short 10029F56h
0x10029f41: movzx edx,word ptr [esi+14h]
0x10029f45: mov eax,[esi+80h]
0x10029f4b: push edx
0x10029f4c: push eax
0x10029f4d: push ebp
0x10029f4e: call 10035F80h
0x10029f53: add esp,0Ch
0x10029f56: test dword ptr [esi+8],100h
0x10029f5d: je short 10029F75h
0x10029f5f: movzx ecx,byte ptr [esi+70h]
0x10029f63: mov edx,[esi+6Ch]
0x10029f66: mov eax,[esi+68h]
0x10029f69: push ecx
0x10029f6a: push edx
0x10029f6b: push eax
0x10029f6c: push ebp
0x10029f6d: call 10037AB0h
0x10029f72: add esp,10h
```

```
0x10029f75: test dword ptr [esi+8],400h
0x10029f7c: je short 10029FBAh
0x10029f7e: mov ecx,[esi+0B4h]
0x10029f84: mov edx,[esi+0B0h]
0x10029f8a: movzx eax,byte ptr [esi+0B9h]
0x10029f91: push ecx
0x10029f92: movzx ecx,byte ptr [esi+0B8h]
0x10029f99: push edx
0x10029f9a: mov edx,[esi+0ACh]
0x10029fa0: push eax
0x10029fa1: mov eax,[esi+0A8h]
0x10029fa7: push ecx
0x10029fa8: mov ecx,[esi+0A4h]
0x10029fae: push edx
0x10029faf: push eax
0x10029fb0: push ecx
0x10029fb1: push ebp
0x10029fb2: call 10036610h
0x10029fb7: add esp,20h
0x10029fba: test dword ptr [esi+8],4000h
0x10029fc1: je short 10029FE2h
0x10029fc3: mov edx,[esi+0E8h]
0x10029fc9: mov eax,[esi+0E4h]
0x10029fcf: movzx ecx,byte ptr [esi+0E0h]
0x10029fd6: push edx
0x10029fd7: push eax
0x10029fd8: push ecx
0x10029fd9: push ebp
0x10029fda: call 10037B40h
0x10029fdf: add esp,10h
0x10029fe2: test byte ptr [esi+8],80h
0x10029fe6: je short 10029FFEh
0x10029fe8: movzx edx,byte ptr [esi+7Ch]
0x10029fec: mov eax,[esi+78h]
0x10029fef: mov ecx,[esi+74h]
0x10029ff2: push edx
0x10029ff3: push eax
0x10029ff4: push ecx
0x10029ff5: push ebp
0x10029ff6: call 10037C20h
0x10029ffb: add esp,10h
0x10029ffe: push edi
0x10029fff: mov edi,200h
0x1002a004: test [esi+8],edi
0x1002a007: je short 1002A019h
0x1002a009: lea edx,[esi+3Ch]
0x1002a00c: push edx
0x1002a00d: push ebp
0x1002a00e: call 10037CB0h
0x1002a013: add esp,8
0x1002a016: or [ebp+6Ch],edi
0x1002a019: test dword ptr [esi+8],2000h
0x1002a020: push ebx
0x1002a021: je short 1002A050h
0x1002a023: xor edi,edi
0x1002a025: cmp [esi+0DCh],edi
0x1002a02b: jle short 1002A050h
0x1002a02d: xor ebx,ebx
0x1002a02f: nop
0x1002a030: mov eax,[esi+0D8h]
0x1002a036: add eax,ebx
0x1002a038: push eax
0x1002a039: push ebp
0x1002a03a: call 10037430h
0x1002a03f: add edi,1
0x1002a042: add esp,8
0x1002a045: add ebx,10h
```

```
0x1002a048: cmp edi,[esi+0DCh]
0x1002a04e: jl short 1002A030h
0x1002a050: xor ebx,ebx
0x1002a052: cmp [esi+30h],ebx
0x1002a055: jle near ptr 1002A10Eh
0x1002a05b: xor edi,edi
0x1002a05d: lea ecx,[ecx]
0x1002a060: mov ecx,[esi+38h]
0x1002a063: mov eax,[edi+ecx]
0x1002a066: test eax,eax
0x1002a068: jle short 1002A0ADh
0x1002a06a: mov edx,ecx
0x1002a06c: mov ecx,[edi+edx+8]
0x1002a070: lea eax,[edi+edx]
0x1002a073: mov edx,[eax+18h]
0x1002a076: push ecx
0x1002a077: mov ecx,[eax+14h]
0x1002a07a: push edx
0x1002a07b: mov edx,[eax+4]
0x1002a07e: mov eax,[eax]
0x1002a080: push ecx
0x1002a081: push edx
0x1002a082: push eax
0x1002a083: push ebp
0x1002a084: call 10036450h
0x1002a089: mov ecx,[esi+38h]
0x1002a08c: add esp,18h
0x1002a08f: mov dword ptr [edi+ecx],0FFFFFFFDh
0x1002a096: jmp short 1002A0FFh
0x1002a098: cmp byte ptr [esi+19h],3
0x1002a09c: jne near ptr 10029ED0h
0x1002a0a2: push 101002C8h
0x1002a0a7: push ebp
0x1002a0a8: call 100231E0h
0x1002a0ad: jne short 1002A0D7h
0x1002a0af: mov edx,ecx
0x1002a0b1: mov ecx,[edi+edx]
0x1002a0b4: lea eax,[edi+edx]
0x1002a0b7: mov edx,[eax+8]
0x1002a0ba: mov eax,[eax+4]
0x1002a0bd: push ecx
0x1002a0be: push 0
0x1002a0c0: push edx
0x1002a0c1: push eax
0x1002a0c2: push ebp
0x1002a0c3: call 10036310h
0x1002a0c8: mov ecx,[esi+38h]
0x1002a0cb: add esp,14h
0x1002a0ce: mov dword ptr [edi+ecx],0FFFFFFFEh
0x1002a0d5: jmp short 1002A0FFh
0x1002a0d7: cmp eax,0FFFFFFFFh
0x1002a0da: jne short 1002A0FFh
0x1002a0dc: mov edx,ecx
0x1002a0de: mov ecx,[edi+edx+8]
0x1002a0e2: lea eax,[edi+edx]
0x1002a0e5: mov edx,[eax+4]
0x1002a0e8: push 0
0x1002a0ea: push ecx
0x1002a0eb: push edx
0x1002a0ec: push ebp
0x1002a0ed: call 10036250h
0x1002a0f2: mov eax,[esi+38h]
0x1002a0f5: add esp,10h
0x1002a0f8: mov dword ptr [edi+eax],0FFFFFFFDh
0x1002a0ff: add ebx,1
0x1002a102: add edi,1Ch
0x1002a105: cmp ebx,[esi+30h]
```

```
0x1002a108: jl near ptr 1002A060h
0x1002a10e: mov eax,[esi+0C4h]
0x1002a114: test eax,eax
0x1002a116: je short 1002A18Bh
0x1002a118: mov edi,[esi+0C0h]
0x1002a11e: lea ecx,[eax+eax*4]
0x1002a121: lea edx,[edi+ecx*4]
0x1002a124: cmp edi,edx
0x1002a126: jae short 1002A18Bh
0x1002a128: mov ebx,10000h
0x1002a12d: lea ecx,[ecx]
0x1002a130: push edi
0x1002a131: push ebp
0x1002a132: call 10023B10h
0x1002a137: add esp,8
0x1002a13a: cmp eax,1
0x1002a13d: je short 1002A172h
0x1002a13f: mov cl,[edi+10h]
0x1002a142: test cl,cl
0x1002a144: je short 1002A172h
0x1002a146: test cl,2
0x1002a149: je short 1002A172h
0x1002a14b: test cl,4
0x1002a14e: jne short 1002A172h
0x1002a150: test byte ptr [edi+3],20h
0x1002a154: jne short 1002A160h
0x1002a156: cmp eax,3
0x1002a159: je short 1002A160h
0x1002a15b: test [ebp+70h],ebx
0x1002a15e: je short 1002A172h
0x1002a160: mov eax,[edi+0Ch]
0x1002a163: mov ecx,[edi+8]
0x1002a166: push eax
0x1002a167: push ecx
0x1002a168: push edi
0x1002a169: push ebp
0x1002a16a: call 10036BF0h
0x1002a16f: add esp,10h
0x1002a172: mov eax,[esi+0C4h]
0x1002a178: lea edx,[eax+eax*4]
0x1002a17b: mov eax,[esi+0C0h]
0x1002a181: add edi,14h
0x1002a184: lea ecx,[eax+edx*4]
0x1002a187: cmp edi,ecx
0x1002a189: jb short 1002A130h
0x1002a18b: pop ebx
0x1002a18c: pop edi
0x1002a18d: pop esi
0x1002a18e: pop ebp
0x1002a18f: ret
```

`png_write_info` is called by png_write_png. It works with the text "Valid palette required for paletted images". It delegates to png_error, png_handle_as_unknown, png_write_info_before_PLTE (17 callees).

12.663 png_write_end at 0x1002a190 (135 instructions)

```
0x1002a190: push ebp
0x1002a191: mov ebp,[esp+8]
0x1002a195: test ebp,ebp
0x1002a197: je near ptr 1002A306h
0x1002a19d: mov eax,[ebp+6Ch]
0x1002a1a0: test al,4
0x1002a1a2: jne short 1002A1AFh
0x1002a1a4: push 101002F4h
0x1002a1a9: push ebp
0x1002a1aa: call 100231E0h
```

```
0x1002a1af: push esi
0x1002a1b0: mov esi,[esp+10h]
0x1002a1b4: test esi,esi
0x1002a1b6: je near ptr 1002A2F8h
0x1002a1bc: test dword ptr [esi+8],200h
0x1002a1c3: je short 1002A1D9h
0x1002a1c5: test eax,200h
0x1002a1ca: jne short 1002A1D9h
0x1002a1cc: lea eax,[esi+3Ch]
0x1002a1cf: push eax
0x1002a1d0: push ebp
0x1002a1d1: call 10037CB0h
0x1002a1d6: add esp,8
0x1002a1d9: push ebx
0x1002a1da: xor ebx,ebx
0x1002a1dc: cmp [esi+30h],ebx
0x1002a1df: push edi
0x1002a1e0: jle near ptr 1002A281h
0x1002a1e6: xor edi,edi
0x1002a1e8: mov ecx,[esi+38h]
0x1002a1eb: mov eax,[edi+ecx]
0x1002a1ee: test eax,eax
0x1002a1f0: jle short 1002A220h
0x1002a1f2: mov edx,ecx
0x1002a1f4: mov ecx,[edi+edx+8]
0x1002a1f8: lea eax,[edi+edx]
0x1002a1fb: mov edx,[eax+18h]
0x1002a1fe: push ecx
0x1002a1ff: mov ecx,[eax+14h]
0x1002a202: push edx
0x1002a203: mov edx,[eax+4]
0x1002a206: mov eax,[eax]
0x1002a208: push ecx
0x1002a209: push edx
0x1002a20a: push eax
0x1002a20b: push ebp
0x1002a20c: call 10036450h
0x1002a211: mov ecx,[esi+38h]
0x1002a214: add esp,18h
0x1002a217: mov dword ptr [edi+ecx],0FFFFFFFDh
0x1002a21e: jmp short 1002A272h
0x1002a220: jl short 1002A24Ah
0x1002a222: mov edx,ecx
0x1002a224: mov ecx,[edi+edx]
0x1002a227: lea eax,[edi+edx]
0x1002a22a: mov edx,[eax+8]
0x1002a22d: mov eax,[eax+4]
0x1002a230: push ecx
0x1002a231: push 0
0x1002a233: push edx
0x1002a234: push eax
0x1002a235: push ebp
0x1002a236: call 10036310h
0x1002a23b: mov ecx,[esi+38h]
0x1002a23e: add esp,14h
0x1002a241: mov dword ptr [edi+ecx],0FFFFFFFEh
0x1002a248: jmp short 1002A272h
0x1002a24a: cmp eax,0FFFFFFFFh
0x1002a24d: jne short 1002A272h
0x1002a24f: mov edx,ecx
0x1002a251: mov ecx,[edi+edx+8]
0x1002a255: lea eax,[edi+edx]
0x1002a258: mov edx,[eax+4]
0x1002a25b: push 0
0x1002a25d: push ecx
0x1002a25e: push edx
0x1002a25f: push ebp
```

```
0x1002a260: call 10036250h
0x1002a265: mov eax,[esi+38h]
0x1002a268: add esp,10h
0x1002a26b: mov dword ptr [edi+eax],0FFFFFFFDh
0x1002a272: add ebx,1
0x1002a275: add edi,1Ch
0x1002a278: cmp ebx,[esi+30h]
0x1002a27b: jl near ptr 1002A1E8h
0x1002a281: mov eax,[esi+0C4h]
0x1002a287: test eax,eax
0x1002a289: je short 1002A2F6h
0x1002a28b: mov edi,[esi+0C0h]
0x1002a291: lea ecx,[eax+eax*4]
0x1002a294: lea edx,[edi+ecx*4]
0x1002a297: cmp edi,edx
0x1002a299: jae short 1002A2F6h
0x1002a29b: mov ebx,10000h
0x1002a2a0: push edi
0x1002a2a1: push ebp
0x1002a2a2: call 10023B10h
0x1002a2a7: add esp,8
0x1002a2aa: cmp eax,1
0x1002a2ad: je short 1002A2DDh
0x1002a2af: mov cl,[edi+10h]
0x1002a2b2: test cl,cl
0x1002a2b4: je short 1002A2DDh
0x1002a2b6: test cl,8
0x1002a2b9: je short 1002A2DDh
0x1002a2bb: test byte ptr [edi+3],20h
0x1002a2bf: jne short 1002A2CBh
0x1002a2c1: cmp eax,3
0x1002a2c4: je short 1002A2CBh
0x1002a2c6: test [ebp+70h],ebx
0x1002a2c9: je short 1002A2DDh
0x1002a2cb: mov eax,[edi+0Ch]
0x1002a2ce: mov ecx,[edi+8]
0x1002a2d1: push eax
0x1002a2d2: push ecx
0x1002a2d3: push edi
0x1002a2d4: push ebp
0x1002a2d5: call 10036BF0h
0x1002a2da: add esp,10h
0x1002a2dd: mov eax,[esi+0C4h]
0x1002a2e3: lea edx,[eax+eax*4]
0x1002a2e6: mov eax,[esi+0C0h]
0x1002a2ec: add edi,14h
0x1002a2ef: lea ecx,[eax+edx*4]
0x1002a2f2: cmp edi,ecx
0x1002a2f4: jb short 1002A2A0h
0x1002a2f6: pop edi
0x1002a2f7: pop ebx
0x1002a2f8: or dword ptr [ebp+6Ch],8
0x1002a2fc: push ebp
0x1002a2fd: call 10037160h
0x1002a302: add esp,4
0x1002a305: pop esi
0x1002a306: pop ebp
0x1002a307: ret
```

`png_write_end` is called by png_write_png. It works with the text "No IDATs written into file". It delegates to png_error, png_handle_as_unknown, sub_10036250 (8 callees).

12.664 png_convert_from_struct_tm at 0x1002a310 (17 instructions)

```
0x1002a310: mov eax,[esp+8]
0x1002a314: mov dx,[eax+14h]
```

```
0x1002a318: mov ecx,[esp+4]
0x1002a31c: add dx,76Ch
0x1002a321: mov [ecx],dx
0x1002a324: movzx edx,byte ptr [eax+10h]
0x1002a328: add dl,1
0x1002a32b: mov [ecx+2],dl
0x1002a32e: movzx edx,byte ptr [eax+0Ch]
0x1002a332: mov [ecx+3],dl
0x1002a335: movzx edx,byte ptr [eax+8]
0x1002a339: mov [ecx+4],dl
0x1002a33c: movzx edx,byte ptr [eax+4]
0x1002a340: mov [ecx+5],dl
0x1002a343: mov al,[eax]
0x1002a345: mov [ecx+6],al
0x1002a348: ret
```

`png_convert_from_struct_tm` is called by png_convert_from_time_t. It calls nothing else.

12.665 png_convert_from_time_t at 0x1002a350 (9 instructions)

```
0x1002a350: lea eax,[esp+8]
0x1002a354: push eax
0x1002a355: call dword ptr ds:[100FE4B0h]
0x1002a35b: mov ecx,[esp+8]
0x1002a35f: push eax
0x1002a360: push ecx
0x1002a361: call 1002A310h
0x1002a366: add esp,0Ch
0x1002a369: ret
```

`png_convert_from_time_t` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_convert_from_struct_tm (1 callees).

12.666 png_write_row at 0x1002a370 (109 instructions)

```
0x1002a370: push esi
0x1002a371: mov esi,[esp+8]
0x1002a375: test esi,esi
0x1002a377: je near ptr 1002A59Eh
0x1002a37d: cmp dword ptr [esi+0E4h],0
0x1002a384: jne short 1002A3ACh
0x1002a386: cmp byte ptr [esi+124h],0
0x1002a38d: jne short 1002A3ACh
0x1002a38f: test dword ptr [esi+6Ch],400h
0x1002a396: jne short 1002A3A3h
0x1002a398: push 10100310h
0x1002a39d: push esi
0x1002a39e: call 100231E0h
0x1002a3a3: push esi
0x1002a3a4: call 100367C0h
0x1002a3a9: add esp,4
0x1002a3ac: cmp byte ptr [esi+123h],0
0x1002a3b3: je near ptr 1002A48Dh
0x1002a3b9: test byte ptr [esi+74h],2
0x1002a3bd: je near ptr 1002A48Dh
0x1002a3c3: movzx eax,byte ptr [esi+124h]
0x1002a3ca: cmp eax,6
0x1002a3cd: ja near ptr 1002A48Dh
0x1002a3d3: jmp dword ptr [eax*4+1002A5A0h]
0x1002a48d: movzx edx,byte ptr [esi+126h]
0x1002a494: mov al,[esi+128h]
0x1002a49a: mov ecx,[esi+0D8h]
0x1002a4a0: mov [esi+108h],dl
0x1002a4a6: mov dl,[esi+12Bh]
0x1002a4ac: mov [esi+109h],al
```

```
0x1002a4b2: imul dl
0x1002a4b4: push edi
0x1002a4b5: lea edi,[esi+100h]
0x1002a4bb: mov [esi+10Bh],al
0x1002a4c1: cmp al,8
0x1002a4c3: mov [edi],ecx
0x1002a4c5: mov [esi+10Ah],dl
0x1002a4cb: movzx eax,al
0x1002a4ce: jb short 1002A4D8h
0x1002a4d0: shr eax,3
0x1002a4d3: imul eax,ecx
0x1002a4d6: jmp short 1002A4E1h
0x1002a4d8: imul eax,ecx
0x1002a4db: add eax,7
0x1002a4de: shr eax,3
0x1002a4e1: mov ecx,[esi+0ECh]
0x1002a4e7: push eax
0x1002a4e8: mov [esi+104h],eax
0x1002a4ee: mov eax,[esp+14h]
0x1002a4f2: push eax
0x1002a4f3: add ecx,1
0x1002a4f6: push ecx
0x1002a4f7: call 100228C6h
0x1002a4fc: add esp,0Ch
0x1002a4ff: cmp byte ptr [esi+123h],0
0x1002a506: je short 1002A540h
0x1002a508: mov al,[esi+124h]
0x1002a50e: cmp al,6
0x1002a510: jae short 1002A540h
0x1002a512: test byte ptr [esi+74h],2
0x1002a516: je short 1002A540h
0x1002a518: movzx edx,al
0x1002a51b: mov eax,[esi+0ECh]
0x1002a521: push edx
0x1002a522: add eax,1
0x1002a525: push eax
0x1002a526: push edi
0x1002a527: call 10036940h
0x1002a52c: add esp,0Ch
0x1002a52f: cmp dword ptr [edi],0
0x1002a532: jne short 1002A540h
0x1002a534: push esi
0x1002a535: call 10037D80h
0x1002a53a: add esp,4
0x1002a53d: pop edi
0x1002a53e: pop esi
0x1002a53f: ret
0x1002a540: cmp dword ptr [esi+74h],0
0x1002a544: je short 1002A54Fh
0x1002a546: push esi
0x1002a547: call 10039380h
0x1002a54c: add esp,4
0x1002a54f: test byte ptr [esi+230h],4
0x1002a556: je short 1002A574h
0x1002a558: cmp byte ptr [esi+234h],40h
0x1002a55f: jne short 1002A574h
0x1002a561: mov ecx,[esi+0ECh]
0x1002a567: add ecx,1
0x1002a56a: push ecx
0x1002a56b: push edi
0x1002a56c: call 100392B0h
0x1002a571: add esp,8
0x1002a574: push edi
0x1002a575: push esi
0x1002a576: call 100380E0h
0x1002a57b: mov eax,[esi+19Ch]
0x1002a581: add esp,8
```

```
0x1002a584: test eax,eax
0x1002a586: je short 1002A59Dh
0x1002a588: movzx edx,byte ptr [esi+124h]
0x1002a58f: mov ecx,[esi+0E4h]
0x1002a595: push edx
0x1002a596: push ecx
0x1002a597: push esi
0x1002a598: call eax
0x1002a59a: add esp,0Ch
0x1002a59d: pop edi
0x1002a59e: pop esi
0x1002a59f: ret
```

`png_write_row` is called by png_write_rows, png_write_image. It works with the text "png_write_info was never called before png_write_row". It delegates to j_memcpy, png_error, sub_100367c0 (8 callees).

12.667 png_set_flush at 0x1002a5c0 (11 instructions)

```
0x1002a5c0: mov ecx,[esp+4]
0x1002a5c4: test ecx,ecx
0x1002a5c6: je short 1002A5DEh
0x1002a5c8: mov eax,[esp+8]
0x1002a5cc: xor edx,edx
0x1002a5ce: test eax,eax
0x1002a5d0: setl dl
0x1002a5d3: sub edx,1
0x1002a5d6: and edx,eax
0x1002a5d8: mov [ecx+150h],edx
0x1002a5de: ret
```

`png_set_flush` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.668 png_write_flush at 0x1002a5e0 (60 instructions)

```
0x1002a5e0: push esi
0x1002a5e1: mov esi,[esp+8]
0x1002a5e5: test esi,esi
0x1002a5e7: je near ptr 1002A6B6h
0x1002a5ed: mov eax,[esi+0E4h]
0x1002a5f3: cmp eax,[esi+0D4h]
0x1002a5f9: jae near ptr 1002A6B6h
0x1002a5ff: push edi
0x1002a600: lea edi,[esi+78h]
0x1002a603: push 2
0x1002a605: push edi
0x1002a606: call 10062A80h
0x1002a60b: add esp,8
0x1002a60e: test eax,eax
0x1002a610: jne short 1002A64Bh
0x1002a612: cmp [esi+88h],eax
0x1002a618: jne short 1002A667h
0x1002a61a: mov ecx,[esi+0B4h]
0x1002a620: mov edx,[esi+0B0h]
0x1002a626: push ecx
0x1002a627: push edx
0x1002a628: push esi
0x1002a629: call 10037020h
0x1002a62e: mov eax,[esi+0B0h]
0x1002a634: mov ecx,[esi+0B4h]
0x1002a63a: add esp,0Ch
0x1002a63d: mov [esi+84h],eax
0x1002a643: mov [esi+88h],ecx
0x1002a649: jmp short 1002A603h
0x1002a64b: mov eax,[esi+90h]
0x1002a651: test eax,eax
```

```
0x1002a653: je short 1002A65Ch
0x1002a655: push eax
0x1002a656: push esi
0x1002a657: call 100231E0h
0x1002a65c: push 10100348h
0x1002a661: push esi
0x1002a662: call 100231E0h
0x1002a667: mov eax,[esi+0B4h]
0x1002a66d: mov ecx,[esi+88h]
0x1002a673: cmp eax,ecx
0x1002a675: je short 1002A6A2h
0x1002a677: mov edx,[esi+0B0h]
0x1002a67d: sub eax,ecx
0x1002a67f: push eax
0x1002a680: push edx
0x1002a681: push esi
0x1002a682: call 10037020h
0x1002a687: mov eax,[esi+0B0h]
0x1002a68d: mov ecx,[esi+0B4h]
0x1002a693: add esp,0Ch
0x1002a696: mov [esi+84h],eax
0x1002a69c: mov [esi+88h],ecx
0x1002a6a2: push esi
0x1002a6a3: mov dword ptr [esi+154h],0
0x1002a6ad: call 1002CA90h
0x1002a6b2: add esp,4
0x1002a6b5: pop edi
0x1002a6b6: pop esi
0x1002a6b7: ret
```

`png_write_flush` is called by sub_10037fe0. It works with the text "zlib error". It delegates to png_error, sub_1002ca90, sub_10037020 (4 callees).

12.669 png_set_filter at 0x1002a6c0 (111 instructions)

```
0x1002a6c0: push esi
0x1002a6c1: mov esi,[esp+8]
0x1002a6c5: test esi,esi
0x1002a6c7: je near ptr 1002A874h
0x1002a6cd: test byte ptr [esi+230h],4
0x1002a6d4: mov eax,[esp+0Ch]
0x1002a6d8: je short 1002A6DFh
0x1002a6da: cmp eax,40h
0x1002a6dd: je short 1002A6E7h
0x1002a6df: test eax,eax
0x1002a6e1: jne near ptr 1002A876h
0x1002a6e7: mov ecx,[esp+10h]
0x1002a6eb: mov eax,ecx
0x1002a6ed: and eax,0FFh
0x1002a6f2: cmp eax,7
0x1002a6f5: ja short 1002A739h
0x1002a6f7: jmp dword ptr [eax*4+1002A884h]
0x1002a739: mov [esi+125h],cl
0x1002a73f: cmp dword ptr [esi+0ECh],0
0x1002a746: je near ptr 1002A874h
0x1002a74c: test byte ptr [esi+125h],10h
0x1002a753: je short 1002A77Ah
0x1002a755: cmp dword ptr [esi+0F0h],0
0x1002a75c: jne short 1002A77Ah
0x1002a75e: mov eax,[esi+0DCh]
0x1002a764: add eax,1
0x1002a767: push eax
0x1002a768: push esi
0x1002a769: call 1002CD20h
0x1002a76e: mov [esi+0F0h],eax
0x1002a774: add esp,8
```

```
0x1002a777: mov byte ptr [eax],1
0x1002a77a: test byte ptr [esi+125h],20h
0x1002a781: je short 1002A7C8h
0x1002a783: cmp dword ptr [esi+0F4h],0
0x1002a78a: jne short 1002A7C8h
0x1002a78c: cmp dword ptr [esi+0E8h],0
0x1002a793: jne short 1002A7ACh
0x1002a795: push 101003C4h
0x1002a79a: push esi
0x1002a79b: call 100230F0h
0x1002a7a0: add esp,8
0x1002a7a3: and byte ptr [esi+125h],0DFh
0x1002a7aa: jmp short 1002A7C8h
0x1002a7ac: mov ecx,[esi+0DCh]
0x1002a7b2: add ecx,1
0x1002a7b5: push ecx
0x1002a7b6: push esi
0x1002a7b7: call 1002CD20h
0x1002a7bc: mov [esi+0F4h],eax
0x1002a7c2: add esp,8
0x1002a7c5: mov byte ptr [eax],2
0x1002a7c8: test byte ptr [esi+125h],40h
0x1002a7cf: je short 1002A816h
0x1002a7d1: cmp dword ptr [esi+0F8h],0
0x1002a7d8: jne short 1002A816h
0x1002a7da: cmp dword ptr [esi+0E8h],0
0x1002a7e1: jne short 1002A7FAh
0x1002a7e3: push 1010039Ch
0x1002a7e8: push esi
0x1002a7e9: call 100230F0h
0x1002a7ee: add esp,8
0x1002a7f1: and byte ptr [esi+125h],0BFh
0x1002a7f8: jmp short 1002A816h
0x1002a7fa: mov edx,[esi+0DCh]
0x1002a800: add edx,1
0x1002a803: push edx
0x1002a804: push esi
0x1002a805: call 1002CD20h
0x1002a80a: mov [esi+0F8h],eax
0x1002a810: add esp,8
0x1002a813: mov byte ptr [eax],3
0x1002a816: test byte ptr [esi+125h],80h
0x1002a81d: je short 1002A864h
0x1002a81f: cmp dword ptr [esi+0FCh],0
0x1002a826: jne short 1002A864h
0x1002a828: cmp dword ptr [esi+0E8h],0
0x1002a82f: jne short 1002A848h
0x1002a831: push 10100374h
0x1002a836: push esi
0x1002a837: call 100230F0h
0x1002a83c: add esp,8
0x1002a83f: and byte ptr [esi+125h],7Fh
0x1002a846: jmp short 1002A864h
0x1002a848: mov eax,[esi+0DCh]
0x1002a84e: add eax,1
0x1002a851: push eax
0x1002a852: push esi
0x1002a853: call 1002CD20h
0x1002a858: mov [esi+0FCh],eax
0x1002a85e: add esp,8
0x1002a861: mov byte ptr [eax],4
0x1002a864: cmp byte ptr [esi+125h],0
0x1002a86b: jne short 1002A874h
0x1002a86d: mov byte ptr [esi+125h],8
0x1002a874: pop esi
0x1002a875: ret
0x1002a876: push 10100354h
```

```
0x1002a87b: push esi
0x1002a87c: call 100231E0h
0x1002a881: lea ecx,[ecx]
0x1002a884: or al,0A7h
0x1002a886: add dl,[eax]
0x1002a888: adc eax,1E1002A7h
0x1002a88d: cmpsd
0x1002a88e: add dl,[eax]
0x1002a890: daa
0x1002a891: cmpsd
0x1002a892: add dl,[eax]
0x1002a894: xor [edi-5901EFFEh],ah
0x1002a89a: add dl,[eax]
```

`png_set_filter` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "Unknown custom filter method". It delegates to png_warning, png_error, png_malloc (3 callees).

12.670 sub_1002a8b0 at 0x1002a8b0 (30 instructions)

```
0x1002a8b0: mov eax,[esi+1FCh]
0x1002a8b6: push ebx
0x1002a8b7: xor ebx,ebx
0x1002a8b9: cmp eax,ebx
0x1002a8bb: mov [esi+1F9h],bl
0x1002a8c1: mov byte ptr [esi+1F8h],1
0x1002a8c8: je short 1002A8DAh
0x1002a8ca: push eax
0x1002a8cb: push esi
0x1002a8cc: mov [esi+1FCh],ebx
0x1002a8d2: call 1002CD70h
0x1002a8d7: add esp,8
0x1002a8da: mov eax,[esi+200h]
0x1002a8e0: cmp eax,ebx
0x1002a8e2: je short 1002A8F4h
0x1002a8e4: push eax
0x1002a8e5: push esi
0x1002a8e6: mov [esi+200h],ebx
0x1002a8ec: call 1002CD70h
0x1002a8f1: add esp,8
0x1002a8f4: mov eax,[esi+204h]
0x1002a8fa: cmp eax,ebx
0x1002a8fc: je short 1002A90Eh
0x1002a8fe: push eax
0x1002a8ff: push esi
0x1002a900: mov [esi+204h],ebx
0x1002a906: call 1002CD70h
0x1002a90b: add esp,8
0x1002a90e: pop ebx
0x1002a90f: ret
```

`sub_1002a8b0` is called by sub_1002a910, png_create_write_struct_2, sub_1002b220. It delegates to png_free (1 callees).

12.671 sub_1002a910 at 0x1002a910 (96 instructions)

```
0x1002a910: push esi
0x1002a911: mov esi,ecx
0x1002a913: test esi,esi
0x1002a915: push edi
0x1002a916: mov edi,eax
0x1002a918: je near ptr 1002AA6Ah
0x1002a91e: call 1002A8B0h
0x1002a923: cmp edi,2
0x1002a926: jne near ptr 1002AA53h
0x1002a92c: test ebx,ebx
```

```
0x1002a92e: jle near ptr 1002A9B5h
0x1002a934: push ebx
0x1002a935: push esi
0x1002a936: call 1002CD20h
0x1002a93b: mov [esi+1FCh],eax
0x1002a941: add esp,8
0x1002a944: xor eax,eax
0x1002a946: test ebx,ebx
0x1002a948: jle short 1002A961h
0x1002a94a: lea ebx,[ebx]
0x1002a950: mov ecx,[esi+1FCh]
0x1002a956: mov byte ptr [eax+ecx],0FFh
0x1002a95a: add eax,1
0x1002a95d: cmp eax,ebx
0x1002a95f: jl short 1002A950h
0x1002a961: lea edi,[ebx+ebx]
0x1002a964: push edi
0x1002a965: push esi
0x1002a966: call 1002CD20h
0x1002a96b: push edi
0x1002a96c: push esi
0x1002a96d: mov [esi+200h],eax
0x1002a973: call 1002CD20h
0x1002a978: mov [esi+204h],eax
0x1002a97e: add esp,10h
0x1002a981: xor eax,eax
0x1002a983: test ebx,ebx
0x1002a985: jle short 1002A9AFh
0x1002a987: jmp short 1002A990h
0x1002a990: mov edx,[esi+200h]
0x1002a996: mov word ptr [edx+eax*2],100h
0x1002a99c: mov ecx,[esi+204h]
0x1002a9a2: mov word ptr [ecx+eax*2],100h
0x1002a9a8: add eax,1
0x1002a9ab: cmp eax,ebx
0x1002a9ad: jl short 1002A990h
0x1002a9af: mov [esi+1F9h],bl
0x1002a9b5: cmp dword ptr [esi+208h],0
0x1002a9bc: jne short 1002A9DDh
0x1002a9be: push 0Ah
0x1002a9c0: push esi
0x1002a9c1: call 1002CD20h
0x1002a9c6: push 0Ah
0x1002a9c8: push esi
0x1002a9c9: mov [esi+208h],eax
0x1002a9cf: call 1002CD20h
0x1002a9d4: add esp,10h
0x1002a9d7: mov [esi+20Ch],eax
0x1002a9dd: mov edx,[esi+208h]
0x1002a9e3: mov eax,8
0x1002a9e8: mov [edx],ax
0x1002a9eb: mov ecx,[esi+20Ch]
0x1002a9f1: mov [ecx],ax
0x1002a9f4: mov edx,[esi+208h]
0x1002a9fa: mov [edx+2],ax
0x1002a9fe: mov ecx,[esi+20Ch]
0x1002aa04: mov [ecx+2],ax
0x1002aa08: mov edx,[esi+208h]
0x1002aa0e: mov [edx+4],ax
0x1002aa12: mov ecx,[esi+20Ch]
0x1002aa18: mov [ecx+4],ax
0x1002aa1c: mov edx,[esi+208h]
0x1002aa22: mov [edx+6],ax
0x1002aa26: mov ecx,[esi+20Ch]
0x1002aa2c: mov [ecx+6],ax
0x1002aa30: mov edx,[esi+208h]
0x1002aa36: mov [edx+8],ax
```

```
0x1002aa3a: mov ecx,[esi+20Ch]
0x1002aa40: mov [ecx+8],ax
0x1002aa44: mov byte ptr [esi+1F8h],2
0x1002aa4b: pop edi
0x1002aa4c: mov eax,1
0x1002aa51: pop esi
0x1002aa52: ret
0x1002aa53: test edi,edi
0x1002aa55: je short 1002AA4Bh
0x1002aa57: cmp edi,1
0x1002aa5a: je short 1002AA4Bh
0x1002aa5c: push 10100408h
0x1002aa61: push esi
0x1002aa62: call 100230F0h
0x1002aa67: add esp,8
0x1002aa6a: pop edi
0x1002aa6b: xor eax,eax
0x1002aa6d: pop esi
0x1002aa6e: ret
```

`sub_1002a910` is called by png_set_filter_heuristics, png_set_filter_heuristics_fixed. It works with the text "Unknown filter heuristic method". It delegates to png_warning, sub_1002a8b0, png_malloc (3 callees).

12.672 png_set_filter_heuristics at 0x1002aa70 (113 instructions)

```
0x1002aa70: sub esp,8
0x1002aa73: push ebx
0x1002aa74: mov ebx,[esp+18h]
0x1002aa78: push esi
0x1002aa79: mov esi,[esp+14h]
0x1002aa7d: push edi
0x1002aa7e: mov edi,[esp+1Ch]
0x1002aa82: mov eax,edi
0x1002aa84: mov ecx,esi
0x1002aa86: call 1002A910h
0x1002aa8b: test eax,eax
0x1002aa8d: je near ptr 1002ABEDh
0x1002aa93: cmp edi,2
0x1002aa96: jne near ptr 1002ABEDh
0x1002aa9c: fld qword ptr ds:[100FE7C8h]
0x1002aaa2: xor ecx,ecx
0x1002aaa4: test ebx,ebx
0x1002aaa6: jle near ptr 1002AB59h
0x1002aaac: fldz
0x1002aaae: mov edx,[esp+24h]
0x1002aab2: fld qword ptr ds:[100FE920h]
0x1002aab8: jmp short 1002AABCh
0x1002aaba: fxch
0x1002aabc: fxch
0x1002aabe: fcom qword ptr [edx+ecx*8]
0x1002aac1: fnstsw ax
0x1002aac3: test ah,1
0x1002aac6: jne short 1002AAE2h
0x1002aac8: mov eax,[esi+200h]
0x1002aace: mov word ptr [eax+ecx*2],100h
0x1002aad4: mov eax,[esi+204h]
0x1002aada: mov word ptr [eax+ecx*2],100h
0x1002aae0: jmp short 1002AB4Ah
0x1002aae2: fld qword ptr [edx+ecx*8]
0x1002aae5: mov edi,[esi+204h]
0x1002aaeb: fnstcw [esp+0Eh]
0x1002aaef: movzx eax,word ptr [esp+0Eh]
0x1002aaf4: fmul st,st(2)
0x1002aaf6: or eax,0C00h
0x1002aafb: mov [esp+10h],eax
0x1002aaff: fadd st,st(3)
```

```
0x1002ab01: fldcw [esp+10h]
0x1002ab05: fistp dword ptr [esp+10h]
0x1002ab09: movzx eax,word ptr [esp+10h]
0x1002ab0e: mov [edi+ecx*2],ax
0x1002ab12: mov edi,[esi+200h]
0x1002ab18: fldcw [esp+0Eh]
0x1002ab1c: fld st(1)
0x1002ab1e: fdiv qword ptr [edx+ecx*8]
0x1002ab21: fnstcw [esp+0Eh]
0x1002ab25: fadd st,st(3)
0x1002ab27: movzx eax,word ptr [esp+0Eh]
0x1002ab2c: or eax,0C00h
0x1002ab31: mov [esp+10h],eax
0x1002ab35: fldcw [esp+10h]
0x1002ab39: fistp dword ptr [esp+10h]
0x1002ab3d: movzx eax,word ptr [esp+10h]
0x1002ab42: mov [edi+ecx*2],ax
0x1002ab46: fldcw [esp+0Eh]
0x1002ab4a: add ecx,1
0x1002ab4d: cmp ecx,ebx
0x1002ab4f: jl near ptr 1002AABAh
0x1002ab55: fstp st(1)
0x1002ab57: fstp st(0)
0x1002ab59: fld1
0x1002ab5b: mov edx,[esp+28h]
0x1002ab5f: fld qword ptr ds:[100FF128h]
0x1002ab65: xor ecx,ecx
0x1002ab67: jmp short 1002AB6Bh
0x1002ab69: fxch
0x1002ab6b: fxch
0x1002ab6d: fcom qword ptr [edx]
0x1002ab6f: fnstsw ax
0x1002ab71: test ah,41h
0x1002ab74: jp short 1002ABDCh
0x1002ab76: fld st(1)
0x1002ab78: mov edi,[esi+20Ch]
0x1002ab7e: fdiv qword ptr [edx]
0x1002ab80: fnstcw [esp+0Eh]
0x1002ab84: fadd st,st(3)
0x1002ab86: movzx eax,word ptr [esp+0Eh]
0x1002ab8b: or eax,0C00h
0x1002ab90: mov [esp+10h],eax
0x1002ab94: fldcw [esp+10h]
0x1002ab98: fistp dword ptr [esp+10h]
0x1002ab9c: movzx eax,word ptr [esp+10h]
0x1002aba1: mov [ecx+edi],ax
0x1002aba5: mov edi,[esi+208h]
0x1002abab: fldcw [esp+0Eh]
0x1002abaf: fld qword ptr [edx]
0x1002abb1: fnstcw [esp+0Eh]
0x1002abb5: fmul st,st(2)
0x1002abb7: movzx eax,word ptr [esp+0Eh]
0x1002abbc: or eax,0C00h
0x1002abc1: mov [esp+10h],eax
0x1002abc5: fadd st,st(3)
0x1002abc7: fldcw [esp+10h]
0x1002abcb: fistp dword ptr [esp+10h]
0x1002abcf: movzx eax,word ptr [esp+10h]
0x1002abd4: mov [ecx+edi],ax
0x1002abd8: fldcw [esp+0Eh]
0x1002abdc: add ecx,2
0x1002abdf: add edx,8
0x1002abe2: cmp ecx,0Ah
0x1002abe5: jl short 1002AB69h
0x1002abe7: fstp st(2)
0x1002abe9: fstp st(0)
0x1002abeb: fstp st(0)
```

```
0x1002abed: pop edi
0x1002abee: pop esi
0x1002abef: pop ebx
0x1002abf0: add esp,8
0x1002abf3: ret
```

`png_set_filter_heuristics` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_1002a910 (1 callees).

12.673 png_set_filter_heuristics_fixed at 0x1002ac00 (165 instructions)

```
0x1002ac00: push ebx
0x1002ac01: mov ebx,[esp+10h]
0x1002ac05: push esi
0x1002ac06: mov esi,[esp+0Ch]
0x1002ac0a: push edi
0x1002ac0b: mov edi,[esp+14h]
0x1002ac0f: mov eax,edi
0x1002ac11: mov ecx,esi
0x1002ac13: call 1002A910h
0x1002ac18: test eax,eax
0x1002ac1a: je near ptr 1002AE0Fh
0x1002ac20: cmp edi,2
0x1002ac23: jne near ptr 1002AE0Fh
0x1002ac29: xor ecx,ecx
0x1002ac2b: test ebx,ebx
0x1002ac2d: jle short 1002ACA2h
0x1002ac2f: push ebp
0x1002ac30: mov ebp,[esp+20h]
0x1002ac34: cmp dword ptr [ebp+ecx*4],0
0x1002ac39: jg short 1002AC55h
0x1002ac3b: mov eax,[esi+200h]
0x1002ac41: mov word ptr [eax+ecx*2],100h
0x1002ac47: mov edx,[esi+204h]
0x1002ac4d: mov word ptr [edx+ecx*2],100h
0x1002ac53: jmp short 1002AC9Ah
0x1002ac55: mov edx,[ebp+ecx*4]
0x1002ac59: shl edx,8
0x1002ac5c: add edx,0C350h
0x1002ac62: mov eax,14F8B589h
0x1002ac67: imul edx
0x1002ac69: sar edx,0Dh
0x1002ac6c: mov eax,edx
0x1002ac6e: shr eax,1Fh
0x1002ac71: add eax,edx
0x1002ac73: mov edx,[esi+204h]
0x1002ac79: mov [edx+ecx*2],ax
0x1002ac7d: mov edi,[ebp+ecx*4]
0x1002ac81: mov eax,edi
0x1002ac83: cdq
0x1002ac84: sub eax,edx
0x1002ac86: sar eax,1
0x1002ac88: add eax,186A000h
0x1002ac8d: cdq
0x1002ac8e: idiv edi
0x1002ac90: mov edx,[esi+200h]
0x1002ac96: mov [edx+ecx*2],ax
0x1002ac9a: add ecx,1
0x1002ac9d: cmp ecx,ebx
0x1002ac9f: jl short 1002AC34h
0x1002aca1: pop ebp
0x1002aca2: mov ecx,[esp+20h]
0x1002aca6: mov edi,[ecx]
0x1002aca8: cmp edi,186A0h
0x1002acae: jl short 1002ACEBh
0x1002acb0: mov eax,edi
```

```
0x1002acb2: cdq
0x1002acb3: sub eax,edx
0x1002acb5: sar eax,1
0x1002acb7: add eax,0C3500h
0x1002acbc: xor edx,edx
0x1002acbe: div edi
0x1002acc0: mov edx,[esi+20Ch]
0x1002acc6: mov [edx],ax
0x1002acc9: mov edi,[ecx]
0x1002accb: lea edi,[edi*8+0C350h]
0x1002acd2: mov eax,4F8B588Fh
0x1002acd7: mul edi
0x1002acd9: mov eax,[esi+208h]
0x1002acdf: sub edi,edx
0x1002ace1: shr edi,1
0x1002ace3: add edi,edx
0x1002ace5: shr edi,10h
0x1002ace8: mov [eax],di
0x1002aceb: mov edi,[ecx+4]
0x1002acee: cmp edi,186A0h
0x1002acf4: jl short 1002AD34h
0x1002acf6: mov eax,edi
0x1002acf8: cdq
0x1002acf9: sub eax,edx
0x1002acfb: sar eax,1
0x1002acfd: add eax,0C3500h
0x1002ad02: xor edx,edx
0x1002ad04: div edi
0x1002ad06: mov edx,[esi+20Ch]
0x1002ad0c: mov [edx+2],ax
0x1002ad10: mov edi,[ecx+4]
0x1002ad13: lea edi,[edi*8+0C350h]
0x1002ad1a: mov eax,4F8B588Fh
0x1002ad1f: mul edi
0x1002ad21: mov eax,[esi+208h]
0x1002ad27: sub edi,edx
0x1002ad29: shr edi,1
0x1002ad2b: add edi,edx
0x1002ad2d: shr edi,10h
0x1002ad30: mov [eax+2],di
0x1002ad34: mov edi,[ecx+8]
0x1002ad37: cmp edi,186A0h
0x1002ad3d: jl short 1002AD7Dh
0x1002ad3f: mov eax,edi
0x1002ad41: cdq
0x1002ad42: sub eax,edx
0x1002ad44: sar eax,1
0x1002ad46: add eax,0C3500h
0x1002ad4b: xor edx,edx
0x1002ad4d: div edi
0x1002ad4f: mov edx,[esi+20Ch]
0x1002ad55: mov [edx+4],ax
0x1002ad59: mov edi,[ecx+8]
0x1002ad5c: lea edi,[edi*8+0C350h]
0x1002ad63: mov eax,4F8B588Fh
0x1002ad68: mul edi
0x1002ad6a: mov eax,[esi+208h]
0x1002ad70: sub edi,edx
0x1002ad72: shr edi,1
0x1002ad74: add edi,edx
0x1002ad76: shr edi,10h
0x1002ad79: mov [eax+4],di
0x1002ad7d: mov edi,[ecx+0Ch]
0x1002ad80: cmp edi,186A0h
0x1002ad86: jl short 1002ADC6h
0x1002ad88: mov eax,edi
0x1002ad8a: cdq
```

```
0x1002ad8b: sub eax,edx
0x1002ad8d: sar eax,1
0x1002ad8f: add eax,0C3500h
0x1002ad94: xor edx,edx
0x1002ad96: div edi
0x1002ad98: mov edx,[esi+20Ch]
0x1002ad9e: mov [edx+6],ax
0x1002ada2: mov edi,[ecx+0Ch]
0x1002ada5: lea edi,[edi*8+0C350h]
0x1002adac: mov eax,4F8B588Fh
0x1002adb1: mul edi
0x1002adb3: mov eax,[esi+208h]
0x1002adb9: sub edi,edx
0x1002adbb: shr edi,1
0x1002adbd: add edi,edx
0x1002adbf: shr edi,10h
0x1002adc2: mov [eax+6],di
0x1002adc6: mov edi,[ecx+10h]
0x1002adc9: cmp edi,186A0h
0x1002adcf: jl short 1002AE0Fh
0x1002add1: mov eax,edi
0x1002add3: cdq
0x1002add4: sub eax,edx
0x1002add6: sar eax,1
0x1002add8: add eax,0C3500h
0x1002addd: xor edx,edx
0x1002addf: div edi
0x1002ade1: mov edx,[esi+20Ch]
0x1002ade7: mov [edx+8],ax
0x1002adeb: mov ecx,[ecx+10h]
0x1002adee: lea ecx,[ecx*8+0C350h]
0x1002adf5: mov eax,4F8B588Fh
0x1002adfa: mul ecx
0x1002adfc: mov eax,[esi+208h]
0x1002ae02: sub ecx,edx
0x1002ae04: shr ecx,1
0x1002ae06: add ecx,edx
0x1002ae08: shr ecx,10h
0x1002ae0b: mov [eax+8],cx
0x1002ae0f: pop edi
0x1002ae10: pop esi
0x1002ae11: pop ebx
0x1002ae12: ret
```

`png_set_filter_heuristics_fixed` has no recorded callers; it is likely reached through the vector table or an indirect jump. It accesses memory-mapped peripheral registers at 0x4f8b588f. It delegates to sub_1002a910 (1 callees).

12.674 png_set_compression_level at 0x1002ae20 (7 instructions)

```
0x1002ae20: mov eax,[esp+4]
0x1002ae24: test eax,eax
0x1002ae26: je short 1002AE36h
0x1002ae28: mov ecx,[esp+8]
0x1002ae2c: or dword ptr [eax+70h],2
0x1002ae30: mov [eax+0B8h],ecx
0x1002ae36: ret
```

`png_set_compression_level` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.675 png_set_compression_mem_level at 0x1002ae40 (7 instructions)

```
0x1002ae40: mov eax,[esp+4]
0x1002ae44: test eax,eax
0x1002ae46: je short 1002AE56h
```

```
0x1002ae48: mov ecx,[esp+8]
0x1002ae4c: or dword ptr [eax+70h],4
0x1002ae50: mov [eax+0C4h],ecx
0x1002ae56: ret
```

`png_set_compression_mem_level` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.676 png_set_compression_strategy at 0x1002ae60 (7 instructions)

```
0x1002ae60: mov eax,[esp+4]
0x1002ae64: test eax,eax
0x1002ae66: je short 1002AE76h
0x1002ae68: mov ecx,[esp+8]
0x1002ae6c: or dword ptr [eax+70h],1
0x1002ae70: mov [eax+0C8h],ecx
0x1002ae76: ret
```

`png_set_compression_strategy` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.677 png_set_compression_window_bits at 0x1002ae80 (28 instructions)

```
0x1002ae80: push esi
0x1002ae81: mov esi,[esp+8]
0x1002ae85: test esi,esi
0x1002ae87: je short 1002AED0h
0x1002ae89: push edi
0x1002ae8a: mov edi,[esp+10h]
0x1002ae8e: cmp edi,0Fh
0x1002ae91: jle short 1002AE9Ah
0x1002ae93: push 10100488h
0x1002ae98: jmp short 1002AEA4h
0x1002ae9a: cmp edi,8
0x1002ae9d: jge short 1002AEB0h
0x1002ae9f: push 10100454h
0x1002aea4: push esi
0x1002aea5: call 100230F0h
0x1002aeaa: add esp,8
0x1002aead: cmp edi,8
0x1002aeb0: jne short 1002AEC5h
0x1002aeb2: push 10100428h
0x1002aeb7: push esi
0x1002aeb8: call 100230F0h
0x1002aebd: add esp,8
0x1002aec0: mov edi,9
0x1002aec5: or dword ptr [esi+70h],8
0x1002aec9: mov [esi+0C0h],edi
0x1002aecf: pop edi
0x1002aed0: pop esi
0x1002aed1: ret
```

`png_set_compression_window_bits` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "Compression window is being reset to 512". It delegates to png_warning (1 callees).

12.678 png_set_compression_method at 0x1002aee0 (17 instructions)

```
0x1002aee0: push esi
0x1002aee1: mov esi,[esp+8]
0x1002aee5: test esi,esi
0x1002aee7: je short 1002AF0Ch
0x1002aee9: push edi
0x1002aeea: mov edi,[esp+10h]
```

```
0x1002aeee: cmp edi,8
0x1002aef1: je short 1002AF01h
0x1002aef3: push 101004BCh
0x1002aef8: push esi
0x1002aef9: call 100230F0h
0x1002aefe: add esp,8
0x1002af01: or dword ptr [esi+70h],10h
0x1002af05: mov [esi+0BCh],edi
0x1002af0b: pop edi
0x1002af0c: pop esi
0x1002af0d: ret
```

`png_set_compression_method` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "Only compression method 8 is supported by PNG". It delegates to png_warning (1 callees).

12.679 png_set_write_status_fn at 0x1002af10 (6 instructions)

```
0x1002af10: mov eax,[esp+4]
0x1002af14: test eax,eax
0x1002af16: je short 1002AF22h
0x1002af18: mov ecx,[esp+8]
0x1002af1c: mov [eax+19Ch],ecx
0x1002af22: ret
```

`png_set_write_status_fn` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.680 png_set_write_user_transform_fn at 0x1002af30 (7 instructions)

```
0x1002af30: mov eax,[esp+4]
0x1002af34: test eax,eax
0x1002af36: je short 1002AF46h
0x1002af38: mov ecx,[esp+8]
0x1002af3c: or dword ptr [eax+74h],100000h
0x1002af43: mov [eax+60h],ecx
0x1002af46: ret
```

`png_set_write_user_transform_fn` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.681 png_create_write_struct_2 at 0x1002af50 (192 instructions)

```
0x1002af50: push ebp
0x1002af51: mov ebp,esp
0x1002af53: sub esp,78h
0x1002af56: mov eax,ds:[10136220h]
0x1002af5b: xor eax,ebp
0x1002af5d: mov [ebp-4],eax
0x1002af60: mov eax,[ebp+8]
0x1002af63: mov ecx,[ebp+0Ch]
0x1002af66: mov edx,[ebp+10h]
0x1002af69: mov [ebp-60h],eax
0x1002af6c: mov eax,[ebp+14h]
0x1002af6f: mov [ebp-78h],eax
0x1002af72: mov eax,[ebp+18h]
0x1002af75: push esi
0x1002af76: mov [ebp-70h],ecx
0x1002af79: mov ecx,[ebp+1Ch]
0x1002af7c: push eax
0x1002af7d: mov [ebp-6Ch],edx
0x1002af80: mov edx,[ebp+20h]
0x1002af83: push ecx
```

```
0x1002af84: mov [ebp-68h],eax
0x1002af87: mov [ebp-74h],ecx
0x1002af8a: mov [ebp-64h],edx
0x1002af8d: mov dword ptr [ebp-5Ch],0
0x1002af94: push 1
0x1002af96: call 1002CB30h
0x1002af9b: mov [ebp-58h],eax
0x1002af9e: mov eax,[ebp-58h]
0x1002afa1: add esp,0Ch
0x1002afa4: test eax,eax
0x1002afa6: je near ptr 1002B14Eh
0x1002afac: mov ecx,[ebp-58h]
0x1002afaf: mov edx,[ebp-58h]
0x1002afb2: mov eax,0F4240h
0x1002afb7: mov [ecx+258h],eax
0x1002afbd: mov ecx,[ebp-58h]
0x1002afc0: mov [edx+25Ch],eax
0x1002afc6: mov eax,ds:[100FE4E0h]
0x1002afcb: push 40h
0x1002afcd: push eax
0x1002afce: push ecx
0x1002afcf: call 10023030h
0x1002afd4: push 0
0x1002afd6: push eax
0x1002afd7: call 100229DCh
0x1002afdc: add esp,14h
0x1002afdf: test eax,eax
0x1002afe1: je short 1002AFEBh
0x1002afe3: push 0
0x1002afe5: call dword ptr ds:[100FE0ACh]
0x1002afeb: mov edx,[ebp-64h]
0x1002afee: mov eax,[ebp-74h]
0x1002aff1: mov ecx,[ebp-68h]
0x1002aff4: push edx
0x1002aff5: mov edx,[ebp-58h]
0x1002aff8: push eax
0x1002aff9: push ecx
0x1002affa: push edx
0x1002affb: call 1002CCA0h
0x1002b000: mov eax,[ebp-78h]
0x1002b003: mov ecx,[ebp-6Ch]
0x1002b006: mov edx,[ebp-70h]
0x1002b009: push eax
0x1002b00a: mov eax,[ebp-58h]
0x1002b00d: push ecx
0x1002b00e: push edx
0x1002b00f: push eax
0x1002b010: call 100230C0h
0x1002b015: add esp,20h
0x1002b018: cmp dword ptr [ebp-60h],0
0x1002b01c: je short 1002B053h
0x1002b01e: xor esi,esi
0x1002b020: push 0
0x1002b022: call 10023AF0h
0x1002b027: mov ecx,[ebp-60h]
0x1002b02a: mov dl,[ecx+esi]
0x1002b02d: add esp,4
0x1002b030: cmp dl,[eax+esi]
0x1002b033: je short 1002B03Fh
0x1002b035: mov eax,[ebp-58h]
0x1002b038: or dword ptr [eax+70h],20000h
0x1002b03f: push 0
0x1002b041: call 10023AF0h
0x1002b046: mov al,[eax+esi]
0x1002b049: add esp,4
0x1002b04c: add esi,1
0x1002b04f: test al,al
```

```
0x1002b051: jne short 1002B020h
0x1002b053: mov ecx,[ebp-58h]
0x1002b056: test dword ptr [ecx+70h],20000h
0x1002b05d: je short 1002B0CDh
0x1002b05f: mov esi,[ebp-60h]
0x1002b062: test esi,esi
0x1002b064: je short 1002B0C6h
0x1002b066: push 0
0x1002b068: call 10023AF0h
0x1002b06d: mov cl,[esi]
0x1002b06f: add esp,4
0x1002b072: cmp cl,[eax]
0x1002b074: jne short 1002B098h
0x1002b076: cmp cl,31h
0x1002b079: jne short 1002B08Dh
0x1002b07b: push 0
0x1002b07d: call 10023AF0h
0x1002b082: mov dl,[esi+2]
0x1002b085: add esp,4
0x1002b088: cmp dl,[eax+2]
0x1002b08b: jne short 1002B098h
0x1002b08d: cmp byte ptr [esi],30h
0x1002b090: jne short 1002B0CDh
0x1002b092: cmp byte ptr [esi+2],39h
0x1002b096: jge short 1002B0CDh
0x1002b098: test esi,esi
0x1002b09a: je short 1002B0C6h
0x1002b09c: push 0
0x1002b09e: call 10023AF0h
0x1002b0a3: push eax
0x1002b0a4: push esi
0x1002b0a5: push 100FFD18h
0x1002b0aa: lea eax,[ebp-54h]
0x1002b0ad: push 50h
0x1002b0af: push eax
0x1002b0b0: call dword ptr ds:[100FE4C0h]
0x1002b0b6: mov edx,[ebp-58h]
0x1002b0b9: lea ecx,[ebp-54h]
0x1002b0bc: push ecx
0x1002b0bd: push edx
0x1002b0be: call 100230F0h
0x1002b0c3: add esp,20h
0x1002b0c6: mov dword ptr [ebp-5Ch],1
0x1002b0cd: mov eax,[ebp-58h]
0x1002b0d0: mov ecx,[ebp-5Ch]
0x1002b0d3: test ecx,ecx
0x1002b0d5: mov dword ptr [eax+0B4h],2000h
0x1002b0df: jne short 1002B113h
0x1002b0e1: mov edx,[ebp-58h]
0x1002b0e4: mov eax,[edx+0B4h]
0x1002b0ea: mov ecx,[ebp-58h]
0x1002b0ed: push eax
0x1002b0ee: push ecx
0x1002b0ef: call 1002CDA0h
0x1002b0f4: mov edx,[ebp-58h]
0x1002b0f7: mov [edx+0B0h],eax
0x1002b0fd: mov eax,[ebp-58h]
0x1002b100: add esp,8
0x1002b103: cmp dword ptr [eax+0B0h],0
0x1002b10a: jne short 1002B113h
0x1002b10c: mov dword ptr [ebp-5Ch],1
0x1002b113: mov ecx,[ebp-5Ch]
0x1002b116: test ecx,ecx
0x1002b118: je short 1002B15Fh
0x1002b11a: mov edx,[ebp-58h]
0x1002b11d: mov eax,[edx+0B0h]
0x1002b123: mov ecx,[ebp-58h]
```

```
0x1002b126: push eax
0x1002b127: push ecx
0x1002b128: call 1002CD70h
0x1002b12d: mov edx,[ebp-58h]
0x1002b130: mov eax,[ebp-68h]
0x1002b133: mov ecx,[ebp-64h]
0x1002b136: push eax
0x1002b137: mov dword ptr [edx+0B0h],0
0x1002b141: mov edx,[ebp-58h]
0x1002b144: push ecx
0x1002b145: push edx
0x1002b146: call 1002CBE0h
0x1002b14b: add esp,14h
0x1002b14e: xor eax,eax
0x1002b150: pop esi
0x1002b151: mov ecx,[ebp-4]
0x1002b154: xor ecx,ebp
0x1002b156: call 10022326h
0x1002b15b: mov esp,ebp
0x1002b15d: pop ebp
0x1002b15e: ret
0x1002b15f: mov eax,[ebp-58h]
0x1002b162: push 0
0x1002b164: push 0
0x1002b166: push 0
0x1002b168: push eax
0x1002b169: call 1002CAD0h
0x1002b16e: mov esi,[ebp-58h]
0x1002b171: add esp,10h
0x1002b174: call 1002A8B0h
0x1002b179: mov eax,[ebp-58h]
0x1002b17c: mov ecx,[ebp-4]
0x1002b17f: xor ecx,ebp
0x1002b181: pop esi
0x1002b182: call 10022326h
0x1002b187: mov esp,ebp
0x1002b189: pop ebp
0x1002b18a: ret
```

`png_create_write_struct_2` is called by png_create_write_struct. It works with the text "Application built with libpng-%.20s but running with %.20s". It delegates to sub_10022326, j__setjmp3, png_set_longjmp_fn (13 callees).

12.682 png_write_rows at 0x1002b190 (22 instructions)

```
0x1002b190: push ebx
0x1002b191: mov ebx,[esp+8]
0x1002b195: test ebx,ebx
0x1002b197: je short 1002B1BDh
0x1002b199: push esi
0x1002b19a: mov esi,[esp+10h]
0x1002b19e: push edi
0x1002b19f: mov edi,[esp+18h]
0x1002b1a3: test edi,edi
0x1002b1a5: jbe short 1002B1BBh
0x1002b1a7: mov eax,[esi]
0x1002b1a9: push eax
0x1002b1aa: push ebx
0x1002b1ab: call 1002A370h
0x1002b1b0: add esp,8
0x1002b1b3: add esi,4
0x1002b1b6: sub edi,1
0x1002b1b9: jne short 1002B1A7h
0x1002b1bb: pop edi
0x1002b1bc: pop esi
0x1002b1bd: pop ebx
0x1002b1be: ret
```

`png_write_rows` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_write_row (1 callees).

12.683 png_write_image at 0x1002b1c0 (37 instructions)

```
0x1002b1c0: push ebx
0x1002b1c1: mov ebx,[esp+8]
0x1002b1c5: test ebx,ebx
0x1002b1c7: je short 1002B216h
0x1002b1c9: push ebx
0x1002b1ca: call 10028710h
0x1002b1cf: add esp,4
0x1002b1d2: test eax,eax
0x1002b1d4: jle short 1002B216h
0x1002b1d6: mov ecx,[ebx+0D0h]
0x1002b1dc: push ebp
0x1002b1dd: mov ebp,[esp+10h]
0x1002b1e1: push esi
0x1002b1e2: push edi
0x1002b1e3: mov [esp+14h],eax
0x1002b1e7: xor esi,esi
0x1002b1e9: test ecx,ecx
0x1002b1eb: mov edi,ebp
0x1002b1ed: jbe short 1002B20Ch
0x1002b1ef: nop
0x1002b1f0: mov eax,[edi]
0x1002b1f2: push eax
0x1002b1f3: push ebx
0x1002b1f4: call 1002A370h
0x1002b1f9: mov ecx,[ebx+0D0h]
0x1002b1ff: add esi,1
0x1002b202: add esp,8
0x1002b205: add edi,4
0x1002b208: cmp esi,ecx
0x1002b20a: jb short 1002B1F0h
0x1002b20c: sub dword ptr [esp+14h],1
0x1002b211: jne short 1002B1E7h
0x1002b213: pop edi
0x1002b214: pop esi
0x1002b215: pop ebp
0x1002b216: pop ebx
0x1002b217: ret
```

`png_write_image` is called by png_write_png. It delegates to png_set_interlace_handling, png_write_row (2 callees).

12.684 sub_1002b220 at 0x1002b220 (80 instructions)

```
0x1002b220: sub esp,40h
0x1002b223: push ebx
0x1002b224: mov ebx,[esp+48h]
0x1002b228: push ebp
0x1002b229: push esi
0x1002b22a: lea eax,[ebx+78h]
0x1002b22d: push edi
0x1002b22e: push eax
0x1002b22f: call 100619A0h
0x1002b234: mov ecx,[ebx+0B0h]
0x1002b23a: push ecx
0x1002b23b: push ebx
0x1002b23c: call 1002CD70h
0x1002b241: mov edx,[ebx+0ECh]
0x1002b247: push edx
0x1002b248: push ebx
0x1002b249: call 1002CD70h
0x1002b24e: mov eax,[ebx+0E8h]
0x1002b254: push eax
```

```
0x1002b255: push ebx
0x1002b256: call 1002CD70h
0x1002b25b: mov ecx,[ebx+0F0h]
0x1002b261: push ecx
0x1002b262: push ebx
0x1002b263: call 1002CD70h
0x1002b268: mov edx,[ebx+0F4h]
0x1002b26e: push edx
0x1002b26f: push ebx
0x1002b270: call 1002CD70h
0x1002b275: mov eax,[ebx+0F8h]
0x1002b27b: push eax
0x1002b27c: push ebx
0x1002b27d: call 1002CD70h
0x1002b282: mov ecx,[ebx+0FCh]
0x1002b288: push ecx
0x1002b289: push ebx
0x1002b28a: call 1002CD70h
0x1002b28f: mov edx,[ebx+210h]
0x1002b295: push edx
0x1002b296: push ebx
0x1002b297: call 1002CD70h
0x1002b29c: add esp,44h
0x1002b29f: mov esi,ebx
0x1002b2a1: call 1002A8B0h
0x1002b2a6: mov eax,[ebx+208h]
0x1002b2ac: push eax
0x1002b2ad: push ebx
0x1002b2ae: call 1002CD70h
0x1002b2b3: mov ecx,[ebx+20Ch]
0x1002b2b9: push ecx
0x1002b2ba: push ebx
0x1002b2bb: call 1002CD70h
0x1002b2c0: mov edx,[ebx+240h]
0x1002b2c6: mov ebp,[ebx+4Ch]
0x1002b2c9: push 28Ch
0x1002b2ce: mov ecx,10h
0x1002b2d3: lea edi,[esp+24h]
0x1002b2d7: rep movsd
0x1002b2d9: mov esi,[ebx+44h]
0x1002b2dc: mov edi,[ebx+48h]
0x1002b2df: push 0
0x1002b2e1: push ebx
0x1002b2e2: mov [esp+70h],edx
0x1002b2e6: call 100228C0h
0x1002b2eb: mov eax,[esp+70h]
0x1002b2ef: mov [ebx+44h],esi
0x1002b2f2: mov [ebx+48h],edi
0x1002b2f5: add esp,1Ch
0x1002b2f8: mov edi,ebx
0x1002b2fa: mov ecx,10h
0x1002b2ff: lea esi,[esp+10h]
0x1002b303: rep movsd
0x1002b305: pop edi
0x1002b306: pop esi
0x1002b307: mov [ebx+4Ch],ebp
0x1002b30a: pop ebp
0x1002b30b: mov [ebx+240h],eax
0x1002b311: pop ebx
0x1002b312: add esp,40h
0x1002b315: ret
```

`sub_1002b220` is called by png_destroy_write_struct. It delegates to j_memset, sub_1002a8b0, png_free (4 callees).

12.685 png_write_png at 0x1002b320 (84 instructions)

```
0x1002b320: push esi
0x1002b321: mov esi,[esp+8]
0x1002b325: test esi,esi
0x1002b327: je near ptr 1002B411h
0x1002b32d: push edi
0x1002b32e: mov edi,[esp+10h]
0x1002b332: test edi,edi
0x1002b334: je near ptr 1002B410h
0x1002b33a: push ebx
0x1002b33b: push edi
0x1002b33c: push esi
0x1002b33d: call 10029E90h
0x1002b342: mov ebx,[esp+20h]
0x1002b346: add esp,8
0x1002b349: test bl,20h
0x1002b34c: je short 1002B357h
0x1002b34e: push esi
0x1002b34f: call 100287F0h
0x1002b354: add esp,4
0x1002b357: test bl,40h
0x1002b35a: je short 1002B36Fh
0x1002b35c: test byte ptr [edi+8],2
0x1002b360: je short 1002B36Fh
0x1002b362: lea eax,[edi+44h]
0x1002b365: push eax
0x1002b366: push esi
0x1002b367: call 100286E0h
0x1002b36c: add esp,8
0x1002b36f: test bl,4
0x1002b372: je short 1002B37Dh
0x1002b374: push esi
0x1002b375: call 100286A0h
0x1002b37a: add esp,4
0x1002b37d: test ebx,100h
0x1002b383: je short 1002B38Eh
0x1002b385: push esi
0x1002b386: call 100287D0h
0x1002b38b: add esp,4
0x1002b38e: test ebx,1000h
0x1002b394: je short 1002B39Ah
0x1002b396: push 1
0x1002b398: jmp short 1002B3A4h
0x1002b39a: test ebx,800h
0x1002b3a0: je short 1002B3AFh
0x1002b3a2: push 0
0x1002b3a4: push 0
0x1002b3a6: push esi
0x1002b3a7: call 10028740h
0x1002b3ac: add esp,0Ch
0x1002b3af: test bl,bl
0x1002b3b1: jns short 1002B3BCh
0x1002b3b3: push esi
0x1002b3b4: call 10028670h
0x1002b3b9: add esp,4
0x1002b3bc: test ebx,200h
0x1002b3c2: je short 1002B3CDh
0x1002b3c4: push esi
0x1002b3c5: call 10028680h
0x1002b3ca: add esp,4
0x1002b3cd: test bl,8
0x1002b3d0: je short 1002B3DBh
0x1002b3d2: push esi
0x1002b3d3: call 100286C0h
0x1002b3d8: add esp,4
0x1002b3db: test ebx,400h
0x1002b3e1: pop ebx
```

```
0x1002b3e2: je short 1002B3EDh
0x1002b3e4: push esi
0x1002b3e5: call 100287E0h
0x1002b3ea: add esp,4
0x1002b3ed: test dword ptr [edi+8],8000h
0x1002b3f4: je short 1002B406h
0x1002b3f6: mov ecx,[edi+0ECh]
0x1002b3fc: push ecx
0x1002b3fd: push esi
0x1002b3fe: call 1002B1C0h
0x1002b403: add esp,8
0x1002b406: push edi
0x1002b407: push esi
0x1002b408: call 1002A190h
0x1002b40d: add esp,8
0x1002b410: pop edi
0x1002b411: pop esi
0x1002b412: ret
```

`png_write_png` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_set_bgr, png_set_swap, png_set_packing (12 callees).

12.686 png_create_write_struct at 0x1002b420 (14 instructions)

```
0x1002b420: mov eax,[esp+10h]
0x1002b424: mov ecx,[esp+0Ch]
0x1002b428: mov edx,[esp+8]
0x1002b42c: push 0
0x1002b42e: push 0
0x1002b430: push 0
0x1002b432: push eax
0x1002b433: mov eax,[esp+14h]
0x1002b437: push ecx
0x1002b438: push edx
0x1002b439: push eax
0x1002b43a: call 1002AF50h
0x1002b43f: add esp,1Ch
0x1002b442: ret
```

`png_create_write_struct` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_create_write_struct_2 (1 callees).

12.687 png_destroy_write_struct at 0x1002b450 (58 instructions)

```
0x1002b450: mov eax,[esp+4]
0x1002b454: push ebx
0x1002b455: push ebp
0x1002b456: push esi
0x1002b457: xor esi,esi
0x1002b459: xor ebx,ebx
0x1002b45b: xor ebp,ebp
0x1002b45d: test eax,eax
0x1002b45f: je short 1002B46Fh
0x1002b461: mov esi,[eax]
0x1002b463: mov ebx,[esi+240h]
0x1002b469: mov ebp,[esi+238h]
0x1002b46f: mov eax,[esp+14h]
0x1002b473: test eax,eax
0x1002b475: je short 1002B4CCh
0x1002b477: push edi
0x1002b478: mov edi,[eax]
0x1002b47a: test edi,edi
0x1002b47c: je short 1002B4CBh
0x1002b47e: test esi,esi
0x1002b480: je short 1002B4B6h
```

```
0x1002b482: push 0FFFFFFFFh
0x1002b484: push 7FFFh
0x1002b489: push edi
0x1002b48a: push esi
0x1002b48b: call 100235E0h
0x1002b490: add esp,10h
0x1002b493: cmp dword ptr [esi+220h],0
0x1002b49a: je short 1002B4B6h
0x1002b49c: mov eax,[esi+224h]
0x1002b4a2: push eax
0x1002b4a3: push esi
0x1002b4a4: call 1002CD70h
0x1002b4a9: add esp,8
0x1002b4ac: mov dword ptr [esi+220h],0
0x1002b4b6: push ebp
0x1002b4b7: push ebx
0x1002b4b8: push edi
0x1002b4b9: call 1002CBE0h
0x1002b4be: mov ecx,[esp+24h]
0x1002b4c2: add esp,0Ch
0x1002b4c5: mov dword ptr [ecx],0
0x1002b4cb: pop edi
0x1002b4cc: test esi,esi
0x1002b4ce: je short 1002B4EBh
0x1002b4d0: push esi
0x1002b4d1: call 1002B220h
0x1002b4d6: push ebp
0x1002b4d7: push ebx
0x1002b4d8: push esi
0x1002b4d9: call 1002CBE0h
0x1002b4de: mov edx,[esp+20h]
0x1002b4e2: add esp,10h
0x1002b4e5: mov dword ptr [edx],0
0x1002b4eb: pop esi
0x1002b4ec: pop ebp
0x1002b4ed: pop ebx
0x1002b4ee: ret
```

`png_destroy_write_struct` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_free_data, sub_1002b220, sub_1002cbe0 (4 callees).

12.688 png_set_bKGD at 0x1002b4f0 (14 instructions)

```
0x1002b4f0: cmp dword ptr [esp+4],0
0x1002b4f5: je short 1002B51Ah
0x1002b4f7: mov eax,[esp+8]
0x1002b4fb: test eax,eax
0x1002b4fd: je short 1002B51Ah
0x1002b4ff: mov ecx,[esp+0Ch]
0x1002b503: mov edx,[ecx]
0x1002b505: mov [eax+5Eh],edx
0x1002b508: mov edx,[ecx+4]
0x1002b50b: mov [eax+62h],edx
0x1002b50e: mov cx,[ecx+8]
0x1002b512: or dword ptr [eax+8],20h
0x1002b516: mov [eax+66h],cx
0x1002b51a: ret
```

`png_set_bKGD` is called by sub_1002f400. It calls nothing else.

12.689 png_set_cHRM_fixed at 0x1002b520 (50 instructions)

```
0x1002b520: mov eax,[esp+4]
0x1002b524: test eax,eax
0x1002b526: je near ptr 1002B5BDh
```

```
0x1002b52c: push esi
0x1002b52d: mov esi,[esp+0Ch]
0x1002b531: test esi,esi
0x1002b533: je near ptr 1002B5BCh
0x1002b539: mov ecx,[esp+20h]
0x1002b53d: mov edx,[esp+1Ch]
0x1002b541: push ebx
0x1002b542: mov ebx,[esp+2Ch]
0x1002b546: push ebp
0x1002b547: mov ebp,[esp+2Ch]
0x1002b54b: push edi
0x1002b54c: mov edi,[esp+38h]
0x1002b550: push edi
0x1002b551: push ebx
0x1002b552: push ebp
0x1002b553: push ecx
0x1002b554: mov ecx,[esp+34h]
0x1002b558: push edx
0x1002b559: mov edx,[esp+34h]
0x1002b55d: push ecx
0x1002b55e: mov ecx,[esp+34h]
0x1002b562: push edx
0x1002b563: push ecx
0x1002b564: push eax
0x1002b565: call 100250A0h
0x1002b56a: add esp,24h
0x1002b56d: test eax,eax
0x1002b56f: je short 1002B5B9h
0x1002b571: mov edx,[esp+1Ch]
0x1002b575: mov eax,[esp+20h]
0x1002b579: mov ecx,[esp+24h]
0x1002b57d: or dword ptr [esi+8],4
0x1002b581: mov [esi+84h],edx
0x1002b587: mov edx,[esp+28h]
0x1002b58b: mov [esi+88h],eax
0x1002b591: mov eax,[esp+2Ch]
0x1002b595: mov [esi+8Ch],ecx
0x1002b59b: mov [esi+90h],edx
0x1002b5a1: mov [esi+94h],eax
0x1002b5a7: mov [esi+98h],ebp
0x1002b5ad: mov [esi+9Ch],ebx
0x1002b5b3: mov [esi+0A0h],edi
0x1002b5b9: pop edi
0x1002b5ba: pop ebp
0x1002b5bb: pop ebx
0x1002b5bc: pop esi
0x1002b5bd: ret
```

`png_set_cHRM_fixed` is called by png_set_cHRM, png_set_sRGB_gAMA_and_cHRM, sub_1002e700. It delegates to sub_100250a0 (1 callees).

12.690 png_set_cHRM at 0x1002b5c0 (73 instructions)

```
0x1002b5c0: fld qword ptr [esp+44h]
0x1002b5c4: push esi
0x1002b5c5: mov esi,[esp+8]
0x1002b5c9: push 10100550h
0x1002b5ce: sub esp,8
0x1002b5d1: fstp qword ptr [esp]
0x1002b5d4: push esi
0x1002b5d5: call 10024710h
0x1002b5da: fld qword ptr [esp+50h]
0x1002b5de: add esp,10h
0x1002b5e1: push eax
0x1002b5e2: push 10100544h
0x1002b5e7: sub esp,8
```

```
0x1002b5ea: fstp qword ptr [esp]
0x1002b5ed: push esi
0x1002b5ee: call 10024710h
0x1002b5f3: fld qword ptr [esp+4Ch]
0x1002b5f7: add esp,10h
0x1002b5fa: push eax
0x1002b5fb: push 10100534h
0x1002b600: sub esp,8
0x1002b603: fstp qword ptr [esp]
0x1002b606: push esi
0x1002b607: call 10024710h
0x1002b60c: fld qword ptr [esp+48h]
0x1002b610: add esp,10h
0x1002b613: push eax
0x1002b614: push 10100524h
0x1002b619: sub esp,8
0x1002b61c: fstp qword ptr [esp]
0x1002b61f: push esi
0x1002b620: call 10024710h
0x1002b625: fld qword ptr [esp+44h]
0x1002b629: add esp,10h
0x1002b62c: push eax
0x1002b62d: push 10100518h
0x1002b632: sub esp,8
0x1002b635: fstp qword ptr [esp]
0x1002b638: push esi
0x1002b639: call 10024710h
0x1002b63e: fld qword ptr [esp+40h]
0x1002b642: add esp,10h
0x1002b645: push eax
0x1002b646: push 1010050Ch
0x1002b64b: sub esp,8
0x1002b64e: fstp qword ptr [esp]
0x1002b651: push esi
0x1002b652: call 10024710h
0x1002b657: fld qword ptr [esp+3Ch]
0x1002b65b: add esp,10h
0x1002b65e: push eax
0x1002b65f: push 101004FCh
0x1002b664: sub esp,8
0x1002b667: fstp qword ptr [esp]
0x1002b66a: push esi
0x1002b66b: call 10024710h
0x1002b670: fld qword ptr [esp+38h]
0x1002b674: add esp,10h
0x1002b677: push eax
0x1002b678: push 101004ECh
0x1002b67d: sub esp,8
0x1002b680: fstp qword ptr [esp]
0x1002b683: push esi
0x1002b684: call 10024710h
0x1002b689: add esp,10h
0x1002b68c: push eax
0x1002b68d: mov eax,[esp+2Ch]
0x1002b691: push eax
0x1002b692: push esi
0x1002b693: call 1002B520h
0x1002b698: add esp,28h
0x1002b69b: pop esi
0x1002b69c: ret
```

`png_set_cHRM` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10024710, png_set_cHRM_fixed (2 callees).

12.691 png_set_gAMA_fixed at 0x1002b6a0 (24 instructions)

```
0x1002b6a0: mov edx,[esp+4]
0x1002b6a4: test edx,edx
0x1002b6a6: je short 1002B6E5h
0x1002b6a8: mov eax,[esp+8]
0x1002b6ac: test eax,eax
0x1002b6ae: je short 1002B6E5h
0x1002b6b0: mov ecx,[esp+0Ch]
0x1002b6b4: cmp ecx,7FFFFFFFh
0x1002b6ba: jle short 1002B6CBh
0x1002b6bc: push 1010057Ch
0x1002b6c1: push edx
0x1002b6c2: call 100230F0h
0x1002b6c7: add esp,8
0x1002b6ca: ret
0x1002b6cb: test ecx,ecx
0x1002b6cd: jg short 1002B6DEh
0x1002b6cf: push 1010055Ch
0x1002b6d4: push edx
0x1002b6d5: call 100230F0h
0x1002b6da: add esp,8
0x1002b6dd: ret
0x1002b6de: or dword ptr [eax+8],1
0x1002b6e2: mov [eax+28h],ecx
0x1002b6e5: ret
```

`png_set_gAMA_fixed` is called by png_set_gAMA, png_set_sRGB_gAMA_and_cHRM, sub_1002e450. It works with the text "Negative or zero gamma ignored", "Gamma too large, ignored". It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to png_warning (1 callees).

12.692 png_set_gAMA at 0x1002b6f0 (16 instructions)

```
0x1002b6f0: fld qword ptr [esp+0Ch]
0x1002b6f4: push esi
0x1002b6f5: mov esi,[esp+8]
0x1002b6f9: push 10100598h
0x1002b6fe: sub esp,8
0x1002b701: fstp qword ptr [esp]
0x1002b704: push esi
0x1002b705: call 10024710h
0x1002b70a: push eax
0x1002b70b: mov eax,[esp+20h]
0x1002b70f: push eax
0x1002b710: push esi
0x1002b711: call 1002B6A0h
0x1002b716: add esp,1Ch
0x1002b719: pop esi
0x1002b71a: ret
```

`png_set_gAMA` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "png_set_gAMA". It delegates to sub_10024710, png_set_gAMA_fixed (2 callees).

12.693 png_set_hIST at 0x1002b720 (60 instructions)

```
0x1002b720: push edi
0x1002b721: mov edi,[esp+8]
0x1002b725: test edi,edi
0x1002b727: je near ptr 1002B7D3h
0x1002b72d: push esi
0x1002b72e: mov esi,[esp+10h]
0x1002b732: test esi,esi
0x1002b734: je near ptr 1002B7D2h
0x1002b73a: movzx eax,word ptr [esi+14h]
0x1002b73e: test ax,ax
```

```
0x1002b741: je near ptr 1002B7C4h
0x1002b747: cmp ax,100h
0x1002b74b: ja short 1002B7C4h
0x1002b74d: push 0
0x1002b74f: push 8
0x1002b751: push esi
0x1002b752: push edi
0x1002b753: call 100235E0h
0x1002b758: push 200h
0x1002b75d: push edi
0x1002b75e: call 1002CDA0h
0x1002b763: add esp,18h
0x1002b766: test eax,eax
0x1002b768: mov [edi+1F4h],eax
0x1002b76e: jne short 1002B781h
0x1002b770: push 101005D8h
0x1002b775: push edi
0x1002b776: call 100230F0h
0x1002b77b: add esp,8
0x1002b77e: pop esi
0x1002b77f: pop edi
0x1002b780: ret
0x1002b781: xor eax,eax
0x1002b783: cmp [esi+14h],ax
0x1002b787: jbe short 1002B7AAh
0x1002b789: mov ecx,[esp+14h]
0x1002b78d: push ebx
0x1002b78e: mov edi,edi
0x1002b790: mov edx,[edi+1F4h]
0x1002b796: mov bx,[ecx+eax*2]
0x1002b79a: mov [edx+eax*2],bx
0x1002b79e: movzx edx,word ptr [esi+14h]
0x1002b7a2: add eax,1
0x1002b7a5: cmp eax,edx
0x1002b7a7: jl short 1002B790h
0x1002b7a9: pop ebx
0x1002b7aa: mov eax,[edi+1F4h]
0x1002b7b0: or dword ptr [esi+8],40h
0x1002b7b4: or dword ptr [esi+0BCh],8
0x1002b7bb: mov [esi+80h],eax
0x1002b7c1: pop esi
0x1002b7c2: pop edi
0x1002b7c3: ret
0x1002b7c4: push 101005A8h
0x1002b7c9: push edi
0x1002b7ca: call 100230F0h
0x1002b7cf: add esp,8
0x1002b7d2: pop esi
0x1002b7d3: pop edi
0x1002b7d4: ret
```

`png_set_hIST` is called by sub_1002f650. It works with the text "Invalid palette size, hIST allocation skipped", "Insufficient memory for hIST chunk data". It delegates to png_warning, png_free_data, png_malloc_warn (3 callees).

12.694 png_set_IHDR at 0x1002b7e0 (82 instructions)

```
0x1002b7e0: push ebp
0x1002b7e1: mov ebp,[esp+8]
0x1002b7e5: test ebp,ebp
0x1002b7e7: je near ptr 1002B8AFh
0x1002b7ed: push esi
0x1002b7ee: mov esi,[esp+10h]
0x1002b7f2: test esi,esi
0x1002b7f4: je near ptr 1002B8AEh
0x1002b7fa: push ebx
0x1002b7fb: mov bl,[esp+2Ch]
```

```
0x1002b7ff: mov dl,[esp+24h]
0x1002b803: mov cl,[esp+20h]
0x1002b807: mov eax,[esp+1Ch]
0x1002b80b: mov [esi+1Ah],bl
0x1002b80e: mov bl,[esp+30h]
0x1002b812: mov [esi+1Bh],bl
0x1002b815: mov bl,[esp+28h]
0x1002b819: push edi
0x1002b81a: mov edi,[esp+1Ch]
0x1002b81e: mov [esi+1Ch],bl
0x1002b821: movzx ebx,byte ptr [esp+34h]
0x1002b826: push ebx
0x1002b827: movzx ebx,byte ptr [esp+34h]
0x1002b82c: push ebx
0x1002b82d: movzx ebx,byte ptr [esp+34h]
0x1002b832: mov [esi+19h],dl
0x1002b835: push ebx
0x1002b836: movzx edx,dl
0x1002b839: mov [esi+18h],cl
0x1002b83c: push edx
0x1002b83d: movzx ecx,cl
0x1002b840: push ecx
0x1002b841: push eax
0x1002b842: push edi
0x1002b843: push ebp
0x1002b844: mov [esi],edi
0x1002b846: mov [esi+4],eax
0x1002b849: call 10023C30h
0x1002b84e: mov al,[esi+19h]
0x1002b851: add esp,20h
0x1002b854: cmp al,3
0x1002b856: mov cl,1
0x1002b858: je short 1002B864h
0x1002b85a: test al,2
0x1002b85c: je short 1002B864h
0x1002b85e: mov byte ptr [esi+1Dh],3
0x1002b862: jmp short 1002B867h
0x1002b864: mov [esi+1Dh],cl
0x1002b867: test al,4
0x1002b869: je short 1002B86Eh
0x1002b86b: add [esi+1Dh],cl
0x1002b86e: mov al,[esi+18h]
0x1002b871: imul byte ptr [esi+1Dh]
0x1002b874: cmp edi,1FFFFF8Eh
0x1002b87a: mov [esi+1Eh],al
0x1002b87d: jbe short 1002B88Bh
0x1002b87f: pop edi
0x1002b880: pop ebx
0x1002b881: mov dword ptr [esi+0Ch],0
0x1002b888: pop esi
0x1002b889: pop ebp
0x1002b88a: ret
0x1002b88b: cmp al,8
0x1002b88d: movzx eax,al
0x1002b890: jb short 1002B8A0h
0x1002b892: shr eax,3
0x1002b895: imul eax,edi
0x1002b898: pop edi
0x1002b899: pop ebx
0x1002b89a: mov [esi+0Ch],eax
0x1002b89d: pop esi
0x1002b89e: pop ebp
0x1002b89f: ret
0x1002b8a0: imul eax,edi
0x1002b8a3: add eax,7
0x1002b8a6: shr eax,3
0x1002b8a9: pop edi
```

```
0x1002b8aa: mov [esi+0Ch],eax
0x1002b8ad: pop ebx
0x1002b8ae: pop esi
0x1002b8af: pop ebp
0x1002b8b0: ret
```

`png_set_IHDR` is called by sub_1002e100. It delegates to sub_10023c30 (1 callees).

12.695 png_set_oFFs at 0x1002b8c0 (13 instructions)

```
0x1002b8c0: cmp dword ptr [esp+4],0
0x1002b8c5: je short 1002B8EBh
0x1002b8c7: mov eax,[esp+8]
0x1002b8cb: test eax,eax
0x1002b8cd: je short 1002B8EBh
0x1002b8cf: mov ecx,[esp+0Ch]
0x1002b8d3: mov edx,[esp+10h]
0x1002b8d7: or dword ptr [eax+8],100h
0x1002b8de: mov [eax+68h],ecx
0x1002b8e1: mov cl,[esp+14h]
0x1002b8e5: mov [eax+6Ch],edx
0x1002b8e8: mov [eax+70h],cl
0x1002b8eb: ret
```

`png_set_oFFs` is called by sub_1002f930. It calls nothing else.

12.696 png_set_pCAL at 0x1002b8f0 (172 instructions)

```
0x1002b8f0: push edi
0x1002b8f1: mov edi,[esp+8]
0x1002b8f5: test edi,edi
0x1002b8f7: je near ptr 1002B984h
0x1002b8fd: push esi
0x1002b8fe: mov esi,[esp+10h]
0x1002b902: test esi,esi
0x1002b904: je near ptr 1002B983h
0x1002b90a: mov eax,[esp+14h]
0x1002b90e: push ebp
0x1002b90f: push eax
0x1002b910: call dword ptr ds:[100FE0B0h]
0x1002b916: mov ebp,eax
0x1002b918: add ebp,1
0x1002b91b: cmp dword ptr [esp+24h],3
0x1002b920: ja near ptr 1002BAD3h
0x1002b926: push ebx
0x1002b927: mov ebx,[esp+2Ch]
0x1002b92b: xor edi,edi
0x1002b92d: test ebx,ebx
0x1002b92f: jle short 1002B95Bh
0x1002b931: mov ecx,[esp+34h]
0x1002b935: mov edx,[ecx+edi*4]
0x1002b938: push edx
0x1002b939: call dword ptr ds:[100FE0B0h]
0x1002b93f: push eax
0x1002b940: mov eax,[esp+38h]
0x1002b944: mov ecx,[eax+edi*4]
0x1002b947: push ecx
0x1002b948: call 10023FA0h
0x1002b94d: add esp,8
0x1002b950: test eax,eax
0x1002b952: je short 1002B986h
0x1002b954: add edi,1
0x1002b957: cmp edi,ebx
0x1002b959: jl short 1002B931h
0x1002b95b: mov edi,[esp+14h]
```

```
0x1002b95f: push ebp
0x1002b960: push edi
0x1002b961: call 1002CDA0h
0x1002b966: add esp,8
0x1002b969: test eax,eax
0x1002b96b: mov [esi+0A4h],eax
0x1002b971: jne short 1002B995h
0x1002b973: push 101006B0h
0x1002b978: push edi
0x1002b979: call 100230F0h
0x1002b97e: add esp,8
0x1002b981: pop ebx
0x1002b982: pop ebp
0x1002b983: pop esi
0x1002b984: pop edi
0x1002b985: ret
0x1002b986: mov edx,[esp+14h]
0x1002b98a: push 1010068Ch
0x1002b98f: push edx
0x1002b990: call 100231E0h
0x1002b995: mov ecx,[esp+1Ch]
0x1002b999: push ebp
0x1002b99a: push ecx
0x1002b99b: push eax
0x1002b99c: call 100228C6h
0x1002b9a1: mov edx,[esp+2Ch]
0x1002b9a5: mov eax,[esp+30h]
0x1002b9a9: mov cl,[esp+34h]
0x1002b9ad: mov ebp,[esp+3Ch]
0x1002b9b1: add esp,0Ch
0x1002b9b4: push ebp
0x1002b9b5: mov [esi+0A8h],edx
0x1002b9bb: mov [esi+0ACh],eax
0x1002b9c1: mov [esi+0B8h],cl
0x1002b9c7: mov [esi+0B9h],bl
0x1002b9cd: call dword ptr ds:[100FE0B0h]
0x1002b9d3: mov edx,[esp+14h]
0x1002b9d7: mov edi,eax
0x1002b9d9: add edi,1
0x1002b9dc: push edi
0x1002b9dd: push edx
0x1002b9de: call 1002CDA0h
0x1002b9e3: add esp,8
0x1002b9e6: test eax,eax
0x1002b9e8: mov [esi+0B0h],eax
0x1002b9ee: jne short 1002BA07h
0x1002b9f0: mov eax,[esp+14h]
0x1002b9f4: push 10100668h
0x1002b9f9: push eax
0x1002b9fa: call 100230F0h
0x1002b9ff: add esp,8
0x1002ba02: pop ebx
0x1002ba03: pop ebp
0x1002ba04: pop esi
0x1002ba05: pop edi
0x1002ba06: ret
0x1002ba07: push edi
0x1002ba08: push ebp
0x1002ba09: push eax
0x1002ba0a: call 100228C6h
0x1002ba0f: mov ebp,[esp+20h]
0x1002ba13: lea edi,[ebx*4+4]
0x1002ba1a: push edi
0x1002ba1b: push ebp
0x1002ba1c: call 1002CDA0h
0x1002ba21: add esp,14h
0x1002ba24: test eax,eax
```

```
0x1002ba26: mov [esi+0B4h],eax
0x1002ba2c: jne short 1002BA41h
0x1002ba2e: push 10100644h
0x1002ba33: push ebp
0x1002ba34: call 100230F0h
0x1002ba39: add esp,8
0x1002ba3c: pop ebx
0x1002ba3d: pop ebp
0x1002ba3e: pop esi
0x1002ba3f: pop edi
0x1002ba40: ret
0x1002ba41: push edi
0x1002ba42: push 0
0x1002ba44: push eax
0x1002ba45: call 100228C0h
0x1002ba4a: add esp,0Ch
0x1002ba4d: xor edi,edi
0x1002ba4f: test ebx,ebx
0x1002ba51: jle short 1002BAA6h
0x1002ba53: mov ecx,[esp+34h]
0x1002ba57: mov edx,[ecx+edi*4]
0x1002ba5a: push edx
0x1002ba5b: call dword ptr ds:[100FE0B0h]
0x1002ba61: mov ebp,eax
0x1002ba63: mov eax,[esp+14h]
0x1002ba67: add ebp,1
0x1002ba6a: push ebp
0x1002ba6b: push eax
0x1002ba6c: call 1002CDA0h
0x1002ba71: mov ecx,[esi+0B4h]
0x1002ba77: mov [ecx+edi*4],eax
0x1002ba7a: mov edx,[esi+0B4h]
0x1002ba80: lea eax,[edx+edi*4]
0x1002ba83: add esp,8
0x1002ba86: cmp dword ptr [eax],0
0x1002ba89: je short 1002BABCh
0x1002ba8b: mov ecx,[esp+34h]
0x1002ba8f: mov edx,[ecx+edi*4]
0x1002ba92: mov eax,[eax]
0x1002ba94: push ebp
0x1002ba95: push edx
0x1002ba96: push eax
0x1002ba97: call 100228C6h
0x1002ba9c: add edi,1
0x1002ba9f: add esp,0Ch
0x1002baa2: cmp edi,ebx
0x1002baa4: jl short 1002BA53h
0x1002baa6: or dword ptr [esi+8],400h
0x1002baad: or dword ptr [esi+0BCh],80h
0x1002bab7: pop ebx
0x1002bab8: pop ebp
0x1002bab9: pop esi
0x1002baba: pop edi
0x1002babb: ret
0x1002babc: mov ecx,[esp+14h]
0x1002bac0: push 1010061Ch
0x1002bac5: push ecx
0x1002bac6: call 100230F0h
0x1002bacb: add esp,8
0x1002bace: pop ebx
0x1002bacf: pop ebp
0x1002bad0: pop esi
0x1002bad1: pop edi
0x1002bad2: ret
0x1002bad3: push 10100600h
0x1002bad8: push edi
```

```
0x1002bad9: call 100231E0h
0x1002bade: int 3
```

`png_set_pCAL` is called by sub_1002faa0. It works with the text "Invalid pCAL equation type", "Insufficient memory for pCAL parameter", "Insufficient memory for pCAL params". It delegates to j_memset, j_memcpy, png_warning (6 callees).

12.697 png_set_sCAL_s at 0x1002bae0 (117 instructions)

```
0x1002bae0: push ebx
0x1002bae1: mov ebx,[esp+8]
0x1002bae5: test ebx,ebx
0x1002bae7: je near ptr 1002BBBBh
0x1002baed: push esi
0x1002baee: mov esi,[esp+10h]
0x1002baf2: test esi,esi
0x1002baf4: je near ptr 1002BBBAh
0x1002bafa: mov eax,[esp+14h]
0x1002bafe: cmp eax,1
0x1002bb01: je short 1002BB13h
0x1002bb03: cmp eax,2
0x1002bb06: je short 1002BB13h
0x1002bb08: push 10100730h
0x1002bb0d: push ebx
0x1002bb0e: call 100231E0h
0x1002bb13: push ebp
0x1002bb14: push edi
0x1002bb15: mov edi,[esp+20h]
0x1002bb19: test edi,edi
0x1002bb1b: je near ptr 1002BC3Ah
0x1002bb21: push edi
0x1002bb22: call dword ptr ds:[100FE0B0h]
0x1002bb28: mov ebp,eax
0x1002bb2a: test ebp,ebp
0x1002bb2c: jbe near ptr 1002BC3Ah
0x1002bb32: cmp byte ptr [edi],2Dh
0x1002bb35: je near ptr 1002BC3Ah
0x1002bb3b: push ebp
0x1002bb3c: push edi
0x1002bb3d: call 10023FA0h
0x1002bb42: add esp,8
0x1002bb45: test eax,eax
0x1002bb47: je near ptr 1002BC3Ah
0x1002bb4d: mov eax,[esp+24h]
0x1002bb51: test eax,eax
0x1002bb53: je near ptr 1002BC2Fh
0x1002bb59: push eax
0x1002bb5a: call dword ptr ds:[100FE0B0h]
0x1002bb60: mov edi,eax
0x1002bb62: test edi,edi
0x1002bb64: jbe near ptr 1002BC2Fh
0x1002bb6a: mov eax,[esp+24h]
0x1002bb6e: cmp byte ptr [eax],2Dh
0x1002bb71: je near ptr 1002BC2Fh
0x1002bb77: push edi
0x1002bb78: push eax
0x1002bb79: call 10023FA0h
0x1002bb7e: add esp,8
0x1002bb81: test eax,eax
0x1002bb83: je near ptr 1002BC2Fh
0x1002bb89: mov al,[esp+1Ch]
0x1002bb8d: add ebp,1
0x1002bb90: push ebp
0x1002bb91: push ebx
0x1002bb92: mov [esi+0E0h],al
0x1002bb98: call 1002CDA0h
```

```
0x1002bb9d: add esp,8
0x1002bba0: test eax,eax
0x1002bba2: mov [esi+0E4h],eax
0x1002bba8: jne short 1002BBBDh
0x1002bbaa: push 10100700h
0x1002bbaf: push ebx
0x1002bbb0: call 100230F0h
0x1002bbb5: add esp,8
0x1002bbb8: pop edi
0x1002bbb9: pop ebp
0x1002bbba: pop esi
0x1002bbbb: pop ebx
0x1002bbbc: ret
0x1002bbbd: mov ecx,[esp+20h]
0x1002bbc1: push ebp
0x1002bbc2: push ecx
0x1002bbc3: push eax
0x1002bbc4: call 100228C6h
0x1002bbc9: add edi,1
0x1002bbcc: push edi
0x1002bbcd: push ebx
0x1002bbce: call 1002CDA0h
0x1002bbd3: add esp,14h
0x1002bbd6: test eax,eax
0x1002bbd8: mov [esi+0E8h],eax
0x1002bbde: jne short 1002BC0Ah
0x1002bbe0: mov edx,[esi+0E4h]
0x1002bbe6: push edx
0x1002bbe7: push ebx
0x1002bbe8: call 1002CD70h
0x1002bbed: push 10100700h
0x1002bbf2: push ebx
0x1002bbf3: mov dword ptr [esi+0E4h],0
0x1002bbfd: call 100230F0h
0x1002bc02: add esp,10h
0x1002bc05: pop edi
0x1002bc06: pop ebp
0x1002bc07: pop esi
0x1002bc08: pop ebx
0x1002bc09: ret
0x1002bc0a: mov ecx,[esp+24h]
0x1002bc0e: push edi
0x1002bc0f: push ecx
0x1002bc10: push eax
0x1002bc11: call 100228C6h
0x1002bc16: or dword ptr [esi+8],4000h
0x1002bc1d: or dword ptr [esi+0BCh],100h
0x1002bc27: add esp,0Ch
0x1002bc2a: pop edi
0x1002bc2b: pop ebp
0x1002bc2c: pop esi
0x1002bc2d: pop ebx
0x1002bc2e: ret
0x1002bc2f: push 101006ECh
0x1002bc34: push ebx
0x1002bc35: call 100231E0h
0x1002bc3a: push 101006D8h
0x1002bc3f: push ebx
0x1002bc40: call 100231E0h
0x1002bc45: int 3
```

`png_set_sCAL_s` is called by png_set_sCAL, png_set_sCAL_fixed, sub_1002fdd0. It works with the text "Invalid sCAL width", "Invalid sCAL height", "Memory allocation failed while processing sCAL". It delegates to j_memcpy, png_warning, png_error (6 callees).

12.698 png_set_sCAL at 0x1002bc50 (79 instructions)

```
0x1002bc50: sub esp,2Ch
0x1002bc53: mov eax,ds:[10136220h]
0x1002bc58: xor eax,esp
0x1002bc5a: mov [esp+28h],eax
0x1002bc5e: fldz
0x1002bc60: push esi
0x1002bc61: fld qword ptr [esp+40h]
0x1002bc65: mov esi,[esp+34h]
0x1002bc69: fcom
0x1002bc6b: push edi
0x1002bc6c: mov edi,[esp+3Ch]
0x1002bc70: fnstsw ax
0x1002bc72: test ah,41h
0x1002bc75: jp short 1002BC9Ah
0x1002bc77: push 10100760h
0x1002bc7c: fstp st(1)
0x1002bc7e: push esi
0x1002bc7f: fstp st(0)
0x1002bc81: call 100230F0h
0x1002bc86: add esp,8
0x1002bc89: pop edi
0x1002bc8a: pop esi
0x1002bc8b: mov ecx,[esp+28h]
0x1002bc8f: xor ecx,esp
0x1002bc91: call 10022326h
0x1002bc96: add esp,2Ch
0x1002bc99: ret
0x1002bc9a: fxch
0x1002bc9c: fcomp qword ptr [esp+4Ch]
0x1002bca0: fnstsw ax
0x1002bca2: test ah,1
0x1002bca5: jne short 1002BCC8h
0x1002bca7: push 10100744h
0x1002bcac: fstp st(0)
0x1002bcae: push esi
0x1002bcaf: call 100230F0h
0x1002bcb4: add esp,8
0x1002bcb7: pop edi
0x1002bcb8: pop esi
0x1002bcb9: mov ecx,[esp+28h]
0x1002bcbd: xor ecx,esp
0x1002bcbf: call 10022326h
0x1002bcc4: add esp,2Ch
0x1002bcc7: ret
0x1002bcc8: push 5
0x1002bcca: sub esp,8
0x1002bccd: fstp qword ptr [esp]
0x1002bcd0: push 12h
0x1002bcd2: lea eax,[esp+2Ch]
0x1002bcd6: push eax
0x1002bcd7: push esi
0x1002bcd8: call 10024050h
0x1002bcdd: fld qword ptr [esp+64h]
0x1002bce1: add esp,18h
0x1002bce4: push 5
0x1002bce6: sub esp,8
0x1002bce9: fstp qword ptr [esp]
0x1002bcec: push 12h
0x1002bcee: lea ecx,[esp+18h]
0x1002bcf2: push ecx
0x1002bcf3: push esi
0x1002bcf4: call 10024050h
0x1002bcf9: mov ecx,[esp+58h]
0x1002bcfd: lea edx,[esp+20h]
0x1002bd01: push edx
0x1002bd02: lea eax,[esp+38h]
```

```
0x1002bd06: push eax
0x1002bd07: push ecx
0x1002bd08: push edi
0x1002bd09: push esi
0x1002bd0a: call 1002BAE0h
0x1002bd0f: mov ecx,[esp+5Ch]
0x1002bd13: add esp,2Ch
0x1002bd16: pop edi
0x1002bd17: pop esi
0x1002bd18: xor ecx,esp
0x1002bd1a: call 10022326h
0x1002bd1f: add esp,2Ch
0x1002bd22: ret
```

`png_set_sCAL` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "Invalid sCAL height ignored", "Invalid sCAL width ignored". It delegates to sub_10022326, png_warning, sub_10024050 (4 callees).

12.699 png_set_sCAL_fixed at 0x1002bd30 (68 instructions)

```
0x1002bd30: sub esp,2Ch
0x1002bd33: mov eax,ds:[10136220h]
0x1002bd38: xor eax,esp
0x1002bd3a: mov [esp+28h],eax
0x1002bd3e: mov eax,[esp+3Ch]
0x1002bd42: test eax,eax
0x1002bd44: push esi
0x1002bd45: mov esi,[esp+34h]
0x1002bd49: push edi
0x1002bd4a: mov edi,[esp+3Ch]
0x1002bd4e: jg short 1002BD6Fh
0x1002bd50: push 10100760h
0x1002bd55: push esi
0x1002bd56: call 100230F0h
0x1002bd5b: add esp,8
0x1002bd5e: pop edi
0x1002bd5f: pop esi
0x1002bd60: mov ecx,[esp+28h]
0x1002bd64: xor ecx,esp
0x1002bd66: call 10022326h
0x1002bd6b: add esp,2Ch
0x1002bd6e: ret
0x1002bd6f: push ebx
0x1002bd70: mov ebx,[esp+4Ch]
0x1002bd74: test ebx,ebx
0x1002bd76: jg short 1002BD98h
0x1002bd78: push 10100744h
0x1002bd7d: push esi
0x1002bd7e: call 100230F0h
0x1002bd83: add esp,8
0x1002bd86: pop ebx
0x1002bd87: pop edi
0x1002bd88: pop esi
0x1002bd89: mov ecx,[esp+28h]
0x1002bd8d: xor ecx,esp
0x1002bd8f: call 10022326h
0x1002bd94: add esp,2Ch
0x1002bd97: ret
0x1002bd98: push eax
0x1002bd99: push 12h
0x1002bd9b: lea eax,[esp+28h]
0x1002bd9f: push eax
0x1002bda0: push esi
0x1002bda1: call 100245F0h
0x1002bda6: push ebx
0x1002bda7: push 12h
```

```
0x1002bda9: lea ecx,[esp+24h]
0x1002bdad: push ecx
0x1002bdae: push esi
0x1002bdaf: call 100245F0h
0x1002bdb4: mov ecx,[esp+64h]
0x1002bdb8: lea edx,[esp+2Ch]
0x1002bdbc: push edx
0x1002bdbd: lea eax,[esp+44h]
0x1002bdc1: push eax
0x1002bdc2: push ecx
0x1002bdc3: push edi
0x1002bdc4: push esi
0x1002bdc5: call 1002BAE0h
0x1002bdca: mov ecx,[esp+68h]
0x1002bdce: add esp,34h
0x1002bdd1: pop ebx
0x1002bdd2: pop edi
0x1002bdd3: pop esi
0x1002bdd4: xor ecx,esp
0x1002bdd6: call 10022326h
0x1002bddb: add esp,2Ch
0x1002bdde: ret
```

`png_set_sCAL_fixed` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "Invalid sCAL height ignored", "Invalid sCAL width ignored". It delegates to sub_10022326, png_warning, sub_100245f0 (4 callees).

12.700 png_set_pHYs at 0x1002bde0 (13 instructions)

```
0x1002bde0: cmp dword ptr [esp+4],0
0x1002bde5: je short 1002BE0Bh
0x1002bde7: mov eax,[esp+8]
0x1002bdeb: test eax,eax
0x1002bded: je short 1002BE0Bh
0x1002bdef: mov ecx,[esp+0Ch]
0x1002bdf3: mov edx,[esp+10h]
0x1002bdf7: or dword ptr [eax+8],80h
0x1002bdfe: mov [eax+74h],ecx
0x1002be01: mov cl,[esp+14h]
0x1002be05: mov [eax+78h],edx
0x1002be08: mov [eax+7Ch],cl
0x1002be0b: ret
```

`png_set_pHYs` is called by sub_1002f7e0. It calls nothing else.

12.701 png_set_PLTE at 0x1002be10 (49 instructions)

```
0x1002be10: push edi
0x1002be11: mov edi,[esp+8]
0x1002be15: test edi,edi
0x1002be17: je near ptr 1002BEA5h
0x1002be1d: push esi
0x1002be1e: mov esi,[esp+10h]
0x1002be22: test esi,esi
0x1002be24: je short 1002BEA4h
0x1002be26: push ebx
0x1002be27: mov ebx,[esp+1Ch]
0x1002be2b: cmp ebx,100h
0x1002be31: ja short 1002BE8Ah
0x1002be33: push 0
0x1002be35: push 1000h
0x1002be3a: push esi
0x1002be3b: push edi
0x1002be3c: call 100235E0h
0x1002be41: push 300h
```

```
0x1002be46: push edi
0x1002be47: call 1002CDD0h
0x1002be4c: mov edx,[esp+30h]
0x1002be50: lea ecx,[ebx+ebx*2]
0x1002be53: push ecx
0x1002be54: push edx
0x1002be55: push eax
0x1002be56: mov [edi+114h],eax
0x1002be5c: call 100228C6h
0x1002be61: mov eax,[edi+114h]
0x1002be67: add esp,24h
0x1002be6a: mov [esi+10h],eax
0x1002be6d: mov [edi+118h],bx
0x1002be74: or dword ptr [esi+0BCh],1000h
0x1002be7e: or dword ptr [esi+8],8
0x1002be82: mov [esi+14h],bx
0x1002be86: pop ebx
0x1002be87: pop esi
0x1002be88: pop edi
0x1002be89: ret
0x1002be8a: cmp byte ptr [esi+19h],3
0x1002be8e: push 1010077Ch
0x1002be93: push edi
0x1002be94: jne short 1002BE9Bh
0x1002be96: call 100231E0h
0x1002be9b: call 100230F0h
0x1002bea0: add esp,8
0x1002bea3: pop ebx
0x1002bea4: pop esi
0x1002bea5: pop edi
0x1002bea6: ret
```

`png_set_PLTE` is called by sub_1002e260. It delegates to j_memcpy, png_warning, png_error (5 callees).

12.702 png_set_sBIT at 0x1002beb0 (12 instructions)

```
0x1002beb0: cmp dword ptr [esp+4],0
0x1002beb5: je short 1002BED2h
0x1002beb7: mov eax,[esp+8]
0x1002bebb: test eax,eax
0x1002bebd: je short 1002BED2h
0x1002bebf: mov ecx,[esp+0Ch]
0x1002bec3: mov edx,[ecx]
0x1002bec5: mov [eax+44h],edx
0x1002bec8: mov cl,[ecx+4]
0x1002becb: or dword ptr [eax+8],2
0x1002becf: mov [eax+48h],cl
0x1002bed2: ret
```

`png_set_sBIT` is called by sub_1002e590. It calls nothing else.

12.703 png_set_sRGB at 0x1002bee0 (9 instructions)

```
0x1002bee0: cmp dword ptr [esp+4],0
0x1002bee5: je short 1002BEFDh
0x1002bee7: mov eax,[esp+8]
0x1002beeb: test eax,eax
0x1002beed: je short 1002BEFDh
0x1002beef: mov cl,[esp+0Ch]
0x1002bef3: or dword ptr [eax+8],800h
0x1002befa: mov [eax+2Ch],cl
0x1002befd: ret
```

`png_set_sRGB` is called by png_set_sRGB_gAMA_and_cHRM. It calls nothing else.

12.704 png_set_sRGB_gAMA_and_cHRM at 0x1002bf00 (32 instructions)

```
0x1002bf00: push edi
0x1002bf01: mov edi,[esp+8]
0x1002bf05: test edi,edi
0x1002bf07: je short 1002BF5Dh
0x1002bf09: push esi
0x1002bf0a: mov esi,[esp+10h]
0x1002bf0e: test esi,esi
0x1002bf10: je short 1002BF5Ch
0x1002bf12: mov eax,[esp+14h]
0x1002bf16: push eax
0x1002bf17: push esi
0x1002bf18: push edi
0x1002bf19: call 1002BEE0h
0x1002bf1e: push 0B18Fh
0x1002bf23: push esi
0x1002bf24: push edi
0x1002bf25: call 1002B6A0h
0x1002bf2a: push 1770h
0x1002bf2f: push 3A98h
0x1002bf34: push 0EA60h
0x1002bf39: push 7530h
0x1002bf3e: push 80E8h
0x1002bf43: push 0FA00h
0x1002bf48: push 8084h
0x1002bf4d: push 7A26h
0x1002bf52: push esi
0x1002bf53: push edi
0x1002bf54: call 1002B520h
0x1002bf59: add esp,40h
0x1002bf5c: pop esi
0x1002bf5d: pop edi
0x1002bf5e: ret
```

`png_set_sRGB_gAMA_and_cHRM` is called by sub_1002ea20. It delegates to png_set_cHRM_fixed, png_set_gAMA_fixed, png_set_sRGB (3 callees).

12.705 png_set_iCCP at 0x1002bf60 (85 instructions)

```
0x1002bf60: push ebp
0x1002bf61: mov ebp,[esp+8]
0x1002bf65: test ebp,ebp
0x1002bf67: je near ptr 1002C04Bh
0x1002bf6d: push esi
0x1002bf6e: mov esi,[esp+10h]
0x1002bf72: test esi,esi
0x1002bf74: je near ptr 1002C04Ah
0x1002bf7a: mov eax,[esp+14h]
0x1002bf7e: test eax,eax
0x1002bf80: je near ptr 1002C04Ah
0x1002bf86: cmp dword ptr [esp+1Ch],0
0x1002bf8b: je near ptr 1002C04Ah
0x1002bf91: push ebx
0x1002bf92: push edi
0x1002bf93: push eax
0x1002bf94: call dword ptr ds:[100FE0B0h]
0x1002bf9a: mov edi,eax
0x1002bf9c: add edi,1
0x1002bf9f: push edi
0x1002bfa0: push ebp
0x1002bfa1: call 1002CDA0h
0x1002bfa6: mov ebx,eax
0x1002bfa8: add esp,8
0x1002bfab: test ebx,ebx
0x1002bfad: jne short 1002BFC2h
```

```
0x1002bfaf: push 101007C0h
0x1002bfb4: push ebp
0x1002bfb5: call 100230F0h
0x1002bfba: add esp,8
0x1002bfbd: pop edi
0x1002bfbe: pop ebx
0x1002bfbf: pop esi
0x1002bfc0: pop ebp
0x1002bfc1: ret
0x1002bfc2: mov eax,[esp+1Ch]
0x1002bfc6: push edi
0x1002bfc7: push eax
0x1002bfc8: push ebx
0x1002bfc9: call 100228C6h
0x1002bfce: mov ecx,[esp+34h]
0x1002bfd2: push ecx
0x1002bfd3: push ebp
0x1002bfd4: call 1002CDA0h
0x1002bfd9: mov edi,eax
0x1002bfdb: add esp,14h
0x1002bfde: test edi,edi
0x1002bfe0: jne short 1002BFFCh
0x1002bfe2: push ebx
0x1002bfe3: push ebp
0x1002bfe4: call 1002CD70h
0x1002bfe9: push 10100794h
0x1002bfee: push ebp
0x1002bfef: call 100230F0h
0x1002bff4: add esp,10h
0x1002bff7: pop edi
0x1002bff8: pop ebx
0x1002bff9: pop esi
0x1002bffa: pop ebp
0x1002bffb: ret
0x1002bffc: mov edx,[esp+28h]
0x1002c000: mov eax,[esp+24h]
0x1002c004: push edx
0x1002c005: push eax
0x1002c006: push edi
0x1002c007: call 100228C6h
0x1002c00c: push 0
0x1002c00e: push 10h
0x1002c010: push esi
0x1002c011: push ebp
0x1002c012: call 100235E0h
0x1002c017: mov ecx,[esp+44h]
0x1002c01b: mov dl,[esp+3Ch]
0x1002c01f: or dword ptr [esi+0BCh],10h
0x1002c026: add esp,1Ch
0x1002c029: or dword ptr [esi+8],1000h
0x1002c030: mov [esi+0CCh],edi
0x1002c036: pop edi
0x1002c037: mov [esi+0C8h],ebx
0x1002c03d: mov [esi+0D0h],ecx
0x1002c043: mov [esi+0D4h],dl
0x1002c049: pop ebx
0x1002c04a: pop esi
0x1002c04b: pop ebp
0x1002c04c: ret
```

`png_set_iCCP` is called by sub_1002ec10. It works with the text "Insufficient memory to process iCCP profile", "Insufficient memory to process iCCP chunk". It delegates to j_memcpy, png_warning, png_free_data (5 callees).

12.706 sub_1002c050 at 0x1002c050 (241 instructions)

```
0x1002c050: mov ecx,[esp+4]
0x1002c054: sub esp,0Ch
0x1002c057: test ecx,ecx
0x1002c059: push ebx
0x1002c05a: push ebp
0x1002c05b: je near ptr 1002C2FEh
0x1002c061: mov ebx,[esp+1Ch]
0x1002c065: test ebx,ebx
0x1002c067: je near ptr 1002C2FEh
0x1002c06d: mov ebp,[esp+24h]
0x1002c071: test ebp,ebp
0x1002c073: je near ptr 1002C2FEh
0x1002c079: mov eax,[ebx+30h]
0x1002c07c: push esi
0x1002c07d: push edi
0x1002c07e: mov edi,[ebx+34h]
0x1002c081: add eax,ebp
0x1002c083: cmp eax,edi
0x1002c085: jle near ptr 1002C12Bh
0x1002c08b: mov esi,[ebx+38h]
0x1002c08e: test esi,esi
0x1002c090: je short 1002C0F6h
0x1002c092: add eax,8
0x1002c095: lea edx,[eax*8]
0x1002c09c: sub edx,eax
0x1002c09e: add edx,edx
0x1002c0a0: add edx,edx
0x1002c0a2: push edx
0x1002c0a3: push ecx
0x1002c0a4: mov [ebx+34h],eax
0x1002c0a7: call 1002CDA0h
0x1002c0ac: add esp,8
0x1002c0af: test eax,eax
0x1002c0b1: mov [ebx+38h],eax
0x1002c0b4: jne short 1002C0D1h
0x1002c0b6: mov eax,[esp+20h]
0x1002c0ba: push esi
0x1002c0bb: push eax
0x1002c0bc: call 1002CD70h
0x1002c0c1: add esp,8
0x1002c0c4: pop edi
0x1002c0c5: pop esi
0x1002c0c6: pop ebp
0x1002c0c7: mov eax,1
0x1002c0cc: pop ebx
0x1002c0cd: add esp,0Ch
0x1002c0d0: ret
0x1002c0d1: lea ecx,[edi*8]
0x1002c0d8: sub ecx,edi
0x1002c0da: add ecx,ecx
0x1002c0dc: add ecx,ecx
0x1002c0de: push ecx
0x1002c0df: push esi
0x1002c0e0: push eax
0x1002c0e1: call 100228C6h
0x1002c0e6: mov edx,[esp+2Ch]
0x1002c0ea: push esi
0x1002c0eb: push edx
0x1002c0ec: call 1002CD70h
0x1002c0f1: add esp,14h
0x1002c0f4: jmp short 1002C12Bh
0x1002c0f6: lea eax,[ebp+8]
0x1002c0f9: lea edx,[eax*8]
0x1002c100: sub edx,eax
0x1002c102: add edx,edx
0x1002c104: add edx,edx
```

```
0x1002c106: push edx
0x1002c107: push ecx
0x1002c108: mov [ebx+34h],eax
0x1002c10b: mov dword ptr [ebx+30h],0
0x1002c112: call 1002CDA0h
0x1002c117: add esp,8
0x1002c11a: test eax,eax
0x1002c11c: mov [ebx+38h],eax
0x1002c11f: je short 1002C0C4h
0x1002c121: or dword ptr [ebx+0BCh],4000h
0x1002c12b: test ebp,ebp
0x1002c12d: mov dword ptr [esp+18h],0
0x1002c135: jle near ptr 1002C2F4h
0x1002c13b: mov edi,[esp+28h]
0x1002c13f: add edi,14h
0x1002c142: mov eax,[ebx+30h]
0x1002c145: mov edx,[ebx+38h]
0x1002c148: lea ecx,[eax*8]
0x1002c14f: sub ecx,eax
0x1002c151: mov eax,[edi-10h]
0x1002c154: test eax,eax
0x1002c156: lea esi,[edx+ecx*4]
0x1002c159: je near ptr 1002C2DEh
0x1002c15f: mov ecx,[edi-14h]
0x1002c162: cmp ecx,0FFFFFFFFh
0x1002c165: jl near ptr 1002C2CCh
0x1002c16b: cmp ecx,3
0x1002c16e: jge near ptr 1002C2CCh
0x1002c174: mov ebp,ds:[100FE0B0h]
0x1002c17a: push eax
0x1002c17b: call ebp
0x1002c17d: cmp dword ptr [edi-14h],0
0x1002c181: mov [esp+10h],eax
0x1002c185: jg short 1002C1A9h
0x1002c187: xor ebx,ebx
0x1002c189: xor ebp,ebp
0x1002c18b: mov eax,[edi-0Ch]
0x1002c18e: test eax,eax
0x1002c190: je short 1002C1C6h
0x1002c192: cmp byte ptr [eax],0
0x1002c195: je short 1002C1C6h
0x1002c197: push eax
0x1002c198: call dword ptr ds:[100FE0B0h]
0x1002c19e: mov [esp+14h],eax
0x1002c1a2: mov eax,[edi-14h]
0x1002c1a5: mov [esi],eax
0x1002c1a7: jmp short 1002C1DCh
0x1002c1a9: mov eax,[edi]
0x1002c1ab: test eax,eax
0x1002c1ad: je short 1002C1B6h
0x1002c1af: push eax
0x1002c1b0: call ebp
0x1002c1b2: mov ebx,eax
0x1002c1b4: jmp short 1002C1B8h
0x1002c1b6: xor ebx,ebx
0x1002c1b8: mov eax,[edi+4]
0x1002c1bb: test eax,eax
0x1002c1bd: je short 1002C189h
0x1002c1bf: push eax
0x1002c1c0: call ebp
0x1002c1c2: mov ebp,eax
0x1002c1c4: jmp short 1002C18Bh
0x1002c1c6: xor ecx,ecx
0x1002c1c8: cmp [edi-14h],ecx
0x1002c1cb: mov dword ptr [esp+14h],0
0x1002c1d3: setg cl
0x1002c1d6: lea ecx,[ecx+ecx-1]
```

```
0x1002c1da: mov [esi],ecx
0x1002c1dc: mov edx,[esp+14h]
0x1002c1e0: mov eax,[esp+10h]
0x1002c1e4: add edx,ebp
0x1002c1e6: add edx,ebx
0x1002c1e8: lea ecx,[edx+eax+4]
0x1002c1ec: mov edx,[esp+20h]
0x1002c1f0: push ecx
0x1002c1f1: push edx
0x1002c1f2: call 1002CDA0h
0x1002c1f7: add esp,8
0x1002c1fa: test eax,eax
0x1002c1fc: mov [esi+4],eax
0x1002c1ff: je near ptr 1002C0C4h
0x1002c205: mov ecx,[esp+10h]
0x1002c209: mov edx,[edi-10h]
0x1002c20c: push ecx
0x1002c20d: push edx
0x1002c20e: push eax
0x1002c20f: call 100228C6h
0x1002c214: mov ecx,[esi+4]
0x1002c217: mov eax,[esp+1Ch]
0x1002c21b: mov byte ptr [eax+ecx],0
0x1002c21f: mov edx,[esi+4]
0x1002c222: xor ecx,ecx
0x1002c224: add esp,0Ch
0x1002c227: cmp [edi-14h],ecx
0x1002c22a: lea eax,[edx+eax+1]
0x1002c22e: jle short 1002C271h
0x1002c230: mov [esi+14h],eax
0x1002c233: mov ecx,[edi]
0x1002c235: push ebx
0x1002c236: push ecx
0x1002c237: push eax
0x1002c238: call 100228C6h
0x1002c23d: mov edx,[esi+14h]
0x1002c240: mov byte ptr [ebx+edx],0
0x1002c244: mov eax,[esi+14h]
0x1002c247: lea ebx,[eax+ebx+1]
0x1002c24b: mov [esi+18h],ebx
0x1002c24e: mov ecx,[edi+4]
0x1002c251: push ebp
0x1002c252: push ecx
0x1002c253: push ebx
0x1002c254: call 100228C6h
0x1002c259: mov edx,[esi+18h]
0x1002c25c: mov byte ptr [edx+ebp],0
0x1002c260: mov eax,[esi+18h]
0x1002c263: lea ecx,[eax+ebp+1]
0x1002c267: add esp,18h
0x1002c26a: mov [esi+8],ecx
0x1002c26d: xor ecx,ecx
0x1002c26f: jmp short 1002C27Ah
0x1002c271: mov [esi+14h],ecx
0x1002c274: mov [esi+18h],ecx
0x1002c277: mov [esi+8],eax
0x1002c27a: mov ebx,[esp+14h]
0x1002c27e: cmp ebx,ecx
0x1002c280: je short 1002C295h
0x1002c282: mov ecx,[edi-0Ch]
0x1002c285: mov edx,[esi+8]
0x1002c288: push ebx
0x1002c289: push ecx
0x1002c28a: push edx
0x1002c28b: call 100228C6h
0x1002c290: add esp,0Ch
0x1002c293: xor ecx,ecx
```

```
0x1002c295: mov eax,[esi+8]
0x1002c298: mov byte ptr [ebx+eax],0
0x1002c29c: cmp [esi],ecx
0x1002c29e: jle short 1002C2B6h
0x1002c2a0: mov eax,[esp+24h]
0x1002c2a4: mov ebp,[esp+2Ch]
0x1002c2a8: mov [esi+10h],ebx
0x1002c2ab: mov [esi+0Ch],ecx
0x1002c2ae: add dword ptr [eax+30h],1
0x1002c2b2: mov ebx,eax
0x1002c2b4: jmp short 1002C2DEh
0x1002c2b6: mov eax,[esp+24h]
0x1002c2ba: mov ebp,[esp+2Ch]
0x1002c2be: mov [esi+0Ch],ebx
0x1002c2c1: mov [esi+10h],ecx
0x1002c2c4: add dword ptr [eax+30h],1
0x1002c2c8: mov ebx,eax
0x1002c2ca: jmp short 1002C2DEh
0x1002c2cc: mov ecx,[esp+20h]
0x1002c2d0: push 101007ECh
0x1002c2d5: push ecx
0x1002c2d6: call 100230F0h
0x1002c2db: add esp,8
0x1002c2de: mov eax,[esp+18h]
0x1002c2e2: add eax,1
0x1002c2e5: add edi,1Ch
0x1002c2e8: cmp eax,ebp
0x1002c2ea: mov [esp+18h],eax
0x1002c2ee: jl near ptr 1002C142h
0x1002c2f4: pop edi
0x1002c2f5: pop esi
0x1002c2f6: pop ebp
0x1002c2f7: xor eax,eax
0x1002c2f9: pop ebx
0x1002c2fa: add esp,0Ch
0x1002c2fd: ret
0x1002c2fe: pop ebp
0x1002c2ff: xor eax,eax
0x1002c301: pop ebx
0x1002c302: add esp,0Ch
0x1002c305: ret
```

`sub_1002c050` is called by png_set_text, sub_100300d0, sub_10030280, sub_10030490. It works with the text "text compression mode is out of range". It delegates to j_memcpy, png_warning, png_free (4 callees).

12.707 png_set_tIME at 0x1002c310 (18 instructions)

```
0x1002c310: mov ecx,[esp+4]
0x1002c314: test ecx,ecx
0x1002c316: je short 1002C33Eh
0x1002c318: mov eax,[esp+8]
0x1002c31c: test eax,eax
0x1002c31e: je short 1002C33Eh
0x1002c320: mov edx,200h
0x1002c325: test [ecx+6Ch],edx
0x1002c328: jne short 1002C33Eh
0x1002c32a: mov ecx,[esp+0Ch]
0x1002c32e: push esi
0x1002c32f: mov esi,[ecx]
0x1002c331: mov [eax+3Ch],esi
0x1002c334: mov ecx,[ecx+4]
0x1002c337: or [eax+8],edx
0x1002c33a: mov [eax+40h],ecx
0x1002c33d: pop esi
0x1002c33e: ret
```

`png_set_tIME` is called by sub_1002ffb0. It calls nothing else.

12.708 png_set_tRNS at 0x1002c340 (80 instructions)

```
0x1002c340: push ebp
0x1002c341: mov ebp,[esp+8]
0x1002c345: test ebp,ebp
0x1002c347: je near ptr 1002C427h
0x1002c34d: push esi
0x1002c34e: mov esi,[esp+10h]
0x1002c352: test esi,esi
0x1002c354: je near ptr 1002C426h
0x1002c35a: push ebx
0x1002c35b: mov ebx,[esp+1Ch]
0x1002c35f: push edi
0x1002c360: mov edi,[esp+1Ch]
0x1002c364: test edi,edi
0x1002c366: je short 1002C3A5h
0x1002c368: push 0
0x1002c36a: push 2000h
0x1002c36f: push esi
0x1002c370: push ebp
0x1002c371: call 100235E0h
0x1002c376: push 100h
0x1002c37b: push ebp
0x1002c37c: call 1002CD20h
0x1002c381: mov [esi+50h],eax
0x1002c384: mov [ebp+188h],eax
0x1002c38a: lea eax,[ebx-1]
0x1002c38d: add esp,18h
0x1002c390: cmp eax,0FFh
0x1002c395: ja short 1002C3A5h
0x1002c397: mov ecx,[esi+50h]
0x1002c39a: push ebx
0x1002c39b: push edi
0x1002c39c: push ecx
0x1002c39d: call 100228C6h
0x1002c3a2: add esp,0Ch
0x1002c3a5: mov edi,[esp+24h]
0x1002c3a9: test edi,edi
0x1002c3ab: je short 1002C40Eh
0x1002c3ad: movzx ecx,byte ptr [esi+18h]
0x1002c3b1: mov eax,1
0x1002c3b6: shl eax,cl
0x1002c3b8: mov cl,[esi+19h]
0x1002c3bb: test cl,cl
0x1002c3bd: jne short 1002C3C7h
0x1002c3bf: movzx edx,word ptr [edi+8]
0x1002c3c3: cmp edx,eax
0x1002c3c5: jg short 1002C3E4h
0x1002c3c7: cmp cl,2
0x1002c3ca: jne short 1002C3F2h
0x1002c3cc: movzx ecx,word ptr [edi+2]
0x1002c3d0: cmp ecx,eax
0x1002c3d2: jg short 1002C3E4h
0x1002c3d4: movzx edx,word ptr [edi+4]
0x1002c3d8: cmp edx,eax
0x1002c3da: jg short 1002C3E4h
0x1002c3dc: movzx ecx,word ptr [edi+6]
0x1002c3e0: cmp ecx,eax
0x1002c3e2: jle short 1002C3F2h
0x1002c3e4: push 10100814h
0x1002c3e9: push ebp
0x1002c3ea: call 100230F0h
0x1002c3ef: add esp,8
0x1002c3f2: test ebx,ebx
0x1002c3f4: mov edx,[edi]
```

```
0x1002c3f6: mov [esi+54h],edx
0x1002c3f9: mov eax,[edi+4]
0x1002c3fc: mov [esi+58h],eax
0x1002c3ff: mov cx,[edi+8]
0x1002c403: mov [esi+5Ch],cx
0x1002c407: jne short 1002C40Eh
0x1002c409: mov ebx,1
0x1002c40e: pop edi
0x1002c40f: mov [esi+16h],bx
0x1002c413: test ebx,ebx
0x1002c415: pop ebx
0x1002c416: je short 1002C426h
0x1002c418: or dword ptr [esi+8],10h
0x1002c41c: or dword ptr [esi+0BCh],2000h
0x1002c426: pop esi
0x1002c427: pop ebp
0x1002c428: ret
```

`png_set_tRNS` is called by sub_1002f170. It delegates to j_memcpy, png_warning, png_free_data (4 callees).

12.709 png_set_sPLT at 0x1002c430 (131 instructions)

```
0x1002c430: push ecx
0x1002c431: push ebp
0x1002c432: mov ebp,[esp+0Ch]
0x1002c436: test ebp,ebp
0x1002c438: je near ptr 1002C5B1h
0x1002c43e: push esi
0x1002c43f: mov esi,[esp+14h]
0x1002c443: test esi,esi
0x1002c445: je near ptr 1002C5B0h
0x1002c44b: mov eax,[esi+0DCh]
0x1002c451: push ebx
0x1002c452: mov ebx,[esp+20h]
0x1002c456: add eax,ebx
0x1002c458: push edi
0x1002c459: shl eax,4
0x1002c45c: push eax
0x1002c45d: push ebp
0x1002c45e: call 1002CDA0h
0x1002c463: mov edi,eax
0x1002c465: add esp,8
0x1002c468: test edi,edi
0x1002c46a: mov [esp+10h],edi
0x1002c46e: jne short 1002C484h
0x1002c470: push 10100874h
0x1002c475: push ebp
0x1002c476: call 100230F0h
0x1002c47b: add esp,8
0x1002c47e: pop edi
0x1002c47f: pop ebx
0x1002c480: pop esi
0x1002c481: pop ebp
0x1002c482: pop ecx
0x1002c483: ret
0x1002c484: mov ecx,[esi+0DCh]
0x1002c48a: mov edx,[esi+0D8h]
0x1002c490: shl ecx,4
0x1002c493: push ecx
0x1002c494: push edx
0x1002c495: push edi
0x1002c496: call 100228C6h
0x1002c49b: mov eax,[esi+0D8h]
0x1002c4a1: push eax
0x1002c4a2: push ebp
0x1002c4a3: call 1002CD70h
```

```
0x1002c4a8: xor eax,eax
0x1002c4aa: add esp,14h
0x1002c4ad: cmp ebx,eax
0x1002c4af: mov [esi+0D8h],eax
0x1002c4b5: mov [esp+18h],eax
0x1002c4b9: jle near ptr 1002C594h
0x1002c4bf: mov ebx,[esp+20h]
0x1002c4c3: add ebx,0Ch
0x1002c4c6: jmp short 1002C4D0h
0x1002c4d0: mov esi,[esi+0DCh]
0x1002c4d6: add esi,[esp+18h]
0x1002c4da: mov ecx,[ebx-0Ch]
0x1002c4dd: shl esi,4
0x1002c4e0: push ecx
0x1002c4e1: add esi,edi
0x1002c4e3: call dword ptr ds:[100FE0B0h]
0x1002c4e9: mov edi,eax
0x1002c4eb: add edi,1
0x1002c4ee: push edi
0x1002c4ef: push ebp
0x1002c4f0: call 1002CDA0h
0x1002c4f5: add esp,8
0x1002c4f8: test eax,eax
0x1002c4fa: mov [esi],eax
0x1002c4fc: jne short 1002C50Eh
0x1002c4fe: push 10100848h
0x1002c503: push ebp
0x1002c504: call 100230F0h
0x1002c509: add esp,8
0x1002c50c: jmp short 1002C570h
0x1002c50e: mov edx,[ebx-0Ch]
0x1002c511: push edi
0x1002c512: push edx
0x1002c513: push eax
0x1002c514: call 100228C6h
0x1002c519: mov eax,[ebx]
0x1002c51b: lea eax,[eax+eax*4]
0x1002c51e: add eax,eax
0x1002c520: push eax
0x1002c521: push ebp
0x1002c522: call 1002CDA0h
0x1002c527: add esp,14h
0x1002c52a: test eax,eax
0x1002c52c: mov [esi+8],eax
0x1002c52f: jne short 1002C550h
0x1002c531: push 10100848h
0x1002c536: push ebp
0x1002c537: call 100230F0h
0x1002c53c: mov ecx,[esi]
0x1002c53e: push ecx
0x1002c53f: push ebp
0x1002c540: call 1002CD70h
0x1002c545: add esp,10h
0x1002c548: mov dword ptr [esi],0
0x1002c54e: jmp short 1002C570h
0x1002c550: mov ecx,[ebx]
0x1002c552: lea edx,[ecx+ecx*4]
0x1002c555: mov ecx,[ebx-4]
0x1002c558: add edx,edx
0x1002c55a: push edx
0x1002c55b: push ecx
0x1002c55c: push eax
0x1002c55d: call 100228C6h
0x1002c562: mov edx,[ebx]
0x1002c564: mov [esi+0Ch],edx
0x1002c567: mov al,[ebx-8]
0x1002c56a: add esp,0Ch
```

```
0x1002c56d: mov [esi+4],al
0x1002c570: mov eax,[esp+18h]
0x1002c574: mov edi,[esp+10h]
0x1002c578: mov esi,[esp+1Ch]
0x1002c57c: add eax,1
0x1002c57f: add ebx,10h
0x1002c582: cmp eax,[esp+24h]
0x1002c586: mov [esp+18h],eax
0x1002c58a: jl near ptr 1002C4D0h
0x1002c590: mov ebx,[esp+24h]
0x1002c594: add [esi+0DCh],ebx
0x1002c59a: or dword ptr [esi+8],2000h
0x1002c5a1: or dword ptr [esi+0BCh],20h
0x1002c5a8: mov [esi+0D8h],edi
0x1002c5ae: pop edi
0x1002c5af: pop ebx
0x1002c5b0: pop esi
0x1002c5b1: pop ebp
0x1002c5b2: pop ecx
0x1002c5b3: ret
```

`png_set_sPLT` is called by sub_1002ee90. It delegates to j_memcpy, png_warning, png_free (4 callees).

12.710 png_set_unknown_chunks at 0x1002c5c0 (112 instructions)

```
0x1002c5c0: push ecx
0x1002c5c1: push ebx
0x1002c5c2: mov ebx,[esp+0Ch]
0x1002c5c6: test ebx,ebx
0x1002c5c8: je near ptr 1002C713h
0x1002c5ce: push ebp
0x1002c5cf: mov ebp,[esp+14h]
0x1002c5d3: test ebp,ebp
0x1002c5d5: je near ptr 1002C712h
0x1002c5db: push edi
0x1002c5dc: mov edi,[esp+20h]
0x1002c5e0: test edi,edi
0x1002c5e2: je near ptr 1002C711h
0x1002c5e8: mov eax,[ebp+0C4h]
0x1002c5ee: add eax,edi
0x1002c5f0: lea eax,[eax+eax*4]
0x1002c5f3: add eax,eax
0x1002c5f5: push esi
0x1002c5f6: add eax,eax
0x1002c5f8: push eax
0x1002c5f9: push ebx
0x1002c5fa: call 1002CDA0h
0x1002c5ff: mov esi,eax
0x1002c601: add esp,8
0x1002c604: test esi,esi
0x1002c606: mov [esp+10h],esi
0x1002c60a: jne short 1002C620h
0x1002c60c: push 10100890h
0x1002c611: push ebx
0x1002c612: call 100230F0h
0x1002c617: add esp,8
0x1002c61a: pop esi
0x1002c61b: pop edi
0x1002c61c: pop ebp
0x1002c61d: pop ebx
0x1002c61e: pop ecx
0x1002c61f: ret
0x1002c620: mov eax,[ebp+0C4h]
0x1002c626: mov edx,[ebp+0C0h]
0x1002c62c: lea ecx,[eax+eax*4]
0x1002c62f: add ecx,ecx
```

```
0x1002c631: add ecx,ecx
0x1002c633: push ecx
0x1002c634: push edx
0x1002c635: push esi
0x1002c636: call 100228C6h
0x1002c63b: mov eax,[ebp+0C0h]
0x1002c641: push eax
0x1002c642: push ebx
0x1002c643: call 1002CD70h
0x1002c648: xor ecx,ecx
0x1002c64a: add esp,14h
0x1002c64d: cmp edi,ecx
0x1002c64f: mov [ebp+0C0h],ecx
0x1002c655: mov [esp+18h],ecx
0x1002c659: jle near ptr 1002C6FAh
0x1002c65f: mov edi,[esp+20h]
0x1002c663: add edi,0Ch
0x1002c666: jmp short 1002C670h
0x1002c670: mov eax,[ebp+0C4h]
0x1002c676: mov edx,[edi-0Ch]
0x1002c679: add eax,ecx
0x1002c67b: lea ecx,[eax+eax*4]
0x1002c67e: lea esi,[esi+ecx*4]
0x1002c681: mov [esi],edx
0x1002c683: mov byte ptr [esi+4],0
0x1002c687: mov ecx,[edi]
0x1002c689: mov [esi+0Ch],ecx
0x1002c68c: mov dl,[ebx+6Ch]
0x1002c68f: mov [esi+10h],dl
0x1002c692: cmp dword ptr [edi],0
0x1002c695: jne short 1002C6A0h
0x1002c697: mov dword ptr [esi+8],0
0x1002c69e: jmp short 1002C6DAh
0x1002c6a0: mov eax,[edi]
0x1002c6a2: push eax
0x1002c6a3: push ebx
0x1002c6a4: call 1002CDA0h
0x1002c6a9: add esp,8
0x1002c6ac: test eax,eax
0x1002c6ae: mov [esi+8],eax
0x1002c6b1: jne short 1002C6CAh
0x1002c6b3: push 10100890h
0x1002c6b8: push ebx
0x1002c6b9: call 100230F0h
0x1002c6be: add esp,8
0x1002c6c1: mov dword ptr [esi+0Ch],0
0x1002c6c8: jmp short 1002C6DAh
0x1002c6ca: mov ecx,[edi]
0x1002c6cc: mov edx,[edi-4]
0x1002c6cf: push ecx
0x1002c6d0: push edx
0x1002c6d1: push eax
0x1002c6d2: call 100228C6h
0x1002c6d7: add esp,0Ch
0x1002c6da: mov ecx,[esp+18h]
0x1002c6de: mov esi,[esp+10h]
0x1002c6e2: add ecx,1
0x1002c6e5: add edi,14h
0x1002c6e8: cmp ecx,[esp+24h]
0x1002c6ec: mov [esp+18h],ecx
0x1002c6f0: jl near ptr 1002C670h
0x1002c6f6: mov edi,[esp+24h]
0x1002c6fa: add [ebp+0C4h],edi
0x1002c700: or dword ptr [ebp+0BCh],200h
0x1002c70a: mov [ebp+0C0h],esi
0x1002c710: pop esi
0x1002c711: pop edi
```

```
0x1002c712: pop ebp
0x1002c713: pop ebx
0x1002c714: pop ecx
0x1002c715: ret
```

`png_set_unknown_chunks` is called by sub_10030750. It works with the text "Out of memory while processing unknown chunk". It delegates to j_memcpy, png_warning, png_free (4 callees).

12.711 png_set_unknown_chunk_location at 0x1002c720 (15 instructions)

```
0x1002c720: cmp dword ptr [esp+4],0
0x1002c725: je short 1002C750h
0x1002c727: mov ecx,[esp+8]
0x1002c72b: test ecx,ecx
0x1002c72d: je short 1002C750h
0x1002c72f: mov eax,[esp+0Ch]
0x1002c733: test eax,eax
0x1002c735: jl short 1002C750h
0x1002c737: cmp eax,[ecx+0C4h]
0x1002c73d: jge short 1002C750h
0x1002c73f: mov ecx,[ecx+0C0h]
0x1002c745: mov dl,[esp+10h]
0x1002c749: lea eax,[eax+eax*4]
0x1002c74c: mov [ecx+eax*4+10h],dl
0x1002c750: ret
```

`png_set_unknown_chunk_location` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.712 png_permit_mng_features at 0x1002c760 (9 instructions)

```
0x1002c760: mov ecx,[esp+4]
0x1002c764: test ecx,ecx
0x1002c766: jne short 1002C76Bh
0x1002c768: xor eax,eax
0x1002c76a: ret
0x1002c76b: mov eax,[esp+8]
0x1002c76f: and eax,5
0x1002c772: mov [ecx+230h],eax
0x1002c778: ret
```

`png_permit_mng_features` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.713 png_set_keep_unknown_chunks at 0x1002c780 (79 instructions)

```
0x1002c780: push esi
0x1002c781: mov esi,[esp+8]
0x1002c785: test esi,esi
0x1002c787: je near ptr 1002C86Fh
0x1002c78d: push ebx
0x1002c78e: mov ebx,[esp+18h]
0x1002c792: test ebx,ebx
0x1002c794: jne short 1002C7CDh
0x1002c796: mov eax,[esp+10h]
0x1002c79a: cmp eax,3
0x1002c79d: je short 1002C7ADh
0x1002c79f: cmp eax,2
0x1002c7a2: je short 1002C7ADh
0x1002c7a4: and dword ptr [esi+70h],0FFFF7FFFh
0x1002c7ab: jmp short 1002C7B4h
0x1002c7ad: or dword ptr [esi+70h],8000h
0x1002c7b4: cmp eax,3
```

```
0x1002c7b7: jne short 1002C7C3h
0x1002c7b9: or dword ptr [esi+70h],10000h
0x1002c7c0: pop ebx
0x1002c7c1: pop esi
0x1002c7c2: ret
0x1002c7c3: and dword ptr [esi+70h],0FFFEFFFFh
0x1002c7ca: pop ebx
0x1002c7cb: pop esi
0x1002c7cc: ret
0x1002c7cd: cmp dword ptr [esp+14h],0
0x1002c7d2: je near ptr 1002C86Eh
0x1002c7d8: push ebp
0x1002c7d9: push edi
0x1002c7da: mov edi,[esi+220h]
0x1002c7e0: lea eax,[edi+ebx]
0x1002c7e3: mov [esp+14h],eax
0x1002c7e7: lea eax,[eax+eax*4]
0x1002c7ea: push eax
0x1002c7eb: push esi
0x1002c7ec: call 1002CD20h
0x1002c7f1: mov ebp,eax
0x1002c7f3: mov eax,[esi+224h]
0x1002c7f9: add esp,8
0x1002c7fc: test eax,eax
0x1002c7fe: je short 1002C825h
0x1002c800: lea ecx,[edi+edi*4]
0x1002c803: push ecx
0x1002c804: push eax
0x1002c805: push ebp
0x1002c806: call 100228C6h
0x1002c80b: mov edx,[esi+224h]
0x1002c811: push edx
0x1002c812: push esi
0x1002c813: call 1002CD70h
0x1002c818: add esp,14h
0x1002c81b: mov dword ptr [esi+224h],0
0x1002c825: mov edx,[esp+1Ch]
0x1002c829: lea ecx,[ebx+ebx*4]
0x1002c82c: push ecx
0x1002c82d: lea eax,[ebp+edi*4]
0x1002c831: add edi,eax
0x1002c833: push edx
0x1002c834: push edi
0x1002c835: call 100228C6h
0x1002c83a: add esp,0Ch
0x1002c83d: test ebx,ebx
0x1002c83f: lea eax,[edi+4]
0x1002c842: jle short 1002C852h
0x1002c844: mov cl,[esp+18h]
0x1002c848: mov [eax],cl
0x1002c84a: add eax,5
0x1002c84d: sub ebx,1
0x1002c850: jne short 1002C848h
0x1002c852: mov eax,[esp+14h]
0x1002c856: or dword ptr [esi+214h],400h
0x1002c860: pop edi
0x1002c861: mov [esi+224h],ebp
0x1002c867: mov [esi+220h],eax
0x1002c86d: pop ebp
0x1002c86e: pop ebx
0x1002c86f: pop esi
0x1002c870: ret
```

`png_set_keep_unknown_chunks` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to j_memcpy, png_malloc, png_free (3 callees).

12.714 png_set_read_user_chunk_fn at 0x1002c880 (8 instructions)

```
0x1002c880: mov eax,[esp+4]
0x1002c884: test eax,eax
0x1002c886: je short 1002C89Ch
0x1002c888: mov ecx,[esp+0Ch]
0x1002c88c: mov edx,[esp+8]
0x1002c890: mov [eax+21Ch],ecx
0x1002c896: mov [eax+218h],edx
0x1002c89c: ret
```

`png_set_read_user_chunk_fn` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.715 png_set_rows at 0x1002c8a0 (27 instructions)

```
0x1002c8a0: mov ecx,[esp+4]
0x1002c8a4: test ecx,ecx
0x1002c8a6: je short 1002C8E5h
0x1002c8a8: push esi
0x1002c8a9: mov esi,[esp+0Ch]
0x1002c8ad: test esi,esi
0x1002c8af: je short 1002C8E4h
0x1002c8b1: mov eax,[esi+0ECh]
0x1002c8b7: test eax,eax
0x1002c8b9: push edi
0x1002c8ba: mov edi,[esp+14h]
0x1002c8be: je short 1002C8D2h
0x1002c8c0: cmp eax,edi
0x1002c8c2: je short 1002C8D2h
0x1002c8c4: push 0
0x1002c8c6: push 40h
0x1002c8c8: push esi
0x1002c8c9: push ecx
0x1002c8ca: call 100235E0h
0x1002c8cf: add esp,10h
0x1002c8d2: mov [esi+0ECh],edi
0x1002c8d8: test edi,edi
0x1002c8da: pop edi
0x1002c8db: je short 1002C8E4h
0x1002c8dd: or dword ptr [esi+8],8000h
0x1002c8e4: pop esi
0x1002c8e5: ret
```

`png_set_rows` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to png_free_data (1 callees).

12.716 png_set_compression_buffer_size at 0x1002c8f0 (30 instructions)

```
0x1002c8f0: push esi
0x1002c8f1: mov esi,[esp+8]
0x1002c8f5: test esi,esi
0x1002c8f7: je short 1002C95Ch
0x1002c8f9: mov eax,[esi+0B0h]
0x1002c8ff: push eax
0x1002c900: push esi
0x1002c901: call 1002CD70h
0x1002c906: mov eax,[esp+14h]
0x1002c90a: add esp,8
0x1002c90d: cmp eax,0FFFFFFFFh
0x1002c910: jbe short 1002C92Fh
0x1002c912: push 101008C0h
0x1002c917: push esi
0x1002c918: call 100230F0h
0x1002c91d: add esp,8
```

```
0x1002c920: mov dword ptr [esi+0B4h],0FFFFFFFFh
0x1002c92a: or eax,0FFFFFFFFh
0x1002c92d: jmp short 1002C935h
0x1002c92f: mov [esi+0B4h],eax
0x1002c935: push eax
0x1002c936: push esi
0x1002c937: call 1002CD20h
0x1002c93c: add esp,8
0x1002c93f: mov [esi+0B0h],eax
0x1002c945: mov [esi+84h],eax
0x1002c94b: mov dword ptr [esi+88h],0
0x1002c955: mov dword ptr [esi+7Ch],0
0x1002c95c: pop esi
0x1002c95d: ret
```

`png_set_compression_buffer_size` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "Attempt to set buffer size beyond max ignored". It delegates to png_warning, png_malloc, png_free (3 callees).

12.717 png_set_invalid at 0x1002c960 (9 instructions)

```
0x1002c960: cmp dword ptr [esp+4],0
0x1002c965: je short 1002C978h
0x1002c967: mov eax,[esp+8]
0x1002c96b: test eax,eax
0x1002c96d: je short 1002C978h
0x1002c96f: mov ecx,[esp+0Ch]
0x1002c973: not ecx
0x1002c975: and [eax+8],ecx
0x1002c978: ret
```

`png_set_invalid` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.718 png_set_user_limits at 0x1002c980 (8 instructions)

```
0x1002c980: mov eax,[esp+4]
0x1002c984: test eax,eax
0x1002c986: je short 1002C99Ch
0x1002c988: mov ecx,[esp+8]
0x1002c98c: mov edx,[esp+0Ch]
0x1002c990: mov [eax+258h],ecx
0x1002c996: mov [eax+25Ch],edx
0x1002c99c: ret
```

`png_set_user_limits` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.719 png_set_chunk_cache_max at 0x1002c9a0 (6 instructions)

```
0x1002c9a0: mov eax,[esp+4]
0x1002c9a4: test eax,eax
0x1002c9a6: je short 1002C9B2h
0x1002c9a8: mov ecx,[esp+8]
0x1002c9ac: mov [eax+260h],ecx
0x1002c9b2: ret
```

`png_set_chunk_cache_max` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.720 png_set_chunk_malloc_max at 0x1002c9c0 (6 instructions)

```
0x1002c9c0: mov eax,[esp+4]
0x1002c9c4: test eax,eax
0x1002c9c6: je short 1002C9D2h
0x1002c9c8: mov ecx,[esp+8]
0x1002c9cc: mov [eax+264h],ecx
0x1002c9d2: ret
```

`png_set_chunk_malloc_max` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.721 png_set_benign_errors at 0x1002c9e0 (7 instructions)

```
0x1002c9e0: cmp dword ptr [esp+8],0
0x1002c9e5: mov eax,[esp+4]
0x1002c9e9: je short 1002C9F3h
0x1002c9eb: or dword ptr [eax+70h],800000h
0x1002c9f2: ret
0x1002c9f3: and dword ptr [eax+70h],0FF7FFFFFh
0x1002c9fa: ret
```

`png_set_benign_errors` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.722 png_set_text at 0x1002ca00 (18 instructions)

```
0x1002ca00: mov eax,[esp+10h]
0x1002ca04: mov ecx,[esp+0Ch]
0x1002ca08: mov edx,[esp+8]
0x1002ca0c: push esi
0x1002ca0d: mov esi,[esp+8]
0x1002ca11: push eax
0x1002ca12: push ecx
0x1002ca13: push edx
0x1002ca14: push esi
0x1002ca15: call 1002C050h
0x1002ca1a: add esp,10h
0x1002ca1d: test eax,eax
0x1002ca1f: je short 1002CA2Ch
0x1002ca21: push 101008F0h
0x1002ca26: push esi
0x1002ca27: call 100231E0h
0x1002ca2c: pop esi
0x1002ca2d: ret
```

`png_set_text` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "Insufficient memory to store text". It delegates to png_error, sub_1002c050 (2 callees).

12.723 sub_1002ca30 at 0x1002ca30 (10 instructions)

```
0x1002ca30: mov ecx,[esp+4]
0x1002ca34: mov eax,[ecx+50h]
0x1002ca37: test eax,eax
0x1002ca39: je short 1002CA41h
0x1002ca3b: mov [esp+4],ecx
0x1002ca3f: jmp eax
0x1002ca41: push 10100914h
0x1002ca46: push ecx
0x1002ca47: call 100231E0h
0x1002ca4c: int 3
```

`sub_1002ca30` is called by png_write_sig, png_write_chunk_start, png_write_chunk_data, png_write_chunk_end. It works with the text "Call to NULL write function". It delegates to png_error (1 callees).

12.724 sub_1002ca90 at 0x1002ca90 (7 instructions)

```
0x1002ca90: mov ecx,[esp+4]
0x1002ca94: mov eax,[ecx+14Ch]
0x1002ca9a: test eax,eax
0x1002ca9c: je short 1002CAA4h
0x1002ca9e: mov [esp+4],ecx
0x1002caa2: jmp eax
0x1002caa4: ret
```

`sub_1002ca90` is called by png_write_flush. It calls nothing else.

12.725 png_set_write_fn at 0x1002cad0 (25 instructions)

```
0x1002cad0: mov eax,[esp+4]
0x1002cad4: test eax,eax
0x1002cad6: je short 1002CB28h
0x1002cad8: mov ecx,[esp+8]
0x1002cadc: mov [eax+58h],ecx
0x1002cadf: mov ecx,[esp+0Ch]
0x1002cae3: test ecx,ecx
0x1002cae5: je short 1002CAECh
0x1002cae7: mov [eax+50h],ecx
0x1002caea: jmp short 1002CAF3h
0x1002caec: mov dword ptr [eax+50h],1002CA50h
0x1002caf3: mov ecx,[esp+10h]
0x1002caf7: test ecx,ecx
0x1002caf9: je short 1002CB03h
0x1002cafb: mov [eax+14Ch],ecx
0x1002cb01: jmp short 1002CB0Dh
0x1002cb03: mov dword ptr [eax+14Ch],1002CAB0h
0x1002cb0d: cmp dword ptr [eax+54h],0
0x1002cb11: je short 1002CB28h
0x1002cb13: push 10100260h
0x1002cb18: push eax
0x1002cb19: mov dword ptr [eax+54h],0
0x1002cb20: call 100230F0h
0x1002cb25: add esp,8
0x1002cb28: ret
```

`png_set_write_fn` is called by png_create_write_struct_2. It delegates to png_warning (1 callees).

12.726 sub_1002cb30 at 0x1002cb30 (50 instructions)

```
0x1002cb30: sub esp,290h
0x1002cb36: mov eax,ds:[10136220h]
0x1002cb3b: xor eax,esp
0x1002cb3d: mov [esp+28Ch],eax
0x1002cb44: mov ecx,[esp+294h]
0x1002cb4b: cmp ecx,2
0x1002cb4e: mov eax,[esp+298h]
0x1002cb55: push esi
0x1002cb56: jne short 1002CB5Fh
0x1002cb58: mov esi,0F0h
0x1002cb5d: jmp short 1002CB69h
0x1002cb5f: cmp ecx,1
0x1002cb62: jne short 1002CBBDh
0x1002cb64: mov esi,28Ch
0x1002cb69: test eax,eax
0x1002cb6b: push edi
0x1002cb6c: push esi
0x1002cb6d: je short 1002CB89h
0x1002cb6f: mov ecx,[esp+2A8h]
0x1002cb76: lea edx,[esp+0Ch]
0x1002cb7a: push edx
0x1002cb7b: mov [esp+248h],ecx
```

```
0x1002cb82: call eax
0x1002cb84: add esp,8
0x1002cb87: jmp short 1002CB92h
0x1002cb89: call dword ptr ds:[100FE550h]
0x1002cb8f: add esp,4
0x1002cb92: mov edi,eax
0x1002cb94: test edi,edi
0x1002cb96: je short 1002CBA4h
0x1002cb98: push esi
0x1002cb99: push 0
0x1002cb9b: push edi
0x1002cb9c: call 100228C0h
0x1002cba1: add esp,0Ch
0x1002cba4: mov eax,edi
0x1002cba6: pop edi
0x1002cba7: pop esi
0x1002cba8: mov ecx,[esp+28Ch]
0x1002cbaf: xor ecx,esp
0x1002cbb1: call 10022326h
0x1002cbb6: add esp,290h
0x1002cbbc: ret
0x1002cbbd: mov ecx,[esp+290h]
0x1002cbc4: pop esi
0x1002cbc5: xor ecx,esp
0x1002cbc7: xor eax,eax
0x1002cbc9: call 10022326h
0x1002cbce: add esp,290h
0x1002cbd4: ret
```

`sub_1002cb30` is called by png_create_info_struct, png_create_read_struct_2, png_create_write_struct_2, sub_1002cce0. It delegates to sub_10022326, j_memset (2 callees).

12.727 sub_1002cbe0 at 0x1002cbe0 (29 instructions)

```
0x1002cbe0: sub esp,290h
0x1002cbe6: mov eax,ds:[10136220h]
0x1002cbeb: xor eax,esp
0x1002cbed: mov [esp+28Ch],eax
0x1002cbf4: mov eax,[esp+294h]
0x1002cbfb: test eax,eax
0x1002cbfd: mov ecx,[esp+298h]
0x1002cc04: je short 1002CC41h
0x1002cc06: test ecx,ecx
0x1002cc08: push eax
0x1002cc09: je short 1002CC38h
0x1002cc0b: mov edx,[esp+2A0h]
0x1002cc12: lea eax,[esp+4]
0x1002cc16: push eax
0x1002cc17: mov [esp+240h],edx
0x1002cc1e: call ecx
0x1002cc20: add esp,8
0x1002cc23: mov ecx,[esp+28Ch]
0x1002cc2a: xor ecx,esp
0x1002cc2c: call 10022326h
0x1002cc31: add esp,290h
0x1002cc37: ret
0x1002cc38: call dword ptr ds:[100FE554h]
0x1002cc3e: add esp,4
0x1002cc41: mov ecx,[esp+28Ch]
0x1002cc48: xor ecx,esp
0x1002cc4a: call 10022326h
0x1002cc4f: add esp,290h
0x1002cc55: ret
```

`sub_1002cbe0` is called by png_destroy_info_struct, png_create_read_struct_2, png_destroy_read_struct, png_create_write_struct_2, png_destroy_write_struct, sub_1002cd00. It delegates to sub_10022326 (1 callees).

12.728 png_malloc_default at 0x1002cc60 (11 instructions)

```
0x1002cc60: cmp dword ptr [esp+4],0
0x1002cc65: je short 1002CC7Ah
0x1002cc67: mov eax,[esp+8]
0x1002cc6b: test eax,eax
0x1002cc6d: je short 1002CC7Ah
0x1002cc6f: push eax
0x1002cc70: call dword ptr ds:[100FE550h]
0x1002cc76: add esp,4
0x1002cc79: ret
0x1002cc7a: xor eax,eax
0x1002cc7c: ret
```

`png_malloc_default` is called by png_malloc. It calls nothing else.

12.729 png_free_default at 0x1002cc80 (9 instructions)

```
0x1002cc80: cmp dword ptr [esp+4],0
0x1002cc85: je short 1002CC97h
0x1002cc87: mov eax,[esp+8]
0x1002cc8b: test eax,eax
0x1002cc8d: je short 1002CC97h
0x1002cc8f: push eax
0x1002cc90: call dword ptr ds:[100FE554h]
0x1002cc96: pop ecx
0x1002cc97: ret
```

`png_free_default` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.730 png_set_mem_fn at 0x1002cca0 (10 instructions)

```
0x1002cca0: mov eax,[esp+4]
0x1002cca4: test eax,eax
0x1002cca6: je short 1002CCC6h
0x1002cca8: mov ecx,[esp+8]
0x1002ccac: mov edx,[esp+0Ch]
0x1002ccb0: mov [eax+238h],ecx
0x1002ccb6: mov ecx,[esp+10h]
0x1002ccba: mov [eax+23Ch],edx
0x1002ccc0: mov [eax+240h],ecx
0x1002ccc6: ret
```

`png_set_mem_fn` is called by png_create_read_struct_2, png_create_write_struct_2. It calls nothing else.

12.731 png_get_mem_ptr at 0x1002ccd0 (6 instructions)

```
0x1002ccd0: mov eax,[esp+4]
0x1002ccd4: test eax,eax
0x1002ccd6: jne short 1002CCD9h
0x1002ccd8: ret
0x1002ccd9: mov eax,[eax+238h]
0x1002ccdf: ret
```

`png_get_mem_ptr` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.732 sub_1002cce0 at 0x1002cce0 (7 instructions)

```
0x1002cce0: mov eax,[esp+4]
0x1002cce4: push 0
0x1002cce6: push 0
0x1002cce8: push eax
```

```
0x1002cce9: call 1002CB30h
0x1002ccee: add esp,0Ch
0x1002ccf1: ret
```

`sub_1002cce0` is called by png_info_init_3. It delegates to sub_1002cb30 (1 callees).

12.733 sub_1002cd00 at 0x1002cd00 (7 instructions)

```
0x1002cd00: mov eax,[esp+4]
0x1002cd04: push 0
0x1002cd06: push 0
0x1002cd08: push eax
0x1002cd09: call 1002CBE0h
0x1002cd0e: add esp,0Ch
0x1002cd11: ret
```

`sub_1002cd00` is called by png_info_init_3. It delegates to sub_1002cbe0 (1 callees).

12.734 png_malloc at 0x1002cd20 (26 instructions)

```
0x1002cd20: push esi
0x1002cd21: mov esi,[esp+8]
0x1002cd25: test esi,esi
0x1002cd27: je short 1002CD61h
0x1002cd29: mov ecx,[esp+0Ch]
0x1002cd2d: test ecx,ecx
0x1002cd2f: je short 1002CD61h
0x1002cd31: mov eax,[esi+23Ch]
0x1002cd37: test eax,eax
0x1002cd39: push ecx
0x1002cd3a: push esi
0x1002cd3b: je short 1002CD41h
0x1002cd3d: call eax
0x1002cd3f: jmp short 1002CD46h
0x1002cd41: call 1002CC60h
0x1002cd46: add esp,8
0x1002cd49: test eax,eax
0x1002cd4b: jne short 1002CD63h
0x1002cd4d: test dword ptr [esi+70h],100000h
0x1002cd54: jne short 1002CD63h
0x1002cd56: push 10100930h
0x1002cd5b: push esi
0x1002cd5c: call 100231E0h
0x1002cd61: xor eax,eax
0x1002cd63: pop esi
0x1002cd64: ret
```

`png_malloc` is called by png_convert_to_rfc1123, sub_10024b00, sub_10024c90, sub_10024de0, png_start_read_image, png_read_png, png_set_filter, sub_1002a910, png_set_tRNS, png_set_keep_unknown_chunks, png_set_compression_buffer_size, png_malloc_warn, png_calloc, sub_1002dcc0, sub_1002ec10, sub_1002ee90, sub_10030750, png_set_quantize, sub_10035b60, sub_10036610, sub_100367c0. It works with the text "Out of Memory". It delegates to png_error, png_malloc_default (2 callees).

12.735 png_free at 0x1002cd70 (23 instructions)

```
0x1002cc80: cmp dword ptr [esp+4],0
0x1002cc85: je short 1002CC97h
0x1002cc87: mov eax,[esp+8]
0x1002cc8b: test eax,eax
0x1002cc8d: je short 1002CC97h
0x1002cc8f: push eax
0x1002cc90: call dword ptr ds:[100FE554h]
0x1002cc96: pop ecx
```

```
0x1002cc97: ret
0x1002cd70: mov ecx,[esp+4]
0x1002cd74: test ecx,ecx
0x1002cd76: je short 1002CD99h
0x1002cd78: mov edx,[esp+8]
0x1002cd7c: test edx,edx
0x1002cd7e: je short 1002CD99h
0x1002cd80: mov eax,[ecx+240h]
0x1002cd86: test eax,eax
0x1002cd88: mov [esp+8],edx
0x1002cd8c: mov [esp+4],ecx
0x1002cd90: je short 1002CD94h
0x1002cd92: jmp eax
0x1002cd94: jmp near ptr 1002CC80h
0x1002cd99: ret
```

`png_free` is called by png_free_data, sub_100239c0, png_create_read_struct_2, png_start_read_image, sub_100280e0, sub_1002a8b0, png_create_write_struct_2, sub_1002b220, png_destroy_write_struct, png_set_sCAL_s, png_set_iCCP, sub_1002c050, png_set_sPLT, png_set_unknown_chunks, png_set_keep_unknown_chunks, png_set_compression_buffer_size, sub_1002d150, sub_1002dcc0, sub_1002ec10, sub_1002ee90, sub_1002faa0, sub_1002fdd0, sub_100300d0, sub_10030280, sub_10030490, sub_10030750, png_set_quantize, sub_10035b60, sub_10035de0, sub_10036050, sub_10036250, sub_10036310, sub_10036450, sub_10036610, sub_10037290, sub_10037430. It calls nothing else.

12.736 png_malloc_warn at 0x1002cda0 (21 instructions)

```
0x1002cda0: push esi
0x1002cda1: mov esi,[esp+8]
0x1002cda5: test esi,esi
0x1002cda7: jne short 1002CDADh
0x1002cda9: xor eax,eax
0x1002cdab: pop esi
0x1002cdac: ret
0x1002cdad: mov ecx,[esp+0Ch]
0x1002cdb1: push edi
0x1002cdb2: mov edi,[esi+70h]
0x1002cdb5: mov eax,edi
0x1002cdb7: push ecx
0x1002cdb8: or eax,100000h
0x1002cdbd: push esi
0x1002cdbe: mov [esi+70h],eax
0x1002cdc1: call 1002CD20h
0x1002cdc6: add esp,8
0x1002cdc9: mov [esi+70h],edi
0x1002cdcc: pop edi
0x1002cdcd: pop esi
0x1002cdce: ret
```

`png_malloc_warn` is called by png_create_read_struct_2, png_create_write_struct_2, png_set_hIST, png_set_pCAL, png_set_sCAL_s, png_set_iCCP, sub_1002c050, png_set_sPLT, png_set_unknown_chunks, sub_1002d150, sub_1002ee90, sub_1002faa0, sub_1002fdd0, sub_100300d0, sub_10030280, sub_10030490, png_set_quantize, sub_10036050. It delegates to png_malloc (1 callees).

12.737 png_calloc at 0x1002cdd0 (20 instructions)

```
0x1002cdd0: mov eax,[esp+4]
0x1002cdd4: push esi
0x1002cdd5: push edi
0x1002cdd6: mov edi,[esp+10h]
0x1002cdda: push edi
0x1002cddb: push eax
0x1002cddc: call 1002CD20h
0x1002cde1: mov esi,eax
```

```
0x1002cde3: add esp,8
0x1002cde6: test esi,esi
0x1002cde8: je short 1002CDF8h
0x1002cdea: push edi
0x1002cdeb: push 0
0x1002cded: push esi
0x1002cdee: call 100228C0h
0x1002cdf3: add esp,0Ch
0x1002cdf6: mov eax,esi
0x1002cdf8: pop edi
0x1002cdf9: pop esi
0x1002cdfa: ret
```

`png_calloc` is called by sub_10024b00, sub_10024c90, png_start_read_image, png_set_PLTE, sub_1002dcc0, png_set_quantize, sub_100367c0. It delegates to j_memset, png_malloc (2 callees).

12.738 png_get_uint_31 at 0x1002ce00 (18 instructions)

```
0x1002ce00: mov ecx,[esp+8]
0x1002ce04: movzx eax,byte ptr [ecx]
0x1002ce07: movzx edx,byte ptr [ecx+1]
0x1002ce0b: shl eax,8
0x1002ce0e: add eax,edx
0x1002ce10: movzx edx,byte ptr [ecx+2]
0x1002ce14: movzx ecx,byte ptr [ecx+3]
0x1002ce18: shl eax,8
0x1002ce1b: add eax,edx
0x1002ce1d: shl eax,8
0x1002ce20: add eax,ecx
0x1002ce22: cmp eax,7FFFFFFFh
0x1002ce27: jbe short 1002CE38h
0x1002ce29: mov edx,[esp+4]
0x1002ce2d: push 10100940h
0x1002ce32: push edx
0x1002ce33: call 100231E0h
0x1002ce38: ret
```

`png_get_uint_31` is called by sub_1002dfd0, sub_1002e100. It works with the text "PNG unsigned integer out of range". It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to png_error (1 callees).

12.739 sub_1002ce40 at 0x1002ce40 (22 instructions)

```
0x1002ce40: movzx eax,byte ptr [ecx]
0x1002ce43: push esi
0x1002ce44: movzx esi,byte ptr [ecx+1]
0x1002ce48: shl eax,8
0x1002ce4b: add eax,esi
0x1002ce4d: movzx esi,byte ptr [ecx+2]
0x1002ce51: movzx ecx,byte ptr [ecx+3]
0x1002ce55: shl eax,8
0x1002ce58: add eax,esi
0x1002ce5a: shl eax,8
0x1002ce5d: add eax,ecx
0x1002ce5f: cmp eax,7FFFFFFFh
0x1002ce64: pop esi
0x1002ce65: jbe short 1002CE7Ch
0x1002ce67: test edx,edx
0x1002ce69: je short 1002CE79h
0x1002ce6b: push 10100964h
0x1002ce70: push edx
0x1002ce71: call 100230F0h
0x1002ce76: add esp,8
0x1002ce79: or eax,0FFFFFFFFh
0x1002ce7c: ret
```

`sub_1002ce40` is called by sub_1002e450, sub_1002e700. It works with the text "PNG fixed point integer out of range". It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to png_warning (1 callees).

12.740 png_get_uint_16 at 0x1002ce80 (6 instructions)

```
0x1002ce80: mov ecx,[esp+4]
0x1002ce84: movzx ax,byte ptr [ecx]
0x1002ce88: movzx cx,byte ptr [ecx+1]
0x1002ce8d: shl ax,8
0x1002ce91: add ax,cx
0x1002ce94: ret
```

`png_get_uint_16` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.741 sub_1002cea0 at 0x1002cea0 (51 instructions)

```
0x1002cea0: push ebx
0x1002cea1: mov ebx,[esp+8]
0x1002cea5: mov al,[ebx+12Ch]
0x1002ceab: cmp al,8
0x1002cead: jae short 1002CF2Bh
0x1002ceaf: push ebp
0x1002ceb0: mov ebp,[esp+10h]
0x1002ceb4: push esi
0x1002ceb5: push edi
0x1002ceb6: movzx edi,al
0x1002ceb9: mov esi,8
0x1002cebe: sub esi,edi
0x1002cec0: push esi
0x1002cec1: lea eax,[edi+ebp+20h]
0x1002cec5: push eax
0x1002cec6: push ebx
0x1002cec7: mov dword ptr [ebx+288h],11h
0x1002ced1: call 10029C10h
0x1002ced6: push esi
0x1002ced7: add ebp,20h
0x1002ceda: push edi
0x1002cedb: push ebp
0x1002cedc: mov byte ptr [ebx+12Ch],8
0x1002cee3: call 10023350h
0x1002cee8: add esp,18h
0x1002ceeb: test eax,eax
0x1002ceed: je short 1002CF1Ch
0x1002ceef: cmp edi,4
0x1002cef2: jae short 1002CF11h
0x1002cef4: add esi,0FFFFFFFCh
0x1002cef7: push esi
0x1002cef8: push edi
0x1002cef9: push ebp
0x1002cefa: call 10023350h
0x1002ceff: add esp,0Ch
0x1002cf02: test eax,eax
0x1002cf04: je short 1002CF11h
0x1002cf06: push 101009B4h
0x1002cf0b: push ebx
0x1002cf0c: call 100231E0h
0x1002cf11: push 1010098Ch
0x1002cf16: push ebx
0x1002cf17: call 100231E0h
0x1002cf1c: cmp edi,3
0x1002cf1f: jae short 1002CF28h
0x1002cf21: or dword ptr [ebx+6Ch],1000h
0x1002cf28: pop edi
0x1002cf29: pop esi
0x1002cf2a: pop ebp
```

```
0x1002cf2b: pop ebx
0x1002cf2c: ret
```

`sub_1002cea0` is called by png_read_info. It works with the text "PNG file corrupted by ASCII conversion", "Not a PNG file". It delegates to png_error, png_sig_cmp, sub_10029c10 (3 callees).

12.742 sub_1002cf30 at 0x1002cf30 (21 instructions)

```
0x1002cf30: push esi
0x1002cf31: mov esi,[esp+8]
0x1002cf35: test esi,esi
0x1002cf37: je short 1002CF58h
0x1002cf39: push ebx
0x1002cf3a: mov ebx,[esp+10h]
0x1002cf3e: push edi
0x1002cf3f: mov edi,[esp+18h]
0x1002cf43: push edi
0x1002cf44: push ebx
0x1002cf45: push esi
0x1002cf46: call 10029C10h
0x1002cf4b: push edi
0x1002cf4c: push ebx
0x1002cf4d: push esi
0x1002cf4e: call 10023500h
0x1002cf53: add esp,18h
0x1002cf56: pop edi
0x1002cf57: pop ebx
0x1002cf58: pop esi
0x1002cf59: ret
```

`sub_1002cf30` is called by png_read_row, sub_1002e050, sub_1002e100, sub_1002e260, sub_1002e450, sub_1002e590, sub_1002e700, sub_1002ea20, sub_1002ec10, sub_1002ee90, sub_1002f170, sub_1002f400, sub_1002f650, sub_1002f7e0, sub_1002f930, sub_1002faa0, sub_1002fdd0, sub_1002ffb0, sub_100300d0, sub_10030280, sub_10030490, sub_10030750, sub_100309c0. It delegates to sub_10023500, sub_10029c10 (2 callees).

12.743 sub_1002cf60 at 0x1002cf60 (44 instructions)

```
0x1002cf60: push esi
0x1002cf61: mov esi,[esp+8]
0x1002cf65: test byte ptr [esi+11Ch],20h
0x1002cf6c: push edi
0x1002cf6d: mov edi,1
0x1002cf72: je short 1002CF85h
0x1002cf74: mov eax,[esi+70h]
0x1002cf77: and eax,300h
0x1002cf7c: cmp eax,300h
0x1002cf81: jne short 1002CF90h
0x1002cf83: jmp short 1002CF8Eh
0x1002cf85: test dword ptr [esi+70h],800h
0x1002cf8c: je short 1002CF90h
0x1002cf8e: xor edi,edi
0x1002cf90: push 4
0x1002cf92: lea ecx,[esp+10h]
0x1002cf96: push ecx
0x1002cf97: push esi
0x1002cf98: mov dword ptr [esi+288h],81h
0x1002cfa2: call 10029C10h
0x1002cfa7: add esp,0Ch
0x1002cfaa: test edi,edi
0x1002cfac: je short 1002CFDFh
0x1002cfae: mov eax,[esp+0Ch]
0x1002cfb2: movzx ecx,byte ptr [esp+0Eh]
0x1002cfb7: movzx edx,al
0x1002cfba: shl edx,8
```

```
0x1002cfbd: movzx eax,ah
0x1002cfc0: add edx,eax
0x1002cfc2: movzx eax,byte ptr [esp+0Fh]
0x1002cfc7: shl edx,8
0x1002cfca: add edx,ecx
0x1002cfcc: shl edx,8
0x1002cfcf: add edx,eax
0x1002cfd1: xor eax,eax
0x1002cfd3: cmp edx,[esi+110h]
0x1002cfd9: pop edi
0x1002cfda: setne al
0x1002cfdd: pop esi
0x1002cfde: ret
0x1002cfdf: pop edi
0x1002cfe0: xor eax,eax
0x1002cfe2: pop esi
0x1002cfe3: ret
```

`sub_1002cf60` is called by sub_1002e050. It delegates to sub_10029c10 (1 callees).

12.744 sub_1002cff0 at 0x1002cff0 (113 instructions)

```
0x1002cff0: sub esp,3Ch
0x1002cff3: mov eax,ds:[10136220h]
0x1002cff8: xor eax,esp
0x1002cffa: mov [esp+38h],eax
0x1002cffe: mov eax,[esp+44h]
0x1002d002: push ebx
0x1002d003: push ebp
0x1002d004: xor ebp,ebp
0x1002d006: push edi
0x1002d007: mov [esp+0Ch],eax
0x1002d00b: mov [esi+78h],ecx
0x1002d00e: mov [esi+7Ch],ebp
0x1002d011: cmp dword ptr [esi+7Ch],0
0x1002d015: jne short 1002D03Fh
0x1002d017: mov eax,[esp+4Ch]
0x1002d01b: test eax,eax
0x1002d01d: jbe short 1002D03Fh
0x1002d01f: cmp eax,0FFFFFFFFh
0x1002d022: ja short 1002D031h
0x1002d024: mov [esi+7Ch],eax
0x1002d027: mov dword ptr [esp+4Ch],0
0x1002d02f: jmp short 1002D03Fh
0x1002d031: add eax,1
0x1002d034: mov dword ptr [esi+7Ch],0FFFFFFFFh
0x1002d03b: mov [esp+4Ch],eax
0x1002d03f: mov eax,[esi+0B4h]
0x1002d045: mov edx,[esi+0B0h]
0x1002d04b: mov [esi+88h],eax
0x1002d051: lea eax,[esi+78h]
0x1002d054: push 0
0x1002d056: push eax
0x1002d057: mov [esi+84h],edx
0x1002d05d: call 10060100h
0x1002d062: mov edi,[esi+0B4h]
0x1002d068: sub edi,[esi+88h]
0x1002d06e: mov ebx,eax
0x1002d070: add esp,8
0x1002d073: test ebx,ebx
0x1002d075: je short 1002D07Ch
0x1002d077: cmp ebx,1
0x1002d07a: jne short 1002D0ADh
0x1002d07c: test edi,edi
0x1002d07e: jle short 1002D0ADh
0x1002d080: mov ecx,[esp+0Ch]
```

```
0x1002d084: test ecx,ecx
0x1002d086: je short 1002D0ABh
0x1002d088: mov eax,[esp+54h]
0x1002d08c: cmp eax,ebp
0x1002d08e: jbe short 1002D0ABh
0x1002d090: sub eax,ebp
0x1002d092: cmp edi,eax
0x1002d094: jae short 1002D098h
0x1002d096: mov eax,edi
0x1002d098: mov edx,[esi+0B0h]
0x1002d09e: push eax
0x1002d09f: push edx
0x1002d0a0: add ecx,ebp
0x1002d0a2: push ecx
0x1002d0a3: call 100228C6h
0x1002d0a8: add esp,0Ch
0x1002d0ab: add ebp,edi
0x1002d0ad: test ebx,ebx
0x1002d0af: je near ptr 1002D011h
0x1002d0b5: lea eax,[esi+78h]
0x1002d0b8: push eax
0x1002d0b9: mov dword ptr [esi+7Ch],0
0x1002d0c0: call 1005FE10h
0x1002d0c5: add esp,4
0x1002d0c8: cmp ebx,1
0x1002d0cb: jne short 1002D0E1h
0x1002d0cd: pop edi
0x1002d0ce: mov eax,ebp
0x1002d0d0: pop ebp
0x1002d0d1: pop ebx
0x1002d0d2: mov ecx,[esp+38h]
0x1002d0d6: xor ecx,esp
0x1002d0d8: call 10022326h
0x1002d0dd: add esp,3Ch
0x1002d0e0: ret
0x1002d0e1: mov eax,[esi+90h]
0x1002d0e7: test eax,eax
0x1002d0e9: jne short 1002D124h
0x1002d0eb: cmp ebx,0FFFFFFFBh
0x1002d0ee: je short 1002D103h
0x1002d0f0: cmp ebx,0FFFFFFFDh
0x1002d0f3: je short 1002D0FCh
0x1002d0f5: mov eax,10100A28h
0x1002d0fa: jmp short 1002D108h
0x1002d0fc: mov eax,101009F8h
0x1002d101: jmp short 1002D108h
0x1002d103: mov eax,101009C4h
0x1002d108: lea ecx,[esi+11Ch]
0x1002d10e: push ecx
0x1002d10f: push eax
0x1002d110: lea edx,[esp+18h]
0x1002d114: push 34h
0x1002d116: push edx
0x1002d117: call dword ptr ds:[100FE4C0h]
0x1002d11d: add esp,10h
0x1002d120: lea eax,[esp+10h]
0x1002d124: push eax
0x1002d125: push esi
0x1002d126: call 100230F0h
0x1002d12b: mov ecx,[esp+4Ch]
0x1002d12f: add esp,8
0x1002d132: pop edi
0x1002d133: pop ebp
0x1002d134: pop ebx
0x1002d135: xor ecx,esp
0x1002d137: xor eax,eax
0x1002d139: call 10022326h
```

```
0x1002d13e: add esp,3Ch
0x1002d141: ret
```

`sub_1002cff0` is called by sub_1002d150. It works with the text "Buffer error in compressed datastream in %s chunk", "Data error in compressed datastream in %s chunk", "Incomplete compressed datastream in %s chunk". It delegates to sub_10022326, j_memcpy, png_warning (5 callees).

12.745 sub_1002d150 at 0x1002d150 (152 instructions)

```
0x1002d150: sub esp,44h
0x1002d153: mov eax,ds:[10136220h]
0x1002d158: xor eax,esp
0x1002d15a: mov [esp+40h],eax
0x1002d15e: mov eax,[esp+58h]
0x1002d162: push ebx
0x1002d163: push ebp
0x1002d164: push esi
0x1002d165: mov esi,[esp+54h]
0x1002d169: push edi
0x1002d16a: mov edi,[esp+64h]
0x1002d16e: mov [esp+10h],eax
0x1002d172: mov eax,[esp+60h]
0x1002d176: cmp edi,eax
0x1002d178: jbe short 1002D18Fh
0x1002d17a: push 10100AE8h
0x1002d17f: push esi
0x1002d180: call 100230F0h
0x1002d185: add esp,8
0x1002d188: xor edi,edi
0x1002d18a: jmp near ptr 1002D2ADh
0x1002d18f: mov ecx,[esp+5Ch]
0x1002d193: test ecx,ecx
0x1002d195: jne near ptr 1002D28Ch
0x1002d19b: push ecx
0x1002d19c: push ecx
0x1002d19d: mov ecx,[esi+284h]
0x1002d1a3: sub eax,edi
0x1002d1a5: push eax
0x1002d1a6: add ecx,edi
0x1002d1a8: mov [esp+20h],eax
0x1002d1ac: call 1002CFF0h
0x1002d1b1: mov ebx,eax
0x1002d1b3: mov eax,[esi+264h]
0x1002d1b9: add esp,0Ch
0x1002d1bc: test eax,eax
0x1002d1be: je short 1002D1DDh
0x1002d1c0: lea ecx,[edi+ebx]
0x1002d1c3: add eax,0FFFFFFFFh
0x1002d1c6: cmp ecx,eax
0x1002d1c8: jb short 1002D1DDh
0x1002d1ca: push 10100ABCh
0x1002d1cf: push esi
0x1002d1d0: call 100230F0h
0x1002d1d5: add esp,8
0x1002d1d8: jmp near ptr 1002D2ADh
0x1002d1dd: test ebx,ebx
0x1002d1df: jbe near ptr 1002D2ADh
0x1002d1e5: lea eax,[edi+ebx]
0x1002d1e8: mov [esp+18h],eax
0x1002d1ec: add eax,1
0x1002d1ef: push eax
0x1002d1f0: push esi
0x1002d1f1: call 1002CDA0h
0x1002d1f6: mov ebp,eax
0x1002d1f8: add esp,8
0x1002d1fb: test ebp,ebp
```

```
0x1002d1fd: je short 1002D27Ch
0x1002d1ff: mov edx,[esi+284h]
0x1002d205: push edi
0x1002d206: push edx
0x1002d207: push ebp
0x1002d208: call 100228C6h
0x1002d20d: mov ecx,[esi+284h]
0x1002d213: lea eax,[edi+ebp]
0x1002d216: push ebx
0x1002d217: push eax
0x1002d218: mov eax,[esp+28h]
0x1002d21c: push eax
0x1002d21d: add ecx,edi
0x1002d21f: call 1002CFF0h
0x1002d224: add esp,18h
0x1002d227: cmp eax,ebx
0x1002d229: lea ecx,[edi+ebp]
0x1002d22c: mov byte ptr [ecx+ebx],0
0x1002d230: jne short 1002D265h
0x1002d232: mov ecx,[esi+284h]
0x1002d238: push ecx
0x1002d239: push esi
0x1002d23a: call 1002CD70h
0x1002d23f: mov edx,[esp+20h]
0x1002d243: mov eax,[esp+18h]
0x1002d247: add esp,8
0x1002d24a: pop edi
0x1002d24b: mov [esi+284h],ebp
0x1002d251: pop esi
0x1002d252: pop ebp
0x1002d253: mov [eax],edx
0x1002d255: pop ebx
0x1002d256: mov ecx,[esp+40h]
0x1002d25a: xor ecx,esp
0x1002d25c: call 10022326h
0x1002d261: add esp,44h
0x1002d264: ret
0x1002d265: push 10100AA4h
0x1002d26a: push esi
0x1002d26b: call 100230F0h
0x1002d270: push ebp
0x1002d271: push esi
0x1002d272: call 1002CD70h
0x1002d277: add esp,10h
0x1002d27a: jmp short 1002D2ADh
0x1002d27c: push 10100A7Ch
0x1002d281: push esi
0x1002d282: call 100230F0h
0x1002d287: add esp,8
0x1002d28a: jmp short 1002D2ADh
0x1002d28c: push ecx
0x1002d28d: push 10100A58h
0x1002d292: lea ecx,[esp+24h]
0x1002d296: push 32h
0x1002d298: push ecx
0x1002d299: call dword ptr ds:[100FE4C0h]
0x1002d29f: lea edx,[esp+2Ch]
0x1002d2a3: push edx
0x1002d2a4: push esi
0x1002d2a5: call 100230F0h
0x1002d2aa: add esp,18h
0x1002d2ad: lea eax,[edi+1]
0x1002d2b0: push eax
0x1002d2b1: push esi
0x1002d2b2: call 1002CDA0h
0x1002d2b7: mov ebx,eax
0x1002d2b9: add esp,8
```

```
0x1002d2bc: test ebx,ebx
0x1002d2be: je short 1002D2EFh
0x1002d2c0: test edi,edi
0x1002d2c2: jbe short 1002D2D5h
0x1002d2c4: mov ecx,[esi+284h]
0x1002d2ca: push edi
0x1002d2cb: push ecx
0x1002d2cc: push ebx
0x1002d2cd: call 100228C6h
0x1002d2d2: add esp,0Ch
0x1002d2d5: mov edx,[esi+284h]
0x1002d2db: push edx
0x1002d2dc: push esi
0x1002d2dd: call 1002CD70h
0x1002d2e2: mov [esi+284h],ebx
0x1002d2e8: add esp,8
0x1002d2eb: mov byte ptr [edi+ebx],0
0x1002d2ef: mov eax,[esp+10h]
0x1002d2f3: mov ecx,[esp+50h]
0x1002d2f7: mov [eax],edi
0x1002d2f9: pop edi
0x1002d2fa: pop esi
0x1002d2fb: pop ebp
0x1002d2fc: pop ebx
0x1002d2fd: xor ecx,esp
0x1002d2ff: call 10022326h
0x1002d304: add esp,44h
0x1002d307: ret
```

`sub_1002d150` is called by sub_1002ec10, sub_10030280, sub_10030490. It works with the text "Unknown zTXt compression type %d", "Not enough memory to decompress chunk", "png_inflate logic error". It delegates to sub_10022326, j_memcpy, png_warning (6 callees).

12.746 sub_1002d310 at 0x1002d310 (42 instructions)

```
0x1002d310: mov ecx,[esp+8]
0x1002d314: mov al,[ecx]
0x1002d316: cmp al,41h
0x1002d318: jb short 1002D35Fh
0x1002d31a: cmp al,7Ah
0x1002d31c: ja short 1002D35Fh
0x1002d31e: cmp al,5Ah
0x1002d320: jbe short 1002D326h
0x1002d322: cmp al,61h
0x1002d324: jb short 1002D35Fh
0x1002d326: mov al,[ecx+1]
0x1002d329: cmp al,41h
0x1002d32b: jb short 1002D35Fh
0x1002d32d: cmp al,7Ah
0x1002d32f: ja short 1002D35Fh
0x1002d331: cmp al,5Ah
0x1002d333: jbe short 1002D339h
0x1002d335: cmp al,61h
0x1002d337: jb short 1002D35Fh
0x1002d339: mov al,[ecx+2]
0x1002d33c: cmp al,41h
0x1002d33e: jb short 1002D35Fh
0x1002d340: cmp al,7Ah
0x1002d342: ja short 1002D35Fh
0x1002d344: cmp al,5Ah
0x1002d346: jbe short 1002D34Ch
0x1002d348: cmp al,61h
0x1002d34a: jb short 1002D35Fh
0x1002d34c: mov al,[ecx+3]
0x1002d34f: cmp al,41h
0x1002d351: jb short 1002D35Fh
```

```
0x1002d353: cmp al,7Ah
0x1002d355: ja short 1002D35Fh
0x1002d357: cmp al,5Ah
0x1002d359: jbe short 1002D36Eh
0x1002d35b: cmp al,61h
0x1002d35d: jae short 1002D36Eh
0x1002d35f: mov eax,[esp+4]
0x1002d363: push 10100AFCh
0x1002d368: push eax
0x1002d369: call 10023240h
0x1002d36e: ret
```

`sub_1002d310` is called by sub_1002dfd0. It works with the text "invalid chunk type". It delegates to png_chunk_error (1 callees).

12.747 sub_1002d370 at 0x1002d370 (231 instructions)

```
0x1002d370: sub esp,10h
0x1002d373: cmp dword ptr [esp+1Ch],0FFh
0x1002d37b: jne short 1002D3C3h
0x1002d37d: mov ecx,[esp+14h]
0x1002d381: mov al,[ecx+10Bh]
0x1002d387: cmp al,8
0x1002d389: movzx eax,al
0x1002d38c: jb short 1002D39Ah
0x1002d38e: shr eax,3
0x1002d391: imul eax,[ecx+0CCh]
0x1002d398: jmp short 1002D3A7h
0x1002d39a: imul eax,[ecx+0CCh]
0x1002d3a1: add eax,7
0x1002d3a4: shr eax,3
0x1002d3a7: push eax
0x1002d3a8: mov eax,[ecx+0ECh]
0x1002d3ae: mov ecx,[esp+1Ch]
0x1002d3b2: add eax,1
0x1002d3b5: push eax
0x1002d3b6: push ecx
0x1002d3b7: call 100228C6h
0x1002d3bc: add esp,0Ch
0x1002d3bf: add esp,10h
0x1002d3c2: ret
0x1002d3c3: mov eax,[esp+14h]
0x1002d3c7: movzx ecx,byte ptr [eax+10Bh]
0x1002d3ce: push ebx
0x1002d3cf: push ebp
0x1002d3d0: push esi
0x1002d3d1: mov esi,[eax+0ECh]
0x1002d3d7: mov edx,ecx
0x1002d3d9: add esi,1
0x1002d3dc: sub edx,1
0x1002d3df: push edi
0x1002d3e0: je near ptr 1002D5B7h
0x1002d3e6: sub edx,1
0x1002d3e9: je near ptr 1002D4F8h
0x1002d3ef: sub edx,2
0x1002d3f2: je short 1002D442h
0x1002d3f4: mov eax,[eax+0CCh]
0x1002d3fa: mov ebp,[esp+28h]
0x1002d3fe: shr ecx,3
0x1002d401: test eax,eax
0x1002d403: mov edi,ecx
0x1002d405: mov bl,80h
0x1002d407: jbe near ptr 1002D664h
0x1002d40d: mov [esp+10h],eax
0x1002d411: mov dl,[esp+2Ch]
0x1002d415: test dl,bl
```

```
0x1002d417: je short 1002D424h
0x1002d419: push edi
0x1002d41a: push esi
0x1002d41b: push ebp
0x1002d41c: call 100228C6h
0x1002d421: add esp,0Ch
0x1002d424: add esi,edi
0x1002d426: add ebp,edi
0x1002d428: cmp bl,1
0x1002d42b: jne short 1002D431h
0x1002d42d: mov bl,80h
0x1002d42f: jmp short 1002D433h
0x1002d431: shr bl,1
0x1002d433: sub dword ptr [esp+10h],1
0x1002d438: jne short 1002D411h
0x1002d43a: pop edi
0x1002d43b: pop esi
0x1002d43c: pop ebp
0x1002d43d: pop ebx
0x1002d43e: add esp,10h
0x1002d441: ret
0x1002d442: test dword ptr [eax+74h],10000h
0x1002d449: mov edi,[esp+28h]
0x1002d44d: mov ecx,[eax+0CCh]
0x1002d453: mov dword ptr [esp+10h],80h
0x1002d45b: je short 1002D46Eh
0x1002d45d: mov eax,4
0x1002d462: xor ebp,ebp
0x1002d464: mov [esp+14h],eax
0x1002d468: mov [esp+18h],eax
0x1002d46c: jmp short 1002D483h
0x1002d46e: mov ebp,4
0x1002d473: mov dword ptr [esp+14h],0
0x1002d47b: mov dword ptr [esp+18h],0FFFFFFFCh
0x1002d483: test ecx,ecx
0x1002d485: mov edx,ebp
0x1002d487: jbe near ptr 1002D664h
0x1002d48d: mov [esp+1Ch],ecx
0x1002d491: mov eax,[esp+2Ch]
0x1002d495: test [esp+10h],eax
0x1002d499: je short 1002D4BDh
0x1002d49b: movzx eax,byte ptr [esi]
0x1002d49e: mov cl,dl
0x1002d4a0: shr eax,cl
0x1002d4a2: mov ecx,4
0x1002d4a7: sub ecx,edx
0x1002d4a9: mov ebx,0F0Fh
0x1002d4ae: sar ebx,cl
0x1002d4b0: and eax,0Fh
0x1002d4b3: mov ecx,edx
0x1002d4b5: shl al,cl
0x1002d4b7: and bl,[edi]
0x1002d4b9: or bl,al
0x1002d4bb: mov [edi],bl
0x1002d4bd: cmp edx,[esp+14h]
0x1002d4c1: jne short 1002D4CDh
0x1002d4c3: add esi,1
0x1002d4c6: mov edx,ebp
0x1002d4c8: add edi,1
0x1002d4cb: jmp short 1002D4D1h
0x1002d4cd: add edx,[esp+18h]
0x1002d4d1: mov eax,1
0x1002d4d6: cmp [esp+10h],eax
0x1002d4da: jne short 1002D4E6h
0x1002d4dc: mov dword ptr [esp+10h],80h
0x1002d4e4: jmp short 1002D4EAh
0x1002d4e6: sar dword ptr [esp+10h],1
```

```
0x1002d4ea: sub [esp+1Ch],eax
0x1002d4ee: jne short 1002D491h
0x1002d4f0: pop edi
0x1002d4f1: pop esi
0x1002d4f2: pop ebp
0x1002d4f3: pop ebx
0x1002d4f4: add esp,10h
0x1002d4f7: ret
0x1002d4f8: test dword ptr [eax+74h],10000h
0x1002d4ff: mov edi,[esp+28h]
0x1002d503: mov ecx,[eax+0CCh]
0x1002d509: mov dword ptr [esp+10h],80h
0x1002d511: je short 1002D527h
0x1002d513: xor ebp,ebp
0x1002d515: mov dword ptr [esp+1Ch],6
0x1002d51d: mov dword ptr [esp+18h],2
0x1002d525: jmp short 1002D53Ch
0x1002d527: mov ebp,6
0x1002d52c: mov dword ptr [esp+1Ch],0
0x1002d534: mov dword ptr [esp+18h],0FFFFFFFEh
0x1002d53c: test ecx,ecx
0x1002d53e: mov edx,ebp
0x1002d540: jbe near ptr 1002D664h
0x1002d546: mov [esp+14h],ecx
0x1002d54a: lea ebx,[ebx]
0x1002d550: mov ecx,[esp+2Ch]
0x1002d554: test [esp+10h],ecx
0x1002d558: je short 1002D57Ch
0x1002d55a: movzx eax,byte ptr [esi]
0x1002d55d: mov cl,dl
0x1002d55f: shr eax,cl
0x1002d561: mov ecx,6
0x1002d566: sub ecx,edx
0x1002d568: mov ebx,3F3Fh
0x1002d56d: sar ebx,cl
0x1002d56f: and eax,3
0x1002d572: mov ecx,edx
0x1002d574: shl al,cl
0x1002d576: and bl,[edi]
0x1002d578: or bl,al
0x1002d57a: mov [edi],bl
0x1002d57c: cmp edx,[esp+1Ch]
0x1002d580: jne short 1002D58Ch
0x1002d582: add esi,1
0x1002d585: mov edx,ebp
0x1002d587: add edi,1
0x1002d58a: jmp short 1002D590h
0x1002d58c: add edx,[esp+18h]
0x1002d590: mov eax,1
0x1002d595: cmp [esp+10h],eax
0x1002d599: jne short 1002D5A5h
0x1002d59b: mov dword ptr [esp+10h],80h
0x1002d5a3: jmp short 1002D5A9h
0x1002d5a5: sar dword ptr [esp+10h],1
0x1002d5a9: sub [esp+14h],eax
0x1002d5ad: jne short 1002D550h
0x1002d5af: pop edi
0x1002d5b0: pop esi
0x1002d5b1: pop ebp
0x1002d5b2: pop ebx
0x1002d5b3: add esp,10h
0x1002d5b6: ret
0x1002d5b7: test dword ptr [eax+74h],10000h
0x1002d5be: mov edi,[esp+28h]
0x1002d5c2: mov ecx,[eax+0CCh]
0x1002d5c8: mov dword ptr [esp+10h],80h
0x1002d5d0: je short 1002D5E6h
```

```
0x1002d5d2: xor ebp,ebp
0x1002d5d4: mov dword ptr [esp+1Ch],7
0x1002d5dc: mov dword ptr [esp+18h],1
0x1002d5e4: jmp short 1002D5FBh
0x1002d5e6: mov ebp,7
0x1002d5eb: mov dword ptr [esp+1Ch],0
0x1002d5f3: mov dword ptr [esp+18h],0FFFFFFFFh
0x1002d5fb: test ecx,ecx
0x1002d5fd: mov edx,ebp
0x1002d5ff: jbe short 1002D664h
0x1002d601: mov [esp+14h],ecx
0x1002d605: mov eax,[esp+2Ch]
0x1002d609: test [esp+10h],eax
0x1002d60d: je short 1002D631h
0x1002d60f: movzx eax,byte ptr [esi]
0x1002d612: mov cl,dl
0x1002d614: shr eax,cl
0x1002d616: mov ecx,7
0x1002d61b: sub ecx,edx
0x1002d61d: mov ebx,7F7Fh
0x1002d622: sar ebx,cl
0x1002d624: and eax,1
0x1002d627: mov ecx,edx
0x1002d629: shl al,cl
0x1002d62b: and bl,[edi]
0x1002d62d: or bl,al
0x1002d62f: mov [edi],bl
0x1002d631: cmp edx,[esp+1Ch]
0x1002d635: jne short 1002D641h
0x1002d637: add esi,1
0x1002d63a: mov edx,ebp
0x1002d63c: add edi,1
0x1002d63f: jmp short 1002D645h
0x1002d641: add edx,[esp+18h]
0x1002d645: mov eax,1
0x1002d64a: cmp [esp+10h],eax
0x1002d64e: jne short 1002D65Ah
0x1002d650: mov dword ptr [esp+10h],80h
0x1002d658: jmp short 1002D65Eh
0x1002d65a: sar dword ptr [esp+10h],1
0x1002d65e: sub [esp+14h],eax
0x1002d662: jne short 1002D605h
0x1002d664: pop edi
0x1002d665: pop esi
0x1002d666: pop ebp
0x1002d667: pop ebx
0x1002d668: add esp,10h
0x1002d66b: ret
```

`sub_1002d370` is called by png_read_row. It delegates to j_memcpy (1 callees).

12.748 sub_1002d670 at 0x1002d670 (335 instructions)

```
0x1002d670: sub esp,4Ch
0x1002d673: mov eax,ds:[10136220h]
0x1002d678: xor eax,esp
0x1002d67a: mov [esp+48h],eax
0x1002d67e: mov eax,[esp+50h]
0x1002d682: mov ecx,[eax+0ECh]
0x1002d688: add ecx,1
0x1002d68b: movzx edx,byte ptr [eax+124h]
0x1002d692: push ebx
0x1002d693: mov ebx,[eax+74h]
0x1002d696: push esi
0x1002d697: lea esi,[eax+100h]
0x1002d69d: mov eax,8
```

```
0x1002d6a2: push edi
0x1002d6a3: mov [esp+30h],eax
0x1002d6a7: mov [esp+34h],eax
0x1002d6ab: mov eax,4
0x1002d6b0: mov edi,2
0x1002d6b5: mov [esp+28h],esi
0x1002d6b9: mov [esp+38h],eax
0x1002d6bd: mov [esp+3Ch],eax
0x1002d6c1: mov [esp+40h],edi
0x1002d6c5: mov [esp+44h],edi
0x1002d6c9: mov dword ptr [esp+48h],1
0x1002d6d1: je near ptr 1002DAC6h
0x1002d6d7: test esi,esi
0x1002d6d9: je near ptr 1002DAC6h
0x1002d6df: mov edx,[esp+edx*4+30h]
0x1002d6e3: mov eax,[esi]
0x1002d6e5: movzx esi,byte ptr [esi+0Bh]
0x1002d6e9: push ebp
0x1002d6ea: mov ebp,eax
0x1002d6ec: imul ebp,edx
0x1002d6ef: mov [esp+30h],edx
0x1002d6f3: mov edx,esi
0x1002d6f5: sub edx,1
0x1002d6f8: mov [esp+24h],ebp
0x1002d6fc: je near ptr 1002D9AFh
0x1002d702: sub edx,1
0x1002d705: je near ptr 1002D8ADh
0x1002d70b: sub edx,edi
0x1002d70d: lea edi,[ebp-1]
0x1002d710: je short 1002D784h
0x1002d712: shr esi,3
0x1002d715: lea ebx,[eax-1]
0x1002d718: imul edi,esi
0x1002d71b: imul ebx,esi
0x1002d71e: add ebx,ecx
0x1002d720: add edi,ecx
0x1002d722: test eax,eax
0x1002d724: mov dword ptr [esp+14h],0
0x1002d72c: jbe near ptr 1002D88Eh
0x1002d732: push esi
0x1002d733: lea eax,[esp+54h]
0x1002d737: push ebx
0x1002d738: push eax
0x1002d739: call 100228C6h
0x1002d73e: mov eax,[esp+3Ch]
0x1002d742: add esp,0Ch
0x1002d745: test eax,eax
0x1002d747: jle short 1002D76Ah
0x1002d749: mov ebp,eax
0x1002d74b: jmp short 1002D750h
0x1002d750: push esi
0x1002d751: lea ecx,[esp+54h]
0x1002d755: push ecx
0x1002d756: push edi
0x1002d757: call 100228C6h
0x1002d75c: add esp,0Ch
0x1002d75f: sub edi,esi
0x1002d761: sub ebp,1
0x1002d764: jne short 1002D750h
0x1002d766: mov ebp,[esp+24h]
0x1002d76a: mov eax,[esp+14h]
0x1002d76e: mov edx,[esp+2Ch]
0x1002d772: add eax,1
0x1002d775: sub ebx,esi
0x1002d777: cmp eax,[edx]
0x1002d779: mov [esp+14h],eax
0x1002d77d: jb short 1002D732h
```

```
0x1002d77f: jmp near ptr 1002D88Eh
0x1002d784: lea edx,[eax-1]
0x1002d787: shr edx,1
0x1002d789: shr edi,1
0x1002d78b: add edx,ecx
0x1002d78d: add edi,ecx
0x1002d78f: test ebx,10000h
0x1002d795: mov [esp+18h],edx
0x1002d799: je short 1002D7CDh
0x1002d79b: or edx,0FFFFFFFFh
0x1002d79e: lea ecx,[eax*4]
0x1002d7a5: sub edx,ecx
0x1002d7a7: or esi,0FFFFFFFFh
0x1002d7aa: lea ecx,[ebp*4]
0x1002d7b1: sub esi,ecx
0x1002d7b3: and edx,4
0x1002d7b6: and esi,4
0x1002d7b9: mov dword ptr [esp+20h],4
0x1002d7c1: xor ebp,ebp
0x1002d7c3: mov dword ptr [esp+1Ch],0FFFFFFFCh
0x1002d7cb: jmp short 1002D802h
0x1002d7cd: lea edx,[eax-1]
0x1002d7d0: and edx,1
0x1002d7d3: add ebp,0FFFFFFFFh
0x1002d7d6: add edx,edx
0x1002d7d8: add edx,edx
0x1002d7da: and ebp,1
0x1002d7dd: add ebp,ebp
0x1002d7df: mov ecx,edx
0x1002d7e1: mov edx,4
0x1002d7e6: add ebp,ebp
0x1002d7e8: mov esi,4
0x1002d7ed: sub edx,ecx
0x1002d7ef: sub esi,ebp
0x1002d7f1: mov ebp,4
0x1002d7f6: mov dword ptr [esp+20h],0
0x1002d7fe: mov [esp+1Ch],ebp
0x1002d802: test eax,eax
0x1002d804: mov dword ptr [esp+28h],0
0x1002d80c: jbe near ptr 1002D88Ah
0x1002d812: mov eax,[esp+18h]
0x1002d816: mov al,[eax]
0x1002d818: mov cl,dl
0x1002d81a: shr al,cl
0x1002d81c: mov ecx,[esp+30h]
0x1002d820: and al,0Fh
0x1002d822: test ecx,ecx
0x1002d824: mov [esp+13h],al
0x1002d828: jle short 1002D864h
0x1002d82a: mov [esp+14h],ecx
0x1002d82e: jmp short 1002D834h
0x1002d830: mov al,[esp+13h]
0x1002d834: mov ecx,4
0x1002d839: sub ecx,esi
0x1002d83b: mov ebx,0F0Fh
0x1002d840: sar ebx,cl
0x1002d842: mov ecx,esi
0x1002d844: shl al,cl
0x1002d846: and bl,[edi]
0x1002d848: or bl,al
0x1002d84a: cmp esi,ebp
0x1002d84c: mov [edi],bl
0x1002d84e: jne short 1002D859h
0x1002d850: mov esi,[esp+20h]
0x1002d854: sub edi,1
0x1002d857: jmp short 1002D85Dh
0x1002d859: add esi,[esp+1Ch]
```

```
0x1002d85d: sub dword ptr [esp+14h],1
0x1002d862: jne short 1002D830h
0x1002d864: cmp edx,ebp
0x1002d866: jne short 1002D873h
0x1002d868: mov edx,[esp+20h]
0x1002d86c: sub dword ptr [esp+18h],1
0x1002d871: jmp short 1002D877h
0x1002d873: add edx,[esp+1Ch]
0x1002d877: mov eax,[esp+28h]
0x1002d87b: mov ecx,[esp+2Ch]
0x1002d87f: add eax,1
0x1002d882: cmp eax,[ecx]
0x1002d884: mov [esp+28h],eax
0x1002d888: jb short 1002D812h
0x1002d88a: mov ebp,[esp+24h]
0x1002d88e: mov ecx,[esp+2Ch]
0x1002d892: mov al,[ecx+0Bh]
0x1002d895: cmp al,8
0x1002d897: mov [ecx],ebp
0x1002d899: movzx eax,al
0x1002d89c: jb near ptr 1002DAB9h
0x1002d8a2: shr eax,3
0x1002d8a5: imul eax,ebp
0x1002d8a8: jmp near ptr 1002DAC2h
0x1002d8ad: lea edx,[eax-1]
0x1002d8b0: lea edi,[ebp-1]
0x1002d8b3: shr edx,2
0x1002d8b6: shr edi,2
0x1002d8b9: add edx,ecx
0x1002d8bb: add edi,ecx
0x1002d8bd: test ebx,10000h
0x1002d8c3: mov [esp+18h],edx
0x1002d8c7: je short 1002D8EBh
0x1002d8c9: lea esi,[ebp+ebp-1]
0x1002d8cd: lea edx,[eax+eax-1]
0x1002d8d1: and edx,6
0x1002d8d4: and esi,6
0x1002d8d7: mov dword ptr [esp+1Ch],6
0x1002d8df: xor ebp,ebp
0x1002d8e1: mov dword ptr [esp+20h],0FFFFFFFEh
0x1002d8e9: jmp short 1002D91Eh
0x1002d8eb: add ebp,0FFFFFFFFh
0x1002d8ee: lea ecx,[eax-1]
0x1002d8f1: and ecx,3
0x1002d8f4: and ebp,3
0x1002d8f7: mov edx,3
0x1002d8fc: sub edx,ecx
0x1002d8fe: mov esi,3
0x1002d903: sub esi,ebp
0x1002d905: add edx,edx
0x1002d907: add esi,esi
0x1002d909: mov dword ptr [esp+1Ch],0
0x1002d911: mov ebp,6
0x1002d916: mov dword ptr [esp+20h],2
0x1002d91e: test eax,eax
0x1002d920: mov dword ptr [esp+14h],0
0x1002d928: jbe near ptr 1002D88Ah
0x1002d92e: mov edi,edi
0x1002d930: mov eax,[esp+18h]
0x1002d934: mov al,[eax]
0x1002d936: mov cl,dl
0x1002d938: shr al,cl
0x1002d93a: mov ecx,[esp+30h]
0x1002d93e: and al,3
0x1002d940: test ecx,ecx
0x1002d942: mov [esp+13h],al
0x1002d946: jle short 1002D984h
```

```
0x1002d948: mov [esp+28h],ecx
0x1002d94c: jmp short 1002D954h
0x1002d950: mov al,[esp+13h]
0x1002d954: mov ecx,6
0x1002d959: sub ecx,esi
0x1002d95b: mov ebx,3F3Fh
0x1002d960: sar ebx,cl
0x1002d962: mov ecx,esi
0x1002d964: shl al,cl
0x1002d966: and bl,[edi]
0x1002d968: or bl,al
0x1002d96a: cmp esi,ebp
0x1002d96c: mov [edi],bl
0x1002d96e: jne short 1002D979h
0x1002d970: mov esi,[esp+1Ch]
0x1002d974: sub edi,1
0x1002d977: jmp short 1002D97Dh
0x1002d979: add esi,[esp+20h]
0x1002d97d: sub dword ptr [esp+28h],1
0x1002d982: jne short 1002D950h
0x1002d984: cmp edx,ebp
0x1002d986: jne short 1002D993h
0x1002d988: mov edx,[esp+1Ch]
0x1002d98c: sub dword ptr [esp+18h],1
0x1002d991: jmp short 1002D997h
0x1002d993: add edx,[esp+20h]
0x1002d997: mov eax,[esp+14h]
0x1002d99b: mov ecx,[esp+2Ch]
0x1002d99f: add eax,1
0x1002d9a2: cmp eax,[ecx]
0x1002d9a4: mov [esp+14h],eax
0x1002d9a8: jb short 1002D930h
0x1002d9aa: jmp near ptr 1002D88Ah
0x1002d9af: lea edx,[eax-1]
0x1002d9b2: lea edi,[ebp-1]
0x1002d9b5: shr edx,3
0x1002d9b8: shr edi,3
0x1002d9bb: add edx,ecx
0x1002d9bd: add edi,ecx
0x1002d9bf: test ebx,10000h
0x1002d9c5: mov [esp+20h],edx
0x1002d9c9: je short 1002D9F1h
0x1002d9cb: lea edx,[eax-1]
0x1002d9ce: lea esi,[ebp-1]
0x1002d9d1: and edx,7
0x1002d9d4: and esi,7
0x1002d9d7: mov dword ptr [esp+18h],7
0x1002d9df: mov dword ptr [esp+1Ch],0
0x1002d9e7: mov dword ptr [esp+14h],0FFFFFFFFh
0x1002d9ef: jmp short 1002DA23h
0x1002d9f1: lea ecx,[eax-1]
0x1002d9f4: and ecx,7
0x1002d9f7: mov edx,7
0x1002d9fc: sub edx,ecx
0x1002d9fe: lea ecx,[ebp-1]
0x1002da01: and ecx,7
0x1002da04: mov esi,7
0x1002da09: sub esi,ecx
0x1002da0b: mov dword ptr [esp+18h],0
0x1002da13: mov dword ptr [esp+1Ch],7
0x1002da1b: mov dword ptr [esp+14h],1
0x1002da23: test eax,eax
0x1002da25: mov [esp+24h],edx
0x1002da29: mov dword ptr [esp+50h],0
0x1002da31: jbe near ptr 1002D88Eh
0x1002da37: mov eax,[esp+20h]
0x1002da3b: mov al,[eax]
```

```
0x1002da3d: mov cl,dl
0x1002da3f: shr al,cl
0x1002da41: mov ecx,[esp+30h]
0x1002da45: and al,1
0x1002da47: test ecx,ecx
0x1002da49: jle short 1002DA88h
0x1002da4b: mov [esp+28h],ecx
0x1002da4f: nop
0x1002da50: mov ecx,7
0x1002da55: sub ecx,esi
0x1002da57: mov edx,7F7Fh
0x1002da5c: sar edx,cl
0x1002da5e: mov bl,al
0x1002da60: mov ecx,esi
0x1002da62: shl bl,cl
0x1002da64: and dl,[edi]
0x1002da66: or dl,bl
0x1002da68: cmp esi,[esp+1Ch]
0x1002da6c: mov [edi],dl
0x1002da6e: jne short 1002DA79h
0x1002da70: mov esi,[esp+18h]
0x1002da74: sub edi,1
0x1002da77: jmp short 1002DA7Dh
0x1002da79: add esi,[esp+14h]
0x1002da7d: sub dword ptr [esp+28h],1
0x1002da82: jne short 1002DA50h
0x1002da84: mov edx,[esp+24h]
0x1002da88: cmp edx,[esp+1Ch]
0x1002da8c: jne short 1002DA99h
0x1002da8e: mov edx,[esp+18h]
0x1002da92: sub dword ptr [esp+20h],1
0x1002da97: jmp short 1002DA9Dh
0x1002da99: add edx,[esp+14h]
0x1002da9d: mov eax,[esp+50h]
0x1002daa1: mov ecx,[esp+2Ch]
0x1002daa5: add eax,1
0x1002daa8: cmp eax,[ecx]
0x1002daaa: mov [esp+24h],edx
0x1002daae: mov [esp+50h],eax
0x1002dab2: jb short 1002DA37h
0x1002dab4: jmp near ptr 1002D88Eh
0x1002dab9: imul eax,ebp
0x1002dabc: add eax,7
0x1002dabf: shr eax,3
0x1002dac2: mov [ecx+4],eax
0x1002dac5: pop ebp
0x1002dac6: mov ecx,[esp+54h]
0x1002daca: pop edi
0x1002dacb: pop esi
0x1002dacc: pop ebx
0x1002dacd: xor ecx,esp
0x1002dacf: call 10022326h
0x1002dad4: add esp,4Ch
0x1002dad7: ret
```

`sub_1002d670` is called by png_read_row. It delegates to sub_10022326, j_memcpy (2 callees).

12.749 sub_1002dae0 at 0x1002dae0 (22 instructions)

```
0x1002dae0: mov eax,[esp+14h]
0x1002dae4: sub esp,10h
0x1002dae7: cmp eax,4
0x1002daea: push ebx
0x1002daeb: push ebp
0x1002daec: push esi
0x1002daed: push edi
```

```
0x1002daee: ja near ptr 1002DC9Dh
0x1002daf4: jmp dword ptr [eax*4+1002DCACh]
0x1002dc9d: mov eax,[esp+24h]
0x1002dca1: push 10100B10h
0x1002dca6: push eax
0x1002dca7: call 100231E0h
0x1002dcac: das
0x1002dcad: fild dword ptr [edx]
0x1002dcaf: adc bl,bh
0x1002dcb1: fiadd dword ptr [edx]
0x1002dcb3: adc [edi],dh
0x1002dcb5: fild dword ptr [edx]
0x1002dcb7: adc [edi-25h],ah
0x1002dcba: add dl,[eax]
0x1002dcbc: int 3
```

`sub_1002dae0` is called by png_read_row. It works with the text "Ignoring bad adaptive filter type". It delegates to png_error (1 callees).

12.750 sub_1002dcc0 at 0x1002dcc0 (232 instructions)

```
0x1002dcc0: sub esp,38h
0x1002dcc3: push ebp
0x1002dcc4: push esi
0x1002dcc5: mov esi,1
0x1002dcca: xor edx,edx
0x1002dccc: mov eax,4
0x1002dcd1: mov ecx,2
0x1002dcd6: push edi
0x1002dcd7: mov [esp+3Ch],esi
0x1002dcdb: mov [esp+24h],esi
0x1002dcdf: mov esi,[esp+48h]
0x1002dce3: mov edi,8
0x1002dce8: push esi
0x1002dce9: mov [esp+2Ch],edx
0x1002dced: mov [esp+30h],eax
0x1002dcf1: mov [esp+34h],edx
0x1002dcf5: mov [esp+38h],ecx
0x1002dcf9: mov [esp+3Ch],edx
0x1002dcfd: mov [esp+44h],edx
0x1002dd01: mov [esp+10h],edi
0x1002dd05: mov [esp+14h],edi
0x1002dd09: mov [esp+18h],eax
0x1002dd0d: mov [esp+1Ch],eax
0x1002dd11: mov [esp+20h],ecx
0x1002dd15: mov [esp+24h],ecx
0x1002dd19: mov [esi+7Ch],edx
0x1002dd1c: call 10031A20h
0x1002dd21: add esp,4
0x1002dd24: cmp byte ptr [esi+123h],0
0x1002dd2b: je short 1002DD7Eh
0x1002dd2d: test byte ptr [esi+74h],2
0x1002dd31: jne short 1002DD47h
0x1002dd33: mov eax,[esi+0D0h]
0x1002dd39: add eax,7
0x1002dd3c: shr eax,3
0x1002dd3f: mov [esi+0D4h],eax
0x1002dd45: jmp short 1002DD53h
0x1002dd47: mov ecx,[esi+0D0h]
0x1002dd4d: mov [esi+0D4h],ecx
0x1002dd53: movzx eax,byte ptr [esi+124h]
0x1002dd5a: mov ebp,[esi+0CCh]
0x1002dd60: add eax,eax
0x1002dd62: add eax,eax
0x1002dd64: mov ecx,[esp+eax+0Ch]
0x1002dd68: mov edx,ebp
```

```
0x1002dd6a: sub edx,[esp+eax+28h]
0x1002dd6e: lea eax,[edx+ecx-1]
0x1002dd72: xor edx,edx
0x1002dd74: div ecx
0x1002dd76: mov [esi+0E0h],eax
0x1002dd7c: jmp short 1002DD96h
0x1002dd7e: mov eax,[esi+0D0h]
0x1002dd84: mov ebp,[esi+0CCh]
0x1002dd8a: mov [esi+0D4h],eax
0x1002dd90: mov [esi+0E0h],ebp
0x1002dd96: movzx eax,byte ptr [esi+129h]
0x1002dd9d: push ebx
0x1002dd9e: mov ebx,[esi+74h]
0x1002dda1: test bl,4
0x1002dda4: je short 1002DDB1h
0x1002dda6: cmp byte ptr [esi+127h],8
0x1002ddad: jae short 1002DDB1h
0x1002ddaf: mov eax,edi
0x1002ddb1: mov edi,ebx
0x1002ddb3: and edi,1000h
0x1002ddb9: je short 1002DE1Dh
0x1002ddbb: mov cl,[esi+126h]
0x1002ddc1: cmp cl,3
0x1002ddc4: jne short 1002DDDBh
0x1002ddc6: xor eax,eax
0x1002ddc8: cmp [esi+11Ah],ax
0x1002ddcf: setne al
0x1002ddd2: lea eax,[eax*8+18h]
0x1002ddd9: jmp short 1002DE1Dh
0x1002dddb: test cl,cl
0x1002dddd: jne short 1002DDF7h
0x1002dddf: cmp eax,8
0x1002dde2: jge short 1002DDE9h
0x1002dde4: mov eax,8
0x1002dde9: cmp word ptr [esi+11Ah],0
0x1002ddf1: je short 1002DE1Dh
0x1002ddf3: add eax,eax
0x1002ddf5: jmp short 1002DE1Dh
0x1002ddf7: cmp cl,2
0x1002ddfa: jne short 1002DE1Dh
0x1002ddfc: cmp word ptr [esi+11Ah],0
0x1002de04: je short 1002DE1Dh
0x1002de06: lea ecx,[eax*4]
0x1002de0d: mov eax,55555556h
0x1002de12: imul ecx
0x1002de14: mov ecx,edx
0x1002de16: shr ecx,1Fh
0x1002de19: add ecx,edx
0x1002de1b: mov eax,ecx
0x1002de1d: mov edx,ebx
0x1002de1f: and edx,8000h
0x1002de25: je short 1002DE68h
0x1002de27: mov cl,[esi+126h]
0x1002de2d: cmp cl,3
0x1002de30: jne short 1002DE39h
0x1002de32: mov eax,20h
0x1002de37: jmp short 1002DE68h
0x1002de39: test cl,cl
0x1002de3b: jne short 1002DE50h
0x1002de3d: xor ecx,ecx
0x1002de3f: cmp eax,8
0x1002de42: setg cl
0x1002de45: sub ecx,1
0x1002de48: and ecx,0FFFFFFF0h
0x1002de4b: add ecx,20h
0x1002de4e: jmp short 1002DE66h
0x1002de50: cmp cl,2
```

```
0x1002de53: jne short 1002DE68h
0x1002de55: xor ecx,ecx
0x1002de57: cmp eax,20h
0x1002de5a: setg cl
0x1002de5d: sub ecx,1
0x1002de60: and ecx,0FFFFFFE0h
0x1002de63: add ecx,40h
0x1002de66: mov eax,ecx
0x1002de68: test ebx,4000h
0x1002de6e: je short 1002DEC7h
0x1002de70: cmp word ptr [esi+11Ah],0
0x1002de78: je short 1002DE7Eh
0x1002de7a: test edi,edi
0x1002de7c: jne short 1002DEB4h
0x1002de7e: test edx,edx
0x1002de80: jne short 1002DEB4h
0x1002de82: mov cl,[esi+126h]
0x1002de88: cmp cl,4
0x1002de8b: je short 1002DEB4h
0x1002de8d: cmp eax,8
0x1002de90: jg short 1002DEA3h
0x1002de92: xor eax,eax
0x1002de94: cmp cl,6
0x1002de97: sete al
0x1002de9a: lea eax,[eax*8+18h]
0x1002dea1: jmp short 1002DEC7h
0x1002dea3: sub cl,6
0x1002dea6: neg cl
0x1002dea8: sbb ecx,ecx
0x1002deaa: and ecx,0FFFFFFF0h
0x1002dead: add ecx,40h
0x1002deb0: mov eax,ecx
0x1002deb2: jmp short 1002DEC7h
0x1002deb4: xor edx,edx
0x1002deb6: cmp eax,10h
0x1002deb9: setg dl
0x1002debc: sub edx,1
0x1002debf: and edx,0FFFFFFE0h
0x1002dec2: add edx,40h
0x1002dec5: mov eax,edx
0x1002dec7: test ebx,100000h
0x1002decd: pop ebx
0x1002dece: je short 1002DEE1h
0x1002ded0: movzx ecx,byte ptr [esi+69h]
0x1002ded4: movzx edx,byte ptr [esi+68h]
0x1002ded8: imul ecx,edx
0x1002dedb: cmp ecx,eax
0x1002dedd: jle short 1002DEE1h
0x1002dedf: mov eax,ecx
0x1002dee1: lea ecx,[ebp+7]
0x1002dee4: and ecx,0FFFFFFF8h
0x1002dee7: cmp eax,8
0x1002deea: jl short 1002DEF6h
0x1002deec: mov edx,eax
0x1002deee: shr edx,3
0x1002def1: imul edx,ecx
0x1002def4: jmp short 1002DF01h
0x1002def6: imul ecx,eax
0x1002def9: add ecx,7
0x1002defc: shr ecx,3
0x1002deff: mov edx,ecx
0x1002df01: add eax,7
0x1002df04: sar eax,3
0x1002df07: lea eax,[eax+edx+1]
0x1002df0b: lea edi,[eax+30h]
0x1002df0e: cmp edi,[esi+27Ch]
0x1002df14: jbe short 1002DF57h
```

```
0x1002df16: mov eax,[esi+244h]
0x1002df1c: push eax
0x1002df1d: push esi
0x1002df1e: call 1002CD70h
0x1002df23: add esp,8
0x1002df26: cmp byte ptr [esi+123h],0
0x1002df2d: push edi
0x1002df2e: push esi
0x1002df2f: je short 1002DF38h
0x1002df31: call 1002CDD0h
0x1002df36: jmp short 1002DF3Dh
0x1002df38: call 1002CD20h
0x1002df3d: mov ecx,eax
0x1002df3f: add esp,8
0x1002df42: add ecx,20h
0x1002df45: mov [esi+244h],eax
0x1002df4b: mov [esi+0ECh],ecx
0x1002df51: mov [esi+27Ch],edi
0x1002df57: mov eax,[esi+0DCh]
0x1002df5d: cmp eax,0FFFFFFFEh
0x1002df60: jbe short 1002DF6Dh
0x1002df62: push 10100B34h
0x1002df67: push esi
0x1002df68: call 100231E0h
0x1002df6d: add eax,1
0x1002df70: cmp eax,[esi+280h]
0x1002df76: jbe short 1002DFADh
0x1002df78: mov edx,[esi+0E8h]
0x1002df7e: push edx
0x1002df7f: push esi
0x1002df80: call 1002CD70h
0x1002df85: mov eax,[esi+0DCh]
0x1002df8b: add eax,1
0x1002df8e: push eax
0x1002df8f: push esi
0x1002df90: call 1002CD20h
0x1002df95: mov [esi+0E8h],eax
0x1002df9b: mov eax,[esi+0DCh]
0x1002dfa1: add esp,10h
0x1002dfa4: add eax,1
0x1002dfa7: mov [esi+280h],eax
0x1002dfad: mov ecx,[esi+0E8h]
0x1002dfb3: push eax
0x1002dfb4: push 0
0x1002dfb6: push ecx
0x1002dfb7: call 100228C0h
0x1002dfbc: add esp,0Ch
0x1002dfbf: or dword ptr [esi+70h],40h
0x1002dfc3: pop edi
0x1002dfc4: pop esi
0x1002dfc5: pop ebp
0x1002dfc6: add esp,38h
0x1002dfc9: ret
```

`sub_1002dcc0` is called by png_read_update_info, png_read_row. It works with the text "Row has too many bytes to allocate in memory". It accesses memory-mapped peripheral registers at 0x55555556. It delegates to j_memset, png_error, png_malloc (6 callees).

12.751 sub_1002dfd0 at 0x1002dfd0 (42 instructions)

```
0x1002dfd0: sub esp,0Ch
0x1002dfd3: mov eax,ds:[10136220h]
0x1002dfd8: xor eax,esp
0x1002dfda: mov [esp+8],eax
0x1002dfde: push ebx
0x1002dfdf: push esi
```

```
0x1002dfe0: mov esi,[esp+18h]
0x1002dfe4: push edi
0x1002dfe5: push 8
0x1002dfe7: lea eax,[esp+10h]
0x1002dfeb: push eax
0x1002dfec: push esi
0x1002dfed: mov dword ptr [esi+288h],21h
0x1002dff7: call 10029C10h
0x1002dffc: lea ecx,[esp+18h]
0x1002e000: push ecx
0x1002e001: push esi
0x1002e002: call 1002CE00h
0x1002e007: mov edx,[esp+24h]
0x1002e00b: lea edi,[esi+11Ch]
0x1002e011: push esi
0x1002e012: mov ebx,eax
0x1002e014: mov [edi],edx
0x1002e016: call 100234E0h
0x1002e01b: push 4
0x1002e01d: push edi
0x1002e01e: push esi
0x1002e01f: call 10023500h
0x1002e024: push edi
0x1002e025: push esi
0x1002e026: call 1002D310h
0x1002e02b: mov ecx,[esp+40h]
0x1002e02f: add esp,2Ch
0x1002e032: pop edi
0x1002e033: mov dword ptr [esi+288h],41h
0x1002e03d: pop esi
0x1002e03e: mov eax,ebx
0x1002e040: pop ebx
0x1002e041: xor ecx,esp
0x1002e043: call 10022326h
0x1002e048: add esp,0Ch
0x1002e04b: ret
```

`sub_1002dfd0` is called by png_read_info, png_read_row, png_read_end, sub_100309c0. It delegates to sub_10022326, sub_100234e0, sub_10023500 (6 callees).

12.752 sub_1002e050 at 0x1002e050 (59 instructions)

```
0x1002e050: push ebx
0x1002e051: push esi
0x1002e052: mov esi,[esp+0Ch]
0x1002e056: mov ebx,[esi+0B4h]
0x1002e05c: push edi
0x1002e05d: mov edi,[esp+14h]
0x1002e061: cmp edi,ebx
0x1002e063: jbe short 1002E08Dh
0x1002e065: jmp short 1002E070h
0x1002e070: mov eax,[esi+0B4h]
0x1002e076: mov ecx,[esi+0B0h]
0x1002e07c: push eax
0x1002e07d: push ecx
0x1002e07e: push esi
0x1002e07f: call 1002CF30h
0x1002e084: sub edi,ebx
0x1002e086: add esp,0Ch
0x1002e089: cmp edi,ebx
0x1002e08b: ja short 1002E070h
0x1002e08d: test edi,edi
0x1002e08f: je short 1002E0A2h
0x1002e091: mov edx,[esi+0B0h]
0x1002e097: push edi
0x1002e098: push edx
```

```
0x1002e099: push esi
0x1002e09a: call 1002CF30h
0x1002e09f: add esp,0Ch
0x1002e0a2: push esi
0x1002e0a3: call 1002CF60h
0x1002e0a8: add esp,4
0x1002e0ab: test eax,eax
0x1002e0ad: je short 1002E0F4h
0x1002e0af: mov al,[esi+11Ch]
0x1002e0b5: and al,20h
0x1002e0b7: je short 1002E0C2h
0x1002e0b9: test dword ptr [esi+70h],200h
0x1002e0c0: je short 1002E0CFh
0x1002e0c2: test al,al
0x1002e0c4: jne short 1002E0E6h
0x1002e0c6: test dword ptr [esi+70h],400h
0x1002e0cd: je short 1002E0E6h
0x1002e0cf: push 10100B64h
0x1002e0d4: push esi
0x1002e0d5: call 10023150h
0x1002e0da: add esp,8
0x1002e0dd: pop edi
0x1002e0de: pop esi
0x1002e0df: mov eax,1
0x1002e0e4: pop ebx
0x1002e0e5: ret
0x1002e0e6: push 10100B64h
0x1002e0eb: push esi
0x1002e0ec: call 10023270h
0x1002e0f1: add esp,8
0x1002e0f4: pop edi
0x1002e0f5: pop esi
0x1002e0f6: xor eax,eax
0x1002e0f8: pop ebx
0x1002e0f9: ret
```

`sub_1002e050` is called by png_read_row, png_read_end, sub_1002e100, sub_1002e260, sub_1002e400, sub_1002e450, sub_1002e590, sub_1002e700, sub_1002ea20, sub_1002ec10, sub_1002ee90, sub_1002f170, sub_1002f400, sub_1002f650, sub_1002f7e0, sub_1002f930, sub_1002faa0, sub_1002fdd0, sub_1002ffb0, sub_100300d0, sub_10030280, sub_10030490, sub_10030750, sub_100309c0. It delegates to png_chunk_warning, png_chunk_benign_error, sub_1002cf30 (4 callees).

12.753 sub_1002e100 at 0x1002e100 (108 instructions)

```
0x1002e100: sub esp,20h
0x1002e103: mov eax,ds:[10136220h]
0x1002e108: xor eax,esp
0x1002e10a: mov [esp+1Ch],eax
0x1002e10e: mov eax,[esp+28h]
0x1002e112: push esi
0x1002e113: mov esi,[esp+28h]
0x1002e117: mov [esp+0Ch],eax
0x1002e11b: mov eax,[esi+6Ch]
0x1002e11e: test al,1
0x1002e120: je short 1002E12Dh
0x1002e122: push 10100B84h
0x1002e127: push esi
0x1002e128: call 100231E0h
0x1002e12d: cmp dword ptr [esp+30h],0Dh
0x1002e132: je short 1002E13Fh
0x1002e134: push 10100B70h
0x1002e139: push esi
0x1002e13a: call 100231E0h
0x1002e13f: push ebx
0x1002e140: push ebp
```

```
0x1002e141: push edi
0x1002e142: push 0Dh
0x1002e144: lea ecx,[esp+20h]
0x1002e148: push ecx
0x1002e149: or eax,1
0x1002e14c: push esi
0x1002e14d: mov [esi+6Ch],eax
0x1002e150: call 1002CF30h
0x1002e155: push 0
0x1002e157: push esi
0x1002e158: call 1002E050h
0x1002e15d: lea edx,[esp+30h]
0x1002e161: push edx
0x1002e162: push esi
0x1002e163: call 1002CE00h
0x1002e168: mov edi,eax
0x1002e16a: lea eax,[esp+3Ch]
0x1002e16e: push eax
0x1002e16f: push esi
0x1002e170: call 1002CE00h
0x1002e175: movzx ecx,byte ptr [esp+49h]
0x1002e17a: movzx edx,byte ptr [esp+48h]
0x1002e17f: movzx ebx,byte ptr [esp+4Ah]
0x1002e184: mov ebp,eax
0x1002e186: movzx eax,byte ptr [esp+4Bh]
0x1002e18b: mov [esp+34h],eax
0x1002e18f: movzx eax,byte ptr [esp+4Ch]
0x1002e194: mov [esp+38h],eax
0x1002e198: mov [esi+123h],al
0x1002e19e: mov al,[esp+34h]
0x1002e1a2: mov [esi+234h],al
0x1002e1a8: movzx eax,cl
0x1002e1ab: add esp,24h
0x1002e1ae: sub eax,2
0x1002e1b1: mov [esi+0CCh],edi
0x1002e1b7: mov [esi+0D0h],ebp
0x1002e1bd: mov [esi+127h],dl
0x1002e1c3: mov [esi+126h],cl
0x1002e1c9: mov [esi+254h],bl
0x1002e1cf: je short 1002E1F6h
0x1002e1d1: sub eax,2
0x1002e1d4: je short 1002E1EDh
0x1002e1d6: sub eax,2
0x1002e1d9: je short 1002E1E4h
0x1002e1db: mov byte ptr [esi+12Ah],1
0x1002e1e2: jmp short 1002E1FDh
0x1002e1e4: mov byte ptr [esi+12Ah],4
0x1002e1eb: jmp short 1002E1FDh
0x1002e1ed: mov byte ptr [esi+12Ah],2
0x1002e1f4: jmp short 1002E1FDh
0x1002e1f6: mov byte ptr [esi+12Ah],3
0x1002e1fd: mov al,dl
0x1002e1ff: imul byte ptr [esi+12Ah]
0x1002e205: mov [esi+129h],al
0x1002e20b: cmp al,8
0x1002e20d: movzx eax,al
0x1002e210: jb short 1002E21Ah
0x1002e212: shr eax,3
0x1002e215: imul eax,edi
0x1002e218: jmp short 1002E223h
0x1002e21a: imul eax,edi
0x1002e21d: add eax,7
0x1002e220: shr eax,3
0x1002e223: mov [esi+0DCh],eax
0x1002e229: mov eax,[esp+10h]
0x1002e22d: push eax
0x1002e22e: mov eax,[esp+18h]
```

```
0x1002e232: push ebx
0x1002e233: push eax
0x1002e234: push ecx
0x1002e235: mov ecx,[esp+28h]
0x1002e239: push edx
0x1002e23a: push ebp
0x1002e23b: push edi
0x1002e23c: push ecx
0x1002e23d: push esi
0x1002e23e: call 1002B7E0h
0x1002e243: mov ecx,[esp+50h]
0x1002e247: add esp,24h
0x1002e24a: pop edi
0x1002e24b: pop ebp
0x1002e24c: pop ebx
0x1002e24d: pop esi
0x1002e24e: xor ecx,esp
0x1002e250: call 10022326h
0x1002e255: add esp,20h
0x1002e258: ret
```

`sub_1002e100` is called by png_read_info, png_read_end. It works with the text "Invalid IHDR chunk", "Out of place IHDR". It delegates to sub_10022326, png_error, png_set_IHDR (6 callees).

12.754 sub_1002e260 at 0x1002e260 (132 instructions)

```
0x1002e260: sub esp,304h
0x1002e266: push esi
0x1002e267: mov esi,[esp+30Ch]
0x1002e26e: mov eax,[esi+6Ch]
0x1002e271: test al,1
0x1002e273: jne short 1002E280h
0x1002e275: push 10100C5Ch
0x1002e27a: push esi
0x1002e27b: call 100231E0h
0x1002e280: test al,4
0x1002e282: je short 1002E2A8h
0x1002e284: push 10100C44h
0x1002e289: push esi
0x1002e28a: call 100230F0h
0x1002e28f: mov eax,[esp+31Ch]
0x1002e296: push eax
0x1002e297: push esi
0x1002e298: call 1002E050h
0x1002e29d: add esp,10h
0x1002e2a0: pop esi
0x1002e2a1: add esp,304h
0x1002e2a7: ret
0x1002e2a8: test al,2
0x1002e2aa: je short 1002E2B7h
0x1002e2ac: push 10100C2Ch
0x1002e2b1: push esi
0x1002e2b2: call 100231E0h
0x1002e2b7: mov cl,[esi+126h]
0x1002e2bd: or eax,2
0x1002e2c0: test cl,2
0x1002e2c3: mov [esi+6Ch],eax
0x1002e2c6: jne short 1002E2ECh
0x1002e2c8: push 10100C04h
0x1002e2cd: push esi
0x1002e2ce: call 100230F0h
0x1002e2d3: mov ecx,[esp+31Ch]
0x1002e2da: push ecx
0x1002e2db: push esi
0x1002e2dc: call 1002E050h
0x1002e2e1: add esp,10h
```

```
0x1002e2e4: pop esi
0x1002e2e5: add esp,304h
0x1002e2eb: ret
0x1002e2ec: push edi
0x1002e2ed: mov edi,[esp+318h]
0x1002e2f4: cmp edi,300h
0x1002e2fa: ja near ptr 1002E3CFh
0x1002e300: mov eax,0AAAAAAABh
0x1002e305: mul edi
0x1002e307: shr edx,1
0x1002e309: lea edx,[edx+edx*2]
0x1002e30c: mov eax,edi
0x1002e30e: sub eax,edx
0x1002e310: jne near ptr 1002E3CFh
0x1002e316: mov eax,55555556h
0x1002e31b: imul edi
0x1002e31d: push ebx
0x1002e31e: mov ebx,edx
0x1002e320: shr ebx,1Fh
0x1002e323: add ebx,edx
0x1002e325: test ebx,ebx
0x1002e327: jle short 1002E35Dh
0x1002e329: push ebp
0x1002e32a: lea edi,[esp+16h]
0x1002e32e: mov ebp,ebx
0x1002e330: push 3
0x1002e332: lea ecx,[esp+14h]
0x1002e336: push ecx
0x1002e337: push esi
0x1002e338: call 1002CF30h
0x1002e33d: mov dl,[esp+1Ch]
0x1002e341: mov al,[esp+1Dh]
0x1002e345: mov cl,[esp+1Eh]
0x1002e349: mov [edi-2],dl
0x1002e34c: mov [edi-1],al
0x1002e34f: mov [edi],cl
0x1002e351: add esp,0Ch
0x1002e354: add edi,3
0x1002e357: sub ebp,1
0x1002e35a: jne short 1002E330h
0x1002e35c: pop ebp
0x1002e35d: push 0
0x1002e35f: push esi
0x1002e360: call 1002E050h
0x1002e365: mov edi,[esp+320h]
0x1002e36c: push ebx
0x1002e36d: lea edx,[esp+1Ch]
0x1002e371: push edx
0x1002e372: push edi
0x1002e373: push esi
0x1002e374: call 1002BE10h
0x1002e379: add esp,18h
0x1002e37c: cmp byte ptr [esi+126h],3
0x1002e383: jne short 1002E3C5h
0x1002e385: test edi,edi
0x1002e387: je short 1002E3C5h
0x1002e389: test byte ptr [edi+8],10h
0x1002e38d: je short 1002E3C5h
0x1002e38f: cmp [esi+11Ah],bx
0x1002e396: jbe short 1002E3ADh
0x1002e398: push 10100BDCh
0x1002e39d: push esi
0x1002e39e: call 100230F0h
0x1002e3a3: add esp,8
0x1002e3a6: mov [esi+11Ah],bx
0x1002e3ad: cmp [edi+16h],bx
0x1002e3b1: jbe short 1002E3C5h
```

```
0x1002e3b3: push 10100BB0h
0x1002e3b8: push esi
0x1002e3b9: call 100230F0h
0x1002e3be: add esp,8
0x1002e3c1: mov [edi+16h],bx
0x1002e3c5: pop ebx
0x1002e3c6: pop edi
0x1002e3c7: pop esi
0x1002e3c8: add esp,304h
0x1002e3ce: ret
0x1002e3cf: cmp cl,3
0x1002e3d2: push 10100B98h
0x1002e3d7: push esi
0x1002e3d8: je short 1002E3F2h
0x1002e3da: call 100230F0h
0x1002e3df: push edi
0x1002e3e0: push esi
0x1002e3e1: call 1002E050h
0x1002e3e6: add esp,10h
0x1002e3e9: pop edi
0x1002e3ea: pop esi
0x1002e3eb: add esp,304h
0x1002e3f1: ret
0x1002e3f2: call 100231E0h
0x1002e3f7: int 3
```

`sub_1002e260` is called by png_read_info, png_read_end. It works with the text "Invalid palette chunk", "Ignoring PLTE chunk in grayscale PNG", "Duplicate PLTE chunk". It accesses memory-mapped peripheral registers at 0x55555556. It delegates to png_warning, png_error, png_set_PLTE (5 callees).

12.755 sub_1002e400 at 0x1002e400 (28 instructions)

```
0x1002e400: push esi
0x1002e401: mov esi,[esp+8]
0x1002e405: mov eax,[esi+6Ch]
0x1002e408: test al,1
0x1002e40a: je short 1002E43Ah
0x1002e40c: test al,4
0x1002e40e: je short 1002E43Ah
0x1002e410: push edi
0x1002e411: mov edi,[esp+14h]
0x1002e415: or eax,18h
0x1002e418: test edi,edi
0x1002e41a: mov [esi+6Ch],eax
0x1002e41d: je short 1002E42Dh
0x1002e41f: push 10100C8Ch
0x1002e424: push esi
0x1002e425: call 100230F0h
0x1002e42a: add esp,8
0x1002e42d: push edi
0x1002e42e: push esi
0x1002e42f: call 1002E050h
0x1002e434: add esp,8
0x1002e437: pop edi
0x1002e438: pop esi
0x1002e439: ret
0x1002e43a: push 10100C78h
0x1002e43f: push esi
0x1002e440: call 100231E0h
0x1002e445: int 3
```

`sub_1002e400` is called by png_read_info, png_read_end. It works with the text "No image in file". It delegates to png_warning, png_error, sub_1002e050 (3 callees).

12.756 sub_1002e450 at 0x1002e450 (119 instructions)

```
0x1002e450: push esi
0x1002e451: mov esi,[esp+8]
0x1002e455: mov eax,[esi+6Ch]
0x1002e458: test al,1
0x1002e45a: jne short 1002E467h
0x1002e45c: push 10100D84h
0x1002e461: push esi
0x1002e462: call 100231E0h
0x1002e467: test al,4
0x1002e469: je short 1002E486h
0x1002e46b: push 10100D6Ch
0x1002e470: push esi
0x1002e471: call 100230F0h
0x1002e476: mov eax,[esp+18h]
0x1002e47a: push eax
0x1002e47b: push esi
0x1002e47c: call 1002E050h
0x1002e481: add esp,10h
0x1002e484: pop esi
0x1002e485: ret
0x1002e486: test al,2
0x1002e488: je short 1002E498h
0x1002e48a: push 10100D54h
0x1002e48f: push esi
0x1002e490: call 100230F0h
0x1002e495: add esp,8
0x1002e498: push ebx
0x1002e499: mov ebx,[esp+10h]
0x1002e49d: test ebx,ebx
0x1002e49f: je short 1002E4CBh
0x1002e4a1: mov eax,[ebx+8]
0x1002e4a4: test al,1
0x1002e4a6: je short 1002E4CBh
0x1002e4a8: test eax,800h
0x1002e4ad: jne short 1002E4CBh
0x1002e4af: push 10100D3Ch
0x1002e4b4: push esi
0x1002e4b5: call 100230F0h
0x1002e4ba: mov ecx,[esp+1Ch]
0x1002e4be: push ecx
0x1002e4bf: push esi
0x1002e4c0: call 1002E050h
0x1002e4c5: add esp,10h
0x1002e4c8: pop ebx
0x1002e4c9: pop esi
0x1002e4ca: ret
0x1002e4cb: push edi
0x1002e4cc: mov edi,[esp+18h]
0x1002e4d0: cmp edi,4
0x1002e4d3: je short 1002E4EEh
0x1002e4d5: push 10100D20h
0x1002e4da: push esi
0x1002e4db: call 100230F0h
0x1002e4e0: push edi
0x1002e4e1: push esi
0x1002e4e2: call 1002E050h
0x1002e4e7: add esp,10h
0x1002e4ea: pop edi
0x1002e4eb: pop ebx
0x1002e4ec: pop esi
0x1002e4ed: ret
0x1002e4ee: push 4
0x1002e4f0: lea edx,[esp+14h]
0x1002e4f4: push edx
0x1002e4f5: push esi
0x1002e4f6: call 1002CF30h
```

```
0x1002e4fb: push 0
0x1002e4fd: push esi
0x1002e4fe: call 1002E050h
0x1002e503: add esp,14h
0x1002e506: test eax,eax
0x1002e508: jne short 1002E586h
0x1002e50a: lea ecx,[esp+10h]
0x1002e50e: xor edx,edx
0x1002e510: call 1002CE40h
0x1002e515: mov edi,eax
0x1002e517: test edi,edi
0x1002e519: jg short 1002E52Dh
0x1002e51b: push 10100CF4h
0x1002e520: push esi
0x1002e521: call 100230F0h
0x1002e526: add esp,8
0x1002e529: pop edi
0x1002e52a: pop ebx
0x1002e52b: pop esi
0x1002e52c: ret
0x1002e52d: test ebx,ebx
0x1002e52f: je short 1002E575h
0x1002e531: test dword ptr [ebx+8],800h
0x1002e538: je short 1002E575h
0x1002e53a: cmp edi,0AFC8h
0x1002e540: jl short 1002E54Ah
0x1002e542: cmp edi,0B3B0h
0x1002e548: jle short 1002E575h
0x1002e54a: push 10100CBCh
0x1002e54f: push esi
0x1002e550: call 100230F0h
0x1002e555: add esp,8
0x1002e558: push edi
0x1002e559: push 10100CA8h
0x1002e55e: call dword ptr ds:[100FE4C4h]
0x1002e564: add eax,40h
0x1002e567: push eax
0x1002e568: call dword ptr ds:[100FE4C8h]
0x1002e56e: add esp,0Ch
0x1002e571: pop edi
0x1002e572: pop ebx
0x1002e573: pop esi
0x1002e574: ret
0x1002e575: push edi
0x1002e576: push ebx
0x1002e577: push esi
0x1002e578: mov [esi+15Ch],edi
0x1002e57e: call 1002B6A0h
0x1002e583: add esp,0Ch
0x1002e586: pop edi
0x1002e587: pop ebx
0x1002e588: pop esi
0x1002e589: ret
```

`sub_1002e450` is called by png_read_info, png_read_end. It works with the text "Ignoring gAMA chunk with out of range gamma", "Duplicate gAMA chunk", "Out of place gAMA chunk". It delegates to png_warning, png_error, png_set_gAMA_fixed (6 callees).

12.757 sub_1002e590 at 0x1002e590 (128 instructions)

```
0x1002e590: push ecx
0x1002e591: push ebx
0x1002e592: push esi
0x1002e593: mov esi,[esp+10h]
0x1002e597: mov eax,[esi+6Ch]
0x1002e59a: xor ebx,ebx
```

```
0x1002e59c: test al,1
0x1002e59e: mov [esp+0Bh],bl
0x1002e5a2: mov [esp+0Ah],bl
0x1002e5a6: mov [esp+9],bl
0x1002e5aa: mov [esp+8],bl
0x1002e5ae: jne short 1002E5BBh
0x1002e5b0: push 10100E04h
0x1002e5b5: push esi
0x1002e5b6: call 100231E0h
0x1002e5bb: test al,4
0x1002e5bd: je short 1002E5DCh
0x1002e5bf: push 10100DECh
0x1002e5c4: push esi
0x1002e5c5: call 100230F0h
0x1002e5ca: mov eax,[esp+20h]
0x1002e5ce: push eax
0x1002e5cf: push esi
0x1002e5d0: call 1002E050h
0x1002e5d5: add esp,10h
0x1002e5d8: pop esi
0x1002e5d9: pop ebx
0x1002e5da: pop ecx
0x1002e5db: ret
0x1002e5dc: test al,2
0x1002e5de: je short 1002E5EEh
0x1002e5e0: push 10100DD4h
0x1002e5e5: push esi
0x1002e5e6: call 100230F0h
0x1002e5eb: add esp,8
0x1002e5ee: push ebp
0x1002e5ef: mov ebp,[esp+18h]
0x1002e5f3: cmp ebp,ebx
0x1002e5f5: je short 1002E61Bh
0x1002e5f7: test byte ptr [ebp+8],2
0x1002e5fb: je short 1002E61Bh
0x1002e5fd: push 10100DBCh
0x1002e602: push esi
0x1002e603: call 100230F0h
0x1002e608: mov ecx,[esp+24h]
0x1002e60c: push ecx
0x1002e60d: push esi
0x1002e60e: call 1002E050h
0x1002e613: add esp,10h
0x1002e616: pop ebp
0x1002e617: pop esi
0x1002e618: pop ebx
0x1002e619: pop ecx
0x1002e61a: ret
0x1002e61b: mov eax,3
0x1002e620: cmp [esi+126h],al
0x1002e626: je short 1002E62Fh
0x1002e628: movzx eax,byte ptr [esi+12Ah]
0x1002e62f: push edi
0x1002e630: mov edi,[esp+20h]
0x1002e634: cmp edi,eax
0x1002e636: jne near ptr 1002E6DFh
0x1002e63c: cmp edi,4
0x1002e63f: ja near ptr 1002E6DFh
0x1002e645: push eax
0x1002e646: lea edx,[esp+14h]
0x1002e64a: push edx
0x1002e64b: push esi
0x1002e64c: call 1002CF30h
0x1002e651: push ebx
0x1002e652: push esi
0x1002e653: call 1002E050h
0x1002e658: add esp,14h
```

```
0x1002e65b: test eax,eax
0x1002e65d: jne near ptr 1002E6F4h
0x1002e663: test byte ptr [esi+126h],2
0x1002e66a: lea eax,[esi+17Ch]
0x1002e670: je short 1002E6ABh
0x1002e672: movzx ecx,byte ptr [esp+10h]
0x1002e677: movzx edx,byte ptr [esp+11h]
0x1002e67c: push eax
0x1002e67d: mov [eax],cl
0x1002e67f: movzx ecx,byte ptr [esp+16h]
0x1002e684: mov [esi+17Dh],dl
0x1002e68a: movzx edx,byte ptr [esp+17h]
0x1002e68f: push ebp
0x1002e690: push esi
0x1002e691: mov [esi+17Eh],cl
0x1002e697: mov [esi+180h],dl
0x1002e69d: call 1002BEB0h
0x1002e6a2: add esp,0Ch
0x1002e6a5: pop edi
0x1002e6a6: pop ebp
0x1002e6a7: pop esi
0x1002e6a8: pop ebx
0x1002e6a9: pop ecx
0x1002e6aa: ret
0x1002e6ab: mov cl,[esp+10h]
0x1002e6af: push eax
0x1002e6b0: mov [esi+17Fh],cl
0x1002e6b6: mov [eax],cl
0x1002e6b8: mov [esi+17Dh],cl
0x1002e6be: mov [esi+17Eh],cl
0x1002e6c4: movzx ecx,byte ptr [esp+15h]
0x1002e6c9: push ebp
0x1002e6ca: push esi
0x1002e6cb: mov [esi+180h],cl
0x1002e6d1: call 1002BEB0h
0x1002e6d6: add esp,0Ch
0x1002e6d9: pop edi
0x1002e6da: pop ebp
0x1002e6db: pop esi
0x1002e6dc: pop ebx
0x1002e6dd: pop ecx
0x1002e6de: ret
0x1002e6df: push 10100DA0h
0x1002e6e4: push esi
0x1002e6e5: call 100230F0h
0x1002e6ea: push edi
0x1002e6eb: push esi
0x1002e6ec: call 1002E050h
0x1002e6f1: add esp,10h
0x1002e6f4: pop edi
0x1002e6f5: pop ebp
0x1002e6f6: pop esi
0x1002e6f7: pop ebx
0x1002e6f8: pop ecx
0x1002e6f9: ret
```

`sub_1002e590` is called by png_read_info, png_read_end. It works with the text "Duplicate sBIT chunk", "Out of place sBIT chunk", "Invalid sBIT after IDAT". It delegates to png_warning, png_error, png_set_sBIT (5 callees).

12.758 sub_1002e700 at 0x1002e700 (250 instructions)

```
0x1002e700: sub esp,3Ch
0x1002e703: mov eax,ds:[10136220h]
0x1002e708: xor eax,esp
0x1002e70a: mov [esp+38h],eax
0x1002e70e: push esi
```

```
0x1002e70f: mov esi,[esp+44h]
0x1002e713: mov eax,[esi+6Ch]
0x1002e716: test al,1
0x1002e718: push edi
0x1002e719: mov edi,[esp+4Ch]
0x1002e71d: mov [esp+10h],esi
0x1002e721: mov [esp+1Ch],edi
0x1002e725: jne short 1002E732h
0x1002e727: push 10100F2Ch
0x1002e72c: push esi
0x1002e72d: call 100231E0h
0x1002e732: test al,4
0x1002e734: je short 1002E760h
0x1002e736: push 10100F14h
0x1002e73b: push esi
0x1002e73c: call 100230F0h
0x1002e741: mov eax,[esp+58h]
0x1002e745: push eax
0x1002e746: push esi
0x1002e747: call 1002E050h
0x1002e74c: add esp,10h
0x1002e74f: pop edi
0x1002e750: pop esi
0x1002e751: mov ecx,[esp+38h]
0x1002e755: xor ecx,esp
0x1002e757: call 10022326h
0x1002e75c: add esp,3Ch
0x1002e75f: ret
0x1002e760: test al,2
0x1002e762: je short 1002E772h
0x1002e764: push 10100EF8h
0x1002e769: push esi
0x1002e76a: call 100230F0h
0x1002e76f: add esp,8
0x1002e772: test edi,edi
0x1002e774: je short 1002E7AEh
0x1002e776: mov eax,[edi+8]
0x1002e779: test al,4
0x1002e77b: je short 1002E7AEh
0x1002e77d: test eax,800h
0x1002e782: jne short 1002E7AEh
0x1002e784: push 10100EE0h
0x1002e789: push esi
0x1002e78a: call 100230F0h
0x1002e78f: mov ecx,[esp+58h]
0x1002e793: push ecx
0x1002e794: push esi
0x1002e795: call 1002E050h
0x1002e79a: add esp,10h
0x1002e79d: pop edi
0x1002e79e: pop esi
0x1002e79f: mov ecx,[esp+38h]
0x1002e7a3: xor ecx,esp
0x1002e7a5: call 10022326h
0x1002e7aa: add esp,3Ch
0x1002e7ad: ret
0x1002e7ae: mov edi,[esp+50h]
0x1002e7b2: cmp edi,20h
0x1002e7b5: je short 1002E7DDh
0x1002e7b7: push 10100EC4h
0x1002e7bc: push esi
0x1002e7bd: call 100230F0h
0x1002e7c2: push edi
0x1002e7c3: push esi
0x1002e7c4: call 1002E050h
0x1002e7c9: add esp,10h
0x1002e7cc: pop edi
```

```
0x1002e7cd: pop esi
0x1002e7ce: mov ecx,[esp+38h]
0x1002e7d2: xor ecx,esp
0x1002e7d4: call 10022326h
0x1002e7d9: add esp,3Ch
0x1002e7dc: ret
0x1002e7dd: push 20h
0x1002e7df: lea edx,[esp+24h]
0x1002e7e3: push edx
0x1002e7e4: push esi
0x1002e7e5: call 1002CF30h
0x1002e7ea: push 0
0x1002e7ec: push esi
0x1002e7ed: call 1002E050h
0x1002e7f2: add esp,14h
0x1002e7f5: test eax,eax
0x1002e7f7: jne near ptr 1002EA02h
0x1002e7fd: push ebx
0x1002e7fe: push ebp
0x1002e7ff: lea ecx,[esp+28h]
0x1002e803: xor edx,edx
0x1002e805: call 1002CE40h
0x1002e80a: lea ecx,[esp+2Ch]
0x1002e80e: xor edx,edx
0x1002e810: mov esi,eax
0x1002e812: call 1002CE40h
0x1002e817: lea ecx,[esp+30h]
0x1002e81b: xor edx,edx
0x1002e81d: mov edi,eax
0x1002e81f: call 1002CE40h
0x1002e824: lea ecx,[esp+34h]
0x1002e828: xor edx,edx
0x1002e82a: mov ebx,eax
0x1002e82c: call 1002CE40h
0x1002e831: lea ecx,[esp+38h]
0x1002e835: xor edx,edx
0x1002e837: mov ebp,eax
0x1002e839: call 1002CE40h
0x1002e83e: lea ecx,[esp+3Ch]
0x1002e842: xor edx,edx
0x1002e844: mov [esp+1Ch],eax
0x1002e848: call 1002CE40h
0x1002e84d: lea ecx,[esp+40h]
0x1002e851: xor edx,edx
0x1002e853: mov [esp+10h],eax
0x1002e857: call 1002CE40h
0x1002e85c: lea ecx,[esp+44h]
0x1002e860: xor edx,edx
0x1002e862: mov [esp+14h],eax
0x1002e866: call 1002CE40h
0x1002e86b: cmp esi,0FFFFFFFFh
0x1002e86e: mov [esp+20h],eax
0x1002e872: je near ptr 1002E9EEh
0x1002e878: cmp edi,0FFFFFFFFh
0x1002e87b: je near ptr 1002E9EEh
0x1002e881: cmp ebx,0FFFFFFFFh
0x1002e884: je near ptr 1002E9EEh
0x1002e88a: cmp ebp,0FFFFFFFFh
0x1002e88d: je near ptr 1002E9EEh
0x1002e893: mov edx,[esp+1Ch]
0x1002e897: cmp edx,0FFFFFFFFh
0x1002e89a: je near ptr 1002E9EEh
0x1002e8a0: cmp dword ptr [esp+10h],0FFFFFFFFh
0x1002e8a5: je near ptr 1002E9EEh
0x1002e8ab: cmp dword ptr [esp+14h],0FFFFFFFFh
0x1002e8b0: je near ptr 1002E9EEh
0x1002e8b6: cmp eax,0FFFFFFFFh
```

```
0x1002e8b9: je near ptr 1002E9EEh
0x1002e8bf: mov ecx,[esp+24h]
0x1002e8c3: test ecx,ecx
0x1002e8c5: je near ptr 1002E9BDh
0x1002e8cb: test dword ptr [ecx+8],800h
0x1002e8d2: je near ptr 1002E9BDh
0x1002e8d8: lea ecx,[esi-763Eh]
0x1002e8de: cmp ecx,7D0h
0x1002e8e4: ja short 1002E954h
0x1002e8e6: lea ecx,[edi-7C9Ch]
0x1002e8ec: cmp ecx,7D0h
0x1002e8f2: ja short 1002E954h
0x1002e8f4: lea ecx,[ebx-0F618h]
0x1002e8fa: cmp ecx,7D0h
0x1002e900: ja short 1002E954h
0x1002e902: lea ecx,[ebp-7D00h]
0x1002e908: cmp ecx,7D0h
0x1002e90e: ja short 1002E954h
0x1002e910: add edx,0FFFF8EB8h
0x1002e916: cmp edx,7D0h
0x1002e91c: ja short 1002E954h
0x1002e91e: mov edx,[esp+10h]
0x1002e922: add edx,0FFFF1988h
0x1002e928: cmp edx,7D0h
0x1002e92e: ja short 1002E954h
0x1002e930: mov ecx,[esp+14h]
0x1002e934: add ecx,0FFFFC950h
0x1002e93a: cmp ecx,7D0h
0x1002e940: ja short 1002E954h
0x1002e942: cmp eax,1388h
0x1002e947: jl short 1002E954h
0x1002e949: cmp eax,1B58h
0x1002e94e: jle near ptr 1002EA00h
0x1002e954: mov edx,[esp+18h]
0x1002e958: push 10100E8Ch
0x1002e95d: push edx
0x1002e95e: call 100230F0h
0x1002e963: add esp,8
0x1002e966: push ebp
0x1002e967: push ebx
0x1002e968: push edi
0x1002e969: push esi
0x1002e96a: mov esi,ds:[100FE4C4h]
0x1002e970: push 10100E70h
0x1002e975: call esi
0x1002e977: mov edi,ds:[100FE4C8h]
0x1002e97d: add eax,40h
0x1002e980: push eax
0x1002e981: call edi
0x1002e983: mov eax,[esp+38h]
0x1002e987: mov ecx,[esp+2Ch]
0x1002e98b: mov edx,[esp+28h]
0x1002e98f: add esp,18h
0x1002e992: push eax
0x1002e993: mov eax,[esp+20h]
0x1002e997: push ecx
0x1002e998: push edx
0x1002e999: push eax
0x1002e99a: push 10100E54h
0x1002e99f: call esi
0x1002e9a1: add eax,40h
0x1002e9a4: push eax
0x1002e9a5: call edi
0x1002e9a7: add esp,18h
0x1002e9aa: pop ebp
0x1002e9ab: pop ebx
0x1002e9ac: pop edi
```

```
0x1002e9ad: pop esi
0x1002e9ae: mov ecx,[esp+38h]
0x1002e9b2: xor ecx,esp
0x1002e9b4: call 10022326h
0x1002e9b9: add esp,3Ch
0x1002e9bc: ret
0x1002e9bd: push eax
0x1002e9be: mov eax,[esp+18h]
0x1002e9c2: push eax
0x1002e9c3: mov eax,[esp+18h]
0x1002e9c7: push eax
0x1002e9c8: push edx
0x1002e9c9: push ebp
0x1002e9ca: push ebx
0x1002e9cb: push edi
0x1002e9cc: push esi
0x1002e9cd: push ecx
0x1002e9ce: mov ecx,[esp+3Ch]
0x1002e9d2: push ecx
0x1002e9d3: call 1002B520h
0x1002e9d8: add esp,28h
0x1002e9db: pop ebp
0x1002e9dc: pop ebx
0x1002e9dd: pop edi
0x1002e9de: pop esi
0x1002e9df: mov ecx,[esp+38h]
0x1002e9e3: xor ecx,esp
0x1002e9e5: call 10022326h
0x1002e9ea: add esp,3Ch
0x1002e9ed: ret
0x1002e9ee: mov edx,[esp+18h]
0x1002e9f2: push 10100E20h
0x1002e9f7: push edx
0x1002e9f8: call 100230F0h
0x1002e9fd: add esp,8
0x1002ea00: pop ebp
0x1002ea01: pop ebx
0x1002ea02: mov ecx,[esp+40h]
0x1002ea06: pop edi
0x1002ea07: pop esi
0x1002ea08: xor ecx,esp
0x1002ea0a: call 10022326h
0x1002ea0f: add esp,3Ch
0x1002ea12: ret
```

`sub_1002e700` is called by png_read_info, png_read_end. It delegates to sub_10022326, png_warning, png_error (7 callees).

12.759 sub_1002ea20 at 0x1002ea20 (156 instructions)

```
0x1002ea20: push edi
0x1002ea21: mov edi,[esp+8]
0x1002ea25: mov eax,[edi+6Ch]
0x1002ea28: test al,1
0x1002ea2a: jne short 1002EA37h
0x1002ea2c: push 10100FE0h
0x1002ea31: push edi
0x1002ea32: call 100231E0h
0x1002ea37: test al,4
0x1002ea39: je short 1002EA56h
0x1002ea3b: push 10100FC8h
0x1002ea40: push edi
0x1002ea41: call 100230F0h
0x1002ea46: mov eax,[esp+18h]
0x1002ea4a: push eax
0x1002ea4b: push edi
```

```
0x1002ea4c: call 1002E050h
0x1002ea51: add esp,10h
0x1002ea54: pop edi
0x1002ea55: ret
0x1002ea56: test al,2
0x1002ea58: je short 1002EA68h
0x1002ea5a: push 10100FB0h
0x1002ea5f: push edi
0x1002ea60: call 100230F0h
0x1002ea65: add esp,8
0x1002ea68: push esi
0x1002ea69: mov esi,[esp+10h]
0x1002ea6d: test esi,esi
0x1002ea6f: je short 1002EA96h
0x1002ea71: test dword ptr [esi+8],800h
0x1002ea78: je short 1002EA96h
0x1002ea7a: push 10100F98h
0x1002ea7f: push edi
0x1002ea80: call 100230F0h
0x1002ea85: mov ecx,[esp+1Ch]
0x1002ea89: push ecx
0x1002ea8a: push edi
0x1002ea8b: call 1002E050h
0x1002ea90: add esp,10h
0x1002ea93: pop esi
0x1002ea94: pop edi
0x1002ea95: ret
0x1002ea96: push ebx
0x1002ea97: mov ebx,[esp+18h]
0x1002ea9b: cmp ebx,1
0x1002ea9e: je short 1002EAB9h
0x1002eaa0: push 10100F7Ch
0x1002eaa5: push edi
0x1002eaa6: call 100230F0h
0x1002eaab: push ebx
0x1002eaac: push edi
0x1002eaad: call 1002E050h
0x1002eab2: add esp,10h
0x1002eab5: pop ebx
0x1002eab6: pop esi
0x1002eab7: pop edi
0x1002eab8: ret
0x1002eab9: push 1
0x1002eabb: lea edx,[esp+14h]
0x1002eabf: push edx
0x1002eac0: push edi
0x1002eac1: call 1002CF30h
0x1002eac6: push 0
0x1002eac8: push edi
0x1002eac9: call 1002E050h
0x1002eace: add esp,14h
0x1002ead1: test eax,eax
0x1002ead3: jne near ptr 1002EC09h
0x1002ead9: movzx ebx,byte ptr [esp+10h]
0x1002eade: cmp ebx,4
0x1002eae1: jl short 1002EAF5h
0x1002eae3: push 10100F68h
0x1002eae8: push edi
0x1002eae9: call 100230F0h
0x1002eaee: add esp,8
0x1002eaf1: pop ebx
0x1002eaf2: pop esi
0x1002eaf3: pop edi
0x1002eaf4: ret
0x1002eaf5: test esi,esi
0x1002eaf7: je near ptr 1002EBFEh
0x1002eafd: test byte ptr [esi+8],1
```

```
0x1002eb01: je short 1002EB3Eh
0x1002eb03: mov eax,[esi+28h]
0x1002eb06: cmp eax,0AFC8h
0x1002eb0b: jl short 1002EB14h
0x1002eb0d: cmp eax,0B3B0h
0x1002eb12: jle short 1002EB3Eh
0x1002eb14: push 10100CBCh
0x1002eb19: push edi
0x1002eb1a: call 100230F0h
0x1002eb1f: mov eax,[esi+28h]
0x1002eb22: add esp,8
0x1002eb25: push eax
0x1002eb26: push 10100F48h
0x1002eb2b: call dword ptr ds:[100FE4C4h]
0x1002eb31: add eax,40h
0x1002eb34: push eax
0x1002eb35: call dword ptr ds:[100FE4C8h]
0x1002eb3b: add esp,0Ch
0x1002eb3e: test byte ptr [esi+8],4
0x1002eb42: je near ptr 1002EBFEh
0x1002eb48: mov eax,[esi+84h]
0x1002eb4e: cmp eax,763Eh
0x1002eb53: jl near ptr 1002EBF0h
0x1002eb59: cmp eax,7E0Eh
0x1002eb5e: jg near ptr 1002EBF0h
0x1002eb64: mov eax,[esi+88h]
0x1002eb6a: cmp eax,7C9Ch
0x1002eb6f: jl short 1002EBF0h
0x1002eb71: cmp eax,846Ch
0x1002eb76: jg short 1002EBF0h
0x1002eb78: mov eax,[esi+8Ch]
0x1002eb7e: cmp eax,0F618h
0x1002eb83: jl short 1002EBF0h
0x1002eb85: cmp eax,0FDE8h
0x1002eb8a: jg short 1002EBF0h
0x1002eb8c: mov eax,[esi+90h]
0x1002eb92: cmp eax,7D00h
0x1002eb97: jl short 1002EBF0h
0x1002eb99: cmp eax,84D0h
0x1002eb9e: jg short 1002EBF0h
0x1002eba0: mov eax,[esi+94h]
0x1002eba6: cmp eax,7148h
0x1002ebab: jl short 1002EBF0h
0x1002ebad: cmp eax,7918h
0x1002ebb2: jg short 1002EBF0h
0x1002ebb4: mov eax,[esi+98h]
0x1002ebba: cmp eax,0E678h
0x1002ebbf: jl short 1002EBF0h
0x1002ebc1: cmp eax,0EE48h
0x1002ebc6: jg short 1002EBF0h
0x1002ebc8: mov eax,[esi+9Ch]
0x1002ebce: cmp eax,36B0h
0x1002ebd3: jl short 1002EBF0h
0x1002ebd5: cmp eax,3E80h
0x1002ebda: jg short 1002EBF0h
0x1002ebdc: mov eax,[esi+0A0h]
0x1002ebe2: cmp eax,1388h
0x1002ebe7: jl short 1002EBF0h
0x1002ebe9: cmp eax,1B58h
0x1002ebee: jle short 1002EBFEh
0x1002ebf0: push 10100E8Ch
0x1002ebf5: push edi
0x1002ebf6: call 100230F0h
0x1002ebfb: add esp,8
0x1002ebfe: push ebx
0x1002ebff: push esi
0x1002ec00: push edi
```

```
0x1002ec01: call 1002BF00h
0x1002ec06: add esp,0Ch
0x1002ec09: pop ebx
0x1002ec0a: pop esi
0x1002ec0b: pop edi
0x1002ec0c: ret
```

`sub_1002ea20` is called by png_read_info, png_read_end. It delegates to png_warning, png_error, png_set_sRGB_gAMA_and_cHRM (5 callees).

12.760 sub_1002ec10 at 0x1002ec10 (220 instructions)

```
0x1002ec10: sub esp,5Ch
0x1002ec13: mov eax,ds:[10136220h]
0x1002ec18: xor eax,esp
0x1002ec1a: mov [esp+58h],eax
0x1002ec1e: push esi
0x1002ec1f: mov esi,[esp+64h]
0x1002ec23: mov eax,[esi+6Ch]
0x1002ec26: test al,1
0x1002ec28: push edi
0x1002ec29: mov edi,[esp+6Ch]
0x1002ec2d: mov [esp+8],edi
0x1002ec31: jne short 1002EC3Eh
0x1002ec33: push 101010FCh
0x1002ec38: push esi
0x1002ec39: call 100231E0h
0x1002ec3e: test al,4
0x1002ec40: je short 1002EC6Ch
0x1002ec42: push 101010E4h
0x1002ec47: push esi
0x1002ec48: call 100230F0h
0x1002ec4d: mov eax,[esp+78h]
0x1002ec51: push eax
0x1002ec52: push esi
0x1002ec53: call 1002E050h
0x1002ec58: add esp,10h
0x1002ec5b: pop edi
0x1002ec5c: pop esi
0x1002ec5d: mov ecx,[esp+58h]
0x1002ec61: xor ecx,esp
0x1002ec63: call 10022326h
0x1002ec68: add esp,5Ch
0x1002ec6b: ret
0x1002ec6c: test al,2
0x1002ec6e: je short 1002EC7Eh
0x1002ec70: push 101010CCh
0x1002ec75: push esi
0x1002ec76: call 100230F0h
0x1002ec7b: add esp,8
0x1002ec7e: push ebx
0x1002ec7f: xor ebx,ebx
0x1002ec81: cmp edi,ebx
0x1002ec83: je short 1002ECB9h
0x1002ec85: test dword ptr [edi+8],1000h
0x1002ec8c: je short 1002ECB9h
0x1002ec8e: push 101010B4h
0x1002ec93: push esi
0x1002ec94: call 100230F0h
0x1002ec99: mov ecx,[esp+7Ch]
0x1002ec9d: push ecx
0x1002ec9e: push esi
0x1002ec9f: call 1002E050h
0x1002eca4: add esp,10h
0x1002eca7: pop ebx
0x1002eca8: pop edi
```

```
0x1002eca9: pop esi
0x1002ecaa: mov ecx,[esp+58h]
0x1002ecae: xor ecx,esp
0x1002ecb0: call 10022326h
0x1002ecb5: add esp,5Ch
0x1002ecb8: ret
0x1002ecb9: mov edx,[esi+284h]
0x1002ecbf: push ebp
0x1002ecc0: push edx
0x1002ecc1: push esi
0x1002ecc2: call 1002CD70h
0x1002ecc7: mov ebp,[esp+80h]
0x1002ecce: lea eax,[ebp+1]
0x1002ecd1: push eax
0x1002ecd2: push esi
0x1002ecd3: call 1002CD20h
0x1002ecd8: push ebp
0x1002ecd9: push eax
0x1002ecda: push esi
0x1002ecdb: mov [esi+284h],eax
0x1002ece1: call 1002CF30h
0x1002ece6: push ebx
0x1002ece7: push esi
0x1002ece8: call 1002E050h
0x1002eced: add esp,24h
0x1002ecf0: test eax,eax
0x1002ecf2: je short 1002ED1Dh
0x1002ecf4: mov ecx,[esi+284h]
0x1002ecfa: push ecx
0x1002ecfb: push esi
0x1002ecfc: call 1002CD70h
0x1002ed01: add esp,8
0x1002ed04: pop ebp
0x1002ed05: mov [esi+284h],ebx
0x1002ed0b: pop ebx
0x1002ed0c: pop edi
0x1002ed0d: pop esi
0x1002ed0e: mov ecx,[esp+58h]
0x1002ed12: xor ecx,esp
0x1002ed14: call 10022326h
0x1002ed19: add esp,5Ch
0x1002ed1c: ret
0x1002ed1d: mov edx,[esi+284h]
0x1002ed23: mov [edx+ebp],bl
0x1002ed26: mov eax,[esi+284h]
0x1002ed2c: mov edi,eax
0x1002ed2e: cmp [edi],bl
0x1002ed30: je short 1002ED39h
0x1002ed32: add edi,1
0x1002ed35: cmp [edi],bl
0x1002ed37: jne short 1002ED32h
0x1002ed39: add edi,1
0x1002ed3c: lea ecx,[eax+ebp-1]
0x1002ed40: cmp edi,ecx
0x1002ed42: jb short 1002ED5Bh
0x1002ed44: push eax
0x1002ed45: push esi
0x1002ed46: call 1002CD70h
0x1002ed4b: mov [esi+284h],ebx
0x1002ed51: push 1010109Ch
0x1002ed56: jmp near ptr 1002EE6Bh
0x1002ed5b: mov al,[edi]
0x1002ed5d: add edi,1
0x1002ed60: test al,al
0x1002ed62: je short 1002ED74h
0x1002ed64: push 1010106Ch
0x1002ed69: push esi
```

```
0x1002ed6a: call 100230F0h
0x1002ed6f: add esp,8
0x1002ed72: xor al,al
0x1002ed74: sub edi,[esi+284h]
0x1002ed7a: lea edx,[esp+14h]
0x1002ed7e: push edx
0x1002ed7f: push edi
0x1002ed80: movzx ebx,al
0x1002ed83: push ebp
0x1002ed84: push ebx
0x1002ed85: push esi
0x1002ed86: call 1002D150h
0x1002ed8b: mov eax,[esp+28h]
0x1002ed8f: mov ebp,eax
0x1002ed91: add esp,14h
0x1002ed94: sub ebp,edi
0x1002ed96: cmp edi,eax
0x1002ed98: ja near ptr 1002EE4Fh
0x1002ed9e: cmp ebp,4
0x1002eda1: jb near ptr 1002EE4Fh
0x1002eda7: mov ecx,[esi+284h]
0x1002edad: lea eax,[ecx+edi]
0x1002edb0: movzx edi,byte ptr [eax+2]
0x1002edb4: xor edx,edx
0x1002edb6: mov dh,[eax]
0x1002edb8: mov dl,[eax+1]
0x1002edbb: shl edx,8
0x1002edbe: or edx,edi
0x1002edc0: movzx edi,byte ptr [eax+3]
0x1002edc4: shl edx,8
0x1002edc7: or edi,edx
0x1002edc9: cmp edi,ebp
0x1002edcb: jbe short 1002EE13h
0x1002edcd: push ecx
0x1002edce: push esi
0x1002edcf: call 1002CD70h
0x1002edd4: push ebp
0x1002edd5: push edi
0x1002edd6: push 10101028h
0x1002eddb: lea eax,[esp+2Ch]
0x1002eddf: push 50h
0x1002ede1: push eax
0x1002ede2: mov dword ptr [esi+284h],0
0x1002edec: call dword ptr ds:[100FE4C0h]
0x1002edf2: lea ecx,[esp+34h]
0x1002edf6: push ecx
0x1002edf7: push esi
0x1002edf8: call 100230F0h
0x1002edfd: add esp,24h
0x1002ee00: pop ebp
0x1002ee01: pop ebx
0x1002ee02: pop edi
0x1002ee03: pop esi
0x1002ee04: mov ecx,[esp+58h]
0x1002ee08: xor ecx,esp
0x1002ee0a: call 10022326h
0x1002ee0f: add esp,5Ch
0x1002ee12: ret
0x1002ee13: mov edx,[esp+10h]
0x1002ee17: push edi
0x1002ee18: push eax
0x1002ee19: push ebx
0x1002ee1a: push ecx
0x1002ee1b: push edx
0x1002ee1c: push esi
0x1002ee1d: call 1002BF60h
0x1002ee22: mov eax,[esi+284h]
```

```
0x1002ee28: push eax
0x1002ee29: push esi
0x1002ee2a: call 1002CD70h
0x1002ee2f: add esp,20h
0x1002ee32: pop ebp
0x1002ee33: pop ebx
0x1002ee34: pop edi
0x1002ee35: mov dword ptr [esi+284h],0
0x1002ee3f: pop esi
0x1002ee40: mov ecx,[esp+58h]
0x1002ee44: xor ecx,esp
0x1002ee46: call 10022326h
0x1002ee4b: add esp,5Ch
0x1002ee4e: ret
0x1002ee4f: mov ecx,[esi+284h]
0x1002ee55: push ecx
0x1002ee56: push esi
0x1002ee57: call 1002CD70h
0x1002ee5c: mov dword ptr [esi+284h],0
0x1002ee66: push 10100FFCh
0x1002ee6b: push esi
0x1002ee6c: call 100230F0h
0x1002ee71: mov ecx,[esp+78h]
0x1002ee75: add esp,10h
0x1002ee78: pop ebp
0x1002ee79: pop ebx
0x1002ee7a: pop edi
0x1002ee7b: pop esi
0x1002ee7c: xor ecx,esp
0x1002ee7e: call 10022326h
0x1002ee83: add esp,5Ch
0x1002ee86: ret
```

`sub_1002ec10` is called by png_read_info, png_read_end. It works with the text "Profile size field missing from iCCP chunk", "Ignoring iCCP chunk with declared size = %u and actual length = %u", "Ignoring nonzero compression type in iCCP chunk". It delegates to sub_10022326, png_warning, png_error (9 callees).

12.761 sub_1002ee90 at 0x1002ee90 (255 instructions)

```
0x1002ee90: sub esp,10h
0x1002ee93: push ebx
0x1002ee94: push edi
0x1002ee95: mov edi,[esp+1Ch]
0x1002ee99: mov eax,[edi+260h]
0x1002ee9f: xor ebx,ebx
0x1002eea1: cmp eax,ebx
0x1002eea3: je short 1002EEEBh
0x1002eea5: cmp eax,1
0x1002eea8: jne short 1002EEBEh
0x1002eeaa: mov eax,[esp+24h]
0x1002eeae: push eax
0x1002eeaf: push edi
0x1002eeb0: call 1002E050h
0x1002eeb5: add esp,8
0x1002eeb8: pop edi
0x1002eeb9: pop ebx
0x1002eeba: add esp,10h
0x1002eebd: ret
0x1002eebe: add eax,0FFFFFFFFh
0x1002eec1: cmp eax,1
0x1002eec4: mov [edi+260h],eax
0x1002eeca: jne short 1002EEEBh
0x1002eecc: push 101011B8h
0x1002eed1: push edi
0x1002eed2: call 100230F0h
0x1002eed7: mov ecx,[esp+2Ch]
```

```
0x1002eedb: push ecx
0x1002eedc: push edi
0x1002eedd: call 1002E050h
0x1002eee2: add esp,10h
0x1002eee5: pop edi
0x1002eee6: pop ebx
0x1002eee7: add esp,10h
0x1002eeea: ret
0x1002eeeb: mov eax,[edi+6Ch]
0x1002eeee: test al,1
0x1002eef0: jne short 1002EEFDh
0x1002eef2: push 1010119Ch
0x1002eef7: push edi
0x1002eef8: call 100231E0h
0x1002eefd: test al,4
0x1002eeff: je short 1002EF20h
0x1002ef01: push 10101184h
0x1002ef06: push edi
0x1002ef07: call 100230F0h
0x1002ef0c: mov edx,[esp+2Ch]
0x1002ef10: push edx
0x1002ef11: push edi
0x1002ef12: call 1002E050h
0x1002ef17: add esp,10h
0x1002ef1a: pop edi
0x1002ef1b: pop ebx
0x1002ef1c: add esp,10h
0x1002ef1f: ret
0x1002ef20: mov eax,[edi+284h]
0x1002ef26: push ebp
0x1002ef27: push eax
0x1002ef28: push edi
0x1002ef29: call 1002CD70h
0x1002ef2e: mov ebp,[esp+30h]
0x1002ef32: lea ecx,[ebp+1]
0x1002ef35: push ecx
0x1002ef36: push edi
0x1002ef37: call 1002CD20h
0x1002ef3c: push ebp
0x1002ef3d: push eax
0x1002ef3e: push edi
0x1002ef3f: mov [edi+284h],eax
0x1002ef45: call 1002CF30h
0x1002ef4a: push ebx
0x1002ef4b: push edi
0x1002ef4c: call 1002E050h
0x1002ef51: add esp,24h
0x1002ef54: test eax,eax
0x1002ef56: je short 1002EF75h
0x1002ef58: mov edx,[edi+284h]
0x1002ef5e: push edx
0x1002ef5f: push edi
0x1002ef60: call 1002CD70h
0x1002ef65: add esp,8
0x1002ef68: pop ebp
0x1002ef69: mov [edi+284h],ebx
0x1002ef6f: pop edi
0x1002ef70: pop ebx
0x1002ef71: add esp,10h
0x1002ef74: ret
0x1002ef75: mov eax,[edi+284h]
0x1002ef7b: mov [eax+ebp],bl
0x1002ef7e: mov ecx,[edi+284h]
0x1002ef84: push esi
0x1002ef85: mov esi,ecx
0x1002ef87: cmp [esi],bl
0x1002ef89: je short 1002EF97h
```

```
0x1002ef8b: jmp short 1002EF90h
0x1002ef90: add esi,1
0x1002ef93: cmp [esi],bl
0x1002ef95: jne short 1002EF90h
0x1002ef97: add esi,1
0x1002ef9a: lea edx,[ecx+ebp-2]
0x1002ef9e: cmp esi,edx
0x1002efa0: jbe short 1002EFC5h
0x1002efa2: push ecx
0x1002efa3: push edi
0x1002efa4: call 1002CD70h
0x1002efa9: push 1010116Ch
0x1002efae: push edi
0x1002efaf: mov [edi+284h],ebx
0x1002efb5: call 100230F0h
0x1002efba: add esp,10h
0x1002efbd: pop esi
0x1002efbe: pop ebp
0x1002efbf: pop edi
0x1002efc0: pop ebx
0x1002efc1: add esp,10h
0x1002efc4: ret
0x1002efc5: mov al,[esi]
0x1002efc7: xor ebx,ebx
0x1002efc9: add esi,1
0x1002efcc: cmp al,8
0x1002efce: setne bl
0x1002efd1: mov [esp+14h],al
0x1002efd5: mov eax,ecx
0x1002efd7: sub eax,esi
0x1002efd9: add eax,ebp
0x1002efdb: xor edx,edx
0x1002efdd: lea ebx,[ebx*4+6]
0x1002efe4: div ebx
0x1002efe6: test edx,edx
0x1002efe8: je short 1002F011h
0x1002efea: push ecx
0x1002efeb: push edi
0x1002efec: call 1002CD70h
0x1002eff1: push 10101150h
0x1002eff6: push edi
0x1002eff7: mov dword ptr [edi+284h],0
0x1002f001: call 100230F0h
0x1002f006: add esp,10h
0x1002f009: pop esi
0x1002f00a: pop ebp
0x1002f00b: pop edi
0x1002f00c: pop ebx
0x1002f00d: add esp,10h
0x1002f010: ret
0x1002f011: cmp eax,19999999h
0x1002f016: jbe short 1002F02Eh
0x1002f018: push 1010113Ch
0x1002f01d: push edi
0x1002f01e: call 100230F0h
0x1002f023: add esp,8
0x1002f026: pop esi
0x1002f027: pop ebp
0x1002f028: pop edi
0x1002f029: pop ebx
0x1002f02a: add esp,10h
0x1002f02d: ret
0x1002f02e: mov [esp+1Ch],eax
0x1002f032: lea eax,[eax+eax*4]
0x1002f035: add eax,eax
0x1002f037: push eax
0x1002f038: push edi
```

```
0x1002f039: call 1002CDA0h
0x1002f03e: add esp,8
0x1002f041: test eax,eax
0x1002f043: mov [esp+18h],eax
0x1002f047: jne short 1002F05Fh
0x1002f049: push 10101118h
0x1002f04e: push edi
0x1002f04f: call 100230F0h
0x1002f054: add esp,8
0x1002f057: pop esi
0x1002f058: pop ebp
0x1002f059: pop edi
0x1002f05a: pop ebx
0x1002f05b: add esp,10h
0x1002f05e: ret
0x1002f05f: xor ecx,ecx
0x1002f061: cmp [esp+1Ch],ecx
0x1002f065: jle near ptr 1002F120h
0x1002f06b: xor edx,edx
0x1002f06d: jmp short 1002F074h
0x1002f070: mov eax,[esp+18h]
0x1002f074: movzx ebx,byte ptr [esi]
0x1002f077: add eax,edx
0x1002f079: cmp byte ptr [esp+14h],8
0x1002f07e: jne short 1002F0A2h
0x1002f080: add esi,1
0x1002f083: mov [eax],bx
0x1002f086: movzx ebx,byte ptr [esi]
0x1002f089: add esi,1
0x1002f08c: mov [eax+2],bx
0x1002f090: movzx ebx,byte ptr [esi]
0x1002f093: add esi,1
0x1002f096: mov [eax+4],bx
0x1002f09a: movzx ebx,byte ptr [esi]
0x1002f09d: add esi,1
0x1002f0a0: jmp short 1002F0F5h
0x1002f0a2: movzx bp,byte ptr [esi+1]
0x1002f0a7: shl bx,8
0x1002f0ab: add bx,bp
0x1002f0ae: mov [eax],bx
0x1002f0b1: movzx ebx,byte ptr [esi+2]
0x1002f0b5: movzx bp,byte ptr [esi+3]
0x1002f0ba: add esi,2
0x1002f0bd: shl bx,8
0x1002f0c1: add bx,bp
0x1002f0c4: mov [eax+2],bx
0x1002f0c8: movzx ebx,byte ptr [esi+2]
0x1002f0cc: movzx bp,byte ptr [esi+3]
0x1002f0d1: add esi,2
0x1002f0d4: shl bx,8
0x1002f0d8: add bx,bp
0x1002f0db: mov [eax+4],bx
0x1002f0df: movzx ebx,byte ptr [esi+2]
0x1002f0e3: movzx bp,byte ptr [esi+3]
0x1002f0e8: add esi,2
0x1002f0eb: shl bx,8
0x1002f0ef: add bx,bp
0x1002f0f2: add esi,2
0x1002f0f5: mov [eax+6],bx
0x1002f0f9: movzx bx,byte ptr [esi]
0x1002f0fd: movzx bp,byte ptr [esi+1]
0x1002f102: shl bx,8
0x1002f106: add bx,bp
0x1002f109: add ecx,1
0x1002f10c: mov [eax+8],bx
0x1002f110: add esi,2
0x1002f113: add edx,0Ah
```

```
0x1002f116: cmp ecx,[esp+1Ch]
0x1002f11a: jl near ptr 1002F070h
0x1002f120: mov eax,[esp+28h]
0x1002f124: mov ecx,[edi+284h]
0x1002f12a: push 1
0x1002f12c: lea edx,[esp+14h]
0x1002f130: push edx
0x1002f131: push eax
0x1002f132: push edi
0x1002f133: mov [esp+20h],ecx
0x1002f137: call 1002C430h
0x1002f13c: mov ecx,[edi+284h]
0x1002f142: push ecx
0x1002f143: push edi
0x1002f144: call 1002CD70h
0x1002f149: mov edx,[esp+30h]
0x1002f14d: push edx
0x1002f14e: push edi
0x1002f14f: mov dword ptr [edi+284h],0
0x1002f159: call 1002CD70h
0x1002f15e: add esp,20h
0x1002f161: pop esi
0x1002f162: pop ebp
0x1002f163: pop edi
0x1002f164: pop ebx
0x1002f165: add esp,10h
0x1002f168: ret
```

`sub_1002ee90` is called by png_read_info, png_read_end. It delegates to png_warning, png_error, png_set_sPLT (8 callees).

12.762 sub_1002f170 at 0x1002f170 (181 instructions)

```
0x1002f170: sub esp,110h
0x1002f176: mov eax,ds:[10136220h]
0x1002f17b: xor eax,esp
0x1002f17d: mov [esp+10Ch],eax
0x1002f184: push ebp
0x1002f185: mov ebp,[esp+11Ch]
0x1002f18c: push esi
0x1002f18d: mov esi,[esp+11Ch]
0x1002f194: mov ecx,[esi+6Ch]
0x1002f197: test cl,1
0x1002f19a: jne short 1002F1A7h
0x1002f19c: push 10101288h
0x1002f1a1: push esi
0x1002f1a2: call 100231E0h
0x1002f1a7: test cl,4
0x1002f1aa: je short 1002F1DFh
0x1002f1ac: push 10101270h
0x1002f1b1: push esi
0x1002f1b2: call 100230F0h
0x1002f1b7: mov eax,[esp+12Ch]
0x1002f1be: push eax
0x1002f1bf: push esi
0x1002f1c0: call 1002E050h
0x1002f1c5: add esp,10h
0x1002f1c8: pop esi
0x1002f1c9: pop ebp
0x1002f1ca: mov ecx,[esp+10Ch]
0x1002f1d1: xor ecx,esp
0x1002f1d3: call 10022326h
0x1002f1d8: add esp,110h
0x1002f1de: ret
0x1002f1df: test ebp,ebp
0x1002f1e1: je short 1002F21Ch
```

```
0x1002f1e3: test byte ptr [ebp+8],10h
0x1002f1e7: je short 1002F21Ch
0x1002f1e9: push 10101258h
0x1002f1ee: push esi
0x1002f1ef: call 100230F0h
0x1002f1f4: mov ecx,[esp+12Ch]
0x1002f1fb: push ecx
0x1002f1fc: push esi
0x1002f1fd: call 1002E050h
0x1002f202: add esp,10h
0x1002f205: pop esi
0x1002f206: pop ebp
0x1002f207: mov ecx,[esp+10Ch]
0x1002f20e: xor ecx,esp
0x1002f210: call 10022326h
0x1002f215: add esp,110h
0x1002f21b: ret
0x1002f21c: mov al,[esi+126h]
0x1002f222: test al,al
0x1002f224: push edi
0x1002f225: jne short 1002F27Dh
0x1002f227: mov edi,[esp+128h]
0x1002f22e: cmp edi,2
0x1002f231: je short 1002F244h
0x1002f233: push 1010123Ch
0x1002f238: push esi
0x1002f239: call 100230F0h
0x1002f23e: push edi
0x1002f23f: jmp near ptr 1002F3D0h
0x1002f244: push 2
0x1002f246: lea edx,[esp+10h]
0x1002f24a: push edx
0x1002f24b: push esi
0x1002f24c: call 1002CF30h
0x1002f251: mov ax,[esp+18h]
0x1002f256: movzx cx,al
0x1002f25a: movzx dx,ah
0x1002f25e: shl cx,8
0x1002f262: add esp,0Ch
0x1002f265: add cx,dx
0x1002f268: mov word ptr [esi+11Ah],1
0x1002f271: mov [esi+194h],cx
0x1002f278: jmp near ptr 1002F357h
0x1002f27d: cmp al,2
0x1002f27f: jne short 1002F2EFh
0x1002f281: mov edi,[esp+128h]
0x1002f288: cmp edi,6
0x1002f28b: jne short 1002F233h
0x1002f28d: push edi
0x1002f28e: lea eax,[esp+14h]
0x1002f292: push eax
0x1002f293: push esi
0x1002f294: call 1002CF30h
0x1002f299: movzx ecx,byte ptr [esp+1Ch]
0x1002f29e: movzx edx,byte ptr [esp+1Dh]
0x1002f2a3: movzx eax,byte ptr [esp+1Eh]
0x1002f2a8: shl cx,8
0x1002f2ac: add cx,dx
0x1002f2af: movzx edx,byte ptr [esp+20h]
0x1002f2b4: mov [esi+18Eh],cx
0x1002f2bb: movzx ecx,byte ptr [esp+1Fh]
0x1002f2c0: shl ax,8
0x1002f2c4: add ax,cx
0x1002f2c7: mov [esi+190h],ax
0x1002f2ce: movzx eax,byte ptr [esp+21h]
0x1002f2d3: shl dx,8
0x1002f2d7: add esp,0Ch
```

```
0x1002f2da: add dx,ax
0x1002f2dd: mov word ptr [esi+11Ah],1
0x1002f2e6: mov [esi+192h],dx
0x1002f2ed: jmp short 1002F357h
0x1002f2ef: cmp al,3
0x1002f2f1: jne near ptr 1002F3BDh
0x1002f2f7: test cl,2
0x1002f2fa: jne short 1002F30Ah
0x1002f2fc: push 10101220h
0x1002f301: push esi
0x1002f302: call 100230F0h
0x1002f307: add esp,8
0x1002f30a: movzx ecx,word ptr [esi+118h]
0x1002f311: mov edi,[esp+128h]
0x1002f318: cmp edi,ecx
0x1002f31a: ja near ptr 1002F233h
0x1002f320: cmp edi,100h
0x1002f326: ja near ptr 1002F233h
0x1002f32c: test edi,edi
0x1002f32e: jne short 1002F341h
0x1002f330: push 10101208h
0x1002f335: push esi
0x1002f336: call 100230F0h
0x1002f33b: push edi
0x1002f33c: jmp near ptr 1002F3D0h
0x1002f341: push edi
0x1002f342: lea edx,[esp+1Ch]
0x1002f346: push edx
0x1002f347: push esi
0x1002f348: call 1002CF30h
0x1002f34d: add esp,0Ch
0x1002f350: mov [esi+11Ah],di
0x1002f357: push 0
0x1002f359: push esi
0x1002f35a: call 1002E050h
0x1002f35f: add esp,8
0x1002f362: test eax,eax
0x1002f364: je short 1002F387h
0x1002f366: pop edi
0x1002f367: mov word ptr [esi+11Ah],0
0x1002f370: pop esi
0x1002f371: pop ebp
0x1002f372: mov ecx,[esp+10Ch]
0x1002f379: xor ecx,esp
0x1002f37b: call 10022326h
0x1002f380: add esp,110h
0x1002f386: ret
0x1002f387: movzx ecx,word ptr [esi+11Ah]
0x1002f38e: lea eax,[esi+18Ch]
0x1002f394: push eax
0x1002f395: push ecx
0x1002f396: lea edx,[esp+20h]
0x1002f39a: push edx
0x1002f39b: push ebp
0x1002f39c: push esi
0x1002f39d: call 1002C340h
0x1002f3a2: add esp,14h
0x1002f3a5: pop edi
0x1002f3a6: pop esi
0x1002f3a7: pop ebp
0x1002f3a8: mov ecx,[esp+10Ch]
0x1002f3af: xor ecx,esp
0x1002f3b1: call 10022326h
0x1002f3b6: add esp,110h
0x1002f3bc: ret
0x1002f3bd: push 101011DCh
0x1002f3c2: push esi
```

```
0x1002f3c3: call 100230F0h
0x1002f3c8: mov eax,[esp+130h]
0x1002f3cf: push eax
0x1002f3d0: push esi
0x1002f3d1: call 1002E050h
0x1002f3d6: mov ecx,[esp+128h]
0x1002f3dd: add esp,10h
0x1002f3e0: pop edi
0x1002f3e1: pop esi
0x1002f3e2: pop ebp
0x1002f3e3: xor ecx,esp
0x1002f3e5: call 10022326h
0x1002f3ea: add esp,110h
0x1002f3f0: ret
```

`sub_1002f170` is called by png_read_info, png_read_end. It delegates to sub_10022326, png_warning, png_error (6 callees).

12.763 sub_1002f400 at 0x1002f400 (183 instructions)

```
0x1002f400: sub esp,0Ch
0x1002f403: mov eax,ds:[10136220h]
0x1002f408: xor eax,esp
0x1002f40a: mov [esp+8],eax
0x1002f40e: push ebx
0x1002f40f: mov ebx,[esp+18h]
0x1002f413: push esi
0x1002f414: mov esi,[esp+18h]
0x1002f418: mov ecx,[esi+6Ch]
0x1002f41b: test cl,1
0x1002f41e: jne short 1002F42Bh
0x1002f420: push 10101330h
0x1002f425: push esi
0x1002f426: call 100231E0h
0x1002f42b: test cl,4
0x1002f42e: je short 1002F45Ah
0x1002f430: push 10101318h
0x1002f435: push esi
0x1002f436: call 100230F0h
0x1002f43b: mov eax,[esp+28h]
0x1002f43f: push eax
0x1002f440: push esi
0x1002f441: call 1002E050h
0x1002f446: add esp,10h
0x1002f449: pop esi
0x1002f44a: pop ebx
0x1002f44b: mov ecx,[esp+8]
0x1002f44f: xor ecx,esp
0x1002f451: call 10022326h
0x1002f456: add esp,0Ch
0x1002f459: ret
0x1002f45a: mov al,[esi+126h]
0x1002f460: cmp al,3
0x1002f462: jne short 1002F493h
0x1002f464: test cl,2
0x1002f467: jne short 1002F493h
0x1002f469: push 101012FCh
0x1002f46e: push esi
0x1002f46f: call 100230F0h
0x1002f474: mov ecx,[esp+28h]
0x1002f478: push ecx
0x1002f479: push esi
0x1002f47a: call 1002E050h
0x1002f47f: add esp,10h
0x1002f482: pop esi
0x1002f483: pop ebx
```

```
0x1002f484: mov ecx,[esp+8]
0x1002f488: xor ecx,esp
0x1002f48a: call 10022326h
0x1002f48f: add esp,0Ch
0x1002f492: ret
0x1002f493: test ebx,ebx
0x1002f495: je short 1002F4C7h
0x1002f497: test byte ptr [ebx+8],20h
0x1002f49b: je short 1002F4C7h
0x1002f49d: push 101012E4h
0x1002f4a2: push esi
0x1002f4a3: call 100230F0h
0x1002f4a8: mov edx,[esp+28h]
0x1002f4ac: push edx
0x1002f4ad: push esi
0x1002f4ae: call 1002E050h
0x1002f4b3: add esp,10h
0x1002f4b6: pop esi
0x1002f4b7: pop ebx
0x1002f4b8: mov ecx,[esp+8]
0x1002f4bc: xor ecx,esp
0x1002f4be: call 10022326h
0x1002f4c3: add esp,0Ch
0x1002f4c6: ret
0x1002f4c7: cmp al,3
0x1002f4c9: jne short 1002F4D2h
0x1002f4cb: mov eax,1
0x1002f4d0: jmp short 1002F4DDh
0x1002f4d2: movzx eax,al
0x1002f4d5: and eax,2
0x1002f4d8: or eax,1
0x1002f4db: add eax,eax
0x1002f4dd: push edi
0x1002f4de: mov edi,[esp+24h]
0x1002f4e2: cmp edi,eax
0x1002f4e4: je short 1002F50Dh
0x1002f4e6: push 101012C8h
0x1002f4eb: push esi
0x1002f4ec: call 100230F0h
0x1002f4f1: push edi
0x1002f4f2: push esi
0x1002f4f3: call 1002E050h
0x1002f4f8: add esp,10h
0x1002f4fb: pop edi
0x1002f4fc: pop esi
0x1002f4fd: pop ebx
0x1002f4fe: mov ecx,[esp+8]
0x1002f502: xor ecx,esp
0x1002f504: call 10022326h
0x1002f509: add esp,0Ch
0x1002f50c: ret
0x1002f50d: push eax
0x1002f50e: lea eax,[esp+10h]
0x1002f512: push eax
0x1002f513: push esi
0x1002f514: call 1002CF30h
0x1002f519: push 0
0x1002f51b: push esi
0x1002f51c: call 1002E050h
0x1002f521: add esp,14h
0x1002f524: test eax,eax
0x1002f526: jne near ptr 1002F637h
0x1002f52c: mov al,[esi+126h]
0x1002f532: cmp al,3
0x1002f534: jne short 1002F5B0h
0x1002f536: test ebx,ebx
0x1002f538: mov al,[esp+0Ch]
```

```
0x1002f53c: mov [esi+138h],al
0x1002f542: je near ptr 1002F626h
0x1002f548: movzx ecx,word ptr [ebx+14h]
0x1002f54c: test cx,cx
0x1002f54f: je near ptr 1002F626h
0x1002f555: movzx dx,al
0x1002f559: cmp dx,cx
0x1002f55c: jb short 1002F57Eh
0x1002f55e: push 101012A4h
0x1002f563: push esi
0x1002f564: call 100230F0h
0x1002f569: add esp,8
0x1002f56c: pop edi
0x1002f56d: pop esi
0x1002f56e: pop ebx
0x1002f56f: mov ecx,[esp+8]
0x1002f573: xor ecx,esp
0x1002f575: call 10022326h
0x1002f57a: add esp,0Ch
0x1002f57d: ret
0x1002f57e: mov ecx,[esi+114h]
0x1002f584: movzx eax,al
0x1002f587: lea eax,[eax+eax*2]
0x1002f58a: movzx edx,byte ptr [eax+ecx]
0x1002f58e: add eax,ecx
0x1002f590: mov [esi+13Ah],dx
0x1002f597: movzx cx,byte ptr [eax+1]
0x1002f59c: mov [esi+13Ch],cx
0x1002f5a3: movzx edx,byte ptr [eax+2]
0x1002f5a7: mov [esi+13Eh],dx
0x1002f5ae: jmp short 1002F626h
0x1002f5b0: test al,2
0x1002f5b2: jne short 1002F5DEh
0x1002f5b4: movzx ax,byte ptr [esp+0Ch]
0x1002f5ba: movzx cx,byte ptr [esp+0Dh]
0x1002f5c0: shl ax,8
0x1002f5c4: add ax,cx
0x1002f5c7: mov [esi+140h],ax
0x1002f5ce: mov [esi+13Ch],ax
0x1002f5d5: mov [esi+13Ah],ax
0x1002f5dc: jmp short 1002F61Fh
0x1002f5de: movzx edx,byte ptr [esp+0Ch]
0x1002f5e3: movzx eax,byte ptr [esp+0Dh]
0x1002f5e8: movzx ecx,byte ptr [esp+0Eh]
0x1002f5ed: shl dx,8
0x1002f5f1: add dx,ax
0x1002f5f4: movzx eax,byte ptr [esp+10h]
0x1002f5f9: mov [esi+13Ah],dx
0x1002f600: movzx edx,byte ptr [esp+0Fh]
0x1002f605: shl cx,8
0x1002f609: add cx,dx
0x1002f60c: mov [esi+13Ch],cx
0x1002f613: movzx ecx,byte ptr [esp+11h]
0x1002f618: shl ax,8
0x1002f61c: add ax,cx
0x1002f61f: mov [esi+13Eh],ax
0x1002f626: lea edx,[esi+138h]
0x1002f62c: push edx
0x1002f62d: push ebx
0x1002f62e: push esi
0x1002f62f: call 1002B4F0h
0x1002f634: add esp,0Ch
0x1002f637: mov ecx,[esp+14h]
0x1002f63b: pop edi
0x1002f63c: pop esi
0x1002f63d: pop ebx
0x1002f63e: xor ecx,esp
```

```
0x1002f640: call 10022326h
0x1002f645: add esp,0Ch
0x1002f648: ret
```

`sub_1002f400` is called by png_read_info, png_read_end. It delegates to sub_10022326, png_warning, png_error (6 callees).

12.764 sub_1002f650 at 0x1002f650 (124 instructions)

```
0x1002f650: sub esp,208h
0x1002f656: mov eax,ds:[10136220h]
0x1002f65b: xor eax,esp
0x1002f65d: mov [esp+204h],eax
0x1002f664: push ebp
0x1002f665: mov ebp,[esp+214h]
0x1002f66c: push edi
0x1002f66d: mov edi,[esp+214h]
0x1002f674: mov eax,[edi+6Ch]
0x1002f677: test al,1
0x1002f679: jne short 1002F686h
0x1002f67b: push 101013B4h
0x1002f680: push edi
0x1002f681: call 100231E0h
0x1002f686: test al,4
0x1002f688: je short 1002F6BDh
0x1002f68a: push 1010139Ch
0x1002f68f: push edi
0x1002f690: call 100230F0h
0x1002f695: mov eax,[esp+224h]
0x1002f69c: push eax
0x1002f69d: push edi
0x1002f69e: call 1002E050h
0x1002f6a3: add esp,10h
0x1002f6a6: pop edi
0x1002f6a7: pop ebp
0x1002f6a8: mov ecx,[esp+204h]
0x1002f6af: xor ecx,esp
0x1002f6b1: call 10022326h
0x1002f6b6: add esp,208h
0x1002f6bc: ret
0x1002f6bd: test al,2
0x1002f6bf: jne short 1002F6F4h
0x1002f6c1: push 10101380h
0x1002f6c6: push edi
0x1002f6c7: call 100230F0h
0x1002f6cc: mov ecx,[esp+224h]
0x1002f6d3: push ecx
0x1002f6d4: push edi
0x1002f6d5: call 1002E050h
0x1002f6da: add esp,10h
0x1002f6dd: pop edi
0x1002f6de: pop ebp
0x1002f6df: mov ecx,[esp+204h]
0x1002f6e6: xor ecx,esp
0x1002f6e8: call 10022326h
0x1002f6ed: add esp,208h
0x1002f6f3: ret
0x1002f6f4: test ebp,ebp
0x1002f6f6: je short 1002F731h
0x1002f6f8: test byte ptr [ebp+8],40h
0x1002f6fc: je short 1002F731h
0x1002f6fe: push 10101368h
0x1002f703: push edi
0x1002f704: call 100230F0h
0x1002f709: mov edx,[esp+224h]
0x1002f710: push edx
```

```
0x1002f711: push edi
0x1002f712: call 1002E050h
0x1002f717: add esp,10h
0x1002f71a: pop edi
0x1002f71b: pop ebp
0x1002f71c: mov ecx,[esp+204h]
0x1002f723: xor ecx,esp
0x1002f725: call 10022326h
0x1002f72a: add esp,208h
0x1002f730: ret
0x1002f731: movzx eax,word ptr [edi+118h]
0x1002f738: push ebx
0x1002f739: push esi
0x1002f73a: mov esi,[esp+224h]
0x1002f741: mov ebx,esi
0x1002f743: shr ebx,1
0x1002f745: cmp ebx,eax
0x1002f747: jne short 1002F7A7h
0x1002f749: cmp ebx,100h
0x1002f74f: ja short 1002F7A7h
0x1002f751: xor esi,esi
0x1002f753: test ebx,ebx
0x1002f755: jbe short 1002F787h
0x1002f757: push 2
0x1002f759: lea ecx,[esp+14h]
0x1002f75d: push ecx
0x1002f75e: push edi
0x1002f75f: call 1002CF30h
0x1002f764: mov ax,[esp+1Ch]
0x1002f769: movzx dx,al
0x1002f76d: movzx ax,ah
0x1002f771: shl dx,8
0x1002f775: add dx,ax
0x1002f778: mov [esp+esi*2+20h],dx
0x1002f77d: add esi,1
0x1002f780: add esp,0Ch
0x1002f783: cmp esi,ebx
0x1002f785: jb short 1002F757h
0x1002f787: push 0
0x1002f789: push edi
0x1002f78a: call 1002E050h
0x1002f78f: add esp,8
0x1002f792: test eax,eax
0x1002f794: jne short 1002F7BCh
0x1002f796: lea ecx,[esp+14h]
0x1002f79a: push ecx
0x1002f79b: push ebp
0x1002f79c: push edi
0x1002f79d: call 1002B720h
0x1002f7a2: add esp,0Ch
0x1002f7a5: jmp short 1002F7BCh
0x1002f7a7: push 1010134Ch
0x1002f7ac: push edi
0x1002f7ad: call 100230F0h
0x1002f7b2: push esi
0x1002f7b3: push edi
0x1002f7b4: call 1002E050h
0x1002f7b9: add esp,10h
0x1002f7bc: mov ecx,[esp+214h]
0x1002f7c3: pop esi
0x1002f7c4: pop ebx
0x1002f7c5: pop edi
0x1002f7c6: pop ebp
0x1002f7c7: xor ecx,esp
0x1002f7c9: call 10022326h
0x1002f7ce: add esp,208h
0x1002f7d4: ret
```

`sub_1002f650` is called by png_read_info, png_read_end. It works with the text "Duplicate hIST chunk", "Missing PLTE before hIST", "Invalid hIST after IDAT". It delegates to sub_10022326, png_warning, png_error (6 callees).

12.765 sub_1002f7e0 at 0x1002f7e0 (116 instructions)

```
0x1002f7e0: sub esp,10h
0x1002f7e3: mov eax,ds:[10136220h]
0x1002f7e8: xor eax,esp
0x1002f7ea: mov [esp+0Ch],eax
0x1002f7ee: push esi
0x1002f7ef: mov esi,[esp+18h]
0x1002f7f3: mov eax,[esi+6Ch]
0x1002f7f6: test al,1
0x1002f7f8: push edi
0x1002f7f9: mov edi,[esp+20h]
0x1002f7fd: jne short 1002F80Ah
0x1002f7ff: push 1010141Ch
0x1002f804: push esi
0x1002f805: call 100231E0h
0x1002f80a: test al,4
0x1002f80c: je short 1002F838h
0x1002f80e: push 10101404h
0x1002f813: push esi
0x1002f814: call 100230F0h
0x1002f819: mov eax,[esp+2Ch]
0x1002f81d: push eax
0x1002f81e: push esi
0x1002f81f: call 1002E050h
0x1002f824: add esp,10h
0x1002f827: pop edi
0x1002f828: pop esi
0x1002f829: mov ecx,[esp+0Ch]
0x1002f82d: xor ecx,esp
0x1002f82f: call 10022326h
0x1002f834: add esp,10h
0x1002f837: ret
0x1002f838: test edi,edi
0x1002f83a: je short 1002F86Ch
0x1002f83c: test byte ptr [edi+8],80h
0x1002f840: je short 1002F86Ch
0x1002f842: push 101013ECh
0x1002f847: push esi
0x1002f848: call 100230F0h
0x1002f84d: mov ecx,[esp+2Ch]
0x1002f851: push ecx
0x1002f852: push esi
0x1002f853: call 1002E050h
0x1002f858: add esp,10h
0x1002f85b: pop edi
0x1002f85c: pop esi
0x1002f85d: mov ecx,[esp+0Ch]
0x1002f861: xor ecx,esp
0x1002f863: call 10022326h
0x1002f868: add esp,10h
0x1002f86b: ret
0x1002f86c: push ebx
0x1002f86d: mov ebx,[esp+28h]
0x1002f871: cmp ebx,9
0x1002f874: je short 1002F89Dh
0x1002f876: push 101013D0h
0x1002f87b: push esi
0x1002f87c: call 100230F0h
0x1002f881: push ebx
0x1002f882: push esi
0x1002f883: call 1002E050h
0x1002f888: add esp,10h
0x1002f88b: pop ebx
```

```
0x1002f88c: pop edi
0x1002f88d: pop esi
0x1002f88e: mov ecx,[esp+0Ch]
0x1002f892: xor ecx,esp
0x1002f894: call 10022326h
0x1002f899: add esp,10h
0x1002f89c: ret
0x1002f89d: push 9
0x1002f89f: lea edx,[esp+10h]
0x1002f8a3: push edx
0x1002f8a4: push esi
0x1002f8a5: call 1002CF30h
0x1002f8aa: push 0
0x1002f8ac: push esi
0x1002f8ad: call 1002E050h
0x1002f8b2: add esp,14h
0x1002f8b5: test eax,eax
0x1002f8b7: jne short 1002F911h
0x1002f8b9: movzx eax,byte ptr [esp+14h]
0x1002f8be: movzx ecx,byte ptr [esp+10h]
0x1002f8c3: movzx edx,byte ptr [esp+11h]
0x1002f8c8: push eax
0x1002f8c9: movzx eax,byte ptr [esp+16h]
0x1002f8ce: shl ecx,8
0x1002f8d1: add ecx,edx
0x1002f8d3: movzx edx,byte ptr [esp+17h]
0x1002f8d8: shl ecx,8
0x1002f8db: add ecx,eax
0x1002f8dd: movzx eax,byte ptr [esp+10h]
0x1002f8e2: shl ecx,8
0x1002f8e5: add ecx,edx
0x1002f8e7: movzx edx,byte ptr [esp+12h]
0x1002f8ec: push ecx
0x1002f8ed: movzx ecx,byte ptr [esp+15h]
0x1002f8f2: shl eax,8
0x1002f8f5: add eax,ecx
0x1002f8f7: movzx ecx,byte ptr [esp+17h]
0x1002f8fc: shl eax,8
0x1002f8ff: add eax,edx
0x1002f901: shl eax,8
0x1002f904: add eax,ecx
0x1002f906: push eax
0x1002f907: push edi
0x1002f908: push esi
0x1002f909: call 1002BDE0h
0x1002f90e: add esp,14h
0x1002f911: mov ecx,[esp+18h]
0x1002f915: pop ebx
0x1002f916: pop edi
0x1002f917: pop esi
0x1002f918: xor ecx,esp
0x1002f91a: call 10022326h
0x1002f91f: add esp,10h
0x1002f922: ret
```

`sub_1002f7e0` is called by png_read_info, png_read_end. It delegates to sub_10022326, png_warning, png_error (6 callees).

12.766 sub_1002f930 at 0x1002f930 (130 instructions)

```
0x1002f930: sub esp,10h
0x1002f933: mov eax,ds:[10136220h]
0x1002f938: xor eax,esp
0x1002f93a: mov [esp+0Ch],eax
0x1002f93e: push ebx
0x1002f93f: mov ebx,[esp+1Ch]
```

```
0x1002f943: push esi
0x1002f944: mov esi,[esp+1Ch]
0x1002f948: mov eax,[esi+6Ch]
0x1002f94b: test al,1
0x1002f94d: jne short 1002F95Ah
0x1002f94f: push 10101484h
0x1002f954: push esi
0x1002f955: call 100231E0h
0x1002f95a: test al,4
0x1002f95c: je short 1002F988h
0x1002f95e: push 1010146Ch
0x1002f963: push esi
0x1002f964: call 100230F0h
0x1002f969: mov eax,[esp+2Ch]
0x1002f96d: push eax
0x1002f96e: push esi
0x1002f96f: call 1002E050h
0x1002f974: add esp,10h
0x1002f977: pop esi
0x1002f978: pop ebx
0x1002f979: mov ecx,[esp+0Ch]
0x1002f97d: xor ecx,esp
0x1002f97f: call 10022326h
0x1002f984: add esp,10h
0x1002f987: ret
0x1002f988: test ebx,ebx
0x1002f98a: je short 1002F9BFh
0x1002f98c: test dword ptr [ebx+8],100h
0x1002f993: je short 1002F9BFh
0x1002f995: push 10101454h
0x1002f99a: push esi
0x1002f99b: call 100230F0h
0x1002f9a0: mov ecx,[esp+2Ch]
0x1002f9a4: push ecx
0x1002f9a5: push esi
0x1002f9a6: call 1002E050h
0x1002f9ab: add esp,10h
0x1002f9ae: pop esi
0x1002f9af: pop ebx
0x1002f9b0: mov ecx,[esp+0Ch]
0x1002f9b4: xor ecx,esp
0x1002f9b6: call 10022326h
0x1002f9bb: add esp,10h
0x1002f9be: ret
0x1002f9bf: push edi
0x1002f9c0: mov edi,[esp+28h]
0x1002f9c4: cmp edi,9
0x1002f9c7: je short 1002F9F0h
0x1002f9c9: push 10101438h
0x1002f9ce: push esi
0x1002f9cf: call 100230F0h
0x1002f9d4: push edi
0x1002f9d5: push esi
0x1002f9d6: call 1002E050h
0x1002f9db: add esp,10h
0x1002f9de: pop edi
0x1002f9df: pop esi
0x1002f9e0: pop ebx
0x1002f9e1: mov ecx,[esp+0Ch]
0x1002f9e5: xor ecx,esp
0x1002f9e7: call 10022326h
0x1002f9ec: add esp,10h
0x1002f9ef: ret
0x1002f9f0: push 9
0x1002f9f2: lea edx,[esp+10h]
0x1002f9f6: push edx
0x1002f9f7: push esi
```

```
0x1002f9f8: call 1002CF30h
0x1002f9fd: push 0
0x1002f9ff: push esi
0x1002fa00: call 1002E050h
0x1002fa05: add esp,14h
0x1002fa08: test eax,eax
0x1002fa0a: jne short 1002FA8Bh
0x1002fa0c: mov al,[esp+0Ch]
0x1002fa10: movzx ecx,al
0x1002fa13: shl ecx,8
0x1002fa16: test al,al
0x1002fa18: jns short 1002FA37h
0x1002fa1a: movzx eax,byte ptr [esp+0Dh]
0x1002fa1f: movzx edx,byte ptr [esp+0Eh]
0x1002fa24: add ecx,eax
0x1002fa26: movzx eax,byte ptr [esp+0Fh]
0x1002fa2b: shl ecx,8
0x1002fa2e: add ecx,edx
0x1002fa30: shl ecx,8
0x1002fa33: add ecx,eax
0x1002fa35: jmp short 1002FA52h
0x1002fa37: movzx edx,byte ptr [esp+0Dh]
0x1002fa3c: movzx eax,byte ptr [esp+0Eh]
0x1002fa41: add ecx,edx
0x1002fa43: movzx edx,byte ptr [esp+0Fh]
0x1002fa48: shl ecx,8
0x1002fa4b: add ecx,eax
0x1002fa4d: shl ecx,8
0x1002fa50: add ecx,edx
0x1002fa52: mov al,[esp+10h]
0x1002fa56: movzx edx,byte ptr [esp+11h]
0x1002fa5b: test al,al
0x1002fa5d: movzx eax,al
0x1002fa60: shl eax,8
0x1002fa63: add eax,edx
0x1002fa65: movzx edx,byte ptr [esp+12h]
0x1002fa6a: shl eax,8
0x1002fa6d: add eax,edx
0x1002fa6f: movzx edx,byte ptr [esp+13h]
0x1002fa74: shl eax,8
0x1002fa77: add eax,edx
0x1002fa79: movzx edx,byte ptr [esp+14h]
0x1002fa7e: push edx
0x1002fa7f: push eax
0x1002fa80: push ecx
0x1002fa81: push ebx
0x1002fa82: push esi
0x1002fa83: call 1002B8C0h
0x1002fa88: add esp,14h
0x1002fa8b: mov ecx,[esp+18h]
0x1002fa8f: pop edi
0x1002fa90: pop esi
0x1002fa91: pop ebx
0x1002fa92: xor ecx,esp
0x1002fa94: call 10022326h
0x1002fa99: add esp,10h
0x1002fa9c: ret
```

`sub_1002f930` is called by png_read_info, png_read_end. It works with the text "Duplicate oFFs chunk", "Invalid oFFs after IDAT", "Missing IHDR before oFFs". It delegates to sub_10022326, png_warning, png_error (6 callees).

12.767 sub_1002faa0 at 0x1002faa0 (295 instructions)

```
0x1002faa0: sub esp,10h
0x1002faa3: push esi
0x1002faa4: mov esi,[esp+18h]
```

```
0x1002faa8: mov eax,[esi+6Ch]
0x1002faab: test al,1
0x1002faad: jne short 1002FABAh
0x1002faaf: push 10101574h
0x1002fab4: push esi
0x1002fab5: call 100231E0h
0x1002faba: test al,4
0x1002fabc: je short 1002FADCh
0x1002fabe: push 1010155Ch
0x1002fac3: push esi
0x1002fac4: call 100230F0h
0x1002fac9: mov eax,[esp+28h]
0x1002facd: push eax
0x1002face: push esi
0x1002facf: call 1002E050h
0x1002fad4: add esp,10h
0x1002fad7: pop esi
0x1002fad8: add esp,10h
0x1002fadb: ret
0x1002fadc: mov eax,[esp+1Ch]
0x1002fae0: push ebp
0x1002fae1: xor ebp,ebp
0x1002fae3: cmp eax,ebp
0x1002fae5: je short 1002FB0Fh
0x1002fae7: test dword ptr [eax+8],400h
0x1002faee: je short 1002FB0Fh
0x1002faf0: push 10101544h
0x1002faf5: push esi
0x1002faf6: call 100230F0h
0x1002fafb: mov ecx,[esp+2Ch]
0x1002faff: push ecx
0x1002fb00: push esi
0x1002fb01: call 1002E050h
0x1002fb06: add esp,10h
0x1002fb09: pop ebp
0x1002fb0a: pop esi
0x1002fb0b: add esp,10h
0x1002fb0e: ret
0x1002fb0f: mov edx,[esi+284h]
0x1002fb15: push edi
0x1002fb16: push edx
0x1002fb17: push esi
0x1002fb18: call 1002CD70h
0x1002fb1d: mov edi,[esp+30h]
0x1002fb21: lea eax,[edi+1]
0x1002fb24: push eax
0x1002fb25: push esi
0x1002fb26: call 1002CDA0h
0x1002fb2b: add esp,10h
0x1002fb2e: cmp eax,ebp
0x1002fb30: mov [esi+284h],eax
0x1002fb36: jne short 1002FB4Dh
0x1002fb38: push 10101528h
0x1002fb3d: push esi
0x1002fb3e: call 100230F0h
0x1002fb43: add esp,8
0x1002fb46: pop edi
0x1002fb47: pop ebp
0x1002fb48: pop esi
0x1002fb49: add esp,10h
0x1002fb4c: ret
0x1002fb4d: push edi
0x1002fb4e: push eax
0x1002fb4f: push esi
0x1002fb50: call 1002CF30h
0x1002fb55: push ebp
0x1002fb56: push esi
```

```
0x1002fb57: call 1002E050h
0x1002fb5c: add esp,14h
0x1002fb5f: test eax,eax
0x1002fb61: je short 1002FB80h
0x1002fb63: mov ecx,[esi+284h]
0x1002fb69: push ecx
0x1002fb6a: push esi
0x1002fb6b: call 1002CD70h
0x1002fb70: add esp,8
0x1002fb73: pop edi
0x1002fb74: mov [esi+284h],ebp
0x1002fb7a: pop ebp
0x1002fb7b: pop esi
0x1002fb7c: add esp,10h
0x1002fb7f: ret
0x1002fb80: mov edx,[esi+284h]
0x1002fb86: mov byte ptr [edi+edx],0
0x1002fb8a: mov ecx,[esi+284h]
0x1002fb90: mov eax,ecx
0x1002fb92: cmp byte ptr [eax],0
0x1002fb95: je short 1002FB9Fh
0x1002fb97: add eax,1
0x1002fb9a: cmp byte ptr [eax],0
0x1002fb9d: jne short 1002FB97h
0x1002fb9f: add ecx,edi
0x1002fba1: lea edx,[eax+0Ch]
0x1002fba4: cmp ecx,edx
0x1002fba6: mov [esp+14h],ecx
0x1002fbaa: ja short 1002FBD4h
0x1002fbac: push 10101514h
0x1002fbb1: push esi
0x1002fbb2: call 100230F0h
0x1002fbb7: mov eax,[esi+284h]
0x1002fbbd: push eax
0x1002fbbe: push esi
0x1002fbbf: call 1002CD70h
0x1002fbc4: add esp,10h
0x1002fbc7: pop edi
0x1002fbc8: mov [esi+284h],ebp
0x1002fbce: pop ebp
0x1002fbcf: pop esi
0x1002fbd0: add esp,10h
0x1002fbd3: ret
0x1002fbd4: mov cl,[eax+1]
0x1002fbd7: movzx edx,byte ptr [eax+2]
0x1002fbdb: test cl,cl
0x1002fbdd: movzx ecx,cl
0x1002fbe0: shl ecx,8
0x1002fbe3: add ecx,edx
0x1002fbe5: movzx edx,byte ptr [eax+3]
0x1002fbe9: shl ecx,8
0x1002fbec: add ecx,edx
0x1002fbee: movzx edx,byte ptr [eax+4]
0x1002fbf2: shl ecx,8
0x1002fbf5: add ecx,edx
0x1002fbf7: mov [esp+10h],ecx
0x1002fbfb: mov cl,[eax+5]
0x1002fbfe: push ebx
0x1002fbff: movzx ebx,cl
0x1002fc02: shl ebx,8
0x1002fc05: test cl,cl
0x1002fc07: jns short 1002FC23h
0x1002fc09: movzx ecx,byte ptr [eax+6]
0x1002fc0d: movzx edx,byte ptr [eax+7]
0x1002fc11: add ebx,ecx
0x1002fc13: movzx ecx,byte ptr [eax+8]
0x1002fc17: shl ebx,8
```

```
0x1002fc1a: add ebx,edx
0x1002fc1c: shl ebx,8
0x1002fc1f: add ebx,ecx
0x1002fc21: jmp short 1002FC3Bh
0x1002fc23: movzx edx,byte ptr [eax+6]
0x1002fc27: movzx ecx,byte ptr [eax+7]
0x1002fc2b: add ebx,edx
0x1002fc2d: movzx edx,byte ptr [eax+8]
0x1002fc31: shl ebx,8
0x1002fc34: add ebx,ecx
0x1002fc36: shl ebx,8
0x1002fc39: add ebx,edx
0x1002fc3b: mov cl,[eax+9]
0x1002fc3e: test cl,cl
0x1002fc40: mov dl,[eax+0Ah]
0x1002fc43: lea edi,[eax+0Bh]
0x1002fc46: mov [esp+13h],cl
0x1002fc4a: mov [esp+24h],dl
0x1002fc4e: mov [esp+1Ch],edi
0x1002fc52: jne short 1002FC5Bh
0x1002fc54: cmp dl,2
0x1002fc57: jne short 1002FC7Dh
0x1002fc59: jmp short 1002FCBDh
0x1002fc5b: cmp cl,1
0x1002fc5e: jne short 1002FC67h
0x1002fc60: cmp dl,3
0x1002fc63: jne short 1002FC7Dh
0x1002fc65: jmp short 1002FCBDh
0x1002fc67: cmp cl,2
0x1002fc6a: jne short 1002FC73h
0x1002fc6c: cmp dl,3
0x1002fc6f: jne short 1002FC7Dh
0x1002fc71: jmp short 1002FCBDh
0x1002fc73: cmp cl,3
0x1002fc76: jne short 1002FCA6h
0x1002fc78: cmp dl,4
0x1002fc7b: je short 1002FCBDh
0x1002fc7d: push 101014E8h
0x1002fc82: push esi
0x1002fc83: call 100230F0h
0x1002fc88: mov eax,[esi+284h]
0x1002fc8e: push eax
0x1002fc8f: push esi
0x1002fc90: call 1002CD70h
0x1002fc95: add esp,10h
0x1002fc98: pop ebx
0x1002fc99: pop edi
0x1002fc9a: mov [esi+284h],ebp
0x1002fca0: pop ebp
0x1002fca1: pop esi
0x1002fca2: add esp,10h
0x1002fca5: ret
0x1002fca6: cmp cl,4
0x1002fca9: jb short 1002FCBDh
0x1002fcab: push 101014BCh
0x1002fcb0: push esi
0x1002fcb1: call 100230F0h
0x1002fcb6: mov dl,[esp+2Ch]
0x1002fcba: add esp,8
0x1002fcbd: cmp byte ptr [edi],0
0x1002fcc0: je short 1002FCCAh
0x1002fcc2: add edi,1
0x1002fcc5: cmp byte ptr [edi],0
0x1002fcc8: jne short 1002FCC2h
0x1002fcca: movzx eax,dl
0x1002fccd: lea ecx,[eax*4]
0x1002fcd4: push ecx
```

```
0x1002fcd5: push esi
0x1002fcd6: mov [esp+2Ch],eax
0x1002fcda: call 1002CDA0h
0x1002fcdf: mov ebp,eax
0x1002fce1: add esp,8
0x1002fce4: test ebp,ebp
0x1002fce6: jne short 1002FD11h
0x1002fce8: mov edx,[esi+284h]
0x1002fcee: push edx
0x1002fcef: push esi
0x1002fcf0: call 1002CD70h
0x1002fcf5: push 101014A0h
0x1002fcfa: push esi
0x1002fcfb: mov [esi+284h],ebp
0x1002fd01: call 100230F0h
0x1002fd06: add esp,10h
0x1002fd09: pop ebx
0x1002fd0a: pop edi
0x1002fd0b: pop ebp
0x1002fd0c: pop esi
0x1002fd0d: add esp,10h
0x1002fd10: ret
0x1002fd11: mov edx,[esp+24h]
0x1002fd15: xor ecx,ecx
0x1002fd17: test edx,edx
0x1002fd19: jle short 1002FD7Bh
0x1002fd1b: jmp short 1002FD20h
0x1002fd20: mov eax,[esp+18h]
0x1002fd24: add edi,1
0x1002fd27: cmp edi,eax
0x1002fd29: mov [ebp+ecx*4],edi
0x1002fd2d: ja short 1002FD3Ch
0x1002fd2f: nop
0x1002fd30: cmp byte ptr [edi],0
0x1002fd33: je short 1002FD70h
0x1002fd35: add edi,1
0x1002fd38: cmp edi,eax
0x1002fd3a: jbe short 1002FD30h
0x1002fd3c: push 10101514h
0x1002fd41: push esi
0x1002fd42: call 100230F0h
0x1002fd47: mov eax,[esi+284h]
0x1002fd4d: push eax
0x1002fd4e: push esi
0x1002fd4f: call 1002CD70h
0x1002fd54: push ebp
0x1002fd55: push esi
0x1002fd56: mov dword ptr [esi+284h],0
0x1002fd60: call 1002CD70h
0x1002fd65: add esp,18h
0x1002fd68: pop ebx
0x1002fd69: pop edi
0x1002fd6a: pop ebp
0x1002fd6b: pop esi
0x1002fd6c: add esp,10h
0x1002fd6f: ret
0x1002fd70: cmp edi,eax
0x1002fd72: ja short 1002FD3Ch
0x1002fd74: add ecx,1
0x1002fd77: cmp ecx,edx
0x1002fd79: jl short 1002FD20h
0x1002fd7b: mov ecx,[esp+1Ch]
0x1002fd7f: mov eax,[esp+14h]
0x1002fd83: push ebp
0x1002fd84: push ecx
0x1002fd85: mov ecx,[esi+284h]
0x1002fd8b: push edx
```

```
0x1002fd8c: movzx edx,byte ptr [esp+1Fh]
0x1002fd91: push edx
0x1002fd92: mov edx,[esp+38h]
0x1002fd96: push ebx
0x1002fd97: push eax
0x1002fd98: push ecx
0x1002fd99: push edx
0x1002fd9a: push esi
0x1002fd9b: call 1002B8F0h
0x1002fda0: mov eax,[esi+284h]
0x1002fda6: push eax
0x1002fda7: push esi
0x1002fda8: call 1002CD70h
0x1002fdad: push ebp
0x1002fdae: push esi
0x1002fdaf: mov dword ptr [esi+284h],0
0x1002fdb9: call 1002CD70h
0x1002fdbe: add esp,34h
0x1002fdc1: pop ebx
0x1002fdc2: pop edi
0x1002fdc3: pop ebp
0x1002fdc4: pop esi
0x1002fdc5: add esp,10h
0x1002fdc8: ret
```

`sub_1002faa0` is called by png_read_info, png_read_end. It works with the text "No memory for pCAL params", "Unrecognized equation type for pCAL chunk", "Invalid pCAL parameters for equation type". It delegates to png_warning, png_error, png_set_pCAL (7 callees).

12.768 sub_1002fdd0 at 0x1002fdd0 (176 instructions)

```
0x1002fdd0: push ecx
0x1002fdd1: push esi
0x1002fdd2: mov esi,[esp+0Ch]
0x1002fdd6: mov eax,[esi+6Ch]
0x1002fdd9: test al,1
0x1002fddb: jne short 1002FDE8h
0x1002fddd: push 10101670h
0x1002fde2: push esi
0x1002fde3: call 100231E0h
0x1002fde8: test al,4
0x1002fdea: je short 1002FE08h
0x1002fdec: push 10101658h
0x1002fdf1: push esi
0x1002fdf2: call 100230F0h
0x1002fdf7: mov eax,[esp+1Ch]
0x1002fdfb: push eax
0x1002fdfc: push esi
0x1002fdfd: call 1002E050h
0x1002fe02: add esp,10h
0x1002fe05: pop esi
0x1002fe06: pop ecx
0x1002fe07: ret
0x1002fe08: mov eax,[esp+10h]
0x1002fe0c: push ebx
0x1002fe0d: xor ebx,ebx
0x1002fe0f: cmp eax,ebx
0x1002fe11: je short 1002FE39h
0x1002fe13: test dword ptr [eax+8],4000h
0x1002fe1a: je short 1002FE39h
0x1002fe1c: push 10101640h
0x1002fe21: push esi
0x1002fe22: call 100230F0h
0x1002fe27: mov ecx,[esp+20h]
0x1002fe2b: push ecx
0x1002fe2c: push esi
```

```
0x1002fe2d: call 1002E050h
0x1002fe32: add esp,10h
0x1002fe35: pop ebx
0x1002fe36: pop esi
0x1002fe37: pop ecx
0x1002fe38: ret
0x1002fe39: push edi
0x1002fe3a: mov edi,[esp+1Ch]
0x1002fe3e: lea edx,[edi+1]
0x1002fe41: push edx
0x1002fe42: push esi
0x1002fe43: call 1002CDA0h
0x1002fe48: add esp,8
0x1002fe4b: cmp eax,ebx
0x1002fe4d: mov [esi+284h],eax
0x1002fe53: jne short 1002FE6Fh
0x1002fe55: push 10101614h
0x1002fe5a: push esi
0x1002fe5b: call 100230F0h
0x1002fe60: push edi
0x1002fe61: push esi
0x1002fe62: call 1002E050h
0x1002fe67: add esp,10h
0x1002fe6a: pop edi
0x1002fe6b: pop ebx
0x1002fe6c: pop esi
0x1002fe6d: pop ecx
0x1002fe6e: ret
0x1002fe6f: push edi
0x1002fe70: push eax
0x1002fe71: push esi
0x1002fe72: call 1002CF30h
0x1002fe77: mov eax,[esi+284h]
0x1002fe7d: push ebx
0x1002fe7e: push esi
0x1002fe7f: mov [edi+eax],bl
0x1002fe82: call 1002E050h
0x1002fe87: add esp,14h
0x1002fe8a: test eax,eax
0x1002fe8c: je short 1002FEA9h
0x1002fe8e: mov ecx,[esi+284h]
0x1002fe94: push ecx
0x1002fe95: push esi
0x1002fe96: call 1002CD70h
0x1002fe9b: add esp,8
0x1002fe9e: pop edi
0x1002fe9f: mov [esi+284h],ebx
0x1002fea5: pop ebx
0x1002fea6: pop esi
0x1002fea7: pop ecx
0x1002fea8: ret
0x1002fea9: mov eax,[esi+284h]
0x1002feaf: mov cl,[eax]
0x1002feb1: cmp cl,1
0x1002feb4: je short 1002FEE1h
0x1002feb6: cmp cl,2
0x1002feb9: je short 1002FEE1h
0x1002febb: push 101015F0h
0x1002fec0: push esi
0x1002fec1: call 100230F0h
0x1002fec6: mov edx,[esi+284h]
0x1002fecc: push edx
0x1002fecd: push esi
0x1002fece: call 1002CD70h
0x1002fed3: add esp,10h
0x1002fed6: pop edi
0x1002fed7: mov [esi+284h],ebx
```

```
0x1002fedd: pop ebx
0x1002fede: pop esi
0x1002fedf: pop ecx
0x1002fee0: ret
0x1002fee1: mov [esp+14h],ebx
0x1002fee5: mov [esp+0Ch],ebx
0x1002fee9: cmp byte ptr [eax+1],2Dh
0x1002feed: push ebp
0x1002feee: je near ptr 1002FF77h
0x1002fef4: lea ecx,[esp+18h]
0x1002fef8: push ecx
0x1002fef9: lea edx,[esp+14h]
0x1002fefd: push edx
0x1002fefe: push edi
0x1002feff: push eax
0x1002ff00: call 10023E50h
0x1002ff05: add esp,10h
0x1002ff08: test eax,eax
0x1002ff0a: je short 1002FF77h
0x1002ff0c: mov eax,[esp+18h]
0x1002ff10: cmp eax,edi
0x1002ff12: jae short 1002FF77h
0x1002ff14: mov ecx,[esi+284h]
0x1002ff1a: mov dl,[ecx+eax]
0x1002ff1d: add eax,1
0x1002ff20: test dl,dl
0x1002ff22: mov [esp+18h],eax
0x1002ff26: jne short 1002FF77h
0x1002ff28: cmp byte ptr [ecx+eax],2Dh
0x1002ff2c: mov ebp,eax
0x1002ff2e: je short 1002FF70h
0x1002ff30: lea eax,[esp+18h]
0x1002ff34: push eax
0x1002ff35: lea edx,[esp+14h]
0x1002ff39: push edx
0x1002ff3a: push edi
0x1002ff3b: push ecx
0x1002ff3c: call 10023E50h
0x1002ff41: add esp,10h
0x1002ff44: test eax,eax
0x1002ff46: je short 1002FF70h
0x1002ff48: cmp [esp+18h],edi
0x1002ff4c: jne short 1002FF70h
0x1002ff4e: mov eax,[esi+284h]
0x1002ff54: lea ecx,[eax+ebp]
0x1002ff57: push ecx
0x1002ff58: mov ecx,[esp+20h]
0x1002ff5c: lea edx,[eax+1]
0x1002ff5f: movsx eax,byte ptr [eax]
0x1002ff62: push edx
0x1002ff63: push eax
0x1002ff64: push ecx
0x1002ff65: push esi
0x1002ff66: call 1002BAE0h
0x1002ff6b: add esp,14h
0x1002ff6e: jmp short 1002FF85h
0x1002ff70: push 101015C0h
0x1002ff75: jmp short 1002FF7Ch
0x1002ff77: push 10101590h
0x1002ff7c: push esi
0x1002ff7d: call 100230F0h
0x1002ff82: add esp,8
0x1002ff85: mov edx,[esi+284h]
0x1002ff8b: push edx
0x1002ff8c: push esi
0x1002ff8d: call 1002CD70h
0x1002ff92: add esp,8
```

```
0x1002ff95: pop ebp
0x1002ff96: pop edi
0x1002ff97: mov [esi+284h],ebx
0x1002ff9d: pop ebx
0x1002ff9e: pop esi
0x1002ff9f: pop ecx
0x1002ffa0: ret
```

`sub_1002fdd0` is called by png_read_info, png_read_end. It works with the text "Invalid sCAL chunk ignored: bad width format", "Invalid sCAL chunk ignored: bad height format", "Invalid sCAL ignored: invalid unit". It delegates to png_warning, png_error, sub_10023e50 (8 callees).

12.769 sub_1002ffb0 at 0x1002ffb0 (96 instructions)

```
0x1002ffb0: sub esp,14h
0x1002ffb3: mov eax,ds:[10136220h]
0x1002ffb8: xor eax,esp
0x1002ffba: mov [esp+10h],eax
0x1002ffbe: push ebx
0x1002ffbf: mov ebx,[esp+20h]
0x1002ffc3: push esi
0x1002ffc4: mov esi,[esp+20h]
0x1002ffc8: mov eax,[esi+6Ch]
0x1002ffcb: test al,1
0x1002ffcd: jne short 1002FFDAh
0x1002ffcf: push 101016C0h
0x1002ffd4: push esi
0x1002ffd5: call 100231E0h
0x1002ffda: test ebx,ebx
0x1002ffdc: je short 10030011h
0x1002ffde: test dword ptr [ebx+8],200h
0x1002ffe5: je short 10030011h
0x1002ffe7: push 101016A8h
0x1002ffec: push esi
0x1002ffed: call 100230F0h
0x1002fff2: mov eax,[esp+30h]
0x1002fff6: push eax
0x1002fff7: push esi
0x1002fff8: call 1002E050h
0x1002fffd: add esp,10h
0x10030000: pop esi
0x10030001: pop ebx
0x10030002: mov ecx,[esp+10h]
0x10030006: xor ecx,esp
0x10030008: call 10022326h
0x1003000d: add esp,14h
0x10030010: ret
0x10030011: test al,4
0x10030013: je short 1003001Bh
0x10030015: or eax,8
0x10030018: mov [esi+6Ch],eax
0x1003001b: push edi
0x1003001c: mov edi,[esp+2Ch]
0x10030020: cmp edi,7
0x10030023: je short 1003004Ch
0x10030025: push 1010168Ch
0x1003002a: push esi
0x1003002b: call 100230F0h
0x10030030: push edi
0x10030031: push esi
0x10030032: call 1002E050h
0x10030037: add esp,10h
0x1003003a: pop edi
0x1003003b: pop esi
0x1003003c: pop ebx
0x1003003d: mov ecx,[esp+10h]
```

```
0x10030041: xor ecx,esp
0x10030043: call 10022326h
0x10030048: add esp,14h
0x1003004b: ret
0x1003004c: push 7
0x1003004e: lea ecx,[esp+18h]
0x10030052: push ecx
0x10030053: push esi
0x10030054: call 1002CF30h
0x10030059: push 0
0x1003005b: push esi
0x1003005c: call 1002E050h
0x10030061: add esp,14h
0x10030064: test eax,eax
0x10030066: jne short 100300BBh
0x10030068: movzx eax,byte ptr [esp+19h]
0x1003006d: movzx edx,byte ptr [esp+1Ah]
0x10030072: mov cl,[esp+18h]
0x10030076: mov [esp+11h],al
0x1003007a: movzx eax,byte ptr [esp+16h]
0x1003007f: mov [esp+12h],dl
0x10030083: movzx edx,byte ptr [esp+17h]
0x10030088: mov [esp+10h],cl
0x1003008c: movzx cx,byte ptr [esp+14h]
0x10030092: mov [esp+0Eh],al
0x10030096: mov [esp+0Fh],dl
0x1003009a: movzx dx,byte ptr [esp+15h]
0x100300a0: lea eax,[esp+0Ch]
0x100300a4: push eax
0x100300a5: shl cx,8
0x100300a9: add cx,dx
0x100300ac: push ebx
0x100300ad: push esi
0x100300ae: mov [esp+18h],cx
0x100300b3: call 1002C310h
0x100300b8: add esp,0Ch
0x100300bb: mov ecx,[esp+1Ch]
0x100300bf: pop edi
0x100300c0: pop esi
0x100300c1: pop ebx
0x100300c2: xor ecx,esp
0x100300c4: call 10022326h
0x100300c9: add esp,14h
0x100300cc: ret
```

`sub_1002ffb0` is called by png_read_info, png_read_end. It works with the text "Duplicate tIME chunk", "Out of place tIME chunk". It delegates to sub_10022326, png_warning, png_error (6 callees).

12.770 sub_100300d0 at 0x100300d0 (151 instructions)

```
0x100300d0: push esi
0x100300d1: mov esi,[esp+8]
0x100300d5: mov eax,[esi+260h]
0x100300db: test eax,eax
0x100300dd: je short 1003011Dh
0x100300df: cmp eax,1
0x100300e2: jne short 100300F4h
0x100300e4: mov eax,[esp+10h]
0x100300e8: push eax
0x100300e9: push esi
0x100300ea: call 1002E050h
0x100300ef: add esp,8
0x100300f2: pop esi
0x100300f3: ret
0x100300f4: add eax,0FFFFFFFFh
0x100300f7: cmp eax,1
```

```
0x100300fa: mov [esi+260h],eax
0x10030100: jne short 1003011Dh
0x10030102: push 10101768h
0x10030107: push esi
0x10030108: call 100230F0h
0x1003010d: mov ecx,[esp+18h]
0x10030111: push ecx
0x10030112: push esi
0x10030113: call 1002E050h
0x10030118: add esp,10h
0x1003011b: pop esi
0x1003011c: ret
0x1003011d: mov eax,[esi+6Ch]
0x10030120: test al,1
0x10030122: jne short 1003012Fh
0x10030124: push 1010174Ch
0x10030129: push esi
0x1003012a: call 100231E0h
0x1003012f: test al,4
0x10030131: je short 10030139h
0x10030133: or eax,8
0x10030136: mov [esi+6Ch],eax
0x10030139: mov edx,[esi+284h]
0x1003013f: push edi
0x10030140: push edx
0x10030141: push esi
0x10030142: call 1002CD70h
0x10030147: mov edi,[esp+1Ch]
0x1003014b: lea eax,[edi+1]
0x1003014e: push eax
0x1003014f: push esi
0x10030150: call 1002CDA0h
0x10030155: add esp,10h
0x10030158: test eax,eax
0x1003015a: mov [esi+284h],eax
0x10030160: jne short 10030173h
0x10030162: push 1010172Ch
0x10030167: push esi
0x10030168: call 100230F0h
0x1003016d: add esp,8
0x10030170: pop edi
0x10030171: pop esi
0x10030172: ret
0x10030173: push edi
0x10030174: push eax
0x10030175: push esi
0x10030176: call 1002CF30h
0x1003017b: push 0
0x1003017d: push esi
0x1003017e: call 1002E050h
0x10030183: add esp,14h
0x10030186: test eax,eax
0x10030188: je short 100301A7h
0x1003018a: mov ecx,[esi+284h]
0x10030190: push ecx
0x10030191: push esi
0x10030192: call 1002CD70h
0x10030197: add esp,8
0x1003019a: pop edi
0x1003019b: mov dword ptr [esi+284h],0
0x100301a5: pop esi
0x100301a6: ret
0x100301a7: push ebx
0x100301a8: mov ebx,[esi+284h]
0x100301ae: lea eax,[ebx+edi]
0x100301b1: mov byte ptr [eax],0
0x100301b4: cmp byte ptr [ebx],0
```

```
0x100301b7: push ebp
0x100301b8: mov ebp,ebx
0x100301ba: je short 100301C9h
0x100301bc: lea esp,[esp]
0x100301c0: add ebp,1
0x100301c3: cmp byte ptr [ebp],0
0x100301c7: jne short 100301C0h
0x100301c9: cmp ebp,eax
0x100301cb: je short 100301D0h
0x100301cd: add ebp,1
0x100301d0: push 1Ch
0x100301d2: push esi
0x100301d3: call 1002CDA0h
0x100301d8: mov edi,eax
0x100301da: xor eax,eax
0x100301dc: add esp,8
0x100301df: cmp edi,eax
0x100301e1: jne short 1003020Dh
0x100301e3: push 10101704h
0x100301e8: push esi
0x100301e9: call 100230F0h
0x100301ee: mov edx,[esi+284h]
0x100301f4: push edx
0x100301f5: push esi
0x100301f6: call 1002CD70h
0x100301fb: add esp,10h
0x100301fe: pop ebp
0x100301ff: pop ebx
0x10030200: pop edi
0x10030201: mov dword ptr [esi+284h],0
0x1003020b: pop esi
0x1003020c: ret
0x1003020d: push ebp
0x1003020e: mov dword ptr [edi],0FFFFFFFFh
0x10030214: mov [edi+4],ebx
0x10030217: mov [edi+14h],eax
0x1003021a: mov [edi+18h],eax
0x1003021d: mov [edi+10h],eax
0x10030220: mov [edi+8],ebp
0x10030223: call dword ptr ds:[100FE0B0h]
0x10030229: push 1
0x1003022b: mov [edi+0Ch],eax
0x1003022e: mov eax,[esp+1Ch]
0x10030232: push edi
0x10030233: push eax
0x10030234: push esi
0x10030235: call 1002C050h
0x1003023a: mov ecx,[esi+284h]
0x10030240: push ecx
0x10030241: push esi
0x10030242: mov ebx,eax
0x10030244: call 1002CD70h
0x10030249: push edi
0x1003024a: push esi
0x1003024b: mov dword ptr [esi+284h],0
0x10030255: call 1002CD70h
0x1003025a: add esp,20h
0x1003025d: test ebx,ebx
0x1003025f: je short 1003026Fh
0x10030261: push 101016D8h
0x10030266: push esi
0x10030267: call 100230F0h
0x1003026c: add esp,8
0x1003026f: pop ebp
0x10030270: pop ebx
0x10030271: pop edi
```

```
0x10030272: pop esi
0x10030273: ret
```

`sub_100300d0` is called by png_read_info, png_read_end. It works with the text "Insufficient memory to process text chunk", "Not enough memory to process text chunk", "No memory to process text chunk". It delegates to png_warning, png_error, sub_1002c050 (7 callees).

12.771 sub_10030280 at 0x10030280 (187 instructions)

```
0x10030280: push ebx
0x10030281: push esi
0x10030282: mov esi,[esp+0Ch]
0x10030286: mov eax,[esi+260h]
0x1003028c: xor ebx,ebx
0x1003028e: cmp eax,ebx
0x10030290: je short 100302D2h
0x10030292: cmp eax,1
0x10030295: jne short 100302A8h
0x10030297: mov eax,[esp+14h]
0x1003029b: push eax
0x1003029c: push esi
0x1003029d: call 1002E050h
0x100302a2: add esp,8
0x100302a5: pop esi
0x100302a6: pop ebx
0x100302a7: ret
0x100302a8: add eax,0FFFFFFFFh
0x100302ab: cmp eax,1
0x100302ae: mov [esi+260h],eax
0x100302b4: jne short 100302D2h
0x100302b6: push 1010185Ch
0x100302bb: push esi
0x100302bc: call 100230F0h
0x100302c1: mov ecx,[esp+1Ch]
0x100302c5: push ecx
0x100302c6: push esi
0x100302c7: call 1002E050h
0x100302cc: add esp,10h
0x100302cf: pop esi
0x100302d0: pop ebx
0x100302d1: ret
0x100302d2: mov eax,[esi+6Ch]
0x100302d5: test al,1
0x100302d7: jne short 100302E4h
0x100302d9: push 10101840h
0x100302de: push esi
0x100302df: call 100231E0h
0x100302e4: test al,4
0x100302e6: je short 100302EEh
0x100302e8: or eax,8
0x100302eb: mov [esi+6Ch],eax
0x100302ee: mov edx,[esi+284h]
0x100302f4: push ebp
0x100302f5: push edx
0x100302f6: push esi
0x100302f7: call 1002CD70h
0x100302fc: mov ebp,[esp+20h]
0x10030300: lea eax,[ebp+1]
0x10030303: push eax
0x10030304: push esi
0x10030305: call 1002CDA0h
0x1003030a: add esp,10h
0x1003030d: cmp eax,ebx
0x1003030f: mov [esi+284h],eax
0x10030315: jne short 10030329h
0x10030317: push 1010181Ch
```

```
0x1003031c: push esi
0x1003031d: call 100230F0h
0x10030322: add esp,8
0x10030325: pop ebp
0x10030326: pop esi
0x10030327: pop ebx
0x10030328: ret
0x10030329: push ebp
0x1003032a: push eax
0x1003032b: push esi
0x1003032c: call 1002CF30h
0x10030331: push ebx
0x10030332: push esi
0x10030333: call 1002E050h
0x10030338: add esp,14h
0x1003033b: test eax,eax
0x1003033d: je short 10030359h
0x1003033f: mov ecx,[esi+284h]
0x10030345: push ecx
0x10030346: push esi
0x10030347: call 1002CD70h
0x1003034c: add esp,8
0x1003034f: pop ebp
0x10030350: mov [esi+284h],ebx
0x10030356: pop esi
0x10030357: pop ebx
0x10030358: ret
0x10030359: mov edx,[esi+284h]
0x1003035f: mov [edx+ebp],bl
0x10030362: mov eax,[esi+284h]
0x10030368: push edi
0x10030369: mov edi,eax
0x1003036b: cmp [edi],bl
0x1003036d: je short 10030377h
0x1003036f: nop
0x10030370: add edi,1
0x10030373: cmp [edi],bl
0x10030375: jne short 10030370h
0x10030377: lea eax,[eax+ebp-2]
0x1003037b: cmp edi,eax
0x1003037d: jb short 100303A5h
0x1003037f: push 10101804h
0x10030384: push esi
0x10030385: call 100230F0h
0x1003038a: mov ecx,[esi+284h]
0x10030390: push ecx
0x10030391: push esi
0x10030392: call 1002CD70h
0x10030397: add esp,10h
0x1003039a: mov [esi+284h],ebx
0x100303a0: pop edi
0x100303a1: pop ebp
0x100303a2: pop esi
0x100303a3: pop ebx
0x100303a4: ret
0x100303a5: movsx ebp,byte ptr [edi+1]
0x100303a9: add edi,1
0x100303ac: cmp ebp,ebx
0x100303ae: je short 100303C0h
0x100303b0: push 101017DCh
0x100303b5: push esi
0x100303b6: call 100230F0h
0x100303bb: add esp,8
0x100303be: xor ebp,ebp
0x100303c0: sub edi,[esi+284h]
0x100303c6: mov eax,[esp+1Ch]
0x100303ca: lea edx,[esp+14h]
```

```
0x100303ce: push edx
0x100303cf: add edi,1
0x100303d2: mov ebx,edi
0x100303d4: push ebx
0x100303d5: push eax
0x100303d6: push ebp
0x100303d7: push esi
0x100303d8: call 1002D150h
0x100303dd: push 1Ch
0x100303df: push esi
0x100303e0: call 1002CDA0h
0x100303e5: mov edi,eax
0x100303e7: xor eax,eax
0x100303e9: add esp,1Ch
0x100303ec: cmp edi,eax
0x100303ee: jne short 1003041Ah
0x100303f0: push 101017B4h
0x100303f5: push esi
0x100303f6: call 100230F0h
0x100303fb: mov ecx,[esi+284h]
0x10030401: push ecx
0x10030402: push esi
0x10030403: call 1002CD70h
0x10030408: add esp,10h
0x1003040b: pop edi
0x1003040c: pop ebp
0x1003040d: mov dword ptr [esi+284h],0
0x10030417: pop esi
0x10030418: pop ebx
0x10030419: ret
0x1003041a: mov ecx,[esp+14h]
0x1003041e: mov [edi],ebp
0x10030420: mov edx,[esi+284h]
0x10030426: push 1
0x10030428: mov [edi+4],edx
0x1003042b: mov edx,[esp+1Ch]
0x1003042f: mov [edi+14h],eax
0x10030432: mov [edi+18h],eax
0x10030435: mov [edi+10h],eax
0x10030438: mov eax,[esi+284h]
0x1003043e: push edi
0x1003043f: push edx
0x10030440: add eax,ebx
0x10030442: push esi
0x10030443: mov [edi+8],eax
0x10030446: mov [edi+0Ch],ecx
0x10030449: call 1002C050h
0x1003044e: push edi
0x1003044f: push esi
0x10030450: mov ebx,eax
0x10030452: call 1002CD70h
0x10030457: mov eax,[esi+284h]
0x1003045d: push eax
0x1003045e: push esi
0x1003045f: call 1002CD70h
0x10030464: add esp,20h
0x10030467: mov dword ptr [esi+284h],0
0x10030471: test ebx,ebx
0x10030473: je near ptr 100303A0h
0x10030479: push 1010178Ch
0x1003047e: push esi
0x1003047f: call 100231E0h
0x10030484: int 3
```

`sub_10030280` is called by png_read_info, png_read_end. It delegates to png_warning, png_error, sub_1002c050 (8 callees).

12.772 sub_10030490 at 0x10030490 (240 instructions)

```
0x10030490: sub esp,0Ch
0x10030493: push ebx
0x10030494: push esi
0x10030495: mov esi,[esp+18h]
0x10030499: mov eax,[esi+260h]
0x1003049f: xor ebx,ebx
0x100304a1: cmp eax,ebx
0x100304a3: je short 100304EBh
0x100304a5: cmp eax,1
0x100304a8: jne short 100304BEh
0x100304aa: mov eax,[esp+20h]
0x100304ae: push eax
0x100304af: push esi
0x100304b0: call 1002E050h
0x100304b5: add esp,8
0x100304b8: pop esi
0x100304b9: pop ebx
0x100304ba: add esp,0Ch
0x100304bd: ret
0x100304be: add eax,0FFFFFFFFh
0x100304c1: cmp eax,1
0x100304c4: mov [esi+260h],eax
0x100304ca: jne short 100304EBh
0x100304cc: push 1010193Ch
0x100304d1: push esi
0x100304d2: call 100230F0h
0x100304d7: mov ecx,[esp+28h]
0x100304db: push ecx
0x100304dc: push esi
0x100304dd: call 1002E050h
0x100304e2: add esp,10h
0x100304e5: pop esi
0x100304e6: pop ebx
0x100304e7: add esp,0Ch
0x100304ea: ret
0x100304eb: mov eax,[esi+6Ch]
0x100304ee: test al,1
0x100304f0: jne short 100304FDh
0x100304f2: push 10101920h
0x100304f7: push esi
0x100304f8: call 100231E0h
0x100304fd: test al,4
0x100304ff: je short 10030507h
0x10030501: or eax,8
0x10030504: mov [esi+6Ch],eax
0x10030507: mov edx,[esi+284h]
0x1003050d: push edi
0x1003050e: push edx
0x1003050f: push esi
0x10030510: call 1002CD70h
0x10030515: mov edi,[esp+2Ch]
0x10030519: lea eax,[edi+1]
0x1003051c: push eax
0x1003051d: push esi
0x1003051e: call 1002CDA0h
0x10030523: add esp,10h
0x10030526: cmp eax,ebx
0x10030528: mov [esi+284h],eax
0x1003052e: jne short 10030545h
0x10030530: push 10101900h
0x10030535: push esi
0x10030536: call 100230F0h
0x1003053b: add esp,8
0x1003053e: pop edi
0x1003053f: pop esi
0x10030540: pop ebx
```

```
0x10030541: add esp,0Ch
0x10030544: ret
0x10030545: push edi
0x10030546: push eax
0x10030547: push esi
0x10030548: call 1002CF30h
0x1003054d: push ebx
0x1003054e: push esi
0x1003054f: call 1002E050h
0x10030554: add esp,14h
0x10030557: test eax,eax
0x10030559: je short 10030578h
0x1003055b: mov ecx,[esi+284h]
0x10030561: push ecx
0x10030562: push esi
0x10030563: call 1002CD70h
0x10030568: add esp,8
0x1003056b: pop edi
0x1003056c: mov [esi+284h],ebx
0x10030572: pop esi
0x10030573: pop ebx
0x10030574: add esp,0Ch
0x10030577: ret
0x10030578: mov edx,[esi+284h]
0x1003057e: mov [edi+edx],bl
0x10030581: mov ecx,[esi+284h]
0x10030587: push ebp
0x10030588: mov ebp,ecx
0x1003058a: cmp [ebp],bl
0x1003058d: mov [esp+14h],ecx
0x10030591: je short 1003059Bh
0x10030593: add ebp,1
0x10030596: cmp [ebp],bl
0x10030599: jne short 10030593h
0x1003059b: lea edx,[ecx+edi]
0x1003059e: add ebp,1
0x100305a1: lea eax,[edx-3]
0x100305a4: cmp ebp,eax
0x100305a6: jb short 100305D1h
0x100305a8: push 101018E8h
0x100305ad: push esi
0x100305ae: call 100230F0h
0x100305b3: mov ecx,[esi+284h]
0x100305b9: push ecx
0x100305ba: push esi
0x100305bb: call 1002CD70h
0x100305c0: add esp,10h
0x100305c3: mov [esi+284h],ebx
0x100305c9: pop ebp
0x100305ca: pop edi
0x100305cb: pop esi
0x100305cc: pop ebx
0x100305cd: add esp,0Ch
0x100305d0: ret
0x100305d1: movsx eax,byte ptr [ebp]
0x100305d5: add ebp,1
0x100305d8: mov [esp+20h],eax
0x100305dc: movsx eax,byte ptr [ebp]
0x100305e0: add ebp,1
0x100305e3: cmp byte ptr [ebp],0
0x100305e7: mov [esp+10h],eax
0x100305eb: mov ebx,ebp
0x100305ed: je short 100305F8h
0x100305ef: nop
0x100305f0: add ebx,1
0x100305f3: cmp byte ptr [ebx],0
0x100305f6: jne short 100305F0h
```

```
0x100305f8: add ebx,1
0x100305fb: cmp ebx,edx
0x100305fd: jb short 10030616h
0x100305ff: push 101018E8h
0x10030604: push esi
0x10030605: call 100230F0h
0x1003060a: mov ecx,[esi+284h]
0x10030610: push ecx
0x10030611: jmp near ptr 10030697h
0x10030616: cmp byte ptr [ebx],0
0x10030619: mov eax,ebx
0x1003061b: je short 10030628h
0x1003061d: lea ecx,[ecx]
0x10030620: add eax,1
0x10030623: cmp byte ptr [eax],0
0x10030626: jne short 10030620h
0x10030628: add eax,1
0x1003062b: cmp eax,edx
0x1003062d: jb short 10030643h
0x1003062f: push 101018D0h
0x10030634: push esi
0x10030635: call 100230F0h
0x1003063a: mov edx,[esi+284h]
0x10030640: push edx
0x10030641: jmp short 10030697h
0x10030643: sub eax,ecx
0x10030645: cmp dword ptr [esp+20h],0
0x1003064a: mov [esp+18h],eax
0x1003064e: je short 10030667h
0x10030650: mov edx,[esp+10h]
0x10030654: lea ecx,[esp+10h]
0x10030658: push ecx
0x10030659: push eax
0x1003065a: push edi
0x1003065b: push edx
0x1003065c: push esi
0x1003065d: call 1002D150h
0x10030662: add esp,14h
0x10030665: jmp short 10030674h
0x10030667: add ecx,eax
0x10030669: push ecx
0x1003066a: call dword ptr ds:[100FE0B0h]
0x10030670: mov [esp+10h],eax
0x10030674: push 1Ch
0x10030676: push esi
0x10030677: call 1002CDA0h
0x1003067c: mov edi,eax
0x1003067e: add esp,8
0x10030681: test edi,edi
0x10030683: jne short 100306B2h
0x10030685: push 101018A8h
0x1003068a: push esi
0x1003068b: call 100230F0h
0x10030690: mov eax,[esi+284h]
0x10030696: push eax
0x10030697: push esi
0x10030698: call 1002CD70h
0x1003069d: add esp,10h
0x100306a0: pop ebp
0x100306a1: pop edi
0x100306a2: mov dword ptr [esi+284h],0
0x100306ac: pop esi
0x100306ad: pop ebx
0x100306ae: add esp,0Ch
0x100306b1: ret
0x100306b2: mov ecx,[esp+20h]
0x100306b6: mov eax,[esp+14h]
```

```
0x100306ba: add ecx,1
0x100306bd: mov [edi],ecx
0x100306bf: mov edx,[esi+284h]
0x100306c5: sub edx,eax
0x100306c7: add edx,ebx
0x100306c9: mov [edi+18h],edx
0x100306cc: mov ecx,[esi+284h]
0x100306d2: mov edx,[esp+10h]
0x100306d6: sub ecx,eax
0x100306d8: add ecx,ebp
0x100306da: mov [edi+14h],ecx
0x100306dd: mov [edi+10h],edx
0x100306e0: mov edx,[esp+24h]
0x100306e4: mov dword ptr [edi+0Ch],0
0x100306eb: mov eax,[esi+284h]
0x100306f1: push 1
0x100306f3: mov [edi+4],eax
0x100306f6: mov ecx,[esi+284h]
0x100306fc: add ecx,[esp+1Ch]
0x10030700: push edi
0x10030701: push edx
0x10030702: push esi
0x10030703: mov [edi+8],ecx
0x10030706: call 1002C050h
0x1003070b: push edi
0x1003070c: push esi
0x1003070d: mov ebx,eax
0x1003070f: call 1002CD70h
0x10030714: mov eax,[esi+284h]
0x1003071a: push eax
0x1003071b: push esi
0x1003071c: call 1002CD70h
0x10030721: add esp,20h
0x10030724: test ebx,ebx
0x10030726: mov dword ptr [esi+284h],0
0x10030730: je near ptr 100305C9h
0x10030736: push 10101880h
0x1003073b: push esi
0x1003073c: call 100231E0h
0x10030741: int 3
```

`sub_10030490` is called by png_read_info, png_read_end. It works with the text "Insufficient memory to store iTXt chunk", "Not enough memory to process iTXt chunk", "Malformed iTXt chunk". It delegates to png_warning, png_error, sub_1002c050 (8 callees).

12.773 sub_10030750 at 0x10030750 (206 instructions)

```
0x10030750: sub esp,14h
0x10030753: mov eax,ds:[10136220h]
0x10030758: xor eax,esp
0x1003075a: mov [esp+10h],eax
0x1003075e: mov eax,[esp+1Ch]
0x10030762: push esi
0x10030763: mov esi,[esp+1Ch]
0x10030767: mov [esp+4],eax
0x1003076b: mov eax,[esi+260h]
0x10030771: test eax,eax
0x10030773: mov dword ptr [esp+8],0
0x1003077b: je short 100307D7h
0x1003077d: cmp eax,1
0x10030780: jne short 100307A0h
0x10030782: mov ecx,[esp+24h]
0x10030786: push ecx
0x10030787: push esi
0x10030788: call 1002E050h
0x1003078d: add esp,8
```

```
0x10030790: pop esi
0x10030791: mov ecx,[esp+10h]
0x10030795: xor ecx,esp
0x10030797: call 10022326h
0x1003079c: add esp,14h
0x1003079f: ret
0x100307a0: add eax,0FFFFFFFFh
0x100307a3: cmp eax,1
0x100307a6: mov [esi+260h],eax
0x100307ac: jne short 100307D7h
0x100307ae: push 1010198Ch
0x100307b3: push esi
0x100307b4: call 100230F0h
0x100307b9: mov edx,[esp+2Ch]
0x100307bd: push edx
0x100307be: push esi
0x100307bf: call 1002E050h
0x100307c4: add esp,10h
0x100307c7: pop esi
0x100307c8: mov ecx,[esp+10h]
0x100307cc: xor ecx,esp
0x100307ce: call 10022326h
0x100307d3: add esp,14h
0x100307d6: ret
0x100307d7: push ebx
0x100307d8: mov ebx,[esi+6Ch]
0x100307db: test bl,4
0x100307de: push ebp
0x100307df: push edi
0x100307e0: je near ptr 10030891h
0x100307e6: mov byte ptr [esp+18h],49h
0x100307eb: mov byte ptr [esp+19h],44h
0x100307f0: mov byte ptr [esp+1Ah],41h
0x100307f5: mov byte ptr [esp+1Bh],54h
0x100307fa: mov byte ptr [esp+1Ch],0
0x100307ff: mov eax,4
0x10030804: lea ecx,[esp+18h]
0x10030808: lea edx,[esi+11Ch]
0x1003080e: mov edi,edi
0x10030810: mov edi,[edx]
0x10030812: cmp edi,[ecx]
0x10030814: jne short 10030828h
0x10030816: sub eax,4
0x10030819: add ecx,4
0x1003081c: add edx,4
0x1003081f: cmp eax,4
0x10030822: jae short 10030810h
0x10030824: test eax,eax
0x10030826: je short 10030885h
0x10030828: movzx ebp,byte ptr [ecx]
0x1003082b: movzx edi,byte ptr [edx]
0x1003082e: sub edi,ebp
0x10030830: jne short 10030877h
0x10030832: sub eax,1
0x10030835: add ecx,1
0x10030838: add edx,1
0x1003083b: test eax,eax
0x1003083d: je short 10030885h
0x1003083f: movzx ebp,byte ptr [ecx]
0x10030842: movzx edi,byte ptr [edx]
0x10030845: sub edi,ebp
0x10030847: jne short 10030877h
0x10030849: sub eax,1
0x1003084c: add ecx,1
0x1003084f: add edx,1
0x10030852: test eax,eax
0x10030854: je short 10030885h
```

```
0x10030856: movzx ebp,byte ptr [ecx]
0x10030859: movzx edi,byte ptr [edx]
0x1003085c: sub edi,ebp
0x1003085e: jne short 10030877h
0x10030860: sub eax,1
0x10030863: add ecx,1
0x10030866: add edx,1
0x10030869: test eax,eax
0x1003086b: je short 10030885h
0x1003086d: movzx eax,byte ptr [ecx]
0x10030870: movzx edi,byte ptr [edx]
0x10030873: sub edi,eax
0x10030875: je short 10030885h
0x10030877: test edi,edi
0x10030879: mov eax,1
0x1003087e: jg short 10030887h
0x10030880: or eax,0FFFFFFFFh
0x10030883: jmp short 10030887h
0x10030885: xor eax,eax
0x10030887: test eax,eax
0x10030889: je short 10030891h
0x1003088b: or ebx,8
0x1003088e: mov [esi+6Ch],ebx
0x10030891: test byte ptr [esi+11Ch],20h
0x10030898: lea ebx,[esi+11Ch]
0x1003089e: jne short 100308C3h
0x100308a0: push ebx
0x100308a1: push esi
0x100308a2: call 10023B10h
0x100308a7: add esp,8
0x100308aa: cmp eax,3
0x100308ad: je short 100308C3h
0x100308af: cmp dword ptr [esi+21Ch],0
0x100308b6: jne short 100308C3h
0x100308b8: push 10101974h
0x100308bd: push esi
0x100308be: call 10023240h
0x100308c3: test dword ptr [esi+70h],8000h
0x100308ca: jne short 100308DEh
0x100308cc: cmp dword ptr [esi+21Ch],0
0x100308d3: jne short 100308DEh
0x100308d5: mov eax,[esp+30h]
0x100308d9: jmp near ptr 1003099Dh
0x100308de: mov ecx,[ebx]
0x100308e0: mov ebp,[esp+30h]
0x100308e4: test ebp,ebp
0x100308e6: mov dl,[ebx+4]
0x100308e9: lea edi,[esi+268h]
0x100308ef: mov [edi],ecx
0x100308f1: mov [edi+4],dl
0x100308f4: mov byte ptr [esi+26Ch],0
0x100308fb: mov [esi+274h],ebp
0x10030901: jne short 1003090Bh
0x10030903: mov [esi+270h],ebp
0x10030909: jmp short 10030923h
0x1003090b: push ebp
0x1003090c: push esi
0x1003090d: call 1002CD20h
0x10030912: push ebp
0x10030913: push eax
0x10030914: push esi
0x10030915: mov [esi+270h],eax
0x1003091b: call 1002CF30h
0x10030920: add esp,14h
0x10030923: mov eax,[esi+21Ch]
0x10030929: test eax,eax
0x1003092b: je short 1003096Eh
```

```
0x1003092d: push edi
0x1003092e: push esi
0x1003092f: call eax
0x10030931: add esp,8
0x10030934: test eax,eax
0x10030936: jge short 10030943h
0x10030938: push 10101960h
0x1003093d: push esi
0x1003093e: call 10023240h
0x10030943: jne short 1003097Fh
0x10030945: test byte ptr [ebx],20h
0x10030948: jne short 10030964h
0x1003094a: push ebx
0x1003094b: push esi
0x1003094c: call 10023B10h
0x10030951: add esp,8
0x10030954: cmp eax,3
0x10030957: je short 10030964h
0x10030959: push 10101974h
0x1003095e: push esi
0x1003095f: call 10023240h
0x10030964: mov eax,[esp+10h]
0x10030968: push 1
0x1003096a: push edi
0x1003096b: push eax
0x1003096c: jmp short 10030976h
0x1003096e: mov ecx,[esp+10h]
0x10030972: push 1
0x10030974: push edi
0x10030975: push ecx
0x10030976: push esi
0x10030977: call 1002C5C0h
0x1003097c: add esp,10h
0x1003097f: mov edx,[esi+270h]
0x10030985: push edx
0x10030986: push esi
0x10030987: call 1002CD70h
0x1003098c: mov eax,[esp+1Ch]
0x10030990: add esp,8
0x10030993: mov dword ptr [esi+270h],0
0x1003099d: push eax
0x1003099e: push esi
0x1003099f: call 1002E050h
0x100309a4: mov ecx,[esp+28h]
0x100309a8: add esp,8
0x100309ab: pop edi
0x100309ac: pop ebp
0x100309ad: pop ebx
0x100309ae: pop esi
0x100309af: xor ecx,esp
0x100309b1: call 10022326h
0x100309b6: add esp,14h
0x100309b9: ret
```

`sub_10030750` is called by png_read_info, png_read_end. It works with the text "error in user chunk", "unknown critical chunk", "No space in chunk cache for unknown chunk". It delegates to sub_10022326, png_warning, png_chunk_error (9 callees).

12.774 sub_100309c0 at 0x100309c0 (240 instructions)

```
0x100309c0: sub esp,80h
0x100309c6: mov eax,ds:[10136220h]
0x100309cb: xor eax,esp
0x100309cd: mov [esp+7Ch],eax
0x100309d1: push ebp
0x100309d2: mov eax,2
```

```
0x100309d7: push esi
0x100309d8: mov esi,[esp+8Ch]
0x100309df: xor ebp,ebp
0x100309e1: push edi
0x100309e2: mov edi,1
0x100309e7: add [esi+0E4h],edi
0x100309ed: mov ecx,4
0x100309f2: mov edx,8
0x100309f7: mov [esp+38h],eax
0x100309fb: mov [esp+20h],eax
0x100309ff: mov [esp+24h],eax
0x10030a03: mov [esp+74h],eax
0x10030a07: mov [esp+5Ch],eax
0x10030a0b: mov [esp+60h],eax
0x10030a0f: mov eax,[esi+0E4h]
0x10030a15: cmp eax,[esi+0D4h]
0x10030a1b: mov [esp+2Ch],ebp
0x10030a1f: mov [esp+30h],ecx
0x10030a23: mov [esp+34h],ebp
0x10030a27: mov [esp+3Ch],ebp
0x10030a2b: mov [esp+40h],edi
0x10030a2f: mov [esp+44h],ebp
0x10030a33: mov [esp+10h],edx
0x10030a37: mov [esp+14h],edx
0x10030a3b: mov [esp+18h],ecx
0x10030a3f: mov [esp+1Ch],ecx
0x10030a43: mov [esp+28h],edi
0x10030a47: mov [esp+64h],ebp
0x10030a4b: mov [esp+68h],ebp
0x10030a4f: mov [esp+6Ch],ecx
0x10030a53: mov [esp+70h],ebp
0x10030a57: mov [esp+78h],ebp
0x10030a5b: mov [esp+7Ch],edi
0x10030a5f: mov [esp+48h],edx
0x10030a63: mov [esp+4Ch],edx
0x10030a67: mov [esp+50h],edx
0x10030a6b: mov [esp+54h],ecx
0x10030a6f: mov [esp+58h],ecx
0x10030a73: jb near ptr 10030CD8h
0x10030a79: cmp byte ptr [esi+123h],0
0x10030a80: push ebx
0x10030a81: je near ptr 10030B1Eh
0x10030a87: mov eax,[esi+0DCh]
0x10030a8d: mov ecx,[esi+0E8h]
0x10030a93: add eax,edi
0x10030a95: push eax
0x10030a96: push ebp
0x10030a97: push ecx
0x10030a98: mov [esi+0E4h],ebp
0x10030a9e: call 100228C0h
0x10030aa3: add esp,0Ch
0x10030aa6: jmp short 10030AB2h
0x10030ab0: xor ebp,ebp
0x10030ab2: add byte ptr [esi+124h],1
0x10030ab9: mov bl,[esi+124h]
0x10030abf: cmp bl,7
0x10030ac2: jae short 10030B1Eh
0x10030ac4: mov edx,[esi+0CCh]
0x10030aca: movzx ecx,bl
0x10030acd: add ecx,ecx
0x10030acf: add ecx,ecx
0x10030ad1: sub edx,[esp+ecx+30h]
0x10030ad5: mov edi,[esp+ecx+14h]
0x10030ad9: lea eax,[edx+edi-1]
0x10030add: xor edx,edx
0x10030adf: div edi
0x10030ae1: test byte ptr [esi+74h],2
```

```
0x10030ae5: mov ebp,eax
0x10030ae7: mov [esi+0E0h],ebp
0x10030aed: jne short 10030B13h
0x10030aef: mov eax,[esi+0D0h]
0x10030af5: sub eax,[esp+ecx+68h]
0x10030af9: mov edi,[esp+ecx+4Ch]
0x10030afd: xor edx,edx
0x10030aff: lea eax,[eax+edi-1]
0x10030b03: div edi
0x10030b05: test eax,eax
0x10030b07: mov [esi+0D4h],eax
0x10030b0d: je short 10030AB0h
0x10030b0f: test ebp,ebp
0x10030b11: je short 10030AB0h
0x10030b13: cmp bl,7
0x10030b16: jb near ptr 10030CD7h
0x10030b1c: xor ebp,ebp
0x10030b1e: test byte ptr [esi+70h],20h
0x10030b22: jne near ptr 10030CACh
0x10030b28: lea ecx,[esp+13h]
0x10030b2c: mov byte ptr [esp+84h],49h
0x10030b34: mov byte ptr [esp+85h],44h
0x10030b3c: mov byte ptr [esp+86h],41h
0x10030b44: mov byte ptr [esp+87h],54h
0x10030b4c: mov byte ptr [esp+88h],0
0x10030b54: mov [esi+84h],ecx
0x10030b5a: mov dword ptr [esi+88h],1
0x10030b64: cmp [esi+7Ch],ebp
0x10030b67: jne near ptr 10030C69h
0x10030b6d: cmp [esi+10Ch],ebp
0x10030b73: jne near ptr 10030C33h
0x10030b79: lea esp,[esp]
0x10030b80: push ebp
0x10030b81: push esi
0x10030b82: call 1002E050h
0x10030b87: push esi
0x10030b88: call 1002DFD0h
0x10030b8d: mov [esi+10Ch],eax
0x10030b93: add esp,0Ch
0x10030b96: mov eax,4
0x10030b9b: lea ecx,[esp+84h]
0x10030ba2: lea edx,[esi+11Ch]
0x10030ba8: mov edi,[edx]
0x10030baa: cmp edi,[ecx]
0x10030bac: jne short 10030BC0h
0x10030bae: sub eax,4
0x10030bb1: add ecx,4
0x10030bb4: add edx,4
0x10030bb7: cmp eax,4
0x10030bba: jae short 10030BA8h
0x10030bbc: cmp eax,ebp
0x10030bbe: je short 10030C1Dh
0x10030bc0: movzx edi,byte ptr [edx]
0x10030bc3: movzx ebx,byte ptr [ecx]
0x10030bc6: sub edi,ebx
0x10030bc8: jne short 10030C0Fh
0x10030bca: sub eax,1
0x10030bcd: add ecx,1
0x10030bd0: add edx,1
0x10030bd3: cmp eax,ebp
0x10030bd5: je short 10030C1Dh
0x10030bd7: movzx edi,byte ptr [edx]
0x10030bda: movzx ebx,byte ptr [ecx]
0x10030bdd: sub edi,ebx
0x10030bdf: jne short 10030C0Fh
0x10030be1: sub eax,1
0x10030be4: add ecx,1
```

```
0x10030be7: add edx,1
0x10030bea: cmp eax,ebp
0x10030bec: je short 10030C1Dh
0x10030bee: movzx edi,byte ptr [edx]
0x10030bf1: movzx ebx,byte ptr [ecx]
0x10030bf4: sub edi,ebx
0x10030bf6: jne short 10030C0Fh
0x10030bf8: sub eax,1
0x10030bfb: add ecx,1
0x10030bfe: add edx,1
0x10030c01: cmp eax,ebp
0x10030c03: je short 10030C1Dh
0x10030c05: movzx edi,byte ptr [edx]
0x10030c08: movzx edx,byte ptr [ecx]
0x10030c0b: sub edi,edx
0x10030c0d: je short 10030C1Dh
0x10030c0f: cmp edi,ebp
0x10030c11: mov eax,1
0x10030c16: jg short 10030C1Fh
0x10030c18: or eax,0FFFFFFFFh
0x10030c1b: jmp short 10030C1Fh
0x10030c1d: xor eax,eax
0x10030c1f: cmp eax,ebp
0x10030c21: jne near ptr 10030CF0h
0x10030c27: cmp [esi+10Ch],ebp
0x10030c2d: je near ptr 10030B80h
0x10030c33: mov eax,[esi+0B4h]
0x10030c39: mov ecx,[esi+10Ch]
0x10030c3f: cmp eax,ecx
0x10030c41: mov edx,[esi+0B0h]
0x10030c47: mov [esi+7Ch],eax
0x10030c4a: mov [esi+78h],edx
0x10030c4d: jbe short 10030C52h
0x10030c4f: mov [esi+7Ch],ecx
0x10030c52: mov eax,[esi+7Ch]
0x10030c55: push eax
0x10030c56: push edx
0x10030c57: push esi
0x10030c58: call 1002CF30h
0x10030c5d: mov ecx,[esi+7Ch]
0x10030c60: add esp,0Ch
0x10030c63: sub [esi+10Ch],ecx
0x10030c69: lea edx,[esi+78h]
0x10030c6c: push 1
0x10030c6e: push edx
0x10030c6f: call 10060100h
0x10030c74: add esp,8
0x10030c77: cmp eax,1
0x10030c7a: je short 10030CFBh
0x10030c7c: cmp eax,ebp
0x10030c7e: jne near ptr 10030D15h
0x10030c84: cmp [esi+88h],ebp
0x10030c8a: jne near ptr 10030B64h
0x10030c90: push 100FFE34h
0x10030c95: push esi
0x10030c96: call 100230F0h
0x10030c9b: add esp,8
0x10030c9e: or dword ptr [esi+6Ch],8
0x10030ca2: or dword ptr [esi+70h],20h
0x10030ca6: mov [esi+88h],ebp
0x10030cac: cmp [esi+10Ch],ebp
0x10030cb2: jne short 10030CB9h
0x10030cb4: cmp [esi+7Ch],ebp
0x10030cb7: je short 10030CC7h
0x10030cb9: push 101019CCh
0x10030cbe: push esi
0x10030cbf: call 100230F0h
```

```
0x10030cc4: add esp,8
0x10030cc7: lea eax,[esi+78h]
0x10030cca: push eax
0x10030ccb: call 1005FE10h
0x10030cd0: add esp,4
0x10030cd3: or dword ptr [esi+6Ch],8
0x10030cd7: pop ebx
0x10030cd8: mov ecx,[esp+88h]
0x10030cdf: pop edi
0x10030ce0: pop esi
0x10030ce1: pop ebp
0x10030ce2: xor ecx,esp
0x10030ce4: call 10022326h
0x10030ce9: add esp,80h
0x10030cef: ret
0x10030cf0: push 100FFE4Ch
0x10030cf5: push esi
0x10030cf6: call 100231E0h
0x10030cfb: cmp [esi+88h],ebp
0x10030d01: je short 10030C90h
0x10030d03: cmp [esi+7Ch],ebp
0x10030d06: jne short 10030C90h
0x10030d08: cmp [esi+10Ch],ebp
0x10030d0e: je short 10030C9Eh
0x10030d10: jmp near ptr 10030C90h
0x10030d15: mov eax,[esi+90h]
0x10030d1b: cmp eax,ebp
0x10030d1d: jne short 10030D24h
0x10030d1f: mov eax,101019B8h
0x10030d24: push eax
0x10030d25: push esi
0x10030d26: call 100231E0h
0x10030d2b: int 3
```

`sub_100309c0` is called by png_read_row. It works with the text "Extra compressed data", "Not enough image data", "Decompression Error". It delegates to sub_10022326, j_memset, png_warning (9 callees).

12.775 png_set_crc_action at 0x10030d30 (18 instructions)

```
0x10030d30: push esi
0x10030d31: mov esi,[esp+8]
0x10030d35: test esi,esi
0x10030d37: je near ptr 10030DCBh
0x10030d3d: mov eax,[esp+0Ch]
0x10030d41: add eax,0FFFFFFFEh
0x10030d44: cmp eax,3
0x10030d47: ja short 10030D79h
0x10030d49: jmp dword ptr [eax*4+10030DD0h]
0x10030d79: and dword ptr [esi+70h],0FFFFF3FFh
0x10030d80: mov eax,[esp+10h]
0x10030d84: add eax,0FFFFFFFFh
0x10030d87: cmp eax,4
0x10030d8a: ja short 10030DC4h
0x10030d8c: jmp dword ptr [eax*4+10030DE0h]
0x10030dc4: and dword ptr [esi+70h],0FFFFFCFFh
0x10030dcb: pop esi
0x10030dcc: ret
```

`png_set_crc_action` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.776 png_set_background_fixed at 0x10030e00 (35 instructions)

```
0x10030e00: mov eax,[esp+4]
0x10030e04: test eax,eax
```

```
0x10030e06: je short 10030E71h
0x10030e08: push ebx
0x10030e09: mov ebx,[esp+10h]
0x10030e0d: test ebx,ebx
0x10030e0f: jne short 10030E21h
0x10030e11: push 10101A10h
0x10030e16: push eax
0x10030e17: call 100230F0h
0x10030e1c: add esp,8
0x10030e1f: pop ebx
0x10030e20: ret
0x10030e21: or dword ptr [eax+74h],80h
0x10030e28: mov ecx,[esp+0Ch]
0x10030e2c: mov edx,[eax+74h]
0x10030e2f: push esi
0x10030e30: mov esi,[ecx]
0x10030e32: mov [eax+138h],esi
0x10030e38: mov esi,[ecx+4]
0x10030e3b: mov [eax+13Ch],esi
0x10030e41: mov cx,[ecx+8]
0x10030e45: mov [eax+140h],cx
0x10030e4c: mov ecx,[esp+1Ch]
0x10030e50: mov [eax+134h],ecx
0x10030e56: mov ecx,[esp+18h]
0x10030e5a: neg ecx
0x10030e5c: sbb ecx,ecx
0x10030e5e: and ecx,100h
0x10030e64: or ecx,edx
0x10030e66: pop esi
0x10030e67: mov [eax+130h],bl
0x10030e6d: mov [eax+74h],ecx
0x10030e70: pop ebx
0x10030e71: ret
```

`png_set_background_fixed` is called by png_set_background. It delegates to png_warning (1 callees).

12.777 png_set_background at 0x10030e80 (20 instructions)

```
0x10030e80: fld qword ptr [esp+14h]
0x10030e84: push esi
0x10030e85: mov esi,[esp+8]
0x10030e89: push 10101A44h
0x10030e8e: sub esp,8
0x10030e91: fstp qword ptr [esp]
0x10030e94: push esi
0x10030e95: call 10024710h
0x10030e9a: mov ecx,[esp+20h]
0x10030e9e: mov edx,[esp+1Ch]
0x10030ea2: push eax
0x10030ea3: mov eax,[esp+28h]
0x10030ea7: push eax
0x10030ea8: push ecx
0x10030ea9: push edx
0x10030eaa: push esi
0x10030eab: call 10030E00h
0x10030eb0: add esp,24h
0x10030eb3: pop esi
0x10030eb4: ret
```

`png_set_background` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10024710, png_set_background_fixed (2 callees).

12.778 png_set_strip_16 at 0x10030ec0 (5 instructions)

```
0x10030ec0: mov eax,[esp+4]
0x10030ec4: test eax,eax
0x10030ec6: je short 10030ECFh
0x10030ec8: or dword ptr [eax+74h],400h
0x10030ecf: ret
```

`png_set_strip_16` is called by png_read_png. It calls nothing else.

12.779 png_set_strip_alpha at 0x10030ed0 (5 instructions)

```
0x10030ed0: mov eax,[esp+4]
0x10030ed4: test eax,eax
0x10030ed6: je short 10030EDFh
0x10030ed8: or dword ptr [eax+70h],400000h
0x10030edf: ret
```

`png_set_strip_alpha` is called by png_read_png. It calls nothing else.

12.780 png_set_quantize at 0x10030ee0 (631 instructions)

```
0x10030ee0: sub esp,48h
0x10030ee3: push edi
0x10030ee4: mov edi,[esp+50h]
0x10030ee8: test edi,edi
0x10030eea: je near ptr 1003177Bh
0x10030ef0: or dword ptr [edi+74h],40h
0x10030ef4: cmp dword ptr [esp+64h],0
0x10030ef9: push esi
0x10030efa: mov esi,[esp+5Ch]
0x10030efe: jne short 10030F30h
0x10030f00: push esi
0x10030f01: push edi
0x10030f02: call 1002CD20h
0x10030f07: mov [edi+1F0h],eax
0x10030f0d: add esp,8
0x10030f10: xor eax,eax
0x10030f12: test esi,esi
0x10030f14: jle short 10030F30h
0x10030f16: jmp short 10030F20h
0x10030f20: mov ecx,[edi+1F0h]
0x10030f26: mov [eax+ecx],al
0x10030f29: add eax,1
0x10030f2c: cmp eax,esi
0x10030f2e: jl short 10030F20h
0x10030f30: push ebx
0x10030f31: mov ebx,[esp+5Ch]
0x10030f35: push ebp
0x10030f36: mov ebp,[esp+68h]
0x10030f3a: cmp esi,ebp
0x10030f3c: jle near ptr 1003158Fh
0x10030f42: cmp dword ptr [esp+6Ch],0
0x10030f47: push esi
0x10030f48: push edi
0x10030f49: je near ptr 1003121Ch
0x10030f4f: call 1002CD20h
0x10030f54: mov [edi+248h],eax
0x10030f5a: add esp,8
0x10030f5d: xor eax,eax
0x10030f5f: test esi,esi
0x10030f61: jle short 10030F73h
0x10030f63: mov edx,[edi+248h]
0x10030f69: mov [eax+edx],al
0x10030f6c: add eax,1
0x10030f6f: cmp eax,esi
```

```
0x10030f71: jl short 10030F63h
0x10030f73: lea eax,[esi-1]
0x10030f76: cmp eax,ebp
0x10030f78: mov [esp+14h],eax
0x10030f7c: jl short 10030FF8h
0x10030f7e: mov esi,[esp+6Ch]
0x10030f82: jmp short 10030F88h
0x10030f84: mov ebx,[esp+60h]
0x10030f88: xor ecx,ecx
0x10030f8a: test eax,eax
0x10030f8c: mov dword ptr [esp+34h],1
0x10030f94: jle short 10030FF8h
0x10030f96: jmp short 10030FA0h
0x10030fa0: mov eax,[edi+248h]
0x10030fa6: mov dl,[eax+ecx+1]
0x10030faa: movzx ebx,byte ptr [eax+ecx]
0x10030fae: mov bx,[esi+ebx*2]
0x10030fb2: add eax,ecx
0x10030fb4: movzx edi,dl
0x10030fb7: cmp bx,[esi+edi*2]
0x10030fbb: mov edi,[esp+5Ch]
0x10030fbf: jae short 10030FD7h
0x10030fc1: mov bl,[eax]
0x10030fc3: mov [eax],dl
0x10030fc5: mov edx,[edi+248h]
0x10030fcb: mov [ecx+edx+1],bl
0x10030fcf: mov dword ptr [esp+34h],0
0x10030fd7: mov eax,[esp+14h]
0x10030fdb: add ecx,1
0x10030fde: cmp ecx,eax
0x10030fe0: jl short 10030FA0h
0x10030fe2: cmp dword ptr [esp+34h],0
0x10030fe7: jne short 10030FF4h
0x10030fe9: sub eax,1
0x10030fec: cmp eax,ebp
0x10030fee: mov [esp+14h],eax
0x10030ff2: jge short 10030F84h
0x10030ff4: mov ebx,[esp+60h]
0x10030ff8: cmp dword ptr [esp+70h],0
0x10030ffd: mov eax,[esp+64h]
0x10031001: je short 1003106Bh
0x10031003: xor esi,esi
0x10031005: test ebp,ebp
0x10031007: mov [esp+14h],esi
0x1003100b: jle near ptr 100311FDh
0x10031011: mov [esp+1Ch],ebx
0x10031015: jmp short 10031020h
0x10031020: mov ecx,[edi+248h]
0x10031026: movzx edx,byte ptr [ecx+esi]
0x1003102a: cmp edx,ebp
0x1003102c: jl short 10031056h
0x1003102e: mov edi,edi
0x10031030: movzx edx,byte ptr [ecx+eax-1]
0x10031035: sub eax,1
0x10031038: cmp edx,ebp
0x1003103a: jge short 10031030h
0x1003103c: mov ecx,[esp+1Ch]
0x10031040: lea esi,[ebx+eax*2]
0x10031043: mov dx,[esi+eax]
0x10031047: add esi,eax
0x10031049: mov [ecx],dx
0x1003104c: mov dl,[esi+2]
0x1003104f: mov esi,[esp+14h]
0x10031053: mov [ecx+2],dl
0x10031056: add dword ptr [esp+1Ch],3
0x1003105b: add esi,1
0x1003105e: cmp esi,ebp
```

```
0x10031060: mov [esp+14h],esi
0x10031064: jl short 10031020h
0x10031066: jmp near ptr 100311FDh
0x1003106b: xor ecx,ecx
0x1003106d: test ebp,ebp
0x1003106f: mov [esp+14h],ecx
0x10031073: jle near ptr 100310F9h
0x10031079: mov esi,ebx
0x1003107b: jmp short 10031080h
0x10031080: mov edx,[edi+248h]
0x10031086: movzx edx,byte ptr [edx+ecx]
0x1003108a: cmp edx,ebp
0x1003108c: jl short 100310EBh
0x1003108e: mov edi,edi
0x10031090: mov ecx,[edi+248h]
0x10031096: movzx ecx,byte ptr [ecx+eax-1]
0x1003109b: sub eax,1
0x1003109e: cmp ecx,ebp
0x100310a0: jge short 10031090h
0x100310a2: lea edx,[ebx+eax*2]
0x100310a5: lea ecx,[eax+edx]
0x100310a8: movzx edx,word ptr [ecx]
0x100310ab: mov [esp+10h],dx
0x100310b0: movzx edx,byte ptr [ecx+2]
0x100310b4: mov [esp+12h],dl
0x100310b8: movzx edx,word ptr [esi]
0x100310bb: mov [ecx],dx
0x100310be: movzx edx,byte ptr [esi+2]
0x100310c2: mov [ecx+2],dl
0x100310c5: movzx edx,byte ptr [esp+12h]
0x100310ca: mov cx,[esp+10h]
0x100310cf: mov [esi],cx
0x100310d2: mov ecx,[esp+14h]
0x100310d6: mov [esi+2],dl
0x100310d9: mov edx,[edi+1F0h]
0x100310df: mov [eax+edx],cl
0x100310e2: mov edx,[edi+1F0h]
0x100310e8: mov [ecx+edx],al
0x100310eb: add ecx,1
0x100310ee: add esi,3
0x100310f1: cmp ecx,ebp
0x100310f3: mov [esp+14h],ecx
0x100310f7: jl short 10031080h
0x100310f9: xor ecx,ecx
0x100310fb: cmp [esp+64h],ecx
0x100310ff: mov [esp+14h],ecx
0x10031103: jle near ptr 100311FDh
0x10031109: lea esp,[esp]
0x10031110: mov eax,[edi+1F0h]
0x10031116: add eax,ecx
0x10031118: mov [esp+30h],eax
0x1003111c: movzx eax,byte ptr [eax]
0x1003111f: cmp eax,ebp
0x10031121: jl near ptr 100311ECh
0x10031127: movzx esi,byte ptr [ebx+1]
0x1003112b: lea ecx,[ebx+eax*2]
0x1003112e: movzx edx,byte ptr [eax+ecx+1]
0x10031133: movzx edi,byte ptr [eax+ecx]
0x10031137: add eax,ecx
0x10031139: movzx ecx,byte ptr [eax+2]
0x1003113d: mov eax,edx
0x1003113f: sub eax,esi
0x10031141: mov [esp+38h],edx
0x10031145: cdq
0x10031146: mov esi,eax
0x10031148: xor esi,edx
0x1003114a: sub esi,edx
```

```
0x1003114c: movzx edx,byte ptr [ebx+2]
0x10031150: mov eax,ecx
0x10031152: sub eax,edx
0x10031154: cdq
0x10031155: xor eax,edx
0x10031157: sub eax,edx
0x10031159: add esi,eax
0x1003115b: mov [esp+3Ch],ecx
0x1003115f: movzx ecx,byte ptr [ebx]
0x10031162: mov eax,edi
0x10031164: sub eax,ecx
0x10031166: cdq
0x10031167: xor eax,edx
0x10031169: sub eax,edx
0x1003116b: mov ebx,1
0x10031170: add esi,eax
0x10031172: cmp ebp,ebx
0x10031174: mov [esp+40h],edi
0x10031178: mov dword ptr [esp+34h],0
0x10031180: jle short 100311D6h
0x10031182: mov edi,[esp+60h]
0x10031186: add edi,4
0x10031189: lea esp,[esp]
0x10031190: movzx edx,byte ptr [edi-1]
0x10031194: mov eax,[esp+40h]
0x10031198: sub eax,edx
0x1003119a: cdq
0x1003119b: mov ecx,eax
0x1003119d: mov eax,[esp+3Ch]
0x100311a1: xor ecx,edx
0x100311a3: sub ecx,edx
0x100311a5: movzx edx,byte ptr [edi+1]
0x100311a9: sub eax,edx
0x100311ab: cdq
0x100311ac: xor eax,edx
0x100311ae: sub eax,edx
0x100311b0: movzx edx,byte ptr [edi]
0x100311b3: add ecx,eax
0x100311b5: mov eax,[esp+38h]
0x100311b9: sub eax,edx
0x100311bb: cdq
0x100311bc: xor eax,edx
0x100311be: sub eax,edx
0x100311c0: add ecx,eax
0x100311c2: cmp ecx,esi
0x100311c4: jge short 100311CCh
0x100311c6: mov esi,ecx
0x100311c8: mov [esp+34h],ebx
0x100311cc: add ebx,1
0x100311cf: add edi,3
0x100311d2: cmp ebx,ebp
0x100311d4: jl short 10031190h
0x100311d6: mov cl,[esp+34h]
0x100311da: mov eax,[esp+30h]
0x100311de: mov ebx,[esp+60h]
0x100311e2: mov edi,[esp+5Ch]
0x100311e6: mov [eax],cl
0x100311e8: mov ecx,[esp+14h]
0x100311ec: add ecx,1
0x100311ef: cmp ecx,[esp+64h]
0x100311f3: mov [esp+14h],ecx
0x100311f7: jl near ptr 10031110h
0x100311fd: mov edx,[edi+248h]
0x10031203: push edx
0x10031204: push edi
0x10031205: call 1002CD70h
0x1003120a: add esp,8
```

```
0x1003120d: mov dword ptr [edi+248h],0
0x10031217: jmp near ptr 10031589h
0x1003121c: mov dword ptr [esp+18h],0
0x10031224: call 1002CD20h
0x10031229: push esi
0x1003122a: push edi
0x1003122b: mov [edi+24Ch],eax
0x10031231: call 1002CD20h
0x10031236: mov [edi+250h],eax
0x1003123c: add esp,10h
0x1003123f: xor eax,eax
0x10031241: test esi,esi
0x10031243: jle short 10031269h
0x10031245: jmp short 10031250h
0x10031250: mov ecx,[edi+24Ch]
0x10031256: mov [eax+ecx],al
0x10031259: mov edx,[edi+250h]
0x1003125f: mov [eax+edx],al
0x10031262: add eax,1
0x10031265: cmp eax,esi
0x10031267: jl short 10031250h
0x10031269: push 0C04h
0x1003126e: push edi
0x1003126f: call 1002CDD0h
0x10031274: add esp,8
0x10031277: mov [esp+1Ch],eax
0x1003127b: mov ebx,esi
0x1003127d: mov dword ptr [esp+18h],60h
0x10031285: jmp short 10031290h
0x10031290: lea eax,[ebx-1]
0x10031293: test eax,eax
0x10031295: mov dword ptr [esp+14h],0
0x1003129d: jle near ptr 1003137Dh
0x100312a3: mov eax,[esp+60h]
0x100312a7: add eax,2
0x100312aa: mov [esp+24h],eax
0x100312ae: mov ebp,eax
0x100312b0: mov eax,[esp+14h]
0x100312b4: mov ecx,[esp+10h]
0x100312b8: add eax,1
0x100312bb: cmp eax,ebx
0x100312bd: mov [esp+40h],eax
0x100312c1: mov [esp+20h],eax
0x100312c5: jge near ptr 1003135Fh
0x100312cb: lea edi,[ebp+2]
0x100312ce: mov edi,edi
0x100312d0: movzx edx,byte ptr [edi+1]
0x100312d4: movzx eax,byte ptr [ebp]
0x100312d8: sub eax,edx
0x100312da: cdq
0x100312db: mov esi,eax
0x100312dd: movzx eax,byte ptr [ebp-2]
0x100312e1: xor esi,edx
0x100312e3: sub esi,edx
0x100312e5: movzx edx,byte ptr [edi-1]
0x100312e9: sub eax,edx
0x100312eb: cdq
0x100312ec: xor eax,edx
0x100312ee: sub eax,edx
0x100312f0: movzx edx,byte ptr [edi]
0x100312f3: add esi,eax
0x100312f5: movzx eax,byte ptr [ebp-1]
0x100312f9: sub eax,edx
0x100312fb: cdq
0x100312fc: xor eax,edx
0x100312fe: sub eax,edx
0x10031300: add esi,eax
```

```
0x10031302: cmp esi,[esp+18h]
0x10031306: jg short 10031345h
0x10031308: mov eax,[esp+5Ch]
0x1003130c: push 8
0x1003130e: push eax
0x1003130f: call 1002CDA0h
0x10031314: add esp,8
0x10031317: test eax,eax
0x10031319: mov [esp+10h],eax
0x1003131d: je near ptr 100314FEh
0x10031323: mov ebp,[esp+1Ch]
0x10031327: mov edx,[ebp+esi*4]
0x1003132b: mov ecx,eax
0x1003132d: mov al,[esp+14h]
0x10031331: mov [ecx],edx
0x10031333: mov dl,[esp+20h]
0x10031337: mov [ecx+4],al
0x1003133a: mov [ecx+5],dl
0x1003133d: mov [ebp+esi*4],ecx
0x10031341: mov ebp,[esp+24h]
0x10031345: mov eax,[esp+20h]
0x10031349: add eax,1
0x1003134c: add edi,3
0x1003134f: cmp eax,ebx
0x10031351: mov [esp+20h],eax
0x10031355: jl near ptr 100312D0h
0x1003135b: mov eax,[esp+40h]
0x1003135f: test ecx,ecx
0x10031361: je near ptr 100314FEh
0x10031367: add ebp,3
0x1003136a: lea ecx,[ebx-1]
0x1003136d: cmp eax,ecx
0x1003136f: mov [esp+14h],eax
0x10031373: mov [esp+24h],ebp
0x10031377: jl near ptr 100312B0h
0x1003137d: cmp dword ptr [esp+10h],0
0x10031382: je near ptr 100314FEh
0x10031388: xor eax,eax
0x1003138a: cmp [esp+18h],eax
0x1003138e: mov [esp+14h],eax
0x10031392: jl near ptr 100314FEh
0x10031398: jmp short 100313A0h
0x100313a0: mov ecx,[esp+1Ch]
0x100313a4: mov ecx,[ecx+eax*4]
0x100313a7: test ecx,ecx
0x100313a9: je near ptr 100314EDh
0x100313af: mov edx,[esp+60h]
0x100313b3: lea esi,[edx+ebx*2]
0x100313b6: mov [esp+30h],ecx
0x100313ba: add esi,ebx
0x100313bc: lea esp,[esp]
0x100313c0: mov eax,[esp+5Ch]
0x100313c4: movzx ebp,byte ptr [ecx+4]
0x100313c8: mov edx,[eax+24Ch]
0x100313ce: movzx eax,byte ptr [edx+ebp]
0x100313d2: cmp eax,ebx
0x100313d4: jge near ptr 100314CFh
0x100313da: movzx edi,byte ptr [ecx+5]
0x100313de: movzx eax,byte ptr [edx+edi]
0x100313e2: cmp eax,ebx
0x100313e4: jge near ptr 100314CFh
0x100313ea: test bl,1
0x100313ed: je short 100313F7h
0x100313ef: mov ecx,ebp
0x100313f1: mov [esp+34h],edi
0x100313f5: jmp short 100313FDh
0x100313f7: mov ecx,edi
```

```
0x100313f9: mov [esp+34h],ebp
0x100313fd: movzx edx,byte ptr [edx+ecx]
0x10031401: mov eax,[esp+60h]
0x10031405: sub esi,3
0x10031408: lea eax,[eax+edx*2]
0x1003140b: add eax,edx
0x1003140d: mov dx,[esi]
0x10031410: mov [eax],dx
0x10031413: mov dl,[esi+2]
0x10031416: sub ebx,1
0x10031419: cmp dword ptr [esp+70h],0
0x1003141e: mov [esp+40h],esi
0x10031422: mov [eax+2],dl
0x10031425: jne short 10031485h
0x10031427: xor esi,esi
0x10031429: cmp [esp+64h],esi
0x1003142d: jle short 10031485h
0x1003142f: mov eax,[esp+5Ch]
0x10031433: jmp short 10031440h
0x10031440: mov edx,[eax+1F0h]
0x10031446: mov edi,[eax+24Ch]
0x1003144c: lea ebp,[edx+esi]
0x1003144f: mov dl,[ebp]
0x10031452: cmp dl,[edi+ecx]
0x10031455: jne short 10031461h
0x10031457: mov edx,[esp+34h]
0x1003145b: mov dl,[edi+edx]
0x1003145e: mov [ebp],dl
0x10031461: mov edx,[eax+1F0h]
0x10031467: lea edi,[edx+esi]
0x1003146a: movzx edx,byte ptr [edi]
0x1003146d: cmp edx,ebx
0x1003146f: jne short 1003147Ch
0x10031471: mov edx,[eax+24Ch]
0x10031477: mov dl,[ecx+edx]
0x1003147a: mov [edi],dl
0x1003147c: add esi,1
0x1003147f: cmp esi,[esp+64h]
0x10031483: jl short 10031440h
0x10031485: mov esi,[esp+5Ch]
0x10031489: mov eax,[esi+250h]
0x1003148f: movzx edx,byte ptr [eax+ebx]
0x10031493: mov edi,[esi+24Ch]
0x10031499: mov al,[edi+ecx]
0x1003149c: mov [edx+edi],al
0x1003149f: mov edx,[esi+24Ch]
0x100314a5: movzx eax,byte ptr [edx+ecx]
0x100314a9: mov edi,[esi+250h]
0x100314af: mov dl,[edi+ebx]
0x100314b2: mov [eax+edi],dl
0x100314b5: mov eax,[esi+24Ch]
0x100314bb: mov [ecx+eax],bl
0x100314be: mov edx,[esi+250h]
0x100314c4: mov esi,[esp+40h]
0x100314c8: mov [ebx+edx],cl
0x100314cb: mov ecx,[esp+30h]
0x100314cf: mov eax,[esp+68h]
0x100314d3: cmp ebx,eax
0x100314d5: jle short 100314FEh
0x100314d7: mov ecx,[ecx]
0x100314d9: test ecx,ecx
0x100314db: mov [esp+30h],ecx
0x100314df: jne near ptr 100313C0h
0x100314e5: cmp ebx,eax
0x100314e7: jle short 100314FEh
0x100314e9: mov eax,[esp+14h]
0x100314ed: add eax,1
```

```
0x100314f0: cmp eax,[esp+18h]
0x100314f4: mov [esp+14h],eax
0x100314f8: jle near ptr 100313A0h
0x100314fe: mov edi,[esp+5Ch]
0x10031502: xor ebp,ebp
0x10031504: mov ecx,[esp+1Ch]
0x10031508: mov eax,[ecx+ebp*4]
0x1003150b: test eax,eax
0x1003150d: je short 1003152Ah
0x1003150f: nop
0x10031510: mov esi,[eax]
0x10031512: push eax
0x10031513: push edi
0x10031514: mov [esp+18h],esi
0x10031518: call 1002CD70h
0x1003151d: add esp,8
0x10031520: test esi,esi
0x10031522: mov eax,esi
0x10031524: jne short 10031510h
0x10031526: mov ecx,[esp+1Ch]
0x1003152a: mov dword ptr [ecx+ebp*4],0
0x10031531: add ebp,1
0x10031534: cmp ebp,301h
0x1003153a: jl short 10031504h
0x1003153c: mov ebp,[esp+68h]
0x10031540: add dword ptr [esp+18h],60h
0x10031545: cmp ebx,ebp
0x10031547: jg near ptr 10031290h
0x1003154d: push ecx
0x1003154e: push edi
0x1003154f: call 1002CD70h
0x10031554: mov eax,[edi+250h]
0x1003155a: push eax
0x1003155b: push edi
0x1003155c: call 1002CD70h
0x10031561: mov ecx,[edi+24Ch]
0x10031567: push ecx
0x10031568: push edi
0x10031569: call 1002CD70h
0x1003156e: mov ebx,[esp+78h]
0x10031572: add esp,18h
0x10031575: mov dword ptr [edi+250h],0
0x1003157f: mov dword ptr [edi+24Ch],0
0x10031589: mov [esp+64h],ebp
0x1003158d: mov esi,ebp
0x1003158f: cmp dword ptr [edi+114h],0
0x10031596: jne short 1003159Eh
0x10031598: mov [edi+114h],ebx
0x1003159e: cmp dword ptr [esp+70h],0
0x100315a3: mov [edi+118h],si
0x100315aa: je near ptr 10031778h
0x100315b0: push 8000h
0x100315b5: push edi
0x100315b6: call 1002CDD0h
0x100315bb: push 8000h
0x100315c0: push edi
0x100315c1: mov [edi+1ECh],eax
0x100315c7: call 1002CD20h
0x100315cc: push 8000h
0x100315d1: mov edi,eax
0x100315d3: push 0FFh
0x100315d8: push edi
0x100315d9: mov [esp+50h],edi
0x100315dd: call 100228C0h
0x100315e2: add esp,1Ch
0x100315e5: test esi,esi
0x100315e7: mov dword ptr [esp+2Ch],0
```

```
0x100315ef: jle near ptr 1003176Ah
0x100315f5: mov edx,[esp+60h]
0x100315f9: add edx,2
0x100315fc: mov [esp+30h],edx
0x10031600: mov edx,[esp+30h]
0x10031604: movzx ecx,byte ptr [edx-1]
0x10031608: movzx eax,byte ptr [edx-2]
0x1003160c: movzx edx,byte ptr [edx]
0x1003160f: shr ecx,3
0x10031612: mov [esp+38h],ecx
0x10031616: shr eax,3
0x10031619: neg ecx
0x1003161b: shr edx,3
0x1003161e: mov [esp+48h],ecx
0x10031622: mov [esp+40h],edx
0x10031626: neg edx
0x10031628: mov ecx,eax
0x1003162a: neg ecx
0x1003162c: mov [esp+44h],eax
0x10031630: mov dword ptr [esp+14h],0
0x10031638: mov [esp+50h],edx
0x1003163c: mov [esp+10h],ecx
0x10031640: mov [esp+28h],eax
0x10031644: mov eax,[esp+14h]
0x10031648: cmp eax,[esp+44h]
0x1003164c: jle short 10031658h
0x1003164e: mov ecx,[esp+10h]
0x10031652: mov [esp+3Ch],ecx
0x10031656: jmp short 10031660h
0x10031658: mov edx,[esp+28h]
0x1003165c: mov [esp+3Ch],edx
0x10031660: mov ecx,[esp+38h]
0x10031664: shl eax,0Ah
0x10031667: mov [esp+4Ch],eax
0x1003166b: mov eax,[esp+48h]
0x1003166f: mov dword ptr [esp+1Ch],0
0x10031677: mov [esp+20h],eax
0x1003167b: mov [esp+18h],ecx
0x1003167f: nop
0x10031680: mov edx,[esp+38h]
0x10031684: cmp [esp+1Ch],edx
0x10031688: mov eax,[esp+20h]
0x1003168c: jg short 10031692h
0x1003168e: mov eax,[esp+18h]
0x10031692: mov ecx,[esp+3Ch]
0x10031696: cmp ecx,eax
0x10031698: lea edx,[eax+ecx]
0x1003169b: mov [esp+54h],edx
0x1003169f: mov [esp+24h],ecx
0x100316a3: jg short 100316A9h
0x100316a5: mov [esp+24h],eax
0x100316a9: mov edi,[esp+1Ch]
0x100316ad: mov ebp,[esp+50h]
0x100316b1: mov ebx,[esp+40h]
0x100316b5: shl edi,5
0x100316b8: or edi,[esp+4Ch]
0x100316bc: xor esi,esi
0x100316be: mov edi,edi
0x100316c0: mov ecx,esi
0x100316c2: or ecx,edi
0x100316c4: cmp esi,[esp+40h]
0x100316c8: mov edx,ebp
0x100316ca: jg short 100316CEh
0x100316cc: mov edx,ebx
0x100316ce: cmp [esp+24h],edx
0x100316d2: mov eax,[esp+24h]
0x100316d6: jg short 100316DAh
```

```
0x100316d8: mov eax,edx
0x100316da: add eax,edx
0x100316dc: mov edx,[esp+34h]
0x100316e0: movzx edx,byte ptr [ecx+edx]
0x100316e4: add eax,[esp+54h]
0x100316e8: cmp eax,edx
0x100316ea: jge short 10031704h
0x100316ec: mov edx,[esp+34h]
0x100316f0: mov [ecx+edx],al
0x100316f3: mov eax,[esp+5Ch]
0x100316f7: mov edx,[eax+1ECh]
0x100316fd: mov al,[esp+2Ch]
0x10031701: mov [ecx+edx],al
0x10031704: add esi,1
0x10031707: add ebp,1
0x1003170a: sub ebx,1
0x1003170d: cmp esi,20h
0x10031710: jl short 100316C0h
0x10031712: mov eax,[esp+1Ch]
0x10031716: mov ecx,1
0x1003171b: add [esp+20h],ecx
0x1003171f: sub [esp+18h],ecx
0x10031723: add eax,ecx
0x10031725: cmp eax,20h
0x10031728: mov [esp+1Ch],eax
0x1003172c: jl near ptr 10031680h
0x10031732: mov eax,[esp+14h]
0x10031736: add [esp+10h],ecx
0x1003173a: sub [esp+28h],ecx
0x1003173e: add eax,ecx
0x10031740: cmp eax,20h
0x10031743: mov [esp+14h],eax
0x10031747: jl near ptr 10031644h
0x1003174d: mov eax,[esp+2Ch]
0x10031751: add dword ptr [esp+30h],3
0x10031756: add eax,ecx
0x10031758: cmp eax,[esp+64h]
0x1003175c: mov [esp+2Ch],eax
0x10031760: jl near ptr 10031600h
0x10031766: mov edi,[esp+34h]
0x1003176a: mov ecx,[esp+5Ch]
0x1003176e: push edi
0x1003176f: push ecx
0x10031770: call 1002CD70h
0x10031775: add esp,8
0x10031778: pop ebp
0x10031779: pop ebx
0x1003177a: pop esi
0x1003177b: pop edi
0x1003177c: add esp,48h
0x1003177f: ret
```

`png_set_quantize` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to j_memset, png_malloc, png_free (5 callees).

12.781 sub_10031780 at 0x10031780 (21 instructions)

```
0x10031780: push ecx
0x10031781: push 186A0h
0x10031786: push eax
0x10031787: push ecx
0x10031788: lea edx,[esp+0Ch]
0x1003178c: push edx
0x1003178d: call 10024770h
0x10031792: add esp,10h
0x10031795: test eax,eax
```

```
0x10031797: je short 100317ABh
0x10031799: mov eax,[esp]
0x1003179c: push eax
0x1003179d: call 10024AE0h
0x100317a2: add esp,4
0x100317a5: test eax,eax
0x100317a7: jne short 100317ABh
0x100317a9: pop ecx
0x100317aa: ret
0x100317ab: mov eax,1
0x100317b0: pop ecx
0x100317b1: ret
```

`sub_10031780` is called by png_set_gamma_fixed, sub_10031a20. It delegates to sub_10024770, sub_10024ae0 (2 callees).

12.782 png_set_gamma_fixed at 0x100317c0 (25 instructions)

```
0x100317c0: push esi
0x100317c1: mov esi,[esp+8]
0x100317c5: test esi,esi
0x100317c7: je short 10031803h
0x100317c9: mov al,[esi+126h]
0x100317cf: test al,4
0x100317d1: push ebx
0x100317d2: mov ebx,[esp+10h]
0x100317d6: push edi
0x100317d7: mov edi,[esp+18h]
0x100317db: jne short 100317EEh
0x100317dd: cmp al,3
0x100317df: je short 100317EEh
0x100317e1: mov eax,edi
0x100317e3: mov ecx,ebx
0x100317e5: call 10031780h
0x100317ea: test eax,eax
0x100317ec: je short 100317F5h
0x100317ee: or dword ptr [esi+74h],2000h
0x100317f5: mov [esi+15Ch],edi
0x100317fb: pop edi
0x100317fc: mov [esi+160h],ebx
0x10031802: pop ebx
0x10031803: pop esi
0x10031804: ret
```

`png_set_gamma_fixed` is called by png_set_gamma. It delegates to sub_10031780 (1 callees).

12.783 png_set_gamma at 0x10031810 (23 instructions)

```
0x10031810: fld qword ptr [esp+10h]
0x10031814: push esi
0x10031815: mov esi,[esp+8]
0x10031819: push 10101A74h
0x1003181e: sub esp,8
0x10031821: fstp qword ptr [esp]
0x10031824: push esi
0x10031825: call 10024710h
0x1003182a: fld qword ptr [esp+1Ch]
0x1003182e: add esp,10h
0x10031831: push eax
0x10031832: push 10101A58h
0x10031837: sub esp,8
0x1003183a: fstp qword ptr [esp]
0x1003183d: push esi
0x1003183e: call 10024710h
0x10031843: add esp,10h
```

```
0x10031846: push eax
0x10031847: push esi
0x10031848: call 100317C0h
0x1003184d: add esp,0Ch
0x10031850: pop esi
0x10031851: ret
```

`png_set_gamma` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "png_set_gamma screen gamma", "png_set_gamma file gamma". It delegates to sub_10024710, png_set_gamma_fixed (2 callees).

12.784 png_set_palette_to_rgb at 0x10031860 (6 instructions)

```
0x10031860: mov eax,[esp+4]
0x10031864: test eax,eax
0x10031866: je short 10031873h
0x10031868: or dword ptr [eax+74h],2001000h
0x1003186f: and dword ptr [eax+70h],0FFFFFFBFh
0x10031873: ret
```

`png_set_palette_to_rgb` is called by png_read_png. It calls nothing else.

12.785 png_set_expand_gray_1_2_4_to_8 at 0x10031880 (6 instructions)

```
0x10031880: mov eax,[esp+4]
0x10031884: test eax,eax
0x10031886: je short 10031893h
0x10031888: or dword ptr [eax+74h],1000h
0x1003188f: and dword ptr [eax+70h],0FFFFFFBFh
0x10031893: ret
```

`png_set_expand_gray_1_2_4_to_8` is called by png_set_gray_to_rgb. It calls nothing else.

12.786 png_set_tRNS_to_alpha at 0x100318a0 (4 instructions)

```
0x100318a0: mov eax,[esp+4]
0x100318a4: or dword ptr [eax+74h],2001000h
0x100318ab: and dword ptr [eax+70h],0FFFFFFBFh
0x100318af: ret
```

`png_set_tRNS_to_alpha` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.787 png_set_gray_to_rgb at 0x100318b0 (9 instructions)

```
0x100318b0: mov ecx,[esp+4]
0x100318b4: test ecx,ecx
0x100318b6: je short 100318CCh
0x100318b8: push ecx
0x100318b9: call 10031880h
0x100318be: or dword ptr [ecx+74h],4000h
0x100318c5: add esp,4
0x100318c8: and dword ptr [ecx+70h],0FFFFFFBFh
0x100318cc: ret
```

`png_set_gray_to_rgb` is called by png_read_png. It delegates to png_set_expand_gray_1_2_4_to_8 (1 callees).

12.788 png_set_rgb_to_gray_fixed at 0x100318d0 (67 instructions)

```
0x100318d0: push ebx
0x100318d1: mov ebx,[esp+8]
```

```
0x100318d5: test ebx,ebx
0x100318d7: je near ptr 100319A8h
0x100318dd: mov eax,[esp+0Ch]
0x100318e1: sub eax,1
0x100318e4: je short 1003190Dh
0x100318e6: sub eax,1
0x100318e9: je short 10031904h
0x100318eb: sub eax,1
0x100318ee: je short 100318FBh
0x100318f0: push 10101AC0h
0x100318f5: push ebx
0x100318f6: call 100231E0h
0x100318fb: or dword ptr [ebx+74h],200000h
0x10031902: jmp short 10031914h
0x10031904: or dword ptr [ebx+74h],400000h
0x1003190b: jmp short 10031914h
0x1003190d: or dword ptr [ebx+74h],600000h
0x10031914: cmp byte ptr [ebx+126h],3
0x1003191b: jne short 10031924h
0x1003191d: or dword ptr [ebx+74h],1000h
0x10031924: mov ecx,[esp+10h]
0x10031928: test ecx,ecx
0x1003192a: push esi
0x1003192b: push edi
0x1003192c: jl short 1003197Eh
0x1003192e: mov esi,[esp+1Ch]
0x10031932: test esi,esi
0x10031934: jl short 1003197Eh
0x10031936: lea eax,[ecx+esi]
0x10031939: cmp eax,186A0h
0x1003193e: jge short 10031970h
0x10031940: shl ecx,0Fh
0x10031943: mov eax,4F8B588Fh
0x10031948: mul ecx
0x1003194a: sub ecx,edx
0x1003194c: shr ecx,1
0x1003194e: add ecx,edx
0x10031950: shr ecx,10h
0x10031953: movzx edi,cx
0x10031956: shl esi,0Fh
0x10031959: mov ecx,esi
0x1003195b: mov eax,4F8B588Fh
0x10031960: mul ecx
0x10031962: sub ecx,edx
0x10031964: shr ecx,1
0x10031966: add ecx,edx
0x10031968: shr ecx,10h
0x1003196b: movzx eax,cx
0x1003196e: jmp short 10031988h
0x10031970: push 10101A90h
0x10031975: push ebx
0x10031976: call 100230F0h
0x1003197b: add esp,8
0x1003197e: mov eax,5B8Ah
0x10031983: mov edi,1B38h
0x10031988: mov ecx,8000h
0x1003198d: sub ecx,eax
0x1003198f: sub ecx,edi
0x10031991: mov [ebx+22Ah],di
0x10031998: pop edi
0x10031999: mov [ebx+22Ch],ax
0x100319a0: mov [ebx+22Eh],cx
0x100319a7: pop esi
0x100319a8: pop ebx
0x100319a9: ret
```

`png_set_rgb_to_gray_fixed` is called by png_set_rgb_to_gray. It works with the text "ignoring out of range rgb_to_gray coefficients", "invalid error action to rgb_to_gray". It accesses memory-mapped peripheral registers at 0x4f8b588f. It delegates to png_warning, png_error (2 callees).

12.789 png_set_rgb_to_gray at 0x100319b0 (27 instructions)

```
0x100319b0: push esi
0x100319b1: mov esi,[esp+8]
0x100319b5: test esi,esi
0x100319b7: je short 100319F9h
0x100319b9: fld qword ptr [esp+18h]
0x100319bd: push 10101B00h
0x100319c2: sub esp,8
0x100319c5: fstp qword ptr [esp]
0x100319c8: push esi
0x100319c9: call 10024710h
0x100319ce: fld qword ptr [esp+20h]
0x100319d2: add esp,10h
0x100319d5: push eax
0x100319d6: push 10101AE4h
0x100319db: sub esp,8
0x100319de: fstp qword ptr [esp]
0x100319e1: push esi
0x100319e2: call 10024710h
0x100319e7: add esp,10h
0x100319ea: push eax
0x100319eb: mov eax,[esp+14h]
0x100319ef: push eax
0x100319f0: push esi
0x100319f1: call 100318D0h
0x100319f6: add esp,10h
0x100319f9: pop esi
0x100319fa: ret
```

`png_set_rgb_to_gray` has no recorded callers; it is likely reached through the vector table or an indirect jump. It works with the text "rgb to gray red coefficient", "rgb to gray green coefficient". It delegates to sub_10024710, png_set_rgb_to_gray_fixed (2 callees).

12.790 png_set_read_user_transform_fn at 0x10031a00 (7 instructions)

```
0x10031a00: mov eax,[esp+4]
0x10031a04: test eax,eax
0x10031a06: je short 10031A16h
0x10031a08: mov ecx,[esp+8]
0x10031a0c: or dword ptr [eax+74h],100000h
0x10031a13: mov [eax+5Ch],ecx
0x10031a16: ret
```

`png_set_read_user_transform_fn` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.791 sub_10031a20 at 0x10031a20 (640 instructions)

```
0x10031a20: sub esp,10h
0x10031a23: push ebx
0x10031a24: push ebp
0x10031a25: push esi
0x10031a26: mov esi,[esp+20h]
0x10031a2a: mov ecx,[esi+74h]
0x10031a2d: movzx edx,byte ptr [esi+126h]
0x10031a34: mov eax,ecx
0x10031a36: and eax,100h
0x10031a3b: push edi
0x10031a3c: mov [esp+14h],edx
```

```
0x10031a40: je short 10031A47h
0x10031a42: test dl,2
0x10031a45: je short 10031A77h
0x10031a47: test cl,cl
0x10031a49: jns short 10031A7Eh
0x10031a4b: test eax,eax
0x10031a4d: jne short 10031A86h
0x10031a4f: test ecx,4000h
0x10031a55: je short 10031A7Eh
0x10031a57: movzx edx,word ptr [esi+13Ah]
0x10031a5e: cmp dx,[esi+13Ch]
0x10031a65: jne short 10031A7Eh
0x10031a67: cmp dx,[esi+13Eh]
0x10031a6e: jne short 10031A7Eh
0x10031a70: mov [esi+140h],dx
0x10031a77: or dword ptr [esi+6Ch],800h
0x10031a7e: test eax,eax
0x10031a80: je near ptr 10031C28h
0x10031a86: test ecx,1000h
0x10031a8c: je near ptr 10031C28h
0x10031a92: mov edx,[esp+14h]
0x10031a96: test dl,2
0x10031a99: jne near ptr 10031BBBh
0x10031a9f: movzx eax,byte ptr [esi+127h]
0x10031aa6: sub eax,1
0x10031aa9: je near ptr 10031B5Dh
0x10031aaf: sub eax,1
0x10031ab2: je short 10031B1Dh
0x10031ab4: sub eax,2
0x10031ab7: je short 10031ADAh
0x10031ab9: movzx eax,word ptr [esi+140h]
0x10031ac0: mov [esi+13Eh],ax
0x10031ac7: mov [esi+13Ch],ax
0x10031ace: mov [esi+13Ah],ax
0x10031ad5: jmp near ptr 10031C2Ch
0x10031ada: mov ax,[esi+140h]
0x10031ae1: imul ax,11h
0x10031ae5: test ecx,2000000h
0x10031aeb: mov [esi+140h],ax
0x10031af2: mov [esi+13Eh],ax
0x10031af9: mov [esi+13Ch],ax
0x10031b00: mov [esi+13Ah],ax
0x10031b07: jne near ptr 10031C2Ch
0x10031b0d: mov ax,[esi+194h]
0x10031b14: imul ax,11h
0x10031b18: jmp near ptr 10031B9Dh
0x10031b1d: mov ax,[esi+140h]
0x10031b24: imul ax,55h
0x10031b28: test ecx,2000000h
0x10031b2e: mov [esi+140h],ax
0x10031b35: mov [esi+13Eh],ax
0x10031b3c: mov [esi+13Ch],ax
0x10031b43: mov [esi+13Ah],ax
0x10031b4a: jne near ptr 10031C2Ch
0x10031b50: mov ax,[esi+194h]
0x10031b57: imul ax,55h
0x10031b5b: jmp short 10031B9Dh
0x10031b5d: mov ax,[esi+140h]
0x10031b64: imul ax,0FFh
0x10031b69: test ecx,2000000h
0x10031b6f: mov [esi+140h],ax
0x10031b76: mov [esi+13Eh],ax
0x10031b7d: mov [esi+13Ch],ax
0x10031b84: mov [esi+13Ah],ax
0x10031b8b: jne near ptr 10031C2Ch
0x10031b91: mov ax,[esi+194h]
0x10031b98: imul ax,0FFh
```

```
0x10031b9d: mov [esi+194h],ax
0x10031ba4: mov [esi+192h],ax
0x10031bab: mov [esi+190h],ax
0x10031bb2: mov [esi+18Eh],ax
0x10031bb9: jmp short 10031C2Ch
0x10031bbb: cmp edx,3
0x10031bbe: jne short 10031C2Ch
0x10031bc0: movzx eax,byte ptr [esi+138h]
0x10031bc7: mov edi,[esi+114h]
0x10031bcd: lea eax,[eax+eax*2]
0x10031bd0: add eax,edi
0x10031bd2: test ecx,80000h
0x10031bd8: movzx edi,byte ptr [eax]
0x10031bdb: mov [esi+13Ah],di
0x10031be2: movzx edi,byte ptr [eax+1]
0x10031be6: mov [esi+13Ch],di
0x10031bed: movzx ax,byte ptr [eax+2]
0x10031bf2: mov [esi+13Eh],ax
0x10031bf9: je short 10031C2Ch
0x10031bfb: test ecx,2000000h
0x10031c01: jne short 10031C2Ch
0x10031c03: movzx edi,word ptr [esi+11Ah]
0x10031c0a: xor eax,eax
0x10031c0c: test edi,edi
0x10031c0e: jle short 10031C2Ch
0x10031c10: mov ecx,[esi+188h]
0x10031c16: add ecx,eax
0x10031c18: or bl,0FFh
0x10031c1b: sub bl,[ecx]
0x10031c1d: add eax,1
0x10031c20: cmp eax,edi
0x10031c22: mov [ecx],bl
0x10031c24: jl short 10031C10h
0x10031c26: jmp short 10031C2Ch
0x10031c28: mov edx,[esp+14h]
0x10031c2c: cmp edx,3
0x10031c2f: mov eax,[esi+138h]
0x10031c35: mov ecx,[esi+13Ch]
0x10031c3b: mov [esi+142h],eax
0x10031c41: mov ax,[esi+140h]
0x10031c48: mov [esi+146h],ecx
0x10031c4e: mov [esi+14Ah],ax
0x10031c55: mov ebx,0FFFFDFFFh
0x10031c5a: jne short 10031CB3h
0x10031c5c: cmp word ptr [esi+11Ah],0
0x10031c64: je short 10031CB3h
0x10031c66: mov eax,[esi+15Ch]
0x10031c6c: mov ecx,[esi+160h]
0x10031c72: call 10031780h
0x10031c77: test eax,eax
0x10031c79: je short 10031CB3h
0x10031c7b: xor ecx,ecx
0x10031c7d: xor ebp,ebp
0x10031c7f: cmp [esi+11Ah],cx
0x10031c86: jbe short 10031CB0h
0x10031c88: movzx edi,word ptr [esi+11Ah]
0x10031c8f: mov edx,[esi+188h]
0x10031c95: mov al,[edx+ecx]
0x10031c98: test al,al
0x10031c9a: je short 10031CA5h
0x10031c9c: cmp al,0FFh
0x10031c9e: je short 10031CA5h
0x10031ca0: mov ebp,1
0x10031ca5: add ecx,1
0x10031ca8: cmp ecx,edi
0x10031caa: jl short 10031C95h
0x10031cac: test ebp,ebp
```

```
0x10031cae: jne short 10031CB3h
0x10031cb0: and [esi+74h],ebx
0x10031cb3: mov eax,[esi+74h]
0x10031cb6: test eax,602000h
0x10031cbb: je near ptr 100321F3h
0x10031cc1: cmp dword ptr [esi+15Ch],0
0x10031cc8: je near ptr 100321F3h
0x10031cce: movzx ecx,byte ptr [esi+127h]
0x10031cd5: push ecx
0x10031cd6: push esi
0x10031cd7: call 10024E30h
0x10031cdc: add esp,8
0x10031cdf: test byte ptr [esi+74h],80h
0x10031ce3: je near ptr 10032193h
0x10031ce9: cmp dword ptr [esp+14h],3
0x10031cee: jne near ptr 1003200Ch
0x10031cf4: movzx ebp,word ptr [esi+118h]
0x10031cfb: mov al,[esi+130h]
0x10031d01: cmp al,2
0x10031d03: mov edx,[esi+114h]
0x10031d09: mov [esp+18h],edx
0x10031d0d: mov [esp+1Ch],ebp
0x10031d11: jne short 10031D62h
0x10031d13: movzx ecx,word ptr [esi+13Ah]
0x10031d1a: mov eax,[esi+164h]
0x10031d20: movzx edx,byte ptr [eax+ecx]
0x10031d24: movzx edi,word ptr [esi+13Ch]
0x10031d2b: mov [esp+24h],dl
0x10031d2f: movzx edx,byte ptr [edi+eax]
0x10031d33: mov [esp+25h],dl
0x10031d37: movzx edx,word ptr [esi+13Eh]
0x10031d3e: mov al,[edx+eax]
0x10031d41: mov [esp+26h],al
0x10031d45: mov eax,[esi+16Ch]
0x10031d4b: movzx edx,byte ptr [eax+edx]
0x10031d4f: mov bl,[eax+ecx]
0x10031d52: mov cl,[eax+edi]
0x10031d55: mov [esp+11h],cl
0x10031d59: mov [esp+12h],dl
0x10031d5d: jmp near ptr 10031E75h
0x10031d62: movzx eax,al
0x10031d65: sub eax,1
0x10031d68: je short 10031DCDh
0x10031d6a: sub eax,1
0x10031d6d: je short 10031DA5h
0x10031d6f: sub eax,1
0x10031d72: je short 10031D7Dh
0x10031d74: mov edi,186A0h
0x10031d79: mov ebx,edi
0x10031d7b: jmp short 10031DD8h
0x10031d7d: mov eax,[esi+134h]
0x10031d83: push eax
0x10031d84: call 10024830h
0x10031d89: mov ecx,[esi+160h]
0x10031d8f: mov edx,[esi+134h]
0x10031d95: push ecx
0x10031d96: push edx
0x10031d97: mov edi,eax
0x10031d99: call 100248D0h
0x10031d9e: add esp,0Ch
0x10031da1: mov ebx,eax
0x10031da3: jmp short 10031DD8h
0x10031da5: mov eax,[esi+15Ch]
0x10031dab: push eax
0x10031dac: call 10024830h
0x10031db1: mov ecx,[esi+160h]
0x10031db7: mov edx,[esi+15Ch]
```

```
0x10031dbd: push ecx
0x10031dbe: push edx
0x10031dbf: mov edi,eax
0x10031dc1: call 100248D0h
0x10031dc6: add esp,0Ch
0x10031dc9: mov ebx,eax
0x10031dcb: jmp short 10031DD8h
0x10031dcd: mov edi,[esi+160h]
0x10031dd3: mov ebx,186A0h
0x10031dd8: push ebx
0x10031dd9: call 10024AE0h
0x10031dde: add esp,4
0x10031de1: test eax,eax
0x10031de3: je short 10031E05h
0x10031de5: mov al,[esi+13Ah]
0x10031deb: mov cl,[esi+13Ch]
0x10031df1: mov dl,[esi+13Eh]
0x10031df7: mov [esp+24h],al
0x10031dfb: mov [esp+25h],cl
0x10031dff: mov [esp+26h],dl
0x10031e03: jmp short 10031E3Eh
0x10031e05: movzx eax,word ptr [esi+13Ah]
0x10031e0c: push ebx
0x10031e0d: push eax
0x10031e0e: call 100249A0h
0x10031e13: movzx ecx,word ptr [esi+13Ch]
0x10031e1a: push ebx
0x10031e1b: push ecx
0x10031e1c: mov [esp+34h],al
0x10031e20: call 100249A0h
0x10031e25: movzx edx,word ptr [esi+13Eh]
0x10031e2c: push ebx
0x10031e2d: push edx
0x10031e2e: mov [esp+3Dh],al
0x10031e32: call 100249A0h
0x10031e37: add esp,18h
0x10031e3a: mov [esp+26h],al
0x10031e3e: movzx eax,word ptr [esi+13Ah]
0x10031e45: push edi
0x10031e46: push eax
0x10031e47: call 100249A0h
0x10031e4c: movzx ecx,word ptr [esi+13Ch]
0x10031e53: push edi
0x10031e54: push ecx
0x10031e55: mov bl,al
0x10031e57: call 100249A0h
0x10031e5c: movzx edx,word ptr [esi+13Eh]
0x10031e63: push edi
0x10031e64: push edx
0x10031e65: mov [esp+29h],al
0x10031e69: call 100249A0h
0x10031e6e: add esp,18h
0x10031e71: mov [esp+12h],al
0x10031e75: xor edi,edi
0x10031e77: test ebp,ebp
0x10031e79: jle near ptr 10032000h
0x10031e7f: mov ecx,[esp+18h]
0x10031e83: add ecx,2
0x10031e86: jmp short 10031E90h
0x10031e90: movzx eax,word ptr [esi+11Ah]
0x10031e97: cmp edi,eax
0x10031e99: jge near ptr 10031FC2h
0x10031e9f: mov edx,[esi+188h]
0x10031ea5: mov al,[edx+edi]
0x10031ea8: add edx,edi
0x10031eaa: cmp al,0FFh
0x10031eac: je near ptr 10031FC2h
```

```
0x10031eb2: test al,al
0x10031eb4: jne short 10031ECAh
0x10031eb6: mov ax,[esp+24h]
0x10031ebb: mov dl,[esp+26h]
0x10031ebf: mov [ecx-2],ax
0x10031ec3: mov [ecx],dl
0x10031ec5: jmp near ptr 10031FF2h
0x10031eca: movzx dx,byte ptr [edx]
0x10031ece: mov ebp,[esi+16Ch]
0x10031ed4: movzx eax,byte ptr [ecx-2]
0x10031ed8: movzx eax,byte ptr [eax+ebp]
0x10031edc: imul ax,dx
0x10031ee0: mov ebp,0FFh
0x10031ee5: sub ebp,edx
0x10031ee7: movzx dx,bl
0x10031eeb: imul bp,dx
0x10031eef: add ax,bp
0x10031ef2: add ax,80h
0x10031ef6: movzx edx,ax
0x10031ef9: movzx edx,dx
0x10031efc: mov eax,edx
0x10031efe: shr eax,8
0x10031f01: add eax,edx
0x10031f03: sar eax,8
0x10031f06: movzx edx,al
0x10031f09: mov eax,[esi+168h]
0x10031f0f: movzx edx,byte ptr [edx+eax]
0x10031f13: mov [ecx-2],dl
0x10031f16: mov eax,[esi+188h]
0x10031f1c: movzx dx,byte ptr [eax+edi]
0x10031f21: movzx eax,byte ptr [ecx-1]
0x10031f25: mov ebp,[esi+16Ch]
0x10031f2b: movzx eax,byte ptr [eax+ebp]
0x10031f2f: imul ax,dx
0x10031f33: mov ebp,0FFh
0x10031f38: sub ebp,edx
0x10031f3a: movzx edx,byte ptr [esp+11h]
0x10031f3f: imul bp,dx
0x10031f43: add ax,bp
0x10031f46: add ax,80h
0x10031f4a: movzx edx,ax
0x10031f4d: movzx edx,dx
0x10031f50: mov eax,edx
0x10031f52: shr eax,8
0x10031f55: add eax,edx
0x10031f57: sar eax,8
0x10031f5a: movzx edx,al
0x10031f5d: mov eax,[esi+168h]
0x10031f63: movzx edx,byte ptr [edx+eax]
0x10031f67: mov [ecx-1],dl
0x10031f6a: mov eax,[esi+188h]
0x10031f70: movzx dx,byte ptr [eax+edi]
0x10031f75: movzx eax,byte ptr [ecx]
0x10031f78: mov ebp,[esi+16Ch]
0x10031f7e: movzx eax,byte ptr [eax+ebp]
0x10031f82: imul ax,dx
0x10031f86: mov ebp,0FFh
0x10031f8b: sub ebp,edx
0x10031f8d: movzx edx,byte ptr [esp+12h]
0x10031f92: imul bp,dx
0x10031f96: add ax,bp
0x10031f99: mov ebp,[esp+1Ch]
0x10031f9d: add ax,80h
0x10031fa1: movzx edx,ax
0x10031fa4: movzx edx,dx
0x10031fa7: mov eax,edx
0x10031fa9: shr eax,8
```

```
0x10031fac: add eax,edx
0x10031fae: sar eax,8
0x10031fb1: movzx edx,al
0x10031fb4: mov eax,[esi+168h]
0x10031fba: movzx edx,byte ptr [edx+eax]
0x10031fbe: mov [ecx],dl
0x10031fc0: jmp short 10031FF2h
0x10031fc2: mov edx,[esi+164h]
0x10031fc8: movzx eax,byte ptr [ecx-2]
0x10031fcc: movzx eax,byte ptr [eax+edx]
0x10031fd0: movzx edx,byte ptr [ecx-1]
0x10031fd4: mov [ecx-2],al
0x10031fd7: mov eax,[esi+164h]
0x10031fdd: mov dl,[edx+eax]
0x10031fe0: movzx eax,byte ptr [ecx]
0x10031fe3: mov [ecx-1],dl
0x10031fe6: mov edx,[esi+164h]
0x10031fec: movzx eax,byte ptr [eax+edx]
0x10031ff0: mov [ecx],al
0x10031ff2: add edi,1
0x10031ff5: add ecx,3
0x10031ff8: cmp edi,ebp
0x10031ffa: jl near ptr 10031E90h
0x10032000: and dword ptr [esi+74h],0FFFFDF7Fh
0x10032007: jmp near ptr 10032336h
0x1003200c: movzx eax,byte ptr [esi+130h]
0x10032013: sub eax,1
0x10032016: mov edi,186A0h
0x1003201b: je short 10032082h
0x1003201d: sub eax,1
0x10032020: je short 1003205Ah
0x10032022: sub eax,1
0x10032025: je short 10032032h
0x10032027: push 10101B20h
0x1003202c: push esi
0x1003202d: call 100231E0h
0x10032032: mov ecx,[esi+134h]
0x10032038: push ecx
0x10032039: call 10024830h
0x1003203e: mov edx,[esi+160h]
0x10032044: mov ebp,eax
0x10032046: mov eax,[esi+134h]
0x1003204c: push edx
0x1003204d: push eax
0x1003204e: call 100248D0h
0x10032053: add esp,0Ch
0x10032056: mov edi,eax
0x10032058: jmp short 10032088h
0x1003205a: mov ecx,[esi+15Ch]
0x10032060: push ecx
0x10032061: call 10024830h
0x10032066: mov edx,[esi+160h]
0x1003206c: mov ebp,eax
0x1003206e: mov eax,[esi+15Ch]
0x10032074: push edx
0x10032075: push eax
0x10032076: call 100248D0h
0x1003207b: add esp,0Ch
0x1003207e: mov edi,eax
0x10032080: jmp short 10032088h
0x10032082: mov ebp,[esi+160h]
0x10032088: movzx ecx,word ptr [esi+140h]
0x1003208f: push ebp
0x10032090: push ecx
0x10032091: push esi
0x10032092: call 10024AA0h
0x10032097: movzx edx,word ptr [esi+140h]
```

```
0x1003209e: push edi
0x1003209f: push edx
0x100320a0: push esi
0x100320a1: mov [esi+14Ah],ax
0x100320a8: call 10024AA0h
0x100320ad: movzx ecx,word ptr [esi+13Ah]
0x100320b4: add esp,18h
0x100320b7: cmp cx,[esi+13Ch]
0x100320be: mov [esi+140h],ax
0x100320c5: jne short 1003210Bh
0x100320c7: cmp cx,[esi+13Eh]
0x100320ce: jne short 1003210Bh
0x100320d0: cmp cx,ax
0x100320d3: jne short 1003210Bh
0x100320d5: movzx ecx,word ptr [esi+14Ah]
0x100320dc: mov [esi+148h],cx
0x100320e3: mov [esi+146h],cx
0x100320ea: mov [esi+144h],cx
0x100320f1: mov [esi+13Eh],ax
0x100320f8: mov [esi+13Ch],ax
0x100320ff: mov [esi+13Ah],ax
0x10032106: jmp near ptr 1003233Dh
0x1003210b: movzx eax,cx
0x1003210e: push ebp
0x1003210f: push eax
0x10032110: push esi
0x10032111: call 10024AA0h
0x10032116: movzx ecx,word ptr [esi+13Ch]
0x1003211d: push ebp
0x1003211e: push ecx
0x1003211f: push esi
0x10032120: mov [esi+144h],ax
0x10032127: call 10024AA0h
0x1003212c: movzx edx,word ptr [esi+13Eh]
0x10032133: push ebp
0x10032134: push edx
0x10032135: push esi
0x10032136: mov [esi+146h],ax
0x1003213d: call 10024AA0h
0x10032142: mov [esi+148h],ax
0x10032149: movzx eax,word ptr [esi+13Ah]
0x10032150: push edi
0x10032151: push eax
0x10032152: push esi
0x10032153: call 10024AA0h
0x10032158: movzx ecx,word ptr [esi+13Ch]
0x1003215f: push edi
0x10032160: push ecx
0x10032161: push esi
0x10032162: mov [esi+13Ah],ax
0x10032169: call 10024AA0h
0x1003216e: movzx edx,word ptr [esi+13Eh]
0x10032175: push edi
0x10032176: push edx
0x10032177: push esi
0x10032178: mov [esi+13Ch],ax
0x1003217f: call 10024AA0h
0x10032184: add esp,48h
0x10032187: mov [esi+13Eh],ax
0x1003218e: jmp near ptr 1003233Dh
0x10032193: cmp dword ptr [esp+14h],3
0x10032198: jne near ptr 1003233Dh
0x1003219e: movzx ecx,word ptr [esi+118h]
0x100321a5: test ecx,ecx
0x100321a7: mov eax,[esi+114h]
0x100321ad: jle short 100321EBh
0x100321af: add eax,2
```

```
0x100321b2: movzx edx,byte ptr [eax-2]
0x100321b6: mov edi,[esi+164h]
0x100321bc: movzx edx,byte ptr [edx+edi]
0x100321c0: mov [eax-2],dl
0x100321c3: movzx edx,byte ptr [eax-1]
0x100321c7: mov edi,[esi+164h]
0x100321cd: movzx edx,byte ptr [edx+edi]
0x100321d1: mov [eax-1],dl
0x100321d4: movzx edx,byte ptr [eax]
0x100321d7: mov edi,[esi+164h]
0x100321dd: movzx edx,byte ptr [edx+edi]
0x100321e1: mov [eax],dl
0x100321e3: add eax,3
0x100321e6: sub ecx,1
0x100321e9: jne short 100321B2h
0x100321eb: and [esi+74h],ebx
0x100321ee: jmp near ptr 1003233Dh
0x100321f3: test al,al
0x100321f5: jns near ptr 1003233Dh
0x100321fb: cmp dword ptr [esp+14h],3
0x10032200: jne near ptr 1003233Dh
0x10032206: movzx ebp,word ptr [esi+11Ah]
0x1003220d: movzx edx,byte ptr [esi+13Ch]
0x10032214: mov cl,[esi+13Ah]
0x1003221a: mov eax,[esi+114h]
0x10032220: xor edi,edi
0x10032222: test ebp,ebp
0x10032224: mov [esp+25h],dl
0x10032228: mov dl,[esi+13Eh]
0x1003222e: mov [esp+1Ch],ebp
0x10032232: mov [esp+24h],cl
0x10032236: jle near ptr 1003232Fh
0x1003223c: add eax,2
0x1003223f: nop
0x10032240: mov ecx,[esi+188h]
0x10032246: mov bl,[ecx+edi]
0x10032249: add ecx,edi
0x1003224b: test bl,bl
0x1003224d: jne short 1003225Fh
0x1003224f: mov cx,[esp+24h]
0x10032254: mov [eax-2],cx
0x10032258: mov [eax],dl
0x1003225a: jmp near ptr 10032321h
0x1003225f: cmp bl,0FFh
0x10032262: je near ptr 10032321h
0x10032268: movzx cx,byte ptr [ecx]
0x1003226c: movzx bp,byte ptr [esp+24h]
0x10032272: mov ebx,0FFh
0x10032277: sub ebx,ecx
0x10032279: imul bx,bp
0x1003227d: movzx bp,byte ptr [eax-2]
0x10032282: imul bp,cx
0x10032286: add bx,bp
0x10032289: add bx,80h
0x1003228e: movzx ecx,bx
0x10032291: movzx bp,byte ptr [esp+25h]
0x10032297: movzx ecx,cx
0x1003229a: mov ebx,ecx
0x1003229c: shr ebx,8
0x1003229f: add ebx,ecx
0x100322a1: sar ebx,8
0x100322a4: mov [eax-2],bl
0x100322a7: mov ecx,[esi+188h]
0x100322ad: movzx cx,byte ptr [ecx+edi]
0x100322b2: mov ebx,0FFh
0x100322b7: sub ebx,ecx
0x100322b9: imul bx,bp
```

```
0x100322bd: movzx bp,byte ptr [eax-1]
0x100322c2: imul bp,cx
0x100322c6: add bx,bp
0x100322c9: add bx,80h
0x100322ce: movzx ecx,bx
0x100322d1: movzx ecx,cx
0x100322d4: mov ebx,ecx
0x100322d6: shr ebx,8
0x100322d9: add ebx,ecx
0x100322db: sar ebx,8
0x100322de: mov [eax-1],bl
0x100322e1: mov ecx,[esi+188h]
0x100322e7: movzx cx,byte ptr [ecx+edi]
0x100322ec: movzx bp,dl
0x100322f0: mov ebx,0FFh
0x100322f5: sub ebx,ecx
0x100322f7: imul bx,bp
0x100322fb: movzx bp,byte ptr [eax]
0x100322ff: imul bp,cx
0x10032303: add bx,bp
0x10032306: mov ebp,[esp+1Ch]
0x1003230a: add bx,80h
0x1003230f: movzx ecx,bx
0x10032312: movzx ecx,cx
0x10032315: mov ebx,ecx
0x10032317: shr ebx,8
0x1003231a: add ebx,ecx
0x1003231c: sar ebx,8
0x1003231f: mov [eax],bl
0x10032321: add edi,1
0x10032324: add eax,3
0x10032327: cmp edi,ebp
0x10032329: jl near ptr 10032240h
0x1003232f: and dword ptr [esi+74h],0FFFFFF7Fh
0x10032336: or dword ptr [esi+70h],400000h
0x1003233d: test byte ptr [esi+74h],8
0x10032341: je near ptr 100323F8h
0x10032347: cmp dword ptr [esp+14h],3
0x1003234c: jne near ptr 100323F8h
0x10032352: movzx eax,byte ptr [esi+17Ch]
0x10032359: movzx ecx,byte ptr [esi+17Dh]
0x10032360: movzx edi,byte ptr [esi+17Eh]
0x10032367: movzx edx,word ptr [esi+118h]
0x1003236e: mov ebx,8
0x10032373: sub ebx,eax
0x10032375: mov eax,8
0x1003237a: sub eax,ecx
0x1003237c: mov ecx,8
0x10032381: sub ecx,edi
0x10032383: xor edi,edi
0x10032385: cmp ebx,edi
0x10032387: mov [esp+24h],eax
0x1003238b: mov [esp+14h],ecx
0x1003238f: jl short 10032396h
0x10032391: cmp ebx,8
0x10032394: jle short 10032398h
0x10032396: xor ebx,ebx
0x10032398: cmp eax,edi
0x1003239a: jl short 100323A1h
0x1003239c: cmp eax,8
0x1003239f: jle short 100323A5h
0x100323a1: mov [esp+24h],edi
0x100323a5: cmp ecx,edi
0x100323a7: jl short 100323AEh
0x100323a9: cmp ecx,8
0x100323ac: jle short 100323B2h
0x100323ae: mov [esp+14h],edi
```

```
0x100323b2: cmp dx,di
0x100323b5: jbe short 100323F8h
0x100323b7: xor eax,eax
0x100323b9: movzx edi,dx
0x100323bc: lea esp,[esp]
0x100323c0: mov edx,[esi+114h]
0x100323c6: add edx,eax
0x100323c8: mov cl,bl
0x100323ca: shr byte ptr [edx],cl
0x100323cc: mov ecx,[esi+114h]
0x100323d2: lea edx,[eax+ecx+1]
0x100323d6: movzx ecx,byte ptr [esp+24h]
0x100323db: shr byte ptr [edx],cl
0x100323dd: mov edx,[esi+114h]
0x100323e3: movzx ecx,byte ptr [esp+14h]
0x100323e8: shr byte ptr [eax+edx+2],cl
0x100323ec: lea edx,[eax+edx+2]
0x100323f0: add eax,3
0x100323f3: sub edi,1
0x100323f6: jne short 100323C0h
0x100323f8: pop edi
0x100323f9: pop esi
0x100323fa: pop ebp
0x100323fb: pop ebx
0x100323fc: add esp,10h
0x100323ff: ret
```

`sub_10031a20` is called by png_start_read_image, sub_1002dcc0. It delegates to png_error, sub_10024830, sub_100248d0 (8 callees).

12.792 sub_10032400 at 0x10032400 (123 instructions)

```
0x10032400: mov ecx,[esp+8]
0x10032404: push esi
0x10032405: mov esi,[esp+8]
0x10032409: mov eax,[esi+74h]
0x1003240c: test eax,1000h
0x10032411: je short 1003245Fh
0x10032413: mov dl,[ecx+19h]
0x10032416: cmp dl,3
0x10032419: jne short 10032438h
0x1003241b: cmp word ptr [esi+11Ah],0
0x10032423: je short 10032432h
0x10032425: test eax,2000000h
0x1003242a: je short 10032432h
0x1003242c: mov byte ptr [ecx+19h],6
0x10032430: jmp short 10032455h
0x10032432: mov byte ptr [ecx+19h],2
0x10032436: jmp short 10032455h
0x10032438: cmp word ptr [esi+11Ah],0
0x10032440: je short 1003244Fh
0x10032442: test eax,2000000h
0x10032447: je short 1003244Fh
0x10032449: or dl,4
0x1003244c: mov [ecx+19h],dl
0x1003244f: cmp byte ptr [ecx+18h],8
0x10032453: jae short 10032459h
0x10032455: mov byte ptr [ecx+18h],8
0x10032459: mov word ptr [ecx+16h],0
0x1003245f: test byte ptr [esi+74h],80h
0x10032463: je short 1003248Ch
0x10032465: and byte ptr [ecx+19h],0FBh
0x10032469: mov word ptr [ecx+16h],0
0x1003246f: mov eax,[esi+138h]
0x10032475: mov [ecx+5Eh],eax
0x10032478: mov edx,[esi+13Ch]
```

```
0x1003247e: mov [ecx+62h],edx
0x10032481: mov ax,[esi+140h]
0x10032488: mov [ecx+66h],ax
0x1003248c: test dword ptr [esi+74h],2000h
0x10032493: je short 1003249Eh
0x10032495: mov edx,[esi+15Ch]
0x1003249b: mov [ecx+28h],edx
0x1003249e: test dword ptr [esi+74h],400h
0x100324a5: je short 100324B1h
0x100324a7: cmp byte ptr [ecx+18h],10h
0x100324ab: jne short 100324B1h
0x100324ad: mov byte ptr [ecx+18h],8
0x100324b1: test dword ptr [esi+74h],4000h
0x100324b8: je short 100324BEh
0x100324ba: or byte ptr [ecx+19h],2
0x100324be: test dword ptr [esi+74h],600000h
0x100324c5: je short 100324CBh
0x100324c7: and byte ptr [ecx+19h],0FDh
0x100324cb: test byte ptr [esi+74h],40h
0x100324cf: je short 100324EFh
0x100324d1: mov al,[ecx+19h]
0x100324d4: cmp al,2
0x100324d6: je short 100324DCh
0x100324d8: cmp al,6
0x100324da: jne short 100324EFh
0x100324dc: cmp dword ptr [esi+1ECh],0
0x100324e3: je short 100324EFh
0x100324e5: cmp byte ptr [ecx+18h],8
0x100324e9: jne short 100324EFh
0x100324eb: mov byte ptr [ecx+19h],3
0x100324ef: test byte ptr [esi+74h],4
0x100324f3: je short 100324FFh
0x100324f5: cmp byte ptr [ecx+18h],8
0x100324f9: jae short 100324FFh
0x100324fb: mov byte ptr [ecx+18h],8
0x100324ff: mov al,[ecx+19h]
0x10032502: cmp al,3
0x10032504: mov dl,1
0x10032506: je short 10032512h
0x10032508: test al,2
0x1003250a: je short 10032512h
0x1003250c: mov byte ptr [ecx+1Dh],3
0x10032510: jmp short 10032515h
0x10032512: mov [ecx+1Dh],dl
0x10032515: test dword ptr [esi+70h],400000h
0x1003251c: je short 10032523h
0x1003251e: and al,0FBh
0x10032520: mov [ecx+19h],al
0x10032523: mov al,[ecx+19h]
0x10032526: test al,4
0x10032528: je short 1003252Dh
0x1003252a: add [ecx+1Dh],dl
0x1003252d: test dword ptr [esi+74h],8000h
0x10032534: je short 1003254Fh
0x10032536: cmp al,2
0x10032538: je short 1003253Eh
0x1003253a: test al,al
0x1003253c: jne short 1003254Fh
0x1003253e: add [ecx+1Dh],dl
0x10032541: test dword ptr [esi+74h],1000000h
0x10032548: je short 1003254Fh
0x1003254a: or al,4
0x1003254c: mov [ecx+19h],al
0x1003254f: test dword ptr [esi+74h],100000h
0x10032556: je short 1003256Eh
0x10032558: mov al,[esi+68h]
0x1003255b: cmp [ecx+18h],al
```

```
0x1003255e: jae short 10032563h
0x10032560: mov [ecx+18h],al
0x10032563: mov al,[esi+69h]
0x10032566: cmp [ecx+1Dh],al
0x10032569: jae short 1003256Eh
0x1003256b: mov [ecx+1Dh],al
0x1003256e: mov al,[ecx+18h]
0x10032571: imul byte ptr [ecx+1Dh]
0x10032574: mov [ecx+1Eh],al
0x10032577: cmp al,8
0x10032579: pop esi
0x1003257a: movzx eax,al
0x1003257d: jb short 10032589h
0x1003257f: shr eax,3
0x10032582: imul eax,[ecx]
0x10032585: mov [ecx+0Ch],eax
0x10032588: ret
0x10032589: imul eax,[ecx]
0x1003258c: add eax,7
0x1003258f: shr eax,3
0x10032592: mov [ecx+0Ch],eax
0x10032595: ret
```

`sub_10032400` is called by png_read_update_info. It calls nothing else.

12.793 sub_100325a0 at 0x100325a0 (116 instructions)

```
0x100325a0: mov edx,[esp+4]
0x100325a4: mov al,[edx+9]
0x100325a7: cmp al,8
0x100325a9: jae near ptr 100326DDh
0x100325af: push ebx
0x100325b0: push ebp
0x100325b1: movzx eax,al
0x100325b4: sub eax,1
0x100325b7: push esi
0x100325b8: push edi
0x100325b9: mov edi,[edx]
0x100325bb: je near ptr 10032675h
0x100325c1: sub eax,1
0x100325c4: je short 1003262Ch
0x100325c6: sub eax,2
0x100325c9: jne near ptr 100326BEh
0x100325cf: mov eax,[esp+18h]
0x100325d3: lea ecx,[edi-1]
0x100325d6: mov esi,ecx
0x100325d8: shr esi,1
0x100325da: add esi,eax
0x100325dc: lea ebp,[edi+eax-1]
0x100325e0: and ecx,1
0x100325e3: mov eax,1
0x100325e8: sub eax,ecx
0x100325ea: add eax,eax
0x100325ec: add eax,eax
0x100325ee: test edi,edi
0x100325f0: jbe near ptr 100326BEh
0x100325f6: mov [esp+14h],edi
0x100325fa: lea ebx,[ebx]
0x10032600: mov bl,[esi]
0x10032602: mov cl,al
0x10032604: shr bl,cl
0x10032606: and bl,0Fh
0x10032609: cmp eax,4
0x1003260c: mov [ebp],bl
0x1003260f: jne short 10032618h
0x10032611: xor eax,eax
```

```
0x10032613: sub esi,1
0x10032616: jmp short 1003261Dh
0x10032618: mov eax,4
0x1003261d: sub ebp,1
0x10032620: sub dword ptr [esp+14h],1
0x10032625: jne short 10032600h
0x10032627: jmp near ptr 100326BEh
0x1003262c: mov eax,[esp+18h]
0x10032630: lea ecx,[edi-1]
0x10032633: mov esi,ecx
0x10032635: shr esi,2
0x10032638: add esi,eax
0x1003263a: lea ebp,[edi+eax-1]
0x1003263e: and ecx,3
0x10032641: mov eax,3
0x10032646: sub eax,ecx
0x10032648: add eax,eax
0x1003264a: test edi,edi
0x1003264c: jbe short 100326BEh
0x1003264e: mov edx,edi
0x10032650: mov bl,[esi]
0x10032652: mov cl,al
0x10032654: shr bl,cl
0x10032656: and bl,3
0x10032659: cmp eax,6
0x1003265c: mov [ebp],bl
0x1003265f: jne short 10032668h
0x10032661: xor eax,eax
0x10032663: sub esi,1
0x10032666: jmp short 1003266Bh
0x10032668: add eax,2
0x1003266b: sub ebp,1
0x1003266e: sub edx,1
0x10032671: jne short 10032650h
0x10032673: jmp short 100326BAh
0x10032675: mov eax,[esp+18h]
0x10032679: lea ecx,[edi-1]
0x1003267c: mov esi,ecx
0x1003267e: shr esi,3
0x10032681: add esi,eax
0x10032683: lea ebp,[edi+eax-1]
0x10032687: and ecx,7
0x1003268a: mov eax,7
0x1003268f: sub eax,ecx
0x10032691: test edi,edi
0x10032693: jbe short 100326BEh
0x10032695: mov edx,edi
0x10032697: mov bl,[esi]
0x10032699: mov cl,al
0x1003269b: shr bl,cl
0x1003269d: and bl,1
0x100326a0: cmp eax,7
0x100326a3: mov [ebp],bl
0x100326a6: jne short 100326AFh
0x100326a8: xor eax,eax
0x100326aa: sub esi,1
0x100326ad: jmp short 100326B2h
0x100326af: add eax,1
0x100326b2: sub ebp,1
0x100326b5: sub edx,1
0x100326b8: jne short 10032697h
0x100326ba: mov edx,[esp+14h]
0x100326be: mov al,[edx+0Ah]
0x100326c1: mov cl,al
0x100326c3: add cl,cl
0x100326c5: add cl,cl
0x100326c7: movzx eax,al
```

```
0x100326ca: add cl,cl
0x100326cc: imul eax,edi
0x100326cf: pop edi
0x100326d0: pop esi
0x100326d1: pop ebp
0x100326d2: mov byte ptr [edx+9],8
0x100326d6: mov [edx+0Bh],cl
0x100326d9: mov [edx+4],eax
0x100326dc: pop ebx
0x100326dd: ret
```

`sub_100325a0` is called by sub_100354b0. It calls nothing else.

12.794 sub_100326e0 at 0x100326e0 (69 instructions)

```
0x100326e0: mov edx,[esp+4]
0x100326e4: sub esp,10h
0x100326e7: push ebx
0x100326e8: mov bl,[edx+8]
0x100326eb: cmp bl,3
0x100326ee: je near ptr 1003288Ah
0x100326f4: mov ecx,[esp+20h]
0x100326f8: push ebp
0x100326f9: mov ebp,[edx]
0x100326fb: push esi
0x100326fc: xor esi,esi
0x100326fe: test bl,2
0x10032701: push edi
0x10032702: mov [esp+24h],esi
0x10032706: je short 10032738h
0x10032708: movzx eax,byte ptr [edx+9]
0x1003270c: movzx esi,byte ptr [ecx]
0x1003270f: mov edi,eax
0x10032711: sub edi,esi
0x10032713: movzx esi,byte ptr [ecx+1]
0x10032717: mov [esp+10h],edi
0x1003271b: mov edi,eax
0x1003271d: sub edi,esi
0x1003271f: movzx esi,byte ptr [ecx+2]
0x10032723: sub eax,esi
0x10032725: mov esi,[esp+24h]
0x10032729: mov [esp+14h],edi
0x1003272d: mov [esp+18h],eax
0x10032731: mov edi,3
0x10032736: jmp short 1003274Bh
0x10032738: movzx eax,byte ptr [ecx+3]
0x1003273c: movzx edi,byte ptr [edx+9]
0x10032740: sub edi,eax
0x10032742: mov [esp+10h],edi
0x10032746: mov edi,1
0x1003274b: test bl,4
0x1003274e: je short 10032761h
0x10032750: movzx ecx,byte ptr [ecx+4]
0x10032754: movzx eax,byte ptr [edx+9]
0x10032758: sub eax,ecx
0x1003275a: mov [esp+edi*4+10h],eax
0x1003275e: add edi,1
0x10032761: xor ecx,ecx
0x10032763: xor eax,eax
0x10032765: cmp edi,ecx
0x10032767: jle near ptr 10032887h
0x1003276d: lea ecx,[ecx]
0x10032770: cmp [esp+eax*4+10h],ecx
0x10032774: jg short 1003277Ch
0x10032776: mov [esp+eax*4+10h],ecx
0x1003277a: jmp short 10032781h
```

```
0x1003277c: mov esi,1
0x10032781: add eax,1
0x10032784: cmp eax,edi
0x10032786: jl short 10032770h
0x10032788: cmp si,cx
0x1003278b: je near ptr 10032887h
0x10032791: movzx eax,byte ptr [edx+9]
0x10032795: add eax,0FFFFFFFEh
0x10032798: cmp eax,0Eh
0x1003279b: ja near ptr 10032887h
0x100327a1: movzx eax,byte ptr [eax+100328A4h]
0x100327a8: jmp dword ptr [eax*4+10032890h]
0x10032887: pop edi
0x10032888: pop esi
0x10032889: pop ebp
0x1003288a: pop ebx
0x1003288b: add esp,10h
0x1003288e: ret
```

`sub_100326e0` is called by sub_100354b0. It calls nothing else.

12.795 sub_100328c0 at 0x100328c0 (32 instructions)

```
0x100328c0: push esi
0x100328c1: mov esi,[esp+8]
0x100328c5: cmp byte ptr [esi+9],10h
0x100328c9: jne short 1003290Bh
0x100328cb: movzx edx,byte ptr [esi+0Ah]
0x100328cf: imul edx,[esi]
0x100328d2: test edx,edx
0x100328d4: mov eax,[esp+0Ch]
0x100328d8: mov ecx,eax
0x100328da: jbe short 100328F0h
0x100328dc: push edi
0x100328dd: mov edi,edx
0x100328df: nop
0x100328e0: mov dl,[eax]
0x100328e2: mov [ecx],dl
0x100328e4: add eax,2
0x100328e7: add ecx,1
0x100328ea: sub edi,1
0x100328ed: jne short 100328E0h
0x100328ef: pop edi
0x100328f0: mov al,[esi+0Ah]
0x100328f3: mov cl,al
0x100328f5: add cl,cl
0x100328f7: add cl,cl
0x100328f9: movzx edx,al
0x100328fc: add cl,cl
0x100328fe: imul edx,[esi]
0x10032901: mov byte ptr [esi+9],8
0x10032905: mov [esi+0Bh],cl
0x10032908: mov [esi+4],edx
0x1003290b: pop esi
0x1003290c: ret
```

`sub_100328c0` is called by sub_100354b0. It calls nothing else.

12.796 sub_10032910 at 0x10032910 (134 instructions)

```
0x10032910: mov eax,[esp+4]
0x10032914: mov cl,[eax+8]
0x10032917: cmp cl,6
0x1003291a: mov edx,[eax]
0x1003291c: push ebx
```

```
0x1003291d: push esi
0x1003291e: jne near ptr 10032A03h
0x10032924: cmp byte ptr [eax+9],8
0x10032928: mov eax,[eax+4]
0x1003292b: jne short 10032978h
0x1003292d: add eax,[esp+10h]
0x10032931: test edx,edx
0x10032933: mov ecx,eax
0x10032935: jbe near ptr 10032A8Ch
0x1003293b: mov esi,edx
0x1003293d: lea ecx,[ecx]
0x10032940: mov dl,[eax-1]
0x10032943: sub eax,1
0x10032946: movzx ebx,byte ptr [eax-1]
0x1003294a: sub eax,1
0x1003294d: mov [ecx-1],bl
0x10032950: movzx ebx,byte ptr [eax-1]
0x10032954: sub ecx,1
0x10032957: sub eax,1
0x1003295a: sub ecx,1
0x1003295d: mov [ecx],bl
0x1003295f: movzx ebx,byte ptr [eax-1]
0x10032963: sub eax,1
0x10032966: sub ecx,1
0x10032969: mov [ecx],bl
0x1003296b: sub ecx,1
0x1003296e: sub esi,1
0x10032971: mov [ecx],dl
0x10032973: jne short 10032940h
0x10032975: pop esi
0x10032976: pop ebx
0x10032977: ret
0x10032978: add eax,[esp+10h]
0x1003297c: test edx,edx
0x1003297e: mov ecx,eax
0x10032980: jbe near ptr 10032A8Ch
0x10032986: mov esi,edx
0x10032988: jmp short 10032990h
0x10032990: mov dl,[eax-1]
0x10032993: sub eax,1
0x10032996: movzx ebx,byte ptr [eax-1]
0x1003299a: sub eax,1
0x1003299d: sub eax,1
0x100329a0: mov [esp+0Dh],bl
0x100329a4: movzx ebx,byte ptr [eax]
0x100329a7: mov [ecx-1],bl
0x100329aa: movzx ebx,byte ptr [eax-1]
0x100329ae: sub ecx,1
0x100329b1: mov [ecx-1],bl
0x100329b4: sub eax,1
0x100329b7: movzx ebx,byte ptr [eax-1]
0x100329bb: sub ecx,1
0x100329be: mov [ecx-1],bl
0x100329c1: sub eax,1
0x100329c4: movzx ebx,byte ptr [eax-1]
0x100329c8: sub ecx,1
0x100329cb: sub eax,1
0x100329ce: mov [ecx-1],bl
0x100329d1: movzx ebx,byte ptr [eax-1]
0x100329d5: sub ecx,1
0x100329d8: sub eax,1
0x100329db: sub ecx,1
0x100329de: mov [ecx],bl
0x100329e0: movzx ebx,byte ptr [eax-1]
0x100329e4: sub eax,1
0x100329e7: sub ecx,1
0x100329ea: mov [ecx],bl
```

```
0x100329ec: sub ecx,1
0x100329ef: mov [ecx],dl
0x100329f1: movzx edx,byte ptr [esp+0Dh]
0x100329f6: sub ecx,1
0x100329f9: sub esi,1
0x100329fc: mov [ecx],dl
0x100329fe: jne short 10032990h
0x10032a00: pop esi
0x10032a01: pop ebx
0x10032a02: ret
0x10032a03: cmp cl,4
0x10032a06: jne near ptr 10032A8Ch
0x10032a0c: cmp byte ptr [eax+9],8
0x10032a10: mov eax,[eax+4]
0x10032a13: jne short 10032A3Fh
0x10032a15: add eax,[esp+10h]
0x10032a19: test edx,edx
0x10032a1b: mov ecx,eax
0x10032a1d: jbe short 10032A8Ch
0x10032a1f: mov esi,edx
0x10032a21: mov dl,[eax-1]
0x10032a24: sub eax,1
0x10032a27: mov bl,[eax-1]
0x10032a2a: sub eax,1
0x10032a2d: sub ecx,1
0x10032a30: mov [ecx],bl
0x10032a32: sub ecx,1
0x10032a35: sub esi,1
0x10032a38: mov [ecx],dl
0x10032a3a: jne short 10032A21h
0x10032a3c: pop esi
0x10032a3d: pop ebx
0x10032a3e: ret
0x10032a3f: add eax,[esp+10h]
0x10032a43: test edx,edx
0x10032a45: mov ecx,eax
0x10032a47: jbe short 10032A8Ch
0x10032a49: mov esi,edx
0x10032a4b: jmp short 10032A50h
0x10032a50: mov dl,[eax-1]
0x10032a53: sub eax,1
0x10032a56: movzx ebx,byte ptr [eax-1]
0x10032a5a: sub eax,1
0x10032a5d: sub eax,1
0x10032a60: mov [esp+0Dh],bl
0x10032a64: movzx ebx,byte ptr [eax]
0x10032a67: sub ecx,1
0x10032a6a: mov [ecx],bl
0x10032a6c: movzx ebx,byte ptr [eax-1]
0x10032a70: sub eax,1
0x10032a73: sub ecx,1
0x10032a76: mov [ecx],bl
0x10032a78: sub ecx,1
0x10032a7b: mov [ecx],dl
0x10032a7d: movzx edx,byte ptr [esp+0Dh]
0x10032a82: sub ecx,1
0x10032a85: sub esi,1
0x10032a88: mov [ecx],dl
0x10032a8a: jne short 10032A50h
0x10032a8c: pop esi
0x10032a8d: pop ebx
0x10032a8e: ret
```

`sub_10032910` is called by sub_100354b0. It calls nothing else.

12.797 sub_10032a90 at 0x10032a90 (81 instructions)

```
0x10032a90: mov eax,[esp+4]
0x10032a94: mov cl,[eax+8]
0x10032a97: cmp cl,6
0x10032a9a: mov edx,[eax]
0x10032a9c: push esi
0x10032a9d: jne short 10032AF8h
0x10032a9f: cmp byte ptr [eax+9],8
0x10032aa3: mov eax,[eax+4]
0x10032aa6: jne short 10032AC9h
0x10032aa8: add eax,[esp+0Ch]
0x10032aac: test edx,edx
0x10032aae: jbe near ptr 10032B5Fh
0x10032ab4: sub eax,1
0x10032ab7: or cl,0FFh
0x10032aba: sub cl,[eax]
0x10032abc: sub eax,3
0x10032abf: sub edx,1
0x10032ac2: mov [eax+3],cl
0x10032ac5: jne short 10032AB4h
0x10032ac7: pop esi
0x10032ac8: ret
0x10032ac9: add eax,[esp+0Ch]
0x10032acd: test edx,edx
0x10032acf: jbe near ptr 10032B5Fh
0x10032ad5: mov esi,edx
0x10032ad7: sub eax,1
0x10032ada: or dl,0FFh
0x10032add: sub dl,[eax]
0x10032adf: mov ecx,eax
0x10032ae1: mov [ecx],dl
0x10032ae3: sub eax,1
0x10032ae6: or dl,0FFh
0x10032ae9: sub dl,[eax]
0x10032aeb: sub eax,6
0x10032aee: sub esi,1
0x10032af1: mov [ecx-1],dl
0x10032af4: jne short 10032AD7h
0x10032af6: pop esi
0x10032af7: ret
0x10032af8: cmp cl,4
0x10032afb: jne short 10032B5Fh
0x10032afd: cmp byte ptr [eax+9],8
0x10032b01: mov eax,[eax+4]
0x10032b04: jne short 10032B31h
0x10032b06: add eax,[esp+0Ch]
0x10032b0a: test edx,edx
0x10032b0c: mov ecx,eax
0x10032b0e: jbe short 10032B5Fh
0x10032b10: mov esi,edx
0x10032b12: sub eax,1
0x10032b15: or dl,0FFh
0x10032b18: sub dl,[eax]
0x10032b1a: sub ecx,1
0x10032b1d: mov [ecx],dl
0x10032b1f: mov dl,[eax-1]
0x10032b22: sub eax,1
0x10032b25: sub ecx,1
0x10032b28: sub esi,1
0x10032b2b: mov [ecx],dl
0x10032b2d: jne short 10032B12h
0x10032b2f: pop esi
0x10032b30: ret
0x10032b31: add eax,[esp+0Ch]
0x10032b35: test edx,edx
0x10032b37: jbe short 10032B5Fh
0x10032b39: mov esi,edx
```

```
0x10032b3b: jmp short 10032B40h
0x10032b40: sub eax,1
0x10032b43: or dl,0FFh
0x10032b46: sub dl,[eax]
0x10032b48: mov ecx,eax
0x10032b4a: mov [ecx],dl
0x10032b4c: sub eax,1
0x10032b4f: or dl,0FFh
0x10032b52: sub dl,[eax]
0x10032b54: sub eax,2
0x10032b57: sub esi,1
0x10032b5a: mov [ecx-1],dl
0x10032b5d: jne short 10032B40h
0x10032b5f: pop esi
0x10032b60: ret
```

`sub_10032a90` is called by sub_100354b0. It calls nothing else.

12.798 sub_10032b70 at 0x10032b70 (276 instructions)

```
0x10032b70: mov edx,[esp+0Ch]
0x10032b74: push ebx
0x10032b75: push ebp
0x10032b76: mov ebp,[esp+0Ch]
0x10032b7a: mov al,[ebp+8]
0x10032b7d: mov ebx,edx
0x10032b7f: push esi
0x10032b80: mov esi,[ebp]
0x10032b83: shr ebx,8
0x10032b86: test al,al
0x10032b88: push edi
0x10032b89: mov [esp+14h],bl
0x10032b8d: jne near ptr 10032CC3h
0x10032b93: mov al,[ebp+9]
0x10032b96: cmp al,8
0x10032b98: jne short 10032C0Ch
0x10032b9a: test byte ptr [esp+20h],80h
0x10032b9f: mov eax,[esp+18h]
0x10032ba3: lea edi,[esi+eax]
0x10032ba6: lea eax,[edi+esi]
0x10032ba9: je short 10032BDEh
0x10032bab: cmp esi,1
0x10032bae: jbe short 10032BC8h
0x10032bb0: lea ecx,[esi-1]
0x10032bb3: sub eax,1
0x10032bb6: mov [eax],dl
0x10032bb8: mov bl,[edi-1]
0x10032bbb: sub edi,1
0x10032bbe: sub eax,1
0x10032bc1: sub ecx,1
0x10032bc4: mov [eax],bl
0x10032bc6: jne short 10032BB3h
0x10032bc8: pop edi
0x10032bc9: mov [eax-1],dl
0x10032bcc: lea ecx,[esi+esi]
0x10032bcf: pop esi
0x10032bd0: mov byte ptr [ebp+0Ah],2
0x10032bd4: mov byte ptr [ebp+0Bh],10h
0x10032bd8: mov [ebp+4],ecx
0x10032bdb: pop ebp
0x10032bdc: pop ebx
0x10032bdd: ret
0x10032bde: test esi,esi
0x10032be0: jbe short 10032BF9h
0x10032be2: mov ecx,esi
0x10032be4: mov bl,[edi-1]
```

```
0x10032be7: sub edi,1
0x10032bea: sub eax,1
0x10032bed: mov [eax],bl
0x10032bef: sub eax,1
0x10032bf2: sub ecx,1
0x10032bf5: mov [eax],dl
0x10032bf7: jne short 10032BE4h
0x10032bf9: pop edi
0x10032bfa: lea ecx,[esi+esi]
0x10032bfd: pop esi
0x10032bfe: mov byte ptr [ebp+0Ah],2
0x10032c02: mov byte ptr [ebp+0Bh],10h
0x10032c06: mov [ebp+4],ecx
0x10032c09: pop ebp
0x10032c0a: pop ebx
0x10032c0b: ret
0x10032c0c: cmp al,10h
0x10032c0e: jne near ptr 10032E84h
0x10032c14: test byte ptr [esp+20h],80h
0x10032c19: mov eax,[esp+18h]
0x10032c1d: lea edi,[eax+esi*2]
0x10032c20: lea eax,[edi+esi*2]
0x10032c23: je short 10032C7Bh
0x10032c25: cmp esi,1
0x10032c28: jbe short 10032C5Bh
0x10032c2a: lea ecx,[esi-1]
0x10032c2d: mov [esp+14h],ecx
0x10032c31: mov [eax-1],bl
0x10032c34: sub eax,1
0x10032c37: sub eax,1
0x10032c3a: mov [eax],dl
0x10032c3c: movzx ecx,byte ptr [edi-1]
0x10032c40: sub edi,1
0x10032c43: sub eax,1
0x10032c46: mov [eax],cl
0x10032c48: movzx ecx,byte ptr [edi-1]
0x10032c4c: sub edi,1
0x10032c4f: sub eax,1
0x10032c52: sub dword ptr [esp+14h],1
0x10032c57: mov [eax],cl
0x10032c59: jne short 10032C31h
0x10032c5b: mov [eax-1],bl
0x10032c5e: sub eax,1
0x10032c61: mov [eax-1],dl
0x10032c64: pop edi
0x10032c65: lea edx,[esi*4]
0x10032c6c: pop esi
0x10032c6d: mov byte ptr [ebp+0Ah],2
0x10032c71: mov byte ptr [ebp+0Bh],20h
0x10032c75: mov [ebp+4],edx
0x10032c78: pop ebp
0x10032c79: pop ebx
0x10032c7a: ret
0x10032c7b: test esi,esi
0x10032c7d: jbe short 10032CACh
0x10032c7f: mov [esp+14h],esi
0x10032c83: movzx ecx,byte ptr [edi-1]
0x10032c87: sub edi,1
0x10032c8a: sub eax,1
0x10032c8d: mov [eax],cl
0x10032c8f: movzx ecx,byte ptr [edi-1]
0x10032c93: sub edi,1
0x10032c96: sub eax,1
0x10032c99: mov [eax],cl
0x10032c9b: sub eax,1
0x10032c9e: mov [eax],bl
0x10032ca0: sub eax,1
```

```
0x10032ca3: sub dword ptr [esp+14h],1
0x10032ca8: mov [eax],dl
0x10032caa: jne short 10032C83h
0x10032cac: pop edi
0x10032cad: lea edx,[esi*4]
0x10032cb4: pop esi
0x10032cb5: mov byte ptr [ebp+0Ah],2
0x10032cb9: mov byte ptr [ebp+0Bh],20h
0x10032cbd: mov [ebp+4],edx
0x10032cc0: pop ebp
0x10032cc1: pop ebx
0x10032cc2: ret
0x10032cc3: cmp al,2
0x10032cc5: jne near ptr 10032E84h
0x10032ccb: mov al,[ebp+9]
0x10032cce: cmp al,8
0x10032cd0: jne near ptr 10032D76h
0x10032cd6: mov eax,[esp+18h]
0x10032cda: lea edi,[eax+esi*2]
0x10032cdd: add edi,esi
0x10032cdf: test byte ptr [esp+20h],80h
0x10032ce4: lea eax,[edi+esi]
0x10032ce7: je short 10032D25h
0x10032ce9: cmp esi,1
0x10032cec: jbe short 10032D20h
0x10032cee: lea ecx,[esi-1]
0x10032cf1: mov [eax-1],dl
0x10032cf4: movzx ebx,byte ptr [edi-1]
0x10032cf8: sub eax,1
0x10032cfb: sub edi,1
0x10032cfe: sub eax,1
0x10032d01: mov [eax],bl
0x10032d03: movzx ebx,byte ptr [edi-1]
0x10032d07: sub edi,1
0x10032d0a: sub eax,1
0x10032d0d: mov [eax],bl
0x10032d0f: movzx ebx,byte ptr [edi-1]
0x10032d13: sub edi,1
0x10032d16: sub eax,1
0x10032d19: sub ecx,1
0x10032d1c: mov [eax],bl
0x10032d1e: jne short 10032CF1h
0x10032d20: mov [eax-1],dl
0x10032d23: jmp short 10032D5Fh
0x10032d25: test esi,esi
0x10032d27: jbe short 10032D5Fh
0x10032d29: mov ecx,esi
0x10032d2b: jmp short 10032D30h
0x10032d30: movzx ebx,byte ptr [edi-1]
0x10032d34: sub edi,1
0x10032d37: mov [eax-1],bl
0x10032d3a: movzx ebx,byte ptr [edi-1]
0x10032d3e: sub eax,1
0x10032d41: sub edi,1
0x10032d44: sub eax,1
0x10032d47: mov [eax],bl
0x10032d49: movzx ebx,byte ptr [edi-1]
0x10032d4d: sub edi,1
0x10032d50: sub eax,1
0x10032d53: mov [eax],bl
0x10032d55: sub eax,1
0x10032d58: sub ecx,1
0x10032d5b: mov [eax],dl
0x10032d5d: jne short 10032D30h
0x10032d5f: pop edi
0x10032d60: lea ecx,[esi*4]
0x10032d67: pop esi
```

```
0x10032d68: mov byte ptr [ebp+0Ah],4
0x10032d6c: mov byte ptr [ebp+0Bh],20h
0x10032d70: mov [ebp+4],ecx
0x10032d73: pop ebp
0x10032d74: pop ebx
0x10032d75: ret
0x10032d76: cmp al,10h
0x10032d78: jne near ptr 10032E84h
0x10032d7e: test byte ptr [esp+20h],80h
0x10032d83: mov ecx,[esp+18h]
0x10032d87: lea eax,[esi+esi*2]
0x10032d8a: lea ecx,[ecx+eax*2]
0x10032d8d: lea eax,[ecx+esi*2]
0x10032d90: je near ptr 10032E0Bh
0x10032d96: cmp esi,1
0x10032d99: jbe short 10032E01h
0x10032d9b: lea edi,[esi-1]
0x10032d9e: mov edi,edi
0x10032da0: mov [eax-1],bl
0x10032da3: sub eax,1
0x10032da6: mov [eax-1],dl
0x10032da9: movzx ebx,byte ptr [ecx-1]
0x10032dad: sub eax,1
0x10032db0: mov [eax-1],bl
0x10032db3: sub ecx,1
0x10032db6: movzx ebx,byte ptr [ecx-1]
0x10032dba: sub eax,1
0x10032dbd: mov [eax-1],bl
0x10032dc0: sub ecx,1
0x10032dc3: movzx ebx,byte ptr [ecx-1]
0x10032dc7: sub eax,1
0x10032dca: sub ecx,1
0x10032dcd: mov [eax-1],bl
0x10032dd0: movzx ebx,byte ptr [ecx-1]
0x10032dd4: sub eax,1
0x10032dd7: sub ecx,1
0x10032dda: mov [eax-1],bl
0x10032ddd: movzx ebx,byte ptr [ecx-1]
0x10032de1: sub eax,1
0x10032de4: sub ecx,1
0x10032de7: sub eax,1
0x10032dea: mov [eax],bl
0x10032dec: movzx ebx,byte ptr [ecx-1]
0x10032df0: sub ecx,1
0x10032df3: sub eax,1
0x10032df6: sub edi,1
0x10032df9: mov [eax],bl
0x10032dfb: mov bl,[esp+14h]
0x10032dff: jne short 10032DA0h
0x10032e01: sub eax,1
0x10032e04: mov [eax],bl
0x10032e06: mov [eax-1],dl
0x10032e09: jmp short 10032E72h
0x10032e0b: test esi,esi
0x10032e0d: jbe short 10032E72h
0x10032e0f: mov edi,esi
0x10032e11: movzx ebx,byte ptr [ecx-1]
0x10032e15: mov [eax-1],bl
0x10032e18: sub ecx,1
0x10032e1b: movzx ebx,byte ptr [ecx-1]
0x10032e1f: sub eax,1
0x10032e22: mov [eax-1],bl
0x10032e25: sub ecx,1
0x10032e28: movzx ebx,byte ptr [ecx-1]
0x10032e2c: sub eax,1
0x10032e2f: mov [eax-1],bl
0x10032e32: sub ecx,1
```

```
0x10032e35: movzx ebx,byte ptr [ecx-1]
0x10032e39: sub eax,1
0x10032e3c: mov [eax-1],bl
0x10032e3f: sub ecx,1
0x10032e42: movzx ebx,byte ptr [ecx-1]
0x10032e46: sub eax,1
0x10032e49: sub ecx,1
0x10032e4c: mov [eax-1],bl
0x10032e4f: movzx ebx,byte ptr [ecx-1]
0x10032e53: sub eax,1
0x10032e56: sub ecx,1
0x10032e59: sub eax,1
0x10032e5c: mov [eax],bl
0x10032e5e: movzx ebx,byte ptr [esp+14h]
0x10032e63: sub eax,1
0x10032e66: mov [eax],bl
0x10032e68: sub eax,1
0x10032e6b: sub edi,1
0x10032e6e: mov [eax],dl
0x10032e70: jne short 10032E11h
0x10032e72: lea edx,[esi*8]
0x10032e79: mov byte ptr [ebp+0Bh],40h
0x10032e7d: mov byte ptr [ebp+0Ah],4
0x10032e81: mov [ebp+4],edx
0x10032e84: pop edi
0x10032e85: pop esi
0x10032e86: pop ebp
0x10032e87: pop ebx
0x10032e88: ret
```

`sub_10032b70` is called by sub_100354b0. It calls nothing else.

12.799 sub_10032e90 at 0x10032e90 (152 instructions)

```
0x10032e90: push ebp
0x10032e91: mov ebp,[esp+8]
0x10032e95: mov cl,[ebp+9]
0x10032e98: cmp cl,8
0x10032e9b: push esi
0x10032e9c: mov esi,[ebp]
0x10032e9f: jb near ptr 10033042h
0x10032ea5: mov al,[ebp+8]
0x10032ea8: mov dl,2
0x10032eaa: test dl,al
0x10032eac: jne near ptr 10033042h
0x10032eb2: test al,al
0x10032eb4: push ebx
0x10032eb5: push edi
0x10032eb6: jne near ptr 10032F55h
0x10032ebc: cmp cl,8
0x10032ebf: jne short 10032EFBh
0x10032ec1: test esi,esi
0x10032ec3: mov eax,[esp+18h]
0x10032ec7: lea ecx,[esi+eax-1]
0x10032ecb: lea eax,[ecx+esi*2]
0x10032ece: jbe near ptr 10033010h
0x10032ed4: mov edi,esi
0x10032ed6: movzx ebx,byte ptr [ecx]
0x10032ed9: mov [eax],bl
0x10032edb: movzx ebx,byte ptr [ecx]
0x10032ede: sub eax,1
0x10032ee1: mov [eax],bl
0x10032ee3: movzx ebx,byte ptr [ecx]
0x10032ee6: sub eax,1
0x10032ee9: mov [eax],bl
0x10032eeb: sub eax,1
```

```
0x10032eee: sub ecx,1
0x10032ef1: sub edi,1
0x10032ef4: jne short 10032ED6h
0x10032ef6: jmp near ptr 10033010h
0x10032efb: test esi,esi
0x10032efd: mov ecx,[esp+18h]
0x10032f01: lea ecx,[ecx+esi*2-1]
0x10032f05: lea eax,[ecx+esi*4]
0x10032f08: jbe near ptr 10033010h
0x10032f0e: mov edi,esi
0x10032f10: movzx ebx,byte ptr [ecx]
0x10032f13: mov [eax],bl
0x10032f15: movzx ebx,byte ptr [ecx-1]
0x10032f19: mov [eax-1],bl
0x10032f1c: movzx ebx,byte ptr [ecx]
0x10032f1f: sub eax,1
0x10032f22: mov [eax-1],bl
0x10032f25: movzx ebx,byte ptr [ecx-1]
0x10032f29: sub eax,1
0x10032f2c: sub eax,1
0x10032f2f: mov [eax],bl
0x10032f31: movzx ebx,byte ptr [ecx]
0x10032f34: sub eax,1
0x10032f37: mov [eax],bl
0x10032f39: movzx ebx,byte ptr [ecx-1]
0x10032f3d: add ecx,0FFFFFFFFh
0x10032f40: sub eax,1
0x10032f43: mov [eax],bl
0x10032f45: sub eax,1
0x10032f48: sub ecx,1
0x10032f4b: sub edi,1
0x10032f4e: jne short 10032F10h
0x10032f50: jmp near ptr 10033010h
0x10032f55: cmp al,4
0x10032f57: jne near ptr 10033010h
0x10032f5d: cmp cl,8
0x10032f60: jne short 10032FA5h
0x10032f62: test esi,esi
0x10032f64: mov eax,[esp+18h]
0x10032f68: lea ecx,[eax+esi*2-1]
0x10032f6c: lea eax,[ecx+esi*2]
0x10032f6f: jbe near ptr 10033010h
0x10032f75: mov edi,esi
0x10032f77: movzx ebx,byte ptr [ecx]
0x10032f7a: mov [eax],bl
0x10032f7c: movzx ebx,byte ptr [ecx-1]
0x10032f80: sub ecx,1
0x10032f83: sub eax,1
0x10032f86: mov [eax],bl
0x10032f88: movzx ebx,byte ptr [ecx]
0x10032f8b: sub eax,1
0x10032f8e: mov [eax],bl
0x10032f90: movzx ebx,byte ptr [ecx]
0x10032f93: sub eax,1
0x10032f96: mov [eax],bl
0x10032f98: sub eax,1
0x10032f9b: sub ecx,1
0x10032f9e: sub edi,1
0x10032fa1: jne short 10032F77h
0x10032fa3: jmp short 10033010h
0x10032fa5: test esi,esi
0x10032fa7: mov ecx,[esp+18h]
0x10032fab: lea ecx,[ecx+esi*4-1]
0x10032faf: lea eax,[ecx+esi*4]
0x10032fb2: jbe short 10033010h
0x10032fb4: mov edi,esi
0x10032fb6: movzx ebx,byte ptr [ecx]
```

```
0x10032fb9: mov [eax],bl
0x10032fbb: movzx ebx,byte ptr [ecx-1]
0x10032fbf: mov [eax-1],bl
0x10032fc2: sub eax,1
0x10032fc5: sub ecx,1
0x10032fc8: movzx ebx,byte ptr [ecx-1]
0x10032fcc: mov [eax-1],bl
0x10032fcf: movzx ebx,byte ptr [ecx-2]
0x10032fd3: sub ecx,1
0x10032fd6: sub eax,1
0x10032fd9: mov [eax-1],bl
0x10032fdc: movzx ebx,byte ptr [ecx]
0x10032fdf: sub eax,1
0x10032fe2: mov [eax-1],bl
0x10032fe5: movzx ebx,byte ptr [ecx-1]
0x10032fe9: sub eax,1
0x10032fec: sub eax,1
0x10032fef: mov [eax],bl
0x10032ff1: movzx ebx,byte ptr [ecx]
0x10032ff4: sub eax,1
0x10032ff7: mov [eax],bl
0x10032ff9: movzx ebx,byte ptr [ecx-1]
0x10032ffd: add ecx,0FFFFFFFFh
0x10033000: sub eax,1
0x10033003: mov [eax],bl
0x10033005: sub eax,1
0x10033008: sub ecx,1
0x1003300b: sub edi,1
0x1003300e: jne short 10032FB6h
0x10033010: add [ebp+0Ah],dl
0x10033013: mov al,[ebp+9]
0x10033016: mov cl,[ebp+0Ah]
0x10033019: or [ebp+8],dl
0x1003301c: imul cl
0x1003301e: pop edi
0x1003301f: mov [ebp+0Bh],al
0x10033022: cmp al,8
0x10033024: pop ebx
0x10033025: movzx eax,al
0x10033028: jb short 10033036h
0x1003302a: shr eax,3
0x1003302d: imul eax,esi
0x10033030: pop esi
0x10033031: mov [ebp+4],eax
0x10033034: pop ebp
0x10033035: ret
0x10033036: imul eax,esi
0x10033039: add eax,7
0x1003303c: shr eax,3
0x1003303f: mov [ebp+4],eax
0x10033042: pop esi
0x10033043: pop ebp
0x10033044: ret
```

`sub_10032e90` is called by sub_100354b0. It calls nothing else.

12.800 sub_10033050 at 0x10033050 (510 instructions)

```
0x10033050: sub esp,28h
0x10033053: mov edx,[esp+30h]
0x10033057: mov ecx,[edx]
0x10033059: mov dl,[edx+8]
0x1003305c: xor eax,eax
0x1003305e: test dl,1
0x10033061: mov [esp+8],ecx
0x10033065: mov [esp+4],eax
```

```
0x10033069: jne near ptr 10033750h
0x1003306f: test dl,2
0x10033072: je near ptr 10033750h
0x10033078: cmp dl,2
0x1003307b: mov eax,[esp+2Ch]
0x1003307f: push ebx
0x10033080: movzx ebx,word ptr [eax+22Eh]
0x10033087: push ebp
0x10033088: push esi
0x10033089: movzx esi,word ptr [eax+22Ch]
0x10033090: push edi
0x10033091: movzx edi,word ptr [eax+22Ah]
0x10033098: mov [esp+30h],edi
0x1003309c: mov [esp+1Ch],esi
0x100330a0: mov [esp+2Ch],ebx
0x100330a4: jne near ptr 100333CFh
0x100330aa: mov edx,[esp+40h]
0x100330ae: cmp byte ptr [edx+9],8
0x100330b2: jne near ptr 1003320Eh
0x100330b8: cmp dword ptr [eax+168h],0
0x100330bf: je near ptr 10033189h
0x100330c5: cmp dword ptr [eax+16Ch],0
0x100330cc: je near ptr 10033189h
0x100330d2: test ecx,ecx
0x100330d4: mov esi,[esp+44h]
0x100330d8: mov [esp+20h],esi
0x100330dc: jbe near ptr 100333D3h
0x100330e2: mov [esp+24h],ecx
0x100330e6: jmp short 100330F0h
0x100330f0: movzx ecx,byte ptr [esi]
0x100330f3: mov ebp,[eax+16Ch]
0x100330f9: movzx edx,byte ptr [ecx+ebp]
0x100330fd: movzx ecx,byte ptr [esi+1]
0x10033101: mov cl,[ecx+ebp]
0x10033104: add esi,1
0x10033107: add esi,1
0x1003310a: mov [esp+12h],dl
0x1003310e: movzx edx,byte ptr [esi]
0x10033111: movzx edx,byte ptr [edx+ebp]
0x10033115: mov [esp+11h],dl
0x10033119: mov dl,[esp+12h]
0x1003311d: add esi,1
0x10033120: cmp dl,cl
0x10033122: mov [esp+13h],cl
0x10033126: jne short 1003313Eh
0x10033128: cmp dl,[esp+11h]
0x1003312c: jne short 1003313Ah
0x1003312e: mov cl,[esi-1]
0x10033131: mov ebp,[esp+20h]
0x10033135: mov [ebp],cl
0x10033138: jmp short 10033172h
0x1003313a: mov cl,[esp+13h]
0x1003313e: movzx edx,byte ptr [esp+11h]
0x10033143: movzx ecx,cl
0x10033146: imul edx,ebx
0x10033149: imul ecx,[esp+1Ch]
0x1003314e: or dword ptr [esp+14h],1
0x10033153: mov ebp,[esp+20h]
0x10033157: add edx,ecx
0x10033159: movzx ecx,byte ptr [esp+12h]
0x1003315e: imul ecx,edi
0x10033161: add edx,ecx
0x10033163: mov ecx,[eax+168h]
0x10033169: shr edx,0Fh
0x1003316c: mov dl,[edx+ecx]
0x1003316f: mov [ebp],dl
0x10033172: add ebp,1
```

```
0x10033175: sub dword ptr [esp+24h],1
0x1003317a: mov [esp+20h],ebp
0x1003317e: jne near ptr 100330F0h
0x10033184: jmp near ptr 100333CBh
0x10033189: test ecx,ecx
0x1003318b: mov esi,[esp+44h]
0x1003318f: mov ebp,esi
0x10033191: jbe near ptr 100333CFh
0x10033197: mov [esp+24h],ecx
0x1003319b: jmp short 100331A0h
0x100331a0: movzx ecx,byte ptr [esi]
0x100331a3: add esi,1
0x100331a6: movzx edx,byte ptr [esi+1]
0x100331aa: mov [esp+11h],cl
0x100331ae: mov cl,[esi]
0x100331b0: add esi,1
0x100331b3: mov [esp+12h],dl
0x100331b7: mov dl,[esp+11h]
0x100331bb: add esi,1
0x100331be: cmp dl,cl
0x100331c0: mov [esp+13h],cl
0x100331c4: jne short 100331D8h
0x100331c6: cmp dl,[esp+12h]
0x100331ca: jne short 100331D4h
0x100331cc: mov cl,[esi-1]
0x100331cf: mov [ebp],cl
0x100331d2: jmp short 100331FFh
0x100331d4: mov cl,[esp+13h]
0x100331d8: movzx edx,byte ptr [esp+12h]
0x100331dd: movzx ecx,cl
0x100331e0: imul edx,ebx
0x100331e3: imul ecx,[esp+1Ch]
0x100331e8: or dword ptr [esp+14h],1
0x100331ed: add edx,ecx
0x100331ef: movzx ecx,byte ptr [esp+11h]
0x100331f4: imul ecx,edi
0x100331f7: add edx,ecx
0x100331f9: shr edx,0Fh
0x100331fc: mov [ebp],dl
0x100331ff: add ebp,1
0x10033202: sub dword ptr [esp+24h],1
0x10033207: jne short 100331A0h
0x10033209: jmp near ptr 100333CBh
0x1003320e: cmp dword ptr [eax+178h],0
0x10033215: je near ptr 1003333Ah
0x1003321b: cmp dword ptr [eax+174h],0
0x10033222: je near ptr 1003333Ah
0x10033228: test ecx,ecx
0x1003322a: mov edx,[esp+44h]
0x1003322e: mov [esp+20h],edx
0x10033232: jbe near ptr 100333CFh
0x10033238: mov [esp+28h],ecx
0x1003323c: lea esp,[esp]
0x10033240: xor ecx,ecx
0x10033242: mov ch,[edx]
0x10033244: add edx,2
0x10033247: add edx,2
0x1003324a: add edx,2
0x1003324d: mov [esp+34h],edx
0x10033251: mov cl,[edx-5]
0x10033254: movzx ebp,cx
0x10033257: xor ecx,ecx
0x10033259: mov ch,[edx-4]
0x1003325c: mov cl,[edx-3]
0x1003325f: movzx ecx,cx
0x10033262: mov [esp+24h],ecx
0x10033266: xor ecx,ecx
```

```
0x10033268: cmp bp,[esp+24h]
0x1003326d: mov ch,[edx-2]
0x10033270: mov cl,[edx-1]
0x10033273: movzx esi,cx
0x10033276: jne short 10033285h
0x10033278: cmp bp,si
0x1003327b: jne short 10033285h
0x1003327d: movzx ecx,bp
0x10033280: jmp near ptr 10033311h
0x10033285: movzx ebx,word ptr [esp+24h]
0x1003328a: movzx ecx,word ptr [eax+158h]
0x10033291: mov [esp+24h],ebx
0x10033295: movzx ebx,bp
0x10033298: movzx ebp,bl
0x1003329b: shr ebp,cl
0x1003329d: movzx edx,si
0x100332a0: mov esi,[eax+178h]
0x100332a6: mov ebp,[esi+ebp*4]
0x100332a9: shr ebx,8
0x100332ac: movzx ebp,word ptr [ebp+ebx*2]
0x100332b1: mov ebx,[esp+24h]
0x100332b5: imul ebp,edi
0x100332b8: movzx edi,bl
0x100332bb: shr edi,cl
0x100332bd: shr ebx,8
0x100332c0: mov edi,[esi+edi*4]
0x100332c3: movzx edi,word ptr [edi+ebx*2]
0x100332c7: imul edi,[esp+1Ch]
0x100332cc: mov ebx,[esp+2Ch]
0x100332d0: add ebp,edi
0x100332d2: movzx edi,dl
0x100332d5: shr edi,cl
0x100332d7: shr edx,8
0x100332da: mov esi,[esi+edi*4]
0x100332dd: movzx edx,word ptr [esi+edx*2]
0x100332e1: imul edx,ebx
0x100332e4: mov edi,[esp+30h]
0x100332e8: add ebp,edx
0x100332ea: shr ebp,0Fh
0x100332ed: movzx edx,bp
0x100332f0: movzx edx,dx
0x100332f3: movzx esi,dl
0x100332f6: shr esi,cl
0x100332f8: mov ecx,[eax+174h]
0x100332fe: shr edx,8
0x10033301: or dword ptr [esp+14h],1
0x10033306: mov ecx,[ecx+esi*4]
0x10033309: movzx ecx,word ptr [ecx+edx*2]
0x1003330d: mov edx,[esp+34h]
0x10033311: mov esi,[esp+20h]
0x10033315: add dword ptr [esp+20h],1
0x1003331a: mov [esi],ch
0x1003331c: mov esi,[esp+20h]
0x10033320: mov [esi],cl
0x10033322: mov ecx,1
0x10033327: add [esp+20h],ecx
0x1003332b: sub [esp+28h],ecx
0x1003332f: jne near ptr 10033240h
0x10033335: jmp near ptr 100333CBh
0x1003333a: test ecx,ecx
0x1003333c: mov edx,[esp+44h]
0x10033340: mov esi,edx
0x10033342: jbe near ptr 100333CFh
0x10033348: mov [esp+20h],ecx
0x1003334c: lea esp,[esp]
0x10033350: xor ecx,ecx
0x10033352: mov ch,[edx]
```

```
0x10033354: add edx,2
0x10033357: add edx,2
0x1003335a: add edx,2
0x1003335d: mov cl,[edx-5]
0x10033360: movzx ebp,cx
0x10033363: xor ecx,ecx
0x10033365: mov ch,[edx-4]
0x10033368: mov [esp+34h],ebp
0x1003336c: mov cl,[edx-3]
0x1003336f: movzx ecx,cx
0x10033372: mov [esp+24h],ecx
0x10033376: xor ecx,ecx
0x10033378: cmp bp,[esp+24h]
0x1003337d: mov ch,[edx-2]
0x10033380: mov cl,[edx-1]
0x10033383: movzx ecx,cx
0x10033386: mov [esp+28h],ecx
0x1003338a: jne short 10033391h
0x1003338c: cmp bp,cx
0x1003338f: je short 10033396h
0x10033391: or dword ptr [esp+14h],1
0x10033396: movzx ecx,word ptr [esp+28h]
0x1003339b: movzx ebp,word ptr [esp+24h]
0x100333a0: imul ecx,ebx
0x100333a3: imul ebp,[esp+1Ch]
0x100333a8: add ecx,ebp
0x100333aa: movzx ebp,word ptr [esp+34h]
0x100333af: imul ebp,edi
0x100333b2: add ecx,ebp
0x100333b4: shr ecx,0Fh
0x100333b7: movzx ecx,cx
0x100333ba: mov [esi],ch
0x100333bc: add esi,1
0x100333bf: mov [esi],cl
0x100333c1: add esi,1
0x100333c4: sub dword ptr [esp+20h],1
0x100333c9: jne short 10033350h
0x100333cb: mov ecx,[esp+18h]
0x100333cf: mov esi,[esp+44h]
0x100333d3: mov edx,[esp+40h]
0x100333d7: cmp byte ptr [edx+8],6
0x100333db: jne near ptr 10033709h
0x100333e1: cmp byte ptr [edx+9],8
0x100333e5: jne near ptr 10033529h
0x100333eb: cmp dword ptr [eax+168h],0
0x100333f2: je near ptr 100334B0h
0x100333f8: cmp dword ptr [eax+16Ch],0
0x100333ff: je near ptr 100334B0h
0x10033405: cmp dword ptr [esp+18h],0
0x1003340a: mov ebp,esi
0x1003340c: jbe near ptr 10033705h
0x10033412: mov edx,[esp+18h]
0x10033416: mov [esp+28h],edx
0x1003341a: lea ebx,[ebx]
0x10033420: movzx edx,byte ptr [esi]
0x10033423: mov ecx,[eax+16Ch]
0x10033429: movzx edx,byte ptr [edx+ecx]
0x1003342d: add esi,1
0x10033430: mov [esp+13h],dl
0x10033434: movzx edx,byte ptr [esi]
0x10033437: movzx edx,byte ptr [edx+ecx]
0x1003343b: add esi,1
0x1003343e: mov [esp+11h],dl
0x10033442: movzx edx,byte ptr [esi]
0x10033445: movzx ecx,byte ptr [edx+ecx]
0x10033449: mov [esp+12h],cl
0x1003344d: mov cl,[esp+13h]
```

```
0x10033451: add esi,1
0x10033454: cmp cl,[esp+11h]
0x10033458: jne short 10033460h
0x1003345a: cmp cl,[esp+12h]
0x1003345e: je short 10033465h
0x10033460: or dword ptr [esp+14h],1
0x10033465: movzx edx,byte ptr [esp+12h]
0x1003346a: movzx ecx,byte ptr [esp+11h]
0x1003346f: imul edx,ebx
0x10033472: imul ecx,[esp+1Ch]
0x10033477: add edx,ecx
0x10033479: movzx ecx,byte ptr [esp+13h]
0x1003347e: imul ecx,edi
0x10033481: add edx,ecx
0x10033483: mov ecx,[eax+168h]
0x10033489: shr edx,0Fh
0x1003348c: mov dl,[edx+ecx]
0x1003348f: mov [ebp],dl
0x10033492: mov cl,[esi]
0x10033494: add ebp,1
0x10033497: mov [ebp],cl
0x1003349a: add ebp,1
0x1003349d: add esi,1
0x100334a0: sub dword ptr [esp+28h],1
0x100334a5: jne near ptr 10033420h
0x100334ab: jmp near ptr 10033705h
0x100334b0: test ecx,ecx
0x100334b2: mov eax,esi
0x100334b4: jbe near ptr 10033709h
0x100334ba: mov [esp+28h],ecx
0x100334be: mov edi,edi
0x100334c0: mov cl,[eax]
0x100334c2: movzx edx,byte ptr [eax+1]
0x100334c6: add eax,1
0x100334c9: add eax,1
0x100334cc: mov [esp+12h],dl
0x100334d0: movzx edx,byte ptr [eax]
0x100334d3: add eax,1
0x100334d6: cmp cl,[esp+12h]
0x100334da: mov [esp+11h],cl
0x100334de: mov [esp+13h],dl
0x100334e2: jne short 100334E8h
0x100334e4: cmp cl,dl
0x100334e6: je short 100334EDh
0x100334e8: or dword ptr [esp+14h],1
0x100334ed: movzx ecx,byte ptr [esp+13h]
0x100334f2: movzx edx,byte ptr [esp+12h]
0x100334f7: imul ecx,ebx
0x100334fa: imul edx,[esp+1Ch]
0x100334ff: add ecx,edx
0x10033501: movzx edx,byte ptr [esp+11h]
0x10033506: imul edx,edi
0x10033509: add ecx,edx
0x1003350b: shr ecx,0Fh
0x1003350e: mov [esi],cl
0x10033510: mov cl,[eax]
0x10033512: add esi,1
0x10033515: mov [esi],cl
0x10033517: add esi,1
0x1003351a: add eax,1
0x1003351d: sub dword ptr [esp+28h],1
0x10033522: jne short 100334C0h
0x10033524: jmp near ptr 10033705h
0x10033529: cmp dword ptr [eax+178h],0
0x10033530: je near ptr 10033666h
0x10033536: cmp dword ptr [eax+174h],0
0x1003353d: je near ptr 10033666h
```

```
0x10033543: cmp dword ptr [esp+18h],0
0x10033548: mov ecx,esi
0x1003354a: mov [esp+28h],esi
0x1003354e: jbe near ptr 10033705h
0x10033554: mov edx,[esp+18h]
0x10033558: mov [esp+24h],edx
0x1003355c: lea esp,[esp]
0x10033560: xor edx,edx
0x10033562: mov dh,[ecx]
0x10033564: add ecx,2
0x10033567: add ecx,2
0x1003356a: add ecx,2
0x1003356d: mov [esp+20h],ecx
0x10033571: mov dl,[ecx-5]
0x10033574: movzx ebp,dx
0x10033577: xor edx,edx
0x10033579: mov dh,[ecx-4]
0x1003357c: mov dl,[ecx-3]
0x1003357f: movzx ebx,dx
0x10033582: xor edx,edx
0x10033584: cmp bp,bx
0x10033587: mov dh,[ecx-2]
0x1003358a: mov [esp+34h],ebx
0x1003358e: mov dl,[ecx-1]
0x10033591: movzx edx,dx
0x10033594: jne short 100335A3h
0x10033596: cmp bp,dx
0x10033599: jne short 100335A3h
0x1003359b: movzx edx,bp
0x1003359e: jmp near ptr 10033631h
0x100335a3: movzx ebx,word ptr [esp+34h]
0x100335a8: movzx ecx,word ptr [eax+158h]
0x100335af: mov esi,[eax+178h]
0x100335b5: mov [esp+34h],ebx
0x100335b9: movzx ebx,bp
0x100335bc: movzx ebp,bl
0x100335bf: shr ebp,cl
0x100335c1: shr ebx,8
0x100335c4: movzx edx,dx
0x100335c7: mov ebp,[esi+ebp*4]
0x100335ca: movzx ebp,word ptr [ebp+ebx*2]
0x100335cf: mov ebx,[esp+34h]
0x100335d3: imul ebp,edi
0x100335d6: movzx edi,bl
0x100335d9: shr edi,cl
0x100335db: shr ebx,8
0x100335de: mov edi,[esi+edi*4]
0x100335e1: movzx edi,word ptr [edi+ebx*2]
0x100335e5: imul edi,[esp+1Ch]
0x100335ea: add ebp,edi
0x100335ec: movzx edi,dl
0x100335ef: shr edi,cl
0x100335f1: shr edx,8
0x100335f4: mov esi,[esi+edi*4]
0x100335f7: movzx edx,word ptr [esi+edx*2]
0x100335fb: imul edx,[esp+2Ch]
0x10033600: mov edi,[esp+30h]
0x10033604: add ebp,edx
0x10033606: shr ebp,0Fh
0x10033609: movzx edx,bp
0x1003360c: movzx edx,dx
0x1003360f: movzx esi,dl
0x10033612: shr esi,cl
0x10033614: mov ecx,[eax+174h]
0x1003361a: shr edx,8
0x1003361d: or dword ptr [esp+14h],1
0x10033622: mov ecx,[ecx+esi*4]
```

```
0x10033625: movzx edx,word ptr [ecx+edx*2]
0x10033629: mov esi,[esp+28h]
0x1003362d: mov ecx,[esp+20h]
0x10033631: mov [esi],dh
0x10033633: add esi,1
0x10033636: mov [esi],dl
0x10033638: movzx edx,byte ptr [ecx]
0x1003363b: add esi,1
0x1003363e: mov [esi],dl
0x10033640: movzx edx,byte ptr [ecx+1]
0x10033644: add ecx,1
0x10033647: add esi,1
0x1003364a: mov [esi],dl
0x1003364c: add esi,1
0x1003364f: add ecx,1
0x10033652: sub dword ptr [esp+24h],1
0x10033657: mov [esp+28h],esi
0x1003365b: jne near ptr 10033560h
0x10033661: jmp near ptr 10033705h
0x10033666: cmp dword ptr [esp+18h],0
0x1003366b: mov eax,esi
0x1003366d: mov ecx,esi
0x1003366f: jbe near ptr 10033705h
0x10033675: mov edx,[esp+18h]
0x10033679: mov [esp+30h],edx
0x1003367d: lea ecx,[ecx]
0x10033680: movzx edx,byte ptr [eax+1]
0x10033684: mov dh,[eax]
0x10033686: add eax,2
0x10033689: add eax,2
0x1003368c: add eax,2
0x1003368f: movzx ebp,dx
0x10033692: movzx edx,byte ptr [eax-3]
0x10033696: mov dh,[eax-4]
0x10033699: mov [esp+34h],ebp
0x1003369d: movzx esi,dx
0x100336a0: cmp bp,si
0x100336a3: movzx edx,byte ptr [eax-1]
0x100336a7: mov dh,[eax-2]
0x100336aa: movzx edx,dx
0x100336ad: jne short 100336B4h
0x100336af: cmp bp,dx
0x100336b2: je short 100336B9h
0x100336b4: or dword ptr [esp+14h],1
0x100336b9: movzx edx,dx
0x100336bc: imul edx,ebx
0x100336bf: movzx esi,si
0x100336c2: imul esi,[esp+1Ch]
0x100336c7: add edx,esi
0x100336c9: movzx esi,word ptr [esp+34h]
0x100336ce: imul esi,edi
0x100336d1: add edx,esi
0x100336d3: shr edx,0Fh
0x100336d6: movzx edx,dx
0x100336d9: mov [ecx],dh
0x100336db: add ecx,1
0x100336de: mov [ecx],dl
0x100336e0: movzx edx,byte ptr [eax]
0x100336e3: add ecx,1
0x100336e6: mov [ecx],dl
0x100336e8: movzx edx,byte ptr [eax+1]
0x100336ec: add eax,1
0x100336ef: add ecx,1
0x100336f2: mov [ecx],dl
0x100336f4: add ecx,1
0x100336f7: add eax,1
0x100336fa: sub dword ptr [esp+30h],1
```

```
0x100336ff: jne near ptr 10033680h
0x10033705: mov ecx,[esp+18h]
0x10033709: mov esi,[esp+40h]
0x1003370d: add byte ptr [esi+0Ah],0FEh
0x10033711: mov al,[esi+9]
0x10033714: mov dl,[esi+0Ah]
0x10033717: and byte ptr [esi+8],0FDh
0x1003371b: imul dl
0x1003371d: mov [esi+0Bh],al
0x10033720: cmp al,8
0x10033722: movzx eax,al
0x10033725: jb short 1003373Ch
0x10033727: shr eax,3
0x1003372a: imul eax,ecx
0x1003372d: pop edi
0x1003372e: mov [esi+4],eax
0x10033731: mov eax,[esp+10h]
0x10033735: pop esi
0x10033736: pop ebp
0x10033737: pop ebx
0x10033738: add esp,28h
0x1003373b: ret
0x1003373c: imul eax,ecx
0x1003373f: add eax,7
0x10033742: shr eax,3
0x10033745: pop edi
0x10033746: mov [esi+4],eax
0x10033749: mov eax,[esp+10h]
0x1003374d: pop esi
0x1003374e: pop ebp
0x1003374f: pop ebx
0x10033750: add esp,28h
0x10033753: ret
```

`sub_10033050` is called by sub_100354b0. It calls nothing else.

12.801 png_build_grayscale_palette at 0x10033760 (26 instructions)

```
0x10033760: push edi
0x10033761: mov edi,[esp+0Ch]
0x10033765: test edi,edi
0x10033767: je short 100337C9h
0x10033769: mov eax,[esp+8]
0x1003376d: add eax,0FFFFFFFFh
0x10033770: cmp eax,7
0x10033773: push esi
0x10033774: ja short 100337A9h
0x10033776: jmp dword ptr [eax*4+100337CCh]
0x100337a9: xor esi,esi
0x100337ab: xor edx,edx
0x100337ad: xor ecx,ecx
0x100337af: test esi,esi
0x100337b1: jle short 100337C8h
0x100337b3: lea eax,[edi+2]
0x100337b6: mov [eax-2],cl
0x100337b9: mov [eax-1],cl
0x100337bc: mov [eax],cl
0x100337be: add eax,3
0x100337c1: add ecx,edx
0x100337c3: sub esi,1
0x100337c6: jne short 100337B6h
0x100337c8: pop esi
0x100337c9: pop edi
0x100337ca: ret
```

`png_build_grayscale_palette` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.802 sub_100337f0 at 0x100337f0 (51 instructions)

```
0x100337f0: sub esp,10h
0x100337f3: cmp dword ptr [esp+20h],0
0x100337f8: mov eax,[esp+14h]
0x100337fc: mov ecx,[eax]
0x100337fe: mov [esp],ecx
0x10033801: je near ptr 100347A2h
0x10033807: mov al,[eax+8]
0x1003380a: test al,4
0x1003380c: push ebp
0x1003380d: je short 10033825h
0x1003380f: cmp al,3
0x10033811: je near ptr 100347A1h
0x10033817: mov ebp,[esp+20h]
0x1003381b: test ebp,ebp
0x1003381d: je near ptr 100347A1h
0x10033823: jmp short 10033829h
0x10033825: mov ebp,[esp+20h]
0x10033829: movzx eax,al
0x1003382c: cmp eax,6
0x1003382f: ja near ptr 1003475Dh
0x10033835: push ebx
0x10033836: push esi
0x10033837: push edi
0x10033838: jmp dword ptr [eax*4+100347A8h]
0x1003475d: mov edx,[esp+18h]
0x10034761: mov al,[edx+8]
0x10034764: test al,4
0x10034766: je short 100347A1h
0x10034768: add byte ptr [edx+0Ah],0FFh
0x1003476c: mov cl,[edx+0Ah]
0x1003476f: and al,0FBh
0x10034771: mov [edx+8],al
0x10034774: mov al,[edx+9]
0x10034777: imul cl
0x10034779: mov [edx+0Bh],al
0x1003477c: cmp al,8
0x1003477e: movzx eax,al
0x10034781: jb short 10034793h
0x10034783: shr eax,3
0x10034786: imul eax,[esp+4]
0x1003478b: mov [edx+4],eax
0x1003478e: pop ebp
0x1003478f: add esp,10h
0x10034792: ret
0x10034793: imul eax,[esp+4]
0x10034798: add eax,7
0x1003479b: shr eax,3
0x1003479e: mov [edx+4],eax
0x100347a1: pop ebp
0x100347a2: add esp,10h
0x100347a5: ret
```

`sub_100337f0` is called by sub_100354b0. It calls nothing else.

12.803 sub_100347f0 at 0x100347f0 (28 instructions)

```
0x100347f0: sub esp,0Ch
0x100347f3: mov edx,[esp+1Ch]
0x100347f7: push ebp
0x100347f8: mov ebp,[esp+14h]
0x100347fc: mov al,[ebp+9]
```

```
0x100347ff: cmp al,8
0x10034801: push esi
0x10034802: mov esi,[ebp]
0x10034805: push edi
0x10034806: mov edi,[esp+24h]
0x1003480a: mov [esp+14h],esi
0x1003480e: ja short 10034814h
0x10034810: test edi,edi
0x10034812: jne short 10034824h
0x10034814: cmp al,10h
0x10034816: jne near ptr 10034B80h
0x1003481c: test edx,edx
0x1003481e: je near ptr 10034B80h
0x10034824: movzx ecx,byte ptr [ebp+8]
0x10034828: cmp ecx,6
0x1003482b: ja near ptr 10034B80h
0x10034831: push ebx
0x10034832: jmp dword ptr [ecx*4+10034B88h]
0x10034b80: pop edi
0x10034b81: pop esi
0x10034b82: pop ebp
0x10034b83: add esp,0Ch
0x10034b86: ret
```

`sub_100347f0` is called by sub_100354b0. It calls nothing else.

12.804 sub_10034bb0 at 0x10034bb0 (196 instructions)

```
0x10034bb0: push ecx
0x10034bb1: push ebx
0x10034bb2: mov ebx,[esp+0Ch]
0x10034bb6: cmp byte ptr [ebx+8],3
0x10034bba: mov edx,[ebx]
0x10034bbc: jne near ptr 10034DF5h
0x10034bc2: mov al,[ebx+9]
0x10034bc5: cmp al,8
0x10034bc7: push ebp
0x10034bc8: mov ebp,[esp+14h]
0x10034bcc: push esi
0x10034bcd: push edi
0x10034bce: jae near ptr 10034CDAh
0x10034bd4: movzx eax,al
0x10034bd7: sub eax,1
0x10034bda: je near ptr 10034C86h
0x10034be0: sub eax,1
0x10034be3: je short 10034C39h
0x10034be5: sub eax,2
0x10034be8: jne near ptr 10034CCFh
0x10034bee: mov eax,edx
0x10034bf0: and eax,1
0x10034bf3: lea esi,[edx-1]
0x10034bf6: shr esi,1
0x10034bf8: add eax,eax
0x10034bfa: add esi,ebp
0x10034bfc: add eax,eax
0x10034bfe: test edx,edx
0x10034c00: lea edi,[edx+ebp-1]
0x10034c04: jbe near ptr 10034CCFh
0x10034c0a: mov [esp+10h],edx
0x10034c0e: mov edi,edi
0x10034c10: mov bl,[esi]
0x10034c12: mov cl,al
0x10034c14: shr bl,cl
0x10034c16: and bl,0Fh
0x10034c19: cmp eax,4
0x10034c1c: mov [edi],bl
```

```
0x10034c1e: jne short 10034C27h
0x10034c20: xor eax,eax
0x10034c22: sub esi,1
0x10034c25: jmp short 10034C2Ah
0x10034c27: add eax,4
0x10034c2a: sub edi,1
0x10034c2d: sub dword ptr [esp+10h],1
0x10034c32: jne short 10034C10h
0x10034c34: jmp near ptr 10034CCBh
0x10034c39: lea ecx,[edx-1]
0x10034c3c: and ecx,3
0x10034c3f: lea esi,[edx-1]
0x10034c42: mov eax,3
0x10034c47: shr esi,2
0x10034c4a: sub eax,ecx
0x10034c4c: add esi,ebp
0x10034c4e: add eax,eax
0x10034c50: test edx,edx
0x10034c52: lea edi,[edx+ebp-1]
0x10034c56: jbe short 10034CCFh
0x10034c58: mov [esp+10h],edx
0x10034c5c: lea esp,[esp]
0x10034c60: mov bl,[esi]
0x10034c62: mov cl,al
0x10034c64: shr bl,cl
0x10034c66: and bl,3
0x10034c69: cmp eax,6
0x10034c6c: mov [edi],bl
0x10034c6e: jne short 10034C77h
0x10034c70: xor eax,eax
0x10034c72: sub esi,1
0x10034c75: jmp short 10034C7Ah
0x10034c77: add eax,2
0x10034c7a: sub edi,1
0x10034c7d: sub dword ptr [esp+10h],1
0x10034c82: jne short 10034C60h
0x10034c84: jmp short 10034CCBh
0x10034c86: lea esi,[edx-1]
0x10034c89: lea ecx,[edx-1]
0x10034c8c: shr esi,3
0x10034c8f: and ecx,7
0x10034c92: mov eax,7
0x10034c97: add esi,ebp
0x10034c99: sub eax,ecx
0x10034c9b: test edx,edx
0x10034c9d: lea edi,[edx+ebp-1]
0x10034ca1: jbe short 10034CCFh
0x10034ca3: mov [esp+10h],edx
0x10034ca7: mov bl,[esi]
0x10034ca9: mov cl,al
0x10034cab: shr bl,cl
0x10034cad: and bl,1
0x10034cb0: cmp eax,7
0x10034cb3: mov [edi],bl
0x10034cb5: jne short 10034CBEh
0x10034cb7: xor eax,eax
0x10034cb9: sub esi,1
0x10034cbc: jmp short 10034CC1h
0x10034cbe: add eax,1
0x10034cc1: sub edi,1
0x10034cc4: sub dword ptr [esp+10h],1
0x10034cc9: jne short 10034CA7h
0x10034ccb: mov ebx,[esp+18h]
0x10034ccf: mov byte ptr [ebx+9],8
0x10034cd3: mov byte ptr [ebx+0Bh],8
0x10034cd7: mov [ebx+4],edx
0x10034cda: cmp byte ptr [ebx+9],8
```

```
0x10034cde: jne near ptr 10034DF2h
0x10034ce4: cmp dword ptr [esp+24h],0
0x10034ce9: lea ecx,[edx+ebp-1]
0x10034ced: je near ptr 10034D81h
0x10034cf3: test edx,edx
0x10034cf5: lea esi,[edx*4]
0x10034cfc: mov [esp+10h],esi
0x10034d00: lea eax,[esi+ebp-1]
0x10034d04: jbe short 10034D68h
0x10034d06: mov edi,[esp+20h]
0x10034d0a: mov ebp,edx
0x10034d0c: mov edx,[esp+28h]
0x10034d10: movzx esi,byte ptr [ecx]
0x10034d13: cmp esi,edx
0x10034d15: jl short 10034D1Ch
0x10034d17: mov byte ptr [eax],0FFh
0x10034d1a: jmp short 10034D25h
0x10034d1c: mov ebx,[esp+24h]
0x10034d20: mov bl,[esi+ebx]
0x10034d23: mov [eax],bl
0x10034d25: movzx esi,byte ptr [ecx]
0x10034d28: lea ebx,[edi+esi*2]
0x10034d2b: movzx ebx,byte ptr [esi+ebx+2]
0x10034d30: mov [eax-1],bl
0x10034d33: movzx esi,byte ptr [ecx]
0x10034d36: sub eax,1
0x10034d39: lea ebx,[edi+esi*2]
0x10034d3c: movzx ebx,byte ptr [esi+ebx+1]
0x10034d41: sub eax,1
0x10034d44: mov [eax],bl
0x10034d46: movzx esi,byte ptr [ecx]
0x10034d49: lea ebx,[edi+esi*2]
0x10034d4c: movzx ebx,byte ptr [esi+ebx]
0x10034d50: sub eax,1
0x10034d53: mov [eax],bl
0x10034d55: sub eax,1
0x10034d58: sub ecx,1
0x10034d5b: sub ebp,1
0x10034d5e: jne short 10034D10h
0x10034d60: mov ebx,[esp+18h]
0x10034d64: mov esi,[esp+10h]
0x10034d68: pop edi
0x10034d69: mov [ebx+4],esi
0x10034d6c: pop esi
0x10034d6d: pop ebp
0x10034d6e: mov byte ptr [ebx+0Bh],20h
0x10034d72: mov byte ptr [ebx+8],6
0x10034d76: mov byte ptr [ebx+0Ah],4
0x10034d7a: mov byte ptr [ebx+9],8
0x10034d7e: pop ebx
0x10034d7f: pop ecx
0x10034d80: ret
0x10034d81: test edx,edx
0x10034d83: lea esi,[edx+edx*2]
0x10034d86: mov [esp+10h],esi
0x10034d8a: lea eax,[esi+ebp-1]
0x10034d8e: jbe short 10034DDFh
0x10034d90: mov edi,[esp+20h]
0x10034d94: mov ebp,edx
0x10034d96: jmp short 10034DA0h
0x10034da0: movzx esi,byte ptr [ecx]
0x10034da3: lea edx,[edi+esi*2]
0x10034da6: movzx edx,byte ptr [esi+edx+2]
0x10034dab: mov [eax],dl
0x10034dad: movzx esi,byte ptr [ecx]
0x10034db0: lea edx,[edi+esi*2]
0x10034db3: movzx edx,byte ptr [esi+edx+1]
```

```
0x10034db8: sub eax,1
0x10034dbb: mov [eax],dl
0x10034dbd: movzx esi,byte ptr [ecx]
0x10034dc0: lea edx,[edi+esi*2]
0x10034dc3: movzx edx,byte ptr [esi+edx]
0x10034dc7: sub eax,1
0x10034dca: mov [eax],dl
0x10034dcc: sub eax,1
0x10034dcf: sub ecx,1
0x10034dd2: sub ebp,1
0x10034dd5: jne short 10034DA0h
0x10034dd7: mov ebx,[esp+18h]
0x10034ddb: mov esi,[esp+10h]
0x10034ddf: mov byte ptr [ebx+0Bh],18h
0x10034de3: mov byte ptr [ebx+8],2
0x10034de7: mov byte ptr [ebx+0Ah],3
0x10034deb: mov [ebx+4],esi
0x10034dee: mov byte ptr [ebx+9],8
0x10034df2: pop edi
0x10034df3: pop esi
0x10034df4: pop ebp
0x10034df5: pop ebx
0x10034df6: pop ecx
0x10034df7: ret
```

`sub_10034bb0` is called by sub_100354b0. It calls nothing else.

12.805 sub_10034e00 at 0x10034e00 (381 instructions)

```
0x10034e00: sub esp,10h
0x10034e03: push ebx
0x10034e04: mov ebx,[esp+18h]
0x10034e08: mov al,[ebx+8]
0x10034e0b: test al,al
0x10034e0d: push esi
0x10034e0e: mov esi,[ebx]
0x10034e10: push edi
0x10034e11: jne near ptr 1003509Ch
0x10034e17: mov ecx,[esp+28h]
0x10034e1b: test ecx,ecx
0x10034e1d: je short 10034E25h
0x10034e1f: movzx edx,word ptr [ecx+8]
0x10034e23: jmp short 10034E31h
0x10034e25: mov dword ptr [esp+14h],0
0x10034e2d: mov edx,[esp+14h]
0x10034e31: mov al,[ebx+9]
0x10034e34: cmp al,8
0x10034e36: push ebp
0x10034e37: mov ebp,[esp+28h]
0x10034e3b: jae near ptr 10034FB5h
0x10034e41: movzx eax,al
0x10034e44: sub eax,1
0x10034e47: je near ptr 10034F3Bh
0x10034e4d: sub eax,1
0x10034e50: je short 10034EC9h
0x10034e52: sub eax,2
0x10034e55: jne near ptr 10034FA6h
0x10034e5b: and edx,0Fh
0x10034e5e: mov eax,edx
0x10034e60: mov ecx,edx
0x10034e62: shl ecx,4
0x10034e65: add ecx,eax
0x10034e67: movzx edx,cx
0x10034e6a: lea eax,[esi-1]
0x10034e6d: and eax,1
0x10034e70: add eax,eax
```

```
0x10034e72: mov [esp+18h],edx
0x10034e76: lea edi,[esi-1]
0x10034e79: shr edi,1
0x10034e7b: add eax,eax
0x10034e7d: mov edx,4
0x10034e82: add edi,ebp
0x10034e84: sub edx,eax
0x10034e86: test esi,esi
0x10034e88: lea ebx,[esi+ebp-1]
0x10034e8c: jbe near ptr 10034F9Eh
0x10034e92: mov [esp+1Ch],esi
0x10034e96: movzx eax,byte ptr [edi]
0x10034e99: mov cl,dl
0x10034e9b: shr eax,cl
0x10034e9d: and eax,0Fh
0x10034ea0: mov cl,al
0x10034ea2: shl cl,4
0x10034ea5: or cl,al
0x10034ea7: cmp edx,4
0x10034eaa: mov [ebx],cl
0x10034eac: jne short 10034EB5h
0x10034eae: xor edx,edx
0x10034eb0: sub edi,1
0x10034eb3: jmp short 10034EBAh
0x10034eb5: mov edx,4
0x10034eba: sub ebx,1
0x10034ebd: sub dword ptr [esp+1Ch],1
0x10034ec2: jne short 10034E96h
0x10034ec4: jmp near ptr 10034F9Eh
0x10034ec9: and edx,3
0x10034ecc: imul edx,55h
0x10034ecf: movzx edx,dx
0x10034ed2: mov [esp+18h],edx
0x10034ed6: lea eax,[esi-1]
0x10034ed9: and eax,3
0x10034edc: lea edi,[esi-1]
0x10034edf: mov edx,3
0x10034ee4: shr edi,2
0x10034ee7: sub edx,eax
0x10034ee9: add edi,ebp
0x10034eeb: add edx,edx
0x10034eed: test esi,esi
0x10034eef: lea ebx,[esi+ebp-1]
0x10034ef3: jbe near ptr 10034F9Eh
0x10034ef9: mov [esp+1Ch],esi
0x10034efd: lea ecx,[ecx]
0x10034f00: movzx eax,byte ptr [edi]
0x10034f03: mov cl,dl
0x10034f05: shr eax,cl
0x10034f07: and eax,3
0x10034f0a: mov cl,al
0x10034f0c: add cl,cl
0x10034f0e: add cl,cl
0x10034f10: or cl,al
0x10034f12: add cl,cl
0x10034f14: add cl,cl
0x10034f16: or cl,al
0x10034f18: add cl,cl
0x10034f1a: add cl,cl
0x10034f1c: or cl,al
0x10034f1e: cmp edx,6
0x10034f21: mov [ebx],cl
0x10034f23: jne short 10034F2Ch
0x10034f25: xor edx,edx
0x10034f27: sub edi,1
0x10034f2a: jmp short 10034F2Fh
0x10034f2c: add edx,2
```

```
0x10034f2f: sub ebx,1
0x10034f32: sub dword ptr [esp+1Ch],1
0x10034f37: jne short 10034F00h
0x10034f39: jmp short 10034F9Eh
0x10034f3b: and edx,1
0x10034f3e: imul edx,0FFh
0x10034f44: movzx edx,dx
0x10034f47: lea edi,[esi-1]
0x10034f4a: lea ecx,[esi-1]
0x10034f4d: shr edi,3
0x10034f50: and ecx,7
0x10034f53: mov eax,7
0x10034f58: add edi,ebp
0x10034f5a: sub eax,ecx
0x10034f5c: test esi,esi
0x10034f5e: mov [esp+18h],edx
0x10034f62: lea edx,[esi+ebp-1]
0x10034f66: jbe short 10034FA2h
0x10034f68: mov [esp+1Ch],esi
0x10034f6c: lea esp,[esp]
0x10034f70: mov bl,[edi]
0x10034f72: mov cl,al
0x10034f74: shr bl,cl
0x10034f76: and bl,1
0x10034f79: neg bl
0x10034f7b: sbb bl,bl
0x10034f7d: and ebx,0FFh
0x10034f83: cmp eax,7
0x10034f86: mov [edx],bl
0x10034f88: jne short 10034F91h
0x10034f8a: xor eax,eax
0x10034f8c: sub edi,1
0x10034f8f: jmp short 10034F94h
0x10034f91: add eax,1
0x10034f94: sub edx,1
0x10034f97: sub dword ptr [esp+1Ch],1
0x10034f9c: jne short 10034F70h
0x10034f9e: mov ebx,[esp+24h]
0x10034fa2: mov edx,[esp+18h]
0x10034fa6: mov ecx,[esp+2Ch]
0x10034faa: mov byte ptr [ebx+9],8
0x10034fae: mov byte ptr [ebx+0Bh],8
0x10034fb2: mov [ebx+4],esi
0x10034fb5: test ecx,ecx
0x10034fb7: je near ptr 10035094h
0x10034fbd: mov al,[ebx+9]
0x10034fc0: cmp al,8
0x10034fc2: jne short 10035005h
0x10034fc4: and edx,0FFh
0x10034fca: test esi,esi
0x10034fcc: lea ecx,[esi+ebp-1]
0x10034fd0: lea eax,[ebp+esi*2-1]
0x10034fd4: jbe near ptr 10035060h
0x10034fda: mov edi,esi
0x10034fdc: lea esp,[esp]
0x10034fe0: movzx bx,byte ptr [ecx]
0x10034fe4: cmp bx,dx
0x10034fe7: jne short 10034FEEh
0x10034fe9: mov byte ptr [eax],0
0x10034fec: jmp short 10034FF1h
0x10034fee: mov byte ptr [eax],0FFh
0x10034ff1: mov bl,[ecx]
0x10034ff3: sub eax,1
0x10034ff6: mov [eax],bl
0x10034ff8: sub eax,1
0x10034ffb: sub ecx,1
0x10034ffe: sub edi,1
```

```
0x10035001: jne short 10034FE0h
0x10035003: jmp short 1003505Ch
0x10035005: cmp al,10h
0x10035007: jne short 10035060h
0x10035009: test esi,esi
0x1003500b: mov eax,[ebx+4]
0x1003500e: lea ecx,[eax+ebp-1]
0x10035012: lea eax,[ebp+eax*2-1]
0x10035016: jbe short 10035060h
0x10035018: mov edi,esi
0x1003501a: lea ebx,[ebx]
0x10035020: cmp [ecx-1],dh
0x10035023: jne short 10035034h
0x10035025: cmp [ecx],dl
0x10035027: jne short 10035034h
0x10035029: mov byte ptr [eax],0
0x1003502c: sub eax,1
0x1003502f: mov byte ptr [eax],0
0x10035032: jmp short 1003503Dh
0x10035034: mov byte ptr [eax],0FFh
0x10035037: sub eax,1
0x1003503a: mov byte ptr [eax],0FFh
0x1003503d: movzx ebx,byte ptr [ecx]
0x10035040: sub eax,1
0x10035043: mov [eax],bl
0x10035045: movzx ebx,byte ptr [ecx-1]
0x10035049: add ecx,0FFFFFFFFh
0x1003504c: sub eax,1
0x1003504f: mov [eax],bl
0x10035051: sub eax,1
0x10035054: sub ecx,1
0x10035057: sub edi,1
0x1003505a: jne short 10035020h
0x1003505c: mov ebx,[esp+24h]
0x10035060: mov al,[ebx+9]
0x10035063: add al,al
0x10035065: mov [ebx+0Bh],al
0x10035068: cmp al,8
0x1003506a: mov byte ptr [ebx+8],4
0x1003506e: mov byte ptr [ebx+0Ah],2
0x10035072: movzx eax,al
0x10035075: jb short 10035088h
0x10035077: shr eax,3
0x1003507a: imul eax,esi
0x1003507d: pop ebp
0x1003507e: pop edi
0x1003507f: pop esi
0x10035080: mov [ebx+4],eax
0x10035083: pop ebx
0x10035084: add esp,10h
0x10035087: ret
0x10035088: imul eax,esi
0x1003508b: add eax,7
0x1003508e: shr eax,3
0x10035091: mov [ebx+4],eax
0x10035094: pop ebp
0x10035095: pop edi
0x10035096: pop esi
0x10035097: pop ebx
0x10035098: add esp,10h
0x1003509b: ret
0x1003509c: cmp al,2
0x1003509e: jne short 10035095h
0x100350a0: mov eax,[esp+28h]
0x100350a4: test eax,eax
0x100350a6: je short 10035095h
0x100350a8: mov cl,[ebx+9]
```

```
0x100350ab: cmp cl,8
0x100350ae: jne near ptr 10035136h
0x100350b4: test esi,esi
0x100350b6: movzx edx,byte ptr [eax+2]
0x100350ba: mov cl,[eax+4]
0x100350bd: mov [esp+20h],dl
0x100350c1: movzx edx,byte ptr [eax+6]
0x100350c5: mov eax,[ebx+4]
0x100350c8: mov [esp+0Fh],cl
0x100350cc: mov ecx,[esp+24h]
0x100350d0: mov [esp+10h],dl
0x100350d4: lea edx,[eax+ecx-1]
0x100350d8: lea ecx,[ecx+esi*4-1]
0x100350dc: jbe near ptr 1003522Fh
0x100350e2: mov edi,esi
0x100350e4: mov al,[esp+20h]
0x100350e8: cmp [edx-2],al
0x100350eb: jne short 10035103h
0x100350ed: mov al,[esp+0Fh]
0x100350f1: cmp [edx-1],al
0x100350f4: jne short 10035103h
0x100350f6: mov al,[esp+10h]
0x100350fa: cmp [edx],al
0x100350fc: jne short 10035103h
0x100350fe: mov byte ptr [ecx],0
0x10035101: jmp short 10035106h
0x10035103: mov byte ptr [ecx],0FFh
0x10035106: movzx eax,byte ptr [edx]
0x10035109: sub ecx,1
0x1003510c: mov [ecx],al
0x1003510e: movzx eax,byte ptr [edx-1]
0x10035112: sub edx,1
0x10035115: sub ecx,1
0x10035118: mov [ecx],al
0x1003511a: movzx eax,byte ptr [edx-1]
0x1003511e: sub edx,1
0x10035121: sub ecx,1
0x10035124: mov [ecx],al
0x10035126: sub ecx,1
0x10035129: sub edx,1
0x1003512c: sub edi,1
0x1003512f: jne short 100350E4h
0x10035131: jmp near ptr 1003522Fh
0x10035136: cmp cl,10h
0x10035139: jne near ptr 1003522Fh
0x1003513f: test esi,esi
0x10035141: movzx ecx,byte ptr [eax+3]
0x10035145: movzx edx,byte ptr [eax+5]
0x10035149: mov [esp+20h],cl
0x1003514d: movzx ecx,byte ptr [eax+7]
0x10035151: mov [esp+0Fh],dl
0x10035155: movzx edx,byte ptr [eax+2]
0x10035159: mov [esp+12h],cl
0x1003515d: movzx ecx,byte ptr [eax+4]
0x10035161: mov [esp+10h],dl
0x10035165: movzx edx,byte ptr [eax+6]
0x10035169: mov eax,[esp+24h]
0x1003516d: mov [esp+11h],cl
0x10035171: mov ecx,[ebx+4]
0x10035174: lea ecx,[ecx+eax-1]
0x10035178: mov [esp+13h],dl
0x1003517c: lea eax,[eax+esi*8-1]
0x10035180: jbe near ptr 1003522Fh
0x10035186: mov edi,esi
0x10035188: jmp short 10035190h
0x10035190: mov dl,[esp+20h]
0x10035194: cmp [ecx-5],dl
```

```
0x10035197: jne short 100351D0h
0x10035199: mov dl,[esp+10h]
0x1003519d: cmp [ecx-4],dl
0x100351a0: jne short 100351D0h
0x100351a2: mov dl,[esp+0Fh]
0x100351a6: cmp [ecx-3],dl
0x100351a9: jne short 100351D0h
0x100351ab: mov dl,[esp+11h]
0x100351af: cmp [ecx-2],dl
0x100351b2: jne short 100351D0h
0x100351b4: mov dl,[esp+12h]
0x100351b8: cmp [ecx-1],dl
0x100351bb: jne short 100351D0h
0x100351bd: mov dl,[esp+13h]
0x100351c1: cmp [ecx],dl
0x100351c3: jne short 100351D0h
0x100351c5: mov byte ptr [eax],0
0x100351c8: sub eax,1
0x100351cb: mov byte ptr [eax],0
0x100351ce: jmp short 100351D9h
0x100351d0: mov byte ptr [eax],0FFh
0x100351d3: sub eax,1
0x100351d6: mov byte ptr [eax],0FFh
0x100351d9: movzx edx,byte ptr [ecx]
0x100351dc: mov [eax-1],dl
0x100351df: movzx edx,byte ptr [ecx-1]
0x100351e3: sub eax,1
0x100351e6: sub ecx,1
0x100351e9: mov [eax-1],dl
0x100351ec: movzx edx,byte ptr [ecx-1]
0x100351f0: sub eax,1
0x100351f3: sub ecx,1
0x100351f6: mov [eax-1],dl
0x100351f9: movzx edx,byte ptr [ecx-1]
0x100351fd: sub eax,1
0x10035200: sub ecx,1
0x10035203: sub eax,1
0x10035206: mov [eax],dl
0x10035208: movzx edx,byte ptr [ecx-1]
0x1003520c: sub ecx,1
0x1003520f: sub eax,1
0x10035212: mov [eax],dl
0x10035214: movzx edx,byte ptr [ecx-1]
0x10035218: sub ecx,1
0x1003521b: sub eax,1
0x1003521e: mov [eax],dl
0x10035220: sub eax,1
0x10035223: sub ecx,1
0x10035226: sub edi,1
0x10035229: jne near ptr 10035190h
0x1003522f: mov al,[ebx+9]
0x10035232: add al,al
0x10035234: add al,al
0x10035236: mov [ebx+0Bh],al
0x10035239: cmp al,8
0x1003523b: mov byte ptr [ebx+8],6
0x1003523f: mov byte ptr [ebx+0Ah],4
0x10035243: movzx eax,al
0x10035246: jb short 10035258h
0x10035248: shr eax,3
0x1003524b: imul eax,esi
0x1003524e: pop edi
0x1003524f: pop esi
0x10035250: mov [ebx+4],eax
0x10035253: pop ebx
0x10035254: add esp,10h
0x10035257: ret
```

```
0x10035258: imul eax,esi
0x1003525b: add eax,7
0x1003525e: pop edi
0x1003525f: shr eax,3
0x10035262: pop esi
0x10035263: mov [ebx+4],eax
0x10035266: pop ebx
0x10035267: add esp,10h
0x1003526a: ret
```

`sub_10034e00` is called by sub_100354b0. It calls nothing else.

12.806 sub_10035270 at 0x10035270 (136 instructions)

```
0x10035270: push ecx
0x10035271: mov edx,[esp+8]
0x10035275: cmp byte ptr [edx+9],8
0x10035279: push ebp
0x1003527a: mov ebp,[edx]
0x1003527c: mov [esp+4],ebp
0x10035280: jne near ptr 100353D3h
0x10035286: mov al,[edx+8]
0x10035289: cmp al,2
0x1003528b: push ebx
0x1003528c: mov ebx,[esp+18h]
0x10035290: push esi
0x10035291: push edi
0x10035292: jne short 1003530Fh
0x10035294: test ebx,ebx
0x10035296: je short 1003530Fh
0x10035298: test ebp,ebp
0x1003529a: mov ecx,[esp+1Ch]
0x1003529e: mov edi,ecx
0x100352a0: jbe short 100352E7h
0x100352a2: movzx eax,byte ptr [ecx]
0x100352a5: movzx esi,byte ptr [ecx+1]
0x100352a9: add ecx,1
0x100352ac: movzx edx,byte ptr [ecx+1]
0x100352b0: and eax,0F8h
0x100352b5: shl eax,5
0x100352b8: add ecx,1
0x100352bb: and esi,0F8h
0x100352c1: or eax,esi
0x100352c3: add eax,eax
0x100352c5: sar edx,3
0x100352c8: add eax,eax
0x100352ca: and edx,1Fh
0x100352cd: or eax,edx
0x100352cf: mov al,[eax+ebx]
0x100352d2: mov [edi],al
0x100352d4: add ecx,1
0x100352d7: add edi,1
0x100352da: sub ebp,1
0x100352dd: jne short 100352A2h
0x100352df: mov ebp,[esp+10h]
0x100352e3: mov edx,[esp+18h]
0x100352e7: mov al,[edx+9]
0x100352ea: mov [edx+0Bh],al
0x100352ed: cmp al,8
0x100352ef: mov byte ptr [edx+8],3
0x100352f3: mov byte ptr [edx+0Ah],1
0x100352f7: movzx eax,al
0x100352fa: jb near ptr 10035399h
0x10035300: shr eax,3
0x10035303: imul eax,ebp
0x10035306: pop edi
```

```
0x10035307: pop esi
0x10035308: pop ebx
0x10035309: mov [edx+4],eax
0x1003530c: pop ebp
0x1003530d: pop ecx
0x1003530e: ret
0x1003530f: cmp al,6
0x10035311: jne near ptr 100353ABh
0x10035317: test ebx,ebx
0x10035319: je near ptr 100353ABh
0x1003531f: test ebp,ebp
0x10035321: mov ecx,[esp+1Ch]
0x10035325: mov edi,ecx
0x10035327: jbe short 10035375h
0x10035329: lea esp,[esp]
0x10035330: movzx eax,byte ptr [ecx]
0x10035333: movzx esi,byte ptr [ecx+1]
0x10035337: add ecx,1
0x1003533a: movzx edx,byte ptr [ecx+1]
0x1003533e: and eax,0F8h
0x10035343: add ecx,1
0x10035346: shl eax,5
0x10035349: and esi,0F8h
0x1003534f: or eax,esi
0x10035351: add eax,eax
0x10035353: sar edx,3
0x10035356: and edx,1Fh
0x10035359: add eax,eax
0x1003535b: or eax,edx
0x1003535d: mov dl,[eax+ebx]
0x10035360: mov [edi],dl
0x10035362: add ecx,2
0x10035365: add edi,1
0x10035368: sub ebp,1
0x1003536b: jne short 10035330h
0x1003536d: mov ebp,[esp+10h]
0x10035371: mov edx,[esp+18h]
0x10035375: mov al,[edx+9]
0x10035378: mov [edx+0Bh],al
0x1003537b: cmp al,8
0x1003537d: mov byte ptr [edx+8],3
0x10035381: mov byte ptr [edx+0Ah],1
0x10035385: movzx eax,al
0x10035388: jb short 10035399h
0x1003538a: shr eax,3
0x1003538d: imul eax,ebp
0x10035390: pop edi
0x10035391: pop esi
0x10035392: pop ebx
0x10035393: mov [edx+4],eax
0x10035396: pop ebp
0x10035397: pop ecx
0x10035398: ret
0x10035399: imul eax,ebp
0x1003539c: pop edi
0x1003539d: pop esi
0x1003539e: add eax,7
0x100353a1: shr eax,3
0x100353a4: pop ebx
0x100353a5: mov [edx+4],eax
0x100353a8: pop ebp
0x100353a9: pop ecx
0x100353aa: ret
0x100353ab: cmp al,3
0x100353ad: jne short 100353D0h
0x100353af: mov ecx,[esp+24h]
0x100353b3: test ecx,ecx
```

```
0x100353b5: je short 100353D0h
0x100353b7: test ebp,ebp
0x100353b9: mov eax,[esp+1Ch]
0x100353bd: jbe short 100353D0h
0x100353bf: nop
0x100353c0: movzx edx,byte ptr [eax]
0x100353c3: mov dl,[edx+ecx]
0x100353c6: mov [eax],dl
0x100353c8: add eax,1
0x100353cb: sub ebp,1
0x100353ce: jne short 100353C0h
0x100353d0: pop edi
0x100353d1: pop esi
0x100353d2: pop ebx
0x100353d3: pop ebp
0x100353d4: pop ecx
0x100353d5: ret
```

`sub_10035270` is called by sub_100354b0. It calls nothing else.

12.807 sub_100353e0 at 0x100353e0 (74 instructions)

```
0x100353e0: mov eax,[esp+4]
0x100353e4: mov cl,[eax+8]
0x100353e7: test cl,2
0x100353ea: je near ptr 100354A8h
0x100353f0: mov edx,[eax]
0x100353f2: mov al,[eax+9]
0x100353f5: cmp al,8
0x100353f7: push edi
0x100353f8: jne short 10035434h
0x100353fa: cmp cl,2
0x100353fd: jne short 10035406h
0x100353ff: mov edi,3
0x10035404: jmp short 10035414h
0x10035406: cmp cl,6
0x10035409: jne near ptr 100354A7h
0x1003540f: mov edi,4
0x10035414: test edx,edx
0x10035416: jbe near ptr 100354A7h
0x1003541c: mov eax,[esp+0Ch]
0x10035420: add eax,2
0x10035423: mov cl,[eax-1]
0x10035426: add [eax-2],cl
0x10035429: add [eax],cl
0x1003542b: add eax,edi
0x1003542d: sub edx,1
0x10035430: jne short 10035423h
0x10035432: pop edi
0x10035433: ret
0x10035434: cmp al,10h
0x10035436: jne short 100354A7h
0x10035438: cmp cl,2
0x1003543b: jne short 10035444h
0x1003543d: mov edi,6
0x10035442: jmp short 1003544Eh
0x10035444: cmp cl,6
0x10035447: jne short 100354A7h
0x10035449: mov edi,8
0x1003544e: test edx,edx
0x10035450: jbe short 100354A7h
0x10035452: mov eax,[esp+0Ch]
0x10035456: push ebx
0x10035457: push esi
0x10035458: add eax,1
0x1003545b: mov esi,edx
```

```
0x1003545d: lea ecx,[ecx]
0x10035460: xor ebx,ebx
0x10035462: mov bh,[eax+3]
0x10035465: xor ecx,ecx
0x10035467: mov ch,[eax+1]
0x1003546a: xor edx,edx
0x1003546c: mov dh,[eax-1]
0x1003546f: mov bl,[eax+4]
0x10035472: mov cl,[eax+2]
0x10035475: mov dl,[eax]
0x10035477: add ebx,ecx
0x10035479: and ebx,0FFFFh
0x1003547f: add edx,ecx
0x10035481: and edx,0FFFFh
0x10035487: mov ecx,ebx
0x10035489: mov ebx,edx
0x1003548b: mov [eax],dl
0x1003548d: mov edx,ecx
0x1003548f: shr ebx,8
0x10035492: shr edx,8
0x10035495: mov [eax-1],bl
0x10035498: mov [eax+3],dl
0x1003549b: mov [eax+4],cl
0x1003549e: add eax,edi
0x100354a0: sub esi,1
0x100354a3: jne short 10035460h
0x100354a5: pop esi
0x100354a6: pop ebx
0x100354a7: pop edi
0x100354a8: ret
```

`sub_100353e0` is called by png_read_row. It calls nothing else.

12.808 sub_100354b0 at 0x100354b0 (332 instructions)

```
0x100354b0: sub esp,38h
0x100354b3: mov eax,ds:[10136220h]
0x100354b8: xor eax,esp
0x100354ba: mov [esp+34h],eax
0x100354be: push esi
0x100354bf: mov esi,[esp+40h]
0x100354c3: mov eax,[esi+0ECh]
0x100354c9: test eax,eax
0x100354cb: jne short 100354FCh
0x100354cd: movzx eax,byte ptr [esi+124h]
0x100354d4: mov ecx,[esi+0E4h]
0x100354da: push eax
0x100354db: push ecx
0x100354dc: push 10101B8Ch
0x100354e1: lea edx,[esp+10h]
0x100354e5: push 32h
0x100354e7: push edx
0x100354e8: call dword ptr ds:[100FE4C0h]
0x100354ee: add esp,14h
0x100354f1: lea eax,[esp+4]
0x100354f5: push eax
0x100354f6: push esi
0x100354f7: call 100231E0h
0x100354fc: mov ecx,[esi+74h]
0x100354ff: test ecx,1000h
0x10035505: je short 10035576h
0x10035507: cmp byte ptr [esi+108h],3
0x1003550e: jne short 1003553Bh
0x10035510: movzx ecx,word ptr [esi+11Ah]
0x10035517: mov edx,[esi+188h]
0x1003551d: push ecx
```

```
0x1003551e: mov ecx,[esi+114h]
0x10035524: push edx
0x10035525: push ecx
0x10035526: add eax,1
0x10035529: push eax
0x1003552a: lea edx,[esi+100h]
0x10035530: push edx
0x10035531: call 10034BB0h
0x10035536: add esp,14h
0x10035539: jmp short 10035576h
0x1003553b: cmp word ptr [esi+11Ah],0
0x10035543: je short 10035561h
0x10035545: test ecx,2000000h
0x1003554b: je short 10035561h
0x1003554d: lea ecx,[esi+18Ch]
0x10035553: push ecx
0x10035554: add eax,1
0x10035557: push eax
0x10035558: lea edx,[esi+100h]
0x1003555e: push edx
0x1003555f: jmp short 1003556Eh
0x10035561: push 0
0x10035563: add eax,1
0x10035566: push eax
0x10035567: lea eax,[esi+100h]
0x1003556d: push eax
0x1003556e: call 10034E00h
0x10035573: add esp,0Ch
0x10035576: mov eax,[esi+70h]
0x10035579: and eax,400000h
0x1003557e: je short 1003559Fh
0x10035580: mov ecx,[esi+0ECh]
0x10035586: or eax,80h
0x1003558b: push eax
0x1003558c: add ecx,1
0x1003558f: push ecx
0x10035590: lea edx,[esi+100h]
0x10035596: push edx
0x10035597: call 10028930h
0x1003559c: add esp,0Ch
0x1003559f: test dword ptr [esi+74h],600000h
0x100355a6: je short 10035606h
0x100355a8: mov eax,[esi+0ECh]
0x100355ae: add eax,1
0x100355b1: push eax
0x100355b2: lea ecx,[esi+100h]
0x100355b8: push ecx
0x100355b9: push esi
0x100355ba: call 10033050h
0x100355bf: add esp,0Ch
0x100355c2: test eax,eax
0x100355c4: je short 10035606h
0x100355c6: mov edx,[esi+74h]
0x100355c9: and edx,600000h
0x100355cf: cmp edx,400000h
0x100355d5: mov byte ptr [esi+228h],1
0x100355dc: jne short 100355ECh
0x100355de: push 10101B64h
0x100355e3: push esi
0x100355e4: call 100230F0h
0x100355e9: add esp,8
0x100355ec: mov eax,[esi+74h]
0x100355ef: and eax,600000h
0x100355f4: cmp eax,200000h
0x100355f9: jne short 10035606h
0x100355fb: push 10101B64h
0x10035600: push esi
```

```
0x10035601: call 100231E0h
0x10035606: test dword ptr [esi+74h],4000h
0x1003560d: je short 10035631h
0x1003560f: test dword ptr [esi+6Ch],800h
0x10035616: jne short 10035631h
0x10035618: mov ecx,[esi+0ECh]
0x1003561e: add ecx,1
0x10035621: push ecx
0x10035622: lea edx,[esi+100h]
0x10035628: push edx
0x10035629: call 10032E90h
0x1003562e: add esp,8
0x10035631: test byte ptr [esi+74h],80h
0x10035635: je short 100356A9h
0x10035637: cmp word ptr [esi+11Ah],0
0x1003563f: jne short 1003564Ah
0x10035641: test byte ptr [esi+126h],4
0x10035648: je short 100356A9h
0x1003564a: mov eax,[esi+158h]
0x10035650: mov ecx,[esi+178h]
0x10035656: mov edx,[esi+174h]
0x1003565c: push eax
0x1003565d: mov eax,[esi+170h]
0x10035663: push ecx
0x10035664: mov ecx,[esi+16Ch]
0x1003566a: push edx
0x1003566b: mov edx,[esi+168h]
0x10035671: push eax
0x10035672: mov eax,[esi+164h]
0x10035678: push ecx
0x10035679: push edx
0x1003567a: push eax
0x1003567b: lea ecx,[esi+142h]
0x10035681: push ecx
0x10035682: mov ecx,[esi+0ECh]
0x10035688: lea edx,[esi+138h]
0x1003568e: push edx
0x1003568f: lea eax,[esi+18Ch]
0x10035695: push eax
0x10035696: add ecx,1
0x10035699: push ecx
0x1003569a: lea edx,[esi+100h]
0x100356a0: push edx
0x100356a1: call 100337F0h
0x100356a6: add esp,30h
0x100356a9: mov eax,[esi+74h]
0x100356ac: test eax,2000h
0x100356b1: je short 10035701h
0x100356b3: test al,al
0x100356b5: jns short 100356CAh
0x100356b7: cmp word ptr [esi+11Ah],0
0x100356bf: jne short 10035701h
0x100356c1: test byte ptr [esi+126h],4
0x100356c8: jne short 10035701h
0x100356ca: cmp byte ptr [esi+126h],3
0x100356d1: je short 10035701h
0x100356d3: mov eax,[esi+158h]
0x100356d9: mov ecx,[esi+170h]
0x100356df: mov edx,[esi+164h]
0x100356e5: push eax
0x100356e6: mov eax,[esi+0ECh]
0x100356ec: push ecx
0x100356ed: push edx
0x100356ee: add eax,1
0x100356f1: push eax
0x100356f2: lea ecx,[esi+100h]
0x100356f8: push ecx
```

```
0x100356f9: call 100347F0h
0x100356fe: add esp,14h
0x10035701: test dword ptr [esi+74h],400h
0x10035708: je short 10035723h
0x1003570a: mov edx,[esi+0ECh]
0x10035710: add edx,1
0x10035713: push edx
0x10035714: lea eax,[esi+100h]
0x1003571a: push eax
0x1003571b: call 100328C0h
0x10035720: add esp,8
0x10035723: test byte ptr [esi+74h],40h
0x10035727: je short 10035764h
0x10035729: mov ecx,[esi+1F0h]
0x1003572f: mov edx,[esi+1ECh]
0x10035735: mov eax,[esi+0ECh]
0x1003573b: push ecx
0x1003573c: push edx
0x1003573d: add eax,1
0x10035740: push eax
0x10035741: lea ecx,[esi+100h]
0x10035747: push ecx
0x10035748: call 10035270h
0x1003574d: add esp,10h
0x10035750: cmp dword ptr [esi+104h],0
0x10035757: jne short 10035764h
0x10035759: push 10101B40h
0x1003575e: push esi
0x1003575f: call 100231E0h
0x10035764: test byte ptr [esi+74h],20h
0x10035768: je short 10035783h
0x1003576a: mov edx,[esi+0ECh]
0x10035770: add edx,1
0x10035773: push edx
0x10035774: lea eax,[esi+100h]
0x1003577a: push eax
0x1003577b: call 10028800h
0x10035780: add esp,8
0x10035783: test byte ptr [esi+74h],8
0x10035787: je short 100357A9h
0x10035789: mov edx,[esi+0ECh]
0x1003578f: lea ecx,[esi+181h]
0x10035795: push ecx
0x10035796: add edx,1
0x10035799: push edx
0x1003579a: lea eax,[esi+100h]
0x100357a0: push eax
0x100357a1: call 100326E0h
0x100357a6: add esp,0Ch
0x100357a9: test byte ptr [esi+74h],4
0x100357ad: je short 100357C8h
0x100357af: mov ecx,[esi+0ECh]
0x100357b5: add ecx,1
0x100357b8: push ecx
0x100357b9: lea edx,[esi+100h]
0x100357bf: push edx
0x100357c0: call 100325A0h
0x100357c5: add esp,8
0x100357c8: test byte ptr [esi+74h],1
0x100357cc: je short 100357E7h
0x100357ce: mov eax,[esi+0ECh]
0x100357d4: add eax,1
0x100357d7: push eax
0x100357d8: lea ecx,[esi+100h]
0x100357de: push ecx
0x100357df: call 10028BE0h
0x100357e4: add esp,8
```

```
0x100357e7: test dword ptr [esi+74h],10000h
0x100357ee: je short 10035809h
0x100357f0: mov edx,[esi+0ECh]
0x100357f6: add edx,1
0x100357f9: push edx
0x100357fa: lea eax,[esi+100h]
0x10035800: push eax
0x10035801: call 100288D0h
0x10035806: add esp,8
0x10035809: test dword ptr [esi+74h],4000h
0x10035810: je short 10035834h
0x10035812: test dword ptr [esi+6Ch],800h
0x10035819: je short 10035834h
0x1003581b: mov ecx,[esi+0ECh]
0x10035821: add ecx,1
0x10035824: push ecx
0x10035825: lea edx,[esi+100h]
0x1003582b: push edx
0x1003582c: call 10032E90h
0x10035831: add esp,8
0x10035834: test dword ptr [esi+74h],8000h
0x1003583b: je short 10035862h
0x1003583d: mov eax,[esi+70h]
0x10035840: movzx ecx,word ptr [esi+12Eh]
0x10035847: mov edx,[esi+0ECh]
0x1003584d: push eax
0x1003584e: push ecx
0x1003584f: add edx,1
0x10035852: push edx
0x10035853: lea eax,[esi+100h]
0x10035859: push eax
0x1003585a: call 10032B70h
0x1003585f: add esp,10h
0x10035862: test dword ptr [esi+74h],80000h
0x10035869: je short 10035884h
0x1003586b: mov ecx,[esi+0ECh]
0x10035871: add ecx,1
0x10035874: push ecx
0x10035875: lea edx,[esi+100h]
0x1003587b: push edx
0x1003587c: call 10032A90h
0x10035881: add esp,8
0x10035884: test dword ptr [esi+74h],20000h
0x1003588b: je short 100358A6h
0x1003588d: mov eax,[esi+0ECh]
0x10035893: add eax,1
0x10035896: push eax
0x10035897: lea ecx,[esi+100h]
0x1003589d: push ecx
0x1003589e: call 10032910h
0x100358a3: add esp,8
0x100358a6: test byte ptr [esi+74h],10h
0x100358aa: je short 100358C5h
0x100358ac: mov edx,[esi+0ECh]
0x100358b2: add edx,1
0x100358b5: push edx
0x100358b6: lea eax,[esi+100h]
0x100358bc: push eax
0x100358bd: call 10028890h
0x100358c2: add esp,8
0x100358c5: test dword ptr [esi+74h],100000h
0x100358cc: je short 1003593Eh
0x100358ce: mov eax,[esi+5Ch]
0x100358d1: test eax,eax
0x100358d3: je short 100358ECh
0x100358d5: mov ecx,[esi+0ECh]
0x100358db: add ecx,1
```

```
0x100358de: push ecx
0x100358df: lea edx,[esi+100h]
0x100358e5: push edx
0x100358e6: push esi
0x100358e7: call eax
0x100358e9: add esp,0Ch
0x100358ec: mov al,[esi+68h]
0x100358ef: test al,al
0x100358f1: je short 100358F9h
0x100358f3: mov [esi+109h],al
0x100358f9: mov al,[esi+69h]
0x100358fc: test al,al
0x100358fe: je short 10035906h
0x10035900: mov [esi+10Ah],al
0x10035906: mov al,[esi+10Ah]
0x1003590c: imul byte ptr [esi+109h]
0x10035912: mov [esi+10Bh],al
0x10035918: cmp al,8
0x1003591a: movzx eax,al
0x1003591d: jb short 1003592Bh
0x1003591f: shr eax,3
0x10035922: imul eax,[esi+100h]
0x10035929: jmp short 10035938h
0x1003592b: imul eax,[esi+100h]
0x10035932: add eax,7
0x10035935: shr eax,3
0x10035938: mov [esi+104h],eax
0x1003593e: mov ecx,[esp+38h]
0x10035942: pop esi
0x10035943: xor ecx,esp
0x10035945: call 10022326h
0x1003594a: add esp,38h
0x1003594d: ret
```

`sub_100354b0` is called by png_read_row. It works with the text "png_do_rgb_to_gray found nongray pixel". It delegates to sub_10022326, png_warning, png_error (21 callees).

12.809 png_save_uint_32 at 0x10035950 (13 instructions)

```
0x10035950: mov eax,[esp+8]
0x10035954: mov ecx,[esp+4]
0x10035958: mov edx,eax
0x1003595a: shr edx,18h
0x1003595d: mov [ecx],dl
0x1003595f: mov edx,eax
0x10035961: shr edx,10h
0x10035964: mov [ecx+1],dl
0x10035967: mov edx,eax
0x10035969: shr edx,8
0x1003596c: mov [ecx+2],dl
0x1003596f: mov [ecx+3],al
0x10035972: ret
```

`png_save_uint_32` is called by png_write_chunk_start, png_write_chunk_end, sub_100371c0, sub_100376c0, sub_10037c20. It calls nothing else.

12.810 png_save_int_32 at 0x10035980 (13 instructions)

```
0x10035980: mov eax,[esp+8]
0x10035984: mov ecx,[esp+4]
0x10035988: mov edx,eax
0x1003598a: sar edx,18h
0x1003598d: mov [ecx],dl
0x1003598f: mov edx,eax
0x10035991: sar edx,10h
```

```
0x10035994: mov [ecx+1],dl
0x10035997: mov edx,eax
0x10035999: sar edx,8
0x1003599c: mov [ecx+2],dl
0x1003599f: mov [ecx+3],al
0x100359a2: ret
```

`png_save_int_32` is called by sub_10036610, sub_10037ab0. It calls nothing else.

12.811 png_save_uint_16 at 0x100359b0 (7 instructions)

```
0x100359b0: mov eax,[esp+8]
0x100359b4: mov ecx,[esp+4]
0x100359b8: mov edx,eax
0x100359ba: shr edx,8
0x100359bd: mov [ecx],dl
0x100359bf: mov [ecx+1],al
0x100359c2: ret
```

`png_save_uint_16` is called by sub_10035f80, sub_10037430, sub_100377c0, sub_10037930, sub_10037cb0. It calls nothing else.

12.812 png_write_sig at 0x100359d0 (34 instructions)

```
0x100359d0: sub esp,0Ch
0x100359d3: mov eax,ds:[10136220h]
0x100359d8: xor eax,esp
0x100359da: mov [esp+8],eax
0x100359de: push esi
0x100359df: mov esi,[esp+14h]
0x100359e3: mov al,0Ah
0x100359e5: mov [esp+9],al
0x100359e9: mov [esp+0Bh],al
0x100359ed: movzx eax,byte ptr [esi+12Ch]
0x100359f4: mov ecx,8
0x100359f9: sub ecx,eax
0x100359fb: push ecx
0x100359fc: lea edx,[esp+eax+8]
0x10035a00: push edx
0x10035a01: push esi
0x10035a02: mov byte ptr [esp+10h],89h
0x10035a07: mov byte ptr [esp+11h],50h
0x10035a0c: mov byte ptr [esp+12h],4Eh
0x10035a11: mov byte ptr [esp+13h],47h
0x10035a16: mov byte ptr [esp+14h],0Dh
0x10035a1b: mov byte ptr [esp+16h],1Ah
0x10035a20: mov dword ptr [esi+288h],12h
0x10035a2a: call 1002CA30h
0x10035a2f: add esp,0Ch
0x10035a32: cmp byte ptr [esi+12Ch],3
0x10035a39: jae short 10035A42h
0x10035a3b: or dword ptr [esi+6Ch],1000h
0x10035a42: mov ecx,[esp+0Ch]
0x10035a46: pop esi
0x10035a47: xor ecx,esp
0x10035a49: call 10022326h
0x10035a4e: add esp,0Ch
0x10035a51: ret
```

`png_write_sig` is called by png_write_info_before_PLTE. It delegates to sub_10022326, sub_1002ca30 (2 callees).

12.813 png_write_chunk_start at 0x10035a60 (40 instructions)

```
0x10035a60: sub esp,0Ch
0x10035a63: mov eax,ds:[10136220h]
0x10035a68: xor eax,esp
0x10035a6a: mov [esp+8],eax
0x10035a6e: push esi
0x10035a6f: mov esi,[esp+14h]
0x10035a73: test esi,esi
0x10035a75: push edi
0x10035a76: mov edi,[esp+1Ch]
0x10035a7a: je short 10035ACCh
0x10035a7c: mov eax,[esp+20h]
0x10035a80: push eax
0x10035a81: lea ecx,[esp+0Ch]
0x10035a85: push ecx
0x10035a86: mov dword ptr [esi+288h],22h
0x10035a90: call 10035950h
0x10035a95: mov edx,[edi]
0x10035a97: push 8
0x10035a99: lea eax,[esp+14h]
0x10035a9d: push eax
0x10035a9e: push esi
0x10035a9f: mov [esp+20h],edx
0x10035aa3: call 1002CA30h
0x10035aa8: mov ecx,[edi]
0x10035aaa: push esi
0x10035aab: mov [esi+11Ch],ecx
0x10035ab1: call 100234E0h
0x10035ab6: push 4
0x10035ab8: push edi
0x10035ab9: push esi
0x10035aba: call 10023500h
0x10035abf: add esp,24h
0x10035ac2: mov dword ptr [esi+288h],42h
0x10035acc: mov ecx,[esp+10h]
0x10035ad0: pop edi
0x10035ad1: pop esi
0x10035ad2: xor ecx,esp
0x10035ad4: call 10022326h
0x10035ad9: add esp,0Ch
0x10035adc: ret
```

`png_write_chunk_start` is called by sub_10035e80, sub_10035f80, sub_10036250, sub_10036310, sub_10036450, sub_10036610, png_write_chunk, sub_10037290, sub_10037430. It delegates to sub_10022326, sub_100234e0, sub_10023500 (5 callees).

12.814 png_write_chunk_data at 0x10035ae0 (25 instructions)

```
0x10035ae0: push ebx
0x10035ae1: mov ebx,[esp+8]
0x10035ae5: test ebx,ebx
0x10035ae7: je short 10035B10h
0x10035ae9: push edi
0x10035aea: mov edi,[esp+10h]
0x10035aee: test edi,edi
0x10035af0: je short 10035B0Fh
0x10035af2: push esi
0x10035af3: mov esi,[esp+18h]
0x10035af7: test esi,esi
0x10035af9: jbe short 10035B0Eh
0x10035afb: push esi
0x10035afc: push edi
0x10035afd: push ebx
0x10035afe: call 1002CA30h
0x10035b03: push esi
```

```
0x10035b04: push edi
0x10035b05: push ebx
0x10035b06: call 10023500h
0x10035b0b: add esp,18h
0x10035b0e: pop esi
0x10035b0f: pop edi
0x10035b10: pop ebx
0x10035b11: ret
```

`png_write_chunk_data` is called by sub_10035de0, sub_10035e80, sub_10035f80, sub_10036250, sub_10036310, sub_10036450, sub_10036610, png_write_chunk, sub_10037290, sub_10037430. It delegates to sub_10023500, sub_1002ca30 (2 callees).

12.815 png_write_chunk_end at 0x10035b20 (18 instructions)

```
0x10035b20: push esi
0x10035b21: mov esi,[esp+8]
0x10035b25: test esi,esi
0x10035b27: je short 10035B54h
0x10035b29: mov eax,[esi+110h]
0x10035b2f: push eax
0x10035b30: lea ecx,[esp+0Ch]
0x10035b34: push ecx
0x10035b35: mov dword ptr [esi+288h],82h
0x10035b3f: call 10035950h
0x10035b44: push 4
0x10035b46: lea edx,[esp+14h]
0x10035b4a: push edx
0x10035b4b: push esi
0x10035b4c: call 1002CA30h
0x10035b51: add esp,14h
0x10035b54: pop esi
0x10035b55: ret
```

`png_write_chunk_end` is called by sub_10035e80, sub_10035f80, sub_10036250, sub_10036310, sub_10036450, sub_10036610, png_write_chunk, sub_10037290, sub_10037430. It delegates to sub_1002ca30, png_save_uint_32 (2 callees).

12.816 sub_10035b60 at 0x10035b60 (204 instructions)

```
0x10035b60: sub esp,38h
0x10035b63: mov eax,ds:[10136220h]
0x10035b68: xor eax,esp
0x10035b6a: mov [esp+34h],eax
0x10035b6e: mov eax,[esp+40h]
0x10035b72: push ebx
0x10035b73: push ebp
0x10035b74: mov ebp,[esp+44h]
0x10035b78: mov ebx,ecx
0x10035b7a: xor ecx,ecx
0x10035b7c: cmp eax,0FFFFFFFFh
0x10035b7f: push edi
0x10035b80: mov edi,edx
0x10035b82: mov [esi+8],ecx
0x10035b85: mov [esi+0Ch],ecx
0x10035b88: mov [esi+10h],ecx
0x10035b8b: mov [esi],ecx
0x10035b8d: mov [esi+4],ecx
0x10035b90: jne short 10035BABh
0x10035b92: pop edi
0x10035b93: mov [esi],ebp
0x10035b95: pop ebp
0x10035b96: mov [esi+4],ebx
0x10035b99: mov eax,ebx
```

```
0x10035b9b: pop ebx
0x10035b9c: mov ecx,[esp+34h]
0x10035ba0: xor ecx,esp
0x10035ba2: call 10022326h
0x10035ba7: add esp,38h
0x10035baa: ret
0x10035bab: cmp eax,3
0x10035bae: jl short 10035BD1h
0x10035bb0: push eax
0x10035bb1: push 10101BB4h
0x10035bb6: lea eax,[esp+14h]
0x10035bba: push 32h
0x10035bbc: push eax
0x10035bbd: call dword ptr ds:[100FE4C0h]
0x10035bc3: lea ecx,[esp+1Ch]
0x10035bc7: push ecx
0x10035bc8: push edi
0x10035bc9: call 100230F0h
0x10035bce: add esp,18h
0x10035bd1: mov edx,[edi+0B4h]
0x10035bd7: mov eax,[edi+0B0h]
0x10035bdd: mov [edi+7Ch],ebx
0x10035be0: mov [edi+78h],ebp
0x10035be3: mov [edi+88h],edx
0x10035be9: mov [edi+84h],eax
0x10035bef: nop
0x10035bf0: lea eax,[edi+78h]
0x10035bf3: push 0
0x10035bf5: push eax
0x10035bf6: call 10062A80h
0x10035bfb: add esp,8
0x10035bfe: test eax,eax
0x10035c00: jne near ptr 10035D9Bh
0x10035c06: cmp [edi+88h],eax
0x10035c0c: jne near ptr 10035CB8h
0x10035c12: mov eax,[esi+8]
0x10035c15: mov ebp,[esi+0Ch]
0x10035c18: cmp eax,ebp
0x10035c1a: jl short 10035C66h
0x10035c1c: mov ebx,[esi+10h]
0x10035c1f: add eax,4
0x10035c22: test ebx,ebx
0x10035c24: mov [esi+0Ch],eax
0x10035c27: je short 10035C55h
0x10035c29: lea ecx,[eax*4]
0x10035c30: push ecx
0x10035c31: push edi
0x10035c32: call 1002CD20h
0x10035c37: lea edx,[ebp*4]
0x10035c3e: push edx
0x10035c3f: push ebx
0x10035c40: push eax
0x10035c41: mov [esi+10h],eax
0x10035c44: call 100228C6h
0x10035c49: push ebx
0x10035c4a: push edi
0x10035c4b: call 1002CD70h
0x10035c50: add esp,1Ch
0x10035c53: jmp short 10035C66h
0x10035c55: add eax,eax
0x10035c57: add eax,eax
0x10035c59: push eax
0x10035c5a: push edi
0x10035c5b: call 1002CD20h
0x10035c60: add esp,8
0x10035c63: mov [esi+10h],eax
0x10035c66: mov ecx,[edi+0B4h]
```

```
0x10035c6c: push ecx
0x10035c6d: push edi
0x10035c6e: call 1002CD20h
0x10035c73: mov edx,[esi+8]
0x10035c76: mov ecx,[esi+10h]
0x10035c79: mov [ecx+edx*4],eax
0x10035c7c: mov edx,[edi+0B4h]
0x10035c82: mov eax,[edi+0B0h]
0x10035c88: mov ecx,[esi+8]
0x10035c8b: push edx
0x10035c8c: mov edx,[esi+10h]
0x10035c8f: push eax
0x10035c90: mov eax,[edx+ecx*4]
0x10035c93: push eax
0x10035c94: call 100228C6h
0x10035c99: add esp,14h
0x10035c9c: add dword ptr [esi+8],1
0x10035ca0: mov ecx,[edi+0B4h]
0x10035ca6: mov edx,[edi+0B0h]
0x10035cac: mov [edi+88h],ecx
0x10035cb2: mov [edi+84h],edx
0x10035cb8: cmp dword ptr [edi+7Ch],0
0x10035cbc: jne near ptr 10035BF0h
0x10035cc2: lea eax,[edi+78h]
0x10035cc5: push 4
0x10035cc7: push eax
0x10035cc8: call 10062A80h
0x10035ccd: add esp,8
0x10035cd0: test eax,eax
0x10035cd2: jne near ptr 10035D96h
0x10035cd8: cmp [edi+88h],eax
0x10035cde: jne short 10035CC2h
0x10035ce0: mov eax,[esi+8]
0x10035ce3: mov ebp,[esi+0Ch]
0x10035ce6: cmp eax,ebp
0x10035ce8: jl short 10035D3Fh
0x10035cea: mov ebx,[esi+10h]
0x10035ced: add eax,4
0x10035cf0: test ebx,ebx
0x10035cf2: mov [esi+0Ch],eax
0x10035cf5: je short 10035D2Bh
0x10035cf7: add eax,eax
0x10035cf9: add eax,eax
0x10035cfb: push eax
0x10035cfc: push edi
0x10035cfd: call 1002CD20h
0x10035d02: lea ecx,[ebp*4]
0x10035d09: push ecx
0x10035d0a: push ebx
0x10035d0b: push eax
0x10035d0c: mov [esi+10h],eax
0x10035d0f: call 100228C6h
0x10035d14: push ebx
0x10035d15: push edi
0x10035d16: call 1002CD70h
0x10035d1b: add esp,1Ch
0x10035d1e: jmp short 10035D3Fh
0x10035d20: push 10100348h
0x10035d25: push edi
0x10035d26: call 100231E0h
0x10035d2b: lea edx,[eax*4]
0x10035d32: push edx
0x10035d33: push edi
0x10035d34: call 1002CD20h
0x10035d39: add esp,8
0x10035d3c: mov [esi+10h],eax
0x10035d3f: mov eax,[edi+0B4h]
```

```
0x10035d45: push eax
0x10035d46: push edi
0x10035d47: call 1002CD20h
0x10035d4c: mov ecx,[esi+8]
0x10035d4f: mov edx,[esi+10h]
0x10035d52: mov [edx+ecx*4],eax
0x10035d55: mov eax,[edi+0B4h]
0x10035d5b: mov ecx,[edi+0B0h]
0x10035d61: mov edx,[esi+8]
0x10035d64: push eax
0x10035d65: mov eax,[esi+10h]
0x10035d68: push ecx
0x10035d69: mov ecx,[eax+edx*4]
0x10035d6c: push ecx
0x10035d6d: call 100228C6h
0x10035d72: add esp,14h
0x10035d75: add dword ptr [esi+8],1
0x10035d79: mov edx,[edi+0B4h]
0x10035d7f: mov eax,[edi+0B0h]
0x10035d85: mov [edi+88h],edx
0x10035d8b: mov [edi+84h],eax
0x10035d91: jmp near ptr 10035CC2h
0x10035d96: cmp eax,1
0x10035d99: je short 10035DB0h
0x10035d9b: mov eax,[edi+90h]
0x10035da1: test eax,eax
0x10035da3: je near ptr 10035D20h
0x10035da9: push eax
0x10035daa: push edi
0x10035dab: call 100231E0h
0x10035db0: mov ecx,[edi+0B4h]
0x10035db6: mov eax,[esi+8]
0x10035db9: mov edi,[edi+88h]
0x10035dbf: imul eax,ecx
0x10035dc2: cmp edi,ecx
0x10035dc4: jae short 10035DCAh
0x10035dc6: sub ecx,edi
0x10035dc8: add eax,ecx
0x10035dca: mov ecx,[esp+40h]
0x10035dce: pop edi
0x10035dcf: pop ebp
0x10035dd0: pop ebx
0x10035dd1: xor ecx,esp
0x10035dd3: call 10022326h
0x10035dd8: add esp,38h
0x10035ddb: ret
```

`sub_10035b60` is called by sub_10036310, sub_10036450, sub_10037290. It works with the text "zlib error", "Unknown compression type %d". It delegates to sub_10022326, j_memcpy, png_warning (7 callees).

12.817 sub_10035de0 at 0x10035de0 (56 instructions)

```
0x10035de0: mov eax,[edi]
0x10035de2: test eax,eax
0x10035de4: je short 10035DF5h
0x10035de6: mov ecx,[edi+4]
0x10035de9: push ecx
0x10035dea: push eax
0x10035deb: push esi
0x10035dec: call 10035AE0h
0x10035df1: add esp,0Ch
0x10035df4: ret
0x10035df5: push ebx
0x10035df6: xor ebx,ebx
0x10035df8: cmp [edi+8],ebx
0x10035dfb: jle short 10035E2Ch
```

```
0x10035dfd: lea ecx,[ecx]
0x10035e00: mov edx,[esi+0B4h]
0x10035e06: mov eax,[edi+10h]
0x10035e09: mov ecx,[eax+ebx*4]
0x10035e0c: push edx
0x10035e0d: push ecx
0x10035e0e: push esi
0x10035e0f: call 10035AE0h
0x10035e14: mov edx,[edi+10h]
0x10035e17: mov eax,[edx+ebx*4]
0x10035e1a: push eax
0x10035e1b: push esi
0x10035e1c: call 1002CD70h
0x10035e21: add ebx,1
0x10035e24: add esp,14h
0x10035e27: cmp ebx,[edi+8]
0x10035e2a: jl short 10035E00h
0x10035e2c: cmp dword ptr [edi+0Ch],0
0x10035e30: pop ebx
0x10035e31: je short 10035E40h
0x10035e33: mov ecx,[edi+10h]
0x10035e36: push ecx
0x10035e37: push esi
0x10035e38: call 1002CD70h
0x10035e3d: add esp,8
0x10035e40: mov ecx,[esi+88h]
0x10035e46: mov eax,[esi+0B4h]
0x10035e4c: cmp ecx,eax
0x10035e4e: jae short 10035E63h
0x10035e50: mov edx,[esi+0B0h]
0x10035e56: sub eax,ecx
0x10035e58: push eax
0x10035e59: push edx
0x10035e5a: push esi
0x10035e5b: call 10035AE0h
0x10035e60: add esp,0Ch
0x10035e63: lea eax,[esi+78h]
0x10035e66: push eax
0x10035e67: call 100629F0h
0x10035e6c: add esp,4
0x10035e6f: mov dword ptr [esi+0A4h],0
0x10035e79: ret
```

`sub_10035de0` is called by sub_10036310, sub_10036450, sub_10037290. It delegates to png_free, png_write_chunk_data, sub_100629f0 (3 callees).

12.818 sub_10035e80 at 0x10035e80 (79 instructions)

```
0x10035e80: sub esp,10h
0x10035e83: mov eax,ds:[10136220h]
0x10035e88: xor eax,esp
0x10035e8a: mov [esp+0Ch],eax
0x10035e8e: push ebp
0x10035e8f: mov ebp,[esp+20h]
0x10035e93: push esi
0x10035e94: mov esi,[esp+20h]
0x10035e98: push edi
0x10035e99: mov edi,[esp+20h]
0x10035e9d: test byte ptr [edi+230h],1
0x10035ea4: mov byte ptr [esp+10h],50h
0x10035ea9: mov byte ptr [esp+11h],4Ch
0x10035eae: mov byte ptr [esp+12h],54h
0x10035eb3: mov byte ptr [esp+13h],45h
0x10035eb8: mov byte ptr [esp+14h],0
0x10035ebd: jne short 10035EC3h
0x10035ebf: test ebp,ebp
```

```
0x10035ec1: je short 10035ECBh
0x10035ec3: cmp ebp,100h
0x10035ec9: jbe short 10035EDFh
0x10035ecb: cmp byte ptr [edi+126h],3
0x10035ed2: push 10101C08h
0x10035ed7: push edi
0x10035ed8: jne short 10035EEEh
0x10035eda: call 100231E0h
0x10035edf: test byte ptr [edi+126h],2
0x10035ee6: jne short 10035F08h
0x10035ee8: push 10101BD0h
0x10035eed: push edi
0x10035eee: call 100230F0h
0x10035ef3: add esp,8
0x10035ef6: pop edi
0x10035ef7: pop esi
0x10035ef8: pop ebp
0x10035ef9: mov ecx,[esp+0Ch]
0x10035efd: xor ecx,esp
0x10035eff: call 10022326h
0x10035f04: add esp,10h
0x10035f07: ret
0x10035f08: lea eax,[ebp+ebp*2]
0x10035f0c: push eax
0x10035f0d: lea ecx,[esp+14h]
0x10035f11: push ecx
0x10035f12: push edi
0x10035f13: mov [edi+118h],bp
0x10035f1a: call 10035A60h
0x10035f1f: add esp,0Ch
0x10035f22: test ebp,ebp
0x10035f24: jbe short 10035F5Ch
0x10035f26: add esi,2
0x10035f29: lea esp,[esp]
0x10035f30: mov dl,[esi-2]
0x10035f33: mov al,[esi-1]
0x10035f36: mov cl,[esi]
0x10035f38: mov [esp+0Ch],dl
0x10035f3c: push 3
0x10035f3e: lea edx,[esp+10h]
0x10035f42: push edx
0x10035f43: push edi
0x10035f44: mov [esp+19h],al
0x10035f48: mov [esp+1Ah],cl
0x10035f4c: call 10035AE0h
0x10035f51: add esp,0Ch
0x10035f54: add esi,3
0x10035f57: sub ebp,1
0x10035f5a: jne short 10035F30h
0x10035f5c: push edi
0x10035f5d: call 10035B20h
0x10035f62: or dword ptr [edi+6Ch],2
0x10035f66: mov ecx,[esp+1Ch]
0x10035f6a: add esp,4
0x10035f6d: pop edi
0x10035f6e: pop esi
0x10035f6f: pop ebp
0x10035f70: xor ecx,esp
0x10035f72: call 10022326h
0x10035f77: add esp,10h
0x10035f7a: ret
```

`sub_10035e80` is called by png_write_info. It works with the text "Ignoring request to write a PLTE chunk in grayscale PNG", "Invalid number of colors in palette". It delegates to sub_10022326, png_warning, png_error (6 callees).

12.819 sub_10035f80 at 0x10035f80 (67 instructions)

```
0x10035f80: sub esp,10h
0x10035f83: mov eax,ds:[10136220h]
0x10035f88: xor eax,esp
0x10035f8a: mov [esp+0Ch],eax
0x10035f8e: push ebx
0x10035f8f: mov ebx,[esp+20h]
0x10035f93: push edi
0x10035f94: mov edi,[esp+1Ch]
0x10035f98: movzx eax,word ptr [edi+118h]
0x10035f9f: cmp ebx,eax
0x10035fa1: mov byte ptr [esp+0Ch],68h
0x10035fa6: mov byte ptr [esp+0Dh],49h
0x10035fab: mov byte ptr [esp+0Eh],53h
0x10035fb0: mov byte ptr [esp+0Fh],54h
0x10035fb5: mov byte ptr [esp+10h],0
0x10035fba: jle short 10035FDBh
0x10035fbc: push 10101C2Ch
0x10035fc1: push edi
0x10035fc2: call 100230F0h
0x10035fc7: add esp,8
0x10035fca: pop edi
0x10035fcb: pop ebx
0x10035fcc: mov ecx,[esp+0Ch]
0x10035fd0: xor ecx,esp
0x10035fd2: call 10022326h
0x10035fd7: add esp,10h
0x10035fda: ret
0x10035fdb: push esi
0x10035fdc: lea ecx,[ebx+ebx]
0x10035fdf: push ecx
0x10035fe0: lea edx,[esp+14h]
0x10035fe4: push edx
0x10035fe5: push edi
0x10035fe6: call 10035A60h
0x10035feb: add esp,0Ch
0x10035fee: xor esi,esi
0x10035ff0: test ebx,ebx
0x10035ff2: jle short 10036028h
0x10035ff4: push ebp
0x10035ff5: mov ebp,[esp+28h]
0x10035ff9: lea esp,[esp]
0x10036000: movzx eax,word ptr [ebp+esi*2]
0x10036005: push eax
0x10036006: lea ecx,[esp+14h]
0x1003600a: push ecx
0x1003600b: call 100359B0h
0x10036010: push 2
0x10036012: lea edx,[esp+1Ch]
0x10036016: push edx
0x10036017: push edi
0x10036018: call 10035AE0h
0x1003601d: add esi,1
0x10036020: add esp,14h
0x10036023: cmp esi,ebx
0x10036025: jl short 10036000h
0x10036027: pop ebp
0x10036028: push edi
0x10036029: call 10035B20h
0x1003602e: mov ecx,[esp+1Ch]
0x10036032: add esp,4
0x10036035: pop esi
0x10036036: pop edi
0x10036037: pop ebx
0x10036038: xor ecx,esp
0x1003603a: call 10022326h
```

```
0x1003603f: add esp,10h
0x10036042: ret
```

`sub_10035f80` is called by png_write_info. It works with the text "Invalid number of histogram entries specified". It delegates to sub_10022326, png_warning, png_save_uint_16 (6 callees).

12.820 sub_10036050 at 0x10036050 (178 instructions)

```
0x10036050: sub esp,38h
0x10036053: mov eax,ds:[10136220h]
0x10036058: xor eax,esp
0x1003605a: mov [esp+34h],eax
0x1003605e: push ebx
0x1003605f: mov ebx,[esp+48h]
0x10036063: push ebp
0x10036064: push esi
0x10036065: mov esi,[esp+48h]
0x10036069: push edi
0x1003606a: mov edi,[esp+50h]
0x1003606e: test edi,edi
0x10036070: mov [esp+10h],esi
0x10036074: mov [esp+18h],ebx
0x10036078: mov dword ptr [esp+14h],0
0x10036080: mov dword ptr [ebx],0
0x10036086: je near ptr 10036222h
0x1003608c: push edi
0x1003608d: call dword ptr ds:[100FE0B0h]
0x10036093: mov ebp,eax
0x10036095: test ebp,ebp
0x10036097: je near ptr 10036222h
0x1003609d: lea eax,[ebp+2]
0x100360a0: push eax
0x100360a1: push esi
0x100360a2: call 1002CDA0h
0x100360a7: add esp,8
0x100360aa: test eax,eax
0x100360ac: mov [ebx],eax
0x100360ae: jne short 100360BAh
0x100360b0: push 10101D4Ch
0x100360b5: jmp near ptr 10036227h
0x100360ba: cmp byte ptr [edi],0
0x100360bd: mov esi,eax
0x100360bf: je short 1003610Fh
0x100360c1: mov ebx,ds:[100FE4C0h]
0x100360c7: mov al,[edi]
0x100360c9: cmp al,20h
0x100360cb: jb short 100360D9h
0x100360cd: cmp al,7Eh
0x100360cf: jbe short 100360D5h
0x100360d1: cmp al,0A1h
0x100360d3: jb short 100360D9h
0x100360d5: mov [esi],al
0x100360d7: jmp short 10036100h
0x100360d9: movzx ecx,al
0x100360dc: push ecx
0x100360dd: push 10101D28h
0x100360e2: lea edx,[esp+24h]
0x100360e6: push 28h
0x100360e8: push edx
0x100360e9: call ebx
0x100360eb: mov ecx,[esp+20h]
0x100360ef: lea eax,[esp+2Ch]
0x100360f3: push eax
0x100360f4: push ecx
0x100360f5: call 100230F0h
0x100360fa: add esp,18h
```

```
0x100360fd: mov byte ptr [esi],20h
0x10036100: add edi,1
0x10036103: add esi,1
0x10036106: cmp byte ptr [edi],0
0x10036109: jne short 100360C7h
0x1003610b: mov ebx,[esp+18h]
0x1003610f: mov edi,[esp+10h]
0x10036113: mov byte ptr [esi],0
0x10036116: mov edx,[ebx]
0x10036118: cmp byte ptr [edx+ebp-1],20h
0x1003611d: lea esi,[edx+ebp-1]
0x10036121: jne short 10036144h
0x10036123: push 10101D00h
0x10036128: push edi
0x10036129: call 100230F0h
0x1003612e: add esp,8
0x10036131: cmp byte ptr [esi],20h
0x10036134: jne short 10036144h
0x10036136: mov byte ptr [esi],0
0x10036139: sub esi,1
0x1003613c: sub ebp,1
0x1003613f: cmp byte ptr [esi],20h
0x10036142: je short 10036136h
0x10036144: mov esi,[ebx]
0x10036146: cmp byte ptr [esi],20h
0x10036149: jne short 1003616Bh
0x1003614b: push 10101CDCh
0x10036150: push edi
0x10036151: call 100230F0h
0x10036156: add esp,8
0x10036159: cmp byte ptr [esi],20h
0x1003615c: jne short 1003616Bh
0x1003615e: mov edi,edi
0x10036160: add esi,1
0x10036163: sub ebp,1
0x10036166: cmp byte ptr [esi],20h
0x10036169: je short 10036160h
0x1003616b: mov al,[esi]
0x1003616d: mov ecx,[ebx]
0x1003616f: xor edx,edx
0x10036171: test al,al
0x10036173: je short 100361A7h
0x10036175: cmp al,20h
0x10036177: jne short 10036196h
0x10036179: test edx,edx
0x1003617b: jne short 10036189h
0x1003617d: mov [ecx],al
0x1003617f: add ecx,1
0x10036182: mov edx,1
0x10036187: jmp short 1003619Dh
0x10036189: sub ebp,1
0x1003618c: mov dword ptr [esp+14h],1
0x10036194: jmp short 1003619Dh
0x10036196: mov [ecx],al
0x10036198: add ecx,1
0x1003619b: xor edx,edx
0x1003619d: mov al,[esi+1]
0x100361a0: add esi,1
0x100361a3: test al,al
0x100361a5: jne short 10036175h
0x100361a7: cmp dword ptr [esp+14h],0
0x100361ac: mov byte ptr [ecx],0
0x100361af: je short 100361BFh
0x100361b1: push 10101CB0h
0x100361b6: push edi
0x100361b7: call 100230F0h
0x100361bc: add esp,8
```

```
0x100361bf: test ebp,ebp
0x100361c1: jne short 100361EFh
0x100361c3: mov eax,[ebx]
0x100361c5: push eax
0x100361c6: push edi
0x100361c7: call 1002CD70h
0x100361cc: push 10101C9Ch
0x100361d1: push edi
0x100361d2: call 100230F0h
0x100361d7: add esp,10h
0x100361da: pop edi
0x100361db: pop esi
0x100361dc: mov eax,ebp
0x100361de: pop ebp
0x100361df: pop ebx
0x100361e0: mov ecx,[esp+34h]
0x100361e4: xor ecx,esp
0x100361e6: call 10022326h
0x100361eb: add esp,38h
0x100361ee: ret
0x100361ef: cmp ebp,4Fh
0x100361f2: jbe short 1003620Dh
0x100361f4: push 10101C70h
0x100361f9: push edi
0x100361fa: call 100230F0h
0x100361ff: mov ecx,[ebx]
0x10036201: add esp,8
0x10036204: mov byte ptr [ecx+4Fh],0
0x10036208: mov ebp,4Fh
0x1003620d: pop edi
0x1003620e: pop esi
0x1003620f: mov eax,ebp
0x10036211: pop ebp
0x10036212: pop ebx
0x10036213: mov ecx,[esp+34h]
0x10036217: xor ecx,esp
0x10036219: call 10022326h
0x1003621e: add esp,38h
0x10036221: ret
0x10036222: push 10101C5Ch
0x10036227: push esi
0x10036228: call 100230F0h
0x1003622d: mov ecx,[esp+4Ch]
0x10036231: add esp,8
0x10036234: pop edi
0x10036235: pop esi
0x10036236: pop ebp
0x10036237: pop ebx
0x10036238: xor ecx,esp
0x1003623a: xor eax,eax
0x1003623c: call 10022326h
0x10036241: add esp,38h
0x10036244: ret
```

`sub_10036050` is called by sub_10036250, sub_10036310, sub_10036450, sub_10036610, sub_10037290, sub_10037430. It works with the text "extra interior spaces removed from keyword", "leading spaces removed from keyword", "trailing spaces removed from keyword". It delegates to sub_10022326, png_warning, png_free (4 callees).

12.821 sub_10036250 at 0x10036250 (70 instructions)

```
0x10036250: sub esp,10h
0x10036253: mov eax,ds:[10136220h]
0x10036258: xor eax,esp
0x1003625a: mov [esp+0Ch],eax
0x1003625e: mov eax,[esp+18h]
0x10036262: push ebp
```

```
0x10036263: mov ebp,[esp+20h]
0x10036267: push esi
0x10036268: mov esi,[esp+1Ch]
0x1003626c: mov cl,74h
0x1003626e: push edi
0x1003626f: mov [esp+10h],cl
0x10036273: mov [esp+13h],cl
0x10036277: lea ecx,[esp+0Ch]
0x1003627b: push ecx
0x1003627c: push eax
0x1003627d: push esi
0x1003627e: mov byte ptr [esp+1Dh],45h
0x10036283: mov byte ptr [esp+1Eh],58h
0x10036288: mov byte ptr [esp+20h],0
0x1003628d: call 10036050h
0x10036292: mov edi,eax
0x10036294: add esp,0Ch
0x10036297: test edi,edi
0x10036299: je short 100362F5h
0x1003629b: test ebp,ebp
0x1003629d: push ebx
0x1003629e: je short 100362B1h
0x100362a0: cmp byte ptr [ebp],0
0x100362a4: je short 100362B1h
0x100362a6: push ebp
0x100362a7: call dword ptr ds:[100FE0B0h]
0x100362ad: mov ebx,eax
0x100362af: jmp short 100362B3h
0x100362b1: xor ebx,ebx
0x100362b3: lea edx,[edi+ebx+1]
0x100362b7: push edx
0x100362b8: lea eax,[esp+18h]
0x100362bc: push eax
0x100362bd: push esi
0x100362be: call 10035A60h
0x100362c3: add edi,1
0x100362c6: push edi
0x100362c7: mov edi,[esp+20h]
0x100362cb: push edi
0x100362cc: push esi
0x100362cd: call 10035AE0h
0x100362d2: add esp,18h
0x100362d5: test ebx,ebx
0x100362d7: je short 100362E4h
0x100362d9: push ebx
0x100362da: push ebp
0x100362db: push esi
0x100362dc: call 10035AE0h
0x100362e1: add esp,0Ch
0x100362e4: push esi
0x100362e5: call 10035B20h
0x100362ea: push edi
0x100362eb: push esi
0x100362ec: call 1002CD70h
0x100362f1: add esp,0Ch
0x100362f4: pop ebx
0x100362f5: mov ecx,[esp+18h]
0x100362f9: pop edi
0x100362fa: pop esi
0x100362fb: pop ebp
0x100362fc: xor ecx,esp
0x100362fe: call 10022326h
0x10036303: add esp,10h
0x10036306: ret
```

`sub_10036250` is called by png_write_info, png_write_end, sub_10036310. It delegates to sub_10022326, png_free, png_write_chunk_start (6 callees).

12.822 sub_10036310 at 0x10036310 (116 instructions)

```
0x10036310: sub esp,28h
0x10036313: mov eax,ds:[10136220h]
0x10036318: xor eax,esp
0x1003631a: mov [esp+24h],eax
0x1003631e: mov eax,[esp+30h]
0x10036322: push ebx
0x10036323: push ebp
0x10036324: mov ebp,[esp+34h]
0x10036328: push esi
0x10036329: mov esi,[esp+40h]
0x1003632d: push edi
0x1003632e: lea ecx,[esp+14h]
0x10036332: push ecx
0x10036333: xor ebx,ebx
0x10036335: push eax
0x10036336: push ebp
0x10036337: mov byte ptr [esp+38h],7Ah
0x1003633c: mov byte ptr [esp+39h],54h
0x10036341: mov byte ptr [esp+3Ah],58h
0x10036346: mov byte ptr [esp+3Bh],74h
0x1003634b: mov byte ptr [esp+3Ch],0
0x10036350: mov [esp+2Ch],ebx
0x10036354: mov [esp+30h],ebx
0x10036358: mov [esp+34h],ebx
0x1003635c: mov [esp+24h],ebx
0x10036360: mov [esp+28h],ebx
0x10036364: call 10036050h
0x10036369: mov edi,eax
0x1003636b: add esp,0Ch
0x1003636e: cmp edi,ebx
0x10036370: jne short 10036393h
0x10036372: mov edx,[esp+14h]
0x10036376: push edx
0x10036377: push ebp
0x10036378: call 1002CD70h
0x1003637d: add esp,8
0x10036380: pop edi
0x10036381: pop esi
0x10036382: pop ebp
0x10036383: pop ebx
0x10036384: mov ecx,[esp+24h]
0x10036388: xor ecx,esp
0x1003638a: call 10022326h
0x1003638f: add esp,28h
0x10036392: ret
0x10036393: cmp esi,ebx
0x10036395: je near ptr 1003641Eh
0x1003639b: cmp [esi],bl
0x1003639d: je short 1003641Eh
0x1003639f: mov ebx,[esp+4Ch]
0x100363a3: cmp ebx,0FFFFFFFFh
0x100363a6: je short 1003641Ch
0x100363a8: push esi
0x100363a9: call dword ptr ds:[100FE0B0h]
0x100363af: push ebx
0x100363b0: push esi
0x100363b1: lea esi,[esp+20h]
0x100363b5: mov ecx,eax
0x100363b7: mov edx,ebp
0x100363b9: call 10035B60h
0x100363be: lea eax,[edi+eax+2]
0x100363c2: push eax
0x100363c3: lea ecx,[esp+38h]
0x100363c7: push ecx
0x100363c8: push ebp
0x100363c9: call 10035A60h
```

```
0x100363ce: mov esi,[esp+28h]
0x100363d2: add edi,1
0x100363d5: push edi
0x100363d6: push esi
0x100363d7: push ebp
0x100363d8: call 10035AE0h
0x100363dd: push esi
0x100363de: push ebp
0x100363df: call 1002CD70h
0x100363e4: push 1
0x100363e6: lea edx,[esp+3Fh]
0x100363ea: push edx
0x100363eb: push ebp
0x100363ec: mov [esp+47h],bl
0x100363f0: call 10035AE0h
0x100363f5: lea edi,[esp+4Ch]
0x100363f9: mov esi,ebp
0x100363fb: call 10035DE0h
0x10036400: push ebp
0x10036401: call 10035B20h
0x10036406: add esp,38h
0x10036409: pop edi
0x1003640a: pop esi
0x1003640b: pop ebp
0x1003640c: pop ebx
0x1003640d: mov ecx,[esp+24h]
0x10036411: xor ecx,esp
0x10036413: call 10022326h
0x10036418: add esp,28h
0x1003641b: ret
0x1003641c: xor ebx,ebx
0x1003641e: push ebx
0x1003641f: push esi
0x10036420: mov esi,[esp+1Ch]
0x10036424: push esi
0x10036425: push ebp
0x10036426: call 10036250h
0x1003642b: push esi
0x1003642c: push ebp
0x1003642d: call 1002CD70h
0x10036432: mov ecx,[esp+4Ch]
0x10036436: add esp,18h
0x10036439: pop edi
0x1003643a: pop esi
0x1003643b: pop ebp
0x1003643c: pop ebx
0x1003643d: xor ecx,esp
0x1003643f: call 10022326h
0x10036444: add esp,28h
0x10036447: ret
```

`sub_10036310` is called by png_write_info, png_write_end. It delegates to sub_10022326, png_free, png_write_chunk_start (9 callees).

12.823 sub_10036450 at 0x10036450 (147 instructions)

```
0x10036450: sub esp,38h
0x10036453: mov eax,ds:[10136220h]
0x10036458: xor eax,esp
0x1003645a: mov [esp+34h],eax
0x1003645e: mov ecx,[esp+48h]
0x10036462: mov eax,[esp+44h]
0x10036466: mov edx,[esp+4Ch]
0x1003646a: push ebx
0x1003646b: push ebp
0x1003646c: mov ebp,[esp+44h]
```

```
0x10036470: push esi
0x10036471: mov esi,[esp+5Ch]
0x10036475: push edi
0x10036476: mov [esp+14h],ecx
0x1003647a: lea ecx,[esp+1Ch]
0x1003647e: push ecx
0x1003647f: xor ebx,ebx
0x10036481: push eax
0x10036482: push ebp
0x10036483: mov [esp+30h],edx
0x10036487: mov byte ptr [esp+48h],69h
0x1003648c: mov byte ptr [esp+49h],54h
0x10036491: mov byte ptr [esp+4Ah],58h
0x10036496: mov byte ptr [esp+4Bh],74h
0x1003649b: mov byte ptr [esp+4Ch],0
0x100364a0: mov [esp+28h],ebx
0x100364a4: mov [esp+3Ch],ebx
0x100364a8: mov [esp+40h],ebx
0x100364ac: mov [esp+44h],ebx
0x100364b0: mov [esp+34h],ebx
0x100364b4: call 10036050h
0x100364b9: mov edi,eax
0x100364bb: add esp,0Ch
0x100364be: cmp edi,ebx
0x100364c0: je near ptr 100365EEh
0x100364c6: mov eax,[esp+14h]
0x100364ca: lea edx,[esp+14h]
0x100364ce: push edx
0x100364cf: push eax
0x100364d0: push ebp
0x100364d1: call 10036050h
0x100364d6: add esp,0Ch
0x100364d9: cmp eax,ebx
0x100364db: mov [esp+20h],eax
0x100364df: jne short 100364F7h
0x100364e1: push 10101D74h
0x100364e6: push ebp
0x100364e7: call 100230F0h
0x100364ec: add esp,8
0x100364ef: mov [esp+14h],ebx
0x100364f3: mov [esp+20h],ebx
0x100364f7: mov eax,[esp+24h]
0x100364fb: cmp eax,ebx
0x100364fd: jne short 10036505h
0x100364ff: mov [esp+18h],ebx
0x10036503: jmp short 10036510h
0x10036505: push eax
0x10036506: call dword ptr ds:[100FE0B0h]
0x1003650c: mov [esp+18h],eax
0x10036510: cmp esi,ebx
0x10036512: jne short 10036518h
0x10036514: xor eax,eax
0x10036516: jmp short 1003651Fh
0x10036518: push esi
0x10036519: call dword ptr ds:[100FE0B0h]
0x1003651f: mov ebx,[esp+50h]
0x10036523: lea ecx,[ebx-2]
0x10036526: push ecx
0x10036527: push esi
0x10036528: lea esi,[esp+30h]
0x1003652c: mov ecx,eax
0x1003652e: mov edx,ebp
0x10036530: call 10035B60h
0x10036535: mov edx,[esp+20h]
0x10036539: mov esi,[esp+28h]
0x1003653d: add eax,edx
0x1003653f: add eax,esi
```

```
0x10036541: lea eax,[eax+edi+5]
0x10036545: push eax
0x10036546: lea ecx,[esp+48h]
0x1003654a: push ecx
0x1003654b: push ebp
0x1003654c: call 10035A60h
0x10036551: mov edx,[esp+30h]
0x10036555: add edi,1
0x10036558: push edi
0x10036559: push edx
0x1003655a: push ebp
0x1003655b: call 10035AE0h
0x10036560: add esp,20h
0x10036563: cmp ebx,1
0x10036566: je short 10036572h
0x10036568: cmp ebx,0FFFFFFFFh
0x1003656b: mov byte ptr [esp+10h],1
0x10036570: jne short 10036577h
0x10036572: mov byte ptr [esp+10h],0
0x10036577: push 2
0x10036579: lea eax,[esp+14h]
0x1003657d: push eax
0x1003657e: push ebp
0x1003657f: mov byte ptr [esp+1Dh],0
0x10036584: call 10035AE0h
0x10036589: mov ebx,[esp+20h]
0x1003658d: add esp,0Ch
0x10036590: test ebx,ebx
0x10036592: mov byte ptr [esp+10h],0
0x10036597: mov eax,ebx
0x10036599: jne short 1003659Fh
0x1003659b: lea eax,[esp+10h]
0x1003659f: add esi,1
0x100365a2: push esi
0x100365a3: push eax
0x100365a4: push ebp
0x100365a5: call 10035AE0h
0x100365aa: mov eax,[esp+30h]
0x100365ae: add esp,0Ch
0x100365b1: test eax,eax
0x100365b3: jne short 100365B9h
0x100365b5: lea eax,[esp+10h]
0x100365b9: mov ecx,[esp+18h]
0x100365bd: add ecx,1
0x100365c0: push ecx
0x100365c1: push eax
0x100365c2: push ebp
0x100365c3: call 10035AE0h
0x100365c8: lea edi,[esp+34h]
0x100365cc: mov esi,ebp
0x100365ce: call 10035DE0h
0x100365d3: push ebp
0x100365d4: call 10035B20h
0x100365d9: mov edx,[esp+2Ch]
0x100365dd: push edx
0x100365de: push ebp
0x100365df: call 1002CD70h
0x100365e4: push ebx
0x100365e5: push ebp
0x100365e6: call 1002CD70h
0x100365eb: add esp,20h
0x100365ee: mov ecx,[esp+44h]
0x100365f2: pop edi
0x100365f3: pop esi
0x100365f4: pop ebp
0x100365f5: pop ebx
0x100365f6: xor ecx,esp
```

```
0x100365f8: call 10022326h
0x100365fd: add esp,38h
0x10036600: ret
```

`sub_10036450` is called by png_write_info, png_write_end. It delegates to sub_10022326, png_warning, png_free (9 callees).

12.824 sub_10036610 at 0x10036610 (145 instructions)

```
0x10036610: sub esp,38h
0x10036613: mov eax,ds:[10136220h]
0x10036618: xor eax,esp
0x1003661a: mov [esp+34h],eax
0x1003661e: cmp dword ptr [esp+4Ch],4
0x10036623: mov eax,[esp+58h]
0x10036627: push ebx
0x10036628: mov ebx,[esp+58h]
0x1003662c: push ebp
0x1003662d: push esi
0x1003662e: mov esi,[esp+48h]
0x10036632: push edi
0x10036633: mov edi,[esp+50h]
0x10036637: mov [esp+24h],ebx
0x1003663b: mov [esp+14h],eax
0x1003663f: mov byte ptr [esp+30h],70h
0x10036644: mov byte ptr [esp+31h],43h
0x10036649: mov byte ptr [esp+32h],41h
0x1003664e: mov byte ptr [esp+33h],4Ch
0x10036653: mov byte ptr [esp+34h],0
0x10036658: jl short 10036668h
0x1003665a: push 101014BCh
0x1003665f: push esi
0x10036660: call 100230F0h
0x10036665: add esp,8
0x10036668: lea ecx,[esp+28h]
0x1003666c: push ecx
0x1003666d: push edi
0x1003666e: push esi
0x1003666f: call 10036050h
0x10036674: mov ebp,eax
0x10036676: add esp,0Ch
0x10036679: add ebp,1
0x1003667c: push ebx
0x1003667d: mov [esp+20h],ebp
0x10036681: call dword ptr ds:[100FE0B0h]
0x10036687: mov ebx,[esp+60h]
0x1003668b: xor edx,edx
0x1003668d: test ebx,ebx
0x1003668f: setne dl
0x10036692: lea ecx,[ebx*4]
0x10036699: push ecx
0x1003669a: push esi
0x1003669b: add eax,edx
0x1003669d: mov [esp+28h],eax
0x100366a1: lea eax,[eax+ebp+0Ah]
0x100366a5: mov [esp+18h],eax
0x100366a9: call 1002CD20h
0x100366ae: add esp,8
0x100366b1: xor edi,edi
0x100366b3: test ebx,ebx
0x100366b5: mov [esp+18h],eax
0x100366b9: jle short 100366FFh
0x100366bb: mov ecx,[esp+14h]
0x100366bf: sub ecx,eax
0x100366c1: mov ebp,eax
0x100366c3: mov [esp+2Ch],ecx
```

```
0x100366c7: jmp short 100366D4h
0x100366d0: mov ecx,[esp+2Ch]
0x100366d4: mov edx,[ecx+ebp]
0x100366d7: push edx
0x100366d8: call dword ptr ds:[100FE0B0h]
0x100366de: xor edx,edx
0x100366e0: lea ecx,[ebx-1]
0x100366e3: cmp edi,ecx
0x100366e5: setne dl
0x100366e8: add edi,1
0x100366eb: add ebp,4
0x100366ee: add eax,edx
0x100366f0: add [esp+10h],eax
0x100366f4: cmp edi,ebx
0x100366f6: mov [ebp-4],eax
0x100366f9: jl short 100366D0h
0x100366fb: mov ebp,[esp+1Ch]
0x100366ff: mov eax,[esp+10h]
0x10036703: push eax
0x10036704: lea ecx,[esp+34h]
0x10036708: push ecx
0x10036709: push esi
0x1003670a: call 10035A60h
0x1003670f: mov edi,[esp+34h]
0x10036713: push ebp
0x10036714: push edi
0x10036715: push esi
0x10036716: call 10035AE0h
0x1003671b: mov edx,[esp+6Ch]
0x1003671f: push edx
0x10036720: lea eax,[esp+54h]
0x10036724: push eax
0x10036725: call 10035980h
0x1003672a: mov ecx,[esp+78h]
0x1003672e: push ecx
0x1003672f: lea edx,[esp+60h]
0x10036733: push edx
0x10036734: call 10035980h
0x10036739: mov al,[esp+84h]
0x10036740: push 0Ah
0x10036742: lea ecx,[esp+64h]
0x10036746: push ecx
0x10036747: push esi
0x10036748: mov [esp+74h],al
0x1003674c: mov [esp+75h],bl
0x10036750: call 10035AE0h
0x10036755: mov edx,[esp+54h]
0x10036759: mov eax,[esp+58h]
0x1003675d: push edx
0x1003675e: push eax
0x1003675f: push esi
0x10036760: call 10035AE0h
0x10036765: add esp,40h
0x10036768: push edi
0x10036769: push esi
0x1003676a: call 1002CD70h
0x1003676f: add esp,8
0x10036772: test ebx,ebx
0x10036774: jle short 10036798h
0x10036776: mov edi,[esp+14h]
0x1003677a: mov ebp,[esp+18h]
0x1003677e: sub ebp,edi
0x10036780: mov ecx,[edi+ebp]
0x10036783: mov edx,[edi]
0x10036785: push ecx
0x10036786: push edx
0x10036787: push esi
```

```
0x10036788: call 10035AE0h
0x1003678d: add esp,0Ch
0x10036790: add edi,4
0x10036793: sub ebx,1
0x10036796: jne short 10036780h
0x10036798: mov eax,[esp+18h]
0x1003679c: push eax
0x1003679d: push esi
0x1003679e: call 1002CD70h
0x100367a3: push esi
0x100367a4: call 10035B20h
0x100367a9: mov ecx,[esp+50h]
0x100367ad: add esp,0Ch
0x100367b0: pop edi
0x100367b1: pop esi
0x100367b2: pop ebp
0x100367b3: pop ebx
0x100367b4: xor ecx,esp
0x100367b6: call 10022326h
0x100367bb: add esp,38h
0x100367be: ret
```

`sub_10036610` is called by png_write_info. It works with the text "Unrecognized equation type for pCAL chunk". It delegates to sub_10022326, png_warning, png_malloc (9 callees).

12.825 sub_100367c0 at 0x100367c0 (99 instructions)

```
0x100367c0: push esi
0x100367c1: mov esi,[esp+8]
0x100367c5: movzx ecx,byte ptr [esi+12Bh]
0x100367cc: movzx eax,byte ptr [esi+128h]
0x100367d3: imul ecx,eax
0x100367d6: cmp ecx,8
0x100367d9: jl short 100367E9h
0x100367db: shr ecx,3
0x100367de: imul ecx,[esi+0CCh]
0x100367e5: mov eax,ecx
0x100367e7: jmp short 100367F8h
0x100367e9: mov eax,[esi+0CCh]
0x100367ef: imul eax,ecx
0x100367f2: add eax,7
0x100367f5: shr eax,3
0x100367f8: push edi
0x100367f9: lea edi,[eax+1]
0x100367fc: push edi
0x100367fd: push esi
0x100367fe: call 1002CD20h
0x10036803: mov [esi+0ECh],eax
0x10036809: add esp,8
0x1003680c: mov byte ptr [eax],0
0x1003680f: test byte ptr [esi+125h],10h
0x10036816: je short 10036834h
0x10036818: mov ecx,[esi+0DCh]
0x1003681e: add ecx,1
0x10036821: push ecx
0x10036822: push esi
0x10036823: call 1002CD20h
0x10036828: mov [esi+0F0h],eax
0x1003682e: add esp,8
0x10036831: mov byte ptr [eax],1
0x10036834: test byte ptr [esi+125h],0E0h
0x1003683b: je short 100368BCh
0x1003683d: push edi
0x1003683e: push esi
0x1003683f: call 1002CDD0h
0x10036844: add esp,8
```

```
0x10036847: test byte ptr [esi+125h],20h
0x1003684e: mov [esi+0E8h],eax
0x10036854: je short 10036872h
0x10036856: mov edx,[esi+0DCh]
0x1003685c: add edx,1
0x1003685f: push edx
0x10036860: push esi
0x10036861: call 1002CD20h
0x10036866: mov [esi+0F4h],eax
0x1003686c: add esp,8
0x1003686f: mov byte ptr [eax],2
0x10036872: test byte ptr [esi+125h],40h
0x10036879: je short 10036897h
0x1003687b: mov eax,[esi+0DCh]
0x10036881: add eax,1
0x10036884: push eax
0x10036885: push esi
0x10036886: call 1002CD20h
0x1003688b: mov [esi+0F8h],eax
0x10036891: add esp,8
0x10036894: mov byte ptr [eax],3
0x10036897: test byte ptr [esi+125h],80h
0x1003689e: je short 100368BCh
0x100368a0: mov ecx,[esi+0DCh]
0x100368a6: add ecx,1
0x100368a9: push ecx
0x100368aa: push esi
0x100368ab: call 1002CD20h
0x100368b0: mov [esi+0FCh],eax
0x100368b6: add esp,8
0x100368b9: mov byte ptr [eax],4
0x100368bc: cmp byte ptr [esi+123h],0
0x100368c3: pop edi
0x100368c4: je short 1003690Ch
0x100368c6: test byte ptr [esi+74h],2
0x100368ca: jne short 100368F2h
0x100368cc: mov edx,[esi+0D0h]
0x100368d2: mov eax,[esi+0CCh]
0x100368d8: add edx,7
0x100368db: shr edx,3
0x100368de: add eax,7
0x100368e1: shr eax,3
0x100368e4: mov [esi+0D4h],edx
0x100368ea: mov [esi+0D8h],eax
0x100368f0: jmp short 10036924h
0x100368f2: mov ecx,[esi+0D0h]
0x100368f8: mov edx,[esi+0CCh]
0x100368fe: mov [esi+0D4h],ecx
0x10036904: mov [esi+0D8h],edx
0x1003690a: jmp short 10036924h
0x1003690c: mov eax,[esi+0D0h]
0x10036912: mov ecx,[esi+0CCh]
0x10036918: mov [esi+0D4h],eax
0x1003691e: mov [esi+0D8h],ecx
0x10036924: mov edx,[esi+0B4h]
0x1003692a: mov eax,[esi+0B0h]
0x10036930: mov [esi+88h],edx
0x10036936: mov [esi+84h],eax
0x1003693c: pop esi
0x1003693d: ret
```

`sub_100367c0` is called by png_write_row. It delegates to png_malloc, png_calloc (2 callees).

12.826 sub_10036940 at 0x10036940 (219 instructions)

```
0x10036940: sub esp,44h
0x10036943: mov eax,4
0x10036948: push ebp
0x10036949: xor edx,edx
0x1003694b: mov [esp+30h],eax
0x1003694f: mov ebp,2
0x10036954: mov [esp+18h],eax
0x10036958: mov [esp+1Ch],eax
0x1003695c: mov eax,[esp+54h]
0x10036960: cmp eax,6
0x10036963: push esi
0x10036964: mov esi,1
0x10036969: mov ecx,8
0x1003696e: mov [esp+30h],edx
0x10036972: mov [esp+38h],edx
0x10036976: mov [esp+3Ch],ebp
0x1003697a: mov [esp+40h],edx
0x1003697e: mov [esp+44h],esi
0x10036982: mov [esp+48h],edx
0x10036986: mov [esp+14h],ecx
0x1003698a: mov [esp+18h],ecx
0x1003698e: mov [esp+24h],ebp
0x10036992: mov [esp+28h],ebp
0x10036996: mov [esp+2Ch],esi
0x1003699a: jge near ptr 10036BE2h
0x100369a0: push ebx
0x100369a1: push edi
0x100369a2: mov edi,[esp+58h]
0x100369a6: movzx ebx,byte ptr [edi+0Bh]
0x100369aa: mov ecx,ebx
0x100369ac: sub ecx,esi
0x100369ae: je near ptr 10036B28h
0x100369b4: sub ecx,esi
0x100369b6: je near ptr 10036AA1h
0x100369bc: sub ecx,ebp
0x100369be: lea ebp,[eax*4]
0x100369c5: je short 10036A2Eh
0x100369c7: mov edx,[edi]
0x100369c9: mov edi,[esp+ebp+38h]
0x100369cd: mov ecx,[esp+5Ch]
0x100369d1: shr ebx,3
0x100369d4: cmp edi,edx
0x100369d6: mov [esp+18h],edx
0x100369da: mov [esp+10h],ecx
0x100369de: mov [esp+60h],edi
0x100369e2: jae near ptr 10036BA0h
0x100369e8: mov eax,[esp+ebp+1Ch]
0x100369ec: mov esi,edi
0x100369ee: imul eax,ebx
0x100369f1: imul esi,ebx
0x100369f4: mov [esp+14h],eax
0x100369f8: add esi,ecx
0x100369fa: lea ebx,[ebx]
0x10036a00: cmp ecx,esi
0x10036a02: je short 10036A17h
0x10036a04: push ebx
0x10036a05: push esi
0x10036a06: push ecx
0x10036a07: call 100228C6h
0x10036a0c: mov ecx,[esp+1Ch]
0x10036a10: mov eax,[esp+20h]
0x10036a14: add esp,0Ch
0x10036a17: add edi,[esp+ebp+1Ch]
0x10036a1b: add ecx,ebx
0x10036a1d: add esi,eax
0x10036a1f: cmp edi,[esp+18h]
```

```
0x10036a23: mov [esp+10h],ecx
0x10036a27: jb short 10036A00h
0x10036a29: jmp near ptr 10036BA0h
0x10036a2e: mov ecx,[edi]
0x10036a30: mov eax,[esp+ebp+38h]
0x10036a34: cmp eax,ecx
0x10036a36: mov edi,[esp+5Ch]
0x10036a3a: mov [esp+18h],ecx
0x10036a3e: mov [esp+10h],edi
0x10036a42: mov esi,4
0x10036a47: mov [esp+60h],eax
0x10036a4b: jae near ptr 10036BA0h
0x10036a51: mov bl,al
0x10036a53: and bl,1
0x10036a56: add bl,bl
0x10036a58: add bl,bl
0x10036a5a: mov ecx,4
0x10036a5f: sub cl,bl
0x10036a61: mov ebx,eax
0x10036a63: shr ebx,1
0x10036a65: movzx ebx,byte ptr [ebx+edi]
0x10036a69: shr ebx,cl
0x10036a6b: mov ecx,esi
0x10036a6d: and ebx,0Fh
0x10036a70: shl ebx,cl
0x10036a72: or edx,ebx
0x10036a74: test esi,esi
0x10036a76: jne short 10036A8Ch
0x10036a78: lea esi,[ecx+4]
0x10036a7b: mov ecx,[esp+10h]
0x10036a7f: mov [ecx],dl
0x10036a81: add ecx,1
0x10036a84: mov [esp+10h],ecx
0x10036a88: xor edx,edx
0x10036a8a: jmp short 10036A8Fh
0x10036a8c: sub esi,4
0x10036a8f: add eax,[esp+ebp+1Ch]
0x10036a93: cmp eax,[esp+18h]
0x10036a97: jb short 10036A51h
0x10036a99: cmp esi,4
0x10036a9c: jmp near ptr 10036B98h
0x10036aa1: mov ecx,[edi]
0x10036aa3: mov edi,[esp+5Ch]
0x10036aa7: lea ebp,[eax*4]
0x10036aae: mov eax,[esp+ebp+38h]
0x10036ab2: cmp eax,ecx
0x10036ab4: mov [esp+18h],ecx
0x10036ab8: mov [esp+10h],edi
0x10036abc: mov esi,6
0x10036ac1: mov [esp+60h],eax
0x10036ac5: jae near ptr 10036BA0h
0x10036acb: jmp short 10036AD0h
0x10036ad0: mov bl,al
0x10036ad2: and bl,3
0x10036ad5: mov ecx,3
0x10036ada: sub cl,bl
0x10036adc: mov ebx,eax
0x10036ade: shr ebx,2
0x10036ae1: movzx ebx,byte ptr [ebx+edi]
0x10036ae5: add cl,cl
0x10036ae7: shr ebx,cl
0x10036ae9: mov ecx,esi
0x10036aeb: and ebx,3
0x10036aee: shl ebx,cl
0x10036af0: or edx,ebx
0x10036af2: test esi,esi
0x10036af4: jne short 10036B0Ah
```

```
0x10036af6: lea esi,[ecx+6]
0x10036af9: mov ecx,[esp+10h]
0x10036afd: mov [ecx],dl
0x10036aff: add ecx,1
0x10036b02: mov [esp+10h],ecx
0x10036b06: xor edx,edx
0x10036b08: jmp short 10036B0Dh
0x10036b0a: sub esi,2
0x10036b0d: add eax,[esp+ebp+1Ch]
0x10036b11: cmp eax,[esp+18h]
0x10036b15: jb short 10036AD0h
0x10036b17: cmp esi,6
0x10036b1a: je near ptr 10036BA0h
0x10036b20: mov ecx,[esp+10h]
0x10036b24: mov [ecx],dl
0x10036b26: jmp short 10036BA0h
0x10036b28: mov ecx,[edi]
0x10036b2a: mov edi,[esp+5Ch]
0x10036b2e: lea ebp,[eax*4]
0x10036b35: mov eax,[esp+ebp+38h]
0x10036b39: cmp eax,ecx
0x10036b3b: mov [esp+18h],ecx
0x10036b3f: mov [esp+10h],edi
0x10036b43: mov esi,7
0x10036b48: mov [esp+60h],eax
0x10036b4c: jae short 10036BA0h
0x10036b4e: mov edi,edi
0x10036b50: mov bl,al
0x10036b52: and bl,7
0x10036b55: mov ecx,7
0x10036b5a: sub cl,bl
0x10036b5c: mov ebx,eax
0x10036b5e: shr ebx,3
0x10036b61: movzx ebx,byte ptr [ebx+edi]
0x10036b65: shr ebx,cl
0x10036b67: mov ecx,esi
0x10036b69: and ebx,1
0x10036b6c: shl ebx,cl
0x10036b6e: or edx,ebx
0x10036b70: test esi,esi
0x10036b72: jne short 10036B88h
0x10036b74: lea esi,[ecx+7]
0x10036b77: mov ecx,[esp+10h]
0x10036b7b: mov [ecx],dl
0x10036b7d: add ecx,1
0x10036b80: mov [esp+10h],ecx
0x10036b84: xor edx,edx
0x10036b86: jmp short 10036B8Bh
0x10036b88: sub esi,1
0x10036b8b: add eax,[esp+ebp+1Ch]
0x10036b8f: cmp eax,[esp+18h]
0x10036b93: jb short 10036B50h
0x10036b95: cmp esi,7
0x10036b98: je short 10036BA0h
0x10036b9a: mov eax,[esp+10h]
0x10036b9e: mov [eax],dl
0x10036ba0: mov ebp,[esp+ebp+1Ch]
0x10036ba4: mov esi,[esp+58h]
0x10036ba8: mov edx,[esi]
0x10036baa: mov ecx,ebp
0x10036bac: sub ecx,[esp+60h]
0x10036bb0: pop edi
0x10036bb1: lea eax,[ecx+edx-1]
0x10036bb5: xor edx,edx
0x10036bb7: div ebp
0x10036bb9: mov cl,[esi+0Bh]
0x10036bbc: cmp cl,8
```

```
0x10036bbf: pop ebx
0x10036bc0: movzx ecx,cl
0x10036bc3: mov [esi],eax
0x10036bc5: jb short 10036BD6h
0x10036bc7: shr ecx,3
0x10036bca: imul ecx,eax
0x10036bcd: mov [esi+4],ecx
0x10036bd0: pop esi
0x10036bd1: pop ebp
0x10036bd2: add esp,44h
0x10036bd5: ret
0x10036bd6: imul ecx,eax
0x10036bd9: add ecx,7
0x10036bdc: shr ecx,3
0x10036bdf: mov [esi+4],ecx
0x10036be2: pop esi
0x10036be3: pop ebp
0x10036be4: add esp,44h
0x10036be7: ret
```

`sub_10036940` is called by png_write_row. It delegates to j_memcpy (1 callees).

12.827 png_write_chunk at 0x10036bf0 (22 instructions)

```
0x10036bf0: push esi
0x10036bf1: mov esi,[esp+8]
0x10036bf5: test esi,esi
0x10036bf7: je short 10036C20h
0x10036bf9: mov eax,[esp+0Ch]
0x10036bfd: push edi
0x10036bfe: mov edi,[esp+18h]
0x10036c02: push edi
0x10036c03: push eax
0x10036c04: push esi
0x10036c05: call 10035A60h
0x10036c0a: mov ecx,[esp+20h]
0x10036c0e: push edi
0x10036c0f: push ecx
0x10036c10: push esi
0x10036c11: call 10035AE0h
0x10036c16: push esi
0x10036c17: call 10035B20h
0x10036c1c: add esp,1Ch
0x10036c1f: pop edi
0x10036c20: pop esi
0x10036c21: ret
```

`png_write_chunk` is called by png_write_info_before_PLTE, png_write_info, png_write_end, sub_10036c30, sub_10037020, sub_10037160, sub_100371c0, sub_10037220, sub_100375c0, sub_100376c0, sub_100377c0, sub_10037930, sub_10037ab0, sub_10037b40, sub_10037c20, sub_10037cb0. It delegates to png_write_chunk_start, png_write_chunk_data, png_write_chunk_end (3 callees).

12.828 sub_10036c30 at 0x10036c30 (106 instructions)

```
0x10036c30: sub esp,1Ch
0x10036c33: mov eax,ds:[10136220h]
0x10036c38: xor eax,esp
0x10036c3a: mov [esp+18h],eax
0x10036c3e: mov ecx,[esp+30h]
0x10036c42: cmp ecx,6
0x10036c45: push ebx
0x10036c46: push esi
0x10036c47: mov esi,[esp+28h]
0x10036c4b: mov byte ptr [esp+8],49h
```

```
0x10036c50: mov byte ptr [esp+9],48h
0x10036c55: mov byte ptr [esp+0Ah],44h
0x10036c5a: mov byte ptr [esp+0Bh],52h
0x10036c5f: mov byte ptr [esp+0Ch],0
0x10036c64: ja near ptr 10036FCBh
0x10036c6a: jmp dword ptr [ecx*4+10036FD8h]
0x10036fcb: push 10101D98h
0x10036fd0: push esi
0x10036fd1: call 100231E0h
0x10036fd6: mov edi,edi
0x10036fd8: jno short 10037046h
0x10036fda: add edx,[eax]
0x10036fdc: retf
0x10037046: inc esp
0x10037047: mov byte ptr [esp+0Eh],41h
0x1003704c: mov byte ptr [esp+0Fh],54h
0x10037051: mov byte ptr [esp+10h],0
0x10037056: jne near ptr 10037125h
0x1003705c: cmp byte ptr [edi+254h],0
0x10037063: jne near ptr 10037125h
0x10037069: movzx ebp,byte ptr [ebx]
0x1003706c: mov eax,ebp
0x1003706e: and al,0Fh
0x10037070: cmp al,8
0x10037072: jne near ptr 1003714Fh
0x10037078: mov ecx,ebp
0x1003707a: and ecx,0F0h
0x10037080: cmp ecx,70h
0x10037083: ja near ptr 1003714Fh
0x10037089: cmp dword ptr [esp+24h],2
0x1003708e: jb near ptr 10037125h
0x10037094: mov edx,[edi+0D0h]
0x1003709a: cmp edx,4000h
0x100370a0: jae near ptr 10037125h
0x100370a6: mov ecx,[edi+0CCh]
0x100370ac: cmp ecx,4000h
0x100370b2: jae short 10037125h
0x100370b4: movzx eax,byte ptr [edi+12Ah]
0x100370bb: push esi
0x100370bc: movzx esi,byte ptr [edi+127h]
0x100370c3: imul eax,esi
0x100370c6: imul eax,ecx
0x100370c9: add eax,0Fh
0x100370cc: shr eax,3
0x100370cf: imul eax,edx
0x100370d2: mov edx,ebp
0x100370d4: shr edx,4
0x100370d7: lea ecx,[edx+7]
0x100370da: mov esi,1
0x100370df: shl esi,cl
0x100370e1: cmp eax,esi
0x100370e3: ja short 100370F6h
0x100370e5: cmp esi,100h
0x100370eb: jb short 100370F6h
0x100370ed: shr esi,1
0x100370ef: sub edx,1
0x100370f2: cmp eax,esi
0x100370f4: jbe short 100370E5h
0x100370f6: shl edx,4
0x100370f9: or edx,8
0x100370fc: mov eax,edx
0x100370fe: cmp ebp,eax
0x10037100: je short 10037124h
0x10037102: movzx ecx,byte ptr [ebx+1]
0x10037106: mov [ebx],al
0x10037108: and ecx,0E0h
0x1003710e: shl eax,8
```

```
0x10037111: add eax,ecx
0x10037113: xor edx,edx
0x10037115: mov esi,1Fh
0x1003711a: div esi
0x1003711c: sub cl,dl
0x1003711e: add cl,1Fh
0x10037121: mov [ebx+1],cl
0x10037124: pop esi
0x10037125: mov edx,[esp+24h]
0x10037129: push edx
0x1003712a: push ebx
0x1003712b: lea eax,[esp+14h]
0x1003712f: push eax
0x10037130: push edi
0x10037131: call 10036BF0h
0x10037136: or dword ptr [edi+6Ch],4
0x1003713a: mov ecx,[esp+24h]
0x1003713e: add esp,10h
0x10037141: pop edi
0x10037142: pop ebp
0x10037143: pop ebx
0x10037144: xor ecx,esp
0x10037146: call 10022326h
0x1003714b: add esp,0Ch
0x1003714e: ret
0x1003714f: push 10101FB0h
0x10037154: push edi
0x10037155: call 100231E0h
0x1003715a: int 3
```

`sub_10036c30` is called by png_write_info_before_PLTE. It works with the text "Invalid image color type specified", "Invalid zlib compression method or flags in IDAT". It delegates to sub_10022326, png_error, png_write_chunk (3 callees).

12.829 sub_10037020 at 0x10037020 (94 instructions)

```
0x10037020: sub esp,0Ch
0x10037023: mov eax,ds:[10136220h]
0x10037028: xor eax,esp
0x1003702a: mov [esp+8],eax
0x1003702e: push ebx
0x1003702f: mov ebx,[esp+18h]
0x10037033: push ebp
0x10037034: push edi
0x10037035: mov edi,[esp+1Ch]
0x10037039: test byte ptr [edi+6Ch],4
0x1003703d: mov byte ptr [esp+0Ch],49h
0x10037042: mov byte ptr [esp+0Dh],44h
0x10037047: mov byte ptr [esp+0Eh],41h
0x1003704c: mov byte ptr [esp+0Fh],54h
0x10037051: mov byte ptr [esp+10h],0
0x10037056: jne near ptr 10037125h
0x1003705c: cmp byte ptr [edi+254h],0
0x10037063: jne near ptr 10037125h
0x10037069: movzx ebp,byte ptr [ebx]
0x1003706c: mov eax,ebp
0x1003706e: and al,0Fh
0x10037070: cmp al,8
0x10037072: jne near ptr 1003714Fh
0x10037078: mov ecx,ebp
0x1003707a: and ecx,0F0h
0x10037080: cmp ecx,70h
0x10037083: ja near ptr 1003714Fh
0x10037089: cmp dword ptr [esp+24h],2
0x1003708e: jb near ptr 10037125h
0x10037094: mov edx,[edi+0D0h]
```

```
0x1003709a: cmp edx,4000h
0x100370a0: jae near ptr 10037125h
0x100370a6: mov ecx,[edi+0CCh]
0x100370ac: cmp ecx,4000h
0x100370b2: jae short 10037125h
0x100370b4: movzx eax,byte ptr [edi+12Ah]
0x100370bb: push esi
0x100370bc: movzx esi,byte ptr [edi+127h]
0x100370c3: imul eax,esi
0x100370c6: imul eax,ecx
0x100370c9: add eax,0Fh
0x100370cc: shr eax,3
0x100370cf: imul eax,edx
0x100370d2: mov edx,ebp
0x100370d4: shr edx,4
0x100370d7: lea ecx,[edx+7]
0x100370da: mov esi,1
0x100370df: shl esi,cl
0x100370e1: cmp eax,esi
0x100370e3: ja short 100370F6h
0x100370e5: cmp esi,100h
0x100370eb: jb short 100370F6h
0x100370ed: shr esi,1
0x100370ef: sub edx,1
0x100370f2: cmp eax,esi
0x100370f4: jbe short 100370E5h
0x100370f6: shl edx,4
0x100370f9: or edx,8
0x100370fc: mov eax,edx
0x100370fe: cmp ebp,eax
0x10037100: je short 10037124h
0x10037102: movzx ecx,byte ptr [ebx+1]
0x10037106: mov [ebx],al
0x10037108: and ecx,0E0h
0x1003710e: shl eax,8
0x10037111: add eax,ecx
0x10037113: xor edx,edx
0x10037115: mov esi,1Fh
0x1003711a: div esi
0x1003711c: sub cl,dl
0x1003711e: add cl,1Fh
0x10037121: mov [ebx+1],cl
0x10037124: pop esi
0x10037125: mov edx,[esp+24h]
0x10037129: push edx
0x1003712a: push ebx
0x1003712b: lea eax,[esp+14h]
0x1003712f: push eax
0x10037130: push edi
0x10037131: call 10036BF0h
0x10037136: or dword ptr [edi+6Ch],4
0x1003713a: mov ecx,[esp+24h]
0x1003713e: add esp,10h
0x10037141: pop edi
0x10037142: pop ebp
0x10037143: pop ebx
0x10037144: xor ecx,esp
0x10037146: call 10022326h
0x1003714b: add esp,0Ch
0x1003714e: ret
0x1003714f: push 10101FB0h
0x10037154: push edi
0x10037155: call 100231E0h
0x1003715a: int 3
```

`sub_10037020` is called by png_write_flush, sub_10037d80, sub_10037fe0. It works with the text "Invalid zlib compression method or flags in IDAT". It delegates to sub_10022326, png_error, png_write_chunk (3 callees).

12.830 sub_10037160 at 0x10037160 (25 instructions)

```
0x10037160: sub esp,0Ch
0x10037163: mov eax,ds:[10136220h]
0x10037168: xor eax,esp
0x1003716a: mov [esp+8],eax
0x1003716e: push esi
0x1003716f: mov esi,[esp+14h]
0x10037173: push 0
0x10037175: push 0
0x10037177: lea eax,[esp+0Ch]
0x1003717b: push eax
0x1003717c: push esi
0x1003717d: mov byte ptr [esp+14h],49h
0x10037182: mov byte ptr [esp+15h],45h
0x10037187: mov byte ptr [esp+16h],4Eh
0x1003718c: mov byte ptr [esp+17h],44h
0x10037191: mov byte ptr [esp+18h],0
0x10037196: call 10036BF0h
0x1003719b: or dword ptr [esi+6Ch],10h
0x1003719f: mov ecx,[esp+1Ch]
0x100371a3: add esp,10h
0x100371a6: pop esi
0x100371a7: xor ecx,esp
0x100371a9: call 10022326h
0x100371ae: add esp,0Ch
0x100371b1: ret
```

`sub_10037160` is called by png_write_end. It delegates to sub_10022326, png_write_chunk (2 callees).

12.831 sub_100371c0 at 0x100371c0 (31 instructions)

```
0x100371c0: sub esp,10h
0x100371c3: mov eax,ds:[10136220h]
0x100371c8: xor eax,esp
0x100371ca: mov [esp+0Ch],eax
0x100371ce: mov al,41h
0x100371d0: push esi
0x100371d1: mov esi,[esp+18h]
0x100371d5: mov [esp+9],al
0x100371d9: mov [esp+0Bh],al
0x100371dd: mov eax,[esp+1Ch]
0x100371e1: push eax
0x100371e2: lea ecx,[esp+8]
0x100371e6: push ecx
0x100371e7: mov byte ptr [esp+10h],67h
0x100371ec: mov byte ptr [esp+12h],4Dh
0x100371f1: mov byte ptr [esp+14h],0
0x100371f6: call 10035950h
0x100371fb: push 4
0x100371fd: lea edx,[esp+10h]
0x10037201: push edx
0x10037202: lea eax,[esp+18h]
0x10037206: push eax
0x10037207: push esi
0x10037208: call 10036BF0h
0x1003720d: mov ecx,[esp+28h]
0x10037211: add esp,18h
0x10037214: pop esi
0x10037215: xor ecx,esp
0x10037217: call 10022326h
0x1003721c: add esp,10h
0x1003721f: ret
```

`sub_100371c0` is called by png_write_info_before_PLTE. It delegates to sub_10022326, png_save_uint_32, png_write_chunk (3 callees).

12.832 sub_10037220 at 0x10037220 (35 instructions)

```
0x10037220: sub esp,10h
0x10037223: mov eax,ds:[10136220h]
0x10037228: xor eax,esp
0x1003722a: mov [esp+0Ch],eax
0x1003722e: push ebx
0x1003722f: mov ebx,[esp+1Ch]
0x10037233: cmp ebx,4
0x10037236: push esi
0x10037237: mov esi,[esp+1Ch]
0x1003723b: mov byte ptr [esp+0Ch],73h
0x10037240: mov byte ptr [esp+0Dh],52h
0x10037245: mov byte ptr [esp+0Eh],47h
0x1003724a: mov byte ptr [esp+0Fh],42h
0x1003724f: mov byte ptr [esp+10h],0
0x10037254: jl short 10037264h
0x10037256: push 10101FE4h
0x1003725b: push esi
0x1003725c: call 100230F0h
0x10037261: add esp,8
0x10037264: push 1
0x10037266: lea eax,[esp+0Fh]
0x1003726a: push eax
0x1003726b: lea ecx,[esp+14h]
0x1003726f: push ecx
0x10037270: push esi
0x10037271: mov [esp+1Bh],bl
0x10037275: call 10036BF0h
0x1003727a: mov ecx,[esp+24h]
0x1003727e: add esp,10h
0x10037281: pop esi
0x10037282: pop ebx
0x10037283: xor ecx,esp
0x10037285: call 10022326h
0x1003728a: add esp,10h
0x1003728d: ret
```

`sub_10037220` is called by png_write_info_before_PLTE. It delegates to sub_10022326, png_warning, png_write_chunk (3 callees).

12.833 sub_10037290 at 0x10037290 (144 instructions)

```
0x10037290: sub esp,24h
0x10037293: mov eax,ds:[10136220h]
0x10037298: xor eax,esp
0x1003729a: mov [esp+20h],eax
0x1003729e: mov eax,[esp+2Ch]
0x100372a2: push ebx
0x100372a3: push ebp
0x100372a4: mov ebp,[esp+30h]
0x100372a8: mov cl,43h
0x100372aa: push esi
0x100372ab: mov [esp+25h],cl
0x100372af: mov [esp+26h],cl
0x100372b3: lea ecx,[esp+0Ch]
0x100372b7: push ecx
0x100372b8: xor esi,esi
0x100372ba: push eax
0x100372bb: push ebp
0x100372bc: mov byte ptr [esp+30h],69h
0x100372c1: mov byte ptr [esp+33h],50h
0x100372c6: mov byte ptr [esp+34h],0
0x100372cb: mov [esp+24h],esi
0x100372cf: mov [esp+28h],esi
0x100372d3: mov [esp+2Ch],esi
```

```
0x100372d7: mov [esp+1Ch],esi
0x100372db: mov [esp+20h],esi
0x100372df: call 10036050h
0x100372e4: mov ebx,eax
0x100372e6: add esp,0Ch
0x100372e9: cmp ebx,esi
0x100372eb: je near ptr 10037411h
0x100372f1: cmp [esp+3Ch],esi
0x100372f5: je short 10037305h
0x100372f7: push 101020A4h
0x100372fc: push ebp
0x100372fd: call 100230F0h
0x10037302: add esp,8
0x10037305: mov eax,[esp+40h]
0x10037309: cmp eax,esi
0x1003730b: push edi
0x1003730c: jne short 10037315h
0x1003730e: xor edi,edi
0x10037310: jmp near ptr 100373C6h
0x10037315: mov edi,[esp+48h]
0x10037319: cmp edi,3
0x1003731c: jle short 10037367h
0x1003731e: movzx ecx,byte ptr [eax+2]
0x10037322: xor edx,edx
0x10037324: mov dh,[eax]
0x10037326: mov dl,[eax+1]
0x10037329: shl edx,8
0x1003732c: or edx,ecx
0x1003732e: movzx ecx,byte ptr [eax+3]
0x10037332: shl edx,8
0x10037335: or edx,ecx
0x10037337: mov esi,edx
0x10037339: jge short 10037367h
0x1003733b: push 10102070h
0x10037340: push ebp
0x10037341: call 100230F0h
0x10037346: mov edx,[esp+18h]
0x1003734a: push edx
0x1003734b: push ebp
0x1003734c: call 1002CD70h
0x10037351: add esp,10h
0x10037354: pop edi
0x10037355: pop esi
0x10037356: pop ebp
0x10037357: pop ebx
0x10037358: mov ecx,[esp+20h]
0x1003735c: xor ecx,esp
0x1003735e: call 10022326h
0x10037363: add esp,24h
0x10037366: ret
0x10037367: cmp edi,esi
0x10037369: jge short 10037397h
0x1003736b: push 10102040h
0x10037370: push ebp
0x10037371: call 100230F0h
0x10037376: mov eax,[esp+18h]
0x1003737a: push eax
0x1003737b: push ebp
0x1003737c: call 1002CD70h
0x10037381: add esp,10h
0x10037384: pop edi
0x10037385: pop esi
0x10037386: pop ebp
0x10037387: pop ebx
0x10037388: mov ecx,[esp+20h]
0x1003738c: xor ecx,esp
0x1003738e: call 10022326h
```

```
0x10037393: add esp,24h
0x10037396: ret
0x10037397: jle short 100373ADh
0x10037399: push 1010200Ch
0x1003739e: push ebp
0x1003739f: call 100230F0h
0x100373a4: mov eax,[esp+4Ch]
0x100373a8: add esp,8
0x100373ab: mov edi,esi
0x100373ad: test edi,edi
0x100373af: je short 100373C6h
0x100373b1: push 0
0x100373b3: push eax
0x100373b4: lea esi,[esp+1Ch]
0x100373b8: mov ecx,edi
0x100373ba: mov edx,ebp
0x100373bc: call 10035B60h
0x100373c1: add esp,8
0x100373c4: mov edi,eax
0x100373c6: lea ecx,[ebx+edi+2]
0x100373ca: push ecx
0x100373cb: lea edx,[esp+2Ch]
0x100373cf: push edx
0x100373d0: push ebp
0x100373d1: call 10035A60h
0x100373d6: mov esi,[esp+1Ch]
0x100373da: mov byte ptr [esi+ebx+1],0
0x100373df: add ebx,2
0x100373e2: push ebx
0x100373e3: push esi
0x100373e4: push ebp
0x100373e5: call 10035AE0h
0x100373ea: add esp,18h
0x100373ed: test edi,edi
0x100373ef: je short 10037400h
0x100373f1: lea edi,[esp+14h]
0x100373f5: mov esi,ebp
0x100373f7: call 10035DE0h
0x100373fc: mov esi,[esp+10h]
0x10037400: push ebp
0x10037401: call 10035B20h
0x10037406: push esi
0x10037407: push ebp
0x10037408: call 1002CD70h
0x1003740d: add esp,0Ch
0x10037410: pop edi
0x10037411: mov ecx,[esp+2Ch]
0x10037415: pop esi
0x10037416: pop ebp
0x10037417: pop ebx
0x10037418: xor ecx,esp
0x1003741a: call 10022326h
0x1003741f: add esp,24h
0x10037422: ret
```

`sub_10037290` is called by png_write_info_before_PLTE. It works with the text "Unknown compression type in iCCP chunk". It delegates to sub_10022326, png_warning, png_free (9 callees).

12.834 sub_10037430 at 0x10037430 (130 instructions)

```
0x10037430: sub esp,20h
0x10037433: mov eax,ds:[10136220h]
0x10037438: xor eax,esp
0x1003743a: mov [esp+1Ch],eax
0x1003743e: push ebx
0x1003743f: mov ebx,[esp+28h]
```

```
0x10037443: push ebp
0x10037444: push esi
0x10037445: mov esi,[esp+34h]
0x10037449: mov ecx,[esi]
0x1003744b: xor eax,eax
0x1003744d: cmp byte ptr [esi+4],8
0x10037451: push edi
0x10037452: mov edi,[esi+0Ch]
0x10037455: setne al
0x10037458: mov byte ptr [esp+18h],73h
0x1003745d: mov byte ptr [esp+19h],50h
0x10037462: mov byte ptr [esp+1Ah],4Ch
0x10037467: mov byte ptr [esp+1Bh],54h
0x1003746c: mov byte ptr [esp+1Ch],0
0x10037471: lea eax,[eax*4+6]
0x10037478: mov [esp+14h],eax
0x1003747c: imul edi,eax
0x1003747f: lea eax,[esp+10h]
0x10037483: push eax
0x10037484: push ecx
0x10037485: push ebx
0x10037486: call 10036050h
0x1003748b: mov ebp,eax
0x1003748d: add esp,0Ch
0x10037490: test ebp,ebp
0x10037492: je near ptr 100375A2h
0x10037498: lea edx,[edi+ebp+2]
0x1003749c: push edx
0x1003749d: lea eax,[esp+1Ch]
0x100374a1: push eax
0x100374a2: push ebx
0x100374a3: call 10035A60h
0x100374a8: mov ecx,[esp+1Ch]
0x100374ac: add ebp,1
0x100374af: push ebp
0x100374b0: push ecx
0x100374b1: push ebx
0x100374b2: call 10035AE0h
0x100374b7: push 1
0x100374b9: lea ebp,[esi+4]
0x100374bc: push ebp
0x100374bd: push ebx
0x100374be: call 10035AE0h
0x100374c3: mov eax,[esi+0Ch]
0x100374c6: mov edi,[esi+8]
0x100374c9: lea edx,[eax+eax*4]
0x100374cc: lea eax,[edi+edx*2]
0x100374cf: add esp,24h
0x100374d2: cmp edi,eax
0x100374d4: jae near ptr 1003758Eh
0x100374da: lea ebx,[ebx]
0x100374e0: cmp byte ptr [ebp],8
0x100374e4: jne short 10037517h
0x100374e6: movzx ecx,byte ptr [edi]
0x100374e9: mov [esp+20h],cl
0x100374ed: mov dl,[edi+2]
0x100374f0: mov [esp+21h],dl
0x100374f4: mov al,[edi+4]
0x100374f7: mov [esp+22h],al
0x100374fb: movzx ecx,byte ptr [edi+6]
0x100374ff: mov [esp+23h],cl
0x10037503: movzx edx,word ptr [edi+8]
0x10037507: push edx
0x10037508: lea eax,[esp+28h]
0x1003750c: push eax
0x1003750d: call 100359B0h
0x10037512: add esp,8
```

```
0x10037515: jmp short 10037564h
0x10037517: movzx ecx,word ptr [edi]
0x1003751a: push ecx
0x1003751b: lea edx,[esp+24h]
0x1003751f: push edx
0x10037520: call 100359B0h
0x10037525: movzx eax,word ptr [edi+2]
0x10037529: push eax
0x1003752a: lea ecx,[esp+2Eh]
0x1003752e: push ecx
0x1003752f: call 100359B0h
0x10037534: movzx edx,word ptr [edi+4]
0x10037538: push edx
0x10037539: lea eax,[esp+38h]
0x1003753d: push eax
0x1003753e: call 100359B0h
0x10037543: movzx ecx,word ptr [edi+6]
0x10037547: push ecx
0x10037548: lea edx,[esp+42h]
0x1003754c: push edx
0x1003754d: call 100359B0h
0x10037552: movzx eax,word ptr [edi+8]
0x10037556: push eax
0x10037557: lea ecx,[esp+4Ch]
0x1003755b: push ecx
0x1003755c: call 100359B0h
0x10037561: add esp,28h
0x10037564: mov edx,[esp+14h]
0x10037568: push edx
0x10037569: lea eax,[esp+24h]
0x1003756d: push eax
0x1003756e: push ebx
0x1003756f: call 10035AE0h
0x10037574: mov eax,[esi+0Ch]
0x10037577: mov edx,[esi+8]
0x1003757a: lea ecx,[eax+eax*4]
0x1003757d: add edi,0Ah
0x10037580: lea eax,[edx+ecx*2]
0x10037583: add esp,0Ch
0x10037586: cmp edi,eax
0x10037588: jb near ptr 100374E0h
0x1003758e: push ebx
0x1003758f: call 10035B20h
0x10037594: mov ecx,[esp+14h]
0x10037598: push ecx
0x10037599: push ebx
0x1003759a: call 1002CD70h
0x1003759f: add esp,0Ch
0x100375a2: mov ecx,[esp+2Ch]
0x100375a6: pop edi
0x100375a7: pop esi
0x100375a8: pop ebp
0x100375a9: pop ebx
0x100375aa: xor ecx,esp
0x100375ac: call 10022326h
0x100375b1: add esp,20h
0x100375b4: ret
```

`sub_10037430` is called by png_write_info. It delegates to sub_10022326, png_free, png_save_uint_16 (7 callees).

12.835 sub_100375c0 at 0x100375c0 (86 instructions)

```
0x100375c0: sub esp,10h
0x100375c3: mov eax,ds:[10136220h]
0x100375c8: xor eax,esp
0x100375ca: mov [esp+0Ch],eax
```

```
0x100375ce: mov ecx,[esp+1Ch]
0x100375d2: test cl,2
0x100375d5: push ebx
0x100375d6: push esi
0x100375d7: mov esi,[esp+20h]
0x100375db: push edi
0x100375dc: mov edi,[esp+20h]
0x100375e0: mov byte ptr [esp+10h],73h
0x100375e5: mov byte ptr [esp+11h],42h
0x100375ea: mov byte ptr [esp+12h],49h
0x100375ef: mov byte ptr [esp+13h],54h
0x100375f4: mov byte ptr [esp+14h],0
0x100375f9: je short 10037647h
0x100375fb: cmp ecx,3
0x100375fe: mov bl,8
0x10037600: je short 10037608h
0x10037602: mov bl,[edi+128h]
0x10037608: mov dl,[esi]
0x1003760a: test dl,dl
0x1003760c: je near ptr 100376A0h
0x10037612: cmp dl,bl
0x10037614: ja near ptr 100376A0h
0x1003761a: mov cl,[esi+1]
0x1003761d: test cl,cl
0x1003761f: je short 100376A0h
0x10037621: cmp cl,bl
0x10037623: ja short 100376A0h
0x10037625: mov al,[esi+2]
0x10037628: test al,al
0x1003762a: je short 100376A0h
0x1003762c: cmp al,bl
0x1003762e: ja short 100376A0h
0x10037630: mov [esp+0Dh],cl
0x10037634: mov ecx,[esp+28h]
0x10037638: mov [esp+0Eh],al
0x1003763c: mov [esp+0Ch],dl
0x10037640: mov eax,3
0x10037645: jmp short 1003765Fh
0x10037647: mov al,[esi+3]
0x1003764a: test al,al
0x1003764c: je short 100376A0h
0x1003764e: cmp al,[edi+128h]
0x10037654: ja short 100376A0h
0x10037656: mov [esp+0Ch],al
0x1003765a: mov eax,1
0x1003765f: test cl,4
0x10037662: je short 1003767Ah
0x10037664: mov cl,[esi+4]
0x10037667: test cl,cl
0x10037669: je short 100376A0h
0x1003766b: cmp cl,[edi+128h]
0x10037671: ja short 100376A0h
0x10037673: mov [esp+eax+0Ch],cl
0x10037677: add eax,1
0x1003767a: push eax
0x1003767b: lea eax,[esp+10h]
0x1003767f: push eax
0x10037680: lea ecx,[esp+18h]
0x10037684: push ecx
0x10037685: push edi
0x10037686: call 10036BF0h
0x1003768b: add esp,10h
0x1003768e: pop edi
0x1003768f: pop esi
0x10037690: pop ebx
0x10037691: mov ecx,[esp+0Ch]
0x10037695: xor ecx,esp
```

```
0x10037697: call 10022326h
0x1003769c: add esp,10h
0x1003769f: ret
0x100376a0: push 101020CCh
0x100376a5: push edi
0x100376a6: call 100230F0h
0x100376ab: mov ecx,[esp+20h]
0x100376af: add esp,8
0x100376b2: pop edi
0x100376b3: pop esi
0x100376b4: pop ebx
0x100376b5: xor ecx,esp
0x100376b7: call 10022326h
0x100376bc: add esp,10h
0x100376bf: ret
```

`sub_100375c0` is called by png_write_info_before_PLTE. It works with the text "Invalid sBIT depth specified". It delegates to sub_10022326, png_warning, png_write_chunk (3 callees).

12.836 sub_100376c0 at 0x100376c0 (90 instructions)

```
0x100376c0: sub esp,2Ch
0x100376c3: mov eax,ds:[10136220h]
0x100376c8: xor eax,esp
0x100376ca: mov [esp+28h],eax
0x100376ce: mov eax,[esp+44h]
0x100376d2: mov ecx,[esp+40h]
0x100376d6: mov edx,[esp+3Ch]
0x100376da: push ebx
0x100376db: mov ebx,[esp+50h]
0x100376df: push ebp
0x100376e0: mov ebp,[esp+50h]
0x100376e4: push esi
0x100376e5: mov esi,[esp+3Ch]
0x100376e9: push edi
0x100376ea: mov edi,[esp+60h]
0x100376ee: push edi
0x100376ef: push ebx
0x100376f0: push ebp
0x100376f1: push eax
0x100376f2: mov eax,[esp+58h]
0x100376f6: push ecx
0x100376f7: mov ecx,[esp+58h]
0x100376fb: push edx
0x100376fc: push eax
0x100376fd: push ecx
0x100376fe: push esi
0x100376ff: mov byte ptr [esp+34h],63h
0x10037704: mov byte ptr [esp+35h],48h
0x10037709: mov byte ptr [esp+36h],52h
0x1003770e: mov byte ptr [esp+37h],4Dh
0x10037713: mov byte ptr [esp+38h],0
0x10037718: call 100250A0h
0x1003771d: add esp,24h
0x10037720: test eax,eax
0x10037722: je near ptr 100377ACh
0x10037728: mov edx,[esp+44h]
0x1003772c: push edx
0x1003772d: lea eax,[esp+1Ch]
0x10037731: push eax
0x10037732: call 10035950h
0x10037737: mov ecx,[esp+50h]
0x1003773b: push ecx
0x1003773c: lea edx,[esp+28h]
0x10037740: push edx
0x10037741: call 10035950h
```

```
0x10037746: mov eax,[esp+5Ch]
0x1003774a: push eax
0x1003774b: lea ecx,[esp+34h]
0x1003774f: push ecx
0x10037750: call 10035950h
0x10037755: mov edx,[esp+68h]
0x10037759: push edx
0x1003775a: lea eax,[esp+40h]
0x1003775e: push eax
0x1003775f: call 10035950h
0x10037764: mov ecx,[esp+74h]
0x10037768: push ecx
0x10037769: lea edx,[esp+4Ch]
0x1003776d: push edx
0x1003776e: call 10035950h
0x10037773: lea eax,[esp+54h]
0x10037777: push ebp
0x10037778: push eax
0x10037779: call 10035950h
0x1003777e: lea ecx,[esp+60h]
0x10037782: push ebx
0x10037783: push ecx
0x10037784: call 10035950h
0x10037789: lea edx,[esp+6Ch]
0x1003778d: push edi
0x1003778e: push edx
0x1003778f: call 10035950h
0x10037794: add esp,40h
0x10037797: push 20h
0x10037799: lea eax,[esp+1Ch]
0x1003779d: push eax
0x1003779e: lea ecx,[esp+18h]
0x100377a2: push ecx
0x100377a3: push esi
0x100377a4: call 10036BF0h
0x100377a9: add esp,10h
0x100377ac: mov ecx,[esp+38h]
0x100377b0: pop edi
0x100377b1: pop esi
0x100377b2: pop ebp
0x100377b3: pop ebx
0x100377b4: xor ecx,esp
0x100377b6: call 10022326h
0x100377bb: add esp,2Ch
0x100377be: ret
```

`sub_100376c0` is called by png_write_info_before_PLTE. It delegates to sub_10022326, sub_100250a0, png_save_uint_32 (4 callees).

12.837 sub_100377c0 at 0x100377c0 (118 instructions)

```
0x100377c0: sub esp,14h
0x100377c3: mov eax,ds:[10136220h]
0x100377c8: xor eax,esp
0x100377ca: mov [esp+10h],eax
0x100377ce: mov eax,[esp+28h]
0x100377d2: cmp eax,3
0x100377d5: mov ecx,[esp+1Ch]
0x100377d9: push esi
0x100377da: mov esi,[esp+1Ch]
0x100377de: mov byte ptr [esp+4],74h
0x100377e3: mov byte ptr [esp+5],52h
0x100377e8: mov byte ptr [esp+6],4Eh
0x100377ed: mov byte ptr [esp+7],53h
0x100377f2: mov byte ptr [esp+8],0
0x100377f7: jne short 10037836h
```

```
0x100377f9: mov eax,[esp+28h]
0x100377fd: test eax,eax
0x100377ff: jle short 1003782Ch
0x10037801: movzx edx,word ptr [esi+118h]
0x10037808: cmp eax,edx
0x1003780a: jg short 1003782Ch
0x1003780c: push eax
0x1003780d: push ecx
0x1003780e: lea eax,[esp+0Ch]
0x10037812: push eax
0x10037813: push esi
0x10037814: call 10036BF0h
0x10037819: add esp,10h
0x1003781c: pop esi
0x1003781d: mov ecx,[esp+10h]
0x10037821: xor ecx,esp
0x10037823: call 10022326h
0x10037828: add esp,14h
0x1003782b: ret
0x1003782c: push 10102198h
0x10037831: jmp near ptr 10037910h
0x10037836: test eax,eax
0x10037838: jne short 1003788Dh
0x1003783a: mov ecx,[esp+24h]
0x1003783e: movzx eax,word ptr [ecx+8]
0x10037842: mov cl,[esi+127h]
0x10037848: mov edx,1
0x1003784d: shl edx,cl
0x1003784f: cmp eax,edx
0x10037851: jl short 1003785Dh
0x10037853: push 10102158h
0x10037858: jmp near ptr 10037910h
0x1003785d: push eax
0x1003785e: lea eax,[esp+10h]
0x10037862: push eax
0x10037863: call 100359B0h
0x10037868: push 2
0x1003786a: lea ecx,[esp+18h]
0x1003786e: push ecx
0x1003786f: lea edx,[esp+14h]
0x10037873: push edx
0x10037874: push esi
0x10037875: call 10036BF0h
0x1003787a: add esp,18h
0x1003787d: pop esi
0x1003787e: mov ecx,[esp+10h]
0x10037882: xor ecx,esp
0x10037884: call 10022326h
0x10037889: add esp,14h
0x1003788c: ret
0x1003788d: cmp eax,2
0x10037890: jne short 1003790Bh
0x10037892: push edi
0x10037893: mov edi,[esp+28h]
0x10037897: movzx eax,word ptr [edi+2]
0x1003789b: push eax
0x1003789c: lea ecx,[esp+14h]
0x100378a0: push ecx
0x100378a1: call 100359B0h
0x100378a6: movzx edx,word ptr [edi+4]
0x100378aa: push edx
0x100378ab: lea eax,[esp+1Eh]
0x100378af: push eax
0x100378b0: call 100359B0h
0x100378b5: movzx ecx,word ptr [edi+6]
0x100378b9: push ecx
0x100378ba: lea edx,[esp+28h]
```

```
0x100378be: push edx
0x100378bf: call 100359B0h
0x100378c4: add esp,18h
0x100378c7: cmp byte ptr [esi+127h],8
0x100378ce: pop edi
0x100378cf: jne short 100378E6h
0x100378d1: mov al,[esp+10h]
0x100378d5: or al,[esp+0Eh]
0x100378d9: or al,[esp+0Ch]
0x100378dd: je short 100378E6h
0x100378df: push 10102118h
0x100378e4: jmp short 10037910h
0x100378e6: push 6
0x100378e8: lea ecx,[esp+10h]
0x100378ec: push ecx
0x100378ed: lea edx,[esp+0Ch]
0x100378f1: push edx
0x100378f2: push esi
0x100378f3: call 10036BF0h
0x100378f8: add esp,10h
0x100378fb: pop esi
0x100378fc: mov ecx,[esp+10h]
0x10037900: xor ecx,esp
0x10037902: call 10022326h
0x10037907: add esp,14h
0x1003790a: ret
0x1003790b: push 101020ECh
0x10037910: push esi
0x10037911: call 100230F0h
0x10037916: mov ecx,[esp+1Ch]
0x1003791a: add esp,8
0x1003791d: pop esi
0x1003791e: xor ecx,esp
0x10037920: call 10022326h
0x10037925: add esp,14h
0x10037928: ret
```

`sub_100377c0` is called by png_write_info. It works with the text "Invalid number of transparent colors specified". It delegates to sub_10022326, png_warning, png_save_uint_16 (4 callees).

12.838 sub_10037930 at 0x10037930 (125 instructions)

```
0x10037930: sub esp,14h
0x10037933: mov eax,ds:[10136220h]
0x10037938: xor eax,esp
0x1003793a: mov [esp+10h],eax
0x1003793e: mov eax,[esp+20h]
0x10037942: cmp eax,3
0x10037945: push esi
0x10037946: mov esi,[esp+1Ch]
0x1003794a: push edi
0x1003794b: mov edi,[esp+24h]
0x1003794f: mov byte ptr [esp+10h],62h
0x10037954: mov byte ptr [esp+11h],4Bh
0x10037959: mov byte ptr [esp+12h],47h
0x1003795e: mov byte ptr [esp+13h],44h
0x10037963: mov byte ptr [esp+14h],0
0x10037968: jne short 100379B1h
0x1003796a: movzx eax,word ptr [esi+118h]
0x10037971: test ax,ax
0x10037974: jne short 1003797Fh
0x10037976: test byte ptr [esi+230h],1
0x1003797d: jne short 100379A7h
0x1003797f: movzx cx,byte ptr [edi]
0x10037983: cmp cx,ax
0x10037986: jb short 100379A7h
```

```
0x10037988: push 10102248h
0x1003798d: push esi
0x1003798e: call 100230F0h
0x10037993: add esp,8
0x10037996: pop edi
0x10037997: pop esi
0x10037998: mov ecx,[esp+10h]
0x1003799c: xor ecx,esp
0x1003799e: call 10022326h
0x100379a3: add esp,14h
0x100379a6: ret
0x100379a7: mov dl,[edi]
0x100379a9: mov [esp+8],dl
0x100379ad: push 1
0x100379af: jmp short 10037A21h
0x100379b1: test al,2
0x100379b3: je near ptr 10037A45h
0x100379b9: movzx edx,word ptr [edi+2]
0x100379bd: push edx
0x100379be: lea eax,[esp+0Ch]
0x100379c2: push eax
0x100379c3: call 100359B0h
0x100379c8: movzx ecx,word ptr [edi+4]
0x100379cc: push ecx
0x100379cd: lea edx,[esp+16h]
0x100379d1: push edx
0x100379d2: call 100359B0h
0x100379d7: movzx eax,word ptr [edi+6]
0x100379db: push eax
0x100379dc: lea ecx,[esp+20h]
0x100379e0: push ecx
0x100379e1: call 100359B0h
0x100379e6: add esp,18h
0x100379e9: cmp byte ptr [esi+127h],8
0x100379f0: jne short 10037A1Fh
0x100379f2: mov dl,[esp+0Ch]
0x100379f6: or dl,[esp+0Ah]
0x100379fa: or dl,[esp+8]
0x100379fe: je short 10037A1Fh
0x10037a00: push 10102208h
0x10037a05: push esi
0x10037a06: call 100230F0h
0x10037a0b: add esp,8
0x10037a0e: pop edi
0x10037a0f: pop esi
0x10037a10: mov ecx,[esp+10h]
0x10037a14: xor ecx,esp
0x10037a16: call 10022326h
0x10037a1b: add esp,14h
0x10037a1e: ret
0x10037a1f: push 6
0x10037a21: lea eax,[esp+0Ch]
0x10037a25: push eax
0x10037a26: lea ecx,[esp+18h]
0x10037a2a: push ecx
0x10037a2b: push esi
0x10037a2c: call 10036BF0h
0x10037a31: add esp,10h
0x10037a34: pop edi
0x10037a35: pop esi
0x10037a36: mov ecx,[esp+10h]
0x10037a3a: xor ecx,esp
0x10037a3c: call 10022326h
0x10037a41: add esp,14h
0x10037a44: ret
0x10037a45: mov cl,[esi+127h]
0x10037a4b: movzx eax,word ptr [edi+8]
```

```
0x10037a4f: mov edx,1
0x10037a54: shl edx,cl
0x10037a56: cmp eax,edx
0x10037a58: jl short 10037A79h
0x10037a5a: push 101021C8h
0x10037a5f: push esi
0x10037a60: call 100230F0h
0x10037a65: add esp,8
0x10037a68: pop edi
0x10037a69: pop esi
0x10037a6a: mov ecx,[esp+10h]
0x10037a6e: xor ecx,esp
0x10037a70: call 10022326h
0x10037a75: add esp,14h
0x10037a78: ret
0x10037a79: push eax
0x10037a7a: lea eax,[esp+0Ch]
0x10037a7e: push eax
0x10037a7f: call 100359B0h
0x10037a84: push 2
0x10037a86: lea ecx,[esp+14h]
0x10037a8a: push ecx
0x10037a8b: lea edx,[esp+20h]
0x10037a8f: push edx
0x10037a90: push esi
0x10037a91: call 10036BF0h
0x10037a96: mov ecx,[esp+30h]
0x10037a9a: add esp,18h
0x10037a9d: pop edi
0x10037a9e: pop esi
0x10037a9f: xor ecx,esp
0x10037aa1: call 10022326h
0x10037aa6: add esp,14h
0x10037aa9: ret
```

`sub_10037930` is called by png_write_info. It delegates to sub_10022326, png_warning, png_save_uint_16 (4 callees).

12.839 sub_10037ab0 at 0x10037ab0 (46 instructions)

```
0x10037ab0: sub esp,18h
0x10037ab3: mov eax,ds:[10136220h]
0x10037ab8: xor eax,esp
0x10037aba: mov [esp+14h],eax
0x10037abe: push ebx
0x10037abf: mov ebx,[esp+2Ch]
0x10037ac3: cmp ebx,2
0x10037ac6: mov al,46h
0x10037ac8: push esi
0x10037ac9: mov esi,[esp+24h]
0x10037acd: mov byte ptr [esp+8],6Fh
0x10037ad2: mov [esp+9],al
0x10037ad6: mov [esp+0Ah],al
0x10037ada: mov byte ptr [esp+0Bh],73h
0x10037adf: mov byte ptr [esp+0Ch],0
0x10037ae4: jl short 10037AF4h
0x10037ae6: push 1010226Ch
0x10037aeb: push esi
0x10037aec: call 100230F0h
0x10037af1: add esp,8
0x10037af4: mov eax,[esp+28h]
0x10037af8: push eax
0x10037af9: lea ecx,[esp+14h]
0x10037afd: push ecx
0x10037afe: call 10035980h
0x10037b03: mov edx,[esp+34h]
0x10037b07: push edx
```

```
0x10037b08: lea eax,[esp+20h]
0x10037b0c: push eax
0x10037b0d: call 10035980h
0x10037b12: push 9
0x10037b14: lea ecx,[esp+24h]
0x10037b18: push ecx
0x10037b19: lea edx,[esp+20h]
0x10037b1d: push edx
0x10037b1e: push esi
0x10037b1f: mov [esp+38h],bl
0x10037b23: call 10036BF0h
0x10037b28: mov ecx,[esp+3Ch]
0x10037b2c: add esp,20h
0x10037b2f: pop esi
0x10037b30: pop ebx
0x10037b31: xor ecx,esp
0x10037b33: call 10022326h
0x10037b38: add esp,18h
0x10037b3b: ret
```

`sub_10037ab0` is called by png_write_info. It works with the text "Unrecognized unit type for oFFs chunk". It delegates to sub_10022326, png_warning, png_save_int_32 (4 callees).

12.840 sub_10037b40 at 0x10037b40 (74 instructions)

```
0x10037b40: sub esp,54h
0x10037b43: mov eax,ds:[10136220h]
0x10037b48: xor eax,esp
0x10037b4a: mov [esp+50h],eax
0x10037b4e: mov eax,[esp+58h]
0x10037b52: push ebx
0x10037b53: mov ebx,[esp+68h]
0x10037b57: push ebp
0x10037b58: mov ebp,ds:[100FE0B0h]
0x10037b5e: push esi
0x10037b5f: push edi
0x10037b60: mov edi,[esp+70h]
0x10037b64: push edi
0x10037b65: mov [esp+14h],eax
0x10037b69: mov byte ptr [esp+1Ch],73h
0x10037b6e: mov byte ptr [esp+1Dh],43h
0x10037b73: mov byte ptr [esp+1Eh],41h
0x10037b78: mov byte ptr [esp+1Fh],4Ch
0x10037b7d: mov byte ptr [esp+20h],0
0x10037b82: call ebp
0x10037b84: push ebx
0x10037b85: mov esi,eax
0x10037b87: call ebp
0x10037b89: mov ebp,eax
0x10037b8b: lea eax,[esi+ebp+2]
0x10037b8f: cmp eax,40h
0x10037b92: mov [esp+14h],eax
0x10037b96: jbe short 10037BBDh
0x10037b98: mov ecx,[esp+10h]
0x10037b9c: push 10102294h
0x10037ba1: push ecx
0x10037ba2: call 100230F0h
0x10037ba7: add esp,8
0x10037baa: pop edi
0x10037bab: pop esi
0x10037bac: pop ebp
0x10037bad: pop ebx
0x10037bae: mov ecx,[esp+50h]
0x10037bb2: xor ecx,esp
0x10037bb4: call 10022326h
0x10037bb9: add esp,54h
```

```
0x10037bbc: ret
0x10037bbd: mov dl,[esp+6Ch]
0x10037bc1: lea eax,[esi+1]
0x10037bc4: push eax
0x10037bc5: lea ecx,[esp+25h]
0x10037bc9: push edi
0x10037bca: push ecx
0x10037bcb: mov [esp+2Ch],dl
0x10037bcf: call 100228C6h
0x10037bd4: push ebp
0x10037bd5: lea edx,[esp+esi+32h]
0x10037bd9: push ebx
0x10037bda: push edx
0x10037bdb: call 100228C6h
0x10037be0: mov eax,[esp+2Ch]
0x10037be4: push eax
0x10037be5: mov eax,[esp+2Ch]
0x10037be9: lea ecx,[esp+3Ch]
0x10037bed: push ecx
0x10037bee: lea edx,[esp+38h]
0x10037bf2: push edx
0x10037bf3: push eax
0x10037bf4: call 10036BF0h
0x10037bf9: mov ecx,[esp+88h]
0x10037c00: add esp,28h
0x10037c03: pop edi
0x10037c04: pop esi
0x10037c05: pop ebp
0x10037c06: pop ebx
0x10037c07: xor ecx,esp
0x10037c09: call 10022326h
0x10037c0e: add esp,54h
0x10037c11: ret
```

`sub_10037b40` is called by png_write_info. It delegates to sub_10022326, j_memcpy, png_warning (4 callees).

12.841 sub_10037c20 at 0x10037c20 (45 instructions)

```
0x10037c20: sub esp,18h
0x10037c23: mov eax,ds:[10136220h]
0x10037c28: xor eax,esp
0x10037c2a: mov [esp+14h],eax
0x10037c2e: push ebx
0x10037c2f: mov ebx,[esp+2Ch]
0x10037c33: cmp ebx,2
0x10037c36: push esi
0x10037c37: mov esi,[esp+24h]
0x10037c3b: mov byte ptr [esp+8],70h
0x10037c40: mov byte ptr [esp+9],48h
0x10037c45: mov byte ptr [esp+0Ah],59h
0x10037c4a: mov byte ptr [esp+0Bh],73h
0x10037c4f: mov byte ptr [esp+0Ch],0
0x10037c54: jl short 10037C64h
0x10037c56: push 101022B8h
0x10037c5b: push esi
0x10037c5c: call 100230F0h
0x10037c61: add esp,8
0x10037c64: mov eax,[esp+28h]
0x10037c68: push eax
0x10037c69: lea ecx,[esp+14h]
0x10037c6d: push ecx
0x10037c6e: call 10035950h
0x10037c73: mov edx,[esp+34h]
0x10037c77: push edx
0x10037c78: lea eax,[esp+20h]
0x10037c7c: push eax
```

```
0x10037c7d: call 10035950h
0x10037c82: push 9
0x10037c84: lea ecx,[esp+24h]
0x10037c88: push ecx
0x10037c89: lea edx,[esp+20h]
0x10037c8d: push edx
0x10037c8e: push esi
0x10037c8f: mov [esp+38h],bl
0x10037c93: call 10036BF0h
0x10037c98: mov ecx,[esp+3Ch]
0x10037c9c: add esp,20h
0x10037c9f: pop esi
0x10037ca0: pop ebx
0x10037ca1: xor ecx,esp
0x10037ca3: call 10022326h
0x10037ca8: add esp,18h
0x10037cab: ret
```

`sub_10037c20` is called by png_write_info. It delegates to sub_10022326, png_warning, png_save_uint_32 (4 callees).

12.842 sub_10037cb0 at 0x10037cb0 (71 instructions)

```
0x10037cb0: sub esp,14h
0x10037cb3: mov eax,ds:[10136220h]
0x10037cb8: xor eax,esp
0x10037cba: mov [esp+10h],eax
0x10037cbe: push ebx
0x10037cbf: push esi
0x10037cc0: mov esi,[esp+24h]
0x10037cc4: mov al,[esi+2]
0x10037cc7: cmp al,0Ch
0x10037cc9: push edi
0x10037cca: mov edi,[esp+24h]
0x10037cce: mov byte ptr [esp+0Ch],74h
0x10037cd3: mov byte ptr [esp+0Dh],49h
0x10037cd8: mov byte ptr [esp+0Eh],4Dh
0x10037cdd: mov byte ptr [esp+0Fh],45h
0x10037ce2: mov byte ptr [esp+10h],0
0x10037ce7: ja short 10037D5Bh
0x10037ce9: cmp al,1
0x10037ceb: jb short 10037D5Bh
0x10037ced: mov al,[esi+3]
0x10037cf0: cmp al,1Fh
0x10037cf2: ja short 10037D5Bh
0x10037cf4: cmp al,1
0x10037cf6: jb short 10037D5Bh
0x10037cf8: cmp byte ptr [esi+4],17h
0x10037cfc: ja short 10037D5Bh
0x10037cfe: mov bl,[esi+6]
0x10037d01: cmp bl,3Ch
0x10037d04: ja short 10037D5Bh
0x10037d06: movzx eax,word ptr [esi]
0x10037d09: push eax
0x10037d0a: lea ecx,[esp+18h]
0x10037d0e: push ecx
0x10037d0f: call 100359B0h
0x10037d14: mov al,[esi+2]
0x10037d17: mov dl,[esi+5]
0x10037d1a: mov [esp+1Eh],al
0x10037d1e: mov al,[esi+3]
0x10037d21: mov [esp+1Fh],al
0x10037d25: mov al,[esi+4]
0x10037d28: mov [esp+20h],al
0x10037d2c: push 7
0x10037d2e: lea eax,[esp+20h]
0x10037d32: push eax
```

```
0x10037d33: lea ecx,[esp+1Ch]
0x10037d37: push ecx
0x10037d38: push edi
0x10037d39: mov [esp+31h],dl
0x10037d3d: mov [esp+32h],bl
0x10037d41: call 10036BF0h
0x10037d46: add esp,18h
0x10037d49: pop edi
0x10037d4a: pop esi
0x10037d4b: pop ebx
0x10037d4c: mov ecx,[esp+10h]
0x10037d50: xor ecx,esp
0x10037d52: call 10022326h
0x10037d57: add esp,14h
0x10037d5a: ret
0x10037d5b: push 101022E0h
0x10037d60: push edi
0x10037d61: call 100230F0h
0x10037d66: mov ecx,[esp+24h]
0x10037d6a: add esp,8
0x10037d6d: pop edi
0x10037d6e: pop esi
0x10037d6f: pop ebx
0x10037d70: xor ecx,esp
0x10037d72: call 10022326h
0x10037d77: add esp,14h
0x10037d7a: ret
```

`sub_10037cb0` is called by png_write_info, png_write_end. It works with the text "Invalid time specified for tIME chunk". It delegates to sub_10022326, png_warning, png_save_uint_16 (4 callees).

12.843 sub_10037d80 at 0x10037d80 (176 instructions)

```
0x10037d80: sub esp,70h
0x10037d83: push ebp
0x10037d84: mov ecx,1
0x10037d89: push esi
0x10037d8a: mov esi,[esp+7Ch]
0x10037d8e: add [esi+0E4h],ecx
0x10037d94: mov ebp,4
0x10037d99: mov eax,2
0x10037d9e: mov edx,8
0x10037da3: mov [esp+38h],ecx
0x10037da7: mov [esp+20h],ecx
0x10037dab: mov [esp+74h],ecx
0x10037daf: mov ecx,[esi+0E4h]
0x10037db5: push edi
0x10037db6: xor edi,edi
0x10037db8: cmp ecx,[esi+0D4h]
0x10037dbe: mov [esp+28h],edi
0x10037dc2: mov [esp+2Ch],ebp
0x10037dc6: mov [esp+30h],edi
0x10037dca: mov [esp+34h],eax
0x10037dce: mov [esp+38h],edi
0x10037dd2: mov [esp+40h],edi
0x10037dd6: mov [esp+0Ch],edx
0x10037dda: mov [esp+10h],edx
0x10037dde: mov [esp+14h],ebp
0x10037de2: mov [esp+18h],ebp
0x10037de6: mov [esp+1Ch],eax
0x10037dea: mov [esp+20h],eax
0x10037dee: mov [esp+60h],edi
0x10037df2: mov [esp+64h],edi
0x10037df6: mov [esp+68h],ebp
0x10037dfa: mov [esp+6Ch],edi
0x10037dfe: mov [esp+70h],eax
```

```
0x10037e02: mov [esp+74h],edi
0x10037e06: mov [esp+44h],edx
0x10037e0a: mov [esp+48h],edx
0x10037e0e: mov [esp+4Ch],edx
0x10037e12: mov [esp+50h],ebp
0x10037e16: mov [esp+54h],ebp
0x10037e1a: mov [esp+58h],eax
0x10037e1e: mov [esp+5Ch],eax
0x10037e22: jb near ptr 10037FCDh
0x10037e28: cmp byte ptr [esi+123h],0
0x10037e2f: push ebx
0x10037e30: je near ptr 10037F2Fh
0x10037e36: test [esi+74h],al
0x10037e39: mov [esi+0E4h],edi
0x10037e3f: je short 10037E57h
0x10037e41: add byte ptr [esi+124h],1
0x10037e48: mov bl,[esi+124h]
0x10037e4e: jmp short 10037EC0h
0x10037e50: mov ebp,4
0x10037e55: xor edi,edi
0x10037e57: add byte ptr [esi+124h],1
0x10037e5e: mov bl,[esi+124h]
0x10037e64: cmp bl,7
0x10037e67: jae near ptr 10037F2Fh
0x10037e6d: mov eax,[esi+0CCh]
0x10037e73: movzx ecx,bl
0x10037e76: add ecx,ecx
0x10037e78: add ecx,ecx
0x10037e7a: sub eax,[esp+ecx+2Ch]
0x10037e7e: mov edi,[esp+ecx+10h]
0x10037e82: xor edx,edx
0x10037e84: lea eax,[eax+edi-1]
0x10037e88: div edi
0x10037e8a: mov edx,[esi+0D0h]
0x10037e90: sub edx,[esp+ecx+64h]
0x10037e94: mov ebp,[esp+ecx+48h]
0x10037e98: mov edi,eax
0x10037e9a: lea eax,[edx+ebp-1]
0x10037e9e: xor edx,edx
0x10037ea0: div ebp
0x10037ea2: test edi,edi
0x10037ea4: mov [esi+0D8h],edi
0x10037eaa: mov [esi+0D4h],eax
0x10037eb0: je short 10037E50h
0x10037eb2: test eax,eax
0x10037eb4: je short 10037E50h
0x10037eb6: mov edx,8
0x10037ebb: lea ebp,[edx-4]
0x10037ebe: xor edi,edi
0x10037ec0: cmp bl,7
0x10037ec3: jae short 10037F2Fh
0x10037ec5: mov ecx,[esi+0E8h]
0x10037ecb: cmp ecx,edi
0x10037ecd: je near ptr 10037FCCh
0x10037ed3: movzx eax,byte ptr [esi+12Bh]
0x10037eda: movzx ebx,byte ptr [esi+128h]
0x10037ee1: imul eax,ebx
0x10037ee4: cmp eax,edx
0x10037ee6: jl short 10037F0Ah
0x10037ee8: shr eax,3
0x10037eeb: imul eax,[esi+0CCh]
0x10037ef2: mov esi,eax
0x10037ef4: add esi,1
0x10037ef7: push esi
0x10037ef8: push edi
0x10037ef9: push ecx
0x10037efa: call 100228C0h
```

```
0x10037eff: add esp,0Ch
0x10037f02: pop ebx
0x10037f03: pop edi
0x10037f04: pop esi
0x10037f05: pop ebp
0x10037f06: add esp,70h
0x10037f09: ret
0x10037f0a: mov esi,[esi+0CCh]
0x10037f10: imul esi,eax
0x10037f13: add esi,7
0x10037f16: shr esi,3
0x10037f19: add esi,1
0x10037f1c: push esi
0x10037f1d: push edi
0x10037f1e: push ecx
0x10037f1f: call 100228C0h
0x10037f24: add esp,0Ch
0x10037f27: pop ebx
0x10037f28: pop edi
0x10037f29: pop esi
0x10037f2a: pop ebp
0x10037f2b: add esp,70h
0x10037f2e: ret
0x10037f2f: lea ebx,[esi+78h]
0x10037f32: push ebp
0x10037f33: push ebx
0x10037f34: call 10062A80h
0x10037f39: add esp,8
0x10037f3c: cmp eax,edi
0x10037f3e: jne short 10037F79h
0x10037f40: cmp [esi+88h],edi
0x10037f46: jne short 10037F32h
0x10037f48: mov eax,[esi+0B4h]
0x10037f4e: mov ecx,[esi+0B0h]
0x10037f54: push eax
0x10037f55: push ecx
0x10037f56: push esi
0x10037f57: call 10037020h
0x10037f5c: mov edx,[esi+0B0h]
0x10037f62: mov eax,[esi+0B4h]
0x10037f68: add esp,0Ch
0x10037f6b: mov [esi+84h],edx
0x10037f71: mov [esi+88h],eax
0x10037f77: jmp short 10037F32h
0x10037f79: cmp eax,1
0x10037f7c: je short 10037F9Ah
0x10037f7e: mov eax,[esi+90h]
0x10037f84: cmp eax,edi
0x10037f86: je short 10037F8Fh
0x10037f88: push eax
0x10037f89: push esi
0x10037f8a: call 100231E0h
0x10037f8f: push 10100348h
0x10037f94: push esi
0x10037f95: call 100231E0h
0x10037f9a: mov ecx,[esi+88h]
0x10037fa0: mov eax,[esi+0B4h]
0x10037fa6: cmp ecx,eax
0x10037fa8: jae short 10037FBDh
0x10037faa: sub eax,ecx
0x10037fac: mov ecx,[esi+0B0h]
0x10037fb2: push eax
0x10037fb3: push ecx
0x10037fb4: push esi
0x10037fb5: call 10037020h
0x10037fba: add esp,0Ch
0x10037fbd: push ebx
```

```
0x10037fbe: call 100629F0h
0x10037fc3: add esp,4
0x10037fc6: mov [esi+0A4h],edi
0x10037fcc: pop ebx
0x10037fcd: pop edi
0x10037fce: pop esi
0x10037fcf: pop ebp
0x10037fd0: add esp,70h
0x10037fd3: ret
```

`sub_10037d80` is called by png_write_row, sub_10037fe0. It works with the text "zlib error". It delegates to j_memset, png_error, sub_10037020 (5 callees).

12.844 sub_10037fe0 at 0x10037fe0 (75 instructions)

```
0x10037fe0: mov eax,[esp+8]
0x10037fe4: push ebx
0x10037fe5: push esi
0x10037fe6: mov esi,[esp+0Ch]
0x10037fea: push edi
0x10037feb: mov edi,[esi+104h]
0x10037ff1: lea ebx,[esi+78h]
0x10037ff4: mov [ebx],eax
0x10037ff6: mov dword ptr [esi+7Ch],0
0x10037ffd: add edi,1
0x10038000: cmp dword ptr [esi+7Ch],0
0x10038004: jne short 1003801Ch
0x10038006: cmp edi,0FFFFFFFFh
0x10038009: jbe short 10038017h
0x1003800b: mov dword ptr [esi+7Ch],0FFFFFFFFh
0x10038012: add edi,1
0x10038015: jmp short 1003801Ch
0x10038017: mov [esi+7Ch],edi
0x1003801a: xor edi,edi
0x1003801c: push 0
0x1003801e: push ebx
0x1003801f: call 10062A80h
0x10038024: add esp,8
0x10038027: test eax,eax
0x10038029: jne near ptr 100380BDh
0x1003802f: cmp [esi+88h],eax
0x10038035: jne short 10038066h
0x10038037: mov ecx,[esi+0B4h]
0x1003803d: mov edx,[esi+0B0h]
0x10038043: push ecx
0x10038044: push edx
0x10038045: push esi
0x10038046: call 10037020h
0x1003804b: mov eax,[esi+0B0h]
0x10038051: mov ecx,[esi+0B4h]
0x10038057: add esp,0Ch
0x1003805a: mov [esi+84h],eax
0x10038060: mov [esi+88h],ecx
0x10038066: test edi,edi
0x10038068: ja short 10038000h
0x1003806a: cmp dword ptr [esi+7Ch],0
0x1003806e: ja short 1003801Ch
0x10038070: mov eax,[esi+0E8h]
0x10038076: test eax,eax
0x10038078: je short 1003808Ch
0x1003807a: mov edx,[esi+0ECh]
0x10038080: mov [esi+0E8h],edx
0x10038086: mov [esi+0ECh],eax
0x1003808c: push esi
0x1003808d: call 10037D80h
0x10038092: add dword ptr [esi+154h],1
```

```
0x10038099: mov eax,[esi+150h]
0x1003809f: mov ecx,[esi+154h]
0x100380a5: add esp,4
0x100380a8: test eax,eax
0x100380aa: jbe short 100380B9h
0x100380ac: cmp ecx,eax
0x100380ae: jb short 100380B9h
0x100380b0: push esi
0x100380b1: call 1002A5E0h
0x100380b6: add esp,4
0x100380b9: pop edi
0x100380ba: pop esi
0x100380bb: pop ebx
0x100380bc: ret
0x100380bd: mov eax,[esi+90h]
0x100380c3: test eax,eax
0x100380c5: je short 100380CEh
0x100380c7: push eax
0x100380c8: push esi
0x100380c9: call 100231E0h
0x100380ce: push 10100348h
0x100380d3: push esi
0x100380d4: call 100231E0h
0x100380d9: int 3
```

`sub_10037fe0` is called by sub_100380e0. It works with the text "zlib error". It delegates to png_error, png_write_flush, sub_10037020 (5 callees).

12.845 sub_100380e0 at 0x100380e0 (899 instructions)

```
0x100380e0: sub esp,30h
0x100380e3: mov eax,[esp+38h]
0x100380e7: push ebx
0x100380e8: mov ebx,[esp+38h]
0x100380ec: mov cl,[ebx+125h]
0x100380f2: movzx edx,byte ptr [ebx+1F9h]
0x100380f9: push ebp
0x100380fa: mov ebp,[eax+4]
0x100380fd: push esi
0x100380fe: push edi
0x100380ff: movzx edi,byte ptr [eax+0Bh]
0x10038103: mov eax,[ebx+0E8h]
0x10038109: add edi,7
0x1003810c: sar edi,3
0x1003810f: test cl,8
0x10038112: mov [esp+2Ch],eax
0x10038116: mov eax,[ebx+0ECh]
0x1003811c: mov [esp+13h],cl
0x10038120: mov [esp+14h],ebp
0x10038124: mov [esp+48h],edx
0x10038128: mov [esp+24h],edi
0x1003812c: mov [esp+18h],eax
0x10038130: mov [esp+28h],eax
0x10038134: mov dword ptr [esp+1Ch],7FFFFFFFh
0x1003813c: je near ptr 10038203h
0x10038142: cmp cl,8
0x10038145: je near ptr 10038203h
0x1003814b: xor eax,eax
0x1003814d: xor edx,edx
0x1003814f: test ebp,ebp
0x10038151: jbe short 10038185h
0x10038153: jmp short 10038160h
0x10038160: mov ecx,[esp+18h]
0x10038164: movzx esi,byte ptr [ecx+edx+1]
0x10038169: cmp esi,80h
0x1003816f: jge short 10038175h
```

```
0x10038171: mov ecx,esi
0x10038173: jmp short 1003817Ch
0x10038175: mov ecx,100h
0x1003817a: sub ecx,esi
0x1003817c: add edx,1
0x1003817f: add eax,ecx
0x10038181: cmp edx,ebp
0x10038183: jb short 10038160h
0x10038185: cmp byte ptr [ebx+1F8h],2
0x1003818c: jne short 100381FFh
0x1003818e: movzx esi,ax
0x10038191: shr eax,0Ah
0x10038194: and eax,3FFFC0h
0x10038199: xor ecx,ecx
0x1003819b: cmp [esp+48h],ecx
0x1003819f: mov edx,eax
0x100381a1: jle short 100381D4h
0x100381a3: mov eax,[ebx+1FCh]
0x100381a9: cmp byte ptr [eax+ecx],0
0x100381ad: jne short 100381CBh
0x100381af: mov eax,[ebx+200h]
0x100381b5: movzx eax,word ptr [eax+ecx*2]
0x100381b9: mov ebp,eax
0x100381bb: imul eax,edx
0x100381be: imul ebp,esi
0x100381c1: shr ebp,8
0x100381c4: shr eax,8
0x100381c7: mov esi,ebp
0x100381c9: mov edx,eax
0x100381cb: add ecx,1
0x100381ce: cmp ecx,[esp+48h]
0x100381d2: jl short 100381A3h
0x100381d4: mov ecx,[ebx+208h]
0x100381da: movzx eax,word ptr [ecx]
0x100381dd: mov ecx,eax
0x100381df: imul ecx,edx
0x100381e2: shr ecx,3
0x100381e5: cmp ecx,3FFFC0h
0x100381eb: jbe short 100381F4h
0x100381ed: mov eax,7FFFFFFFh
0x100381f2: jmp short 100381FFh
0x100381f4: imul eax,esi
0x100381f7: shr eax,3
0x100381fa: shl ecx,0Ah
0x100381fd: add eax,ecx
0x100381ff: mov [esp+1Ch],eax
0x10038203: mov al,[esp+13h]
0x10038207: cmp al,10h
0x10038209: jne near ptr 10038309h
0x1003820f: mov esi,[esp+18h]
0x10038213: mov eax,[ebx+0F0h]
0x10038219: add esi,1
0x1003821c: xor ebp,ebp
0x1003821e: add eax,1
0x10038221: test edi,edi
0x10038223: mov ecx,esi
0x10038225: jbe short 1003823Fh
0x10038227: mov ebp,edi
0x10038229: lea esp,[esp]
0x10038230: mov dl,[ecx]
0x10038232: mov [eax],dl
0x10038234: add ecx,1
0x10038237: add eax,1
0x1003823a: sub edi,1
0x1003823d: jne short 10038230h
0x1003823f: cmp ebp,[esp+14h]
0x10038243: jae short 10038264h
```

```
0x10038245: mov edi,[esp+14h]
0x10038249: sub edi,ebp
0x1003824b: jmp short 10038250h
0x10038250: mov dl,[ecx]
0x10038252: sub dl,[esi]
0x10038254: add ecx,1
0x10038257: mov [eax],dl
0x10038259: add esi,1
0x1003825c: add eax,1
0x1003825f: sub edi,1
0x10038262: jne short 10038250h
0x10038264: mov eax,[ebx+0F0h]
0x1003826a: mov [esp+18h],eax
0x1003826e: test byte ptr [esp+13h],20h
0x10038273: je near ptr 100386BFh
0x10038279: mov ecx,[esp+1Ch]
0x1003827d: xor edi,edi
0x1003827f: cmp byte ptr [ebx+1F8h],2
0x10038286: mov [esp+30h],ecx
0x1003828a: jne near ptr 100385C8h
0x10038290: movzx esi,cx
0x10038293: shr ecx,0Ah
0x10038296: xor edx,edx
0x10038298: and ecx,3FFFC0h
0x1003829e: cmp [esp+48h],edx
0x100382a2: mov [esp+34h],esi
0x100382a6: jle short 100382DFh
0x100382a8: mov ebp,[ebx+1FCh]
0x100382ae: mov edi,edi
0x100382b0: cmp byte ptr [edx+ebp],2
0x100382b4: jne short 100382D6h
0x100382b6: mov eax,[ebx+204h]
0x100382bc: movzx eax,word ptr [eax+edx*2]
0x100382c0: mov esi,eax
0x100382c2: imul eax,ecx
0x100382c5: imul esi,[esp+34h]
0x100382ca: shr esi,8
0x100382cd: shr eax,8
0x100382d0: mov [esp+34h],esi
0x100382d4: mov ecx,eax
0x100382d6: add edx,1
0x100382d9: cmp edx,[esp+48h]
0x100382dd: jl short 100382B0h
0x100382df: mov edx,[ebx+20Ch]
0x100382e5: movzx edx,word ptr [edx+4]
0x100382e9: mov eax,edx
0x100382eb: imul eax,ecx
0x100382ee: shr eax,3
0x100382f1: cmp eax,3FFFC0h
0x100382f6: jbe near ptr 100385B9h
0x100382fc: mov dword ptr [esp+30h],7FFFFFFFh
0x10038304: jmp near ptr 100385C8h
0x10038309: test al,10h
0x1003830b: je near ptr 100384D4h
0x10038311: mov ecx,[esp+1Ch]
0x10038315: xor edi,edi
0x10038317: cmp byte ptr [ebx+1F8h],2
0x1003831e: mov [esp+30h],ecx
0x10038322: jne short 100383A3h
0x10038324: movzx esi,cx
0x10038327: shr ecx,0Ah
0x1003832a: and ecx,3FFFC0h
0x10038330: xor edx,edx
0x10038332: cmp [esp+48h],edi
0x10038336: jle short 10038371h
0x10038338: jmp short 10038340h
0x10038340: mov eax,[ebx+1FCh]
```

```
0x10038346: cmp byte ptr [eax+edx],1
0x1003834a: jne short 10038368h
0x1003834c: mov eax,[ebx+204h]
0x10038352: movzx eax,word ptr [eax+edx*2]
0x10038356: mov ebp,eax
0x10038358: imul eax,ecx
0x1003835b: imul ebp,esi
0x1003835e: shr ebp,8
0x10038361: shr eax,8
0x10038364: mov esi,ebp
0x10038366: mov ecx,eax
0x10038368: add edx,1
0x1003836b: cmp edx,[esp+48h]
0x1003836f: jl short 10038340h
0x10038371: mov edx,[ebx+20Ch]
0x10038377: movzx edx,word ptr [edx+2]
0x1003837b: mov eax,edx
0x1003837d: imul eax,ecx
0x10038380: shr eax,3
0x10038383: cmp eax,3FFFC0h
0x10038388: jbe short 10038394h
0x1003838a: mov dword ptr [esp+30h],7FFFFFFFh
0x10038392: jmp short 100383A3h
0x10038394: imul edx,esi
0x10038397: shr edx,3
0x1003839a: shl eax,0Ah
0x1003839d: add edx,eax
0x1003839f: mov [esp+30h],edx
0x100383a3: mov ebp,[esp+18h]
0x100383a7: mov ecx,[ebx+0F0h]
0x100383ad: mov eax,[esp+24h]
0x100383b1: add ebp,1
0x100383b4: add ecx,1
0x100383b7: test eax,eax
0x100383b9: mov [esp+20h],edi
0x100383bd: mov [esp+38h],ebp
0x100383c1: mov edx,ebp
0x100383c3: jbe short 100383FBh
0x100383c5: mov ebp,eax
0x100383c7: mov [esp+20h],eax
0x100383cb: jmp short 100383D0h
0x100383d0: mov al,[edx]
0x100383d2: movzx esi,al
0x100383d5: cmp esi,80h
0x100383db: mov [ecx],al
0x100383dd: jge short 100383E3h
0x100383df: mov eax,esi
0x100383e1: jmp short 100383EAh
0x100383e3: mov eax,100h
0x100383e8: sub eax,esi
0x100383ea: add edi,eax
0x100383ec: add edx,1
0x100383ef: add ecx,1
0x100383f2: sub ebp,1
0x100383f5: jne short 100383D0h
0x100383f7: mov ebp,[esp+38h]
0x100383fb: mov eax,[esp+14h]
0x100383ff: cmp [esp+20h],eax
0x10038403: jae short 10038443h
0x10038405: mov al,[edx]
0x10038407: sub al,[ebp]
0x1003840a: mov [ecx],al
0x1003840c: movzx eax,al
0x1003840f: cmp eax,80h
0x10038414: jge short 1003841Ah
0x10038416: mov esi,eax
0x10038418: jmp short 10038421h
```

```
0x1003841a: mov esi,100h
0x1003841f: sub esi,eax
0x10038421: add edi,esi
0x10038423: cmp edi,[esp+30h]
0x10038427: ja short 10038443h
0x10038429: mov eax,[esp+20h]
0x1003842d: add eax,1
0x10038430: add edx,1
0x10038433: add ebp,1
0x10038436: add ecx,1
0x10038439: cmp eax,[esp+14h]
0x1003843d: mov [esp+20h],eax
0x10038441: jb short 10038405h
0x10038443: cmp byte ptr [ebx+1F8h],2
0x1003844a: jne short 100384C0h
0x1003844c: movzx esi,di
0x1003844f: shr edi,0Ah
0x10038452: and edi,3FFFC0h
0x10038458: xor ecx,ecx
0x1003845a: cmp [esp+48h],ecx
0x1003845e: mov edx,edi
0x10038460: jle short 10038493h
0x10038462: mov edi,[ebx+1FCh]
0x10038468: cmp byte ptr [edi+ecx],1
0x1003846c: jne short 1003848Ah
0x1003846e: mov eax,[ebx+204h]
0x10038474: movzx eax,word ptr [eax+ecx*2]
0x10038478: mov ebp,eax
0x1003847a: imul eax,edx
0x1003847d: imul ebp,esi
0x10038480: shr ebp,8
0x10038483: shr eax,8
0x10038486: mov esi,ebp
0x10038488: mov edx,eax
0x1003848a: add ecx,1
0x1003848d: cmp ecx,[esp+48h]
0x10038491: jl short 10038468h
0x10038493: mov ecx,[ebx+20Ch]
0x10038499: movzx ecx,word ptr [ecx+2]
0x1003849d: mov eax,ecx
0x1003849f: imul eax,edx
0x100384a2: shr eax,3
0x100384a5: cmp eax,3FFFC0h
0x100384aa: jbe short 100384B3h
0x100384ac: mov edi,7FFFFFFFh
0x100384b1: jmp short 100384C0h
0x100384b3: imul ecx,esi
0x100384b6: shr ecx,3
0x100384b9: shl eax,0Ah
0x100384bc: add ecx,eax
0x100384be: mov edi,ecx
0x100384c0: cmp edi,[esp+1Ch]
0x100384c4: jae short 100384D4h
0x100384c6: mov edx,[ebx+0F0h]
0x100384cc: mov [esp+1Ch],edi
0x100384d0: mov [esp+18h],edx
0x100384d4: cmp byte ptr [esp+13h],20h
0x100384d9: jne near ptr 1003826Eh
0x100384df: mov eax,[ebx+0F4h]
0x100384e5: mov edx,[esp+28h]
0x100384e9: xor ecx,ecx
0x100384eb: add eax,1
0x100384ee: cmp [esp+14h],ecx
0x100384f2: lea edi,[edx+1]
0x100384f5: jbe short 10038516h
0x100384f7: mov esi,edi
0x100384f9: sub esi,edx
```

```
0x100384fb: add esi,[esp+2Ch]
0x100384ff: nop
0x10038500: mov dl,[ecx+edi]
0x10038503: sub dl,[esi]
0x10038505: add ecx,1
0x10038508: mov [eax],dl
0x1003850a: add esi,1
0x1003850d: add eax,1
0x10038510: cmp ecx,[esp+14h]
0x10038514: jb short 10038500h
0x10038516: mov eax,[ebx+0F4h]
0x1003851c: mov [esp+18h],eax
0x10038520: test byte ptr [esp+13h],40h
0x10038525: je near ptr 10038965h
0x1003852b: mov ecx,[esp+1Ch]
0x1003852f: xor edx,edx
0x10038531: cmp byte ptr [ebx+1F8h],2
0x10038538: mov [esp+20h],edx
0x1003853c: mov [esp+34h],ecx
0x10038540: jne near ptr 100387FAh
0x10038546: mov edi,[esp+48h]
0x1003854a: movzx esi,cx
0x1003854d: shr ecx,0Ah
0x10038550: and ecx,3FFFC0h
0x10038556: cmp edi,edx
0x10038558: jle short 1003858Fh
0x1003855a: lea ebx,[ebx]
0x10038560: mov eax,[ebx+1FCh]
0x10038566: cmp byte ptr [edx+eax],3
0x1003856a: jne short 10038588h
0x1003856c: mov eax,[ebx+204h]
0x10038572: movzx eax,word ptr [eax+edx*2]
0x10038576: mov ebp,eax
0x10038578: imul eax,ecx
0x1003857b: imul ebp,esi
0x1003857e: shr ebp,8
0x10038581: shr eax,8
0x10038584: mov esi,ebp
0x10038586: mov ecx,eax
0x10038588: add edx,1
0x1003858b: cmp edx,edi
0x1003858d: jl short 10038560h
0x1003858f: mov edx,[ebx+20Ch]
0x10038595: movzx edx,word ptr [edx+6]
0x10038599: mov eax,edx
0x1003859b: imul eax,ecx
0x1003859e: shr eax,3
0x100385a1: cmp eax,3FFFC0h
0x100385a6: jbe near ptr 100387EBh
0x100385ac: mov dword ptr [esp+34h],7FFFFFFFh
0x100385b4: jmp near ptr 100387FAh
0x100385b9: imul edx,esi
0x100385bc: shr edx,3
0x100385bf: shl eax,0Ah
0x100385c2: add edx,eax
0x100385c4: mov [esp+30h],edx
0x100385c8: mov edx,[esp+28h]
0x100385cc: mov ecx,[ebx+0F4h]
0x100385d2: mov eax,[esp+2Ch]
0x100385d6: add edx,1
0x100385d9: add ecx,1
0x100385dc: add eax,1
0x100385df: cmp dword ptr [esp+14h],0
0x100385e4: mov dword ptr [esp+34h],0
0x100385ec: jbe short 1003862Eh
0x100385ee: mov ebp,eax
0x100385f0: sub ebp,edx
```

```
0x100385f2: mov al,[edx]
0x100385f4: sub al,[edx+ebp]
0x100385f7: add ecx,1
0x100385fa: mov [ecx-1],al
0x100385fd: movzx eax,al
0x10038600: add edx,1
0x10038603: cmp eax,80h
0x10038608: jge short 1003860Eh
0x1003860a: mov esi,eax
0x1003860c: jmp short 10038615h
0x1003860e: mov esi,100h
0x10038613: sub esi,eax
0x10038615: add edi,esi
0x10038617: cmp edi,[esp+30h]
0x1003861b: ja short 1003862Eh
0x1003861d: mov eax,[esp+34h]
0x10038621: add eax,1
0x10038624: cmp eax,[esp+14h]
0x10038628: mov [esp+34h],eax
0x1003862c: jb short 100385F2h
0x1003862e: cmp byte ptr [ebx+1F8h],2
0x10038635: jne short 100386ABh
0x10038637: movzx esi,di
0x1003863a: shr edi,0Ah
0x1003863d: and edi,3FFFC0h
0x10038643: xor ecx,ecx
0x10038645: cmp [esp+48h],ecx
0x10038649: mov edx,edi
0x1003864b: jle short 1003867Eh
0x1003864d: mov edi,[ebx+1FCh]
0x10038653: cmp byte ptr [edi+ecx],2
0x10038657: jne short 10038675h
0x10038659: mov eax,[ebx+200h]
0x1003865f: movzx eax,word ptr [eax+ecx*2]
0x10038663: mov ebp,eax
0x10038665: imul eax,edx
0x10038668: imul ebp,esi
0x1003866b: shr ebp,8
0x1003866e: shr eax,8
0x10038671: mov esi,ebp
0x10038673: mov edx,eax
0x10038675: add ecx,1
0x10038678: cmp ecx,[esp+48h]
0x1003867c: jl short 10038653h
0x1003867e: mov ecx,[ebx+208h]
0x10038684: movzx ecx,word ptr [ecx+4]
0x10038688: mov eax,ecx
0x1003868a: imul eax,edx
0x1003868d: shr eax,3
0x10038690: cmp eax,3FFFC0h
0x10038695: jbe short 1003869Eh
0x10038697: mov edi,7FFFFFFFh
0x1003869c: jmp short 100386ABh
0x1003869e: imul ecx,esi
0x100386a1: shr ecx,3
0x100386a4: shl eax,0Ah
0x100386a7: add ecx,eax
0x100386a9: mov edi,ecx
0x100386ab: cmp edi,[esp+1Ch]
0x100386af: jae short 100386BFh
0x100386b1: mov edx,[ebx+0F4h]
0x100386b7: mov [esp+1Ch],edi
0x100386bb: mov [esp+18h],edx
0x100386bf: cmp byte ptr [esp+13h],40h
0x100386c4: jne near ptr 10038520h
0x100386ca: mov ebp,[esp+28h]
0x100386ce: mov ecx,[ebx+0F8h]
```

```
0x100386d4: mov esi,[esp+2Ch]
0x100386d8: mov edx,[esp+24h]
0x100386dc: add ebp,1
0x100386df: xor eax,eax
0x100386e1: add ecx,1
0x100386e4: add esi,1
0x100386e7: test edx,edx
0x100386e9: mov edi,ebp
0x100386eb: jbe short 10038713h
0x100386ed: mov [esp+44h],edx
0x100386f1: mov [esp+38h],edx
0x100386f5: mov al,[esi]
0x100386f7: mov dl,[edi]
0x100386f9: shr al,1
0x100386fb: sub dl,al
0x100386fd: mov [ecx],dl
0x100386ff: add ecx,1
0x10038702: add esi,1
0x10038705: add edi,1
0x10038708: sub dword ptr [esp+44h],1
0x1003870d: jne short 100386F5h
0x1003870f: mov eax,[esp+38h]
0x10038713: cmp eax,[esp+14h]
0x10038717: jae short 1003874Ah
0x10038719: mov edx,[esp+14h]
0x1003871d: sub edx,eax
0x1003871f: mov [esp+44h],edx
0x10038723: movzx edx,byte ptr [ebp]
0x10038727: movzx eax,byte ptr [esi]
0x1003872a: add eax,edx
0x1003872c: cdq
0x1003872d: sub eax,edx
0x1003872f: mov dl,[edi]
0x10038731: sar eax,1
0x10038733: sub dl,al
0x10038735: mov [ecx],dl
0x10038737: add ecx,1
0x1003873a: add ebp,1
0x1003873d: add esi,1
0x10038740: add edi,1
0x10038743: sub dword ptr [esp+44h],1
0x10038748: jne short 10038723h
0x1003874a: mov eax,[ebx+0F8h]
0x10038750: mov [esp+18h],eax
0x10038754: test byte ptr [esp+13h],80h
0x10038759: je near ptr 10038C4Ch
0x1003875f: mov ecx,[esp+1Ch]
0x10038763: xor ebp,ebp
0x10038765: cmp byte ptr [ebx+1F8h],2
0x1003876c: mov [esp+30h],ebp
0x10038770: mov [esp+20h],ecx
0x10038774: jne near ptr 10038A6Fh
0x1003877a: movzx esi,cx
0x1003877d: shr ecx,0Ah
0x10038780: xor edx,edx
0x10038782: and ecx,3FFFC0h
0x10038788: cmp [esp+48h],edx
0x1003878c: jle short 100387C1h
0x1003878e: mov edi,edi
0x10038790: mov eax,[ebx+1FCh]
0x10038796: cmp byte ptr [edx+eax],4
0x1003879a: jne short 100387B8h
0x1003879c: mov eax,[ebx+204h]
0x100387a2: movzx eax,word ptr [eax+edx*2]
0x100387a6: mov edi,eax
0x100387a8: imul eax,ecx
0x100387ab: imul edi,esi
```

```
0x100387ae: shr edi,8
0x100387b1: shr eax,8
0x100387b4: mov esi,edi
0x100387b6: mov ecx,eax
0x100387b8: add edx,1
0x100387bb: cmp edx,[esp+48h]
0x100387bf: jl short 10038790h
0x100387c1: mov edx,[ebx+20Ch]
0x100387c7: movzx edx,word ptr [edx+8]
0x100387cb: mov eax,edx
0x100387cd: imul eax,ecx
0x100387d0: shr eax,3
0x100387d3: cmp eax,3FFFC0h
0x100387d8: jbe near ptr 10038A60h
0x100387de: mov dword ptr [esp+20h],7FFFFFFFh
0x100387e6: jmp near ptr 10038A6Fh
0x100387eb: imul edx,esi
0x100387ee: shr edx,3
0x100387f1: shl eax,0Ah
0x100387f4: add edx,eax
0x100387f6: mov [esp+34h],edx
0x100387fa: mov ebp,[esp+28h]
0x100387fe: mov ecx,[ebx+0F8h]
0x10038804: mov esi,[esp+2Ch]
0x10038808: mov eax,[esp+24h]
0x1003880c: add ebp,1
0x1003880f: add ecx,1
0x10038812: add esi,1
0x10038815: test eax,eax
0x10038817: mov dword ptr [esp+30h],0
0x1003881f: mov edi,ebp
0x10038821: jbe short 10038863h
0x10038823: mov [esp+38h],eax
0x10038827: mov [esp+30h],eax
0x1003882b: jmp short 10038830h
0x10038830: mov dl,[esi]
0x10038832: mov al,[edi]
0x10038834: shr dl,1
0x10038836: sub al,dl
0x10038838: mov [ecx],al
0x1003883a: movzx eax,al
0x1003883d: add ecx,1
0x10038840: add esi,1
0x10038843: add edi,1
0x10038846: cmp eax,80h
0x1003884b: jge short 10038851h
0x1003884d: mov edx,eax
0x1003884f: jmp short 10038858h
0x10038851: mov edx,100h
0x10038856: sub edx,eax
0x10038858: add [esp+20h],edx
0x1003885c: sub dword ptr [esp+38h],1
0x10038861: jne short 10038830h
0x10038863: mov eax,[esp+14h]
0x10038867: cmp [esp+30h],eax
0x1003886b: jae short 100388C8h
0x1003886d: lea ecx,[ecx]
0x10038870: movzx edx,byte ptr [esi]
0x10038873: movzx eax,byte ptr [ebp]
0x10038877: add eax,edx
0x10038879: cdq
0x1003887a: sub eax,edx
0x1003887c: mov edx,eax
0x1003887e: mov al,[edi]
0x10038880: sar edx,1
0x10038882: sub al,dl
0x10038884: mov [ecx],al
```

```
0x10038886: movzx eax,al
0x10038889: add ecx,1
0x1003888c: add ebp,1
0x1003888f: add esi,1
0x10038892: add edi,1
0x10038895: cmp eax,80h
0x1003889a: jge short 100388A0h
0x1003889c: mov edx,eax
0x1003889e: jmp short 100388A7h
0x100388a0: mov edx,100h
0x100388a5: sub edx,eax
0x100388a7: mov eax,[esp+20h]
0x100388ab: add eax,edx
0x100388ad: cmp eax,[esp+34h]
0x100388b1: mov [esp+20h],eax
0x100388b5: ja short 100388C8h
0x100388b7: mov eax,[esp+30h]
0x100388bb: add eax,1
0x100388be: cmp eax,[esp+14h]
0x100388c2: mov [esp+30h],eax
0x100388c6: jb short 10038870h
0x100388c8: cmp byte ptr [ebx+1F8h],2
0x100388cf: jne short 1003894Dh
0x100388d1: mov edx,[esp+20h]
0x100388d5: movzx esi,dx
0x100388d8: shr edx,0Ah
0x100388db: xor ecx,ecx
0x100388dd: and edx,3FFFC0h
0x100388e3: cmp [esp+48h],ecx
0x100388e7: jle short 1003891Bh
0x100388e9: mov edi,[ebx+1FCh]
0x100388ef: nop
0x100388f0: cmp byte ptr [ecx+edi],0
0x100388f4: jne short 10038912h
0x100388f6: mov eax,[ebx+200h]
0x100388fc: movzx eax,word ptr [eax+ecx*2]
0x10038900: mov ebp,eax
0x10038902: imul eax,edx
0x10038905: imul ebp,esi
0x10038908: shr ebp,8
0x1003890b: shr eax,8
0x1003890e: mov esi,ebp
0x10038910: mov edx,eax
0x10038912: add ecx,1
0x10038915: cmp ecx,[esp+48h]
0x10038919: jl short 100388F0h
0x1003891b: mov ecx,[ebx+208h]
0x10038921: movzx ecx,word ptr [ecx+6]
0x10038925: mov eax,ecx
0x10038927: imul eax,edx
0x1003892a: shr eax,3
0x1003892d: cmp eax,3FFFC0h
0x10038932: jbe short 1003893Eh
0x10038934: mov dword ptr [esp+20h],7FFFFFFFh
0x1003893c: jmp short 1003894Dh
0x1003893e: imul ecx,esi
0x10038941: shr ecx,3
0x10038944: shl eax,0Ah
0x10038947: add ecx,eax
0x10038949: mov [esp+20h],ecx
0x1003894d: mov eax,[esp+20h]
0x10038951: cmp eax,[esp+1Ch]
0x10038955: jae short 10038965h
0x10038957: mov edx,[ebx+0F8h]
0x1003895d: mov [esp+1Ch],eax
0x10038961: mov [esp+18h],edx
0x10038965: cmp byte ptr [esp+13h],80h
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```
0x1003896a: jne near ptr 10038754h
0x10038970: mov eax,[esp+28h]
0x10038974: mov edi,[ebx+0FCh]
0x1003897a: mov edx,[esp+2Ch]
0x1003897e: add eax,1
0x10038981: add edx,1
0x10038984: add edi,1
0x10038987: cmp dword ptr [esp+24h],0
0x1003898c: mov dword ptr [esp+34h],0
0x10038994: mov ecx,eax
0x10038996: mov [esp+30h],eax
0x1003899a: mov esi,edx
0x1003899c: jbe short 100389BFh
0x1003899e: mov ebp,[esp+24h]
0x100389a2: mov [esp+34h],ebp
0x100389a6: mov bl,[ecx]
0x100389a8: sub bl,[esi]
0x100389aa: add edi,1
0x100389ad: mov [edi-1],bl
0x100389b0: add esi,1
0x100389b3: add ecx,1
0x100389b6: sub ebp,1
0x100389b9: jne short 100389A6h
0x100389bb: mov [esp+30h],ecx
0x100389bf: mov ebx,[esp+14h]
0x100389c3: cmp [esp+34h],ebx
0x100389c7: mov ebp,eax
0x100389c9: jae near ptr 10038A57h
0x100389cf: mov ecx,edx
0x100389d1: sub ecx,eax
0x100389d3: sub ebx,[esp+34h]
0x100389d7: mov [esp+2Ch],ecx
0x100389db: mov [esp+34h],ebx
0x100389df: jmp short 100389E5h
0x100389e1: mov ecx,[esp+2Ch]
0x100389e5: movzx edx,byte ptr [ecx+ebp]
0x100389e9: movzx eax,byte ptr [esi]
0x100389ec: movzx ecx,byte ptr [ebp]
0x100389f0: mov [esp+24h],eax
0x100389f4: mov [esp+28h],ecx
0x100389f8: sub eax,edx
0x100389fa: add esi,1
0x100389fd: add ebp,1
0x10038a00: sub ecx,edx
0x10038a02: test eax,eax
0x10038a04: jge short 10038A10h
0x10038a06: mov ebx,eax
0x10038a08: neg ebx
0x10038a0a: mov [esp+38h],ebx
0x10038a0e: jmp short 10038A14h
0x10038a10: mov [esp+38h],eax
0x10038a14: test ecx,ecx
0x10038a16: mov ebx,ecx
0x10038a18: jge short 10038A1Ch
0x10038a1a: neg ebx
0x10038a1c: add eax,ecx
0x10038a1e: jns short 10038A22h
0x10038a20: neg eax
0x10038a22: mov ecx,[esp+38h]
0x10038a26: cmp ecx,ebx
0x10038a28: jg short 10038A34h
0x10038a2a: cmp ecx,eax
0x10038a2c: jg short 10038A34h
0x10038a2e: mov edx,[esp+28h]
0x10038a32: jmp short 10038A3Ch
0x10038a34: cmp ebx,eax
0x10038a36: jg short 10038A3Ch
```

```
0x10038a38: mov edx,[esp+24h]
0x10038a3c: mov eax,[esp+30h]
0x10038a40: mov cl,[eax]
0x10038a42: sub cl,dl
0x10038a44: mov [edi],cl
0x10038a46: add eax,1
0x10038a49: add edi,1
0x10038a4c: sub dword ptr [esp+34h],1
0x10038a51: mov [esp+30h],eax
0x10038a55: jne short 100389E1h
0x10038a57: mov ebx,[esp+44h]
0x10038a5b: jmp near ptr 10038C42h
0x10038a60: imul edx,esi
0x10038a63: shr edx,3
0x10038a66: shl eax,0Ah
0x10038a69: add edx,eax
0x10038a6b: mov [esp+20h],edx
0x10038a6f: mov edi,[esp+28h]
0x10038a73: mov edx,[esp+2Ch]
0x10038a77: mov esi,[ebx+0FCh]
0x10038a7d: mov eax,[esp+24h]
0x10038a81: add edi,1
0x10038a84: add edx,1
0x10038a87: add esi,1
0x10038a8a: test eax,eax
0x10038a8c: mov dword ptr [esp+44h],0
0x10038a94: mov [esp+28h],edi
0x10038a98: mov [esp+34h],edi
0x10038a9c: mov [esp+2Ch],edx
0x10038aa0: jbe short 10038AE6h
0x10038aa2: mov [esp+38h],eax
0x10038aa6: mov [esp+44h],eax
0x10038aaa: lea ebx,[ebx]
0x10038ab0: mov al,[edi]
0x10038ab2: sub al,[edx]
0x10038ab4: add esi,1
0x10038ab7: mov [esi-1],al
0x10038aba: movzx eax,al
0x10038abd: add edx,1
0x10038ac0: add edi,1
0x10038ac3: cmp eax,80h
0x10038ac8: jge short 10038ACEh
0x10038aca: mov ecx,eax
0x10038acc: jmp short 10038AD5h
0x10038ace: mov ecx,100h
0x10038ad3: sub ecx,eax
0x10038ad5: add ebp,ecx
0x10038ad7: sub dword ptr [esp+38h],1
0x10038adc: jne short 10038AB0h
0x10038ade: mov [esp+30h],ebp
0x10038ae2: mov [esp+34h],edi
0x10038ae6: mov eax,[esp+44h]
0x10038aea: cmp eax,[esp+14h]
0x10038aee: mov ecx,[esp+28h]
0x10038af2: jae near ptr 10038BB8h
0x10038af8: mov edi,[esp+2Ch]
0x10038afc: sub edi,ecx
0x10038afe: mov [esp+2Ch],edi
0x10038b02: jmp short 10038B10h
0x10038b04: mov edx,[esp+28h]
0x10038b08: mov ecx,[esp+24h]
0x10038b0c: mov edi,[esp+2Ch]
0x10038b10: movzx eax,byte ptr [edx]
0x10038b13: add edx,1
0x10038b16: mov [esp+28h],edx
0x10038b1a: movzx edx,byte ptr [edi+ecx]
0x10038b1e: movzx edi,byte ptr [ecx]
```

```
0x10038b21: add ecx,1
0x10038b24: mov [esp+24h],ecx
0x10038b28: mov [esp+3Ch],eax
0x10038b2c: mov ecx,edi
0x10038b2e: sub eax,edx
0x10038b30: sub ecx,edx
0x10038b32: test eax,eax
0x10038b34: jge short 10038B40h
0x10038b36: mov ebp,eax
0x10038b38: neg ebp
0x10038b3a: mov [esp+38h],ebp
0x10038b3e: jmp short 10038B44h
0x10038b40: mov [esp+38h],eax
0x10038b44: test ecx,ecx
0x10038b46: mov ebp,ecx
0x10038b48: jge short 10038B4Ch
0x10038b4a: neg ebp
0x10038b4c: add eax,ecx
0x10038b4e: jns short 10038B52h
0x10038b50: neg eax
0x10038b52: mov ecx,[esp+38h]
0x10038b56: cmp ecx,ebp
0x10038b58: jg short 10038B62h
0x10038b5a: cmp ecx,eax
0x10038b5c: jg short 10038B62h
0x10038b5e: mov edx,edi
0x10038b60: jmp short 10038B6Ah
0x10038b62: cmp ebp,eax
0x10038b64: jg short 10038B6Ah
0x10038b66: mov edx,[esp+3Ch]
0x10038b6a: mov ecx,[esp+34h]
0x10038b6e: mov al,[ecx]
0x10038b70: sub al,dl
0x10038b72: mov [esi],al
0x10038b74: movzx eax,al
0x10038b77: add ecx,1
0x10038b7a: add esi,1
0x10038b7d: cmp eax,80h
0x10038b82: mov [esp+34h],ecx
0x10038b86: jge short 10038B8Ch
0x10038b88: mov ecx,eax
0x10038b8a: jmp short 10038B93h
0x10038b8c: mov ecx,100h
0x10038b91: sub ecx,eax
0x10038b93: mov ebp,[esp+30h]
0x10038b97: add ebp,ecx
0x10038b99: cmp ebp,[esp+20h]
0x10038b9d: mov [esp+30h],ebp
0x10038ba1: ja short 10038BB8h
0x10038ba3: mov eax,[esp+44h]
0x10038ba7: add eax,1
0x10038baa: cmp eax,[esp+14h]
0x10038bae: mov [esp+44h],eax
0x10038bb2: jb near ptr 10038B04h
0x10038bb8: cmp byte ptr [ebx+1F8h],2
0x10038bbf: jne short 10038C3Ch
0x10038bc1: mov edi,[esp+48h]
0x10038bc5: movzx esi,bp
0x10038bc8: shr ebp,0Ah
0x10038bcb: and ebp,3FFFC0h
0x10038bd1: xor ecx,ecx
0x10038bd3: test edi,edi
0x10038bd5: mov edx,ebp
0x10038bd7: jle short 10038C0Fh
0x10038bd9: lea esp,[esp]
0x10038be0: mov eax,[ebx+1FCh]
0x10038be6: cmp byte ptr [ecx+eax],4
```

```
0x10038bea: jne short 10038C08h
0x10038bec: mov eax,[ebx+200h]
0x10038bf2: movzx eax,word ptr [eax+ecx*2]
0x10038bf6: mov ebp,eax
0x10038bf8: imul eax,edx
0x10038bfb: imul ebp,esi
0x10038bfe: shr ebp,8
0x10038c01: shr eax,8
0x10038c04: mov esi,ebp
0x10038c06: mov edx,eax
0x10038c08: add ecx,1
0x10038c0b: cmp ecx,edi
0x10038c0d: jl short 10038BE0h
0x10038c0f: mov ecx,[ebx+208h]
0x10038c15: movzx ecx,word ptr [ecx+8]
0x10038c19: mov eax,ecx
0x10038c1b: imul eax,edx
0x10038c1e: shr eax,3
0x10038c21: cmp eax,3FFFC0h
0x10038c26: jbe short 10038C2Fh
0x10038c28: mov ebp,7FFFFFFFh
0x10038c2d: jmp short 10038C3Ch
0x10038c2f: imul ecx,esi
0x10038c32: shr ecx,3
0x10038c35: shl eax,0Ah
0x10038c38: add ecx,eax
0x10038c3a: mov ebp,ecx
0x10038c3c: cmp ebp,[esp+1Ch]
0x10038c40: jae short 10038C4Ch
0x10038c42: mov edx,[ebx+0FCh]
0x10038c48: mov [esp+18h],edx
0x10038c4c: mov eax,[esp+18h]
0x10038c50: push eax
0x10038c51: push ebx
0x10038c52: call 10037FE0h
0x10038c57: add esp,8
0x10038c5a: cmp byte ptr [ebx+1F9h],0
0x10038c61: jbe short 10038C96h
0x10038c63: mov eax,1
0x10038c68: cmp [esp+48h],eax
0x10038c6c: jle short 10038C87h
0x10038c6e: mov edi,edi
0x10038c70: mov ecx,[ebx+1FCh]
0x10038c76: mov dl,[ecx+eax-1]
0x10038c7a: add ecx,eax
0x10038c7c: add eax,1
0x10038c7f: cmp eax,[esp+48h]
0x10038c83: mov [ecx],dl
0x10038c85: jl short 10038C70h
0x10038c87: mov edx,[esp+18h]
0x10038c8b: mov ecx,[ebx+1FCh]
0x10038c91: mov dl,[edx]
0x10038c93: mov [eax+ecx],dl
0x10038c96: pop edi
0x10038c97: pop esi
0x10038c98: pop ebp
0x10038c99: pop ebx
0x10038c9a: add esp,30h
0x10038c9d: ret
```

`sub_100380e0` is called by png_write_row. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to sub_10037fe0 (1 callees).

12.846 sub_10038ca0 at 0x10038ca0 (110 instructions)

```
0x10038ca0: push ebx
0x10038ca1: mov ebx,[esp+8]
0x10038ca5: cmp byte ptr [ebx+9],8
0x10038ca9: jne near ptr 10038DD0h
0x10038caf: cmp byte ptr [ebx+0Ah],1
0x10038cb3: jne near ptr 10038DD0h
0x10038cb9: mov ecx,[esp+10h]
0x10038cbd: sub ecx,1
0x10038cc0: push ebp
0x10038cc1: push esi
0x10038cc2: push edi
0x10038cc3: je near ptr 10038D60h
0x10038cc9: sub ecx,1
0x10038ccc: je short 10038D20h
0x10038cce: sub ecx,2
0x10038cd1: jne near ptr 10038DA2h
0x10038cd7: mov edi,[ebx]
0x10038cd9: mov esi,[esp+18h]
0x10038cdd: xor edx,edx
0x10038cdf: test edi,edi
0x10038ce1: mov ebp,esi
0x10038ce3: mov ecx,4
0x10038ce8: jbe near ptr 10038DA2h
0x10038cee: mov edi,edi
0x10038cf0: movzx eax,byte ptr [ebp]
0x10038cf4: and eax,0Fh
0x10038cf7: shl eax,cl
0x10038cf9: or edx,eax
0x10038cfb: test ecx,ecx
0x10038cfd: jne short 10038D0Dh
0x10038cff: mov [esi],dl
0x10038d01: add esi,1
0x10038d04: mov ecx,4
0x10038d09: xor edx,edx
0x10038d0b: jmp short 10038D10h
0x10038d0d: sub ecx,4
0x10038d10: add ebp,1
0x10038d13: sub edi,1
0x10038d16: jne short 10038CF0h
0x10038d18: cmp ecx,4
0x10038d1b: jmp near ptr 10038D9Eh
0x10038d20: mov edi,[ebx]
0x10038d22: mov esi,[esp+18h]
0x10038d26: xor edx,edx
0x10038d28: test edi,edi
0x10038d2a: mov ebp,esi
0x10038d2c: mov ecx,6
0x10038d31: jbe short 10038DA2h
0x10038d33: movzx eax,byte ptr [ebp]
0x10038d37: and eax,3
0x10038d3a: shl eax,cl
0x10038d3c: or edx,eax
0x10038d3e: test ecx,ecx
0x10038d40: jne short 10038D50h
0x10038d42: mov [esi],dl
0x10038d44: add esi,1
0x10038d47: mov ecx,6
0x10038d4c: xor edx,edx
0x10038d4e: jmp short 10038D53h
0x10038d50: sub ecx,2
0x10038d53: add ebp,1
0x10038d56: sub edi,1
0x10038d59: jne short 10038D33h
0x10038d5b: cmp ecx,6
0x10038d5e: jmp short 10038D9Eh
0x10038d60: mov edi,[ebx]
```

```
0x10038d62: mov esi,[esp+18h]
0x10038d66: xor edx,edx
0x10038d68: test edi,edi
0x10038d6a: mov ebp,esi
0x10038d6c: mov ecx,80h
0x10038d71: jbe short 10038DA2h
0x10038d73: cmp byte ptr [ebp],0
0x10038d77: je short 10038D7Bh
0x10038d79: or edx,ecx
0x10038d7b: add ebp,1
0x10038d7e: cmp ecx,1
0x10038d81: jle short 10038D87h
0x10038d83: sar ecx,1
0x10038d85: jmp short 10038D93h
0x10038d87: mov [esi],dl
0x10038d89: add esi,1
0x10038d8c: mov ecx,80h
0x10038d91: xor edx,edx
0x10038d93: sub edi,1
0x10038d96: jne short 10038D73h
0x10038d98: cmp ecx,80h
0x10038d9e: je short 10038DA2h
0x10038da0: mov [esi],dl
0x10038da2: mov al,[esp+1Ch]
0x10038da6: mov [ebx+9],al
0x10038da9: imul byte ptr [ebx+0Ah]
0x10038dac: pop edi
0x10038dad: pop esi
0x10038dae: mov [ebx+0Bh],al
0x10038db1: cmp al,8
0x10038db3: pop ebp
0x10038db4: movzx eax,al
0x10038db7: jb short 10038DC4h
0x10038db9: shr eax,3
0x10038dbc: imul eax,[ebx]
0x10038dbf: mov [ebx+4],eax
0x10038dc2: pop ebx
0x10038dc3: ret
0x10038dc4: imul eax,[ebx]
0x10038dc7: add eax,7
0x10038dca: shr eax,3
0x10038dcd: mov [ebx+4],eax
0x10038dd0: pop ebx
0x10038dd1: ret
```

`sub_10038ca0` is called by sub_10039380. It calls nothing else.

12.847 sub_10038de0 at 0x10038de0 (221 instructions)

```
0x10038de0: mov edx,[esp+4]
0x10038de4: sub esp,34h
0x10038de7: push ebx
0x10038de8: mov bl,[edx+8]
0x10038deb: cmp bl,3
0x10038dee: je near ptr 1003906Ah
0x10038df4: test bl,2
0x10038df7: mov ecx,[esp+44h]
0x10038dfb: push ebp
0x10038dfc: push esi
0x10038dfd: push edi
0x10038dfe: je short 10038E34h
0x10038e00: movzx eax,byte ptr [edx+9]
0x10038e04: movzx esi,byte ptr [ecx]
0x10038e07: mov edi,eax
0x10038e09: sub edi,esi
0x10038e0b: mov [esp+24h],esi
```

```
0x10038e0f: movzx esi,byte ptr [ecx+1]
0x10038e13: mov ebp,eax
0x10038e15: sub ebp,esi
0x10038e17: mov [esp+28h],esi
0x10038e1b: movzx esi,byte ptr [ecx+2]
0x10038e1f: sub eax,esi
0x10038e21: mov [esp+38h],ebp
0x10038e25: mov [esp+3Ch],eax
0x10038e29: mov [esp+2Ch],esi
0x10038e2d: mov ebp,3
0x10038e32: jmp short 10038E47h
0x10038e34: movzx eax,byte ptr [ecx+3]
0x10038e38: movzx edi,byte ptr [edx+9]
0x10038e3c: sub edi,eax
0x10038e3e: mov [esp+24h],eax
0x10038e42: mov ebp,1
0x10038e47: test bl,4
0x10038e4a: mov [esp+48h],ebp
0x10038e4e: mov [esp+34h],edi
0x10038e52: je short 10038E71h
0x10038e54: movzx eax,byte ptr [ecx+4]
0x10038e58: movzx esi,byte ptr [edx+9]
0x10038e5c: sub esi,eax
0x10038e5e: mov [esp+ebp*4+34h],esi
0x10038e62: mov edi,[esp+34h]
0x10038e66: mov [esp+ebp*4+24h],eax
0x10038e6a: add ebp,1
0x10038e6d: mov [esp+48h],ebp
0x10038e71: mov bl,[edx+9]
0x10038e74: cmp bl,8
0x10038e77: jae near ptr 10038F1Ah
0x10038e7d: mov cl,[ecx+3]
0x10038e80: cmp cl,1
0x10038e83: mov eax,[esp+4Ch]
0x10038e87: mov esi,[edx+4]
0x10038e8a: jne short 10038E98h
0x10038e8c: cmp bl,2
0x10038e8f: jne short 10038E98h
0x10038e91: mov byte ptr [esp+48h],55h
0x10038e96: jmp short 10038EACh
0x10038e98: cmp bl,4
0x10038e9b: jne short 10038EA7h
0x10038e9d: cmp cl,3
0x10038ea0: mov byte ptr [esp+48h],11h
0x10038ea5: je short 10038EACh
0x10038ea7: mov byte ptr [esp+48h],0FFh
0x10038eac: test esi,esi
0x10038eae: jbe near ptr 10039067h
0x10038eb4: mov edx,[esp+24h]
0x10038eb8: neg edx
0x10038eba: mov [esp+10h],esi
0x10038ebe: mov edi,edi
0x10038ec0: cmp edi,edx
0x10038ec2: movzx cx,byte ptr [eax]
0x10038ec6: movzx ecx,cx
0x10038ec9: mov [esp+14h],ecx
0x10038ecd: mov byte ptr [eax],0
0x10038ed0: mov esi,edi
0x10038ed2: jle short 10038F08h
0x10038ed4: mov ebp,edi
0x10038ed6: neg ebp
0x10038ed8: jmp short 10038EE4h
0x10038ee0: mov ecx,[esp+14h]
0x10038ee4: test esi,esi
0x10038ee6: jle short 10038EF0h
0x10038ee8: mov bl,cl
0x10038eea: mov ecx,esi
```

```
0x10038eec: shl bl,cl
0x10038eee: jmp short 10038EFCh
0x10038ef0: mov ebx,ecx
0x10038ef2: mov cx,bp
0x10038ef5: shr bx,cl
0x10038ef8: and bl,[esp+48h]
0x10038efc: sub esi,[esp+24h]
0x10038f00: or [eax],bl
0x10038f02: sub ebp,edx
0x10038f04: cmp esi,edx
0x10038f06: jg short 10038EE0h
0x10038f08: add eax,1
0x10038f0b: sub dword ptr [esp+10h],1
0x10038f10: jne short 10038EC0h
0x10038f12: pop edi
0x10038f13: pop esi
0x10038f14: pop ebp
0x10038f15: pop ebx
0x10038f16: add esp,34h
0x10038f19: ret
0x10038f1a: mov esi,[esp+4Ch]
0x10038f1e: mov edx,[edx]
0x10038f20: jne near ptr 10038FB6h
0x10038f26: imul edx,ebp
0x10038f29: xor ebx,ebx
0x10038f2b: test edx,edx
0x10038f2d: mov [esp+1Ch],edx
0x10038f31: mov [esp+14h],ebx
0x10038f35: jbe near ptr 10039067h
0x10038f3b: jmp short 10038F40h
0x10038f40: xor edx,edx
0x10038f42: mov eax,ebx
0x10038f44: div ebp
0x10038f46: movzx ax,byte ptr [esi]
0x10038f4a: movzx ecx,ax
0x10038f4d: mov [esp+10h],ecx
0x10038f51: mov byte ptr [esi],0
0x10038f54: mov eax,[esp+edx*4+34h]
0x10038f58: lea ecx,[esp+edx*4+24h]
0x10038f5c: mov edx,[ecx]
0x10038f5e: mov edi,edx
0x10038f60: neg edi
0x10038f62: cmp eax,edi
0x10038f64: jle short 10038F9Eh
0x10038f66: mov ecx,edx
0x10038f68: neg ecx
0x10038f6a: mov ebp,eax
0x10038f6c: mov [esp+18h],ecx
0x10038f70: neg ebp
0x10038f72: test eax,eax
0x10038f74: jle short 10038F80h
0x10038f76: mov bl,[esp+10h]
0x10038f7a: mov ecx,eax
0x10038f7c: shl bl,cl
0x10038f7e: jmp short 10038F8Ah
0x10038f80: mov ebx,[esp+10h]
0x10038f84: mov cx,bp
0x10038f87: shr bx,cl
0x10038f8a: or [esi],bl
0x10038f8c: sub eax,edx
0x10038f8e: sub ebp,edi
0x10038f90: cmp eax,[esp+18h]
0x10038f94: jg short 10038F72h
0x10038f96: mov ebp,[esp+48h]
0x10038f9a: mov ebx,[esp+14h]
0x10038f9e: add ebx,1
0x10038fa1: add esi,1
```

```
0x10038fa4: cmp ebx,[esp+1Ch]
0x10038fa8: mov [esp+14h],ebx
0x10038fac: jb short 10038F40h
0x10038fae: pop edi
0x10038faf: pop esi
0x10038fb0: pop ebp
0x10038fb1: pop ebx
0x10038fb2: add esp,34h
0x10038fb5: ret
0x10038fb6: imul edx,ebp
0x10038fb9: xor edi,edi
0x10038fbb: test edx,edx
0x10038fbd: mov [esp+20h],edx
0x10038fc1: mov [esp+14h],edi
0x10038fc5: jbe near ptr 10039067h
0x10038fcb: jmp short 10038FD0h
0x10038fd0: xor edx,edx
0x10038fd2: mov eax,edi
0x10038fd4: div ebp
0x10038fd6: movzx ax,byte ptr [esi]
0x10038fda: movzx cx,byte ptr [esi+1]
0x10038fdf: shl ax,8
0x10038fe3: add ax,cx
0x10038fe6: movzx ebx,ax
0x10038fe9: mov [esp+1Ch],ebx
0x10038fed: mov dword ptr [esp+48h],0
0x10038ff5: mov eax,[esp+edx*4+34h]
0x10038ff9: lea ecx,[esp+edx*4+24h]
0x10038ffd: mov edx,[ecx]
0x10038fff: mov [esp+18h],edx
0x10039003: neg edx
0x10039005: cmp eax,edx
0x10039007: jle short 10039044h
0x10039009: mov ecx,[ecx]
0x1003900b: neg ecx
0x1003900d: mov edi,eax
0x1003900f: mov [esp+10h],ecx
0x10039013: neg edi
0x10039015: jmp short 1003901Bh
0x10039017: mov ebx,[esp+1Ch]
0x1003901b: test eax,eax
0x1003901d: jle short 10039029h
0x1003901f: mov ecx,eax
0x10039021: shl ebx,cl
0x10039023: or [esp+48h],ebx
0x10039027: jmp short 10039034h
0x10039029: mov cx,di
0x1003902c: shr bx,cl
0x1003902f: or [esp+48h],bx
0x10039034: sub eax,[esp+18h]
0x10039038: sub edi,edx
0x1003903a: cmp eax,[esp+10h]
0x1003903e: jg short 10039017h
0x10039040: mov edi,[esp+14h]
0x10039044: mov dl,[esp+49h]
0x10039048: mov al,[esp+48h]
0x1003904c: mov [esi],dl
0x1003904e: add esi,1
0x10039051: add edi,1
0x10039054: mov [esi],al
0x10039056: add esi,1
0x10039059: cmp edi,[esp+20h]
0x1003905d: mov [esp+14h],edi
0x10039061: jb near ptr 10038FD0h
0x10039067: pop edi
0x10039068: pop esi
0x10039069: pop ebp
```

```
0x1003906a: pop ebx
0x1003906b: add esp,34h
0x1003906e: ret
```

`sub_10038de0` is called by sub_10039380. It calls nothing else.

12.848 sub_10039070 at 0x10039070 (132 instructions)

```
0x10039070: mov eax,[esp+4]
0x10039074: mov cl,[eax+8]
0x10039077: cmp cl,6
0x1003907a: push ebx
0x1003907b: push esi
0x1003907c: jne near ptr 10039157h
0x10039082: cmp byte ptr [eax+9],8
0x10039086: jne short 100390D6h
0x10039088: mov edx,[eax]
0x1003908a: test edx,edx
0x1003908c: mov eax,[esp+10h]
0x10039090: mov ecx,eax
0x10039092: jbe near ptr 100391DBh
0x10039098: mov esi,edx
0x1003909a: lea ebx,[ebx]
0x100390a0: mov dl,[eax]
0x100390a2: movzx ebx,byte ptr [eax+1]
0x100390a6: add eax,1
0x100390a9: mov [ecx],bl
0x100390ab: movzx ebx,byte ptr [eax+1]
0x100390af: add eax,1
0x100390b2: add ecx,1
0x100390b5: mov [ecx],bl
0x100390b7: movzx ebx,byte ptr [eax+1]
0x100390bb: add eax,1
0x100390be: add ecx,1
0x100390c1: mov [ecx],bl
0x100390c3: add ecx,1
0x100390c6: mov [ecx],dl
0x100390c8: add eax,1
0x100390cb: add ecx,1
0x100390ce: sub esi,1
0x100390d1: jne short 100390A0h
0x100390d3: pop esi
0x100390d4: pop ebx
0x100390d5: ret
0x100390d6: mov esi,[eax]
0x100390d8: test esi,esi
0x100390da: mov eax,[esp+10h]
0x100390de: mov ecx,eax
0x100390e0: jbe near ptr 100391DBh
0x100390e6: mov dl,[eax]
0x100390e8: movzx ebx,byte ptr [eax+1]
0x100390ec: add eax,1
0x100390ef: add eax,1
0x100390f2: mov [esp+0Dh],bl
0x100390f6: movzx ebx,byte ptr [eax]
0x100390f9: mov [ecx],bl
0x100390fb: movzx ebx,byte ptr [eax+1]
0x100390ff: mov [ecx+1],bl
0x10039102: add eax,1
0x10039105: movzx ebx,byte ptr [eax+1]
0x10039109: add ecx,1
0x1003910c: mov [ecx+1],bl
0x1003910f: add eax,1
0x10039112: movzx ebx,byte ptr [eax+1]
0x10039116: add ecx,1
0x10039119: add eax,1
```

```
0x1003911c: mov [ecx+1],bl
0x1003911f: movzx ebx,byte ptr [eax+1]
0x10039123: add ecx,1
0x10039126: add eax,1
0x10039129: add ecx,1
0x1003912c: mov [ecx],bl
0x1003912e: movzx ebx,byte ptr [eax+1]
0x10039132: add eax,1
0x10039135: add ecx,1
0x10039138: mov [ecx],bl
0x1003913a: add ecx,1
0x1003913d: mov [ecx],dl
0x1003913f: movzx edx,byte ptr [esp+0Dh]
0x10039144: add ecx,1
0x10039147: mov [ecx],dl
0x10039149: add eax,1
0x1003914c: add ecx,1
0x1003914f: sub esi,1
0x10039152: jne short 100390E6h
0x10039154: pop esi
0x10039155: pop ebx
0x10039156: ret
0x10039157: cmp cl,4
0x1003915a: jne short 100391DBh
0x1003915c: cmp byte ptr [eax+9],8
0x10039160: jne short 1003918Dh
0x10039162: mov edx,[eax]
0x10039164: test edx,edx
0x10039166: mov eax,[esp+10h]
0x1003916a: mov ecx,eax
0x1003916c: jbe short 100391DBh
0x1003916e: mov esi,edx
0x10039170: mov dl,[eax]
0x10039172: mov bl,[eax+1]
0x10039175: add eax,1
0x10039178: mov [ecx],bl
0x1003917a: add ecx,1
0x1003917d: mov [ecx],dl
0x1003917f: add eax,1
0x10039182: add ecx,1
0x10039185: sub esi,1
0x10039188: jne short 10039170h
0x1003918a: pop esi
0x1003918b: pop ebx
0x1003918c: ret
0x1003918d: mov esi,[eax]
0x1003918f: test esi,esi
0x10039191: mov eax,[esp+10h]
0x10039195: mov ecx,eax
0x10039197: jbe short 100391DBh
0x10039199: lea esp,[esp]
0x100391a0: mov dl,[eax]
0x100391a2: movzx ebx,byte ptr [eax+1]
0x100391a6: add eax,1
0x100391a9: add eax,1
0x100391ac: mov [esp+0Dh],bl
0x100391b0: movzx ebx,byte ptr [eax]
0x100391b3: mov [ecx],bl
0x100391b5: movzx ebx,byte ptr [eax+1]
0x100391b9: add eax,1
0x100391bc: add ecx,1
0x100391bf: mov [ecx],bl
0x100391c1: add ecx,1
0x100391c4: mov [ecx],dl
0x100391c6: movzx edx,byte ptr [esp+0Dh]
0x100391cb: add ecx,1
0x100391ce: mov [ecx],dl
```

```
0x100391d0: add eax,1
0x100391d3: add ecx,1
0x100391d6: sub esi,1
0x100391d9: jne short 100391A0h
0x100391db: pop esi
0x100391dc: pop ebx
0x100391dd: ret
```

`sub_10039070` is called by sub_10039380. It calls nothing else.

12.849 sub_100391e0 at 0x100391e0 (74 instructions)

```
0x100391e0: mov eax,[esp+4]
0x100391e4: mov cl,[eax+8]
0x100391e7: cmp cl,6
0x100391ea: jne short 1003923Fh
0x100391ec: mov ecx,[eax]
0x100391ee: cmp byte ptr [eax+9],8
0x100391f2: mov eax,[esp+8]
0x100391f6: jne short 10039214h
0x100391f8: test ecx,ecx
0x100391fa: jbe near ptr 100392A6h
0x10039200: add eax,3
0x10039203: or dl,0FFh
0x10039206: sub dl,[eax]
0x10039208: add eax,1
0x1003920b: sub ecx,1
0x1003920e: mov [eax-1],dl
0x10039211: jne short 10039200h
0x10039213: ret
0x10039214: test ecx,ecx
0x10039216: jbe near ptr 100392A6h
0x1003921c: lea esp,[esp]
0x10039220: add eax,6
0x10039223: or dl,0FFh
0x10039226: sub dl,[eax]
0x10039228: add eax,1
0x1003922b: mov [eax-1],dl
0x1003922e: or dl,0FFh
0x10039231: sub dl,[eax]
0x10039233: add eax,1
0x10039236: sub ecx,1
0x10039239: mov [eax-1],dl
0x1003923c: jne short 10039220h
0x1003923e: ret
0x1003923f: cmp cl,4
0x10039242: jne short 100392A6h
0x10039244: cmp byte ptr [eax+9],8
0x10039248: jne short 1003927Eh
0x1003924a: mov edx,[eax]
0x1003924c: test edx,edx
0x1003924e: mov eax,[esp+8]
0x10039252: mov ecx,eax
0x10039254: jbe short 100392A6h
0x10039256: push esi
0x10039257: mov esi,edx
0x10039259: lea esp,[esp]
0x10039260: mov dl,[ecx]
0x10039262: mov [eax],dl
0x10039264: add ecx,1
0x10039267: or dl,0FFh
0x1003926a: sub dl,[ecx]
0x1003926c: add eax,1
0x1003926f: mov [eax],dl
0x10039271: add eax,1
0x10039274: add ecx,1
```

```
0x10039277: sub esi,1
0x1003927a: jne short 10039260h
0x1003927c: pop esi
0x1003927d: ret
0x1003927e: mov ecx,[eax]
0x10039280: test ecx,ecx
0x10039282: mov eax,[esp+8]
0x10039286: jbe short 100392A6h
0x10039288: add eax,2
0x1003928b: or dl,0FFh
0x1003928e: sub dl,[eax]
0x10039290: add eax,1
0x10039293: mov [eax-1],dl
0x10039296: or dl,0FFh
0x10039299: sub dl,[eax]
0x1003929b: add eax,1
0x1003929e: sub ecx,1
0x100392a1: mov [eax-1],dl
0x100392a4: jne short 10039288h
0x100392a6: ret
```

`sub_100391e0` is called by sub_10039380. It calls nothing else.

12.850 sub_100392b0 at 0x100392b0 (74 instructions)

```
0x100392b0: mov eax,[esp+4]
0x100392b4: mov cl,[eax+8]
0x100392b7: test cl,2
0x100392ba: je near ptr 10039378h
0x100392c0: mov edx,[eax]
0x100392c2: mov al,[eax+9]
0x100392c5: cmp al,8
0x100392c7: push edi
0x100392c8: jne short 10039304h
0x100392ca: cmp cl,2
0x100392cd: jne short 100392D6h
0x100392cf: mov edi,3
0x100392d4: jmp short 100392E4h
0x100392d6: cmp cl,6
0x100392d9: jne near ptr 10039377h
0x100392df: mov edi,4
0x100392e4: test edx,edx
0x100392e6: jbe near ptr 10039377h
0x100392ec: mov eax,[esp+0Ch]
0x100392f0: add eax,2
0x100392f3: mov cl,[eax-1]
0x100392f6: sub [eax-2],cl
0x100392f9: sub [eax],cl
0x100392fb: add eax,edi
0x100392fd: sub edx,1
0x10039300: jne short 100392F3h
0x10039302: pop edi
0x10039303: ret
0x10039304: cmp al,10h
0x10039306: jne short 10039377h
0x10039308: cmp cl,2
0x1003930b: jne short 10039314h
0x1003930d: mov edi,6
0x10039312: jmp short 1003931Eh
0x10039314: cmp cl,6
0x10039317: jne short 10039377h
0x10039319: mov edi,8
0x1003931e: test edx,edx
0x10039320: jbe short 10039377h
0x10039322: mov eax,[esp+0Ch]
0x10039326: push ebx
```

```
0x10039327: push esi
0x10039328: add eax,1
0x1003932b: mov esi,edx
0x1003932d: lea ecx,[ecx]
0x10039330: xor ebx,ebx
0x10039332: mov bh,[eax+3]
0x10039335: xor ecx,ecx
0x10039337: mov ch,[eax+1]
0x1003933a: xor edx,edx
0x1003933c: mov dh,[eax-1]
0x1003933f: mov bl,[eax+4]
0x10039342: mov cl,[eax+2]
0x10039345: mov dl,[eax]
0x10039347: sub ebx,ecx
0x10039349: and ebx,0FFFFh
0x1003934f: sub edx,ecx
0x10039351: and edx,0FFFFh
0x10039357: mov ecx,ebx
0x10039359: mov ebx,edx
0x1003935b: mov [eax],dl
0x1003935d: mov edx,ecx
0x1003935f: shr ebx,8
0x10039362: shr edx,8
0x10039365: mov [eax-1],bl
0x10039368: mov [eax+3],dl
0x1003936b: mov [eax+4],cl
0x1003936e: add eax,edi
0x10039370: sub esi,1
0x10039373: jne short 10039330h
0x10039375: pop esi
0x10039376: pop ebx
0x10039377: pop edi
0x10039378: ret
```

`sub_100392b0` is called by png_write_row. It calls nothing else.

12.851 sub_10039380 at 0x10039380 (106 instructions)

```
0x10039380: push esi
0x10039381: mov esi,[esp+8]
0x10039385: test esi,esi
0x10039387: je near ptr 100394EAh
0x1003938d: test dword ptr [esi+74h],100000h
0x10039394: je short 100393B4h
0x10039396: mov eax,[esi+60h]
0x10039399: test eax,eax
0x1003939b: je short 100393B4h
0x1003939d: mov ecx,[esi+0ECh]
0x100393a3: add ecx,1
0x100393a6: push ecx
0x100393a7: lea edx,[esi+100h]
0x100393ad: push edx
0x100393ae: push esi
0x100393af: call eax
0x100393b1: add esp,0Ch
0x100393b4: test dword ptr [esi+74h],8000h
0x100393bb: je short 100393DAh
0x100393bd: mov eax,[esi+70h]
0x100393c0: mov ecx,[esi+0ECh]
0x100393c6: push eax
0x100393c7: add ecx,1
0x100393ca: push ecx
0x100393cb: lea edx,[esi+100h]
0x100393d1: push edx
0x100393d2: call 10028930h
0x100393d7: add esp,0Ch
```

```
0x100393da: test dword ptr [esi+74h],10000h
0x100393e1: je short 100393FCh
0x100393e3: mov eax,[esi+0ECh]
0x100393e9: add eax,1
0x100393ec: push eax
0x100393ed: lea ecx,[esi+100h]
0x100393f3: push ecx
0x100393f4: call 100288D0h
0x100393f9: add esp,8
0x100393fc: test byte ptr [esi+74h],4
0x10039400: je short 10039423h
0x10039402: movzx edx,byte ptr [esi+127h]
0x10039409: mov eax,[esi+0ECh]
0x1003940f: push edx
0x10039410: add eax,1
0x10039413: push eax
0x10039414: lea ecx,[esi+100h]
0x1003941a: push ecx
0x1003941b: call 10038CA0h
0x10039420: add esp,0Ch
0x10039423: test byte ptr [esi+74h],10h
0x10039427: je short 10039442h
0x10039429: mov edx,[esi+0ECh]
0x1003942f: add edx,1
0x10039432: push edx
0x10039433: lea eax,[esi+100h]
0x10039439: push eax
0x1003943a: call 10028890h
0x1003943f: add esp,8
0x10039442: test byte ptr [esi+74h],8
0x10039446: je short 10039468h
0x10039448: mov edx,[esi+0ECh]
0x1003944e: lea ecx,[esi+181h]
0x10039454: push ecx
0x10039455: add edx,1
0x10039458: push edx
0x10039459: lea eax,[esi+100h]
0x1003945f: push eax
0x10039460: call 10038DE0h
0x10039465: add esp,0Ch
0x10039468: test dword ptr [esi+74h],20000h
0x1003946f: je short 1003948Ah
0x10039471: mov ecx,[esi+0ECh]
0x10039477: add ecx,1
0x1003947a: push ecx
0x1003947b: lea edx,[esi+100h]
0x10039481: push edx
0x10039482: call 10039070h
0x10039487: add esp,8
0x1003948a: test dword ptr [esi+74h],80000h
0x10039491: je short 100394ACh
0x10039493: mov eax,[esi+0ECh]
0x10039499: add eax,1
0x1003949c: push eax
0x1003949d: lea ecx,[esi+100h]
0x100394a3: push ecx
0x100394a4: call 100391E0h
0x100394a9: add esp,8
0x100394ac: test byte ptr [esi+74h],1
0x100394b0: je short 100394CBh
0x100394b2: mov edx,[esi+0ECh]
0x100394b8: add edx,1
0x100394bb: push edx
0x100394bc: lea eax,[esi+100h]
0x100394c2: push eax
0x100394c3: call 10028BE0h
0x100394c8: add esp,8
```

```
0x100394cb: test byte ptr [esi+74h],20h
0x100394cf: je short 100394EAh
0x100394d1: mov ecx,[esi+0ECh]
0x100394d7: add ecx,1
0x100394da: push ecx
0x100394db: add esi,100h
0x100394e1: push esi
0x100394e2: call 10028800h
0x100394e7: add esp,8
0x100394ea: pop esi
0x100394eb: ret
```

`sub_10039380` is called by png_write_row. It delegates to sub_10028800, sub_10028890, sub_100288d0 (9 callees).

12.852 sub_1003b170 at 0x1003b170 (11 instructions)

```
0x1003b170: mov ecx,[esp+4]
0x1003b174: mov eax,[ecx+1A4h]
0x1003b17a: xor edx,edx
0x1003b17c: mov [ecx+0C4h],edx
0x1003b182: mov [ecx+7Ch],edx
0x1003b185: mov [ecx+18Ch],edx
0x1003b18b: mov [eax+0Ch],dl
0x1003b18e: mov [eax+0Dh],dl
0x1003b191: mov [eax+14h],edx
0x1003b194: mov [eax+0A0h],edx
0x1003b19a: ret
```

`sub_1003b170` is called by sub_1003b1a0. It calls nothing else.

12.853 sub_1003b1a0 at 0x1003b1a0 (31 instructions)

```
0x1003b1a0: push esi
0x1003b1a1: mov esi,[esp+8]
0x1003b1a5: mov eax,[esi+4]
0x1003b1a8: mov ecx,[eax]
0x1003b1aa: push 0A8h
0x1003b1af: push 0
0x1003b1b1: push esi
0x1003b1b2: call ecx
0x1003b1b4: mov [esi+1A4h],eax
0x1003b1ba: add esp,0Ch
0x1003b1bd: mov dword ptr [eax],1003B170h
0x1003b1c3: mov dword ptr [eax+4],1003AC90h
0x1003b1ca: mov dword ptr [eax+8],1003AFE0h
0x1003b1d1: mov dword ptr [eax+18h],1003AA00h
0x1003b1d8: mov dword ptr [eax+5Ch],0
0x1003b1df: lea ecx,[eax+60h]
0x1003b1e2: mov edx,10h
0x1003b1e7: jmp short 1003B1F0h
0x1003b1f0: mov dword ptr [ecx-44h],1003AA00h
0x1003b1f7: mov dword ptr [ecx],0
0x1003b1fd: add ecx,4
0x1003b200: sub edx,1
0x1003b203: jne short 1003B1F0h
0x1003b205: mov ecx,1003A860h
0x1003b20a: push esi
0x1003b20b: mov [eax+1Ch],ecx
0x1003b20e: mov [eax+54h],ecx
0x1003b211: call 1003B170h
0x1003b216: add esp,4
0x1003b219: pop esi
0x1003b21a: ret
```

`sub_1003b1a0` is called by sub_1003b220. It delegates to sub_1003b170 (1 callees).

12.854 sub_1003b220 at 0x1003b220 (72 instructions)

```
0x1003b220: mov eax,[esp+8]
0x1003b224: push ebx
0x1003b225: push ebp
0x1003b226: push esi
0x1003b227: mov esi,[esp+10h]
0x1003b22b: push edi
0x1003b22c: xor edi,edi
0x1003b22e: cmp eax,50h
0x1003b231: mov [esi+4],edi
0x1003b234: je short 1003B257h
0x1003b236: mov ecx,[esi]
0x1003b238: mov dword ptr [ecx+14h],0Dh
0x1003b23f: mov edx,[esi]
0x1003b241: mov dword ptr [edx+18h],50h
0x1003b248: mov ecx,[esi]
0x1003b24a: mov [ecx+1Ch],eax
0x1003b24d: mov edx,[esi]
0x1003b24f: mov eax,[edx]
0x1003b251: push esi
0x1003b252: call eax
0x1003b254: add esp,4
0x1003b257: mov eax,[esp+1Ch]
0x1003b25b: cmp eax,1C0h
0x1003b260: je short 1003B283h
0x1003b262: mov ecx,[esi]
0x1003b264: mov dword ptr [ecx+14h],16h
0x1003b26b: mov edx,[esi]
0x1003b26d: mov dword ptr [edx+18h],1C0h
0x1003b274: mov ecx,[esi]
0x1003b276: mov [ecx+1Ch],eax
0x1003b279: mov edx,[esi]
0x1003b27b: mov eax,[edx]
0x1003b27d: push esi
0x1003b27e: call eax
0x1003b280: add esp,4
0x1003b283: mov ebx,[esi]
0x1003b285: mov ebp,[esi+0Ch]
0x1003b288: push 1C0h
0x1003b28d: push edi
0x1003b28e: push esi
0x1003b28f: call 100228C0h
0x1003b294: push esi
0x1003b295: mov [esi],ebx
0x1003b297: mov [esi+0Ch],ebp
0x1003b29a: mov byte ptr [esi+10h],1
0x1003b29e: call 1003EB30h
0x1003b2a3: mov [esi+8],edi
0x1003b2a6: mov [esi+18h],edi
0x1003b2a9: mov [esi+90h],edi
0x1003b2af: mov [esi+94h],edi
0x1003b2b5: mov [esi+98h],edi
0x1003b2bb: mov [esi+9Ch],edi
0x1003b2c1: mov [esi+0A0h],edi
0x1003b2c7: mov [esi+0B0h],edi
0x1003b2cd: mov [esi+0A4h],edi
0x1003b2d3: mov [esi+0B4h],edi
0x1003b2d9: mov [esi+0A8h],edi
0x1003b2df: mov [esi+0B8h],edi
0x1003b2e5: mov [esi+0ACh],edi
0x1003b2eb: mov [esi+0BCh],edi
0x1003b2f1: push esi
0x1003b2f2: mov [esi+10Ch],edi
0x1003b2f8: call 1003B1A0h
0x1003b2fd: push esi
0x1003b2fe: call 1003DFF0h
0x1003b303: add esp,18h
```

```
0x1003b306: pop edi
0x1003b307: mov dword ptr [esi+14h],0C8h
0x1003b30e: pop esi
0x1003b30f: pop ebp
0x1003b310: pop ebx
0x1003b311: ret
```

`sub_1003b220` is called by sub_1001b350. It delegates to j_memset, sub_1003b1a0, sub_1003dff0 (4 callees).

12.855 sub_1003b4a0 at 0x1003b4a0 (24 instructions)

```
0x1003b4a0: push esi
0x1003b4a1: mov esi,[esp+8]
0x1003b4a5: mov eax,[esi+14h]
0x1003b4a8: push edi
0x1003b4a9: add eax,0FFFFFF38h
0x1003b4ae: xor edi,edi
0x1003b4b0: cmp eax,0Ah
0x1003b4b3: ja short 1003B521h
0x1003b4b5: movzx eax,byte ptr [eax+1003B558h]
0x1003b4bc: jmp dword ptr [eax*4+1003B544h]
0x1003b521: mov eax,[esi]
0x1003b523: mov dword ptr [eax+14h],15h
0x1003b52a: mov ecx,[esi]
0x1003b52c: mov edx,[esi+14h]
0x1003b52f: mov [ecx+18h],edx
0x1003b532: mov eax,[esi]
0x1003b534: mov ecx,[eax]
0x1003b536: push esi
0x1003b537: call ecx
0x1003b539: add esp,4
0x1003b53c: mov eax,edi
0x1003b53e: pop edi
0x1003b53f: pop esi
0x1003b540: ret
```

`sub_1003b4a0` is called by sub_1003b6e0. It calls nothing else.

12.856 sub_1003b610 at 0x1003b610 (68 instructions)

```
0x1003b610: push esi
0x1003b611: mov esi,[esp+8]
0x1003b615: mov eax,[esi+14h]
0x1003b618: cmp eax,0CDh
0x1003b61d: je short 1003B626h
0x1003b61f: cmp eax,0CEh
0x1003b624: jne short 1003B65Ch
0x1003b626: cmp byte ptr [esi+40h],0
0x1003b62a: jne short 1003B65Ch
0x1003b62c: mov eax,[esi+78h]
0x1003b62f: cmp eax,[esi+60h]
0x1003b632: jae short 1003B647h
0x1003b634: mov ecx,[esi]
0x1003b636: mov dword ptr [ecx+14h],45h
0x1003b63d: mov edx,[esi]
0x1003b63f: mov eax,[edx]
0x1003b641: push esi
0x1003b642: call eax
0x1003b644: add esp,4
0x1003b647: mov ecx,[esi+190h]
0x1003b64d: mov edx,[ecx+4]
0x1003b650: push esi
0x1003b651: call edx
0x1003b653: mov dword ptr [esi+14h],0D2h
0x1003b65a: jmp short 1003B68Bh
```

```
0x1003b65c: cmp eax,0CFh
0x1003b661: jne short 1003B66Ch
0x1003b663: mov dword ptr [esi+14h],0D2h
0x1003b66a: jmp short 1003B68Eh
0x1003b66c: cmp eax,0D2h
0x1003b671: je short 1003B68Eh
0x1003b673: mov eax,[esi]
0x1003b675: mov dword ptr [eax+14h],15h
0x1003b67c: mov ecx,[esi]
0x1003b67e: mov edx,[esi+14h]
0x1003b681: mov [ecx+18h],edx
0x1003b684: mov eax,[esi]
0x1003b686: mov ecx,[eax]
0x1003b688: push esi
0x1003b689: call ecx
0x1003b68b: add esp,4
0x1003b68e: mov edx,[esi+1A0h]
0x1003b694: cmp byte ptr [edx+11h],0
0x1003b698: jne short 1003B6BEh
0x1003b69a: lea ebx,[ebx]
0x1003b6a0: mov eax,[esi+1A0h]
0x1003b6a6: mov ecx,[eax]
0x1003b6a8: push esi
0x1003b6a9: call ecx
0x1003b6ab: add esp,4
0x1003b6ae: test eax,eax
0x1003b6b0: je short 1003B6D4h
0x1003b6b2: mov edx,[esi+1A0h]
0x1003b6b8: cmp byte ptr [edx+11h],0
0x1003b6bc: je short 1003B6A0h
0x1003b6be: mov eax,[esi+18h]
0x1003b6c1: mov ecx,[eax+18h]
0x1003b6c4: push esi
0x1003b6c5: call ecx
0x1003b6c7: push esi
0x1003b6c8: call 1003D150h
0x1003b6cd: add esp,8
0x1003b6d0: mov al,1
0x1003b6d2: pop esi
0x1003b6d3: ret
0x1003b6d4: xor al,al
0x1003b6d6: pop esi
0x1003b6d7: ret
```

`sub_1003b610` is called by sub_1001b350. It delegates to sub_1003d150 (1 callees).

12.857 sub_1003b6e0 at 0x1003b6e0 (43 instructions)

```
0x1003b6e0: push esi
0x1003b6e1: mov esi,[esp+8]
0x1003b6e5: mov eax,[esi+14h]
0x1003b6e8: cmp eax,0C8h
0x1003b6ed: je short 1003B711h
0x1003b6ef: cmp eax,0C9h
0x1003b6f4: je short 1003B711h
0x1003b6f6: mov eax,[esi]
0x1003b6f8: mov dword ptr [eax+14h],15h
0x1003b6ff: mov ecx,[esi]
0x1003b701: mov edx,[esi+14h]
0x1003b704: mov [ecx+18h],edx
0x1003b707: mov eax,[esi]
0x1003b709: mov ecx,[eax]
0x1003b70b: push esi
0x1003b70c: call ecx
0x1003b70e: add esp,4
0x1003b711: push esi
```

```
0x1003b712: call 1003B4A0h
0x1003b717: mov ecx,eax
0x1003b719: add esp,4
0x1003b71c: sub ecx,1
0x1003b71f: je short 1003B74Fh
0x1003b721: sub ecx,1
0x1003b724: jne short 1003B754h
0x1003b726: cmp [esp+0Ch],cl
0x1003b72a: je short 1003B73Fh
0x1003b72c: mov edx,[esi]
0x1003b72e: mov dword ptr [edx+14h],35h
0x1003b735: mov eax,[esi]
0x1003b737: mov ecx,[eax]
0x1003b739: push esi
0x1003b73a: call ecx
0x1003b73c: add esp,4
0x1003b73f: push esi
0x1003b740: call 1003D150h
0x1003b745: add esp,4
0x1003b748: mov eax,2
0x1003b74d: pop esi
0x1003b74e: ret
0x1003b74f: mov eax,1
0x1003b754: pop esi
0x1003b755: ret
```

`sub_1003b6e0` is called by sub_1001b350. It delegates to sub_1003b4a0, sub_1003d150 (2 callees).

12.858 sub_1003b760 at 0x1003b760 (73 instructions)

```
0x1003b760: push ebx
0x1003b761: push edi
0x1003b762: mov edi,0CCh
0x1003b767: cmp [esi+14h],edi
0x1003b76a: je short 1003B784h
0x1003b76c: mov eax,[esi+190h]
0x1003b772: mov ecx,[eax]
0x1003b774: push esi
0x1003b775: call ecx
0x1003b777: add esp,4
0x1003b77a: mov dword ptr [esi+78h],0
0x1003b781: mov [esi+14h],edi
0x1003b784: mov edx,[esi+190h]
0x1003b78a: cmp byte ptr [edx+8],0
0x1003b78e: je short 1003B80Dh
0x1003b790: lea edi,[esi+78h]
0x1003b793: mov eax,[edi]
0x1003b795: cmp eax,[esi+60h]
0x1003b798: jae short 1003B7E1h
0x1003b79a: lea ebx,[ebx]
0x1003b7a0: mov ecx,[esi+8]
0x1003b7a3: test ecx,ecx
0x1003b7a5: je short 1003B7BEh
0x1003b7a7: mov [ecx+4],eax
0x1003b7aa: mov eax,[esi+8]
0x1003b7ad: mov ecx,[esi+60h]
0x1003b7b0: mov [eax+8],ecx
0x1003b7b3: mov edx,[esi+8]
0x1003b7b6: mov eax,[edx]
0x1003b7b8: push esi
0x1003b7b9: call eax
0x1003b7bb: add esp,4
0x1003b7be: mov ecx,[esi+194h]
0x1003b7c4: mov edx,[ecx+4]
0x1003b7c7: mov ebx,[edi]
0x1003b7c9: push 0
```

```
0x1003b7cb: push edi
0x1003b7cc: push 0
0x1003b7ce: push esi
0x1003b7cf: call edx
0x1003b7d1: mov eax,[edi]
0x1003b7d3: add esp,10h
0x1003b7d6: cmp eax,ebx
0x1003b7d8: je short 1003B823h
0x1003b7da: mov ecx,eax
0x1003b7dc: cmp ecx,[esi+60h]
0x1003b7df: jb short 1003B7A0h
0x1003b7e1: mov edx,[esi+190h]
0x1003b7e7: mov eax,[edx+4]
0x1003b7ea: push esi
0x1003b7eb: call eax
0x1003b7ed: mov ecx,[esi+190h]
0x1003b7f3: mov edx,[ecx]
0x1003b7f5: push esi
0x1003b7f6: call edx
0x1003b7f8: mov dword ptr [edi],0
0x1003b7fe: mov eax,[esi+190h]
0x1003b804: add esp,8
0x1003b807: cmp byte ptr [eax+8],0
0x1003b80b: jne short 1003B793h
0x1003b80d: xor ecx,ecx
0x1003b80f: cmp [esi+41h],cl
0x1003b812: pop edi
0x1003b813: setne cl
0x1003b816: mov al,1
0x1003b818: pop ebx
0x1003b819: add ecx,0CDh
0x1003b81f: mov [esi+14h],ecx
0x1003b822: ret
0x1003b823: pop edi
0x1003b824: xor al,al
0x1003b826: pop ebx
0x1003b827: ret
```

`sub_1003b760` is called by sub_1003ba90. It calls nothing else.

12.859 sub_1003b830 at 0x1003b830 (56 instructions)

```
0x1003b830: push esi
0x1003b831: mov esi,[esp+8]
0x1003b835: cmp dword ptr [esi+14h],0CDh
0x1003b83c: je short 1003B859h
0x1003b83e: mov eax,[esi]
0x1003b840: mov dword ptr [eax+14h],15h
0x1003b847: mov ecx,[esi]
0x1003b849: mov edx,[esi+14h]
0x1003b84c: mov [ecx+18h],edx
0x1003b84f: mov eax,[esi]
0x1003b851: mov ecx,[eax]
0x1003b853: push esi
0x1003b854: call ecx
0x1003b856: add esp,4
0x1003b859: mov ecx,[esi+78h]
0x1003b85c: cmp ecx,[esi+60h]
0x1003b85f: jb short 1003B87Bh
0x1003b861: mov edx,[esi]
0x1003b863: mov dword ptr [edx+14h],7Eh
0x1003b86a: mov eax,[esi]
0x1003b86c: mov ecx,[eax+4]
0x1003b86f: push 0FFFFFFFFh
0x1003b871: push esi
0x1003b872: call ecx
```

```
0x1003b874: add esp,8
0x1003b877: xor eax,eax
0x1003b879: pop esi
0x1003b87a: ret
0x1003b87b: mov eax,[esi+8]
0x1003b87e: test eax,eax
0x1003b880: je short 1003B899h
0x1003b882: mov [eax+4],ecx
0x1003b885: mov edx,[esi+8]
0x1003b888: mov eax,[esi+60h]
0x1003b88b: mov [edx+8],eax
0x1003b88e: mov ecx,[esi+8]
0x1003b891: mov edx,[ecx]
0x1003b893: push esi
0x1003b894: call edx
0x1003b896: add esp,4
0x1003b899: mov ecx,[esp+10h]
0x1003b89d: mov eax,[esi+194h]
0x1003b8a3: push ecx
0x1003b8a4: mov ecx,[esp+10h]
0x1003b8a8: lea edx,[esp+0Ch]
0x1003b8ac: push edx
0x1003b8ad: push ecx
0x1003b8ae: mov dword ptr [esp+14h],0
0x1003b8b6: mov edx,[eax+4]
0x1003b8b9: push esi
0x1003b8ba: call edx
0x1003b8bc: mov eax,[esp+18h]
0x1003b8c0: add [esi+78h],eax
0x1003b8c3: add esp,10h
0x1003b8c6: pop esi
0x1003b8c7: ret
```

`sub_1003b830` is called by sub_1001b350. It calls nothing else.

12.860 sub_1003ba90 at 0x1003ba90 (74 instructions)

```
0x1003ba90: push esi
0x1003ba91: mov esi,[esp+8]
0x1003ba95: cmp dword ptr [esi+14h],0CAh
0x1003ba9c: jne short 1003BABFh
0x1003ba9e: push esi
0x1003ba9f: call 1003C200h
0x1003baa4: add esp,4
0x1003baa7: cmp byte ptr [esi+40h],0
0x1003baab: je short 1003BAB8h
0x1003baad: mov dword ptr [esi+14h],0CFh
0x1003bab4: mov al,1
0x1003bab6: pop esi
0x1003bab7: ret
0x1003bab8: mov dword ptr [esi+14h],0CBh
0x1003babf: mov eax,[esi+14h]
0x1003bac2: cmp eax,0CBh
0x1003bac7: jne short 1003BB3Ah
0x1003bac9: mov eax,[esi+1A0h]
0x1003bacf: cmp byte ptr [eax+10h],0
0x1003bad3: je short 1003BB2Ah
0x1003bad5: mov eax,[esi+8]
0x1003bad8: test eax,eax
0x1003bada: je short 1003BAE4h
0x1003badc: mov ecx,[eax]
0x1003bade: push esi
0x1003badf: call ecx
0x1003bae1: add esp,4
0x1003bae4: mov edx,[esi+1A0h]
0x1003baea: mov eax,[edx]
```

```
0x1003baec: push esi
0x1003baed: call eax
0x1003baef: add esp,4
0x1003baf2: test eax,eax
0x1003baf4: je short 1003BB26h
0x1003baf6: cmp eax,2
0x1003baf9: je short 1003BB2Ah
0x1003bafb: mov ecx,[esi+8]
0x1003bafe: test ecx,ecx
0x1003bb00: je short 1003BAD5h
0x1003bb02: cmp eax,3
0x1003bb05: je short 1003BB0Ch
0x1003bb07: cmp eax,1
0x1003bb0a: jne short 1003BAD5h
0x1003bb0c: add dword ptr [ecx+4],1
0x1003bb10: mov eax,[esi+8]
0x1003bb13: mov ecx,[eax+4]
0x1003bb16: cmp ecx,[eax+8]
0x1003bb19: jl short 1003BAD5h
0x1003bb1b: mov edx,[esi+120h]
0x1003bb21: add [eax+8],edx
0x1003bb24: jmp short 1003BAD5h
0x1003bb26: xor al,al
0x1003bb28: pop esi
0x1003bb29: ret
0x1003bb2a: mov eax,[esi+7Ch]
0x1003bb2d: mov [esi+84h],eax
0x1003bb33: call 1003B760h
0x1003bb38: pop esi
0x1003bb39: ret
0x1003bb3a: cmp eax,0CCh
0x1003bb3f: je short 1003BB5Ch
0x1003bb41: mov ecx,[esi]
0x1003bb43: mov dword ptr [ecx+14h],15h
0x1003bb4a: mov edx,[esi]
0x1003bb4c: mov eax,[esi+14h]
0x1003bb4f: mov [edx+18h],eax
0x1003bb52: mov ecx,[esi]
0x1003bb54: mov edx,[ecx]
0x1003bb56: push esi
0x1003bb57: call edx
0x1003bb59: add esp,4
0x1003bb5c: call 1003B760h
0x1003bb61: pop esi
0x1003bb62: ret
```

`sub_1003ba90` is called by sub_1001b350. It delegates to sub_1003b760, sub_1003c200 (2 callees).

12.861 sub_1003bb70 at 0x1003bb70 (48 instructions)

```
0x1003bb70: cmp byte ptr [edx+48h],0
0x1003bb74: jne near ptr 1003BC0Ch
0x1003bb7a: cmp byte ptr [edx+10Ah],0
0x1003bb81: jne near ptr 1003BC0Ch
0x1003bb87: mov eax,3
0x1003bb8c: cmp [edx+28h],eax
0x1003bb8f: push esi
0x1003bb90: jne short 1003BC08h
0x1003bb92: cmp [edx+24h],eax
0x1003bb95: jne short 1003BC08h
0x1003bb97: mov esi,2
0x1003bb9c: cmp [edx+2Ch],esi
0x1003bb9f: jne short 1003BC08h
0x1003bba1: cmp [edx+64h],eax
0x1003bba4: jne short 1003BC08h
0x1003bba6: mov ecx,[edx+0C4h]
```

```
0x1003bbac: cmp [ecx+8],esi
0x1003bbaf: jne short 1003BC08h
0x1003bbb1: mov eax,1
0x1003bbb6: cmp [ecx+60h],eax
0x1003bbb9: jne short 1003BC08h
0x1003bbbb: cmp [ecx+0B8h],eax
0x1003bbc1: jne short 1003BC08h
0x1003bbc3: cmp [ecx+0Ch],esi
0x1003bbc6: jg short 1003BC08h
0x1003bbc8: cmp [ecx+64h],eax
0x1003bbcb: jne short 1003BC08h
0x1003bbcd: cmp [ecx+0BCh],eax
0x1003bbd3: jne short 1003BC08h
0x1003bbd5: mov esi,[edx+118h]
0x1003bbdb: cmp [ecx+24h],esi
0x1003bbde: jne short 1003BC08h
0x1003bbe0: cmp [ecx+7Ch],esi
0x1003bbe3: jne short 1003BC08h
0x1003bbe5: cmp [ecx+0D4h],esi
0x1003bbeb: jne short 1003BC08h
0x1003bbed: mov edx,[edx+11Ch]
0x1003bbf3: cmp [ecx+28h],edx
0x1003bbf6: jne short 1003BC08h
0x1003bbf8: cmp [ecx+80h],edx
0x1003bbfe: jne short 1003BC08h
0x1003bc00: cmp [ecx+0D8h],edx
0x1003bc06: je short 1003BC0Ah
0x1003bc08: xor al,al
0x1003bc0a: pop esi
0x1003bc0b: ret
0x1003bc0c: xor al,al
0x1003bc0e: ret
```

`sub_1003bb70` is called by sub_1003bc10, sub_1003beb0. It calls nothing else.

12.862 sub_1003bc10 at 0x1003bc10 (149 instructions)

```
0x1003bc10: push esi
0x1003bc11: mov esi,[esp+8]
0x1003bc15: cmp dword ptr [esi+14h],0CAh
0x1003bc1c: je short 1003BC39h
0x1003bc1e: mov eax,[esi]
0x1003bc20: mov dword ptr [eax+14h],15h
0x1003bc27: mov ecx,[esi]
0x1003bc29: mov edx,[esi+14h]
0x1003bc2c: mov [ecx+18h],edx
0x1003bc2f: mov eax,[esi]
0x1003bc31: mov ecx,[eax]
0x1003bc33: push esi
0x1003bc34: call ecx
0x1003bc36: add esp,4
0x1003bc39: push ebx
0x1003bc3a: push edi
0x1003bc3b: push esi
0x1003bc3c: call 1003D2C0h
0x1003bc41: mov ebx,[esi+0C4h]
0x1003bc47: add esp,4
0x1003bc4a: cmp dword ptr [esi+24h],0
0x1003bc4e: mov dword ptr [esp+10h],0
0x1003bc56: jle near ptr 1003BD33h
0x1003bc5c: add ebx,24h
0x1003bc5f: push ebp
0x1003bc60: mov edi,[esi+118h]
0x1003bc66: xor edx,edx
0x1003bc68: cmp [esi+48h],dl
0x1003bc6b: mov ecx,1
```

```
0x1003bc70: setne dl
0x1003bc73: lea edx,[edx*4+4]
0x1003bc7a: cmp edi,edx
0x1003bc7c: jg short 1003BCADh
0x1003bc7e: mov edi,edi
0x1003bc80: mov ebp,[ebx-1Ch]
0x1003bc83: mov eax,[esi+110h]
0x1003bc89: imul ebp,ecx
0x1003bc8c: add ebp,ebp
0x1003bc8e: cdq
0x1003bc8f: idiv ebp
0x1003bc91: test edx,edx
0x1003bc93: jne short 1003BCADh
0x1003bc95: add ecx,ecx
0x1003bc97: mov eax,edi
0x1003bc99: imul eax,ecx
0x1003bc9c: cmp [esi+48h],dl
0x1003bc9f: setne dl
0x1003bca2: lea edx,[edx*4+4]
0x1003bca9: cmp eax,edx
0x1003bcab: jle short 1003BC80h
0x1003bcad: imul edi,ecx
0x1003bcb0: xor eax,eax
0x1003bcb2: mov [ebx],edi
0x1003bcb4: cmp [esi+48h],al
0x1003bcb7: mov edi,[esi+11Ch]
0x1003bcbd: setne al
0x1003bcc0: mov ecx,1
0x1003bcc5: lea eax,[eax*4+4]
0x1003bccc: cmp edi,eax
0x1003bcce: jg short 1003BCFDh
0x1003bcd0: mov ebp,[ebx-18h]
0x1003bcd3: mov eax,[esi+114h]
0x1003bcd9: imul ebp,ecx
0x1003bcdc: add ebp,ebp
0x1003bcde: cdq
0x1003bcdf: idiv ebp
0x1003bce1: test edx,edx
0x1003bce3: jne short 1003BCFDh
0x1003bce5: add ecx,ecx
0x1003bce7: cmp [esi+48h],dl
0x1003bcea: mov eax,edi
0x1003bcec: setne dl
0x1003bcef: imul eax,ecx
0x1003bcf2: lea edx,[edx*4+4]
0x1003bcf9: cmp eax,edx
0x1003bcfb: jle short 1003BCD0h
0x1003bcfd: mov eax,edi
0x1003bcff: imul eax,ecx
0x1003bd02: mov ecx,[ebx]
0x1003bd04: lea edx,[eax+eax]
0x1003bd07: cmp ecx,edx
0x1003bd09: mov [ebx+4],eax
0x1003bd0c: jle short 1003BD12h
0x1003bd0e: mov [ebx],edx
0x1003bd10: jmp short 1003BD1Bh
0x1003bd12: add ecx,ecx
0x1003bd14: cmp eax,ecx
0x1003bd16: jle short 1003BD1Bh
0x1003bd18: mov [ebx+4],ecx
0x1003bd1b: mov eax,[esp+14h]
0x1003bd1f: add eax,1
0x1003bd22: add ebx,58h
0x1003bd25: cmp eax,[esi+24h]
0x1003bd28: mov [esp+14h],eax
0x1003bd2c: jl near ptr 1003BC60h
0x1003bd32: pop ebp
```

```
0x1003bd33: mov edi,[esi+0C4h]
0x1003bd39: xor ebx,ebx
0x1003bd3b: cmp [esi+24h],ebx
0x1003bd3e: jle short 1003BD94h
0x1003bd40: add edi,8
0x1003bd43: mov edx,[edi+1Ch]
0x1003bd46: imul edx,[edi]
0x1003bd49: mov ecx,[esi+110h]
0x1003bd4f: imul edx,[esi+1Ch]
0x1003bd53: imul ecx,[esi+180h]
0x1003bd5a: push ecx
0x1003bd5b: push edx
0x1003bd5c: call 1003D200h
0x1003bd61: mov ecx,[edi+20h]
0x1003bd64: imul ecx,[edi+4]
0x1003bd68: mov [edi+24h],eax
0x1003bd6b: imul ecx,[esi+20h]
0x1003bd6f: mov eax,[esi+114h]
0x1003bd75: imul eax,[esi+180h]
0x1003bd7c: push eax
0x1003bd7d: push ecx
0x1003bd7e: call 1003D200h
0x1003bd83: mov [edi+28h],eax
0x1003bd86: add ebx,1
0x1003bd89: add esp,10h
0x1003bd8c: add edi,58h
0x1003bd8f: cmp ebx,[esi+24h]
0x1003bd92: jl short 1003BD43h
0x1003bd94: mov eax,[esi+2Ch]
0x1003bd97: add eax,0FFFFFFFFh
0x1003bd9a: cmp eax,4
0x1003bd9d: pop edi
0x1003bd9e: pop ebx
0x1003bd9f: ja short 1003BDC3h
0x1003bda1: jmp dword ptr [eax*4+1003BDFCh]
0x1003bdc3: mov edx,[esi+24h]
0x1003bdc6: mov [esi+64h],edx
0x1003bdc9: cmp byte ptr [esi+4Ah],0
0x1003bdcd: mov eax,1
0x1003bdd2: jne short 1003BDD7h
0x1003bdd4: mov eax,[esi+64h]
0x1003bdd7: mov edx,esi
0x1003bdd9: mov [esi+68h],eax
0x1003bddc: call 1003BB70h
0x1003bde1: test al,al
0x1003bde3: je short 1003BDF0h
0x1003bde5: mov eax,[esi+114h]
0x1003bdeb: mov [esi+6Ch],eax
0x1003bdee: pop esi
0x1003bdef: ret
0x1003bdf0: mov dword ptr [esi+6Ch],1
0x1003bdf7: pop esi
0x1003bdf8: ret
```

`sub_1003bc10` is called by sub_1001b350, sub_1003beb0. It delegates to sub_1003bb70, sub_1003d200, sub_1003d2c0 (3 callees).

12.863 sub_1003be10 at 0x1003be10 (42 instructions)

```
0x1003be10: mov eax,[ebx+4]
0x1003be13: mov ecx,[eax]
0x1003be15: push esi
0x1003be16: push edi
0x1003be17: push 580h
0x1003be1c: push 1
0x1003be1e: push ebx
```

```
0x1003be1f: call ecx
0x1003be21: mov esi,eax
0x1003be23: add esi,100h
0x1003be29: push 100h
0x1003be2e: lea edx,[esi-100h]
0x1003be34: push 0
0x1003be36: push edx
0x1003be37: mov [ebx+124h],esi
0x1003be3d: call 100228C0h
0x1003be42: add esp,18h
0x1003be45: xor eax,eax
0x1003be47: jmp short 1003BE50h
0x1003be50: mov [eax+esi],al
0x1003be53: add eax,1
0x1003be56: cmp eax,0FFh
0x1003be5b: jle short 1003BE50h
0x1003be5d: push 180h
0x1003be62: add esi,80h
0x1003be68: lea eax,[esi+80h]
0x1003be6e: push 0FFh
0x1003be73: push eax
0x1003be74: call 100228C0h
0x1003be79: push 180h
0x1003be7e: lea eax,[esi+200h]
0x1003be84: push 0
0x1003be86: push eax
0x1003be87: call 100228C0h
0x1003be8c: lea edi,[esi+380h]
0x1003be92: mov esi,[ebx+124h]
0x1003be98: add esp,18h
0x1003be9b: mov ecx,20h
0x1003bea0: rep movsd
0x1003bea2: pop edi
0x1003bea3: pop esi
0x1003bea4: ret
```

`sub_1003be10` is called by sub_1003beb0. It delegates to j_memset (1 callees).

12.864 sub_1003beb0 at 0x1003beb0 (149 instructions)

```
0x1003beb0: push ecx
0x1003beb1: push ebx
0x1003beb2: push edi
0x1003beb3: mov edi,[esi+190h]
0x1003beb9: push esi
0x1003beba: call 1003BC10h
0x1003bebf: add esp,4
0x1003bec2: mov ebx,esi
0x1003bec4: call 1003BE10h
0x1003bec9: xor ebx,ebx
0x1003becb: mov edx,esi
0x1003becd: mov [edi+0Ch],ebx
0x1003bed0: call 1003BB70h
0x1003bed5: mov [edi+10h],al
0x1003bed8: mov [edi+14h],ebx
0x1003bedb: mov [edi+18h],ebx
0x1003bede: mov al,[esi+4Ah]
0x1003bee1: cmp al,bl
0x1003bee3: je short 1003BEEAh
0x1003bee5: cmp [esi+40h],bl
0x1003bee8: jne short 1003BEF3h
0x1003beea: mov [esi+58h],bl
0x1003beed: mov [esi+59h],bl
0x1003bef0: mov [esi+5Ah],bl
0x1003bef3: cmp al,bl
0x1003bef5: je short 1003BF55h
```

```
0x1003bef7: cmp [esi+41h],bl
0x1003befa: je short 1003BF0Fh
0x1003befc: mov eax,[esi]
0x1003befe: mov dword ptr [eax+14h],30h
0x1003bf05: mov ecx,[esi]
0x1003bf07: mov edx,[ecx]
0x1003bf09: push esi
0x1003bf0a: call edx
0x1003bf0c: add esp,4
0x1003bf0f: cmp dword ptr [esi+64h],3
0x1003bf13: je short 1003BF6Ah
0x1003bf15: mov [esi+59h],bl
0x1003bf18: mov [esi+5Ah],bl
0x1003bf1b: mov [esi+74h],ebx
0x1003bf1e: mov byte ptr [esi+58h],1
0x1003bf22: cmp [esi+58h],bl
0x1003bf25: je short 1003BF39h
0x1003bf27: push esi
0x1003bf28: call 10046A70h
0x1003bf2d: mov eax,[esi+1B8h]
0x1003bf33: add esp,4
0x1003bf36: mov [edi+14h],eax
0x1003bf39: cmp [esi+5Ah],bl
0x1003bf3c: jne short 1003BF43h
0x1003bf3e: cmp [esi+59h],bl
0x1003bf41: je short 1003BF55h
0x1003bf43: push esi
0x1003bf44: call 10045DA0h
0x1003bf49: mov ecx,[esi+1B8h]
0x1003bf4f: add esp,4
0x1003bf52: mov [edi+18h],ecx
0x1003bf55: cmp [esi+41h],bl
0x1003bf58: jne short 1003BF9Ch
0x1003bf5a: cmp [edi+10h],bl
0x1003bf5d: push esi
0x1003bf5e: je short 1003BF80h
0x1003bf60: call 10044AB0h
0x1003bf65: add esp,4
0x1003bf68: jmp short 1003BF8Eh
0x1003bf6a: cmp [esi+74h],ebx
0x1003bf6d: je short 1003BF75h
0x1003bf6f: mov byte ptr [esi+59h],1
0x1003bf73: jmp short 1003BF22h
0x1003bf75: cmp [esi+50h],bl
0x1003bf78: je short 1003BF1Eh
0x1003bf7a: mov byte ptr [esi+5Ah],1
0x1003bf7e: jmp short 1003BF22h
0x1003bf80: call 100443B0h
0x1003bf85: push esi
0x1003bf86: call 10043DA0h
0x1003bf8b: add esp,8
0x1003bf8e: movzx edx,byte ptr [esi+5Ah]
0x1003bf92: push edx
0x1003bf93: push esi
0x1003bf94: call 10043A40h
0x1003bf99: add esp,8
0x1003bf9c: push esi
0x1003bf9d: call 100436F0h
0x1003bfa2: add esp,4
0x1003bfa5: cmp [esi+0CAh],bl
0x1003bfab: push esi
0x1003bfac: je short 1003BFB5h
0x1003bfae: call 100430C0h
0x1003bfb3: jmp short 1003BFBAh
0x1003bfb5: call 10041FD0h
0x1003bfba: mov eax,[esi+1A0h]
0x1003bfc0: add esp,4
```

```
0x1003bfc3: cmp [eax+10h],bl
0x1003bfc6: jne short 1003BFD1h
0x1003bfc8: cmp [esi+40h],bl
0x1003bfcb: mov [esp+8],bl
0x1003bfcf: je short 1003BFD6h
0x1003bfd1: mov byte ptr [esp+8],1
0x1003bfd6: mov ecx,[esp+8]
0x1003bfda: push ecx
0x1003bfdb: push esi
0x1003bfdc: call 10040160h
0x1003bfe1: add esp,8
0x1003bfe4: cmp [esi+41h],bl
0x1003bfe7: jne short 1003BFF3h
0x1003bfe9: push ebx
0x1003bfea: push esi
0x1003bfeb: call 1003F250h
0x1003bff0: add esp,8
0x1003bff3: mov edx,[esi+4]
0x1003bff6: mov eax,[edx+18h]
0x1003bff9: push esi
0x1003bffa: call eax
0x1003bffc: mov ecx,[esi+1A0h]
0x1003c002: mov edx,[ecx+8]
0x1003c005: push esi
0x1003c006: call edx
0x1003c008: mov ecx,[esi+8]
0x1003c00b: add esp,8
0x1003c00e: cmp ecx,ebx
0x1003c010: je short 1003C05Eh
0x1003c012: cmp [esi+40h],bl
0x1003c015: jne short 1003C05Eh
0x1003c017: mov eax,[esi+1A0h]
0x1003c01d: cmp [eax+10h],bl
0x1003c020: je short 1003C05Eh
0x1003c022: cmp [esi+0C9h],bl
0x1003c028: mov eax,[esi+24h]
0x1003c02b: je short 1003C031h
0x1003c02d: lea eax,[eax+eax*2+2]
0x1003c031: mov [ecx+4],ebx
0x1003c034: mov ecx,[esi+120h]
0x1003c03a: mov edx,[esi+8]
0x1003c03d: imul ecx,eax
0x1003c040: mov [edx+8],ecx
0x1003c043: mov eax,[esi+8]
0x1003c046: mov [eax+0Ch],ebx
0x1003c049: mov edx,[esi+8]
0x1003c04c: xor ecx,ecx
0x1003c04e: cmp [esi+5Ah],bl
0x1003c051: setne cl
0x1003c054: add ecx,2
0x1003c057: mov [edx+10h],ecx
0x1003c05a: add dword ptr [edi+0Ch],1
0x1003c05e: pop edi
0x1003c05f: pop ebx
0x1003c060: pop ecx
0x1003c061: ret
```

`sub_1003beb0` is called by sub_1003c200. It delegates to sub_1003bb70, sub_1003bc10, sub_1003be10 (14 callees).

12.865 sub_1003c200 at 0x1003c200 (16 instructions)

```
0x1003c200: push esi
0x1003c201: mov esi,[esp+8]
0x1003c205: mov eax,[esi+4]
0x1003c208: mov ecx,[eax]
0x1003c20a: push 1Ch
```

```
0x1003c20c: push 1
0x1003c20e: push esi
0x1003c20f: call ecx
0x1003c211: mov [esi+190h],eax
0x1003c217: add esp,0Ch
0x1003c21a: mov dword ptr [eax],1003C070h
0x1003c220: mov dword ptr [eax+4],1003C1D0h
0x1003c227: mov byte ptr [eax+8],0
0x1003c22b: call 1003BEB0h
0x1003c230: pop esi
0x1003c231: ret
```

`sub_1003c200` is called by sub_1003ba90. It delegates to sub_1003beb0 (1 callees).

12.866 sub_1003c3d0 at 0x1003c3d0 (16 instructions)

```
0x1003c3d0: mov eax,[esp+4]
0x1003c3d4: xor ecx,ecx
0x1003c3d6: mov dword ptr [eax],1003C240h
0x1003c3dc: mov dword ptr [eax+4],1003C2C0h
0x1003c3e3: mov dword ptr [eax+8],1003C260h
0x1003c3ea: mov dword ptr [eax+0Ch],1003C300h
0x1003c3f1: mov dword ptr [eax+10h],1003C3B0h
0x1003c3f8: mov [eax+68h],ecx
0x1003c3fb: mov [eax+6Ch],ecx
0x1003c3fe: mov [eax+14h],ecx
0x1003c401: mov dword ptr [eax+70h],10103618h
0x1003c408: mov dword ptr [eax+74h],7Eh
0x1003c40f: mov [eax+78h],ecx
0x1003c412: mov [eax+7Ch],ecx
0x1003c415: mov [eax+80h],ecx
0x1003c41b: ret
```

`sub_1003c3d0` is called by sub_1001b350. It calls nothing else.

12.867 sub_1003c530 at 0x1003c530 (17 instructions)

```
0x1003c530: jmp near ptr 1003D190h
0x1003d190: push esi
0x1003d191: mov esi,[esp+8]
0x1003d195: mov eax,[esi+4]
0x1003d198: push edi
0x1003d199: xor edi,edi
0x1003d19b: cmp eax,edi
0x1003d19d: je short 1003D1A8h
0x1003d19f: mov eax,[eax+28h]
0x1003d1a2: push esi
0x1003d1a3: call eax
0x1003d1a5: add esp,4
0x1003d1a8: mov [esi+14h],edi
0x1003d1ab: mov [esi+4],edi
0x1003d1ae: pop edi
0x1003d1af: pop esi
0x1003d1b0: ret
```

`sub_1003c530` is called by sub_1001b350, sub_100a0020. It calls nothing else.

12.868 sub_1003d150 at 0x1003d150 (19 instructions)

```
0x1003d150: push esi
0x1003d151: mov esi,[esp+8]
0x1003d155: mov eax,[esi+4]
0x1003d158: test eax,eax
0x1003d15a: je short 1003D187h
```

```
0x1003d15c: mov eax,[eax+24h]
0x1003d15f: push 1
0x1003d161: push esi
0x1003d162: call eax
0x1003d164: add esp,8
0x1003d167: cmp byte ptr [esi+10h],0
0x1003d16b: je short 1003D180h
0x1003d16d: mov dword ptr [esi+14h],0C8h
0x1003d174: mov dword ptr [esi+10Ch],0
0x1003d17e: pop esi
0x1003d17f: ret
0x1003d180: mov dword ptr [esi+14h],64h
0x1003d187: pop esi
0x1003d188: ret
```

`sub_1003d150` is called by sub_1003b610, sub_1003b6e0. It calls nothing else.

12.869 sub_1003d200 at 0x1003d200 (6 instructions)

```
0x1003d200: mov ecx,[esp+8]
0x1003d204: mov eax,[esp+4]
0x1003d208: lea eax,[eax+ecx-1]
0x1003d20c: cdq
0x1003d20d: idiv ecx
0x1003d20f: ret
```

`sub_1003d200` is called by sub_1003bc10, sub_1003d2c0. It calls nothing else.

12.870 sub_1003d210 at 0x1003d210 (11 instructions)

```
0x1003d210: mov eax,[esp+4]
0x1003d214: push esi
0x1003d215: mov esi,[esp+0Ch]
0x1003d219: lea ecx,[eax+esi-1]
0x1003d21d: mov eax,ecx
0x1003d21f: cdq
0x1003d220: idiv esi
0x1003d222: mov eax,ecx
0x1003d224: pop esi
0x1003d225: sub eax,edx
0x1003d227: ret
```

`sub_1003d210` is called by sub_10040160, sub_10043a40, sub_10043da0. It calls nothing else.

12.871 sub_1003d2c0 at 0x1003d2c0 (331 instructions)

```
0x1003d2c0: push esi
0x1003d2c1: mov esi,[esp+8]
0x1003d2c5: mov eax,[esi+180h]
0x1003d2cb: mov edx,[esi+30h]
0x1003d2ce: mov ecx,[esi+34h]
0x1003d2d1: imul edx,eax
0x1003d2d4: cmp edx,ecx
0x1003d2d6: ja short 1003D314h
0x1003d2d8: push eax
0x1003d2d9: mov eax,[esi+1Ch]
0x1003d2dc: push eax
0x1003d2dd: call 1003D200h
0x1003d2e2: mov ecx,[esi+180h]
0x1003d2e8: mov edx,[esi+20h]
0x1003d2eb: push ecx
0x1003d2ec: push edx
0x1003d2ed: mov [esi+5Ch],eax
0x1003d2f0: call 1003D200h
```

```
0x1003d2f5: add esp,10h
0x1003d2f8: mov [esi+60h],eax
0x1003d2fb: mov dword ptr [esi+118h],1
0x1003d305: mov dword ptr [esi+11Ch],1
0x1003d30f: jmp near ptr 1003D69Eh
0x1003d314: push ebx
0x1003d315: lea ebx,[ecx+ecx]
0x1003d318: cmp edx,ebx
0x1003d31a: push eax
0x1003d31b: ja short 1003D34Ah
0x1003d31d: mov eax,[esi+1Ch]
0x1003d320: add eax,eax
0x1003d322: push eax
0x1003d323: call 1003D200h
0x1003d328: mov ecx,[esi+180h]
0x1003d32e: mov edx,[esi+20h]
0x1003d331: push ecx
0x1003d332: add edx,edx
0x1003d334: push edx
0x1003d335: mov [esi+5Ch],eax
0x1003d338: call 1003D200h
0x1003d33d: mov [esi+60h],eax
0x1003d340: mov eax,2
0x1003d345: jmp near ptr 1003D68Eh
0x1003d34a: lea ebx,[ecx+ecx*2]
0x1003d34d: cmp edx,ebx
0x1003d34f: ja short 1003D380h
0x1003d351: mov eax,[esi+1Ch]
0x1003d354: lea eax,[eax+eax*2]
0x1003d357: push eax
0x1003d358: call 1003D200h
0x1003d35d: mov ecx,[esi+180h]
0x1003d363: mov [esi+5Ch],eax
0x1003d366: mov eax,[esi+20h]
0x1003d369: push ecx
0x1003d36a: lea edx,[eax+eax*2]
0x1003d36d: push edx
0x1003d36e: call 1003D200h
0x1003d373: mov [esi+60h],eax
0x1003d376: mov eax,3
0x1003d37b: jmp near ptr 1003D68Eh
0x1003d380: lea ebx,[ecx*4]
0x1003d387: cmp edx,ebx
0x1003d389: ja short 1003D3BCh
0x1003d38b: mov eax,[esi+1Ch]
0x1003d38e: add eax,eax
0x1003d390: add eax,eax
0x1003d392: push eax
0x1003d393: call 1003D200h
0x1003d398: mov edx,[esi+20h]
0x1003d39b: mov ecx,[esi+180h]
0x1003d3a1: add edx,edx
0x1003d3a3: push ecx
0x1003d3a4: add edx,edx
0x1003d3a6: push edx
0x1003d3a7: mov [esi+5Ch],eax
0x1003d3aa: call 1003D200h
0x1003d3af: mov [esi+60h],eax
0x1003d3b2: mov eax,4
0x1003d3b7: jmp near ptr 1003D68Eh
0x1003d3bc: lea ebx,[ecx+ecx*4]
0x1003d3bf: cmp edx,ebx
0x1003d3c1: ja short 1003D3F2h
0x1003d3c3: mov eax,[esi+1Ch]
0x1003d3c6: lea eax,[eax+eax*4]
0x1003d3c9: push eax
0x1003d3ca: call 1003D200h
```

```
0x1003d3cf: mov ecx,[esi+180h]
0x1003d3d5: mov [esi+5Ch],eax
0x1003d3d8: mov eax,[esi+20h]
0x1003d3db: push ecx
0x1003d3dc: lea edx,[eax+eax*4]
0x1003d3df: push edx
0x1003d3e0: call 1003D200h
0x1003d3e5: mov [esi+60h],eax
0x1003d3e8: mov eax,5
0x1003d3ed: jmp near ptr 1003D68Eh
0x1003d3f2: lea ebx,[ecx+ecx*2]
0x1003d3f5: add ebx,ebx
0x1003d3f7: cmp edx,ebx
0x1003d3f9: ja short 1003D42Eh
0x1003d3fb: mov eax,[esi+1Ch]
0x1003d3fe: lea eax,[eax+eax*2]
0x1003d401: add eax,eax
0x1003d403: push eax
0x1003d404: call 1003D200h
0x1003d409: mov ecx,[esi+180h]
0x1003d40f: mov [esi+5Ch],eax
0x1003d412: mov eax,[esi+20h]
0x1003d415: lea edx,[eax+eax*2]
0x1003d418: push ecx
0x1003d419: add edx,edx
0x1003d41b: push edx
0x1003d41c: call 1003D200h
0x1003d421: mov [esi+60h],eax
0x1003d424: mov eax,6
0x1003d429: jmp near ptr 1003D68Eh
0x1003d42e: lea ebx,[ecx*8]
0x1003d435: sub ebx,ecx
0x1003d437: cmp edx,ebx
0x1003d439: ja short 1003D476h
0x1003d43b: mov eax,[esi+1Ch]
0x1003d43e: lea ecx,[eax*8]
0x1003d445: sub ecx,eax
0x1003d447: push ecx
0x1003d448: call 1003D200h
0x1003d44d: mov edx,[esi+180h]
0x1003d453: mov [esi+5Ch],eax
0x1003d456: mov eax,[esi+20h]
0x1003d459: lea ecx,[eax*8]
0x1003d460: push edx
0x1003d461: sub ecx,eax
0x1003d463: push ecx
0x1003d464: call 1003D200h
0x1003d469: mov [esi+60h],eax
0x1003d46c: mov eax,7
0x1003d471: jmp near ptr 1003D68Eh
0x1003d476: lea ebx,[ecx*8]
0x1003d47d: cmp edx,ebx
0x1003d47f: ja short 1003D4B6h
0x1003d481: mov edx,[esi+1Ch]
0x1003d484: add edx,edx
0x1003d486: add edx,edx
0x1003d488: add edx,edx
0x1003d48a: push edx
0x1003d48b: call 1003D200h
0x1003d490: mov ecx,[esi+20h]
0x1003d493: add ecx,ecx
0x1003d495: mov [esi+5Ch],eax
0x1003d498: mov eax,[esi+180h]
0x1003d49e: add ecx,ecx
0x1003d4a0: push eax
0x1003d4a1: add ecx,ecx
0x1003d4a3: push ecx
```

```
0x1003d4a4: call 1003D200h
0x1003d4a9: mov [esi+60h],eax
0x1003d4ac: mov eax,8
0x1003d4b1: jmp near ptr 1003D68Eh
0x1003d4b6: lea ebx,[ecx+ecx*8]
0x1003d4b9: cmp edx,ebx
0x1003d4bb: ja short 1003D4ECh
0x1003d4bd: mov eax,[esi+1Ch]
0x1003d4c0: lea edx,[eax+eax*8]
0x1003d4c3: push edx
0x1003d4c4: call 1003D200h
0x1003d4c9: mov [esi+5Ch],eax
0x1003d4cc: mov eax,[esi+180h]
0x1003d4d2: push eax
0x1003d4d3: mov eax,[esi+20h]
0x1003d4d6: lea ecx,[eax+eax*8]
0x1003d4d9: push ecx
0x1003d4da: call 1003D200h
0x1003d4df: mov [esi+60h],eax
0x1003d4e2: mov eax,9
0x1003d4e7: jmp near ptr 1003D68Eh
0x1003d4ec: lea ebx,[ecx+ecx*4]
0x1003d4ef: add ebx,ebx
0x1003d4f1: cmp edx,ebx
0x1003d4f3: ja short 1003D528h
0x1003d4f5: mov eax,[esi+1Ch]
0x1003d4f8: lea edx,[eax+eax*4]
0x1003d4fb: add edx,edx
0x1003d4fd: push edx
0x1003d4fe: call 1003D200h
0x1003d503: mov [esi+5Ch],eax
0x1003d506: mov eax,[esi+180h]
0x1003d50c: push eax
0x1003d50d: mov eax,[esi+20h]
0x1003d510: lea ecx,[eax+eax*4]
0x1003d513: add ecx,ecx
0x1003d515: push ecx
0x1003d516: call 1003D200h
0x1003d51b: mov [esi+60h],eax
0x1003d51e: mov eax,0Ah
0x1003d523: jmp near ptr 1003D68Eh
0x1003d528: mov ebx,ecx
0x1003d52a: imul ebx,0Bh
0x1003d52d: cmp edx,ebx
0x1003d52f: ja short 1003D560h
0x1003d531: mov edx,[esi+1Ch]
0x1003d534: imul edx,0Bh
0x1003d537: push edx
0x1003d538: call 1003D200h
0x1003d53d: mov ecx,[esi+20h]
0x1003d540: imul ecx,0Bh
0x1003d543: mov [esi+5Ch],eax
0x1003d546: mov eax,[esi+180h]
0x1003d54c: push eax
0x1003d54d: push ecx
0x1003d54e: call 1003D200h
0x1003d553: mov [esi+60h],eax
0x1003d556: mov eax,0Bh
0x1003d55b: jmp near ptr 1003D68Eh
0x1003d560: lea ebx,[ecx+ecx*2]
0x1003d563: add ebx,ebx
0x1003d565: add ebx,ebx
0x1003d567: cmp edx,ebx
0x1003d569: ja short 1003D5A2h
0x1003d56b: mov eax,[esi+1Ch]
0x1003d56e: lea edx,[eax+eax*2]
0x1003d571: add edx,edx
```

```
0x1003d573: add edx,edx
0x1003d575: push edx
0x1003d576: call 1003D200h
0x1003d57b: mov [esi+5Ch],eax
0x1003d57e: mov eax,[esi+180h]
0x1003d584: push eax
0x1003d585: mov eax,[esi+20h]
0x1003d588: lea ecx,[eax+eax*2]
0x1003d58b: add ecx,ecx
0x1003d58d: add ecx,ecx
0x1003d58f: push ecx
0x1003d590: call 1003D200h
0x1003d595: mov [esi+60h],eax
0x1003d598: mov eax,0Ch
0x1003d59d: jmp near ptr 1003D68Eh
0x1003d5a2: mov ebx,ecx
0x1003d5a4: imul ebx,0Dh
0x1003d5a7: cmp edx,ebx
0x1003d5a9: ja short 1003D5DAh
0x1003d5ab: mov edx,[esi+1Ch]
0x1003d5ae: imul edx,0Dh
0x1003d5b1: push edx
0x1003d5b2: call 1003D200h
0x1003d5b7: mov ecx,[esi+20h]
0x1003d5ba: imul ecx,0Dh
0x1003d5bd: mov [esi+5Ch],eax
0x1003d5c0: mov eax,[esi+180h]
0x1003d5c6: push eax
0x1003d5c7: push ecx
0x1003d5c8: call 1003D200h
0x1003d5cd: mov [esi+60h],eax
0x1003d5d0: mov eax,0Dh
0x1003d5d5: jmp near ptr 1003D68Eh
0x1003d5da: lea ebx,[ecx*8]
0x1003d5e1: sub ebx,ecx
0x1003d5e3: add ebx,ebx
0x1003d5e5: cmp edx,ebx
0x1003d5e7: ja short 1003D625h
0x1003d5e9: mov eax,[esi+1Ch]
0x1003d5ec: lea edx,[eax*8]
0x1003d5f3: sub edx,eax
0x1003d5f5: add edx,edx
0x1003d5f7: push edx
0x1003d5f8: call 1003D200h
0x1003d5fd: mov [esi+5Ch],eax
0x1003d600: mov eax,[esi+180h]
0x1003d606: push eax
0x1003d607: mov eax,[esi+20h]
0x1003d60a: lea ecx,[eax*8]
0x1003d611: sub ecx,eax
0x1003d613: add ecx,ecx
0x1003d615: push ecx
0x1003d616: call 1003D200h
0x1003d61b: mov [esi+60h],eax
0x1003d61e: mov eax,0Eh
0x1003d623: jmp short 1003D68Eh
0x1003d625: mov ebx,ecx
0x1003d627: shl ebx,4
0x1003d62a: sub ebx,ecx
0x1003d62c: cmp edx,ebx
0x1003d62e: ja short 1003D664h
0x1003d630: mov eax,[esi+1Ch]
0x1003d633: mov edx,eax
0x1003d635: shl edx,4
0x1003d638: sub edx,eax
0x1003d63a: push edx
0x1003d63b: call 1003D200h
```

```
0x1003d640: mov [esi+5Ch],eax
0x1003d643: mov eax,[esi+180h]
0x1003d649: push eax
0x1003d64a: mov eax,[esi+20h]
0x1003d64d: mov ecx,eax
0x1003d64f: shl ecx,4
0x1003d652: sub ecx,eax
0x1003d654: push ecx
0x1003d655: call 1003D200h
0x1003d65a: mov [esi+60h],eax
0x1003d65d: mov eax,0Fh
0x1003d662: jmp short 1003D68Eh
0x1003d664: mov edx,[esi+1Ch]
0x1003d667: shl edx,4
0x1003d66a: push edx
0x1003d66b: call 1003D200h
0x1003d670: mov ecx,[esi+20h]
0x1003d673: mov [esi+5Ch],eax
0x1003d676: mov eax,[esi+180h]
0x1003d67c: push eax
0x1003d67d: shl ecx,4
0x1003d680: push ecx
0x1003d681: call 1003D200h
0x1003d686: mov [esi+60h],eax
0x1003d689: mov eax,10h
0x1003d68e: add esp,10h
0x1003d691: mov [esi+118h],eax
0x1003d697: mov [esi+11Ch],eax
0x1003d69d: pop ebx
0x1003d69e: mov eax,[esi+0C4h]
0x1003d6a4: xor ecx,ecx
0x1003d6a6: cmp [esi+24h],ecx
0x1003d6a9: jle short 1003D6CCh
0x1003d6ab: add eax,28h
0x1003d6ae: mov edi,edi
0x1003d6b0: mov edx,[esi+118h]
0x1003d6b6: mov [eax-4],edx
0x1003d6b9: mov edx,[esi+11Ch]
0x1003d6bf: mov [eax],edx
0x1003d6c1: add ecx,1
0x1003d6c4: add eax,58h
0x1003d6c7: cmp ecx,[esi+24h]
0x1003d6ca: jl short 1003D6B0h
0x1003d6cc: pop esi
0x1003d6cd: ret
```

`sub_1003d2c0` is called by sub_1003bc10. It delegates to sub_1003d200 (1 callees).

12.872 sub_1003dff0 at 0x1003dff0 (19 instructions)

```
0x1003dff0: push esi
0x1003dff1: mov esi,[esp+8]
0x1003dff5: mov eax,[esi+4]
0x1003dff8: mov ecx,[eax]
0x1003dffa: push 18h
0x1003dffc: push 0
0x1003dffe: push esi
0x1003dfff: call ecx
0x1003e001: mov [esi+1A0h],eax
0x1003e007: add esp,0Ch
0x1003e00a: mov dword ptr [eax],1003DE90h
0x1003e010: mov dword ptr [eax+4],1003DF80h
0x1003e017: mov dword ptr [eax+8],1003DE50h
0x1003e01e: mov dword ptr [eax+0Ch],1003DFD0h
0x1003e025: mov byte ptr [eax+10h],0
0x1003e029: mov byte ptr [eax+11h],0
```

```
0x1003e02d: mov dword ptr [eax+14h],1
0x1003e034: pop esi
0x1003e035: ret
```

`sub_1003dff0` is called by sub_1003b220. It calls nothing else.

12.873 sub_1003eb30 at 0x1003eb30 (88 instructions)

```
0x1003eb30: push ecx
0x1003eb31: push ebx
0x1003eb32: push esi
0x1003eb33: push edi
0x1003eb34: mov edi,[esp+14h]
0x1003eb38: xor ebx,ebx
0x1003eb3a: push edi
0x1003eb3b: mov [edi+4],ebx
0x1003eb3e: call 100B8550h
0x1003eb43: push 54h
0x1003eb45: push edi
0x1003eb46: mov [esp+18h],eax
0x1003eb4a: call 10047610h
0x1003eb4f: mov esi,eax
0x1003eb51: add esp,0Ch
0x1003eb54: cmp esi,ebx
0x1003eb56: jne short 1003EB76h
0x1003eb58: push edi
0x1003eb59: call 100B3980h
0x1003eb5e: mov eax,[edi]
0x1003eb60: mov dword ptr [eax+14h],38h
0x1003eb67: mov ecx,[edi]
0x1003eb69: mov [ecx+18h],ebx
0x1003eb6c: mov edx,[edi]
0x1003eb6e: mov eax,[edx]
0x1003eb70: push edi
0x1003eb71: call eax
0x1003eb73: add esp,8
0x1003eb76: mov dword ptr [esi],1003E060h
0x1003eb7c: mov dword ptr [esi+4],1003E190h
0x1003eb83: mov dword ptr [esi+8],1003E230h
0x1003eb8a: mov dword ptr [esi+0Ch],1003E2E0h
0x1003eb91: mov dword ptr [esi+10h],1003E390h
0x1003eb98: mov dword ptr [esi+14h],1003E400h
0x1003eb9f: mov dword ptr [esi+18h],1003E470h
0x1003eba6: mov dword ptr [esi+1Ch],1003E750h
0x1003ebad: mov dword ptr [esi+20h],1003E890h
0x1003ebb4: mov dword ptr [esi+24h],1003E9E0h
0x1003ebbb: mov dword ptr [esi+28h],1003EAF0h
0x1003ebc2: mov dword ptr [esi+30h],3B9ACA00h
0x1003ebc9: mov ecx,[esp+0Ch]
0x1003ebcd: mov [esi+2Ch],ecx
0x1003ebd0: mov [esi+38h],ebx
0x1003ebd3: mov [esi+40h],ebx
0x1003ebd6: mov [esi+34h],ebx
0x1003ebd9: mov [esi+3Ch],ebx
0x1003ebdc: mov [esi+44h],ebx
0x1003ebdf: mov [esi+48h],ebx
0x1003ebe2: mov dword ptr [esi+4Ch],54h
0x1003ebe9: push 101040E8h
0x1003ebee: mov [edi+4],esi
0x1003ebf1: call dword ptr ds:[100FE4A4h]
0x1003ebf7: add esp,4
0x1003ebfa: cmp eax,ebx
0x1003ebfc: je short 1003EC51h
0x1003ebfe: lea edx,[esp+14h]
0x1003ec02: push edx
0x1003ec03: lea ecx,[esp+10h]
```

```
0x1003ec07: push ecx
0x1003ec08: push 101040E0h
0x1003ec0d: push eax
0x1003ec0e: mov byte ptr [esp+24h],78h
0x1003ec13: call dword ptr ds:[100FE4A8h]
0x1003ec19: add esp,10h
0x1003ec1c: test eax,eax
0x1003ec1e: jle short 1003EC51h
0x1003ec20: mov al,[esp+14h]
0x1003ec24: cmp al,6Dh
0x1003ec26: je short 1003EC2Ch
0x1003ec28: cmp al,4Dh
0x1003ec2a: jne short 1003EC44h
0x1003ec2c: mov eax,[esp+0Ch]
0x1003ec30: imul eax,3E8h
0x1003ec36: imul eax,3E8h
0x1003ec3c: pop edi
0x1003ec3d: mov [esi+2Ch],eax
0x1003ec40: pop esi
0x1003ec41: pop ebx
0x1003ec42: pop ecx
0x1003ec43: ret
0x1003ec44: mov eax,[esp+0Ch]
0x1003ec48: imul eax,3E8h
0x1003ec4e: mov [esi+2Ch],eax
0x1003ec51: pop edi
0x1003ec52: pop esi
0x1003ec53: pop ebx
0x1003ec54: pop ecx
0x1003ec55: ret
```

`sub_1003eb30` is called by sub_1003b220. It works with the text "JPEGMEM". It delegates to sub_10047610, sub_100b3980, sub_100b8550 (3 callees).

12.874 sub_1003ec60 at 0x1003ec60 (61 instructions)

```
0x1003ec60: sub esp,0Ch
0x1003ec63: mov ecx,[esi+24h]
0x1003ec66: mov eax,[esi+4]
0x1003ec69: mov edx,[eax]
0x1003ec6b: push ebx
0x1003ec6c: push ebp
0x1003ec6d: mov ebp,[esi+11Ch]
0x1003ec73: add ecx,ecx
0x1003ec75: push edi
0x1003ec76: mov edi,[esi+194h]
0x1003ec7c: add ecx,ecx
0x1003ec7e: add ecx,ecx
0x1003ec80: push ecx
0x1003ec81: push 1
0x1003ec83: push esi
0x1003ec84: mov [esp+20h],edi
0x1003ec88: call edx
0x1003ec8a: mov [edi+38h],eax
0x1003ec8d: mov ecx,[esi+24h]
0x1003ec90: lea edx,[eax+ecx*4]
0x1003ec93: mov [edi+3Ch],edx
0x1003ec96: mov eax,[esi+0C4h]
0x1003ec9c: xor ebx,ebx
0x1003ec9e: add esp,0Ch
0x1003eca1: cmp [esi+24h],ebx
0x1003eca4: jle short 1003ED03h
0x1003eca6: add ebp,4
0x1003eca9: mov [esp+0Ch],ebp
0x1003ecad: lea ebp,[eax+0Ch]
0x1003ecb0: mov eax,[ebp+1Ch]
```

```
0x1003ecb3: imul eax,[ebp]
0x1003ecb7: cdq
0x1003ecb8: idiv dword ptr [esi+11Ch]
0x1003ecbe: mov edi,[esp+0Ch]
0x1003ecc2: imul edi,eax
0x1003ecc5: lea ecx,[edi*8]
0x1003eccc: push ecx
0x1003eccd: mov [esp+14h],eax
0x1003ecd1: mov eax,[esi+4]
0x1003ecd4: mov edx,[eax]
0x1003ecd6: push 1
0x1003ecd8: push esi
0x1003ecd9: call edx
0x1003ecdb: mov ecx,[esp+1Ch]
0x1003ecdf: lea eax,[eax+ecx*4]
0x1003ece2: mov ecx,[esp+20h]
0x1003ece6: mov edx,[ecx+38h]
0x1003ece9: mov [edx+ebx*4],eax
0x1003ecec: mov ecx,[ecx+3Ch]
0x1003ecef: lea eax,[eax+edi*4]
0x1003ecf2: mov [ecx+ebx*4],eax
0x1003ecf5: add ebx,1
0x1003ecf8: add esp,0Ch
0x1003ecfb: add ebp,58h
0x1003ecfe: cmp ebx,[esi+24h]
0x1003ed01: jl short 1003ECB0h
0x1003ed03: pop edi
0x1003ed04: pop ebp
0x1003ed05: pop ebx
0x1003ed06: add esp,0Ch
0x1003ed09: ret
```

`sub_1003ec60` is called by sub_1003f250. It calls nothing else.

12.875 sub_1003f250 at 0x1003f250 (78 instructions)

```
0x1003f250: push ebx
0x1003f251: push ebp
0x1003f252: push esi
0x1003f253: mov esi,[esp+10h]
0x1003f257: mov eax,[esi+4]
0x1003f25a: mov ecx,[eax]
0x1003f25c: push edi
0x1003f25d: push 50h
0x1003f25f: push 1
0x1003f261: push esi
0x1003f262: call ecx
0x1003f264: mov ebx,eax
0x1003f266: add esp,0Ch
0x1003f269: cmp byte ptr [esp+18h],0
0x1003f26e: mov [esi+194h],ebx
0x1003f274: mov dword ptr [ebx],1003F1D0h
0x1003f27a: je short 1003F28Fh
0x1003f27c: mov edx,[esi]
0x1003f27e: mov dword ptr [edx+14h],3
0x1003f285: mov eax,[esi]
0x1003f287: mov ecx,[eax]
0x1003f289: push esi
0x1003f28a: call ecx
0x1003f28c: add esp,4
0x1003f28f: mov edx,[esi+1B0h]
0x1003f295: cmp byte ptr [edx+8],0
0x1003f299: je short 1003F2CBh
0x1003f29b: cmp dword ptr [esi+11Ch],2
0x1003f2a2: jge short 1003F2B7h
0x1003f2a4: mov eax,[esi]
```

```
0x1003f2a6: mov dword ptr [eax+14h],30h
0x1003f2ad: mov ecx,[esi]
0x1003f2af: mov edx,[ecx]
0x1003f2b1: push esi
0x1003f2b2: call edx
0x1003f2b4: add esp,4
0x1003f2b7: call 1003EC60h
0x1003f2bc: mov ecx,[esi+11Ch]
0x1003f2c2: add ecx,2
0x1003f2c5: mov [esp+18h],ecx
0x1003f2c9: jmp short 1003F2D7h
0x1003f2cb: mov eax,[esi+11Ch]
0x1003f2d1: mov [esp+18h],eax
0x1003f2d5: mov ecx,eax
0x1003f2d7: mov edi,[esi+0C4h]
0x1003f2dd: xor ebp,ebp
0x1003f2df: cmp [esi+24h],ebp
0x1003f2e2: jle short 1003F32Bh
0x1003f2e4: add edi,0Ch
0x1003f2e7: add ebx,8
0x1003f2ea: jmp short 1003F2F4h
0x1003f2f0: mov ecx,[esp+18h]
0x1003f2f4: mov eax,[edi+1Ch]
0x1003f2f7: imul eax,[edi]
0x1003f2fa: cdq
0x1003f2fb: idiv dword ptr [esi+11Ch]
0x1003f301: mov edx,[esi+4]
0x1003f304: imul eax,ecx
0x1003f307: mov ecx,[edx+8]
0x1003f30a: push eax
0x1003f30b: mov eax,[edi+18h]
0x1003f30e: imul eax,[edi+10h]
0x1003f312: push eax
0x1003f313: push 1
0x1003f315: push esi
0x1003f316: call ecx
0x1003f318: mov [ebx],eax
0x1003f31a: add ebp,1
0x1003f31d: add esp,10h
0x1003f320: add ebx,4
0x1003f323: add edi,58h
0x1003f326: cmp ebp,[esi+24h]
0x1003f329: jl short 1003F2F0h
0x1003f32b: pop edi
0x1003f32c: pop esi
0x1003f32d: pop ebp
0x1003f32e: pop ebx
0x1003f32f: ret
```

`sub_1003f250` is called by sub_1003beb0. It delegates to sub_1003ec60 (1 callees).

12.876 sub_10040160 at 0x10040160 (105 instructions)

```
0x10040160: push ebx
0x10040161: push ebp
0x10040162: push esi
0x10040163: mov esi,[esp+10h]
0x10040167: mov eax,[esi+4]
0x1004016a: mov ecx,[eax]
0x1004016c: push 74h
0x1004016e: push 1
0x10040170: push esi
0x10040171: call ecx
0x10040173: mov ebp,eax
0x10040175: xor ebx,ebx
0x10040177: mov [esi+198h],ebp
```

```
0x1004017d: add esp,0Ch
0x10040180: cmp [esp+14h],bl
0x10040184: mov dword ptr [ebp],1003F380h
0x1004018b: mov dword ptr [ebp+8],10040110h
0x10040192: mov [ebp+70h],ebx
0x10040195: je near ptr 1004022Eh
0x1004019b: cmp [esi+24h],ebx
0x1004019e: push edi
0x1004019f: mov edi,[esi+0C4h]
0x100401a5: mov [esp+18h],ebx
0x100401a9: jle short 10040215h
0x100401ab: lea edx,[ebp+48h]
0x100401ae: add edi,0Ch
0x100401b1: mov [esp+14h],edx
0x100401b5: cmp byte ptr [esi+0C9h],0
0x100401bc: mov eax,[edi]
0x100401be: mov ecx,eax
0x100401c0: je short 100401C5h
0x100401c2: lea ecx,[ecx+ecx*2]
0x100401c5: mov ebx,[esi+4]
0x100401c8: push ecx
0x100401c9: push eax
0x100401ca: mov eax,[edi+14h]
0x100401cd: push eax
0x100401ce: call 1003D210h
0x100401d3: mov ecx,[edi-4]
0x100401d6: mov edx,[edi+10h]
0x100401d9: add esp,8
0x100401dc: push eax
0x100401dd: push ecx
0x100401de: push edx
0x100401df: call 1003D210h
0x100401e4: add esp,8
0x100401e7: push eax
0x100401e8: mov eax,[ebx+14h]
0x100401eb: push 1
0x100401ed: push 1
0x100401ef: push esi
0x100401f0: call eax
0x100401f2: mov ecx,[esp+2Ch]
0x100401f6: mov [ecx],eax
0x100401f8: mov eax,[esp+30h]
0x100401fc: add eax,1
0x100401ff: add ecx,4
0x10040202: add esp,18h
0x10040205: add edi,58h
0x10040208: cmp eax,[esi+24h]
0x1004020b: mov [esp+18h],eax
0x1004020f: mov [esp+14h],ecx
0x10040213: jl short 100401B5h
0x10040215: pop edi
0x10040216: lea ecx,[ebp+48h]
0x10040219: pop esi
0x1004021a: mov dword ptr [ebp+4],1003F5E0h
0x10040221: mov dword ptr [ebp+0Ch],1003F7D0h
0x10040228: mov [ebp+10h],ecx
0x1004022b: pop ebp
0x1004022c: pop ebx
0x1004022d: ret
0x1004022e: mov edx,[esi+4]
0x10040231: mov eax,[edx+4]
0x10040234: push 500h
0x10040239: push 1
0x1004023b: push esi
0x1004023c: call eax
0x1004023e: lea ecx,[eax+80h]
0x10040244: mov [ebp+24h],ecx
```

```
0x10040247: lea edx,[eax+100h]
0x1004024d: mov [ebp+28h],edx
0x10040250: lea ecx,[eax+180h]
0x10040256: mov [ebp+2Ch],ecx
0x10040259: lea edx,[eax+200h]
0x1004025f: mov [ebp+30h],edx
0x10040262: lea ecx,[eax+280h]
0x10040268: mov [ebp+34h],ecx
0x1004026b: lea edx,[eax+300h]
0x10040271: mov [ebp+38h],edx
0x10040274: mov [ebp+20h],eax
0x10040277: add esp,0Ch
0x1004027a: lea ecx,[eax+380h]
0x10040280: lea edx,[eax+400h]
0x10040286: mov [ebp+3Ch],ecx
0x10040289: mov [ebp+40h],edx
0x1004028c: add eax,480h
0x10040291: mov [ebp+44h],eax
0x10040294: pop esi
0x10040295: mov [ebp+10h],ebx
0x10040298: mov dword ptr [ebp+4],100B8550h
0x1004029f: mov dword ptr [ebp+0Ch],1003F3A0h
0x100402a6: pop ebp
0x100402a7: pop ebx
0x100402a8: ret
```

`sub_10040160` is called by sub_1003beb0. It delegates to sub_1003d210 (1 callees).

12.877 sub_10041fd0 at 0x10041fd0 (61 instructions)

```
0x10041fd0: push ebx
0x10041fd1: push esi
0x10041fd2: mov esi,[esp+0Ch]
0x10041fd6: mov eax,[esi+4]
0x10041fd9: mov ecx,[eax]
0x10041fdb: push 0D8h
0x10041fe0: push 1
0x10041fe2: push esi
0x10041fe3: call ecx
0x10041fe5: mov ebx,eax
0x10041fe7: mov [esi+1A8h],ebx
0x10041fed: add esp,0Ch
0x10041ff0: mov dword ptr [ebx],10041A10h
0x10041ff6: cmp byte ptr [esi+0C9h],0
0x10041ffd: je short 10042051h
0x10041fff: mov eax,[esi+24h]
0x10042002: mov edx,[esi+4]
0x10042005: mov ecx,[edx]
0x10042007: push ebp
0x10042008: shl eax,8
0x1004200b: push eax
0x1004200c: push 1
0x1004200e: push esi
0x1004200f: call ecx
0x10042011: xor ebp,ebp
0x10042013: add esp,0Ch
0x10042016: cmp [esi+24h],ebp
0x10042019: mov [esi+8Ch],eax
0x1004201f: mov edx,eax
0x10042021: jle short 10042041h
0x10042023: push edi
0x10042024: or eax,0FFFFFFFFh
0x10042027: mov edi,edx
0x10042029: mov ecx,40h
0x1004202e: add ebp,1
0x10042031: rep stosd
```

```
0x10042033: add edx,100h
0x10042039: cmp ebp,[esi+24h]
0x1004203c: jl short 10042024h
0x1004203e: xor ebp,ebp
0x10042040: pop edi
0x10042041: mov [ebx+2Ch],ebp
0x10042044: mov [ebx+30h],ebp
0x10042047: mov [ebx+34h],ebp
0x1004204a: mov [ebx+38h],ebp
0x1004204d: pop ebp
0x1004204e: pop esi
0x1004204f: pop ebx
0x10042050: ret
0x10042051: xor eax,eax
0x10042053: pop esi
0x10042054: mov [ebx+50h],eax
0x10042057: mov [ebx+40h],eax
0x1004205a: mov [ebx+54h],eax
0x1004205d: mov [ebx+44h],eax
0x10042060: mov [ebx+58h],eax
0x10042063: mov [ebx+48h],eax
0x10042066: mov [ebx+5Ch],eax
0x10042069: mov [ebx+4Ch],eax
0x1004206c: pop ebx
0x1004206d: ret
```

`sub_10041fd0` is called by sub_1003beb0. It calls nothing else.

12.878 sub_100430c0 at 0x100430c0 (51 instructions)

```
0x100430c0: push ebx
0x100430c1: push esi
0x100430c2: mov esi,[esp+0Ch]
0x100430c6: mov eax,[esi+4]
0x100430c9: mov ecx,[eax]
0x100430cb: push 0BCh
0x100430d0: push 1
0x100430d2: push esi
0x100430d3: call ecx
0x100430d5: add esp,0Ch
0x100430d8: mov [esi+1A8h],eax
0x100430de: mov dword ptr [eax],10042D30h
0x100430e4: lea ecx,[eax+78h]
0x100430e7: mov edx,10h
0x100430ec: xor ebx,ebx
0x100430ee: mov edi,edi
0x100430f0: mov [ecx-40h],ebx
0x100430f3: mov [ecx],ebx
0x100430f5: add ecx,4
0x100430f8: sub edx,1
0x100430fb: jne short 100430F0h
0x100430fd: mov byte ptr [eax+0B8h],71h
0x10043104: cmp [esi+0C9h],bl
0x1004310a: je short 1004314Bh
0x1004310c: mov eax,[esi+24h]
0x1004310f: mov edx,[esi+4]
0x10043112: mov ecx,[edx]
0x10043114: shl eax,8
0x10043117: push eax
0x10043118: push 1
0x1004311a: push esi
0x1004311b: call ecx
0x1004311d: add esp,0Ch
0x10043120: cmp [esi+24h],ebx
0x10043123: mov [esi+8Ch],eax
0x10043129: mov edx,eax
```

```
0x1004312b: jle short 1004314Bh
0x1004312d: push edi
0x1004312e: mov edi,edi
0x10043130: or eax,0FFFFFFFFh
0x10043133: mov edi,edx
0x10043135: mov ecx,40h
0x1004313a: add ebx,1
0x1004313d: rep stosd
0x1004313f: add edx,100h
0x10043145: cmp ebx,[esi+24h]
0x10043148: jl short 10043130h
0x1004314a: pop edi
0x1004314b: pop esi
0x1004314c: pop ebx
0x1004314d: ret
```

`sub_100430c0` is called by sub_1003beb0. It calls nothing else.

12.879 sub_100436f0 at 0x100436f0 (43 instructions)

```
0x100436f0: push esi
0x100436f1: mov esi,[esp+8]
0x100436f5: mov eax,[esi+4]
0x100436f8: mov ecx,[eax]
0x100436fa: push edi
0x100436fb: push 54h
0x100436fd: push 1
0x100436ff: push esi
0x10043700: call ecx
0x10043702: mov [esi+1ACh],eax
0x10043708: mov dword ptr [eax],10043150h
0x1004370e: mov ecx,[esi+0C4h]
0x10043714: xor edi,edi
0x10043716: add esp,0Ch
0x10043719: cmp [esi+24h],edi
0x1004371c: jle short 1004375Eh
0x1004371e: push ebx
0x1004371f: push ebp
0x10043720: lea ebp,[ecx+54h]
0x10043723: lea ebx,[eax+2Ch]
0x10043726: mov edx,[esi+4]
0x10043729: mov eax,[edx]
0x1004372b: push 100h
0x10043730: push 1
0x10043732: push esi
0x10043733: call eax
0x10043735: push 100h
0x1004373a: push 0
0x1004373c: push eax
0x1004373d: mov [ebp],eax
0x10043740: call 100228C0h
0x10043745: mov dword ptr [ebx],0FFFFFFFFh
0x1004374b: add edi,1
0x1004374e: add esp,18h
0x10043751: add ebx,4
0x10043754: add ebp,58h
0x10043757: cmp edi,[esi+24h]
0x1004375a: jl short 10043726h
0x1004375c: pop ebp
0x1004375d: pop ebx
0x1004375e: pop edi
0x1004375f: pop esi
0x10043760: ret
```

`sub_100436f0` is called by sub_1003beb0. It delegates to j_memset (1 callees).

12.880 sub_10043a40 at 0x10043a40 (58 instructions)

```
0x10043a40: push esi
0x10043a41: mov esi,[esp+8]
0x10043a45: mov eax,[esi+4]
0x10043a48: mov ecx,[eax]
0x10043a4a: push edi
0x10043a4b: push 1Ch
0x10043a4d: push 1
0x10043a4f: push esi
0x10043a50: call ecx
0x10043a52: mov edi,eax
0x10043a54: mov [esi+19Ch],edi
0x10043a5a: add esp,0Ch
0x10043a5d: mov dword ptr [edi],10043940h
0x10043a63: mov dword ptr [edi+8],0
0x10043a6a: mov dword ptr [edi+0Ch],0
0x10043a71: cmp byte ptr [esi+4Ah],0
0x10043a75: je short 10043AD0h
0x10043a77: cmp byte ptr [esp+10h],0
0x10043a7c: mov eax,[esi+114h]
0x10043a82: mov [edi+10h],eax
0x10043a85: je short 10043AB6h
0x10043a87: mov edx,[esi+60h]
0x10043a8a: push ebp
0x10043a8b: mov ebp,[esi+4]
0x10043a8e: push eax
0x10043a8f: push eax
0x10043a90: push edx
0x10043a91: call 1003D210h
0x10043a96: mov ecx,[ebp+10h]
0x10043a99: add esp,8
0x10043a9c: push eax
0x10043a9d: mov eax,[esi+64h]
0x10043aa0: imul eax,[esi+5Ch]
0x10043aa4: push eax
0x10043aa5: push 0
0x10043aa7: push 1
0x10043aa9: push esi
0x10043aaa: call ecx
0x10043aac: add esp,18h
0x10043aaf: pop ebp
0x10043ab0: mov [edi+8],eax
0x10043ab3: pop edi
0x10043ab4: pop esi
0x10043ab5: ret
0x10043ab6: mov edx,[esi+4]
0x10043ab9: mov ecx,[edx+8]
0x10043abc: push eax
0x10043abd: mov eax,[esi+64h]
0x10043ac0: imul eax,[esi+5Ch]
0x10043ac4: push eax
0x10043ac5: push 1
0x10043ac7: push esi
0x10043ac8: call ecx
0x10043aca: add esp,10h
0x10043acd: mov [edi+0Ch],eax
0x10043ad0: pop edi
0x10043ad1: pop esi
0x10043ad2: ret
```

`sub_10043a40` is called by sub_1003beb0. It delegates to sub_1003d210 (1 callees).

12.881 sub_10043da0 at 0x10043da0 (127 instructions)

```
0x10043da0: sub esp,0Ch
0x10043da3: push ebx
```

```
0x10043da4: push esi
0x10043da5: mov esi,[esp+18h]
0x10043da9: mov eax,[esi+4]
0x10043dac: mov ecx,[eax]
0x10043dae: push edi
0x10043daf: push 0A0h
0x10043db4: push 1
0x10043db6: push esi
0x10043db7: call ecx
0x10043db9: mov ebx,eax
0x10043dbb: mov [esi+1B0h],ebx
0x10043dc1: add esp,0Ch
0x10043dc4: mov dword ptr [ebx],10043AE0h
0x10043dca: mov dword ptr [ebx+4],10043B00h
0x10043dd1: mov byte ptr [ebx+8],0
0x10043dd5: cmp byte ptr [esi+10Ah],0
0x10043ddc: mov [esp+14h],ebx
0x10043de0: je short 10043DF5h
0x10043de2: mov edx,[esi]
0x10043de4: mov dword ptr [edx+14h],1Ah
0x10043deb: mov eax,[esi]
0x10043ded: mov ecx,[eax]
0x10043def: push esi
0x10043df0: call ecx
0x10043df2: add esp,4
0x10043df5: cmp dword ptr [esi+24h],0
0x10043df9: mov edi,[esi+0C4h]
0x10043dff: mov dword ptr [esp+1Ch],0
0x10043e07: jle near ptr 10043F2Ch
0x10043e0d: add edi,8
0x10043e10: add ebx,34h
0x10043e13: push ebp
0x10043e14: mov eax,[edi+1Ch]
0x10043e17: imul eax,[edi]
0x10043e1a: cdq
0x10043e1b: idiv dword ptr [esi+118h]
0x10043e21: mov ebp,eax
0x10043e23: mov eax,[edi+20h]
0x10043e26: imul eax,[edi+4]
0x10043e2a: cdq
0x10043e2b: idiv dword ptr [esi+11Ch]
0x10043e31: mov edx,[esi+114h]
0x10043e37: mov [esp+14h],ebp
0x10043e3b: mov [esp+10h],edx
0x10043e3f: mov ecx,eax
0x10043e41: mov eax,[esi+110h]
0x10043e47: mov [ebx+30h],ecx
0x10043e4a: cmp byte ptr [edi+2Ch],0
0x10043e4e: jne short 10043E5Bh
0x10043e50: mov dword ptr [ebx],10043BE0h
0x10043e56: jmp near ptr 10043F11h
0x10043e5b: cmp ebp,eax
0x10043e5d: jne short 10043E6Eh
0x10043e5f: cmp ecx,edx
0x10043e61: jne short 10043E6Eh
0x10043e63: mov dword ptr [ebx],10043BD0h
0x10043e69: jmp near ptr 10043F11h
0x10043e6e: add ebp,ebp
0x10043e70: cmp ebp,eax
0x10043e72: jne short 10043E95h
0x10043e74: cmp ecx,edx
0x10043e76: jne short 10043E80h
0x10043e78: mov dword ptr [ebx],10043CD0h
0x10043e7e: jmp short 10043EE2h
0x10043e80: cmp ebp,eax
0x10043e82: jne short 10043E95h
0x10043e84: lea edx,[ecx+ecx]
```

```
0x10043e87: cmp edx,[esp+10h]
0x10043e8b: jne short 10043E95h
0x10043e8d: mov dword ptr [ebx],10043D30h
0x10043e93: jmp short 10043EE2h
0x10043e95: cdq
0x10043e96: idiv dword ptr [esp+14h]
0x10043e9a: test edx,edx
0x10043e9c: mov [esp+14h],eax
0x10043ea0: jne short 10043ECFh
0x10043ea2: mov eax,[esp+10h]
0x10043ea6: cdq
0x10043ea7: idiv ecx
0x10043ea9: test edx,edx
0x10043eab: jne short 10043ECFh
0x10043ead: mov ebp,[esp+20h]
0x10043eb1: mov ecx,[esp+18h]
0x10043eb5: mov dl,[esp+14h]
0x10043eb9: mov dword ptr [ebx],10043BF0h
0x10043ebf: mov [ecx+ebp+8Ch],dl
0x10043ec6: mov [ecx+ebp+96h],al
0x10043ecd: jmp short 10043EE2h
0x10043ecf: mov eax,[esi]
0x10043ed1: mov dword ptr [eax+14h],27h
0x10043ed8: mov ecx,[esi]
0x10043eda: mov edx,[ecx]
0x10043edc: push esi
0x10043edd: call edx
0x10043edf: add esp,4
0x10043ee2: mov eax,[esi+114h]
0x10043ee8: mov ecx,[esi+110h]
0x10043eee: mov edx,[esi+5Ch]
0x10043ef1: mov ebp,[esi+4]
0x10043ef4: push eax
0x10043ef5: push ecx
0x10043ef6: push edx
0x10043ef7: add ebp,8
0x10043efa: call 1003D210h
0x10043eff: add esp,8
0x10043f02: push eax
0x10043f03: mov eax,[ebp]
0x10043f06: push 1
0x10043f08: push esi
0x10043f09: call eax
0x10043f0b: add esp,10h
0x10043f0e: mov [ebx-28h],eax
0x10043f11: mov eax,[esp+20h]
0x10043f15: add eax,1
0x10043f18: add ebx,4
0x10043f1b: add edi,58h
0x10043f1e: cmp eax,[esi+24h]
0x10043f21: mov [esp+20h],eax
0x10043f25: jl near ptr 10043E14h
0x10043f2b: pop ebp
0x10043f2c: pop edi
0x10043f2d: pop esi
0x10043f2e: pop ebx
0x10043f2f: add esp,0Ch
0x10043f32: ret
```

`sub_10043da0` is called by sub_1003beb0. It delegates to sub_1003d210 (1 callees).

12.882 sub_10043f40 at 0x10043f40 (68 instructions)

```
0x10043f40: push ecx
0x10043f41: push ebx
0x10043f42: push ebp
```

```
0x10043f43: push esi
0x10043f44: push edi
0x10043f45: mov edi,eax
0x10043f47: mov eax,[edi+4]
0x10043f4a: mov ecx,[eax]
0x10043f4c: mov esi,[edi+1B4h]
0x10043f52: push 400h
0x10043f57: push 1
0x10043f59: push edi
0x10043f5a: call ecx
0x10043f5c: mov [esi+8],eax
0x10043f5f: mov edx,[edi+4]
0x10043f62: mov eax,[edx]
0x10043f64: push 400h
0x10043f69: push 1
0x10043f6b: push edi
0x10043f6c: call eax
0x10043f6e: mov [esi+0Ch],eax
0x10043f71: mov ecx,[edi+4]
0x10043f74: mov edx,[ecx]
0x10043f76: push 400h
0x10043f7b: push 1
0x10043f7d: push edi
0x10043f7e: call edx
0x10043f80: mov [esi+10h],eax
0x10043f83: mov eax,[edi+4]
0x10043f86: mov ecx,[eax]
0x10043f88: push 400h
0x10043f8d: push 1
0x10043f8f: push edi
0x10043f90: call ecx
0x10043f92: add esp,30h
0x10043f95: mov [esi+14h],eax
0x10043f98: xor eax,eax
0x10043f9a: mov dword ptr [esp+10h],5B6900h
0x10043fa2: mov edi,0FF1DAF00h
0x10043fa7: mov edx,0FF4D0B80h
0x10043fac: mov ecx,2C8D00h
0x10043fb1: mov ebp,[esi+8]
0x10043fb4: mov ebx,edx
0x10043fb6: sar ebx,10h
0x10043fb9: mov [eax+ebp],ebx
0x10043fbc: mov ebp,[esi+0Ch]
0x10043fbf: mov ebx,edi
0x10043fc1: sar ebx,10h
0x10043fc4: mov [eax+ebp],ebx
0x10043fc7: mov ebp,[esi+10h]
0x10043fca: mov ebx,[esp+10h]
0x10043fce: mov [eax+ebp],ebx
0x10043fd1: mov ebp,[esi+14h]
0x10043fd4: mov [eax+ebp],ecx
0x10043fd7: sub ebx,0B6D2h
0x10043fdd: sub ecx,581Ah
0x10043fe3: add edx,166E9h
0x10043fe9: add edi,1C5A2h
0x10043fef: add eax,4
0x10043ff2: cmp ecx,0FFD4CB1Ah
0x10043ff8: mov [esp+10h],ebx
0x10043ffc: jge short 10043FB1h
0x10043ffe: pop edi
0x10043fff: pop esi
0x10044000: pop ebp
0x10044001: pop ebx
0x10044002: pop ecx
0x10044003: ret
```

`sub_10043f40` is called by sub_100443b0. It calls nothing else.

12.883 sub_100443b0 at 0x100443b0 (106 instructions)

```
0x100443b0: push esi
0x100443b1: mov esi,[esp+8]
0x100443b5: mov eax,[esi+4]
0x100443b8: mov ecx,[eax]
0x100443ba: push edi
0x100443bb: push 18h
0x100443bd: push 1
0x100443bf: push esi
0x100443c0: call ecx
0x100443c2: mov edi,eax
0x100443c4: mov [esi+1B4h],edi
0x100443ca: mov dword ptr [edi],100B3980h
0x100443d0: mov eax,[esi+28h]
0x100443d3: add eax,0FFFFFFFFh
0x100443d6: add esp,0Ch
0x100443d9: cmp eax,4
0x100443dc: ja short 100443FDh
0x100443de: jmp dword ptr [eax*4+10044520h]
0x100443fd: cmp dword ptr [esi+24h],1
0x10044401: jge short 10044416h
0x10044403: mov edx,[esi]
0x10044405: mov dword ptr [edx+14h],0Bh
0x1004440c: mov eax,[esi]
0x1004440e: mov ecx,[eax]
0x10044410: push esi
0x10044411: call ecx
0x10044413: add esp,4
0x10044416: mov ecx,[esi+2Ch]
0x10044419: mov eax,ecx
0x1004441b: mov edx,1
0x10044420: sub eax,edx
0x10044422: je near ptr 100444B9h
0x10044428: sub eax,edx
0x1004442a: je short 1004447Fh
0x1004442c: sub eax,2
0x1004442f: je short 1004444Ch
0x10044431: cmp ecx,[esi+28h]
0x10044434: jne near ptr 100444C8h
0x1004443a: mov edx,[esi+24h]
0x1004443d: mov [esi+64h],edx
0x10044440: mov dword ptr [edi+4],10044130h
0x10044447: jmp near ptr 10044505h
0x1004444c: mov eax,[esi+28h]
0x1004444f: cmp eax,5
0x10044452: mov dword ptr [esi+64h],4
0x10044459: jne short 1004446Eh
0x1004445b: mov eax,esi
0x1004445d: mov dword ptr [edi+4],10044250h
0x10044464: call 10043F40h
0x10044469: jmp near ptr 10044505h
0x1004446e: cmp eax,4
0x10044471: jne short 100444C8h
0x10044473: mov dword ptr [edi+4],10044130h
0x1004447a: jmp near ptr 10044505h
0x1004447f: mov eax,[esi+28h]
0x10044482: cmp eax,3
0x10044485: mov dword ptr [esi+64h],3
0x1004448c: jne short 1004449Eh
0x1004448e: mov eax,esi
0x10044490: mov dword ptr [edi+4],10044010h
0x10044497: call 10043F40h
0x1004449c: jmp short 10044505h
0x1004449e: cmp eax,edx
0x100444a0: jne short 100444ABh
0x100444a2: mov dword ptr [edi+4],100441F0h
0x100444a9: jmp short 10044505h
```

```
0x100444ab: cmp eax,2
0x100444ae: jne short 100444C8h
0x100444b0: mov dword ptr [edi+4],10044130h
0x100444b7: jmp short 10044505h
0x100444b9: mov eax,[esi+28h]
0x100444bc: cmp eax,edx
0x100444be: mov [esi+64h],edx
0x100444c1: je short 100444DDh
0x100444c3: cmp eax,3
0x100444c6: je short 100444DDh
0x100444c8: mov eax,[esi]
0x100444ca: mov dword ptr [eax+14h],1Ch
0x100444d1: mov ecx,[esi]
0x100444d3: mov edx,[ecx]
0x100444d5: push esi
0x100444d6: call edx
0x100444d8: add esp,4
0x100444db: jmp short 10044505h
0x100444dd: mov dword ptr [edi+4],100441C0h
0x100444e4: cmp [esi+24h],edx
0x100444e7: mov eax,edx
0x100444e9: jle short 10044505h
0x100444eb: mov ecx,58h
0x100444f0: mov edi,[esi+0C4h]
0x100444f6: mov byte ptr [edi+ecx+34h],0
0x100444fb: add eax,edx
0x100444fd: add ecx,58h
0x10044500: cmp eax,[esi+24h]
0x10044503: jl short 100444F0h
0x10044505: cmp byte ptr [esi+4Ah],0
0x10044509: je short 10044515h
0x1004450b: pop edi
0x1004450c: mov dword ptr [esi+68h],1
0x10044513: pop esi
0x10044514: ret
0x10044515: mov eax,[esi+64h]
0x10044518: pop edi
0x10044519: mov [esi+68h],eax
0x1004451c: pop esi
0x1004451d: ret
```

`sub_100443b0` is called by sub_1003beb0. It delegates to sub_10043f40 (1 callees).

12.884 sub_10044ab0 at 0x10044ab0 (110 instructions)

```
0x10044540: push ecx
0x10044541: push ebx
0x10044542: push ebp
0x10044543: push esi
0x10044544: push edi
0x10044545: mov edi,eax
0x10044547: mov eax,[edi+4]
0x1004454a: mov ecx,[eax]
0x1004454c: mov esi,[edi+1B0h]
0x10044552: push 400h
0x10044557: push 1
0x10044559: push edi
0x1004455a: call ecx
0x1004455c: mov [esi+10h],eax
0x1004455f: mov edx,[edi+4]
0x10044562: mov eax,[edx]
0x10044564: push 400h
0x10044569: push 1
0x1004456b: push edi
0x1004456c: call eax
0x1004456e: mov [esi+14h],eax
```

```
0x10044571: mov ecx,[edi+4]
0x10044574: mov edx,[ecx]
0x10044576: push 400h
0x1004457b: push 1
0x1004457d: push edi
0x1004457e: call edx
0x10044580: mov [esi+18h],eax
0x10044583: mov eax,[edi+4]
0x10044586: mov ecx,[eax]
0x10044588: push 400h
0x1004458d: push 1
0x1004458f: push edi
0x10044590: call ecx
0x10044592: add esp,30h
0x10044595: mov [esi+1Ch],eax
0x10044598: xor eax,eax
0x1004459a: mov dword ptr [esp+10h],5B6900h
0x100445a2: mov edi,0FF1DAF00h
0x100445a7: mov edx,0FF4D0B80h
0x100445ac: mov ecx,2C8D00h
0x100445b1: mov ebp,[esi+10h]
0x100445b4: mov ebx,edx
0x100445b6: sar ebx,10h
0x100445b9: mov [eax+ebp],ebx
0x100445bc: mov ebp,[esi+14h]
0x100445bf: mov ebx,edi
0x100445c1: sar ebx,10h
0x100445c4: mov [eax+ebp],ebx
0x100445c7: mov ebp,[esi+18h]
0x100445ca: mov ebx,[esp+10h]
0x100445ce: mov [eax+ebp],ebx
0x100445d1: mov ebp,[esi+1Ch]
0x100445d4: mov [eax+ebp],ecx
0x100445d7: sub ebx,0B6D2h
0x100445dd: sub ecx,581Ah
0x100445e3: add edx,166E9h
0x100445e9: add edi,1C5A2h
0x100445ef: add eax,4
0x100445f2: cmp ecx,0FFD4CB1Ah
0x100445f8: mov [esp+10h],ebx
0x100445fc: jge short 100445B1h
0x100445fe: pop edi
0x100445ff: pop esi
0x10044600: pop ebp
0x10044601: pop ebx
0x10044602: pop ecx
0x10044603: ret
0x10044ab0: push esi
0x10044ab1: push edi
0x10044ab2: mov edi,[esp+0Ch]
0x10044ab6: mov eax,[edi+4]
0x10044ab9: mov ecx,[eax]
0x10044abb: push 30h
0x10044abd: push 1
0x10044abf: push edi
0x10044ac0: call ecx
0x10044ac2: mov esi,eax
0x10044ac4: mov [edi+1B0h],esi
0x10044aca: xor eax,eax
0x10044acc: mov dword ptr [esi],10044610h
0x10044ad2: mov [esi+8],al
0x10044ad5: mov edx,[edi+64h]
0x10044ad8: imul edx,[edi+5Ch]
0x10044adc: add esp,0Ch
0x10044adf: mov [esi+28h],edx
0x10044ae2: cmp dword ptr [edi+114h],2
0x10044ae9: jne short 10044B16h
```

```
0x10044aeb: mov ecx,edx
0x10044aed: push ecx
0x10044aee: mov dword ptr [esi+4],10044630h
0x10044af5: mov dword ptr [esi+0Ch],10044890h
0x10044afc: mov eax,[edi+4]
0x10044aff: mov edx,[eax+4]
0x10044b02: push 1
0x10044b04: push edi
0x10044b05: call edx
0x10044b07: add esp,0Ch
0x10044b0a: mov [esi+20h],eax
0x10044b0d: mov eax,edi
0x10044b0f: pop edi
0x10044b10: pop esi
0x10044b11: jmp near ptr 10044540h
0x10044b16: mov [esi+20h],eax
0x10044b19: mov eax,edi
0x10044b1b: pop edi
0x10044b1c: mov dword ptr [esi+4],10044700h
0x10044b23: mov dword ptr [esi+0Ch],10044740h
0x10044b2a: pop esi
0x10044b2b: jmp near ptr 10044540h
```

`sub_10044ab0` is called by sub_1003beb0. It calls nothing else.

12.885 sub_10045da0 at 0x10045da0 (174 instructions)

```
0x10045b90: mov ecx,[eax+4]
0x10045b93: mov edx,[ecx]
0x10045b95: push esi
0x10045b96: mov esi,[eax+1B8h]
0x10045b9c: push edi
0x10045b9d: push 7FCh
0x10045ba2: push 1
0x10045ba4: push eax
0x10045ba5: call edx
0x10045ba7: add eax,3FCh
0x10045bac: add esp,0Ch
0x10045baf: mov [esi+28h],eax
0x10045bb2: xor edx,edx
0x10045bb4: xor ecx,ecx
0x10045bb6: xor edi,edi
0x10045bb8: mov esi,eax
0x10045bba: lea ebx,[ebx]
0x10045bc0: mov [eax+ecx*4],edx
0x10045bc3: mov [esi],edi
0x10045bc5: add ecx,1
0x10045bc8: sub esi,4
0x10045bcb: add edx,1
0x10045bce: sub edi,1
0x10045bd1: cmp ecx,10h
0x10045bd4: jl short 10045BC0h
0x10045bd6: cmp ecx,30h
0x10045bd9: jge short 10045C05h
0x10045bdb: lea esi,[ecx*4]
0x10045be2: mov edi,esi
0x10045be4: mov esi,eax
0x10045be6: sub esi,edi
0x10045be8: mov edi,edx
0x10045bea: neg edi
0x10045bec: mov [eax+ecx*4],edx
0x10045bef: mov [esi],edi
0x10045bf1: add ecx,1
0x10045bf4: mov edi,ecx
0x10045bf6: not edi
0x10045bf8: and edi,1
```

```
0x10045bfb: sub esi,4
0x10045bfe: add edx,edi
0x10045c00: cmp ecx,30h
0x10045c03: jl short 10045BE8h
0x10045c05: cmp ecx,0FFh
0x10045c0b: jg short 10045C33h
0x10045c0d: lea esi,[ecx*4]
0x10045c14: push ebx
0x10045c15: mov ebx,esi
0x10045c17: mov edi,edx
0x10045c19: mov esi,eax
0x10045c1b: neg edi
0x10045c1d: sub esi,ebx
0x10045c1f: pop ebx
0x10045c20: mov [eax+ecx*4],edx
0x10045c23: mov [esi],edi
0x10045c25: add ecx,1
0x10045c28: sub esi,4
0x10045c2b: cmp ecx,0FFh
0x10045c31: jle short 10045C20h
0x10045c33: pop edi
0x10045c34: pop esi
0x10045c35: ret
0x10045da0: push esi
0x10045da1: mov esi,[esp+8]
0x10045da5: mov eax,[esi+4]
0x10045da8: mov ecx,[eax]
0x10045daa: push edi
0x10045dab: push 2Ch
0x10045dad: push 1
0x10045daf: push esi
0x10045db0: call ecx
0x10045db2: mov edi,eax
0x10045db4: mov [esi+1B8h],edi
0x10045dba: add esp,0Ch
0x10045dbd: mov dword ptr [edi],10045C70h
0x10045dc3: mov dword ptr [edi+0Ch],10045D90h
0x10045dca: mov dword ptr [edi+20h],0
0x10045dd1: mov dword ptr [edi+28h],0
0x10045dd8: cmp dword ptr [esi+64h],3
0x10045ddc: je short 10045DF1h
0x10045dde: mov edx,[esi]
0x10045de0: mov dword ptr [edx+14h],30h
0x10045de7: mov eax,[esi]
0x10045de9: mov ecx,[eax]
0x10045deb: push esi
0x10045dec: call ecx
0x10045dee: add esp,4
0x10045df1: mov edx,[esi+4]
0x10045df4: mov eax,[edx]
0x10045df6: push ebp
0x10045df7: push 80h
0x10045dfc: push 1
0x10045dfe: push esi
0x10045dff: call eax
0x10045e01: add esp,0Ch
0x10045e04: mov [edi+18h],eax
0x10045e07: xor ebp,ebp
0x10045e09: lea esp,[esp]
0x10045e10: mov ecx,[esi+4]
0x10045e13: mov edx,[ecx+4]
0x10045e16: push 1000h
0x10045e1b: push 1
0x10045e1d: push esi
0x10045e1e: call edx
0x10045e20: mov ecx,[edi+18h]
0x10045e23: mov [ecx+ebp],eax
```

```
0x10045e26: add ebp,4
0x10045e29: add esp,0Ch
0x10045e2c: cmp ebp,80h
0x10045e32: jl short 10045E10h
0x10045e34: mov byte ptr [edi+1Ch],1
0x10045e38: cmp byte ptr [esi+5Ah],0
0x10045e3c: je short 10045E9Fh
0x10045e3e: mov ebp,[esi+54h]
0x10045e41: cmp ebp,8
0x10045e44: jge short 10045E62h
0x10045e46: mov edx,[esi]
0x10045e48: mov dword ptr [edx+14h],3Ah
0x10045e4f: mov eax,[esi]
0x10045e51: mov dword ptr [eax+18h],8
0x10045e58: mov ecx,[esi]
0x10045e5a: mov edx,[ecx]
0x10045e5c: push esi
0x10045e5d: call edx
0x10045e5f: add esp,4
0x10045e62: cmp ebp,100h
0x10045e68: jle short 10045E86h
0x10045e6a: mov eax,[esi]
0x10045e6c: mov dword ptr [eax+14h],3Bh
0x10045e73: mov ecx,[esi]
0x10045e75: mov dword ptr [ecx+18h],100h
0x10045e7c: mov edx,[esi]
0x10045e7e: mov eax,[edx]
0x10045e80: push esi
0x10045e81: call eax
0x10045e83: add esp,4
0x10045e86: mov ecx,[esi+4]
0x10045e89: mov edx,[ecx+8]
0x10045e8c: push 3
0x10045e8e: push ebp
0x10045e8f: push 1
0x10045e91: push esi
0x10045e92: call edx
0x10045e94: add esp,10h
0x10045e97: mov [edi+10h],eax
0x10045e9a: mov [edi+14h],ebp
0x10045e9d: jmp short 10045EA6h
0x10045e9f: mov dword ptr [edi+10h],0
0x10045ea6: cmp dword ptr [esi+4Ch],0
0x10045eaa: mov ecx,2
0x10045eaf: pop ebp
0x10045eb0: je short 10045EB5h
0x10045eb2: mov [esi+4Ch],ecx
0x10045eb5: cmp [esi+4Ch],ecx
0x10045eb8: jne short 10045EDFh
0x10045eba: mov eax,[esi+5Ch]
0x10045ebd: mov edx,[esi+4]
0x10045ec0: add eax,ecx
0x10045ec2: mov ecx,[edx+4]
0x10045ec5: lea eax,[eax+eax*2]
0x10045ec8: add eax,eax
0x10045eca: push eax
0x10045ecb: push 1
0x10045ecd: push esi
0x10045ece: call ecx
0x10045ed0: add esp,0Ch
0x10045ed3: mov [edi+20h],eax
0x10045ed6: pop edi
0x10045ed7: mov eax,esi
0x10045ed9: pop esi
0x10045eda: jmp near ptr 10045B90h
0x10045edf: pop edi
```

```
0x10045ee0: pop esi
0x10045ee1: ret
```

`sub_10045da0` is called by sub_1003beb0. It calls nothing else.

12.886 sub_10045ef0 at 0x10045ef0 (85 instructions)

```
0x10045ef0: sub esp,0Ch
0x10045ef3: push ebx
0x10045ef4: push ebp
0x10045ef5: push esi
0x10045ef6: push edi
0x10045ef7: mov edi,[esp+20h]
0x10045efb: mov esi,[edi+64h]
0x10045efe: mov edx,[edi+54h]
0x10045f01: mov [esp+14h],esi
0x10045f05: mov [esp+18h],edx
0x10045f09: mov ebp,1
0x10045f0e: mov edi,edi
0x10045f10: add ebp,1
0x10045f13: cmp esi,1
0x10045f16: mov eax,ebp
0x10045f18: jle short 10045F28h
0x10045f1a: lea ecx,[esi-1]
0x10045f1d: lea ecx,[ecx]
0x10045f20: imul eax,ebp
0x10045f23: sub ecx,1
0x10045f26: jne short 10045F20h
0x10045f28: cmp eax,edx
0x10045f2a: jle short 10045F10h
0x10045f2c: sub ebp,1
0x10045f2f: cmp ebp,2
0x10045f32: jge short 10045F4Ch
0x10045f34: mov ecx,[edi]
0x10045f36: mov dword ptr [ecx+14h],3Ah
0x10045f3d: mov edx,[edi]
0x10045f3f: mov [edx+18h],eax
0x10045f42: mov eax,[edi]
0x10045f44: mov ecx,[eax]
0x10045f46: push edi
0x10045f47: call ecx
0x10045f49: add esp,4
0x10045f4c: test esi,esi
0x10045f4e: mov ebx,1
0x10045f53: jle short 10045F74h
0x10045f55: mov edi,[esp+24h]
0x10045f59: mov ecx,esi
0x10045f5b: mov eax,ebp
0x10045f5d: mov edx,esi
0x10045f5f: rep stosd
0x10045f61: imul ebx,ebp
0x10045f64: sub edx,1
0x10045f67: jne short 10045F61h
0x10045f69: jmp short 10045F74h
0x10045f70: mov esi,[esp+14h]
0x10045f74: xor ebp,ebp
0x10045f76: test esi,esi
0x10045f78: mov byte ptr [esp+13h],0
0x10045f7d: jle short 10045FCDh
0x10045f7f: nop
0x10045f80: mov edx,[esp+20h]
0x10045f84: cmp dword ptr [edx+2Ch],2
0x10045f88: jne short 10045F93h
0x10045f8a: mov edi,[ebp*4+10104620h]
0x10045f91: jmp short 10045F95h
0x10045f93: mov edi,ebp
```

```
0x10045f95: mov eax,[esp+24h]
0x10045f99: mov esi,[eax+edi*4]
0x10045f9c: mov eax,ebx
0x10045f9e: cdq
0x10045f9f: idiv esi
0x10045fa1: lea ecx,[esi+1]
0x10045fa4: imul eax,ecx
0x10045fa7: cmp eax,[esp+18h]
0x10045fab: jg short 10045FC6h
0x10045fad: mov edx,[esp+24h]
0x10045fb1: add ebp,1
0x10045fb4: cmp ebp,[esp+14h]
0x10045fb8: mov [edx+edi*4],ecx
0x10045fbb: mov ebx,eax
0x10045fbd: mov byte ptr [esp+13h],1
0x10045fc2: jl short 10045F80h
0x10045fc4: jmp short 10045F70h
0x10045fc6: cmp byte ptr [esp+13h],0
0x10045fcb: jne short 10045F70h
0x10045fcd: pop edi
0x10045fce: pop esi
0x10045fcf: pop ebp
0x10045fd0: mov eax,ebx
0x10045fd2: pop ebx
0x10045fd3: add esp,0Ch
0x10045fd6: ret
```

`sub_10045ef0` is called by sub_10046020. It calls nothing else.

12.887 sub_10045fe0 at 0x10045fe0 (10 instructions)

```
0x10045fe0: imul ecx,0FFh
0x10045fe6: mov eax,esi
0x10045fe8: cdq
0x10045fe9: sub eax,edx
0x10045feb: sar eax,1
0x10045fed: add ecx,eax
0x10045fef: mov eax,ecx
0x10045ff1: cdq
0x10045ff2: idiv esi
0x10045ff4: ret
```

`sub_10045fe0` is called by sub_10046020. It calls nothing else.

12.888 sub_10046000 at 0x10046000 (6 instructions)

```
0x10046000: imul eax,1FEh
0x10046006: lea eax,[eax+ecx+0FFh]
0x1004600d: cdq
0x1004600e: add ecx,ecx
0x10046010: idiv ecx
0x10046012: ret
```

`sub_10046000` is called by sub_10046190. It calls nothing else.

12.889 sub_10046020 at 0x10046020 (126 instructions)

```
0x10046020: sub esp,20h
0x10046023: push ebx
0x10046024: push ebp
0x10046025: mov ebp,[esp+2Ch]
0x10046029: mov ebx,[ebp+1B8h]
0x1004602f: push esi
0x10046030: push edi
```

```
0x10046031: lea esi,[ebx+20h]
0x10046034: push esi
0x10046035: push ebp
0x10046036: mov [esp+34h],ebx
0x1004603a: call 10045EF0h
0x1004603f: add esp,8
0x10046042: cmp dword ptr [ebp+64h],3
0x10046046: mov edi,eax
0x10046048: push 1
0x1004604a: mov [esp+20h],edi
0x1004604e: push ebp
0x1004604f: jne short 1004607Eh
0x10046051: mov eax,[ebp]
0x10046054: add eax,18h
0x10046057: mov [eax],edi
0x10046059: mov ecx,[esi]
0x1004605b: mov [eax+4],ecx
0x1004605e: mov edx,[ebx+24h]
0x10046061: mov [eax+8],edx
0x10046064: mov ecx,[ebx+28h]
0x10046067: mov [eax+0Ch],ecx
0x1004606a: mov edx,[ebp]
0x1004606d: mov dword ptr [edx+14h],60h
0x10046074: mov eax,[ebp]
0x10046077: mov ecx,[eax+4]
0x1004607a: call ecx
0x1004607c: jmp short 10046096h
0x1004607e: mov edx,[ebp]
0x10046081: mov dword ptr [edx+14h],61h
0x10046088: mov eax,[ebp]
0x1004608b: mov [eax+18h],edi
0x1004608e: mov ecx,[ebp]
0x10046091: mov edx,[ecx+4]
0x10046094: call edx
0x10046096: mov ecx,[ebp+64h]
0x10046099: mov eax,[ebp+4]
0x1004609c: mov edx,[eax+8]
0x1004609f: add esp,8
0x100460a2: push ecx
0x100460a3: push edi
0x100460a4: push 1
0x100460a6: push ebp
0x100460a7: call edx
0x100460a9: add esp,10h
0x100460ac: cmp dword ptr [ebp+64h],0
0x100460b0: mov ecx,eax
0x100460b2: mov [esp+28h],ecx
0x100460b6: mov [esp+14h],edi
0x100460ba: mov dword ptr [esp+24h],0
0x100460c2: jle near ptr 1004617Fh
0x100460c8: mov ebx,ecx
0x100460ca: mov [esp+18h],esi
0x100460ce: mov edi,edi
0x100460d0: mov eax,[esp+18h]
0x100460d4: mov esi,[eax]
0x100460d6: mov eax,[esp+14h]
0x100460da: cdq
0x100460db: idiv esi
0x100460dd: test esi,esi
0x100460df: mov [esp+20h],esi
0x100460e3: mov dword ptr [esp+10h],0
0x100460eb: mov edi,eax
0x100460ed: jle short 10046149h
0x100460ef: xor ebp,ebp
0x100460f1: jmp short 100460F7h
0x100460f3: mov esi,[esp+20h]
0x100460f7: mov ecx,[esp+10h]
```

```
0x100460fb: add esi,0FFFFFFFFh
0x100460fe: call 10045FE0h
0x10046103: cmp ebp,[esp+1Ch]
0x10046107: mov edx,ebp
0x10046109: jge short 1004612Eh
0x1004610b: jmp short 10046110h
0x10046110: xor ecx,ecx
0x10046112: test edi,edi
0x10046114: jle short 10046124h
0x10046116: mov esi,[ebx]
0x10046118: add esi,ecx
0x1004611a: add ecx,1
0x1004611d: cmp ecx,edi
0x1004611f: mov [esi+edx],al
0x10046122: jl short 10046116h
0x10046124: add edx,[esp+14h]
0x10046128: cmp edx,[esp+1Ch]
0x1004612c: jl short 10046110h
0x1004612e: mov eax,[esp+10h]
0x10046132: add eax,1
0x10046135: add ebp,edi
0x10046137: cmp eax,[esp+20h]
0x1004613b: mov [esp+10h],eax
0x1004613f: jl short 100460F3h
0x10046141: mov ebp,[esp+34h]
0x10046145: mov ecx,[esp+28h]
0x10046149: mov eax,[esp+24h]
0x1004614d: add dword ptr [esp+18h],4
0x10046152: add eax,1
0x10046155: add ebx,4
0x10046158: cmp eax,[ebp+64h]
0x1004615b: mov [esp+14h],edi
0x1004615f: mov [esp+24h],eax
0x10046163: jl near ptr 100460D0h
0x10046169: mov eax,[esp+2Ch]
0x1004616d: pop edi
0x1004616e: pop esi
0x1004616f: mov [eax+10h],ecx
0x10046172: mov ecx,[esp+14h]
0x10046176: pop ebp
0x10046177: mov [eax+14h],ecx
0x1004617a: pop ebx
0x1004617b: add esp,20h
0x1004617e: ret
0x1004617f: mov [ebx+14h],edi
0x10046182: pop edi
0x10046183: pop esi
0x10046184: pop ebp
0x10046185: mov [ebx+10h],ecx
0x10046188: pop ebx
0x10046189: add esp,20h
0x1004618c: ret
```

`sub_10046020` is called by sub_10046a70. It delegates to sub_10045ef0, sub_10045fe0 (2 callees).

12.890 sub_10046190 at 0x10046190 (99 instructions)

```
0x10046190: sub esp,14h
0x10046193: push esi
0x10046194: push edi
0x10046195: mov edi,[esp+20h]
0x10046199: cmp dword ptr [edi+4Ch],1
0x1004619d: mov esi,[edi+1B8h]
0x100461a3: mov [esp+18h],esi
0x100461a7: jne short 100461B7h
0x100461a9: mov dword ptr [esp+8],1FEh
```

```
0x100461b1: mov byte ptr [esi+1Ch],1
0x100461b5: jmp short 100461C3h
0x100461b7: mov dword ptr [esp+8],0
0x100461bf: mov byte ptr [esi+1Ch],0
0x100461c3: mov ecx,[edi+64h]
0x100461c6: mov edx,[esp+8]
0x100461ca: mov eax,[edi+4]
0x100461cd: mov eax,[eax+8]
0x100461d0: push ecx
0x100461d1: add edx,100h
0x100461d7: push edx
0x100461d8: push 1
0x100461da: push edi
0x100461db: call eax
0x100461dd: xor ecx,ecx
0x100461df: mov [esi+18h],eax
0x100461e2: mov eax,[esi+14h]
0x100461e5: add esp,10h
0x100461e8: cmp [edi+64h],ecx
0x100461eb: mov [esp+0Ch],ecx
0x100461ef: jle near ptr 100462CFh
0x100461f5: push ebx
0x100461f6: lea edx,[esi+20h]
0x100461f9: push ebp
0x100461fa: mov [esp+18h],edx
0x100461fe: jmp short 1004620Ch
0x10046200: mov eax,[esp+1Ch]
0x10046204: mov esi,[esp+20h]
0x10046208: mov ecx,[esp+14h]
0x1004620c: mov edx,[esp+18h]
0x10046210: mov ebp,[edx]
0x10046212: cdq
0x10046213: idiv ebp
0x10046215: cmp dword ptr [esp+10h],0
0x1004621a: mov [esp+1Ch],eax
0x1004621e: je short 1004622Dh
0x10046220: mov eax,[esi+18h]
0x10046223: add dword ptr [eax+ecx*4],0FFh
0x1004622a: lea eax,[eax+ecx*4]
0x1004622d: mov edx,[esi+18h]
0x10046230: mov edi,[edx+ecx*4]
0x10046233: add ebp,0FFFFFFFFh
0x10046236: mov ecx,ebp
0x10046238: xor eax,eax
0x1004623a: xor ebx,ebx
0x1004623c: call 10046000h
0x10046241: mov ecx,eax
0x10046243: xor esi,esi
0x10046245: cmp esi,ecx
0x10046247: jle short 10046262h
0x10046249: lea esp,[esp]
0x10046250: add ebx,1
0x10046253: mov ecx,ebp
0x10046255: mov eax,ebx
0x10046257: call 10046000h
0x1004625c: mov ecx,eax
0x1004625e: cmp esi,ecx
0x10046260: jg short 10046250h
0x10046262: mov al,[esp+1Ch]
0x10046266: imul bl
0x10046268: mov [esi+edi],al
0x1004626b: add esi,1
0x1004626e: cmp esi,0FFh
0x10046274: jle short 10046245h
0x10046276: cmp dword ptr [esp+10h],0
0x1004627b: je short 100462B0h
0x1004627d: mov eax,1
```

```
0x10046282: lea ecx,[edi-1]
0x10046285: jmp short 10046290h
0x10046290: movzx edx,byte ptr [edi]
0x10046293: mov [ecx],dl
0x10046295: movzx edx,byte ptr [edi+0FFh]
0x1004629c: mov [edi+eax+0FFh],dl
0x100462a3: add eax,1
0x100462a6: sub ecx,1
0x100462a9: cmp eax,0FFh
0x100462ae: jle short 10046290h
0x100462b0: mov eax,[esp+14h]
0x100462b4: mov ecx,[esp+28h]
0x100462b8: add dword ptr [esp+18h],4
0x100462bd: add eax,1
0x100462c0: cmp eax,[ecx+64h]
0x100462c3: mov [esp+14h],eax
0x100462c7: jl near ptr 10046200h
0x100462cd: pop ebp
0x100462ce: pop ebx
0x100462cf: pop edi
0x100462d0: pop esi
0x100462d1: add esp,14h
0x100462d4: ret
```

`sub_10046190` is called by sub_10046a70. It delegates to sub_10046000 (1 callees).

12.891 sub_10046920 at 0x10046920 (27 instructions)

```
0x10046920: mov eax,[esi+1B8h]
0x10046926: push ebp
0x10046927: mov ebp,[esi+5Ch]
0x1004692a: push edi
0x1004692b: xor edi,edi
0x1004692d: cmp [esi+64h],edi
0x10046930: lea ebp,[ebp+ebp+4]
0x10046934: jle short 1004695Dh
0x10046936: push ebx
0x10046937: lea ebx,[eax+44h]
0x1004693a: lea ebx,[ebx]
0x10046940: mov eax,[esi+4]
0x10046943: mov ecx,[eax+4]
0x10046946: push ebp
0x10046947: push 1
0x10046949: push esi
0x1004694a: call ecx
0x1004694c: mov [ebx],eax
0x1004694e: add edi,1
0x10046951: add esp,0Ch
0x10046954: add ebx,4
0x10046957: cmp edi,[esi+64h]
0x1004695a: jl short 10046940h
0x1004695c: pop ebx
0x1004695d: pop edi
0x1004695e: pop ebp
0x1004695f: ret
```

`sub_10046920` is called by sub_10046a70. It calls nothing else.

12.892 sub_10046a70 at 0x10046a70 (50 instructions)

```
0x10046a70: push esi
0x10046a71: mov esi,[esp+8]
0x10046a75: mov eax,[esi+4]
0x10046a78: mov ecx,[eax]
0x10046a7a: push 58h
```

```
0x10046a7c: push 1
0x10046a7e: push esi
0x10046a7f: call ecx
0x10046a81: mov [esi+1B8h],eax
0x10046a87: xor ecx,ecx
0x10046a89: mov dword ptr [eax],10046960h
0x10046a8f: mov dword ptr [eax+8],100B3980h
0x10046a96: mov dword ptr [eax+0Ch],10046A50h
0x10046a9d: mov [eax+44h],ecx
0x10046aa0: mov [eax+34h],ecx
0x10046aa3: mov eax,4
0x10046aa8: add esp,0Ch
0x10046aab: cmp [esi+64h],eax
0x10046aae: jle short 10046AC8h
0x10046ab0: mov edx,[esi]
0x10046ab2: mov dword ptr [edx+14h],39h
0x10046ab9: mov ecx,[esi]
0x10046abb: mov [ecx+18h],eax
0x10046abe: mov edx,[esi]
0x10046ac0: mov eax,[edx]
0x10046ac2: push esi
0x10046ac3: call eax
0x10046ac5: add esp,4
0x10046ac8: mov eax,100h
0x10046acd: cmp [esi+54h],eax
0x10046ad0: jle short 10046AEAh
0x10046ad2: mov ecx,[esi]
0x10046ad4: mov dword ptr [ecx+14h],3Bh
0x10046adb: mov edx,[esi]
0x10046add: mov [edx+18h],eax
0x10046ae0: mov eax,[esi]
0x10046ae2: mov ecx,[eax]
0x10046ae4: push esi
0x10046ae5: call ecx
0x10046ae7: add esp,4
0x10046aea: push esi
0x10046aeb: call 10046020h
0x10046af0: push esi
0x10046af1: call 10046190h
0x10046af6: add esp,8
0x10046af9: cmp dword ptr [esi+4Ch],2
0x10046afd: jne short 10046B04h
0x10046aff: call 10046920h
0x10046b04: pop esi
0x10046b05: ret
```

`sub_10046a70` is called by sub_1003beb0. It delegates to sub_10046020, sub_10046190, sub_10046920 (3 callees).

12.893 sub_10047610 at 0x10047610 (5 instructions)

```
0x10047610: mov eax,[esp+8]
0x10047614: push eax
0x10047615: call dword ptr ds:[100FE550h]
0x1004761b: add esp,4
0x1004761e: ret
```

`sub_10047610` is called by sub_1003eb30. It calls nothing else.

12.894 sub_1005fdf0 at 0x1005fdf0 (174 instructions)

```
0x1005fb50: push esi
0x1005fb51: mov esi,edx
0x1005fb53: test esi,esi
0x1005fb55: not eax
0x1005fb57: je short 1005FB82h
```

```
0x1005fb59: lea esp,[esp]
0x1005fb60: test cl,3
0x1005fb63: je short 1005FB82h
0x1005fb65: movzx edx,byte ptr [ecx]
0x1005fb68: xor edx,eax
0x1005fb6a: and edx,0FFh
0x1005fb70: shr eax,8
0x1005fb73: xor eax,[edx*4+10104908h]
0x1005fb7a: add ecx,1
0x1005fb7d: sub esi,1
0x1005fb80: jne short 1005FB60h
0x1005fb82: cmp esi,20h
0x1005fb85: push ebx
0x1005fb86: push edi
0x1005fb87: jb near ptr 1005FD6Ch
0x1005fb8d: mov edi,esi
0x1005fb8f: shr edi,5
0x1005fb92: xor eax,[ecx]
0x1005fb94: add ecx,4
0x1005fb97: mov edx,eax
0x1005fb99: shr edx,10h
0x1005fb9c: movzx ebx,ah
0x1005fb9f: and edx,0FFh
0x1005fba5: mov edx,[edx*4+10104D08h]
0x1005fbac: xor edx,[ebx*4+10105108h]
0x1005fbb3: mov ebx,eax
0x1005fbb5: and eax,0FFh
0x1005fbba: shr ebx,18h
0x1005fbbd: xor edx,[ebx*4+10104908h]
0x1005fbc4: add ecx,4
0x1005fbc7: xor edx,[eax*4+10105508h]
0x1005fbce: add ecx,4
0x1005fbd1: xor edx,[ecx-8]
0x1005fbd4: add ecx,4
0x1005fbd7: mov eax,edx
0x1005fbd9: shr eax,10h
0x1005fbdc: and eax,0FFh
0x1005fbe1: mov eax,[eax*4+10104D08h]
0x1005fbe8: movzx ebx,dh
0x1005fbeb: xor eax,[ebx*4+10105108h]
0x1005fbf2: mov ebx,edx
0x1005fbf4: and edx,0FFh
0x1005fbfa: shr ebx,18h
0x1005fbfd: xor eax,[ebx*4+10104908h]
0x1005fc04: add ecx,4
0x1005fc07: xor eax,[edx*4+10105508h]
0x1005fc0e: add ecx,4
0x1005fc11: xor eax,[ecx-10h]
0x1005fc14: mov edx,eax
0x1005fc16: shr edx,10h
0x1005fc19: movzx ebx,ah
0x1005fc1c: and edx,0FFh
0x1005fc22: mov edx,[edx*4+10104D08h]
0x1005fc29: xor edx,[ebx*4+10105108h]
0x1005fc30: mov ebx,eax
0x1005fc32: and eax,0FFh
0x1005fc37: shr ebx,18h
0x1005fc3a: xor edx,[ebx*4+10104908h]
0x1005fc41: xor edx,[eax*4+10105508h]
0x1005fc48: xor edx,[ecx-0Ch]
0x1005fc4b: mov eax,edx
0x1005fc4d: shr eax,10h
0x1005fc50: and eax,0FFh
0x1005fc55: mov eax,[eax*4+10104D08h]
0x1005fc5c: movzx ebx,dh
0x1005fc5f: xor eax,[ebx*4+10105108h]
0x1005fc66: mov ebx,edx
```

```
0x1005fc68: and edx,0FFh
0x1005fc6e: shr ebx,18h
0x1005fc71: xor eax,[ebx*4+10104908h]
0x1005fc78: xor eax,[edx*4+10105508h]
0x1005fc7f: xor eax,[ecx-8]
0x1005fc82: mov edx,eax
0x1005fc84: shr edx,10h
0x1005fc87: movzx ebx,ah
0x1005fc8a: and edx,0FFh
0x1005fc90: mov edx,[edx*4+10104D08h]
0x1005fc97: xor edx,[ebx*4+10105108h]
0x1005fc9e: mov ebx,eax
0x1005fca0: and eax,0FFh
0x1005fca5: shr ebx,18h
0x1005fca8: xor edx,[ebx*4+10104908h]
0x1005fcaf: xor edx,[eax*4+10105508h]
0x1005fcb6: xor edx,[ecx-4]
0x1005fcb9: mov eax,edx
0x1005fcbb: shr eax,10h
0x1005fcbe: and eax,0FFh
0x1005fcc3: mov eax,[eax*4+10104D08h]
0x1005fcca: movzx ebx,dh
0x1005fccd: xor eax,[ebx*4+10105108h]
0x1005fcd4: mov ebx,edx
0x1005fcd6: shr ebx,18h
0x1005fcd9: xor eax,[ebx*4+10104908h]
0x1005fce0: and edx,0FFh
0x1005fce6: xor eax,[edx*4+10105508h]
0x1005fced: xor eax,[ecx]
0x1005fcef: add ecx,4
0x1005fcf2: mov edx,eax
0x1005fcf4: shr edx,10h
0x1005fcf7: movzx ebx,ah
0x1005fcfa: and edx,0FFh
0x1005fd00: mov edx,[edx*4+10104D08h]
0x1005fd07: xor edx,[ebx*4+10105108h]
0x1005fd0e: mov ebx,eax
0x1005fd10: and eax,0FFh
0x1005fd15: shr ebx,18h
0x1005fd18: xor edx,[ebx*4+10104908h]
0x1005fd1f: add ecx,4
0x1005fd22: xor edx,[eax*4+10105508h]
0x1005fd29: sub esi,20h
0x1005fd2c: xor edx,[ecx-4]
0x1005fd2f: mov eax,edx
0x1005fd31: shr eax,10h
0x1005fd34: and eax,0FFh
0x1005fd39: mov eax,[eax*4+10104D08h]
0x1005fd40: movzx ebx,dh
0x1005fd43: xor eax,[ebx*4+10105108h]
0x1005fd4a: mov ebx,edx
0x1005fd4c: shr ebx,18h
0x1005fd4f: xor eax,[ebx*4+10104908h]
0x1005fd56: and edx,0FFh
0x1005fd5c: xor eax,[edx*4+10105508h]
0x1005fd63: sub edi,1
0x1005fd66: jne near ptr 1005FB92h
0x1005fd6c: cmp esi,4
0x1005fd6f: jb short 1005FDB9h
0x1005fd71: mov edx,esi
0x1005fd73: shr edx,2
0x1005fd76: xor eax,[ecx]
0x1005fd78: add ecx,4
0x1005fd7b: mov edi,eax
0x1005fd7d: shr edi,10h
0x1005fd80: and edi,0FFh
0x1005fd86: mov edi,[edi*4+10104D08h]
```

```
0x1005fd8d: movzx ebx,ah
0x1005fd90: xor edi,[ebx*4+10105108h]
0x1005fd97: mov ebx,eax
0x1005fd99: shr ebx,18h
0x1005fd9c: xor edi,[ebx*4+10104908h]
0x1005fda3: and eax,0FFh
0x1005fda8: xor edi,[eax*4+10105508h]
0x1005fdaf: sub esi,4
0x1005fdb2: sub edx,1
0x1005fdb5: mov eax,edi
0x1005fdb7: jne short 1005FD76h
0x1005fdb9: test esi,esi
0x1005fdbb: pop edi
0x1005fdbc: pop ebx
0x1005fdbd: je short 1005FDDDh
0x1005fdbf: nop
0x1005fdc0: movzx edx,byte ptr [ecx]
0x1005fdc3: xor edx,eax
0x1005fdc5: and edx,0FFh
0x1005fdcb: shr eax,8
0x1005fdce: xor eax,[edx*4+10104908h]
0x1005fdd5: add ecx,1
0x1005fdd8: sub esi,1
0x1005fddb: jne short 1005FDC0h
0x1005fddd: not eax
0x1005fddf: pop esi
0x1005fde0: ret
0x1005fdf0: mov ecx,[esp+8]
0x1005fdf4: test ecx,ecx
0x1005fdf6: jne short 1005FDFBh
0x1005fdf8: xor eax,eax
0x1005fdfa: ret
0x1005fdfb: mov edx,[esp+0Ch]
0x1005fdff: mov eax,[esp+4]
0x1005fe03: jmp near ptr 1005FB50h
```

`sub_1005fdf0` is called by sub_100234e0, sub_10023500, sub_10061a80, sub_100629f0, sub_10062a80. It calls nothing else.

12.895 sub_1005fe10 at 0x1005fe10 (32 instructions)

```
0x1005fe10: mov edx,[esp+4]
0x1005fe14: xor ecx,ecx
0x1005fe16: cmp edx,ecx
0x1005fe18: je short 1005FE7Bh
0x1005fe1a: mov eax,[edx+1Ch]
0x1005fe1d: cmp eax,ecx
0x1005fe1f: je short 1005FE7Bh
0x1005fe21: mov [eax+1Ch],ecx
0x1005fe24: mov [edx+14h],ecx
0x1005fe27: mov [edx+8],ecx
0x1005fe2a: mov [edx+18h],ecx
0x1005fe2d: mov dword ptr [edx+30h],1
0x1005fe34: mov [eax],ecx
0x1005fe36: mov [eax+4],ecx
0x1005fe39: mov [eax+0Ch],ecx
0x1005fe3c: mov [eax+20h],ecx
0x1005fe3f: mov [eax+28h],ecx
0x1005fe42: mov [eax+2Ch],ecx
0x1005fe45: mov [eax+30h],ecx
0x1005fe48: mov [eax+38h],ecx
0x1005fe4b: mov [eax+3Ch],ecx
0x1005fe4e: lea ecx,[eax+530h]
0x1005fe54: mov dword ptr [eax+14h],8000h
0x1005fe5b: mov [eax+6Ch],ecx
0x1005fe5e: mov [eax+50h],ecx
```

```
0x1005fe61: mov [eax+4Ch],ecx
0x1005fe64: mov dword ptr [eax+1BC0h],1
0x1005fe6e: mov dword ptr [eax+1BC4h],0FFFFFFFFh
0x1005fe78: xor eax,eax
0x1005fe7a: ret
0x1005fe7b: mov eax,0FFFFFFFEh
0x1005fe80: ret
```

`sub_1005fe10` is called by sub_1002cff0, sub_100309c0, sub_1005fe90. It calls nothing else.

12.896 sub_1005fe90 at 0x1005fe90 (60 instructions)

```
0x1005fe90: push ebp
0x1005fe91: mov ebp,[esp+8]
0x1005fe95: test ebp,ebp
0x1005fe97: push edi
0x1005fe98: je short 1005FF0Dh
0x1005fe9a: mov edi,[ebp+1Ch]
0x1005fe9d: test edi,edi
0x1005fe9f: je short 1005FF0Dh
0x1005fea1: push ebx
0x1005fea2: push esi
0x1005fea3: mov esi,[esp+18h]
0x1005fea7: test esi,esi
0x1005fea9: jge short 1005FEB1h
0x1005feab: xor ebx,ebx
0x1005fead: neg esi
0x1005feaf: jmp short 1005FEC1h
0x1005feb1: mov ebx,esi
0x1005feb3: sar ebx,4
0x1005feb6: add ebx,1
0x1005feb9: cmp esi,30h
0x1005febc: jge short 1005FEC1h
0x1005febe: and esi,0Fh
0x1005fec1: test esi,esi
0x1005fec3: je short 1005FED9h
0x1005fec5: cmp esi,8
0x1005fec8: jl short 1005FECFh
0x1005feca: cmp esi,0Fh
0x1005fecd: jle short 1005FED9h
0x1005fecf: pop esi
0x1005fed0: pop ebx
0x1005fed1: pop edi
0x1005fed2: mov eax,0FFFFFFFEh
0x1005fed7: pop ebp
0x1005fed8: ret
0x1005fed9: mov eax,[edi+34h]
0x1005fedc: test eax,eax
0x1005fede: je short 1005FEF9h
0x1005fee0: cmp [edi+24h],esi
0x1005fee3: je short 1005FEF9h
0x1005fee5: mov ecx,[ebp+24h]
0x1005fee8: push eax
0x1005fee9: mov eax,[ebp+28h]
0x1005feec: push eax
0x1005feed: call ecx
0x1005feef: add esp,8
0x1005fef2: mov dword ptr [edi+34h],0
0x1005fef9: push ebp
0x1005fefa: mov [edi+8],ebx
0x1005fefd: mov [edi+24h],esi
0x1005ff00: call 1005FE10h
0x1005ff05: add esp,4
0x1005ff08: pop esi
0x1005ff09: pop ebx
0x1005ff0a: pop edi
```

```
0x1005ff0b: pop ebp
0x1005ff0c: ret
0x1005ff0d: pop edi
0x1005ff0e: mov eax,0FFFFFFFEh
0x1005ff13: pop ebp
0x1005ff14: ret
```

`sub_1005fe90` is called by sub_1005ff20. It delegates to sub_1005fe10 (1 callees).

12.897 sub_1005ff20 at 0x1005ff20 (68 instructions)

```
0x1005ff20: mov eax,[esp+0Ch]
0x1005ff24: push ebp
0x1005ff25: xor ebp,ebp
0x1005ff27: cmp eax,ebp
0x1005ff29: je near ptr 1005FFC5h
0x1005ff2f: cmp byte ptr [eax],31h
0x1005ff32: jne near ptr 1005FFC5h
0x1005ff38: cmp dword ptr [esp+14h],38h
0x1005ff3d: jne near ptr 1005FFC5h
0x1005ff43: push esi
0x1005ff44: mov esi,[esp+0Ch]
0x1005ff48: cmp esi,ebp
0x1005ff4a: jne short 1005FF52h
0x1005ff4c: pop esi
0x1005ff4d: lea eax,[ebp-2]
0x1005ff50: pop ebp
0x1005ff51: ret
0x1005ff52: cmp [esi+20h],ebp
0x1005ff55: mov [esi+18h],ebp
0x1005ff58: jne short 1005FF64h
0x1005ff5a: mov dword ptr [esi+20h],100635F0h
0x1005ff61: mov [esi+28h],ebp
0x1005ff64: cmp [esi+24h],ebp
0x1005ff67: jne short 1005FF70h
0x1005ff69: mov dword ptr [esi+24h],10047600h
0x1005ff70: mov eax,[esi+28h]
0x1005ff73: mov ecx,[esi+20h]
0x1005ff76: push edi
0x1005ff77: push 1BCCh
0x1005ff7c: push 1
0x1005ff7e: push eax
0x1005ff7f: call ecx
0x1005ff81: mov edi,eax
0x1005ff83: add esp,0Ch
0x1005ff86: cmp edi,ebp
0x1005ff88: jne short 1005FF93h
0x1005ff8a: pop edi
0x1005ff8b: pop esi
0x1005ff8c: mov eax,0FFFFFFFCh
0x1005ff91: pop ebp
0x1005ff92: ret
0x1005ff93: mov edx,[esp+14h]
0x1005ff97: push ebx
0x1005ff98: push edx
0x1005ff99: mov [esi+1Ch],edi
0x1005ff9c: push esi
0x1005ff9d: mov [edi+34h],ebp
0x1005ffa0: call 1005FE90h
0x1005ffa5: mov ebx,eax
0x1005ffa7: add esp,8
0x1005ffaa: cmp ebx,ebp
0x1005ffac: je short 1005FFBEh
0x1005ffae: mov eax,[esi+28h]
0x1005ffb1: mov ecx,[esi+24h]
0x1005ffb4: push edi
```

```
0x1005ffb5: push eax
0x1005ffb6: call ecx
0x1005ffb8: add esp,8
0x1005ffbb: mov [esi+1Ch],ebp
0x1005ffbe: mov eax,ebx
0x1005ffc0: pop ebx
0x1005ffc1: pop edi
0x1005ffc2: pop esi
0x1005ffc3: pop ebp
0x1005ffc4: ret
0x1005ffc5: mov eax,0FFFFFFFAh
0x1005ffca: pop ebp
0x1005ffcb: ret
```

`sub_1005ff20` is called by sub_1005ffd0. It delegates to sub_1005fe90 (1 callees).

12.898 sub_1005ffd0 at 0x1005ffd0 (10 instructions)

```
0x1005ffd0: mov eax,[esp+0Ch]
0x1005ffd4: mov ecx,[esp+8]
0x1005ffd8: mov edx,[esp+4]
0x1005ffdc: push eax
0x1005ffdd: push ecx
0x1005ffde: push 0Fh
0x1005ffe0: push edx
0x1005ffe1: call 1005FF20h
0x1005ffe6: add esp,10h
0x1005ffe9: ret
```

`sub_1005ffd0` is called by png_create_read_struct_2. It delegates to sub_1005ff20 (1 callees).

12.899 sub_10060100 at 0x10060100 (49 instructions)

```
0x10060100: mov eax,[esp+4]
0x10060104: sub esp,2Ch
0x10060107: test eax,eax
0x10060109: push edi
0x1006010a: je near ptr 100616F3h
0x10060110: mov edi,[eax+1Ch]
0x10060113: test edi,edi
0x10060115: je near ptr 100616F3h
0x1006011b: cmp dword ptr [eax+0Ch],0
0x1006011f: je near ptr 100616F3h
0x10060125: cmp dword ptr [eax],0
0x10060128: jne short 10060134h
0x1006012a: cmp dword ptr [eax+4],0
0x1006012e: jne near ptr 100616F3h
0x10060134: cmp dword ptr [edi],0Bh
0x10060137: jne short 1006013Fh
0x10060139: mov dword ptr [edi],0Ch
0x1006013f: mov ecx,[eax+0Ch]
0x10060142: mov edx,[eax+4]
0x10060145: push ebx
0x10060146: mov ebx,[edi+38h]
0x10060149: mov [esp+20h],ecx
0x1006014d: mov ecx,[eax+10h]
0x10060150: push ebp
0x10060151: mov ebp,[eax]
0x10060153: mov eax,ecx
0x10060155: mov [esp+18h],ecx
0x10060159: mov ecx,[edi]
0x1006015b: cmp ecx,1Eh
0x1006015e: push esi
0x1006015f: mov esi,[edi+3Ch]
0x10060162: mov [esp+10h],edx
```

```
0x10060166: mov [esp+18h],ebx
0x1006016a: mov [esp+38h],edx
0x1006016e: mov [esp+2Ch],eax
0x10060172: mov dword ptr [esp+30h],0
0x1006017a: jbe short 10060198h
0x1006017c: mov eax,0FFFFFFFEh
0x10060181: pop esi
0x10060182: pop ebp
0x10060183: pop ebx
0x10060184: pop edi
0x10060185: add esp,2Ch
0x10060188: ret
0x10060198: jmp dword ptr [ecx*4+10061700h]
0x100616f3: mov eax,0FFFFFFFEh
0x100616f8: pop edi
0x100616f9: add esp,2Ch
0x100616fc: ret
```

`sub_10060100` is called by png_read_row, sub_1002cff0, sub_100309c0. It calls nothing else.

12.900 sub_10061790 at 0x10061790 (32 instructions)

```
0x10061790: push esi
0x10061791: mov esi,[esp+8]
0x10061795: test esi,esi
0x10061797: je short 100617D3h
0x10061799: mov eax,[esi+1Ch]
0x1006179c: test eax,eax
0x1006179e: je short 100617D3h
0x100617a0: mov ecx,[esi+24h]
0x100617a3: test ecx,ecx
0x100617a5: je short 100617D3h
0x100617a7: mov eax,[eax+34h]
0x100617aa: test eax,eax
0x100617ac: je short 100617B8h
0x100617ae: push eax
0x100617af: mov eax,[esi+28h]
0x100617b2: push eax
0x100617b3: call ecx
0x100617b5: add esp,8
0x100617b8: mov ecx,[esi+1Ch]
0x100617bb: mov edx,[esi+28h]
0x100617be: mov eax,[esi+24h]
0x100617c1: push ecx
0x100617c2: push edx
0x100617c3: call eax
0x100617c5: add esp,8
0x100617c8: mov dword ptr [esi+1Ch],0
0x100617cf: xor eax,eax
0x100617d1: pop esi
0x100617d2: ret
0x100617d3: mov eax,0FFFFFFFEh
0x100617d8: pop esi
0x100617d9: ret
```

`sub_10061790` is called by sub_100280e0, sub_1008d0a0. It calls nothing else.

12.901 sub_10061920 at 0x10061920 (16 instructions)

```
0x10061920: push esi
0x10061921: mov esi,[eax+8]
0x10061924: push edi
0x10061925: mov edi,[eax+14h]
0x10061928: mov edx,ecx
0x1006192a: shr edx,8
```

```
0x1006192d: mov [esi+edi],dl
0x10061930: mov edi,[eax+8]
0x10061933: mov esi,1
0x10061938: add [eax+14h],esi
0x1006193b: mov edx,[eax+14h]
0x1006193e: mov [edx+edi],cl
0x10061941: add [eax+14h],esi
0x10061944: pop edi
0x10061945: pop esi
0x10061946: ret
```

`sub_10061920` is called by sub_10062a80. It calls nothing else.

12.902 sub_10061950 at 0x10061950 (33 instructions)

```
0x10061950: push esi
0x10061951: mov esi,eax
0x10061953: mov eax,[esi+1Ch]
0x10061956: mov ecx,[esi+10h]
0x10061959: push edi
0x1006195a: mov edi,[eax+14h]
0x1006195d: cmp edi,ecx
0x1006195f: jbe short 10061963h
0x10061961: mov edi,ecx
0x10061963: test edi,edi
0x10061965: je short 1006199Ch
0x10061967: mov eax,[eax+10h]
0x1006196a: mov ecx,[esi+0Ch]
0x1006196d: push edi
0x1006196e: push eax
0x1006196f: push ecx
0x10061970: call 100228C6h
0x10061975: mov eax,[esi+1Ch]
0x10061978: add [esi+0Ch],edi
0x1006197b: add [eax+10h],edi
0x1006197e: add [esi+14h],edi
0x10061981: sub [esi+10h],edi
0x10061984: mov eax,[esi+1Ch]
0x10061987: sub [eax+14h],edi
0x1006198a: mov esi,[esi+1Ch]
0x1006198d: add esp,0Ch
0x10061990: cmp dword ptr [esi+14h],0
0x10061994: jne short 1006199Ch
0x10061996: mov edx,[esi+8]
0x10061999: mov [esi+10h],edx
0x1006199c: pop edi
0x1006199d: pop esi
0x1006199e: ret
```

`sub_10061950` is called by sub_10062640, sub_100628c0, sub_10062a80. It delegates to j_memcpy (1 callees).

12.903 sub_100619a0 at 0x100619a0 (85 instructions)

```
0x100619a0: push esi
0x100619a1: mov esi,[esp+8]
0x100619a5: test esi,esi
0x100619a7: je near ptr 10061A6Bh
0x100619ad: mov eax,[esi+1Ch]
0x100619b0: test eax,eax
0x100619b2: je near ptr 10061A6Bh
0x100619b8: push edi
0x100619b9: mov edi,[eax+4]
0x100619bc: cmp edi,2Ah
0x100619bf: je short 100619EAh
0x100619c1: cmp edi,45h
```

```
0x100619c4: je short 100619EAh
0x100619c6: cmp edi,49h
0x100619c9: je short 100619EAh
0x100619cb: cmp edi,5Bh
0x100619ce: je short 100619EAh
0x100619d0: cmp edi,67h
0x100619d3: je short 100619EAh
0x100619d5: cmp edi,71h
0x100619d8: je short 100619EAh
0x100619da: cmp edi,29Ah
0x100619e0: je short 100619EAh
0x100619e2: pop edi
0x100619e3: mov eax,0FFFFFFFEh
0x100619e8: pop esi
0x100619e9: ret
0x100619ea: mov eax,[eax+8]
0x100619ed: test eax,eax
0x100619ef: je short 100619FEh
0x100619f1: mov ecx,[esi+24h]
0x100619f4: push eax
0x100619f5: mov eax,[esi+28h]
0x100619f8: push eax
0x100619f9: call ecx
0x100619fb: add esp,8
0x100619fe: mov edx,[esi+1Ch]
0x10061a01: mov eax,[edx+44h]
0x10061a04: test eax,eax
0x10061a06: je short 10061A15h
0x10061a08: mov ecx,[esi+24h]
0x10061a0b: push eax
0x10061a0c: mov eax,[esi+28h]
0x10061a0f: push eax
0x10061a10: call ecx
0x10061a12: add esp,8
0x10061a15: mov edx,[esi+1Ch]
0x10061a18: mov eax,[edx+40h]
0x10061a1b: test eax,eax
0x10061a1d: je short 10061A2Ch
0x10061a1f: mov ecx,[esi+24h]
0x10061a22: push eax
0x10061a23: mov eax,[esi+28h]
0x10061a26: push eax
0x10061a27: call ecx
0x10061a29: add esp,8
0x10061a2c: mov edx,[esi+1Ch]
0x10061a2f: mov eax,[edx+38h]
0x10061a32: test eax,eax
0x10061a34: je short 10061A43h
0x10061a36: mov ecx,[esi+24h]
0x10061a39: push eax
0x10061a3a: mov eax,[esi+28h]
0x10061a3d: push eax
0x10061a3e: call ecx
0x10061a40: add esp,8
0x10061a43: mov edx,[esi+1Ch]
0x10061a46: mov eax,[esi+28h]
0x10061a49: mov ecx,[esi+24h]
0x10061a4c: push edx
0x10061a4d: push eax
0x10061a4e: call ecx
0x10061a50: add esp,8
0x10061a53: xor eax,eax
0x10061a55: cmp edi,71h
0x10061a58: setne al
0x10061a5b: pop edi
0x10061a5c: mov dword ptr [esi+1Ch],0
0x10061a63: pop esi
```

```
0x10061a64: sub eax,1
0x10061a67: and eax,0FFFFFFFDh
0x10061a6a: ret
0x10061a6b: mov eax,0FFFFFFFEh
0x10061a70: pop esi
0x10061a71: ret
```

`sub_100619a0` is called by sub_1002b220, sub_1008d1b0. It calls nothing else.

12.904 sub_10061a80 at 0x10061a80 (46 instructions)

```
0x10061a80: mov eax,[esi+4]
0x10061a83: push edi
0x10061a84: mov edi,eax
0x10061a86: cmp edi,ecx
0x10061a88: jbe short 10061A8Ch
0x10061a8a: mov edi,ecx
0x10061a8c: test edi,edi
0x10061a8e: jne short 10061A94h
0x10061a90: xor eax,eax
0x10061a92: pop edi
0x10061a93: ret
0x10061a94: sub eax,edi
0x10061a96: mov [esi+4],eax
0x10061a99: mov eax,[esi+1Ch]
0x10061a9c: mov eax,[eax+18h]
0x10061a9f: cmp eax,1
0x10061aa2: jne short 10061AB3h
0x10061aa4: mov ecx,[esi]
0x10061aa6: mov edx,[esi+30h]
0x10061aa9: push edi
0x10061aaa: push ecx
0x10061aab: push edx
0x10061aac: call 10063FB0h
0x10061ab1: jmp short 10061AC5h
0x10061ab3: cmp eax,2
0x10061ab6: jne short 10061ACBh
0x10061ab8: mov eax,[esi]
0x10061aba: mov ecx,[esi+30h]
0x10061abd: push edi
0x10061abe: push eax
0x10061abf: push ecx
0x10061ac0: call 1005FDF0h
0x10061ac5: mov [esi+30h],eax
0x10061ac8: add esp,0Ch
0x10061acb: mov edx,[esi]
0x10061acd: mov eax,[esp+8]
0x10061ad1: push edi
0x10061ad2: push edx
0x10061ad3: push eax
0x10061ad4: call 100228C6h
0x10061ad9: add [esi],edi
0x10061adb: add [esi+8],edi
0x10061ade: add esp,0Ch
0x10061ae1: mov eax,edi
0x10061ae3: pop edi
0x10061ae4: ret
```

`sub_10061a80` is called by sub_10061d00. It delegates to j_memcpy, sub_1005fdf0, sub_10063fb0 (3 callees).

12.905 sub_10061af0 at 0x10061af0 (38 instructions)

```
0x10061af0: mov eax,[esi+2Ch]
0x10061af3: mov ecx,[esi+4Ch]
0x10061af6: mov edx,[esi+44h]
```

```
0x10061af9: push edi
0x10061afa: add eax,eax
0x10061afc: mov [esi+3Ch],eax
0x10061aff: xor edi,edi
0x10061b01: mov [edx+ecx*2-2],di
0x10061b06: mov eax,[esi+4Ch]
0x10061b09: mov edx,[esi+44h]
0x10061b0c: lea ecx,[eax+eax-2]
0x10061b10: push ecx
0x10061b11: push edi
0x10061b12: push edx
0x10061b13: call 100228C0h
0x10061b18: mov eax,[esi+84h]
0x10061b1e: lea eax,[eax+eax*2]
0x10061b21: add eax,eax
0x10061b23: movzx ecx,word ptr [eax+eax+101073E2h]
0x10061b2b: add eax,eax
0x10061b2d: mov [esi+80h],ecx
0x10061b33: movzx edx,word ptr [eax+101073E0h]
0x10061b3a: mov [esi+8Ch],edx
0x10061b40: movzx ecx,word ptr [eax+101073E4h]
0x10061b47: mov [esi+90h],ecx
0x10061b4d: movzx edx,word ptr [eax+101073E6h]
0x10061b54: mov eax,2
0x10061b59: add esp,0Ch
0x10061b5c: mov [esi+6Ch],edi
0x10061b5f: mov [esi+5Ch],edi
0x10061b62: mov [esi+74h],edi
0x10061b65: mov [esi+68h],edi
0x10061b68: mov [esi+48h],edi
0x10061b6b: mov [esi+7Ch],edx
0x10061b6e: mov [esi+78h],eax
0x10061b71: mov [esi+60h],eax
0x10061b74: pop edi
0x10061b75: ret
```

`sub_10061af0` is called by sub_100629f0. It delegates to j_memset (1 callees).

12.906 sub_10061d00 at 0x10061d00 (126 instructions)

```
0x10061d00: push ebx
0x10061d01: mov ebx,[edi+2Ch]
0x10061d04: push ebp
0x10061d05: push esi
0x10061d06: mov ebp,[edi+3Ch]
0x10061d09: sub ebp,[edi+74h]
0x10061d0c: mov eax,[edi+6Ch]
0x10061d0f: mov ecx,[edi+2Ch]
0x10061d12: lea edx,[ebx+ecx-106h]
0x10061d19: sub ebp,eax
0x10061d1b: cmp eax,edx
0x10061d1d: jb short 10061D7Eh
0x10061d1f: mov eax,[edi+38h]
0x10061d22: push ebx
0x10061d23: lea ecx,[eax+ebx]
0x10061d26: push ecx
0x10061d27: push eax
0x10061d28: call 100228C6h
0x10061d2d: mov edx,[edi+4Ch]
0x10061d30: mov eax,[edi+44h]
0x10061d33: sub [edi+70h],ebx
0x10061d36: sub [edi+6Ch],ebx
0x10061d39: add esp,0Ch
0x10061d3c: sub [edi+5Ch],ebx
0x10061d3f: lea ecx,[eax+edx*2]
0x10061d42: movzx eax,word ptr [ecx-2]
```

```
0x10061d46: sub ecx,2
0x10061d49: cmp eax,ebx
0x10061d4b: jb short 10061D51h
0x10061d4d: sub eax,ebx
0x10061d4f: jmp short 10061D53h
0x10061d51: xor eax,eax
0x10061d53: sub edx,1
0x10061d56: mov [ecx],ax
0x10061d59: jne short 10061D42h
0x10061d5b: mov ecx,[edi+40h]
0x10061d5e: mov edx,ebx
0x10061d60: lea ecx,[ecx+ebx*2]
0x10061d63: movzx eax,word ptr [ecx-2]
0x10061d67: sub ecx,2
0x10061d6a: cmp eax,ebx
0x10061d6c: jb short 10061D72h
0x10061d6e: sub eax,ebx
0x10061d70: jmp short 10061D74h
0x10061d72: xor eax,eax
0x10061d74: sub edx,1
0x10061d77: mov [ecx],ax
0x10061d7a: jne short 10061D63h
0x10061d7c: add ebp,ebx
0x10061d7e: mov esi,[edi]
0x10061d80: cmp dword ptr [esi+4],0
0x10061d84: je near ptr 10061E52h
0x10061d8a: mov edx,[edi+74h]
0x10061d8d: add edx,[edi+6Ch]
0x10061d90: mov ecx,ebp
0x10061d92: add edx,[edi+38h]
0x10061d95: push edx
0x10061d96: call 10061A80h
0x10061d9b: add [edi+74h],eax
0x10061d9e: mov edx,[edi+74h]
0x10061da1: add esp,4
0x10061da4: cmp edx,3
0x10061da7: jb short 10061DC9h
0x10061da9: mov eax,[edi+6Ch]
0x10061dac: mov ecx,[edi+38h]
0x10061daf: lea esi,[eax+ecx]
0x10061db2: movzx eax,byte ptr [esi]
0x10061db5: mov ecx,[edi+58h]
0x10061db8: mov [edi+48h],eax
0x10061dbb: shl eax,cl
0x10061dbd: movzx ecx,byte ptr [esi+1]
0x10061dc1: xor eax,ecx
0x10061dc3: and eax,[edi+54h]
0x10061dc6: mov [edi+48h],eax
0x10061dc9: cmp edx,106h
0x10061dcf: jae short 10061DDDh
0x10061dd1: mov eax,[edi]
0x10061dd3: cmp dword ptr [eax+4],0
0x10061dd7: jne near ptr 10061D06h
0x10061ddd: mov eax,[edi+16C0h]
0x10061de3: mov ecx,[edi+3Ch]
0x10061de6: cmp eax,ecx
0x10061de8: jae short 10061E52h
0x10061dea: mov esi,[edi+6Ch]
0x10061ded: add esi,edx
0x10061def: cmp eax,esi
0x10061df1: jae short 10061E21h
0x10061df3: sub ecx,esi
0x10061df5: mov ebx,ecx
0x10061df7: cmp ebx,102h
0x10061dfd: jbe short 10061E04h
0x10061dff: mov ebx,102h
0x10061e04: mov ecx,[edi+38h]
```

```
0x10061e07: push ebx
0x10061e08: add ecx,esi
0x10061e0a: push 0
0x10061e0c: push ecx
0x10061e0d: call 100228C0h
0x10061e12: add esp,0Ch
0x10061e15: add ebx,esi
0x10061e17: pop esi
0x10061e18: pop ebp
0x10061e19: mov [edi+16C0h],ebx
0x10061e1f: pop ebx
0x10061e20: ret
0x10061e21: lea edx,[esi+102h]
0x10061e27: cmp eax,edx
0x10061e29: jae short 10061E52h
0x10061e2b: sub esi,eax
0x10061e2d: add esi,102h
0x10061e33: sub ecx,eax
0x10061e35: cmp esi,ecx
0x10061e37: jbe short 10061E3Bh
0x10061e39: mov esi,ecx
0x10061e3b: mov ecx,[edi+38h]
0x10061e3e: push esi
0x10061e3f: add ecx,eax
0x10061e41: push 0
0x10061e43: push ecx
0x10061e44: call 100228C0h
0x10061e49: add esp,0Ch
0x10061e4c: add [edi+16C0h],esi
0x10061e52: pop esi
0x10061e53: pop ebp
0x10061e54: pop ebx
0x10061e55: ret
```

`sub_10061d00` is called by sub_10062640, sub_100628c0. It delegates to j_memset, j_memcpy, sub_10061a80 (3 callees).

12.907 sub_10062640 at 0x10062640 (203 instructions)

```
0x10062640: push ebx
0x10062641: push ebp
0x10062642: push esi
0x10062643: push edi
0x10062644: mov edi,eax
0x10062646: mov esi,1
0x1006264b: jmp short 10062650h
0x10062650: mov ebp,[edi+74h]
0x10062653: cmp ebp,102h
0x10062659: jae short 1006267Fh
0x1006265b: call 10061D00h
0x10062660: mov ebp,[edi+74h]
0x10062663: cmp ebp,102h
0x10062669: mov ebx,[esp+14h]
0x1006266d: jae short 10062677h
0x1006266f: test ebx,ebx
0x10062671: je near ptr 1006284Dh
0x10062677: test ebp,ebp
0x10062679: je near ptr 10062854h
0x1006267f: cmp ebp,3
0x10062682: mov dword ptr [edi+60h],0
0x10062689: jb near ptr 10062735h
0x1006268f: mov eax,[edi+6Ch]
0x10062692: test eax,eax
0x10062694: jbe near ptr 10062735h
0x1006269a: mov ecx,[edi+38h]
0x1006269d: lea edx,[ecx+eax]
```

```
0x100626a0: movzx ecx,byte ptr [edx-1]
0x100626a4: lea eax,[edx-1]
0x100626a7: movzx ebx,byte ptr [eax+esi]
0x100626ab: add eax,esi
0x100626ad: cmp ecx,ebx
0x100626af: jne near ptr 10062735h
0x100626b5: movzx ebx,byte ptr [eax+esi]
0x100626b9: add eax,esi
0x100626bb: cmp ecx,ebx
0x100626bd: jne short 10062735h
0x100626bf: movzx ebx,byte ptr [eax+esi]
0x100626c3: add eax,esi
0x100626c5: cmp ecx,ebx
0x100626c7: jne short 10062735h
0x100626c9: add edx,102h
0x100626cf: nop
0x100626d0: movzx ebx,byte ptr [eax+esi]
0x100626d4: add eax,esi
0x100626d6: cmp ecx,ebx
0x100626d8: jne short 10062724h
0x100626da: movzx ebx,byte ptr [eax+esi]
0x100626de: add eax,esi
0x100626e0: cmp ecx,ebx
0x100626e2: jne short 10062724h
0x100626e4: movzx ebx,byte ptr [eax+esi]
0x100626e8: add eax,esi
0x100626ea: cmp ecx,ebx
0x100626ec: jne short 10062724h
0x100626ee: movzx ebx,byte ptr [eax+esi]
0x100626f2: add eax,esi
0x100626f4: cmp ecx,ebx
0x100626f6: jne short 10062724h
0x100626f8: movzx ebx,byte ptr [eax+esi]
0x100626fc: add eax,esi
0x100626fe: cmp ecx,ebx
0x10062700: jne short 10062724h
0x10062702: movzx ebx,byte ptr [eax+esi]
0x10062706: add eax,esi
0x10062708: cmp ecx,ebx
0x1006270a: jne short 10062724h
0x1006270c: movzx ebx,byte ptr [eax+esi]
0x10062710: add eax,esi
0x10062712: cmp ecx,ebx
0x10062714: jne short 10062724h
0x10062716: movzx ebx,byte ptr [eax+esi]
0x1006271a: add eax,esi
0x1006271c: cmp ecx,ebx
0x1006271e: jne short 10062724h
0x10062720: cmp eax,edx
0x10062722: jb short 100626D0h
0x10062724: sub eax,edx
0x10062726: add eax,102h
0x1006272b: cmp eax,ebp
0x1006272d: mov [edi+60h],eax
0x10062730: jbe short 10062735h
0x10062732: mov [edi+60h],ebp
0x10062735: cmp dword ptr [edi+60h],3
0x10062739: mov edx,[edi+16A0h]
0x1006273f: jb short 100627B4h
0x10062741: mov al,[edi+60h]
0x10062744: mov ecx,[edi+16A4h]
0x1006274a: mov [ecx+edx*2],si
0x1006274e: mov ecx,[edi+16A0h]
0x10062754: mov edx,[edi+1698h]
0x1006275a: sub al,3
0x1006275c: mov [edx+ecx],al
0x1006275f: add [edi+16A0h],esi
```

```
0x10062765: movzx edx,al
0x10062768: movzx eax,byte ptr [edx+10107E80h]
0x1006276f: add [edi+eax*4+498h],si
0x10062777: lea eax,[edi+eax*4+498h]
0x1006277e: movzx eax,byte ptr ds:[10107C80h]
0x10062785: add [edi+eax*4+988h],si
0x1006278d: mov ecx,[edi+169Ch]
0x10062793: mov eax,[edi+60h]
0x10062796: sub ecx,esi
0x10062798: xor edx,edx
0x1006279a: cmp [edi+16A0h],ecx
0x100627a0: mov dword ptr [edi+60h],0
0x100627a7: sete dl
0x100627aa: sub [edi+74h],eax
0x100627ad: add [edi+6Ch],eax
0x100627b0: mov ecx,edx
0x100627b2: jmp short 1006280Ah
0x100627b4: mov ecx,[edi+6Ch]
0x100627b7: mov eax,[edi+38h]
0x100627ba: mov al,[eax+ecx]
0x100627bd: mov ecx,[edi+16A4h]
0x100627c3: mov word ptr [ecx+edx*2],0
0x100627c9: mov ecx,[edi+16A0h]
0x100627cf: mov edx,[edi+1698h]
0x100627d5: mov [edx+ecx],al
0x100627d8: add [edi+16A0h],esi
0x100627de: movzx edx,al
0x100627e1: add [edi+edx*4+94h],si
0x100627e9: lea eax,[edi+edx*4+94h]
0x100627f0: mov eax,[edi+169Ch]
0x100627f6: sub eax,esi
0x100627f8: xor ecx,ecx
0x100627fa: cmp [edi+16A0h],eax
0x10062800: sete cl
0x10062803: add dword ptr [edi+74h],0FFFFFFFFh
0x10062807: add [edi+6Ch],esi
0x1006280a: test ecx,ecx
0x1006280c: je near ptr 10062650h
0x10062812: mov ecx,[edi+5Ch]
0x10062815: test ecx,ecx
0x10062817: jl short 10062820h
0x10062819: mov eax,[edi+38h]
0x1006281c: add eax,ecx
0x1006281e: jmp short 10062822h
0x10062820: xor eax,eax
0x10062822: mov edx,[edi+6Ch]
0x10062825: push 0
0x10062827: sub edx,ecx
0x10062829: push edx
0x1006282a: push eax
0x1006282b: push edi
0x1006282c: call 10065A70h
0x10062831: mov eax,[edi+6Ch]
0x10062834: mov [edi+5Ch],eax
0x10062837: mov eax,[edi]
0x10062839: add esp,10h
0x1006283c: call 10061950h
0x10062841: mov ecx,[edi]
0x10062843: cmp dword ptr [ecx+10h],0
0x10062847: jne near ptr 10062650h
0x1006284d: pop edi
0x1006284e: pop esi
0x1006284f: pop ebp
0x10062850: xor eax,eax
0x10062852: pop ebx
0x10062853: ret
0x10062854: mov ecx,[edi+5Ch]
```

```
0x10062857: test ecx,ecx
0x10062859: jl short 10062862h
0x1006285b: mov eax,[edi+38h]
0x1006285e: add eax,ecx
0x10062860: jmp short 10062864h
0x10062862: xor eax,eax
0x10062864: xor edx,edx
0x10062866: cmp ebx,4
0x10062869: sete dl
0x1006286c: push edx
0x1006286d: mov edx,[edi+6Ch]
0x10062870: sub edx,ecx
0x10062872: push edx
0x10062873: push eax
0x10062874: push edi
0x10062875: call 10065A70h
0x1006287a: mov eax,[edi+6Ch]
0x1006287d: mov [edi+5Ch],eax
0x10062880: mov eax,[edi]
0x10062882: add esp,10h
0x10062885: call 10061950h
0x1006288a: mov ecx,[edi]
0x1006288c: xor eax,eax
0x1006288e: cmp [ecx+10h],eax
0x10062891: jne short 100628A4h
0x10062893: cmp ebx,4
0x10062896: setne al
0x10062899: pop edi
0x1006289a: pop esi
0x1006289b: pop ebp
0x1006289c: pop ebx
0x1006289d: sub eax,1
0x100628a0: and eax,2
0x100628a3: ret
0x100628a4: cmp ebx,4
0x100628a7: sete al
0x100628aa: pop edi
0x100628ab: pop esi
0x100628ac: pop ebp
0x100628ad: pop ebx
0x100628ae: lea eax,[eax+eax+1]
0x100628b2: ret
```

`sub_10062640` is called by sub_10062a80. It delegates to sub_10061950, sub_10061d00, sub_10065a70 (3 callees).

12.908 sub_100628c0 at 0x100628c0 (103 instructions)

```
0x100628c0: push ebx
0x100628c1: push ebp
0x100628c2: push edi
0x100628c3: mov edi,eax
0x100628c5: mov ebx,1
0x100628ca: or ebp,0FFFFFFFFh
0x100628cd: lea ecx,[ecx]
0x100628d0: cmp dword ptr [edi+74h],0
0x100628d4: jne short 100628E5h
0x100628d6: call 10061D00h
0x100628db: cmp dword ptr [edi+74h],0
0x100628df: je near ptr 10062985h
0x100628e5: mov eax,[edi+6Ch]
0x100628e8: mov ecx,[edi+38h]
0x100628eb: mov edx,[edi+16A0h]
0x100628f1: mov dword ptr [edi+60h],0
0x100628f8: mov al,[eax+ecx]
0x100628fb: mov ecx,[edi+16A4h]
0x10062901: mov word ptr [ecx+edx*2],0
```

```
0x10062907: mov ecx,[edi+16A0h]
0x1006290d: mov edx,[edi+1698h]
0x10062913: mov [edx+ecx],al
0x10062916: add [edi+16A0h],ebx
0x1006291c: movzx edx,al
0x1006291f: add [edi+edx*4+94h],bx
0x10062927: mov ecx,[edi+169Ch]
0x1006292d: add [edi+6Ch],ebx
0x10062930: add [edi+74h],ebp
0x10062933: lea eax,[edi+edx*4+94h]
0x1006293a: mov eax,[edi+6Ch]
0x1006293d: sub ecx,ebx
0x1006293f: cmp [edi+16A0h],ecx
0x10062945: jne short 100628D0h
0x10062947: mov edx,[edi+5Ch]
0x1006294a: test edx,edx
0x1006294c: jl short 10062955h
0x1006294e: mov ecx,[edi+38h]
0x10062951: add ecx,edx
0x10062953: jmp short 10062957h
0x10062955: xor ecx,ecx
0x10062957: push 0
0x10062959: sub eax,edx
0x1006295b: push eax
0x1006295c: push ecx
0x1006295d: push edi
0x1006295e: call 10065A70h
0x10062963: mov edx,[edi+6Ch]
0x10062966: mov eax,[edi]
0x10062968: add esp,10h
0x1006296b: mov [edi+5Ch],edx
0x1006296e: call 10061950h
0x10062973: mov eax,[edi]
0x10062975: cmp dword ptr [eax+10h],0
0x10062979: jne near ptr 100628D0h
0x1006297f: pop edi
0x10062980: pop ebp
0x10062981: xor eax,eax
0x10062983: pop ebx
0x10062984: ret
0x10062985: mov ebp,[esp+10h]
0x10062989: test ebp,ebp
0x1006298b: je short 1006297Fh
0x1006298d: mov ecx,[edi+5Ch]
0x10062990: test ecx,ecx
0x10062992: jl short 1006299Bh
0x10062994: mov eax,[edi+38h]
0x10062997: add eax,ecx
0x10062999: jmp short 1006299Dh
0x1006299b: xor eax,eax
0x1006299d: xor edx,edx
0x1006299f: cmp ebp,4
0x100629a2: sete dl
0x100629a5: push edx
0x100629a6: mov edx,[edi+6Ch]
0x100629a9: sub edx,ecx
0x100629ab: push edx
0x100629ac: push eax
0x100629ad: push edi
0x100629ae: call 10065A70h
0x100629b3: mov eax,[edi+6Ch]
0x100629b6: mov [edi+5Ch],eax
0x100629b9: mov eax,[edi]
0x100629bb: add esp,10h
0x100629be: call 10061950h
0x100629c3: mov ecx,[edi]
0x100629c5: xor eax,eax
```

```
0x100629c7: cmp [ecx+10h],eax
0x100629ca: jne short 100629DCh
0x100629cc: cmp ebp,4
0x100629cf: setne al
0x100629d2: pop edi
0x100629d3: pop ebp
0x100629d4: pop ebx
0x100629d5: sub eax,1
0x100629d8: and eax,2
0x100629db: ret
0x100629dc: cmp ebp,4
0x100629df: sete al
0x100629e2: pop edi
0x100629e3: pop ebp
0x100629e4: pop ebx
0x100629e5: lea eax,[eax+eax+1]
0x100629e9: ret
```

`sub_100628c0` is called by sub_10062a80. It delegates to sub_10061950, sub_10061d00, sub_10065a70 (3 callees).

12.909 sub_100629f0 at 0x100629f0 (58 instructions)

```
0x100629f0: push ebx
0x100629f1: push esi
0x100629f2: push edi
0x100629f3: mov edi,[esp+10h]
0x100629f7: xor ebx,ebx
0x100629f9: cmp edi,ebx
0x100629fb: je short 10062A76h
0x100629fd: mov esi,[edi+1Ch]
0x10062a00: cmp esi,ebx
0x10062a02: je short 10062A76h
0x10062a04: cmp [edi+20h],ebx
0x10062a07: je short 10062A76h
0x10062a09: cmp [edi+24h],ebx
0x10062a0c: je short 10062A76h
0x10062a0e: mov [edi+14h],ebx
0x10062a11: mov [edi+8],ebx
0x10062a14: mov [edi+18h],ebx
0x10062a17: mov dword ptr [edi+2Ch],2
0x10062a1e: mov eax,[esi+8]
0x10062a21: mov [esi+10h],eax
0x10062a24: mov eax,[esi+18h]
0x10062a27: cmp eax,ebx
0x10062a29: mov [esi+14h],ebx
0x10062a2c: jge short 10062A33h
0x10062a2e: neg eax
0x10062a30: mov [esi+18h],eax
0x10062a33: mov eax,[esi+18h]
0x10062a36: mov ecx,eax
0x10062a38: neg ecx
0x10062a3a: sbb ecx,ecx
0x10062a3c: and ecx,0FFFFFFB9h
0x10062a3f: push ebx
0x10062a40: add ecx,71h
0x10062a43: cmp eax,2
0x10062a46: push ebx
0x10062a47: mov [esi+4],ecx
0x10062a4a: push ebx
0x10062a4b: jne short 10062A54h
0x10062a4d: call 1005FDF0h
0x10062a52: jmp short 10062A59h
0x10062a54: call 10063FB0h
0x10062a59: add esp,0Ch
0x10062a5c: mov [edi+30h],eax
0x10062a5f: push esi
```

```
0x10062a60: mov [esi+28h],ebx
0x10062a63: call 10065420h
0x10062a68: add esp,4
0x10062a6b: call 10061AF0h
0x10062a70: pop edi
0x10062a71: pop esi
0x10062a72: xor eax,eax
0x10062a74: pop ebx
0x10062a75: ret
0x10062a76: pop edi
0x10062a77: pop esi
0x10062a78: mov eax,0FFFFFFFEh
0x10062a7d: pop ebx
0x10062a7e: ret
```

`sub_100629f0` is called by sub_10035de0, sub_10037d80. It delegates to sub_1005fdf0, sub_10061af0, sub_10063fb0 (4 callees).

12.910 sub_10062a80 at 0x10062a80 (690 instructions)

```
0x10062a80: push esi
0x10062a81: push edi
0x10062a82: mov edi,[esp+0Ch]
0x10062a86: test edi,edi
0x10062a88: je near ptr 100632A8h
0x10062a8e: mov esi,[edi+1Ch]
0x10062a91: test esi,esi
0x10062a93: je near ptr 100632A8h
0x10062a99: mov ecx,[esp+10h]
0x10062a9d: cmp ecx,5
0x10062aa0: ja near ptr 100632A8h
0x10062aa6: cmp dword ptr [edi+0Ch],0
0x10062aaa: je near ptr 100632A0h
0x10062ab0: cmp dword ptr [edi],0
0x10062ab3: jne short 10062ABFh
0x10062ab5: cmp dword ptr [edi+4],0
0x10062ab9: jne near ptr 100632A0h
0x10062abf: mov eax,[esi+4]
0x10062ac2: cmp eax,29Ah
0x10062ac7: jne short 10062AD2h
0x10062ac9: cmp ecx,4
0x10062acc: jne near ptr 100632A0h
0x10062ad2: cmp dword ptr [edi+10h],0
0x10062ad6: jne short 10062AE8h
0x10062ad8: mov eax,ds:[101074FCh]
0x10062add: mov [edi+18h],eax
0x10062ae0: pop edi
0x10062ae1: mov eax,0FFFFFFFBh
0x10062ae6: pop esi
0x10062ae7: ret
0x10062ae8: cmp eax,2Ah
0x10062aeb: mov edx,[esi+28h]
0x10062aee: push ebx
0x10062aef: push ebp
0x10062af0: mov [esi],edi
0x10062af2: mov [esp+14h],edx
0x10062af6: mov [esi+28h],ecx
0x10062af9: mov ebx,1
0x10062afe: jne near ptr 10062DA6h
0x10062b04: mov eax,2
0x10062b09: cmp [esi+18h],eax
0x10062b0c: jne near ptr 10062D10h
0x10062b12: push 0
0x10062b14: push 0
0x10062b16: push 0
0x10062b18: call 1005FDF0h
```

```
0x10062b1d: mov [edi+30h],eax
0x10062b20: mov ecx,[esi+14h]
0x10062b23: mov eax,[esi+8]
0x10062b26: mov byte ptr [eax+ecx],1Fh
0x10062b2a: add [esi+14h],ebx
0x10062b2d: mov eax,[esi+14h]
0x10062b30: mov edx,[esi+8]
0x10062b33: mov byte ptr [eax+edx],8Bh
0x10062b37: add [esi+14h],ebx
0x10062b3a: mov eax,[esi+14h]
0x10062b3d: mov ecx,[esi+8]
0x10062b40: mov byte ptr [eax+ecx],8
0x10062b44: add [esi+14h],ebx
0x10062b47: mov eax,[esi+1Ch]
0x10062b4a: mov ebp,[esi+14h]
0x10062b4d: add esp,0Ch
0x10062b50: test eax,eax
0x10062b52: jne near ptr 10062BE1h
0x10062b58: mov edx,[esi+8]
0x10062b5b: mov [edx+ebp],al
0x10062b5e: add [esi+14h],ebx
0x10062b61: mov eax,[esi+14h]
0x10062b64: mov ecx,[esi+8]
0x10062b67: mov byte ptr [eax+ecx],0
0x10062b6b: add [esi+14h],ebx
0x10062b6e: mov eax,[esi+14h]
0x10062b71: mov edx,[esi+8]
0x10062b74: mov byte ptr [eax+edx],0
0x10062b78: add [esi+14h],ebx
0x10062b7b: mov eax,[esi+14h]
0x10062b7e: mov ecx,[esi+8]
0x10062b81: mov byte ptr [eax+ecx],0
0x10062b85: add [esi+14h],ebx
0x10062b88: mov eax,[esi+14h]
0x10062b8b: mov edx,[esi+8]
0x10062b8e: mov byte ptr [eax+edx],0
0x10062b92: add [esi+14h],ebx
0x10062b95: mov eax,[esi+84h]
0x10062b9b: cmp eax,9
0x10062b9e: mov ecx,[esi+14h]
0x10062ba1: jne short 10062BA8h
0x10062ba3: lea eax,[ebx+1]
0x10062ba6: jmp short 10062BBFh
0x10062ba8: cmp dword ptr [esi+88h],2
0x10062baf: jge short 10062BBAh
0x10062bb1: cmp eax,2
0x10062bb4: jl short 10062BBAh
0x10062bb6: xor eax,eax
0x10062bb8: jmp short 10062BBFh
0x10062bba: mov eax,4
0x10062bbf: mov edx,[esi+8]
0x10062bc2: mov [ecx+edx],al
0x10062bc5: add [esi+14h],ebx
0x10062bc8: mov eax,[esi+14h]
0x10062bcb: mov ecx,[esi+8]
0x10062bce: mov byte ptr [eax+ecx],0Bh
0x10062bd2: add [esi+14h],ebx
0x10062bd5: mov dword ptr [esi+4],71h
0x10062bdc: jmp near ptr 10062DA6h
0x10062be1: mov edx,[eax+24h]
0x10062be4: mov ecx,[eax+2Ch]
0x10062be7: neg edx
0x10062be9: sbb dl,dl
0x10062beb: and edx,10h
0x10062bee: neg ecx
0x10062bf0: sbb cl,cl
0x10062bf2: and ecx,2
```

```
0x10062bf5: add dl,cl
0x10062bf7: mov ecx,[eax+1Ch]
0x10062bfa: neg ecx
0x10062bfc: sbb cl,cl
0x10062bfe: and ecx,8
0x10062c01: add dl,cl
0x10062c03: mov ecx,[eax+10h]
0x10062c06: neg ecx
0x10062c08: sbb cl,cl
0x10062c0a: and ecx,4
0x10062c0d: add dl,cl
0x10062c0f: cmp dword ptr [eax],0
0x10062c12: mov ecx,[esi+8]
0x10062c15: setne al
0x10062c18: add dl,al
0x10062c1a: mov [ecx+ebp],dl
0x10062c1d: add [esi+14h],ebx
0x10062c20: mov eax,[esi+14h]
0x10062c23: mov edx,[esi+1Ch]
0x10062c26: movzx edx,byte ptr [edx+4]
0x10062c2a: mov ecx,[esi+8]
0x10062c2d: mov [eax+ecx],dl
0x10062c30: add [esi+14h],ebx
0x10062c33: mov ecx,[esi+1Ch]
0x10062c36: movzx ecx,byte ptr [ecx+5]
0x10062c3a: mov eax,[esi+14h]
0x10062c3d: mov edx,[esi+8]
0x10062c40: mov [eax+edx],cl
0x10062c43: add [esi+14h],ebx
0x10062c46: mov edx,[esi+1Ch]
0x10062c49: movzx edx,byte ptr [edx+6]
0x10062c4d: mov eax,[esi+14h]
0x10062c50: mov ecx,[esi+8]
0x10062c53: mov [eax+ecx],dl
0x10062c56: add [esi+14h],ebx
0x10062c59: mov ecx,[esi+1Ch]
0x10062c5c: movzx ecx,byte ptr [ecx+7]
0x10062c60: mov eax,[esi+14h]
0x10062c63: mov edx,[esi+8]
0x10062c66: mov [eax+edx],cl
0x10062c69: add [esi+14h],ebx
0x10062c6c: mov eax,[esi+84h]
0x10062c72: cmp eax,9
0x10062c75: mov ecx,[esi+14h]
0x10062c78: jne short 10062C81h
0x10062c7a: mov eax,2
0x10062c7f: jmp short 10062C98h
0x10062c81: cmp dword ptr [esi+88h],2
0x10062c88: jge short 10062C93h
0x10062c8a: cmp eax,2
0x10062c8d: jl short 10062C93h
0x10062c8f: xor eax,eax
0x10062c91: jmp short 10062C98h
0x10062c93: mov eax,4
0x10062c98: mov edx,[esi+8]
0x10062c9b: mov [ecx+edx],al
0x10062c9e: add [esi+14h],ebx
0x10062ca1: mov ecx,[esi+1Ch]
0x10062ca4: mov eax,[esi+14h]
0x10062ca7: mov cl,[ecx+0Ch]
0x10062caa: mov edx,[esi+8]
0x10062cad: mov [eax+edx],cl
0x10062cb0: add [esi+14h],ebx
0x10062cb3: mov eax,[esi+1Ch]
0x10062cb6: cmp dword ptr [eax+10h],0
0x10062cba: mov ecx,[esi+14h]
0x10062cbd: je short 10062CE0h
```

```
0x10062cbf: mov al,[eax+14h]
0x10062cc2: mov edx,[esi+8]
0x10062cc5: mov [ecx+edx],al
0x10062cc8: add [esi+14h],ebx
0x10062ccb: mov ecx,[esi+1Ch]
0x10062cce: mov cl,[ecx+15h]
0x10062cd1: mov eax,[esi+14h]
0x10062cd4: mov edx,[esi+8]
0x10062cd7: mov [eax+edx],cl
0x10062cda: add [esi+14h],ebx
0x10062cdd: mov ecx,[esi+14h]
0x10062ce0: mov edx,[esi+1Ch]
0x10062ce3: cmp dword ptr [edx+2Ch],0
0x10062ce7: je short 10062CFDh
0x10062ce9: mov eax,[esi+8]
0x10062cec: push ecx
0x10062ced: mov ecx,[edi+30h]
0x10062cf0: push eax
0x10062cf1: push ecx
0x10062cf2: call 1005FDF0h
0x10062cf7: add esp,0Ch
0x10062cfa: mov [edi+30h],eax
0x10062cfd: mov dword ptr [esi+20h],0
0x10062d04: mov dword ptr [esi+4],45h
0x10062d0b: jmp near ptr 10062DA6h
0x10062d10: mov ecx,[esi+30h]
0x10062d13: sub ecx,8
0x10062d16: shl ecx,0Ch
0x10062d19: add ecx,800h
0x10062d1f: cmp [esi+88h],eax
0x10062d25: jge short 10062D46h
0x10062d27: mov ebp,[esi+84h]
0x10062d2d: cmp ebp,eax
0x10062d2f: jl short 10062D46h
0x10062d31: cmp ebp,6
0x10062d34: jge short 10062D3Ah
0x10062d36: mov eax,ebx
0x10062d38: jmp short 10062D48h
0x10062d3a: xor edx,edx
0x10062d3c: cmp ebp,6
0x10062d3f: setne dl
0x10062d42: add eax,edx
0x10062d44: jmp short 10062D48h
0x10062d46: xor eax,eax
0x10062d48: shl eax,6
0x10062d4b: or ecx,eax
0x10062d4d: cmp dword ptr [esi+6Ch],0
0x10062d51: je short 10062D56h
0x10062d53: or ecx,20h
0x10062d56: mov eax,8421085h
0x10062d5b: mul ecx
0x10062d5d: sub ecx,edx
0x10062d5f: shr ecx,1
0x10062d61: add ecx,edx
0x10062d63: shr ecx,4
0x10062d66: add ecx,ebx
0x10062d68: mov eax,ecx
0x10062d6a: shl ecx,5
0x10062d6d: sub ecx,eax
0x10062d6f: mov eax,esi
0x10062d71: mov dword ptr [esi+4],71h
0x10062d78: call 10061920h
0x10062d7d: cmp dword ptr [esi+6Ch],0
0x10062d81: je short 10062D95h
0x10062d83: movzx ecx,word ptr [edi+32h]
0x10062d87: call 10061920h
0x10062d8c: movzx ecx,word ptr [edi+30h]
```

```
0x10062d90: call 10061920h
0x10062d95: push 0
0x10062d97: push 0
0x10062d99: push 0
0x10062d9b: call 10063FB0h
0x10062da0: add esp,0Ch
0x10062da3: mov [edi+30h],eax
0x10062da6: cmp dword ptr [esi+4],45h
0x10062daa: jne near ptr 10062E78h
0x10062db0: mov eax,[esi+1Ch]
0x10062db3: cmp dword ptr [eax+10h],0
0x10062db7: je near ptr 10062E71h
0x10062dbd: movzx eax,word ptr [eax+14h]
0x10062dc1: cmp [esi+20h],eax
0x10062dc4: mov ecx,[esi+14h]
0x10062dc7: jae short 10062E37h
0x10062dc9: lea esp,[esp]
0x10062dd0: mov eax,[esi+14h]
0x10062dd3: cmp eax,[esi+0Ch]
0x10062dd6: jne short 10062E0Eh
0x10062dd8: mov edx,[esi+1Ch]
0x10062ddb: cmp dword ptr [edx+2Ch],0
0x10062ddf: je short 10062DFDh
0x10062de1: cmp eax,ecx
0x10062de3: jbe short 10062DFDh
0x10062de5: sub eax,ecx
0x10062de7: push eax
0x10062de8: mov eax,[esi+8]
0x10062deb: add eax,ecx
0x10062ded: mov ecx,[edi+30h]
0x10062df0: push eax
0x10062df1: push ecx
0x10062df2: call 1005FDF0h
0x10062df7: add esp,0Ch
0x10062dfa: mov [edi+30h],eax
0x10062dfd: mov eax,edi
0x10062dff: call 10061950h
0x10062e04: mov eax,[esi+14h]
0x10062e07: cmp eax,[esi+0Ch]
0x10062e0a: mov ecx,eax
0x10062e0c: je short 10062E37h
0x10062e0e: mov edx,[esi+1Ch]
0x10062e11: mov edx,[edx+10h]
0x10062e14: mov ebx,[esi+20h]
0x10062e17: mov dl,[edx+ebx]
0x10062e1a: mov ebp,[esi+8]
0x10062e1d: mov [eax+ebp],dl
0x10062e20: mov eax,1
0x10062e25: add [esi+14h],eax
0x10062e28: add [esi+20h],eax
0x10062e2b: mov eax,[esi+1Ch]
0x10062e2e: movzx edx,word ptr [eax+14h]
0x10062e32: cmp [esi+20h],edx
0x10062e35: jb short 10062DD0h
0x10062e37: mov eax,[esi+1Ch]
0x10062e3a: cmp dword ptr [eax+2Ch],0
0x10062e3e: je short 10062E5Fh
0x10062e40: mov eax,[esi+14h]
0x10062e43: cmp eax,ecx
0x10062e45: jbe short 10062E5Fh
0x10062e47: mov edx,[esi+8]
0x10062e4a: sub eax,ecx
0x10062e4c: push eax
0x10062e4d: mov eax,[edi+30h]
0x10062e50: add edx,ecx
0x10062e52: push edx
0x10062e53: push eax
```

```
0x10062e54: call 1005FDF0h
0x10062e59: add esp,0Ch
0x10062e5c: mov [edi+30h],eax
0x10062e5f: mov ecx,[esi+1Ch]
0x10062e62: mov edx,[esi+20h]
0x10062e65: cmp edx,[ecx+14h]
0x10062e68: jne short 10062E78h
0x10062e6a: mov dword ptr [esi+20h],0
0x10062e71: mov dword ptr [esi+4],49h
0x10062e78: cmp dword ptr [esi+4],49h
0x10062e7c: mov ebp,5Bh
0x10062e81: jne near ptr 10062F2Fh
0x10062e87: mov eax,[esi+1Ch]
0x10062e8a: cmp dword ptr [eax+1Ch],0
0x10062e8e: je near ptr 10062F2Ch
0x10062e94: mov edx,[esi+14h]
0x10062e97: mov eax,[esi+14h]
0x10062e9a: cmp eax,[esi+0Ch]
0x10062e9d: jne short 10062ED5h
0x10062e9f: mov ecx,[esi+1Ch]
0x10062ea2: cmp dword ptr [ecx+2Ch],0
0x10062ea6: je short 10062EC4h
0x10062ea8: cmp eax,edx
0x10062eaa: jbe short 10062EC4h
0x10062eac: mov ecx,[edi+30h]
0x10062eaf: sub eax,edx
0x10062eb1: push eax
0x10062eb2: mov eax,[esi+8]
0x10062eb5: add eax,edx
0x10062eb7: push eax
0x10062eb8: push ecx
0x10062eb9: call 1005FDF0h
0x10062ebe: add esp,0Ch
0x10062ec1: mov [edi+30h],eax
0x10062ec4: mov eax,edi
0x10062ec6: call 10061950h
0x10062ecb: mov eax,[esi+14h]
0x10062ece: cmp eax,[esi+0Ch]
0x10062ed1: mov edx,eax
0x10062ed3: je short 10062EF8h
0x10062ed5: mov ecx,[esi+20h]
0x10062ed8: mov ebx,[esi+1Ch]
0x10062edb: mov ebx,[ebx+1Ch]
0x10062ede: movzx ebx,byte ptr [ebx+ecx]
0x10062ee2: add ecx,1
0x10062ee5: mov [esi+20h],ecx
0x10062ee8: mov ecx,[esi+8]
0x10062eeb: mov [eax+ecx],bl
0x10062eee: add dword ptr [esi+14h],1
0x10062ef2: test ebx,ebx
0x10062ef4: jne short 10062E97h
0x10062ef6: jmp short 10062EFDh
0x10062ef8: mov ebx,1
0x10062efd: mov eax,[esi+1Ch]
0x10062f00: cmp dword ptr [eax+2Ch],0
0x10062f04: je short 10062F25h
0x10062f06: mov eax,[esi+14h]
0x10062f09: cmp eax,edx
0x10062f0b: jbe short 10062F25h
0x10062f0d: mov ecx,[esi+8]
0x10062f10: sub eax,edx
0x10062f12: push eax
0x10062f13: add ecx,edx
0x10062f15: mov edx,[edi+30h]
0x10062f18: push ecx
0x10062f19: push edx
0x10062f1a: call 1005FDF0h
```

```
0x10062f1f: add esp,0Ch
0x10062f22: mov [edi+30h],eax
0x10062f25: test ebx,ebx
0x10062f27: jne short 10062F2Fh
0x10062f29: mov [esi+20h],ebx
0x10062f2c: mov [esi+4],ebp
0x10062f2f: cmp [esi+4],ebp
0x10062f32: jne near ptr 10062FE1h
0x10062f38: mov eax,[esi+1Ch]
0x10062f3b: cmp dword ptr [eax+24h],0
0x10062f3f: je near ptr 10062FDAh
0x10062f45: mov edx,[esi+14h]
0x10062f48: mov eax,[esi+14h]
0x10062f4b: cmp eax,[esi+0Ch]
0x10062f4e: jne short 10062F86h
0x10062f50: mov ecx,[esi+1Ch]
0x10062f53: cmp dword ptr [ecx+2Ch],0
0x10062f57: je short 10062F75h
0x10062f59: cmp eax,edx
0x10062f5b: jbe short 10062F75h
0x10062f5d: mov ecx,[edi+30h]
0x10062f60: sub eax,edx
0x10062f62: push eax
0x10062f63: mov eax,[esi+8]
0x10062f66: add eax,edx
0x10062f68: push eax
0x10062f69: push ecx
0x10062f6a: call 1005FDF0h
0x10062f6f: add esp,0Ch
0x10062f72: mov [edi+30h],eax
0x10062f75: mov eax,edi
0x10062f77: call 10061950h
0x10062f7c: mov eax,[esi+14h]
0x10062f7f: cmp eax,[esi+0Ch]
0x10062f82: mov edx,eax
0x10062f84: je short 10062FA9h
0x10062f86: mov ecx,[esi+20h]
0x10062f89: mov ebx,[esi+1Ch]
0x10062f8c: mov ebx,[ebx+24h]
0x10062f8f: movzx ebx,byte ptr [ebx+ecx]
0x10062f93: add ecx,1
0x10062f96: mov [esi+20h],ecx
0x10062f99: mov ecx,[esi+8]
0x10062f9c: mov [eax+ecx],bl
0x10062f9f: add dword ptr [esi+14h],1
0x10062fa3: test ebx,ebx
0x10062fa5: jne short 10062F48h
0x10062fa7: jmp short 10062FAEh
0x10062fa9: mov ebx,1
0x10062fae: mov eax,[esi+1Ch]
0x10062fb1: cmp dword ptr [eax+2Ch],0
0x10062fb5: je short 10062FD6h
0x10062fb7: mov eax,[esi+14h]
0x10062fba: cmp eax,edx
0x10062fbc: jbe short 10062FD6h
0x10062fbe: mov ecx,[esi+8]
0x10062fc1: sub eax,edx
0x10062fc3: push eax
0x10062fc4: add ecx,edx
0x10062fc6: mov edx,[edi+30h]
0x10062fc9: push ecx
0x10062fca: push edx
0x10062fcb: call 1005FDF0h
0x10062fd0: add esp,0Ch
0x10062fd3: mov [edi+30h],eax
0x10062fd6: test ebx,ebx
0x10062fd8: jne short 10062FE1h
```

```
0x10062fda: mov dword ptr [esi+4],67h
0x10062fe1: cmp dword ptr [esi+4],67h
0x10062fe5: jne short 10063050h
0x10062fe7: mov eax,[esi+1Ch]
0x10062fea: cmp dword ptr [eax+2Ch],0
0x10062fee: je short 10063049h
0x10062ff0: mov ecx,[esi+14h]
0x10062ff3: add ecx,2
0x10062ff6: cmp ecx,[esi+0Ch]
0x10062ff9: jbe short 10063002h
0x10062ffb: mov eax,edi
0x10062ffd: call 10061950h
0x10063002: mov eax,[esi+14h]
0x10063005: lea edx,[eax+2]
0x10063008: cmp edx,[esi+0Ch]
0x1006300b: mov ebp,1
0x10063010: ja short 10063055h
0x10063012: movzx edx,byte ptr [edi+30h]
0x10063016: mov ecx,[esi+8]
0x10063019: mov [eax+ecx],dl
0x1006301c: add [esi+14h],ebp
0x1006301f: movzx edx,byte ptr [edi+31h]
0x10063023: mov eax,[esi+14h]
0x10063026: mov ecx,[esi+8]
0x10063029: push 0
0x1006302b: push 0
0x1006302d: mov [eax+ecx],dl
0x10063030: add [esi+14h],ebp
0x10063033: push 0
0x10063035: call 1005FDF0h
0x1006303a: mov [edi+30h],eax
0x1006303d: add esp,0Ch
0x10063040: mov dword ptr [esi+4],71h
0x10063047: jmp short 10063055h
0x10063049: mov dword ptr [esi+4],71h
0x10063050: mov ebp,1
0x10063055: cmp dword ptr [esi+14h],0
0x10063059: je short 10063076h
0x1006305b: mov eax,edi
0x1006305d: call 10061950h
0x10063062: cmp dword ptr [edi+10h],0
0x10063066: jne short 1006309Dh
0x10063068: mov dword ptr [esi+28h],0FFFFFFFFh
0x1006306f: pop ebp
0x10063070: pop ebx
0x10063071: pop edi
0x10063072: xor eax,eax
0x10063074: pop esi
0x10063075: ret
0x10063076: cmp dword ptr [edi+4],0
0x1006307a: mov ebx,[esp+18h]
0x1006307e: jne short 100630A1h
0x10063080: cmp ebx,[esp+14h]
0x10063084: jg short 100630A1h
0x10063086: cmp ebx,4
0x10063089: je short 100630A1h
0x1006308b: mov eax,ds:[101074FCh]
0x10063090: pop ebp
0x10063091: pop ebx
0x10063092: mov [edi+18h],eax
0x10063095: pop edi
0x10063096: mov eax,0FFFFFFFBh
0x1006309b: pop esi
0x1006309c: ret
0x1006309d: mov ebx,[esp+18h]
0x100630a1: mov eax,[esi+4]
0x100630a4: cmp eax,29Ah
```

```
0x100630a9: jne short 100630C4h
0x100630ab: cmp dword ptr [edi+4],0
0x100630af: je short 100630CAh
0x100630b1: mov ecx,ds:[101074FCh]
0x100630b7: pop ebp
0x100630b8: pop ebx
0x100630b9: mov [edi+18h],ecx
0x100630bc: pop edi
0x100630bd: mov eax,0FFFFFFFBh
0x100630c2: pop esi
0x100630c3: ret
0x100630c4: cmp dword ptr [edi+4],0
0x100630c8: jne short 100630E3h
0x100630ca: cmp dword ptr [esi+74h],0
0x100630ce: jne short 100630E3h
0x100630d0: test ebx,ebx
0x100630d2: je near ptr 100631B2h
0x100630d8: cmp eax,29Ah
0x100630dd: je near ptr 100631B2h
0x100630e3: mov eax,[esi+88h]
0x100630e9: cmp eax,2
0x100630ec: push ebx
0x100630ed: jne short 100630FBh
0x100630ef: mov eax,esi
0x100630f1: call 100628C0h
0x100630f6: add esp,4
0x100630f9: jmp short 10063122h
0x100630fb: cmp eax,3
0x100630fe: jne short 1006310Ch
0x10063100: mov eax,esi
0x10063102: call 10062640h
0x10063107: add esp,4
0x1006310a: jmp short 10063122h
0x1006310c: mov eax,[esi+84h]
0x10063112: lea edx,[eax+eax*2]
0x10063115: mov eax,[edx*4+101073E8h]
0x1006311c: push esi
0x1006311d: call eax
0x1006311f: add esp,8
0x10063122: cmp eax,2
0x10063125: je short 1006312Ch
0x10063127: cmp eax,3
0x1006312a: jne short 10063133h
0x1006312c: mov dword ptr [esi+4],29Ah
0x10063133: test eax,eax
0x10063135: je near ptr 10063288h
0x1006313b: cmp eax,2
0x1006313e: je near ptr 10063288h
0x10063144: cmp eax,ebp
0x10063146: jne short 100631B2h
0x10063148: cmp ebx,ebp
0x1006314a: jne short 10063157h
0x1006314c: push esi
0x1006314d: call 10065880h
0x10063152: add esp,4
0x10063155: jmp short 100631A1h
0x10063157: cmp ebx,5
0x1006315a: je short 100631A1h
0x1006315c: push 0
0x1006315e: push 0
0x10063160: push 0
0x10063162: push esi
0x10063163: call 100657E0h
0x10063168: add esp,10h
0x1006316b: cmp ebx,3
0x1006316e: jne short 100631A1h
0x10063170: mov ecx,[esi+4Ch]
```

```
0x10063173: mov edx,[esi+44h]
0x10063176: mov word ptr [edx+ecx*2-2],0
0x1006317d: mov eax,[esi+4Ch]
0x10063180: mov edx,[esi+44h]
0x10063183: lea ecx,[eax+eax-2]
0x10063187: push ecx
0x10063188: push 0
0x1006318a: push edx
0x1006318b: call 100228C0h
0x10063190: add esp,0Ch
0x10063193: cmp dword ptr [esi+74h],0
0x10063197: jne short 100631A1h
0x10063199: xor eax,eax
0x1006319b: mov [esi+6Ch],eax
0x1006319e: mov [esi+5Ch],eax
0x100631a1: mov eax,edi
0x100631a3: call 10061950h
0x100631a8: cmp dword ptr [edi+10h],0
0x100631ac: je near ptr 10063068h
0x100631b2: cmp ebx,4
0x100631b5: jne near ptr 1006306Fh
0x100631bb: mov eax,[esi+18h]
0x100631be: test eax,eax
0x100631c0: jg short 100631C9h
0x100631c2: mov eax,ebp
0x100631c4: pop ebp
0x100631c5: pop ebx
0x100631c6: pop edi
0x100631c7: pop esi
0x100631c8: ret
0x100631c9: cmp eax,2
0x100631cc: jne near ptr 10063254h
0x100631d2: movzx edx,byte ptr [edi+30h]
0x100631d6: mov ecx,[esi+14h]
0x100631d9: mov eax,[esi+8]
0x100631dc: mov [eax+ecx],dl
0x100631df: add [esi+14h],ebp
0x100631e2: movzx edx,byte ptr [edi+31h]
0x100631e6: mov eax,[esi+14h]
0x100631e9: mov ecx,[esi+8]
0x100631ec: mov [eax+ecx],dl
0x100631ef: add [esi+14h],ebp
0x100631f2: movzx edx,byte ptr [edi+32h]
0x100631f6: mov eax,[esi+14h]
0x100631f9: mov ecx,[esi+8]
0x100631fc: mov [eax+ecx],dl
0x100631ff: add [esi+14h],ebp
0x10063202: movzx edx,byte ptr [edi+33h]
0x10063206: mov eax,[esi+14h]
0x10063209: mov ecx,[esi+8]
0x1006320c: mov [eax+ecx],dl
0x1006320f: add [esi+14h],ebp
0x10063212: movzx edx,byte ptr [edi+8]
0x10063216: mov eax,[esi+14h]
0x10063219: mov ecx,[esi+8]
0x1006321c: mov [eax+ecx],dl
0x1006321f: add [esi+14h],ebp
0x10063222: movzx edx,byte ptr [edi+9]
0x10063226: mov eax,[esi+14h]
0x10063229: mov ecx,[esi+8]
0x1006322c: mov [eax+ecx],dl
0x1006322f: add [esi+14h],ebp
0x10063232: movzx edx,byte ptr [edi+0Ah]
0x10063236: mov eax,[esi+14h]
0x10063239: mov ecx,[esi+8]
0x1006323c: mov [eax+ecx],dl
0x1006323f: add [esi+14h],ebp
```

```
0x10063242: movzx edx,byte ptr [edi+0Bh]
0x10063246: mov eax,[esi+14h]
0x10063249: mov ecx,[esi+8]
0x1006324c: mov [eax+ecx],dl
0x1006324f: add [esi+14h],ebp
0x10063252: jmp short 10063268h
0x10063254: movzx ecx,word ptr [edi+32h]
0x10063258: mov eax,esi
0x1006325a: call 10061920h
0x1006325f: movzx ecx,word ptr [edi+30h]
0x10063263: call 10061920h
0x10063268: mov eax,edi
0x1006326a: call 10061950h
0x1006326f: mov eax,[esi+18h]
0x10063272: test eax,eax
0x10063274: jle short 1006327Bh
0x10063276: neg eax
0x10063278: mov [esi+18h],eax
0x1006327b: pop ebp
0x1006327c: xor eax,eax
0x1006327e: cmp [esi+14h],eax
0x10063281: pop ebx
0x10063282: pop edi
0x10063283: sete al
0x10063286: pop esi
0x10063287: ret
0x10063288: cmp dword ptr [edi+10h],0
0x1006328c: jne near ptr 1006306Fh
0x10063292: pop ebp
0x10063293: pop ebx
0x10063294: pop edi
0x10063295: mov dword ptr [esi+28h],0FFFFFFFFh
0x1006329c: xor eax,eax
0x1006329e: pop esi
0x1006329f: ret
0x100632a0: mov eax,ds:[101074F0h]
0x100632a5: mov [edi+18h],eax
0x100632a8: pop edi
0x100632a9: mov eax,0FFFFFFFEh
0x100632ae: pop esi
0x100632af: ret
```

`sub_10062a80` is called by png_write_flush, sub_10035b60, sub_10037d80, sub_10037fe0. It delegates to j_memset, sub_1005fdf0, sub_10061920 (9 callees).

12.911 sub_10063fb0 at 0x10063fb0 (221 instructions)

```
0x10063fb0: mov ecx,[esp+4]
0x10063fb4: push ebx
0x10063fb5: mov ebx,[esp+10h]
0x10063fb9: push edi
0x10063fba: mov edi,ecx
0x10063fbc: shr edi,10h
0x10063fbf: and ecx,0FFFFh
0x10063fc5: cmp ebx,1
0x10063fc8: jne short 10063FFBh
0x10063fca: mov eax,[esp+10h]
0x10063fce: movzx edx,byte ptr [eax]
0x10063fd1: add ecx,edx
0x10063fd3: cmp ecx,0FFF1h
0x10063fd9: jb short 10063FE1h
0x10063fdb: sub ecx,0FFF1h
0x10063fe1: add edi,ecx
0x10063fe3: cmp edi,0FFF1h
0x10063fe9: jb short 10063FF1h
0x10063feb: sub edi,0FFF1h
```

```
0x10063ff1: mov eax,edi
0x10063ff3: shl eax,10h
0x10063ff6: pop edi
0x10063ff7: or eax,ecx
0x10063ff9: pop ebx
0x10063ffa: ret
0x10063ffb: push esi
0x10063ffc: mov esi,[esp+14h]
0x10064000: test esi,esi
0x10064002: jne short 1006400Bh
0x10064004: lea eax,[esi+1]
0x10064007: pop esi
0x10064008: pop edi
0x10064009: pop ebx
0x1006400a: ret
0x1006400b: cmp ebx,10h
0x1006400e: jae short 1006404Fh
0x10064010: test ebx,ebx
0x10064012: je short 10064025h
0x10064014: movzx eax,byte ptr [esi]
0x10064017: add ecx,eax
0x10064019: sub ebx,1
0x1006401c: add esi,1
0x1006401f: add edi,ecx
0x10064021: test ebx,ebx
0x10064023: jne short 10064014h
0x10064025: cmp ecx,0FFF1h
0x1006402b: jb short 10064033h
0x1006402d: sub ecx,0FFF1h
0x10064033: mov eax,80078071h
0x10064038: mul edi
0x1006403a: shr edx,0Fh
0x1006403d: mov eax,edx
0x1006403f: shl eax,4
0x10064042: sub eax,edx
0x10064044: add eax,edi
0x10064046: pop esi
0x10064047: shl eax,10h
0x1006404a: pop edi
0x1006404b: or eax,ecx
0x1006404d: pop ebx
0x1006404e: ret
0x1006404f: cmp ebx,15B0h
0x10064055: jb near ptr 10064139h
0x1006405b: mov eax,5E6EA9AFh
0x10064060: mul ebx
0x10064062: push ebp
0x10064063: mov ebp,edx
0x10064065: shr ebp,0Bh
0x10064068: jmp short 10064070h
0x10064070: sub ebx,15B0h
0x10064076: mov eax,15Bh
0x1006407b: jmp short 10064080h
0x10064080: movzx edx,byte ptr [esi]
0x10064083: add ecx,edx
0x10064085: movzx edx,byte ptr [esi+1]
0x10064089: add edi,ecx
0x1006408b: add ecx,edx
0x1006408d: movzx edx,byte ptr [esi+2]
0x10064091: add edi,ecx
0x10064093: add ecx,edx
0x10064095: movzx edx,byte ptr [esi+3]
0x10064099: add edi,ecx
0x1006409b: add ecx,edx
0x1006409d: movzx edx,byte ptr [esi+4]
0x100640a1: add edi,ecx
0x100640a3: add ecx,edx
```

```
0x100640a5: movzx edx,byte ptr [esi+5]
0x100640a9: add edi,ecx
0x100640ab: add ecx,edx
0x100640ad: movzx edx,byte ptr [esi+6]
0x100640b1: add edi,ecx
0x100640b3: add ecx,edx
0x100640b5: movzx edx,byte ptr [esi+7]
0x100640b9: add edi,ecx
0x100640bb: add ecx,edx
0x100640bd: movzx edx,byte ptr [esi+8]
0x100640c1: add edi,ecx
0x100640c3: add ecx,edx
0x100640c5: movzx edx,byte ptr [esi+9]
0x100640c9: add edi,ecx
0x100640cb: add ecx,edx
0x100640cd: movzx edx,byte ptr [esi+0Ah]
0x100640d1: add edi,ecx
0x100640d3: add ecx,edx
0x100640d5: movzx edx,byte ptr [esi+0Bh]
0x100640d9: add edi,ecx
0x100640db: add ecx,edx
0x100640dd: movzx edx,byte ptr [esi+0Ch]
0x100640e1: add edi,ecx
0x100640e3: add ecx,edx
0x100640e5: movzx edx,byte ptr [esi+0Dh]
0x100640e9: add edi,ecx
0x100640eb: add ecx,edx
0x100640ed: movzx edx,byte ptr [esi+0Eh]
0x100640f1: add edi,ecx
0x100640f3: add ecx,edx
0x100640f5: movzx edx,byte ptr [esi+0Fh]
0x100640f9: add edi,ecx
0x100640fb: add ecx,edx
0x100640fd: add edi,ecx
0x100640ff: add esi,10h
0x10064102: sub eax,1
0x10064105: jne near ptr 10064080h
0x1006410b: mov eax,80078071h
0x10064110: mul ecx
0x10064112: shr edx,0Fh
0x10064115: imul edx,0FFFF000Fh
0x1006411b: add ecx,edx
0x1006411d: mov eax,80078071h
0x10064122: mul edi
0x10064124: shr edx,0Fh
0x10064127: imul edx,0FFFF000Fh
0x1006412d: add edi,edx
0x1006412f: sub ebp,1
0x10064132: jne near ptr 10064070h
0x10064138: pop ebp
0x10064139: test ebx,ebx
0x1006413b: je near ptr 10064217h
0x10064141: cmp ebx,10h
0x10064144: jb near ptr 100641DEh
0x1006414a: mov eax,ebx
0x1006414c: shr eax,4
0x1006414f: nop
0x10064150: movzx edx,byte ptr [esi]
0x10064153: add ecx,edx
0x10064155: movzx edx,byte ptr [esi+1]
0x10064159: add edi,ecx
0x1006415b: add ecx,edx
0x1006415d: movzx edx,byte ptr [esi+2]
0x10064161: add edi,ecx
0x10064163: add ecx,edx
0x10064165: movzx edx,byte ptr [esi+3]
0x10064169: add edi,ecx
```

```
0x1006416b: add ecx,edx
0x1006416d: movzx edx,byte ptr [esi+4]
0x10064171: add edi,ecx
0x10064173: add ecx,edx
0x10064175: movzx edx,byte ptr [esi+5]
0x10064179: add edi,ecx
0x1006417b: add ecx,edx
0x1006417d: movzx edx,byte ptr [esi+6]
0x10064181: add edi,ecx
0x10064183: add ecx,edx
0x10064185: movzx edx,byte ptr [esi+7]
0x10064189: add edi,ecx
0x1006418b: add ecx,edx
0x1006418d: movzx edx,byte ptr [esi+8]
0x10064191: add edi,ecx
0x10064193: add ecx,edx
0x10064195: movzx edx,byte ptr [esi+9]
0x10064199: add edi,ecx
0x1006419b: add ecx,edx
0x1006419d: movzx edx,byte ptr [esi+0Ah]
0x100641a1: add edi,ecx
0x100641a3: add ecx,edx
0x100641a5: movzx edx,byte ptr [esi+0Bh]
0x100641a9: add edi,ecx
0x100641ab: add ecx,edx
0x100641ad: movzx edx,byte ptr [esi+0Ch]
0x100641b1: add edi,ecx
0x100641b3: add ecx,edx
0x100641b5: movzx edx,byte ptr [esi+0Dh]
0x100641b9: add edi,ecx
0x100641bb: add ecx,edx
0x100641bd: movzx edx,byte ptr [esi+0Eh]
0x100641c1: add edi,ecx
0x100641c3: add ecx,edx
0x100641c5: movzx edx,byte ptr [esi+0Fh]
0x100641c9: add edi,ecx
0x100641cb: add ecx,edx
0x100641cd: sub ebx,10h
0x100641d0: add edi,ecx
0x100641d2: add esi,10h
0x100641d5: sub eax,1
0x100641d8: jne near ptr 10064150h
0x100641de: test ebx,ebx
0x100641e0: je short 100641F3h
0x100641e2: movzx eax,byte ptr [esi]
0x100641e5: add ecx,eax
0x100641e7: sub ebx,1
0x100641ea: add esi,1
0x100641ed: add edi,ecx
0x100641ef: test ebx,ebx
0x100641f1: jne short 100641E2h
0x100641f3: mov eax,80078071h
0x100641f8: mul ecx
0x100641fa: shr edx,0Fh
0x100641fd: imul edx,0FFFF000Fh
0x10064203: add ecx,edx
0x10064205: mov eax,80078071h
0x1006420a: mul edi
0x1006420c: shr edx,0Fh
0x1006420f: imul edx,0FFFF000Fh
0x10064215: add edi,edx
0x10064217: mov eax,edi
0x10064219: pop esi
0x1006421a: shl eax,10h
0x1006421d: pop edi
0x1006421e: or eax,ecx
```

```
0x10064220: pop ebx
0x10064221: ret
```

`sub_10063fb0` is called by sub_10061a80, sub_100629f0, sub_10062a80. It accesses memory-mapped peripheral registers at 0x5e6ea9af. It calls nothing else.

12.912 sub_10064230 at 0x10064230 (29 instructions)

```
0x10064230: push esi
0x10064231: lea eax,[edx+94h]
0x10064237: mov ecx,11Eh
0x1006423c: xor esi,esi
0x1006423e: mov edi,edi
0x10064240: mov [eax],si
0x10064243: add eax,4
0x10064246: sub ecx,1
0x10064249: jne short 10064240h
0x1006424b: lea eax,[edx+988h]
0x10064251: mov ecx,1Eh
0x10064256: mov [eax],si
0x10064259: add eax,4
0x1006425c: sub ecx,1
0x1006425f: jne short 10064256h
0x10064261: lea eax,[edx+0A7Ch]
0x10064267: mov ecx,13h
0x1006426c: lea esp,[esp]
0x10064270: mov [eax],si
0x10064273: add eax,4
0x10064276: sub ecx,1
0x10064279: jne short 10064270h
0x1006427b: mov [edx+16ACh],esi
0x10064281: mov [edx+16A8h],esi
0x10064287: mov [edx+16B0h],esi
0x1006428d: mov [edx+16A0h],esi
0x10064293: mov word ptr [edx+494h],1
0x1006429c: pop esi
0x1006429d: ret
```

`sub_10064230` is called by sub_10065a70. It calls nothing else.

12.913 sub_100642a0 at 0x100642a0 (59 instructions)

```
0x100642a0: push ecx
0x100642a1: mov edx,[eax+1450h]
0x100642a7: push ebp
0x100642a8: push esi
0x100642a9: mov esi,[esp+10h]
0x100642ad: mov ebp,[eax+esi*4+0B5Ch]
0x100642b4: lea ecx,[esi+esi]
0x100642b7: cmp ecx,edx
0x100642b9: mov [esp+8],ebp
0x100642bd: jg near ptr 1006435Bh
0x100642c3: push ebx
0x100642c4: jge short 100642FAh
0x100642c6: mov esi,[eax+ecx*4+0B60h]
0x100642cd: mov ebp,[eax+ecx*4+0B5Ch]
0x100642d4: movzx edx,word ptr [edi+esi*4]
0x100642d8: movzx ebx,word ptr [edi+ebp*4]
0x100642dc: cmp dx,bx
0x100642df: jb short 100642F3h
0x100642e1: jne short 100642F6h
0x100642e3: mov dl,[esi+eax+1458h]
0x100642ea: cmp dl,[eax+ebp+1458h]
0x100642f1: ja short 100642F6h
0x100642f3: add ecx,1
```

```
0x100642f6: mov ebp,[esp+0Ch]
0x100642fa: mov esi,[eax+ecx*4+0B5Ch]
0x10064301: movzx edx,word ptr [edi+ebp*4]
0x10064305: movzx ebx,word ptr [edi+esi*4]
0x10064309: cmp dx,bx
0x1006430c: jb short 1006433Bh
0x1006430e: jne short 10064320h
0x10064310: mov dl,[eax+ebp+1458h]
0x10064317: cmp dl,[esi+eax+1458h]
0x1006431e: jbe short 1006434Bh
0x10064320: mov edx,[esp+14h]
0x10064324: mov [eax+edx*4+0B5Ch],esi
0x1006432b: mov edx,[eax+1450h]
0x10064331: mov [esp+14h],ecx
0x10064335: add ecx,ecx
0x10064337: cmp ecx,edx
0x10064339: jle short 100642C4h
0x1006433b: mov ecx,[esp+14h]
0x1006433f: pop ebx
0x10064340: pop esi
0x10064341: mov [eax+ecx*4+0B5Ch],ebp
0x10064348: pop ebp
0x10064349: pop ecx
0x1006434a: ret
0x1006434b: mov edx,[esp+14h]
0x1006434f: pop ebx
0x10064350: pop esi
0x10064351: mov [eax+edx*4+0B5Ch],ebp
0x10064358: pop ebp
0x10064359: pop ecx
0x1006435a: ret
0x1006435b: mov [eax+esi*4+0B5Ch],ebp
0x10064362: pop esi
0x10064363: pop ebp
0x10064364: pop ecx
0x10064365: ret
```

`sub_100642a0` is called by sub_10065510. It calls nothing else.

12.914 sub_10064370 at 0x10064370 (144 instructions)

```
0x10064370: sub esp,20h
0x10064373: mov edx,[ecx+4]
0x10064376: push ebx
0x10064377: mov ebx,[ecx]
0x10064379: mov ecx,[ecx+8]
0x1006437c: mov [esp+18h],edx
0x10064380: mov edx,[ecx+4]
0x10064383: push ebp
0x10064384: mov ebp,[ecx]
0x10064386: mov [esp+24h],edx
0x1006438a: mov edx,[ecx+8]
0x1006438d: push esi
0x1006438e: mov esi,[ecx+10h]
0x10064391: xor ecx,ecx
0x10064393: mov [eax+0B3Ch],ecx
0x10064399: mov [eax+0B40h],ecx
0x1006439f: mov [eax+0B44h],ecx
0x100643a5: mov [eax+0B48h],ecx
0x100643ab: mov [eax+0B4Ch],ecx
0x100643b1: mov [eax+0B50h],ecx
0x100643b7: mov [eax+0B54h],ecx
0x100643bd: mov [eax+0B58h],ecx
0x100643c3: mov [esp+24h],edx
0x100643c7: mov edx,[eax+1454h]
0x100643cd: mov edx,[eax+edx*4+0B5Ch]
```

```
0x100643d4: push edi
0x100643d5: mov [ebx+edx*4+2],cx
0x100643da: mov edi,[eax+1454h]
0x100643e0: add edi,1
0x100643e3: cmp edi,23Dh
0x100643e9: mov [esp+1Ch],esi
0x100643ed: mov [esp+18h],ecx
0x100643f1: jge near ptr 1006457Fh
0x100643f7: lea ecx,[eax+edi*4+0B5Ch]
0x100643fe: mov [esp+14h],ecx
0x10064402: mov ecx,23Dh
0x10064407: sub ecx,edi
0x10064409: add edi,ecx
0x1006440b: mov [esp+20h],ecx
0x1006440f: mov [esp+10h],edi
0x10064413: jmp short 10064419h
0x10064415: mov esi,[esp+1Ch]
0x10064419: mov edx,[esp+14h]
0x1006441d: mov edx,[edx]
0x1006441f: movzx ecx,word ptr [ebx+edx*4+2]
0x10064424: movzx ecx,word ptr [ebx+ecx*4+2]
0x10064429: add ecx,1
0x1006442c: cmp ecx,esi
0x1006442e: jle short 10064437h
0x10064430: add dword ptr [esp+18h],1
0x10064435: mov ecx,esi
0x10064437: cmp edx,[esp+24h]
0x1006443b: mov [ebx+edx*4+2],cx
0x10064440: jg short 1006448Ah
0x10064442: mov edi,[esp+28h]
0x10064446: add word ptr [eax+ecx*2+0B3Ch],1
0x1006444f: xor esi,esi
0x10064451: cmp edx,edi
0x10064453: jl short 10064460h
0x10064455: mov esi,edx
0x10064457: sub esi,edi
0x10064459: mov edi,[esp+2Ch]
0x1006445d: mov esi,[edi+esi*4]
0x10064460: movzx edi,word ptr [ebx+edx*4]
0x10064464: movzx edi,di
0x10064467: add ecx,esi
0x10064469: imul ecx,edi
0x1006446c: add [eax+16A8h],ecx
0x10064472: test ebp,ebp
0x10064474: je short 10064486h
0x10064476: movzx edx,word ptr [ebp+edx*4+2]
0x1006447b: add edx,esi
0x1006447d: imul edx,edi
0x10064480: add [eax+16ACh],edx
0x10064486: mov edi,[esp+10h]
0x1006448a: add dword ptr [esp+14h],4
0x1006448f: sub dword ptr [esp+20h],1
0x10064494: jne near ptr 10064415h
0x1006449a: mov ebp,[esp+18h]
0x1006449e: test ebp,ebp
0x100644a0: je near ptr 1006457Fh
0x100644a6: mov ecx,[esp+1Ch]
0x100644aa: lea edx,[ecx-1]
0x100644ad: mov [esp+2Ch],edx
0x100644b1: lea esi,[eax+ecx*2+0B3Ch]
0x100644b8: jmp short 100644C0h
0x100644c0: mov ecx,[esp+2Ch]
0x100644c4: cmp word ptr [eax+ecx*2+0B3Ch],0
0x100644cd: lea edx,[eax+ecx*2+0B3Ch]
0x100644d4: jne short 100644E2h
0x100644d6: sub edx,2
0x100644d9: sub ecx,1
```

```
0x100644dc: cmp word ptr [edx],0
0x100644e0: je short 100644D6h
0x100644e2: add word ptr [eax+ecx*2+0B3Ch],0FFFFh
0x100644ec: add word ptr [eax+ecx*2+0B3Eh],2
0x100644f5: add word ptr [esi],0FFFFh
0x100644fa: sub ebp,2
0x100644fd: test ebp,ebp
0x100644ff: jg short 100644C0h
0x10064501: mov edx,[esp+1Ch]
0x10064505: test edx,edx
0x10064507: je short 1006457Fh
0x10064509: mov [esp+20h],esi
0x1006450d: lea ecx,[ecx]
0x10064510: movzx esi,word ptr [esi]
0x10064513: test esi,esi
0x10064515: mov [esp+1Ch],esi
0x10064519: je short 1006456Dh
0x1006451b: lea ebp,[eax+edi*4+0B5Ch]
0x10064522: mov ecx,[ebp-4]
0x10064525: sub dword ptr [esp+10h],1
0x1006452a: sub ebp,4
0x1006452d: cmp ecx,[esp+24h]
0x10064531: mov [esp+2Ch],ebp
0x10064535: jg short 10064565h
0x10064537: movzx edi,word ptr [ebx+ecx*4+2]
0x1006453c: cmp edi,edx
0x1006453e: lea esi,[ebx+ecx*4+2]
0x10064542: je short 1006455Ch
0x10064544: movzx ecx,word ptr [ebx+ecx*4]
0x10064548: mov ebp,edx
0x1006454a: sub ebp,edi
0x1006454c: imul ebp,ecx
0x1006454f: add [eax+16A8h],ebp
0x10064555: mov ebp,[esp+2Ch]
0x10064559: mov [esi],dx
0x1006455c: sub dword ptr [esp+1Ch],1
0x10064561: mov esi,[esp+1Ch]
0x10064565: test esi,esi
0x10064567: jne short 10064522h
0x10064569: mov edi,[esp+10h]
0x1006456d: mov esi,[esp+20h]
0x10064571: sub edx,1
0x10064574: sub esi,2
0x10064577: test edx,edx
0x10064579: mov [esp+20h],esi
0x1006457d: jne short 10064510h
0x1006457f: pop edi
0x10064580: pop esi
0x10064581: pop ebp
0x10064582: pop ebx
0x10064583: add esp,20h
0x10064586: ret
```

`sub_10064370` is called by sub_10065510. It calls nothing else.

12.915 sub_10064590 at 0x10064590 (72 instructions)

```
0x10064590: sub esp,0Ch
0x10064593: push ebx
0x10064594: push ebp
0x10064595: mov ebp,[esp+18h]
0x10064599: push esi
0x1006459a: push edi
0x1006459b: movzx edi,word ptr [eax+2]
0x1006459f: xor edx,edx
0x100645a1: test edi,edi
```

```
0x100645a3: mov ebx,ecx
0x100645a5: mov dword ptr [esp+14h],0FFFFFFFFh
0x100645ad: lea ecx,[edx+7]
0x100645b0: lea esi,[edx+4]
0x100645b3: jne short 100645BDh
0x100645b5: mov ecx,8Ah
0x100645ba: lea esi,[edx+3]
0x100645bd: test ebx,ebx
0x100645bf: mov word ptr [eax+ebx*4+6],0FFFFh
0x100645c6: jl near ptr 1006466Ah
0x100645cc: add eax,6
0x100645cf: add ebx,1
0x100645d2: mov [esp+18h],ebx
0x100645d6: mov [esp+10h],eax
0x100645da: mov ebx,1
0x100645df: nop
0x100645e0: mov eax,edi
0x100645e2: mov edi,[esp+10h]
0x100645e6: movzx edi,word ptr [edi]
0x100645e9: add edx,ebx
0x100645eb: cmp edx,ecx
0x100645ed: jge short 100645F3h
0x100645ef: cmp eax,edi
0x100645f1: je short 1006465Bh
0x100645f3: cmp edx,esi
0x100645f5: jge short 10064601h
0x100645f7: add [ebp+eax*4+0A7Ch],dx
0x100645ff: jmp short 10064631h
0x10064601: test eax,eax
0x10064603: je short 1006461Ch
0x10064605: cmp eax,[esp+14h]
0x10064609: je short 10064613h
0x1006460b: add [ebp+eax*4+0A7Ch],bx
0x10064613: add [ebp+0ABCh],bx
0x1006461a: jmp short 10064631h
0x1006461c: cmp edx,0Ah
0x1006461f: jg short 1006462Ah
0x10064621: add [ebp+0AC0h],bx
0x10064628: jmp short 10064631h
0x1006462a: add [ebp+0AC4h],bx
0x10064631: xor edx,edx
0x10064633: test edi,edi
0x10064635: mov [esp+14h],eax
0x10064639: jne short 10064645h
0x1006463b: mov ecx,8Ah
0x10064640: lea esi,[edx+3]
0x10064643: jmp short 1006465Bh
0x10064645: cmp eax,edi
0x10064647: jne short 10064653h
0x10064649: mov ecx,6
0x1006464e: lea esi,[ecx-3]
0x10064651: jmp short 1006465Bh
0x10064653: mov ecx,7
0x10064658: lea esi,[ecx-3]
0x1006465b: add dword ptr [esp+10h],4
0x10064660: sub [esp+18h],ebx
0x10064664: jne near ptr 100645E0h
0x1006466a: pop edi
0x1006466b: pop esi
0x1006466c: pop ebp
0x1006466d: pop ebx
0x1006466e: add esp,0Ch
0x10064671: ret
```

`sub_10064590` is called by sub_10065710. It calls nothing else.

12.916 sub_10064680 at 0x10064680 (334 instructions)

```
0x10064680: sub esp,18h
0x10064683: push ebx
0x10064684: push ebp
0x10064685: movzx ebp,word ptr [edx+2]
0x10064689: push esi
0x1006468a: xor esi,esi
0x1006468c: test ebp,ebp
0x1006468e: push edi
0x1006468f: mov ebx,ecx
0x10064691: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x10064699: mov [esp+14h],ebp
0x1006469d: lea ecx,[esi+7]
0x100646a0: lea edi,[esi+4]
0x100646a3: jne short 100646ADh
0x100646a5: mov ecx,8Ah
0x100646aa: lea edi,[ebp+3]
0x100646ad: test ebx,ebx
0x100646af: jl near ptr 10064B83h
0x100646b5: add edx,6
0x100646b8: add ebx,1
0x100646bb: mov [esp+18h],edx
0x100646bf: mov [esp+20h],ebx
0x100646c3: mov ebp,1
0x100646c8: jmp short 100646D0h
0x100646d0: mov ebx,[esp+18h]
0x100646d4: movzx ebx,word ptr [ebx]
0x100646d7: mov edx,[esp+14h]
0x100646db: add esi,ebp
0x100646dd: cmp esi,ecx
0x100646df: mov [esp+24h],edx
0x100646e3: mov [esp+14h],ebx
0x100646e7: mov [esp+10h],esi
0x100646eb: jge short 100646F5h
0x100646ed: cmp edx,ebx
0x100646ef: je near ptr 10064B74h
0x100646f5: cmp esi,edi
0x100646f7: jge near ptr 1006479Fh
0x100646fd: lea ecx,[ecx]
0x10064700: movzx edi,word ptr [eax+edx*4+0A7Eh]
0x10064708: mov ecx,[eax+16BCh]
0x1006470e: mov ebx,10h
0x10064713: sub ebx,edi
0x10064715: cmp ecx,ebx
0x10064717: jle short 10064774h
0x10064719: movzx esi,word ptr [eax+edx*4+0A7Ch]
0x10064721: mov edx,esi
0x10064723: shl edx,cl
0x10064725: mov ecx,[eax+8]
0x10064728: or [eax+16B8h],dx
0x1006472f: movzx ebx,byte ptr [eax+16B8h]
0x10064736: mov edx,[eax+14h]
0x10064739: mov [ecx+edx],bl
0x1006473c: add [eax+14h],ebp
0x1006473f: mov ecx,[eax+14h]
0x10064742: movzx ebx,byte ptr [eax+16B9h]
0x10064749: mov edx,[eax+8]
0x1006474c: mov [ecx+edx],bl
0x1006474f: mov edx,[eax+16BCh]
0x10064755: add [eax+14h],ebp
0x10064758: mov cl,10h
0x1006475a: sub cl,dl
0x1006475c: shr si,cl
0x1006475f: lea ecx,[edx+edi-10h]
0x10064763: mov edx,[esp+24h]
0x10064767: mov [eax+16B8h],si
0x1006476e: mov esi,[esp+10h]
```

```
0x10064772: jmp short 10064788h
0x10064774: mov bx,[eax+edx*4+0A7Ch]
0x1006477c: shl bx,cl
0x1006477f: or [eax+16B8h],bx
0x10064786: add ecx,edi
0x10064788: sub esi,ebp
0x1006478a: mov [eax+16BCh],ecx
0x10064790: mov [esp+10h],esi
0x10064794: jne near ptr 10064700h
0x1006479a: jmp near ptr 10064B46h
0x1006479f: test edx,edx
0x100647a1: je near ptr 1006494Ch
0x100647a7: cmp edx,[esp+1Ch]
0x100647ab: je near ptr 1006484Dh
0x100647b1: movzx edi,word ptr [eax+edx*4+0A7Eh]
0x100647b9: mov ecx,[eax+16BCh]
0x100647bf: mov ebx,10h
0x100647c4: sub ebx,edi
0x100647c6: cmp ecx,ebx
0x100647c8: mov [esp+1Ch],edi
0x100647cc: jle short 10064829h
0x100647ce: movzx esi,word ptr [eax+edx*4+0A7Ch]
0x100647d6: mov edi,esi
0x100647d8: shl edi,cl
0x100647da: mov ecx,[eax+8]
0x100647dd: or [eax+16B8h],di
0x100647e4: movzx ebx,byte ptr [eax+16B8h]
0x100647eb: mov edi,[eax+14h]
0x100647ee: mov [ecx+edi],bl
0x100647f1: add [eax+14h],ebp
0x100647f4: movzx ebx,byte ptr [eax+16B9h]
0x100647fb: mov ecx,[eax+8]
0x100647fe: mov edi,[eax+14h]
0x10064801: mov [edi+ecx],bl
0x10064804: mov ebx,[eax+16BCh]
0x1006480a: add [eax+14h],ebp
0x1006480d: mov cl,10h
0x1006480f: sub cl,bl
0x10064811: shr si,cl
0x10064814: mov ecx,[esp+1Ch]
0x10064818: lea ecx,[ebx+ecx-10h]
0x1006481c: mov [eax+16B8h],si
0x10064823: mov esi,[esp+10h]
0x10064827: jmp short 10064841h
0x10064829: mov di,[eax+edx*4+0A7Ch]
0x10064831: shl di,cl
0x10064834: or [eax+16B8h],di
0x1006483b: mov edi,[esp+1Ch]
0x1006483f: add ecx,edi
0x10064841: sub esi,ebp
0x10064843: mov [eax+16BCh],ecx
0x10064849: mov [esp+10h],esi
0x1006484d: movzx edi,word ptr [eax+0ABEh]
0x10064854: mov ecx,[eax+16BCh]
0x1006485a: mov ebx,10h
0x1006485f: sub ebx,edi
0x10064861: cmp ecx,ebx
0x10064863: mov [esp+1Ch],edi
0x10064867: jle short 100648C3h
0x10064869: movzx esi,word ptr [eax+0ABCh]
0x10064870: mov edi,esi
0x10064872: shl edi,cl
0x10064874: mov ecx,[eax+8]
0x10064877: or [eax+16B8h],di
0x1006487e: movzx ebx,byte ptr [eax+16B8h]
0x10064885: mov edi,[eax+14h]
0x10064888: mov [ecx+edi],bl
```

```
0x1006488b: add [eax+14h],ebp
0x1006488e: movzx ebx,byte ptr [eax+16B9h]
0x10064895: mov ecx,[eax+8]
0x10064898: mov edi,[eax+14h]
0x1006489b: mov [edi+ecx],bl
0x1006489e: mov ebx,[eax+16BCh]
0x100648a4: add [eax+14h],ebp
0x100648a7: mov cl,10h
0x100648a9: sub cl,bl
0x100648ab: shr si,cl
0x100648ae: mov ecx,[esp+1Ch]
0x100648b2: lea ecx,[ebx+ecx-10h]
0x100648b6: mov [eax+16B8h],si
0x100648bd: mov esi,[esp+10h]
0x100648c1: jmp short 100648DAh
0x100648c3: mov di,[eax+0ABCh]
0x100648ca: shl di,cl
0x100648cd: or [eax+16B8h],di
0x100648d4: mov edi,[esp+1Ch]
0x100648d8: add ecx,edi
0x100648da: add esi,0FFFFFFFDh
0x100648dd: cmp ecx,0Eh
0x100648e0: mov [eax+16BCh],ecx
0x100648e6: jle short 1006493Bh
0x100648e8: mov edi,esi
0x100648ea: shl edi,cl
0x100648ec: mov ecx,[eax+8]
0x100648ef: or [eax+16B8h],di
0x100648f6: movzx ebx,byte ptr [eax+16B8h]
0x100648fd: mov edi,[eax+14h]
0x10064900: mov [ecx+edi],bl
0x10064903: add [eax+14h],ebp
0x10064906: movzx ebx,byte ptr [eax+16B9h]
0x1006490d: mov edi,[eax+14h]
0x10064910: mov ecx,[eax+8]
0x10064913: mov [edi+ecx],bl
0x10064916: mov ebx,[eax+16BCh]
0x1006491c: add [eax+14h],ebp
0x1006491f: mov cl,10h
0x10064921: sub cl,bl
0x10064923: shr si,cl
0x10064926: add ebx,0FFFFFFF2h
0x10064929: mov [eax+16BCh],ebx
0x1006492f: mov [eax+16B8h],si
0x10064936: jmp near ptr 10064B46h
0x1006493b: shl esi,cl
0x1006493d: or [eax+16B8h],si
0x10064944: add ecx,2
0x10064947: jmp near ptr 10064B40h
0x1006494c: cmp esi,0Ah
0x1006494f: mov ecx,[eax+16BCh]
0x10064955: mov ebx,10h
0x1006495a: jg near ptr 10064A54h
0x10064960: movzx edi,word ptr [eax+0AC2h]
0x10064967: sub ebx,edi
0x10064969: cmp ecx,ebx
0x1006496b: mov [esp+1Ch],edi
0x1006496f: jle short 100649CBh
0x10064971: movzx esi,word ptr [eax+0AC0h]
0x10064978: mov edi,esi
0x1006497a: shl edi,cl
0x1006497c: mov ecx,[eax+8]
0x1006497f: or [eax+16B8h],di
0x10064986: movzx ebx,byte ptr [eax+16B8h]
0x1006498d: mov edi,[eax+14h]
0x10064990: mov [ecx+edi],bl
0x10064993: add [eax+14h],ebp
```

```
0x10064996: movzx ebx,byte ptr [eax+16B9h]
0x1006499d: mov ecx,[eax+8]
0x100649a0: mov edi,[eax+14h]
0x100649a3: mov [edi+ecx],bl
0x100649a6: mov ebx,[eax+16BCh]
0x100649ac: add [eax+14h],ebp
0x100649af: mov cl,10h
0x100649b1: sub cl,bl
0x100649b3: shr si,cl
0x100649b6: mov ecx,[esp+1Ch]
0x100649ba: lea ecx,[ebx+ecx-10h]
0x100649be: mov [eax+16B8h],si
0x100649c5: mov esi,[esp+10h]
0x100649c9: jmp short 100649E2h
0x100649cb: mov di,[eax+0AC0h]
0x100649d2: shl di,cl
0x100649d5: or [eax+16B8h],di
0x100649dc: mov edi,[esp+1Ch]
0x100649e0: add ecx,edi
0x100649e2: add esi,0FFFFFFFDh
0x100649e5: cmp ecx,0Dh
0x100649e8: mov [eax+16BCh],ecx
0x100649ee: jle short 10064A43h
0x100649f0: mov edi,esi
0x100649f2: shl edi,cl
0x100649f4: mov ecx,[eax+8]
0x100649f7: or [eax+16B8h],di
0x100649fe: movzx ebx,byte ptr [eax+16B8h]
0x10064a05: mov edi,[eax+14h]
0x10064a08: mov [ecx+edi],bl
0x10064a0b: add [eax+14h],ebp
0x10064a0e: movzx ebx,byte ptr [eax+16B9h]
0x10064a15: mov edi,[eax+14h]
0x10064a18: mov ecx,[eax+8]
0x10064a1b: mov [edi+ecx],bl
0x10064a1e: mov ebx,[eax+16BCh]
0x10064a24: add [eax+14h],ebp
0x10064a27: mov cl,10h
0x10064a29: sub cl,bl
0x10064a2b: shr si,cl
0x10064a2e: add ebx,0FFFFFFF3h
0x10064a31: mov [eax+16BCh],ebx
0x10064a37: mov [eax+16B8h],si
0x10064a3e: jmp near ptr 10064B46h
0x10064a43: shl esi,cl
0x10064a45: or [eax+16B8h],si
0x10064a4c: add ecx,3
0x10064a4f: jmp near ptr 10064B40h
0x10064a54: movzx edi,word ptr [eax+0AC6h]
0x10064a5b: sub ebx,edi
0x10064a5d: cmp ecx,ebx
0x10064a5f: mov [esp+1Ch],edi
0x10064a63: jle short 10064ABFh
0x10064a65: movzx esi,word ptr [eax+0AC4h]
0x10064a6c: mov edi,esi
0x10064a6e: shl edi,cl
0x10064a70: mov ecx,[eax+8]
0x10064a73: or [eax+16B8h],di
0x10064a7a: movzx ebx,byte ptr [eax+16B8h]
0x10064a81: mov edi,[eax+14h]
0x10064a84: mov [ecx+edi],bl
0x10064a87: add [eax+14h],ebp
0x10064a8a: movzx ebx,byte ptr [eax+16B9h]
0x10064a91: mov ecx,[eax+8]
0x10064a94: mov edi,[eax+14h]
0x10064a97: mov [edi+ecx],bl
0x10064a9a: mov ebx,[eax+16BCh]
```

```
0x10064aa0: add [eax+14h],ebp
0x10064aa3: mov cl,10h
0x10064aa5: sub cl,bl
0x10064aa7: shr si,cl
0x10064aaa: mov ecx,[esp+1Ch]
0x10064aae: lea ecx,[ebx+ecx-10h]
0x10064ab2: mov [eax+16B8h],si
0x10064ab9: mov esi,[esp+10h]
0x10064abd: jmp short 10064AD6h
0x10064abf: mov di,[eax+0AC4h]
0x10064ac6: shl di,cl
0x10064ac9: or [eax+16B8h],di
0x10064ad0: mov edi,[esp+1Ch]
0x10064ad4: add ecx,edi
0x10064ad6: add esi,0FFFFFFF5h
0x10064ad9: cmp ecx,9
0x10064adc: mov [eax+16BCh],ecx
0x10064ae2: jle short 10064B34h
0x10064ae4: mov edi,esi
0x10064ae6: shl edi,cl
0x10064ae8: mov ecx,[eax+8]
0x10064aeb: or [eax+16B8h],di
0x10064af2: movzx ebx,byte ptr [eax+16B8h]
0x10064af9: mov edi,[eax+14h]
0x10064afc: mov [ecx+edi],bl
0x10064aff: add [eax+14h],ebp
0x10064b02: movzx ebx,byte ptr [eax+16B9h]
0x10064b09: mov edi,[eax+14h]
0x10064b0c: mov ecx,[eax+8]
0x10064b0f: mov [edi+ecx],bl
0x10064b12: mov ebx,[eax+16BCh]
0x10064b18: add [eax+14h],ebp
0x10064b1b: mov cl,10h
0x10064b1d: sub cl,bl
0x10064b1f: shr si,cl
0x10064b22: add ebx,0FFFFFFF7h
0x10064b25: mov [eax+16BCh],ebx
0x10064b2b: mov [eax+16B8h],si
0x10064b32: jmp short 10064B46h
0x10064b34: shl esi,cl
0x10064b36: or [eax+16B8h],si
0x10064b3d: add ecx,7
0x10064b40: mov [eax+16BCh],ecx
0x10064b46: mov ecx,[esp+14h]
0x10064b4a: xor esi,esi
0x10064b4c: test ecx,ecx
0x10064b4e: mov [esp+1Ch],edx
0x10064b52: jne short 10064B5Eh
0x10064b54: mov ecx,8Ah
0x10064b59: lea edi,[esi+3]
0x10064b5c: jmp short 10064B74h
0x10064b5e: cmp edx,ecx
0x10064b60: jne short 10064B6Ch
0x10064b62: mov ecx,6
0x10064b67: lea edi,[ecx-3]
0x10064b6a: jmp short 10064B74h
0x10064b6c: mov ecx,7
0x10064b71: lea edi,[ecx-3]
0x10064b74: add dword ptr [esp+18h],4
0x10064b79: sub [esp+20h],ebp
0x10064b7d: jne near ptr 100646D0h
0x10064b83: pop edi
0x10064b84: pop esi
0x10064b85: pop ebp
0x10064b86: pop ebx
0x10064b87: add esp,18h
0x10064b8a: ret
```

`sub_10064680` is called by sub_10064b90. It calls nothing else.

12.917 sub_10064b90 at 0x10064b90 (492 instructions)

```
0x10064680: sub esp,18h
0x10064683: push ebx
0x10064684: push ebp
0x10064685: movzx ebp,word ptr [edx+2]
0x10064689: push esi
0x1006468a: xor esi,esi
0x1006468c: test ebp,ebp
0x1006468e: push edi
0x1006468f: mov ebx,ecx
0x10064691: mov dword ptr [esp+1Ch],0FFFFFFFFh
0x10064699: mov [esp+14h],ebp
0x1006469d: lea ecx,[esi+7]
0x100646a0: lea edi,[esi+4]
0x100646a3: jne short 100646ADh
0x100646a5: mov ecx,8Ah
0x100646aa: lea edi,[ebp+3]
0x100646ad: test ebx,ebx
0x100646af: jl near ptr 10064B83h
0x100646b5: add edx,6
0x100646b8: add ebx,1
0x100646bb: mov [esp+18h],edx
0x100646bf: mov [esp+20h],ebx
0x100646c3: mov ebp,1
0x100646c8: jmp short 100646D0h
0x100646d0: mov ebx,[esp+18h]
0x100646d4: movzx ebx,word ptr [ebx]
0x100646d7: mov edx,[esp+14h]
0x100646db: add esi,ebp
0x100646dd: cmp esi,ecx
0x100646df: mov [esp+24h],edx
0x100646e3: mov [esp+14h],ebx
0x100646e7: mov [esp+10h],esi
0x100646eb: jge short 100646F5h
0x100646ed: cmp edx,ebx
0x100646ef: je near ptr 10064B74h
0x100646f5: cmp esi,edi
0x100646f7: jge near ptr 1006479Fh
0x100646fd: lea ecx,[ecx]
0x10064700: movzx edi,word ptr [eax+edx*4+0A7Eh]
0x10064708: mov ecx,[eax+16BCh]
0x1006470e: mov ebx,10h
0x10064713: sub ebx,edi
0x10064715: cmp ecx,ebx
0x10064717: jle short 10064774h
0x10064719: movzx esi,word ptr [eax+edx*4+0A7Ch]
0x10064721: mov edx,esi
0x10064723: shl edx,cl
0x10064725: mov ecx,[eax+8]
0x10064728: or [eax+16B8h],dx
0x1006472f: movzx ebx,byte ptr [eax+16B8h]
0x10064736: mov edx,[eax+14h]
0x10064739: mov [ecx+edx],bl
0x1006473c: add [eax+14h],ebp
0x1006473f: mov ecx,[eax+14h]
0x10064742: movzx ebx,byte ptr [eax+16B9h]
0x10064749: mov edx,[eax+8]
0x1006474c: mov [ecx+edx],bl
0x1006474f: mov edx,[eax+16BCh]
0x10064755: add [eax+14h],ebp
0x10064758: mov cl,10h
0x1006475a: sub cl,dl
0x1006475c: shr si,cl
0x1006475f: lea ecx,[edx+edi-10h]
```

```
0x10064763: mov edx,[esp+24h]
0x10064767: mov [eax+16B8h],si
0x1006476e: mov esi,[esp+10h]
0x10064772: jmp short 10064788h
0x10064774: mov bx,[eax+edx*4+0A7Ch]
0x1006477c: shl bx,cl
0x1006477f: or [eax+16B8h],bx
0x10064786: add ecx,edi
0x10064788: sub esi,ebp
0x1006478a: mov [eax+16BCh],ecx
0x10064790: mov [esp+10h],esi
0x10064794: jne near ptr 10064700h
0x1006479a: jmp near ptr 10064B46h
0x1006479f: test edx,edx
0x100647a1: je near ptr 1006494Ch
0x100647a7: cmp edx,[esp+1Ch]
0x100647ab: je near ptr 1006484Dh
0x100647b1: movzx edi,word ptr [eax+edx*4+0A7Eh]
0x100647b9: mov ecx,[eax+16BCh]
0x100647bf: mov ebx,10h
0x100647c4: sub ebx,edi
0x100647c6: cmp ecx,ebx
0x100647c8: mov [esp+1Ch],edi
0x100647cc: jle short 10064829h
0x100647ce: movzx esi,word ptr [eax+edx*4+0A7Ch]
0x100647d6: mov edi,esi
0x100647d8: shl edi,cl
0x100647da: mov ecx,[eax+8]
0x100647dd: or [eax+16B8h],di
0x100647e4: movzx ebx,byte ptr [eax+16B8h]
0x100647eb: mov edi,[eax+14h]
0x100647ee: mov [ecx+edi],bl
0x100647f1: add [eax+14h],ebp
0x100647f4: movzx ebx,byte ptr [eax+16B9h]
0x100647fb: mov ecx,[eax+8]
0x100647fe: mov edi,[eax+14h]
0x10064801: mov [edi+ecx],bl
0x10064804: mov ebx,[eax+16BCh]
0x1006480a: add [eax+14h],ebp
0x1006480d: mov cl,10h
0x1006480f: sub cl,bl
0x10064811: shr si,cl
0x10064814: mov ecx,[esp+1Ch]
0x10064818: lea ecx,[ebx+ecx-10h]
0x1006481c: mov [eax+16B8h],si
0x10064823: mov esi,[esp+10h]
0x10064827: jmp short 10064841h
0x10064829: mov di,[eax+edx*4+0A7Ch]
0x10064831: shl di,cl
0x10064834: or [eax+16B8h],di
0x1006483b: mov edi,[esp+1Ch]
0x1006483f: add ecx,edi
0x10064841: sub esi,ebp
0x10064843: mov [eax+16BCh],ecx
0x10064849: mov [esp+10h],esi
0x1006484d: movzx edi,word ptr [eax+0ABEh]
0x10064854: mov ecx,[eax+16BCh]
0x1006485a: mov ebx,10h
0x1006485f: sub ebx,edi
0x10064861: cmp ecx,ebx
0x10064863: mov [esp+1Ch],edi
0x10064867: jle short 100648C3h
0x10064869: movzx esi,word ptr [eax+0ABCh]
0x10064870: mov edi,esi
0x10064872: shl edi,cl
0x10064874: mov ecx,[eax+8]
0x10064877: or [eax+16B8h],di
```

```
0x1006487e: movzx ebx,byte ptr [eax+16B8h]
0x10064885: mov edi,[eax+14h]
0x10064888: mov [ecx+edi],bl
0x1006488b: add [eax+14h],ebp
0x1006488e: movzx ebx,byte ptr [eax+16B9h]
0x10064895: mov ecx,[eax+8]
0x10064898: mov edi,[eax+14h]
0x1006489b: mov [edi+ecx],bl
0x1006489e: mov ebx,[eax+16BCh]
0x100648a4: add [eax+14h],ebp
0x100648a7: mov cl,10h
0x100648a9: sub cl,bl
0x100648ab: shr si,cl
0x100648ae: mov ecx,[esp+1Ch]
0x100648b2: lea ecx,[ebx+ecx-10h]
0x100648b6: mov [eax+16B8h],si
0x100648bd: mov esi,[esp+10h]
0x100648c1: jmp short 100648DAh
0x100648c3: mov di,[eax+0ABCh]
0x100648ca: shl di,cl
0x100648cd: or [eax+16B8h],di
0x100648d4: mov edi,[esp+1Ch]
0x100648d8: add ecx,edi
0x100648da: add esi,0FFFFFFFDh
0x100648dd: cmp ecx,0Eh
0x100648e0: mov [eax+16BCh],ecx
0x100648e6: jle short 1006493Bh
0x100648e8: mov edi,esi
0x100648ea: shl edi,cl
0x100648ec: mov ecx,[eax+8]
0x100648ef: or [eax+16B8h],di
0x100648f6: movzx ebx,byte ptr [eax+16B8h]
0x100648fd: mov edi,[eax+14h]
0x10064900: mov [ecx+edi],bl
0x10064903: add [eax+14h],ebp
0x10064906: movzx ebx,byte ptr [eax+16B9h]
0x1006490d: mov edi,[eax+14h]
0x10064910: mov ecx,[eax+8]
0x10064913: mov [edi+ecx],bl
0x10064916: mov ebx,[eax+16BCh]
0x1006491c: add [eax+14h],ebp
0x1006491f: mov cl,10h
0x10064921: sub cl,bl
0x10064923: shr si,cl
0x10064926: add ebx,0FFFFFFF2h
0x10064929: mov [eax+16BCh],ebx
0x1006492f: mov [eax+16B8h],si
0x10064936: jmp near ptr 10064B46h
0x1006493b: shl esi,cl
0x1006493d: or [eax+16B8h],si
0x10064944: add ecx,2
0x10064947: jmp near ptr 10064B40h
0x1006494c: cmp esi,0Ah
0x1006494f: mov ecx,[eax+16BCh]
0x10064955: mov ebx,10h
0x1006495a: jg near ptr 10064A54h
0x10064960: movzx edi,word ptr [eax+0AC2h]
0x10064967: sub ebx,edi
0x10064969: cmp ecx,ebx
0x1006496b: mov [esp+1Ch],edi
0x1006496f: jle short 100649CBh
0x10064971: movzx esi,word ptr [eax+0AC0h]
0x10064978: mov edi,esi
0x1006497a: shl edi,cl
0x1006497c: mov ecx,[eax+8]
0x1006497f: or [eax+16B8h],di
0x10064986: movzx ebx,byte ptr [eax+16B8h]
```

```
0x1006498d: mov edi,[eax+14h]
0x10064990: mov [ecx+edi],bl
0x10064993: add [eax+14h],ebp
0x10064996: movzx ebx,byte ptr [eax+16B9h]
0x1006499d: mov ecx,[eax+8]
0x100649a0: mov edi,[eax+14h]
0x100649a3: mov [edi+ecx],bl
0x100649a6: mov ebx,[eax+16BCh]
0x100649ac: add [eax+14h],ebp
0x100649af: mov cl,10h
0x100649b1: sub cl,bl
0x100649b3: shr si,cl
0x100649b6: mov ecx,[esp+1Ch]
0x100649ba: lea ecx,[ebx+ecx-10h]
0x100649be: mov [eax+16B8h],si
0x100649c5: mov esi,[esp+10h]
0x100649c9: jmp short 100649E2h
0x100649cb: mov di,[eax+0AC0h]
0x100649d2: shl di,cl
0x100649d5: or [eax+16B8h],di
0x100649dc: mov edi,[esp+1Ch]
0x100649e0: add ecx,edi
0x100649e2: add esi,0FFFFFFFDh
0x100649e5: cmp ecx,0Dh
0x100649e8: mov [eax+16BCh],ecx
0x100649ee: jle short 10064A43h
0x100649f0: mov edi,esi
0x100649f2: shl edi,cl
0x100649f4: mov ecx,[eax+8]
0x100649f7: or [eax+16B8h],di
0x100649fe: movzx ebx,byte ptr [eax+16B8h]
0x10064a05: mov edi,[eax+14h]
0x10064a08: mov [ecx+edi],bl
0x10064a0b: add [eax+14h],ebp
0x10064a0e: movzx ebx,byte ptr [eax+16B9h]
0x10064a15: mov edi,[eax+14h]
0x10064a18: mov ecx,[eax+8]
0x10064a1b: mov [edi+ecx],bl
0x10064a1e: mov ebx,[eax+16BCh]
0x10064a24: add [eax+14h],ebp
0x10064a27: mov cl,10h
0x10064a29: sub cl,bl
0x10064a2b: shr si,cl
0x10064a2e: add ebx,0FFFFFFF3h
0x10064a31: mov [eax+16BCh],ebx
0x10064a37: mov [eax+16B8h],si
0x10064a3e: jmp near ptr 10064B46h
0x10064a43: shl esi,cl
0x10064a45: or [eax+16B8h],si
0x10064a4c: add ecx,3
0x10064a4f: jmp near ptr 10064B40h
0x10064a54: movzx edi,word ptr [eax+0AC6h]
0x10064a5b: sub ebx,edi
0x10064a5d: cmp ecx,ebx
0x10064a5f: mov [esp+1Ch],edi
0x10064a63: jle short 10064ABFh
0x10064a65: movzx esi,word ptr [eax+0AC4h]
0x10064a6c: mov edi,esi
0x10064a6e: shl edi,cl
0x10064a70: mov ecx,[eax+8]
0x10064a73: or [eax+16B8h],di
0x10064a7a: movzx ebx,byte ptr [eax+16B8h]
0x10064a81: mov edi,[eax+14h]
0x10064a84: mov [ecx+edi],bl
0x10064a87: add [eax+14h],ebp
0x10064a8a: movzx ebx,byte ptr [eax+16B9h]
0x10064a91: mov ecx,[eax+8]
```

```
0x10064a94: mov edi,[eax+14h]
0x10064a97: mov [edi+ecx],bl
0x10064a9a: mov ebx,[eax+16BCh]
0x10064aa0: add [eax+14h],ebp
0x10064aa3: mov cl,10h
0x10064aa5: sub cl,bl
0x10064aa7: shr si,cl
0x10064aaa: mov ecx,[esp+1Ch]
0x10064aae: lea ecx,[ebx+ecx-10h]
0x10064ab2: mov [eax+16B8h],si
0x10064ab9: mov esi,[esp+10h]
0x10064abd: jmp short 10064AD6h
0x10064abf: mov di,[eax+0AC4h]
0x10064ac6: shl di,cl
0x10064ac9: or [eax+16B8h],di
0x10064ad0: mov edi,[esp+1Ch]
0x10064ad4: add ecx,edi
0x10064ad6: add esi,0FFFFFFF5h
0x10064ad9: cmp ecx,9
0x10064adc: mov [eax+16BCh],ecx
0x10064ae2: jle short 10064B34h
0x10064ae4: mov edi,esi
0x10064ae6: shl edi,cl
0x10064ae8: mov ecx,[eax+8]
0x10064aeb: or [eax+16B8h],di
0x10064af2: movzx ebx,byte ptr [eax+16B8h]
0x10064af9: mov edi,[eax+14h]
0x10064afc: mov [ecx+edi],bl
0x10064aff: add [eax+14h],ebp
0x10064b02: movzx ebx,byte ptr [eax+16B9h]
0x10064b09: mov edi,[eax+14h]
0x10064b0c: mov ecx,[eax+8]
0x10064b0f: mov [edi+ecx],bl
0x10064b12: mov ebx,[eax+16BCh]
0x10064b18: add [eax+14h],ebp
0x10064b1b: mov cl,10h
0x10064b1d: sub cl,bl
0x10064b1f: shr si,cl
0x10064b22: add ebx,0FFFFFFF7h
0x10064b25: mov [eax+16BCh],ebx
0x10064b2b: mov [eax+16B8h],si
0x10064b32: jmp short 10064B46h
0x10064b34: shl esi,cl
0x10064b36: or [eax+16B8h],si
0x10064b3d: add ecx,7
0x10064b40: mov [eax+16BCh],ecx
0x10064b46: mov ecx,[esp+14h]
0x10064b4a: xor esi,esi
0x10064b4c: test ecx,ecx
0x10064b4e: mov [esp+1Ch],edx
0x10064b52: jne short 10064B5Eh
0x10064b54: mov ecx,8Ah
0x10064b59: lea edi,[esi+3]
0x10064b5c: jmp short 10064B74h
0x10064b5e: cmp edx,ecx
0x10064b60: jne short 10064B6Ch
0x10064b62: mov ecx,6
0x10064b67: lea edi,[ecx-3]
0x10064b6a: jmp short 10064B74h
0x10064b6c: mov ecx,7
0x10064b71: lea edi,[ecx-3]
0x10064b74: add dword ptr [esp+18h],4
0x10064b79: sub [esp+20h],ebp
0x10064b7d: jne near ptr 100646D0h
0x10064b83: pop edi
0x10064b84: pop esi
0x10064b85: pop ebp
```

```
0x10064b86: pop ebx
0x10064b87: add esp,18h
0x10064b8a: ret
0x10064b90: push ecx
0x10064b91: mov ecx,[eax+16BCh]
0x10064b97: cmp ecx,0Bh
0x10064b9a: push ebx
0x10064b9b: mov ebx,[esp+14h]
0x10064b9f: push ebp
0x10064ba0: push esi
0x10064ba1: push edi
0x10064ba2: mov ebp,1
0x10064ba7: jle short 10064C07h
0x10064ba9: mov esi,[esp+18h]
0x10064bad: add esi,0FFFFFEFFh
0x10064bb3: mov edx,esi
0x10064bb5: shl edx,cl
0x10064bb7: mov ecx,[eax+14h]
0x10064bba: or [eax+16B8h],dx
0x10064bc1: movzx ebx,byte ptr [eax+16B8h]
0x10064bc8: mov edx,[eax+8]
0x10064bcb: mov [ecx+edx],bl
0x10064bce: add [eax+14h],ebp
0x10064bd1: movzx ebx,byte ptr [eax+16B9h]
0x10064bd8: mov ecx,[eax+14h]
0x10064bdb: mov edx,[eax+8]
0x10064bde: mov [ecx+edx],bl
0x10064be1: mov edx,[eax+16BCh]
0x10064be7: add [eax+14h],ebp
0x10064bea: mov ebx,[esp+20h]
0x10064bee: mov cl,10h
0x10064bf0: sub cl,dl
0x10064bf2: shr si,cl
0x10064bf5: add edx,0FFFFFFF5h
0x10064bf8: mov [eax+16BCh],edx
0x10064bfe: mov [eax+16B8h],si
0x10064c05: jmp short 10064C23h
0x10064c07: mov edx,[esp+18h]
0x10064c0b: add edx,0FFFFFEFFh
0x10064c11: shl edx,cl
0x10064c13: or [eax+16B8h],dx
0x10064c1a: add ecx,5
0x10064c1d: mov [eax+16BCh],ecx
0x10064c23: mov ecx,[eax+16BCh]
0x10064c29: cmp ecx,0Bh
0x10064c2c: jle short 10064C8Dh
0x10064c2e: mov esi,[esp+1Ch]
0x10064c32: add esi,0FFFFFFFFh
0x10064c35: mov edx,esi
0x10064c37: shl edx,cl
0x10064c39: mov ecx,[eax+14h]
0x10064c3c: mov [esp+10h],ebx
0x10064c40: or [eax+16B8h],dx
0x10064c47: movzx ebx,byte ptr [eax+16B8h]
0x10064c4e: mov edx,[eax+8]
0x10064c51: mov [ecx+edx],bl
0x10064c54: add [eax+14h],ebp
0x10064c57: movzx ebx,byte ptr [eax+16B9h]
0x10064c5e: mov ecx,[eax+14h]
0x10064c61: mov edx,[eax+8]
0x10064c64: mov [ecx+edx],bl
0x10064c67: mov edx,[eax+16BCh]
0x10064c6d: add [eax+14h],ebp
0x10064c70: mov ebx,[esp+10h]
0x10064c74: mov cl,10h
0x10064c76: sub cl,dl
0x10064c78: shr si,cl
```

```
0x10064c7b: add edx,0FFFFFFF5h
0x10064c7e: mov [eax+16BCh],edx
0x10064c84: mov [eax+16B8h],si
0x10064c8b: jmp short 10064CA6h
0x10064c8d: mov edx,[esp+1Ch]
0x10064c91: add edx,0FFFFFFFFh
0x10064c94: shl edx,cl
0x10064c96: or [eax+16B8h],dx
0x10064c9d: add ecx,5
0x10064ca0: mov [eax+16BCh],ecx
0x10064ca6: mov ecx,[eax+16BCh]
0x10064cac: cmp ecx,0Ch
0x10064caf: jle short 10064D0Ch
0x10064cb1: lea esi,[ebx-4]
0x10064cb4: mov edx,esi
0x10064cb6: shl edx,cl
0x10064cb8: mov ecx,[eax+14h]
0x10064cbb: mov [esp+10h],ebx
0x10064cbf: or [eax+16B8h],dx
0x10064cc6: movzx ebx,byte ptr [eax+16B8h]
0x10064ccd: mov edx,[eax+8]
0x10064cd0: mov [ecx+edx],bl
0x10064cd3: add [eax+14h],ebp
0x10064cd6: movzx ebx,byte ptr [eax+16B9h]
0x10064cdd: mov ecx,[eax+14h]
0x10064ce0: mov edx,[eax+8]
0x10064ce3: mov [ecx+edx],bl
0x10064ce6: mov edx,[eax+16BCh]
0x10064cec: add [eax+14h],ebp
0x10064cef: mov ebx,[esp+10h]
0x10064cf3: mov cl,10h
0x10064cf5: sub cl,dl
0x10064cf7: shr si,cl
0x10064cfa: add edx,0FFFFFFF4h
0x10064cfd: mov [eax+16BCh],edx
0x10064d03: mov [eax+16B8h],si
0x10064d0a: jmp short 10064D21h
0x10064d0c: lea edx,[ebx-4]
0x10064d0f: shl edx,cl
0x10064d11: or [eax+16B8h],dx
0x10064d18: add ecx,4
0x10064d1b: mov [eax+16BCh],ecx
0x10064d21: xor edi,edi
0x10064d23: test ebx,ebx
0x10064d25: jle near ptr 10064DC3h
0x10064d2b: jmp short 10064D30h
0x10064d30: mov ecx,[eax+16BCh]
0x10064d36: cmp ecx,0Dh
0x10064d39: movzx edx,byte ptr [edi+101076ACh]
0x10064d40: jle short 10064D9Eh
0x10064d42: movzx esi,word ptr [eax+edx*4+0A7Eh]
0x10064d4a: mov edx,esi
0x10064d4c: shl edx,cl
0x10064d4e: mov ecx,[eax+14h]
0x10064d51: or [eax+16B8h],dx
0x10064d58: movzx ebx,byte ptr [eax+16B8h]
0x10064d5f: mov edx,[eax+8]
0x10064d62: mov [ecx+edx],bl
0x10064d65: add [eax+14h],ebp
0x10064d68: movzx ebx,byte ptr [eax+16B9h]
0x10064d6f: mov ecx,[eax+14h]
0x10064d72: mov edx,[eax+8]
0x10064d75: mov [ecx+edx],bl
0x10064d78: mov edx,[eax+16BCh]
0x10064d7e: add [eax+14h],ebp
0x10064d81: mov ebx,[esp+20h]
0x10064d85: mov cl,10h
```

```
0x10064d87: sub cl,dl
0x10064d89: shr si,cl
0x10064d8c: add edx,0FFFFFFF3h
0x10064d8f: mov [eax+16BCh],edx
0x10064d95: mov [eax+16B8h],si
0x10064d9c: jmp short 10064DB9h
0x10064d9e: mov dx,[eax+edx*4+0A7Eh]
0x10064da6: shl dx,cl
0x10064da9: or [eax+16B8h],dx
0x10064db0: add ecx,3
0x10064db3: mov [eax+16BCh],ecx
0x10064db9: add edi,ebp
0x10064dbb: cmp edi,ebx
0x10064dbd: jl near ptr 10064D30h
0x10064dc3: mov ecx,[esp+18h]
0x10064dc7: add ecx,0FFFFFFFFh
0x10064dca: lea edx,[eax+94h]
0x10064dd0: call 10064680h
0x10064dd5: mov ecx,[esp+1Ch]
0x10064dd9: pop edi
0x10064dda: pop esi
0x10064ddb: pop ebp
0x10064ddc: add ecx,0FFFFFFFFh
0x10064ddf: lea edx,[eax+988h]
0x10064de5: pop ebx
0x10064de6: add esp,4
0x10064de9: jmp near ptr 10064680h
```

`sub_10064b90` is called by sub_10065a70. It delegates to sub_10064680 (1 callees).

12.918 sub_10064df0 at 0x10064df0 (261 instructions)

```
0x10064df0: sub esp,0Ch
0x10064df3: push ebx
0x10064df4: mov ebx,[esp+14h]
0x10064df8: push ebp
0x10064df9: xor ecx,ecx
0x10064dfb: cmp [eax+16A0h],ecx
0x10064e01: push esi
0x10064e02: push edi
0x10064e03: je near ptr 10065142h
0x10064e09: lea esp,[esp]
0x10064e10: mov edx,[eax+16A4h]
0x10064e16: movzx ebp,word ptr [edx+ecx*2]
0x10064e1a: mov edx,[eax+1698h]
0x10064e20: movzx esi,byte ptr [ecx+edx]
0x10064e24: add ecx,1
0x10064e27: test ebp,ebp
0x10064e29: mov [esp+18h],ecx
0x10064e2d: mov ecx,[eax+16BCh]
0x10064e33: jne near ptr 10064EBAh
0x10064e39: movzx edi,word ptr [ebx+esi*4+2]
0x10064e3e: mov edx,10h
0x10064e43: sub edx,edi
0x10064e45: cmp ecx,edx
0x10064e47: jle short 10064EA5h
0x10064e49: movzx esi,word ptr [ebx+esi*4]
0x10064e4d: mov edx,esi
0x10064e4f: shl edx,cl
0x10064e51: mov ecx,[eax+8]
0x10064e54: mov [esp+10h],ebx
0x10064e58: or [eax+16B8h],dx
0x10064e5f: movzx ebx,byte ptr [eax+16B8h]
0x10064e66: mov edx,[eax+14h]
0x10064e69: mov [ecx+edx],bl
0x10064e6c: add dword ptr [eax+14h],1
```

```
0x10064e70: movzx ebx,byte ptr [eax+16B9h]
0x10064e77: mov ecx,[eax+14h]
0x10064e7a: mov edx,[eax+8]
0x10064e7d: mov [ecx+edx],bl
0x10064e80: mov edx,[eax+16BCh]
0x10064e86: add dword ptr [eax+14h],1
0x10064e8a: mov ebx,[esp+10h]
0x10064e8e: mov cl,10h
0x10064e90: sub cl,dl
0x10064e92: shr si,cl
0x10064e95: lea ecx,[edx+edi-10h]
0x10064e99: mov [eax+16B8h],si
0x10064ea0: jmp near ptr 1006512Ch
0x10064ea5: mov dx,[ebx+esi*4]
0x10064ea9: shl dx,cl
0x10064eac: or [eax+16B8h],dx
0x10064eb3: add ecx,edi
0x10064eb5: jmp near ptr 1006512Ch
0x10064eba: movzx edx,byte ptr [esi+10107E80h]
0x10064ec1: movzx edi,word ptr [ebx+edx*4+406h]
0x10064ec9: mov ebx,10h
0x10064ece: sub ebx,edi
0x10064ed0: cmp ecx,ebx
0x10064ed2: mov [esp+10h],edi
0x10064ed6: mov edi,[esp+20h]
0x10064eda: mov [esp+14h],edx
0x10064ede: jle short 10064F43h
0x10064ee0: movzx edi,word ptr [edi+edx*4+404h]
0x10064ee8: mov edx,edi
0x10064eea: shl edx,cl
0x10064eec: mov ecx,[eax+8]
0x10064eef: or [eax+16B8h],dx
0x10064ef6: movzx ebx,byte ptr [eax+16B8h]
0x10064efd: mov edx,[eax+14h]
0x10064f00: mov [ecx+edx],bl
0x10064f03: add dword ptr [eax+14h],1
0x10064f07: mov ecx,[eax+14h]
0x10064f0a: movzx ebx,byte ptr [eax+16B9h]
0x10064f11: mov edx,[eax+8]
0x10064f14: mov [ecx+edx],bl
0x10064f17: mov edx,[eax+16BCh]
0x10064f1d: add dword ptr [eax+14h],1
0x10064f21: mov cl,10h
0x10064f23: sub cl,dl
0x10064f25: shr di,cl
0x10064f28: mov ecx,[esp+10h]
0x10064f2c: lea edx,[edx+ecx-10h]
0x10064f30: mov [eax+16BCh],edx
0x10064f36: mov edx,[esp+14h]
0x10064f3a: mov [eax+16B8h],di
0x10064f41: jmp short 10064F61h
0x10064f43: mov di,[edi+edx*4+404h]
0x10064f4b: shl di,cl
0x10064f4e: or [eax+16B8h],di
0x10064f55: mov edi,[esp+10h]
0x10064f59: add ecx,edi
0x10064f5b: mov [eax+16BCh],ecx
0x10064f61: mov edi,[edx*4+10107638h]
0x10064f68: test edi,edi
0x10064f6a: mov ebx,[esp+20h]
0x10064f6e: je short 10064FEAh
0x10064f70: sub esi,[edx*4+10107F80h]
0x10064f77: mov ecx,[eax+16BCh]
0x10064f7d: mov edx,10h
0x10064f82: sub edx,edi
0x10064f84: cmp ecx,edx
0x10064f86: jle short 10064FD9h
```

```
0x10064f88: mov edx,esi
0x10064f8a: shl edx,cl
0x10064f8c: mov ecx,[eax+8]
0x10064f8f: or [eax+16B8h],dx
0x10064f96: movzx ebx,byte ptr [eax+16B8h]
0x10064f9d: mov edx,[eax+14h]
0x10064fa0: mov [ecx+edx],bl
0x10064fa3: add dword ptr [eax+14h],1
0x10064fa7: movzx ebx,byte ptr [eax+16B9h]
0x10064fae: mov ecx,[eax+14h]
0x10064fb1: mov edx,[eax+8]
0x10064fb4: mov [ecx+edx],bl
0x10064fb7: mov edx,[eax+16BCh]
0x10064fbd: add dword ptr [eax+14h],1
0x10064fc1: mov ebx,[esp+20h]
0x10064fc5: mov cl,10h
0x10064fc7: sub cl,dl
0x10064fc9: shr si,cl
0x10064fcc: lea ecx,[edx+edi-10h]
0x10064fd0: mov [eax+16B8h],si
0x10064fd7: jmp short 10064FE4h
0x10064fd9: shl esi,cl
0x10064fdb: or [eax+16B8h],si
0x10064fe2: add ecx,edi
0x10064fe4: mov [eax+16BCh],ecx
0x10064fea: sub ebp,1
0x10064fed: cmp ebp,100h
0x10064ff3: jae short 10064FFEh
0x10064ff5: movzx edi,byte ptr [ebp+10107C80h]
0x10064ffc: jmp short 1006500Ah
0x10064ffe: mov edx,ebp
0x10065000: shr edx,7
0x10065003: movzx edi,byte ptr [edx+10107D80h]
0x1006500a: mov ecx,[esp+24h]
0x1006500e: movzx edx,word ptr [ecx+edi*4+2]
0x10065013: mov ecx,[eax+16BCh]
0x10065019: mov esi,10h
0x1006501e: sub esi,edx
0x10065020: cmp ecx,esi
0x10065022: mov [esp+14h],edx
0x10065026: jle short 1006508Bh
0x10065028: mov edx,[esp+24h]
0x1006502c: movzx esi,word ptr [edx+edi*4]
0x10065030: mov edx,esi
0x10065032: shl edx,cl
0x10065034: mov ecx,[eax+8]
0x10065037: or [eax+16B8h],dx
0x1006503e: movzx ebx,byte ptr [eax+16B8h]
0x10065045: mov edx,[eax+14h]
0x10065048: mov [ecx+edx],bl
0x1006504b: add dword ptr [eax+14h],1
0x1006504f: movzx ebx,byte ptr [eax+16B9h]
0x10065056: mov ecx,[eax+14h]
0x10065059: mov edx,[eax+8]
0x1006505c: mov [ecx+edx],bl
0x1006505f: mov edx,[eax+16BCh]
0x10065065: add dword ptr [eax+14h],1
0x10065069: mov ebx,[esp+20h]
0x1006506d: mov cl,10h
0x1006506f: sub cl,dl
0x10065071: shr si,cl
0x10065074: mov ecx,[esp+14h]
0x10065078: lea edx,[edx+ecx-10h]
0x1006507c: mov [eax+16BCh],edx
0x10065082: mov [eax+16B8h],si
0x10065089: jmp short 100650A5h
0x1006508b: mov esi,[esp+24h]
```

```
0x1006508f: mov si,[esi+edi*4]
0x10065093: shl si,cl
0x10065096: or [eax+16B8h],si
0x1006509d: add ecx,edx
0x1006509f: mov [eax+16BCh],ecx
0x100650a5: mov esi,[edi*4+101076C0h]
0x100650ac: test esi,esi
0x100650ae: je near ptr 10065132h
0x100650b4: mov ecx,[eax+16BCh]
0x100650ba: sub ebp,[edi*4+10107FF8h]
0x100650c1: mov edx,10h
0x100650c6: sub edx,esi
0x100650c8: cmp ecx,edx
0x100650ca: jle short 10065121h
0x100650cc: mov edx,ebp
0x100650ce: shl edx,cl
0x100650d0: mov ecx,[eax+8]
0x100650d3: mov [esp+14h],ebx
0x100650d7: or [eax+16B8h],dx
0x100650de: movzx ebx,byte ptr [eax+16B8h]
0x100650e5: mov edx,[eax+14h]
0x100650e8: mov [ecx+edx],bl
0x100650eb: add dword ptr [eax+14h],1
0x100650ef: movzx ebx,byte ptr [eax+16B9h]
0x100650f6: mov ecx,[eax+14h]
0x100650f9: mov edx,[eax+8]
0x100650fc: mov [ecx+edx],bl
0x100650ff: mov edx,[eax+16BCh]
0x10065105: add dword ptr [eax+14h],1
0x10065109: mov ebx,[esp+14h]
0x1006510d: mov cl,10h
0x1006510f: sub cl,dl
0x10065111: shr bp,cl
0x10065114: lea ecx,[edx+esi-10h]
0x10065118: mov [eax+16B8h],bp
0x1006511f: jmp short 1006512Ch
0x10065121: shl ebp,cl
0x10065123: or [eax+16B8h],bp
0x1006512a: add ecx,esi
0x1006512c: mov [eax+16BCh],ecx
0x10065132: mov ecx,[esp+18h]
0x10065136: cmp ecx,[eax+16A0h]
0x1006513c: jb near ptr 10064E10h
0x10065142: movzx edi,word ptr [ebx+402h]
0x10065149: mov ecx,[eax+16BCh]
0x1006514f: mov edx,10h
0x10065154: sub edx,edi
0x10065156: cmp ecx,edx
0x10065158: jle short 100651CDh
0x1006515a: movzx esi,word ptr [ebx+400h]
0x10065161: mov edx,esi
0x10065163: shl edx,cl
0x10065165: mov ecx,[eax+8]
0x10065168: mov [esp+20h],ebx
0x1006516c: or [eax+16B8h],dx
0x10065173: mov edx,[eax+14h]
0x10065176: mov bl,[eax+16B8h]
0x1006517c: mov [ecx+edx],bl
0x1006517f: add dword ptr [eax+14h],1
0x10065183: mov ebp,[eax+14h]
0x10065186: mov ecx,[eax+8]
0x10065189: mov dl,[eax+16B9h]
0x1006518f: mov ebx,[esp+20h]
0x10065193: mov [ecx+ebp],dl
0x10065196: mov edx,[eax+16BCh]
0x1006519c: add dword ptr [eax+14h],1
0x100651a0: mov cl,10h
```

```
0x100651a2: sub cl,dl
0x100651a4: shr si,cl
0x100651a7: lea ecx,[edx+edi-10h]
0x100651ab: pop edi
0x100651ac: mov [eax+16BCh],ecx
0x100651b2: mov [eax+16B8h],si
0x100651b9: movzx edx,word ptr [ebx+402h]
0x100651c0: pop esi
0x100651c1: pop ebp
0x100651c2: mov [eax+16B4h],edx
0x100651c8: pop ebx
0x100651c9: add esp,0Ch
0x100651cc: ret
0x100651cd: mov dx,[ebx+400h]
0x100651d4: shl dx,cl
0x100651d7: or [eax+16B8h],dx
0x100651de: add ecx,edi
0x100651e0: pop edi
0x100651e1: mov [eax+16BCh],ecx
0x100651e7: movzx ecx,word ptr [ebx+402h]
0x100651ee: pop esi
0x100651ef: pop ebp
0x100651f0: mov [eax+16B4h],ecx
0x100651f6: pop ebx
0x100651f7: add esp,0Ch
0x100651fa: ret
```

`sub_10064df0` is called by sub_10065a70. It calls nothing else.

12.919 sub_10065200 at 0x10065200 (31 instructions)

```
0x10065200: mov edx,0F3FFC07Fh
0x10065205: xor eax,eax
0x10065207: lea ecx,[esi+94h]
0x1006520d: lea ecx,[ecx]
0x10065210: test dl,1
0x10065213: je short 1006521Bh
0x10065215: cmp word ptr [ecx],0
0x10065219: jne short 10065264h
0x1006521b: add eax,1
0x1006521e: add ecx,4
0x10065221: shr edx,1
0x10065223: cmp eax,1Fh
0x10065226: jle short 10065210h
0x10065228: cmp word ptr [esi+0B8h],0
0x10065230: jne short 10065267h
0x10065232: cmp word ptr [esi+0BCh],0
0x1006523a: jne short 10065267h
0x1006523c: cmp word ptr [esi+0C8h],0
0x10065244: jne short 10065267h
0x10065246: mov eax,20h
0x1006524b: lea ecx,[esi+114h]
0x10065251: cmp word ptr [ecx],0
0x10065255: jne short 10065267h
0x10065257: add eax,1
0x1006525a: add ecx,4
0x1006525d: cmp eax,100h
0x10065262: jl short 10065251h
0x10065264: xor eax,eax
0x10065266: ret
0x10065267: mov eax,1
0x1006526c: ret
```

`sub_10065200` is called by sub_10065a70. It calls nothing else.

12.920 sub_10065270 at 0x10065270 (13 instructions)

```
0x10065270: xor eax,eax
0x10065272: push esi
0x10065273: mov esi,ecx
0x10065275: and esi,1
0x10065278: or eax,esi
0x1006527a: sub edx,1
0x1006527d: shr ecx,1
0x1006527f: add eax,eax
0x10065281: test edx,edx
0x10065283: jg short 10065273h
0x10065285: shr eax,1
0x10065287: pop esi
0x10065288: ret
```

`sub_10065270` is called by sub_10065490. It calls nothing else.

12.921 sub_10065290 at 0x10065290 (31 instructions)

```
0x10065290: mov ecx,[eax+16BCh]
0x10065296: cmp ecx,10h
0x10065299: push ebx
0x1006529a: jne short 100652D5h
0x1006529c: movzx ebx,byte ptr [eax+16B8h]
0x100652a3: mov edx,[eax+14h]
0x100652a6: mov ecx,[eax+8]
0x100652a9: mov [ecx+edx],bl
0x100652ac: add dword ptr [eax+14h],1
0x100652b0: movzx ebx,byte ptr [eax+16B9h]
0x100652b7: mov ecx,[eax+14h]
0x100652ba: mov edx,[eax+8]
0x100652bd: mov [ecx+edx],bl
0x100652c0: add dword ptr [eax+14h],1
0x100652c4: xor ecx,ecx
0x100652c6: mov [eax+16BCh],ecx
0x100652cc: mov [eax+16B8h],cx
0x100652d3: pop ebx
0x100652d4: ret
0x100652d5: cmp ecx,8
0x100652d8: jl short 10065303h
0x100652da: mov ecx,[eax+8]
0x100652dd: mov edx,[eax+14h]
0x100652e0: mov bl,[eax+16B8h]
0x100652e6: mov [ecx+edx],bl
0x100652e9: movzx cx,byte ptr [eax+16B9h]
0x100652f1: add dword ptr [eax+14h],1
0x100652f5: add dword ptr [eax+16BCh],0FFFFFFF8h
0x100652fc: mov [eax+16B8h],cx
0x10065303: pop ebx
0x10065304: ret
```

`sub_10065290` is called by sub_10065880. It calls nothing else.

12.922 sub_10065310 at 0x10065310 (35 instructions)

```
0x10065310: mov edx,[eax+16BCh]
0x10065316: cmp edx,8
0x10065319: push ebx
0x1006531a: push esi
0x1006531b: jle short 1006535Ah
0x1006531d: movzx ebx,byte ptr [eax+16B8h]
0x10065324: mov edx,[eax+14h]
0x10065327: mov ecx,[eax+8]
0x1006532a: mov [ecx+edx],bl
0x1006532d: movzx ebx,byte ptr [eax+16B9h]
```

```
0x10065334: mov edx,[eax+8]
0x10065337: mov esi,1
0x1006533c: add [eax+14h],esi
0x1006533f: mov ecx,[eax+14h]
0x10065342: mov [ecx+edx],bl
0x10065345: add [eax+14h],esi
0x10065348: xor ecx,ecx
0x1006534a: pop esi
0x1006534b: mov [eax+16B8h],cx
0x10065352: mov [eax+16BCh],ecx
0x10065358: pop ebx
0x10065359: ret
0x1006535a: xor ecx,ecx
0x1006535c: cmp edx,ecx
0x1006535e: jle short 10065373h
0x10065360: mov esi,[eax+14h]
0x10065363: mov edx,[eax+8]
0x10065366: mov bl,[eax+16B8h]
0x1006536c: mov [edx+esi],bl
0x1006536f: add dword ptr [eax+14h],1
0x10065373: pop esi
0x10065374: mov [eax+16B8h],cx
0x1006537b: mov [eax+16BCh],ecx
0x10065381: pop ebx
0x10065382: ret
```

`sub_10065310` is called by sub_10065390. It calls nothing else.

12.923 sub_10065390 at 0x10065390 (49 instructions)

```
0x10065390: push ebx
0x10065391: push esi
0x10065392: push edi
0x10065393: mov ebx,ecx
0x10065395: mov esi,edx
0x10065397: call 10065310h
0x1006539c: cmp dword ptr [esp+10h],0
0x100653a1: mov dword ptr [eax+16B4h],8
0x100653ab: mov edi,1
0x100653b0: je short 100653ECh
0x100653b2: mov ecx,[eax+14h]
0x100653b5: mov edx,[eax+8]
0x100653b8: mov [ecx+edx],bl
0x100653bb: add [eax+14h],edi
0x100653be: mov ecx,[eax+14h]
0x100653c1: mov edx,[eax+8]
0x100653c4: mov [ecx+edx],bh
0x100653c7: add [eax+14h],edi
0x100653ca: mov edx,[eax+8]
0x100653cd: push ebp
0x100653ce: mov ebp,[eax+14h]
0x100653d1: mov cl,bl
0x100653d3: not cl
0x100653d5: mov [edx+ebp],cl
0x100653d8: add [eax+14h],edi
0x100653db: mov ebp,[eax+14h]
0x100653de: mov edx,[eax+8]
0x100653e1: mov ecx,ebx
0x100653e3: not ecx
0x100653e5: mov [edx+ebp],ch
0x100653e8: add [eax+14h],edi
0x100653eb: pop ebp
0x100653ec: test ebx,ebx
0x100653ee: je short 1006540Eh
0x100653f0: mov ecx,[eax+14h]
0x100653f3: mov edx,[eax+8]
```

```
0x100653f6: sub ebx,edi
0x100653f8: mov [esp+10h],ebx
0x100653fc: mov bl,[esi]
0x100653fe: mov [ecx+edx],bl
0x10065401: mov ebx,[esp+10h]
0x10065405: add [eax+14h],edi
0x10065408: add esi,edi
0x1006540a: test ebx,ebx
0x1006540c: jne short 100653F0h
0x1006540e: pop edi
0x1006540f: pop esi
0x10065410: pop ebx
0x10065411: ret
```

`sub_10065390` is called by sub_100657e0. It delegates to sub_10065310 (1 callees).

12.924 sub_10065420 at 0x10065420 (44 instructions)

```
0x10064230: push esi
0x10064231: lea eax,[edx+94h]
0x10064237: mov ecx,11Eh
0x1006423c: xor esi,esi
0x1006423e: mov edi,edi
0x10064240: mov [eax],si
0x10064243: add eax,4
0x10064246: sub ecx,1
0x10064249: jne short 10064240h
0x1006424b: lea eax,[edx+988h]
0x10064251: mov ecx,1Eh
0x10064256: mov [eax],si
0x10064259: add eax,4
0x1006425c: sub ecx,1
0x1006425f: jne short 10064256h
0x10064261: lea eax,[edx+0A7Ch]
0x10064267: mov ecx,13h
0x1006426c: lea esp,[esp]
0x10064270: mov [eax],si
0x10064273: add eax,4
0x10064276: sub ecx,1
0x10064279: jne short 10064270h
0x1006427b: mov [edx+16ACh],esi
0x10064281: mov [edx+16A8h],esi
0x10064287: mov [edx+16B0h],esi
0x1006428d: mov [edx+16A0h],esi
0x10064293: mov word ptr [edx+494h],1
0x1006429c: pop esi
0x1006429d: ret
0x10065420: mov edx,[esp+4]
0x10065424: lea eax,[edx+94h]
0x1006542a: mov [edx+0B18h],eax
0x10065430: lea eax,[edx+0A7Ch]
0x10065436: mov [edx+0B30h],eax
0x1006543c: xor eax,eax
0x1006543e: lea ecx,[edx+988h]
0x10065444: mov dword ptr [edx+0B20h],1013623Ch
0x1006544e: mov [edx+0B24h],ecx
0x10065454: mov dword ptr [edx+0B2Ch],10136250h
0x1006545e: mov dword ptr [edx+0B38h],10136264h
0x10065468: mov [edx+16B8h],ax
0x1006546f: mov [edx+16BCh],eax
0x10065475: mov dword ptr [edx+16B4h],8
0x1006547f: jmp near ptr 10064230h
```

`sub_10065420` is called by sub_100629f0. It calls nothing else.

12.925 sub_10065490 at 0x10065490 (40 instructions)

```
0x10065490: sub esp,24h
0x10065493: mov eax,ds:[10136220h]
0x10065498: xor eax,esp
0x1006549a: mov [esp+20h],eax
0x1006549e: push esi
0x1006549f: lea esi,[esp+6]
0x100654a3: xor ecx,ecx
0x100654a5: mov eax,1
0x100654aa: sub edx,esi
0x100654ac: lea esp,[esp]
0x100654b0: lea esi,[edx+eax*2]
0x100654b3: mov si,[esp+esi+4]
0x100654b8: add si,cx
0x100654bb: add si,si
0x100654be: movzx ecx,si
0x100654c1: mov [esp+eax*2+4],cx
0x100654c6: add eax,1
0x100654c9: cmp eax,0Fh
0x100654cc: jle short 100654B0h
0x100654ce: xor esi,esi
0x100654d0: test ebx,ebx
0x100654d2: jl short 100654FDh
0x100654d4: movzx edx,word ptr [edi+esi*4+2]
0x100654d9: test edx,edx
0x100654db: je short 100654F6h
0x100654dd: movzx eax,word ptr [esp+edx*2+4]
0x100654e2: movzx ecx,ax
0x100654e5: add eax,1
0x100654e8: mov [esp+edx*2+4],ax
0x100654ed: call 10065270h
0x100654f2: mov [edi+esi*4],ax
0x100654f6: add esi,1
0x100654f9: cmp esi,ebx
0x100654fb: jle short 100654D4h
0x100654fd: mov ecx,[esp+24h]
0x10065501: pop esi
0x10065502: xor ecx,esp
0x10065504: call 10022326h
0x10065509: add esp,24h
0x1006550c: ret
```

`sub_10065490` is called by sub_10065510. It delegates to sub_10022326, sub_10065270 (2 callees).

12.926 sub_10065510 at 0x10065510 (126 instructions)

```
0x10065510: sub esp,8
0x10065513: mov eax,[esp+0Ch]
0x10065517: push ebx
0x10065518: push ebp
0x10065519: push edi
0x1006551a: mov edi,[eax]
0x1006551c: mov eax,[eax+8]
0x1006551f: mov ecx,[eax+0Ch]
0x10065522: mov ebp,[eax]
0x10065524: xor edx,edx
0x10065526: or ebx,0FFFFFFFFh
0x10065529: xor eax,eax
0x1006552b: cmp ecx,edx
0x1006552d: mov [esp+10h],ecx
0x10065531: mov [esp+0Ch],ebx
0x10065535: mov [esi+1450h],edx
0x1006553b: mov dword ptr [esi+1454h],23Dh
0x10065545: jle short 1006557Eh
0x10065547: cmp [edi+eax*4],dx
0x1006554b: je short 10065570h
```

```
0x1006554d: add dword ptr [esi+1450h],1
0x10065554: mov ecx,[esi+1450h]
0x1006555a: mov [esi+ecx*4+0B5Ch],eax
0x10065561: mov [esp+0Ch],eax
0x10065565: mov [eax+esi+1458h],dl
0x1006556c: mov ebx,eax
0x1006556e: jmp short 10065575h
0x10065570: mov [edi+eax*4+2],dx
0x10065575: add eax,1
0x10065578: cmp eax,[esp+10h]
0x1006557c: jl short 10065547h
0x1006557e: cmp dword ptr [esi+1450h],2
0x10065585: jge short 100655D9h
0x10065587: cmp ebx,2
0x1006558a: jge short 10065593h
0x1006558c: add ebx,1
0x1006558f: mov eax,ebx
0x10065591: jmp short 10065595h
0x10065593: xor eax,eax
0x10065595: add dword ptr [esi+1450h],1
0x1006559c: mov ecx,[esi+1450h]
0x100655a2: mov [esi+ecx*4+0B5Ch],eax
0x100655a9: mov word ptr [edi+eax*4],1
0x100655af: mov [esi+eax+1458h],dl
0x100655b6: add dword ptr [esi+16A8h],0FFFFFFFFh
0x100655bd: cmp ebp,edx
0x100655bf: je short 100655CCh
0x100655c1: movzx eax,word ptr [ebp+eax*4+2]
0x100655c6: sub [esi+16ACh],eax
0x100655cc: cmp dword ptr [esi+1450h],2
0x100655d3: jl short 10065587h
0x100655d5: mov [esp+0Ch],ebx
0x100655d9: mov ecx,[esp+18h]
0x100655dd: mov [ecx+4],ebx
0x100655e0: mov eax,[esi+1450h]
0x100655e6: cdq
0x100655e7: sub eax,edx
0x100655e9: mov ebp,eax
0x100655eb: sar ebp,1
0x100655ed: cmp ebp,1
0x100655f0: jl short 10065605h
0x100655f2: push ebp
0x100655f3: mov eax,esi
0x100655f5: call 100642A0h
0x100655fa: sub ebp,1
0x100655fd: add esp,4
0x10065600: cmp ebp,1
0x10065603: jge short 100655F2h
0x10065605: mov ebx,[esp+10h]
0x10065609: lea esp,[esp]
0x10065610: mov eax,[esi+1450h]
0x10065616: mov edx,[esi+eax*4+0B5Ch]
0x1006561d: mov ebp,[esi+0B60h]
0x10065623: add eax,0FFFFFFFFh
0x10065626: mov [esi+1450h],eax
0x1006562c: push 1
0x1006562e: mov eax,esi
0x10065630: mov [esi+0B60h],edx
0x10065636: call 100642A0h
0x1006563b: mov eax,[esi+0B60h]
0x10065641: or edx,0FFFFFFFFh
0x10065644: add [esi+1454h],edx
0x1006564a: mov ecx,[esi+1454h]
0x10065650: mov [esi+ecx*4+0B5Ch],ebp
0x10065657: add [esi+1454h],edx
0x1006565d: mov ecx,[esi+1454h]
0x10065663: mov [esi+ecx*4+0B5Ch],eax
```

```
0x1006566a: mov cx,[edi+eax*4]
0x1006566e: add cx,[edi+ebp*4]
0x10065672: add esp,4
0x10065675: mov [edi+ebx*4],cx
0x10065679: mov cl,[esi+eax+1458h]
0x10065680: mov dl,[esi+ebp+1458h]
0x10065687: cmp dl,cl
0x10065689: jb short 10065690h
0x1006568b: movzx ecx,dl
0x1006568e: jmp short 10065693h
0x10065690: movzx ecx,cl
0x10065693: add cl,1
0x10065696: mov [esi+ebx+1458h],cl
0x1006569d: movzx ecx,bx
0x100656a0: mov [edi+eax*4+2],cx
0x100656a5: mov [edi+ebp*4+2],cx
0x100656aa: mov [esi+0B60h],ebx
0x100656b0: push 1
0x100656b2: mov eax,esi
0x100656b4: add ebx,1
0x100656b7: call 100642A0h
0x100656bc: add esp,4
0x100656bf: cmp dword ptr [esi+1450h],2
0x100656c6: jge near ptr 10065610h
0x100656cc: add dword ptr [esi+1454h],0FFFFFFFFh
0x100656d3: mov eax,[esi+1454h]
0x100656d9: mov edx,[esi+0B60h]
0x100656df: mov ecx,[esp+18h]
0x100656e3: mov [esi+eax*4+0B5Ch],edx
0x100656ea: mov eax,esi
0x100656ec: call 10064370h
0x100656f1: mov ebx,[esp+0Ch]
0x100656f5: lea edx,[esi+0B3Ch]
0x100656fb: call 10065490h
0x10065700: pop edi
0x10065701: pop ebp
0x10065702: pop ebx
0x10065703: add esp,8
0x10065706: ret
```

`sub_10065510` is called by sub_10065710, sub_10065a70. It delegates to sub_100642a0, sub_10064370, sub_10065490 (3 callees).

12.927 sub_10065710 at 0x10065710 (50 instructions)

```
0x10065710: push esi
0x10065711: mov esi,eax
0x10065713: mov ecx,[esi+0B1Ch]
0x10065719: lea eax,[esi+94h]
0x1006571f: push esi
0x10065720: call 10064590h
0x10065725: mov ecx,[esi+0B28h]
0x1006572b: lea eax,[esi+988h]
0x10065731: push esi
0x10065732: call 10064590h
0x10065737: lea eax,[esi+0B30h]
0x1006573d: push eax
0x1006573e: call 10065510h
0x10065743: add esp,0Ch
0x10065746: mov eax,12h
0x1006574b: jmp short 10065750h
0x10065750: movzx ecx,byte ptr [eax+101076ACh]
0x10065757: cmp word ptr [esi+ecx*4+0A7Eh],0
0x10065760: jne short 100657CDh
0x10065762: movzx edx,byte ptr [eax+101076ABh]
0x10065769: cmp word ptr [esi+edx*4+0A7Eh],0
```

```
0x10065772: jne short 100657ACh
0x10065774: movzx ecx,byte ptr [eax+101076AAh]
0x1006577b: cmp word ptr [esi+ecx*4+0A7Eh],0
0x10065784: jne short 100657BBh
0x10065786: movzx edx,byte ptr [eax+101076A9h]
0x1006578d: cmp word ptr [esi+edx*4+0A7Eh],0
0x10065796: jne short 100657CAh
0x10065798: sub eax,4
0x1006579b: cmp eax,3
0x1006579e: jge short 10065750h
0x100657a0: lea ecx,[eax+eax*2+11h]
0x100657a4: add [esi+16A8h],ecx
0x100657aa: pop esi
0x100657ab: ret
0x100657ac: sub eax,1
0x100657af: lea ecx,[eax+eax*2+11h]
0x100657b3: add [esi+16A8h],ecx
0x100657b9: pop esi
0x100657ba: ret
0x100657bb: sub eax,2
0x100657be: lea ecx,[eax+eax*2+11h]
0x100657c2: add [esi+16A8h],ecx
0x100657c8: pop esi
0x100657c9: ret
0x100657ca: sub eax,3
0x100657cd: lea ecx,[eax+eax*2+11h]
0x100657d1: add [esi+16A8h],ecx
0x100657d7: pop esi
0x100657d8: ret
```

`sub_10065710` is called by sub_10065a70. It delegates to sub_10064590, sub_10065510 (2 callees).

12.928 sub_100657e0 at 0x100657e0 (46 instructions)

```
0x100657e0: mov eax,[esp+4]
0x100657e4: mov ecx,[eax+16BCh]
0x100657ea: cmp ecx,0Dh
0x100657ed: jle short 10065858h
0x100657ef: push ebx
0x100657f0: push esi
0x100657f1: mov esi,[esp+18h]
0x100657f5: mov edx,esi
0x100657f7: shl edx,cl
0x100657f9: mov ecx,[eax+8]
0x100657fc: or [eax+16B8h],dx
0x10065803: movzx ebx,byte ptr [eax+16B8h]
0x1006580a: mov edx,[eax+14h]
0x1006580d: mov [ecx+edx],bl
0x10065810: movzx ebx,byte ptr [eax+16B9h]
0x10065817: add dword ptr [eax+14h],1
0x1006581b: mov ecx,[eax+14h]
0x1006581e: mov edx,[eax+8]
0x10065821: mov [ecx+edx],bl
0x10065824: mov edx,[eax+16BCh]
0x1006582a: add dword ptr [eax+14h],1
0x1006582e: mov cl,10h
0x10065830: sub cl,dl
0x10065832: shr si,cl
0x10065835: mov ecx,[esp+14h]
0x10065839: add edx,0FFFFFFF3h
0x1006583c: mov [eax+16BCh],edx
0x10065842: mov edx,[esp+10h]
0x10065846: mov [eax+16B8h],si
0x1006584d: pop esi
0x1006584e: pop ebx
0x1006584f: push 1
```

```
0x10065851: call 10065390h
0x10065856: pop ecx
0x10065857: ret
0x10065858: mov edx,[esp+10h]
0x1006585c: shl edx,cl
0x1006585e: add ecx,3
0x10065861: mov [eax+16BCh],ecx
0x10065867: mov ecx,[esp+0Ch]
0x1006586b: or [eax+16B8h],dx
0x10065872: mov edx,[esp+8]
0x10065876: push 1
0x10065878: call 10065390h
0x1006587d: pop ecx
0x1006587e: ret
```

`sub_100657e0` is called by sub_10062a80, sub_10065a70. It delegates to sub_10065390 (1 callees).

12.929 sub_10065880 at 0x10065880 (130 instructions)

```
0x10065880: mov eax,[esp+4]
0x10065884: mov ecx,[eax+16BCh]
0x1006588a: mov edx,2
0x1006588f: shl edx,cl
0x10065891: push ebx
0x10065892: push esi
0x10065893: push edi
0x10065894: or [eax+16B8h],dx
0x1006589b: cmp ecx,0Dh
0x1006589e: mov esi,1
0x100658a3: jle short 100658EFh
0x100658a5: movzx ebx,byte ptr [eax+16B8h]
0x100658ac: mov edx,[eax+14h]
0x100658af: mov ecx,[eax+8]
0x100658b2: mov [ecx+edx],bl
0x100658b5: add [eax+14h],esi
0x100658b8: movzx ebx,byte ptr [eax+16B9h]
0x100658bf: mov ecx,[eax+14h]
0x100658c2: mov edx,[eax+8]
0x100658c5: mov [ecx+edx],bl
0x100658c8: mov edx,[eax+16BCh]
0x100658ce: add [eax+14h],esi
0x100658d1: mov cl,10h
0x100658d3: sub cl,dl
0x100658d5: mov edi,2
0x100658da: shr di,cl
0x100658dd: add edx,0FFFFFFF3h
0x100658e0: mov [eax+16BCh],edx
0x100658e6: mov [eax+16B8h],di
0x100658ed: jmp short 100658F8h
0x100658ef: add ecx,3
0x100658f2: mov [eax+16BCh],ecx
0x100658f8: mov ecx,[eax+16BCh]
0x100658fe: xor edx,edx
0x10065900: shl edx,cl
0x10065902: or [eax+16B8h],dx
0x10065909: cmp ecx,9
0x1006590c: jle short 10065955h
0x1006590e: movzx ebx,byte ptr [eax+16B8h]
0x10065915: mov edx,[eax+14h]
0x10065918: mov ecx,[eax+8]
0x1006591b: mov [ecx+edx],bl
0x1006591e: add [eax+14h],esi
0x10065921: movzx ebx,byte ptr [eax+16B9h]
0x10065928: mov ecx,[eax+14h]
0x1006592b: mov edx,[eax+8]
0x1006592e: mov [ecx+edx],bl
```

```
0x10065931: mov edx,[eax+16BCh]
0x10065937: add [eax+14h],esi
0x1006593a: mov cl,10h
0x1006593c: sub cl,dl
0x1006593e: xor edi,edi
0x10065940: shr di,cl
0x10065943: add edx,0FFFFFFF7h
0x10065946: mov [eax+16BCh],edx
0x1006594c: mov [eax+16B8h],di
0x10065953: jmp short 1006595Eh
0x10065955: add ecx,7
0x10065958: mov [eax+16BCh],ecx
0x1006595e: call 10065290h
0x10065963: mov ecx,[eax+16BCh]
0x10065969: mov edx,[eax+16B4h]
0x1006596f: sub edx,ecx
0x10065971: add edx,0Bh
0x10065974: cmp edx,9
0x10065977: jge near ptr 10065A5Fh
0x1006597d: mov edx,2
0x10065982: shl edx,cl
0x10065984: or [eax+16B8h],dx
0x1006598b: cmp ecx,0Dh
0x1006598e: jle short 100659DAh
0x10065990: movzx ebx,byte ptr [eax+16B8h]
0x10065997: mov edx,[eax+14h]
0x1006599a: mov ecx,[eax+8]
0x1006599d: mov [ecx+edx],bl
0x100659a0: add [eax+14h],esi
0x100659a3: movzx ebx,byte ptr [eax+16B9h]
0x100659aa: mov ecx,[eax+14h]
0x100659ad: mov edx,[eax+8]
0x100659b0: mov [ecx+edx],bl
0x100659b3: mov edx,[eax+16BCh]
0x100659b9: add [eax+14h],esi
0x100659bc: mov cl,10h
0x100659be: sub cl,dl
0x100659c0: mov edi,2
0x100659c5: shr di,cl
0x100659c8: add edx,0FFFFFFF3h
0x100659cb: mov [eax+16BCh],edx
0x100659d1: mov [eax+16B8h],di
0x100659d8: jmp short 100659E3h
0x100659da: add ecx,3
0x100659dd: mov [eax+16BCh],ecx
0x100659e3: mov ecx,[eax+16BCh]
0x100659e9: xor edx,edx
0x100659eb: shl edx,cl
0x100659ed: or [eax+16B8h],dx
0x100659f4: cmp ecx,9
0x100659f7: jle short 10065A51h
0x100659f9: movzx ebx,byte ptr [eax+16B8h]
0x10065a00: mov edx,[eax+14h]
0x10065a03: mov ecx,[eax+8]
0x10065a06: mov [ecx+edx],bl
0x10065a09: add [eax+14h],esi
0x10065a0c: movzx ebx,byte ptr [eax+16B9h]
0x10065a13: mov ecx,[eax+14h]
0x10065a16: mov edx,[eax+8]
0x10065a19: mov [ecx+edx],bl
0x10065a1c: mov edx,[eax+16BCh]
0x10065a22: add [eax+14h],esi
0x10065a25: mov cl,10h
0x10065a27: sub cl,dl
0x10065a29: xor esi,esi
0x10065a2b: shr si,cl
0x10065a2e: add edx,0FFFFFFF7h
```

```
0x10065a31: mov [eax+16BCh],edx
0x10065a37: mov [eax+16B8h],si
0x10065a3e: call 10065290h
0x10065a43: pop edi
0x10065a44: pop esi
0x10065a45: mov dword ptr [eax+16B4h],7
0x10065a4f: pop ebx
0x10065a50: ret
0x10065a51: add ecx,7
0x10065a54: mov [eax+16BCh],ecx
0x10065a5a: call 10065290h
0x10065a5f: pop edi
0x10065a60: pop esi
0x10065a61: mov dword ptr [eax+16B4h],7
0x10065a6b: pop ebx
0x10065a6c: ret
```

`sub_10065880` is called by sub_10062a80. It delegates to sub_10065290 (1 callees).

12.930 sub_10065a70 at 0x10065a70 (186 instructions)

```
0x10065310: mov edx,[eax+16BCh]
0x10065316: cmp edx,8
0x10065319: push ebx
0x1006531a: push esi
0x1006531b: jle short 1006535Ah
0x1006531d: movzx ebx,byte ptr [eax+16B8h]
0x10065324: mov edx,[eax+14h]
0x10065327: mov ecx,[eax+8]
0x1006532a: mov [ecx+edx],bl
0x1006532d: movzx ebx,byte ptr [eax+16B9h]
0x10065334: mov edx,[eax+8]
0x10065337: mov esi,1
0x1006533c: add [eax+14h],esi
0x1006533f: mov ecx,[eax+14h]
0x10065342: mov [ecx+edx],bl
0x10065345: add [eax+14h],esi
0x10065348: xor ecx,ecx
0x1006534a: pop esi
0x1006534b: mov [eax+16B8h],cx
0x10065352: mov [eax+16BCh],ecx
0x10065358: pop ebx
0x10065359: ret
0x1006535a: xor ecx,ecx
0x1006535c: cmp edx,ecx
0x1006535e: jle short 10065373h
0x10065360: mov esi,[eax+14h]
0x10065363: mov edx,[eax+8]
0x10065366: mov bl,[eax+16B8h]
0x1006536c: mov [edx+esi],bl
0x1006536f: add dword ptr [eax+14h],1
0x10065373: pop esi
0x10065374: mov [eax+16B8h],cx
0x1006537b: mov [eax+16BCh],ecx
0x10065381: pop ebx
0x10065382: ret
0x10065a70: push ecx
0x10065a71: push ebx
0x10065a72: mov ebx,[esp+14h]
0x10065a76: push esi
0x10065a77: mov esi,[esp+10h]
0x10065a7b: cmp dword ptr [esi+84h],0
0x10065a82: push edi
0x10065a83: mov dword ptr [esp+0Ch],0
0x10065a8b: jle short 10065AE1h
0x10065a8d: mov edi,[esi]
```

```
0x10065a8f: cmp dword ptr [edi+2Ch],2
0x10065a93: jne short 10065A9Dh
0x10065a95: call 10065200h
0x10065a9a: mov [edi+2Ch],eax
0x10065a9d: lea eax,[esi+0B18h]
0x10065aa3: push eax
0x10065aa4: call 10065510h
0x10065aa9: lea ecx,[esi+0B24h]
0x10065aaf: push ecx
0x10065ab0: call 10065510h
0x10065ab5: add esp,8
0x10065ab8: mov eax,esi
0x10065aba: call 10065710h
0x10065abf: mov edx,[esi+16A8h]
0x10065ac5: mov ecx,[esi+16ACh]
0x10065acb: add edx,0Ah
0x10065ace: add ecx,0Ah
0x10065ad1: shr edx,3
0x10065ad4: shr ecx,3
0x10065ad7: cmp ecx,edx
0x10065ad9: mov [esp+0Ch],eax
0x10065add: ja short 10065AE6h
0x10065adf: jmp short 10065AE4h
0x10065ae1: lea ecx,[ebx+5]
0x10065ae4: mov edx,ecx
0x10065ae6: lea eax,[ebx+4]
0x10065ae9: cmp eax,edx
0x10065aeb: ja short 10065B0Ah
0x10065aed: mov eax,[esp+18h]
0x10065af1: test eax,eax
0x10065af3: je short 10065B0Ah
0x10065af5: mov edi,[esp+20h]
0x10065af9: push edi
0x10065afa: push ebx
0x10065afb: push eax
0x10065afc: push esi
0x10065afd: call 100657E0h
0x10065b02: add esp,10h
0x10065b05: jmp near ptr 10065C5Fh
0x10065b0a: cmp dword ptr [esi+88h],4
0x10065b11: je near ptr 10065BD5h
0x10065b17: cmp ecx,edx
0x10065b19: je near ptr 10065BD5h
0x10065b1f: mov ecx,[esi+16BCh]
0x10065b25: cmp ecx,0Dh
0x10065b28: mov edi,[esp+20h]
0x10065b2c: lea edx,[edi+4]
0x10065b2f: jle short 10065B83h
0x10065b31: mov eax,edx
0x10065b33: shl eax,cl
0x10065b35: mov ecx,[esi+14h]
0x10065b38: or [esi+16B8h],ax
0x10065b3f: movzx ebx,byte ptr [esi+16B8h]
0x10065b46: mov eax,[esi+8]
0x10065b49: mov [ecx+eax],bl
0x10065b4c: add dword ptr [esi+14h],1
0x10065b50: movzx ebx,byte ptr [esi+16B9h]
0x10065b57: mov ecx,[esi+14h]
0x10065b5a: mov eax,[esi+8]
0x10065b5d: mov [ecx+eax],bl
0x10065b60: mov ebx,[esi+16BCh]
0x10065b66: add dword ptr [esi+14h],1
0x10065b6a: mov cl,10h
0x10065b6c: sub cl,bl
0x10065b6e: shr dx,cl
0x10065b71: add ebx,0FFFFFFF3h
0x10065b74: mov [esi+16BCh],ebx
```

```
0x10065b7a: mov [esi+16B8h],dx
0x10065b81: jmp short 10065B95h
0x10065b83: shl edx,cl
0x10065b85: or [esi+16B8h],dx
0x10065b8c: add ecx,3
0x10065b8f: mov [esi+16BCh],ecx
0x10065b95: mov eax,[esp+0Ch]
0x10065b99: mov ecx,[esi+0B28h]
0x10065b9f: mov edx,[esi+0B1Ch]
0x10065ba5: add eax,1
0x10065ba8: push eax
0x10065ba9: add ecx,1
0x10065bac: push ecx
0x10065bad: add edx,1
0x10065bb0: push edx
0x10065bb1: mov eax,esi
0x10065bb3: call 10064B90h
0x10065bb8: lea eax,[esi+988h]
0x10065bbe: push eax
0x10065bbf: lea ecx,[esi+94h]
0x10065bc5: push ecx
0x10065bc6: mov eax,esi
0x10065bc8: call 10064DF0h
0x10065bcd: add esp,14h
0x10065bd0: jmp near ptr 10065C5Fh
0x10065bd5: mov ecx,[esi+16BCh]
0x10065bdb: cmp ecx,0Dh
0x10065bde: mov edi,[esp+20h]
0x10065be2: lea eax,[edi+2]
0x10065be5: jle short 10065C39h
0x10065be7: mov edx,eax
0x10065be9: shl edx,cl
0x10065beb: mov ecx,[esi+14h]
0x10065bee: or [esi+16B8h],dx
0x10065bf5: movzx ebx,byte ptr [esi+16B8h]
0x10065bfc: mov edx,[esi+8]
0x10065bff: mov [ecx+edx],bl
0x10065c02: add dword ptr [esi+14h],1
0x10065c06: movzx ebx,byte ptr [esi+16B9h]
0x10065c0d: mov ecx,[esi+14h]
0x10065c10: mov edx,[esi+8]
0x10065c13: mov [ecx+edx],bl
0x10065c16: mov edx,[esi+16BCh]
0x10065c1c: add dword ptr [esi+14h],1
0x10065c20: mov cl,10h
0x10065c22: sub cl,dl
0x10065c24: shr ax,cl
0x10065c27: add edx,0FFFFFFF3h
0x10065c2a: mov [esi+16BCh],edx
0x10065c30: mov [esi+16B8h],ax
0x10065c37: jmp short 10065C4Bh
0x10065c39: shl eax,cl
0x10065c3b: or [esi+16B8h],ax
0x10065c42: add ecx,3
0x10065c45: mov [esi+16BCh],ecx
0x10065c4b: push 10107C08h
0x10065c50: push 10107788h
0x10065c55: mov eax,esi
0x10065c57: call 10064DF0h
0x10065c5c: add esp,8
0x10065c5f: mov edx,esi
0x10065c61: call 10064230h
0x10065c66: test edi,edi
0x10065c68: pop edi
0x10065c69: je short 10065C77h
0x10065c6b: mov eax,esi
0x10065c6d: pop esi
```

```
0x10065c6e: pop ebx
0x10065c6f: add esp,4
0x10065c72: jmp near ptr 10065310h
0x10065c77: pop esi
0x10065c78: pop ebx
0x10065c79: pop ecx
0x10065c7a: ret
```

`sub_10065a70` is called by sub_10062640, sub_100628c0. It delegates to sub_10064230, sub_10064b90, sub_10064df0 (7 callees).

12.931 sub_10065c7b at 0x10065c7b (81 instructions)

```
0x10065c7b: push ebx
0x10065c7c: push esi
0x10065c7d: mov esi,[esp+0Ch]
0x10065c81: xor ebx,ebx
0x10065c83: cmp [esi+8],ebx
0x10065c86: push edi
0x10065c87: je short 10065C90h
0x10065c89: push esi
0x10065c8a: call 1006FAC1h
0x10065c8f: pop ecx
0x10065c90: push esi
0x10065c91: call dword ptr [esi+23Ch]
0x10065c97: push esi
0x10065c98: call 10067617h
0x10065c9d: mov eax,[esi+18h]
0x10065ca0: cmp eax,ebx
0x10065ca2: pop ecx
0x10065ca3: pop ecx
0x10065ca4: je short 10065CCCh
0x10065ca6: push eax
0x10065ca7: call 100209E0h
0x10065cac: jmp short 10065CCBh
0x10065cae: mov edi,[esi+2A8h]
0x10065cb4: mov eax,[edi]
0x10065cb6: mov [esi+2A8h],eax
0x10065cbc: push dword ptr [edi+8]
0x10065cbf: call 100209E0h
0x10065cc4: push edi
0x10065cc5: call 100209E0h
0x10065cca: pop ecx
0x10065ccb: pop ecx
0x10065ccc: cmp [esi+2A8h],ebx
0x10065cd2: jne short 10065CAEh
0x10065cd4: mov eax,[esi+254h]
0x10065cda: cmp eax,ebx
0x10065cdc: je short 10065CEDh
0x10065cde: test word ptr [esi+0Ch],200h
0x10065ce4: je short 10065CEDh
0x10065ce6: push eax
0x10065ce7: call 100209E0h
0x10065cec: pop ecx
0x10065ced: test word ptr [esi+0Ch],800h
0x10065cf3: je short 10065D10h
0x10065cf5: push dword ptr [esi+268h]
0x10065cfb: push dword ptr [esi+264h]
0x10065d01: push dword ptr [esi+274h]
0x10065d07: call dword ptr [esi+270h]
0x10065d0d: add esp,0Ch
0x10065d10: mov eax,[esi+294h]
0x10065d16: cmp eax,ebx
0x10065d18: jbe short 10065D66h
0x10065d1a: mov eax,[esi+290h]
0x10065d20: mov edi,[eax+ebx*4]
```

```
0x10065d23: cmp word ptr [edi+0Ch],41h
0x10065d28: jne short 10065D51h
0x10065d2a: push 4
0x10065d2c: push dword ptr [edi+10h]
0x10065d2f: push 10108070h
0x10065d34: call dword ptr ds:[100FE414h]
0x10065d3a: add esp,0Ch
0x10065d3d: test eax,eax
0x10065d3f: jne short 10065D51h
0x10065d41: push dword ptr [edi+10h]
0x10065d44: call 100209E0h
0x10065d49: push edi
0x10065d4a: call 100209E0h
0x10065d4f: pop ecx
0x10065d50: pop ecx
0x10065d51: inc ebx
0x10065d52: cmp ebx,[esi+294h]
0x10065d58: jb short 10065D1Ah
0x10065d5a: push dword ptr [esi+290h]
0x10065d60: call 100209E0h
0x10065d65: pop ecx
0x10065d66: push esi
0x10065d67: call 100209E0h
0x10065d6c: pop ecx
0x10065d6d: pop edi
0x10065d6e: pop esi
0x10065d6f: pop ebx
0x10065d70: ret
```

`sub_10065c7b` is called by sub_10065d71, sub_1006891d. It delegates to j_free, sub_10067617, sub_1006fac1 (3 callees).

12.932 sub_10065d71 at 0x10065d71 (14 instructions)

```
0x10065d71: mov eax,[esp+4]
0x10065d75: push esi
0x10065d76: mov esi,[eax+284h]
0x10065d7c: push edi
0x10065d7d: mov edi,[eax+274h]
0x10065d83: push eax
0x10065d84: call 10065C7Bh
0x10065d89: push edi
0x10065d8a: call esi
0x10065d8c: pop ecx
0x10065d8d: pop ecx
0x10065d8e: pop edi
0x10065d8f: pop esi
0x10065d90: ret
```

`sub_10065d71` is called by sub_100143f0. It delegates to sub_10065c7b (1 callees).

12.933 sub_10065d91 at 0x10065d91 (65 instructions)

```
0x10065d91: push edi
0x10065d92: mov edi,eax
0x10065d94: mov eax,[esi+0B8h]
0x10065d9a: shl edi,2
0x10065d9d: mov eax,[edi+eax]
0x10065da0: xor edx,edx
0x10065da2: cmp eax,edx
0x10065da4: je short 10065DB2h
0x10065da6: cmp [esi+1E4h],edx
0x10065dac: jne near ptr 10065E39h
0x10065db2: mov ecx,[esi+0BCh]
0x10065db8: mov ecx,[edi+ecx]
```

```
0x10065dbb: cmp ecx,edx
0x10065dbd: je short 10065E05h
0x10065dbf: cmp eax,edx
0x10065dc1: je short 10065E05h
0x10065dc3: cmp ecx,ebx
0x10065dc5: jb short 10065E05h
0x10065dc7: push edx
0x10065dc8: push eax
0x10065dc9: push dword ptr [esi+274h]
0x10065dcf: call dword ptr [esi+280h]
0x10065dd5: mov ecx,[esi+0B8h]
0x10065ddb: add esp,0Ch
0x10065dde: cmp eax,[edi+ecx]
0x10065de1: je short 10065E20h
0x10065de3: push dword ptr [esi+1D8h]
0x10065de9: push 1010813Ch
0x10065dee: push 101080FCh
0x10065df3: push dword ptr [esi+274h]
0x10065df9: call 1006FAF1h
0x10065dfe: add esp,10h
0x10065e01: xor eax,eax
0x10065e03: pop edi
0x10065e04: ret
0x10065e05: push 2
0x10065e07: push edx
0x10065e08: push dword ptr [esi+274h]
0x10065e0e: call dword ptr [esi+280h]
0x10065e14: mov ecx,[esi+0B8h]
0x10065e1a: add esp,0Ch
0x10065e1d: mov [edi+ecx],eax
0x10065e20: mov eax,[esi+0B8h]
0x10065e26: mov eax,[edi+eax]
0x10065e29: mov [esi+1E4h],eax
0x10065e2f: mov eax,[esi+0BCh]
0x10065e35: and dword ptr [edi+eax],0
0x10065e39: push ebx
0x10065e3a: push dword ptr [esp+0Ch]
0x10065e3e: push dword ptr [esi+274h]
0x10065e44: call dword ptr [esi+27Ch]
0x10065e4a: add esp,0Ch
0x10065e4d: cmp eax,ebx
0x10065e4f: je short 10065E5Eh
0x10065e51: push dword ptr [esi+1D8h]
0x10065e57: push 10108120h
0x10065e5c: jmp short 10065DEEh
0x10065e5e: add [esi+1E4h],ebx
0x10065e64: mov eax,[esi+0BCh]
0x10065e6a: add eax,edi
0x10065e6c: add [eax],ebx
0x10065e6e: xor eax,eax
0x10065e70: inc eax
0x10065e71: pop edi
0x10065e72: ret
```

`sub_10065d91` is called by sub_10065e73. It works with the text "TIFFAppendToStrip". It delegates to sub_1006faf1 (1 callees).

12.934 sub_10065e73 at 0x10065e73 (38 instructions)

```
0x10065e73: push esi
0x10065e74: mov esi,[esp+8]
0x10065e78: mov eax,[esi+260h]
0x10065e7e: test eax,eax
0x10065e80: jle short 10065EE5h
0x10065e82: movzx edx,word ptr [esi+5Eh]
0x10065e86: mov ecx,[esi+0Ch]
```

```
0x10065e89: test ecx,edx
0x10065e8b: jne short 10065EA2h
0x10065e8d: test cx,100h
0x10065e92: jne short 10065EA2h
0x10065e94: push eax
0x10065e95: push dword ptr [esi+254h]
0x10065e9b: call 1006FC09h
0x10065ea0: pop ecx
0x10065ea1: pop ecx
0x10065ea2: test word ptr [esi+0Ch],400h
0x10065ea8: je short 10065EB2h
0x10065eaa: mov eax,[esi+1F8h]
0x10065eb0: jmp short 10065EB8h
0x10065eb2: mov eax,[esi+1E0h]
0x10065eb8: push ebx
0x10065eb9: push dword ptr [esi+254h]
0x10065ebf: mov ebx,[esi+260h]
0x10065ec5: call 10065D91h
0x10065eca: test eax,eax
0x10065ecc: pop ecx
0x10065ecd: pop ebx
0x10065ece: jne short 10065ED2h
0x10065ed0: pop esi
0x10065ed1: ret
0x10065ed2: mov eax,[esi+254h]
0x10065ed8: and dword ptr [esi+260h],0
0x10065edf: mov [esi+25Ch],eax
0x10065ee5: xor eax,eax
0x10065ee7: inc eax
0x10065ee8: pop esi
0x10065ee9: ret
```

`sub_10065e73` is called by sub_1006f391, sub_1006fa8e. It delegates to sub_10065d91, sub_1006fc09 (2 callees).

12.935 sub_10065eea at 0x10065eea (15 instructions)

```
0x10065eea: push edi
0x10065eeb: lea eax,[edx+ecx]
0x10065eee: mov edi,eax
0x10065ef0: sub edi,edx
0x10065ef2: cmp edi,ecx
0x10065ef4: pop edi
0x10065ef5: je short 10065F12h
0x10065ef7: push dword ptr [esp+4]
0x10065efb: push 10108158h
0x10065f00: push dword ptr [esi]
0x10065f02: push dword ptr [esi+274h]
0x10065f08: call 1006FAF1h
0x10065f0d: add esp,10h
0x10065f10: xor eax,eax
0x10065f12: ret
```

`sub_10065eea` is called by sub_10065faf, sub_10066210, sub_100686da. It works with the text "Integer overflow in %s". It delegates to sub_1006faf1 (1 callees).

12.936 sub_10065f13 at 0x10065f13 (28 instructions)

```
0x10065f13: push ebp
0x10065f14: mov ebp,esp
0x10065f16: mov ecx,[ebp+8]
0x10065f19: mov eax,[ebp+0Ch]
0x10065f1c: xor edx,edx
0x10065f1e: div dword ptr [ecx+64h]
0x10065f21: cmp word ptr [ecx+8Ah],2
0x10065f29: jne short 10065F64h
```

```
0x10065f2b: movzx edx,word ptr [ecx+62h]
0x10065f2f: cmp [ebp+10h],dx
0x10065f33: jb short 10065F57h
0x10065f35: movzx eax,dx
0x10065f38: push eax
0x10065f39: movzx eax,word ptr [ebp+10h]
0x10065f3d: push eax
0x10065f3e: push 10108170h
0x10065f43: push dword ptr [ecx]
0x10065f45: push dword ptr [ecx+274h]
0x10065f4b: call 1006FAF1h
0x10065f50: add esp,14h
0x10065f53: xor eax,eax
0x10065f55: pop ebp
0x10065f56: ret
0x10065f57: movzx edx,word ptr [ebp+10h]
0x10065f5b: imul edx,[ecx+0B0h]
0x10065f62: add eax,edx
0x10065f64: pop ebp
0x10065f65: ret
```

`sub_10065f13` is called by sub_100143f0. It delegates to sub_1006faf1 (1 callees).

12.937 sub_10065f66 at 0x10065f66 (28 instructions)

```
0x10065f66: push esi
0x10065f67: mov esi,[esp+8]
0x10065f6b: mov eax,[esi+64h]
0x10065f6e: cmp eax,0FFFFFFFFh
0x10065f71: jne short 10065F78h
0x10065f73: xor eax,eax
0x10065f75: inc eax
0x10065f76: jmp short 10065F90h
0x10065f78: mov edx,[esi+3Ch]
0x10065f7b: neg eax
0x10065f7d: cmp edx,eax
0x10065f7f: jae short 10065F8Eh
0x10065f81: mov ecx,[esi+64h]
0x10065f84: lea eax,[ecx+edx-1]
0x10065f88: xor edx,edx
0x10065f8a: div ecx
0x10065f8c: jmp short 10065F90h
0x10065f8e: xor eax,eax
0x10065f90: cmp word ptr [esi+8Ah],2
0x10065f98: jne short 10065FADh
0x10065f9a: movzx ecx,word ptr [esi+62h]
0x10065f9e: push 10108194h
0x10065fa3: push ecx
0x10065fa4: push eax
0x10065fa5: call 10068414h
0x10065faa: add esp,0Ch
0x10065fad: pop esi
0x10065fae: ret
```

`sub_10065f66` is called by sub_1007171b. It delegates to sub_10068414 (1 callees).

12.938 sub_10065faf at 0x10065faf (116 instructions)

```
0x10065faf: push ebp
0x10065fb0: mov ebp,esp
0x10065fb2: push ebx
0x10065fb3: push esi
0x10065fb4: mov esi,[ebp+8]
0x10065fb7: cmp word ptr [esi+8Ah],1
0x10065fbf: push edi
```

```
0x10065fc0: mov edi,101081D4h
0x10065fc5: jne near ptr 100660B3h
0x10065fcb: cmp word ptr [esi+5Ah],6
0x10065fd0: jne near ptr 1006609Eh
0x10065fd6: test word ptr [esi+0Ch],4000h
0x10065fdc: jne near ptr 1006609Eh
0x10065fe2: lea eax,[ebp+0Ah]
0x10065fe5: push eax
0x10065fe6: lea eax,[ebp+8]
0x10065fe9: push eax
0x10065fea: push 212h
0x10065fef: push esi
0x10065ff0: call 10067985h
0x10065ff5: add esp,10h
0x10065ff8: cmp word ptr [ebp+8],0
0x10065ffd: jne short 1006601Bh
0x10065fff: push 101081B8h
0x10066004: push dword ptr [esi]
0x10066006: push dword ptr [esi+274h]
0x1006600c: call 1006FAF1h
0x10066011: add esp,0Ch
0x10066014: xor eax,eax
0x10066016: jmp near ptr 100660E9h
0x1006601b: movzx ecx,word ptr [ebp+8]
0x1006601f: mov eax,[esi+38h]
0x10066022: mov edx,ecx
0x10066024: neg edx
0x10066026: cmp eax,edx
0x10066028: jae short 10066034h
0x1006602a: lea eax,[eax+ecx-1]
0x1006602e: xor edx,edx
0x10066030: div ecx
0x10066032: jmp short 10066036h
0x10066034: xor eax,eax
0x10066036: imul ecx,eax
0x10066039: movzx eax,word ptr [esi+54h]
0x1006603d: mov edi,101081D4h
0x10066042: push edi
0x10066043: mov ebx,ecx
0x10066045: push eax
0x10066046: push ebx
0x10066047: call 10068414h
0x1006604c: add esp,0Ch
0x1006604f: test al,7
0x10066051: movzx eax,word ptr [esi+54h]
0x10066055: push edi
0x10066056: push eax
0x10066057: push ebx
0x10066058: je short 1006606Ah
0x1006605a: call 10068414h
0x1006605f: mov edi,eax
0x10066061: shr edi,3
0x10066064: add esp,0Ch
0x10066067: inc edi
0x10066068: jmp short 10066077h
0x1006606a: call 10068414h
0x1006606f: mov edi,eax
0x10066071: add esp,0Ch
0x10066074: shr edi,3
0x10066077: movzx ecx,word ptr [ebp+8]
0x1006607b: mov eax,101081A8h
0x10066080: push eax
0x10066081: push eax
0x10066082: mov eax,edi
0x10066084: cdq
0x10066085: idiv ecx
0x10066087: push eax
```

```
0x10066088: push 2
0x1006608a: call 10068414h
0x1006608f: add esp,0Ch
0x10066092: mov ecx,eax
0x10066094: mov edx,edi
0x10066096: call 10065EEAh
0x1006609b: pop ecx
0x1006609c: jmp short 100660E9h
0x1006609e: movzx eax,word ptr [esi+62h]
0x100660a2: push edi
0x100660a3: push eax
0x100660a4: push dword ptr [esi+38h]
0x100660a7: call 10068414h
0x100660ac: add esp,0Ch
0x100660af: mov ebx,eax
0x100660b1: jmp short 100660B6h
0x100660b3: mov ebx,[esi+38h]
0x100660b6: movzx eax,word ptr [esi+54h]
0x100660ba: push edi
0x100660bb: push eax
0x100660bc: push ebx
0x100660bd: call 10068414h
0x100660c2: add esp,0Ch
0x100660c5: test al,7
0x100660c7: movzx eax,word ptr [esi+54h]
0x100660cb: push edi
0x100660cc: push eax
0x100660cd: push ebx
0x100660ce: je short 100660DEh
0x100660d0: call 10068414h
0x100660d5: shr eax,3
0x100660d8: add esp,0Ch
0x100660db: inc eax
0x100660dc: jmp short 100660E9h
0x100660de: call 10068414h
0x100660e3: add esp,0Ch
0x100660e6: shr eax,3
0x100660e9: pop edi
0x100660ea: pop esi
0x100660eb: pop ebx
0x100660ec: pop ebp
0x100660ed: ret
```

`sub_10065faf` is called by sub_100143f0, sub_10066210, sub_100702f4, sub_1007171b. It delegates to sub_10065eea, sub_10067985, sub_10068414 (4 callees).

12.939 sub_10066210 at 0x10066210 (117 instructions)

```
0x10066210: push ebp
0x10066211: mov ebp,esp
0x10066213: push ecx
0x10066214: cmp dword ptr [ebp+0Ch],0FFFFFFFFh
0x10066218: push esi
0x10066219: mov esi,[ebp+8]
0x1006621c: jne short 10066224h
0x1006621e: mov eax,[esi+3Ch]
0x10066221: mov [ebp+0Ch],eax
0x10066224: cmp word ptr [esi+8Ah],1
0x1006622c: jne near ptr 10066331h
0x10066232: cmp word ptr [esi+5Ah],6
0x10066237: jne near ptr 10066331h
0x1006623d: test word ptr [esi+0Ch],4000h
0x10066243: jne near ptr 10066331h
0x10066249: lea eax,[ebp+0Ah]
0x1006624c: push eax
0x1006624d: lea eax,[ebp+8]
```

```
0x10066250: push eax
0x10066251: push 212h
0x10066256: push esi
0x10066257: call 10067985h
0x1006625c: movzx ecx,word ptr [ebp+8]
0x10066260: movzx eax,word ptr [ebp+0Ah]
0x10066264: imul eax,ecx
0x10066267: add esp,10h
0x1006626a: test eax,eax
0x1006626c: mov [ebp-4],eax
0x1006626f: jne short 1006628Dh
0x10066271: push 101081B8h
0x10066276: push dword ptr [esi]
0x10066278: push dword ptr [esi+274h]
0x1006627e: call 1006FAF1h
0x10066283: add esp,0Ch
0x10066286: xor eax,eax
0x10066288: jmp near ptr 10066349h
0x1006628d: mov eax,[esi+38h]
0x10066290: mov edx,ecx
0x10066292: neg edx
0x10066294: cmp eax,edx
0x10066296: jae short 100662A2h
0x10066298: lea eax,[eax+ecx-1]
0x1006629c: xor edx,edx
0x1006629e: div ecx
0x100662a0: jmp short 100662A4h
0x100662a2: xor eax,eax
0x100662a4: imul ecx,eax
0x100662a7: movzx eax,word ptr [esi+54h]
0x100662ab: push ebx
0x100662ac: push edi
0x100662ad: mov edi,101081A8h
0x100662b2: push edi
0x100662b3: mov ebx,ecx
0x100662b5: push eax
0x100662b6: push ebx
0x100662b7: call 10068414h
0x100662bc: add esp,0Ch
0x100662bf: test al,7
0x100662c1: movzx eax,word ptr [esi+54h]
0x100662c5: push edi
0x100662c6: push eax
0x100662c7: push ebx
0x100662c8: je short 100662DAh
0x100662ca: call 10068414h
0x100662cf: mov ebx,eax
0x100662d1: shr ebx,3
0x100662d4: add esp,0Ch
0x100662d7: inc ebx
0x100662d8: jmp short 100662E7h
0x100662da: call 10068414h
0x100662df: mov ebx,eax
0x100662e1: add esp,0Ch
0x100662e4: shr ebx,3
0x100662e7: movzx ecx,word ptr [ebp+0Ah]
0x100662eb: mov eax,[ebp+0Ch]
0x100662ee: mov edx,ecx
0x100662f0: neg edx
0x100662f2: cmp eax,edx
0x100662f4: jae short 10066300h
0x100662f6: lea eax,[ecx+eax-1]
0x100662fa: xor edx,edx
0x100662fc: div ecx
0x100662fe: jmp short 10066302h
0x10066300: xor eax,eax
0x10066302: imul ecx,eax
```

```
0x10066305: push edi
0x10066306: push ebx
0x10066307: push ecx
0x10066308: call 10068414h
0x1006630d: mov ebx,eax
0x1006630f: cdq
0x10066310: idiv dword ptr [ebp-4]
0x10066313: add esp,0Ch
0x10066316: push edi
0x10066317: push edi
0x10066318: push eax
0x10066319: push 2
0x1006631b: call 10068414h
0x10066320: add esp,0Ch
0x10066323: mov ecx,eax
0x10066325: mov edx,ebx
0x10066327: call 10065EEAh
0x1006632c: pop ecx
0x1006632d: pop edi
0x1006632e: pop ebx
0x1006632f: jmp short 10066349h
0x10066331: push 101081A8h
0x10066336: push esi
0x10066337: call 10065FAFh
0x1006633c: pop ecx
0x1006633d: push eax
0x1006633e: push dword ptr [ebp+0Ch]
0x10066341: call 10068414h
0x10066346: add esp,0Ch
0x10066349: pop esi
0x1006634a: leave
0x1006634b: ret
```

`sub_10066210` is called by sub_1006634c, sub_10068247. It delegates to sub_10065eea, sub_10065faf, sub_10067985 (5 callees).

12.940 sub_1006634c at 0x1006634c (12 instructions)

```
0x1006634c: mov ecx,[esp+4]
0x10066350: mov eax,[ecx+64h]
0x10066353: mov edx,[ecx+3Ch]
0x10066356: cmp eax,edx
0x10066358: jbe short 1006635Ch
0x1006635a: mov eax,edx
0x1006635c: push eax
0x1006635d: push ecx
0x1006635e: call 10066210h
0x10066363: pop ecx
0x10066364: pop ecx
0x10066365: ret
```

`sub_1006634c` is called by sub_100143f0, sub_1007171b. It delegates to sub_10066210 (1 callees).

12.941 sub_10066f46 at 0x10066f46 (38 instructions)

```
0x10066f46: push 0
0x10066f48: push edi
0x10066f49: push esi
0x10066f4a: call 1006FDD7h
0x10066f4f: add esp,0Ch
0x10066f52: test eax,eax
0x10066f54: jne short 10066F87h
0x10066f56: cmp edi,0FFFFh
0x10066f5c: mov eax,101082C0h
0x10066f61: ja short 10066F68h
```

```
0x10066f63: mov eax,100FE92Bh
0x10066f68: push edi
0x10066f69: push eax
0x10066f6a: push dword ptr [esi]
0x10066f6c: push 101083F4h
0x10066f71: push 10108224h
0x10066f76: push dword ptr [esi+274h]
0x10066f7c: call 1006FAF1h
0x10066f81: add esp,18h
0x10066f84: xor eax,eax
0x10066f86: ret
0x10066f87: cmp edi,101h
0x10066f8d: je short 10066FBAh
0x10066f8f: test byte ptr [esi+0Ch],40h
0x10066f93: je short 10066FBAh
0x10066f95: cmp byte ptr [eax+0Eh],0
0x10066f99: jne short 10066FBAh
0x10066f9b: push dword ptr [eax+10h]
0x10066f9e: push dword ptr [esi]
0x10066fa0: push 101083C8h
0x10066fa5: push 10108224h
0x10066faa: push dword ptr [esi+274h]
0x10066fb0: call 1006FAF1h
0x10066fb5: add esp,14h
0x10066fb8: jmp short 10066F84h
0x10066fba: xor eax,eax
0x10066fbc: inc eax
0x10066fbd: ret
```

`sub_10066f46` is called by sub_10066fbe. It works with the text "TIFFSetField", "pseudo-", "%s: Unknown %stag %u". It delegates to sub_1006faf1, sub_1006fdd7 (2 callees).

12.942 sub_10066fbe at 0x10066fbe (17 instructions)

```
0x10066fbe: push esi
0x10066fbf: mov esi,[esp+8]
0x10066fc3: push edi
0x10066fc4: mov edi,[esp+10h]
0x10066fc8: call 10066F46h
0x10066fcd: test eax,eax
0x10066fcf: je short 10066FE2h
0x10066fd1: push dword ptr [esp+14h]
0x10066fd5: push edi
0x10066fd6: push esi
0x10066fd7: call dword ptr [esi+29Ch]
0x10066fdd: add esp,0Ch
0x10066fe0: jmp short 10066FE4h
0x10066fe2: xor eax,eax
0x10066fe4: pop edi
0x10066fe5: pop esi
0x10066fe6: ret
```

`sub_10066fbe` is called by sub_1006796f. It delegates to sub_10066f46 (1 callees).

12.943 sub_100675c7 at 0x100675c7 (32 instructions)

```
0x100675c7: push ebp
0x100675c8: mov ebp,esp
0x100675ca: push esi
0x100675cb: mov esi,[ebp+8]
0x100675ce: push 0
0x100675d0: push dword ptr [ebp+0Ch]
0x100675d3: push esi
0x100675d4: call 1006FDD7h
0x100675d9: add esp,0Ch
```

```
0x100675dc: test eax,eax
0x100675de: je short 10067612h
0x100675e0: cmp dword ptr [ebp+0Ch],0FFFFh
0x100675e7: ja short 10067600h
0x100675e9: movzx eax,word ptr [eax+0Ch]
0x100675ed: xor edx,edx
0x100675ef: mov ecx,eax
0x100675f1: and ecx,1Fh
0x100675f4: inc edx
0x100675f5: shl edx,cl
0x100675f7: shr eax,5
0x100675fa: test [esi+eax*4+28h],edx
0x100675fe: je short 10067612h
0x10067600: push dword ptr [ebp+10h]
0x10067603: push dword ptr [ebp+0Ch]
0x10067606: push esi
0x10067607: call dword ptr [esi+2A0h]
0x1006760d: add esp,0Ch
0x10067610: jmp short 10067614h
0x10067612: xor eax,eax
0x10067614: pop esi
0x10067615: pop ebp
0x10067616: ret
```

`sub_100675c7` is called by sub_10067985, sub_1006e289. It delegates to sub_1006fdd7 (1 callees).

12.944 sub_10067617 at 0x10067617 (135 instructions)

```
0x10067617: push ebx
0x10067618: push ebp
0x10067619: mov ebp,[esp+0Ch]
0x1006761d: push esi
0x1006761e: push edi
0x1006761f: push 4
0x10067621: xor edi,edi
0x10067623: lea esi,[ebp+28h]
0x10067626: push edi
0x10067627: push esi
0x10067628: call 100209F0h
0x1006762d: mov eax,[esi+70h]
0x10067630: add esp,0Ch
0x10067633: cmp eax,edi
0x10067635: je short 10067641h
0x10067637: push eax
0x10067638: call 100209E0h
0x1006763d: pop ecx
0x1006763e: mov [esi+70h],edi
0x10067641: mov eax,[esi+74h]
0x10067644: cmp eax,edi
0x10067646: je short 10067652h
0x10067648: push eax
0x10067649: call 100209E0h
0x1006764e: pop ecx
0x1006764f: mov [esi+74h],edi
0x10067652: mov eax,[esi+78h]
0x10067655: cmp eax,edi
0x10067657: je short 10067663h
0x10067659: push eax
0x1006765a: call 100209E0h
0x1006765f: pop ecx
0x10067660: mov [esi+78h],edi
0x10067663: lea ebx,[esi+84h]
0x10067669: mov eax,[ebx]
0x1006766b: cmp eax,edi
0x1006766d: je short 10067678h
0x1006766f: push eax
```

```
0x10067670: call 100209E0h
0x10067675: pop ecx
0x10067676: mov [ebx],edi
0x10067678: lea ebx,[esi+0A0h]
0x1006767e: mov eax,[ebx]
0x10067680: cmp eax,edi
0x10067682: je short 1006768Dh
0x10067684: push eax
0x10067685: call 100209E0h
0x1006768a: pop ecx
0x1006768b: mov [ebx],edi
0x1006768d: lea ebx,[esi+0C0h]
0x10067693: mov eax,[ebx]
0x10067695: cmp eax,edi
0x10067697: je short 100676A2h
0x10067699: push eax
0x1006769a: call 100209E0h
0x1006769f: pop ecx
0x100676a0: mov [ebx],edi
0x100676a2: lea ebx,[esi+0ACh]
0x100676a8: mov eax,[ebx]
0x100676aa: cmp eax,edi
0x100676ac: je short 100676B7h
0x100676ae: push eax
0x100676af: call 100209E0h
0x100676b4: pop ecx
0x100676b5: mov [ebx],edi
0x100676b7: lea ebx,[esi+0B0h]
0x100676bd: mov eax,[ebx]
0x100676bf: cmp eax,edi
0x100676c1: je short 100676CCh
0x100676c3: push eax
0x100676c4: call 100209E0h
0x100676c9: pop ecx
0x100676ca: mov [ebx],edi
0x100676cc: lea ebx,[esi+0B4h]
0x100676d2: mov eax,[ebx]
0x100676d4: cmp eax,edi
0x100676d6: je short 100676E1h
0x100676d8: push eax
0x100676d9: call 100209E0h
0x100676de: pop ecx
0x100676df: mov [ebx],edi
0x100676e1: lea ebx,[esi+0B8h]
0x100676e7: mov eax,[ebx]
0x100676e9: cmp eax,edi
0x100676eb: je short 100676F6h
0x100676ed: push eax
0x100676ee: call 100209E0h
0x100676f3: pop ecx
0x100676f4: mov [ebx],edi
0x100676f6: lea ebx,[esi+90h]
0x100676fc: mov eax,[ebx]
0x100676fe: cmp eax,edi
0x10067700: je short 1006770Bh
0x10067702: push eax
0x10067703: call 100209E0h
0x10067708: pop ecx
0x10067709: mov [ebx],edi
0x1006770b: lea ebx,[esi+94h]
0x10067711: mov eax,[ebx]
0x10067713: cmp eax,edi
0x10067715: je short 10067720h
0x10067717: push eax
0x10067718: call 100209E0h
0x1006771d: pop ecx
0x1006771e: mov [ebx],edi
```

```
0x10067720: and dword ptr [ebp+2Ch],0FFFFFE7Fh
0x10067727: xor ebp,ebp
0x10067729: cmp [esi+0C4h],edi
0x1006772f: jle short 10067754h
0x10067731: xor ebx,ebx
0x10067733: mov eax,[esi+0C8h]
0x10067739: mov eax,[eax+ebx+8]
0x1006773d: cmp eax,edi
0x1006773f: je short 10067748h
0x10067741: push eax
0x10067742: call 100209E0h
0x10067747: pop ecx
0x10067748: inc ebp
0x10067749: add ebx,0Ch
0x1006774c: cmp ebp,[esi+0C4h]
0x10067752: jl short 10067733h
0x10067754: mov [esi+0C4h],edi
0x1006775a: add esi,0C8h
0x10067760: mov eax,[esi]
0x10067762: cmp eax,edi
0x10067764: je short 1006776Fh
0x10067766: push eax
0x10067767: call 100209E0h
0x1006776c: pop ecx
0x1006776d: mov [esi],edi
0x1006776f: pop edi
0x10067770: pop esi
0x10067771: pop ebp
0x10067772: pop ebx
0x10067773: ret
```

`sub_10067617` is called by sub_10065c7b, sub_1006f391, sub_1007171b. It delegates to j_free, j_memset (2 callees).

12.945 sub_10067774 at 0x10067774 (111 instructions)

```
0x10067774: push ebp
0x10067775: mov ebp,esp
0x10067777: push ecx
0x10067778: test word ptr [esi+0Ch],800h
0x1006777e: push edi
0x1006777f: je near ptr 10067821h
0x10067785: mov edi,[ebx]
0x10067787: lea eax,[edi+2]
0x1006778a: cmp eax,[esi+268h]
0x10067790: jbe short 100677A9h
0x10067792: push dword ptr [esi]
0x10067794: push 10108440h
0x10067799: push 101081F8h
0x1006779e: push dword ptr [esi+274h]
0x100677a4: jmp near ptr 100678C2h
0x100677a9: mov eax,[esi+264h]
0x100677af: add eax,edi
0x100677b1: push 2
0x100677b3: push eax
0x100677b4: lea eax,[ebp-4]
0x100677b7: push eax
0x100677b8: call 10020A00h
0x100677bd: add esp,0Ch
0x100677c0: test byte ptr [esi+0Ch],80h
0x100677c4: je short 100677D0h
0x100677c6: lea eax,[ebp-4]
0x100677c9: push eax
0x100677ca: call 1006FB29h
0x100677cf: pop ecx
0x100677d0: movzx eax,word ptr [ebp-4]
0x100677d4: mov ecx,[ebp+8]
```

```
0x100677d7: imul eax,0Ch
0x100677da: test ecx,ecx
0x100677dc: lea eax,[edi+eax+2]
0x100677e0: je short 100677E4h
0x100677e2: mov [ecx],eax
0x100677e4: lea ecx,[eax+4]
0x100677e7: cmp ecx,[esi+268h]
0x100677ed: jbe short 100677F8h
0x100677ef: push dword ptr [esi]
0x100677f1: push 1010841Ch
0x100677f6: jmp short 10067799h
0x100677f8: mov ecx,[esi+264h]
0x100677fe: push 4
0x10067800: add ecx,eax
0x10067802: push ecx
0x10067803: push ebx
0x10067804: call 10020A00h
0x10067809: add esp,0Ch
0x1006780c: test byte ptr [esi+0Ch],80h
0x10067810: je short 10067819h
0x10067812: push ebx
0x10067813: call 1006FB38h
0x10067818: pop ecx
0x10067819: xor eax,eax
0x1006781b: inc eax
0x1006781c: jmp near ptr 100678CCh
0x10067821: push 0
0x10067823: push dword ptr [ebx]
0x10067825: push dword ptr [esi+274h]
0x1006782b: call dword ptr [esi+280h]
0x10067831: add esp,0Ch
0x10067834: cmp eax,[ebx]
0x10067836: jne short 100678B0h
0x10067838: push 2
0x1006783a: lea eax,[ebp-4]
0x1006783d: push eax
0x1006783e: push dword ptr [esi+274h]
0x10067844: call dword ptr [esi+278h]
0x1006784a: add esp,0Ch
0x1006784d: cmp eax,2
0x10067850: jne short 100678B0h
0x10067852: test byte ptr [esi+0Ch],80h
0x10067856: je short 10067862h
0x10067858: lea eax,[ebp-4]
0x1006785b: push eax
0x1006785c: call 1006FB29h
0x10067861: pop ecx
0x10067862: movzx eax,word ptr [ebp-4]
0x10067866: mov edi,[ebp+8]
0x10067869: imul eax,0Ch
0x1006786c: test edi,edi
0x1006786e: push 1
0x10067870: push eax
0x10067871: push dword ptr [esi+274h]
0x10067877: je short 10067883h
0x10067879: call dword ptr [esi+280h]
0x1006787f: mov [edi],eax
0x10067881: jmp short 10067889h
0x10067883: call dword ptr [esi+280h]
0x10067889: add esp,0Ch
0x1006788c: push 4
0x1006788e: push ebx
0x1006788f: push dword ptr [esi+274h]
0x10067895: call dword ptr [esi+278h]
0x1006789b: add esp,0Ch
0x1006789e: cmp eax,4
0x100678a1: je near ptr 1006780Ch
```

```
0x100678a7: push dword ptr [esi]
0x100678a9: push 1010841Ch
0x100678ae: jmp short 100678B7h
0x100678b0: push dword ptr [esi]
0x100678b2: push 10108440h
0x100678b7: push 101081F8h
0x100678bc: push dword ptr [esi+274h]
0x100678c2: call 1006FAF1h
0x100678c7: add esp,10h
0x100678ca: xor eax,eax
0x100678cc: pop edi
0x100678cd: leave
0x100678ce: ret
```

`sub_10067774` is called by sub_100678cf, sub_10067905. It works with the text "TIFFAdvanceDirectory", "%s: Error fetching directory link", "%s: Error fetching directory count". It delegates to j_memcpy, sub_1006faf1, sub_1006fb29 (4 callees).

12.946 sub_100678cf at 0x100678cf (26 instructions)

```
0x100678cf: push ebp
0x100678d0: mov ebp,esp
0x100678d2: push esi
0x100678d3: mov esi,[ebp+8]
0x100678d6: mov eax,[esi+1CCh]
0x100678dc: push edi
0x100678dd: xor edi,edi
0x100678df: test eax,eax
0x100678e1: mov [ebp+8],eax
0x100678e4: je short 100678FEh
0x100678e6: push ebx
0x100678e7: push 0
0x100678e9: lea ebx,[ebp+8]
0x100678ec: call 10067774h
0x100678f1: test eax,eax
0x100678f3: pop ecx
0x100678f4: je short 100678FDh
0x100678f6: inc edi
0x100678f7: cmp dword ptr [ebp+8],0
0x100678fb: jne short 100678E7h
0x100678fd: pop ebx
0x100678fe: mov ax,di
0x10067901: pop edi
0x10067902: pop esi
0x10067903: pop ebp
0x10067904: ret
```

`sub_100678cf` is called by sub_100143f0. It delegates to sub_10067774 (1 callees).

12.947 sub_10067905 at 0x10067905 (41 instructions)

```
0x10067905: push ebp
0x10067906: mov ebp,esp
0x10067908: push ecx
0x10067909: push ebx
0x1006790a: push esi
0x1006790b: mov esi,[ebp+8]
0x1006790e: mov eax,[esi+1CCh]
0x10067914: push edi
0x10067915: mov edi,[ebp+0Ch]
0x10067918: test di,di
0x1006791b: mov [ebp-4],eax
0x1006791e: movzx eax,di
0x10067921: mov [ebp+8],eax
0x10067924: jbe short 10067949h
```

```
0x10067926: cmp dword ptr [ebp-4],0
0x1006792a: je short 10067949h
0x1006792c: push 0
0x1006792e: lea ebx,[ebp-4]
0x10067931: call 10067774h
0x10067936: test eax,eax
0x10067938: pop ecx
0x10067939: je short 1006796Bh
0x1006793b: add dword ptr [ebp+8],0FFFFh
0x10067942: cmp word ptr [ebp+8],0
0x10067947: ja short 10067926h
0x10067949: sub edi,[ebp+8]
0x1006794c: mov eax,[ebp-4]
0x1006794f: and word ptr [esi+20h],0
0x10067954: dec edi
0x10067955: push esi
0x10067956: mov [esi+14h],eax
0x10067959: mov [esi+1DCh],di
0x10067960: call 1007171Bh
0x10067965: pop ecx
0x10067966: pop edi
0x10067967: pop esi
0x10067968: pop ebx
0x10067969: leave
0x1006796a: ret
0x1006796b: xor eax,eax
0x1006796d: jmp short 10067966h
```

`sub_10067905` is called by sub_100143f0. It delegates to sub_10067774, sub_1007171b (2 callees).

12.948 sub_1006796f at 0x1006796f (7 instructions)

```
0x1006796f: lea eax,[esp+0Ch]
0x10067973: push eax
0x10067974: push dword ptr [esp+0Ch]
0x10067978: push dword ptr [esp+0Ch]
0x1006797c: call 10066FBEh
0x10067981: add esp,0Ch
0x10067984: ret
```

`sub_1006796f` is called by sub_100143f0, sub_1006799b, sub_1006daaf, sub_10070aeb, sub_10071523, sub_100715da, sub_1007171b. It delegates to sub_10066fbe (1 callees).

12.949 sub_10067985 at 0x10067985 (7 instructions)

```
0x10067985: lea eax,[esp+0Ch]
0x10067989: push eax
0x1006798a: push dword ptr [esp+0Ch]
0x1006798e: push dword ptr [esp+0Ch]
0x10067992: call 100675C7h
0x10067997: add esp,0Ch
0x1006799a: ret
```

`sub_10067985` is called by sub_100143f0, sub_10065faf, sub_10066210, sub_10068d67, sub_1006901a, sub_1006daaf, sub_1006f20b, sub_1006f2c7, sub_1006f35a. It delegates to sub_100675c7 (1 callees).

12.950 sub_1006799b at 0x1006799b (63 instructions)

```
0x1006799b: push ebp
0x1006799c: mov ebp,esp
0x1006799e: push ebx
0x1006799f: push esi
0x100679a0: push edi
0x100679a1: mov edi,[ebp+8]
```

```
0x100679a4: lea eax,[ebp+8]
0x100679a7: push eax
0x100679a8: lea esi,[edi+28h]
0x100679ab: call 1006FCA6h
0x100679b0: push dword ptr [ebp+8]
0x100679b3: push eax
0x100679b4: push edi
0x100679b5: call 1006FFE0h
0x100679ba: push 0D0h
0x100679bf: push 0
0x100679c1: push esi
0x100679c2: call 100209F0h
0x100679c7: or dword ptr [esi+3Ch],0FFFFFFFFh
0x100679cb: mov eax,ds:[1013939Ch]
0x100679d0: xor ecx,ecx
0x100679d2: xor ebx,ebx
0x100679d4: inc ebx
0x100679d5: mov [esi+36h],bx
0x100679d9: mov [esi+2Ch],bx
0x100679dd: mov [esi+34h],bx
0x100679e1: mov [esi+38h],bx
0x100679e5: mov [esi+3Ah],bx
0x100679e9: mov [esi+1Ch],ecx
0x100679ec: mov [esi+20h],ecx
0x100679ef: mov [esi+24h],ebx
0x100679f2: mov [esi+98h],ebx
0x100679f8: mov word ptr [esi+60h],2
0x100679fe: mov [esi+2Eh],bx
0x10067a02: mov [esi+18h],ebx
0x10067a05: mov word ptr [esi+0A4h],2
0x10067a0e: mov word ptr [esi+0A6h],2
0x10067a17: mov [esi+0A8h],bx
0x10067a1e: add esp,1Ch
0x10067a21: cmp eax,ecx
0x10067a23: mov dword ptr [edi+28Ch],1007B5B1h
0x10067a2d: mov [edi+298h],ecx
0x10067a33: mov dword ptr [edi+29Ch],1006650Eh
0x10067a3d: mov dword ptr [edi+2A0h],10066FE7h
0x10067a47: mov [edi+2A4h],ecx
0x10067a4d: je short 10067A53h
0x10067a4f: push edi
0x10067a50: call eax
0x10067a52: pop ecx
0x10067a53: push ebx
0x10067a54: push 103h
0x10067a59: push edi
0x10067a5a: call 1006796Fh
0x10067a5f: and dword ptr [edi+0Ch],0FFFFFBF7h
0x10067a66: or dword ptr [edi+1FCh],0FFFFFFFFh
0x10067a6d: add esp,0Ch
0x10067a70: or dword ptr [edi+24Ch],0FFFFFFFFh
0x10067a77: pop edi
0x10067a78: pop esi
0x10067a79: mov eax,ebx
0x10067a7b: pop ebx
0x10067a7c: pop ebp
0x10067a7d: ret
```

`sub_1006799b` is called by sub_10067a7e, sub_1006891d, sub_1007171b. It delegates to j_memset, sub_1006796f, sub_1006fca6 (4 callees).

12.951 sub_10067a7e at 0x10067a7e (13 instructions)

```
0x10067a7e: push esi
0x10067a7f: mov esi,[esp+8]
0x10067a83: push esi
```

```
0x10067a84: call 1006799Bh
0x10067a89: or dword ptr [esi+1D8h],0FFFFFFFFh
0x10067a90: or dword ptr [esi+1E0h],0FFFFFFFFh
0x10067a97: xor eax,eax
0x10067a99: pop ecx
0x10067a9a: mov [esi+10h],eax
0x10067a9d: mov [esi+14h],eax
0x10067aa0: mov [esi+1E4h],eax
0x10067aa6: pop esi
0x10067aa7: ret
```

`sub_10067a7e` is called by sub_1006f391. It delegates to sub_1006799b (1 callees).

12.952 sub_10067aa8 at 0x10067aa8 (72 instructions)

```
0x10067aa8: push ebp
0x10067aa9: mov ebp,esp
0x10067aab: test word ptr [esi+0Ch],800h
0x10067ab1: push ebx
0x10067ab2: jne near ptr 10067B40h
0x10067ab8: lea ebx,[esi+0B8h]
0x10067abe: mov eax,[ebx]
0x10067ac0: push 0
0x10067ac2: push dword ptr [eax+edi*4]
0x10067ac5: push dword ptr [esi+274h]
0x10067acb: call dword ptr [esi+280h]
0x10067ad1: mov ecx,[ebx]
0x10067ad3: add esp,0Ch
0x10067ad6: cmp eax,[ecx+edi*4]
0x10067ad9: je short 10067B02h
0x10067adb: push edi
0x10067adc: push dword ptr [esi+1D8h]
0x10067ae2: push dword ptr [esi]
0x10067ae4: push 10108564h
0x10067ae9: push dword ptr [ebp+10h]
0x10067aec: push dword ptr [esi+274h]
0x10067af2: call 1006FAF1h
0x10067af7: add esp,18h
0x10067afa: or eax,0FFFFFFFFh
0x10067afd: jmp near ptr 10067B9Bh
0x10067b02: push dword ptr [ebp+0Ch]
0x10067b05: push dword ptr [ebp+8]
0x10067b08: push dword ptr [esi+274h]
0x10067b0e: call dword ptr [esi+278h]
0x10067b14: add esp,0Ch
0x10067b17: cmp eax,[ebp+0Ch]
0x10067b1a: je short 10067B98h
0x10067b1c: push dword ptr [ebp+0Ch]
0x10067b1f: push eax
0x10067b20: push dword ptr [esi+1D8h]
0x10067b26: push dword ptr [esi]
0x10067b28: push 10108528h
0x10067b2d: push dword ptr [ebp+10h]
0x10067b30: push dword ptr [esi+274h]
0x10067b36: call 1006FAF1h
0x10067b3b: add esp,1Ch
0x10067b3e: jmp short 10067AFAh
0x10067b40: mov eax,[esi+0B8h]
0x10067b46: mov ebx,[ebp+0Ch]
0x10067b49: push dword ptr [ebp+0Ch]
0x10067b4c: lea ecx,[eax+edi*4]
0x10067b4f: mov edx,[ecx]
0x10067b51: mov eax,[esi+268h]
0x10067b57: add ebx,edx
0x10067b59: cmp ebx,eax
0x10067b5b: jbe short 10067B84h
```

```
0x10067b5d: sub eax,edx
0x10067b5f: push eax
0x10067b60: push edi
0x10067b61: push dword ptr [esi+1D8h]
0x10067b67: push dword ptr [esi]
0x10067b69: push 101084E0h
0x10067b6e: push dword ptr [ebp+10h]
0x10067b71: push dword ptr [esi+274h]
0x10067b77: call 1006FAF1h
0x10067b7c: add esp,20h
0x10067b7f: jmp near ptr 10067AFAh
0x10067b84: mov eax,[esi+264h]
0x10067b8a: add eax,edx
0x10067b8c: push eax
0x10067b8d: push dword ptr [ebp+8]
0x10067b90: call 10020A00h
0x10067b95: add esp,0Ch
0x10067b98: mov eax,[ebp+0Ch]
0x10067b9b: pop ebx
0x10067b9c: pop ebp
0x10067b9d: ret
```

`sub_10067aa8` is called by sub_10067f4a. It works with the text "%s: Seek error at scanline %lu, strip %lu". It delegates to j_memcpy, sub_1006faf1 (2 callees).

12.953 sub_10067b9e at 0x10067b9e (75 instructions)

```
0x10067b9e: push ebp
0x10067b9f: mov ebp,esp
0x10067ba1: test word ptr [esi+0Ch],800h
0x10067ba7: push ebx
0x10067ba8: jne near ptr 10067C46h
0x10067bae: lea ebx,[esi+0B8h]
0x10067bb4: mov eax,[ebx]
0x10067bb6: push 0
0x10067bb8: push dword ptr [eax+edi*4]
0x10067bbb: push dword ptr [esi+274h]
0x10067bc1: call dword ptr [esi+280h]
0x10067bc7: mov ecx,[ebx]
0x10067bc9: add esp,0Ch
0x10067bcc: cmp eax,[ecx+edi*4]
0x10067bcf: je short 10067BFEh
0x10067bd1: push edi
0x10067bd2: push dword ptr [esi+1F4h]
0x10067bd8: push dword ptr [esi+1D8h]
0x10067bde: push dword ptr [esi]
0x10067be0: push 10108620h
0x10067be5: push dword ptr [ebp+10h]
0x10067be8: push dword ptr [esi+274h]
0x10067bee: call 1006FAF1h
0x10067bf3: add esp,1Ch
0x10067bf6: or eax,0FFFFFFFFh
0x10067bf9: jmp near ptr 10067CA7h
0x10067bfe: push dword ptr [ebp+0Ch]
0x10067c01: push dword ptr [ebp+8]
0x10067c04: push dword ptr [esi+274h]
0x10067c0a: call dword ptr [esi+278h]
0x10067c10: add esp,0Ch
0x10067c13: cmp eax,[ebp+0Ch]
0x10067c16: je near ptr 10067CA4h
0x10067c1c: push dword ptr [ebp+0Ch]
0x10067c1f: push eax
0x10067c20: push dword ptr [esi+1F4h]
0x10067c26: push dword ptr [esi+1D8h]
0x10067c2c: push dword ptr [esi]
0x10067c2e: push 101085E0h
```

```
0x10067c33: push dword ptr [ebp+10h]
0x10067c36: push dword ptr [esi+274h]
0x10067c3c: call 1006FAF1h
0x10067c41: add esp,20h
0x10067c44: jmp short 10067BF6h
0x10067c46: mov eax,[esi+0B8h]
0x10067c4c: mov ebx,[ebp+0Ch]
0x10067c4f: push dword ptr [ebp+0Ch]
0x10067c52: lea ecx,[eax+edi*4]
0x10067c55: mov edx,[ecx]
0x10067c57: mov eax,[esi+268h]
0x10067c5d: add ebx,edx
0x10067c5f: cmp ebx,eax
0x10067c61: jbe short 10067C90h
0x10067c63: sub eax,edx
0x10067c65: push eax
0x10067c66: push edi
0x10067c67: push dword ptr [esi+1F4h]
0x10067c6d: push dword ptr [esi+1D8h]
0x10067c73: push dword ptr [esi]
0x10067c75: push 10108590h
0x10067c7a: push dword ptr [ebp+10h]
0x10067c7d: push dword ptr [esi+274h]
0x10067c83: call 1006FAF1h
0x10067c88: add esp,24h
0x10067c8b: jmp near ptr 10067BF6h
0x10067c90: mov eax,[esi+264h]
0x10067c96: add eax,edx
0x10067c98: push eax
0x10067c99: push dword ptr [ebp+8]
0x10067c9c: call 10020A00h
0x10067ca1: add esp,0Ch
0x10067ca4: mov eax,[ebp+0Ch]
0x10067ca7: pop ebx
0x10067ca8: pop ebp
0x10067ca9: ret
```

`sub_10067b9e` is called by sub_100680ec. It works with the text "%s: Seek error at row %ld, col %ld, tile %ld". It delegates to j_memcpy, sub_1006faf1 (2 callees).

12.954 sub_10067caa at 0x10067caa (56 instructions)

```
0x10067caa: push esi
0x10067cab: mov esi,[esp+8]
0x10067caf: mov eax,[esi+254h]
0x10067cb5: test eax,eax
0x10067cb7: push edi
0x10067cb8: mov edi,200h
0x10067cbd: je short 10067CD2h
0x10067cbf: test [esi+0Ch],edi
0x10067cc2: je short 10067CCBh
0x10067cc4: push eax
0x10067cc5: call 100209E0h
0x10067cca: pop ecx
0x10067ccb: and dword ptr [esi+254h],0
0x10067cd2: mov eax,[esp+10h]
0x10067cd6: test eax,eax
0x10067cd8: je short 10067CF3h
0x10067cda: mov ecx,[esp+14h]
0x10067cde: and dword ptr [esi+0Ch],0FFFFFDFFh
0x10067ce5: mov [esi+258h],ecx
0x10067ceb: mov [esi+254h],eax
0x10067cf1: jmp short 10067D27h
0x10067cf3: mov eax,[esp+14h]
0x10067cf7: cmp eax,0FFFFFC00h
0x10067cfc: jae short 10067D08h
```

```
0x10067cfe: add eax,3FFh
0x10067d03: shr eax,0Ah
0x10067d06: jmp short 10067D0Ah
0x10067d08: xor eax,eax
0x10067d0a: shl eax,0Ah
0x10067d0d: test eax,eax
0x10067d0f: mov [esi+258h],eax
0x10067d15: jle short 10067D24h
0x10067d17: push eax
0x10067d18: call 100EFDD0h
0x10067d1d: pop ecx
0x10067d1e: mov [esi+254h],eax
0x10067d24: or [esi+0Ch],edi
0x10067d27: cmp dword ptr [esi+254h],0
0x10067d2e: je short 10067D3Eh
0x10067d30: cmp dword ptr [esi+258h],0
0x10067d37: je short 10067D3Eh
0x10067d39: xor eax,eax
0x10067d3b: inc eax
0x10067d3c: jmp short 10067D67h
0x10067d3e: push dword ptr [esi+1D8h]
0x10067d44: push dword ptr [esi]
0x10067d46: push 10108650h
0x10067d4b: push 101084C8h
0x10067d50: push dword ptr [esi+274h]
0x10067d56: call 1006FAF1h
0x10067d5b: and dword ptr [esi+258h],0
0x10067d62: add esp,14h
0x10067d65: xor eax,eax
0x10067d67: pop edi
0x10067d68: pop esi
0x10067d69: ret
```

`sub_10067caa` is called by sub_10067f4a, sub_100680ec. It works with the text "TIFFReadBufferSetup", "%s: No space for data buffer at scanline %ld". It delegates to j_free, sub_1006faf1, j_malloc (3 callees).

12.955 sub_10067d6a at 0x10067d6a (34 instructions)

```
0x10067d6a: test byte ptr [esi+0Ch],20h
0x10067d6e: jne short 10067D81h
0x10067d70: push esi
0x10067d71: call dword ptr [esi+204h]
0x10067d77: test eax,eax
0x10067d79: pop ecx
0x10067d7a: jne short 10067D7Dh
0x10067d7c: ret
0x10067d7d: or dword ptr [esi+0Ch],20h
0x10067d81: xor edx,edx
0x10067d83: mov [esi+1E0h],edi
0x10067d89: mov eax,edi
0x10067d8b: div dword ptr [esi+0B0h]
0x10067d91: imul edx,[esi+64h]
0x10067d95: test dword ptr [esi+0Ch],20000h
0x10067d9c: mov [esi+1D8h],edx
0x10067da2: je short 10067DB4h
0x10067da4: and dword ptr [esi+25Ch],0
0x10067dab: and dword ptr [esi+260h],0
0x10067db2: jmp short 10067DCFh
0x10067db4: mov eax,[esi+254h]
0x10067dba: mov [esi+25Ch],eax
0x10067dc0: mov eax,[esi+0BCh]
0x10067dc6: mov eax,[eax+edi*4]
0x10067dc9: mov [esi+260h],eax
0x10067dcf: xor edx,edx
0x10067dd1: mov eax,edi
0x10067dd3: div dword ptr [esi+0B0h]
```

```
0x10067dd9: push eax
0x10067dda: push esi
0x10067ddb: call dword ptr [esi+208h]
0x10067de1: pop ecx
0x10067de2: pop ecx
0x10067de3: ret
```

`sub_10067d6a` is called by sub_10067f4a. It calls nothing else.

12.956 sub_10067de4 at 0x10067de4 (63 instructions)

```
0x10067de4: test byte ptr [esi+0Ch],20h
0x10067de8: jne short 10067DFBh
0x10067dea: push esi
0x10067deb: call dword ptr [esi+204h]
0x10067df1: test eax,eax
0x10067df3: pop ecx
0x10067df4: jne short 10067DF7h
0x10067df6: ret
0x10067df7: or dword ptr [esi+0Ch],20h
0x10067dfb: mov [esi+1F8h],edi
0x10067e01: mov ecx,[esi+44h]
0x10067e04: mov eax,[esi+38h]
0x10067e07: neg ecx
0x10067e09: cmp eax,ecx
0x10067e0b: jae short 10067E1Ch
0x10067e0d: mov ecx,[esi+44h]
0x10067e10: lea eax,[ecx+eax-1]
0x10067e14: xor edx,edx
0x10067e16: div ecx
0x10067e18: mov ecx,eax
0x10067e1a: jmp short 10067E1Eh
0x10067e1c: xor ecx,ecx
0x10067e1e: xor edx,edx
0x10067e20: mov eax,edi
0x10067e22: div ecx
0x10067e24: imul edx,[esi+48h]
0x10067e28: mov [esi+1D8h],edx
0x10067e2e: mov ecx,[esi+48h]
0x10067e31: mov eax,[esi+3Ch]
0x10067e34: neg ecx
0x10067e36: cmp eax,ecx
0x10067e38: jae short 10067E49h
0x10067e3a: mov ecx,[esi+48h]
0x10067e3d: lea eax,[ecx+eax-1]
0x10067e41: xor edx,edx
0x10067e43: div ecx
0x10067e45: mov ecx,eax
0x10067e47: jmp short 10067E4Bh
0x10067e49: xor ecx,ecx
0x10067e4b: xor edx,edx
0x10067e4d: mov eax,edi
0x10067e4f: div ecx
0x10067e51: imul edx,[esi+44h]
0x10067e55: test dword ptr [esi+0Ch],20000h
0x10067e5c: mov [esi+1F4h],edx
0x10067e62: je short 10067E74h
0x10067e64: and dword ptr [esi+25Ch],0
0x10067e6b: and dword ptr [esi+260h],0
0x10067e72: jmp short 10067E8Fh
0x10067e74: mov eax,[esi+254h]
0x10067e7a: mov [esi+25Ch],eax
0x10067e80: mov eax,[esi+0BCh]
0x10067e86: mov eax,[eax+edi*4]
0x10067e89: mov [esi+260h],eax
0x10067e8f: xor edx,edx
```

```
0x10067e91: mov eax,edi
0x10067e93: div dword ptr [esi+0B0h]
0x10067e99: push eax
0x10067e9a: push esi
0x10067e9b: call dword ptr [esi+208h]
0x10067ea1: pop ecx
0x10067ea2: pop ecx
0x10067ea3: ret
```

`sub_10067de4` is called by sub_100680ec. It calls nothing else.

12.957 sub_10067ea4 at 0x10067ea4 (23 instructions)

```
0x10067ea4: cmp dword ptr [eax+8],1
0x10067ea8: jne short 10067EC2h
0x10067eaa: push 101086D8h
0x10067eaf: push dword ptr [eax]
0x10067eb1: push dword ptr [eax+274h]
0x10067eb7: call 1006FAF1h
0x10067ebc: add esp,0Ch
0x10067ebf: xor eax,eax
0x10067ec1: ret
0x10067ec2: mov ecx,[eax+0Ch]
0x10067ec5: shr ecx,0Ah
0x10067ec8: and ecx,1
0x10067ecb: xor ecx,[esp+4]
0x10067ecf: je short 10067EE5h
0x10067ed1: cmp dword ptr [esp+4],0
0x10067ed6: mov ecx,101086ACh
0x10067edb: jne short 10067EE2h
0x10067edd: mov ecx,10108680h
0x10067ee2: push ecx
0x10067ee3: jmp short 10067EAFh
0x10067ee5: xor eax,eax
0x10067ee7: inc eax
0x10067ee8: ret
```

`sub_10067ea4` is called by sub_10068247, sub_1006831d, sub_100683ba. It works with the text "Can not read scanlines from a tiled image", "Can not read tiles from a stripped image", "File not open for reading". It delegates to sub_1006faf1 (1 callees).

12.958 sub_10067f4a at 0x10067f4a (122 instructions)

```
0x10067f4a: push ebp
0x10067f4b: mov ebp,esp
0x10067f4d: push ebx
0x10067f4e: push esi
0x10067f4f: mov esi,[ebp+8]
0x10067f52: mov eax,[esi+0Ch]
0x10067f55: test eax,20000h
0x10067f5a: push edi
0x10067f5b: jne near ptr 100680DFh
0x10067f61: mov ecx,[ebp+0Ch]
0x10067f64: mov edx,[esi+0BCh]
0x10067f6a: mov edi,ecx
0x10067f6c: shl edi,2
0x10067f6f: mov ebx,[edi+edx]
0x10067f72: test ebx,ebx
0x10067f74: ja short 10067F99h
0x10067f76: push ecx
0x10067f77: push ebx
0x10067f78: push dword ptr [esi]
0x10067f7a: push 1010875Ch
0x10067f7f: push 10108498h
0x10067f84: push dword ptr [esi+274h]
```

```
0x10067f8a: call 1006FAF1h
0x10067f8f: add esp,18h
0x10067f92: xor eax,eax
0x10067f94: jmp near ptr 100680E7h
0x10067f99: test ax,800h
0x10067f9d: je near ptr 10068040h
0x10067fa3: movzx edx,word ptr [esi+5Eh]
0x10067fa7: test eax,edx
0x10067fa9: jne short 10067FB5h
0x10067fab: test ax,100h
0x10067faf: je near ptr 10068040h
0x10067fb5: test ax,200h
0x10067fb9: je short 10067FCCh
0x10067fbb: mov eax,[esi+254h]
0x10067fc1: test eax,eax
0x10067fc3: je short 10067FCCh
0x10067fc5: push eax
0x10067fc6: call 100209E0h
0x10067fcb: pop ecx
0x10067fcc: mov eax,[esi+268h]
0x10067fd2: and dword ptr [esi+0Ch],0FFFFFDFFh
0x10067fd9: cmp ebx,eax
0x10067fdb: ja short 1006800Ch
0x10067fdd: mov edx,[esi+0B8h]
0x10067fe3: mov ecx,eax
0x10067fe5: sub ecx,ebx
0x10067fe7: cmp [edi+edx],ecx
0x10067fea: ja short 1006800Ch
0x10067fec: mov [esi+258h],ebx
0x10067ff2: mov eax,[esi+0B8h]
0x10067ff8: mov eax,[edi+eax]
0x10067ffb: add eax,[esi+264h]
0x10068001: mov [esi+254h],eax
0x10068007: jmp near ptr 100680DFh
0x1006800c: mov ecx,[esi+0B8h]
0x10068012: sub eax,[ecx+edi]
0x10068015: push ebx
0x10068016: push eax
0x10068017: push dword ptr [ebp+0Ch]
0x1006801a: push dword ptr [esi]
0x1006801c: push 10108720h
0x10068021: push 10108498h
0x10068026: push dword ptr [esi+274h]
0x1006802c: call 1006FAF1h
0x10068031: add esp,1Ch
0x10068034: or dword ptr [esi+1E0h],0FFFFFFFFh
0x1006803b: jmp near ptr 10067F92h
0x10068040: cmp ebx,[esi+258h]
0x10068046: jbe short 100680A1h
0x10068048: or dword ptr [esi+1E0h],0FFFFFFFFh
0x1006804f: test ax,200h
0x10068053: jne short 10068075h
0x10068055: push ecx
0x10068056: push dword ptr [esi]
0x10068058: push 101086F4h
0x1006805d: push 10108498h
0x10068062: push dword ptr [esi+274h]
0x10068068: call 1006FAF1h
0x1006806d: add esp,14h
0x10068070: jmp near ptr 10067F92h
0x10068075: cmp ebx,0FFFFFC00h
0x1006807b: jae short 10068088h
0x1006807d: lea eax,[ebx+3FFh]
0x10068083: shr eax,0Ah
0x10068086: jmp short 1006808Ah
0x10068088: xor eax,eax
0x1006808a: shl eax,0Ah
```

```
0x1006808d: push eax
0x1006808e: push 0
0x10068090: push esi
0x10068091: call 10067CAAh
0x10068096: add esp,0Ch
0x10068099: test eax,eax
0x1006809b: je near ptr 10067F92h
0x100680a1: mov edi,[ebp+0Ch]
0x100680a4: push 10108498h
0x100680a9: push ebx
0x100680aa: push dword ptr [esi+254h]
0x100680b0: call 10067AA8h
0x100680b5: add esp,0Ch
0x100680b8: cmp eax,ebx
0x100680ba: jne near ptr 10067F92h
0x100680c0: movzx ecx,word ptr [esi+5Eh]
0x100680c4: mov eax,[esi+0Ch]
0x100680c7: test eax,ecx
0x100680c9: jne short 100680DFh
0x100680cb: test ax,100h
0x100680cf: jne short 100680DFh
0x100680d1: push ebx
0x100680d2: push dword ptr [esi+254h]
0x100680d8: call 1006FC09h
0x100680dd: pop ecx
0x100680de: pop ecx
0x100680df: mov edi,[ebp+0Ch]
0x100680e2: call 10067D6Ah
0x100680e7: pop edi
0x100680e8: pop esi
0x100680e9: pop ebx
0x100680ea: pop ebp
0x100680eb: ret
```

`sub_10067f4a` is called by sub_10068247. It works with the text "%s: Data buffer too small to hold strip %lu", "%s: Invalid strip byte count %lu, strip %lu". It delegates to j_free, sub_10067aa8, sub_10067caa (6 callees).

12.959 sub_100680ec at 0x100680ec (108 instructions)

```
0x100680ec: push ebp
0x100680ed: mov ebp,esp
0x100680ef: push ebx
0x100680f0: push esi
0x100680f1: mov esi,[ebp+8]
0x100680f4: mov eax,[esi+0Ch]
0x100680f7: test eax,20000h
0x100680fc: push edi
0x100680fd: mov edi,[ebp+0Ch]
0x10068100: jne near ptr 1006823Dh
0x10068106: mov ecx,[esi+0BCh]
0x1006810c: mov edx,edi
0x1006810e: shl edx,2
0x10068111: mov ebx,[edx+ecx]
0x10068114: test ebx,ebx
0x10068116: mov [ebp+8],edx
0x10068119: ja short 10068139h
0x1006811b: push edi
0x1006811c: push ebx
0x1006811d: push 101087B4h
0x10068122: push dword ptr [esi]
0x10068124: push dword ptr [esi+274h]
0x1006812a: call 1006FAF1h
0x1006812f: add esp,14h
0x10068132: xor eax,eax
0x10068134: jmp near ptr 10068242h
0x10068139: test ax,800h
```

```
0x1006813d: je short 100681AFh
0x1006813f: movzx ecx,word ptr [esi+5Eh]
0x10068143: test eax,ecx
0x10068145: jne short 1006814Dh
0x10068147: test ax,100h
0x1006814b: je short 100681AFh
0x1006814d: test ax,200h
0x10068151: je short 10068167h
0x10068153: mov eax,[esi+254h]
0x10068159: test eax,eax
0x1006815b: je short 10068167h
0x1006815d: push eax
0x1006815e: call 100209E0h
0x10068163: mov edx,[ebp+8]
0x10068166: pop ecx
0x10068167: mov eax,[esi+268h]
0x1006816d: and dword ptr [esi+0Ch],0FFFFFDFFh
0x10068174: cmp ebx,eax
0x10068176: ja short 100681A6h
0x10068178: lea ecx,[esi+0B8h]
0x1006817e: mov edi,[ecx]
0x10068180: sub eax,ebx
0x10068182: cmp [edx+edi],eax
0x10068185: ja short 100681A6h
0x10068187: mov edi,[ebp+0Ch]
0x1006818a: mov [esi+258h],ebx
0x10068190: mov eax,[ecx]
0x10068192: mov eax,[edx+eax]
0x10068195: add eax,[esi+264h]
0x1006819b: mov [esi+254h],eax
0x100681a1: jmp near ptr 1006823Dh
0x100681a6: or dword ptr [esi+1F8h],0FFFFFFFFh
0x100681ad: jmp short 10068132h
0x100681af: cmp ebx,[esi+258h]
0x100681b5: jbe short 10068202h
0x100681b7: or dword ptr [esi+1F8h],0FFFFFFFFh
0x100681be: test ax,200h
0x100681c2: jne short 100681D6h
0x100681c4: push edi
0x100681c5: push dword ptr [esi]
0x100681c7: push 10108788h
0x100681cc: push 101084B8h
0x100681d1: jmp near ptr 10068124h
0x100681d6: cmp ebx,0FFFFFC00h
0x100681dc: jae short 100681E9h
0x100681de: lea eax,[ebx+3FFh]
0x100681e4: shr eax,0Ah
0x100681e7: jmp short 100681EBh
0x100681e9: xor eax,eax
0x100681eb: shl eax,0Ah
0x100681ee: push eax
0x100681ef: push 0
0x100681f1: push esi
0x100681f2: call 10067CAAh
0x100681f7: add esp,0Ch
0x100681fa: test eax,eax
0x100681fc: je near ptr 10068132h
0x10068202: push 101084B8h
0x10068207: push ebx
0x10068208: push dword ptr [esi+254h]
0x1006820e: call 10067B9Eh
0x10068213: add esp,0Ch
0x10068216: cmp eax,ebx
0x10068218: jne near ptr 10068132h
0x1006821e: movzx ecx,word ptr [esi+5Eh]
0x10068222: mov eax,[esi+0Ch]
0x10068225: test eax,ecx
```

```
0x10068227: jne short 1006823Dh
0x10068229: test ax,100h
0x1006822d: jne short 1006823Dh
0x1006822f: push ebx
0x10068230: push dword ptr [esi+254h]
0x10068236: call 1006FC09h
0x1006823b: pop ecx
0x1006823c: pop ecx
0x1006823d: call 10067DE4h
0x10068242: pop edi
0x10068243: pop esi
0x10068244: pop ebx
0x10068245: pop ebp
0x10068246: ret
```

`sub_100680ec` is called by sub_1006831d. It works with the text "TIFFFillTile", "%s: Data buffer too small to hold tile %ld". It delegates to j_free, sub_10067b9e, sub_10067caa (6 callees).

12.960 sub_10068247 at 0x10068247 (91 instructions)

```
0x10068247: push ebp
0x10068248: mov ebp,esp
0x1006824a: push esi
0x1006824b: mov esi,[ebp+8]
0x1006824e: push 0
0x10068250: mov eax,esi
0x10068252: call 10067EA4h
0x10068257: test eax,eax
0x10068259: pop ecx
0x1006825a: jne short 10068264h
0x1006825c: or eax,0FFFFFFFFh
0x1006825f: jmp near ptr 1006831Ah
0x10068264: mov eax,[esi+0B4h]
0x1006826a: cmp [ebp+0Ch],eax
0x1006826d: jb short 1006828Ah
0x1006826f: push eax
0x10068270: push dword ptr [ebp+0Ch]
0x10068273: push 101087FCh
0x10068278: push dword ptr [esi]
0x1006827a: push dword ptr [esi+274h]
0x10068280: call 1006FAF1h
0x10068285: add esp,14h
0x10068288: jmp short 1006825Ch
0x1006828a: push ebx
0x1006828b: mov ebx,[esi+64h]
0x1006828e: push edi
0x1006828f: mov edi,[esi+3Ch]
0x10068292: cmp ebx,edi
0x10068294: jb short 1006829Bh
0x10068296: xor ecx,ecx
0x10068298: inc ecx
0x10068299: jmp short 100682A7h
0x1006829b: mov ecx,ebx
0x1006829d: lea eax,[ecx+edi-1]
0x100682a1: xor edx,edx
0x100682a3: div ecx
0x100682a5: mov ecx,eax
0x100682a7: mov eax,[ebp+0Ch]
0x100682aa: xor edx,edx
0x100682ac: div ecx
0x100682ae: dec ecx
0x100682af: cmp edx,ecx
0x100682b1: jne short 100682BDh
0x100682b3: xor edx,edx
0x100682b5: mov eax,edi
0x100682b7: div ebx
```

```
0x100682b9: test edx,edx
0x100682bb: jne short 100682BFh
0x100682bd: mov edx,ebx
0x100682bf: push edx
0x100682c0: push esi
0x100682c1: call 10066210h
0x100682c6: mov edi,[ebp+14h]
0x100682c9: cmp edi,0FFFFFFFFh
0x100682cc: pop ecx
0x100682cd: pop ecx
0x100682ce: je short 100682D4h
0x100682d0: cmp edi,eax
0x100682d2: jle short 100682D6h
0x100682d4: mov edi,eax
0x100682d6: push dword ptr [ebp+0Ch]
0x100682d9: push esi
0x100682da: call 10067F4Ah
0x100682df: test eax,eax
0x100682e1: pop ecx
0x100682e2: pop ecx
0x100682e3: je short 10068315h
0x100682e5: mov eax,[ebp+0Ch]
0x100682e8: xor edx,edx
0x100682ea: div dword ptr [esi+0B0h]
0x100682f0: push eax
0x100682f1: push edi
0x100682f2: push dword ptr [ebp+10h]
0x100682f5: push esi
0x100682f6: call dword ptr [esi+224h]
0x100682fc: add esp,10h
0x100682ff: test eax,eax
0x10068301: jle short 10068315h
0x10068303: push edi
0x10068304: push dword ptr [ebp+10h]
0x10068307: push esi
0x10068308: call dword ptr [esi+28Ch]
0x1006830e: add esp,0Ch
0x10068311: mov eax,edi
0x10068313: jmp short 10068318h
0x10068315: or eax,0FFFFFFFFh
0x10068318: pop edi
0x10068319: pop ebx
0x1006831a: pop esi
0x1006831b: pop ebp
0x1006831c: ret
```

`sub_10068247` is called by sub_100143f0. It delegates to sub_10066210, sub_10067ea4, sub_10067f4a (4 callees).

12.961 sub_1006831d at 0x1006831d (63 instructions)

```
0x1006831d: push ebp
0x1006831e: mov ebp,esp
0x10068320: push ebx
0x10068321: push esi
0x10068322: mov esi,[ebp+8]
0x10068325: mov ebx,[esi+1FCh]
0x1006832b: push 1
0x1006832d: mov eax,esi
0x1006832f: call 10067EA4h
0x10068334: test eax,eax
0x10068336: pop ecx
0x10068337: jne short 1006833Eh
0x10068339: or eax,0FFFFFFFFh
0x1006833c: jmp short 100683B6h
0x1006833e: mov eax,[esi+0B4h]
0x10068344: cmp [ebp+0Ch],eax
```

```
0x10068347: jb short 10068364h
0x10068349: push eax
0x1006834a: push dword ptr [ebp+0Ch]
0x1006834d: push 10108820h
0x10068352: push dword ptr [esi]
0x10068354: push dword ptr [esi+274h]
0x1006835a: call 1006FAF1h
0x1006835f: add esp,14h
0x10068362: jmp short 10068339h
0x10068364: push edi
0x10068365: mov edi,[ebp+14h]
0x10068368: cmp edi,0FFFFFFFFh
0x1006836b: je short 10068371h
0x1006836d: cmp edi,ebx
0x1006836f: jle short 10068373h
0x10068371: mov edi,ebx
0x10068373: push dword ptr [ebp+0Ch]
0x10068376: push esi
0x10068377: call 100680ECh
0x1006837c: test eax,eax
0x1006837e: pop ecx
0x1006837f: pop ecx
0x10068380: je short 100683B2h
0x10068382: mov eax,[ebp+0Ch]
0x10068385: xor edx,edx
0x10068387: div dword ptr [esi+0B0h]
0x1006838d: push eax
0x1006838e: push edi
0x1006838f: push dword ptr [ebp+10h]
0x10068392: push esi
0x10068393: call dword ptr [esi+22Ch]
0x10068399: add esp,10h
0x1006839c: test eax,eax
0x1006839e: je short 100683B2h
0x100683a0: push edi
0x100683a1: push dword ptr [ebp+10h]
0x100683a4: push esi
0x100683a5: call dword ptr [esi+28Ch]
0x100683ab: add esp,0Ch
0x100683ae: mov eax,edi
0x100683b0: jmp short 100683B5h
0x100683b2: or eax,0FFFFFFFFh
0x100683b5: pop edi
0x100683b6: pop esi
0x100683b7: pop ebx
0x100683b8: pop ebp
0x100683b9: ret
```

`sub_1006831d` is called by sub_100683ba. It delegates to sub_10067ea4, sub_100680ec, sub_1006faf1 (3 callees).

12.962 sub_100683ba at 0x100683ba (37 instructions)

```
0x100683ba: push ebp
0x100683bb: mov ebp,esp
0x100683bd: push esi
0x100683be: mov esi,[ebp+8]
0x100683c1: push 1
0x100683c3: mov eax,esi
0x100683c5: call 10067EA4h
0x100683ca: test eax,eax
0x100683cc: pop ecx
0x100683cd: je short 1006840Eh
0x100683cf: push dword ptr [ebp+1Ch]
0x100683d2: push dword ptr [ebp+18h]
0x100683d5: push dword ptr [ebp+14h]
0x100683d8: push dword ptr [ebp+10h]
```

```
0x100683db: push esi
0x100683dc: call 1006853Eh
0x100683e1: add esp,14h
0x100683e4: test eax,eax
0x100683e6: je short 1006840Eh
0x100683e8: push 0FFFFFFFFh
0x100683ea: push dword ptr [ebp+0Ch]
0x100683ed: push dword ptr [ebp+1Ch]
0x100683f0: push dword ptr [ebp+18h]
0x100683f3: push dword ptr [ebp+14h]
0x100683f6: push dword ptr [ebp+10h]
0x100683f9: push esi
0x100683fa: call 1006844Eh
0x100683ff: add esp,14h
0x10068402: push eax
0x10068403: push esi
0x10068404: call 1006831Dh
0x10068409: add esp,10h
0x1006840c: jmp short 10068411h
0x1006840e: or eax,0FFFFFFFFh
0x10068411: pop esi
0x10068412: pop ebp
0x10068413: ret
```

`sub_100683ba` is called by sub_100143f0. It delegates to sub_10067ea4, sub_1006831d, sub_1006844e (4 callees).

12.963 sub_10068414 at 0x10068414 (21 instructions)

```
0x10068414: push ebp
0x10068415: mov ebp,esp
0x10068417: mov ecx,[ebp+8]
0x1006841a: imul ecx,[ebp+0Ch]
0x1006841e: cmp dword ptr [ebp+0Ch],0
0x10068422: je short 1006844Ah
0x10068424: xor edx,edx
0x10068426: mov eax,ecx
0x10068428: div dword ptr [ebp+0Ch]
0x1006842b: cmp eax,[ebp+8]
0x1006842e: je short 1006844Ah
0x10068430: push dword ptr [ebp+10h]
0x10068433: push 10108158h
0x10068438: push dword ptr [esi]
0x1006843a: push dword ptr [esi+274h]
0x10068440: call 1006FAF1h
0x10068445: add esp,10h
0x10068448: xor ecx,ecx
0x1006844a: mov eax,ecx
0x1006844c: pop ebp
0x1006844d: ret
```

`sub_10068414` is called by sub_10065f66, sub_10065faf, sub_10066210, sub_100685bc, sub_10068686, sub_100686da. It works with the text "Integer overflow in %s". It delegates to sub_1006faf1 (1 callees).

12.964 sub_1006844e at 0x1006844e (94 instructions)

```
0x1006844e: push ebp
0x1006844f: mov ebp,esp
0x10068451: push ecx
0x10068452: push ecx
0x10068453: push ebx
0x10068454: push esi
0x10068455: mov esi,[ebp+8]
0x10068458: mov eax,[esi+4Ch]
0x1006845b: mov ebx,[esi+44h]
0x1006845e: xor ecx,ecx
```

```
0x10068460: mov [ebp+8],eax
0x10068463: mov eax,[esi+40h]
0x10068466: inc ecx
0x10068467: cmp eax,ecx
0x10068469: push edi
0x1006846a: mov edi,[esi+48h]
0x1006846d: jne short 10068473h
0x1006846f: and dword ptr [ebp+14h],0
0x10068473: cmp ebx,0FFFFFFFFh
0x10068476: jne short 1006847Bh
0x10068478: mov ebx,[esi+38h]
0x1006847b: cmp edi,0FFFFFFFFh
0x1006847e: jne short 10068483h
0x10068480: mov edi,[esi+3Ch]
0x10068483: cmp dword ptr [ebp+8],0FFFFFFFFh
0x10068487: jne short 1006848Ch
0x10068489: mov [ebp+8],eax
0x1006848c: test ebx,ebx
0x1006848e: je near ptr 10068537h
0x10068494: test edi,edi
0x10068496: je near ptr 10068537h
0x1006849c: cmp dword ptr [ebp+8],0
0x100684a0: je near ptr 10068537h
0x100684a6: mov eax,[esi+38h]
0x100684a9: mov ecx,ebx
0x100684ab: neg ecx
0x100684ad: cmp eax,ecx
0x100684af: jae short 100684BEh
0x100684b1: lea eax,[eax+ebx-1]
0x100684b5: xor edx,edx
0x100684b7: div ebx
0x100684b9: mov [ebp-8],eax
0x100684bc: jmp short 100684C2h
0x100684be: and dword ptr [ebp-8],0
0x100684c2: mov eax,[esi+3Ch]
0x100684c5: mov ecx,edi
0x100684c7: neg ecx
0x100684c9: cmp eax,ecx
0x100684cb: jae short 100684DAh
0x100684cd: lea eax,[eax+edi-1]
0x100684d1: xor edx,edx
0x100684d3: div edi
0x100684d5: mov [ebp-4],eax
0x100684d8: jmp short 100684DEh
0x100684da: and dword ptr [ebp-4],0
0x100684de: mov ecx,[ebp+8]
0x100684e1: mov eax,[esi+40h]
0x100684e4: mov edx,ecx
0x100684e6: neg edx
0x100684e8: cmp eax,edx
0x100684ea: jae short 100684F6h
0x100684ec: lea eax,[eax+ecx-1]
0x100684f0: xor edx,edx
0x100684f2: div ecx
0x100684f4: jmp short 100684F8h
0x100684f6: xor eax,eax
0x100684f8: xor edx,edx
0x100684fa: cmp word ptr [esi+8Ah],2
0x10068502: jne short 10068515h
0x10068504: movzx ecx,word ptr [ebp+18h]
0x10068508: imul ecx,eax
0x1006850b: mov eax,[ebp+14h]
0x1006850e: div dword ptr [ebp+8]
0x10068511: add ecx,eax
0x10068513: jmp short 1006851Dh
0x10068515: mov eax,[ebp+14h]
0x10068518: div dword ptr [ebp+8]
```

```
0x1006851b: mov ecx,eax
0x1006851d: mov eax,[ebp+10h]
0x10068520: imul ecx,[ebp-4]
0x10068524: xor edx,edx
0x10068526: div edi
0x10068528: xor edx,edx
0x1006852a: add ecx,eax
0x1006852c: mov eax,[ebp+0Ch]
0x1006852f: imul ecx,[ebp-8]
0x10068533: div ebx
0x10068535: add ecx,eax
0x10068537: pop edi
0x10068538: pop esi
0x10068539: mov eax,ecx
0x1006853b: pop ebx
0x1006853c: leave
0x1006853d: ret
```

`sub_1006844e` is called by sub_100683ba. It calls nothing else.

12.965 sub_1006853e at 0x1006853e (49 instructions)

```
0x1006853e: push ebp
0x1006853f: mov ebp,esp
0x10068541: mov eax,[ebp+8]
0x10068544: mov ecx,[eax+38h]
0x10068547: cmp [ebp+0Ch],ecx
0x1006854a: jb short 1006856Ah
0x1006854c: dec ecx
0x1006854d: push ecx
0x1006854e: push dword ptr [ebp+0Ch]
0x10068551: push 10108864h
0x10068556: push dword ptr [eax]
0x10068558: push dword ptr [eax+274h]
0x1006855e: call 1006FAF1h
0x10068563: add esp,14h
0x10068566: xor eax,eax
0x10068568: pop ebp
0x10068569: ret
0x1006856a: mov ecx,[eax+3Ch]
0x1006856d: cmp [ebp+10h],ecx
0x10068570: jb short 1006857Eh
0x10068572: dec ecx
0x10068573: push ecx
0x10068574: push dword ptr [ebp+10h]
0x10068577: push 101087DCh
0x1006857c: jmp short 10068556h
0x1006857e: mov ecx,[eax+40h]
0x10068581: cmp [ebp+14h],ecx
0x10068584: jb short 10068592h
0x10068586: dec ecx
0x10068587: push ecx
0x10068588: push dword ptr [ebp+14h]
0x1006858b: push 10108840h
0x10068590: jmp short 10068556h
0x10068592: cmp word ptr [eax+8Ah],2
0x1006859a: jne short 100685B7h
0x1006859c: movzx ecx,word ptr [eax+62h]
0x100685a0: cmp [ebp+18h],cx
0x100685a4: jb short 100685B7h
0x100685a6: movzx ecx,cx
0x100685a9: dec ecx
0x100685aa: push ecx
0x100685ab: movzx ecx,word ptr [ebp+18h]
0x100685af: push ecx
0x100685b0: push 10108170h
```

```
0x100685b5: jmp short 10068556h
0x100685b7: xor eax,eax
0x100685b9: inc eax
0x100685ba: pop ebp
0x100685bb: ret
```

`sub_1006853e` is called by sub_100683ba. It delegates to sub_1006faf1 (1 callees).

12.966 sub_100685bc at 0x100685bc (85 instructions)

```
0x100685bc: push ebp
0x100685bd: mov ebp,esp
0x100685bf: push ecx
0x100685c0: push ebx
0x100685c1: push esi
0x100685c2: mov esi,[ebp+8]
0x100685c5: mov eax,[esi+44h]
0x100685c8: cmp eax,0FFFFFFFFh
0x100685cb: mov ebx,[esi+48h]
0x100685ce: mov ecx,[esi+4Ch]
0x100685d1: push edi
0x100685d2: mov [ebp+8],eax
0x100685d5: jne short 100685DDh
0x100685d7: mov eax,[esi+38h]
0x100685da: mov [ebp+8],eax
0x100685dd: cmp ebx,0FFFFFFFFh
0x100685e0: jne short 100685E5h
0x100685e2: mov ebx,[esi+3Ch]
0x100685e5: cmp ecx,0FFFFFFFFh
0x100685e8: jne short 100685EDh
0x100685ea: mov ecx,[esi+40h]
0x100685ed: xor eax,eax
0x100685ef: cmp [ebp+8],eax
0x100685f2: mov edi,10108884h
0x100685f7: je short 10068668h
0x100685f9: cmp ebx,eax
0x100685fb: je short 10068668h
0x100685fd: cmp ecx,eax
0x100685ff: je short 10068668h
0x10068601: mov eax,[esi+40h]
0x10068604: mov edx,ecx
0x10068606: neg edx
0x10068608: cmp eax,edx
0x1006860a: jae short 10068619h
0x1006860c: lea eax,[eax+ecx-1]
0x10068610: xor edx,edx
0x10068612: div ecx
0x10068614: mov [ebp-4],eax
0x10068617: jmp short 1006861Dh
0x10068619: and dword ptr [ebp-4],0
0x1006861d: mov eax,[esi+3Ch]
0x10068620: mov ecx,ebx
0x10068622: neg ecx
0x10068624: cmp eax,ecx
0x10068626: jae short 10068634h
0x10068628: lea eax,[eax+ebx-1]
0x1006862c: xor edx,edx
0x1006862e: div ebx
0x10068630: mov ecx,eax
0x10068632: jmp short 10068636h
0x10068634: xor ecx,ecx
0x10068636: mov ebx,[ebp+8]
0x10068639: mov eax,[esi+38h]
0x1006863c: mov edx,ebx
0x1006863e: neg edx
0x10068640: cmp eax,edx
```

```
0x10068642: jae short 1006864Eh
0x10068644: lea eax,[eax+ebx-1]
0x10068648: xor edx,edx
0x1006864a: div ebx
0x1006864c: jmp short 10068650h
0x1006864e: xor eax,eax
0x10068650: push edi
0x10068651: push dword ptr [ebp-4]
0x10068654: push edi
0x10068655: push ecx
0x10068656: push eax
0x10068657: call 10068414h
0x1006865c: add esp,0Ch
0x1006865f: push eax
0x10068660: call 10068414h
0x10068665: add esp,0Ch
0x10068668: cmp word ptr [esi+8Ah],2
0x10068670: jne short 10068681h
0x10068672: movzx ecx,word ptr [esi+62h]
0x10068676: push edi
0x10068677: push ecx
0x10068678: push eax
0x10068679: call 10068414h
0x1006867e: add esp,0Ch
0x10068681: pop edi
0x10068682: pop esi
0x10068683: pop ebx
0x10068684: leave
0x10068685: ret
```

`sub_100685bc` is called by sub_1007171b. It works with the text "TIFFNumberOfTiles". It delegates to sub_10068414 (1 callees).

12.967 sub_10068686 at 0x10068686 (36 instructions)

```
0x10068686: push esi
0x10068687: mov esi,[esp+8]
0x1006868b: cmp dword ptr [esi+48h],0
0x1006868f: je short 100686D6h
0x10068691: mov eax,[esi+44h]
0x10068694: test eax,eax
0x10068696: je short 100686D6h
0x10068698: push edi
0x10068699: mov edi,10108898h
0x1006869e: push edi
0x1006869f: push eax
0x100686a0: movzx eax,word ptr [esi+54h]
0x100686a4: push eax
0x100686a5: call 10068414h
0x100686aa: add esp,0Ch
0x100686ad: cmp word ptr [esi+8Ah],1
0x100686b5: jne short 100686C6h
0x100686b7: movzx ecx,word ptr [esi+62h]
0x100686bb: push edi
0x100686bc: push ecx
0x100686bd: push eax
0x100686be: call 10068414h
0x100686c3: add esp,0Ch
0x100686c6: test al,7
0x100686c8: pop edi
0x100686c9: je short 100686D1h
0x100686cb: shr eax,3
0x100686ce: inc eax
0x100686cf: pop esi
0x100686d0: ret
0x100686d1: shr eax,3
```

```
0x100686d4: pop esi
0x100686d5: ret
0x100686d6: xor eax,eax
0x100686d8: pop esi
0x100686d9: ret
```

`sub_10068686` is called by sub_100143f0, sub_100686da. It works with the text "TIFFTileRowSize". It delegates to sub_10068414 (1 callees).

12.968 sub_100686da at 0x100686da (121 instructions)

```
0x100686da: push ebp
0x100686db: mov ebp,esp
0x100686dd: push ecx
0x100686de: push ebx
0x100686df: push esi
0x100686e0: mov esi,[ebp+8]
0x100686e3: cmp dword ptr [esi+48h],0
0x100686e7: push edi
0x100686e8: je near ptr 100687B7h
0x100686ee: mov eax,[esi+44h]
0x100686f1: test eax,eax
0x100686f3: je near ptr 100687B7h
0x100686f9: cmp dword ptr [esi+4Ch],0
0x100686fd: je near ptr 100687B7h
0x10068703: cmp word ptr [esi+8Ah],1
0x1006870b: jne near ptr 10068813h
0x10068711: cmp word ptr [esi+5Ah],6
0x10068716: jne near ptr 10068813h
0x1006871c: test word ptr [esi+0Ch],4000h
0x10068722: jne near ptr 10068813h
0x10068728: movzx ecx,word ptr [esi+0CCh]
0x1006872f: mov edx,ecx
0x10068731: neg edx
0x10068733: cmp eax,edx
0x10068735: jae short 10068748h
0x10068737: movzx edi,word ptr [esi+0CCh]
0x1006873e: lea eax,[edi+eax-1]
0x10068742: xor edx,edx
0x10068744: div edi
0x10068746: jmp short 1006874Ah
0x10068748: xor eax,eax
0x1006874a: imul ecx,eax
0x1006874d: movzx eax,word ptr [esi+54h]
0x10068751: mov edi,101088A8h
0x10068756: push edi
0x10068757: mov ebx,ecx
0x10068759: push eax
0x1006875a: push ebx
0x1006875b: call 10068414h
0x10068760: add esp,0Ch
0x10068763: test al,7
0x10068765: movzx eax,word ptr [esi+54h]
0x10068769: push edi
0x1006876a: push eax
0x1006876b: push ebx
0x1006876c: je short 1006877Ch
0x1006876e: call 10068414h
0x10068773: shr eax,3
0x10068776: add esp,0Ch
0x10068779: inc eax
0x1006877a: jmp short 10068787h
0x1006877c: call 10068414h
0x10068781: add esp,0Ch
0x10068784: shr eax,3
0x10068787: movzx ecx,word ptr [esi+0CEh]
```

```
0x1006878e: movzx ebx,word ptr [esi+0CCh]
0x10068795: imul ebx,ecx
0x10068798: test ebx,ebx
0x1006879a: mov [ebp+8],eax
0x1006879d: mov [ebp-4],ebx
0x100687a0: jne short 100687BEh
0x100687a2: push 101081B8h
0x100687a7: push dword ptr [esi]
0x100687a9: push dword ptr [esi+274h]
0x100687af: call 1006FAF1h
0x100687b4: add esp,0Ch
0x100687b7: xor eax,eax
0x100687b9: pop edi
0x100687ba: pop esi
0x100687bb: pop ebx
0x100687bc: leave
0x100687bd: ret
0x100687be: mov edx,ecx
0x100687c0: neg edx
0x100687c2: cmp [ebp+0Ch],edx
0x100687c5: jae short 100687E3h
0x100687c7: movzx ebx,word ptr [esi+0CEh]
0x100687ce: mov eax,[ebp+0Ch]
0x100687d1: lea eax,[ebx+eax-1]
0x100687d5: xor edx,edx
0x100687d7: div ebx
0x100687d9: mov ebx,[ebp-4]
0x100687dc: mov edx,eax
0x100687de: mov eax,[ebp+8]
0x100687e1: jmp short 100687E5h
0x100687e3: xor edx,edx
0x100687e5: imul ecx,edx
0x100687e8: push edi
0x100687e9: push eax
0x100687ea: push ecx
0x100687eb: call 10068414h
0x100687f0: mov [ebp+8],eax
0x100687f3: cdq
0x100687f4: idiv ebx
0x100687f6: add esp,0Ch
0x100687f9: push edi
0x100687fa: push edi
0x100687fb: push eax
0x100687fc: push 2
0x100687fe: call 10068414h
0x10068803: mov edx,[ebp+8]
0x10068806: add esp,0Ch
0x10068809: mov ecx,eax
0x1006880b: call 10065EEAh
0x10068810: pop ecx
0x10068811: jmp short 1006882Ch
0x10068813: mov edi,101088A8h
0x10068818: push edi
0x10068819: push esi
0x1006881a: call 10068686h
0x1006881f: pop ecx
0x10068820: push eax
0x10068821: push dword ptr [ebp+0Ch]
0x10068824: call 10068414h
0x10068829: add esp,0Ch
0x1006882c: push edi
0x1006882d: push dword ptr [esi+4Ch]
0x10068830: push eax
0x10068831: call 10068414h
0x10068836: add esp,0Ch
0x10068839: jmp near ptr 100687B9h
```

`sub_100686da` is called by sub_1006883e, sub_100715da. It delegates to sub_10065eea, sub_10068414, sub_10068686 (4 callees).

12.969 sub_1006883e at 0x1006883e (7 instructions)

```
0x1006883e: mov eax,[esp+4]
0x10068842: push dword ptr [eax+48h]
0x10068845: push eax
0x10068846: call 100686DAh
0x1006884b: pop ecx
0x1006884c: pop ecx
0x1006884d: ret
```

`sub_1006883e` is called by sub_100143f0, sub_100702f4, sub_1007171b. It delegates to sub_100686da (1 callees).

12.970 sub_100688a1 at 0x100688a1 (8 instructions)

```
0x100688a1: cmp dword ptr [esp+4],4D4Dh
0x100688a9: mov dword ptr [eax+1D4h],101088B8h
0x100688b3: jne short 100688C7h
0x100688b5: mov dword ptr [eax+1D0h],101088ECh
0x100688bf: or dword ptr [eax+0Ch],80h
0x100688c6: ret
0x100688c7: mov dword ptr [eax+1D0h],10108920h
0x100688d1: ret
```

`sub_100688a1` is called by sub_1006891d. It calls nothing else.

12.971 sub_100688d2 at 0x100688d2 (30 instructions)

```
0x100688d2: mov ecx,[esp+4]
0x100688d6: mov al,[ecx]
0x100688d8: push esi
0x100688d9: or esi,0FFFFFFFFh
0x100688dc: cmp al,61h
0x100688de: je short 1006890Bh
0x100688e0: cmp al,72h
0x100688e2: je short 100688FEh
0x100688e4: cmp al,77h
0x100688e6: je short 1006890Bh
0x100688e8: push ecx
0x100688e9: push 10108964h
0x100688ee: push dword ptr [esp+14h]
0x100688f2: push 0
0x100688f4: call 1006FAF1h
0x100688f9: add esp,10h
0x100688fc: jmp short 10068919h
0x100688fe: xor esi,esi
0x10068900: cmp byte ptr [ecx+1],2Bh
0x10068904: jne short 10068919h
0x10068906: push 2
0x10068908: pop esi
0x10068909: jmp short 10068919h
0x1006890b: cmp al,77h
0x1006890d: mov esi,102h
0x10068912: jne short 10068919h
0x10068914: mov esi,302h
0x10068919: mov eax,esi
0x1006891b: pop esi
0x1006891c: ret
```

`sub_100688d2` is called by sub_1006891d. It delegates to sub_1006faf1 (1 callees).

12.972 sub_1006891d at 0x1006891d (321 instructions)

```
0x1006891d: push ebp
0x1006891e: mov ebp,esp
0x10068920: push ecx
0x10068921: push ebx
0x10068922: push esi
0x10068923: push edi
0x10068924: mov edi,10108954h
0x10068929: push edi
0x1006892a: push dword ptr [ebp+0Ch]
0x1006892d: call 100688D2h
0x10068932: cmp eax,0FFFFFFFFh
0x10068935: pop ecx
0x10068936: pop ecx
0x10068937: mov [ebp-4],eax
0x1006893a: je near ptr 10068D4Bh
0x10068940: push dword ptr [ebp+8]
0x10068943: call 100FC10Eh
0x10068948: add eax,2B1h
0x1006894d: push eax
0x1006894e: call 100EFDD0h
0x10068953: mov esi,eax
0x10068955: test esi,esi
0x10068957: pop ecx
0x10068958: pop ecx
0x10068959: jne short 10068974h
0x1006895b: push dword ptr [ebp+8]
0x1006895e: push 10108A84h
0x10068963: push edi
0x10068964: push dword ptr [ebp+10h]
0x10068967: call 1006FAF1h
0x1006896c: add esp,10h
0x1006896f: jmp near ptr 10068D4Bh
0x10068974: push 2B0h
0x10068979: push 0
0x1006897b: push esi
0x1006897c: call 100209F0h
0x10068981: push dword ptr [ebp+8]
0x10068984: lea eax,[esi+2B0h]
0x1006898a: push eax
0x1006898b: mov [esi],eax
0x1006898d: call 100FC114h
0x10068992: mov eax,[ebp-4]
0x10068995: mov edx,[ebp+14h]
0x10068998: or word ptr [esi+1DCh],0FFFFh
0x100689a1: and dword ptr [esi+1E4h],0
0x100689a8: or dword ptr [esi+1E0h],0FFFFFFFFh
0x100689af: or dword ptr [esi+1D8h],0FFFFFFFFh
0x100689b6: and eax,0FFFFFCFFh
0x100689bb: mov [esi+8],eax
0x100689be: mov eax,[ebp+10h]
0x100689c1: add esp,14h
0x100689c4: test edx,edx
0x100689c6: mov [esi+274h],eax
0x100689cc: je near ptr 10068D3Ah
0x100689d2: mov ebx,[ebp+18h]
0x100689d5: test ebx,ebx
0x100689d7: je near ptr 10068D3Ah
0x100689dd: mov edi,[ebp+1Ch]
0x100689e0: test edi,edi
0x100689e2: je near ptr 10068D35h
0x100689e8: mov ecx,[ebp+20h]
0x100689eb: test ecx,ecx
0x100689ed: je near ptr 10068D35h
0x100689f3: mov eax,[ebp+24h]
0x100689f6: test eax,eax
0x100689f8: je near ptr 10068D35h
```

```
0x100689fe: mov [esi+288h],eax
0x10068a04: mov eax,[ebp+28h]
0x10068a07: test eax,eax
0x10068a09: mov [esi+278h],edx
0x10068a0f: mov [esi+27Ch],ebx
0x10068a15: mov [esi+280h],edi
0x10068a1b: mov [esi+284h],ecx
0x10068a21: je short 10068A2Bh
0x10068a23: mov [esi+26Ch],eax
0x10068a29: jmp short 10068A35h
0x10068a2b: mov dword ptr [esi+26Ch],1006889Eh
0x10068a35: mov eax,[ebp+2Ch]
0x10068a38: test eax,eax
0x10068a3a: je short 10068A44h
0x10068a3c: mov [esi+270h],eax
0x10068a42: jmp short 10068A4Eh
0x10068a44: mov dword ptr [esi+270h],1007B5B1h
0x10068a4e: push esi
0x10068a4f: call 100701F4h
0x10068a54: mov edx,[ebp-4]
0x10068a57: xor edi,edi
0x10068a59: inc edi
0x10068a5a: test edx,edx
0x10068a5c: pop ecx
0x10068a5d: mov [esi+0Ch],edi
0x10068a60: mov ebx,8000h
0x10068a65: jne short 10068A70h
0x10068a67: mov dword ptr [esi+0Ch],801h
0x10068a6e: jmp short 10068A75h
0x10068a70: cmp edx,2
0x10068a73: jne short 10068A78h
0x10068a75: or [esi+0Ch],ebx
0x10068a78: mov eax,[ebp+0Ch]
0x10068a7b: cmp byte ptr [eax],0
0x10068a7e: mov ecx,eax
0x10068a80: je near ptr 10068B1Dh
0x10068a86: movsx eax,byte ptr [ecx]
0x10068a89: cmp eax,4Dh
0x10068a8c: jg short 10068AD0h
0x10068a8e: je short 10068AC3h
0x10068a90: sub eax,42h
0x10068a93: je short 10068AB6h
0x10068a95: dec eax
0x10068a96: je short 10068AADh
0x10068a98: sub eax,5
0x10068a9b: je short 10068AA2h
0x10068a9d: sub eax,4
0x10068aa0: jne short 10068B13h
0x10068aa2: mov eax,[esi+0Ch]
0x10068aa5: and eax,0FFFFFFFEh
0x10068aa8: or eax,2
0x10068aab: jmp short 10068ABEh
0x10068aad: test edx,edx
0x10068aaf: jne short 10068B13h
0x10068ab1: or [esi+0Ch],ebx
0x10068ab4: jmp short 10068B13h
0x10068ab6: mov eax,[esi+0Ch]
0x10068ab9: and eax,0FFFFFFFDh
0x10068abc: or eax,edi
0x10068abe: mov [esi+0Ch],eax
0x10068ac1: jmp short 10068B13h
0x10068ac3: test edx,edx
0x10068ac5: jne short 10068B13h
0x10068ac7: or dword ptr [esi+0Ch],800h
0x10068ace: jmp short 10068B13h
0x10068ad0: sub eax,62h
0x10068ad3: je short 10068B05h
```

```
0x10068ad5: dec eax
0x10068ad6: je short 10068AF8h
0x10068ad8: sub eax,5
0x10068adb: je short 10068AEFh
0x10068add: sub eax,5
0x10068ae0: jne short 10068B13h
0x10068ae2: test edx,edx
0x10068ae4: jne short 10068B13h
0x10068ae6: and dword ptr [esi+0Ch],0FFFFF7FFh
0x10068aed: jmp short 10068B13h
0x10068aef: or dword ptr [esi+0Ch],10000h
0x10068af6: jmp short 10068B13h
0x10068af8: test edx,edx
0x10068afa: jne short 10068B13h
0x10068afc: and dword ptr [esi+0Ch],0FFFF7FFFh
0x10068b03: jmp short 10068B13h
0x10068b05: test dx,100h
0x10068b0a: je short 10068B13h
0x10068b0c: or dword ptr [esi+0Ch],80h
0x10068b13: inc ecx
0x10068b14: cmp byte ptr [ecx],0
0x10068b17: jne near ptr 10068A86h
0x10068b1d: mov edi,200h
0x10068b22: test [esi+8],edi
0x10068b25: jne near ptr 10068C74h
0x10068b2b: push 8
0x10068b2d: lea ebx,[esi+1C8h]
0x10068b33: push ebx
0x10068b34: push dword ptr [esi+274h]
0x10068b3a: call dword ptr [esi+278h]
0x10068b40: add esp,0Ch
0x10068b43: cmp eax,8
0x10068b46: jne near ptr 10068C74h
0x10068b4c: movzx eax,word ptr [ebx]
0x10068b4f: cmp ax,4D4Dh
0x10068b53: je short 10068B7Bh
0x10068b55: cmp ax,4949h
0x10068b59: je short 10068B7Bh
0x10068b5b: movzx eax,ax
0x10068b5e: push eax
0x10068b5f: push eax
0x10068b60: push 10108A58h
0x10068b65: push dword ptr [ebp+8]
0x10068b68: push dword ptr [esi+274h]
0x10068b6e: call 1006FAF1h
0x10068b73: add esp,14h
0x10068b76: jmp near ptr 10068D17h
0x10068b7b: movzx eax,ax
0x10068b7e: push eax
0x10068b7f: mov eax,esi
0x10068b81: call 100688A1h
0x10068b86: add esp,4
0x10068b89: test byte ptr [esi+0Ch],80h
0x10068b8d: je short 10068BA9h
0x10068b8f: lea eax,[esi+1CAh]
0x10068b95: push eax
0x10068b96: call 1006FB29h
0x10068b9b: lea eax,[esi+1CCh]
0x10068ba1: push eax
0x10068ba2: call 1006FB38h
0x10068ba7: pop ecx
0x10068ba8: pop ecx
0x10068ba9: movzx eax,word ptr [esi+1CAh]
0x10068bb0: cmp ax,2Bh
0x10068bb4: jne short 10068BC0h
0x10068bb6: push 10108A08h
0x10068bbb: jmp near ptr 10068C80h
```

```
0x10068bc0: cmp ax,2Ah
0x10068bc4: je short 10068BD2h
0x10068bc6: movzx eax,ax
0x10068bc9: push eax
0x10068bca: push eax
0x10068bcb: push 101089D8h
0x10068bd0: jmp short 10068B65h
0x10068bd2: or [esi+0Ch],edi
0x10068bd5: mov eax,[esi+0Ch]
0x10068bd8: xor ecx,ecx
0x10068bda: test eax,10000h
0x10068bdf: mov [esi+254h],ecx
0x10068be5: mov [esi+25Ch],ecx
0x10068beb: mov [esi+258h],ecx
0x10068bf1: jne near ptr 10068D31h
0x10068bf7: mov ecx,[ebp+0Ch]
0x10068bfa: mov cl,[ecx]
0x10068bfc: cmp cl,61h
0x10068bff: je short 10068C60h
0x10068c01: cmp cl,72h
0x10068c04: jne near ptr 10068D17h
0x10068c0a: test ax,800h
0x10068c0e: mov ecx,[esi+1CCh]
0x10068c14: mov [esi+14h],ecx
0x10068c17: je short 10068C41h
0x10068c19: lea eax,[esi+268h]
0x10068c1f: push eax
0x10068c20: lea eax,[esi+264h]
0x10068c26: push eax
0x10068c27: push dword ptr [esi+274h]
0x10068c2d: call dword ptr [esi+26Ch]
0x10068c33: add esp,0Ch
0x10068c36: test eax,eax
0x10068c38: jne short 10068C41h
0x10068c3a: and dword ptr [esi+0Ch],0FFFFF7FFh
0x10068c41: push esi
0x10068c42: call 1007171Bh
0x10068c47: test eax,eax
0x10068c49: pop ecx
0x10068c4a: je near ptr 10068D17h
0x10068c50: or dword ptr [esi+260h],0FFFFFFFFh
0x10068c57: or dword ptr [esi+0Ch],10h
0x10068c5b: jmp near ptr 10068D31h
0x10068c60: push esi
0x10068c61: call 1006799Bh
0x10068c66: test eax,eax
0x10068c68: pop ecx
0x10068c69: je near ptr 10068D17h
0x10068c6f: jmp near ptr 10068D31h
0x10068c74: xor ebx,ebx
0x10068c76: cmp [esi+8],ebx
0x10068c79: jne short 10068C96h
0x10068c7b: push 101089C0h
0x10068c80: push dword ptr [ebp+8]
0x10068c83: push dword ptr [esi+274h]
0x10068c89: call 1006FAF1h
0x10068c8e: add esp,0Ch
0x10068c91: jmp near ptr 10068D17h
0x10068c96: mov ecx,[esi+0Ch]
0x10068c99: and ecx,80h
0x10068c9f: mov eax,ecx
0x10068ca1: neg eax
0x10068ca3: sbb eax,eax
0x10068ca5: and eax,404h
0x10068caa: add eax,4949h
0x10068caf: cmp ecx,ebx
0x10068cb1: lea edi,[esi+1C8h]
```

```
0x10068cb7: mov [edi],ax
0x10068cba: lea eax,[esi+1CAh]
0x10068cc0: mov word ptr [eax],2Ah
0x10068cc5: je short 10068CCEh
0x10068cc7: push eax
0x10068cc8: call 1006FB29h
0x10068ccd: pop ecx
0x10068cce: push ebx
0x10068ccf: push ebx
0x10068cd0: push dword ptr [esi+274h]
0x10068cd6: mov [esi+1CCh],ebx
0x10068cdc: call dword ptr [esi+280h]
0x10068ce2: push 8
0x10068ce4: push edi
0x10068ce5: push dword ptr [esi+274h]
0x10068ceb: call dword ptr [esi+27Ch]
0x10068cf1: add esp,18h
0x10068cf4: cmp eax,8
0x10068cf7: je short 10068D00h
0x10068cf9: push 101089A4h
0x10068cfe: jmp short 10068C80h
0x10068d00: movzx eax,word ptr [edi]
0x10068d03: push eax
0x10068d04: mov eax,esi
0x10068d06: call 100688A1h
0x10068d0b: push esi
0x10068d0c: call 1006799Bh
0x10068d11: test eax,eax
0x10068d13: pop ecx
0x10068d14: pop ecx
0x10068d15: jne short 10068D24h
0x10068d17: and dword ptr [esi+8],0
0x10068d1b: push esi
0x10068d1c: call 10065C7Bh
0x10068d21: pop ecx
0x10068d22: jmp short 10068D4Bh
0x10068d24: mov [esi+10h],ebx
0x10068d27: mov [esi+18h],ebx
0x10068d2a: mov [esi+1Ch],ebx
0x10068d2d: mov [esi+20h],bx
0x10068d31: mov eax,esi
0x10068d33: jmp short 10068D4Dh
0x10068d35: mov edi,10108954h
0x10068d3a: push 10108974h
0x10068d3f: push edi
0x10068d40: push dword ptr [ebp+10h]
0x10068d43: call 1006FAF1h
0x10068d48: add esp,0Ch
0x10068d4b: xor eax,eax
0x10068d4d: pop edi
0x10068d4e: pop esi
0x10068d4f: pop ebx
0x10068d50: leave
0x10068d51: ret
```

`sub_1006891d` is called by sub_10020970. It works with the text "One of the client procedures is NULL pointer.", "Error writing TIFF header", "Cannot read TIFF header". It delegates to j_memset, sub_10065c7b, sub_1006799b (13 callees).

12.973 sub_10068d52 at 0x10068d52 (3 instructions)

```
0x10068d52: mov eax,[esp+4]
0x10068d56: mov eax,[eax]
0x10068d58: ret
```

`sub_10068d52` is called by sub_10069057, sub_1006cff7, sub_1006d21a, sub_1006d370, sub_1006d6a9, sub_1006df2c. It calls nothing else.

12.974 sub_10068d59 at 0x10068d59 (5 instructions)

```
0x10068d59: mov eax,[esp+4]
0x10068d5d: mov eax,[eax+0Ch]
0x10068d60: shr eax,0Ah
0x10068d63: and eax,1
0x10068d66: ret
```

`sub_10068d59` is called by sub_100143f0, sub_1006d727, sub_1006d9b2. It calls nothing else.

12.975 sub_10068d67 at 0x10068d67 (169 instructions)

```
0x10068d67: push ebp
0x10068d68: mov ebp,esp
0x10068d6a: push ecx
0x10068d6b: push esi
0x10068d6c: mov esi,[ebp+8]
0x10068d6f: cmp dword ptr [esi+200h],0
0x10068d76: push edi
0x10068d77: jne short 10068D90h
0x10068d79: push 10108CB0h
0x10068d7e: push dword ptr [ebp+0Ch]
0x10068d81: call dword ptr ds:[100FE4F8h]
0x10068d87: pop ecx
0x10068d88: pop ecx
0x10068d89: xor eax,eax
0x10068d8b: jmp near ptr 10068F96h
0x10068d90: movzx eax,word ptr [esi+54h]
0x10068d94: test eax,eax
0x10068d96: push ebx
0x10068d97: jle near ptr 10068F81h
0x10068d9d: push 2
0x10068d9f: pop ebx
0x10068da0: cmp eax,ebx
0x10068da2: jle short 10068DB7h
0x10068da4: cmp eax,4
0x10068da7: je short 10068DB7h
0x10068da9: cmp eax,8
0x10068dac: je short 10068DB7h
0x10068dae: cmp eax,10h
0x10068db1: jne near ptr 10068F81h
0x10068db7: movzx eax,word ptr [esi+0A8h]
0x10068dbe: movzx edi,word ptr [esi+62h]
0x10068dc2: sub edi,eax
0x10068dc4: lea eax,[ebp-4]
0x10068dc7: push eax
0x10068dc8: push 106h
0x10068dcd: push esi
0x10068dce: call 10067985h
0x10068dd3: add esp,0Ch
0x10068dd6: test eax,eax
0x10068dd8: jne short 10068DFEh
0x10068dda: mov eax,edi
0x10068ddc: dec eax
0x10068ddd: je short 10068DF7h
0x10068ddf: sub eax,ebx
0x10068de1: je short 10068DF2h
0x10068de3: push 10108AA8h
0x10068de8: push 10108C98h
0x10068ded: jmp near ptr 10068F87h
0x10068df2: mov [ebp-4],ebx
0x10068df5: jmp short 10068DFEh
0x10068df7: mov dword ptr [ebp-4],1
```

```
0x10068dfe: movzx eax,word ptr [ebp-4]
0x10068e02: cmp eax,6
0x10068e05: jg near ptr 10068EEBh
0x10068e0b: je near ptr 10068F7Ch
0x10068e11: test eax,eax
0x10068e13: jl near ptr 10068F01h
0x10068e19: cmp eax,1
0x10068e1c: jle short 10068E98h
0x10068e1e: cmp eax,ebx
0x10068e20: je short 10068E82h
0x10068e22: cmp eax,3
0x10068e25: je short 10068E98h
0x10068e27: cmp eax,5
0x10068e2a: jne near ptr 10068F01h
0x10068e30: lea eax,[ebp+8]
0x10068e33: push eax
0x10068e34: push 14Ch
0x10068e39: push esi
0x10068e3a: call 1006E579h
0x10068e3f: add esp,0Ch
0x10068e42: cmp word ptr [ebp+8],1
0x10068e47: je short 10068E69h
0x10068e49: movzx eax,word ptr [ebp+8]
0x10068e4d: push eax
0x10068e4e: push 10108C90h
0x10068e53: push 10108C5Ch
0x10068e58: push dword ptr [ebp+0Ch]
0x10068e5b: call dword ptr ds:[100FE4F8h]
0x10068e61: add esp,10h
0x10068e64: jmp near ptr 10068F93h
0x10068e69: movzx eax,word ptr [esi+62h]
0x10068e6d: cmp ax,4
0x10068e71: jae near ptr 10068F7Ch
0x10068e77: movzx eax,ax
0x10068e7a: push eax
0x10068e7b: push 10108C4Ch
0x10068e80: jmp short 10068E53h
0x10068e82: cmp edi,3
0x10068e85: jge near ptr 10068F7Ch
0x10068e8b: push edi
0x10068e8c: push 10108C3Ch
0x10068e91: push 10108C10h
0x10068e96: jmp short 10068E58h
0x10068e98: cmp word ptr [esi+8Ah],1
0x10068ea0: jne near ptr 10068F7Ch
0x10068ea6: movzx edx,word ptr [esi+62h]
0x10068eaa: cmp dx,1
0x10068eae: je near ptr 10068F7Ch
0x10068eb4: movzx ecx,word ptr [esi+54h]
0x10068eb8: cmp cx,8
0x10068ebc: jae near ptr 10068F7Ch
0x10068ec2: movzx ecx,cx
0x10068ec5: push ecx
0x10068ec6: movzx ecx,dx
0x10068ec9: push ecx
0x10068eca: push 10108C4Ch
0x10068ecf: push eax
0x10068ed0: push 10108AA8h
0x10068ed5: push 10108BC0h
0x10068eda: push dword ptr [ebp+0Ch]
0x10068edd: call dword ptr ds:[100FE4F8h]
0x10068ee3: add esp,1Ch
0x10068ee6: jmp near ptr 10068F93h
0x10068eeb: mov ecx,eax
0x10068eed: sub ecx,8
0x10068ef0: je near ptr 10068F7Ch
0x10068ef6: sub ecx,8044h
```

```
0x10068efc: je short 10068F61h
0x10068efe: dec ecx
0x10068eff: je short 10068F11h
0x10068f01: push eax
0x10068f02: push 10108AA8h
0x10068f07: push 10108B98h
0x10068f0c: jmp near ptr 10068E58h
0x10068f11: movzx ecx,word ptr [esi+58h]
0x10068f15: mov eax,8774h
0x10068f1a: cmp cx,ax
0x10068f1d: je short 10068F41h
0x10068f1f: lea edx,[eax+1]
0x10068f22: cmp cx,dx
0x10068f25: je short 10068F41h
0x10068f27: push edx
0x10068f28: push eax
0x10068f29: push 10108B8Ch
0x10068f2e: push 10108B60h
0x10068f33: push dword ptr [ebp+0Ch]
0x10068f36: call dword ptr ds:[100FE4F8h]
0x10068f3c: add esp,14h
0x10068f3f: jmp short 10068F93h
0x10068f41: movzx eax,word ptr [esi+8Ah]
0x10068f48: cmp ax,1
0x10068f4c: je short 10068F7Ch
0x10068f4e: movzx eax,ax
0x10068f51: push eax
0x10068f52: push 10108B4Ch
0x10068f57: push 10108B1Ch
0x10068f5c: jmp near ptr 10068E58h
0x10068f61: mov eax,8774h
0x10068f66: cmp [esi+58h],ax
0x10068f6a: je short 10068F7Ch
0x10068f6c: push eax
0x10068f6d: push 10108B8Ch
0x10068f72: push 10108AF8h
0x10068f77: jmp near ptr 10068E58h
0x10068f7c: xor eax,eax
0x10068f7e: inc eax
0x10068f7f: jmp short 10068F95h
0x10068f81: push eax
0x10068f82: push 10108AC4h
0x10068f87: push dword ptr [ebp+0Ch]
0x10068f8a: call dword ptr ds:[100FE4F8h]
0x10068f90: add esp,0Ch
0x10068f93: xor eax,eax
0x10068f95: pop ebx
0x10068f96: pop edi
0x10068f97: pop esi
0x10068f98: leave
0x10068f99: ret
```

`sub_10068d67` is called by sub_1006df2c. It works with the text "PhotometricInterpretation", "Sorry, can not handle images with %d-bit samples", "Sorry, LogL data must have %s=%d". It delegates to sub_10067985, sub_1006e579 (2 callees).

12.976 sub_10068f9a at 0x10068f9a (52 instructions)

```
0x10068f9a: push esi
0x10068f9b: mov esi,[esp+8]
0x10068f9f: mov eax,[esi+38h]
0x10068fa2: push edi
0x10068fa3: xor edi,edi
0x10068fa5: cmp eax,edi
0x10068fa7: je short 10068FB3h
0x10068fa9: push eax
```

```
0x10068faa: call 100209E0h
0x10068faf: pop ecx
0x10068fb0: mov [esi+38h],edi
0x10068fb3: mov eax,[esi+3Ch]
0x10068fb6: cmp eax,edi
0x10068fb8: je short 10068FC4h
0x10068fba: push eax
0x10068fbb: call 100209E0h
0x10068fc0: pop ecx
0x10068fc1: mov [esi+3Ch],edi
0x10068fc4: mov eax,[esi+40h]
0x10068fc7: cmp eax,edi
0x10068fc9: je short 10068FD5h
0x10068fcb: push eax
0x10068fcc: call 100209E0h
0x10068fd1: pop ecx
0x10068fd2: mov [esi+40h],edi
0x10068fd5: mov eax,[esi+44h]
0x10068fd8: cmp eax,edi
0x10068fda: je short 10068FE6h
0x10068fdc: push eax
0x10068fdd: call 100209E0h
0x10068fe2: pop ecx
0x10068fe3: mov [esi+44h],edi
0x10068fe6: mov eax,[esi+48h]
0x10068fe9: cmp eax,edi
0x10068feb: je short 10068FF7h
0x10068fed: push eax
0x10068fee: call 100209E0h
0x10068ff3: pop ecx
0x10068ff4: mov [esi+48h],edi
0x10068ff7: mov eax,[esi+24h]
0x10068ffa: cmp eax,edi
0x10068ffc: je short 10069017h
0x10068ffe: push eax
0x10068fff: call 100209E0h
0x10069004: push dword ptr [esi+28h]
0x10069007: call 100209E0h
0x1006900c: push dword ptr [esi+2Ch]
0x1006900f: call 100209E0h
0x10069014: add esp,0Ch
0x10069017: pop edi
0x10069018: pop esi
0x10069019: ret
```

`sub_10068f9a` is called by sub_1006df2c. It delegates to j_free (1 callees).

12.977 sub_1006901a at 0x1006901a (25 instructions)

```
0x1006901a: push ebp
0x1006901b: mov ebp,esp
0x1006901d: push ecx
0x1006901e: lea eax,[ebp-4]
0x10069021: push eax
0x10069022: push 103h
0x10069027: push dword ptr [ebp+8]
0x1006902a: call 10067985h
0x1006902f: mov ax,[ebp-4]
0x10069033: add esp,0Ch
0x10069036: cmp ax,3
0x1006903a: je short 10069052h
0x1006903c: cmp ax,4
0x10069040: je short 10069052h
0x10069042: cmp ax,2
0x10069046: je short 10069052h
0x10069048: cmp ax,8003h
```

```
0x1006904c: je short 10069052h
0x1006904e: xor eax,eax
0x10069050: leave
0x10069051: ret
0x10069052: xor eax,eax
0x10069054: inc eax
0x10069055: leave
0x10069056: ret
```

`sub_1006901a` is called by sub_1006daaf. It delegates to sub_10067985 (1 callees).

12.978 sub_10069057 at 0x10069057 (29 instructions)

```
0x10069057: push esi
0x10069058: mov esi,[esp+8]
0x1006905c: mov eax,[esi+30h]
0x1006905f: test eax,eax
0x10069061: jne short 10069085h
0x10069063: push 10108D28h
0x10069068: push dword ptr [esi]
0x1006906a: call 10068D52h
0x1006906f: pop ecx
0x10069070: push eax
0x10069071: mov eax,[esi]
0x10069073: push dword ptr [eax+274h]
0x10069079: call 1006FAF1h
0x1006907e: add esp,0Ch
0x10069081: xor eax,eax
0x10069083: pop esi
0x10069084: ret
0x10069085: cmp dword ptr [esi+34h],0
0x10069089: jne short 10069092h
0x1006908b: push 10108CE8h
0x10069090: jmp short 10069068h
0x10069092: push dword ptr [esp+14h]
0x10069096: push dword ptr [esp+14h]
0x1006909a: push dword ptr [esp+14h]
0x1006909e: push esi
0x1006909f: call eax
0x100690a1: add esp,10h
0x100690a4: pop esi
0x100690a5: ret
```

`sub_10069057` is called by sub_1006df2c. It delegates to sub_10068d52, sub_1006faf1 (2 callees).

12.979 sub_1006cec2 at 0x1006cec2 (42 instructions)

```
0x1006cec2: push ebp
0x1006cec3: mov ebp,esp
0x1006cec5: push ecx
0x1006cec6: push ecx
0x1006cec7: cmp dword ptr [esi+44h],0
0x1006cecb: jne short 1006CEFDh
0x1006cecd: push 1818h
0x1006ced2: call 100EFDD0h
0x1006ced7: test eax,eax
0x1006ced9: pop ecx
0x1006ceda: mov [esi+44h],eax
0x1006cedd: jne short 1006CEFDh
0x1006cedf: mov eax,[esi]
0x1006cee1: push 10108D78h
0x1006cee6: push 101362CCh
0x1006ceeb: push dword ptr [eax+274h]
0x1006cef1: call 1006FAF1h
0x1006cef6: add esp,0Ch
```

```
0x1006cef9: xor eax,eax
0x1006cefb: leave
0x1006cefc: ret
0x1006cefd: lea eax,[ebp-8]
0x1006cf00: push eax
0x1006cf01: push 211h
0x1006cf06: push dword ptr [esi]
0x1006cf08: call 1006E579h
0x1006cf0d: lea eax,[ebp-4]
0x1006cf10: push eax
0x1006cf11: push 214h
0x1006cf16: push dword ptr [esi]
0x1006cf18: call 1006E579h
0x1006cf1d: push dword ptr [ebp-4]
0x1006cf20: push dword ptr [ebp-8]
0x1006cf23: push dword ptr [esi+44h]
0x1006cf26: call 100729B6h
0x1006cf2b: xor ecx,ecx
0x1006cf2d: add esp,24h
0x1006cf30: test eax,eax
0x1006cf32: setge cl
0x1006cf35: mov eax,ecx
0x1006cf37: leave
0x1006cf38: ret
```

`sub_1006cec2` is called by sub_1006d727, sub_1006d9b2. It delegates to sub_1006e579, sub_1006faf1, sub_100729b6 (4 callees).

12.980 sub_1006cf39 at 0x1006cf39 (61 instructions)

```
0x1006cf39: push ebp
0x1006cf3a: mov ebp,esp
0x1006cf3c: sub esp,10h
0x1006cf3f: cmp dword ptr [esi+48h],0
0x1006cf43: jne short 1006CF75h
0x1006cf45: push 46CCh
0x1006cf4a: call 100EFDD0h
0x1006cf4f: test eax,eax
0x1006cf51: pop ecx
0x1006cf52: mov [esi+48h],eax
0x1006cf55: jne short 1006CF75h
0x1006cf57: mov eax,[esi]
0x1006cf59: push 10108DE0h
0x1006cf5e: push 101362E0h
0x1006cf63: push dword ptr [eax+274h]
0x1006cf69: call 1006FAF1h
0x1006cf6e: add esp,0Ch
0x1006cf71: xor eax,eax
0x1006cf73: leave
0x1006cf74: ret
0x1006cf75: lea eax,[ebp-4]
0x1006cf78: push eax
0x1006cf79: push 13Eh
0x1006cf7e: push dword ptr [esi]
0x1006cf80: call 1006E579h
0x1006cf85: fld dword ptr ds:[10108DDCh]
0x1006cf8b: mov eax,[ebp-4]
0x1006cf8e: fstp dword ptr [ebp-0Ch]
0x1006cf91: fld dword ptr [eax]
0x1006cf93: fdiv dword ptr [eax+4]
0x1006cf96: fld qword ptr ds:[100FEC40h]
0x1006cf9c: fmul st(1),st
0x1006cf9e: fxch
0x1006cfa0: fstp dword ptr [ebp-10h]
0x1006cfa3: fld dword ptr [eax]
0x1006cfa5: fld1
```

```
0x1006cfa7: fsubrp
0x1006cfa9: fsub dword ptr [eax+4]
0x1006cfac: fdiv dword ptr [eax+4]
0x1006cfaf: lea eax,[ebp-10h]
0x1006cfb2: push eax
0x1006cfb3: push 10136278h
0x1006cfb8: push dword ptr [esi+48h]
0x1006cfbb: fmulp
0x1006cfbd: fstp dword ptr [ebp-8]
0x1006cfc0: call 100727AAh
0x1006cfc5: add esp,18h
0x1006cfc8: test eax,eax
0x1006cfca: jge short 1006CFF0h
0x1006cfcc: mov eax,[esi]
0x1006cfce: push 10108DA4h
0x1006cfd3: push 101362E0h
0x1006cfd8: push dword ptr [eax+274h]
0x1006cfde: call 1006FAF1h
0x1006cfe3: push dword ptr [esi+48h]
0x1006cfe6: call 100209E0h
0x1006cfeb: add esp,10h
0x1006cfee: jmp short 1006CF71h
0x1006cff0: mov eax,1006B7E5h
0x1006cff5: leave
0x1006cff6: ret
```

`sub_1006cf39` is called by sub_1006d727. It works with the text "initCIELabConversion". It delegates to j_free, sub_1006e579, sub_1006faf1 (5 callees).

12.981 sub_1006cff7 at 0x1006cff7 (212 instructions)

```
0x1006cff7: push ebp
0x1006cff8: mov ebp,esp
0x1006cffa: push ecx
0x1006cffb: push esi
0x1006cffc: push edi
0x1006cffd: mov edi,[ebp+8]
0x1006d000: movzx ecx,word ptr [edi+18h]
0x1006d004: push 8
0x1006d006: pop eax
0x1006d007: cdq
0x1006d008: idiv ecx
0x1006d00a: mov esi,[edi+38h]
0x1006d00d: mov [ebp-4],ecx
0x1006d010: test eax,eax
0x1006d012: jne short 1006D015h
0x1006d014: inc eax
0x1006d015: inc eax
0x1006d016: shl eax,0Ah
0x1006d019: push eax
0x1006d01a: call 100EFDD0h
0x1006d01f: test eax,eax
0x1006d021: pop ecx
0x1006d022: mov [edi+3Ch],eax
0x1006d025: jne short 1006D04Ch
0x1006d027: push 10108E10h
0x1006d02c: push dword ptr [edi]
0x1006d02e: call 10068D52h
0x1006d033: pop ecx
0x1006d034: push eax
0x1006d035: mov eax,[edi]
0x1006d037: push dword ptr [eax+274h]
0x1006d03d: call 1006FAF1h
0x1006d042: add esp,0Ch
0x1006d045: xor eax,eax
0x1006d047: jmp near ptr 1006D216h
```

```
0x1006d04c: add eax,400h
0x1006d051: xor edx,edx
0x1006d053: mov ecx,0FFFFFF00h
0x1006d058: push ebx
0x1006d059: mov ebx,[edi+3Ch]
0x1006d05c: mov [ebx+edx*4],eax
0x1006d05f: mov ebx,[ebp-4]
0x1006d062: dec ebx
0x1006d063: je near ptr 1006D111h
0x1006d069: dec ebx
0x1006d06a: je short 1006D0ADh
0x1006d06c: dec ebx
0x1006d06d: dec ebx
0x1006d06e: je short 1006D087h
0x1006d070: sub ebx,4
0x1006d073: je short 1006D07Eh
0x1006d075: sub ebx,8
0x1006d078: jne near ptr 1006D205h
0x1006d07e: movzx edi,byte ptr [edx+esi]
0x1006d082: jmp near ptr 1006D1EFh
0x1006d087: mov edi,edx
0x1006d089: sar edi,4
0x1006d08c: movzx edi,byte ptr [edi+esi]
0x1006d090: mov ebx,edi
0x1006d092: or ebx,ecx
0x1006d094: shl ebx,8
0x1006d097: or ebx,edi
0x1006d099: shl ebx,8
0x1006d09c: or ebx,edi
0x1006d09e: mov [eax],ebx
0x1006d0a0: mov edi,edx
0x1006d0a2: add eax,4
0x1006d0a5: and edi,0Fh
0x1006d0a8: jmp near ptr 1006D1EBh
0x1006d0ad: mov edi,edx
0x1006d0af: sar edi,6
0x1006d0b2: movzx edi,byte ptr [edi+esi]
0x1006d0b6: mov ebx,edi
0x1006d0b8: or ebx,ecx
0x1006d0ba: shl ebx,8
0x1006d0bd: or ebx,edi
0x1006d0bf: shl ebx,8
0x1006d0c2: or ebx,edi
0x1006d0c4: mov [eax],ebx
0x1006d0c6: mov edi,edx
0x1006d0c8: sar edi,4
0x1006d0cb: and edi,3
0x1006d0ce: movzx edi,byte ptr [edi+esi]
0x1006d0d2: mov ebx,edi
0x1006d0d4: or ebx,ecx
0x1006d0d6: shl ebx,8
0x1006d0d9: or ebx,edi
0x1006d0db: shl ebx,8
0x1006d0de: or ebx,edi
0x1006d0e0: add eax,4
0x1006d0e3: mov [eax],ebx
0x1006d0e5: mov edi,edx
0x1006d0e7: sar edi,2
0x1006d0ea: and edi,3
0x1006d0ed: movzx edi,byte ptr [edi+esi]
0x1006d0f1: mov ebx,edi
0x1006d0f3: or ebx,ecx
0x1006d0f5: shl ebx,8
0x1006d0f8: or ebx,edi
0x1006d0fa: shl ebx,8
0x1006d0fd: add eax,4
0x1006d100: or ebx,edi
```

```
0x1006d102: mov [eax],ebx
0x1006d104: mov edi,edx
0x1006d106: add eax,4
0x1006d109: and edi,3
0x1006d10c: jmp near ptr 1006D1EBh
0x1006d111: mov edi,edx
0x1006d113: sar edi,7
0x1006d116: movzx edi,byte ptr [edi+esi]
0x1006d11a: mov ebx,edi
0x1006d11c: or ebx,ecx
0x1006d11e: shl ebx,8
0x1006d121: or ebx,edi
0x1006d123: shl ebx,8
0x1006d126: or ebx,edi
0x1006d128: mov [eax],ebx
0x1006d12a: mov edi,edx
0x1006d12c: sar edi,6
0x1006d12f: and edi,1
0x1006d132: movzx edi,byte ptr [edi+esi]
0x1006d136: mov ebx,edi
0x1006d138: or ebx,ecx
0x1006d13a: shl ebx,8
0x1006d13d: or ebx,edi
0x1006d13f: shl ebx,8
0x1006d142: or ebx,edi
0x1006d144: add eax,4
0x1006d147: mov [eax],ebx
0x1006d149: mov edi,edx
0x1006d14b: sar edi,5
0x1006d14e: and edi,1
0x1006d151: movzx edi,byte ptr [edi+esi]
0x1006d155: mov ebx,edi
0x1006d157: or ebx,ecx
0x1006d159: shl ebx,8
0x1006d15c: or ebx,edi
0x1006d15e: shl ebx,8
0x1006d161: or ebx,edi
0x1006d163: add eax,4
0x1006d166: mov [eax],ebx
0x1006d168: mov edi,edx
0x1006d16a: sar edi,4
0x1006d16d: and edi,1
0x1006d170: movzx edi,byte ptr [edi+esi]
0x1006d174: mov ebx,edi
0x1006d176: or ebx,ecx
0x1006d178: shl ebx,8
0x1006d17b: or ebx,edi
0x1006d17d: shl ebx,8
0x1006d180: or ebx,edi
0x1006d182: add eax,4
0x1006d185: mov [eax],ebx
0x1006d187: mov edi,edx
0x1006d189: sar edi,3
0x1006d18c: and edi,1
0x1006d18f: movzx edi,byte ptr [edi+esi]
0x1006d193: mov ebx,edi
0x1006d195: or ebx,ecx
0x1006d197: shl ebx,8
0x1006d19a: or ebx,edi
0x1006d19c: shl ebx,8
0x1006d19f: or ebx,edi
0x1006d1a1: add eax,4
0x1006d1a4: mov [eax],ebx
0x1006d1a6: mov edi,edx
0x1006d1a8: sar edi,2
0x1006d1ab: and edi,1
0x1006d1ae: movzx edi,byte ptr [edi+esi]
```

```
0x1006d1b2: mov ebx,edi
0x1006d1b4: or ebx,ecx
0x1006d1b6: shl ebx,8
0x1006d1b9: or ebx,edi
0x1006d1bb: shl ebx,8
0x1006d1be: or ebx,edi
0x1006d1c0: mov edi,edx
0x1006d1c2: sar edi,1
0x1006d1c4: add eax,4
0x1006d1c7: mov [eax],ebx
0x1006d1c9: and edi,1
0x1006d1cc: movzx edi,byte ptr [edi+esi]
0x1006d1d0: mov ebx,edi
0x1006d1d2: or ebx,ecx
0x1006d1d4: shl ebx,8
0x1006d1d7: or ebx,edi
0x1006d1d9: shl ebx,8
0x1006d1dc: add eax,4
0x1006d1df: or ebx,edi
0x1006d1e1: mov [eax],ebx
0x1006d1e3: mov edi,edx
0x1006d1e5: add eax,4
0x1006d1e8: and edi,1
0x1006d1eb: movzx edi,byte ptr [edi+esi]
0x1006d1ef: mov ebx,edi
0x1006d1f1: or ebx,ecx
0x1006d1f3: shl ebx,8
0x1006d1f6: or ebx,edi
0x1006d1f8: shl ebx,8
0x1006d1fb: or ebx,edi
0x1006d1fd: mov edi,[ebp+8]
0x1006d200: mov [eax],ebx
0x1006d202: add eax,4
0x1006d205: inc edx
0x1006d206: cmp edx,100h
0x1006d20c: jl near ptr 1006D059h
0x1006d212: xor eax,eax
0x1006d214: inc eax
0x1006d215: pop ebx
0x1006d216: pop edi
0x1006d217: pop esi
0x1006d218: leave
0x1006d219: ret
```

`sub_1006cff7` is called by sub_1006d21a. It delegates to sub_10068d52, sub_1006faf1, j_malloc (3 callees).

12.982 sub_1006d21a at 0x1006d21a (82 instructions)

```
0x1006d21a: movzx eax,word ptr [edi+18h]
0x1006d21e: push ebp
0x1006d21f: push esi
0x1006d220: xor esi,esi
0x1006d222: inc esi
0x1006d223: mov cl,al
0x1006d225: shl esi,cl
0x1006d227: mov ebp,0FFh
0x1006d22c: dec esi
0x1006d22d: cmp ax,10h
0x1006d231: jne short 1006D235h
0x1006d233: mov esi,ebp
0x1006d235: lea eax,[esi+1]
0x1006d238: push eax
0x1006d239: call 100EFDD0h
0x1006d23e: pop ecx
0x1006d23f: xor ecx,ecx
0x1006d241: cmp eax,ecx
```

```
0x1006d243: mov [edi+38h],eax
0x1006d246: jne short 1006D26Ah
0x1006d248: push 10108E30h
0x1006d24d: push dword ptr [edi]
0x1006d24f: call 10068D52h
0x1006d254: pop ecx
0x1006d255: push eax
0x1006d256: mov eax,[edi]
0x1006d258: push dword ptr [eax+274h]
0x1006d25e: call 1006FAF1h
0x1006d263: add esp,0Ch
0x1006d266: xor eax,eax
0x1006d268: jmp short 1006D2DBh
0x1006d26a: cmp [edi+20h],cx
0x1006d26e: push ebx
0x1006d26f: jne short 1006D291h
0x1006d271: cmp esi,ecx
0x1006d273: jl short 1006D2A9h
0x1006d275: mov ebx,esi
0x1006d277: imul ebx,0FFh
0x1006d27d: mov eax,ebx
0x1006d27f: cdq
0x1006d280: idiv esi
0x1006d282: mov edx,[edi+38h]
0x1006d285: sub ebx,ebp
0x1006d287: mov [ecx+edx],al
0x1006d28a: inc ecx
0x1006d28b: cmp ecx,esi
0x1006d28d: jle short 1006D27Dh
0x1006d28f: jmp short 1006D2A9h
0x1006d291: cmp esi,ecx
0x1006d293: jl short 1006D2A9h
0x1006d295: xor ebx,ebx
0x1006d297: mov eax,ebx
0x1006d299: cdq
0x1006d29a: idiv esi
0x1006d29c: mov edx,[edi+38h]
0x1006d29f: add ebx,ebp
0x1006d2a1: mov [ecx+edx],al
0x1006d2a4: inc ecx
0x1006d2a5: cmp ecx,esi
0x1006d2a7: jle short 1006D297h
0x1006d2a9: cmp word ptr [edi+18h],10h
0x1006d2ae: pop ebx
0x1006d2af: ja short 1006D2D8h
0x1006d2b1: movzx eax,word ptr [edi+20h]
0x1006d2b5: cmp ax,1
0x1006d2b9: je short 1006D2C0h
0x1006d2bb: test ax,ax
0x1006d2be: jne short 1006D2D8h
0x1006d2c0: push edi
0x1006d2c1: call 1006CFF7h
0x1006d2c6: test eax,eax
0x1006d2c8: pop ecx
0x1006d2c9: je short 1006D266h
0x1006d2cb: push dword ptr [edi+38h]
0x1006d2ce: call 100209E0h
0x1006d2d3: and dword ptr [edi+38h],0
0x1006d2d7: pop ecx
0x1006d2d8: xor eax,eax
0x1006d2da: inc eax
0x1006d2db: pop esi
0x1006d2dc: pop ebp
0x1006d2dd: ret
```

`sub_1006d21a` is called by sub_1006d6a9. It works with the text "No space for photometric conversion table". It delegates to j_free, sub_10068d52, sub_1006cff7 (5 callees).

12.983 sub_1006d2de at 0x1006d2de (39 instructions)

```
0x1006d2de: mov cl,[eax+18h]
0x1006d2e1: xor edx,edx
0x1006d2e3: inc edx
0x1006d2e4: push ebx
0x1006d2e5: mov ebx,[eax+28h]
0x1006d2e8: shl edx,cl
0x1006d2ea: push esi
0x1006d2eb: mov esi,[eax+24h]
0x1006d2ee: push edi
0x1006d2ef: test edx,edx
0x1006d2f1: mov edi,[eax+2Ch]
0x1006d2f4: jle short 1006D31Eh
0x1006d2f6: mov eax,100h
0x1006d2fb: mov cx,[esi]
0x1006d2fe: dec edx
0x1006d2ff: inc esi
0x1006d300: inc esi
0x1006d301: cmp cx,ax
0x1006d304: jae short 1006D325h
0x1006d306: mov cx,[ebx]
0x1006d309: inc ebx
0x1006d30a: inc ebx
0x1006d30b: cmp cx,ax
0x1006d30e: jae short 1006D325h
0x1006d310: mov cx,[edi]
0x1006d313: inc edi
0x1006d314: inc edi
0x1006d315: cmp cx,ax
0x1006d318: jae short 1006D325h
0x1006d31a: test edx,edx
0x1006d31c: jg short 1006D2FBh
0x1006d31e: push 8
0x1006d320: pop eax
0x1006d321: pop edi
0x1006d322: pop esi
0x1006d323: pop ebx
0x1006d324: ret
0x1006d325: push 10h
0x1006d327: jmp short 1006D320h
```

`sub_1006d2de` is called by sub_1006d6a9. It calls nothing else.

12.984 sub_1006d329 at 0x1006d329 (31 instructions)

```
0x1006d329: mov cl,[eax+18h]
0x1006d32c: xor edx,edx
0x1006d32e: inc edx
0x1006d32f: push ebx
0x1006d330: mov ebx,[eax+24h]
0x1006d333: shl edx,cl
0x1006d335: push esi
0x1006d336: mov esi,[eax+28h]
0x1006d339: push edi
0x1006d33a: dec edx
0x1006d33b: mov edi,[eax+2Ch]
0x1006d33e: js short 1006D36Ch
0x1006d340: mov ecx,ebx
0x1006d342: sub ecx,esi
0x1006d344: lea eax,[esi+edx*2]
0x1006d347: sub edi,esi
0x1006d349: movzx si,byte ptr [ecx+eax+1]
0x1006d34f: mov [ecx+eax],si
0x1006d353: movzx si,byte ptr [eax+1]
0x1006d358: mov [eax],si
0x1006d35b: movzx si,byte ptr [edi+eax+1]
```

```
0x1006d361: mov [edi+eax],si
0x1006d365: dec edx
0x1006d366: dec eax
0x1006d367: dec eax
0x1006d368: test edx,edx
0x1006d36a: jge short 1006D349h
0x1006d36c: pop edi
0x1006d36d: pop esi
0x1006d36e: pop ebx
0x1006d36f: ret
```

`sub_1006d329` is called by sub_1006d6a9. It calls nothing else.

12.985 sub_1006d370 at 0x1006d370 (292 instructions)

```
0x1006d370: push ebp
0x1006d371: mov ebp,esp
0x1006d373: sub esp,10h
0x1006d376: push ebx
0x1006d377: mov ebx,[ebp+8]
0x1006d37a: mov eax,[ebx+2Ch]
0x1006d37d: movzx ecx,word ptr [ebx+18h]
0x1006d381: push esi
0x1006d382: push edi
0x1006d383: mov [ebp-0Ch],eax
0x1006d386: push 8
0x1006d388: pop eax
0x1006d389: cdq
0x1006d38a: idiv ecx
0x1006d38c: mov esi,[ebx+24h]
0x1006d38f: mov edi,[ebx+28h]
0x1006d392: mov [ebp-10h],ecx
0x1006d395: inc eax
0x1006d396: shl eax,0Ah
0x1006d399: push eax
0x1006d39a: call 100EFDD0h
0x1006d39f: test eax,eax
0x1006d3a1: pop ecx
0x1006d3a2: mov [ebx+40h],eax
0x1006d3a5: jne short 1006D3CCh
0x1006d3a7: push 10108E5Ch
0x1006d3ac: push dword ptr [ebx]
0x1006d3ae: call 10068D52h
0x1006d3b3: pop ecx
0x1006d3b4: push eax
0x1006d3b5: mov eax,[ebx]
0x1006d3b7: push dword ptr [eax+274h]
0x1006d3bd: call 1006FAF1h
0x1006d3c2: add esp,0Ch
0x1006d3c5: xor eax,eax
0x1006d3c7: jmp near ptr 1006D6A4h
0x1006d3cc: and dword ptr [ebp-8],0
0x1006d3d0: lea ecx,[eax+400h]
0x1006d3d6: mov edx,0FFFFFF00h
0x1006d3db: mov eax,[ebx+40h]
0x1006d3de: mov ebx,[ebp-8]
0x1006d3e1: mov [eax+ebx*4],ecx
0x1006d3e4: mov eax,[ebp-10h]
0x1006d3e7: dec eax
0x1006d3e8: je near ptr 1006D4FCh
0x1006d3ee: dec eax
0x1006d3ef: je short 1006D45Bh
0x1006d3f1: dec eax
0x1006d3f2: dec eax
0x1006d3f3: je short 1006D420h
0x1006d3f5: sub eax,4
```

```
0x1006d3f8: jne near ptr 1006D68Eh
0x1006d3fe: movzx eax,bl
0x1006d401: mov ebx,[ebp-0Ch]
0x1006d404: add eax,eax
0x1006d406: movzx ebx,word ptr [eax+ebx]
0x1006d40a: mov [ebp-4],eax
0x1006d40d: movzx eax,byte ptr [eax+edi]
0x1006d411: or ebx,edx
0x1006d413: shl ebx,8
0x1006d416: or ebx,eax
0x1006d418: mov eax,[ebp-4]
0x1006d41b: jmp near ptr 1006D680h
0x1006d420: mov al,[ebp-8]
0x1006d423: mov ebx,[ebp-0Ch]
0x1006d426: shr al,4
0x1006d429: movzx eax,al
0x1006d42c: add eax,eax
0x1006d42e: movzx ebx,word ptr [eax+ebx]
0x1006d432: mov [ebp-4],eax
0x1006d435: movzx eax,byte ptr [eax+edi]
0x1006d439: or ebx,edx
0x1006d43b: shl ebx,8
0x1006d43e: or ebx,eax
0x1006d440: mov eax,[ebp-4]
0x1006d443: movzx eax,byte ptr [eax+esi]
0x1006d447: shl ebx,8
0x1006d44a: or ebx,eax
0x1006d44c: mov al,[ebp-8]
0x1006d44f: mov [ecx],ebx
0x1006d451: add ecx,4
0x1006d454: and al,0Fh
0x1006d456: jmp near ptr 1006D663h
0x1006d45b: mov al,[ebp-8]
0x1006d45e: mov ebx,[ebp-0Ch]
0x1006d461: shr al,6
0x1006d464: movzx eax,al
0x1006d467: add eax,eax
0x1006d469: movzx ebx,word ptr [eax+ebx]
0x1006d46d: mov [ebp-4],eax
0x1006d470: movzx eax,byte ptr [eax+edi]
0x1006d474: or ebx,edx
0x1006d476: shl ebx,8
0x1006d479: or ebx,eax
0x1006d47b: mov eax,[ebp-4]
0x1006d47e: movzx eax,byte ptr [eax+esi]
0x1006d482: shl ebx,8
0x1006d485: or ebx,eax
0x1006d487: mov eax,[ebp-8]
0x1006d48a: mov [ecx],ebx
0x1006d48c: mov ebx,[ebp-0Ch]
0x1006d48f: sar eax,4
0x1006d492: and al,3
0x1006d494: movzx eax,al
0x1006d497: add eax,eax
0x1006d499: movzx ebx,word ptr [eax+ebx]
0x1006d49d: mov [ebp-4],eax
0x1006d4a0: movzx eax,byte ptr [eax+edi]
0x1006d4a4: or ebx,edx
0x1006d4a6: shl ebx,8
0x1006d4a9: or ebx,eax
0x1006d4ab: mov eax,[ebp-4]
0x1006d4ae: movzx eax,byte ptr [eax+esi]
0x1006d4b2: shl ebx,8
0x1006d4b5: or ebx,eax
0x1006d4b7: mov eax,[ebp-8]
0x1006d4ba: sar eax,2
0x1006d4bd: and al,3
```

```
0x1006d4bf: movzx eax,al
0x1006d4c2: add ecx,4
0x1006d4c5: mov [ecx],ebx
0x1006d4c7: mov ebx,[ebp-0Ch]
0x1006d4ca: add eax,eax
0x1006d4cc: movzx ebx,word ptr [eax+ebx]
0x1006d4d0: mov [ebp-4],eax
0x1006d4d3: movzx eax,byte ptr [eax+edi]
0x1006d4d7: or ebx,edx
0x1006d4d9: shl ebx,8
0x1006d4dc: or ebx,eax
0x1006d4de: mov eax,[ebp-4]
0x1006d4e1: movzx eax,byte ptr [eax+esi]
0x1006d4e5: shl ebx,8
0x1006d4e8: add ecx,4
0x1006d4eb: or ebx,eax
0x1006d4ed: mov al,[ebp-8]
0x1006d4f0: mov [ecx],ebx
0x1006d4f2: add ecx,4
0x1006d4f5: and al,3
0x1006d4f7: jmp near ptr 1006D663h
0x1006d4fc: mov al,[ebp-8]
0x1006d4ff: shr al,7
0x1006d502: movzx eax,al
0x1006d505: add eax,eax
0x1006d507: mov ebx,[ebp-0Ch]
0x1006d50a: movzx ebx,word ptr [eax+ebx]
0x1006d50e: mov [ebp-4],eax
0x1006d511: movzx eax,byte ptr [eax+edi]
0x1006d515: or ebx,edx
0x1006d517: shl ebx,8
0x1006d51a: or ebx,eax
0x1006d51c: mov eax,[ebp-4]
0x1006d51f: movzx eax,byte ptr [eax+esi]
0x1006d523: shl ebx,8
0x1006d526: or ebx,eax
0x1006d528: mov eax,[ebp-8]
0x1006d52b: mov [ecx],ebx
0x1006d52d: sar eax,6
0x1006d530: and al,1
0x1006d532: mov ebx,[ebp-0Ch]
0x1006d535: movzx eax,al
0x1006d538: add eax,eax
0x1006d53a: movzx ebx,word ptr [eax+ebx]
0x1006d53e: mov [ebp-4],eax
0x1006d541: movzx eax,byte ptr [eax+edi]
0x1006d545: or ebx,edx
0x1006d547: shl ebx,8
0x1006d54a: or ebx,eax
0x1006d54c: mov eax,[ebp-4]
0x1006d54f: movzx eax,byte ptr [eax+esi]
0x1006d553: shl ebx,8
0x1006d556: or ebx,eax
0x1006d558: mov eax,[ebp-8]
0x1006d55b: sar eax,5
0x1006d55e: and al,1
0x1006d560: movzx eax,al
0x1006d563: add eax,eax
0x1006d565: mov [ebp-4],eax
0x1006d568: add ecx,4
0x1006d56b: mov [ecx],ebx
0x1006d56d: mov ebx,[ebp-0Ch]
0x1006d570: movzx ebx,word ptr [eax+ebx]
0x1006d574: movzx eax,byte ptr [eax+edi]
0x1006d578: or ebx,edx
0x1006d57a: shl ebx,8
0x1006d57d: or ebx,eax
```

```
0x1006d57f: mov eax,[ebp-4]
0x1006d582: movzx eax,byte ptr [eax+esi]
0x1006d586: shl ebx,8
0x1006d589: or ebx,eax
0x1006d58b: mov eax,[ebp-8]
0x1006d58e: sar eax,4
0x1006d591: and al,1
0x1006d593: movzx eax,al
0x1006d596: add eax,eax
0x1006d598: add ecx,4
0x1006d59b: mov [ecx],ebx
0x1006d59d: mov ebx,[ebp-0Ch]
0x1006d5a0: movzx ebx,word ptr [eax+ebx]
0x1006d5a4: mov [ebp-4],eax
0x1006d5a7: movzx eax,byte ptr [eax+edi]
0x1006d5ab: or ebx,edx
0x1006d5ad: shl ebx,8
0x1006d5b0: or ebx,eax
0x1006d5b2: mov eax,[ebp-4]
0x1006d5b5: movzx eax,byte ptr [eax+esi]
0x1006d5b9: shl ebx,8
0x1006d5bc: or ebx,eax
0x1006d5be: mov eax,[ebp-8]
0x1006d5c1: sar eax,3
0x1006d5c4: and al,1
0x1006d5c6: add ecx,4
0x1006d5c9: mov [ecx],ebx
0x1006d5cb: mov ebx,[ebp-0Ch]
0x1006d5ce: movzx eax,al
0x1006d5d1: add eax,eax
0x1006d5d3: movzx ebx,word ptr [eax+ebx]
0x1006d5d7: or ebx,edx
0x1006d5d9: add ecx,4
0x1006d5dc: mov [ebp-4],eax
0x1006d5df: shl ebx,8
0x1006d5e2: movzx eax,byte ptr [eax+edi]
0x1006d5e6: or ebx,eax
0x1006d5e8: mov eax,[ebp-4]
0x1006d5eb: movzx eax,byte ptr [eax+esi]
0x1006d5ef: shl ebx,8
0x1006d5f2: or ebx,eax
0x1006d5f4: mov eax,[ebp-8]
0x1006d5f7: mov [ecx],ebx
0x1006d5f9: mov ebx,[ebp-0Ch]
0x1006d5fc: sar eax,2
0x1006d5ff: and al,1
0x1006d601: movzx eax,al
0x1006d604: add eax,eax
0x1006d606: movzx ebx,word ptr [eax+ebx]
0x1006d60a: mov [ebp-4],eax
0x1006d60d: movzx eax,byte ptr [eax+edi]
0x1006d611: or ebx,edx
0x1006d613: shl ebx,8
0x1006d616: or ebx,eax
0x1006d618: mov eax,[ebp-4]
0x1006d61b: movzx eax,byte ptr [eax+esi]
0x1006d61f: shl ebx,8
0x1006d622: or ebx,eax
0x1006d624: mov eax,[ebp-8]
0x1006d627: sar eax,1
0x1006d629: and al,1
0x1006d62b: movzx eax,al
0x1006d62e: add ecx,4
0x1006d631: mov [ecx],ebx
0x1006d633: mov ebx,[ebp-0Ch]
0x1006d636: add eax,eax
0x1006d638: movzx ebx,word ptr [eax+ebx]
```

```
0x1006d63c: mov [ebp-4],eax
0x1006d63f: movzx eax,byte ptr [eax+edi]
0x1006d643: or ebx,edx
0x1006d645: shl ebx,8
0x1006d648: or ebx,eax
0x1006d64a: mov eax,[ebp-4]
0x1006d64d: movzx eax,byte ptr [eax+esi]
0x1006d651: shl ebx,8
0x1006d654: add ecx,4
0x1006d657: or ebx,eax
0x1006d659: mov al,[ebp-8]
0x1006d65c: mov [ecx],ebx
0x1006d65e: add ecx,4
0x1006d661: and al,1
0x1006d663: mov ebx,[ebp-0Ch]
0x1006d666: movzx eax,al
0x1006d669: add eax,eax
0x1006d66b: movzx ebx,word ptr [eax+ebx]
0x1006d66f: mov [ebp-4],eax
0x1006d672: movzx eax,byte ptr [eax+edi]
0x1006d676: or ebx,edx
0x1006d678: shl ebx,8
0x1006d67b: or ebx,eax
0x1006d67d: mov eax,[ebp-4]
0x1006d680: movzx eax,byte ptr [eax+esi]
0x1006d684: shl ebx,8
0x1006d687: or ebx,eax
0x1006d689: mov [ecx],ebx
0x1006d68b: add ecx,4
0x1006d68e: inc dword ptr [ebp-8]
0x1006d691: cmp dword ptr [ebp-8],100h
0x1006d698: mov ebx,[ebp+8]
0x1006d69b: jl near ptr 1006D3DBh
0x1006d6a1: xor eax,eax
0x1006d6a3: inc eax
0x1006d6a4: pop edi
0x1006d6a5: pop esi
0x1006d6a6: pop ebx
0x1006d6a7: leave
0x1006d6a8: ret
```

`sub_1006d370` is called by sub_1006d6a9. It works with the text "No space for Palette mapping table". It delegates to sub_10068d52, sub_1006faf1, j_malloc (3 callees).

12.986 sub_1006d6a9 at 0x1006d6a9 (48 instructions)

```
0x1006d6a9: push edi
0x1006d6aa: mov edi,eax
0x1006d6ac: movzx eax,word ptr [edi+20h]
0x1006d6b0: test eax,eax
0x1006d6b2: jl short 1006D6CDh
0x1006d6b4: cmp eax,1
0x1006d6b7: jle short 1006D71Ch
0x1006d6b9: cmp eax,2
0x1006d6bc: je short 1006D715h
0x1006d6be: cmp eax,3
0x1006d6c1: je short 1006D6D2h
0x1006d6c3: cmp eax,4
0x1006d6c6: jle short 1006D6CDh
0x1006d6c8: cmp eax,6
0x1006d6cb: jle short 1006D715h
0x1006d6cd: xor eax,eax
0x1006d6cf: inc eax
0x1006d6d0: pop edi
0x1006d6d1: ret
0x1006d6d2: mov eax,edi
```

```
0x1006d6d4: call 1006D2DEh
0x1006d6d9: cmp eax,10h
0x1006d6dc: jne short 1006D6E7h
0x1006d6de: mov eax,edi
0x1006d6e0: call 1006D329h
0x1006d6e5: jmp short 1006D705h
0x1006d6e7: push 10108E80h
0x1006d6ec: push dword ptr [edi]
0x1006d6ee: call 10068D52h
0x1006d6f3: pop ecx
0x1006d6f4: push eax
0x1006d6f5: mov eax,[edi]
0x1006d6f7: push dword ptr [eax+274h]
0x1006d6fd: call 1007007Dh
0x1006d702: add esp,0Ch
0x1006d705: cmp word ptr [edi+18h],8
0x1006d70a: ja short 1006D6CDh
0x1006d70c: push edi
0x1006d70d: call 1006D370h
0x1006d712: pop ecx
0x1006d713: jmp short 1006D721h
0x1006d715: cmp word ptr [edi+18h],8
0x1006d71a: je short 1006D6CDh
0x1006d71c: call 1006D21Ah
0x1006d721: test eax,eax
0x1006d723: jne short 1006D6CDh
0x1006d725: pop edi
0x1006d726: ret
```

`sub_1006d6a9` is called by sub_1006d727. It works with the text "Assuming 8-bit colormap". It delegates to sub_10068d52, sub_1006d21a, sub_1006d2de (6 callees).

12.987 sub_1006d727 at 0x1006d727 (180 instructions)

```
0x1006d727: push ebp
0x1006d728: mov ebp,esp
0x1006d72a: push ecx
0x1006d72b: push ecx
0x1006d72c: push esi
0x1006d72d: mov esi,eax
0x1006d72f: push dword ptr [esi]
0x1006d731: call 10068D59h
0x1006d736: test eax,eax
0x1006d738: pop ecx
0x1006d739: mov eax,1006918Ah
0x1006d73e: jne short 1006D745h
0x1006d740: mov eax,10069686h
0x1006d745: and dword ptr [esi+34h],0
0x1006d749: mov [esi+30h],eax
0x1006d74c: movzx eax,word ptr [esi+20h]
0x1006d750: test eax,eax
0x1006d752: jl near ptr 1006D9A2h
0x1006d758: cmp eax,1
0x1006d75b: jle near ptr 1006D954h
0x1006d761: cmp eax,2
0x1006d764: je near ptr 1006D8EFh
0x1006d76a: cmp eax,3
0x1006d76d: je near ptr 1006D899h
0x1006d773: cmp eax,5
0x1006d776: je near ptr 1006D861h
0x1006d77c: cmp eax,6
0x1006d77f: je short 1006D7B1h
0x1006d781: cmp eax,8
0x1006d784: jne near ptr 1006D9A2h
0x1006d78a: mov eax,esi
0x1006d78c: call 1006D6A9h
```

```
0x1006d791: test eax,eax
0x1006d793: je near ptr 1006D9A2h
0x1006d799: cmp word ptr [esi+18h],8
0x1006d79e: jne near ptr 1006D9A2h
0x1006d7a4: call 1006CF39h
0x1006d7a9: mov [esi+34h],eax
0x1006d7ac: jmp near ptr 1006D9A2h
0x1006d7b1: cmp word ptr [esi+18h],8
0x1006d7b6: jne near ptr 1006D9A2h
0x1006d7bc: call 1006CEC2h
0x1006d7c1: test eax,eax
0x1006d7c3: je near ptr 1006D9A2h
0x1006d7c9: lea eax,[ebp-8]
0x1006d7cc: push eax
0x1006d7cd: lea eax,[ebp-4]
0x1006d7d0: push eax
0x1006d7d1: push 212h
0x1006d7d6: push dword ptr [esi]
0x1006d7d8: call 1006E579h
0x1006d7dd: movzx eax,word ptr [ebp-4]
0x1006d7e1: movzx ecx,word ptr [ebp-8]
0x1006d7e5: shl eax,4
0x1006d7e8: or eax,ecx
0x1006d7ea: add esp,10h
0x1006d7ed: sub eax,11h
0x1006d7f0: je short 1006D855h
0x1006d7f2: dec eax
0x1006d7f3: je short 1006D849h
0x1006d7f5: sub eax,0Fh
0x1006d7f8: je short 1006D83Dh
0x1006d7fa: dec eax
0x1006d7fb: je short 1006D831h
0x1006d7fd: sub eax,1Fh
0x1006d800: je short 1006D825h
0x1006d802: dec eax
0x1006d803: je short 1006D819h
0x1006d805: dec eax
0x1006d806: dec eax
0x1006d807: jne near ptr 1006D9A2h
0x1006d80d: mov dword ptr [esi+34h],1006B8ADh
0x1006d814: jmp near ptr 1006D9A2h
0x1006d819: mov dword ptr [esi+34h],1006C16Dh
0x1006d820: jmp near ptr 1006D9A2h
0x1006d825: mov dword ptr [esi+34h],1006C645h
0x1006d82c: jmp near ptr 1006D9A2h
0x1006d831: mov dword ptr [esi+34h],1006C856h
0x1006d838: jmp near ptr 1006D9A2h
0x1006d83d: mov dword ptr [esi+34h],1006CB49h
0x1006d844: jmp near ptr 1006D9A2h
0x1006d849: mov dword ptr [esi+34h],1006CC6Bh
0x1006d850: jmp near ptr 1006D9A2h
0x1006d855: mov dword ptr [esi+34h],1006CDB5h
0x1006d85c: jmp near ptr 1006D9A2h
0x1006d861: mov eax,esi
0x1006d863: call 1006D6A9h
0x1006d868: test eax,eax
0x1006d86a: je near ptr 1006D9A2h
0x1006d870: cmp word ptr [esi+18h],8
0x1006d875: jne near ptr 1006D9A2h
0x1006d87b: cmp dword ptr [esi+38h],0
0x1006d87f: jne short 1006D88Dh
0x1006d881: mov dword ptr [esi+34h],1006A855h
0x1006d888: jmp near ptr 1006D9A2h
0x1006d88d: mov dword ptr [esi+34h],1006AF0Fh
0x1006d894: jmp near ptr 1006D9A2h
0x1006d899: mov eax,esi
0x1006d89b: call 1006D6A9h
```

```
0x1006d8a0: test eax,eax
0x1006d8a2: je near ptr 1006D9A2h
0x1006d8a8: movzx eax,word ptr [esi+18h]
0x1006d8ac: dec eax
0x1006d8ad: je short 1006D8E3h
0x1006d8af: dec eax
0x1006d8b0: je short 1006D8D7h
0x1006d8b2: dec eax
0x1006d8b3: dec eax
0x1006d8b4: je short 1006D8CBh
0x1006d8b6: sub eax,4
0x1006d8b9: jne near ptr 1006D9A2h
0x1006d8bf: mov dword ptr [esi+34h],10069B65h
0x1006d8c6: jmp near ptr 1006D9A2h
0x1006d8cb: mov dword ptr [esi+34h],10069BB7h
0x1006d8d2: jmp near ptr 1006D9A2h
0x1006d8d7: mov dword ptr [esi+34h],10069C38h
0x1006d8de: jmp near ptr 1006D9A2h
0x1006d8e3: mov dword ptr [esi+34h],10069CF8h
0x1006d8ea: jmp near ptr 1006D9A2h
0x1006d8ef: movzx eax,word ptr [esi+18h]
0x1006d8f3: cmp eax,8
0x1006d8f6: je short 1006D92Ch
0x1006d8f8: cmp eax,10h
0x1006d8fb: jne near ptr 1006D9A2h
0x1006d901: mov eax,[esi+0Ch]
0x1006d904: cmp eax,1
0x1006d907: jne short 1006D915h
0x1006d909: mov dword ptr [esi+34h],1006A71Fh
0x1006d910: jmp near ptr 1006D9A2h
0x1006d915: cmp eax,2
0x1006d918: jne short 1006D923h
0x1006d91a: mov dword ptr [esi+34h],1006A79Eh
0x1006d921: jmp short 1006D9A2h
0x1006d923: mov dword ptr [esi+34h],1006A6A5h
0x1006d92a: jmp short 1006D9A2h
0x1006d92c: mov eax,[esi+0Ch]
0x1006d92f: cmp eax,1
0x1006d932: jne short 1006D93Dh
0x1006d934: mov dword ptr [esi+34h],1006A383h
0x1006d93b: jmp short 1006D9A2h
0x1006d93d: cmp eax,2
0x1006d940: jne short 1006D94Bh
0x1006d942: mov dword ptr [esi+34h],1006A5F2h
0x1006d949: jmp short 1006D9A2h
0x1006d94b: mov dword ptr [esi+34h],1006A11Fh
0x1006d952: jmp short 1006D9A2h
0x1006d954: mov eax,esi
0x1006d956: call 1006D6A9h
0x1006d95b: test eax,eax
0x1006d95d: je short 1006D9A2h
0x1006d95f: movzx eax,word ptr [esi+18h]
0x1006d963: dec eax
0x1006d964: je short 1006D99Bh
0x1006d966: dec eax
0x1006d967: je short 1006D992h
0x1006d969: dec eax
0x1006d96a: dec eax
0x1006d96b: je short 1006D989h
0x1006d96d: sub eax,4
0x1006d970: je short 1006D980h
0x1006d972: sub eax,8
0x1006d975: jne short 1006D9A2h
0x1006d977: mov dword ptr [esi+34h],10069E5Ch
0x1006d97e: jmp short 1006D9A2h
0x1006d980: mov dword ptr [esi+34h],10069E0Ah
0x1006d987: jmp short 1006D9A2h
```

```
0x1006d989: mov dword ptr [esi+34h],1006A09Eh
0x1006d990: jmp short 1006D9A2h
0x1006d992: mov dword ptr [esi+34h],10069FDEh
0x1006d999: jmp short 1006D9A2h
0x1006d99b: mov dword ptr [esi+34h],10069ECCh
0x1006d9a2: xor eax,eax
0x1006d9a4: cmp [esi+30h],eax
0x1006d9a7: je short 1006D9AFh
0x1006d9a9: cmp [esi+34h],eax
0x1006d9ac: je short 1006D9AFh
0x1006d9ae: inc eax
0x1006d9af: pop esi
0x1006d9b0: leave
0x1006d9b1: ret
```

`sub_1006d727` is called by sub_1006daaf. It delegates to sub_10068d59, sub_1006cec2, sub_1006cf39 (5 callees).

12.988 sub_1006d9b2 at 0x1006d9b2 (81 instructions)

```
0x1006d9b2: push ebp
0x1006d9b3: mov ebp,esp
0x1006d9b5: push ecx
0x1006d9b6: push ecx
0x1006d9b7: push esi
0x1006d9b8: mov esi,eax
0x1006d9ba: push dword ptr [esi]
0x1006d9bc: call 10068D59h
0x1006d9c1: test eax,eax
0x1006d9c3: pop ecx
0x1006d9c4: mov eax,1006938Ah
0x1006d9c9: jne short 1006D9D0h
0x1006d9cb: mov eax,1006987Fh
0x1006d9d0: and dword ptr [esi+34h],0
0x1006d9d4: mov [esi+30h],eax
0x1006d9d7: movzx eax,word ptr [esi+20h]
0x1006d9db: dec eax
0x1006d9dc: dec eax
0x1006d9dd: je short 1006DA3Dh
0x1006d9df: sub eax,4
0x1006d9e2: jne near ptr 1006DA99h
0x1006d9e8: cmp word ptr [esi+18h],8
0x1006d9ed: jne near ptr 1006DA99h
0x1006d9f3: cmp word ptr [esi+1Ah],3
0x1006d9f8: jne near ptr 1006DA99h
0x1006d9fe: call 1006CEC2h
0x1006da03: test eax,eax
0x1006da05: je near ptr 1006DA99h
0x1006da0b: lea eax,[ebp-8]
0x1006da0e: push eax
0x1006da0f: lea eax,[ebp-4]
0x1006da12: push eax
0x1006da13: push 212h
0x1006da18: push dword ptr [esi]
0x1006da1a: call 1006E579h
0x1006da1f: movzx eax,word ptr [ebp-4]
0x1006da23: movzx ecx,word ptr [ebp-8]
0x1006da27: shl eax,4
0x1006da2a: or eax,ecx
0x1006da2c: add esp,10h
0x1006da2f: cmp eax,11h
0x1006da32: jne short 1006DA99h
0x1006da34: mov dword ptr [esi+34h],1006CE2Fh
0x1006da3b: jmp short 1006DA99h
0x1006da3d: movzx eax,word ptr [esi+18h]
0x1006da41: cmp eax,8
0x1006da44: je short 1006DA73h
```

```
0x1006da46: cmp eax,10h
0x1006da49: jne short 1006DA99h
0x1006da4b: mov eax,[esi+0Ch]
0x1006da4e: cmp eax,1
0x1006da51: jne short 1006DA5Ch
0x1006da53: mov dword ptr [esi+34h],1006B670h
0x1006da5a: jmp short 1006DA99h
0x1006da5c: cmp eax,2
0x1006da5f: jne short 1006DA6Ah
0x1006da61: mov dword ptr [esi+34h],1006B707h
0x1006da68: jmp short 1006DA99h
0x1006da6a: mov dword ptr [esi+34h],1006B5EFh
0x1006da71: jmp short 1006DA99h
0x1006da73: mov eax,[esi+0Ch]
0x1006da76: cmp eax,1
0x1006da79: jne short 1006DA84h
0x1006da7b: mov dword ptr [esi+34h],1006B285h
0x1006da82: jmp short 1006DA99h
0x1006da84: cmp eax,2
0x1006da87: jne short 1006DA92h
0x1006da89: mov dword ptr [esi+34h],1006B533h
0x1006da90: jmp short 1006DA99h
0x1006da92: mov dword ptr [esi+34h],1006AFFCh
0x1006da99: cmp dword ptr [esi+30h],0
0x1006da9d: je short 1006DAAAh
0x1006da9f: cmp dword ptr [esi+34h],0
0x1006daa3: je short 1006DAAAh
0x1006daa5: xor eax,eax
0x1006daa7: inc eax
0x1006daa8: jmp short 1006DAACh
0x1006daaa: xor eax,eax
0x1006daac: pop esi
0x1006daad: leave
0x1006daae: ret
```

`sub_1006d9b2` is called by sub_1006daaf. It delegates to sub_10068d59, sub_1006cec2, sub_1006e579 (3 callees).

12.989 sub_1006daaf at 0x1006daaf (362 instructions)

```
0x1006daaf: push ebp
0x1006dab0: mov ebp,esp
0x1006dab2: mov eax,[ebp+0Ch]
0x1006dab5: mov ecx,[ebp+10h]
0x1006dab8: sub esp,18h
0x1006dabb: push ebx
0x1006dabc: push esi
0x1006dabd: mov esi,[ebp+8]
0x1006dac0: push edi
0x1006dac1: lea edi,[esi+18h]
0x1006dac4: xor ebx,ebx
0x1006dac6: push edi
0x1006dac7: push 102h
0x1006dacc: push eax
0x1006dacd: mov [esi+4Ch],ebx
0x1006dad0: mov [esi+50h],ebx
0x1006dad3: mov [esi+24h],ebx
0x1006dad6: mov [esi+28h],ebx
0x1006dad9: mov [esi+2Ch],ebx
0x1006dadc: mov word ptr [esi+1Eh],4
0x1006dae2: mov [esi],eax
0x1006dae4: mov [esi+4],ecx
0x1006dae7: call 1006E579h
0x1006daec: movzx eax,word ptr [edi]
0x1006daef: add esp,0Ch
0x1006daf2: cmp eax,ebx
0x1006daf4: jle near ptr 1006DF13h
```

```
0x1006dafa: cmp eax,2
0x1006dafd: jle short 1006DB12h
0x1006daff: cmp eax,4
0x1006db02: je short 1006DB12h
0x1006db04: cmp eax,8
0x1006db07: je short 1006DB12h
0x1006db09: cmp eax,10h
0x1006db0c: jne near ptr 1006DF13h
0x1006db12: lea edi,[esi+1Ah]
0x1006db15: push edi
0x1006db16: push 115h
0x1006db1b: push dword ptr [ebp+0Ch]
0x1006db1e: mov [esi+0Ch],ebx
0x1006db21: call 1006E579h
0x1006db26: lea eax,[ebp-0Ch]
0x1006db29: push eax
0x1006db2a: lea eax,[ebp-4]
0x1006db2d: push eax
0x1006db2e: push 152h
0x1006db33: push dword ptr [ebp+0Ch]
0x1006db36: call 1006E579h
0x1006db3b: xor ecx,ecx
0x1006db3d: inc ecx
0x1006db3e: add esp,1Ch
0x1006db41: cmp [ebp-4],cx
0x1006db45: jb short 1006DB66h
0x1006db47: mov eax,[ebp-0Ch]
0x1006db4a: movzx eax,word ptr [eax]
0x1006db4d: cmp eax,ebx
0x1006db4f: je short 1006DB5Dh
0x1006db51: lea edx,[eax-1]
0x1006db54: cmp edx,ecx
0x1006db56: ja short 1006DB66h
0x1006db58: mov [esi+0Ch],eax
0x1006db5b: jmp short 1006DB66h
0x1006db5d: cmp word ptr [edi],3
0x1006db61: jbe short 1006DB66h
0x1006db63: mov [esi+0Ch],ecx
0x1006db66: lea edi,[esi+20h]
0x1006db69: push edi
0x1006db6a: mov ebx,106h
0x1006db6f: push ebx
0x1006db70: push dword ptr [ebp+0Ch]
0x1006db73: call 10067985h
0x1006db78: add esp,0Ch
0x1006db7b: test eax,eax
0x1006db7d: jne short 1006DB82h
0x1006db7f: and [edi],ax
0x1006db82: cmp word ptr [ebp-4],0
0x1006db87: jne short 1006DB9Fh
0x1006db89: cmp word ptr [esi+1Ah],4
0x1006db8e: jne short 1006DB9Fh
0x1006db90: cmp word ptr [edi],2
0x1006db94: jne short 1006DB9Fh
0x1006db96: xor eax,eax
0x1006db98: inc eax
0x1006db99: mov [esi+0Ch],eax
0x1006db9c: mov [ebp-4],eax
0x1006db9f: movzx eax,word ptr [esi+1Ah]
0x1006dba3: movzx ecx,word ptr [ebp-4]
0x1006dba7: sub eax,ecx
0x1006dba9: mov [ebp-8],eax
0x1006dbac: lea eax,[ebp+10h]
0x1006dbaf: push eax
0x1006dbb0: push 103h
0x1006dbb5: push dword ptr [ebp+0Ch]
0x1006dbb8: call 1006E579h
```

```
0x1006dbbd: lea eax,[ebp+8]
0x1006dbc0: push eax
0x1006dbc1: push 11Ch
0x1006dbc6: push dword ptr [ebp+0Ch]
0x1006dbc9: call 1006E579h
0x1006dbce: push edi
0x1006dbcf: push ebx
0x1006dbd0: mov ebx,[ebp+0Ch]
0x1006dbd3: push ebx
0x1006dbd4: call 10067985h
0x1006dbd9: add esp,24h
0x1006dbdc: test eax,eax
0x1006dbde: jne short 1006DC0Fh
0x1006dbe0: mov eax,[ebp-8]
0x1006dbe3: dec eax
0x1006dbe4: je short 1006DC00h
0x1006dbe6: dec eax
0x1006dbe7: dec eax
0x1006dbe8: je short 1006DBF9h
0x1006dbea: push 10108AA8h
0x1006dbef: push 10108C98h
0x1006dbf4: jmp near ptr 1006DF19h
0x1006dbf9: mov word ptr [edi],2
0x1006dbfe: jmp short 1006DC0Fh
0x1006dc00: push ebx
0x1006dc01: call 1006901Ah
0x1006dc06: neg eax
0x1006dc08: sbb eax,eax
0x1006dc0a: inc eax
0x1006dc0b: pop ecx
0x1006dc0c: mov [edi],ax
0x1006dc0f: movzx eax,word ptr [edi]
0x1006dc12: cmp eax,6
0x1006dc15: jg near ptr 1006DDDAh
0x1006dc1b: je near ptr 1006DDADh
0x1006dc21: test eax,eax
0x1006dc23: jl near ptr 1006DDF0h
0x1006dc29: cmp eax,1
0x1006dc2c: jle near ptr 1006DD34h
0x1006dc32: cmp eax,2
0x1006dc35: je near ptr 1006DD91h
0x1006dc3b: cmp eax,3
0x1006dc3e: je short 1006DC9Bh
0x1006dc40: cmp eax,5
0x1006dc43: jne near ptr 1006DDF0h
0x1006dc49: lea eax,[ebp+0Ch]
0x1006dc4c: push eax
0x1006dc4d: push 14Ch
0x1006dc52: push ebx
0x1006dc53: call 1006E579h
0x1006dc58: add esp,0Ch
0x1006dc5b: cmp word ptr [ebp+0Ch],1
0x1006dc60: je short 1006DC82h
0x1006dc62: movzx eax,word ptr [ebp+0Ch]
0x1006dc66: push eax
0x1006dc67: push 10108C90h
0x1006dc6c: push 10108C5Ch
0x1006dc71: push dword ptr [ebp+14h]
0x1006dc74: call dword ptr ds:[100FE4F8h]
0x1006dc7a: add esp,10h
0x1006dc7d: jmp near ptr 1006DF25h
0x1006dc82: movzx eax,word ptr [esi+1Ah]
0x1006dc86: cmp ax,4
0x1006dc8a: jae near ptr 1006DE96h
0x1006dc90: movzx eax,ax
0x1006dc93: push eax
0x1006dc94: push 10108C4Ch
```

```
0x1006dc99: jmp short 1006DC6Ch
0x1006dc9b: lea eax,[ebp-18h]
0x1006dc9e: push eax
0x1006dc9f: lea eax,[ebp-14h]
0x1006dca2: push eax
0x1006dca3: lea eax,[ebp-10h]
0x1006dca6: push eax
0x1006dca7: push 140h
0x1006dcac: push ebx
0x1006dcad: call 10067985h
0x1006dcb2: add esp,14h
0x1006dcb5: test eax,eax
0x1006dcb7: jne short 1006DCCEh
0x1006dcb9: push 10108ED4h
0x1006dcbe: push dword ptr [ebp+14h]
0x1006dcc1: call dword ptr ds:[100FE4F8h]
0x1006dcc7: pop ecx
0x1006dcc8: pop ecx
0x1006dcc9: jmp near ptr 1006DF25h
0x1006dcce: mov cl,[esi+18h]
0x1006dcd1: xor eax,eax
0x1006dcd3: inc eax
0x1006dcd4: shl eax,cl
0x1006dcd6: lea ebx,[eax+eax]
0x1006dcd9: push ebx
0x1006dcda: call 100EFDD0h
0x1006dcdf: push ebx
0x1006dce0: mov [esi+24h],eax
0x1006dce3: call 100EFDD0h
0x1006dce8: push ebx
0x1006dce9: mov [esi+28h],eax
0x1006dcec: call 100EFDD0h
0x1006dcf1: mov ecx,[esi+24h]
0x1006dcf4: add esp,0Ch
0x1006dcf7: test ecx,ecx
0x1006dcf9: mov [esi+2Ch],eax
0x1006dcfc: je near ptr 1006DD87h
0x1006dd02: cmp dword ptr [esi+28h],0
0x1006dd06: je short 1006DD87h
0x1006dd08: test eax,eax
0x1006dd0a: je short 1006DD87h
0x1006dd0c: push ebx
0x1006dd0d: push dword ptr [ebp-10h]
0x1006dd10: push ecx
0x1006dd11: call 10020A00h
0x1006dd16: push ebx
0x1006dd17: push dword ptr [ebp-14h]
0x1006dd1a: push dword ptr [esi+28h]
0x1006dd1d: call 10020A00h
0x1006dd22: push ebx
0x1006dd23: push dword ptr [ebp-18h]
0x1006dd26: push dword ptr [esi+2Ch]
0x1006dd29: call 10020A00h
0x1006dd2e: mov ebx,[ebp+0Ch]
0x1006dd31: add esp,24h
0x1006dd34: cmp word ptr [ebp+8],1
0x1006dd39: jne near ptr 1006DE96h
0x1006dd3f: movzx ecx,word ptr [esi+1Ah]
0x1006dd43: cmp cx,1
0x1006dd47: je near ptr 1006DE96h
0x1006dd4d: movzx eax,word ptr [esi+18h]
0x1006dd51: cmp ax,8
0x1006dd55: jae near ptr 1006DE96h
0x1006dd5b: movzx eax,ax
0x1006dd5e: push eax
0x1006dd5f: movzx eax,cx
0x1006dd62: push eax
```

```
0x1006dd63: movzx eax,word ptr [edi]
0x1006dd66: push 10108C4Ch
0x1006dd6b: push eax
0x1006dd6c: push 10108AA8h
0x1006dd71: push 10108BC0h
0x1006dd76: push dword ptr [ebp+14h]
0x1006dd79: call dword ptr ds:[100FE4F8h]
0x1006dd7f: add esp,1Ch
0x1006dd82: jmp near ptr 1006DF25h
0x1006dd87: push 10108EB4h
0x1006dd8c: jmp near ptr 1006DCBEh
0x1006dd91: cmp dword ptr [ebp-8],3
0x1006dd95: jge near ptr 1006DE96h
0x1006dd9b: push dword ptr [ebp-8]
0x1006dd9e: push 10108C3Ch
0x1006dda3: push 10108C10h
0x1006dda8: jmp near ptr 1006DC71h
0x1006ddad: cmp word ptr [ebp+8],1
0x1006ddb2: jne near ptr 1006DE96h
0x1006ddb8: cmp word ptr [ebp+10h],7
0x1006ddbd: jne near ptr 1006DE96h
0x1006ddc3: push 1
0x1006ddc5: push 10002h
0x1006ddca: push ebx
0x1006ddcb: call 1006796Fh
0x1006ddd0: mov word ptr [edi],2
0x1006ddd5: jmp near ptr 1006DE93h
0x1006ddda: mov ecx,eax
0x1006dddc: sub ecx,8
0x1006dddf: je near ptr 1006DE96h
0x1006dde5: sub ecx,8044h
0x1006ddeb: je short 1006DE60h
0x1006dded: dec ecx
0x1006ddee: je short 1006DE00h
0x1006ddf0: push eax
0x1006ddf1: push 10108AA8h
0x1006ddf6: push 10108B98h
0x1006ddfb: jmp near ptr 1006DC71h
0x1006de00: mov eax,8774h
0x1006de05: cmp [ebp+10h],ax
0x1006de09: je short 1006DE31h
0x1006de0b: lea ecx,[eax+1]
0x1006de0e: cmp [ebp+10h],cx
0x1006de12: je short 1006DE31h
0x1006de14: push ecx
0x1006de15: push eax
0x1006de16: push 10108B8Ch
0x1006de1b: push 10108B60h
0x1006de20: push dword ptr [ebp+14h]
0x1006de23: call dword ptr ds:[100FE4F8h]
0x1006de29: add esp,14h
0x1006de2c: jmp near ptr 1006DF25h
0x1006de31: cmp word ptr [ebp+8],1
0x1006de36: je short 1006DE4Ch
0x1006de38: movzx eax,word ptr [ebp+8]
0x1006de3c: push eax
0x1006de3d: push 10108B4Ch
0x1006de42: push 10108B1Ch
0x1006de47: jmp near ptr 1006DC71h
0x1006de4c: push 3
0x1006de4e: push 10018h
0x1006de53: push ebx
0x1006de54: call 1006796Fh
0x1006de59: mov word ptr [edi],2
0x1006de5e: jmp short 1006DE8Dh
0x1006de60: mov eax,8774h
0x1006de65: cmp [ebp+10h],ax
```

```
0x1006de69: je short 1006DE7Bh
0x1006de6b: push eax
0x1006de6c: push 10108B8Ch
0x1006de71: push 10108AF8h
0x1006de76: jmp near ptr 1006DC71h
0x1006de7b: push 3
0x1006de7d: push 10018h
0x1006de82: push ebx
0x1006de83: call 1006796Fh
0x1006de88: mov word ptr [edi],1
0x1006de8d: mov word ptr [esi+18h],8
0x1006de93: add esp,0Ch
0x1006de96: xor eax,eax
0x1006de98: mov [esi+38h],eax
0x1006de9b: mov [esi+3Ch],eax
0x1006de9e: mov [esi+40h],eax
0x1006dea1: mov [esi+44h],eax
0x1006dea4: mov [esi+48h],eax
0x1006dea7: lea eax,[esi+10h]
0x1006deaa: push eax
0x1006deab: push 100h
0x1006deb0: push ebx
0x1006deb1: call 10067985h
0x1006deb6: lea eax,[esi+14h]
0x1006deb9: push eax
0x1006deba: push 101h
0x1006debf: push ebx
0x1006dec0: call 10067985h
0x1006dec5: lea eax,[esi+1Ch]
0x1006dec8: push eax
0x1006dec9: push 112h
0x1006dece: push ebx
0x1006decf: call 1006E579h
0x1006ded4: add esp,24h
0x1006ded7: cmp word ptr [ebp+8],2
0x1006dedc: jne short 1006DEE8h
0x1006dede: cmp dword ptr [ebp-8],1
0x1006dee2: jle short 1006DEE8h
0x1006dee4: xor eax,eax
0x1006dee6: jmp short 1006DEEBh
0x1006dee8: xor eax,eax
0x1006deea: inc eax
0x1006deeb: mov [esi+8],eax
0x1006deee: test eax,eax
0x1006def0: mov eax,esi
0x1006def2: je short 1006DF07h
0x1006def4: call 1006D727h
0x1006def9: test eax,eax
0x1006defb: jne short 1006DF0Eh
0x1006defd: push 10108E98h
0x1006df02: jmp near ptr 1006DCBEh
0x1006df07: call 1006D9B2h
0x1006df0c: jmp short 1006DEF9h
0x1006df0e: xor eax,eax
0x1006df10: inc eax
0x1006df11: jmp short 1006DF27h
0x1006df13: push eax
0x1006df14: push 10108AC4h
0x1006df19: push dword ptr [ebp+14h]
0x1006df1c: call dword ptr ds:[100FE4F8h]
0x1006df22: add esp,0Ch
0x1006df25: xor eax,eax
0x1006df27: pop edi
0x1006df28: pop esi
0x1006df29: pop ebx
0x1006df2a: leave
0x1006df2b: ret
```

`sub_1006daaf` is called by sub_1006df2c. It works with the text "PhotometricInterpretation", "Sorry, can not handle images with %d-bit samples", "Sorry, LogL data must have %s=%d". It delegates to j_memcpy, sub_1006796f, sub_10067985 (8 callees).

12.990 sub_1006df2c at 0x1006df2c (70 instructions)

```
0x1006df2c: push ebp
0x1006df2d: mov ebp,esp
0x1006df2f: sub esp,458h
0x1006df35: mov eax,ds:[10136220h]
0x1006df3a: xor eax,ebp
0x1006df3c: mov [ebp-4],eax
0x1006df3f: push esi
0x1006df40: mov esi,[ebp+8]
0x1006df43: push edi
0x1006df44: mov edi,[ebp+14h]
0x1006df47: push 3FFh
0x1006df4c: lea eax,[ebp-403h]
0x1006df52: push 0
0x1006df54: push eax
0x1006df55: mov byte ptr [ebp-404h],0
0x1006df5c: call 100228C0h
0x1006df61: lea eax,[ebp-404h]
0x1006df67: push eax
0x1006df68: push esi
0x1006df69: call 10068D67h
0x1006df6e: add esp,14h
0x1006df71: test eax,eax
0x1006df73: je short 1006DFD7h
0x1006df75: lea eax,[ebp-404h]
0x1006df7b: push eax
0x1006df7c: push dword ptr [ebp+1Ch]
0x1006df7f: lea eax,[ebp-458h]
0x1006df85: push esi
0x1006df86: push eax
0x1006df87: call 1006DAAFh
0x1006df8c: add esp,10h
0x1006df8f: test eax,eax
0x1006df91: je short 1006DFD7h
0x1006df93: mov ax,[ebp+18h]
0x1006df97: push dword ptr [ebp-444h]
0x1006df9d: mov [ebp-43Ah],ax
0x1006dfa4: mov eax,[ebp+10h]
0x1006dfa7: sub eax,[ebp-444h]
0x1006dfad: push dword ptr [ebp+0Ch]
0x1006dfb0: imul eax,[ebp+0Ch]
0x1006dfb4: lea eax,[edi+eax*4]
0x1006dfb7: push eax
0x1006dfb8: lea eax,[ebp-458h]
0x1006dfbe: push eax
0x1006dfbf: call 10069057h
0x1006dfc4: mov esi,eax
0x1006dfc6: lea eax,[ebp-458h]
0x1006dfcc: push eax
0x1006dfcd: call 10068F9Ah
0x1006dfd2: add esp,14h
0x1006dfd5: jmp short 1006DFFBh
0x1006dfd7: lea eax,[ebp-404h]
0x1006dfdd: push eax
0x1006dfde: push 10108EF4h
0x1006dfe3: push esi
0x1006dfe4: call 10068D52h
0x1006dfe9: pop ecx
0x1006dfea: push eax
0x1006dfeb: push dword ptr [esi+274h]
0x1006dff1: call 1006FAF1h
0x1006dff6: add esp,10h
```

```
0x1006dff9: xor esi,esi
0x1006dffb: mov ecx,[ebp-4]
0x1006dffe: pop edi
0x1006dfff: mov eax,esi
0x1006e001: xor ecx,ebp
0x1006e003: pop esi
0x1006e004: call 10022326h
0x1006e009: leave
0x1006e00a: ret
```

`sub_1006df2c` is called by sub_1006e00b. It delegates to sub_10022326, j_memset, sub_10068d52 (8 callees).

12.991 sub_1006e00b at 0x1006e00b (12 instructions)

```
0x1006e00b: push ebp
0x1006e00c: mov ebp,esp
0x1006e00e: push dword ptr [ebp+18h]
0x1006e011: push 4
0x1006e013: push dword ptr [ebp+14h]
0x1006e016: push dword ptr [ebp+10h]
0x1006e019: push dword ptr [ebp+0Ch]
0x1006e01c: push dword ptr [ebp+8]
0x1006e01f: call 1006DF2Ch
0x1006e024: add esp,18h
0x1006e027: pop ebp
0x1006e028: ret
```

`sub_1006e00b` is called by sub_100143f0. It delegates to sub_1006df2c (1 callees).

12.992 sub_1006e029 at 0x1006e029 (42 instructions)

```
0x1006e029: push ebp
0x1006e02a: mov ebp,esp
0x1006e02c: push ebx
0x1006e02d: push esi
0x1006e02e: mov esi,[ebp+10h]
0x1006e031: push edi
0x1006e032: mov edi,[ebp+14h]
0x1006e035: mov ecx,esi
0x1006e037: xor ebx,ebx
0x1006e039: imul ecx,edi
0x1006e03c: test esi,esi
0x1006e03e: je short 1006E05Fh
0x1006e040: test edi,edi
0x1006e042: je short 1006E05Fh
0x1006e044: xor edx,edx
0x1006e046: mov eax,ecx
0x1006e048: div edi
0x1006e04a: cmp eax,esi
0x1006e04c: jne short 1006E05Fh
0x1006e04e: push ecx
0x1006e04f: push dword ptr [ebp+0Ch]
0x1006e052: call 100AD4B0h
0x1006e057: mov ebx,eax
0x1006e059: test ebx,ebx
0x1006e05b: pop ecx
0x1006e05c: pop ecx
0x1006e05d: jne short 1006E07Ch
0x1006e05f: mov eax,[ebp+8]
0x1006e062: push edi
0x1006e063: push esi
0x1006e064: push dword ptr [ebp+18h]
0x1006e067: push 10108EF8h
0x1006e06c: push dword ptr [eax]
0x1006e06e: push dword ptr [eax+274h]
```

```
0x1006e074: call 1006FAF1h
0x1006e079: add esp,18h
0x1006e07c: pop edi
0x1006e07d: pop esi
0x1006e07e: mov eax,ebx
0x1006e080: pop ebx
0x1006e081: pop ebp
0x1006e082: ret
```

`sub_1006e029` is called by sub_1006e083, sub_1006ff15, sub_100704ba. It delegates to sub_1006faf1, j_realloc (2 callees).

12.993 sub_1006e083 at 0x1006e083 (8 instructions)

```
0x1006e083: push dword ptr [esp+10h]
0x1006e087: push dword ptr [esp+10h]
0x1006e08b: push dword ptr [esp+10h]
0x1006e08f: push 0
0x1006e091: push dword ptr [esp+14h]
0x1006e095: call 1006E029h
0x1006e09a: add esp,14h
0x1006e09d: ret
```

`sub_1006e083` is called by sub_1006ff15, sub_100702f4, sub_10070584, sub_10071192, sub_1007127c, sub_10071346, sub_10071411, sub_10071523, sub_100715da, sub_1007171b. It delegates to sub_1006e029 (1 callees).

12.994 sub_1006e09e at 0x1006e09e (111 instructions)

```
0x1006e09e: push ebp
0x1006e09f: mov ebp,esp
0x1006e0a1: xor eax,eax
0x1006e0a3: mov [esi+0B8h],eax
0x1006e0a9: mov [esi+0B4h],eax
0x1006e0af: mov [esi+0B0h],eax
0x1006e0b5: movzx ecx,word ptr [esi+2Ch]
0x1006e0b9: sub esp,0Ch
0x1006e0bc: cmp cx,1Eh
0x1006e0c0: jb short 1006E0C4h
0x1006e0c2: leave
0x1006e0c3: ret
0x1006e0c4: push ebx
0x1006e0c5: push edi
0x1006e0c6: xor edi,edi
0x1006e0c8: inc edi
0x1006e0c9: shl edi,cl
0x1006e0cb: lea ebx,[edi+edi]
0x1006e0ce: push ebx
0x1006e0cf: mov [ebp-8],edi
0x1006e0d2: call 100EFDD0h
0x1006e0d7: test eax,eax
0x1006e0d9: pop ecx
0x1006e0da: mov [esi+0B0h],eax
0x1006e0e0: je near ptr 1006E1DDh
0x1006e0e6: and word ptr [eax],0
0x1006e0ea: xor eax,eax
0x1006e0ec: inc eax
0x1006e0ed: cmp edi,eax
0x1006e0ef: mov [ebp-4],eax
0x1006e0f2: jle short 1006E14Fh
0x1006e0f4: fild dword ptr [ebp-8]
0x1006e0f7: fsub qword ptr ds:[100FE8F8h]
0x1006e0fd: fstp qword ptr [ebp-0Ch]
0x1006e100: fld qword ptr ds:[10108F40h]
0x1006e106: sub esp,10h
0x1006e109: fstp qword ptr [esp+8]
```

```
0x1006e10d: fild dword ptr [ebp-4]
0x1006e110: fdiv qword ptr [ebp-0Ch]
0x1006e113: fstp qword ptr [esp]
0x1006e116: call 100FC11Ah
0x1006e11b: fmul qword ptr ds:[100FFBA0h]
0x1006e121: add esp,8
0x1006e124: fadd qword ptr ds:[100FE7C8h]
0x1006e12a: fstp qword ptr [esp]
0x1006e12d: call dword ptr ds:[100FE548h]
0x1006e133: pop ecx
0x1006e134: pop ecx
0x1006e135: call 10022906h
0x1006e13a: mov edx,[ebp-4]
0x1006e13d: inc dword ptr [ebp-4]
0x1006e140: cmp [ebp-4],edi
0x1006e143: mov ecx,[esi+0B0h]
0x1006e149: mov [ecx+edx*2],ax
0x1006e14d: jl short 1006E100h
0x1006e14f: movzx eax,word ptr [esi+80h]
0x1006e156: movzx ecx,word ptr [esi+3Ah]
0x1006e15a: sub ecx,eax
0x1006e15c: cmp ecx,1
0x1006e15f: jle near ptr 1006E1F1h
0x1006e165: push ebx
0x1006e166: call 100EFDD0h
0x1006e16b: xor edi,edi
0x1006e16d: cmp eax,edi
0x1006e16f: pop ecx
0x1006e170: mov [esi+0B4h],eax
0x1006e176: je short 1006E198h
0x1006e178: push ebx
0x1006e179: push dword ptr [esi+0B0h]
0x1006e17f: push eax
0x1006e180: call 10020A00h
0x1006e185: push ebx
0x1006e186: call 100EFDD0h
0x1006e18b: add esp,10h
0x1006e18e: cmp eax,edi
0x1006e190: mov [esi+0B8h],eax
0x1006e196: jne short 1006E1E1h
0x1006e198: mov eax,[esi+0B0h]
0x1006e19e: cmp eax,edi
0x1006e1a0: je short 1006E1A9h
0x1006e1a2: push eax
0x1006e1a3: call 100209E0h
0x1006e1a8: pop ecx
0x1006e1a9: mov eax,[esi+0B4h]
0x1006e1af: cmp eax,edi
0x1006e1b1: je short 1006E1BAh
0x1006e1b3: push eax
0x1006e1b4: call 100209E0h
0x1006e1b9: pop ecx
0x1006e1ba: mov eax,[esi+0B8h]
0x1006e1c0: cmp eax,edi
0x1006e1c2: je short 1006E1CBh
0x1006e1c4: push eax
0x1006e1c5: call 100209E0h
0x1006e1ca: pop ecx
0x1006e1cb: mov [esi+0B8h],edi
0x1006e1d1: mov [esi+0B4h],edi
0x1006e1d7: mov [esi+0B0h],edi
0x1006e1dd: xor eax,eax
0x1006e1df: jmp short 1006E1F4h
0x1006e1e1: push ebx
0x1006e1e2: push dword ptr [esi+0B0h]
0x1006e1e8: push eax
0x1006e1e9: call 10020A00h
```

```
0x1006e1ee: add esp,0Ch
0x1006e1f1: xor eax,eax
0x1006e1f3: inc eax
0x1006e1f4: pop edi
0x1006e1f5: pop ebx
0x1006e1f6: leave
0x1006e1f7: ret
```

`sub_1006e09e` is called by sub_1006e289. It delegates to j_free, j_memcpy, sub_10022906 (5 callees).

12.995 sub_1006e1f8 at 0x1006e1f8 (48 instructions)

```
0x1006e1f8: push ebp
0x1006e1f9: mov ebp,esp
0x1006e1fb: push ecx
0x1006e1fc: push 18h
0x1006e1fe: call 100EFDD0h
0x1006e203: test eax,eax
0x1006e205: pop ecx
0x1006e206: mov [esi+0ACh],eax
0x1006e20c: jne short 1006E210h
0x1006e20e: leave
0x1006e20f: ret
0x1006e210: cmp word ptr [esi+32h],6
0x1006e215: jne short 1006E256h
0x1006e217: fldz
0x1006e219: fstp dword ptr [eax]
0x1006e21b: mov eax,[esi+0ACh]
0x1006e221: fld dword ptr ds:[100FF364h]
0x1006e227: fst dword ptr [eax+14h]
0x1006e22a: mov eax,[esi+0ACh]
0x1006e230: fst dword ptr [eax+0Ch]
0x1006e233: mov eax,[esi+0ACh]
0x1006e239: fstp dword ptr [eax+4]
0x1006e23c: mov eax,[esi+0ACh]
0x1006e242: fld dword ptr ds:[10108F48h]
0x1006e248: fst dword ptr [eax+10h]
0x1006e24b: mov eax,[esi+0ACh]
0x1006e251: fstp dword ptr [eax+8]
0x1006e254: jmp short 1006E284h
0x1006e256: xor eax,eax
0x1006e258: mov ecx,[esi+0ACh]
0x1006e25e: fldz
0x1006e260: fstp dword ptr [eax+ecx]
0x1006e263: xor edx,edx
0x1006e265: mov cl,[esi+2Ch]
0x1006e268: inc edx
0x1006e269: shl edx,cl
0x1006e26b: mov ecx,[esi+0ACh]
0x1006e271: add eax,8
0x1006e274: dec edx
0x1006e275: cmp eax,18h
0x1006e278: mov [ebp-4],edx
0x1006e27b: fild dword ptr [ebp-4]
0x1006e27e: fstp dword ptr [eax+ecx-4]
0x1006e282: jl short 1006E258h
0x1006e284: xor eax,eax
0x1006e286: inc eax
0x1006e287: leave
0x1006e288: ret
```

`sub_1006e1f8` is called by sub_1006e289. It delegates to j_malloc (1 callees).

12.996 sub_1006e289 at 0x1006e289 (222 instructions)

```
0x1006e289: push ebp
0x1006e28a: mov ebp,esp
0x1006e28c: mov eax,[ebp+8]
0x1006e28f: push ebx
0x1006e290: mov ebx,[ebp+0Ch]
0x1006e293: push esi
0x1006e294: push edi
0x1006e295: mov edi,[ebp+10h]
0x1006e298: push edi
0x1006e299: push ebx
0x1006e29a: push eax
0x1006e29b: lea esi,[eax+28h]
0x1006e29e: call 100675C7h
0x1006e2a3: add esp,0Ch
0x1006e2a6: test eax,eax
0x1006e2a8: jne near ptr 1006E571h
0x1006e2ae: mov eax,13Eh
0x1006e2b3: cmp ebx,eax
0x1006e2b5: ja near ptr 1006E42Dh
0x1006e2bb: je near ptr 1006E408h
0x1006e2c1: add eax,0FFFFFFD8h
0x1006e2c4: cmp ebx,eax
0x1006e2c6: ja short 1006E32Dh
0x1006e2c8: je short 1006E328h
0x1006e2ca: sub ebx,0FEh
0x1006e2d0: je short 1006E31Ch
0x1006e2d2: sub ebx,4
0x1006e2d5: je short 1006E313h
0x1006e2d7: sub ebx,5
0x1006e2da: je short 1006E30Ah
0x1006e2dc: sub ebx,3
0x1006e2df: je short 1006E301h
0x1006e2e1: sub ebx,8
0x1006e2e4: je short 1006E2F8h
0x1006e2e6: sub ebx,3
0x1006e2e9: jne near ptr 1006E39Ch
0x1006e2ef: mov cx,[esi+3Ah]
0x1006e2f3: jmp near ptr 1006E56Ch
0x1006e2f8: mov cx,[esi+38h]
0x1006e2fc: jmp near ptr 1006E56Ch
0x1006e301: mov cx,[esi+36h]
0x1006e305: jmp near ptr 1006E56Ch
0x1006e30a: mov cx,[esi+34h]
0x1006e30e: jmp near ptr 1006E56Ch
0x1006e313: mov cx,[esi+2Ch]
0x1006e317: jmp near ptr 1006E56Ch
0x1006e31c: mov ecx,[esi+28h]
0x1006e31f: mov eax,[edi]
0x1006e321: mov [eax],ecx
0x1006e323: jmp near ptr 1006E571h
0x1006e328: mov ecx,[esi+3Ch]
0x1006e32b: jmp short 1006E31Fh
0x1006e32d: sub ebx,118h
0x1006e333: je near ptr 1006E3FFh
0x1006e339: dec ebx
0x1006e33a: je near ptr 1006E3F6h
0x1006e340: sub ebx,3
0x1006e343: je near ptr 1006E3EDh
0x1006e349: sub ebx,0Ch
0x1006e34c: je near ptr 1006E3E4h
0x1006e352: sub ebx,5
0x1006e355: je short 1006E372h
0x1006e357: sub ebx,10h
0x1006e35a: jne short 1006E39Ch
0x1006e35c: mov eax,[ebp+8]
0x1006e35f: mov eax,[eax+248h]
```

```
0x1006e365: mov ax,[eax]
0x1006e368: mov ecx,[edi]
0x1006e36a: mov [ecx],ax
0x1006e36d: jmp near ptr 1006E571h
0x1006e372: cmp dword ptr [esi+0B0h],0
0x1006e379: jne short 1006E3A3h
0x1006e37b: call 1006E09Eh
0x1006e380: test eax,eax
0x1006e382: jne short 1006E3A3h
0x1006e384: mov eax,[ebp+8]
0x1006e387: push 10108F54h
0x1006e38c: push dword ptr [eax]
0x1006e38e: push dword ptr [eax+274h]
0x1006e394: call 1006FAF1h
0x1006e399: add esp,0Ch
0x1006e39c: xor eax,eax
0x1006e39e: jmp near ptr 1006E574h
0x1006e3a3: mov eax,[edi]
0x1006e3a5: mov ecx,[esi+0B0h]
0x1006e3ab: add edi,4
0x1006e3ae: mov [eax],ecx
0x1006e3b0: movzx eax,word ptr [esi+80h]
0x1006e3b7: movzx ecx,word ptr [esi+3Ah]
0x1006e3bb: sub ecx,eax
0x1006e3bd: xor eax,eax
0x1006e3bf: inc eax
0x1006e3c0: cmp ecx,eax
0x1006e3c2: jle near ptr 1006E574h
0x1006e3c8: mov ecx,[edi]
0x1006e3ca: mov edx,[esi+0B4h]
0x1006e3d0: add edi,4
0x1006e3d3: mov [ecx],edx
0x1006e3d5: mov ecx,[edi]
0x1006e3d7: mov edx,[esi+0B8h]
0x1006e3dd: mov [ecx],edx
0x1006e3df: jmp near ptr 1006E574h
0x1006e3e4: mov cx,[esi+60h]
0x1006e3e8: jmp near ptr 1006E56Ch
0x1006e3ed: mov cx,[esi+62h]
0x1006e3f1: jmp near ptr 1006E56Ch
0x1006e3f6: mov cx,[esi+42h]
0x1006e3fa: jmp near ptr 1006E56Ch
0x1006e3ff: mov cx,[esi+40h]
0x1006e403: jmp near ptr 1006E56Ch
0x1006e408: fld dword ptr ds:[10108F50h]
0x1006e40e: fstp dword ptr ds:[101393A4h]
0x1006e414: fld dword ptr ds:[10108F4Ch]
0x1006e41a: fstp dword ptr ds:[101393A8h]
0x1006e420: mov eax,[edi]
0x1006e422: mov dword ptr [eax],101393A4h
0x1006e428: jmp near ptr 1006E571h
0x1006e42d: mov eax,212h
0x1006e432: cmp ebx,eax
0x1006e434: ja near ptr 1006E4E5h
0x1006e43a: je near ptr 1006E4CAh
0x1006e440: sub ebx,14Ch
0x1006e446: je short 1006E4BDh
0x1006e448: dec ebx
0x1006e449: dec ebx
0x1006e44a: je short 1006E4B1h
0x1006e44c: dec ebx
0x1006e44d: dec ebx
0x1006e44e: je short 1006E493h
0x1006e450: dec ebx
0x1006e451: dec ebx
0x1006e452: je short 1006E479h
0x1006e454: dec ebx
```

```
0x1006e455: je short 1006E470h
0x1006e457: sub ebx,0BEh
0x1006e45d: jne near ptr 1006E39Ch
0x1006e463: mov eax,[edi]
0x1006e465: mov dword ptr [eax],101362F8h
0x1006e46b: jmp near ptr 1006E571h
0x1006e470: mov cx,[esi+2Eh]
0x1006e474: jmp near ptr 1006E56Ch
0x1006e479: mov cx,[esi+80h]
0x1006e480: mov eax,[edi]
0x1006e482: add edi,4
0x1006e485: mov [eax],cx
0x1006e488: mov ecx,[esi+84h]
0x1006e48e: jmp near ptr 1006E31Fh
0x1006e493: mov eax,[edi]
0x1006e495: and word ptr [eax],0
0x1006e499: mov cl,[esi+2Ch]
0x1006e49c: add edi,4
0x1006e49f: xor eax,eax
0x1006e4a1: inc eax
0x1006e4a2: mov edx,eax
0x1006e4a4: shl edx,cl
0x1006e4a6: mov ecx,[edi]
0x1006e4a8: dec edx
0x1006e4a9: mov [ecx],dx
0x1006e4ac: jmp near ptr 1006E574h
0x1006e4b1: mov eax,[edi]
0x1006e4b3: mov word ptr [eax],4
0x1006e4b8: jmp near ptr 1006E571h
0x1006e4bd: mov ecx,[edi]
0x1006e4bf: xor eax,eax
0x1006e4c1: inc eax
0x1006e4c2: mov [ecx],ax
0x1006e4c5: jmp near ptr 1006E574h
0x1006e4ca: mov cx,[esi+0A4h]
0x1006e4d1: mov eax,[edi]
0x1006e4d3: add edi,4
0x1006e4d6: mov [eax],cx
0x1006e4d9: mov cx,[esi+0A6h]
0x1006e4e0: jmp near ptr 1006E56Ch
0x1006e4e5: sub ebx,213h
0x1006e4eb: je short 1006E565h
0x1006e4ed: dec ebx
0x1006e4ee: je short 1006E544h
0x1006e4f0: sub ebx,7ECFh
0x1006e4f6: je short 1006E520h
0x1006e4f8: dec ebx
0x1006e4f9: je short 1006E515h
0x1006e4fb: dec ebx
0x1006e4fc: je short 1006E50Dh
0x1006e4fe: dec ebx
0x1006e4ff: jne near ptr 1006E39Ch
0x1006e505: mov ecx,[esi+24h]
0x1006e508: jmp near ptr 1006E31Fh
0x1006e50d: mov ecx,[esi+18h]
0x1006e510: jmp near ptr 1006E31Fh
0x1006e515: mov ax,[esi+2Eh]
0x1006e519: dec ax
0x1006e51b: jmp near ptr 1006E368h
0x1006e520: xor eax,eax
0x1006e522: inc eax
0x1006e523: cmp [esi+80h],ax
0x1006e52a: jne short 1006E53Bh
0x1006e52c: mov ecx,[esi+84h]
0x1006e532: cmp [ecx],ax
0x1006e535: jne short 1006E53Bh
0x1006e537: mov ecx,eax
```

```
0x1006e539: jmp short 1006E53Dh
0x1006e53b: xor ecx,ecx
0x1006e53d: mov edx,[edi]
0x1006e53f: mov [edx],cx
0x1006e542: jmp short 1006E574h
0x1006e544: cmp dword ptr [esi+0ACh],0
0x1006e54b: jne short 1006E55Ah
0x1006e54d: call 1006E1F8h
0x1006e552: test eax,eax
0x1006e554: je near ptr 1006E39Ch
0x1006e55a: mov ecx,[esi+0ACh]
0x1006e560: jmp near ptr 1006E31Fh
0x1006e565: mov cx,[esi+0A8h]
0x1006e56c: mov eax,[edi]
0x1006e56e: mov [eax],cx
0x1006e571: xor eax,eax
0x1006e573: inc eax
0x1006e574: pop edi
0x1006e575: pop esi
0x1006e576: pop ebx
0x1006e577: pop ebp
0x1006e578: ret
```

`sub_1006e289` is called by sub_1006e579. It delegates to sub_100675c7, sub_1006e09e, sub_1006e1f8 (4 callees).

12.997 sub_1006e579 at 0x1006e579 (7 instructions)

```
0x1006e579: lea eax,[esp+0Ch]
0x1006e57d: push eax
0x1006e57e: push dword ptr [esp+0Ch]
0x1006e582: push dword ptr [esp+0Ch]
0x1006e586: call 1006E289h
0x1006e58b: add esp,0Ch
0x1006e58e: ret
```

`sub_1006e579` is called by sub_100143f0, sub_10068d67, sub_1006cec2, sub_1006cf39, sub_1006d727, sub_1006d9b2, sub_1006daaf. It delegates to sub_1006e289 (1 callees).

12.998 sub_1006e58f at 0x1006e58f (19 instructions)

```
0x1006e58f: cmp esi,0FFFFh
0x1006e595: mov ax,[esp+4]
0x1006e59a: mov [edx],ax
0x1006e59d: mov dword ptr [edx+4],1
0x1006e5a4: jbe short 1006E5B0h
0x1006e5a6: mov word ptr [edx+2],4
0x1006e5ac: mov [edx+8],esi
0x1006e5af: ret
0x1006e5b0: mov word ptr [edx+2],3
0x1006e5b6: mov eax,[ecx+1D4h]
0x1006e5bc: mov eax,[eax+0Ch]
0x1006e5bf: and eax,esi
0x1006e5c1: cmp word ptr [ecx+1C8h],4D4Dh
0x1006e5ca: jne short 1006E5D7h
0x1006e5cc: mov ecx,[ecx+1D0h]
0x1006e5d2: mov ecx,[ecx+0Ch]
0x1006e5d5: shl eax,cl
0x1006e5d7: mov [edx+8],eax
0x1006e5da: ret
```

`sub_1006e58f` is called by sub_1006f391. It calls nothing else.

12.999 sub_1006e5db at 0x1006e5db (20 instructions)

```
0x1006e5db: mov ax,[esp+4]
0x1006e5e0: mov [edx],ax
0x1006e5e3: mov dword ptr [edx+4],1
0x1006e5ea: mov word ptr [edx+2],3
0x1006e5f0: cmp word ptr [ecx+1C8h],4D4Dh
0x1006e5f9: mov eax,[ecx+1D4h]
0x1006e5ff: mov eax,[eax+0Ch]
0x1006e602: jne short 1006E61Ah
0x1006e604: mov ecx,[ecx+1D0h]
0x1006e60a: mov ecx,[ecx+0Ch]
0x1006e60d: push esi
0x1006e60e: movzx esi,word ptr [esp+0Ch]
0x1006e613: and eax,esi
0x1006e615: shl eax,cl
0x1006e617: pop esi
0x1006e618: jmp short 1006E621h
0x1006e61a: movzx ecx,word ptr [esp+8]
0x1006e61f: and eax,ecx
0x1006e621: mov [edx+8],eax
0x1006e624: ret
```

`sub_1006e5db` is called by sub_1006f391. It calls nothing else.

12.1000 sub_1006e625 at 0x1006e625 (52 instructions)

```
0x1006e625: push ebp
0x1006e626: mov ebp,esp
0x1006e628: test byte ptr [edi+0Ch],80h
0x1006e62c: je short 1006E677h
0x1006e62e: movzx eax,word ptr [esi+2]
0x1006e632: add eax,0FFFFFFFDh
0x1006e635: cmp eax,9
0x1006e638: ja short 1006E677h
0x1006e63a: movzx eax,byte ptr [eax+1006E70Dh]
0x1006e641: jmp dword ptr [eax*4+1006E6F9h]
0x1006e677: mov eax,[edi+1E8h]
0x1006e67d: mov [esi+8],eax
0x1006e680: movzx eax,word ptr [esi+2]
0x1006e684: push ebx
0x1006e685: push eax
0x1006e686: call 1006FCE1h
0x1006e68b: push 0
0x1006e68d: push dword ptr [esi+8]
0x1006e690: mov ebx,eax
0x1006e692: push dword ptr [edi+274h]
0x1006e698: imul ebx,[esi+4]
0x1006e69c: call dword ptr [edi+280h]
0x1006e6a2: add esp,10h
0x1006e6a5: cmp eax,[esi+8]
0x1006e6a8: jne short 1006E6D0h
0x1006e6aa: push ebx
0x1006e6ab: push dword ptr [ebp+8]
0x1006e6ae: push dword ptr [edi+274h]
0x1006e6b4: call dword ptr [edi+27Ch]
0x1006e6ba: add esp,0Ch
0x1006e6bd: cmp eax,ebx
0x1006e6bf: jne short 1006E6D0h
0x1006e6c1: inc ebx
0x1006e6c2: and ebx,0FFFFFFFEh
0x1006e6c5: add [edi+1E8h],ebx
0x1006e6cb: xor eax,eax
0x1006e6cd: inc eax
0x1006e6ce: jmp short 1006E6F4h
0x1006e6d0: movzx eax,word ptr [esi]
0x1006e6d3: push eax
```

```
0x1006e6d4: push edi
0x1006e6d5: call 1006FE65h
0x1006e6da: push dword ptr [eax+10h]
0x1006e6dd: push 10109010h
0x1006e6e2: push dword ptr [edi]
0x1006e6e4: push dword ptr [edi+274h]
0x1006e6ea: call 1006FAF1h
0x1006e6ef: add esp,18h
0x1006e6f2: xor eax,eax
0x1006e6f4: pop ebx
0x1006e6f5: pop ebp
0x1006e6f6: ret
```

`sub_1006e625` is called by sub_1006e90e, sub_1006e977, sub_1006ea05, sub_1006ea55, sub_1006eb83, sub_1006ee28. It delegates to sub_1006faf1, sub_1006fce1, sub_1006fe65 (3 callees).

12.1001 sub_1006e717 at 0x1006e717 (147 instructions)

```
0x1006e717: push ebp
0x1006e718: mov ebp,esp
0x1006e71a: sub esp,0Ch
0x1006e71d: push ebx
0x1006e71e: push 2
0x1006e720: xor ebx,ebx
0x1006e722: push ebx
0x1006e723: push dword ptr [esi+274h]
0x1006e729: call dword ptr [esi+280h]
0x1006e72f: add esp,0Ch
0x1006e732: inc eax
0x1006e733: and eax,0FFFFFFFEh
0x1006e736: test byte ptr [esi+0Ch],80h
0x1006e73a: mov [esi+10h],eax
0x1006e73d: mov [ebp-0Ch],eax
0x1006e740: je short 1006E74Ch
0x1006e742: lea eax,[ebp-0Ch]
0x1006e745: push eax
0x1006e746: call 1006FB38h
0x1006e74b: pop ecx
0x1006e74c: test word ptr [esi+0Ch],2000h
0x1006e752: je short 1006E7C9h
0x1006e754: push ebx
0x1006e755: push dword ptr [esi+1F0h]
0x1006e75b: push dword ptr [esi+274h]
0x1006e761: call dword ptr [esi+280h]
0x1006e767: push 4
0x1006e769: lea eax,[ebp-0Ch]
0x1006e76c: push eax
0x1006e76d: push dword ptr [esi+274h]
0x1006e773: call dword ptr [esi+27Ch]
0x1006e779: add esp,18h
0x1006e77c: cmp eax,4
0x1006e77f: je short 1006E7A0h
0x1006e781: push dword ptr [esi]
0x1006e783: push 10109074h
0x1006e788: push 10108F90h
0x1006e78d: push dword ptr [esi+274h]
0x1006e793: call 1006FAF1h
0x1006e798: add esp,10h
0x1006e79b: jmp near ptr 1006E909h
0x1006e7a0: lea eax,[esi+1ECh]
0x1006e7a6: dec word ptr [eax]
0x1006e7a9: movzx eax,word ptr [eax]
0x1006e7ac: cmp ax,bx
0x1006e7af: je short 1006E7BAh
0x1006e7b1: add dword ptr [esi+1F0h],4
0x1006e7b8: jmp short 1006E7C1h
```

```
0x1006e7ba: and dword ptr [esi+0Ch],0FFFFDFFFh
0x1006e7c1: xor eax,eax
0x1006e7c3: inc eax
0x1006e7c4: jmp near ptr 1006E90Bh
0x1006e7c9: lea eax,[esi+1CCh]
0x1006e7cf: mov ecx,[eax]
0x1006e7d1: cmp ecx,ebx
0x1006e7d3: jne short 1006E80Fh
0x1006e7d5: mov ecx,[esi+10h]
0x1006e7d8: push ebx
0x1006e7d9: push 4
0x1006e7db: push dword ptr [esi+274h]
0x1006e7e1: mov [eax],ecx
0x1006e7e3: call dword ptr [esi+280h]
0x1006e7e9: push 4
0x1006e7eb: lea eax,[ebp-0Ch]
0x1006e7ee: push eax
0x1006e7ef: push dword ptr [esi+274h]
0x1006e7f5: call dword ptr [esi+27Ch]
0x1006e7fb: add esp,18h
0x1006e7fe: cmp eax,4
0x1006e801: je short 1006E7C1h
0x1006e803: push 101089A4h
0x1006e808: push dword ptr [esi]
0x1006e80a: jmp near ptr 1006E8FBh
0x1006e80f: mov [ebp-8],ecx
0x1006e812: push ebx
0x1006e813: push dword ptr [ebp-8]
0x1006e816: push dword ptr [esi+274h]
0x1006e81c: call dword ptr [esi+280h]
0x1006e822: add esp,0Ch
0x1006e825: cmp eax,[ebp-8]
0x1006e828: jne near ptr 1006E8F1h
0x1006e82e: push 2
0x1006e830: lea eax,[ebp-4]
0x1006e833: push eax
0x1006e834: push dword ptr [esi+274h]
0x1006e83a: call dword ptr [esi+278h]
0x1006e840: add esp,0Ch
0x1006e843: cmp eax,2
0x1006e846: jne near ptr 1006E8F1h
0x1006e84c: test byte ptr [esi+0Ch],80h
0x1006e850: je short 1006E85Ch
0x1006e852: lea eax,[ebp-4]
0x1006e855: push eax
0x1006e856: call 1006FB29h
0x1006e85b: pop ecx
0x1006e85c: movzx eax,word ptr [ebp-4]
0x1006e860: imul eax,0Ch
0x1006e863: push 1
0x1006e865: push eax
0x1006e866: push dword ptr [esi+274h]
0x1006e86c: call dword ptr [esi+280h]
0x1006e872: push 4
0x1006e874: lea eax,[ebp-8]
0x1006e877: push eax
0x1006e878: push dword ptr [esi+274h]
0x1006e87e: call dword ptr [esi+278h]
0x1006e884: add esp,18h
0x1006e887: cmp eax,4
0x1006e88a: jne short 1006E8EAh
0x1006e88c: test byte ptr [esi+0Ch],80h
0x1006e890: je short 1006E89Ch
0x1006e892: lea eax,[ebp-8]
0x1006e895: push eax
0x1006e896: call 1006FB38h
0x1006e89b: pop ecx
```

```
0x1006e89c: cmp [ebp-8],ebx
0x1006e89f: jne near ptr 1006E812h
0x1006e8a5: push 1
0x1006e8a7: push ebx
0x1006e8a8: push dword ptr [esi+274h]
0x1006e8ae: call dword ptr [esi+280h]
0x1006e8b4: push ebx
0x1006e8b5: add eax,0FFFFFFFCh
0x1006e8b8: push eax
0x1006e8b9: push dword ptr [esi+274h]
0x1006e8bf: call dword ptr [esi+280h]
0x1006e8c5: push 4
0x1006e8c7: lea eax,[ebp-0Ch]
0x1006e8ca: push eax
0x1006e8cb: push dword ptr [esi+274h]
0x1006e8d1: call dword ptr [esi+27Ch]
0x1006e8d7: add esp,24h
0x1006e8da: cmp eax,4
0x1006e8dd: je near ptr 1006E7C1h
0x1006e8e3: push 10108464h
0x1006e8e8: jmp short 1006E8F6h
0x1006e8ea: push 10109054h
0x1006e8ef: jmp short 1006E8F6h
0x1006e8f1: push 10109034h
0x1006e8f6: push 10108F90h
0x1006e8fb: push dword ptr [esi+274h]
0x1006e901: call 1006FAF1h
0x1006e906: add esp,0Ch
0x1006e909: xor eax,eax
0x1006e90b: pop ebx
0x1006e90c: leave
0x1006e90d: ret
```

`sub_1006e717` is called by sub_1006f391. It works with the text "Error writing directory link", "Error writing TIFF header", "TIFFLinkDirectory". It delegates to sub_1006faf1, sub_1006fb29, sub_1006fb38 (3 callees).

12.1002 sub_1006e90e at 0x1006e90e (43 instructions)

```
0x1006e90e: push ebp
0x1006e90f: mov ebp,esp
0x1006e911: push ebx
0x1006e912: push esi
0x1006e913: mov esi,eax
0x1006e915: mov ax,[ebp+8]
0x1006e919: and dword ptr [ebp+8],0
0x1006e91d: mov [esi],ax
0x1006e920: xor eax,eax
0x1006e922: push edi
0x1006e923: mov edi,ecx
0x1006e925: mov word ptr [esi+2],3
0x1006e92b: mov cl,[edi+54h]
0x1006e92e: inc eax
0x1006e92f: shl eax,cl
0x1006e931: cmp dword ptr [ebp+0Ch],0
0x1006e935: mov [esi+4],eax
0x1006e938: mov ebx,[edi+1E8h]
0x1006e93e: jbe short 1006E95Eh
0x1006e940: mov eax,[ebp+10h]
0x1006e943: mov ecx,[ebp+8]
0x1006e946: push dword ptr [eax+ecx*4]
0x1006e949: call 1006E625h
0x1006e94e: test eax,eax
0x1006e950: pop ecx
0x1006e951: je short 1006E973h
0x1006e953: inc dword ptr [ebp+8]
0x1006e956: mov eax,[ebp+8]
```

```
0x1006e959: cmp eax,[ebp+0Ch]
0x1006e95c: jb short 1006E940h
0x1006e95e: mov eax,[esi+4]
0x1006e961: imul eax,[ebp+0Ch]
0x1006e965: mov [esi+4],eax
0x1006e968: xor eax,eax
0x1006e96a: mov [esi+8],ebx
0x1006e96d: inc eax
0x1006e96e: pop edi
0x1006e96f: pop esi
0x1006e970: pop ebx
0x1006e971: pop ebp
0x1006e972: ret
0x1006e973: xor eax,eax
0x1006e975: jmp short 1006E96Eh
```

`sub_1006e90e` is called by sub_1006edb9, sub_1006f391. It delegates to sub_1006e625 (1 callees).

12.1003 sub_1006e977 at 0x1006e977 (53 instructions)

```
0x1006e977: push esi
0x1006e978: push edi
0x1006e979: mov esi,ecx
0x1006e97b: mov edi,edx
0x1006e97d: mov edx,[esi+4]
0x1006e980: cmp edx,4
0x1006e983: ja short 1006E9FBh
0x1006e985: cmp word ptr [edi+1C8h],4D4Dh
0x1006e98e: movsx ecx,byte ptr [eax]
0x1006e991: jne short 1006E9C4h
0x1006e993: shl ecx,18h
0x1006e996: cmp edx,2
0x1006e999: mov [esi+8],ecx
0x1006e99c: jb short 1006E9AAh
0x1006e99e: movsx edi,byte ptr [eax+1]
0x1006e9a2: shl edi,10h
0x1006e9a5: or edi,ecx
0x1006e9a7: mov [esi+8],edi
0x1006e9aa: cmp edx,3
0x1006e9ad: jb short 1006E9B9h
0x1006e9af: movsx ecx,byte ptr [eax+2]
0x1006e9b3: shl ecx,8
0x1006e9b6: or [esi+8],ecx
0x1006e9b9: cmp edx,4
0x1006e9bc: jne short 1006E9F6h
0x1006e9be: movsx eax,byte ptr [eax+3]
0x1006e9c2: jmp short 1006E9F3h
0x1006e9c4: cmp edx,2
0x1006e9c7: mov [esi+8],ecx
0x1006e9ca: jb short 1006E9D8h
0x1006e9cc: movsx edi,byte ptr [eax+1]
0x1006e9d0: shl edi,8
0x1006e9d3: or edi,ecx
0x1006e9d5: mov [esi+8],edi
0x1006e9d8: cmp edx,3
0x1006e9db: jb short 1006E9E7h
0x1006e9dd: movsx ecx,byte ptr [eax+2]
0x1006e9e1: shl ecx,10h
0x1006e9e4: or [esi+8],ecx
0x1006e9e7: cmp edx,4
0x1006e9ea: jne short 1006E9F6h
0x1006e9ec: movsx eax,byte ptr [eax+3]
0x1006e9f0: shl eax,18h
0x1006e9f3: or [esi+8],eax
0x1006e9f6: xor eax,eax
0x1006e9f8: inc eax
```

```
0x1006e9f9: jmp short 1006EA02h
0x1006e9fb: push eax
0x1006e9fc: call 1006E625h
0x1006ea01: pop ecx
0x1006ea02: pop edi
0x1006ea03: pop esi
0x1006ea04: ret
```

`sub_1006e977` is called by sub_1006eba4. It delegates to sub_1006e625 (1 callees).

12.1004 sub_1006ea05 at 0x1006ea05 (32 instructions)

```
0x1006ea05: push esi
0x1006ea06: push edi
0x1006ea07: mov esi,ecx
0x1006ea09: mov edi,edx
0x1006ea0b: mov edx,[esi+4]
0x1006ea0e: cmp edx,2
0x1006ea11: ja short 1006EA4Bh
0x1006ea13: cmp word ptr [edi+1C8h],4D4Dh
0x1006ea1c: movzx ecx,word ptr [eax]
0x1006ea1f: jne short 1006EA32h
0x1006ea21: shl ecx,10h
0x1006ea24: cmp edx,2
0x1006ea27: mov [esi+8],ecx
0x1006ea2a: jne short 1006EA46h
0x1006ea2c: movzx eax,word ptr [eax+2]
0x1006ea30: jmp short 1006EA41h
0x1006ea32: cmp edx,2
0x1006ea35: mov [esi+8],ecx
0x1006ea38: jne short 1006EA46h
0x1006ea3a: movzx eax,word ptr [eax+2]
0x1006ea3e: shl eax,10h
0x1006ea41: or eax,ecx
0x1006ea43: mov [esi+8],eax
0x1006ea46: xor eax,eax
0x1006ea48: inc eax
0x1006ea49: jmp short 1006EA52h
0x1006ea4b: push eax
0x1006ea4c: call 1006E625h
0x1006ea51: pop ecx
0x1006ea52: pop edi
0x1006ea53: pop esi
0x1006ea54: ret
```

`sub_1006ea05` is called by sub_1006eba4, sub_1006f20b, sub_1006f35a. It delegates to sub_1006e625 (1 callees).

12.1005 sub_1006ea55 at 0x1006ea55 (105 instructions)

```
0x1006ea55: push ebp
0x1006ea56: mov ebp,esp
0x1006ea58: sub esp,0Ch
0x1006ea5b: push ebx
0x1006ea5c: push esi
0x1006ea5d: mov esi,eax
0x1006ea5f: mov eax,[esi+4]
0x1006ea62: shl eax,3
0x1006ea65: push edi
0x1006ea66: push eax
0x1006ea67: call 100EFDD0h
0x1006ea6c: mov ebx,eax
0x1006ea6e: xor edi,edi
0x1006ea70: cmp ebx,edi
0x1006ea72: pop ecx
0x1006ea73: jne short 1006EA94h
```

```
0x1006ea75: mov eax,[ebp+8]
0x1006ea78: push 101090E4h
0x1006ea7d: push dword ptr [eax]
0x1006ea7f: push dword ptr [eax+274h]
0x1006ea85: call 1006FAF1h
0x1006ea8a: add esp,0Ch
0x1006ea8d: xor eax,eax
0x1006ea8f: jmp near ptr 1006EB7Eh
0x1006ea94: cmp [esi+4],edi
0x1006ea97: jbe near ptr 1006EB69h
0x1006ea9d: mov eax,[ebp+0Ch]
0x1006eaa0: fld dword ptr [eax+edi*4]
0x1006eaa3: mov dword ptr [ebp-0Ch],1
0x1006eaaa: fstp dword ptr [ebp-4]
0x1006eaad: fldz
0x1006eaaf: fcomp dword ptr [ebp-4]
0x1006eab2: fnstsw ax
0x1006eab4: test ah,41h
0x1006eab7: jne short 1006EB01h
0x1006eab9: cmp word ptr [esi+2],5
0x1006eabe: jne short 1006EAF5h
0x1006eac0: movzx eax,word ptr [esi]
0x1006eac3: fld dword ptr [ebp-4]
0x1006eac6: push ecx
0x1006eac7: push ecx
0x1006eac8: fstp qword ptr [esp]
0x1006eacb: push eax
0x1006eacc: push dword ptr [ebp+8]
0x1006eacf: call 1006FE65h
0x1006ead4: pop ecx
0x1006ead5: pop ecx
0x1006ead6: push dword ptr [eax+10h]
0x1006ead9: mov eax,[ebp+8]
0x1006eadc: push 101090A0h
0x1006eae1: push dword ptr [eax]
0x1006eae3: push dword ptr [eax+274h]
0x1006eae9: call 1007007Dh
0x1006eaee: fldz
0x1006eaf0: add esp,18h
0x1006eaf3: jmp short 1006EAFEh
0x1006eaf5: fld dword ptr [ebp-4]
0x1006eaf8: or dword ptr [ebp-0Ch],0FFFFFFFFh
0x1006eafc: fchs
0x1006eafe: fstp dword ptr [ebp-4]
0x1006eb01: fldz
0x1006eb03: mov dword ptr [ebp-8],1
0x1006eb0a: fcomp dword ptr [ebp-4]
0x1006eb0d: fnstsw ax
0x1006eb0f: test ah,5
0x1006eb12: jp short 1006EB44h
0x1006eb14: fld dword ptr ds:[1010909Ch]
0x1006eb1a: fcom dword ptr [ebp-4]
0x1006eb1d: fnstsw ax
0x1006eb1f: test ah,41h
0x1006eb22: jne short 1006EB42h
0x1006eb24: mov eax,[ebp-8]
0x1006eb27: cmp eax,10000000h
0x1006eb2c: jae short 1006EB42h
0x1006eb2e: fld dword ptr [ebp-4]
0x1006eb31: shl eax,3
0x1006eb34: fmul qword ptr ds:[100FF128h]
0x1006eb3a: mov [ebp-8],eax
0x1006eb3d: fstp dword ptr [ebp-4]
0x1006eb40: jmp short 1006EB1Ah
0x1006eb42: fstp st(0)
0x1006eb44: fld dword ptr [ebp-4]
0x1006eb47: fadd qword ptr ds:[100FE7C8h]
```

```
0x1006eb4d: fimul dword ptr [ebp-0Ch]
0x1006eb50: call 10022906h
0x1006eb55: mov [ebx+edi*8],eax
0x1006eb58: mov eax,[ebp-8]
0x1006eb5b: mov [ebx+edi*8+4],eax
0x1006eb5f: inc edi
0x1006eb60: cmp edi,[esi+4]
0x1006eb63: jb near ptr 1006EA9Dh
0x1006eb69: mov edi,[ebp+8]
0x1006eb6c: push ebx
0x1006eb6d: call 1006E625h
0x1006eb72: push ebx
0x1006eb73: mov esi,eax
0x1006eb75: call 100209E0h
0x1006eb7a: pop ecx
0x1006eb7b: pop ecx
0x1006eb7c: mov eax,esi
0x1006eb7e: pop edi
0x1006eb7f: pop esi
0x1006eb80: pop ebx
0x1006eb81: leave
0x1006eb82: ret
```

`sub_1006ea55` is called by sub_1006f391. It works with the text "No space to write RATIONAL array". It delegates to j_free, sub_10022906, sub_1006e625 (7 callees).

12.1006 sub_1006eb83 at 0x1006eb83 (15 instructions)

```
0x1006eb83: push esi
0x1006eb84: mov esi,ecx
0x1006eb86: cmp dword ptr [esi+4],1
0x1006eb8a: jne short 1006EB96h
0x1006eb8c: mov eax,[eax]
0x1006eb8e: mov [esi+8],eax
0x1006eb91: xor eax,eax
0x1006eb93: inc eax
0x1006eb94: pop esi
0x1006eb95: ret
0x1006eb96: push eax
0x1006eb97: call 1006E625h
0x1006eb9c: pop ecx
0x1006eb9d: pop esi
0x1006eb9e: ret
```

`sub_1006eb83` is called by sub_1006eba4, sub_1006f391. It delegates to sub_1006e625 (1 callees).

12.1007 sub_1006eb9f at 0x1006eb9f (53 instructions)

```
0x1006e625: push ebp
0x1006e626: mov ebp,esp
0x1006e628: test byte ptr [edi+0Ch],80h
0x1006e62c: je short 1006E677h
0x1006e62e: movzx eax,word ptr [esi+2]
0x1006e632: add eax,0FFFFFFFDh
0x1006e635: cmp eax,9
0x1006e638: ja short 1006E677h
0x1006e63a: movzx eax,byte ptr [eax+1006E70Dh]
0x1006e641: jmp dword ptr [eax*4+1006E6F9h]
0x1006e677: mov eax,[edi+1E8h]
0x1006e67d: mov [esi+8],eax
0x1006e680: movzx eax,word ptr [esi+2]
0x1006e684: push ebx
0x1006e685: push eax
0x1006e686: call 1006FCE1h
0x1006e68b: push 0
```

```
0x1006e68d: push dword ptr [esi+8]
0x1006e690: mov ebx,eax
0x1006e692: push dword ptr [edi+274h]
0x1006e698: imul ebx,[esi+4]
0x1006e69c: call dword ptr [edi+280h]
0x1006e6a2: add esp,10h
0x1006e6a5: cmp eax,[esi+8]
0x1006e6a8: jne short 1006E6D0h
0x1006e6aa: push ebx
0x1006e6ab: push dword ptr [ebp+8]
0x1006e6ae: push dword ptr [edi+274h]
0x1006e6b4: call dword ptr [edi+27Ch]
0x1006e6ba: add esp,0Ch
0x1006e6bd: cmp eax,ebx
0x1006e6bf: jne short 1006E6D0h
0x1006e6c1: inc ebx
0x1006e6c2: and ebx,0FFFFFFFEh
0x1006e6c5: add [edi+1E8h],ebx
0x1006e6cb: xor eax,eax
0x1006e6cd: inc eax
0x1006e6ce: jmp short 1006E6F4h
0x1006e6d0: movzx eax,word ptr [esi]
0x1006e6d3: push eax
0x1006e6d4: push edi
0x1006e6d5: call 1006FE65h
0x1006e6da: push dword ptr [eax+10h]
0x1006e6dd: push 10109010h
0x1006e6e2: push dword ptr [edi]
0x1006e6e4: push dword ptr [edi+274h]
0x1006e6ea: call 1006FAF1h
0x1006e6ef: add esp,18h
0x1006e6f2: xor eax,eax
0x1006e6f4: pop ebx
0x1006e6f5: pop ebp
0x1006e6f6: ret
0x1006eb9f: jmp near ptr 1006E625h
```

`sub_1006eb9f` is called by sub_1006eba4. It delegates to sub_1006faf1, sub_1006fce1, sub_1006fe65 (3 callees).

12.1008 sub_1006eba4 at 0x1006eba4 (188 instructions)

```
0x1006eba4: push ebp
0x1006eba5: mov ebp,esp
0x1006eba7: sub esp,64h
0x1006ebaa: mov eax,ds:[10136220h]
0x1006ebaf: xor eax,ebp
0x1006ebb1: mov [ebp-4],eax
0x1006ebb4: mov eax,[ebp+8]
0x1006ebb7: and dword ptr [ebp-64h],0
0x1006ebbb: push ebx
0x1006ebbc: push esi
0x1006ebbd: push edi
0x1006ebbe: push dword ptr [ebp+0Ch]
0x1006ebc1: mov [ebp-5Ch],eax
0x1006ebc4: mov eax,[ebp+14h]
0x1006ebc7: mov esi,ecx
0x1006ebc9: mov [ebp-60h],eax
0x1006ebcc: mov edi,edx
0x1006ebce: lea ebx,[ebp-54h]
0x1006ebd1: call 1006FCE1h
0x1006ebd6: imul eax,esi
0x1006ebd9: cmp eax,50h
0x1006ebdc: pop ecx
0x1006ebdd: jbe short 1006EC19h
0x1006ebdf: push dword ptr [ebp+0Ch]
0x1006ebe2: call 1006FCE1h
```

```
0x1006ebe7: imul eax,esi
0x1006ebea: push eax
0x1006ebeb: call 100EFDD0h
0x1006ebf0: mov ebx,eax
0x1006ebf2: xor edx,edx
0x1006ebf4: cmp ebx,edx
0x1006ebf6: pop ecx
0x1006ebf7: pop ecx
0x1006ebf8: jne short 1006EC1Bh
0x1006ebfa: mov eax,[ebp-5Ch]
0x1006ebfd: push 10109108h
0x1006ec02: push dword ptr [eax]
0x1006ec04: push dword ptr [eax+274h]
0x1006ec0a: call 1006FAF1h
0x1006ec0f: add esp,0Ch
0x1006ec12: xor eax,eax
0x1006ec14: jmp near ptr 1006EDAAh
0x1006ec19: xor edx,edx
0x1006ec1b: mov ecx,[ebp-60h]
0x1006ec1e: mov ax,[ebp+10h]
0x1006ec22: mov [ecx],ax
0x1006ec25: mov eax,[ebp+0Ch]
0x1006ec28: cmp eax,8
0x1006ec2b: mov [ecx+2],ax
0x1006ec2f: mov [ecx+4],esi
0x1006ec32: jg near ptr 1006ED2Ah
0x1006ec38: je near ptr 1006ED07h
0x1006ec3e: dec eax
0x1006ec3f: je near ptr 1006ECE2h
0x1006ec45: dec eax
0x1006ec46: dec eax
0x1006ec47: je short 1006ECAFh
0x1006ec49: dec eax
0x1006ec4a: je short 1006EC86h
0x1006ec4c: dec eax
0x1006ec4d: dec eax
0x1006ec4e: jne near ptr 1006ED99h
0x1006ec54: cmp esi,edx
0x1006ec56: mov [ebp-58h],edx
0x1006ec59: jle short 1006EC74h
0x1006ec5b: mov eax,[ebp-58h]
0x1006ec5e: fld qword ptr [edi+eax*8]
0x1006ec61: call 100228D0h
0x1006ec66: mov ecx,[ebp-58h]
0x1006ec69: inc dword ptr [ebp-58h]
0x1006ec6c: cmp [ebp-58h],esi
0x1006ec6f: mov [ecx+ebx],al
0x1006ec72: jl short 1006EC5Bh
0x1006ec74: mov ecx,[ebp-60h]
0x1006ec77: mov edx,[ebp-5Ch]
0x1006ec7a: mov eax,ebx
0x1006ec7c: call 1006E977h
0x1006ec81: jmp near ptr 1006ED8Eh
0x1006ec86: cmp esi,edx
0x1006ec88: mov [ebp-58h],edx
0x1006ec8b: jle near ptr 1006ED81h
0x1006ec91: mov eax,[ebp-58h]
0x1006ec94: fld qword ptr [edi+eax*8]
0x1006ec97: call 10022906h
0x1006ec9c: mov ecx,[ebp-58h]
0x1006ec9f: inc dword ptr [ebp-58h]
0x1006eca2: cmp [ebp-58h],esi
0x1006eca5: mov [ebx+ecx*4],eax
0x1006eca8: jl short 1006EC91h
0x1006ecaa: jmp near ptr 1006ED81h
0x1006ecaf: cmp esi,edx
0x1006ecb1: mov [ebp-58h],edx
```

```
0x1006ecb4: jle short 1006ECD0h
0x1006ecb6: mov eax,[ebp-58h]
0x1006ecb9: fld qword ptr [edi+eax*8]
0x1006ecbc: call 10022906h
0x1006ecc1: mov ecx,[ebp-58h]
0x1006ecc4: inc dword ptr [ebp-58h]
0x1006ecc7: cmp [ebp-58h],esi
0x1006ecca: mov [ebx+ecx*2],ax
0x1006ecce: jl short 1006ECB6h
0x1006ecd0: mov ecx,[ebp-60h]
0x1006ecd3: mov edx,[ebp-5Ch]
0x1006ecd6: mov eax,ebx
0x1006ecd8: call 1006EA05h
0x1006ecdd: jmp near ptr 1006ED8Eh
0x1006ece2: cmp esi,edx
0x1006ece4: mov [ebp-58h],edx
0x1006ece7: jle short 1006EC74h
0x1006ece9: mov eax,[ebp-58h]
0x1006ecec: fld qword ptr [edi+eax*8]
0x1006ecef: call 10022906h
0x1006ecf4: mov ecx,[ebp-58h]
0x1006ecf7: inc dword ptr [ebp-58h]
0x1006ecfa: cmp [ebp-58h],esi
0x1006ecfd: mov [ecx+ebx],al
0x1006ed00: jl short 1006ECE9h
0x1006ed02: jmp near ptr 1006EC74h
0x1006ed07: cmp esi,edx
0x1006ed09: mov [ebp-58h],edx
0x1006ed0c: jle short 1006ECD0h
0x1006ed0e: mov eax,[ebp-58h]
0x1006ed11: fld qword ptr [edi+eax*8]
0x1006ed14: call 100228D0h
0x1006ed19: mov ecx,[ebp-58h]
0x1006ed1c: inc dword ptr [ebp-58h]
0x1006ed1f: cmp [ebp-58h],esi
0x1006ed22: mov [ebx+ecx*2],ax
0x1006ed26: jl short 1006ED0Eh
0x1006ed28: jmp short 1006ECD0h
0x1006ed2a: sub eax,9
0x1006ed2d: je short 1006ED61h
0x1006ed2f: dec eax
0x1006ed30: dec eax
0x1006ed31: je short 1006ED44h
0x1006ed33: dec eax
0x1006ed34: jne short 1006ED99h
0x1006ed36: push edi
0x1006ed37: mov edi,[ebp-5Ch]
0x1006ed3a: mov esi,ecx
0x1006ed3c: call 1006EB9Fh
0x1006ed41: pop ecx
0x1006ed42: jmp short 1006ED8Eh
0x1006ed44: xor eax,eax
0x1006ed46: cmp esi,edx
0x1006ed48: jle short 1006ED55h
0x1006ed4a: fld qword ptr [edi+eax*8]
0x1006ed4d: fstp dword ptr [ebx+eax*4]
0x1006ed50: inc eax
0x1006ed51: cmp eax,esi
0x1006ed53: jl short 1006ED4Ah
0x1006ed55: mov edi,[ebp-5Ch]
0x1006ed58: mov eax,ebx
0x1006ed5a: call 1006EB83h
0x1006ed5f: jmp short 1006ED8Eh
0x1006ed61: cmp esi,edx
0x1006ed63: mov [ebp-58h],edx
0x1006ed66: jle short 1006ED81h
0x1006ed68: mov eax,[ebp-58h]
```

```
0x1006ed6b: fld qword ptr [edi+eax*8]
0x1006ed6e: call 100228D0h
0x1006ed73: mov ecx,[ebp-58h]
0x1006ed76: inc dword ptr [ebp-58h]
0x1006ed79: cmp [ebp-58h],esi
0x1006ed7c: mov [ebx+ecx*4],eax
0x1006ed7f: jl short 1006ED68h
0x1006ed81: mov ecx,[ebp-60h]
0x1006ed84: mov edi,[ebp-5Ch]
0x1006ed87: mov eax,ebx
0x1006ed89: call 1006EB83h
0x1006ed8e: test eax,eax
0x1006ed90: je short 1006ED99h
0x1006ed92: mov dword ptr [ebp-64h],1
0x1006ed99: lea eax,[ebp-54h]
0x1006ed9c: cmp ebx,eax
0x1006ed9e: je short 1006EDA7h
0x1006eda0: push ebx
0x1006eda1: call 100209E0h
0x1006eda6: pop ecx
0x1006eda7: mov eax,[ebp-64h]
0x1006edaa: mov ecx,[ebp-4]
0x1006edad: pop edi
0x1006edae: pop esi
0x1006edaf: xor ecx,ebp
0x1006edb1: pop ebx
0x1006edb2: call 10022326h
0x1006edb7: leave
0x1006edb8: ret
```

`sub_1006eba4` is called by sub_1006f2c7. It works with the text "No space to write array". It delegates to j_free, sub_10022326, sub_100228d0 (11 callees).

12.1009 sub_1006edb9 at 0x1006edb9 (45 instructions)

```
0x1006edb9: mov cl,[esi+54h]
0x1006edbc: movzx eax,word ptr [esi+62h]
0x1006edc0: push ebx
0x1006edc1: push edi
0x1006edc2: xor edi,edi
0x1006edc4: inc edi
0x1006edc5: shl edi,cl
0x1006edc7: movzx ecx,word ptr [esi+0A8h]
0x1006edce: add edi,edi
0x1006edd0: sub eax,ecx
0x1006edd2: lea ebx,[esi+0D8h]
0x1006edd8: js short 1006EDE4h
0x1006edda: cmp eax,1
0x1006eddd: jle short 1006EE0Dh
0x1006eddf: cmp eax,2
0x1006ede2: je short 1006EDFBh
0x1006ede4: push edi
0x1006ede5: push dword ptr [ebx+8]
0x1006ede8: push dword ptr [ebx]
0x1006edea: call 10020A10h
0x1006edef: add esp,0Ch
0x1006edf2: test eax,eax
0x1006edf4: je short 1006EDFBh
0x1006edf6: push 3
0x1006edf8: pop eax
0x1006edf9: jmp short 1006EE10h
0x1006edfb: push edi
0x1006edfc: push dword ptr [ebx+4]
0x1006edff: push dword ptr [ebx]
0x1006ee01: call 10020A10h
0x1006ee06: add esp,0Ch
```

```
0x1006ee09: test eax,eax
0x1006ee0b: jne short 1006EDF6h
0x1006ee0d: xor eax,eax
0x1006ee0f: inc eax
0x1006ee10: push ebx
0x1006ee11: push eax
0x1006ee12: mov eax,[esp+14h]
0x1006ee16: push 12Dh
0x1006ee1b: mov ecx,esi
0x1006ee1d: call 1006E90Eh
0x1006ee22: add esp,0Ch
0x1006ee25: pop edi
0x1006ee26: pop ebx
0x1006ee27: ret
```

`sub_1006edb9` is called by sub_1006f391. It delegates to sub_10020a10, sub_1006e90e (2 callees).

12.1010 sub_1006ee28 at 0x1006ee28 (59 instructions)

```
0x1006e977: push esi
0x1006e978: push edi
0x1006e979: mov esi,ecx
0x1006e97b: mov edi,edx
0x1006e97d: mov edx,[esi+4]
0x1006e980: cmp edx,4
0x1006e983: ja short 1006E9FBh
0x1006e985: cmp word ptr [edi+1C8h],4D4Dh
0x1006e98e: movsx ecx,byte ptr [eax]
0x1006e991: jne short 1006E9C4h
0x1006e993: shl ecx,18h
0x1006e996: cmp edx,2
0x1006e999: mov [esi+8],ecx
0x1006e99c: jb short 1006E9AAh
0x1006e99e: movsx edi,byte ptr [eax+1]
0x1006e9a2: shl edi,10h
0x1006e9a5: or edi,ecx
0x1006e9a7: mov [esi+8],edi
0x1006e9aa: cmp edx,3
0x1006e9ad: jb short 1006E9B9h
0x1006e9af: movsx ecx,byte ptr [eax+2]
0x1006e9b3: shl ecx,8
0x1006e9b6: or [esi+8],ecx
0x1006e9b9: cmp edx,4
0x1006e9bc: jne short 1006E9F6h
0x1006e9be: movsx eax,byte ptr [eax+3]
0x1006e9c2: jmp short 1006E9F3h
0x1006e9c4: cmp edx,2
0x1006e9c7: mov [esi+8],ecx
0x1006e9ca: jb short 1006E9D8h
0x1006e9cc: movsx edi,byte ptr [eax+1]
0x1006e9d0: shl edi,8
0x1006e9d3: or edi,ecx
0x1006e9d5: mov [esi+8],edi
0x1006e9d8: cmp edx,3
0x1006e9db: jb short 1006E9E7h
0x1006e9dd: movsx ecx,byte ptr [eax+2]
0x1006e9e1: shl ecx,10h
0x1006e9e4: or [esi+8],ecx
0x1006e9e7: cmp edx,4
0x1006e9ea: jne short 1006E9F6h
0x1006e9ec: movsx eax,byte ptr [eax+3]
0x1006e9f0: shl eax,18h
0x1006e9f3: or [esi+8],eax
0x1006e9f6: xor eax,eax
0x1006e9f8: inc eax
0x1006e9f9: jmp short 1006EA02h
```

```
0x1006e9fb: push eax
0x1006e9fc: call 1006E625h
0x1006ea01: pop ecx
0x1006ea02: pop edi
0x1006ea03: pop esi
0x1006ea04: ret
0x1006ee28: mov word ptr [ecx],14Dh
0x1006ee2d: mov word ptr [ecx+2],2
0x1006ee33: mov eax,[edx+0E4h]
0x1006ee39: mov [ecx+4],eax
0x1006ee3c: mov eax,[edx+0E8h]
0x1006ee42: jmp near ptr 1006E977h
```

`sub_1006ee28` is called by sub_1006f391. It delegates to sub_1006e625 (1 callees).

12.1011 sub_1006ee47 at 0x1006ee47 (29 instructions)

```
0x1006ee47: push ebp
0x1006ee48: mov ebp,esp
0x1006ee4a: sub esp,10h
0x1006ee4d: push ebx
0x1006ee4e: push esi
0x1006ee4f: mov esi,ecx
0x1006ee51: mov cx,[eax]
0x1006ee54: push edi
0x1006ee55: mov edi,edx
0x1006ee57: movzx edx,word ptr [eax+6]
0x1006ee5b: mov [esi],cx
0x1006ee5e: mov cx,[eax+8]
0x1006ee62: mov [esi+2],cx
0x1006ee66: movzx ecx,dx
0x1006ee69: mov [esi+4],ecx
0x1006ee6c: mov ecx,[eax+8]
0x1006ee6f: xor ebx,ebx
0x1006ee71: dec ecx
0x1006ee72: inc ebx
0x1006ee73: cmp ecx,0Ch
0x1006ee76: mov [ebp-0Ch],edx
0x1006ee79: ja near ptr 1006EF24h
0x1006ee7f: jmp dword ptr [ecx*4+1006F1D7h]
0x1006ef24: mov eax,ebx
0x1006ef26: pop edi
0x1006ef27: pop esi
0x1006ef28: pop ebx
0x1006ef29: leave
0x1006ef2a: ret
```

`sub_1006ee47` is called by sub_1006f391. It calls nothing else.

12.1012 sub_1006f20b at 0x1006f20b (73 instructions)

```
0x1006f20b: push ebp
0x1006f20c: mov ebp,esp
0x1006f20e: sub esp,20h
0x1006f211: mov eax,ds:[10136220h]
0x1006f216: xor eax,ebp
0x1006f218: mov [ebp-4],eax
0x1006f21b: mov eax,[ebp+8]
0x1006f21e: push ebx
0x1006f21f: push esi
0x1006f220: push edi
0x1006f221: movzx edi,word ptr [eax+62h]
0x1006f225: cmp di,0Ah
0x1006f229: mov [ebp-1Ch],eax
0x1006f22c: mov esi,ecx
```

```
0x1006f22e: lea ebx,[ebp-18h]
0x1006f231: jbe short 1006F261h
0x1006f233: movzx eax,di
0x1006f236: add eax,eax
0x1006f238: push eax
0x1006f239: call 100EFDD0h
0x1006f23e: mov ebx,eax
0x1006f240: test ebx,ebx
0x1006f242: mov eax,[ebp-1Ch]
0x1006f245: pop ecx
0x1006f246: jne short 1006F261h
0x1006f248: push 10109120h
0x1006f24d: push dword ptr [eax]
0x1006f24f: push dword ptr [eax+274h]
0x1006f255: call 1006FAF1h
0x1006f25a: add esp,0Ch
0x1006f25d: xor eax,eax
0x1006f25f: jmp short 1006F2B8h
0x1006f261: lea ecx,[ebp-20h]
0x1006f264: push ecx
0x1006f265: push dword ptr [ebp+0Ch]
0x1006f268: push eax
0x1006f269: call 10067985h
0x1006f26e: add esp,0Ch
0x1006f271: test di,di
0x1006f274: jbe short 1006F287h
0x1006f276: mov ecx,ebx
0x1006f278: movzx eax,di
0x1006f27b: mov dx,[ebp-20h]
0x1006f27f: mov [ecx],dx
0x1006f282: inc ecx
0x1006f283: inc ecx
0x1006f284: dec eax
0x1006f285: jne short 1006F27Bh
0x1006f287: mov ax,[ebp+0Ch]
0x1006f28b: mov edx,[ebp-1Ch]
0x1006f28e: mov [esi],ax
0x1006f291: movzx eax,di
0x1006f294: mov [esi+4],eax
0x1006f297: mov eax,ebx
0x1006f299: mov ecx,esi
0x1006f29b: mov word ptr [esi+2],3
0x1006f2a1: call 1006EA05h
0x1006f2a6: mov esi,eax
0x1006f2a8: lea eax,[ebp-18h]
0x1006f2ab: cmp ebx,eax
0x1006f2ad: je short 1006F2B6h
0x1006f2af: push ebx
0x1006f2b0: call 100209E0h
0x1006f2b5: pop ecx
0x1006f2b6: mov eax,esi
0x1006f2b8: mov ecx,[ebp-4]
0x1006f2bb: pop edi
0x1006f2bc: pop esi
0x1006f2bd: xor ecx,ebp
0x1006f2bf: pop ebx
0x1006f2c0: call 10022326h
0x1006f2c5: leave
0x1006f2c6: ret
```

`sub_1006f20b` is called by sub_1006f391. It works with the text "No space to write per-sample shorts". It delegates to j_free, sub_10022326, sub_10067985 (6 callees).

12.1013 sub_1006f2c7 at 0x1006f2c7 (59 instructions)

```
0x1006f2c7: push ebp
0x1006f2c8: mov ebp,esp
0x1006f2ca: sub esp,58h
0x1006f2cd: push esi
0x1006f2ce: movzx esi,word ptr [ebx+62h]
0x1006f2d2: cmp si,0Ah
0x1006f2d6: push edi
0x1006f2d7: lea edi,[ebp-58h]
0x1006f2da: jbe short 1006F308h
0x1006f2dc: movzx eax,si
0x1006f2df: shl eax,3
0x1006f2e2: push eax
0x1006f2e3: call 100EFDD0h
0x1006f2e8: mov edi,eax
0x1006f2ea: test edi,edi
0x1006f2ec: pop ecx
0x1006f2ed: jne short 1006F308h
0x1006f2ef: push 10109144h
0x1006f2f4: push dword ptr [ebx]
0x1006f2f6: push dword ptr [ebx+274h]
0x1006f2fc: call 1006FAF1h
0x1006f301: add esp,0Ch
0x1006f304: xor eax,eax
0x1006f306: jmp short 1006F356h
0x1006f308: lea eax,[ebp-8]
0x1006f30b: push eax
0x1006f30c: push dword ptr [ebp+0Ch]
0x1006f30f: push ebx
0x1006f310: call 10067985h
0x1006f315: add esp,0Ch
0x1006f318: test si,si
0x1006f31b: jbe short 1006F32Dh
0x1006f31d: mov ecx,edi
0x1006f31f: movzx eax,si
0x1006f322: fld qword ptr [ebp-8]
0x1006f325: fstp qword ptr [ecx]
0x1006f327: add ecx,8
0x1006f32a: dec eax
0x1006f32b: jne short 1006F322h
0x1006f32d: push dword ptr [ebp+10h]
0x1006f330: movzx ecx,si
0x1006f333: push dword ptr [ebp+0Ch]
0x1006f336: mov edx,edi
0x1006f338: push dword ptr [ebp+8]
0x1006f33b: push ebx
0x1006f33c: call 1006EBA4h
0x1006f341: mov esi,eax
0x1006f343: lea eax,[ebp-58h]
0x1006f346: add esp,10h
0x1006f349: cmp edi,eax
0x1006f34b: je short 1006F354h
0x1006f34d: push edi
0x1006f34e: call 100209E0h
0x1006f353: pop ecx
0x1006f354: mov eax,esi
0x1006f356: pop edi
0x1006f357: pop esi
0x1006f358: leave
0x1006f359: ret
```

`sub_1006f2c7` is called by sub_1006f391. It works with the text "No space to write per-sample values". It delegates to j_free, sub_10067985, sub_1006eba4 (5 callees).

12.1014 sub_1006f35a at 0x1006f35a (20 instructions)

```
0x1006f35a: push ebp
0x1006f35b: mov ebp,esp
0x1006f35d: push ecx
0x1006f35e: lea eax,[ebp-2]
0x1006f361: push eax
0x1006f362: lea eax,[ebp-4]
0x1006f365: push eax
0x1006f366: push edi
0x1006f367: push dword ptr [ebp+8]
0x1006f36a: call 10067985h
0x1006f36f: mov edx,[ebp+8]
0x1006f372: add esp,10h
0x1006f375: lea eax,[ebp-4]
0x1006f378: mov ecx,esi
0x1006f37a: mov [esi],di
0x1006f37d: mov word ptr [esi+2],3
0x1006f383: mov dword ptr [esi+4],2
0x1006f38a: call 1006EA05h
0x1006f38f: leave
0x1006f390: ret
```

`sub_1006f35a` is called by sub_1006f391. It delegates to sub_10067985, sub_1006ea05 (2 callees).

12.1015 sub_1006f391 at 0x1006f391 (526 instructions)

```
0x1006f391: push ebp
0x1006f392: mov ebp,esp
0x1006f394: sub esp,48h
0x1006f397: push ebx
0x1006f398: push esi
0x1006f399: mov ebx,eax
0x1006f39b: xor esi,esi
0x1006f39d: cmp [ebx+8],esi
0x1006f3a0: jne short 1006F3AAh
0x1006f3a2: xor eax,eax
0x1006f3a4: inc eax
0x1006f3a5: jmp near ptr 1006FA7Dh
0x1006f3aa: cmp [ebp+8],esi
0x1006f3ad: push edi
0x1006f3ae: je short 1006F42Fh
0x1006f3b0: mov eax,[ebx+0Ch]
0x1006f3b3: test ax,1000h
0x1006f3b7: je short 1006F3D7h
0x1006f3b9: and eax,0FFFFEFFFh
0x1006f3be: push ebx
0x1006f3bf: mov [ebx+0Ch],eax
0x1006f3c2: call dword ptr [ebx+218h]
0x1006f3c8: test eax,eax
0x1006f3ca: pop ecx
0x1006f3cb: jne short 1006F3D7h
0x1006f3cd: push 10109194h
0x1006f3d2: jmp near ptr 1006F48Bh
0x1006f3d7: push ebx
0x1006f3d8: call dword ptr [ebx+234h]
0x1006f3de: cmp [ebx+260h],esi
0x1006f3e4: pop ecx
0x1006f3e5: jle short 1006F402h
0x1006f3e7: test byte ptr [ebx+0Ch],40h
0x1006f3eb: je short 1006F402h
0x1006f3ed: push ebx
0x1006f3ee: call 10065E73h
0x1006f3f3: test eax,eax
0x1006f3f5: pop ecx
0x1006f3f6: jne short 1006F402h
0x1006f3f8: push 10109168h
```

```
0x1006f3fd: jmp near ptr 1006F48Bh
0x1006f402: test word ptr [ebx+0Ch],200h
0x1006f408: je short 1006F42Bh
0x1006f40a: lea edi,[ebx+254h]
0x1006f410: mov eax,[edi]
0x1006f412: cmp eax,esi
0x1006f414: je short 1006F42Bh
0x1006f416: push eax
0x1006f417: call 100209E0h
0x1006f41c: pop ecx
0x1006f41d: mov [edi],esi
0x1006f41f: mov [ebx+260h],esi
0x1006f425: mov [ebx+258h],esi
0x1006f42b: and dword ptr [ebx+0Ch],0FFFFFFAFh
0x1006f42f: lea edi,[ebx+28h]
0x1006f432: mov [ebp-18h],edi
0x1006f435: mov [ebp-0Ch],esi
0x1006f438: xor eax,eax
0x1006f43a: xor edx,edx
0x1006f43c: mov ecx,eax
0x1006f43e: and ecx,1Fh
0x1006f441: inc edx
0x1006f442: shl edx,cl
0x1006f444: mov ecx,eax
0x1006f446: shr ecx,5
0x1006f449: test [ebx+ecx*4+28h],edx
0x1006f44d: je short 1006F460h
0x1006f44f: cmp eax,41h
0x1006f452: je short 1006F460h
0x1006f454: xor ecx,ecx
0x1006f456: cmp eax,5
0x1006f459: setb cl
0x1006f45c: inc ecx
0x1006f45d: add [ebp-0Ch],ecx
0x1006f460: inc eax
0x1006f461: cmp eax,7Fh
0x1006f464: jbe short 1006F43Ah
0x1006f466: mov esi,[ebp-0Ch]
0x1006f469: add esi,[edi+0C4h]
0x1006f46f: mov [ebp-0Ch],esi
0x1006f472: imul esi,0Ch
0x1006f475: push esi
0x1006f476: mov [ebp-20h],esi
0x1006f479: call 100EFDD0h
0x1006f47e: test eax,eax
0x1006f480: pop ecx
0x1006f481: mov [ebp-10h],eax
0x1006f484: jne short 1006F4A2h
0x1006f486: push 10108FE8h
0x1006f48b: push dword ptr [ebx]
0x1006f48d: push dword ptr [ebx+274h]
0x1006f493: call 1006FAF1h
0x1006f498: add esp,0Ch
0x1006f49b: xor eax,eax
0x1006f49d: jmp near ptr 1006FA7Ch
0x1006f4a2: cmp dword ptr [ebx+10h],0
0x1006f4a6: jne short 1006F4BAh
0x1006f4a8: mov esi,ebx
0x1006f4aa: call 1006E717h
0x1006f4af: test eax,eax
0x1006f4b1: je near ptr 1006FA44h
0x1006f4b7: mov esi,[ebp-20h]
0x1006f4ba: mov eax,[ebx+10h]
0x1006f4bd: lea ecx,[eax+esi+6]
0x1006f4c1: test cl,1
0x1006f4c4: lea eax,[ebx+1E8h]
0x1006f4ca: mov [eax],ecx
```

```
0x1006f4cc: je short 1006F4D1h
0x1006f4ce: inc ecx
0x1006f4cf: mov [eax],ecx
0x1006f4d1: push 0
0x1006f4d3: push dword ptr [eax]
0x1006f4d5: push dword ptr [ebx+274h]
0x1006f4db: call dword ptr [ebx+280h]
0x1006f4e1: mov eax,[ebp-10h]
0x1006f4e4: inc word ptr [ebx+1DCh]
0x1006f4eb: push 10h
0x1006f4ed: mov [ebp-4],eax
0x1006f4f0: lea eax,[ebp-48h]
0x1006f4f3: push edi
0x1006f4f4: push eax
0x1006f4f5: call 10020A00h
0x1006f4fa: add esp,18h
0x1006f4fd: test dword ptr [ebp-48h],80000000h
0x1006f504: je short 1006F520h
0x1006f506: cmp word ptr [edi+80h],0
0x1006f50e: jne short 1006F520h
0x1006f510: and dword ptr [ebp-48h],7FFFFFFFh
0x1006f517: dec dword ptr [ebp-0Ch]
0x1006f51a: sub esi,0Ch
0x1006f51d: mov [ebp-20h],esi
0x1006f520: mov eax,[ebx+294h]
0x1006f526: test eax,eax
0x1006f528: mov [ebp-30h],eax
0x1006f52b: jle near ptr 1006F786h
0x1006f531: mov eax,[ebp-10h]
0x1006f534: add eax,0Ch
0x1006f537: and dword ptr [ebp-2Ch],0
0x1006f53b: mov [ebp-8],eax
0x1006f53e: jmp short 1006F543h
0x1006f540: mov edi,[ebp-18h]
0x1006f543: mov ecx,[ebp-2Ch]
0x1006f546: mov eax,[ebx+290h]
0x1006f54c: mov eax,[ecx+eax]
0x1006f54f: movzx ecx,word ptr [eax+0Ch]
0x1006f553: cmp cx,41h
0x1006f557: mov esi,[ebp-4]
0x1006f55a: mov [ebp-1Ch],eax
0x1006f55d: jne short 1006F599h
0x1006f55f: mov ecx,[edi+0C4h]
0x1006f565: and dword ptr [ebp-28h],0
0x1006f569: test ecx,ecx
0x1006f56b: jle near ptr 1006F775h
0x1006f571: mov edx,[edi+0C8h]
0x1006f577: mov [ebp-24h],edx
0x1006f57a: mov [ebp-34h],ecx
0x1006f57d: mov edx,[ebp-24h]
0x1006f580: xor ecx,ecx
0x1006f582: cmp [edx],eax
0x1006f584: sete cl
0x1006f587: add dword ptr [ebp-24h],0Ch
0x1006f58b: or [ebp-28h],ecx
0x1006f58e: dec dword ptr [ebp-34h]
0x1006f591: jne short 1006F57Dh
0x1006f593: cmp dword ptr [ebp-28h],0
0x1006f597: jmp short 1006F5B2h
0x1006f599: movzx ecx,word ptr [eax+0Ch]
0x1006f59d: xor edx,edx
0x1006f59f: mov [ebp-34h],ecx
0x1006f5a2: and ecx,1Fh
0x1006f5a5: inc edx
0x1006f5a6: shl edx,cl
0x1006f5a8: mov ecx,[ebp-34h]
0x1006f5ab: shr ecx,5
```

```
0x1006f5ae: test [ebp+ecx*4-48h],edx
0x1006f5b2: je near ptr 1006F775h
0x1006f5b8: movzx ecx,word ptr [eax+0Ch]
0x1006f5bc: movzx ecx,cx
0x1006f5bf: cmp ecx,18h
0x1006f5c2: jg near ptr 1006F88Dh
0x1006f5c8: je near ptr 1006F845h
0x1006f5ce: cmp ecx,7
0x1006f5d1: jg near ptr 1006F709h
0x1006f5d7: je near ptr 1006F6F0h
0x1006f5dd: dec ecx
0x1006f5de: je near ptr 1006F6C7h
0x1006f5e4: dec ecx
0x1006f5e5: je near ptr 1006F69Eh
0x1006f5eb: dec ecx
0x1006f5ec: je short 1006F669h
0x1006f5ee: dec ecx
0x1006f5ef: je short 1006F609h
0x1006f5f1: dec ecx
0x1006f5f2: dec ecx
0x1006f5f3: jne near ptr 1006F949h
0x1006f5f9: push dword ptr [eax]
0x1006f5fb: mov ecx,esi
0x1006f5fd: push ebx
0x1006f5fe: call 1006F20Bh
0x1006f603: pop ecx
0x1006f604: jmp near ptr 1006F741h
0x1006f609: lea eax,[edi+64h]
0x1006f60c: push eax
0x1006f60d: push ebx
0x1006f60e: mov eax,esi
0x1006f610: mov word ptr [esi],11Eh
0x1006f615: mov word ptr [esi+2],5
0x1006f61b: mov dword ptr [esi+4],1
0x1006f622: call 1006EA55h
0x1006f627: test eax,eax
0x1006f629: pop ecx
0x1006f62a: pop ecx
0x1006f62b: je near ptr 1006FA44h
0x1006f631: mov eax,[ebp-8]
0x1006f634: mov word ptr [eax],11Fh
0x1006f639: add edi,68h
0x1006f63c: push edi
0x1006f63d: push ebx
0x1006f63e: mov dword ptr [esi+10h],1
0x1006f645: mov word ptr [esi+0Eh],5
0x1006f64b: call 1006EA55h
0x1006f650: test eax,eax
0x1006f652: pop ecx
0x1006f653: pop ecx
0x1006f654: je near ptr 1006FA44h
0x1006f65a: add esi,0Ch
0x1006f65d: add dword ptr [ebp-8],0Ch
0x1006f661: mov [ebp-4],esi
0x1006f664: jmp near ptr 1006F74Ah
0x1006f669: lea eax,[edi+58h]
0x1006f66c: push eax
0x1006f66d: push ebx
0x1006f66e: mov eax,esi
0x1006f670: mov word ptr [esi],11Ah
0x1006f675: mov word ptr [esi+2],5
0x1006f67b: mov dword ptr [esi+4],1
0x1006f682: call 1006EA55h
0x1006f687: test eax,eax
0x1006f689: pop ecx
0x1006f68a: pop ecx
0x1006f68b: je near ptr 1006FA44h
```

```
0x1006f691: mov eax,[ebp-8]
0x1006f694: mov word ptr [eax],11Bh
0x1006f699: add edi,5Ch
0x1006f69c: jmp short 1006F63Ch
0x1006f69e: mov esi,[edi+1Ch]
0x1006f6a1: mov edx,[ebp-4]
0x1006f6a4: push 142h
0x1006f6a9: mov ecx,ebx
0x1006f6ab: call 1006E58Fh
0x1006f6b0: add dword ptr [ebp-4],0Ch
0x1006f6b4: add dword ptr [ebp-8],0Ch
0x1006f6b8: mov esi,[edi+20h]
0x1006f6bb: mov dword ptr [esp],143h
0x1006f6c2: jmp near ptr 1006F826h
0x1006f6c7: mov esi,[edi+10h]
0x1006f6ca: mov edx,[ebp-4]
0x1006f6cd: push 100h
0x1006f6d2: mov ecx,ebx
0x1006f6d4: call 1006E58Fh
0x1006f6d9: add dword ptr [ebp-4],0Ch
0x1006f6dd: add dword ptr [ebp-8],0Ch
0x1006f6e1: mov esi,[edi+14h]
0x1006f6e4: mov dword ptr [esp],101h
0x1006f6eb: jmp near ptr 1006F826h
0x1006f6f0: movzx eax,word ptr [edi+30h]
0x1006f6f4: push eax
0x1006f6f5: push 103h
0x1006f6fa: mov edx,esi
0x1006f6fc: mov ecx,ebx
0x1006f6fe: call 1006E5DBh
0x1006f703: pop ecx
0x1006f704: jmp near ptr 1006F830h
0x1006f709: cmp ecx,8
0x1006f70c: je near ptr 1006F836h
0x1006f712: cmp ecx,11h
0x1006f715: je near ptr 1006F81Eh
0x1006f71b: jle near ptr 1006F949h
0x1006f721: cmp ecx,13h
0x1006f724: jle near ptr 1006F5F9h
0x1006f72a: cmp ecx,17h
0x1006f72d: jne near ptr 1006F949h
0x1006f733: mov eax,[ebp-1Ch]
0x1006f736: mov edi,[eax]
0x1006f738: mov esi,[ebp-4]
0x1006f73b: push ebx
0x1006f73c: call 1006F35Ah
0x1006f741: pop ecx
0x1006f742: test eax,eax
0x1006f744: je near ptr 1006FA44h
0x1006f74a: mov eax,[ebp-1Ch]
0x1006f74d: add dword ptr [ebp-4],0Ch
0x1006f751: add dword ptr [ebp-8],0Ch
0x1006f755: cmp word ptr [eax+0Ch],41h
0x1006f75a: je short 1006F775h
0x1006f75c: movzx ecx,word ptr [eax+0Ch]
0x1006f760: xor edx,edx
0x1006f762: mov eax,ecx
0x1006f764: inc edx
0x1006f765: and ecx,1Fh
0x1006f768: shl edx,cl
0x1006f76a: shr eax,5
0x1006f76d: lea eax,[ebp+eax*4-48h]
0x1006f771: not edx
0x1006f773: and [eax],edx
0x1006f775: dec dword ptr [ebp-30h]
0x1006f778: add dword ptr [ebp-2Ch],4
0x1006f77c: cmp dword ptr [ebp-30h],0
```

```
0x1006f780: jg near ptr 1006F540h
0x1006f786: test byte ptr [ebx+0Ch],80h
0x1006f78a: movzx eax,word ptr [ebp-0Ch]
0x1006f78e: mov [ebp-14h],eax
0x1006f791: mov eax,[ebx+14h]
0x1006f794: mov [ebp-38h],eax
0x1006f797: je short 1006F7E5h
0x1006f799: cmp word ptr [ebp-0Ch],0
0x1006f79e: mov esi,[ebp-10h]
0x1006f7a1: je short 1006F7CAh
0x1006f7a3: push 2
0x1006f7a5: push esi
0x1006f7a6: call 1006FB53h
0x1006f7ab: lea eax,[esi+4]
0x1006f7ae: push 2
0x1006f7b0: push eax
0x1006f7b1: call 1006FB9Ah
0x1006f7b6: add dword ptr [ebp-14h],0FFFFh
0x1006f7bd: add esp,10h
0x1006f7c0: add esi,0Ch
0x1006f7c3: cmp word ptr [ebp-14h],0
0x1006f7c8: jne short 1006F7A3h
0x1006f7ca: movzx eax,word ptr [ebp-0Ch]
0x1006f7ce: mov [ebp-14h],eax
0x1006f7d1: lea eax,[ebp-14h]
0x1006f7d4: push eax
0x1006f7d5: call 1006FB29h
0x1006f7da: lea eax,[ebp-38h]
0x1006f7dd: push eax
0x1006f7de: call 1006FB38h
0x1006f7e3: pop ecx
0x1006f7e4: pop ecx
0x1006f7e5: push 0
0x1006f7e7: push dword ptr [ebx+10h]
0x1006f7ea: push dword ptr [ebx+274h]
0x1006f7f0: call dword ptr [ebx+280h]
0x1006f7f6: push 2
0x1006f7f8: lea eax,[ebp-14h]
0x1006f7fb: push eax
0x1006f7fc: push dword ptr [ebx+274h]
0x1006f802: call dword ptr [ebx+27Ch]
0x1006f808: add esp,18h
0x1006f80b: cmp eax,2
0x1006f80e: je near ptr 1006F9F4h
0x1006f814: push 10108FC8h
0x1006f819: jmp near ptr 1006FA34h
0x1006f81e: mov esi,[edi+3Ch]
0x1006f821: push 116h
0x1006f826: mov edx,[ebp-4]
0x1006f829: mov ecx,ebx
0x1006f82b: call 1006E58Fh
0x1006f830: pop ecx
0x1006f831: jmp near ptr 1006F74Ah
0x1006f836: movzx eax,word ptr [edi+32h]
0x1006f83a: push eax
0x1006f83b: push 106h
0x1006f840: jmp near ptr 1006F6FAh
0x1006f845: mov edx,[ebx+0Ch]
0x1006f848: and edx,400h
0x1006f84e: neg edx
0x1006f850: sbb edx,edx
0x1006f852: and edx,2Eh
0x1006f855: add edx,117h
0x1006f85b: cmp edx,[eax]
0x1006f85d: jne near ptr 1006F775h
0x1006f863: mov ecx,[ebp-4]
0x1006f866: mov eax,[ebp-18h]
```

```
0x1006f869: mov [ecx],dx
0x1006f86c: mov word ptr [ecx+2],4
0x1006f872: mov edx,[eax+8Ch]
0x1006f878: mov [ecx+4],edx
0x1006f87b: mov eax,[eax+94h]
0x1006f881: mov edi,ebx
0x1006f883: call 1006EB83h
0x1006f888: jmp near ptr 1006F742h
0x1006f88d: cmp ecx,25h
0x1006f890: jg near ptr 1006F92Ah
0x1006f896: je near ptr 1006F733h
0x1006f89c: cmp ecx,19h
0x1006f89f: je short 1006F8E9h
0x1006f8a1: cmp ecx,1Ah
0x1006f8a4: je short 1006F8D3h
0x1006f8a6: cmp ecx,20h
0x1006f8a9: je near ptr 1006F5F9h
0x1006f8af: add ecx,0FFFFFFDFh
0x1006f8b2: cmp ecx,1
0x1006f8b5: ja near ptr 1006F949h
0x1006f8bb: push esi
0x1006f8bc: push dword ptr [eax]
0x1006f8be: push ebx
0x1006f8bf: call 1006FD74h
0x1006f8c4: pop ecx
0x1006f8c5: push eax
0x1006f8c6: call 1006F2C7h
0x1006f8cb: add esp,0Ch
0x1006f8ce: jmp near ptr 1006F742h
0x1006f8d3: add edi,70h
0x1006f8d6: push edi
0x1006f8d7: push 3
0x1006f8d9: push 140h
0x1006f8de: mov eax,esi
0x1006f8e0: mov ecx,ebx
0x1006f8e2: call 1006E90Eh
0x1006f8e7: jmp short 1006F8CBh
0x1006f8e9: mov edx,[ebx+0Ch]
0x1006f8ec: and edx,400h
0x1006f8f2: neg edx
0x1006f8f4: sbb edx,edx
0x1006f8f6: and edx,33h
0x1006f8f9: add edx,111h
0x1006f8ff: cmp edx,[eax]
0x1006f901: jne near ptr 1006F775h
0x1006f907: mov ecx,[ebp-4]
0x1006f90a: mov eax,[ebp-18h]
0x1006f90d: mov [ecx],dx
0x1006f910: mov word ptr [ecx+2],4
0x1006f916: mov edx,[eax+8Ch]
0x1006f91c: mov [ecx+4],edx
0x1006f91f: mov eax,[eax+90h]
0x1006f925: jmp near ptr 1006F881h
0x1006f92a: sub ecx,27h
0x1006f92d: je near ptr 1006F733h
0x1006f933: sub ecx,5
0x1006f936: je near ptr 1006F9E5h
0x1006f93c: dec ecx
0x1006f93d: dec ecx
0x1006f93e: je near ptr 1006F9D7h
0x1006f944: sub ecx,3
0x1006f947: je short 1006F964h
0x1006f949: mov edi,150h
0x1006f94e: cmp [eax],edi
0x1006f950: je near ptr 1006F738h
0x1006f956: mov ecx,esi
0x1006f958: mov edx,ebx
```

```
0x1006f95a: call 1006EE47h
0x1006f95f: jmp near ptr 1006F742h
0x1006f964: mov eax,[ebp-1Ch]
0x1006f967: mov ax,[eax]
0x1006f96a: mov ecx,[ebp-4]
0x1006f96d: mov [ecx],ax
0x1006f970: mov eax,[ebp-18h]
0x1006f973: mov word ptr [ecx+2],4
0x1006f979: movzx edx,word ptr [eax+9Ch]
0x1006f980: mov [ecx+4],edx
0x1006f983: mov eax,[eax+0A0h]
0x1006f989: mov edi,ebx
0x1006f98b: call 1006EB83h
0x1006f990: test eax,eax
0x1006f992: je near ptr 1006FA44h
0x1006f998: mov eax,[ebp-4]
0x1006f99b: cmp dword ptr [eax+4],0
0x1006f99f: jbe near ptr 1006F74Ah
0x1006f9a5: or dword ptr [ebx+0Ch],2000h
0x1006f9ac: mov cx,[eax+4]
0x1006f9b0: mov [ebx+1ECh],cx
0x1006f9b7: cmp dword ptr [eax+4],1
0x1006f9bb: jbe short 1006F9C2h
0x1006f9bd: mov eax,[eax+8]
0x1006f9c0: jmp short 1006F9CCh
0x1006f9c2: mov ecx,[ebx+10h]
0x1006f9c5: sub ecx,[ebp-10h]
0x1006f9c8: lea eax,[ecx+eax+0Ah]
0x1006f9cc: mov [ebx+1F0h],eax
0x1006f9d2: jmp near ptr 1006F74Ah
0x1006f9d7: mov ecx,esi
0x1006f9d9: mov edx,ebx
0x1006f9db: call 1006EE28h
0x1006f9e0: jmp near ptr 1006F742h
0x1006f9e5: push dword ptr [ebp-4]
0x1006f9e8: mov esi,ebx
0x1006f9ea: call 1006EDB9h
0x1006f9ef: jmp near ptr 1006F741h
0x1006f9f4: push dword ptr [ebp-20h]
0x1006f9f7: push dword ptr [ebp-10h]
0x1006f9fa: push dword ptr [ebx+274h]
0x1006fa00: call dword ptr [ebx+27Ch]
0x1006fa06: add esp,0Ch
0x1006fa09: cmp eax,[ebp-20h]
0x1006fa0c: je short 1006FA15h
0x1006fa0e: push 10108FA4h
0x1006fa13: jmp short 1006FA34h
0x1006fa15: push 4
0x1006fa17: lea eax,[ebp-38h]
0x1006fa1a: push eax
0x1006fa1b: push dword ptr [ebx+274h]
0x1006fa21: call dword ptr [ebx+27Ch]
0x1006fa27: add esp,0Ch
0x1006fa2a: cmp eax,4
0x1006fa2d: je short 1006FA50h
0x1006fa2f: push 10108464h
0x1006fa34: push dword ptr [ebx]
0x1006fa36: push dword ptr [ebx+274h]
0x1006fa3c: call 1006FAF1h
0x1006fa41: add esp,0Ch
0x1006fa44: push dword ptr [ebp-10h]
0x1006fa47: call 100209E0h
0x1006fa4c: xor eax,eax
0x1006fa4e: jmp short 1006FA7Bh
0x1006fa50: cmp dword ptr [ebp+8],0
0x1006fa54: je short 1006FA70h
0x1006fa56: push ebx
```

```
0x1006fa57: call 10067617h
0x1006fa5c: and dword ptr [ebx+0Ch],0FFFFFFF7h
0x1006fa60: push ebx
0x1006fa61: call dword ptr [ebx+23Ch]
0x1006fa67: push ebx
0x1006fa68: call 10067A7Eh
0x1006fa6d: add esp,0Ch
0x1006fa70: push dword ptr [ebp-10h]
0x1006fa73: call 100209E0h
0x1006fa78: xor eax,eax
0x1006fa7a: inc eax
0x1006fa7b: pop ecx
0x1006fa7c: pop edi
0x1006fa7d: pop esi
0x1006fa7e: pop ebx
0x1006fa7f: leave
0x1006fa80: ret
```

`sub_1006f391` is called by sub_1006fa81. It works with the text "Error writing directory link", "Error writing directory contents", "Error writing directory count". It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to j_free, j_memcpy, sub_10065e73 (24 callees).

12.1016 sub_1006fa81 at 0x1006fa81 (5 instructions)

```
0x1006fa81: mov eax,[esp+4]
0x1006fa85: push 1
0x1006fa87: call 1006F391h
0x1006fa8c: pop ecx
0x1006fa8d: ret
```

`sub_1006fa81` is called by sub_1006fac1. It delegates to sub_1006f391 (1 callees).

12.1017 sub_1006fa8e at 0x1006fa8e (22 instructions)

```
0x1006fa8e: push esi
0x1006fa8f: mov esi,[esp+8]
0x1006fa93: mov eax,[esi+0Ch]
0x1006fa96: test al,40h
0x1006fa98: jne short 1006FA9Eh
0x1006fa9a: xor eax,eax
0x1006fa9c: pop esi
0x1006fa9d: ret
0x1006fa9e: test ax,1000h
0x1006faa2: je short 1006FAB8h
0x1006faa4: and eax,0FFFFEFFFh
0x1006faa9: push esi
0x1006faaa: mov [esi+0Ch],eax
0x1006faad: call dword ptr [esi+218h]
0x1006fab3: test eax,eax
0x1006fab5: pop ecx
0x1006fab6: je short 1006FA9Ah
0x1006fab8: push esi
0x1006fab9: call 10065E73h
0x1006fabe: pop ecx
0x1006fabf: pop esi
0x1006fac0: ret
```

`sub_1006fa8e` is called by sub_1006fac1. It delegates to sub_10065e73 (1 callees).

12.1018 sub_1006fac1 at 0x1006fac1 (23 instructions)

```
0x1006fac1: push esi
0x1006fac2: mov esi,[esp+8]
0x1006fac6: cmp dword ptr [esi+8],0
```

```
0x1006faca: je short 1006FAECh
0x1006facc: push esi
0x1006facd: call 1006FA8Eh
0x1006fad2: test eax,eax
0x1006fad4: pop ecx
0x1006fad5: jne short 1006FADBh
0x1006fad7: xor eax,eax
0x1006fad9: pop esi
0x1006fada: ret
0x1006fadb: test byte ptr [esi+0Ch],8
0x1006fadf: je short 1006FAECh
0x1006fae1: push esi
0x1006fae2: call 1006FA81h
0x1006fae7: test eax,eax
0x1006fae9: pop ecx
0x1006faea: je short 1006FAD7h
0x1006faec: xor eax,eax
0x1006faee: inc eax
0x1006faef: pop esi
0x1006faf0: ret
```

`sub_1006fac1` is called by sub_10065c7b. It delegates to sub_1006fa81, sub_1006fa8e (2 callees).

12.1019 sub_1006faf1 at 0x1006faf1 (23 instructions)

```
0x1006faf1: push ebp
0x1006faf2: mov ebp,esp
0x1006faf4: mov eax,ds:[101361FCh]
0x1006faf9: test eax,eax
0x1006fafb: je short 1006FB0Ch
0x1006fafd: lea ecx,[ebp+14h]
0x1006fb00: push ecx
0x1006fb01: push dword ptr [ebp+10h]
0x1006fb04: push dword ptr [ebp+0Ch]
0x1006fb07: call eax
0x1006fb09: add esp,0Ch
0x1006fb0c: mov eax,ds:[101393ACh]
0x1006fb11: test eax,eax
0x1006fb13: je short 1006FB27h
0x1006fb15: lea ecx,[ebp+14h]
0x1006fb18: push ecx
0x1006fb19: push dword ptr [ebp+10h]
0x1006fb1c: push dword ptr [ebp+0Ch]
0x1006fb1f: push dword ptr [ebp+8]
0x1006fb22: call eax
0x1006fb24: add esp,10h
0x1006fb27: pop ebp
0x1006fb28: ret
```

`sub_1006faf1` is called by sub_10065d91, sub_10065eea, sub_10065f13, sub_10065faf, sub_10066210, sub_10066f46, sub_10067774, sub_10067aa8, sub_10067b9e, sub_10067caa, sub_10067ea4, sub_10067f4a, sub_100680ec, sub_10068247, sub_1006831d, sub_10068414, sub_1006853e, sub_100686da, sub_100688d2, sub_1006891d, sub_10069057, sub_1006cec2, sub_1006cf39, sub_1006cff7, sub_1006d21a, sub_1006d370, sub_1006df2c, sub_1006e029, sub_1006e289, sub_1006e625, sub_1006e717, sub_1006ea55, sub_1006eb9f, sub_1006eba4, sub_1006f20b, sub_1006f2c7, sub_1006f391, sub_1006fe65, sub_1006ff15, sub_1006ffe0, sub_100702f4, sub_1007049b, sub_10070584, sub_10070799, sub_10070c4a, sub_10071192, sub_1007127c, sub_10071346, sub_1007171b. It calls nothing else.

12.1020 sub_1006fb29 at 0x1006fb29 (6 instructions)

```
0x1006fb29: mov eax,[esp+4]
0x1006fb2d: mov cl,[eax+1]
0x1006fb30: mov dl,[eax]
```

```
0x1006fb32: mov [eax+1],dl
0x1006fb35: mov [eax],cl
0x1006fb37: ret
```

`sub_1006fb29` is called by sub_10067774, sub_1006891d, sub_1006e717, sub_1006f391, sub_10070584. It calls nothing else.

12.1021 sub_1006fb38 at 0x1006fb38 (10 instructions)

```
0x1006fb38: mov eax,[esp+4]
0x1006fb3c: mov cl,[eax+3]
0x1006fb3f: mov dl,[eax]
0x1006fb41: mov [eax+3],dl
0x1006fb44: mov dl,[eax+1]
0x1006fb47: mov [eax],cl
0x1006fb49: mov cl,[eax+2]
0x1006fb4c: mov [eax+2],dl
0x1006fb4f: mov [eax+1],cl
0x1006fb52: ret
```

`sub_1006fb38` is called by sub_10067774, sub_1006891d, sub_1006e717, sub_1006f391, sub_10070584. It calls nothing else.

12.1022 sub_1006fb53 at 0x1006fb53 (13 instructions)

```
0x1006fb53: cmp dword ptr [esp+8],0
0x1006fb58: jbe short 1006FB75h
0x1006fb5a: mov eax,[esp+4]
0x1006fb5e: mov cl,[eax+1]
0x1006fb61: mov dl,[eax]
0x1006fb63: dec dword ptr [esp+8]
0x1006fb67: mov [eax+1],dl
0x1006fb6a: mov [eax],cl
0x1006fb6c: inc eax
0x1006fb6d: inc eax
0x1006fb6e: cmp dword ptr [esp+8],0
0x1006fb73: ja short 1006FB5Eh
0x1006fb75: ret
```

`sub_1006fb53` is called by sub_1006f391, sub_1007171b. It calls nothing else.

12.1023 sub_1006fb9a at 0x1006fb9a (16 instructions)

```
0x1006fb9a: cmp dword ptr [esp+8],0
0x1006fb9f: jbe short 1006FBC9h
0x1006fba1: mov eax,[esp+4]
0x1006fba5: mov cl,[eax+3]
0x1006fba8: mov dl,[eax]
0x1006fbaa: dec dword ptr [esp+8]
0x1006fbae: mov [eax+3],dl
0x1006fbb1: mov dl,[eax+1]
0x1006fbb4: mov [eax],cl
0x1006fbb6: mov cl,[eax+2]
0x1006fbb9: mov [eax+2],dl
0x1006fbbc: mov [eax+1],cl
0x1006fbbf: add eax,4
0x1006fbc2: cmp dword ptr [esp+8],0
0x1006fbc7: ja short 1006FBA5h
0x1006fbc9: ret
```

`sub_1006fb9a` is called by sub_1006f391, sub_1007171b. It calls nothing else.

12.1024 sub_1006fc09 at 0x1006fc09 (46 instructions)

```
0x1006fc09: mov eax,[esp+4]
0x1006fc0d: push esi
0x1006fc0e: mov esi,[esp+0Ch]
0x1006fc12: cmp esi,8
0x1006fc15: jbe near ptr 1006FCA0h
0x1006fc1b: lea ecx,[esi-9]
0x1006fc1e: shr ecx,3
0x1006fc21: inc ecx
0x1006fc22: movzx edx,byte ptr [eax]
0x1006fc25: mov dl,[edx+101091C0h]
0x1006fc2b: mov [eax],dl
0x1006fc2d: movzx edx,byte ptr [eax+1]
0x1006fc31: mov dl,[edx+101091C0h]
0x1006fc37: mov [eax+1],dl
0x1006fc3a: movzx edx,byte ptr [eax+2]
0x1006fc3e: mov dl,[edx+101091C0h]
0x1006fc44: mov [eax+2],dl
0x1006fc47: movzx edx,byte ptr [eax+3]
0x1006fc4b: mov dl,[edx+101091C0h]
0x1006fc51: mov [eax+3],dl
0x1006fc54: movzx edx,byte ptr [eax+4]
0x1006fc58: mov dl,[edx+101091C0h]
0x1006fc5e: mov [eax+4],dl
0x1006fc61: movzx edx,byte ptr [eax+5]
0x1006fc65: mov dl,[edx+101091C0h]
0x1006fc6b: mov [eax+5],dl
0x1006fc6e: movzx edx,byte ptr [eax+6]
0x1006fc72: mov dl,[edx+101091C0h]
0x1006fc78: mov [eax+6],dl
0x1006fc7b: movzx edx,byte ptr [eax+7]
0x1006fc7f: mov dl,[edx+101091C0h]
0x1006fc85: mov [eax+7],dl
0x1006fc88: add eax,8
0x1006fc8b: sub esi,8
0x1006fc8e: dec ecx
0x1006fc8f: jne short 1006FC22h
0x1006fc91: jmp short 1006FCA0h
0x1006fc93: movzx ecx,byte ptr [eax]
0x1006fc96: mov cl,[ecx+101091C0h]
0x1006fc9c: dec esi
0x1006fc9d: mov [eax],cl
0x1006fc9f: inc eax
0x1006fca0: test esi,esi
0x1006fca2: ja short 1006FC93h
0x1006fca4: pop esi
0x1006fca5: ret
```

`sub_1006fc09` is called by sub_10065e73, sub_10067f4a, sub_100680ec. It calls nothing else.

12.1025 sub_1006fca6 at 0x1006fca6 (4 instructions)

```
0x1006fca6: mov eax,[esp+4]
0x1006fcaa: mov dword ptr [eax],0A6h
0x1006fcb0: mov eax,10109FC8h
0x1006fcb5: ret
```

`sub_1006fca6` is called by sub_1006799b. It calls nothing else.

12.1026 sub_1006fcb6 at 0x1006fcb6 (22 instructions)

```
0x1006fcb6: mov eax,[esp+4]
0x1006fcba: mov ecx,[eax]
0x1006fcbc: mov eax,[esp+8]
0x1006fcc0: mov edx,[eax]
```

```
0x1006fcc2: mov eax,[ecx]
0x1006fcc4: push esi
0x1006fcc5: mov esi,[edx]
0x1006fcc7: cmp eax,esi
0x1006fcc9: je short 1006FCCFh
0x1006fccb: sub eax,esi
0x1006fccd: pop esi
0x1006fcce: ret
0x1006fccf: mov ecx,[ecx+8]
0x1006fcd2: test ecx,ecx
0x1006fcd4: jne short 1006FCDAh
0x1006fcd6: xor eax,eax
0x1006fcd8: pop esi
0x1006fcd9: ret
0x1006fcda: mov eax,[edx+8]
0x1006fcdd: sub eax,ecx
0x1006fcdf: pop esi
0x1006fce0: ret
```

`sub_1006fcb6` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.1027 sub_1006fce1 at 0x1006fce1 (7 instructions)

```
0x1006fce1: mov eax,[esp+4]
0x1006fce5: cmp eax,0Dh
0x1006fce8: ja short 1006FD08h
0x1006fcea: movzx eax,byte ptr [eax+1006FD1Dh]
0x1006fcf1: jmp dword ptr [eax*4+1006FD0Dh]
0x1006fd08: xor eax,eax
0x1006fd0a: ret
```

`sub_1006fce1` is called by sub_1006e625, sub_1006eb9f, sub_1006eba4, sub_100702f4, sub_10070799. It calls nothing else.

12.1028 sub_1006fd74 at 0x1006fd74 (49 instructions)

```
0x1006fd74: mov ecx,[esp+4]
0x1006fd78: movzx eax,word ptr [ecx+54h]
0x1006fd7c: test al,7
0x1006fd7e: movzx eax,ax
0x1006fd81: je short 1006FD89h
0x1006fd83: shr eax,3
0x1006fd86: inc eax
0x1006fd87: jmp short 1006FD8Ch
0x1006fd89: shr eax,3
0x1006fd8c: movzx ecx,word ptr [ecx+56h]
0x1006fd90: dec ecx
0x1006fd91: je short 1006FDC1h
0x1006fd93: dec ecx
0x1006fd94: je short 1006FDABh
0x1006fd96: dec ecx
0x1006fd97: je short 1006FD9Dh
0x1006fd99: push 7
0x1006fd9b: pop eax
0x1006fd9c: ret
0x1006fd9d: xor ecx,ecx
0x1006fd9f: cmp eax,4
0x1006fda2: setne cl
0x1006fda5: add ecx,0Bh
0x1006fda8: mov eax,ecx
0x1006fdaa: ret
0x1006fdab: cmp eax,1
0x1006fdae: ja short 1006FDB4h
0x1006fdb0: push 6
0x1006fdb2: pop eax
```

```
0x1006fdb3: ret
0x1006fdb4: push 2
0x1006fdb6: pop ecx
0x1006fdb7: cmp ecx,eax
0x1006fdb9: sbb eax,eax
0x1006fdbb: neg eax
0x1006fdbd: add eax,8
0x1006fdc0: ret
0x1006fdc1: cmp eax,1
0x1006fdc4: ja short 1006FDCAh
0x1006fdc6: xor eax,eax
0x1006fdc8: inc eax
0x1006fdc9: ret
0x1006fdca: push 2
0x1006fdcc: pop ecx
0x1006fdcd: cmp ecx,eax
0x1006fdcf: sbb eax,eax
0x1006fdd1: neg eax
0x1006fdd3: add eax,3
0x1006fdd6: ret
```

`sub_1006fd74` is called by sub_1006f391. It calls nothing else.

12.1029 sub_1006fdd7 at 0x1006fdd7 (53 instructions)

```
0x1006fdd7: push ebp
0x1006fdd8: mov ebp,esp
0x1006fdda: sub esp,18h
0x1006fddd: mov ecx,[ebp+10h]
0x1006fde0: mov edx,[ebp+0Ch]
0x1006fde3: push ebx
0x1006fde4: xor ebx,ebx
0x1006fde6: lea eax,[ebp-18h]
0x1006fde9: push esi
0x1006fdea: mov esi,[ebp+8]
0x1006fded: mov [ebp-4],eax
0x1006fdf0: mov eax,[esi+298h]
0x1006fdf6: cmp eax,ebx
0x1006fdf8: mov [ebp-18h],ebx
0x1006fdfb: mov [ebp-14h],bx
0x1006fdff: mov [ebp-12h],bx
0x1006fe03: mov [ebp-10h],ebx
0x1006fe06: mov [ebp-0Ch],bx
0x1006fe0a: mov [ebp-0Ah],bl
0x1006fe0d: mov [ebp-9],bl
0x1006fe10: mov [ebp-8],ebx
0x1006fe13: je short 1006FE22h
0x1006fe15: cmp [eax],edx
0x1006fe17: jne short 1006FE22h
0x1006fe19: cmp ecx,ebx
0x1006fe1b: je short 1006FE61h
0x1006fe1d: cmp ecx,[eax+8]
0x1006fe20: je short 1006FE61h
0x1006fe22: mov eax,[esi+290h]
0x1006fe28: cmp eax,ebx
0x1006fe2a: jne short 1006FE30h
0x1006fe2c: xor eax,eax
0x1006fe2e: jmp short 1006FE61h
0x1006fe30: push 1006FCB6h
0x1006fe35: push 4
0x1006fe37: push dword ptr [esi+294h]
0x1006fe3d: mov [ebp-18h],edx
0x1006fe40: push eax
0x1006fe41: lea eax,[ebp-4]
0x1006fe44: push eax
0x1006fe45: mov [ebp-10h],ecx
```

```
0x1006fe48: call dword ptr ds:[100FE490h]
0x1006fe4e: add esp,14h
0x1006fe51: cmp eax,ebx
0x1006fe53: je short 1006FE59h
0x1006fe55: mov eax,[eax]
0x1006fe57: jmp short 1006FE5Bh
0x1006fe59: xor eax,eax
0x1006fe5b: mov [esi+298h],eax
0x1006fe61: pop esi
0x1006fe62: pop ebx
0x1006fe63: leave
0x1006fe64: ret
```

`sub_1006fdd7` is called by sub_10066f46, sub_100675c7, sub_1006fe65, sub_1006ff15. It calls nothing else.

12.1030 sub_1006fe65 at 0x1006fe65 (21 instructions)

```
0x1006fe65: push esi
0x1006fe66: mov esi,[esp+8]
0x1006fe6a: push edi
0x1006fe6b: push 0
0x1006fe6d: push dword ptr [esp+14h]
0x1006fe71: push esi
0x1006fe72: call 1006FDD7h
0x1006fe77: mov edi,eax
0x1006fe79: add esp,0Ch
0x1006fe7c: test edi,edi
0x1006fe7e: jne short 1006FE9Ch
0x1006fe80: push dword ptr [esp+10h]
0x1006fe84: push 1010B188h
0x1006fe89: push 1010B174h
0x1006fe8e: push dword ptr [esi+274h]
0x1006fe94: call 1006FAF1h
0x1006fe99: add esp,10h
0x1006fe9c: mov eax,edi
0x1006fe9e: pop edi
0x1006fe9f: pop esi
0x1006fea0: ret
```

`sub_1006fe65` is called by sub_1006e625, sub_1006ea55, sub_1006eb9f, sub_10070528, sub_10070799, sub_10070aeb, sub_10070c4a, sub_10070dd6, sub_10071192, sub_1007127c, sub_10071346, sub_1007171b. It works with the text "TIFFFieldWithTag", "Internal error, unknown tag 0x%x". It delegates to sub_1006faf1, sub_1006fdd7 (2 callees).

12.1031 sub_1006fea1 at 0x1006fea1 (43 instructions)

```
0x1006fea1: push esi
0x1006fea2: push 14h
0x1006fea4: call 100EFDD0h
0x1006fea9: mov esi,eax
0x1006feab: test esi,esi
0x1006fead: pop ecx
0x1006feae: jne short 1006FEB2h
0x1006feb0: pop esi
0x1006feb1: ret
0x1006feb2: push edi
0x1006feb3: push 14h
0x1006feb5: push 0
0x1006feb7: push esi
0x1006feb8: call 100209F0h
0x1006febd: mov eax,[esp+20h]
0x1006fec1: mov edi,[esp+1Ch]
0x1006fec5: push 20h
0x1006fec7: mov [esi],edi
0x1006fec9: mov word ptr [esi+4],0FFFDh
0x1006fecf: mov word ptr [esi+6],0FFFDh
```

```
0x1006fed5: mov [esi+8],eax
0x1006fed8: mov word ptr [esi+0Ch],41h
0x1006fede: mov byte ptr [esi+0Eh],1
0x1006fee2: mov byte ptr [esi+0Fh],1
0x1006fee6: call 100EFDD0h
0x1006feeb: add esp,10h
0x1006feee: test eax,eax
0x1006fef0: mov [esi+10h],eax
0x1006fef3: jne short 1006FF00h
0x1006fef5: push esi
0x1006fef6: call 100209E0h
0x1006fefb: pop ecx
0x1006fefc: xor eax,eax
0x1006fefe: jmp short 1006FF12h
0x1006ff00: push edi
0x1006ff01: push 1010B1ACh
0x1006ff06: push eax
0x1006ff07: call dword ptr ds:[100FE4F8h]
0x1006ff0d: add esp,0Ch
0x1006ff10: mov eax,esi
0x1006ff12: pop edi
0x1006ff13: pop esi
0x1006ff14: ret
```

`sub_1006fea1` is called by sub_1007171b. It works with the text "Tag %d". It delegates to j_free, j_memset, j_malloc (3 callees).

12.1032 sub_1006ff15 at 0x1006ff15 (66 instructions)

```
0x1006ff15: push ebp
0x1006ff16: mov ebp,esp
0x1006ff18: push esi
0x1006ff19: mov esi,[ebp+8]
0x1006ff1c: mov eax,[esi+294h]
0x1006ff22: and dword ptr [esi+298h],0
0x1006ff29: test eax,eax
0x1006ff2b: push 1010B15Ch
0x1006ff30: push 4
0x1006ff32: jbe short 1006FF4Bh
0x1006ff34: mov ecx,[ebp+10h]
0x1006ff37: add eax,ecx
0x1006ff39: push eax
0x1006ff3a: push dword ptr [esi+290h]
0x1006ff40: push esi
0x1006ff41: call 1006E029h
0x1006ff46: add esp,14h
0x1006ff49: jmp short 1006FF57h
0x1006ff4b: push dword ptr [ebp+10h]
0x1006ff4e: push esi
0x1006ff4f: call 1006E083h
0x1006ff54: add esp,10h
0x1006ff57: test eax,eax
0x1006ff59: mov [esi+290h],eax
0x1006ff5f: jne short 1006FF7Dh
0x1006ff61: push 1010B1B4h
0x1006ff66: push 1010B148h
0x1006ff6b: push dword ptr [esi+274h]
0x1006ff71: call 1006FAF1h
0x1006ff76: add esp,0Ch
0x1006ff79: xor eax,eax
0x1006ff7b: jmp short 1006FFDDh
0x1006ff7d: cmp dword ptr [ebp+10h],0
0x1006ff81: mov ecx,[esi+294h]
0x1006ff87: push ebx
0x1006ff88: lea ebx,[eax+ecx*4]
0x1006ff8b: jle short 1006FFBDh
```

```
0x1006ff8d: mov eax,[ebp+10h]
0x1006ff90: push edi
0x1006ff91: mov edi,[ebp+0Ch]
0x1006ff94: mov [ebp+8],eax
0x1006ff97: push dword ptr [edi+8]
0x1006ff9a: push dword ptr [edi]
0x1006ff9c: push esi
0x1006ff9d: call 1006FDD7h
0x1006ffa2: add esp,0Ch
0x1006ffa5: test eax,eax
0x1006ffa7: jne short 1006FFB4h
0x1006ffa9: mov [ebx],edi
0x1006ffab: add ebx,4
0x1006ffae: inc dword ptr [esi+294h]
0x1006ffb4: add edi,14h
0x1006ffb7: dec dword ptr [ebp+8]
0x1006ffba: jne short 1006FF97h
0x1006ffbc: pop edi
0x1006ffbd: push 1006FCB6h
0x1006ffc2: push 4
0x1006ffc4: push dword ptr [esi+294h]
0x1006ffca: push dword ptr [esi+290h]
0x1006ffd0: call dword ptr ds:[100FE408h]
0x1006ffd6: mov eax,[ebp+10h]
0x1006ffd9: add esp,10h
0x1006ffdc: pop ebx
0x1006ffdd: pop esi
0x1006ffde: pop ebp
0x1006ffdf: ret
```

`sub_1006ff15` is called by sub_1006ffe0, sub_1007171b. It works with the text "_TIFFMergeFieldInfo", "for field info array", "Failed to allocate field info array". It delegates to sub_1006e029, sub_1006e083, sub_1006faf1 (4 callees).

12.1033 sub_1006ffe0 at 0x1006ffe0 (49 instructions)

```
0x1006ffe0: push ebx
0x1006ffe1: push esi
0x1006ffe2: mov esi,[esp+0Ch]
0x1006ffe6: xor ebx,ebx
0x1006ffe8: cmp [esi+290h],ebx
0x1006ffee: je short 1007004Dh
0x1006fff0: cmp [esi+294h],ebx
0x1006fff6: jbe short 1007003Ah
0x1006fff8: push edi
0x1006fff9: mov eax,[esi+290h]
0x1006ffff: mov edi,[eax+ebx*4]
0x10070002: cmp word ptr [edi+0Ch],41h
0x10070007: jne short 10070030h
0x10070009: push 4
0x1007000b: push dword ptr [edi+10h]
0x1007000e: push 10108070h
0x10070013: call dword ptr ds:[100FE414h]
0x10070019: add esp,0Ch
0x1007001c: test eax,eax
0x1007001e: jne short 10070030h
0x10070020: push dword ptr [edi+10h]
0x10070023: call 100209E0h
0x10070028: push edi
0x10070029: call 100209E0h
0x1007002e: pop ecx
0x1007002f: pop ecx
0x10070030: inc ebx
0x10070031: cmp ebx,[esi+294h]
0x10070037: jb short 1006FFF9h
0x10070039: pop edi
0x1007003a: push dword ptr [esi+290h]
```

```
0x10070040: call 100209E0h
0x10070045: and dword ptr [esi+294h],0
0x1007004c: pop ecx
0x1007004d: push dword ptr [esp+14h]
0x10070051: push dword ptr [esp+14h]
0x10070055: push esi
0x10070056: call 1006FF15h
0x1007005b: add esp,0Ch
0x1007005e: test eax,eax
0x10070060: jne short 1007007Ah
0x10070062: push 1010B1ECh
0x10070067: push 1010B1D8h
0x1007006c: push dword ptr [esi+274h]
0x10070072: call 1006FAF1h
0x10070077: add esp,0Ch
0x1007007a: pop esi
0x1007007b: pop ebx
0x1007007c: ret
```

`sub_1006ffe0` is called by sub_1006799b. It works with the text "_TIFFSetupFieldInfo", "Setting up field info failed". It delegates to j_free, sub_1006faf1, sub_1006ff15 (3 callees).

12.1034 sub_1007007d at 0x1007007d (23 instructions)

```
0x1007007d: push ebp
0x1007007e: mov ebp,esp
0x10070080: mov eax,ds:[101361F8h]
0x10070085: test eax,eax
0x10070087: je short 10070098h
0x10070089: lea ecx,[ebp+14h]
0x1007008c: push ecx
0x1007008d: push dword ptr [ebp+10h]
0x10070090: push dword ptr [ebp+0Ch]
0x10070093: call eax
0x10070095: add esp,0Ch
0x10070098: mov eax,ds:[101393B0h]
0x1007009d: test eax,eax
0x1007009f: je short 100700B3h
0x100700a1: lea ecx,[ebp+14h]
0x100700a4: push ecx
0x100700a5: push dword ptr [ebp+10h]
0x100700a8: push dword ptr [ebp+0Ch]
0x100700ab: push dword ptr [ebp+8]
0x100700ae: call eax
0x100700b0: add esp,10h
0x100700b3: pop ebp
0x100700b4: ret
```

`sub_1007007d` is called by sub_1006d6a9, sub_1006ea55, sub_10070528, sub_10070aeb, sub_1007171b. It calls nothing else.

12.1035 sub_100701f4 at 0x100701f4 (28 instructions)

```
0x100701f4: mov eax,[esp+4]
0x100701f8: and dword ptr [eax+0Ch],0FFFDFEFFh
0x100701ff: push esi
0x10070200: mov ecx,100700D1h
0x10070205: xor esi,esi
0x10070207: inc esi
0x10070208: mov [eax+204h],ecx
0x1007020e: mov edx,100700D1h
0x10070213: mov [eax+20Ch],ecx
0x10070219: mov [eax+218h],ecx
0x1007021f: mov ecx,1007B5B1h
0x10070224: mov [eax+200h],esi
```

```
0x1007022a: mov [eax+210h],esi
0x10070230: mov [eax+208h],edx
0x10070236: mov dword ptr [eax+21Ch],100701BEh
0x10070240: mov dword ptr [eax+224h],100701D0h
0x1007024a: mov dword ptr [eax+22Ch],100701E2h
0x10070254: mov [eax+214h],edx
0x1007025a: mov dword ptr [eax+220h],1007014Dh
0x10070264: mov dword ptr [eax+228h],1007015Fh
0x1007026e: mov dword ptr [eax+230h],10070171h
0x10070278: mov [eax+234h],ecx
0x1007027e: mov dword ptr [eax+238h],100700B5h
0x10070288: mov [eax+23Ch],ecx
0x1007028e: mov dword ptr [eax+240h],10066366h
0x10070298: mov dword ptr [eax+244h],1006884Eh
0x100702a2: pop esi
0x100702a3: ret
```

`sub_100701f4` is called by sub_1006891d. It calls nothing else.

12.1036 sub_100702d3 at 0x100702d3 (12 instructions)

```
0x100702d3: xor ecx,ecx
0x100702d5: cmp [esp+4],cx
0x100702da: jbe short 100702F1h
0x100702dc: mov dx,[eax]
0x100702df: cmp dx,[esp+8]
0x100702e4: je short 100702F3h
0x100702e6: add eax,0Ch
0x100702e9: inc ecx
0x100702ea: cmp cx,[esp+4]
0x100702ef: jb short 100702DCh
0x100702f1: xor eax,eax
0x100702f3: ret
```

`sub_100702d3` is called by sub_1007171b. It calls nothing else.

12.1037 sub_100702f4 at 0x100702f4 (131 instructions)

```
0x100702f4: push ebp
0x100702f5: mov ebp,esp
0x100702f7: push ecx
0x100702f8: push ecx
0x100702f9: mov eax,[esi+0BCh]
0x100702ff: test eax,eax
0x10070301: je short 1007030Ah
0x10070303: push eax
0x10070304: call 100209E0h
0x10070309: pop ecx
0x1007030a: push 1010B3D8h
0x1007030f: push 4
0x10070311: push dword ptr [esi+0B4h]
0x10070317: push esi
0x10070318: call 1006E083h
0x1007031d: add esp,10h
0x10070320: test eax,eax
0x10070322: mov [esi+0BCh],eax
0x10070328: jne short 1007032Fh
0x1007032a: or eax,0FFFFFFFFh
0x1007032d: leave
0x1007032e: ret
0x1007032f: cmp word ptr [esi+58h],1
0x10070334: push ebx
0x10070335: push edi
0x10070336: je near ptr 1007041Fh
0x1007033c: movzx edi,word ptr [ebp+0Ch]
```

```
0x10070340: push dword ptr [esi+274h]
0x10070346: imul edi,0Ch
0x10070349: add edi,0Eh
0x1007034c: call dword ptr [esi+288h]
0x10070352: cmp word ptr [ebp+0Ch],0
0x10070357: mov ebx,[ebp+8]
0x1007035a: mov [ebp-8],eax
0x1007035d: movzx eax,word ptr [ebp+0Ch]
0x10070361: pop ecx
0x10070362: mov [ebp-4],eax
0x10070365: jbe short 10070398h
0x10070367: movzx eax,word ptr [ebx+2]
0x1007036b: push eax
0x1007036c: call 1006FCE1h
0x10070371: test eax,eax
0x10070373: pop ecx
0x10070374: je near ptr 100703FBh
0x1007037a: mov ecx,[ebx+4]
0x1007037d: imul ecx,eax
0x10070380: cmp ecx,4
0x10070383: jbe short 10070387h
0x10070385: add edi,ecx
0x10070387: add dword ptr [ebp-4],0FFFFh
0x1007038e: add ebx,0Ch
0x10070391: cmp word ptr [ebp-4],0
0x10070396: ja short 10070367h
0x10070398: mov eax,[ebp-8]
0x1007039b: sub eax,edi
0x1007039d: cmp word ptr [esi+8Ah],2
0x100703a5: jne short 100703AFh
0x100703a7: movzx ecx,word ptr [esi+62h]
0x100703ab: xor edx,edx
0x100703ad: div ecx
0x100703af: xor ecx,ecx
0x100703b1: cmp [esi+0B4h],ecx
0x100703b7: jbe short 100703CBh
0x100703b9: mov edx,[esi+0BCh]
0x100703bf: mov [edx+ecx*4],eax
0x100703c2: inc ecx
0x100703c3: cmp ecx,[esi+0B4h]
0x100703c9: jb short 100703B9h
0x100703cb: mov edx,[esi+0B8h]
0x100703d1: lea eax,[ecx*4-4]
0x100703d8: mov ecx,[esi+0BCh]
0x100703de: add ecx,eax
0x100703e0: mov eax,[eax+edx]
0x100703e3: mov edi,[ecx]
0x100703e5: mov edx,[ebp-8]
0x100703e8: add edi,eax
0x100703ea: cmp edi,edx
0x100703ec: jbe near ptr 1007047Eh
0x100703f2: sub edx,eax
0x100703f4: mov [ecx],edx
0x100703f6: jmp near ptr 1007047Eh
0x100703fb: movzx eax,word ptr [ebx+2]
0x100703ff: push eax
0x10070400: push dword ptr [esi]
0x10070402: push 1010B3A4h
0x10070407: push 1010B334h
0x1007040c: push dword ptr [esi+274h]
0x10070412: call 1006FAF1h
0x10070417: add esp,14h
0x1007041a: or eax,0FFFFFFFFh
0x1007041d: jmp short 10070497h
0x1007041f: test word ptr [esi+0Ch],400h
0x10070425: push esi
0x10070426: je short 1007044Ch
```

```
0x10070428: call 1006883Eh
0x1007042d: pop ecx
0x1007042e: xor ecx,ecx
0x10070430: cmp [esi+0B4h],ecx
0x10070436: jbe short 1007047Eh
0x10070438: mov edx,[esi+0BCh]
0x1007043e: mov [edx+ecx*4],eax
0x10070441: inc ecx
0x10070442: cmp ecx,[esi+0B4h]
0x10070448: jb short 10070438h
0x1007044a: jmp short 1007047Eh
0x1007044c: call 10065FAFh
0x10070451: mov edi,eax
0x10070453: mov eax,[esi+3Ch]
0x10070456: xor edx,edx
0x10070458: div dword ptr [esi+0B0h]
0x1007045e: pop ecx
0x1007045f: xor ecx,ecx
0x10070461: cmp [esi+0B4h],ecx
0x10070467: jbe short 1007047Eh
0x10070469: imul eax,edi
0x1007046c: mov edx,[esi+0BCh]
0x10070472: mov [edx+ecx*4],eax
0x10070475: inc ecx
0x10070476: cmp ecx,[esi+0B4h]
0x1007047c: jb short 1007046Ch
0x1007047e: or dword ptr [esi+28h],1000000h
0x10070485: test dword ptr [esi+28h],20000h
0x1007048c: jne short 10070494h
0x1007048e: mov eax,[esi+3Ch]
0x10070491: mov [esi+64h],eax
0x10070494: xor eax,eax
0x10070496: inc eax
0x10070497: pop edi
0x10070498: pop ebx
0x10070499: leave
0x1007049a: ret
```

`sub_100702f4` is called by sub_1007171b. It works with the text "%s: Cannot determine size of unknown tag type %d". It delegates to j_free, sub_10065faf, sub_1006883e (6 callees).

12.1038 sub_1007049b at 0x1007049b (8 instructions)

```
0x1007049b: push dword ptr [esp+4]
0x1007049f: push dword ptr [eax]
0x100704a1: push 1010B3F4h
0x100704a6: push 1010B34Ch
0x100704ab: push dword ptr [eax+274h]
0x100704b1: call 1006FAF1h
0x100704b6: add esp,14h
0x100704b9: ret
```

`sub_1007049b` is called by sub_1007171b. It works with the text "MissingRequired". It delegates to sub_1006faf1 (1 callees).

12.1039 sub_100704ba at 0x100704ba (46 instructions)

```
0x100704ba: test edi,edi
0x100704bc: jne short 100704C1h
0x100704be: xor eax,eax
0x100704c0: ret
0x100704c1: movzx eax,word ptr [esi+20h]
0x100704c5: xor edx,edx
0x100704c7: test ax,ax
0x100704ca: push ebx
```

```
0x100704cb: jbe short 100704E3h
0x100704cd: mov ecx,[esi+18h]
0x100704d0: test ecx,ecx
0x100704d2: je short 100704E3h
0x100704d4: movzx ebx,dx
0x100704d7: cmp [ecx+ebx*4],edi
0x100704da: je short 10070508h
0x100704dc: inc edx
0x100704dd: cmp dx,[esi+20h]
0x100704e1: jb short 100704D0h
0x100704e3: inc eax
0x100704e4: mov [esi+20h],ax
0x100704e8: movzx eax,ax
0x100704eb: cmp eax,[esi+1Ch]
0x100704ee: jle short 10070518h
0x100704f0: push 1010B428h
0x100704f5: push 8
0x100704f7: push eax
0x100704f8: push dword ptr [esi+18h]
0x100704fb: push esi
0x100704fc: call 1006E029h
0x10070501: add esp,14h
0x10070504: test eax,eax
0x10070506: jne short 1007050Ch
0x10070508: xor eax,eax
0x1007050a: pop ebx
0x1007050b: ret
0x1007050c: movzx ecx,word ptr [esi+20h]
0x10070510: add ecx,ecx
0x10070512: mov [esi+1Ch],ecx
0x10070515: mov [esi+18h],eax
0x10070518: movzx eax,word ptr [esi+20h]
0x1007051c: mov ecx,[esi+18h]
0x1007051f: mov [ecx+eax*4-4],edi
0x10070523: xor eax,eax
0x10070525: inc eax
0x10070526: pop ebx
0x10070527: ret
```

`sub_100704ba` is called by sub_1007171b. It works with the text "for IFD list". It delegates to sub_1006e029 (1 callees).

12.1040 sub_10070528 at 0x10070528 (37 instructions)

```
0x10070528: mov ecx,[eax+4]
0x1007052b: cmp edx,ecx
0x1007052d: jbe short 10070558h
0x1007052f: movzx eax,word ptr [eax]
0x10070532: push edx
0x10070533: push ecx
0x10070534: push eax
0x10070535: push esi
0x10070536: call 1006FE65h
0x1007053b: pop ecx
0x1007053c: pop ecx
0x1007053d: push dword ptr [eax+10h]
0x10070540: push 1010B478h
0x10070545: push dword ptr [esi]
0x10070547: push dword ptr [esi+274h]
0x1007054d: call 1007007Dh
0x10070552: add esp,18h
0x10070555: xor eax,eax
0x10070557: ret
0x10070558: jae short 10070580h
0x1007055a: movzx eax,word ptr [eax]
0x1007055d: push edx
0x1007055e: push ecx
```

```
0x1007055f: push eax
0x10070560: push esi
0x10070561: call 1006FE65h
0x10070566: pop ecx
0x10070567: pop ecx
0x10070568: push dword ptr [eax+10h]
0x1007056b: push 1010B438h
0x10070570: push dword ptr [esi]
0x10070572: push dword ptr [esi+274h]
0x10070578: call 1007007Dh
0x1007057d: add esp,18h
0x10070580: xor eax,eax
0x10070582: inc eax
0x10070583: ret
```

`sub_10070528` is called by sub_10070dd6, sub_10071192, sub_1007127c, sub_10071346, sub_10071411, sub_1007171b. It delegates to sub_1006fe65, sub_1007007d (2 callees).

12.1041 sub_10070584 at 0x10070584 (163 instructions)

```
0x10070584: push ebp
0x10070585: mov ebp,esp
0x10070587: push ecx
0x10070588: mov [esi+10h],eax
0x1007058b: mov eax,[ebp+0Ch]
0x1007058e: test eax,eax
0x10070590: je short 10070595h
0x10070592: and dword ptr [eax],0
0x10070595: test word ptr [esi+0Ch],800h
0x1007059b: push ebx
0x1007059c: push edi
0x1007059d: jne near ptr 1007068Dh
0x100705a3: push 0
0x100705a5: push dword ptr [esi+10h]
0x100705a8: push dword ptr [esi+274h]
0x100705ae: call dword ptr [esi+280h]
0x100705b4: add esp,0Ch
0x100705b7: cmp eax,[esi+10h]
0x100705ba: je short 100705D3h
0x100705bc: push dword ptr [esi]
0x100705be: push 1010B53Ch
0x100705c3: push 1010B35Ch
0x100705c8: push dword ptr [esi+274h]
0x100705ce: jmp near ptr 1007078Ah
0x100705d3: push 2
0x100705d5: lea eax,[ebp-4]
0x100705d8: push eax
0x100705d9: push dword ptr [esi+274h]
0x100705df: call dword ptr [esi+278h]
0x100705e5: add esp,0Ch
0x100705e8: cmp eax,2
0x100705eb: je short 100705F6h
0x100705ed: push dword ptr [esi]
0x100705ef: push 1010B514h
0x100705f4: jmp short 100705C3h
0x100705f6: test byte ptr [esi+0Ch],80h
0x100705fa: je short 10070606h
0x100705fc: lea eax,[ebp-4]
0x100705ff: push eax
0x10070600: call 1006FB29h
0x10070605: pop ecx
0x10070606: movzx eax,word ptr [ebp-4]
0x1007060a: push 1010B4FCh
0x1007060f: push 0Ch
0x10070611: push eax
0x10070612: push esi
```

```
0x10070613: call 1006E083h
0x10070618: mov ebx,eax
0x1007061a: add esp,10h
0x1007061d: test ebx,ebx
0x1007061f: je near ptr 10070792h
0x10070625: movzx eax,word ptr [ebp-4]
0x10070629: imul eax,0Ch
0x1007062c: push eax
0x1007062d: push ebx
0x1007062e: push dword ptr [esi+274h]
0x10070634: call dword ptr [esi+278h]
0x1007063a: movzx ecx,word ptr [ebp-4]
0x1007063e: imul ecx,0Ch
0x10070641: add esp,0Ch
0x10070644: cmp eax,ecx
0x10070646: je short 1007066Dh
0x10070648: push dword ptr [esi]
0x1007064a: push 1010B4D8h
0x1007064f: push 1010B35Ch
0x10070654: push dword ptr [esi+274h]
0x1007065a: call 1006FAF1h
0x1007065f: push ebx
0x10070660: call 100209E0h
0x10070665: add esp,14h
0x10070668: jmp near ptr 10070792h
0x1007066d: cmp dword ptr [ebp+0Ch],0
0x10070671: je near ptr 1007076Dh
0x10070677: push 4
0x10070679: push dword ptr [ebp+0Ch]
0x1007067c: push dword ptr [esi+274h]
0x10070682: call dword ptr [esi+278h]
0x10070688: jmp near ptr 1007075Bh
0x1007068d: mov eax,[esi+268h]
0x10070693: cmp eax,2
0x10070696: mov edi,[esi+10h]
0x10070699: jb near ptr 10070778h
0x1007069f: add eax,0FFFFFFFEh
0x100706a2: cmp edi,eax
0x100706a4: ja near ptr 10070778h
0x100706aa: mov eax,[esi+264h]
0x100706b0: add eax,edi
0x100706b2: push 2
0x100706b4: push eax
0x100706b5: lea eax,[ebp-4]
0x100706b8: push eax
0x100706b9: call 10020A00h
0x100706be: add esp,0Ch
0x100706c1: inc edi
0x100706c2: inc edi
0x100706c3: test byte ptr [esi+0Ch],80h
0x100706c7: je short 100706D3h
0x100706c9: lea eax,[ebp-4]
0x100706cc: push eax
0x100706cd: call 1006FB29h
0x100706d2: pop ecx
0x100706d3: movzx eax,word ptr [ebp-4]
0x100706d7: push 1010B4FCh
0x100706dc: push 0Ch
0x100706de: push eax
0x100706df: push esi
0x100706e0: call 1006E083h
0x100706e5: mov ebx,eax
0x100706e7: add esp,10h
0x100706ea: test ebx,ebx
0x100706ec: je near ptr 10070792h
0x100706f2: movzx eax,word ptr [ebp-4]
0x100706f6: imul eax,0Ch
```

```
0x100706f9: lea ecx,[eax+edi]
0x100706fc: cmp ecx,[esi+268h]
0x10070702: jbe short 1007071Bh
0x10070704: push dword ptr [esi]
0x10070706: push 1010B4B8h
0x1007070b: push 1010B35Ch
0x10070710: push dword ptr [esi+274h]
0x10070716: jmp near ptr 1007065Ah
0x1007071b: push eax
0x1007071c: mov eax,[esi+264h]
0x10070722: add eax,edi
0x10070724: push eax
0x10070725: push ebx
0x10070726: call 10020A00h
0x1007072b: add esp,0Ch
0x1007072e: cmp dword ptr [ebp+0Ch],0
0x10070732: je short 1007076Dh
0x10070734: movzx eax,word ptr [ebp-4]
0x10070738: imul eax,0Ch
0x1007073b: add edi,eax
0x1007073d: lea eax,[edi+4]
0x10070740: cmp eax,[esi+268h]
0x10070746: ja short 1007075Eh
0x10070748: mov eax,[esi+264h]
0x1007074e: push 4
0x10070750: add eax,edi
0x10070752: push eax
0x10070753: push dword ptr [ebp+0Ch]
0x10070756: call 10020A00h
0x1007075b: add esp,0Ch
0x1007075e: test byte ptr [esi+0Ch],80h
0x10070762: je short 1007076Dh
0x10070764: push dword ptr [ebp+0Ch]
0x10070767: call 1006FB38h
0x1007076c: pop ecx
0x1007076d: mov eax,[ebp+8]
0x10070770: mov [eax],ebx
0x10070772: mov ax,[ebp-4]
0x10070776: jmp short 10070795h
0x10070778: push dword ptr [esi]
0x1007077a: push 1010B514h
0x1007077f: push 1010B35Ch
0x10070784: push dword ptr [esi+274h]
0x1007078a: call 1006FAF1h
0x1007078f: add esp,10h
0x10070792: xor ax,ax
0x10070795: pop edi
0x10070796: pop ebx
0x10070797: leave
0x10070798: ret
```

`sub_10070584` is called by sub_1007171b. It works with the text "%s: Can not read TIFF directory", "%.100s: Can not read TIFF directory", "to read TIFF directory". It delegates to j_free, j_memcpy, sub_1006e083 (6 callees).

12.1042 sub_10070799 at 0x10070799 (76 instructions)

```
0x10070799: push ebp
0x1007079a: mov ebp,esp
0x1007079c: movzx eax,word ptr [esi+2]
0x100707a0: push edi
0x100707a1: push eax
0x100707a2: call 1006FCE1h
0x100707a7: pop ecx
0x100707a8: mov ecx,eax
0x100707aa: mov eax,[esi+4]
0x100707ad: mov edi,eax
```

```
0x100707af: imul edi,ecx
0x100707b2: test eax,eax
0x100707b4: je short 100707FDh
0x100707b6: test ecx,ecx
0x100707b8: je short 100707FDh
0x100707ba: xor edx,edx
0x100707bc: mov eax,edi
0x100707be: div ecx
0x100707c0: cmp eax,[esi+4]
0x100707c3: jne short 100707FDh
0x100707c5: test word ptr [ebx+0Ch],800h
0x100707cb: jne short 10070824h
0x100707cd: push 0
0x100707cf: push dword ptr [esi+8]
0x100707d2: push dword ptr [ebx+274h]
0x100707d8: call dword ptr [ebx+280h]
0x100707de: add esp,0Ch
0x100707e1: cmp eax,[esi+8]
0x100707e4: jne short 100707FDh
0x100707e6: push edi
0x100707e7: push dword ptr [ebp+8]
0x100707ea: push dword ptr [ebx+274h]
0x100707f0: call dword ptr [ebx+278h]
0x100707f6: add esp,0Ch
0x100707f9: cmp eax,edi
0x100707fb: je short 10070852h
0x100707fd: movzx eax,word ptr [esi]
0x10070800: push eax
0x10070801: push ebx
0x10070802: call 1006FE65h
0x10070807: push dword ptr [eax+10h]
0x1007080a: push 1010B564h
0x1007080f: push dword ptr [ebx]
0x10070811: push dword ptr [ebx+274h]
0x10070817: call 1006FAF1h
0x1007081c: add esp,18h
0x1007081f: xor eax,eax
0x10070821: pop edi
0x10070822: pop ebp
0x10070823: ret
0x10070824: mov eax,[esi+8]
0x10070827: lea ecx,[eax+edi]
0x1007082a: cmp ecx,eax
0x1007082c: jb short 100707FDh
0x1007082e: lea ecx,[eax+edi]
0x10070831: cmp ecx,edi
0x10070833: jb short 100707FDh
0x10070835: cmp ecx,[ebx+268h]
0x1007083b: ja short 100707FDh
0x1007083d: mov ecx,[ebx+264h]
0x10070843: push edi
0x10070844: add ecx,eax
0x10070846: push ecx
0x10070847: push dword ptr [ebp+8]
0x1007084a: call 10020A00h
0x1007084f: add esp,0Ch
0x10070852: test byte ptr [ebx+0Ch],80h
0x10070856: je short 100708A1h
0x10070858: movzx eax,word ptr [esi+2]
0x1007085c: add eax,0FFFFFFFDh
0x1007085f: cmp eax,9
0x10070862: ja short 100708A1h
0x10070864: movzx eax,byte ptr [eax+100708BDh]
0x1007086b: jmp dword ptr [eax*4+100708A9h]
0x100708a1: mov eax,edi
0x100708a3: jmp near ptr 10070821h
```

`sub_10070799` is called by sub_10070a20, sub_10070a94, sub_10070b7f, sub_1007171b. It delegates to j_memcpy, sub_1006faf1, sub_1006fce1 (4 callees).

12.1043 sub_10070a20 at 0x10070a20 (53 instructions)

```
0x10070a20: push ebx
0x10070a21: push esi
0x10070a22: mov esi,ecx
0x10070a24: mov ecx,[esi+4]
0x10070a27: cmp ecx,4
0x10070a2a: mov ebx,edx
0x10070a2c: ja short 10070A84h
0x10070a2e: cmp word ptr [ebx+1C8h],4D4Dh
0x10070a37: jne short 10070A5Ch
0x10070a39: dec ecx
0x10070a3a: je short 10070A57h
0x10070a3c: dec ecx
0x10070a3d: je short 10070A51h
0x10070a3f: dec ecx
0x10070a40: je short 10070A4Bh
0x10070a42: dec ecx
0x10070a43: jne short 10070A7Fh
0x10070a45: mov cl,[esi+8]
0x10070a48: mov [eax+3],cl
0x10070a4b: mov cl,[esi+9]
0x10070a4e: mov [eax+2],cl
0x10070a51: mov cl,[esi+0Ah]
0x10070a54: mov [eax+1],cl
0x10070a57: mov cl,[esi+0Bh]
0x10070a5a: jmp short 10070A7Dh
0x10070a5c: dec ecx
0x10070a5d: je short 10070A7Ah
0x10070a5f: dec ecx
0x10070a60: je short 10070A74h
0x10070a62: dec ecx
0x10070a63: je short 10070A6Eh
0x10070a65: dec ecx
0x10070a66: jne short 10070A7Fh
0x10070a68: mov cl,[esi+0Bh]
0x10070a6b: mov [eax+3],cl
0x10070a6e: mov cl,[esi+0Ah]
0x10070a71: mov [eax+2],cl
0x10070a74: mov cl,[esi+9]
0x10070a77: mov [eax+1],cl
0x10070a7a: mov cl,[esi+8]
0x10070a7d: mov [eax],cl
0x10070a7f: xor eax,eax
0x10070a81: inc eax
0x10070a82: jmp short 10070A91h
0x10070a84: push eax
0x10070a85: call 10070799h
0x10070a8a: neg eax
0x10070a8c: sbb eax,eax
0x10070a8e: pop ecx
0x10070a8f: neg eax
0x10070a91: pop esi
0x10070a92: pop ebx
0x10070a93: ret
```

`sub_10070a20` is called by sub_10070aeb. It delegates to sub_10070799 (1 callees).

12.1044 sub_10070a94 at 0x10070a94 (37 instructions)

```
0x10070a94: push ebx
0x10070a95: push esi
0x10070a96: mov esi,ecx
```

```
0x10070a98: mov ecx,[esi+4]
0x10070a9b: cmp ecx,2
0x10070a9e: mov ebx,edx
0x10070aa0: ja short 10070ADBh
0x10070aa2: cmp word ptr [ebx+1C8h],4D4Dh
0x10070aab: jne short 10070AC1h
0x10070aad: dec ecx
0x10070aae: je short 10070ABBh
0x10070ab0: dec ecx
0x10070ab1: jne short 10070AD6h
0x10070ab3: mov cx,[esi+8]
0x10070ab7: mov [eax+2],cx
0x10070abb: mov cx,[esi+0Ah]
0x10070abf: jmp short 10070AD3h
0x10070ac1: dec ecx
0x10070ac2: je short 10070ACFh
0x10070ac4: dec ecx
0x10070ac5: jne short 10070AD6h
0x10070ac7: mov cx,[esi+0Ah]
0x10070acb: mov [eax+2],cx
0x10070acf: mov cx,[esi+8]
0x10070ad3: mov [eax],cx
0x10070ad6: xor eax,eax
0x10070ad8: inc eax
0x10070ad9: jmp short 10070AE8h
0x10070adb: push eax
0x10070adc: call 10070799h
0x10070ae1: neg eax
0x10070ae3: sbb eax,eax
0x10070ae5: pop ecx
0x10070ae6: neg eax
0x10070ae8: pop esi
0x10070ae9: pop ebx
0x10070aea: ret
```

`sub_10070a94` is called by sub_10070aeb, sub_10071192, sub_10071411. It delegates to sub_10070799 (1 callees).

12.1045 sub_10070aeb at 0x10070aeb (64 instructions)

```
0x10070aeb: push ebp
0x10070aec: mov ebp,esp
0x10070aee: push ecx
0x10070aef: mov eax,[esi+4]
0x10070af2: cmp eax,2
0x10070af5: jbe short 10070B20h
0x10070af7: push eax
0x10070af8: movzx eax,word ptr [esi]
0x10070afb: push eax
0x10070afc: push edi
0x10070afd: call 1006FE65h
0x10070b02: pop ecx
0x10070b03: pop ecx
0x10070b04: push dword ptr [eax+10h]
0x10070b07: push 1010B5B8h
0x10070b0c: push dword ptr [edi]
0x10070b0e: push dword ptr [edi+274h]
0x10070b14: call 1007007Dh
0x10070b19: add esp,14h
0x10070b1c: xor eax,eax
0x10070b1e: leave
0x10070b1f: ret
0x10070b20: movzx eax,word ptr [esi+2]
0x10070b24: dec eax
0x10070b25: je short 10070B4Fh
0x10070b27: dec eax
0x10070b28: dec eax
```

```
0x10070b29: je short 10070B34h
0x10070b2b: sub eax,3
0x10070b2e: je short 10070B4Fh
0x10070b30: dec eax
0x10070b31: dec eax
0x10070b32: jne short 10070B1Ch
0x10070b34: lea eax,[ebp-4]
0x10070b37: mov ecx,esi
0x10070b39: mov edx,edi
0x10070b3b: call 10070A94h
0x10070b40: test eax,eax
0x10070b42: je short 10070B1Ch
0x10070b44: movzx eax,word ptr [ebp-2]
0x10070b48: push eax
0x10070b49: movzx eax,word ptr [ebp-4]
0x10070b4d: jmp short 10070B68h
0x10070b4f: lea eax,[ebp-4]
0x10070b52: mov ecx,esi
0x10070b54: mov edx,edi
0x10070b56: call 10070A20h
0x10070b5b: test eax,eax
0x10070b5d: je short 10070B1Ch
0x10070b5f: movzx eax,byte ptr [ebp-3]
0x10070b63: push eax
0x10070b64: movzx eax,byte ptr [ebp-4]
0x10070b68: push eax
0x10070b69: movzx eax,word ptr [esi]
0x10070b6c: push eax
0x10070b6d: push edi
0x10070b6e: call 1006796Fh
0x10070b73: add esp,10h
0x10070b76: test eax,eax
0x10070b78: je short 10070B1Ch
0x10070b7a: xor eax,eax
0x10070b7c: inc eax
0x10070b7d: leave
0x10070b7e: ret
```

`sub_10070aeb` is called by sub_1007171b. It delegates to sub_1006796f, sub_1006fe65, sub_1007007d (5 callees).

12.1046 sub_10070b7f at 0x10070b7f (18 instructions)

```
0x10070b7f: push esi
0x10070b80: mov esi,ecx
0x10070b82: cmp dword ptr [esi+4],1
0x10070b86: jne short 10070B92h
0x10070b88: mov ecx,[esi+8]
0x10070b8b: mov [eax],ecx
0x10070b8d: xor eax,eax
0x10070b8f: inc eax
0x10070b90: pop esi
0x10070b91: ret
0x10070b92: push eax
0x10070b93: call 10070799h
0x10070b98: neg eax
0x10070b9a: pop ecx
0x10070b9b: sbb eax,eax
0x10070b9d: neg eax
0x10070b9f: pop esi
0x10070ba0: ret
```

`sub_10070b7f` is called by sub_1007127c, sub_10071411, sub_10071523. It delegates to sub_10070799 (1 callees).

12.1047 sub_10070c4a at 0x10070c4a (33 instructions)

```
0x10070c4a: push ebp
0x10070c4b: mov ebp,esp
0x10070c4d: push ecx
0x10070c4e: push ebx
0x10070c4f: push esi
0x10070c50: push edi
0x10070c51: mov esi,eax
0x10070c53: mov edi,ecx
0x10070c55: movzx eax,word ptr [edi+2]
0x10070c59: dec eax
0x10070c5a: cmp eax,0Bh
0x10070c5d: mov ebx,edx
0x10070c5f: ja near ptr 10070D7Eh
0x10070c65: movzx eax,byte ptr [eax+10070DCAh]
0x10070c6c: jmp dword ptr [eax*4+10070DAEh]
0x10070d7e: movzx eax,word ptr [edi]
0x10070d81: push eax
0x10070d82: push ebx
0x10070d83: call 1006FE65h
0x10070d88: push dword ptr [eax+10h]
0x10070d8b: movzx eax,word ptr [edi+2]
0x10070d8f: push eax
0x10070d90: push 1010B610h
0x10070d95: push dword ptr [ebx]
0x10070d97: push dword ptr [ebx+274h]
0x10070d9d: call 1006FAF1h
0x10070da2: add esp,1Ch
0x10070da5: xor eax,eax
0x10070da7: pop edi
0x10070da8: pop esi
0x10070da9: pop ebx
0x10070daa: leave
0x10070dab: ret
```

`sub_10070c4a` is called by sub_10071346. It delegates to sub_1006faf1, sub_1006fe65 (2 callees).

12.1048 sub_10070dd6 at 0x10070dd6 (45 instructions)

```
0x10070dd6: push ebp
0x10070dd7: mov ebp,esp
0x10070dd9: sub esp,10h
0x10070ddc: push ebx
0x10070ddd: push esi
0x10070dde: push edi
0x10070ddf: mov edi,eax
0x10070de1: movzx eax,word ptr [edi]
0x10070de4: mov ebx,ecx
0x10070de6: push eax
0x10070de7: xor esi,esi
0x10070de9: push ebx
0x10070dea: mov [ebp-8],esi
0x10070ded: call 1006FE65h
0x10070df2: xor edx,edx
0x10070df4: mov [ebp-10h],eax
0x10070df7: mov eax,[edi+4]
0x10070dfa: inc edx
0x10070dfb: cmp eax,edx
0x10070dfd: pop ecx
0x10070dfe: pop ecx
0x10070dff: jbe near ptr 10070F83h
0x10070e05: movzx ecx,word ptr [edi+2]
0x10070e09: dec ecx
0x10070e0a: cmp ecx,0Bh
0x10070e0d: ja near ptr 1007113Ah
0x10070e13: jmp dword ptr [ecx*4+10071142h]
```

```
0x10070f83: mov eax,edi
0x10070f85: mov esi,ebx
0x10070f87: call 10070528h
0x10070f8c: test eax,eax
0x10070f8e: je near ptr 1007113Ah
0x10070f94: movzx edx,word ptr [edi+2]
0x10070f98: movzx eax,dx
0x10070f9b: lea ecx,[eax-1]
0x10070f9e: cmp ecx,0Bh
0x10070fa1: ja near ptr 1007113Ah
0x10070fa7: movzx ecx,byte ptr [ecx+10071186h]
0x10070fae: jmp dword ptr [ecx*4+10071172h]
0x1007113a: mov eax,[ebp-8]
0x1007113d: pop edi
0x1007113e: pop esi
0x1007113f: pop ebx
0x10071140: leave
0x10071141: ret
```

`sub_10070dd6` is called by sub_10071523, sub_1007171b. It delegates to sub_1006fe65, sub_10070528 (2 callees).

12.1049 sub_10071192 at 0x10071192 (90 instructions)

```
0x10071192: push ebp
0x10071193: mov ebp,esp
0x10071195: sub esp,28h
0x10071198: mov eax,ds:[10136220h]
0x1007119d: xor eax,ebp
0x1007119f: mov [ebp-4],eax
0x100711a2: mov eax,[ebp+8]
0x100711a5: and dword ptr [ebp-24h],0
0x100711a9: push ebx
0x100711aa: push esi
0x100711ab: mov esi,edx
0x100711ad: mov [ebp-28h],eax
0x100711b0: movzx eax,word ptr [esi+62h]
0x100711b4: movzx ebx,ax
0x100711b7: mov edx,ebx
0x100711b9: mov eax,ecx
0x100711bb: mov [ebp-1Ch],ecx
0x100711be: call 10070528h
0x100711c3: test eax,eax
0x100711c5: je short 10071244h
0x100711c7: mov eax,[ebp-1Ch]
0x100711ca: mov eax,[eax+4]
0x100711cd: cmp eax,0Ah
0x100711d0: push edi
0x100711d1: lea edi,[ebp-18h]
0x100711d4: jbe short 100711EDh
0x100711d6: push 1010B678h
0x100711db: push 2
0x100711dd: push eax
0x100711de: push esi
0x100711df: call 1006E083h
0x100711e4: mov edi,eax
0x100711e6: add esp,10h
0x100711e9: test edi,edi
0x100711eb: je short 10071243h
0x100711ed: mov ecx,[ebp-1Ch]
0x100711f0: mov eax,edi
0x100711f2: mov edx,esi
0x100711f4: call 10070A94h
0x100711f9: test eax,eax
0x100711fb: je short 10071235h
0x100711fd: mov eax,[ebp-1Ch]
0x10071200: mov edx,[eax+4]
```

```
0x10071203: cmp ebx,edx
0x10071205: jge short 10071209h
0x10071207: mov edx,ebx
0x10071209: xor ebx,ebx
0x1007120b: inc ebx
0x1007120c: cmp edx,ebx
0x1007120e: mov [ebp-20h],ebx
0x10071211: jle short 10071229h
0x10071213: movzx ecx,word ptr [edi]
0x10071216: mov eax,ebx
0x10071218: cmp [edi+eax*2],cx
0x1007121c: jne short 10071255h
0x1007121e: inc dword ptr [ebp-20h]
0x10071221: movzx eax,word ptr [ebp-20h]
0x10071225: cmp eax,edx
0x10071227: jl short 10071218h
0x10071229: mov ax,[edi]
0x1007122c: mov ecx,[ebp-28h]
0x1007122f: mov [ecx],ax
0x10071232: mov [ebp-24h],ebx
0x10071235: lea eax,[ebp-18h]
0x10071238: cmp edi,eax
0x1007123a: je short 10071243h
0x1007123c: push edi
0x1007123d: call 100209E0h
0x10071242: pop ecx
0x10071243: pop edi
0x10071244: mov ecx,[ebp-4]
0x10071247: mov eax,[ebp-24h]
0x1007124a: pop esi
0x1007124b: xor ecx,ebp
0x1007124d: pop ebx
0x1007124e: call 10022326h
0x10071253: leave
0x10071254: ret
0x10071255: mov eax,[ebp-1Ch]
0x10071258: movzx eax,word ptr [eax]
0x1007125b: push eax
0x1007125c: push esi
0x1007125d: call 1006FE65h
0x10071262: push dword ptr [eax+10h]
0x10071265: push 1010B63Ch
0x1007126a: push dword ptr [esi]
0x1007126c: push dword ptr [esi+274h]
0x10071272: call 1006FAF1h
0x10071277: add esp,18h
0x1007127a: jmp short 10071235h
```

`sub_10071192` is called by sub_1007171b. It works with the text "to fetch per-sample values". It delegates to j_free, sub_10022326, sub_1006e083 (7 callees).

12.1050 sub_1007127c at 0x1007127c (81 instructions)

```
0x1007127c: push ebp
0x1007127d: mov ebp,esp
0x1007127f: sub esp,30h
0x10071282: and dword ptr [ebp-8],0
0x10071286: push ebx
0x10071287: mov ebx,eax
0x10071289: movzx eax,word ptr [ebx+62h]
0x1007128d: push esi
0x1007128e: push edi
0x1007128f: movzx edi,ax
0x10071292: mov eax,[ebp+8]
0x10071295: mov edx,edi
0x10071297: mov esi,ebx
```

```
0x10071299: call 10070528h
0x1007129e: test eax,eax
0x100712a0: je short 10071317h
0x100712a2: mov eax,[ebp+8]
0x100712a5: mov eax,[eax+4]
0x100712a8: cmp eax,0Ah
0x100712ab: lea esi,[ebp-30h]
0x100712ae: jbe short 100712C7h
0x100712b0: push 1010B678h
0x100712b5: push 4
0x100712b7: push eax
0x100712b8: push ebx
0x100712b9: call 1006E083h
0x100712be: mov esi,eax
0x100712c0: add esp,10h
0x100712c3: test esi,esi
0x100712c5: je short 10071317h
0x100712c7: mov ecx,[ebp+8]
0x100712ca: mov eax,esi
0x100712cc: call 10070B7Fh
0x100712d1: test eax,eax
0x100712d3: je short 10071309h
0x100712d5: mov eax,[ebp+8]
0x100712d8: mov edx,[eax+4]
0x100712db: cmp edi,edx
0x100712dd: jge short 100712E1h
0x100712df: mov edx,edi
0x100712e1: xor edi,edi
0x100712e3: inc edi
0x100712e4: cmp edx,edi
0x100712e6: mov [ebp-4],edi
0x100712e9: jle short 100712FFh
0x100712eb: mov ecx,[esi]
0x100712ed: mov eax,edi
0x100712ef: cmp [esi+eax*4],ecx
0x100712f2: jne short 1007131Fh
0x100712f4: inc dword ptr [ebp-4]
0x100712f7: movzx eax,word ptr [ebp-4]
0x100712fb: cmp eax,edx
0x100712fd: jl short 100712EFh
0x100712ff: mov eax,[esi]
0x10071301: mov ecx,[ebp+0Ch]
0x10071304: mov [ecx],eax
0x10071306: mov [ebp-8],edi
0x10071309: lea eax,[ebp-30h]
0x1007130c: cmp esi,eax
0x1007130e: je short 10071317h
0x10071310: push esi
0x10071311: call 100209E0h
0x10071316: pop ecx
0x10071317: mov eax,[ebp-8]
0x1007131a: pop edi
0x1007131b: pop esi
0x1007131c: pop ebx
0x1007131d: leave
0x1007131e: ret
0x1007131f: mov eax,[ebp+8]
0x10071322: movzx eax,word ptr [eax]
0x10071325: push eax
0x10071326: push ebx
0x10071327: call 1006FE65h
0x1007132c: push dword ptr [eax+10h]
0x1007132f: push 1010B63Ch
0x10071334: push dword ptr [ebx]
0x10071336: push dword ptr [ebx+274h]
0x1007133c: call 1006FAF1h
```

```
0x10071341: add esp,18h
0x10071344: jmp short 10071309h
```

`sub_1007127c` is called by sub_1007171b. It works with the text "to fetch per-sample values". It delegates to j_free, sub_1006e083, sub_1006faf1 (6 callees).

12.1051 sub_10071346 at 0x10071346 (83 instructions)

```
0x10071346: push ebp
0x10071347: mov ebp,esp
0x10071349: sub esp,54h
0x1007134c: and dword ptr [ebp-4],0
0x10071350: push ebx
0x10071351: push esi
0x10071352: mov esi,eax
0x10071354: movzx eax,word ptr [esi+62h]
0x10071358: movzx ebx,ax
0x1007135b: mov eax,[ebp+8]
0x1007135e: mov edx,ebx
0x10071360: call 10070528h
0x10071365: test eax,eax
0x10071367: je short 100713E3h
0x10071369: mov eax,[ebp+8]
0x1007136c: mov eax,[eax+4]
0x1007136f: cmp eax,0Ah
0x10071372: push edi
0x10071373: lea edi,[ebp-54h]
0x10071376: jbe short 1007138Fh
0x10071378: push 1010B678h
0x1007137d: push 8
0x1007137f: push eax
0x10071380: push esi
0x10071381: call 1006E083h
0x10071386: mov edi,eax
0x10071388: add esp,10h
0x1007138b: test edi,edi
0x1007138d: je short 100713E2h
0x1007138f: mov ecx,[ebp+8]
0x10071392: mov eax,edi
0x10071394: mov edx,esi
0x10071396: call 10070C4Ah
0x1007139b: test eax,eax
0x1007139d: je short 100713D4h
0x1007139f: mov eax,[ebp+8]
0x100713a2: mov ecx,[eax+4]
0x100713a5: cmp ebx,ecx
0x100713a7: jge short 100713ABh
0x100713a9: mov ecx,ebx
0x100713ab: xor ebx,ebx
0x100713ad: inc ebx
0x100713ae: cmp ecx,ebx
0x100713b0: mov edx,ebx
0x100713b2: jle short 100713CAh
0x100713b4: mov eax,ebx
0x100713b6: fld qword ptr [edi]
0x100713b8: fcomp qword ptr [edi+eax*8]
0x100713bb: fnstsw ax
0x100713bd: test ah,44h
0x100713c0: jp short 100713EAh
0x100713c2: inc edx
0x100713c3: movzx eax,dx
0x100713c6: cmp eax,ecx
0x100713c8: jl short 100713B6h
0x100713ca: fld qword ptr [edi]
0x100713cc: mov eax,[ebp+0Ch]
0x100713cf: fstp qword ptr [eax]
```

```
0x100713d1: mov [ebp-4],ebx
0x100713d4: lea eax,[ebp-54h]
0x100713d7: cmp edi,eax
0x100713d9: je short 100713E2h
0x100713db: push edi
0x100713dc: call 100209E0h
0x100713e1: pop ecx
0x100713e2: pop edi
0x100713e3: mov eax,[ebp-4]
0x100713e6: pop esi
0x100713e7: pop ebx
0x100713e8: leave
0x100713e9: ret
0x100713ea: mov eax,[ebp+8]
0x100713ed: movzx eax,word ptr [eax]
0x100713f0: push eax
0x100713f1: push esi
0x100713f2: call 1006FE65h
0x100713f7: push dword ptr [eax+10h]
0x100713fa: push 1010B63Ch
0x100713ff: push dword ptr [esi]
0x10071401: push dword ptr [esi+274h]
0x10071407: call 1006FAF1h
0x1007140c: add esp,18h
0x1007140f: jmp short 100713D4h
```

`sub_10071346` is called by sub_1007171b. It works with the text "to fetch per-sample values". It delegates to j_free, sub_1006e083, sub_1006faf1 (6 callees).

12.1052 sub_10071411 at 0x10071411 (109 instructions)

```
0x10071411: push ebp
0x10071412: mov ebp,esp
0x10071414: push ecx
0x10071415: push ecx
0x10071416: mov edx,[ebp+8]
0x10071419: push ebx
0x1007141a: push esi
0x1007141b: mov ebx,eax
0x1007141d: mov esi,ecx
0x1007141f: mov eax,edi
0x10071421: call 10070528h
0x10071426: cmp dword ptr [ebx],0
0x10071429: jne short 1007144Bh
0x1007142b: push 1010B6A8h
0x10071430: push 4
0x10071432: push dword ptr [ebp+8]
0x10071435: push esi
0x10071436: call 1006E083h
0x1007143b: add esp,10h
0x1007143e: test eax,eax
0x10071440: mov [ebx],eax
0x10071442: jne short 1007144Bh
0x10071444: xor eax,eax
0x10071446: jmp near ptr 1007151Fh
0x1007144b: mov eax,[ebp+8]
0x1007144e: mov ebx,[ebx]
0x10071450: shl eax,2
0x10071453: push eax
0x10071454: push 0
0x10071456: push ebx
0x10071457: mov [ebp-4],ebx
0x1007145a: call 100209F0h
0x1007145f: add esp,0Ch
0x10071462: cmp word ptr [edi+2],3
0x10071467: jne short 100714B0h
```

```
0x10071469: push 1010B694h
0x1007146e: push 2
0x10071470: push dword ptr [edi+4]
0x10071473: push esi
0x10071474: call 1006E083h
0x10071479: mov ebx,eax
0x1007147b: add esp,10h
0x1007147e: test ebx,ebx
0x10071480: je short 10071444h
0x10071482: mov ecx,edi
0x10071484: mov edx,esi
0x10071486: call 10070A94h
0x1007148b: mov esi,eax
0x1007148d: xor eax,eax
0x1007148f: cmp esi,eax
0x10071491: je short 100714ADh
0x10071493: cmp [ebp+8],eax
0x10071496: jle short 100714ADh
0x10071498: cmp eax,[edi+4]
0x1007149b: jge short 100714ADh
0x1007149d: movzx ecx,word ptr [ebx+eax*2]
0x100714a1: mov edx,[ebp-4]
0x100714a4: mov [edx+eax*4],ecx
0x100714a7: inc eax
0x100714a8: cmp eax,[ebp+8]
0x100714ab: jl short 10071498h
0x100714ad: push ebx
0x100714ae: jmp short 10071508h
0x100714b0: mov eax,[edi+4]
0x100714b3: cmp [ebp+8],eax
0x100714b6: je short 10071510h
0x100714b8: push 1010B694h
0x100714bd: push 4
0x100714bf: push eax
0x100714c0: push esi
0x100714c1: call 1006E083h
0x100714c6: add esp,10h
0x100714c9: test eax,eax
0x100714cb: mov [ebp-8],eax
0x100714ce: je near ptr 10071444h
0x100714d4: mov ecx,edi
0x100714d6: mov ebx,esi
0x100714d8: call 10070B7Fh
0x100714dd: mov esi,eax
0x100714df: xor edx,edx
0x100714e1: cmp esi,edx
0x100714e3: je short 10071505h
0x100714e5: cmp [ebp+8],edx
0x100714e8: jle short 10071505h
0x100714ea: mov eax,[ebp-4]
0x100714ed: mov ecx,[ebp-8]
0x100714f0: sub ecx,eax
0x100714f2: cmp edx,[edi+4]
0x100714f5: jge short 10071505h
0x100714f7: mov ebx,[ecx+eax]
0x100714fa: mov [eax],ebx
0x100714fc: inc edx
0x100714fd: add eax,4
0x10071500: cmp edx,[ebp+8]
0x10071503: jl short 100714F2h
0x10071505: push dword ptr [ebp-8]
0x10071508: call 100209E0h
0x1007150d: pop ecx
0x1007150e: jmp short 1007151Dh
0x10071510: mov eax,ebx
0x10071512: mov ecx,edi
0x10071514: mov ebx,esi
```

```
0x10071516: call 10070B7Fh
0x1007151b: mov esi,eax
0x1007151d: mov eax,esi
0x1007151f: pop esi
0x10071520: pop ebx
0x10071521: leave
0x10071522: ret
```

`sub_10071411` is called by sub_1007171b. It works with the text "to fetch strip tag", "for strip array". It delegates to j_free, j_memset, sub_1006e083 (6 callees).

12.1053 sub_10071523 at 0x10071523 (76 instructions)

```
0x10071523: push ebp
0x10071524: mov ebp,esp
0x10071526: push ecx
0x10071527: push ecx
0x10071528: cmp word ptr [esi+2],5
0x1007152d: jne short 1007153Bh
0x1007152f: mov ecx,[ebp+8]
0x10071532: mov eax,esi
0x10071534: call 10070DD6h
0x10071539: leave
0x1007153a: ret
0x1007153b: push ebx
0x1007153c: mov ebx,[ebp+8]
0x1007153f: push edi
0x10071540: mov edi,1010B384h
0x10071545: push edi
0x10071546: push 4
0x10071548: push dword ptr [esi+4]
0x1007154b: push ebx
0x1007154c: call 1006E083h
0x10071551: add esp,10h
0x10071554: test eax,eax
0x10071556: mov [ebp-4],eax
0x10071559: je short 10071566h
0x1007155b: mov ecx,esi
0x1007155d: call 10070B7Fh
0x10071562: test eax,eax
0x10071564: jne short 1007156Ah
0x10071566: xor ebx,ebx
0x10071568: jmp short 100715C5h
0x1007156a: push edi
0x1007156b: push 4
0x1007156d: push dword ptr [esi+4]
0x10071570: push ebx
0x10071571: call 1006E083h
0x10071576: xor ebx,ebx
0x10071578: add esp,10h
0x1007157b: mov edi,eax
0x1007157d: test edi,edi
0x1007157f: setne bl
0x10071582: test ebx,ebx
0x10071584: je short 100715C5h
0x10071586: xor edx,edx
0x10071588: cmp [esi+4],edx
0x1007158b: jbe short 100715ADh
0x1007158d: mov ecx,[ebp-4]
0x10071590: sub ecx,edi
0x10071592: mov ebx,[ecx+eax]
0x10071595: fild dword ptr [ecx+eax]
0x10071598: test ebx,ebx
0x1007159a: jge short 100715A2h
0x1007159c: fadd dword ptr ds:[100FE7C0h]
0x100715a2: fstp dword ptr [eax]
```

```
0x100715a4: inc edx
0x100715a5: add eax,4
0x100715a8: cmp edx,[esi+4]
0x100715ab: jb short 10071592h
0x100715ad: movzx eax,word ptr [esi]
0x100715b0: push edi
0x100715b1: push eax
0x100715b2: push dword ptr [ebp+8]
0x100715b5: call 1006796Fh
0x100715ba: push edi
0x100715bb: mov ebx,eax
0x100715bd: call 100209E0h
0x100715c2: add esp,10h
0x100715c5: cmp dword ptr [ebp-4],0
0x100715c9: je short 100715D4h
0x100715cb: push dword ptr [ebp-4]
0x100715ce: call 100209E0h
0x100715d3: pop ecx
0x100715d4: pop edi
0x100715d5: mov eax,ebx
0x100715d7: pop ebx
0x100715d8: leave
0x100715d9: ret
```

`sub_10071523` is called by sub_1007171b. It delegates to j_free, sub_1006796f, sub_1006e083 (5 callees).

12.1054 sub_100715da at 0x100715da (107 instructions)

```
0x100715da: push ebp
0x100715db: mov ebp,esp
0x100715dd: sub esp,18h
0x100715e0: mov eax,[esi+0BCh]
0x100715e6: push ebx
0x100715e7: mov ebx,[eax]
0x100715e9: mov eax,[esi+0B8h]
0x100715ef: mov eax,[eax]
0x100715f1: push edi
0x100715f2: push 1
0x100715f4: push esi
0x100715f5: mov [ebp-4],ebx
0x100715f8: mov [ebp-8],eax
0x100715fb: call 100686DAh
0x10071600: pop ecx
0x10071601: pop ecx
0x10071602: mov ecx,eax
0x10071604: mov eax,2000h
0x10071609: cmp ecx,eax
0x1007160b: jle short 10071618h
0x1007160d: mov edi,ecx
0x1007160f: mov dword ptr [ebp-0Ch],1
0x10071616: jmp short 1007162Bh
0x10071618: test ecx,ecx
0x1007161a: jle near ptr 10071717h
0x10071620: cdq
0x10071621: idiv ecx
0x10071623: mov edi,eax
0x10071625: mov [ebp-0Ch],eax
0x10071628: imul edi,ecx
0x1007162b: mov eax,[ebp-0Ch]
0x1007162e: cmp eax,[esi+64h]
0x10071631: jae near ptr 10071717h
0x10071637: mov eax,edi
0x10071639: neg eax
0x1007163b: cmp ebx,eax
0x1007163d: jae short 1007164Bh
0x1007163f: lea eax,[edi+ebx-1]
```

```
0x10071643: xor edx,edx
0x10071645: div edi
0x10071647: mov ebx,eax
0x10071649: jmp short 1007164Dh
0x1007164b: xor ebx,ebx
0x1007164d: test ebx,ebx
0x1007164f: je near ptr 10071717h
0x10071655: push 1010B6DCh
0x1007165a: push 4
0x1007165c: push ebx
0x1007165d: push esi
0x1007165e: call 1006E083h
0x10071663: push 1010B6B8h
0x10071668: push 4
0x1007166a: push ebx
0x1007166b: push esi
0x1007166c: mov [ebp-18h],eax
0x1007166f: call 1006E083h
0x10071674: mov ecx,[ebp-18h]
0x10071677: add esp,20h
0x1007167a: test ecx,ecx
0x1007167c: mov [ebp-10h],eax
0x1007167f: je near ptr 10071708h
0x10071685: test eax,eax
0x10071687: je short 10071701h
0x10071689: test ebx,ebx
0x1007168b: jbe short 100716B0h
0x1007168d: sub ecx,eax
0x1007168f: mov [ebp-14h],ebx
0x10071692: cmp edi,[ebp-4]
0x10071695: jbe short 1007169Ah
0x10071697: mov edi,[ebp-4]
0x1007169a: mov edx,[ebp-8]
0x1007169d: add [ebp-8],edi
0x100716a0: sub [ebp-4],edi
0x100716a3: mov [ecx+eax],edi
0x100716a6: mov [eax],edx
0x100716a8: add eax,4
0x100716ab: dec dword ptr [ebp-14h]
0x100716ae: jne short 10071692h
0x100716b0: push dword ptr [ebp-0Ch]
0x100716b3: mov [esi+0B4h],ebx
0x100716b9: push 116h
0x100716be: push esi
0x100716bf: mov [esi+0B0h],ebx
0x100716c5: call 1006796Fh
0x100716ca: push dword ptr [esi+0BCh]
0x100716d0: call 100209E0h
0x100716d5: push dword ptr [esi+0B8h]
0x100716db: call 100209E0h
0x100716e0: mov eax,[ebp-18h]
0x100716e3: mov [esi+0BCh],eax
0x100716e9: mov eax,[ebp-10h]
0x100716ec: add esp,14h
0x100716ef: mov [esi+0B8h],eax
0x100716f5: mov dword ptr [esi+0C0h],1
0x100716ff: jmp short 10071717h
0x10071701: push ecx
0x10071702: call 100209E0h
0x10071707: pop ecx
0x10071708: cmp dword ptr [ebp-10h],0
0x1007170c: je short 10071717h
0x1007170e: push dword ptr [ebp-10h]
0x10071711: call 100209E0h
0x10071716: pop ecx
0x10071717: pop edi
0x10071718: pop ebx
```

```
0x10071719: leave
0x1007171a: ret
```

`sub_100715da` is called by sub_1007171b. It delegates to j_free, sub_1006796f, sub_100686da (4 callees).

12.1055 sub_1007171b at 0x1007171b (1041 instructions)

```
0x1007171b: push ebp
0x1007171c: mov ebp,esp
0x1007171e: sub esp,28h
0x10071721: push ebx
0x10071722: mov ebx,[ebp+8]
0x10071725: push esi
0x10071726: xor eax,eax
0x10071728: push edi
0x10071729: mov edi,[ebx+14h]
0x1007172c: mov esi,ebx
0x1007172e: mov [ebp-8],eax
0x10071731: mov [ebp-1Ch],eax
0x10071734: mov [ebp-24h],eax
0x10071737: mov [ebx+10h],edi
0x1007173a: call 100704BAh
0x1007173f: test eax,eax
0x10071741: jne short 1007174Ah
0x10071743: xor eax,eax
0x10071745: jmp near ptr 100724EAh
0x1007174a: push ebx
0x1007174b: call dword ptr [ebx+23Ch]
0x10071751: inc word ptr [ebx+1DCh]
0x10071758: lea edi,[ebx+14h]
0x1007175b: lea eax,[ebp-8]
0x1007175e: push edi
0x1007175f: push eax
0x10071760: mov eax,[edi]
0x10071762: call 10070584h
0x10071767: movzx eax,ax
0x1007176a: add esp,0Ch
0x1007176d: test ax,ax
0x10071770: mov [ebp-14h],eax
0x10071773: jne short 10071793h
0x10071775: push dword ptr [edi]
0x10071777: push dword ptr [ebx]
0x10071779: push 1010BB44h
0x1007177e: push 1010B308h
0x10071783: push dword ptr [ebx+274h]
0x10071789: call 1006FAF1h
0x1007178e: add esp,14h
0x10071791: jmp short 10071743h
0x10071793: and dword ptr [ebx+0Ch],0FFFFFFBFh
0x10071797: push ebx
0x10071798: call 10067617h
0x1007179d: push ebx
0x1007179e: call 1006799Bh
0x100717a3: push 1
0x100717a5: push 11Ch
0x100717aa: push ebx
0x100717ab: call 1006796Fh
0x100717b0: movzx eax,word ptr [ebp-14h]
0x100717b4: mov edi,[ebp-8]
0x100717b7: add esp,14h
0x100717ba: mov esi,eax
0x100717bc: jmp short 100717FAh
0x100717be: test byte ptr [ebx+0Ch],80h
0x100717c2: je short 100717DAh
0x100717c4: push 2
0x100717c6: push edi
```

```
0x100717c7: call 1006FB53h
0x100717cc: lea eax,[edi+4]
0x100717cf: push 2
0x100717d1: push eax
0x100717d2: call 1006FB9Ah
0x100717d7: add esp,10h
0x100717da: cmp word ptr [edi],115h
0x100717df: jne short 100717F6h
0x100717e1: mov eax,edi
0x100717e3: mov ecx,ebx
0x100717e5: call 10070DD6h
0x100717ea: test eax,eax
0x100717ec: je near ptr 10071A38h
0x100717f2: and word ptr [edi],0
0x100717f6: dec esi
0x100717f7: add edi,0Ch
0x100717fa: test esi,esi
0x100717fc: jg short 100717BEh
0x100717fe: movzx eax,word ptr [ebp-14h]
0x10071802: and dword ptr [ebp-10h],0
0x10071806: test eax,eax
0x10071808: mov edi,[ebp-8]
0x1007180b: mov [ebp-0Ch],eax
0x1007180e: jle near ptr 10071C94h
0x10071814: movzx eax,word ptr [edi]
0x10071817: test ax,ax
0x1007181a: je near ptr 10071AE6h
0x10071820: mov ecx,[ebp-10h]
0x10071823: cmp ecx,[ebx+294h]
0x10071829: jb short 10071830h
0x1007182b: xor ecx,ecx
0x1007182d: mov [ebp-10h],ecx
0x10071830: mov edx,[ebx+290h]
0x10071836: mov edx,[edx+ecx*4]
0x10071839: movzx eax,ax
0x1007183c: cmp eax,[edx]
0x1007183e: jae short 1007186Ch
0x10071840: cmp dword ptr [ebp-1Ch],0
0x10071844: jne short 10071867h
0x10071846: push dword ptr [ebx]
0x10071848: push 1010BB00h
0x1007184d: push 1010B308h
0x10071852: push dword ptr [ebx+274h]
0x10071858: call 1007007Dh
0x1007185d: add esp,10h
0x10071860: mov dword ptr [ebp-1Ch],1
0x10071867: xor ecx,ecx
0x10071869: mov [ebp-10h],ecx
0x1007186c: cmp ecx,[ebx+294h]
0x10071872: jae near ptr 10071ADFh
0x10071878: mov eax,[ebx+290h]
0x1007187e: movzx edx,word ptr [edi]
0x10071881: lea eax,[eax+ecx*4]
0x10071884: mov esi,[eax]
0x10071886: cmp [esi],edx
0x10071888: jae short 10071899h
0x1007188a: inc ecx
0x1007188b: add eax,4
0x1007188e: cmp ecx,[ebx+294h]
0x10071894: mov [ebp-10h],ecx
0x10071897: jb short 10071884h
0x10071899: cmp ecx,[ebx+294h]
0x1007189f: jae near ptr 10071ADFh
0x100718a5: mov eax,[ebp-10h]
0x100718a8: mov ecx,[ebx+290h]
0x100718ae: movzx esi,word ptr [edi]
0x100718b1: lea edx,[ecx+eax*4]
```

```
0x100718b4: mov eax,[edx]
0x100718b6: cmp [eax],esi
0x100718b8: jne near ptr 10071ADFh
0x100718be: cmp word ptr [eax+0Ch],0
0x100718c3: je near ptr 10071960h
0x100718c9: movzx esi,word ptr [edi+2]
0x100718cd: cmp si,[eax+8]
0x100718d1: mov [ebp-20h],eax
0x100718d4: je short 1007190Ah
0x100718d6: mov [ebp-18h],edx
0x100718d9: cmp dword ptr [eax+8],0
0x100718dd: je short 1007190Ah
0x100718df: add dword ptr [ebp-18h],4
0x100718e3: mov eax,[ebp-18h]
0x100718e6: inc dword ptr [ebp-10h]
0x100718e9: mov edx,[ebx+294h]
0x100718ef: cmp [ebp-10h],edx
0x100718f2: mov eax,[eax]
0x100718f4: mov [ebp-20h],eax
0x100718f7: jae short 10071926h
0x100718f9: movzx edx,word ptr [edi]
0x100718fc: cmp [eax],edx
0x100718fe: jne short 10071926h
0x10071900: mov dx,[edi+2]
0x10071904: cmp dx,[eax+8]
0x10071908: jne short 100718D9h
0x1007190a: movzx ecx,word ptr [eax+4]
0x1007190e: cmp cx,0FFFFh
0x10071912: je short 1007196Ch
0x10071914: cmp cx,0FFFDh
0x10071918: je short 1007196Ch
0x1007191a: cmp cx,0FFFEh
0x1007191e: jne short 10071950h
0x10071920: movzx edx,word ptr [ebx+62h]
0x10071924: jmp short 10071953h
0x10071926: mov eax,[ebp-10h]
0x10071929: mov eax,[ecx+eax*4-4]
0x1007192d: push dword ptr [eax+10h]
0x10071930: movzx eax,si
0x10071933: push eax
0x10071934: push dword ptr [ebx]
0x10071936: push 1010BAD0h
0x1007193b: push 1010B308h
0x10071940: push dword ptr [ebx+274h]
0x10071946: call 1007007Dh
0x1007194b: add esp,18h
0x1007194e: jmp short 10071960h
0x10071950: movsx edx,cx
0x10071953: mov eax,edi
0x10071955: mov esi,ebx
0x10071957: call 10070528h
0x1007195c: test eax,eax
0x1007195e: jne short 10071969h
0x10071960: and word ptr [edi],0
0x10071964: jmp near ptr 10071AE6h
0x10071969: mov eax,[ebp-20h]
0x1007196c: movzx esi,word ptr [edi]
0x1007196f: mov ecx,11Ch
0x10071974: cmp esi,ecx
0x10071976: jg near ptr 10071A95h
0x1007197c: je near ptr 10071AC9h
0x10071982: cmp esi,100h
0x10071988: jl near ptr 10071AE6h
0x1007198e: cmp esi,101h
0x10071994: jle near ptr 10071AC9h
0x1007199a: cmp esi,103h
0x100719a0: je short 100719DEh
```

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0x100719a2: cmp esi,111h
0x100719a8: je short 100719C2h
0x100719aa: cmp esi,116h
0x100719b0: je near ptr 10071AC9h
0x100719b6: cmp esi,117h
0x100719bc: jne near ptr 10071AE6h
0x100719c2: movzx ecx,word ptr [eax+0Ch]
0x100719c6: xor edx,edx
0x100719c8: mov eax,ecx
0x100719ca: and ecx,1Fh
0x100719cd: inc edx
0x100719ce: shr eax,5
0x100719d1: shl edx,cl
0x100719d3: lea eax,[ebx+eax*4+28h]
0x100719d7: or [eax],edx
0x100719d9: jmp near ptr 10071AE6h
0x100719de: cmp dword ptr [edi+4],1
0x100719e2: jne short 10071A50h
0x100719e4: cmp word ptr [ebx+1C8h],4D4Dh
0x100719ed: jne short 10071A0Fh
0x100719ef: movzx edx,word ptr [edi+2]
0x100719f3: mov ecx,[ebx+1D0h]
0x100719f9: mov eax,[edi+8]
0x100719fc: shl edx,2
0x100719ff: mov ecx,[ecx+edx]
0x10071a02: shr eax,cl
0x10071a04: mov ecx,[ebx+1D4h]
0x10071a0a: and eax,[ecx+edx]
0x10071a0d: jmp short 10071A1Fh
0x10071a0f: movzx eax,word ptr [edi+2]
0x10071a13: mov ecx,[ebx+1D4h]
0x10071a19: mov eax,[ecx+eax*4]
0x10071a1c: and eax,[edi+8]
0x10071a1f: mov [ebp-4],eax
0x10071a22: movzx eax,ax
0x10071a25: push eax
0x10071a26: push esi
0x10071a27: push ebx
0x10071a28: call 1006796Fh
0x10071a2d: add esp,0Ch
0x10071a30: test eax,eax
0x10071a32: jne near ptr 10071AE6h
0x10071a38: cmp dword ptr [ebp-8],0
0x10071a3c: je near ptr 10071743h
0x10071a42: push dword ptr [ebp-8]
0x10071a45: call 100209E0h
0x10071a4a: pop ecx
0x10071a4b: jmp near ptr 10071743h
0x10071a50: cmp word ptr [edi+2],4
0x10071a55: jne short 10071A6Fh
0x10071a57: lea eax,[ebp-4]
0x10071a5a: push eax
0x10071a5b: push edi
0x10071a5c: mov eax,ebx
0x10071a5e: call 1007127Ch
0x10071a63: test eax,eax
0x10071a65: pop ecx
0x10071a66: pop ecx
0x10071a67: je short 10071A38h
0x10071a69: movzx eax,word ptr [ebp-4]
0x10071a6d: jmp short 10071A85h
0x10071a6f: lea eax,[ebp+8]
0x10071a72: push eax
0x10071a73: mov ecx,edi
0x10071a75: mov edx,ebx
0x10071a77: call 10071192h
0x10071a7c: test eax,eax
```

```
0x10071a7e: pop ecx
0x10071a7f: je short 10071A38h
0x10071a81: movzx eax,word ptr [ebp+8]
0x10071a85: push eax
0x10071a86: movzx eax,word ptr [edi]
0x10071a89: push eax
0x10071a8a: push ebx
0x10071a8b: call 1006796Fh
0x10071a90: add esp,0Ch
0x10071a93: jmp short 10071AD2h
0x10071a95: cmp esi,142h
0x10071a9b: jl short 10071AE6h
0x10071a9d: cmp esi,143h
0x10071aa3: jle short 10071AC9h
0x10071aa5: cmp esi,145h
0x10071aab: jle near ptr 100719C2h
0x10071ab1: cmp esi,152h
0x10071ab7: je short 10071AC9h
0x10071ab9: cmp esi,80E4h
0x10071abf: jle short 10071AE6h
0x10071ac1: cmp esi,80E6h
0x10071ac7: jg short 10071AE6h
0x10071ac9: mov eax,edi
0x10071acb: mov ecx,ebx
0x10071acd: call 10070DD6h
0x10071ad2: test eax,eax
0x10071ad4: je near ptr 10071A38h
0x10071ada: jmp near ptr 10071960h
0x10071adf: mov dword ptr [ebp-24h],1
0x10071ae6: dec dword ptr [ebp-0Ch]
0x10071ae9: add edi,0Ch
0x10071aec: cmp dword ptr [ebp-0Ch],0
0x10071af0: jg near ptr 10071814h
0x10071af6: cmp dword ptr [ebp-24h],0
0x10071afa: je near ptr 10071C94h
0x10071b00: movzx eax,word ptr [ebp-14h]
0x10071b04: mov esi,[ebp-8]
0x10071b07: xor edi,edi
0x10071b09: test eax,eax
0x10071b0b: mov [ebp-0Ch],eax
0x10071b0e: jle near ptr 10071C94h
0x10071b14: movzx eax,word ptr [esi]
0x10071b17: test ax,ax
0x10071b1a: je near ptr 10071C84h
0x10071b20: cmp edi,[ebx+294h]
0x10071b26: jae short 10071B38h
0x10071b28: mov ecx,[ebx+290h]
0x10071b2e: mov ecx,[ecx+edi*4]
0x10071b31: movzx edx,ax
0x10071b34: cmp edx,[ecx]
0x10071b36: jae short 10071B3Ah
0x10071b38: xor edi,edi
0x10071b3a: cmp edi,[ebx+294h]
0x10071b40: jae short 10071B7Fh
0x10071b42: movzx ecx,ax
0x10071b45: mov eax,[ebx+290h]
0x10071b4b: lea eax,[eax+edi*4]
0x10071b4e: mov edx,[eax]
0x10071b50: cmp [edx],ecx
0x10071b52: jae short 10071B60h
0x10071b54: inc edi
0x10071b55: add eax,4
0x10071b58: cmp edi,[ebx+294h]
0x10071b5e: jb short 10071B4Eh
0x10071b60: cmp edi,[ebx+294h]
0x10071b66: jae short 10071B7Fh
0x10071b68: mov eax,[ebx+290h]
```

```
0x10071b6e: mov ecx,[eax+edi*4]
0x10071b71: movzx eax,word ptr [esi]
0x10071b74: movzx eax,ax
0x10071b77: cmp [ecx],eax
0x10071b79: je near ptr 10071C0Ah
0x10071b7f: movzx eax,word ptr [esi]
0x10071b82: movzx eax,ax
0x10071b85: push eax
0x10071b86: push eax
0x10071b87: push dword ptr [ebx]
0x10071b89: push 1010BA9Ch
0x10071b8e: push 1010B308h
0x10071b93: push dword ptr [ebx+274h]
0x10071b99: call 1007007Dh
0x10071b9e: movzx eax,word ptr [esi+2]
0x10071ba2: add esp,18h
0x10071ba5: push 1
0x10071ba7: push eax
0x10071ba8: movzx eax,word ptr [esi]
0x10071bab: push eax
0x10071bac: push ebx
0x10071bad: call 1006FEA1h
0x10071bb2: add esp,0Ch
0x10071bb5: push eax
0x10071bb6: push ebx
0x10071bb7: call 1006FF15h
0x10071bbc: add esp,0Ch
0x10071bbf: test eax,eax
0x10071bc1: jne short 10071BE5h
0x10071bc3: movzx eax,word ptr [esi]
0x10071bc6: push eax
0x10071bc7: push eax
0x10071bc8: push 1010BA64h
0x10071bcd: push 1010B308h
0x10071bd2: push dword ptr [ebx+274h]
0x10071bd8: call 1007007Dh
0x10071bdd: add esp,14h
0x10071be0: jmp near ptr 10071C80h
0x10071be5: xor edi,edi
0x10071be7: cmp [ebx+294h],edi
0x10071bed: jbe short 10071C0Ah
0x10071bef: movzx ecx,word ptr [esi]
0x10071bf2: mov eax,[ebx+290h]
0x10071bf8: mov edx,[eax]
0x10071bfa: cmp [edx],ecx
0x10071bfc: jae short 10071C0Ah
0x10071bfe: inc edi
0x10071bff: add eax,4
0x10071c02: cmp edi,[ebx+294h]
0x10071c08: jb short 10071BF8h
0x10071c0a: mov ecx,[ebx+290h]
0x10071c10: mov eax,[ecx+edi*4]
0x10071c13: movzx edx,word ptr [esi+2]
0x10071c17: add eax,8
0x10071c1a: cmp dx,[eax]
0x10071c1d: je short 10071C84h
0x10071c1f: lea edx,[ecx+edi*4]
0x10071c22: mov [ebp-18h],edx
0x10071c25: mov edx,[ebx+294h]
0x10071c2b: cmp edi,edx
0x10071c2d: jae short 10071C84h
0x10071c2f: cmp dword ptr [eax],0
0x10071c32: je short 10071C84h
0x10071c34: add dword ptr [ebp-18h],4
0x10071c38: mov eax,[ebp-18h]
0x10071c3b: mov eax,[eax]
0x10071c3d: inc edi
```

```
0x10071c3e: cmp edi,edx
0x10071c40: jae short 10071C57h
0x10071c42: movzx edx,word ptr [esi]
0x10071c45: cmp [eax],edx
0x10071c47: jne short 10071C57h
0x10071c49: mov dx,[esi+2]
0x10071c4d: add eax,8
0x10071c50: cmp dx,[eax]
0x10071c53: jne short 10071C25h
0x10071c55: jmp short 10071C84h
0x10071c57: mov eax,[ecx+edi*4-4]
0x10071c5b: push dword ptr [eax+10h]
0x10071c5e: movzx eax,word ptr [esi+2]
0x10071c62: movzx eax,ax
0x10071c65: push eax
0x10071c66: push dword ptr [ebx]
0x10071c68: push 1010BAD0h
0x10071c6d: push 1010B308h
0x10071c72: push dword ptr [ebx+274h]
0x10071c78: call 1007007Dh
0x10071c7d: add esp,18h
0x10071c80: and word ptr [esi],0
0x10071c84: dec dword ptr [ebp-0Ch]
0x10071c87: add esi,0Ch
0x10071c8a: cmp dword ptr [ebp-0Ch],0
0x10071c8e: jg near ptr 10071B14h
0x10071c94: cmp word ptr [ebx+58h],6
0x10071c99: jne short 10071CFEh
0x10071c9b: cmp word ptr [ebx+8Ah],2
0x10071ca3: jne short 10071CFEh
0x10071ca5: mov eax,[ebp-8]
0x10071ca8: push 111h
0x10071cad: push dword ptr [ebp-14h]
0x10071cb0: call 100702D3h
0x10071cb5: test eax,eax
0x10071cb7: pop ecx
0x10071cb8: pop ecx
0x10071cb9: je short 10071CFEh
0x10071cbb: cmp dword ptr [eax+4],1
0x10071cbf: jne short 10071CFEh
0x10071cc1: mov eax,[ebp-8]
0x10071cc4: push 117h
0x10071cc9: push dword ptr [ebp-14h]
0x10071ccc: call 100702D3h
0x10071cd1: test eax,eax
0x10071cd3: pop ecx
0x10071cd4: pop ecx
0x10071cd5: je short 10071CFEh
0x10071cd7: cmp dword ptr [eax+4],1
0x10071cdb: jne short 10071CFEh
0x10071cdd: push 1010BA10h
0x10071ce2: mov word ptr [ebx+8Ah],1
0x10071ceb: push 1010B9F8h
0x10071cf0: push dword ptr [ebx+274h]
0x10071cf6: call 1007007Dh
0x10071cfb: add esp,0Ch
0x10071cfe: mov eax,[ebx+28h]
0x10071d01: test al,2
0x10071d03: jne short 10071D17h
0x10071d05: push 10109F90h
0x10071d0a: mov eax,ebx
0x10071d0c: call 1007049Bh
0x10071d11: pop ecx
0x10071d12: jmp near ptr 10071A38h
0x10071d17: test al,4
0x10071d19: push ebx
0x10071d1a: jne short 10071D47h
```

```
0x10071d1c: call 10065F66h
0x10071d21: mov [ebx+0B4h],eax
0x10071d27: mov eax,[ebx+38h]
0x10071d2a: mov [ebx+44h],eax
0x10071d2d: mov eax,[ebx+64h]
0x10071d30: mov [ebx+48h],eax
0x10071d33: mov eax,[ebx+40h]
0x10071d36: mov [ebx+4Ch],eax
0x10071d39: and dword ptr [ebx+0Ch],0FFFFFBFFh
0x10071d40: mov edx,400h
0x10071d45: jmp short 10071D5Ah
0x10071d47: call 100685BCh
0x10071d4c: mov edx,400h
0x10071d51: mov [ebx+0B4h],eax
0x10071d57: or [ebx+0Ch],edx
0x10071d5a: mov esi,[ebx+0B4h]
0x10071d60: test esi,esi
0x10071d62: pop ecx
0x10071d63: jne short 10071D94h
0x10071d65: test [ebx+0Ch],edx
0x10071d68: mov eax,1010B9F0h
0x10071d6d: jne short 10071D74h
0x10071d6f: mov eax,1010B9E8h
0x10071d74: push eax
0x10071d75: push dword ptr [ebx]
0x10071d77: push 1010B9C4h
0x10071d7c: push 1010B308h
0x10071d81: push dword ptr [ebx+274h]
0x10071d87: call 1006FAF1h
0x10071d8c: add esp,14h
0x10071d8f: jmp near ptr 10071A38h
0x10071d94: cmp word ptr [ebx+8Ah],2
0x10071d9c: lea ecx,[ebx+0B0h]
0x10071da2: mov [ecx],esi
0x10071da4: jne short 10071DB7h
0x10071da6: movzx edi,word ptr [ebx+62h]
0x10071daa: xor edx,edx
0x10071dac: mov eax,esi
0x10071dae: div edi
0x10071db0: mov edx,400h
0x10071db5: mov [ecx],eax
0x10071db7: mov eax,[ebx+28h]
0x10071dba: mov ecx,2000000h
0x10071dbf: test ecx,eax
0x10071dc1: jne short 10071DD9h
0x10071dc3: cmp word ptr [ebx+58h],6
0x10071dc8: jne short 10071DEAh
0x10071dca: test [ebx+0Ch],edx
0x10071dcd: jne short 10071DEFh
0x10071dcf: cmp esi,1
0x10071dd2: jne short 10071DEAh
0x10071dd4: or eax,ecx
0x10071dd6: mov [ebx+28h],eax
0x10071dd9: movzx eax,word ptr [ebp-14h]
0x10071ddd: mov edi,[ebp-8]
0x10071de0: mov [ebp-0Ch],eax
0x10071de3: test eax,eax
0x10071de5: jmp near ptr 10072113h
0x10071dea: test [ebx+0Ch],edx
0x10071ded: je short 10071DF6h
0x10071def: mov eax,10109D24h
0x10071df4: jmp short 10071DFBh
0x10071df6: mov eax,10109EECh
0x10071dfb: push eax
0x10071dfc: jmp near ptr 10071D0Ah
0x10071e01: movzx edx,word ptr [edi]
0x10071e04: test dx,dx
```

```
0x10071e07: je near ptr 10072109h
0x10071e0d: movzx esi,dx
0x10071e10: mov eax,141h
0x10071e15: cmp esi,eax
0x10071e17: jg near ptr 10071FB2h
0x10071e1d: je near ptr 100720FCh
0x10071e23: cmp esi,119h
0x10071e29: jg near ptr 10071EF3h
0x10071e2f: cmp esi,118h
0x10071e35: jge near ptr 1007204Ah
0x10071e3b: mov eax,esi
0x10071e3d: sub eax,0FFh
0x10071e42: je short 10071E83h
0x10071e44: sub eax,3
0x10071e47: je near ptr 1007204Ah
0x10071e4d: sub eax,0Fh
0x10071e50: je near ptr 10071FE0h
0x10071e56: sub eax,6
0x10071e59: jne near ptr 1007203Ch
0x10071e5f: lea eax,[ebx+0BCh]
0x10071e65: push dword ptr [ebx+0B4h]
0x10071e6b: mov edi,[ebp-10h]
0x10071e6e: mov ecx,ebx
0x10071e70: call 10071411h
0x10071e75: test eax,eax
0x10071e77: pop ecx
0x10071e78: je near ptr 10071A38h
0x10071e7e: jmp near ptr 10072106h
0x10071e83: and dword ptr [ebp-4],0
0x10071e87: cmp word ptr [ebx+1C8h],4D4Dh
0x10071e90: jne short 10071EB2h
0x10071e92: movzx edx,word ptr [edi+2]
0x10071e96: mov ecx,[ebx+1D0h]
0x10071e9c: mov eax,[edi+8]
0x10071e9f: shl edx,2
0x10071ea2: mov ecx,[ecx+edx]
0x10071ea5: shr eax,cl
0x10071ea7: mov ecx,[ebx+1D4h]
0x10071ead: and eax,[ecx+edx]
0x10071eb0: jmp short 10071EC2h
0x10071eb2: movzx eax,word ptr [edi+2]
0x10071eb6: mov ecx,[ebx+1D4h]
0x10071ebc: mov eax,[ecx+eax*4]
0x10071ebf: and eax,[edi+8]
0x10071ec2: dec eax
0x10071ec3: dec eax
0x10071ec4: je short 10071ED6h
0x10071ec6: dec eax
0x10071ec7: jne near ptr 10072109h
0x10071ecd: mov dword ptr [ebp-4],2
0x10071ed4: jmp short 10071EDDh
0x10071ed6: mov dword ptr [ebp-4],1
0x10071edd: push dword ptr [ebp-4]
0x10071ee0: push 0FEh
0x10071ee5: push ebx
0x10071ee6: call 1006796Fh
0x10071eeb: add esp,0Ch
0x10071eee: jmp near ptr 10072109h
0x10071ef3: mov eax,esi
0x10071ef5: sub eax,129h
0x10071efa: je near ptr 100720FCh
0x10071f00: sub eax,4
0x10071f03: je short 10071F0Eh
0x10071f05: sub eax,13h
0x10071f08: jne near ptr 1007203Ch
0x10071f0e: mov cl,[ebx+54h]
0x10071f11: xor eax,eax
```

```
0x10071f13: inc eax
0x10071f14: shl eax,cl
0x10071f16: cmp dx,140h
0x10071f1b: mov [ebp-4],eax
0x10071f1e: je short 10071F25h
0x10071f20: cmp [edi+4],eax
0x10071f23: je short 10071F3Fh
0x10071f25: mov edx,[ebp-4]
0x10071f28: imul edx,3
0x10071f2b: mov eax,edi
0x10071f2d: mov esi,ebx
0x10071f2f: call 10070528h
0x10071f34: test eax,eax
0x10071f36: je near ptr 10072109h
0x10071f3c: mov eax,[ebp-4]
0x10071f3f: push 1010B9A4h
0x10071f44: push 2
0x10071f46: push dword ptr [edi+4]
0x10071f49: add eax,eax
0x10071f4b: push ebx
0x10071f4c: mov [ebp-4],eax
0x10071f4f: call 1006E083h
0x10071f54: add esp,10h
0x10071f57: test eax,eax
0x10071f59: mov [ebp-18h],eax
0x10071f5c: je near ptr 10072109h
0x10071f62: push eax
0x10071f63: mov esi,edi
0x10071f65: call 10070799h
0x10071f6a: test eax,eax
0x10071f6c: pop ecx
0x10071f6d: je short 10071FA3h
0x10071f6f: mov cl,[ebx+54h]
0x10071f72: xor eax,eax
0x10071f74: inc eax
0x10071f75: shl eax,cl
0x10071f77: mov ecx,[ebp-10h]
0x10071f7a: cmp [ecx+4],eax
0x10071f7d: jne short 10071F86h
0x10071f7f: xor eax,eax
0x10071f81: mov [ebp-4],eax
0x10071f84: jmp short 10071F89h
0x10071f86: mov eax,[ebp-4]
0x10071f89: mov esi,[ebp-18h]
0x10071f8c: lea edx,[esi+eax*2]
0x10071f8f: push edx
0x10071f90: add eax,esi
0x10071f92: push eax
0x10071f93: movzx eax,word ptr [ecx]
0x10071f96: push esi
0x10071f97: push eax
0x10071f98: push ebx
0x10071f99: call 1006796Fh
0x10071f9e: add esp,14h
0x10071fa1: jmp short 10071FA6h
0x10071fa3: mov esi,[ebp-18h]
0x10071fa6: push esi
0x10071fa7: call 100209E0h
0x10071fac: pop ecx
0x10071fad: jmp near ptr 10072106h
0x10071fb2: cmp esi,155h
0x10071fb8: jg short 10072020h
0x10071fba: cmp esi,154h
0x10071fc0: jge short 10071FEBh
0x10071fc2: mov eax,esi
0x10071fc4: sub eax,144h
0x10071fc9: je short 10071FE0h
```

```
0x10071fcb: dec eax
0x10071fcc: je near ptr 10071E5Fh
0x10071fd2: sub eax,0Bh
0x10071fd5: je near ptr 100720FCh
0x10071fdb: sub eax,3
0x10071fde: jmp short 1007203Ah
0x10071fe0: lea eax,[ebx+0B8h]
0x10071fe6: jmp near ptr 10071E65h
0x10071feb: fldz
0x10071fed: lea eax,[ebp-28h]
0x10071ff0: push eax
0x10071ff1: fstp qword ptr [ebp-28h]
0x10071ff4: push edi
0x10071ff5: mov eax,ebx
0x10071ff7: call 10071346h
0x10071ffc: test eax,eax
0x10071ffe: pop ecx
0x10071fff: pop ecx
0x10072000: je near ptr 10071A38h
0x10072006: movzx eax,word ptr [edi]
0x10072009: fld qword ptr [ebp-28h]
0x1007200c: push ecx
0x1007200d: push ecx
0x1007200e: fstp qword ptr [esp]
0x10072011: push eax
0x10072012: push ebx
0x10072013: call 1006796Fh
0x10072018: add esp,10h
0x1007201b: jmp near ptr 100720E7h
0x10072020: mov eax,esi
0x10072022: sub eax,212h
0x10072027: je near ptr 100720FCh
0x1007202d: dec eax
0x1007202e: dec eax
0x1007202f: je near ptr 100720F1h
0x10072035: sub eax,7ED0h
0x1007203a: je short 1007204Ah
0x1007203c: mov eax,edi
0x1007203e: mov ecx,ebx
0x10072040: call 10070DD6h
0x10072045: jmp near ptr 10072109h
0x1007204a: cmp dword ptr [edi+4],1
0x1007204e: jne short 10072095h
0x10072050: cmp word ptr [ebx+1C8h],4D4Dh
0x10072059: jne short 1007207Bh
0x1007205b: movzx edx,word ptr [edi+2]
0x1007205f: mov ecx,[ebx+1D0h]
0x10072065: mov eax,[edi+8]
0x10072068: shl edx,2
0x1007206b: mov ecx,[ecx+edx]
0x1007206e: shr eax,cl
0x10072070: mov ecx,[ebx+1D4h]
0x10072076: and eax,[ecx+edx]
0x10072079: jmp short 1007208Bh
0x1007207b: movzx eax,word ptr [edi+2]
0x1007207f: mov ecx,[ebx+1D4h]
0x10072085: mov eax,[ecx+eax*4]
0x10072088: and eax,[edi+8]
0x1007208b: mov [ebp-4],eax
0x1007208e: movzx eax,ax
0x10072091: push eax
0x10072092: push esi
0x10072093: jmp short 100720DEh
0x10072095: cmp dx,102h
0x1007209a: jne short 100720BFh
0x1007209c: cmp word ptr [edi+2],4
0x100720a1: jne short 100720BFh
```

```
0x100720a3: lea eax,[ebp-4]
0x100720a6: push eax
0x100720a7: push edi
0x100720a8: mov eax,ebx
0x100720aa: call 1007127Ch
0x100720af: test eax,eax
0x100720b1: pop ecx
0x100720b2: pop ecx
0x100720b3: je near ptr 10071A38h
0x100720b9: movzx eax,word ptr [ebp-4]
0x100720bd: jmp short 100720D9h
0x100720bf: lea eax,[ebp+8]
0x100720c2: push eax
0x100720c3: mov ecx,edi
0x100720c5: mov edx,ebx
0x100720c7: call 10071192h
0x100720cc: test eax,eax
0x100720ce: pop ecx
0x100720cf: je near ptr 10071A38h
0x100720d5: movzx eax,word ptr [ebp+8]
0x100720d9: push eax
0x100720da: movzx eax,word ptr [edi]
0x100720dd: push eax
0x100720de: push ebx
0x100720df: call 1006796Fh
0x100720e4: add esp,0Ch
0x100720e7: test eax,eax
0x100720e9: je near ptr 10071A38h
0x100720ef: jmp short 10072109h
0x100720f1: push ebx
0x100720f2: mov esi,edi
0x100720f4: call 10071523h
0x100720f9: pop ecx
0x100720fa: jmp short 10072109h
0x100720fc: mov esi,[ebp-10h]
0x100720ff: mov edi,ebx
0x10072101: call 10070AEBh
0x10072106: mov edi,[ebp-10h]
0x10072109: dec dword ptr [ebp-0Ch]
0x1007210c: add edi,0Ch
0x1007210f: cmp dword ptr [ebp-0Ch],0
0x10072113: mov [ebp-10h],edi
0x10072116: jg near ptr 10071E01h
0x1007211c: cmp word ptr [ebx+58h],6
0x10072121: jne near ptr 100721FCh
0x10072127: test word ptr [ebx+28h],100h
0x1007212d: mov esi,1010B9F8h
0x10072132: jne short 1007215Fh
0x10072134: push 1010B970h
0x10072139: push esi
0x1007213a: push dword ptr [ebx+274h]
0x10072140: call 1007007Dh
0x10072145: push 6
0x10072147: push 106h
0x1007214c: push ebx
0x1007214d: call 1006796Fh
0x10072152: add esp,18h
0x10072155: test eax,eax
0x10072157: je near ptr 10071A38h
0x1007215d: jmp short 10072180h
0x1007215f: cmp word ptr [ebx+5Ah],2
0x10072164: jne short 10072180h
0x10072166: push 1010B920h
0x1007216b: mov word ptr [ebx+5Ah],6
0x10072171: push esi
0x10072172: push dword ptr [ebx+274h]
0x10072178: call 1007007Dh
```

```
0x1007217d: add esp,0Ch
0x10072180: test byte ptr [ebx+28h],40h
0x10072184: jne short 100721AFh
0x10072186: push 1010B8E4h
0x1007218b: push esi
0x1007218c: push dword ptr [ebx+274h]
0x10072192: call 1007007Dh
0x10072197: push 8
0x10072199: push 102h
0x1007219e: push ebx
0x1007219f: call 1006796Fh
0x100721a4: add esp,18h
0x100721a7: test eax,eax
0x100721a9: je near ptr 10071A38h
0x100721af: test dword ptr [ebx+28h],10000h
0x100721b6: jne short 100721FCh
0x100721b8: movzx eax,word ptr [ebx+5Ah]
0x100721bc: cmp ax,2
0x100721c0: je short 10072216h
0x100721c2: cmp ax,6
0x100721c6: je short 10072216h
0x100721c8: test ax,ax
0x100721cb: je short 100721D3h
0x100721cd: cmp ax,1
0x100721d1: jne short 100721FCh
0x100721d3: push 1010B898h
0x100721d8: push esi
0x100721d9: push dword ptr [ebx+274h]
0x100721df: call 1007007Dh
0x100721e4: push 1
0x100721e6: push 115h
0x100721eb: push ebx
0x100721ec: call 1006796Fh
0x100721f1: add esp,18h
0x100721f4: test eax,eax
0x100721f6: je near ptr 10071A38h
0x100721fc: cmp word ptr [ebx+5Ah],3
0x10072201: jne short 1007222Bh
0x10072203: test dword ptr [ebx+28h],4000000h
0x1007220a: jne short 1007222Bh
0x1007220c: push 1010B88Ch
0x10072211: jmp near ptr 10071D0Ah
0x10072216: push 1010B840h
0x1007221b: push esi
0x1007221c: push dword ptr [ebx+274h]
0x10072222: call 1007007Dh
0x10072227: push 3
0x10072229: jmp short 100721E6h
0x1007222b: movzx eax,word ptr [ebx+58h]
0x1007222f: cmp ax,6
0x10072233: je near ptr 1007236Dh
0x10072239: test dword ptr [ebx+28h],1000000h
0x10072240: jne short 100722B7h
0x10072242: movzx eax,word ptr [ebx+8Ah]
0x10072249: cmp ax,1
0x1007224d: jne short 10072258h
0x1007224f: cmp dword ptr [ebx+0B4h],1
0x10072256: ja short 1007226Ah
0x10072258: cmp ax,2
0x1007225c: jne short 10072274h
0x1007225e: movzx eax,word ptr [ebx+62h]
0x10072262: cmp [ebx+0B4h],eax
0x10072268: je short 10072274h
0x1007226a: push 10109EB0h
0x1007226f: jmp near ptr 10071D0Ah
0x10072274: push 117h
0x10072279: push ebx
```

```
0x1007227a: call 1006FE65h
0x1007227f: push dword ptr [eax+10h]
0x10072282: mov edi,1010B308h
0x10072287: push dword ptr [ebx]
0x10072289: push 1010B7F0h
0x1007228e: push edi
0x1007228f: push dword ptr [ebx+274h]
0x10072295: call 1007007Dh
0x1007229a: push dword ptr [ebp-14h]
0x1007229d: mov esi,ebx
0x1007229f: push dword ptr [ebp-8]
0x100722a2: call 100702F4h
0x100722a7: add esp,24h
0x100722aa: test eax,eax
0x100722ac: jl near ptr 10071A38h
0x100722b2: jmp near ptr 10072372h
0x100722b7: xor edi,edi
0x100722b9: inc edi
0x100722ba: cmp [ebx+0B4h],edi
0x100722c0: jne near ptr 100723CDh
0x100722c6: mov ecx,[ebx+0B8h]
0x100722cc: xor esi,esi
0x100722ce: cmp [ecx],esi
0x100722d0: je near ptr 100723CDh
0x100722d6: mov ecx,[ebx+0BCh]
0x100722dc: cmp [ecx],esi
0x100722de: je short 10072330h
0x100722e0: cmp ax,di
0x100722e3: jne short 10072304h
0x100722e5: push dword ptr [ebx+274h]
0x100722eb: call dword ptr [ebx+288h]
0x100722f1: pop ecx
0x100722f2: mov ecx,[ebx+0B8h]
0x100722f8: sub eax,[ecx]
0x100722fa: mov ecx,[ebx+0BCh]
0x10072300: cmp [ecx],eax
0x10072302: ja short 10072330h
0x10072304: cmp [ebx+8],esi
0x10072307: jne near ptr 100723CDh
0x1007230d: cmp [ebx+58h],di
0x10072311: jne near ptr 100723CDh
0x10072317: push ebx
0x10072318: call 10065FAFh
0x1007231d: imul eax,[ebx+3Ch]
0x10072321: pop ecx
0x10072322: mov ecx,[ebx+0BCh]
0x10072328: cmp [ecx],eax
0x1007232a: jae near ptr 100723CDh
0x10072330: push 117h
0x10072335: push ebx
0x10072336: call 1006FE65h
0x1007233b: push dword ptr [eax+10h]
0x1007233e: push dword ptr [ebx]
0x10072340: push 1010B7B0h
0x10072345: push 1010B308h
0x1007234a: push dword ptr [ebx+274h]
0x10072350: call 1007007Dh
0x10072355: push dword ptr [ebp-14h]
0x10072358: mov esi,ebx
0x1007235a: push dword ptr [ebp-8]
0x1007235d: call 100702F4h
0x10072362: add esp,24h
0x10072365: test eax,eax
0x10072367: jl near ptr 10071A38h
0x1007236d: mov edi,1010B308h
0x10072372: cmp dword ptr [ebp-8],0
0x10072376: je short 10072381h
```

```
0x10072378: push dword ptr [ebp-8]
0x1007237b: call 100209E0h
0x10072380: pop ecx
0x10072381: xor esi,esi
0x10072383: inc esi
0x10072384: test dword ptr [ebx+28h],80000h
0x1007238b: jne short 10072399h
0x1007238d: mov cl,[ebx+54h]
0x10072390: mov eax,esi
0x10072392: shl eax,cl
0x10072394: dec eax
0x10072395: mov [ebx+6Ah],ax
0x10072399: mov eax,[ebx+0B4h]
0x1007239f: cmp eax,esi
0x100723a1: jbe near ptr 1007242Ah
0x100723a7: mov eax,[ebx+0B8h]
0x100723ad: mov [ebx+0C0h],esi
0x100723b3: mov ecx,esi
0x100723b5: add eax,4
0x100723b8: mov edx,[eax-4]
0x100723bb: cmp edx,[eax]
0x100723bd: ja short 10072423h
0x100723bf: inc ecx
0x100723c0: add eax,4
0x100723c3: cmp ecx,[ebx+0B4h]
0x100723c9: jb short 100723B8h
0x100723cb: jmp short 1007242Ah
0x100723cd: cmp [ebx+8Ah],di
0x100723d4: jne short 1007236Dh
0x100723d6: cmp dword ptr [ebx+0B4h],2
0x100723dd: jbe short 1007236Dh
0x100723df: cmp word ptr [ebx+58h],1
0x100723e4: jne short 1007236Dh
0x100723e6: mov eax,[ebx+0BCh]
0x100723ec: mov ecx,[eax]
0x100723ee: cmp ecx,[eax+4]
0x100723f1: je near ptr 1007236Dh
0x100723f7: test ecx,ecx
0x100723f9: je near ptr 1007236Dh
0x100723ff: cmp dword ptr [eax+4],0
0x10072403: je near ptr 1007236Dh
0x10072409: push 117h
0x1007240e: push ebx
0x1007240f: call 1006FE65h
0x10072414: push dword ptr [eax+10h]
0x10072417: push dword ptr [ebx]
0x10072419: push 1010B770h
0x1007241e: jmp near ptr 10072345h
0x10072423: and dword ptr [ebx+0C0h],0
0x1007242a: test byte ptr [ebx+28h],80h
0x1007242e: jne short 1007243Fh
0x10072430: push esi
0x10072431: push 103h
0x10072436: push ebx
0x10072437: call 1006796Fh
0x1007243c: add esp,0Ch
0x1007243f: cmp [ebx+0B4h],esi
0x10072445: jne short 10072466h
0x10072447: cmp [ebx+58h],si
0x1007244b: jne short 10072466h
0x1007244d: mov eax,[ebx+0Ch]
0x10072450: and eax,8400h
0x10072455: cmp eax,8000h
0x1007245a: jne short 10072466h
0x1007245c: mov esi,ebx
0x1007245e: call 100715DAh
0x10072463: xor esi,esi
```

```
0x10072465: inc esi
0x10072466: or eax,0FFFFFFFFh
0x10072469: push ebx
0x1007246a: mov [ebx+1D8h],eax
0x10072470: mov [ebx+1E0h],eax
0x10072476: mov [ebx+1F4h],eax
0x1007247c: mov [ebx+1F8h],eax
0x10072482: mov [ebx+1FCh],eax
0x10072488: call 10065FAFh
0x1007248d: test eax,eax
0x1007248f: pop ecx
0x10072490: mov [ebx+24Ch],eax
0x10072496: jne short 100724B3h
0x10072498: push dword ptr [ebx]
0x1007249a: push 1010B748h
0x1007249f: push edi
0x100724a0: push dword ptr [ebx+274h]
0x100724a6: call 1006FAF1h
0x100724ab: add esp,10h
0x100724ae: jmp near ptr 10071743h
0x100724b3: test word ptr [ebx+0Ch],400h
0x100724b9: push ebx
0x100724ba: je short 100724D5h
0x100724bc: call 1006883Eh
0x100724c1: test eax,eax
0x100724c3: pop ecx
0x100724c4: mov [ebx+1FCh],eax
0x100724ca: jne short 100724E8h
0x100724cc: push dword ptr [ebx]
0x100724ce: push 1010B724h
0x100724d3: jmp short 1007249Fh
0x100724d5: call 1006634Ch
0x100724da: test eax,eax
0x100724dc: pop ecx
0x100724dd: jne short 100724E8h
0x100724df: push dword ptr [ebx]
0x100724e1: push 1010B700h
0x100724e6: jmp short 1007249Fh
0x100724e8: mov eax,esi
0x100724ea: pop edi
0x100724eb: pop esi
0x100724ec: pop ebx
0x100724ed: leave
0x100724ee: ret
```

`sub_1007171b` is called by sub_10067905, sub_1006891d. It works with the text "TileOffsets", "StripOffsets", "TIFFReadDirectory". It delegates to j_free, sub_10065f66, sub_10065faf (32 callees).

12.1056 sub_100727aa at 0x100727aa (133 instructions)

```
0x100727aa: push ebp
0x100727ab: mov ebp,esp
0x100727ad: sub esp,0Ch
0x100727b0: push esi
0x100727b1: mov esi,[ebp+8]
0x100727b4: push edi
0x100727b5: push 54h
0x100727b7: push dword ptr [ebp+0Ch]
0x100727ba: lea eax,[esi+1Ch]
0x100727bd: push eax
0x100727be: mov dword ptr [esi],5DCh
0x100727c4: call 10020A00h
0x100727c9: fld dword ptr [esi+64h]
0x100727cc: fld1
0x100727ce: mov eax,[esi]
0x100727d0: fdivrp
```

```
0x100727d2: mov [ebp+8],eax
0x100727d5: add esp,0Ch
0x100727d8: fstp qword ptr [ebp-0Ch]
0x100727db: fld dword ptr [esi+40h]
0x100727de: fsub dword ptr [esi+58h]
0x100727e1: fidiv dword ptr [ebp+8]
0x100727e4: and dword ptr [ebp+8],0
0x100727e8: test eax,eax
0x100727ea: fstp dword ptr [esi+4]
0x100727ed: jl short 10072833h
0x100727ef: lea edi,[esi+70h]
0x100727f2: fld qword ptr [ebp-0Ch]
0x100727f5: sub esp,10h
0x100727f8: fstp qword ptr [esp+8]
0x100727fc: fild dword ptr [ebp+8]
0x100727ff: fidiv dword ptr [esi]
0x10072801: fstp qword ptr [esp]
0x10072804: call 100FC11Ah
0x10072809: fstp dword ptr [ebp-4]
0x1007280c: mov eax,[esi+4Ch]
0x1007280f: fld dword ptr [ebp-4]
0x10072812: fild dword ptr [esi+4Ch]
0x10072815: add esp,10h
0x10072818: test eax,eax
0x1007281a: jge short 10072822h
0x1007281c: fadd dword ptr ds:[100FE7C0h]
0x10072822: inc dword ptr [ebp+8]
0x10072825: fmulp
0x10072827: mov eax,[ebp+8]
0x1007282a: fstp dword ptr [edi]
0x1007282c: add edi,4
0x1007282f: cmp eax,[esi]
0x10072831: jle short 100727F2h
0x10072833: fld dword ptr [esi+68h]
0x10072836: mov eax,[esi]
0x10072838: fld1
0x1007283a: mov [ebp+8],eax
0x1007283d: fdivrp
0x1007283f: fstp qword ptr [ebp-0Ch]
0x10072842: fld dword ptr [esi+40h]
0x10072845: fsub dword ptr [esi+58h]
0x10072848: fidiv dword ptr [ebp+8]
0x1007284b: and dword ptr [ebp+8],0
0x1007284f: test eax,eax
0x10072851: fstp dword ptr [esi+8]
0x10072854: jl short 1007289Dh
0x10072856: lea edi,[esi+17E4h]
0x1007285c: fld qword ptr [ebp-0Ch]
0x1007285f: sub esp,10h
0x10072862: fstp qword ptr [esp+8]
0x10072866: fild dword ptr [ebp+8]
0x10072869: fidiv dword ptr [esi]
0x1007286b: fstp qword ptr [esp]
0x1007286e: call 100FC11Ah
0x10072873: fstp dword ptr [ebp-4]
0x10072876: mov eax,[esi+50h]
0x10072879: fld dword ptr [ebp-4]
0x1007287c: fild dword ptr [esi+50h]
0x1007287f: add esp,10h
0x10072882: test eax,eax
0x10072884: jge short 1007288Ch
0x10072886: fadd dword ptr ds:[100FE7C0h]
0x1007288c: inc dword ptr [ebp+8]
0x1007288f: fmulp
0x10072891: mov eax,[ebp+8]
0x10072894: fstp dword ptr [edi]
0x10072896: add edi,4
```

```
0x10072899: cmp eax,[esi]
0x1007289b: jle short 1007285Ch
0x1007289d: fld dword ptr [esi+6Ch]
0x100728a0: mov eax,[esi]
0x100728a2: fld1
0x100728a4: mov [ebp+8],eax
0x100728a7: fdivrp
0x100728a9: fstp qword ptr [ebp-0Ch]
0x100728ac: fld dword ptr [esi+40h]
0x100728af: fsub dword ptr [esi+58h]
0x100728b2: fidiv dword ptr [ebp+8]
0x100728b5: and dword ptr [ebp+8],0
0x100728b9: test eax,eax
0x100728bb: fstp dword ptr [esi+0Ch]
0x100728be: jl short 10072907h
0x100728c0: lea edi,[esi+2F58h]
0x100728c6: fld qword ptr [ebp-0Ch]
0x100728c9: sub esp,10h
0x100728cc: fstp qword ptr [esp+8]
0x100728d0: fild dword ptr [ebp+8]
0x100728d3: fidiv dword ptr [esi]
0x100728d5: fstp qword ptr [esp]
0x100728d8: call 100FC11Ah
0x100728dd: fstp dword ptr [ebp-4]
0x100728e0: mov eax,[esi+54h]
0x100728e3: fld dword ptr [ebp-4]
0x100728e6: fild dword ptr [esi+54h]
0x100728e9: add esp,10h
0x100728ec: test eax,eax
0x100728ee: jge short 100728F6h
0x100728f0: fadd dword ptr ds:[100FE7C0h]
0x100728f6: inc dword ptr [ebp+8]
0x100728f9: fmulp
0x100728fb: mov eax,[ebp+8]
0x100728fe: fstp dword ptr [edi]
0x10072900: add edi,4
0x10072903: cmp eax,[esi]
0x10072905: jle short 100728C6h
0x10072907: mov eax,[ebp+10h]
0x1007290a: fld dword ptr [eax]
0x1007290c: pop edi
0x1007290d: fstp dword ptr [esi+10h]
0x10072910: fld dword ptr [eax+4]
0x10072913: fstp dword ptr [esi+14h]
0x10072916: fld dword ptr [eax+8]
0x10072919: xor eax,eax
0x1007291b: fstp dword ptr [esi+18h]
0x1007291e: pop esi
0x1007291f: leave
0x10072920: ret
```

`sub_100727aa` is called by sub_1006cf39. It delegates to j_memcpy, j_pow (2 callees).

12.1057 sub_100729b6 at 0x100729b6 (192 instructions)

```
0x100729b6: push ebp
0x100729b7: mov ebp,esp
0x100729b9: sub esp,18h
0x100729bc: push ebx
0x100729bd: push esi
0x100729be: mov esi,[ebp+8]
0x100729c1: push edi
0x100729c2: mov ebx,100h
0x100729c7: push ebx
0x100729c8: lea edi,[esi+18h]
0x100729cb: push 0
```

```
0x100729cd: push edi
0x100729ce: call 100209F0h
0x100729d3: add edi,ebx
0x100729d5: add esp,0Ch
0x100729d8: mov [esi],edi
0x100729da: xor eax,eax
0x100729dc: mov [eax+edi],al
0x100729df: inc eax
0x100729e0: cmp eax,ebx
0x100729e2: jl short 100729DCh
0x100729e4: push 200h
0x100729e9: lea eax,[edi+100h]
0x100729ef: push 0FFh
0x100729f4: push eax
0x100729f5: call 100209F0h
0x100729fa: lea eax,[edi+300h]
0x10072a00: mov edi,[ebp+0Ch]
0x10072a03: fld dword ptr [edi]
0x10072a05: mov [esi+4],eax
0x10072a08: fld qword ptr ds:[100FF110h]
0x10072a0e: add eax,400h
0x10072a13: fmul st(1),st
0x10072a15: mov [esi+8],eax
0x10072a18: fld st(0)
0x10072a1a: add eax,400h
0x10072a1f: mov [esi+0Ch],eax
0x10072a22: fsubrp st(2),st
0x10072a24: add eax,400h
0x10072a29: fxch
0x10072a2b: mov [esi+10h],eax
0x10072a2e: add eax,400h
0x10072a33: fstp dword ptr [ebp+8]
0x10072a36: add esp,0Ch
0x10072a39: fld dword ptr [ebp+8]
0x10072a3c: mov [esi+14h],eax
0x10072a3f: fld qword ptr ds:[1010BBB8h]
0x10072a45: fmul st(1),st
0x10072a47: fld qword ptr ds:[100FE7C8h]
0x10072a4d: fadd st(2),st
0x10072a4f: fxch st(2)
0x10072a51: call 100228D0h
0x10072a56: fld dword ptr [edi]
0x10072a58: fmul dword ptr [ebp+8]
0x10072a5b: mov [ebp-0Ch],eax
0x10072a5e: fdiv dword ptr [edi+4]
0x10072a61: fstp dword ptr [ebp+8]
0x10072a64: fld dword ptr [ebp+8]
0x10072a67: fmul st,st(1)
0x10072a69: fadd st,st(2)
0x10072a6b: call 100228D0h
0x10072a70: fld dword ptr [edi+8]
0x10072a73: fmul st,st(3)
0x10072a75: neg eax
0x10072a77: mov [ebp-14h],eax
0x10072a7a: fsubp st(3),st
0x10072a7c: fxch st(2)
0x10072a7e: fstp dword ptr [ebp+8]
0x10072a81: fld dword ptr [ebp+8]
0x10072a84: fmul st,st(2)
0x10072a86: fadd st,st(1)
0x10072a88: call 100228D0h
0x10072a8d: fld dword ptr [edi+8]
0x10072a90: fmul dword ptr [ebp+8]
0x10072a93: mov [ebp-10h],eax
0x10072a96: fdiv dword ptr [edi+4]
0x10072a99: fstp dword ptr [ebp+8]
0x10072a9c: fld dword ptr [ebp+8]
```

```
0x10072a9f: fmulp st(2),st
0x10072aa1: faddp
0x10072aa3: call 100228D0h
0x10072aa8: fld qword ptr ds:[100FF370h]
0x10072aae: fldz
0x10072ab0: neg eax
0x10072ab2: fld1
0x10072ab4: mov [ebp-4],ebx
0x10072ab7: fld qword ptr ds:[1010BBB0h]
0x10072abd: mov [ebp-18h],eax
0x10072ac0: xor edi,edi
0x10072ac2: mov dword ptr [ebp+8],0FFFFFF80h
0x10072ac9: mov ebx,8000h
0x10072ace: mov eax,[ebp+10h]
0x10072ad1: fld dword ptr [eax+10h]
0x10072ad4: fsub st,st(4)
0x10072ad6: fld dword ptr [eax+14h]
0x10072ad9: fsub st,st(5)
0x10072adb: fsub st,st(1)
0x10072add: fld st(4)
0x10072adf: fucomp
0x10072ae1: fnstsw ax
0x10072ae3: test ah,44h
0x10072ae6: jp short 10072AECh
0x10072ae8: fstp st(0)
0x10072aea: fld st(2)
0x10072aec: fxch
0x10072aee: call 100228D0h
0x10072af3: mov ecx,[ebp+8]
0x10072af6: sub ecx,eax
0x10072af8: mov [ebp+0Ch],ecx
0x10072afb: fild dword ptr [ebp+0Ch]
0x10072afe: fmul st,st(2)
0x10072b00: fdivrp
0x10072b02: call 100228D0h
0x10072b07: mov [ebp+0Ch],eax
0x10072b0a: mov eax,[ebp+10h]
0x10072b0d: fld dword ptr [eax+8]
0x10072b10: fsub st,st(4)
0x10072b12: fld dword ptr [eax+0Ch]
0x10072b15: fsub st,st(5)
0x10072b17: fsub st,st(1)
0x10072b19: fld st(4)
0x10072b1b: fucomp
0x10072b1d: fnstsw ax
0x10072b1f: test ah,44h
0x10072b22: jp short 10072B28h
0x10072b24: fstp st(0)
0x10072b26: fld st(2)
0x10072b28: fxch
0x10072b2a: call 100228D0h
0x10072b2f: mov ecx,[ebp+8]
0x10072b32: sub ecx,eax
0x10072b34: mov [ebp-8],ecx
0x10072b37: fild dword ptr [ebp-8]
0x10072b3a: fmul st,st(2)
0x10072b3c: fdivrp
0x10072b3e: call 100228D0h
0x10072b43: mov ecx,[ebp+0Ch]
0x10072b46: imul ecx,[ebp-0Ch]
0x10072b4a: mov edx,[esi+4]
0x10072b4d: add ecx,ebx
0x10072b4f: sar ecx,10h
0x10072b52: mov [edi+edx],ecx
0x10072b55: mov edx,[esi+8]
0x10072b58: mov ecx,eax
0x10072b5a: imul eax,[ebp-18h]
```

```
0x10072b5e: imul ecx,[ebp-10h]
0x10072b62: add ecx,ebx
0x10072b64: sar ecx,10h
0x10072b67: mov [edi+edx],ecx
0x10072b6a: mov ecx,[ebp+0Ch]
0x10072b6d: imul ecx,[ebp-14h]
0x10072b71: mov edx,[esi+0Ch]
0x10072b74: mov [edi+edx],ecx
0x10072b77: mov ecx,[esi+10h]
0x10072b7a: add eax,ebx
0x10072b7c: mov [edi+ecx],eax
0x10072b7f: mov ecx,[ebp+10h]
0x10072b82: fld dword ptr [ecx+4]
0x10072b85: fsub dword ptr [ecx]
0x10072b87: fld st(3)
0x10072b89: fucomp
0x10072b8b: fnstsw ax
0x10072b8d: test ah,44h
0x10072b90: jp short 10072B96h
0x10072b92: fstp st(0)
0x10072b94: fld st(1)
0x10072b96: fld dword ptr [ecx]
0x10072b98: call 100228D0h
0x10072b9d: mov ecx,[ebp+8]
0x10072ba0: sub ecx,eax
0x10072ba2: add ecx,80h
0x10072ba8: mov [ebp+0Ch],ecx
0x10072bab: fild dword ptr [ebp+0Ch]
0x10072bae: fmul qword ptr ds:[100FE918h]
0x10072bb4: fdivrp
0x10072bb6: call 100228D0h
0x10072bbb: mov ecx,[esi+14h]
0x10072bbe: inc dword ptr [ebp+8]
0x10072bc1: mov [edi+ecx],eax
0x10072bc4: add edi,4
0x10072bc7: dec dword ptr [ebp-4]
0x10072bca: jne near ptr 10072ACEh
0x10072bd0: fstp st(3)
0x10072bd2: pop edi
0x10072bd3: fstp st(1)
0x10072bd5: pop esi
0x10072bd6: fstp st(1)
0x10072bd8: xor eax,eax
0x10072bda: fstp st(0)
0x10072bdc: pop ebx
0x10072bdd: leave
0x10072bde: ret
```

`sub_100729b6` is called by sub_1006cec2. It delegates to j_memset, sub_100228d0 (2 callees).

12.1058 sub_1007f1a0 at 0x1007f1a0 (55 instructions)

```
0x1007f1a0: mov edx,[esp+4]
0x1007f1a4: test edx,edx
0x1007f1a6: je near ptr 1007F250h
0x1007f1ac: cmp dword ptr [edx],52530A0Ah
0x1007f1b2: jne near ptr 1007F250h
0x1007f1b8: mov eax,[esp+8]
0x1007f1bc: cmp eax,1000h
0x1007f1c1: ja short 1007F1D8h
0x1007f1c3: je short 1007F20Ch
0x1007f1c5: cmp eax,8
0x1007f1c8: ja short 1007F238h
0x1007f1ca: movzx ecx,byte ptr [eax+1007F260h]
0x1007f1d1: jmp dword ptr [ecx*4+1007F258h]
0x1007f1d8: cmp eax,5000h
```

```
0x1007f1dd: ja short 1007F212h
0x1007f1df: je short 1007F20Ch
0x1007f1e1: cmp eax,3000h
0x1007f1e6: ja short 1007F1FEh
0x1007f1e8: je short 1007F20Ch
0x1007f1ea: mov ecx,eax
0x1007f1ec: sub ecx,1001h
0x1007f1f2: je short 1007F20Ch
0x1007f1f4: sub ecx,4
0x1007f1f7: je short 1007F20Ch
0x1007f1f9: sub ecx,1
0x1007f1fc: jmp short 1007F236h
0x1007f1fe: cmp eax,3001h
0x1007f203: je short 1007F20Ch
0x1007f205: cmp eax,4006h
0x1007f20a: jne short 1007F238h
0x1007f20c: mov [edx+34h],eax
0x1007f20f: xor eax,eax
0x1007f211: ret
0x1007f212: cmp eax,0B000h
0x1007f217: ja short 1007F22Ah
0x1007f219: je short 1007F20Ch
0x1007f21b: mov ecx,eax
0x1007f21d: sub ecx,9000h
0x1007f223: je short 1007F20Ch
0x1007f225: sub ecx,1
0x1007f228: jmp short 1007F236h
0x1007f22a: cmp eax,0B001h
0x1007f22f: je short 1007F20Ch
0x1007f231: cmp eax,10001h
0x1007f236: je short 1007F20Ch
0x1007f238: push 0
0x1007f23a: push 0
0x1007f23c: push 801h
0x1007f241: push edx
0x1007f242: call 10080800h
0x1007f247: add esp,10h
0x1007f24a: mov eax,801h
0x1007f24f: ret
0x1007f250: mov eax,2
0x1007f255: ret
```

`sub_1007f1a0` is called by sub_10018540. It accesses memory-mapped peripheral registers at 0x52530a0a. It delegates to sub_10080800 (1 callees).

12.1059 sub_1007f270 at 0x1007f270 (16 instructions)

```
0x1007f270: mov eax,[esp+4]
0x1007f274: test eax,eax
0x1007f276: je short 1007F2A2h
0x1007f278: cmp dword ptr [eax],52530A0Ah
0x1007f27e: jne short 1007F2A2h
0x1007f280: mov cx,[esp+8]
0x1007f285: mov dx,[esp+0Ch]
0x1007f28a: mov [eax+50h],cx
0x1007f28e: mov cx,[esp+10h]
0x1007f293: mov [eax+52h],dx
0x1007f297: mov [eax+54h],cx
0x1007f29b: mov byte ptr [eax+56h],0
0x1007f29f: xor eax,eax
0x1007f2a1: ret
0x1007f2a2: mov eax,2
0x1007f2a7: ret
```

`sub_1007f270` is called by sub_100187f0. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1060 sub_1007f2b0 at 0x1007f2b0 (11 instructions)

```
0x1007f2b0: mov eax,[esp+4]
0x1007f2b4: test eax,eax
0x1007f2b6: je short 1007F2CAh
0x1007f2b8: cmp dword ptr [eax],52530A0Ah
0x1007f2be: jne short 1007F2CAh
0x1007f2c0: mov cl,[esp+8]
0x1007f2c4: mov [eax+56h],cl
0x1007f2c7: xor eax,eax
0x1007f2c9: ret
0x1007f2ca: mov eax,2
0x1007f2cf: ret
```

`sub_1007f2b0` is called by sub_100187f0. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1061 sub_1007f2d0 at 0x1007f2d0 (11 instructions)

```
0x1007f2d0: mov eax,[esp+4]
0x1007f2d4: test eax,eax
0x1007f2d6: je short 1007F2EAh
0x1007f2d8: cmp dword ptr [eax],52530A0Ah
0x1007f2de: jne short 1007F2EAh
0x1007f2e0: mov cl,[esp+8]
0x1007f2e4: mov [eax+70h],cl
0x1007f2e7: xor eax,eax
0x1007f2e9: ret
0x1007f2ea: mov eax,2
0x1007f2ef: ret
```

`sub_1007f2d0` is called by sub_100187f0. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1062 sub_1007f2f0 at 0x1007f2f0 (11 instructions)

```
0x1007f2f0: mov eax,[esp+4]
0x1007f2f4: test eax,eax
0x1007f2f6: je short 1007F30Ah
0x1007f2f8: cmp dword ptr [eax],52530A0Ah
0x1007f2fe: jne short 1007F30Ah
0x1007f300: mov cl,[esp+8]
0x1007f304: mov [eax+57h],cl
0x1007f307: xor eax,eax
0x1007f309: ret
0x1007f30a: mov eax,2
0x1007f30f: ret
```

`sub_1007f2f0` is called by sub_100187f0. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1063 sub_1007f310 at 0x1007f310 (21 instructions)

```
0x1007f310: mov eax,[esp+4]
0x1007f314: test eax,eax
0x1007f316: je short 1007F34Bh
0x1007f318: cmp dword ptr [eax],52530A0Ah
0x1007f31e: jne short 1007F34Bh
0x1007f320: cmp byte ptr [eax+158h],0
0x1007f327: je short 1007F33Eh
0x1007f329: push 0
0x1007f32b: push 0
0x1007f32d: push 0Bh
0x1007f32f: push eax
```

```
0x1007f330: call 10080800h
0x1007f335: add esp,10h
0x1007f338: mov eax,0Bh
0x1007f33d: ret
0x1007f33e: mov cl,[esp+8]
0x1007f342: mov [eax+172h],cl
0x1007f348: xor eax,eax
0x1007f34a: ret
0x1007f34b: mov eax,2
0x1007f350: ret
```

`sub_1007f310` is called by sub_100187f0. It accesses memory-mapped peripheral registers at 0x52530a0a. It delegates to sub_10080800 (1 callees).

12.1064 sub_1007f360 at 0x1007f360 (9 instructions)

```
0x1007f360: mov eax,[esp+4]
0x1007f364: test eax,eax
0x1007f366: je short 1007F374h
0x1007f368: cmp dword ptr [eax],52530A0Ah
0x1007f36e: jne short 1007F374h
0x1007f370: mov eax,[eax+4]
0x1007f373: ret
0x1007f374: xor eax,eax
0x1007f376: ret
```

`sub_1007f360` is called by sub_10018510, sub_10018540, sub_100185d0, sub_10018600, sub_10018640. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1065 sub_1007f380 at 0x1007f380 (39 instructions)

```
0x1007f380: mov eax,[esi+2B0h]
0x1007f386: push ebx
0x1007f387: xor ebx,ebx
0x1007f389: cmp eax,ebx
0x1007f38b: push edi
0x1007f38c: je short 1007F3A2h
0x1007f38e: mov edi,edi
0x1007f390: mov edi,[eax+8]
0x1007f393: push eax
0x1007f394: mov eax,[eax]
0x1007f396: push esi
0x1007f397: call eax
0x1007f399: add esp,8
0x1007f39c: cmp edi,ebx
0x1007f39e: mov eax,edi
0x1007f3a0: jne short 1007F390h
0x1007f3a2: cmp [esp+0Ch],bl
0x1007f3a6: mov [esi+2B0h],ebx
0x1007f3ac: mov [esi+2B4h],ebx
0x1007f3b2: je short 1007F3DEh
0x1007f3b4: mov eax,[esi+2B8h]
0x1007f3ba: cmp eax,ebx
0x1007f3bc: je short 1007F3D2h
0x1007f3be: mov edi,edi
0x1007f3c0: mov ecx,[eax]
0x1007f3c2: mov edi,[eax+8]
0x1007f3c5: push eax
0x1007f3c6: push esi
0x1007f3c7: call ecx
0x1007f3c9: add esp,8
0x1007f3cc: cmp edi,ebx
0x1007f3ce: mov eax,edi
0x1007f3d0: jne short 1007F3C0h
0x1007f3d2: mov [esi+2B8h],ebx
```

```
0x1007f3d8: mov [esi+2BCh],ebx
0x1007f3de: pop edi
0x1007f3df: xor eax,eax
0x1007f3e1: pop ebx
0x1007f3e2: ret
```

`sub_1007f380` is called by sub_1007f450, sub_1007f790. It calls nothing else.

12.1066 sub_1007f3f0 at 0x1007f3f0 (25 instructions)

```
0x1007f3f0: mov eax,[esi+224h]
0x1007f3f6: test eax,eax
0x1007f3f8: je short 1007F440h
0x1007f3fa: mov ecx,[eax+460h]
0x1007f400: test ecx,ecx
0x1007f402: je short 1007F41Bh
0x1007f404: mov eax,[eax+464h]
0x1007f40a: test eax,eax
0x1007f40c: je short 1007F41Bh
0x1007f40e: push ecx
0x1007f40f: push eax
0x1007f410: mov eax,[esi+9Ch]
0x1007f416: call eax
0x1007f418: add esp,8
0x1007f41b: mov eax,[esi+224h]
0x1007f421: test eax,eax
0x1007f423: je short 1007F440h
0x1007f425: mov ecx,[esi+9Ch]
0x1007f42b: push 470h
0x1007f430: push eax
0x1007f431: call ecx
0x1007f433: add esp,8
0x1007f436: mov dword ptr [esi+224h],0
0x1007f440: xor eax,eax
0x1007f442: ret
```

`sub_1007f3f0` is called by sub_1007f450, sub_1007f790. It calls nothing else.

12.1067 sub_1007f450 at 0x1007f450 (142 instructions)

```
0x1007f450: push ebx
0x1007f451: push esi
0x1007f452: mov esi,eax
0x1007f454: push esi
0x1007f455: call 10086E70h
0x1007f45a: call 1007F3F0h
0x1007f45f: push esi
0x1007f460: call 10082DF0h
0x1007f465: xor ebx,ebx
0x1007f467: push ebx
0x1007f468: call 1007F380h
0x1007f46d: mov eax,1
0x1007f472: mov ecx,7FFFFFFFh
0x1007f477: add esp,0Ch
0x1007f47a: mov [esi+1B1h],bl
0x1007f480: mov [esi+1B4h],ebx
0x1007f486: mov [esi+1B8h],ebx
0x1007f48c: mov [esi+1BCh],ebx
0x1007f492: mov [esi+1CCh],bl
0x1007f498: mov [esi+1D0h],ebx
0x1007f49e: mov [esi+1D4h],ebx
0x1007f4a4: mov [esi+1D8h],ebx
0x1007f4aa: mov [esi+1DCh],bl
0x1007f4b0: mov [esi+1E0h],ebx
0x1007f4b6: mov [esi+1E4h],ebx
```

```
0x1007f4bc: mov [esi+1E8h],ebx
0x1007f4c2: mov [esi+1ECh],ebx
0x1007f4c8: mov [esi+1F0h],bl
0x1007f4ce: mov [esi+1F1h],bl
0x1007f4d4: mov [esi+1F2h],bl
0x1007f4da: mov [esi+1F3h],bl
0x1007f4e0: mov [esi+1F4h],bl
0x1007f4e6: mov [esi+1F5h],bl
0x1007f4ec: mov [esi+1F6h],bl
0x1007f4f2: mov [esi+1F8h],bl
0x1007f4f8: mov [esi+1FCh],ebx
0x1007f4fe: mov [esi+20Ch],ebx
0x1007f504: mov [esi+204h],ebx
0x1007f50a: mov [esi+228h],ebx
0x1007f510: mov [esi+22Ch],ebx
0x1007f516: mov [esi+230h],ebx
0x1007f51c: mov [esi+234h],ebx
0x1007f522: mov [esi+14Ch],ebx
0x1007f528: mov [esi+3F8h],bx
0x1007f52f: mov [esi+3FAh],bl
0x1007f535: mov [esi+3FBh],bl
0x1007f53b: mov [esi+3FCh],bl
0x1007f541: mov [esi+3FDh],bl
0x1007f547: mov [esi+3FEh],bl
0x1007f54d: mov [esi+400h],ebx
0x1007f553: mov [esi+404h],ebx
0x1007f559: mov [esi+408h],bl
0x1007f55f: mov [esi+40Ch],ebx
0x1007f565: mov [esi+410h],ebx
0x1007f56b: mov [esi+414h],ebx
0x1007f571: mov [esi+418h],ebx
0x1007f577: mov [esi+41Ch],bx
0x1007f57e: mov [esi+41Eh],bx
0x1007f585: mov [esi+420h],bx
0x1007f58c: mov [esi+422h],bl
0x1007f592: mov [esi+424h],bx
0x1007f599: mov [esi+426h],bl
0x1007f59f: mov [esi+438h],al
0x1007f5a5: mov [esi+43Ch],eax
0x1007f5ab: mov [esi+440h],ecx
0x1007f5b1: mov [esi+444h],bl
0x1007f5b7: mov [esi+448h],ebx
0x1007f5bd: mov [esi+44Ch],ebx
0x1007f5c3: mov [esi+450h],ebx
0x1007f5c9: mov [esi+454h],ebx
0x1007f5cf: mov [esi+458h],al
0x1007f5d5: mov [esi+45Ch],eax
0x1007f5db: mov [esi+460h],ecx
0x1007f5e1: mov [esi+464h],bl
0x1007f5e7: mov [esi+468h],ebx
0x1007f5ed: mov [esi+46Ch],ebx
0x1007f5f3: mov [esi+470h],ebx
0x1007f5f9: mov [esi+474h],ebx
0x1007f5ff: mov [esi+478h],eax
0x1007f605: mov [esi+47Ch],bl
0x1007f60b: mov [esi+47Eh],bx
0x1007f612: mov eax,[esi+8B8h]
0x1007f618: cmp eax,ebx
0x1007f61a: mov [esi+480h],bx
0x1007f621: mov [esi+482h],bx
0x1007f628: mov [esi+484h],bx
0x1007f62f: mov [esi+488h],ebx
0x1007f635: mov [esi+78Ch],ebx
0x1007f63b: mov [esi+890h],ebx
0x1007f641: mov [esi+894h],ebx
0x1007f647: mov [esi+898h],ebx
0x1007f64d: mov [esi+89Ch],ebx
```

```
0x1007f653: mov [esi+8A0h],ebx
0x1007f659: mov [esi+8A4h],ebx
0x1007f65f: mov [esi+8A8h],ebx
0x1007f665: mov [esi+8ACh],ebx
0x1007f66b: mov [esi+8B0h],ebx
0x1007f671: mov [esi+8B4h],bl
0x1007f677: je short 1007F696h
0x1007f679: mov ecx,[esi+8BCh]
0x1007f67f: cmp ecx,ebx
0x1007f681: je short 1007F696h
0x1007f683: push eax
0x1007f684: mov eax,[esi+9Ch]
0x1007f68a: push ecx
0x1007f68b: call eax
0x1007f68d: add esp,8
0x1007f690: mov [esi+8BCh],ebx
0x1007f696: mov [esi+8B8h],ebx
0x1007f69c: mov [esi+8C0h],bx
0x1007f6a3: mov [esi+8C2h],bx
0x1007f6aa: mov [esi+8C4h],bx
0x1007f6b1: mov [esi+8C8h],ebx
0x1007f6b7: mov [esi+8CCh],bl
0x1007f6bd: mov [esi+8CDh],bl
0x1007f6c3: mov [esi+8D0h],ebx
0x1007f6c9: mov [esi+8D4h],ebx
0x1007f6cf: mov [esi+8D8h],ebx
0x1007f6d5: mov [esi+8DCh],ebx
0x1007f6db: mov [esi+8E0h],bl
0x1007f6e1: mov [esi+8E4h],ebx
0x1007f6e7: mov [esi+8E8h],ebx
0x1007f6ed: mov [esi+8ECh],ebx
0x1007f6f3: mov [esi+8F0h],ebx
0x1007f6f9: mov [esi+8F4h],ebx
0x1007f6ff: mov [esi+8F8h],ebx
0x1007f705: mov [esi+8FCh],ebx
0x1007f70b: mov [esi+900h],bl
0x1007f711: mov [esi+901h],bl
0x1007f717: mov [esi+902h],bl
0x1007f71d: mov [esi+904h],ebx
0x1007f723: mov [esi+908h],ebx
0x1007f729: mov [esi+90Ch],ebx
0x1007f72f: mov [esi+910h],bx
0x1007f736: mov [esi+914h],bx
0x1007f73d: mov [esi+918h],ebx
0x1007f743: mov [esi+91Ch],ebx
0x1007f749: mov [esi+920h],ebx
0x1007f74f: pop esi
0x1007f750: xor eax,eax
0x1007f752: pop ebx
0x1007f753: ret
```

`sub_1007f450` is called by sub_10080150. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to sub_1007f380, sub_1007f3f0, sub_10082df0 (4 callees).

12.1068 sub_1007f760 at 0x1007f760 (7 instructions)

```
0x1007f760: xor ecx,ecx
0x1007f762: mov [eax+84h],ecx
0x1007f768: mov [eax+88h],cl
0x1007f76e: mov [eax+8Ch],ecx
0x1007f774: mov [eax+90h],ecx
0x1007f77a: mov [eax+94h],ecx
0x1007f780: ret
```

`sub_1007f760` is called by sub_1007ff70, sub_10080150. It calls nothing else.

12.1069 sub_1007f790 at 0x1007f790 (373 instructions)

```
0x1007f790: push ebx
0x1007f791: push esi
0x1007f792: mov esi,[esp+0Ch]
0x1007f796: xor ebx,ebx
0x1007f798: cmp esi,ebx
0x1007f79a: je near ptr 1007FEFEh
0x1007f7a0: cmp dword ptr [esi],52530A0Ah
0x1007f7a6: jne near ptr 1007FEFEh
0x1007f7ac: call 1007F3F0h
0x1007f7b1: push esi
0x1007f7b2: call 100A0020h
0x1007f7b7: add esp,4
0x1007f7ba: cmp [esi+974h],bl
0x1007f7c0: je short 1007F7D1h
0x1007f7c2: push esi
0x1007f7c3: call 10098B40h
0x1007f7c8: push esi
0x1007f7c9: call 1008D0A0h
0x1007f7ce: add esp,8
0x1007f7d1: cmp [esi+158h],bl
0x1007f7d7: je short 1007F7EAh
0x1007f7d9: cmp [esi+15Ah],bl
0x1007f7df: jne short 1007F7EAh
0x1007f7e1: push esi
0x1007f7e2: call 1008AD30h
0x1007f7e7: add esp,4
0x1007f7ea: mov eax,[esi+160h]
0x1007f7f0: cmp eax,ebx
0x1007f7f2: je short 1007F80Dh
0x1007f7f4: mov ecx,[esi+15Ch]
0x1007f7fa: mov edx,[esi+9Ch]
0x1007f800: push ecx
0x1007f801: push eax
0x1007f802: call edx
0x1007f804: add esp,8
0x1007f807: mov [esi+160h],ebx
0x1007f80d: mov eax,[esi+168h]
0x1007f813: cmp eax,ebx
0x1007f815: je short 1007F830h
0x1007f817: mov ecx,[esi+164h]
0x1007f81d: mov edx,[esi+9Ch]
0x1007f823: push ecx
0x1007f824: push eax
0x1007f825: call edx
0x1007f827: add esp,8
0x1007f82a: mov [esi+168h],ebx
0x1007f830: mov eax,[esi+178h]
0x1007f836: cmp eax,ebx
0x1007f838: je short 1007F853h
0x1007f83a: mov ecx,[esi+174h]
0x1007f840: mov edx,[esi+9Ch]
0x1007f846: push ecx
0x1007f847: push eax
0x1007f848: call edx
0x1007f84a: add esp,8
0x1007f84d: mov [esi+178h],ebx
0x1007f853: cmp [esi+198h],ebx
0x1007f859: je short 1007F871h
0x1007f85b: jmp short 1007F860h
0x1007f860: push esi
0x1007f861: call 1008AD80h
0x1007f866: add esp,4
0x1007f869: cmp [esi+198h],ebx
0x1007f86f: jne short 1007F860h
0x1007f871: cmp [esi+190h],ebx
0x1007f877: je short 1007F891h
```

```
0x1007f879: lea esp,[esp]
0x1007f880: push esi
0x1007f881: call 1008AE00h
0x1007f886: add esp,4
0x1007f889: cmp [esi+190h],ebx
0x1007f88f: jne short 1007F880h
0x1007f891: mov eax,[esi+1ACh]
0x1007f897: cmp eax,ebx
0x1007f899: je short 1007F8B4h
0x1007f89b: mov ecx,[esi+1A8h]
0x1007f8a1: mov edx,[esi+9Ch]
0x1007f8a7: push ecx
0x1007f8a8: push eax
0x1007f8a9: call edx
0x1007f8ab: add esp,8
0x1007f8ae: mov [esi+1ACh],ebx
0x1007f8b4: push edi
0x1007f8b5: push esi
0x1007f8b6: call 1008ACF0h
0x1007f8bb: mov edi,1
0x1007f8c0: push edi
0x1007f8c1: call 1007F380h
0x1007f8c6: mov eax,[esi+8B8h]
0x1007f8cc: add esp,8
0x1007f8cf: cmp eax,ebx
0x1007f8d1: je short 1007F8EAh
0x1007f8d3: mov ecx,[esi+8BCh]
0x1007f8d9: cmp ecx,ebx
0x1007f8db: je short 1007F8EAh
0x1007f8dd: push eax
0x1007f8de: mov eax,[esi+9Ch]
0x1007f8e4: push ecx
0x1007f8e5: call eax
0x1007f8e7: add esp,8
0x1007f8ea: mov ecx,[esi+178h]
0x1007f8f0: mov [esi+8],ebx
0x1007f8f3: mov [esi+0Ch],ebx
0x1007f8f6: mov [esi+10h],ebx
0x1007f8f9: mov [esi+14h],ebx
0x1007f8fc: mov [esi+18h],ebx
0x1007f8ff: mov [esi+1Ch],ebx
0x1007f902: mov [esi+20h],ebx
0x1007f905: mov [esi+24h],ebx
0x1007f908: mov [esi+28h],ebx
0x1007f90b: mov byte ptr [esi+2Ch],10h
0x1007f90f: mov [esi+30h],ebx
0x1007f912: mov [esi+3Ch],bl
0x1007f915: mov [esi+40h],ebx
0x1007f918: mov [esi+44h],ebx
0x1007f91b: mov [esi+48h],ebx
0x1007f91e: mov [esi+4Ch],ebx
0x1007f921: mov [esi+84h],ebx
0x1007f927: mov [esi+88h],bl
0x1007f92d: mov [esi+8Ch],ebx
0x1007f933: mov [esi+90h],ebx
0x1007f939: mov [esi+94h],ebx
0x1007f93f: mov [esi+108h],bl
0x1007f945: mov dword ptr [esi+10Ch],40404040h
0x1007f94f: mov [esi+110h],ebx
0x1007f955: mov [esi+114h],ebx
0x1007f95b: mov [esi+118h],ebx
0x1007f961: mov [esi+11Ch],bl
0x1007f967: mov [esi+11Dh],bl
0x1007f96d: mov [esi+11Eh],bl
0x1007f973: mov [esi+11Fh],bl
0x1007f979: mov [esi+120h],bl
0x1007f97f: mov [esi+121h],bl
```

```
0x1007f985: mov [esi+122h],bl
0x1007f98b: mov [esi+123h],bl
0x1007f991: mov [esi+124h],bl
0x1007f997: mov [esi+125h],bl
0x1007f99d: mov [esi+126h],bl
0x1007f9a3: mov [esi+127h],bl
0x1007f9a9: mov [esi+128h],bl
0x1007f9af: mov [esi+129h],bl
0x1007f9b5: mov [esi+12Ah],bl
0x1007f9bb: mov [esi+12Bh],bl
0x1007f9c1: mov [esi+12Ch],bl
0x1007f9c7: mov [esi+12Dh],bl
0x1007f9cd: mov [esi+12Eh],bl
0x1007f9d3: mov [esi+12Fh],bl
0x1007f9d9: mov [esi+130h],bl
0x1007f9df: mov [esi+131h],bl
0x1007f9e5: mov [esi+132h],bl
0x1007f9eb: mov [esi+133h],bl
0x1007f9f1: mov [esi+134h],bl
0x1007f9f7: mov [esi+135h],bl
0x1007f9fd: mov [esi+136h],bl
0x1007fa03: mov [esi+137h],bl
0x1007fa09: mov [esi+13Ch],ebx
0x1007fa0f: mov [esi+140h],ebx
0x1007fa15: mov [esi+144h],bl
0x1007fa1b: mov [esi+145h],bl
0x1007fa21: mov [esi+146h],bl
0x1007fa27: mov [esi+147h],bl
0x1007fa2d: mov [esi+148h],bl
0x1007fa33: mov [esi+150h],bl
0x1007fa39: mov [esi+151h],bl
0x1007fa3f: mov [esi+152h],bl
0x1007fa45: mov [esi+153h],bl
0x1007fa4b: mov [esi+154h],bl
0x1007fa51: mov [esi+155h],bl
0x1007fa57: mov [esi+156h],bl
0x1007fa5d: mov [esi+157h],bl
0x1007fa63: mov [esi+158h],bl
0x1007fa69: mov [esi+159h],bl
0x1007fa6f: mov [esi+15Ah],bl
0x1007fa75: mov [esi+15Ch],ebx
0x1007fa7b: mov [esi+160h],ebx
0x1007fa81: mov [esi+164h],ebx
0x1007fa87: mov [esi+168h],ebx
0x1007fa8d: mov [esi+16Ch],ebx
0x1007fa93: mov [esi+170h],bl
0x1007fa99: mov [esi+171h],bl
0x1007fa9f: mov [esi+17Ch],ecx
0x1007faa5: mov [esi+180h],ebx
0x1007faab: mov [esi+1A0h],bl
0x1007fab1: mov [esi+1A1h],bl
0x1007fab7: mov [esi+1A4h],ebx
0x1007fabd: mov [esi+1A8h],ebx
0x1007fac3: mov [esi+1ACh],ebx
0x1007fac9: mov [esi+1B0h],bl
0x1007facf: mov [esi+1B4h],ebx
0x1007fad5: mov [esi+1B8h],ebx
0x1007fadb: mov [esi+1BCh],ebx
0x1007fae1: mov [esi+1C4h],ebx
0x1007fae7: mov [esi+1C0h],ebx
0x1007faed: mov [esi+1C8h],ebx
0x1007faf3: mov [esi+1CCh],bl
0x1007faf9: mov [esi+1D0h],ebx
0x1007faff: mov [esi+1D4h],ebx
0x1007fb05: mov [esi+1D8h],ebx
0x1007fb0b: mov [esi+1DCh],bl
0x1007fb11: mov [esi+1DDh],bl
```

```
0x1007fb17: mov [esi+1E0h],ebx
0x1007fb1d: mov [esi+1E4h],ebx
0x1007fb23: mov [esi+1E8h],ebx
0x1007fb29: mov [esi+1ECh],ebx
0x1007fb2f: mov [esi+1F0h],bl
0x1007fb35: mov [esi+1F1h],bl
0x1007fb3b: mov [esi+1F2h],bl
0x1007fb41: mov [esi+1F3h],bl
0x1007fb47: mov [esi+1F4h],bl
0x1007fb4d: mov [esi+1F5h],bl
0x1007fb53: mov [esi+1F6h],bl
0x1007fb59: mov [esi+1F7h],bl
0x1007fb5f: mov [esi+1F8h],bl
0x1007fb65: mov [esi+1FCh],ebx
0x1007fb6b: mov [esi+200h],ebx
0x1007fb71: mov [esi+204h],ebx
0x1007fb77: mov [esi+208h],ebx
0x1007fb7d: mov [esi+214h],ebx
0x1007fb83: mov [esi+218h],ebx
0x1007fb89: mov [esi+21Ch],ebx
0x1007fb8f: mov [esi+220h],ebx
0x1007fb95: mov [esi+224h],ebx
0x1007fb9b: mov [esi+228h],ebx
0x1007fba1: mov [esi+22Ch],ebx
0x1007fba7: mov [esi+230h],ebx
0x1007fbad: mov [esi+234h],ebx
0x1007fbb3: mov byte ptr [esi+238h],0FFh
0x1007fbba: mov [esi+23Ch],ebx
0x1007fbc0: mov [esi+240h],edi
0x1007fbc6: mov [esi+244h],ebx
0x1007fbcc: mov [esi+248h],edi
0x1007fbd2: mov [esi+24Ch],ebx
0x1007fbd8: mov [esi+250h],ebx
0x1007fbde: mov [esi+254h],ebx
0x1007fbe4: mov [esi+258h],ebx
0x1007fbea: mov [esi+25Ch],ebx
0x1007fbf0: mov [esi+260h],ebx
0x1007fbf6: mov [esi+264h],ebx
0x1007fbfc: mov [esi+268h],edi
0x1007fc02: mov [esi+26Ch],ebx
0x1007fc08: mov [esi+270h],ebx
0x1007fc0e: mov [esi+274h],ebx
0x1007fc14: mov [esi+278h],ebx
0x1007fc1a: mov [esi+27Ch],ebx
0x1007fc20: mov [esi+280h],ebx
0x1007fc26: mov [esi+284h],ebx
0x1007fc2c: mov byte ptr [esi+288h],1
0x1007fc33: mov byte ptr [esi+289h],1
0x1007fc3a: mov [esi+28Ch],edi
0x1007fc40: mov [esi+290h],ebx
0x1007fc46: mov [esi+294h],ebx
0x1007fc4c: mov [esi+298h],ebx
0x1007fc52: mov [esi+29Ch],ebx
0x1007fc58: mov [esi+2A0h],ebx
0x1007fc5e: mov [esi+2A4h],ebx
0x1007fc64: mov [esi+2A8h],ebx
0x1007fc6a: mov [esi+2ACh],ebx
0x1007fc70: mov [esi+2B0h],ebx
0x1007fc76: mov [esi+2B4h],ebx
0x1007fc7c: mov [esi+2B8h],ebx
0x1007fc82: mov [esi+2BCh],ebx
0x1007fc88: mov [esi+3C8h],ebx
0x1007fc8e: mov [esi+3CCh],ebx
0x1007fc94: mov [esi+3D0h],ebx
0x1007fc9a: mov [esi+3D4h],ebx
0x1007fca0: mov [esi+3D8h],ebx
0x1007fca6: mov [esi+3DCh],ebx
```

```
0x1007fcac: mov [esi+3E0h],ebx
0x1007fcb2: mov [esi+3E4h],ebx
0x1007fcb8: mov [esi+3E8h],ebx
0x1007fcbe: mov [esi+3ECh],ebx
0x1007fcc4: mov [esi+3F0h],ebx
0x1007fcca: mov [esi+3F4h],ebx
0x1007fcd0: mov [esi+14Ch],ebx
0x1007fcd6: mov [esi+3F8h],bx
0x1007fcdd: mov [esi+3FAh],bl
0x1007fce3: mov [esi+3FBh],bl
0x1007fce9: mov [esi+3FCh],bl
0x1007fcef: mov [esi+3FDh],bl
0x1007fcf5: mov [esi+3FEh],bl
0x1007fcfb: mov [esi+400h],ebx
0x1007fd01: mov [esi+404h],ebx
0x1007fd07: mov [esi+408h],bl
0x1007fd0d: mov [esi+40Ch],ebx
0x1007fd13: mov [esi+410h],ebx
0x1007fd19: mov [esi+414h],ebx
0x1007fd1f: mov [esi+418h],ebx
0x1007fd25: mov [esi+41Ch],bx
0x1007fd2c: mov [esi+41Eh],bx
0x1007fd33: mov [esi+420h],bx
0x1007fd3a: mov [esi+422h],bl
0x1007fd40: mov [esi+424h],bx
0x1007fd47: mov [esi+426h],bl
0x1007fd4d: mov byte ptr [esi+438h],1
0x1007fd54: mov [esi+43Ch],edi
0x1007fd5a: mov eax,7FFFFFFFh
0x1007fd5f: mov [esi+440h],eax
0x1007fd65: mov [esi+444h],bl
0x1007fd6b: mov [esi+448h],ebx
0x1007fd71: mov [esi+44Ch],ebx
0x1007fd77: mov [esi+450h],ebx
0x1007fd7d: mov [esi+454h],ebx
0x1007fd83: mov byte ptr [esi+458h],1
0x1007fd8a: mov [esi+45Ch],edi
0x1007fd90: mov [esi+460h],eax
0x1007fd96: mov [esi+464h],bl
0x1007fd9c: mov [esi+468h],ebx
0x1007fda2: mov [esi+46Ch],ebx
0x1007fda8: mov [esi+470h],ebx
0x1007fdae: mov [esi+474h],ebx
0x1007fdb4: mov [esi+478h],edi
0x1007fdba: mov [esi+47Ch],bl
0x1007fdc0: mov [esi+47Eh],bx
0x1007fdc7: mov [esi+480h],bx
0x1007fdce: mov [esi+482h],bx
0x1007fdd5: mov [esi+484h],bx
0x1007fddc: mov [esi+488h],ebx
0x1007fde2: mov [esi+78Ch],ebx
0x1007fde8: mov [esi+890h],ebx
0x1007fdee: mov [esi+894h],ebx
0x1007fdf4: mov [esi+898h],ebx
0x1007fdfa: mov [esi+89Ch],ebx
0x1007fe00: mov [esi+8A0h],ebx
0x1007fe06: mov [esi+8A4h],ebx
0x1007fe0c: mov [esi+8A8h],ebx
0x1007fe12: mov [esi+8ACh],ebx
0x1007fe18: mov [esi+8B0h],ebx
0x1007fe1e: mov [esi+8B4h],bl
0x1007fe24: mov [esi+8B8h],ebx
0x1007fe2a: mov [esi+8BCh],ebx
0x1007fe30: mov [esi+8C0h],bx
0x1007fe37: mov [esi+8C2h],bx
0x1007fe3e: mov [esi+8C4h],bx
0x1007fe45: mov [esi+8C8h],ebx
```

```
0x1007fe4b: mov [esi+8CCh],bl
0x1007fe51: mov [esi+8CDh],bl
0x1007fe57: mov [esi+8D0h],ebx
0x1007fe5d: mov [esi+8D4h],ebx
0x1007fe63: mov [esi+8D8h],ebx
0x1007fe69: mov [esi+8DCh],ebx
0x1007fe6f: mov [esi+8E0h],bl
0x1007fe75: mov [esi+8E4h],ebx
0x1007fe7b: mov [esi+8E8h],ebx
0x1007fe81: mov [esi+8ECh],ebx
0x1007fe87: mov [esi+8F0h],ebx
0x1007fe8d: mov [esi+8F4h],ebx
0x1007fe93: mov [esi+8F8h],ebx
0x1007fe99: mov [esi+8FCh],ebx
0x1007fe9f: mov [esi+900h],bl
0x1007fea5: mov [esi+901h],bl
0x1007feab: mov [esi+902h],bl
0x1007feb1: mov [esi+904h],ebx
0x1007feb7: mov [esi+908h],ebx
0x1007febd: mov [esi+90Ch],ebx
0x1007fec3: mov [esi+910h],bx
0x1007feca: mov [esi+914h],bx
0x1007fed1: mov [esi+918h],ebx
0x1007fed7: mov [esi+91Ch],ebx
0x1007fedd: mov [esi+920h],ebx
0x1007fee3: mov [esi+974h],bl
0x1007fee9: push esi
0x1007feea: mov [esi+975h],bl
0x1007fef0: call 10082DF0h
0x1007fef5: add esp,4
0x1007fef8: pop edi
0x1007fef9: pop esi
0x1007fefa: xor eax,eax
0x1007fefc: pop ebx
0x1007fefd: ret
0x1007fefe: pop esi
0x1007feff: mov eax,2
0x1007ff04: pop ebx
0x1007ff05: ret
```

`sub_1007f790` is called by sub_1007ff10, sub_10080300. It accesses memory-mapped peripheral registers at 0x40404040. It delegates to sub_1007f380, sub_1007f3f0, sub_10082df0 (10 callees).

12.1070 sub_1007ff10 at 0x1007ff10 (32 instructions)

```
0x1007ff10: push esi
0x1007ff11: push edi
0x1007ff12: mov edi,[esp+0Ch]
0x1007ff16: mov esi,[edi]
0x1007ff18: test esi,esi
0x1007ff1a: je short 1007FF61h
0x1007ff1c: cmp dword ptr [esi],52530A0Ah
0x1007ff22: jne short 1007FF61h
0x1007ff24: push esi
0x1007ff25: call 1007F790h
0x1007ff2a: mov eax,[esi+210h]
0x1007ff30: push eax
0x1007ff31: push esi
0x1007ff32: call 10087020h
0x1007ff37: push esi
0x1007ff38: call 1008D1F0h
0x1007ff3d: mov dword ptr [esi],0
0x1007ff43: mov eax,[edi]
0x1007ff45: mov ecx,[eax+9Ch]
0x1007ff4b: push 0E50h
0x1007ff50: push eax
```

```
0x1007ff51: call ecx
0x1007ff53: add esp,18h
0x1007ff56: mov dword ptr [edi],0
0x1007ff5c: pop edi
0x1007ff5d: xor eax,eax
0x1007ff5f: pop esi
0x1007ff60: ret
0x1007ff61: pop edi
0x1007ff62: mov eax,2
0x1007ff67: pop esi
0x1007ff68: ret
```

`sub_1007ff10` is called by sub_10018660. It accesses memory-mapped peripheral registers at 0x52530a0a. It delegates to sub_1007f790, sub_10087020, sub_1008d1f0 (3 callees).

12.1071 sub_1007ff70 at 0x1007ff70 (144 instructions)

```
0x1007ff70: push ebx
0x1007ff71: push esi
0x1007ff72: mov esi,[esp+0Ch]
0x1007ff76: xor ebx,ebx
0x1007ff78: cmp esi,ebx
0x1007ff7a: je near ptr 10080144h
0x1007ff80: cmp dword ptr [esi],52530A0Ah
0x1007ff86: jne near ptr 10080144h
0x1007ff8c: cmp [esi+98h],ebx
0x1007ff92: jne short 1007FFA9h
0x1007ff94: push ebx
0x1007ff95: push ebx
0x1007ff96: push 3
0x1007ff98: push esi
0x1007ff99: call 10080800h
0x1007ff9e: add esp,10h
0x1007ffa1: pop esi
0x1007ffa2: mov eax,3
0x1007ffa7: pop ebx
0x1007ffa8: ret
0x1007ffa9: cmp [esi+9Ch],ebx
0x1007ffaf: je short 1007FF94h
0x1007ffb1: cmp [esi+0ACh],ebx
0x1007ffb7: je short 1007FF94h
0x1007ffb9: cmp [esi+0DCh],ebx
0x1007ffbf: je short 1007FF94h
0x1007ffc1: cmp [esi+0E8h],ebx
0x1007ffc7: je short 1007FF94h
0x1007ffc9: cmp [esi+0ECh],ebx
0x1007ffcf: je short 1007FF94h
0x1007ffd1: cmp [esi+0F0h],ebx
0x1007ffd7: je short 1007FF94h
0x1007ffd9: cmp [esi+158h],bl
0x1007ffdf: jne near ptr 1008012Fh
0x1007ffe5: cmp [esi+1B0h],bl
0x1007ffeb: jne near ptr 1008012Fh
0x1007fff1: cmp [esi+1A1h],bl
0x1007fff7: jne near ptr 1008012Fh
0x1007fffd: cmp [esi+1A0h],bl
0x10080003: jne near ptr 1008012Fh
0x10080009: push edi
0x1008000a: mov eax,esi
0x1008000c: call 1007F760h
0x10080011: mov eax,[esi+0ECh]
0x10080017: push esi
0x10080018: mov byte ptr [esi+158h],1
0x1008001f: mov byte ptr [esi+1B0h],1
0x10080026: mov byte ptr [esi+1F0h],1
0x1008002d: mov [esi+1B4h],ebx
```

```
0x10080033: mov [esi+1B8h],ebx
0x10080039: mov [esi+1BCh],ebx
0x1008003f: mov [esi+1D0h],ebx
0x10080045: mov [esi+1D4h],ebx
0x1008004b: mov [esi+1D8h],ebx
0x10080051: mov [esi+1DCh],bl
0x10080057: mov [esi+1E0h],ebx
0x1008005d: call eax
0x1008005f: mov [esi+1E4h],eax
0x10080065: mov [esi+1E8h],eax
0x1008006b: mov eax,[esi+0A4h]
0x10080071: add esp,4
0x10080074: cmp eax,ebx
0x10080076: mov [esi+16Ch],ebx
0x1008007c: mov [esi+1ECh],ebx
0x10080082: je short 10080095h
0x10080084: push esi
0x10080085: call eax
0x10080087: add esp,4
0x1008008a: test al,al
0x1008008c: jne short 10080095h
0x1008008e: mov edi,385h
0x10080093: jmp short 100800A0h
0x10080095: push esi
0x10080096: call 1008CB40h
0x1008009b: add esp,4
0x1008009e: mov edi,eax
0x100800a0: mov al,[esi+15Ah]
0x100800a6: cmp al,1
0x100800a8: jne short 100800E2h
0x100800aa: push esi
0x100800ab: mov [esi+158h],bl
0x100800b1: call 10086E70h
0x100800b6: add esp,4
0x100800b9: cmp edi,ebx
0x100800bb: jne short 10080129h
0x100800bd: cmp [esi+170h],bl
0x100800c3: je short 100800EFh
0x100800c5: mov ecx,[esi+0ECh]
0x100800cb: push esi
0x100800cc: mov edi,0Eh
0x100800d1: call ecx
0x100800d3: add esp,4
0x100800d6: mov [esi+16Ch],eax
0x100800dc: mov eax,edi
0x100800de: pop edi
0x100800df: pop esi
0x100800e0: pop ebx
0x100800e1: ret
0x100800e2: cmp al,bl
0x100800e4: je short 100800B9h
0x100800e6: pop edi
0x100800e7: pop esi
0x100800e8: mov eax,4
0x100800ed: pop ebx
0x100800ee: ret
0x100800ef: cmp [esi+1F1h],bl
0x100800f5: je short 10080102h
0x100800f7: mov edi,0Fh
0x100800fc: mov eax,edi
0x100800fe: pop edi
0x100800ff: pop esi
0x10080100: pop ebx
0x10080101: ret
0x10080102: cmp [esi+1F3h],bl
0x10080108: je short 10080115h
0x1008010a: mov edi,10h
```

```
0x1008010f: mov eax,edi
0x10080111: pop edi
0x10080112: pop esi
0x10080113: pop ebx
0x10080114: ret
0x10080115: cmp [esi+1F4h],bl
0x1008011b: mov [esi+1F0h],bl
0x10080121: je short 10080129h
0x10080123: mov [esi+1F4h],bl
0x10080129: mov eax,edi
0x1008012b: pop edi
0x1008012c: pop esi
0x1008012d: pop ebx
0x1008012e: ret
0x1008012f: push ebx
0x10080130: push ebx
0x10080131: push 0Bh
0x10080133: push esi
0x10080134: call 10080800h
0x10080139: add esp,10h
0x1008013c: pop esi
0x1008013d: mov eax,0Bh
0x10080142: pop ebx
0x10080143: ret
0x10080144: pop esi
0x10080145: mov eax,2
0x1008014a: pop ebx
0x1008014b: ret
```

`sub_1007ff70` is called by sub_100187f0. It accesses memory-mapped peripheral registers at 0x52530a0a. It delegates to sub_1007f760, sub_10080800, sub_10086e70 (4 callees).

12.1072 sub_10080150 at 0x10080150 (135 instructions)

```
0x10080150: push ebx
0x10080151: push esi
0x10080152: mov esi,[esp+0Ch]
0x10080156: xor ebx,ebx
0x10080158: cmp esi,ebx
0x1008015a: je near ptr 100802ECh
0x10080160: cmp dword ptr [esi],52530A0Ah
0x10080166: jne near ptr 100802ECh
0x1008016c: cmp [esi+1B0h],bl
0x10080172: jne short 10080187h
0x10080174: push ebx
0x10080175: push ebx
0x10080176: push 0Bh
0x10080178: push esi
0x10080179: call 10080800h
0x1008017e: add esp,10h
0x10080181: pop esi
0x10080182: lea eax,[ebx+0Bh]
0x10080185: pop ebx
0x10080186: ret
0x10080187: push edi
0x10080188: mov eax,esi
0x1008018a: call 1007F760h
0x1008018f: cmp [esi+1F0h],bl
0x10080195: jne short 100801B4h
0x10080197: cmp [esi+158h],bl
0x1008019d: jne short 100801B4h
0x1008019f: mov eax,[esi+0ECh]
0x100801a5: push esi
0x100801a6: call eax
0x100801a8: mov byte ptr [esi+1F0h],1
0x100801af: jmp near ptr 1008025Bh
```

```
0x100801b4: cmp [esi+1F1h],bl
0x100801ba: jne short 100801E2h
0x100801bc: cmp [esi+170h],bl
0x100801c2: jne short 100801E2h
0x100801c4: cmp [esi+1F3h],bl
0x100801ca: jne short 100801E2h
0x100801cc: push ebx
0x100801cd: push ebx
0x100801ce: push 0Bh
0x100801d0: push esi
0x100801d1: call 10080800h
0x100801d6: add esp,10h
0x100801d9: pop edi
0x100801da: pop esi
0x100801db: mov eax,0Bh
0x100801e0: pop ebx
0x100801e1: ret
0x100801e2: cmp [esi+158h],bl
0x100801e8: mov [esi+1F1h],bl
0x100801ee: mov [esi+1F3h],bl
0x100801f4: push esi
0x100801f5: je short 10080253h
0x100801f7: cmp [esi+170h],bl
0x100801fd: je short 10080218h
0x100801ff: mov ecx,[esi+0ECh]
0x10080205: call ecx
0x10080207: sub eax,[esi+16Ch]
0x1008020d: add esp,4
0x10080210: add [esi+1E4h],eax
0x10080216: jmp short 10080229h
0x10080218: mov edx,[esi+0ECh]
0x1008021e: call edx
0x10080220: add esp,4
0x10080223: mov [esi+1E4h],eax
0x10080229: push esi
0x1008022a: mov [esi+170h],bl
0x10080230: call 1008CB40h
0x10080235: add esp,4
0x10080238: cmp [esi+15Ah],bl
0x1008023e: mov edi,eax
0x10080240: je short 1008026Ch
0x10080242: push esi
0x10080243: mov [esi+158h],bl
0x10080249: call 10086E70h
0x1008024e: add esp,4
0x10080251: jmp short 1008026Ch
0x10080253: mov eax,[esi+0ECh]
0x10080259: call eax
0x1008025b: push esi
0x1008025c: mov [esi+1E4h],eax
0x10080262: call 10086C00h
0x10080267: mov edi,eax
0x10080269: add esp,8
0x1008026c: cmp edi,ebx
0x1008026e: jne short 100802E6h
0x10080270: cmp [esi+170h],bl
0x10080276: je short 10080295h
0x10080278: mov ecx,[esi+0ECh]
0x1008027e: push esi
0x1008027f: mov edi,0Eh
0x10080284: call ecx
0x10080286: add esp,4
0x10080289: mov [esi+16Ch],eax
0x1008028f: mov eax,edi
0x10080291: pop edi
0x10080292: pop esi
0x10080293: pop ebx
```

```
0x10080294: ret
0x10080295: cmp [esi+1F1h],bl
0x1008029b: je short 100802A8h
0x1008029d: mov edi,0Fh
0x100802a2: mov eax,edi
0x100802a4: pop edi
0x100802a5: pop esi
0x100802a6: pop ebx
0x100802a7: ret
0x100802a8: cmp [esi+1F3h],bl
0x100802ae: je short 100802BBh
0x100802b0: mov edi,10h
0x100802b5: mov eax,edi
0x100802b7: pop edi
0x100802b8: pop esi
0x100802b9: pop ebx
0x100802ba: ret
0x100802bb: cmp [esi+1F4h],bl
0x100802c1: mov [esi+1F0h],bl
0x100802c7: je short 100802CFh
0x100802c9: mov [esi+1F4h],bl
0x100802cf: cmp [esi+1F5h],bl
0x100802d5: je short 100802E6h
0x100802d7: mov eax,esi
0x100802d9: mov [esi+1B0h],bl
0x100802df: call 1007F450h
0x100802e4: mov edi,eax
0x100802e6: mov eax,edi
0x100802e8: pop edi
0x100802e9: pop esi
0x100802ea: pop ebx
0x100802eb: ret
0x100802ec: pop esi
0x100802ed: mov eax,2
0x100802f2: pop ebx
0x100802f3: ret
```

`sub_10080150` is called by sub_100187f0. It accesses memory-mapped peripheral registers at 0x52530a0a. It delegates to sub_1007f450, sub_1007f760, sub_10080800 (6 callees).

12.1073 sub_10080300 at 0x10080300 (141 instructions)

```
0x10080300: push ebx
0x10080301: push esi
0x10080302: push edi
0x10080303: mov edi,[esp+14h]
0x10080307: push 0E50h
0x1008030c: call edi
0x1008030e: mov esi,eax
0x10080310: xor ebx,ebx
0x10080312: add esp,4
0x10080315: cmp esi,ebx
0x10080317: je near ptr 1008046Ch
0x1008031d: mov ecx,[esp+1Ch]
0x10080321: fld1
0x10080323: mov eax,[esp+10h]
0x10080327: fstp qword ptr [esi+58h]
0x1008032a: mov [esi+0B8h],ecx
0x10080330: fld qword ptr ds:[10108F40h]
0x10080336: mov ecx,2710h
0x1008033b: fstp qword ptr [esi+60h]
0x1008033e: mov [esi+7Ch],ecx
0x10080341: fld qword ptr ds:[10126AC8h]
0x10080347: mov [esi+80h],ecx
0x1008034d: fstp qword ptr [esi+68h]
0x10080350: lea ecx,[esp+14h]
```

```
0x10080354: fldz
0x10080356: push ecx
0x10080357: fstp qword ptr [esi+3C0h]
0x1008035d: push ebx
0x1008035e: push ebx
0x1008035f: push ebx
0x10080360: push ebx
0x10080361: push ebx
0x10080362: push ebx
0x10080363: push ebx
0x10080364: mov edx,[esp+38h]
0x10080368: push ebx
0x10080369: push ebx
0x1008036a: push ebx
0x1008036b: push ebx
0x1008036c: push ebx
0x1008036d: push ebx
0x1008036e: push ebx
0x1008036f: mov [esi+4],eax
0x10080372: mov eax,1
0x10080377: push eax
0x10080378: push eax
0x10080379: push eax
0x1008037a: push ebx
0x1008037b: push esi
0x1008037c: mov dword ptr [esi],52530A0Ah
0x10080382: mov [esi+34h],ebx
0x10080385: mov [esi+38h],ebx
0x10080388: mov [esi+50h],bx
0x1008038c: mov [esi+52h],bx
0x10080390: mov [esi+54h],bx
0x10080394: mov [esi+56h],al
0x10080397: mov [esi+70h],al
0x1008039a: mov [esi+71h],bl
0x1008039d: mov [esi+72h],al
0x100803a0: mov [esi+73h],al
0x100803a3: mov dword ptr [esi+74h],2121h
0x100803aa: mov [esi+78h],ebx
0x100803ad: mov [esi+98h],edi
0x100803b3: mov [esi+9Ch],edx
0x100803b9: mov [esi+0A0h],ebx
0x100803bf: mov [esi+0A4h],ebx
0x100803c5: mov [esi+0A8h],ebx
0x100803cb: mov [esi+0ACh],ebx
0x100803d1: mov [esi+0B0h],ebx
0x100803d7: mov [esi+0B4h],ebx
0x100803dd: mov [esi+0BCh],ebx
0x100803e3: mov [esi+0C0h],ebx
0x100803e9: mov [esi+0C4h],ebx
0x100803ef: mov [esi+0C8h],ebx
0x100803f5: mov [esi+0CCh],ebx
0x100803fb: mov [esi+0D0h],ebx
0x10080401: mov [esi+0D4h],ebx
0x10080407: mov [esi+0D8h],ebx
0x1008040d: mov [esi+0DCh],ebx
0x10080413: mov [esi+0E0h],ebx
0x10080419: mov [esi+0E4h],ebx
0x1008041f: mov [esi+0E8h],ebx
0x10080425: mov [esi+0ECh],ebx
0x1008042b: mov [esi+0F0h],ebx
0x10080431: mov [esi+0F4h],ebx
0x10080437: mov [esi+0F8h],ebx
0x1008043d: mov [esi+0FCh],ebx
0x10080443: mov [esi+100h],ebx
0x10080449: mov [esi+104h],ebx
0x1008044f: call 10089910h
0x10080454: add esp,50h
```

```
0x10080457: test eax,eax
0x10080459: je short 10080472h
0x1008045b: mov edx,[esi+9Ch]
0x10080461: push 0E50h
0x10080466: push esi
0x10080467: call edx
0x10080469: add esp,8
0x1008046c: pop edi
0x1008046d: pop esi
0x1008046e: xor eax,eax
0x10080470: pop ebx
0x10080471: ret
0x10080472: mov eax,[esp+14h]
0x10080476: push esi
0x10080477: mov [esi+210h],eax
0x1008047d: mov [esi+172h],bl
0x10080483: mov [esi+174h],ebx
0x10080489: mov [esi+178h],ebx
0x1008048f: mov [esi+17Ch],ebx
0x10080495: mov [esi+180h],ebx
0x1008049b: mov [esi+184h],ebx
0x100804a1: mov [esi+188h],ebx
0x100804a7: mov [esi+18Ch],ebx
0x100804ad: mov [esi+190h],ebx
0x100804b3: mov [esi+194h],ebx
0x100804b9: mov [esi+198h],ebx
0x100804bf: mov [esi+19Ch],ebx
0x100804c5: call 1008CD60h
0x100804ca: mov eax,9
0x100804cf: mov [esi+95Ch],eax
0x100804d5: mov [esi+968h],eax
0x100804db: mov eax,1000h
0x100804e0: push esi
0x100804e1: mov dword ptr [esi+960h],8
0x100804eb: mov dword ptr [esi+964h],0Fh
0x100804f5: mov [esi+96Ch],ebx
0x100804fb: mov [esi+970h],eax
0x10080501: mov [esi+978h],ebx
0x10080507: mov dword ptr [esi+97Ch],64h
0x10080511: mov [esi+980h],ebx
0x10080517: mov [esi+984h],bl
0x1008051d: mov [esi+985h],bl
0x10080523: mov [esi+988h],eax
0x10080529: call 1007F790h
0x1008052e: add esp,8
0x10080531: pop edi
0x10080532: mov eax,esi
0x10080534: pop esi
0x10080535: pop ebx
0x10080536: ret
```

`sub_10080300` is called by sub_100187f0. It accesses memory-mapped peripheral registers at 0x52530a0a. It delegates to sub_1007f790, sub_10089910, sub_1008cd60 (3 callees).

12.1074 sub_10080540 at 0x10080540 (11 instructions)

```
0x10080540: mov eax,[esp+4]
0x10080544: test eax,eax
0x10080546: je short 1008055Dh
0x10080548: cmp dword ptr [eax],52530A0Ah
0x1008054e: jne short 1008055Dh
0x10080550: mov ecx,[esp+8]
0x10080554: mov [eax+0A4h],ecx
0x1008055a: xor eax,eax
0x1008055c: ret
```

```
0x1008055d: mov eax,2
0x10080562: ret
```

`sub_10080540` is called by sub_10018760. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1075 sub_10080570 at 0x10080570 (11 instructions)

```
0x10080570: mov eax,[esp+4]
0x10080574: test eax,eax
0x10080576: je short 1008058Dh
0x10080578: cmp dword ptr [eax],52530A0Ah
0x1008057e: jne short 1008058Dh
0x10080580: mov ecx,[esp+8]
0x10080584: mov [eax+0A8h],ecx
0x1008058a: xor eax,eax
0x1008058c: ret
0x1008058d: mov eax,2
0x10080592: ret
```

`sub_10080570` is called by sub_10018760. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1076 sub_100805a0 at 0x100805a0 (11 instructions)

```
0x100805a0: mov eax,[esp+4]
0x100805a4: test eax,eax
0x100805a6: je short 100805BDh
0x100805a8: cmp dword ptr [eax],52530A0Ah
0x100805ae: jne short 100805BDh
0x100805b0: mov ecx,[esp+8]
0x100805b4: mov [eax+0ACh],ecx
0x100805ba: xor eax,eax
0x100805bc: ret
0x100805bd: mov eax,2
0x100805c2: ret
```

`sub_100805a0` is called by sub_10018760. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1077 sub_100805d0 at 0x100805d0 (11 instructions)

```
0x100805d0: mov eax,[esp+4]
0x100805d4: test eax,eax
0x100805d6: je short 100805EDh
0x100805d8: cmp dword ptr [eax],52530A0Ah
0x100805de: jne short 100805EDh
0x100805e0: mov ecx,[esp+8]
0x100805e4: mov [eax+0B4h],ecx
0x100805ea: xor eax,eax
0x100805ec: ret
0x100805ed: mov eax,2
0x100805f2: ret
```

`sub_100805d0` is called by sub_10018760. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1078 sub_10080600 at 0x10080600 (11 instructions)

```
0x10080600: mov eax,[esp+4]
0x10080604: test eax,eax
0x10080606: je short 1008061Dh
```

```
0x10080608: cmp dword ptr [eax],52530A0Ah
0x1008060e: jne short 1008061Dh
0x10080610: mov ecx,[esp+8]
0x10080614: mov [eax+0BCh],ecx
0x1008061a: xor eax,eax
0x1008061c: ret
0x1008061d: mov eax,2
0x10080622: ret
```

`sub_10080600` is called by sub_10018760. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1079 sub_10080630 at 0x10080630 (11 instructions)

```
0x10080630: mov eax,[esp+4]
0x10080634: test eax,eax
0x10080636: je short 1008064Dh
0x10080638: cmp dword ptr [eax],52530A0Ah
0x1008063e: jne short 1008064Dh
0x10080640: mov ecx,[esp+8]
0x10080644: mov [eax+0DCh],ecx
0x1008064a: xor eax,eax
0x1008064c: ret
0x1008064d: mov eax,2
0x10080652: ret
```

`sub_10080630` is called by sub_10018760. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1080 sub_10080660 at 0x10080660 (11 instructions)

```
0x10080660: mov eax,[esp+4]
0x10080664: test eax,eax
0x10080666: je short 1008067Dh
0x10080668: cmp dword ptr [eax],52530A0Ah
0x1008066e: jne short 1008067Dh
0x10080670: mov ecx,[esp+8]
0x10080674: mov [eax+0E4h],ecx
0x1008067a: xor eax,eax
0x1008067c: ret
0x1008067d: mov eax,2
0x10080682: ret
```

`sub_10080660` is called by sub_10018760. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1081 sub_10080690 at 0x10080690 (11 instructions)

```
0x10080690: mov eax,[esp+4]
0x10080694: test eax,eax
0x10080696: je short 100806ADh
0x10080698: cmp dword ptr [eax],52530A0Ah
0x1008069e: jne short 100806ADh
0x100806a0: mov ecx,[esp+8]
0x100806a4: mov [eax+0E8h],ecx
0x100806aa: xor eax,eax
0x100806ac: ret
0x100806ad: mov eax,2
0x100806b2: ret
```

`sub_10080690` is called by sub_10018760. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1082 sub_100806c0 at 0x100806c0 (11 instructions)

```
0x100806c0: mov eax,[esp+4]
0x100806c4: test eax,eax
0x100806c6: je short 100806DDh
0x100806c8: cmp dword ptr [eax],52530A0Ah
0x100806ce: jne short 100806DDh
0x100806d0: mov ecx,[esp+8]
0x100806d4: mov [eax+0ECh],ecx
0x100806da: xor eax,eax
0x100806dc: ret
0x100806dd: mov eax,2
0x100806e2: ret
```

`sub_100806c0` is called by sub_10018760. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1083 sub_100806f0 at 0x100806f0 (11 instructions)

```
0x100806f0: mov eax,[esp+4]
0x100806f4: test eax,eax
0x100806f6: je short 1008070Dh
0x100806f8: cmp dword ptr [eax],52530A0Ah
0x100806fe: jne short 1008070Dh
0x10080700: mov ecx,[esp+8]
0x10080704: mov [eax+0F0h],ecx
0x1008070a: xor eax,eax
0x1008070c: ret
0x1008070d: mov eax,2
0x10080712: ret
```

`sub_100806f0` is called by sub_10018760. It accesses memory-mapped peripheral registers at 0x52530a0a. It calls nothing else.

12.1084 sub_10080720 at 0x10080720 (68 instructions)

```
0x10080720: push ebx
0x10080721: mov ebx,[esp+8]
0x10080725: test ebx,ebx
0x10080727: je near ptr 100807F7h
0x1008072d: mov eax,[esp+10h]
0x10080731: mov ecx,[esp+14h]
0x10080735: push esi
0x10080736: push edi
0x10080737: mov edi,[esp+14h]
0x1008073b: xor esi,esi
0x1008073d: mov [ebx+8Ch],eax
0x10080743: mov [ebx+90h],ecx
0x10080749: mov [ebx+84h],edi
0x1008074f: mov eax,2Ch
0x10080754: lea ecx,[esi+58h]
0x10080757: jmp short 10080760h
0x10080760: mov edx,[eax*8+10127608h]
0x10080767: cmp edx,edi
0x10080769: jge short 10080770h
0x1008076b: lea esi,[eax+1]
0x1008076e: jmp short 10080775h
0x10080770: jle short 10080799h
0x10080772: lea ecx,[eax-1]
0x10080775: lea eax,[ecx+esi]
0x10080778: sar eax,1
0x1008077a: cmp esi,ecx
0x1008077c: jle short 10080760h
0x1008077e: mov dword ptr [ebx+94h],101278D0h
0x10080788: test edi,edi
```

```
0x1008078a: jne short 100807AFh
0x1008078c: pop edi
0x1008078d: pop esi
0x1008078e: mov byte ptr [ebx+88h],0
0x10080795: mov al,1
0x10080797: pop ebx
0x10080798: ret
0x10080799: lea eax,[eax*8+10127608h]
0x100807a0: test eax,eax
0x100807a2: je short 1008077Eh
0x100807a4: mov edx,[eax+4]
0x100807a7: mov [ebx+94h],edx
0x100807ad: jmp short 10080788h
0x100807af: and edi,3C00h
0x100807b5: cmp edi,800h
0x100807bb: je short 100807EEh
0x100807bd: cmp edi,1000h
0x100807c3: je short 100807E1h
0x100807c5: cmp edi,2000h
0x100807cb: setne al
0x100807ce: pop edi
0x100807cf: pop esi
0x100807d0: lea eax,[eax*8+1]
0x100807d7: mov [ebx+88h],al
0x100807dd: mov al,1
0x100807df: pop ebx
0x100807e0: ret
0x100807e1: pop edi
0x100807e2: pop esi
0x100807e3: mov byte ptr [ebx+88h],2
0x100807ea: mov al,1
0x100807ec: pop ebx
0x100807ed: ret
0x100807ee: pop edi
0x100807ef: mov byte ptr [ebx+88h],5
0x100807f6: pop esi
0x100807f7: mov al,1
0x100807f9: pop ebx
0x100807fa: ret
```

`sub_10080720` is called by sub_100187f0, sub_10080800. It works with the text "Unknown error". It calls nothing else.

12.1085 sub_10080800 at 0x10080800 (42 instructions)

```
0x10080800: mov eax,[esp+10h]
0x10080804: mov ecx,[esp+0Ch]
0x10080808: push esi
0x10080809: mov esi,[esp+8]
0x1008080d: push edi
0x1008080e: mov edi,[esp+10h]
0x10080812: push eax
0x10080813: push ecx
0x10080814: push edi
0x10080815: push esi
0x10080816: call 10080720h
0x1008081b: add esp,10h
0x1008081e: test esi,esi
0x10080820: je short 10080869h
0x10080822: cmp dword ptr [esi],52530A0Ah
0x10080828: jne short 10080869h
0x1008082a: mov eax,[esi+0B4h]
0x10080830: test eax,eax
0x10080832: je short 10080869h
0x10080834: mov edx,[esi+94h]
0x1008083a: mov ecx,[esi+90h]
0x10080840: push edx
```

```
0x10080841: mov edx,[esi+8Ch]
0x10080847: push ecx
0x10080848: mov ecx,[esi+110h]
0x1008084e: push edx
0x1008084f: mov edx,[esi+10Ch]
0x10080855: push ecx
0x10080856: movzx ecx,byte ptr [esi+88h]
0x1008085d: push edx
0x1008085e: push ecx
0x1008085f: push edi
0x10080860: push esi
0x10080861: call eax
0x10080863: add esp,20h
0x10080866: pop edi
0x10080867: pop esi
0x10080868: ret
0x10080869: pop edi
0x1008086a: mov al,1
0x1008086c: pop esi
0x1008086d: ret
```

`sub_10080800` is called by sub_1007f1a0, sub_1007f310, sub_1007ff70, sub_10080150, sub_10080870, sub_10080920, sub_100809d0, sub_10081660, sub_10082e60, sub_10083b00, sub_10086c00, sub_10087190, sub_10089620, sub_10089910, sub_1008ad30, sub_1008ae80, sub_1008af60, sub_1008c420, sub_1008c510, sub_1008cb40, sub_1008d0a0, sub_1008d1b0, sub_100a0020. It accesses memory-mapped peripheral registers at 0x52530a0a. It delegates to sub_10080720 (1 callees).

12.1086 sub_10080870 at 0x10080870 (25 instructions)

```
0x10080870: test eax,eax
0x10080872: jne short 10080879h
0x10080874: mov eax,1
0x10080879: cmp byte ptr [esi+1F0h],0
0x10080880: je short 100808BBh
0x10080882: push eax
0x10080883: mov eax,[esi+0F0h]
0x10080889: push esi
0x1008088a: call eax
0x1008088c: add esp,8
0x1008088f: test al,al
0x10080891: jne short 100808ABh
0x10080893: push 0
0x10080895: push 0
0x10080897: push 386h
0x1008089c: push esi
0x1008089d: call 10080800h
0x100808a2: add esp,10h
0x100808a5: mov eax,386h
0x100808aa: ret
0x100808ab: cmp byte ptr [esi+1F0h],0
0x100808b2: je short 100808BBh
0x100808b4: mov byte ptr [esi+1F1h],1
0x100808bb: xor eax,eax
0x100808bd: ret
```

`sub_10080870` is called by sub_10080920, sub_100809d0, sub_10083b00. It delegates to sub_10080800 (1 callees).

12.1087 sub_100808c0 at 0x100808c0 (38 instructions)

```
0x100808c0: mov ecx,[edx+18h]
0x100808c3: test ecx,ecx
0x100808c5: push esi
0x100808c6: mov esi,1
0x100808cb: jne short 100808D8h
```

```
0x100808cd: cmp dword ptr [edx+0Ch],2
0x100808d1: jne short 10080917h
0x100808d3: mov ecx,3E8h
0x100808d8: mov edx,[edx+78h]
0x100808db: sub edx,esi
0x100808dd: je short 1008090Bh
0x100808df: sub edx,esi
0x100808e1: je short 100808FFh
0x100808e3: sub edx,esi
0x100808e5: je short 100808F3h
0x100808e7: imul eax,3E8h
0x100808ed: xor edx,edx
0x100808ef: div ecx
0x100808f1: pop esi
0x100808f2: ret
0x100808f3: imul eax,1F40h
0x100808f9: xor edx,edx
0x100808fb: div ecx
0x100808fd: pop esi
0x100808fe: ret
0x100808ff: imul eax,0BB8h
0x10080905: xor edx,edx
0x10080907: div ecx
0x10080909: pop esi
0x1008090a: ret
0x1008090b: imul eax,1F4h
0x10080911: xor edx,edx
0x10080913: div ecx
0x10080915: pop esi
0x10080916: ret
0x10080917: mov eax,esi
0x10080919: pop esi
0x1008091a: ret
```

`sub_100808c0` is called by sub_100809d0. It calls nothing else.

12.1088 sub_10080920 at 0x10080920 (58 instructions)

```
0x10080920: push ebx
0x10080921: push esi
0x10080922: mov esi,[esp+0Ch]
0x10080926: xor ebx,ebx
0x10080928: cmp [esi+1F0h],bl
0x1008092e: push edi
0x1008092f: je near ptr 100809C9h
0x10080935: cmp [esi+1CCh],bl
0x1008093b: jne near ptr 100809C9h
0x10080941: mov edi,[esi+230h]
0x10080947: mov ecx,[esi+234h]
0x1008094d: cmp edi,ecx
0x1008094f: jae short 100809C9h
0x10080951: mov edx,[esi+228h]
0x10080957: mov eax,[esi+22Ch]
0x1008095d: cmp edx,eax
0x1008095f: jae short 100809C9h
0x10080961: sub ecx,edi
0x10080963: push ecx
0x10080964: sub eax,edx
0x10080966: push eax
0x10080967: mov eax,[esi+0E8h]
0x1008096d: push edi
0x1008096e: push edx
0x1008096f: push esi
0x10080970: call eax
0x10080972: add esp,14h
0x10080975: test al,al
```

```
0x10080977: jne short 10080992h
0x10080979: push ebx
0x1008097a: push ebx
0x1008097b: push 388h
0x10080980: push esi
0x10080981: call 10080800h
0x10080986: add esp,10h
0x10080989: pop edi
0x1008098a: pop esi
0x1008098b: mov eax,388h
0x10080990: pop ebx
0x10080991: ret
0x10080992: cmp [esi+1F4h],bl
0x10080998: mov [esi+228h],ebx
0x1008099e: mov [esi+22Ch],ebx
0x100809a4: mov [esi+230h],ebx
0x100809aa: mov [esi+234h],ebx
0x100809b0: mov [esi+1F6h],bl
0x100809b6: jne short 100809C9h
0x100809b8: mov eax,[esp+14h]
0x100809bc: cmp eax,ebx
0x100809be: je short 100809C9h
0x100809c0: call 10080870h
0x100809c5: cmp eax,ebx
0x100809c7: jne short 100809CBh
0x100809c9: xor eax,eax
0x100809cb: pop edi
0x100809cc: pop esi
0x100809cd: pop ebx
0x100809ce: ret
```

`sub_10080920` is called by sub_10086c00, sub_1008c510, sub_1008c980. It delegates to sub_10080800, sub_10080870 (2 callees).

12.1089 sub_100809d0 at 0x100809d0 (121 instructions)

```
0x100809d0: push ebx
0x100809d1: push ebp
0x100809d2: push esi
0x100809d3: mov esi,eax
0x100809d5: xor ebx,ebx
0x100809d7: xor ebp,ebp
0x100809d9: cmp [esi+45Ch],ebx
0x100809df: push edi
0x100809e0: jbe near ptr 10080B29h
0x100809e6: cmp [esi+1F0h],bl
0x100809ec: je short 10080A48h
0x100809ee: cmp [esi+1CCh],bl
0x100809f4: jne short 10080A48h
0x100809f6: mov edi,[esi+230h]
0x100809fc: mov ecx,[esi+234h]
0x10080a02: cmp edi,ecx
0x10080a04: jae short 10080A48h
0x10080a06: mov edx,[esi+228h]
0x10080a0c: mov eax,[esi+22Ch]
0x10080a12: cmp edx,eax
0x10080a14: jae short 10080A48h
0x10080a16: sub ecx,edi
0x10080a18: push ecx
0x10080a19: sub eax,edx
0x10080a1b: push eax
0x10080a1c: mov eax,[esi+0E8h]
0x10080a22: push edi
0x10080a23: push edx
0x10080a24: push esi
0x10080a25: call eax
```

```
0x10080a27: add esp,14h
0x10080a2a: test al,al
0x10080a2c: jne short 10080A48h
0x10080a2e: push ebx
0x10080a2f: push ebx
0x10080a30: push 388h
0x10080a35: push esi
0x10080a36: call 10080800h
0x10080a3b: add esp,10h
0x10080a3e: pop edi
0x10080a3f: pop esi
0x10080a40: pop ebp
0x10080a41: mov eax,388h
0x10080a46: pop ebx
0x10080a47: ret
0x10080a48: cmp [esi+1F8h],bl
0x10080a4e: mov [esi+228h],ebx
0x10080a54: mov [esi+22Ch],ebx
0x10080a5a: mov [esi+230h],ebx
0x10080a60: mov [esi+234h],ebx
0x10080a66: mov [esi+1F6h],bl
0x10080a6c: je short 10080A94h
0x10080a6e: mov eax,1
0x10080a73: add [esi+1FCh],eax
0x10080a79: cmp [esi+1FCh],eax
0x10080a7f: jbe short 10080A94h
0x10080a81: pop edi
0x10080a82: mov [esi+204h],ebx
0x10080a88: mov [esi+1F0h],bl
0x10080a8e: pop esi
0x10080a8f: pop ebp
0x10080a90: xor eax,eax
0x10080a92: pop ebx
0x10080a93: ret
0x10080a94: mov eax,[esi+0ECh]
0x10080a9a: cmp eax,ebx
0x10080a9c: je short 10080AE2h
0x10080a9e: push esi
0x10080a9f: call eax
0x10080aa1: mov ecx,[esi+1E4h]
0x10080aa7: add esp,4
0x10080aaa: cmp eax,ecx
0x10080aac: mov [esi+1E0h],eax
0x10080ab2: jae short 10080ABCh
0x10080ab4: not ecx
0x10080ab6: lea edi,[ecx+eax+1]
0x10080aba: jmp short 10080AC0h
0x10080abc: mov edi,eax
0x10080abe: sub edi,ecx
0x10080ac0: mov ecx,[esi+1E8h]
0x10080ac6: cmp eax,ecx
0x10080ac8: jae short 10080AD8h
0x10080aca: not ecx
0x10080acc: lea ecx,[ecx+eax+1]
0x10080ad0: mov [esi+1E0h],ecx
0x10080ad6: jmp short 10080AE4h
0x10080ad8: sub eax,ecx
0x10080ada: mov [esi+1E0h],eax
0x10080ae0: jmp short 10080AE4h
0x10080ae2: xor edi,edi
0x10080ae4: mov eax,[esi+45Ch]
0x10080aea: mov edx,esi
0x10080aec: call 100808C0h
0x10080af1: mov ebp,eax
0x10080af3: cmp ebp,edi
0x10080af5: jbe short 10080AFBh
0x10080af7: sub eax,edi
```

```
0x10080af9: jmp short 10080B00h
0x10080afb: mov eax,1
0x10080b00: cmp [esi+1F0h],bl
0x10080b06: jne short 10080B18h
0x10080b08: cmp [esi+1DCh],bl
0x10080b0e: jne short 10080B18h
0x10080b10: cmp [esi+158h],bl
0x10080b16: je short 10080B29h
0x10080b18: cmp [esi+1CCh],bl
0x10080b1e: jne short 10080B37h
0x10080b20: call 10080870h
0x10080b25: cmp eax,ebx
0x10080b27: jne short 10080B45h
0x10080b29: cmp [esi+1CCh],bl
0x10080b2f: jne short 10080B37h
0x10080b31: add [esi+1BCh],ebp
0x10080b37: mov edx,[esi+478h]
0x10080b3d: mov [esi+45Ch],edx
0x10080b43: xor eax,eax
0x10080b45: pop edi
0x10080b46: pop esi
0x10080b47: pop ebp
0x10080b48: pop ebx
0x10080b49: ret
```

`sub_100809d0` is called by sub_10083b00. It delegates to sub_10080800, sub_10080870, sub_100808c0 (3 callees).

12.1090 sub_10080b50 at 0x10080b50 (66 instructions)

```
0x10080b50: cmp byte ptr [ecx+1F0h],0
0x10080b57: jne short 10080B66h
0x10080b59: cmp byte ptr [ecx+1DCh],0
0x10080b60: je near ptr 10080CB7h
0x10080b66: cmp byte ptr [ecx+1CCh],0
0x10080b6d: jne near ptr 10080CB7h
0x10080b73: mov eax,[ecx+34h]
0x10080b76: cmp eax,1000h
0x10080b7b: ja short 10080BDCh
0x10080b7d: je short 10080BD1h
0x10080b7f: cmp eax,8
0x10080b82: ja near ptr 10080CB7h
0x10080b88: jmp dword ptr [eax*4+10080CB8h]
0x10080bd1: mov dword ptr [ecx+3C8h],1008D6B0h
0x10080bdb: ret
0x10080bdc: cmp eax,5000h
0x10080be1: ja short 10080C5Eh
0x10080be3: je short 10080C53h
0x10080be5: cmp eax,3000h
0x10080bea: ja short 10080C2Fh
0x10080bec: je short 10080C24h
0x10080bee: sub eax,1001h
0x10080bf3: je short 10080C19h
0x10080bf5: sub eax,4
0x10080bf8: je short 10080C0Eh
0x10080bfa: sub eax,1
0x10080bfd: jne near ptr 10080CB7h
0x10080c03: mov dword ptr [ecx+3C8h],100912F0h
0x10080c0d: ret
0x10080c0e: mov dword ptr [ecx+3C8h],100919D0h
0x10080c18: ret
0x10080c19: mov dword ptr [ecx+3C8h],1008F6F0h
0x10080c23: ret
0x10080c24: mov dword ptr [ecx+3C8h],1008E050h
0x10080c2e: ret
0x10080c2f: cmp eax,3001h
0x10080c34: je short 10080C48h
```

```
0x10080c36: cmp eax,4006h
0x10080c3b: jne short 10080CB7h
0x10080c3d: mov dword ptr [ecx+3C8h],100920C0h
0x10080c47: ret
0x10080c48: mov dword ptr [ecx+3C8h],100900A0h
0x10080c52: ret
0x10080c53: mov dword ptr [ecx+3C8h],1008E960h
0x10080c5d: ret
0x10080c5e: cmp eax,0B000h
0x10080c63: ja short 10080C94h
0x10080c65: je short 10080C89h
0x10080c67: sub eax,9000h
0x10080c6c: je short 10080C7Eh
0x10080c6e: sub eax,1
0x10080c71: jne short 10080CB7h
0x10080c73: mov dword ptr [ecx+3C8h],1008FCC0h
0x10080c7d: ret
0x10080c7e: mov dword ptr [ecx+3C8h],1008DC70h
0x10080c88: ret
0x10080c89: mov dword ptr [ecx+3C8h],1008E580h
0x10080c93: ret
0x10080c94: cmp eax,0B001h
0x10080c99: je short 10080CADh
0x10080c9b: cmp eax,10001h
0x10080ca0: jne short 10080CB7h
0x10080ca2: mov dword ptr [ecx+3C8h],1008F340h
0x10080cac: ret
0x10080cad: mov dword ptr [ecx+3C8h],10090660h
0x10080cb7: ret
```

`sub_10080b50` is called by sub_10081660. It calls nothing else.

12.1091 sub_10081660 at 0x10081660 (79 instructions)

```
0x10081660: mov eax,[esi+10h]
0x10081663: mov ecx,[esi+14h]
0x10081666: mov [esi+2A4h],eax
0x1008166c: mov [esi+294h],eax
0x10081672: mov [esi+24Ch],eax
0x10081678: push ebx
0x10081679: xor ebx,ebx
0x1008167b: add eax,eax
0x1008167d: push edi
0x1008167e: mov [esi+2ACh],ecx
0x10081684: mov [esi+29Ch],ecx
0x1008168a: mov edi,1
0x1008168f: add eax,eax
0x10081691: mov ecx,esi
0x10081693: mov [esi+2A0h],ebx
0x10081699: mov [esi+2A8h],ebx
0x1008169f: mov [esi+290h],ebx
0x100816a5: mov [esi+298h],ebx
0x100816ab: mov byte ptr [esi+238h],0FFh
0x100816b2: mov [esi+23Ch],ebx
0x100816b8: mov [esi+240h],edi
0x100816be: mov [esi+244h],ebx
0x100816c4: mov [esi+248h],edi
0x100816ca: mov [esi+25Ch],eax
0x100816d0: mov [esi+288h],bl
0x100816d6: mov byte ptr [esi+289h],1
0x100816dd: call 10080B50h
0x100816e2: mov ecx,[esi+25Ch]
0x100816e8: mov edx,[esi+98h]
0x100816ee: push ecx
0x100816ef: call edx
0x100816f1: add esp,4
```

```
0x100816f4: cmp eax,ebx
0x100816f6: mov [esi+284h],eax
0x100816fc: jne short 1008170Fh
0x100816fe: push ebx
0x100816ff: push ebx
0x10081700: push edi
0x10081701: push esi
0x10081702: call 10080800h
0x10081707: add esp,10h
0x1008170a: mov eax,edi
0x1008170c: pop edi
0x1008170d: pop ebx
0x1008170e: ret
0x1008170f: push ebp
0x10081710: mov ebp,[esi+2A8h]
0x10081716: xor edi,edi
0x10081718: cmp ebp,[esi+2ACh]
0x1008171e: jge short 10081744h
0x10081720: mov eax,[esi+3C8h]
0x10081726: push esi
0x10081727: call eax
0x10081729: mov edi,eax
0x1008172b: add esp,4
0x1008172e: cmp edi,ebx
0x10081730: jne short 1008173Dh
0x10081732: push esi
0x10081733: call 10098A00h
0x10081738: add esp,4
0x1008173b: mov edi,eax
0x1008173d: add ebp,1
0x10081740: cmp edi,ebx
0x10081742: je short 10081718h
0x10081744: mov eax,[esi+284h]
0x1008174a: cmp eax,ebx
0x1008174c: pop ebp
0x1008174d: je short 10081768h
0x1008174f: mov ecx,[esi+25Ch]
0x10081755: mov edx,[esi+9Ch]
0x1008175b: push ecx
0x1008175c: push eax
0x1008175d: call edx
0x1008175f: add esp,8
0x10081762: mov [esi+284h],ebx
0x10081768: mov eax,edi
0x1008176a: pop edi
0x1008176b: pop ebx
0x1008176c: ret
```

`sub_10081660` is called by sub_10086c00. It delegates to sub_10080800, sub_10080b50, sub_10098a00 (3 callees).

12.1092 sub_10082df0 at 0x10082df0 (40 instructions)

```
0x10082df0: mov eax,[esp+4]
0x10082df4: push ebx
0x10082df5: push esi
0x10082df6: mov esi,[eax+210h]
0x10082dfc: push 1
0x10082dfe: xor ebx,ebx
0x10082e00: push ebx
0x10082e01: push ebx
0x10082e02: push ebx
0x10082e03: push ebx
0x10082e04: push ebx
0x10082e05: push ebx
0x10082e06: push ebx
0x10082e07: push esi
```

```
0x10082e08: push eax
0x10082e09: call 10087190h
0x10082e0e: add esp,28h
0x10082e11: cmp eax,ebx
0x10082e13: jne short 10082E52h
0x10082e15: mov byte ptr [esi+1Fh],1
0x10082e19: mov byte ptr [esi+20h],1
0x10082e1d: mov [esi+24h],ebx
0x10082e20: mov [esi+28h],ebx
0x10082e23: mov [esi+2Ch],bl
0x10082e26: mov [esi+30h],ebx
0x10082e29: mov [esi+34h],ebx
0x10082e2c: mov [esi+38h],ebx
0x10082e2f: mov [esi+3Ch],ebx
0x10082e32: mov [esi+40h],bl
0x10082e35: mov [esi+41h],bl
0x10082e38: mov [esi+42h],bx
0x10082e3c: mov [esi+44h],bx
0x10082e40: mov [esi+46h],bx
0x10082e44: mov [esi+48h],bx
0x10082e48: mov [esi+4Ah],bx
0x10082e4c: mov [esi+4Ch],bx
0x10082e50: xor eax,eax
0x10082e52: pop esi
0x10082e53: pop ebx
0x10082e54: ret
```

`sub_10082df0` is called by sub_1007f450, sub_1007f790, sub_10082e60. It delegates to sub_10087190 (1 callees).

12.1093 sub_10082e60 at 0x10082e60 (251 instructions)

```
0x10082e60: push ebp
0x10082e61: mov ebp,[esp+8]
0x10082e65: push ebp
0x10082e66: call 10082DF0h
0x10082e6b: xor edx,edx
0x10082e6d: add esp,4
0x10082e70: cmp eax,edx
0x10082e72: jne near ptr 10083245h
0x10082e78: push ebx
0x10082e79: mov ebx,[ebp+224h]
0x10082e7f: cmp ebx,edx
0x10082e81: push esi
0x10082e82: push edi
0x10082e83: mov [ebp+1B1h],dl
0x10082e89: je near ptr 10083097h
0x10082e8f: movzx eax,byte ptr [ebx]
0x10082e92: mov [ebp+132h],al
0x10082e98: movzx ecx,byte ptr [ebx+1]
0x10082e9c: mov [ebp+133h],cl
0x10082ea2: movzx eax,byte ptr [ebx+2]
0x10082ea6: mov [ebp+134h],al
0x10082eac: movzx ecx,byte ptr [ebx+3]
0x10082eb0: mov [ebp+135h],cl
0x10082eb6: movzx eax,byte ptr [ebx+4]
0x10082eba: mov [ebp+136h],al
0x10082ec0: movzx ecx,byte ptr [ebx+5]
0x10082ec4: mov [ebp+137h],cl
0x10082eca: movzx eax,byte ptr [ebx+6]
0x10082ece: mov [ebp+138h],al
0x10082ed4: movzx ecx,word ptr [ebx+8]
0x10082ed8: mov [ebp+41Ch],cx
0x10082edf: mov ax,[ebx+0Ah]
0x10082ee3: mov [ebp+41Eh],ax
0x10082eea: movzx ecx,word ptr [ebx+0Ch]
0x10082eee: mov [ebp+420h],cx
```

```
0x10082ef5: movzx eax,byte ptr [ebx+0Eh]
0x10082ef9: mov [ebp+422h],al
0x10082eff: movzx ecx,word ptr [ebx+10h]
0x10082f03: mov [ebp+424h],cx
0x10082f0a: movzx eax,byte ptr [ebx+12h]
0x10082f0e: mov [ebp+426h],al
0x10082f14: movzx ecx,byte ptr [ebx+13h]
0x10082f18: mov [ebp+438h],cl
0x10082f1e: mov eax,[ebx+18h]
0x10082f21: mov [ebp+440h],eax
0x10082f27: movzx ecx,byte ptr [ebx+1Ch]
0x10082f2b: mov [ebp+444h],cl
0x10082f31: mov eax,[ebx+20h]
0x10082f34: mov [ebp+448h],eax
0x10082f3a: mov ecx,[ebx+24h]
0x10082f3d: mov [ebp+44Ch],ecx
0x10082f43: mov eax,[ebx+28h]
0x10082f46: mov [ebp+450h],eax
0x10082f4c: mov ecx,[ebx+2Ch]
0x10082f4f: mov eax,[ebp+45Ch]
0x10082f55: mov [ebp+454h],ecx
0x10082f5b: mov [ebp+478h],eax
0x10082f61: mov ecx,[ebx+30h]
0x10082f64: mov [ebp+488h],ecx
0x10082f6a: lea esi,[ebx+34h]
0x10082f6d: lea edi,[ebp+48Ch]
0x10082f73: mov ecx,0C0h
0x10082f78: rep movsd
0x10082f7a: mov eax,[ebx+334h]
0x10082f80: mov [ebp+78Ch],eax
0x10082f86: lea esi,[ebx+338h]
0x10082f8c: lea edi,[ebp+790h]
0x10082f92: mov ecx,40h
0x10082f97: rep movsd
0x10082f99: mov ecx,[ebx+438h]
0x10082f9f: mov [ebp+890h],ecx
0x10082fa5: mov eax,[ebx+43Ch]
0x10082fab: mov [ebp+894h],eax
0x10082fb1: mov ecx,[ebx+440h]
0x10082fb7: mov [ebp+898h],ecx
0x10082fbd: mov eax,[ebx+444h]
0x10082fc3: mov [ebp+89Ch],eax
0x10082fc9: mov ecx,[ebx+448h]
0x10082fcf: mov [ebp+8A0h],ecx
0x10082fd5: mov eax,[ebx+44Ch]
0x10082fdb: mov [ebp+8A4h],eax
0x10082fe1: mov ecx,[ebx+450h]
0x10082fe7: mov [ebp+8A8h],ecx
0x10082fed: mov eax,[ebx+454h]
0x10082ff3: mov [ebp+8ACh],eax
0x10082ff9: mov ecx,[ebx+458h]
0x10082fff: mov [ebp+8B0h],ecx
0x10083005: movzx eax,byte ptr [ebx+45Ch]
0x1008300c: mov [ebp+8B4h],al
0x10083012: mov eax,[ebx+460h]
0x10083018: cmp eax,edx
0x1008301a: mov [ebp+8B8h],eax
0x10083020: je short 10083068h
0x10083022: mov ecx,[ebp+98h]
0x10083028: push eax
0x10083029: call ecx
0x1008302b: add esp,4
0x1008302e: test eax,eax
0x10083030: mov [ebp+8BCh],eax
0x10083036: jne short 1008304Fh
0x10083038: push eax
0x10083039: push eax
```

```
0x1008303a: push 1
0x1008303c: push ebp
0x1008303d: call 10080800h
0x10083042: add esp,10h
0x10083045: pop edi
0x10083046: pop esi
0x10083047: pop ebx
0x10083048: mov eax,1
0x1008304d: pop ebp
0x1008304e: ret
0x1008304f: mov edx,[ebp+8B8h]
0x10083055: mov ecx,[ebx+464h]
0x1008305b: push edx
0x1008305c: push ecx
0x1008305d: push eax
0x1008305e: call 100228C6h
0x10083063: add esp,0Ch
0x10083066: xor edx,edx
0x10083068: movzx eax,word ptr [ebx+468h]
0x1008306f: mov [ebp+8C0h],ax
0x10083076: mov cx,[ebx+46Ah]
0x1008307d: mov [ebp+8C2h],cx
0x10083084: movzx eax,word ptr [ebx+46Ch]
0x1008308b: mov [ebp+8C4h],ax
0x10083092: jmp near ptr 100831BCh
0x10083097: cmp [ebp+1F7h],dl
0x1008309d: mov [ebp+132h],dl
0x100830a3: mov [ebp+133h],dl
0x100830a9: mov [ebp+134h],dl
0x100830af: mov [ebp+135h],dl
0x100830b5: mov [ebp+136h],dl
0x100830bb: mov [ebp+137h],dl
0x100830c1: mov [ebp+138h],dl
0x100830c7: jne short 100830F1h
0x100830c9: mov [ebp+41Ch],dx
0x100830d0: mov [ebp+41Eh],dx
0x100830d7: mov [ebp+420h],dx
0x100830de: mov [ebp+422h],dl
0x100830e4: mov [ebp+424h],dx
0x100830eb: mov [ebp+426h],dl
0x100830f1: mov ecx,[ebp+45Ch]
0x100830f7: mov [ebp+478h],ecx
0x100830fd: mov ecx,[ebp+8B8h]
0x10083103: cmp ecx,edx
0x10083105: mov byte ptr [ebp+438h],1
0x1008310c: mov dword ptr [ebp+440h],7FFFFFFFh
0x10083116: mov [ebp+444h],dl
0x1008311c: mov [ebp+448h],edx
0x10083122: mov [ebp+44Ch],edx
0x10083128: mov [ebp+450h],edx
0x1008312e: mov [ebp+454h],edx
0x10083134: mov [ebp+488h],edx
0x1008313a: mov [ebp+78Ch],edx
0x10083140: mov [ebp+890h],edx
0x10083146: mov [ebp+894h],edx
0x1008314c: mov [ebp+898h],edx
0x10083152: mov [ebp+89Ch],edx
0x10083158: mov [ebp+8A0h],edx
0x1008315e: mov [ebp+8A4h],edx
0x10083164: mov [ebp+8A8h],edx
0x1008316a: mov [ebp+8ACh],edx
0x10083170: mov [ebp+8B0h],edx
0x10083176: mov [ebp+8B4h],dl
0x1008317c: je short 100831A1h
0x1008317e: mov eax,[ebp+8BCh]
0x10083184: cmp eax,edx
0x10083186: je short 100831A1h
```

```
0x10083188: mov edx,[ebp+9Ch]
0x1008318e: push ecx
0x1008318f: push eax
0x10083190: call edx
0x10083192: add esp,8
0x10083195: mov dword ptr [ebp+8BCh],0
0x1008319f: xor edx,edx
0x100831a1: mov [ebp+8B8h],edx
0x100831a7: mov [ebp+8C0h],dx
0x100831ae: mov [ebp+8C2h],dx
0x100831b5: mov [ebp+8C4h],dx
0x100831bc: cmp byte ptr [ebp+1F7h],0
0x100831c3: jne short 10083240h
0x100831c5: mov esi,[ebp+2B0h]
0x100831cb: cmp esi,edx
0x100831cd: je short 10083240h
0x100831cf: nop
0x100831d0: cmp byte ptr [esi+1Eh],0
0x100831d4: mov ebx,[esi+8]
0x100831d7: jne short 1008323Ah
0x100831d9: mov eax,[esi+0Ch]
0x100831dc: cmp eax,edx
0x100831de: je short 100831E5h
0x100831e0: mov [eax+8],ebx
0x100831e3: jmp short 100831EBh
0x100831e5: mov [ebp+2B0h],ebx
0x100831eb: cmp ebx,edx
0x100831ed: je short 100831F4h
0x100831ef: mov [ebx+0Ch],eax
0x100831f2: jmp short 100831FAh
0x100831f4: mov [ebp+2B4h],eax
0x100831fa: mov eax,[esi+58h]
0x100831fd: cmp byte ptr [eax+20h],0
0x10083201: je short 1008321Dh
0x10083203: cmp dword ptr [eax+1Ch],2
0x10083207: jb short 10083247h
0x10083209: add dword ptr [eax+1Ch],0FFFFFFFFh
0x1008320d: mov eax,[ebp+9Ch]
0x10083213: push 5Ch
0x10083215: push esi
0x10083216: call eax
0x10083218: add esp,8
0x1008321b: jmp short 10083238h
0x1008321d: push eax
0x1008321e: push ebp
0x1008321f: call 10086EA0h
0x10083224: mov ecx,[ebp+9Ch]
0x1008322a: push 5Ch
0x1008322c: push esi
0x1008322d: mov edi,eax
0x1008322f: call ecx
0x10083231: add esp,10h
0x10083234: test edi,edi
0x10083236: jne short 10083261h
0x10083238: xor edx,edx
0x1008323a: cmp ebx,edx
0x1008323c: mov esi,ebx
0x1008323e: jne short 100831D0h
0x10083240: pop edi
0x10083241: pop esi
0x10083242: xor eax,eax
0x10083244: pop ebx
0x10083245: pop ebp
0x10083246: ret
0x10083247: push edx
0x10083248: push edx
0x10083249: push 3E7h
```

```
0x1008324e: push ebp
0x1008324f: call 10080800h
0x10083254: add esp,10h
0x10083257: pop edi
0x10083258: pop esi
0x10083259: pop ebx
0x1008325a: mov eax,3E7h
0x1008325f: pop ebp
0x10083260: ret
0x10083261: mov eax,edi
0x10083263: pop edi
0x10083264: pop esi
0x10083265: pop ebx
0x10083266: pop ebp
0x10083267: ret
```

`sub_10082e60` is called by sub_10083b00. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to j_memcpy, sub_10080800, sub_10082df0 (4 callees).

12.1094 sub_10083b00 at 0x10083b00 (159 instructions)

```
0x10083b00: push ebp
0x10083b01: push esi
0x10083b02: mov esi,[esp+0Ch]
0x10083b06: cmp byte ptr [esi+1B0h],0
0x10083b0d: push edi
0x10083b0e: je near ptr 10083D04h
0x10083b14: cmp byte ptr [esi+1F0h],0
0x10083b1b: je near ptr 10083CFBh
0x10083b21: cmp byte ptr [esi+130h],0
0x10083b28: je near ptr 10083CFBh
0x10083b2e: mov edi,[esi+208h]
0x10083b34: test edi,edi
0x10083b36: je near ptr 10083CFBh
0x10083b3c: mov ebp,1
0x10083b41: add [esi+20Ch],ebp
0x10083b47: movzx eax,byte ptr [edi+1Ch]
0x10083b4b: sub eax,ebp
0x10083b4d: je near ptr 10083CB6h
0x10083b53: sub eax,ebp
0x10083b55: je near ptr 10083C6Dh
0x10083b5b: sub eax,ebp
0x10083b5d: jne near ptr 10083CFBh
0x10083b63: mov eax,[edi+24h]
0x10083b66: test eax,eax
0x10083b68: je short 10083B77h
0x10083b6a: cmp eax,7FFFFFFFh
0x10083b6f: jae short 10083B77h
0x10083b71: add eax,0FFFFFFFFh
0x10083b74: mov [edi+24h],eax
0x10083b77: cmp dword ptr [edi+24h],0
0x10083b7b: je short 10083BEBh
0x10083b7d: push esi
0x10083b7e: call 10082E60h
0x10083b83: add esp,4
0x10083b86: test eax,eax
0x10083b88: jne near ptr 10083D3Bh
0x10083b8e: mov eax,[esi+0D0h]
0x10083b94: test eax,eax
0x10083b96: je short 10083BB1h
0x10083b98: mov ecx,[edi+24h]
0x10083b9b: mov edx,[esi+20Ch]
0x10083ba1: push ecx
0x10083ba2: push edx
0x10083ba3: push esi
0x10083ba4: call eax
```

```
0x10083ba6: add esp,0Ch
0x10083ba9: test al,al
0x10083bab: je near ptr 10083C9Bh
0x10083bb1: mov [esi+204h],edi
0x10083bb7: mov edi,[edi+20h]
0x10083bba: test edi,edi
0x10083bbc: je near ptr 10083CFBh
0x10083bc2: cmp edi,[esi+45Ch]
0x10083bc8: jbe short 10083BD0h
0x10083bca: mov [esi+45Ch],edi
0x10083bd0: cmp dword ptr [esi+45Ch],0
0x10083bd7: jne short 10083BDFh
0x10083bd9: mov [esi+45Ch],ebp
0x10083bdf: mov byte ptr [esi+1F6h],1
0x10083be6: jmp near ptr 10083CFBh
0x10083beb: movzx eax,byte ptr [edi+1Dh]
0x10083bef: sub eax,ebp
0x10083bf1: je near ptr 10083CB6h
0x10083bf7: sub eax,ebp
0x10083bf9: jne near ptr 10083CFBh
0x10083bff: push esi
0x10083c00: call 10082E60h
0x10083c05: add esp,4
0x10083c08: test eax,eax
0x10083c0a: jne near ptr 10083D3Bh
0x10083c10: mov eax,[esi+0D0h]
0x10083c16: test eax,eax
0x10083c18: je short 10083C2Dh
0x10083c1a: mov ecx,[esi+20Ch]
0x10083c20: push 0
0x10083c22: push ecx
0x10083c23: push esi
0x10083c24: call eax
0x10083c26: add esp,0Ch
0x10083c29: test al,al
0x10083c2b: je short 10083C9Bh
0x10083c2d: mov [esi+204h],edi
0x10083c33: mov byte ptr [esi+1F8h],1
0x10083c3a: mov dword ptr [esi+1FCh],0
0x10083c44: mov edi,[edi+20h]
0x10083c47: cmp edi,[esi+45Ch]
0x10083c4d: jbe short 10083C55h
0x10083c4f: mov [esi+45Ch],edi
0x10083c55: cmp dword ptr [esi+45Ch],0
0x10083c5c: jne near ptr 10083CFBh
0x10083c62: mov [esi+45Ch],ebp
0x10083c68: jmp near ptr 10083CFBh
0x10083c6d: push esi
0x10083c6e: call 10082E60h
0x10083c73: add esp,4
0x10083c76: test eax,eax
0x10083c78: jne near ptr 10083D3Bh
0x10083c7e: mov eax,[esi+0D0h]
0x10083c84: test eax,eax
0x10083c86: je short 10083C2Dh
0x10083c88: mov edx,[esi+20Ch]
0x10083c8e: push 0
0x10083c90: push edx
0x10083c91: push esi
0x10083c92: call eax
0x10083c94: add esp,0Ch
0x10083c97: test al,al
0x10083c99: jne short 10083C2Dh
0x10083c9b: push 0
0x10083c9d: push 0
0x10083c9f: push 388h
0x10083ca4: push esi
```

```
0x10083ca5: call 10080800h
0x10083caa: add esp,10h
0x10083cad: pop edi
0x10083cae: pop esi
0x10083caf: mov eax,388h
0x10083cb4: pop ebp
0x10083cb5: ret
0x10083cb6: mov edi,[edi+20h]
0x10083cb9: cmp edi,[esi+45Ch]
0x10083cbf: jbe short 10083CC7h
0x10083cc1: mov [esi+45Ch],edi
0x10083cc7: cmp dword ptr [esi+45Ch],0
0x10083cce: jne short 10083CD6h
0x10083cd0: mov [esi+45Ch],ebp
0x10083cd6: mov eax,esi
0x10083cd8: call 100809D0h
0x10083cdd: test eax,eax
0x10083cdf: jne short 10083D3Bh
0x10083ce1: cmp [esi+1F1h],al
0x10083ce7: jne short 10083CF4h
0x10083ce9: mov eax,ebp
0x10083ceb: call 10080870h
0x10083cf0: test eax,eax
0x10083cf2: jne short 10083D3Bh
0x10083cf4: mov byte ptr [esi+1F2h],0Ah
0x10083cfb: cmp byte ptr [esi+1B0h],0
0x10083d02: jne short 10083D29h
0x10083d04: cmp byte ptr [esi+158h],0
0x10083d0b: je short 10083D29h
0x10083d0d: mov eax,[esi+0D0h]
0x10083d13: test eax,eax
0x10083d15: je short 10083D29h
0x10083d17: push 0
0x10083d19: push 0
0x10083d1b: push esi
0x10083d1c: call eax
0x10083d1e: add esp,0Ch
0x10083d21: test al,al
0x10083d23: je near ptr 10083C9Bh
0x10083d29: cmp dword ptr [esi+204h],0
0x10083d30: jne short 10083D39h
0x10083d32: mov byte ptr [esi+1F6h],1
0x10083d39: xor eax,eax
0x10083d3b: pop edi
0x10083d3c: pop esi
0x10083d3d: pop ebp
0x10083d3e: ret
```

`sub_10083b00` is called by sub_10086c00, sub_1008c510, sub_1008c980. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to sub_10080800, sub_10080870, sub_100809d0 (4 callees).

12.1095 sub_10086c00 at 0x10086c00 (142 instructions)

```
0x10086c00: push ebx
0x10086c01: push esi
0x10086c02: mov esi,[esp+0Ch]
0x10086c06: xor ebx,ebx
0x10086c08: xor eax,eax
0x10086c0a: cmp [esi+1F2h],bl
0x10086c10: jne short 10086C50h
0x10086c12: cmp [esi+1D0h],ebx
0x10086c18: jne short 10086C2Ah
0x10086c1a: cmp [esi+1D4h],ebx
0x10086c20: jne short 10086C2Ah
0x10086c22: cmp [esi+1D8h],ebx
0x10086c28: je short 10086C50h
```

```
0x10086c2a: mov byte ptr [esi+1DCh],1
0x10086c31: call 10081660h
0x10086c36: cmp eax,ebx
0x10086c38: jne near ptr 10086E34h
0x10086c3e: mov ecx,[esi+2B8h]
0x10086c44: mov [esi+204h],ecx
0x10086c4a: lea ebx,[ebx]
0x10086c50: cmp [esi+1DCh],bl
0x10086c56: je short 10086C5Eh
0x10086c58: mov [esi+1F1h],bl
0x10086c5e: mov cl,[esi+1F2h]
0x10086c64: cmp cl,bl
0x10086c66: je short 10086CC5h
0x10086c68: cmp cl,63h
0x10086c6b: jae short 10086CC5h
0x10086c6d: movzx eax,cl
0x10086c70: add eax,0FFFFFFFFh
0x10086c73: cmp eax,0Ah
0x10086c76: ja near ptr 10086DFEh
0x10086c7c: jmp dword ptr [eax*4+10086E38h]
0x10086cc5: mov ecx,[esi+204h]
0x10086ccb: cmp ecx,ebx
0x10086ccd: je short 10086CD9h
0x10086ccf: mov edx,[ecx+4]
0x10086cd2: push ecx
0x10086cd3: push esi
0x10086cd4: call edx
0x10086cd6: add esp,8
0x10086cd9: mov dl,[esi+1F1h]
0x10086cdf: cmp dl,bl
0x10086ce1: jne short 10086CE9h
0x10086ce3: mov [esi+1F2h],bl
0x10086ce9: cmp eax,ebx
0x10086ceb: jne short 10086D2Eh
0x10086ced: mov ecx,[esi+204h]
0x10086cf3: cmp ecx,ebx
0x10086cf5: je short 10086D2Eh
0x10086cf7: cmp dl,bl
0x10086cf9: jne short 10086D2Eh
0x10086cfb: cmp [esi+1F3h],bl
0x10086d01: jne short 10086D2Eh
0x10086d03: cmp dword ptr [esi+0Ch],2
0x10086d07: mov ecx,[ecx+8]
0x10086d0a: mov [esi+204h],ecx
0x10086d10: jne short 10086D1Fh
0x10086d12: cmp ecx,ebx
0x10086d14: jne short 10086D1Fh
0x10086d16: push esi
0x10086d17: call 10083B00h
0x10086d1c: add esp,4
0x10086d1f: cmp [esi+204h],ebx
0x10086d25: jne short 10086D2Eh
0x10086d27: mov byte ptr [esi+1F6h],1
0x10086d2e: cmp [esi+1DCh],bl
0x10086d34: je short 10086D88h
0x10086d36: mov ecx,[esi+1D0h]
0x10086d3c: cmp ecx,ebx
0x10086d3e: je short 10086D50h
0x10086d40: cmp ecx,[esi+1B4h]
0x10086d46: ja short 10086D50h
0x10086d48: mov [esi+1D0h],ebx
0x10086d4e: jmp short 10086D82h
0x10086d50: mov ecx,[esi+1D4h]
0x10086d56: cmp ecx,ebx
0x10086d58: je short 10086D6Ah
0x10086d5a: cmp ecx,[esi+1B8h]
0x10086d60: ja short 10086D6Ah
```

```
0x10086d62: mov [esi+1D4h],ebx
0x10086d68: jmp short 10086D82h
0x10086d6a: mov ecx,[esi+1D8h]
0x10086d70: cmp ecx,ebx
0x10086d72: je short 10086D88h
0x10086d74: cmp ecx,[esi+1BCh]
0x10086d7a: ja short 10086D88h
0x10086d7c: mov [esi+1D8h],ebx
0x10086d82: mov [esi+1DCh],bl
0x10086d88: cmp eax,ebx
0x10086d8a: jne near ptr 10086E34h
0x10086d90: cmp [esi+204h],ebx
0x10086d96: je short 10086DC4h
0x10086d98: cmp [esi+1F0h],bl
0x10086d9e: je short 10086DA8h
0x10086da0: cmp [esi+1F1h],bl
0x10086da6: je short 10086DB0h
0x10086da8: cmp [esi+1DCh],bl
0x10086dae: je short 10086DC4h
0x10086db0: cmp [esi+1F3h],bl
0x10086db6: jne short 10086DC4h
0x10086db8: cmp [esi+1F4h],bl
0x10086dbe: je near ptr 10086C50h
0x10086dc4: cmp [esi+1F1h],bl
0x10086dca: jne short 10086DE3h
0x10086dcc: cmp [esi+1F6h],bl
0x10086dd2: je short 10086DE3h
0x10086dd4: push 1
0x10086dd6: push esi
0x10086dd7: call 10080920h
0x10086ddc: add esp,8
0x10086ddf: cmp eax,ebx
0x10086de1: jne short 10086E34h
0x10086de3: mov al,[esi+1F1h]
0x10086de9: cmp al,bl
0x10086deb: je short 10086E1Ah
0x10086ded: cmp [esi+1F2h],bl
0x10086df3: jne short 10086E16h
0x10086df5: mov byte ptr [esi+1F2h],63h
0x10086dfc: jmp short 10086E20h
0x10086dfe: push ebx
0x10086dff: push ebx
0x10086e00: push 3E7h
0x10086e05: push esi
0x10086e06: call 10080800h
0x10086e0b: add esp,10h
0x10086e0e: pop esi
0x10086e0f: mov eax,3E7h
0x10086e14: pop ebx
0x10086e15: ret
0x10086e16: cmp al,bl
0x10086e18: jne short 10086E32h
0x10086e1a: mov [esi+1F2h],bl
0x10086e20: cmp al,bl
0x10086e22: jne short 10086E32h
0x10086e24: cmp [esi+204h],ebx
0x10086e2a: jne short 10086E32h
0x10086e2c: mov [esi+1F0h],bl
0x10086e32: xor eax,eax
0x10086e34: pop esi
0x10086e35: pop ebx
0x10086e36: ret
```

`sub_10086c00` is called by sub_10080150. It delegates to sub_10080800, sub_10080920, sub_10081660 (4 callees).

12.1096 sub_10086e70 at 0x10086e70 (21 instructions)

```
0x10086e70: push edi
0x10086e71: mov edi,[esp+8]
0x10086e75: mov eax,[edi+2B0h]
0x10086e7b: test eax,eax
0x10086e7d: je short 10086E99h
0x10086e7f: push esi
0x10086e80: cmp byte ptr [eax+21h],0
0x10086e84: mov esi,[eax+8]
0x10086e87: jne short 10086E92h
0x10086e89: push eax
0x10086e8a: mov eax,[eax]
0x10086e8c: push edi
0x10086e8d: call eax
0x10086e8f: add esp,8
0x10086e92: test esi,esi
0x10086e94: mov eax,esi
0x10086e96: jne short 10086E80h
0x10086e98: pop esi
0x10086e99: xor eax,eax
0x10086e9b: pop edi
0x10086e9c: ret
```

`sub_10086e70` is called by sub_1007f450, sub_1007ff70, sub_10080150. It calls nothing else.

12.1097 sub_10086ea0 at 0x10086ea0 (42 instructions)

```
0x10086ea0: push esi
0x10086ea1: mov esi,[esp+0Ch]
0x10086ea5: mov eax,[esi+1Ch]
0x10086ea8: test eax,eax
0x10086eaa: je short 10086EB2h
0x10086eac: add eax,0FFFFFFFFh
0x10086eaf: mov [esi+1Ch],eax
0x10086eb2: cmp dword ptr [esi+1Ch],0
0x10086eb6: jne short 10086F11h
0x10086eb8: mov eax,[esi+478h]
0x10086ebe: test eax,eax
0x10086ec0: push edi
0x10086ec1: mov edi,[esp+0Ch]
0x10086ec5: je short 10086EDEh
0x10086ec7: mov ecx,[esi+47Ch]
0x10086ecd: test ecx,ecx
0x10086ecf: je short 10086EDEh
0x10086ed1: push eax
0x10086ed2: mov eax,[edi+9Ch]
0x10086ed8: push ecx
0x10086ed9: call eax
0x10086edb: add esp,8
0x10086ede: mov ecx,[esi+494h]
0x10086ee4: test ecx,ecx
0x10086ee6: je short 10086EFFh
0x10086ee8: mov eax,[esi+498h]
0x10086eee: test eax,eax
0x10086ef0: je short 10086EFFh
0x10086ef2: push ecx
0x10086ef3: mov ecx,[edi+9Ch]
0x10086ef9: push eax
0x10086efa: call ecx
0x10086efc: add esp,8
0x10086eff: mov edx,[edi+9Ch]
0x10086f05: push 49Ch
0x10086f0a: push esi
0x10086f0b: call edx
0x10086f0d: add esp,8
0x10086f10: pop edi
```

```
0x10086f11: xor eax,eax
0x10086f13: pop esi
0x10086f14: ret
```

`sub_10086ea0` is called by sub_10082e60, sub_10087020. It calls nothing else.

12.1098 sub_10087020 at 0x10087020 (36 instructions)

```
0x10087020: push ebx
0x10087021: mov ebx,[esp+8]
0x10087025: push esi
0x10087026: mov esi,[esp+10h]
0x1008702a: cmp word ptr [esi+1Ch],0
0x1008702f: mov ecx,[esi+0Ch]
0x10087032: mov eax,[esi+8]
0x10087035: mov edx,[esi+58h]
0x10087038: push edi
0x10087039: je short 1008705Fh
0x1008703b: test ecx,ecx
0x1008703d: je short 10087044h
0x1008703f: mov [ecx+8],eax
0x10087042: jmp short 1008704Ah
0x10087044: mov [ebx+2B0h],eax
0x1008704a: test eax,eax
0x1008704c: je short 10087056h
0x1008704e: mov ecx,[esi+0Ch]
0x10087051: mov [eax+0Ch],ecx
0x10087054: jmp short 1008705Fh
0x10087056: mov eax,[esi+0Ch]
0x10087059: mov [ebx+2B4h],eax
0x1008705f: push edx
0x10087060: push ebx
0x10087061: call 10086EA0h
0x10087066: mov ecx,[ebx+9Ch]
0x1008706c: push 5Ch
0x1008706e: push esi
0x1008706f: mov edi,eax
0x10087071: call ecx
0x10087073: add esp,10h
0x10087076: mov eax,edi
0x10087078: pop edi
0x10087079: pop esi
0x1008707a: pop ebx
0x1008707b: ret
```

`sub_10087020` is called by sub_1007ff10. It delegates to sub_10086ea0 (1 callees).

12.1099 sub_10087190 at 0x10087190 (225 instructions)

```
0x10087190: push ecx
0x10087191: mov ecx,[esp+0Ch]
0x10087195: mov eax,[esp+10h]
0x10087199: mov edx,[esp+14h]
0x1008719d: push ebx
0x1008719e: mov bl,[esp+24h]
0x100871a2: push ebp
0x100871a3: push esi
0x100871a4: mov esi,[ecx+58h]
0x100871a7: mov [esi+24h],eax
0x100871aa: mov al,[esp+28h]
0x100871ae: mov [esi+2Eh],bl
0x100871b1: mov bl,[esp+30h]
0x100871b5: mov [esi+2Dh],al
0x100871b8: movzx eax,al
0x100871bb: cmp eax,0Eh
```

```
0x100871be: mov [esi+28h],edx
0x100871c1: mov dl,[esp+24h]
0x100871c5: mov [esi+2Fh],bl
0x100871c8: mov bl,[esp+34h]
0x100871cc: push edi
0x100871cd: mov dword ptr [esp+10h],0
0x100871d5: mov [esi+2Ch],dl
0x100871d8: mov [esi+30h],bl
0x100871db: mov byte ptr [esi+31h],0
0x100871df: mov byte ptr [esi+32h],0
0x100871e3: ja short 10087242h
0x100871e5: movzx eax,byte ptr [eax+10087544h]
0x100871ec: jmp dword ptr [eax*4+1008752Ch]
0x10087242: mov edi,[esp+10h]
0x10087246: imul edi,[esp+20h]
0x1008724b: mov eax,[esi+494h]
0x10087251: mov ebp,[esp+18h]
0x10087255: mov ebx,edi
0x10087257: imul ebx,[esp+24h]
0x1008725c: cmp ebx,eax
0x1008725e: mov [esp+30h],edi
0x10087262: je short 100872C1h
0x10087264: mov edx,[esi+498h]
0x1008726a: test edx,edx
0x1008726c: je short 10087289h
0x1008726e: mov ecx,[ebp+9Ch]
0x10087274: push eax
0x10087275: push edx
0x10087276: call ecx
0x10087278: mov ecx,[esp+24h]
0x1008727c: add esp,8
0x1008727f: mov dword ptr [esi+498h],0
0x10087289: test ebx,ebx
0x1008728b: je near ptr 10087318h
0x10087291: mov edx,[ebp+98h]
0x10087297: push ebx
0x10087298: call edx
0x1008729a: add esp,4
0x1008729d: test eax,eax
0x1008729f: mov [esi+498h],eax
0x100872a5: jne short 10087314h
0x100872a7: push 0
0x100872a9: push 0
0x100872ab: push 1
0x100872ad: push ebp
0x100872ae: call 10080800h
0x100872b3: add esp,10h
0x100872b6: pop edi
0x100872b7: pop esi
0x100872b8: pop ebp
0x100872b9: mov eax,1
0x100872be: pop ebx
0x100872bf: pop ecx
0x100872c0: ret
0x100872c1: test ebx,ebx
0x100872c3: je short 1008731Ch
0x100872c5: mov edx,[esi+498h]
0x100872cb: mov eax,ebx
0x100872cd: and eax,0FFFFFFFCh
0x100872d0: jbe short 100872F4h
0x100872d2: lea ecx,[eax-1]
0x100872d5: shr ecx,2
0x100872d8: add ecx,1
0x100872db: mov [esp+2Ch],ecx
0x100872df: xor eax,eax
0x100872e1: mov edi,edx
0x100872e3: rep stosd
```

```
0x100872e5: mov eax,[esp+2Ch]
0x100872e9: mov edi,[esp+30h]
0x100872ed: mov ecx,[esp+1Ch]
0x100872f1: lea edx,[edx+eax*4]
0x100872f4: mov eax,[esi+498h]
0x100872fa: add eax,ebx
0x100872fc: cmp edx,eax
0x100872fe: jae short 10087318h
0x10087300: mov byte ptr [edx],0
0x10087303: mov eax,[esi+498h]
0x10087309: add edx,1
0x1008730c: add eax,ebx
0x1008730e: cmp edx,eax
0x10087310: jb short 10087300h
0x10087312: jmp short 10087318h
0x10087314: mov ecx,[esp+1Ch]
0x10087318: mov dl,[esp+28h]
0x1008731c: cmp byte ptr [esi+35h],0
0x10087320: mov eax,[esp+10h]
0x10087324: mov [esi+48Ch],eax
0x1008732a: mov [esi+490h],edi
0x10087330: mov [esi+494h],ebx
0x10087336: jne short 1008733Bh
0x10087338: mov [esi+35h],dl
0x1008733b: cmp byte ptr [esi+36h],0
0x1008733f: jne short 10087344h
0x10087341: mov [esi+36h],dl
0x10087344: mov edi,[esp+20h]
0x10087348: xor edx,edx
0x1008734a: cmp edi,edx
0x1008734c: je short 10087367h
0x1008734e: mov eax,[esp+24h]
0x10087352: cmp eax,edx
0x10087354: je short 10087367h
0x10087356: cmp [ecx+2Ch],dl
0x10087359: jne short 10087367h
0x1008735b: mov [ecx+30h],edx
0x1008735e: mov [ecx+34h],edi
0x10087361: mov [ecx+38h],edx
0x10087364: mov [ecx+3Ch],eax
0x10087367: cmp [ecx+1Ch],dx
0x1008736b: je short 1008738Dh
0x1008736d: mov byte ptr [ecx+40h],0
0x10087371: mov byte ptr [ecx+41h],0
0x10087375: mov [ecx+42h],dx
0x10087379: mov [ecx+44h],dx
0x1008737d: mov [ecx+46h],dx
0x10087381: mov [ecx+48h],dx
0x10087385: mov [ecx+4Ah],dx
0x10087389: mov [ecx+4Ch],dx
0x1008738d: cmp byte ptr [esp+3Ch],0
0x10087392: je near ptr 10087521h
0x10087398: mov [ecx+50h],edx
0x1008739b: mov [ecx+54h],edx
0x1008739e: mov byte ptr [esi+37h],0
0x100873a2: mov byte ptr [esi+38h],0
0x100873a6: movzx ecx,byte ptr [ebp+134h]
0x100873ad: mov [esi+39h],cl
0x100873b0: movzx eax,byte ptr [ebp+135h]
0x100873b7: mov [esi+3Ah],al
0x100873ba: movzx ecx,byte ptr [ebp+136h]
0x100873c1: mov [esi+3Bh],cl
0x100873c4: movzx eax,byte ptr [ebp+137h]
0x100873cb: mov [esi+3Ch],al
0x100873ce: movzx ecx,byte ptr [ebp+138h]
0x100873d5: mov eax,[esi+478h]
0x100873db: cmp eax,edx
```

```
0x100873dd: mov [esi+3Dh],cl
0x100873e0: je short 1008740Bh
0x100873e2: mov ecx,[esi+47Ch]
0x100873e8: cmp ecx,edx
0x100873ea: je short 10087405h
0x100873ec: mov edx,[ebp+9Ch]
0x100873f2: push eax
0x100873f3: push ecx
0x100873f4: call edx
0x100873f6: add esp,8
0x100873f9: mov dword ptr [esi+47Ch],0
0x10087403: xor edx,edx
0x10087405: mov [esi+478h],edx
0x1008740b: cmp byte ptr [ebp+134h],0
0x10087412: je short 10087420h
0x10087414: mov eax,[ebp+890h]
0x1008741a: mov [esi+450h],eax
0x10087420: cmp byte ptr [ebp+135h],0
0x10087427: je short 10087489h
0x10087429: mov ecx,[ebp+894h]
0x1008742f: mov [esi+454h],ecx
0x10087435: mov eax,[ebp+898h]
0x1008743b: mov [esi+458h],eax
0x10087441: mov ecx,[ebp+89Ch]
0x10087447: mov [esi+45Ch],ecx
0x1008744d: mov eax,[ebp+8A0h]
0x10087453: mov [esi+460h],eax
0x10087459: mov ecx,[ebp+8A4h]
0x1008745f: mov [esi+464h],ecx
0x10087465: mov eax,[ebp+8A8h]
0x1008746b: mov [esi+468h],eax
0x10087471: mov ecx,[ebp+8ACh]
0x10087477: mov [esi+46Ch],ecx
0x1008747d: mov eax,[ebp+8B0h]
0x10087483: mov [esi+470h],eax
0x10087489: cmp byte ptr [ebp+136h],0
0x10087490: je short 1008749Eh
0x10087492: mov cl,[ebp+8B4h]
0x10087498: mov [esi+474h],cl
0x1008749e: cmp byte ptr [ebp+137h],0
0x100874a5: je short 100874EEh
0x100874a7: mov eax,[ebp+8B8h]
0x100874ad: cmp eax,edx
0x100874af: je short 100874E2h
0x100874b1: mov edx,[ebp+98h]
0x100874b7: push eax
0x100874b8: call edx
0x100874ba: add esp,4
0x100874bd: test eax,eax
0x100874bf: mov [esi+47Ch],eax
0x100874c5: je near ptr 100872A7h
0x100874cb: mov ecx,[ebp+8B8h]
0x100874d1: mov edx,[ebp+8BCh]
0x100874d7: push ecx
0x100874d8: push edx
0x100874d9: push eax
0x100874da: call 100228C6h
0x100874df: add esp,0Ch
0x100874e2: mov eax,[ebp+8B8h]
0x100874e8: mov [esi+478h],eax
0x100874ee: cmp byte ptr [ebp+138h],0
0x100874f5: je short 10087521h
0x100874f7: mov cx,[ebp+8C0h]
0x100874fe: mov [esi+484h],cx
0x10087505: mov dx,[ebp+8C2h]
0x1008750c: mov [esi+486h],dx
0x10087513: mov ax,[ebp+8C4h]
```

```
0x1008751a: mov [esi+488h],ax
0x10087521: pop edi
0x10087522: pop esi
0x10087523: pop ebp
0x10087524: xor eax,eax
0x10087526: pop ebx
0x10087527: pop ecx
0x10087528: ret
```

`sub_10087190` is called by sub_10082df0. It delegates to j_memcpy, sub_10080800 (2 callees).

12.1100 sub_10089620 at 0x10089620 (176 instructions)

```
0x10089620: push ebx
0x10089621: push ebp
0x10089622: push esi
0x10089623: push edi
0x10089624: mov edi,[esp+14h]
0x10089628: mov eax,[edi+98h]
0x1008962e: push 49Ch
0x10089633: xor ebx,ebx
0x10089635: xor ebp,ebp
0x10089637: call eax
0x10089639: mov esi,eax
0x1008963b: add esp,4
0x1008963e: cmp esi,ebx
0x10089640: jne short 10089657h
0x10089642: push ebx
0x10089643: push ebx
0x10089644: push 1
0x10089646: push edi
0x10089647: call 10080800h
0x1008964c: add esp,10h
0x1008964f: pop edi
0x10089650: pop esi
0x10089651: pop ebp
0x10089652: lea eax,[ebx+1]
0x10089655: pop ebx
0x10089656: ret
0x10089657: mov cl,[esp+18h]
0x1008965b: mov dl,[esp+1Ch]
0x1008965f: mov eax,[esp+20h]
0x10089663: mov [esi+21h],cl
0x10089666: mov ecx,[esp+24h]
0x1008966a: mov [esi+22h],dl
0x1008966d: mov dl,[esp+30h]
0x10089671: mov [esi+28h],ecx
0x10089674: mov cl,[esp+2Ch]
0x10089678: mov [esi+2Eh],dl
0x1008967b: mov dl,[esp+34h]
0x1008967f: mov [esi+2Dh],cl
0x10089682: movzx ecx,cl
0x10089685: cmp ecx,0Eh
0x10089688: mov [esi+24h],eax
0x1008968b: mov al,[esp+28h]
0x1008968f: mov [esi+2Fh],dl
0x10089692: mov dl,[esp+38h]
0x10089696: mov dword ptr [esi],10086EA0h
0x1008969c: mov [esi+4],ebx
0x1008969f: mov dword ptr [esi+1Ch],1
0x100896a6: mov [esi+20h],bl
0x100896a9: mov [esi+2Ch],al
0x100896ac: mov [esi+30h],dl
0x100896af: mov [esi+31h],bl
0x100896b2: mov [esi+32h],bl
0x100896b5: mov [esi+33h],bl
```

```
0x100896b8: mov [esi+34h],bl
0x100896bb: mov [esi+35h],al
0x100896be: mov [esi+36h],al
0x100896c1: ja short 1008970Dh
0x100896c3: movzx ecx,byte ptr [ecx+100898F8h]
0x100896ca: jmp dword ptr [ecx*4+100898E0h]
0x1008970d: mov eax,ebp
0x1008970f: imul eax,[esp+20h]
0x10089714: mov [esi+490h],eax
0x1008971a: imul eax,[esp+24h]
0x1008971f: cmp eax,ebx
0x10089721: mov [esi+48Ch],ebp
0x10089727: mov [esi+494h],eax
0x1008972d: je short 1008976Ah
0x1008972f: mov edx,[edi+98h]
0x10089735: push eax
0x10089736: call edx
0x10089738: add esp,4
0x1008973b: cmp eax,ebx
0x1008973d: mov [esi+498h],eax
0x10089743: jne short 1008976Ah
0x10089745: mov eax,[edi+9Ch]
0x1008974b: push 49Ch
0x10089750: push esi
0x10089751: call eax
0x10089753: push ebx
0x10089754: push ebx
0x10089755: push 1
0x10089757: push edi
0x10089758: call 10080800h
0x1008975d: add esp,18h
0x10089760: pop edi
0x10089761: pop esi
0x10089762: pop ebp
0x10089763: mov eax,1
0x10089768: pop ebx
0x10089769: ret
0x1008976a: movzx ecx,byte ptr [edi+134h]
0x10089771: mov [esi+39h],cl
0x10089774: movzx edx,byte ptr [edi+135h]
0x1008977b: mov [esi+3Ah],dl
0x1008977e: mov al,[edi+136h]
0x10089784: mov [esi+3Bh],al
0x10089787: movzx ecx,byte ptr [edi+137h]
0x1008978e: mov [esi+3Ch],cl
0x10089791: movzx edx,byte ptr [edi+138h]
0x10089798: mov [esi+3Dh],dl
0x1008979b: cmp [edi+134h],bl
0x100897a1: je short 100897AFh
0x100897a3: mov eax,[edi+890h]
0x100897a9: mov [esi+450h],eax
0x100897af: cmp [edi+135h],bl
0x100897b5: je short 10089817h
0x100897b7: mov ecx,[edi+894h]
0x100897bd: mov [esi+454h],ecx
0x100897c3: mov edx,[edi+898h]
0x100897c9: mov [esi+458h],edx
0x100897cf: mov eax,[edi+89Ch]
0x100897d5: mov [esi+45Ch],eax
0x100897db: mov ecx,[edi+8A0h]
0x100897e1: mov [esi+460h],ecx
0x100897e7: mov edx,[edi+8A4h]
0x100897ed: mov [esi+464h],edx
0x100897f3: mov eax,[edi+8A8h]
0x100897f9: mov [esi+468h],eax
0x100897ff: mov ecx,[edi+8ACh]
0x10089805: mov [esi+46Ch],ecx
```

```
0x1008980b: mov edx,[edi+8B0h]
0x10089811: mov [esi+470h],edx
0x10089817: cmp [edi+136h],bl
0x1008981d: je short 1008982Bh
0x1008981f: mov al,[edi+8B4h]
0x10089825: mov [esi+474h],al
0x1008982b: cmp [edi+137h],bl
0x10089831: je short 100898A0h
0x10089833: mov eax,[edi+8B8h]
0x10089839: cmp eax,ebx
0x1008983b: mov [esi+478h],eax
0x10089841: je short 100898A0h
0x10089843: mov ecx,[edi+98h]
0x10089849: push eax
0x1008984a: call ecx
0x1008984c: add esp,4
0x1008984f: cmp eax,ebx
0x10089851: mov [esi+47Ch],eax
0x10089857: jne short 10089889h
0x10089859: mov eax,[esi+498h]
0x1008985f: cmp eax,ebx
0x10089861: je short 10089876h
0x10089863: mov edx,[esi+494h]
0x10089869: push edx
0x1008986a: push eax
0x1008986b: mov eax,[edi+9Ch]
0x10089871: call eax
0x10089873: add esp,8
0x10089876: mov ecx,[edi+9Ch]
0x1008987c: push 49Ch
0x10089881: push esi
0x10089882: call ecx
0x10089884: jmp near ptr 10089753h
0x10089889: mov edx,[esi+478h]
0x1008988f: mov ecx,[edi+8BCh]
0x10089895: push edx
0x10089896: push ecx
0x10089897: push eax
0x10089898: call 100228C6h
0x1008989d: add esp,0Ch
0x100898a0: cmp [edi+138h],bl
0x100898a6: je short 100898D2h
0x100898a8: mov dx,[edi+8C0h]
0x100898af: mov [esi+484h],dx
0x100898b6: mov ax,[edi+8C2h]
0x100898bd: mov [esi+486h],ax
0x100898c4: mov cx,[edi+8C4h]
0x100898cb: mov [esi+488h],cx
0x100898d2: mov edx,[esp+3Ch]
0x100898d6: pop edi
0x100898d7: mov [edx],esi
0x100898d9: pop esi
0x100898da: pop ebp
0x100898db: xor eax,eax
0x100898dd: pop ebx
0x100898de: ret
```

`sub_10089620` is called by sub_10089910. It delegates to j_memcpy, sub_10080800 (2 callees).

12.1101 sub_10089910 at 0x10089910 (148 instructions)

```
0x10089910: push ebx
0x10089911: push esi
0x10089912: push edi
0x10089913: mov edi,[esp+10h]
0x10089917: mov eax,[edi+98h]
```

```
0x1008991d: push 5Ch
0x1008991f: call eax
0x10089921: mov esi,eax
0x10089923: xor ebx,ebx
0x10089925: add esp,4
0x10089928: cmp esi,ebx
0x1008992a: jne short 10089940h
0x1008992c: push ebx
0x1008992d: push ebx
0x1008992e: push 1
0x10089930: push edi
0x10089931: call 10080800h
0x10089936: add esp,10h
0x10089939: pop edi
0x1008993a: pop esi
0x1008993b: lea eax,[ebx+1]
0x1008993e: pop ebx
0x1008993f: ret
0x10089940: mov edx,[esp+3Ch]
0x10089944: mov eax,[esp+38h]
0x10089948: push ebp
0x10089949: lea ecx,[esp+14h]
0x1008994d: push ecx
0x1008994e: mov ecx,[esp+3Ch]
0x10089952: push edx
0x10089953: mov edx,[esp+3Ch]
0x10089957: push eax
0x10089958: mov eax,[esp+3Ch]
0x1008995c: push ecx
0x1008995d: mov ecx,[esp+3Ch]
0x10089961: push edx
0x10089962: mov edx,[esp+3Ch]
0x10089966: push eax
0x10089967: mov eax,[esp+3Ch]
0x1008996b: push ecx
0x1008996c: mov ecx,[esp+38h]
0x10089970: push edx
0x10089971: push eax
0x10089972: push ecx
0x10089973: push edi
0x10089974: call 10089620h
0x10089979: mov ebp,eax
0x1008997b: add esp,2Ch
0x1008997e: cmp ebp,ebx
0x10089980: je short 10089997h
0x10089982: mov edx,[edi+9Ch]
0x10089988: push 5Ch
0x1008998a: push esi
0x1008998b: call edx
0x1008998d: add esp,8
0x10089990: mov eax,ebp
0x10089992: pop ebp
0x10089993: pop edi
0x10089994: pop esi
0x10089995: pop ebx
0x10089996: ret
0x10089997: mov cx,[esp+18h]
0x1008999c: mov al,[esp+20h]
0x100899a0: mov dl,[esp+24h]
0x100899a4: mov dword ptr [esi],10087020h
0x100899aa: mov [esi+4],ebx
0x100899ad: mov [esi+1Ch],cx
0x100899b1: mov [esi+1Eh],bl
0x100899b4: mov [esi+1Fh],al
0x100899b7: mov [esi+20h],dl
0x100899ba: cmp [edi+1B0h],bl
0x100899c0: je short 100899DEh
```

```
0x100899c2: cmp [edi+1F0h],bl
0x100899c8: jne short 100899D2h
0x100899ca: cmp [edi+1DCh],bl
0x100899d0: je short 100899DEh
0x100899d2: cmp [edi+1F4h],bl
0x100899d8: jne short 100899DEh
0x100899da: mov al,1
0x100899dc: jmp short 100899E0h
0x100899de: xor al,al
0x100899e0: cmp cx,bx
0x100899e3: mov edx,[esp+48h]
0x100899e7: mov [esi+21h],al
0x100899ea: mov eax,[esp+44h]
0x100899ee: mov [esi+24h],eax
0x100899f1: mov al,[esp+4Ch]
0x100899f5: mov [esi+28h],edx
0x100899f8: mov edx,[esp+50h]
0x100899fc: mov [esi+2Ch],al
0x100899ff: mov eax,[esp+54h]
0x10089a03: mov [esi+30h],edx
0x10089a06: mov edx,[esp+58h]
0x10089a0a: mov [esi+34h],eax
0x10089a0d: mov eax,[esp+5Ch]
0x10089a11: mov [esi+38h],edx
0x10089a14: mov edx,[esp+14h]
0x10089a18: mov [esi+3Ch],eax
0x10089a1b: mov [esi+40h],bl
0x10089a1e: mov [esi+41h],bl
0x10089a21: mov [esi+42h],bx
0x10089a25: mov [esi+44h],bx
0x10089a29: mov [esi+46h],bx
0x10089a2d: mov [esi+48h],bx
0x10089a31: mov [esi+4Ah],bx
0x10089a35: mov [esi+4Ch],bx
0x10089a39: mov [esi+50h],ebx
0x10089a3c: mov [esi+54h],ebx
0x10089a3f: mov [esi+58h],edx
0x10089a42: je short 10089A97h
0x10089a44: mov eax,[edi+2B4h]
0x10089a4a: cmp eax,ebx
0x10089a4c: je short 10089A5Dh
0x10089a4e: mov edi,edi
0x10089a50: cmp [eax+1Ch],cx
0x10089a54: jbe short 10089A83h
0x10089a56: mov eax,[eax+0Ch]
0x10089a59: cmp eax,ebx
0x10089a5b: jne short 10089A50h
0x10089a5d: mov edx,[edi+2B0h]
0x10089a63: mov [esi+8],edx
0x10089a66: mov [edi+2B0h],esi
0x10089a6c: mov eax,[esi+8]
0x10089a6f: cmp eax,ebx
0x10089a71: je short 10089A91h
0x10089a73: mov [eax+0Ch],esi
0x10089a76: mov eax,[esp+60h]
0x10089a7a: pop ebp
0x10089a7b: pop edi
0x10089a7c: mov [eax],esi
0x10089a7e: pop esi
0x10089a7f: xor eax,eax
0x10089a81: pop ebx
0x10089a82: ret
0x10089a83: mov [esi+0Ch],eax
0x10089a86: mov ecx,[eax+8]
0x10089a89: mov [esi+8],ecx
0x10089a8c: mov [eax+8],esi
0x10089a8f: jmp short 10089A6Ch
```

```
0x10089a91: mov [edi+2B4h],esi
0x10089a97: mov eax,[esp+60h]
0x10089a9b: pop ebp
0x10089a9c: pop edi
0x10089a9d: mov [eax],esi
0x10089a9f: pop esi
0x10089aa0: xor eax,eax
0x10089aa2: pop ebx
0x10089aa3: ret
```

`sub_10089910` is called by sub_10080300. It delegates to sub_10080800, sub_10089620 (2 callees).

12.1102 sub_1008acf0 at 0x1008acf0 (21 instructions)

```
0x1008acf0: push edi
0x1008acf1: mov edi,[esp+8]
0x1008acf5: mov eax,[edi+114h]
0x1008acfb: test eax,eax
0x1008acfd: je short 1008AD14h
0x1008acff: push esi
0x1008ad00: mov esi,[eax+18h]
0x1008ad03: push eax
0x1008ad04: mov eax,[eax+8]
0x1008ad07: push edi
0x1008ad08: call eax
0x1008ad0a: add esp,8
0x1008ad0d: test esi,esi
0x1008ad0f: mov eax,esi
0x1008ad11: jne short 1008AD00h
0x1008ad13: pop esi
0x1008ad14: mov dword ptr [edi+114h],0
0x1008ad1e: mov dword ptr [edi+118h],0
0x1008ad28: xor eax,eax
0x1008ad2a: pop edi
0x1008ad2b: ret
```

`sub_1008acf0` is called by sub_1007f790. It calls nothing else.

12.1103 sub_1008ad30 at 0x1008ad30 (25 instructions)

```
0x1008ad30: push esi
0x1008ad31: mov esi,[esp+8]
0x1008ad35: cmp byte ptr [esi+15Ah],0
0x1008ad3c: jne short 1008AD72h
0x1008ad3e: mov eax,[esi+0A8h]
0x1008ad44: test eax,eax
0x1008ad46: mov byte ptr [esi+15Ah],1
0x1008ad4d: je short 1008AD72h
0x1008ad4f: push esi
0x1008ad50: call eax
0x1008ad52: add esp,4
0x1008ad55: test al,al
0x1008ad57: jne short 1008AD72h
0x1008ad59: push 0
0x1008ad5b: push 0
0x1008ad5d: push 385h
0x1008ad62: push esi
0x1008ad63: call 10080800h
0x1008ad68: add esp,10h
0x1008ad6b: mov eax,385h
0x1008ad70: pop esi
0x1008ad71: ret
0x1008ad72: xor eax,eax
0x1008ad74: pop esi
0x1008ad75: ret
```

`sub_1008ad30` is called by sub_1007f790, sub_1008af60, sub_1008b280, sub_1008c510. It delegates to sub_10080800 (1 callees).

12.1104 sub_1008ad80 at 0x1008ad80 (48 instructions)

```
0x1008ad80: push esi
0x1008ad81: push edi
0x1008ad82: mov edi,[esp+0Ch]
0x1008ad86: mov esi,[edi+198h]
0x1008ad8c: mov eax,[esi]
0x1008ad8e: test eax,eax
0x1008ad90: mov [edi+198h],eax
0x1008ad96: jne short 1008AD9Eh
0x1008ad98: mov [edi+19Ch],eax
0x1008ad9e: cmp byte ptr [esi+0Ch],0
0x1008ada2: je short 1008ADCFh
0x1008ada4: mov eax,[edi+0A0h]
0x1008adaa: test eax,eax
0x1008adac: je short 1008ADCFh
0x1008adae: mov ecx,[esi+8]
0x1008adb1: mov edx,[esi+4]
0x1008adb4: push ecx
0x1008adb5: push edx
0x1008adb6: push edi
0x1008adb7: call eax
0x1008adb9: mov eax,[edi+9Ch]
0x1008adbf: add esp,0Ch
0x1008adc2: push 18h
0x1008adc4: push esi
0x1008adc5: call eax
0x1008adc7: add esp,8
0x1008adca: pop edi
0x1008adcb: xor eax,eax
0x1008adcd: pop esi
0x1008adce: ret
0x1008adcf: mov eax,[esi+4]
0x1008add2: test eax,eax
0x1008add4: je short 1008ADE6h
0x1008add6: mov ecx,[esi+8]
0x1008add9: mov edx,[edi+9Ch]
0x1008addf: push ecx
0x1008ade0: push eax
0x1008ade1: call edx
0x1008ade3: add esp,8
0x1008ade6: mov eax,[edi+9Ch]
0x1008adec: push 18h
0x1008adee: push esi
0x1008adef: call eax
0x1008adf1: add esp,8
0x1008adf4: pop edi
0x1008adf5: xor eax,eax
0x1008adf7: pop esi
0x1008adf8: ret
```

`sub_1008ad80` is called by sub_1007f790, sub_1008ae80, sub_1008c510. It calls nothing else.

12.1105 sub_1008ae00 at 0x1008ae00 (48 instructions)

```
0x1008ae00: push esi
0x1008ae01: push edi
0x1008ae02: mov edi,[esp+0Ch]
0x1008ae06: mov esi,[edi+190h]
0x1008ae0c: mov eax,[esi]
0x1008ae0e: test eax,eax
0x1008ae10: mov [edi+190h],eax
0x1008ae16: jne short 1008AE1Eh
```

```
0x1008ae18: mov [edi+194h],eax
0x1008ae1e: cmp byte ptr [esi+0Ch],0
0x1008ae22: je short 1008AE4Fh
0x1008ae24: mov eax,[edi+0A0h]
0x1008ae2a: test eax,eax
0x1008ae2c: je short 1008AE4Fh
0x1008ae2e: mov ecx,[esi+8]
0x1008ae31: mov edx,[esi+4]
0x1008ae34: push ecx
0x1008ae35: push edx
0x1008ae36: push edi
0x1008ae37: call eax
0x1008ae39: mov eax,[edi+9Ch]
0x1008ae3f: add esp,0Ch
0x1008ae42: push 18h
0x1008ae44: push esi
0x1008ae45: call eax
0x1008ae47: add esp,8
0x1008ae4a: pop edi
0x1008ae4b: xor eax,eax
0x1008ae4d: pop esi
0x1008ae4e: ret
0x1008ae4f: mov eax,[esi+4]
0x1008ae52: test eax,eax
0x1008ae54: je short 1008AE66h
0x1008ae56: mov ecx,[esi+8]
0x1008ae59: mov edx,[edi+9Ch]
0x1008ae5f: push ecx
0x1008ae60: push eax
0x1008ae61: call edx
0x1008ae63: add esp,8
0x1008ae66: mov eax,[edi+9Ch]
0x1008ae6c: push 18h
0x1008ae6e: push esi
0x1008ae6f: call eax
0x1008ae71: add esp,8
0x1008ae74: pop edi
0x1008ae75: xor eax,eax
0x1008ae77: pop esi
0x1008ae78: ret
```

`sub_1008ae00` is called by sub_1007f790, sub_1008c980. It calls nothing else.

12.1106 sub_1008ae80 at 0x1008ae80 (86 instructions)

```
0x1008ae80: mov ecx,[esp+8]
0x1008ae84: push ebp
0x1008ae85: mov ebp,[esp+8]
0x1008ae89: push esi
0x1008ae8a: push edi
0x1008ae8b: mov edi,eax
0x1008ae8d: mov eax,[ebp+198h]
0x1008ae93: xor edx,edx
0x1008ae95: test eax,eax
0x1008ae97: mov [esp+10h],ecx
0x1008ae9b: mov [ebx],edx
0x1008ae9d: je short 1008AEA9h
0x1008ae9f: nop
0x1008aea0: add edx,[eax+14h]
0x1008aea3: mov eax,[eax]
0x1008aea5: test eax,eax
0x1008aea7: jne short 1008AEA0h
0x1008aea9: cmp edi,edx
0x1008aeab: ja short 1008AF0Ch
0x1008aead: test edi,edi
0x1008aeaf: je near ptr 1008AF4Ch
```

```
0x1008aeb5: mov esi,[ebp+198h]
0x1008aebb: mov eax,[esi+14h]
0x1008aebe: cmp eax,edi
0x1008aec0: ja short 1008AEF0h
0x1008aec2: push eax
0x1008aec3: mov eax,[esi+10h]
0x1008aec6: push eax
0x1008aec7: push ecx
0x1008aec8: call 100228C6h
0x1008aecd: mov eax,[esi+14h]
0x1008aed0: add [ebx],eax
0x1008aed2: add [esp+1Ch],eax
0x1008aed6: sub edi,[esi+14h]
0x1008aed9: push ebp
0x1008aeda: call 1008AD80h
0x1008aedf: add esp,10h
0x1008aee2: test eax,eax
0x1008aee4: jne short 1008AF4Eh
0x1008aee6: test edi,edi
0x1008aee8: je short 1008AF4Ch
0x1008aeea: mov ecx,[esp+10h]
0x1008aeee: jmp short 1008AEB5h
0x1008aef0: mov edx,[esi+10h]
0x1008aef3: push edi
0x1008aef4: push edx
0x1008aef5: push ecx
0x1008aef6: call 100228C6h
0x1008aefb: sub [esi+14h],edi
0x1008aefe: add [esi+10h],edi
0x1008af01: add [ebx],edi
0x1008af03: add esp,0Ch
0x1008af06: pop edi
0x1008af07: pop esi
0x1008af08: xor eax,eax
0x1008af0a: pop ebp
0x1008af0b: ret
0x1008af0c: mov edx,[ebp+0ACh]
0x1008af12: lea eax,[esp+10h]
0x1008af16: push eax
0x1008af17: push edi
0x1008af18: push ecx
0x1008af19: push ebp
0x1008af1a: mov dword ptr [esp+20h],0
0x1008af22: call edx
0x1008af24: add esp,10h
0x1008af27: test al,al
0x1008af29: jne short 1008AF46h
0x1008af2b: push 0
0x1008af2d: push 0
0x1008af2f: push 385h
0x1008af34: push ebp
0x1008af35: call 10080800h
0x1008af3a: add esp,10h
0x1008af3d: pop edi
0x1008af3e: pop esi
0x1008af3f: mov eax,385h
0x1008af44: pop ebp
0x1008af45: ret
0x1008af46: mov eax,[esp+10h]
0x1008af4a: add [ebx],eax
0x1008af4c: xor eax,eax
0x1008af4e: pop edi
0x1008af4f: pop esi
0x1008af50: pop ebp
0x1008af51: ret
```

`sub_1008ae80` is called by sub_1008af60. It delegates to j_memcpy, sub_10080800, sub_1008ad80 (3 callees).

12.1107 sub_1008af60 at 0x1008af60 (231 instructions)

```
0x1008af60: push ecx
0x1008af61: cmp byte ptr [esi+172h],0
0x1008af68: push ebx
0x1008af69: push ebp
0x1008af6a: mov ebp,[esp+10h]
0x1008af6e: push edi
0x1008af6f: mov edi,eax
0x1008af71: je near ptr 1008B245h
0x1008af77: mov eax,[esp+1Ch]
0x1008af7b: mov dword ptr [eax],0
0x1008af81: cmp dword ptr [esi+178h],0
0x1008af88: jne short 1008AFD5h
0x1008af8a: mov ecx,[esi+98h]
0x1008af90: push 8000h
0x1008af95: mov dword ptr [esi+174h],8000h
0x1008af9f: call ecx
0x1008afa1: add esp,4
0x1008afa4: test eax,eax
0x1008afa6: mov [esi+178h],eax
0x1008afac: jne short 1008AFC5h
0x1008afae: push eax
0x1008afaf: push eax
0x1008afb0: push 1
0x1008afb2: push esi
0x1008afb3: call 10080800h
0x1008afb8: add esp,10h
0x1008afbb: pop edi
0x1008afbc: pop ebp
0x1008afbd: mov eax,1
0x1008afc2: pop ebx
0x1008afc3: pop ecx
0x1008afc4: ret
0x1008afc5: mov dword ptr [esi+180h],0
0x1008afcf: mov [esi+17Ch],eax
0x1008afd5: mov ebx,[esp+18h]
0x1008afd9: cmp ebx,[esi+174h]
0x1008afdf: jbe near ptr 1008B0AAh
0x1008afe5: cmp dword ptr [ebp],0
0x1008afe9: jne short 1008B02Ch
0x1008afeb: mov eax,[esi+180h]
0x1008aff1: test eax,eax
0x1008aff3: je short 1008B029h
0x1008aff5: mov edx,[esi+17Ch]
0x1008affb: push eax
0x1008affc: push edx
0x1008affd: push edi
0x1008affe: call 100228C6h
0x1008b003: mov eax,[esi+180h]
0x1008b009: add esp,0Ch
0x1008b00c: add eax,edi
0x1008b00e: mov [ebp],eax
0x1008b011: mov ecx,[esi+178h]
0x1008b017: mov [esi+17Ch],ecx
0x1008b01d: mov dword ptr [esi+180h],0
0x1008b027: jmp short 1008B02Ch
0x1008b029: mov [ebp],edi
0x1008b02c: mov eax,[ebp]
0x1008b02f: sub edi,eax
0x1008b031: push eax
0x1008b032: add edi,ebx
0x1008b034: push esi
0x1008b035: lea ebx,[esp+14h]
0x1008b039: mov eax,edi
0x1008b03b: call 1008AE80h
0x1008b040: add esp,8
0x1008b043: test eax,eax
```

```
0x1008b045: jne near ptr 1008B26Fh
0x1008b04b: cmp [esi+171h],al
0x1008b051: mov eax,[esp+0Ch]
0x1008b055: je short 1008B086h
0x1008b057: test eax,eax
0x1008b059: jne short 1008B086h
0x1008b05b: push esi
0x1008b05c: call 1008AD30h
0x1008b061: add esp,4
0x1008b064: test eax,eax
0x1008b066: jne near ptr 1008B26Fh
0x1008b06c: mov eax,[esp+18h]
0x1008b070: mov edx,[esp+1Ch]
0x1008b074: sub eax,edi
0x1008b076: pop edi
0x1008b077: mov [edx],eax
0x1008b079: pop ebp
0x1008b07a: mov byte ptr [esi+171h],0
0x1008b081: xor eax,eax
0x1008b083: pop ebx
0x1008b084: pop ecx
0x1008b085: ret
0x1008b086: cmp eax,edi
0x1008b088: jae short 1008B092h
0x1008b08a: add [ebp],eax
0x1008b08d: jmp near ptr 1008B25Fh
0x1008b092: mov eax,[esp+18h]
0x1008b096: mov ecx,[esp+1Ch]
0x1008b09a: pop edi
0x1008b09b: mov [ecx],eax
0x1008b09d: pop ebp
0x1008b09e: mov byte ptr [esi+171h],0
0x1008b0a5: xor eax,eax
0x1008b0a7: pop ebx
0x1008b0a8: pop ecx
0x1008b0a9: ret
0x1008b0aa: cmp byte ptr [esi+170h],0
0x1008b0b1: jne near ptr 1008B266h
0x1008b0b7: jmp short 1008B0C0h
0x1008b0c0: cmp byte ptr [esi+15Ah],0
0x1008b0c7: jne near ptr 1008B1E6h
0x1008b0cd: mov eax,[esi+180h]
0x1008b0d3: cmp ebx,eax
0x1008b0d5: jbe near ptr 1008B1E6h
0x1008b0db: mov ecx,[esi+178h]
0x1008b0e1: mov edx,[esi+17Ch]
0x1008b0e7: mov ebx,ecx
0x1008b0e9: sub ebx,edx
0x1008b0eb: sub ebx,eax
0x1008b0ed: add ebx,[esi+174h]
0x1008b0f3: cmp ebx,400h
0x1008b0f9: jae short 1008B116h
0x1008b0fb: test eax,eax
0x1008b0fd: je short 1008B10Ah
0x1008b0ff: push eax
0x1008b100: push edx
0x1008b101: push ecx
0x1008b102: call 100228C6h
0x1008b107: add esp,0Ch
0x1008b10a: mov edx,[esi+178h]
0x1008b110: mov [esi+17Ch],edx
0x1008b116: mov ecx,[esi+180h]
0x1008b11c: mov eax,[esi+174h]
0x1008b122: sub eax,ecx
0x1008b124: cmp eax,400h
0x1008b129: jb near ptr 1008B229h
0x1008b12f: mov eax,[esi+17Ch]
```

```
0x1008b135: add eax,ecx
0x1008b137: push eax
0x1008b138: push esi
0x1008b139: lea ebx,[esp+14h]
0x1008b13d: mov eax,400h
0x1008b142: call 1008AE80h
0x1008b147: xor ebx,ebx
0x1008b149: add esp,8
0x1008b14c: cmp eax,ebx
0x1008b14e: jne near ptr 1008B26Fh
0x1008b154: mov eax,[esp+0Ch]
0x1008b158: add [esi+180h],eax
0x1008b15e: cmp [esi+171h],bl
0x1008b164: mov ecx,[esi+180h]
0x1008b16a: je short 1008B1BCh
0x1008b16c: cmp eax,ebx
0x1008b16e: jne short 1008B1BCh
0x1008b170: push esi
0x1008b171: call 1008AD30h
0x1008b176: add esp,4
0x1008b179: cmp eax,ebx
0x1008b17b: jne near ptr 1008B26Fh
0x1008b181: mov eax,[esi+180h]
0x1008b187: cmp eax,ebx
0x1008b189: je short 1008B19Ch
0x1008b18b: mov ecx,[esi+17Ch]
0x1008b191: push eax
0x1008b192: push ecx
0x1008b193: push edi
0x1008b194: call 100228C6h
0x1008b199: add esp,0Ch
0x1008b19c: mov edx,[esi+180h]
0x1008b1a2: mov eax,[esp+1Ch]
0x1008b1a6: mov [eax],edx
0x1008b1a8: mov ecx,[esi+178h]
0x1008b1ae: mov [esi+17Ch],ecx
0x1008b1b4: mov [esi+180h],ebx
0x1008b1ba: jmp short 1008B1D0h
0x1008b1bc: cmp [esp+18h],ecx
0x1008b1c0: jbe short 1008B1D0h
0x1008b1c2: cmp eax,400h
0x1008b1c7: jae short 1008B1D0h
0x1008b1c9: mov byte ptr [esi+170h],1
0x1008b1d0: mov [esi+171h],bl
0x1008b1d6: cmp [esi+170h],bl
0x1008b1dc: mov ebx,[esp+18h]
0x1008b1e0: je near ptr 1008B0C0h
0x1008b1e6: cmp byte ptr [esi+170h],0
0x1008b1ed: jne short 1008B266h
0x1008b1ef: cmp byte ptr [esi+15Ah],0
0x1008b1f6: jne short 1008B266h
0x1008b1f8: mov edx,[esi+17Ch]
0x1008b1fe: push ebx
0x1008b1ff: push edx
0x1008b200: push edi
0x1008b201: call 100228C6h
0x1008b206: mov eax,[esp+28h]
0x1008b20a: add esp,0Ch
0x1008b20d: mov [eax],ebx
0x1008b20f: add [esi+17Ch],ebx
0x1008b215: sub [esi+180h],ebx
0x1008b21b: pop edi
0x1008b21c: pop ebp
0x1008b21d: mov byte ptr [esi+171h],0
0x1008b224: xor eax,eax
0x1008b226: pop ebx
0x1008b227: pop ecx
```

```
0x1008b228: ret
0x1008b229: push 0
0x1008b22b: push 0
0x1008b22d: push 3E7h
0x1008b232: push esi
0x1008b233: call 10080800h
0x1008b238: add esp,10h
0x1008b23b: pop edi
0x1008b23c: pop ebp
0x1008b23d: mov eax,3E7h
0x1008b242: pop ebx
0x1008b243: pop ecx
0x1008b244: ret
0x1008b245: mov ebx,[esp+1Ch]
0x1008b249: mov eax,[esp+18h]
0x1008b24d: push edi
0x1008b24e: push esi
0x1008b24f: call 1008AE80h
0x1008b254: add esp,8
0x1008b257: test eax,eax
0x1008b259: jne short 1008B26Fh
0x1008b25b: cmp [ebx],eax
0x1008b25d: jne short 1008B266h
0x1008b25f: mov byte ptr [esi+170h],1
0x1008b266: mov byte ptr [esi+171h],0
0x1008b26d: xor eax,eax
0x1008b26f: pop edi
0x1008b270: pop ebp
0x1008b271: pop ebx
0x1008b272: pop ecx
0x1008b273: ret
```

`sub_1008af60` is called by sub_1008c510, sub_1008cb40. It delegates to j_memcpy, sub_10080800, sub_1008ad30 (4 callees).

12.1108 sub_1008b280 at 0x1008b280 (532 instructions)

```
0x1008b280: sub esp,6C8h
0x1008b286: push ebx
0x1008b287: xor ebx,ebx
0x1008b289: push ebp
0x1008b28a: push edi
0x1008b28b: mov dword ptr [esp+14h],40404040h
0x1008b293: mov dword ptr [esp+18h],100A10B0h
0x1008b29b: mov dword ptr [esp+1Ch],100A1B40h
0x1008b2a3: mov dword ptr [esp+20h],100AAB30h
0x1008b2ab: mov dword ptr [esp+24h],100ACAB0h
0x1008b2b3: mov dword ptr [esp+28h],100A3510h
0x1008b2bb: mov [esp+2Ch],ebx
0x1008b2bf: mov [esp+30h],ebx
0x1008b2c3: mov dword ptr [esp+34h],4241434Bh
0x1008b2cb: mov dword ptr [esp+38h],100A1330h
0x1008b2d3: mov dword ptr [esp+3Ch],100A1880h
0x1008b2db: mov dword ptr [esp+40h],100A7AC0h
0x1008b2e3: mov dword ptr [esp+44h],100ABE70h
0x1008b2eb: mov dword ptr [esp+48h],100A2970h
0x1008b2f3: mov [esp+4Ch],ebx
0x1008b2f7: mov [esp+50h],ebx
0x1008b2fb: mov dword ptr [esp+54h],42415349h
0x1008b303: mov dword ptr [esp+58h],100A1290h
0x1008b30b: mov dword ptr [esp+5Ch],100A1900h
0x1008b313: mov dword ptr [esp+60h],100A7150h
0x1008b31b: mov dword ptr [esp+64h],100ABAC0h
0x1008b323: mov dword ptr [esp+68h],100A2740h
0x1008b32b: mov [esp+6Ch],ebx
0x1008b32f: mov [esp+70h],ebx
```

```
0x1008b333: mov dword ptr [esp+74h],434C4950h
0x1008b33b: mov dword ptr [esp+78h],100A1290h
0x1008b343: mov dword ptr [esp+7Ch],100A1900h
0x1008b34b: mov dword ptr [esp+80h],100A8220h
0x1008b356: mov dword ptr [esp+84h],100AC110h
0x1008b361: mov dword ptr [esp+88h],100A2B60h
0x1008b36c: mov [esp+8Ch],ebx
0x1008b373: mov [esp+90h],ebx
0x1008b37a: mov dword ptr [esp+94h],434C4F4Eh
0x1008b385: mov dword ptr [esp+98h],100A1420h
0x1008b390: mov dword ptr [esp+9Ch],100A1A40h
0x1008b39b: mov dword ptr [esp+0A0h],100A74B0h
0x1008b3a6: mov dword ptr [esp+0A4h],100ABBC0h
0x1008b3b1: mov dword ptr [esp+0A8h],100A27D0h
0x1008b3bc: mov [esp+0ACh],ebx
0x1008b3c3: mov [esp+0B0h],ebx
0x1008b3ca: mov dword ptr [esp+0B4h],4442594Bh
0x1008b3d5: mov dword ptr [esp+0B8h],100A1150h
0x1008b3e0: mov dword ptr [esp+0BCh],100A1AC0h
0x1008b3eb: mov dword ptr [esp+0C0h],100AA470h
0x1008b3f6: mov dword ptr [esp+0C4h],100AC8C0h
0x1008b401: mov dword ptr [esp+0C8h],100A3370h
0x1008b40c: mov [esp+0CCh],ebx
0x1008b413: mov [esp+0D0h],ebx
0x1008b41a: mov dword ptr [esp+0D4h],44454649h
0x1008b425: mov dword ptr [esp+0D8h],100A1240h
0x1008b430: mov dword ptr [esp+0DCh],100A14F0h
0x1008b43b: mov dword ptr [esp+0E0h],100A6EA0h
0x1008b446: mov dword ptr [esp+0E4h],100AB9D0h
0x1008b451: mov dword ptr [esp+0E8h],100A26B0h
0x1008b45c: mov [esp+0ECh],ebx
0x1008b463: mov [esp+0F0h],ebx
0x1008b46a: mov dword ptr [esp+0F4h],44484452h
0x1008b475: mov dword ptr [esp+0F8h],100A1420h
0x1008b480: mov dword ptr [esp+0FCh],100A1A40h
0x1008b48b: mov dword ptr [esp+100h],100A9970h
0x1008b496: mov dword ptr [esp+104h],100AC640h
0x1008b4a1: mov dword ptr [esp+108h],100A3160h
0x1008b4ac: mov [esp+10Ch],ebx
0x1008b4b3: mov [esp+110h],ebx
0x1008b4ba: mov dword ptr [esp+114h],44495343h
0x1008b4c5: mov dword ptr [esp+118h],100A10B0h
0x1008b4d0: mov dword ptr [esp+11Ch],100A1840h
0x1008b4db: mov dword ptr [esp+120h],100A7960h
0x1008b4e6: mov dword ptr [esp+124h],100ABE00h
0x1008b4f1: mov dword ptr [esp+128h],100A28E0h
0x1008b4fc: mov [esp+12Ch],ebx
0x1008b503: mov [esp+130h],ebx
0x1008b50a: mov dword ptr [esp+134h],44524F50h
0x1008b515: mov dword ptr [esp+138h],100A10B0h
0x1008b520: mov dword ptr [esp+13Ch],100A1A80h
0x1008b52b: mov dword ptr [esp+140h],100AA380h
0x1008b536: mov dword ptr [esp+144h],100AC840h
0x1008b541: mov ebp,100A1330h
0x1008b546: mov edi,100A1490h
0x1008b54b: mov edx,100A96D0h
0x1008b550: mov ecx,100AC5E0h
0x1008b555: mov eax,100A3010h
0x1008b55a: mov dword ptr [esp+148h],100A32E0h
0x1008b565: mov [esp+14Ch],ebx
0x1008b56c: mov [esp+150h],ebx
0x1008b573: mov dword ptr [esp+154h],454E444Ch
0x1008b57e: mov dword ptr [esp+158h],100A1100h
0x1008b589: mov dword ptr [esp+15Ch],100A1510h
0x1008b594: mov dword ptr [esp+160h],100A6DC0h
0x1008b59f: mov dword ptr [esp+164h],100AB9A0h
0x1008b5aa: mov dword ptr [esp+168h],100A2670h
```

```
0x1008b5b5: mov [esp+16Ch],ebx
0x1008b5bc: mov [esp+170h],ebx
0x1008b5c3: mov dword ptr [esp+174h],4652414Dh
0x1008b5ce: mov dword ptr [esp+178h],100A12E0h
0x1008b5d9: mov dword ptr [esp+17Ch],100A18A0h
0x1008b5e4: mov dword ptr [esp+180h],100A7C90h
0x1008b5ef: mov dword ptr [esp+184h],100ABF10h
0x1008b5fa: mov dword ptr [esp+188h],100A29D0h
0x1008b605: mov [esp+18Ch],ebx
0x1008b60c: mov [esp+190h],ebx
0x1008b613: mov dword ptr [esp+194h],49444154h
0x1008b61e: mov dword ptr [esp+198h],100A1330h
0x1008b629: mov dword ptr [esp+19Ch],100A1490h
0x1008b634: mov dword ptr [esp+1A0h],100A41D0h
0x1008b63f: mov dword ptr [esp+1A4h],100AC5E0h
0x1008b64a: mov dword ptr [esp+1A8h],100A1C50h
0x1008b655: mov [esp+1ACh],ebx
0x1008b65c: mov [esp+1B0h],ebx
0x1008b663: mov dword ptr [esp+1B4h],49454E44h
0x1008b66e: mov dword ptr [esp+1B8h],100A13D0h
0x1008b679: mov dword ptr [esp+1BCh],100A17A0h
0x1008b684: mov dword ptr [esp+1C0h],100A4350h
0x1008b68f: mov dword ptr [esp+1C4h],100AC620h
0x1008b69a: mov dword ptr [esp+1C8h],100A1CE0h
0x1008b6a5: mov [esp+1CCh],ebx
0x1008b6ac: mov [esp+1D0h],ebx
0x1008b6b3: mov dword ptr [esp+1D4h],49484452h
0x1008b6be: mov dword ptr [esp+1D8h],100A1150h
0x1008b6c9: mov dword ptr [esp+1DCh],100A1A20h
0x1008b6d4: mov dword ptr [esp+1E0h],100A3AB0h
0x1008b6df: mov dword ptr [esp+1E4h],100AAC60h
0x1008b6ea: mov dword ptr [esp+1E8h],100A1B80h
0x1008b6f5: mov [esp+1ECh],ebx
0x1008b6fc: mov [esp+1F0h],ebx
0x1008b703: mov dword ptr [esp+1F4h],494A4E47h
0x1008b70e: mov dword ptr [esp+1F8h],100A13D0h
0x1008b719: mov dword ptr [esp+1FCh],100A17A0h
0x1008b724: mov dword ptr [esp+200h],100AA2F0h
0x1008b72f: mov dword ptr [esp+204h],100AC620h
0x1008b73a: mov dword ptr [esp+208h],100A32B0h
0x1008b745: mov [esp+20Ch],ebx
0x1008b74c: mov [esp+210h],ebx
0x1008b753: mov dword ptr [esp+214h],49504E47h
0x1008b75e: mov dword ptr [esp+218h],100A13D0h
0x1008b769: mov dword ptr [esp+21Ch],100A17A0h
0x1008b774: mov dword ptr [esp+220h],100A9D20h
0x1008b77f: mov dword ptr [esp+224h],100AC620h
0x1008b78a: mov dword ptr [esp+228h],100A3210h
0x1008b795: mov [esp+22Ch],ebx
0x1008b79c: mov [esp+230h],ebx
0x1008b7a3: mov dword ptr [esp+234h],4A444141h
0x1008b7ae: mov [esp+238h],ebp
0x1008b7b5: mov [esp+23Ch],edi
0x1008b7bc: mov [esp+240h],edx
0x1008b7c3: mov [esp+244h],ecx
0x1008b7ca: mov [esp+248h],eax
0x1008b7d1: mov [esp+24Ch],ebx
0x1008b7d8: mov [esp+250h],ebx
0x1008b7df: mov dword ptr [esp+254h],4A444154h
0x1008b7ea: mov dword ptr [esp+258h],100A1330h
0x1008b7f5: mov dword ptr [esp+25Ch],100A1490h
0x1008b800: mov dword ptr [esp+260h],100A97F0h
0x1008b80b: mov dword ptr [esp+264h],100AC5E0h
0x1008b816: mov dword ptr [esp+268h],100A30A0h
0x1008b821: mov [esp+26Ch],ebx
0x1008b828: mov [esp+270h],ebx
0x1008b82f: mov dword ptr [esp+274h],4A484452h
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0x1008b83a: mov dword ptr [esp+278h],100A1150h
0x1008b845: mov dword ptr [esp+27Ch],100A1A20h
0x1008b850: mov dword ptr [esp+280h],100A9330h
0x1008b85b: mov dword ptr [esp+284h],100AC560h
0x1008b866: mov dword ptr [esp+288h],100A2F90h
0x1008b871: mov [esp+28Ch],ebx
0x1008b878: mov [esp+290h],ebx
0x1008b87f: mov dword ptr [esp+294h],4A534550h
0x1008b88a: mov dword ptr [esp+298h],100A13D0h
0x1008b895: mov dword ptr [esp+29Ch],100A17A0h
0x1008b8a0: mov dword ptr [esp+2A0h],100A9900h
0x1008b8ab: mov dword ptr [esp+2A4h],100AC620h
0x1008b8b6: mov dword ptr [esp+2A8h],100A3130h
0x1008b8c1: mov [esp+2ACh],ebx
0x1008b8c8: mov [esp+2B0h],ebx
0x1008b8cf: mov dword ptr [esp+2B4h],4A644141h
0x1008b8da: mov [esp+2B8h],ebp
0x1008b8e1: mov [esp+2BCh],edi
0x1008b8e8: mov [esp+2C0h],edx
0x1008b8ef: mov [esp+2C4h],ecx
0x1008b8f6: mov [esp+2C8h],eax
0x1008b8fd: mov [esp+2CCh],ebx
0x1008b904: mov [esp+2D0h],ebx
0x1008b90b: mov dword ptr [esp+2D4h],4C4F4F50h
0x1008b916: mov dword ptr [esp+2D8h],100A11F0h
0x1008b921: mov dword ptr [esp+2DCh],100A17C0h
0x1008b92c: mov dword ptr [esp+2E0h],100A6B20h
0x1008b937: mov dword ptr [esp+2E4h],100AB8C0h
0x1008b942: mov dword ptr [esp+2E8h],100A25C0h
0x1008b94d: mov [esp+2ECh],ebx
0x1008b954: mov [esp+2F0h],ebx
0x1008b95b: mov dword ptr [esp+2F4h],4D41474Eh
0x1008b966: mov dword ptr [esp+2F8h],100A1420h
0x1008b971: mov dword ptr [esp+2FCh],100A1A40h
0x1008b97c: mov dword ptr [esp+300h],100AA670h
0x1008b987: mov dword ptr [esp+304h],100AC990h
0x1008b992: mov dword ptr [esp+308h],100A3490h
0x1008b99d: mov [esp+30Ch],ebx
0x1008b9a4: mov [esp+310h],ebx
0x1008b9ab: mov dword ptr [esp+314h],4D454E44h
0x1008b9b6: mov dword ptr [esp+318h],100A13D0h
0x1008b9c1: mov dword ptr [esp+31Ch],100A17A0h
0x1008b9cc: mov dword ptr [esp+320h],100A6A60h
0x1008b9d7: mov dword ptr [esp+324h],100AC620h
0x1008b9e2: mov dword ptr [esp+328h],100A2590h
0x1008b9ed: mov [esp+32Ch],ebx
0x1008b9f4: mov [esp+330h],ebx
0x1008b9fb: mov dword ptr [esp+334h],4D484452h
0x1008ba06: mov dword ptr [esp+338h],100A11F0h
0x1008ba11: mov dword ptr [esp+33Ch],100A1780h
0x1008ba1c: mov dword ptr [esp+340h],100A6830h
0x1008ba27: mov dword ptr [esp+344h],100AB840h
0x1008ba32: mov dword ptr [esp+348h],100A2530h
0x1008ba3d: mov [esp+34Ch],ebx
0x1008ba44: mov [esp+350h],ebx
0x1008ba4b: mov dword ptr [esp+354h],4D4F5645h
0x1008ba56: mov dword ptr [esp+358h],100A1150h
0x1008ba61: mov dword ptr [esp+35Ch],100A1A20h
0x1008ba6c: mov dword ptr [esp+360h],100A8100h
0x1008ba77: mov dword ptr [esp+364h],100AC0A0h
0x1008ba82: mov dword ptr [esp+368h],100A2B00h
0x1008ba8d: mov [esp+36Ch],ebx
0x1008ba94: mov [esp+370h],ebx
0x1008ba9b: mov dword ptr [esp+374h],4F524452h
0x1008baa6: mov dword ptr [esp+378h],100A10B0h
0x1008bab1: mov dword ptr [esp+37Ch],100A1B00h
0x1008babc: mov dword ptr [esp+380h],100AA570h
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0x1008bac7: mov dword ptr [esp+384h],100AC920h
0x1008bad2: mov dword ptr [esp+388h],100A3400h
0x1008badd: mov [esp+38Ch],ebx
0x1008bae4: mov [esp+390h],ebx
0x1008baeb: mov dword ptr [esp+394h],50415354h
0x1008baf6: mov dword ptr [esp+398h],100A1420h
0x1008bb01: mov dword ptr [esp+39Ch],100A1800h
0x1008bb0c: mov dword ptr [esp+3A0h],100A76D0h
0x1008bb17: mov dword ptr [esp+3A4h],100ABC90h
0x1008bb22: mov dword ptr [esp+3A8h],100A2840h
0x1008bb2d: mov [esp+3ACh],ebx
0x1008bb34: mov [esp+3B0h],ebx
0x1008bb3b: mov dword ptr [esp+3B4h],504C5445h
0x1008bb46: mov dword ptr [esp+3B8h],100A1010h
0x1008bb51: mov dword ptr [esp+3BCh],100A1470h
0x1008bb5c: mov dword ptr [esp+3C0h],100A3E40h
0x1008bb67: mov dword ptr [esp+3C4h],100AACD0h
0x1008bb72: mov dword ptr [esp+3C8h],100A1BE0h
0x1008bb7d: mov [esp+3CCh],ebx
0x1008bb84: mov [esp+3D0h],ebx
0x1008bb8b: mov dword ptr [esp+3D4h],50504C54h
0x1008bb96: mov dword ptr [esp+3D8h],100A1380h
0x1008bba1: mov dword ptr [esp+3DCh],100A1A60h
0x1008bbac: mov dword ptr [esp+3E0h],100A9DB0h
0x1008bbb7: mov dword ptr [esp+3E4h],100AC720h
0x1008bbc2: mov dword ptr [esp+3E8h],100A3240h
0x1008bbcd: mov [esp+3ECh],ebx
0x1008bbd4: mov [esp+3F0h],ebx
0x1008bbdb: mov dword ptr [esp+3F4h],50524F4Dh
0x1008bbe6: mov dword ptr [esp+3F8h],100A1100h
0x1008bbf1: mov dword ptr [esp+3FCh],100A1510h
0x1008bbfc: mov dword ptr [esp+400h],100A9BB0h
0x1008bc07: mov dword ptr [esp+404h],100AC6E0h
0x1008bc12: mov dword ptr [esp+408h],100A31C0h
0x1008bc1d: mov [esp+40Ch],ebx
0x1008bc24: mov [esp+410h],ebx
0x1008bc2b: mov dword ptr [esp+414h],53415645h
0x1008bc36: mov dword ptr [esp+418h],100A1330h
0x1008bc41: mov dword ptr [esp+41Ch],100A1920h
0x1008bc4c: mov dword ptr [esp+420h],100A86E0h
0x1008bc57: mov dword ptr [esp+424h],100AC270h
0x1008bc62: mov dword ptr [esp+428h],100A2C60h
0x1008bc6d: mov [esp+42Ch],ebx
0x1008bc74: mov [esp+430h],ebx
0x1008bc7b: mov dword ptr [esp+434h],5345454Bh
0x1008bc86: mov dword ptr [esp+438h],100A10B0h
0x1008bc91: mov dword ptr [esp+43Ch],100A19E0h
0x1008bc9c: mov dword ptr [esp+440h],100A8AA0h
0x1008bca7: mov dword ptr [esp+444h],100AC3E0h
0x1008bcb2: mov dword ptr [esp+448h],100A2D70h
0x1008bcbd: mov [esp+44Ch],ebx
0x1008bcc4: mov [esp+450h],ebx
0x1008bccb: mov dword ptr [esp+454h],53484F57h
0x1008bcd6: mov dword ptr [esp+458h],100A10B0h
0x1008bce1: mov dword ptr [esp+45Ch],10089160h
0x1008bcec: mov dword ptr [esp+460h],100A8390h
0x1008bcf7: mov dword ptr [esp+464h],100AC190h
0x1008bd02: mov dword ptr [esp+468h],100A2BC0h
0x1008bd0d: mov [esp+46Ch],ebx
0x1008bd14: mov [esp+470h],ebx
0x1008bd1b: mov dword ptr [esp+474h],5445524Dh
0x1008bd26: mov dword ptr [esp+478h],100A1330h
0x1008bd31: mov dword ptr [esp+47Ch],100A1880h
0x1008bd3c: mov dword ptr [esp+480h],100A8510h
0x1008bd47: mov dword ptr [esp+484h],100AC210h
0x1008bd52: mov dword ptr [esp+488h],100A2C10h
0x1008bd5d: mov [esp+48Ch],ebx
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0x1008bd64: mov [esp+490h],ebx
0x1008bd6b: mov dword ptr [esp+494h],624B4744h
0x1008bd76: mov dword ptr [esp+498h],100A1330h
0x1008bd81: mov dword ptr [esp+49Ch],100A1880h
0x1008bd8c: mov dword ptr [esp+4A0h],100A5E60h
0x1008bd97: mov dword ptr [esp+4A4h],100AB470h
0x1008bda2: mov dword ptr [esp+4A8h],100A2260h
0x1008bdad: mov [esp+4ACh],ebx
0x1008bdb4: mov [esp+4B0h],ebx
0x1008bdbb: mov dword ptr [esp+4B4h],6348524Dh
0x1008bdc6: mov dword ptr [esp+4B8h],100A1240h
0x1008bdd1: mov dword ptr [esp+4BCh],100A14F0h
0x1008bddc: mov dword ptr [esp+4C0h],100A4B20h
0x1008bde7: mov dword ptr [esp+4C4h],100AAED0h
0x1008bdf2: mov dword ptr [esp+4C8h],100A1E40h
0x1008bdfd: mov [esp+4CCh],ebx
0x1008be04: mov [esp+4D0h],ebx
0x1008be0b: mov dword ptr [esp+4D4h],65585049h
0x1008be16: mov dword ptr [esp+4D8h],100A1330h
0x1008be21: mov dword ptr [esp+4DCh],100A19A0h
0x1008be2c: mov dword ptr [esp+4E0h],100A8C20h
0x1008be37: mov dword ptr [esp+4E4h],100AC420h
0x1008be42: mov dword ptr [esp+4E8h],100A2DF0h
0x1008be4d: mov [esp+4ECh],ebx
0x1008be54: mov [esp+4F0h],ebx
0x1008be5b: mov dword ptr [esp+4F4h],66505249h
0x1008be66: mov dword ptr [esp+4F8h],100A1100h
0x1008be71: mov dword ptr [esp+4FCh],100A1510h
0x1008be7c: mov dword ptr [esp+500h],100A8D30h
0x1008be87: mov dword ptr [esp+504h],100AC480h
0x1008be92: mov dword ptr [esp+508h],100A2E80h
0x1008be9d: mov [esp+50Ch],ebx
0x1008bea4: mov [esp+510h],ebx
0x1008beab: mov dword ptr [esp+514h],67414D41h
0x1008beb6: mov dword ptr [esp+518h],100A10B0h
0x1008bec1: mov dword ptr [esp+51Ch],10089160h
0x1008becc: mov dword ptr [esp+520h],100A4960h
0x1008bed7: mov dword ptr [esp+524h],100AAE80h
0x1008bee2: mov dword ptr [esp+528h],100A1E00h
0x1008beed: mov [esp+52Ch],ebx
0x1008bef4: mov [esp+530h],ebx
0x1008befb: mov dword ptr [esp+534h],68495354h
0x1008bf06: mov dword ptr [esp+538h],100A11A0h
0x1008bf11: mov dword ptr [esp+53Ch],100A1760h
0x1008bf1c: mov dword ptr [esp+540h],100A6690h
0x1008bf27: mov dword ptr [esp+544h],100AB770h
0x1008bf32: mov dword ptr [esp+548h],100A2470h
0x1008bf3d: mov [esp+54Ch],ebx
0x1008bf44: mov [esp+550h],ebx
0x1008bf4b: mov dword ptr [esp+554h],69434350h
0x1008bf56: mov dword ptr [esp+558h],100A1290h
0x1008bf61: mov dword ptr [esp+55Ch],100A1530h
0x1008bf6c: mov dword ptr [esp+560h],100ACAD0h
0x1008bf77: mov dword ptr [esp+564h],100AAFC0h
0x1008bf82: mov dword ptr [esp+568h],100A1EF0h
0x1008bf8d: mov [esp+56Ch],ebx
0x1008bf94: mov [esp+570h],ebx
0x1008bf9b: mov dword ptr [esp+574h],69545874h
0x1008bfa6: mov dword ptr [esp+578h],100A1240h
0x1008bfb1: mov dword ptr [esp+57Ch],100A1650h
0x1008bfbc: mov dword ptr [esp+580h],100A56C0h
0x1008bfc7: mov dword ptr [esp+584h],100AB2E0h
0x1008bfd2: mov dword ptr [esp+588h],100A2130h
0x1008bfdd: mov [esp+58Ch],ebx
0x1008bfe4: mov [esp+590h],ebx
0x1008bfeb: mov dword ptr [esp+594h],6E454544h
0x1008bff6: mov dword ptr [esp+598h],100A10B0h
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0x1008c001: mov dword ptr [esp+59Ch],100A19E0h
0x1008c00c: mov dword ptr [esp+5A0h],100A90C0h
0x1008c017: mov dword ptr [esp+5A4h],100AC4C0h
0x1008c022: mov dword ptr [esp+5A8h],100A2EC0h
0x1008c02d: mov [esp+5ACh],ebx
0x1008c034: mov [esp+5B0h],ebx
0x1008c03b: mov dword ptr [esp+5B4h],70444167h
0x1008c046: mov dword ptr [esp+5B8h],100A1150h
0x1008c051: mov dword ptr [esp+5BCh],100A1A20h
0x1008c05c: mov dword ptr [esp+5C0h],100A9260h
0x1008c067: mov dword ptr [esp+5C4h],100AC500h
0x1008c072: mov dword ptr [esp+5C8h],100A2F40h
0x1008c07d: mov [esp+5CCh],ebx
0x1008c084: mov [esp+5D0h],ebx
0x1008c08b: mov dword ptr [esp+5D4h],70485973h
0x1008c096: mov dword ptr [esp+5D8h],100A1150h
0x1008c0a1: mov dword ptr [esp+5DCh],100A1A20h
0x1008c0ac: mov dword ptr [esp+5E0h],100A61D0h
0x1008c0b7: mov dword ptr [esp+5E4h],100AC500h
0x1008c0c2: mov dword ptr [esp+5E8h],100A22D0h
0x1008c0cd: mov [esp+5ECh],ebx
0x1008c0d4: mov [esp+5F0h],ebx
0x1008c0db: mov dword ptr [esp+5F4h],73424954h
0x1008c0e6: mov dword ptr [esp+5F8h],100A10B0h
0x1008c0f1: mov dword ptr [esp+5FCh],10089160h
0x1008c0fc: mov dword ptr [esp+600h],100A62C0h
0x1008c107: mov dword ptr [esp+604h],100AB530h
0x1008c112: mov dword ptr [esp+608h],100A2320h
0x1008c11d: mov [esp+60Ch],ebx
0x1008c124: mov [esp+610h],ebx
0x1008c12b: mov dword ptr [esp+614h],73504C54h
0x1008c136: mov dword ptr [esp+618h],100A1290h
0x1008c141: mov dword ptr [esp+61Ch],100A16F0h
0x1008c14c: mov dword ptr [esp+620h],100A6480h
0x1008c157: mov dword ptr [esp+624h],100AB690h
0x1008c162: mov dword ptr [esp+628h],100A2380h
0x1008c16d: mov [esp+62Ch],ebx
0x1008c174: mov [esp+630h],ebx
0x1008c17b: mov dword ptr [esp+634h],73524742h
0x1008c186: mov dword ptr [esp+638h],100A1100h
0x1008c191: mov dword ptr [esp+63Ch],100A1510h
0x1008c19c: mov dword ptr [esp+640h],100A4E80h
0x1008c1a7: cmp [esi+158h],bl
0x1008c1ad: mov dword ptr [esp+644h],100AAF80h
0x1008c1b8: mov dword ptr [esp+648h],100A1EB0h
0x1008c1c3: mov [esp+64Ch],ebx
0x1008c1ca: mov [esp+650h],ebx
0x1008c1d1: mov dword ptr [esp+654h],74455874h
0x1008c1dc: mov dword ptr [esp+658h],100A1150h
0x1008c1e7: mov dword ptr [esp+65Ch],100A1590h
0x1008c1f2: mov dword ptr [esp+660h],100A5030h
0x1008c1fd: mov dword ptr [esp+664h],100AB100h
0x1008c208: mov dword ptr [esp+668h],100A1FB0h
0x1008c213: mov [esp+66Ch],ebx
0x1008c21a: mov [esp+670h],ebx
0x1008c221: mov dword ptr [esp+674h],74494D45h
0x1008c22c: mov dword ptr [esp+678h],100A10B0h
0x1008c237: mov dword ptr [esp+67Ch],10089160h
0x1008c242: mov dword ptr [esp+680h],100A6760h
0x1008c24d: mov dword ptr [esp+684h],100AB7E0h
0x1008c258: mov dword ptr [esp+688h],100A24D0h
0x1008c263: mov [esp+68Ch],ebx
0x1008c26a: mov [esp+690h],ebx
0x1008c271: mov dword ptr [esp+694h],74524E53h
0x1008c27c: mov dword ptr [esp+698h],100A1060h
0x1008c287: mov dword ptr [esp+69Ch],100A14D0h
0x1008c292: mov dword ptr [esp+6A0h],100A4480h
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0x1008c29d: mov dword ptr [esp+6A4h],100AAD50h
0x1008c2a8: mov dword ptr [esp+6A8h],100A1D10h
0x1008c2b3: mov [esp+6ACh],ebx
0x1008c2ba: mov [esp+6B0h],ebx
0x1008c2c1: mov dword ptr [esp+6B4h],7A545874h
0x1008c2cc: mov dword ptr [esp+6B8h],100A1420h
0x1008c2d7: mov dword ptr [esp+6BCh],100A15F0h
0x1008c2e2: mov dword ptr [esp+6C0h],100A52C0h
0x1008c2ed: mov dword ptr [esp+6C4h],100AB1C0h
0x1008c2f8: mov dword ptr [esp+6C8h],100A2070h
0x1008c303: mov [esp+6CCh],ebx
0x1008c30a: mov [esp+6D0h],ebx
0x1008c311: je short 1008C321h
0x1008c313: cmp [esi+1B0h],bl
0x1008c319: jne short 1008C321h
0x1008c31b: mov [esi+1F1h],bl
0x1008c321: mov ebp,[esp+6D8h]
0x1008c328: push ebp
0x1008c329: call 100A3680h
0x1008c32e: sub dword ptr [esp+6E0h],4
0x1008c336: add esp,4
0x1008c339: add ebp,4
0x1008c33c: xor edx,edx
0x1008c33e: mov [esp+10h],eax
0x1008c342: mov [esp+0Ch],ebx
0x1008c346: mov eax,0Bh
0x1008c34b: lea ecx,[edx+34h]
0x1008c34e: mov edi,edi
0x1008c350: mov edi,eax
0x1008c352: shl edi,5
0x1008c355: mov edi,[esp+edi+34h]
0x1008c359: cmp edi,[esp+10h]
0x1008c35d: jge short 1008C364h
0x1008c35f: lea edx,[eax+1]
0x1008c362: jmp short 1008C369h
0x1008c364: jle short 1008C374h
0x1008c366: lea ecx,[eax-1]
0x1008c369: lea eax,[ecx+edx]
0x1008c36c: sar eax,1
0x1008c36e: cmp edx,ecx
0x1008c370: jle short 1008C350h
0x1008c372: jmp short 1008C37Fh
0x1008c374: shl eax,5
0x1008c377: lea edi,[esp+eax+34h]
0x1008c37b: cmp edi,ebx
0x1008c37d: jne short 1008C383h
0x1008c37f: lea edi,[esp+14h]
0x1008c383: mov eax,[esp+10h]
0x1008c387: add dword ptr [esi+110h],1
0x1008c38e: mov [esi+10Ch],eax
0x1008c394: mov eax,[edi+0Ch]
0x1008c397: cmp eax,ebx
0x1008c399: je short 1008C409h
0x1008c39b: mov edx,[esp+6DCh]
0x1008c3a2: lea ecx,[esp+0Ch]
0x1008c3a6: push ecx
0x1008c3a7: push ebp
0x1008c3a8: push edx
0x1008c3a9: push edi
0x1008c3aa: push esi
0x1008c3ab: call eax
0x1008c3ad: mov ebp,eax
0x1008c3af: mov eax,[esp+20h]
0x1008c3b3: add esp,14h
0x1008c3b6: cmp ebp,ebx
0x1008c3b8: jne short 1008C3D0h
0x1008c3ba: cmp eax,ebx
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0x1008c3bc: je short 1008C3DEh
0x1008c3be: cmp dword ptr [edi],40404040h
0x1008c3c4: jne short 1008C3D0h
0x1008c3c6: mov ecx,[esp+10h]
0x1008c3ca: mov [eax],ecx
0x1008c3cc: mov eax,[esp+0Ch]
0x1008c3d0: cmp eax,ebx
0x1008c3d2: je short 1008C3DEh
0x1008c3d4: push eax
0x1008c3d5: push esi
0x1008c3d6: call 100A0F80h
0x1008c3db: add esp,8
0x1008c3de: cmp [esi+11Dh],bl
0x1008c3e4: jne short 1008C40Dh
0x1008c3e6: cmp [esi+11Eh],bl
0x1008c3ec: jne short 1008C40Dh
0x1008c3ee: cmp [esi+121h],bl
0x1008c3f4: jne short 1008C40Dh
0x1008c3f6: push esi
0x1008c3f7: call 1008AD30h
0x1008c3fc: add esp,4
0x1008c3ff: pop edi
0x1008c400: pop ebp
0x1008c401: pop ebx
0x1008c402: add esp,6C8h
0x1008c408: ret
0x1008c409: xor ebp,ebp
0x1008c40b: jmp short 1008C3DEh
0x1008c40d: pop edi
0x1008c40e: mov eax,ebp
0x1008c410: pop ebp
0x1008c411: pop ebx
0x1008c412: add esp,6C8h
0x1008c418: ret
```

`sub_1008b280` is called by sub_1008c420, sub_1008c980. It accesses memory-mapped peripheral registers at 0x40404040. It delegates to sub_1008ad30, sub_100a0f80, png_get_uint_32 (3 callees).

12.1109 sub_1008c420 at 0x1008c420 (84 instructions)

```
0x1008c420: sub esp,8
0x1008c423: push ebx
0x1008c424: push ebp
0x1008c425: push esi
0x1008c426: mov esi,eax
0x1008c428: mov eax,[esi+74h]
0x1008c42b: test al,0Fh
0x1008c42d: mov dword ptr [esp+0Ch],0
0x1008c435: je near ptr 1008C4F4h
0x1008c43b: mov bl,[edi]
0x1008c43d: shr bl,5
0x1008c440: not bl
0x1008c442: and bl,1
0x1008c445: lea ebp,[ecx-4]
0x1008c448: je short 1008C457h
0x1008c44a: test eax,0F000h
0x1008c44f: jne short 1008C45Eh
0x1008c451: push ebp
0x1008c452: jmp near ptr 1008C4F5h
0x1008c457: test eax,0F00h
0x1008c45c: je short 1008C451h
0x1008c45e: push ebp
0x1008c45f: push edi
0x1008c460: push esi
0x1008c461: call 100A3660h
0x1008c466: mov [esp+1Ch],eax
```

```
0x1008c46a: lea eax,[edi+ebp]
0x1008c46d: push eax
0x1008c46e: call 100A3680h
0x1008c473: add esp,10h
0x1008c476: cmp [esp+10h],eax
0x1008c47a: je short 1008C451h
0x1008c47c: test bl,bl
0x1008c47e: mov eax,[esi+74h]
0x1008c481: je short 1008C4B4h
0x1008c483: and eax,0F000h
0x1008c488: cmp eax,2000h
0x1008c48d: je short 1008C4CEh
0x1008c48f: cmp eax,3000h
0x1008c494: jne short 1008C451h
0x1008c496: push 0
0x1008c498: push 0
0x1008c49a: push 403h
0x1008c49f: push esi
0x1008c4a0: call 10080800h
0x1008c4a5: add esp,10h
0x1008c4a8: pop esi
0x1008c4a9: pop ebp
0x1008c4aa: mov eax,403h
0x1008c4af: pop ebx
0x1008c4b0: add esp,8
0x1008c4b3: ret
0x1008c4b4: and eax,0F00h
0x1008c4b9: cmp eax,100h
0x1008c4be: je short 1008C502h
0x1008c4c0: cmp eax,200h
0x1008c4c5: je short 1008C4CEh
0x1008c4c7: cmp eax,300h
0x1008c4cc: jmp short 1008C494h
0x1008c4ce: push 0
0x1008c4d0: push 0
0x1008c4d2: push 403h
0x1008c4d7: push esi
0x1008c4d8: call 10080800h
0x1008c4dd: add esp,10h
0x1008c4e0: test al,al
0x1008c4e2: jne near ptr 1008C451h
0x1008c4e8: pop esi
0x1008c4e9: pop ebp
0x1008c4ea: mov eax,403h
0x1008c4ef: pop ebx
0x1008c4f0: add esp,8
0x1008c4f3: ret
0x1008c4f4: push ecx
0x1008c4f5: push edi
0x1008c4f6: call 1008B280h
0x1008c4fb: mov [esp+14h],eax
0x1008c4ff: add esp,8
0x1008c502: mov eax,[esp+0Ch]
0x1008c506: pop esi
0x1008c507: pop ebp
0x1008c508: pop ebx
0x1008c509: add esp,8
0x1008c50c: ret
```

`sub_1008c420` is called by sub_1008c510. It delegates to sub_10080800, sub_1008b280, sub_100a3660 (4 callees).

12.1110 sub_1008c510 at 0x1008c510 (294 instructions)

```
0x1008c510: sub esp,0Ch
0x1008c513: push ebp
0x1008c514: push esi
```

```
0x1008c515: mov esi,eax
0x1008c517: mov eax,[esi+15Ch]
0x1008c51d: mov [esp+0Ch],eax
0x1008c521: mov eax,[esi+204h]
0x1008c527: xor ebp,ebp
0x1008c529: cmp eax,ebp
0x1008c52b: push edi
0x1008c52c: mov edi,[esi+160h]
0x1008c532: mov [esp+14h],edi
0x1008c536: mov [esp+0Ch],ebp
0x1008c53a: je near ptr 1008C5EEh
0x1008c540: mov ecx,[eax+4]
0x1008c543: push eax
0x1008c544: push esi
0x1008c545: call ecx
0x1008c547: add esp,8
0x1008c54a: cmp eax,ebp
0x1008c54c: jne near ptr 1008C943h
0x1008c552: cmp [esi+204h],ebp
0x1008c558: je short 1008C5B3h
0x1008c55a: cmp byte ptr [esi+1F1h],0
0x1008c561: jne short 1008C5B3h
0x1008c563: cmp byte ptr [esi+1F3h],0
0x1008c56a: jne short 1008C5B3h
0x1008c56c: cmp byte ptr [esi+158h],0
0x1008c573: je short 1008C585h
0x1008c575: cmp byte ptr [esi+1B0h],0
0x1008c57c: jne short 1008C585h
0x1008c57e: mov byte ptr [esi+1F1h],0
0x1008c585: mov edx,[esi+204h]
0x1008c58b: mov ecx,[edx+8]
0x1008c58e: cmp ecx,ebp
0x1008c590: mov [esi+204h],ecx
0x1008c596: jne short 1008C5B3h
0x1008c598: cmp byte ptr [esi+130h],0
0x1008c59f: je short 1008C5B3h
0x1008c5a1: cmp byte ptr [esi+11Dh],0
0x1008c5a8: jne short 1008C5B3h
0x1008c5aa: push esi
0x1008c5ab: call 10083B00h
0x1008c5b0: add esp,4
0x1008c5b3: cmp eax,ebp
0x1008c5b5: jne near ptr 1008C943h
0x1008c5bb: mov eax,[esi+204h]
0x1008c5c1: cmp eax,ebp
0x1008c5c3: je near ptr 1008C65Bh
0x1008c5c9: cmp byte ptr [esi+1F1h],0
0x1008c5d0: jne near ptr 1008C65Bh
0x1008c5d6: cmp byte ptr [esi+1F3h],0
0x1008c5dd: jne short 1008C65Bh
0x1008c5df: cmp byte ptr [esi+1F4h],0
0x1008c5e6: je near ptr 1008C540h
0x1008c5ec: jmp short 1008C65Bh
0x1008c5ee: mov al,[esi+1F2h]
0x1008c5f4: test al,al
0x1008c5f6: je short 1008C65Bh
0x1008c5f8: movzx eax,al
0x1008c5fb: add eax,0FFFFFFFFh
0x1008c5fe: cmp eax,0Ah
0x1008c601: ja short 1008C65Bh
0x1008c603: jmp dword ptr [eax*4+1008C94Ch]
0x1008c65b: cmp byte ptr [esi+1F1h],0
0x1008c662: push ebx
0x1008c663: jne near ptr 1008C940h
0x1008c669: cmp byte ptr [esi+1F3h],0
0x1008c670: jne near ptr 1008C916h
0x1008c676: cmp byte ptr [esi+15Ah],0
```

```
0x1008c67d: jne near ptr 1008C916h
0x1008c683: cmp byte ptr [esi+1F4h],0
0x1008c68a: je short 1008C69Ch
0x1008c68c: cmp byte ptr [esi+171h],0
0x1008c693: jne short 1008C69Ch
0x1008c695: mov byte ptr [esi+1F0h],0
0x1008c69c: mov bl,2
0x1008c69e: cmp [esi+171h],bl
0x1008c6a4: ja short 1008C6D8h
0x1008c6a6: lea eax,[esp+10h]
0x1008c6aa: push eax
0x1008c6ab: mov ebp,4
0x1008c6b0: lea ecx,[esi+188h]
0x1008c6b6: push ebp
0x1008c6b7: push ecx
0x1008c6b8: mov eax,edi
0x1008c6ba: call 1008AF60h
0x1008c6bf: add esp,0Ch
0x1008c6c2: test eax,eax
0x1008c6c4: jne near ptr 1008C942h
0x1008c6ca: cmp [esi+170h],al
0x1008c6d0: je short 1008C744h
0x1008c6d2: mov [esi+171h],bl
0x1008c6d8: cmp byte ptr [esi+170h],0
0x1008c6df: jne near ptr 1008C916h
0x1008c6e5: mov al,[esi+171h]
0x1008c6eb: cmp al,bl
0x1008c6ed: ja short 1008C76Bh
0x1008c6ef: mov ebx,[esp+10h]
0x1008c6f3: cmp ebx,ebp
0x1008c6f5: je short 1008C76Bh
0x1008c6f7: push esi
0x1008c6f8: call 1008AD30h
0x1008c6fd: add esp,4
0x1008c700: test eax,eax
0x1008c702: jne near ptr 1008C942h
0x1008c708: test ebx,ebx
0x1008c70a: jne short 1008C728h
0x1008c70c: cmp [esi+11Eh],al
0x1008c712: jne short 1008C728h
0x1008c714: cmp [esi+11Dh],al
0x1008c71a: jne short 1008C728h
0x1008c71c: cmp [esi+121h],al
0x1008c722: je near ptr 1008C916h
0x1008c728: push 0
0x1008c72a: push 0
0x1008c72c: push 4
0x1008c72e: push esi
0x1008c72f: call 10080800h
0x1008c734: add esp,10h
0x1008c737: pop ebx
0x1008c738: pop edi
0x1008c739: pop esi
0x1008c73a: mov eax,4
0x1008c73f: pop ebp
0x1008c740: add esp,0Ch
0x1008c743: ret
0x1008c744: push edi
0x1008c745: call 100A3680h
0x1008c74a: add esp,4
0x1008c74d: cmp eax,7FFFFFFh
0x1008c752: mov [esi+184h],eax
0x1008c758: jbe near ptr 1008C6D8h
0x1008c75e: pop ebx
0x1008c75f: pop edi
0x1008c760: pop esi
0x1008c761: mov eax,404h
```

```
0x1008c766: pop ebp
0x1008c767: add esp,0Ch
0x1008c76a: ret
0x1008c76b: test byte ptr [esi+74h],0Fh
0x1008c76f: mov ebp,[esi+184h]
0x1008c775: je short 1008C77Ch
0x1008c777: add ebp,8
0x1008c77a: jmp short 1008C77Fh
0x1008c77c: add ebp,4
0x1008c77f: mov ebx,[esi+198h]
0x1008c785: test ebx,ebx
0x1008c787: je short 1008C7CDh
0x1008c789: cmp ebp,[ebx+14h]
0x1008c78c: ja short 1008C7CDh
0x1008c78e: mov edi,[ebx+10h]
0x1008c791: sub [ebx+14h],ebp
0x1008c794: lea edx,[edi+ebp]
0x1008c797: mov [ebx+10h],edx
0x1008c79a: mov ecx,ebp
0x1008c79c: mov eax,esi
0x1008c79e: mov byte ptr [esi+171h],0
0x1008c7a5: call 1008C420h
0x1008c7aa: mov edi,eax
0x1008c7ac: test edi,edi
0x1008c7ae: jne near ptr 1008C90Ch
0x1008c7b4: cmp [ebx+14h],eax
0x1008c7b7: jne near ptr 1008C908h
0x1008c7bd: push esi
0x1008c7be: call 1008AD80h
0x1008c7c3: add esp,4
0x1008c7c6: mov edi,eax
0x1008c7c8: jmp near ptr 1008C908h
0x1008c7cd: cmp ebp,[esp+14h]
0x1008c7d1: jae short 1008C826h
0x1008c7d3: lea eax,[esp+10h]
0x1008c7d7: push eax
0x1008c7d8: lea ecx,[esi+188h]
0x1008c7de: push ebp
0x1008c7df: push ecx
0x1008c7e0: mov eax,edi
0x1008c7e2: call 1008AF60h
0x1008c7e7: mov edi,eax
0x1008c7e9: add esp,0Ch
0x1008c7ec: test edi,edi
0x1008c7ee: jne near ptr 1008C90Ch
0x1008c7f4: cmp [esi+170h],al
0x1008c7fa: je short 1008C808h
0x1008c7fc: mov byte ptr [esi+171h],3
0x1008c803: jmp near ptr 1008C908h
0x1008c808: cmp [esp+10h],ebp
0x1008c80c: jne near ptr 1008C728h
0x1008c812: mov edi,[esp+18h]
0x1008c816: mov ecx,ebp
0x1008c818: mov eax,esi
0x1008c81a: call 1008C420h
0x1008c81f: mov edi,eax
0x1008c821: jmp near ptr 1008C908h
0x1008c826: cmp ebp,1000000h
0x1008c82c: jbe short 1008C84Dh
0x1008c82e: push 0
0x1008c830: push 0
0x1008c832: push 429h
0x1008c837: push esi
0x1008c838: call 10080800h
0x1008c83d: add esp,10h
0x1008c840: pop ebx
0x1008c841: pop edi
```

```
0x1008c842: pop esi
0x1008c843: mov eax,429h
0x1008c848: pop ebp
0x1008c849: add esp,0Ch
0x1008c84c: ret
0x1008c84d: test al,al
0x1008c84f: jne short 1008C894h
0x1008c851: mov edx,[esi+98h]
0x1008c857: lea eax,[ebp+1]
0x1008c85a: push eax
0x1008c85b: mov [esi+164h],eax
0x1008c861: mov dword ptr [esi+18Ch],0
0x1008c86b: call edx
0x1008c86d: add esp,4
0x1008c870: test eax,eax
0x1008c872: mov [esi+168h],eax
0x1008c878: jne short 1008C894h
0x1008c87a: push eax
0x1008c87b: push eax
0x1008c87c: push 1
0x1008c87e: push esi
0x1008c87f: call 10080800h
0x1008c884: add esp,10h
0x1008c887: pop ebx
0x1008c888: pop edi
0x1008c889: pop esi
0x1008c88a: mov eax,1
0x1008c88f: pop ebp
0x1008c890: add esp,0Ch
0x1008c893: ret
0x1008c894: lea eax,[esp+10h]
0x1008c898: push eax
0x1008c899: mov eax,[esi+168h]
0x1008c89f: lea ecx,[esi+18Ch]
0x1008c8a5: push ebp
0x1008c8a6: push ecx
0x1008c8a7: call 1008AF60h
0x1008c8ac: mov edi,eax
0x1008c8ae: add esp,0Ch
0x1008c8b1: test edi,edi
0x1008c8b3: jne short 1008C90Ch
0x1008c8b5: cmp [esi+170h],al
0x1008c8bb: je short 1008C8C6h
0x1008c8bd: mov byte ptr [esi+171h],4
0x1008c8c4: jmp short 1008C908h
0x1008c8c6: cmp [esp+10h],ebp
0x1008c8ca: jne near ptr 1008C728h
0x1008c8d0: mov edi,[esi+168h]
0x1008c8d6: mov ecx,ebp
0x1008c8d8: mov eax,esi
0x1008c8da: call 1008C420h
0x1008c8df: mov edi,eax
0x1008c8e1: mov eax,[esi+168h]
0x1008c8e7: test eax,eax
0x1008c8e9: je short 1008C908h
0x1008c8eb: mov edx,[esi+164h]
0x1008c8f1: push edx
0x1008c8f2: push eax
0x1008c8f3: mov eax,[esi+9Ch]
0x1008c8f9: call eax
0x1008c8fb: add esp,8
0x1008c8fe: mov dword ptr [esi+168h],0
0x1008c908: test edi,edi
0x1008c90a: je short 1008C916h
0x1008c90c: pop ebx
0x1008c90d: mov eax,edi
0x1008c90f: pop edi
```

```
0x1008c910: pop esi
0x1008c911: pop ebp
0x1008c912: add esp,0Ch
0x1008c915: ret
0x1008c916: cmp byte ptr [esi+1F1h],0
0x1008c91d: jne short 1008C940h
0x1008c91f: cmp byte ptr [esi+170h],0
0x1008c926: jne short 1008C940h
0x1008c928: cmp byte ptr [esi+1F6h],0
0x1008c92f: je short 1008C940h
0x1008c931: push 1
0x1008c933: push esi
0x1008c934: call 10080920h
0x1008c939: add esp,8
0x1008c93c: test eax,eax
0x1008c93e: jne short 1008C942h
0x1008c940: xor eax,eax
0x1008c942: pop ebx
0x1008c943: pop edi
0x1008c944: pop esi
0x1008c945: pop ebp
0x1008c946: add esp,0Ch
0x1008c949: ret
```

`sub_1008c510` is called by sub_1008cb40. It delegates to sub_10080800, sub_10080920, sub_10083b00 (8 callees).

12.1111 sub_1008c980 at 0x1008c980 (88 instructions)

```
0x1008c980: push esi
0x1008c981: mov esi,eax
0x1008c983: mov eax,[esi+204h]
0x1008c989: test eax,eax
0x1008c98b: je near ptr 1008CA38h
0x1008c991: push eax
0x1008c992: mov eax,[eax+4]
0x1008c995: push esi
0x1008c996: call eax
0x1008c998: add esp,8
0x1008c99b: test eax,eax
0x1008c99d: jne near ptr 1008CB0Dh
0x1008c9a3: cmp [esi+204h],eax
0x1008c9a9: je short 1008C9FDh
0x1008c9ab: cmp [esi+1F1h],al
0x1008c9b1: jne short 1008C9FDh
0x1008c9b3: cmp [esi+1F3h],al
0x1008c9b9: jne short 1008C9FDh
0x1008c9bb: cmp [esi+158h],al
0x1008c9c1: je short 1008C9D1h
0x1008c9c3: cmp [esi+1B0h],al
0x1008c9c9: jne short 1008C9D1h
0x1008c9cb: mov [esi+1F1h],al
0x1008c9d1: mov ecx,[esi+204h]
0x1008c9d7: mov ecx,[ecx+8]
0x1008c9da: test ecx,ecx
0x1008c9dc: mov [esi+204h],ecx
0x1008c9e2: jne short 1008C9FDh
0x1008c9e4: cmp [esi+130h],cl
0x1008c9ea: je short 1008C9FDh
0x1008c9ec: cmp [esi+11Dh],cl
0x1008c9f2: jne short 1008C9FDh
0x1008c9f4: push esi
0x1008c9f5: call 10083B00h
0x1008c9fa: add esp,4
0x1008c9fd: test eax,eax
0x1008c9ff: jne near ptr 1008CB0Dh
0x1008ca05: mov eax,[esi+204h]
```

```
0x1008ca0b: test eax,eax
0x1008ca0d: je near ptr 1008CAA1h
0x1008ca13: cmp byte ptr [esi+1F1h],0
0x1008ca1a: jne near ptr 1008CB04h
0x1008ca20: cmp byte ptr [esi+1F3h],0
0x1008ca27: jne short 1008CAA1h
0x1008ca29: cmp byte ptr [esi+1F4h],0
0x1008ca30: je near ptr 1008C991h
0x1008ca36: jmp short 1008CAA1h
0x1008ca38: mov al,[esi+1F2h]
0x1008ca3e: test al,al
0x1008ca40: je short 1008CAA1h
0x1008ca42: movzx eax,al
0x1008ca45: add eax,0FFFFFFFFh
0x1008ca48: cmp eax,0Ah
0x1008ca4b: ja short 1008CAA1h
0x1008ca4d: jmp dword ptr [eax*4+1008CB10h]
0x1008caa1: cmp byte ptr [esi+1F1h],0
0x1008caa8: jne short 1008CB04h
0x1008caaa: cmp byte ptr [esi+1F3h],0
0x1008cab1: jne short 1008CB04h
0x1008cab3: cmp byte ptr [esi+15Ah],0
0x1008caba: jne short 1008CB04h
0x1008cabc: mov eax,[esi+190h]
0x1008cac2: mov byte ptr [esi+171h],0
0x1008cac9: mov edx,[eax+8]
0x1008cacc: mov eax,[eax+4]
0x1008cacf: push edx
0x1008cad0: push eax
0x1008cad1: call 1008B280h
0x1008cad6: add esp,8
0x1008cad9: test eax,eax
0x1008cadb: jne short 1008CB0Dh
0x1008cadd: cmp [esi+1F1h],al
0x1008cae3: jne short 1008CB04h
0x1008cae5: cmp [esi+170h],al
0x1008caeb: jne short 1008CB04h
0x1008caed: cmp [esi+1F6h],al
0x1008caf3: je short 1008CB04h
0x1008caf5: push 1
0x1008caf7: push esi
0x1008caf8: call 10080920h
0x1008cafd: add esp,8
0x1008cb00: test eax,eax
0x1008cb02: jne short 1008CB0Dh
0x1008cb04: push esi
0x1008cb05: call 1008AE00h
0x1008cb0a: add esp,4
0x1008cb0d: pop esi
0x1008cb0e: ret
```

`sub_1008c980` is called by sub_1008cb40. It delegates to sub_10080920, sub_10083b00, sub_1008ae00 (4 callees).

12.1112 sub_1008cb40 at 0x1008cb40 (123 instructions)

```
0x1008cb40: push esi
0x1008cb41: mov esi,[esp+8]
0x1008cb45: cmp dword ptr [esi+160h],0
0x1008cb4c: jne short 1008CB86h
0x1008cb4e: mov eax,[esi+98h]
0x1008cb54: push 1068h
0x1008cb59: mov dword ptr [esi+15Ch],1068h
0x1008cb63: call eax
0x1008cb65: add esp,4
0x1008cb68: test eax,eax
0x1008cb6a: mov [esi+160h],eax
```

```
0x1008cb70: jne short 1008CB86h
0x1008cb72: push eax
0x1008cb73: push eax
0x1008cb74: push 1
0x1008cb76: push esi
0x1008cb77: call 10080800h
0x1008cb7c: add esp,10h
0x1008cb7f: mov eax,1
0x1008cb84: pop esi
0x1008cb85: ret
0x1008cb86: cmp byte ptr [esi+159h],0
0x1008cb8d: je short 1008CB9Ch
0x1008cb8f: cmp byte ptr [esi+171h],1
0x1008cb96: jne near ptr 1008CC74h
0x1008cb9c: mov eax,[esi+160h]
0x1008cba2: lea ecx,[esp+8]
0x1008cba6: push ecx
0x1008cba7: lea edx,[esi+188h]
0x1008cbad: push 8
0x1008cbaf: push edx
0x1008cbb0: call 1008AF60h
0x1008cbb5: add esp,0Ch
0x1008cbb8: test eax,eax
0x1008cbba: jne near ptr 1008CD0Ch
0x1008cbc0: cmp [esi+170h],al
0x1008cbc6: je short 1008CBD4h
0x1008cbc8: mov byte ptr [esi+171h],1
0x1008cbcf: jmp near ptr 1008CC74h
0x1008cbd4: cmp dword ptr [esp+8],8
0x1008cbd9: je short 1008CBF1h
0x1008cbdb: push 0
0x1008cbdd: push 0
0x1008cbdf: push 4
0x1008cbe1: push esi
0x1008cbe2: call 10080800h
0x1008cbe7: add esp,10h
0x1008cbea: mov eax,4
0x1008cbef: pop esi
0x1008cbf0: ret
0x1008cbf1: mov eax,[esi+160h]
0x1008cbf7: push eax
0x1008cbf8: call 100A3680h
0x1008cbfd: add esp,4
0x1008cc00: cmp eax,89504E47h
0x1008cc05: jne short 1008CC10h
0x1008cc07: mov dword ptr [esi+8],1
0x1008cc0e: jmp short 1008CC50h
0x1008cc10: mov ecx,[esi+160h]
0x1008cc16: push ecx
0x1008cc17: call 100A3680h
0x1008cc1c: add esp,4
0x1008cc1f: cmp eax,8B4A4E47h
0x1008cc24: jne short 1008CC2Fh
0x1008cc26: mov dword ptr [esi+8],3
0x1008cc2d: jmp short 1008CC50h
0x1008cc2f: mov edx,[esi+160h]
0x1008cc35: push edx
0x1008cc36: call 100A3680h
0x1008cc3b: add esp,4
0x1008cc3e: cmp eax,8A4D4E47h
0x1008cc43: jne near ptr 1008CCF5h
0x1008cc49: mov dword ptr [esi+8],2
0x1008cc50: mov eax,[esi+160h]
0x1008cc56: add eax,4
0x1008cc59: push eax
0x1008cc5a: call 100A3680h
0x1008cc5f: add esp,4
```

```
0x1008cc62: cmp eax,0D0A1A0Ah
0x1008cc67: jne near ptr 1008CCF5h
0x1008cc6d: mov byte ptr [esi+159h],1
0x1008cc74: cmp byte ptr [esi+170h],0
0x1008cc7b: jne short 1008CCF1h
0x1008cc7d: lea ecx,[ecx]
0x1008cc80: cmp byte ptr [esi+158h],0
0x1008cc87: je short 1008CC99h
0x1008cc89: cmp byte ptr [esi+1B0h],0
0x1008cc90: jne short 1008CC99h
0x1008cc92: mov byte ptr [esi+1F1h],0
0x1008cc99: cmp dword ptr [esi+190h],0
0x1008cca0: mov eax,esi
0x1008cca2: je short 1008CCABh
0x1008cca4: call 1008C980h
0x1008cca9: jmp short 1008CCB0h
0x1008ccab: call 1008C510h
0x1008ccb0: test eax,eax
0x1008ccb2: jne short 1008CD0Ch
0x1008ccb4: cmp [esi+15Ah],al
0x1008ccba: je short 1008CCC4h
0x1008ccbc: cmp [esi+204h],eax
0x1008ccc2: je short 1008CCF1h
0x1008ccc4: cmp byte ptr [esi+170h],0
0x1008cccb: jne short 1008CCF1h
0x1008cccd: cmp byte ptr [esi+1F3h],0
0x1008ccd4: jne short 1008CCF1h
0x1008ccd6: cmp byte ptr [esi+1F1h],0
0x1008ccdd: je short 1008CC80h
0x1008ccdf: cmp byte ptr [esi+158h],0
0x1008cce6: je short 1008CCF1h
0x1008cce8: cmp byte ptr [esi+1B0h],0
0x1008ccef: je short 1008CC80h
0x1008ccf1: xor eax,eax
0x1008ccf3: pop esi
0x1008ccf4: ret
0x1008ccf5: push 0
0x1008ccf7: push 0
0x1008ccf9: push 401h
0x1008ccfe: push esi
0x1008ccff: call 10080800h
0x1008cd04: mov eax,401h
0x1008cd09: add esp,10h
0x1008cd0c: pop esi
0x1008cd0d: ret
```

`sub_1008cb40` is called by sub_1007ff70, sub_10080150. It accesses memory-mapped peripheral registers at 0x89504e47. It delegates to sub_10080800, sub_1008af60, sub_1008c510 (5 callees).

12.1113 sub_1008cd60 at 0x1008cd60 (9 instructions)

```
0x1008cd60: mov eax,[esp+4]
0x1008cd64: xor cl,cl
0x1008cd66: mov dword ptr [eax+944h],1008CD10h
0x1008cd70: mov dword ptr [eax+948h],1008CD40h
0x1008cd7a: mov [eax+94Ch],eax
0x1008cd80: mov [eax+974h],cl
0x1008cd86: mov [eax+975h],cl
0x1008cd8c: xor eax,eax
0x1008cd8e: ret
```

`sub_1008cd60` is called by sub_10080300. It calls nothing else.

12.1114 sub_1008d0a0 at 0x1008d0a0 (21 instructions)

```
0x1008d0a0: push esi
0x1008d0a1: mov esi,[esp+8]
0x1008d0a5: lea eax,[esi+924h]
0x1008d0ab: push eax
0x1008d0ac: mov byte ptr [esi+974h],0
0x1008d0b3: call 10061790h
0x1008d0b8: add esp,4
0x1008d0bb: test eax,eax
0x1008d0bd: je short 1008D0D4h
0x1008d0bf: push 0
0x1008d0c1: push eax
0x1008d0c2: push 5
0x1008d0c4: push esi
0x1008d0c5: call 10080800h
0x1008d0ca: add esp,10h
0x1008d0cd: mov eax,5
0x1008d0d2: pop esi
0x1008d0d3: ret
0x1008d0d4: xor eax,eax
0x1008d0d6: pop esi
0x1008d0d7: ret
```

`sub_1008d0a0` is called by sub_1007f790, sub_1008d1f0. It delegates to sub_10061790, sub_10080800 (2 callees).

12.1115 sub_1008d1b0 at 0x1008d1b0 (21 instructions)

```
0x1008d1b0: push esi
0x1008d1b1: mov esi,[esp+8]
0x1008d1b5: lea eax,[esi+924h]
0x1008d1bb: push eax
0x1008d1bc: call 100619A0h
0x1008d1c1: add esp,4
0x1008d1c4: test eax,eax
0x1008d1c6: je short 1008D1DDh
0x1008d1c8: push 0
0x1008d1ca: push eax
0x1008d1cb: push 5
0x1008d1cd: push esi
0x1008d1ce: call 10080800h
0x1008d1d3: add esp,10h
0x1008d1d6: mov eax,5
0x1008d1db: pop esi
0x1008d1dc: ret
0x1008d1dd: mov byte ptr [esi+975h],0
0x1008d1e4: xor eax,eax
0x1008d1e6: pop esi
0x1008d1e7: ret
```

`sub_1008d1b0` is called by sub_1008d1f0. It delegates to sub_100619a0, sub_10080800 (2 callees).

12.1116 sub_1008d1f0 at 0x1008d1f0 (15 instructions)

```
0x1008d1f0: push esi
0x1008d1f1: mov esi,[esp+8]
0x1008d1f5: cmp byte ptr [esi+974h],0
0x1008d1fc: je short 1008D207h
0x1008d1fe: push esi
0x1008d1ff: call 1008D0A0h
0x1008d204: add esp,4
0x1008d207: cmp byte ptr [esi+975h],0
0x1008d20e: je short 1008D219h
0x1008d210: push esi
0x1008d211: call 1008D1B0h
0x1008d216: add esp,4
```

```
0x1008d219: xor eax,eax
0x1008d21b: pop esi
0x1008d21c: ret
```

`sub_1008d1f0` is called by sub_1007ff10. It delegates to sub_1008d0a0, sub_1008d1b0 (2 callees).

12.1117 sub_10098a00 at 0x10098a00 (78 instructions)

```
0x10098a00: mov eax,[esp+4]
0x10098a04: mov ecx,[eax+240h]
0x10098a0a: add [eax+23Ch],ecx
0x10098a10: mov cl,[eax+238h]
0x10098a16: test cl,cl
0x10098a18: jl near ptr 10098B33h
0x10098a1e: cmp cl,7
0x10098a21: jge near ptr 10098B33h
0x10098a27: push ebx
0x10098a28: push ebp
0x10098a29: push esi
0x10098a2a: push edi
0x10098a2b: jmp short 10098A30h
0x10098a30: mov ebp,[eax+140h]
0x10098a36: cmp [eax+23Ch],ebp
0x10098a3c: jge short 10098A50h
0x10098a3e: mov edx,[eax+244h]
0x10098a44: cmp edx,[eax+13Ch]
0x10098a4a: jl near ptr 10098B10h
0x10098a50: add byte ptr [eax+238h],1
0x10098a57: mov cl,[eax+238h]
0x10098a5d: cmp cl,7
0x10098a60: jge near ptr 10098B03h
0x10098a66: mov edi,[eax+13Ch]
0x10098a6c: movsx ecx,cl
0x10098a6f: add ecx,ecx
0x10098a71: mov edx,[ecx+ecx+101278E0h]
0x10098a78: add ecx,ecx
0x10098a7a: mov [eax+23Ch],edx
0x10098a80: mov edx,[ecx+101278FCh]
0x10098a86: mov [eax+240h],edx
0x10098a8c: mov edx,[ecx+10127918h]
0x10098a92: mov [eax+244h],edx
0x10098a98: mov edx,[ecx+10127934h]
0x10098a9e: mov esi,[eax+244h]
0x10098aa4: mov [eax+248h],edx
0x10098aaa: mov edx,[ecx+10127950h]
0x10098ab0: mov ecx,[ecx+1012796Ch]
0x10098ab6: sub edx,esi
0x10098ab8: add edx,edi
0x10098aba: shr edx,cl
0x10098abc: mov ecx,[eax+250h]
0x10098ac2: cmp ecx,1
0x10098ac5: mov [eax+24Ch],edx
0x10098acb: jle short 10098B17h
0x10098acd: imul edx,ecx
0x10098ad0: mov [eax+25Ch],edx
0x10098ad6: cmp [eax+23Ch],ebp
0x10098adc: jge short 10098B03h
0x10098ade: cmp esi,edi
0x10098ae0: jge short 10098B03h
0x10098ae2: mov ecx,[eax+280h]
0x10098ae8: xor edx,edx
0x10098aea: cmp [eax+25Ch],edx
0x10098af0: jle short 10098B03h
0x10098af2: mov byte ptr [ecx],0
0x10098af5: add edx,1
0x10098af8: add ecx,1
```

```
0x10098afb: cmp edx,[eax+25Ch]
0x10098b01: jl short 10098AF2h
0x10098b03: cmp byte ptr [eax+238h],7
0x10098b0a: jl near ptr 10098A30h
0x10098b10: pop edi
0x10098b11: pop esi
0x10098b12: pop ebp
0x10098b13: xor eax,eax
0x10098b15: pop ebx
0x10098b16: ret
0x10098b17: mov ecx,[eax+258h]
0x10098b1d: test ecx,ecx
0x10098b1f: jle short 10098AD0h
0x10098b21: mov ebx,[eax+254h]
0x10098b27: add ebx,edx
0x10098b29: sar ebx,cl
0x10098b2b: mov [eax+25Ch],ebx
0x10098b31: jmp short 10098AD6h
0x10098b33: xor eax,eax
0x10098b35: ret
```

`sub_10098a00` is called by sub_10081660. It calls nothing else.

12.1118 sub_10098b40 at 0x10098b40 (38 instructions)

```
0x10098b40: push esi
0x10098b41: mov esi,[esp+8]
0x10098b45: mov eax,[esi+284h]
0x10098b4b: test eax,eax
0x10098b4d: je short 10098B68h
0x10098b4f: mov ecx,[esi+13Ch]
0x10098b55: mov edx,[esi+9Ch]
0x10098b5b: add ecx,ecx
0x10098b5d: add ecx,ecx
0x10098b5f: add ecx,ecx
0x10098b61: push ecx
0x10098b62: push eax
0x10098b63: call edx
0x10098b65: add esp,8
0x10098b68: mov eax,[esi+280h]
0x10098b6e: test eax,eax
0x10098b70: je short 10098B85h
0x10098b72: mov ecx,[esi+260h]
0x10098b78: mov edx,[esi+9Ch]
0x10098b7e: push ecx
0x10098b7f: push eax
0x10098b80: call edx
0x10098b82: add esp,8
0x10098b85: mov eax,[esi+27Ch]
0x10098b8b: test eax,eax
0x10098b8d: je short 10098BA2h
0x10098b8f: mov ecx,[esi+260h]
0x10098b95: mov edx,[esi+9Ch]
0x10098b9b: push ecx
0x10098b9c: push eax
0x10098b9d: call edx
0x10098b9f: add esp,8
0x10098ba2: mov dword ptr [esi+27Ch],0
0x10098bac: mov dword ptr [esi+280h],0
0x10098bb6: mov dword ptr [esi+284h],0
0x10098bc0: xor eax,eax
0x10098bc2: pop esi
0x10098bc3: ret
```

`sub_10098b40` is called by sub_1007f790. It calls nothing else.

12.1119 sub_100a0020 at 0x100a0020 (171 instructions)

```
0x100a0020: push ebp
0x100a0021: mov ebp,esp
0x100a0023: mov eax,[ebp+8]
0x100a0026: push ebx
0x100a0027: push esi
0x100a0028: xor ebx,ebx
0x100a002a: add eax,0A08h
0x100a002f: push ebx
0x100a0030: push eax
0x100a0031: call 100229DCh
0x100a0036: add esp,8
0x100a0039: cmp eax,ebx
0x100a003b: je short 100A0054h
0x100a003d: mov ecx,[ebp+8]
0x100a0040: push ebx
0x100a0041: push eax
0x100a0042: push 6
0x100a0044: push ecx
0x100a0045: call 10080800h
0x100a004a: add esp,10h
0x100a004d: lea eax,[ebx+6]
0x100a0050: pop esi
0x100a0051: pop ebx
0x100a0052: pop ebp
0x100a0053: ret
0x100a0054: mov esi,[ebp+8]
0x100a0057: cmp [esi+9E5h],bl
0x100a005d: je short 100A006Eh
0x100a005f: mov edx,[esi+994h]
0x100a0065: push edx
0x100a0066: call 1003C530h
0x100a006b: add esp,4
0x100a006e: cmp [esi+9EBh],bl
0x100a0074: je short 100A0085h
0x100a0076: mov eax,[esi+9A0h]
0x100a007c: push eax
0x100a007d: call 1003C530h
0x100a0082: add esp,4
0x100a0085: cmp [esi+9E4h],bl
0x100a008b: je short 100A009Ch
0x100a008d: mov ecx,[esi+98Ch]
0x100a0093: push ecx
0x100a0094: call 1003C530h
0x100a0099: add esp,4
0x100a009c: mov eax,[esi+9C0h]
0x100a00a2: cmp eax,ebx
0x100a00a4: je short 100A00BFh
0x100a00a6: mov edx,[esi+9C4h]
0x100a00ac: push edx
0x100a00ad: push eax
0x100a00ae: mov eax,[esi+9Ch]
0x100a00b4: call eax
0x100a00b6: add esp,8
0x100a00b9: mov [esi+9C0h],ebx
0x100a00bf: mov eax,[esi+9ACh]
0x100a00c5: cmp eax,ebx
0x100a00c7: je short 100A00E2h
0x100a00c9: mov ecx,[esi+9B0h]
0x100a00cf: mov edx,[esi+9Ch]
0x100a00d5: push ecx
0x100a00d6: push eax
0x100a00d7: call edx
0x100a00d9: add esp,8
0x100a00dc: mov [esi+9ACh],ebx
0x100a00e2: mov eax,[esi+98Ch]
0x100a00e8: cmp eax,ebx
```

```
0x100a00ea: je short 100A0103h
0x100a00ec: push 1A0h
0x100a00f1: push eax
0x100a00f2: mov eax,[esi+9Ch]
0x100a00f8: call eax
0x100a00fa: add esp,8
0x100a00fd: mov [esi+98Ch],ebx
0x100a0103: mov eax,[esi+990h]
0x100a0109: cmp eax,ebx
0x100a010b: je short 100A0124h
0x100a010d: mov ecx,[esi+9Ch]
0x100a0113: push 84h
0x100a0118: push eax
0x100a0119: call ecx
0x100a011b: add esp,8
0x100a011e: mov [esi+990h],ebx
0x100a0124: mov eax,[esi+994h]
0x100a012a: cmp eax,ebx
0x100a012c: je short 100A0145h
0x100a012e: mov edx,[esi+9Ch]
0x100a0134: push 1C0h
0x100a0139: push eax
0x100a013a: call edx
0x100a013c: add esp,8
0x100a013f: mov [esi+994h],ebx
0x100a0145: mov eax,[esi+99Ch]
0x100a014b: cmp eax,ebx
0x100a014d: je short 100A0163h
0x100a014f: push 1Ch
0x100a0151: push eax
0x100a0152: mov eax,[esi+9Ch]
0x100a0158: call eax
0x100a015a: add esp,8
0x100a015d: mov [esi+99Ch],ebx
0x100a0163: mov eax,[esi+998h]
0x100a0169: cmp eax,ebx
0x100a016b: je short 100A0184h
0x100a016d: mov ecx,[esi+9Ch]
0x100a0173: push 84h
0x100a0178: push eax
0x100a0179: call ecx
0x100a017b: add esp,8
0x100a017e: mov [esi+998h],ebx
0x100a0184: mov eax,[esi+9A0h]
0x100a018a: cmp eax,ebx
0x100a018c: je short 100A01A5h
0x100a018e: mov edx,[esi+9Ch]
0x100a0194: push 1C0h
0x100a0199: push eax
0x100a019a: call edx
0x100a019c: add esp,8
0x100a019f: mov [esi+9A0h],ebx
0x100a01a5: mov eax,[esi+9A8h]
0x100a01ab: cmp eax,ebx
0x100a01ad: je short 100A01C3h
0x100a01af: push 1Ch
0x100a01b1: push eax
0x100a01b2: mov eax,[esi+9Ch]
0x100a01b8: call eax
0x100a01ba: add esp,8
0x100a01bd: mov [esi+9A8h],ebx
0x100a01c3: mov eax,[esi+9A4h]
0x100a01c9: cmp eax,ebx
0x100a01cb: je short 100A01E4h
0x100a01cd: mov ecx,[esi+9Ch]
0x100a01d3: push 84h
0x100a01d8: push eax
```

```
0x100a01d9: call ecx
0x100a01db: add esp,8
0x100a01de: mov [esi+9A4h],ebx
0x100a01e4: mov eax,[esi+9DCh]
0x100a01ea: cmp eax,ebx
0x100a01ec: je short 100A0207h
0x100a01ee: mov edx,[esi+9E0h]
0x100a01f4: push edx
0x100a01f5: push eax
0x100a01f6: mov eax,[esi+9Ch]
0x100a01fc: call eax
0x100a01fe: add esp,8
0x100a0201: mov [esi+9DCh],ebx
0x100a0207: mov eax,[esi+9D4h]
0x100a020d: cmp eax,ebx
0x100a020f: je short 100A022Ah
0x100a0211: mov ecx,[esi+9D8h]
0x100a0217: mov edx,[esi+9Ch]
0x100a021d: push ecx
0x100a021e: push eax
0x100a021f: call edx
0x100a0221: add esp,8
0x100a0224: mov [esi+9D4h],ebx
0x100a022a: mov [esi+9E4h],bl
0x100a0230: mov [esi+9E5h],bl
0x100a0236: mov [esi+9E6h],bl
0x100a023c: mov [esi+9E7h],bl
0x100a0242: mov [esi+9E8h],bl
0x100a0248: mov [esi+9E9h],bl
0x100a024e: mov [esi+9EBh],bl
0x100a0254: mov [esi+9ECh],bl
0x100a025a: mov [esi+9EDh],bl
0x100a0260: mov [esi+9EEh],bl
0x100a0266: pop esi
0x100a0267: xor eax,eax
0x100a0269: pop ebx
0x100a026a: pop ebp
0x100a026b: ret
```

`sub_100a0020` is called by sub_1007f790. It delegates to j__setjmp3, sub_1003c530, sub_10080800 (3 callees).

12.1120 sub_100a0f80 at 0x100a0f80 (35 instructions)

```
0x100a0f80: mov eax,[esp+4]
0x100a0f84: cmp dword ptr [eax+114h],0
0x100a0f8b: mov ecx,[esp+8]
0x100a0f8f: jne short 100A0FF6h
0x100a0f91: cmp dword ptr [eax+1A4h],0
0x100a0f98: mov [eax+114h],ecx
0x100a0f9e: jne short 100A1008h
0x100a0fa0: mov edx,[ecx]
0x100a0fa2: mov [eax+1A4h],edx
0x100a0fa8: mov edx,[ecx]
0x100a0faa: cmp edx,49484452h
0x100a0fb0: jne short 100A0FC6h
0x100a0fb2: mov dword ptr [eax+0Ch],1
0x100a0fb9: mov edx,[eax+0Ch]
0x100a0fbc: mov [eax+8],edx
0x100a0fbf: mov [eax+118h],ecx
0x100a0fc5: ret
0x100a0fc6: cmp edx,4A484452h
0x100a0fcc: jne short 100A0FE2h
0x100a0fce: mov dword ptr [eax+0Ch],3
0x100a0fd5: mov edx,[eax+0Ch]
0x100a0fd8: mov [eax+8],edx
0x100a0fdb: mov [eax+118h],ecx
```

```
0x100a0fe1: ret
0x100a0fe2: mov dword ptr [eax+0Ch],2
0x100a0fe9: mov edx,[eax+0Ch]
0x100a0fec: mov [eax+8],edx
0x100a0fef: mov [eax+118h],ecx
0x100a0ff5: ret
0x100a0ff6: mov edx,[eax+118h]
0x100a0ffc: mov [ecx+1Ch],edx
0x100a0fff: mov edx,[eax+118h]
0x100a1005: mov [edx+18h],ecx
0x100a1008: mov [eax+118h],ecx
0x100a100e: ret
```

`sub_100a0f80` is called by sub_1008b280. It accesses memory-mapped peripheral registers at 0x49484452. It calls nothing else.

12.1121 sub_100a3570 at 0x100a3570 (56 instructions)

```
0x100a3570: xor ecx,ecx
0x100a3572: lea edx,[esi+0A48h]
0x100a3578: mov eax,ecx
0x100a357a: shr eax,1
0x100a357c: test cl,1
0x100a357f: je short 100A3586h
0x100a3581: xor eax,0EDB88320h
0x100a3586: test al,1
0x100a3588: je short 100A3593h
0x100a358a: shr eax,1
0x100a358c: xor eax,0EDB88320h
0x100a3591: jmp short 100A3595h
0x100a3593: shr eax,1
0x100a3595: test al,1
0x100a3597: je short 100A35A2h
0x100a3599: shr eax,1
0x100a359b: xor eax,0EDB88320h
0x100a35a0: jmp short 100A35A4h
0x100a35a2: shr eax,1
0x100a35a4: test al,1
0x100a35a6: je short 100A35B1h
0x100a35a8: shr eax,1
0x100a35aa: xor eax,0EDB88320h
0x100a35af: jmp short 100A35B3h
0x100a35b1: shr eax,1
0x100a35b3: test al,1
0x100a35b5: je short 100A35C0h
0x100a35b7: shr eax,1
0x100a35b9: xor eax,0EDB88320h
0x100a35be: jmp short 100A35C2h
0x100a35c0: shr eax,1
0x100a35c2: test al,1
0x100a35c4: je short 100A35CFh
0x100a35c6: shr eax,1
0x100a35c8: xor eax,0EDB88320h
0x100a35cd: jmp short 100A35D1h
0x100a35cf: shr eax,1
0x100a35d1: test al,1
0x100a35d3: je short 100A35DEh
0x100a35d5: shr eax,1
0x100a35d7: xor eax,0EDB88320h
0x100a35dc: jmp short 100A35E0h
0x100a35de: shr eax,1
0x100a35e0: test al,1
0x100a35e2: je short 100A35EDh
0x100a35e4: shr eax,1
0x100a35e6: xor eax,0EDB88320h
0x100a35eb: jmp short 100A35EFh
```

```
0x100a35ed: shr eax,1
0x100a35ef: mov [edx],eax
0x100a35f1: add ecx,1
0x100a35f4: add edx,4
0x100a35f7: cmp ecx,100h
0x100a35fd: jl near ptr 100A3578h
0x100a3603: mov byte ptr [esi+0E48h],1
0x100a360a: ret
```

`sub_100a3570` is called by sub_100a3610. It accesses memory-mapped peripheral registers at 0xedb88320. It calls nothing else.

12.1122 sub_100a3610 at 0x100a3610 (26 instructions)

```
0x100a3610: push ebp
0x100a3611: mov ebp,[esp+8]
0x100a3615: push esi
0x100a3616: mov esi,eax
0x100a3618: cmp byte ptr [esi+0E48h],0
0x100a361f: push edi
0x100a3620: mov edi,ecx
0x100a3622: jne short 100A3629h
0x100a3624: call 100A3570h
0x100a3629: xor eax,eax
0x100a362b: test ebx,ebx
0x100a362d: jle short 100A364Dh
0x100a362f: nop
0x100a3630: movzx ecx,byte ptr [eax+ebp]
0x100a3634: xor ecx,edi
0x100a3636: and ecx,0FFh
0x100a363c: shr edi,8
0x100a363f: xor edi,[esi+ecx*4+0A48h]
0x100a3646: add eax,1
0x100a3649: cmp eax,ebx
0x100a364b: jl short 100A3630h
0x100a364d: mov eax,edi
0x100a364f: pop edi
0x100a3650: pop esi
0x100a3651: pop ebp
0x100a3652: ret
```

`sub_100a3610` is called by sub_100a3660. It delegates to sub_100a3570 (1 callees).

12.1123 sub_100a3660 at 0x100a3660 (11 instructions)

```
0x100a3660: mov eax,[esp+8]
0x100a3664: push ebx
0x100a3665: mov ebx,[esp+10h]
0x100a3669: push eax
0x100a366a: mov eax,[esp+0Ch]
0x100a366e: or ecx,0FFFFFFFFh
0x100a3671: call 100A3610h
0x100a3676: add esp,4
0x100a3679: not eax
0x100a367b: pop ebx
0x100a367c: ret
```

`sub_100a3660` is called by sub_1008c420. It delegates to sub_100a3610 (1 callees).

12.1124 png_get_uint_32 at 0x100a3680 (12 instructions)

```
0x100a3680: mov ecx,[esp+4]
0x100a3684: movzx eax,byte ptr [ecx]
0x100a3687: movzx edx,byte ptr [ecx+1]
```

```
0x100a368b: shl eax,8
0x100a368e: add eax,edx
0x100a3690: movzx edx,byte ptr [ecx+2]
0x100a3694: movzx ecx,byte ptr [ecx+3]
0x100a3698: shl eax,8
0x100a369b: add eax,edx
0x100a369d: shl eax,8
0x100a36a0: add eax,ecx
0x100a36a2: ret
```

`png_get_uint_32` is called by sub_1008b280, sub_1008c420, sub_1008c510, sub_1008cb40. It calls nothing else.

12.1125 j_realloc at 0x100ad4b0 (1 instructions)

```
0x100ad4b0: jmp dword ptr ds:[100FE410h]
```

`j_realloc` is called by sub_1006e029, sub_100aec00, sub_100b0530, sub_100bff00. It calls nothing else.

12.1126 sub_100ad4c0 at 0x100ad4c0 (21 instructions)

```
0x100ad4c0: push 44h
0x100ad4c2: call 100EFDD0h
0x100ad4c7: xor ecx,ecx
0x100ad4c9: add esp,4
0x100ad4cc: cmp eax,ecx
0x100ad4ce: jne short 100AD4D3h
0x100ad4d0: xor eax,eax
0x100ad4d2: ret
0x100ad4d3: mov [eax],ecx
0x100ad4d5: mov [eax+4],ecx
0x100ad4d8: mov [eax+8],ecx
0x100ad4db: mov [eax+0Ch],ecx
0x100ad4de: mov [eax+10h],ecx
0x100ad4e1: mov [eax+14h],ecx
0x100ad4e4: mov [eax+18h],ecx
0x100ad4e7: mov [eax+1Ch],ecx
0x100ad4ea: mov [eax+34h],ecx
0x100ad4ed: mov [eax+38h],ecx
0x100ad4f0: mov [eax+3Ch],ecx
0x100ad4f3: mov dword ptr [eax+40h],0FFFFFFFFh
0x100ad4fa: ret
```

`sub_100ad4c0` is called by sub_100adc00, sub_100adca0, sub_100ae0a0. It delegates to j_malloc (1 callees).

12.1127 sub_100ad500 at 0x100ad500 (13 instructions)

```
0x100ad500: test byte ptr [esi+4],8
0x100ad504: je short 100AD51Dh
0x100ad506: mov eax,[esi+0Ch]
0x100ad509: test eax,eax
0x100ad50b: je short 100AD51Dh
0x100ad50d: push eax
0x100ad50e: call 100209E0h
0x100ad513: add esp,4
0x100ad516: mov dword ptr [esi+0Ch],0
0x100ad51d: push esi
0x100ad51e: call 100209E0h
0x100ad523: pop ecx
0x100ad524: ret
```

`sub_100ad500` is called by sub_100adc00, sub_100adca0, sub_100adf90, sub_100ae0a0. It delegates to j_free (1 callees).

12.1128 sub_100ad5b0 at 0x100ad5b0 (35 instructions)

```
0x100ad5b0: push edi
0x100ad5b1: mov edi,ecx
0x100ad5b3: test edi,edi
0x100ad5b5: je short 100AD5F3h
0x100ad5b7: test eax,eax
0x100ad5b9: jne short 100AD5E4h
0x100ad5bb: push 2010h
0x100ad5c0: call 100EFDD0h
0x100ad5c5: add esp,4
0x100ad5c8: test eax,eax
0x100ad5ca: mov [esi+0Ch],eax
0x100ad5cd: je short 100AD5DCh
0x100ad5cf: or dword ptr [esi+4],8
0x100ad5d3: mov dword ptr [esi+14h],2000h
0x100ad5da: jmp short 100AD600h
0x100ad5dc: lea eax,[esi+20h]
0x100ad5df: mov [esi+0Ch],eax
0x100ad5e2: jmp short 100AD5F9h
0x100ad5e4: mov ecx,[esp+8]
0x100ad5e8: add ecx,0FFFFFFF0h
0x100ad5eb: mov [esi+0Ch],eax
0x100ad5ee: mov [esi+14h],ecx
0x100ad5f1: jmp short 100AD600h
0x100ad5f3: lea edx,[esi+20h]
0x100ad5f6: mov [esi+0Ch],edx
0x100ad5f9: mov dword ptr [esi+14h],1
0x100ad600: mov eax,[esi+0Ch]
0x100ad603: and edi,0Fh
0x100ad606: or [esi+4],edi
0x100ad609: add eax,10h
0x100ad60c: mov [esi+10h],eax
0x100ad60f: mov [esi+18h],eax
0x100ad612: mov dword ptr [esi+1Ch],0
0x100ad619: pop edi
0x100ad61a: ret
```

`sub_100ad5b0` is called by sub_100adc00, sub_100adca0, sub_100ae0a0. It delegates to j_malloc (1 callees).

12.1129 sub_100ad620 at 0x100ad620 (54 instructions)

```
0x100ad620: push esi
0x100ad621: mov esi,[esp+8]
0x100ad625: test byte ptr [esi+8],7
0x100ad629: jne short 100AD671h
0x100ad62b: test byte ptr [esi],1
0x100ad62e: je short 100AD671h
0x100ad630: mov eax,[esi+10h]
0x100ad633: mov edx,[esi+14h]
0x100ad636: mov ecx,[esi+34h]
0x100ad639: or dword ptr [esi+4],10h
0x100ad63d: push edx
0x100ad63e: mov [esi+18h],eax
0x100ad641: mov ecx,[ecx]
0x100ad643: push eax
0x100ad644: mov eax,[esi+38h]
0x100ad647: push eax
0x100ad648: call ecx
0x100ad64a: add esp,0Ch
0x100ad64d: test eax,eax
0x100ad64f: mov [esi+1Ch],eax
0x100ad652: jg short 100AD676h
0x100ad654: jge short 100AD666h
0x100ad656: or dword ptr [esi+8],2
0x100ad65a: mov dword ptr [esi+1Ch],0
0x100ad661: or eax,0FFFFFFFFh
```

```
0x100ad664: pop esi
0x100ad665: ret
0x100ad666: or dword ptr [esi+8],1
0x100ad66a: mov dword ptr [esi+1Ch],0
0x100ad671: or eax,0FFFFFFFFh
0x100ad674: pop esi
0x100ad675: ret
0x100ad676: cmp dword ptr [esp+0Ch],0
0x100ad67b: je short 100AD6A4h
0x100ad67d: add eax,0FFFFFFFFh
0x100ad680: mov [esi+1Ch],eax
0x100ad683: jns short 100AD692h
0x100ad685: push 1
0x100ad687: push esi
0x100ad688: call 100AD620h
0x100ad68d: add esp,8
0x100ad690: pop esi
0x100ad691: ret
0x100ad692: mov ecx,[esi+18h]
0x100ad695: add dword ptr [esi+3Ch],1
0x100ad699: movzx eax,byte ptr [ecx]
0x100ad69c: add ecx,1
0x100ad69f: mov [esi+18h],ecx
0x100ad6a2: pop esi
0x100ad6a3: ret
0x100ad6a4: mov edx,[esi+18h]
0x100ad6a7: movzx eax,byte ptr [edx]
0x100ad6aa: pop esi
0x100ad6ab: ret
```

`sub_100ad620` is called by sub_100ad620, sub_100ad810, sub_100ade80, sub_100aef60, sub_100af330, sub_100c0430. It delegates to sub_100ad620 (1 callees).

12.1130 sub_100ad6b0 at 0x100ad6b0 (65 instructions)

```
0x100ad6b0: push esi
0x100ad6b1: mov esi,[esp+8]
0x100ad6b5: test byte ptr [esi+8],7
0x100ad6b9: je short 100AD6C0h
0x100ad6bb: or eax,0FFFFFFFFh
0x100ad6be: pop esi
0x100ad6bf: ret
0x100ad6c0: test byte ptr [esi],6
0x100ad6c3: je short 100AD6BBh
0x100ad6c5: mov eax,[esi+10h]
0x100ad6c8: push edi
0x100ad6c9: mov edi,[esi+18h]
0x100ad6cc: sub edi,eax
0x100ad6ce: test edi,edi
0x100ad6d0: jle short 100AD6F1h
0x100ad6d2: mov ecx,[esi+34h]
0x100ad6d5: mov edx,[esi+38h]
0x100ad6d8: push edi
0x100ad6d9: push eax
0x100ad6da: mov eax,[ecx+4]
0x100ad6dd: push edx
0x100ad6de: call eax
0x100ad6e0: add esp,0Ch
0x100ad6e3: cmp eax,edi
0x100ad6e5: je short 100AD6F1h
0x100ad6e7: or dword ptr [esi+8],2
0x100ad6eb: pop edi
0x100ad6ec: or eax,0FFFFFFFFh
0x100ad6ef: pop esi
0x100ad6f0: ret
0x100ad6f1: or dword ptr [esi+4],20h
```

```
0x100ad6f5: mov edx,[esp+10h]
0x100ad6f9: cmp edx,0FFFFFFFFh
0x100ad6fc: mov ecx,[esi+14h]
0x100ad6ff: mov edi,[esi+10h]
0x100ad702: mov eax,[esi+4]
0x100ad705: mov [esi+1Ch],ecx
0x100ad708: mov [esi+18h],edi
0x100ad70b: je short 100AD740h
0x100ad70d: or eax,20h
0x100ad710: mov [esi+4],eax
0x100ad713: lea eax,[ecx-1]
0x100ad716: test eax,eax
0x100ad718: mov [esi+1Ch],eax
0x100ad71b: jge short 100AD72Dh
0x100ad71d: movzx ecx,dl
0x100ad720: push ecx
0x100ad721: push esi
0x100ad722: call 100AD6B0h
0x100ad727: add esp,8
0x100ad72a: pop edi
0x100ad72b: pop esi
0x100ad72c: ret
0x100ad72d: mov ecx,1
0x100ad732: add [esi+3Ch],ecx
0x100ad735: mov [edi],dl
0x100ad737: add [esi+18h],ecx
0x100ad73a: pop edi
0x100ad73b: movzx eax,dl
0x100ad73e: pop esi
0x100ad73f: ret
0x100ad740: pop edi
0x100ad741: xor eax,eax
0x100ad743: pop esi
0x100ad744: ret
```

`sub_100ad6b0` is called by sub_100ad6b0, sub_100ad810, sub_100adf70, sub_100af000, sub_100af150. It delegates to sub_100ad6b0 (1 callees).

12.1131 sub_100ad750 at 0x100ad750 (16 instructions)

```
0x100ad750: mov cl,[edx]
0x100ad752: xor eax,eax
0x100ad754: test cl,cl
0x100ad756: je short 100AD79Ah
0x100ad758: jmp short 100AD760h
0x100ad760: movsx ecx,cl
0x100ad763: add ecx,0FFFFFFD5h
0x100ad766: cmp ecx,4Ch
0x100ad769: ja short 100AD790h
0x100ad76b: movzx ecx,byte ptr [ecx+100AD7B4h]
0x100ad772: jmp dword ptr [ecx*4+100AD79Ch]
0x100ad790: mov cl,[edx+1]
0x100ad793: add edx,1
0x100ad796: test cl,cl
0x100ad798: jne short 100AD760h
0x100ad79a: ret
```

`sub_100ad750` is called by sub_100adca0. It calls nothing else.

12.1132 sub_100ad810 at 0x100ad810 (87 instructions)

```
0x100ad810: mov eax,[esp+0Ch]
0x100ad814: xor ecx,ecx
0x100ad816: push ebx
0x100ad817: test eax,eax
```

```
0x100ad819: setl cl
0x100ad81c: push ebp
0x100ad81d: push esi
0x100ad81e: mov esi,[esp+10h]
0x100ad822: push edi
0x100ad823: mov edi,[esp+18h]
0x100ad827: mov ebx,eax
0x100ad829: mov [esp+1Ch],ecx
0x100ad82d: mov ebp,1
0x100ad832: cmp dword ptr [esp+1Ch],0
0x100ad837: jne short 100AD841h
0x100ad839: test ebx,ebx
0x100ad83b: jle near ptr 100AD8E2h
0x100ad841: test byte ptr [edi+8],7
0x100ad845: jne short 100AD8C1h
0x100ad847: mov eax,[edi+40h]
0x100ad84a: test eax,eax
0x100ad84c: jl short 100AD853h
0x100ad84e: cmp [edi+3Ch],eax
0x100ad851: jge short 100AD8BDh
0x100ad853: add dword ptr [edi+1Ch],0FFFFFFFFh
0x100ad857: jns short 100AD865h
0x100ad859: push ebp
0x100ad85a: push edi
0x100ad85b: call 100AD620h
0x100ad860: add esp,8
0x100ad863: jmp short 100AD873h
0x100ad865: mov ecx,[edi+18h]
0x100ad868: add [edi+3Ch],ebp
0x100ad86b: movzx eax,byte ptr [ecx]
0x100ad86e: add ecx,ebp
0x100ad870: mov [edi+18h],ecx
0x100ad873: cmp eax,0FFFFFFFFh
0x100ad876: je short 100AD8C1h
0x100ad878: test byte ptr [esi+8],7
0x100ad87c: jne short 100AD8CEh
0x100ad87e: mov ecx,[esi+40h]
0x100ad881: test ecx,ecx
0x100ad883: jl short 100AD88Ah
0x100ad885: cmp [esi+3Ch],ecx
0x100ad888: jge short 100AD8D6h
0x100ad88a: or dword ptr [esi+4],20h
0x100ad88e: add dword ptr [esi+1Ch],0FFFFFFFFh
0x100ad892: jns short 100AD8A3h
0x100ad894: movzx edx,al
0x100ad897: push edx
0x100ad898: push esi
0x100ad899: call 100AD6B0h
0x100ad89e: add esp,8
0x100ad8a1: jmp short 100AD8B1h
0x100ad8a3: mov ecx,[esi+18h]
0x100ad8a6: add [esi+3Ch],ebp
0x100ad8a9: mov [ecx],al
0x100ad8ab: add [esi+18h],ebp
0x100ad8ae: movzx eax,al
0x100ad8b1: cmp eax,0FFFFFFFFh
0x100ad8b4: je short 100AD8CEh
0x100ad8b6: sub ebx,ebp
0x100ad8b8: jmp near ptr 100AD832h
0x100ad8bd: or dword ptr [edi+8],4
0x100ad8c1: cmp dword ptr [esp+1Ch],0
0x100ad8c6: je short 100AD8CEh
0x100ad8c8: test byte ptr [edi+8],2
0x100ad8cc: je short 100AD8E2h
0x100ad8ce: pop edi
0x100ad8cf: pop esi
0x100ad8d0: pop ebp
```

```
0x100ad8d1: or eax,0FFFFFFFFh
0x100ad8d4: pop ebx
0x100ad8d5: ret
0x100ad8d6: or dword ptr [esi+8],4
0x100ad8da: pop edi
0x100ad8db: pop esi
0x100ad8dc: pop ebp
0x100ad8dd: or eax,0FFFFFFFFh
0x100ad8e0: pop ebx
0x100ad8e1: ret
0x100ad8e2: pop edi
0x100ad8e3: pop esi
0x100ad8e4: pop ebp
0x100ad8e5: xor eax,eax
0x100ad8e7: pop ebx
0x100ad8e8: ret
```

`sub_100ad810` is called by sub_100ae7e0. It delegates to sub_100ad620, sub_100ad6b0 (2 callees).

12.1133 sub_100adc00 at 0x100adc00 (59 instructions)

```
0x100adc00: push esi
0x100adc01: call 100AD4C0h
0x100adc06: mov esi,eax
0x100adc08: test esi,esi
0x100adc0a: jne short 100ADC0Eh
0x100adc0c: pop esi
0x100adc0d: ret
0x100adc0e: push edi
0x100adc0f: push 18h
0x100adc11: mov dword ptr [esi],0Bh
0x100adc17: call 100EFDD0h
0x100adc1c: mov edi,eax
0x100adc1e: add esp,4
0x100adc21: test edi,edi
0x100adc23: jne short 100ADC2Fh
0x100adc25: call 100AD500h
0x100adc2a: pop edi
0x100adc2b: xor eax,eax
0x100adc2d: pop esi
0x100adc2e: ret
0x100adc2f: push ebx
0x100adc30: lea ebx,[edi+8]
0x100adc33: mov dword ptr [edi],0FFFFFFFFh
0x100adc39: mov dword ptr [edi+4],0
0x100adc40: mov byte ptr [ebx],0
0x100adc43: push ebx
0x100adc44: mov [esi+38h],edi
0x100adc47: call dword ptr ds:[100FE46Ch]
0x100adc4d: push 1B6h
0x100adc52: push 8702h
0x100adc57: push ebx
0x100adc58: call dword ptr ds:[100FE3ECh]
0x100adc5e: add esp,10h
0x100adc61: test eax,eax
0x100adc63: mov [edi],eax
0x100adc65: jge short 100ADC72h
0x100adc67: call 100AD500h
0x100adc6c: pop ebx
0x100adc6d: pop edi
0x100adc6e: xor eax,eax
0x100adc70: pop esi
0x100adc71: ret
0x100adc72: push ebx
0x100adc73: call dword ptr ds:[100FE3F4h]
0x100adc79: add esp,4
```

```
0x100adc7c: test eax,eax
0x100adc7e: je short 100ADC84h
0x100adc80: or dword ptr [edi+4],1
0x100adc84: xor eax,eax
0x100adc86: push 0
0x100adc88: lea ecx,[eax+2]
0x100adc8b: call 100AD5B0h
0x100adc90: add esp,4
0x100adc93: pop ebx
0x100adc94: pop edi
0x100adc95: mov dword ptr [esi+34h],101368F8h
0x100adc9c: mov eax,esi
0x100adc9e: pop esi
0x100adc9f: ret
```

`sub_100adc00` is called by sub_100af150. It delegates to sub_100ad4c0, sub_100ad500, sub_100ad5b0 (4 callees).

12.1134 sub_100adca0 at 0x100adca0 (43 instructions)

```
0x100adca0: push esi
0x100adca1: call 100AD4C0h
0x100adca6: mov esi,eax
0x100adca8: test esi,esi
0x100adcaa: jne short 100ADCAEh
0x100adcac: pop esi
0x100adcad: ret
0x100adcae: mov edx,[esp+0Ch]
0x100adcb2: push edi
0x100adcb3: call 100AD750h
0x100adcb8: test al,8
0x100adcba: mov edi,[esp+0Ch]
0x100adcbe: mov [esi],eax
0x100adcc0: je short 100ADCD1h
0x100adcc2: push 8000h
0x100adcc7: push edi
0x100adcc8: call dword ptr ds:[100FE590h]
0x100adcce: add esp,8
0x100adcd1: push 18h
0x100adcd3: call 100EFDD0h
0x100adcd8: add esp,4
0x100adcdb: test eax,eax
0x100adcdd: jne short 100ADCE9h
0x100adcdf: call 100AD500h
0x100adce4: pop edi
0x100adce5: xor eax,eax
0x100adce7: pop esi
0x100adce8: ret
0x100adce9: mov dword ptr [eax+4],0
0x100adcf0: mov [eax],edi
0x100adcf2: mov byte ptr [eax+8],0
0x100adcf6: mov [esi+38h],eax
0x100adcf9: mov ecx,2
0x100adcfe: or [eax+4],ecx
0x100add01: push 0
0x100add03: xor eax,eax
0x100add05: call 100AD5B0h
0x100add0a: add esp,4
0x100add0d: pop edi
0x100add0e: mov dword ptr [esi+34h],101368F8h
0x100add15: mov eax,esi
0x100add17: pop esi
0x100add18: ret
```

`sub_100adca0` is called by sub_100156b0. It delegates to sub_100ad4c0, sub_100ad500, sub_100ad5b0 (5 callees).

12.1135 sub_100ade80 at 0x100ade80 (44 instructions)

```
0x100ade80: push ebx
0x100ade81: mov ebx,[esp+0Ch]
0x100ade85: test ebx,ebx
0x100ade87: push esi
0x100ade88: push edi
0x100ade89: mov edi,ebx
0x100ade8b: jle short 100ADED2h
0x100ade8d: mov esi,[esp+10h]
0x100ade91: test byte ptr [esi+8],7
0x100ade95: jne short 100ADEDCh
0x100ade97: mov eax,[esi+40h]
0x100ade9a: test eax,eax
0x100ade9c: jl short 100ADEA3h
0x100ade9e: cmp [esi+3Ch],eax
0x100adea1: jge short 100ADED8h
0x100adea3: add dword ptr [esi+1Ch],0FFFFFFFFh
0x100adea7: jns short 100ADEB6h
0x100adea9: push 1
0x100adeab: push esi
0x100adeac: call 100AD620h
0x100adeb1: add esp,8
0x100adeb4: jmp short 100ADEC6h
0x100adeb6: mov ecx,[esi+18h]
0x100adeb9: add dword ptr [esi+3Ch],1
0x100adebd: movzx eax,byte ptr [ecx]
0x100adec0: add ecx,1
0x100adec3: mov [esi+18h],ecx
0x100adec6: cmp eax,0FFFFFFFFh
0x100adec9: je short 100ADEDCh
0x100adecb: sub edi,1
0x100adece: test edi,edi
0x100aded0: jg short 100ADE91h
0x100aded2: pop edi
0x100aded3: pop esi
0x100aded4: mov eax,ebx
0x100aded6: pop ebx
0x100aded7: ret
0x100aded8: or dword ptr [esi+8],4
0x100adedc: mov eax,ebx
0x100adede: sub eax,edi
0x100adee0: pop edi
0x100adee1: pop esi
0x100adee2: pop ebx
0x100adee3: ret
```

`sub_100ade80` is called by sub_100c1dc0, sub_100c2390. It delegates to sub_100ad620 (1 callees).

12.1136 sub_100adf70 at 0x100adf70 (10 instructions)

```
0x100adf70: mov eax,[esp+4]
0x100adf74: test byte ptr [eax+4],10h
0x100adf78: je short 100ADF7Dh
0x100adf7a: xor eax,eax
0x100adf7c: ret
0x100adf7d: push 0FFFFFFFFh
0x100adf7f: push eax
0x100adf80: call 100AD6B0h
0x100adf85: add esp,8
0x100adf88: ret
```

`sub_100adf70` is called by sub_100adf90, sub_100ae030. It delegates to sub_100ad6b0 (1 callees).

12.1137 sub_100adf90 at 0x100adf90 (14 instructions)

```
0x100adf90: push esi
0x100adf91: mov esi,[esp+8]
0x100adf95: push esi
0x100adf96: call 100ADF70h
0x100adf9b: mov eax,[esi+34h]
0x100adf9e: mov ecx,[esi+38h]
0x100adfa1: mov edx,[eax+0Ch]
0x100adfa4: push ecx
0x100adfa5: call edx
0x100adfa7: add esp,8
0x100adfaa: call 100AD500h
0x100adfaf: xor eax,eax
0x100adfb1: pop esi
0x100adfb2: ret
```

`sub_100adf90` is called by sub_100156b0, sub_100ae0a0, sub_100ae880, sub_100c25a0. It delegates to sub_100ad500, sub_100adf70 (2 callees).

12.1138 sub_100ae030 at 0x100ae030 (39 instructions)

```
0x100ae030: push esi
0x100ae031: mov esi,[esp+8]
0x100ae035: mov eax,[esi+4]
0x100ae038: and dword ptr [esi+8],0FFFFFFFEh
0x100ae03c: test al,10h
0x100ae03e: push edi
0x100ae03f: mov edi,[esp+14h]
0x100ae043: je short 100AE053h
0x100ae045: cmp edi,1
0x100ae048: jne short 100AE064h
0x100ae04a: mov eax,[esp+10h]
0x100ae04e: sub eax,[esi+1Ch]
0x100ae051: jmp short 100AE068h
0x100ae053: test al,20h
0x100ae055: je short 100AE064h
0x100ae057: push esi
0x100ae058: call 100ADF70h
0x100ae05d: add esp,4
0x100ae060: test eax,eax
0x100ae062: jne short 100AE08Eh
0x100ae064: mov eax,[esp+10h]
0x100ae068: mov ecx,[esi+10h]
0x100ae06b: mov edx,[esi+34h]
0x100ae06e: and dword ptr [esi+4],0FFFFFFCFh
0x100ae072: push edi
0x100ae073: push eax
0x100ae074: mov eax,[esi+38h]
0x100ae077: mov dword ptr [esi+1Ch],0
0x100ae07e: mov [esi+18h],ecx
0x100ae081: mov ecx,[edx+8]
0x100ae084: push eax
0x100ae085: call ecx
0x100ae087: add esp,0Ch
0x100ae08a: test eax,eax
0x100ae08c: jge short 100AE091h
0x100ae08e: or eax,0FFFFFFFFh
0x100ae091: pop edi
0x100ae092: pop esi
0x100ae093: ret
```

`sub_100ae030` is called by sub_100ae170, sub_100ae7e0, sub_100af150, sub_100af330, sub_100af960, sub_100afa70, sub_100afb80. It delegates to sub_100adf70 (1 callees).

12.1139 sub_100ae0a0 at 0x100ae0a0 (81 instructions)

```
0x100ae0a0: push ebp
0x100ae0a1: push esi
0x100ae0a2: call 100AD4C0h
0x100ae0a7: mov esi,eax
0x100ae0a9: xor ebp,ebp
0x100ae0ab: xor eax,eax
0x100ae0ad: cmp esi,ebp
0x100ae0af: je near ptr 100AE164h
0x100ae0b5: push edi
0x100ae0b6: push ebp
0x100ae0b7: lea ecx,[ebp+2]
0x100ae0ba: mov dword ptr [esi],0Bh
0x100ae0c0: call 100AD5B0h
0x100ae0c5: push 18h
0x100ae0c7: mov dword ptr [esi+34h],10136918h
0x100ae0ce: call 100EFDD0h
0x100ae0d3: mov edi,eax
0x100ae0d5: add esp,8
0x100ae0d8: cmp edi,ebp
0x100ae0da: jne short 100AE0E7h
0x100ae0dc: call 100AD500h
0x100ae0e1: pop edi
0x100ae0e2: pop esi
0x100ae0e3: xor eax,eax
0x100ae0e5: pop ebp
0x100ae0e6: ret
0x100ae0e7: push ebx
0x100ae0e8: mov ebx,[esp+18h]
0x100ae0ec: cmp ebx,ebp
0x100ae0ee: mov [esi+38h],edi
0x100ae0f1: mov [edi+14h],ebp
0x100ae0f4: mov [edi],ebp
0x100ae0f6: jg short 100AE108h
0x100ae0f8: mov dword ptr [edi+4],400h
0x100ae0ff: mov dword ptr [edi+10h],1
0x100ae106: jmp short 100AE10Eh
0x100ae108: mov [edi+4],ebx
0x100ae10b: mov [edi+10h],ebp
0x100ae10e: mov ebp,[esp+14h]
0x100ae112: test ebp,ebp
0x100ae114: je short 100AE11Ah
0x100ae116: mov [edi],ebp
0x100ae118: jmp short 100AE12Fh
0x100ae11a: mov eax,[edi+4]
0x100ae11d: push eax
0x100ae11e: call 100EFDD0h
0x100ae123: add esp,4
0x100ae126: mov [edi],eax
0x100ae128: mov dword ptr [edi+14h],1
0x100ae12f: xor eax,eax
0x100ae131: cmp [edi],eax
0x100ae133: jne short 100AE145h
0x100ae135: push esi
0x100ae136: call 100ADF90h
0x100ae13b: add esp,4
0x100ae13e: pop ebx
0x100ae13f: pop edi
0x100ae140: pop esi
0x100ae141: xor eax,eax
0x100ae143: pop ebp
0x100ae144: ret
0x100ae145: cmp ebx,eax
0x100ae147: jle short 100AE15Ah
0x100ae149: cmp ebp,eax
0x100ae14b: je short 100AE15Ah
0x100ae14d: mov [edi+8],ebx
```

```
0x100ae150: pop ebx
0x100ae151: mov [edi+0Ch],eax
0x100ae154: pop edi
0x100ae155: mov eax,esi
0x100ae157: pop esi
0x100ae158: pop ebp
0x100ae159: ret
0x100ae15a: mov [edi+8],eax
0x100ae15d: mov [edi+0Ch],eax
0x100ae160: pop ebx
0x100ae161: mov eax,esi
0x100ae163: pop edi
0x100ae164: pop esi
0x100ae165: pop ebp
0x100ae166: ret
```

`sub_100ae0a0` is called by sub_100ae7e0, sub_100af150, sub_100c25a0. It delegates to sub_100ad4c0, sub_100ad500, sub_100ad5b0 (5 callees).

12.1140 sub_100ae170 at 0x100ae170 (7 instructions)

```
0x100ae170: mov eax,[esp+4]
0x100ae174: push 0
0x100ae176: push 0
0x100ae178: push eax
0x100ae179: call 100AE030h
0x100ae17e: add esp,0Ch
0x100ae181: ret
```

`sub_100ae170` is called by sub_100afb80. It delegates to sub_100ae030 (1 callees).

12.1141 sub_100ae190 at 0x100ae190 (39 instructions)

```
0x100ae190: jmp near ptr 100AE8A0h
0x100ae8a0: push ebx
0x100ae8a1: push edi
0x100ae8a2: xor edi,edi
0x100ae8a4: xor ebx,ebx
0x100ae8a6: cmp ds:[101398D0h],edi
0x100ae8ac: jle short 100AE8FAh
0x100ae8ae: push esi
0x100ae8af: mov esi,10139558h
0x100ae8b4: mov eax,[esi-4]
0x100ae8b7: cmp eax,edi
0x100ae8b9: je short 100AE8C7h
0x100ae8bb: push eax
0x100ae8bc: call 100209E0h
0x100ae8c1: add esp,4
0x100ae8c4: mov [esi-4],edi
0x100ae8c7: mov eax,[esi]
0x100ae8c9: cmp eax,edi
0x100ae8cb: je short 100AE8D8h
0x100ae8cd: push eax
0x100ae8ce: call 100209E0h
0x100ae8d3: add esp,4
0x100ae8d6: mov [esi],edi
0x100ae8d8: mov eax,[esi+4]
0x100ae8db: cmp eax,edi
0x100ae8dd: je short 100AE8EBh
0x100ae8df: push eax
0x100ae8e0: call 100209E0h
0x100ae8e5: add esp,4
0x100ae8e8: mov [esi+4],edi
0x100ae8eb: add ebx,1
0x100ae8ee: add esi,1Ch
```

```
0x100ae8f1: cmp ebx,ds:[101398D0h]
0x100ae8f7: jl short 100AE8B4h
0x100ae8f9: pop esi
0x100ae8fa: mov ds:[101398D0h],edi
0x100ae900: pop edi
0x100ae901: pop ebx
0x100ae902: ret
```

`sub_100ae190` is called by sub_100156b0. It delegates to j_free (1 callees).

12.1142 sub_100ae1a0 at 0x100ae1a0 (92 instructions)

```
0x100ae1a0: sub esp,0Ch
0x100ae1a3: lea eax,[esp]
0x100ae1a6: push eax
0x100ae1a7: push 10127A8Ch
0x100ae1ac: push 10127A88h
0x100ae1b1: push 10127A88h
0x100ae1b6: push 0
0x100ae1b8: mov dword ptr [esp+14h],100BF960h
0x100ae1c0: mov dword ptr [esp+18h],100BFC00h
0x100ae1c8: mov dword ptr [esp+1Ch],100BF0A0h
0x100ae1d0: call 100AE910h
0x100ae1d5: lea ecx,[esp+14h]
0x100ae1d9: push ecx
0x100ae1da: push 10127A68h
0x100ae1df: push 100FEBB4h
0x100ae1e4: push 100FEBB4h
0x100ae1e9: push 1
0x100ae1eb: mov dword ptr [esp+28h],100BE800h
0x100ae1f3: mov dword ptr [esp+2Ch],100BEE10h
0x100ae1fb: mov dword ptr [esp+30h],100BE070h
0x100ae203: call 100AE910h
0x100ae208: lea edx,[esp+28h]
0x100ae20c: push edx
0x100ae20d: push 10127A68h
0x100ae212: push 10127A64h
0x100ae217: push 100FEBB4h
0x100ae21c: push 1
0x100ae21e: call 100AE910h
0x100ae223: lea eax,[esp+3Ch]
0x100ae227: push eax
0x100ae228: push 10127A68h
0x100ae22d: push 10127A60h
0x100ae232: push 100FEBB4h
0x100ae237: push 1
0x100ae239: call 100AE910h
0x100ae23e: add esp,50h
0x100ae241: lea ecx,[esp]
0x100ae244: push ecx
0x100ae245: push 10127A48h
0x100ae24a: push 10127A44h
0x100ae24f: push 10127A44h
0x100ae254: push 2
0x100ae256: mov dword ptr [esp+14h],100BD4D0h
0x100ae25e: mov dword ptr [esp+18h],100BDDD0h
0x100ae266: mov dword ptr [esp+1Ch],100BCB50h
0x100ae26e: call 100AE910h
0x100ae273: lea edx,[esp+14h]
0x100ae277: push edx
0x100ae278: push 10127A2Ch
0x100ae27d: push 100FEBB8h
0x100ae282: push 100FEBB8h
0x100ae287: push 3
0x100ae289: mov dword ptr [esp+28h],100BC2F0h
0x100ae291: mov dword ptr [esp+2Ch],100BC930h
```

```
0x100ae299: mov dword ptr [esp+30h],100BBD20h
0x100ae2a1: call 100AE910h
0x100ae2a6: lea eax,[esp+28h]
0x100ae2aa: push eax
0x100ae2ab: push 101279F8h
0x100ae2b0: push 100FEBC0h
0x100ae2b5: push 100FEBC0h
0x100ae2ba: push 4
0x100ae2bc: mov dword ptr [esp+3Ch],100BB030h
0x100ae2c4: mov dword ptr [esp+40h],100BB7E0h
0x100ae2cc: mov dword ptr [esp+44h],100BADE0h
0x100ae2d4: call 100AE910h
0x100ae2d9: lea ecx,[esp+3Ch]
0x100ae2dd: push ecx
0x100ae2de: push 101279C8h
0x100ae2e3: push 100FEBBCh
0x100ae2e8: push 100FEBBCh
0x100ae2ed: push 5
0x100ae2ef: mov dword ptr [esp+50h],100BA560h
0x100ae2f7: mov dword ptr [esp+54h],100B82B0h
0x100ae2ff: mov dword ptr [esp+58h],100B3410h
0x100ae307: call 100AE910h
0x100ae30c: add esp,50h
0x100ae30f: lea edx,[esp]
0x100ae312: mov dword ptr [esp],100B24E0h
0x100ae319: mov dword ptr [esp+4],100B2780h
0x100ae321: mov dword ptr [esp+8],100B1F90h
0x100ae329: push edx
0x100ae32a: push 101279ACh
0x100ae32f: push 100FEBB0h
0x100ae334: push 100FEBB0h
0x100ae339: push 6
0x100ae33b: call 100AE910h
0x100ae340: push 100AE190h
0x100ae345: call 100224F7h
0x100ae34a: xor eax,eax
0x100ae34c: add esp,24h
0x100ae34f: ret
```

`sub_100ae1a0` is called by sub_100156b0. It delegates to sub_100224f7, sub_100ae910 (2 callees).

12.1143 sub_100ae350 at 0x100ae350 (21 instructions)

```
0x100ae350: push esi
0x100ae351: mov esi,[esp+8]
0x100ae355: mov eax,[esi+24h]
0x100ae358: test eax,eax
0x100ae35a: je short 100AE36Ch
0x100ae35c: push eax
0x100ae35d: call 100209E0h
0x100ae362: add esp,4
0x100ae365: mov dword ptr [esi+24h],0
0x100ae36c: mov eax,[esi+1Ch]
0x100ae36f: test eax,eax
0x100ae371: je short 100AE383h
0x100ae373: push eax
0x100ae374: call 100209E0h
0x100ae379: add esp,4
0x100ae37c: mov dword ptr [esi+1Ch],0
0x100ae383: push esi
0x100ae384: call 100209E0h
0x100ae389: add esp,4
0x100ae38c: pop esi
0x100ae38d: ret
```

`sub_100ae350` is called by sub_100156b0, sub_100ae620. It delegates to j_free (1 callees).

12.1144 sub_100ae5c0 at 0x100ae5c0 (39 instructions)

```
0x100ae5c0: mov ecx,[esp+4]
0x100ae5c4: push esi
0x100ae5c5: mov esi,[esp+0Ch]
0x100ae5c9: push edi
0x100ae5ca: mov edi,[esp+14h]
0x100ae5ce: mov eax,esi
0x100ae5d0: imul eax,edi
0x100ae5d3: cmp eax,[ecx+28h]
0x100ae5d6: jg short 100AE60Dh
0x100ae5d8: cmp esi,[ecx+20h]
0x100ae5db: jg short 100AE60Dh
0x100ae5dd: xor eax,eax
0x100ae5df: test esi,esi
0x100ae5e1: mov [ecx+14h],esi
0x100ae5e4: mov [ecx+18h],edi
0x100ae5e7: jle short 100AE608h
0x100ae5e9: add edi,edi
0x100ae5eb: push ebx
0x100ae5ec: xor edx,edx
0x100ae5ee: add edi,edi
0x100ae5f0: push ebp
0x100ae5f1: mov ebx,[ecx+24h]
0x100ae5f4: mov ebp,[ecx+1Ch]
0x100ae5f7: add ebx,edx
0x100ae5f9: mov [ebp+eax*4],ebx
0x100ae5fd: add eax,1
0x100ae600: add edx,edi
0x100ae602: cmp eax,esi
0x100ae604: jl short 100AE5F1h
0x100ae606: pop ebp
0x100ae607: pop ebx
0x100ae608: pop edi
0x100ae609: xor eax,eax
0x100ae60b: pop esi
0x100ae60c: ret
0x100ae60d: pop edi
0x100ae60e: or eax,0FFFFFFFFh
0x100ae611: pop esi
0x100ae612: ret
```

`sub_100ae5c0` is called by sub_100af330. It calls nothing else.

12.1145 sub_100ae620 at 0x100ae620 (84 instructions)

```
0x100ae620: push ebx
0x100ae621: push esi
0x100ae622: push 2Ch
0x100ae624: call 100EFDD0h
0x100ae629: mov esi,eax
0x100ae62b: xor ebx,ebx
0x100ae62d: add esp,4
0x100ae630: cmp esi,ebx
0x100ae632: jne short 100AE639h
0x100ae634: pop esi
0x100ae635: xor eax,eax
0x100ae637: pop ebx
0x100ae638: ret
0x100ae639: mov eax,[esp+10h]
0x100ae63d: push edi
0x100ae63e: mov edi,[esp+10h]
0x100ae642: mov ecx,edi
0x100ae644: imul ecx,eax
0x100ae647: cmp edi,ebx
0x100ae649: mov [esi],ebx
0x100ae64b: mov [esi+14h],edi
```

```
0x100ae64e: mov [esi+18h],eax
0x100ae651: mov [esi+1Ch],ebx
0x100ae654: mov [esi+20h],edi
0x100ae657: mov [esi+24h],ebx
0x100ae65a: mov [esi+28h],ecx
0x100ae65d: jle short 100AE676h
0x100ae65f: lea edx,[edi*4]
0x100ae666: push edx
0x100ae667: call 100EFDD0h
0x100ae66c: add esp,4
0x100ae66f: cmp eax,ebx
0x100ae671: mov [esi+1Ch],eax
0x100ae674: je short 100AE691h
0x100ae676: mov eax,[esi+28h]
0x100ae679: cmp eax,ebx
0x100ae67b: jle short 100AE6A0h
0x100ae67d: add eax,eax
0x100ae67f: add eax,eax
0x100ae681: push eax
0x100ae682: call 100EFDD0h
0x100ae687: add esp,4
0x100ae68a: cmp eax,ebx
0x100ae68c: mov [esi+24h],eax
0x100ae68f: jne short 100AE6A0h
0x100ae691: push esi
0x100ae692: call 100AE350h
0x100ae697: add esp,4
0x100ae69a: pop edi
0x100ae69b: pop esi
0x100ae69c: xor eax,eax
0x100ae69e: pop ebx
0x100ae69f: ret
0x100ae6a0: xor eax,eax
0x100ae6a2: cmp edi,ebx
0x100ae6a4: jle short 100AE6BFh
0x100ae6a6: mov ecx,[esi+18h]
0x100ae6a9: mov edx,[esi+24h]
0x100ae6ac: imul ecx,eax
0x100ae6af: lea ecx,[edx+ecx*4]
0x100ae6b2: mov edx,[esi+1Ch]
0x100ae6b5: mov [edx+eax*4],ecx
0x100ae6b8: add eax,1
0x100ae6bb: cmp eax,edi
0x100ae6bd: jl short 100AE6A6h
0x100ae6bf: xor eax,eax
0x100ae6c1: cmp [esi+28h],ebx
0x100ae6c4: jle short 100AE6D4h
0x100ae6c6: mov ecx,[esi+24h]
0x100ae6c9: mov [ecx+eax*4],ebx
0x100ae6cc: add eax,1
0x100ae6cf: cmp eax,[esi+28h]
0x100ae6d2: jl short 100AE6C6h
0x100ae6d4: mov eax,[esi+14h]
0x100ae6d7: mov edx,[esi+18h]
0x100ae6da: mov [esi+10h],eax
0x100ae6dd: pop edi
0x100ae6de: mov [esi+4],ebx
0x100ae6e1: mov [esi+8],ebx
0x100ae6e4: mov [esi+0Ch],edx
0x100ae6e7: mov eax,esi
0x100ae6e9: pop esi
0x100ae6ea: pop ebx
0x100ae6eb: ret
```

`sub_100ae620` is called by sub_100156b0. It delegates to sub_100ae350, j_malloc (2 callees).

12.1146 sub_100ae770 at 0x100ae770 (19 instructions)

```
0x100ae770: push 28h
0x100ae772: call 100EFDD0h
0x100ae777: xor ecx,ecx
0x100ae779: add esp,4
0x100ae77c: cmp eax,ecx
0x100ae77e: jne short 100AE783h
0x100ae780: xor eax,eax
0x100ae782: ret
0x100ae783: mov [eax],ecx
0x100ae785: mov [eax+4],ecx
0x100ae788: mov [eax+8],ecx
0x100ae78b: mov [eax+0Ch],ecx
0x100ae78e: mov dword ptr [eax+1Ch],4000h
0x100ae795: mov [eax+10h],ecx
0x100ae798: mov [eax+14h],ecx
0x100ae79b: mov [eax+18h],ecx
0x100ae79e: mov dword ptr [eax+24h],1
0x100ae7a5: mov [eax+20h],ecx
0x100ae7a8: ret
```

`sub_100ae770` is called by sub_100affb0, sub_100b0040. It delegates to j_malloc (1 callees).

12.1147 sub_100ae7b0 at 0x100ae7b0 (18 instructions)

```
0x100ae7b0: push esi
0x100ae7b1: push 2Ch
0x100ae7b3: call 100EFDD0h
0x100ae7b8: mov esi,eax
0x100ae7ba: add esp,4
0x100ae7bd: test esi,esi
0x100ae7bf: jne short 100AE7C3h
0x100ae7c1: pop esi
0x100ae7c2: ret
0x100ae7c3: push 2Ch
0x100ae7c5: push 0
0x100ae7c7: push esi
0x100ae7c8: call 100228C0h
0x100ae7cd: add esp,0Ch
0x100ae7d0: mov dword ptr [esi+28h],10000h
0x100ae7d7: mov eax,esi
0x100ae7d9: pop esi
0x100ae7da: ret
```

`sub_100ae7b0` is called by sub_100ae7e0. It delegates to j_memset, j_malloc (2 callees).

12.1148 sub_100ae7e0 at 0x100ae7e0 (64 instructions)

```
0x100ae7e0: push esi
0x100ae7e1: call 100AE7B0h
0x100ae7e6: mov esi,eax
0x100ae7e8: test esi,esi
0x100ae7ea: jne short 100AE7F0h
0x100ae7ec: xor eax,eax
0x100ae7ee: pop esi
0x100ae7ef: ret
0x100ae7f0: mov eax,[edi]
0x100ae7f2: mov [esi],eax
0x100ae7f4: mov ecx,[edi+4]
0x100ae7f7: mov [esi+4],ecx
0x100ae7fa: mov edx,[edi+8]
0x100ae7fd: mov [esi+8],edx
0x100ae800: mov eax,[edi+0Ch]
0x100ae803: mov [esi+0Ch],eax
0x100ae806: mov ecx,[edi+10h]
```

```
0x100ae809: mov [esi+10h],ecx
0x100ae80c: mov edx,[edi+14h]
0x100ae80f: mov [esi+14h],edx
0x100ae812: mov eax,[edi+18h]
0x100ae815: mov [esi+18h],eax
0x100ae818: mov ecx,[edi+1Ch]
0x100ae81b: mov [esi+1Ch],ecx
0x100ae81e: mov edx,[edi+24h]
0x100ae821: mov [esi+24h],edx
0x100ae824: mov eax,[edi+28h]
0x100ae827: push 0
0x100ae829: push 0
0x100ae82b: mov [esi+28h],eax
0x100ae82e: call 100AE0A0h
0x100ae833: add esp,8
0x100ae836: test eax,eax
0x100ae838: mov [esi+20h],eax
0x100ae83b: je short 100AE7ECh
0x100ae83d: mov ecx,[edi+20h]
0x100ae840: push 0
0x100ae842: push 0
0x100ae844: push ecx
0x100ae845: call 100AE030h
0x100ae84a: add esp,0Ch
0x100ae84d: test eax,eax
0x100ae84f: jne short 100AE7ECh
0x100ae851: mov edx,[edi+20h]
0x100ae854: mov eax,[esi+20h]
0x100ae857: push 0FFFFFFFFh
0x100ae859: push edx
0x100ae85a: push eax
0x100ae85b: call 100AD810h
0x100ae860: add esp,0Ch
0x100ae863: test eax,eax
0x100ae865: jne short 100AE7ECh
0x100ae867: mov ecx,[esi+20h]
0x100ae86a: push eax
0x100ae86b: push eax
0x100ae86c: push ecx
0x100ae86d: call 100AE030h
0x100ae872: add esp,0Ch
0x100ae875: neg eax
0x100ae877: sbb eax,eax
0x100ae879: not eax
0x100ae87b: and eax,esi
0x100ae87d: pop esi
0x100ae87e: ret
```

`sub_100ae7e0` is called by sub_100affb0. It delegates to sub_100ad810, sub_100ae030, sub_100ae0a0 (4 callees).

12.1149 sub_100ae880 at 0x100ae880 (10 instructions)

```
0x100ae880: mov eax,[esi+20h]
0x100ae883: test eax,eax
0x100ae885: je short 100AE890h
0x100ae887: push eax
0x100ae888: call 100ADF90h
0x100ae88d: add esp,4
0x100ae890: push esi
0x100ae891: call 100209E0h
0x100ae896: pop ecx
0x100ae897: ret
```

`sub_100ae880` is called by sub_100af0f0, sub_100af150, sub_100af770. It delegates to j_free, sub_100adf90 (2 callees).

12.1150 sub_100ae910 at 0x100ae910 (60 instructions)

```
0x100ae910: mov eax,ds:[101398D0h]
0x100ae915: cmp eax,20h
0x100ae918: jl short 100AE91Eh
0x100ae91a: or eax,0FFFFFFFFh
0x100ae91d: ret
0x100ae91e: mov ecx,[esp+8]
0x100ae922: push esi
0x100ae923: lea esi,[eax*8]
0x100ae92a: sub esi,eax
0x100ae92c: mov eax,[esp+8]
0x100ae930: lea esi,[esi*4+10139550h]
0x100ae937: push ecx
0x100ae938: mov [esi],eax
0x100ae93a: call 100BFE40h
0x100ae93f: add esp,4
0x100ae942: test eax,eax
0x100ae944: mov [esi+4],eax
0x100ae947: je short 100AE997h
0x100ae949: mov edx,[esp+10h]
0x100ae94d: push edx
0x100ae94e: call 100BFE40h
0x100ae953: add esp,4
0x100ae956: test eax,eax
0x100ae958: mov [esi+8],eax
0x100ae95b: jne short 100AE96Eh
0x100ae95d: mov eax,[esi+4]
0x100ae960: push eax
0x100ae961: call 100209E0h
0x100ae966: add esp,4
0x100ae969: or eax,0FFFFFFFFh
0x100ae96c: pop esi
0x100ae96d: ret
0x100ae96e: mov ecx,[esp+14h]
0x100ae972: push ecx
0x100ae973: call 100BFE40h
0x100ae978: add esp,4
0x100ae97b: test eax,eax
0x100ae97d: mov [esi+0Ch],eax
0x100ae980: jne short 100AE99Ch
0x100ae982: mov edx,[esi+4]
0x100ae985: push edx
0x100ae986: call 100209E0h
0x100ae98b: mov eax,[esi+8]
0x100ae98e: push eax
0x100ae98f: call 100209E0h
0x100ae994: add esp,8
0x100ae997: or eax,0FFFFFFFFh
0x100ae99a: pop esi
0x100ae99b: ret
0x100ae99c: mov eax,[esp+18h]
0x100ae9a0: mov ecx,[eax]
0x100ae9a2: add dword ptr ds:[101398D0h],1
0x100ae9a9: mov [esi+10h],ecx
0x100ae9ac: mov edx,[eax+4]
0x100ae9af: mov [esi+14h],edx
0x100ae9b2: mov eax,[eax+8]
0x100ae9b5: mov [esi+18h],eax
0x100ae9b8: xor eax,eax
0x100ae9ba: pop esi
0x100ae9bb: ret
```

`sub_100ae910` is called by sub_100ae1a0. It delegates to j_free, sub_100bfe40 (2 callees).

12.1151 sub_100ae9c0 at 0x100ae9c0 (30 instructions)

```
0x100ae9c0: push ebx
0x100ae9c1: push esi
0x100ae9c2: push edi
0x100ae9c3: xor edi,edi
0x100ae9c5: cmp ds:[101398D0h],edi
0x100ae9cb: mov esi,10139550h
0x100ae9d0: jle short 100AE9F5h
0x100ae9d2: mov ebx,[esp+10h]
0x100ae9d6: mov eax,[esi+18h]
0x100ae9d9: test eax,eax
0x100ae9db: je short 100AE9E7h
0x100ae9dd: push ebx
0x100ae9de: call eax
0x100ae9e0: add esp,4
0x100ae9e3: test eax,eax
0x100ae9e5: je short 100AE9FCh
0x100ae9e7: add edi,1
0x100ae9ea: add esi,1Ch
0x100ae9ed: cmp edi,ds:[101398D0h]
0x100ae9f3: jl short 100AE9D6h
0x100ae9f5: pop edi
0x100ae9f6: pop esi
0x100ae9f7: or eax,0FFFFFFFFh
0x100ae9fa: pop ebx
0x100ae9fb: ret
0x100ae9fc: mov eax,[esi]
0x100ae9fe: pop edi
0x100ae9ff: pop esi
0x100aea00: pop ebx
0x100aea01: ret
```

`sub_100ae9c0` is called by sub_100156b0, sub_100af270. It calls nothing else.

12.1152 sub_100aea50 at 0x100aea50 (16 instructions)

```
0x100aea50: mov edx,ds:[101398D0h]
0x100aea56: xor ecx,ecx
0x100aea58: test edx,edx
0x100aea5a: push esi
0x100aea5b: mov eax,10139550h
0x100aea60: jle short 100AEA74h
0x100aea62: mov esi,[esp+8]
0x100aea66: cmp [eax],esi
0x100aea68: je short 100AEA76h
0x100aea6a: add ecx,1
0x100aea6d: add eax,1Ch
0x100aea70: cmp ecx,edx
0x100aea72: jl short 100AEA66h
0x100aea74: xor eax,eax
0x100aea76: pop esi
0x100aea77: ret
```

`sub_100aea50` is called by sub_100af270, sub_100af750. It calls nothing else.

12.1153 sub_100aeb10 at 0x100aeb10 (15 instructions)

```
0x100aeb10: mov edx,1
0x100aeb15: shl edx,cl
0x100aeb17: push esi
0x100aeb18: lea esi,[edx-1]
0x100aeb1b: and eax,esi
0x100aeb1d: cmp dword ptr [esp+8],0
0x100aeb22: je short 100AEB34h
0x100aeb24: add ecx,0FFFFFFFFh
```

```
0x100aeb27: mov esi,1
0x100aeb2c: shl esi,cl
0x100aeb2e: test eax,esi
0x100aeb30: je short 100AEB34h
0x100aeb32: sub eax,edx
0x100aeb34: pop esi
0x100aeb35: ret
```

`sub_100aeb10` is called by sub_100af330. It calls nothing else.

12.1154 sub_100aeb40 at 0x100aeb40 (68 instructions)

```
0x100aeb40: push esi
0x100aeb41: mov esi,[eax+10h]
0x100aeb44: xor ecx,ecx
0x100aeb46: cmp esi,ecx
0x100aeb48: jle near ptr 100AEBE7h
0x100aeb4e: mov edx,[eax+18h]
0x100aeb51: mov ecx,[edx]
0x100aeb53: push ebx
0x100aeb54: push edi
0x100aeb55: mov edi,[ecx]
0x100aeb57: mov [eax],edi
0x100aeb59: mov edi,[ecx+4]
0x100aeb5c: mov [eax+4],edi
0x100aeb5f: mov edi,[ecx+10h]
0x100aeb62: mov ebx,[ecx]
0x100aeb64: sub edi,1
0x100aeb67: imul edi,[ecx+8]
0x100aeb6b: lea edi,[edi+ebx+1]
0x100aeb6f: mov [eax+8],edi
0x100aeb72: mov edi,[ecx+14h]
0x100aeb75: sub edi,1
0x100aeb78: imul edi,[ecx+0Ch]
0x100aeb7c: mov ecx,[ecx+4]
0x100aeb7f: lea ecx,[edi+ecx+1]
0x100aeb83: mov edi,1
0x100aeb88: cmp esi,edi
0x100aeb8a: mov [eax+0Ch],ecx
0x100aeb8d: jle short 100AEBE3h
0x100aeb8f: lea esi,[edx+4]
0x100aeb92: mov ecx,[esi]
0x100aeb94: mov edx,[ecx]
0x100aeb96: cmp [eax],edx
0x100aeb98: jle short 100AEB9Ch
0x100aeb9a: mov [eax],edx
0x100aeb9c: mov edx,[ecx+4]
0x100aeb9f: cmp [eax+4],edx
0x100aeba2: jle short 100AEBA7h
0x100aeba4: mov [eax+4],edx
0x100aeba7: mov edx,[ecx+10h]
0x100aebaa: mov ebx,[ecx]
0x100aebac: sub edx,1
0x100aebaf: imul edx,[ecx+8]
0x100aebb3: lea edx,[edx+ebx+1]
0x100aebb7: cmp [eax+8],edx
0x100aebba: jge short 100AEBBFh
0x100aebbc: mov [eax+8],edx
0x100aebbf: mov edx,[ecx+14h]
0x100aebc2: sub edx,1
0x100aebc5: imul edx,[ecx+0Ch]
0x100aebc9: mov ecx,[ecx+4]
0x100aebcc: lea ecx,[edx+ecx+1]
0x100aebd0: cmp [eax+0Ch],ecx
0x100aebd3: jge short 100AEBD8h
0x100aebd5: mov [eax+0Ch],ecx
```

```
0x100aebd8: add edi,1
0x100aebdb: add esi,4
0x100aebde: cmp edi,[eax+10h]
0x100aebe1: jl short 100AEB92h
0x100aebe3: pop edi
0x100aebe4: pop ebx
0x100aebe5: pop esi
0x100aebe6: ret
0x100aebe7: mov [eax],ecx
0x100aebe9: mov [eax+4],ecx
0x100aebec: mov [eax+8],ecx
0x100aebef: mov [eax+0Ch],ecx
0x100aebf2: pop esi
0x100aebf3: ret
```

`sub_100aeb40` is called by sub_100af7d0, sub_100affb0. It calls nothing else.

12.1155 sub_100aec00 at 0x100aec00 (30 instructions)

```
0x100aec00: mov eax,[esi+18h]
0x100aec03: test eax,eax
0x100aec05: jne short 100AEC19h
0x100aec07: lea eax,[edi*4]
0x100aec0e: push eax
0x100aec0f: call 100EFDD0h
0x100aec14: add esp,4
0x100aec17: jmp short 100AEC2Ah
0x100aec19: lea ecx,[edi*4]
0x100aec20: push ecx
0x100aec21: push eax
0x100aec22: call 100AD4B0h
0x100aec27: add esp,8
0x100aec2a: test eax,eax
0x100aec2c: jne short 100AEC32h
0x100aec2e: or eax,0FFFFFFFFh
0x100aec31: ret
0x100aec32: mov [esi+18h],eax
0x100aec35: mov eax,[esi+10h]
0x100aec38: cmp eax,edi
0x100aec3a: mov [esi+14h],edi
0x100aec3d: jge short 100AEC52h
0x100aec3f: nop
0x100aec40: mov edx,[esi+18h]
0x100aec43: mov dword ptr [edx+eax*4],0
0x100aec4a: add eax,1
0x100aec4d: cmp eax,[esi+14h]
0x100aec50: jl short 100AEC40h
0x100aec52: xor eax,eax
0x100aec54: ret
```

`sub_100aec00` is called by sub_100af7d0, sub_100affb0. It delegates to j_realloc, j_malloc (2 callees).

12.1156 sub_100aee20 at 0x100aee20 (20 instructions)

```
0x100aee20: mov ecx,[esp+4]
0x100aee24: mov edx,[ecx+10h]
0x100aee27: push esi
0x100aee28: xor eax,eax
0x100aee2a: test edx,edx
0x100aee2c: push edi
0x100aee2d: jle short 100AEE47h
0x100aee2f: mov ecx,[ecx+18h]
0x100aee32: mov esi,[esp+10h]
0x100aee36: mov edi,[ecx]
0x100aee38: cmp [edi+28h],esi
```

```
0x100aee3b: je short 100AEE4Ah
0x100aee3d: add eax,1
0x100aee40: add ecx,4
0x100aee43: cmp eax,edx
0x100aee45: jl short 100AEE36h
0x100aee47: or eax,0FFFFFFFFh
0x100aee4a: pop edi
0x100aee4b: pop esi
0x100aee4c: ret
```

`sub_100aee20` is called by sub_100b0040. It calls nothing else.

12.1157 sub_100aee50 at 0x100aee50 (35 instructions)

```
0x100aee50: push ebx
0x100aee51: push ebp
0x100aee52: push esi
0x100aee53: mov esi,[esp+10h]
0x100aee57: mov eax,[esi+18h]
0x100aee5a: mov eax,[eax]
0x100aee5c: mov edx,[esi+10h]
0x100aee5f: mov ebx,[eax+8]
0x100aee62: mov ebp,[eax+0Ch]
0x100aee65: xor ecx,ecx
0x100aee67: test edx,edx
0x100aee69: mov eax,1
0x100aee6e: jle short 100AEE91h
0x100aee70: mov esi,[esi+18h]
0x100aee73: push edi
0x100aee74: mov edi,[esi]
0x100aee76: cmp [edi+8],ebx
0x100aee79: jne short 100AEE8Eh
0x100aee7b: cmp [edi+0Ch],ebp
0x100aee7e: jne short 100AEE8Eh
0x100aee80: add ecx,eax
0x100aee82: add esi,4
0x100aee85: cmp ecx,edx
0x100aee87: jl short 100AEE74h
0x100aee89: pop edi
0x100aee8a: pop esi
0x100aee8b: pop ebp
0x100aee8c: pop ebx
0x100aee8d: ret
0x100aee8e: xor eax,eax
0x100aee90: pop edi
0x100aee91: pop esi
0x100aee92: pop ebp
0x100aee93: pop ebx
0x100aee94: ret
```

`sub_100aee50` is called by sub_100b0040. It calls nothing else.

12.1158 sub_100aeea0 at 0x100aeea0 (72 instructions)

```
0x100aeea0: sub esp,0Ch
0x100aeea3: push ebx
0x100aeea4: push ebp
0x100aeea5: push esi
0x100aeea6: mov esi,[eax+10h]
0x100aeea9: xor ebx,ebx
0x100aeeab: cmp esi,ebx
0x100aeead: push edi
0x100aeeae: jle short 100AEF2Fh
0x100aeeb0: mov edi,[eax+18h]
0x100aeeb3: mov eax,[edi]
```

```
0x100aeeb5: mov ecx,[eax+10h]
0x100aeeb8: mov edx,[eax+14h]
0x100aeebb: mov ebp,[eax+4]
0x100aeebe: mov ebx,[eax]
0x100aeec0: sub ecx,1
0x100aeec3: imul ecx,[eax+8]
0x100aeec7: sub edx,1
0x100aeeca: imul edx,[eax+0Ch]
0x100aeece: add ecx,ebx
0x100aeed0: add edx,ebp
0x100aeed2: test esi,esi
0x100aeed4: mov [esp+10h],ebp
0x100aeed8: jle short 100AEF38h
0x100aeeda: mov [esp+14h],edi
0x100aeede: mov [esp+18h],esi
0x100aeee2: mov eax,[esp+14h]
0x100aeee6: mov esi,[eax]
0x100aeee8: mov edi,[esi]
0x100aeeea: cmp edi,ebx
0x100aeeec: jge short 100AEEF0h
0x100aeeee: mov ebx,edi
0x100aeef0: mov ebp,[esi+4]
0x100aeef3: cmp ebp,[esp+10h]
0x100aeef7: jge short 100AEEFDh
0x100aeef9: mov [esp+10h],ebp
0x100aeefd: mov eax,[esi+10h]
0x100aef00: sub eax,1
0x100aef03: imul eax,[esi+8]
0x100aef07: add eax,edi
0x100aef09: cmp eax,ecx
0x100aef0b: jle short 100AEF0Fh
0x100aef0d: mov ecx,eax
0x100aef0f: mov eax,[esi+14h]
0x100aef12: sub eax,1
0x100aef15: imul eax,[esi+0Ch]
0x100aef19: add eax,ebp
0x100aef1b: cmp eax,edx
0x100aef1d: jle short 100AEF21h
0x100aef1f: mov edx,eax
0x100aef21: add dword ptr [esp+14h],4
0x100aef26: sub dword ptr [esp+18h],1
0x100aef2b: jne short 100AEEE2h
0x100aef2d: jmp short 100AEF38h
0x100aef2f: or ecx,0FFFFFFFFh
0x100aef32: mov [esp+10h],ebx
0x100aef36: or edx,ecx
0x100aef38: mov eax,[esp+20h]
0x100aef3c: mov esi,[esp+10h]
0x100aef40: mov [eax],ebx
0x100aef42: mov eax,[esp+24h]
0x100aef46: mov [eax],esi
0x100aef48: mov eax,[esp+28h]
0x100aef4c: pop edi
0x100aef4d: pop esi
0x100aef4e: mov [eax],ecx
0x100aef50: mov ecx,[esp+24h]
0x100aef54: pop ebp
0x100aef55: mov [ecx],edx
0x100aef57: pop ebx
0x100aef58: add esp,0Ch
0x100aef5b: ret
```

`sub_100aeea0` is called by sub_100afb80. It calls nothing else.

12.1159 sub_100aef60 at 0x100aef60 (61 instructions)

```
0x100aef60: mov eax,[esp+8]
0x100aef64: add eax,7
0x100aef67: cdq
0x100aef68: and edx,7
0x100aef6b: push ebp
0x100aef6c: push edi
0x100aef6d: add eax,edx
0x100aef6f: mov edi,eax
0x100aef71: sar edi,3
0x100aef74: xor ebp,ebp
0x100aef76: sub edi,1
0x100aef79: js short 100AEFC4h
0x100aef7b: jmp short 100AEF80h
0x100aef80: test byte ptr [esi+8],7
0x100aef84: jne short 100AEFDAh
0x100aef86: mov eax,[esi+40h]
0x100aef89: test eax,eax
0x100aef8b: jl short 100AEF92h
0x100aef8d: cmp [esi+3Ch],eax
0x100aef90: jge short 100AEFD6h
0x100aef92: add dword ptr [esi+1Ch],0FFFFFFFFh
0x100aef96: jns short 100AEFA5h
0x100aef98: push 1
0x100aef9a: push esi
0x100aef9b: call 100AD620h
0x100aefa0: add esp,8
0x100aefa3: jmp short 100AEFB5h
0x100aefa5: mov ecx,[esi+18h]
0x100aefa8: add dword ptr [esi+3Ch],1
0x100aefac: movzx eax,byte ptr [ecx]
0x100aefaf: add ecx,1
0x100aefb2: mov [esi+18h],ecx
0x100aefb5: cmp eax,0FFFFFFFFh
0x100aefb8: je short 100AEFDAh
0x100aefba: shl ebp,8
0x100aefbd: or ebp,eax
0x100aefbf: sub edi,1
0x100aefc2: jns short 100AEF80h
0x100aefc4: cmp dword ptr [esp+0Ch],0
0x100aefc9: je short 100AEFE0h
0x100aefcb: call dword ptr ds:[100FE474h]
0x100aefd1: pop edi
0x100aefd2: xor eax,eax
0x100aefd4: pop ebp
0x100aefd5: ret
0x100aefd6: or dword ptr [esi+8],4
0x100aefda: pop edi
0x100aefdb: or eax,0FFFFFFFFh
0x100aefde: pop ebp
0x100aefdf: ret
0x100aefe0: mov ecx,[esp+10h]
0x100aefe4: mov eax,1
0x100aefe9: shl eax,cl
0x100aefeb: mov ecx,[esp+14h]
0x100aefef: pop edi
0x100aeff0: sub eax,1
0x100aeff3: and eax,ebp
0x100aeff5: mov [ecx],eax
0x100aeff7: xor eax,eax
0x100aeff9: pop ebp
0x100aeffa: ret
```

`sub_100aef60` is called by sub_100af960, sub_100afb80. It delegates to sub_100ad620 (1 callees).

12.1160 sub_100af000 at 0x100af000 (69 instructions)

```
0x100af000: cmp dword ptr [esp+4],0
0x100af005: push ebx
0x100af006: push ebp
0x100af007: push edi
0x100af008: mov edi,eax
0x100af00a: je short 100AF012h
0x100af00c: call dword ptr ds:[100FE474h]
0x100af012: mov ecx,edi
0x100af014: mov eax,1
0x100af019: shl eax,cl
0x100af01b: sub eax,1
0x100af01e: and [esp+14h],eax
0x100af022: lea eax,[edi+7]
0x100af025: cdq
0x100af026: and edx,7
0x100af029: add eax,edx
0x100af02b: mov edi,eax
0x100af02d: sar edi,3
0x100af030: sub edi,1
0x100af033: js short 100AF078h
0x100af035: lea ebp,[edi*8]
0x100af03c: lea esp,[esp]
0x100af040: mov ebx,[esp+14h]
0x100af044: mov eax,[esi+8]
0x100af047: mov ecx,ebp
0x100af049: sar ebx,cl
0x100af04b: and ebx,0FFh
0x100af051: test al,7
0x100af053: jne short 100AF067h
0x100af055: mov ecx,[esi+40h]
0x100af058: test ecx,ecx
0x100af05a: jl short 100AF07Eh
0x100af05c: cmp [esi+3Ch],ecx
0x100af05f: jl short 100AF07Eh
0x100af061: or eax,4
0x100af064: mov [esi+8],eax
0x100af067: or eax,0FFFFFFFFh
0x100af06a: cmp eax,ebx
0x100af06c: jne short 100AF0ACh
0x100af06e: sub edi,1
0x100af071: sub ebp,8
0x100af074: test edi,edi
0x100af076: jge short 100AF040h
0x100af078: pop edi
0x100af079: pop ebp
0x100af07a: xor eax,eax
0x100af07c: pop ebx
0x100af07d: ret
0x100af07e: or dword ptr [esi+4],20h
0x100af082: add dword ptr [esi+1Ch],0FFFFFFFFh
0x100af086: jns short 100AF097h
0x100af088: movzx ecx,bl
0x100af08b: push ecx
0x100af08c: push esi
0x100af08d: call 100AD6B0h
0x100af092: add esp,8
0x100af095: jmp short 100AF06Ah
0x100af097: mov edx,[esi+18h]
0x100af09a: mov ecx,1
0x100af09f: add [esi+3Ch],ecx
0x100af0a2: mov [edx],bl
0x100af0a4: add [esi+18h],ecx
0x100af0a7: movzx eax,bl
0x100af0aa: jmp short 100AF06Ah
0x100af0ac: pop edi
0x100af0ad: pop ebp
```

```
0x100af0ae: or eax,0FFFFFFFFh
0x100af0b1: pop ebx
0x100af0b2: ret
```

`sub_100af000` is called by sub_100afa70, sub_100afb80. It delegates to sub_100ad6b0 (1 callees).

12.1161 sub_100af0c0 at 0x100af0c0 (13 instructions)

```
0x100af0c0: cmp edx,esi
0x100af0c2: je short 100AF0D7h
0x100af0c4: jle short 100AF0CDh
0x100af0c6: mov ecx,edx
0x100af0c8: sub ecx,esi
0x100af0ca: shl eax,cl
0x100af0cc: ret
0x100af0cd: cmp esi,edx
0x100af0cf: jle short 100AF0D7h
0x100af0d1: mov ecx,esi
0x100af0d3: sub ecx,edx
0x100af0d5: sar eax,cl
0x100af0d7: ret
```

`sub_100af0c0` is called by sub_100afb80. It calls nothing else.

12.1162 sub_100af0e0 at 0x100af0e0 (4 instructions)

```
0x100af0e0: cdq
0x100af0e1: idiv ecx
0x100af0e3: imul eax,ecx
0x100af0e6: ret
```

`sub_100af0e0` is called by sub_100afb80. It calls nothing else.

12.1163 sub_100af0f0 at 0x100af0f0 (34 instructions)

```
0x100af0f0: push ebx
0x100af0f1: mov ebx,[esp+8]
0x100af0f5: cmp dword ptr [ebx+18h],0
0x100af0f9: je short 100AF12Fh
0x100af0fb: push edi
0x100af0fc: xor edi,edi
0x100af0fe: cmp [ebx+10h],edi
0x100af101: jle short 100AF122h
0x100af103: push esi
0x100af104: mov eax,[ebx+18h]
0x100af107: mov esi,[eax+edi*4]
0x100af10a: call 100AE880h
0x100af10f: mov ecx,[ebx+18h]
0x100af112: mov dword ptr [ecx+edi*4],0
0x100af119: add edi,1
0x100af11c: cmp edi,[ebx+10h]
0x100af11f: jl short 100AF104h
0x100af121: pop esi
0x100af122: mov edx,[ebx+18h]
0x100af125: push edx
0x100af126: call 100209E0h
0x100af12b: add esp,4
0x100af12e: pop edi
0x100af12f: mov eax,[ebx+20h]
0x100af132: test eax,eax
0x100af134: je short 100AF13Fh
0x100af136: push eax
0x100af137: call 100B1420h
0x100af13c: add esp,4
```

```
0x100af13f: push ebx
0x100af140: call 100209E0h
0x100af145: add esp,4
0x100af148: pop ebx
0x100af149: ret
```

`sub_100af0f0` is called by sub_100156b0, sub_100af270, sub_100affb0, sub_100b0040. It delegates to j_free, sub_100ae880, sub_100b1420 (3 callees).

12.1164 sub_100af150 at 0x100af150 (99 instructions)

```
0x100af150: push esi
0x100af151: push edi
0x100af152: push 2Ch
0x100af154: mov edi,eax
0x100af156: call 100EFDD0h
0x100af15b: mov esi,eax
0x100af15d: add esp,4
0x100af160: test esi,esi
0x100af162: je near ptr 100AF20Eh
0x100af168: mov eax,[esp+0Ch]
0x100af16c: mov edx,[esp+14h]
0x100af170: mov [esi],eax
0x100af172: mov eax,[esp+18h]
0x100af176: mov [esi+0Ch],eax
0x100af179: movzx eax,word ptr [esp+1Ch]
0x100af17e: mov [esi+18h],eax
0x100af181: add eax,7
0x100af184: mov ecx,[esp+10h]
0x100af188: mov [esi+8],edx
0x100af18b: cdq
0x100af18c: and edx,7
0x100af18f: add eax,edx
0x100af191: sar eax,3
0x100af194: mov [esi+24h],eax
0x100af197: imul eax,ebx
0x100af19a: imul eax,edi
0x100af19d: cmp dword ptr [esp+24h],0
0x100af1a2: mov [esi+4],ecx
0x100af1a5: mov ecx,[esp+20h]
0x100af1a9: mov [esi+14h],edi
0x100af1ac: mov dword ptr [esi+28h],10000h
0x100af1b3: mov [esi+10h],ebx
0x100af1b6: mov [esi+1Ch],ecx
0x100af1b9: mov dword ptr [esi+20h],0
0x100af1c0: mov edi,eax
0x100af1c2: je short 100AF1D1h
0x100af1c4: push edi
0x100af1c5: push 0
0x100af1c7: call 100AE0A0h
0x100af1cc: add esp,8
0x100af1cf: jmp short 100AF1D6h
0x100af1d1: call 100ADC00h
0x100af1d6: test eax,eax
0x100af1d8: mov [esi+20h],eax
0x100af1db: je short 100AF209h
0x100af1dd: push 0
0x100af1df: add edi,0FFFFFFFFh
0x100af1e2: push edi
0x100af1e3: push eax
0x100af1e4: call 100AE030h
0x100af1e9: add esp,0Ch
0x100af1ec: test eax,eax
0x100af1ee: jl short 100AF209h
0x100af1f0: mov eax,[esi+20h]
0x100af1f3: test byte ptr [eax+8],7
```

```
0x100af1f7: jne short 100AF209h
0x100af1f9: mov ecx,[eax+40h]
0x100af1fc: test ecx,ecx
0x100af1fe: jl short 100AF213h
0x100af200: cmp [eax+3Ch],ecx
0x100af203: jl short 100AF213h
0x100af205: or dword ptr [eax+8],4
0x100af209: call 100AE880h
0x100af20e: pop edi
0x100af20f: xor eax,eax
0x100af211: pop esi
0x100af212: ret
0x100af213: or dword ptr [eax+4],20h
0x100af217: mov eax,[esi+20h]
0x100af21a: add dword ptr [eax+1Ch],0FFFFFFFFh
0x100af21e: mov eax,[esi+20h]
0x100af221: cmp dword ptr [eax+1Ch],0
0x100af225: jge short 100AF250h
0x100af227: push 0
0x100af229: push eax
0x100af22a: call 100AD6B0h
0x100af22f: add esp,8
0x100af232: cmp eax,0FFFFFFFFh
0x100af235: je short 100AF209h
0x100af237: mov ecx,[esi+20h]
0x100af23a: push 0
0x100af23c: push 0
0x100af23e: push ecx
0x100af23f: call 100AE030h
0x100af244: add esp,0Ch
0x100af247: test eax,eax
0x100af249: jl short 100AF209h
0x100af24b: pop edi
0x100af24c: mov eax,esi
0x100af24e: pop esi
0x100af24f: ret
0x100af250: mov ecx,1
0x100af255: add [eax+3Ch],ecx
0x100af258: mov edx,[esi+20h]
0x100af25b: mov eax,[edx+18h]
0x100af25e: mov byte ptr [eax],0
0x100af261: mov eax,[esi+20h]
0x100af264: add [eax+18h],ecx
0x100af267: jmp short 100AF237h
```

`sub_100af150` is called by sub_100af7d0. It delegates to sub_100ad6b0, sub_100adc00, sub_100ae030 (6 callees).

12.1165 sub_100af270 at 0x100af270 (48 instructions)

```
0x100af270: mov eax,[esp+8]
0x100af274: test eax,eax
0x100af276: push esi
0x100af277: mov esi,[esp+8]
0x100af27b: jge short 100AF28Ah
0x100af27d: push esi
0x100af27e: call 100AE9C0h
0x100af283: add esp,4
0x100af286: test eax,eax
0x100af288: jl short 100AF2DCh
0x100af28a: push eax
0x100af28b: call 100AEA50h
0x100af290: add esp,4
0x100af293: test eax,eax
0x100af295: je short 100AF2DCh
0x100af297: mov eax,[eax+10h]
0x100af29a: test eax,eax
```

```
0x100af29c: je short 100AF2DCh
0x100af29e: mov ecx,[esp+10h]
0x100af2a2: push ecx
0x100af2a3: push esi
0x100af2a4: call eax
0x100af2a6: mov esi,eax
0x100af2a8: add esp,8
0x100af2ab: test esi,esi
0x100af2ad: je short 100AF2DCh
0x100af2af: mov eax,[esi+1Ch]
0x100af2b2: test eax,4000h
0x100af2b7: jne short 100AF2E0h
0x100af2b9: test al,al
0x100af2bb: je short 100AF2E0h
0x100af2bd: cmp dword ptr [esi+20h],0
0x100af2c1: jne short 100AF2E0h
0x100af2c3: push eax
0x100af2c4: call 100B1DF0h
0x100af2c9: add esp,4
0x100af2cc: test eax,eax
0x100af2ce: mov [esi+20h],eax
0x100af2d1: jne short 100AF2E0h
0x100af2d3: push esi
0x100af2d4: call 100AF0F0h
0x100af2d9: add esp,4
0x100af2dc: xor eax,eax
0x100af2de: pop esi
0x100af2df: ret
0x100af2e0: mov eax,esi
0x100af2e2: pop esi
0x100af2e3: ret
```

`sub_100af270` is called by sub_100156b0. It delegates to sub_100ae9c0, sub_100aea50, sub_100af0f0 (4 callees).

12.1166 sub_100af330 at 0x100af330 (166 instructions)

```
0x100af330: mov eax,[esp+8]
0x100af334: sub esp,10h
0x100af337: test eax,eax
0x100af339: jl near ptr 100AF51Dh
0x100af33f: mov ecx,[esp+14h]
0x100af343: cmp eax,[ecx+10h]
0x100af346: jge near ptr 100AF51Dh
0x100af34c: mov ecx,[ecx+18h]
0x100af34f: mov edx,[esp+1Ch]
0x100af353: push ebx
0x100af354: push esi
0x100af355: mov esi,[ecx+eax*4]
0x100af358: mov eax,[esi+10h]
0x100af35b: cmp edx,eax
0x100af35d: push edi
0x100af35e: jge near ptr 100AF513h
0x100af364: mov ecx,[esi+14h]
0x100af367: mov edi,[esp+2Ch]
0x100af36b: cmp edi,ecx
0x100af36d: jge near ptr 100AF513h
0x100af373: mov ebx,[esp+30h]
0x100af377: add edx,ebx
0x100af379: cmp edx,eax
0x100af37b: jg near ptr 100AF513h
0x100af381: mov eax,[esp+34h]
0x100af385: add edi,eax
0x100af387: cmp edi,ecx
0x100af389: jg near ptr 100AF513h
0x100af38f: mov edi,[esp+38h]
0x100af393: cmp [edi+14h],eax
```

```
0x100af396: jne short 100AF39Dh
0x100af398: cmp [edi+18h],ebx
0x100af39b: je short 100AF3B0h
0x100af39d: push ebx
0x100af39e: push eax
0x100af39f: push edi
0x100af3a0: call 100AE5C0h
0x100af3a5: add esp,0Ch
0x100af3a8: test eax,eax
0x100af3aa: jne near ptr 100AF513h
0x100af3b0: mov edx,[edi+1Ch]
0x100af3b3: mov eax,[edx]
0x100af3b5: mov ebx,1
0x100af3ba: cmp [edi+14h],ebx
0x100af3bd: push ebp
0x100af3be: mov [esp+10h],eax
0x100af3c2: jle short 100AF3D4h
0x100af3c4: mov edi,edx
0x100af3c6: mov ebp,[edi+4]
0x100af3c9: sub ebp,[edi]
0x100af3cb: sar ebp,2
0x100af3ce: mov [esp+28h],ebp
0x100af3d2: jmp short 100AF3E0h
0x100af3d4: mov dword ptr [esp+28h],0
0x100af3dc: mov ebp,[esp+28h]
0x100af3e0: xor edi,edi
0x100af3e2: cmp [esp+38h],edi
0x100af3e6: mov [esp+1Ch],edi
0x100af3ea: jle near ptr 100AF4F8h
0x100af3f0: mov edx,[esp+30h]
0x100af3f4: mov ecx,[esp+10h]
0x100af3f8: lea eax,[edi+edx]
0x100af3fb: imul eax,[esi+10h]
0x100af3ff: add eax,[esp+2Ch]
0x100af403: push 0
0x100af405: imul eax,[esi+24h]
0x100af409: mov [esp+1Ch],ecx
0x100af40d: mov ecx,[esi+20h]
0x100af410: push eax
0x100af411: push ecx
0x100af412: call 100AE030h
0x100af417: add esp,0Ch
0x100af41a: test eax,eax
0x100af41c: jl near ptr 100AF508h
0x100af422: mov eax,[esp+34h]
0x100af426: test eax,eax
0x100af428: mov [esp+14h],eax
0x100af42c: jle near ptr 100AF4DDh
0x100af432: mov ebp,[esi+24h]
0x100af435: xor edi,edi
0x100af437: test ebp,ebp
0x100af439: jle short 100AF4A5h
0x100af43b: jmp short 100AF440h
0x100af440: mov eax,[esi+20h]
0x100af443: test byte ptr [eax+8],7
0x100af447: jne near ptr 100AF508h
0x100af44d: cmp dword ptr [eax+40h],0
0x100af451: jl short 100AF45Fh
0x100af453: mov edx,[eax+3Ch]
0x100af456: cmp edx,[eax+40h]
0x100af459: jge near ptr 100AF502h
0x100af45f: mov eax,[esi+20h]
0x100af462: add dword ptr [eax+1Ch],0FFFFFFFFh
0x100af466: mov eax,[esi+20h]
0x100af469: cmp dword ptr [eax+1Ch],0
0x100af46d: jge short 100AF47Dh
0x100af46f: mov ecx,eax
```

```
0x100af471: push ebx
0x100af472: push ecx
0x100af473: call 100AD620h
0x100af478: add esp,8
0x100af47b: jmp short 100AF48Eh
0x100af47d: add [eax+3Ch],ebx
0x100af480: mov ecx,[esi+20h]
0x100af483: mov edx,[ecx+18h]
0x100af486: movzx eax,byte ptr [edx]
0x100af489: add edx,ebx
0x100af48b: mov [ecx+18h],edx
0x100af48e: cmp eax,0FFFFFFFFh
0x100af491: je short 100AF508h
0x100af493: shl edi,8
0x100af496: and eax,0FFh
0x100af49b: or eax,edi
0x100af49d: sub ebp,ebx
0x100af49f: test ebp,ebp
0x100af4a1: mov edi,eax
0x100af4a3: jg short 100AF440h
0x100af4a5: mov edx,[esi+1Ch]
0x100af4a8: mov ecx,[esi+18h]
0x100af4ab: push edx
0x100af4ac: mov eax,edi
0x100af4ae: call 100AEB10h
0x100af4b3: mov ecx,[esp+1Ch]
0x100af4b7: mov [ecx],eax
0x100af4b9: mov eax,[esp+18h]
0x100af4bd: sub eax,ebx
0x100af4bf: add ecx,4
0x100af4c2: add esp,4
0x100af4c5: test eax,eax
0x100af4c7: mov [esp+14h],eax
0x100af4cb: mov [esp+18h],ecx
0x100af4cf: jg near ptr 100AF432h
0x100af4d5: mov edi,[esp+1Ch]
0x100af4d9: mov ebp,[esp+28h]
0x100af4dd: mov eax,[esp+10h]
0x100af4e1: add edi,ebx
0x100af4e3: cmp edi,[esp+38h]
0x100af4e7: lea ecx,[eax+ebp*4]
0x100af4ea: mov [esp+1Ch],edi
0x100af4ee: mov [esp+10h],ecx
0x100af4f2: jl near ptr 100AF3F0h
0x100af4f8: pop ebp
0x100af4f9: pop edi
0x100af4fa: pop esi
0x100af4fb: xor eax,eax
0x100af4fd: pop ebx
0x100af4fe: add esp,10h
0x100af501: ret
0x100af502: mov esi,eax
0x100af504: or dword ptr [esi+8],4
0x100af508: pop ebp
0x100af509: pop edi
0x100af50a: pop esi
0x100af50b: or eax,0FFFFFFFFh
0x100af50e: pop ebx
0x100af50f: add esp,10h
0x100af512: ret
0x100af513: pop edi
0x100af514: pop esi
0x100af515: or eax,0FFFFFFFFh
0x100af518: pop ebx
0x100af519: add esp,10h
0x100af51c: ret
0x100af51d: or eax,0FFFFFFFFh
```

```
0x100af520: add esp,10h
0x100af523: ret
```

`sub_100af330` is called by sub_100156b0. It delegates to sub_100ad620, sub_100ae030, sub_100ae5c0 (4 callees).

12.1167 sub_100af750 at 0x100af750 (9 instructions)

```
0x100af750: mov eax,[esp+4]
0x100af754: push eax
0x100af755: call 100AEA50h
0x100af75a: add esp,4
0x100af75d: test eax,eax
0x100af75f: jne short 100AF762h
0x100af761: ret
0x100af762: mov eax,[eax+4]
0x100af765: ret
```

`sub_100af750` is called by sub_100156b0. It delegates to sub_100aea50 (1 callees).

12.1168 sub_100af770 at 0x100af770 (100 instructions)

```
0x100aeb40: push esi
0x100aeb41: mov esi,[eax+10h]
0x100aeb44: xor ecx,ecx
0x100aeb46: cmp esi,ecx
0x100aeb48: jle near ptr 100AEBE7h
0x100aeb4e: mov edx,[eax+18h]
0x100aeb51: mov ecx,[edx]
0x100aeb53: push ebx
0x100aeb54: push edi
0x100aeb55: mov edi,[ecx]
0x100aeb57: mov [eax],edi
0x100aeb59: mov edi,[ecx+4]
0x100aeb5c: mov [eax+4],edi
0x100aeb5f: mov edi,[ecx+10h]
0x100aeb62: mov ebx,[ecx]
0x100aeb64: sub edi,1
0x100aeb67: imul edi,[ecx+8]
0x100aeb6b: lea edi,[edi+ebx+1]
0x100aeb6f: mov [eax+8],edi
0x100aeb72: mov edi,[ecx+14h]
0x100aeb75: sub edi,1
0x100aeb78: imul edi,[ecx+0Ch]
0x100aeb7c: mov ecx,[ecx+4]
0x100aeb7f: lea ecx,[edi+ecx+1]
0x100aeb83: mov edi,1
0x100aeb88: cmp esi,edi
0x100aeb8a: mov [eax+0Ch],ecx
0x100aeb8d: jle short 100AEBE3h
0x100aeb8f: lea esi,[edx+4]
0x100aeb92: mov ecx,[esi]
0x100aeb94: mov edx,[ecx]
0x100aeb96: cmp [eax],edx
0x100aeb98: jle short 100AEB9Ch
0x100aeb9a: mov [eax],edx
0x100aeb9c: mov edx,[ecx+4]
0x100aeb9f: cmp [eax+4],edx
0x100aeba2: jle short 100AEBA7h
0x100aeba4: mov [eax+4],edx
0x100aeba7: mov edx,[ecx+10h]
0x100aebaa: mov ebx,[ecx]
0x100aebac: sub edx,1
0x100aebaf: imul edx,[ecx+8]
0x100aebb3: lea edx,[edx+ebx+1]
0x100aebb7: cmp [eax+8],edx
```

```
0x100aebba: jge short 100AEBBFh
0x100aebbc: mov [eax+8],edx
0x100aebbf: mov edx,[ecx+14h]
0x100aebc2: sub edx,1
0x100aebc5: imul edx,[ecx+0Ch]
0x100aebc9: mov ecx,[ecx+4]
0x100aebcc: lea ecx,[edx+ecx+1]
0x100aebd0: cmp [eax+0Ch],ecx
0x100aebd3: jge short 100AEBD8h
0x100aebd5: mov [eax+0Ch],ecx
0x100aebd8: add edi,1
0x100aebdb: add esi,4
0x100aebde: cmp edi,[eax+10h]
0x100aebe1: jl short 100AEB92h
0x100aebe3: pop edi
0x100aebe4: pop ebx
0x100aebe5: pop esi
0x100aebe6: ret
0x100aebe7: mov [eax],ecx
0x100aebe9: mov [eax+4],ecx
0x100aebec: mov [eax+8],ecx
0x100aebef: mov [eax+0Ch],ecx
0x100aebf2: pop esi
0x100aebf3: ret
0x100af770: push ebx
0x100af771: mov ebx,[esp+8]
0x100af775: push edi
0x100af776: mov edi,[esp+10h]
0x100af77a: cmp edi,[ebx+10h]
0x100af77d: jge short 100AF7BEh
0x100af77f: mov eax,[ebx+18h]
0x100af782: push esi
0x100af783: mov esi,[eax+edi*4]
0x100af786: call 100AE880h
0x100af78b: mov eax,[ebx+10h]
0x100af78e: cmp edi,eax
0x100af790: pop esi
0x100af791: jge short 100AF7B1h
0x100af793: mov ecx,[ebx+18h]
0x100af796: sub eax,edi
0x100af798: lea ecx,[ecx+edi*4]
0x100af79b: lea edx,[eax*4-4]
0x100af7a2: push edx
0x100af7a3: lea eax,[ecx+4]
0x100af7a6: push eax
0x100af7a7: push ecx
0x100af7a8: call dword ptr ds:[100FE498h]
0x100af7ae: add esp,0Ch
0x100af7b1: add dword ptr [ebx+10h],0FFFFFFFFh
0x100af7b5: pop edi
0x100af7b6: mov eax,ebx
0x100af7b8: pop ebx
0x100af7b9: jmp near ptr 100AEB40h
0x100af7be: pop edi
0x100af7bf: pop ebx
0x100af7c0: ret
```

`sub_100af770` is called by sub_100b0040. It delegates to sub_100ae880 (1 callees).

12.1169 sub_100af7d0 at 0x100af7d0 (67 instructions)

```
0x100af7d0: push ebp
0x100af7d1: mov ebp,[esp+0Ch]
0x100af7d5: test ebp,ebp
0x100af7d7: push esi
0x100af7d8: mov esi,[esp+0Ch]
```

```
0x100af7dc: push edi
0x100af7dd: jge short 100AF7E2h
0x100af7df: mov ebp,[esi+10h]
0x100af7e2: mov eax,[esi+14h]
0x100af7e5: cmp [esi+10h],eax
0x100af7e8: jl short 100AF800h
0x100af7ea: lea edi,[eax+80h]
0x100af7f0: call 100AEC00h
0x100af7f5: test eax,eax
0x100af7f7: je short 100AF800h
0x100af7f9: pop edi
0x100af7fa: pop esi
0x100af7fb: or eax,0FFFFFFFFh
0x100af7fe: pop ebp
0x100af7ff: ret
0x100af800: mov ecx,[esp+18h]
0x100af804: mov eax,[ecx+1Ch]
0x100af807: movzx edx,word ptr [ecx+18h]
0x100af80b: push ebx
0x100af80c: mov ebx,[ecx+10h]
0x100af80f: push 1
0x100af811: push eax
0x100af812: mov eax,[ecx+0Ch]
0x100af815: push edx
0x100af816: mov edx,[ecx+8]
0x100af819: push eax
0x100af81a: mov eax,[ecx+4]
0x100af81d: push edx
0x100af81e: mov edx,[ecx]
0x100af820: push eax
0x100af821: mov eax,[ecx+14h]
0x100af824: push edx
0x100af825: call 100AF150h
0x100af82a: add esp,1Ch
0x100af82d: mov edi,eax
0x100af82f: test edi,edi
0x100af831: pop ebx
0x100af832: je short 100AF7F9h
0x100af834: mov eax,[esi+10h]
0x100af837: cmp ebp,eax
0x100af839: jge short 100AF856h
0x100af83b: mov ecx,[esi+18h]
0x100af83e: sub eax,ebp
0x100af840: add eax,eax
0x100af842: add eax,eax
0x100af844: lea ecx,[ecx+ebp*4]
0x100af847: push eax
0x100af848: push ecx
0x100af849: add ecx,4
0x100af84c: push ecx
0x100af84d: call dword ptr ds:[100FE498h]
0x100af853: add esp,0Ch
0x100af856: mov edx,[esi+18h]
0x100af859: mov [edx+ebp*4],edi
0x100af85c: add dword ptr [esi+10h],1
0x100af860: mov eax,esi
0x100af862: call 100AEB40h
0x100af867: pop edi
0x100af868: pop esi
0x100af869: xor eax,eax
0x100af86b: pop ebp
0x100af86c: ret
```

`sub_100af7d0` is called by sub_100afb80, sub_100b0040. It delegates to sub_100aeb40, sub_100aec00, sub_100af150 (3 callees).

12.1170 sub_100af960 at 0x100af960 (96 instructions)

```
0x100af960: push ecx
0x100af961: mov eax,[esp+0Ch]
0x100af965: test eax,eax
0x100af967: push ebx
0x100af968: push ebp
0x100af969: push esi
0x100af96a: push edi
0x100af96b: jl near ptr 100AFA67h
0x100af971: mov ecx,[esp+18h]
0x100af975: cmp eax,[ecx+10h]
0x100af978: jge near ptr 100AFA67h
0x100af97e: mov ecx,[ecx+18h]
0x100af981: mov ebp,[ecx+eax*4]
0x100af984: mov eax,[esp+20h]
0x100af988: test eax,eax
0x100af98a: jl near ptr 100AFA67h
0x100af990: mov edi,[ebp+10h]
0x100af993: cmp eax,edi
0x100af995: jge near ptr 100AFA67h
0x100af99b: mov ecx,[esp+24h]
0x100af99f: test ecx,ecx
0x100af9a1: jl near ptr 100AFA67h
0x100af9a7: mov esi,[ebp+14h]
0x100af9aa: cmp ecx,esi
0x100af9ac: jge near ptr 100AFA67h
0x100af9b2: mov ebx,[esp+28h]
0x100af9b6: test ebx,ebx
0x100af9b8: jl near ptr 100AFA67h
0x100af9be: mov edx,[esp+2Ch]
0x100af9c2: test edx,edx
0x100af9c4: jl near ptr 100AFA67h
0x100af9ca: add ebx,eax
0x100af9cc: cmp ebx,edi
0x100af9ce: jg near ptr 100AFA67h
0x100af9d4: lea edi,[ecx+edx]
0x100af9d7: cmp edi,esi
0x100af9d9: jg near ptr 100AFA67h
0x100af9df: mov ebx,[esp+30h]
0x100af9e3: xor esi,esi
0x100af9e5: test edx,edx
0x100af9e7: mov [esp+1Ch],esi
0x100af9eb: jle short 100AFA5Fh
0x100af9ed: jmp short 100AF9F8h
0x100af9f0: mov ecx,[esp+24h]
0x100af9f4: mov eax,[esp+20h]
0x100af9f8: lea edx,[esi+ecx]
0x100af9fb: imul edx,[ebp+10h]
0x100af9ff: add edx,eax
0x100afa01: imul edx,[ebp+24h]
0x100afa05: mov eax,[ebp+20h]
0x100afa08: push 0
0x100afa0a: push edx
0x100afa0b: push eax
0x100afa0c: call 100AE030h
0x100afa11: add esp,0Ch
0x100afa14: test eax,eax
0x100afa16: jl short 100AFA67h
0x100afa18: xor edi,edi
0x100afa1a: cmp [esp+28h],edi
0x100afa1e: jle short 100AFA52h
0x100afa20: mov edx,[ebp+18h]
0x100afa23: mov eax,[ebp+1Ch]
0x100afa26: mov esi,[ebp+20h]
0x100afa29: lea ecx,[esp+10h]
0x100afa2d: push ecx
0x100afa2e: push edx
```

```
0x100afa2f: push eax
0x100afa30: call 100AEF60h
0x100afa35: add esp,0Ch
0x100afa38: test eax,eax
0x100afa3a: jne short 100AFA67h
0x100afa3c: mov ecx,[esp+10h]
0x100afa40: mov [ebx],ecx
0x100afa42: add edi,1
0x100afa45: add ebx,4
0x100afa48: cmp edi,[esp+28h]
0x100afa4c: jl short 100AFA20h
0x100afa4e: mov esi,[esp+1Ch]
0x100afa52: add esi,1
0x100afa55: cmp esi,[esp+2Ch]
0x100afa59: mov [esp+1Ch],esi
0x100afa5d: jl short 100AF9F0h
0x100afa5f: pop edi
0x100afa60: pop esi
0x100afa61: pop ebp
0x100afa62: xor eax,eax
0x100afa64: pop ebx
0x100afa65: pop ecx
0x100afa66: ret
0x100afa67: pop edi
0x100afa68: pop esi
0x100afa69: pop ebp
0x100afa6a: or eax,0FFFFFFFFh
0x100afa6d: pop ebx
0x100afa6e: pop ecx
0x100afa6f: ret
```

`sub_100af960` is called by sub_100b0040. It delegates to sub_100ae030, sub_100aef60 (2 callees).

12.1171 sub_100afa70 at 0x100afa70 (90 instructions)

```
0x100afa70: mov eax,[esp+8]
0x100afa74: test eax,eax
0x100afa76: push ebx
0x100afa77: push ebp
0x100afa78: push esi
0x100afa79: push edi
0x100afa7a: jl near ptr 100AFB6Eh
0x100afa80: mov ecx,[esp+14h]
0x100afa84: cmp eax,[ecx+10h]
0x100afa87: jge near ptr 100AFB6Eh
0x100afa8d: mov ecx,[ecx+18h]
0x100afa90: mov edi,[ecx+eax*4]
0x100afa93: mov eax,[esp+1Ch]
0x100afa97: test eax,eax
0x100afa99: jl near ptr 100AFB6Eh
0x100afa9f: mov ebx,[edi+10h]
0x100afaa2: cmp eax,ebx
0x100afaa4: jge near ptr 100AFB6Eh
0x100afaaa: mov ecx,[esp+20h]
0x100afaae: test ecx,ecx
0x100afab0: jl near ptr 100AFB6Eh
0x100afab6: mov esi,[edi+14h]
0x100afab9: cmp ecx,esi
0x100afabb: jge near ptr 100AFB6Eh
0x100afac1: mov ebp,[esp+24h]
0x100afac5: test ebp,ebp
0x100afac7: jl near ptr 100AFB6Eh
0x100afacd: mov edx,[esp+28h]
0x100afad1: test edx,edx
0x100afad3: jl near ptr 100AFB6Eh
0x100afad9: add ebp,eax
```

```
0x100afadb: cmp ebp,ebx
0x100afadd: jg near ptr 100AFB6Eh
0x100afae3: lea ebx,[ecx+edx]
0x100afae6: cmp ebx,esi
0x100afae8: jg near ptr 100AFB6Eh
0x100afaee: mov ebp,[esp+2Ch]
0x100afaf2: xor esi,esi
0x100afaf4: test edx,edx
0x100afaf6: mov [esp+18h],esi
0x100afafa: jle short 100AFB67h
0x100afafc: jmp short 100AFB08h
0x100afb00: mov ecx,[esp+20h]
0x100afb04: mov eax,[esp+1Ch]
0x100afb08: lea edx,[esi+ecx]
0x100afb0b: imul edx,[edi+10h]
0x100afb0f: add edx,eax
0x100afb11: imul edx,[edi+24h]
0x100afb15: mov eax,[edi+20h]
0x100afb18: push 0
0x100afb1a: push edx
0x100afb1b: push eax
0x100afb1c: call 100AE030h
0x100afb21: add esp,0Ch
0x100afb24: test eax,eax
0x100afb26: jl short 100AFB6Eh
0x100afb28: xor ebx,ebx
0x100afb2a: cmp [esp+24h],ebx
0x100afb2e: jle short 100AFB5Ah
0x100afb30: mov eax,[ebp]
0x100afb33: mov ecx,[edi+1Ch]
0x100afb36: mov esi,[edi+20h]
0x100afb39: push eax
0x100afb3a: mov eax,[edi+18h]
0x100afb3d: push ecx
0x100afb3e: add ebp,4
0x100afb41: call 100AF000h
0x100afb46: add esp,8
0x100afb49: test eax,eax
0x100afb4b: jne short 100AFB6Eh
0x100afb4d: add ebx,1
0x100afb50: cmp ebx,[esp+24h]
0x100afb54: jl short 100AFB30h
0x100afb56: mov esi,[esp+18h]
0x100afb5a: add esi,1
0x100afb5d: cmp esi,[esp+28h]
0x100afb61: mov [esp+18h],esi
0x100afb65: jl short 100AFB00h
0x100afb67: pop edi
0x100afb68: pop esi
0x100afb69: pop ebp
0x100afb6a: xor eax,eax
0x100afb6c: pop ebx
0x100afb6d: ret
0x100afb6e: pop edi
0x100afb6f: pop esi
0x100afb70: pop ebp
0x100afb71: or eax,0FFFFFFFFh
0x100afb74: pop ebx
0x100afb75: ret
```

`sub_100afa70` is called by sub_100b0040. It delegates to sub_100ae030, sub_100af000 (2 callees).

12.1172 sub_100afb80 at 0x100afb80 (263 instructions)

```
0x100afb80: sub esp,38h
0x100afb83: mov ecx,[esp+40h]
```

```
0x100afb87: push ebx
0x100afb88: push ebp
0x100afb89: push esi
0x100afb8a: mov esi,[esp+48h]
0x100afb8e: mov eax,[esi+18h]
0x100afb91: mov ebp,[eax+ecx*4]
0x100afb94: push edi
0x100afb95: lea edx,[esp+14h]
0x100afb99: push edx
0x100afb9a: lea eax,[esp+54h]
0x100afb9e: push eax
0x100afb9f: lea ecx,[esp+2Ch]
0x100afba3: push ecx
0x100afba4: lea edx,[esp+2Ch]
0x100afba8: push edx
0x100afba9: mov eax,esi
0x100afbab: mov [esp+5Ch],ebp
0x100afbaf: call 100AEEA0h
0x100afbb4: mov eax,[esp+60h]
0x100afbb8: sub eax,[esp+68h]
0x100afbbc: mov ecx,[esp+70h]
0x100afbc0: add eax,ecx
0x100afbc2: cdq
0x100afbc3: idiv ecx
0x100afbc5: mov edi,[esp+74h]
0x100afbc9: mov [esp+44h],edi
0x100afbcd: mov [esp+40h],ecx
0x100afbd1: mov ecx,[esp+7Ch]
0x100afbd5: mov [esp+50h],ecx
0x100afbd9: mov [esp+20h],eax
0x100afbdd: mov eax,[esp+24h]
0x100afbe1: sub eax,[esp+6Ch]
0x100afbe5: add eax,edi
0x100afbe7: cdq
0x100afbe8: idiv edi
0x100afbea: mov edx,[esp+6Ch]
0x100afbee: mov edi,[esp+64h]
0x100afbf2: mov [esp+3Ch],edx
0x100afbf6: mov edx,[esp+78h]
0x100afbfa: mov [esp+54h],edx
0x100afbfe: mov ebx,eax
0x100afc00: mov eax,[esp+68h]
0x100afc04: mov [esp+38h],eax
0x100afc08: mov eax,[esp+20h]
0x100afc0c: mov [esp+48h],eax
0x100afc10: lea eax,[esp+38h]
0x100afc14: push eax
0x100afc15: push edi
0x100afc16: push esi
0x100afc17: mov [esp+40h],ebx
0x100afc1b: mov [esp+58h],ebx
0x100afc1f: call 100AF7D0h
0x100afc24: add esp,1Ch
0x100afc27: test eax,eax
0x100afc29: jne near ptr 100AFE79h
0x100afc2f: mov eax,[ebp+10h]
0x100afc32: sub eax,1
0x100afc35: imul eax,[ebp+8]
0x100afc39: add eax,[ebp]
0x100afc3c: mov ecx,[esi+18h]
0x100afc3f: mov edi,[ecx+edi*4]
0x100afc42: mov edx,[edi+20h]
0x100afc45: mov [esp+18h],eax
0x100afc49: mov eax,[ebp+14h]
0x100afc4c: sub eax,1
0x100afc4f: imul eax,[ebp+0Ch]
0x100afc53: add eax,[ebp+4]
```

```
0x100afc56: push edx
0x100afc57: mov [esp+24h],eax
0x100afc5b: mov [esp+70h],edi
0x100afc5f: call 100AE170h
0x100afc64: xor ecx,ecx
0x100afc66: add esp,4
0x100afc69: test ebx,ebx
0x100afc6b: mov [esp+68h],ecx
0x100afc6f: jle near ptr 100AFE6Fh
0x100afc75: mov eax,[edi+0Ch]
0x100afc78: imul eax,ecx
0x100afc7b: add eax,[edi+4]
0x100afc7e: cmp dword ptr [esp+10h],0
0x100afc83: mov [esp+5Ch],eax
0x100afc87: mov dword ptr [esp+64h],0
0x100afc8f: jle near ptr 100AFE60h
0x100afc95: mov esi,[edi+8]
0x100afc98: imul esi,[esp+64h]
0x100afc9d: add esi,[edi]
0x100afc9f: mov ebx,[ebp]
0x100afca2: mov ecx,[ebp+8]
0x100afca5: mov edi,esi
0x100afca7: sub edi,ebx
0x100afca9: mov eax,edi
0x100afcab: mov [esp+14h],ecx
0x100afcaf: call 100AF0E0h
0x100afcb4: mov ecx,[ebp+0Ch]
0x100afcb7: mov ebp,[ebp+4]
0x100afcba: add eax,ebx
0x100afcbc: mov [esp+50h],eax
0x100afcc0: mov eax,[esp+5Ch]
0x100afcc4: sub eax,ebp
0x100afcc6: mov [esp+58h],ebp
0x100afcca: mov [esp+1Ch],eax
0x100afcce: call 100AF0E0h
0x100afcd3: mov ecx,[esp+14h]
0x100afcd7: mov ebp,eax
0x100afcd9: add ebp,[esp+58h]
0x100afcdd: mov eax,edi
0x100afcdf: call 100C06F0h
0x100afce4: mov edi,eax
0x100afce6: mov eax,[esp+18h]
0x100afcea: add edi,ebx
0x100afcec: cmp edi,eax
0x100afcee: jle short 100AFCF2h
0x100afcf0: mov edi,eax
0x100afcf2: mov eax,[esp+4Ch]
0x100afcf6: mov ecx,[eax+0Ch]
0x100afcf9: mov eax,[esp+1Ch]
0x100afcfd: call 100C06F0h
0x100afd02: mov ecx,[esp+4Ch]
0x100afd06: add eax,[ecx+4]
0x100afd09: mov ecx,[esp+20h]
0x100afd0d: cmp eax,ecx
0x100afd0f: mov [esp+58h],eax
0x100afd13: jle short 100AFD19h
0x100afd15: mov [esp+58h],ecx
0x100afd19: mov eax,[esp+50h]
0x100afd1d: sub eax,esi
0x100afd1f: mov ecx,ebp
0x100afd21: imul eax,eax
0x100afd24: sub ecx,[esp+5Ch]
0x100afd28: imul ecx,ecx
0x100afd2b: lea edx,[ecx+eax]
0x100afd2e: mov [esp+1Ch],edx
0x100afd32: mov edx,edi
0x100afd34: sub edx,esi
```

```
0x100afd36: imul edx,edx
0x100afd39: add ecx,edx
0x100afd3b: mov [esp+14h],ecx
0x100afd3f: mov ecx,[esp+58h]
0x100afd43: sub ecx,[esp+5Ch]
0x100afd47: mov esi,[esp+14h]
0x100afd4b: imul ecx,ecx
0x100afd4e: add edx,ecx
0x100afd50: add ecx,eax
0x100afd52: mov eax,[esp+1Ch]
0x100afd56: cmp esi,eax
0x100afd58: jl short 100AFD7Ah
0x100afd5a: cmp eax,edx
0x100afd5c: jg short 100AFD76h
0x100afd5e: cmp eax,ecx
0x100afd60: jg short 100AFD76h
0x100afd62: mov eax,[esp+50h]
0x100afd66: mov esi,[esp+4Ch]
0x100afd6a: sub eax,ebx
0x100afd6c: cdq
0x100afd6d: idiv dword ptr [esi+8]
0x100afd70: mov ecx,eax
0x100afd72: mov eax,ebp
0x100afd74: jmp short 100AFDB6h
0x100afd76: cmp esi,eax
0x100afd78: jg short 100AFD94h
0x100afd7a: cmp esi,edx
0x100afd7c: jg short 100AFD94h
0x100afd7e: cmp esi,ecx
0x100afd80: jg short 100AFD94h
0x100afd82: mov esi,[esp+4Ch]
0x100afd86: mov eax,edi
0x100afd88: sub eax,ebx
0x100afd8a: cdq
0x100afd8b: idiv dword ptr [esi+8]
0x100afd8e: mov ecx,eax
0x100afd90: mov eax,ebp
0x100afd92: jmp short 100AFDB6h
0x100afd94: cmp edx,eax
0x100afd96: jg short 100AFDA2h
0x100afd98: cmp edx,esi
0x100afd9a: jg short 100AFDA2h
0x100afd9c: cmp esi,ecx
0x100afd9e: mov eax,edi
0x100afda0: jle short 100AFDA6h
0x100afda2: mov eax,[esp+50h]
0x100afda6: mov esi,[esp+4Ch]
0x100afdaa: sub eax,ebx
0x100afdac: cdq
0x100afdad: idiv dword ptr [esi+8]
0x100afdb0: mov ecx,eax
0x100afdb2: mov eax,[esp+58h]
0x100afdb6: sub eax,[esi+4]
0x100afdb9: mov edi,[esp+4Ch]
0x100afdbd: cdq
0x100afdbe: idiv dword ptr [esi+0Ch]
0x100afdc1: mov edx,[edi+10h]
0x100afdc4: push 0
0x100afdc6: imul edx,eax
0x100afdc9: mov eax,[edi+20h]
0x100afdcc: add edx,ecx
0x100afdce: imul edx,[edi+24h]
0x100afdd2: push edx
0x100afdd3: push eax
0x100afdd4: call 100AE030h
0x100afdd9: add esp,0Ch
0x100afddc: test eax,eax
```

```
0x100afdde: jl near ptr 100AFE79h
0x100afde4: mov edx,[edi+18h]
0x100afde7: mov eax,[edi+1Ch]
0x100afdea: mov esi,[edi+20h]
0x100afded: lea ecx,[esp+60h]
0x100afdf1: push ecx
0x100afdf2: push edx
0x100afdf3: push eax
0x100afdf4: call 100AEF60h
0x100afdf9: add esp,0Ch
0x100afdfc: test eax,eax
0x100afdfe: jne short 100AFE79h
0x100afe00: mov ebx,[esp+6Ch]
0x100afe04: mov edx,[ebx+18h]
0x100afe07: mov esi,[edi+18h]
0x100afe0a: cmp edx,esi
0x100afe0c: jne short 100AFE16h
0x100afe0e: mov ecx,[ebx+1Ch]
0x100afe11: cmp ecx,[edi+1Ch]
0x100afe14: je short 100AFE23h
0x100afe16: mov eax,[esp+60h]
0x100afe1a: call 100AF0C0h
0x100afe1f: mov [esp+60h],eax
0x100afe23: mov eax,[esp+60h]
0x100afe27: mov ecx,[ebx+1Ch]
0x100afe2a: mov esi,[ebx+20h]
0x100afe2d: push eax
0x100afe2e: push ecx
0x100afe2f: mov eax,edx
0x100afe31: call 100AF000h
0x100afe36: add esp,8
0x100afe39: test eax,eax
0x100afe3b: jne short 100AFE79h
0x100afe3d: mov eax,[esp+64h]
0x100afe41: mov ebp,[esp+4Ch]
0x100afe45: add eax,1
0x100afe48: cmp eax,[esp+10h]
0x100afe4c: mov [esp+64h],eax
0x100afe50: mov edi,ebx
0x100afe52: jl near ptr 100AFC95h
0x100afe58: mov ecx,[esp+68h]
0x100afe5c: mov ebx,[esp+24h]
0x100afe60: add ecx,1
0x100afe63: cmp ecx,ebx
0x100afe65: mov [esp+68h],ecx
0x100afe69: jl near ptr 100AFC75h
0x100afe6f: pop edi
0x100afe70: pop esi
0x100afe71: pop ebp
0x100afe72: xor eax,eax
0x100afe74: pop ebx
0x100afe75: add esp,38h
0x100afe78: ret
0x100afe79: pop edi
0x100afe7a: pop esi
0x100afe7b: pop ebp
0x100afe7c: or eax,0FFFFFFFFh
0x100afe7f: pop ebx
0x100afe80: add esp,38h
0x100afe83: ret
```

`sub_100afb80` is called by sub_100b0040. It delegates to sub_100ae030, sub_100ae170, sub_100aeea0 (9 callees).

12.1173 sub_100affb0 at 0x100affb0 (54 instructions)

```
0x100affb0: push ebx
0x100affb1: push ebp
0x100affb2: push esi
0x100affb3: push edi
0x100affb4: call 100AE770h
0x100affb9: mov ebp,[esp+14h]
0x100affbd: mov edi,[ebp+10h]
0x100affc0: mov esi,eax
0x100affc2: call 100AEC00h
0x100affc7: test eax,eax
0x100affc9: jne short 100B0016h
0x100affcb: xor ebx,ebx
0x100affcd: cmp [ebp+10h],ebx
0x100affd0: jle short 100AFFF8h
0x100affd2: mov eax,[ebp+18h]
0x100affd5: mov edi,[eax+ebx*4]
0x100affd8: call 100AE7E0h
0x100affdd: mov ecx,[esi+18h]
0x100affe0: mov [ecx+ebx*4],eax
0x100affe3: mov edx,[esi+18h]
0x100affe6: cmp dword ptr [edx+ebx*4],0
0x100affea: je short 100B0016h
0x100affec: add dword ptr [esi+10h],1
0x100afff0: add ebx,1
0x100afff3: cmp ebx,[ebp+10h]
0x100afff6: jl short 100AFFD2h
0x100afff8: mov eax,esi
0x100afffa: call 100AEB40h
0x100affff: mov ebp,[ebp+20h]
0x100b0002: test ebp,ebp
0x100b0004: je short 100B002Ah
0x100b0006: push ebp
0x100b0007: call 100B1AC0h
0x100b000c: add esp,4
0x100b000f: test eax,eax
0x100b0011: mov [esi+20h],eax
0x100b0014: jne short 100B002Ah
0x100b0016: test esi,esi
0x100b0018: je short 100B0023h
0x100b001a: push esi
0x100b001b: call 100AF0F0h
0x100b0020: add esp,4
0x100b0023: pop edi
0x100b0024: pop esi
0x100b0025: pop ebp
0x100b0026: xor eax,eax
0x100b0028: pop ebx
0x100b0029: ret
0x100b002a: pop edi
0x100b002b: mov eax,esi
0x100b002d: pop esi
0x100b002e: pop ebp
0x100b002f: pop ebx
0x100b0030: ret
```

`sub_100affb0` is called by sub_100b0040. It delegates to sub_100ae770, sub_100ae7e0, sub_100aeb40 (6 callees).

12.1174 sub_100b0040 at 0x100b0040 (371 instructions)

```
0x100b0040: mov eax,[esp+4]
0x100b0044: sub esp,54h
0x100b0047: push ebx
0x100b0048: push ebp
0x100b0049: push esi
0x100b004a: push edi
```

```
0x100b004b: push eax
0x100b004c: call 100AFFB0h
0x100b0051: mov edi,eax
0x100b0053: add esp,4
0x100b0056: test edi,edi
0x100b0058: je near ptr 100B049Bh
0x100b005e: push edi
0x100b005f: call 100AEE50h
0x100b0064: add esp,4
0x100b0067: test eax,eax
0x100b0069: jne near ptr 100B0111h
0x100b006f: mov ecx,[edi+18h]
0x100b0072: mov eax,[ecx]
0x100b0074: mov edx,[eax+8]
0x100b0077: mov ebx,[edi+10h]
0x100b007a: cmp ebx,1
0x100b007d: mov eax,[eax+0Ch]
0x100b0080: mov [esp+1Ch],edx
0x100b0084: mov [esp+20h],eax
0x100b0088: mov [esp+2Ch],ebx
0x100b008c: jle short 100B00B8h
0x100b008e: add ecx,4
0x100b0091: lea esi,[ebx-1]
0x100b0094: mov eax,[ecx]
0x100b0096: mov edx,[eax+8]
0x100b0099: cmp edx,[esp+1Ch]
0x100b009d: mov eax,[eax+0Ch]
0x100b00a0: jge short 100B00A6h
0x100b00a2: mov [esp+1Ch],edx
0x100b00a6: cmp eax,[esp+20h]
0x100b00aa: jge short 100B00B0h
0x100b00ac: mov [esp+20h],eax
0x100b00b0: add ecx,4
0x100b00b3: sub esi,1
0x100b00b6: jne short 100B0094h
0x100b00b8: xor esi,esi
0x100b00ba: test ebx,ebx
0x100b00bc: jle short 100B0111h
0x100b00be: mov edi,edi
0x100b00c0: mov ecx,[edi+18h]
0x100b00c3: mov eax,[ecx+esi*4]
0x100b00c6: mov edx,[eax+18h]
0x100b00c9: mov ebp,[eax+28h]
0x100b00cc: mov eax,[eax+1Ch]
0x100b00cf: mov ecx,[esp+20h]
0x100b00d3: push edx
0x100b00d4: mov edx,[esp+20h]
0x100b00d8: push eax
0x100b00d9: push ecx
0x100b00da: push edx
0x100b00db: push 0
0x100b00dd: push 0
0x100b00df: lea ebx,[esi+1]
0x100b00e2: push ebx
0x100b00e3: push esi
0x100b00e4: push edi
0x100b00e5: call 100AFB80h
0x100b00ea: add esp,24h
0x100b00ed: test eax,eax
0x100b00ef: jne near ptr 100B049Bh
0x100b00f5: mov eax,[edi+18h]
0x100b00f8: mov ecx,[eax+esi*4+4]
0x100b00fc: push esi
0x100b00fd: push edi
0x100b00fe: mov [ecx+28h],ebp
0x100b0101: call 100AF770h
0x100b0106: mov esi,ebx
```

```
0x100b0108: add esp,8
0x100b010b: cmp esi,[esp+2Ch]
0x100b010f: jl short 100B00C0h
0x100b0111: mov edx,[edi+18h]
0x100b0114: mov eax,[edx]
0x100b0116: mov ecx,[eax+10h]
0x100b0119: mov edx,[eax+14h]
0x100b011c: mov [esp+10h],ecx
0x100b0120: mov ecx,[eax+8]
0x100b0123: mov [esp+24h],edx
0x100b0127: mov edx,[eax+0Ch]
0x100b012a: mov eax,[edi+20h]
0x100b012d: mov [esp+20h],eax
0x100b0131: mov eax,[eax]
0x100b0133: push eax
0x100b0134: mov [esp+30h],ecx
0x100b0138: mov [esp+2Ch],edx
0x100b013c: call 100B0C30h
0x100b0141: mov ecx,[esp+70h]
0x100b0145: mov edx,[ecx]
0x100b0147: push edx
0x100b0148: mov [esp+20h],eax
0x100b014c: call 100B0C30h
0x100b0151: mov esi,eax
0x100b0153: add esp,8
0x100b0156: mov [esp+14h],esi
0x100b015a: call 100AE770h
0x100b015f: mov ebx,eax
0x100b0161: test ebx,ebx
0x100b0163: je near ptr 100B049Bh
0x100b0169: xor eax,eax
0x100b016b: xor ebp,ebp
0x100b016d: cmp esi,eax
0x100b016f: jle short 100B01D5h
0x100b0171: mov ecx,[esp+2Ch]
0x100b0175: mov edx,[esp+28h]
0x100b0179: mov [esp+4Ch],ecx
0x100b017d: mov ecx,[esp+10h]
0x100b0181: mov [esp+50h],edx
0x100b0185: mov edx,[esp+24h]
0x100b0189: mov [esp+44h],eax
0x100b018d: mov [esp+48h],eax
0x100b0191: mov [esp+54h],ecx
0x100b0195: mov [esp+58h],edx
0x100b0199: mov dword ptr [esp+5Ch],8
0x100b01a1: mov [esp+60h],eax
0x100b01a5: lea eax,[esp+44h]
0x100b01a9: push eax
0x100b01aa: push 0FFFFFFFFh
0x100b01ac: push ebx
0x100b01ad: call 100AF7D0h
0x100b01b2: add esp,0Ch
0x100b01b5: test eax,eax
0x100b01b7: jne near ptr 100B049Bh
0x100b01bd: mov edx,[ebx+18h]
0x100b01c0: mov eax,[edx+ebp*4]
0x100b01c3: mov ecx,ebp
0x100b01c5: and ecx,7FFFh
0x100b01cb: add ebp,1
0x100b01ce: cmp ebp,esi
0x100b01d0: mov [eax+28h],ecx
0x100b01d3: jl short 100B01A5h
0x100b01d5: mov ebp,[esp+6Ch]
0x100b01d9: push ebp
0x100b01da: call 100B1AC0h
0x100b01df: add esp,4
0x100b01e2: test eax,eax
```

```
0x100b01e4: je near ptr 100B049Bh
0x100b01ea: mov edx,[esp+70h]
0x100b01ee: push 0
0x100b01f0: push edx
0x100b01f1: push 0
0x100b01f3: mov [ebx+20h],eax
0x100b01f6: mov eax,[esp+2Ch]
0x100b01fa: mov ecx,[ebp]
0x100b01fd: push 0
0x100b01ff: push ebp
0x100b0200: push eax
0x100b0201: mov [ebx+1Ch],ecx
0x100b0204: call 100B1B50h
0x100b0209: add esp,18h
0x100b020c: test eax,eax
0x100b020e: mov [esp+2Ch],eax
0x100b0212: je near ptr 100B049Bh
0x100b0218: mov eax,[esp+18h]
0x100b021c: lea ecx,[eax+eax*4]
0x100b021f: add ecx,ecx
0x100b0221: add ecx,ecx
0x100b0223: push ecx
0x100b0224: mov [esp+40h],eax
0x100b0228: call dword ptr ds:[100FE550h]
0x100b022e: xor ebp,ebp
0x100b0230: add esp,4
0x100b0233: cmp [esp+18h],ebp
0x100b0237: mov [esp+28h],eax
0x100b023b: mov [esp+40h],eax
0x100b023f: jle short 100B02ADh
0x100b0241: lea esi,[eax+8]
0x100b0244: mov edx,ebp
0x100b0246: and edx,7FFFh
0x100b024c: push edx
0x100b024d: push edi
0x100b024e: call 100AEE20h
0x100b0253: mov [esp+24h],eax
0x100b0257: mov eax,[esp+18h]
0x100b025b: add eax,eax
0x100b025d: add eax,eax
0x100b025f: push eax
0x100b0260: call dword ptr ds:[100FE550h]
0x100b0266: add esp,0Ch
0x100b0269: test eax,eax
0x100b026b: mov [esi-8],eax
0x100b026e: je near ptr 100B049Bh
0x100b0274: mov ecx,[edi+18h]
0x100b0277: mov eax,[esp+1Ch]
0x100b027b: mov edx,[ecx+eax*4]
0x100b027e: mov ecx,[edx+18h]
0x100b0281: mov [esi-4],ecx
0x100b0284: mov edx,[edi+18h]
0x100b0287: mov eax,[edx+eax*4]
0x100b028a: mov ecx,[eax+1Ch]
0x100b028d: mov edx,[esp+10h]
0x100b0291: mov [esi],ecx
0x100b0293: mov [esi+4],edx
0x100b0296: mov dword ptr [esi+8],1
0x100b029d: add ebp,1
0x100b02a0: add esi,14h
0x100b02a3: cmp ebp,[esp+18h]
0x100b02a7: jl short 100B0244h
0x100b02a9: mov esi,[esp+14h]
0x100b02ad: lea eax,[esi+esi*4]
0x100b02b0: add eax,eax
0x100b02b2: add eax,eax
0x100b02b4: push eax
```

```
0x100b02b5: mov [esp+38h],esi
0x100b02b9: call dword ptr ds:[100FE550h]
0x100b02bf: add esp,4
0x100b02c2: xor ebp,ebp
0x100b02c4: test esi,esi
0x100b02c6: mov [esp+20h],eax
0x100b02ca: mov [esp+38h],eax
0x100b02ce: jle near ptr 100B034Eh
0x100b02d4: mov ecx,[esp+10h]
0x100b02d8: lea edx,[ecx*4]
0x100b02df: mov [esp+30h],edx
0x100b02e3: lea esi,[eax+8]
0x100b02e6: mov eax,ebp
0x100b02e8: and eax,7FFFh
0x100b02ed: push eax
0x100b02ee: push ebx
0x100b02ef: call 100AEE20h
0x100b02f4: mov ecx,[esp+38h]
0x100b02f8: push ecx
0x100b02f9: mov [esp+28h],eax
0x100b02fd: call dword ptr ds:[100FE550h]
0x100b0303: add esp,0Ch
0x100b0306: test eax,eax
0x100b0308: mov [esi-8],eax
0x100b030b: je near ptr 100B049Bh
0x100b0311: mov edx,[ebx+18h]
0x100b0314: mov eax,[esp+1Ch]
0x100b0318: mov ecx,[edx+eax*4]
0x100b031b: mov edx,[ecx+18h]
0x100b031e: mov [esi-4],edx
0x100b0321: mov ecx,[ebx+18h]
0x100b0324: mov edx,[ecx+eax*4]
0x100b0327: mov eax,[edx+1Ch]
0x100b032a: mov ecx,[esp+10h]
0x100b032e: mov [esi],eax
0x100b0330: mov [esi+4],ecx
0x100b0333: mov dword ptr [esi+8],1
0x100b033a: add ebp,1
0x100b033d: add esi,14h
0x100b0340: cmp ebp,[esp+14h]
0x100b0344: jl short 100B02E6h
0x100b0346: mov esi,[esp+14h]
0x100b034a: mov eax,[esp+20h]
0x100b034e: cmp dword ptr [esp+24h],0
0x100b0353: mov dword ptr [esp+1Ch],0
0x100b035b: jle near ptr 100B042Eh
0x100b0361: xor esi,esi
0x100b0363: cmp [esp+18h],esi
0x100b0367: jle short 100B03AFh
0x100b0369: mov ebp,[esp+28h]
0x100b036d: lea ecx,[ecx]
0x100b0370: mov edx,esi
0x100b0372: and edx,7FFFh
0x100b0378: push edx
0x100b0379: push edi
0x100b037a: call 100AEE20h
0x100b037f: mov ecx,[ebp]
0x100b0382: mov edx,[esp+18h]
0x100b0386: push ecx
0x100b0387: mov ecx,[esp+28h]
0x100b038b: push 1
0x100b038d: push edx
0x100b038e: push ecx
0x100b038f: push 0
0x100b0391: push eax
0x100b0392: push edi
0x100b0393: call 100AF960h
```

```
0x100b0398: add esp,24h
0x100b039b: test eax,eax
0x100b039d: jne near ptr 100B049Bh
0x100b03a3: add esi,1
0x100b03a6: add ebp,14h
0x100b03a9: cmp esi,[esp+18h]
0x100b03ad: jl short 100B0370h
0x100b03af: mov ecx,[esp+2Ch]
0x100b03b3: lea edx,[esp+34h]
0x100b03b7: push edx
0x100b03b8: lea eax,[esp+40h]
0x100b03bc: push eax
0x100b03bd: push ecx
0x100b03be: call 100B0C80h
0x100b03c3: xor esi,esi
0x100b03c5: add esp,0Ch
0x100b03c8: cmp [esp+14h],esi
0x100b03cc: jle short 100B0411h
0x100b03ce: mov ebp,[esp+20h]
0x100b03d2: mov edx,esi
0x100b03d4: and edx,7FFFh
0x100b03da: push edx
0x100b03db: push ebx
0x100b03dc: call 100AEE20h
0x100b03e1: mov ecx,[ebp]
0x100b03e4: mov edx,[esp+18h]
0x100b03e8: push ecx
0x100b03e9: mov ecx,[esp+28h]
0x100b03ed: push 1
0x100b03ef: push edx
0x100b03f0: push ecx
0x100b03f1: push 0
0x100b03f3: push eax
0x100b03f4: push ebx
0x100b03f5: call 100AFA70h
0x100b03fa: add esp,24h
0x100b03fd: test eax,eax
0x100b03ff: jne near ptr 100B049Bh
0x100b0405: add esi,1
0x100b0408: add ebp,14h
0x100b040b: cmp esi,[esp+14h]
0x100b040f: jl short 100B03D2h
0x100b0411: mov eax,[esp+1Ch]
0x100b0415: add eax,1
0x100b0418: cmp eax,[esp+24h]
0x100b041c: mov [esp+1Ch],eax
0x100b0420: jl near ptr 100B0361h
0x100b0426: mov esi,[esp+14h]
0x100b042a: mov eax,[esp+20h]
0x100b042e: test esi,esi
0x100b0430: jle short 100B044Ch
0x100b0432: mov ebp,eax
0x100b0434: mov edx,[ebp]
0x100b0437: push edx
0x100b0438: call 100209E0h
0x100b043d: add esp,4
0x100b0440: add ebp,14h
0x100b0443: sub esi,1
0x100b0446: jne short 100B0434h
0x100b0448: mov eax,[esp+20h]
0x100b044c: push eax
0x100b044d: call 100209E0h
0x100b0452: mov ebp,[esp+1Ch]
0x100b0456: add esp,4
0x100b0459: test ebp,ebp
0x100b045b: jle short 100B0474h
0x100b045d: mov esi,[esp+28h]
```

```
0x100b0461: mov eax,[esi]
0x100b0463: push eax
0x100b0464: call 100209E0h
0x100b0469: add esp,4
0x100b046c: add esi,14h
0x100b046f: sub ebp,1
0x100b0472: jne short 100B0461h
0x100b0474: mov ecx,[esp+28h]
0x100b0478: push ecx
0x100b0479: call 100209E0h
0x100b047e: mov edx,[esp+30h]
0x100b0482: push edx
0x100b0483: call 100B1470h
0x100b0488: push edi
0x100b0489: call 100AF0F0h
0x100b048e: add esp,0Ch
0x100b0491: pop edi
0x100b0492: pop esi
0x100b0493: pop ebp
0x100b0494: mov eax,ebx
0x100b0496: pop ebx
0x100b0497: add esp,54h
0x100b049a: ret
0x100b049b: pop edi
0x100b049c: pop esi
0x100b049d: pop ebp
0x100b049e: xor eax,eax
0x100b04a0: pop ebx
0x100b04a1: add esp,54h
0x100b04a4: ret
```

`sub_100b0040` is called by sub_100156b0. It delegates to j_free, sub_100ae770, sub_100aee20 (16 callees).

12.1175 sub_100b04b0 at 0x100b04b0 (32 instructions)

```
0x100b04b0: push esi
0x100b04b1: push 48h
0x100b04b3: call 100EFDD0h
0x100b04b8: mov esi,eax
0x100b04ba: add esp,4
0x100b04bd: test esi,esi
0x100b04bf: jne short 100B04C3h
0x100b04c1: pop esi
0x100b04c2: ret
0x100b04c3: push 48h
0x100b04c5: push 0
0x100b04c7: push esi
0x100b04c8: call 100228C0h
0x100b04cd: xor eax,eax
0x100b04cf: mov dword ptr [esi+10h],0
0x100b04d6: mov [esi+14h],eax
0x100b04d9: mov [esi+18h],eax
0x100b04dc: mov [esi+1Ch],eax
0x100b04df: mov [esi+20h],eax
0x100b04e2: mov [esi+24h],eax
0x100b04e5: mov [esi+28h],eax
0x100b04e8: mov [esi+2Ch],eax
0x100b04eb: mov [esi+30h],eax
0x100b04ee: mov [esi+34h],eax
0x100b04f1: mov [esi+38h],eax
0x100b04f4: mov [esi+3Ch],eax
0x100b04f7: mov [esi+40h],eax
0x100b04fa: mov [esi+44h],eax
0x100b04fd: add esp,0Ch
0x100b0500: mov eax,esi
```

```
0x100b0502: pop esi
0x100b0503: ret
```

`sub_100b04b0` is called by sub_100b1990, sub_100b1ac0. It delegates to j_memset, j_malloc (2 callees).

12.1176 sub_100b0530 at 0x100b0530 (21 instructions)

```
0x100b0530: mov eax,[edi+8]
0x100b0533: test eax,eax
0x100b0535: jne short 100B0549h
0x100b0537: lea eax,[esi*4]
0x100b053e: push eax
0x100b053f: call 100EFDD0h
0x100b0544: add esp,4
0x100b0547: jmp short 100B055Ah
0x100b0549: lea ecx,[esi*4]
0x100b0550: push ecx
0x100b0551: push eax
0x100b0552: call 100AD4B0h
0x100b0557: add esp,8
0x100b055a: test eax,eax
0x100b055c: jne short 100B0562h
0x100b055e: or eax,0FFFFFFFFh
0x100b0561: ret
0x100b0562: mov [edi+8],eax
0x100b0565: mov [edi+4],esi
0x100b0568: xor eax,eax
0x100b056a: ret
```

`sub_100b0530` is called by sub_100b1030, sub_100b10a0, sub_100b1490. It delegates to j_realloc, j_malloc (2 callees).

12.1177 sub_100b0570 at 0x100b0570 (19 instructions)

```
0x100b0570: push esi
0x100b0571: push 98h
0x100b0576: call 100EFDD0h
0x100b057b: mov esi,eax
0x100b057d: add esp,4
0x100b0580: test esi,esi
0x100b0582: jne short 100B0586h
0x100b0584: pop esi
0x100b0585: ret
0x100b0586: push 98h
0x100b058b: push 0
0x100b058d: push esi
0x100b058e: call 100228C0h
0x100b0593: add esp,0Ch
0x100b0596: mov dword ptr [esi],0
0x100b059c: mov dword ptr [esi+4],0
0x100b05a3: mov eax,esi
0x100b05a5: pop esi
0x100b05a6: ret
```

`sub_100b0570` is called by sub_100b10f0. It delegates to j_memset, j_malloc (2 callees).

12.1178 sub_100b05b0 at 0x100b05b0 (11 instructions)

```
0x100b05b0: add dword ptr [esi],0FFFFFFFFh
0x100b05b3: cmp dword ptr [esi],0
0x100b05b6: jg short 100B05C9h
0x100b05b8: mov eax,[esi+4]
0x100b05bb: mov ecx,[eax]
0x100b05bd: push esi
0x100b05be: call ecx
```

```
0x100b05c0: push esi
0x100b05c1: call 100209E0h
0x100b05c6: add esp,8
0x100b05c9: ret
```

`sub_100b05b0` is called by sub_100b1000, sub_100b14f0, sub_100b1620, sub_100b1e50. It delegates to j_free (1 callees).

12.1179 sub_100b05d0 at 0x100b05d0 (2 instructions)

```
0x100b05d0: add dword ptr [eax],1
0x100b05d3: ret
```

`sub_100b05d0` is called by sub_100b1030, sub_100b10a0. It calls nothing else.

12.1180 sub_100b05e0 at 0x100b05e0 (3 instructions)

```
0x100b05e0: mov dword ptr [eax],0
0x100b05e6: mov dword ptr [eax+4],0
0x100b05ed: ret
```

`sub_100b05e0` is called by sub_100b10f0, sub_100b14f0, sub_100b1620. It calls nothing else.

12.1181 sub_100b05f0 at 0x100b05f0 (9 instructions)

```
0x100b05f0: mov eax,[esi]
0x100b05f2: test eax,eax
0x100b05f4: je short 100B0605h
0x100b05f6: push eax
0x100b05f7: call 100209E0h
0x100b05fc: add esp,4
0x100b05ff: mov dword ptr [esi],0
0x100b0605: mov dword ptr [esi+4],0
0x100b060c: ret
```

`sub_100b05f0` is called by sub_100b0640, sub_100b14f0, sub_100b1620. It delegates to j_free (1 callees).

12.1182 sub_100b0610 at 0x100b0610 (14 instructions)

```
0x100229a0: jmp dword ptr ds:[100FE4F4h]
0x100b0610: fldz
0x100b0612: fld st(0)
0x100b0614: fld qword ptr [esp+4]
0x100b0618: fucom
0x100b061a: fnstsw ax
0x100b061c: fstp st(1)
0x100b061e: test ah,44h
0x100b0621: jp short 100B0626h
0x100b0623: fstp st(0)
0x100b0625: ret
0x100b0626: fstp st(1)
0x100b0628: fld qword ptr [esp+0Ch]
0x100b062c: jmp near ptr 100229A0h
```

`sub_100b0610` is called by sub_100b0640. It calls nothing else.

12.1183 sub_100b0640 at 0x100b0640 (108 instructions)

```
0x100b0640: push ebp
0x100b0641: mov ebp,esp
0x100b0643: and esp,0FFFFFFC0h
0x100b0646: sub esp,38h
```

```
0x100b0649: push esi
0x100b064a: push edi
0x100b064b: mov esi,ecx
0x100b064d: mov edi,eax
0x100b064f: call 100B05F0h
0x100b0654: mov eax,[edi]
0x100b0656: test eax,eax
0x100b0658: jne short 100B0688h
0x100b065a: push 10h
0x100b065c: mov dword ptr [esi+4],2
0x100b0663: call 100EFDD0h
0x100b0668: add esp,4
0x100b066b: test eax,eax
0x100b066d: mov [esi],eax
0x100b066f: je near ptr 100B0732h
0x100b0675: fldz
0x100b0677: fstp qword ptr [eax]
0x100b0679: mov eax,[esi]
0x100b067b: fld1
0x100b067d: fstp qword ptr [eax+8]
0x100b0680: xor eax,eax
0x100b0682: pop edi
0x100b0683: pop esi
0x100b0684: mov esp,ebp
0x100b0686: pop ebp
0x100b0687: ret
0x100b0688: cmp eax,1
0x100b068b: jne near ptr 100B0719h
0x100b0691: push 800h
0x100b0696: mov dword ptr [esi+4],100h
0x100b069d: call 100EFDD0h
0x100b06a2: add esp,4
0x100b06a5: test eax,eax
0x100b06a7: mov [esi],eax
0x100b06a9: je near ptr 100B0732h
0x100b06af: mov ecx,[edi+4]
0x100b06b2: movzx edx,word ptr [ecx]
0x100b06b5: mov [esp+34h],edx
0x100b06b9: xor edi,edi
0x100b06bb: cmp [esi+4],edi
0x100b06be: fild dword ptr [esp+34h]
0x100b06c2: mov [esp+34h],edi
0x100b06c6: fmul qword ptr ds:[10127AA8h]
0x100b06cc: fstp qword ptr [esp+38h]
0x100b06d0: jle near ptr 100B0769h
0x100b06d6: fld qword ptr [esp+38h]
0x100b06da: mov eax,[esi+4]
0x100b06dd: sub esp,10h
0x100b06e0: fstp qword ptr [esp+8]
0x100b06e4: sub eax,1
0x100b06e7: fild dword ptr [esp+44h]
0x100b06eb: mov [esp+44h],eax
0x100b06ef: fild dword ptr [esp+44h]
0x100b06f3: fdivp
0x100b06f5: fstp qword ptr [esp]
0x100b06f8: call 100B0610h
0x100b06fd: mov ecx,[esi]
0x100b06ff: fstp qword ptr [ecx+edi*8]
0x100b0702: add edi,1
0x100b0705: add esp,10h
0x100b0708: cmp edi,[esi+4]
0x100b070b: mov [esp+34h],edi
0x100b070f: jl short 100B06D6h
0x100b0711: xor eax,eax
0x100b0713: pop edi
0x100b0714: pop esi
0x100b0715: mov esp,ebp
```

```
0x100b0717: pop ebp
0x100b0718: ret
0x100b0719: lea edx,[eax*8]
0x100b0720: push edx
0x100b0721: mov [esi+4],eax
0x100b0724: call 100EFDD0h
0x100b0729: add esp,4
0x100b072c: test eax,eax
0x100b072e: mov [esi],eax
0x100b0730: jne short 100B073Bh
0x100b0732: or eax,0FFFFFFFFh
0x100b0735: pop edi
0x100b0736: pop esi
0x100b0737: mov esp,ebp
0x100b0739: pop ebp
0x100b073a: ret
0x100b073b: xor eax,eax
0x100b073d: cmp [esi+4],eax
0x100b0740: jle short 100B0769h
0x100b0742: fld qword ptr ds:[100FFBA0h]
0x100b0748: mov ecx,[edi+4]
0x100b074b: movzx edx,word ptr [ecx+eax*2]
0x100b074f: mov [esp+34h],edx
0x100b0753: mov ecx,[esi]
0x100b0755: add eax,1
0x100b0758: fild dword ptr [esp+34h]
0x100b075c: fdiv st,st(1)
0x100b075e: fstp qword ptr [ecx+eax*8-8]
0x100b0762: cmp eax,[esi+4]
0x100b0765: jl short 100B0748h
0x100b0767: fstp st(0)
0x100b0769: pop edi
0x100b076a: xor eax,eax
0x100b076c: pop esi
0x100b076d: mov esp,ebp
0x100b076f: pop ebp
0x100b0770: ret
```

`sub_100b0640` is called by sub_100b14f0, sub_100b1620. It delegates to sub_100b05f0, sub_100b0610, j_malloc (3 callees).

12.1184 sub_100b0810 at 0x100b0810 (192 instructions)

```
0x100b0810: sub esp,8
0x100b0813: push ebp
0x100b0814: mov ebp,[esp+10h]
0x100b0818: push esi
0x100b0819: push edi
0x100b081a: mov edi,eax
0x100b081c: mov eax,[ebx]
0x100b081e: test eax,eax
0x100b0820: je short 100B0831h
0x100b0822: push eax
0x100b0823: call 100209E0h
0x100b0828: add esp,4
0x100b082b: mov dword ptr [ebx],0
0x100b0831: mov esi,[ebp+4]
0x100b0834: mov edx,1
0x100b0839: cmp esi,edx
0x100b083b: jle short 100B085Dh
0x100b083d: mov ecx,[ebp]
0x100b0840: add ecx,8
0x100b0843: fld qword ptr [ecx]
0x100b0845: fcomp qword ptr [ecx-8]
0x100b0848: fnstsw ax
0x100b084a: test ah,5
```

```
0x100b084d: jnp near ptr 100B0917h
0x100b0853: add edx,1
0x100b0856: add ecx,8
0x100b0859: cmp edx,esi
0x100b085b: jl short 100B0843h
0x100b085d: lea eax,[edi*8]
0x100b0864: push eax
0x100b0865: call 100EFDD0h
0x100b086a: add esp,4
0x100b086d: test eax,eax
0x100b086f: mov [ebx],eax
0x100b0871: je near ptr 100B0917h
0x100b0877: test edi,edi
0x100b0879: mov [ebx+4],edi
0x100b087c: mov dword ptr [esp+0Ch],0
0x100b0884: jle near ptr 100B090Eh
0x100b088a: fld1
0x100b088c: fld qword ptr ds:[100FE7C8h]
0x100b0892: mov ecx,[ebx+4]
0x100b0895: fild dword ptr [esp+0Ch]
0x100b0899: mov edx,[ebp+4]
0x100b089c: sub ecx,1
0x100b089f: mov [esp+18h],ecx
0x100b08a3: fild dword ptr [esp+18h]
0x100b08a7: xor ecx,ecx
0x100b08a9: test edx,edx
0x100b08ab: fdivp
0x100b08ad: fld st(2)
0x100b08af: jle short 100B08F3h
0x100b08b1: mov edi,[ebp]
0x100b08b4: mov esi,edi
0x100b08b6: fld qword ptr [esi]
0x100b08b8: fld st(0)
0x100b08ba: fucomp st(3)
0x100b08bc: fnstsw ax
0x100b08be: test ah,44h
0x100b08c1: jnp short 100B0925h
0x100b08c3: lea eax,[edx-1]
0x100b08c6: cmp ecx,eax
0x100b08c8: jge short 100B0921h
0x100b08ca: fld qword ptr [esi+8]
0x100b08cd: fxch st(3)
0x100b08cf: fcom
0x100b08d1: fnstsw ax
0x100b08d3: test ah,41h
0x100b08d6: jne short 100B08E5h
0x100b08d8: fcom st(3)
0x100b08da: fnstsw ax
0x100b08dc: test ah,5
0x100b08df: jnp near ptr 100B09D5h
0x100b08e5: fstp st(1)
0x100b08e7: fstp st(2)
0x100b08e9: add ecx,1
0x100b08ec: add esi,8
0x100b08ef: cmp ecx,edx
0x100b08f1: jl short 100B08B6h
0x100b08f3: fstp st(1)
0x100b08f5: mov eax,[esp+0Ch]
0x100b08f9: mov ecx,[ebx]
0x100b08fb: fstp qword ptr [ecx+eax*8]
0x100b08fe: add eax,1
0x100b0901: cmp eax,[ebx+4]
0x100b0904: mov [esp+0Ch],eax
0x100b0908: jl short 100B0892h
0x100b090a: fstp st(1)
0x100b090c: fstp st(0)
0x100b090e: pop edi
```

```
0x100b090f: pop esi
0x100b0910: xor eax,eax
0x100b0912: pop ebp
0x100b0913: add esp,8
0x100b0916: ret
0x100b0917: pop edi
0x100b0918: pop esi
0x100b0919: or eax,0FFFFFFFFh
0x100b091c: pop ebp
0x100b091d: add esp,8
0x100b0920: ret
0x100b0921: fstp st(0)
0x100b0923: jmp short 100B08E9h
0x100b0925: mov [esp+18h],ecx
0x100b0929: fstp st(0)
0x100b092b: add ecx,1
0x100b092e: mov eax,edx
0x100b0930: sub eax,ecx
0x100b0932: cmp eax,4
0x100b0935: jl short 100B0977h
0x100b0937: lea esi,[edi+ecx*8+10h]
0x100b093b: fld st(1)
0x100b093d: fcomp qword ptr [esi-10h]
0x100b0940: fnstsw ax
0x100b0942: test ah,44h
0x100b0945: jp short 100B09A2h
0x100b0947: fld st(1)
0x100b0949: fcomp qword ptr [esi-8]
0x100b094c: fnstsw ax
0x100b094e: test ah,44h
0x100b0951: jp short 100B0995h
0x100b0953: fld st(1)
0x100b0955: fcomp qword ptr [esi]
0x100b0957: fnstsw ax
0x100b0959: test ah,44h
0x100b095c: jp short 100B099Ah
0x100b095e: fld st(1)
0x100b0960: fcomp qword ptr [esi+8]
0x100b0963: fnstsw ax
0x100b0965: test ah,44h
0x100b0968: jp short 100B099Fh
0x100b096a: add ecx,4
0x100b096d: lea eax,[edx-3]
0x100b0970: add esi,20h
0x100b0973: cmp ecx,eax
0x100b0975: jl short 100B093Bh
0x100b0977: cmp ecx,edx
0x100b0979: jge short 100B09A2h
0x100b097b: lea esi,[edi+ecx*8]
0x100b097e: fld st(1)
0x100b0980: fcomp qword ptr [esi]
0x100b0982: fnstsw ax
0x100b0984: test ah,44h
0x100b0987: jp short 100B09A2h
0x100b0989: add ecx,1
0x100b098c: add esi,8
0x100b098f: cmp ecx,edx
0x100b0991: jl short 100B097Eh
0x100b0993: jmp short 100B09A2h
0x100b0995: add ecx,1
0x100b0998: jmp short 100B09A2h
0x100b099a: add ecx,2
0x100b099d: jmp short 100B09A2h
0x100b099f: add ecx,3
0x100b09a2: cmp ecx,edx
0x100b09a4: fstp st(1)
0x100b09a6: jge near ptr 100B08F5h
```

```
0x100b09ac: fstp st(0)
0x100b09ae: sub edx,1
0x100b09b1: mov [esp+10h],edx
0x100b09b5: fild dword ptr [esp+10h]
0x100b09b9: add ecx,0FFFFFFFFh
0x100b09bc: mov [esp+10h],ecx
0x100b09c0: fild dword ptr [esp+10h]
0x100b09c4: fdiv st,st(1)
0x100b09c6: fild dword ptr [esp+18h]
0x100b09ca: fdivrp st(2),st
0x100b09cc: faddp
0x100b09ce: fmul st,st(1)
0x100b09d0: jmp near ptr 100B08F5h
0x100b09d5: fstp st(2)
0x100b09d7: mov [esp+18h],ecx
0x100b09db: fild dword ptr [esp+18h]
0x100b09df: mov eax,edx
0x100b09e1: sub eax,1
0x100b09e4: mov [esp+18h],eax
0x100b09e8: fild dword ptr [esp+18h]
0x100b09ec: fld st(1)
0x100b09ee: fdiv st,st(1)
0x100b09f0: fxch st(4)
0x100b09f2: fsub st,st(3)
0x100b09f4: fxch st(5)
0x100b09f6: fsubrp st(3),st
0x100b09f8: fxch st(4)
0x100b09fa: fdivrp st(2),st
0x100b09fc: fadd st,st(5)
0x100b09fe: fdivrp st(3),st
0x100b0a00: fxch st(2)
0x100b0a02: fsub st,st(1)
0x100b0a04: fmulp st(2),st
0x100b0a06: faddp
0x100b0a08: jmp near ptr 100B08F5h
```

`sub_100b0810` is called by sub_100b14f0, sub_100b1620. It delegates to j_free, j_malloc (2 callees).

12.1185 sub_100b0a10 at 0x100b0a10 (112 instructions)

```
0x100b0a10: fld qword ptr [ecx+28h]
0x100b0a13: fmul qword ptr [ecx+50h]
0x100b0a16: fld qword ptr [ecx+48h]
0x100b0a19: fmul qword ptr [ecx+30h]
0x100b0a1c: fsubp
0x100b0a1e: fmul qword ptr [ecx]
0x100b0a20: fld qword ptr [ecx+20h]
0x100b0a23: fmul qword ptr [ecx+50h]
0x100b0a26: fld qword ptr [ecx+40h]
0x100b0a29: fmul qword ptr [ecx+30h]
0x100b0a2c: fsubp
0x100b0a2e: fmul qword ptr [ecx+8]
0x100b0a31: fsubp
0x100b0a33: fld qword ptr [ecx+48h]
0x100b0a36: fmul qword ptr [ecx+20h]
0x100b0a39: fld qword ptr [ecx+40h]
0x100b0a3c: fmul qword ptr [ecx+28h]
0x100b0a3f: fsubp
0x100b0a41: fmul qword ptr [ecx+10h]
0x100b0a44: faddp
0x100b0a46: fldz
0x100b0a48: fcomp
0x100b0a4a: fnstsw ax
0x100b0a4c: fld st(0)
0x100b0a4e: test ah,41h
0x100b0a51: jnp short 100B0A55h
```

```
0x100b0a53: fchs
0x100b0a55: fcomp qword ptr ds:[10127AB0h]
0x100b0a5b: fnstsw ax
0x100b0a5d: test ah,5
0x100b0a60: jp short 100B0A68h
0x100b0a62: fstp st(0)
0x100b0a64: or eax,0FFFFFFFFh
0x100b0a67: ret
0x100b0a68: fld qword ptr [ecx+28h]
0x100b0a6b: xor eax,eax
0x100b0a6d: fmul qword ptr [ecx+50h]
0x100b0a70: fld qword ptr [ecx+48h]
0x100b0a73: fmul qword ptr [ecx+30h]
0x100b0a76: fsubp
0x100b0a78: fdiv st,st(1)
0x100b0a7a: fstp qword ptr [edx]
0x100b0a7c: fld qword ptr [ecx+20h]
0x100b0a7f: fmul qword ptr [ecx+50h]
0x100b0a82: fld qword ptr [ecx+40h]
0x100b0a85: fmul qword ptr [ecx+30h]
0x100b0a88: fsubp
0x100b0a8a: fdiv st,st(1)
0x100b0a8c: fchs
0x100b0a8e: fstp qword ptr [edx+20h]
0x100b0a91: fld qword ptr [ecx+48h]
0x100b0a94: fmul qword ptr [ecx+20h]
0x100b0a97: fld qword ptr [ecx+40h]
0x100b0a9a: fmul qword ptr [ecx+28h]
0x100b0a9d: fsubp
0x100b0a9f: fdiv st,st(1)
0x100b0aa1: fstp qword ptr [edx+40h]
0x100b0aa4: fld qword ptr [ecx+8]
0x100b0aa7: fmul qword ptr [ecx+50h]
0x100b0aaa: fld qword ptr [ecx+48h]
0x100b0aad: fmul qword ptr [ecx+10h]
0x100b0ab0: fsubp
0x100b0ab2: fdiv st,st(1)
0x100b0ab4: fchs
0x100b0ab6: fstp qword ptr [edx+8]
0x100b0ab9: fld qword ptr [ecx]
0x100b0abb: fmul qword ptr [ecx+50h]
0x100b0abe: fld qword ptr [ecx+40h]
0x100b0ac1: fmul qword ptr [ecx+10h]
0x100b0ac4: fsubp
0x100b0ac6: fdiv st,st(1)
0x100b0ac8: fstp qword ptr [edx+28h]
0x100b0acb: fld qword ptr [ecx+48h]
0x100b0ace: fmul qword ptr [ecx]
0x100b0ad0: fld qword ptr [ecx+40h]
0x100b0ad3: fmul qword ptr [ecx+8]
0x100b0ad6: fsubp
0x100b0ad8: fdiv st,st(1)
0x100b0ada: fchs
0x100b0adc: fstp qword ptr [edx+48h]
0x100b0adf: fld qword ptr [ecx+8]
0x100b0ae2: fmul qword ptr [ecx+30h]
0x100b0ae5: fld qword ptr [ecx+10h]
0x100b0ae8: fmul qword ptr [ecx+28h]
0x100b0aeb: fsubp
0x100b0aed: fdiv st,st(1)
0x100b0aef: fstp qword ptr [edx+10h]
0x100b0af2: fld qword ptr [ecx]
0x100b0af4: fmul qword ptr [ecx+30h]
0x100b0af7: fld qword ptr [ecx+20h]
0x100b0afa: fmul qword ptr [ecx+10h]
0x100b0afd: fsubp
0x100b0aff: fdiv st,st(1)
```

```
0x100b0b01: fchs
0x100b0b03: fstp qword ptr [edx+30h]
0x100b0b06: fld qword ptr [ecx+28h]
0x100b0b09: fmul qword ptr [ecx]
0x100b0b0b: fld qword ptr [ecx+20h]
0x100b0b0e: fmul qword ptr [ecx+8]
0x100b0b11: fsubp
0x100b0b13: fdivrp
0x100b0b15: fstp qword ptr [edx+50h]
0x100b0b18: fld qword ptr [ecx+18h]
0x100b0b1b: fchs
0x100b0b1d: fstp qword ptr [edx+18h]
0x100b0b20: fld qword ptr [ecx+38h]
0x100b0b23: fchs
0x100b0b25: fstp qword ptr [edx+38h]
0x100b0b28: fld qword ptr [ecx+58h]
0x100b0b2b: fchs
0x100b0b2d: fstp qword ptr [edx+58h]
0x100b0b30: ret
```

`sub_100b0a10` is called by sub_100b1620, sub_100b1e50. It calls nothing else.

12.1186 sub_100b0b40 at 0x100b0b40 (25 instructions)

```
0x100b0b40: cmp dword ptr [esp+4],0
0x100b0b45: je short 100B0B61h
0x100b0b47: cmp eax,4C616220h
0x100b0b4c: je short 100B0B5Bh
0x100b0b4e: cmp eax,58595A20h
0x100b0b53: jne short 100B0B76h
0x100b0b55: mov eax,101h
0x100b0b5a: ret
0x100b0b5b: mov eax,201h
0x100b0b60: ret
0x100b0b61: cmp eax,47524159h
0x100b0b66: je short 100B0B8Bh
0x100b0b68: cmp eax,52474220h
0x100b0b6d: je short 100B0B85h
0x100b0b6f: cmp eax,59436272h
0x100b0b74: je short 100B0B7Fh
0x100b0b76: call dword ptr ds:[100FE474h]
0x100b0b7c: xor eax,eax
0x100b0b7e: ret
0x100b0b7f: mov eax,500h
0x100b0b84: ret
0x100b0b85: mov eax,400h
0x100b0b8a: ret
0x100b0b8b: mov eax,300h
0x100b0b90: ret
```

`sub_100b0b40` is called by sub_100b1990. It accesses memory-mapped peripheral registers at 0x47524159. It calls nothing else.

12.1187 sub_100b0ba0 at 0x100b0ba0 (30 instructions)

```
0x100b0ba0: cmp dword ptr [esp+4],0
0x100b0ba5: push esi
0x100b0ba6: mov esi,[edi]
0x100b0ba8: mov eax,[esi]
0x100b0baa: je short 100B0BC7h
0x100b0bac: add ecx,0FFFFFFFFh
0x100b0baf: mov edx,1
0x100b0bb4: shl edx,cl
0x100b0bb6: mov ecx,edx
0x100b0bb8: neg ecx
```

```
0x100b0bba: cmp eax,ecx
0x100b0bbc: jl short 100B0BC2h
0x100b0bbe: cmp eax,edx
0x100b0bc0: jl short 100B0BD6h
0x100b0bc2: or eax,0FFFFFFFFh
0x100b0bc5: pop esi
0x100b0bc6: ret
0x100b0bc7: test eax,eax
0x100b0bc9: jl short 100B0BC2h
0x100b0bcb: mov edx,1
0x100b0bd0: shl edx,cl
0x100b0bd2: cmp eax,edx
0x100b0bd4: jge short 100B0BC2h
0x100b0bd6: mov ecx,[esp+0Ch]
0x100b0bda: add esi,4
0x100b0bdd: mov [edi],esi
0x100b0bdf: mov [ecx],eax
0x100b0be1: xor eax,eax
0x100b0be3: pop esi
0x100b0be4: ret
```

`sub_100b0ba0` is called by sub_100b0c80. It calls nothing else.

12.1188 sub_100b0bf0 at 0x100b0bf0 (24 instructions)

```
0x100b0bf0: cmp dword ptr [esp+4],0
0x100b0bf5: je short 100B0C11h
0x100b0bf7: add ecx,0FFFFFFFFh
0x100b0bfa: mov eax,1
0x100b0bff: shl eax,cl
0x100b0c01: mov ecx,eax
0x100b0c03: neg ecx
0x100b0c05: cmp edx,ecx
0x100b0c07: jl short 100B0C0Dh
0x100b0c09: cmp edx,eax
0x100b0c0b: jl short 100B0C20h
0x100b0c0d: or eax,0FFFFFFFFh
0x100b0c10: ret
0x100b0c11: test edx,edx
0x100b0c13: jl short 100B0C0Dh
0x100b0c15: mov eax,1
0x100b0c1a: shl eax,cl
0x100b0c1c: cmp edx,eax
0x100b0c1e: jge short 100B0C0Dh
0x100b0c20: mov ecx,[esi]
0x100b0c22: mov [ecx],edx
0x100b0c24: add dword ptr [esi],4
0x100b0c27: xor eax,eax
0x100b0c29: ret
```

`sub_100b0bf0` is called by sub_100b0c80. It calls nothing else.

12.1189 sub_100b0c30 at 0x100b0c30 (9 instructions)

```
0x100b0c30: mov eax,[esp+4]
0x100b0c34: sar eax,8
0x100b0c37: add eax,0FFFFFFFFh
0x100b0c3a: cmp eax,4
0x100b0c3d: ja short 100B0C52h
0x100b0c3f: jmp dword ptr [eax*4+100B0C5Ch]
0x100b0c52: call dword ptr ds:[100FE474h]
0x100b0c58: xor eax,eax
0x100b0c5a: ret
```

`sub_100b0c30` is called by sub_100b0040, sub_100b1990. It calls nothing else.

12.1190 sub_100b0c80 at 0x100b0c80 (286 instructions)

```
0x100b0c80: push ebp
0x100b0c81: mov ebp,esp
0x100b0c83: and esp,0FFFFFFF8h
0x100b0c86: mov eax,803Ch
0x100b0c8b: call 100229B0h
0x100b0c90: mov ecx,[ebp+8]
0x100b0c93: mov edx,[ecx]
0x100b0c95: mov eax,[ebp+0Ch]
0x100b0c98: cmp edx,[eax]
0x100b0c9a: push ebx
0x100b0c9b: push esi
0x100b0c9c: push edi
0x100b0c9d: jg near ptr 100B0FE7h
0x100b0ca3: mov esi,[ecx+4]
0x100b0ca6: mov ecx,[ebp+10h]
0x100b0ca9: cmp esi,[ecx]
0x100b0cab: jg near ptr 100B0FE7h
0x100b0cb1: mov eax,[eax+4]
0x100b0cb4: mov ebx,[eax+0Ch]
0x100b0cb7: mov edi,[eax+10h]
0x100b0cba: mov ecx,1
0x100b0cbf: cmp edx,ecx
0x100b0cc1: mov [esp+2Ch],ebx
0x100b0cc5: jle short 100B0CEBh
0x100b0cc7: add eax,24h
0x100b0cca: lea ebx,[ebx]
0x100b0cd0: cmp [eax-4],ebx
0x100b0cd3: jne near ptr 100B0FE7h
0x100b0cd9: cmp [eax],edi
0x100b0cdb: jne near ptr 100B0FE7h
0x100b0ce1: add ecx,1
0x100b0ce4: add eax,14h
0x100b0ce7: cmp ecx,edx
0x100b0ce9: jl short 100B0CD0h
0x100b0ceb: xor ecx,ecx
0x100b0ced: test esi,esi
0x100b0cef: jle short 100B0D1Bh
0x100b0cf1: mov edx,[ebp+10h]
0x100b0cf4: mov eax,[edx+4]
0x100b0cf7: add eax,10h
0x100b0cfa: lea ebx,[ebx]
0x100b0d00: cmp [eax-4],ebx
0x100b0d03: jne near ptr 100B0FE7h
0x100b0d09: cmp [eax],edi
0x100b0d0b: jne near ptr 100B0FE7h
0x100b0d11: add ecx,1
0x100b0d14: add eax,14h
0x100b0d17: cmp ecx,esi
0x100b0d19: jl short 100B0D00h
0x100b0d1b: mov eax,[ebp+8]
0x100b0d1e: mov edx,[eax+8]
0x100b0d21: mov eax,[edx]
0x100b0d23: xor ecx,ecx
0x100b0d25: test eax,eax
0x100b0d27: mov [esp+38h],edx
0x100b0d2b: jle short 100B0D52h
0x100b0d2d: mov esi,[edx+8]
0x100b0d30: mov ebx,eax
0x100b0d32: mov eax,[esi]
0x100b0d34: mov edx,[eax+8]
0x100b0d37: cmp edx,ecx
0x100b0d39: jle short 100B0D3Dh
0x100b0d3b: mov ecx,edx
0x100b0d3d: mov eax,[eax+0Ch]
0x100b0d40: cmp eax,ecx
0x100b0d42: jle short 100B0D46h
```

```
0x100b0d44: mov ecx,eax
0x100b0d46: add esi,4
0x100b0d49: sub ebx,1
0x100b0d4c: jne short 100B0D32h
0x100b0d4e: mov ebx,[esp+2Ch]
0x100b0d52: mov eax,800h
0x100b0d57: imul edi,ebx
0x100b0d5a: cdq
0x100b0d5b: idiv ecx
0x100b0d5d: xor edx,edx
0x100b0d5f: mov ecx,eax
0x100b0d61: mov eax,edi
0x100b0d63: xor edi,edi
0x100b0d65: cmp eax,edx
0x100b0d67: mov [esp+44h],ecx
0x100b0d6b: mov [esp+40h],eax
0x100b0d6f: mov [esp+30h],edi
0x100b0d73: jle near ptr 100B0FDAh
0x100b0d79: jmp short 100B0D84h
0x100b0d80: mov ecx,[esp+44h]
0x100b0d84: sub eax,edi
0x100b0d86: cmp eax,ecx
0x100b0d88: mov [esp+10h],eax
0x100b0d8c: jl short 100B0D92h
0x100b0d8e: mov [esp+10h],ecx
0x100b0d92: mov ecx,[ebp+8]
0x100b0d95: cmp [ecx],edx
0x100b0d97: mov [esp+24h],edx
0x100b0d9b: jle near ptr 100B0E7Ch
0x100b0da1: lea esi,[esp+48h]
0x100b0da5: mov [esp+14h],edx
0x100b0da9: mov [esp+18h],esi
0x100b0dad: jmp short 100B0DB4h
0x100b0db0: mov edi,[esp+30h]
0x100b0db4: mov eax,[ebp+0Ch]
0x100b0db7: mov ebx,[eax+4]
0x100b0dba: add ebx,[esp+14h]
0x100b0dbe: mov eax,1
0x100b0dc3: mov ecx,[ebx+4]
0x100b0dc6: shl eax,cl
0x100b0dc8: sub eax,1
0x100b0dcb: cmp [ebx+8],edx
0x100b0dce: mov [esp+20h],eax
0x100b0dd2: fild dword ptr [esp+20h]
0x100b0dd6: je short 100B0DE8h
0x100b0dd8: add ecx,0FFFFFFFFh
0x100b0ddb: mov eax,1
0x100b0de0: shl eax,cl
0x100b0de2: mov [esp+1Ch],eax
0x100b0de6: jmp short 100B0DECh
0x100b0de8: mov [esp+1Ch],edx
0x100b0dec: mov ecx,[ebx]
0x100b0dee: lea eax,[edi*4]
0x100b0df5: add ecx,eax
0x100b0df7: cmp [esp+10h],edx
0x100b0dfb: mov [esp+20h],ecx
0x100b0dff: mov [esp+28h],edx
0x100b0e03: jle short 100B0E54h
0x100b0e05: mov eax,[ebx+8]
0x100b0e08: mov ecx,[ebx+4]
0x100b0e0b: lea edx,[esp+2Ch]
0x100b0e0f: push edx
0x100b0e10: push eax
0x100b0e11: lea edi,[esp+28h]
0x100b0e15: call 100B0BA0h
0x100b0e1a: add esp,8
0x100b0e1d: test eax,eax
```

```
0x100b0e1f: jne near ptr 100B0FE5h
0x100b0e25: mov ecx,[esp+2Ch]
0x100b0e29: sub ecx,[esp+1Ch]
0x100b0e2d: mov edx,[ebp+8]
0x100b0e30: mov [esp+34h],ecx
0x100b0e34: fild dword ptr [esp+34h]
0x100b0e38: fdiv st,st(1)
0x100b0e3a: fstp qword ptr [esi]
0x100b0e3c: mov eax,[edx]
0x100b0e3e: lea esi,[esi+eax*8]
0x100b0e41: mov eax,[esp+28h]
0x100b0e45: add eax,1
0x100b0e48: cmp eax,[esp+10h]
0x100b0e4c: mov [esp+28h],eax
0x100b0e50: jl short 100B0E05h
0x100b0e52: xor edx,edx
0x100b0e54: mov eax,[esp+24h]
0x100b0e58: fstp st(0)
0x100b0e5a: mov esi,[esp+18h]
0x100b0e5e: mov ecx,[ebp+8]
0x100b0e61: add dword ptr [esp+14h],14h
0x100b0e66: add eax,1
0x100b0e69: add esi,8
0x100b0e6c: cmp eax,[ecx]
0x100b0e6e: mov [esp+24h],eax
0x100b0e72: mov [esp+18h],esi
0x100b0e76: jl near ptr 100B0DB0h
0x100b0e7c: mov edi,[esp+38h]
0x100b0e80: lea ecx,[esp+48h]
0x100b0e84: xor ebx,ebx
0x100b0e86: cmp [edi],edx
0x100b0e88: mov esi,ecx
0x100b0e8a: jle short 100B0ED8h
0x100b0e8c: mov edx,[edi+8]
0x100b0e8f: mov eax,[edx+ebx*4]
0x100b0e92: mov edx,[eax+0Ch]
0x100b0e95: cmp edx,[eax+8]
0x100b0e98: jle short 100B0EB1h
0x100b0e9a: lea edx,[esp+48h]
0x100b0e9e: cmp ecx,edx
0x100b0ea0: jne short 100B0EABh
0x100b0ea2: lea esi,[esp+4048h]
0x100b0ea9: jmp short 100B0EB3h
0x100b0eab: lea esi,[esp+48h]
0x100b0eaf: jmp short 100B0EB3h
0x100b0eb1: mov esi,ecx
0x100b0eb3: push dword ptr [esp+10h]
0x100b0eb7: mov edx,[eax+4]
0x100b0eba: push esi
0x100b0ebb: push ecx
0x100b0ebc: push eax
0x100b0ebd: mov eax,[edx+4]
0x100b0ec0: call eax
0x100b0ec2: add esp,10h
0x100b0ec5: test eax,eax
0x100b0ec7: jne near ptr 100B0FE7h
0x100b0ecd: add ebx,1
0x100b0ed0: cmp ebx,[edi]
0x100b0ed2: mov ecx,esi
0x100b0ed4: jl short 100B0E8Ch
0x100b0ed6: xor edx,edx
0x100b0ed8: mov ecx,[ebp+8]
0x100b0edb: cmp [ecx+4],edx
0x100b0ede: mov [esp+24h],edx
0x100b0ee2: jle near ptr 100B0FC2h
0x100b0ee8: mov eax,[esp+30h]
0x100b0eec: lea ecx,[eax*4]
```

```
0x100b0ef3: mov [esp+3Ch],ecx
0x100b0ef7: mov [esp+14h],edx
0x100b0efb: mov ebx,esi
0x100b0efd: mov [esp+18h],esi
0x100b0f01: mov edx,[ebp+10h]
0x100b0f04: mov edi,[edx+4]
0x100b0f07: add edi,[esp+14h]
0x100b0f0b: mov eax,1
0x100b0f10: mov ecx,[edi+4]
0x100b0f13: shl eax,cl
0x100b0f15: sub eax,1
0x100b0f18: mov [esp+34h],eax
0x100b0f1c: fild dword ptr [esp+34h]
0x100b0f20: xor eax,eax
0x100b0f22: cmp [edi+8],eax
0x100b0f25: je short 100B0F37h
0x100b0f27: add ecx,0FFFFFFFFh
0x100b0f2a: mov edx,1
0x100b0f2f: shl edx,cl
0x100b0f31: mov [esp+1Ch],edx
0x100b0f35: jmp short 100B0F3Bh
0x100b0f37: mov [esp+1Ch],eax
0x100b0f3b: mov ecx,[edi]
0x100b0f3d: add ecx,[esp+3Ch]
0x100b0f41: cmp [esp+10h],eax
0x100b0f45: mov [esp+20h],ecx
0x100b0f49: mov [esp+28h],eax
0x100b0f4d: jle short 100B0F97h
0x100b0f4f: fild dword ptr [esp+1Ch]
0x100b0f53: fld qword ptr [ebx]
0x100b0f55: fmul st,st(2)
0x100b0f57: fadd st,st(1)
0x100b0f59: call 100228D0h
0x100b0f5e: mov edx,[ebp+8]
0x100b0f61: mov ecx,[edx+4]
0x100b0f64: mov edx,[edi+8]
0x100b0f67: lea ebx,[ebx+ecx*8]
0x100b0f6a: mov ecx,[edi+4]
0x100b0f6d: push edx
0x100b0f6e: mov edx,eax
0x100b0f70: lea esi,[esp+24h]
0x100b0f74: mov [esp+30h],eax
0x100b0f78: call 100B0BF0h
0x100b0f7d: add esp,4
0x100b0f80: test eax,eax
0x100b0f82: jne short 100B0FE3h
0x100b0f84: mov eax,[esp+28h]
0x100b0f88: add eax,1
0x100b0f8b: cmp eax,[esp+10h]
0x100b0f8f: mov [esp+28h],eax
0x100b0f93: jl short 100B0F53h
0x100b0f95: fstp st(0)
0x100b0f97: mov eax,[esp+24h]
0x100b0f9b: fstp st(0)
0x100b0f9d: mov ebx,[esp+18h]
0x100b0fa1: mov ecx,[ebp+8]
0x100b0fa4: add dword ptr [esp+14h],14h
0x100b0fa9: add eax,1
0x100b0fac: add ebx,8
0x100b0faf: cmp eax,[ecx+4]
0x100b0fb2: mov [esp+24h],eax
0x100b0fb6: mov [esp+18h],ebx
0x100b0fba: jl near ptr 100B0F01h
0x100b0fc0: xor edx,edx
0x100b0fc2: mov edi,[esp+30h]
0x100b0fc6: add edi,[esp+10h]
0x100b0fca: mov eax,[esp+40h]
```

```
0x100b0fce: cmp edi,eax
0x100b0fd0: mov [esp+30h],edi
0x100b0fd4: jl near ptr 100B0D80h
0x100b0fda: xor eax,eax
0x100b0fdc: pop edi
0x100b0fdd: pop esi
0x100b0fde: pop ebx
0x100b0fdf: mov esp,ebp
0x100b0fe1: pop ebp
0x100b0fe2: ret
0x100b0fe3: fstp st(0)
0x100b0fe5: fstp st(0)
0x100b0fe7: pop edi
0x100b0fe8: pop esi
0x100b0fe9: or eax,0FFFFFFFFh
0x100b0fec: pop ebx
0x100b0fed: mov esp,ebp
0x100b0fef: pop ebp
0x100b0ff0: ret
```

`sub_100b0c80` is called by sub_100b0040. It delegates to sub_100228d0, sub_100229b0, sub_100b0ba0 (4 callees).

12.1191 sub_100b1000 at 0x100b1000 (15 instructions)

```
0x100b1000: mov eax,[edi]
0x100b1002: sub eax,1
0x100b1005: cmp ebx,eax
0x100b1007: je short 100B100Fh
0x100b1009: call dword ptr ds:[100FE474h]
0x100b100f: mov ecx,[edi+8]
0x100b1012: push esi
0x100b1013: mov esi,[ecx+ebx*4]
0x100b1016: call 100B05B0h
0x100b101b: mov edx,[edi+8]
0x100b101e: mov dword ptr [edx+ebx*4],0
0x100b1025: add dword ptr [edi],0FFFFFFFFh
0x100b1028: xor eax,eax
0x100b102a: pop esi
0x100b102b: ret
```

`sub_100b1000` is called by sub_100b13e0. It delegates to sub_100b05b0 (1 callees).

12.1192 sub_100b1030 at 0x100b1030 (46 instructions)

```
0x100b1030: push ebx
0x100b1031: push esi
0x100b1032: mov ebx,eax
0x100b1034: test ebx,ebx
0x100b1036: push edi
0x100b1037: mov edi,ecx
0x100b1039: jge short 100B103Dh
0x100b103b: mov ebx,[edi]
0x100b103d: mov eax,[edi]
0x100b103f: cmp eax,[edi+4]
0x100b1042: jl short 100B1050h
0x100b1044: lea esi,[eax+10h]
0x100b1047: call 100B0530h
0x100b104c: test eax,eax
0x100b104e: jne short 100B105Fh
0x100b1050: mov eax,[esp+10h]
0x100b1054: call 100B05D0h
0x100b1059: mov esi,eax
0x100b105b: test esi,esi
0x100b105d: jne short 100B1066h
0x100b105f: pop edi
```

```
0x100b1060: pop esi
0x100b1061: or eax,0FFFFFFFFh
0x100b1064: pop ebx
0x100b1065: ret
0x100b1066: mov eax,[edi]
0x100b1068: sub eax,ebx
0x100b106a: test eax,eax
0x100b106c: jle short 100B108Ah
0x100b106e: mov ecx,[edi+8]
0x100b1071: lea edx,[eax*4]
0x100b1078: lea ecx,[ecx+ebx*4]
0x100b107b: push edx
0x100b107c: push ecx
0x100b107d: add ecx,4
0x100b1080: push ecx
0x100b1081: call dword ptr ds:[100FE498h]
0x100b1087: add esp,0Ch
0x100b108a: mov eax,[edi+8]
0x100b108d: mov [eax+ebx*4],esi
0x100b1090: add dword ptr [edi],1
0x100b1093: pop edi
0x100b1094: pop esi
0x100b1095: xor eax,eax
0x100b1097: pop ebx
0x100b1098: ret
```

`sub_100b1030` is called by sub_100b14f0, sub_100b1620, sub_100b1e50. It delegates to sub_100b0530, sub_100b05d0 (2 callees).

12.1193 sub_100b10a0 at 0x100b10a0 (34 instructions)

```
0x100b10a0: push esi
0x100b10a1: push edi
0x100b10a2: mov edi,eax
0x100b10a4: mov esi,[edi]
0x100b10a6: add esi,[ebx]
0x100b10a8: cmp esi,[edi+4]
0x100b10ab: jle short 100B10B6h
0x100b10ad: call 100B0530h
0x100b10b2: test eax,eax
0x100b10b4: jne short 100B10E6h
0x100b10b6: xor ecx,ecx
0x100b10b8: cmp [ebx],ecx
0x100b10ba: jle short 100B10E1h
0x100b10bc: lea esp,[esp]
0x100b10c0: mov eax,[ebx+8]
0x100b10c3: mov eax,[eax+ecx*4]
0x100b10c6: call 100B05D0h
0x100b10cb: test eax,eax
0x100b10cd: je short 100B10E6h
0x100b10cf: mov edx,[edi]
0x100b10d1: mov esi,[edi+8]
0x100b10d4: mov [esi+edx*4],eax
0x100b10d7: add dword ptr [edi],1
0x100b10da: add ecx,1
0x100b10dd: cmp ecx,[ebx]
0x100b10df: jl short 100B10C0h
0x100b10e1: pop edi
0x100b10e2: xor eax,eax
0x100b10e4: pop esi
0x100b10e5: ret
0x100b10e6: pop edi
0x100b10e7: or eax,0FFFFFFFFh
0x100b10ea: pop esi
0x100b10eb: ret
```

`sub_100b10a0` is called by sub_100b14d0. It delegates to sub_100b0530, sub_100b05d0 (2 callees).

12.1194 sub_100b10f0 at 0x100b10f0 (50 instructions)

```
0x100b10f0: push esi
0x100b10f1: call 100B0570h
0x100b10f6: mov ecx,eax
0x100b10f8: xor esi,esi
0x100b10fa: cmp ecx,esi
0x100b10fc: jne short 100B1102h
0x100b10fe: xor eax,eax
0x100b1100: pop esi
0x100b1101: ret
0x100b1102: mov dword ptr [ecx+4],10136928h
0x100b1109: mov [ecx+10h],esi
0x100b110c: mov [ecx+14h],esi
0x100b110f: mov [ecx+18h],esi
0x100b1112: mov [ecx+1Ch],esi
0x100b1115: lea eax,[ecx+20h]
0x100b1118: mov edx,3
0x100b111d: lea ecx,[ecx]
0x100b1120: call 100B05E0h
0x100b1125: add eax,8
0x100b1128: sub edx,1
0x100b112b: jne short 100B1120h
0x100b112d: xor eax,eax
0x100b112f: mov [ecx+38h],eax
0x100b1132: mov [ecx+3Ch],eax
0x100b1135: mov [ecx+40h],eax
0x100b1138: mov [ecx+44h],eax
0x100b113b: mov [ecx+48h],eax
0x100b113e: mov [ecx+4Ch],eax
0x100b1141: mov [ecx+50h],eax
0x100b1144: mov [ecx+54h],eax
0x100b1147: mov [ecx+58h],eax
0x100b114a: mov [ecx+5Ch],eax
0x100b114d: mov [ecx+60h],eax
0x100b1150: mov [ecx+64h],eax
0x100b1153: mov [ecx+68h],eax
0x100b1156: mov [ecx+6Ch],eax
0x100b1159: mov [ecx+70h],eax
0x100b115c: mov [ecx+74h],eax
0x100b115f: mov [ecx+78h],eax
0x100b1162: mov [ecx+7Ch],eax
0x100b1165: mov [ecx+80h],eax
0x100b116b: mov [ecx+84h],eax
0x100b1171: mov [ecx+88h],eax
0x100b1177: mov [ecx+8Ch],eax
0x100b117d: mov [ecx+90h],eax
0x100b1183: mov [ecx+94h],eax
0x100b1189: add dword ptr [ecx],1
0x100b118c: mov eax,ecx
0x100b118e: pop esi
0x100b118f: ret
```

`sub_100b10f0` is called by sub_100b14f0, sub_100b1620, sub_100b1e50. It delegates to sub_100b0570, sub_100b05e0 (2 callees).

12.1195 sub_100b13e0 at 0x100b13e0 (24 instructions)

```
0x100b13e0: push edi
0x100b13e1: mov edi,eax
0x100b13e3: mov eax,[edi]
0x100b13e5: test eax,eax
0x100b13e7: jle short 100B13FFh
0x100b13e9: push ebx
```

```
0x100b13ea: lea ebx,[ebx]
0x100b13f0: lea ebx,[eax-1]
0x100b13f3: call 100B1000h
0x100b13f8: mov eax,[edi]
0x100b13fa: test eax,eax
0x100b13fc: jg short 100B13F0h
0x100b13fe: pop ebx
0x100b13ff: mov eax,[edi+8]
0x100b1402: test eax,eax
0x100b1404: je short 100B140Fh
0x100b1406: push eax
0x100b1407: call 100209E0h
0x100b140c: add esp,4
0x100b140f: push edi
0x100b1410: call 100209E0h
0x100b1415: add esp,4
0x100b1418: pop edi
0x100b1419: ret
```

`sub_100b13e0` is called by sub_100b1420, sub_100b1470, sub_100b1490, sub_100b1620, sub_100b1990. It delegates to j_free, sub_100b1000 (2 callees).

12.1196 sub_100b1420 at 0x100b1420 (28 instructions)

```
0x100b1420: push ebx
0x100b1421: mov ebx,[esp+8]
0x100b1425: push esi
0x100b1426: push edi
0x100b1427: lea esi,[ebx+14h]
0x100b142a: mov edi,0Dh
0x100b142f: nop
0x100b1430: mov eax,[esi]
0x100b1432: test eax,eax
0x100b1434: je short 100B1441h
0x100b1436: call 100B13E0h
0x100b143b: mov dword ptr [esi],0
0x100b1441: add esi,4
0x100b1444: sub edi,1
0x100b1447: jne short 100B1430h
0x100b1449: mov eax,[ebx+10h]
0x100b144c: test eax,eax
0x100b144e: je short 100B1459h
0x100b1450: push eax
0x100b1451: call 100C1FB0h
0x100b1456: add esp,4
0x100b1459: push ebx
0x100b145a: call 100209E0h
0x100b145f: add esp,4
0x100b1462: pop edi
0x100b1463: pop esi
0x100b1464: pop ebx
0x100b1465: ret
```

`sub_100b1420` is called by sub_100156b0, sub_100af0f0, sub_100b1990. It delegates to j_free, sub_100b13e0, sub_100c1fb0 (3 callees).

12.1197 sub_100b1470 at 0x100b1470 (11 instructions)

```
0x100b1470: push esi
0x100b1471: mov esi,[esp+8]
0x100b1475: mov eax,[esi+8]
0x100b1478: test eax,eax
0x100b147a: je short 100B1481h
0x100b147c: call 100B13E0h
0x100b1481: push esi
```

```
0x100b1482: call 100209E0h
0x100b1487: add esp,4
0x100b148a: pop esi
0x100b148b: ret
```

`sub_100b1470` is called by sub_100b0040. It delegates to j_free, sub_100b13e0 (2 callees).

12.1198 sub_100b1490 at 0x100b1490 (22 instructions)

```
0x100b1490: push edi
0x100b1491: push 0Ch
0x100b1493: call 100EFDD0h
0x100b1498: mov edi,eax
0x100b149a: xor eax,eax
0x100b149c: add esp,4
0x100b149f: cmp edi,eax
0x100b14a1: je short 100B14C0h
0x100b14a3: push esi
0x100b14a4: lea esi,[eax+10h]
0x100b14a7: mov [edi+8],eax
0x100b14aa: mov [edi],eax
0x100b14ac: mov [edi+4],eax
0x100b14af: call 100B0530h
0x100b14b4: test eax,eax
0x100b14b6: pop esi
0x100b14b7: mov eax,edi
0x100b14b9: je short 100B14C2h
0x100b14bb: call 100B13E0h
0x100b14c0: xor eax,eax
0x100b14c2: pop edi
0x100b14c3: ret
```

`sub_100b1490` is called by sub_100b14d0, sub_100b14f0, sub_100b1620, sub_100b1b50. It delegates to sub_100b0530, sub_100b13e0, j_malloc (3 callees).

12.1199 sub_100b14d0 at 0x100b14d0 (14 instructions)

```
0x100b14d0: push esi
0x100b14d1: call 100B1490h
0x100b14d6: mov esi,eax
0x100b14d8: test esi,esi
0x100b14da: je short 100B14E5h
0x100b14dc: call 100B10A0h
0x100b14e1: test eax,eax
0x100b14e3: je short 100B14E9h
0x100b14e5: xor eax,eax
0x100b14e7: pop esi
0x100b14e8: ret
0x100b14e9: mov eax,esi
0x100b14eb: pop esi
0x100b14ec: ret
```

`sub_100b14d0` is called by sub_100b1ac0. It delegates to sub_100b10a0, sub_100b1490 (2 callees).

12.1200 sub_100b14f0 at 0x100b14f0 (97 instructions)

```
0x100b14f0: sub esp,0Ch
0x100b14f3: push ebp
0x100b14f4: push esi
0x100b14f5: push edi
0x100b14f6: lea eax,[esp+10h]
0x100b14fa: call 100B05E0h
0x100b14ff: mov eax,[esp+1Ch]
0x100b1503: push 6B545243h
```

```
0x100b1508: push eax
0x100b1509: call 100C0860h
0x100b150e: mov ebp,eax
0x100b1510: add esp,8
0x100b1513: test ebp,ebp
0x100b1515: je short 100B1596h
0x100b1517: cmp dword ptr [ebp+4],63757276h
0x100b151e: jne short 100B1596h
0x100b1520: call 100B10F0h
0x100b1525: mov edi,eax
0x100b1527: test edi,edi
0x100b1529: je short 100B1596h
0x100b152b: call 100B1490h
0x100b1530: mov ecx,eax
0x100b1532: test ecx,ecx
0x100b1534: mov [esp+0Ch],ecx
0x100b1538: je short 100B1596h
0x100b153a: push edi
0x100b153b: or eax,0FFFFFFFFh
0x100b153e: call 100B1030h
0x100b1543: add esp,4
0x100b1546: test eax,eax
0x100b1548: jne short 100B1596h
0x100b154a: cmp dword ptr [esp+20h],0
0x100b154f: mov eax,1
0x100b1554: mov [edi+8],eax
0x100b1557: mov dword ptr [edi+0Ch],3
0x100b155e: mov [edi+10h],eax
0x100b1561: mov [edi+18h],eax
0x100b1564: mov [edi+1Ch],eax
0x100b1567: jne short 100B15A0h
0x100b1569: fld qword ptr ds:[10127AD0h]
0x100b156f: lea eax,[ebp+0Ch]
0x100b1572: fstp qword ptr [edi+38h]
0x100b1575: lea ecx,[edi+20h]
0x100b1578: fld1
0x100b157a: mov dword ptr [edi+14h],0
0x100b1581: fstp qword ptr [edi+58h]
0x100b1584: fld qword ptr ds:[10127AC8h]
0x100b158a: fstp qword ptr [edi+78h]
0x100b158d: call 100B0640h
0x100b1592: test eax,eax
0x100b1594: je short 100B15F6h
0x100b1596: pop edi
0x100b1597: pop esi
0x100b1598: or eax,0FFFFFFFFh
0x100b159b: pop ebp
0x100b159c: add esp,0Ch
0x100b159f: ret
0x100b15a0: fld qword ptr ds:[10127AC0h]
0x100b15a6: mov [edi+14h],eax
0x100b15a9: fstp qword ptr [edi+38h]
0x100b15ac: lea eax,[esp+10h]
0x100b15b0: fld1
0x100b15b2: fstp qword ptr [edi+58h]
0x100b15b5: fld qword ptr ds:[10127AB8h]
0x100b15bb: fstp qword ptr [edi+78h]
0x100b15be: call 100B05E0h
0x100b15c3: lea eax,[ebp+0Ch]
0x100b15c6: lea ecx,[esp+10h]
0x100b15ca: call 100B0640h
0x100b15cf: test eax,eax
0x100b15d1: jne short 100B1596h
0x100b15d3: mov eax,[esp+14h]
0x100b15d7: push ebx
0x100b15d8: lea ecx,[esp+14h]
0x100b15dc: push ecx
```

```
0x100b15dd: lea ebx,[edi+20h]
0x100b15e0: call 100B0810h
0x100b15e5: add esp,4
0x100b15e8: test eax,eax
0x100b15ea: pop ebx
0x100b15eb: jne short 100B1596h
0x100b15ed: lea esi,[esp+10h]
0x100b15f1: call 100B05F0h
0x100b15f6: push ebp
0x100b15f7: call 100BFF80h
0x100b15fc: add esp,4
0x100b15ff: mov esi,edi
0x100b1601: call 100B05B0h
0x100b1606: mov eax,[esp+0Ch]
0x100b160a: mov edx,[esp+24h]
0x100b160e: pop edi
0x100b160f: pop esi
0x100b1610: mov [edx],eax
0x100b1612: xor eax,eax
0x100b1614: pop ebp
0x100b1615: add esp,0Ch
0x100b1618: ret
```

`sub_100b14f0` is called by sub_100b1990. It accesses memory-mapped peripheral registers at 0x63757276. It delegates to sub_100b05b0, sub_100b05e0, sub_100b05f0 (10 callees).

12.1201 sub_100b1620 at 0x100b1620 (266 instructions)

```
0x100b1620: push ebp
0x100b1621: mov ebp,esp
0x100b1623: and esp,0FFFFFFF8h
0x100b1626: sub esp,8Ch
0x100b162c: push ebx
0x100b162d: push esi
0x100b162e: push edi
0x100b162f: xor edi,edi
0x100b1631: mov ebx,eax
0x100b1633: lea eax,[esp+30h]
0x100b1637: mov [esp+10h],edi
0x100b163b: mov [esp+18h],edi
0x100b163f: mov [esp+28h],edi
0x100b1643: mov [esp+1Ch],edi
0x100b1647: mov [esp+2Ch],edi
0x100b164b: mov [esp+20h],edi
0x100b164f: call 100B05E0h
0x100b1654: push 72545243h
0x100b1659: push ebx
0x100b165a: call 100C0860h
0x100b165f: add esp,8
0x100b1662: cmp eax,edi
0x100b1664: mov [esp+24h],eax
0x100b1668: je near ptr 100B1938h
0x100b166e: push 67545243h
0x100b1673: push ebx
0x100b1674: call 100C0860h
0x100b1679: add esp,8
0x100b167c: cmp eax,edi
0x100b167e: mov [esp+28h],eax
0x100b1682: je near ptr 100B1938h
0x100b1688: push 62545243h
0x100b168d: push ebx
0x100b168e: call 100C0860h
0x100b1693: add esp,8
0x100b1696: cmp eax,edi
0x100b1698: mov [esp+2Ch],eax
0x100b169c: je near ptr 100B1938h
```

```
0x100b16a2: push 7258595Ah
0x100b16a7: push ebx
0x100b16a8: call 100C0860h
0x100b16ad: add esp,8
0x100b16b0: cmp eax,edi
0x100b16b2: mov [esp+18h],eax
0x100b16b6: je near ptr 100B1938h
0x100b16bc: push 6758595Ah
0x100b16c1: push ebx
0x100b16c2: call 100C0860h
0x100b16c7: mov esi,eax
0x100b16c9: add esp,8
0x100b16cc: test esi,esi
0x100b16ce: mov [esp+1Ch],esi
0x100b16d2: je near ptr 100B1938h
0x100b16d8: push 6258595Ah
0x100b16dd: push ebx
0x100b16de: call 100C0860h
0x100b16e3: mov ebx,eax
0x100b16e5: add esp,8
0x100b16e8: test ebx,ebx
0x100b16ea: mov [esp+20h],ebx
0x100b16ee: je near ptr 100B1938h
0x100b16f4: xor eax,eax
0x100b16f6: mov ecx,58595A20h
0x100b16fb: jmp short 100B1700h
0x100b1700: mov edx,[esp+eax+24h]
0x100b1704: cmp dword ptr [edx+4],63757276h
0x100b170b: jne near ptr 100B1938h
0x100b1711: mov edx,[esp+eax+18h]
0x100b1715: cmp [edx+4],ecx
0x100b1718: jne near ptr 100B1938h
0x100b171e: add eax,4
0x100b1721: cmp eax,0Ch
0x100b1724: jl short 100B1700h
0x100b1726: call 100B10F0h
0x100b172b: mov edi,eax
0x100b172d: test edi,edi
0x100b172f: je near ptr 100B1938h
0x100b1735: mov eax,3
0x100b173a: mov [edi+8],eax
0x100b173d: mov [edi+0Ch],eax
0x100b1740: call 100B1490h
0x100b1745: test eax,eax
0x100b1747: mov [esp+10h],eax
0x100b174b: je near ptr 100B1938h
0x100b1751: mov ecx,[esp+10h]
0x100b1755: push edi
0x100b1756: or eax,0FFFFFFFFh
0x100b1759: call 100B1030h
0x100b175e: add esp,4
0x100b1761: test eax,eax
0x100b1763: jne near ptr 100B1938h
0x100b1769: xor ecx,ecx
0x100b176b: cmp [ebp+8],ecx
0x100b176e: mov eax,1
0x100b1773: mov [edi+10h],ecx
0x100b1776: mov [edi+18h],eax
0x100b1779: mov [edi+1Ch],eax
0x100b177c: jne near ptr 100B1816h
0x100b1782: mov eax,[esp+18h]
0x100b1786: mov [edi+14h],ecx
0x100b1789: fild dword ptr [eax+0Ch]
0x100b178c: fld qword ptr ds:[10127AD8h]
0x100b1792: fmul st(1),st
0x100b1794: fxch
0x100b1796: fstp qword ptr [edi+38h]
```

```
0x100b1799: fild dword ptr [eax+10h]
0x100b179c: fmul st,st(1)
0x100b179e: fstp qword ptr [edi+58h]
0x100b17a1: fild dword ptr [eax+14h]
0x100b17a4: fmul st,st(1)
0x100b17a6: fstp qword ptr [edi+78h]
0x100b17a9: fild dword ptr [esi+0Ch]
0x100b17ac: fmul st,st(1)
0x100b17ae: fstp qword ptr [edi+40h]
0x100b17b1: fild dword ptr [esi+10h]
0x100b17b4: fmul st,st(1)
0x100b17b6: fstp qword ptr [edi+60h]
0x100b17b9: fild dword ptr [esi+14h]
0x100b17bc: lea esi,[edi+20h]
0x100b17bf: fmul st,st(1)
0x100b17c1: fstp qword ptr [edi+80h]
0x100b17c7: fild dword ptr [ebx+0Ch]
0x100b17ca: fmul st,st(1)
0x100b17cc: fstp qword ptr [edi+48h]
0x100b17cf: fild dword ptr [ebx+10h]
0x100b17d2: fmul st,st(1)
0x100b17d4: fstp qword ptr [edi+68h]
0x100b17d7: fimul dword ptr [ebx+14h]
0x100b17da: xor ebx,ebx
0x100b17dc: fstp qword ptr [edi+88h]
0x100b17e2: fldz
0x100b17e4: fst qword ptr [edi+50h]
0x100b17e7: fst qword ptr [edi+70h]
0x100b17ea: fstp qword ptr [edi+90h]
0x100b17f0: mov eax,[esp+ebx*4+24h]
0x100b17f4: add eax,0Ch
0x100b17f7: mov ecx,esi
0x100b17f9: call 100B0640h
0x100b17fe: test eax,eax
0x100b1800: jne near ptr 100B1938h
0x100b1806: add ebx,1
0x100b1809: add esi,8
0x100b180c: cmp ebx,3
0x100b180f: jl short 100B17F0h
0x100b1811: jmp near ptr 100B18F8h
0x100b1816: mov [edi+14h],eax
0x100b1819: mov eax,[esp+18h]
0x100b181d: fild dword ptr [eax+0Ch]
0x100b1820: lea edx,[edi+38h]
0x100b1823: fld qword ptr ds:[10127AD8h]
0x100b1829: lea ecx,[esp+38h]
0x100b182d: fmul st(1),st
0x100b182f: fxch
0x100b1831: fstp qword ptr [esp+38h]
0x100b1835: fild dword ptr [eax+10h]
0x100b1838: fmul st,st(1)
0x100b183a: fstp qword ptr [esp+58h]
0x100b183e: fild dword ptr [eax+14h]
0x100b1841: fmul st,st(1)
0x100b1843: fstp qword ptr [esp+78h]
0x100b1847: fild dword ptr [esi+0Ch]
0x100b184a: fmul st,st(1)
0x100b184c: fstp qword ptr [esp+40h]
0x100b1850: fild dword ptr [esi+10h]
0x100b1853: fmul st,st(1)
0x100b1855: fstp qword ptr [esp+60h]
0x100b1859: fild dword ptr [esi+14h]
0x100b185c: fmul st,st(1)
0x100b185e: fstp qword ptr [esp+80h]
0x100b1865: fild dword ptr [ebx+0Ch]
0x100b1868: fmul st,st(1)
0x100b186a: fstp qword ptr [esp+48h]
```

```
0x100b186e: fild dword ptr [ebx+10h]
0x100b1871: fmul st,st(1)
0x100b1873: fstp qword ptr [esp+68h]
0x100b1877: fimul dword ptr [ebx+14h]
0x100b187a: fstp qword ptr [esp+88h]
0x100b1881: fldz
0x100b1883: fst qword ptr [esp+50h]
0x100b1887: fst qword ptr [esp+70h]
0x100b188b: fstp qword ptr [esp+90h]
0x100b1892: call 100B0A10h
0x100b1897: test eax,eax
0x100b1899: jne near ptr 100B1938h
0x100b189f: mov [esp+14h],eax
0x100b18a3: lea ebx,[edi+20h]
0x100b18a6: lea eax,[esp+30h]
0x100b18aa: call 100B05E0h
0x100b18af: mov eax,[esp+14h]
0x100b18b3: mov eax,[esp+eax*4+24h]
0x100b18b7: add eax,0Ch
0x100b18ba: lea ecx,[esp+30h]
0x100b18be: call 100B0640h
0x100b18c3: test eax,eax
0x100b18c5: jne short 100B1938h
0x100b18c7: mov eax,[esp+34h]
0x100b18cb: lea ecx,[esp+30h]
0x100b18cf: push ecx
0x100b18d0: call 100B0810h
0x100b18d5: add esp,4
0x100b18d8: test eax,eax
0x100b18da: jne short 100B1938h
0x100b18dc: lea esi,[esp+30h]
0x100b18e0: call 100B05F0h
0x100b18e5: mov eax,[esp+14h]
0x100b18e9: add eax,1
0x100b18ec: add ebx,8
0x100b18ef: cmp eax,3
0x100b18f2: mov [esp+14h],eax
0x100b18f6: jl short 100B18A6h
0x100b18f8: xor esi,esi
0x100b18fa: lea ebx,[ebx]
0x100b1900: mov edx,[esp+esi+24h]
0x100b1904: push edx
0x100b1905: call 100BFF80h
0x100b190a: mov eax,[esp+esi+1Ch]
0x100b190e: push eax
0x100b190f: call 100BFF80h
0x100b1914: add esi,4
0x100b1917: add esp,8
0x100b191a: cmp esi,0Ch
0x100b191d: jl short 100B1900h
0x100b191f: mov esi,edi
0x100b1921: call 100B05B0h
0x100b1926: mov ecx,[ebp+0Ch]
0x100b1929: mov edx,[esp+10h]
0x100b192d: mov [ecx],edx
0x100b192f: xor eax,eax
0x100b1931: pop edi
0x100b1932: pop esi
0x100b1933: pop ebx
0x100b1934: mov esp,ebp
0x100b1936: pop ebp
0x100b1937: ret
0x100b1938: xor esi,esi
0x100b193a: lea ebx,[ebx]
0x100b1940: mov eax,[esp+esi+24h]
0x100b1944: test eax,eax
0x100b1946: je short 100B1951h
```

```
0x100b1948: push eax
0x100b1949: call 100BFF80h
0x100b194e: add esp,4
0x100b1951: mov eax,[esp+esi+18h]
0x100b1955: test eax,eax
0x100b1957: je short 100B1962h
0x100b1959: push eax
0x100b195a: call 100BFF80h
0x100b195f: add esp,4
0x100b1962: add esi,4
0x100b1965: cmp esi,0Ch
0x100b1968: jl short 100B1940h
0x100b196a: mov eax,[esp+10h]
0x100b196e: test eax,eax
0x100b1970: je short 100B1977h
0x100b1972: call 100B13E0h
0x100b1977: test edi,edi
0x100b1979: je short 100B1982h
0x100b197b: mov esi,edi
0x100b197d: call 100B05B0h
0x100b1982: pop edi
0x100b1983: pop esi
0x100b1984: or eax,0FFFFFFFFh
0x100b1987: pop ebx
0x100b1988: mov esp,ebp
0x100b198a: pop ebp
0x100b198b: ret
```

`sub_100b1620` is called by sub_100b1990. It accesses memory-mapped peripheral registers at 0x58595a20. It delegates to sub_100b05b0, sub_100b05e0, sub_100b05f0 (12 callees).

12.1202 sub_100b1990 at 0x100b1990 (100 instructions)

```
0x100b1990: sub esp,60h
0x100b1993: push esi
0x100b1994: mov dword ptr [esp+4],0
0x100b199c: mov dword ptr [esp+8],0
0x100b19a4: call 100B04B0h
0x100b19a9: mov esi,eax
0x100b19ab: test esi,esi
0x100b19ad: je near ptr 100B1AACh
0x100b19b3: push edi
0x100b19b4: mov edi,[esp+6Ch]
0x100b19b8: lea eax,[esp+10h]
0x100b19bc: push eax
0x100b19bd: push edi
0x100b19be: call 100BFE90h
0x100b19c3: push edi
0x100b19c4: call 100C2330h
0x100b19c9: add esp,0Ch
0x100b19cc: test eax,eax
0x100b19ce: mov [esi+10h],eax
0x100b19d1: pop edi
0x100b19d2: je near ptr 100B1AA3h
0x100b19d8: mov eax,[esp+1Ch]
0x100b19dc: push 0
0x100b19de: call 100B0B40h
0x100b19e3: mov [esi],eax
0x100b19e5: mov eax,[esp+24h]
0x100b19e9: push 1
0x100b19eb: call 100B0B40h
0x100b19f0: mov ecx,[esi]
0x100b19f2: push ecx
0x100b19f3: mov [esi+8],eax
0x100b19f6: call 100B0C30h
0x100b19fb: mov edx,[esi+8]
```

```
0x100b19fe: push edx
0x100b19ff: mov [esi+4],eax
0x100b1a02: call 100B0C30h
0x100b1a07: mov [esi+0Ch],eax
0x100b1a0a: mov eax,[esi+4]
0x100b1a0d: add esp,10h
0x100b1a10: cmp eax,1
0x100b1a13: jne short 100B1A58h
0x100b1a15: mov ecx,[esi+10h]
0x100b1a18: lea eax,[esp+4]
0x100b1a1c: push eax
0x100b1a1d: push 0
0x100b1a1f: push ecx
0x100b1a20: call 100B14F0h
0x100b1a25: add esp,0Ch
0x100b1a28: test eax,eax
0x100b1a2a: jne short 100B1A89h
0x100b1a2c: mov eax,[esi+10h]
0x100b1a2f: lea edx,[esp+8]
0x100b1a33: push edx
0x100b1a34: push 1
0x100b1a36: push eax
0x100b1a37: call 100B14F0h
0x100b1a3c: add esp,0Ch
0x100b1a3f: test eax,eax
0x100b1a41: jne short 100B1A89h
0x100b1a43: mov eax,[esp+4]
0x100b1a47: mov ecx,[esp+8]
0x100b1a4b: mov [esi+14h],eax
0x100b1a4e: mov [esi+24h],ecx
0x100b1a51: mov eax,esi
0x100b1a53: pop esi
0x100b1a54: add esp,60h
0x100b1a57: ret
0x100b1a58: cmp eax,3
0x100b1a5b: jne short 100B1A43h
0x100b1a5d: mov eax,[esi+10h]
0x100b1a60: lea ecx,[esp+4]
0x100b1a64: push ecx
0x100b1a65: push 0
0x100b1a67: call 100B1620h
0x100b1a6c: add esp,8
0x100b1a6f: test eax,eax
0x100b1a71: jne short 100B1A89h
0x100b1a73: mov eax,[esi+10h]
0x100b1a76: lea edx,[esp+8]
0x100b1a7a: push edx
0x100b1a7b: push 1
0x100b1a7d: call 100B1620h
0x100b1a82: add esp,8
0x100b1a85: test eax,eax
0x100b1a87: je short 100B1A43h
0x100b1a89: mov eax,[esp+4]
0x100b1a8d: test eax,eax
0x100b1a8f: je short 100B1A96h
0x100b1a91: call 100B13E0h
0x100b1a96: mov eax,[esp+8]
0x100b1a9a: test eax,eax
0x100b1a9c: je short 100B1AA3h
0x100b1a9e: call 100B13E0h
0x100b1aa3: push esi
0x100b1aa4: call 100B1420h
0x100b1aa9: add esp,4
0x100b1aac: xor eax,eax
0x100b1aae: pop esi
0x100b1aaf: add esp,60h
0x100b1ab2: ret
```

`sub_100b1990` is called by sub_100b1df0. It delegates to sub_100b04b0, sub_100b0b40, sub_100b0c30 (9 callees).

12.1203 sub_100b1ac0 at 0x100b1ac0 (55 instructions)

```
0x100b1ac0: sub esp,8
0x100b1ac3: push ebx
0x100b1ac4: push ebp
0x100b1ac5: push edi
0x100b1ac6: call 100B04B0h
0x100b1acb: mov ebp,eax
0x100b1acd: test ebp,ebp
0x100b1acf: je short 100B1B42h
0x100b1ad1: push esi
0x100b1ad2: mov esi,[esp+1Ch]
0x100b1ad6: mov eax,[esi]
0x100b1ad8: mov [ebp],eax
0x100b1adb: mov ecx,[esi+4]
0x100b1ade: mov [ebp+4],ecx
0x100b1ae1: mov edx,[esi+8]
0x100b1ae4: mov [ebp+8],edx
0x100b1ae7: mov eax,[esi+0Ch]
0x100b1aea: mov [ebp+0Ch],eax
0x100b1aed: mov ecx,[esi+10h]
0x100b1af0: push ecx
0x100b1af1: call 100C2330h
0x100b1af6: mov [ebp+10h],eax
0x100b1af9: add esp,4
0x100b1afc: mov eax,ebp
0x100b1afe: sub eax,esi
0x100b1b00: lea edi,[esi+14h]
0x100b1b03: mov dword ptr [esp+10h],0
0x100b1b0b: mov [esp+14h],eax
0x100b1b0f: pop esi
0x100b1b10: mov ebx,[edi]
0x100b1b12: test ebx,ebx
0x100b1b14: je short 100B1B26h
0x100b1b16: call 100B14D0h
0x100b1b1b: test eax,eax
0x100b1b1d: mov edx,[esp+10h]
0x100b1b21: mov [edx+edi],eax
0x100b1b24: je short 100B1B42h
0x100b1b26: mov eax,[esp+0Ch]
0x100b1b2a: add eax,1
0x100b1b2d: add edi,4
0x100b1b30: cmp eax,0Dh
0x100b1b33: mov [esp+0Ch],eax
0x100b1b37: jl short 100B1B10h
0x100b1b39: pop edi
0x100b1b3a: mov eax,ebp
0x100b1b3c: pop ebp
0x100b1b3d: pop ebx
0x100b1b3e: add esp,8
0x100b1b41: ret
0x100b1b42: pop edi
0x100b1b43: pop ebp
0x100b1b44: xor eax,eax
0x100b1b46: pop ebx
0x100b1b47: add esp,8
0x100b1b4a: ret
```

`sub_100b1ac0` is called by sub_100affb0, sub_100b0040. It delegates to sub_100b04b0, sub_100b14d0, sub_100c2330 (3 callees).

12.1204 sub_100b1b50 at 0x100b1b50 (34 instructions)

```
0x100b1b50: push ecx
0x100b1b51: push ebx
0x100b1b52: push ebp
0x100b1b53: push esi
0x100b1b54: push edi
0x100b1b55: push 0Ch
0x100b1b57: call 100EFDD0h
0x100b1b5c: mov esi,eax
0x100b1b5e: add esp,4
0x100b1b61: test esi,esi
0x100b1b63: mov [esp+10h],esi
0x100b1b67: je near ptr 100B1CF3h
0x100b1b6d: call 100B1490h
0x100b1b72: test eax,eax
0x100b1b74: mov [esi+8],eax
0x100b1b77: je near ptr 100B1CF3h
0x100b1b7d: mov ecx,[esp+24h]
0x100b1b81: cmp ecx,3
0x100b1b84: ja near ptr 100B1C1Ah
0x100b1b8a: jmp dword ptr [ecx*4+100B1DE0h]
0x100b1c1a: mov eax,[esp+10h]
0x100b1c1e: pop edi
0x100b1c1f: pop esi
0x100b1c20: pop ebp
0x100b1c21: pop ebx
0x100b1c22: pop ecx
0x100b1c23: ret
0x100b1cf3: pop edi
0x100b1cf4: pop esi
0x100b1cf5: pop ebp
0x100b1cf6: xor eax,eax
0x100b1cf8: pop ebx
0x100b1cf9: pop ecx
0x100b1cfa: ret
```

`sub_100b1b50` is called by sub_100b0040. It delegates to sub_100b1490, j_malloc (2 callees).

12.1205 sub_100b1df0 at 0x100b1df0 (38 instructions)

```
0x100b1df0: push ebx
0x100b1df1: mov ebx,[esp+8]
0x100b1df5: cmp ebx,501h
0x100b1dfb: push esi
0x100b1dfc: push edi
0x100b1dfd: je short 100B1E32h
0x100b1dff: push ebx
0x100b1e00: call 100C25E0h
0x100b1e05: mov esi,eax
0x100b1e07: add esp,4
0x100b1e0a: test esi,esi
0x100b1e0c: je short 100B1E3Dh
0x100b1e0e: push esi
0x100b1e0f: call 100B1990h
0x100b1e14: add esp,4
0x100b1e17: push esi
0x100b1e18: mov edi,eax
0x100b1e1a: call 100C1FB0h
0x100b1e1f: add esp,4
0x100b1e22: test edi,edi
0x100b1e24: je short 100B1E3Dh
0x100b1e26: test bl,bl
0x100b1e28: je short 100B1E2Ch
0x100b1e2a: mov [edi],ebx
0x100b1e2c: mov eax,edi
0x100b1e2e: pop edi
```

```
0x100b1e2f: pop esi
0x100b1e30: pop ebx
0x100b1e31: ret
0x100b1e32: call 100B1E50h
0x100b1e37: mov edi,eax
0x100b1e39: test edi,edi
0x100b1e3b: jne short 100B1E2Ch
0x100b1e3d: pop edi
0x100b1e3e: pop esi
0x100b1e3f: xor eax,eax
0x100b1e41: pop ebx
0x100b1e42: ret
```

`sub_100b1df0` is called by sub_100156b0, sub_100af270, sub_100b1e50. It delegates to sub_100b1990, sub_100b1e50, sub_100c1fb0 (4 callees).

12.1206 sub_100b1e50 at 0x100b1e50 (104 instructions)

```
0x100b1e50: push ecx
0x100b1e51: push ebx
0x100b1e52: push ebp
0x100b1e53: push esi
0x100b1e54: push edi
0x100b1e55: push 401h
0x100b1e5a: call 100B1DF0h
0x100b1e5f: xor edi,edi
0x100b1e61: add esp,4
0x100b1e64: cmp eax,edi
0x100b1e66: mov [esp+10h],eax
0x100b1e6a: je near ptr 100B1F84h
0x100b1e70: mov dword ptr [eax],501h
0x100b1e76: call 100B10F0h
0x100b1e7b: mov esi,eax
0x100b1e7d: cmp esi,edi
0x100b1e7f: je near ptr 100B1F84h
0x100b1e85: fld1
0x100b1e87: mov ebp,3
0x100b1e8c: mov [esi+8],ebp
0x100b1e8f: mov [esi+0Ch],ebp
0x100b1e92: fst qword ptr [esi+38h]
0x100b1e95: fldz
0x100b1e97: lea ebx,[esi+38h]
0x100b1e9a: fst qword ptr [esi+40h]
0x100b1e9d: mov [esi+10h],edi
0x100b1ea0: fld qword ptr ds:[10127B10h]
0x100b1ea6: mov [esi+14h],edi
0x100b1ea9: fstp qword ptr [esi+48h]
0x100b1eac: mov [esi+18h],edi
0x100b1eaf: fxch
0x100b1eb1: mov dword ptr [esi+1Ch],1
0x100b1eb8: fst qword ptr [esi+58h]
0x100b1ebb: fld qword ptr ds:[10127B08h]
0x100b1ec1: fstp qword ptr [esi+60h]
0x100b1ec4: fld qword ptr ds:[10127B00h]
0x100b1eca: fstp qword ptr [esi+68h]
0x100b1ecd: fstp qword ptr [esi+78h]
0x100b1ed0: fld qword ptr ds:[10127AF8h]
0x100b1ed6: fstp qword ptr [esi+80h]
0x100b1edc: fstp qword ptr [esi+88h]
0x100b1ee2: fld qword ptr ds:[10127AF0h]
0x100b1ee8: fstp qword ptr [esi+50h]
0x100b1eeb: fld qword ptr ds:[10127AE8h]
0x100b1ef1: fstp qword ptr [esi+70h]
0x100b1ef4: fld qword ptr ds:[10127AE0h]
0x100b1efa: fstp qword ptr [esi+90h]
0x100b1f00: call 100B10F0h
```

```
0x100b1f05: mov edi,eax
0x100b1f07: xor eax,eax
0x100b1f09: cmp edi,eax
0x100b1f0b: je short 100B1F84h
0x100b1f0d: mov [edi+8],ebp
0x100b1f10: mov [edi+0Ch],ebp
0x100b1f13: mov ecx,1
0x100b1f18: mov [edi+14h],ecx
0x100b1f1b: mov [edi+1Ch],ecx
0x100b1f1e: lea edx,[edi+38h]
0x100b1f21: mov ecx,ebx
0x100b1f23: mov [edi+10h],eax
0x100b1f26: mov [edi+18h],eax
0x100b1f29: call 100B0A10h
0x100b1f2e: mov ebx,[esp+10h]
0x100b1f32: xor ebp,ebp
0x100b1f34: add ebx,14h
0x100b1f37: mov ecx,[ebx]
0x100b1f39: test ecx,ecx
0x100b1f3b: je short 100B1F4Ch
0x100b1f3d: push esi
0x100b1f3e: xor eax,eax
0x100b1f40: call 100B1030h
0x100b1f45: add esp,4
0x100b1f48: test eax,eax
0x100b1f4a: jne short 100B1F84h
0x100b1f4c: mov ecx,[ebx+10h]
0x100b1f4f: test ecx,ecx
0x100b1f51: je short 100B1F63h
0x100b1f53: push edi
0x100b1f54: or eax,0FFFFFFFFh
0x100b1f57: call 100B1030h
0x100b1f5c: add esp,4
0x100b1f5f: test eax,eax
0x100b1f61: jne short 100B1F84h
0x100b1f63: add ebp,1
0x100b1f66: add ebx,4
0x100b1f69: cmp ebp,4
0x100b1f6c: jl short 100B1F37h
0x100b1f6e: call 100B05B0h
0x100b1f73: mov esi,edi
0x100b1f75: call 100B05B0h
0x100b1f7a: mov eax,[esp+10h]
0x100b1f7e: pop edi
0x100b1f7f: pop esi
0x100b1f80: pop ebp
0x100b1f81: pop ebx
0x100b1f82: pop ecx
0x100b1f83: ret
0x100b1f84: pop edi
0x100b1f85: pop esi
0x100b1f86: pop ebp
0x100b1f87: xor eax,eax
0x100b1f89: pop ebx
0x100b1f8a: pop ecx
0x100b1f8b: ret
```

`sub_100b1e50` is called by sub_100b1df0. It delegates to sub_100b05b0, sub_100b0a10, sub_100b1030 (5 callees).

12.1207 sub_100b3980 at 0x100b3980 (1 instructions)

```
0x100b3980: ret
```

`sub_100b3980` is called by sub_1003eb30. It calls nothing else.

12.1208 sub_100b8550 at 0x100b8550 (2 instructions)

```
0x100b8550: xor eax,eax
0x100b8552: ret
```

`sub_100b8550` is called by sub_1003eb30. It calls nothing else.

12.1209 sub_100bfe40 at 0x100bfe40 (30 instructions)

```
0x100bfe40: push edi
0x100bfe41: mov edi,[esp+8]
0x100bfe45: mov eax,edi
0x100bfe47: lea edx,[eax+1]
0x100bfe4a: lea ebx,[ebx]
0x100bfe50: mov cl,[eax]
0x100bfe52: add eax,1
0x100bfe55: test cl,cl
0x100bfe57: jne short 100BFE50h
0x100bfe59: sub eax,edx
0x100bfe5b: add eax,1
0x100bfe5e: push eax
0x100bfe5f: call 100EFDD0h
0x100bfe64: add esp,4
0x100bfe67: test eax,eax
0x100bfe69: jne short 100BFE6Dh
0x100bfe6b: pop edi
0x100bfe6c: ret
0x100bfe6d: push esi
0x100bfe6e: mov esi,eax
0x100bfe70: mov ecx,edi
0x100bfe72: sub esi,edi
0x100bfe74: mov dl,[ecx]
0x100bfe76: mov [esi+ecx],dl
0x100bfe79: add ecx,1
0x100bfe7c: test dl,dl
0x100bfe7e: jne short 100BFE74h
0x100bfe80: pop esi
0x100bfe81: pop edi
0x100bfe82: ret
```

`sub_100bfe40` is called by sub_100ae910. It delegates to j_malloc (1 callees).

12.1210 sub_100bfe90 at 0x100bfe90 (10 instructions)

```
0x100bfe90: push esi
0x100bfe91: mov esi,[esp+8]
0x100bfe95: push edi
0x100bfe96: mov edi,[esp+10h]
0x100bfe9a: mov ecx,16h
0x100bfe9f: rep movsd
0x100bfea1: pop edi
0x100bfea2: xor eax,eax
0x100bfea4: pop esi
0x100bfea5: ret
```

`sub_100bfe90` is called by sub_100b1990. It calls nothing else.

12.1211 sub_100bfeb0 at 0x100bfeb0 (10 instructions)

```
0x100bfeb0: mov eax,[esp+4]
0x100bfeb4: mov ecx,[esp+8]
0x100bfeb8: mov eax,[eax+4]
0x100bfebb: mov ecx,[ecx+4]
0x100bfebe: cmp eax,ecx
0x100bfec0: jbe short 100BFEC8h
```

```
0x100bfec2: mov eax,1
0x100bfec7: ret
0x100bfec8: sbb eax,eax
0x100bfeca: ret
```

`sub_100bfeb0` has no recorded callers; it is a leaf stub, likely reached through the vector table. It calls nothing else.

12.1212 sub_100bfed0 at 0x100bfed0 (13 instructions)

```
0x100bfed0: mov ecx,ds:[10136DD0h]
0x100bfed6: test ecx,ecx
0x100bfed8: mov eax,10136DD0h
0x100bfedd: je short 100BFEEEh
0x100bfedf: nop
0x100bfee0: cmp ecx,edx
0x100bfee2: je short 100BFEF0h
0x100bfee4: mov ecx,[eax+1Ch]
0x100bfee7: add eax,1Ch
0x100bfeea: test ecx,ecx
0x100bfeec: jne short 100BFEE0h
0x100bfeee: xor eax,eax
0x100bfef0: ret
```

`sub_100bfed0` is called by sub_100c0b60, sub_100c2390. It calls nothing else.

12.1213 sub_100bff00 at 0x100bff00 (21 instructions)

```
0x100bff00: mov eax,[edi+8]
0x100bff03: test eax,eax
0x100bff05: je short 100BFF1Ah
0x100bff07: lea ecx,[esi*8]
0x100bff0e: push ecx
0x100bff0f: push eax
0x100bff10: call 100AD4B0h
0x100bff15: add esp,8
0x100bff18: jmp short 100BFF2Ah
0x100bff1a: lea edx,[esi*8]
0x100bff21: push edx
0x100bff22: call 100EFDD0h
0x100bff27: add esp,4
0x100bff2a: test eax,eax
0x100bff2c: jne short 100BFF32h
0x100bff2e: or eax,0FFFFFFFFh
0x100bff31: ret
0x100bff32: mov [edi+8],eax
0x100bff35: mov [edi+4],esi
0x100bff38: xor eax,eax
0x100bff3a: ret
```

`sub_100bff00` is called by sub_100c0a20, sub_100c2250. It delegates to j_realloc, j_malloc (2 callees).

12.1214 sub_100bff40 at 0x100bff40 (14 instructions)

```
0x100bff40: mov ecx,[edx]
0x100bff42: xor eax,eax
0x100bff44: test ecx,ecx
0x100bff46: jle short 100BFF5Eh
0x100bff48: mov edx,[edx+8]
0x100bff4b: jmp short 100BFF50h
0x100bff50: cmp [edx],esi
0x100bff52: je short 100BFF61h
0x100bff54: add eax,1
0x100bff57: add edx,8
0x100bff5a: cmp eax,ecx
```

```
0x100bff5c: jl short 100BFF50h
0x100bff5e: or eax,0FFFFFFFFh
0x100bff61: ret
```

`sub_100bff40` is called by sub_100c0860, sub_100c1f00. It calls nothing else.

12.1215 sub_100bff70 at 0x100bff70 (3 instructions)

```
0x100bff70: mov eax,[esp+4]
0x100bff74: add dword ptr [eax],1
0x100bff77: ret
```

`sub_100bff70` is called by sub_100c0860, sub_100c0a20, sub_100c0aa0, sub_100c2390. It calls nothing else.

12.1216 sub_100bff80 at 0x100bff80 (18 instructions)

```
0x100bff80: push esi
0x100bff81: mov esi,[esp+8]
0x100bff85: add dword ptr [esi],0FFFFFFFFh
0x100bff88: mov eax,[esi]
0x100bff8a: test eax,eax
0x100bff8c: jg short 100BFFA7h
0x100bff8e: mov eax,[esi+8]
0x100bff91: cmp dword ptr [eax],0
0x100bff94: je short 100BFF9Eh
0x100bff96: mov eax,[eax]
0x100bff98: push esi
0x100bff99: call eax
0x100bff9b: add esp,4
0x100bff9e: push esi
0x100bff9f: call 100209E0h
0x100bffa4: add esp,4
0x100bffa7: pop esi
0x100bffa8: ret
```

`sub_100bff80` is called by sub_100b14f0, sub_100b1620, sub_100c0aa0, sub_100c0ae0, sub_100c2390. It delegates to j_free (1 callees).

12.1217 sub_100bffb0 at 0x100bffb0 (20 instructions)

```
0x100bffb0: push esi
0x100bffb1: push 68h
0x100bffb3: call 100EFDD0h
0x100bffb8: mov esi,eax
0x100bffba: add esp,4
0x100bffbd: test esi,esi
0x100bffbf: jne short 100BFFC3h
0x100bffc1: pop esi
0x100bffc2: ret
0x100bffc3: push 68h
0x100bffc5: push 0
0x100bffc7: push esi
0x100bffc8: call 100228C0h
0x100bffcd: add esp,0Ch
0x100bffd0: mov dword ptr [esi],0
0x100bffd6: mov dword ptr [esi+8],0
0x100bffdd: mov dword ptr [esi+4],0
0x100bffe4: mov eax,esi
0x100bffe6: pop esi
0x100bffe7: ret
```

`sub_100bffb0` is called by sub_100c0b60. It delegates to j_memset, j_malloc (2 callees).

12.1218 sub_100c0430 at 0x100c0430 (52 instructions)

```
0x100c0430: push ebx
0x100c0431: push ebp
0x100c0432: push edi
0x100c0433: xor edi,edi
0x100c0435: xor ebp,ebp
0x100c0437: test eax,eax
0x100c0439: mov ebx,eax
0x100c043b: jle short 100C048Dh
0x100c043d: lea ecx,[ecx]
0x100c0440: test byte ptr [esi+8],7
0x100c0444: jne short 100C04A0h
0x100c0446: mov eax,[esi+40h]
0x100c0449: test eax,eax
0x100c044b: jl short 100C0452h
0x100c044d: cmp [esi+3Ch],eax
0x100c0450: jge short 100C049Ch
0x100c0452: add dword ptr [esi+1Ch],0FFFFFFFFh
0x100c0456: jns short 100C0465h
0x100c0458: push 1
0x100c045a: push esi
0x100c045b: call 100AD620h
0x100c0460: add esp,8
0x100c0463: jmp short 100C0475h
0x100c0465: mov ecx,[esi+18h]
0x100c0468: add dword ptr [esi+3Ch],1
0x100c046c: movzx eax,byte ptr [ecx]
0x100c046f: add ecx,1
0x100c0472: mov [esi+18h],ecx
0x100c0475: cmp eax,0FFFFFFFFh
0x100c0478: je short 100C04A0h
0x100c047a: shld ebp,edi,8
0x100c047e: cdq
0x100c047f: shl edi,8
0x100c0482: sub ebx,1
0x100c0485: or edi,eax
0x100c0487: or ebp,edx
0x100c0489: test ebx,ebx
0x100c048b: jg short 100C0440h
0x100c048d: mov eax,[esp+10h]
0x100c0491: mov [eax],edi
0x100c0493: pop edi
0x100c0494: mov [eax+4],ebp
0x100c0497: pop ebp
0x100c0498: xor eax,eax
0x100c049a: pop ebx
0x100c049b: ret
0x100c049c: or dword ptr [esi+8],4
0x100c04a0: pop edi
0x100c04a1: pop ebp
0x100c04a2: or eax,0FFFFFFFFh
0x100c04a5: pop ebx
0x100c04a6: ret
```

`sub_100c0430` is called by sub_100c0510, sub_100c0550, sub_100c05c0, sub_100c05f0. It delegates to sub_100ad620 (1 callees).

12.1219 sub_100c0510 at 0x100c0510 (17 instructions)

```
0x100c0510: lea eax,[esp-8]
0x100c0514: sub esp,8
0x100c0517: push eax
0x100c0518: mov eax,2
0x100c051d: call 100C0430h
0x100c0522: add esp,4
```

```
0x100c0525: test eax,eax
0x100c0527: je short 100C0530h
0x100c0529: or eax,0FFFFFFFFh
0x100c052c: add esp,8
0x100c052f: ret
0x100c0530: mov cx,[esp]
0x100c0534: mov edx,[esp+0Ch]
0x100c0538: mov [edx],cx
0x100c053b: xor eax,eax
0x100c053d: add esp,8
0x100c0540: ret
```

`sub_100c0510` is called by sub_100c08c0. It delegates to sub_100c0430 (1 callees).

12.1220 sub_100c0550 at 0x100c0550 (37 instructions)

```
0x100c0550: lea eax,[esp-8]
0x100c0554: sub esp,8
0x100c0557: push eax
0x100c0558: mov eax,4
0x100c055d: call 100C0430h
0x100c0562: add esp,4
0x100c0565: test eax,eax
0x100c0567: je short 100C0570h
0x100c0569: or eax,0FFFFFFFFh
0x100c056c: add esp,8
0x100c056f: ret
0x100c0570: mov ecx,[esp]
0x100c0573: mov eax,ecx
0x100c0575: and eax,80000000h
0x100c057a: xor edx,edx
0x100c057c: or eax,edx
0x100c057e: je short 100C05A4h
0x100c0580: not ecx
0x100c0582: and ecx,7FFFFFFFh
0x100c0588: or eax,0FFFFFFFFh
0x100c058b: sub eax,ecx
0x100c058d: mov ecx,0FFFFFFFFh
0x100c0592: sbb ecx,edx
0x100c0594: mov [esp+4],ecx
0x100c0598: mov ecx,[esp+0Ch]
0x100c059c: mov [ecx],eax
0x100c059e: xor eax,eax
0x100c05a0: add esp,8
0x100c05a3: ret
0x100c05a4: mov edx,[esp+4]
0x100c05a8: mov eax,ecx
0x100c05aa: mov ecx,[esp+0Ch]
0x100c05ae: mov [ecx],eax
0x100c05b0: mov [esp+4],edx
0x100c05b4: xor eax,eax
0x100c05b6: add esp,8
0x100c05b9: ret
```

`sub_100c0550` is called by sub_100c0930. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to sub_100c0430 (1 callees).

12.1221 sub_100c05c0 at 0x100c05c0 (17 instructions)

```
0x100c05c0: lea eax,[esp-8]
0x100c05c4: sub esp,8
0x100c05c7: push eax
0x100c05c8: mov eax,4
0x100c05cd: call 100C0430h
0x100c05d2: add esp,4
```

```
0x100c05d5: test eax,eax
0x100c05d7: je short 100C05E0h
0x100c05d9: or eax,0FFFFFFFFh
0x100c05dc: add esp,8
0x100c05df: ret
0x100c05e0: mov ecx,[esp]
0x100c05e3: mov edx,[esp+0Ch]
0x100c05e7: mov [edx],ecx
0x100c05e9: xor eax,eax
0x100c05eb: add esp,8
0x100c05ee: ret
```

`sub_100c05c0` is called by sub_100c07b0, sub_100c1dc0, sub_100c2390. It delegates to sub_100c0430 (1 callees).

12.1222 sub_100c05f0 at 0x100c05f0 (18 instructions)

```
0x100c05f0: lea eax,[esp-8]
0x100c05f4: sub esp,8
0x100c05f7: push eax
0x100c05f8: mov eax,8
0x100c05fd: call 100C0430h
0x100c0602: add esp,4
0x100c0605: test eax,eax
0x100c0607: je short 100C0610h
0x100c0609: or eax,0FFFFFFFFh
0x100c060c: add esp,8
0x100c060f: ret
0x100c0610: mov ecx,[esp]
0x100c0613: mov edx,[esp+4]
0x100c0617: mov [edi],ecx
0x100c0619: mov [edi+4],edx
0x100c061c: xor eax,eax
0x100c061e: add esp,8
0x100c0621: ret
```

`sub_100c05f0` is called by sub_100c1dc0. It delegates to sub_100c0430 (1 callees).

12.1223 sub_100c06f0 at 0x100c06f0 (5 instructions)

```
0x100c06f0: lea eax,[eax+ecx-1]
0x100c06f4: cdq
0x100c06f5: idiv ecx
0x100c06f7: imul eax,ecx
0x100c06fa: ret
```

`sub_100c06f0` is called by sub_100afb80. It calls nothing else.

12.1224 sub_100c07b0 at 0x100c07b0 (68 instructions)

```
0x100c07b0: push ebx
0x100c07b1: push ebp
0x100c07b2: push esi
0x100c07b3: mov esi,eax
0x100c07b5: mov eax,[edi+4]
0x100c07b8: xor ebp,ebp
0x100c07ba: cmp eax,ebp
0x100c07bc: je short 100C07CAh
0x100c07be: push eax
0x100c07bf: call 100209E0h
0x100c07c4: add esp,4
0x100c07c7: mov [edi+4],ebp
0x100c07ca: push edi
0x100c07cb: call 100C05C0h
0x100c07d0: add esp,4
```

```
0x100c07d3: test eax,eax
0x100c07d5: jne short 100C0833h
0x100c07d7: mov eax,[edi]
0x100c07d9: lea eax,[eax+eax*4]
0x100c07dc: add eax,eax
0x100c07de: add eax,eax
0x100c07e0: push eax
0x100c07e1: call 100EFDD0h
0x100c07e6: add esp,4
0x100c07e9: cmp eax,ebp
0x100c07eb: mov [edi+4],eax
0x100c07ee: je short 100C0833h
0x100c07f0: cmp [edi],ebp
0x100c07f2: mov ebx,eax
0x100c07f4: jle short 100C082Dh
0x100c07f6: push ebx
0x100c07f7: call 100C05C0h
0x100c07fc: add esp,4
0x100c07ff: test eax,eax
0x100c0801: jne short 100C0833h
0x100c0803: lea ecx,[ebx+4]
0x100c0806: push ecx
0x100c0807: call 100C05C0h
0x100c080c: add esp,4
0x100c080f: test eax,eax
0x100c0811: jne short 100C0833h
0x100c0813: lea edx,[ebx+8]
0x100c0816: push edx
0x100c0817: call 100C05C0h
0x100c081c: add esp,4
0x100c081f: test eax,eax
0x100c0821: jne short 100C0833h
0x100c0823: add ebp,1
0x100c0826: add ebx,14h
0x100c0829: cmp ebp,[edi]
0x100c082b: jl short 100C07F6h
0x100c082d: pop esi
0x100c082e: pop ebp
0x100c082f: xor eax,eax
0x100c0831: pop ebx
0x100c0832: ret
0x100c0833: mov eax,[edi+4]
0x100c0836: test eax,eax
0x100c0838: je short 100C084Ah
0x100c083a: push eax
0x100c083b: call 100209E0h
0x100c0840: add esp,4
0x100c0843: mov dword ptr [edi+4],0
0x100c084a: pop esi
0x100c084b: pop ebp
0x100c084c: or eax,0FFFFFFFFh
0x100c084f: pop ebx
0x100c0850: ret
```

`sub_100c07b0` is called by sub_100c2390. It delegates to j_free, sub_100c05c0, j_malloc (3 callees).

12.1225 sub_100c0860 at 0x100c0860 (20 instructions)

```
0x100c0860: mov eax,[esp+4]
0x100c0864: push esi
0x100c0865: mov esi,[esp+0Ch]
0x100c0869: push edi
0x100c086a: mov edi,[eax+60h]
0x100c086d: mov edx,edi
0x100c086f: call 100BFF40h
0x100c0874: test eax,eax
```

```
0x100c0876: jl short 100C088Ch
0x100c0878: mov ecx,[edi+8]
0x100c087b: mov edx,[ecx+eax*8+4]
0x100c087f: push edx
0x100c0880: call 100BFF70h
0x100c0885: add esp,4
0x100c0888: test eax,eax
0x100c088a: jne short 100C088Eh
0x100c088c: xor eax,eax
0x100c088e: pop edi
0x100c088f: pop esi
0x100c0890: ret
```

`sub_100c0860` is called by sub_100b14f0, sub_100b1620. It delegates to sub_100bff40, sub_100bff70 (2 callees).

12.1226 sub_100c08a0 at 0x100c08a0 (9 instructions)

```
0x100c08a0: mov ecx,[eax]
0x100c08a2: mov edx,[eax+4]
0x100c08a5: push 100BFEB0h
0x100c08aa: push 14h
0x100c08ac: push ecx
0x100c08ad: push edx
0x100c08ae: call dword ptr ds:[100FE408h]
0x100c08b4: add esp,10h
0x100c08b7: ret
```

`sub_100c08a0` is called by sub_100c2390. It calls nothing else.

12.1227 sub_100c08c0 at 0x100c08c0 (46 instructions)

```
0x100c08c0: push esi
0x100c08c1: push edi
0x100c08c2: mov edi,eax
0x100c08c4: mov esi,ecx
0x100c08c6: push edi
0x100c08c7: call 100C0510h
0x100c08cc: add esp,4
0x100c08cf: test eax,eax
0x100c08d1: jne short 100C0926h
0x100c08d3: lea eax,[edi+2]
0x100c08d6: push eax
0x100c08d7: call 100C0510h
0x100c08dc: add esp,4
0x100c08df: test eax,eax
0x100c08e1: jne short 100C0926h
0x100c08e3: lea ecx,[edi+4]
0x100c08e6: push ecx
0x100c08e7: call 100C0510h
0x100c08ec: add esp,4
0x100c08ef: test eax,eax
0x100c08f1: jne short 100C0926h
0x100c08f3: lea edx,[edi+6]
0x100c08f6: push edx
0x100c08f7: call 100C0510h
0x100c08fc: add esp,4
0x100c08ff: test eax,eax
0x100c0901: jne short 100C0926h
0x100c0903: lea eax,[edi+8]
0x100c0906: push eax
0x100c0907: call 100C0510h
0x100c090c: add esp,4
0x100c090f: test eax,eax
0x100c0911: jne short 100C0926h
0x100c0913: add edi,0Ah
```

```
0x100c0916: push edi
0x100c0917: call 100C0510h
0x100c091c: add esp,4
0x100c091f: test eax,eax
0x100c0921: jne short 100C0926h
0x100c0923: pop edi
0x100c0924: pop esi
0x100c0925: ret
0x100c0926: pop edi
0x100c0927: or eax,0FFFFFFFFh
0x100c092a: pop esi
0x100c092b: ret
```

`sub_100c08c0` is called by sub_100c1dc0. It delegates to sub_100c0510 (1 callees).

12.1228 sub_100c0930 at 0x100c0930 (28 instructions)

```
0x100c0930: push esi
0x100c0931: push edi
0x100c0932: mov edi,eax
0x100c0934: mov esi,ecx
0x100c0936: push edi
0x100c0937: call 100C0550h
0x100c093c: add esp,4
0x100c093f: test eax,eax
0x100c0941: jne short 100C0966h
0x100c0943: lea eax,[edi+4]
0x100c0946: push eax
0x100c0947: call 100C0550h
0x100c094c: add esp,4
0x100c094f: test eax,eax
0x100c0951: jne short 100C0966h
0x100c0953: add edi,8
0x100c0956: push edi
0x100c0957: call 100C0550h
0x100c095c: add esp,4
0x100c095f: test eax,eax
0x100c0961: jne short 100C0966h
0x100c0963: pop edi
0x100c0964: pop esi
0x100c0965: ret
0x100c0966: pop edi
0x100c0967: or eax,0FFFFFFFFh
0x100c096a: pop esi
0x100c096b: ret
```

`sub_100c0930` is called by sub_100c1dc0. It delegates to sub_100c0550 (1 callees).

12.1229 sub_100c0a20 at 0x100c0a20 (51 instructions)

```
0x100c0a20: push ebx
0x100c0a21: push esi
0x100c0a22: mov ebx,eax
0x100c0a24: test ebx,ebx
0x100c0a26: push edi
0x100c0a27: mov edi,ecx
0x100c0a29: jge short 100C0A2Dh
0x100c0a2b: mov ebx,[edi]
0x100c0a2d: mov eax,[edi]
0x100c0a2f: cmp eax,[edi+4]
0x100c0a32: jl short 100C0A40h
0x100c0a34: lea esi,[eax+20h]
0x100c0a37: call 100BFF00h
0x100c0a3c: test eax,eax
0x100c0a3e: jne short 100C0A53h
```

```
0x100c0a40: mov eax,[esp+14h]
0x100c0a44: push eax
0x100c0a45: call 100BFF70h
0x100c0a4a: mov esi,eax
0x100c0a4c: add esp,4
0x100c0a4f: test esi,esi
0x100c0a51: jne short 100C0A5Ah
0x100c0a53: pop edi
0x100c0a54: pop esi
0x100c0a55: or eax,0FFFFFFFFh
0x100c0a58: pop ebx
0x100c0a59: ret
0x100c0a5a: mov eax,[edi]
0x100c0a5c: sub eax,ebx
0x100c0a5e: test eax,eax
0x100c0a60: jle short 100C0A7Eh
0x100c0a62: mov ecx,[edi+8]
0x100c0a65: lea edx,[eax*8]
0x100c0a6c: lea ecx,[ecx+ebx*8]
0x100c0a6f: push edx
0x100c0a70: push ecx
0x100c0a71: add ecx,8
0x100c0a74: push ecx
0x100c0a75: call dword ptr ds:[100FE498h]
0x100c0a7b: add esp,0Ch
0x100c0a7e: mov eax,[edi+8]
0x100c0a81: mov ecx,[esp+10h]
0x100c0a85: lea eax,[eax+ebx*8]
0x100c0a88: mov [eax+4],esi
0x100c0a8b: mov [eax],ecx
0x100c0a8d: add dword ptr [edi],1
0x100c0a90: pop edi
0x100c0a91: pop esi
0x100c0a92: xor eax,eax
0x100c0a94: pop ebx
0x100c0a95: ret
```

`sub_100c0a20` is called by sub_100c1f00, sub_100c2290. It delegates to sub_100bff00, sub_100bff70 (2 callees).

12.1230 sub_100c0aa0 at 0x100c0aa0 (25 instructions)

```
0x100c0aa0: push edi
0x100c0aa1: push eax
0x100c0aa2: call 100BFF70h
0x100c0aa7: mov edi,eax
0x100c0aa9: add esp,4
0x100c0aac: test edi,edi
0x100c0aae: jne short 100C0AB5h
0x100c0ab0: or eax,0FFFFFFFFh
0x100c0ab3: pop edi
0x100c0ab4: ret
0x100c0ab5: mov edx,[ecx+8]
0x100c0ab8: mov eax,[esp+8]
0x100c0abc: mov ecx,[edx+eax*8+4]
0x100c0ac0: push esi
0x100c0ac1: lea esi,[edx+eax*8]
0x100c0ac4: push ecx
0x100c0ac5: call 100BFF80h
0x100c0aca: mov edx,[esp+14h]
0x100c0ace: add esp,4
0x100c0ad1: mov [esi+4],edi
0x100c0ad4: mov [esi],edx
0x100c0ad6: pop esi
0x100c0ad7: xor eax,eax
0x100c0ad9: pop edi
0x100c0ada: ret
```

`sub_100c0aa0` is called by sub_100c1f00. It delegates to sub_100bff70, sub_100bff80 (2 callees).

12.1231 sub_100c0ae0 at 0x100c0ae0 (23 instructions)

```
0x100c0ae0: mov eax,[esi+8]
0x100c0ae3: mov ecx,[eax+edi*8+4]
0x100c0ae7: push ecx
0x100c0ae8: call 100BFF80h
0x100c0aed: mov eax,[esi]
0x100c0aef: sub eax,edi
0x100c0af1: sub eax,1
0x100c0af4: add esp,4
0x100c0af7: test eax,eax
0x100c0af9: jle short 100C0B16h
0x100c0afb: mov edx,[esi+8]
0x100c0afe: add eax,eax
0x100c0b00: add eax,eax
0x100c0b02: lea ecx,[edx+edi*8]
0x100c0b05: add eax,eax
0x100c0b07: push eax
0x100c0b08: lea edx,[ecx+8]
0x100c0b0b: push edx
0x100c0b0c: push ecx
0x100c0b0d: call dword ptr ds:[100FE498h]
0x100c0b13: add esp,0Ch
0x100c0b16: add dword ptr [esi],0FFFFFFFFh
0x100c0b19: ret
```

`sub_100c0ae0` is called by sub_100c1f00, sub_100c1f70. It delegates to sub_100bff80 (1 callees).

12.1232 sub_100c0b60 at 0x100c0b60 (33 instructions)

```
0x100c0b60: push ebx
0x100c0b61: mov ebx,[esp+8]
0x100c0b65: push esi
0x100c0b66: push edi
0x100c0b67: mov edx,ebx
0x100c0b69: call 100BFED0h
0x100c0b6e: mov edi,eax
0x100c0b70: test edi,edi
0x100c0b72: je short 100C0B7Fh
0x100c0b74: call 100BFFB0h
0x100c0b79: mov esi,eax
0x100c0b7b: test esi,esi
0x100c0b7d: jne short 100C0B85h
0x100c0b7f: pop edi
0x100c0b80: pop esi
0x100c0b81: xor eax,eax
0x100c0b83: pop ebx
0x100c0b84: ret
0x100c0b85: add dword ptr [esi],1
0x100c0b88: push 5Ch
0x100c0b8a: lea eax,[esi+0Ch]
0x100c0b8d: push 0
0x100c0b8f: add edi,4
0x100c0b92: push eax
0x100c0b93: mov [esi+8],edi
0x100c0b96: mov [esi+4],ebx
0x100c0b99: call 100228C0h
0x100c0b9e: add esp,0Ch
0x100c0ba1: pop edi
0x100c0ba2: mov eax,esi
0x100c0ba4: pop esi
0x100c0ba5: pop ebx
0x100c0ba6: ret
```

`sub_100c0b60` is called by sub_100c2390. It delegates to j_memset, sub_100bfed0, sub_100bffb0 (3 callees).

12.1233 sub_100c1dc0 at 0x100c1dc0 (111 instructions)

```
0x100c1dc0: push ebx
0x100c1dc1: push esi
0x100c1dc2: mov ebx,eax
0x100c1dc4: mov esi,ecx
0x100c1dc6: push ebx
0x100c1dc7: call 100C05C0h
0x100c1dcc: add esp,4
0x100c1dcf: test eax,eax
0x100c1dd1: jne near ptr 100C1EF6h
0x100c1dd7: lea eax,[ebx+4]
0x100c1dda: push eax
0x100c1ddb: call 100C05C0h
0x100c1de0: add esp,4
0x100c1de3: test eax,eax
0x100c1de5: jne near ptr 100C1EF6h
0x100c1deb: lea ecx,[ebx+8]
0x100c1dee: push ecx
0x100c1def: call 100C05C0h
0x100c1df4: add esp,4
0x100c1df7: test eax,eax
0x100c1df9: jne near ptr 100C1EF6h
0x100c1dff: lea edx,[ebx+0Ch]
0x100c1e02: push edx
0x100c1e03: call 100C05C0h
0x100c1e08: add esp,4
0x100c1e0b: test eax,eax
0x100c1e0d: jne near ptr 100C1EF6h
0x100c1e13: lea eax,[ebx+10h]
0x100c1e16: push eax
0x100c1e17: call 100C05C0h
0x100c1e1c: add esp,4
0x100c1e1f: test eax,eax
0x100c1e21: jne near ptr 100C1EF6h
0x100c1e27: lea ecx,[ebx+14h]
0x100c1e2a: push ecx
0x100c1e2b: call 100C05C0h
0x100c1e30: add esp,4
0x100c1e33: test eax,eax
0x100c1e35: jne near ptr 100C1EF6h
0x100c1e3b: lea eax,[ebx+18h]
0x100c1e3e: mov ecx,esi
0x100c1e40: call 100C08C0h
0x100c1e45: test eax,eax
0x100c1e47: jne near ptr 100C1EF6h
0x100c1e4d: lea edx,[ebx+24h]
0x100c1e50: push edx
0x100c1e51: call 100C05C0h
0x100c1e56: add esp,4
0x100c1e59: test eax,eax
0x100c1e5b: jne near ptr 100C1EF6h
0x100c1e61: lea eax,[ebx+28h]
0x100c1e64: push eax
0x100c1e65: call 100C05C0h
0x100c1e6a: add esp,4
0x100c1e6d: test eax,eax
0x100c1e6f: jne near ptr 100C1EF6h
0x100c1e75: lea ecx,[ebx+2Ch]
0x100c1e78: push ecx
0x100c1e79: call 100C05C0h
0x100c1e7e: add esp,4
0x100c1e81: test eax,eax
0x100c1e83: jne short 100C1EF6h
0x100c1e85: lea edx,[ebx+30h]
0x100c1e88: push edx
0x100c1e89: call 100C05C0h
0x100c1e8e: add esp,4
```

```
0x100c1e91: test eax,eax
0x100c1e93: jne short 100C1EF6h
0x100c1e95: lea eax,[ebx+34h]
0x100c1e98: push eax
0x100c1e99: call 100C05C0h
0x100c1e9e: add esp,4
0x100c1ea1: test eax,eax
0x100c1ea3: jne short 100C1EF6h
0x100c1ea5: push edi
0x100c1ea6: lea edi,[ebx+38h]
0x100c1ea9: call 100C05F0h
0x100c1eae: test eax,eax
0x100c1eb0: pop edi
0x100c1eb1: jne short 100C1EF6h
0x100c1eb3: lea ecx,[ebx+40h]
0x100c1eb6: push ecx
0x100c1eb7: call 100C05C0h
0x100c1ebc: add esp,4
0x100c1ebf: test eax,eax
0x100c1ec1: jne short 100C1EF6h
0x100c1ec3: lea eax,[ebx+44h]
0x100c1ec6: mov ecx,esi
0x100c1ec8: call 100C0930h
0x100c1ecd: test eax,eax
0x100c1ecf: jne short 100C1EF6h
0x100c1ed1: add ebx,50h
0x100c1ed4: push ebx
0x100c1ed5: call 100C05C0h
0x100c1eda: add esp,4
0x100c1edd: test eax,eax
0x100c1edf: jne short 100C1EF6h
0x100c1ee1: push 2Ch
0x100c1ee3: push esi
0x100c1ee4: call 100ADE80h
0x100c1ee9: add esp,8
0x100c1eec: cmp eax,2Ch
0x100c1eef: jne short 100C1EF6h
0x100c1ef1: pop esi
0x100c1ef2: xor eax,eax
0x100c1ef4: pop ebx
0x100c1ef5: ret
0x100c1ef6: pop esi
0x100c1ef7: or eax,0FFFFFFFFh
0x100c1efa: pop ebx
0x100c1efb: ret
```

`sub_100c1dc0` is called by sub_100c2390. It delegates to sub_100ade80, sub_100c05c0, sub_100c05f0 (5 callees).

12.1234 sub_100c1f00 at 0x100c1f00 (48 instructions)

```
0x100c1f00: mov eax,[esp+4]
0x100c1f04: push ebx
0x100c1f05: mov ebx,[eax+60h]
0x100c1f08: push esi
0x100c1f09: mov esi,[esp+10h]
0x100c1f0d: push edi
0x100c1f0e: mov edx,ebx
0x100c1f10: call 100BFF40h
0x100c1f15: mov edi,eax
0x100c1f17: test edi,edi
0x100c1f19: mov eax,[esp+18h]
0x100c1f1d: jl short 100C1F46h
0x100c1f1f: test eax,eax
0x100c1f21: je short 100C1F39h
0x100c1f23: push esi
0x100c1f24: push edi
```

```
0x100c1f25: mov ecx,ebx
0x100c1f27: call 100C0AA0h
0x100c1f2c: add esp,8
0x100c1f2f: test eax,eax
0x100c1f31: jne short 100C1F5Dh
0x100c1f33: pop edi
0x100c1f34: pop esi
0x100c1f35: xor eax,eax
0x100c1f37: pop ebx
0x100c1f38: ret
0x100c1f39: mov esi,ebx
0x100c1f3b: call 100C0AE0h
0x100c1f40: pop edi
0x100c1f41: pop esi
0x100c1f42: xor eax,eax
0x100c1f44: pop ebx
0x100c1f45: ret
0x100c1f46: test eax,eax
0x100c1f48: je short 100C1F33h
0x100c1f4a: push eax
0x100c1f4b: push esi
0x100c1f4c: or eax,0FFFFFFFFh
0x100c1f4f: mov ecx,ebx
0x100c1f51: call 100C0A20h
0x100c1f56: add esp,8
0x100c1f59: test eax,eax
0x100c1f5b: je short 100C1F33h
0x100c1f5d: pop edi
0x100c1f5e: pop esi
0x100c1f5f: or eax,0FFFFFFFFh
0x100c1f62: pop ebx
0x100c1f63: ret
```

`sub_100c1f00` is called by sub_100c2390. It delegates to sub_100bff40, sub_100c0a20, sub_100c0aa0 (4 callees).

12.1235 sub_100c1f70 at 0x100c1f70 (22 instructions)

```
0x100c1f70: push esi
0x100c1f71: mov esi,eax
0x100c1f73: cmp dword ptr [esi+8],0
0x100c1f77: je short 100C1F98h
0x100c1f79: cmp dword ptr [esi],0
0x100c1f7c: jle short 100C1F8Ch
0x100c1f7e: push edi
0x100c1f7f: nop
0x100c1f80: xor edi,edi
0x100c1f82: call 100C0AE0h
0x100c1f87: cmp [esi],edi
0x100c1f89: jg short 100C1F80h
0x100c1f8b: pop edi
0x100c1f8c: mov eax,[esi+8]
0x100c1f8f: push eax
0x100c1f90: call 100209E0h
0x100c1f95: add esp,4
0x100c1f98: push esi
0x100c1f99: call 100209E0h
0x100c1f9e: add esp,4
0x100c1fa1: pop esi
0x100c1fa2: ret
```

`sub_100c1f70` is called by sub_100c1fb0, sub_100c2250, sub_100c2330. It delegates to j_free, sub_100c0ae0 (2 callees).

12.1236 sub_100c1fb0 at 0x100c1fb0 (17 instructions)

```
0x100c1fb0: push esi
0x100c1fb1: mov esi,[esp+8]
0x100c1fb5: mov eax,[esi+60h]
0x100c1fb8: test eax,eax
0x100c1fba: je short 100C1FC1h
0x100c1fbc: call 100C1F70h
0x100c1fc1: mov eax,[esi+5Ch]
0x100c1fc4: test eax,eax
0x100c1fc6: je short 100C1FD1h
0x100c1fc8: push eax
0x100c1fc9: call 100209E0h
0x100c1fce: add esp,4
0x100c1fd1: push esi
0x100c1fd2: call 100209E0h
0x100c1fd7: add esp,4
0x100c1fda: pop esi
0x100c1fdb: ret
```

`sub_100c1fb0` is called by sub_100b1420, sub_100b1df0, sub_100c22e0, sub_100c2330, sub_100c2390. It delegates to j_free, sub_100c1f70 (2 callees).

12.1237 sub_100c2250 at 0x100c2250 (22 instructions)

```
0x100c2250: push edi
0x100c2251: push 0Ch
0x100c2253: call 100EFDD0h
0x100c2258: mov edi,eax
0x100c225a: xor eax,eax
0x100c225c: add esp,4
0x100c225f: cmp edi,eax
0x100c2261: je short 100C2280h
0x100c2263: push esi
0x100c2264: lea esi,[eax+20h]
0x100c2267: mov [edi+4],eax
0x100c226a: mov [edi],eax
0x100c226c: mov [edi+8],eax
0x100c226f: call 100BFF00h
0x100c2274: test eax,eax
0x100c2276: pop esi
0x100c2277: mov eax,edi
0x100c2279: je short 100C2282h
0x100c227b: call 100C1F70h
0x100c2280: xor eax,eax
0x100c2282: pop edi
0x100c2283: ret
```

`sub_100c2250` is called by sub_100c2290, sub_100c22e0. It delegates to sub_100bff00, sub_100c1f70, j_malloc (3 callees).

12.1238 sub_100c2290 at 0x100c2290 (32 instructions)

```
0x100c2290: push ebx
0x100c2291: push esi
0x100c2292: call 100C2250h
0x100c2297: mov ebx,eax
0x100c2299: test ebx,ebx
0x100c229b: je short 100C22CEh
0x100c229d: xor esi,esi
0x100c229f: cmp [edi],esi
0x100c22a1: jle short 100C22C9h
0x100c22a3: mov eax,[edi+8]
0x100c22a6: mov ecx,[eax+esi*8+4]
0x100c22aa: mov edx,[eax+esi*8]
```

```
0x100c22ad: lea eax,[eax+esi*8]
0x100c22b0: push ecx
0x100c22b1: push edx
0x100c22b2: mov eax,esi
0x100c22b4: mov ecx,ebx
0x100c22b6: call 100C0A20h
0x100c22bb: add esp,8
0x100c22be: test eax,eax
0x100c22c0: jne short 100C22CEh
0x100c22c2: add esi,1
0x100c22c5: cmp esi,[edi]
0x100c22c7: jl short 100C22A3h
0x100c22c9: pop esi
0x100c22ca: mov eax,ebx
0x100c22cc: pop ebx
0x100c22cd: ret
0x100c22ce: pop esi
0x100c22cf: xor eax,eax
0x100c22d1: pop ebx
0x100c22d2: ret
```

`sub_100c2290` is called by sub_100c2330. It delegates to sub_100c0a20, sub_100c2250 (2 callees).

12.1239 sub_100c22e0 at 0x100c22e0 (27 instructions)

```
0x100c22e0: push esi
0x100c22e1: push 68h
0x100c22e3: call 100EFDD0h
0x100c22e8: mov esi,eax
0x100c22ea: add esp,4
0x100c22ed: test esi,esi
0x100c22ef: je short 100C2306h
0x100c22f1: call 100C2250h
0x100c22f6: test eax,eax
0x100c22f8: mov [esi+60h],eax
0x100c22fb: jne short 100C230Ah
0x100c22fd: push esi
0x100c22fe: call 100C1FB0h
0x100c2303: add esp,4
0x100c2306: xor eax,eax
0x100c2308: pop esi
0x100c2309: ret
0x100c230a: push 58h
0x100c230c: push 0
0x100c230e: push esi
0x100c230f: call 100228C0h
0x100c2314: add esp,0Ch
0x100c2317: mov dword ptr [esi+58h],0
0x100c231e: mov dword ptr [esi+5Ch],0
0x100c2325: mov eax,esi
0x100c2327: pop esi
0x100c2328: ret
```

`sub_100c22e0` is called by sub_100c2330, sub_100c2390. It delegates to j_memset, sub_100c1fb0, sub_100c2250 (4 callees).

12.1240 sub_100c2330 at 0x100c2330 (35 instructions)

```
0x100c2330: push ebx
0x100c2331: call 100C22E0h
0x100c2336: mov ebx,eax
0x100c2338: xor eax,eax
0x100c233a: cmp ebx,eax
0x100c233c: je short 100C2379h
0x100c233e: push ebp
```

```
0x100c233f: mov ebp,[esp+0Ch]
0x100c2343: push esi
0x100c2344: push edi
0x100c2345: mov ecx,16h
0x100c234a: mov esi,ebp
0x100c234c: mov edi,ebx
0x100c234e: rep movsd
0x100c2350: mov [ebx+58h],eax
0x100c2353: mov [ebx+5Ch],eax
0x100c2356: mov eax,[ebx+60h]
0x100c2359: call 100C1F70h
0x100c235e: mov edi,[ebp+60h]
0x100c2361: call 100C2290h
0x100c2366: test eax,eax
0x100c2368: pop edi
0x100c2369: pop esi
0x100c236a: mov [ebx+60h],eax
0x100c236d: pop ebp
0x100c236e: jne short 100C237Dh
0x100c2370: push ebx
0x100c2371: call 100C1FB0h
0x100c2376: add esp,4
0x100c2379: xor eax,eax
0x100c237b: pop ebx
0x100c237c: ret
0x100c237d: mov eax,ebx
0x100c237f: pop ebx
0x100c2380: ret
```

`sub_100c2330` is called by sub_100b1990, sub_100b1ac0. It delegates to sub_100c1f70, sub_100c1fb0, sub_100c2290 (4 callees).

12.1241 sub_100c2390 at 0x100c2390 (174 instructions)

```
0x100c2390: sub esp,20h
0x100c2393: push ebx
0x100c2394: push ebp
0x100c2395: push esi
0x100c2396: push edi
0x100c2397: xor ebx,ebx
0x100c2399: call 100C22E0h
0x100c239e: mov ebp,eax
0x100c23a0: test ebp,ebp
0x100c23a2: mov [esp+14h],ebp
0x100c23a6: je near ptr 100C258Ah
0x100c23ac: mov esi,[esp+34h]
0x100c23b0: mov ecx,esi
0x100c23b2: call 100C1DC0h
0x100c23b7: test eax,eax
0x100c23b9: je short 100C23CDh
0x100c23bb: push 10128F4Ch
0x100c23c0: call 100C2630h
0x100c23c5: add esp,4
0x100c23c8: jmp near ptr 100C2574h
0x100c23cd: lea edi,[ebp+58h]
0x100c23d0: mov eax,esi
0x100c23d2: call 100C07B0h
0x100c23d7: test eax,eax
0x100c23d9: je short 100C23EDh
0x100c23db: push 10129458h
0x100c23e0: call 100C2630h
0x100c23e5: add esp,4
0x100c23e8: jmp near ptr 100C2574h
0x100c23ed: mov eax,edi
0x100c23ef: call 100C08A0h
0x100c23f4: mov edi,[edi]
```

```
0x100c23f6: lea esi,[edi+edi*2+21h]
0x100c23fa: xor ecx,ecx
0x100c23fc: add esi,esi
0x100c23fe: add esi,esi
0x100c2400: cmp edi,ecx
0x100c2402: mov [esp+24h],edi
0x100c2406: mov [esp+10h],esi
0x100c240a: mov [esp+20h],ecx
0x100c240e: mov [esp+18h],ecx
0x100c2412: mov [esp+28h],ecx
0x100c2416: jle near ptr 100C2566h
0x100c241c: mov [esp+1Ch],ecx
0x100c2420: jmp short 100C2426h
0x100c2422: mov ecx,[esp+18h]
0x100c2426: mov ebp,[ebp+5Ch]
0x100c2429: add ebp,[esp+1Ch]
0x100c242d: mov eax,[ebp+4]
0x100c2430: cmp eax,[esp+20h]
0x100c2434: jne short 100C2479h
0x100c2436: test ecx,ecx
0x100c2438: je near ptr 100C254Ah
0x100c243e: push ecx
0x100c243f: call 100BFF70h
0x100c2444: mov ebx,eax
0x100c2446: add esp,4
0x100c2449: test ebx,ebx
0x100c244b: je near ptr 100C2570h
0x100c2451: mov eax,[ebp]
0x100c2454: mov ecx,[esp+14h]
0x100c2458: push ebx
0x100c2459: push eax
0x100c245a: push ecx
0x100c245b: call 100C1F00h
0x100c2460: add esp,0Ch
0x100c2463: test eax,eax
0x100c2465: jne near ptr 100C2570h
0x100c246b: push ebx
0x100c246c: call 100BFF80h
0x100c2471: add esp,4
0x100c2474: jmp near ptr 100C254Ah
0x100c2479: sub eax,esi
0x100c247b: mov edi,eax
0x100c247d: test edi,edi
0x100c247f: jle short 100C249Fh
0x100c2481: mov edx,[esp+34h]
0x100c2485: push edi
0x100c2486: push edx
0x100c2487: call 100ADE80h
0x100c248c: add esp,8
0x100c248f: cmp eax,edi
0x100c2491: jne near ptr 100C2570h
0x100c2497: add esi,edi
0x100c2499: mov [esp+10h],esi
0x100c249d: jmp short 100C24A7h
0x100c249f: jge short 100C24A7h
0x100c24a1: call dword ptr ds:[100FE474h]
0x100c24a7: lea eax,[esp+2Ch]
0x100c24ab: mov [esp+20h],esi
0x100c24af: mov esi,[esp+34h]
0x100c24b3: push eax
0x100c24b4: call 100C05C0h
0x100c24b9: add esp,4
0x100c24bc: test eax,eax
0x100c24be: jne near ptr 100C2570h
0x100c24c4: push 4
0x100c24c6: push esi
0x100c24c7: call 100ADE80h
```

```
0x100c24cc: add esp,8
0x100c24cf: cmp eax,4
0x100c24d2: jne near ptr 100C2570h
0x100c24d8: mov edx,[esp+2Ch]
0x100c24dc: add dword ptr [esp+10h],8
0x100c24e1: call 100BFED0h
0x100c24e6: test eax,eax
0x100c24e8: jne short 100C24F0h
0x100c24ea: mov [esp+18h],eax
0x100c24ee: jmp short 100C2542h
0x100c24f0: push edx
0x100c24f1: call 100C0B60h
0x100c24f6: mov ebx,eax
0x100c24f8: add esp,4
0x100c24fb: test ebx,ebx
0x100c24fd: je short 100C2570h
0x100c24ff: mov edi,[ebp+8]
0x100c2502: mov ecx,[ebx+8]
0x100c2505: mov eax,[ecx+8]
0x100c2508: sub edi,8
0x100c250b: push edi
0x100c250c: mov edx,esi
0x100c250e: push edx
0x100c250f: push ebx
0x100c2510: call eax
0x100c2512: add esp,0Ch
0x100c2515: test eax,eax
0x100c2517: jne short 100C2570h
0x100c2519: mov ecx,[ebp]
0x100c251c: mov edx,[esp+14h]
0x100c2520: add [esp+10h],edi
0x100c2524: push ebx
0x100c2525: push ecx
0x100c2526: push edx
0x100c2527: call 100C1F00h
0x100c252c: add esp,0Ch
0x100c252f: test eax,eax
0x100c2531: jne short 100C2570h
0x100c2533: push ebx
0x100c2534: mov [esp+1Ch],ebx
0x100c2538: call 100BFF80h
0x100c253d: add esp,4
0x100c2540: xor ebx,ebx
0x100c2542: mov edi,[esp+24h]
0x100c2546: mov esi,[esp+10h]
0x100c254a: mov eax,[esp+28h]
0x100c254e: add dword ptr [esp+1Ch],14h
0x100c2553: mov ebp,[esp+14h]
0x100c2557: add eax,1
0x100c255a: cmp eax,edi
0x100c255c: mov [esp+28h],eax
0x100c2560: jl near ptr 100C2422h
0x100c2566: pop edi
0x100c2567: pop esi
0x100c2568: mov eax,ebp
0x100c256a: pop ebp
0x100c256b: pop ebx
0x100c256c: add esp,20h
0x100c256f: ret
0x100c2570: mov ebp,[esp+14h]
0x100c2574: push ebp
0x100c2575: call 100C1FB0h
0x100c257a: add esp,4
0x100c257d: test ebx,ebx
0x100c257f: je short 100C258Ah
0x100c2581: push ebx
0x100c2582: call 100BFF80h
```

```
0x100c2587: add esp,4
0x100c258a: pop edi
0x100c258b: pop esi
0x100c258c: pop ebp
0x100c258d: xor eax,eax
0x100c258f: pop ebx
0x100c2590: add esp,20h
0x100c2593: ret
```

`sub_100c2390` is called by sub_100c25a0. It works with the text "cannot get header", "cannot get tab table". It delegates to sub_100ade80, sub_100bfed0, sub_100bff70 (13 callees).

12.1242 sub_100c25a0 at 0x100c25a0 (28 instructions)

```
0x100c25a0: mov eax,[esp+8]
0x100c25a4: mov ecx,[esp+4]
0x100c25a8: push esi
0x100c25a9: push edi
0x100c25aa: push eax
0x100c25ab: push ecx
0x100c25ac: call 100AE0A0h
0x100c25b1: mov esi,eax
0x100c25b3: add esp,8
0x100c25b6: test esi,esi
0x100c25b8: je short 100C25C9h
0x100c25ba: push esi
0x100c25bb: call 100C2390h
0x100c25c0: mov edi,eax
0x100c25c2: add esp,4
0x100c25c5: test edi,edi
0x100c25c7: jne short 100C25CEh
0x100c25c9: pop edi
0x100c25ca: xor eax,eax
0x100c25cc: pop esi
0x100c25cd: ret
0x100c25ce: push esi
0x100c25cf: call 100ADF90h
0x100c25d4: add esp,4
0x100c25d7: mov eax,edi
0x100c25d9: pop edi
0x100c25da: pop esi
0x100c25db: ret
```

`sub_100c25a0` is called by sub_100c25e0. It delegates to sub_100adf90, sub_100ae0a0, sub_100c2390 (3 callees).

12.1243 sub_100c25e0 at 0x100c25e0 (19 instructions)

```
0x100c25e0: mov eax,[esp+4]
0x100c25e4: cmp eax,301h
0x100c25e9: je short 100C2609h
0x100c25eb: cmp eax,401h
0x100c25f0: je short 100C25F5h
0x100c25f2: xor eax,eax
0x100c25f4: ret
0x100c25f5: mov eax,ds:[10137D90h]
0x100c25fa: push eax
0x100c25fb: push 10137148h
0x100c2600: call 100C25A0h
0x100c2605: add esp,8
0x100c2608: ret
0x100c2609: mov ecx,ds:[10137D94h]
0x100c260f: push ecx
0x100c2610: push 10137D98h
0x100c2615: call 100C25A0h
```

```
0x100c261a: add esp,8
0x100c261d: ret
```

`sub_100c25e0` is called by sub_100b1df0. It delegates to sub_100c25a0 (1 callees).

12.1244 sub_100c2630 at 0x100c2630 (10 instructions)

```
0x100c2630: mov ecx,[esp+4]
0x100c2634: lea eax,[esp+8]
0x100c2638: push eax
0x100c2639: push ecx
0x100c263a: call dword ptr ds:[100FE4C4h]
0x100c2640: add eax,40h
0x100c2643: push eax
0x100c2644: call dword ptr ds:[100FE44Ch]
0x100c264a: add esp,0Ch
0x100c264d: ret
```

`sub_100c2630` is called by sub_100c2390. It calls nothing else.

12.1245 sub_100d2d10 at 0x100d2d10 (22 instructions)

```
0x100d2d10: mov ecx,[esp+4]
0x100d2d14: mov eax,[ecx+65B0h]
0x100d2d1a: cmp eax,1
0x100d2d1d: je short 100D2D38h
0x100d2d1f: mov ecx,[esp+8]
0x100d2d23: mov edx,[esp+0Ch]
0x100d2d27: and ecx,7
0x100d2d2a: and edx,1
0x100d2d2d: lea ecx,[edx+ecx*2]
0x100d2d30: add ecx,ecx
0x100d2d32: shr eax,cl
0x100d2d34: and eax,3
0x100d2d37: ret
0x100d2d38: movzx eax,word ptr [ecx+66B8h]
0x100d2d3f: movzx ecx,word ptr [ecx+66BAh]
0x100d2d46: add eax,[esp+8]
0x100d2d4a: add ecx,[esp+0Ch]
0x100d2d4e: and eax,0Fh
0x100d2d51: shl eax,4
0x100d2d54: and ecx,0Fh
0x100d2d57: movsx eax,byte ptr [eax+ecx+1012A818h]
0x100d2d5f: ret
```

`sub_100d2d10` is called by sub_100dd260, sub_100df640, sub_100df840, sub_100df9e0, sub_100dfca0. It calls nothing else.

12.1246 sub_100d2d60 at 0x100d2d60 (76 instructions)

```
0x100d2d60: mov eax,[esp+4]
0x100d2d64: push ebx
0x100d2d65: mov ebx,[esp+14h]
0x100d2d69: mov ecx,eax
0x100d2d6b: sub ecx,ebx
0x100d2d6d: add ecx,[esp+0Ch]
0x100d2d71: push ebp
0x100d2d72: cmp eax,ecx
0x100d2d74: push esi
0x100d2d75: push edi
0x100d2d76: mov [esp+14h],ecx
0x100d2d7a: ja near ptr 100D2E18h
0x100d2d80: cmp ebx,4
0x100d2d83: mov edx,[esp+1Ch]
```

```
0x100d2d87: mov ecx,ebx
0x100d2d89: mov esi,eax
0x100d2d8b: jb short 100D2DA4h
0x100d2d8d: lea ecx,[ecx]
0x100d2d90: mov edi,[esi]
0x100d2d92: cmp edi,[edx]
0x100d2d94: jne short 100D2DA8h
0x100d2d96: sub ecx,4
0x100d2d99: add edx,4
0x100d2d9c: add esi,4
0x100d2d9f: cmp ecx,4
0x100d2da2: jae short 100D2D90h
0x100d2da4: test ecx,ecx
0x100d2da6: je short 100D2E05h
0x100d2da8: movzx ebp,byte ptr [edx]
0x100d2dab: movzx edi,byte ptr [esi]
0x100d2dae: sub edi,ebp
0x100d2db0: jne short 100D2DF7h
0x100d2db2: sub ecx,1
0x100d2db5: add edx,1
0x100d2db8: add esi,1
0x100d2dbb: test ecx,ecx
0x100d2dbd: je short 100D2E05h
0x100d2dbf: movzx ebp,byte ptr [edx]
0x100d2dc2: movzx edi,byte ptr [esi]
0x100d2dc5: sub edi,ebp
0x100d2dc7: jne short 100D2DF7h
0x100d2dc9: sub ecx,1
0x100d2dcc: add edx,1
0x100d2dcf: add esi,1
0x100d2dd2: test ecx,ecx
0x100d2dd4: je short 100D2E05h
0x100d2dd6: movzx ebp,byte ptr [edx]
0x100d2dd9: movzx edi,byte ptr [esi]
0x100d2ddc: sub edi,ebp
0x100d2dde: jne short 100D2DF7h
0x100d2de0: sub ecx,1
0x100d2de3: add edx,1
0x100d2de6: add esi,1
0x100d2de9: test ecx,ecx
0x100d2deb: je short 100D2E05h
0x100d2ded: movzx ecx,byte ptr [edx]
0x100d2df0: movzx edi,byte ptr [esi]
0x100d2df3: sub edi,ecx
0x100d2df5: je short 100D2E05h
0x100d2df7: test edi,edi
0x100d2df9: mov ecx,1
0x100d2dfe: jg short 100D2E07h
0x100d2e00: or ecx,0FFFFFFFFh
0x100d2e03: jmp short 100D2E07h
0x100d2e05: xor ecx,ecx
0x100d2e07: test ecx,ecx
0x100d2e09: je short 100D2E1Ah
0x100d2e0b: add eax,1
0x100d2e0e: cmp eax,[esp+14h]
0x100d2e12: jbe near ptr 100D2D80h
0x100d2e18: xor eax,eax
0x100d2e1a: pop edi
0x100d2e1b: pop esi
0x100d2e1c: pop ebp
0x100d2e1d: pop ebx
0x100d2e1e: ret
```

`sub_100d2d60` is called by sub_100eb620. It calls nothing else.

12.1247 sub_100d2e20 at 0x100d2e20 (31 instructions)

```
0x100d2e20: cmp dword ptr [esp+8],0
0x100d2e25: jne short 100D2E84h
0x100d2e27: push esi
0x100d2e28: mov esi,[esp+8]
0x100d2e2c: mov eax,[esi+2E844h]
0x100d2e32: test eax,eax
0x100d2e34: je short 100D2E50h
0x100d2e36: mov ecx,[esp+10h]
0x100d2e3a: mov edx,[esi+6278h]
0x100d2e40: push ecx
0x100d2e41: push edx
0x100d2e42: push 1012DF98h
0x100d2e47: push eax
0x100d2e48: call dword ptr ds:[100FE4F8h]
0x100d2e4e: jmp short 100D2E71h
0x100d2e50: mov eax,[esp+10h]
0x100d2e54: mov ecx,[esi+6278h]
0x100d2e5a: push eax
0x100d2e5b: push ecx
0x100d2e5c: push 1012DF98h
0x100d2e61: call dword ptr ds:[100FE4C4h]
0x100d2e67: add eax,40h
0x100d2e6a: push eax
0x100d2e6b: call dword ptr ds:[100FE4C8h]
0x100d2e71: add esp,10h
0x100d2e74: push 1
0x100d2e76: add esi,2E804h
0x100d2e7c: push esi
0x100d2e7d: call dword ptr ds:[100FE4E0h]
0x100d2e83: pop esi
0x100d2e84: ret
```

`sub_100d2e20` is called by sub_10016040, sub_100d4ce0, sub_100dd690, sub_100de200, sub_100dea30, sub_100df640, sub_100dfca0, sub_100e0950, sub_100e2010, sub_100e67b0, sub_100e6ff0, sub_100e7350, sub_100eadb0. It works with the text "%s: Out of memory in %s". It calls nothing else.

12.1248 sub_100d2e90 at 0x100d2e90 (54 instructions)

```
0x100d2e90: push esi
0x100d2e91: mov esi,[esp+8]
0x100d2e95: cmp dword ptr [esi+6620h],0
0x100d2e9c: jne short 100D2F13h
0x100d2e9e: mov eax,[esi+6278h]
0x100d2ea4: push ebx
0x100d2ea5: push edi
0x100d2ea6: mov edi,ds:[100FE4C4h]
0x100d2eac: push eax
0x100d2ead: push 1012DFE8h
0x100d2eb2: call edi
0x100d2eb4: mov ebx,ds:[100FE4C8h]
0x100d2eba: add eax,40h
0x100d2ebd: push eax
0x100d2ebe: call ebx
0x100d2ec0: mov ecx,[esi]
0x100d2ec2: mov edx,[esi+4]
0x100d2ec5: mov eax,[ecx+14h]
0x100d2ec8: push edx
0x100d2ec9: call eax
0x100d2ecb: add esp,10h
0x100d2ece: test eax,eax
0x100d2ed0: je short 100D2EF0h
0x100d2ed2: push 1012DFD0h
0x100d2ed7: call edi
0x100d2ed9: add eax,40h
```

```
0x100d2edc: push eax
0x100d2edd: call ebx
0x100d2edf: add esp,8
0x100d2ee2: pop edi
0x100d2ee3: pop ebx
0x100d2ee4: mov dword ptr [esi+6620h],1
0x100d2eee: pop esi
0x100d2eef: ret
0x100d2ef0: mov ecx,[esi]
0x100d2ef2: mov edx,[esi+4]
0x100d2ef5: mov eax,[ecx+18h]
0x100d2ef8: push edx
0x100d2ef9: call eax
0x100d2efb: cdq
0x100d2efc: add esp,4
0x100d2eff: push edx
0x100d2f00: push eax
0x100d2f01: push 1012DFB4h
0x100d2f06: call edi
0x100d2f08: add eax,40h
0x100d2f0b: push eax
0x100d2f0c: call ebx
0x100d2f0e: add esp,10h
0x100d2f11: pop edi
0x100d2f12: pop ebx
0x100d2f13: mov dword ptr [esi+6620h],1
0x100d2f1d: pop esi
0x100d2f1e: ret
```

`sub_100d2e90` is called by sub_100d3290. It works with the text "Unexpected end of file". It calls nothing else.

12.1249 sub_100d2f20 at 0x100d2f20 (14 instructions)

```
0x100d2f20: mov eax,[esp+4]
0x100d2f24: cmp word ptr [eax+6274h],4949h
0x100d2f2d: mov eax,[esp+8]
0x100d2f31: jne short 100D2F3Eh
0x100d2f33: xor ecx,ecx
0x100d2f35: mov ch,[eax+1]
0x100d2f38: mov cl,[eax]
0x100d2f3a: mov ax,cx
0x100d2f3d: ret
0x100d2f3e: xor edx,edx
0x100d2f40: mov dh,[eax]
0x100d2f42: mov dl,[eax+1]
0x100d2f45: mov ax,dx
0x100d2f48: ret
```

`sub_100d2f20` is called by sub_100d2f50, sub_100e2b50. It calls nothing else.

12.1250 sub_100d2f50 at 0x100d2f50 (23 instructions)

```
0x100d2f50: push ecx
0x100d2f51: push esi
0x100d2f52: mov esi,[esp+0Ch]
0x100d2f56: mov edx,[esi+4]
0x100d2f59: push 2
0x100d2f5b: mov al,0FFh
0x100d2f5d: mov [esp+8],al
0x100d2f61: mov [esp+9],al
0x100d2f65: mov eax,[esi]
0x100d2f67: mov eax,[eax]
0x100d2f69: push 1
0x100d2f6b: lea ecx,[esp+0Ch]
0x100d2f6f: push ecx
```

```
0x100d2f70: push edx
0x100d2f71: call eax
0x100d2f73: lea ecx,[esp+14h]
0x100d2f77: push ecx
0x100d2f78: push esi
0x100d2f79: call 100D2F20h
0x100d2f7e: add esp,18h
0x100d2f81: pop esi
0x100d2f82: pop ecx
0x100d2f83: ret
```

`sub_100d2f50` is called by sub_100d3020, sub_100d5e70, sub_100e2a20, sub_100e2ab0, sub_100e2b50, sub_100e3e60, sub_100e40d0, sub_100e4940, sub_100e4a30, sub_100e53f0, sub_100e5ad0, sub_100e5c30, sub_100e6090, sub_100e6190, sub_100e6460, sub_100e8a00, sub_100ea5a0, sub_100eaba0, sub_100eb010, sub_100eb620. It delegates to sub_100d2f20 (1 callees).

12.1251 sub_100d2f90 at 0x100d2f90 (23 instructions)

```
0x100d2f90: mov eax,[esp+4]
0x100d2f94: xor ecx,ecx
0x100d2f96: cmp word ptr [eax+6274h],4949h
0x100d2f9f: mov eax,[esp+8]
0x100d2fa3: jne short 100D2FBDh
0x100d2fa5: mov ch,[eax+3]
0x100d2fa8: movzx edx,byte ptr [eax+1]
0x100d2fac: mov cl,[eax+2]
0x100d2faf: movzx eax,byte ptr [eax]
0x100d2fb2: shl ecx,8
0x100d2fb5: or ecx,edx
0x100d2fb7: shl ecx,8
0x100d2fba: or eax,ecx
0x100d2fbc: ret
0x100d2fbd: mov ch,[eax]
0x100d2fbf: movzx edx,byte ptr [eax+2]
0x100d2fc3: mov cl,[eax+1]
0x100d2fc6: movzx eax,byte ptr [eax+3]
0x100d2fca: shl ecx,8
0x100d2fcd: or ecx,edx
0x100d2fcf: shl ecx,8
0x100d2fd2: or eax,ecx
0x100d2fd4: ret
```

`sub_100d2f90` is called by sub_100d2fe0. It calls nothing else.

12.1252 sub_100d2fe0 at 0x100d2fe0 (25 instructions)

```
0x100d2fe0: push ecx
0x100d2fe1: push esi
0x100d2fe2: mov esi,[esp+0Ch]
0x100d2fe6: mov edx,[esi+4]
0x100d2fe9: mov al,0FFh
0x100d2feb: push 4
0x100d2fed: mov [esp+8],al
0x100d2ff1: mov [esp+9],al
0x100d2ff5: mov [esp+0Ah],al
0x100d2ff9: mov [esp+0Bh],al
0x100d2ffd: mov eax,[esi]
0x100d2fff: mov eax,[eax]
0x100d3001: push 1
0x100d3003: lea ecx,[esp+0Ch]
0x100d3007: push ecx
0x100d3008: push edx
0x100d3009: call eax
0x100d300b: lea ecx,[esp+14h]
```

```
0x100d300f: push ecx
0x100d3010: push esi
0x100d3011: call 100D2F90h
0x100d3016: add esp,18h
0x100d3019: pop esi
0x100d301a: pop ecx
0x100d301b: ret
```

`sub_100d2fe0` is called by sub_100d3020, sub_100e2a20, sub_100e2ab0, sub_100e2b50, sub_100e3e60, sub_100e42b0, sub_100e4940, sub_100e4a30, sub_100e53f0, sub_100e55a0, sub_100e5ad0, sub_100e5c30, sub_100e6090, sub_100e6190, sub_100e8a00, sub_100ea5a0, sub_100eaba0, sub_100eb010, sub_100eb170, sub_100eb620. It delegates to sub_100d2f90 (1 callees).

12.1253 sub_100d3020 at 0x100d3020 (13 instructions)

```
0x100d3020: cmp dword ptr [esp+8],3
0x100d3025: jne short 100D3038h
0x100d3027: mov eax,[esp+4]
0x100d302b: push eax
0x100d302c: call 100D2F50h
0x100d3031: movzx eax,ax
0x100d3034: add esp,4
0x100d3037: ret
0x100d3038: mov ecx,[esp+4]
0x100d303c: push ecx
0x100d303d: call 100D2FE0h
0x100d3042: add esp,4
0x100d3045: ret
```

`sub_100d3020` is called by sub_100e8a00. It delegates to sub_100d2f50, sub_100d2fe0 (2 callees).

12.1254 sub_100d3050 at 0x100d3050 (4 instructions)

```
0x100d3050: mov eax,[esp+4]
0x100d3054: mov [esp+4],eax
0x100d3058: fld dword ptr [esp+4]
0x100d305c: ret
```

`sub_100d3050` is called by sub_100e42b0, sub_100e4a30, sub_100e55a0. It calls nothing else.

12.1255 sub_100d3060 at 0x100d3060 (25 instructions)

```
0x100d3060: sub esp,10h
0x100d3063: mov eax,ds:[10136220h]
0x100d3068: xor eax,esp
0x100d306a: mov [esp+0Ch],eax
0x100d306e: mov eax,[esp+18h]
0x100d3072: add eax,0FFFFFFFDh
0x100d3075: cmp eax,9
0x100d3078: push esi
0x100d3079: mov esi,[esp+18h]
0x100d307d: ja near ptr 100D3234h
0x100d3083: jmp dword ptr [eax*4+100D325Ch]
0x100d3234: mov eax,[esi]
0x100d3236: mov ecx,[esi+4]
0x100d3239: mov edx,[eax+1Ch]
0x100d323c: push ecx
0x100d323d: call edx
0x100d323f: mov [esp+8],eax
0x100d3243: fild dword ptr [esp+8]
0x100d3247: add esp,4
0x100d324a: mov ecx,[esp+10h]
0x100d324e: pop esi
```

```
0x100d324f: xor ecx,esp
0x100d3251: call 10022326h
0x100d3256: add esp,10h
0x100d3259: ret
```

`sub_100d3060` is called by sub_100e2b50, sub_100e3e60, sub_100e6190, sub_100e8a00. It delegates to sub_10022326 (1 callees).

12.1256 sub_100d3290 at 0x100d3290 (40 instructions)

```
0x100d3290: push ebx
0x100d3291: mov ebx,[esp+0Ch]
0x100d3295: push esi
0x100d3296: mov esi,[esp+0Ch]
0x100d329a: mov eax,[esi]
0x100d329c: mov ecx,[esi+4]
0x100d329f: mov edx,[eax]
0x100d32a1: push edi
0x100d32a2: mov edi,[esp+18h]
0x100d32a6: push edi
0x100d32a7: push 2
0x100d32a9: push ebx
0x100d32aa: push ecx
0x100d32ab: call edx
0x100d32ad: add esp,10h
0x100d32b0: cmp eax,edi
0x100d32b2: jge short 100D32BDh
0x100d32b4: push esi
0x100d32b5: call 100D2E90h
0x100d32ba: add esp,4
0x100d32bd: push 1234h
0x100d32c2: call dword ptr ds:[100FE6B8h]
0x100d32c8: xor ecx,ecx
0x100d32ca: cmp ax,1234h
0x100d32ce: sete cl
0x100d32d1: xor edx,edx
0x100d32d3: cmp word ptr [esi+6274h],4949h
0x100d32dc: sete dl
0x100d32df: cmp edx,ecx
0x100d32e1: jne short 100D32F2h
0x100d32e3: lea eax,[edi+edi]
0x100d32e6: push eax
0x100d32e7: push ebx
0x100d32e8: push ebx
0x100d32e9: call dword ptr ds:[100FE4ECh]
0x100d32ef: add esp,0Ch
0x100d32f2: pop edi
0x100d32f3: pop esi
0x100d32f4: pop ebx
0x100d32f5: ret
```

`sub_100d3290` is called by sub_100eb620. It delegates to sub_100d2e90 (1 callees).

12.1257 sub_100d40c0 at 0x100d40c0 (160 instructions)

```
0x100d40c0: sub esp,20h
0x100d40c3: push ebp
0x100d40c4: push edi
0x100d40c5: mov edi,[esp+2Ch]
0x100d40c9: xor ebp,ebp
0x100d40cb: cmp [edi+66B4h],bp
0x100d40d2: mov [esp+8],ebp
0x100d40d6: jbe near ptr 100D42EFh
0x100d40dc: push ebx
0x100d40dd: push esi
```

```
0x100d40de: mov edi,edi
0x100d40e0: movzx eax,word ptr [edi+66B6h]
0x100d40e7: test eax,eax
0x100d40e9: mov dword ptr [esp+34h],0
0x100d40f1: jbe near ptr 100D42D7h
0x100d40f7: mov esi,ebp
0x100d40f9: and esi,7
0x100d40fc: add esi,esi
0x100d40fe: mov [esp+1Ch],esi
0x100d4102: jmp short 100D4114h
0x100d4110: mov esi,[esp+1Ch]
0x100d4114: mov ecx,[esp+34h]
0x100d4118: movzx eax,word ptr [edi+66BCh]
0x100d411f: mov ebx,[edi+65B0h]
0x100d4125: movzx edx,word ptr [edi+66C0h]
0x100d412c: and ecx,1
0x100d412f: add ecx,esi
0x100d4131: add ecx,ecx
0x100d4133: mov esi,[edi+66E0h]
0x100d4139: shr ebx,cl
0x100d413b: mov [esp+24h],ecx
0x100d413f: mov ecx,eax
0x100d4141: shr ebp,cl
0x100d4143: and ebx,3
0x100d4146: mov [esp+2Ch],ebx
0x100d414a: imul ebp,edx
0x100d414d: mov edx,[esp+34h]
0x100d4151: mov ecx,edx
0x100d4153: mov [esp+18h],ecx
0x100d4157: mov ecx,eax
0x100d4159: mov eax,edx
0x100d415b: shr eax,cl
0x100d415d: add ebp,eax
0x100d415f: lea ecx,[ebx+ebp*4]
0x100d4162: xor eax,eax
0x100d4164: cmp [esi+ecx*2],ax
0x100d4168: jne near ptr 100D42B9h
0x100d416e: mov esi,[esp+10h]
0x100d4172: lea ecx,[esi+2]
0x100d4175: mov [esp+14h],eax
0x100d4179: mov [esp+18h],eax
0x100d417d: lea eax,[ecx-4]
0x100d4180: cmp eax,ecx
0x100d4182: ja near ptr 100D42B9h
0x100d4188: lea ecx,[edx+2]
0x100d418b: lea edx,[ecx-4]
0x100d418e: mov [esp+20h],edx
0x100d4192: mov ebp,[esp+20h]
0x100d4196: cmp ebp,ecx
0x100d4198: ja near ptr 100D4273h
0x100d419e: mov edi,edi
0x100d41a0: movzx ecx,word ptr [edi+66B4h]
0x100d41a7: cmp eax,ecx
0x100d41a9: jae near ptr 100D4259h
0x100d41af: movzx ecx,word ptr [edi+66B6h]
0x100d41b6: cmp ebp,ecx
0x100d41b8: jae near ptr 100D4259h
0x100d41be: mov ecx,eax
0x100d41c0: and ecx,7
0x100d41c3: mov edx,ebp
0x100d41c5: and edx,1
0x100d41c8: lea esi,[edx+ecx*2]
0x100d41cb: mov edx,[edi+65B0h]
0x100d41d1: add esi,esi
0x100d41d3: mov ebx,edx
0x100d41d5: mov ecx,esi
0x100d41d7: shr ebx,cl
```

```
0x100d41d9: mov ecx,[esp+24h]
0x100d41dd: shr edx,cl
0x100d41df: xor ebx,edx
0x100d41e1: test bl,3
0x100d41e4: jne short 100D4259h
0x100d41e6: movzx ebx,word ptr [edi+66BCh]
0x100d41ed: mov edx,[edi+65B0h]
0x100d41f3: mov ecx,esi
0x100d41f5: shr edx,cl
0x100d41f7: mov ecx,ebx
0x100d41f9: mov esi,eax
0x100d41fb: shr esi,cl
0x100d41fd: movzx ecx,word ptr [edi+66C0h]
0x100d4204: imul esi,ecx
0x100d4207: mov ecx,ebp
0x100d4209: mov [esp+28h],ecx
0x100d420d: mov ecx,ebx
0x100d420f: mov ebx,ebp
0x100d4211: shr ebx,cl
0x100d4213: mov ecx,[edi+66E0h]
0x100d4219: and edx,3
0x100d421c: add esi,ebx
0x100d421e: lea esi,[edx+esi*4]
0x100d4221: cmp word ptr [ecx+esi*2],0
0x100d4226: je short 100D4259h
0x100d4228: movzx ecx,word ptr [edi+66BCh]
0x100d422f: movzx esi,word ptr [edi+66C0h]
0x100d4236: add dword ptr [esp+14h],1
0x100d423b: mov ebx,eax
0x100d423d: shr ebx,cl
0x100d423f: imul ebx,esi
0x100d4242: mov esi,ebp
0x100d4244: shr esi,cl
0x100d4246: mov ecx,[edi+66E0h]
0x100d424c: add ebx,esi
0x100d424e: lea edx,[edx+ebx*4]
0x100d4251: movzx ecx,word ptr [ecx+edx*2]
0x100d4255: add [esp+18h],ecx
0x100d4259: mov ecx,[esp+34h]
0x100d425d: add ebp,1
0x100d4260: add ecx,2
0x100d4263: cmp ebp,ecx
0x100d4265: jbe near ptr 100D41A0h
0x100d426b: mov ebx,[esp+2Ch]
0x100d426f: mov esi,[esp+10h]
0x100d4273: add eax,1
0x100d4276: lea edx,[esi+2]
0x100d4279: cmp eax,edx
0x100d427b: jbe near ptr 100D4192h
0x100d4281: mov ebp,[esp+14h]
0x100d4285: test ebp,ebp
0x100d4287: je short 100D42B9h
0x100d4289: mov eax,[esp+18h]
0x100d428d: movzx ecx,word ptr [edi+66BCh]
0x100d4294: xor edx,edx
0x100d4296: div ebp
0x100d4298: movzx edx,word ptr [edi+66C0h]
0x100d429f: shr esi,cl
0x100d42a1: imul esi,edx
0x100d42a4: mov edx,[esp+34h]
0x100d42a8: shr edx,cl
0x100d42aa: mov ecx,[edi+66E0h]
0x100d42b0: add esi,edx
0x100d42b2: lea edx,[ebx+esi*4]
0x100d42b5: mov [ecx+edx*2],ax
0x100d42b9: mov ecx,[esp+34h]
0x100d42bd: movzx eax,word ptr [edi+66B6h]
```

```
0x100d42c4: mov ebp,[esp+10h]
0x100d42c8: add ecx,1
0x100d42cb: cmp ecx,eax
0x100d42cd: mov [esp+34h],ecx
0x100d42d1: jb near ptr 100D4110h
0x100d42d7: movzx eax,word ptr [edi+66B4h]
0x100d42de: add ebp,1
0x100d42e1: cmp ebp,eax
0x100d42e3: mov [esp+10h],ebp
0x100d42e7: jb near ptr 100D40E0h
0x100d42ed: pop esi
0x100d42ee: pop ebx
0x100d42ef: pop edi
0x100d42f0: pop ebp
0x100d42f1: add esp,20h
0x100d42f4: ret
```

`sub_100d40c0` is called by sub_10016040. It calls nothing else.

12.1258 sub_100d4300 at 0x100d4300 (31 instructions)

```
0x100d4300: push esi
0x100d4301: mov esi,[esp+8]
0x100d4305: push edi
0x100d4306: mov edi,0CD4h
0x100d430b: jmp short 100D4310h
0x100d4310: mov eax,[esi]
0x100d4312: mov ecx,[esi+4]
0x100d4315: mov edx,[eax+8]
0x100d4318: push 0
0x100d431a: push edi
0x100d431b: push ecx
0x100d431c: call edx
0x100d431e: mov eax,[esi]
0x100d4320: mov ecx,[esi+4]
0x100d4323: mov edx,[eax+1Ch]
0x100d4326: push ecx
0x100d4327: call edx
0x100d4329: add esp,10h
0x100d432c: cmp eax,0Fh
0x100d432f: jg short 100D4344h
0x100d4331: add edi,0D0Ch
0x100d4337: cmp edi,52584h
0x100d433d: jb short 100D4310h
0x100d433f: pop edi
0x100d4340: xor eax,eax
0x100d4342: pop esi
0x100d4343: ret
0x100d4344: pop edi
0x100d4345: mov eax,1
0x100d434a: pop esi
0x100d434b: ret
```

`sub_100d4300` is called by sub_100eb620. It calls nothing else.

12.1259 sub_100d4660 at 0x100d4660 (13 instructions)

```
0x100d4660: push esi
0x100d4661: push edi
0x100d4662: mov edi,[esp+0Ch]
0x100d4666: push 6000h
0x100d466b: lea esi,[edi+0E8h]
0x100d4671: push 0
0x100d4673: push esi
0x100d4674: call 100228C0h
```

```
0x100d4679: add esp,0Ch
0x100d467c: mov [edi+60ECh],esi
0x100d4682: pop edi
0x100d4683: pop esi
0x100d4684: ret
```

`sub_100d4660` is called by sub_100d4ce0, sub_100d5e70. It delegates to j_memset (1 callees).

12.1260 sub_100d4ce0 at 0x100d4ce0 (142 instructions)

```
0x100d4ce0: mov eax,10004h
0x100d4ce5: call 100229B0h
0x100d4cea: mov eax,ds:[10136220h]
0x100d4cef: xor eax,esp
0x100d4cf1: mov [esp+10000h],eax
0x100d4cf8: cmp dword ptr [esp+10010h],0
0x100d4d00: push esi
0x100d4d01: mov esi,[esp+10010h]
0x100d4d08: push edi
0x100d4d09: mov edi,[esp+10010h]
0x100d4d10: jne short 100D4D1Bh
0x100d4d12: push edi
0x100d4d13: call 100D4660h
0x100d4d18: add esp,4
0x100d4d1b: push 50h
0x100d4d1d: push 0
0x100d4d1f: push esi
0x100d4d20: call 100228C0h
0x100d4d25: mov eax,[edi+60ECh]
0x100d4d2b: mov [esi+34h],eax
0x100d4d2e: mov ecx,[edi+60ECh]
0x100d4d34: mov [esi+38h],ecx
0x100d4d37: mov edx,[edi+60ECh]
0x100d4d3d: mov [esi+3Ch],edx
0x100d4d40: mov eax,[edi+60ECh]
0x100d4d46: mov [esi+40h],eax
0x100d4d49: mov ecx,[edi+60ECh]
0x100d4d4f: mov [esi+44h],ecx
0x100d4d52: mov edx,[edi+60ECh]
0x100d4d58: mov [esi+48h],edx
0x100d4d5b: push 1
0x100d4d5d: mov dword ptr [esi+18h],7FFFFFFFh
0x100d4d64: mov eax,[edi]
0x100d4d66: mov edx,[edi+4]
0x100d4d69: mov eax,[eax]
0x100d4d6b: push 2
0x100d4d6d: lea ecx,[esp+1Ch]
0x100d4d71: push ecx
0x100d4d72: push edx
0x100d4d73: call eax
0x100d4d75: add esp,1Ch
0x100d4d78: cmp byte ptr [esp+9],0D8h
0x100d4d7d: je short 100D4D98h
0x100d4d7f: pop edi
0x100d4d80: xor eax,eax
0x100d4d82: pop esi
0x100d4d83: mov ecx,[esp+10000h]
0x100d4d8a: xor ecx,esp
0x100d4d8c: call 10022326h
0x100d4d91: add esp,10004h
0x100d4d97: ret
0x100d4d98: push ebx
0x100d4d99: push ebp
0x100d4d9a: lea ebx,[ebx]
0x100d4da0: mov ecx,[edi]
0x100d4da2: mov eax,[edi+4]
```

```
0x100d4da5: mov ecx,[ecx]
0x100d4da7: push 2
0x100d4da9: push 2
0x100d4dab: lea edx,[esp+18h]
0x100d4daf: push edx
0x100d4db0: push eax
0x100d4db1: call ecx
0x100d4db3: movzx ebx,byte ptr [esp+22h]
0x100d4db8: movzx eax,byte ptr [esp+21h]
0x100d4dbd: movzx ecx,byte ptr [esp+23h]
0x100d4dc2: xor edx,edx
0x100d4dc4: mov dh,[esp+20h]
0x100d4dc8: shl ebx,8
0x100d4dcb: or ebx,ecx
0x100d4dcd: add esp,10h
0x100d4dd0: sub ebx,2
0x100d4dd3: or edx,eax
0x100d4dd5: mov ebp,edx
0x100d4dd7: cmp ebp,0FF00h
0x100d4ddd: jle near ptr 100D4F11h
0x100d4de3: mov edx,[edi]
0x100d4de5: mov ecx,[edi+4]
0x100d4de8: mov edx,[edx]
0x100d4dea: push ebx
0x100d4deb: push 1
0x100d4ded: lea eax,[esp+18h]
0x100d4df1: push eax
0x100d4df2: push ecx
0x100d4df3: call edx
0x100d4df5: lea eax,[ebp-0FFC0h]
0x100d4dfb: add esp,10h
0x100d4dfe: cmp eax,1Dh
0x100d4e01: ja near ptr 100D4EF4h
0x100d4e07: movzx eax,byte ptr [eax+100D4FB0h]
0x100d4e0e: jmp dword ptr [eax*4+100D4F98h]
0x100d4ef4: cmp ebp,0FFDAh
0x100d4efa: jne near ptr 100D4DA0h
0x100d4f00: cmp dword ptr [esp+10020h],0
0x100d4f08: je short 100D4F15h
0x100d4f0a: mov eax,1
0x100d4f0f: jmp short 100D4F7Dh
0x100d4f11: xor eax,eax
0x100d4f13: jmp short 100D4F7Dh
0x100d4f15: cmp dword ptr [esi+10h],0
0x100d4f19: je short 100D4F50h
0x100d4f1b: mov edx,[esi+38h]
0x100d4f1e: lea ecx,[esi+38h]
0x100d4f21: mov [esi+3Ch],edx
0x100d4f24: mov eax,[ecx]
0x100d4f26: mov [esi+40h],eax
0x100d4f29: mov edx,[ecx]
0x100d4f2b: mov [esi+44h],edx
0x100d4f2e: mov eax,[ecx]
0x100d4f30: mov [esi+48h],eax
0x100d4f33: xor eax,eax
0x100d4f35: cmp [esi+10h],eax
0x100d4f38: jle short 100D4F50h
0x100d4f3a: lea ebx,[ebx]
0x100d4f40: mov edx,[esi+34h]
0x100d4f43: mov [ecx],edx
0x100d4f45: add eax,1
0x100d4f48: add ecx,4
0x100d4f4b: cmp eax,[esi+10h]
0x100d4f4e: jl short 100D4F40h
0x100d4f50: mov eax,[esi+0Ch]
0x100d4f53: imul eax,[esi+8]
0x100d4f57: push 4
```

```
0x100d4f59: push eax
0x100d4f5a: call dword ptr ds:[100FE4FCh]
0x100d4f60: push 1012E040h
0x100d4f65: push eax
0x100d4f66: push edi
0x100d4f67: mov [esi+4Ch],eax
0x100d4f6a: call 100D2E20h
0x100d4f6f: mov eax,1
0x100d4f74: add esp,14h
0x100d4f77: mov [edi+6610h],eax
0x100d4f7d: mov ecx,[esp+10010h]
0x100d4f84: pop ebp
0x100d4f85: pop ebx
0x100d4f86: pop edi
0x100d4f87: pop esi
0x100d4f88: xor ecx,esp
0x100d4f8a: call 10022326h
0x100d4f8f: add esp,10004h
0x100d4f95: ret
```

`sub_100d4ce0` is called by sub_100ea5a0, sub_100eb620. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to sub_10022326, j_memset, sub_100229b0 (5 callees).

12.1261 sub_100d5e70 at 0x100d5e70 (67 instructions)

```
0x100d5e70: sub esp,3Ch
0x100d5e73: mov eax,ds:[10136220h]
0x100d5e78: xor eax,esp
0x100d5e7a: mov [esp+38h],eax
0x100d5e7e: push ebx
0x100d5e7f: push ebp
0x100d5e80: push esi
0x100d5e81: mov esi,[esp+4Ch]
0x100d5e85: push edi
0x100d5e86: push esi
0x100d5e87: call 100D4660h
0x100d5e8c: add esp,4
0x100d5e8f: xor edi,edi
0x100d5e91: push esi
0x100d5e92: call 100D2F50h
0x100d5e97: mov [esp+edi*2+18h],ax
0x100d5e9c: add edi,1
0x100d5e9f: add esp,4
0x100d5ea2: cmp edi,0Dh
0x100d5ea5: jl short 100D5E91h
0x100d5ea7: xor edi,edi
0x100d5ea9: lea esp,[esp]
0x100d5eb0: mov eax,[esi]
0x100d5eb2: mov ecx,[esi+4]
0x100d5eb5: mov edx,[eax+1Ch]
0x100d5eb8: push ecx
0x100d5eb9: call edx
0x100d5ebb: and eax,0Fh
0x100d5ebe: mov [esp+edi*2+32h],ax
0x100d5ec3: add edi,1
0x100d5ec6: add esp,4
0x100d5ec9: cmp edi,0Dh
0x100d5ecc: jl short 100D5EB0h
0x100d5ece: xor edi,edi
0x100d5ed0: cmp word ptr [esp+edi*2+2Eh],0
0x100d5ed6: lea edx,[esi+0E8h]
0x100d5edc: jbe short 100D5F1Dh
0x100d5ede: movzx eax,word ptr [esp+edi*2+2Eh]
0x100d5ee3: movzx ebp,word ptr [esp+edi*2+14h]
0x100d5ee8: mov ebx,0Bh
0x100d5eed: mov [esp+10h],eax
```

```
0x100d5ef1: mov eax,ebp
0x100d5ef3: mov cl,bl
0x100d5ef5: shr eax,cl
0x100d5ef7: and eax,1
0x100d5efa: cmp dword ptr [edx+eax*4],0
0x100d5efe: jne short 100D5F10h
0x100d5f00: add dword ptr [esi+60ECh],0Ch
0x100d5f07: mov ecx,[esi+60ECh]
0x100d5f0d: mov [edx+eax*4],ecx
0x100d5f10: mov edx,[edx+eax*4]
0x100d5f13: sub ebx,1
0x100d5f16: sub dword ptr [esp+10h],1
0x100d5f1b: jne short 100D5EF1h
0x100d5f1d: mov [edx+8],edi
0x100d5f20: add edi,1
0x100d5f23: cmp edi,0Dh
0x100d5f26: jl short 100D5ED0h
0x100d5f28: mov ecx,[esp+48h]
0x100d5f2c: pop edi
0x100d5f2d: pop esi
0x100d5f2e: pop ebp
0x100d5f2f: pop ebx
0x100d5f30: xor ecx,esp
0x100d5f32: call 10022326h
0x100d5f37: add esp,3Ch
0x100d5f3a: ret
```

`sub_100d5e70` is called by sub_100e2b50. It delegates to sub_10022326, sub_100d2f50, sub_100d4660 (3 callees).

12.1262 sub_100d64f0 at 0x100d64f0 (44 instructions)

```
0x100d64f0: sub esp,104h
0x100d64f6: mov eax,ds:[10136220h]
0x100d64fb: xor eax,esp
0x100d64fd: mov [esp+100h],eax
0x100d6504: push esi
0x100d6505: mov esi,[esp+10Ch]
0x100d650c: mov ecx,[esi+65C0h]
0x100d6512: mov eax,[esi]
0x100d6514: mov edx,[esi+4]
0x100d6517: mov eax,[eax+8]
0x100d651a: push 0
0x100d651c: push ecx
0x100d651d: push edx
0x100d651e: call eax
0x100d6520: mov ecx,[esi]
0x100d6522: mov eax,[esi+4]
0x100d6525: mov ecx,[ecx]
0x100d6527: push 100h
0x100d652c: push 1
0x100d652e: lea edx,[esp+18h]
0x100d6532: push edx
0x100d6533: push eax
0x100d6534: call ecx
0x100d6536: add esp,1Ch
0x100d6539: mov eax,0Fh
0x100d653e: pop esi
0x100d653f: nop
0x100d6540: cmp byte ptr [esp+eax],0
0x100d6544: jne short 100D6567h
0x100d6546: add eax,10h
0x100d6549: cmp eax,100h
0x100d654e: jl short 100D6540h
0x100d6550: xor eax,eax
0x100d6552: mov ecx,[esp+100h]
0x100d6559: xor ecx,esp
```

```
0x100d655b: call 10022326h
0x100d6560: add esp,104h
0x100d6566: ret
0x100d6567: mov ecx,[esp+100h]
0x100d656e: xor ecx,esp
0x100d6570: mov eax,1
0x100d6575: call 10022326h
0x100d657a: add esp,104h
0x100d6580: ret
```

`sub_100d64f0` is called by sub_100eb620. It delegates to sub_10022326 (1 callees).

12.1263 sub_100d6590 at 0x100d6590 (49 instructions)

```
0x100d6590: sub esp,404h
0x100d6596: push esi
0x100d6597: push edi
0x100d6598: push 400h
0x100d659d: lea eax,[esp+10h]
0x100d65a1: push 0
0x100d65a3: push eax
0x100d65a4: mov byte ptr [esp+14h],0
0x100d65a9: mov byte ptr [esp+15h],55h
0x100d65ae: mov byte ptr [esp+16h],0AAh
0x100d65b3: mov byte ptr [esp+17h],0FFh
0x100d65b8: call 100228C0h
0x100d65bd: mov esi,[esp+41Ch]
0x100d65c4: mov ecx,[esi]
0x100d65c6: mov edx,[esi+4]
0x100d65c9: mov eax,[ecx+8]
0x100d65cc: push 2
0x100d65ce: push 0FFFFF830h
0x100d65d3: push edx
0x100d65d4: call eax
0x100d65d6: add esp,18h
0x100d65d9: mov edi,7D0h
0x100d65de: mov edi,edi
0x100d65e0: mov ecx,[esi]
0x100d65e2: mov edx,[esi+4]
0x100d65e5: mov eax,[ecx+1Ch]
0x100d65e8: push edx
0x100d65e9: call eax
0x100d65eb: add dword ptr [esp+eax*4+10h],1
0x100d65f0: lea eax,[esp+eax*4+10h]
0x100d65f4: add esp,4
0x100d65f7: sub edi,1
0x100d65fa: jne short 100D65E0h
0x100d65fc: pop edi
0x100d65fd: xor eax,eax
0x100d65ff: mov ecx,0C8h
0x100d6604: pop esi
0x100d6605: movzx edx,byte ptr [esp+eax]
0x100d6609: cmp [esp+edx*4+4],ecx
0x100d660d: jl short 100D6623h
0x100d660f: add eax,1
0x100d6612: cmp eax,4
0x100d6615: jl short 100D6605h
0x100d6617: mov eax,1
0x100d661c: add esp,404h
0x100d6622: ret
0x100d6623: xor eax,eax
0x100d6625: add esp,404h
0x100d662b: ret
```

`sub_100d6590` is called by sub_100eb620. It delegates to j_memset (1 callees).

12.1264 sub_100d6630 at 0x100d6630 (63 instructions)

```
0x100d6630: sub esp,10h
0x100d6633: mov eax,ds:[10136220h]
0x100d6638: xor eax,esp
0x100d663a: mov [esp+0Ch],eax
0x100d663e: push esi
0x100d663f: mov esi,[esp+18h]
0x100d6643: mov eax,[esi]
0x100d6645: mov ecx,[esi+4]
0x100d6648: mov edx,[eax+8]
0x100d664b: push edi
0x100d664c: push 0
0x100d664e: push 0
0x100d6650: push ecx
0x100d6651: call edx
0x100d6653: add esp,0Ch
0x100d6656: xor edi,edi
0x100d6658: mov eax,[esi]
0x100d665a: mov edx,[esi+4]
0x100d665d: mov eax,[eax]
0x100d665f: push 0Ch
0x100d6661: push 1
0x100d6663: lea ecx,[esp+10h]
0x100d6667: push ecx
0x100d6668: push edx
0x100d6669: call eax
0x100d666b: movzx edx,byte ptr [esp+1Fh]
0x100d6670: mov cl,[esp+21h]
0x100d6674: mov al,[esp+1Ch]
0x100d6678: shr cl,4
0x100d667b: shr dl,4
0x100d667e: and cl,dl
0x100d6680: movzx edx,byte ptr [esp+1Ah]
0x100d6685: shr al,4
0x100d6688: and cl,al
0x100d668a: shr dl,4
0x100d668d: and cl,dl
0x100d668f: and cl,[esp+23h]
0x100d6693: add esp,10h
0x100d6696: and cl,[esp+10h]
0x100d669a: and cl,[esp+0Eh]
0x100d669e: and cl,[esp+9]
0x100d66a2: and cl,3
0x100d66a5: cmp cl,3
0x100d66a8: jne short 100D66CBh
0x100d66aa: add edi,1
0x100d66ad: cmp edi,400h
0x100d66b3: jl short 100D6658h
0x100d66b5: pop edi
0x100d66b6: mov eax,1
0x100d66bb: pop esi
0x100d66bc: mov ecx,[esp+0Ch]
0x100d66c0: xor ecx,esp
0x100d66c2: call 10022326h
0x100d66c7: add esp,10h
0x100d66ca: ret
0x100d66cb: mov ecx,[esp+14h]
0x100d66cf: pop edi
0x100d66d0: pop esi
0x100d66d1: xor ecx,esp
0x100d66d3: xor eax,eax
0x100d66d5: call 10022326h
0x100d66da: add esp,10h
0x100d66dd: ret
```

`sub_100d6630` is called by sub_100eb620. It delegates to sub_10022326 (1 callees).

12.1265 sub_100d66e0 at 0x100d66e0 (130 instructions)

```
0x100d66e0: sub esp,0A0h
0x100d66e6: mov eax,ds:[10136220h]
0x100d66eb: xor eax,esp
0x100d66ed: mov [esp+9Ch],eax
0x100d66f4: push ebp
0x100d66f5: push esi
0x100d66f6: push edi
0x100d66f7: mov eax,ds:[10129888h]
0x100d66fc: mov dl,ds:[1012988Eh]
0x100d6702: mov cx,ds:[1012988Ch]
0x100d6709: mov [esp+14h],eax
0x100d670d: xor eax,eax
0x100d670f: mov [esp+1Bh],eax
0x100d6713: mov [esp+1Fh],al
0x100d6717: mov eax,ds:[1012E090h]
0x100d671c: mov [esp+20h],eax
0x100d6720: movzx eax,byte ptr ds:[1012E09Ah]
0x100d6727: mov [esp+2Ah],al
0x100d672b: xor eax,eax
0x100d672d: mov [esp+3Ah],eax
0x100d6731: mov [esp+3Eh],ax
0x100d6736: mov eax,ds:[1012E088h]
0x100d673b: mov [esp+40h],eax
0x100d673f: movzx eax,word ptr ds:[1012E08Ch]
0x100d6746: mov [esp+18h],cx
0x100d674b: mov ecx,ds:[1012E094h]
0x100d6751: mov [esp+1Ah],dl
0x100d6755: movzx edx,word ptr ds:[1012E098h]
0x100d675c: mov [esp+44h],ax
0x100d6761: xor eax,eax
0x100d6763: mov [esp+24h],ecx
0x100d6767: mov ecx,ds:[101298B0h]
0x100d676d: mov [esp+46h],eax
0x100d6771: mov [esp+4Ah],eax
0x100d6775: mov [esp+4Eh],al
0x100d6779: mov [esp+5Ah],eax
0x100d677d: mov [esp+5Eh],ax
0x100d6782: mov [esp+66h],eax
0x100d6786: mov [esp+6Ah],eax
0x100d678a: mov [esp+6Eh],al
0x100d678e: mov eax,ds:[10129890h]
0x100d6793: mov [esp+28h],dx
0x100d6798: mov dx,ds:[101298B4h]
0x100d679f: mov [esp+34h],ecx
0x100d67a3: mov [esp+54h],ecx
0x100d67a7: mov ecx,ds:[1012E080h]
0x100d67ad: mov [esp+74h],eax
0x100d67b1: movzx eax,word ptr ds:[1012E07Ch]
0x100d67b8: mov [esp+38h],dx
0x100d67bd: mov [esp+58h],dx
0x100d67c2: movzx edx,word ptr ds:[1012E084h]
0x100d67c9: mov esi,[esp+0B0h]
0x100d67d0: mov [esp+60h],ecx
0x100d67d4: mov ecx,ds:[10129894h]
0x100d67da: mov [esp+84h],ax
0x100d67e2: xor eax,eax
0x100d67e4: mov [esp+64h],dx
0x100d67e9: mov edx,ds:[1012E078h]
0x100d67ef: mov [esp+78h],ecx
0x100d67f3: movzx ecx,byte ptr ds:[1012E07Eh]
0x100d67fa: push eax
0x100d67fb: mov [esp+84h],edx
0x100d6802: mov edx,[esi]
0x100d6804: mov [esp+8Bh],eax
0x100d680b: mov [esp+8Fh],eax
0x100d6812: mov eax,[esi+4]
```

```
0x100d6815: mov [esp+8Ah],cl
0x100d681c: mov ecx,[edx+8]
0x100d681f: push 0C00h
0x100d6824: push eax
0x100d6825: mov dword ptr [esp+1Ch],0
0x100d682d: mov dword ptr [esp+37h],0
0x100d6835: mov dword ptr [esp+3Ch],3
0x100d683d: mov dword ptr [esp+5Ch],32h
0x100d6845: mov dword ptr [esp+7Ch],33h
0x100d684d: mov dword ptr [esp+88h],0
0x100d6858: call ecx
0x100d685a: mov edx,[esi]
0x100d685c: mov ecx,[esi+4]
0x100d685f: mov edx,[edx]
0x100d6861: push 18h
0x100d6863: push 1
0x100d6865: lea eax,[esp+0A4h]
0x100d686c: push eax
0x100d686d: push ecx
0x100d686e: call edx
0x100d6870: movzx eax,byte ptr [esp+0B4h]
0x100d6878: movzx ecx,byte ptr [esp+0C0h]
0x100d6880: and eax,3
0x100d6883: shl eax,4
0x100d6886: and ecx,3
0x100d6889: add esp,1Ch
0x100d688c: or eax,ecx
0x100d688e: mov [esp+0Ch],eax
0x100d6892: lea edx,[esp+14h]
0x100d6896: mov ebp,4
0x100d689b: jmp short 100D68A0h
0x100d68a0: mov eax,[esp+0Ch]
0x100d68a4: cmp eax,[edx-4]
0x100d68a7: jne short 100D68DCh
0x100d68a9: mov ecx,esi
0x100d68ab: sub ecx,edx
0x100d68ad: mov eax,edx
0x100d68af: lea edi,[ecx+6485h]
0x100d68b5: mov cl,[eax]
0x100d68b7: mov [edi+eax],cl
0x100d68ba: add eax,1
0x100d68bd: test cl,cl
0x100d68bf: jne short 100D68B5h
0x100d68c1: lea eax,[edx+0Ch]
0x100d68c4: mov ecx,esi
0x100d68c6: sub ecx,eax
0x100d68c8: lea edi,[ecx+64C5h]
0x100d68ce: mov edi,edi
0x100d68d0: mov cl,[eax]
0x100d68d2: mov [edi+eax],cl
0x100d68d5: add eax,1
0x100d68d8: test cl,cl
0x100d68da: jne short 100D68D0h
0x100d68dc: add edx,20h
0x100d68df: sub ebp,1
0x100d68e2: jne short 100D68A0h
0x100d68e4: mov ecx,[esp+0A8h]
0x100d68eb: pop edi
0x100d68ec: pop esi
0x100d68ed: pop ebp
0x100d68ee: xor ecx,esp
0x100d68f0: call 10022326h
0x100d68f5: add esp,0A0h
0x100d68fb: ret
```

`sub_100d66e0` is called by sub_100eb620. It works with the text "PENTAX", "NIKON", "C740UZ". It delegates to sub_10022326 (1 callees).

12.1266 sub_100d6900 at 0x100d6900 (51 instructions)

```
0x100d6900: sub esp,1ACh
0x100d6906: mov eax,ds:[10136220h]
0x100d690b: xor eax,esp
0x100d690d: mov [esp+1A8h],eax
0x100d6914: push esi
0x100d6915: mov esi,[esp+1B4h]
0x100d691c: mov eax,[esi]
0x100d691e: mov ecx,[esi+4]
0x100d6921: mov edx,[eax+8]
0x100d6924: push 2
0x100d6926: push 0FFFFFE58h
0x100d692b: push ecx
0x100d692c: call edx
0x100d692e: mov eax,[esi]
0x100d6930: mov edx,[esi+4]
0x100d6933: mov eax,[eax]
0x100d6935: push 1A8h
0x100d693a: push 1
0x100d693c: lea ecx,[esp+18h]
0x100d6940: push ecx
0x100d6941: push edx
0x100d6942: call eax
0x100d6944: add esp,1Ch
0x100d6947: xor ecx,ecx
0x100d6949: xor eax,eax
0x100d694b: pop esi
0x100d694c: lea esp,[esp]
0x100d6950: cmp byte ptr [esp+ecx],0
0x100d6954: je short 100D6959h
0x100d6956: add eax,1
0x100d6959: cmp byte ptr [esp+ecx+1],0
0x100d695e: je short 100D6963h
0x100d6960: add eax,1
0x100d6963: cmp byte ptr [esp+ecx+2],0
0x100d6968: je short 100D696Dh
0x100d696a: add eax,1
0x100d696d: cmp byte ptr [esp+ecx+3],0
0x100d6972: je short 100D6977h
0x100d6974: add eax,1
0x100d6977: add ecx,4
0x100d697a: cmp ecx,1A8h
0x100d6980: jb short 100D6950h
0x100d6982: xor ecx,ecx
0x100d6984: cmp eax,14h
0x100d6987: setg cl
0x100d698a: mov eax,ecx
0x100d698c: mov ecx,[esp+1A8h]
0x100d6993: xor ecx,esp
0x100d6995: call 10022326h
0x100d699a: add esp,1ACh
0x100d69a0: ret
```

`sub_100d6900` is called by sub_100eb620. It delegates to sub_10022326 (1 callees).

12.1267 sub_100dbdd0 at 0x100dbdd0 (76 instructions)

```
0x100dbdd0: cmp dword ptr [esp+0Ch],0
0x100dbdd5: push esi
0x100dbdd6: push edi
0x100dbdd7: je near ptr 100DBE87h
0x100dbddd: mov ecx,[esp+18h]
0x100dbde1: xor eax,eax
0x100dbde3: jmp short 100DBDF0h
0x100dbdf0: imul ecx,2E90EDDh
0x100dbdf6: add ecx,1
```

```
0x100dbdf9: mov [eax+1013D928h],ecx
0x100dbdff: add eax,4
0x100dbe02: cmp eax,10h
0x100dbe05: jb short 100DBDF0h
0x100dbe07: mov eax,ds:[1013D930h]
0x100dbe0c: xor eax,ds:[1013D928h]
0x100dbe12: mov ecx,ds:[1013D934h]
0x100dbe18: shr eax,1Fh
0x100dbe1b: lea edx,[ecx+ecx]
0x100dbe1e: or eax,edx
0x100dbe20: mov ds:[1013D934h],eax
0x100dbe25: mov eax,1013D92Ch
0x100dbe2a: mov ecx,7Bh
0x100dbe2f: nop
0x100dbe30: mov edx,[eax+8]
0x100dbe33: xor edx,[eax]
0x100dbe35: mov esi,[eax-4]
0x100dbe38: xor esi,[eax+4]
0x100dbe3b: shr edx,1Fh
0x100dbe3e: add esi,esi
0x100dbe40: or edx,esi
0x100dbe42: mov [eax+0Ch],edx
0x100dbe45: add eax,4
0x100dbe48: sub ecx,1
0x100dbe4b: jne short 100DBE30h
0x100dbe4d: mov esi,ds:[100FE6C0h]
0x100dbe53: mov ds:[10139920h],ecx
0x100dbe59: lea esp,[esp]
0x100dbe60: mov eax,[ecx*4+1013D928h]
0x100dbe67: push eax
0x100dbe68: call esi
0x100dbe6a: mov ecx,ds:[10139920h]
0x100dbe70: mov [ecx*4+1013D928h],eax
0x100dbe77: add ecx,1
0x100dbe7a: cmp ecx,7Fh
0x100dbe7d: mov ds:[10139920h],ecx
0x100dbe83: jb short 100DBE60h
0x100dbe85: jmp short 100DBE8Dh
0x100dbe87: mov ecx,ds:[10139920h]
0x100dbe8d: mov edi,[esp+10h]
0x100dbe91: test edi,edi
0x100dbe93: je short 100DBEDEh
0x100dbe95: mov esi,[esp+0Ch]
0x100dbe99: lea edx,[ecx+1]
0x100dbe9c: push ebx
0x100dbe9d: lea ecx,[ecx]
0x100dbea0: lea eax,[edx-40h]
0x100dbea3: and eax,7Fh
0x100dbea6: mov eax,[eax*4+1013D928h]
0x100dbead: mov ebx,edx
0x100dbeaf: and ebx,7Fh
0x100dbeb2: xor eax,[ebx*4+1013D928h]
0x100dbeb9: mov ebx,ecx
0x100dbebb: and ebx,7Fh
0x100dbebe: mov [ebx*4+1013D928h],eax
0x100dbec5: xor [esi],eax
0x100dbec7: sub edi,1
0x100dbeca: add ecx,1
0x100dbecd: add edx,1
0x100dbed0: add esi,4
0x100dbed3: test edi,edi
0x100dbed5: jne short 100DBEA0h
0x100dbed7: mov ds:[10139920h],ecx
0x100dbedd: pop ebx
0x100dbede: pop edi
0x100dbedf: pop esi
0x100dbee0: ret
```

`sub_100dbdd0` is called by sub_100e8a00. It calls nothing else.

12.1268 sub_100dd260 at 0x100dd260 (316 instructions)

```
0x100dd260: sub esp,0BCh
0x100dd266: mov eax,ds:[10136220h]
0x100dd26b: xor eax,esp
0x100dd26d: mov [esp+0B8h],eax
0x100dd274: mov eax,[esp+0C4h]
0x100dd27b: push ebp
0x100dd27c: xor ebp,ebp
0x100dd27e: push edi
0x100dd27f: mov edi,[esp+0C8h]
0x100dd286: cmp [edi+65B0h],ebp
0x100dd28c: mov [esp+0Ch],ebp
0x100dd290: mov [esp+1Ch],ebp
0x100dd294: je near ptr 100DD676h
0x100dd29a: cmp eax,ebp
0x100dd29c: push ebx
0x100dd29d: push esi
0x100dd29e: je short 100DD2B8h
0x100dd2a0: push 10129514h
0x100dd2a5: push eax
0x100dd2a6: call dword ptr ds:[100FE430h]
0x100dd2ac: add esp,8
0x100dd2af: mov [esp+14h],eax
0x100dd2b3: jmp near ptr 100DD3BAh
0x100dd2b8: mov esi,20h
0x100dd2bd: push esi
0x100dd2be: call dword ptr ds:[100FE550h]
0x100dd2c4: mov ebx,eax
0x100dd2c6: add esp,4
0x100dd2c9: cmp ebx,ebp
0x100dd2cb: je near ptr 100DD674h
0x100dd2d1: mov ebp,ds:[100FE4D8h]
0x100dd2d7: lea eax,[esi-10h]
0x100dd2da: push eax
0x100dd2db: push ebx
0x100dd2dc: call dword ptr ds:[100FE438h]
0x100dd2e2: add esp,8
0x100dd2e5: test eax,eax
0x100dd2e7: jne short 100DD315h
0x100dd2e9: push ebx
0x100dd2ea: call dword ptr ds:[100FE554h]
0x100dd2f0: add esp,4
0x100dd2f3: call ebp
0x100dd2f5: cmp dword ptr [eax],22h
0x100dd2f8: jne near ptr 100DD674h
0x100dd2fe: add esi,esi
0x100dd300: push esi
0x100dd301: call dword ptr ds:[100FE550h]
0x100dd307: mov ebx,eax
0x100dd309: add esp,4
0x100dd30c: test ebx,ebx
0x100dd30e: jne short 100DD2D7h
0x100dd310: jmp near ptr 100DD674h
0x100dd315: cmp byte ptr [ebx+1],3Ah
0x100dd319: jne short 100DD32Dh
0x100dd31b: add esi,0FFFFFFFEh
0x100dd31e: push esi
0x100dd31f: lea ecx,[ebx+2]
0x100dd322: push ecx
0x100dd323: push ebx
0x100dd324: call dword ptr ds:[100FE498h]
0x100dd32a: add esp,0Ch
0x100dd32d: cmp byte ptr [ebx],0
0x100dd330: mov eax,ebx
```

```
0x100dd332: je short 100DD344h
0x100dd334: cmp byte ptr [eax],5Ch
0x100dd337: jne short 100DD33Ch
0x100dd339: mov byte ptr [eax],2Fh
0x100dd33c: add eax,1
0x100dd33f: cmp byte ptr [eax],0
0x100dd342: jne short 100DD334h
0x100dd344: mov eax,ebx
0x100dd346: lea edx,[eax+1]
0x100dd349: lea esp,[esp]
0x100dd350: mov cl,[eax]
0x100dd352: add eax,1
0x100dd355: test cl,cl
0x100dd357: jne short 100DD350h
0x100dd359: sub eax,edx
0x100dd35b: mov esi,eax
0x100dd35d: add esi,ebx
0x100dd35f: cmp byte ptr [esi-1],2Fh
0x100dd363: jne short 100DD368h
0x100dd365: add esi,0FFFFFFFFh
0x100dd368: cmp byte ptr [ebx],2Fh
0x100dd36b: jne short 100DD3B0h
0x100dd36d: mov ebp,ds:[100FE430h]
0x100dd373: mov edx,ds:[1012E284h]
0x100dd379: mov [esi],edx
0x100dd37b: mov eax,ds:[1012E288h]
0x100dd380: mov [esi+4],eax
0x100dd383: mov ecx,ds:[1012E28Ch]
0x100dd389: push 10129514h
0x100dd38e: push ebx
0x100dd38f: mov [esi+8],ecx
0x100dd392: call ebp
0x100dd394: add esp,8
0x100dd397: test eax,eax
0x100dd399: mov [esp+14h],eax
0x100dd39d: jne short 100DD3B0h
0x100dd39f: cmp esi,ebx
0x100dd3a1: je short 100DD3B0h
0x100dd3a3: sub esi,1
0x100dd3a6: cmp byte ptr [esi],2Fh
0x100dd3a9: jne short 100DD3A3h
0x100dd3ab: cmp byte ptr [ebx],2Fh
0x100dd3ae: je short 100DD373h
0x100dd3b0: push ebx
0x100dd3b1: call dword ptr ds:[100FE554h]
0x100dd3b7: add esp,4
0x100dd3ba: mov esi,[esp+14h]
0x100dd3be: test esi,esi
0x100dd3c0: je near ptr 100DD674h
0x100dd3c6: push esi
0x100dd3c7: lea edx,[esp+4Ch]
0x100dd3cb: push 80h
0x100dd3d0: push edx
0x100dd3d1: call dword ptr ds:[100FE518h]
0x100dd3d7: add esp,0Ch
0x100dd3da: test eax,eax
0x100dd3dc: je near ptr 100DD66Ah
0x100dd3e2: lea eax,[esp+48h]
0x100dd3e6: push 23h
0x100dd3e8: push eax
0x100dd3e9: call dword ptr ds:[100FE454h]
0x100dd3ef: add esp,8
0x100dd3f2: test eax,eax
0x100dd3f4: je short 100DD3F9h
0x100dd3f6: mov byte ptr [eax],0
0x100dd3f9: lea ecx,[esp+44h]
0x100dd3fd: push ecx
```

```
0x100dd3fe: lea edx,[esp+20h]
0x100dd402: push edx
0x100dd403: lea eax,[esp+18h]
0x100dd407: push eax
0x100dd408: lea ecx,[esp+54h]
0x100dd40c: push 1012E278h
0x100dd411: push ecx
0x100dd412: call dword ptr ds:[100FE4A8h]
0x100dd418: add esp,14h
0x100dd41b: cmp eax,3
0x100dd41e: jne near ptr 100DD62Fh
0x100dd424: movzx eax,word ptr [edi+66B6h]
0x100dd42b: mov esi,[esp+10h]
0x100dd42f: cmp esi,eax
0x100dd431: jae near ptr 100DD62Fh
0x100dd437: movzx eax,word ptr [edi+66B4h]
0x100dd43e: mov ebp,[esp+1Ch]
0x100dd442: cmp ebp,eax
0x100dd444: jae near ptr 100DD62Fh
0x100dd44a: mov eax,[esp+44h]
0x100dd44e: cdq
0x100dd44f: cmp edx,[edi+65A4h]
0x100dd455: jg near ptr 100DD62Fh
0x100dd45b: jl short 100DD469h
0x100dd45d: cmp eax,[edi+65A0h]
0x100dd463: ja near ptr 100DD62Fh
0x100dd469: xor eax,eax
0x100dd46b: mov ecx,esi
0x100dd46d: mov [esp+18h],eax
0x100dd471: mov [esp+28h],eax
0x100dd475: lea eax,[ebp+1]
0x100dd478: lea edx,[ebp-1]
0x100dd47b: sub ecx,ebp
0x100dd47d: mov dword ptr [esp+20h],1
0x100dd485: mov [esp+30h],eax
0x100dd489: mov [esp+34h],edx
0x100dd48d: mov [esp+40h],ecx
0x100dd491: cmp dword ptr [esp+18h],0
0x100dd496: jne near ptr 100DD5A1h
0x100dd49c: cmp edx,eax
0x100dd49e: mov ebx,edx
0x100dd4a0: jg near ptr 100DD583h
0x100dd4a6: lea esi,[ecx+edx]
0x100dd4a9: add ecx,eax
0x100dd4ab: mov [esp+3Ch],esi
0x100dd4af: mov [esp+38h],ecx
0x100dd4b3: mov esi,[esp+3Ch]
0x100dd4b7: cmp esi,ecx
0x100dd4b9: jg near ptr 100DD570h
0x100dd4bf: nop
0x100dd4c0: movzx eax,word ptr [edi+66B4h]
0x100dd4c7: cmp ebx,eax
0x100dd4c9: jae near ptr 100DD559h
0x100dd4cf: movzx eax,word ptr [edi+66B6h]
0x100dd4d6: cmp esi,eax
0x100dd4d8: jae near ptr 100DD559h
0x100dd4de: cmp ebx,ebp
0x100dd4e0: jne short 100DD4ECh
0x100dd4e2: cmp esi,[esp+10h]
0x100dd4e6: je near ptr 100DD559h
0x100dd4ec: push esi
0x100dd4ed: push ebx
0x100dd4ee: push edi
0x100dd4ef: call 100D2D10h
0x100dd4f4: mov edx,[esp+1Ch]
0x100dd4f8: push edx
0x100dd4f9: push ebp
```

```
0x100dd4fa: push edi
0x100dd4fb: mov [esp+44h],eax
0x100dd4ff: call 100D2D10h
0x100dd504: mov ecx,[esp+44h]
0x100dd508: add esp,18h
0x100dd50b: cmp ecx,eax
0x100dd50d: jne short 100DD555h
0x100dd50f: movzx ebp,word ptr [edi+66BCh]
0x100dd516: push esi
0x100dd517: push ebx
0x100dd518: push edi
0x100dd519: call 100D2D10h
0x100dd51e: mov ecx,ebp
0x100dd520: mov edx,ebx
0x100dd522: sar edx,cl
0x100dd524: movzx ecx,word ptr [edi+66C0h]
0x100dd52b: imul edx,ecx
0x100dd52e: mov ecx,esi
0x100dd530: mov [esp+38h],ecx
0x100dd534: mov ecx,ebp
0x100dd536: mov ebp,esi
0x100dd538: sar ebp,cl
0x100dd53a: add esp,0Ch
0x100dd53d: add edx,ebp
0x100dd53f: lea edx,[eax+edx*4]
0x100dd542: mov eax,[edi+66E0h]
0x100dd548: movzx ecx,word ptr [eax+edx*2]
0x100dd54c: add [esp+28h],ecx
0x100dd550: add dword ptr [esp+18h],1
0x100dd555: mov ebp,[esp+1Ch]
0x100dd559: mov ecx,[esp+38h]
0x100dd55d: add esi,1
0x100dd560: cmp esi,ecx
0x100dd562: jle near ptr 100DD4C0h
0x100dd568: mov edx,[esp+34h]
0x100dd56c: mov eax,[esp+30h]
0x100dd570: add ebx,1
0x100dd573: cmp ebx,eax
0x100dd575: jle near ptr 100DD4B3h
0x100dd57b: mov esi,[esp+10h]
0x100dd57f: mov ecx,[esp+40h]
0x100dd583: add dword ptr [esp+20h],1
0x100dd588: add eax,1
0x100dd58b: sub edx,1
0x100dd58e: cmp dword ptr [esp+20h],3
0x100dd593: mov [esp+30h],eax
0x100dd597: mov [esp+34h],edx
0x100dd59b: jl near ptr 100DD491h
0x100dd5a1: mov eax,[esp+28h]
0x100dd5a5: cdq
0x100dd5a6: idiv dword ptr [esp+18h]
0x100dd5aa: push esi
0x100dd5ab: push ebp
0x100dd5ac: push edi
0x100dd5ad: mov ebx,eax
0x100dd5af: call 100D2D10h
0x100dd5b4: movzx ecx,word ptr [edi+66BCh]
0x100dd5bb: movzx edx,word ptr [edi+66C0h]
0x100dd5c2: sar ebp,cl
0x100dd5c4: sar esi,cl
0x100dd5c6: mov ecx,[edi+66E0h]
0x100dd5cc: add esp,0Ch
0x100dd5cf: imul ebp,edx
0x100dd5d2: add ebp,esi
0x100dd5d4: lea eax,[eax+ebp*4]
0x100dd5d7: mov [ecx+eax*2],bx
0x100dd5db: cmp dword ptr [edi+0BCh],0
```

```
0x100dd5e2: je short 100DD62Fh
0x100dd5e4: mov eax,[esp+24h]
0x100dd5e8: mov edx,eax
0x100dd5ea: add eax,1
0x100dd5ed: test edx,edx
0x100dd5ef: mov [esp+24h],eax
0x100dd5f3: jne short 100DD60Dh
0x100dd5f5: push 1012E260h
0x100dd5fa: call dword ptr ds:[100FE4C4h]
0x100dd600: add eax,40h
0x100dd603: push eax
0x100dd604: call dword ptr ds:[100FE4C8h]
0x100dd60a: add esp,8
0x100dd60d: mov eax,[esp+1Ch]
0x100dd611: mov ecx,[esp+10h]
0x100dd615: push eax
0x100dd616: push ecx
0x100dd617: push 1012E258h
0x100dd61c: call dword ptr ds:[100FE4C4h]
0x100dd622: add eax,40h
0x100dd625: push eax
0x100dd626: call dword ptr ds:[100FE4C8h]
0x100dd62c: add esp,10h
0x100dd62f: mov esi,[esp+14h]
0x100dd633: push esi
0x100dd634: lea edx,[esp+4Ch]
0x100dd638: push 80h
0x100dd63d: push edx
0x100dd63e: call dword ptr ds:[100FE518h]
0x100dd644: add esp,0Ch
0x100dd647: test eax,eax
0x100dd649: jne near ptr 100DD3E2h
0x100dd64f: cmp [esp+24h],eax
0x100dd653: je short 100DD66Ah
0x100dd655: call dword ptr ds:[100FE4C4h]
0x100dd65b: add eax,40h
0x100dd65e: push eax
0x100dd65f: push 0Ah
0x100dd661: call dword ptr ds:[100FE520h]
0x100dd667: add esp,8
0x100dd66a: push esi
0x100dd66b: call dword ptr ds:[100FE540h]
0x100dd671: add esp,4
0x100dd674: pop esi
0x100dd675: pop ebx
0x100dd676: mov ecx,[esp+0C0h]
0x100dd67d: pop edi
0x100dd67e: pop ebp
0x100dd67f: xor ecx,esp
0x100dd681: call 10022326h
0x100dd686: add esp,0BCh
0x100dd68c: ret
```

`sub_100dd260` is called by sub_10016040. It works with the text " %d,%d", "Fixed dead pixels at:", "%d %d %d". It delegates to sub_10022326, sub_100d2d10 (2 callees).

12.1269 sub_100dd690 at 0x100dd690 (221 instructions)

```
0x100dd690: sub esp,1Ch
0x100dd693: push ebx
0x100dd694: push ebp
0x100dd695: push esi
0x100dd696: mov esi,[esp+30h]
0x100dd69a: push edi
0x100dd69b: xor edi,edi
0x100dd69d: push 100FE928h
```

```
0x100dd6a2: push esi
0x100dd6a3: mov [esp+28h],edi
0x100dd6a7: mov [esp+2Ch],edi
0x100dd6ab: mov [esp+30h],edi
0x100dd6af: mov [esp+18h],edi
0x100dd6b3: xor ebp,ebp
0x100dd6b5: call dword ptr ds:[100FE430h]
0x100dd6bb: mov ebx,eax
0x100dd6bd: add esp,8
0x100dd6c0: cmp ebx,edi
0x100dd6c2: mov [esp+18h],ebx
0x100dd6c6: jne short 100DD6DAh
0x100dd6c8: push esi
0x100dd6c9: call dword ptr ds:[100FE428h]
0x100dd6cf: add esp,4
0x100dd6d2: pop edi
0x100dd6d3: pop esi
0x100dd6d4: pop ebp
0x100dd6d5: pop ebx
0x100dd6d6: add esp,1Ch
0x100dd6d9: ret
0x100dd6da: push ebx
0x100dd6db: call dword ptr ds:[100FE42Ch]
0x100dd6e1: add esp,4
0x100dd6e4: cmp eax,50h
0x100dd6e7: jne near ptr 100DD94Ah
0x100dd6ed: push ebx
0x100dd6ee: call dword ptr ds:[100FE42Ch]
0x100dd6f4: add esp,4
0x100dd6f7: cmp eax,35h
0x100dd6fa: jne near ptr 100DD94Ah
0x100dd700: cmp edi,3
0x100dd703: jge near ptr 100DD79Ch
0x100dd709: push ebx
0x100dd70a: call dword ptr ds:[100FE42Ch]
0x100dd710: mov esi,eax
0x100dd712: add esp,4
0x100dd715: cmp esi,0FFFFFFFFh
0x100dd718: je short 100DD793h
0x100dd71a: cmp esi,23h
0x100dd71d: jne short 100DD729h
0x100dd71f: mov dword ptr [esp+10h],1
0x100dd727: jmp short 100DD700h
0x100dd729: cmp esi,0Ah
0x100dd72c: jne short 100DD738h
0x100dd72e: mov dword ptr [esp+10h],0
0x100dd736: jmp short 100DD73Fh
0x100dd738: cmp dword ptr [esp+10h],0
0x100dd73d: jne short 100DD700h
0x100dd73f: push esi
0x100dd740: call dword ptr ds:[100FE45Ch]
0x100dd746: add esp,4
0x100dd749: test eax,eax
0x100dd74b: je short 100DD754h
0x100dd74d: mov ebp,1
0x100dd752: jmp short 100DD758h
0x100dd754: test ebp,ebp
0x100dd756: je short 100DD700h
0x100dd758: push esi
0x100dd759: call dword ptr ds:[100FE45Ch]
0x100dd75f: add esp,4
0x100dd762: test eax,eax
0x100dd764: je short 100DD777h
0x100dd766: mov eax,[esp+edi*4+20h]
0x100dd76a: lea eax,[eax+eax*4]
0x100dd76d: lea ecx,[esi+eax*2-30h]
0x100dd771: mov [esp+edi*4+20h],ecx
```

```
0x100dd775: jmp short 100DD700h
0x100dd777: push esi
0x100dd778: call dword ptr ds:[100FE460h]
0x100dd77e: add esp,4
0x100dd781: test eax,eax
0x100dd783: je near ptr 100DD946h
0x100dd789: xor ebp,ebp
0x100dd78b: add edi,1
0x100dd78e: jmp near ptr 100DD700h
0x100dd793: cmp edi,3
0x100dd796: jl near ptr 100DD946h
0x100dd79c: mov esi,[esp+30h]
0x100dd7a0: movzx eax,word ptr [esi+66B6h]
0x100dd7a7: cmp [esp+20h],eax
0x100dd7ab: jne near ptr 100DD93Ah
0x100dd7b1: movzx edx,word ptr [esi+66B4h]
0x100dd7b8: cmp [esp+24h],edx
0x100dd7bc: jne near ptr 100DD93Ah
0x100dd7c2: cmp dword ptr [esp+28h],0FFFFh
0x100dd7ca: jne near ptr 100DD93Ah
0x100dd7d0: push 2
0x100dd7d2: push eax
0x100dd7d3: call dword ptr ds:[100FE4FCh]
0x100dd7d9: mov edi,eax
0x100dd7db: push 1012E2D0h
0x100dd7e0: push edi
0x100dd7e1: push esi
0x100dd7e2: mov [esp+28h],edi
0x100dd7e6: call 100D2E20h
0x100dd7eb: xor ebp,ebp
0x100dd7ed: add esp,14h
0x100dd7f0: cmp [esi+66B4h],bp
0x100dd7f7: mov [esp+10h],ebp
0x100dd7fb: jbe near ptr 100DD917h
0x100dd801: movzx eax,word ptr [esi+66B6h]
0x100dd808: push ebx
0x100dd809: push eax
0x100dd80a: push 2
0x100dd80c: push edi
0x100dd80d: call dword ptr ds:[100FE53Ch]
0x100dd813: xor ebx,ebx
0x100dd815: add esp,10h
0x100dd818: cmp [esi+66B6h],bx
0x100dd81f: jbe near ptr 100DD8FDh
0x100dd825: mov eax,ebp
0x100dd827: and eax,7
0x100dd82a: add eax,eax
0x100dd82c: mov [esp+1Ch],eax
0x100dd830: jmp short 100DD836h
0x100dd832: mov eax,[esp+1Ch]
0x100dd836: movzx ecx,word ptr [esi+66BCh]
0x100dd83d: movzx edx,word ptr [esi+66C0h]
0x100dd844: sar ebp,cl
0x100dd846: mov edi,ebx
0x100dd848: and edi,1
0x100dd84b: add edi,eax
0x100dd84d: imul ebp,edx
0x100dd850: mov edx,[esi+65B0h]
0x100dd856: mov eax,ebx
0x100dd858: sar eax,cl
0x100dd85a: add edi,edi
0x100dd85c: mov ecx,edi
0x100dd85e: shr edx,cl
0x100dd860: mov ecx,[esi+66E0h]
0x100dd866: add ebp,eax
0x100dd868: and edx,3
0x100dd86b: lea eax,[edx+ebp*4]
```

```
0x100dd86e: mov edx,[esp+14h]
0x100dd872: lea ebp,[ecx+eax*2]
0x100dd875: movzx eax,word ptr [edx+ebx*2]
0x100dd879: push eax
0x100dd87a: call dword ptr ds:[100FE6B8h]
0x100dd880: movzx edx,word ptr [ebp]
0x100dd884: movzx ecx,ax
0x100dd887: sub edx,ecx
0x100dd889: test edx,edx
0x100dd88b: jle short 100DD8DDh
0x100dd88d: movzx ecx,word ptr [esi+66BCh]
0x100dd894: mov eax,[esp+10h]
0x100dd898: movzx edx,word ptr [esi+66C0h]
0x100dd89f: sar eax,cl
0x100dd8a1: imul eax,edx
0x100dd8a4: mov edx,ebx
0x100dd8a6: sar edx,cl
0x100dd8a8: mov ecx,edi
0x100dd8aa: add eax,edx
0x100dd8ac: mov edx,[esi+65B0h]
0x100dd8b2: shr edx,cl
0x100dd8b4: mov ecx,[esi+66E0h]
0x100dd8ba: and edx,3
0x100dd8bd: lea eax,[edx+eax*4]
0x100dd8c0: mov edx,[esp+14h]
0x100dd8c4: lea ebp,[ecx+eax*2]
0x100dd8c7: movzx eax,word ptr [edx+ebx*2]
0x100dd8cb: push eax
0x100dd8cc: call dword ptr ds:[100FE6B8h]
0x100dd8d2: movzx ecx,ax
0x100dd8d5: movzx eax,word ptr [ebp]
0x100dd8d9: sub eax,ecx
0x100dd8db: jmp short 100DD8DFh
0x100dd8dd: xor eax,eax
0x100dd8df: mov [ebp],ax
0x100dd8e3: movzx edx,word ptr [esi+66B6h]
0x100dd8ea: mov ebp,[esp+10h]
0x100dd8ee: add ebx,1
0x100dd8f1: cmp ebx,edx
0x100dd8f3: jl near ptr 100DD832h
0x100dd8f9: mov edi,[esp+14h]
0x100dd8fd: movzx eax,word ptr [esi+66B4h]
0x100dd904: mov ebx,[esp+18h]
0x100dd908: add ebp,1
0x100dd90b: cmp ebp,eax
0x100dd90d: mov [esp+10h],ebp
0x100dd911: jl near ptr 100DD801h
0x100dd917: push ebx
0x100dd918: call dword ptr ds:[100FE540h]
0x100dd91e: push edi
0x100dd91f: call dword ptr ds:[100FE554h]
0x100dd925: add esp,8
0x100dd928: pop edi
0x100dd929: mov dword ptr [esi+65F8h],0
0x100dd933: pop esi
0x100dd934: pop ebp
0x100dd935: pop ebx
0x100dd936: add esp,1Ch
0x100dd939: ret
0x100dd93a: mov ecx,[esp+34h]
0x100dd93e: push ecx
0x100dd93f: push 1012E2B0h
0x100dd944: jmp short 100DD950h
0x100dd946: mov esi,[esp+34h]
0x100dd94a: push esi
0x100dd94b: push 1012E290h
0x100dd950: call dword ptr ds:[100FE4C4h]
```

```
0x100dd956: add eax,40h
0x100dd959: push eax
0x100dd95a: call dword ptr ds:[100FE4C8h]
0x100dd960: push ebx
0x100dd961: call dword ptr ds:[100FE540h]
0x100dd967: add esp,10h
0x100dd96a: pop edi
0x100dd96b: pop esi
0x100dd96c: pop ebp
0x100dd96d: pop ebx
0x100dd96e: add esp,1Ch
0x100dd971: ret
```

`sub_100dd690` is called by sub_10016040. It delegates to sub_100d2e20 (1 callees).

12.1270 sub_100dd980 at 0x100dd980 (223 instructions)

```
0x100dd980: push ebp
0x100dd981: mov ebp,esp
0x100dd983: and esp,0FFFFFFF8h
0x100dd986: sub esp,0A4h
0x100dd98c: mov eax,[ebp+8]
0x100dd98f: push ebx
0x100dd990: add eax,30h
0x100dd993: mov [esp+8],eax
0x100dd997: push esi
0x100dd998: lea ecx,[esp+1Ch]
0x100dd99c: lea eax,[esp+24h]
0x100dd9a0: push edi
0x100dd9a1: mov ebx,3
0x100dd9a6: mov [esp+1Ch],ecx
0x100dd9aa: mov [esp+14h],eax
0x100dd9ae: xor edx,edx
0x100dd9b0: test ebx,ebx
0x100dd9b2: sete dl
0x100dd9b5: xor ecx,ecx
0x100dd9b7: cmp ebx,1
0x100dd9ba: sete cl
0x100dd9bd: mov esi,[esp+1Ch]
0x100dd9c1: mov dword ptr [esp+18h],0
0x100dd9c9: mov [esp+0Ch],edx
0x100dd9cd: fild dword ptr [esp+0Ch]
0x100dd9d1: xor edx,edx
0x100dd9d3: cmp ebx,2
0x100dd9d6: sete dl
0x100dd9d9: mov [esp+0Ch],ecx
0x100dd9dd: fstp qword ptr [eax-8]
0x100dd9e0: fild dword ptr [esp+0Ch]
0x100dd9e4: xor ecx,ecx
0x100dd9e6: cmp ebx,3
0x100dd9e9: sete cl
0x100dd9ec: fstp qword ptr [eax]
0x100dd9ee: mov [esp+0Ch],edx
0x100dd9f2: fild dword ptr [esp+0Ch]
0x100dd9f6: xor edx,edx
0x100dd9f8: cmp ebx,4
0x100dd9fb: sete dl
0x100dd9fe: mov [esp+0Ch],ecx
0x100dda02: fstp qword ptr [eax+8]
0x100dda05: fild dword ptr [esp+0Ch]
0x100dda09: xor ecx,ecx
0x100dda0b: cmp ebx,5
0x100dda0e: sete cl
0x100dda11: fstp qword ptr [eax+10h]
0x100dda14: mov [esp+0Ch],edx
0x100dda18: fild dword ptr [esp+0Ch]
```

```
0x100dda1c: mov [esp+0Ch],ecx
0x100dda20: fstp qword ptr [eax+18h]
0x100dda23: fild dword ptr [esp+0Ch]
0x100dda27: mov ecx,[ebp+8]
0x100dda2a: add ecx,30h
0x100dda2d: mov [esp+0Ch],ecx
0x100dda31: fstp qword ptr [eax+20h]
0x100dda34: mov edx,[ebp+10h]
0x100dda37: xor edi,edi
0x100dda39: cmp edx,4
0x100dda3c: jl short 100DDA8Ah
0x100dda3e: mov ecx,[esp+10h]
0x100dda42: mov eax,[esp+0Ch]
0x100dda46: add edx,0FFFFFFFCh
0x100dda49: shr edx,2
0x100dda4c: add edx,1
0x100dda4f: lea edi,[edx*4]
0x100dda56: fld qword ptr [eax-30h]
0x100dda59: add eax,60h
0x100dda5c: fmul qword ptr [ecx-30h]
0x100dda5f: add ecx,60h
0x100dda62: sub edx,1
0x100dda65: fadd qword ptr [esi]
0x100dda67: fld qword ptr [eax-78h]
0x100dda6a: fmul qword ptr [ecx-78h]
0x100dda6d: faddp
0x100dda6f: fld qword ptr [ecx-60h]
0x100dda72: fmul qword ptr [eax-60h]
0x100dda75: faddp
0x100dda77: fld qword ptr [eax-48h]
0x100dda7a: fmul qword ptr [ecx-48h]
0x100dda7d: faddp
0x100dda7f: fstp qword ptr [esi]
0x100dda81: jne short 100DDA56h
0x100dda83: mov edx,[ebp+10h]
0x100dda86: mov eax,[esp+14h]
0x100dda8a: cmp edi,edx
0x100dda8c: jge short 100DDAC8h
0x100dda8e: mov ecx,[ebp+8]
0x100dda91: lea eax,[edi+edi*2]
0x100dda94: lea edx,[ebx+eax]
0x100dda97: lea edx,[ecx+edx*8-18h]
0x100dda9b: mov ecx,[esp+18h]
0x100dda9f: add eax,ecx
0x100ddaa1: mov ecx,[ebp+8]
0x100ddaa4: lea ecx,[ecx+eax*8]
0x100ddaa7: mov eax,[ebp+10h]
0x100ddaaa: sub eax,edi
0x100ddaac: lea esp,[esp]
0x100ddab0: fld qword ptr [edx]
0x100ddab2: add ecx,18h
0x100ddab5: fmul qword ptr [ecx-18h]
0x100ddab8: add edx,18h
0x100ddabb: sub eax,1
0x100ddabe: fadd qword ptr [esi]
0x100ddac0: fstp qword ptr [esi]
0x100ddac2: jne short 100DDAB0h
0x100ddac4: mov eax,[esp+14h]
0x100ddac8: mov ecx,[esp+18h]
0x100ddacc: add dword ptr [esp+0Ch],8
0x100ddad1: add ecx,1
0x100ddad4: add esi,8
0x100ddad7: cmp ecx,3
0x100ddada: mov [esp+18h],ecx
0x100ddade: jl near ptr 100DDA34h
0x100ddae4: add dword ptr [esp+1Ch],30h
0x100ddae9: add dword ptr [esp+10h],8
```

```
0x100ddaee: add ebx,1
0x100ddaf1: add eax,30h
0x100ddaf4: lea edx,[ebx-3]
0x100ddaf7: cmp edx,3
0x100ddafa: mov [esp+14h],eax
0x100ddafe: jl near ptr 100DD9AEh
0x100ddb04: lea eax,[esp+20h]
0x100ddb08: mov [esp+10h],eax
0x100ddb0c: mov ebx,eax
0x100ddb0e: xor edi,edi
0x100ddb10: lea eax,[esp+28h]
0x100ddb14: fld qword ptr [ebx]
0x100ddb16: mov esi,[esp+10h]
0x100ddb1a: fld qword ptr [eax-8]
0x100ddb1d: xor edx,edx
0x100ddb1f: fdiv st,st(1)
0x100ddb21: lea ecx,[esp+28h]
0x100ddb25: fstp qword ptr [eax-8]
0x100ddb28: fld qword ptr [eax]
0x100ddb2a: fdiv st,st(1)
0x100ddb2c: fstp qword ptr [eax]
0x100ddb2e: fld qword ptr [eax+8]
0x100ddb31: fdiv st,st(1)
0x100ddb33: fstp qword ptr [eax+8]
0x100ddb36: fld qword ptr [eax+10h]
0x100ddb39: fdiv st,st(1)
0x100ddb3b: fstp qword ptr [eax+10h]
0x100ddb3e: fld qword ptr [eax+18h]
0x100ddb41: fdiv st,st(1)
0x100ddb43: fstp qword ptr [eax+18h]
0x100ddb46: fdivr qword ptr [eax+20h]
0x100ddb49: fstp qword ptr [eax+20h]
0x100ddb4c: lea esp,[esp]
0x100ddb50: cmp edx,edi
0x100ddb52: je short 100DDB93h
0x100ddb54: fld qword ptr [esi]
0x100ddb56: fld qword ptr [eax-8]
0x100ddb59: fmul st,st(1)
0x100ddb5b: fsubr qword ptr [ecx-8]
0x100ddb5e: fstp qword ptr [ecx-8]
0x100ddb61: fld qword ptr [eax]
0x100ddb63: fmul st,st(1)
0x100ddb65: fsubr qword ptr [ecx]
0x100ddb67: fstp qword ptr [ecx]
0x100ddb69: fld qword ptr [eax+8]
0x100ddb6c: fmul st,st(1)
0x100ddb6e: fsubr qword ptr [ecx+8]
0x100ddb71: fstp qword ptr [ecx+8]
0x100ddb74: fld qword ptr [eax+10h]
0x100ddb77: fmul st,st(1)
0x100ddb79: fsubr qword ptr [ecx+10h]
0x100ddb7c: fstp qword ptr [ecx+10h]
0x100ddb7f: fld qword ptr [eax+18h]
0x100ddb82: fmul st,st(1)
0x100ddb84: fsubr qword ptr [ecx+18h]
0x100ddb87: fstp qword ptr [ecx+18h]
0x100ddb8a: fmul qword ptr [eax+20h]
0x100ddb8d: fsubr qword ptr [ecx+20h]
0x100ddb90: fstp qword ptr [ecx+20h]
0x100ddb93: add edx,1
0x100ddb96: add esi,30h
0x100ddb99: add ecx,30h
0x100ddb9c: cmp edx,3
0x100ddb9f: jl short 100DDB50h
0x100ddba1: add dword ptr [esp+10h],8
0x100ddba6: add edi,1
0x100ddba9: add ebx,38h
```

```
0x100ddbac: add eax,30h
0x100ddbaf: cmp edi,3
0x100ddbb2: jl near ptr 100DDB14h
0x100ddbb8: mov ecx,[ebp+10h]
0x100ddbbb: test ecx,ecx
0x100ddbbd: jle short 100DDC0Ch
0x100ddbbf: mov edx,[ebp+8]
0x100ddbc2: fldz
0x100ddbc4: mov eax,[ebp+0Ch]
0x100ddbc7: add edx,10h
0x100ddbca: mov edi,ecx
0x100ddbcc: lea ecx,[esp+40h]
0x100ddbd0: mov esi,3
0x100ddbd5: fst qword ptr [eax]
0x100ddbd7: add eax,8
0x100ddbda: fld qword ptr [ecx-8]
0x100ddbdd: add ecx,30h
0x100ddbe0: sub esi,1
0x100ddbe3: fmul qword ptr [edx-10h]
0x100ddbe6: fadd st,st(1)
0x100ddbe8: fst qword ptr [eax-8]
0x100ddbeb: fld qword ptr [edx-8]
0x100ddbee: fmul qword ptr [ecx-30h]
0x100ddbf1: faddp
0x100ddbf3: fst qword ptr [eax-8]
0x100ddbf6: fld qword ptr [ecx-28h]
0x100ddbf9: fmul qword ptr [edx]
0x100ddbfb: faddp
0x100ddbfd: fstp qword ptr [eax-8]
0x100ddc00: jne short 100DDBD5h
0x100ddc02: add edx,18h
0x100ddc05: sub edi,1
0x100ddc08: jne short 100DDBCCh
0x100ddc0a: fstp st(0)
0x100ddc0c: pop edi
0x100ddc0d: pop esi
0x100ddc0e: pop ebx
0x100ddc0f: mov esp,ebp
0x100ddc11: pop ebp
0x100ddc12: ret
```

`sub_100dd980` is called by sub_100ddc20, sub_100e67b0. It calls nothing else.

12.1271 sub_100ddc20 at 0x100ddc20 (100 instructions)

```
0x100ddc20: push ebp
0x100ddc21: mov ebp,esp
0x100ddc23: and esp,0FFFFFFF8h
0x100ddc26: fldz
0x100ddc28: sub esp,0C4h
0x100ddc2e: push ebx
0x100ddc2f: push esi
0x100ddc30: mov esi,[ebp+8]
0x100ddc33: mov ebx,[esi+66D0h]
0x100ddc39: test ebx,ebx
0x100ddc3b: push edi
0x100ddc3c: jle short 100DDC7Eh
0x100ddc3e: mov ecx,[ebp+0Ch]
0x100ddc41: lea edx,[esp+10h]
0x100ddc45: add ecx,10h
0x100ddc48: mov edi,ebx
0x100ddc4a: mov eax,1012A7A8h
0x100ddc4f: fld qword ptr [eax-18h]
0x100ddc52: add eax,8
0x100ddc55: fmul qword ptr [ecx-10h]
0x100ddc58: add edx,8
```

```
0x100ddc5b: cmp eax,1012A7C0h
0x100ddc60: fadd st,st(1)
0x100ddc62: fld qword ptr [ecx-8]
0x100ddc65: fmul qword ptr [eax-8]
0x100ddc68: faddp
0x100ddc6a: fld qword ptr [eax+10h]
0x100ddc6d: fmul qword ptr [ecx]
0x100ddc6f: faddp
0x100ddc71: fstp qword ptr [edx-8]
0x100ddc74: jl short 100DDC4Fh
0x100ddc76: add ecx,18h
0x100ddc79: sub edi,1
0x100ddc7c: jne short 100DDC4Ah
0x100ddc7e: xor edx,edx
0x100ddc80: test ebx,ebx
0x100ddc82: jle short 100DDCD3h
0x100ddc84: fld1
0x100ddc86: lea ecx,[esi+0E784h]
0x100ddc8c: lea eax,[esp+18h]
0x100ddc90: fld qword ptr [eax-8]
0x100ddc93: add edx,1
0x100ddc96: fadd st,st(2)
0x100ddc98: add ecx,4
0x100ddc9b: add eax,18h
0x100ddc9e: fadd qword ptr [eax-18h]
0x100ddca1: fadd qword ptr [eax-10h]
0x100ddca4: fld qword ptr [eax-20h]
0x100ddca7: fdiv st,st(1)
0x100ddca9: fstp qword ptr [eax-20h]
0x100ddcac: fld qword ptr [eax-18h]
0x100ddcaf: fdiv st,st(1)
0x100ddcb1: fstp qword ptr [eax-18h]
0x100ddcb4: fld qword ptr [eax-10h]
0x100ddcb7: fdiv st,st(1)
0x100ddcb9: fstp qword ptr [eax-10h]
0x100ddcbc: fstp dword ptr [esp+0Ch]
0x100ddcc0: fld dword ptr [esp+0Ch]
0x100ddcc4: fdivr st,st(1)
0x100ddcc6: fstp dword ptr [ecx-4]
0x100ddcc9: cmp edx,[esi+66D0h]
0x100ddccf: jl short 100DDC90h
0x100ddcd1: fstp st(1)
0x100ddcd3: mov eax,[esi+66D0h]
0x100ddcd9: fstp st(0)
0x100ddcdb: push eax
0x100ddcdc: lea ecx,[esp+74h]
0x100ddce0: push ecx
0x100ddce1: lea edx,[esp+18h]
0x100ddce5: push edx
0x100ddce6: call 100DD980h
0x100ddceb: add esp,0Ch
0x100ddcee: mov dword ptr [esi+6604h],0
0x100ddcf8: lea ebx,[esp+70h]
0x100ddcfc: lea edi,[esi+0E7C4h]
0x100ddd02: mov dword ptr [esp+0Ch],3
0x100ddd0a: lea ebx,[ebx]
0x100ddd10: xor edx,edx
0x100ddd12: cmp [esi+66D0h],edx
0x100ddd18: jle short 100DDD35h
0x100ddd1a: mov eax,edi
0x100ddd1c: mov ecx,ebx
0x100ddd1e: mov edi,edi
0x100ddd20: fld qword ptr [ecx]
0x100ddd22: add edx,1
0x100ddd25: fstp dword ptr [eax]
0x100ddd27: add eax,4
0x100ddd2a: add ecx,18h
```

```
0x100ddd2d: cmp edx,[esi+66D0h]
0x100ddd33: jl short 100DDD20h
0x100ddd35: add edi,10h
0x100ddd38: add ebx,8
0x100ddd3b: sub dword ptr [esp+0Ch],1
0x100ddd40: jne short 100DDD10h
0x100ddd42: pop edi
0x100ddd43: pop esi
0x100ddd44: pop ebx
0x100ddd45: mov esp,ebp
0x100ddd47: pop ebp
0x100ddd48: ret
```

`sub_100ddc20` is called by sub_100e64c0, sub_100e8a00. It delegates to sub_100dd980 (1 callees).

12.1272 sub_100ddd50 at 0x100ddd50 (369 instructions)

```
0x100ddd50: mov eax,[esp+0Ch]
0x100ddd54: fld qword ptr ds:[100FF110h]
0x100ddd5a: mov ecx,[esp+8]
0x100ddd5e: sub esp,34h
0x100ddd61: push ebx
0x100ddd62: mov ebx,[esp+3Ch]
0x100ddd66: push ebp
0x100ddd67: push esi
0x100ddd68: push edi
0x100ddd69: mov edi,[esp+58h]
0x100ddd6d: xor edx,edx
0x100ddd6f: cmp edi,4
0x100ddd72: jl near ptr 100DDED7h
0x100ddd78: lea ebp,[edi+3]
0x100ddd7b: imul ebp,eax
0x100ddd7e: lea ebp,[ecx+ebp*4]
0x100ddd81: mov [esp+2Ch],ebp
0x100ddd85: lea ebp,[eax+eax*2]
0x100ddd88: lea ebp,[ecx+ebp*4]
0x100ddd8b: mov [esp+28h],ebp
0x100ddd8f: lea ebp,[edi+2]
0x100ddd92: imul ebp,eax
0x100ddd95: lea ebp,[ecx+ebp*4]
0x100ddd98: mov [esp+24h],ebp
0x100ddd9c: lea ebp,[ecx+eax*8]
0x100ddd9f: mov [esp+20h],ebp
0x100ddda3: lea ebp,[edi+1]
0x100ddda6: imul ebp,eax
0x100ddda9: lea ebp,[ecx+ebp*4]
0x100dddac: mov [esp+1Ch],ebp
0x100dddb0: lea ebp,[ecx+eax*4]
0x100dddb3: mov [esp+18h],ebp
0x100dddb7: mov ebp,eax
0x100dddb9: imul ebp,edi
0x100dddbc: lea ebp,[ecx+ebp*4]
0x100dddbf: mov [esp+14h],ebp
0x100dddc3: mov [esp+30h],ebp
0x100dddc7: lea edx,[edi-3]
0x100dddca: mov ebp,edx
0x100dddcc: imul ebp,eax
0x100dddcf: lea ebp,[ecx+ebp*4]
0x100dddd2: mov [esp+3Ch],ebp
0x100dddd6: lea ebp,[edi-2]
0x100dddd9: imul ebp,eax
0x100ddddc: lea ebp,[ecx+ebp*4]
0x100ddddf: mov [esp+38h],ebp
0x100ddde3: lea ebp,[edi*4-4]
0x100dddea: imul ebp,eax
0x100ddded: add ebx,8
```

```
0x100dddf0: add edx,0FFFFFFFFh
0x100dddf3: shr edx,2
0x100dddf6: mov [esp+58h],ebx
0x100dddfa: add edx,1
0x100dddfd: mov ebx,eax
0x100dddff: neg ebx
0x100dde01: mov [esp+40h],edx
0x100dde05: mov esi,eax
0x100dde07: add ebp,ecx
0x100dde09: add edx,edx
0x100dde0b: shl esi,4
0x100dde0e: shl ebx,4
0x100dde11: mov [esp+10h],ecx
0x100dde15: mov [esp+34h],ebp
0x100dde19: add edx,edx
0x100dde1b: mov ebp,[esp+10h]
0x100dde1f: fld dword ptr [ebp]
0x100dde22: mov ebp,[esp+30h]
0x100dde26: fmul st,st(1)
0x100dde28: add [esp+10h],esi
0x100dde2c: add [esp+30h],ebx
0x100dde30: fadd dword ptr [ebp]
0x100dde33: mov ebp,[esp+14h]
0x100dde37: add [esp+14h],esi
0x100dde3b: fadd dword ptr [ebp]
0x100dde3e: mov ebp,[esp+58h]
0x100dde42: fstp dword ptr [ebp-8]
0x100dde45: mov ebp,[esp+18h]
0x100dde49: fld dword ptr [ebp]
0x100dde4c: mov ebp,[esp+34h]
0x100dde50: fmul st,st(1)
0x100dde52: add [esp+18h],esi
0x100dde56: add [esp+34h],ebx
0x100dde5a: fadd dword ptr [ebp]
0x100dde5d: mov ebp,[esp+1Ch]
0x100dde61: add [esp+1Ch],esi
0x100dde65: fadd dword ptr [ebp]
0x100dde68: mov ebp,[esp+58h]
0x100dde6c: fstp dword ptr [ebp-4]
0x100dde6f: mov ebp,[esp+20h]
0x100dde73: fld dword ptr [ebp]
0x100dde76: mov ebp,[esp+38h]
0x100dde7a: fmul st,st(1)
0x100dde7c: add [esp+20h],esi
0x100dde80: add [esp+38h],ebx
0x100dde84: fadd dword ptr [ebp]
0x100dde87: mov ebp,[esp+24h]
0x100dde8b: add [esp+24h],esi
0x100dde8f: fadd dword ptr [ebp]
0x100dde92: mov ebp,[esp+58h]
0x100dde96: fstp dword ptr [ebp]
0x100dde99: mov ebp,[esp+28h]
0x100dde9d: fld dword ptr [ebp]
0x100ddea0: mov ebp,[esp+3Ch]
0x100ddea4: add [esp+28h],esi
0x100ddea8: fmul st,st(1)
0x100ddeaa: add [esp+3Ch],ebx
0x100ddeae: fadd dword ptr [ebp]
0x100ddeb1: mov ebp,[esp+2Ch]
0x100ddeb5: add [esp+2Ch],esi
0x100ddeb9: fadd dword ptr [ebp]
0x100ddebc: mov ebp,[esp+58h]
0x100ddec0: add dword ptr [esp+58h],10h
0x100ddec5: sub dword ptr [esp+40h],1
0x100ddeca: fstp dword ptr [ebp+4]
0x100ddecd: jne near ptr 100DDE1Bh
0x100dded3: mov ebx,[esp+48h]
```

```
0x100dded7: cmp edx,edi
0x100dded9: jge short 100DDF49h
0x100ddedb: mov esi,edx
0x100ddedd: imul esi,eax
0x100ddee0: lea ebp,[edx+edi]
0x100ddee3: imul ebp,eax
0x100ddee6: lea esi,[ecx+esi*4]
0x100ddee9: mov [esp+3Ch],esi
0x100ddeed: lea ebp,[ecx+ebp*4]
0x100ddef0: mov [esp+38h],ebp
0x100ddef4: mov esi,eax
0x100ddef6: neg esi
0x100ddef8: add esi,esi
0x100ddefa: mov ebp,edi
0x100ddefc: sub ebp,edx
0x100ddefe: add esi,esi
0x100ddf00: imul ebp,eax
0x100ddf03: lea ebp,[ecx+ebp*4]
0x100ddf06: mov [esp+40h],esi
0x100ddf0a: mov [esp+58h],ebp
0x100ddf0e: jmp short 100DDF14h
0x100ddf10: mov esi,[esp+40h]
0x100ddf14: mov ebp,[esp+3Ch]
0x100ddf18: fld dword ptr [ebp]
0x100ddf1b: mov ebp,[esp+58h]
0x100ddf1f: add [esp+58h],esi
0x100ddf23: fmul st,st(1)
0x100ddf25: lea esi,[eax*4]
0x100ddf2c: add [esp+3Ch],esi
0x100ddf30: fadd dword ptr [ebp]
0x100ddf33: mov ebp,[esp+38h]
0x100ddf37: add [esp+38h],esi
0x100ddf3b: add edx,1
0x100ddf3e: cmp edx,edi
0x100ddf40: fadd dword ptr [ebp]
0x100ddf43: fstp dword ptr [ebx+edx*4-4]
0x100ddf47: jl short 100DDF10h
0x100ddf49: mov ebp,[esp+54h]
0x100ddf4d: lea esi,[edx+edi]
0x100ddf50: cmp esi,ebp
0x100ddf52: jge short 100DDFBAh
0x100ddf54: mov [esp+38h],esi
0x100ddf58: imul esi,eax
0x100ddf5b: lea esi,[ecx+esi*4]
0x100ddf5e: mov [esp+3Ch],esi
0x100ddf62: mov ebp,edx
0x100ddf64: imul ebp,eax
0x100ddf67: mov esi,edx
0x100ddf69: sub esi,edi
0x100ddf6b: imul esi,eax
0x100ddf6e: lea ebp,[ecx+ebp*4]
0x100ddf71: mov [esp+58h],ebp
0x100ddf75: lea ebp,[ecx+esi*4]
0x100ddf78: jmp short 100DDF7Eh
0x100ddf7a: mov ebp,[esp+40h]
0x100ddf7e: mov esi,[esp+58h]
0x100ddf82: fld dword ptr [esi]
0x100ddf84: mov esi,[esp+3Ch]
0x100ddf88: add dword ptr [esp+38h],1
0x100ddf8d: fmul st,st(1)
0x100ddf8f: add edx,1
0x100ddf92: fadd dword ptr [ebp]
0x100ddf95: fadd dword ptr [esi]
0x100ddf97: lea esi,[eax*4]
0x100ddf9e: add [esp+58h],esi
0x100ddfa2: add [esp+3Ch],esi
0x100ddfa6: add ebp,esi
```

```
0x100ddfa8: fstp dword ptr [ebx+edx*4-4]
0x100ddfac: mov [esp+40h],ebp
0x100ddfb0: mov ebp,[esp+54h]
0x100ddfb4: cmp [esp+38h],ebp
0x100ddfb8: jl short 100DDF7Ah
0x100ddfba: mov esi,ebp
0x100ddfbc: sub esi,edx
0x100ddfbe: cmp esi,4
0x100ddfc1: jl near ptr 100DE183h
0x100ddfc7: lea ebp,[edx+2]
0x100ddfca: lea ebx,[ebx+ebp*4]
0x100ddfcd: mov [esp+40h],ebp
0x100ddfd1: lea ebp,[edx+3]
0x100ddfd4: imul ebp,eax
0x100ddfd7: lea ebp,[ecx+ebp*4]
0x100ddfda: mov [esp+20h],ebp
0x100ddfde: mov ebp,[esp+40h]
0x100ddfe2: imul ebp,eax
0x100ddfe5: lea ebp,[ecx+ebp*4]
0x100ddfe8: mov [esp+30h],ebp
0x100ddfec: lea ebp,[edx+1]
0x100ddfef: imul ebp,eax
0x100ddff2: lea ebp,[ecx+ebp*4]
0x100ddff5: mov [esp+34h],ebp
0x100ddff9: mov ebp,edx
0x100ddffb: sub ebp,edi
0x100ddffd: mov [esp+3Ch],ebp
0x100de001: imul ebp,eax
0x100de004: lea ebp,[ecx+ebp*4]
0x100de007: mov [esp+2Ch],ebp
0x100de00b: mov ebp,[esp+3Ch]
0x100de00f: add ebp,3
0x100de012: imul ebp,eax
0x100de015: lea ebp,[ecx+ebp*4]
0x100de018: mov [esp+24h],ebp
0x100de01c: mov ebp,[esp+3Ch]
0x100de020: add ebp,2
0x100de023: imul ebp,eax
0x100de026: lea ebp,[ecx+ebp*4]
0x100de029: mov [esp+28h],ebp
0x100de02d: mov ebp,[esp+3Ch]
0x100de031: add ebp,1
0x100de034: imul ebp,eax
0x100de037: lea ebp,[ecx+ebp*4]
0x100de03a: mov [esp+3Ch],ebp
0x100de03e: mov ebp,[esp+54h]
0x100de042: lea ebp,[ebp+ebp-4]
0x100de046: sub ebp,edx
0x100de048: sub ebp,edi
0x100de04a: imul ebp,eax
0x100de04d: lea ebp,[ecx+ebp*4]
0x100de050: mov [esp+1Ch],ebp
0x100de054: mov ebp,[esp+54h]
0x100de058: add ebp,ebp
0x100de05a: sub ebp,edx
0x100de05c: sub ebp,edi
0x100de05e: mov [esp+40h],ebp
0x100de062: add ebp,0FFFFFFFBh
0x100de065: imul ebp,eax
0x100de068: lea ebp,[ecx+ebp*4]
0x100de06b: mov [esp+10h],ebp
0x100de06f: mov ebp,[esp+40h]
0x100de073: add ebp,0FFFFFFFDh
0x100de076: imul ebp,eax
0x100de079: lea ebp,[ecx+ebp*4]
0x100de07c: mov [esp+14h],ebp
0x100de080: mov ebp,[esp+40h]
```

```
0x100de084: add ebp,0FFFFFFFEh
0x100de087: mov [esp+58h],ebx
0x100de08b: imul ebp,eax
0x100de08e: mov ebx,edx
0x100de090: imul ebx,eax
0x100de093: lea ebp,[ecx+ebp*4]
0x100de096: mov [esp+18h],ebp
0x100de09a: mov ebp,[esp+54h]
0x100de09e: lea ebx,[ecx+ebx*4]
0x100de0a1: sub ebp,edx
0x100de0a3: mov [esp+38h],ebx
0x100de0a7: mov ebx,eax
0x100de0a9: sub ebp,4
0x100de0ac: neg ebx
0x100de0ae: mov esi,eax
0x100de0b0: shr ebp,2
0x100de0b3: shl esi,4
0x100de0b6: shl ebx,4
0x100de0b9: add ebp,1
0x100de0bc: mov [esp+40h],ebp
0x100de0c0: lea edx,[edx+ebp*4]
0x100de0c3: mov ebp,[esp+38h]
0x100de0c7: fld dword ptr [ebp]
0x100de0ca: mov ebp,[esp+2Ch]
0x100de0ce: fmul st,st(1)
0x100de0d0: add [esp+38h],esi
0x100de0d4: add [esp+2Ch],esi
0x100de0d8: fadd dword ptr [ebp]
0x100de0db: mov ebp,[esp+18h]
0x100de0df: add [esp+18h],ebx
0x100de0e3: fadd dword ptr [ebp]
0x100de0e6: mov ebp,[esp+58h]
0x100de0ea: fstp dword ptr [ebp-8]
0x100de0ed: mov ebp,[esp+34h]
0x100de0f1: fld dword ptr [ebp]
0x100de0f4: mov ebp,[esp+3Ch]
0x100de0f8: fmul st,st(1)
0x100de0fa: add [esp+34h],esi
0x100de0fe: add [esp+3Ch],esi
0x100de102: fadd dword ptr [ebp]
0x100de105: mov ebp,[esp+14h]
0x100de109: add [esp+14h],ebx
0x100de10d: fadd dword ptr [ebp]
0x100de110: mov ebp,[esp+58h]
0x100de114: fstp dword ptr [ebp-4]
0x100de117: mov ebp,[esp+30h]
0x100de11b: fld dword ptr [ebp]
0x100de11e: mov ebp,[esp+28h]
0x100de122: fmul st,st(1)
0x100de124: add [esp+30h],esi
0x100de128: add [esp+28h],esi
0x100de12c: fadd dword ptr [ebp]
0x100de12f: mov ebp,[esp+1Ch]
0x100de133: add [esp+1Ch],ebx
0x100de137: fadd dword ptr [ebp]
0x100de13a: mov ebp,[esp+58h]
0x100de13e: fstp dword ptr [ebp]
0x100de141: mov ebp,[esp+20h]
0x100de145: fld dword ptr [ebp]
0x100de148: mov ebp,[esp+24h]
0x100de14c: add [esp+24h],esi
0x100de150: fmul st,st(1)
0x100de152: add [esp+20h],esi
0x100de156: fadd dword ptr [ebp]
0x100de159: mov ebp,[esp+10h]
0x100de15d: add [esp+10h],ebx
0x100de161: fadd dword ptr [ebp]
```

```
0x100de164: mov ebp,[esp+58h]
0x100de168: add dword ptr [esp+58h],10h
0x100de16d: sub dword ptr [esp+40h],1
0x100de172: fstp dword ptr [ebp+4]
0x100de175: jne near ptr 100DE0C3h
0x100de17b: mov ebp,[esp+54h]
0x100de17f: mov ebx,[esp+48h]
0x100de183: cmp edx,ebp
0x100de185: jge short 100DE1EEh
0x100de187: lea esi,[eax*4]
0x100de18e: mov [esp+3Ch],esi
0x100de192: mov ebp,edx
0x100de194: sub ebp,edi
0x100de196: imul ebp,eax
0x100de199: mov esi,edx
0x100de19b: imul esi,eax
0x100de19e: lea ebp,[ecx+ebp*4]
0x100de1a1: mov [esp+58h],ebp
0x100de1a5: mov ebp,[esp+54h]
0x100de1a9: lea esi,[ecx+esi*4]
0x100de1ac: mov [esp+40h],esi
0x100de1b0: add ebp,ebp
0x100de1b2: sub ebp,edx
0x100de1b4: mov esi,eax
0x100de1b6: neg esi
0x100de1b8: sub ebp,edi
0x100de1ba: mov edi,[esp+40h]
0x100de1be: add esi,esi
0x100de1c0: sub ebp,2
0x100de1c3: add esi,esi
0x100de1c5: imul ebp,eax
0x100de1c8: mov eax,[esp+58h]
0x100de1cc: lea ecx,[ecx+ebp*4]
0x100de1cf: mov ebp,[esp+3Ch]
0x100de1d3: fld dword ptr [edi]
0x100de1d5: add edx,1
0x100de1d8: fmul st,st(1)
0x100de1da: add edi,ebp
0x100de1dc: fadd dword ptr [eax]
0x100de1de: add eax,ebp
0x100de1e0: fadd dword ptr [ecx]
0x100de1e2: add ecx,esi
0x100de1e4: cmp edx,[esp+54h]
0x100de1e8: fstp dword ptr [ebx+edx*4-4]
0x100de1ec: jl short 100DE1D3h
0x100de1ee: pop edi
0x100de1ef: fstp st(0)
0x100de1f1: pop esi
0x100de1f2: pop ebp
0x100de1f3: pop ebx
0x100de1f4: add esp,34h
0x100de1f7: ret
```

`sub_100ddd50` is called by sub_100de200. It calls nothing else.

12.1273 sub_100de200 at 0x100de200 (604 instructions)

```
0x100de200: sub esp,44h
0x100de203: push ebx
0x100de204: push ebp
0x100de205: push esi
0x100de206: mov esi,[esp+54h]
0x100de20a: xor ebp,ebp
0x100de20c: cmp [esi+0BCh],ebp
0x100de212: push edi
0x100de213: lea edi,[ebp+1]
```

```
0x100de216: je short 100DE230h
0x100de218: push 1012E2FCh
0x100de21d: call dword ptr ds:[100FE4C4h]
0x100de223: add eax,40h
0x100de226: push eax
0x100de227: call dword ptr ds:[100FE4C8h]
0x100de22d: add esp,8
0x100de230: mov eax,[esi+65FCh]
0x100de236: mov ecx,eax
0x100de238: and ecx,7FFFFFFFh
0x100de23e: cmp ecx,8000h
0x100de244: jae short 100DE259h
0x100de246: mov edx,eax
0x100de248: add edi,1
0x100de24b: mov ebx,edx
0x100de24d: mov ecx,edi
0x100de24f: shl ebx,cl
0x100de251: cmp ebx,10000h
0x100de257: jb short 100DE248h
0x100de259: sub edi,1
0x100de25c: mov ecx,edi
0x100de25e: shl dword ptr [esi+65F8h],cl
0x100de264: shl eax,cl
0x100de266: movzx ecx,word ptr [esi+66BEh]
0x100de26d: mov [esi+65FCh],eax
0x100de273: movzx eax,word ptr [esi+66C0h]
0x100de27a: mov [esp+2Ch],edi
0x100de27e: mov edi,ecx
0x100de280: imul edi,eax
0x100de283: cmp edi,15550000h
0x100de289: mov [esp+10h],edi
0x100de28d: jge short 100DE2A6h
0x100de28f: lea edx,[ecx+edi*2]
0x100de292: add edx,edi
0x100de294: add edx,eax
0x100de296: add edx,edx
0x100de298: add edx,edx
0x100de29a: push edx
0x100de29b: call dword ptr ds:[100FE550h]
0x100de2a1: add esp,4
0x100de2a4: mov ebp,eax
0x100de2a6: push 1012E2E8h
0x100de2ab: push ebp
0x100de2ac: push esi
0x100de2ad: call 100D2E20h
0x100de2b2: lea eax,[edi+edi*2]
0x100de2b5: lea ecx,[ebp+eax*4]
0x100de2b9: mov eax,[esi+66D0h]
0x100de2bf: add esp,0Ch
0x100de2c2: cmp eax,3
0x100de2c5: mov [esp+18h],ecx
0x100de2c9: jne short 100DE2D9h
0x100de2cb: cmp dword ptr [esi+65B0h],0
0x100de2d2: je short 100DE2D9h
0x100de2d4: mov eax,4
0x100de2d9: test eax,eax
0x100de2db: fldz
0x100de2dd: fld qword ptr ds:[100FFBA0h]
0x100de2e3: jle near ptr 100DE8C1h
0x100de2e9: fstp st(1)
0x100de2eb: mov dword ptr [esp+20h],0
0x100de2f3: fld qword ptr ds:[10127AD8h]
0x100de2f9: mov [esp+28h],eax
0x100de2fd: jmp short 100DE301h
0x100de2ff: fstp st(0)
0x100de301: xor edi,edi
0x100de303: cmp [esp+10h],edi
```

```
0x100de307: jle short 100DE358h
0x100de309: mov ebx,[esp+20h]
0x100de30d: fstp st(0)
0x100de30f: fstp st(0)
0x100de311: mov edx,[esi+66E0h]
0x100de317: movzx eax,word ptr [edx+ebx]
0x100de31b: mov ecx,[esp+2Ch]
0x100de31f: shl eax,cl
0x100de321: mov [esp+58h],eax
0x100de325: fild dword ptr [esp+58h]
0x100de329: call 1002299Ah
0x100de32e: fstp dword ptr [esp+58h]
0x100de332: fld dword ptr [esp+58h]
0x100de336: add edi,1
0x100de339: fmul qword ptr ds:[100FE920h]
0x100de33f: add ebx,8
0x100de342: cmp edi,[esp+10h]
0x100de346: fstp dword ptr [ebp+edi*4-4]
0x100de34a: jl short 100DE311h
0x100de34c: fld qword ptr ds:[100FFBA0h]
0x100de352: fld qword ptr ds:[10127AD8h]
0x100de358: xor eax,eax
0x100de35a: mov [esp+58h],eax
0x100de35e: mov [esp+1Ch],eax
0x100de362: jmp short 100DE366h
0x100de364: fstp st(0)
0x100de366: mov ebx,[esp+58h]
0x100de36a: and ebx,1
0x100de36d: add ebx,1
0x100de370: imul ebx,[esp+10h]
0x100de375: xor edi,edi
0x100de377: cmp [esi+66BEh],di
0x100de37e: jbe near ptr 100DE426h
0x100de384: mov ecx,[esp+58h]
0x100de388: fstp st(0)
0x100de38a: mov eax,1
0x100de38f: fstp st(0)
0x100de391: shl eax,cl
0x100de393: mov [esp+14h],eax
0x100de397: jmp short 100DE3A0h
0x100de3a0: movzx eax,word ptr [esi+66C0h]
0x100de3a7: mov ecx,[esp+14h]
0x100de3ab: push ecx
0x100de3ac: push eax
0x100de3ad: imul eax,edi
0x100de3b0: add eax,[esp+24h]
0x100de3b4: push 1
0x100de3b6: lea edx,[ebp+eax*4]
0x100de3ba: mov eax,[esp+24h]
0x100de3be: push edx
0x100de3bf: push eax
0x100de3c0: call 100DDD50h
0x100de3c5: xor eax,eax
0x100de3c7: add esp,14h
0x100de3ca: cmp [esi+66C0h],ax
0x100de3d1: jbe short 100DE40Ch
0x100de3d3: fld qword ptr ds:[100FF378h]
0x100de3d9: mov ecx,[esp+18h]
0x100de3dd: movzx edx,word ptr [esi+66C0h]
0x100de3e4: fld dword ptr [ecx+eax*4]
0x100de3e7: imul edx,edi
0x100de3ea: fmul st,st(1)
0x100de3ec: fstp dword ptr [esp+24h]
0x100de3f0: fld dword ptr [esp+24h]
0x100de3f4: add edx,eax
0x100de3f6: add edx,ebx
0x100de3f8: add eax,1
```

```
0x100de3fb: fstp dword ptr [ebp+edx*4]
0x100de3ff: movzx edx,word ptr [esi+66C0h]
0x100de406: cmp eax,edx
0x100de408: jl short 100DE3DDh
0x100de40a: fstp st(0)
0x100de40c: movzx eax,word ptr [esi+66BEh]
0x100de413: add edi,1
0x100de416: cmp edi,eax
0x100de418: jl short 100DE3A0h
0x100de41a: fld qword ptr ds:[100FFBA0h]
0x100de420: fld qword ptr ds:[10127AD8h]
0x100de426: xor edi,edi
0x100de428: cmp [esi+66C0h],di
0x100de42f: jbe near ptr 100DE4DAh
0x100de435: mov ecx,[esp+58h]
0x100de439: fstp st(0)
0x100de43b: mov eax,1
0x100de440: fstp st(0)
0x100de442: shl eax,cl
0x100de444: lea ecx,[ebp+ebx*4]
0x100de448: mov [esp+24h],ecx
0x100de44c: mov [esp+14h],eax
0x100de450: movzx eax,word ptr [esi+66BEh]
0x100de457: mov edx,[esp+14h]
0x100de45b: movzx ecx,word ptr [esi+66C0h]
0x100de462: push edx
0x100de463: mov edx,[esp+28h]
0x100de467: push eax
0x100de468: mov eax,[esp+20h]
0x100de46c: push ecx
0x100de46d: push edx
0x100de46e: push eax
0x100de46f: call 100DDD50h
0x100de474: xor eax,eax
0x100de476: add esp,14h
0x100de479: cmp [esi+66BEh],ax
0x100de480: jbe short 100DE4BBh
0x100de482: fld qword ptr ds:[100FF378h]
0x100de488: mov ecx,[esp+18h]
0x100de48c: movzx edx,word ptr [esi+66C0h]
0x100de493: fld dword ptr [ecx+eax*4]
0x100de496: imul edx,eax
0x100de499: fmul st,st(1)
0x100de49b: fstp dword ptr [esp+30h]
0x100de49f: fld dword ptr [esp+30h]
0x100de4a3: add edx,edi
0x100de4a5: add edx,ebx
0x100de4a7: add eax,1
0x100de4aa: fstp dword ptr [ebp+edx*4]
0x100de4ae: movzx edx,word ptr [esi+66BEh]
0x100de4b5: cmp eax,edx
0x100de4b7: jl short 100DE48Ch
0x100de4b9: fstp st(0)
0x100de4bb: movzx eax,word ptr [esi+66C0h]
0x100de4c2: add dword ptr [esp+24h],4
0x100de4c7: add edi,1
0x100de4ca: cmp edi,eax
0x100de4cc: jl short 100DE450h
0x100de4ce: fld qword ptr ds:[100FFBA0h]
0x100de4d4: fld qword ptr ds:[10127AD8h]
0x100de4da: mov ecx,[esp+58h]
0x100de4de: fld dword ptr [ecx*4+1012B3F8h]
0x100de4e5: xor edx,edx
0x100de4e7: cmp [esp+10h],edx
0x100de4eb: fmul dword ptr [esi+34h]
0x100de4ee: fstp dword ptr [esp+14h]
0x100de4f2: jle near ptr 100DE578h
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```
0x100de4f8: fld dword ptr [esp+14h]
0x100de4fc: mov eax,[esp+1Ch]
0x100de500: fld st(0)
0x100de502: lea ecx,[ebp+eax*4]
0x100de506: fchs
0x100de508: lea edi,[ebp+ebx*4]
0x100de50c: fstp dword ptr [esp+24h]
0x100de510: fld dword ptr [esp+24h]
0x100de514: fldz
0x100de516: fld dword ptr [ecx]
0x100de518: fsub dword ptr [edi]
0x100de51a: fstp dword ptr [esp+30h]
0x100de51e: fld dword ptr [esp+30h]
0x100de522: fst dword ptr [esp+24h]
0x100de526: fst dword ptr [ecx]
0x100de528: fcomp st(2)
0x100de52a: fnstsw ax
0x100de52c: test ah,5
0x100de52f: jp short 100DE53Bh
0x100de531: fld dword ptr [esp+24h]
0x100de535: fadd st,st(3)
0x100de537: fstp dword ptr [ecx]
0x100de539: jmp short 100DE552h
0x100de53b: fld dword ptr [esp+24h]
0x100de53f: fcom st(3)
0x100de541: fnstsw ax
0x100de543: test ah,41h
0x100de546: jne short 100DE54Eh
0x100de548: fsub st,st(3)
0x100de54a: fstp dword ptr [ecx]
0x100de54c: jmp short 100DE552h
0x100de54e: fstp st(0)
0x100de550: fst dword ptr [ecx]
0x100de552: cmp dword ptr [esp+1Ch],0
0x100de557: je short 100DE563h
0x100de559: fld dword ptr [ecx]
0x100de55b: fadd dword ptr [ebp+edx*4]
0x100de55f: fstp dword ptr [ebp+edx*4]
0x100de563: add edx,1
0x100de566: add edi,4
0x100de569: add ecx,4
0x100de56c: cmp edx,[esp+10h]
0x100de570: jl short 100DE516h
0x100de572: fstp st(2)
0x100de574: fstp st(0)
0x100de576: jmp short 100DE57Ah
0x100de578: fldz
0x100de57a: mov eax,[esp+58h]
0x100de57e: add eax,1
0x100de581: cmp eax,5
0x100de584: mov [esp+1Ch],ebx
0x100de588: mov [esp+58h],eax
0x100de58c: jl near ptr 100DE364h
0x100de592: xor ecx,ecx
0x100de594: cmp [esp+10h],ecx
0x100de598: jle near ptr 100DE636h
0x100de59e: mov edx,[esp+20h]
0x100de5a2: fldz
0x100de5a4: lea ebx,[ebp+ebx*4]
0x100de5a8: fld dword ptr [ebx]
0x100de5aa: fadd dword ptr [ebp+ecx*4]
0x100de5ae: fmul st(0),st
0x100de5b0: fmul st,st(3)
0x100de5b2: fcomp st(4)
0x100de5b4: fnstsw ax
0x100de5b6: test ah,5
0x100de5b9: jp short 100DE5D2h
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0x100de5bb: fld dword ptr [ebx]
0x100de5bd: fadd dword ptr [ebp+ecx*4]
0x100de5c1: fmul st(0),st
0x100de5c3: fmul st,st(3)
0x100de5c5: fcomp
0x100de5c7: fnstsw ax
0x100de5c9: test ah,5
0x100de5cc: jp short 100DE5D2h
0x100de5ce: fld st(0)
0x100de5d0: jmp short 100DE5F3h
0x100de5d2: fld dword ptr [ebx]
0x100de5d4: fadd dword ptr [ebp+ecx*4]
0x100de5d8: fmul st(0),st
0x100de5da: fmul st,st(3)
0x100de5dc: fcomp st(4)
0x100de5de: fnstsw ax
0x100de5e0: test ah,5
0x100de5e3: jp short 100DE5F1h
0x100de5e5: fld dword ptr [ebx]
0x100de5e7: fadd dword ptr [ebp+ecx*4]
0x100de5eb: fmul st(0),st
0x100de5ed: fmul st,st(3)
0x100de5ef: jmp short 100DE5F3h
0x100de5f1: fld st(3)
0x100de5f3: mov edi,[esi+66E0h]
0x100de5f9: fnstcw [esp+58h]
0x100de5fd: movzx eax,word ptr [esp+58h]
0x100de602: or eax,0C00h
0x100de607: mov [esp+30h],eax
0x100de60b: add ecx,1
0x100de60e: add edx,8
0x100de611: fldcw [esp+30h]
0x100de615: add ebx,4
0x100de618: cmp ecx,[esp+10h]
0x100de61c: fistp dword ptr [esp+30h]
0x100de620: mov ax,[esp+30h]
0x100de625: mov [edx+edi-8],ax
0x100de62a: fldcw [esp+58h]
0x100de62e: jl near ptr 100DE5A8h
0x100de634: fstp st(0)
0x100de636: add dword ptr [esp+20h],2
0x100de63b: sub dword ptr [esp+28h],1
0x100de640: jne near ptr 100DE2FFh
0x100de646: fstp st(1)
0x100de648: fstp st(1)
0x100de64a: mov eax,[esi+65B0h]
0x100de650: test eax,eax
0x100de652: je near ptr 100DEA1Ah
0x100de658: cmp dword ptr [esi+66D0h],3
0x100de65f: jne near ptr 100DEA1Ah
0x100de665: mov ecx,eax
0x100de667: and ecx,20h
0x100de66a: or ecx,10h
0x100de66d: mov edx,eax
0x100de66f: shr ecx,4
0x100de672: fld dword ptr [esi+ecx*4+0E784h]
0x100de679: lea ecx,[esi+ecx*4+0E784h]
0x100de680: fld qword ptr ds:[100FE7B8h]
0x100de686: and edx,2
0x100de689: fmul st(1),st
0x100de68b: or edx,1
0x100de68e: fld dword ptr [esi+edx*4+0E784h]
0x100de695: and eax,200h
0x100de69a: or eax,100h
0x100de69f: fdivp st(2),st
0x100de6a1: shr eax,8
0x100de6a4: mov [esp+44h],ebp
```

```
0x100de6a8: mov dword ptr [esp+10h],0FFFFFFFFh
0x100de6b0: mov dword ptr [esp+18h],1
0x100de6b8: fxch
0x100de6ba: fstp dword ptr [esp+3Ch]
0x100de6be: fmul dword ptr [esi+eax*4+0E784h]
0x100de6c5: movzx eax,word ptr [esi+66B6h]
0x100de6cc: add eax,eax
0x100de6ce: lea edx,[ebp+eax*2]
0x100de6d2: fdiv dword ptr [ecx]
0x100de6d4: lea ecx,[eax+ebp]
0x100de6d7: movzx eax,word ptr [esi+66B4h]
0x100de6de: sub eax,1
0x100de6e1: cmp eax,1
0x100de6e4: mov [esp+48h],ecx
0x100de6e8: mov [esp+4Ch],edx
0x100de6ec: fstp dword ptr [esp+40h]
0x100de6f0: jle near ptr 100DEA1Ah
0x100de6f6: fld qword ptr ds:[100FE7C8h]
0x100de6fc: mov eax,[esp+18h]
0x100de700: add eax,1
0x100de703: cmp [esp+10h],eax
0x100de707: mov [esp+28h],eax
0x100de70b: jge near ptr 100DE7BCh
0x100de711: mov edi,[esp+10h]
0x100de715: mov ecx,[esp+48h]
0x100de719: mov eax,[esp+4Ch]
0x100de71d: mov edx,[esp+44h]
0x100de721: add edi,1
0x100de724: mov [esp+10h],edi
0x100de728: mov [esp+48h],eax
0x100de72c: mov eax,[esi+65B0h]
0x100de732: and edi,7
0x100de735: mov [esp+44h],ecx
0x100de739: lea ecx,[edi*4+2]
0x100de740: shr eax,cl
0x100de742: movzx ecx,word ptr [esi+66B6h]
0x100de749: and eax,1
0x100de74c: cmp eax,ecx
0x100de74e: mov [esp+4Ch],edx
0x100de752: jge short 100DE7AEh
0x100de754: lea ecx,[edi+edi]
0x100de757: mov [esp+58h],ecx
0x100de75b: movzx ecx,word ptr [esi+66BCh]
0x100de762: mov edi,[esp+10h]
0x100de766: movzx ebx,word ptr [esi+66C0h]
0x100de76d: sar edi,cl
0x100de76f: imul edi,ebx
0x100de772: mov ebx,eax
0x100de774: sar ebx,cl
0x100de776: mov ecx,eax
0x100de778: and ecx,1
0x100de77b: add ecx,[esp+58h]
0x100de77f: add edi,ebx
0x100de781: mov ebx,[esi+65B0h]
0x100de787: add ecx,ecx
0x100de789: shr ebx,cl
0x100de78b: add eax,2
0x100de78e: and ebx,3
0x100de791: lea ecx,[ebx+edi*4]
0x100de794: mov edi,[esi+66E0h]
0x100de79a: mov cx,[edi+ecx*2]
0x100de79e: mov [edx+eax*2-4],cx
0x100de7a3: movzx ecx,word ptr [esi+66B6h]
0x100de7aa: cmp eax,ecx
0x100de7ac: jl short 100DE75Bh
0x100de7ae: mov edx,[esp+28h]
0x100de7b2: cmp [esp+10h],edx
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0x100de7b6: jl near ptr 100DE711h
0x100de7bc: mov eax,[esp+18h]
0x100de7c0: fld dword ptr [esi+34h]
0x100de7c3: mov edi,[esi+65B0h]
0x100de7c9: fmul qword ptr ds:[1012E2E0h]
0x100de7cf: and eax,7
0x100de7d2: lea ecx,[eax*4]
0x100de7d9: shr edi,cl
0x100de7db: fstp dword ptr [esp+14h]
0x100de7df: movzx ecx,word ptr [esi+66B6h]
0x100de7e6: and edi,1
0x100de7e9: add edi,1
0x100de7ec: sub ecx,1
0x100de7ef: cmp edi,ecx
0x100de7f1: jge near ptr 100DE9FEh
0x100de7f7: mov edx,[esp+18h]
0x100de7fb: fld dword ptr [esp+14h]
0x100de7ff: and edx,1
0x100de802: fchs
0x100de804: lea ecx,[esp+edx*4+3Ch]
0x100de808: fstp dword ptr [esp+24h]
0x100de80c: lea edx,[eax+eax]
0x100de80f: mov eax,[esp+44h]
0x100de813: mov [esp+2Ch],ecx
0x100de817: lea ecx,[eax+edi*2+2]
0x100de81b: mov [esp+20h],ecx
0x100de81f: mov ecx,[esp+4Ch]
0x100de823: sub eax,ecx
0x100de825: mov [esp+1Ch],edx
0x100de829: lea ebx,[ecx+edi*2-2]
0x100de82d: mov [esp+30h],eax
0x100de831: mov edx,[esp+30h]
0x100de835: movzx ecx,word ptr [edx+ebx]
0x100de839: movzx edx,word ptr [ebx+4]
0x100de83d: mov eax,[esi+65F8h]
0x100de843: add ecx,edx
0x100de845: lea edx,[eax*4]
0x100de84c: sub ecx,edx
0x100de84e: movzx edx,word ptr [ebx]
0x100de851: add ecx,edx
0x100de853: mov edx,[esp+20h]
0x100de857: movzx edx,word ptr [edx]
0x100de85a: add ecx,edx
0x100de85c: test ecx,ecx
0x100de85e: mov [esp+58h],ecx
0x100de862: fild dword ptr [esp+58h]
0x100de866: jge short 100DE86Eh
0x100de868: fadd dword ptr ds:[100FE7C0h]
0x100de86e: mov ecx,[esp+2Ch]
0x100de872: fmul dword ptr [ecx]
0x100de874: mov edx,[esp+48h]
0x100de878: movzx ecx,word ptr [edx+edi*2]
0x100de87c: sub ecx,eax
0x100de87e: test ecx,ecx
0x100de880: mov [esp+58h],ecx
0x100de884: fild dword ptr [esp+58h]
0x100de888: jge short 100DE890h
0x100de88a: fadd dword ptr ds:[100FE7C0h]
0x100de890: test eax,eax
0x100de892: fmulp st(2),st
0x100de894: mov [esp+58h],eax
0x100de898: faddp
0x100de89a: fild dword ptr [esp+58h]
0x100de89e: jge short 100DE8A6h
0x100de8a0: fadd dword ptr ds:[100FE7C0h]
0x100de8a6: faddp
0x100de8a8: fstp dword ptr [esp+58h]
```

```
0x100de8ac: fld dword ptr [esp+58h]
0x100de8b0: fcom
0x100de8b2: fnstsw ax
0x100de8b4: test ah,5
0x100de8b7: jp short 100DE8C8h
0x100de8b9: fstp st(0)
0x100de8bb: fstp dword ptr [esp+58h]
0x100de8bf: jmp short 100DE8D3h
0x100de8c1: fstp st(0)
0x100de8c3: jmp near ptr 100DE64Ah
0x100de8c8: fstp st(1)
0x100de8ca: call 1002299Ah
0x100de8cf: fstp dword ptr [esp+58h]
0x100de8d3: movzx ecx,word ptr [esi+66BCh]
0x100de8da: mov edx,[esp+18h]
0x100de8de: movzx eax,word ptr [esi+66C0h]
0x100de8e5: sar edx,cl
0x100de8e7: imul edx,eax
0x100de8ea: mov eax,edi
0x100de8ec: sar eax,cl
0x100de8ee: mov ecx,edi
0x100de8f0: and ecx,1
0x100de8f3: add ecx,[esp+1Ch]
0x100de8f7: add edx,eax
0x100de8f9: mov eax,[esi+65B0h]
0x100de8ff: add ecx,ecx
0x100de901: shr eax,cl
0x100de903: and eax,3
0x100de906: lea ecx,[eax+edx*4]
0x100de909: mov edx,[esi+66E0h]
0x100de90f: lea eax,[edx+ecx*2]
0x100de912: mov [esp+38h],eax
0x100de916: movzx eax,word ptr [eax]
0x100de919: mov [esp+34h],eax
0x100de91d: fild dword ptr [esp+34h]
0x100de921: call 1002299Ah
0x100de926: fstp dword ptr [esp+34h]
0x100de92a: fld dword ptr [esp+34h]
0x100de92e: fld dword ptr [esp+58h]
0x100de932: fld st(0)
0x100de934: fsubp st(2),st
0x100de936: fxch
0x100de938: fstp dword ptr [esp+58h]
0x100de93c: fld dword ptr [esp+58h]
0x100de940: fld dword ptr [esp+24h]
0x100de944: fcomp
0x100de946: fnstsw ax
0x100de948: test ah,41h
0x100de94b: jne short 100DE953h
0x100de94d: fadd dword ptr [esp+14h]
0x100de951: jmp short 100DE96Ah
0x100de953: fld dword ptr [esp+14h]
0x100de957: fcom
0x100de959: fnstsw ax
0x100de95b: test ah,5
0x100de95e: jp short 100DE964h
0x100de960: fsubp
0x100de962: jmp short 100DE96Ah
0x100de964: fstp st(1)
0x100de966: fstp st(0)
0x100de968: fldz
0x100de96a: fstp dword ptr [esp+58h]
0x100de96e: fadd dword ptr [esp+58h]
0x100de972: fmul st(0),st
0x100de974: fld qword ptr ds:[100FE7C8h]
0x100de97a: fadd st(1),st
0x100de97c: fld qword ptr ds:[100FFBA0h]
```

```
0x100de982: fcom st(2)
0x100de984: fnstsw ax
0x100de986: test ah,41h
0x100de989: jne short 100DE9A2h
0x100de98b: fld st(2)
0x100de98d: fldz
0x100de98f: fcom
0x100de991: fnstsw ax
0x100de993: fstp st(1)
0x100de995: test ah,41h
0x100de998: jne short 100DE9A0h
0x100de99a: fstp st(3)
0x100de99c: fstp st(0)
0x100de99e: jmp short 100DE9B1h
0x100de9a0: fstp st(0)
0x100de9a2: fcom st(2)
0x100de9a4: fnstsw ax
0x100de9a6: test ah,41h
0x100de9a9: jne short 100DE9AFh
0x100de9ab: fstp st(0)
0x100de9ad: jmp short 100DE9B1h
0x100de9af: fstp st(2)
0x100de9b1: mov edx,[esp+38h]
0x100de9b5: fnstcw [esp+58h]
0x100de9b9: movzx eax,word ptr [esp+58h]
0x100de9be: fxch
0x100de9c0: add dword ptr [esp+20h],4
0x100de9c5: or eax,0C00h
0x100de9ca: mov [esp+34h],eax
0x100de9ce: add edi,2
0x100de9d1: add ebx,4
0x100de9d4: fldcw [esp+34h]
0x100de9d8: fistp dword ptr [esp+34h]
0x100de9dc: mov cx,[esp+34h]
0x100de9e1: mov [edx],cx
0x100de9e4: movzx eax,word ptr [esi+66B6h]
0x100de9eb: fldcw [esp+58h]
0x100de9ef: sub eax,1
0x100de9f2: cmp edi,eax
0x100de9f4: fldz
0x100de9f6: fxch
0x100de9f8: jl near ptr 100DE831h
0x100de9fe: movzx ecx,word ptr [esi+66B4h]
0x100dea05: mov eax,[esp+28h]
0x100dea09: sub ecx,1
0x100dea0c: cmp eax,ecx
0x100dea0e: mov [esp+18h],eax
0x100dea12: jl near ptr 100DE6FCh
0x100dea18: fstp st(0)
0x100dea1a: push ebp
0x100dea1b: fstp st(0)
0x100dea1d: call dword ptr ds:[100FE554h]
0x100dea23: add esp,4
0x100dea26: pop edi
0x100dea27: pop esi
0x100dea28: pop ebp
0x100dea29: pop ebx
0x100dea2a: add esp,44h
0x100dea2d: ret
```

`sub_100de200` is called by sub_100dea30. It works with the text "Wavelet denoising...". It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to j__CIsqrt, sub_100d2e20, sub_100ddd50 (3 callees).

12.1274 sub_100dea30 at 0x100dea30 (876 instructions)

```
0x100dea30: push ebp
0x100dea31: mov ebp,esp
0x100dea33: and esp,0FFFFFFC0h
0x100dea36: fldz
0x100dea38: sub esp,0B4h
0x100dea3e: push ebx
0x100dea3f: push esi
0x100dea40: mov esi,[ebp+8]
0x100dea43: fcom dword ptr [esi+3Ch]
0x100dea46: push edi
0x100dea47: fnstsw ax
0x100dea49: test ah,44h
0x100dea4c: jnp short 100DEA72h
0x100dea4e: mov eax,[esi+3Ch]
0x100dea51: mov ecx,[esi+40h]
0x100dea54: mov edx,[esi+44h]
0x100dea57: mov [esi+0E784h],eax
0x100dea5d: mov eax,[esi+48h]
0x100dea60: mov [esi+0E788h],ecx
0x100dea66: mov [esi+0E78Ch],edx
0x100dea6c: mov [esi+0E790h],eax
0x100dea72: cmp dword ptr [esi+0D8h],0
0x100dea79: fldz
0x100dea7b: jne short 100DEAA1h
0x100dea7d: cmp dword ptr [esi+0DCh],0
0x100dea84: je near ptr 100DF071h
0x100dea8a: fld dword ptr ds:[100FF11Ch]
0x100dea90: fcomp dword ptr [esi+0E774h]
0x100dea96: fnstsw ax
0x100dea98: test ah,44h
0x100dea9b: jp near ptr 100DEE02h
0x100deaa1: push 40h
0x100deaa3: fstp st(1)
0x100deaa5: lea ecx,[esp+84h]
0x100deaac: fstp st(0)
0x100deaae: push 0
0x100deab0: push ecx
0x100deab1: call 100228C0h
0x100deab6: mov edx,[esi+0CCh]
0x100deabc: mov eax,[esi+0D4h]
0x100deac2: movzx ecx,word ptr [esi+66B4h]
0x100deac9: add eax,edx
0x100deacb: add esp,0Ch
0x100deace: cmp eax,ecx
0x100dead0: mov [esp+50h],eax
0x100dead4: jb short 100DEADAh
0x100dead6: mov [esp+50h],ecx
0x100deada: mov eax,[esi+0C8h]
0x100deae0: mov ecx,[esi+0D0h]
0x100deae6: add ecx,eax
0x100deae8: movzx eax,word ptr [esi+66B6h]
0x100deaef: cmp ecx,eax
0x100deaf1: jae short 100DEAFBh
0x100deaf3: mov edi,ecx
0x100deaf5: mov [esp+44h],edi
0x100deaf9: jmp short 100DEB01h
0x100deafb: mov edi,eax
0x100deafd: mov [esp+44h],eax
0x100deb01: cmp edx,[esp+50h]
0x100deb05: fld qword ptr [esp+0A8h]
0x100deb0c: fld qword ptr [esp+0A0h]
0x100deb13: mov [esp+3Ch],edx
0x100deb17: fld qword ptr [esp+98h]
0x100deb1e: fld qword ptr [esp+90h]
0x100deb25: fld qword ptr [esp+88h]
0x100deb2c: fld qword ptr [esp+80h]
```

```
0x100deb33: jae near ptr 100DED86h
0x100deb39: mov ecx,eax
0x100deb3b: imul ecx,edx
0x100deb3e: add eax,eax
0x100deb40: add eax,eax
0x100deb42: add eax,eax
0x100deb44: mov [esp+54h],ecx
0x100deb48: mov [esp+30h],eax
0x100deb4c: mov ecx,[esi+0C8h]
0x100deb52: cmp ecx,edi
0x100deb54: mov [esp+4Ch],ecx
0x100deb58: jae near ptr 100DED6Dh
0x100deb5e: xor eax,eax
0x100deb60: mov [esp+5Ch],eax
0x100deb64: mov [esp+60h],eax
0x100deb68: mov [esp+64h],eax
0x100deb6c: mov [esp+68h],eax
0x100deb70: mov [esp+6Ch],eax
0x100deb74: mov [esp+70h],eax
0x100deb78: mov [esp+74h],eax
0x100deb7c: mov [esp+78h],eax
0x100deb80: lea eax,[edx+8]
0x100deb83: cmp edx,eax
0x100deb85: mov ebx,edx
0x100deb87: mov [esp+48h],ebx
0x100deb8b: jae near ptr 100DEC98h
0x100deb91: mov edx,[esp+54h]
0x100deb95: mov [esp+58h],edx
0x100deb99: cmp ebx,[esp+50h]
0x100deb9d: jae near ptr 100DEC98h
0x100deba3: lea eax,[ecx+8]
0x100deba6: cmp ecx,eax
0x100deba8: mov edx,ecx
0x100debaa: mov [esp+7Ch],edx
0x100debae: jae near ptr 100DEC77h
0x100debb4: cmp edx,[esp+44h]
0x100debb8: jae near ptr 100DEC77h
0x100debbe: mov ecx,[esi+65FCh]
0x100debc4: mov edi,[esi+65B0h]
0x100debca: xor eax,eax
0x100debcc: add ecx,0FFFFFFE7h
0x100debcf: mov [esp+40h],ecx
0x100debd3: test edi,edi
0x100debd5: je short 100DEC1Bh
0x100debd7: mov eax,ebx
0x100debd9: mov ecx,edx
0x100debdb: and ecx,1
0x100debde: and eax,7
0x100debe1: lea ecx,[ecx+eax*2]
0x100debe4: add ecx,ecx
0x100debe6: mov eax,edi
0x100debe8: movzx edi,word ptr [esi+66C0h]
0x100debef: shr eax,cl
0x100debf1: movzx ecx,word ptr [esi+66BCh]
0x100debf8: shr ebx,cl
0x100debfa: shr edx,cl
0x100debfc: mov ecx,[esi+66E0h]
0x100dec02: and eax,3
0x100dec05: imul ebx,edi
0x100dec08: add ebx,edx
0x100dec0a: lea edx,[eax+ebx*4]
0x100dec0d: movzx ecx,word ptr [ecx+edx*2]
0x100dec11: mov edx,[esp+7Ch]
0x100dec15: mov ebx,[esp+48h]
0x100dec19: jmp short 100DEC2Eh
0x100dec1b: mov ecx,[esp+58h]
0x100dec1f: mov edi,[esi+66E0h]
```

```
0x100dec25: add ecx,edx
0x100dec27: lea ecx,[eax+ecx*4]
0x100dec2a: movzx ecx,word ptr [edi+ecx*2]
0x100dec2e: cmp ecx,[esp+40h]
0x100dec32: jg near ptr 100DED52h
0x100dec38: sub ecx,[esi+65F8h]
0x100dec3e: jns short 100DEC42h
0x100dec40: xor ecx,ecx
0x100dec42: mov edi,[esi+65B0h]
0x100dec48: add [esp+eax*4+5Ch],ecx
0x100dec4c: add dword ptr [esp+eax*4+6Ch],1
0x100dec51: test edi,edi
0x100dec53: jne short 100DEC61h
0x100dec55: add eax,1
0x100dec58: cmp eax,4
0x100dec5b: jb near ptr 100DEBD3h
0x100dec61: mov ecx,[esp+4Ch]
0x100dec65: add edx,1
0x100dec68: lea eax,[ecx+8]
0x100dec6b: cmp edx,eax
0x100dec6d: mov [esp+7Ch],edx
0x100dec71: jb near ptr 100DEBB4h
0x100dec77: movzx eax,word ptr [esi+66B6h]
0x100dec7e: add [esp+58h],eax
0x100dec82: mov eax,[esp+3Ch]
0x100dec86: add ebx,1
0x100dec89: add eax,8
0x100dec8c: cmp ebx,eax
0x100dec8e: mov [esp+48h],ebx
0x100dec92: jb near ptr 100DEB99h
0x100dec98: mov edx,[esp+5Ch]
0x100dec9c: fild dword ptr [esp+5Ch]
0x100deca0: test edx,edx
0x100deca2: jge short 100DECAAh
0x100deca4: fadd qword ptr ds:[100FEFC8h]
0x100decaa: mov eax,[esp+60h]
0x100decae: faddp
0x100decb0: test eax,eax
0x100decb2: fild dword ptr [esp+60h]
0x100decb6: jge short 100DECBEh
0x100decb8: fadd qword ptr ds:[100FEFC8h]
0x100decbe: mov edx,[esp+64h]
0x100decc2: faddp st(2),st
0x100decc4: test edx,edx
0x100decc6: fild dword ptr [esp+64h]
0x100decca: jge short 100DECD2h
0x100deccc: fadd qword ptr ds:[100FEFC8h]
0x100decd2: mov eax,[esp+68h]
0x100decd6: faddp st(3),st
0x100decd8: test eax,eax
0x100decda: fild dword ptr [esp+68h]
0x100decde: jge short 100DECE6h
0x100dece0: fadd qword ptr ds:[100FEFC8h]
0x100dece6: mov edx,[esp+6Ch]
0x100decea: faddp st(4),st
0x100decec: test edx,edx
0x100decee: fild dword ptr [esp+6Ch]
0x100decf2: jge short 100DECFAh
0x100decf4: fadd qword ptr ds:[100FEFC8h]
0x100decfa: mov eax,[esp+70h]
0x100decfe: faddp st(5),st
0x100ded00: test eax,eax
0x100ded02: fild dword ptr [esp+70h]
0x100ded06: jge short 100DED0Eh
0x100ded08: fadd qword ptr ds:[100FEFC8h]
0x100ded0e: mov edx,[esp+74h]
0x100ded12: faddp st(6),st
```

```
0x100ded14: test edx,edx
0x100ded16: fild dword ptr [esp+74h]
0x100ded1a: jge short 100DED22h
0x100ded1c: fadd qword ptr ds:[100FEFC8h]
0x100ded22: fadd qword ptr [esp+0B0h]
0x100ded29: mov eax,[esp+78h]
0x100ded2d: test eax,eax
0x100ded2f: fstp qword ptr [esp+0B0h]
0x100ded36: fild dword ptr [esp+78h]
0x100ded3a: jge short 100DED42h
0x100ded3c: fadd qword ptr ds:[100FEFC8h]
0x100ded42: fadd qword ptr [esp+0B8h]
0x100ded49: fstp qword ptr [esp+0B8h]
0x100ded50: jmp short 100DED56h
0x100ded52: mov ecx,[esp+4Ch]
0x100ded56: mov edi,[esp+44h]
0x100ded5a: mov edx,[esp+3Ch]
0x100ded5e: add ecx,8
0x100ded61: cmp ecx,edi
0x100ded63: mov [esp+4Ch],ecx
0x100ded67: jb near ptr 100DEB5Eh
0x100ded6d: mov ecx,[esp+30h]
0x100ded71: add [esp+54h],ecx
0x100ded75: add edx,8
0x100ded78: cmp edx,[esp+50h]
0x100ded7c: mov [esp+3Ch],edx
0x100ded80: jb near ptr 100DEB4Ch
0x100ded86: fldz
0x100ded88: fld st(0)
0x100ded8a: fucomp st(2)
0x100ded8c: fnstsw ax
0x100ded8e: test ah,44h
0x100ded91: jnp short 100DED9Fh
0x100ded93: fxch st(5)
0x100ded95: fdivrp
0x100ded97: fstp dword ptr [esi+0E784h]
0x100ded9d: jmp short 100DEDA3h
0x100ded9f: fstp st(1)
0x100deda1: fstp st(4)
0x100deda3: fld st(3)
0x100deda5: fucomp
0x100deda7: fnstsw ax
0x100deda9: test ah,44h
0x100dedac: jnp short 100DEDBAh
0x100dedae: fdivp st(4),st
0x100dedb0: fxch st(3)
0x100dedb2: fstp dword ptr [esi+0E788h]
0x100dedb8: jmp short 100DEDBEh
0x100dedba: fstp st(0)
0x100dedbc: fstp st(3)
0x100dedbe: fld st(1)
0x100dedc0: fucomp st(3)
0x100dedc2: fnstsw ax
0x100dedc4: test ah,44h
0x100dedc7: jnp short 100DEDDCh
0x100dedc9: fld qword ptr [esp+0B0h]
0x100dedd0: fdivrp st(3),st
0x100dedd2: fxch st(2)
0x100dedd4: fstp dword ptr [esi+0E78Ch]
0x100dedda: jmp short 100DEDDEh
0x100deddc: fstp st(2)
0x100dedde: fld st(0)
0x100dede0: fucomp st(2)
0x100dede2: fnstsw ax
0x100dede4: test ah,44h
0x100dede7: jnp short 100DEDFCh
0x100dede9: fld qword ptr [esp+0B8h]
```

```
0x100dedf0: fdivrp st(2),st
0x100dedf2: fxch
0x100dedf4: fstp dword ptr [esi+0E790h]
0x100dedfa: jmp short 100DEDFEh
0x100dedfc: fstp st(1)
0x100dedfe: fldz
0x100dee00: fxch
0x100dee02: cmp dword ptr [esi+0DCh],0
0x100dee09: je near ptr 100DF071h
0x100dee0f: fld dword ptr ds:[100FF11Ch]
0x100dee15: fcomp dword ptr [esi+0E774h]
0x100dee1b: fnstsw ax
0x100dee1d: test ah,44h
0x100dee20: jnp near ptr 100DF071h
0x100dee26: mov ebx,[esi+65F8h]
0x100dee2c: xor eax,eax
0x100dee2e: xor edx,edx
0x100dee30: mov [esp+5Ch],eax
0x100dee34: mov [esp+60h],eax
0x100dee38: mov [esp+64h],eax
0x100dee3c: mov [esp+68h],eax
0x100dee40: mov [esp+6Ch],eax
0x100dee44: mov [esp+70h],eax
0x100dee48: mov [esp+74h],eax
0x100dee4c: mov [esp+78h],eax
0x100dee50: mov [esp+3Ch],edx
0x100dee54: lea edi,[esi+66E6h]
0x100dee5a: xor eax,eax
0x100dee5c: and edx,7
0x100dee5f: add edx,edx
0x100dee61: mov [esp+4Ch],eax
0x100dee65: mov [esp+48h],edx
0x100dee69: jmp short 100DEE6Fh
0x100dee6b: mov edx,[esp+48h]
0x100dee6f: mov ecx,eax
0x100dee71: and ecx,1
0x100dee74: add ecx,edx
0x100dee76: mov edx,[esi+65B0h]
0x100dee7c: add ecx,ecx
0x100dee7e: shr edx,cl
0x100dee80: movzx ecx,word ptr [edi-2]
0x100dee84: sub ecx,ebx
0x100dee86: and edx,3
0x100dee89: test ecx,ecx
0x100dee8b: jle short 100DEE91h
0x100dee8d: add [esp+edx*4+5Ch],ecx
0x100dee91: add dword ptr [esp+edx*4+6Ch],1
0x100dee96: lea ecx,[eax-1]
0x100dee99: mov eax,[esi+65B0h]
0x100dee9f: and ecx,1
0x100deea2: add ecx,[esp+48h]
0x100deea6: add ecx,ecx
0x100deea8: shr eax,cl
0x100deeaa: movzx ecx,word ptr [edi]
0x100deead: sub ecx,ebx
0x100deeaf: and eax,3
0x100deeb2: test ecx,ecx
0x100deeb4: jle short 100DEEBAh
0x100deeb6: add [esp+eax*4+5Ch],ecx
0x100deeba: movzx ecx,word ptr [edi+2]
0x100deebe: add dword ptr [esp+eax*4+6Ch],1
0x100deec3: sub ecx,ebx
0x100deec5: test ecx,ecx
0x100deec7: jle short 100DEECDh
0x100deec9: add [esp+edx*4+5Ch],ecx
0x100deecd: movzx ecx,word ptr [edi+4]
0x100deed1: add dword ptr [esp+edx*4+6Ch],1
```

```
0x100deed6: lea edx,[esp+edx*4+6Ch]
0x100deeda: sub ecx,ebx
0x100deedc: test ecx,ecx
0x100deede: jle short 100DEEE4h
0x100deee0: add [esp+eax*4+5Ch],ecx
0x100deee4: add dword ptr [esp+eax*4+6Ch],1
0x100deee9: lea eax,[esp+eax*4+6Ch]
0x100deeed: mov eax,[esp+4Ch]
0x100deef1: add eax,4
0x100deef4: add edi,8
0x100deef7: cmp eax,8
0x100deefa: mov [esp+4Ch],eax
0x100deefe: jb near ptr 100DEE6Bh
0x100def04: mov edx,[esp+3Ch]
0x100def08: add edx,1
0x100def0b: cmp edx,8
0x100def0e: mov [esp+3Ch],edx
0x100def12: jb near ptr 100DEE5Ah
0x100def18: mov edi,[esp+5Ch]
0x100def1c: test edi,edi
0x100def1e: je near ptr 100DEFFAh
0x100def24: mov edx,[esp+60h]
0x100def28: test edx,edx
0x100def2a: je near ptr 100DEFFAh
0x100def30: mov ecx,[esp+64h]
0x100def34: test ecx,ecx
0x100def36: je near ptr 100DEFFAh
0x100def3c: mov eax,[esp+68h]
0x100def40: test eax,eax
0x100def42: je near ptr 100DEFFAh
0x100def48: mov ebx,[esp+6Ch]
0x100def4c: fild dword ptr [esp+6Ch]
0x100def50: test ebx,ebx
0x100def52: jge short 100DEF5Ah
0x100def54: fadd dword ptr ds:[100FE7C0h]
0x100def5a: test edi,edi
0x100def5c: mov [esp+30h],edi
0x100def60: fild dword ptr [esp+30h]
0x100def64: jge short 100DEF6Ch
0x100def66: fadd dword ptr ds:[100FE7C0h]
0x100def6c: fdivp
0x100def6e: mov edi,[esp+70h]
0x100def72: test edi,edi
0x100def74: fstp dword ptr [esi+0E784h]
0x100def7a: fild dword ptr [esp+70h]
0x100def7e: jge short 100DEF86h
0x100def80: fadd dword ptr ds:[100FE7C0h]
0x100def86: test edx,edx
0x100def88: mov [esp+30h],edx
0x100def8c: fild dword ptr [esp+30h]
0x100def90: jge short 100DEF98h
0x100def92: fadd dword ptr ds:[100FE7C0h]
0x100def98: fdivp
0x100def9a: mov edx,[esp+74h]
0x100def9e: test edx,edx
0x100defa0: fstp dword ptr [esi+0E788h]
0x100defa6: fild dword ptr [esp+74h]
0x100defaa: jge short 100DEFB2h
0x100defac: fadd dword ptr ds:[100FE7C0h]
0x100defb2: test ecx,ecx
0x100defb4: mov [esp+30h],ecx
0x100defb8: fild dword ptr [esp+30h]
0x100defbc: jge short 100DEFC4h
0x100defbe: fadd dword ptr ds:[100FE7C0h]
0x100defc4: fdivp
0x100defc6: mov ecx,[esp+78h]
0x100defca: test ecx,ecx
```

```
0x100defcc: fstp dword ptr [esi+0E78Ch]
0x100defd2: fild dword ptr [esp+78h]
0x100defd6: jge short 100DEFDEh
0x100defd8: fadd dword ptr ds:[100FE7C0h]
0x100defde: test eax,eax
0x100defe0: mov [esp+30h],eax
0x100defe4: fild dword ptr [esp+30h]
0x100defe8: jge short 100DEFF0h
0x100defea: fadd dword ptr ds:[100FE7C0h]
0x100deff0: fdivp
0x100deff2: fstp dword ptr [esi+0E790h]
0x100deff8: jmp short 100DF071h
0x100deffa: fld st(1)
0x100deffc: fcomp dword ptr [esi+0E774h]
0x100df002: fnstsw ax
0x100df004: test ah,44h
0x100df007: jnp short 100DF04Ah
0x100df009: fld st(1)
0x100df00b: fcomp dword ptr [esi+0E77Ch]
0x100df011: fnstsw ax
0x100df013: test ah,44h
0x100df016: jnp short 100DF04Ah
0x100df018: mov edx,[esi+0E774h]
0x100df01e: mov eax,[esi+0E778h]
0x100df024: mov ecx,[esi+0E77Ch]
0x100df02a: mov [esi+0E784h],edx
0x100df030: mov edx,[esi+0E780h]
0x100df036: mov [esi+0E788h],eax
0x100df03c: mov [esi+0E78Ch],ecx
0x100df042: mov [esi+0E790h],edx
0x100df048: jmp short 100DF071h
0x100df04a: mov eax,[esi+6278h]
0x100df050: fstp st(1)
0x100df052: push eax
0x100df053: fstp st(0)
0x100df055: push 1012E388h
0x100df05a: call dword ptr ds:[100FE4C4h]
0x100df060: add eax,40h
0x100df063: push eax
0x100df064: call dword ptr ds:[100FE4C8h]
0x100df06a: fldz
0x100df06c: fldz
0x100df06e: add esp,0Ch
0x100df071: fld st(1)
0x100df073: fcomp dword ptr [esi+0E790h]
0x100df079: fnstsw ax
0x100df07b: test ah,44h
0x100df07e: jp short 100DF0A1h
0x100df080: cmp dword ptr [esi+66D0h],4
0x100df087: jge short 100DF091h
0x100df089: fld dword ptr [esi+0E788h]
0x100df08f: jmp short 100DF093h
0x100df091: fld1
0x100df093: fstp dword ptr [esp+48h]
0x100df097: fld dword ptr [esp+48h]
0x100df09b: fstp dword ptr [esi+0E790h]
0x100df0a1: mov ecx,[esi+65FCh]
0x100df0a7: fxch
0x100df0a9: fcomp dword ptr [esi+34h]
0x100df0ac: mov ebx,[esi+65F8h]
0x100df0b2: mov [esp+40h],ecx
0x100df0b6: fnstsw ax
0x100df0b8: test ah,44h
0x100df0bb: jnp short 100DF0CAh
0x100df0bd: push esi
0x100df0be: fstp st(0)
0x100df0c0: call 100DE200h
```

```
0x100df0c5: fldz
0x100df0c7: add esp,4
0x100df0ca: fld qword ptr ds:[100FFB40h]
0x100df0d0: mov edx,[esi+65F8h]
0x100df0d6: sub [esi+65FCh],edx
0x100df0dc: fld st(0)
0x100df0de: fld st(2)
0x100df0e0: lea edi,[esi+0E784h]
0x100df0e6: fld dword ptr [edi]
0x100df0e8: fcomp st(3)
0x100df0ea: fnstsw ax
0x100df0ec: fstp st(2)
0x100df0ee: test ah,5
0x100df0f1: jp short 100DF0F7h
0x100df0f3: fstp st(0)
0x100df0f5: fld dword ptr [edi]
0x100df0f7: fld dword ptr [edi]
0x100df0f9: fcomp st(3)
0x100df0fb: fnstsw ax
0x100df0fd: fstp st(2)
0x100df0ff: test ah,41h
0x100df102: jne short 100DF108h
0x100df104: fstp st(0)
0x100df106: fld dword ptr [edi]
0x100df108: fld dword ptr [esi+0E788h]
0x100df10e: fcomp st(2)
0x100df110: fnstsw ax
0x100df112: test ah,5
0x100df115: jp short 100DF121h
0x100df117: fstp st(1)
0x100df119: fld dword ptr [esi+0E788h]
0x100df11f: fxch
0x100df121: fld dword ptr [esi+0E788h]
0x100df127: fcomp
0x100df129: fnstsw ax
0x100df12b: test ah,41h
0x100df12e: jne short 100DF138h
0x100df130: fstp st(0)
0x100df132: fld dword ptr [esi+0E788h]
0x100df138: fld dword ptr [esi+0E78Ch]
0x100df13e: fcomp st(2)
0x100df140: fnstsw ax
0x100df142: test ah,5
0x100df145: jp short 100DF151h
0x100df147: fstp st(1)
0x100df149: fld dword ptr [esi+0E78Ch]
0x100df14f: fxch
0x100df151: fld dword ptr [esi+0E78Ch]
0x100df157: fcomp
0x100df159: fnstsw ax
0x100df15b: test ah,41h
0x100df15e: jne short 100DF168h
0x100df160: fstp st(0)
0x100df162: fld dword ptr [esi+0E78Ch]
0x100df168: fld dword ptr [esi+0E790h]
0x100df16e: fcomp st(2)
0x100df170: fnstsw ax
0x100df172: test ah,5
0x100df175: jp short 100DF181h
0x100df177: fstp st(1)
0x100df179: fld dword ptr [esi+0E790h]
0x100df17f: fxch
0x100df181: fld dword ptr [esi+0E790h]
0x100df187: fcomp
0x100df189: fnstsw ax
0x100df18b: test ah,41h
0x100df18e: jne short 100DF198h
```

```
0x100df190: fstp st(0)
0x100df192: fld dword ptr [esi+0E790h]
0x100df198: cmp dword ptr [esi+9Ch],0
0x100df19f: jne short 100DF1A5h
0x100df1a1: fstp st(0)
0x100df1a3: jmp short 100DF1A7h
0x100df1a5: fstp st(1)
0x100df1a7: fstp dword ptr [esp+30h]
0x100df1ab: fld dword ptr [edi]
0x100df1ad: fld dword ptr [esp+30h]
0x100df1b1: fld st(0)
0x100df1b3: fdivp st(2),st
0x100df1b5: fxch
0x100df1b7: fstp dword ptr [esp+30h]
0x100df1bb: fld dword ptr [esp+30h]
0x100df1bf: fst dword ptr [edi]
0x100df1c1: fld qword ptr ds:[100FFBA0h]
0x100df1c7: mov eax,[esi+65FCh]
0x100df1cd: fmul st(1),st
0x100df1cf: test eax,eax
0x100df1d1: fild dword ptr [esi+65FCh]
0x100df1d7: jge short 100DF1DFh
0x100df1d9: fadd dword ptr ds:[100FE7C0h]
0x100df1df: fdivp st(2),st
0x100df1e1: fxch
0x100df1e3: fstp dword ptr [esp+5Ch]
0x100df1e7: fld dword ptr [esi+0E788h]
0x100df1ed: fdiv st,st(2)
0x100df1ef: fstp dword ptr [esp+30h]
0x100df1f3: fld dword ptr [esp+30h]
0x100df1f7: fst dword ptr [esi+0E788h]
0x100df1fd: fmul st,st(1)
0x100df1ff: mov ecx,[esi+65FCh]
0x100df205: fild dword ptr [esi+65FCh]
0x100df20b: test ecx,ecx
0x100df20d: jge short 100DF215h
0x100df20f: fadd dword ptr ds:[100FE7C0h]
0x100df215: fdivp
0x100df217: fstp dword ptr [esp+60h]
0x100df21b: fld dword ptr [esi+0E78Ch]
0x100df221: fdiv st,st(2)
0x100df223: fstp dword ptr [esp+30h]
0x100df227: fld dword ptr [esp+30h]
0x100df22b: fst dword ptr [esi+0E78Ch]
0x100df231: fmul st,st(1)
0x100df233: mov edx,[esi+65FCh]
0x100df239: fild dword ptr [esi+65FCh]
0x100df23f: test edx,edx
0x100df241: jge short 100DF249h
0x100df243: fadd dword ptr ds:[100FE7C0h]
0x100df249: fdivp
0x100df24b: fstp dword ptr [esp+64h]
0x100df24f: fld dword ptr [esi+0E790h]
0x100df255: fdivrp st(2),st
0x100df257: fxch
0x100df259: fstp dword ptr [esp+30h]
0x100df25d: fld dword ptr [esp+30h]
0x100df261: fst dword ptr [esi+0E790h]
0x100df267: fmulp
0x100df269: mov eax,[esi+65FCh]
0x100df26f: fild dword ptr [esi+65FCh]
0x100df275: test eax,eax
0x100df277: jge short 100DF27Fh
0x100df279: fadd dword ptr ds:[100FE7C0h]
0x100df27f: cmp dword ptr [esi+0BCh],0
0x100df286: fdivp
0x100df288: fstp dword ptr [esp+68h]
```

```
0x100df28c: je short 100DF2EEh
0x100df28e: mov ecx,[esp+40h]
0x100df292: push ecx
0x100df293: push ebx
0x100df294: push 1012E34Ch
0x100df299: call dword ptr ds:[100FE4C4h]
0x100df29f: add eax,40h
0x100df2a2: push eax
0x100df2a3: call dword ptr ds:[100FE4C8h]
0x100df2a9: add esp,10h
0x100df2ac: mov ebx,4
0x100df2b1: fld dword ptr [edi]
0x100df2b3: sub esp,8
0x100df2b6: fstp qword ptr [esp]
0x100df2b9: push 1012E348h
0x100df2be: call dword ptr ds:[100FE4C4h]
0x100df2c4: add eax,40h
0x100df2c7: push eax
0x100df2c8: call dword ptr ds:[100FE4C8h]
0x100df2ce: add esp,10h
0x100df2d1: add edi,4
0x100df2d4: sub ebx,1
0x100df2d7: jne short 100DF2B1h
0x100df2d9: call dword ptr ds:[100FE4C4h]
0x100df2df: add eax,40h
0x100df2e2: push eax
0x100df2e3: push 0Ah
0x100df2e5: call dword ptr ds:[100FE520h]
0x100df2eb: add esp,8
0x100df2ee: movzx eax,word ptr [esi+66BEh]
0x100df2f5: movzx edx,word ptr [esi+66C0h]
0x100df2fc: imul eax,edx
0x100df2ff: lea ecx,[eax*4]
0x100df306: xor edi,edi
0x100df308: test ecx,ecx
0x100df30a: mov [esp+54h],eax
0x100df30e: mov [esp+40h],ecx
0x100df312: jbe short 100DF368h
0x100df314: mov eax,[esi+66E0h]
0x100df31a: lea ebx,[eax+edi*2]
0x100df31d: movzx eax,word ptr [ebx]
0x100df320: test eax,eax
0x100df322: je short 100DF361h
0x100df324: sub eax,[esi+65F8h]
0x100df32a: mov ecx,edi
0x100df32c: and ecx,3
0x100df32f: fld dword ptr [esp+ecx*4+5Ch]
0x100df333: mov [esp+30h],eax
0x100df337: fimul dword ptr [esp+30h]
0x100df33b: call 100228D0h
0x100df340: cmp eax,0FFFFh
0x100df345: jge short 100DF355h
0x100df347: xor edx,edx
0x100df349: test eax,eax
0x100df34b: setl dl
0x100df34e: sub edx,1
0x100df351: and eax,edx
0x100df353: jmp short 100DF35Ah
0x100df355: mov eax,0FFFFh
0x100df35a: mov ecx,[esp+40h]
0x100df35e: mov [ebx],ax
0x100df361: add edi,1
0x100df364: cmp edi,ecx
0x100df366: jb short 100DF314h
0x100df368: fld1
0x100df36a: lea ebx,[esi+50h]
0x100df36d: fcom qword ptr [ebx]
```

```
0x100df36f: fnstsw ax
0x100df371: test ah,44h
0x100df374: jp short 100DF386h
0x100df376: fcomp qword ptr [esi+60h]
0x100df379: fnstsw ax
0x100df37b: test ah,44h
0x100df37e: jnp near ptr 100DF639h
0x100df384: jmp short 100DF388h
0x100df386: fstp st(0)
0x100df388: cmp dword ptr [esi+66D0h],3
0x100df38f: jne near ptr 100DF639h
0x100df395: cmp dword ptr [esi+0BCh],0
0x100df39c: je short 100DF3B6h
0x100df39e: push 1012E324h
0x100df3a3: call dword ptr ds:[100FE4C4h]
0x100df3a9: add eax,40h
0x100df3ac: push eax
0x100df3ad: call dword ptr ds:[100FE4C8h]
0x100df3b3: add esp,8
0x100df3b6: xor eax,eax
0x100df3b8: mov [esp+58h],eax
0x100df3bc: mov [esp+4Ch],eax
0x100df3c0: mov [esp+50h],ebx
0x100df3c4: fld1
0x100df3c6: fcomp qword ptr [ebx]
0x100df3c8: fnstsw ax
0x100df3ca: test ah,44h
0x100df3cd: jnp near ptr 100DF619h
0x100df3d3: mov eax,[esp+54h]
0x100df3d7: lea ecx,[eax+eax]
0x100df3da: push ecx
0x100df3db: call dword ptr ds:[100FE550h]
0x100df3e1: mov edi,eax
0x100df3e3: push 1012E314h
0x100df3e8: push edi
0x100df3e9: push esi
0x100df3ea: mov [esp+58h],edi
0x100df3ee: call 100D2E20h
0x100df3f3: xor eax,eax
0x100df3f5: add esp,10h
0x100df3f8: cmp [esp+54h],eax
0x100df3fc: jbe short 100DF41Ch
0x100df3fe: mov ecx,[esp+4Ch]
0x100df402: mov edx,[esi+66E0h]
0x100df408: mov dx,[ecx+edx]
0x100df40c: mov [edi+eax*2],dx
0x100df410: add eax,1
0x100df413: add ecx,8
0x100df416: cmp eax,[esp+54h]
0x100df41a: jb short 100DF402h
0x100df41c: xor edi,edi
0x100df41e: cmp [esi+66BEh],di
0x100df425: jbe near ptr 100DF60Bh
0x100df42b: fld qword ptr ds:[100FE7C8h]
0x100df431: fld1
0x100df433: movzx ecx,word ptr [esi+66BEh]
0x100df43a: mov [esp+30h],ecx
0x100df43e: mov eax,edi
0x100df440: test eax,eax
0x100df442: fild dword ptr [esp+30h]
0x100df446: mov [esp+30h],eax
0x100df44a: fmul st,st(2)
0x100df44c: fild dword ptr [esp+30h]
0x100df450: jge short 100DF458h
0x100df452: fadd qword ptr ds:[100FEFC8h]
0x100df458: fsub st,st(1)
0x100df45a: add ecx,0FFFFFFFEh
```

```
0x100df45d: fnstcw [esp+3Ch]
0x100df461: fmul qword ptr [ebx]
0x100df463: movzx eax,word ptr [esp+3Ch]
0x100df468: or eax,0C00h
0x100df46d: mov [esp+30h],eax
0x100df471: faddp
0x100df473: fstp dword ptr [esp+44h]
0x100df477: fld dword ptr [esp+44h]
0x100df47b: fld st(0)
0x100df47d: fldcw [esp+30h]
0x100df481: fistp qword ptr [esp+30h]
0x100df485: mov eax,[esp+30h]
0x100df489: cmp eax,ecx
0x100df48b: mov [esp+40h],eax
0x100df48f: fldcw [esp+3Ch]
0x100df493: jg near ptr 100DF5F3h
0x100df499: test eax,eax
0x100df49b: mov [esp+30h],eax
0x100df49f: fild dword ptr [esp+30h]
0x100df4a3: jge short 100DF4ABh
0x100df4a5: fadd dword ptr ds:[100FE7C0h]
0x100df4ab: fsubp
0x100df4ad: xor edx,edx
0x100df4af: cmp [esi+66C0h],dx
0x100df4b6: fstp dword ptr [esp+44h]
0x100df4ba: jbe near ptr 100DF5F5h
0x100df4c0: fld dword ptr [esp+44h]
0x100df4c4: movzx ecx,word ptr [esi+66C0h]
0x100df4cb: mov [esp+30h],ecx
0x100df4cf: mov eax,edx
0x100df4d1: test eax,eax
0x100df4d3: fild dword ptr [esp+30h]
0x100df4d7: mov [esp+30h],eax
0x100df4db: fmul st,st(3)
0x100df4dd: fild dword ptr [esp+30h]
0x100df4e1: jge short 100DF4E9h
0x100df4e3: fadd qword ptr ds:[100FEFC8h]
0x100df4e9: fsub st,st(1)
0x100df4eb: fnstcw [esp+3Ch]
0x100df4ef: fmul qword ptr [ebx]
0x100df4f1: movzx eax,word ptr [esp+3Ch]
0x100df4f6: or eax,0C00h
0x100df4fb: mov [esp+30h],eax
0x100df4ff: lea ebx,[ecx-2]
0x100df502: faddp
0x100df504: fstp dword ptr [esp+44h]
0x100df508: fld dword ptr [esp+44h]
0x100df50c: fld st(0)
0x100df50e: fldcw [esp+30h]
0x100df512: fistp qword ptr [esp+30h]
0x100df516: mov eax,[esp+30h]
0x100df51a: cmp eax,ebx
0x100df51c: fldcw [esp+3Ch]
0x100df520: jg near ptr 100DF5DBh
0x100df526: mov ebx,eax
0x100df528: test ebx,ebx
0x100df52a: mov [esp+30h],ebx
0x100df52e: fild dword ptr [esp+30h]
0x100df532: jge short 100DF53Ah
0x100df534: fadd dword ptr ds:[100FE7C0h]
0x100df53a: fsubp
0x100df53c: mov ebx,ecx
0x100df53e: imul ebx,[esp+40h]
0x100df543: fnstcw [esp+3Ch]
0x100df547: fstp dword ptr [esp+44h]
0x100df54b: fld dword ptr [esp+44h]
0x100df54f: fld st(0)
```

```
0x100df551: fsubr st,st(3)
0x100df553: add ebx,eax
0x100df555: mov eax,[esp+48h]
0x100df559: lea eax,[eax+ebx*2]
0x100df55c: movzx ebx,word ptr [eax+2]
0x100df560: mov [esp+30h],ebx
0x100df564: movzx ebx,word ptr [eax]
0x100df567: fild dword ptr [esp+30h]
0x100df56b: mov [esp+30h],ebx
0x100df56f: movzx ebx,word ptr [eax+ecx*2+2]
0x100df574: fmul st,st(2)
0x100df576: fild dword ptr [esp+30h]
0x100df57a: movzx eax,word ptr [eax+ecx*2]
0x100df57e: imul ecx,edi
0x100df581: fmul st,st(2)
0x100df583: faddp
0x100df585: fld st(3)
0x100df587: fsubr st,st(5)
0x100df589: fmulp
0x100df58b: mov [esp+30h],ebx
0x100df58f: mov ebx,[esp+58h]
0x100df593: fild dword ptr [esp+30h]
0x100df597: mov [esp+30h],eax
0x100df59b: movzx eax,word ptr [esp+3Ch]
0x100df5a0: fmulp st(3),st
0x100df5a2: or eax,0C00h
0x100df5a7: fild dword ptr [esp+30h]
0x100df5ab: mov [esp+30h],eax
0x100df5af: add ecx,edx
0x100df5b1: lea ecx,[ebx+ecx*4]
0x100df5b4: mov ebx,[esi+66E0h]
0x100df5ba: fmulp st(2),st
0x100df5bc: fxch st(2)
0x100df5be: faddp
0x100df5c0: fmul st,st(2)
0x100df5c2: faddp
0x100df5c4: fldcw [esp+30h]
0x100df5c8: fistp dword ptr [esp+30h]
0x100df5cc: mov ax,[esp+30h]
0x100df5d1: mov [ebx+ecx*2],ax
0x100df5d5: fldcw [esp+3Ch]
0x100df5d9: jmp short 100DF5DDh
0x100df5db: fstp st(0)
0x100df5dd: movzx eax,word ptr [esi+66C0h]
0x100df5e4: mov ebx,[esp+50h]
0x100df5e8: add edx,1
0x100df5eb: cmp edx,eax
0x100df5ed: jb near ptr 100DF4C4h
0x100df5f3: fstp st(0)
0x100df5f5: movzx ecx,word ptr [esi+66BEh]
0x100df5fc: add edi,1
0x100df5ff: cmp edi,ecx
0x100df601: jb near ptr 100DF433h
0x100df607: fstp st(0)
0x100df609: fstp st(0)
0x100df60b: mov edx,[esp+48h]
0x100df60f: push edx
0x100df610: call dword ptr ds:[100FE554h]
0x100df616: add esp,4
0x100df619: mov eax,[esp+4Ch]
0x100df61d: add dword ptr [esp+58h],2
0x100df622: add eax,4
0x100df625: add ebx,10h
0x100df628: cmp eax,8
0x100df62b: mov [esp+50h],ebx
0x100df62f: mov [esp+4Ch],eax
0x100df633: jb near ptr 100DF3C4h
```

```
0x100df639: pop edi
0x100df63a: pop esi
0x100df63b: pop ebx
0x100df63c: mov esp,ebp
0x100df63e: pop ebp
0x100df63f: ret
```

`sub_100dea30` is called by sub_10016040. It works with the text "Correcting chromatic aberration...", "Scaling with darkness %d, saturation %d, and". It delegates to j_memset, sub_100228d0, sub_100d2e20 (4 callees).

12.1275 sub_100df640 at 0x100df640 (129 instructions)

```
0x100df640: push ecx
0x100df641: push esi
0x100df642: mov esi,[esp+0Ch]
0x100df646: cmp word ptr [esi+66BCh],0
0x100df64e: push edi
0x100df64f: je near ptr 100DF759h
0x100df655: cmp dword ptr [esi+0B4h],0
0x100df65c: je short 100DF67Fh
0x100df65e: mov ax,[esi+66BEh]
0x100df665: mov cx,[esi+66C0h]
0x100df66c: mov [esi+66B4h],ax
0x100df673: mov [esi+66B6h],cx
0x100df67a: jmp near ptr 100DF759h
0x100df67f: movzx edx,word ptr [esi+66B6h]
0x100df686: movzx eax,word ptr [esi+66B4h]
0x100df68d: imul edx,eax
0x100df690: push ebp
0x100df691: push 8
0x100df693: push edx
0x100df694: call dword ptr ds:[100FE4FCh]
0x100df69a: mov ebp,eax
0x100df69c: push 1012E3B4h
0x100df6a1: push ebp
0x100df6a2: push esi
0x100df6a3: mov [esp+20h],ebp
0x100df6a7: call 100D2E20h
0x100df6ac: xor eax,eax
0x100df6ae: add esp,14h
0x100df6b1: cmp [esi+66B4h],ax
0x100df6b8: mov [esp+14h],eax
0x100df6bc: jbe near ptr 100DF739h
0x100df6c2: push ebx
0x100df6c3: xor edi,edi
0x100df6c5: cmp [esi+66B6h],di
0x100df6cc: jbe short 100DF726h
0x100df6ce: mov ebx,eax
0x100df6d0: sar ebx,1
0x100df6d2: push edi
0x100df6d3: push eax
0x100df6d4: push esi
0x100df6d5: call 100D2D10h
0x100df6da: movzx ecx,word ptr [esi+66C0h]
0x100df6e1: movzx ebp,word ptr [esi+66B6h]
0x100df6e8: imul ecx,ebx
0x100df6eb: imul ebp,[esp+24h]
0x100df6f0: mov edx,edi
0x100df6f2: sar edx,1
0x100df6f4: add ecx,edx
0x100df6f6: mov edx,[esi+66E0h]
0x100df6fc: add ebp,edi
0x100df6fe: lea ecx,[eax+ecx*4]
0x100df701: mov cx,[edx+ecx*2]
0x100df705: lea eax,[eax+ebp*4]
0x100df708: mov ebp,[esp+1Ch]
```

```
0x100df70c: mov [ebp+eax*2],cx
0x100df711: movzx edx,word ptr [esi+66B6h]
0x100df718: mov eax,[esp+24h]
0x100df71c: add edi,1
0x100df71f: add esp,0Ch
0x100df722: cmp edi,edx
0x100df724: jl short 100DF6D2h
0x100df726: movzx ecx,word ptr [esi+66B4h]
0x100df72d: add eax,1
0x100df730: cmp eax,ecx
0x100df732: mov [esp+18h],eax
0x100df736: jl short 100DF6C3h
0x100df738: pop ebx
0x100df739: mov edx,[esi+66E0h]
0x100df73f: push edx
0x100df740: call dword ptr ds:[100FE554h]
0x100df746: add esp,4
0x100df749: mov [esi+66E0h],ebp
0x100df74f: mov word ptr [esi+66BCh],0
0x100df758: pop ebp
0x100df759: mov edx,[esi+65B0h]
0x100df75f: test edx,edx
0x100df761: je near ptr 100DF823h
0x100df767: cmp dword ptr [esi+66D0h],3
0x100df76e: jne near ptr 100DF823h
0x100df774: mov eax,[esi+0B8h]
0x100df77a: test eax,eax
0x100df77c: mov [esi+6600h],eax
0x100df782: je short 100DF793h
0x100df784: mov dword ptr [esi+66D0h],4
0x100df78e: jmp near ptr 100DF823h
0x100df793: movzx eax,word ptr [esi+66B4h]
0x100df79a: shr edx,5
0x100df79d: and edx,1
0x100df7a0: cmp edx,eax
0x100df7a2: jge short 100DF809h
0x100df7a4: mov eax,[esi+65B0h]
0x100df7aa: mov ecx,edx
0x100df7ac: and ecx,7
0x100df7af: lea ecx,[ecx*4+2]
0x100df7b6: shr eax,cl
0x100df7b8: movzx ecx,word ptr [esi+66B6h]
0x100df7bf: and eax,1
0x100df7c2: cmp eax,ecx
0x100df7c4: jge short 100DF7FBh
0x100df7c6: jmp short 100DF7D0h
0x100df7d0: movzx ecx,word ptr [esi+66B6h]
0x100df7d7: mov edi,[esi+66E0h]
0x100df7dd: imul ecx,edx
0x100df7e0: add ecx,eax
0x100df7e2: lea ecx,[edi+ecx*8]
0x100df7e5: mov di,[ecx+6]
0x100df7e9: mov [ecx+2],di
0x100df7ed: movzx ecx,word ptr [esi+66B6h]
0x100df7f4: add eax,2
0x100df7f7: cmp eax,ecx
0x100df7f9: jl short 100DF7D0h
0x100df7fb: movzx eax,word ptr [esi+66B4h]
0x100df802: add edx,2
0x100df805: cmp edx,eax
0x100df807: jl short 100DF7A4h
0x100df809: mov eax,[esi+65B0h]
0x100df80f: mov ecx,eax
0x100df811: and ecx,0D5555555h
0x100df817: add ecx,ecx
0x100df819: not ecx
0x100df81b: and ecx,eax
```

```
0x100df81d: mov [esi+65B0h],ecx
0x100df823: cmp dword ptr [esi+0B4h],0
0x100df82a: je short 100DF836h
0x100df82c: mov dword ptr [esi+65B0h],0
0x100df836: pop edi
0x100df837: pop esi
0x100df838: pop ecx
0x100df839: ret
```

`sub_100df640` is called by sub_10016040. It accesses memory-mapped peripheral registers at 0xd5555555. It delegates to sub_100d2d10, sub_100d2e20 (2 callees).

12.1276 sub_100df840 at 0x100df840 (126 instructions)

```
0x100df840: sub esp,28h
0x100df843: push ebp
0x100df844: push edi
0x100df845: mov edi,[esp+34h]
0x100df849: xor ebp,ebp
0x100df84b: cmp [edi+66B4h],bp
0x100df852: mov [esp+8],ebp
0x100df856: jbe near ptr 100DF9DAh
0x100df85c: push ebx
0x100df85d: push esi
0x100df85e: mov edi,edi
0x100df860: movzx eax,word ptr [edi+66B6h]
0x100df867: xor esi,esi
0x100df869: test eax,eax
0x100df86b: mov [esp+3Ch],esi
0x100df86f: jbe near ptr 100DF9C2h
0x100df875: lea eax,[ebp-1]
0x100df878: mov [esp+14h],eax
0x100df87c: lea esp,[esp]
0x100df880: mov ecx,[esp+40h]
0x100df884: cmp esi,ecx
0x100df886: jne short 100DF8A8h
0x100df888: cmp ebp,ecx
0x100df88a: jb short 100DF8A8h
0x100df88c: movzx edx,word ptr [edi+66B4h]
0x100df893: sub edx,ecx
0x100df895: cmp ebp,edx
0x100df897: jae short 100DF8A8h
0x100df899: movzx eax,word ptr [edi+66B6h]
0x100df8a0: sub eax,ecx
0x100df8a2: mov [esp+3Ch],eax
0x100df8a6: mov esi,eax
0x100df8a8: mov ebx,[esp+14h]
0x100df8ac: xor eax,eax
0x100df8ae: mov [esp+18h],eax
0x100df8b2: mov [esp+1Ch],eax
0x100df8b6: mov [esp+20h],eax
0x100df8ba: mov [esp+24h],eax
0x100df8be: mov [esp+28h],eax
0x100df8c2: mov [esp+2Ch],eax
0x100df8c6: mov [esp+30h],eax
0x100df8ca: mov [esp+34h],eax
0x100df8ce: lea eax,[ebp+2]
0x100df8d1: cmp ebx,eax
0x100df8d3: je near ptr 100DF959h
0x100df8d9: movzx ebp,word ptr [edi+66B6h]
0x100df8e0: lea eax,[esi+2]
0x100df8e3: imul ebp,ebx
0x100df8e6: add esi,0FFFFFFFFh
0x100df8e9: cmp esi,eax
0x100df8eb: je short 100DF93Ah
0x100df8ed: lea ecx,[ecx]
```

```
0x100df8f0: movzx eax,word ptr [edi+66B4h]
0x100df8f7: cmp ebx,eax
0x100df8f9: jae short 100DF92Ch
0x100df8fb: movzx eax,word ptr [edi+66B6h]
0x100df902: cmp esi,eax
0x100df904: jae short 100DF92Ch
0x100df906: push esi
0x100df907: push ebx
0x100df908: push edi
0x100df909: call 100D2D10h
0x100df90e: lea ecx,[esi+ebp]
0x100df911: lea edx,[eax+ecx*4]
0x100df914: mov ecx,[edi+66E0h]
0x100df91a: movzx edx,word ptr [ecx+edx*2]
0x100df91e: add [esp+eax*4+24h],edx
0x100df922: lea eax,[esp+eax*4+34h]
0x100df926: add esp,0Ch
0x100df929: add dword ptr [eax],1
0x100df92c: mov eax,[esp+3Ch]
0x100df930: add esi,1
0x100df933: add eax,2
0x100df936: cmp esi,eax
0x100df938: jne short 100DF8F0h
0x100df93a: movzx ecx,word ptr [edi+66B6h]
0x100df941: mov esi,[esp+3Ch]
0x100df945: add ebp,ecx
0x100df947: mov ecx,[esp+10h]
0x100df94b: add ebx,1
0x100df94e: add ecx,2
0x100df951: cmp ebx,ecx
0x100df953: jne short 100DF8E6h
0x100df955: mov ebp,[esp+10h]
0x100df959: push esi
0x100df95a: push ebp
0x100df95b: push edi
0x100df95c: call 100D2D10h
0x100df961: xor ecx,ecx
0x100df963: add esp,0Ch
0x100df966: cmp [edi+66D0h],ecx
0x100df96c: mov [esp+3Ch],eax
0x100df970: jle short 100DF9ACh
0x100df972: cmp ecx,[esp+3Ch]
0x100df976: je short 100DF9A1h
0x100df978: mov ebx,[esp+ecx*4+28h]
0x100df97c: test ebx,ebx
0x100df97e: je short 100DF9A1h
0x100df980: mov eax,[esp+ecx*4+18h]
0x100df984: xor edx,edx
0x100df986: div ebx
0x100df988: movzx edx,word ptr [edi+66B6h]
0x100df98f: mov ebx,[edi+66E0h]
0x100df995: imul edx,ebp
0x100df998: add edx,esi
0x100df99a: lea edx,[ecx+edx*4]
0x100df99d: mov [ebx+edx*2],ax
0x100df9a1: add ecx,1
0x100df9a4: cmp ecx,[edi+66D0h]
0x100df9aa: jl short 100DF972h
0x100df9ac: movzx eax,word ptr [edi+66B6h]
0x100df9b3: add esi,1
0x100df9b6: cmp esi,eax
0x100df9b8: mov [esp+3Ch],esi
0x100df9bc: jb near ptr 100DF880h
0x100df9c2: movzx eax,word ptr [edi+66B4h]
0x100df9c9: add ebp,1
0x100df9cc: cmp ebp,eax
0x100df9ce: mov [esp+10h],ebp
```

```
0x100df9d2: jb near ptr 100DF860h
0x100df9d8: pop esi
0x100df9d9: pop ebx
0x100df9da: pop edi
0x100df9db: pop ebp
0x100df9dc: add esp,28h
0x100df9df: ret
```

`sub_100df840` is called by sub_100df9e0, sub_100e03b0, sub_100e0950. It delegates to sub_100d2d10 (1 callees).

12.1277 sub_100df9e0 at 0x100df9e0 (190 instructions)

```
0x100df9e0: mov eax,8024h
0x100df9e5: call 100229B0h
0x100df9ea: push ebx
0x100df9eb: push ebp
0x100df9ec: push esi
0x100df9ed: mov esi,[esp+8034h]
0x100df9f4: cmp dword ptr [esi+0BCh],0
0x100df9fb: push edi
0x100df9fc: je short 100DFA16h
0x100df9fe: push 1012E3C8h
0x100dfa03: call dword ptr ds:[100FE4C4h]
0x100dfa09: add eax,40h
0x100dfa0c: push eax
0x100dfa0d: call dword ptr ds:[100FE4C8h]
0x100dfa13: add esp,8
0x100dfa16: push 1
0x100dfa18: push esi
0x100dfa19: call 100DF840h
0x100dfa1e: mov eax,[esi+66D0h]
0x100dfa24: mov [esp+38h],eax
0x100dfa28: add esp,8
0x100dfa2b: mov dword ptr [esp+10h],0
0x100dfa33: lea eax,[esp+34h]
0x100dfa37: jmp short 100DFA40h
0x100dfa40: mov dword ptr [esp+14h],0
0x100dfa48: mov [esp+1Ch],eax
0x100dfa4c: lea esp,[esp]
0x100dfa50: mov esi,eax
0x100dfa52: xor eax,eax
0x100dfa54: mov [esp+20h],eax
0x100dfa58: mov [esp+24h],eax
0x100dfa5c: mov [esp+28h],eax
0x100dfa60: mov [esp+2Ch],eax
0x100dfa64: or ebp,0FFFFFFFFh
0x100dfa67: xor ecx,ecx
0x100dfa69: or edi,0FFFFFFFFh
0x100dfa6c: test ebp,ebp
0x100dfa6e: sete cl
0x100dfa71: mov [esp+18h],ecx
0x100dfa75: xor ebx,ebx
0x100dfa77: test edi,edi
0x100dfa79: sete bl
0x100dfa7c: add ebx,[esp+18h]
0x100dfa80: cmp ebx,2
0x100dfa83: je short 100DFAD9h
0x100dfa85: mov edx,[esp+14h]
0x100dfa89: mov ecx,[esp+10h]
0x100dfa8d: lea eax,[edi+edx]
0x100dfa90: push eax
0x100dfa91: mov eax,[esp+803Ch]
0x100dfa98: lea edx,[ecx+ebp]
0x100dfa9b: push edx
0x100dfa9c: push eax
0x100dfa9d: call 100D2D10h
```

```
0x100dfaa2: mov ecx,[esp+8044h]
0x100dfaa9: movzx edx,word ptr [ecx+66B6h]
0x100dfab0: imul edx,ebp
0x100dfab3: add edx,edi
0x100dfab5: lea ecx,[eax+edx*4]
0x100dfab8: mov [esi],ecx
0x100dfaba: add esi,4
0x100dfabd: mov [esi],ebx
0x100dfabf: add esi,4
0x100dfac2: mov edx,1
0x100dfac7: mov ecx,ebx
0x100dfac9: shl edx,cl
0x100dfacb: mov [esi],eax
0x100dfacd: lea eax,[esp+eax*4+2Ch]
0x100dfad1: add esp,0Ch
0x100dfad4: add esi,4
0x100dfad7: add [eax],edx
0x100dfad9: add edi,1
0x100dfadc: cmp edi,1
0x100dfadf: jle short 100DFA75h
0x100dfae1: add ebp,1
0x100dfae4: cmp ebp,1
0x100dfae7: jle near ptr 100DFA67h
0x100dfaed: xor edi,edi
0x100dfaef: cmp [esp+30h],edi
0x100dfaf3: jle short 100DFB3Ch
0x100dfaf5: jmp short 100DFB00h
0x100dfb00: mov eax,[esp+14h]
0x100dfb04: mov ecx,[esp+10h]
0x100dfb08: mov edx,[esp+8038h]
0x100dfb0f: push eax
0x100dfb10: push ecx
0x100dfb11: push edx
0x100dfb12: call 100D2D10h
0x100dfb17: add esp,0Ch
0x100dfb1a: cmp edi,eax
0x100dfb1c: je short 100DFB33h
0x100dfb1e: mov eax,100h
0x100dfb23: cdq
0x100dfb24: idiv dword ptr [esp+edi*4+20h]
0x100dfb28: mov [esi],edi
0x100dfb2a: add esi,4
0x100dfb2d: add esi,4
0x100dfb30: mov [esi-4],eax
0x100dfb33: add edi,1
0x100dfb36: cmp edi,[esp+30h]
0x100dfb3a: jl short 100DFB00h
0x100dfb3c: mov ecx,[esp+14h]
0x100dfb40: mov eax,[esp+1Ch]
0x100dfb44: add ecx,1
0x100dfb47: add eax,80h
0x100dfb4c: cmp ecx,10h
0x100dfb4f: mov [esp+14h],ecx
0x100dfb53: mov [esp+1Ch],eax
0x100dfb57: jl near ptr 100DFA50h
0x100dfb5d: mov ecx,[esp+10h]
0x100dfb61: add ecx,1
0x100dfb64: cmp ecx,10h
0x100dfb67: mov [esp+10h],ecx
0x100dfb6b: jl near ptr 100DFA40h
0x100dfb71: mov esi,[esp+8038h]
0x100dfb78: movzx eax,word ptr [esi+66B4h]
0x100dfb7f: mov edx,1
0x100dfb84: sub eax,edx
0x100dfb86: cmp eax,edx
0x100dfb88: mov [esp+10h],edx
0x100dfb8c: jle near ptr 100DFC8Bh
```

```
0x100dfb92: jmp short 100DFBA0h
0x100dfba0: movzx eax,word ptr [esi+66B6h]
0x100dfba7: mov ebx,1
0x100dfbac: lea ecx,[eax-1]
0x100dfbaf: cmp ecx,ebx
0x100dfbb1: jle near ptr 100DFC72h
0x100dfbb7: mov ecx,edx
0x100dfbb9: and ecx,0Fh
0x100dfbbc: shl ecx,4
0x100dfbbf: mov [esp+18h],ecx
0x100dfbc3: jmp short 100DFBD4h
0x100dfbd0: mov ecx,[esp+18h]
0x100dfbd4: imul eax,edx
0x100dfbd7: mov edx,[esi+66E0h]
0x100dfbdd: add eax,ebx
0x100dfbdf: lea edi,[edx+eax*8]
0x100dfbe2: mov eax,ebx
0x100dfbe4: and eax,0Fh
0x100dfbe7: add eax,ecx
0x100dfbe9: shl eax,7
0x100dfbec: xor ecx,ecx
0x100dfbee: lea eax,[esp+eax+34h]
0x100dfbf2: mov [esp+20h],ecx
0x100dfbf6: mov [esp+24h],ecx
0x100dfbfa: mov [esp+28h],ecx
0x100dfbfe: mov [esp+2Ch],ecx
0x100dfc02: mov esi,8
0x100dfc07: mov ecx,[eax]
0x100dfc09: movzx ebp,word ptr [edi+ecx*2]
0x100dfc0d: mov ecx,[eax+4]
0x100dfc10: mov edx,[eax+8]
0x100dfc13: shl ebp,cl
0x100dfc15: lea edx,[esp+edx*4+20h]
0x100dfc19: sub esi,1
0x100dfc1c: add eax,0Ch
0x100dfc1f: add [edx],ebp
0x100dfc21: test esi,esi
0x100dfc23: jne short 100DFC07h
0x100dfc25: mov esi,[esp+8038h]
0x100dfc2c: mov edx,[esi+66D0h]
0x100dfc32: sub edx,1
0x100dfc35: je short 100DFC59h
0x100dfc37: jmp short 100DFC40h
0x100dfc40: mov ecx,[eax]
0x100dfc42: mov ebp,[esp+ecx*4+20h]
0x100dfc46: imul ebp,[eax+4]
0x100dfc4a: sar ebp,8
0x100dfc4d: add eax,8
0x100dfc50: sub edx,1
0x100dfc53: mov [edi+ecx*2],bp
0x100dfc57: jne short 100DFC40h
0x100dfc59: movzx eax,word ptr [esi+66B6h]
0x100dfc60: lea edx,[eax-1]
0x100dfc63: add ebx,1
0x100dfc66: cmp ebx,edx
0x100dfc68: mov edx,[esp+10h]
0x100dfc6c: jl near ptr 100DFBD0h
0x100dfc72: movzx eax,word ptr [esi+66B4h]
0x100dfc79: add edx,1
0x100dfc7c: sub eax,1
0x100dfc7f: cmp edx,eax
0x100dfc81: mov [esp+10h],edx
0x100dfc85: jl near ptr 100DFBA0h
0x100dfc8b: pop edi
0x100dfc8c: pop esi
0x100dfc8d: pop ebp
0x100dfc8e: pop ebx
```

```
0x100dfc8f: add esp,8024h
0x100dfc95: ret
```

`sub_100df9e0` is called by sub_10016040, sub_100dfca0. It works with the text "Bilinear interpolation...". It delegates to sub_100229b0, sub_100d2d10, sub_100df840 (3 callees).

12.1278 sub_100dfca0 at 0x100dfca0 (521 instructions)

```
0x100dfca0: sub esp,474h
0x100dfca6: push ebx
0x100dfca7: push ebp
0x100dfca8: mov ebp,[esp+480h]
0x100dfcaf: push esi
0x100dfcb0: push edi
0x100dfcb1: mov edi,7
0x100dfcb6: mov esi,1
0x100dfcbb: push ebp
0x100dfcbc: mov [esp+30h],edi
0x100dfcc0: mov [esp+2Ch],esi
0x100dfcc4: call 100DF9E0h
0x100dfcc9: add esp,4
0x100dfccc: cmp dword ptr [ebp+0BCh],0
0x100dfcd3: je short 100DFCEDh
0x100dfcd5: push 1012E3F8h
0x100dfcda: call dword ptr ds:[100FE4C4h]
0x100dfce0: add eax,40h
0x100dfce3: push eax
0x100dfce4: call dword ptr ds:[100FE4C8h]
0x100dfcea: add esp,8
0x100dfced: cmp [ebp+65B0h],esi
0x100dfcf3: jne short 100DFD04h
0x100dfcf5: mov esi,0Fh
0x100dfcfa: mov edi,esi
0x100dfcfc: mov [esp+28h],esi
0x100dfd00: mov [esp+2Ch],edi
0x100dfd04: lea eax,[edi+1]
0x100dfd07: add esi,1
0x100dfd0a: imul esi,eax
0x100dfd0d: push 500h
0x100dfd12: push esi
0x100dfd13: call dword ptr ds:[100FE4FCh]
0x100dfd19: mov esi,eax
0x100dfd1b: push 1012E3E4h
0x100dfd20: push esi
0x100dfd21: push ebp
0x100dfd22: call 100D2E20h
0x100dfd27: add esp,14h
0x100dfd2a: test edi,edi
0x100dfd2c: mov dword ptr [esp+10h],0
0x100dfd34: jl near ptr 100E0013h
0x100dfd3a: lea ebx,[ebx]
0x100dfd40: cmp dword ptr [esp+28h],0
0x100dfd45: mov dword ptr [esp+14h],0
0x100dfd4d: jl near ptr 100DFFFEh
0x100dfd53: mov eax,[esp+10h]
0x100dfd57: shl eax,4
0x100dfd5a: mov [esp+60h],eax
0x100dfd5e: mov edi,edi
0x100dfd60: mov ecx,[esp+60h]
0x100dfd64: mov edx,[esp+14h]
0x100dfd68: add ecx,edx
0x100dfd6a: mov [esp+ecx*4+84h],esi
0x100dfd71: mov edi,1012B418h
0x100dfd76: mov dword ptr [esp+20h],40h
0x100dfd7e: mov edi,edi
0x100dfd80: movsx eax,byte ptr [edi]
```

```
0x100dfd83: movsx ecx,byte ptr [edi+1]
0x100dfd87: add edi,1
0x100dfd8a: movsx edx,byte ptr [edi+1]
0x100dfd8e: add edi,1
0x100dfd91: mov [esp+1Ch],edx
0x100dfd95: movsx edx,byte ptr [edi+1]
0x100dfd99: add edi,1
0x100dfd9c: mov [esp+18h],edx
0x100dfda0: movsx edx,byte ptr [edi+1]
0x100dfda4: add edi,1
0x100dfda7: mov [esp+34h],edx
0x100dfdab: movsx edx,byte ptr [edi+1]
0x100dfdaf: add edi,1
0x100dfdb2: mov [esp+30h],edx
0x100dfdb6: mov edx,[esp+14h]
0x100dfdba: mov [esp+38h],ecx
0x100dfdbe: add ecx,edx
0x100dfdc0: push ecx
0x100dfdc1: mov ecx,[esp+14h]
0x100dfdc5: mov [esp+28h],eax
0x100dfdc9: add eax,ecx
0x100dfdcb: push eax
0x100dfdcc: push ebp
0x100dfdcd: add edi,1
0x100dfdd0: call 100D2D10h
0x100dfdd5: mov edx,[esp+24h]
0x100dfdd9: mov ecx,[esp+28h]
0x100dfddd: mov ebx,eax
0x100dfddf: mov eax,[esp+20h]
0x100dfde3: add edx,eax
0x100dfde5: push edx
0x100dfde6: mov edx,[esp+20h]
0x100dfdea: add ecx,edx
0x100dfdec: push ecx
0x100dfded: push ebp
0x100dfdee: call 100D2D10h
0x100dfdf3: add esp,18h
0x100dfdf6: cmp eax,ebx
0x100dfdf8: jne near ptr 100DFF11h
0x100dfdfe: mov eax,[esp+14h]
0x100dfe02: mov ecx,[esp+10h]
0x100dfe06: add eax,1
0x100dfe09: push eax
0x100dfe0a: push ecx
0x100dfe0b: push ebp
0x100dfe0c: call 100D2D10h
0x100dfe11: add esp,0Ch
0x100dfe14: cmp eax,ebx
0x100dfe16: jne short 100DFE37h
0x100dfe18: mov edx,[esp+14h]
0x100dfe1c: mov eax,[esp+10h]
0x100dfe20: push edx
0x100dfe21: add eax,1
0x100dfe24: push eax
0x100dfe25: push ebp
0x100dfe26: call 100D2D10h
0x100dfe2b: add esp,0Ch
0x100dfe2e: cmp eax,ebx
0x100dfe30: mov ecx,2
0x100dfe35: je short 100DFE3Ch
0x100dfe37: mov ecx,1
0x100dfe3c: mov eax,[esp+24h]
0x100dfe40: sub eax,[esp+1Ch]
0x100dfe44: cdq
0x100dfe45: xor eax,edx
0x100dfe47: sub eax,edx
0x100dfe49: cmp eax,ecx
```

```
0x100dfe4b: jne short 100DFE62h
0x100dfe4d: mov eax,[esp+38h]
0x100dfe51: sub eax,[esp+18h]
0x100dfe55: cdq
0x100dfe56: xor eax,edx
0x100dfe58: sub eax,edx
0x100dfe5a: cmp eax,ecx
0x100dfe5c: je near ptr 100DFF11h
0x100dfe62: movzx ecx,word ptr [ebp+66B6h]
0x100dfe69: imul ecx,[esp+24h]
0x100dfe6e: add ecx,[esp+38h]
0x100dfe72: add esi,4
0x100dfe75: lea edx,[ebx+ecx*4]
0x100dfe78: mov [esi-4],edx
0x100dfe7b: movzx eax,word ptr [ebp+66B6h]
0x100dfe82: imul eax,[esp+1Ch]
0x100dfe87: add eax,[esp+18h]
0x100dfe8b: mov edx,[esp+34h]
0x100dfe8f: lea ecx,[ebx+eax*4]
0x100dfe92: mov al,[esp+30h]
0x100dfe96: mov [esi],ecx
0x100dfe98: add esi,4
0x100dfe9b: mov [esi],edx
0x100dfe9d: add esi,4
0x100dfea0: test al,1
0x100dfea2: je short 100DFEADh
0x100dfea4: mov dword ptr [esi],0
0x100dfeaa: add esi,4
0x100dfead: test al,2
0x100dfeaf: je short 100DFEBAh
0x100dfeb1: mov dword ptr [esi],1
0x100dfeb7: add esi,4
0x100dfeba: test al,4
0x100dfebc: je short 100DFEC7h
0x100dfebe: mov dword ptr [esi],2
0x100dfec4: add esi,4
0x100dfec7: test al,8
0x100dfec9: je short 100DFED4h
0x100dfecb: mov dword ptr [esi],3
0x100dfed1: add esi,4
0x100dfed4: test al,10h
0x100dfed6: je short 100DFEE1h
0x100dfed8: mov dword ptr [esi],4
0x100dfede: add esi,4
0x100dfee1: test al,20h
0x100dfee3: je short 100DFEEEh
0x100dfee5: mov dword ptr [esi],5
0x100dfeeb: add esi,4
0x100dfeee: test al,40h
0x100dfef0: je short 100DFEFBh
0x100dfef2: mov dword ptr [esi],6
0x100dfef8: add esi,4
0x100dfefb: test al,al
0x100dfefd: jns short 100DFF08h
0x100dfeff: mov dword ptr [esi],7
0x100dff05: add esi,4
0x100dff08: mov dword ptr [esi],0FFFFFFFFh
0x100dff0e: add esi,4
0x100dff11: sub dword ptr [esp+20h],1
0x100dff16: jne near ptr 100DFD80h
0x100dff1c: mov dword ptr [esi],7FFFFFFFh
0x100dff22: add esi,4
0x100dff25: mov dword ptr ds:[1013DB34h],1012B598h
0x100dff2f: mov dword ptr [esp+20h],8
0x100dff37: jmp short 100DFF40h
0x100dff40: mov eax,ds:[1013DB34h]
0x100dff45: movsx edi,byte ptr [eax]
```

```
0x100dff48: movsx ebx,byte ptr [eax+1]
0x100dff4c: mov ecx,[esp+14h]
0x100dff50: mov edx,[esp+10h]
0x100dff54: add eax,1
0x100dff57: add eax,1
0x100dff5a: mov ds:[1013DB34h],eax
0x100dff5f: movzx eax,word ptr [ebp+66B6h]
0x100dff66: imul eax,edi
0x100dff69: add eax,ebx
0x100dff6b: push ecx
0x100dff6c: add eax,eax
0x100dff6e: add eax,eax
0x100dff70: push edx
0x100dff71: mov [esi],eax
0x100dff73: push ebp
0x100dff74: add esi,4
0x100dff77: call 100D2D10h
0x100dff7c: mov edx,[esp+1Ch]
0x100dff80: mov [esp+24h],eax
0x100dff84: mov eax,[esp+20h]
0x100dff88: lea ecx,[ebx+eax]
0x100dff8b: push ecx
0x100dff8c: lea eax,[edi+edx]
0x100dff8f: push eax
0x100dff90: push ebp
0x100dff91: call 100D2D10h
0x100dff96: add esp,18h
0x100dff99: cmp eax,[esp+18h]
0x100dff9d: je short 100DFFD5h
0x100dff9f: mov ecx,[esp+14h]
0x100dffa3: mov eax,[esp+10h]
0x100dffa7: lea edx,[ecx+ebx*2]
0x100dffaa: push edx
0x100dffab: lea ecx,[eax+edi*2]
0x100dffae: push ecx
0x100dffaf: push ebp
0x100dffb0: call 100D2D10h
0x100dffb5: add esp,0Ch
0x100dffb8: cmp eax,[esp+18h]
0x100dffbc: jne short 100DFFD5h
0x100dffbe: movzx edx,word ptr [ebp+66B6h]
0x100dffc5: mov eax,[esp+18h]
0x100dffc9: imul edx,edi
0x100dffcc: add edx,ebx
0x100dffce: lea ecx,[eax+edx*8]
0x100dffd1: mov [esi],ecx
0x100dffd3: jmp short 100DFFDBh
0x100dffd5: mov dword ptr [esi],0
0x100dffdb: add esi,4
0x100dffde: sub dword ptr [esp+20h],1
0x100dffe3: jne near ptr 100DFF40h
0x100dffe9: mov eax,[esp+14h]
0x100dffed: add eax,1
0x100dfff0: cmp eax,[esp+28h]
0x100dfff4: mov [esp+14h],eax
0x100dfff8: jle near ptr 100DFD60h
0x100dfffe: mov eax,[esp+10h]
0x100e0002: add eax,1
0x100e0005: cmp eax,[esp+2Ch]
0x100e0009: mov [esp+10h],eax
0x100e000d: jle near ptr 100DFD40h
0x100e0013: movzx eax,word ptr [ebp+66B6h]
0x100e001a: lea edx,[eax+eax*2]
0x100e001d: push 8
0x100e001f: push edx
0x100e0020: call dword ptr ds:[100FE4FCh]
0x100e0026: mov esi,eax
```

```
0x100e0028: push 1012E3E4h
0x100e002d: push esi
0x100e002e: push ebp
0x100e002f: mov [esp+60h],esi
0x100e0033: call 100D2E20h
0x100e0038: movzx eax,word ptr [ebp+66B6h]
0x100e003f: movzx edx,word ptr [ebp+66B4h]
0x100e0046: lea ecx,[esi+eax*8]
0x100e0049: shl eax,4
0x100e004c: add eax,esi
0x100e004e: sub edx,2
0x100e0051: add esp,14h
0x100e0054: cmp edx,2
0x100e0057: mov [esp+3Ch],esi
0x100e005b: mov [esp+40h],ecx
0x100e005f: mov [esp+44h],eax
0x100e0063: mov dword ptr [esp+10h],2
0x100e006b: jle near ptr 100E0329h
0x100e0071: movzx eax,word ptr [ebp+66B6h]
0x100e0078: lea ecx,[eax-2]
0x100e007b: cmp ecx,2
0x100e007e: mov dword ptr [esp+14h],2
0x100e0086: jle near ptr 100E02BFh
0x100e008c: mov ecx,[esp+10h]
0x100e0090: and ecx,[esp+2Ch]
0x100e0094: mov edx,[esp+44h]
0x100e0098: shl ecx,4
0x100e009b: add edx,10h
0x100e009e: mov [esp+30h],ecx
0x100e00a2: mov [esp+1Ch],edx
0x100e00a6: jmp short 100E00B0h
0x100e00b0: imul eax,[esp+10h]
0x100e00b5: mov ecx,[esp+14h]
0x100e00b9: mov edx,[ebp+66E0h]
0x100e00bf: add eax,ecx
0x100e00c1: and ecx,[esp+28h]
0x100e00c5: lea edi,[edx+eax*8]
0x100e00c8: add ecx,[esp+30h]
0x100e00cc: xor eax,eax
0x100e00ce: mov esi,[esp+ecx*4+84h]
0x100e00d5: mov ecx,[esi]
0x100e00d7: cmp ecx,7FFFFFFFh
0x100e00dd: mov [esp+64h],eax
0x100e00e1: mov [esp+68h],eax
0x100e00e5: mov [esp+6Ch],eax
0x100e00e9: mov [esp+70h],eax
0x100e00ed: mov [esp+74h],eax
0x100e00f1: mov [esp+78h],eax
0x100e00f5: mov [esp+7Ch],eax
0x100e00f9: mov [esp+80h],eax
0x100e0100: je short 100E0166h
0x100e0102: mov eax,[esi+4]
0x100e0105: movzx edx,word ptr [edi+eax*2]
0x100e0109: movzx ecx,word ptr [edi+ecx*2]
0x100e010d: sub ecx,edx
0x100e010f: mov edx,ecx
0x100e0111: sar edx,1Fh
0x100e0114: mov eax,edx
0x100e0116: xor eax,ecx
0x100e0118: mov ecx,[esi+8]
0x100e011b: sub eax,edx
0x100e011d: shl eax,cl
0x100e011f: mov ecx,[esi+0Ch]
0x100e0122: lea ecx,[esp+ecx*4+64h]
0x100e0126: add esi,14h
0x100e0129: add [ecx],eax
0x100e012b: mov ecx,[esi-4]
```

```
0x100e012e: cmp ecx,0FFFFFFFFh
0x100e0131: je short 100E0158h
0x100e0133: add [esp+ecx*4+64h],eax
0x100e0137: lea ecx,[esp+ecx*4+64h]
0x100e013b: mov ecx,[esi]
0x100e013d: cmp ecx,0FFFFFFFFh
0x100e0140: je short 100E0155h
0x100e0142: add [esp+ecx*4+64h],eax
0x100e0146: lea ecx,[esp+ecx*4+64h]
0x100e014a: mov ecx,[esi+4]
0x100e014d: add esi,4
0x100e0150: cmp ecx,0FFFFFFFFh
0x100e0153: jne short 100E0142h
0x100e0155: add esi,4
0x100e0158: mov ecx,[esi]
0x100e015a: cmp ecx,7FFFFFFFh
0x100e0160: jne short 100E0102h
0x100e0162: mov eax,[esp+64h]
0x100e0166: add esi,4
0x100e0169: mov edx,eax
0x100e016b: mov ebx,1
0x100e0170: mov ecx,[esp+ebx*4+64h]
0x100e0174: cmp edx,ecx
0x100e0176: jle short 100E017Ah
0x100e0178: mov edx,ecx
0x100e017a: cmp eax,ecx
0x100e017c: jge short 100E0180h
0x100e017e: mov eax,ecx
0x100e0180: add ebx,1
0x100e0183: cmp ebx,8
0x100e0186: jl short 100E0170h
0x100e0188: test eax,eax
0x100e018a: jne short 100E019Fh
0x100e018c: mov ecx,[edi]
0x100e018e: mov eax,[esp+1Ch]
0x100e0192: mov [eax],ecx
0x100e0194: mov edx,[edi+4]
0x100e0197: mov [eax+4],edx
0x100e019a: jmp near ptr 100E029Dh
0x100e019f: mov ecx,[esp+10h]
0x100e01a3: sar eax,1
0x100e01a5: add eax,edx
0x100e01a7: mov [esp+34h],eax
0x100e01ab: xor eax,eax
0x100e01ad: mov [esp+50h],eax
0x100e01b1: mov [esp+54h],eax
0x100e01b5: mov [esp+58h],eax
0x100e01b9: mov [esp+5Ch],eax
0x100e01bd: mov eax,[esp+14h]
0x100e01c1: push eax
0x100e01c2: push ecx
0x100e01c3: push ebp
0x100e01c4: call 100D2D10h
0x100e01c9: add esp,0Ch
0x100e01cc: xor ecx,ecx
0x100e01ce: mov [esp+18h],eax
0x100e01d2: mov [esp+20h],ecx
0x100e01d6: mov [esp+24h],ecx
0x100e01da: lea ebx,[ebx]
0x100e01e0: mov edx,[esp+34h]
0x100e01e4: cmp [esp+ecx*4+64h],edx
0x100e01e8: jg short 100E0231h
0x100e01ea: mov edx,[ebp+66D0h]
0x100e01f0: xor eax,eax
0x100e01f2: test edx,edx
0x100e01f4: jle short 100E022Ch
0x100e01f6: cmp eax,[esp+18h]
```

```
0x100e01fa: jne short 100E0215h
0x100e01fc: mov ecx,[esi+4]
0x100e01ff: test ecx,ecx
0x100e0201: je short 100E0215h
0x100e0203: movzx ebx,word ptr [edi+eax*2]
0x100e0207: movzx ecx,word ptr [edi+ecx*2]
0x100e020b: add ebx,ecx
0x100e020d: sar ebx,1
0x100e020f: add [esp+eax*4+50h],ebx
0x100e0213: jmp short 100E0221h
0x100e0215: mov ecx,[esi]
0x100e0217: add ecx,eax
0x100e0219: movzx ecx,word ptr [edi+ecx*2]
0x100e021d: add [esp+eax*4+50h],ecx
0x100e0221: add eax,1
0x100e0224: cmp eax,edx
0x100e0226: jl short 100E01F6h
0x100e0228: mov ecx,[esp+20h]
0x100e022c: add dword ptr [esp+24h],1
0x100e0231: add ecx,1
0x100e0234: add esi,8
0x100e0237: cmp ecx,8
0x100e023a: mov [esp+20h],ecx
0x100e023e: jl short 100E01E0h
0x100e0240: xor ecx,ecx
0x100e0242: cmp [ebp+66D0h],ecx
0x100e0248: jle short 100E029Dh
0x100e024a: mov ebx,[esp+1Ch]
0x100e024e: mov edx,[esp+18h]
0x100e0252: cmp ecx,edx
0x100e0254: movzx esi,word ptr [edi+edx*2]
0x100e0258: je short 100E0271h
0x100e025a: mov edx,[esp+18h]
0x100e025e: mov eax,[esp+ecx*4+50h]
0x100e0262: sub eax,[esp+edx*4+50h]
0x100e0266: cdq
0x100e0267: idiv dword ptr [esp+24h]
0x100e026b: mov edx,[esp+18h]
0x100e026f: add esi,eax
0x100e0271: cmp esi,0FFFFh
0x100e0277: jge short 100E0287h
0x100e0279: xor eax,eax
0x100e027b: test esi,esi
0x100e027d: setl al
0x100e0280: sub eax,1
0x100e0283: and eax,esi
0x100e0285: jmp short 100E028Ch
0x100e0287: mov eax,0FFFFh
0x100e028c: mov [ebx],ax
0x100e028f: add ecx,1
0x100e0292: add ebx,2
0x100e0295: cmp ecx,[ebp+66D0h]
0x100e029b: jl short 100E0252h
0x100e029d: movzx eax,word ptr [ebp+66B6h]
0x100e02a4: mov ecx,[esp+14h]
0x100e02a8: add dword ptr [esp+1Ch],8
0x100e02ad: add ecx,1
0x100e02b0: lea edx,[eax-2]
0x100e02b3: cmp ecx,edx
0x100e02b5: mov [esp+14h],ecx
0x100e02b9: jl near ptr 100E00B0h
0x100e02bf: mov esi,[esp+10h]
0x100e02c3: cmp esi,3
0x100e02c6: jle short 100E02F8h
0x100e02c8: movzx eax,word ptr [ebp+66B6h]
0x100e02cf: mov edx,[esp+3Ch]
0x100e02d3: lea ecx,[eax*8-20h]
```

```
0x100e02da: push ecx
0x100e02db: lea ecx,[esi-2]
0x100e02de: imul ecx,eax
0x100e02e1: add edx,10h
0x100e02e4: push edx
0x100e02e5: mov edx,[ebp+66E0h]
0x100e02eb: lea eax,[edx+ecx*8+10h]
0x100e02ef: push eax
0x100e02f0: call 100228C6h
0x100e02f5: add esp,0Ch
0x100e02f8: mov eax,[esp+3Ch]
0x100e02fc: mov edx,[esp+44h]
0x100e0300: mov ecx,[esp+40h]
0x100e0304: mov [esp+44h],eax
0x100e0308: movzx eax,word ptr [ebp+66B4h]
0x100e030f: add esi,1
0x100e0312: sub eax,2
0x100e0315: cmp esi,eax
0x100e0317: mov [esp+3Ch],ecx
0x100e031b: mov [esp+40h],edx
0x100e031f: mov [esp+10h],esi
0x100e0323: jl near ptr 100E0071h
0x100e0329: movzx eax,word ptr [ebp+66B6h]
0x100e0330: mov esi,[esp+10h]
0x100e0334: mov edx,[esp+3Ch]
0x100e0338: lea ecx,[eax*8-20h]
0x100e033f: push ecx
0x100e0340: lea ecx,[esi-2]
0x100e0343: imul ecx,eax
0x100e0346: add edx,10h
0x100e0349: push edx
0x100e034a: mov edx,[ebp+66E0h]
0x100e0350: lea eax,[edx+ecx*8+10h]
0x100e0354: push eax
0x100e0355: call 100228C6h
0x100e035a: movzx eax,word ptr [ebp+66B6h]
0x100e0361: mov edx,[esp+4Ch]
0x100e0365: add esi,0FFFFFFFFh
0x100e0368: imul esi,eax
0x100e036b: lea ecx,[eax*8-20h]
0x100e0372: mov eax,[ebp+66E0h]
0x100e0378: push ecx
0x100e0379: add edx,10h
0x100e037c: push edx
0x100e037d: lea ecx,[eax+esi*8+10h]
0x100e0381: push ecx
0x100e0382: call 100228C6h
0x100e0387: mov edx,[esp+64h]
0x100e038b: mov esi,ds:[100FE554h]
0x100e0391: push edx
0x100e0392: call esi
0x100e0394: mov eax,[esp+0A0h]
0x100e039b: push eax
0x100e039c: call esi
0x100e039e: add esp,20h
0x100e03a1: pop edi
0x100e03a2: pop esi
0x100e03a3: pop ebp
0x100e03a4: pop ebx
0x100e03a5: add esp,474h
0x100e03ab: ret
```

`sub_100dfca0` is called by sub_10016040. It works with the text "VNG interpolation...". It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to j_memcpy, sub_100d2d10, sub_100d2e20 (4 callees).

12.1279 sub_100e03b0 at 0x100e03b0 (427 instructions)

```
0x100e03b0: sub esp,50h
0x100e03b3: push ebx
0x100e03b4: push ebp
0x100e03b5: mov ebp,[esp+5Ch]
0x100e03b9: movzx eax,word ptr [ebp+66B6h]
0x100e03c0: push esi
0x100e03c1: push edi
0x100e03c2: mov ecx,1
0x100e03c7: mov [esp+50h],eax
0x100e03cb: neg eax
0x100e03cd: push 3
0x100e03cf: push ebp
0x100e03d0: mov [esp+54h],ecx
0x100e03d4: mov dword ptr [esp+5Ch],0FFFFFFFFh
0x100e03dc: mov [esp+60h],eax
0x100e03e0: mov [esp+64h],ecx
0x100e03e4: call 100DF840h
0x100e03e9: add esp,8
0x100e03ec: cmp dword ptr [ebp+0BCh],0
0x100e03f3: je short 100E040Dh
0x100e03f5: push 1012E410h
0x100e03fa: call dword ptr ds:[100FE4C4h]
0x100e0400: add eax,40h
0x100e0403: push eax
0x100e0404: call dword ptr ds:[100FE4C8h]
0x100e040a: add esp,8
0x100e040d: movzx eax,word ptr [ebp+66B4h]
0x100e0414: mov ebx,3
0x100e0419: sub eax,ebx
0x100e041b: cmp eax,ebx
0x100e041d: mov [esp+10h],ebx
0x100e0421: jle near ptr 100E063Eh
0x100e0427: movzx eax,word ptr [ebp+66B6h]
0x100e042e: mov edi,edi
0x100e0430: mov esi,[ebp+65B0h]
0x100e0436: mov edi,ebx
0x100e0438: and edi,7
0x100e043b: mov edx,esi
0x100e043d: lea ecx,[edi*4+2]
0x100e0444: shr edx,cl
0x100e0446: and edx,1
0x100e0449: add edx,3
0x100e044c: mov ecx,edx
0x100e044e: and ecx,1
0x100e0451: lea ecx,[ecx+edi*2]
0x100e0454: add ecx,ecx
0x100e0456: shr esi,cl
0x100e0458: lea ecx,[eax-3]
0x100e045b: mov [esp+1Ch],edx
0x100e045f: and esi,3
0x100e0462: cmp edx,ecx
0x100e0464: mov [esp+18h],esi
0x100e0468: jge near ptr 100E0625h
0x100e046e: mov edi,edi
0x100e0470: mov esi,[esp+18h]
0x100e0474: imul eax,ebx
0x100e0477: add eax,edx
0x100e0479: mov edx,[ebp+66E0h]
0x100e047f: lea ecx,[edx+eax*8]
0x100e0482: movzx esi,word ptr [ecx+esi*2]
0x100e0486: xor edx,edx
0x100e0488: mov [esp+14h],edx
0x100e048c: mov eax,1
0x100e0491: mov [esp+20h],esi
0x100e0495: jmp short 100E04A0h
0x100e04a0: lea esi,[eax*8]
```

```
0x100e04a7: movzx ebp,word ptr [esi+ecx+2]
0x100e04ac: mov edi,esi
0x100e04ae: neg edi
0x100e04b0: movzx ebx,word ptr [edi+ecx+2]
0x100e04b5: mov [esp+24h],ebp
0x100e04b9: mov ebp,[esp+18h]
0x100e04bd: add esi,ebp
0x100e04bf: add edi,ebp
0x100e04c1: movzx ebp,word ptr [ecx+esi*2]
0x100e04c5: movzx edi,word ptr [ecx+edi*2]
0x100e04c9: mov esi,[esp+20h]
0x100e04cd: add esi,ebx
0x100e04cf: add esi,[esp+24h]
0x100e04d3: lea eax,[eax+eax*2]
0x100e04d6: add esi,esi
0x100e04d8: sub esi,edi
0x100e04da: sub esi,ebp
0x100e04dc: mov [esp+edx+44h],esi
0x100e04e0: mov esi,edi
0x100e04e2: sub esi,[esp+20h]
0x100e04e6: add eax,eax
0x100e04e8: mov edi,esi
0x100e04ea: sar edi,1Fh
0x100e04ed: mov [esp+2Ch],edi
0x100e04f1: mov edi,ebp
0x100e04f3: sub edi,[esp+20h]
0x100e04f7: add eax,eax
0x100e04f9: mov [esp+34h],edi
0x100e04fd: sar edi,1Fh
0x100e0500: mov [esp+30h],edi
0x100e0504: add eax,eax
0x100e0506: mov edi,ebx
0x100e0508: sub edi,[esp+24h]
0x100e050c: mov ebp,edi
0x100e050e: sar ebp,1Fh
0x100e0511: mov [esp+28h],ebp
0x100e0515: mov ebp,ecx
0x100e0517: sub ebp,eax
0x100e0519: movzx eax,word ptr [eax+ecx+2]
0x100e051e: sub eax,[esp+24h]
0x100e0522: movzx ebp,word ptr [ebp+2]
0x100e0526: mov [esp+38h],eax
0x100e052a: sar eax,1Fh
0x100e052d: mov [esp+24h],eax
0x100e0531: mov eax,[esp+28h]
0x100e0535: xor eax,edi
0x100e0537: mov edi,[esp+2Ch]
0x100e053b: xor edi,esi
0x100e053d: mov esi,[esp+30h]
0x100e0541: add eax,edi
0x100e0543: mov edi,esi
0x100e0545: xor edi,[esp+34h]
0x100e0549: sub ebp,ebx
0x100e054b: add eax,edi
0x100e054d: sub eax,[esp+28h]
0x100e0551: mov edi,[esp+24h]
0x100e0555: sub eax,[esp+2Ch]
0x100e0559: xor edi,[esp+38h]
0x100e055d: sub eax,esi
0x100e055f: mov ebx,ebp
0x100e0561: sar ebx,1Fh
0x100e0564: mov esi,ebx
0x100e0566: xor esi,ebp
0x100e0568: add esi,edi
0x100e056a: sub esi,ebx
0x100e056c: sub esi,[esp+24h]
0x100e0570: lea eax,[eax+eax*2]
```

```
0x100e0573: lea eax,[eax+esi*2]
0x100e0576: mov [esp+edx+3Ch],eax
0x100e057a: mov eax,[esp+14h]
0x100e057e: add eax,1
0x100e0581: lea edx,[eax*4]
0x100e0588: mov [esp+14h],eax
0x100e058c: mov eax,[esp+edx+4Ch]
0x100e0590: test eax,eax
0x100e0592: jg near ptr 100E04A0h
0x100e0598: mov edx,[esp+40h]
0x100e059c: xor eax,eax
0x100e059e: cmp [esp+3Ch],edx
0x100e05a2: mov edi,ecx
0x100e05a4: setg al
0x100e05a7: mov edx,[esp+eax*4+4Ch]
0x100e05ab: mov eax,[esp+eax*4+44h]
0x100e05af: add edx,edx
0x100e05b1: add edx,edx
0x100e05b3: add edx,edx
0x100e05b5: movzx esi,word ptr [edx+ecx+2]
0x100e05ba: sub edi,edx
0x100e05bc: movzx edx,word ptr [edi+2]
0x100e05c0: sar eax,2
0x100e05c3: cmp si,dx
0x100e05c6: mov edi,eax
0x100e05c8: jae short 100E05E2h
0x100e05ca: movzx edx,dx
0x100e05cd: cmp eax,edx
0x100e05cf: jl short 100E05D3h
0x100e05d1: mov edi,edx
0x100e05d3: movzx esi,si
0x100e05d6: cmp esi,edi
0x100e05d8: jg short 100E05FAh
0x100e05da: cmp eax,edx
0x100e05dc: jl short 100E05FCh
0x100e05de: mov eax,edx
0x100e05e0: jmp short 100E05FCh
0x100e05e2: movzx esi,si
0x100e05e5: cmp eax,esi
0x100e05e7: jl short 100E05EBh
0x100e05e9: mov edi,esi
0x100e05eb: movzx edx,dx
0x100e05ee: cmp edx,edi
0x100e05f0: jle short 100E05F6h
0x100e05f2: mov eax,edx
0x100e05f4: jmp short 100E05FCh
0x100e05f6: cmp eax,esi
0x100e05f8: jl short 100E05FCh
0x100e05fa: mov eax,esi
0x100e05fc: mov ebp,[esp+64h]
0x100e0600: mov edx,[esp+1Ch]
0x100e0604: mov ebx,[esp+10h]
0x100e0608: mov [ecx+2],ax
0x100e060c: movzx eax,word ptr [ebp+66B6h]
0x100e0613: add edx,2
0x100e0616: lea ecx,[eax-3]
0x100e0619: cmp edx,ecx
0x100e061b: mov [esp+1Ch],edx
0x100e061f: jl near ptr 100E0470h
0x100e0625: movzx ecx,word ptr [ebp+66B4h]
0x100e062c: add ebx,1
0x100e062f: sub ecx,3
0x100e0632: cmp ebx,ecx
0x100e0634: mov [esp+10h],ebx
0x100e0638: jl near ptr 100E0430h
0x100e063e: movzx edx,word ptr [ebp+66B4h]
0x100e0645: mov esi,1
```

```
0x100e064a: sub edx,esi
0x100e064c: cmp edx,esi
0x100e064e: mov [esp+10h],esi
0x100e0652: jle near ptr 100E0764h
0x100e0658: jmp short 100E0660h
0x100e0660: mov edx,[ebp+65B0h]
0x100e0666: mov eax,esi
0x100e0668: and eax,7
0x100e066b: lea ecx,[eax*4]
0x100e0672: mov edi,edx
0x100e0674: shr edi,cl
0x100e0676: and edi,1
0x100e0679: add edi,1
0x100e067c: lea ecx,[edi-1]
0x100e067f: and ecx,1
0x100e0682: lea ecx,[ecx+eax*2]
0x100e0685: movzx eax,word ptr [ebp+66B6h]
0x100e068c: add ecx,ecx
0x100e068e: shr edx,cl
0x100e0690: lea ecx,[eax-1]
0x100e0693: and edx,3
0x100e0696: cmp edi,ecx
0x100e0698: jge near ptr 100E074Bh
0x100e069e: mov edi,edi
0x100e06a0: mov ecx,[ebp+66E0h]
0x100e06a6: imul eax,esi
0x100e06a9: add eax,edi
0x100e06ab: lea esi,[esp+4Ch]
0x100e06af: lea eax,[ecx+eax*8]
0x100e06b2: mov ecx,1
0x100e06b7: mov [esp+2Ch],esi
0x100e06bb: jmp short 100E06C0h
0x100e06c0: lea esi,[ecx*4]
0x100e06c7: add ecx,ecx
0x100e06c9: mov ebx,edx
0x100e06cb: sub ebx,esi
0x100e06cd: movzx ebx,word ptr [eax+ebx*2]
0x100e06d1: add esi,edx
0x100e06d3: movzx esi,word ptr [eax+esi*2]
0x100e06d7: add ecx,ecx
0x100e06d9: add ebx,esi
0x100e06db: add ecx,ecx
0x100e06dd: mov esi,eax
0x100e06df: sub esi,ecx
0x100e06e1: movzx ecx,word ptr [ecx+eax+2]
0x100e06e6: movzx esi,word ptr [esi+2]
0x100e06ea: sub ebx,esi
0x100e06ec: sub ebx,ecx
0x100e06ee: movzx ecx,word ptr [eax+2]
0x100e06f2: lea ecx,[ebx+ecx*2]
0x100e06f5: sar ecx,1
0x100e06f7: cmp ecx,0FFFFh
0x100e06fd: jge short 100E070Fh
0x100e06ff: test ecx,ecx
0x100e0701: jge short 100E0707h
0x100e0703: xor ecx,ecx
0x100e0705: jmp short 100E0714h
0x100e0707: cmp ecx,0FFFFh
0x100e070d: jl short 100E0714h
0x100e070f: mov ecx,0FFFFh
0x100e0714: mov [eax+edx*2],cx
0x100e0718: mov ecx,2
0x100e071d: sub ecx,edx
0x100e071f: mov edx,ecx
0x100e0721: mov ecx,[esp+2Ch]
0x100e0725: add ecx,4
0x100e0728: mov [esp+2Ch],ecx
```

```
0x100e072c: mov ecx,[ecx]
0x100e072e: test ecx,ecx
0x100e0730: jg short 100E06C0h
0x100e0732: movzx eax,word ptr [ebp+66B6h]
0x100e0739: mov esi,[esp+10h]
0x100e073d: add edi,2
0x100e0740: lea ecx,[eax-1]
0x100e0743: cmp edi,ecx
0x100e0745: jl near ptr 100E06A0h
0x100e074b: movzx edx,word ptr [ebp+66B4h]
0x100e0752: add esi,1
0x100e0755: sub edx,1
0x100e0758: cmp esi,edx
0x100e075a: mov [esp+10h],esi
0x100e075e: jl near ptr 100E0660h
0x100e0764: movzx eax,word ptr [ebp+66B4h]
0x100e076b: mov esi,1
0x100e0770: sub eax,esi
0x100e0772: cmp eax,esi
0x100e0774: mov [esp+10h],esi
0x100e0778: jle near ptr 100E0940h
0x100e077e: mov edi,edi
0x100e0780: mov eax,[ebp+65B0h]
0x100e0786: mov edx,esi
0x100e0788: and edx,7
0x100e078b: mov edi,eax
0x100e078d: lea ecx,[edx*4+2]
0x100e0794: shr edi,cl
0x100e0796: mov ebx,2
0x100e079b: and edi,1
0x100e079e: add edi,1
0x100e07a1: mov ecx,edi
0x100e07a3: and ecx,1
0x100e07a6: lea ecx,[ecx+edx*2]
0x100e07a9: add ecx,ecx
0x100e07ab: shr eax,cl
0x100e07ad: mov [esp+1Ch],edi
0x100e07b1: and eax,3
0x100e07b4: sub ebx,eax
0x100e07b6: movzx eax,word ptr [ebp+66B6h]
0x100e07bd: lea edx,[eax-1]
0x100e07c0: cmp edi,edx
0x100e07c2: mov [esp+18h],ebx
0x100e07c6: jge near ptr 100E0927h
0x100e07cc: lea esp,[esp]
0x100e07d0: mov ecx,[ebp+66E0h]
0x100e07d6: imul eax,esi
0x100e07d9: mov edx,[esp+50h]
0x100e07dd: add eax,edi
0x100e07df: lea esi,[ecx+eax*8]
0x100e07e2: xor ecx,ecx
0x100e07e4: add edx,1
0x100e07e7: cmp edx,ecx
0x100e07e9: mov [esp+14h],ecx
0x100e07ed: jle near ptr 100E08C3h
0x100e07f3: movzx eax,word ptr [esi+2]
0x100e07f7: add eax,eax
0x100e07f9: mov [esp+24h],eax
0x100e07fd: lea ecx,[ecx]
0x100e0800: lea edi,[edx*4]
0x100e0807: movzx ebp,word ptr [esi+2]
0x100e080b: mov eax,ebx
0x100e080d: sub eax,edi
0x100e080f: movzx eax,word ptr [esi+eax*2]
0x100e0813: add edi,ebx
0x100e0815: movzx ebx,word ptr [esi+edi*2]
0x100e0819: mov [esp+20h],ebx
```

```
0x100e081d: mov edi,eax
0x100e081f: sub edi,ebx
0x100e0821: add edx,edx
0x100e0823: add edx,edx
0x100e0825: add edx,edx
0x100e0827: mov ebx,edi
0x100e0829: sar ebx,1Fh
0x100e082c: mov [esp+38h],ebx
0x100e0830: mov ebx,esi
0x100e0832: sub ebx,edx
0x100e0834: movzx ebx,word ptr [ebx+2]
0x100e0838: movzx edx,word ptr [edx+esi+2]
0x100e083d: mov [esp+30h],ebx
0x100e0841: sub ebx,ebp
0x100e0843: sub eax,[esp+30h]
0x100e0847: mov ebp,ebx
0x100e0849: sar ebp,1Fh
0x100e084c: mov [esp+2Ch],ebp
0x100e0850: movzx ebp,word ptr [esi+2]
0x100e0854: mov [esp+28h],edx
0x100e0858: sub eax,[esp+28h]
0x100e085c: sub edx,ebp
0x100e085e: add eax,[esp+24h]
0x100e0862: mov ebp,edx
0x100e0864: add eax,[esp+20h]
0x100e0868: sar ebp,1Fh
0x100e086b: mov [esp+34h],ebp
0x100e086f: xor ebp,edx
0x100e0871: mov edx,[esp+2Ch]
0x100e0875: xor edx,ebx
0x100e0877: add ebp,edx
0x100e0879: mov edx,[esp+38h]
0x100e087d: mov ebx,edx
0x100e087f: xor ebx,edi
0x100e0881: add ebp,ebx
0x100e0883: sub ebp,[esp+2Ch]
0x100e0887: mov ebx,[esp+18h]
0x100e088b: sub ebp,edx
0x100e088d: sub ebp,[esp+34h]
0x100e0891: mov [esp+ecx+44h],eax
0x100e0895: mov eax,[esp+14h]
0x100e0899: add eax,1
0x100e089c: mov [esp+ecx+3Ch],ebp
0x100e08a0: lea ecx,[eax*4]
0x100e08a7: mov edx,[esp+ecx+50h]
0x100e08ab: add edx,[esp+ecx+4Ch]
0x100e08af: mov [esp+14h],eax
0x100e08b3: test edx,edx
0x100e08b5: jg near ptr 100E0800h
0x100e08bb: mov ebp,[esp+64h]
0x100e08bf: mov edi,[esp+1Ch]
0x100e08c3: mov eax,[esp+40h]
0x100e08c7: cmp [esp+3Ch],eax
0x100e08cb: je short 100E08DEh
0x100e08cd: xor ecx,ecx
0x100e08cf: cmp [esp+3Ch],eax
0x100e08d3: setg cl
0x100e08d6: mov eax,[esp+ecx*4+44h]
0x100e08da: sar eax,1
0x100e08dc: jmp short 100E08EBh
0x100e08de: mov edx,[esp+44h]
0x100e08e2: mov eax,[esp+48h]
0x100e08e6: add eax,edx
0x100e08e8: sar eax,2
0x100e08eb: cmp eax,0FFFFh
0x100e08f0: jge short 100E0901h
0x100e08f2: test eax,eax
```

```
0x100e08f4: jge short 100E08FAh
0x100e08f6: xor eax,eax
0x100e08f8: jmp short 100E0906h
0x100e08fa: cmp eax,0FFFFh
0x100e08ff: jl short 100E0906h
0x100e0901: mov eax,0FFFFh
0x100e0906: mov [esi+ebx*2],ax
0x100e090a: movzx eax,word ptr [ebp+66B6h]
0x100e0911: mov esi,[esp+10h]
0x100e0915: add edi,2
0x100e0918: lea ecx,[eax-1]
0x100e091b: cmp edi,ecx
0x100e091d: mov [esp+1Ch],edi
0x100e0921: jl near ptr 100E07D0h
0x100e0927: movzx edx,word ptr [ebp+66B4h]
0x100e092e: add esi,1
0x100e0931: sub edx,1
0x100e0934: cmp esi,edx
0x100e0936: mov [esp+10h],esi
0x100e093a: jl near ptr 100E0780h
0x100e0940: pop edi
0x100e0941: pop esi
0x100e0942: pop ebp
0x100e0943: pop ebx
0x100e0944: add esp,50h
0x100e0947: ret
```

`sub_100e03b0` is called by sub_10016040. It works with the text "PPG interpolation...". It delegates to sub_100df840 (1 callees).

12.1280 sub_100e0950 at 0x100e0950 (1062 instructions)

```
0x100e0950: push ebp
0x100e0951: mov ebp,esp
0x100e0953: and esp,0FFFFFFF8h
0x100e0956: mov eax,400BCh
0x100e095b: call 100229B0h
0x100e0960: push ebx
0x100e0961: mov ebx,[ebp+8]
0x100e0964: cmp dword ptr [ebx+0BCh],0
0x100e096b: push esi
0x100e096c: push edi
0x100e096d: je short 100E0987h
0x100e096f: push 1012E45Ch
0x100e0974: call dword ptr ds:[100FE4C4h]
0x100e097a: add eax,40h
0x100e097d: push eax
0x100e097e: call dword ptr ds:[100FE4C8h]
0x100e0984: add esp,8
0x100e0987: xor esi,esi
0x100e0989: mov [esp+10h],esi
0x100e098d: lea ecx,[ecx]
0x100e0990: fild dword ptr [esp+10h]
0x100e0994: fdiv qword ptr ds:[100FFBA0h]
0x100e099a: fstp dword ptr [esp+0Ch]
0x100e099e: fld dword ptr ds:[1012E458h]
0x100e09a4: fld dword ptr [esp+0Ch]
0x100e09a8: fcom
0x100e09aa: fnstsw ax
0x100e09ac: fstp st(1)
0x100e09ae: test ah,41h
0x100e09b1: jne short 100E09C8h
0x100e09b3: fld qword ptr ds:[1012E450h]
0x100e09b9: call 100229A0h
0x100e09be: fstp dword ptr [esp+0Ch]
0x100e09c2: fld dword ptr [esp+0Ch]
```

```
0x100e09c6: jmp short 100E09D4h
0x100e09c8: fmul qword ptr ds:[1010BBA0h]
0x100e09ce: fadd qword ptr ds:[1010BB90h]
0x100e09d4: fstp dword ptr [esp+esi*4+0C8h]
0x100e09db: add esi,1
0x100e09de: cmp esi,10000h
0x100e09e4: mov [esp+10h],esi
0x100e09e8: jl short 100E0990h
0x100e09ea: fldz
0x100e09ec: lea esi,[esp+98h]
0x100e09f3: mov [esp+1Ch],esi
0x100e09f7: mov edx,1012A7D8h
0x100e09fc: mov ecx,1012A798h
0x100e0a01: mov edi,[ebx+66D0h]
0x100e0a07: test edi,edi
0x100e0a09: jle short 100E0A60h
0x100e0a0b: lea eax,[ebx+0E7D4h]
0x100e0a11: fld dword ptr [eax-10h]
0x100e0a14: add eax,4
0x100e0a17: fmul qword ptr [ecx-8]
0x100e0a1a: add esi,4
0x100e0a1d: sub edi,1
0x100e0a20: fdiv dword ptr [edx]
0x100e0a22: fstp dword ptr [esp+0Ch]
0x100e0a26: fld dword ptr [esp+0Ch]
0x100e0a2a: fadd st,st(1)
0x100e0a2c: fstp dword ptr [esp+0Ch]
0x100e0a30: fld dword ptr [esp+0Ch]
0x100e0a34: fld dword ptr [eax-4]
0x100e0a37: fmul qword ptr [ecx]
0x100e0a39: fdiv dword ptr [edx]
0x100e0a3b: fstp dword ptr [esp+0Ch]
0x100e0a3f: fadd dword ptr [esp+0Ch]
0x100e0a43: fstp dword ptr [esp+0Ch]
0x100e0a47: fld dword ptr [esp+0Ch]
0x100e0a4b: fld dword ptr [eax+0Ch]
0x100e0a4e: fmul qword ptr [ecx+8]
0x100e0a51: fdiv dword ptr [edx]
0x100e0a53: fstp dword ptr [esp+0Ch]
0x100e0a57: fadd dword ptr [esp+0Ch]
0x100e0a5b: fstp dword ptr [esi-4]
0x100e0a5e: jne short 100E0A11h
0x100e0a60: mov esi,[esp+1Ch]
0x100e0a64: add ecx,18h
0x100e0a67: add esi,10h
0x100e0a6a: add edx,4
0x100e0a6d: cmp ecx,1012A7E0h
0x100e0a73: mov [esp+1Ch],esi
0x100e0a77: jl short 100E0A01h
0x100e0a79: push 5
0x100e0a7b: fstp st(0)
0x100e0a7d: push ebx
0x100e0a7e: call 100DF840h
0x100e0a83: push 1A0000h
0x100e0a88: call dword ptr ds:[100FE550h]
0x100e0a8e: push 1012E438h
0x100e0a93: push eax
0x100e0a94: push ebx
0x100e0a95: mov [esp+58h],eax
0x100e0a99: call 100D2E20h
0x100e0a9e: movzx eax,word ptr [ebx+66B4h]
0x100e0aa5: sub eax,5
0x100e0aa8: add esp,18h
0x100e0aab: cmp eax,2
0x100e0aae: mov dword ptr [esp+38h],2
0x100e0ab6: jle near ptr 100E183Ah
0x100e0abc: fld qword ptr ds:[100FE9C0h]
```

```
0x100e0ac2: fld qword ptr ds:[100FE9C8h]
0x100e0ac8: fld qword ptr ds:[1010BE00h]
0x100e0ace: fld qword ptr ds:[1012E430h]
0x100e0ad4: fld qword ptr ds:[1012E428h]
0x100e0ada: movzx ecx,word ptr [ebx+66B6h]
0x100e0ae1: sub ecx,5
0x100e0ae4: cmp ecx,2
0x100e0ae7: mov dword ptr [esp+24h],2
0x100e0aef: jle near ptr 100E1811h
0x100e0af5: mov eax,[esp+38h]
0x100e0af9: neg eax
0x100e0afb: mov [esp+48h],eax
0x100e0aff: nop
0x100e0b00: mov ecx,[esp+38h]
0x100e0b04: mov esi,[esp+24h]
0x100e0b08: lea eax,[ecx+100h]
0x100e0b0e: cmp ecx,eax
0x100e0b10: mov [esp+14h],ecx
0x100e0b14: jge near ptr 100E0D0Dh
0x100e0b1a: mov eax,esi
0x100e0b1c: neg eax
0x100e0b1e: mov [esp+18h],eax
0x100e0b22: movzx edx,word ptr [ebx+66B4h]
0x100e0b29: sub edx,2
0x100e0b2c: cmp [esp+14h],edx
0x100e0b30: jge near ptr 100E0D0Dh
0x100e0b36: mov eax,[esp+14h]
0x100e0b3a: mov edi,[ebx+65B0h]
0x100e0b40: and eax,7
0x100e0b43: add eax,eax
0x100e0b45: mov ecx,esi
0x100e0b47: and ecx,1
0x100e0b4a: add ecx,eax
0x100e0b4c: add ecx,ecx
0x100e0b4e: mov edx,edi
0x100e0b50: shr edx,cl
0x100e0b52: and edx,1
0x100e0b55: add edx,esi
0x100e0b57: mov ecx,edx
0x100e0b59: and ecx,1
0x100e0b5c: add ecx,eax
0x100e0b5e: add ecx,ecx
0x100e0b60: shr edi,cl
0x100e0b62: lea eax,[esi+100h]
0x100e0b68: mov [esp+20h],edx
0x100e0b6c: and edi,3
0x100e0b6f: cmp edx,eax
0x100e0b71: mov [esp+10h],edi
0x100e0b75: jge near ptr 100E0CE9h
0x100e0b7b: jmp short 100E0B81h
0x100e0b7d: mov edi,[esp+10h]
0x100e0b81: movzx eax,word ptr [ebx+66B6h]
0x100e0b88: lea ecx,[eax-2]
0x100e0b8b: cmp edx,ecx
0x100e0b8d: jge near ptr 100E0CE9h
0x100e0b93: imul eax,[esp+14h]
0x100e0b98: mov ecx,[ebx+66E0h]
0x100e0b9e: add eax,edx
0x100e0ba0: movzx esi,word ptr [ecx+eax*8-6]
0x100e0ba5: lea eax,[ecx+eax*8]
0x100e0ba8: movzx ecx,word ptr [eax+0Ah]
0x100e0bac: movzx esi,si
0x100e0baf: movzx ecx,cx
0x100e0bb2: add esi,ecx
0x100e0bb4: movzx ecx,word ptr [eax+edi*2]
0x100e0bb8: add esi,ecx
0x100e0bba: add esi,esi
```

```
0x100e0bbc: mov ecx,esi
0x100e0bbe: movzx esi,word ptr [eax+edi*2-10h]
0x100e0bc3: sub ecx,esi
0x100e0bc5: movzx esi,word ptr [eax+edi*2+10h]
0x100e0bca: movzx edi,word ptr [eax-6]
0x100e0bce: sub ecx,esi
0x100e0bd0: movzx esi,word ptr [eax+0Ah]
0x100e0bd4: sar ecx,2
0x100e0bd7: cmp di,si
0x100e0bda: mov edi,ecx
0x100e0bdc: jae short 100E0BF6h
0x100e0bde: movzx esi,word ptr [eax+0Ah]
0x100e0be2: cmp ecx,esi
0x100e0be4: jl short 100E0BE8h
0x100e0be6: mov edi,esi
0x100e0be8: movzx esi,word ptr [eax-6]
0x100e0bec: cmp esi,edi
0x100e0bee: jg short 100E0C10h
0x100e0bf0: movzx esi,word ptr [eax+0Ah]
0x100e0bf4: jmp short 100E0C0Ch
0x100e0bf6: movzx esi,word ptr [eax-6]
0x100e0bfa: cmp ecx,esi
0x100e0bfc: jl short 100E0C00h
0x100e0bfe: mov edi,esi
0x100e0c00: movzx esi,word ptr [eax+0Ah]
0x100e0c04: cmp esi,edi
0x100e0c06: jg short 100E0C10h
0x100e0c08: movzx esi,word ptr [eax-6]
0x100e0c0c: cmp ecx,esi
0x100e0c0e: jl short 100E0C12h
0x100e0c10: mov ecx,esi
0x100e0c12: mov esi,[esp+18h]
0x100e0c16: add edx,esi
0x100e0c18: mov esi,[esp+40h]
0x100e0c1c: lea edx,[edx+edx*2]
0x100e0c1f: lea edx,[esi+edx*2]
0x100e0c22: mov [edx+2],cx
0x100e0c26: movzx ecx,word ptr [ebx+66B6h]
0x100e0c2d: add ecx,ecx
0x100e0c2f: mov [esp+44h],edx
0x100e0c33: add ecx,ecx
0x100e0c35: add ecx,ecx
0x100e0c37: mov edx,eax
0x100e0c39: sub edx,ecx
0x100e0c3b: movzx edi,word ptr [edx+2]
0x100e0c3f: movzx edx,word ptr [ecx+eax+2]
0x100e0c44: mov [esp+0Ch],edx
0x100e0c48: movzx edx,dx
0x100e0c4b: movzx esi,di
0x100e0c4e: add edx,esi
0x100e0c50: mov esi,[esp+10h]
0x100e0c54: movzx esi,word ptr [eax+esi*2]
0x100e0c58: add edx,esi
0x100e0c5a: mov esi,[esp+10h]
0x100e0c5e: add esi,ecx
0x100e0c60: movzx esi,word ptr [eax+esi*2]
0x100e0c64: add edx,edx
0x100e0c66: sub edx,esi
0x100e0c68: mov esi,[esp+10h]
0x100e0c6c: sub esi,ecx
0x100e0c6e: movzx ecx,word ptr [eax+esi*2]
0x100e0c72: sub edx,ecx
0x100e0c74: movzx ecx,word ptr [ebx+66B6h]
0x100e0c7b: add ecx,ecx
0x100e0c7d: add ecx,ecx
0x100e0c7f: sar edx,2
0x100e0c82: add ecx,ecx
```

```
0x100e0c84: cmp di,[esp+0Ch]
0x100e0c89: mov edi,edx
0x100e0c8b: jae short 100E0CA0h
0x100e0c8d: movzx esi,word ptr [ecx+eax+2]
0x100e0c92: cmp edx,esi
0x100e0c94: jl short 100E0C98h
0x100e0c96: mov edi,esi
0x100e0c98: sub eax,ecx
0x100e0c9a: movzx eax,word ptr [eax+2]
0x100e0c9e: jmp short 100E0CB3h
0x100e0ca0: mov esi,eax
0x100e0ca2: sub esi,ecx
0x100e0ca4: movzx esi,word ptr [esi+2]
0x100e0ca8: cmp edx,esi
0x100e0caa: jl short 100E0CAEh
0x100e0cac: mov edi,esi
0x100e0cae: movzx eax,word ptr [ecx+eax+2]
0x100e0cb3: cmp eax,edi
0x100e0cb5: jle short 100E0CBBh
0x100e0cb7: mov edx,eax
0x100e0cb9: jmp short 100E0CC1h
0x100e0cbb: cmp edx,esi
0x100e0cbd: jl short 100E0CC1h
0x100e0cbf: mov edx,esi
0x100e0cc1: mov eax,[esp+44h]
0x100e0cc5: mov esi,[esp+24h]
0x100e0cc9: mov [eax+60002h],dx
0x100e0cd0: mov edx,[esp+20h]
0x100e0cd4: add edx,2
0x100e0cd7: lea eax,[esi+100h]
0x100e0cdd: cmp edx,eax
0x100e0cdf: mov [esp+20h],edx
0x100e0ce3: jl near ptr 100E0B7Dh
0x100e0ce9: mov ecx,[esp+14h]
0x100e0ced: mov eax,[esp+38h]
0x100e0cf1: add dword ptr [esp+18h],100h
0x100e0cf9: add ecx,1
0x100e0cfc: add eax,100h
0x100e0d01: cmp ecx,eax
0x100e0d03: mov [esp+14h],ecx
0x100e0d07: jl near ptr 100E0B22h
0x100e0d0d: mov ecx,[esp+48h]
0x100e0d11: mov [esp+18h],ecx
0x100e0d15: mov dword ptr [esp+3Ch],2
0x100e0d1d: mov eax,[esp+38h]
0x100e0d21: lea ecx,[eax+1]
0x100e0d24: add eax,0FFh
0x100e0d29: cmp ecx,eax
0x100e0d2b: mov [esp+14h],ecx
0x100e0d2f: jge near ptr 100E11A0h
0x100e0d35: lea edx,[ecx*4+4]
0x100e0d3c: mov [esp+10h],edx
0x100e0d40: movzx eax,word ptr [ebx+66B4h]
0x100e0d47: sub eax,3
0x100e0d4a: cmp ecx,eax
0x100e0d4c: jge near ptr 100E11A0h
0x100e0d52: lea edx,[esi+1]
0x100e0d55: lea eax,[esi+0FFh]
0x100e0d5b: cmp edx,eax
0x100e0d5d: mov [esp+20h],edx
0x100e0d61: jge near ptr 100E1183h
0x100e0d67: jmp short 100E0D6Dh
0x100e0d69: mov edx,[esp+20h]
0x100e0d6d: movzx eax,word ptr [ebx+66B6h]
0x100e0d74: lea edi,[eax-3]
0x100e0d77: cmp edx,edi
0x100e0d79: jge near ptr 100E1183h
```

```
0x100e0d7f: mov edi,[ebx+66E0h]
0x100e0d85: imul eax,ecx
0x100e0d88: add eax,edx
0x100e0d8a: lea edi,[edi+eax*8]
0x100e0d8d: mov eax,[esp+18h]
0x100e0d91: add eax,ecx
0x100e0d93: shl eax,8
0x100e0d96: sub eax,esi
0x100e0d98: mov esi,[esp+40h]
0x100e0d9c: add eax,edx
0x100e0d9e: and edx,1
0x100e0da1: lea eax,[eax+eax*2]
0x100e0da4: add eax,eax
0x100e0da6: and ecx,7
0x100e0da9: lea ecx,[edx+ecx*2]
0x100e0dac: add esi,eax
0x100e0dae: add ecx,ecx
0x100e0db0: mov [esp+34h],eax
0x100e0db4: mov eax,[ebx+65B0h]
0x100e0dba: mov [esp+0Ch],edx
0x100e0dbe: mov edx,eax
0x100e0dc0: shr edx,cl
0x100e0dc2: mov [esp+44h],ecx
0x100e0dc6: mov ecx,2
0x100e0dcb: and edx,3
0x100e0dce: sub ecx,edx
0x100e0dd0: cmp ecx,1
0x100e0dd3: jne near ptr 100E0E7Ah
0x100e0dd9: mov ecx,[esp+10h]
0x100e0ddd: mov edx,[esp+0Ch]
0x100e0de1: and ecx,1Ch
0x100e0de4: lea ecx,[ecx+edx*2]
0x100e0de7: shr eax,cl
0x100e0de9: mov edx,edi
0x100e0deb: and eax,3
0x100e0dee: mov ecx,eax
0x100e0df0: neg eax
0x100e0df2: mov [esp+1Ch],eax
0x100e0df6: lea eax,[ecx+ecx+4]
0x100e0dfa: sub edx,eax
0x100e0dfc: movzx eax,word ptr [edx]
0x100e0dff: mov edx,[esp+1Ch]
0x100e0e03: movzx edx,word ptr [edi+edx*2+0Ch]
0x100e0e08: add eax,edx
0x100e0e0a: movzx edx,word ptr [esi+8]
0x100e0e0e: sub eax,edx
0x100e0e10: movzx edx,word ptr [esi-4]
0x100e0e14: sub eax,edx
0x100e0e16: movzx edx,word ptr [edi+2]
0x100e0e1a: sar eax,1
0x100e0e1c: add eax,edx
0x100e0e1e: cmp eax,0FFFFh
0x100e0e23: jge short 100E0E33h
0x100e0e25: xor edx,edx
0x100e0e27: test eax,eax
0x100e0e29: setl dl
0x100e0e2c: sub edx,1
0x100e0e2f: and eax,edx
0x100e0e31: jmp short 100E0E38h
0x100e0e33: mov eax,0FFFFh
0x100e0e38: mov edx,[esp+1Ch]
0x100e0e3c: mov [esi+edx*2+4],ax
0x100e0e41: movzx edx,word ptr [ebx+66B6h]
0x100e0e48: add edx,edx
0x100e0e4a: add edx,edx
0x100e0e4c: lea eax,[edx+ecx]
0x100e0e4f: movzx eax,word ptr [edi+eax*2]
```

```
0x100e0e53: mov ebx,ecx
0x100e0e55: sub ebx,edx
0x100e0e57: movzx edx,word ptr [edi+ebx*2]
0x100e0e5b: mov ebx,[ebp+8]
0x100e0e5e: add eax,edx
0x100e0e60: movzx edx,word ptr [esi+602h]
0x100e0e67: sub eax,edx
0x100e0e69: movzx edx,word ptr [esi-5FEh]
0x100e0e70: sub eax,edx
0x100e0e72: movzx edx,word ptr [edi+2]
0x100e0e76: sar eax,1
0x100e0e78: jmp short 100E0EE0h
0x100e0e7a: movzx eax,word ptr [ebx+66B6h]
0x100e0e81: add eax,eax
0x100e0e83: add eax,eax
0x100e0e85: mov edx,ecx
0x100e0e87: sub edx,eax
0x100e0e89: mov [esp+1Ch],edx
0x100e0e8d: lea edx,[eax+ecx]
0x100e0e90: movzx eax,word ptr [edi+edx*2-8]
0x100e0e95: movzx edx,word ptr [edi+edx*2+8]
0x100e0e9a: add eax,edx
0x100e0e9c: mov edx,[esp+1Ch]
0x100e0ea0: movzx edx,word ptr [edi+edx*2-8]
0x100e0ea5: add eax,edx
0x100e0ea7: mov edx,[esp+1Ch]
0x100e0eab: movzx edx,word ptr [edi+edx*2+8]
0x100e0eb0: add eax,edx
0x100e0eb2: movzx edx,word ptr [esi+608h]
0x100e0eb9: sub eax,edx
0x100e0ebb: movzx edx,word ptr [esi-604h]
0x100e0ec2: sub eax,edx
0x100e0ec4: movzx edx,word ptr [esi+5FCh]
0x100e0ecb: sub eax,edx
0x100e0ecd: movzx edx,word ptr [esi-5F8h]
0x100e0ed4: sub eax,edx
0x100e0ed6: movzx edx,word ptr [esi+2]
0x100e0eda: add eax,1
0x100e0edd: sar eax,2
0x100e0ee0: add eax,edx
0x100e0ee2: cmp eax,0FFFFh
0x100e0ee7: jge short 100E0EF7h
0x100e0ee9: xor edx,edx
0x100e0eeb: test eax,eax
0x100e0eed: setl dl
0x100e0ef0: sub edx,1
0x100e0ef3: and eax,edx
0x100e0ef5: jmp short 100E0EFCh
0x100e0ef7: mov eax,0FFFFh
0x100e0efc: fld dword ptr ds:[100FF140h]
0x100e0f02: mov [esi+ecx*2],ax
0x100e0f06: mov eax,[ebx+65B0h]
0x100e0f0c: fst dword ptr [esp+30h]
0x100e0f10: mov ecx,[esp+44h]
0x100e0f14: fst dword ptr [esp+2Ch]
0x100e0f18: shr eax,cl
0x100e0f1a: fstp dword ptr [esp+28h]
0x100e0f1e: and eax,3
0x100e0f21: mov cx,[edi+eax*2]
0x100e0f25: mov [esi+eax*2],cx
0x100e0f29: mov edx,[ebx+66D0h]
0x100e0f2f: xor eax,eax
0x100e0f31: cmp edx,4
0x100e0f34: jl near ptr 100E1041h
0x100e0f3a: lea ecx,[esi+4]
0x100e0f3d: movzx edi,word ptr [ecx-4]
0x100e0f41: mov [esp+0Ch],edi
```

```
0x100e0f45: movzx edi,word ptr [ecx-2]
0x100e0f49: fild dword ptr [esp+0Ch]
0x100e0f4d: fld dword ptr [esp+eax*4+98h]
0x100e0f54: mov [esp+0Ch],edi
0x100e0f58: movzx edi,word ptr [ecx]
0x100e0f5b: fmul st,st(1)
0x100e0f5d: add eax,4
0x100e0f60: fadd dword ptr [esp+28h]
0x100e0f64: add ecx,8
0x100e0f67: fstp dword ptr [esp+28h]
0x100e0f6b: fld dword ptr [esp+eax*4+98h]
0x100e0f72: fmul st,st(1)
0x100e0f74: fadd dword ptr [esp+2Ch]
0x100e0f78: fstp dword ptr [esp+2Ch]
0x100e0f7c: fmul dword ptr [esp+eax*4+0A8h]
0x100e0f83: fadd dword ptr [esp+30h]
0x100e0f87: fstp dword ptr [esp+30h]
0x100e0f8b: fild dword ptr [esp+0Ch]
0x100e0f8f: mov [esp+0Ch],edi
0x100e0f93: fld dword ptr [esp+eax*4+8Ch]
0x100e0f9a: movzx edi,word ptr [ecx-6]
0x100e0f9e: fmul st,st(1)
0x100e0fa0: fadd dword ptr [esp+28h]
0x100e0fa4: fstp dword ptr [esp+28h]
0x100e0fa8: fld dword ptr [esp+eax*4+9Ch]
0x100e0faf: fmul st,st(1)
0x100e0fb1: fadd dword ptr [esp+2Ch]
0x100e0fb5: fstp dword ptr [esp+2Ch]
0x100e0fb9: fmul dword ptr [esp+eax*4+0ACh]
0x100e0fc0: fadd dword ptr [esp+30h]
0x100e0fc4: fstp dword ptr [esp+30h]
0x100e0fc8: fild dword ptr [esp+0Ch]
0x100e0fcc: mov [esp+0Ch],edi
0x100e0fd0: fld dword ptr [esp+eax*4+90h]
0x100e0fd7: lea edi,[edx-3]
0x100e0fda: cmp eax,edi
0x100e0fdc: fmul st,st(1)
0x100e0fde: fadd dword ptr [esp+28h]
0x100e0fe2: fstp dword ptr [esp+28h]
0x100e0fe6: fld dword ptr [esp+eax*4+0A0h]
0x100e0fed: fmul st,st(1)
0x100e0fef: fadd dword ptr [esp+2Ch]
0x100e0ff3: fstp dword ptr [esp+2Ch]
0x100e0ff7: fmul dword ptr [esp+eax*4+0B0h]
0x100e0ffe: fadd dword ptr [esp+30h]
0x100e1002: fstp dword ptr [esp+30h]
0x100e1006: fild dword ptr [esp+0Ch]
0x100e100a: fld dword ptr [esp+eax*4+94h]
0x100e1011: fmul st,st(1)
0x100e1013: fadd dword ptr [esp+28h]
0x100e1017: fstp dword ptr [esp+28h]
0x100e101b: fld dword ptr [esp+eax*4+0A4h]
0x100e1022: fmul st,st(1)
0x100e1024: fadd dword ptr [esp+2Ch]
0x100e1028: fstp dword ptr [esp+2Ch]
0x100e102c: fmul dword ptr [esp+eax*4+0B4h]
0x100e1033: fadd dword ptr [esp+30h]
0x100e1037: fstp dword ptr [esp+30h]
0x100e103b: jl near ptr 100E0F3Dh
0x100e1041: cmp eax,edx
0x100e1043: jge short 100E1089h
0x100e1045: movzx ecx,word ptr [esi+eax*2]
0x100e1049: mov [esp+0Ch],ecx
0x100e104d: add eax,1
0x100e1050: cmp eax,edx
0x100e1052: fild dword ptr [esp+0Ch]
0x100e1056: fld dword ptr [esp+eax*4+94h]
```

```
0x100e105d: fmul st,st(1)
0x100e105f: fadd dword ptr [esp+28h]
0x100e1063: fstp dword ptr [esp+28h]
0x100e1067: fld dword ptr [esp+eax*4+0A4h]
0x100e106e: fmul st,st(1)
0x100e1070: fadd dword ptr [esp+2Ch]
0x100e1074: fstp dword ptr [esp+2Ch]
0x100e1078: fmul dword ptr [esp+eax*4+0B4h]
0x100e107f: fadd dword ptr [esp+30h]
0x100e1083: fstp dword ptr [esp+30h]
0x100e1087: jl short 100E1045h
0x100e1089: fld dword ptr [esp+28h]
0x100e108d: call 100228D0h
0x100e1092: cmp eax,0FFFFh
0x100e1097: jge short 100E10A8h
0x100e1099: test eax,eax
0x100e109b: jge short 100E10A1h
0x100e109d: xor eax,eax
0x100e109f: jmp short 100E10ADh
0x100e10a1: cmp eax,0FFFFh
0x100e10a6: jl short 100E10ADh
0x100e10a8: mov eax,0FFFFh
0x100e10ad: fld dword ptr [esp+eax*4+0C8h]
0x100e10b4: fstp dword ptr [esp+28h]
0x100e10b8: fld dword ptr [esp+2Ch]
0x100e10bc: call 100228D0h
0x100e10c1: cmp eax,0FFFFh
0x100e10c6: jge short 100E10D7h
0x100e10c8: test eax,eax
0x100e10ca: jge short 100E10D0h
0x100e10cc: xor eax,eax
0x100e10ce: jmp short 100E10DCh
0x100e10d0: cmp eax,0FFFFh
0x100e10d5: jl short 100E10DCh
0x100e10d7: mov eax,0FFFFh
0x100e10dc: fld dword ptr [esp+eax*4+0C8h]
0x100e10e3: fstp dword ptr [esp+2Ch]
0x100e10e7: fld dword ptr [esp+30h]
0x100e10eb: call 100228D0h
0x100e10f0: cmp eax,0FFFFh
0x100e10f5: jge short 100E1106h
0x100e10f7: test eax,eax
0x100e10f9: jge short 100E10FFh
0x100e10fb: xor eax,eax
0x100e10fd: jmp short 100E110Bh
0x100e10ff: cmp eax,0FFFFh
0x100e1104: jl short 100E110Bh
0x100e1106: mov eax,0FFFFh
0x100e110b: fld dword ptr [esp+eax*4+0C8h]
0x100e1112: fstp dword ptr [esp+30h]
0x100e1116: fld dword ptr [esp+2Ch]
0x100e111a: fld st(0)
0x100e111c: fmul st,st(6)
0x100e111e: fsub st,st(5)
0x100e1120: fmul st,st(4)
0x100e1122: call 100228D0h
0x100e1127: fld dword ptr [esp+28h]
0x100e112b: mov esi,[esp+34h]
0x100e112f: fsub st,st(1)
0x100e1131: mov edi,[esp+40h]
0x100e1135: mov [esi+edi+0C0000h],ax
0x100e113d: fmul st,st(3)
0x100e113f: call 100228D0h
0x100e1144: fsub dword ptr [esp+30h]
0x100e1148: mov [esi+edi+0C0002h],ax
0x100e1150: fmul st,st(1)
0x100e1152: call 100228D0h
```

```
0x100e1157: mov ecx,[esp+20h]
0x100e115b: mov [esi+edi+0C0004h],ax
0x100e1163: mov eax,[esp+24h]
0x100e1167: mov esi,[esp+24h]
0x100e116b: add ecx,1
0x100e116e: add eax,0FFh
0x100e1173: mov [esp+20h],ecx
0x100e1177: cmp ecx,eax
0x100e1179: mov ecx,[esp+14h]
0x100e117d: jl near ptr 100E0D69h
0x100e1183: mov eax,[esp+38h]
0x100e1187: add dword ptr [esp+10h],4
0x100e118c: add ecx,1
0x100e118f: add eax,0FFh
0x100e1194: cmp ecx,eax
0x100e1196: mov [esp+14h],ecx
0x100e119a: jl near ptr 100E0D40h
0x100e11a0: add dword ptr [esp+18h],100h
0x100e11a8: sub dword ptr [esp+3Ch],1
0x100e11ad: jne near ptr 100E0D1Dh
0x100e11b3: mov esi,[esp+40h]
0x100e11b7: fstp st(4)
0x100e11b9: fstp st(2)
0x100e11bb: push 20000h
0x100e11c0: fstp st(0)
0x100e11c2: lea eax,[esi+180000h]
0x100e11c8: push 0
0x100e11ca: fstp st(0)
0x100e11cc: push eax
0x100e11cd: fstp st(0)
0x100e11cf: call 100228C0h
0x100e11d4: mov eax,[esp+44h]
0x100e11d8: lea ecx,[eax+2]
0x100e11db: mov edx,ecx
0x100e11dd: lea ecx,[eax+0FEh]
0x100e11e3: add esp,0Ch
0x100e11e6: cmp edx,ecx
0x100e11e8: mov [esp+14h],edx
0x100e11ec: jge near ptr 100E1533h
0x100e11f2: mov ecx,edx
0x100e11f4: sub ecx,eax
0x100e11f6: shl ecx,8
0x100e11f9: mov [esp+3Ch],ecx
0x100e11fd: lea ecx,[ecx]
0x100e1200: movzx eax,word ptr [ebx+66B4h]
0x100e1207: sub eax,4
0x100e120a: cmp edx,eax
0x100e120c: jge near ptr 100E1533h
0x100e1212: mov edx,[esp+24h]
0x100e1216: lea eax,[edx+2]
0x100e1219: add edx,0FEh
0x100e121f: cmp eax,edx
0x100e1221: mov [esp+20h],eax
0x100e1225: jge near ptr 100E1509h
0x100e122b: mov edx,eax
0x100e122d: sub edx,[esp+24h]
0x100e1231: mov [esp+1Ch],edx
0x100e1235: jmp short 100E1240h
0x100e1240: movzx esi,word ptr [ebx+66B6h]
0x100e1247: sub esi,4
0x100e124a: cmp eax,esi
0x100e124c: jge near ptr 100E1509h
0x100e1252: lea eax,[ecx+edx]
0x100e1255: mov edx,[esp+40h]
0x100e1259: lea ecx,[eax+eax*2+3]
0x100e125d: mov [esp+0Ch],eax
0x100e1261: lea eax,[edx+ecx*2+0C0000h]
```

```
0x100e1268: xor edx,edx
0x100e126a: mov [esp+34h],edx
0x100e126e: mov edi,edi
0x100e1270: movsx ecx,word ptr [eax-6]
0x100e1274: movsx edi,word ptr [eax-0Ch]
0x100e1278: mov esi,ecx
0x100e127a: sub esi,edi
0x100e127c: mov edi,esi
0x100e127e: sar edi,1Fh
0x100e1281: mov ebx,edi
0x100e1283: xor ebx,esi
0x100e1285: sub ebx,edi
0x100e1287: movsx esi,word ptr [eax-2]
0x100e128b: movsx edi,word ptr [eax-8]
0x100e128f: mov [esp+edx+50h],ebx
0x100e1293: movsx ebx,word ptr [eax-0Ah]
0x100e1297: mov [esp+18h],esi
0x100e129b: sub esi,edi
0x100e129d: movsx edi,word ptr [eax-4]
0x100e12a1: mov [esp+10h],edi
0x100e12a5: sub edi,ebx
0x100e12a7: mov ebx,edi
0x100e12a9: imul ebx,edi
0x100e12ac: mov edi,esi
0x100e12ae: imul edi,esi
0x100e12b1: movsx esi,word ptr [eax]
0x100e12b4: add ebx,edi
0x100e12b6: mov [esp+edx+70h],ebx
0x100e12ba: mov edi,ecx
0x100e12bc: sub edi,esi
0x100e12be: movsx esi,word ptr [eax]
0x100e12c1: sar edi,1Fh
0x100e12c4: mov ebx,ecx
0x100e12c6: sub ebx,esi
0x100e12c8: movsx esi,word ptr [eax]
0x100e12cb: xor edi,ebx
0x100e12cd: mov ebx,ecx
0x100e12cf: sub ebx,esi
0x100e12d1: movsx esi,word ptr [eax+2]
0x100e12d5: sar ebx,1Fh
0x100e12d8: sub edi,ebx
0x100e12da: mov [esp+edx+54h],edi
0x100e12de: mov edx,[esp+10h]
0x100e12e2: mov edi,edx
0x100e12e4: sub edi,esi
0x100e12e6: movsx esi,word ptr [eax+2]
0x100e12ea: sub edx,esi
0x100e12ec: movsx esi,word ptr [eax+4]
0x100e12f0: imul edi,edx
0x100e12f3: mov edx,[esp+18h]
0x100e12f7: mov ebx,edx
0x100e12f9: sub ebx,esi
0x100e12fb: movsx esi,word ptr [eax+4]
0x100e12ff: sub edx,esi
0x100e1301: movsx esi,word ptr [eax-606h]
0x100e1308: imul ebx,edx
0x100e130b: mov edx,[esp+34h]
0x100e130f: add edi,ebx
0x100e1311: mov [esp+edx+74h],edi
0x100e1315: mov edi,ecx
0x100e1317: sub edi,esi
0x100e1319: movsx esi,word ptr [eax-606h]
0x100e1320: sar edi,1Fh
0x100e1323: mov ebx,ecx
0x100e1325: sub ebx,esi
0x100e1327: movsx esi,word ptr [eax-606h]
0x100e132e: xor edi,ebx
```

```
0x100e1330: mov ebx,ecx
0x100e1332: sub ebx,esi
0x100e1334: movsx esi,word ptr [eax-604h]
0x100e133b: sar ebx,1Fh
0x100e133e: sub edi,ebx
0x100e1340: mov [esp+edx+58h],edi
0x100e1344: mov edx,[esp+10h]
0x100e1348: mov edi,edx
0x100e134a: sub edi,esi
0x100e134c: movsx esi,word ptr [eax-604h]
0x100e1353: sub edx,esi
0x100e1355: movsx esi,word ptr [eax-602h]
0x100e135c: imul edi,edx
0x100e135f: mov edx,[esp+18h]
0x100e1363: mov ebx,edx
0x100e1365: sub ebx,esi
0x100e1367: movsx esi,word ptr [eax-602h]
0x100e136e: sub edx,esi
0x100e1370: movsx esi,word ptr [eax+5FAh]
0x100e1377: imul ebx,edx
0x100e137a: mov edx,[esp+34h]
0x100e137e: add edi,ebx
0x100e1380: mov [esp+edx+78h],edi
0x100e1384: mov edi,ecx
0x100e1386: sub edi,esi
0x100e1388: movsx esi,word ptr [eax+5FAh]
0x100e138f: mov ebx,ecx
0x100e1391: sub ebx,esi
0x100e1393: movsx esi,word ptr [eax+5FAh]
0x100e139a: sub ecx,esi
0x100e139c: movsx esi,word ptr [eax+5FCh]
0x100e13a3: sar edi,1Fh
0x100e13a6: xor edi,ebx
0x100e13a8: sar ecx,1Fh
0x100e13ab: sub edi,ecx
0x100e13ad: mov ecx,[esp+10h]
0x100e13b1: mov [esp+edx+5Ch],edi
0x100e13b5: mov edi,ecx
0x100e13b7: sub edi,esi
0x100e13b9: movsx esi,word ptr [eax+5FCh]
0x100e13c0: sub ecx,esi
0x100e13c2: movsx esi,word ptr [eax+5FEh]
0x100e13c9: imul edi,ecx
0x100e13cc: mov ecx,[esp+18h]
0x100e13d0: mov ebx,ecx
0x100e13d2: sub ebx,esi
0x100e13d4: movsx esi,word ptr [eax+5FEh]
0x100e13db: sub ecx,esi
0x100e13dd: imul ebx,ecx
0x100e13e0: add edi,ebx
0x100e13e2: mov [esp+edx+7Ch],edi
0x100e13e6: add edx,10h
0x100e13e9: add eax,60000h
0x100e13ee: cmp edx,20h
0x100e13f1: mov [esp+34h],edx
0x100e13f5: jl near ptr 100E1270h
0x100e13fb: mov edi,[esp+50h]
0x100e13ff: mov ebx,[esp+54h]
0x100e1403: cmp edi,ebx
0x100e1405: mov esi,edi
0x100e1407: ja short 100E140Bh
0x100e1409: mov esi,ebx
0x100e140b: mov ecx,[esp+68h]
0x100e140f: mov edx,[esp+6Ch]
0x100e1413: cmp ecx,edx
0x100e1415: mov eax,ecx
0x100e1417: ja short 100E141Bh
```

```
0x100e1419: mov eax,edx
0x100e141b: cmp esi,eax
0x100e141d: jae short 100E142Fh
0x100e141f: cmp edi,ebx
0x100e1421: jbe short 100E1429h
0x100e1423: mov [esp+10h],edi
0x100e1427: jmp short 100E143Bh
0x100e1429: mov [esp+10h],ebx
0x100e142d: jmp short 100E143Bh
0x100e142f: cmp ecx,edx
0x100e1431: mov [esp+10h],ecx
0x100e1435: ja short 100E143Bh
0x100e1437: mov [esp+10h],edx
0x100e143b: mov edi,[esp+70h]
0x100e143f: mov ebx,[esp+74h]
0x100e1443: cmp edi,ebx
0x100e1445: mov esi,edi
0x100e1447: ja short 100E144Bh
0x100e1449: mov esi,ebx
0x100e144b: mov ecx,[esp+88h]
0x100e1452: mov edx,[esp+8Ch]
0x100e1459: cmp ecx,edx
0x100e145b: mov eax,ecx
0x100e145d: ja short 100E1461h
0x100e145f: mov eax,edx
0x100e1461: cmp esi,eax
0x100e1463: jae short 100E1471h
0x100e1465: cmp edi,ebx
0x100e1467: jbe short 100E146Dh
0x100e1469: mov edx,edi
0x100e146b: jmp short 100E1477h
0x100e146d: mov edx,ebx
0x100e146f: jmp short 100E1477h
0x100e1471: cmp ecx,edx
0x100e1473: jbe short 100E1477h
0x100e1475: mov edx,ecx
0x100e1477: mov eax,[esp+40h]
0x100e147b: mov ecx,[esp+0Ch]
0x100e147f: mov esi,[esp+10h]
0x100e1483: add eax,180000h
0x100e1488: add ecx,eax
0x100e148a: xor eax,eax
0x100e148c: lea esp,[esp]
0x100e1490: cmp [esp+eax+50h],esi
0x100e1494: ja short 100E149Fh
0x100e1496: cmp [esp+eax+70h],edx
0x100e149a: ja short 100E149Fh
0x100e149c: add byte ptr [ecx],1
0x100e149f: cmp [esp+eax+54h],esi
0x100e14a3: ja short 100E14AEh
0x100e14a5: cmp [esp+eax+74h],edx
0x100e14a9: ja short 100E14AEh
0x100e14ab: add byte ptr [ecx],1
0x100e14ae: cmp [esp+eax+58h],esi
0x100e14b2: ja short 100E14BDh
0x100e14b4: cmp [esp+eax+78h],edx
0x100e14b8: ja short 100E14BDh
0x100e14ba: add byte ptr [ecx],1
0x100e14bd: cmp [esp+eax+5Ch],esi
0x100e14c1: ja short 100E14CCh
0x100e14c3: cmp [esp+eax+7Ch],edx
0x100e14c7: ja short 100E14CCh
0x100e14c9: add byte ptr [ecx],1
0x100e14cc: add eax,10h
0x100e14cf: add ecx,10000h
0x100e14d5: cmp eax,20h
0x100e14d8: jl short 100E1490h
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```
0x100e14da: mov eax,[esp+20h]
0x100e14de: mov edx,[esp+1Ch]
0x100e14e2: mov ecx,[esp+24h]
0x100e14e6: mov ebx,[ebp+8]
0x100e14e9: add eax,1
0x100e14ec: add ecx,0FEh
0x100e14f2: add edx,1
0x100e14f5: cmp eax,ecx
0x100e14f7: mov ecx,[esp+3Ch]
0x100e14fb: mov [esp+20h],eax
0x100e14ff: mov [esp+1Ch],edx
0x100e1503: jl near ptr 100E1240h
0x100e1509: mov edx,[esp+14h]
0x100e150d: mov eax,[esp+38h]
0x100e1511: mov esi,[esp+40h]
0x100e1515: add edx,1
0x100e1518: add ecx,100h
0x100e151e: add eax,0FEh
0x100e1523: cmp edx,eax
0x100e1525: mov [esp+14h],edx
0x100e1529: mov [esp+3Ch],ecx
0x100e152d: jl near ptr 100E1200h
0x100e1533: mov edx,[esp+38h]
0x100e1537: mov edi,[esp+24h]
0x100e153b: lea ecx,[edx+3]
0x100e153e: lea eax,[edx+0FDh]
0x100e1544: cmp ecx,eax
0x100e1546: mov [esp+14h],ecx
0x100e154a: jge near ptr 100E17CFh
0x100e1550: mov eax,ecx
0x100e1552: sub eax,edx
0x100e1554: mov [esp+18h],eax
0x100e1558: shl eax,8
0x100e155b: mov [esp+1Ch],eax
0x100e155f: movzx edx,word ptr [ebx+66B4h]
0x100e1566: sub edx,5
0x100e1569: cmp ecx,edx
0x100e156b: jge near ptr 100E17CFh
0x100e1571: lea eax,[edi+3]
0x100e1574: lea edx,[edi+0FDh]
0x100e157a: cmp eax,edx
0x100e157c: mov [esp+20h],eax
0x100e1580: jge near ptr 100E17AAh
0x100e1586: mov ecx,eax
0x100e1588: sub ecx,edi
0x100e158a: mov [esp+34h],ecx
0x100e158e: movzx edx,word ptr [ebx+66B6h]
0x100e1595: lea ecx,[edx-5]
0x100e1598: cmp eax,ecx
0x100e159a: mov [esp+4Ch],edx
0x100e159e: jge near ptr 100E17A6h
0x100e15a4: mov eax,[esp+18h]
0x100e15a8: lea edi,[esp+90h]
0x100e15af: add eax,1
0x100e15b2: mov dword ptr [esp+3Ch],0
0x100e15ba: mov [esp+0Ch],edi
0x100e15be: mov ecx,[esp+18h]
0x100e15c2: add ecx,0FFFFFFFFh
0x100e15c5: cmp ecx,eax
0x100e15c7: mov dword ptr [edi],0
0x100e15cd: mov [esp+10h],ecx
0x100e15d1: jg short 100E1639h
0x100e15d3: mov eax,[esp+34h]
0x100e15d7: add eax,1
0x100e15da: mov [esp+44h],eax
0x100e15de: mov eax,[esp+34h]
0x100e15e2: add eax,0FFFFFFFFh
```

```
0x100e15e5: cmp eax,[esp+44h]
0x100e15e9: jg short 100E1623h
0x100e15eb: mov edx,[esp+10h]
0x100e15ef: mov ecx,[esp+3Ch]
0x100e15f3: add ecx,edx
0x100e15f5: shl ecx,8
0x100e15f8: lea edx,[ecx+esi+180000h]
0x100e15ff: mov ecx,[edi]
0x100e1601: movsx edi,byte ptr [edx+eax]
0x100e1605: add ecx,edi
0x100e1607: mov edi,[esp+34h]
0x100e160b: add edi,1
0x100e160e: add eax,1
0x100e1611: cmp eax,edi
0x100e1613: mov [esp+44h],edi
0x100e1617: jle short 100E1601h
0x100e1619: mov edi,[esp+0Ch]
0x100e161d: mov edx,[esp+4Ch]
0x100e1621: mov [edi],ecx
0x100e1623: mov ecx,[esp+10h]
0x100e1627: mov eax,[esp+18h]
0x100e162b: add ecx,1
0x100e162e: add eax,1
0x100e1631: cmp ecx,eax
0x100e1633: mov [esp+10h],ecx
0x100e1637: jle short 100E15DEh
0x100e1639: mov ecx,[esp+3Ch]
0x100e163d: add ecx,100h
0x100e1643: add edi,4
0x100e1646: cmp ecx,200h
0x100e164c: mov [esp+0Ch],edi
0x100e1650: mov [esp+3Ch],ecx
0x100e1654: jl near ptr 100E15BEh
0x100e165a: mov edi,[esp+90h]
0x100e1661: mov eax,[esp+94h]
0x100e1668: cmp edi,eax
0x100e166a: je near ptr 100E16F0h
0x100e1670: xor ecx,ecx
0x100e1672: cmp eax,edi
0x100e1674: setg cl
0x100e1677: imul edx,[esp+14h]
0x100e167c: mov edi,[ebx+66E0h]
0x100e1682: shl ecx,8
0x100e1685: add ecx,[esp+18h]
0x100e1689: shl ecx,8
0x100e168c: add ecx,[esp+34h]
0x100e1690: mov eax,ecx
0x100e1692: lea ecx,[eax+eax*2]
0x100e1695: mov eax,[esp+20h]
0x100e1699: add edx,eax
0x100e169b: mov [esp+0Ch],ecx
0x100e169f: mov cx,[esi+ecx*2]
0x100e16a3: mov [edi+edx*8],cx
0x100e16a7: movzx edx,word ptr [ebx+66B6h]
0x100e16ae: imul edx,[esp+14h]
0x100e16b3: mov edi,[esp+0Ch]
0x100e16b7: movzx edi,word ptr [esi+edi*2+2]
0x100e16bc: mov ecx,[ebx+66E0h]
0x100e16c2: add edx,eax
0x100e16c4: mov [ecx+edx*8+2],di
0x100e16c9: movzx edx,word ptr [ebx+66B6h]
0x100e16d0: imul edx,[esp+14h]
0x100e16d5: mov edi,[esp+0Ch]
0x100e16d9: movzx edi,word ptr [esi+edi*2+4]
0x100e16de: mov ecx,[ebx+66E0h]
0x100e16e4: add edx,eax
0x100e16e6: mov [ecx+edx*8+4],di
```

```
0x100e16eb: jmp near ptr 100E1788h
0x100e16f0: imul edx,[esp+14h]
0x100e16f5: mov ecx,[esp+18h]
0x100e16f9: mov edi,[esp+34h]
0x100e16fd: mov eax,[esp+1Ch]
0x100e1701: add edx,[esp+20h]
0x100e1705: add ecx,100h
0x100e170b: shl ecx,8
0x100e170e: add ecx,edi
0x100e1710: add eax,edi
0x100e1712: lea ecx,[ecx+ecx*2]
0x100e1715: movzx ecx,word ptr [esi+ecx*2]
0x100e1719: lea eax,[eax+eax*2]
0x100e171c: movzx edi,word ptr [esi+eax*2]
0x100e1720: add ecx,edi
0x100e1722: mov edi,[ebx+66E0h]
0x100e1728: sar ecx,1
0x100e172a: mov [edi+edx*8],cx
0x100e172e: movzx ecx,word ptr [esi+eax*2+2]
0x100e1733: movzx edx,word ptr [esi+eax*2+60002h]
0x100e173b: mov edi,[ebx+66E0h]
0x100e1741: add edx,ecx
0x100e1743: movzx ecx,word ptr [ebx+66B6h]
0x100e174a: imul ecx,[esp+14h]
0x100e174f: add ecx,[esp+20h]
0x100e1753: sar edx,1
0x100e1755: mov [edi+ecx*8+2],dx
0x100e175a: movzx ecx,word ptr [ebx+66B6h]
0x100e1761: movzx edx,word ptr [esi+eax*2+4]
0x100e1766: imul ecx,[esp+14h]
0x100e176b: movzx eax,word ptr [esi+eax*2+60004h]
0x100e1773: mov edi,[ebx+66E0h]
0x100e1779: add edx,eax
0x100e177b: mov eax,[esp+20h]
0x100e177f: sar edx,1
0x100e1781: add ecx,eax
0x100e1783: mov [edi+ecx*8+4],dx
0x100e1788: mov edi,[esp+24h]
0x100e178c: add dword ptr [esp+34h],1
0x100e1791: add eax,1
0x100e1794: lea ecx,[edi+0FDh]
0x100e179a: cmp eax,ecx
0x100e179c: mov [esp+20h],eax
0x100e17a0: jl near ptr 100E158Eh
0x100e17a6: mov ecx,[esp+14h]
0x100e17aa: mov eax,[esp+38h]
0x100e17ae: add dword ptr [esp+1Ch],100h
0x100e17b6: add dword ptr [esp+18h],1
0x100e17bb: add ecx,1
0x100e17be: add eax,0FDh
0x100e17c3: cmp ecx,eax
0x100e17c5: mov [esp+14h],ecx
0x100e17c9: jl near ptr 100E155Fh
0x100e17cf: fld qword ptr ds:[1012E428h]
0x100e17d5: movzx edx,word ptr [ebx+66B6h]
0x100e17dc: fld qword ptr ds:[1012E430h]
0x100e17e2: add edi,0FAh
0x100e17e8: fld qword ptr ds:[1010BE00h]
0x100e17ee: sub edx,5
0x100e17f1: cmp edi,edx
0x100e17f3: fld qword ptr ds:[100FE9C8h]
0x100e17f9: fld qword ptr ds:[100FE9C0h]
0x100e17ff: mov [esp+24h],edi
0x100e1803: fxch st(4)
0x100e1805: fxch
0x100e1807: fxch st(3)
0x100e1809: fxch
```

```
0x100e180b: jl near ptr 100E0B00h
0x100e1811: movzx ecx,word ptr [ebx+66B4h]
0x100e1818: mov eax,[esp+38h]
0x100e181c: add eax,0FAh
0x100e1821: sub ecx,5
0x100e1824: cmp eax,ecx
0x100e1826: mov [esp+38h],eax
0x100e182a: jl near ptr 100E0ADAh
0x100e1830: fstp st(4)
0x100e1832: fstp st(2)
0x100e1834: fstp st(0)
0x100e1836: fstp st(0)
0x100e1838: fstp st(0)
0x100e183a: mov edx,[esp+40h]
0x100e183e: push edx
0x100e183f: call dword ptr ds:[100FE554h]
0x100e1845: add esp,4
0x100e1848: pop edi
0x100e1849: pop esi
0x100e184a: pop ebx
0x100e184b: mov esp,ebp
0x100e184d: pop ebp
0x100e184e: ret
```

`sub_100e0950` is called by sub_10016040. It delegates to j_memset, sub_100228d0, j__CIpow (6 callees).

12.1281 sub_100e1850 at 0x100e1850 (147 instructions)

```
0x100e1850: sub esp,38h
0x100e1853: push ebp
0x100e1854: mov ebp,[esp+40h]
0x100e1858: mov eax,1
0x100e185d: cmp [ebp+98h],eax
0x100e1863: mov [esp+14h],eax
0x100e1867: jl near ptr 100E1A4Bh
0x100e186d: push ebx
0x100e186e: push esi
0x100e186f: push edi
0x100e1870: cmp dword ptr [ebp+0BCh],0
0x100e1877: je short 100E1892h
0x100e1879: push eax
0x100e187a: push 1012E474h
0x100e187f: call dword ptr ds:[100FE4C4h]
0x100e1885: add eax,40h
0x100e1888: push eax
0x100e1889: call dword ptr ds:[100FE4C8h]
0x100e188f: add esp,0Ch
0x100e1892: xor esi,esi
0x100e1894: mov [esp+14h],esi
0x100e1898: jmp short 100E18A0h
0x100e18a0: movzx ecx,word ptr [ebp+66B6h]
0x100e18a7: movzx edx,word ptr [ebp+66B4h]
0x100e18ae: mov eax,[ebp+66E0h]
0x100e18b4: imul ecx,edx
0x100e18b7: lea ecx,[eax+ecx*8]
0x100e18ba: cmp eax,ecx
0x100e18bc: jae short 100E18E9h
0x100e18be: mov edi,edi
0x100e18c0: mov dx,[esi+eax]
0x100e18c4: mov [eax+6],dx
0x100e18c8: movzx edx,word ptr [ebp+66B4h]
0x100e18cf: movzx ecx,word ptr [ebp+66B6h]
0x100e18d6: imul ecx,edx
0x100e18d9: mov edx,[ebp+66E0h]
0x100e18df: add eax,8
0x100e18e2: lea ecx,[edx+ecx*8]
```

```
0x100e18e5: cmp eax,ecx
0x100e18e7: jb short 100E18C0h
0x100e18e9: movzx eax,word ptr [ebp+66B4h]
0x100e18f0: movzx ecx,word ptr [ebp+66B6h]
0x100e18f7: mov edx,[ebp+66E0h]
0x100e18fd: sub eax,1
0x100e1900: imul eax,ecx
0x100e1903: lea ebx,[edx+ecx*8]
0x100e1906: lea eax,[edx+eax*8]
0x100e1909: cmp ebx,eax
0x100e190b: mov [esp+1Ch],ebx
0x100e190f: jae near ptr 100E1A21h
0x100e1915: mov eax,ebx
0x100e1917: sub eax,edx
0x100e1919: sar eax,3
0x100e191c: add eax,1
0x100e191f: cdq
0x100e1920: idiv ecx
0x100e1922: cmp edx,2
0x100e1925: jl near ptr 100E19F5h
0x100e192b: mov edx,ecx
0x100e192d: neg edx
0x100e192f: xor esi,esi
0x100e1931: cmp edx,ecx
0x100e1933: jg short 100E198Fh
0x100e1935: lea eax,[edx+1]
0x100e1938: lea edi,[ebx+edx*8-6]
0x100e193c: mov [esp+18h],eax
0x100e1940: mov [esp+10h],edi
0x100e1944: lea edi,[eax-2]
0x100e1947: cmp edi,eax
0x100e1949: jg short 100E1974h
0x100e194b: mov eax,[esp+10h]
0x100e194f: mov edi,3
0x100e1954: movzx ebx,word ptr [eax+4]
0x100e1958: movzx ebp,word ptr [eax]
0x100e195b: sub ebx,ebp
0x100e195d: mov [esp+esi*4+24h],ebx
0x100e1961: add esi,1
0x100e1964: add eax,8
0x100e1967: sub edi,1
0x100e196a: jne short 100E1954h
0x100e196c: mov eax,[esp+18h]
0x100e1970: mov ebx,[esp+1Ch]
0x100e1974: lea edi,[ecx*8]
0x100e197b: add [esp+10h],edi
0x100e197f: add edx,ecx
0x100e1981: add eax,ecx
0x100e1983: cmp edx,ecx
0x100e1985: mov [esp+18h],eax
0x100e1989: jle short 100E1944h
0x100e198b: mov ebp,[esp+4Ch]
0x100e198f: xor edi,edi
0x100e1991: movzx eax,byte ptr [edi+1012B5B9h]
0x100e1998: movzx ecx,byte ptr [edi+1012B5B8h]
0x100e199f: mov edx,[esp+eax*4+24h]
0x100e19a3: mov esi,[esp+ecx*4+24h]
0x100e19a7: cmp esi,edx
0x100e19a9: lea eax,[esp+eax*4+24h]
0x100e19ad: lea ecx,[esp+ecx*4+24h]
0x100e19b1: jle short 100E19BDh
0x100e19b3: xor edx,esi
0x100e19b5: mov [ecx],edx
0x100e19b7: xor [eax],edx
0x100e19b9: mov eax,[eax]
0x100e19bb: xor [ecx],eax
0x100e19bd: add edi,2
```

```
0x100e19c0: cmp edi,26h
0x100e19c3: jb short 100E1991h
0x100e19c5: movzx ecx,word ptr [ebx+2]
0x100e19c9: mov edx,[esp+34h]
0x100e19cd: lea eax,[ecx+edx]
0x100e19d0: cmp eax,0FFFFh
0x100e19d5: jge short 100E19E6h
0x100e19d7: test eax,eax
0x100e19d9: jge short 100E19DFh
0x100e19db: xor eax,eax
0x100e19dd: jmp short 100E19EBh
0x100e19df: cmp eax,0FFFFh
0x100e19e4: jl short 100E19EBh
0x100e19e6: mov eax,0FFFFh
0x100e19eb: mov ecx,[esp+14h]
0x100e19ef: mov [ecx+ebx],ax
0x100e19f3: mov esi,ecx
0x100e19f5: movzx eax,word ptr [ebp+66B4h]
0x100e19fc: movzx ecx,word ptr [ebp+66B6h]
0x100e1a03: mov edx,[ebp+66E0h]
0x100e1a09: sub eax,1
0x100e1a0c: imul eax,ecx
0x100e1a0f: add ebx,8
0x100e1a12: lea eax,[edx+eax*8]
0x100e1a15: cmp ebx,eax
0x100e1a17: mov [esp+1Ch],ebx
0x100e1a1b: jb near ptr 100E1915h
0x100e1a21: add esi,4
0x100e1a24: cmp esi,6
0x100e1a27: mov [esp+14h],esi
0x100e1a2b: jl near ptr 100E18A0h
0x100e1a31: mov eax,[esp+20h]
0x100e1a35: add eax,1
0x100e1a38: cmp eax,[ebp+98h]
0x100e1a3e: mov [esp+20h],eax
0x100e1a42: jle near ptr 100E1870h
0x100e1a48: pop edi
0x100e1a49: pop esi
0x100e1a4a: pop ebx
0x100e1a4b: pop ebp
0x100e1a4c: add esp,38h
0x100e1a4f: ret
```

`sub_100e1850` is called by sub_10016040. It works with the text "Median filter pass %d...". It calls nothing else.

12.1282 sub_100e1a50 at 0x100e1a50 (442 instructions)

```
0x100e1a50: sub esp,68h
0x100e1a53: push ebx
0x100e1a54: mov ebx,[esp+70h]
0x100e1a58: mov eax,[ebx+66D0h]
0x100e1a5e: sub eax,3
0x100e1a61: cmp eax,1
0x100e1a64: mov dword ptr [esp+10h],7FFFFFFFh
0x100e1a6c: ja near ptr 100E2005h
0x100e1a72: cmp dword ptr [ebx+0BCh],0
0x100e1a79: je short 100E1A93h
0x100e1a7b: push 1012E490h
0x100e1a80: call dword ptr ds:[100FE4C4h]
0x100e1a86: add eax,40h
0x100e1a89: push eax
0x100e1a8a: call dword ptr ds:[100FE4C8h]
0x100e1a90: add esp,8
0x100e1a93: push ebp
0x100e1a94: push esi
0x100e1a95: mov esi,[ebx+66D0h]
```

```
0x100e1a9b: test esi,esi
0x100e1a9d: push edi
0x100e1a9e: mov [esp+28h],esi
0x100e1aa2: jle short 100E1ACFh
0x100e1aa4: fld qword ptr ds:[100FFBA0h]
0x100e1aaa: lea edi,[ebx+0E784h]
0x100e1ab0: mov ebp,esi
0x100e1ab2: fld dword ptr [edi]
0x100e1ab4: fmul st,st(1)
0x100e1ab6: call 100228D0h
0x100e1abb: cmp [esp+1Ch],eax
0x100e1abf: jle short 100E1AC5h
0x100e1ac1: mov [esp+1Ch],eax
0x100e1ac5: add edi,4
0x100e1ac8: sub ebp,1
0x100e1acb: jne short 100E1AB2h
0x100e1acd: fstp st(0)
0x100e1acf: cmp word ptr [ebx+66B4h],0
0x100e1ad7: mov dword ptr [esp+18h],0
0x100e1adf: jbe near ptr 100E2002h
0x100e1ae5: fldz
0x100e1ae7: jmp short 100E1AEBh
0x100e1ae9: mov ebx,edx
0x100e1aeb: movzx edi,word ptr [ebx+66B6h]
0x100e1af2: test edi,edi
0x100e1af4: mov dword ptr [esp+14h],0
0x100e1afc: jle near ptr 100E1FE2h
0x100e1b02: xor eax,eax
0x100e1b04: test esi,esi
0x100e1b06: jle short 100E1B35h
0x100e1b08: mov edx,[esp+7Ch]
0x100e1b0c: mov edx,[edx+66E0h]
0x100e1b12: mov ecx,edi
0x100e1b14: imul ecx,[esp+18h]
0x100e1b19: add ecx,[esp+14h]
0x100e1b1d: lea ecx,[edx+ecx*8]
0x100e1b20: mov edx,[esp+1Ch]
0x100e1b24: movzx ebx,word ptr [ecx]
0x100e1b27: cmp ebx,edx
0x100e1b29: jg short 100E1B35h
0x100e1b2b: add eax,1
0x100e1b2e: add ecx,2
0x100e1b31: cmp eax,esi
0x100e1b33: jl short 100E1B24h
0x100e1b35: cmp eax,esi
0x100e1b37: je near ptr 100E1FC4h
0x100e1b3d: xor ebp,ebp
0x100e1b3f: xor ecx,ecx
0x100e1b41: cmp esi,4
0x100e1b44: jl near ptr 100E1C55h
0x100e1b4a: fild dword ptr [esp+1Ch]
0x100e1b4e: mov edx,[esp+7Ch]
0x100e1b52: mov edx,[edx+66E0h]
0x100e1b58: mov eax,edi
0x100e1b5a: imul eax,[esp+18h]
0x100e1b5f: fstp dword ptr [esp+10h]
0x100e1b63: fld dword ptr [esp+10h]
0x100e1b67: add eax,[esp+14h]
0x100e1b6b: lea ebx,[esi-3]
0x100e1b6e: lea edx,[edx+eax*8+4]
0x100e1b72: movzx eax,word ptr [edx-4]
0x100e1b76: mov [esp+24h],eax
0x100e1b7a: fild dword ptr [esp+24h]
0x100e1b7e: fstp dword ptr [esp+24h]
0x100e1b82: fld dword ptr [esp+24h]
0x100e1b86: fst dword ptr [esp+ecx*4+58h]
0x100e1b8a: fcom
```

```
0x100e1b8c: fnstsw ax
0x100e1b8e: test ah,5
0x100e1b91: jp short 100E1B99h
0x100e1b93: fstp dword ptr [esp+10h]
0x100e1b97: jmp short 100E1B9Fh
0x100e1b99: fstp st(0)
0x100e1b9b: fst dword ptr [esp+10h]
0x100e1b9f: movzx eax,word ptr [edx-2]
0x100e1ba3: fld dword ptr [esp+10h]
0x100e1ba7: mov [esp+24h],eax
0x100e1bab: fstp dword ptr [esp+ecx*4+68h]
0x100e1baf: fild dword ptr [esp+24h]
0x100e1bb3: fstp dword ptr [esp+24h]
0x100e1bb7: fld dword ptr [esp+24h]
0x100e1bbb: fst dword ptr [esp+ecx*4+5Ch]
0x100e1bbf: fcom
0x100e1bc1: fnstsw ax
0x100e1bc3: test ah,5
0x100e1bc6: jp short 100E1BCEh
0x100e1bc8: fstp dword ptr [esp+10h]
0x100e1bcc: jmp short 100E1BD4h
0x100e1bce: fstp st(0)
0x100e1bd0: fst dword ptr [esp+10h]
0x100e1bd4: movzx eax,word ptr [edx]
0x100e1bd7: fld dword ptr [esp+10h]
0x100e1bdb: mov [esp+24h],eax
0x100e1bdf: fstp dword ptr [esp+ecx*4+6Ch]
0x100e1be3: fild dword ptr [esp+24h]
0x100e1be7: fstp dword ptr [esp+24h]
0x100e1beb: fld dword ptr [esp+24h]
0x100e1bef: fst dword ptr [esp+ecx*4+60h]
0x100e1bf3: fcom
0x100e1bf5: fnstsw ax
0x100e1bf7: test ah,5
0x100e1bfa: jp short 100E1C02h
0x100e1bfc: fstp dword ptr [esp+10h]
0x100e1c00: jmp short 100E1C08h
0x100e1c02: fstp st(0)
0x100e1c04: fst dword ptr [esp+10h]
0x100e1c08: movzx eax,word ptr [edx+2]
0x100e1c0c: fld dword ptr [esp+10h]
0x100e1c10: mov [esp+24h],eax
0x100e1c14: fstp dword ptr [esp+ecx*4+70h]
0x100e1c18: fild dword ptr [esp+24h]
0x100e1c1c: fstp dword ptr [esp+24h]
0x100e1c20: fld dword ptr [esp+24h]
0x100e1c24: fst dword ptr [esp+ecx*4+64h]
0x100e1c28: fcom
0x100e1c2a: fnstsw ax
0x100e1c2c: test ah,5
0x100e1c2f: jp short 100E1C37h
0x100e1c31: fstp dword ptr [esp+10h]
0x100e1c35: jmp short 100E1C3Dh
0x100e1c37: fstp st(0)
0x100e1c39: fst dword ptr [esp+10h]
0x100e1c3d: fld dword ptr [esp+10h]
0x100e1c41: add ecx,4
0x100e1c44: fstp dword ptr [esp+ecx*4+64h]
0x100e1c48: add edx,8
0x100e1c4b: cmp ecx,ebx
0x100e1c4d: jl near ptr 100E1B72h
0x100e1c53: fstp st(0)
0x100e1c55: cmp ecx,esi
0x100e1c57: jge short 100E1CBEh
0x100e1c59: imul edi,[esp+18h]
0x100e1c5e: fild dword ptr [esp+1Ch]
0x100e1c62: fstp dword ptr [esp+10h]
```

```
0x100e1c66: fld dword ptr [esp+10h]
0x100e1c6a: add edi,[esp+14h]
0x100e1c6e: mov eax,[esp+7Ch]
0x100e1c72: mov eax,[eax+66E0h]
0x100e1c78: lea edx,[ecx+edi*4]
0x100e1c7b: lea edx,[eax+edx*2]
0x100e1c7e: movzx eax,word ptr [edx]
0x100e1c81: mov [esp+24h],eax
0x100e1c85: fild dword ptr [esp+24h]
0x100e1c89: fstp dword ptr [esp+24h]
0x100e1c8d: fld dword ptr [esp+24h]
0x100e1c91: fst dword ptr [esp+ecx*4+58h]
0x100e1c95: fcom
0x100e1c97: fnstsw ax
0x100e1c99: test ah,5
0x100e1c9c: jp short 100E1CA4h
0x100e1c9e: fstp dword ptr [esp+10h]
0x100e1ca2: jmp short 100E1CAAh
0x100e1ca4: fstp st(0)
0x100e1ca6: fst dword ptr [esp+10h]
0x100e1caa: fld dword ptr [esp+10h]
0x100e1cae: add ecx,1
0x100e1cb1: fstp dword ptr [esp+ecx*4+64h]
0x100e1cb5: add edx,2
0x100e1cb8: cmp ecx,esi
0x100e1cba: jl short 100E1C7Eh
0x100e1cbc: fstp st(0)
0x100e1cbe: lea ecx,[esp+38h]
0x100e1cc2: mov [esp+20h],ebp
0x100e1cc6: mov [esp+24h],ecx
0x100e1cca: mov [esp+10h],ebp
0x100e1cce: xor ebp,ebp
0x100e1cd0: test esi,esi
0x100e1cd2: jle near ptr 100E1D90h
0x100e1cd8: mov edi,[esp+24h]
0x100e1cdc: xor ebx,ebx
0x100e1cde: fst dword ptr [edi]
0x100e1ce0: cmp esi,4
0x100e1ce3: jl short 100E1D4Fh
0x100e1ce5: mov ecx,[esp+10h]
0x100e1ce9: lea edx,[esi-4]
0x100e1cec: lea eax,[ebp+esi*4]
0x100e1cf0: shl eax,4
0x100e1cf3: shr edx,2
0x100e1cf6: add edx,1
0x100e1cf9: lea ecx,[esp+ecx+5Ch]
0x100e1cfd: lea eax,[eax+1012B528h]
0x100e1d03: lea ebx,[edx*4]
0x100e1d0a: fld dword ptr [eax-8]
0x100e1d0d: add eax,10h
0x100e1d10: fmul dword ptr [ecx-4]
0x100e1d13: add ecx,10h
0x100e1d16: sub edx,1
0x100e1d19: fadd dword ptr [edi]
0x100e1d1b: fstp dword ptr [esp+2Ch]
0x100e1d1f: fld dword ptr [esp+2Ch]
0x100e1d23: fld dword ptr [eax-14h]
0x100e1d26: fmul dword ptr [ecx-10h]
0x100e1d29: faddp
0x100e1d2b: fstp dword ptr [esp+2Ch]
0x100e1d2f: fld dword ptr [esp+2Ch]
0x100e1d33: fld dword ptr [ecx-0Ch]
0x100e1d36: fmul dword ptr [eax-10h]
0x100e1d39: faddp
0x100e1d3b: fstp dword ptr [esp+2Ch]
0x100e1d3f: fld dword ptr [esp+2Ch]
0x100e1d43: fld dword ptr [eax-0Ch]
```

```
0x100e1d46: fmul dword ptr [ecx-8]
0x100e1d49: faddp
0x100e1d4b: fstp dword ptr [edi]
0x100e1d4d: jne short 100E1D0Ah
0x100e1d4f: cmp ebx,esi
0x100e1d51: jge short 100E1D82h
0x100e1d53: mov edx,[esp+20h]
0x100e1d57: lea eax,[ebp+esi*4-0Ch]
0x100e1d5b: lea ecx,[ebx+eax*4]
0x100e1d5e: add edx,ebx
0x100e1d60: mov eax,esi
0x100e1d62: lea edx,[esp+edx*4+58h]
0x100e1d66: lea ecx,[ecx*4+1012B5E0h]
0x100e1d6d: sub eax,ebx
0x100e1d6f: fld dword ptr [ecx]
0x100e1d71: add ecx,4
0x100e1d74: fmul dword ptr [edx]
0x100e1d76: add edx,4
0x100e1d79: sub eax,1
0x100e1d7c: fadd dword ptr [edi]
0x100e1d7e: fstp dword ptr [edi]
0x100e1d80: jne short 100E1D6Fh
0x100e1d82: add ebp,1
0x100e1d85: add edi,4
0x100e1d88: cmp ebp,esi
0x100e1d8a: jl near ptr 100E1CDCh
0x100e1d90: mov edi,[esp+20h]
0x100e1d94: lea ecx,[esi-1]
0x100e1d97: fst dword ptr [esp+edi+30h]
0x100e1d9b: cmp ecx,4
0x100e1d9e: mov edx,1
0x100e1da3: jl short 100E1E08h
0x100e1da5: mov eax,[esp+10h]
0x100e1da9: lea ecx,[esi-5]
0x100e1dac: shr ecx,2
0x100e1daf: add ecx,edx
0x100e1db1: lea eax,[esp+eax+40h]
0x100e1db5: lea edx,[ecx*4+1]
0x100e1dbc: fld dword ptr [eax-4]
0x100e1dbf: add eax,10h
0x100e1dc2: sub ecx,1
0x100e1dc5: fld dword ptr [eax-10h]
0x100e1dc8: fld dword ptr [eax-0Ch]
0x100e1dcb: fld dword ptr [eax-8]
0x100e1dce: fld st(3)
0x100e1dd0: fmulp st(4),st
0x100e1dd2: fld dword ptr [esp+edi+30h]
0x100e1dd6: faddp st(4),st
0x100e1dd8: fxch st(3)
0x100e1dda: fstp dword ptr [esp+2Ch]
0x100e1dde: fld dword ptr [esp+2Ch]
0x100e1de2: fld st(2)
0x100e1de4: fmulp st(3),st
0x100e1de6: faddp st(2),st
0x100e1de8: fxch
0x100e1dea: fstp dword ptr [esp+2Ch]
0x100e1dee: fld st(0)
0x100e1df0: fmulp
0x100e1df2: fadd dword ptr [esp+2Ch]
0x100e1df6: fstp dword ptr [esp+2Ch]
0x100e1dfa: fld st(0)
0x100e1dfc: fmulp
0x100e1dfe: fadd dword ptr [esp+2Ch]
0x100e1e02: fstp dword ptr [esp+edi+30h]
0x100e1e06: jne short 100E1DBCh
0x100e1e08: cmp edx,esi
0x100e1e0a: jge short 100E1E2Bh
```

```
0x100e1e0c: lea eax,[edi+edx]
0x100e1e0f: mov ecx,esi
0x100e1e11: lea eax,[esp+eax*4+38h]
0x100e1e15: sub ecx,edx
0x100e1e17: fld dword ptr [eax]
0x100e1e19: add eax,4
0x100e1e1c: sub ecx,1
0x100e1e1f: fmul st(0),st
0x100e1e21: fadd dword ptr [esp+edi+30h]
0x100e1e25: fstp dword ptr [esp+edi+30h]
0x100e1e29: jne short 100E1E17h
0x100e1e2b: mov eax,[esp+10h]
0x100e1e2f: add dword ptr [esp+24h],10h
0x100e1e34: add eax,10h
0x100e1e37: add edi,4
0x100e1e3a: cmp eax,20h
0x100e1e3d: mov [esp+10h],eax
0x100e1e41: mov [esp+20h],edi
0x100e1e45: jl near ptr 100E1CCEh
0x100e1e4b: fstp st(0)
0x100e1e4d: fld dword ptr [esp+34h]
0x100e1e51: fdiv dword ptr [esp+30h]
0x100e1e55: call 1002299Ah
0x100e1e5a: fstp dword ptr [esp+2Ch]
0x100e1e5e: fld dword ptr [esp+2Ch]
0x100e1e62: lea edx,[esi-1]
0x100e1e65: cmp edx,4
0x100e1e68: mov eax,1
0x100e1e6d: jl short 100E1EA1h
0x100e1e6f: lea ecx,[esi-3]
0x100e1e72: fld st(0)
0x100e1e74: add eax,4
0x100e1e77: cmp eax,ecx
0x100e1e79: fmul dword ptr [esp+eax*4+28h]
0x100e1e7d: fstp dword ptr [esp+eax*4+28h]
0x100e1e81: fld dword ptr [esp+eax*4+2Ch]
0x100e1e85: fmul st,st(1)
0x100e1e87: fstp dword ptr [esp+eax*4+2Ch]
0x100e1e8b: fld dword ptr [esp+eax*4+30h]
0x100e1e8f: fmul st,st(1)
0x100e1e91: fstp dword ptr [esp+eax*4+30h]
0x100e1e95: fld st(0)
0x100e1e97: fmul dword ptr [esp+eax*4+34h]
0x100e1e9b: fstp dword ptr [esp+eax*4+34h]
0x100e1e9f: jl short 100E1E72h
0x100e1ea1: cmp eax,esi
0x100e1ea3: jge short 100E1EB6h
0x100e1ea5: fld dword ptr [esp+eax*4+38h]
0x100e1ea9: add eax,1
0x100e1eac: cmp eax,esi
0x100e1eae: fmul st,st(1)
0x100e1eb0: fstp dword ptr [esp+eax*4+34h]
0x100e1eb4: jl short 100E1EA5h
0x100e1eb6: xor ebx,ebx
0x100e1eb8: fstp st(0)
0x100e1eba: test esi,esi
0x100e1ebc: fldz
0x100e1ebe: jle near ptr 100E1F5Ch
0x100e1ec4: lea edx,[esp+58h]
0x100e1ec8: xor eax,eax
0x100e1eca: fst dword ptr [edx]
0x100e1ecc: cmp esi,4
0x100e1ecf: jl short 100E1F28h
0x100e1ed1: lea ecx,[ebx+esi*4]
0x100e1ed4: shl ecx,4
0x100e1ed7: lea edi,[esi-3]
0x100e1eda: lea ecx,[ecx+1012B5A8h]
```

```
0x100e1ee0: fld dword ptr [esp+eax*4+38h]
0x100e1ee4: add eax,4
0x100e1ee7: fmul dword ptr [ecx-8]
0x100e1eea: add ecx,10h
0x100e1eed: cmp eax,edi
0x100e1eef: fadd dword ptr [edx]
0x100e1ef1: fstp dword ptr [esp+2Ch]
0x100e1ef5: fld dword ptr [esp+2Ch]
0x100e1ef9: fld dword ptr [esp+eax*4+2Ch]
0x100e1efd: fmul dword ptr [ecx-14h]
0x100e1f00: faddp
0x100e1f02: fstp dword ptr [esp+2Ch]
0x100e1f06: fld dword ptr [esp+2Ch]
0x100e1f0a: fld dword ptr [esp+eax*4+30h]
0x100e1f0e: fmul dword ptr [ecx-10h]
0x100e1f11: faddp
0x100e1f13: fstp dword ptr [esp+2Ch]
0x100e1f17: fld dword ptr [esp+2Ch]
0x100e1f1b: fld dword ptr [esp+eax*4+34h]
0x100e1f1f: fmul dword ptr [ecx-0Ch]
0x100e1f22: faddp
0x100e1f24: fstp dword ptr [edx]
0x100e1f26: jl short 100E1EE0h
0x100e1f28: cmp eax,esi
0x100e1f2a: jge short 100E1F4Eh
0x100e1f2c: lea ecx,[ebx+esi*4-0Ch]
0x100e1f30: lea ecx,[eax+ecx*4]
0x100e1f33: lea ecx,[ecx*4+1012B660h]
0x100e1f3a: fld dword ptr [esp+eax*4+38h]
0x100e1f3e: add eax,1
0x100e1f41: fmul dword ptr [ecx]
0x100e1f43: add ecx,4
0x100e1f46: cmp eax,esi
0x100e1f48: fadd dword ptr [edx]
0x100e1f4a: fstp dword ptr [edx]
0x100e1f4c: jl short 100E1F3Ah
0x100e1f4e: add ebx,1
0x100e1f51: add edx,4
0x100e1f54: cmp ebx,esi
0x100e1f56: jl near ptr 100E1EC8h
0x100e1f5c: xor ecx,ecx
0x100e1f5e: test esi,esi
0x100e1f60: jle short 100E1FC4h
0x100e1f62: mov edi,[esp+18h]
0x100e1f66: mov edx,[esp+7Ch]
0x100e1f6a: mov ebx,[esp+14h]
0x100e1f6e: fld dword ptr [esp+ecx*4+58h]
0x100e1f72: movzx esi,word ptr [edx+66B6h]
0x100e1f79: fidiv dword ptr [esp+28h]
0x100e1f7d: mov ebp,[edx+66E0h]
0x100e1f83: imul esi,edi
0x100e1f86: add esi,ebx
0x100e1f88: lea esi,[ecx+esi*4]
0x100e1f8b: add ecx,1
0x100e1f8e: fnstcw [esp+10h]
0x100e1f92: movzx eax,word ptr [esp+10h]
0x100e1f97: or eax,0C00h
0x100e1f9c: mov [esp+2Ch],eax
0x100e1fa0: fldcw [esp+2Ch]
0x100e1fa4: fistp dword ptr [esp+2Ch]
0x100e1fa8: mov ax,[esp+2Ch]
0x100e1fad: mov [ebp+esi*2],ax
0x100e1fb2: mov esi,[edx+66D0h]
0x100e1fb8: cmp ecx,esi
0x100e1fba: fldcw [esp+10h]
0x100e1fbe: mov [esp+28h],esi
0x100e1fc2: jl short 100E1F6Eh
```

```
0x100e1fc4: mov ecx,[esp+7Ch]
0x100e1fc8: mov eax,[esp+14h]
0x100e1fcc: movzx edi,word ptr [ecx+66B6h]
0x100e1fd3: add eax,1
0x100e1fd6: cmp eax,edi
0x100e1fd8: mov [esp+14h],eax
0x100e1fdc: jl near ptr 100E1B02h
0x100e1fe2: mov edx,[esp+7Ch]
0x100e1fe6: mov eax,[esp+18h]
0x100e1fea: movzx ecx,word ptr [edx+66B4h]
0x100e1ff1: add eax,1
0x100e1ff4: cmp eax,ecx
0x100e1ff6: mov [esp+18h],eax
0x100e1ffa: jl near ptr 100E1AE9h
0x100e2000: fstp st(0)
0x100e2002: pop edi
0x100e2003: pop esi
0x100e2004: pop ebp
0x100e2005: pop ebx
0x100e2006: add esp,68h
0x100e2009: ret
```

`sub_100e1a50` is called by sub_10016040. It accesses memory-mapped peripheral registers at 0x7fffffff. It delegates to sub_100228d0, j__CIsqrt (2 callees).

12.1283 sub_100e2010 at 0x100e2010 (733 instructions)

```
0x100e2010: sub esp,5Ch
0x100e2013: push ebx
0x100e2014: push ebp
0x100e2015: push esi
0x100e2016: mov esi,[esp+6Ch]
0x100e201a: cmp dword ptr [esi+0BCh],0
0x100e2021: push edi
0x100e2022: je short 100E203Ch
0x100e2024: push 1012E4C8h
0x100e2029: call dword ptr ds:[100FE4C4h]
0x100e202f: add eax,40h
0x100e2032: push eax
0x100e2033: call dword ptr ds:[100FE4C8h]
0x100e2039: add esp,8
0x100e203c: fld qword ptr ds:[100FF110h]
0x100e2042: mov eax,4
0x100e2047: sub eax,[esi+9Ch]
0x100e204d: mov [esp+44h],eax
0x100e2051: fild dword ptr [esp+44h]
0x100e2055: call 100229A0h
0x100e205a: fstp dword ptr [esp+58h]
0x100e205e: mov ebp,[esi+66D0h]
0x100e2064: xor edi,edi
0x100e2066: test ebp,ebp
0x100e2068: jle short 100E208Fh
0x100e206a: fld qword ptr ds:[1012E430h]
0x100e2070: lea ebx,[esi+0E784h]
0x100e2076: fld dword ptr [ebx]
0x100e2078: fmul st,st(1)
0x100e207a: call 100228D0h
0x100e207f: mov [esp+edi*4+5Ch],eax
0x100e2083: add edi,1
0x100e2086: add ebx,4
0x100e2089: cmp edi,ebp
0x100e208b: jl short 100E2076h
0x100e208d: fstp st(0)
0x100e208f: xor ebx,ebx
0x100e2091: lea edx,[ebp-1]
0x100e2094: cmp edx,4
```

```
0x100e2097: mov [esp+2Ch],ebx
0x100e209b: lea ecx,[ebx+1]
0x100e209e: jl near ptr 100E2163h
0x100e20a4: mov eax,0FFFF187Ch
0x100e20a9: sub eax,esi
0x100e20ab: mov [esp+40h],eax
0x100e20af: mov eax,0FFFF1880h
0x100e20b4: sub eax,esi
0x100e20b6: mov [esp+3Ch],eax
0x100e20ba: mov eax,0FFFF1884h
0x100e20bf: sub eax,esi
0x100e20c1: mov [esp+38h],eax
0x100e20c5: mov eax,0FFFF1878h
0x100e20ca: xor edi,edi
0x100e20cc: sub eax,esi
0x100e20ce: lea edx,[esi+0E78Ch]
0x100e20d4: mov [esp+44h],eax
0x100e20d8: jmp short 100E20E0h
0x100e20e0: fld dword ptr [edi+esi+0E784h]
0x100e20e7: fld dword ptr [edx-4]
0x100e20ea: fcompp
0x100e20ec: fnstsw ax
0x100e20ee: test ah,41h
0x100e20f1: jne short 100E20FCh
0x100e20f3: mov eax,[esp+44h]
0x100e20f7: mov ebx,ecx
0x100e20f9: lea edi,[eax+edx]
0x100e20fc: fld dword ptr [edi+esi+0E784h]
0x100e2103: fld dword ptr [edx]
0x100e2105: fcompp
0x100e2107: fnstsw ax
0x100e2109: test ah,41h
0x100e210c: jne short 100E2118h
0x100e210e: mov eax,[esp+40h]
0x100e2112: lea ebx,[ecx+1]
0x100e2115: lea edi,[eax+edx]
0x100e2118: fld dword ptr [edi+esi+0E784h]
0x100e211f: fld dword ptr [edx+4]
0x100e2122: fcompp
0x100e2124: fnstsw ax
0x100e2126: test ah,41h
0x100e2129: jne short 100E2135h
0x100e212b: mov eax,[esp+3Ch]
0x100e212f: lea ebx,[ecx+2]
0x100e2132: lea edi,[eax+edx]
0x100e2135: fld dword ptr [edi+esi+0E784h]
0x100e213c: fld dword ptr [edx+8]
0x100e213f: fcompp
0x100e2141: fnstsw ax
0x100e2143: test ah,41h
0x100e2146: jne short 100E2152h
0x100e2148: mov eax,[esp+38h]
0x100e214c: lea ebx,[ecx+3]
0x100e214f: lea edi,[eax+edx]
0x100e2152: add ecx,4
0x100e2155: lea eax,[ebp-3]
0x100e2158: add edx,10h
0x100e215b: cmp ecx,eax
0x100e215d: jl short 100E20E0h
0x100e215f: mov [esp+2Ch],ebx
0x100e2163: cmp ecx,ebp
0x100e2165: jge short 100E219Ch
0x100e2167: lea edi,[ebx*4]
0x100e216e: lea edx,[esi+ecx*4+0E784h]
0x100e2175: fld dword ptr [edi+esi+0E784h]
0x100e217c: fld dword ptr [edx]
0x100e217e: fcompp
```

```
0x100e2180: fnstsw ax
0x100e2182: test ah,41h
0x100e2185: jne short 100E2192h
0x100e2187: mov [esp+2Ch],ecx
0x100e218b: lea edi,[ecx*4]
0x100e2192: add ecx,1
0x100e2195: add edx,4
0x100e2198: cmp ecx,ebp
0x100e219a: jl short 100E2175h
0x100e219c: mov cl,[esi+66BCh]
0x100e21a2: movzx eax,word ptr [esi+66B4h]
0x100e21a9: cdq
0x100e21aa: mov edi,4
0x100e21af: sar edi,cl
0x100e21b1: push 4
0x100e21b3: idiv edi
0x100e21b5: mov ecx,eax
0x100e21b7: movzx eax,word ptr [esi+66B6h]
0x100e21be: cdq
0x100e21bf: idiv edi
0x100e21c1: mov [esp+34h],ecx
0x100e21c5: mov [esp+38h],eax
0x100e21c9: imul eax,ecx
0x100e21cc: push eax
0x100e21cd: mov [esp+40h],eax
0x100e21d1: call dword ptr ds:[100FE4FCh]
0x100e21d7: push 1012E4B0h
0x100e21dc: push eax
0x100e21dd: push esi
0x100e21de: mov [esp+3Ch],eax
0x100e21e2: call 100D2E20h
0x100e21e7: add esp,14h
0x100e21ea: cmp dword ptr [esi+66D0h],0
0x100e21f1: mov dword ptr [esp+24h],0
0x100e21f9: jle near ptr 100E2A07h
0x100e21ff: nop
0x100e2200: mov ecx,[esp+2Ch]
0x100e2204: cmp [esp+24h],ecx
0x100e2208: je near ptr 100E29ECh
0x100e220e: mov edx,[esp+38h]
0x100e2212: mov ebp,[esp+28h]
0x100e2216: lea eax,[edx*4]
0x100e221d: push eax
0x100e221e: xor esi,esi
0x100e2220: push esi
0x100e2221: push ebp
0x100e2222: call 100228C0h
0x100e2227: fldz
0x100e2229: add esp,0Ch
0x100e222c: cmp [esp+30h],esi
0x100e2230: mov [esp+14h],esi
0x100e2234: jbe near ptr 100E2522h
0x100e223a: mov [esp+40h],esi
0x100e223e: cmp dword ptr [esp+34h],0
0x100e2243: mov dword ptr [esp+20h],0
0x100e224b: jbe near ptr 100E24FDh
0x100e2251: mov ecx,[esp+70h]
0x100e2255: fst dword ptr [esp+1Ch]
0x100e2259: mov cl,[ecx+66BCh]
0x100e225f: fst dword ptr [esp+10h]
0x100e2263: mov eax,4
0x100e2268: sar eax,cl
0x100e226a: mov ecx,[esp+14h]
0x100e226e: mov dword ptr [esp+18h],0
0x100e2276: mov edi,eax
0x100e2278: imul edi,ecx
0x100e227b: add ecx,1
```

```
0x100e227e: imul ecx,eax
0x100e2281: cmp edi,ecx
0x100e2283: mov [esp+44h],edi
0x100e2287: mov [esp+54h],ecx
0x100e228b: jae near ptr 100E24ABh
0x100e2291: mov ecx,[esp+20h]
0x100e2295: mov edx,eax
0x100e2297: imul edx,ecx
0x100e229a: add ecx,1
0x100e229d: imul ecx,eax
0x100e22a0: mov [esp+3Ch],ecx
0x100e22a4: sub ecx,edx
0x100e22a6: mov [esp+48h],edx
0x100e22aa: mov [esp+4Ch],ecx
0x100e22ae: cmp dword ptr [esp+4Ch],4
0x100e22b3: mov ebx,[esp+48h]
0x100e22b7: jl near ptr 100E241Fh
0x100e22bd: mov edx,[esp+70h]
0x100e22c1: mov eax,[edx+66E0h]
0x100e22c7: movzx edx,word ptr [edx+66B6h]
0x100e22ce: mov ecx,[esp+24h]
0x100e22d2: imul edx,edi
0x100e22d5: mov ebp,[esp+ecx*4+5Ch]
0x100e22d9: lea esi,[ecx+ecx]
0x100e22dc: lea ecx,[edx+ebx]
0x100e22df: lea esi,[esi+ecx*8+10h]
0x100e22e3: lea edi,[eax+ecx*8+8]
0x100e22e7: mov ecx,[esp+3Ch]
0x100e22eb: sub ecx,ebx
0x100e22ed: sub ecx,4
0x100e22f0: shr ecx,2
0x100e22f3: add esi,eax
0x100e22f5: add ecx,1
0x100e22f8: lea ebx,[ebx+ecx*4]
0x100e22fb: mov eax,[esp+24h]
0x100e22ff: add eax,eax
0x100e2301: movzx eax,word ptr [eax+edi-8]
0x100e2306: mov [esp+50h],eax
0x100e230a: cdq
0x100e230b: idiv ebp
0x100e230d: cmp eax,1
0x100e2310: jne short 100E2345h
0x100e2312: mov eax,[esp+2Ch]
0x100e2316: movzx eax,word ptr [edi+eax*2-8]
0x100e231b: cmp ax,5DC0h
0x100e231f: jbe short 100E2345h
0x100e2321: fild dword ptr [esp+50h]
0x100e2325: add dword ptr [esp+18h],1
0x100e232a: movzx edx,ax
0x100e232d: mov [esp+50h],edx
0x100e2331: fadd dword ptr [esp+10h]
0x100e2335: fstp dword ptr [esp+10h]
0x100e2339: fild dword ptr [esp+50h]
0x100e233d: fadd dword ptr [esp+1Ch]
0x100e2341: fstp dword ptr [esp+1Ch]
0x100e2345: movzx eax,word ptr [esi-8]
0x100e2349: mov [esp+50h],eax
0x100e234d: cdq
0x100e234e: idiv ebp
0x100e2350: cmp eax,1
0x100e2353: jne short 100E2387h
0x100e2355: mov eax,[esp+2Ch]
0x100e2359: movzx eax,word ptr [edi+eax*2]
0x100e235d: cmp ax,5DC0h
0x100e2361: jbe short 100E2387h
0x100e2363: fild dword ptr [esp+50h]
0x100e2367: add dword ptr [esp+18h],1
```

```
0x100e236c: movzx edx,ax
0x100e236f: mov [esp+50h],edx
0x100e2373: fadd dword ptr [esp+10h]
0x100e2377: fstp dword ptr [esp+10h]
0x100e237b: fild dword ptr [esp+50h]
0x100e237f: fadd dword ptr [esp+1Ch]
0x100e2383: fstp dword ptr [esp+1Ch]
0x100e2387: movzx eax,word ptr [esi]
0x100e238a: mov [esp+50h],eax
0x100e238e: cdq
0x100e238f: idiv ebp
0x100e2391: cmp eax,1
0x100e2394: jne short 100E23C9h
0x100e2396: mov eax,[esp+2Ch]
0x100e239a: movzx eax,word ptr [edi+eax*2+8]
0x100e239f: cmp ax,5DC0h
0x100e23a3: jbe short 100E23C9h
0x100e23a5: fild dword ptr [esp+50h]
0x100e23a9: add dword ptr [esp+18h],1
0x100e23ae: movzx edx,ax
0x100e23b1: mov [esp+50h],edx
0x100e23b5: fadd dword ptr [esp+10h]
0x100e23b9: fstp dword ptr [esp+10h]
0x100e23bd: fild dword ptr [esp+50h]
0x100e23c1: fadd dword ptr [esp+1Ch]
0x100e23c5: fstp dword ptr [esp+1Ch]
0x100e23c9: movzx eax,word ptr [esi+8]
0x100e23cd: mov [esp+50h],eax
0x100e23d1: cdq
0x100e23d2: idiv ebp
0x100e23d4: cmp eax,1
0x100e23d7: jne short 100E240Ch
0x100e23d9: mov eax,[esp+2Ch]
0x100e23dd: movzx eax,word ptr [edi+eax*2+10h]
0x100e23e2: cmp ax,5DC0h
0x100e23e6: jbe short 100E240Ch
0x100e23e8: fild dword ptr [esp+50h]
0x100e23ec: add dword ptr [esp+18h],1
0x100e23f1: movzx edx,ax
0x100e23f4: mov [esp+50h],edx
0x100e23f8: fadd dword ptr [esp+10h]
0x100e23fc: fstp dword ptr [esp+10h]
0x100e2400: fild dword ptr [esp+50h]
0x100e2404: fadd dword ptr [esp+1Ch]
0x100e2408: fstp dword ptr [esp+1Ch]
0x100e240c: add edi,20h
0x100e240f: add esi,20h
0x100e2412: sub ecx,1
0x100e2415: jne near ptr 100E22FBh
0x100e241b: mov edi,[esp+44h]
0x100e241f: cmp ebx,[esp+3Ch]
0x100e2423: jae short 100E249Ah
0x100e2425: mov eax,[esp+24h]
0x100e2429: mov ebp,[esp+eax*4+5Ch]
0x100e242d: mov eax,[esp+70h]
0x100e2431: movzx ecx,word ptr [eax+66B6h]
0x100e2438: mov esi,[esp+3Ch]
0x100e243c: imul ecx,edi
0x100e243f: mov edx,[eax+66E0h]
0x100e2445: add ecx,ebx
0x100e2447: sub esi,ebx
0x100e2449: mov ebx,[esp+24h]
0x100e244d: lea ecx,[edx+ecx*8]
0x100e2450: movzx eax,word ptr [ecx+ebx*2]
0x100e2454: mov [esp+50h],eax
0x100e2458: cdq
0x100e2459: idiv ebp
```

```
0x100e245b: cmp eax,1
0x100e245e: jne short 100E2492h
0x100e2460: mov eax,[esp+2Ch]
0x100e2464: movzx eax,word ptr [ecx+eax*2]
0x100e2468: cmp ax,5DC0h
0x100e246c: jbe short 100E2492h
0x100e246e: fild dword ptr [esp+50h]
0x100e2472: add dword ptr [esp+18h],1
0x100e2477: movzx edx,ax
0x100e247a: mov [esp+50h],edx
0x100e247e: fadd dword ptr [esp+10h]
0x100e2482: fstp dword ptr [esp+10h]
0x100e2486: fild dword ptr [esp+50h]
0x100e248a: fadd dword ptr [esp+1Ch]
0x100e248e: fstp dword ptr [esp+1Ch]
0x100e2492: add ecx,8
0x100e2495: sub esi,1
0x100e2498: jne short 100E2450h
0x100e249a: add edi,1
0x100e249d: cmp edi,[esp+54h]
0x100e24a1: mov [esp+44h],edi
0x100e24a5: jb near ptr 100E22AEh
0x100e24ab: mov ecx,[esp+70h]
0x100e24af: mov cl,[ecx+66BCh]
0x100e24b5: mov eax,4
0x100e24ba: sar eax,cl
0x100e24bc: mov edx,eax
0x100e24be: imul edx,eax
0x100e24c1: cmp [esp+18h],edx
0x100e24c5: jne short 100E24E8h
0x100e24c7: fld dword ptr [esp+10h]
0x100e24cb: mov eax,[esp+40h]
0x100e24cf: fdiv dword ptr [esp+1Ch]
0x100e24d3: mov ecx,[esp+20h]
0x100e24d7: mov edx,[esp+28h]
0x100e24db: add eax,ecx
0x100e24dd: fstp dword ptr [esp+54h]
0x100e24e1: fld dword ptr [esp+54h]
0x100e24e5: fstp dword ptr [edx+eax*4]
0x100e24e8: mov eax,[esp+20h]
0x100e24ec: add eax,1
0x100e24ef: cmp eax,[esp+34h]
0x100e24f3: mov [esp+20h],eax
0x100e24f7: jb near ptr 100E2251h
0x100e24fd: mov eax,[esp+14h]
0x100e2501: mov ecx,[esp+34h]
0x100e2505: add [esp+40h],ecx
0x100e2509: add eax,1
0x100e250c: cmp eax,[esp+30h]
0x100e2510: mov [esp+14h],eax
0x100e2514: jb near ptr 100E223Eh
0x100e251a: mov ebp,[esp+28h]
0x100e251e: mov ebx,ecx
0x100e2520: jmp short 100E2526h
0x100e2522: mov ebx,[esp+34h]
0x100e2526: fld dword ptr [esp+58h]
0x100e252a: fld st(0)
0x100e252c: fdivr qword ptr ds:[1012E4A8h]
0x100e2532: call 100228D0h
0x100e2537: test eax,eax
0x100e2539: mov [esp+3Ch],eax
0x100e253d: je near ptr 100E283Dh
0x100e2543: sub dword ptr [esp+3Ch],1
0x100e2548: cmp dword ptr [esp+30h],0
0x100e254d: mov dword ptr [esp+14h],0
0x100e2555: jbe near ptr 100E2778h
0x100e255b: mov [esp+40h],ebp
```

```
0x100e255f: xor edi,edi
0x100e2561: test ebx,ebx
0x100e2563: jbe near ptr 100E2758h
0x100e2569: mov edx,[esp+40h]
0x100e256d: mov [esp+20h],edx
0x100e2571: mov eax,[esp+20h]
0x100e2575: fld st(1)
0x100e2577: fcomp dword ptr [eax]
0x100e2579: fnstsw ax
0x100e257b: test ah,44h
0x100e257e: jp near ptr 100E2748h
0x100e2584: fxch
0x100e2586: xor esi,esi
0x100e2588: fst dword ptr [esp+10h]
0x100e258c: xor ecx,ecx
0x100e258e: movsx eax,byte ptr [ecx*2+1012B6E0h]
0x100e2596: movsx edx,byte ptr [ecx*2+1012B6E1h]
0x100e259e: add eax,[esp+14h]
0x100e25a2: mov ebp,[esp+30h]
0x100e25a6: add edx,edi
0x100e25a8: cmp eax,ebp
0x100e25aa: jae short 100E25F1h
0x100e25ac: cmp edx,ebx
0x100e25ae: jae short 100E25F1h
0x100e25b0: imul eax,ebx
0x100e25b3: add eax,edx
0x100e25b5: mov edx,[esp+28h]
0x100e25b9: lea edx,[edx+eax*4]
0x100e25bc: fcom dword ptr [edx]
0x100e25be: fnstsw ax
0x100e25c0: test ah,5
0x100e25c3: jp short 100E25F1h
0x100e25c5: mov eax,ecx
0x100e25c7: and eax,1
0x100e25ca: lea ebp,[eax+1]
0x100e25cd: test ebp,ebp
0x100e25cf: mov [esp+54h],ebp
0x100e25d3: fild dword ptr [esp+54h]
0x100e25d7: jge short 100E25DFh
0x100e25d9: fadd dword ptr ds:[100FE7C0h]
0x100e25df: fmul dword ptr [edx]
0x100e25e1: mov ebp,[esp+30h]
0x100e25e5: lea esi,[esi+eax+1]
0x100e25e9: fadd dword ptr [esp+10h]
0x100e25ed: fstp dword ptr [esp+10h]
0x100e25f1: movsx eax,byte ptr [ecx*2+1012B6E2h]
0x100e25f9: movsx edx,byte ptr [ecx*2+1012B6E3h]
0x100e2601: add eax,[esp+14h]
0x100e2605: add edx,edi
0x100e2607: cmp eax,ebp
0x100e2609: jae short 100E2653h
0x100e260b: cmp edx,ebx
0x100e260d: jae short 100E2653h
0x100e260f: imul eax,ebx
0x100e2612: add eax,edx
0x100e2614: mov edx,[esp+28h]
0x100e2618: lea edx,[edx+eax*4]
0x100e261b: fcom dword ptr [edx]
0x100e261d: fnstsw ax
0x100e261f: test ah,5
0x100e2622: jp short 100E2653h
0x100e2624: lea eax,[ecx-1]
0x100e2627: and eax,1
0x100e262a: add eax,1
0x100e262d: test eax,eax
0x100e262f: mov [esp+54h],eax
0x100e2633: fild dword ptr [esp+54h]
```

```
0x100e2637: jge short 100E263Fh
0x100e2639: fadd dword ptr ds:[100FE7C0h]
0x100e263f: fmul dword ptr [edx]
0x100e2641: lea edx,[ecx-1]
0x100e2644: and edx,1
0x100e2647: lea esi,[esi+edx+1]
0x100e264b: fadd dword ptr [esp+10h]
0x100e264f: fstp dword ptr [esp+10h]
0x100e2653: movsx eax,byte ptr [ecx*2+1012B6E4h]
0x100e265b: movsx edx,byte ptr [ecx*2+1012B6E5h]
0x100e2663: add eax,[esp+14h]
0x100e2667: add edx,edi
0x100e2669: cmp eax,ebp
0x100e266b: jae short 100E26B2h
0x100e266d: cmp edx,ebx
0x100e266f: jae short 100E26B2h
0x100e2671: imul eax,ebx
0x100e2674: add eax,edx
0x100e2676: mov edx,[esp+28h]
0x100e267a: lea edx,[edx+eax*4]
0x100e267d: fcom dword ptr [edx]
0x100e267f: fnstsw ax
0x100e2681: test ah,5
0x100e2684: jp short 100E26B2h
0x100e2686: mov eax,ecx
0x100e2688: and eax,1
0x100e268b: lea ebp,[eax+1]
0x100e268e: test ebp,ebp
0x100e2690: mov [esp+54h],ebp
0x100e2694: fild dword ptr [esp+54h]
0x100e2698: jge short 100E26A0h
0x100e269a: fadd dword ptr ds:[100FE7C0h]
0x100e26a0: fmul dword ptr [edx]
0x100e26a2: mov ebp,[esp+30h]
0x100e26a6: lea esi,[esi+eax+1]
0x100e26aa: fadd dword ptr [esp+10h]
0x100e26ae: fstp dword ptr [esp+10h]
0x100e26b2: movsx eax,byte ptr [ecx*2+1012B6E6h]
0x100e26ba: movsx edx,byte ptr [ecx*2+1012B6E7h]
0x100e26c2: add eax,[esp+14h]
0x100e26c6: add edx,edi
0x100e26c8: cmp eax,ebp
0x100e26ca: jae short 100E2717h
0x100e26cc: cmp edx,ebx
0x100e26ce: jae short 100E2717h
0x100e26d0: mov ebp,[esp+28h]
0x100e26d4: imul eax,ebx
0x100e26d7: add eax,edx
0x100e26d9: lea edx,[ebp+eax*4]
0x100e26dd: fcom dword ptr [edx]
0x100e26df: fnstsw ax
0x100e26e1: test ah,5
0x100e26e4: jp short 100E271Bh
0x100e26e6: lea eax,[ecx-1]
0x100e26e9: and eax,1
0x100e26ec: add eax,1
0x100e26ef: test eax,eax
0x100e26f1: mov [esp+54h],eax
0x100e26f5: fild dword ptr [esp+54h]
0x100e26f9: jge short 100E2701h
0x100e26fb: fadd dword ptr ds:[100FE7C0h]
0x100e2701: fmul dword ptr [edx]
0x100e2703: lea edx,[ecx-1]
0x100e2706: and edx,1
0x100e2709: lea esi,[esi+edx+1]
0x100e270d: fadd dword ptr [esp+10h]
0x100e2711: fstp dword ptr [esp+10h]
```

```
0x100e2715: jmp short 100E271Bh
0x100e2717: mov ebp,[esp+28h]
0x100e271b: add ecx,4
0x100e271e: cmp ecx,8
0x100e2721: jb near ptr 100E258Eh
0x100e2727: cmp esi,3
0x100e272a: mov [esp+18h],esi
0x100e272e: jle short 100E2746h
0x100e2730: fld dword ptr [esp+10h]
0x100e2734: mov eax,[esp+20h]
0x100e2738: fadd st,st(2)
0x100e273a: fild dword ptr [esp+18h]
0x100e273e: fadd st,st(3)
0x100e2740: fdivp
0x100e2742: fchs
0x100e2744: fstp dword ptr [eax]
0x100e2746: fxch
0x100e2748: add dword ptr [esp+20h],4
0x100e274d: add edi,1
0x100e2750: cmp edi,ebx
0x100e2752: jb near ptr 100E2571h
0x100e2758: mov ecx,[esp+14h]
0x100e275c: add ecx,1
0x100e275f: lea eax,[ebx*4]
0x100e2766: add [esp+40h],eax
0x100e276a: cmp ecx,[esp+30h]
0x100e276e: mov [esp+14h],ecx
0x100e2772: jb near ptr 100E255Fh
0x100e2778: xor edx,edx
0x100e277a: xor esi,esi
0x100e277c: cmp dword ptr [esp+38h],4
0x100e2781: jl short 100E27FFh
0x100e2783: mov edi,[esp+38h]
0x100e2787: add edi,0FFFFFFFCh
0x100e278a: shr edi,2
0x100e278d: add edi,1
0x100e2790: lea ecx,[ebp+8]
0x100e2793: lea edx,[edi*4]
0x100e279a: fxch
0x100e279c: fcom dword ptr [ecx-8]
0x100e279f: fnstsw ax
0x100e27a1: test ah,41h
0x100e27a4: jne short 100E27B3h
0x100e27a6: fld dword ptr [ecx-8]
0x100e27a9: mov esi,1
0x100e27ae: fchs
0x100e27b0: fstp dword ptr [ecx-8]
0x100e27b3: fcom dword ptr [ecx-4]
0x100e27b6: fnstsw ax
0x100e27b8: test ah,41h
0x100e27bb: jne short 100E27CAh
0x100e27bd: fld dword ptr [ecx-4]
0x100e27c0: mov esi,1
0x100e27c5: fchs
0x100e27c7: fstp dword ptr [ecx-4]
0x100e27ca: fcom dword ptr [ecx]
0x100e27cc: fnstsw ax
0x100e27ce: test ah,41h
0x100e27d1: jne short 100E27DEh
0x100e27d3: fld dword ptr [ecx]
0x100e27d5: mov esi,1
0x100e27da: fchs
0x100e27dc: fstp dword ptr [ecx]
0x100e27de: fcom dword ptr [ecx+4]
0x100e27e1: fnstsw ax
0x100e27e3: test ah,41h
0x100e27e6: jne short 100E27F5h
```

```
0x100e27e8: fld dword ptr [ecx+4]
0x100e27eb: mov esi,1
0x100e27f0: fchs
0x100e27f2: fstp dword ptr [ecx+4]
0x100e27f5: add ecx,10h
0x100e27f8: fxch
0x100e27fa: sub edi,1
0x100e27fd: jne short 100E279Ah
0x100e27ff: cmp edx,[esp+38h]
0x100e2803: jge short 100E282Eh
0x100e2805: mov ecx,[esp+38h]
0x100e2809: fxch
0x100e280b: fcom dword ptr [ebp+edx*4]
0x100e280f: fnstsw ax
0x100e2811: test ah,41h
0x100e2814: jne short 100E2825h
0x100e2816: fld dword ptr [ebp+edx*4]
0x100e281a: mov esi,1
0x100e281f: fchs
0x100e2821: fstp dword ptr [ebp+edx*4]
0x100e2825: add edx,1
0x100e2828: fxch
0x100e282a: cmp edx,ecx
0x100e282c: jl short 100E2809h
0x100e282e: test esi,esi
0x100e2830: je short 100E283Dh
0x100e2832: cmp dword ptr [esp+3Ch],0
0x100e2837: jne near ptr 100E2543h
0x100e283d: mov edx,[esp+38h]
0x100e2841: fstp st(0)
0x100e2843: xor ecx,ecx
0x100e2845: test edx,edx
0x100e2847: jle short 100E2865h
0x100e2849: fld1
0x100e284b: fld st(1)
0x100e284d: fcomp dword ptr [ebp+ecx*4]
0x100e2851: fnstsw ax
0x100e2853: test ah,44h
0x100e2856: jp short 100E285Ch
0x100e2858: fst dword ptr [ebp+ecx*4]
0x100e285c: add ecx,1
0x100e285f: cmp ecx,edx
0x100e2861: jl short 100E284Bh
0x100e2863: fstp st(0)
0x100e2865: cmp dword ptr [esp+30h],0
0x100e286a: fstp st(0)
0x100e286c: mov dword ptr [esp+14h],0
0x100e2874: jbe near ptr 100E29ECh
0x100e287a: mov dword ptr [esp+40h],0
0x100e2882: cmp dword ptr [esp+34h],0
0x100e2887: mov dword ptr [esp+20h],0
0x100e288f: jbe near ptr 100E29CFh
0x100e2895: jmp short 100E28A0h
0x100e28a0: mov ecx,[esp+70h]
0x100e28a4: mov cl,[ecx+66BCh]
0x100e28aa: mov eax,4
0x100e28af: sar eax,cl
0x100e28b1: mov ecx,[esp+14h]
0x100e28b5: mov ebp,eax
0x100e28b7: imul ebp,ecx
0x100e28ba: add ecx,1
0x100e28bd: imul ecx,eax
0x100e28c0: cmp ebp,ecx
0x100e28c2: jae near ptr 100E29BAh
0x100e28c8: jmp short 100E28D0h
0x100e28d0: mov ecx,[esp+20h]
0x100e28d4: mov ebx,eax
```

```
0x100e28d6: imul ebx,ecx
0x100e28d9: add ecx,1
0x100e28dc: imul ecx,eax
0x100e28df: cmp ebx,ecx
0x100e28e1: jae near ptr 100E2994h
0x100e28e7: mov edx,[esp+24h]
0x100e28eb: mov eax,[esp+edx*4+5Ch]
0x100e28ef: mov [esp+54h],eax
0x100e28f3: mov eax,[esp+70h]
0x100e28f7: movzx ecx,word ptr [eax+66B6h]
0x100e28fe: mov edx,[eax+66E0h]
0x100e2904: imul ecx,ebp
0x100e2907: mov eax,[esp+24h]
0x100e290b: add ecx,ebx
0x100e290d: lea edi,[edx+ecx*8]
0x100e2910: movzx esi,word ptr [edi+eax*2]
0x100e2914: mov eax,esi
0x100e2916: cdq
0x100e2917: idiv dword ptr [esp+54h]
0x100e291b: cmp eax,1
0x100e291e: jle short 100E296Eh
0x100e2920: mov edx,[esp+20h]
0x100e2924: mov ecx,[esp+40h]
0x100e2928: mov eax,[esp+28h]
0x100e292c: add ecx,edx
0x100e292e: fld dword ptr [eax+ecx*4]
0x100e2931: mov ecx,[esp+2Ch]
0x100e2935: movzx edx,word ptr [edi+ecx*2]
0x100e2939: mov [esp+50h],edx
0x100e293d: fild dword ptr [esp+50h]
0x100e2941: fmulp
0x100e2943: call 100228D0h
0x100e2948: cmp esi,eax
0x100e294a: jge short 100E296Eh
0x100e294c: cmp eax,0FFFFh
0x100e2951: jge short 100E2961h
0x100e2953: xor ecx,ecx
0x100e2955: test eax,eax
0x100e2957: setl cl
0x100e295a: sub ecx,1
0x100e295d: and eax,ecx
0x100e295f: jmp short 100E2966h
0x100e2961: mov eax,0FFFFh
0x100e2966: mov edx,[esp+24h]
0x100e296a: mov [edi+edx*2],ax
0x100e296e: mov eax,[esp+70h]
0x100e2972: mov cl,[eax+66BCh]
0x100e2978: mov eax,[esp+20h]
0x100e297c: mov edx,4
0x100e2981: sar edx,cl
0x100e2983: add eax,1
0x100e2986: add ebx,1
0x100e2989: imul edx,eax
0x100e298c: cmp ebx,edx
0x100e298e: jb near ptr 100E28F3h
0x100e2994: mov ecx,[esp+70h]
0x100e2998: mov cl,[ecx+66BCh]
0x100e299e: mov eax,4
0x100e29a3: sar eax,cl
0x100e29a5: mov ecx,[esp+14h]
0x100e29a9: add ecx,1
0x100e29ac: add ebp,1
0x100e29af: imul ecx,eax
0x100e29b2: cmp ebp,ecx
0x100e29b4: jb near ptr 100E28D0h
0x100e29ba: mov eax,[esp+20h]
0x100e29be: add eax,1
```

```
0x100e29c1: cmp eax,[esp+34h]
0x100e29c5: mov [esp+20h],eax
0x100e29c9: jb near ptr 100E28A0h
0x100e29cf: mov eax,[esp+14h]
0x100e29d3: mov edx,[esp+34h]
0x100e29d7: add [esp+40h],edx
0x100e29db: add eax,1
0x100e29de: cmp eax,[esp+30h]
0x100e29e2: mov [esp+14h],eax
0x100e29e6: jb near ptr 100E2882h
0x100e29ec: mov eax,[esp+24h]
0x100e29f0: mov ecx,[esp+70h]
0x100e29f4: add eax,1
0x100e29f7: cmp eax,[ecx+66D0h]
0x100e29fd: mov [esp+24h],eax
0x100e2a01: jl near ptr 100E2200h
0x100e2a07: mov edx,[esp+28h]
0x100e2a0b: pop edi
0x100e2a0c: pop esi
0x100e2a0d: pop ebp
0x100e2a0e: pop ebx
0x100e2a0f: add esp,5Ch
0x100e2a12: mov [esp+4],edx
0x100e2a16: jmp dword ptr ds:[100FE554h]
```

`sub_100e2010` is called by sub_10016040. It delegates to j_memset, sub_100228d0, j__CIpow (4 callees).

12.1284 sub_100e2a20 at 0x100e2a20 (52 instructions)

```
0x100e2a20: push ebx
0x100e2a21: push esi
0x100e2a22: mov esi,[esp+0Ch]
0x100e2a26: push edi
0x100e2a27: push esi
0x100e2a28: call 100D2F50h
0x100e2a2d: mov ecx,[esp+1Ch]
0x100e2a31: movzx eax,ax
0x100e2a34: push esi
0x100e2a35: mov [ecx],eax
0x100e2a37: call 100D2F50h
0x100e2a3c: mov edi,[esp+24h]
0x100e2a40: movzx edx,ax
0x100e2a43: push esi
0x100e2a44: mov [edi],edx
0x100e2a46: call 100D2FE0h
0x100e2a4b: mov ebx,[esp+2Ch]
0x100e2a4f: mov [ebx],eax
0x100e2a51: mov eax,[esi]
0x100e2a53: mov ecx,[esi+4]
0x100e2a56: mov edx,[eax+18h]
0x100e2a59: push ecx
0x100e2a5a: call edx
0x100e2a5c: mov ecx,[esp+34h]
0x100e2a60: add eax,4
0x100e2a63: mov [ecx],eax
0x100e2a65: mov ecx,[edi]
0x100e2a67: add esp,10h
0x100e2a6a: cmp ecx,0Eh
0x100e2a6d: sbb eax,eax
0x100e2a6f: and eax,ecx
0x100e2a71: movsx edx,byte ptr [eax+1012E4E4h]
0x100e2a78: sub edx,30h
0x100e2a7b: imul edx,[ebx]
0x100e2a7e: cmp edx,4
0x100e2a81: jbe short 100E2AA1h
0x100e2a83: mov edi,[esi]
```

```
0x100e2a85: push 0
0x100e2a87: push esi
0x100e2a88: call 100D2FE0h
0x100e2a8d: add eax,[esp+1Ch]
0x100e2a91: mov ecx,[edi+8]
0x100e2a94: add esp,4
0x100e2a97: push eax
0x100e2a98: mov eax,[esi+4]
0x100e2a9b: push eax
0x100e2a9c: call ecx
0x100e2a9e: add esp,0Ch
0x100e2aa1: pop edi
0x100e2aa2: pop esi
0x100e2aa3: pop ebx
0x100e2aa4: ret
```

`sub_100e2a20` is called by sub_100e2ab0, sub_100e2b50, sub_100e3e60, sub_100e40d0, sub_100e8a00. It delegates to sub_100d2f50, sub_100d2fe0 (2 callees).

12.1285 sub_100e2ab0 at 0x100e2ab0 (58 instructions)

```
0x100e2ab0: sub esp,10h
0x100e2ab3: push esi
0x100e2ab4: mov esi,[esp+18h]
0x100e2ab8: push edi
0x100e2ab9: push esi
0x100e2aba: call 100D2F50h
0x100e2abf: movzx edi,ax
0x100e2ac2: add esp,4
0x100e2ac5: test edi,edi
0x100e2ac7: je short 100E2B3Ch
0x100e2ac9: push ebx
0x100e2aca: push ebp
0x100e2acb: mov ebp,[esp+28h]
0x100e2acf: nop
0x100e2ad0: lea eax,[esp+14h]
0x100e2ad4: push eax
0x100e2ad5: lea ecx,[esp+1Ch]
0x100e2ad9: push ecx
0x100e2ada: lea edx,[esp+24h]
0x100e2ade: push edx
0x100e2adf: lea eax,[esp+1Ch]
0x100e2ae3: push eax
0x100e2ae4: push ebp
0x100e2ae5: push esi
0x100e2ae6: sub edi,1
0x100e2ae9: call 100E2A20h
0x100e2aee: mov ebx,[esp+28h]
0x100e2af2: add esp,18h
0x100e2af5: cmp ebx,[esp+2Ch]
0x100e2af9: jne short 100E2B0Ch
0x100e2afb: push esi
0x100e2afc: call 100D2FE0h
0x100e2b01: add esp,4
0x100e2b04: add eax,ebp
0x100e2b06: mov [esi+65C4h],eax
0x100e2b0c: cmp ebx,[esp+30h]
0x100e2b10: jne short 100E2B21h
0x100e2b12: push esi
0x100e2b13: call 100D2FE0h
0x100e2b18: add esp,4
0x100e2b1b: mov [esi+65D0h],eax
0x100e2b21: mov edx,[esp+14h]
0x100e2b25: mov ecx,[esi]
0x100e2b27: mov eax,[esi+4]
0x100e2b2a: mov ecx,[ecx+8]
```

```
0x100e2b2d: push 0
0x100e2b2f: push edx
0x100e2b30: push eax
0x100e2b31: call ecx
0x100e2b33: add esp,0Ch
0x100e2b36: test edi,edi
0x100e2b38: jne short 100E2AD0h
0x100e2b3a: pop ebp
0x100e2b3b: pop ebx
0x100e2b3c: pop edi
0x100e2b3d: pop esi
0x100e2b3e: add esp,10h
0x100e2b41: ret
```

`sub_100e2ab0` is called by sub_100e2b50. It delegates to sub_100d2f50, sub_100d2fe0, sub_100e2a20 (3 callees).

12.1286 sub_100e2b50 at 0x100e2b50 (1345 instructions)

```
0x100e2b50: sub esp,198h
0x100e2b56: mov eax,ds:[10136220h]
0x100e2b5b: xor eax,esp
0x100e2b5d: mov [esp+194h],eax
0x100e2b64: push ebx
0x100e2b65: push ebp
0x100e2b66: mov ebp,[esp+1A4h]
0x100e2b6d: movzx eax,word ptr [ebp+6274h]
0x100e2b74: mov ecx,[ebp]
0x100e2b77: mov ecx,[ecx]
0x100e2b79: push esi
0x100e2b7a: push edi
0x100e2b7b: push 0Ah
0x100e2b7d: push 1
0x100e2b7f: lea edx,[esp+5Ch]
0x100e2b83: mov [esp+44h],eax
0x100e2b87: mov eax,[ebp+4]
0x100e2b8a: xor esi,esi
0x100e2b8c: push edx
0x100e2b8d: push eax
0x100e2b8e: mov [esp+38h],esi
0x100e2b92: mov [esp+40h],esi
0x100e2b96: mov [esp+3Ch],esi
0x100e2b9a: xor edi,edi
0x100e2b9c: xor ebx,ebx
0x100e2b9e: call ecx
0x100e2ba0: push 3
0x100e2ba2: lea edx,[esp+68h]
0x100e2ba6: push 1012E5ACh
0x100e2bab: push edx
0x100e2bac: call dword ptr ds:[100FE414h]
0x100e2bb2: add esp,1Ch
0x100e2bb5: test eax,eax
0x100e2bb7: je near ptr 100E3D46h
0x100e2bbd: push 3
0x100e2bbf: lea eax,[esp+58h]
0x100e2bc3: push 1012E5A8h
0x100e2bc8: push eax
0x100e2bc9: call dword ptr ds:[100FE414h]
0x100e2bcf: add esp,0Ch
0x100e2bd2: test eax,eax
0x100e2bd4: je near ptr 100E3D46h
0x100e2bda: push 4
0x100e2bdc: lea ecx,[esp+58h]
0x100e2be0: push 1012E5A0h
0x100e2be5: push ecx
0x100e2be6: call dword ptr ds:[100FE414h]
0x100e2bec: add esp,0Ch
```

```
0x100e2bef: test eax,eax
0x100e2bf1: je near ptr 100E3D46h
0x100e2bf7: push 4
0x100e2bf9: lea edx,[esp+58h]
0x100e2bfd: push 1012E598h
0x100e2c02: push edx
0x100e2c03: call dword ptr ds:[100FE414h]
0x100e2c09: add esp,0Ch
0x100e2c0c: test eax,eax
0x100e2c0e: je near ptr 100E3D46h
0x100e2c14: push 2
0x100e2c16: lea eax,[esp+58h]
0x100e2c1a: push 1012E594h
0x100e2c1f: push eax
0x100e2c20: call dword ptr ds:[100FE414h]
0x100e2c26: add esp,0Ch
0x100e2c29: test eax,eax
0x100e2c2b: je near ptr 100E3C6Bh
0x100e2c31: push 3
0x100e2c33: lea ecx,[esp+58h]
0x100e2c37: push 1012E590h
0x100e2c3c: push ecx
0x100e2c3d: call dword ptr ds:[100FE414h]
0x100e2c43: add esp,0Ch
0x100e2c46: test eax,eax
0x100e2c48: je near ptr 100E3C6Bh
0x100e2c4e: mov edi,1012E588h
0x100e2c53: lea esi,[esp+54h]
0x100e2c57: mov ecx,6
0x100e2c5c: xor edx,edx
0x100e2c5e: repe cmpsb
0x100e2c60: jne short 100E2CB7h
0x100e2c62: mov eax,[ebp]
0x100e2c65: mov ecx,[ebp+4]
0x100e2c68: mov edx,[eax+18h]
0x100e2c6b: push ecx
0x100e2c6c: call edx
0x100e2c6e: mov esi,eax
0x100e2c70: push ebp
0x100e2c71: mov [esp+1B8h],esi
0x100e2c78: call 100D2F50h
0x100e2c7d: push ebp
0x100e2c7e: mov [ebp+6274h],ax
0x100e2c85: call 100D2F50h
0x100e2c8a: add esp,0Ch
0x100e2c8d: cmp ax,2Ah
0x100e2c91: jne near ptr 100E3D3Ah
0x100e2c97: push ebp
0x100e2c98: call 100D2FE0h
0x100e2c9d: mov ecx,[ebp]
0x100e2ca0: mov edx,[ebp+4]
0x100e2ca3: add eax,0FFFFFFF8h
0x100e2ca6: push 1
0x100e2ca8: push eax
0x100e2ca9: mov eax,[ecx+8]
0x100e2cac: push edx
0x100e2cad: call eax
0x100e2caf: add esp,10h
0x100e2cb2: jmp near ptr 100E2E0Bh
0x100e2cb7: mov edi,10129890h
0x100e2cbc: lea esi,[esp+54h]
0x100e2cc0: mov ecx,8
0x100e2cc5: xor edx,edx
0x100e2cc7: repe cmpsb
0x100e2cc9: jne short 100E2D0Eh
0x100e2ccb: mov eax,[ebp]
0x100e2cce: mov ecx,[ebp+4]
```

```
0x100e2cd1: mov edx,[eax+18h]
0x100e2cd4: push ecx
0x100e2cd5: call edx
0x100e2cd7: mov ecx,[ebp+4]
0x100e2cda: mov esi,eax
0x100e2cdc: mov eax,[ebp]
0x100e2cdf: mov edx,[eax+8]
0x100e2ce2: push 1
0x100e2ce4: push 0FFFFFFFEh
0x100e2ce6: sub esi,0Ah
0x100e2ce9: push ecx
0x100e2cea: mov [esp+1C0h],esi
0x100e2cf1: call edx
0x100e2cf3: push ebp
0x100e2cf4: call 100D2F50h
0x100e2cf9: push ebp
0x100e2cfa: mov [ebp+6274h],ax
0x100e2d01: call 100D2F50h
0x100e2d06: add esp,18h
0x100e2d09: jmp near ptr 100E2E0Bh
0x100e2d0e: mov esi,ds:[100FE414h]
0x100e2d14: push 8
0x100e2d16: lea eax,[esp+58h]
0x100e2d1a: push 1012E57Ch
0x100e2d1f: push eax
0x100e2d20: call esi
0x100e2d22: add esp,0Ch
0x100e2d25: test eax,eax
0x100e2d27: je near ptr 100E2DE8h
0x100e2d2d: push 4
0x100e2d2f: lea ecx,[esp+58h]
0x100e2d33: push 1012E574h
0x100e2d38: push ecx
0x100e2d39: call esi
0x100e2d3b: add esp,0Ch
0x100e2d3e: test eax,eax
0x100e2d40: je near ptr 100E2DE8h
0x100e2d46: mov edi,1012E568h
0x100e2d4b: lea esi,[esp+54h]
0x100e2d4f: mov ecx,0Ah
0x100e2d54: xor edx,edx
0x100e2d56: repe cmpsb
0x100e2d58: je near ptr 100E2DE8h
0x100e2d5e: mov edi,1012E560h
0x100e2d63: lea esi,[esp+54h]
0x100e2d67: mov ecx,6
0x100e2d6c: xor eax,eax
0x100e2d6e: repe cmpsb
0x100e2d70: je short 100E2DE2h
0x100e2d72: mov edi,1012E558h
0x100e2d77: lea esi,[esp+54h]
0x100e2d7b: mov ecx,6
0x100e2d80: xor edx,edx
0x100e2d82: repe cmpsb
0x100e2d84: je short 100E2DE2h
0x100e2d86: mov edi,1012E550h
0x100e2d8b: lea esi,[esp+54h]
0x100e2d8f: mov ecx,6
0x100e2d94: xor eax,eax
0x100e2d96: repe cmpsb
0x100e2d98: je short 100E2DE2h
0x100e2d9a: mov edi,101298A8h
0x100e2d9f: lea esi,[esp+54h]
0x100e2da3: mov ecx,6
0x100e2da8: xor edx,edx
0x100e2daa: repe cmpsb
0x100e2dac: je short 100E2DE2h
```

```
0x100e2dae: mov edi,1012E54Ch
0x100e2db3: lea esi,[esp+54h]
0x100e2db7: mov ecx,4
0x100e2dbc: xor eax,eax
0x100e2dbe: repe cmpsb
0x100e2dc0: je short 100E2DDCh
0x100e2dc2: mov edi,1012E548h
0x100e2dc7: lea esi,[esp+54h]
0x100e2dcb: mov ecx,4
0x100e2dd0: xor edx,edx
0x100e2dd2: repe cmpsb
0x100e2dd4: je short 100E2DDCh
0x100e2dd6: push 1
0x100e2dd8: push 0FFFFFFF6h
0x100e2dda: jmp short 100E2DF5h
0x100e2ddc: push 1
0x100e2dde: push 0FFFFFFFCh
0x100e2de0: jmp short 100E2DF5h
0x100e2de2: push 1
0x100e2de4: push 0FFFFFFFEh
0x100e2de6: jmp short 100E2DF5h
0x100e2de8: push 1
0x100e2dea: mov word ptr [ebp+6274h],4949h
0x100e2df3: push 2
0x100e2df5: mov ecx,[ebp+4]
0x100e2df8: mov eax,[ebp]
0x100e2dfb: mov edx,[eax+8]
0x100e2dfe: push ecx
0x100e2dff: call edx
0x100e2e01: mov esi,[esp+1BCh]
0x100e2e08: add esp,0Ch
0x100e2e0b: push ebp
0x100e2e0c: call 100D2F50h
0x100e2e11: movzx eax,ax
0x100e2e14: add esp,4
0x100e2e17: cmp eax,3E8h
0x100e2e1c: ja near ptr 100E3D46h
0x100e2e22: test eax,eax
0x100e2e24: je near ptr 100E3D3Ah
0x100e2e2a: mov edi,[esp+1B4h]
0x100e2e31: shl edi,10h
0x100e2e34: mov [esp+38h],edi
0x100e2e38: jmp short 100E2E4Bh
0x100e2e40: mov esi,[esp+1B0h]
0x100e2e47: mov edi,[esp+38h]
0x100e2e4b: sub eax,1
0x100e2e4e: mov [esp+34h],eax
0x100e2e52: lea eax,[esp+40h]
0x100e2e56: push eax
0x100e2e57: lea ecx,[esp+1Ch]
0x100e2e5b: push ecx
0x100e2e5c: lea edx,[esp+1Ch]
0x100e2e60: push edx
0x100e2e61: lea eax,[esp+2Ch]
0x100e2e65: push eax
0x100e2e66: push esi
0x100e2e67: push ebp
0x100e2e68: call 100E2A20h
0x100e2e6d: mov ebx,[esp+38h]
0x100e2e71: or ebx,edi
0x100e2e73: add esp,18h
0x100e2e76: cmp ebx,2
0x100e2e79: mov [esp+20h],ebx
0x100e2e7d: jne short 100E2EC1h
0x100e2e7f: lea ecx,[ebp+6485h]
0x100e2e85: push 101298B0h
0x100e2e8a: push ecx
```

```
0x100e2e8b: call dword ptr ds:[100FE424h]
0x100e2e91: add esp,8
0x100e2e94: test eax,eax
0x100e2e96: je near ptr 100E31ADh
0x100e2e9c: push ebp
0x100e2e9d: call 100D2F50h
0x100e2ea2: push ebp
0x100e2ea3: call 100D2F50h
0x100e2ea8: movzx edx,ax
0x100e2eab: mov [esp+18h],edx
0x100e2eaf: add esp,8
0x100e2eb2: fild dword ptr [esp+10h]
0x100e2eb6: fstp dword ptr [ebp+6590h]
0x100e2ebc: jmp near ptr 100E31ADh
0x100e2ec1: mov eax,4
0x100e2ec6: cmp ebx,eax
0x100e2ec8: jne near ptr 100E300Eh
0x100e2ece: mov eax,[esp+18h]
0x100e2ed2: add eax,0FFFFFFE5h
0x100e2ed5: cmp eax,7
0x100e2ed8: ja near ptr 100E31ADh
0x100e2ede: push ebp
0x100e2edf: call 100D2FE0h
0x100e2ee4: push ebp
0x100e2ee5: call 100D2F50h
0x100e2eea: movzx ecx,ax
0x100e2eed: add esp,8
0x100e2ef0: cmp ecx,7FFFh
0x100e2ef6: je short 100E2F46h
0x100e2ef8: fldz
0x100e2efa: fcomp dword ptr [ebp+6590h]
0x100e2f00: fnstsw ax
0x100e2f02: test ah,44h
0x100e2f05: jp short 100E2F46h
0x100e2f07: test ecx,ecx
0x100e2f09: mov [esp+1Ch],ecx
0x100e2f0d: fild dword ptr [esp+1Ch]
0x100e2f11: jge short 100E2F19h
0x100e2f13: fadd qword ptr ds:[100FEFC8h]
0x100e2f19: fmul qword ptr ds:[1012E540h]
0x100e2f1f: fsub qword ptr ds:[100FF108h]
0x100e2f25: fld qword ptr ds:[100FF110h]
0x100e2f2b: fxch
0x100e2f2d: call 100229A0h
0x100e2f32: fstp dword ptr [esp+10h]
0x100e2f36: fld dword ptr [esp+10h]
0x100e2f3a: fmul qword ptr ds:[100FF380h]
0x100e2f40: fstp dword ptr [ebp+6590h]
0x100e2f46: push ebp
0x100e2f47: call 100D2F50h
0x100e2f4c: push ebp
0x100e2f4d: call 100D2F50h
0x100e2f52: movzx ecx,ax
0x100e2f55: add esp,8
0x100e2f58: cmp ecx,7FFFh
0x100e2f5e: je short 100E2F9Ah
0x100e2f60: fldz
0x100e2f62: fcomp dword ptr [ebp+6598h]
0x100e2f68: fnstsw ax
0x100e2f6a: test ah,44h
0x100e2f6d: jp short 100E2F9Ah
0x100e2f6f: test ecx,ecx
0x100e2f71: mov [esp+1Ch],ecx
0x100e2f75: fild dword ptr [esp+1Ch]
0x100e2f79: jge short 100E2F81h
0x100e2f7b: fadd qword ptr ds:[100FEFC8h]
0x100e2f81: fmul qword ptr ds:[1012E538h]
```

```
0x100e2f87: fld qword ptr ds:[100FF110h]
0x100e2f8d: fxch
0x100e2f8f: call 100229A0h
0x100e2f94: fstp dword ptr [ebp+6598h]
0x100e2f9a: push ebp
0x100e2f9b: call 100D2F50h
0x100e2fa0: movzx ecx,ax
0x100e2fa3: add esp,4
0x100e2fa6: cmp ecx,0FFFFh
0x100e2fac: je short 100E2FE1h
0x100e2fae: fldz
0x100e2fb0: fcomp dword ptr [ebp+6594h]
0x100e2fb6: fnstsw ax
0x100e2fb8: test ah,44h
0x100e2fbb: jp short 100E2FE1h
0x100e2fbd: movsx ecx,cx
0x100e2fc0: mov [esp+10h],ecx
0x100e2fc4: fild dword ptr [esp+10h]
0x100e2fc8: fmul qword ptr ds:[1012E530h]
0x100e2fce: fld qword ptr ds:[100FF110h]
0x100e2fd4: fxch
0x100e2fd6: call 100229A0h
0x100e2fdb: fstp dword ptr [ebp+6594h]
0x100e2fe1: push ebp
0x100e2fe2: call 100D2F50h
0x100e2fe7: push ebp
0x100e2fe8: call 100D2F50h
0x100e2fed: movzx edx,ax
0x100e2ff0: push ebp
0x100e2ff1: mov [esp+38h],edx
0x100e2ff5: call 100D2F50h
0x100e2ffa: push ebp
0x100e2ffb: call 100D2F50h
0x100e3000: movzx eax,ax
0x100e3003: mov [ebp+65A8h],eax
0x100e3009: jmp near ptr 100E31AAh
0x100e300e: cmp ebx,7
0x100e3011: jne short 100E3046h
0x100e3013: cmp dword ptr [esp+14h],2
0x100e3018: jne near ptr 100E31ADh
0x100e301e: cmp dword ptr [esp+18h],14h
0x100e3023: jbe near ptr 100E31ADh
0x100e3029: mov ecx,[ebp]
0x100e302c: mov eax,[ebp+4]
0x100e302f: mov ecx,[ecx+10h]
0x100e3032: push 40h
0x100e3034: lea edx,[ebp+6505h]
0x100e303a: push edx
0x100e303b: push eax
0x100e303c: call ecx
0x100e303e: add esp,0Ch
0x100e3041: jmp near ptr 100E31ADh
0x100e3046: cmp ebx,8
0x100e3049: jne short 100E3069h
0x100e304b: cmp [esp+14h],eax
0x100e304f: jne near ptr 100E31ADh
0x100e3055: push ebp
0x100e3056: call 100D2FE0h
0x100e305b: add esp,4
0x100e305e: mov [ebp+65A8h],eax
0x100e3064: jmp near ptr 100E31ADh
0x100e3069: cmp ebx,9
0x100e306c: jne short 100E30A3h
0x100e306e: mov edi,101298B8h
0x100e3073: lea esi,[ebp+6485h]
0x100e3079: mov ecx,6
0x100e307e: xor edx,edx
```

```
0x100e3080: repe cmpsb
0x100e3082: jne near ptr 100E31ADh
0x100e3088: mov eax,[ebp]
0x100e308b: mov edx,[ebp+4]
0x100e308e: mov eax,[eax]
0x100e3090: push 1
0x100e3092: push 40h
0x100e3094: lea ecx,[ebp+6545h]
0x100e309a: push ecx
0x100e309b: push edx
0x100e309c: call eax
0x100e309e: jmp near ptr 100E31AAh
0x100e30a3: cmp ebx,0Ch
0x100e30a6: jne short 100E30D5h
0x100e30a8: cmp [esp+18h],eax
0x100e30ac: jne near ptr 100E31ADh
0x100e30b2: mov esi,[esp+14h]
0x100e30b6: push esi
0x100e30b7: push ebp
0x100e30b8: call 100D3060h
0x100e30bd: fstp dword ptr [ebp+0E774h]
0x100e30c3: push esi
0x100e30c4: push ebp
0x100e30c5: call 100D3060h
0x100e30ca: fstp dword ptr [ebp+0E77Ch]
0x100e30d0: jmp near ptr 100E31AAh
0x100e30d5: cmp ebx,10h
0x100e30d8: jne short 100E30F8h
0x100e30da: cmp [esp+14h],eax
0x100e30de: jne near ptr 100E31ADh
0x100e30e4: push ebp
0x100e30e5: call 100D2FE0h
0x100e30ea: add esp,4
0x100e30ed: mov [ebp+65B8h],eax
0x100e30f3: jmp near ptr 100E31ADh
0x100e30f8: cmp ebx,11h
0x100e30fb: jne short 100E314Eh
0x100e30fd: cmp dword ptr [ebp+6614h],0
0x100e3104: je near ptr 100E31ADh
0x100e310a: push 5
0x100e310c: lea ecx,[ebp+6485h]
0x100e3112: push 101298B0h
0x100e3117: push ecx
0x100e3118: call dword ptr ds:[100FE414h]
0x100e311e: add esp,0Ch
0x100e3121: test eax,eax
0x100e3123: jne near ptr 100E31ADh
0x100e3129: mov edi,[ebp]
0x100e312c: push eax
0x100e312d: push ebp
0x100e312e: call 100D2FE0h
0x100e3133: mov edx,[ebp+4]
0x100e3136: add eax,esi
0x100e3138: add esp,4
0x100e313b: push eax
0x100e313c: mov eax,[edi+8]
0x100e313f: push edx
0x100e3140: call eax
0x100e3142: push esi
0x100e3143: push ebp
0x100e3144: call 100E8A00h
0x100e3149: add esp,14h
0x100e314c: jmp short 100E31ADh
0x100e314e: cmp ebx,14h
0x100e3151: jne short 100E317Fh
0x100e3153: cmp dword ptr [esp+18h],0A00h
0x100e315b: jne short 100E31ADh
```

```
0x100e315d: cmp dword ptr [esp+14h],7
0x100e3162: jne short 100E31ADh
0x100e3164: mov ecx,[ebp]
0x100e3167: mov edx,[ebp+4]
0x100e316a: mov eax,[ecx+8]
0x100e316d: push 1
0x100e316f: push 4E0h
0x100e3174: push edx
0x100e3175: call eax
0x100e3177: add esp,0Ch
0x100e317a: jmp near ptr 100E3A6Ah
0x100e317f: cmp ebx,15h
0x100e3182: jne short 100E31ADh
0x100e3184: cmp dword ptr [esp+14h],2
0x100e3189: jne short 100E31ADh
0x100e318b: cmp dword ptr [ebp+6614h],0
0x100e3192: je short 100E31ADh
0x100e3194: mov ecx,[ebp]
0x100e3197: mov eax,[ebp+4]
0x100e319a: mov ecx,[ecx]
0x100e319c: push 1
0x100e319e: push 40h
0x100e31a0: lea edx,[ebp+64C5h]
0x100e31a6: push edx
0x100e31a7: push eax
0x100e31a8: call ecx
0x100e31aa: add esp,10h
0x100e31ad: lea edx,[ebp+6485h]
0x100e31b3: push 10129888h
0x100e31b8: push edx
0x100e31b9: call dword ptr ds:[100FE424h]
0x100e31bf: add esp,8
0x100e31c2: test eax,eax
0x100e31c4: je short 100E31F2h
0x100e31c6: cmp ebx,1Bh
0x100e31c9: jne short 100E31DCh
0x100e31cb: mov dword ptr [esp+20h],1018h
0x100e31d3: mov edi,[esp+20h]
0x100e31d7: jmp near ptr 100E3A3Bh
0x100e31dc: cmp ebx,1Ch
0x100e31df: jne short 100E31F2h
0x100e31e1: mov dword ptr [esp+20h],1017h
0x100e31e9: mov edi,[esp+20h]
0x100e31ed: jmp near ptr 100E3A0Bh
0x100e31f2: cmp ebx,1Dh
0x100e31f5: jne short 100E3265h
0x100e31f7: mov eax,[ebp]
0x100e31fa: mov ecx,[ebp+4]
0x100e31fd: mov edx,[eax+1Ch]
0x100e3200: push ecx
0x100e3201: call edx
0x100e3203: mov esi,eax
0x100e3205: add esp,4
0x100e3208: test esi,esi
0x100e320a: je near ptr 100E3AB0h
0x100e3210: cmp esi,0FFFFFFFFh
0x100e3213: je near ptr 100E3AB0h
0x100e3219: push esi
0x100e321a: call dword ptr ds:[100FE45Ch]
0x100e3220: add esp,4
0x100e3223: test eax,eax
0x100e3225: je short 100E322Ch
0x100e3227: add esi,0FFFFFFD0h
0x100e322a: jmp short 100E323Dh
0x100e322c: mov eax,0CCCCCCCDh
0x100e3231: mul esi
0x100e3233: shr edx,3
```

```
0x100e3236: lea eax,[edx+edx*4]
0x100e3239: add eax,eax
0x100e323b: sub esi,eax
0x100e323d: mov eax,[esp+30h]
0x100e3241: lea ecx,[eax+eax*4]
0x100e3244: mov eax,[ebp]
0x100e3247: lea edx,[esi+ecx*2]
0x100e324a: mov ecx,[ebp+4]
0x100e324d: mov [esp+30h],edx
0x100e3251: mov edx,[eax+1Ch]
0x100e3254: push ecx
0x100e3255: call edx
0x100e3257: mov esi,eax
0x100e3259: add esp,4
0x100e325c: test esi,esi
0x100e325e: jne short 100E3210h
0x100e3260: jmp near ptr 100E3AB0h
0x100e3265: cmp ebx,81h
0x100e326b: jne short 100E32C7h
0x100e326d: mov eax,[esp+14h]
0x100e3271: cmp eax,4
0x100e3274: jne near ptr 100E3340h
0x100e327a: push ebp
0x100e327b: call 100D2FE0h
0x100e3280: mov ecx,[ebp]
0x100e3283: mov edx,[ebp+4]
0x100e3286: mov [ebp+65C0h],eax
0x100e328c: add eax,29h
0x100e328f: push 0
0x100e3291: push eax
0x100e3292: mov eax,[ecx+8]
0x100e3295: push edx
0x100e3296: call eax
0x100e3298: push ebp
0x100e3299: call 100D2F50h
0x100e329e: add ax,ax
0x100e32a1: push ebp
0x100e32a2: mov [ebp+66B0h],ax
0x100e32a9: call 100D2F50h
0x100e32ae: add esp,18h
0x100e32b1: mov [ebp+66B2h],ax
0x100e32b8: mov dword ptr [ebp+65B0h],61616161h
0x100e32c2: jmp near ptr 100E3AB0h
0x100e32c7: cmp ebx,29h
0x100e32ca: jne short 100E334Ah
0x100e32cc: cmp dword ptr [esp+14h],1
0x100e32d1: jne near ptr 100E3AB0h
0x100e32d7: mov eax,[esp+2Ch]
0x100e32db: cmp eax,12h
0x100e32de: jae short 100E32ECh
0x100e32e0: movsx eax,byte ptr [eax+1012E518h]
0x100e32e7: sub eax,30h
0x100e32ea: jmp short 100E32EEh
0x100e32ec: xor eax,eax
0x100e32ee: mov ecx,[ebp]
0x100e32f1: mov edx,[ebp+4]
0x100e32f4: shl eax,5
0x100e32f7: add eax,8
0x100e32fa: push 1
0x100e32fc: push eax
0x100e32fd: mov eax,[ecx+8]
0x100e3300: push edx
0x100e3301: call eax
0x100e3303: add esp,0Ch
0x100e3306: xor esi,esi
0x100e3308: push ebp
0x100e3309: call 100D2FE0h
```

```
0x100e330e: add esp,4
0x100e3311: test eax,eax
0x100e3313: mov [esp+1Ch],eax
0x100e3317: fild dword ptr [esp+1Ch]
0x100e331b: jge short 100E3323h
0x100e331d: fadd dword ptr ds:[100FE7C0h]
0x100e3323: mov ecx,esi
0x100e3325: xor ecx,2
0x100e3328: shr ecx,1
0x100e332a: xor ecx,esi
0x100e332c: add esi,1
0x100e332f: fstp dword ptr [ebp+ecx*4+0E774h]
0x100e3336: cmp esi,4
0x100e3339: jb short 100E3308h
0x100e333b: jmp near ptr 100E3AB0h
0x100e3340: cmp eax,7
0x100e3343: je short 100E3375h
0x100e3345: jmp near ptr 100E3AB0h
0x100e334a: cmp ebx,100h
0x100e3350: jne short 100E335Eh
0x100e3352: cmp dword ptr [esp+14h],7
0x100e3357: je short 100E3375h
0x100e3359: jmp near ptr 100E3AB0h
0x100e335e: cmp ebx,280h
0x100e3364: jne short 100E339Ah
0x100e3366: cmp dword ptr [esp+14h],1
0x100e336b: jne near ptr 100E3AB0h
0x100e3371: mov ebx,[esp+20h]
0x100e3375: mov edx,[ebp]
0x100e3378: mov eax,[ebp+4]
0x100e337b: mov ecx,[edx+18h]
0x100e337e: mov edi,[esp+28h]
0x100e3382: mov esi,[esp+18h]
0x100e3386: push eax
0x100e3387: call ecx
0x100e3389: add esp,4
0x100e338c: mov [ebp+65C4h],eax
0x100e3392: mov [ebp+65D0h],esi
0x100e3398: jmp short 100E33A2h
0x100e339a: mov edi,[esp+28h]
0x100e339e: mov esi,[esp+18h]
0x100e33a2: cmp ebx,88h
0x100e33a8: jne short 100E33E0h
0x100e33aa: cmp dword ptr [esp+14h],4
0x100e33af: jne near ptr 100E3AB0h
0x100e33b5: push ebp
0x100e33b6: call 100D2FE0h
0x100e33bb: add esp,4
0x100e33be: test eax,eax
0x100e33c0: mov [ebp+65C4h],eax
0x100e33c6: je near ptr 100E3AB0h
0x100e33cc: mov edx,[esp+1B0h]
0x100e33d3: add eax,edx
0x100e33d5: mov [ebp+65C4h],eax
0x100e33db: jmp near ptr 100E3AB0h
0x100e33e0: cmp ebx,89h
0x100e33e6: jne short 100E3407h
0x100e33e8: cmp dword ptr [esp+14h],4
0x100e33ed: jne near ptr 100E3AB0h
0x100e33f3: push ebp
0x100e33f4: call 100D2FE0h
0x100e33f9: add esp,4
0x100e33fc: mov [ebp+65D0h],eax
0x100e3402: jmp near ptr 100E3AB0h
0x100e3407: cmp ebx,8Ch
0x100e340d: je short 100E3417h
0x100e340f: cmp ebx,96h
```

```
0x100e3415: jne short 100E342Ch
0x100e3417: mov eax,[ebp]
0x100e341a: mov ecx,[ebp+4]
0x100e341d: mov edx,[eax+18h]
0x100e3420: push ecx
0x100e3421: call edx
0x100e3423: add esp,4
0x100e3426: mov [ebp+65C8h],eax
0x100e342c: cmp ebx,97h
0x100e3432: jne near ptr 100E3560h
0x100e3438: mov esi,4
0x100e343d: lea ecx,[ecx]
0x100e3440: mov eax,[ebp]
0x100e3443: mov ecx,[ebp+4]
0x100e3446: mov edx,[eax+1Ch]
0x100e3449: push ecx
0x100e344a: call edx
0x100e344c: add esp,4
0x100e344f: sub esi,1
0x100e3452: lea ecx,[edi+edi*4]
0x100e3455: lea edi,[eax+ecx*2-30h]
0x100e3459: jne short 100E3440h
0x100e345b: mov eax,edi
0x100e345d: sub eax,64h
0x100e3460: mov [esp+28h],edi
0x100e3464: je short 100E34CBh
0x100e3466: sub eax,2
0x100e3469: je short 100E34B6h
0x100e346b: sub eax,1
0x100e346e: jne near ptr 100E3515h
0x100e3474: mov edx,[ebp]
0x100e3477: mov eax,[ebp+4]
0x100e347a: mov ecx,[edx+8]
0x100e347d: push 1
0x100e347f: push 10h
0x100e3481: push eax
0x100e3482: call ecx
0x100e3484: add esp,0Ch
0x100e3487: lea esi,[ebp+0E774h]
0x100e348d: mov edi,4
0x100e3492: push ebp
0x100e3493: call 100D2F50h
0x100e3498: movzx edx,ax
0x100e349b: mov [esp+14h],edx
0x100e349f: add esp,4
0x100e34a2: add esi,4
0x100e34a5: sub edi,1
0x100e34a8: fild dword ptr [esp+10h]
0x100e34ac: fstp dword ptr [esi-4]
0x100e34af: jne short 100E3492h
0x100e34b1: jmp near ptr 100E3AB0h
0x100e34b6: mov eax,[ebp]
0x100e34b9: mov ecx,[ebp+4]
0x100e34bc: mov edx,[eax+8]
0x100e34bf: push 1
0x100e34c1: push 6
0x100e34c3: push ecx
0x100e34c4: call edx
0x100e34c6: jmp near ptr 100E3BE2h
0x100e34cb: mov eax,[ebp]
0x100e34ce: mov ecx,[ebp+4]
0x100e34d1: mov edx,[eax+8]
0x100e34d4: push 1
0x100e34d6: push 44h
0x100e34d8: push ecx
0x100e34d9: call edx
0x100e34db: add esp,0Ch
```

```
0x100e34de: xor esi,esi
0x100e34e0: push ebp
0x100e34e1: call 100D2F50h
0x100e34e6: movzx eax,ax
0x100e34e9: mov ecx,esi
0x100e34eb: and ecx,1
0x100e34ee: mov edx,esi
0x100e34f0: mov [esp+14h],eax
0x100e34f4: add ecx,ecx
0x100e34f6: shr edx,1
0x100e34f8: fild dword ptr [esp+14h]
0x100e34fc: or ecx,edx
0x100e34fe: add esi,1
0x100e3501: add esp,4
0x100e3504: cmp esi,4
0x100e3507: fstp dword ptr [ebp+ecx*4+0E774h]
0x100e350e: jb short 100E34E0h
0x100e3510: jmp near ptr 100E3AB0h
0x100e3515: cmp edi,0C8h
0x100e351b: jb near ptr 100E3AB0h
0x100e3521: cmp dword ptr [esp+28h],0CDh
0x100e3529: je short 100E3541h
0x100e352b: mov eax,[ebp]
0x100e352e: mov ecx,[ebp+4]
0x100e3531: mov edx,[eax+8]
0x100e3534: push 1
0x100e3536: push 118h
0x100e353b: push ecx
0x100e353c: call edx
0x100e353e: add esp,0Ch
0x100e3541: mov eax,[ebp]
0x100e3544: mov edx,[ebp+4]
0x100e3547: mov eax,[eax]
0x100e3549: push 1
0x100e354b: push 144h
0x100e3550: lea ecx,[esp+68h]
0x100e3554: push ecx
0x100e3555: push edx
0x100e3556: call eax
0x100e3558: add esp,10h
0x100e355b: jmp near ptr 100E3AB0h
0x100e3560: cmp ebx,0A4h
0x100e3566: jne short 100E35C4h
0x100e3568: cmp dword ptr [esp+14h],3
0x100e356d: jne near ptr 100E3AB0h
0x100e3573: mov eax,[esp+2Ch]
0x100e3577: mov ecx,[ebp]
0x100e357a: mov ecx,[ecx+8]
0x100e357d: lea edx,[eax+eax*2]
0x100e3580: mov eax,[ebp+4]
0x100e3583: push 1
0x100e3585: shl edx,4
0x100e3588: push edx
0x100e3589: push eax
0x100e358a: call ecx
0x100e358c: add esp,0Ch
0x100e358f: lea esi,[ebp+0E774h]
0x100e3595: mov edi,3
0x100e359a: lea ebx,[ebx]
0x100e35a0: push ebp
0x100e35a1: call 100D2F50h
0x100e35a6: movzx edx,ax
0x100e35a9: mov [esp+14h],edx
0x100e35ad: add esp,4
0x100e35b0: add esi,4
0x100e35b3: sub edi,1
0x100e35b6: fild dword ptr [esp+10h]
```

```
0x100e35ba: fstp dword ptr [esi-4]
0x100e35bd: jne short 100E35A0h
0x100e35bf: jmp near ptr 100E3AB0h
0x100e35c4: cmp ebx,0A7h
0x100e35ca: jne near ptr 100E36B5h
0x100e35d0: add edi,0FFFFFF38h
0x100e35d6: cmp edi,0Ch
0x100e35d9: jae near ptr 100E3AB0h
0x100e35df: fldz
0x100e35e1: fcomp dword ptr [ebp+0E774h]
0x100e35e7: fnstsw ax
0x100e35e9: test ah,44h
0x100e35ec: jp near ptr 100E3AB0h
0x100e35f2: mov eax,[esp+30h]
0x100e35f6: mov ecx,[ebp]
0x100e35f9: mov edx,[ebp+4]
0x100e35fc: and eax,0FFh
0x100e3601: mov bl,[eax+1012B6F0h]
0x100e3607: mov eax,[ecx+1Ch]
0x100e360a: push edx
0x100e360b: mov [esp+2Bh],bl
0x100e360f: call eax
0x100e3611: mov ecx,[ebp]
0x100e3614: mov edx,[ebp+4]
0x100e3617: mov esi,eax
0x100e3619: mov eax,[ecx+1Ch]
0x100e361c: push edx
0x100e361d: call eax
0x100e361f: mov ecx,[ebp]
0x100e3622: mov edx,[ebp+4]
0x100e3625: xor esi,eax
0x100e3627: mov eax,[ecx+1Ch]
0x100e362a: push edx
0x100e362b: call eax
0x100e362d: mov ecx,[ebp]
0x100e3630: mov edx,[ebp+4]
0x100e3633: mov edi,eax
0x100e3635: mov eax,[ecx+1Ch]
0x100e3638: push edx
0x100e3639: xor edi,esi
0x100e363b: call eax
0x100e363d: xor eax,edi
0x100e363f: mov dl,[eax+1012B7F0h]
0x100e3645: mov al,bl
0x100e3647: mov bl,60h
0x100e3649: add esp,10h
0x100e364c: xor ecx,ecx
0x100e364e: imul bl
0x100e3650: add dl,al
0x100e3652: xor [esp+ecx+60h],dl
0x100e3656: add al,[esp+27h]
0x100e365a: add ecx,1
0x100e365d: cmp ecx,144h
0x100e3663: jb short 100E3650h
0x100e3665: mov ecx,[esp+28h]
0x100e3669: movsx eax,byte ptr [ecx+1012E440h]
0x100e3670: sub eax,30h
0x100e3673: mov ebx,eax
0x100e3675: xor esi,esi
0x100e3677: and ebx,1
0x100e367a: and eax,0FFFFFFFEh
0x100e367d: lea edi,[esp+eax+60h]
0x100e3681: push edi
0x100e3682: push ebp
0x100e3683: call 100D2F20h
0x100e3688: movzx edx,ax
0x100e368b: mov eax,esi
```

```
0x100e368d: shr eax,1
0x100e368f: xor eax,ebx
0x100e3691: mov [esp+18h],edx
0x100e3695: xor eax,esi
0x100e3697: add esi,1
0x100e369a: fild dword ptr [esp+18h]
0x100e369e: add esp,8
0x100e36a1: add edi,2
0x100e36a4: cmp esi,4
0x100e36a7: fstp dword ptr [ebp+eax*4+0E774h]
0x100e36ae: jb short 100E3681h
0x100e36b0: jmp near ptr 100E3AB0h
0x100e36b5: cmp ebx,200h
0x100e36bb: jne short 100E3723h
0x100e36bd: cmp esi,3
0x100e36c0: jne short 100E36D9h
0x100e36c2: push ebp
0x100e36c3: call 100D2FE0h
0x100e36c8: push ebp
0x100e36c9: call 100D2FE0h
0x100e36ce: mov [ebp+65A8h],eax
0x100e36d4: jmp near ptr 100E3AADh
0x100e36d9: cmp esi,4
0x100e36dc: jne near ptr 100E3AB0h
0x100e36e2: push ebp
0x100e36e3: call 100D2F50h
0x100e36e8: push ebp
0x100e36e9: movzx esi,ax
0x100e36ec: call 100D2F50h
0x100e36f1: movzx ecx,ax
0x100e36f4: push ebp
0x100e36f5: add esi,ecx
0x100e36f7: call 100D2F50h
0x100e36fc: movzx edx,ax
0x100e36ff: push ebp
0x100e3700: add esi,edx
0x100e3702: call 100D2F50h
0x100e3707: movzx eax,ax
0x100e370a: add eax,esi
0x100e370c: cdq
0x100e370d: and edx,3
0x100e3710: add eax,edx
0x100e3712: add esp,10h
0x100e3715: sar eax,2
0x100e3718: mov [ebp+65F8h],eax
0x100e371e: jmp near ptr 100E3AB0h
0x100e3723: cmp ebx,201h
0x100e3729: jne short 100E3739h
0x100e372b: cmp esi,4
0x100e372e: je near ptr 100E3BE5h
0x100e3734: jmp near ptr 100E3AB0h
0x100e3739: cmp ebx,220h
0x100e373f: jne short 100E3768h
0x100e3741: cmp esi,35h
0x100e3744: jne near ptr 100E3AB0h
0x100e374a: mov eax,[ebp]
0x100e374d: mov ecx,[ebp+4]
0x100e3750: mov edx,[eax+8]
0x100e3753: push 1
0x100e3755: push 0Eh
0x100e3757: push ecx
0x100e3758: call edx
0x100e375a: push ebp
0x100e375b: call 100D5E70h
0x100e3760: add esp,10h
0x100e3763: jmp near ptr 100E3AB0h
0x100e3768: cmp ebx,401h
```

```
0x100e376e: jne short 100E37AAh
0x100e3770: cmp esi,4
0x100e3773: jne near ptr 100E3AB0h
0x100e3779: push ebp
0x100e377a: call 100D2FE0h
0x100e377f: push ebp
0x100e3780: mov esi,eax
0x100e3782: call 100D2FE0h
0x100e3787: push ebp
0x100e3788: add esi,eax
0x100e378a: call 100D2FE0h
0x100e378f: push ebp
0x100e3790: add esi,eax
0x100e3792: call 100D2FE0h
0x100e3797: add eax,esi
0x100e3799: add esp,10h
0x100e379c: shr eax,2
0x100e379f: mov [ebp+65F8h],eax
0x100e37a5: jmp near ptr 100E3AB0h
0x100e37aa: cmp ebx,0E01h
0x100e37b0: jne near ptr 100E3865h
0x100e37b6: movsx eax,word ptr [ebp+6274h]
0x100e37bd: mov ecx,[ebp]
0x100e37c0: mov edx,[ebp+4]
0x100e37c3: push 1
0x100e37c5: mov [esp+18h],eax
0x100e37c9: mov word ptr [ebp+6274h],4949h
0x100e37d2: mov eax,[ecx+8]
0x100e37d5: push 16h
0x100e37d7: push edx
0x100e37d8: call eax
0x100e37da: add esp,0Ch
0x100e37dd: cmp dword ptr [esp+18h],2Ch
0x100e37e2: mov edi,16h
0x100e37e7: jbe short 100E3855h
0x100e37e9: lea esp,[esp]
0x100e37f0: push ebp
0x100e37f1: call 100D2FE0h
0x100e37f6: mov ecx,[ebp]
0x100e37f9: mov edx,[ebp+4]
0x100e37fc: push 1
0x100e37fe: mov ebx,eax
0x100e3800: mov eax,[ecx+8]
0x100e3803: push 0Eh
0x100e3805: push edx
0x100e3806: mov [esp+30h],ebx
0x100e380a: call eax
0x100e380c: push ebp
0x100e380d: call 100D2FE0h
0x100e3812: mov esi,eax
0x100e3814: add esp,14h
0x100e3817: sub esi,4
0x100e381a: cmp ebx,76A43207h
0x100e3820: jne short 100E3836h
0x100e3822: push ebp
0x100e3823: call 100D2F50h
0x100e3828: movzx ecx,ax
0x100e382b: add esp,4
0x100e382e: mov [ebp+66C8h],ecx
0x100e3834: jmp short 100E3848h
0x100e3836: mov edx,[ebp]
0x100e3839: mov eax,[ebp+4]
0x100e383c: mov ecx,[edx+8]
0x100e383f: push 1
0x100e3841: push esi
0x100e3842: push eax
0x100e3843: call ecx
```

```
0x100e3845: add esp,0Ch
0x100e3848: lea edi,[edi+esi+16h]
0x100e384c: lea edx,[edi+16h]
0x100e384f: cmp edx,[esp+18h]
0x100e3853: jb short 100E37F0h
0x100e3855: mov ax,[esp+14h]
0x100e385a: mov esi,[esp+18h]
0x100e385e: mov [ebp+6274h],ax
0x100e3865: cmp ebx,0E80h
0x100e386b: jne near ptr 100E38F2h
0x100e3871: cmp esi,100h
0x100e3877: jne near ptr 100E3AB0h
0x100e387d: cmp dword ptr [esp+14h],7
0x100e3882: jne near ptr 100E3AB0h
0x100e3888: mov ecx,[ebp]
0x100e388b: mov edx,[ebp+4]
0x100e388e: mov eax,[ecx+8]
0x100e3891: push 1
0x100e3893: push 30h
0x100e3895: push edx
0x100e3896: call eax
0x100e3898: push ebp
0x100e3899: call 100D2F50h
0x100e389e: movzx ecx,ax
0x100e38a1: imul ecx,1FCh
0x100e38a7: mov [esp+20h],ecx
0x100e38ab: push ebp
0x100e38ac: fild dword ptr [esp+24h]
0x100e38b0: fmul qword ptr ds:[1012E500h]
0x100e38b6: fmul qword ptr ds:[10127AD8h]
0x100e38bc: fstp dword ptr [ebp+0E774h]
0x100e38c2: call 100D2F50h
0x100e38c7: movzx edx,ax
0x100e38ca: add esp,14h
0x100e38cd: imul edx,17Eh
0x100e38d3: mov [esp+10h],edx
0x100e38d7: fild dword ptr [esp+10h]
0x100e38db: fmul qword ptr ds:[1012E4F8h]
0x100e38e1: fmul qword ptr ds:[10127AD8h]
0x100e38e7: fstp dword ptr [ebp+0E77Ch]
0x100e38ed: jmp near ptr 100E3AB0h
0x100e38f2: cmp ebx,0F00h
0x100e38f8: jne short 100E3958h
0x100e38fa: cmp dword ptr [esp+14h],7
0x100e38ff: jne near ptr 100E3AB0h
0x100e3905: mov eax,[esp+18h]
0x100e3909: cmp eax,266h
0x100e390e: jne short 100E392Bh
0x100e3910: mov eax,[ebp]
0x100e3913: mov ecx,[ebp+4]
0x100e3916: mov edx,[eax+8]
0x100e3919: push 1
0x100e391b: push 0B0h
0x100e3920: push ecx
0x100e3921: call edx
0x100e3923: add esp,0Ch
0x100e3926: jmp near ptr 100E3A6Ah
0x100e392b: cmp eax,2DEh
0x100e3930: je short 100E393Dh
0x100e3932: cmp eax,5DEh
0x100e3937: jne near ptr 100E3C44h
0x100e393d: mov eax,[ebp]
0x100e3940: mov ecx,[ebp+4]
0x100e3943: mov edx,[eax+8]
0x100e3946: push 1
0x100e3948: push 94h
0x100e394d: push ecx
```

```
0x100e394e: call edx
0x100e3950: add esp,0Ch
0x100e3953: jmp near ptr 100E3A6Ah
0x100e3958: cmp ebx,1011h
0x100e395e: jne short 100E396Ah
0x100e3960: cmp esi,9
0x100e3963: je short 100E3972h
0x100e3965: jmp near ptr 100E3AB0h
0x100e396a: cmp ebx,20400200h
0x100e3970: jne short 100E39B6h
0x100e3972: lea ebx,[ebp+0E794h]
0x100e3978: mov dword ptr [esp+10h],3
0x100e3980: mov esi,ebx
0x100e3982: mov edi,3
0x100e3987: push ebp
0x100e3988: call 100D2F50h
0x100e398d: movsx eax,ax
0x100e3990: mov [esp+20h],eax
0x100e3994: add esp,4
0x100e3997: add esi,4
0x100e399a: sub edi,1
0x100e399d: fild dword ptr [esp+1Ch]
0x100e39a1: fmul qword ptr ds:[10127AA8h]
0x100e39a7: fstp dword ptr [esi-4]
0x100e39aa: jne short 100E3987h
0x100e39ac: add ebx,10h
0x100e39af: sub dword ptr [esp+10h],1
0x100e39b4: jne short 100E3980h
0x100e39b6: mov edi,[esp+20h]
0x100e39ba: cmp edi,1012h
0x100e39c0: je short 100E39CAh
0x100e39c2: cmp edi,20400600h
0x100e39c8: jne short 100E39FBh
0x100e39ca: mov esi,4
0x100e39cf: cmp [esp+18h],esi
0x100e39d3: jne short 100E39FBh
0x100e39d5: mov dword ptr [ebp+65F8h],0
0x100e39df: nop
0x100e39e0: push ebp
0x100e39e1: call 100D2F50h
0x100e39e6: movzx ecx,ax
0x100e39e9: add ecx,ecx
0x100e39eb: add ecx,ecx
0x100e39ed: add [ebp+65F8h],ecx
0x100e39f3: add esp,4
0x100e39f6: sub esi,1
0x100e39f9: jne short 100E39E0h
0x100e39fb: cmp edi,1017h
0x100e3a01: je short 100E3A0Bh
0x100e3a03: cmp edi,20400100h
0x100e3a09: jne short 100E3A2Bh
0x100e3a0b: push ebp
0x100e3a0c: call 100D2F50h
0x100e3a11: movzx edx,ax
0x100e3a14: mov [esp+20h],edx
0x100e3a18: add esp,4
0x100e3a1b: fild dword ptr [esp+1Ch]
0x100e3a1f: fmul qword ptr ds:[10127AA8h]
0x100e3a25: fstp dword ptr [ebp+0E774h]
0x100e3a2b: cmp edi,1018h
0x100e3a31: je short 100E3A3Bh
0x100e3a33: cmp edi,20400100h
0x100e3a39: jne short 100E3A5Bh
0x100e3a3b: push ebp
0x100e3a3c: call 100D2F50h
0x100e3a41: movzx eax,ax
0x100e3a44: mov [esp+20h],eax
```

```
0x100e3a48: add esp,4
0x100e3a4b: fild dword ptr [esp+1Ch]
0x100e3a4f: fmul qword ptr ds:[10127AA8h]
0x100e3a55: fstp dword ptr [ebp+0E77Ch]
0x100e3a5b: cmp edi,2011h
0x100e3a61: jne short 100E3AB0h
0x100e3a63: cmp dword ptr [esp+18h],2
0x100e3a68: jne short 100E3AB0h
0x100e3a6a: push ebp
0x100e3a6b: mov word ptr [ebp+6274h],4D4Dh
0x100e3a74: call 100D2F50h
0x100e3a79: movzx ecx,ax
0x100e3a7c: mov [esp+20h],ecx
0x100e3a80: push ebp
0x100e3a81: fild dword ptr [esp+24h]
0x100e3a85: fmul qword ptr ds:[10127AA8h]
0x100e3a8b: fstp dword ptr [ebp+0E774h]
0x100e3a91: call 100D2F50h
0x100e3a96: movzx edx,ax
0x100e3a99: mov [esp+24h],edx
0x100e3a9d: fild dword ptr [esp+24h]
0x100e3aa1: fmul qword ptr ds:[10127AA8h]
0x100e3aa7: fstp dword ptr [ebp+0E77Ch]
0x100e3aad: add esp,8
0x100e3ab0: mov edi,[esp+20h]
0x100e3ab4: mov eax,edi
0x100e3ab6: or eax,70h
0x100e3ab9: cmp eax,2070h
0x100e3abe: jne short 100E3AEDh
0x100e3ac0: cmp dword ptr [esp+14h],4
0x100e3ac5: jne short 100E3AEDh
0x100e3ac7: mov ebx,[ebp]
0x100e3aca: push 0
0x100e3acc: push ebp
0x100e3acd: call 100D2FE0h
0x100e3ad2: mov esi,[esp+1B8h]
0x100e3ad9: mov ecx,[ebp+4]
0x100e3adc: mov edx,[ebx+8]
0x100e3adf: add esp,4
0x100e3ae2: add eax,esi
0x100e3ae4: push eax
0x100e3ae5: push ecx
0x100e3ae6: call edx
0x100e3ae8: add esp,0Ch
0x100e3aeb: jmp short 100E3AF4h
0x100e3aed: mov esi,[esp+1B0h]
0x100e3af4: cmp edi,2010h
0x100e3afa: jne short 100E3B16h
0x100e3afc: cmp dword ptr [esp+14h],7
0x100e3b01: je near ptr 100E3C44h
0x100e3b07: mov dword ptr [ebp+2E7FCh],100D9680h
0x100e3b11: jmp near ptr 100E3C44h
0x100e3b16: cmp edi,2020h
0x100e3b1c: jne short 100E3B37h
0x100e3b1e: push 102h
0x100e3b23: push 101h
0x100e3b28: push esi
0x100e3b29: push ebp
0x100e3b2a: call 100E2AB0h
0x100e3b2f: add esp,10h
0x100e3b32: jmp near ptr 100E3C44h
0x100e3b37: cmp edi,2040h
0x100e3b3d: jne short 100E3B4Fh
0x100e3b3f: push edi
0x100e3b40: push esi
0x100e3b41: push ebp
0x100e3b42: call 100E2B50h
```

```
0x100e3b47: add esp,0Ch
0x100e3b4a: jmp near ptr 100E3C44h
0x100e3b4f: cmp edi,0B028h
0x100e3b55: jne short 100E3B88h
0x100e3b57: mov edi,[ebp]
0x100e3b5a: push 0
0x100e3b5c: push ebp
0x100e3b5d: call 100D2FE0h
0x100e3b62: mov ecx,[edi+8]
0x100e3b65: add esp,4
0x100e3b68: push eax
0x100e3b69: mov eax,[ebp+4]
0x100e3b6c: push eax
0x100e3b6d: call ecx
0x100e3b6f: push 89h
0x100e3b74: push 88h
0x100e3b79: push esi
0x100e3b7a: push ebp
0x100e3b7b: call 100E2AB0h
0x100e3b80: add esp,1Ch
0x100e3b83: jmp near ptr 100E3C44h
0x100e3b88: cmp edi,4001h
0x100e3b8e: jne near ptr 100E3C44h
0x100e3b94: cmp dword ptr [esp+18h],1F4h
0x100e3b9c: jbe near ptr 100E3C44h
0x100e3ba2: mov eax,[esp+18h]
0x100e3ba6: cmp eax,246h
0x100e3bab: jne short 100E3BB4h
0x100e3bad: mov eax,32h
0x100e3bb2: jmp short 100E3BD3h
0x100e3bb4: cmp eax,28Dh
0x100e3bb9: jne short 100E3BC2h
0x100e3bbb: mov eax,44h
0x100e3bc0: jmp short 100E3BD3h
0x100e3bc2: sub eax,1400h
0x100e3bc7: neg eax
0x100e3bc9: sbb eax,eax
0x100e3bcb: and eax,0FFFFFFF0h
0x100e3bce: add eax,8Eh
0x100e3bd3: mov edx,[ebp]
0x100e3bd6: mov ecx,[edx+8]
0x100e3bd9: push 1
0x100e3bdb: push eax
0x100e3bdc: mov eax,[ebp+4]
0x100e3bdf: push eax
0x100e3be0: call ecx
0x100e3be2: add esp,0Ch
0x100e3be5: xor esi,esi
0x100e3be7: push ebp
0x100e3be8: call 100D2F50h
0x100e3bed: movzx edx,ax
0x100e3bf0: mov eax,esi
0x100e3bf2: shr eax,1
0x100e3bf4: mov [esp+20h],edx
0x100e3bf8: xor eax,esi
0x100e3bfa: add esi,1
0x100e3bfd: fild dword ptr [esp+20h]
0x100e3c01: add esp,4
0x100e3c04: cmp esi,4
0x100e3c07: fstp dword ptr [ebp+eax*4+0E774h]
0x100e3c0e: jb short 100E3BE7h
0x100e3c10: mov ecx,[ebp]
0x100e3c13: mov edx,[ebp+4]
0x100e3c16: mov eax,[ecx+8]
0x100e3c19: push 1
0x100e3c1b: push 16h
0x100e3c1d: push edx
```

```
0x100e3c1e: call eax
0x100e3c20: add esp,0Ch
0x100e3c23: xor esi,esi
0x100e3c25: push ebp
0x100e3c26: call 100D2F50h
0x100e3c2b: mov ecx,esi
0x100e3c2d: shr ecx,1
0x100e3c2f: xor ecx,esi
0x100e3c31: add esi,1
0x100e3c34: add esp,4
0x100e3c37: cmp esi,4
0x100e3c3a: mov [ebp+ecx*2+0E76Ch],ax
0x100e3c42: jb short 100E3C25h
0x100e3c44: mov eax,[esp+40h]
0x100e3c48: mov edx,[ebp]
0x100e3c4b: mov ecx,[ebp+4]
0x100e3c4e: mov edx,[edx+8]
0x100e3c51: push 0
0x100e3c53: push eax
0x100e3c54: push ecx
0x100e3c55: call edx
0x100e3c57: mov eax,[esp+40h]
0x100e3c5b: add esp,0Ch
0x100e3c5e: test eax,eax
0x100e3c60: jne near ptr 100E2E40h
0x100e3c66: jmp near ptr 100E3D3Ah
0x100e3c6b: mov eax,[ebp]
0x100e3c6e: mov ecx,[ebp+4]
0x100e3c71: mov word ptr [ebp+6274h],4D4Dh
0x100e3c7a: mov edx,[eax+18h]
0x100e3c7d: push ecx
0x100e3c7e: call edx
0x100e3c80: add esp,4
0x100e3c83: cmp eax,[ebp+65C0h]
0x100e3c89: jge near ptr 100E3D3Ah
0x100e3c8f: nop
0x100e3c90: cmp eax,4000h
0x100e3c95: jae near ptr 100E3D3Ah
0x100e3c9b: mov [esp+44h],esi
0x100e3c9f: mov esi,edi
0x100e3ca1: push ebp
0x100e3ca2: mov edi,ebx
0x100e3ca4: call 100D2F50h
0x100e3ca9: add esp,4
0x100e3cac: cmp edi,100h
0x100e3cb2: movzx ebx,ax
0x100e3cb5: jne short 100E3D1Fh
0x100e3cb7: cmp ebx,edi
0x100e3cb9: jne short 100E3D1Fh
0x100e3cbb: mov eax,[esp+44h]
0x100e3cbf: lea ecx,[eax-101h]
0x100e3cc5: cmp ecx,17Eh
0x100e3ccb: ja short 100E3D1Fh
0x100e3ccd: lea edx,[esi-101h]
0x100e3cd3: cmp edx,17Eh
0x100e3cd9: ja short 100E3D1Fh
0x100e3cdb: test eax,eax
0x100e3cdd: mov [esp+34h],eax
0x100e3ce1: fild dword ptr [esp+34h]
0x100e3ce5: jge short 100E3CEDh
0x100e3ce7: fadd dword ptr ds:[100FE7C0h]
0x100e3ced: fstp dword ptr [ebp+0E774h]
0x100e3cf3: mov eax,esi
0x100e3cf5: test eax,eax
0x100e3cf7: fld dword ptr ds:[1012E4F4h]
0x100e3cfd: mov [esp+34h],eax
0x100e3d01: fst dword ptr [ebp+0E778h]
```

```
0x100e3d07: fild dword ptr [esp+34h]
0x100e3d0b: jge short 100E3D13h
0x100e3d0d: fadd dword ptr ds:[100FE7C0h]
0x100e3d13: fstp dword ptr [ebp+0E77Ch]
0x100e3d19: fstp dword ptr [ebp+0E780h]
0x100e3d1f: mov ecx,[ebp]
0x100e3d22: mov edx,[ebp+4]
0x100e3d25: mov eax,[ecx+18h]
0x100e3d28: push edx
0x100e3d29: call eax
0x100e3d2b: add esp,4
0x100e3d2e: cmp eax,[ebp+65C0h]
0x100e3d34: jl near ptr 100E3C90h
0x100e3d3a: mov cx,[esp+3Ch]
0x100e3d3f: mov [ebp+6274h],cx
0x100e3d46: mov ecx,[esp+1A4h]
0x100e3d4d: pop edi
0x100e3d4e: pop esi
0x100e3d4f: pop ebp
0x100e3d50: pop ebx
0x100e3d51: xor ecx,esp
0x100e3d53: call 10022326h
0x100e3d58: add esp,198h
0x100e3d5e: ret
```

`sub_100e2b50` is called by sub_100e2b50, sub_100e3e60. It works with the text "PENTAX", "EPSON", "NIKON". It accesses memory-mapped peripheral registers at 0x61616161. It delegates to sub_10022326, j__CIpow, sub_100d2f20 (11 callees).

12.1287 sub_100e3d60 at 0x100e3d60 (86 instructions)

```
0x100e3d60: sub esp,3Ch
0x100e3d63: mov eax,ds:[10136220h]
0x100e3d68: xor eax,esp
0x100e3d6a: mov [esp+38h],eax
0x100e3d6e: cmp dword ptr [esp+44h],0
0x100e3d73: push esi
0x100e3d74: push edi
0x100e3d75: mov edi,[esp+48h]
0x100e3d79: mov byte ptr [esp+3Fh],0
0x100e3d7e: je short 100E3DA0h
0x100e3d80: mov esi,13h
0x100e3d85: mov eax,[edi]
0x100e3d87: mov ecx,[edi+4]
0x100e3d8a: mov edx,[eax+1Ch]
0x100e3d8d: push ecx
0x100e3d8e: sub esi,1
0x100e3d91: call edx
0x100e3d93: add esp,4
0x100e3d96: test esi,esi
0x100e3d98: mov [esp+esi+2Ch],al
0x100e3d9c: jne short 100E3D85h
0x100e3d9e: jmp short 100E3DB6h
0x100e3da0: mov eax,[edi]
0x100e3da2: mov edx,[edi+4]
0x100e3da5: mov eax,[eax]
0x100e3da7: push 1
0x100e3da9: push 13h
0x100e3dab: lea ecx,[esp+34h]
0x100e3daf: push ecx
0x100e3db0: push edx
0x100e3db1: call eax
0x100e3db3: add esp,10h
0x100e3db6: xor eax,eax
0x100e3db8: lea ecx,[esp+8]
0x100e3dbc: push ecx
```

```
0x100e3dbd: mov [esp+0Ch],eax
0x100e3dc1: mov [esp+10h],eax
0x100e3dc5: mov [esp+14h],eax
0x100e3dc9: mov [esp+18h],eax
0x100e3dcd: mov [esp+1Ch],eax
0x100e3dd1: mov [esp+20h],eax
0x100e3dd5: mov [esp+24h],eax
0x100e3dd9: mov [esp+28h],eax
0x100e3ddd: mov [esp+2Ch],eax
0x100e3de1: lea edx,[esp+10h]
0x100e3de5: push edx
0x100e3de6: lea eax,[esp+18h]
0x100e3dea: push eax
0x100e3deb: lea ecx,[esp+20h]
0x100e3def: push ecx
0x100e3df0: lea edx,[esp+28h]
0x100e3df4: push edx
0x100e3df5: lea eax,[esp+30h]
0x100e3df9: push eax
0x100e3dfa: lea ecx,[esp+44h]
0x100e3dfe: push 1012E5B0h
0x100e3e03: push ecx
0x100e3e04: call dword ptr ds:[100FE4A8h]
0x100e3e0a: add esp,20h
0x100e3e0d: cmp eax,6
0x100e3e10: jne short 100E3E4Fh
0x100e3e12: mov esi,ds:[100FE444h]
0x100e3e18: sub dword ptr [esp+1Ch],76Ch
0x100e3e20: sub dword ptr [esp+18h],1
0x100e3e25: lea edx,[esp+8]
0x100e3e29: push edx
0x100e3e2a: call esi
0x100e3e2c: add esp,4
0x100e3e2f: test edx,edx
0x100e3e31: jl short 100E3E4Fh
0x100e3e33: jg short 100E3E39h
0x100e3e35: test eax,eax
0x100e3e37: jbe short 100E3E4Fh
0x100e3e39: lea eax,[esp+8]
0x100e3e3d: push eax
0x100e3e3e: call esi
0x100e3e40: add esp,4
0x100e3e43: mov [edi+65A0h],eax
0x100e3e49: mov [edi+65A4h],edx
0x100e3e4f: mov ecx,[esp+40h]
0x100e3e53: pop edi
0x100e3e54: pop esi
0x100e3e55: xor ecx,esp
0x100e3e57: call 10022326h
0x100e3e5c: add esp,3Ch
0x100e3e5f: ret
```

`sub_100e3d60` is called by sub_100e3e60, sub_100e5c30, sub_100eb620. It works with the text "%d:%d:%d %d:%d:%d". It delegates to sub_10022326 (1 callees).

12.1288 sub_100e3e60 at 0x100e3e60 (186 instructions)

```
0x100e3e60: sub esp,10h
0x100e3e63: push ebx
0x100e3e64: push esi
0x100e3e65: mov esi,[esp+1Ch]
0x100e3e69: push edi
0x100e3e6a: push 7
0x100e3e6c: lea eax,[esi+6485h]
0x100e3e72: push 1012E5C4h
0x100e3e77: push eax
```

```
0x100e3e78: call dword ptr ds:[100FE414h]
0x100e3e7e: mov ebx,eax
0x100e3e80: neg ebx
0x100e3e82: sbb ebx,ebx
0x100e3e84: push esi
0x100e3e85: add ebx,1
0x100e3e88: call 100D2F50h
0x100e3e8d: movzx edi,ax
0x100e3e90: add esp,10h
0x100e3e93: test edi,edi
0x100e3e95: je near ptr 100E40C2h
0x100e3e9b: push ebp
0x100e3e9c: mov ebp,[esp+28h]
0x100e3ea0: lea ecx,[esp+18h]
0x100e3ea4: push ecx
0x100e3ea5: lea edx,[esp+20h]
0x100e3ea9: push edx
0x100e3eaa: lea eax,[esp+2Ch]
0x100e3eae: push eax
0x100e3eaf: lea ecx,[esp+1Ch]
0x100e3eb3: push ecx
0x100e3eb4: push ebp
0x100e3eb5: sub edi,1
0x100e3eb8: push esi
0x100e3eb9: mov [esp+2Ch],edi
0x100e3ebd: call 100E2A20h
0x100e3ec2: mov eax,[esp+28h]
0x100e3ec6: add esp,18h
0x100e3ec9: cmp eax,9202h
0x100e3ece: ja near ptr 100E3FD8h
0x100e3ed4: je near ptr 100E3FACh
0x100e3eda: cmp eax,8827h
0x100e3edf: ja short 100E3F44h
0x100e3ee1: je short 100E3F25h
0x100e3ee3: sub eax,829Ah
0x100e3ee8: je short 100E3F0Ch
0x100e3eea: sub eax,3
0x100e3eed: jne near ptr 100E40A4h
0x100e3ef3: mov edx,[esp+24h]
0x100e3ef7: push edx
0x100e3ef8: push esi
0x100e3ef9: call 100D3060h
0x100e3efe: fstp dword ptr [esi+6598h]
0x100e3f04: add esp,8
0x100e3f07: jmp near ptr 100E40A4h
0x100e3f0c: mov eax,[esp+24h]
0x100e3f10: push eax
0x100e3f11: push esi
0x100e3f12: call 100D3060h
0x100e3f17: fstp dword ptr [esi+6594h]
0x100e3f1d: add esp,8
0x100e3f20: jmp near ptr 100E40A4h
0x100e3f25: push esi
0x100e3f26: call 100D2F50h
0x100e3f2b: movzx ecx,ax
0x100e3f2e: mov [esp+18h],ecx
0x100e3f32: add esp,4
0x100e3f35: fild dword ptr [esp+14h]
0x100e3f39: fstp dword ptr [esi+6590h]
0x100e3f3f: jmp near ptr 100E40A4h
0x100e3f44: cmp eax,9003h
0x100e3f49: jb near ptr 100E40A4h
0x100e3f4f: cmp eax,9004h
0x100e3f54: jbe short 100E3F9Ch
0x100e3f56: cmp eax,9201h
0x100e3f5b: jne near ptr 100E40A4h
0x100e3f61: mov edx,[esp+24h]
```

```
0x100e3f65: push edx
0x100e3f66: push esi
0x100e3f67: call 100D3060h
0x100e3f6c: fchs
0x100e3f6e: fld qword ptr ds:[100FF370h]
0x100e3f74: add esp,8
0x100e3f77: fcomp
0x100e3f79: fnstsw ax
0x100e3f7b: test ah,41h
0x100e3f7e: jne near ptr 100E40A2h
0x100e3f84: fld qword ptr ds:[100FF110h]
0x100e3f8a: fxch
0x100e3f8c: call 100229A0h
0x100e3f91: fstp dword ptr [esi+6594h]
0x100e3f97: jmp near ptr 100E40A4h
0x100e3f9c: push 0
0x100e3f9e: push esi
0x100e3f9f: call 100E3D60h
0x100e3fa4: add esp,8
0x100e3fa7: jmp near ptr 100E40A4h
0x100e3fac: mov eax,[esp+24h]
0x100e3fb0: push eax
0x100e3fb1: push esi
0x100e3fb2: call 100D3060h
0x100e3fb7: fmul qword ptr ds:[100FE7C8h]
0x100e3fbd: fld qword ptr ds:[100FF110h]
0x100e3fc3: add esp,8
0x100e3fc6: fxch
0x100e3fc8: call 100229A0h
0x100e3fcd: fstp dword ptr [esi+6598h]
0x100e3fd3: jmp near ptr 100E40A4h
0x100e3fd8: cmp eax,0A002h
0x100e3fdd: ja short 100E4034h
0x100e3fdf: je short 100E401Ah
0x100e3fe1: sub eax,920Ah
0x100e3fe6: je short 100E4001h
0x100e3fe8: sub eax,72h
0x100e3feb: jne near ptr 100E40A4h
0x100e3ff1: push eax
0x100e3ff2: push ebp
0x100e3ff3: push esi
0x100e3ff4: call 100E2B50h
0x100e3ff9: add esp,0Ch
0x100e3ffc: jmp near ptr 100E40A4h
0x100e4001: mov ecx,[esp+24h]
0x100e4005: push ecx
0x100e4006: push esi
0x100e4007: call 100D3060h
0x100e400c: fstp dword ptr [esi+659Ch]
0x100e4012: add esp,8
0x100e4015: jmp near ptr 100E40A4h
0x100e401a: test ebx,ebx
0x100e401c: je near ptr 100E40A4h
0x100e4022: push esi
0x100e4023: call 100D2FE0h
0x100e4028: add esp,4
0x100e402b: mov [esi+66B2h],ax
0x100e4032: jmp short 100E40A4h
0x100e4034: cmp eax,0A003h
0x100e4039: je short 100E408Ch
0x100e403b: cmp eax,0A302h
0x100e4040: jne short 100E40A4h
0x100e4042: push esi
0x100e4043: call 100D2FE0h
0x100e4048: add esp,4
0x100e404b: cmp eax,20002h
0x100e4050: jne short 100E40A4h
```

```
0x100e4052: xor edi,edi
0x100e4054: mov [esi+65B4h],edi
0x100e405a: lea ebx,[ebx]
0x100e4060: mov edx,[esi]
0x100e4062: mov eax,[esi+4]
0x100e4065: mov ecx,[edx+1Ch]
0x100e4068: push eax
0x100e4069: call ecx
0x100e406b: imul eax,1010101h
0x100e4071: mov ecx,edi
0x100e4073: shl eax,cl
0x100e4075: add edi,2
0x100e4078: add esp,4
0x100e407b: or [esi+65B4h],eax
0x100e4081: cmp edi,8
0x100e4084: jb short 100E4060h
0x100e4086: mov edi,[esp+14h]
0x100e408a: jmp short 100E40A4h
0x100e408c: test ebx,ebx
0x100e408e: je short 100E40A4h
0x100e4090: push esi
0x100e4091: call 100D2FE0h
0x100e4096: add esp,4
0x100e4099: mov [esi+66B0h],ax
0x100e40a0: jmp short 100E40A4h
0x100e40a2: fstp st(0)
0x100e40a4: mov eax,[esp+18h]
0x100e40a8: mov edx,[esi]
0x100e40aa: mov ecx,[esi+4]
0x100e40ad: mov edx,[edx+8]
0x100e40b0: push 0
0x100e40b2: push eax
0x100e40b3: push ecx
0x100e40b4: call edx
0x100e40b6: add esp,0Ch
0x100e40b9: test edi,edi
0x100e40bb: jne near ptr 100E3EA0h
0x100e40c1: pop ebp
0x100e40c2: pop edi
0x100e40c3: pop esi
0x100e40c4: pop ebx
0x100e40c5: add esp,10h
0x100e40c8: ret
```

`sub_100e3e60` is called by sub_100e8a00. It works with the text "EASTMAN". It delegates to j__CIpow, sub_100d2f50, sub_100d2fe0 (7 callees).

12.1289 sub_100e40d0 at 0x100e40d0 (50 instructions)

```
0x100e40d0: sub esp,10h
0x100e40d3: push ebp
0x100e40d4: push esi
0x100e40d5: mov esi,[esp+1Ch]
0x100e40d9: push esi
0x100e40da: call 100D2F50h
0x100e40df: movzx ebp,ax
0x100e40e2: add esp,4
0x100e40e5: test ebp,ebp
0x100e40e7: je near ptr 100E41E6h
0x100e40ed: push ebx
0x100e40ee: push edi
0x100e40ef: nop
0x100e40f0: lea eax,[esp+18h]
0x100e40f4: push eax
0x100e40f5: lea ecx,[esp+18h]
0x100e40f9: push ecx
```

```
0x100e40fa: mov ecx,[esp+30h]
0x100e40fe: lea edx,[esp+24h]
0x100e4102: push edx
0x100e4103: lea eax,[esp+1Ch]
0x100e4107: push eax
0x100e4108: push ecx
0x100e4109: push esi
0x100e410a: sub ebp,1
0x100e410d: call 100E2A20h
0x100e4112: mov edi,[esp+28h]
0x100e4116: lea eax,[edi-1]
0x100e4119: add esp,18h
0x100e411c: cmp eax,1Ch
0x100e411f: ja near ptr 100E41C7h
0x100e4125: movzx edx,byte ptr [eax+100E4200h]
0x100e412c: jmp dword ptr [edx*4+100E41ECh]
0x100e41c7: mov eax,[esp+18h]
0x100e41cb: mov edx,[esi]
0x100e41cd: mov ecx,[esi+4]
0x100e41d0: mov edx,[edx+8]
0x100e41d3: push 0
0x100e41d5: push eax
0x100e41d6: push ecx
0x100e41d7: call edx
0x100e41d9: add esp,0Ch
0x100e41dc: test ebp,ebp
0x100e41de: jne near ptr 100E40F0h
0x100e41e4: pop edi
0x100e41e5: pop ebx
0x100e41e6: pop esi
0x100e41e7: pop ebp
0x100e41e8: add esp,10h
0x100e41eb: ret
```

`sub_100e40d0` is called by sub_100e8a00. It delegates to sub_100d2f50, sub_100e2a20 (2 callees).

12.1290 sub_100e4220 at 0x100e4220 (54 instructions)

```
0x100e4220: push ecx
0x100e4221: fldz
0x100e4223: push ebx
0x100e4224: mov ebx,[esp+10h]
0x100e4228: fldz
0x100e422a: push esi
0x100e422b: push edi
0x100e422c: mov edi,[esp+14h]
0x100e4230: add ebx,18h
0x100e4233: add edi,0E794h
0x100e4239: mov edx,1012AC58h
0x100e423e: mov eax,edi
0x100e4240: mov ecx,ebx
0x100e4242: mov esi,3
0x100e4247: jmp short 100E4258h
0x100e4249: fxch
0x100e424b: mov eax,edi
0x100e424d: mov ecx,ebx
0x100e424f: mov esi,3
0x100e4254: jmp short 100E4258h
0x100e4256: fxch
0x100e4258: fxch
0x100e425a: add ecx,4
0x100e425d: fst dword ptr [eax]
0x100e425f: add eax,4
0x100e4262: sub esi,1
0x100e4265: fld dword ptr [ecx-1Ch]
0x100e4268: fmul dword ptr [edx-4]
```

```
0x100e426b: fadd st,st(2)
0x100e426d: fstp dword ptr [esp+0Ch]
0x100e4271: fld dword ptr [esp+0Ch]
0x100e4275: fst dword ptr [eax-4]
0x100e4278: fld dword ptr [ecx-10h]
0x100e427b: fmul dword ptr [edx]
0x100e427d: faddp
0x100e427f: fstp dword ptr [esp+0Ch]
0x100e4283: fld dword ptr [esp+0Ch]
0x100e4287: fst dword ptr [eax-4]
0x100e428a: fld dword ptr [ecx-4]
0x100e428d: fmul dword ptr [edx+4]
0x100e4290: faddp
0x100e4292: fstp dword ptr [eax-4]
0x100e4295: jne short 100E4256h
0x100e4297: add edx,0Ch
0x100e429a: add edi,10h
0x100e429d: cmp edx,1012AC7Ch
0x100e42a3: jl short 100E4249h
0x100e42a5: pop edi
0x100e42a6: fstp st(0)
0x100e42a8: pop esi
0x100e42a9: fstp st(0)
0x100e42ab: pop ebx
0x100e42ac: pop ecx
0x100e42ad: ret
```

`sub_100e4220` is called by sub_100e42b0. It calls nothing else.

12.1291 sub_100e42b0 at 0x100e42b0 (312 instructions)

```
0x100e42b0: sub esp,70h
0x100e42b3: mov eax,ds:[10136220h]
0x100e42b8: xor eax,esp
0x100e42ba: mov [esp+6Ch],eax
0x100e42be: mov ecx,[esp+78h]
0x100e42c2: push ebp
0x100e42c3: mov ebp,[esp+78h]
0x100e42c7: mov eax,[ebp]
0x100e42ca: mov edx,[ebp+4]
0x100e42cd: push esi
0x100e42ce: push edi
0x100e42cf: xor edi,edi
0x100e42d1: push edi
0x100e42d2: push ecx
0x100e42d3: mov [esp+14h],edi
0x100e42d7: mov [esp+20h],edi
0x100e42db: mov eax,[eax+8]
0x100e42de: push edx
0x100e42df: mov [esp+20h],edi
0x100e42e3: call eax
0x100e42e5: push ebp
0x100e42e6: call 100D2FE0h
0x100e42eb: add esp,10h
0x100e42ee: cmp eax,504B5453h
0x100e42f3: jne near ptr 100E4620h
0x100e42f9: push ebx
0x100e42fa: lea ebx,[ebx]
0x100e4300: push ebp
0x100e4301: call 100D2FE0h
0x100e4306: mov ecx,[ebp]
0x100e4309: mov eax,[ebp+4]
0x100e430c: mov ecx,[ecx]
0x100e430e: push 28h
0x100e4310: push 1
0x100e4312: lea edx,[esp+60h]
```

```
0x100e4316: push edx
0x100e4317: push eax
0x100e4318: call ecx
0x100e431a: push ebp
0x100e431b: call 100D2FE0h
0x100e4320: mov edx,[ebp]
0x100e4323: mov ecx,[edx+18h]
0x100e4326: mov [esp+2Ch],eax
0x100e432a: mov eax,[ebp+4]
0x100e432d: push eax
0x100e432e: call ecx
0x100e4330: add esp,1Ch
0x100e4333: mov edi,1012E6B0h
0x100e4338: lea esi,[esp+54h]
0x100e433c: mov ecx,12h
0x100e4341: xor edx,edx
0x100e4343: repe cmpsb
0x100e4345: mov ebx,eax
0x100e4347: jne short 100E4359h
0x100e4349: mov eax,[esp+14h]
0x100e434d: mov [ebp+65C4h],ebx
0x100e4353: mov [ebp+65D0h],eax
0x100e4359: mov edi,1012E69Ch
0x100e435e: lea esi,[esp+54h]
0x100e4362: mov ecx,13h
0x100e4367: xor edx,edx
0x100e4369: repe cmpsb
0x100e436b: jne short 100E437Dh
0x100e436d: mov eax,[esp+14h]
0x100e4371: mov [ebp+65CCh],ebx
0x100e4377: mov [ebp+65D8h],eax
0x100e437d: mov edi,1012E688h
0x100e4382: lea esi,[esp+54h]
0x100e4386: mov ecx,13h
0x100e438b: xor edx,edx
0x100e438d: repe cmpsb
0x100e438f: jne short 100E43CEh
0x100e4391: mov eax,[ebp]
0x100e4394: mov edx,[ebp+4]
0x100e4397: mov eax,[eax+20h]
0x100e439a: lea ecx,[esp+10h]
0x100e439e: push ecx
0x100e439f: push 1010BCACh
0x100e43a4: push edx
0x100e43a5: call eax
0x100e43a7: mov eax,[esp+1Ch]
0x100e43ab: add esp,0Ch
0x100e43ae: cmp eax,15h
0x100e43b1: jae short 100E43CEh
0x100e43b3: mov ecx,[eax*4+10138988h]
0x100e43ba: lea edx,[ebp+64C5h]
0x100e43c0: mov al,[ecx]
0x100e43c2: mov [edx],al
0x100e43c4: add ecx,1
0x100e43c7: add edx,1
0x100e43ca: test al,al
0x100e43cc: jne short 100E43C0h
0x100e43ce: mov edi,1012E66Ch
0x100e43d3: lea esi,[esp+54h]
0x100e43d7: mov ecx,1Ah
0x100e43dc: xor edx,edx
0x100e43de: repe cmpsb
0x100e43e0: jne short 100E4417h
0x100e43e2: mov [esp+10h],edx
0x100e43e6: push ebp
0x100e43e7: call 100D2FE0h
0x100e43ec: push eax
```

```
0x100e43ed: call 100D3050h
0x100e43f2: mov eax,[esp+18h]
0x100e43f6: fstp dword ptr [esp+eax*4+38h]
0x100e43fa: add eax,1
0x100e43fd: add esp,8
0x100e4400: cmp eax,9
0x100e4403: mov [esp+10h],eax
0x100e4407: jl short 100E43E6h
0x100e4409: lea eax,[esp+30h]
0x100e440d: push eax
0x100e440e: push ebp
0x100e440f: call 100E4220h
0x100e4414: add esp,8
0x100e4417: mov edi,1012E654h
0x100e441c: lea esi,[esp+54h]
0x100e4420: mov ecx,16h
0x100e4425: xor edx,edx
0x100e4427: repe cmpsb
0x100e4429: jne short 100E4468h
0x100e442b: xor eax,eax
0x100e442d: mov [esp+10h],eax
0x100e4431: mov ecx,[ebp]
0x100e4434: mov ecx,[ecx+20h]
0x100e4437: lea edx,[esp+eax*4+30h]
0x100e443b: mov eax,[ebp+4]
0x100e443e: push edx
0x100e443f: push 1012E650h
0x100e4444: push eax
0x100e4445: call ecx
0x100e4447: mov eax,[esp+1Ch]
0x100e444b: add eax,1
0x100e444e: add esp,0Ch
0x100e4451: cmp eax,9
0x100e4454: mov [esp+10h],eax
0x100e4458: jl short 100E4431h
0x100e445a: lea edx,[esp+30h]
0x100e445e: push edx
0x100e445f: push ebp
0x100e4460: call 100E4220h
0x100e4465: add esp,8
0x100e4468: mov edi,1012E634h
0x100e446d: lea esi,[esp+54h]
0x100e4471: mov ecx,1Ah
0x100e4476: xor eax,eax
0x100e4478: repe cmpsb
0x100e447a: jne short 100E4495h
0x100e447c: mov ecx,[ebp]
0x100e447f: mov eax,[ebp+4]
0x100e4482: mov ecx,[ecx+20h]
0x100e4485: lea edx,[esp+1Ch]
0x100e4489: push edx
0x100e448a: push 1010BCACh
0x100e448f: push eax
0x100e4490: call ecx
0x100e4492: add esp,0Ch
0x100e4495: mov edi,1012E618h
0x100e449a: lea esi,[esp+54h]
0x100e449e: mov ecx,1Bh
0x100e44a3: xor edx,edx
0x100e44a5: repe cmpsb
0x100e44a7: jne short 100E44C4h
0x100e44a9: mov eax,[ebp]
0x100e44ac: mov edx,[ebp+4]
0x100e44af: mov eax,[eax+20h]
0x100e44b2: lea ecx,[ebp+66C8h]
0x100e44b8: push ecx
0x100e44b9: push 1010BCACh
```

```
0x100e44be: push edx
0x100e44bf: call eax
0x100e44c1: add esp,0Ch
0x100e44c4: mov edi,1012E600h
0x100e44c9: lea esi,[esp+54h]
0x100e44cd: mov ecx,18h
0x100e44d2: xor edx,edx
0x100e44d4: repe cmpsb
0x100e44d6: jne short 100E4512h
0x100e44d8: xor esi,esi
0x100e44da: lea ebx,[ebx]
0x100e44e0: mov eax,[ebp]
0x100e44e3: mov edx,[ebp+4]
0x100e44e6: mov eax,[eax+20h]
0x100e44e9: lea ecx,[esp+10h]
0x100e44ed: push ecx
0x100e44ee: push 1010BCACh
0x100e44f3: push edx
0x100e44f4: call eax
0x100e44f6: add esp,0Ch
0x100e44f9: cmp dword ptr [esp+10h],1
0x100e44fe: jne short 100E450Ah
0x100e4500: mov eax,esi
0x100e4502: sar eax,1
0x100e4504: xor eax,esi
0x100e4506: mov [esp+18h],eax
0x100e450a: add esi,1
0x100e450d: cmp esi,4
0x100e4510: jl short 100E44E0h
0x100e4512: mov edi,1012E5E8h
0x100e4517: lea esi,[esp+54h]
0x100e451b: mov ecx,17h
0x100e4520: xor edx,edx
0x100e4522: repe cmpsb
0x100e4524: jne short 100E454Fh
0x100e4526: mov eax,[ebp]
0x100e4529: mov edx,[ebp+4]
0x100e452c: mov eax,[eax+20h]
0x100e452f: lea ecx,[esp+10h]
0x100e4533: push ecx
0x100e4534: push 1010BCACh
0x100e4539: push edx
0x100e453a: call eax
0x100e453c: mov ecx,[esp+1Ch]
0x100e4540: add esp,0Ch
0x100e4543: sub ecx,[ebp+66C8h]
0x100e4549: mov [ebp+66C8h],ecx
0x100e454f: mov edi,1012E5D4h
0x100e4554: lea esi,[esp+54h]
0x100e4558: mov ecx,11h
0x100e455d: xor edx,edx
0x100e455f: repe cmpsb
0x100e4561: jne near ptr 100E45EDh
0x100e4567: fldz
0x100e4569: fcomp dword ptr [ebp+0E774h]
0x100e456f: fnstsw ax
0x100e4571: test ah,44h
0x100e4574: jp short 100E45EDh
0x100e4576: lea esi,[esp+20h]
0x100e457a: lea edi,[edx+4]
0x100e457d: lea ecx,[ecx]
0x100e4580: mov eax,[ebp]
0x100e4583: mov ecx,[ebp+4]
0x100e4586: mov edx,[eax+20h]
0x100e4589: push esi
0x100e458a: push 1010BCACh
0x100e458f: push ecx
```

```
0x100e4590: call edx
0x100e4592: add esp,0Ch
0x100e4595: add esi,4
0x100e4598: sub edi,1
0x100e459b: jne short 100E4580h
0x100e459d: cmp [esp+24h],edi
0x100e45a1: je short 100E45AFh
0x100e45a3: fild dword ptr [esp+20h]
0x100e45a7: fidiv dword ptr [esp+24h]
0x100e45ab: fldz
0x100e45ad: jmp short 100E45B5h
0x100e45af: fldz
0x100e45b1: fld st(0)
0x100e45b3: fxch
0x100e45b5: cmp dword ptr [esp+28h],0
0x100e45ba: fxch
0x100e45bc: fstp dword ptr [ebp+0E774h]
0x100e45c2: je short 100E45CEh
0x100e45c4: fild dword ptr [esp+20h]
0x100e45c8: fidiv dword ptr [esp+28h]
0x100e45cc: jmp short 100E45D0h
0x100e45ce: fld st(0)
0x100e45d0: cmp dword ptr [esp+2Ch],0
0x100e45d5: fstp dword ptr [ebp+0E778h]
0x100e45db: je short 100E45E7h
0x100e45dd: fstp st(0)
0x100e45df: fild dword ptr [esp+20h]
0x100e45e3: fidiv dword ptr [esp+2Ch]
0x100e45e7: fstp dword ptr [ebp+0E77Ch]
0x100e45ed: push ebx
0x100e45ee: push ebp
0x100e45ef: call 100E42B0h
0x100e45f4: mov ecx,[esp+1Ch]
0x100e45f8: mov eax,[ebp]
0x100e45fb: mov edx,[ebp+4]
0x100e45fe: mov eax,[eax+8]
0x100e4601: push 0
0x100e4603: add ebx,ecx
0x100e4605: push ebx
0x100e4606: push edx
0x100e4607: call eax
0x100e4609: push ebp
0x100e460a: call 100D2FE0h
0x100e460f: add esp,18h
0x100e4612: cmp eax,504B5453h
0x100e4617: je near ptr 100E4300h
0x100e461d: xor edi,edi
0x100e461f: pop ebx
0x100e4620: mov esi,[esp+18h]
0x100e4624: cmp esi,edi
0x100e4626: je short 100E4666h
0x100e4628: mov ecx,[ebp+66C8h]
0x100e462e: mov eax,0B60B60B7h
0x100e4633: imul ecx
0x100e4635: add edx,ecx
0x100e4637: sar edx,6
0x100e463a: mov ecx,edx
0x100e463c: shr ecx,1Fh
0x100e463f: add ecx,edx
0x100e4641: add ecx,[esp+14h]
0x100e4645: xor eax,eax
0x100e4647: and ecx,3
0x100e464a: movzx edx,byte ptr [ecx+1012E5CCh]
0x100e4651: cmp esi,1
0x100e4654: sete al
0x100e4657: imul edx,eax
0x100e465a: imul edx,1010101h
```

```
0x100e4660: mov [ebp+65B0h],edx
0x100e4666: mov ecx,[esp+78h]
0x100e466a: pop edi
0x100e466b: pop esi
0x100e466c: pop ebp
0x100e466d: xor ecx,esp
0x100e466f: call 10022326h
0x100e4674: add esp,70h
0x100e4677: ret
```

`sub_100e42b0` is called by sub_100e42b0, sub_100e8a00. It works with the text "NeutObj_neutrals", "icc_camera_to_tone_matrix", "ShootObj_back_type". It accesses memory-mapped peripheral registers at 0x504b5453. It delegates to sub_10022326, sub_100d2fe0, sub_100d3050 (5 callees).

12.1292 sub_100e4940 at 0x100e4940 (73 instructions)

```
0x100e4940: sub esp,10h
0x100e4943: push ebx
0x100e4944: push esi
0x100e4945: push edi
0x100e4946: mov edi,[esp+20h]
0x100e494a: push edi
0x100e494b: xor ebx,ebx
0x100e494d: xor esi,esi
0x100e494f: call 100D2F50h
0x100e4954: push edi
0x100e4955: call 100D2FE0h
0x100e495a: add esp,8
0x100e495d: cmp eax,80008h
0x100e4962: jne near ptr 100E4A21h
0x100e4968: push edi
0x100e4969: call 100D2FE0h
0x100e496e: add esp,4
0x100e4971: test eax,eax
0x100e4973: je near ptr 100E4A21h
0x100e4979: push ebp
0x100e497a: push edi
0x100e497b: call 100D2F50h
0x100e4980: movzx ebp,ax
0x100e4983: add esp,4
0x100e4986: cmp ebp,0Ah
0x100e4989: je short 100E4994h
0x100e498b: cmp ebp,0Ch
0x100e498e: jne near ptr 100E4A20h
0x100e4994: mov eax,20h
0x100e4999: sub eax,ebp
0x100e499b: mov [esp+10h],ebx
0x100e499f: mov [esp+1Ch],eax
0x100e49a3: add edi,66E4h
0x100e49a9: mov dword ptr [esp+18h],8
0x100e49b1: mov dword ptr [esp+14h],8
0x100e49b9: lea esp,[esp]
0x100e49c0: cmp esi,ebp
0x100e49c2: jge short 100E49FAh
0x100e49c4: mov eax,[esp+24h]
0x100e49c8: push eax
0x100e49c9: call 100D2F50h
0x100e49ce: movzx ecx,ax
0x100e49d1: mov eax,[esp+14h]
0x100e49d5: mov edx,eax
0x100e49d7: and edx,1
0x100e49da: movzx edx,word ptr [edx*2+1012B8F0h]
0x100e49e2: add eax,1
0x100e49e5: xor ecx,edx
0x100e49e7: shl ebx,10h
0x100e49ea: add esp,4
```

```
0x100e49ed: or ebx,ecx
0x100e49ef: mov [esp+10h],eax
0x100e49f3: mov eax,[esp+1Ch]
0x100e49f7: add esi,10h
0x100e49fa: mov ecx,20h
0x100e49ff: sub ecx,esi
0x100e4a01: mov edx,ebx
0x100e4a03: shl edx,cl
0x100e4a05: mov ecx,eax
0x100e4a07: sub esi,ebp
0x100e4a09: add edi,2
0x100e4a0c: shr edx,cl
0x100e4a0e: sub dword ptr [esp+14h],1
0x100e4a13: mov [edi-2],dx
0x100e4a17: jne short 100E49C0h
0x100e4a19: sub dword ptr [esp+18h],1
0x100e4a1e: jne short 100E49B1h
0x100e4a20: pop ebp
0x100e4a21: pop edi
0x100e4a22: pop esi
0x100e4a23: pop ebx
0x100e4a24: add esp,10h
0x100e4a27: ret
```

`sub_100e4940` is called by sub_100e4a30. It delegates to sub_100d2f50, sub_100d2fe0 (2 callees).

12.1293 sub_100e4a30 at 0x100e4a30 (539 instructions)

```
0x100e4a30: sub esp,0Ch
0x100e4a33: mov ecx,[esp+18h]
0x100e4a37: push esi
0x100e4a38: mov esi,[esp+14h]
0x100e4a3c: mov eax,[esi]
0x100e4a3e: push edi
0x100e4a3f: mov edi,[esp+1Ch]
0x100e4a43: lea edx,[edi+ecx-4]
0x100e4a47: mov ecx,[esi+4]
0x100e4a4a: push 0
0x100e4a4c: push edx
0x100e4a4d: mov edx,[eax+8]
0x100e4a50: push ecx
0x100e4a51: mov dword ptr [esp+14h],0FFFFFFFFh
0x100e4a59: mov word ptr [esp+18h],410h
0x100e4a60: mov word ptr [esp+1Ah],45F3h
0x100e4a67: call edx
0x100e4a69: push esi
0x100e4a6a: call 100D2FE0h
0x100e4a6f: mov ecx,[esi]
0x100e4a71: mov edx,[esi+4]
0x100e4a74: add eax,edi
0x100e4a76: push 0
0x100e4a78: push eax
0x100e4a79: mov eax,[ecx+8]
0x100e4a7c: push edx
0x100e4a7d: call eax
0x100e4a7f: push esi
0x100e4a80: call 100D2F50h
0x100e4a85: movzx eax,ax
0x100e4a88: add esp,20h
0x100e4a8b: cmp eax,64h
0x100e4a8e: jg near ptr 100E5127h
0x100e4a94: test eax,eax
0x100e4a96: je near ptr 100E5127h
0x100e4a9c: push ebx
0x100e4a9d: push ebp
0x100e4a9e: jmp short 100E4AA4h
```

```
0x100e4aa0: mov eax,[esp+20h]
0x100e4aa4: sub eax,1
0x100e4aa7: push esi
0x100e4aa8: mov [esp+24h],eax
0x100e4aac: call 100D2F50h
0x100e4ab1: push esi
0x100e4ab2: movzx edi,ax
0x100e4ab5: call 100D2FE0h
0x100e4aba: mov ecx,[esi]
0x100e4abc: mov edx,[esi+4]
0x100e4abf: mov ebx,eax
0x100e4ac1: mov eax,[ecx+18h]
0x100e4ac4: push edx
0x100e4ac5: call eax
0x100e4ac7: mov ebp,[esi]
0x100e4ac9: add esp,0Ch
0x100e4acc: add eax,4
0x100e4acf: push 0
0x100e4ad1: push esi
0x100e4ad2: mov [esp+20h],eax
0x100e4ad6: add ebp,8
0x100e4ad9: call 100D2FE0h
0x100e4ade: add eax,[esp+2Ch]
0x100e4ae2: mov ecx,[esi+4]
0x100e4ae5: mov edx,[ebp]
0x100e4ae8: add esp,4
0x100e4aeb: push eax
0x100e4aec: push ecx
0x100e4aed: call edx
0x100e4aef: mov eax,edi
0x100e4af1: sar eax,8
0x100e4af4: add eax,8
0x100e4af7: or eax,8
0x100e4afa: add esp,0Ch
0x100e4afd: cmp eax,38h
0x100e4b00: jne short 100E4B1Bh
0x100e4b02: mov ecx,[esi]
0x100e4b04: mov edx,[esi+4]
0x100e4b07: mov eax,[ecx+18h]
0x100e4b0a: push ebx
0x100e4b0b: push edx
0x100e4b0c: call eax
0x100e4b0e: add esp,4
0x100e4b11: push eax
0x100e4b12: push esi
0x100e4b13: call 100E4A30h
0x100e4b18: add esp,0Ch
0x100e4b1b: cmp edi,810h
0x100e4b21: jne short 100E4B40h
0x100e4b23: mov ecx,[esi]
0x100e4b25: mov eax,[esi+4]
0x100e4b28: mov ecx,[ecx]
0x100e4b2a: push 1
0x100e4b2c: push 40h
0x100e4b2e: lea edx,[esi+6545h]
0x100e4b34: push edx
0x100e4b35: push eax
0x100e4b36: call ecx
0x100e4b38: add esp,10h
0x100e4b3b: jmp near ptr 100E5105h
0x100e4b40: cmp edi,80Ah
0x100e4b46: jne short 100E4B9Eh
0x100e4b48: mov edx,[esi]
0x100e4b4a: mov eax,[esi+4]
0x100e4b4d: mov ecx,[edx]
0x100e4b4f: push 1
0x100e4b51: push 40h
```

```
0x100e4b53: lea edi,[esi+6485h]
0x100e4b59: push edi
0x100e4b5a: push eax
0x100e4b5b: call ecx
0x100e4b5d: mov eax,edi
0x100e4b5f: add esp,10h
0x100e4b62: lea edx,[eax+1]
0x100e4b65: mov cl,[eax]
0x100e4b67: add eax,1
0x100e4b6a: test cl,cl
0x100e4b6c: jne short 100E4B65h
0x100e4b6e: sub eax,edx
0x100e4b70: mov edx,[esi]
0x100e4b72: mov ecx,[edx+8]
0x100e4b75: add eax,0FFFFFFC1h
0x100e4b78: push 1
0x100e4b7a: push eax
0x100e4b7b: mov eax,[esi+4]
0x100e4b7e: push eax
0x100e4b7f: call ecx
0x100e4b81: mov edx,[esi]
0x100e4b83: mov ecx,[esi+4]
0x100e4b86: mov edx,[edx]
0x100e4b88: push 1
0x100e4b8a: push 40h
0x100e4b8c: lea eax,[esi+64C5h]
0x100e4b92: push eax
0x100e4b93: push ecx
0x100e4b94: call edx
0x100e4b96: add esp,1Ch
0x100e4b99: jmp near ptr 100E5105h
0x100e4b9e: cmp edi,1810h
0x100e4ba4: jne short 100E4BC9h
0x100e4ba6: mov eax,[esi]
0x100e4ba8: mov ecx,[esi+4]
0x100e4bab: mov edx,[eax+8]
0x100e4bae: push 1
0x100e4bb0: push 0Ch
0x100e4bb2: push ecx
0x100e4bb3: call edx
0x100e4bb5: push esi
0x100e4bb6: call 100D2FE0h
0x100e4bbb: add esp,10h
0x100e4bbe: mov [esi+66C8h],eax
0x100e4bc4: jmp near ptr 100E5105h
0x100e4bc9: cmp edi,1835h
0x100e4bcf: jne short 100E4BE5h
0x100e4bd1: push esi
0x100e4bd2: call 100D2FE0h
0x100e4bd7: add esp,4
0x100e4bda: mov [esi+65F4h],eax
0x100e4be0: jmp near ptr 100E5105h
0x100e4be5: cmp edi,2007h
0x100e4beb: jne short 100E4C0Ch
0x100e4bed: mov eax,[esi]
0x100e4bef: mov ecx,[esi+4]
0x100e4bf2: mov edx,[eax+18h]
0x100e4bf5: push ecx
0x100e4bf6: call edx
0x100e4bf8: add esp,4
0x100e4bfb: mov [esi+65C4h],eax
0x100e4c01: mov [esi+65D0h],ebx
0x100e4c07: jmp near ptr 100E5105h
0x100e4c0c: cmp edi,1818h
0x100e4c12: jne short 100E4C6Bh
0x100e4c14: push esi
0x100e4c15: call 100D2FE0h
```

```
0x100e4c1a: push esi
0x100e4c1b: call 100D2FE0h
0x100e4c20: push eax
0x100e4c21: call 100D3050h
0x100e4c26: fchs
0x100e4c28: fld qword ptr ds:[100FF110h]
0x100e4c2e: add esp,0Ch
0x100e4c31: fxch
0x100e4c33: call 100229A0h
0x100e4c38: fstp dword ptr [esi+6594h]
0x100e4c3e: push esi
0x100e4c3f: call 100D2FE0h
0x100e4c44: push eax
0x100e4c45: call 100D3050h
0x100e4c4a: fmul qword ptr ds:[100FE7C8h]
0x100e4c50: fld qword ptr ds:[100FF110h]
0x100e4c56: add esp,8
0x100e4c59: fxch
0x100e4c5b: call 100229A0h
0x100e4c60: fstp dword ptr [esi+6598h]
0x100e4c66: jmp near ptr 100E5105h
0x100e4c6b: cmp edi,102Ah
0x100e4c71: jne near ptr 100E4D91h
0x100e4c77: push esi
0x100e4c78: call 100D2FE0h
0x100e4c7d: push esi
0x100e4c7e: call 100D2F50h
0x100e4c83: movzx eax,ax
0x100e4c86: mov [esp+30h],eax
0x100e4c8a: add esp,8
0x100e4c8d: fild dword ptr [esp+28h]
0x100e4c91: fmul qword ptr ds:[1012E540h]
0x100e4c97: fsub qword ptr ds:[100FF108h]
0x100e4c9d: fld qword ptr ds:[100FF110h]
0x100e4ca3: fxch
0x100e4ca5: call 100229A0h
0x100e4caa: fstp dword ptr [esp+28h]
0x100e4cae: fld dword ptr [esp+28h]
0x100e4cb2: push esi
0x100e4cb3: fmul qword ptr ds:[100FF380h]
0x100e4cb9: fstp dword ptr [esi+6590h]
0x100e4cbf: call 100D2F50h
0x100e4cc4: push esi
0x100e4cc5: call 100D2F50h
0x100e4cca: movsx ecx,ax
0x100e4ccd: mov [esp+30h],ecx
0x100e4cd1: add esp,8
0x100e4cd4: fild dword ptr [esp+28h]
0x100e4cd8: fmul qword ptr ds:[1012E538h]
0x100e4cde: fld qword ptr ds:[100FF110h]
0x100e4ce4: fxch
0x100e4ce6: call 100229A0h
0x100e4ceb: fstp dword ptr [esi+6598h]
0x100e4cf1: push esi
0x100e4cf2: call 100D2F50h
0x100e4cf7: movsx edx,ax
0x100e4cfa: neg edx
0x100e4cfc: mov [esp+2Ch],edx
0x100e4d00: add esp,4
0x100e4d03: fild dword ptr [esp+28h]
0x100e4d07: fmul qword ptr ds:[1012E540h]
0x100e4d0d: fld qword ptr ds:[100FF110h]
0x100e4d13: fxch
0x100e4d15: call 100229A0h
0x100e4d1a: fstp dword ptr [esi+6594h]
0x100e4d20: push esi
0x100e4d21: call 100D2F50h
```

```
0x100e4d26: push esi
0x100e4d27: call 100D2F50h
0x100e4d2c: movzx eax,ax
0x100e4d2f: add esp,8
0x100e4d32: cmp eax,11h
0x100e4d35: mov [esp+10h],eax
0x100e4d39: jle short 100E4D43h
0x100e4d3b: mov dword ptr [esp+10h],0
0x100e4d43: mov eax,[esi]
0x100e4d45: mov ecx,[esi+4]
0x100e4d48: mov edx,[eax+8]
0x100e4d4b: push 1
0x100e4d4d: push 20h
0x100e4d4f: push ecx
0x100e4d50: call edx
0x100e4d52: fld dword ptr [esi+6594h]
0x100e4d58: fcomp qword ptr ds:[1012E720h]
0x100e4d5e: add esp,0Ch
0x100e4d61: fnstsw ax
0x100e4d63: test ah,41h
0x100e4d66: jne near ptr 100E5105h
0x100e4d6c: push esi
0x100e4d6d: call 100D2F50h
0x100e4d72: movzx eax,ax
0x100e4d75: mov [esp+2Ch],eax
0x100e4d79: add esp,4
0x100e4d7c: fild dword ptr [esp+28h]
0x100e4d80: fdiv qword ptr ds:[100FFB08h]
0x100e4d86: fstp dword ptr [esi+6594h]
0x100e4d8c: jmp near ptr 100E5105h
0x100e4d91: cmp edi,102Ch
0x100e4d97: jne near ptr 100E4E31h
0x100e4d9d: push esi
0x100e4d9e: call 100D2F50h
0x100e4da3: add esp,4
0x100e4da6: cmp ax,200h
0x100e4daa: push 1
0x100e4dac: jbe short 100E4DEDh
0x100e4dae: mov ecx,[esi]
0x100e4db0: mov edx,[esi+4]
0x100e4db3: mov eax,[ecx+8]
0x100e4db6: push 76h
0x100e4db8: push edx
0x100e4db9: call eax
0x100e4dbb: add esp,0Ch
0x100e4dbe: xor edi,edi
0x100e4dc0: push esi
0x100e4dc1: call 100D2F50h
0x100e4dc6: movzx ecx,ax
0x100e4dc9: mov [esp+2Ch],ecx
0x100e4dcd: mov edx,edi
0x100e4dcf: xor edx,2
0x100e4dd2: fild dword ptr [esp+2Ch]
0x100e4dd6: add edi,1
0x100e4dd9: add esp,4
0x100e4ddc: cmp edi,4
0x100e4ddf: fstp dword ptr [esi+edx*4+0E774h]
0x100e4de6: jl short 100E4DC0h
0x100e4de8: jmp near ptr 100E5105h
0x100e4ded: mov eax,[esi]
0x100e4def: mov ecx,[esi+4]
0x100e4df2: mov edx,[eax+8]
0x100e4df5: push 62h
0x100e4df7: push ecx
0x100e4df8: call edx
0x100e4dfa: add esp,0Ch
0x100e4dfd: xor edi,edi
```

```
0x100e4dff: nop
0x100e4e00: push esi
0x100e4e01: call 100D2F50h
0x100e4e06: mov ecx,edi
0x100e4e08: movzx eax,ax
0x100e4e0b: xor ecx,2
0x100e4e0e: sar ecx,1
0x100e4e10: mov [esp+2Ch],eax
0x100e4e14: xor ecx,edi
0x100e4e16: add edi,1
0x100e4e19: fild dword ptr [esp+2Ch]
0x100e4e1d: add esp,4
0x100e4e20: cmp edi,4
0x100e4e23: fstp dword ptr [esi+ecx*4+0E774h]
0x100e4e2a: jl short 100E4E00h
0x100e4e2c: jmp near ptr 100E5105h
0x100e4e31: cmp edi,32h
0x100e4e34: jne near ptr 100E4F6Bh
0x100e4e3a: cmp ebx,300h
0x100e4e40: jne short 100E4EA9h
0x100e4e42: mov edx,[esi]
0x100e4e44: mov eax,[esi+4]
0x100e4e47: mov ecx,[edx+8]
0x100e4e4a: push 1
0x100e4e4c: push 48h
0x100e4e4e: push eax
0x100e4e4f: call ecx
0x100e4e51: add esp,0Ch
0x100e4e54: xor edi,edi
0x100e4e56: push esi
0x100e4e57: call 100D2F50h
0x100e4e5c: movzx edx,ax
0x100e4e5f: mov [esp+2Ch],edx
0x100e4e63: mov eax,edi
0x100e4e65: sar eax,1
0x100e4e67: fild dword ptr [esp+2Ch]
0x100e4e6b: xor eax,edi
0x100e4e6d: add edi,1
0x100e4e70: add esp,4
0x100e4e73: cmp edi,4
0x100e4e76: fdivr qword ptr ds:[101048F8h]
0x100e4e7c: fstp dword ptr [esp+28h]
0x100e4e80: fld dword ptr [esp+28h]
0x100e4e84: fstp dword ptr [esi+eax*4+0E774h]
0x100e4e8b: jl short 100E4E56h
0x100e4e8d: cmp dword ptr [esp+10h],0
0x100e4e92: jne near ptr 100E5105h
0x100e4e98: fld dword ptr ds:[100FF11Ch]
0x100e4e9e: fstp dword ptr [esi+0E774h]
0x100e4ea4: jmp near ptr 100E5105h
0x100e4ea9: fldz
0x100e4eab: fcomp dword ptr [esi+0E774h]
0x100e4eb1: fnstsw ax
0x100e4eb3: test ah,44h
0x100e4eb6: jp near ptr 100E5105h
0x100e4ebc: push esi
0x100e4ebd: call 100D2F50h
0x100e4ec2: add esp,4
0x100e4ec5: cmp ax,[esp+14h]
0x100e4eca: jne short 100E4EFCh
0x100e4ecc: lea ecx,[esi+64C5h]
0x100e4ed2: push 1012E718h
0x100e4ed7: push ecx
0x100e4ed8: call dword ptr ds:[100FE424h]
0x100e4ede: add esp,8
0x100e4ee1: test eax,eax
0x100e4ee3: mov eax,1012E704h
```

```
0x100e4ee8: jne short 100E4EEFh
0x100e4eea: mov eax,1012E6F0h
0x100e4eef: mov edx,[esp+10h]
0x100e4ef3: movsx eax,byte ptr [eax+edx]
0x100e4ef7: sub eax,2Eh
0x100e4efa: jmp short 100E4F16h
0x100e4efc: mov eax,[esp+10h]
0x100e4f00: movsx eax,byte ptr [eax+1012E6DCh]
0x100e4f07: sub eax,30h
0x100e4f0a: xor ecx,ecx
0x100e4f0c: mov [esp+16h],cx
0x100e4f11: mov [esp+14h],cx
0x100e4f16: mov ecx,[esi]
0x100e4f18: mov ecx,[ecx+8]
0x100e4f1b: lea edx,[eax*8+4Eh]
0x100e4f22: mov eax,[esi+4]
0x100e4f25: push 1
0x100e4f27: push edx
0x100e4f28: push eax
0x100e4f29: call ecx
0x100e4f2b: add esp,0Ch
0x100e4f2e: xor edi,edi
0x100e4f30: push esi
0x100e4f31: call 100D2F50h
0x100e4f36: mov edx,edi
0x100e4f38: and edx,1
0x100e4f3b: xor ax,[esp+edx*2+18h]
0x100e4f40: mov ecx,edi
0x100e4f42: movzx eax,ax
0x100e4f45: xor ecx,2
0x100e4f48: sar ecx,1
0x100e4f4a: mov [esp+2Ch],eax
0x100e4f4e: xor ecx,edi
0x100e4f50: add edi,1
0x100e4f53: fild dword ptr [esp+2Ch]
0x100e4f57: add esp,4
0x100e4f5a: cmp edi,4
0x100e4f5d: fstp dword ptr [esi+ecx*4+0E774h]
0x100e4f64: jl short 100E4F30h
0x100e4f66: jmp near ptr 100E4E8Dh
0x100e4f6b: cmp edi,10A9h
0x100e4f71: jne short 100E4FD6h
0x100e4f73: cmp ebx,42h
0x100e4f76: jle short 100E4F8Ah
0x100e4f78: mov edx,[esp+10h]
0x100e4f7c: movsx eax,byte ptr [edx+1012E6D0h]
0x100e4f83: sub eax,30h
0x100e4f86: mov [esp+10h],eax
0x100e4f8a: mov edx,[esp+10h]
0x100e4f8e: mov ecx,[esi]
0x100e4f90: lea eax,[edx*8+2]
0x100e4f97: mov edx,[esi+4]
0x100e4f9a: push 1
0x100e4f9c: push eax
0x100e4f9d: mov eax,[ecx+8]
0x100e4fa0: push edx
0x100e4fa1: call eax
0x100e4fa3: add esp,0Ch
0x100e4fa6: xor edi,edi
0x100e4fa8: push esi
0x100e4fa9: call 100D2F50h
0x100e4fae: movzx ecx,ax
0x100e4fb1: mov edx,edi
0x100e4fb3: sar edx,1
0x100e4fb5: mov [esp+2Ch],ecx
0x100e4fb9: xor edx,edi
0x100e4fbb: add edi,1
```

```
0x100e4fbe: fild dword ptr [esp+2Ch]
0x100e4fc2: add esp,4
0x100e4fc5: cmp edi,4
0x100e4fc8: fstp dword ptr [esi+edx*4+0E774h]
0x100e4fcf: jl short 100E4FA8h
0x100e4fd1: jmp near ptr 100E5105h
0x100e4fd6: cmp edi,1030h
0x100e4fdc: jne short 100E4FFFh
0x100e4fde: mov ecx,[esp+10h]
0x100e4fe2: mov eax,18040h
0x100e4fe7: sar eax,cl
0x100e4fe9: test al,1
0x100e4feb: je near ptr 100E5105h
0x100e4ff1: push esi
0x100e4ff2: call 100E4940h
0x100e4ff7: add esp,4
0x100e4ffa: jmp near ptr 100E5105h
0x100e4fff: cmp edi,1031h
0x100e5005: jne short 100E502Fh
0x100e5007: push esi
0x100e5008: call 100D2F50h
0x100e500d: push esi
0x100e500e: call 100D2F50h
0x100e5013: push esi
0x100e5014: mov [esi+66B2h],ax
0x100e501b: call 100D2F50h
0x100e5020: add esp,0Ch
0x100e5023: mov [esi+66B0h],ax
0x100e502a: jmp near ptr 100E5105h
0x100e502f: cmp edi,5029h
0x100e5035: jne short 100E5072h
0x100e5037: mov ecx,ebx
0x100e5039: sar ecx,10h
0x100e503c: mov [esp+28h],ecx
0x100e5040: fild dword ptr [esp+28h]
0x100e5044: and ebx,0FFFFh
0x100e504a: cmp ebx,2
0x100e504d: fstp dword ptr [esp+28h]
0x100e5051: fld dword ptr [esp+28h]
0x100e5055: fst dword ptr [esi+659Ch]
0x100e505b: jne near ptr 100E5103h
0x100e5061: fmul qword ptr ds:[1012E540h]
0x100e5067: fstp dword ptr [esi+659Ch]
0x100e506d: jmp near ptr 100E5105h
0x100e5072: cmp edi,5813h
0x100e5078: jne short 100E508Eh
0x100e507a: push ebx
0x100e507b: call 100D3050h
0x100e5080: fstp dword ptr [esi+6588h]
0x100e5086: add esp,4
0x100e5089: jmp near ptr 100E5105h
0x100e508e: cmp edi,5814h
0x100e5094: jne short 100E50A7h
0x100e5096: push ebx
0x100e5097: call 100D3050h
0x100e509c: fstp dword ptr [esi+658Ch]
0x100e50a2: add esp,4
0x100e50a5: jmp short 100E5105h
0x100e50a7: cmp edi,5817h
0x100e50ad: jne short 100E50B7h
0x100e50af: mov [esi+65A8h],ebx
0x100e50b5: jmp short 100E5105h
0x100e50b7: cmp edi,5834h
0x100e50bd: jne short 100E50C7h
0x100e50bf: mov [esi+65B8h],ebx
0x100e50c5: jmp short 100E5105h
0x100e50c7: cmp edi,580Eh
```

```
0x100e50cd: jne short 100E50E0h
0x100e50cf: mov eax,ebx
0x100e50d1: cdq
0x100e50d2: mov [esi+65A0h],eax
0x100e50d8: mov [esi+65A4h],edx
0x100e50de: jmp short 100E5105h
0x100e50e0: cmp edi,180Eh
0x100e50e6: jne short 100E5105h
0x100e50e8: push esi
0x100e50e9: call 100D2FE0h
0x100e50ee: add esp,4
0x100e50f1: mov [esi+65A0h],eax
0x100e50f7: mov dword ptr [esi+65A4h],0
0x100e5101: jmp short 100E5105h
0x100e5103: fstp st(0)
0x100e5105: mov eax,[esp+18h]
0x100e5109: mov edx,[esi]
0x100e510b: mov ecx,[esi+4]
0x100e510e: mov edx,[edx+8]
0x100e5111: push 0
0x100e5113: push eax
0x100e5114: push ecx
0x100e5115: call edx
0x100e5117: add esp,0Ch
0x100e511a: cmp dword ptr [esp+20h],0
0x100e511f: jne near ptr 100E4AA0h
0x100e5125: pop ebp
0x100e5126: pop ebx
0x100e5127: pop edi
0x100e5128: pop esi
0x100e5129: add esp,0Ch
0x100e512c: ret
```

`sub_100e4a30` is called by sub_100e4a30, sub_100eb010, sub_100eb620. It delegates to j__CIpow, sub_100d2f50, sub_100d2fe0 (6 callees).

12.1294 sub_100e5130 at 0x100e5130 (192 instructions)

```
0x100e5130: sub esp,0A8h
0x100e5136: mov eax,ds:[10136220h]
0x100e513b: xor eax,esp
0x100e513d: mov [esp+0A4h],eax
0x100e5144: push ebp
0x100e5145: mov ebp,[esp+0B0h]
0x100e514c: mov eax,[ebp]
0x100e514f: mov ecx,[ebp+4]
0x100e5152: mov edx,[eax+8]
0x100e5155: push esi
0x100e5156: push 0
0x100e5158: push 0
0x100e515a: push ecx
0x100e515b: call edx
0x100e515d: add esp,0Ch
0x100e5160: xor eax,eax
0x100e5162: push ebx
0x100e5163: mov [esp+0Ch],eax
0x100e5167: mov [esp+10h],eax
0x100e516b: mov [esp+14h],eax
0x100e516f: mov [esp+18h],eax
0x100e5173: mov [esp+1Ch],eax
0x100e5177: mov [esp+20h],eax
0x100e517b: mov [esp+24h],eax
0x100e517f: mov [esp+28h],eax
0x100e5183: mov [esp+2Ch],eax
0x100e5187: push edi
0x100e5188: jmp short 100E5190h
```

```
0x100e5190: mov eax,[ebp]
0x100e5193: mov edx,[ebp+4]
0x100e5196: mov eax,[eax+10h]
0x100e5199: push 80h
0x100e519e: lea ecx,[esp+38h]
0x100e51a2: push ecx
0x100e51a3: push edx
0x100e51a4: call eax
0x100e51a6: lea ecx,[esp+40h]
0x100e51aa: push 3Dh
0x100e51ac: push ecx
0x100e51ad: call dword ptr ds:[100FE454h]
0x100e51b3: mov ebx,eax
0x100e51b5: add esp,14h
0x100e51b8: test ebx,ebx
0x100e51ba: je short 100E51C4h
0x100e51bc: mov byte ptr [ebx],0
0x100e51bf: add ebx,1
0x100e51c2: jmp short 100E51DFh
0x100e51c4: lea eax,[esp+34h]
0x100e51c8: lea edx,[eax+1]
0x100e51cb: jmp short 100E51D0h
0x100e51d0: mov cl,[eax]
0x100e51d2: add eax,1
0x100e51d5: test cl,cl
0x100e51d7: jne short 100E51D0h
0x100e51d9: sub eax,edx
0x100e51db: lea ebx,[esp+eax+34h]
0x100e51df: mov edi,1012E768h
0x100e51e4: lea esi,[esp+34h]
0x100e51e8: mov ecx,4
0x100e51ed: xor edx,edx
0x100e51ef: repe cmpsb
0x100e51f1: jne short 100E5211h
0x100e51f3: lea eax,[esp+24h]
0x100e51f7: push eax
0x100e51f8: lea ecx,[esp+24h]
0x100e51fc: push ecx
0x100e51fd: lea edx,[esp+24h]
0x100e5201: push edx
0x100e5202: push 1012E054h
0x100e5207: push ebx
0x100e5208: call dword ptr ds:[100FE4A8h]
0x100e520e: add esp,14h
0x100e5211: mov edi,1012E764h
0x100e5216: lea esi,[esp+34h]
0x100e521a: mov ecx,4
0x100e521f: xor eax,eax
0x100e5221: repe cmpsb
0x100e5223: jne short 100E5243h
0x100e5225: lea ecx,[esp+10h]
0x100e5229: push ecx
0x100e522a: lea edx,[esp+18h]
0x100e522e: push edx
0x100e522f: lea eax,[esp+20h]
0x100e5233: push eax
0x100e5234: push 1012E758h
0x100e5239: push ebx
0x100e523a: call dword ptr ds:[100FE4A8h]
0x100e5240: add esp,14h
0x100e5243: mov edi,1012E754h
0x100e5248: lea esi,[esp+34h]
0x100e524c: mov ecx,4
0x100e5251: xor edx,edx
0x100e5253: repe cmpsb
0x100e5255: jne short 100E5267h
0x100e5257: push ebx
```

```
0x100e5258: call dword ptr ds:[100FE458h]
0x100e525e: add esp,4
0x100e5261: mov [ebp+65C4h],eax
0x100e5267: mov edi,1012E750h
0x100e526c: lea esi,[esp+34h]
0x100e5270: mov ecx,4
0x100e5275: xor eax,eax
0x100e5277: repe cmpsb
0x100e5279: jne short 100E528Ch
0x100e527b: push ebx
0x100e527c: call dword ptr ds:[100FE458h]
0x100e5282: add esp,4
0x100e5285: mov [ebp+66B2h],ax
0x100e528c: mov edi,1012E74Ch
0x100e5291: lea esi,[esp+34h]
0x100e5295: mov ecx,4
0x100e529a: xor edx,edx
0x100e529c: repe cmpsb
0x100e529e: jne short 100E52B1h
0x100e52a0: push ebx
0x100e52a1: call dword ptr ds:[100FE458h]
0x100e52a7: add esp,4
0x100e52aa: mov [ebp+66B0h],ax
0x100e52b1: mov edi,1012E748h
0x100e52b6: lea esi,[esp+34h]
0x100e52ba: mov ecx,4
0x100e52bf: xor eax,eax
0x100e52c1: repe cmpsb
0x100e52c3: jne short 100E52D6h
0x100e52c5: push ebx
0x100e52c6: call dword ptr ds:[100FE458h]
0x100e52cc: add esp,4
0x100e52cf: mov [ebp+66C4h],ax
0x100e52d6: mov edi,1012E744h
0x100e52db: lea esi,[esp+34h]
0x100e52df: mov ecx,4
0x100e52e4: xor edx,edx
0x100e52e6: repe cmpsb
0x100e52e8: jne short 100E52FBh
0x100e52ea: push ebx
0x100e52eb: call dword ptr ds:[100FE458h]
0x100e52f1: add esp,4
0x100e52f4: mov [ebp+66C6h],ax
0x100e52fb: push 4
0x100e52fd: lea eax,[esp+38h]
0x100e5301: push 1012E73Ch
0x100e5306: push eax
0x100e5307: call dword ptr ds:[100FE414h]
0x100e530d: add esp,0Ch
0x100e5310: test eax,eax
0x100e5312: jne near ptr 100E5190h
0x100e5318: movzx edx,word ptr [ebp+66C4h]
0x100e531f: movzx ecx,word ptr [ebp+66C6h]
0x100e5326: mov eax,[ebp+65C4h]
0x100e532c: imul ecx,edx
0x100e532f: mov esi,ds:[100FE444h]
0x100e5335: sub dword ptr [esp+24h],76Ch
0x100e533d: sub dword ptr [esp+20h],1
0x100e5342: lea edx,[esp+10h]
0x100e5346: lea ecx,[eax+ecx*2]
0x100e5349: push edx
0x100e534a: mov [ebp+65C0h],ecx
0x100e5350: call esi
0x100e5352: add esp,4
0x100e5355: test edx,edx
0x100e5357: pop edi
0x100e5358: pop ebx
```

```
0x100e5359: jl short 100E5377h
0x100e535b: jg short 100E5361h
0x100e535d: test eax,eax
0x100e535f: jbe short 100E5377h
0x100e5361: lea eax,[esp+8]
0x100e5365: push eax
0x100e5366: call esi
0x100e5368: add esp,4
0x100e536b: mov [ebp+65A0h],eax
0x100e5371: mov [ebp+65A4h],edx
0x100e5377: mov ecx,ds:[1012E734h]
0x100e537d: mov [ebp+6485h],ecx
0x100e5383: mov dx,ds:[1012E738h]
0x100e538a: mov [ebp+6489h],dx
0x100e5391: movzx eax,byte ptr ds:[1012E73Ah]
0x100e5398: mov [ebp+648Bh],al
0x100e539e: mov ecx,ds:[1012E728h]
0x100e53a4: mov [ebp+64C5h],ecx
0x100e53aa: mov edx,ds:[1012E72Ch]
0x100e53b0: mov ecx,[esp+0ACh]
0x100e53b7: mov [ebp+64C9h],edx
0x100e53bd: movzx eax,byte ptr ds:[1012E730h]
0x100e53c4: pop esi
0x100e53c5: mov [ebp+64CDh],al
0x100e53cb: mov dword ptr [ebp+2E7F4h],100D6E10h
0x100e53d5: pop ebp
0x100e53d6: xor ecx,esp
0x100e53d8: call 10022326h
0x100e53dd: add esp,0A8h
0x100e53e3: ret
```

`sub_100e5130` is called by sub_100eb620. It works with the text "%d.%d.%d", "Rollei", "%d:%d:%d". It delegates to sub_10022326 (1 callees).

12.1295 sub_100e53f0 at 0x100e53f0 (137 instructions)

```
0x100e53f0: sub esp,10h
0x100e53f3: mov eax,ds:[10136220h]
0x100e53f8: xor eax,esp
0x100e53fa: mov [esp+0Ch],eax
0x100e53fe: push ebx
0x100e53ff: mov ebx,[esp+18h]
0x100e5403: mov eax,[ebx]
0x100e5405: mov ecx,[ebx+4]
0x100e5408: push esi
0x100e5409: push edi
0x100e540a: push 0
0x100e540c: mov word ptr [ebx+6274h],4949h
0x100e5415: mov edx,[eax+8]
0x100e5418: push 4
0x100e541a: push ecx
0x100e541b: call edx
0x100e541d: push ebx
0x100e541e: call 100D2FE0h
0x100e5423: mov esi,[ebx]
0x100e5425: add esp,10h
0x100e5428: mov edi,eax
0x100e542a: push 0
0x100e542c: push ebx
0x100e542d: mov [esp+14h],edi
0x100e5431: add esi,8
0x100e5434: call 100D2FE0h
0x100e5439: mov ecx,[esi]
0x100e543b: add esp,4
0x100e543e: push eax
0x100e543f: mov eax,[ebx+4]
```

```
0x100e5442: push eax
0x100e5443: call ecx
0x100e5445: add esp,0Ch
0x100e5448: test edi,edi
0x100e544a: je near ptr 100E54CFh
0x100e5450: push ebp
0x100e5451: sub dword ptr [esp+10h],1
0x100e5456: push ebx
0x100e5457: call 100D2FE0h
0x100e545c: push ebx
0x100e545d: mov ebp,eax
0x100e545f: call 100D2FE0h
0x100e5464: mov edx,[ebx]
0x100e5466: mov ecx,[ebx+4]
0x100e5469: mov edx,[edx]
0x100e546b: push 1
0x100e546d: push 8
0x100e546f: lea eax,[esp+24h]
0x100e5473: push eax
0x100e5474: push ecx
0x100e5475: call edx
0x100e5477: add esp,18h
0x100e547a: mov edi,1012E77Ch
0x100e547f: lea esi,[esp+14h]
0x100e5483: mov ecx,5
0x100e5488: xor eax,eax
0x100e548a: repe cmpsb
0x100e548c: jne short 100E5494h
0x100e548e: mov [ebx+65C8h],ebp
0x100e5494: mov edi,1012E774h
0x100e5499: lea esi,[esp+14h]
0x100e549d: mov ecx,6
0x100e54a2: xor edx,edx
0x100e54a4: repe cmpsb
0x100e54a6: jne short 100E54AEh
0x100e54a8: mov [ebx+65C4h],ebp
0x100e54ae: mov edi,1012E76Ch
0x100e54b3: lea esi,[esp+14h]
0x100e54b7: mov ecx,5
0x100e54bc: xor eax,eax
0x100e54be: repe cmpsb
0x100e54c0: jne short 100E54C8h
0x100e54c2: mov [ebx+65C0h],ebp
0x100e54c8: cmp [esp+10h],eax
0x100e54cc: jne short 100E5451h
0x100e54ce: pop ebp
0x100e54cf: mov edx,[ebx+65C8h]
0x100e54d5: mov ecx,[ebx]
0x100e54d7: mov eax,[ebx+4]
0x100e54da: mov ecx,[ecx+8]
0x100e54dd: push 0
0x100e54df: add edx,14h
0x100e54e2: push edx
0x100e54e3: push eax
0x100e54e4: call ecx
0x100e54e6: mov edx,[ebx]
0x100e54e8: mov eax,[ebx+4]
0x100e54eb: mov ecx,[edx]
0x100e54ed: push 1
0x100e54ef: push 40h
0x100e54f1: lea esi,[ebx+6485h]
0x100e54f7: push esi
0x100e54f8: push eax
0x100e54f9: call ecx
0x100e54fb: push 20h
0x100e54fd: push esi
0x100e54fe: mov byte ptr [ebx+64C4h],0
```

```
0x100e5505: call dword ptr ds:[100FE454h]
0x100e550b: add esp,24h
0x100e550e: test eax,eax
0x100e5510: je short 100E552Eh
0x100e5512: lea ecx,[eax+1]
0x100e5515: lea esi,[ebx+64C5h]
0x100e551b: sub esi,ecx
0x100e551d: lea ecx,[ecx]
0x100e5520: mov dl,[ecx]
0x100e5522: mov [esi+ecx],dl
0x100e5525: add ecx,1
0x100e5528: test dl,dl
0x100e552a: jne short 100E5520h
0x100e552c: mov [eax],dl
0x100e552e: push ebx
0x100e552f: call 100D2F50h
0x100e5534: push ebx
0x100e5535: mov [ebx+66B2h],ax
0x100e553c: call 100D2F50h
0x100e5541: push ebx
0x100e5542: mov [ebx+66B0h],ax
0x100e5549: mov dword ptr [ebx+2E7FCh],100D8E20h
0x100e5553: call 100D2FE0h
0x100e5558: push ebx
0x100e5559: call 100D2F50h
0x100e555e: push ebx
0x100e555f: mov [ebx+66C4h],ax
0x100e5566: call 100D2F50h
0x100e556b: mov ecx,[esp+2Ch]
0x100e556f: add esp,14h
0x100e5572: pop edi
0x100e5573: pop esi
0x100e5574: mov [ebx+66C6h],ax
0x100e557b: mov dword ptr [ebx+2E7F4h],100D6C40h
0x100e5585: mov dword ptr [ebx+65FCh],3FFFh
0x100e558f: pop ebx
0x100e5590: xor ecx,esp
0x100e5592: call 10022326h
0x100e5597: add esp,10h
0x100e559a: ret
```

`sub_100e53f0` is called by sub_100eb620. It delegates to sub_10022326, sub_100d2f50, sub_100d2fe0 (3 callees).

12.1296 sub_100e55a0 at 0x100e55a0 (171 instructions)

```
0x100e55a0: sub esp,2Ch
0x100e55a3: xor eax,eax
0x100e55a5: push ebx
0x100e55a6: mov ebx,[esp+38h]
0x100e55aa: push esi
0x100e55ab: mov esi,[esp+38h]
0x100e55af: mov [esi+6258h],eax
0x100e55b5: mov [esi+625Ch],eax
0x100e55bb: mov [esi+6260h],eax
0x100e55c1: mov [esi+6264h],eax
0x100e55c7: mov [esi+6268h],eax
0x100e55cd: mov [esi+626Ch],eax
0x100e55d3: mov [esi+6270h],eax
0x100e55d9: mov eax,[esi]
0x100e55db: mov ecx,[esi+4]
0x100e55de: mov edx,[eax+8]
0x100e55e1: push 0
0x100e55e3: push ebx
0x100e55e4: push ecx
0x100e55e5: call edx
0x100e55e7: push esi
```

```
0x100e55e8: call 100D2FE0h
0x100e55ed: push esi
0x100e55ee: mov [esi+6274h],ax
0x100e55f5: call 100D2FE0h
0x100e55fa: and eax,0FFFFFF00h
0x100e55ff: add esp,14h
0x100e5602: cmp eax,52617700h
0x100e5607: jne near ptr 100E596Fh
0x100e560d: push edi
0x100e560e: mov edi,[esi]
0x100e5610: push 0
0x100e5612: push esi
0x100e5613: add edi,8
0x100e5616: call 100D2FE0h
0x100e561b: mov ecx,[edi]
0x100e561d: add eax,ebx
0x100e561f: add esp,4
0x100e5622: push eax
0x100e5623: mov eax,[esi+4]
0x100e5626: push eax
0x100e5627: call ecx
0x100e5629: push esi
0x100e562a: call 100D2FE0h
0x100e562f: mov edi,eax
0x100e5631: push esi
0x100e5632: mov [esp+50h],edi
0x100e5636: call 100D2FE0h
0x100e563b: add esp,14h
0x100e563e: test edi,edi
0x100e5640: je near ptr 100E5864h
0x100e5646: push ebp
0x100e5647: jmp short 100E5654h
0x100e5650: mov ebx,[esp+44h]
0x100e5654: sub dword ptr [esp+40h],1
0x100e5659: push esi
0x100e565a: call 100D2FE0h
0x100e565f: push esi
0x100e5660: mov ebp,eax
0x100e5662: call 100D2FE0h
0x100e5667: push esi
0x100e5668: call 100D2FE0h
0x100e566d: push esi
0x100e566e: mov [esp+24h],eax
0x100e5672: call 100D2FE0h
0x100e5677: mov edx,[esi]
0x100e5679: mov ecx,[edx+18h]
0x100e567c: mov edi,eax
0x100e567e: mov eax,[esi+4]
0x100e5681: push eax
0x100e5682: call ecx
0x100e5684: mov edx,[esi]
0x100e5686: mov ecx,[edx+8]
0x100e5689: push 0
0x100e568b: mov [esp+28h],eax
0x100e568f: mov eax,[esi+4]
0x100e5692: add ebx,edi
0x100e5694: push ebx
0x100e5695: push eax
0x100e5696: call ecx
0x100e5698: add esp,20h
0x100e569b: cmp ebp,210h
0x100e56a1: ja near ptr 100E57C5h
0x100e56a7: je near ptr 100E57B4h
0x100e56ad: lea eax,[ebp-100h]
0x100e56b3: cmp eax,12h
0x100e56b6: ja near ptr 100E5843h
0x100e56bc: jmp dword ptr [eax*4+100E5978h]
```

```
0x100e57b4: push edi
0x100e57b5: call 100D3050h
0x100e57ba: fstp dword ptr [esi+6270h]
0x100e57c0: add esp,4
0x100e57c3: jmp short 100E5843h
0x100e57c5: lea eax,[ebp-21Ah]
0x100e57cb: cmp eax,0E7h
0x100e57d0: ja short 100E5843h
0x100e57d2: movzx eax,byte ptr [eax+100E59E0h]
0x100e57d9: jmp dword ptr [eax*4+100E59C4h]
0x100e5843: mov eax,[esp+10h]
0x100e5847: mov edx,[esi]
0x100e5849: mov ecx,[esi+4]
0x100e584c: mov edx,[edx+8]
0x100e584f: push 0
0x100e5851: push eax
0x100e5852: push ecx
0x100e5853: call edx
0x100e5855: add esp,0Ch
0x100e5858: cmp dword ptr [esp+40h],0
0x100e585d: jne near ptr 100E5650h
0x100e5863: pop ebp
0x100e5864: cmp dword ptr [esi+6258h],3
0x100e586b: pop edi
0x100e586c: mov eax,100D80F0h
0x100e5871: jl short 100E5878h
0x100e5873: mov eax,100D8350h
0x100e5878: cmp byte ptr [esi+64C5h],0
0x100e587f: mov [esi+2E7FCh],eax
0x100e5885: mov dword ptr [esi+65FCh],0FFFFh
0x100e588f: mov eax,ds:[1012985Ch]
0x100e5894: mov [esi+6485h],eax
0x100e589a: mov ecx,ds:[10129860h]
0x100e58a0: mov [esi+6489h],ecx
0x100e58a6: mov dx,ds:[10129864h]
0x100e58ad: mov [esi+648Dh],dx
0x100e58b4: jne near ptr 100E596Fh
0x100e58ba: movzx eax,word ptr [esi+66B0h]
0x100e58c1: cmp eax,1020h
0x100e58c6: jg near ptr 100E5951h
0x100e58cc: je short 100E5934h
0x100e58ce: cmp eax,80Ch
0x100e58d3: je short 100E58FDh
0x100e58d5: cmp eax,0A7Ah
0x100e58da: jne near ptr 100E596Fh
0x100e58e0: mov eax,ds:[1012E7A0h]
0x100e58e5: mov [esi+64C5h],eax
0x100e58eb: mov cl,ds:[1012E7A4h]
0x100e58f1: mov [esi+64C9h],cl
0x100e58f7: pop esi
0x100e58f8: pop ebx
0x100e58f9: add esp,2Ch
0x100e58fc: ret
0x100e58fd: mov edx,ds:[1012E794h]
0x100e5903: mov [esi+64C5h],edx
0x100e5909: mov eax,ds:[1012E798h]
0x100e590e: mov [esi+64C9h],eax
0x100e5914: mov cx,ds:[1012E79Ch]
0x100e591b: mov [esi+64CDh],cx
0x100e5922: mov dl,ds:[1012E79Eh]
0x100e5928: mov [esi+64CFh],dl
0x100e592e: pop esi
0x100e592f: pop ebx
0x100e5930: add esp,2Ch
0x100e5933: ret
0x100e5934: mov eax,ds:[1012E78Ch]
0x100e5939: mov [esi+64C5h],eax
```

```
0x100e593f: mov cl,ds:[1012E790h]
0x100e5945: mov [esi+64C9h],cl
0x100e594b: pop esi
0x100e594c: pop ebx
0x100e594d: add esp,2Ch
0x100e5950: ret
0x100e5951: cmp eax,1570h
0x100e5956: jne short 100E596Fh
0x100e5958: mov edx,ds:[1012E784h]
0x100e595e: mov [esi+64C5h],edx
0x100e5964: mov al,ds:[1012E788h]
0x100e5969: mov [esi+64C9h],al
0x100e596f: pop esi
0x100e5970: pop ebx
0x100e5971: add esp,2Ch
0x100e5974: ret
```

`sub_100e55a0` is called by sub_100eb620. It works with the text "Phase One". It accesses memory-mapped peripheral registers at 0x52617700. It delegates to sub_100d2fe0, sub_100d3050 (2 callees).

12.1297 sub_100e5ad0 at 0x100e5ad0 (108 instructions)

```
0x100e5ad0: push ecx
0x100e5ad1: mov ecx,[esp+0Ch]
0x100e5ad5: push ebx
0x100e5ad6: push esi
0x100e5ad7: mov esi,[esp+10h]
0x100e5adb: mov eax,[esi]
0x100e5add: mov edx,[esi+4]
0x100e5ae0: mov eax,[eax+8]
0x100e5ae3: push 0
0x100e5ae5: push ecx
0x100e5ae6: push edx
0x100e5ae7: call eax
0x100e5ae9: push esi
0x100e5aea: call 100D2FE0h
0x100e5aef: mov ebx,eax
0x100e5af1: add esp,10h
0x100e5af4: cmp ebx,0FFh
0x100e5afa: ja near ptr 100E5C23h
0x100e5b00: test ebx,ebx
0x100e5b02: je near ptr 100E5C0Bh
0x100e5b08: push ebp
0x100e5b09: push edi
0x100e5b0a: lea ebx,[ebx]
0x100e5b10: push esi
0x100e5b11: sub ebx,1
0x100e5b14: call 100D2F50h
0x100e5b19: push esi
0x100e5b1a: movzx edi,ax
0x100e5b1d: call 100D2F50h
0x100e5b22: mov ecx,[esi]
0x100e5b24: mov edx,[esi+4]
0x100e5b27: movzx ebp,ax
0x100e5b2a: mov eax,[ecx+18h]
0x100e5b2d: push edx
0x100e5b2e: call eax
0x100e5b30: add esp,0Ch
0x100e5b33: cmp edi,100h
0x100e5b39: mov [esp+10h],eax
0x100e5b3d: jne short 100E5B61h
0x100e5b3f: push esi
0x100e5b40: call 100D2F50h
0x100e5b45: push esi
0x100e5b46: mov [esi+66B0h],ax
0x100e5b4d: call 100D2F50h
```

```
0x100e5b52: add esp,8
0x100e5b55: mov [esi+66B2h],ax
0x100e5b5c: jmp near ptr 100E5BEAh
0x100e5b61: cmp edi,121h
0x100e5b67: jne short 100E5B97h
0x100e5b69: push esi
0x100e5b6a: call 100D2F50h
0x100e5b6f: push esi
0x100e5b70: mov [esi+66B4h],ax
0x100e5b77: call 100D2F50h
0x100e5b7c: add esp,8
0x100e5b7f: cmp ax,10BCh
0x100e5b83: mov [esi+66B6h],ax
0x100e5b8a: jne short 100E5BEAh
0x100e5b8c: mov word ptr [esi+66B6h],10BFh
0x100e5b95: jmp short 100E5BEAh
0x100e5b97: cmp edi,130h
0x100e5b9d: jne short 100E5BB8h
0x100e5b9f: mov ecx,[esi]
0x100e5ba1: mov edx,[esi+4]
0x100e5ba4: mov eax,[ecx+1Ch]
0x100e5ba7: push edx
0x100e5ba8: call eax
0x100e5baa: add esp,4
0x100e5bad: sar eax,7
0x100e5bb0: mov [esi+65E4h],eax
0x100e5bb6: jmp short 100E5BEAh
0x100e5bb8: cmp edi,2FF0h
0x100e5bbe: jne short 100E5BEAh
0x100e5bc0: xor edi,edi
0x100e5bc2: push esi
0x100e5bc3: call 100D2F50h
0x100e5bc8: movzx ecx,ax
0x100e5bcb: mov [esp+1Ch],ecx
0x100e5bcf: mov edx,edi
0x100e5bd1: xor edx,1
0x100e5bd4: fild dword ptr [esp+1Ch]
0x100e5bd8: add edi,1
0x100e5bdb: add esp,4
0x100e5bde: cmp edi,4
0x100e5be1: fstp dword ptr [esi+edx*4+0E774h]
0x100e5be8: jb short 100E5BC2h
0x100e5bea: mov ecx,[esp+10h]
0x100e5bee: mov eax,[esi]
0x100e5bf0: mov edx,[esi+4]
0x100e5bf3: mov eax,[eax+8]
0x100e5bf6: push 0
0x100e5bf8: add ecx,ebp
0x100e5bfa: push ecx
0x100e5bfb: push edx
0x100e5bfc: call eax
0x100e5bfe: add esp,0Ch
0x100e5c01: test ebx,ebx
0x100e5c03: jne near ptr 100E5B10h
0x100e5c09: pop edi
0x100e5c0a: pop ebp
0x100e5c0b: movzx eax,word ptr [esi+65E4h]
0x100e5c12: mov cx,ax
0x100e5c15: shl word ptr [esi+66B4h],cl
0x100e5c1c: shr word ptr [esi+66B6h],cl
0x100e5c23: pop esi
0x100e5c24: pop ebx
0x100e5c25: pop ecx
0x100e5c26: ret
```

`sub_100e5ad0` is called by sub_100eb620. It delegates to sub_100d2f50, sub_100d2fe0 (2 callees).

12.1298 sub_100e5c30 at 0x100e5c30 (401 instructions)

```
0x100e5c30: sub esp,0B0h
0x100e5c36: mov eax,ds:[10136220h]
0x100e5c3b: xor eax,esp
0x100e5c3d: mov [esp+0ACh],eax
0x100e5c44: push ebx
0x100e5c45: push ebp
0x100e5c46: push esi
0x100e5c47: push edi
0x100e5c48: mov edi,[esp+0C4h]
0x100e5c4f: mov eax,[edi]
0x100e5c51: mov edx,[edi+4]
0x100e5c54: push 1
0x100e5c56: push 4
0x100e5c58: lea ecx,[esp+18h]
0x100e5c5c: push ecx
0x100e5c5d: mov word ptr [edi+6274h],4949h
0x100e5c66: mov eax,[eax]
0x100e5c68: push edx
0x100e5c69: call eax
0x100e5c6b: push edi
0x100e5c6c: call 100D2FE0h
0x100e5c71: mov ecx,[edi]
0x100e5c73: mov edx,[edi+4]
0x100e5c76: mov esi,eax
0x100e5c78: mov eax,[ecx+18h]
0x100e5c7b: push edx
0x100e5c7c: mov [esp+2Ch],esi
0x100e5c80: call eax
0x100e5c82: mov ebx,eax
0x100e5c84: add esp,18h
0x100e5c87: add ebx,esi
0x100e5c89: mov edx,4
0x100e5c8e: mov eax,1012E7E8h
0x100e5c93: lea ecx,[esp+10h]
0x100e5c97: mov esi,[ecx]
0x100e5c99: cmp esi,[eax]
0x100e5c9b: jne short 100E5CAFh
0x100e5c9d: sub edx,4
0x100e5ca0: add eax,4
0x100e5ca3: add ecx,4
0x100e5ca6: cmp edx,4
0x100e5ca9: jae short 100E5C97h
0x100e5cab: test edx,edx
0x100e5cad: je short 100E5D0Ch
0x100e5caf: movzx ebp,byte ptr [eax]
0x100e5cb2: movzx esi,byte ptr [ecx]
0x100e5cb5: sub esi,ebp
0x100e5cb7: jne short 100E5CFEh
0x100e5cb9: sub edx,1
0x100e5cbc: add eax,1
0x100e5cbf: add ecx,1
0x100e5cc2: test edx,edx
0x100e5cc4: je short 100E5D0Ch
0x100e5cc6: movzx ebp,byte ptr [eax]
0x100e5cc9: movzx esi,byte ptr [ecx]
0x100e5ccc: sub esi,ebp
0x100e5cce: jne short 100E5CFEh
0x100e5cd0: sub edx,1
0x100e5cd3: add eax,1
0x100e5cd6: add ecx,1
0x100e5cd9: test edx,edx
0x100e5cdb: je short 100E5D0Ch
0x100e5cdd: movzx ebp,byte ptr [eax]
0x100e5ce0: movzx esi,byte ptr [ecx]
0x100e5ce3: sub esi,ebp
0x100e5ce5: jne short 100E5CFEh
```

```
0x100e5ce7: sub edx,1
0x100e5cea: add eax,1
0x100e5ced: add ecx,1
0x100e5cf0: test edx,edx
0x100e5cf2: je short 100E5D0Ch
0x100e5cf4: movzx edx,byte ptr [eax]
0x100e5cf7: movzx esi,byte ptr [ecx]
0x100e5cfa: sub esi,edx
0x100e5cfc: je short 100E5D0Ch
0x100e5cfe: test esi,esi
0x100e5d00: mov eax,1
0x100e5d05: jg short 100E5D0Eh
0x100e5d07: or eax,0FFFFFFFFh
0x100e5d0a: jmp short 100E5D0Eh
0x100e5d0c: xor eax,eax
0x100e5d0e: test eax,eax
0x100e5d10: je near ptr 100E602Fh
0x100e5d16: mov eax,4
0x100e5d1b: mov ecx,1012E7E0h
0x100e5d20: lea edx,[esp+10h]
0x100e5d24: mov esi,[edx]
0x100e5d26: cmp esi,[ecx]
0x100e5d28: jne short 100E5D3Ch
0x100e5d2a: sub eax,4
0x100e5d2d: add ecx,4
0x100e5d30: add edx,4
0x100e5d33: cmp eax,4
0x100e5d36: jae short 100E5D24h
0x100e5d38: test eax,eax
0x100e5d3a: je short 100E5D99h
0x100e5d3c: movzx esi,byte ptr [edx]
0x100e5d3f: movzx ebp,byte ptr [ecx]
0x100e5d42: sub esi,ebp
0x100e5d44: jne short 100E5D8Bh
0x100e5d46: sub eax,1
0x100e5d49: add ecx,1
0x100e5d4c: add edx,1
0x100e5d4f: test eax,eax
0x100e5d51: je short 100E5D99h
0x100e5d53: movzx esi,byte ptr [edx]
0x100e5d56: movzx ebp,byte ptr [ecx]
0x100e5d59: sub esi,ebp
0x100e5d5b: jne short 100E5D8Bh
0x100e5d5d: sub eax,1
0x100e5d60: add ecx,1
0x100e5d63: add edx,1
0x100e5d66: test eax,eax
0x100e5d68: je short 100E5D99h
0x100e5d6a: movzx esi,byte ptr [edx]
0x100e5d6d: movzx ebp,byte ptr [ecx]
0x100e5d70: sub esi,ebp
0x100e5d72: jne short 100E5D8Bh
0x100e5d74: sub eax,1
0x100e5d77: add ecx,1
0x100e5d7a: add edx,1
0x100e5d7d: test eax,eax
0x100e5d7f: je short 100E5D99h
0x100e5d81: movzx esi,byte ptr [edx]
0x100e5d84: movzx eax,byte ptr [ecx]
0x100e5d87: sub esi,eax
0x100e5d89: je short 100E5D99h
0x100e5d8b: test esi,esi
0x100e5d8d: mov eax,1
0x100e5d92: jg short 100E5D9Bh
0x100e5d94: or eax,0FFFFFFFFh
0x100e5d97: jmp short 100E5D9Bh
0x100e5d99: xor eax,eax
```

```
0x100e5d9b: test eax,eax
0x100e5d9d: je near ptr 100E602Fh
0x100e5da3: mov eax,4
0x100e5da8: mov ecx,1012E7D8h
0x100e5dad: lea edx,[esp+10h]
0x100e5db1: mov esi,[edx]
0x100e5db3: cmp esi,[ecx]
0x100e5db5: jne short 100E5DC9h
0x100e5db7: sub eax,4
0x100e5dba: add ecx,4
0x100e5dbd: add edx,4
0x100e5dc0: cmp eax,4
0x100e5dc3: jae short 100E5DB1h
0x100e5dc5: test eax,eax
0x100e5dc7: je short 100E5E26h
0x100e5dc9: movzx ebp,byte ptr [ecx]
0x100e5dcc: movzx esi,byte ptr [edx]
0x100e5dcf: sub esi,ebp
0x100e5dd1: jne short 100E5E18h
0x100e5dd3: sub eax,1
0x100e5dd6: add ecx,1
0x100e5dd9: add edx,1
0x100e5ddc: test eax,eax
0x100e5dde: je short 100E5E26h
0x100e5de0: movzx ebp,byte ptr [ecx]
0x100e5de3: movzx esi,byte ptr [edx]
0x100e5de6: sub esi,ebp
0x100e5de8: jne short 100E5E18h
0x100e5dea: sub eax,1
0x100e5ded: add ecx,1
0x100e5df0: add edx,1
0x100e5df3: test eax,eax
0x100e5df5: je short 100E5E26h
0x100e5df7: movzx ebp,byte ptr [ecx]
0x100e5dfa: movzx esi,byte ptr [edx]
0x100e5dfd: sub esi,ebp
0x100e5dff: jne short 100E5E18h
0x100e5e01: sub eax,1
0x100e5e04: add ecx,1
0x100e5e07: add edx,1
0x100e5e0a: test eax,eax
0x100e5e0c: je short 100E5E26h
0x100e5e0e: movzx ecx,byte ptr [ecx]
0x100e5e11: movzx esi,byte ptr [edx]
0x100e5e14: sub esi,ecx
0x100e5e16: je short 100E5E26h
0x100e5e18: test esi,esi
0x100e5e1a: mov eax,1
0x100e5e1f: jg short 100E5E28h
0x100e5e21: or eax,0FFFFFFFFh
0x100e5e24: jmp short 100E5E28h
0x100e5e26: xor eax,eax
0x100e5e28: test eax,eax
0x100e5e2a: jne short 100E5EA1h
0x100e5e2c: mov edx,[edi]
0x100e5e2e: mov eax,[edi+4]
0x100e5e31: mov ecx,[edx+18h]
0x100e5e34: push eax
0x100e5e35: call ecx
0x100e5e37: add eax,7
0x100e5e3a: add esp,4
0x100e5e3d: cmp eax,ebx
0x100e5e3f: jge near ptr 100E606Bh
0x100e5e45: push edi
0x100e5e46: call 100D2F50h
0x100e5e4b: push edi
0x100e5e4c: movzx esi,ax
```

```
0x100e5e4f: call 100D2F50h
0x100e5e54: add esi,1
0x100e5e57: and esi,0FFFFFFFEh
0x100e5e5a: add esp,8
0x100e5e5d: cmp esi,14h
0x100e5e60: movzx eax,ax
0x100e5e63: jne short 100E5E76h
0x100e5e65: cmp eax,esi
0x100e5e67: jne short 100E5E76h
0x100e5e69: push 0
0x100e5e6b: push edi
0x100e5e6c: call 100E3D60h
0x100e5e71: add esp,8
0x100e5e74: jmp short 100E5E87h
0x100e5e76: mov edx,[edi]
0x100e5e78: mov ecx,[edx+8]
0x100e5e7b: push 1
0x100e5e7d: push eax
0x100e5e7e: mov eax,[edi+4]
0x100e5e81: push eax
0x100e5e82: call ecx
0x100e5e84: add esp,0Ch
0x100e5e87: mov edx,[edi]
0x100e5e89: mov eax,[edi+4]
0x100e5e8c: mov ecx,[edx+18h]
0x100e5e8f: push eax
0x100e5e90: call ecx
0x100e5e92: add eax,7
0x100e5e95: add esp,4
0x100e5e98: cmp eax,ebx
0x100e5e9a: jl short 100E5E45h
0x100e5e9c: jmp near ptr 100E606Bh
0x100e5ea1: mov eax,4
0x100e5ea6: mov ecx,1012E7D0h
0x100e5eab: lea edx,[esp+10h]
0x100e5eaf: nop
0x100e5eb0: mov esi,[edx]
0x100e5eb2: cmp esi,[ecx]
0x100e5eb4: jne short 100E5EC8h
0x100e5eb6: sub eax,4
0x100e5eb9: add ecx,4
0x100e5ebc: add edx,4
0x100e5ebf: cmp eax,4
0x100e5ec2: jae short 100E5EB0h
0x100e5ec4: test eax,eax
0x100e5ec6: je short 100E5F25h
0x100e5ec8: movzx esi,byte ptr [edx]
0x100e5ecb: movzx ebx,byte ptr [ecx]
0x100e5ece: sub esi,ebx
0x100e5ed0: jne short 100E5F17h
0x100e5ed2: sub eax,1
0x100e5ed5: add ecx,1
0x100e5ed8: add edx,1
0x100e5edb: test eax,eax
0x100e5edd: je short 100E5F25h
0x100e5edf: movzx esi,byte ptr [edx]
0x100e5ee2: movzx ebx,byte ptr [ecx]
0x100e5ee5: sub esi,ebx
0x100e5ee7: jne short 100E5F17h
0x100e5ee9: sub eax,1
0x100e5eec: add ecx,1
0x100e5eef: add edx,1
0x100e5ef2: test eax,eax
0x100e5ef4: je short 100E5F25h
0x100e5ef6: movzx esi,byte ptr [edx]
0x100e5ef9: movzx ebx,byte ptr [ecx]
0x100e5efc: sub esi,ebx
```

```
0x100e5efe: jne short 100E5F17h
0x100e5f00: sub eax,1
0x100e5f03: add ecx,1
0x100e5f06: add edx,1
0x100e5f09: test eax,eax
0x100e5f0b: je short 100E5F25h
0x100e5f0d: movzx esi,byte ptr [edx]
0x100e5f10: movzx edx,byte ptr [ecx]
0x100e5f13: sub esi,edx
0x100e5f15: je short 100E5F25h
0x100e5f17: test esi,esi
0x100e5f19: mov eax,1
0x100e5f1e: jg short 100E5F27h
0x100e5f20: or eax,0FFFFFFFFh
0x100e5f23: jmp short 100E5F27h
0x100e5f25: xor eax,eax
0x100e5f27: test eax,eax
0x100e5f29: mov esi,[esp+14h]
0x100e5f2d: jne near ptr 100E601Ch
0x100e5f33: cmp esi,40h
0x100e5f36: jae near ptr 100E601Ch
0x100e5f3c: mov eax,[edi]
0x100e5f3e: mov edx,[edi+4]
0x100e5f41: mov eax,[eax]
0x100e5f43: push 1
0x100e5f45: push 40h
0x100e5f47: lea ecx,[esp+44h]
0x100e5f4b: push ecx
0x100e5f4c: push edx
0x100e5f4d: call eax
0x100e5f4f: xor eax,eax
0x100e5f51: lea ecx,[esp+3Ch]
0x100e5f55: push ecx
0x100e5f56: mov [esp+2Ch],eax
0x100e5f5a: mov [esp+30h],eax
0x100e5f5e: mov [esp+34h],eax
0x100e5f62: mov [esp+38h],eax
0x100e5f66: mov [esp+3Ch],eax
0x100e5f6a: mov [esp+40h],eax
0x100e5f6e: mov [esp+44h],eax
0x100e5f72: mov [esp+48h],eax
0x100e5f76: mov [esp+4Ch],eax
0x100e5f7a: lea edx,[esp+2Ch]
0x100e5f7e: push edx
0x100e5f7f: lea eax,[esp+34h]
0x100e5f83: push eax
0x100e5f84: lea ecx,[esp+3Ch]
0x100e5f88: push ecx
0x100e5f89: lea edx,[esp+44h]
0x100e5f8d: push edx
0x100e5f8e: lea eax,[esp+0A0h]
0x100e5f95: push eax
0x100e5f96: lea ecx,[esp+64h]
0x100e5f9a: push 1012E7B8h
0x100e5f9f: push ecx
0x100e5fa0: mov byte ptr [esp+esi+6Ch],0
0x100e5fa5: call dword ptr ds:[100FE4A8h]
0x100e5fab: add esp,30h
0x100e5fae: cmp eax,6
0x100e5fb1: jne near ptr 100E606Bh
0x100e5fb7: mov ebp,ds:[100FE440h]
0x100e5fbd: xor esi,esi
0x100e5fbf: mov ebx,1012B8F4h
0x100e5fc4: lea edx,[esp+7Ch]
0x100e5fc8: push edx
0x100e5fc9: push ebx
0x100e5fca: call ebp
```

```
0x100e5fcc: add esp,8
0x100e5fcf: test eax,eax
0x100e5fd1: je short 100E5FDEh
0x100e5fd3: add esi,1
0x100e5fd6: add ebx,4
0x100e5fd9: cmp esi,0Ch
0x100e5fdc: jb short 100E5FC4h
0x100e5fde: sub dword ptr [esp+2Ch],76Ch
0x100e5fe6: lea eax,[esp+18h]
0x100e5fea: mov [esp+28h],esi
0x100e5fee: mov esi,ds:[100FE444h]
0x100e5ff4: push eax
0x100e5ff5: call esi
0x100e5ff7: add esp,4
0x100e5ffa: test edx,edx
0x100e5ffc: jl short 100E606Bh
0x100e5ffe: jg short 100E6004h
0x100e6000: test eax,eax
0x100e6002: jbe short 100E606Bh
0x100e6004: lea ecx,[esp+18h]
0x100e6008: push ecx
0x100e6009: call esi
0x100e600b: add esp,4
0x100e600e: mov [edi+65A0h],eax
0x100e6014: mov [edi+65A4h],edx
0x100e601a: jmp short 100E606Bh
0x100e601c: mov edx,[edi]
0x100e601e: mov eax,[edi+4]
0x100e6021: mov ecx,[edx+8]
0x100e6024: push 1
0x100e6026: push esi
0x100e6027: push eax
0x100e6028: call ecx
0x100e602a: add esp,0Ch
0x100e602d: jmp short 100E606Bh
0x100e602f: push edi
0x100e6030: call 100D2FE0h
0x100e6035: mov edx,[edi]
0x100e6037: mov eax,[edi+4]
0x100e603a: mov ecx,[edx+18h]
0x100e603d: push eax
0x100e603e: call ecx
0x100e6040: add eax,7
0x100e6043: add esp,8
0x100e6046: cmp eax,ebx
0x100e6048: jge short 100E606Bh
0x100e604a: lea ebx,[ebx]
0x100e6050: push edi
0x100e6051: call 100E5C30h
0x100e6056: mov edx,[edi]
0x100e6058: mov eax,[edi+4]
0x100e605b: mov ecx,[edx+18h]
0x100e605e: push eax
0x100e605f: call ecx
0x100e6061: add eax,7
0x100e6064: add esp,8
0x100e6067: cmp eax,ebx
0x100e6069: jl short 100E6050h
0x100e606b: mov ecx,[esp+0BCh]
0x100e6072: pop edi
0x100e6073: pop esi
0x100e6074: pop ebp
0x100e6075: pop ebx
0x100e6076: xor ecx,esp
0x100e6078: call 10022326h
0x100e607d: add esp,0B0h
0x100e6083: ret
```

`sub_100e5c30` is called by sub_100e5c30, sub_100eb620. It delegates to sub_10022326, sub_100d2f50, sub_100d2fe0 (5 callees).

12.1299 sub_100e6090 at 0x100e6090 (75 instructions)

```
0x100e6090: mov ecx,[esp+8]
0x100e6094: push esi
0x100e6095: mov esi,[esp+8]
0x100e6099: mov eax,[esi]
0x100e609b: mov edx,[esi+4]
0x100e609e: mov eax,[eax+8]
0x100e60a1: push edi
0x100e60a2: push 0
0x100e60a4: add ecx,2
0x100e60a7: push ecx
0x100e60a8: push edx
0x100e60a9: call eax
0x100e60ab: mov ecx,[esi]
0x100e60ad: mov edx,[esi+4]
0x100e60b0: mov word ptr [esi+6274h],4949h
0x100e60b9: mov eax,[ecx+1Ch]
0x100e60bc: push edx
0x100e60bd: call eax
0x100e60bf: mov edi,eax
0x100e60c1: add esp,10h
0x100e60c4: cmp edi,6
0x100e60c7: jne short 100E60DBh
0x100e60c9: mov ecx,[esi]
0x100e60cb: mov edx,[esi+4]
0x100e60ce: mov eax,[ecx+8]
0x100e60d1: push 1
0x100e60d3: push 5
0x100e60d5: push edx
0x100e60d6: call eax
0x100e60d8: add esp,0Ch
0x100e60db: push esi
0x100e60dc: call 100D2FE0h
0x100e60e1: add esp,4
0x100e60e4: cmp eax,[esp+14h]
0x100e60e8: jne near ptr 100E6185h
0x100e60ee: cmp edi,6
0x100e60f1: jle short 100E6102h
0x100e60f3: push esi
0x100e60f4: call 100D2FE0h
0x100e60f9: add esp,4
0x100e60fc: mov [esi+65C0h],eax
0x100e6102: push esi
0x100e6103: call 100D2F50h
0x100e6108: push esi
0x100e6109: mov [esi+66B4h],ax
0x100e6110: mov [esi+66B0h],ax
0x100e6117: call 100D2F50h
0x100e611c: mov [esi+66B6h],ax
0x100e6123: mov [esi+66B2h],ax
0x100e612a: mov ecx,ds:[1012E7FCh]
0x100e6130: mov [esi+6485h],ecx
0x100e6136: mov dl,ds:[1012E800h]
0x100e613c: movzx ecx,word ptr [esi+66B4h]
0x100e6143: mov [esi+6489h],dl
0x100e6149: push ecx
0x100e614a: movzx edx,ax
0x100e614d: push edx
0x100e614e: push edi
0x100e614f: lea eax,[esi+64C5h]
0x100e6155: push 1012E7F0h
0x100e615a: push eax
0x100e615b: call dword ptr ds:[100FE4F8h]
```

```
0x100e6161: add esp,1Ch
0x100e6164: cmp edi,6
0x100e6167: jne short 100E6176h
0x100e6169: pop edi
0x100e616a: mov dword ptr [esi+2E7FCh],100DCC40h
0x100e6174: pop esi
0x100e6175: ret
0x100e6176: cmp edi,9
0x100e6179: jne short 100E6185h
0x100e617b: mov dword ptr [esi+2E7FCh],100DD160h
0x100e6185: pop edi
0x100e6186: pop esi
0x100e6187: ret
```

`sub_100e6090` is called by sub_100eb620. It delegates to sub_100d2f50, sub_100d2fe0 (2 callees).

12.1300 sub_100e6190 at 0x100e6190 (214 instructions)

```
0x100e6190: push ecx
0x100e6191: push ebx
0x100e6192: push esi
0x100e6193: mov esi,[esp+10h]
0x100e6197: mov eax,[esi]
0x100e6199: mov ecx,[esi+4]
0x100e619c: push edi
0x100e619d: push 0
0x100e619f: mov word ptr [esi+6274h],4949h
0x100e61a8: mov edx,[eax+8]
0x100e61ab: push 4
0x100e61ad: push ecx
0x100e61ae: call edx
0x100e61b0: push esi
0x100e61b1: call 100D2F50h
0x100e61b6: mov edx,[esi]
0x100e61b8: xor ecx,ecx
0x100e61ba: cmp ax,2
0x100e61be: mov eax,[esi+4]
0x100e61c1: sete cl
0x100e61c4: push 1
0x100e61c6: push 0Eh
0x100e61c8: push eax
0x100e61c9: mov [esi+6614h],ecx
0x100e61cf: mov ecx,[edx+8]
0x100e61d2: call ecx
0x100e61d4: push esi
0x100e61d5: call 100D2FE0h
0x100e61da: imul eax,[esi+6614h]
0x100e61e1: push esi
0x100e61e2: mov [esi+6614h],eax
0x100e61e8: call 100D2FE0h
0x100e61ed: push esi
0x100e61ee: mov edi,eax
0x100e61f0: call 100D2FE0h
0x100e61f5: push esi
0x100e61f6: mov ebx,eax
0x100e61f8: call 100D2FE0h
0x100e61fd: push esi
0x100e61fe: mov [esp+44h],eax
0x100e6202: call 100D2FE0h
0x100e6207: push esi
0x100e6208: mov [esi+65A0h],eax
0x100e620e: mov dword ptr [esi+65A4h],0
0x100e6218: call 100D2FE0h
0x100e621d: add esp,34h
0x100e6220: test eax,eax
0x100e6222: je short 100E6234h
```

```
0x100e6224: mov [esi+65A0h],eax
0x100e622a: mov dword ptr [esi+65A4h],0
0x100e6234: mov edx,[esi]
0x100e6236: mov eax,[esi+4]
0x100e6239: mov ecx,[edx+8]
0x100e623c: push 0
0x100e623e: add edi,4
0x100e6241: push edi
0x100e6242: push eax
0x100e6243: call ecx
0x100e6245: push esi
0x100e6246: call 100D2FE0h
0x100e624b: push esi
0x100e624c: mov [esi+66B2h],ax
0x100e6253: call 100D2FE0h
0x100e6258: push esi
0x100e6259: mov [esi+66B0h],ax
0x100e6260: call 100D2F50h
0x100e6265: push esi
0x100e6266: call 100D2F50h
0x100e626b: movzx eax,ax
0x100e626e: add esp,1Ch
0x100e6271: cmp eax,8
0x100e6274: je short 100E6287h
0x100e6276: cmp eax,10h
0x100e6279: jne short 100E6291h
0x100e627b: mov dword ptr [esi+2E7FCh],100D8E20h
0x100e6285: jmp short 100E6291h
0x100e6287: mov dword ptr [esi+2E7FCh],100DAEF0h
0x100e6291: mov edx,[esi]
0x100e6293: mov eax,[esi+4]
0x100e6296: mov ecx,[edx+8]
0x100e6299: push 0
0x100e629b: add ebx,318h
0x100e62a1: push ebx
0x100e62a2: push eax
0x100e62a3: call ecx
0x100e62a5: mov edx,ds:[1012A2C0h]
0x100e62ab: mov [esi+6485h],edx
0x100e62b1: mov al,ds:[1012A2C4h]
0x100e62b6: push esi
0x100e62b7: mov [esi+6489h],al
0x100e62bd: call 100D2FE0h
0x100e62c2: push eax
0x100e62c3: lea ecx,[esi+64C5h]
0x100e62c9: push 1010BCACh
0x100e62ce: push ecx
0x100e62cf: call dword ptr ds:[100FE4F8h]
0x100e62d5: mov edx,[esi]
0x100e62d7: mov eax,[esi+4]
0x100e62da: mov ecx,[edx+8]
0x100e62dd: push 1
0x100e62df: push 0Ch
0x100e62e1: push eax
0x100e62e2: call ecx
0x100e62e4: push esi
0x100e62e5: call 100D2FE0h
0x100e62ea: and eax,0FFFFFFh
0x100e62ef: add esp,2Ch
0x100e62f2: sub eax,3
0x100e62f5: je short 100E6314h
0x100e62f7: sub eax,1
0x100e62fa: je short 100E6308h
0x100e62fc: mov dword ptr [esi+6614h],0
0x100e6306: jmp short 100E631Eh
0x100e6308: mov dword ptr [esi+65B0h],49494949h
0x100e6312: jmp short 100E631Eh
```

```
0x100e6314: mov dword ptr [esi+65B0h],94949494h
0x100e631e: mov edx,[esi]
0x100e6320: mov eax,[esi+4]
0x100e6323: mov ecx,[edx+8]
0x100e6326: push 1
0x100e6328: push 48h
0x100e632a: push eax
0x100e632b: call ecx
0x100e632d: push esi
0x100e632e: call 100D2FE0h
0x100e6333: add eax,0E10h
0x100e6338: xor edx,edx
0x100e633a: mov ecx,168h
0x100e633f: div ecx
0x100e6341: add esp,10h
0x100e6344: cmp edx,0B4h
0x100e634a: ja short 100E637Bh
0x100e634c: je short 100E636Fh
0x100e634e: test edx,edx
0x100e6350: je short 100E6363h
0x100e6352: cmp edx,5Ah
0x100e6355: jne short 100E638Dh
0x100e6357: mov dword ptr [esi+66C8h],7
0x100e6361: jmp short 100E638Dh
0x100e6363: mov dword ptr [esi+66C8h],2
0x100e636d: jmp short 100E638Dh
0x100e636f: mov dword ptr [esi+66C8h],1
0x100e6379: jmp short 100E638Dh
0x100e637b: cmp edx,10Eh
0x100e6381: jne short 100E638Dh
0x100e6383: mov dword ptr [esi+66C8h],4
0x100e638d: push 0Bh
0x100e638f: push esi
0x100e6390: call 100D3060h
0x100e6395: fstp dword ptr [esi+0E774h]
0x100e639b: push 0Bh
0x100e639d: push esi
0x100e639e: call 100D3060h
0x100e63a3: fstp dword ptr [esi+0E77Ch]
0x100e63a9: push esi
0x100e63aa: call 100D2FE0h
0x100e63af: or edx,0FFFFFFFFh
0x100e63b2: mov ecx,eax
0x100e63b4: mov eax,[esi]
0x100e63b6: shl edx,cl
0x100e63b8: mov ecx,[esi+4]
0x100e63bb: push 1
0x100e63bd: push 29Ch
0x100e63c2: not edx
0x100e63c4: mov [esi+65FCh],edx
0x100e63ca: mov edx,[eax+8]
0x100e63cd: push ecx
0x100e63ce: call edx
0x100e63d0: push esi
0x100e63d1: call 100D2FE0h
0x100e63d6: test eax,eax
0x100e63d8: mov [esp+30h],eax
0x100e63dc: fild dword ptr [esp+30h]
0x100e63e0: jge short 100E63E8h
0x100e63e2: fadd dword ptr ds:[100FE7C0h]
0x100e63e8: fdiv qword ptr ds:[1012E808h]
0x100e63ee: mov ecx,[esp+38h]
0x100e63f2: mov eax,[esi]
0x100e63f4: mov edx,[esi+4]
0x100e63f7: push 0
0x100e63f9: push ecx
0x100e63fa: push edx
```

```
0x100e63fb: fstp dword ptr [esi+6594h]
0x100e6401: mov eax,[eax+8]
0x100e6404: call eax
0x100e6406: mov eax,[esi+0A0h]
0x100e640c: add esp,30h
0x100e640f: cmp eax,[esi+6614h]
0x100e6415: jae short 100E642Fh
0x100e6417: mov ecx,[esi]
0x100e6419: mov ecx,[ecx+8]
0x100e641c: lea edx,[eax*8]
0x100e6423: mov eax,[esi+4]
0x100e6426: push 1
0x100e6428: push edx
0x100e6429: push eax
0x100e642a: call ecx
0x100e642c: add esp,0Ch
0x100e642f: push esi
0x100e6430: call 100D2FE0h
0x100e6435: add eax,8
0x100e6438: push esi
0x100e6439: mov [esi+65C0h],eax
0x100e643f: call 100D2FE0h
0x100e6444: add esp,8
0x100e6447: xor edx,edx
0x100e6449: mov cl,20h
0x100e644b: call 100FC230h
0x100e6450: add [esi+65C0h],eax
0x100e6456: pop edi
0x100e6457: pop esi
0x100e6458: pop ebx
0x100e6459: pop ecx
0x100e645a: ret
```

`sub_100e6190` is called by sub_100eb620. It accesses memory-mapped peripheral registers at 0x49494949. It delegates to sub_100d2f50, sub_100d2fe0, sub_100d3060 (4 callees).

12.1301 sub_100e6460 at 0x100e6460 (43 instructions)

```
0x100e6460: mov ecx,[esp+8]
0x100e6464: push ebx
0x100e6465: mov ebx,[esp+8]
0x100e6469: mov eax,[ebx]
0x100e646b: mov edx,[ebx+4]
0x100e646e: mov eax,[eax+8]
0x100e6471: push ebp
0x100e6472: push esi
0x100e6473: push 0
0x100e6475: push ecx
0x100e6476: push edx
0x100e6477: call eax
0x100e6479: mov ebp,[esp+28h]
0x100e647d: add ebp,0FFFFFFFFh
0x100e6480: add esp,0Ch
0x100e6483: xor esi,esi
0x100e6485: test ebp,ebp
0x100e6487: jle short 100E64B2h
0x100e6489: push edi
0x100e648a: mov edi,[esp+1Ch]
0x100e648e: mov edi,edi
0x100e6490: push ebx
0x100e6491: call 100D2F50h
0x100e6496: add esp,4
0x100e6499: test al,al
0x100e649b: mov [esi+edi],al
0x100e649e: je short 100E64A7h
0x100e64a0: add esi,1
```

```
0x100e64a3: cmp esi,ebp
0x100e64a5: jl short 100E6490h
0x100e64a7: mov byte ptr [esi+edi],0
0x100e64ab: mov eax,edi
0x100e64ad: pop edi
0x100e64ae: pop esi
0x100e64af: pop ebp
0x100e64b0: pop ebx
0x100e64b1: ret
0x100e64b2: mov eax,[esp+18h]
0x100e64b6: mov byte ptr [esi+eax],0
0x100e64ba: pop esi
0x100e64bb: pop ebp
0x100e64bc: pop ebx
0x100e64bd: ret
```

`sub_100e6460` is called by sub_100eb170. It delegates to sub_100d2f50 (1 callees).

12.1302 sub_100e64c0 at 0x100e64c0 (117 instructions)

```
0x100e64c0: push ebp
0x100e64c1: mov ebp,esp
0x100e64c3: and esp,0FFFFFFF8h
0x100e64c6: sub esp,94h
0x100e64cc: mov eax,ds:[10136220h]
0x100e64d1: xor eax,esp
0x100e64d3: mov [esp+90h],eax
0x100e64da: mov eax,[ebp+8]
0x100e64dd: mov ecx,[ebp+10h]
0x100e64e0: push ebx
0x100e64e1: push esi
0x100e64e2: push edi
0x100e64e3: push ecx
0x100e64e4: mov [esp+10h],eax
0x100e64e8: mov eax,[ebp+0Ch]
0x100e64eb: push eax
0x100e64ec: lea ecx,[esp+18h]
0x100e64f0: push 1012E810h
0x100e64f5: push ecx
0x100e64f6: call dword ptr ds:[100FE4F8h]
0x100e64fc: add esp,10h
0x100e64ff: xor esi,esi
0x100e6501: mov ebx,1012B930h
0x100e6506: mov edx,[ebx]
0x100e6508: mov eax,edx
0x100e650a: lea edi,[eax+1]
0x100e650d: lea ecx,[ecx]
0x100e6510: mov cl,[eax]
0x100e6512: add eax,1
0x100e6515: test cl,cl
0x100e6517: jne short 100E6510h
0x100e6519: sub eax,edi
0x100e651b: push eax
0x100e651c: push edx
0x100e651d: lea edx,[esp+18h]
0x100e6521: push edx
0x100e6522: call dword ptr ds:[100FE414h]
0x100e6528: add esp,0Ch
0x100e652b: test eax,eax
0x100e652d: je short 100E6552h
0x100e652f: add esi,1
0x100e6532: add ebx,20h
0x100e6535: cmp esi,0D3h
0x100e653b: jb short 100E6506h
0x100e653d: pop edi
0x100e653e: pop esi
```

```
0x100e653f: pop ebx
0x100e6540: mov ecx,[esp+90h]
0x100e6547: xor ecx,esp
0x100e6549: call 10022326h
0x100e654e: mov esp,ebp
0x100e6550: pop ebp
0x100e6551: ret
0x100e6552: mov edx,[esp+0Ch]
0x100e6556: shl esi,5
0x100e6559: movzx eax,word ptr [esi+1012B934h]
0x100e6560: test ax,ax
0x100e6563: je short 100E656Eh
0x100e6565: movzx eax,ax
0x100e6568: mov [edx+65F8h],eax
0x100e656e: movzx eax,word ptr [esi+1012B936h]
0x100e6575: test ax,ax
0x100e6578: je short 100E6583h
0x100e657a: movzx ecx,ax
0x100e657d: mov [edx+65FCh],ecx
0x100e6583: fld qword ptr ds:[100FE7D8h]
0x100e6589: xor eax,eax
0x100e658b: lea ecx,[esi+1012B93Ah]
0x100e6591: movsx esi,word ptr [ecx-2]
0x100e6595: mov [esp+0Ch],esi
0x100e6599: movsx esi,word ptr [ecx]
0x100e659c: fild dword ptr [esp+0Ch]
0x100e65a0: mov [esp+0Ch],esi
0x100e65a4: movsx esi,word ptr [ecx+2]
0x100e65a8: fdiv st,st(1)
0x100e65aa: add eax,6
0x100e65ad: add ecx,0Ch
0x100e65b0: cmp eax,0Ch
0x100e65b3: fstp qword ptr [esp+eax*8-20h]
0x100e65b7: fild dword ptr [esp+0Ch]
0x100e65bb: mov [esp+0Ch],esi
0x100e65bf: movsx esi,word ptr [ecx-8]
0x100e65c3: fdiv st,st(1)
0x100e65c5: fstp qword ptr [esp+eax*8-18h]
0x100e65c9: fild dword ptr [esp+0Ch]
0x100e65cd: mov [esp+0Ch],esi
0x100e65d1: movsx esi,word ptr [ecx-6]
0x100e65d5: fdiv st,st(1)
0x100e65d7: fstp qword ptr [esp+eax*8-10h]
0x100e65db: fild dword ptr [esp+0Ch]
0x100e65df: mov [esp+0Ch],esi
0x100e65e3: movsx esi,word ptr [ecx-4]
0x100e65e7: fdiv st,st(1)
0x100e65e9: fstp qword ptr [esp+eax*8-8]
0x100e65ed: fild dword ptr [esp+0Ch]
0x100e65f1: mov [esp+0Ch],esi
0x100e65f5: fdiv st,st(1)
0x100e65f7: fstp qword ptr [esp+eax*8]
0x100e65fa: fild dword ptr [esp+0Ch]
0x100e65fe: fdiv st,st(1)
0x100e6600: fstp qword ptr [esp+eax*8+8]
0x100e6604: jl short 100E6591h
0x100e6606: lea eax,[esp+10h]
0x100e660a: fstp st(0)
0x100e660c: push eax
0x100e660d: push edx
0x100e660e: call 100DDC20h
0x100e6613: mov ecx,[esp+0A4h]
0x100e661a: add esp,8
0x100e661d: pop edi
0x100e661e: pop esi
0x100e661f: pop ebx
0x100e6620: xor ecx,esp
```

```
0x100e6622: call 10022326h
0x100e6627: mov esp,ebp
0x100e6629: pop ebp
0x100e662a: ret
```

`sub_100e64c0` is called by sub_100eb620. It delegates to sub_10022326, sub_100ddc20 (2 callees).

12.1303 sub_100e6630 at 0x100e6630 (37 instructions)

```
0x100e6630: mov edx,[esp+4]
0x100e6634: push ebx
0x100e6635: push ebp
0x100e6636: push esi
0x100e6637: mov esi,[esp+14h]
0x100e663b: push edi
0x100e663c: xor edi,edi
0x100e663e: mov [edx+6604h],edi
0x100e6644: lea ebx,[edx+0E7C4h]
0x100e664a: lea ebx,[ebx]
0x100e6650: xor eax,eax
0x100e6652: cmp [edx+66D0h],eax
0x100e6658: jle short 100E668Ch
0x100e665a: lea esi,[esi+esi*2]
0x100e665d: add esi,esi
0x100e665f: add esi,esi
0x100e6661: mov ecx,ebx
0x100e6663: mov ebp,[edx+66D0h]
0x100e6669: imul ebp,edi
0x100e666c: add ebp,esi
0x100e666e: add ebp,eax
0x100e6670: add eax,1
0x100e6673: add ecx,4
0x100e6676: fld dword ptr [ebp*4+1012D390h]
0x100e667d: fstp dword ptr [ecx-4]
0x100e6680: cmp eax,[edx+66D0h]
0x100e6686: jl short 100E6663h
0x100e6688: mov esi,[esp+18h]
0x100e668c: add edi,1
0x100e668f: add ebx,10h
0x100e6692: cmp edi,3
0x100e6695: jl short 100E6650h
0x100e6697: pop edi
0x100e6698: pop esi
0x100e6699: pop ebp
0x100e669a: pop ebx
0x100e669b: ret
```

`sub_100e6630` is called by sub_100eb620. It calls nothing else.

12.1304 sub_100e66a0 at 0x100e66a0 (93 instructions)

```
0x100e66a0: push ebp
0x100e66a1: mov ebp,esp
0x100e66a3: and esp,0FFFFFFF8h
0x100e66a6: sub esp,2Ch
0x100e66a9: mov eax,ds:[10136220h]
0x100e66ae: xor eax,esp
0x100e66b0: mov [esp+28h],eax
0x100e66b4: fldz
0x100e66b6: push ebx
0x100e66b7: mov ebx,[ebp+8]
0x100e66ba: fst qword ptr [esp+10h]
0x100e66be: mov eax,[ebx]
0x100e66c0: fstp qword ptr [esp+18h]
0x100e66c4: mov edx,[ebx+4]
```

```
0x100e66c7: mov eax,[eax]
0x100e66c9: push esi
0x100e66ca: push edi
0x100e66cb: mov esi,2
0x100e66d0: push esi
0x100e66d1: push esi
0x100e66d2: lea ecx,[esp+34h]
0x100e66d6: push ecx
0x100e66d7: push edx
0x100e66d8: mov [esp+20h],ebx
0x100e66dc: call eax
0x100e66de: add esp,10h
0x100e66e1: sub [ebp+0Ch],esi
0x100e66e4: je near ptr 100E6793h
0x100e66ea: jmp short 100E66F8h
0x100e66f0: mov ebx,[esp+10h]
0x100e66f4: fstp st(0)
0x100e66f6: fstp st(0)
0x100e66f8: mov ecx,[ebx]
0x100e66fa: mov edx,[ebx+4]
0x100e66fd: mov eax,[ecx]
0x100e66ff: sub dword ptr [ebp+0Ch],1
0x100e6703: push 1
0x100e6705: push 2
0x100e6707: lea edi,[esp+esi*2+34h]
0x100e670b: push edi
0x100e670c: push edx
0x100e670d: call eax
0x100e670f: movzx edx,byte ptr [edi]
0x100e6712: movzx edi,byte ptr [esp+esi*2+3Dh]
0x100e6717: mov eax,esi
0x100e6719: mov [esp+1Ch],edx
0x100e671d: shl edx,8
0x100e6720: xor eax,2
0x100e6723: add eax,eax
0x100e6725: movzx ecx,byte ptr [esp+eax+3Ch]
0x100e672a: or edi,edx
0x100e672c: movzx edx,byte ptr [esp+eax+3Dh]
0x100e6731: mov ebx,ecx
0x100e6733: shl ebx,8
0x100e6736: or edx,ebx
0x100e6738: sub edx,edi
0x100e673a: mov [esp+24h],edx
0x100e673e: xor edx,edx
0x100e6740: mov dh,[esp+esi*2+3Dh]
0x100e6744: fild dword ptr [esp+24h]
0x100e6748: xor ebx,ebx
0x100e674a: mov bh,[esp+eax+3Dh]
0x100e674e: add esi,1
0x100e6751: fmul st(0),st
0x100e6753: add esp,10h
0x100e6756: and esi,3
0x100e6759: fadd qword ptr [esp+18h]
0x100e675d: or edx,[esp+0Ch]
0x100e6761: fst qword ptr [esp+18h]
0x100e6765: or ebx,ecx
0x100e6767: sub ebx,edx
0x100e6769: cmp dword ptr [ebp+0Ch],0
0x100e676d: mov [esp+0Ch],ebx
0x100e6771: fild dword ptr [esp+0Ch]
0x100e6775: fmul st(0),st
0x100e6777: fadd qword ptr [esp+20h]
0x100e677b: fst qword ptr [esp+20h]
0x100e677f: jne near ptr 100E66F0h
0x100e6785: fcompp
0x100e6787: fnstsw ax
0x100e6789: test ah,41h
```

```
0x100e678c: mov eax,4D4Dh
0x100e6791: je short 100E6798h
0x100e6793: mov eax,4949h
0x100e6798: mov ecx,[esp+34h]
0x100e679c: pop edi
0x100e679d: pop esi
0x100e679e: pop ebx
0x100e679f: xor ecx,esp
0x100e67a1: call 10022326h
0x100e67a6: mov esp,ebp
0x100e67a8: pop ebp
0x100e67a9: ret
```

`sub_100e66a0` is called by sub_100eb620. It delegates to sub_10022326 (1 callees).

12.1305 sub_100e67b0 at 0x100e67b0 (528 instructions)

```
0x100e67b0: push ebp
0x100e67b1: mov ebp,esp
0x100e67b3: and esp,0FFFFFFF8h
0x100e67b6: sub esp,134h
0x100e67bc: fldz
0x100e67be: push ebx
0x100e67bf: mov ebx,[ebp+8]
0x100e67c2: fst qword ptr [esp+20h]
0x100e67c6: fst qword ptr [esp+28h]
0x100e67ca: xor ecx,ecx
0x100e67cc: fld1
0x100e67ce: mov eax,14h
0x100e67d3: fcom qword ptr [ebx+78h]
0x100e67d6: mov [esp+54h],eax
0x100e67da: mov [esp+60h],eax
0x100e67de: mov [esp+90h],eax
0x100e67e5: mov [esp+9Ch],eax
0x100e67ec: mov [esp+0A8h],eax
0x100e67f3: push esi
0x100e67f4: mov edx,0Eh
0x100e67f9: fnstsw ax
0x100e67fb: mov esi,24h
0x100e6800: test ah,41h
0x100e6803: push edi
0x100e6804: mov dword ptr [esp+38h],0Ah
0x100e680c: mov dword ptr [esp+3Ch],63707274h
0x100e6814: mov [esp+40h],ecx
0x100e6818: mov [esp+44h],esi
0x100e681c: mov dword ptr [esp+48h],64657363h
0x100e6824: mov [esp+4Ch],ecx
0x100e6828: mov dword ptr [esp+50h],28h
0x100e6830: mov dword ptr [esp+54h],77747074h
0x100e6838: mov [esp+58h],ecx
0x100e683c: mov dword ptr [esp+60h],626B7074h
0x100e6844: mov [esp+64h],ecx
0x100e6848: mov dword ptr [esp+6Ch],72545243h
0x100e6850: mov [esp+70h],ecx
0x100e6854: mov [esp+74h],edx
0x100e6858: mov dword ptr [esp+78h],67545243h
0x100e6860: mov [esp+7Ch],ecx
0x100e6864: mov [esp+80h],edx
0x100e686b: mov dword ptr [esp+84h],62545243h
0x100e6876: mov [esp+88h],ecx
0x100e687d: mov [esp+8Ch],edx
0x100e6884: mov dword ptr [esp+90h],7258595Ah
0x100e688f: mov [esp+94h],ecx
0x100e6896: mov dword ptr [esp+9Ch],6758595Ah
0x100e68a1: mov [esp+0A0h],ecx
0x100e68a8: mov dword ptr [esp+0A8h],6258595Ah
```

```
0x100e68b3: mov [esp+0ACh],ecx
0x100e68ba: mov dword ptr [esp+0C4h],1000000h
0x100e68c5: jp short 100E68CCh
0x100e68c7: mov ecx,1
0x100e68cc: fst qword ptr [esp+ecx*8+28h]
0x100e68d0: fld st(1)
0x100e68d2: fcomp qword ptr [ebx+78h]
0x100e68d5: fnstsw ax
0x100e68d7: test ah,44h
0x100e68da: jnp near ptr 100E698Eh
0x100e68e0: fld qword ptr [ebx+70h]
0x100e68e3: fsub st,st(1)
0x100e68e5: fld qword ptr [ebx+78h]
0x100e68e8: fsub st,st(2)
0x100e68ea: fmulp
0x100e68ec: fcomp st(2)
0x100e68ee: fnstsw ax
0x100e68f0: fstp st(1)
0x100e68f2: test ah,41h
0x100e68f5: jp near ptr 100E6990h
0x100e68fb: fstp st(0)
0x100e68fd: fld qword ptr [ebx+70h]
0x100e6900: fchs
0x100e6902: fstp qword ptr [esp+18h]
0x100e6906: jmp short 100E6912h
0x100e6910: fstp st(0)
0x100e6912: fld qword ptr [esp+30h]
0x100e6916: fadd qword ptr [esp+28h]
0x100e691a: fmul qword ptr ds:[100FE7C8h]
0x100e6920: fst qword ptr [esp+20h]
0x100e6924: fst qword ptr [ebx+80h]
0x100e692a: fdiv qword ptr [ebx+78h]
0x100e692d: fld qword ptr [esp+18h]
0x100e6931: call 100229A0h
0x100e6936: fld1
0x100e6938: fsub st(1),st
0x100e693a: fxch
0x100e693c: fdiv qword ptr [ebx+70h]
0x100e693f: fld st(1)
0x100e6941: fld qword ptr [esp+20h]
0x100e6945: fdiv st(1),st
0x100e6947: fxch st(2)
0x100e6949: fsubrp
0x100e694b: fcomp qword ptr ds:[100FF2E0h]
0x100e6951: fnstsw ax
0x100e6953: test ah,41h
0x100e6956: jne short 100E695Fh
0x100e6958: mov eax,1
0x100e695d: jmp short 100E6961h
0x100e695f: xor eax,eax
0x100e6961: sub esi,1
0x100e6964: fstp qword ptr [esp+eax*8+28h]
0x100e6968: jne short 100E6910h
0x100e696a: fld st(0)
0x100e696c: fdiv qword ptr [ebx+70h]
0x100e696f: fsub st,st(1)
0x100e6971: fmul qword ptr [ebx+80h]
0x100e6977: fstp qword ptr [ebx+88h]
0x100e697d: fld qword ptr [ebx+80h]
0x100e6983: fdiv qword ptr [ebx+78h]
0x100e6986: fstp qword ptr [ebx+80h]
0x100e698c: jmp short 100E6990h
0x100e698e: fstp st(1)
0x100e6990: fadd qword ptr [ebx+70h]
0x100e6993: fst qword ptr [esp+18h]
0x100e6997: fld qword ptr [ebx+80h]
0x100e699d: fstp qword ptr [esp+20h]
```

```
0x100e69a1: fld qword ptr [ebx+80h]
0x100e69a7: fxch
0x100e69a9: call 100229A0h
0x100e69ae: fld1
0x100e69b0: cmp dword ptr [ebx+66D0h],1
0x100e69b7: fld st(0)
0x100e69b9: fsubrp st(2),st
0x100e69bb: lea esi,[ebx+0E7C4h]
0x100e69c1: fld qword ptr [ebx+88h]
0x100e69c7: mov ecx,0Ch
0x100e69cc: fadd st,st(1)
0x100e69ce: lea edi,[esp+0C8h]
0x100e69d5: rep movsd
0x100e69d7: fmulp st(2),st
0x100e69d9: fxch
0x100e69db: fdiv qword ptr [esp+18h]
0x100e69df: fld qword ptr [esp+20h]
0x100e69e3: fmul st(0),st
0x100e69e5: fld qword ptr [ebx+78h]
0x100e69e8: fmul qword ptr ds:[100FE7C8h]
0x100e69ee: fmulp
0x100e69f0: fld st(2)
0x100e69f2: fsub qword ptr [ebx+80h]
0x100e69f8: fmul qword ptr [ebx+88h]
0x100e69fe: fsubp
0x100e6a00: faddp
0x100e6a02: fdivr st,st(1)
0x100e6a04: fsubrp
0x100e6a06: fstp qword ptr [ebx+90h]
0x100e6a0c: je short 100E6A2Bh
0x100e6a0e: cmp dword ptr [ebx+0C0h],0
0x100e6a15: jne short 100E6A2Bh
0x100e6a17: mov eax,[ebx+0A8h]
0x100e6a1d: cmp eax,1
0x100e6a20: jl short 100E6A2Bh
0x100e6a22: cmp eax,5
0x100e6a25: jg short 100E6A2Bh
0x100e6a27: xor eax,eax
0x100e6a29: jmp short 100E6A30h
0x100e6a2b: mov eax,1
0x100e6a30: or [ebx+6604h],eax
0x100e6a36: jne near ptr 100E6DE5h
0x100e6a3c: push 1
0x100e6a3e: push 400h
0x100e6a43: call dword ptr ds:[100FE4FCh]
0x100e6a49: push 1012E868h
0x100e6a4e: push eax
0x100e6a4f: push ebx
0x100e6a50: mov [ebx+65E0h],eax
0x100e6a56: call 100D2E20h
0x100e6a5b: mov edi,[ebx+65E0h]
0x100e6a61: mov ecx,14h
0x100e6a66: mov esi,1012DF48h
0x100e6a6b: add esp,14h
0x100e6a6e: rep movsd
0x100e6a70: cmp dword ptr [ebx+0A8h],5
0x100e6a77: jne short 100E6A85h
0x100e6a79: mov eax,[ebx+65E0h]
0x100e6a7f: mov ecx,[eax+14h]
0x100e6a82: mov [eax+10h],ecx
0x100e6a85: mov edx,[ebx+65E0h]
0x100e6a8b: mov dword ptr [edx],0FCh
0x100e6a91: xor ecx,ecx
0x100e6a93: test ecx,ecx
0x100e6a95: je short 100E6AB0h
0x100e6a97: xor edx,edx
0x100e6a99: cmp ecx,1
```

```
0x100e6a9c: setle dl
0x100e6a9f: sub edx,1
0x100e6aa2: and edx,0F3F3E6BDh
0x100e6aa8: add edx,64657363h
0x100e6aae: jmp short 100E6AB5h
0x100e6ab0: mov edx,74657874h
0x100e6ab5: mov eax,[ebx+65E0h]
0x100e6abb: mov esi,[eax]
0x100e6abd: shr esi,2
0x100e6ac0: mov [eax+esi*4],edx
0x100e6ac3: mov edx,[ebx+65E0h]
0x100e6ac9: mov esi,[edx]
0x100e6acb: lea eax,[ecx+ecx*2]
0x100e6ace: add eax,eax
0x100e6ad0: add eax,eax
0x100e6ad2: mov [esp+eax+40h],esi
0x100e6ad6: mov eax,[esp+eax+44h]
0x100e6ada: add eax,3
0x100e6add: and eax,0FFFFFFFCh
0x100e6ae0: add eax,esi
0x100e6ae2: add ecx,1
0x100e6ae5: cmp ecx,[esp+38h]
0x100e6ae9: mov [edx],eax
0x100e6aeb: jl short 100E6A93h
0x100e6aed: mov edi,[ebx+65E0h]
0x100e6af3: add edi,80h
0x100e6af9: mov ecx,1Fh
0x100e6afe: lea esi,[esp+38h]
0x100e6b02: rep movsd
0x100e6b04: mov ecx,[ebx+0A8h]
0x100e6b0a: mov eax,[ecx*4+101389ECh]
0x100e6b11: lea edx,[eax+1]
0x100e6b14: mov cl,[eax]
0x100e6b16: add eax,1
0x100e6b19: test cl,cl
0x100e6b1b: jne short 100E6B14h
0x100e6b1d: mov ecx,[ebx+65E0h]
0x100e6b23: sub eax,edx
0x100e6b25: mov edx,[esp+4Ch]
0x100e6b29: add eax,1
0x100e6b2c: shr edx,2
0x100e6b2f: mov [ecx+edx*4+8],eax
0x100e6b33: mov edx,[ebx+65E0h]
0x100e6b39: mov ecx,ds:[1012B924h]
0x100e6b3f: mov eax,[esp+58h]
0x100e6b43: lea eax,[eax+edx+8]
0x100e6b47: mov [eax],ecx
0x100e6b49: mov edx,ds:[1012B928h]
0x100e6b4f: mov [eax+4],edx
0x100e6b52: mov ecx,ds:[1012B92Ch]
0x100e6b58: mov [eax+8],ecx
0x100e6b5b: cmp dword ptr [ebx+0ACh],8
0x100e6b62: jne short 100E6B83h
0x100e6b64: fld qword ptr ds:[100FE920h]
0x100e6b6a: fdiv qword ptr [ebx+90h]
0x100e6b70: fadd qword ptr ds:[100FE7C8h]
0x100e6b76: call 100228D0h
0x100e6b7b: movsx edi,ax
0x100e6b7e: shl edi,10h
0x100e6b81: jmp short 100E6B8Ah
0x100e6b83: mov edi,[esp+0C4h]
0x100e6b8a: mov eax,[ebx+65E0h]
0x100e6b90: add eax,[esp+70h]
0x100e6b94: mov ecx,63757276h
0x100e6b99: mov [eax],ecx
0x100e6b9b: xor edx,edx
0x100e6b9d: mov [eax+4],edx
```

```
0x100e6ba0: mov esi,1
0x100e6ba5: mov [eax+8],esi
0x100e6ba8: mov [eax+0Ch],edi
0x100e6bab: mov eax,[ebx+65E0h]
0x100e6bb1: add eax,[esp+7Ch]
0x100e6bb5: push 3
0x100e6bb7: mov [eax],ecx
0x100e6bb9: mov [eax+4],edx
0x100e6bbc: mov [eax+8],esi
0x100e6bbf: mov [eax+0Ch],edi
0x100e6bc2: mov eax,[ebx+65E0h]
0x100e6bc8: add eax,[esp+8Ch]
0x100e6bcf: mov [eax],ecx
0x100e6bd1: mov [eax+4],edx
0x100e6bd4: mov [eax+8],esi
0x100e6bd7: mov [eax+0Ch],edi
0x100e6bda: mov eax,[ebx+0A8h]
0x100e6be0: mov ecx,[eax*4+101389D8h]
0x100e6be7: lea edx,[esp+0FCh]
0x100e6bee: push edx
0x100e6bef: push ecx
0x100e6bf0: call 100DD980h
0x100e6bf5: fld qword ptr ds:[1010BBB8h]
0x100e6bfb: fldz
0x100e6bfd: add esp,0Ch
0x100e6c00: fld qword ptr ds:[100FE7C8h]
0x100e6c06: mov dword ptr [esp+10h],0
0x100e6c0e: mov dword ptr [esp+0Ch],1012DDE0h
0x100e6c16: lea esi,[esp+94h]
0x100e6c1d: lea ecx,[esp+0F8h]
0x100e6c24: mov dword ptr [esp+20h],3
0x100e6c2c: mov eax,[esp+0Ch]
0x100e6c30: fld st(1)
0x100e6c32: mov edx,3
0x100e6c37: fld qword ptr [eax]
0x100e6c39: add ecx,8
0x100e6c3c: fmul qword ptr [ecx-8]
0x100e6c3f: add eax,8
0x100e6c42: sub edx,1
0x100e6c45: faddp
0x100e6c47: jne short 100E6C37h
0x100e6c49: mov edi,[ebx+65E0h]
0x100e6c4f: fnstcw [esp+14h]
0x100e6c53: movzx eax,word ptr [esp+14h]
0x100e6c58: fmul st,st(3)
0x100e6c5a: or eax,0C00h
0x100e6c5f: mov [esp+18h],eax
0x100e6c63: mov eax,[esi]
0x100e6c65: fadd st,st(1)
0x100e6c67: shr eax,2
0x100e6c6a: fldcw [esp+18h]
0x100e6c6e: add eax,[esp+10h]
0x100e6c72: add esi,0Ch
0x100e6c75: sub dword ptr [esp+20h],1
0x100e6c7a: fistp qword ptr [esp+18h]
0x100e6c7e: mov edx,[esp+18h]
0x100e6c82: mov [edi+eax*4+8],edx
0x100e6c86: fldcw [esp+14h]
0x100e6c8a: jne short 100E6C2Ch
0x100e6c8c: mov eax,[esp+0Ch]
0x100e6c90: add dword ptr [esp+10h],1
0x100e6c95: add eax,18h
0x100e6c98: cmp eax,1012DE28h
0x100e6c9d: mov [esp+0Ch],eax
0x100e6ca1: jl near ptr 100E6C16h
0x100e6ca7: mov edi,ds:[100FE6C0h]
0x100e6cad: fstp st(2)
```

```
0x100e6caf: fstp st(1)
0x100e6cb1: xor esi,esi
0x100e6cb3: fstp st(0)
0x100e6cb5: mov ecx,[ebx+65E0h]
0x100e6cbb: mov edx,[esi+ecx]
0x100e6cbe: push edx
0x100e6cbf: call edi
0x100e6cc1: mov ecx,[ebx+65E0h]
0x100e6cc7: mov [esi+ecx],eax
0x100e6cca: add esi,4
0x100e6ccd: cmp esi,400h
0x100e6cd3: jl short 100E6CB5h
0x100e6cd5: mov edx,[ebx+65E0h]
0x100e6cdb: mov ecx,ds:[1012E850h]
0x100e6ce1: mov eax,[esp+40h]
0x100e6ce5: lea eax,[eax+edx+8]
0x100e6ce9: mov [eax],ecx
0x100e6ceb: mov edx,ds:[1012E854h]
0x100e6cf1: mov [eax+4],edx
0x100e6cf4: mov ecx,ds:[1012E858h]
0x100e6cfa: mov [eax+8],ecx
0x100e6cfd: mov edx,ds:[1012E85Ch]
0x100e6d03: mov [eax+0Ch],edx
0x100e6d06: mov ecx,ds:[1012E860h]
0x100e6d0c: mov [eax+10h],ecx
0x100e6d0f: mov edx,ds:[1012E864h]
0x100e6d15: mov [eax+14h],edx
0x100e6d18: mov eax,[ebx+0A8h]
0x100e6d1e: mov ecx,[eax*4+101389ECh]
0x100e6d25: mov edx,[ebx+65E0h]
0x100e6d2b: mov eax,[esp+4Ch]
0x100e6d2f: lea edx,[eax+edx+0Ch]
0x100e6d33: mov al,[ecx]
0x100e6d35: mov [edx],al
0x100e6d37: add ecx,1
0x100e6d3a: add edx,1
0x100e6d3d: test al,al
0x100e6d3f: jne short 100E6D33h
0x100e6d41: fldz
0x100e6d43: lea ecx,[esp+0C8h]
0x100e6d4a: mov dword ptr [esp+14h],0
0x100e6d52: mov [esp+10h],ecx
0x100e6d56: xor edi,edi
0x100e6d58: cmp [ebx+66D0h],edi
0x100e6d5e: jle short 100E6DCAh
0x100e6d60: mov esi,[esp+10h]
0x100e6d64: fst dword ptr [esp+0Ch]
0x100e6d68: mov eax,[esp+0Ch]
0x100e6d6c: cmp eax,3
0x100e6d6f: fst dword ptr [esi]
0x100e6d71: jge short 100E6DBCh
0x100e6d73: mov edx,[ebx+0A8h]
0x100e6d79: mov ecx,[esp+14h]
0x100e6d7d: mov edx,[edx*4+101389D8h]
0x100e6d84: add ecx,eax
0x100e6d86: lea eax,[edi+eax*4]
0x100e6d89: lea ecx,[edx+ecx*8]
0x100e6d8c: lea edx,[ebx+eax*4+0E7C4h]
0x100e6d93: mov eax,3
0x100e6d98: sub eax,[esp+0Ch]
0x100e6d9c: fld qword ptr [ecx]
0x100e6d9e: add ecx,8
0x100e6da1: fstp dword ptr [esp+18h]
0x100e6da5: add edx,10h
0x100e6da8: sub eax,1
0x100e6dab: fld dword ptr [esp+18h]
0x100e6daf: fmul dword ptr [edx-10h]
```

```
0x100e6db2: fadd dword ptr [esi]
0x100e6db4: fstp dword ptr [esi]
0x100e6db6: jne short 100E6D9Ch
0x100e6db8: mov eax,[esp+0Ch]
0x100e6dbc: add edi,1
0x100e6dbf: add esi,4
0x100e6dc2: cmp edi,[ebx+66D0h]
0x100e6dc8: jl short 100E6D6Ch
0x100e6dca: mov eax,[esp+14h]
0x100e6dce: add dword ptr [esp+10h],10h
0x100e6dd3: add eax,3
0x100e6dd6: cmp eax,9
0x100e6dd9: mov [esp+14h],eax
0x100e6ddd: jl near ptr 100E6D56h
0x100e6de3: fstp st(0)
0x100e6de5: cmp dword ptr [ebx+0BCh],0
0x100e6dec: je short 100E6E23h
0x100e6dee: cmp dword ptr [ebx+6604h],0
0x100e6df5: mov eax,1012E838h
0x100e6dfa: jne short 100E6E01h
0x100e6dfc: mov eax,1012E818h
0x100e6e01: mov ecx,[ebx+0A8h]
0x100e6e07: mov edx,[ecx*4+101389ECh]
0x100e6e0e: push edx
0x100e6e0f: push eax
0x100e6e10: call dword ptr ds:[100FE4C4h]
0x100e6e16: add eax,40h
0x100e6e19: push eax
0x100e6e1a: call dword ptr ds:[100FE4C8h]
0x100e6e20: add esp,0Ch
0x100e6e23: push 20000h
0x100e6e28: lea eax,[ebx+0E7F4h]
0x100e6e2e: push 0
0x100e6e30: push eax
0x100e6e31: call 100228C0h
0x100e6e36: mov esi,[ebx+66E0h]
0x100e6e3c: add esp,0Ch
0x100e6e3f: cmp word ptr [ebx+66B4h],0
0x100e6e47: mov dword ptr [esp+0Ch],0
0x100e6e4f: jbe near ptr 100E6FA5h
0x100e6e55: fldz
0x100e6e57: xor edi,edi
0x100e6e59: cmp [ebx+66B6h],di
0x100e6e60: mov [esp+20h],edi
0x100e6e64: jbe near ptr 100E6F89h
0x100e6e6a: cmp dword ptr [ebx+6604h],0
0x100e6e71: jne near ptr 100E6F0Fh
0x100e6e77: xor eax,eax
0x100e6e79: fst dword ptr [esp+30h]
0x100e6e7d: cmp [ebx+66D0h],eax
0x100e6e83: fst dword ptr [esp+2Ch]
0x100e6e87: fst dword ptr [esp+28h]
0x100e6e8b: jle short 100E6ED7h
0x100e6e8d: mov ecx,[ebx+66D0h]
0x100e6e93: movzx edx,word ptr [esi+eax*2]
0x100e6e97: mov [esp+18h],edx
0x100e6e9b: add eax,1
0x100e6e9e: cmp eax,ecx
0x100e6ea0: fild dword ptr [esp+18h]
0x100e6ea4: fld dword ptr [esp+eax*4+0C4h]
0x100e6eab: fmul st,st(1)
0x100e6ead: fadd dword ptr [esp+28h]
0x100e6eb1: fstp dword ptr [esp+28h]
0x100e6eb5: fld dword ptr [esp+eax*4+0D4h]
0x100e6ebc: fmul st,st(1)
0x100e6ebe: fadd dword ptr [esp+2Ch]
0x100e6ec2: fstp dword ptr [esp+2Ch]
```

```
0x100e6ec6: fmul dword ptr [esp+eax*4+0E4h]
0x100e6ecd: fadd dword ptr [esp+30h]
0x100e6ed1: fstp dword ptr [esp+30h]
0x100e6ed5: jl short 100E6E93h
0x100e6ed7: xor edi,edi
0x100e6ed9: fld dword ptr [esp+edi*4+28h]
0x100e6edd: call 100228D0h
0x100e6ee2: cmp eax,0FFFFh
0x100e6ee7: jge short 100E6EF8h
0x100e6ee9: test eax,eax
0x100e6eeb: jge short 100E6EF1h
0x100e6eed: xor eax,eax
0x100e6eef: jmp short 100E6EFDh
0x100e6ef1: cmp eax,0FFFFh
0x100e6ef6: jl short 100E6EFDh
0x100e6ef8: mov eax,0FFFFh
0x100e6efd: mov [esi+edi*2],ax
0x100e6f01: add edi,1
0x100e6f04: cmp edi,3
0x100e6f07: jl short 100E6ED9h
0x100e6f09: mov edi,[esp+20h]
0x100e6f0d: jmp short 100E6F3Bh
0x100e6f0f: cmp dword ptr [ebx+0C0h],0
0x100e6f16: je short 100E6F3Bh
0x100e6f18: mov eax,[esp+0Ch]
0x100e6f1c: mov edx,[ebx+65B0h]
0x100e6f22: and eax,7
0x100e6f25: mov ecx,edi
0x100e6f27: and ecx,1
0x100e6f2a: lea ecx,[ecx+eax*2]
0x100e6f2d: add ecx,ecx
0x100e6f2f: shr edx,cl
0x100e6f31: and edx,3
0x100e6f34: mov ax,[esi+edx*2]
0x100e6f38: mov [esi],ax
0x100e6f3b: xor eax,eax
0x100e6f3d: cmp [ebx+66D0h],eax
0x100e6f43: jle short 100E6F70h
0x100e6f45: xor edx,edx
0x100e6f47: movzx ecx,word ptr [esi+eax*2]
0x100e6f4b: shr ecx,3
0x100e6f4e: add ecx,edx
0x100e6f50: add dword ptr [ebx+ecx*4+0E7F4h],1
0x100e6f58: lea ecx,[ebx+ecx*4+0E7F4h]
0x100e6f5f: add eax,1
0x100e6f62: add edx,2000h
0x100e6f68: cmp eax,[ebx+66D0h]
0x100e6f6e: jl short 100E6F47h
0x100e6f70: movzx edx,word ptr [ebx+66B6h]
0x100e6f77: add edi,1
0x100e6f7a: add esi,8
0x100e6f7d: cmp edi,edx
0x100e6f7f: mov [esp+20h],edi
0x100e6f83: jl near ptr 100E6E6Ah
0x100e6f89: mov eax,[esp+0Ch]
0x100e6f8d: movzx ecx,word ptr [ebx+66B4h]
0x100e6f94: add eax,1
0x100e6f97: cmp eax,ecx
0x100e6f99: mov [esp+0Ch],eax
0x100e6f9d: jl near ptr 100E6E57h
0x100e6fa3: fstp st(0)
0x100e6fa5: cmp dword ptr [ebx+66D0h],4
0x100e6fac: jne short 100E6FC1h
0x100e6fae: cmp dword ptr [ebx+0A8h],0
0x100e6fb5: je short 100E6FC1h
0x100e6fb7: mov dword ptr [ebx+66D0h],3
0x100e6fc1: cmp dword ptr [ebx+0C0h],0
```

```
0x100e6fc8: je short 100E6FDDh
0x100e6fca: cmp dword ptr [ebx+65B0h],0
0x100e6fd1: je short 100E6FDDh
0x100e6fd3: mov dword ptr [ebx+66D0h],1
0x100e6fdd: pop edi
0x100e6fde: pop esi
0x100e6fdf: pop ebx
0x100e6fe0: mov esp,ebp
0x100e6fe2: pop ebp
0x100e6fe3: ret
```

`sub_100e67b0` is called by sub_10016040. It works with the text "Converting to %s colorspace...", "Building histograms...", "auto-generated by dcraw". It accesses memory-mapped peripheral registers at 0x62545243. It delegates to j_memset, sub_100228d0, j__CIpow (5 callees).

12.1306 sub_100e6ff0 at 0x100e6ff0 (233 instructions)

```
0x100e6ff0: push ebp
0x100e6ff1: mov ebp,esp
0x100e6ff3: and esp,0FFFFFFF8h
0x100e6ff6: sub esp,34h
0x100e6ff9: push ebx
0x100e6ffa: push esi
0x100e6ffb: mov esi,[ebp+8]
0x100e6ffe: cmp word ptr [esi+66C2h],0
0x100e7006: push edi
0x100e7007: je near ptr 100E7342h
0x100e700d: cmp dword ptr [esi+0BCh],0
0x100e7014: je short 100E702Eh
0x100e7016: push 1012E88Ch
0x100e701b: call dword ptr ds:[100FE4C4h]
0x100e7021: add eax,40h
0x100e7024: push eax
0x100e7025: call dword ptr ds:[100FE4C8h]
0x100e702b: add esp,8
0x100e702e: movzx ecx,word ptr [esi+66BCh]
0x100e7035: fld qword ptr ds:[100FE7C8h]
0x100e703b: movzx eax,word ptr [esi+66C2h]
0x100e7042: lea ebx,[eax+ecx-1]
0x100e7046: sar ebx,cl
0x100e7048: mov [esi+66C2h],bx
0x100e704f: call 1002299Ah
0x100e7054: fst qword ptr [esp+38h]
0x100e7058: movzx ecx,bx
0x100e705b: fnstcw [esp+10h]
0x100e705f: movzx eax,word ptr [esp+10h]
0x100e7064: mov [esp+2Ch],ecx
0x100e7068: or eax,0C00h
0x100e706d: push 8
0x100e706f: fild dword ptr [esp+30h]
0x100e7073: mov [esp+30h],eax
0x100e7077: movzx eax,word ptr [esi+66B4h]
0x100e707e: fdiv st,st(1)
0x100e7080: fldcw [esp+30h]
0x100e7084: fistp dword ptr [esp+30h]
0x100e7088: mov dx,[esp+30h]
0x100e708d: movzx ecx,dx
0x100e7090: movzx edx,bx
0x100e7093: fldcw [esp+14h]
0x100e7097: sub eax,edx
0x100e7099: mov [esp+30h],eax
0x100e709d: movzx ebx,cx
0x100e70a0: fild dword ptr [esp+30h]
0x100e70a4: mov [esp+28h],ecx
0x100e70a8: fnstcw [esp+14h]
0x100e70ac: mov ecx,ebx
```

```
0x100e70ae: movzx eax,word ptr [esp+14h]
0x100e70b3: fdivrp
0x100e70b5: or eax,0C00h
0x100e70ba: mov [esp+30h],eax
0x100e70be: fldcw [esp+30h]
0x100e70c2: fistp dword ptr [esp+30h]
0x100e70c6: mov ax,[esp+30h]
0x100e70cb: movzx eax,ax
0x100e70ce: movzx edi,ax
0x100e70d1: fldcw [esp+14h]
0x100e70d5: mov [esp+2Ch],eax
0x100e70d9: imul ecx,edi
0x100e70dc: push ecx
0x100e70dd: call dword ptr ds:[100FE4FCh]
0x100e70e3: push 1012E87Ch
0x100e70e8: push eax
0x100e70e9: push esi
0x100e70ea: mov [esp+40h],eax
0x100e70ee: call 100D2E20h
0x100e70f3: xor edx,edx
0x100e70f5: add esp,14h
0x100e70f8: cmp edi,edx
0x100e70fa: mov [esp+18h],edx
0x100e70fe: jle near ptr 100E7307h
0x100e7104: fld1
0x100e7106: mov [esp+1Ch],edx
0x100e710a: fld qword ptr [esp+38h]
0x100e710e: xor edi,edi
0x100e7110: test ebx,ebx
0x100e7112: mov [esp+20h],edi
0x100e7116: jle near ptr 100E72EBh
0x100e711c: mov [esp+14h],edx
0x100e7120: fild dword ptr [esp+14h]
0x100e7124: movzx eax,word ptr [esi+66C2h]
0x100e712b: mov [esp+10h],eax
0x100e712f: add edx,edi
0x100e7131: fmul st,st(1)
0x100e7133: fild dword ptr [esp+10h]
0x100e7137: faddp
0x100e7139: fstp dword ptr [esp+10h]
0x100e713d: fld dword ptr [esp+10h]
0x100e7141: fnstcw [esp+10h]
0x100e7145: movzx eax,word ptr [esp+10h]
0x100e714a: fld st(0)
0x100e714c: or eax,0C00h
0x100e7151: mov [esp+30h],eax
0x100e7155: fldcw [esp+30h]
0x100e7159: fistp qword ptr [esp+30h]
0x100e715d: mov ecx,[esp+30h]
0x100e7161: mov [esp+30h],edx
0x100e7165: fldcw [esp+10h]
0x100e7169: fild dword ptr [esp+30h]
0x100e716d: fnstcw [esp+10h]
0x100e7171: fmul st,st(2)
0x100e7173: movzx eax,word ptr [esp+10h]
0x100e7178: or eax,0C00h
0x100e717d: fstp dword ptr [esp+30h]
0x100e7181: fld dword ptr [esp+30h]
0x100e7185: mov [esp+30h],eax
0x100e7189: movzx eax,word ptr [esi+66B4h]
0x100e7190: fld st(0)
0x100e7192: fldcw [esp+30h]
0x100e7196: sub eax,2
0x100e7199: cmp ecx,eax
0x100e719b: fistp qword ptr [esp+30h]
0x100e719f: mov edx,[esp+30h]
0x100e71a3: fldcw [esp+10h]
```

```
0x100e71a7: jg near ptr 100E72CAh
0x100e71ad: movzx eax,word ptr [esi+66B6h]
0x100e71b4: lea ebx,[eax-2]
0x100e71b7: cmp edx,ebx
0x100e71b9: jg near ptr 100E72CAh
0x100e71bf: mov ebx,ecx
0x100e71c1: test ebx,ebx
0x100e71c3: mov [esp+30h],ebx
0x100e71c7: fild dword ptr [esp+30h]
0x100e71cb: jge short 100E71D3h
0x100e71cd: fadd dword ptr ds:[100FE7C0h]
0x100e71d3: fsubp st(2),st
0x100e71d5: mov ebx,edx
0x100e71d7: test ebx,ebx
0x100e71d9: fxch
0x100e71db: mov [esp+30h],ebx
0x100e71df: fstp dword ptr [esp+10h]
0x100e71e3: fild dword ptr [esp+30h]
0x100e71e7: jge short 100E71EFh
0x100e71e9: fadd dword ptr ds:[100FE7C0h]
0x100e71ef: imul eax,ecx
0x100e71f2: fsubp
0x100e71f4: fstp dword ptr [esp+30h]
0x100e71f8: mov ecx,[esi+66E0h]
0x100e71fe: add eax,edx
0x100e7200: lea edx,[ecx+eax*8]
0x100e7203: xor ecx,ecx
0x100e7205: cmp [esi+66D0h],ecx
0x100e720b: jle near ptr 100E72CEh
0x100e7211: mov eax,[esp+1Ch]
0x100e7215: fstp st(0)
0x100e7217: fld dword ptr [esp+30h]
0x100e721b: add eax,edi
0x100e721d: mov edi,[esp+2Ch]
0x100e7221: fld st(0)
0x100e7223: fsubr st,st(2)
0x100e7225: lea edi,[edi+eax*8]
0x100e7228: fld dword ptr [esp+10h]
0x100e722c: fld st(0)
0x100e722e: fsubr st,st(4)
0x100e7230: movzx eax,word ptr [esi+66B6h]
0x100e7237: fnstcw [esp+10h]
0x100e723b: lea eax,[ecx+eax*4]
0x100e723e: movzx ebx,word ptr [edx+eax*2+8]
0x100e7243: movzx eax,word ptr [edx+eax*2]
0x100e7247: mov [esp+30h],ebx
0x100e724b: add ecx,1
0x100e724e: add edi,2
0x100e7251: fild dword ptr [esp+30h]
0x100e7255: mov [esp+30h],eax
0x100e7259: movzx eax,word ptr [edx+ecx*2+6]
0x100e725e: fmul st,st(4)
0x100e7260: fild dword ptr [esp+30h]
0x100e7264: mov [esp+30h],eax
0x100e7268: movzx eax,word ptr [edx+ecx*2-2]
0x100e726d: fmul st,st(4)
0x100e726f: faddp
0x100e7271: fmul st,st(2)
0x100e7273: fild dword ptr [esp+30h]
0x100e7277: mov [esp+30h],eax
0x100e727b: movzx eax,word ptr [esp+10h]
0x100e7280: or eax,0C00h
0x100e7285: fmul st,st(5)
0x100e7287: fild dword ptr [esp+30h]
0x100e728b: mov [esp+30h],eax
0x100e728f: fmul st,st(5)
0x100e7291: faddp
```

```
0x100e7293: fmul st,st(2)
0x100e7295: faddp
0x100e7297: fldcw [esp+30h]
0x100e729b: fistp dword ptr [esp+30h]
0x100e729f: mov ax,[esp+30h]
0x100e72a4: mov [edi-2],ax
0x100e72a8: cmp ecx,[esi+66D0h]
0x100e72ae: fldcw [esp+10h]
0x100e72b2: jl near ptr 100E7230h
0x100e72b8: mov edi,[esp+20h]
0x100e72bc: fstp st(3)
0x100e72be: fstp st(1)
0x100e72c0: fstp st(0)
0x100e72c2: fstp st(0)
0x100e72c4: fld qword ptr [esp+38h]
0x100e72c8: jmp short 100E72CEh
0x100e72ca: fstp st(1)
0x100e72cc: fstp st(0)
0x100e72ce: movzx ebx,word ptr [esp+24h]
0x100e72d3: sub dword ptr [esp+14h],1
0x100e72d8: mov edx,[esp+18h]
0x100e72dc: add edi,1
0x100e72df: cmp edi,ebx
0x100e72e1: mov [esp+20h],edi
0x100e72e5: jl near ptr 100E7120h
0x100e72eb: movzx eax,word ptr [esp+28h]
0x100e72f0: add [esp+1Ch],ebx
0x100e72f4: add edx,1
0x100e72f7: cmp edx,eax
0x100e72f9: mov [esp+18h],edx
0x100e72fd: jl near ptr 100E710Eh
0x100e7303: fstp st(0)
0x100e7305: fstp st(0)
0x100e7307: mov ecx,[esi+66E0h]
0x100e730d: push ecx
0x100e730e: call dword ptr ds:[100FE554h]
0x100e7314: mov dx,[esp+28h]
0x100e7319: mov ax,[esp+2Ch]
0x100e731e: mov ecx,[esp+30h]
0x100e7322: add esp,4
0x100e7325: mov [esi+66B6h],dx
0x100e732c: mov [esi+66B4h],ax
0x100e7333: mov [esi+66E0h],ecx
0x100e7339: mov word ptr [esi+66C2h],0
0x100e7342: pop edi
0x100e7343: pop esi
0x100e7344: pop ebx
0x100e7345: mov esp,ebp
0x100e7347: pop ebp
0x100e7348: ret
```

`sub_100e6ff0` is called by sub_10016040. It works with the text "Rotating image 45 degrees...". It delegates to j__CIsqrt, sub_100d2e20 (2 callees).

12.1307 sub_100e7350 at 0x100e7350 (269 instructions)

```
0x100e7350: fld1
0x100e7352: sub esp,24h
0x100e7355: push esi
0x100e7356: mov esi,[esp+2Ch]
0x100e735a: fcomp qword ptr [esi+66D8h]
0x100e7360: fnstsw ax
0x100e7362: test ah,44h
0x100e7365: jnp near ptr 100E7733h
0x100e736b: cmp dword ptr [esi+0BCh],0
0x100e7372: je short 100E738Ch
```

```
0x100e7374: push 1012E8B8h
0x100e7379: call dword ptr ds:[100FE4C4h]
0x100e737f: add eax,40h
0x100e7382: push eax
0x100e7383: call dword ptr ds:[100FE4C8h]
0x100e7389: add esp,8
0x100e738c: fld1
0x100e738e: push ebx
0x100e738f: fcomp qword ptr [esi+66D8h]
0x100e7395: push ebp
0x100e7396: push edi
0x100e7397: push 8
0x100e7399: fnstsw ax
0x100e739b: test ah,41h
0x100e739e: jne near ptr 100E755Fh
0x100e73a4: movzx eax,word ptr [esi+66B4h]
0x100e73ab: mov [esp+3Ch],eax
0x100e73af: movzx edx,word ptr [esi+66B6h]
0x100e73b6: fild dword ptr [esp+3Ch]
0x100e73ba: fnstcw [esp+3Ch]
0x100e73be: movzx eax,word ptr [esp+3Ch]
0x100e73c3: fdiv qword ptr [esi+66D8h]
0x100e73c9: or eax,0C00h
0x100e73ce: mov [esp+2Ch],eax
0x100e73d2: fadd qword ptr ds:[100FE7C8h]
0x100e73d8: fldcw [esp+2Ch]
0x100e73dc: fistp dword ptr [esp+2Ch]
0x100e73e0: mov cx,[esp+2Ch]
0x100e73e5: movzx ebx,cx
0x100e73e8: movzx edi,bx
0x100e73eb: fldcw [esp+3Ch]
0x100e73ef: mov [esp+1Ch],ebx
0x100e73f3: imul edx,edi
0x100e73f6: push edx
0x100e73f7: call dword ptr ds:[100FE4FCh]
0x100e73fd: push 1012E8ACh
0x100e7402: push eax
0x100e7403: push esi
0x100e7404: mov [esp+34h],eax
0x100e7408: call 100D2E20h
0x100e740d: fldz
0x100e740f: add esp,14h
0x100e7412: test edi,edi
0x100e7414: mov dword ptr [esp+14h],0
0x100e741c: jle near ptr 100E7553h
0x100e7422: fld qword ptr ds:[100FE7C8h]
0x100e7428: fld1
0x100e742a: jmp short 100E742Eh
0x100e742c: fxch st(2)
0x100e742e: fld st(2)
0x100e7430: call 100228D0h
0x100e7435: movzx edx,word ptr [esi+66B6h]
0x100e743c: mov edi,[esi+66E0h]
0x100e7442: mov ecx,edx
0x100e7444: imul ecx,eax
0x100e7447: lea ecx,[edi+ecx*8]
0x100e744a: movzx edi,word ptr [esi+66B4h]
0x100e7451: mov [esp+38h],eax
0x100e7455: add eax,1
0x100e7458: cmp eax,edi
0x100e745a: fild dword ptr [esp+38h]
0x100e745e: mov [esp+1Ch],ecx
0x100e7462: mov [esp+10h],ecx
0x100e7466: fsubr st,st(3)
0x100e7468: jge short 100E7471h
0x100e746a: lea ecx,[ecx+edx*8]
0x100e746d: mov [esp+1Ch],ecx
```

```
0x100e7471: xor edi,edi
0x100e7473: test edx,edx
0x100e7475: jle near ptr 100E752Fh
0x100e747b: xor edx,edx
0x100e747d: cmp [esi+66D0h],edx
0x100e7483: jle near ptr 100E750Dh
0x100e7489: mov eax,[esp+10h]
0x100e748d: fld st(1)
0x100e748f: sub eax,ecx
0x100e7491: fsub st,st(1)
0x100e7493: mov [esp+28h],eax
0x100e7497: jmp short 100E749Dh
0x100e7499: mov eax,[esp+28h]
0x100e749d: movzx eax,word ptr [eax+ecx]
0x100e74a1: movzx ebx,word ptr [esi+66B6h]
0x100e74a8: imul ebx,[esp+14h]
0x100e74ad: mov [esp+38h],eax
0x100e74b1: movzx eax,word ptr [ecx]
0x100e74b4: fild dword ptr [esp+38h]
0x100e74b8: mov ebp,[esp+20h]
0x100e74bc: mov [esp+38h],eax
0x100e74c0: fmul st,st(1)
0x100e74c2: add ebx,edi
0x100e74c4: lea ebx,[edx+ebx*4]
0x100e74c7: fild dword ptr [esp+38h]
0x100e74cb: add edx,1
0x100e74ce: fnstcw [esp+38h]
0x100e74d2: movzx eax,word ptr [esp+38h]
0x100e74d7: fmul st,st(3)
0x100e74d9: or eax,0C00h
0x100e74de: mov [esp+24h],eax
0x100e74e2: faddp
0x100e74e4: add ecx,2
0x100e74e7: fadd st,st(4)
0x100e74e9: fldcw [esp+24h]
0x100e74ed: fistp dword ptr [esp+24h]
0x100e74f1: mov ax,[esp+24h]
0x100e74f6: mov [ebp+ebx*2],ax
0x100e74fb: cmp edx,[esi+66D0h]
0x100e7501: fldcw [esp+38h]
0x100e7505: jl short 100E7499h
0x100e7507: mov ecx,[esp+1Ch]
0x100e750b: fstp st(0)
0x100e750d: movzx edx,word ptr [esi+66B6h]
0x100e7514: add dword ptr [esp+10h],8
0x100e7519: add edi,1
0x100e751c: add ecx,8
0x100e751f: cmp edi,edx
0x100e7521: mov [esp+1Ch],ecx
0x100e7525: jl near ptr 100E747Bh
0x100e752b: mov ebx,[esp+18h]
0x100e752f: mov ecx,[esp+14h]
0x100e7533: fstp st(0)
0x100e7535: movzx eax,bx
0x100e7538: fxch st(2)
0x100e753a: fadd qword ptr [esi+66D8h]
0x100e7540: add ecx,1
0x100e7543: cmp ecx,eax
0x100e7545: mov [esp+14h],ecx
0x100e7549: jl near ptr 100E742Ch
0x100e754f: fstp st(1)
0x100e7551: fstp st(1)
0x100e7553: mov [esi+66B4h],bx
0x100e755a: jmp near ptr 100E7714h
0x100e755f: movzx eax,word ptr [esi+66B6h]
0x100e7566: mov [esp+3Ch],eax
0x100e756a: movzx edx,word ptr [esi+66B4h]
```

```
0x100e7571: fild dword ptr [esp+3Ch]
0x100e7575: fnstcw [esp+3Ch]
0x100e7579: movzx eax,word ptr [esp+3Ch]
0x100e757e: fmul qword ptr [esi+66D8h]
0x100e7584: or eax,0C00h
0x100e7589: mov [esp+2Ch],eax
0x100e758d: fadd qword ptr ds:[100FE7C8h]
0x100e7593: fldcw [esp+2Ch]
0x100e7597: fistp dword ptr [esp+2Ch]
0x100e759b: mov cx,[esp+2Ch]
0x100e75a0: movzx ebp,cx
0x100e75a3: movzx edi,bp
0x100e75a6: fldcw [esp+3Ch]
0x100e75aa: mov [esp+1Ch],ebp
0x100e75ae: imul edx,edi
0x100e75b1: push edx
0x100e75b2: call dword ptr ds:[100FE4FCh]
0x100e75b8: mov ebx,eax
0x100e75ba: push 1012E8ACh
0x100e75bf: push ebx
0x100e75c0: push esi
0x100e75c1: mov [esp+34h],ebx
0x100e75c5: call 100D2E20h
0x100e75ca: fldz
0x100e75cc: add esp,14h
0x100e75cf: test edi,edi
0x100e75d1: jle near ptr 100E770Dh
0x100e75d7: fld qword ptr ds:[100FE7C8h]
0x100e75dd: mov [esp+24h],ebx
0x100e75e1: fld1
0x100e75e3: mov [esp+28h],edi
0x100e75e7: fld st(2)
0x100e75e9: call 100228D0h
0x100e75ee: mov ecx,[esi+66E0h]
0x100e75f4: movzx edx,word ptr [esi+66B6h]
0x100e75fb: mov [esp+38h],eax
0x100e75ff: lea ebp,[ecx+eax*8]
0x100e7602: add eax,1
0x100e7605: cmp eax,edx
0x100e7607: fild dword ptr [esp+38h]
0x100e760b: mov [esp+10h],ebp
0x100e760f: fsubr st,st(3)
0x100e7611: jge short 100E7616h
0x100e7613: add ebp,8
0x100e7616: cmp word ptr [esi+66B4h],0
0x100e761e: mov dword ptr [esp+14h],0
0x100e7626: jbe near ptr 100E76E3h
0x100e762c: movzx eax,word ptr [esp+18h]
0x100e7631: add eax,eax
0x100e7633: add eax,eax
0x100e7635: add eax,eax
0x100e7637: mov [esp+30h],eax
0x100e763b: mov [esp+1Ch],ebx
0x100e763f: xor edi,edi
0x100e7641: cmp [esi+66D0h],edi
0x100e7647: jle short 100E76B2h
0x100e7649: mov ebx,[esp+10h]
0x100e764d: fld st(1)
0x100e764f: mov edx,[esp+1Ch]
0x100e7653: fsub st,st(1)
0x100e7655: mov ecx,ebp
0x100e7657: sub ebx,ebp
0x100e7659: movzx eax,word ptr [ecx+ebx]
0x100e765d: mov [esp+38h],eax
0x100e7661: movzx eax,word ptr [ecx]
0x100e7664: fild dword ptr [esp+38h]
0x100e7668: mov [esp+38h],eax
```

```
0x100e766c: add edi,1
0x100e766f: fmul st,st(1)
0x100e7671: add ecx,2
0x100e7674: fild dword ptr [esp+38h]
0x100e7678: add edx,2
0x100e767b: fnstcw [esp+38h]
0x100e767f: movzx eax,word ptr [esp+38h]
0x100e7684: fmul st,st(3)
0x100e7686: or eax,0C00h
0x100e768b: mov [esp+2Ch],eax
0x100e768f: faddp
0x100e7691: fadd st,st(4)
0x100e7693: fldcw [esp+2Ch]
0x100e7697: fistp dword ptr [esp+2Ch]
0x100e769b: mov ax,[esp+2Ch]
0x100e76a0: mov [edx-2],ax
0x100e76a4: cmp edi,[esi+66D0h]
0x100e76aa: fldcw [esp+38h]
0x100e76ae: jl short 100E7659h
0x100e76b0: fstp st(0)
0x100e76b2: movzx eax,word ptr [esi+66B6h]
0x100e76b9: movzx edx,word ptr [esi+66B4h]
0x100e76c0: mov ecx,[esp+30h]
0x100e76c4: add dword ptr [esp+14h],1
0x100e76c9: add [esp+1Ch],ecx
0x100e76cd: add eax,eax
0x100e76cf: add eax,eax
0x100e76d1: add eax,eax
0x100e76d3: add [esp+10h],eax
0x100e76d7: add ebp,eax
0x100e76d9: cmp [esp+14h],edx
0x100e76dd: jl near ptr 100E763Fh
0x100e76e3: fstp st(0)
0x100e76e5: mov ebx,[esp+24h]
0x100e76e9: fld st(0)
0x100e76eb: add ebx,8
0x100e76ee: sub dword ptr [esp+28h],1
0x100e76f3: fdiv qword ptr [esi+66D8h]
0x100e76f9: mov [esp+24h],ebx
0x100e76fd: faddp st(3),st
0x100e76ff: jne near ptr 100E75E7h
0x100e7705: mov ebp,[esp+18h]
0x100e7709: fstp st(1)
0x100e770b: fstp st(0)
0x100e770d: mov [esi+66B6h],bp
0x100e7714: mov eax,[esi+66E0h]
0x100e771a: fstp st(0)
0x100e771c: push eax
0x100e771d: call dword ptr ds:[100FE554h]
0x100e7723: mov ecx,[esp+24h]
0x100e7727: add esp,4
0x100e772a: pop edi
0x100e772b: pop ebp
0x100e772c: mov [esi+66E0h],ecx
0x100e7732: pop ebx
0x100e7733: pop esi
0x100e7734: add esp,24h
0x100e7737: ret
```

`sub_100e7350` is called by sub_10016040. It works with the text "Stretching the image...". It delegates to sub_100228d0, sub_100d2e20 (2 callees).

12.1308 sub_100e7740 at 0x100e7740 (29 instructions)

```
0x100e7740: mov edx,[esp+8]
0x100e7744: mov ecx,[esp+0Ch]
```

```
0x100e7748: push esi
0x100e7749: mov esi,[esp+8]
0x100e774d: mov eax,[esi+66C8h]
0x100e7753: test al,4
0x100e7755: je short 100E775Dh
0x100e7757: xor edx,ecx
0x100e7759: xor ecx,edx
0x100e775b: xor edx,ecx
0x100e775d: test al,2
0x100e775f: je short 100E7771h
0x100e7761: push edi
0x100e7762: movzx edi,word ptr [esi+66BEh]
0x100e7769: sub edi,edx
0x100e776b: sub edi,1
0x100e776e: mov edx,edi
0x100e7770: pop edi
0x100e7771: test al,1
0x100e7773: je short 100E7783h
0x100e7775: movzx eax,word ptr [esi+66C0h]
0x100e777c: sub eax,ecx
0x100e777e: sub eax,1
0x100e7781: mov ecx,eax
0x100e7783: movzx eax,word ptr [esi+66C0h]
0x100e778a: imul eax,edx
0x100e778d: add eax,ecx
0x100e778f: pop esi
0x100e7790: ret
```

`sub_100e7740` is called by sub_10016040. It calls nothing else.

12.1309 sub_100e77a0 at 0x100e77a0 (104 instructions)

```
0x100e77a0: push ecx
0x100e77a1: fldz
0x100e77a3: push ebx
0x100e77a4: mov ebx,[esp+0Ch]
0x100e77a8: fstp dword ptr [esp+4]
0x100e77ac: movzx ecx,word ptr [ebx+66B4h]
0x100e77b3: movzx eax,word ptr [ebx+66B6h]
0x100e77ba: imul eax,ecx
0x100e77bd: mov [esp+0Ch],eax
0x100e77c1: push esi
0x100e77c2: push edi
0x100e77c3: fild dword ptr [esp+14h]
0x100e77c7: fmul qword ptr ds:[1012E8D8h]
0x100e77cd: call 100228D0h
0x100e77d2: cmp word ptr [ebx+66C2h],0
0x100e77da: mov edi,eax
0x100e77dc: je short 100E77E5h
0x100e77de: cdq
0x100e77df: sub eax,edx
0x100e77e1: sar eax,1
0x100e77e3: mov edi,eax
0x100e77e5: test dword ptr [ebx+9Ch],0FFFFFFFDh
0x100e77ef: jne short 100E77FAh
0x100e77f1: cmp dword ptr [ebx+0C4h],0
0x100e77f8: je short 100E77FDh
0x100e77fa: or edi,0FFFFFFFFh
0x100e77fd: mov eax,[ebx+66D0h]
0x100e7803: test eax,eax
0x100e7805: jle short 100E7864h
0x100e7807: push ebp
0x100e7808: lea esi,[ebx+167F0h]
0x100e780e: mov ebp,eax
0x100e7810: mov eax,1FFFh
0x100e7815: xor edx,edx
```

```
0x100e7817: mov [esp+18h],eax
0x100e781b: mov ecx,esi
0x100e781d: lea ecx,[ecx]
0x100e7820: add edx,[ecx]
0x100e7822: cmp edx,edi
0x100e7824: jg short 100E7831h
0x100e7826: sub eax,1
0x100e7829: sub ecx,4
0x100e782c: cmp eax,20h
0x100e782f: jg short 100E7820h
0x100e7831: mov [esp+18h],eax
0x100e7835: fild dword ptr [esp+18h]
0x100e7839: fstp dword ptr [esp+18h]
0x100e783d: fld dword ptr [esp+10h]
0x100e7841: fld dword ptr [esp+18h]
0x100e7845: fcom
0x100e7847: fnstsw ax
0x100e7849: fstp st(1)
0x100e784b: test ah,41h
0x100e784e: jne short 100E7856h
0x100e7850: fstp dword ptr [esp+10h]
0x100e7854: jmp short 100E7858h
0x100e7856: fstp st(0)
0x100e7858: add esi,8000h
0x100e785e: sub ebp,1
0x100e7861: jne short 100E7810h
0x100e7863: pop ebp
0x100e7864: fld dword ptr [ebx+38h]
0x100e7867: mov edi,[esp+18h]
0x100e786b: fdivr qword ptr ds:[100FF128h]
0x100e7871: xor esi,esi
0x100e7873: mov [esp+14h],esi
0x100e7877: fmul dword ptr [esp+0Ch]
0x100e787b: fstp dword ptr [esp+0Ch]
0x100e787f: nop
0x100e7880: cmp dword ptr [ebx+6608h],0
0x100e7887: fild dword ptr [esp+14h]
0x100e788b: fdiv dword ptr [esp+0Ch]
0x100e788f: fstp dword ptr [esp+14h]
0x100e7893: jne short 100E789Bh
0x100e7895: fld dword ptr [esp+14h]
0x100e7899: jmp short 100E78CDh
0x100e789b: fld dword ptr [esp+14h]
0x100e789f: fcom qword ptr [ebx+80h]
0x100e78a5: fnstsw ax
0x100e78a7: test ah,41h
0x100e78aa: jp short 100E78B1h
0x100e78ac: fmul qword ptr [ebx+78h]
0x100e78af: jmp short 100E78CDh
0x100e78b1: fld qword ptr [ebx+70h]
0x100e78b4: call 100229A0h
0x100e78b9: fld qword ptr [ebx+88h]
0x100e78bf: fadd qword ptr ds:[100FE8F8h]
0x100e78c5: fmulp
0x100e78c7: fsub qword ptr [ebx+88h]
0x100e78cd: fmul qword ptr ds:[100FE920h]
0x100e78d3: call 100228D0h
0x100e78d8: cmp eax,0FFh
0x100e78dd: jle short 100E78E4h
0x100e78df: mov eax,0FFh
0x100e78e4: mov [esi+edi],al
0x100e78e7: add esi,1
0x100e78ea: cmp esi,10000h
0x100e78f0: mov [esp+14h],esi
0x100e78f4: jl short 100E7880h
0x100e78f6: pop edi
0x100e78f7: pop esi
```

```
0x100e78f8: pop ebx
0x100e78f9: pop ecx
0x100e78fa: ret
```

`sub_100e77a0` is called by sub_10016040. It delegates to sub_100228d0, j__CIpow (2 callees).

12.1310 sub_100e80f0 at 0x100e80f0 (48 instructions)

```
0x100e80f0: push esi
0x100e80f1: mov esi,[esp+8]
0x100e80f5: push edi
0x100e80f6: push 2E848h
0x100e80fb: xor edi,edi
0x100e80fd: push edi
0x100e80fe: push esi
0x100e80ff: call 100228C0h
0x100e8104: fld1
0x100e8106: fstp dword ptr [esi+38h]
0x100e8109: or eax,0FFFFFFFFh
0x100e810c: fld1
0x100e810e: add esp,0Ch
0x100e8111: fst qword ptr [esi+68h]
0x100e8114: mov [esi+8],edi
0x100e8117: fst qword ptr [esi+60h]
0x100e811a: mov [esi+0Ch],edi
0x100e811d: fst qword ptr [esi+58h]
0x100e8120: mov [esi+20h],edi
0x100e8123: fstp qword ptr [esi+50h]
0x100e8126: mov [esi+24h],edi
0x100e8129: fld qword ptr ds:[1012E940h]
0x100e812f: mov [esi+28h],edi
0x100e8132: fstp qword ptr [esi+70h]
0x100e8135: mov ecx,1
0x100e813a: fld qword ptr ds:[1012E938h]
0x100e8140: mov [esi+30h],edi
0x100e8143: fstp qword ptr [esi+78h]
0x100e8146: mov [esi+0CCh],edi
0x100e814c: fldz
0x100e814e: mov [esi+0C8h],edi
0x100e8154: pop edi
0x100e8155: fst qword ptr [esi+90h]
0x100e815b: mov dword ptr [esi],10138928h
0x100e8161: fst qword ptr [esi+88h]
0x100e8167: mov [esi+10h],eax
0x100e816a: fstp qword ptr [esi+80h]
0x100e8170: mov [esi+14h],eax
0x100e8173: mov [esi+18h],eax
0x100e8176: mov [esi+1Ch],eax
0x100e8179: mov [esi+2Ch],ecx
0x100e817c: mov [esi+0A8h],ecx
0x100e8182: mov dword ptr [esi+0ACh],8
0x100e818c: mov [esi+0D4h],eax
0x100e8192: mov [esi+0D0h],eax
0x100e8198: mov [esi+0E0h],eax
0x100e819e: pop esi
0x100e819f: ret
```

`sub_100e80f0` is called by sub_10016040. It delegates to j_memset (1 callees).

12.1311 sub_100e81a0 at 0x100e81a0 (25 instructions)

```
0x100e81a0: push esi
0x100e81a1: mov esi,[esp+8]
0x100e81a5: mov eax,[esi+627Ch]
0x100e81ab: test eax,eax
```

```
0x100e81ad: push edi
0x100e81ae: mov edi,ds:[100FE554h]
0x100e81b4: je short 100E81BCh
0x100e81b6: push eax
0x100e81b7: call edi
0x100e81b9: add esp,4
0x100e81bc: mov eax,[esi+65E0h]
0x100e81c2: test eax,eax
0x100e81c4: je short 100E81CCh
0x100e81c6: push eax
0x100e81c7: call edi
0x100e81c9: add esp,4
0x100e81cc: mov eax,[esi+66E0h]
0x100e81d2: test eax,eax
0x100e81d4: je short 100E81DCh
0x100e81d6: push eax
0x100e81d7: call edi
0x100e81d9: add esp,4
0x100e81dc: pop edi
0x100e81dd: pop esi
0x100e81de: ret
```

`sub_100e81a0` is called by sub_10016040. It calls nothing else.

12.1312 sub_100e81e0 at 0x100e81e0 (108 instructions)

```
0x100e81e0: sub esp,8
0x100e81e3: push ebx
0x100e81e4: mov ebx,[esp+18h]
0x100e81e8: test ebx,ebx
0x100e81ea: push ebp
0x100e81eb: push esi
0x100e81ec: push edi
0x100e81ed: je near ptr 100E866Eh
0x100e81f3: mov ebp,[esp+28h]
0x100e81f7: test ebp,ebp
0x100e81f9: je near ptr 100E866Eh
0x100e81ff: mov eax,[esp+20h]
0x100e8203: mov dword ptr [ebx+eax*4],100FE92Bh
0x100e820a: mov edi,1
0x100e820f: mov [ebp],edi
0x100e8212: mov ecx,[ebx+4]
0x100e8215: mov al,[ecx]
0x100e8217: movsx edx,al
0x100e821a: sub edx,2
0x100e821d: or edx,2
0x100e8220: cmp edx,2Bh
0x100e8223: mov [esp+13h],al
0x100e8227: jne near ptr 100E866Eh
0x100e822d: lea ecx,[ecx]
0x100e8230: mov eax,[ebp]
0x100e8233: mov ecx,[ebx+eax*4]
0x100e8236: mov cl,[ecx+1]
0x100e8239: add eax,1
0x100e823c: mov [ebp],eax
0x100e823f: movsx eax,cl
0x100e8242: push eax
0x100e8243: push 1012E994h
0x100e8248: mov [esp+2Ch],cl
0x100e824c: mov [esp+1Ch],eax
0x100e8250: call dword ptr ds:[100FE454h]
0x100e8256: add esp,8
0x100e8259: test eax,eax
0x100e825b: je short 100E82A7h
0x100e825d: sub eax,1012E994h
0x100e8262: movsx edx,byte ptr [eax+1012E984h]
```

```
0x100e8269: lea edi,[eax+1012E984h]
0x100e826f: sub edx,30h
0x100e8272: xor esi,esi
0x100e8274: test edx,edx
0x100e8276: jle short 100E82A2h
0x100e8278: mov eax,[ebp]
0x100e827b: add eax,esi
0x100e827d: mov ecx,[ebx+eax*4]
0x100e8280: movsx edx,byte ptr [ecx]
0x100e8283: push edx
0x100e8284: call dword ptr ds:[100FE45Ch]
0x100e828a: add esp,4
0x100e828d: test eax,eax
0x100e828f: je near ptr 100E8691h
0x100e8295: movsx eax,byte ptr [edi]
0x100e8298: add esi,1
0x100e829b: sub eax,30h
0x100e829e: cmp esi,eax
0x100e82a0: jl short 100E8278h
0x100e82a2: mov edi,1
0x100e82a7: mov eax,[esp+14h]
0x100e82ab: add eax,0FFFFFFCCh
0x100e82ae: cmp eax,46h
0x100e82b1: ja near ptr 100E86BCh
0x100e82b7: movzx ecx,byte ptr [eax+100E8764h]
0x100e82be: jmp dword ptr [ecx*4+100E86E4h]
0x100e866e: mov eax,[esp+1Ch]
0x100e8672: cmp dword ptr [eax+0E0h],0
0x100e8679: jge short 100E8687h
0x100e867b: mov ecx,[eax+0DCh]
0x100e8681: mov [eax+0E0h],ecx
0x100e8687: pop edi
0x100e8688: pop esi
0x100e8689: pop ebp
0x100e868a: xor eax,eax
0x100e868c: pop ebx
0x100e868d: add esp,8
0x100e8690: ret
0x100e8691: movsx ecx,byte ptr [esp+24h]
0x100e8696: push ecx
0x100e8697: push 1012E960h
0x100e869c: call dword ptr ds:[100FE4C4h]
0x100e86a2: add eax,40h
0x100e86a5: push eax
0x100e86a6: call dword ptr ds:[100FE4C8h]
0x100e86ac: add esp,0Ch
0x100e86af: pop edi
0x100e86b0: pop esi
0x100e86b1: pop ebp
0x100e86b2: mov eax,1
0x100e86b7: pop ebx
0x100e86b8: add esp,8
0x100e86bb: ret
0x100e86bc: movsx edx,byte ptr [esp+24h]
0x100e86c1: push edx
0x100e86c2: push 1012E948h
0x100e86c7: call dword ptr ds:[100FE4C4h]
0x100e86cd: add eax,40h
0x100e86d0: push eax
0x100e86d1: call dword ptr ds:[100FE4C8h]
0x100e86d7: add esp,0Ch
0x100e86da: mov eax,edi
0x100e86dc: pop edi
0x100e86dd: pop esi
0x100e86de: pop ebp
0x100e86df: pop ebx
```

```
0x100e86e0: add esp,8
0x100e86e3: ret
```

`sub_100e81e0` is called by sub_10016040. It calls nothing else.

12.1313 sub_100e8a00 at 0x100e8a00 (1116 instructions)

```
0x100e8a00: push ebp
0x100e8a01: mov ebp,esp
0x100e8a03: and esp,0FFFFFFF8h
0x100e8a06: sub esp,34Ch
0x100e8a0c: mov eax,ds:[10136220h]
0x100e8a11: xor eax,esp
0x100e8a13: mov [esp+348h],eax
0x100e8a1a: fld1
0x100e8a1c: xor eax,eax
0x100e8a1e: fst qword ptr [esp+94h]
0x100e8a25: push ebx
0x100e8a26: mov ebx,[ebp+8]
0x100e8a29: fst qword ptr [esp+0A0h]
0x100e8a30: fst qword ptr [esp+0A8h]
0x100e8a37: push esi
0x100e8a38: fst qword ptr [esp+0B4h]
0x100e8a3f: push edi
0x100e8a40: fldz
0x100e8a42: mov edi,[ebx+65E8h]
0x100e8a48: cmp edi,0Ah
0x100e8a4b: fst qword ptr [esp+80h]
0x100e8a52: fst qword ptr [esp+88h]
0x100e8a59: mov dword ptr [esp+3Ch],10h
0x100e8a61: fst qword ptr [esp+90h]
0x100e8a68: mov [esp+44h],eax
0x100e8a6c: fstp qword ptr [esp+98h]
0x100e8a73: mov [esp+2Ch],eax
0x100e8a77: mov [esp+24h],al
0x100e8a7b: mov byte ptr [esp+25h],1
0x100e8a80: mov byte ptr [esp+26h],2
0x100e8a85: fst qword ptr [esp+68h]
0x100e8a89: mov byte ptr [esp+27h],3
0x100e8a8e: fst qword ptr [esp+70h]
0x100e8a92: mov [esp+50h],eax
0x100e8a96: fstp qword ptr [esp+78h]
0x100e8a9a: mov [esp+54h],eax
0x100e8a9e: mov [esp+58h],eax
0x100e8aa2: mov [esp+5Ch],eax
0x100e8aa6: mov [esp+60h],eax
0x100e8aaa: mov dword ptr [esp+64h],0FFFh
0x100e8ab2: mov [esp+30h],eax
0x100e8ab6: mov [esp+34h],eax
0x100e8aba: mov [esp+38h],eax
0x100e8abe: jb short 100E8ADAh
0x100e8ac0: mov eax,1
0x100e8ac5: pop edi
0x100e8ac6: pop esi
0x100e8ac7: pop ebx
0x100e8ac8: mov ecx,[esp+348h]
0x100e8acf: xor ecx,esp
0x100e8ad1: call 10022326h
0x100e8ad6: mov esp,ebp
0x100e8ad8: pop ebp
0x100e8ad9: ret
0x100e8ada: lea ecx,[edi+1]
0x100e8add: mov [esp+40h],edi
0x100e8ae1: mov [ebx+65E8h],ecx
0x100e8ae7: mov [esp+10h],eax
0x100e8aeb: lea esi,[esp+120h]
```

```
0x100e8af2: jmp short 100E8AF6h
0x100e8af4: xor eax,eax
0x100e8af6: xor edx,edx
0x100e8af8: cmp eax,[esp+10h]
0x100e8afc: sete dl
0x100e8aff: add eax,1
0x100e8b02: add esi,8
0x100e8b05: cmp eax,4
0x100e8b08: mov [esp+1Ch],edx
0x100e8b0c: fild dword ptr [esp+1Ch]
0x100e8b10: fstp qword ptr [esi-8]
0x100e8b13: jl short 100E8AF6h
0x100e8b15: mov eax,[esp+10h]
0x100e8b19: add eax,1
0x100e8b1c: cmp eax,4
0x100e8b1f: mov [esp+10h],eax
0x100e8b23: jl short 100E8AF4h
0x100e8b25: push ebx
0x100e8b26: call 100D2F50h
0x100e8b2b: movzx eax,ax
0x100e8b2e: add esp,4
0x100e8b31: cmp eax,200h
0x100e8b36: ja short 100E8AC0h
0x100e8b38: test eax,eax
0x100e8b3a: je near ptr 100E925Ah
0x100e8b40: jmp short 100E8B46h
0x100e8b42: mov edi,[esp+40h]
0x100e8b46: sub eax,1
0x100e8b49: mov [esp+28h],eax
0x100e8b4d: lea eax,[esp+48h]
0x100e8b51: push eax
0x100e8b52: lea ecx,[esp+20h]
0x100e8b56: push ecx
0x100e8b57: mov ecx,[ebp+0Ch]
0x100e8b5a: lea edx,[esp+14h]
0x100e8b5e: push edx
0x100e8b5f: lea eax,[esp+58h]
0x100e8b63: push eax
0x100e8b64: push ecx
0x100e8b65: push ebx
0x100e8b66: call 100E2A20h
0x100e8b6b: mov esi,[esp+64h]
0x100e8b6f: add esp,18h
0x100e8b72: cmp esi,827Dh
0x100e8b78: ja near ptr 100E95A1h
0x100e8b7e: je near ptr 100E9585h
0x100e8b84: cmp esi,26h
0x100e8b87: ja near ptr 100E8C36h
0x100e8b8d: cmp esi,24h
0x100e8b90: jae near ptr 100E8C16h
0x100e8b96: lea eax,[esi-2]
0x100e8b99: cmp eax,15h
0x100e8b9c: ja near ptr 100E9170h
0x100e8ba2: movzx edx,byte ptr [eax+100EA268h]
0x100e8ba9: jmp dword ptr [edx*4+100EA250h]
0x100e8c16: push ebx
0x100e8c17: call 100D2F50h
0x100e8c1c: movzx edx,ax
0x100e8c1f: mov [esp+14h],edx
0x100e8c23: add esp,4
0x100e8c26: fild dword ptr [esp+10h]
0x100e8c2a: fstp dword ptr [ebx+esi*4+0E6E4h]
0x100e8c31: jmp near ptr 100E9170h
0x100e8c36: cmp esi,131h
0x100e8c3c: ja near ptr 100E8FD7h
0x100e8c42: je near ptr 100E8F1Ah
0x100e8c48: add esi,0FFFFFFD9h
```

```
0x100e8c4b: cmp esi,0FCh
0x100e8c51: ja near ptr 100E9170h
0x100e8c57: movzx eax,byte ptr [esi+100EA2C4h]
0x100e8c5e: jmp dword ptr [eax*4+100EA280h]
0x100e8f1a: mov edx,[ebx]
0x100e8f1c: mov ecx,[ebx+4]
0x100e8f1f: mov edx,[edx+10h]
0x100e8f22: push 40h
0x100e8f24: lea eax,[esp+214h]
0x100e8f2b: push eax
0x100e8f2c: push ecx
0x100e8f2d: call edx
0x100e8f2f: mov esi,ds:[100FE414h]
0x100e8f35: push 5
0x100e8f37: lea eax,[esp+220h]
0x100e8f3e: push 1012EA64h
0x100e8f43: push eax
0x100e8f44: call esi
0x100e8f46: add esp,18h
0x100e8f49: test eax,eax
0x100e8f4b: je short 100E8FC8h
0x100e8f4d: push 5
0x100e8f4f: lea ecx,[esp+214h]
0x100e8f56: push 1012EA5Ch
0x100e8f5b: push ecx
0x100e8f5c: call esi
0x100e8f5e: add esp,0Ch
0x100e8f61: test eax,eax
0x100e8f63: je short 100E8FC8h
0x100e8f65: push 5
0x100e8f67: lea edx,[esp+214h]
0x100e8f6e: push 1012EA54h
0x100e8f73: push edx
0x100e8f74: call esi
0x100e8f76: add esp,0Ch
0x100e8f79: test eax,eax
0x100e8f7b: je short 100E8FC8h
0x100e8f7d: push 6
0x100e8f7f: lea eax,[esp+214h]
0x100e8f86: push 1012EA4Ch
0x100e8f8b: push eax
0x100e8f8c: call esi
0x100e8f8e: add esp,0Ch
0x100e8f91: test eax,eax
0x100e8f93: je short 100E8FC8h
0x100e8f95: push 0Ah
0x100e8f97: lea ecx,[esp+214h]
0x100e8f9e: push 1012EA40h
0x100e8fa3: push ecx
0x100e8fa4: call esi
0x100e8fa6: add esp,0Ch
0x100e8fa9: test eax,eax
0x100e8fab: je short 100E8FC8h
0x100e8fad: mov edi,1012EA24h
0x100e8fb2: lea esi,[esp+210h]
0x100e8fb9: mov ecx,1Bh
0x100e8fbe: xor edx,edx
0x100e8fc0: repe cmpsb
0x100e8fc2: jne near ptr 100E9170h
0x100e8fc8: mov dword ptr [ebx+6614h],0
0x100e8fd2: jmp near ptr 100E9170h
0x100e8fd7: cmp esi,202h
0x100e8fdd: ja near ptr 100E941Ah
0x100e8fe3: je near ptr 100E9402h
0x100e8fe9: add esi,0FFFFFECEh
0x100e8fef: cmp esi,0CFh
0x100e8ff5: ja near ptr 100E9170h
```

```
0x100e8ffb: movzx eax,byte ptr [esi+100EA3E8h]
0x100e9002: jmp dword ptr [eax*4+100EA3C4h]
0x100e9170: mov edx,[esp+48h]
0x100e9174: mov ecx,[ebx]
0x100e9176: mov eax,[ebx+4]
0x100e9179: mov ecx,[ecx+8]
0x100e917c: push 0
0x100e917e: push edx
0x100e917f: push eax
0x100e9180: call ecx
0x100e9182: mov eax,[esp+34h]
0x100e9186: add esp,0Ch
0x100e9189: test eax,eax
0x100e918b: jne near ptr 100E8B42h
0x100e9191: mov eax,[esp+34h]
0x100e9195: test eax,eax
0x100e9197: je near ptr 100E925Ah
0x100e919d: push eax
0x100e919e: call dword ptr ds:[100FE550h]
0x100e91a4: mov esi,eax
0x100e91a6: add esp,4
0x100e91a9: test esi,esi
0x100e91ab: je near ptr 100E925Ah
0x100e91b1: mov edi,[esp+30h]
0x100e91b5: mov edx,[ebx]
0x100e91b7: mov eax,[ebx+4]
0x100e91ba: mov ecx,[edx+8]
0x100e91bd: push 0
0x100e91bf: push edi
0x100e91c0: push eax
0x100e91c1: call ecx
0x100e91c3: mov eax,[esp+40h]
0x100e91c7: mov edx,[ebx]
0x100e91c9: mov ecx,[ebx+4]
0x100e91cc: mov edx,[edx]
0x100e91ce: push 1
0x100e91d0: push eax
0x100e91d1: push esi
0x100e91d2: push ecx
0x100e91d3: call edx
0x100e91d5: mov eax,[esp+54h]
0x100e91d9: mov ecx,[esp+50h]
0x100e91dd: push eax
0x100e91de: push 1
0x100e91e0: shr ecx,2
0x100e91e3: push ecx
0x100e91e4: push esi
0x100e91e5: call 100DBDD0h
0x100e91ea: mov edx,[ebx]
0x100e91ec: mov eax,[ebx+4]
0x100e91ef: add esp,2Ch
0x100e91f2: mov [esp+28h],edx
0x100e91f6: mov [esp+48h],eax
0x100e91fa: mov dword ptr [ebx],10138928h
0x100e9200: call dword ptr ds:[100FE420h]
0x100e9206: test eax,eax
0x100e9208: mov [ebx+4],eax
0x100e920b: je short 100E9243h
0x100e920d: mov edx,[esp+34h]
0x100e9211: mov ecx,[ebx]
0x100e9213: push 1
0x100e9215: push edx
0x100e9216: push esi
0x100e9217: push eax
0x100e9218: mov eax,[ecx+4]
0x100e921b: call eax
0x100e921d: mov ecx,[ebx]
```

```
0x100e921f: mov edx,[ebx+4]
0x100e9222: mov eax,[ecx+8]
0x100e9225: push 0
0x100e9227: push 0
0x100e9229: push edx
0x100e922a: call eax
0x100e922c: neg edi
0x100e922e: push edi
0x100e922f: push ebx
0x100e9230: call 100E8A00h
0x100e9235: mov ecx,[ebx]
0x100e9237: mov edx,[ebx+4]
0x100e923a: mov eax,[ecx+0Ch]
0x100e923d: push edx
0x100e923e: call eax
0x100e9240: add esp,28h
0x100e9243: mov ecx,[esp+28h]
0x100e9247: mov edx,[esp+48h]
0x100e924b: push esi
0x100e924c: mov [ebx],ecx
0x100e924e: mov [ebx+4],edx
0x100e9251: call dword ptr ds:[100FE554h]
0x100e9257: add esp,4
0x100e925a: cmp dword ptr [ebx+66D0h],0
0x100e9261: jle short 100E92A2h
0x100e9263: mov edi,[ebx+66D0h]
0x100e9269: lea esi,[esp+120h]
0x100e9270: lea ecx,[esp+0A0h]
0x100e9277: jmp short 100E9280h
0x100e9280: mov edx,[ebx+66D0h]
0x100e9286: mov eax,esi
0x100e9288: fld qword ptr [eax]
0x100e928a: add eax,8
0x100e928d: sub edx,1
0x100e9290: fmul qword ptr [ecx]
0x100e9292: fstp qword ptr [eax-8]
0x100e9295: jne short 100E9288h
0x100e9297: add ecx,8
0x100e929a: add esi,20h
0x100e929d: sub edi,1
0x100e92a0: jne short 100E9280h
0x100e92a2: cmp dword ptr [esp+44h],0
0x100e92a7: je near ptr 100E9339h
0x100e92ad: cmp dword ptr [ebx+66D0h],0
0x100e92b4: jle short 100E9328h
0x100e92b6: mov ecx,[ebx+66D0h]
0x100e92bc: fldz
0x100e92be: lea eax,[esp+120h]
0x100e92c5: mov [esp+2Ch],eax
0x100e92c9: lea eax,[esp+0C0h]
0x100e92d0: mov [esp+30h],ecx
0x100e92d4: xor edx,edx
0x100e92d6: mov [esp+38h],edx
0x100e92da: mov ecx,[esp+2Ch]
0x100e92de: fst qword ptr [eax]
0x100e92e0: mov esi,[ebx+66D0h]
0x100e92e6: lea edi,[esp+edx+68h]
0x100e92ea: lea edx,[esp+edx+1A0h]
0x100e92f1: fld qword ptr [edx]
0x100e92f3: add ecx,8
0x100e92f6: fmul qword ptr [ecx-8]
0x100e92f9: add edx,18h
0x100e92fc: sub esi,1
0x100e92ff: fmul qword ptr [edi]
0x100e9301: fadd qword ptr [eax]
0x100e9303: fstp qword ptr [eax]
0x100e9305: jne short 100E92F1h
```

```
0x100e9307: mov edx,[esp+38h]
0x100e930b: add edx,8
0x100e930e: add eax,8
0x100e9311: cmp edx,18h
0x100e9314: mov [esp+38h],edx
0x100e9318: jl short 100E92DAh
0x100e931a: add dword ptr [esp+2Ch],20h
0x100e931f: sub dword ptr [esp+30h],1
0x100e9324: jne short 100E92D4h
0x100e9326: fstp st(0)
0x100e9328: lea ecx,[esp+0C0h]
0x100e932f: push ecx
0x100e9330: push ebx
0x100e9331: call 100DDC20h
0x100e9336: add esp,8
0x100e9339: fldz
0x100e933b: fcomp qword ptr [esp+80h]
0x100e9342: fnstsw ax
0x100e9344: test ah,44h
0x100e9347: jnp short 100E9387h
0x100e9349: fldz
0x100e934b: xor eax,eax
0x100e934d: cmp [ebx+66D0h],eax
0x100e9353: fstp dword ptr [ebx+0E780h]
0x100e9359: jle short 100E9387h
0x100e935b: fld1
0x100e935d: lea ecx,[ebx+0E774h]
0x100e9363: fld qword ptr [esp+eax*8+80h]
0x100e936a: add eax,1
0x100e936d: fstp dword ptr [esp+28h]
0x100e9371: add ecx,4
0x100e9374: fld dword ptr [esp+28h]
0x100e9378: fdivr st,st(1)
0x100e937a: fstp dword ptr [ecx-4]
0x100e937d: cmp eax,[ebx+66D0h]
0x100e9383: jl short 100E9363h
0x100e9385: fstp st(0)
0x100e9387: cmp dword ptr [esp+44h],0
0x100e938c: jne short 100E93C5h
0x100e938e: xor edx,edx
0x100e9390: cmp [ebx+66D0h],edx
0x100e9396: jle short 100E93C5h
0x100e9398: lea ecx,[esp+120h]
0x100e939f: lea eax,[ebx+0E784h]
0x100e93a5: fld dword ptr [eax]
0x100e93a7: add edx,1
0x100e93aa: fld qword ptr [ecx]
0x100e93ac: add eax,4
0x100e93af: fstp dword ptr [esp+28h]
0x100e93b3: add ecx,28h
0x100e93b6: fdiv dword ptr [esp+28h]
0x100e93ba: fstp dword ptr [eax-4]
0x100e93bd: cmp edx,[ebx+66D0h]
0x100e93c3: jl short 100E93A5h
0x100e93c5: mov ecx,[esp+354h]
0x100e93cc: pop edi
0x100e93cd: pop esi
0x100e93ce: pop ebx
0x100e93cf: xor ecx,esp
0x100e93d1: xor eax,eax
0x100e93d3: call 10022326h
0x100e93d8: mov esp,ebp
0x100e93da: pop ebp
0x100e93db: ret
0x100e9402: push ebx
0x100e9403: call 100D2FE0h
0x100e9408: lea ecx,[edi+edi*8]
```

```
0x100e940b: add esp,4
0x100e940e: mov [ebx+ecx*4+6110h],eax
0x100e9415: jmp near ptr 100E9170h
0x100e941a: cmp esi,7221h
0x100e9420: ja near ptr 100E94E4h
0x100e9426: je near ptr 100E94D2h
0x100e942c: sub esi,7010h
0x100e9432: je short 100E9469h
0x100e9434: sub esi,1F0h
0x100e943a: je short 100E9457h
0x100e943c: sub esi,1
0x100e943f: jne near ptr 100E9170h
0x100e9445: push ebx
0x100e9446: call 100D2FE0h
0x100e944b: add esp,4
0x100e944e: mov [esp+34h],eax
0x100e9452: jmp near ptr 100E9170h
0x100e9457: push ebx
0x100e9458: call 100D2FE0h
0x100e945d: add esp,4
0x100e9460: mov [esp+30h],eax
0x100e9464: jmp near ptr 100E9170h
0x100e9469: xor esi,esi
0x100e946b: jmp short 100E9470h
0x100e9470: push ebx
0x100e9471: call 100D2F50h
0x100e9476: add esi,1
0x100e9479: add esp,4
0x100e947c: shr eax,2
0x100e947f: and eax,0FFFh
0x100e9484: cmp esi,4
0x100e9487: mov [esp+esi*4+50h],eax
0x100e948b: jl short 100E9470h
0x100e948d: xor ecx,ecx
0x100e948f: mov esi,[esp+ecx*4+50h]
0x100e9493: add esi,1
0x100e9496: cmp esi,[esp+ecx*4+54h]
0x100e949a: jg short 100E94C5h
0x100e949c: mov eax,[esp+ecx*4+54h]
0x100e94a0: mov edi,1
0x100e94a5: sub eax,esi
0x100e94a7: shl edi,cl
0x100e94a9: lea edx,[ebx+esi*2+6764h]
0x100e94b0: add eax,1
0x100e94b3: mov si,[edx-2]
0x100e94b7: add si,di
0x100e94ba: mov [edx],si
0x100e94bd: add edx,2
0x100e94c0: sub eax,1
0x100e94c3: jne short 100E94B3h
0x100e94c5: add ecx,1
0x100e94c8: cmp ecx,5
0x100e94cb: jl short 100E948Fh
0x100e94cd: jmp near ptr 100E9170h
0x100e94d2: push ebx
0x100e94d3: call 100D2FE0h
0x100e94d8: add esp,4
0x100e94db: mov [esp+38h],eax
0x100e94df: jmp near ptr 100E9170h
0x100e94e4: sub esi,7250h
0x100e94ea: je short 100E9562h
0x100e94ec: sub esi,0B3h
0x100e94f2: je short 100E952Eh
0x100e94f4: sub esi,10h
0x100e94f7: jne near ptr 100E9170h
0x100e94fd: lea ecx,[ecx]
0x100e9500: push ebx
```

```
0x100e9501: call 100D2F50h
0x100e9506: movzx edx,ax
0x100e9509: mov eax,esi
0x100e950b: sar eax,1
0x100e950d: mov [esp+14h],edx
0x100e9511: xor eax,esi
0x100e9513: add esi,1
0x100e9516: fild dword ptr [esp+14h]
0x100e951a: add esp,4
0x100e951d: cmp esi,4
0x100e9520: fstp dword ptr [ebx+eax*4+0E774h]
0x100e9527: jl short 100E9500h
0x100e9529: jmp near ptr 100E9170h
0x100e952e: xor esi,esi
0x100e9530: push ebx
0x100e9531: call 100D2F50h
0x100e9536: xor edx,edx
0x100e9538: add esp,4
0x100e953b: cmp esi,2
0x100e953e: setl dl
0x100e9541: movzx ecx,ax
0x100e9544: mov [esp+10h],ecx
0x100e9548: fild dword ptr [esp+10h]
0x100e954c: xor edx,esi
0x100e954e: add esi,1
0x100e9551: fstp dword ptr [ebx+edx*4+0E774h]
0x100e9558: cmp esi,4
0x100e955b: jl short 100E9530h
0x100e955d: jmp near ptr 100E9170h
0x100e9562: mov eax,[ebx]
0x100e9564: mov ecx,[ebx+4]
0x100e9567: mov edx,[eax+18h]
0x100e956a: push ecx
0x100e956b: call edx
0x100e956d: push eax
0x100e956e: push ebx
0x100e956f: call 100EABA0h
0x100e9574: add esp,0Ch
0x100e9577: mov word ptr [ebx+66B2h],0
0x100e9580: jmp near ptr 100E9170h
0x100e9585: mov eax,[ebx]
0x100e9587: mov edx,[ebx+4]
0x100e958a: mov eax,[eax+10h]
0x100e958d: push 40h
0x100e958f: lea ecx,[ebx+6505h]
0x100e9595: push ecx
0x100e9596: push edx
0x100e9597: call eax
0x100e9599: add esp,0Ch
0x100e959c: jmp near ptr 100E9170h
0x100e95a1: cmp esi,0C51Ah
0x100e95a7: ja near ptr 100E9C58h
0x100e95ad: je near ptr 100E9C15h
0x100e95b3: cmp esi,8773h
0x100e95b9: ja near ptr 100E97F5h
0x100e95bf: je near ptr 100E9D75h
0x100e95c5: cmp esi,85FFh
0x100e95cb: ja near ptr 100E9653h
0x100e95d1: je near ptr 100E97A0h
0x100e95d7: add esi,0FFFF7D72h
0x100e95dd: cmp esi,0Fh
0x100e95e0: ja near ptr 100E9170h
0x100e95e6: movzx ecx,byte ptr [esi+100EA4CCh]
0x100e95ed: jmp dword ptr [ecx*4+100EA4B8h]
0x100e9653: cmp esi,8606h
0x100e9659: ja near ptr 100E97BCh
0x100e965f: je near ptr 100E978Bh
```

```
0x100e9665: sub esi,8602h
0x100e966b: je near ptr 100E974Eh
0x100e9671: sub esi,1
0x100e9674: jne near ptr 100E9170h
0x100e967a: mov eax,[ebx]
0x100e967c: mov edx,[ebx+4]
0x100e967f: mov eax,[eax]
0x100e9681: push 7
0x100e9683: push 1
0x100e9685: lea ecx,[esp+218h]
0x100e968c: push ecx
0x100e968d: push edx
0x100e968e: call eax
0x100e9690: push 6
0x100e9692: lea ecx,[esp+224h]
0x100e9699: push 1012EA08h
0x100e969e: push ecx
0x100e969f: call dword ptr ds:[100FE414h]
0x100e96a5: add esp,1Ch
0x100e96a8: test eax,eax
0x100e96aa: jne near ptr 100E9170h
0x100e96b0: xor edi,edi
0x100e96b2: lea edx,[ebx+0E7C4h]
0x100e96b8: mov dword ptr [ebx+66D0h],4
0x100e96c2: mov [ebx+6604h],edi
0x100e96c8: mov [esp+10h],edx
0x100e96cc: lea esp,[esp]
0x100e96d0: xor esi,esi
0x100e96d2: mov eax,[ebx]
0x100e96d4: mov ecx,esi
0x100e96d6: xor ecx,1
0x100e96d9: add ecx,edi
0x100e96db: lea edx,[ebx+ecx*4+0E7C4h]
0x100e96e2: mov ecx,[ebx+4]
0x100e96e5: push edx
0x100e96e6: mov edx,[eax+20h]
0x100e96e9: push 1012E650h
0x100e96ee: push ecx
0x100e96ef: call edx
0x100e96f1: add esi,1
0x100e96f4: add esp,0Ch
0x100e96f7: cmp esi,4
0x100e96fa: jl short 100E96D2h
0x100e96fc: cmp dword ptr [ebx+0DCh],0
0x100e9703: je short 100E973Ch
0x100e9705: mov edx,[esp+10h]
0x100e9709: fldz
0x100e970b: mov eax,edx
0x100e970d: mov ecx,4
0x100e9712: fadd dword ptr [eax]
0x100e9714: add eax,4
0x100e9717: sub ecx,1
0x100e971a: jne short 100E9712h
0x100e971c: fstp dword ptr [esp+20h]
0x100e9720: mov eax,edx
0x100e9722: fld dword ptr [esp+20h]
0x100e9726: mov ecx,4
0x100e972b: fld dword ptr [eax]
0x100e972d: add eax,4
0x100e9730: sub ecx,1
0x100e9733: fdiv st,st(1)
0x100e9735: fstp dword ptr [eax-4]
0x100e9738: jne short 100E972Bh
0x100e973a: fstp st(0)
0x100e973c: add dword ptr [esp+10h],10h
0x100e9741: add edi,4
0x100e9744: cmp edi,0Ch
```

```
0x100e9747: jl short 100E96D0h
0x100e9749: jmp near ptr 100E9170h
0x100e974e: xor esi,esi
0x100e9750: push ebx
0x100e9751: call 100D2F50h
0x100e9756: movzx eax,ax
0x100e9759: mov [esp+14h],eax
0x100e975d: mov ecx,esi
0x100e975f: xor ecx,1
0x100e9762: fild dword ptr [esp+14h]
0x100e9766: add esi,1
0x100e9769: add esp,4
0x100e976c: cmp esi,4
0x100e976f: fdivr qword ptr ds:[1012EA00h]
0x100e9775: fstp dword ptr [esp+10h]
0x100e9779: fld dword ptr [esp+10h]
0x100e977d: fstp dword ptr [ebx+ecx*4+0E774h]
0x100e9784: jl short 100E9750h
0x100e9786: jmp near ptr 100E9170h
0x100e978b: mov edx,[ebx]
0x100e978d: mov eax,[ebx+4]
0x100e9790: mov ecx,[edx+18h]
0x100e9793: push eax
0x100e9794: call ecx
0x100e9796: push eax
0x100e9797: push ebx
0x100e9798: call 100E42B0h
0x100e979d: add esp,0Ch
0x100e97a0: mov edx,ds:[10129EE4h]
0x100e97a6: mov [ebx+6485h],edx
0x100e97ac: mov al,ds:[10129EE8h]
0x100e97b1: mov [ebx+6489h],al
0x100e97b7: jmp near ptr 100E9170h
0x100e97bc: cmp esi,8769h
0x100e97c2: jne near ptr 100E9170h
0x100e97c8: mov esi,[ebx]
0x100e97ca: push 0
0x100e97cc: push ebx
0x100e97cd: add esi,8
0x100e97d0: call 100D2FE0h
0x100e97d5: mov edi,[ebp+0Ch]
0x100e97d8: mov ecx,[ebx+4]
0x100e97db: mov edx,[esi]
0x100e97dd: add esp,4
0x100e97e0: add eax,edi
0x100e97e2: push eax
0x100e97e3: push ecx
0x100e97e4: call edx
0x100e97e6: push edi
0x100e97e7: push ebx
0x100e97e8: call 100E3E60h
0x100e97ed: add esp,14h
0x100e97f0: jmp near ptr 100E9170h
0x100e97f5: cmp esi,9218h
0x100e97fb: ja near ptr 100E9909h
0x100e9801: je near ptr 100E98A4h
0x100e9807: cmp esi,920Ah
0x100e980d: ja short 100E987Fh
0x100e980f: je short 100E9866h
0x100e9811: cmp esi,8825h
0x100e9817: je short 100E9839h
0x100e9819: cmp esi,9102h
0x100e981f: jne near ptr 100E9170h
0x100e9825: push ebx
0x100e9826: call 100D2FE0h
0x100e982b: add esp,4
0x100e982e: mov [ebx+65ACh],eax
```

```
0x100e9834: jmp near ptr 100E9170h
0x100e9839: mov esi,[ebx]
0x100e983b: push 0
0x100e983d: push ebx
0x100e983e: add esi,8
0x100e9841: call 100D2FE0h
0x100e9846: mov edi,[ebp+0Ch]
0x100e9849: mov ecx,[esi]
0x100e984b: add eax,edi
0x100e984d: add esp,4
0x100e9850: push eax
0x100e9851: mov eax,[ebx+4]
0x100e9854: push eax
0x100e9855: call ecx
0x100e9857: push edi
0x100e9858: push ebx
0x100e9859: call 100E40D0h
0x100e985e: add esp,14h
0x100e9861: jmp near ptr 100E9170h
0x100e9866: mov edx,[esp+0Ch]
0x100e986a: push edx
0x100e986b: push ebx
0x100e986c: call 100D3060h
0x100e9871: fstp dword ptr [ebx+659Ch]
0x100e9877: add esp,8
0x100e987a: jmp near ptr 100E9170h
0x100e987f: cmp esi,9211h
0x100e9885: jne near ptr 100E9170h
0x100e988b: mov eax,[esp+0Ch]
0x100e988f: push eax
0x100e9890: push ebx
0x100e9891: call 100D3020h
0x100e9896: add esp,8
0x100e9899: mov [ebx+65A8h],eax
0x100e989f: jmp near ptr 100E9170h
0x100e98a4: lea ecx,[ebx+0E7C4h]
0x100e98aa: mov dword ptr [ebx+6604h],0
0x100e98b4: mov [esp+10h],ecx
0x100e98b8: mov dword ptr [esp+20h],3
0x100e98c0: mov edx,[esp+0Ch]
0x100e98c4: push edx
0x100e98c5: push ebx
0x100e98c6: call 100D3060h
0x100e98cb: fstp st(0)
0x100e98cd: mov esi,[esp+18h]
0x100e98d1: add esp,8
0x100e98d4: mov edi,3
0x100e98d9: lea esp,[esp]
0x100e98e0: mov eax,[esp+0Ch]
0x100e98e4: push eax
0x100e98e5: push ebx
0x100e98e6: call 100D3060h
0x100e98eb: fstp dword ptr [esi]
0x100e98ed: add esp,8
0x100e98f0: add esi,4
0x100e98f3: sub edi,1
0x100e98f6: jne short 100E98E0h
0x100e98f8: add dword ptr [esp+10h],10h
0x100e98fd: sub dword ptr [esp+20h],1
0x100e9902: jne short 100E98C0h
0x100e9904: jmp near ptr 100E9170h
0x100e9909: cmp esi,0B4C7h
0x100e990f: ja near ptr 100E9B55h
0x100e9915: je short 100E9969h
0x100e9917: cmp esi,0B4C3h
0x100e991d: jne near ptr 100E9170h
0x100e9923: mov ecx,ds:[1012E9F4h]
```

```
0x100e9929: mov [ebx+6485h],ecx
0x100e992f: mov dx,ds:[1012E9F8h]
0x100e9936: mov ecx,[ebx]
0x100e9938: mov [ebx+6489h],dx
0x100e993f: mov al,ds:[1012E9FAh]
0x100e9944: mov edx,[ebx+4]
0x100e9947: mov [ebx+648Bh],al
0x100e994d: mov eax,[ecx+18h]
0x100e9950: push edx
0x100e9951: call eax
0x100e9953: mov ecx,[esp+20h]
0x100e9957: add esp,4
0x100e995a: mov [ebx+65C0h],eax
0x100e9960: mov [esp+2Ch],ecx
0x100e9964: jmp near ptr 100E9170h
0x100e9969: cmp dword ptr [esp+2Ch],0
0x100e996e: je near ptr 100E9170h
0x100e9974: mov edx,[ebx]
0x100e9976: mov eax,[ebx+4]
0x100e9979: mov ecx,[edx+8]
0x100e997c: push 1
0x100e997e: push 4Eh
0x100e9980: push eax
0x100e9981: call ecx
0x100e9983: push ebx
0x100e9984: call 100D2FE0h
0x100e9989: push ebx
0x100e998a: mov [ebx+66B2h],ax
0x100e9991: call 100D2FE0h
0x100e9996: push ebx
0x100e9997: mov [ebx+66B0h],ax
0x100e999e: call 100D2FE0h
0x100e99a3: and eax,7
0x100e99a6: push ebx
0x100e99a7: mov [ebx+66BAh],ax
0x100e99ae: call 100D2FE0h
0x100e99b3: mov dx,[ebx+66B2h]
0x100e99ba: and eax,7
0x100e99bd: sub dx,ax
0x100e99c0: sub dx,[ebx+66BAh]
0x100e99c7: push ebx
0x100e99c8: mov [ebx+66B6h],dx
0x100e99cf: call 100D2FE0h
0x100e99d4: and eax,7
0x100e99d7: push ebx
0x100e99d8: mov [ebx+66B8h],ax
0x100e99df: call 100D2FE0h
0x100e99e4: mov cx,[ebx+66B0h]
0x100e99eb: and eax,7
0x100e99ee: sub cx,ax
0x100e99f1: sub cx,[ebx+66B8h]
0x100e99f8: add esp,24h
0x100e99fb: cmp word ptr [ebx+66B2h],1C5Eh
0x100e9a04: mov [ebx+66B4h],cx
0x100e9a0b: jne short 100E9A28h
0x100e9a0d: mov word ptr [ebx+66B4h],1544h
0x100e9a16: mov word ptr [ebx+66B6h],1C4Ch
0x100e9a1f: mov word ptr [ebx+66BAh],7
0x100e9a28: mov edx,[ebx]
0x100e9a2a: mov eax,[ebx+4]
0x100e9a2d: mov ecx,[edx+8]
0x100e9a30: push 1
0x100e9a32: push 34h
0x100e9a34: push eax
0x100e9a35: call ecx
0x100e9a37: add esp,0Ch
0x100e9a3a: lea esi,[ebx+0E774h]
```

```
0x100e9a40: mov edi,3
0x100e9a45: push 0Bh
0x100e9a47: push ebx
0x100e9a48: call 100D3060h
0x100e9a4d: fstp dword ptr [esi]
0x100e9a4f: add esp,8
0x100e9a52: add esi,4
0x100e9a55: sub edi,1
0x100e9a58: jne short 100E9A45h
0x100e9a5a: mov edx,[ebx]
0x100e9a5c: mov eax,[ebx+4]
0x100e9a5f: mov ecx,[edx+8]
0x100e9a62: push 1
0x100e9a64: push 72h
0x100e9a66: push eax
0x100e9a67: call ecx
0x100e9a69: push ebx
0x100e9a6a: call 100D2F50h
0x100e9a6f: movzx esi,word ptr [ebx+66B6h]
0x100e9a76: movzx edi,word ptr [ebx+66B4h]
0x100e9a7d: movzx ecx,di
0x100e9a80: movzx edx,si
0x100e9a83: movzx eax,ax
0x100e9a86: imul ecx,edx
0x100e9a89: shr eax,7
0x100e9a8c: imul eax,5Ah
0x100e9a8f: lea ecx,[ecx+ecx*2]
0x100e9a92: add ecx,ecx
0x100e9a94: add esp,10h
0x100e9a97: cmp ecx,[esp+2Ch]
0x100e9a9b: mov [ebx+66C8h],eax
0x100e9aa1: jne short 100E9ADEh
0x100e9aa3: cdq
0x100e9aa4: mov ecx,0B4h
0x100e9aa9: idiv ecx
0x100e9aab: cmp edx,5Ah
0x100e9aae: jne short 100E9AD0h
0x100e9ab0: mov eax,edi
0x100e9ab2: xor eax,esi
0x100e9ab4: xor [ebx+66B4h],ax
0x100e9abb: mov [ebx+66B6h],ax
0x100e9ac2: movzx eax,word ptr [ebx+66B4h]
0x100e9ac9: xor [ebx+66B6h],ax
0x100e9ad0: xor eax,eax
0x100e9ad2: mov [ebx+66C8h],eax
0x100e9ad8: mov [ebx+65B0h],eax
0x100e9ade: movzx edx,word ptr [ebx+66B6h]
0x100e9ae5: movzx ecx,word ptr [ebx+66B4h]
0x100e9aec: imul ecx,edx
0x100e9aef: mov eax,431BDE83h
0x100e9af4: imul ecx
0x100e9af6: sar edx,12h
0x100e9af9: mov eax,edx
0x100e9afb: shr eax,1Fh
0x100e9afe: add eax,edx
0x100e9b00: push eax
0x100e9b01: lea ecx,[ebx+64C5h]
0x100e9b07: push 1012E9E4h
0x100e9b0c: push ecx
0x100e9b0d: call dword ptr ds:[100FE4F8h]
0x100e9b13: add esp,0Ch
0x100e9b16: cmp dword ptr [ebx+65B0h],0
0x100e9b1d: mov dword ptr [ebx+2E7FCh],100D8AE0h
0x100e9b27: je short 100E9B46h
0x100e9b29: test byte ptr [ebx+66BAh],1
0x100e9b30: je short 100E9B3Ch
0x100e9b32: mov dword ptr [ebx+65B0h],61616161h
```

```
0x100e9b3c: mov dword ptr [ebx+2E7FCh],100D8E20h
0x100e9b46: mov dword ptr [ebx+65FCh],0FFFFh
0x100e9b50: jmp near ptr 100E9170h
0x100e9b55: cmp esi,0C516h
0x100e9b5b: jb near ptr 100E9170h
0x100e9b61: cmp esi,0C517h
0x100e9b67: ja near ptr 100E9170h
0x100e9b6d: mov edi,[esp+1Ch]
0x100e9b71: push edi
0x100e9b72: call dword ptr ds:[100FE550h]
0x100e9b78: add esp,4
0x100e9b7b: test eax,eax
0x100e9b7d: mov [esp+10h],eax
0x100e9b81: je near ptr 100E9170h
0x100e9b87: mov edx,[ebx]
0x100e9b89: mov ecx,[edx]
0x100e9b8b: push edi
0x100e9b8c: mov esi,eax
0x100e9b8e: mov eax,[ebx+4]
0x100e9b91: push 1
0x100e9b93: push esi
0x100e9b94: push eax
0x100e9b95: call ecx
0x100e9b97: add esp,10h
0x100e9b9a: add esi,0FFFFFFFFh
0x100e9b9d: je short 100E9C02h
0x100e9b9f: mov edx,[esp+10h]
0x100e9ba3: add edx,edi
0x100e9ba5: mov edi,ds:[100FE414h]
0x100e9bab: mov [esp+20h],edx
0x100e9baf: cmp esi,[esp+20h]
0x100e9bb3: jae short 100E9C02h
0x100e9bb5: push 8
0x100e9bb7: add esi,1
0x100e9bba: push 1012E9D8h
0x100e9bbf: push esi
0x100e9bc0: call edi
0x100e9bc2: add esp,0Ch
0x100e9bc5: test eax,eax
0x100e9bc7: jne short 100E9BF0h
0x100e9bc9: lea eax,[ebx+0E77Ch]
0x100e9bcf: push eax
0x100e9bd0: lea ecx,[ebx+0E778h]
0x100e9bd6: push ecx
0x100e9bd7: lea edx,[ebx+0E774h]
0x100e9bdd: push edx
0x100e9bde: lea eax,[esi+8]
0x100e9be1: push 1012E9CCh
0x100e9be6: push eax
0x100e9be7: call dword ptr ds:[100FE4A8h]
0x100e9bed: add esp,14h
0x100e9bf0: push 0Ah
0x100e9bf2: push esi
0x100e9bf3: call dword ptr ds:[100FE454h]
0x100e9bf9: mov esi,eax
0x100e9bfb: add esp,8
0x100e9bfe: test esi,esi
0x100e9c00: jne short 100E9BAFh
0x100e9c02: mov ecx,[esp+10h]
0x100e9c06: push ecx
0x100e9c07: call dword ptr ds:[100FE554h]
0x100e9c0d: add esp,4
0x100e9c10: jmp near ptr 100E9170h
0x100e9c15: cmp byte ptr [ebx+6485h],0
0x100e9c1c: jne near ptr 100E9170h
0x100e9c22: mov edx,ds:[1012E9C0h]
0x100e9c28: mov [ebx+6485h],edx
```

```
0x100e9c2e: mov eax,ds:[1012E9C4h]
0x100e9c33: mov [ebx+6489h],eax
0x100e9c39: mov cx,ds:[1012E9C8h]
0x100e9c40: mov [ebx+648Dh],cx
0x100e9c47: mov dl,ds:[1012E9CAh]
0x100e9c4d: mov [ebx+648Fh],dl
0x100e9c53: jmp near ptr 100E9170h
0x100e9c58: cmp esi,0C612h
0x100e9c5e: ja near ptr 100E9D46h
0x100e9c64: je near ptr 100E9CF8h
0x100e9c6a: cmp esi,0C51Bh
0x100e9c70: jne near ptr 100E9170h
0x100e9c76: movsx eax,word ptr [ebx+6274h]
0x100e9c7d: mov ecx,[ebx]
0x100e9c7f: mov edx,[ebx+4]
0x100e9c82: mov [esp+20h],eax
0x100e9c86: mov eax,[ecx+18h]
0x100e9c89: push edx
0x100e9c8a: call eax
0x100e9c8c: mov ecx,[ebx+65E8h]
0x100e9c92: push ebx
0x100e9c93: mov edi,eax
0x100e9c95: mov [esp+18h],ecx
0x100e9c99: call 100D2F50h
0x100e9c9e: push ebx
0x100e9c9f: mov [ebx+6274h],ax
0x100e9ca6: call 100D2F50h
0x100e9cab: mov esi,[ebx]
0x100e9cad: add esp,0Ch
0x100e9cb0: push 0
0x100e9cb2: push ebx
0x100e9cb3: add esi,8
0x100e9cb6: call 100D2FE0h
0x100e9cbb: mov edx,[ebx+4]
0x100e9cbe: add eax,edi
0x100e9cc0: add esp,4
0x100e9cc3: push eax
0x100e9cc4: mov eax,[esi]
0x100e9cc6: push edx
0x100e9cc7: call eax
0x100e9cc9: push edi
0x100e9cca: push ebx
0x100e9ccb: call 100E8A00h
0x100e9cd0: mov ecx,[esp+24h]
0x100e9cd4: mov dx,[esp+34h]
0x100e9cd9: add esp,14h
0x100e9cdc: mov dword ptr [ebx+65FCh],0FFFFh
0x100e9ce6: mov [ebx+65E8h],ecx
0x100e9cec: mov [ebx+6274h],dx
0x100e9cf3: jmp near ptr 100E9170h
0x100e9cf8: mov esi,4
0x100e9cfd: lea ecx,[ecx]
0x100e9d00: mov eax,[ebx]
0x100e9d02: mov ecx,[ebx+4]
0x100e9d05: mov edx,[eax+1Ch]
0x100e9d08: push ecx
0x100e9d09: call edx
0x100e9d0b: mov ecx,[ebx+6618h]
0x100e9d11: shl ecx,8
0x100e9d14: add eax,ecx
0x100e9d16: add esp,4
0x100e9d19: sub esi,1
0x100e9d1c: mov [ebx+6618h],eax
0x100e9d22: jne short 100E9D00h
0x100e9d24: cmp byte ptr [ebx+6485h],0
0x100e9d2b: jne short 100E9D37h
0x100e9d2d: mov dword ptr [ebx+6485h],474E44h
```

```
0x100e9d37: mov dword ptr [ebx+6614h],1
0x100e9d41: jmp near ptr 100E9170h
0x100e9d46: cmp esi,0FD04h
0x100e9d4c: ja near ptr 100EA12Eh
0x100e9d52: je near ptr 100EA0E6h
0x100e9d58: add esi,0FFFF39EAh
0x100e9d5e: cmp esi,79h
0x100e9d61: ja near ptr 100E9170h
0x100e9d67: movzx edx,byte ptr [esi+100EA51Ch]
0x100e9d6e: jmp dword ptr [edx*4+100EA4DCh]
0x100e9d75: mov eax,[ebx]
0x100e9d77: mov ecx,[ebx+4]
0x100e9d7a: mov edx,[eax+18h]
0x100e9d7d: push ecx
0x100e9d7e: call edx
0x100e9d80: mov [ebx+65CCh],eax
0x100e9d86: mov eax,[esp+20h]
0x100e9d8a: add esp,4
0x100e9d8d: mov [ebx+65D8h],eax
0x100e9d93: jmp near ptr 100E9170h
0x100ea0e6: mov eax,[ebx]
0x100ea0e8: mov ecx,[ebx+4]
0x100ea0eb: mov edx,[eax+8]
0x100ea0ee: push 1
0x100ea0f0: push 10h
0x100ea0f2: push ecx
0x100ea0f3: call edx
0x100ea0f5: push ebx
0x100ea0f6: call 100D2FE0h
0x100ea0fb: mov ecx,[ebx+4]
0x100ea0fe: mov [ebx+65C0h],eax
0x100ea104: mov eax,[ebx]
0x100ea106: mov edx,[eax+8]
0x100ea109: push 1
0x100ea10b: push 1Ch
0x100ea10d: push ecx
0x100ea10e: call edx
0x100ea110: push ebx
0x100ea111: call 100D2FE0h
0x100ea116: add esp,20h
0x100ea119: add [ebx+65C0h],eax
0x100ea11f: mov dword ptr [ebx+2E7FCh],100D8B50h
0x100ea129: jmp near ptr 100E9170h
0x100ea12e: cmp esi,0FD09h
0x100ea134: jne near ptr 100E9170h
0x100ea13a: mov edi,[esp+1Ch]
0x100ea13e: cmp edi,10h
0x100ea141: mov [esp+3Ch],edi
0x100ea145: jbe short 100EA150h
0x100ea147: mov edi,10h
0x100ea14c: mov [esp+3Ch],edi
0x100ea150: mov eax,[ebx]
0x100ea152: mov edx,[ebx+4]
0x100ea155: mov eax,[eax]
0x100ea157: push edi
0x100ea158: push 1
0x100ea15a: lea ecx,[esp+208h]
0x100ea161: push ecx
0x100ea162: push edx
0x100ea163: call eax
0x100ea165: xor esi,esi
0x100ea167: add esp,10h
0x100ea16a: xor edx,edx
0x100ea16c: cmp edi,esi
0x100ea16e: mov [ebx+66D0h],esi
0x100ea174: jle short 100EA1D5h
0x100ea176: xor edi,edi
```

```
0x100ea178: jmp short 100EA180h
0x100ea180: mov cl,[esp+edx+200h]
0x100ea187: mov eax,1
0x100ea18c: shl eax,cl
0x100ea18e: mov ecx,esi
0x100ea190: and ecx,eax
0x100ea192: neg ecx
0x100ea194: sbb ecx,ecx
0x100ea196: add ecx,1
0x100ea199: add edx,1
0x100ea19c: add edi,ecx
0x100ea19e: or esi,eax
0x100ea1a0: cmp edx,[esp+3Ch]
0x100ea1a4: jl short 100EA180h
0x100ea1a6: cmp esi,38h
0x100ea1a9: mov [ebx+66D0h],edi
0x100ea1af: jne short 100EA1C8h
0x100ea1b1: mov dx,ds:[1012E9BCh]
0x100ea1b8: mov al,ds:[1012E9BEh]
0x100ea1bd: mov [esp+24h],dx
0x100ea1c2: mov [esp+26h],al
0x100ea1c6: jmp short 100EA1D5h
0x100ea1c8: cmp esi,3Ah
0x100ea1cb: jne short 100EA1D5h
0x100ea1cd: mov dword ptr [esp+24h],1040305h
0x100ea1d5: xor eax,eax
0x100ea1d7: cmp [ebx+66D0h],eax
0x100ea1dd: jle short 100EA203h
0x100ea1df: mov ecx,[ebx+66D0h]
0x100ea1e5: jmp short 100EA1F0h
0x100ea1f0: movzx edx,byte ptr [esp+eax+24h]
0x100ea1f5: mov [esp+edx+250h],al
0x100ea1fc: add eax,1
0x100ea1ff: cmp eax,ecx
0x100ea201: jl short 100EA1F0h
0x100ea203: mov edi,[esp+3Ch]
0x100ea207: mov byte ptr [ebx+eax+6280h],0
0x100ea20f: mov esi,[ebx+65B0h]
0x100ea215: mov ecx,10h
0x100ea21a: sub ecx,1
0x100ea21d: xor edx,edx
0x100ea21f: mov eax,ecx
0x100ea221: div edi
0x100ea223: movzx eax,byte ptr [esp+edx+200h]
0x100ea22b: movzx edx,byte ptr [esp+eax+250h]
0x100ea233: lea eax,[esi*4]
0x100ea23a: or edx,eax
0x100ea23c: test ecx,ecx
0x100ea23e: mov esi,edx
0x100ea240: jne short 100EA21Ah
0x100ea242: mov [ebx+65B0h],esi
0x100ea248: jmp near ptr 100E9170h
```

`sub_100e8a00` is called by sub_100e2b50, sub_100e8a00, sub_100ea5a0. It works with the text "Hasselblad", "%f %f %f", "Neutral ". It accesses memory-mapped peripheral registers at 0x431bde83. It delegates to sub_10022326, sub_100d2f50, sub_100d2fe0 (13 callees).

12.1314 sub_100ea5a0 at 0x100ea5a0 (341 instructions)

```
0x100ea5a0: sub esp,5Ch
0x100ea5a3: push ebx
0x100ea5a4: push ebp
0x100ea5a5: push esi
0x100ea5a6: mov esi,[esp+6Ch]
0x100ea5aa: mov eax,[esi]
0x100ea5ac: mov ecx,[esi+4]
```

```
0x100ea5af: mov edx,[eax+8]
0x100ea5b2: push edi
0x100ea5b3: mov edi,[esp+74h]
0x100ea5b7: xor ebx,ebx
0x100ea5b9: push ebx
0x100ea5ba: or ebp,0FFFFFFFFh
0x100ea5bd: push edi
0x100ea5be: push ecx
0x100ea5bf: mov [esp+20h],ebx
0x100ea5c3: mov [esp+1Ch],ebp
0x100ea5c7: mov [esp+24h],ebp
0x100ea5cb: call edx
0x100ea5cd: push esi
0x100ea5ce: call 100D2F50h
0x100ea5d3: add esp,10h
0x100ea5d6: cmp ax,4949h
0x100ea5da: mov [esi+6274h],ax
0x100ea5e1: je short 100EA5EDh
0x100ea5e3: cmp ax,4D4Dh
0x100ea5e7: jne near ptr 100EAB5Dh
0x100ea5ed: push esi
0x100ea5ee: call 100D2F50h
0x100ea5f3: push 168h
0x100ea5f8: lea eax,[esi+60F0h]
0x100ea5fe: push 0
0x100ea600: push eax
0x100ea601: call 100228C0h
0x100ea606: push esi
0x100ea607: mov [esi+65E8h],ebx
0x100ea60d: call 100D2FE0h
0x100ea612: add esp,14h
0x100ea615: test eax,eax
0x100ea617: je short 100EA64Bh
0x100ea619: lea esp,[esp]
0x100ea620: mov ecx,[esi]
0x100ea622: mov edx,[esi+4]
0x100ea625: add eax,edi
0x100ea627: push 0
0x100ea629: push eax
0x100ea62a: mov eax,[ecx+8]
0x100ea62d: push edx
0x100ea62e: call eax
0x100ea630: push edi
0x100ea631: push esi
0x100ea632: call 100E8A00h
0x100ea637: add esp,14h
0x100ea63a: test eax,eax
0x100ea63c: jne short 100EA64Bh
0x100ea63e: push esi
0x100ea63f: call 100D2FE0h
0x100ea644: add esp,4
0x100ea647: test eax,eax
0x100ea649: jne short 100EA620h
0x100ea64b: mov eax,[esi+65C4h]
0x100ea651: test eax,eax
0x100ea653: mov dword ptr [esi+65DCh],10h
0x100ea65d: je short 100EA6A3h
0x100ea65f: mov ecx,[esi]
0x100ea661: mov edx,[esi+4]
0x100ea664: push 0
0x100ea666: push eax
0x100ea667: mov eax,[ecx+8]
0x100ea66a: push edx
0x100ea66b: call eax
0x100ea66d: push 1
0x100ea66f: lea ecx,[esp+2Ch]
0x100ea673: push ecx
```

```
0x100ea674: push esi
0x100ea675: call 100D4CE0h
0x100ea67a: add esp,18h
0x100ea67d: test eax,eax
0x100ea67f: je short 100EA6A3h
0x100ea681: mov edx,[esp+1Ch]
0x100ea685: mov ax,[esp+24h]
0x100ea68a: mov cx,[esp+20h]
0x100ea68f: mov [esi+65DCh],edx
0x100ea695: mov [esi+66C4h],ax
0x100ea69c: mov [esi+66C6h],cx
0x100ea6a3: xor ecx,ecx
0x100ea6a5: cmp [esi+65E8h],ebx
0x100ea6ab: jle near ptr 100EA74Eh
0x100ea6b1: lea eax,[esi+60F4h]
0x100ea6b7: mov edx,[eax+18h]
0x100ea6ba: cmp ebx,edx
0x100ea6bc: jge short 100EA6C0h
0x100ea6be: mov ebx,edx
0x100ea6c0: cmp ebx,3
0x100ea6c3: jle short 100EA6CAh
0x100ea6c5: mov ebx,3
0x100ea6ca: cmp dword ptr [eax+8],6
0x100ea6ce: jne short 100EA6D5h
0x100ea6d0: cmp edx,3
0x100ea6d3: je short 100EA734h
0x100ea6d5: movzx edi,word ptr [esi+66B0h]
0x100ea6dc: movzx edx,word ptr [esi+66B2h]
0x100ea6e3: imul edx,edi
0x100ea6e6: mov edi,[eax]
0x100ea6e8: imul edi,[eax-4]
0x100ea6ec: cmp edi,edx
0x100ea6ee: jle short 100EA734h
0x100ea6f0: movzx edx,word ptr [eax-4]
0x100ea6f4: mov [esi+66B2h],dx
0x100ea6fb: movzx edx,word ptr [eax]
0x100ea6fe: mov [esi+66B0h],dx
0x100ea705: mov edx,[eax+4]
0x100ea708: mov [esi+65F0h],edx
0x100ea70e: mov edx,[eax+8]
0x100ea711: mov [esi+65F4h],edx
0x100ea717: mov edx,[eax+10h]
0x100ea71a: mov [esi+65C0h],edx
0x100ea720: mov edx,[eax+14h]
0x100ea723: mov [esi+66CCh],edx
0x100ea729: mov edx,[eax+18h]
0x100ea72c: mov [esi+65ECh],edx
0x100ea732: mov ebp,ecx
0x100ea734: add ecx,1
0x100ea737: add eax,24h
0x100ea73a: cmp ecx,[esi+65E8h]
0x100ea740: jl near ptr 100EA6B7h
0x100ea746: mov [esp+10h],ebp
0x100ea74a: mov [esp+14h],ebx
0x100ea74e: movzx ecx,word ptr [esi+66B2h]
0x100ea755: lea eax,[ecx+1]
0x100ea758: cdq
0x100ea759: sub eax,edx
0x100ea75b: sar eax,1
0x100ea75d: imul ax,[esi+66C2h]
0x100ea765: mov [esi+66C2h],ax
0x100ea76c: mov eax,[esi+6108h]
0x100ea772: xor edi,edi
0x100ea774: cmp eax,edi
0x100ea776: je short 100EA77Eh
0x100ea778: mov [esi+66CCh],eax
0x100ea77e: cmp ebp,edi
```

```
0x100ea780: mov ebx,8
0x100ea785: jl near ptr 100EA9A2h
0x100ea78b: cmp [esi+2E7FCh],edi
0x100ea791: jne near ptr 100EA9A2h
0x100ea797: mov eax,[esi+65F4h]
0x100ea79d: cmp eax,8001h
0x100ea7a2: ja near ptr 100EA913h
0x100ea7a8: je near ptr 100EA8FEh
0x100ea7ae: cmp eax,63h
0x100ea7b1: ja near ptr 100EA88Dh
0x100ea7b7: je near ptr 100EA87Eh
0x100ea7bd: cmp eax,7
0x100ea7c0: ja near ptr 100EA94Ah
0x100ea7c6: movzx eax,byte ptr [eax+100EAB74h]
0x100ea7cd: jmp dword ptr [eax*4+100EAB68h]
0x100ea87e: mov dword ptr [esi+2E7FCh],100D52A0h
0x100ea888: jmp near ptr 100EA9A2h
0x100ea88d: cmp eax,106h
0x100ea892: je short 100EA8EFh
0x100ea894: cmp eax,7FFFh
0x100ea899: jne near ptr 100EA94Ah
0x100ea89f: movzx eax,word ptr [esi+66B0h]
0x100ea8a6: movzx edx,ax
0x100ea8a9: imul edx,[esi+65F0h]
0x100ea8b0: imul edx,ecx
0x100ea8b3: lea ecx,[ebp+ebp*8]
0x100ea8b7: mov dword ptr [esi+2E7FCh],100DC340h
0x100ea8c1: mov ecx,[esi+ecx*4+6110h]
0x100ea8c8: add ecx,ecx
0x100ea8ca: add ecx,ecx
0x100ea8cc: add ecx,ecx
0x100ea8ce: cmp ecx,edx
0x100ea8d0: je near ptr 100EA9A2h
0x100ea8d6: add eax,8
0x100ea8d9: mov [esi+66B0h],ax
0x100ea8e0: mov dword ptr [esi+2E7FCh],100DC1D0h
0x100ea8ea: jmp near ptr 100EA9A2h
0x100ea8ef: mov dword ptr [esi+2E7FCh],100DB290h
0x100ea8f9: jmp near ptr 100EA9A2h
0x100ea8fe: mov [esi+66ACh],ebx
0x100ea904: mov dword ptr [esi+2E7FCh],100D8B50h
0x100ea90e: jmp near ptr 100EA9A2h
0x100ea913: cmp eax,8799h
0x100ea918: ja short 100EA93Ch
0x100ea91a: je short 100EA930h
0x100ea91c: sub eax,8005h
0x100ea921: je short 100EA904h
0x100ea923: sub eax,5Eh
0x100ea926: je short 100EA9A2h
0x100ea928: mov [esi+6614h],edi
0x100ea92e: jmp short 100EA9A2h
0x100ea930: mov dword ptr [esi+2E7FCh],100D6090h
0x100ea93a: jmp short 100EA9A2h
0x100ea93c: cmp eax,0FDE8h
0x100ea941: je short 100EA95Eh
0x100ea943: cmp eax,0FFFFh
0x100ea948: je short 100EA952h
0x100ea94a: mov [esi+6614h],edi
0x100ea950: jmp short 100EA9A2h
0x100ea952: mov dword ptr [esi+2E7FCh],100D5F40h
0x100ea95c: jmp short 100EA9A2h
0x100ea95e: lea edx,[ebp+ebp*8]
0x100ea962: mov eax,[esi+edx*4+6100h]
0x100ea969: sub eax,2
0x100ea96c: je short 100EA992h
0x100ea96e: sub eax,4
0x100ea971: je short 100EA986h
```

```
0x100ea973: sub eax,801Dh
0x100ea978: jne short 100EA9A2h
0x100ea97a: mov dword ptr [esi+2E7FCh],100DB7E0h
0x100ea984: jmp short 100EA9A2h
0x100ea986: mov dword ptr [esi+2E7FCh],100DB9C0h
0x100ea990: jmp short 100EA99Ch
0x100ea992: mov dword ptr [esi+2E7FCh],100DBC10h
0x100ea99c: mov [esi+65B0h],edi
0x100ea9a2: mov ecx,[esi+6618h]
0x100ea9a8: cmp ecx,edi
0x100ea9aa: jne short 100EAA06h
0x100ea9ac: cmp dword ptr [esi+65ECh],3
0x100ea9b3: jne short 100EA9DAh
0x100ea9b5: lea eax,[ebp+ebp*8]
0x100ea9b9: cmp [esi+eax*4+6110h],edi
0x100ea9c0: je short 100EA9DAh
0x100ea9c2: mov eax,[esi+65F0h]
0x100ea9c8: cmp eax,0Eh
0x100ea9cb: je short 100EA9DAh
0x100ea9cd: cmp eax,800h
0x100ea9d2: je short 100EA9DAh
0x100ea9d4: mov [esi+6614h],edi
0x100ea9da: cmp ecx,edi
0x100ea9dc: jne short 100EAA06h
0x100ea9de: cmp [esi+65F0h],ebx
0x100ea9e4: jne short 100EAA1Dh
0x100ea9e6: mov eax,1
0x100ea9eb: cmp [esi+65F4h],eax
0x100ea9f1: jne short 100EAA06h
0x100ea9f3: lea ecx,[ebp+ebp*8]
0x100ea9f7: cmp [esi+ecx*4+6100h],eax
0x100ea9fe: jne short 100EAA06h
0x100eaa00: mov [esi+6614h],edi
0x100eaa06: cmp [esi+65F0h],ebx
0x100eaa0c: jne short 100EAA1Dh
0x100eaa0e: cmp dword ptr [esi+65ECh],4
0x100eaa15: jne short 100EAA1Dh
0x100eaa17: mov [esi+6614h],edi
0x100eaa1d: xor ebx,ebx
0x100eaa1f: cmp [esi+65E8h],ebx
0x100eaa25: jle near ptr 100EAB5Dh
0x100eaa2b: lea ecx,[esi+60F0h]
0x100eaa31: cmp ebx,ebp
0x100eaa33: je near ptr 100EAAB9h
0x100eaa39: mov edx,[esp+14h]
0x100eaa3d: cmp [ecx+1Ch],edx
0x100eaa40: jne short 100EAAB9h
0x100eaa42: mov eax,[ecx+8]
0x100eaa45: movzx ebp,word ptr [esi+66C6h]
0x100eaa4c: lea edi,[eax+1]
0x100eaa4f: mov eax,[esi+65DCh]
0x100eaa55: lea edx,[eax+1]
0x100eaa58: movzx eax,word ptr [esi+66C4h]
0x100eaa5f: imul eax,ebp
0x100eaa62: mov ebp,edx
0x100eaa64: imul ebp,edx
0x100eaa67: xor edx,edx
0x100eaa69: div ebp
0x100eaa6b: mov ebp,eax
0x100eaa6d: mov eax,edi
0x100eaa6f: imul eax,edi
0x100eaa72: mov edi,eax
0x100eaa74: mov eax,[ecx+4]
0x100eaa77: imul eax,[ecx]
0x100eaa7a: cdq
0x100eaa7b: idiv edi
0x100eaa7d: cmp eax,ebp
```

```
0x100eaa7f: mov ebp,[esp+10h]
0x100eaa83: jle short 100EAAB9h
0x100eaa85: mov ax,[ecx]
0x100eaa88: mov [esi+66C4h],ax
0x100eaa8f: mov dx,[ecx+4]
0x100eaa93: mov [esi+66C6h],dx
0x100eaa9a: mov eax,[ecx+14h]
0x100eaa9d: mov [esi+65C4h],eax
0x100eaaa3: mov edx,[ecx+20h]
0x100eaaa6: mov [esi+65D0h],edx
0x100eaaac: mov eax,[ecx+8]
0x100eaaaf: mov [esi+65DCh],eax
0x100eaab5: mov [esp+18h],ebx
0x100eaab9: add ebx,1
0x100eaabc: add ecx,24h
0x100eaabf: cmp ebx,[esi+65E8h]
0x100eaac5: jl near ptr 100EAA31h
0x100eaacb: mov eax,[esp+18h]
0x100eaacf: test eax,eax
0x100eaad1: jl near ptr 100EAB5Dh
0x100eaad7: lea ecx,[eax+eax*8]
0x100eaada: mov edx,[esi+ecx*4+610Ch]
0x100eaae1: lea eax,[esi+ecx*4]
0x100eaae4: shl edx,5
0x100eaae7: or [esi+65DCh],edx
0x100eaaed: mov ecx,[eax+60FCh]
0x100eaaf3: sub ecx,0
0x100eaaf6: je short 100EAB53h
0x100eaaf8: sub ecx,1
0x100eaafb: je short 100EAB26h
0x100eaafd: sub ecx,0FDE7h
0x100eab03: jne short 100EAB5Dh
0x100eab05: cmp dword ptr [eax+6100h],6
0x100eab0c: mov eax,100DB9C0h
0x100eab11: je short 100EAB18h
0x100eab13: mov eax,100DBC10h
0x100eab18: pop edi
0x100eab19: mov [esi+2E800h],eax
0x100eab1f: pop esi
0x100eab20: pop ebp
0x100eab21: pop ebx
0x100eab22: add esp,5Ch
0x100eab25: ret
0x100eab26: cmp dword ptr [eax+60F8h],8
0x100eab2d: jle short 100EAB41h
0x100eab2f: pop edi
0x100eab30: mov dword ptr [esi+2E800h],100DBD40h
0x100eab3a: pop esi
0x100eab3b: pop ebp
0x100eab3c: pop ebx
0x100eab3d: add esp,5Ch
0x100eab40: ret
0x100eab41: pop edi
0x100eab42: mov dword ptr [esi+2E7F4h],100D6C40h
0x100eab4c: pop esi
0x100eab4d: pop ebp
0x100eab4e: pop ebx
0x100eab4f: add esp,5Ch
0x100eab52: ret
0x100eab53: mov dword ptr [esi+2E7F4h],100D6CD0h
0x100eab5d: pop edi
0x100eab5e: pop esi
0x100eab5f: pop ebp
0x100eab60: pop ebx
0x100eab61: add esp,5Ch
0x100eab64: ret
```

`sub_100ea5a0` is called by sub_100eaba0, sub_100eadb0, sub_100eb010, sub_100eb620. It delegates to j_memset, sub_100d2f50, sub_100d2fe0 (5 callees).

12.1315 sub_100eaba0 at 0x100eaba0 (174 instructions)

```
0x100eaba0: sub esp,14h
0x100eaba3: push ebx
0x100eaba4: mov ebx,[esp+1Ch]
0x100eaba8: movzx ecx,word ptr [ebx+6274h]
0x100eabaf: mov edx,[ebx]
0x100eabb1: xor eax,eax
0x100eabb3: push esi
0x100eabb4: mov esi,[esp+24h]
0x100eabb8: push eax
0x100eabb9: mov [esp+10h],eax
0x100eabbd: mov [esp+14h],eax
0x100eabc1: mov eax,[ebx+4]
0x100eabc4: mov [esp+1Ch],ecx
0x100eabc8: mov ecx,[edx+8]
0x100eabcb: push esi
0x100eabcc: push eax
0x100eabcd: call ecx
0x100eabcf: mov edx,[ebx]
0x100eabd1: mov eax,[ebx+4]
0x100eabd4: mov ecx,[edx+1Ch]
0x100eabd7: push eax
0x100eabd8: call ecx
0x100eabda: add esp,10h
0x100eabdd: test eax,eax
0x100eabdf: jne near ptr 100EADA2h
0x100eabe5: mov edx,[ebx]
0x100eabe7: mov eax,[ebx+4]
0x100eabea: mov ecx,[edx+1Ch]
0x100eabed: push eax
0x100eabee: call ecx
0x100eabf0: add esp,4
0x100eabf3: sub eax,4Dh
0x100eabf6: jne near ptr 100EADA2h
0x100eabfc: mov edx,[ebx]
0x100eabfe: mov eax,[ebx+4]
0x100eac01: mov ecx,[edx+1Ch]
0x100eac04: push eax
0x100eac05: call ecx
0x100eac07: add esp,4
0x100eac0a: sub eax,52h
0x100eac0d: jne near ptr 100EADA2h
0x100eac13: mov edx,[ebx]
0x100eac15: mov eax,[ebx+4]
0x100eac18: mov ecx,[edx+1Ch]
0x100eac1b: push eax
0x100eac1c: call ecx
0x100eac1e: imul eax,101h
0x100eac24: push ebx
0x100eac25: mov [ebx+6274h],ax
0x100eac2c: call 100D2FE0h
0x100eac31: mov edx,[ebx]
0x100eac33: mov ecx,[edx+18h]
0x100eac36: lea esi,[eax+esi+8]
0x100eac3a: mov eax,[ebx+4]
0x100eac3d: push eax
0x100eac3e: mov [esp+14h],esi
0x100eac42: call ecx
0x100eac44: add esp,0Ch
0x100eac47: cmp eax,esi
0x100eac49: mov [esp+20h],eax
0x100eac4d: jge near ptr 100EAD7Eh
0x100eac53: push ebp
```

```
0x100eac54: push edi
0x100eac55: xor esi,esi
0x100eac57: lea edi,[esi+4]
0x100eac5a: lea ebx,[ebx]
0x100eac60: mov edx,[ebx]
0x100eac62: mov eax,[ebx+4]
0x100eac65: mov ecx,[edx+1Ch]
0x100eac68: push eax
0x100eac69: call ecx
0x100eac6b: shl esi,8
0x100eac6e: add esp,4
0x100eac71: or esi,eax
0x100eac73: sub edi,1
0x100eac76: jne short 100EAC60h
0x100eac78: push ebx
0x100eac79: call 100D2FE0h
0x100eac7e: add esp,4
0x100eac81: cmp esi,505244h
0x100eac87: mov ebp,eax
0x100eac89: je near ptr 100EAD1Eh
0x100eac8f: cmp esi,545457h
0x100eac95: je short 100EACFDh
0x100eac97: cmp esi,574247h
0x100eac9d: jne near ptr 100EAD4Ah
0x100eaca3: push ebx
0x100eaca4: call 100D2FE0h
0x100eaca9: add esp,4
0x100eacac: xor edx,edx
0x100eacae: mov edi,1012EA78h
0x100eacb3: lea esi,[ebx+64C5h]
0x100eacb9: mov ecx,0Ch
0x100eacbe: repe cmpsb
0x100eacc0: mov eax,edx
0x100eacc2: setne al
0x100eacc5: sub eax,1
0x100eacc8: and eax,3
0x100eaccb: mov edi,eax
0x100eaccd: xor esi,esi
0x100eaccf: nop
0x100eacd0: push ebx
0x100eacd1: call 100D2F50h
0x100eacd6: movzx ecx,ax
0x100eacd9: mov edx,esi
0x100eacdb: sar edx,1
0x100eacdd: mov [esp+20h],ecx
0x100eace1: xor edx,esi
0x100eace3: xor edx,edi
0x100eace5: fild dword ptr [esp+20h]
0x100eace9: add esi,1
0x100eacec: add esp,4
0x100eacef: cmp esi,4
0x100eacf2: fstp dword ptr [ebx+edx*4+0E774h]
0x100eacf9: jl short 100EACD0h
0x100eacfb: jmp short 100EAD4Ah
0x100eacfd: mov eax,[ebx]
0x100eacff: mov ecx,[ebx+4]
0x100ead02: mov edx,[eax+18h]
0x100ead05: push ecx
0x100ead06: call edx
0x100ead08: push eax
0x100ead09: push ebx
0x100ead0a: call 100EA5A0h
0x100ead0f: mov eax,[esp+1Ch]
0x100ead13: add esp,0Ch
0x100ead16: mov [ebx+65C0h],eax
0x100ead1c: jmp short 100EAD4Ah
0x100ead1e: mov ecx,[ebx]
```

```
0x100ead20: mov edx,[ebx+4]
0x100ead23: mov eax,[ecx+8]
0x100ead26: push 1
0x100ead28: push 8
0x100ead2a: push edx
0x100ead2b: call eax
0x100ead2d: push ebx
0x100ead2e: call 100D2F50h
0x100ead33: movzx ecx,ax
0x100ead36: push ebx
0x100ead37: mov [esp+28h],ecx
0x100ead3b: call 100D2F50h
0x100ead40: movzx edx,ax
0x100ead43: add esp,14h
0x100ead46: mov [esp+18h],edx
0x100ead4a: mov ecx,[esp+28h]
0x100ead4e: mov eax,[ebx]
0x100ead50: lea edx,[ecx+ebp+8]
0x100ead54: mov ecx,[ebx+4]
0x100ead57: push 0
0x100ead59: push edx
0x100ead5a: mov edx,[eax+8]
0x100ead5d: push ecx
0x100ead5e: call edx
0x100ead60: mov eax,[ebx]
0x100ead62: mov ecx,[ebx+4]
0x100ead65: mov edx,[eax+18h]
0x100ead68: push ecx
0x100ead69: call edx
0x100ead6b: add esp,10h
0x100ead6e: cmp eax,[esp+10h]
0x100ead72: mov [esp+28h],eax
0x100ead76: jl near ptr 100EAC55h
0x100ead7c: pop edi
0x100ead7d: pop ebp
0x100ead7e: mov ax,[esp+0Ch]
0x100ead83: mov cx,[esp+10h]
0x100ead88: mov dx,[esp+18h]
0x100ead8d: mov [ebx+66B0h],ax
0x100ead94: mov [ebx+66B2h],cx
0x100ead9b: mov [ebx+6274h],dx
0x100eada2: pop esi
0x100eada3: pop ebx
0x100eada4: add esp,14h
0x100eada7: ret
```

`sub_100eaba0` is called by sub_100e8a00, sub_100eb620. It works with the text "DiMAGE A200". It delegates to sub_100d2f50, sub_100d2fe0, sub_100ea5a0 (3 callees).

12.1316 sub_100eadb0 at 0x100eadb0 (203 instructions)

```
0x100eadb0: push ecx
0x100eadb1: push ebx
0x100eadb2: push ebp
0x100eadb3: push esi
0x100eadb4: mov esi,ds:[100FE464h]
0x100eadba: push edi
0x100eadbb: mov edi,[esp+18h]
0x100eadbf: mov eax,[edi+6278h]
0x100eadc5: push 2Eh
0x100eadc7: push eax
0x100eadc8: call esi
0x100eadca: mov ecx,[edi+6278h]
0x100eadd0: mov ebx,eax
0x100eadd2: push 2Fh
0x100eadd4: push ecx
```

```
0x100eadd5: mov [esp+20h],ebx
0x100eadd9: call esi
0x100eaddb: mov ebp,eax
0x100eaddd: add esp,10h
0x100eade0: test ebp,ebp
0x100eade2: jne short 100EAE01h
0x100eade4: mov edx,[edi+6278h]
0x100eadea: push 5Ch
0x100eadec: push edx
0x100eaded: call esi
0x100eadef: mov ebp,eax
0x100eadf1: add esp,8
0x100eadf4: test ebp,ebp
0x100eadf6: jne short 100EAE01h
0x100eadf8: mov ebp,[edi+6278h]
0x100eadfe: sub ebp,1
0x100eae01: add ebp,1
0x100eae04: test ebx,ebx
0x100eae06: je near ptr 100EB006h
0x100eae0c: mov eax,ebx
0x100eae0e: lea edx,[eax+1]
0x100eae11: mov cl,[eax]
0x100eae13: add eax,1
0x100eae16: test cl,cl
0x100eae18: jne short 100EAE11h
0x100eae1a: sub eax,edx
0x100eae1c: cmp eax,4
0x100eae1f: jne near ptr 100EB006h
0x100eae25: sub ebx,ebp
0x100eae27: cmp ebx,8
0x100eae2a: jne near ptr 100EB006h
0x100eae30: mov eax,[edi+6278h]
0x100eae36: lea edx,[eax+1]
0x100eae39: lea esp,[esp]
0x100eae40: mov cl,[eax]
0x100eae42: add eax,1
0x100eae45: test cl,cl
0x100eae47: jne short 100EAE40h
0x100eae49: sub eax,edx
0x100eae4b: add eax,1
0x100eae4e: push eax
0x100eae4f: call dword ptr ds:[100FE550h]
0x100eae55: mov ebx,eax
0x100eae57: push 1012EAD8h
0x100eae5c: push ebx
0x100eae5d: push edi
0x100eae5e: mov [esp+28h],ebx
0x100eae62: call 100D2E20h
0x100eae67: mov ecx,[edi+6278h]
0x100eae6d: add esp,10h
0x100eae70: mov edx,ebx
0x100eae72: mov al,[ecx]
0x100eae74: mov [edx],al
0x100eae76: add ecx,1
0x100eae79: add edx,1
0x100eae7c: test al,al
0x100eae7e: jne short 100EAE72h
0x100eae80: mov eax,[esp+10h]
0x100eae84: sub ebx,[edi+6278h]
0x100eae8a: push 1012EAD0h
0x100eae8f: push eax
0x100eae90: lea esi,[ebx+eax]
0x100eae93: call dword ptr ds:[100FE440h]
0x100eae99: add esp,8
0x100eae9c: test eax,eax
0x100eae9e: je short 100EAEEFh
0x100eaea0: mov eax,[esp+10h]
```

```
0x100eaea4: movsx ecx,byte ptr [eax+1]
0x100eaea8: push ecx
0x100eaea9: call dword ptr ds:[100FE41Ch]
0x100eaeaf: add esp,4
0x100eaeb2: test eax,eax
0x100eaeb4: mov ecx,1012EAC8h
0x100eaeb9: jne short 100EAEC0h
0x100eaebb: mov ecx,1012EAD0h
0x100eaec0: mov al,[ecx]
0x100eaec2: mov [esi],al
0x100eaec4: add ecx,1
0x100eaec7: add esi,1
0x100eaeca: test al,al
0x100eaecc: jne short 100EAEC0h
0x100eaece: movsx edx,byte ptr [ebp]
0x100eaed2: push edx
0x100eaed3: call dword ptr ds:[100FE45Ch]
0x100eaed9: add esp,4
0x100eaedc: test eax,eax
0x100eaede: je short 100EAF24h
0x100eaee0: mov eax,[ebp+4]
0x100eaee3: mov [ebx+ebp],eax
0x100eaee6: mov ecx,[ebp]
0x100eaee9: mov [ebx+ebp+4],ecx
0x100eaeed: jmp short 100EAF24h
0x100eaeef: movsx edx,byte ptr [esi-1]
0x100eaef3: mov ebx,ds:[100FE45Ch]
0x100eaef9: sub esi,1
0x100eaefc: push edx
0x100eaefd: call ebx
0x100eaeff: add esp,4
0x100eaf02: test eax,eax
0x100eaf04: je short 100EAF24h
0x100eaf06: cmp byte ptr [esi],39h
0x100eaf09: jne short 100EAF21h
0x100eaf0b: mov byte ptr [esi],30h
0x100eaf0e: movsx eax,byte ptr [esi-1]
0x100eaf12: sub esi,1
0x100eaf15: push eax
0x100eaf16: call ebx
0x100eaf18: add esp,4
0x100eaf1b: test eax,eax
0x100eaf1d: jne short 100EAF06h
0x100eaf1f: jmp short 100EAF24h
0x100eaf21: add byte ptr [esi],1
0x100eaf24: mov esi,[esp+18h]
0x100eaf28: mov ecx,[edi+6278h]
0x100eaf2e: mov eax,esi
0x100eaf30: mov dl,[eax]
0x100eaf32: cmp dl,[ecx]
0x100eaf34: jne short 100EAF50h
0x100eaf36: test dl,dl
0x100eaf38: je short 100EAF4Ch
0x100eaf3a: mov dl,[eax+1]
0x100eaf3d: cmp dl,[ecx+1]
0x100eaf40: jne short 100EAF50h
0x100eaf42: add eax,2
0x100eaf45: add ecx,2
0x100eaf48: test dl,dl
0x100eaf4a: jne short 100EAF30h
0x100eaf4c: xor eax,eax
0x100eaf4e: jmp short 100EAF55h
0x100eaf50: sbb eax,eax
0x100eaf52: sbb eax,0FFFFFFFFh
0x100eaf55: test eax,eax
0x100eaf57: mov ebp,ds:[100FE4C8h]
0x100eaf5d: je short 100EAFD9h
```

```
0x100eaf5f: mov ecx,[esp+18h]
0x100eaf63: mov esi,[edi]
0x100eaf65: mov ebx,[edi+4]
0x100eaf68: push 100FE928h
0x100eaf6d: push ecx
0x100eaf6e: mov dword ptr [edi],10138928h
0x100eaf74: call dword ptr ds:[100FE430h]
0x100eaf7a: add esp,8
0x100eaf7d: test eax,eax
0x100eaf7f: mov [edi+4],eax
0x100eaf82: je short 100EAFD0h
0x100eaf84: cmp dword ptr [edi+0BCh],0
0x100eaf8b: je short 100EAFA6h
0x100eaf8d: mov edx,[esp+18h]
0x100eaf91: push edx
0x100eaf92: push 1012EAA8h
0x100eaf97: call dword ptr ds:[100FE4C4h]
0x100eaf9d: add eax,40h
0x100eafa0: push eax
0x100eafa1: call ebp
0x100eafa3: add esp,0Ch
0x100eafa6: push 0Ch
0x100eafa8: push edi
0x100eafa9: call 100EA5A0h
0x100eafae: mov eax,[edi]
0x100eafb0: mov ecx,[edi+4]
0x100eafb3: mov dword ptr [edi+65C4h],0
0x100eafbd: mov dword ptr [edi+6614h],1
0x100eafc7: mov edx,[eax+0Ch]
0x100eafca: push ecx
0x100eafcb: call edx
0x100eafcd: add esp,0Ch
0x100eafd0: mov [edi],esi
0x100eafd2: mov esi,[esp+18h]
0x100eafd6: mov [edi+4],ebx
0x100eafd9: mov eax,[edi+65A0h]
0x100eafdf: or eax,[edi+65A4h]
0x100eafe5: jne short 100EAFFCh
0x100eafe7: push esi
0x100eafe8: push 1012EA84h
0x100eafed: call dword ptr ds:[100FE4C4h]
0x100eaff3: add eax,40h
0x100eaff6: push eax
0x100eaff7: call ebp
0x100eaff9: add esp,0Ch
0x100eaffc: push esi
0x100eaffd: call dword ptr ds:[100FE554h]
0x100eb003: add esp,4
0x100eb006: pop edi
0x100eb007: pop esi
0x100eb008: pop ebp
0x100eb009: pop ebx
0x100eb00a: pop ecx
0x100eb00b: ret
```

`sub_100eadb0` is called by sub_100eb620. It works with the text "Failed to read metadata from %s", "Reading metadata from %s ...". It delegates to sub_100d2e20, sub_100ea5a0 (2 callees).

12.1317 sub_100eb010 at 0x100eb010 (122 instructions)

```
0x100eb010: mov ecx,[esp+8]
0x100eb014: push esi
0x100eb015: mov esi,[esp+8]
0x100eb019: mov eax,[esi]
0x100eb01b: mov edx,[esi+4]
0x100eb01e: mov eax,[eax+8]
```

```
0x100eb021: push 0
0x100eb023: push ecx
0x100eb024: push edx
0x100eb025: call eax
0x100eb027: mov ecx,[esi]
0x100eb029: mov edx,[esi+4]
0x100eb02c: mov eax,[ecx+1Ch]
0x100eb02f: push edx
0x100eb030: call eax
0x100eb032: add esp,10h
0x100eb035: cmp eax,0FFh
0x100eb03a: jne near ptr 100EB161h
0x100eb040: mov ecx,[esi]
0x100eb042: mov edx,[esi+4]
0x100eb045: mov eax,[ecx+1Ch]
0x100eb048: push edx
0x100eb049: call eax
0x100eb04b: add esp,4
0x100eb04e: cmp eax,0D8h
0x100eb053: jne near ptr 100EB161h
0x100eb059: mov ecx,[esi]
0x100eb05b: mov edx,[esi+4]
0x100eb05e: mov eax,[ecx+1Ch]
0x100eb061: push edx
0x100eb062: call eax
0x100eb064: add esp,4
0x100eb067: cmp eax,0FFh
0x100eb06c: jne near ptr 100EB15Ah
0x100eb072: push ebx
0x100eb073: push ebp
0x100eb074: push edi
0x100eb075: mov ecx,[esi]
0x100eb077: mov edx,[esi+4]
0x100eb07a: mov eax,[ecx+1Ch]
0x100eb07d: push edx
0x100eb07e: call eax
0x100eb080: mov ebp,eax
0x100eb082: add esp,4
0x100eb085: cmp ebp,0DAh
0x100eb08b: je near ptr 100EB157h
0x100eb091: push esi
0x100eb092: mov word ptr [esi+6274h],4D4Dh
0x100eb09b: call 100D2F50h
0x100eb0a0: mov ecx,[esi]
0x100eb0a2: mov edx,[esi+4]
0x100eb0a5: movzx edi,ax
0x100eb0a8: mov eax,[ecx+18h]
0x100eb0ab: push edx
0x100eb0ac: sub edi,2
0x100eb0af: call eax
0x100eb0b1: add esp,8
0x100eb0b4: cmp ebp,0C0h
0x100eb0ba: mov ebx,eax
0x100eb0bc: je short 100EB0C6h
0x100eb0be: cmp ebp,0C3h
0x100eb0c4: jne short 100EB0EEh
0x100eb0c6: mov ecx,[esi]
0x100eb0c8: mov edx,[esi+4]
0x100eb0cb: mov eax,[ecx+1Ch]
0x100eb0ce: push edx
0x100eb0cf: call eax
0x100eb0d1: push esi
0x100eb0d2: call 100D2F50h
0x100eb0d7: push esi
0x100eb0d8: mov [esi+66B0h],ax
0x100eb0df: call 100D2F50h
0x100eb0e4: add esp,0Ch
```

```
0x100eb0e7: mov [esi+66B2h],ax
0x100eb0ee: push esi
0x100eb0ef: call 100D2F50h
0x100eb0f4: push esi
0x100eb0f5: mov [esi+6274h],ax
0x100eb0fc: call 100D2FE0h
0x100eb101: push esi
0x100eb102: mov ebp,eax
0x100eb104: call 100D2FE0h
0x100eb109: add esp,0Ch
0x100eb10c: cmp eax,48454150h
0x100eb111: jne short 100EB124h
0x100eb113: mov ecx,edi
0x100eb115: sub ecx,ebp
0x100eb117: push ecx
0x100eb118: add ebp,ebx
0x100eb11a: push ebp
0x100eb11b: push esi
0x100eb11c: call 100E4A30h
0x100eb121: add esp,0Ch
0x100eb124: lea edx,[ebx+6]
0x100eb127: push edx
0x100eb128: push esi
0x100eb129: call 100EA5A0h
0x100eb12e: mov eax,[esi]
0x100eb130: mov ecx,[esi+4]
0x100eb133: mov edx,[eax+8]
0x100eb136: push 0
0x100eb138: add ebx,edi
0x100eb13a: push ebx
0x100eb13b: push ecx
0x100eb13c: call edx
0x100eb13e: mov eax,[esi]
0x100eb140: mov ecx,[esi+4]
0x100eb143: mov edx,[eax+1Ch]
0x100eb146: push ecx
0x100eb147: call edx
0x100eb149: add esp,18h
0x100eb14c: cmp eax,0FFh
0x100eb151: je near ptr 100EB075h
0x100eb157: pop edi
0x100eb158: pop ebp
0x100eb159: pop ebx
0x100eb15a: mov eax,1
0x100eb15f: pop esi
0x100eb160: ret
0x100eb161: xor eax,eax
0x100eb163: pop esi
0x100eb164: ret
```

`sub_100eb010` is called by sub_100eb620. It accesses memory-mapped peripheral registers at 0x48454150. It delegates to sub_100d2f50, sub_100d2fe0, sub_100e4a30 (4 callees).

12.1318 sub_100eb170 at 0x100eb170 (346 instructions)

```
0x100eb170: sub esp,898h
0x100eb176: mov eax,ds:[10136220h]
0x100eb17b: xor eax,esp
0x100eb17d: mov [esp+894h],eax
0x100eb184: push ebx
0x100eb185: mov ebx,[esp+8A0h]
0x100eb18c: mov eax,[ebx]
0x100eb18e: mov ecx,[ebx+4]
0x100eb191: push esi
0x100eb192: push 0
0x100eb194: mov word ptr [ebx+6274h],4949h
```

```
0x100eb19d: mov edx,[eax+8]
0x100eb1a0: push 24h
0x100eb1a2: push ecx
0x100eb1a3: call edx
0x100eb1a5: push ebx
0x100eb1a6: call 100D2FE0h
0x100eb1ab: mov ecx,[ebx+4]
0x100eb1ae: mov [ebx+66C8h],eax
0x100eb1b4: mov eax,[ebx]
0x100eb1b6: mov edx,[eax+8]
0x100eb1b9: push 2
0x100eb1bb: push 0FFFFFFFCh
0x100eb1bd: push ecx
0x100eb1be: call edx
0x100eb1c0: mov esi,[ebx]
0x100eb1c2: add esp,1Ch
0x100eb1c5: push 0
0x100eb1c7: push ebx
0x100eb1c8: add esi,8
0x100eb1cb: call 100D2FE0h
0x100eb1d0: mov ecx,[esi]
0x100eb1d2: add esp,4
0x100eb1d5: push eax
0x100eb1d6: mov eax,[ebx+4]
0x100eb1d9: push eax
0x100eb1da: call ecx
0x100eb1dc: push ebx
0x100eb1dd: call 100D2FE0h
0x100eb1e2: add esp,10h
0x100eb1e5: cmp eax,64434553h
0x100eb1ea: jne near ptr 100EB608h
0x100eb1f0: push ebp
0x100eb1f1: push edi
0x100eb1f2: push ebx
0x100eb1f3: call 100D2FE0h
0x100eb1f8: push ebx
0x100eb1f9: call 100D2FE0h
0x100eb1fe: add esp,8
0x100eb201: test eax,eax
0x100eb203: mov [esp+18h],eax
0x100eb207: je near ptr 100EB5FCh
0x100eb20d: lea ecx,[ecx]
0x100eb210: sub dword ptr [esp+18h],1
0x100eb215: push ebx
0x100eb216: call 100D2FE0h
0x100eb21b: push ebx
0x100eb21c: mov esi,eax
0x100eb21e: call 100D2FE0h
0x100eb223: push ebx
0x100eb224: mov ebp,eax
0x100eb226: call 100D2FE0h
0x100eb22b: mov edx,[ebx]
0x100eb22d: mov ecx,[edx+18h]
0x100eb230: mov edi,eax
0x100eb232: mov eax,[ebx+4]
0x100eb235: push eax
0x100eb236: call ecx
0x100eb238: mov edx,[ebx]
0x100eb23a: mov ecx,[edx+8]
0x100eb23d: push 0
0x100eb23f: mov [esp+34h],eax
0x100eb243: mov eax,[ebx+4]
0x100eb246: push esi
0x100eb247: push eax
0x100eb248: call ecx
0x100eb24a: push ebx
0x100eb24b: call 100D2FE0h
```

```
0x100eb250: mov edx,edi
0x100eb252: shl edx,18h
0x100eb255: or edx,20434553h
0x100eb25b: add esp,20h
0x100eb25e: cmp eax,edx
0x100eb260: jne near ptr 100EB606h
0x100eb266: cmp edi,47414D49h
0x100eb26c: jg near ptr 100EB371h
0x100eb272: je short 100EB2B4h
0x100eb274: cmp edi,32414D49h
0x100eb27a: je short 100EB2B4h
0x100eb27c: cmp edi,464D4143h
0x100eb282: jne near ptr 100EB5DCh
0x100eb288: lea eax,[ebp-1Ch]
0x100eb28b: add esi,18h
0x100eb28e: cmp eax,20000h
0x100eb293: mov [ebx+65C8h],esi
0x100eb299: mov [ebx+65D4h],eax
0x100eb29f: jbe near ptr 100EB5DCh
0x100eb2a5: mov dword ptr [ebx+65D4h],20000h
0x100eb2af: jmp near ptr 100EB5DCh
0x100eb2b4: mov eax,[ebx]
0x100eb2b6: mov ecx,[ebx+4]
0x100eb2b9: mov edx,[eax+8]
0x100eb2bc: push 1
0x100eb2be: push 0Ch
0x100eb2c0: push ecx
0x100eb2c1: call edx
0x100eb2c3: push ebx
0x100eb2c4: call 100D2FE0h
0x100eb2c9: push ebx
0x100eb2ca: mov edi,eax
0x100eb2cc: call 100D2FE0h
0x100eb2d1: movzx ecx,word ptr [ebx+66B2h]
0x100eb2d8: add esp,14h
0x100eb2db: cmp edi,ecx
0x100eb2dd: jle short 100EB301h
0x100eb2df: movzx edx,word ptr [ebx+66B0h]
0x100eb2e6: cmp eax,edx
0x100eb2e8: jle short 100EB301h
0x100eb2ea: mov [ebx+66B0h],ax
0x100eb2f1: lea eax,[esi+18h]
0x100eb2f4: mov [ebx+66B2h],di
0x100eb2fb: mov [ebx+65C0h],eax
0x100eb301: mov ecx,[ebx]
0x100eb303: mov edx,[ebx+4]
0x100eb306: mov eax,[ecx+8]
0x100eb309: push 0
0x100eb30b: add esi,1Ch
0x100eb30e: push esi
0x100eb30f: push edx
0x100eb310: call eax
0x100eb312: mov ecx,[ebx]
0x100eb314: mov edx,[ebx+4]
0x100eb317: mov eax,[ecx+1Ch]
0x100eb31a: push edx
0x100eb31b: call eax
0x100eb31d: add esp,10h
0x100eb320: cmp eax,0FFh
0x100eb325: jne near ptr 100EB5DCh
0x100eb32b: mov ecx,[ebx]
0x100eb32d: mov edx,[ebx+4]
0x100eb330: mov eax,[ecx+1Ch]
0x100eb333: push edx
0x100eb334: call eax
0x100eb336: add esp,4
0x100eb339: cmp eax,0D8h
```

```
0x100eb33e: jne near ptr 100EB5DCh
0x100eb344: lea ecx,[ebp-1Ch]
0x100eb347: cmp [ebx+65D0h],ecx
0x100eb34d: jge near ptr 100EB5DCh
0x100eb353: add ebp,0FFFFFFE4h
0x100eb356: mov [ebx+65C4h],esi
0x100eb35c: mov [ebx+65D0h],ebp
0x100eb362: mov dword ptr [ebx+2E7F4h],100E7F10h
0x100eb36c: jmp near ptr 100EB5DCh
0x100eb371: cmp edi,504F5250h
0x100eb377: jne near ptr 100EB5DCh
0x100eb37d: push ebx
0x100eb37e: call 100D2FE0h
0x100eb383: push ebx
0x100eb384: call 100D2FE0h
0x100eb389: mov edx,[ebx]
0x100eb38b: mov ecx,[edx+8]
0x100eb38e: mov edi,eax
0x100eb390: mov eax,[ebx+4]
0x100eb393: push 1
0x100eb395: push 0Ch
0x100eb397: push eax
0x100eb398: mov [esp+24h],edi
0x100eb39c: call ecx
0x100eb39e: add esp,14h
0x100eb3a1: cmp edi,100h
0x100eb3a7: lea ebp,[esi+edi*8+18h]
0x100eb3ab: jbe short 100EB3B9h
0x100eb3ad: mov dword ptr [esp+10h],100h
0x100eb3b5: mov edi,[esp+10h]
0x100eb3b9: add edi,edi
0x100eb3bb: xor esi,esi
0x100eb3bd: test edi,edi
0x100eb3bf: jle short 100EB3D9h
0x100eb3c1: push ebx
0x100eb3c2: call 100D2FE0h
0x100eb3c7: lea edx,[ebp+eax*2]
0x100eb3cb: mov [esp+esi*4+28h],edx
0x100eb3cf: add esi,1
0x100eb3d2: add esp,4
0x100eb3d5: cmp esi,edi
0x100eb3d7: jl short 100EB3C1h
0x100eb3d9: xor esi,esi
0x100eb3db: cmp [esp+10h],esi
0x100eb3df: mov [esp+1Ch],esi
0x100eb3e3: jle near ptr 100EB5DCh
0x100eb3e9: jmp short 100EB3F4h
0x100eb3f0: mov esi,[esp+1Ch]
0x100eb3f4: mov ecx,[esp+esi*8+24h]
0x100eb3f8: push 40h
0x100eb3fa: lea eax,[esp+868h]
0x100eb401: push eax
0x100eb402: push ecx
0x100eb403: push ebx
0x100eb404: call 100E6460h
0x100eb409: mov eax,[esp+esi*8+38h]
0x100eb40d: push 40h
0x100eb40f: lea edx,[esp+838h]
0x100eb416: push edx
0x100eb417: push eax
0x100eb418: push ebx
0x100eb419: call 100E6460h
0x100eb41e: mov ebp,ds:[100FE458h]
0x100eb424: add esp,20h
0x100eb427: mov edi,1012EB34h
0x100eb42c: lea esi,[esp+864h]
0x100eb433: mov ecx,4
```

```
0x100eb438: xor edx,edx
0x100eb43a: repe cmpsb
0x100eb43c: jne short 100EB459h
0x100eb43e: lea eax,[esp+824h]
0x100eb445: push eax
0x100eb446: call ebp
0x100eb448: mov [esp+18h],eax
0x100eb44c: fild dword ptr [esp+18h]
0x100eb450: add esp,4
0x100eb453: fstp dword ptr [ebx+6590h]
0x100eb459: mov edi,1012EB28h
0x100eb45e: lea esi,[esp+864h]
0x100eb465: mov ecx,9
0x100eb46a: xor edx,edx
0x100eb46c: repe cmpsb
0x100eb46e: jne short 100EB48Fh
0x100eb470: lea eax,[esp+824h]
0x100eb477: mov edx,eax
0x100eb479: mov ecx,ebx
0x100eb47b: sub ecx,edx
0x100eb47d: lea edx,[ecx+6485h]
0x100eb483: mov cl,[eax]
0x100eb485: mov [edx+eax],cl
0x100eb488: add eax,1
0x100eb48b: test cl,cl
0x100eb48d: jne short 100EB483h
0x100eb48f: mov edi,1012EB1Ch
0x100eb494: lea esi,[esp+864h]
0x100eb49b: mov ecx,9
0x100eb4a0: xor eax,eax
0x100eb4a2: repe cmpsb
0x100eb4a4: jne short 100EB4CCh
0x100eb4a6: lea eax,[esp+824h]
0x100eb4ad: mov edx,eax
0x100eb4af: mov ecx,ebx
0x100eb4b1: sub ecx,edx
0x100eb4b3: lea edx,[ecx+64C5h]
0x100eb4b9: lea esp,[esp]
0x100eb4c0: mov cl,[eax]
0x100eb4c2: mov [edx+eax],cl
0x100eb4c5: add eax,1
0x100eb4c8: test cl,cl
0x100eb4ca: jne short 100EB4C0h
0x100eb4cc: mov edi,1012EB14h
0x100eb4d1: lea esi,[esp+864h]
0x100eb4d8: mov ecx,8
0x100eb4dd: xor eax,eax
0x100eb4df: repe cmpsb
0x100eb4e1: jne short 100EB502h
0x100eb4e3: lea eax,[esp+824h]
0x100eb4ea: mov edx,eax
0x100eb4ec: mov ecx,ebx
0x100eb4ee: sub ecx,edx
0x100eb4f0: lea edx,[ecx+6505h]
0x100eb4f6: mov cl,[eax]
0x100eb4f8: mov [edx+eax],cl
0x100eb4fb: add eax,1
0x100eb4fe: test cl,cl
0x100eb500: jne short 100EB4F6h
0x100eb502: mov edi,1012EB0Ch
0x100eb507: lea esi,[esp+864h]
0x100eb50e: mov ecx,5
0x100eb513: xor eax,eax
0x100eb515: repe cmpsb
0x100eb517: jne short 100EB533h
0x100eb519: lea ecx,[esp+824h]
0x100eb520: push ecx
```

```
0x100eb521: call ebp
0x100eb523: cdq
0x100eb524: add esp,4
0x100eb527: mov [ebx+65A0h],eax
0x100eb52d: mov [ebx+65A4h],edx
0x100eb533: mov edi,1012EB04h
0x100eb538: lea esi,[esp+864h]
0x100eb53f: mov ecx,8
0x100eb544: xor edx,edx
0x100eb546: repe cmpsb
0x100eb548: jne short 100EB56Bh
0x100eb54a: lea eax,[esp+824h]
0x100eb551: push eax
0x100eb552: call ebp
0x100eb554: mov [esp+18h],eax
0x100eb558: fild dword ptr [esp+18h]
0x100eb55c: add esp,4
0x100eb55f: fdiv qword ptr ds:[1012E720h]
0x100eb565: fstp dword ptr [ebx+6594h]
0x100eb56b: mov edi,1012EAF8h
0x100eb570: lea esi,[esp+864h]
0x100eb577: mov ecx,9
0x100eb57c: xor edx,edx
0x100eb57e: repe cmpsb
0x100eb580: jne short 100EB599h
0x100eb582: lea eax,[esp+824h]
0x100eb589: push eax
0x100eb58a: call dword ptr ds:[100FE4B4h]
0x100eb590: fstp dword ptr [ebx+6598h]
0x100eb596: add esp,4
0x100eb599: mov edi,1012EAF0h
0x100eb59e: lea esi,[esp+864h]
0x100eb5a5: mov ecx,8
0x100eb5aa: xor edx,edx
0x100eb5ac: repe cmpsb
0x100eb5ae: jne short 100EB5C7h
0x100eb5b0: lea eax,[esp+824h]
0x100eb5b7: push eax
0x100eb5b8: call dword ptr ds:[100FE4B4h]
0x100eb5be: fstp dword ptr [ebx+659Ch]
0x100eb5c4: add esp,4
0x100eb5c7: mov eax,[esp+1Ch]
0x100eb5cb: add eax,1
0x100eb5ce: cmp eax,[esp+10h]
0x100eb5d2: mov [esp+1Ch],eax
0x100eb5d6: jl near ptr 100EB3F0h
0x100eb5dc: mov edx,[esp+20h]
0x100eb5e0: mov ecx,[ebx]
0x100eb5e2: mov eax,[ebx+4]
0x100eb5e5: mov ecx,[ecx+8]
0x100eb5e8: push 0
0x100eb5ea: push edx
0x100eb5eb: push eax
0x100eb5ec: call ecx
0x100eb5ee: add esp,0Ch
0x100eb5f1: cmp dword ptr [esp+18h],0
0x100eb5f6: jne near ptr 100EB210h
0x100eb5fc: mov dword ptr [ebx+661Ch],1
0x100eb606: pop edi
0x100eb607: pop ebp
0x100eb608: mov ecx,[esp+89Ch]
0x100eb60f: pop esi
0x100eb610: pop ebx
0x100eb611: xor ecx,esp
0x100eb613: call 10022326h
0x100eb618: add esp,898h
0x100eb61e: ret
```

`sub_100eb170` is called by sub_100eb620. It works with the text "APERTURE", "EXPTIME", "CAMMODEL". It accesses memory-mapped peripheral registers at 0x464d4143. It delegates to sub_10022326, sub_100d2fe0, sub_100e6460 (3 callees).

12.1319 sub_100eb620 at 0x100eb620 (4010 instructions)

```
0x100eb620: sub esp,7Ch
0x100eb623: mov eax,ds:[10136220h]
0x100eb628: xor eax,esp
0x100eb62a: mov [esp+78h],eax
0x100eb62e: fldz
0x100eb630: push ebx
0x100eb631: xor ebx,ebx
0x100eb633: push ebp
0x100eb634: mov ebp,[esp+88h]
0x100eb63b: or eax,0FFFFFFFFh
0x100eb63e: fst dword ptr [ebp+659Ch]
0x100eb644: push 80h
0x100eb649: fst dword ptr [ebp+6598h]
0x100eb64f: mov [ebp+65B0h],eax
0x100eb655: fst dword ptr [ebp+6594h]
0x100eb65b: mov [ebp+66C8h],eax
0x100eb661: fstp dword ptr [ebp+6590h]
0x100eb667: mov [ebp+66CCh],eax
0x100eb66d: lea eax,[ebp+662Ch]
0x100eb673: push ebx
0x100eb674: push eax
0x100eb675: mov [ebp+0E766h],bx
0x100eb67c: mov [ebp+65E4h],ebx
0x100eb682: mov [ebp+66C2h],bx
0x100eb689: mov [ebp+66B2h],bx
0x100eb690: mov [ebp+66B0h],bx
0x100eb697: mov [ebp+66BAh],bx
0x100eb69e: mov [ebp+66B8h],bx
0x100eb6a5: mov [ebp+66B6h],bx
0x100eb6ac: mov [ebp+66B4h],bx
0x100eb6b3: mov [ebp+65FCh],ebx
0x100eb6b9: mov [ebp+6505h],bl
0x100eb6bf: mov [ebp+64C5h],bl
0x100eb6c5: mov [ebp+6485h],bl
0x100eb6cb: mov [ebp+6545h],bl
0x100eb6d1: mov [ebp+6285h],bl
0x100eb6d7: mov [ebp+6280h],bl
0x100eb6dd: mov [ebp+65B8h],ebx
0x100eb6e3: call 100228C0h
0x100eb6e8: push 80h
0x100eb6ed: lea ecx,[ebp+66E4h]
0x100eb6f3: push ebx
0x100eb6f4: push ecx
0x100eb6f5: call 100228C0h
0x100eb6fa: fld1
0x100eb6fc: mov eax,1
0x100eb701: fstp qword ptr [ebp+66D8h]
0x100eb707: fldz
0x100eb709: add esp,18h
0x100eb70c: mov [ebp+6608h],eax
0x100eb712: mov [ebp+6604h],eax
0x100eb718: mov [ebp+6614h],eax
0x100eb71e: mov eax,7FFFFFFFh
0x100eb723: push esi
0x100eb724: mov [ebp+6628h],eax
0x100eb72a: mov [ebp+6624h],eax
0x100eb730: mov [ebp+66C6h],bx
0x100eb737: mov [ebp+66C4h],bx
0x100eb73e: mov [ebp+65D0h],ebx
0x100eb744: mov [ebp+65C4h],ebx
0x100eb74a: mov [ebp+2E800h],ebx
```

```
0x100eb750: mov [ebp+2E7FCh],ebx
0x100eb756: mov dword ptr [ebp+2E7F4h],100E7F10h
0x100eb760: mov [ebp+65F4h],ebx
0x100eb766: mov [ebp+65F0h],ebx
0x100eb76c: mov [ebp+65D4h],ebx
0x100eb772: mov [ebp+65C0h],ebx
0x100eb778: mov [ebp+66ACh],ebx
0x100eb77e: mov [ebp+6618h],ebx
0x100eb784: mov [ebp+6610h],ebx
0x100eb78a: mov [ebp+65ACh],ebx
0x100eb790: mov [ebp+661Ch],ebx
0x100eb796: mov [ebp+65F8h],ebx
0x100eb79c: mov [ebp+65ECh],ebx
0x100eb7a2: mov [ebp+65A8h],ebx
0x100eb7a8: mov [ebp+65A0h],ebx
0x100eb7ae: mov [ebp+65A4h],ebx
0x100eb7b4: mov [ebp+660Ch],ebx
0x100eb7ba: mov [ebp+6620h],ebx
0x100eb7c0: mov [ebp+65D8h],ebx
0x100eb7c6: mov [ebp+6600h],ebx
0x100eb7cc: xor eax,eax
0x100eb7ce: lea ecx,[ebp+0E784h]
0x100eb7d4: push edi
0x100eb7d5: xor edx,edx
0x100eb7d7: cmp eax,1
0x100eb7da: sete dl
0x100eb7dd: cmp eax,3
0x100eb7e0: mov esi,3
0x100eb7e5: mov [esp+10h],edx
0x100eb7e9: fild dword ptr [esp+10h]
0x100eb7ed: sbb edx,edx
0x100eb7ef: neg edx
0x100eb7f1: mov [esp+10h],edx
0x100eb7f5: fstp dword ptr [ecx-10h]
0x100eb7f8: lea edx,[ecx+10h]
0x100eb7fb: fild dword ptr [esp+10h]
0x100eb7ff: fstp dword ptr [ecx]
0x100eb801: fst dword ptr [edx]
0x100eb803: add edx,10h
0x100eb806: sub esi,1
0x100eb809: jne short 100EB801h
0x100eb80b: xor edx,edx
0x100eb80d: cmp eax,ebx
0x100eb80f: sete dl
0x100eb812: mov [esp+10h],edx
0x100eb816: fild dword ptr [esp+10h]
0x100eb81a: xor edx,edx
0x100eb81c: cmp eax,1
0x100eb81f: sete dl
0x100eb822: fstp dword ptr [ecx+40h]
0x100eb825: mov [esp+10h],edx
0x100eb829: fild dword ptr [esp+10h]
0x100eb82d: xor edx,edx
0x100eb82f: cmp eax,2
0x100eb832: sete dl
0x100eb835: fstp dword ptr [ecx+50h]
0x100eb838: add eax,1
0x100eb83b: add ecx,4
0x100eb83e: cmp eax,4
0x100eb841: mov [esp+10h],edx
0x100eb845: fild dword ptr [esp+10h]
0x100eb849: fstp dword ptr [ecx+5Ch]
0x100eb84c: jb short 100EB7D5h
0x100eb84e: fstp st(0)
0x100eb850: mov dword ptr [ebp+66D0h],3
0x100eb85a: mov dword ptr [ebp+65F0h],0Ch
0x100eb864: xor eax,eax
```

```
0x100eb866: lea ecx,[ebp+6764h]
0x100eb86c: lea esp,[esp]
0x100eb870: mov [ecx],ax
0x100eb873: add eax,1
0x100eb876: add ecx,2
0x100eb879: cmp eax,4000h
0x100eb87e: jb short 100EB870h
0x100eb880: push ebp
0x100eb881: call 100D2F50h
0x100eb886: push ebp
0x100eb887: mov [ebp+6274h],ax
0x100eb88e: call 100D2FE0h
0x100eb893: mov ecx,[ebp+4]
0x100eb896: push ebx
0x100eb897: mov edi,eax
0x100eb899: mov eax,[ebp]
0x100eb89c: mov edx,[eax+8]
0x100eb89f: push ebx
0x100eb8a0: push ecx
0x100eb8a1: call edx
0x100eb8a3: mov eax,[ebp]
0x100eb8a6: mov edx,[ebp+4]
0x100eb8a9: mov eax,[eax]
0x100eb8ab: push 20h
0x100eb8ad: push 1
0x100eb8af: lea ecx,[esp+84h]
0x100eb8b6: push ecx
0x100eb8b7: push edx
0x100eb8b8: call eax
0x100eb8ba: mov ecx,[ebp]
0x100eb8bd: mov edx,[ebp+4]
0x100eb8c0: mov eax,[ecx+8]
0x100eb8c3: push 2
0x100eb8c5: push ebx
0x100eb8c6: push edx
0x100eb8c7: call eax
0x100eb8c9: mov ecx,[ebp]
0x100eb8cc: mov edx,[ebp+4]
0x100eb8cf: mov eax,[ecx+18h]
0x100eb8d2: push edx
0x100eb8d3: call eax
0x100eb8d5: push 4
0x100eb8d7: push 1012E598h
0x100eb8dc: lea ecx,[esp+0A4h]
0x100eb8e3: push 20h
0x100eb8e5: push ecx
0x100eb8e6: mov [esp+58h],eax
0x100eb8ea: call 100D2D60h
0x100eb8ef: add esp,44h
0x100eb8f2: cmp eax,ebx
0x100eb8f4: jne near ptr 100EC3F5h
0x100eb8fa: push 4
0x100eb8fc: push 1012E5A0h
0x100eb901: lea edx,[esp+70h]
0x100eb905: push 20h
0x100eb907: push edx
0x100eb908: call 100D2D60h
0x100eb90d: add esp,10h
0x100eb910: cmp eax,ebx
0x100eb912: jne near ptr 100EC3F5h
0x100eb918: movzx eax,word ptr [ebp+6274h]
0x100eb91f: cmp ax,4949h
0x100eb923: je near ptr 100EC341h
0x100eb929: cmp ax,4D4Dh
0x100eb92d: je near ptr 100EC341h
0x100eb933: mov eax,4
0x100eb938: mov ecx,1012F234h
```

```
0x100eb93d: lea edx,[esp+68h]
0x100eb941: mov esi,[edx]
0x100eb943: cmp esi,[ecx]
0x100eb945: jne short 100EB959h
0x100eb947: sub eax,4
0x100eb94a: add ecx,4
0x100eb94d: add edx,4
0x100eb950: cmp eax,4
0x100eb953: jae short 100EB941h
0x100eb955: cmp eax,ebx
0x100eb957: je short 100EB9B6h
0x100eb959: movzx esi,byte ptr [edx]
0x100eb95c: movzx edi,byte ptr [ecx]
0x100eb95f: sub esi,edi
0x100eb961: jne short 100EB9A8h
0x100eb963: sub eax,1
0x100eb966: add ecx,1
0x100eb969: add edx,1
0x100eb96c: cmp eax,ebx
0x100eb96e: je short 100EB9B6h
0x100eb970: movzx esi,byte ptr [edx]
0x100eb973: movzx edi,byte ptr [ecx]
0x100eb976: sub esi,edi
0x100eb978: jne short 100EB9A8h
0x100eb97a: sub eax,1
0x100eb97d: add ecx,1
0x100eb980: add edx,1
0x100eb983: cmp eax,ebx
0x100eb985: je short 100EB9B6h
0x100eb987: movzx esi,byte ptr [edx]
0x100eb98a: movzx edi,byte ptr [ecx]
0x100eb98d: sub esi,edi
0x100eb98f: jne short 100EB9A8h
0x100eb991: sub eax,1
0x100eb994: add ecx,1
0x100eb997: add edx,1
0x100eb99a: cmp eax,ebx
0x100eb99c: je short 100EB9B6h
0x100eb99e: movzx esi,byte ptr [edx]
0x100eb9a1: movzx eax,byte ptr [ecx]
0x100eb9a4: sub esi,eax
0x100eb9a6: je short 100EB9B6h
0x100eb9a8: cmp esi,ebx
0x100eb9aa: mov eax,1
0x100eb9af: jg short 100EB9B8h
0x100eb9b1: or eax,0FFFFFFFFh
0x100eb9b4: jmp short 100EB9B8h
0x100eb9b6: xor eax,eax
0x100eb9b8: cmp eax,ebx
0x100eb9ba: jne near ptr 100EBAA6h
0x100eb9c0: mov eax,4
0x100eb9c5: mov ecx,100FF61Ch
0x100eb9ca: lea edx,[esp+6Eh]
0x100eb9ce: mov edi,edi
0x100eb9d0: mov esi,[edx]
0x100eb9d2: cmp esi,[ecx]
0x100eb9d4: jne short 100EB9E8h
0x100eb9d6: sub eax,4
0x100eb9d9: add ecx,4
0x100eb9dc: add edx,4
0x100eb9df: cmp eax,4
0x100eb9e2: jae short 100EB9D0h
0x100eb9e4: cmp eax,ebx
0x100eb9e6: je short 100EBA45h
0x100eb9e8: movzx esi,byte ptr [edx]
0x100eb9eb: movzx edi,byte ptr [ecx]
0x100eb9ee: sub esi,edi
```

```
0x100eb9f0: jne short 100EBA37h
0x100eb9f2: sub eax,1
0x100eb9f5: add ecx,1
0x100eb9f8: add edx,1
0x100eb9fb: cmp eax,ebx
0x100eb9fd: je short 100EBA45h
0x100eb9ff: movzx esi,byte ptr [edx]
0x100eba02: movzx edi,byte ptr [ecx]
0x100eba05: sub esi,edi
0x100eba07: jne short 100EBA37h
0x100eba09: sub eax,1
0x100eba0c: add ecx,1
0x100eba0f: add edx,1
0x100eba12: cmp eax,ebx
0x100eba14: je short 100EBA45h
0x100eba16: movzx esi,byte ptr [edx]
0x100eba19: movzx edi,byte ptr [ecx]
0x100eba1c: sub esi,edi
0x100eba1e: jne short 100EBA37h
0x100eba20: sub eax,1
0x100eba23: add ecx,1
0x100eba26: add edx,1
0x100eba29: cmp eax,ebx
0x100eba2b: je short 100EBA45h
0x100eba2d: movzx esi,byte ptr [edx]
0x100eba30: movzx ecx,byte ptr [ecx]
0x100eba33: sub esi,ecx
0x100eba35: je short 100EBA45h
0x100eba37: cmp esi,ebx
0x100eba39: mov eax,1
0x100eba3e: jg short 100EBA47h
0x100eba40: or eax,0FFFFFFFFh
0x100eba43: jmp short 100EBA47h
0x100eba45: xor eax,eax
0x100eba47: cmp eax,ebx
0x100eba49: jne short 100EBAA6h
0x100eba4b: mov edx,[ebp]
0x100eba4e: mov eax,[ebp+4]
0x100eba51: mov ecx,[edx+8]
0x100eba54: push ebx
0x100eba55: push 4
0x100eba57: push eax
0x100eba58: call ecx
0x100eba5a: push ebp
0x100eba5b: call 100D2F50h
0x100eba60: mov edx,[ebp]
0x100eba63: movzx eax,ax
0x100eba66: add eax,4
0x100eba69: push ebx
0x100eba6a: mov [ebp+65C0h],eax
0x100eba70: mov ecx,[edx+8]
0x100eba73: push eax
0x100eba74: mov eax,[ebp+4]
0x100eba77: push eax
0x100eba78: call ecx
0x100eba7a: mov edx,[ebp]
0x100eba7d: mov eax,[ebp+4]
0x100eba80: mov ecx,[edx+1Ch]
0x100eba83: push eax
0x100eba84: call ecx
0x100eba86: add esp,20h
0x100eba89: cmp eax,0FFh
0x100eba8e: je short 100EBA9Bh
0x100eba90: push 0Ch
0x100eba92: push ebp
0x100eba93: call 100EA5A0h
0x100eba98: add esp,8
```

```
0x100eba9b: mov [ebp+65C4h],ebx
0x100ebaa1: jmp near ptr 100EC415h
0x100ebaa6: mov eax,7
0x100ebaab: mov ecx,1012F22Ch
0x100ebab0: lea edx,[esp+81h]
0x100ebab7: mov esi,[edx]
0x100ebab9: cmp esi,[ecx]
0x100ebabb: jne short 100EBACFh
0x100ebabd: sub eax,4
0x100ebac0: add ecx,4
0x100ebac3: add edx,4
0x100ebac6: cmp eax,4
0x100ebac9: jae short 100EBAB7h
0x100ebacb: cmp eax,ebx
0x100ebacd: je short 100EBB2Ch
0x100ebacf: movzx esi,byte ptr [edx]
0x100ebad2: movzx edi,byte ptr [ecx]
0x100ebad5: sub esi,edi
0x100ebad7: jne short 100EBB1Eh
0x100ebad9: sub eax,1
0x100ebadc: add ecx,1
0x100ebadf: add edx,1
0x100ebae2: cmp eax,ebx
0x100ebae4: je short 100EBB2Ch
0x100ebae6: movzx esi,byte ptr [edx]
0x100ebae9: movzx edi,byte ptr [ecx]
0x100ebaec: sub esi,edi
0x100ebaee: jne short 100EBB1Eh
0x100ebaf0: sub eax,1
0x100ebaf3: add ecx,1
0x100ebaf6: add edx,1
0x100ebaf9: cmp eax,ebx
0x100ebafb: je short 100EBB2Ch
0x100ebafd: movzx esi,byte ptr [edx]
0x100ebb00: movzx edi,byte ptr [ecx]
0x100ebb03: sub esi,edi
0x100ebb05: jne short 100EBB1Eh
0x100ebb07: sub eax,1
0x100ebb0a: add ecx,1
0x100ebb0d: add edx,1
0x100ebb10: cmp eax,ebx
0x100ebb12: je short 100EBB2Ch
0x100ebb14: movzx esi,byte ptr [edx]
0x100ebb17: movzx edx,byte ptr [ecx]
0x100ebb1a: sub esi,edx
0x100ebb1c: je short 100EBB2Ch
0x100ebb1e: cmp esi,ebx
0x100ebb20: mov eax,1
0x100ebb25: jg short 100EBB2Eh
0x100ebb27: or eax,0FFFFFFFFh
0x100ebb2a: jmp short 100EBB2Eh
0x100ebb2c: xor eax,eax
0x100ebb2e: cmp eax,ebx
0x100ebb30: jne near ptr 100EBBE5h
0x100ebb36: mov eax,ds:[1012F224h]
0x100ebb3b: mov [ebp+6485h],eax
0x100ebb41: mov cx,ds:[1012F228h]
0x100ebb48: mov [ebp+6489h],cx
0x100ebb4f: mov dl,ds:[1012F22Ah]
0x100ebb55: mov [ebp+648Bh],dl
0x100ebb5b: mov eax,ds:[1012F218h]
0x100ebb60: mov [ebp+64C5h],eax
0x100ebb66: mov ecx,ds:[1012F21Ch]
0x100ebb6c: mov eax,[ebp]
0x100ebb6f: mov [ebp+64C9h],ecx
0x100ebb75: mov dx,ds:[1012F220h]
0x100ebb7c: mov ecx,[ebp+4]
```

```
0x100ebb7f: push ebx
0x100ebb80: mov [ebp+64CDh],dx
0x100ebb87: mov edx,[eax+8]
0x100ebb8a: push 21h
0x100ebb8c: push ecx
0x100ebb8d: call edx
0x100ebb8f: push 1
0x100ebb91: push ebp
0x100ebb92: call 100E3D60h
0x100ebb97: mov eax,[ebp]
0x100ebb9a: mov ecx,[ebp+4]
0x100ebb9d: mov edx,[eax+8]
0x100ebba0: push ebx
0x100ebba1: push 3Ch
0x100ebba3: push ecx
0x100ebba4: call edx
0x100ebba6: add esp,20h
0x100ebba9: xor esi,esi
0x100ebbab: jmp short 100EBBB0h
0x100ebbb0: push ebp
0x100ebbb1: call 100D2FE0h
0x100ebbb6: add esp,4
0x100ebbb9: test eax,eax
0x100ebbbb: mov [esp+10h],eax
0x100ebbbf: fild dword ptr [esp+10h]
0x100ebbc3: jge short 100EBBCBh
0x100ebbc5: fadd dword ptr ds:[100FE7C0h]
0x100ebbcb: mov eax,esi
0x100ebbcd: shr eax,1
0x100ebbcf: xor eax,esi
0x100ebbd1: add esi,1
0x100ebbd4: fstp dword ptr [ebp+eax*4+0E774h]
0x100ebbdb: cmp esi,4
0x100ebbde: jb short 100EBBB0h
0x100ebbe0: jmp near ptr 100EC415h
0x100ebbe5: mov edi,1012F214h
0x100ebbea: lea esi,[esp+68h]
0x100ebbee: mov ecx,4
0x100ebbf3: xor edx,edx
0x100ebbf5: repe cmpsb
0x100ebbf7: jne short 100EBC4Fh
0x100ebbf9: mov eax,ds:[1012F208h]
0x100ebbfe: mov [ebp+6485h],eax
0x100ebc04: mov ecx,ds:[1012F20Ch]
0x100ebc0a: mov [ebp+6489h],ecx
0x100ebc10: mov dl,ds:[1012F210h]
0x100ebc16: mov [ebp+648Dh],dl
0x100ebc1c: mov eax,ds:[1012F1F8h]
0x100ebc21: mov [ebp+64C5h],eax
0x100ebc27: mov ecx,ds:[1012F1FCh]
0x100ebc2d: mov [ebp+64C9h],ecx
0x100ebc33: mov edx,ds:[1012F200h]
0x100ebc39: mov [ebp+64CDh],edx
0x100ebc3f: mov eax,ds:[1012F204h]
0x100ebc44: mov [ebp+64D1h],eax
0x100ebc4a: jmp near ptr 100EC415h
0x100ebc4f: mov edi,1012F1F0h
0x100ebc54: lea esi,[esp+68h]
0x100ebc58: mov ecx,5
0x100ebc5d: xor edx,edx
0x100ebc5f: repe cmpsb
0x100ebc61: jne short 100EBCB2h
0x100ebc63: mov eax,ds:[1012F1E8h]
0x100ebc68: mov [ebp+6485h],eax
0x100ebc6e: mov cx,ds:[1012F1ECh]
0x100ebc75: mov [ebp+6489h],cx
0x100ebc7c: mov edx,ds:[1012F1D8h]
```

```
0x100ebc82: mov [ebp+64C5h],edx
0x100ebc88: mov eax,ds:[1012F1DCh]
0x100ebc8d: mov [ebp+64C9h],eax
0x100ebc93: mov ecx,ds:[1012F1E0h]
0x100ebc99: mov [ebp+64CDh],ecx
0x100ebc9f: mov dx,ds:[1012F1E4h]
0x100ebca6: mov [ebp+64D1h],dx
0x100ebcad: jmp near ptr 100EC415h
0x100ebcb2: mov edi,1012F1D0h
0x100ebcb7: lea esi,[esp+68h]
0x100ebcbb: mov ecx,5
0x100ebcc0: xor eax,eax
0x100ebcc2: repe cmpsb
0x100ebcc4: jne short 100EBD15h
0x100ebcc6: mov ecx,ds:[1012F1E8h]
0x100ebccc: mov [ebp+6485h],ecx
0x100ebcd2: mov dx,ds:[1012F1ECh]
0x100ebcd9: mov [ebp+6489h],dx
0x100ebce0: mov eax,ds:[1012F1C0h]
0x100ebce5: mov [ebp+64C5h],eax
0x100ebceb: mov ecx,ds:[1012F1C4h]
0x100ebcf1: mov [ebp+64C9h],ecx
0x100ebcf7: mov edx,ds:[1012F1C8h]
0x100ebcfd: mov [ebp+64CDh],edx
0x100ebd03: mov ax,ds:[1012F1CCh]
0x100ebd09: mov [ebp+64D1h],ax
0x100ebd10: jmp near ptr 100EC415h
0x100ebd15: mov eax,8
0x100ebd1a: mov ecx,1012E57Ch
0x100ebd1f: lea edx,[esp+68h]
0x100ebd23: mov esi,[edx]
0x100ebd25: cmp esi,[ecx]
0x100ebd27: jne short 100EBD3Bh
0x100ebd29: sub eax,4
0x100ebd2c: add ecx,4
0x100ebd2f: add edx,4
0x100ebd32: cmp eax,4
0x100ebd35: jae short 100EBD23h
0x100ebd37: cmp eax,ebx
0x100ebd39: je short 100EBD98h
0x100ebd3b: movzx esi,byte ptr [edx]
0x100ebd3e: movzx edi,byte ptr [ecx]
0x100ebd41: sub esi,edi
0x100ebd43: jne short 100EBD8Ah
0x100ebd45: sub eax,1
0x100ebd48: add ecx,1
0x100ebd4b: add edx,1
0x100ebd4e: cmp eax,ebx
0x100ebd50: je short 100EBD98h
0x100ebd52: movzx esi,byte ptr [edx]
0x100ebd55: movzx edi,byte ptr [ecx]
0x100ebd58: sub esi,edi
0x100ebd5a: jne short 100EBD8Ah
0x100ebd5c: sub eax,1
0x100ebd5f: add ecx,1
0x100ebd62: add edx,1
0x100ebd65: cmp eax,ebx
0x100ebd67: je short 100EBD98h
0x100ebd69: movzx esi,byte ptr [edx]
0x100ebd6c: movzx edi,byte ptr [ecx]
0x100ebd6f: sub esi,edi
0x100ebd71: jne short 100EBD8Ah
0x100ebd73: sub eax,1
0x100ebd76: add ecx,1
0x100ebd79: add edx,1
0x100ebd7c: cmp eax,ebx
0x100ebd7e: je short 100EBD98h
```

```
0x100ebd80: movzx esi,byte ptr [edx]
0x100ebd83: movzx ecx,byte ptr [ecx]
0x100ebd86: sub esi,ecx
0x100ebd88: je short 100EBD98h
0x100ebd8a: cmp esi,ebx
0x100ebd8c: mov eax,1
0x100ebd91: jg short 100EBD9Ah
0x100ebd93: or eax,0FFFFFFFFh
0x100ebd96: jmp short 100EBD9Ah
0x100ebd98: xor eax,eax
0x100ebd9a: cmp eax,ebx
0x100ebd9c: jne near ptr 100EBE64h
0x100ebda2: mov edx,[ebp]
0x100ebda5: mov eax,[ebp+4]
0x100ebda8: mov ecx,[edx+8]
0x100ebdab: push ebx
0x100ebdac: push 54h
0x100ebdae: push eax
0x100ebdaf: call ecx
0x100ebdb1: push ebp
0x100ebdb2: call 100D2FE0h
0x100ebdb7: push ebp
0x100ebdb8: mov [ebp+65C4h],eax
0x100ebdbe: call 100D2FE0h
0x100ebdc3: mov edx,[ebp]
0x100ebdc6: mov [ebp+65D0h],eax
0x100ebdcc: mov eax,[ebp+4]
0x100ebdcf: mov ecx,[edx+8]
0x100ebdd2: push ebx
0x100ebdd3: push 5Ch
0x100ebdd5: push eax
0x100ebdd6: call ecx
0x100ebdd8: push ebp
0x100ebdd9: call 100D2FE0h
0x100ebdde: push eax
0x100ebddf: push ebp
0x100ebde0: call 100E5AD0h
0x100ebde5: add esp,2Ch
0x100ebde8: cmp dword ptr [ebp+65C4h],78h
0x100ebdef: jle short 100EBE31h
0x100ebdf1: mov edx,[ebp]
0x100ebdf4: mov eax,[ebp+4]
0x100ebdf7: mov ecx,[edx+8]
0x100ebdfa: push ebx
0x100ebdfb: push 78h
0x100ebdfd: push eax
0x100ebdfe: call ecx
0x100ebe00: push ebp
0x100ebe01: call 100D2FE0h
0x100ebe06: xor ecx,ecx
0x100ebe08: add esp,10h
0x100ebe0b: cmp eax,ebx
0x100ebe0d: setne cl
0x100ebe10: add [ebp+6614h],ecx
0x100ebe16: cmp dword ptr [ebp+6614h],2
0x100ebe1d: jne short 100EBE31h
0x100ebe1f: cmp [ebp+0A0h],ebx
0x100ebe25: je short 100EBE31h
0x100ebe27: push eax
0x100ebe28: push ebp
0x100ebe29: call 100E5AD0h
0x100ebe2e: add esp,8
0x100ebe31: mov edx,[ebp]
0x100ebe34: mov eax,[ebp+4]
0x100ebe37: mov ecx,[edx+8]
0x100ebe3a: push ebx
0x100ebe3b: push 64h
```

```
0x100ebe3d: push eax
0x100ebe3e: call ecx
0x100ebe40: push ebp
0x100ebe41: call 100D2FE0h
0x100ebe46: mov edx,[ebp+65C4h]
0x100ebe4c: add edx,0Ch
0x100ebe4f: push edx
0x100ebe50: push ebp
0x100ebe51: mov [ebp+65C0h],eax
0x100ebe57: call 100EA5A0h
0x100ebe5c: add esp,18h
0x100ebe5f: jmp near ptr 100EC415h
0x100ebe64: mov eax,4
0x100ebe69: mov ecx,1012E7E8h
0x100ebe6e: lea edx,[esp+68h]
0x100ebe72: mov esi,[edx]
0x100ebe74: cmp esi,[ecx]
0x100ebe76: jne short 100EBE8Ah
0x100ebe78: sub eax,4
0x100ebe7b: add ecx,4
0x100ebe7e: add edx,4
0x100ebe81: cmp eax,4
0x100ebe84: jae short 100EBE72h
0x100ebe86: cmp eax,ebx
0x100ebe88: je short 100EBEE7h
0x100ebe8a: movzx esi,byte ptr [edx]
0x100ebe8d: movzx edi,byte ptr [ecx]
0x100ebe90: sub esi,edi
0x100ebe92: jne short 100EBED9h
0x100ebe94: sub eax,1
0x100ebe97: add ecx,1
0x100ebe9a: add edx,1
0x100ebe9d: cmp eax,ebx
0x100ebe9f: je short 100EBEE7h
0x100ebea1: movzx esi,byte ptr [edx]
0x100ebea4: movzx edi,byte ptr [ecx]
0x100ebea7: sub esi,edi
0x100ebea9: jne short 100EBED9h
0x100ebeab: sub eax,1
0x100ebeae: add ecx,1
0x100ebeb1: add edx,1
0x100ebeb4: cmp eax,ebx
0x100ebeb6: je short 100EBEE7h
0x100ebeb8: movzx esi,byte ptr [edx]
0x100ebebb: movzx edi,byte ptr [ecx]
0x100ebebe: sub esi,edi
0x100ebec0: jne short 100EBED9h
0x100ebec2: sub eax,1
0x100ebec5: add ecx,1
0x100ebec8: add edx,1
0x100ebecb: cmp eax,ebx
0x100ebecd: je short 100EBEE7h
0x100ebecf: movzx esi,byte ptr [edx]
0x100ebed2: movzx eax,byte ptr [ecx]
0x100ebed5: sub esi,eax
0x100ebed7: je short 100EBEE7h
0x100ebed9: cmp esi,ebx
0x100ebedb: mov eax,1
0x100ebee0: jg short 100EBEE9h
0x100ebee2: or eax,0FFFFFFFFh
0x100ebee5: jmp short 100EBEE9h
0x100ebee7: xor eax,eax
0x100ebee9: cmp eax,ebx
0x100ebeeb: jne short 100EBF09h
0x100ebeed: mov ecx,[ebp]
0x100ebef0: mov edx,[ebp+4]
0x100ebef3: mov eax,[ecx+8]
```

```
0x100ebef6: push ebx
0x100ebef7: push ebx
0x100ebef8: push edx
0x100ebef9: call eax
0x100ebefb: push ebp
0x100ebefc: call 100E5C30h
0x100ebf01: add esp,10h
0x100ebf04: jmp near ptr 100EC415h
0x100ebf09: mov eax,6
0x100ebf0e: mov ecx,1012F1B8h
0x100ebf13: lea edx,[esp+68h]
0x100ebf17: mov esi,[edx]
0x100ebf19: cmp esi,[ecx]
0x100ebf1b: jne short 100EBF2Fh
0x100ebf1d: sub eax,4
0x100ebf20: add ecx,4
0x100ebf23: add edx,4
0x100ebf26: cmp eax,4
0x100ebf29: jae short 100EBF17h
0x100ebf2b: cmp eax,ebx
0x100ebf2d: je short 100EBF8Ch
0x100ebf2f: movzx esi,byte ptr [edx]
0x100ebf32: movzx edi,byte ptr [ecx]
0x100ebf35: sub esi,edi
0x100ebf37: jne short 100EBF7Eh
0x100ebf39: sub eax,1
0x100ebf3c: add ecx,1
0x100ebf3f: add edx,1
0x100ebf42: cmp eax,ebx
0x100ebf44: je short 100EBF8Ch
0x100ebf46: movzx esi,byte ptr [edx]
0x100ebf49: movzx edi,byte ptr [ecx]
0x100ebf4c: sub esi,edi
0x100ebf4e: jne short 100EBF7Eh
0x100ebf50: sub eax,1
0x100ebf53: add ecx,1
0x100ebf56: add edx,1
0x100ebf59: cmp eax,ebx
0x100ebf5b: je short 100EBF8Ch
0x100ebf5d: movzx esi,byte ptr [edx]
0x100ebf60: movzx edi,byte ptr [ecx]
0x100ebf63: sub esi,edi
0x100ebf65: jne short 100EBF7Eh
0x100ebf67: sub eax,1
0x100ebf6a: add ecx,1
0x100ebf6d: add edx,1
0x100ebf70: cmp eax,ebx
0x100ebf72: je short 100EBF8Ch
0x100ebf74: movzx esi,byte ptr [edx]
0x100ebf77: movzx ecx,byte ptr [ecx]
0x100ebf7a: sub esi,ecx
0x100ebf7c: je short 100EBF8Ch
0x100ebf7e: cmp esi,ebx
0x100ebf80: mov eax,1
0x100ebf85: jg short 100EBF8Eh
0x100ebf87: or eax,0FFFFFFFFh
0x100ebf8a: jmp short 100EBF8Eh
0x100ebf8c: xor eax,eax
0x100ebf8e: cmp eax,ebx
0x100ebf90: jne near ptr 100EC035h
0x100ebf96: mov edx,[ebp]
0x100ebf99: mov eax,[ebp+4]
0x100ebf9c: mov ecx,[edx+8]
0x100ebf9f: push ebx
0x100ebfa0: push 6
0x100ebfa2: push eax
0x100ebfa3: call ecx
```

```
0x100ebfa5: mov edx,[ebp]
0x100ebfa8: mov ecx,[edx]
0x100ebfaa: push 8
0x100ebfac: push 1
0x100ebfae: lea eax,[ebp+6485h]
0x100ebfb4: push eax
0x100ebfb5: mov eax,[ebp+4]
0x100ebfb8: push eax
0x100ebfb9: call ecx
0x100ebfbb: mov edx,[ebp]
0x100ebfbe: mov ecx,[edx]
0x100ebfc0: push 8
0x100ebfc2: push 1
0x100ebfc4: lea eax,[ebp+64C5h]
0x100ebfca: push eax
0x100ebfcb: mov eax,[ebp+4]
0x100ebfce: push eax
0x100ebfcf: call ecx
0x100ebfd1: mov edx,[ebp]
0x100ebfd4: mov ecx,[edx]
0x100ebfd6: push 10h
0x100ebfd8: push 1
0x100ebfda: lea eax,[ebp+6505h]
0x100ebfe0: push eax
0x100ebfe1: mov eax,[ebp+4]
0x100ebfe4: push eax
0x100ebfe5: call ecx
0x100ebfe7: push ebp
0x100ebfe8: call 100D2F50h
0x100ebfed: movzx edx,ax
0x100ebff0: add esp,40h
0x100ebff3: push ebp
0x100ebff4: mov [ebp+65C0h],edx
0x100ebffa: call 100D2F50h
0x100ebfff: push ebp
0x100ec000: call 100D2F50h
0x100ec005: push ebp
0x100ec006: mov [ebp+66B2h],ax
0x100ec00d: call 100D2F50h
0x100ec012: add esp,0Ch
0x100ec015: mov [ebp+66B0h],ax
0x100ec01c: mov dword ptr [ebp+2E7FCh],100D8F80h
0x100ec026: mov dword ptr [ebp+65B0h],61616161h
0x100ec030: jmp near ptr 100EC415h
0x100ec035: mov eax,9
0x100ec03a: mov ecx,1012F1ACh
0x100ec03f: lea edx,[esp+68h]
0x100ec043: mov esi,[edx]
0x100ec045: cmp esi,[ecx]
0x100ec047: jne short 100EC05Bh
0x100ec049: sub eax,4
0x100ec04c: add ecx,4
0x100ec04f: add edx,4
0x100ec052: cmp eax,4
0x100ec055: jae short 100EC043h
0x100ec057: cmp eax,ebx
0x100ec059: je short 100EC0B8h
0x100ec05b: movzx esi,byte ptr [edx]
0x100ec05e: movzx edi,byte ptr [ecx]
0x100ec061: sub esi,edi
0x100ec063: jne short 100EC0AAh
0x100ec065: sub eax,1
0x100ec068: add ecx,1
0x100ec06b: add edx,1
0x100ec06e: cmp eax,ebx
0x100ec070: je short 100EC0B8h
0x100ec072: movzx esi,byte ptr [edx]
```

```
0x100ec075: movzx edi,byte ptr [ecx]
0x100ec078: sub esi,edi
0x100ec07a: jne short 100EC0AAh
0x100ec07c: sub eax,1
0x100ec07f: add ecx,1
0x100ec082: add edx,1
0x100ec085: cmp eax,ebx
0x100ec087: je short 100EC0B8h
0x100ec089: movzx esi,byte ptr [edx]
0x100ec08c: movzx edi,byte ptr [ecx]
0x100ec08f: sub esi,edi
0x100ec091: jne short 100EC0AAh
0x100ec093: sub eax,1
0x100ec096: add ecx,1
0x100ec099: add edx,1
0x100ec09c: cmp eax,ebx
0x100ec09e: je short 100EC0B8h
0x100ec0a0: movzx esi,byte ptr [edx]
0x100ec0a3: movzx eax,byte ptr [ecx]
0x100ec0a6: sub esi,eax
0x100ec0a8: je short 100EC0B8h
0x100ec0aa: cmp esi,ebx
0x100ec0ac: mov eax,1
0x100ec0b1: jg short 100EC0BAh
0x100ec0b3: or eax,0FFFFFFFFh
0x100ec0b6: jmp short 100EC0BAh
0x100ec0b8: xor eax,eax
0x100ec0ba: cmp eax,ebx
0x100ec0bc: jne short 100EC0CCh
0x100ec0be: push ebp
0x100ec0bf: call 100E5130h
0x100ec0c4: add esp,4
0x100ec0c7: jmp near ptr 100EC415h
0x100ec0cc: mov eax,4
0x100ec0d1: mov ecx,1012F1A4h
0x100ec0d6: lea edx,[esp+68h]
0x100ec0da: lea ebx,[ebx]
0x100ec0e0: mov esi,[edx]
0x100ec0e2: cmp esi,[ecx]
0x100ec0e4: jne short 100EC0F8h
0x100ec0e6: sub eax,4
0x100ec0e9: add ecx,4
0x100ec0ec: add edx,4
0x100ec0ef: cmp eax,4
0x100ec0f2: jae short 100EC0E0h
0x100ec0f4: cmp eax,ebx
0x100ec0f6: je short 100EC155h
0x100ec0f8: movzx esi,byte ptr [edx]
0x100ec0fb: movzx edi,byte ptr [ecx]
0x100ec0fe: sub esi,edi
0x100ec100: jne short 100EC147h
0x100ec102: sub eax,1
0x100ec105: add ecx,1
0x100ec108: add edx,1
0x100ec10b: cmp eax,ebx
0x100ec10d: je short 100EC155h
0x100ec10f: movzx esi,byte ptr [edx]
0x100ec112: movzx edi,byte ptr [ecx]
0x100ec115: sub esi,edi
0x100ec117: jne short 100EC147h
0x100ec119: sub eax,1
0x100ec11c: add ecx,1
0x100ec11f: add edx,1
0x100ec122: cmp eax,ebx
0x100ec124: je short 100EC155h
0x100ec126: movzx esi,byte ptr [edx]
0x100ec129: movzx edi,byte ptr [ecx]
```

```
0x100ec12c: sub esi,edi
0x100ec12e: jne short 100EC147h
0x100ec130: sub eax,1
0x100ec133: add ecx,1
0x100ec136: add edx,1
0x100ec139: cmp eax,ebx
0x100ec13b: je short 100EC155h
0x100ec13d: movzx esi,byte ptr [edx]
0x100ec140: movzx ecx,byte ptr [ecx]
0x100ec143: sub esi,ecx
0x100ec145: je short 100EC155h
0x100ec147: cmp esi,ebx
0x100ec149: mov eax,1
0x100ec14e: jg short 100EC157h
0x100ec150: or eax,0FFFFFFFFh
0x100ec153: jmp short 100EC157h
0x100ec155: xor eax,eax
0x100ec157: cmp eax,ebx
0x100ec159: jne short 100EC169h
0x100ec15b: push ebp
0x100ec15c: call 100E53F0h
0x100ec161: add esp,4
0x100ec164: jmp near ptr 100EC415h
0x100ec169: mov eax,4
0x100ec16e: mov ecx,1012F19Ch
0x100ec173: lea edx,[esp+68h]
0x100ec177: mov esi,[edx]
0x100ec179: cmp esi,[ecx]
0x100ec17b: jne short 100EC18Fh
0x100ec17d: sub eax,4
0x100ec180: add ecx,4
0x100ec183: add edx,4
0x100ec186: cmp eax,4
0x100ec189: jae short 100EC177h
0x100ec18b: cmp eax,ebx
0x100ec18d: je short 100EC1ECh
0x100ec18f: movzx esi,byte ptr [edx]
0x100ec192: movzx edi,byte ptr [ecx]
0x100ec195: sub esi,edi
0x100ec197: jne short 100EC1DEh
0x100ec199: sub eax,1
0x100ec19c: add ecx,1
0x100ec19f: add edx,1
0x100ec1a2: cmp eax,ebx
0x100ec1a4: je short 100EC1ECh
0x100ec1a6: movzx esi,byte ptr [edx]
0x100ec1a9: movzx edi,byte ptr [ecx]
0x100ec1ac: sub esi,edi
0x100ec1ae: jne short 100EC1DEh
0x100ec1b0: sub eax,1
0x100ec1b3: add ecx,1
0x100ec1b6: add edx,1
0x100ec1b9: cmp eax,ebx
0x100ec1bb: je short 100EC1ECh
0x100ec1bd: movzx esi,byte ptr [edx]
0x100ec1c0: movzx edi,byte ptr [ecx]
0x100ec1c3: sub esi,edi
0x100ec1c5: jne short 100EC1DEh
0x100ec1c7: sub eax,1
0x100ec1ca: add ecx,1
0x100ec1cd: add edx,1
0x100ec1d0: cmp eax,ebx
0x100ec1d2: je short 100EC1ECh
0x100ec1d4: movzx esi,byte ptr [edx]
0x100ec1d7: movzx edx,byte ptr [ecx]
0x100ec1da: sub esi,edx
0x100ec1dc: je short 100EC1ECh
```

```
0x100ec1de: cmp esi,ebx
0x100ec1e0: mov eax,1
0x100ec1e5: jg short 100EC1EEh
0x100ec1e7: or eax,0FFFFFFFFh
0x100ec1ea: jmp short 100EC1EEh
0x100ec1ec: xor eax,eax
0x100ec1ee: cmp eax,ebx
0x100ec1f0: jne short 100EC1FEh
0x100ec1f2: push ebx
0x100ec1f3: push ebp
0x100ec1f4: call 100EABA0h
0x100ec1f9: jmp near ptr 100EC412h
0x100ec1fe: mov eax,4
0x100ec203: mov ecx,1012F194h
0x100ec208: lea edx,[esp+68h]
0x100ec20c: lea esp,[esp]
0x100ec210: mov esi,[edx]
0x100ec212: cmp esi,[ecx]
0x100ec214: jne short 100EC228h
0x100ec216: sub eax,4
0x100ec219: add ecx,4
0x100ec21c: add edx,4
0x100ec21f: cmp eax,4
0x100ec222: jae short 100EC210h
0x100ec224: cmp eax,ebx
0x100ec226: je short 100EC285h
0x100ec228: movzx esi,byte ptr [edx]
0x100ec22b: movzx edi,byte ptr [ecx]
0x100ec22e: sub esi,edi
0x100ec230: jne short 100EC277h
0x100ec232: sub eax,1
0x100ec235: add ecx,1
0x100ec238: add edx,1
0x100ec23b: cmp eax,ebx
0x100ec23d: je short 100EC285h
0x100ec23f: movzx esi,byte ptr [edx]
0x100ec242: movzx edi,byte ptr [ecx]
0x100ec245: sub esi,edi
0x100ec247: jne short 100EC277h
0x100ec249: sub eax,1
0x100ec24c: add ecx,1
0x100ec24f: add edx,1
0x100ec252: cmp eax,ebx
0x100ec254: je short 100EC285h
0x100ec256: movzx esi,byte ptr [edx]
0x100ec259: movzx edi,byte ptr [ecx]
0x100ec25c: sub esi,edi
0x100ec25e: jne short 100EC277h
0x100ec260: sub eax,1
0x100ec263: add ecx,1
0x100ec266: add edx,1
0x100ec269: cmp eax,ebx
0x100ec26b: je short 100EC285h
0x100ec26d: movzx esi,byte ptr [edx]
0x100ec270: movzx eax,byte ptr [ecx]
0x100ec273: sub esi,eax
0x100ec275: je short 100EC285h
0x100ec277: cmp esi,ebx
0x100ec279: mov eax,1
0x100ec27e: jg short 100EC287h
0x100ec280: or eax,0FFFFFFFFh
0x100ec283: jmp short 100EC287h
0x100ec285: xor eax,eax
0x100ec287: cmp eax,ebx
0x100ec289: jne short 100EC299h
0x100ec28b: push ebp
0x100ec28c: call 100EB170h
```

```
0x100ec291: add esp,4
0x100ec294: jmp near ptr 100EC415h
0x100ec299: movzx ecx,byte ptr [esp+68h]
0x100ec29e: sub ecx,43h
0x100ec2a1: jne short 100EC2ADh
0x100ec2a3: movzx ecx,byte ptr [esp+69h]
0x100ec2a8: sub ecx,49h
0x100ec2ab: je short 100EC2BBh
0x100ec2ad: cmp ecx,ebx
0x100ec2af: mov eax,1
0x100ec2b4: jg short 100EC2BDh
0x100ec2b6: or eax,0FFFFFFFFh
0x100ec2b9: jmp short 100EC2BDh
0x100ec2bb: xor eax,eax
0x100ec2bd: cmp eax,ebx
0x100ec2bf: jne short 100EC2CFh
0x100ec2c1: push ebp
0x100ec2c2: call 100E6190h
0x100ec2c7: add esp,4
0x100ec2ca: jmp near ptr 100EC415h
0x100ec2cf: lea edi,[ebp+64C5h]
0x100ec2d5: mov ebx,1012D454h
0x100ec2da: lea esi,[ebp+6485h]
0x100ec2e0: sub edi,1012D460h
0x100ec2e6: sub esi,ebx
0x100ec2e8: mov dword ptr [esp+10h],44h
0x100ec2f0: mov ecx,[esp+14h]
0x100ec2f4: cmp ecx,[ebx-4]
0x100ec2f7: jne short 100EC32Ah
0x100ec2f9: mov eax,ebx
0x100ec2fb: jmp short 100EC300h
0x100ec300: mov cl,[eax]
0x100ec302: mov [esi+eax],cl
0x100ec305: add eax,1
0x100ec308: test cl,cl
0x100ec30a: jne short 100EC300h
0x100ec30c: lea eax,[ebx+0Ch]
0x100ec30f: nop
0x100ec310: mov cl,[eax]
0x100ec312: mov [edi+eax],cl
0x100ec315: add eax,1
0x100ec318: test cl,cl
0x100ec31a: jne short 100EC310h
0x100ec31c: cmp [ebx+1Fh],cl
0x100ec31f: je short 100EC32Ah
0x100ec321: push ebp
0x100ec322: call 100EADB0h
0x100ec327: add esp,4
0x100ec32a: sub esi,24h
0x100ec32d: sub edi,24h
0x100ec330: add ebx,24h
0x100ec333: sub dword ptr [esp+10h],1
0x100ec338: jne short 100EC2F0h
0x100ec33a: xor ebx,ebx
0x100ec33c: jmp near ptr 100EC415h
0x100ec341: mov eax,8
0x100ec346: mov ecx,1012F188h
0x100ec34b: lea edx,[esp+6Eh]
0x100ec34f: nop
0x100ec350: mov esi,[edx]
0x100ec352: cmp esi,[ecx]
0x100ec354: jne short 100EC368h
0x100ec356: sub eax,4
0x100ec359: add ecx,4
0x100ec35c: add edx,4
0x100ec35f: cmp eax,4
0x100ec362: jae short 100EC350h
```

```
0x100ec364: cmp eax,ebx
0x100ec366: je short 100EC3C5h
0x100ec368: movzx esi,byte ptr [edx]
0x100ec36b: movzx ebx,byte ptr [ecx]
0x100ec36e: sub esi,ebx
0x100ec370: jne short 100EC3B7h
0x100ec372: sub eax,1
0x100ec375: add ecx,1
0x100ec378: add edx,1
0x100ec37b: test eax,eax
0x100ec37d: je short 100EC3C5h
0x100ec37f: movzx esi,byte ptr [edx]
0x100ec382: movzx ebx,byte ptr [ecx]
0x100ec385: sub esi,ebx
0x100ec387: jne short 100EC3B7h
0x100ec389: sub eax,1
0x100ec38c: add ecx,1
0x100ec38f: add edx,1
0x100ec392: test eax,eax
0x100ec394: je short 100EC3C5h
0x100ec396: movzx esi,byte ptr [edx]
0x100ec399: movzx ebx,byte ptr [ecx]
0x100ec39c: sub esi,ebx
0x100ec39e: jne short 100EC3B7h
0x100ec3a0: sub eax,1
0x100ec3a3: add ecx,1
0x100ec3a6: add edx,1
0x100ec3a9: test eax,eax
0x100ec3ab: je short 100EC3C5h
0x100ec3ad: movzx esi,byte ptr [edx]
0x100ec3b0: movzx edx,byte ptr [ecx]
0x100ec3b3: sub esi,edx
0x100ec3b5: je short 100EC3C5h
0x100ec3b7: test esi,esi
0x100ec3b9: mov eax,1
0x100ec3be: jg short 100EC3C7h
0x100ec3c0: or eax,0FFFFFFFFh
0x100ec3c3: jmp short 100EC3C7h
0x100ec3c5: xor eax,eax
0x100ec3c7: test eax,eax
0x100ec3c9: jne short 100EC3E6h
0x100ec3cb: mov eax,[esp+14h]
0x100ec3cf: sub eax,edi
0x100ec3d1: push eax
0x100ec3d2: push edi
0x100ec3d3: push ebp
0x100ec3d4: mov [ebp+65C0h],edi
0x100ec3da: call 100E4A30h
0x100ec3df: add esp,0Ch
0x100ec3e2: xor ebx,ebx
0x100ec3e4: jmp short 100EC415h
0x100ec3e6: push 0
0x100ec3e8: push ebp
0x100ec3e9: call 100EA5A0h
0x100ec3ee: add esp,8
0x100ec3f1: xor ebx,ebx
0x100ec3f3: jmp short 100EC415h
0x100ec3f5: mov esi,eax
0x100ec3f7: lea ecx,[esp+68h]
0x100ec3fb: sub esi,ecx
0x100ec3fd: push esi
0x100ec3fe: push ebp
0x100ec3ff: call 100E55A0h
0x100ec404: add esp,8
0x100ec407: cmp esi,ebx
0x100ec409: je short 100EC415h
0x100ec40b: push ebx
```

```
0x100ec40c: push ebp
0x100ec40d: call 100EA5A0h
0x100ec412: add esp,8
0x100ec415: cmp byte ptr [ebp+6485h],0
0x100ec41c: jne short 100EC446h
0x100ec41e: mov edx,[esp+14h]
0x100ec422: push edx
0x100ec423: push ebx
0x100ec424: push ebp
0x100ec425: call 100E6090h
0x100ec42a: add esp,0Ch
0x100ec42d: cmp byte ptr [ebp+6485h],0
0x100ec434: jne short 100EC446h
0x100ec436: push ebx
0x100ec437: push ebp
0x100ec438: mov [ebp+6614h],ebx
0x100ec43e: call 100EB010h
0x100ec443: add esp,8
0x100ec446: mov ebx,ds:[100FE424h]
0x100ec44c: xor esi,esi
0x100ec44e: mov edi,edi
0x100ec450: mov eax,[esi+1013894Ch]
0x100ec456: push eax
0x100ec457: lea edi,[ebp+6485h]
0x100ec45d: push edi
0x100ec45e: call ebx
0x100ec460: add esp,8
0x100ec463: test eax,eax
0x100ec465: je short 100EC47Eh
0x100ec467: mov ecx,[esi+1013894Ch]
0x100ec46d: mov edx,edi
0x100ec46f: nop
0x100ec470: mov al,[ecx]
0x100ec472: mov [edx],al
0x100ec474: add ecx,1
0x100ec477: add edx,1
0x100ec47a: test al,al
0x100ec47c: jne short 100EC470h
0x100ec47e: add esi,4
0x100ec481: cmp esi,3Ch
0x100ec484: jb short 100EC450h
0x100ec486: mov ebx,ds:[100FE414h]
0x100ec48c: push 5
0x100ec48e: push 101298A0h
0x100ec493: lea esi,[ebp+6485h]
0x100ec499: push esi
0x100ec49a: call ebx
0x100ec49c: add esp,0Ch
0x100ec49f: test eax,eax
0x100ec4a1: jne short 100EC4AFh
0x100ec4a3: mov [ebp+64D5h],al
0x100ec4a9: mov [ebp+6495h],al
0x100ec4af: mov eax,esi
0x100ec4b1: lea edx,[eax+1]
0x100ec4b4: mov cl,[eax]
0x100ec4b6: add eax,1
0x100ec4b9: test cl,cl
0x100ec4bb: jne short 100EC4B4h
0x100ec4bd: sub eax,edx
0x100ec4bf: cmp byte ptr [eax+ebp+6484h],20h
0x100ec4c7: lea eax,[eax+ebp+6484h]
0x100ec4ce: jne short 100EC4DBh
0x100ec4d0: mov byte ptr [eax],0
0x100ec4d3: sub eax,1
0x100ec4d6: cmp byte ptr [eax],20h
0x100ec4d9: je short 100EC4D0h
0x100ec4db: lea eax,[ebp+64C5h]
```

```
0x100ec4e1: lea edx,[eax+1]
0x100ec4e4: mov cl,[eax]
0x100ec4e6: add eax,1
0x100ec4e9: test cl,cl
0x100ec4eb: jne short 100EC4E4h
0x100ec4ed: sub eax,edx
0x100ec4ef: cmp byte ptr [eax+ebp+64C4h],20h
0x100ec4f7: lea eax,[eax+ebp+64C4h]
0x100ec4fe: jne short 100EC50Bh
0x100ec500: mov byte ptr [eax],0
0x100ec503: sub eax,1
0x100ec506: cmp byte ptr [eax],20h
0x100ec509: je short 100EC500h
0x100ec50b: lea eax,[ebp+6485h]
0x100ec511: lea edx,[eax+1]
0x100ec514: mov cl,[eax]
0x100ec516: add eax,1
0x100ec519: test cl,cl
0x100ec51b: jne short 100EC514h
0x100ec51d: sub eax,edx
0x100ec51f: mov esi,eax
0x100ec521: push esi
0x100ec522: lea eax,[ebp+6485h]
0x100ec528: push eax
0x100ec529: lea edi,[ebp+64C5h]
0x100ec52f: push edi
0x100ec530: call dword ptr ds:[100FE418h]
0x100ec536: add esp,0Ch
0x100ec539: test eax,eax
0x100ec53b: jne short 100EC566h
0x100ec53d: mov cl,[esi+ebp+64C5h]
0x100ec544: add esi,1
0x100ec547: cmp cl,20h
0x100ec54a: jne short 100EC566h
0x100ec54c: mov edx,40h
0x100ec551: sub edx,esi
0x100ec553: push edx
0x100ec554: lea eax,[esi+ebp+64C5h]
0x100ec55b: push eax
0x100ec55c: push edi
0x100ec55d: call dword ptr ds:[100FE498h]
0x100ec563: add esp,0Ch
0x100ec566: push 0Fh
0x100ec568: push 1012F178h
0x100ec56d: push edi
0x100ec56e: call ebx
0x100ec570: add esp,0Ch
0x100ec573: test eax,eax
0x100ec575: jne short 100EC58Dh
0x100ec577: lea eax,[ebp+64D4h]
0x100ec57d: mov edx,edi
0x100ec57f: sub edx,eax
0x100ec581: mov cl,[eax]
0x100ec583: mov [edx+eax],cl
0x100ec586: add eax,1
0x100ec589: test cl,cl
0x100ec58b: jne short 100EC581h
0x100ec58d: xor ebx,ebx
0x100ec58f: cmp [ebp+6614h],ebx
0x100ec595: mov byte ptr [ebp+6544h],0
0x100ec59c: mov byte ptr [ebp+6504h],0
0x100ec5a3: mov byte ptr [ebp+64C4h],0
0x100ec5aa: mov byte ptr [ebp+6584h],0
0x100ec5b1: mov byte ptr [ebp+6484h],0
0x100ec5b8: je near ptr 100EFB7Dh
0x100ec5be: cmp [ebp+65FCh],ebx
0x100ec5c4: jne short 100EC5DCh
```

```
0x100ec5c6: mov ecx,[ebp+65F0h]
0x100ec5cc: mov edx,1
0x100ec5d1: shl edx,cl
0x100ec5d3: sub edx,1
0x100ec5d6: mov [ebp+65FCh],edx
0x100ec5dc: cmp [ebp+66B4h],bx
0x100ec5e3: jne short 100EC5F3h
0x100ec5e5: mov ax,[ebp+66B0h]
0x100ec5ec: mov [ebp+66B4h],ax
0x100ec5f3: cmp [ebp+66B6h],bx
0x100ec5fa: jne short 100EC60Ah
0x100ec5fc: mov cx,[ebp+66B2h]
0x100ec603: mov [ebp+66B6h],cx
0x100ec60a: movzx ecx,word ptr [ebp+66C2h]
0x100ec611: cmp cx,bx
0x100ec614: je short 100EC639h
0x100ec616: movzx eax,word ptr [ebp+66B4h]
0x100ec61d: fld1
0x100ec61f: add ax,cx
0x100ec622: fstp qword ptr [ebp+66D8h]
0x100ec628: mov [ebp+66B6h],ax
0x100ec62f: add eax,0FFFFFFFFh
0x100ec632: mov [ebp+66B4h],ax
0x100ec639: cmp word ptr [ebp+66B4h],0A40h
0x100ec642: jne short 100EC661h
0x100ec644: cmp word ptr [ebp+66B6h],0F60h
0x100ec64d: jne short 100EC661h
0x100ec64f: mov word ptr [ebp+66B4h],0A38h
0x100ec658: mov word ptr [ebp+66B6h],0F38h
0x100ec661: cmp word ptr [ebp+66B4h],0C40h
0x100ec66a: jne short 100EC689h
0x100ec66c: cmp word ptr [ebp+66B6h],1300h
0x100ec675: jne short 100EC689h
0x100ec677: mov word ptr [ebp+66B4h],0C34h
0x100ec680: mov word ptr [ebp+66B6h],1250h
0x100ec689: cmp word ptr [ebp+66B4h],0BC6h
0x100ec692: jne short 100EC6A8h
0x100ec694: cmp word ptr [ebp+66B6h],1000h
0x100ec69d: jne short 100EC6A8h
0x100ec69f: mov word ptr [ebp+66B6h],0FAEh
0x100ec6a8: cmp [ebp+6618h],ebx
0x100ec6ae: je short 100EC70Eh
0x100ec6b0: cmp dword ptr [ebp+65B0h],0FFFFFFFFh
0x100ec6b7: jne short 100EC6BFh
0x100ec6b9: mov [ebp+65B0h],ebx
0x100ec6bf: cmp [ebp+65B0h],ebx
0x100ec6c5: je short 100EC6D5h
0x100ec6c7: mov edx,[ebp+65ECh]
0x100ec6cd: mov [ebp+6614h],edx
0x100ec6d3: jmp short 100EC6E1h
0x100ec6d5: mov eax,[ebp+65ECh]
0x100ec6db: mov [ebp+66D0h],eax
0x100ec6e1: mov eax,[ebp+65F4h]
0x100ec6e7: cmp eax,1
0x100ec6ea: jne short 100EC6F6h
0x100ec6ec: mov dword ptr [ebp+2E7FCh],100D5D70h
0x100ec6f6: cmp eax,7
0x100ec6f9: jne near ptr 100EFA8Ah
0x100ec6ff: mov dword ptr [ebp+2E7FCh],100D5C00h
0x100ec709: jmp near ptr 100EFA8Ah
0x100ec70e: xor edx,edx
0x100ec710: mov edi,101298B8h
0x100ec715: lea esi,[ebp+6485h]
0x100ec71b: mov ecx,6
0x100ec720: repe cmpsb
0x100ec722: mov eax,ebx
0x100ec724: sete al
```

```
0x100ec727: cmp eax,ebx
0x100ec729: mov [esp+10h],eax
0x100ec72d: je near ptr 100EC7CBh
0x100ec733: lea eax,[edx+8]
0x100ec736: mov ecx,1012F188h
0x100ec73b: lea edx,[esp+6Eh]
0x100ec73f: nop
0x100ec740: mov esi,[edx]
0x100ec742: cmp esi,[ecx]
0x100ec744: jne short 100EC758h
0x100ec746: sub eax,4
0x100ec749: add ecx,4
0x100ec74c: add edx,4
0x100ec74f: cmp eax,4
0x100ec752: jae short 100EC740h
0x100ec754: cmp eax,ebx
0x100ec756: je short 100EC7B5h
0x100ec758: movzx esi,byte ptr [edx]
0x100ec75b: movzx edi,byte ptr [ecx]
0x100ec75e: sub esi,edi
0x100ec760: jne short 100EC7A7h
0x100ec762: sub eax,1
0x100ec765: add ecx,1
0x100ec768: add edx,1
0x100ec76b: cmp eax,ebx
0x100ec76d: je short 100EC7B5h
0x100ec76f: movzx esi,byte ptr [edx]
0x100ec772: movzx edi,byte ptr [ecx]
0x100ec775: sub esi,edi
0x100ec777: jne short 100EC7A7h
0x100ec779: sub eax,1
0x100ec77c: add ecx,1
0x100ec77f: add edx,1
0x100ec782: cmp eax,ebx
0x100ec784: je short 100EC7B5h
0x100ec786: movzx esi,byte ptr [edx]
0x100ec789: movzx edi,byte ptr [ecx]
0x100ec78c: sub esi,edi
0x100ec78e: jne short 100EC7A7h
0x100ec790: sub eax,1
0x100ec793: add ecx,1
0x100ec796: add edx,1
0x100ec799: cmp eax,ebx
0x100ec79b: je short 100EC7B5h
0x100ec79d: movzx esi,byte ptr [edx]
0x100ec7a0: movzx eax,byte ptr [ecx]
0x100ec7a3: sub esi,eax
0x100ec7a5: je short 100EC7B5h
0x100ec7a7: cmp esi,ebx
0x100ec7a9: mov eax,1
0x100ec7ae: jg short 100EC7B7h
0x100ec7b0: or eax,0FFFFFFFFh
0x100ec7b3: jmp short 100EC7B7h
0x100ec7b5: xor eax,eax
0x100ec7b7: cmp eax,ebx
0x100ec7b9: mov eax,100D52A0h
0x100ec7be: jne short 100EC7C5h
0x100ec7c0: mov eax,100D4880h
0x100ec7c5: mov [ebp+2E7FCh],eax
0x100ec7cb: lea eax,[ebp+6485h]
0x100ec7d1: xor edx,edx
0x100ec7d3: mov edi,101298B0h
0x100ec7d8: mov esi,eax
0x100ec7da: mov ecx,6
0x100ec7df: repe cmpsb
0x100ec7e1: mov edx,100D8B50h
0x100ec7e6: jne short 100EC817h
```

```
0x100ec7e8: cmp [ebp+2E7FCh],ebx
0x100ec7ee: jne short 100EC7F6h
0x100ec7f0: mov [ebp+2E7FCh],edx
0x100ec7f6: cmp byte ptr [ebp+64C5h],45h
0x100ec7fd: jne short 100EC817h
0x100ec7ff: xor ecx,ecx
0x100ec801: cmp [ebp+65C0h],ebx
0x100ec807: sete cl
0x100ec80a: add ecx,ecx
0x100ec80c: add ecx,ecx
0x100ec80e: or ecx,2
0x100ec811: or [ebp+66ACh],ecx
0x100ec817: mov esi,eax
0x100ec819: mov edi,10129868h
0x100ec81e: mov ecx,6
0x100ec823: xor eax,eax
0x100ec825: repe cmpsb
0x100ec827: jne short 100EC839h
0x100ec829: mov [ebp+2E7FCh],edx
0x100ec82f: mov dword ptr [ebp+65FCh],0F7Fh
0x100ec839: cmp [ebp+661Ch],ebx
0x100ec83f: jne near ptr 100EF9E0h
0x100ec845: cmp [esp+10h],eax
0x100ec849: je short 100EC898h
0x100ec84b: cmp dword ptr [ebp+65F0h],0Fh
0x100ec852: jne short 100EC898h
0x100ec854: movzx eax,word ptr [ebp+66B6h]
0x100ec85b: movzx ecx,ax
0x100ec85e: cmp ecx,0D10h
0x100ec864: je short 100EC870h
0x100ec866: cmp ecx,0F20h
0x100ec86c: je short 100EC87Ah
0x100ec86e: jmp short 100EC883h
0x100ec870: add eax,0FFFFFFBEh
0x100ec873: mov [ebp+66B6h],ax
0x100ec87a: add word ptr [ebp+66B6h],0FFFAh
0x100ec883: mov [ebp+65B0h],ebx
0x100ec889: mov dword ptr [ebp+2E7FCh],100D5510h
0x100ec893: jmp near ptr 100EF9E0h
0x100ec898: lea ebx,[ebp+64C5h]
0x100ec89e: mov edi,1012F168h
0x100ec8a3: mov esi,ebx
0x100ec8a5: mov ecx,0Eh
0x100ec8aa: xor edx,edx
0x100ec8ac: repe cmpsb
0x100ec8ae: jne short 100EC8EAh
0x100ec8b0: fld qword ptr ds:[1012F160h]
0x100ec8b6: mov word ptr [ebp+66B4h],265h
0x100ec8bf: mov word ptr [ebp+66B6h],356h
0x100ec8c8: mov word ptr [ebp+66B2h],380h
0x100ec8d1: mov dword ptr [ebp+65B0h],0E1E4E1E4h
0x100ec8db: mov dword ptr [ebp+2E7FCh],100D3CD0h
0x100ec8e5: jmp near ptr 100EF9D0h
0x100ec8ea: mov edi,1012F14Ch
0x100ec8ef: mov esi,ebx
0x100ec8f1: mov ecx,0Dh
0x100ec8f6: xor eax,eax
0x100ec8f8: repe cmpsb
0x100ec8fa: je near ptr 100EF99Bh
0x100ec900: mov edi,1012F138h
0x100ec905: mov esi,ebx
0x100ec907: mov ecx,12h
0x100ec90c: xor edx,edx
0x100ec90e: repe cmpsb
0x100ec910: je near ptr 100EF99Bh
0x100ec916: mov edi,1012F128h
0x100ec91b: mov esi,ebx
```

```
0x100ec91d: mov ecx,0Eh
0x100ec922: xor eax,eax
0x100ec924: repe cmpsb
0x100ec926: jne short 100EC95Ch
0x100ec928: mov word ptr [ebp+66B4h],3C8h
0x100ec931: mov word ptr [ebp+66B6h],50Ah
0x100ec93a: mov word ptr [ebp+66B2h],528h
0x100ec943: mov dword ptr [ebp+65B0h],1B4E4B1Eh
0x100ec94d: mov dword ptr [ebp+2E7FCh],100D4350h
0x100ec957: jmp near ptr 100EF9D6h
0x100ec95c: mov edi,1012F118h
0x100ec961: mov esi,ebx
0x100ec963: mov ecx,10h
0x100ec968: xor edx,edx
0x100ec96a: repe cmpsb
0x100ec96c: jne short 100EC999h
0x100ec96e: mov word ptr [ebp+66B4h],400h
0x100ec977: mov word ptr [ebp+66B6h],610h
0x100ec980: mov dword ptr [ebp+65B0h],1E4B4E1Bh
0x100ec98a: mov dword ptr [ebp+2E7FCh],100D4350h
0x100ec994: jmp near ptr 100EF9D6h
0x100ec999: mov edi,1012F108h
0x100ec99e: mov esi,ebx
0x100ec9a0: mov ecx,10h
0x100ec9a5: xor eax,eax
0x100ec9a7: repe cmpsb
0x100ec9a9: jne short 100EC9F1h
0x100ec9ab: mov eax,0Ch
0x100ec9b0: mov word ptr [ebp+66B4h],6D8h
0x100ec9b9: mov word ptr [ebp+66B6h],928h
0x100ec9c2: mov word ptr [ebp+66B0h],6E6h
0x100ec9cb: mov word ptr [ebp+66B2h],960h
0x100ec9d4: mov [ebp+66B8h],ax
0x100ec9db: mov [ebp+66BAh],ax
0x100ec9e2: mov dword ptr [ebp+2E7FCh],100D4350h
0x100ec9ec: jmp near ptr 100EF9E0h
0x100ec9f1: mov edi,1012F0F8h
0x100ec9f6: mov esi,ebx
0x100ec9f8: mov ecx,0Fh
0x100ec9fd: xor edx,edx
0x100ec9ff: repe cmpsb
0x100eca01: jne short 100ECA49h
0x100eca03: mov eax,4
0x100eca08: mov word ptr [ebp+66B4h],7A8h
0x100eca11: mov word ptr [ebp+66B6h],0A38h
0x100eca1a: mov word ptr [ebp+66B0h],7B0h
0x100eca23: mov word ptr [ebp+66B2h],0A68h
0x100eca2c: mov [ebp+66B8h],ax
0x100eca33: mov [ebp+66BAh],ax
0x100eca3a: mov dword ptr [ebp+2E7FCh],100D4350h
0x100eca44: jmp near ptr 100EF9E0h
0x100eca49: mov edi,1012F0E8h
0x100eca4e: mov esi,ebx
0x100eca50: mov ecx,0Fh
0x100eca55: xor eax,eax
0x100eca57: repe cmpsb
0x100eca59: jne short 100ECAA6h
0x100eca5b: mov word ptr [ebp+66B4h],7C0h
0x100eca64: mov word ptr [ebp+66B6h],0A3Ch
0x100eca6d: mov word ptr [ebp+66B0h],7C8h
0x100eca76: mov word ptr [ebp+66B2h],0A70h
0x100eca7f: mov word ptr [ebp+66B8h],6
0x100eca88: mov word ptr [ebp+66BAh],0Ah
0x100eca91: mov dword ptr [ebp+2E7FCh],100D4350h
0x100eca9b: mov [ebp+6604h],eax
0x100ecaa1: jmp near ptr 100EF9E0h
0x100ecaa6: mov edi,1012F0D8h
```

```
0x100ecaab: mov esi,ebx
0x100ecaad: mov ecx,0Fh
0x100ecab2: xor edx,edx
0x100ecab4: repe cmpsb
0x100ecab6: jne short 100ECB10h
0x100ecab8: push ebp
0x100ecab9: call 100D4300h
0x100ecabe: add esp,4
0x100ecac1: test eax,eax
0x100ecac3: je short 100ECADEh
0x100ecac5: mov eax,ds:[1012F0D0h]
0x100ecaca: mov [ebp+64CFh],eax
0x100ecad0: mov cx,ds:[1012F0D4h]
0x100ecad7: mov [ebp+64D3h],cx
0x100ecade: mov word ptr [ebp+66B4h],7A8h
0x100ecae7: mov word ptr [ebp+66B6h],0A38h
0x100ecaf0: mov word ptr [ebp+66B0h],7B0h
0x100ecaf9: mov word ptr [ebp+66B2h],0A70h
0x100ecb02: mov word ptr [ebp+66B8h],8
0x100ecb0b: jmp near ptr 100ECC35h
0x100ecb10: mov edi,1012F0C0h
0x100ecb15: mov esi,ebx
0x100ecb17: mov ecx,0Fh
0x100ecb1c: xor edx,edx
0x100ecb1e: repe cmpsb
0x100ecb20: jne short 100ECB67h
0x100ecb22: mov word ptr [ebp+66B4h],918h
0x100ecb2b: mov word ptr [ebp+66B6h],0C28h
0x100ecb34: mov word ptr [ebp+66B0h],924h
0x100ecb3d: mov word ptr [ebp+66B2h],0C50h
0x100ecb46: mov word ptr [ebp+66B8h],0Ch
0x100ecb4f: mov word ptr [ebp+66BAh],24h
0x100ecb58: mov dword ptr [ebp+2E7FCh],100D4350h
0x100ecb62: jmp near ptr 100EF9E0h
0x100ecb67: mov edi,1012F0B0h
0x100ecb6c: mov esi,ebx
0x100ecb6e: mov ecx,0Fh
0x100ecb73: xor eax,eax
0x100ecb75: repe cmpsb
0x100ecb77: jne short 100ECBBEh
0x100ecb79: mov word ptr [ebp+66B4h],9A8h
0x100ecb82: mov word ptr [ebp+66B6h],0CE2h
0x100ecb8b: mov word ptr [ebp+66B0h],9B0h
0x100ecb94: mov word ptr [ebp+66B2h],0D08h
0x100ecb9d: mov word ptr [ebp+66B8h],5
0x100ecba6: mov word ptr [ebp+66BAh],6
0x100ecbaf: mov dword ptr [ebp+2E7FCh],100D4350h
0x100ecbb9: jmp near ptr 100EF9E0h
0x100ecbbe: mov edi,1012F0A0h
0x100ecbc3: mov esi,ebx
0x100ecbc5: mov ecx,0Fh
0x100ecbca: xor edx,edx
0x100ecbcc: repe cmpsb
0x100ecbce: jne short 100ECBF6h
0x100ecbd0: mov word ptr [ebp+66B4h],9A8h
0x100ecbd9: mov word ptr [ebp+66B6h],0CD8h
0x100ecbe2: mov word ptr [ebp+66B0h],9B4h
0x100ecbeb: mov word ptr [ebp+66B2h],0D10h
0x100ecbf4: jmp short 100ECC2Ch
0x100ecbf6: mov edi,1012F090h
0x100ecbfb: mov esi,ebx
0x100ecbfd: mov ecx,0Fh
0x100ecc02: xor eax,eax
0x100ecc04: repe cmpsb
0x100ecc06: jne short 100ECC4Dh
0x100ecc08: mov word ptr [ebp+66B4h],0AC8h
0x100ecc11: mov word ptr [ebp+66B6h],0E58h
```

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0x100ecc1a: mov word ptr [ebp+66B0h],0AD4h
0x100ecc23: mov word ptr [ebp+66B2h],0E98h
0x100ecc2c: mov word ptr [ebp+66B8h],6
0x100ecc35: mov word ptr [ebp+66BAh],0Ch
0x100ecc3e: mov dword ptr [ebp+2E7FCh],100D4350h
0x100ecc48: jmp near ptr 100EF9E0h
0x100ecc4d: mov edi,1012F080h
0x100ecc52: mov esi,ebx
0x100ecc54: mov ecx,0Fh
0x100ecc59: xor edx,edx
0x100ecc5b: repe cmpsb
0x100ecc5d: jne short 100ECCA4h
0x100ecc5f: mov word ptr [ebp+66B4h],0BD0h
0x100ecc68: mov word ptr [ebp+66B6h],0FC0h
0x100ecc71: mov word ptr [ebp+66B0h],0BE8h
0x100ecc7a: mov word ptr [ebp+66B2h],1008h
0x100ecc83: mov word ptr [ebp+66B8h],0Ch
0x100ecc8c: mov word ptr [ebp+66BAh],30h
0x100ecc95: mov dword ptr [ebp+2E7FCh],100D4350h
0x100ecc9f: jmp near ptr 100EF9E0h
0x100ecca4: mov edi,1012F070h
0x100ecca9: mov esi,ebx
0x100eccab: mov ecx,10h
0x100eccb0: xor eax,eax
0x100eccb2: repe cmpsb
0x100eccb4: jne short 100ECCFBh
0x100eccb6: mov word ptr [ebp+66B4h],850h
0x100eccbf: mov word ptr [ebp+66B6h],0B18h
0x100eccc8: mov word ptr [ebp+66B0h],858h
0x100eccd1: mov word ptr [ebp+66B2h],0B48h
0x100eccda: mov word ptr [ebp+66B8h],8
0x100ecce3: mov word ptr [ebp+66BAh],2Ch
0x100eccec: mov dword ptr [ebp+2E7FCh],100D4350h
0x100eccf6: jmp near ptr 100EF9E0h
0x100eccfb: mov edi,1012F05Ch
0x100ecd00: mov esi,ebx
0x100ecd02: mov ecx,13h
0x100ecd07: xor edx,edx
0x100ecd09: repe cmpsb
0x100ecd0b: jne short 100ECD25h
0x100ecd0d: mov word ptr [ebp+66B6h],768h
0x100ecd16: mov dword ptr [ebp+65B0h],0B4B4B4B4h
0x100ecd20: jmp near ptr 100EF9D6h
0x100ecd25: mov edx,[esp+10h]
0x100ecd29: test edx,edx
0x100ecd2b: je near ptr 100ECEAFh
0x100ecd31: movzx eax,word ptr [ebp+66B2h]
0x100ecd38: cmp ax,860h
0x100ecd3c: jne short 100ECD87h
0x100ecd3e: mov edi,1012F04Ch
0x100ecd43: mov esi,ebx
0x100ecd45: mov ecx,0Dh
0x100ecd4a: xor eax,eax
0x100ecd4c: mov word ptr [ebp+66B4h],60Eh
0x100ecd55: mov word ptr [ebp+66B6h],828h
0x100ecd5e: mov word ptr [ebp+66B8h],8
0x100ecd67: mov word ptr [ebp+66BAh],4
0x100ecd70: repe cmpsb
0x100ecd72: jne near ptr 100EF9E0h
0x100ecd78: mov dword ptr [ebp+65B0h],0B4B4B4B4h
0x100ecd82: jmp near ptr 100EF9D6h
0x100ecd87: cmp ax,8B0h
0x100ecd8b: jne short 100ECDB6h
0x100ecd8d: mov word ptr [ebp+66B4h],5A8h
0x100ecd96: mov word ptr [ebp+66B6h],880h
0x100ecd9f: mov word ptr [ebp+66B8h],6
0x100ecda8: mov word ptr [ebp+66BAh],30h
```

```
0x100ecdb1: jmp near ptr 100EF9E0h
0x100ecdb6: cmp ax,948h
0x100ecdba: jne short 100ECDD0h
0x100ecdbc: mov word ptr [ebp+66B4h],6B8h
0x100ecdc5: mov word ptr [ebp+66B6h],908h
0x100ecdce: jmp short 100ECDE8h
0x100ecdd0: cmp ax,0A70h
0x100ecdd4: jne short 100ECDFFh
0x100ecdd6: mov word ptr [ebp+66B4h],7A8h
0x100ecddf: mov word ptr [ebp+66B6h],0A38h
0x100ecde8: mov word ptr [ebp+66B8h],6
0x100ecdf1: mov word ptr [ebp+66BAh],0Ch
0x100ecdfa: jmp near ptr 100EF9E0h
0x100ecdff: cmp ax,0C50h
0x100ece03: jne short 100ECE51h
0x100ece05: cmp dword ptr [ebp+65B8h],80000170h
0x100ece0f: mov word ptr [ebp+66B4h],808h
0x100ece18: mov word ptr [ebp+66B6h],0C10h
0x100ece21: mov word ptr [ebp+66B8h],0Ch
0x100ece2a: mov word ptr [ebp+66BAh],40h
0x100ece33: jne near ptr 100EF9E0h
0x100ece39: push 1012F040h
0x100ece3e: push 101298B8h
0x100ece43: push ebp
0x100ece44: call 100E64C0h
0x100ece49: add esp,0Ch
0x100ece4c: jmp near ptr 100EF9E0h
0x100ece51: cmp ax,0C58h
0x100ece55: jne short 100ECE80h
0x100ece57: mov word ptr [ebp+66B4h],918h
0x100ece60: mov word ptr [ebp+66B6h],0C28h
0x100ece69: mov word ptr [ebp+66B8h],0Ch
0x100ece72: mov word ptr [ebp+66BAh],2Ch
0x100ece7b: jmp near ptr 100EF9E0h
0x100ece80: cmp ax,0D10h
0x100ece84: jne short 100ECEAFh
0x100ece86: mov word ptr [ebp+66B4h],9A8h
0x100ece8f: mov word ptr [ebp+66B6h],0CD8h
0x100ece98: mov word ptr [ebp+66B8h],6
0x100ecea1: mov word ptr [ebp+66BAh],4
0x100eceaa: jmp near ptr 100EF9E0h
0x100eceaf: mov edi,1012F034h
0x100eceb4: mov esi,ebx
0x100eceb6: mov ecx,0Bh
0x100ecebb: xor eax,eax
0x100ecebd: repe cmpsb
0x100ecebf: jne short 100ECEDDh
0x100ecec1: movzx ecx,word ptr [ebp+68F4h]
0x100ecec8: mov dword ptr [ebp+65B0h],61616161h
0x100eced2: mov [ebp+65F8h],ecx
0x100eced8: jmp near ptr 100EF9E0h
0x100ecedd: test edx,edx
0x100ecedf: je near ptr 100ED1A6h
0x100ecee5: movzx eax,word ptr [ebp+66B2h]
0x100eceec: cmp ax,0DBCh
0x100ecef0: jne short 100ECF2Ch
0x100ecef2: cmp dword ptr [ebp+65B8h],80000189h
0x100ecefc: mov word ptr [ebp+66B8h],0Eh
0x100ecf05: mov word ptr [ebp+66BAh],2Ah
0x100ecf0e: jne near ptr 100ED156h
0x100ecf14: push 1012F028h
0x100ecf19: push 101298B8h
0x100ecf1e: push ebp
0x100ecf1f: call 100E64C0h
0x100ecf24: add esp,0Ch
0x100ecf27: jmp near ptr 100ED156h
0x100ecf2c: cmp ax,0E0Ch
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```
0x100ecf30: jne short 100ECF49h
0x100ecf32: mov word ptr [ebp+66B8h],0Ch
0x100ecf3b: mov word ptr [ebp+66BAh],4Ah
0x100ecf44: jmp near ptr 100ED156h
0x100ecf49: cmp ax,0F68h
0x100ecf4d: jne short 100ECF78h
0x100ecf4f: mov word ptr [ebp+66B4h],0A2Ah
0x100ecf58: mov word ptr [ebp+66B6h],0F44h
0x100ecf61: mov word ptr [ebp+66B8h],12h
0x100ecf6a: mov word ptr [ebp+66BAh],1Eh
0x100ecf73: jmp near ptr 100EF9E0h
0x100ecf78: cmp ax,0F6Ch
0x100ecf7c: jne short 100ECFE0h
0x100ecf7e: add word ptr [ebp+66B4h],0FFFEh
0x100ecf87: cmp dword ptr [ebp+65B8h],80000236h
0x100ecf91: mov word ptr [ebp+66B8h],12h
0x100ecf9a: mov word ptr [ebp+66BAh],2Ah
0x100ecfa3: jne short 100ECFB8h
0x100ecfa5: push 1012F01Ch
0x100ecfaa: push 101298B8h
0x100ecfaf: push ebp
0x100ecfb0: call 100E64C0h
0x100ecfb5: add esp,0Ch
0x100ecfb8: cmp dword ptr [ebp+65B8h],80000254h
0x100ecfc2: jne near ptr 100ED156h
0x100ecfc8: push 1012F010h
0x100ecfcd: push 101298B8h
0x100ecfd2: push ebp
0x100ecfd3: call 100E64C0h
0x100ecfd8: add esp,0Ch
0x100ecfdb: jmp near ptr 100ED156h
0x100ecfe0: cmp ax,0F90h
0x100ecfe4: jne short 100ED006h
0x100ecfe6: add word ptr [ebp+66B4h],0FFFEh
0x100ecfef: mov word ptr [ebp+66B8h],14h
0x100ecff8: mov word ptr [ebp+66BAh],4Ch
0x100ed001: jmp near ptr 100ED156h
0x100ed006: cmp ax,1008h
0x100ed00a: jne short 100ED035h
0x100ed00c: mov word ptr [ebp+66B4h],0BD0h
0x100ed015: mov word ptr [ebp+66B6h],0FC0h
0x100ed01e: mov word ptr [ebp+66B8h],0Ch
0x100ed027: mov word ptr [ebp+66BAh],30h
0x100ed030: jmp near ptr 100EF9E0h
0x100ed035: cmp ax,10D8h
0x100ed039: jne short 100ED07Eh
0x100ed03b: add word ptr [ebp+66B4h],0FFFEh
0x100ed044: cmp dword ptr [ebp+65B8h],80000176h
0x100ed04e: mov word ptr [ebp+66B8h],12h
0x100ed057: mov word ptr [ebp+66BAh],16h
0x100ed060: jne near ptr 100ED156h
0x100ed066: push 1012F004h
0x100ed06b: push 101298B8h
0x100ed070: push ebp
0x100ed071: call 100E64C0h
0x100ed076: add esp,0Ch
0x100ed079: jmp near ptr 100ED156h
0x100ed07e: cmp ax,117Ch
0x100ed082: jne short 100ED09Bh
0x100ed084: mov word ptr [ebp+66B8h],22h
0x100ed08d: mov word ptr [ebp+66BAh],5Ah
0x100ed096: jmp near ptr 100ED156h
0x100ed09b: cmp ax,1180h
0x100ed09f: jne short 100ED0D4h
0x100ed0a1: mov word ptr [ebp+66B4h],0CFEh
0x100ed0aa: mov word ptr [ebp+66B6h],1150h
0x100ed0b3: mov word ptr [ebp+66B8h],0Ah
```

```
0x100ed0bc: mov word ptr [ebp+66BAh],0Ch
0x100ed0c5: mov dword ptr [ebp+65B0h],49494949h
0x100ed0cf: jmp near ptr 100EF9E0h
0x100ed0d4: cmp ax,4B8h
0x100ed0d8: jne short 100ED109h
0x100ed0da: mov ax,[ebp+66B6h]
0x100ed0e1: add ax,ax
0x100ed0e4: add ax,ax
0x100ed0e7: mov word ptr [ebp+66B8h],33h
0x100ed0f0: mov word ptr [ebp+66BAh],3Eh
0x100ed0f9: mov [ebp+66B6h],ax
0x100ed100: mov [ebp+66B2h],ax
0x100ed107: jmp short 100ED156h
0x100ed109: cmp ax,5A8h
0x100ed10d: jne short 100ED13Eh
0x100ed10f: mov ax,[ebp+66B6h]
0x100ed116: add ax,ax
0x100ed119: add ax,ax
0x100ed11c: mov word ptr [ebp+66B8h],33h
0x100ed125: mov word ptr [ebp+66BAh],9Eh
0x100ed12e: mov [ebp+66B6h],ax
0x100ed135: mov [ebp+66B2h],ax
0x100ed13c: jmp short 100ED156h
0x100ed13e: cmp ax,13F4h
0x100ed142: jne short 100ED177h
0x100ed144: mov word ptr [ebp+66B8h],0Dh
0x100ed14d: mov word ptr [ebp+66BAh],62h
0x100ed156: mov dx,[ebp+66B8h]
0x100ed15d: mov ax,[ebp+66BAh]
0x100ed164: sub [ebp+66B4h],dx
0x100ed16b: sub [ebp+66B6h],ax
0x100ed172: jmp near ptr 100EF9E0h
0x100ed177: cmp ax,1650h
0x100ed17b: jne short 100ED1A6h
0x100ed17d: mov word ptr [ebp+66B4h],0EA8h
0x100ed186: mov word ptr [ebp+66B6h],1608h
0x100ed18f: mov word ptr [ebp+66B8h],14h
0x100ed198: mov word ptr [ebp+66BAh],3Eh
0x100ed1a1: jmp near ptr 100EF9E0h
0x100ed1a6: mov edi,1012F000h
0x100ed1ab: mov esi,ebx
0x100ed1ad: mov ecx,3
0x100ed1b2: xor edx,edx
0x100ed1b4: repe cmpsb
0x100ed1b6: jne short 100ED1E1h
0x100ed1b8: fld dword ptr [ebp+0E774h]
0x100ed1be: fmul qword ptr ds:[1012EFF8h]
0x100ed1c4: fstp dword ptr [ebp+0E774h]
0x100ed1ca: fld dword ptr [ebp+0E77Ch]
0x100ed1d0: fmul qword ptr ds:[1012EFF0h]
0x100ed1d6: fstp dword ptr [ebp+0E77Ch]
0x100ed1dc: jmp near ptr 100EF9E0h
0x100ed1e1: mov edi,1012EFECh
0x100ed1e6: mov esi,ebx
0x100ed1e8: mov ecx,4
0x100ed1ed: xor eax,eax
0x100ed1ef: repe cmpsb
0x100ed1f1: jne short 100ED20Dh
0x100ed1f3: fld qword ptr ds:[100FE7C8h]
0x100ed1f9: add word ptr [ebp+66B6h],0FFFCh
0x100ed202: fstp qword ptr [ebp+66D8h]
0x100ed208: jmp near ptr 100EF9E0h
0x100ed20d: mov edi,1012EFE4h
0x100ed212: mov esi,ebx
0x100ed214: mov ecx,5
0x100ed219: xor edx,edx
0x100ed21b: repe cmpsb
```

```
0x100ed21d: je near ptr 100EF987h
0x100ed223: mov edi,1012EFE0h
0x100ed228: mov esi,ebx
0x100ed22a: mov ecx,4
0x100ed22f: xor eax,eax
0x100ed231: repe cmpsb
0x100ed233: je near ptr 100EF987h
0x100ed239: mov edi,1012EFDCh
0x100ed23e: mov esi,ebx
0x100ed240: mov ecx,4
0x100ed245: xor edx,edx
0x100ed247: repe cmpsb
0x100ed249: je near ptr 100EF987h
0x100ed24f: mov edi,1012EFD8h
0x100ed254: mov esi,ebx
0x100ed256: mov ecx,3
0x100ed25b: xor eax,eax
0x100ed25d: repe cmpsb
0x100ed25f: je near ptr 100EF973h
0x100ed265: mov edi,1012EFD0h
0x100ed26a: mov esi,ebx
0x100ed26c: mov ecx,5
0x100ed271: xor edx,edx
0x100ed273: repe cmpsb
0x100ed275: je near ptr 100EF973h
0x100ed27b: mov esi,ds:[100FE414h]
0x100ed281: push 3
0x100ed283: push 1012EFCCh
0x100ed288: push ebx
0x100ed289: call esi
0x100ed28b: add esp,0Ch
0x100ed28e: test eax,eax
0x100ed290: je near ptr 100EF968h
0x100ed296: push 3
0x100ed298: push 1012EFC8h
0x100ed29d: push ebx
0x100ed29e: call esi
0x100ed2a0: add esp,0Ch
0x100ed2a3: test eax,eax
0x100ed2a5: je near ptr 100EF968h
0x100ed2ab: push 3
0x100ed2ad: push 1012EFC4h
0x100ed2b2: push ebx
0x100ed2b3: call esi
0x100ed2b5: add esp,0Ch
0x100ed2b8: test eax,eax
0x100ed2ba: je near ptr 100EF968h
0x100ed2c0: mov edi,1012EFC0h
0x100ed2c5: mov esi,ebx
0x100ed2c7: mov ecx,4
0x100ed2cc: xor eax,eax
0x100ed2ce: repe cmpsb
0x100ed2d0: jne short 100ED2E0h
0x100ed2d2: add word ptr [ebp+66B6h],0FFD6h
0x100ed2db: jmp near ptr 100EF9E0h
0x100ed2e0: mov edi,1012EFB8h
0x100ed2e5: mov esi,ebx
0x100ed2e7: mov ecx,5
0x100ed2ec: xor edx,edx
0x100ed2ee: repe cmpsb
0x100ed2f0: jne short 100ED342h
0x100ed2f2: cmp dword ptr [ebp+65F4h],8799h
0x100ed2fc: jne near ptr 100EF9E0h
0x100ed302: push ebp
0x100ed303: call 100D64F0h
0x100ed308: add esp,4
0x100ed30b: test eax,eax
```

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0x100ed30d: jne near ptr 100EF9E0h
0x100ed313: add word ptr [ebp+66B6h],3
0x100ed31b: movzx eax,word ptr [ebp+66B6h]
0x100ed322: or dword ptr [ebp+66ACh],8
0x100ed329: add eax,3
0x100ed32c: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100ed336: mov [ebp+66B2h],ax
0x100ed33d: jmp near ptr 100EF9E0h
0x100ed342: mov edi,1012EFB0h
0x100ed347: mov esi,ebx
0x100ed349: mov ecx,5
0x100ed34e: xor eax,eax
0x100ed350: repe cmpsb
0x100ed352: jne short 100ED375h
0x100ed354: add word ptr [ebp+66B6h],0FFFCh
0x100ed35d: mov word ptr [ebp+66BAh],1
0x100ed366: mov dword ptr [ebp+65B0h],94949494h
0x100ed370: jmp near ptr 100EF9E0h
0x100ed375: mov esi,ds:[100FE414h]
0x100ed37b: push 3
0x100ed37d: push 1012EFACh
0x100ed382: push ebx
0x100ed383: call esi
0x100ed385: add esp,0Ch
0x100ed388: test eax,eax
0x100ed38a: jne short 100ED3A3h
0x100ed38c: add word ptr [ebp+66B6h],0FFF2h
0x100ed395: mov word ptr [ebp+66BAh],6
0x100ed39e: jmp near ptr 100EF9E0h
0x100ed3a3: push 3
0x100ed3a5: push 1012EFA8h
0x100ed3aa: push ebx
0x100ed3ab: call esi
0x100ed3ad: add esp,0Ch
0x100ed3b0: test eax,eax
0x100ed3b2: jne short 100ED3DEh
0x100ed3b4: movzx eax,word ptr [ebp+66B6h]
0x100ed3bb: cmp ax,0CC0h
0x100ed3bf: jne short 100ED3CFh
0x100ed3c1: mov word ptr [ebp+66B6h],0CA0h
0x100ed3ca: jmp near ptr 100EF9E0h
0x100ed3cf: add eax,0FFFFFFF8h
0x100ed3d2: mov [ebp+66B6h],ax
0x100ed3d9: jmp near ptr 100EF9E0h
0x100ed3de: mov edi,1012EFA0h
0x100ed3e3: mov esi,ebx
0x100ed3e5: mov ecx,5
0x100ed3ea: xor edx,edx
0x100ed3ec: repe cmpsb
0x100ed3ee: jne short 100ED3FEh
0x100ed3f0: add word ptr [ebp+66B6h],0FFE0h
0x100ed3f9: jmp near ptr 100EF9E0h
0x100ed3fe: mov edi,1012EF90h
0x100ed403: mov esi,ebx
0x100ed405: mov ecx,0Eh
0x100ed40a: xor eax,eax
0x100ed40c: repe cmpsb
0x100ed40e: jne short 100ED429h
0x100ed410: mov dword ptr [ebp+66ACh],1
0x100ed41a: mov dword ptr [ebp+65B0h],94949494h
0x100ed424: jmp near ptr 100EF9E0h
0x100ed429: mov eax,[esp+14h]
0x100ed42d: cmp eax,182004h
0x100ed432: jne short 100ED4ABh
0x100ed434: push 3
0x100ed436: push ebp
0x100ed437: mov word ptr [ebp+66B4h],3C3h
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0x100ed440: mov word ptr [ebp+66B6h],507h
0x100ed449: mov word ptr [ebp+66B2h],660h
0x100ed452: mov dword ptr [ebp+2E7FCh],100D69B0h
0x100ed45c: mov dword ptr [ebp+65FCh],3F4h
0x100ed466: mov dword ptr [ebp+66D0h],4
0x100ed470: mov dword ptr [ebp+65B0h],1E1E1E1Eh
0x100ed47a: call 100E6630h
0x100ed47f: fld dword ptr ds:[1012EF8Ch]
0x100ed485: fstp dword ptr [ebp+0E784h]
0x100ed48b: add esp,8
0x100ed48e: fld dword ptr ds:[1012EF88h]
0x100ed494: fstp dword ptr [ebp+0E788h]
0x100ed49a: fld dword ptr ds:[1012EF84h]
0x100ed4a0: fstp dword ptr [ebp+0E790h]
0x100ed4a6: jmp near ptr 100EF9E0h
0x100ed4ab: cmp eax,25A000h
0x100ed4b0: jne short 100ED503h
0x100ed4b2: push 1012EF7Ch
0x100ed4b7: push 101298B0h
0x100ed4bc: push ebp
0x100ed4bd: mov word ptr [ebp+66B4h],4B3h
0x100ed4c6: mov word ptr [ebp+66B6h],650h
0x100ed4cf: mov word ptr [ebp+66B2h],800h
0x100ed4d8: mov dword ptr [ebp+2E7FCh],100D69B0h
0x100ed4e2: mov dword ptr [ebp+66D0h],4
0x100ed4ec: mov dword ptr [ebp+65B0h],4B4B4B4Bh
0x100ed4f6: call 100E64C0h
0x100ed4fb: add esp,0Ch
0x100ed4fe: jmp near ptr 100EF9E0h
0x100ed503: cmp eax,48D000h
0x100ed508: jne near ptr 100ED5C7h
0x100ed50e: mov ecx,[ebp+65A0h]
0x100ed514: or ecx,[ebp+65A4h]
0x100ed51a: mov word ptr [ebp+66B4h],604h
0x100ed523: mov word ptr [ebp+66B6h],810h
0x100ed52c: mov dword ptr [ebp+66D0h],4
0x100ed536: mov dword ptr [ebp+65B0h],0E1E1E1E1h
0x100ed540: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100ed54a: mov dword ptr [ebp+66ACh],6
0x100ed554: jne short 100ED573h
0x100ed556: push ebp
0x100ed557: call 100D6590h
0x100ed55c: add esp,4
0x100ed55f: test eax,eax
0x100ed561: je short 100ED573h
0x100ed563: mov edx,ds:[1012EF74h]
0x100ed569: mov [ebx],edx
0x100ed56b: mov al,ds:[1012EF78h]
0x100ed570: mov [ebx+4],al
0x100ed573: mov edi,1012EF74h
0x100ed578: mov esi,ebx
0x100ed57a: mov ecx,5
0x100ed57f: xor edx,edx
0x100ed581: repe cmpsb
0x100ed583: je near ptr 100EF9E0h
0x100ed589: push 3
0x100ed58b: push ebp
0x100ed58c: mov dword ptr [ebp+65B0h],0B4B4B4B4h
0x100ed596: call 100E6630h
0x100ed59b: fld dword ptr ds:[1012EF70h]
0x100ed5a1: fstp dword ptr [ebp+0E784h]
0x100ed5a7: add esp,8
0x100ed5aa: fld dword ptr ds:[1012EF6Ch]
0x100ed5b0: fstp dword ptr [ebp+0E788h]
0x100ed5b6: fld dword ptr ds:[1012EF68h]
0x100ed5bc: fstp dword ptr [ebp+0E78Ch]
0x100ed5c2: jmp near ptr 100EF9E0h
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0x100ed5c7: mov edi,1012EF60h
0x100ed5cc: mov esi,ebx
0x100ed5ce: mov ecx,6
0x100ed5d3: xor eax,eax
0x100ed5d5: repe cmpsb
0x100ed5d7: jne short 100ED615h
0x100ed5d9: mov ecx,[ebp+65A0h]
0x100ed5df: or ecx,[ebp+65A4h]
0x100ed5e5: jne short 100ED5F4h
0x100ed5e7: push ebp
0x100ed5e8: call 100D6630h
0x100ed5ed: add esp,4
0x100ed5f0: test eax,eax
0x100ed5f2: je short 100ED627h
0x100ed5f4: mov word ptr [ebp+66B4h],4B6h
0x100ed5fd: mov word ptr [ebp+66B6h],650h
0x100ed606: mov dword ptr [ebp+66ACh],7
0x100ed610: jmp near ptr 100EF9E0h
0x100ed615: mov edi,1012EF58h
0x100ed61a: mov esi,ebx
0x100ed61c: mov ecx,6
0x100ed621: xor edx,edx
0x100ed623: repe cmpsb
0x100ed625: jne short 100ED65Ah
0x100ed627: mov eax,ds:[1012EF58h]
0x100ed62c: mov [ebx],eax
0x100ed62e: mov cx,ds:[1012EF5Ch]
0x100ed635: mov [ebx+4],cx
0x100ed639: mov word ptr [ebp+66B4h],4B4h
0x100ed642: mov word ptr [ebp+66B6h],650h
0x100ed64b: mov dword ptr [ebp+65B0h],4B4B4B4Bh
0x100ed655: jmp near ptr 100EF9D6h
0x100ed65a: mov eax,[esp+14h]
0x100ed65e: cmp eax,48E000h
0x100ed663: jne near ptr 100ED72Bh
0x100ed669: fld dword ptr ds:[1012EF54h]
0x100ed66f: mov edx,[ebp+65A0h]
0x100ed675: or edx,[ebp+65A4h]
0x100ed67b: fstp dword ptr [ebp+0E784h]
0x100ed681: fld dword ptr ds:[1012EF50h]
0x100ed687: mov word ptr [ebp+66B4h],606h
0x100ed690: fstp dword ptr [ebp+0E78Ch]
0x100ed696: mov word ptr [ebp+66B6h],810h
0x100ed69f: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100ed6a9: mov dword ptr [ebp+66ACh],7
0x100ed6b3: jne short 100ED6BEh
0x100ed6b5: push ebp
0x100ed6b6: call 100D66E0h
0x100ed6bb: add esp,4
0x100ed6be: cmp byte ptr [ebx],45h
0x100ed6c1: jne short 100ED6E4h
0x100ed6c3: lea eax,[ebp+64C6h]
0x100ed6c9: push eax
0x100ed6ca: call dword ptr ds:[100FE458h]
0x100ed6d0: add esp,4
0x100ed6d3: cmp eax,0E74h
0x100ed6d8: jge short 100ED6E4h
0x100ed6da: mov dword ptr [ebp+65B0h],49494949h
0x100ed6e4: mov edi,1012E090h
0x100ed6e9: mov esi,ebx
0x100ed6eb: mov ecx,0Bh
0x100ed6f0: xor edx,edx
0x100ed6f2: repe cmpsb
0x100ed6f4: jne near ptr 100EF9E0h
0x100ed6fa: fld dword ptr ds:[1012EF4Ch]
0x100ed700: mov dword ptr [ebp+66C8h],1
0x100ed70a: fstp dword ptr [ebp+0E784h]
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0x100ed710: mov dword ptr [ebp+65B0h],16161616h
0x100ed71a: fld dword ptr ds:[1012EF48h]
0x100ed720: fstp dword ptr [ebp+0E78Ch]
0x100ed726: jmp near ptr 100EF9E0h
0x100ed72b: cmp eax,599000h
0x100ed730: jne near ptr 100ED7C4h
0x100ed736: mov eax,[ebp+65A0h]
0x100ed73c: or eax,[ebp+65A4h]
0x100ed742: mov word ptr [ebp+66B4h],6AEh
0x100ed74b: mov word ptr [ebp+66B6h],8F0h
0x100ed754: mov dword ptr [ebp+65B0h],16161616h
0x100ed75e: jne short 100ED7A0h
0x100ed760: push ebp
0x100ed761: call 100D6900h
0x100ed766: add esp,4
0x100ed769: test eax,eax
0x100ed76b: je short 100ED7A0h
0x100ed76d: mov ecx,ds:[10129878h]
0x100ed773: mov [ebp+6485h],ecx
0x100ed779: mov edx,ds:[1012987Ch]
0x100ed77f: mov [ebp+6489h],edx
0x100ed785: mov eax,ds:[1012EF3Ch]
0x100ed78a: mov [ebx],eax
0x100ed78c: mov ecx,ds:[1012EF40h]
0x100ed792: mov [ebx+4],ecx
0x100ed795: mov dx,ds:[1012EF44h]
0x100ed79c: mov [ebx+8],dx
0x100ed7a0: xor eax,eax
0x100ed7a2: cmp byte ptr [ebp+6485h],4Dh
0x100ed7a9: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100ed7b3: sete al
0x100ed7b6: add eax,6
0x100ed7b9: mov [ebp+66ACh],eax
0x100ed7bf: jmp near ptr 100EF9E0h
0x100ed7c4: mov edi,1012EF34h
0x100ed7c9: mov esi,ebx
0x100ed7cb: mov ecx,6
0x100ed7d0: xor edx,edx
0x100ed7d2: repe cmpsb
0x100ed7d4: jne short 100ED7F7h
0x100ed7d6: mov word ptr [ebp+66B4h],6ACh
0x100ed7df: mov word ptr [ebp+66B6h],8F0h
0x100ed7e8: mov dword ptr [ebp+65B0h],0B4B4B4B4h
0x100ed7f2: jmp near ptr 100EF9D6h
0x100ed7f7: cmp eax,718000h
0x100ed7fc: jne short 100ED81Fh
0x100ed7fe: mov word ptr [ebp+66B4h],784h
0x100ed807: mov word ptr [ebp+66B6h],0A10h
0x100ed810: mov dword ptr [ebp+65B0h],0B4B4B4B4h
0x100ed81a: jmp near ptr 100EF9D6h
0x100ed81f: cmp eax,895000h
0x100ed824: jne short 100ED85Bh
0x100ed826: mov word ptr [ebp+66B4h],846h
0x100ed82f: mov word ptr [ebp+66B6h],0B10h
0x100ed838: mov dword ptr [ebp+65FCh],0F83h
0x100ed842: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100ed84c: mov dword ptr [ebp+66ACh],7
0x100ed856: jmp near ptr 100EF9E0h
0x100ed85b: mov edi,1012EF24h
0x100ed860: mov esi,ebx
0x100ed862: mov ecx,0Eh
0x100ed867: xor eax,eax
0x100ed869: repe cmpsb
0x100ed86b: je near ptr 100EE10Ch
0x100ed871: mov edi,1012EF14h
0x100ed876: mov esi,ebx
0x100ed878: mov ecx,0Eh
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0x100ed87d: xor edx,edx
0x100ed87f: repe cmpsb
0x100ed881: je near ptr 100EE10Ch
0x100ed887: mov edi,1012E57Ch
0x100ed88c: lea esi,[ebp+6485h]
0x100ed892: mov ecx,9
0x100ed897: xor eax,eax
0x100ed899: repe cmpsb
0x100ed89b: lea ecx,[edx+6]
0x100ed89e: jne near ptr 100EDA00h
0x100ed8a4: lea eax,[ebp+64CCh]
0x100ed8aa: mov edi,1012EF0Ch
0x100ed8af: mov esi,eax
0x100ed8b1: xor edx,edx
0x100ed8b3: repe cmpsb
0x100ed8b5: jne short 100ED8F1h
0x100ed8b7: mov ecx,ds:[1012EF04h]
0x100ed8bd: mov [eax],ecx
0x100ed8bf: mov dx,ds:[1012EF08h]
0x100ed8c6: mov [eax+4],dx
0x100ed8ca: mov cl,ds:[1012EF0Ah]
0x100ed8d0: mov [eax+6],cl
0x100ed8d3: mov word ptr [ebp+66B4h],860h
0x100ed8dc: mov word ptr [ebp+66B6h],0B40h
0x100ed8e5: mov dword ptr [ebp+66C8h],6
0x100ed8ef: jmp short 100ED8FBh
0x100ed8f1: mov dword ptr [ebp+65FCh],3E00h
0x100ed8fb: cmp dword ptr [ebp+6614h],2
0x100ed902: jne short 100ED917h
0x100ed904: cmp dword ptr [ebp+0A0h],0
0x100ed90b: je short 100ED917h
0x100ed90d: mov dword ptr [ebp+65FCh],2F00h
0x100ed917: movzx edi,word ptr [ebp+66B0h]
0x100ed91e: movzx edx,word ptr [ebp+66B4h]
0x100ed925: movzx esi,word ptr [ebp+66B2h]
0x100ed92c: movzx ebx,word ptr [ebp+66B6h]
0x100ed933: mov eax,edi
0x100ed935: sub eax,edx
0x100ed937: cdq
0x100ed938: sub eax,edx
0x100ed93a: sar eax,1
0x100ed93c: mov [ebp+66B8h],ax
0x100ed943: movzx ecx,si
0x100ed946: movzx edx,bx
0x100ed949: mov eax,ecx
0x100ed94b: sub eax,edx
0x100ed94d: cdq
0x100ed94e: sub eax,edx
0x100ed950: sar eax,1
0x100ed952: cmp dword ptr [ebp+6614h],2
0x100ed959: mov [ebp+66BAh],ax
0x100ed960: jne short 100ED993h
0x100ed962: cmp dword ptr [ebp+65E4h],0
0x100ed969: je short 100ED979h
0x100ed96b: add esi,esi
0x100ed96d: mov [ebp+66B2h],si
0x100ed974: movzx ecx,si
0x100ed977: jmp short 100ED97Eh
0x100ed979: imul ecx,edi
0x100ed97c: add ecx,ecx
0x100ed97e: xor eax,eax
0x100ed980: cmp eax,[ebp+0A0h]
0x100ed986: sbb edx,edx
0x100ed988: neg edx
0x100ed98a: imul edx,ecx
0x100ed98d: add [ebp+65C0h],edx
0x100ed993: movzx eax,word ptr [ebp+66B4h]
```

```
0x100ed99a: xor ecx,ecx
0x100ed99c: cmp [ebp+65E4h],ecx
0x100ed9a2: mov dx,bx
0x100ed9a5: sete cl
0x100ed9a8: shr dx,cl
0x100ed9ab: movzx ecx,word ptr [ebp+65E4h]
0x100ed9b2: shr ax,cl
0x100ed9b5: movzx ecx,word ptr [ebp+66B4h]
0x100ed9bc: add ax,dx
0x100ed9bf: mov [ebp+66B6h],ax
0x100ed9c6: mov [ebp+66C2h],dx
0x100ed9cd: mov [ebp+66B0h],cx
0x100ed9d4: mov dword ptr [ebp+2E7FCh],100D6AC0h
0x100ed9de: add eax,0FFFFFFFFh
0x100ed9e1: test dl,1
0x100ed9e4: mov [ebp+66B4h],ax
0x100ed9eb: jne near ptr 100EF9E0h
0x100ed9f1: mov dword ptr [ebp+65B0h],49494949h
0x100ed9fb: jmp near ptr 100EF9E0h
0x100eda00: mov edi,1012EEFCh
0x100eda05: mov esi,ebx
0x100eda07: xor edx,edx
0x100eda09: repe cmpsb
0x100eda0b: jne short 100EDA42h
0x100eda0d: mov word ptr [ebp+66B4h],3DAh
0x100eda16: mov word ptr [ebp+66B6h],5FEh
0x100eda1f: mov dword ptr [ebp+65C0h],201h
0x100eda29: mov dword ptr [ebp+65B0h],61616161h
0x100eda33: mov dword ptr [ebp+2E7FCh],100D9A60h
0x100eda3d: jmp near ptr 100EF9E0h
0x100eda42: mov edi,1012EEF4h
0x100eda47: mov esi,ebx
0x100eda49: mov ecx,8
0x100eda4e: xor eax,eax
0x100eda50: repe cmpsb
0x100eda52: jne short 100EDA74h
0x100eda54: mov word ptr [ebp+66B4h],6B0h
0x100eda5d: mov word ptr [ebp+66B6h],908h
0x100eda66: mov word ptr [ebp+66B2h],920h
0x100eda6f: jmp near ptr 100EDBADh
0x100eda74: mov edi,1012EEECh
0x100eda79: mov esi,ebx
0x100eda7b: mov ecx,8
0x100eda80: xor edx,edx
0x100eda82: repe cmpsb
0x100eda84: je near ptr 100EDB69h
0x100eda8a: push 10129880h
0x100eda8f: lea edi,[ebp+6485h]
0x100eda95: push edi
0x100eda96: call dword ptr ds:[100FE440h]
0x100eda9c: add esp,8
0x100eda9f: test eax,eax
0x100edaa1: jne near ptr 100EDC0Bh
0x100edaa7: mov esi,ds:[100FE414h]
0x100edaad: push 8
0x100edaaf: push 1012EEE0h
0x100edab4: push ebx
0x100edab5: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100edabf: call esi
0x100edac1: add esp,0Ch
0x100edac4: test eax,eax
0x100edac6: jne short 100EDAF3h
0x100edac8: mov edi,1012EA78h
0x100edacd: mov esi,ebx
0x100edacf: mov ecx,0Ch
0x100edad4: xor eax,eax
0x100edad6: repe cmpsb
```

```
0x100edad8: jne short 100EDAE4h
0x100edada: mov dword ptr [ebp+65B0h],49494949h
0x100edae4: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100edaee: jmp near ptr 100EF9E0h
0x100edaf3: push 5
0x100edaf5: push 1012EED8h
0x100edafa: push ebx
0x100edafb: call esi
0x100edafd: add esp,0Ch
0x100edb00: test eax,eax
0x100edb02: je near ptr 100EDBCFh
0x100edb08: push 5
0x100edb0a: push 1012EED0h
0x100edb0f: push ebx
0x100edb10: call esi
0x100edb12: add esp,0Ch
0x100edb15: test eax,eax
0x100edb17: je near ptr 100EDBCFh
0x100edb1d: push 6
0x100edb1f: push 1012EEC8h
0x100edb24: push ebx
0x100edb25: call esi
0x100edb27: add esp,0Ch
0x100edb2a: test eax,eax
0x100edb2c: je near ptr 100EDBCFh
0x100edb32: push 8
0x100edb34: push 1012EEBCh
0x100edb39: push ebx
0x100edb3a: call esi
0x100edb3c: add esp,0Ch
0x100edb3f: test eax,eax
0x100edb41: jne near ptr 100EF9E0h
0x100edb47: mov al,[ebp+64CDh]
0x100edb4d: cmp al,34h
0x100edb4f: jne short 100EDB65h
0x100edb51: mov word ptr [ebp+66B4h],6B4h
0x100edb5a: mov word ptr [ebp+66B6h],900h
0x100edb63: jmp short 100EDB9Ch
0x100edb65: cmp al,35h
0x100edb67: jne short 100EDB86h
0x100edb69: mov word ptr [ebp+66B4h],7A4h
0x100edb72: mov word ptr [ebp+66B6h],0A2Fh
0x100edb7b: mov word ptr [ebp+66B2h],0A40h
0x100edb84: jmp short 100EDB9Ch
0x100edb86: cmp al,36h
0x100edb88: jne short 100EDB9Ch
0x100edb8a: mov word ptr [ebp+66B4h],858h
0x100edb93: mov word ptr [ebp+66B6h],0B20h
0x100edb9c: add dword ptr [ebp+65C0h],0Eh
0x100edba3: mov dword ptr [ebp+65B0h],61616161h
0x100edbad: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100edbb7: mov dword ptr [ebp+65FCh],3DFh
0x100edbc1: mov word ptr [ebp+6274h],4D4Dh
0x100edbca: jmp near ptr 100EF9E0h
0x100edbcf: xor ecx,ecx
0x100edbd1: cmp byte ptr [ebx],4Dh
0x100edbd4: lea esi,[ebp+64D9h]
0x100edbda: sete cl
0x100edbdd: lea edx,[ecx+ebp+64CBh]
0x100edbe4: push edx
0x100edbe5: push 1012EEB0h
0x100edbea: push esi
0x100edbeb: call dword ptr ds:[100FE4F8h]
0x100edbf1: push esi
0x100edbf2: push edi
0x100edbf3: push ebp
0x100edbf4: call 100E64C0h
```

```
0x100edbf9: add esp,18h
0x100edbfc: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100edc06: jmp near ptr 100EF9E0h
0x100edc0b: mov edi,1012EEA8h
0x100edc10: mov esi,ebx
0x100edc12: mov ecx,8
0x100edc17: xor eax,eax
0x100edc19: repe cmpsb
0x100edc1b: jne short 100EDC2Bh
0x100edc1d: add word ptr [ebp+66B4h],0FFFEh
0x100edc26: jmp near ptr 100EF9E0h
0x100edc2b: mov edi,1012EEA0h
0x100edc30: mov esi,ebx
0x100edc32: mov ecx,5
0x100edc37: xor edx,edx
0x100edc39: repe cmpsb
0x100edc3b: jne short 100EDC4Ch
0x100edc3d: mov dword ptr [ebp+65B0h],16161616h
0x100edc47: jmp near ptr 100EF9E0h
0x100edc4c: mov edi,1012EE98h
0x100edc51: mov esi,ebx
0x100edc53: mov ecx,8
0x100edc58: xor eax,eax
0x100edc5a: repe cmpsb
0x100edc5c: jne near ptr 100EDD13h
0x100edc62: cmp dword ptr [esp+14h],308040h
0x100edc6a: jne short 100EDCC7h
0x100edc6c: fld dword ptr [ebp+0E774h]
0x100edc72: mov word ptr [ebp+66B4h],604h
0x100edc7b: fld qword ptr ds:[100FF108h]
0x100edc81: mov word ptr [ebp+66B6h],810h
0x100edc8a: fmul st(1),st
0x100edc8c: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100edc96: fxch
0x100edc98: fstp dword ptr [ebp+0E774h]
0x100edc9e: fmul dword ptr [ebp+0E77Ch]
0x100edca4: fstp dword ptr [ebp+0E77Ch]
0x100edcaa: fld dword ptr ds:[1012EE94h]
0x100edcb0: fstp dword ptr [ebp+0E784h]
0x100edcb6: fld dword ptr ds:[1012EE90h]
0x100edcbc: fstp dword ptr [ebp+0E78Ch]
0x100edcc2: jmp near ptr 100EF9E0h
0x100edcc7: fld dword ptr ds:[1012EE8Ch]
0x100edccd: mov word ptr [ebp+66B4h],608h
0x100edcd6: fstp dword ptr [ebp+0E784h]
0x100edcdc: mov word ptr [ebp+66B6h],814h
0x100edce5: fld dword ptr ds:[1012EE88h]
0x100edceb: mov word ptr [ebp+66B2h],0C40h
0x100edcf4: fstp dword ptr [ebp+0E78Ch]
0x100edcfa: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100edd04: mov dword ptr [ebp+65FCh],0F7Ch
0x100edd0e: jmp near ptr 100EF9E0h
0x100edd13: mov eax,[esp+14h]
0x100edd17: cmp eax,5D4BC0h
0x100edd1c: jne short 100EDD6Ah
0x100edd1e: fld dword ptr ds:[1012EE84h]
0x100edd24: mov word ptr [ebp+66B4h],6C9h
0x100edd2d: fstp dword ptr [ebp+0E784h]
0x100edd33: mov word ptr [ebp+66B6h],914h
0x100edd3c: fld dword ptr ds:[1012EE80h]
0x100edd42: mov word ptr [ebp+66B2h],0DC0h
0x100edd4b: fstp dword ptr [ebp+0E78Ch]
0x100edd51: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100edd5b: mov dword ptr [ebp+65FCh],0F7Ah
0x100edd65: jmp near ptr 100EF9E0h
0x100edd6a: mov edi,1012EE74h
0x100edd6f: mov esi,ebx
```

```
0x100edd71: mov ecx,0Bh
0x100edd76: xor edx,edx
0x100edd78: repe cmpsb
0x100edd7a: jne short 100EDDA7h
0x100edd7c: mov word ptr [ebp+66B4h],8FEh
0x100edd85: mov word ptr [ebp+66B6h],0C00h
0x100edd8e: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100edd98: mov dword ptr [ebp+66ACh],7
0x100edda2: jmp near ptr 100EF9E0h
0x100edda7: mov edi,1012EE70h
0x100eddac: mov esi,ebx
0x100eddae: mov ecx,4
0x100eddb3: xor edx,edx
0x100eddb5: repe cmpsb
0x100eddb7: jne short 100EDE02h
0x100eddb9: mov eax,0D62B80D7h
0x100eddbe: mul dword ptr [esp+14h]
0x100eddc2: shr edx,0Bh
0x100eddc5: shr edx,1
0x100eddc7: mov word ptr [ebp+66B4h],990h
0x100eddd0: mov word ptr [ebp+66B6h],0CC0h
0x100eddd9: mov [ebp+66B2h],dx
0x100edde0: mov word ptr [ebp+6274h],4D4Dh
0x100edde9: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100eddf3: mov dword ptr [ebp+65FCh],0FFFFh
0x100eddfd: jmp near ptr 100EF9E0h
0x100ede02: mov edi,1012EE64h
0x100ede07: mov esi,ebx
0x100ede09: mov ecx,0Bh
0x100ede0e: xor edx,edx
0x100ede10: repe cmpsb
0x100ede12: jne short 100EDE6Bh
0x100ede14: fld dword ptr ds:[1012EE60h]
0x100ede1a: mov word ptr [ebp+66B4h],1E4h
0x100ede23: fstp dword ptr [ebp+0E784h]
0x100ede29: mov word ptr [ebp+66B6h],284h
0x100ede32: fld dword ptr ds:[1012EE5Ch]
0x100ede38: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100ede42: fstp dword ptr [ebp+0E78Ch]
0x100ede48: mov dword ptr [ebp+66C8h],2
0x100ede52: mov dword ptr [ebp+65B0h],16161616h
0x100ede5c: mov dword ptr [ebp+65F8h],10h
0x100ede66: jmp near ptr 100EF9E0h
0x100ede6b: mov edi,1012EE50h
0x100ede70: mov esi,ebx
0x100ede72: mov ecx,9
0x100ede77: xor edx,edx
0x100ede79: repe cmpsb
0x100ede7b: jne short 100EDEC9h
0x100ede7d: fld dword ptr ds:[1012EE4Ch]
0x100ede83: mov word ptr [ebp+66B4h],1DDh
0x100ede8c: fstp dword ptr [ebp+0E784h]
0x100ede92: mov word ptr [ebp+66B6h],280h
0x100ede9b: fld dword ptr ds:[1012EE48h]
0x100edea1: mov word ptr [ebp+6274h],4949h
0x100edeaa: fstp dword ptr [ebp+0E78Ch]
0x100edeb0: mov dword ptr [ebp+65C0h],0F00h
0x100edeba: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100edec4: jmp near ptr 100EF9E0h
0x100edec9: mov edi,1012EE44h
0x100edece: mov esi,ebx
0x100eded0: mov ecx,4
0x100eded5: xor edx,edx
0x100eded7: repe cmpsb
0x100eded9: jne short 100EDEFBh
0x100ededb: mov ax,[ebp+66B0h]
0x100edee2: sub ax,2
```

```
0x100edee6: mov word ptr [ebp+66B8h],2
0x100edeef: mov [ebp+66B4h],ax
0x100edef6: jmp near ptr 100EF9E0h
0x100edefb: mov edi,1012EE3Ch
0x100edf00: mov esi,ebx
0x100edf02: mov ecx,5
0x100edf07: xor edx,edx
0x100edf09: repe cmpsb
0x100edf0b: jne short 100EDF44h
0x100edf0d: fld dword ptr ds:[1012EE38h]
0x100edf13: mov word ptr [ebp+66B4h],4B0h
0x100edf1c: fstp dword ptr [ebp+0E788h]
0x100edf22: mov word ptr [ebp+66B6h],640h
0x100edf2b: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100edf35: mov dword ptr [ebp+65B0h],49494949h
0x100edf3f: jmp near ptr 100EF9E0h
0x100edf44: mov edi,1012EE30h
0x100edf49: mov esi,ebx
0x100edf4b: mov ecx,7
0x100edf50: xor edx,edx
0x100edf52: repe cmpsb
0x100edf54: jne short 100EDF77h
0x100edf56: mov word ptr [ebp+66B4h],300h
0x100edf5f: mov word ptr [ebp+66B6h],400h
0x100edf68: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100edf72: jmp near ptr 100EF9E0h
0x100edf77: mov edi,1012EE28h
0x100edf7c: mov esi,ebx
0x100edf7e: mov ecx,7
0x100edf83: xor edx,edx
0x100edf85: repe cmpsb
0x100edf87: jne short 100EDFAAh
0x100edf89: mov word ptr [ebp+66B4h],410h
0x100edf92: mov word ptr [ebp+66B6h],570h
0x100edf9b: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100edfa5: jmp near ptr 100EF9E0h
0x100edfaa: mov edi,1012EE20h
0x100edfaf: mov esi,ebx
0x100edfb1: mov ecx,7
0x100edfb6: xor edx,edx
0x100edfb8: repe cmpsb
0x100edfba: jne short 100EDFDDh
0x100edfbc: mov word ptr [ebp+66B4h],4B0h
0x100edfc5: mov word ptr [ebp+66B6h],640h
0x100edfce: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100edfd8: jmp near ptr 100EF9E0h
0x100edfdd: mov edi,1012EE18h
0x100edfe2: mov esi,ebx
0x100edfe4: mov ecx,7
0x100edfe9: xor edx,edx
0x100edfeb: repe cmpsb
0x100edfed: jne short 100EE027h
0x100edfef: cmp eax,7270E0h
0x100edff4: mov word ptr [ebp+66B4h],7A6h
0x100edffd: mov word ptr [ebp+66B6h],0A1Ch
0x100ee006: mov eax,100DAEF0h
0x100ee00b: jb short 100EE012h
0x100ee00d: mov eax,100D8E20h
0x100ee012: mov [ebp+2E7FCh],eax
0x100ee018: mov dword ptr [ebp+65FCh],0FFF0h
0x100ee022: jmp near ptr 100EF9E0h
0x100ee027: mov edi,1012EE10h
0x100ee02c: mov esi,ebx
0x100ee02e: mov ecx,7
0x100ee033: xor edx,edx
0x100ee035: repe cmpsb
0x100ee037: jne short 100EE064h
```

```
0x100ee039: mov word ptr [ebp+66B4h],9A5h
0x100ee042: mov word ptr [ebp+66B6h],0CC8h
0x100ee04b: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100ee055: mov dword ptr [ebp+65FCh],0FFF0h
0x100ee05f: jmp near ptr 100EF9E0h
0x100ee064: mov edi,1012EE04h
0x100ee069: mov esi,ebx
0x100ee06b: mov ecx,0Ch
0x100ee070: xor edx,edx
0x100ee072: repe cmpsb
0x100ee074: jne short 100EE0C1h
0x100ee076: cmp eax,1E8480h
0x100ee07b: mov word ptr [ebp+66B4h],400h
0x100ee084: mov word ptr [ebp+66B6h],55Fh
0x100ee08d: mov word ptr [ebp+66B2h],560h
0x100ee096: mov dword ptr [ebp+65B0h],49494949h
0x100ee0a0: mov dword ptr [ebp+65FCh],3FFh
0x100ee0aa: mov eax,100DAEF0h
0x100ee0af: jb short 100EE0B6h
0x100ee0b1: mov eax,100D8E20h
0x100ee0b6: mov [ebp+2E7FCh],eax
0x100ee0bc: jmp near ptr 100EF9E0h
0x100ee0c1: mov edi,1012EDFCh
0x100ee0c6: mov esi,ebx
0x100ee0c8: mov ecx,5
0x100ee0cd: xor edx,edx
0x100ee0cf: repe cmpsb
0x100ee0d1: jne short 100EE11Bh
0x100ee0d3: mov word ptr [ebp+66B4h],4B7h
0x100ee0dc: mov word ptr [ebp+66B6h],648h
0x100ee0e5: mov word ptr [ebp+6274h],4949h
0x100ee0ee: mov dword ptr [ebp+65B0h],16161616h
0x100ee0f8: mov dword ptr [ebp+65C0h],0C8Ch
0x100ee102: mov dword ptr [ebp+65FCh],3FFh
0x100ee10c: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100ee116: jmp near ptr 100EF9E0h
0x100ee11b: mov edi,1012EDF4h
0x100ee120: mov esi,ebx
0x100ee122: mov ecx,5
0x100ee127: xor edx,edx
0x100ee129: repe cmpsb
0x100ee12b: jne short 100EE16Fh
0x100ee12d: cmp eax,989680h
0x100ee132: mov word ptr [ebp+66B4h],0BB8h
0x100ee13b: mov word ptr [ebp+66B6h],8A0h
0x100ee144: mov dword ptr [ebp+65B0h],61616161h
0x100ee14e: mov eax,100DAEF0h
0x100ee153: jb short 100EE15Ah
0x100ee155: mov eax,100D8E20h
0x100ee15a: mov [ebp+2E7FCh],eax
0x100ee160: mov dword ptr [ebp+65FCh],0FFC0h
0x100ee16a: jmp near ptr 100EF9E0h
0x100ee16f: mov edi,1012EDECh
0x100ee174: mov esi,ebx
0x100ee176: mov ecx,7
0x100ee17b: xor eax,eax
0x100ee17d: repe cmpsb
0x100ee17f: jne near ptr 100EE248h
0x100ee185: fld dword ptr ds:[1012EDE8h]
0x100ee18b: mov ecx,[ebp]
0x100ee18e: fstp dword ptr [ebp+0E784h]
0x100ee194: mov edx,800h
0x100ee199: fld dword ptr ds:[1012EDE4h]
0x100ee19f: push eax
0x100ee1a0: mov [ebp+66B6h],dx
0x100ee1a7: fstp dword ptr [ebp+0E78Ch]
0x100ee1ad: mov [ebp+66B2h],dx
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0x100ee1b4: mov edx,[ebp+4]
0x100ee1b7: mov word ptr [ebp+66B4h],600h
0x100ee1c0: mov dword ptr [ebp+65B0h],61616161h
0x100ee1ca: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100ee1d4: mov dword ptr [ebp+65FCh],3FFh
0x100ee1de: mov eax,[ecx+8]
0x100ee1e1: push 300000h
0x100ee1e6: push edx
0x100ee1e7: call eax
0x100ee1e9: push 10000h
0x100ee1ee: push ebp
0x100ee1ef: call 100E66A0h
0x100ee1f4: add esp,14h
0x100ee1f7: cmp ax,4D4Dh
0x100ee1fb: mov [ebp+6274h],ax
0x100ee202: jne near ptr 100EF9E0h
0x100ee208: add word ptr [ebp+66B4h],0FFF0h
0x100ee211: add word ptr [ebp+66B6h],0FFE4h
0x100ee21a: mov word ptr [ebp+66B8h],10h
0x100ee223: mov word ptr [ebp+66BAh],1Ch
0x100ee22c: mov dword ptr [ebp+65FCh],0F5C0h
0x100ee236: mov dword ptr [ebp+6485h],475349h
0x100ee240: mov byte ptr [ebx],0
0x100ee243: jmp near ptr 100EF9E0h
0x100ee248: lea eax,[ebp+6485h]
0x100ee24e: mov edi,1012E9C0h
0x100ee253: mov esi,eax
0x100ee255: mov ecx,0Bh
0x100ee25a: xor edx,edx
0x100ee25c: repe cmpsb
0x100ee25e: jne near ptr 100EE307h
0x100ee264: cmp dword ptr [ebp+2E7FCh],100D52A0h
0x100ee26e: jne short 100EE27Ah
0x100ee270: mov dword ptr [ebp+2E7FCh],100D8700h
0x100ee27a: movzx eax,word ptr [ebp+66B2h]
0x100ee281: cmp ax,1C5Eh
0x100ee285: jne short 100EE2BAh
0x100ee287: mov word ptr [ebp+66B4h],1544h
0x100ee290: mov word ptr [ebp+66B6h],1C50h
0x100ee299: mov word ptr [ebp+66B8h],4
0x100ee2a2: mov word ptr [ebp+66BAh],7
0x100ee2ab: mov dword ptr [ebp+65B0h],61616161h
0x100ee2b5: jmp near ptr 100EF9E0h
0x100ee2ba: cmp ax,0FFAh
0x100ee2be: jne near ptr 100EF9E0h
0x100ee2c4: mov eax,ds:[1012EDDCh]
0x100ee2c9: mov [ebx],eax
0x100ee2cb: mov cl,ds:[1012EDE0h]
0x100ee2d1: add word ptr [ebp+66B4h],0FFFAh
0x100ee2da: add word ptr [ebp+66B6h],0FFF6h
0x100ee2e3: mov [ebx+4],cl
0x100ee2e6: mov word ptr [ebp+66B8h],6
0x100ee2ef: mov word ptr [ebp+66BAh],3
0x100ee2f8: mov dword ptr [ebp+65B0h],61616161h
0x100ee302: jmp near ptr 100EF9E0h
0x100ee307: mov edi,10129948h
0x100ee30c: mov esi,eax
0x100ee30e: mov ecx,6
0x100ee313: xor edx,edx
0x100ee315: repe cmpsb
0x100ee317: jne near ptr 100EE40Dh
0x100ee31d: lea eax,[edx+4]
0x100ee320: mov ecx,1012EDD4h
0x100ee325: lea edx,[esp+68h]
0x100ee329: lea esp,[esp]
0x100ee330: mov esi,[edx]
0x100ee332: cmp esi,[ecx]
```

```
0x100ee334: jne short 100EE348h
0x100ee336: sub eax,4
0x100ee339: add ecx,4
0x100ee33c: add edx,4
0x100ee33f: cmp eax,4
0x100ee342: jae short 100EE330h
0x100ee344: test eax,eax
0x100ee346: je short 100EE3A5h
0x100ee348: movzx esi,byte ptr [edx]
0x100ee34b: movzx edi,byte ptr [ecx]
0x100ee34e: sub esi,edi
0x100ee350: jne short 100EE397h
0x100ee352: sub eax,1
0x100ee355: add ecx,1
0x100ee358: add edx,1
0x100ee35b: test eax,eax
0x100ee35d: je short 100EE3A5h
0x100ee35f: movzx esi,byte ptr [edx]
0x100ee362: movzx edi,byte ptr [ecx]
0x100ee365: sub esi,edi
0x100ee367: jne short 100EE397h
0x100ee369: sub eax,1
0x100ee36c: add ecx,1
0x100ee36f: add edx,1
0x100ee372: test eax,eax
0x100ee374: je short 100EE3A5h
0x100ee376: movzx esi,byte ptr [edx]
0x100ee379: movzx edi,byte ptr [ecx]
0x100ee37c: sub esi,edi
0x100ee37e: jne short 100EE397h
0x100ee380: sub eax,1
0x100ee383: add ecx,1
0x100ee386: add edx,1
0x100ee389: test eax,eax
0x100ee38b: je short 100EE3A5h
0x100ee38d: movzx esi,byte ptr [edx]
0x100ee390: movzx eax,byte ptr [ecx]
0x100ee393: sub esi,eax
0x100ee395: je short 100EE3A5h
0x100ee397: test esi,esi
0x100ee399: mov eax,1
0x100ee39e: jg short 100EE3A7h
0x100ee3a0: or eax,0FFFFFFFFh
0x100ee3a3: jmp short 100EE3A7h
0x100ee3a5: xor eax,eax
0x100ee3a7: test eax,eax
0x100ee3a9: jne short 100EE3EBh
0x100ee3ab: mov ecx,[ebp]
0x100ee3ae: mov edx,[ebp+4]
0x100ee3b1: push eax
0x100ee3b2: mov eax,[ecx+8]
0x100ee3b5: push 0Eh
0x100ee3b7: push edx
0x100ee3b8: call eax
0x100ee3ba: push ebp
0x100ee3bb: call 100D2FE0h
0x100ee3c0: push ebp
0x100ee3c1: mov [ebp+66B4h],ax
0x100ee3c8: call 100D2FE0h
0x100ee3cd: add esp,14h
0x100ee3d0: mov [ebp+66B6h],ax
0x100ee3d7: mov dword ptr [ebp+65B0h],61616161h
0x100ee3e1: mov dword ptr [ebp+65C0h],44h
0x100ee3eb: cmp dword ptr [ebp+2E7FCh],0
0x100ee3f2: jne short 100EE3FEh
0x100ee3f4: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100ee3fe: mov dword ptr [ebp+65FCh],3FFFh
```

```
0x100ee408: jmp near ptr 100EF9E0h
0x100ee40d: mov edi,10129EE4h
0x100ee412: mov esi,eax
0x100ee414: mov ecx,5
0x100ee419: xor edx,edx
0x100ee41b: repe cmpsb
0x100ee41d: jne near ptr 100EE6B8h
0x100ee423: mov ecx,[ebp+65C0h]
0x100ee429: mov eax,[ebp]
0x100ee42c: push edx
0x100ee42d: mov edx,[ebp+4]
0x100ee430: push ecx
0x100ee431: mov dword ptr [ebp+65FCh],3FFFh
0x100ee43b: mov eax,[eax+8]
0x100ee43e: push edx
0x100ee43f: call eax
0x100ee441: push 1
0x100ee443: lea ecx,[esp+28h]
0x100ee447: push ecx
0x100ee448: push ebp
0x100ee449: call 100D4CE0h
0x100ee44e: add esp,18h
0x100ee451: test eax,eax
0x100ee453: je short 100EE466h
0x100ee455: cmp dword ptr [esp+18h],0Fh
0x100ee45a: jne short 100EE466h
0x100ee45c: mov dword ptr [ebp+65FCh],1FFFh
0x100ee466: mov esi,[ebp+65ECh]
0x100ee46c: cmp esi,1
0x100ee46f: jbe short 100EE47Dh
0x100ee471: mov dword ptr [ebp+65B0h],0
0x100ee47b: jmp short 100EE48Ch
0x100ee47d: movzx edx,word ptr [ebp+66B0h]
0x100ee484: cmp [ebp+6628h],edx
0x100ee48a: jae short 100EE496h
0x100ee48c: mov dword ptr [ebp+2E7FCh],100D88F0h
0x100ee496: movzx ecx,word ptr [ebp+66B6h]
0x100ee49d: movzx eax,word ptr [ebp+66B4h]
0x100ee4a4: mov edi,ecx
0x100ee4a6: or edi,eax
0x100ee4a8: mov edx,800h
0x100ee4ad: cmp di,dx
0x100ee4b0: jne near ptr 100EE55Eh
0x100ee4b6: cmp esi,1
0x100ee4b9: jne short 100EE525h
0x100ee4bb: mov [ebp+65B0h],esi
0x100ee4c1: mov eax,ds:[1012EDCCh]
0x100ee4c6: mov [ebp+6280h],eax
0x100ee4cc: mov cl,ds:[1012EDD0h]
0x100ee4d2: mov [ebp+6284h],cl
0x100ee4d8: mov edx,ds:[1012EDC0h]
0x100ee4de: mov [ebx],edx
0x100ee4e0: mov eax,ds:[1012EDC4h]
0x100ee4e5: mov [ebx+4],eax
0x100ee4e8: mov cx,ds:[1012EDC8h]
0x100ee4ef: mov [ebx+8],cx
0x100ee4f3: mov dl,ds:[1012EDCAh]
0x100ee4f9: mov [ebx+0Ah],dl
0x100ee4fc: mov word ptr [ebp+66B8h],8
0x100ee505: mov word ptr [ebp+66BAh],12h
0x100ee50e: mov word ptr [ebp+66B4h],7F0h
0x100ee517: mov word ptr [ebp+66B6h],7E0h
0x100ee520: jmp near ptr 100EF9E0h
0x100ee525: mov eax,ds:[1012A784h]
0x100ee52a: mov [ebx],eax
0x100ee52c: mov cl,ds:[1012A788h]
0x100ee532: mov [ebx+4],cl
```

```
0x100ee535: mov word ptr [ebp+66B8h],0Ah
0x100ee53e: mov word ptr [ebp+66BAh],10h
0x100ee547: mov word ptr [ebp+66B4h],7ECh
0x100ee550: mov word ptr [ebp+66B6h],7E6h
0x100ee559: jmp near ptr 100EF9E0h
0x100ee55e: movzx esi,cx
0x100ee561: movzx edi,ax
0x100ee564: add esi,edi
0x100ee566: cmp esi,1454h
0x100ee56c: jne near ptr 100EE61Bh
0x100ee572: cmp byte ptr [ebx],0
0x100ee575: jne short 100EE588h
0x100ee577: mov esi,ds:[1012A774h]
0x100ee57d: mov [ebx],esi
0x100ee57f: mov esi,ds:[1012A778h]
0x100ee585: mov [ebx+4],esi
0x100ee588: cmp cx,ax
0x100ee58b: jbe short 100EE5BBh
0x100ee58d: mov word ptr [ebp+66B8h],6
0x100ee596: mov word ptr [ebp+66BAh],20h
0x100ee59f: mov [ebp+66B4h],dx
0x100ee5a6: mov word ptr [ebp+66B6h],0C00h
0x100ee5af: mov dword ptr [ebp+65B0h],61616161h
0x100ee5b9: jmp short 100EE5E7h
0x100ee5bb: mov word ptr [ebp+66BAh],6
0x100ee5c4: mov word ptr [ebp+66B8h],20h
0x100ee5cd: mov [ebp+66B6h],dx
0x100ee5d4: mov word ptr [ebp+66B4h],0C00h
0x100ee5dd: mov dword ptr [ebp+65B0h],16161616h
0x100ee5e7: fldz
0x100ee5e9: fcomp dword ptr [ebp+0E774h]
0x100ee5ef: fnstsw ax
0x100ee5f1: test ah,44h
0x100ee5f4: jnp short 100EE60Ch
0x100ee5f6: cmp byte ptr [ebx],56h
0x100ee5f9: je short 100EE60Ch
0x100ee5fb: mov eax,[ebp+65ECh]
0x100ee601: mov [ebp+6614h],eax
0x100ee607: jmp near ptr 100EF9E0h
0x100ee60c: mov dword ptr [ebp+65B0h],0
0x100ee616: jmp near ptr 100EF9E0h
0x100ee61b: cmp cx,844h
0x100ee620: jne short 100EE667h
0x100ee622: mov edx,ds:[1012A764h]
0x100ee628: mov [ebx],edx
0x100ee62a: mov ecx,ds:[1012A768h]
0x100ee630: add eax,0FFFFFFC4h
0x100ee633: add word ptr [ebp+66B6h],0FF92h
0x100ee63c: mov [ebx+4],ecx
0x100ee63f: mov word ptr [ebp+66B8h],1Eh
0x100ee648: mov [ebp+66B4h],ax
0x100ee64f: mov word ptr [ebp+66BAh],37h
0x100ee658: mov dword ptr [ebp+65B0h],49494949h
0x100ee662: jmp near ptr 100EF9E0h
0x100ee667: cmp cx,0C63h
0x100ee66c: jne near ptr 100EF9E0h
0x100ee672: mov edx,ds:[1012A764h]
0x100ee678: mov [ebx],edx
0x100ee67a: mov ecx,ds:[1012A768h]
0x100ee680: mov [ebx+4],ecx
0x100ee683: mov ecx,18h
0x100ee688: add eax,0FFFFFFD0h
0x100ee68b: add word ptr [ebp+66B6h],0FFD0h
0x100ee694: mov [ebp+66B8h],cx
0x100ee69b: mov [ebp+66B4h],ax
0x100ee6a2: mov [ebp+66BAh],cx
0x100ee6a9: mov dword ptr [ebp+65B0h],16161616h
```

```
0x100ee6b3: jmp near ptr 100EF9E0h
0x100ee6b8: mov edi,1012E558h
0x100ee6bd: mov esi,eax
0x100ee6bf: mov ecx,6
0x100ee6c4: xor edx,edx
0x100ee6c6: repe cmpsb
0x100ee6c8: je near ptr 100EF4CEh
0x100ee6ce: mov edi,1012E568h
0x100ee6d3: mov esi,eax
0x100ee6d5: mov ecx,0Ah
0x100ee6da: xor edx,edx
0x100ee6dc: repe cmpsb
0x100ee6de: je near ptr 100EF4CEh
0x100ee6e4: mov edi,1012EDB8h
0x100ee6e9: mov esi,ebx
0x100ee6eb: mov ecx,7
0x100ee6f0: xor edx,edx
0x100ee6f2: repe cmpsb
0x100ee6f4: jne short 100EE72Bh
0x100ee6f6: mov word ptr [ebp+66B4h],6B6h
0x100ee6ff: mov word ptr [ebp+66B6h],900h
0x100ee708: mov dword ptr [ebp+65B0h],16161616h
0x100ee712: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100ee71c: mov dword ptr [ebp+66ACh],7
0x100ee726: jmp near ptr 100EF9E0h
0x100ee72b: mov edi,10129890h
0x100ee730: mov esi,eax
0x100ee732: mov ecx,8
0x100ee737: xor edx,edx
0x100ee739: repe cmpsb
0x100ee73b: jne near ptr 100EE891h
0x100ee741: movzx eax,word ptr [ebp+66B4h]
0x100ee748: mov edx,[ebp+65B4h]
0x100ee74e: movzx ecx,al
0x100ee751: and ecx,1
0x100ee754: add ecx,eax
0x100ee756: cmp dword ptr [ebp+2E7FCh],100D9680h
0x100ee760: mov [ebp+66B4h],cx
0x100ee767: mov [ebp+65B0h],edx
0x100ee76d: jne short 100EE77Bh
0x100ee76f: shr dword ptr [ebp+65F8h],4
0x100ee776: jmp near ptr 100EF9E0h
0x100ee77b: mov edi,1012EDB0h
0x100ee780: mov esi,ebx
0x100ee782: mov ecx,5
0x100ee787: xor eax,eax
0x100ee789: repe cmpsb
0x100ee78b: je near ptr 100EE885h
0x100ee791: push 4
0x100ee793: push 1012EDA8h
0x100ee798: push ebx
0x100ee799: call dword ptr ds:[100FE414h]
0x100ee79f: add esp,0Ch
0x100ee7a2: test eax,eax
0x100ee7a4: je near ptr 100EE885h
0x100ee7aa: mov edi,1012EDA0h
0x100ee7af: mov esi,ebx
0x100ee7b1: mov ecx,6
0x100ee7b6: xor edx,edx
0x100ee7b8: repe cmpsb
0x100ee7ba: je near ptr 100EE857h
0x100ee7c0: mov edi,1012ED98h
0x100ee7c5: mov esi,ebx
0x100ee7c7: mov ecx,6
0x100ee7cc: xor eax,eax
0x100ee7ce: repe cmpsb
0x100ee7d0: je near ptr 100EE857h
```

```
0x100ee7d6: mov edi,1012ED90h
0x100ee7db: mov esi,ebx
0x100ee7dd: mov ecx,6
0x100ee7e2: xor edx,edx
0x100ee7e4: repe cmpsb
0x100ee7e6: jne short 100EE810h
0x100ee7e8: add word ptr [ebp+66B6h],0FFE2h
0x100ee7f1: cmp dword ptr [ebp+2E7FCh],100D8E20h
0x100ee7fb: jne near ptr 100EF9E0h
0x100ee801: mov dword ptr [ebp+65FCh],0F790h
0x100ee80b: jmp near ptr 100EF9E0h
0x100ee810: mov edi,1012ED88h
0x100ee815: mov esi,ebx
0x100ee817: mov ecx,8
0x100ee81c: xor eax,eax
0x100ee81e: repe cmpsb
0x100ee820: jne near ptr 100EF9E0h
0x100ee826: mov ecx,[esp+14h]
0x100ee82a: add ecx,0FF5C6800h
0x100ee830: mov dword ptr [ebp+65C4h],0A39800h
0x100ee83a: mov [ebp+65D0h],ecx
0x100ee840: mov word ptr [ebp+66C6h],1E0h
0x100ee849: mov word ptr [ebp+66C4h],280h
0x100ee852: jmp near ptr 100EF9E0h
0x100ee857: add word ptr [ebp+66B6h],0FFECh
0x100ee860: cmp dword ptr [ebp+2E7FCh],100D8E20h
0x100ee86a: jne near ptr 100EF9E0h
0x100ee870: mov dword ptr [ebp+65FCh],0FC30h
0x100ee87a: mov [ebp+65F8h],edx
0x100ee880: jmp near ptr 100EF9E0h
0x100ee885: shl dword ptr [ebp+65F8h],2
0x100ee88c: jmp near ptr 100EF9E0h
0x100ee891: mov edi,1012F218h
0x100ee896: mov esi,ebx
0x100ee898: mov ecx,0Ah
0x100ee89d: xor edx,edx
0x100ee89f: repe cmpsb
0x100ee8a1: jne short 100EE8D8h
0x100ee8a3: mov word ptr [ebp+66B4h],7FFh
0x100ee8ac: mov word ptr [ebp+66B6h],0C00h
0x100ee8b5: mov dword ptr [ebp+65B0h],61616161h
0x100ee8bf: mov dword ptr [ebp+65C0h],1A00h
0x100ee8c9: mov dword ptr [ebp+2E7FCh],100D8B50h
0x100ee8d3: jmp near ptr 100EF9E0h
0x100ee8d8: mov edi,1012ED7Ch
0x100ee8dd: mov esi,ebx
0x100ee8df: mov ecx,9
0x100ee8e4: xor edx,edx
0x100ee8e6: repe cmpsb
0x100ee8e8: jne short 100EE940h
0x100ee8ea: mov word ptr [ebp+66B6h],0CD8h
0x100ee8f3: mov word ptr [ebp+66BAh],5
0x100ee8fc: mov dword ptr [ebp+65C0h],0D27C0h
0x100ee906: mov dword ptr [ebp+2E7FCh],100DBEF0h
0x100ee910: mov dword ptr [ebp+65B0h],9C9C9C9Ch
0x100ee91a: mov dword ptr [ebp+66D0h],4
0x100ee924: mov eax,ds:[1012ED74h]
0x100ee929: mov [ebp+6280h],eax
0x100ee92f: mov cl,ds:[1012ED78h]
0x100ee935: mov [ebp+6284h],cl
0x100ee93b: jmp near ptr 100EF9E0h
0x100ee940: mov edi,1012ED6Ch
0x100ee945: mov esi,ebx
0x100ee947: mov ecx,7
0x100ee94c: xor edx,edx
0x100ee94e: repe cmpsb
0x100ee950: jne short 100EE97Dh
```

```
0x100ee952: mov word ptr [ebp+66B6h],0C25h
0x100ee95b: mov word ptr [ebp+66BAh],3Bh
0x100ee964: mov dword ptr [ebp+65C0h],0C03C0h
0x100ee96e: mov dword ptr [ebp+2E7FCh],100DBEF0h
0x100ee978: jmp near ptr 100EF9E0h
0x100ee97d: mov esi,eax
0x100ee97f: mov edi,1012E574h
0x100ee984: mov ecx,5
0x100ee989: xor eax,eax
0x100ee98b: repe cmpsb
0x100ee98d: jne short 100EE9C4h
0x100ee98f: cmp word ptr [ebp+66B2h],0F90h
0x100ee998: jne short 100EE9C4h
0x100ee99a: push 1012ED64h
0x100ee99f: push 1012E574h
0x100ee9a4: push ebp
0x100ee9a5: call 100E64C0h
0x100ee9aa: add esp,0Ch
0x100ee9ad: mov word ptr [ebp+66B6h],0F55h
0x100ee9b6: mov word ptr [ebp+6274h],4D4Dh
0x100ee9bf: jmp near ptr 100EF9E0h
0x100ee9c4: mov edi,1012EA18h
0x100ee9c9: mov esi,ebx
0x100ee9cb: mov ecx,0Ah
0x100ee9d0: xor edx,edx
0x100ee9d2: repe cmpsb
0x100ee9d4: jne short 100EE9E4h
0x100ee9d6: add word ptr [ebp+66B4h],0FFFFh
0x100ee9df: jmp near ptr 100EF9E0h
0x100ee9e4: mov edi,1012ED58h
0x100ee9e9: mov esi,ebx
0x100ee9eb: mov ecx,0Ah
0x100ee9f0: xor eax,eax
0x100ee9f2: repe cmpsb
0x100ee9f4: jne short 100EEA04h
0x100ee9f6: add word ptr [ebp+66B4h],0FFFCh
0x100ee9ff: jmp near ptr 100EF9E0h
0x100eea04: mov edi,1012ED50h
0x100eea09: mov esi,ebx
0x100eea0b: mov ecx,6
0x100eea10: xor edx,edx
0x100eea12: repe cmpsb
0x100eea14: jne short 100EEA2Ah
0x100eea16: mov word ptr [ebp+66B4h],1E0h
0x100eea1f: mov word ptr [ebp+66B6h],280h
0x100eea28: jmp short 100EEA4Eh
0x100eea2a: mov edi,1012ED48h
0x100eea2f: mov esi,ebx
0x100eea31: mov ecx,6
0x100eea36: xor eax,eax
0x100eea38: repe cmpsb
0x100eea3a: jne short 100EEA67h
0x100eea3c: mov word ptr [ebp+66B4h],856h
0x100eea45: mov word ptr [ebp+66B6h],0B20h
0x100eea4e: mov dword ptr [ebp+65B0h],0
0x100eea58: mov dword ptr [ebp+2E7FCh],100DB0D0h
0x100eea62: jmp near ptr 100EF9E0h
0x100eea67: mov edi,1012ED40h
0x100eea6c: mov esi,ebx
0x100eea6e: mov ecx,5
0x100eea73: xor edx,edx
0x100eea75: repe cmpsb
0x100eea77: jne short 100EEAA1h
0x100eea79: mov eax,868h
0x100eea7e: mov [ebp+66B4h],ax
0x100eea85: mov [ebp+66B0h],ax
0x100eea8c: mov eax,0B30h
```

```
0x100eea91: mov [ebp+66B6h],ax
0x100eea98: mov [ebp+66B2h],ax
0x100eea9f: jmp short 100EEAEDh
0x100eeaa1: mov edi,1012ED38h
0x100eeaa6: mov esi,ebx
0x100eeaa8: mov ecx,5
0x100eeaad: xor eax,eax
0x100eeaaf: repe cmpsb
0x100eeab1: jne near ptr 100EEB65h
0x100eeab7: mov word ptr [ebp+66B4h],6D0h
0x100eeac0: mov word ptr [ebp+66B6h],920h
0x100eeac9: mov word ptr [ebp+66B0h],6F3h
0x100eead2: mov word ptr [ebp+66B2h],922h
0x100eeadb: mov word ptr [ebp+66B8h],21h
0x100eeae4: mov word ptr [ebp+66BAh],1
0x100eeaed: movzx ecx,word ptr [ebp+66B0h]
0x100eeaf4: movzx edx,word ptr [ebp+66B2h]
0x100eeafb: mov eax,[esp+14h]
0x100eeaff: imul ecx,edx
0x100eeb02: sub eax,ecx
0x100eeb04: mov word ptr [ebp+6274h],4949h
0x100eeb0d: mov [ebp+65C0h],eax
0x100eeb13: je short 100EEB4Ch
0x100eeb15: mov eax,[ebp]
0x100eeb18: mov ecx,[ebp+4]
0x100eeb1b: mov edx,[eax+8]
0x100eeb1e: push 0
0x100eeb20: push 0A8h
0x100eeb25: push ecx
0x100eeb26: call edx
0x100eeb28: push 100h
0x100eeb2d: lea eax,[ebp+6764h]
0x100eeb33: push eax
0x100eeb34: push ebp
0x100eeb35: call 100D3290h
0x100eeb3a: add esp,18h
0x100eeb3d: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100eeb47: jmp near ptr 100EF9E0h
0x100eeb4c: mov dword ptr [ebp+6608h],0
0x100eeb56: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100eeb60: jmp near ptr 100EF9E0h
0x100eeb65: lea eax,[ebp+6485h]
0x100eeb6b: push 101298A0h
0x100eeb70: push eax
0x100eeb71: call dword ptr ds:[100FE440h]
0x100eeb77: add esp,8
0x100eeb7a: test eax,eax
0x100eeb7c: jne near ptr 100EEF06h
0x100eeb82: cmp dword ptr [ebp+65B0h],0FFFFFFFFh
0x100eeb89: jne short 100EEB95h
0x100eeb8b: mov dword ptr [ebp+65B0h],61616161h
0x100eeb95: push 6
0x100eeb97: push 1012ED30h
0x100eeb9c: push ebx
0x100eeb9d: call dword ptr ds:[100FE414h]
0x100eeba3: add esp,0Ch
0x100eeba6: test eax,eax
0x100eeba8: je short 100EEBF2h
0x100eebaa: mov edi,1012ED24h
0x100eebaf: mov esi,ebx
0x100eebb1: mov ecx,9
0x100eebb6: xor eax,eax
0x100eebb8: repe cmpsb
0x100eebba: je short 100EEBF2h
0x100eebbc: mov edi,1012ED1Ch
0x100eebc1: mov esi,ebx
0x100eebc3: mov ecx,8
```

```
0x100eebc8: xor edx,edx
0x100eebca: repe cmpsb
0x100eebcc: je short 100EEBF2h
0x100eebce: mov edi,1012ED14h
0x100eebd3: mov esi,ebx
0x100eebd5: mov ecx,7
0x100eebda: xor eax,eax
0x100eebdc: repe cmpsb
0x100eebde: je short 100EEBF2h
0x100eebe0: mov edi,1012ED0Ch
0x100eebe5: mov esi,ebx
0x100eebe7: mov ecx,7
0x100eebec: xor edx,edx
0x100eebee: repe cmpsb
0x100eebf0: jne short 100EEC06h
0x100eebf2: add word ptr [ebp+66B6h],0FFFCh
0x100eebfb: mov word ptr [ebp+66BAh],2
0x100eec04: jmp short 100EEC70h
0x100eec06: mov edi,1012ED04h
0x100eec0b: mov esi,ebx
0x100eec0d: mov ecx,8
0x100eec12: xor eax,eax
0x100eec14: repe cmpsb
0x100eec16: jne short 100EEC2Ch
0x100eec18: add word ptr [ebp+66B6h],0FFFCh
0x100eec21: mov word ptr [ebp+66BAh],2
0x100eec2a: jmp short 100EEC5Ch
0x100eec2c: mov edi,1012ECFCh
0x100eec31: mov esi,ebx
0x100eec33: mov ecx,8
0x100eec38: xor edx,edx
0x100eec3a: repe cmpsb
0x100eec3c: jne short 100EEC4Ah
0x100eec3e: mov dword ptr [ebp+65F8h],0D6h
0x100eec48: jmp short 100EEC5Ch
0x100eec4a: mov edi,1012ECF4h
0x100eec4f: mov esi,ebx
0x100eec51: mov ecx,8
0x100eec56: xor eax,eax
0x100eec58: repe cmpsb
0x100eec5a: jne short 100EEC70h
0x100eec5c: mov dword ptr [ebp+66D0h],1
0x100eec66: mov dword ptr [ebp+65B0h],0
0x100eec70: mov edi,1012ECF0h
0x100eec75: lea esi,[ebp+64C9h]
0x100eec7b: mov ecx,4
0x100eec80: xor edx,edx
0x100eec82: repe cmpsb
0x100eec84: jne short 100EEC9Dh
0x100eec86: mov eax,ds:[1012ECE8h]
0x100eec8b: mov [ebp+6280h],eax
0x100eec91: mov cl,ds:[1012ECECh]
0x100eec97: mov [ebp+6284h],cl
0x100eec9d: push 1012ECE0h
0x100eeca2: push ebx
0x100eeca3: call dword ptr ds:[100FE424h]
0x100eeca9: add esp,8
0x100eecac: test eax,eax
0x100eecae: je short 100EECCAh
0x100eecb0: mov edx,ds:[1012ECE0h]
0x100eecb6: mov [ebx],edx
0x100eecb8: mov al,ds:[1012ECE4h]
0x100eecbd: mov [ebx+4],al
0x100eecc0: mov dword ptr [ebp+65C0h],3C40h
0x100eecca: push 3
0x100eeccc: push 1012E1E4h
0x100eecd1: push ebx
```

```
0x100eecd2: call dword ptr ds:[100FE414h]
0x100eecd8: add esp,0Ch
0x100eecdb: test eax,eax
0x100eecdd: jne near ptr 100EED90h
0x100eece3: cmp dword ptr [esp+14h],186A0h
0x100eeceb: mov word ptr [ebp+66B4h],0F2h
0x100eecf4: jae short 100EED10h
0x100eecf6: fld qword ptr ds:[1012ECD8h]
0x100eecfc: mov word ptr [ebp+66B2h],100h
0x100eed05: mov word ptr [ebp+66B6h],0F9h
0x100eed0e: jmp short 100EED28h
0x100eed10: fld qword ptr ds:[1012ECD0h]
0x100eed16: mov word ptr [ebp+66B2h],200h
0x100eed1f: mov word ptr [ebp+66B6h],1F5h
0x100eed28: movzx ecx,word ptr [ebp+66B2h]
0x100eed2f: fstp qword ptr [ebp+66D8h]
0x100eed35: add ecx,1
0x100eed38: add [ebp+65C0h],ecx
0x100eed3e: push 1
0x100eed40: push ebp
0x100eed41: mov dword ptr [ebp+66D0h],4
0x100eed4b: mov dword ptr [ebp+65B0h],8D8D8D8Dh
0x100eed55: call 100E6630h
0x100eed5a: add esp,8
0x100eed5d: fld dword ptr ds:[1012ECCCh]
0x100eed63: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100eed6d: fstp dword ptr [ebp+0E788h]
0x100eed73: fld dword ptr ds:[1012ECC8h]
0x100eed79: fstp dword ptr [ebp+0E78Ch]
0x100eed7f: fld dword ptr ds:[1012ECC4h]
0x100eed85: fstp dword ptr [ebp+0E790h]
0x100eed8b: jmp near ptr 100EF9E0h
0x100eed90: mov edi,1012ECC0h
0x100eed95: mov esi,ebx
0x100eed97: mov ecx,3
0x100eed9c: xor edx,edx
0x100eed9e: repe cmpsb
0x100eeda0: jne short 100EEDDDh
0x100eeda2: mov eax,ds:[1012ECB8h]
0x100eeda7: mov [ebx],eax
0x100eeda9: mov cl,ds:[1012ECBCh]
0x100eedaf: mov [ebx+4],cl
0x100eedb2: mov word ptr [ebp+66B4h],200h
0x100eedbb: mov word ptr [ebp+66B6h],300h
0x100eedc4: mov dword ptr [ebp+65C0h],480h
0x100eedce: mov dword ptr [ebp+2E7FCh],100DA5D0h
0x100eedd8: jmp near ptr 100EF9E0h
0x100eeddd: mov esi,ds:[100FE424h]
0x100eede3: push 1012ECB0h
0x100eede8: push ebx
0x100eede9: call esi
0x100eedeb: add esp,8
0x100eedee: test eax,eax
0x100eedf0: je short 100EEE2Dh
0x100eedf2: mov edx,ds:[1012ECB0h]
0x100eedf8: mov [ebx],edx
0x100eedfa: mov al,ds:[1012ECB4h]
0x100eedff: mov [ebx+4],al
0x100eee02: mov word ptr [ebp+66B4h],200h
0x100eee0b: mov word ptr [ebp+66B6h],300h
0x100eee14: mov dword ptr [ebp+65C0h],4D00h
0x100eee1e: mov dword ptr [ebp+2E7FCh],100DA5D0h
0x100eee28: jmp near ptr 100EF9E0h
0x100eee2d: push 1012ECA8h
0x100eee32: push ebx
0x100eee33: call esi
0x100eee35: add esp,8
```

```
0x100eee38: test eax,eax
0x100eee3a: je short 100EEE8Bh
0x100eee3c: cmp dword ptr [ebp+65F4h],7
0x100eee43: fld qword ptr ds:[1012ECA0h]
0x100eee49: mov ecx,ds:[1012ECA8h]
0x100eee4f: mov [ebx],ecx
0x100eee51: mov dx,ds:[1012ECACh]
0x100eee58: fstp qword ptr [ebp+66D8h]
0x100eee5e: mov [ebx+4],dx
0x100eee62: mov word ptr [ebp+66B4h],3D0h
0x100eee6b: mov word ptr [ebp+66B6h],350h
0x100eee74: mov eax,100B3980h
0x100eee79: je short 100EEE80h
0x100eee7b: mov eax,100DADC0h
0x100eee80: mov [ebp+2E7FCh],eax
0x100eee86: jmp near ptr 100EF9E0h
0x100eee8b: mov edi,1012EC98h
0x100eee90: mov esi,ebx
0x100eee92: mov ecx,7
0x100eee97: xor eax,eax
0x100eee99: repe cmpsb
0x100eee9b: jne near ptr 100EF9E0h
0x100eeea1: mov word ptr [ebp+66C6h],80h
0x100eeeaa: mov word ptr [ebp+66C4h],0C0h
0x100eeeb3: mov dword ptr [ebp+65C4h],1800h
0x100eeebd: mov dword ptr [ebp+65DCh],168h
0x100eeec7: mov dword ptr [ebp+2E7F4h],100D6CD0h
0x100eeed1: mov word ptr [ebp+66B4h],400h
0x100eeeda: mov word ptr [ebp+66B6h],600h
0x100eeee3: mov dword ptr [ebp+65C0h],13800h
0x100eeeed: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100eeef7: mov dword ptr [ebp+65F8h],11h
0x100eef01: jmp near ptr 100EF9E0h
0x100eef06: mov edi,1012F1F8h
0x100eef0b: mov esi,ebx
0x100eef0d: mov ecx,10h
0x100eef12: xor edx,edx
0x100eef14: repe cmpsb
0x100eef16: jne short 100EEF58h
0x100eef18: push 2
0x100eef1a: push ebp
0x100eef1b: mov word ptr [ebp+66B4h],200h
0x100eef24: mov word ptr [ebp+66B6h],300h
0x100eef2d: mov dword ptr [ebp+65C0h],0E30h
0x100eef37: mov dword ptr [ebp+2E7FCh],100DA5D0h
0x100eef41: mov dword ptr [ebp+65B0h],61616161h
0x100eef4b: call 100E6630h
0x100eef50: add esp,8
0x100eef53: jmp near ptr 100EF9E0h
0x100eef58: mov edi,1012F1D8h
0x100eef5d: mov esi,ebx
0x100eef5f: mov ecx,0Eh
0x100eef64: xor eax,eax
0x100eef66: repe cmpsb
0x100eef68: jne near ptr 100EF035h
0x100eef6e: mov ecx,[ebp]
0x100eef71: mov edx,[ebp+4]
0x100eef74: push eax
0x100eef75: mov eax,[ecx+8]
0x100eef78: push 220h
0x100eef7d: push edx
0x100eef7e: call eax
0x100eef80: push ebp
0x100eef81: call 100D2F50h
0x100eef86: push ebp
0x100eef87: mov [ebp+66B4h],ax
0x100eef8e: call 100D2F50h
```

```
0x100eef93: push ebp
0x100eef94: mov [ebp+66B6h],ax
0x100eef9b: call 100D2FE0h
0x100eefa0: push ebp
0x100eefa1: call 100D2F50h
0x100eefa6: movzx edx,word ptr [ebp+66B4h]
0x100eefad: xor ecx,ecx
0x100eefaf: add esp,1Ch
0x100eefb2: cmp ax,1Eh
0x100eefb6: movzx eax,word ptr [ebp+66B6h]
0x100eefbd: sete cl
0x100eefc0: cmp dx,ax
0x100eefc3: lea ecx,[ecx+ecx+2E0h]
0x100eefca: mov [ebp+65C0h],ecx
0x100eefd0: jbe short 100EF01Ch
0x100eefd2: xor eax,edx
0x100eefd4: xor [ebp+66B6h],ax
0x100eefdb: mov edx,[ebp]
0x100eefde: mov [ebp+66B4h],ax
0x100eefe5: movzx eax,word ptr [ebp+66B6h]
0x100eefec: xor [ebp+66B4h],ax
0x100eeff3: mov eax,[ebp+4]
0x100eeff6: add ecx,0FFFFFFFAh
0x100eeff9: push 0
0x100eeffb: push ecx
0x100eeffc: mov ecx,[edx+8]
0x100eefff: push eax
0x100ef000: call ecx
0x100ef002: push ebp
0x100ef003: call 100D2F50h
0x100ef008: add esp,10h
0x100ef00b: not eax
0x100ef00d: and al,3
0x100ef00f: neg al
0x100ef011: sbb eax,eax
0x100ef013: add eax,6
0x100ef016: mov [ebp+66C8h],eax
0x100ef01c: mov dword ptr [ebp+2E7FCh],100D9FF0h
0x100ef026: mov dword ptr [ebp+65B0h],61616161h
0x100ef030: jmp near ptr 100EF9E0h
0x100ef035: mov edi,1012F1C0h
0x100ef03a: mov esi,ebx
0x100ef03c: mov ecx,0Eh
0x100ef041: xor edx,edx
0x100ef043: repe cmpsb
0x100ef045: jne short 100EF094h
0x100ef047: mov al,[esp+6Dh]
0x100ef04b: movsx ecx,al
0x100ef04e: mov edx,2E2h
0x100ef053: sub edx,ecx
0x100ef055: test al,al
0x100ef057: mov [ebp+65C0h],edx
0x100ef05d: je short 100EF069h
0x100ef05f: mov dword ptr [ebp+64CFh],303032h
0x100ef069: mov dword ptr [ebp+2E7FCh],100DA5D0h
0x100ef073: mov word ptr [ebp+66B4h],1E0h
0x100ef07c: mov word ptr [ebp+66B6h],280h
0x100ef085: mov dword ptr [ebp+65B0h],61616161h
0x100ef08f: jmp near ptr 100EF9E0h
0x100ef094: mov edi,1012E734h
0x100ef099: lea esi,[ebp+6485h]
0x100ef09f: mov ecx,7
0x100ef0a4: xor eax,eax
0x100ef0a6: repe cmpsb
0x100ef0a8: jne near ptr 100EF14Ah
0x100ef0ae: cmp [ebp+2E7FCh],eax
0x100ef0b4: jne near ptr 100EF14Ah
```

```
0x100ef0ba: movzx eax,word ptr [ebp+66B2h]
0x100ef0c1: cmp eax,524h
0x100ef0c6: je short 100EF0F5h
0x100ef0c8: cmp eax,0A08h
0x100ef0cd: jne short 100EF119h
0x100ef0cf: mov word ptr [ebp+66B4h],7A8h
0x100ef0d8: mov word ptr [ebp+66B6h],0A00h
0x100ef0e1: mov word ptr [ebp+66B8h],2
0x100ef0ea: mov word ptr [ebp+66BAh],8
0x100ef0f3: jmp short 100EF119h
0x100ef0f5: mov word ptr [ebp+66B4h],406h
0x100ef0fe: mov word ptr [ebp+66B6h],514h
0x100ef107: mov word ptr [ebp+66B8h],1
0x100ef110: mov word ptr [ebp+66BAh],6
0x100ef119: fld dword ptr ds:[1012EC94h]
0x100ef11f: mov dword ptr [ebp+65B0h],16161616h
0x100ef129: fstp dword ptr [ebp+0E784h]
0x100ef12f: mov dword ptr [ebp+2E7FCh],100D6EE0h
0x100ef139: fld dword ptr ds:[1012EC90h]
0x100ef13f: fstp dword ptr [ebp+0E78Ch]
0x100ef145: jmp near ptr 100EF9E0h
0x100ef14a: mov edi,1012EC84h
0x100ef14f: mov esi,ebx
0x100ef151: mov ecx,0Bh
0x100ef156: xor edx,edx
0x100ef158: repe cmpsb
0x100ef15a: jne short 100EF1A8h
0x100ef15c: fld dword ptr ds:[1012EC80h]
0x100ef162: mov eax,400h
0x100ef167: fstp dword ptr [ebp+0E784h]
0x100ef16d: mov word ptr [ebp+66B4h],300h
0x100ef176: fld dword ptr ds:[1012EC7Ch]
0x100ef17c: mov [ebp+66B6h],ax
0x100ef183: fstp dword ptr [ebp+0E78Ch]
0x100ef189: mov [ebp+65C0h],eax
0x100ef18f: mov dword ptr [ebp+65B0h],49494949h
0x100ef199: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100ef1a3: jmp near ptr 100EF9E0h
0x100ef1a8: mov edi,1012EC70h
0x100ef1ad: mov esi,ebx
0x100ef1af: mov ecx,0Ah
0x100ef1b4: xor eax,eax
0x100ef1b6: repe cmpsb
0x100ef1b8: jne short 100EF1E5h
0x100ef1ba: mov word ptr [ebp+66B4h],4B8h
0x100ef1c3: mov word ptr [ebp+66B6h],660h
0x100ef1cc: mov dword ptr [ebp+65C0h],0CC0h
0x100ef1d6: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100ef1e0: jmp near ptr 100EF9E0h
0x100ef1e5: mov eax,[esp+14h]
0x100ef1e9: cmp eax,311960h
0x100ef1ee: jne short 100EF21Ah
0x100ef1f0: mov word ptr [ebp+66B4h],60Ah
0x100ef1f9: mov word ptr [ebp+66B6h],816h
0x100ef202: mov word ptr [ebp+66B2h],820h
0x100ef20b: mov dword ptr [ebp+2E7FCh],100DAEF0h
0x100ef215: jmp near ptr 100EF9E0h
0x100ef21a: mov edi,1012EC68h
0x100ef21f: mov esi,ebx
0x100ef221: mov ecx,8
0x100ef226: xor edx,edx
0x100ef228: repe cmpsb
0x100ef22a: jne short 100EF257h
0x100ef22c: mov word ptr [ebp+66B4h],6A4h
0x100ef235: mov word ptr [ebp+66B6h],8D4h
0x100ef23e: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100ef248: mov dword ptr [ebp+65FCh],0FFFFh
```

```
0x100ef252: jmp near ptr 100EF9E0h
0x100ef257: mov edi,1012EC60h
0x100ef25c: mov esi,ebx
0x100ef25e: mov ecx,8
0x100ef263: xor edx,edx
0x100ef265: repe cmpsb
0x100ef267: jne short 100EF28Ah
0x100ef269: mov word ptr [ebp+66B4h],784h
0x100ef272: mov word ptr [ebp+66B6h],0A10h
0x100ef27b: mov dword ptr [ebp+2E7FCh],100D9E40h
0x100ef285: jmp near ptr 100EF9E0h
0x100ef28a: mov edi,1012EC58h
0x100ef28f: mov esi,ebx
0x100ef291: mov ecx,7
0x100ef296: xor edx,edx
0x100ef298: repe cmpsb
0x100ef29a: jne short 100EF2C5h
0x100ef29c: mov word ptr [ebp+66B4h],6B8h
0x100ef2a5: mov word ptr [ebp+66B6h],908h
0x100ef2ae: mov word ptr [ebp+66B2h],0DC0h
0x100ef2b7: mov word ptr [ebp+66BAh],2
0x100ef2c0: jmp near ptr 100EF9E0h
0x100ef2c5: mov edi,1012EC50h
0x100ef2ca: mov esi,ebx
0x100ef2cc: mov ecx,7
0x100ef2d1: xor edx,edx
0x100ef2d3: repe cmpsb
0x100ef2d5: jne short 100EF30Fh
0x100ef2d7: fld dword ptr ds:[1012EC4Ch]
0x100ef2dd: mov word ptr [ebp+66B4h],786h
0x100ef2e6: fstp dword ptr [ebp+0E784h]
0x100ef2ec: mov word ptr [ebp+66B6h],0A14h
0x100ef2f5: fld dword ptr ds:[1012EC48h]
0x100ef2fb: mov word ptr [ebp+66B2h],0F40h
0x100ef304: fstp dword ptr [ebp+0E78Ch]
0x100ef30a: jmp near ptr 100EF9E0h
0x100ef30f: mov edi,1012EC40h
0x100ef314: mov esi,ebx
0x100ef316: mov ecx,8
0x100ef31b: xor edx,edx
0x100ef31d: repe cmpsb
0x100ef31f: jne short 100EF359h
0x100ef321: fld dword ptr ds:[1012EC3Ch]
0x100ef327: mov word ptr [ebp+66B4h],608h
0x100ef330: fstp dword ptr [ebp+0E784h]
0x100ef336: mov word ptr [ebp+66B6h],80Ah
0x100ef33f: fld dword ptr ds:[1012EC38h]
0x100ef345: mov word ptr [ebp+66B2h],0C40h
0x100ef34e: fstp dword ptr [ebp+0E78Ch]
0x100ef354: jmp near ptr 100EF9E0h
0x100ef359: mov edi,1012EC30h
0x100ef35e: mov esi,ebx
0x100ef360: mov ecx,7
0x100ef365: xor edx,edx
0x100ef367: repe cmpsb
0x100ef369: jne short 100EF3A3h
0x100ef36b: fld dword ptr ds:[1012EC2Ch]
0x100ef371: mov word ptr [ebp+66B4h],78Bh
0x100ef37a: fstp dword ptr [ebp+0E784h]
0x100ef380: mov word ptr [ebp+66B6h],0A0Ah
0x100ef389: fld dword ptr ds:[1012EC28h]
0x100ef38f: mov word ptr [ebp+66B2h],0F40h
0x100ef398: fstp dword ptr [ebp+0E78Ch]
0x100ef39e: jmp near ptr 100EF9E0h
0x100ef3a3: mov edi,1012EC20h
0x100ef3a8: mov esi,ebx
0x100ef3aa: mov ecx,7
```

```
0x100ef3af: xor edx,edx
0x100ef3b1: repe cmpsb
0x100ef3b3: jne short 100EF3EDh
0x100ef3b5: fld dword ptr ds:[1012EC1Ch]
0x100ef3bb: mov word ptr [ebp+66B4h],7A8h
0x100ef3c4: fstp dword ptr [ebp+0E784h]
0x100ef3ca: mov word ptr [ebp+66B6h],0A0Ah
0x100ef3d3: fld dword ptr ds:[1012EC18h]
0x100ef3d9: mov word ptr [ebp+66B2h],0F40h
0x100ef3e2: fstp dword ptr [ebp+0E78Ch]
0x100ef3e8: jmp near ptr 100EF9E0h
0x100ef3ed: mov edi,1012EC10h
0x100ef3f2: mov esi,ebx
0x100ef3f4: mov ecx,8
0x100ef3f9: xor edx,edx
0x100ef3fb: repe cmpsb
0x100ef3fd: jne short 100EF441h
0x100ef3ff: fld dword ptr ds:[1012EC0Ch]
0x100ef405: mov word ptr [ebp+66B4h],788h
0x100ef40e: fstp dword ptr [ebp+0E784h]
0x100ef414: mov word ptr [ebp+66B6h],0A08h
0x100ef41d: fld dword ptr ds:[1012EC08h]
0x100ef423: mov word ptr [ebp+66B2h],0F0Ch
0x100ef42c: fstp dword ptr [ebp+0E78Ch]
0x100ef432: mov dword ptr [ebp+65FCh],0FFFh
0x100ef43c: jmp near ptr 100EF9E0h
0x100ef441: cmp eax,8E1D00h
0x100ef446: jne short 100EF480h
0x100ef448: fld dword ptr ds:[1012EC04h]
0x100ef44e: mov word ptr [ebp+66B4h],85Eh
0x100ef457: fstp dword ptr [ebp+0E784h]
0x100ef45d: mov word ptr [ebp+66B6h],0B1Ch
0x100ef466: fld dword ptr ds:[1012EC00h]
0x100ef46c: mov word ptr [ebp+66B2h],10C0h
0x100ef475: fstp dword ptr [ebp+0E78Ch]
0x100ef47b: jmp near ptr 100EF9E0h
0x100ef480: mov edi,1012EBF8h
0x100ef485: mov esi,ebx
0x100ef487: mov ecx,8
0x100ef48c: xor eax,eax
0x100ef48e: repe cmpsb
0x100ef490: jne near ptr 100EF9E0h
0x100ef496: fld dword ptr ds:[1012EBF4h]
0x100ef49c: mov word ptr [ebp+66B4h],90Eh
0x100ef4a5: fstp dword ptr [ebp+0E784h]
0x100ef4ab: mov word ptr [ebp+66B6h],0C0Ah
0x100ef4b4: fld dword ptr ds:[1012EBF0h]
0x100ef4ba: mov word ptr [ebp+66B2h],1240h
0x100ef4c3: fstp dword ptr [ebp+0E78Ch]
0x100ef4c9: jmp near ptr 100EF9E0h
0x100ef4ce: movzx esi,word ptr [ebp+66B6h]
0x100ef4d5: movzx edi,si
0x100ef4d8: lea ecx,[edi*8]
0x100ef4df: mov eax,92492493h
0x100ef4e4: imul ecx
0x100ef4e6: add edx,ecx
0x100ef4e8: sar edx,2
0x100ef4eb: mov eax,edx
0x100ef4ed: shr eax,1Fh
0x100ef4f0: add eax,edx
0x100ef4f2: cmp eax,1
0x100ef4f5: mov dword ptr [ebp+65FCh],0FFF0h
0x100ef4ff: mov ecx,1
0x100ef504: jl short 100EF508h
0x100ef506: mov ecx,eax
0x100ef508: mov eax,[esp+14h]
0x100ef50c: sub eax,[ebp+65C0h]
```

```
0x100ef512: xor edx,edx
0x100ef514: div ecx
0x100ef516: movzx ebx,word ptr [ebp+66B4h]
0x100ef51d: movzx ecx,bx
0x100ef520: cmp eax,ecx
0x100ef522: jne short 100EF52Eh
0x100ef524: mov dword ptr [ebp+2E7FCh],100D9280h
0x100ef52e: cmp dword ptr [ebp+2E7FCh],0
0x100ef535: jne short 100EF541h
0x100ef537: mov dword ptr [ebp+2E7FCh],100D8E20h
0x100ef541: mov eax,edi
0x100ef543: cmp eax,0E7Eh
0x100ef548: jg near ptr 100EF795h
0x100ef54e: je near ptr 100EF76Ah
0x100ef554: cmp eax,0C8Dh
0x100ef559: jg near ptr 100EF64Bh
0x100ef55f: je near ptr 100EF60Ch
0x100ef565: cmp eax,0C62h
0x100ef56a: jg near ptr 100EF600h
0x100ef570: je short 100EF58Dh
0x100ef572: cmp eax,0A08h
0x100ef577: je short 100EF5E8h
0x100ef579: cmp eax,0C3Ah
0x100ef57e: jne near ptr 100EF9E0h
0x100ef584: mov word ptr [ebp+66BAh],0FFF2h
0x100ef58d: add word ptr [ebp+66BAh],12h
0x100ef595: mov eax,916h
0x100ef59a: cmp [ebp+66B4h],ax
0x100ef5a1: mov word ptr [ebp+66B6h],0C18h
0x100ef5aa: jbe short 100EF5C6h
0x100ef5ac: mov [ebp+66B4h],ax
0x100ef5b3: mov word ptr [ebp+66B8h],0Dh
0x100ef5bc: mov dword ptr [ebp+65B0h],49494949h
0x100ef5c6: push 1012EBE8h
0x100ef5cb: push 1012E568h
0x100ef5d0: push ebp
0x100ef5d1: mov dword ptr [ebp+660Ch],1
0x100ef5db: call 100E64C0h
0x100ef5e0: add esp,0Ch
0x100ef5e3: jmp near ptr 100EF9E0h
0x100ef5e8: push 1012EBE0h
0x100ef5ed: push 1012E568h
0x100ef5f2: push ebp
0x100ef5f3: call 100E64C0h
0x100ef5f8: add esp,0Ch
0x100ef5fb: jmp near ptr 100EF9E0h
0x100ef600: cmp eax,0C69h
0x100ef605: je short 100EF616h
0x100ef607: jmp near ptr 100EF9E0h
0x100ef60c: add esi,0FFFFFFE5h
0x100ef60f: mov [ebp+66B6h],si
0x100ef616: add word ptr [ebp+66B6h],0FFF6h
0x100ef61f: push 1012EBD8h
0x100ef624: push 1012E568h
0x100ef629: push ebp
0x100ef62a: mov dword ptr [ebp+65B0h],49494949h
0x100ef634: mov dword ptr [ebp+660Ch],1
0x100ef63e: call 100E64C0h
0x100ef643: add esp,0Ch
0x100ef646: jmp near ptr 100EF9E0h
0x100ef64b: cmp eax,0D2Ah
0x100ef650: jg near ptr 100EF70Fh
0x100ef656: je short 100EF689h
0x100ef658: sub eax,0CE8h
0x100ef65d: je near ptr 100EF6E3h
0x100ef663: sub eax,1Ah
0x100ef666: jne near ptr 100EF9E0h
```

```
0x100ef66c: add esi,2Bh
0x100ef66f: mov [ebp+66B6h],si
0x100ef676: mov word ptr [ebp+66BAh],0FFFAh
0x100ef67f: mov dword ptr [ebp+65FCh],0F7F0h
0x100ef689: add word ptr [ebp+66B6h],0FFAEh
0x100ef692: add word ptr [ebp+66BAh],0Fh
0x100ef69a: cmp word ptr [ebp+66B4h],9B0h
0x100ef6a3: jbe short 100EF6B7h
0x100ef6a5: mov word ptr [ebp+66B8h],0Ah
0x100ef6ae: mov word ptr [ebp+66B4h],9A6h
0x100ef6b7: push 1012EBCCh
0x100ef6bc: push 1012E568h
0x100ef6c1: push ebp
0x100ef6c2: mov dword ptr [ebp+65B0h],49494949h
0x100ef6cc: mov dword ptr [ebp+660Ch],1
0x100ef6d6: call 100E64C0h
0x100ef6db: add esp,0Ch
0x100ef6de: jmp near ptr 100EF9E0h
0x100ef6e3: push 1012EBC0h
0x100ef6e8: push 1012E568h
0x100ef6ed: add esi,0FFFFFFEFh
0x100ef6f0: push ebp
0x100ef6f1: mov [ebp+66B6h],si
0x100ef6f8: mov dword ptr [ebp+660Ch],1
0x100ef702: call 100E64C0h
0x100ef707: add esp,0Ch
0x100ef70a: jmp near ptr 100EF9E0h
0x100ef70f: cmp eax,0E6Ah
0x100ef714: jne near ptr 100EF9E0h
0x100ef71a: add ebx,0FFFFFFFEh
0x100ef71d: mov [ebp+66B4h],bx
0x100ef724: mov word ptr [ebp+66BAh],0FFF2h
0x100ef72d: mov dword ptr [ebp+65FCh],0F7F0h
0x100ef737: add word ptr [ebp+66B4h],0FFFFh
0x100ef740: movzx eax,word ptr [ebp+66B4h]
0x100ef747: cmp ax,0AEEh
0x100ef74b: mov word ptr [ebp+66B6h],0E58h
0x100ef754: jne short 100EF7C4h
0x100ef756: mov word ptr [ebp+66B4h],0AC8h
0x100ef75f: mov word ptr [ebp+66B8h],0Fh
0x100ef768: jmp short 100EF7CEh
0x100ef76a: push 1012EBB8h
0x100ef76f: push 1012E568h
0x100ef774: push ebp
0x100ef775: mov word ptr [ebp+66B6h],0E62h
0x100ef77e: mov dword ptr [ebp+65B0h],49494949h
0x100ef788: call 100E64C0h
0x100ef78d: add esp,0Ch
0x100ef790: jmp near ptr 100EF9E0h
0x100ef795: cmp eax,0FDCh
0x100ef79a: jg near ptr 100EF8BAh
0x100ef7a0: je near ptr 100EF842h
0x100ef7a6: sub eax,0E8Ch
0x100ef7ab: cmp eax,9Ch
0x100ef7b0: ja near ptr 100EF9E0h
0x100ef7b6: movzx edx,byte ptr [eax+100EFBC8h]
0x100ef7bd: jmp dword ptr [edx*4+100EFBB4h]
0x100ef7c4: mov dword ptr [ebp+65B0h],49494949h
0x100ef7ce: add word ptr [ebp+66BAh],11h
0x100ef7d6: push 1012EBACh
0x100ef7db: push 1012E568h
0x100ef7e0: push ebp
0x100ef7e1: mov dword ptr [ebp+660Ch],1
0x100ef7eb: call 100E64C0h
0x100ef7f0: add esp,0Ch
0x100ef7f3: jmp near ptr 100EF9E0h
0x100ef842: cmp word ptr [ebp+66B4h],8CAh
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0x100ef84b: mov word ptr [ebp+66B6h],0F8Eh
0x100ef854: jne short 100EF885h
0x100ef856: cmp byte ptr [ebp+6485h],50h
0x100ef85d: mov dword ptr [ebp+65B0h],16161616h
0x100ef867: je near ptr 100EF9E0h
0x100ef86d: push 1012EB9Ch
0x100ef872: push 1012E568h
0x100ef877: push ebp
0x100ef878: call 100E64C0h
0x100ef87d: add esp,0Ch
0x100ef880: jmp near ptr 100EF9E0h
0x100ef885: push 1012EB94h
0x100ef88a: push 1012E568h
0x100ef88f: push ebp
0x100ef890: mov word ptr [ebp+66B6h],0FB2h
0x100ef899: mov dword ptr [ebp+65B0h],49494949h
0x100ef8a3: mov dword ptr [ebp+660Ch],1
0x100ef8ad: call 100E64C0h
0x100ef8b2: add esp,0Ch
0x100ef8b5: jmp near ptr 100EF9E0h
0x100ef8ba: sub eax,10C2h
0x100ef8bf: je short 100EF906h
0x100ef8c1: sub eax,28h
0x100ef8c4: je short 100EF923h
0x100ef8c6: sub eax,0B2h
0x100ef8cb: jne near ptr 100EF9E0h
0x100ef8d1: push 1012EB88h
0x100ef8d6: push 1012E568h
0x100ef8db: add ebx,0FFFFFFFAh
0x100ef8de: push ebp
0x100ef8df: mov [ebp+66B4h],bx
0x100ef8e6: mov word ptr [ebp+66B6h],114Dh
0x100ef8ef: mov dword ptr [ebp+65B0h],16161616h
0x100ef8f9: call 100E64C0h
0x100ef8fe: add esp,0Ch
0x100ef901: jmp near ptr 100EF9E0h
0x100ef906: add ebx,26h
0x100ef909: mov [ebp+66B4h],bx
0x100ef910: mov word ptr [ebp+66BAh],0FFF2h
0x100ef919: mov dword ptr [ebp+65B0h],49494949h
0x100ef923: add word ptr [ebp+66B4h],0FFD9h
0x100ef92c: movzx eax,word ptr [ebp+66B4h]
0x100ef933: cmp ax,960h
0x100ef937: mov word ptr [ebp+66B6h],1098h
0x100ef940: jne short 100EF94Bh
0x100ef942: mov word ptr [ebp+66B8h],0Fh
0x100ef94b: add word ptr [ebp+66BAh],11h
0x100ef953: push 1012EB80h
0x100ef958: push 1012E568h
0x100ef95d: push ebp
0x100ef95e: call 100E64C0h
0x100ef963: add esp,0Ch
0x100ef966: jmp short 100EF9E0h
0x100ef968: add word ptr [ebp+66B6h],0FFFFh
0x100ef971: jmp short 100EF9E0h
0x100ef973: add word ptr [ebp+66B6h],0FFFCh
0x100ef97c: mov word ptr [ebp+66BAh],2
0x100ef985: jmp short 100EF9E0h
0x100ef987: add word ptr [ebp+66B4h],0FFFDh
0x100ef990: add word ptr [ebp+66B6h],0FFFCh
0x100ef999: jmp short 100EF9E0h
0x100ef99b: fld qword ptr ds:[1012EB78h]
0x100ef9a1: mov word ptr [ebp+66B4h],305h
0x100ef9aa: mov word ptr [ebp+66B6h],3C0h
0x100ef9b3: mov word ptr [ebp+66B2h],3E0h
0x100ef9bc: mov dword ptr [ebp+65B0h],1E4E1E4Eh
0x100ef9c6: mov dword ptr [ebp+2E7FCh],100D4350h
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```
0x100ef9d0: fstp qword ptr [ebp+66D8h]
0x100ef9d6: mov dword ptr [ebp+66D0h],4
0x100ef9e0: cmp byte ptr [ebp+64C5h],0
0x100ef9e7: lea esi,[ebp+64C5h]
0x100ef9ed: jne short 100EFA0Eh
0x100ef9ef: movzx eax,word ptr [ebp+66B4h]
0x100ef9f6: movzx ecx,word ptr [ebp+66B6h]
0x100ef9fd: push eax
0x100ef9fe: push ecx
0x100ef9ff: push 1012EB6Ch
0x100efa04: push esi
0x100efa05: call dword ptr ds:[100FE4F8h]
0x100efa0b: add esp,10h
0x100efa0e: cmp dword ptr [ebp+65B0h],0FFFFFFFFh
0x100efa15: jne short 100EFA21h
0x100efa17: mov dword ptr [ebp+65B0h],94949494h
0x100efa21: cmp dword ptr [ebp+6604h],0
0x100efa28: je short 100EFA3Bh
0x100efa2a: push esi
0x100efa2b: lea eax,[ebp+6485h]
0x100efa31: push eax
0x100efa32: push ebp
0x100efa33: call 100E64C0h
0x100efa38: add esp,0Ch
0x100efa3b: mov eax,[ebp+65C4h]
0x100efa41: test eax,eax
0x100efa43: je short 100EFA8Ah
0x100efa45: cmp word ptr [ebp+66C6h],0
0x100efa4d: jne short 100EFA8Ah
0x100efa4f: mov edx,[ebp]
0x100efa52: mov ecx,[edx+8]
0x100efa55: push 0
0x100efa57: push eax
0x100efa58: mov eax,[ebp+4]
0x100efa5b: push eax
0x100efa5c: call ecx
0x100efa5e: push 1
0x100efa60: lea edx,[esp+28h]
0x100efa64: push edx
0x100efa65: push ebp
0x100efa66: call 100D4CE0h
0x100efa6b: add esp,18h
0x100efa6e: test eax,eax
0x100efa70: je short 100EFA8Ah
0x100efa72: mov ax,[esp+20h]
0x100efa77: mov cx,[esp+1Ch]
0x100efa7c: mov [ebp+66C4h],ax
0x100efa83: mov [ebp+66C6h],cx
0x100efa8a: mov eax,[ebp+2E7FCh]
0x100efa90: xor ebx,ebx
0x100efa92: cmp eax,ebx
0x100efa94: je short 100EFAA0h
0x100efa96: cmp word ptr [ebp+66B4h],16h
0x100efa9e: jae short 100EFAA6h
0x100efaa0: mov [ebp+6614h],ebx
0x100efaa6: cmp eax,100B3980h
0x100efaab: jne short 100EFAD2h
0x100efaad: mov edx,[ebp+6278h]
0x100efab3: push edx
0x100efab4: push 1012EB44h
0x100efab9: call dword ptr ds:[100FE4C4h]
0x100efabf: add eax,40h
0x100efac2: push eax
0x100efac3: call dword ptr ds:[100FE4C8h]
0x100efac9: add esp,0Ch
0x100efacc: mov [ebp+6614h],ebx
0x100efad2: cmp byte ptr [ebp+6280h],0
```

```
0x100efad9: lea ecx,[ebp+6280h]
0x100efadf: jne short 100EFB02h
0x100efae1: cmp dword ptr [ebp+66D0h],3
0x100efae8: mov edx,1012EB40h
0x100efaed: je short 100EFAF4h
0x100efaef: mov edx,1012EB38h
0x100efaf4: mov al,[edx]
0x100efaf6: mov [ecx],al
0x100efaf8: add edx,1
0x100efafb: add ecx,1
0x100efafe: test al,al
0x100efb00: jne short 100EFAF4h
0x100efb02: cmp [ebp+66B0h],bx
0x100efb09: jne short 100EFB19h
0x100efb0b: mov ax,[ebp+66B4h]
0x100efb12: mov [ebp+66B0h],ax
0x100efb19: cmp [ebp+66B2h],bx
0x100efb20: jne short 100EFB30h
0x100efb22: mov cx,[ebp+66B6h]
0x100efb29: mov [ebp+66B2h],cx
0x100efb30: mov eax,[ebp+65B0h]
0x100efb36: cmp eax,ebx
0x100efb38: je short 100EFB7Dh
0x100efb3a: cmp dword ptr [ebp+66D0h],3
0x100efb41: jne short 100EFB7Dh
0x100efb43: xor ecx,ecx
0x100efb45: mov edx,eax
0x100efb47: shr edx,cl
0x100efb49: and edx,0Fh
0x100efb4c: cmp dl,9
0x100efb4f: jne short 100EFB5Ah
0x100efb51: mov edx,2
0x100efb56: shl edx,cl
0x100efb58: or eax,edx
0x100efb5a: mov edx,eax
0x100efb5c: shr edx,cl
0x100efb5e: and edx,0Fh
0x100efb61: cmp dl,6
0x100efb64: jne short 100EFB6Fh
0x100efb66: mov edx,8
0x100efb6b: shl edx,cl
0x100efb6d: or eax,edx
0x100efb6f: add ecx,4
0x100efb72: cmp ecx,20h
0x100efb75: jb short 100EFB45h
0x100efb77: mov [ebp+65B0h],eax
0x100efb7d: or eax,0FFFFFFFFh
0x100efb80: cmp [ebp+66C8h],eax
0x100efb86: pop edi
0x100efb87: pop esi
0x100efb88: jne short 100EFBA0h
0x100efb8a: mov ecx,[ebp+66CCh]
0x100efb90: cmp ecx,eax
0x100efb92: mov [ebp+66C8h],ecx
0x100efb98: jne short 100EFBA0h
0x100efb9a: mov [ebp+66C8h],ebx
0x100efba0: mov ecx,[esp+80h]
0x100efba7: pop ebp
0x100efba8: pop ebx
0x100efba9: xor ecx,esp
0x100efbab: call 10022326h
0x100efbb0: add esp,7Ch
0x100efbb3: ret
```

`sub_100eb620` is called by sub_10016040. It works with the text "CASIO", "Minolta", "MINOLTA". It accesses memory-mapped peripheral registers at 0x49494949. It delegates to sub_10022326, j_memset, sub_100d2d60 (30 callees).

12.1320 sub_100efc70 at 0x100efc70 (23 instructions)

```
0x100efc70: push esi
0x100efc71: mov esi,[esp+8]
0x100efc75: test esi,esi
0x100efc77: jne short 100EFC7Eh
0x100efc79: or eax,0FFFFFFFFh
0x100efc7c: pop esi
0x100efc7d: ret
0x100efc7e: push esi
0x100efc7f: call 100F3240h
0x100efc84: push esi
0x100efc85: call 100F3040h
0x100efc8a: push esi
0x100efc8b: call 100F1D50h
0x100efc90: push esi
0x100efc91: call 100F16C0h
0x100efc96: push esi
0x100efc97: call 100F0450h
0x100efc9c: push esi
0x100efc9d: call 100F0400h
0x100efca2: add esp,18h
0x100efca5: xor eax,eax
0x100efca7: pop esi
0x100efca8: ret
```

`sub_100efc70` is called by sub_10015d20. It delegates to sub_100f0400, sub_100f0450, sub_100f16c0 (6 callees).

12.1321 sub_100efcb0 at 0x100efcb0 (14 instructions)

```
0x100efcb0: push esi
0x100efcb1: mov esi,[esp+8]
0x100efcb5: cmp dword ptr [esi+18h],0
0x100efcb9: jle near ptr 100EFD99h
0x100efcbf: mov eax,[esi+20h]
0x100efcc2: cmp eax,5
0x100efcc5: ja near ptr 100EFDA4h
0x100efccb: jmp dword ptr [eax*4+100EFDACh]
0x100efd99: mov eax,0FFFFFFFBh
0x100efd9e: pop esi
0x100efd9f: ret
0x100efda4: mov eax,0FFFFFFFDh
0x100efda9: pop esi
0x100efdaa: ret
```

`sub_100efcb0` is called by sub_10015d20. It calls nothing else.

12.1322 j_malloc at 0x100efdd0 (1 instructions)

```
0x100efdd0: jmp dword ptr ds:[100FE550h]
```

`j_malloc` is called by sub_100143f0, sub_10067caa, sub_1006891d, sub_1006cec2, sub_1006cf39, sub_1006cff7, sub_1006d21a, sub_1006d370, sub_1006daaf, sub_1006e09e, sub_1006e1f8, sub_1006ea55, sub_1006eba4, sub_1006f20b, sub_1006f2c7, sub_1006f391, sub_1006fea1, sub_100ad4c0, sub_100ad5b0, sub_100adc00, sub_100adca0, sub_100ae0a0, sub_100ae620, sub_100ae770, sub_100ae7b0, sub_100aec00, sub_100af150, sub_100b04b0, sub_100b0530, sub_100b0570, sub_100b0640, sub_100b0810, sub_100b1490, sub_100b1b50, sub_100bfe40, sub_100bff00, sub_100bffb0, sub_100c07b0, sub_100c2250, sub_100c22e0. It calls nothing else.

12.1323 sub_100efde0 at 0x100efde0 (7 instructions)

```
0x100209e0: jmp dword ptr ds:[100FE554h]
0x100efde0: mov eax,[esp+4]
0x100efde4: test eax,eax
```

```
0x100efde6: je short 100EFDF1h
0x100efde8: mov [esp+4],eax
0x100efdec: jmp near ptr 100209E0h
0x100efdf1: ret
```

`sub_100efde0` is called by sub_100f0450, sub_100f16c0, sub_100f16f0, sub_100f1c20, sub_100f1d50, sub_100f3040, sub_100f6390. It calls nothing else.

12.1324 sub_100f0400 at 0x100f0400 (13 instructions)

```
0x100f0400: push esi
0x100f0401: mov esi,[esp+8]
0x100f0405: mov eax,[esi+0Ch]
0x100f0408: test eax,eax
0x100f040a: je short 100F042Ah
0x100f040c: push eax
0x100f040d: call 100209E0h
0x100f0412: add esp,4
0x100f0415: mov dword ptr [esi+0Ch],0
0x100f041c: mov dword ptr [esi+10h],0
0x100f0423: mov dword ptr [esi+14h],0
0x100f042a: pop esi
0x100f042b: ret
```

`sub_100f0400` is called by sub_100efc70. It delegates to j_free (1 callees).

12.1325 sub_100f0450 at 0x100f0450 (17 instructions)

```
0x100209e0: jmp dword ptr ds:[100FE554h]
0x100efde0: mov eax,[esp+4]
0x100efde4: test eax,eax
0x100efde6: je short 100EFDF1h
0x100efde8: mov [esp+4],eax
0x100efdec: jmp near ptr 100209E0h
0x100efdf1: ret
0x100f0450: push esi
0x100f0451: mov esi,[esp+8]
0x100f0455: mov eax,[esi+2A8h]
0x100f045b: push eax
0x100f045c: call 100EFDE0h
0x100f0461: mov ecx,[esi+2A4h]
0x100f0467: add esp,4
0x100f046a: pop esi
0x100f046b: mov [esp+4],ecx
0x100f046f: jmp near ptr 100EFDE0h
```

`sub_100f0450` is called by sub_100efc70. It delegates to sub_100efde0 (1 callees).

12.1326 sub_100f16c0 at 0x100f16c0 (17 instructions)

```
0x100209e0: jmp dword ptr ds:[100FE554h]
0x100efde0: mov eax,[esp+4]
0x100efde4: test eax,eax
0x100efde6: je short 100EFDF1h
0x100efde8: mov [esp+4],eax
0x100efdec: jmp near ptr 100209E0h
0x100efdf1: ret
0x100f16c0: push esi
0x100f16c1: mov esi,[esp+8]
0x100f16c5: mov eax,[esi+2A0h]
0x100f16cb: push eax
0x100f16cc: call 100EFDE0h
0x100f16d1: mov ecx,[esi+2ACh]
0x100f16d7: add esp,4
```

```
0x100f16da: pop esi
0x100f16db: mov [esp+4],ecx
0x100f16df: jmp near ptr 100EFDE0h
```

`sub_100f16c0` is called by sub_100efc70. It delegates to sub_100efde0 (1 callees).

12.1327 sub_100f16f0 at 0x100f16f0 (54 instructions)

```
0x100f16f0: push ebx
0x100f16f1: push esi
0x100f16f2: mov esi,[esp+0Ch]
0x100f16f6: xor ebx,ebx
0x100f16f8: cmp [esi+168h],ebx
0x100f16fe: je near ptr 100F17CCh
0x100f1704: movzx eax,word ptr [esi+30h]
0x100f1708: push edi
0x100f1709: movsx edi,word ptr [esi+164h]
0x100f1710: cmp edi,eax
0x100f1712: jge near ptr 100F17B6h
0x100f1718: jmp short 100F1720h
0x100f1720: movsx ecx,word ptr [esi+164h]
0x100f1727: mov eax,[esi+168h]
0x100f172d: mov edx,edi
0x100f172f: sub edx,ecx
0x100f1731: imul edx,110h
0x100f1737: mov ecx,[edx+eax+10Ch]
0x100f173e: push ecx
0x100f173f: call 100EFDE0h
0x100f1744: movsx edx,word ptr [esi+164h]
0x100f174b: mov ecx,[esi+168h]
0x100f1751: mov eax,edi
0x100f1753: sub eax,edx
0x100f1755: imul eax,110h
0x100f175b: mov [eax+ecx+10Ch],ebx
0x100f1762: movsx edx,word ptr [esi+164h]
0x100f1769: mov ecx,[esi+168h]
0x100f176f: mov eax,edi
0x100f1771: sub eax,edx
0x100f1773: imul eax,110h
0x100f1779: mov edx,[eax+ecx+104h]
0x100f1780: push edx
0x100f1781: call 100EFDE0h
0x100f1786: movsx eax,word ptr [esi+164h]
0x100f178d: mov edx,[esi+168h]
0x100f1793: mov ecx,edi
0x100f1795: sub ecx,eax
0x100f1797: imul ecx,110h
0x100f179d: mov [ecx+edx+104h],ebx
0x100f17a4: movzx eax,word ptr [esi+30h]
0x100f17a8: add edi,1
0x100f17ab: add esp,8
0x100f17ae: cmp edi,eax
0x100f17b0: jl near ptr 100F1720h
0x100f17b6: mov ecx,[esi+168h]
0x100f17bc: push ecx
0x100f17bd: call 100209E0h
0x100f17c2: add esp,4
0x100f17c5: mov [esi+168h],ebx
0x100f17cb: pop edi
0x100f17cc: pop esi
0x100f17cd: pop ebx
0x100f17ce: ret
```

`sub_100f16f0` is called by sub_100f3040. It delegates to j_free, sub_100efde0 (2 callees).

12.1328 sub_100f1c20 at 0x100f1c20 (67 instructions)

```
0x100f1c20: push ebx
0x100f1c21: push ebp
0x100f1c22: push esi
0x100f1c23: mov esi,[esp+10h]
0x100f1c27: xor ebp,ebp
0x100f1c29: xor ebx,ebx
0x100f1c2b: cmp [esi+178h],ebp
0x100f1c31: jle short 100F1CA1h
0x100f1c33: push edi
0x100f1c34: lea edi,[esi+1CCh]
0x100f1c3a: lea ebx,[ebx]
0x100f1c40: mov eax,[edi]
0x100f1c42: cmp eax,ebp
0x100f1c44: je short 100F1C92h
0x100f1c46: mov ecx,[edi-50h]
0x100f1c49: add ecx,0FFFFFFFFh
0x100f1c4c: cmp ecx,0Fh
0x100f1c4f: ja short 100F1C87h
0x100f1c51: movzx ecx,byte ptr [ecx+100F1D34h]
0x100f1c58: jmp dword ptr [ecx*4+100F1D1Ch]
0x100f1c87: push eax
0x100f1c88: call 100EFDE0h
0x100f1c8d: add esp,4
0x100f1c90: mov [edi],ebp
0x100f1c92: add ebx,1
0x100f1c95: add edi,4
0x100f1c98: cmp ebx,[esi+178h]
0x100f1c9e: jl short 100F1C40h
0x100f1ca0: pop edi
0x100f1ca1: mov edx,[esi+20h]
0x100f1ca4: push edx
0x100f1ca5: mov [esi+178h],ebp
0x100f1cab: call 100EFDE0h
0x100f1cb0: mov eax,[esi+224h]
0x100f1cb6: push eax
0x100f1cb7: mov [esi+20h],ebp
0x100f1cba: call 100EFDE0h
0x100f1cbf: mov ecx,[esi+174h]
0x100f1cc5: push ecx
0x100f1cc6: mov [esi+224h],ebp
0x100f1ccc: call 100EFDE0h
0x100f1cd1: mov edx,[esi+4Ch]
0x100f1cd4: push edx
0x100f1cd5: mov [esi+174h],ebp
0x100f1cdb: call 100EFDE0h
0x100f1ce0: mov eax,[esi+5Ch]
0x100f1ce3: push eax
0x100f1ce4: mov [esi+4Ch],ebp
0x100f1ce7: call 100EFDE0h
0x100f1cec: mov ecx,[esi+60h]
0x100f1cef: push ecx
0x100f1cf0: mov [esi+5Ch],ebp
0x100f1cf3: call 100EFDE0h
0x100f1cf8: mov edx,[esi+64h]
0x100f1cfb: push edx
0x100f1cfc: mov [esi+60h],ebp
0x100f1cff: call 100EFDE0h
0x100f1d04: mov eax,[esi+68h]
0x100f1d07: push eax
0x100f1d08: mov [esi+64h],ebp
0x100f1d0b: call 100EFDE0h
0x100f1d10: add esp,20h
0x100f1d13: mov [esi+68h],ebp
0x100f1d16: pop esi
0x100f1d17: pop ebp
```

```
0x100f1d18: pop ebx
0x100f1d19: ret
```

`sub_100f1c20` is called by sub_100f1d50. It delegates to sub_100efde0 (1 callees).

12.1329 sub_100f1d50 at 0x100f1d50 (31 instructions)

```
0x100f1d50: push esi
0x100f1d51: mov esi,[esp+8]
0x100f1d55: push edi
0x100f1d56: xor edi,edi
0x100f1d58: cmp [esi+278h],di
0x100f1d5f: jle short 100F1D8Ah
0x100f1d61: push ebx
0x100f1d62: xor ebx,ebx
0x100f1d64: mov eax,[esi+27Ch]
0x100f1d6a: add eax,ebx
0x100f1d6c: push eax
0x100f1d6d: call 100F1C20h
0x100f1d72: movsx ecx,word ptr [esi+278h]
0x100f1d79: add edi,1
0x100f1d7c: add esp,4
0x100f1d7f: add ebx,258h
0x100f1d85: cmp edi,ecx
0x100f1d87: jl short 100F1D64h
0x100f1d89: pop ebx
0x100f1d8a: mov edx,[esi+27Ch]
0x100f1d90: push edx
0x100f1d91: call 100EFDE0h
0x100f1d96: mov eax,[esi+2B4h]
0x100f1d9c: push eax
0x100f1d9d: call 100EFDE0h
0x100f1da2: push esi
0x100f1da3: call 100F6390h
0x100f1da8: add esp,0Ch
0x100f1dab: pop edi
0x100f1dac: pop esi
0x100f1dad: ret
```

`sub_100f1d50` is called by sub_100efc70. It delegates to sub_100efde0, sub_100f1c20, sub_100f6390 (3 callees).

12.1330 sub_100f3040 at 0x100f3040 (68 instructions)

```
0x100209e0: jmp dword ptr ds:[100FE554h]
0x100efde0: mov eax,[esp+4]
0x100efde4: test eax,eax
0x100efde6: je short 100EFDF1h
0x100efde8: mov [esp+4],eax
0x100efdec: jmp near ptr 100209E0h
0x100efdf1: ret
0x100f3040: push esi
0x100f3041: mov esi,[esp+8]
0x100f3045: push edi
0x100f3046: push esi
0x100f3047: call 100F16F0h
0x100f304c: mov eax,[esi+174h]
0x100f3052: push eax
0x100f3053: call 100EFDE0h
0x100f3058: mov ecx,[esi+198h]
0x100f305e: push ecx
0x100f305f: call 100EFDE0h
0x100f3064: mov edx,[esi+19Ch]
0x100f306a: push edx
0x100f306b: call 100EFDE0h
0x100f3070: mov eax,[esi+1B0h]
```

```
0x100f3076: push eax
0x100f3077: call 100EFDE0h
0x100f307c: mov ecx,[esi+1B8h]
0x100f3082: push ecx
0x100f3083: call 100EFDE0h
0x100f3088: mov edx,[esi+1F4h]
0x100f308e: push edx
0x100f308f: call 100EFDE0h
0x100f3094: mov eax,[esi+1F8h]
0x100f309a: push eax
0x100f309b: call 100EFDE0h
0x100f30a0: mov ecx,[esi+220h]
0x100f30a6: push ecx
0x100f30a7: call 100EFDE0h
0x100f30ac: mov edx,[esi+204h]
0x100f30b2: push edx
0x100f30b3: call 100EFDE0h
0x100f30b8: xor edi,edi
0x100f30ba: add esp,28h
0x100f30bd: cmp [esi+228h],edi
0x100f30c3: jle short 100F30F2h
0x100f30c5: push ebx
0x100f30c6: xor ebx,ebx
0x100f30c8: jmp short 100F30D0h
0x100f30d0: mov eax,[esi+22Ch]
0x100f30d6: mov ecx,[eax+ebx+0Ch]
0x100f30da: push ecx
0x100f30db: call 100EFDE0h
0x100f30e0: add edi,1
0x100f30e3: add esp,4
0x100f30e6: add ebx,10h
0x100f30e9: cmp edi,[esi+228h]
0x100f30ef: jl short 100F30D0h
0x100f30f1: pop ebx
0x100f30f2: mov edx,[esi+22Ch]
0x100f30f8: push edx
0x100f30f9: call 100EFDE0h
0x100f30fe: mov eax,[esi+234h]
0x100f3104: push eax
0x100f3105: call 100EFDE0h
0x100f310a: mov ecx,[esi+240h]
0x100f3110: add esp,8
0x100f3113: pop edi
0x100f3114: pop esi
0x100f3115: mov [esp+4],ecx
0x100f3119: jmp near ptr 100EFDE0h
```

`sub_100f3040` is called by sub_100efc70. It delegates to sub_100efde0, sub_100f16f0 (2 callees).

12.1331 sub_100f3240 at 0x100f3240 (11 instructions)

```
0x100209e0: jmp dword ptr ds:[100FE554h]
0x100efde0: mov eax,[esp+4]
0x100efde4: test eax,eax
0x100efde6: je short 100EFDF1h
0x100efde8: mov [esp+4],eax
0x100efdec: jmp near ptr 100209E0h
0x100efdf1: ret
0x100f3240: mov eax,[esp+4]
0x100f3244: mov ecx,[eax+3Ch]
0x100f3247: mov [esp+4],ecx
0x100f324b: jmp near ptr 100EFDE0h
```

`sub_100f3240` is called by sub_100efc70. It calls nothing else.

12.1332 sub_100f6390 at 0x100f6390 (31 instructions)

```
0x100209e0: jmp dword ptr ds:[100FE554h]
0x100f6390: push ebx
0x100f6391: mov ebx,[esp+8]
0x100f6395: mov eax,[ebx+290h]
0x100f639b: test eax,eax
0x100f639d: je short 100F63EBh
0x100f639f: push edi
0x100f63a0: xor edi,edi
0x100f63a2: cmp [ebx+28Ch],edi
0x100f63a8: jle short 100F63DAh
0x100f63aa: push esi
0x100f63ab: lea esi,[eax+124h]
0x100f63b1: mov eax,[esi-118h]
0x100f63b7: push eax
0x100f63b8: call 100EFDE0h
0x100f63bd: mov ecx,[esi]
0x100f63bf: push ecx
0x100f63c0: call 100EFDE0h
0x100f63c5: add edi,1
0x100f63c8: add esp,8
0x100f63cb: add esi,128h
0x100f63d1: cmp edi,[ebx+28Ch]
0x100f63d7: jl short 100F63B1h
0x100f63d9: pop esi
0x100f63da: mov edx,[ebx+290h]
0x100f63e0: pop edi
0x100f63e1: pop ebx
0x100f63e2: mov [esp+4],edx
0x100f63e6: jmp near ptr 100209E0h
0x100f63eb: pop ebx
0x100f63ec: ret
```

`sub_100f6390` is called by sub_100f1d50. It delegates to sub_100efde0 (1 callees).

12.1333 j_strlen at 0x100fc10e (1 instructions)

```
0x100fc10e: jmp dword ptr ds:[100FE4A0h]
```

`j_strlen` is called by sub_1006891d. It calls nothing else.

12.1334 j_strcpy at 0x100fc114 (1 instructions)

```
0x100fc114: jmp dword ptr ds:[100FE49Ch]
```

`j_strcpy` is called by sub_1006891d. It calls nothing else.

12.1335 j_pow at 0x100fc11a (1 instructions)

```
0x100fc11a: jmp dword ptr ds:[100FE494h]
```

`j_pow` is called by sub_1006e09e, sub_100727aa. It calls nothing else.

12.1336 sub_100fc230 at 0x100fc230 (15 instructions)

```
0x100fc230: cmp cl,40h
0x100fc233: jae short 100FC24Ah
0x100fc235: cmp cl,20h
0x100fc238: jae short 100FC240h
0x100fc23a: shld edx,eax,cl
0x100fc23d: shl eax,cl
0x100fc23f: ret
0x100fc240: mov edx,eax
```

```
0x100fc242: xor eax,eax
0x100fc244: and cl,1Fh
0x100fc247: shl edx,cl
0x100fc249: ret
0x100fc24a: xor eax,eax
0x100fc24c: xor edx,edx
0x100fc24e: ret
```

`sub_100fc230` is called by sub_100e6190. It calls nothing else.

12.1337 sub_100fc385 at 0x100fc385 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc385: mov edx,[esp+8]
0x100fc389: lea eax,[edx-10h]
0x100fc38c: mov ecx,[edx-14h]
0x100fc38f: xor ecx,eax
0x100fc391: call 10022326h
0x100fc396: mov eax,10130688h
0x100fc39b: jmp near ptr 10022982h
```

`sub_100fc385` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1338 sub_100fc3a8 at 0x100fc3a8 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc3a8: mov edx,[esp+8]
0x100fc3ac: lea eax,[edx-10h]
0x100fc3af: mov ecx,[edx-14h]
0x100fc3b2: xor ecx,eax
0x100fc3b4: call 10022326h
0x100fc3b9: mov eax,1013072Ch
0x100fc3be: jmp near ptr 10022982h
```

`sub_100fc3a8` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1339 sub_100fc3f1 at 0x100fc3f1 (12 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc3f1: mov edx,[esp+8]
0x100fc3f5: lea eax,[edx-7B8h]
0x100fc3fb: mov ecx,[edx-7BCh]
0x100fc401: xor ecx,eax
0x100fc403: call 10022326h
0x100fc408: add eax,0Ch
0x100fc40b: mov ecx,[edx-4]
0x100fc40e: xor ecx,eax
0x100fc410: call 10022326h
0x100fc415: mov eax,10130768h
0x100fc41a: jmp near ptr 10022982h
```

`sub_100fc3f1` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1340 sub_100fc4db at 0x100fc4db (12 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc4db: mov edx,[esp+8]
0x100fc4df: lea eax,[edx-408h]
0x100fc4e5: mov ecx,[edx-40Ch]
```

```
0x100fc4eb: xor ecx,eax
0x100fc4ed: call 10022326h
0x100fc4f2: add eax,10h
0x100fc4f5: mov ecx,[edx-4]
0x100fc4f8: xor ecx,eax
0x100fc4fa: call 10022326h
0x100fc4ff: mov eax,10130818h
0x100fc504: jmp near ptr 10022982h
```

`sub_100fc4db` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1341 sub_100fc510 at 0x100fc510 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc510: mov edx,[esp+8]
0x100fc514: lea eax,[edx+0Ch]
0x100fc517: mov ecx,[edx-68h]
0x100fc51a: xor ecx,eax
0x100fc51c: call 10022326h
0x100fc521: mov eax,10130880h
0x100fc526: jmp near ptr 10022982h
```

`sub_100fc510` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1342 sub_100fc530 at 0x100fc530 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc530: mov edx,[esp+8]
0x100fc534: lea eax,[edx+0Ch]
0x100fc537: mov ecx,[edx-80h]
0x100fc53a: xor ecx,eax
0x100fc53c: call 10022326h
0x100fc541: mov eax,1013091Ch
0x100fc546: jmp near ptr 10022982h
```

`sub_100fc530` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1343 sub_100fc550 at 0x100fc550 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc550: mov edx,[esp+8]
0x100fc554: lea eax,[edx+0Ch]
0x100fc557: mov ecx,[edx-38h]
0x100fc55a: xor ecx,eax
0x100fc55c: call 10022326h
0x100fc561: mov eax,10130974h
0x100fc566: jmp near ptr 10022982h
```

`sub_100fc550` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1344 sub_100fc570 at 0x100fc570 (11 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc570: mov edx,[esp+8]
0x100fc574: lea eax,[edx+0Ch]
0x100fc577: mov ecx,[edx-6Ch]
0x100fc57a: xor ecx,eax
0x100fc57c: call 10022326h
```

```
0x100fc581: mov ecx,[edx+3E860h]
0x100fc587: xor ecx,eax
0x100fc589: call 10022326h
0x100fc58e: mov eax,101309CCh
0x100fc593: jmp near ptr 10022982h
```

`sub_100fc570` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1345 sub_100fc5a0 at 0x100fc5a0 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc5a0: mov edx,[esp+8]
0x100fc5a4: lea eax,[edx+0Ch]
0x100fc5a7: mov ecx,[edx-44h]
0x100fc5aa: xor ecx,eax
0x100fc5ac: call 10022326h
0x100fc5b1: mov eax,10130A24h
0x100fc5b6: jmp near ptr 10022982h
```

`sub_100fc5a0` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1346 sub_100fc5c0 at 0x100fc5c0 (11 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc5c0: mov edx,[esp+8]
0x100fc5c4: lea eax,[edx+0Ch]
0x100fc5c7: mov ecx,[edx-60h]
0x100fc5ca: xor ecx,eax
0x100fc5cc: call 10022326h
0x100fc5d1: mov ecx,[edx+38Ch]
0x100fc5d7: xor ecx,eax
0x100fc5d9: call 10022326h
0x100fc5de: mov eax,10130A7Ch
0x100fc5e3: jmp near ptr 10022982h
```

`sub_100fc5c0` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1347 sub_100fc5f0 at 0x100fc5f0 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc5f0: mov edx,[esp+8]
0x100fc5f4: lea eax,[edx+0Ch]
0x100fc5f7: mov ecx,[edx-370h]
0x100fc5fd: xor ecx,eax
0x100fc5ff: call 10022326h
0x100fc604: mov eax,10130AD4h
0x100fc609: jmp near ptr 10022982h
```

`sub_100fc5f0` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1348 sub_100fc610 at 0x100fc610 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc610: mov edx,[esp+8]
0x100fc614: lea eax,[edx+0Ch]
0x100fc617: mov ecx,[edx-30h]
0x100fc61a: xor ecx,eax
0x100fc61c: call 10022326h
```

```
0x100fc621: mov eax,10130B2Ch
0x100fc626: jmp near ptr 10022982h
```

`sub_100fc610` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1349 sub_100fc630 at 0x100fc630 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc630: mov edx,[esp+8]
0x100fc634: lea eax,[edx+0Ch]
0x100fc637: mov ecx,[edx-74h]
0x100fc63a: xor ecx,eax
0x100fc63c: call 10022326h
0x100fc641: mov eax,10130B84h
0x100fc646: jmp near ptr 10022982h
```

`sub_100fc630` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1350 sub_100fc658 at 0x100fc658 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc658: mov edx,[esp+8]
0x100fc65c: lea eax,[edx-8]
0x100fc65f: mov ecx,[edx-0Ch]
0x100fc662: xor ecx,eax
0x100fc664: call 10022326h
0x100fc669: mov eax,10130BB0h
0x100fc66e: jmp near ptr 10022982h
```

`sub_100fc658` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1351 sub_100fc6d1 at 0x100fc6d1 (12 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc6d1: mov edx,[esp+8]
0x100fc6d5: lea eax,[edx-0DD0h]
0x100fc6db: mov ecx,[edx-0DD4h]
0x100fc6e1: xor ecx,eax
0x100fc6e3: call 10022326h
0x100fc6e8: add eax,10h
0x100fc6eb: mov ecx,[edx-4]
0x100fc6ee: xor ecx,eax
0x100fc6f0: call 10022326h
0x100fc6f5: mov eax,10130BD4h
0x100fc6fa: jmp near ptr 10022982h
```

`sub_100fc6d1` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1352 sub_100fc70b at 0x100fc70b (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc70b: mov edx,[esp+8]
0x100fc70f: lea eax,[edx-10h]
0x100fc712: mov ecx,[edx-14h]
0x100fc715: xor ecx,eax
0x100fc717: call 10022326h
0x100fc71c: mov eax,10130C30h
0x100fc721: jmp near ptr 10022982h
```

`sub_100fc70b` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1353 sub_100fc768 at 0x100fc768 (11 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc768: mov edx,[esp+8]
0x100fc76c: lea eax,[edx+0Ch]
0x100fc76f: mov ecx,[edx-78h]
0x100fc772: xor ecx,eax
0x100fc774: call 10022326h
0x100fc779: mov ecx,[edx+294h]
0x100fc77f: xor ecx,eax
0x100fc781: call 10022326h
0x100fc786: mov eax,10130C88h
0x100fc78b: jmp near ptr 10022982h
```

`sub_100fc768` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1354 sub_100fc7a6 at 0x100fc7a6 (12 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc7a6: mov edx,[esp+8]
0x100fc7aa: lea eax,[edx-944h]
0x100fc7b0: mov ecx,[edx-948h]
0x100fc7b6: xor ecx,eax
0x100fc7b8: call 10022326h
0x100fc7bd: add eax,10h
0x100fc7c0: mov ecx,[edx-4]
0x100fc7c3: xor ecx,eax
0x100fc7c5: call 10022326h
0x100fc7ca: mov eax,10130CBCh
0x100fc7cf: jmp near ptr 10022982h
```

`sub_100fc7a6` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1355 sub_100fc7e0 at 0x100fc7e0 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc7e0: mov edx,[esp+8]
0x100fc7e4: lea eax,[edx+0Ch]
0x100fc7e7: mov ecx,[edx-4Ch]
0x100fc7ea: xor ecx,eax
0x100fc7ec: call 10022326h
0x100fc7f1: mov eax,10130D14h
0x100fc7f6: jmp near ptr 10022982h
```

`sub_100fc7e0` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1356 sub_100fc839 at 0x100fc839 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc839: mov edx,[esp+8]
0x100fc83d: lea eax,[edx-8]
0x100fc840: mov ecx,[edx-0Ch]
0x100fc843: xor ecx,eax
0x100fc845: call 10022326h
0x100fc84a: mov eax,10130F18h
0x100fc84f: jmp near ptr 10022982h
```

`sub_100fc839` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1357 sub_100fc868 at 0x100fc868 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc868: mov edx,[esp+8]
0x100fc86c: lea eax,[edx-4]
0x100fc86f: mov ecx,[edx-8]
0x100fc872: xor ecx,eax
0x100fc874: call 10022326h
0x100fc879: mov eax,10130F44h
0x100fc87e: jmp near ptr 10022982h
```

`sub_100fc868` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1358 sub_100fc899 at 0x100fc899 (11 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc899: mov edx,[esp+8]
0x100fc89d: lea eax,[edx+0Ch]
0x100fc8a0: mov ecx,[edx-1Ch]
0x100fc8a3: xor ecx,eax
0x100fc8a5: call 10022326h
0x100fc8aa: mov ecx,[edx+40Ch]
0x100fc8b0: xor ecx,eax
0x100fc8b2: call 10022326h
0x100fc8b7: mov eax,10130FA4h
0x100fc8bc: jmp near ptr 10022982h
```

`sub_100fc899` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1359 sub_100fc94b at 0x100fc94b (12 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc94b: mov edx,[esp+8]
0x100fc94f: lea eax,[edx-3F8h]
0x100fc955: mov ecx,[edx-3FCh]
0x100fc95b: xor ecx,eax
0x100fc95d: call 10022326h
0x100fc962: add eax,0Ch
0x100fc965: mov ecx,[edx-8]
0x100fc968: xor ecx,eax
0x100fc96a: call 10022326h
0x100fc96f: mov eax,10131038h
0x100fc974: jmp near ptr 10022982h
```

`sub_100fc94b` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1360 sub_100fc998 at 0x100fc998 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc998: mov edx,[esp+8]
0x100fc99c: lea eax,[edx-8]
0x100fc99f: mov ecx,[edx-0Ch]
0x100fc9a2: xor ecx,eax
0x100fc9a4: call 10022326h
0x100fc9a9: mov eax,1013106Ch
0x100fc9ae: jmp near ptr 10022982h
```

`sub_100fc998` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1361 sub_100fc9c5 at 0x100fc9c5 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc9c5: mov edx,[esp+8]
0x100fc9c9: lea eax,[edx+0Ch]
0x100fc9cc: mov ecx,[edx-18h]
0x100fc9cf: xor ecx,eax
0x100fc9d1: call 10022326h
0x100fc9d6: mov eax,101310CCh
0x100fc9db: jmp near ptr 10022982h
```

`sub_100fc9c5` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1362 sub_100fc9e9 at 0x100fc9e9 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fc9e9: mov edx,[esp+8]
0x100fc9ed: lea eax,[edx-34h]
0x100fc9f0: mov ecx,[edx-38h]
0x100fc9f3: xor ecx,eax
0x100fc9f5: call 10022326h
0x100fc9fa: mov eax,101310F8h
0x100fc9ff: jmp near ptr 10022982h
```

`sub_100fc9e9` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1363 sub_100fca18 at 0x100fca18 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fca18: mov edx,[esp+8]
0x100fca1c: lea eax,[edx-28h]
0x100fca1f: mov ecx,[edx-2Ch]
0x100fca22: xor ecx,eax
0x100fca24: call 10022326h
0x100fca29: mov eax,10131124h
0x100fca2e: jmp near ptr 10022982h
```

`sub_100fca18` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1364 sub_100fca58 at 0x100fca58 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fca58: mov edx,[esp+8]
0x100fca5c: lea eax,[edx-4]
0x100fca5f: mov ecx,[edx-8]
0x100fca62: xor ecx,eax
0x100fca64: call 10022326h
0x100fca69: mov eax,10131158h
0x100fca6e: jmp near ptr 10022982h
```

`sub_100fca58` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1365 sub_100fca89 at 0x100fca89 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fca89: mov edx,[esp+8]
0x100fca8d: lea eax,[edx-48h]
0x100fca90: mov ecx,[edx-4Ch]
0x100fca93: xor ecx,eax
0x100fca95: call 10022326h
0x100fca9a: mov eax,101311A4h
0x100fca9f: jmp near ptr 10022982h
```

`sub_100fca89` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1366 sub_100fcb18 at 0x100fcb18 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcb18: mov edx,[esp+8]
0x100fcb1c: lea eax,[edx-40h]
0x100fcb1f: mov ecx,[edx-44h]
0x100fcb22: xor ecx,eax
0x100fcb24: call 10022326h
0x100fcb29: mov eax,10131228h
0x100fcb2e: jmp near ptr 10022982h
```

`sub_100fcb18` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1367 sub_100fcb4b at 0x100fcb4b (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcb4b: mov edx,[esp+8]
0x100fcb4f: lea eax,[edx-1Ch]
0x100fcb52: mov ecx,[edx-20h]
0x100fcb55: xor ecx,eax
0x100fcb57: call 10022326h
0x100fcb5c: mov eax,10131254h
0x100fcb61: jmp near ptr 10022982h
```

`sub_100fcb4b` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1368 sub_100fcb75 at 0x100fcb75 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcb75: mov edx,[esp+8]
0x100fcb79: lea eax,[edx+0Ch]
0x100fcb7c: mov ecx,[edx-18h]
0x100fcb7f: xor ecx,eax
0x100fcb81: call 10022326h
0x100fcb86: mov eax,101312B4h
0x100fcb8b: jmp near ptr 10022982h
```

`sub_100fcb75` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1369 sub_100fcb98 at 0x100fcb98 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcb98: mov edx,[esp+8]
0x100fcb9c: lea eax,[edx+0Ch]
0x100fcb9f: mov ecx,[edx-30h]
```

```
0x100fcba2: xor ecx,eax
0x100fcba4: call 10022326h
0x100fcba9: mov eax,10131348h
0x100fcbae: jmp near ptr 10022982h
```

`sub_100fcb98` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1370 sub_100fcbd9 at 0x100fcbd9 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcbd9: mov edx,[esp+8]
0x100fcbdd: lea eax,[edx-54h]
0x100fcbe0: mov ecx,[edx-58h]
0x100fcbe3: xor ecx,eax
0x100fcbe5: call 10022326h
0x100fcbea: mov eax,10131384h
0x100fcbef: jmp near ptr 10022982h
```

`sub_100fcbd9` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1371 sub_100fcc17 at 0x100fcc17 (12 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcc17: mov edx,[esp+8]
0x100fcc1b: lea eax,[edx-9Ch]
0x100fcc21: mov ecx,[edx-0A0h]
0x100fcc27: xor ecx,eax
0x100fcc29: call 10022326h
0x100fcc2e: add eax,8
0x100fcc31: mov ecx,[edx-8]
0x100fcc34: xor ecx,eax
0x100fcc36: call 10022326h
0x100fcc3b: mov eax,101313B8h
0x100fcc40: jmp near ptr 10022982h
```

`sub_100fcc17` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1372 sub_100fcca8 at 0x100fcca8 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcca8: mov edx,[esp+8]
0x100fccac: lea eax,[edx-10h]
0x100fccaf: mov ecx,[edx-14h]
0x100fccb2: xor ecx,eax
0x100fccb4: call 10022326h
0x100fccb9: mov eax,10131420h
0x100fccbe: jmp near ptr 10022982h
```

`sub_100fcca8` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1373 sub_100fccd8 at 0x100fccd8 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fccd8: mov edx,[esp+8]
0x100fccdc: lea eax,[edx-18h]
0x100fccdf: mov ecx,[edx-1Ch]
0x100fcce2: xor ecx,eax
0x100fcce4: call 10022326h
```

```
0x100fcce9: mov eax,1013144Ch
0x100fccee: jmp near ptr 10022982h
```

`sub_100fccd8` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1374 sub_100fcd60 at 0x100fcd60 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcd60: mov edx,[esp+8]
0x100fcd64: lea eax,[edx-30h]
0x100fcd67: mov ecx,[edx-34h]
0x100fcd6a: xor ecx,eax
0x100fcd6c: call 10022326h
0x100fcd71: mov eax,101314C4h
0x100fcd76: jmp near ptr 10022982h
```

`sub_100fcd60` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1375 sub_100fcd88 at 0x100fcd88 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcd88: mov edx,[esp+8]
0x100fcd8c: lea eax,[edx-30h]
0x100fcd8f: mov ecx,[edx-34h]
0x100fcd92: xor ecx,eax
0x100fcd94: call 10022326h
0x100fcd99: mov eax,101314F0h
0x100fcd9e: jmp near ptr 10022982h
```

`sub_100fcd88` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1376 sub_100fcdb9 at 0x100fcdb9 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcdb9: mov edx,[esp+8]
0x100fcdbd: lea eax,[edx-4Ch]
0x100fcdc0: mov ecx,[edx-50h]
0x100fcdc3: xor ecx,eax
0x100fcdc5: call 10022326h
0x100fcdca: mov eax,1013151Ch
0x100fcdcf: jmp near ptr 10022982h
```

`sub_100fcdb9` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1377 sub_100fcdf1 at 0x100fcdf1 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcdf1: mov edx,[esp+8]
0x100fcdf5: lea eax,[edx-34h]
0x100fcdf8: mov ecx,[edx-38h]
0x100fcdfb: xor ecx,eax
0x100fcdfd: call 10022326h
0x100fce02: mov eax,10131550h
0x100fce07: jmp near ptr 10022982h
```

`sub_100fcdf1` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1378 sub_100fce18 at 0x100fce18 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fce18: mov edx,[esp+8]
0x100fce1c: lea eax,[edx-2Ch]
0x100fce1f: mov ecx,[edx-30h]
0x100fce22: xor ecx,eax
0x100fce24: call 10022326h
0x100fce29: mov eax,1013157Ch
0x100fce2e: jmp near ptr 10022982h
```

`sub_100fce18` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1379 sub_100fcea8 at 0x100fcea8 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcea8: mov edx,[esp+8]
0x100fceac: lea eax,[edx-40h]
0x100fceaf: mov ecx,[edx-44h]
0x100fceb2: xor ecx,eax
0x100fceb4: call 10022326h
0x100fceb9: mov eax,101315FCh
0x100fcebe: jmp near ptr 10022982h
```

`sub_100fcea8` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1380 sub_100fcedb at 0x100fcedb (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcedb: mov edx,[esp+8]
0x100fcedf: lea eax,[edx-0BCh]
0x100fcee5: mov ecx,[edx-0C0h]
0x100fceeb: xor ecx,eax
0x100fceed: call 10022326h
0x100fcef2: mov eax,10131628h
0x100fcef7: jmp near ptr 10022982h
```

`sub_100fcedb` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1381 sub_100fcf69 at 0x100fcf69 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcf69: mov edx,[esp+8]
0x100fcf6d: lea eax,[edx-0Ch]
0x100fcf70: mov ecx,[edx-10h]
0x100fcf73: xor ecx,eax
0x100fcf75: call 10022326h
0x100fcf7a: mov eax,101316ACh
0x100fcf7f: jmp near ptr 10022982h
```

`sub_100fcf69` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1382 sub_100fcf9b at 0x100fcf9b (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcf9b: mov edx,[esp+8]
0x100fcf9f: lea eax,[edx-8]
0x100fcfa2: mov ecx,[edx-0Ch]
```

```
0x100fcfa5: xor ecx,eax
0x100fcfa7: call 10022326h
0x100fcfac: mov eax,101316D8h
0x100fcfb1: jmp near ptr 10022982h
```

`sub_100fcf9b` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1383 sub_100fcfd8 at 0x100fcfd8 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fcfd8: mov edx,[esp+8]
0x100fcfdc: lea eax,[edx-4Ch]
0x100fcfdf: mov ecx,[edx-50h]
0x100fcfe2: xor ecx,eax
0x100fcfe4: call 10022326h
0x100fcfe9: mov eax,10131714h
0x100fcfee: jmp near ptr 10022982h
```

`sub_100fcfd8` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1384 sub_100fd061 at 0x100fd061 (12 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd061: mov edx,[esp+8]
0x100fd065: lea eax,[edx-6Ch]
0x100fd068: mov ecx,[edx-70h]
0x100fd06b: xor ecx,eax
0x100fd06d: call 10022326h
0x100fd072: add eax,8
0x100fd075: mov ecx,[edx-4]
0x100fd078: xor ecx,eax
0x100fd07a: call 10022326h
0x100fd07f: mov eax,1013178Ch
0x100fd084: jmp near ptr 10022982h
```

`sub_100fd061` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1385 sub_100fd0a8 at 0x100fd0a8 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd0a8: mov edx,[esp+8]
0x100fd0ac: lea eax,[edx-40h]
0x100fd0af: mov ecx,[edx-44h]
0x100fd0b2: xor ecx,eax
0x100fd0b4: call 10022326h
0x100fd0b9: mov eax,101317C8h
0x100fd0be: jmp near ptr 10022982h
```

`sub_100fd0a8` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1386 sub_100fd0dc at 0x100fd0dc (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd0dc: mov edx,[esp+8]
0x100fd0e0: lea eax,[edx-0Ch]
0x100fd0e3: mov ecx,[edx-10h]
0x100fd0e6: xor ecx,eax
0x100fd0e8: call 10022326h
```

```
0x100fd0ed: mov eax,101317F4h
0x100fd0f2: jmp near ptr 10022982h
```

`sub_100fd0dc` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1387 sub_100fd10b at 0x100fd10b (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd10b: mov edx,[esp+8]
0x100fd10f: lea eax,[edx-10h]
0x100fd112: mov ecx,[edx-14h]
0x100fd115: xor ecx,eax
0x100fd117: call 10022326h
0x100fd11c: mov eax,10131820h
0x100fd121: jmp near ptr 10022982h
```

`sub_100fd10b` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1388 sub_100fd15c at 0x100fd15c (12 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd15c: mov edx,[esp+8]
0x100fd160: lea eax,[edx-6Ch]
0x100fd163: mov ecx,[edx-70h]
0x100fd166: xor ecx,eax
0x100fd168: call 10022326h
0x100fd16d: add eax,8
0x100fd170: mov ecx,[edx-4]
0x100fd173: xor ecx,eax
0x100fd175: call 10022326h
0x100fd17a: mov eax,10131864h
0x100fd17f: jmp near ptr 10022982h
```

`sub_100fd15c` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1389 sub_100fd198 at 0x100fd198 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd198: mov edx,[esp+8]
0x100fd19c: lea eax,[edx-48h]
0x100fd19f: mov ecx,[edx-4Ch]
0x100fd1a2: xor ecx,eax
0x100fd1a4: call 10022326h
0x100fd1a9: mov eax,10131890h
0x100fd1ae: jmp near ptr 10022982h
```

`sub_100fd198` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1390 sub_100fd205 at 0x100fd205 (12 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd205: mov edx,[esp+8]
0x100fd209: lea eax,[edx-6Ch]
0x100fd20c: mov ecx,[edx-70h]
0x100fd20f: xor ecx,eax
0x100fd211: call 10022326h
0x100fd216: add eax,8
0x100fd219: mov ecx,[edx-4]
```

```
0x100fd21c: xor ecx,eax
0x100fd21e: call 10022326h
0x100fd223: mov eax,101318E4h
0x100fd228: jmp near ptr 10022982h
```

`sub_100fd205` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1391 sub_100fd34a at 0x100fd34a (12 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd34a: mov edx,[esp+8]
0x100fd34e: lea eax,[edx-70h]
0x100fd351: mov ecx,[edx-74h]
0x100fd354: xor ecx,eax
0x100fd356: call 10022326h
0x100fd35b: add eax,0Ch
0x100fd35e: mov ecx,[edx-4]
0x100fd361: xor ecx,eax
0x100fd363: call 10022326h
0x100fd368: mov eax,101319C0h
0x100fd36d: jmp near ptr 10022982h
```

`sub_100fd34a` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1392 sub_100fd38b at 0x100fd38b (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd38b: mov edx,[esp+8]
0x100fd38f: lea eax,[edx-0Ch]
0x100fd392: mov ecx,[edx-10h]
0x100fd395: xor ecx,eax
0x100fd397: call 10022326h
0x100fd39c: mov eax,101319ECh
0x100fd3a1: jmp near ptr 10022982h
```

`sub_100fd38b` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1393 sub_100fd3c0 at 0x100fd3c0 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd3c0: mov edx,[esp+8]
0x100fd3c4: lea eax,[edx-3Ch]
0x100fd3c7: mov ecx,[edx-40h]
0x100fd3ca: xor ecx,eax
0x100fd3cc: call 10022326h
0x100fd3d1: mov eax,10131A20h
0x100fd3d6: jmp near ptr 10022982h
```

`sub_100fd3c0` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1394 sub_100fd418 at 0x100fd418 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd418: mov edx,[esp+8]
0x100fd41c: lea eax,[edx-28h]
0x100fd41f: mov ecx,[edx-2Ch]
0x100fd422: xor ecx,eax
0x100fd424: call 10022326h
```

```
0x100fd429: mov eax,10131A78h
0x100fd42e: jmp near ptr 10022982h
```

`sub_100fd418` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1395 sub_100fd448 at 0x100fd448 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd448: mov edx,[esp+8]
0x100fd44c: lea eax,[edx-24h]
0x100fd44f: mov ecx,[edx-28h]
0x100fd452: xor ecx,eax
0x100fd454: call 10022326h
0x100fd459: mov eax,10131AA4h
0x100fd45e: jmp near ptr 10022982h
```

`sub_100fd448` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

12.1396 sub_100fd478 at 0x100fd478 (8 instructions)

```
0x10022982: jmp dword ptr ds:[100FE544h]
0x100fd478: mov edx,[esp+8]
0x100fd47c: lea eax,[edx-1Ch]
0x100fd47f: mov ecx,[edx-20h]
0x100fd482: xor ecx,eax
0x100fd484: call 10022326h
0x100fd489: mov eax,10131AD0h
0x100fd48e: jmp near ptr 10022982h
```

`sub_100fd478` has no recorded callers; it is likely reached through the vector table or an indirect jump. It delegates to sub_10022326 (1 callees).

13 Dynamic Analysis (Emulated)

Observed: the entry point was executed in a sandboxed, pure-software p-code interpreter. The target's own instructions ran without any native execution on the host, so this reflects the program's real behavior rather than its static form.

Derived: emulation from 0x10022894 executed 67 instructions before stopping at 0x0; control left the mapped image on a return or external call.

14 Packing and Obfuscation

Inferred: the target is not packed (score 0; threshold 3).

No packing signals fired.

15 Methodology and Limitations

- Identified the format from file signatures and fingerprinted the file with MD5, SHA-1, and SHA-256.
- Parsed PE, ELF, and Mach-O headers, sections, imports, exports, symbols, and build metadata.
- Measured Shannon entropy for the file and every section.
- Extracted ASCII and UTF-16LE strings and classified network, path, registry, and command indicators.
- Classified imported APIs into capability categories and evaluated security mitigations from header fields.
- Disassembled x86/x86-64 with iced-x86 and every other supported architecture through an embedded SLEIGH engine, by recursive descent.

- Ranked functions by the capabilities and indicators they touch and listed the most relevant ones.

15.1 Limitations

- Analysis is static: the target is never executed, so run-time behavior is inferred, not observed.
- Imports show what a program can call, not what it does; dynamically resolved APIs are invisible to import analysis.
- Disassembly covers code reachable from the entry point, exports, and symbols; indirect-only code may be missed.
- A few architectures are parsed but not yet disassembled; for those, code analysis is omitted.